$\langle qu(b(i(t|t))ick)\rangle$

The Four-Era Canon

A Machine-Verified, Discrete-Ledger Framework for Foundational Physics

Volume I — The Four-Era Canon

Qubit_era $\longrightarrow$ Bosonic_era $\longrightarrow$ Fermion_era $\longrightarrow$ Baryomesonic_era

axioms	A1–A9
conventions	C1–C13
definitions	D0.1–D4.3, Def-EM.20–22, Def-H.1–43, Def-BM.1–20
theorem and data-cell tags	481
registry tags (total)	623
dependency edges (Req graph)	8014
free dimensionful scales	1 ($\hbar$, P1)
verification suite	80 checkers, exact arithmetic, no floating point

Edition of 2026-10-09

Machine-verified: `verification/run_all.py` — suite state 80/80 green.

The mathematics is transplanted unmodified from the canon files preserved in `working_markdowns/`; this volume is the sanitized presentation layer.

Preface — Guide to this Volume

This volume is the presentation layer of the $\langle qu(b(i(t|t))ick)\rangle$ corpus (read qubitick): a formal corpus deriving, from nine axioms and one free dimensionful scale $\mathfrak{h}$, the qubit, bosonic, fermionic and baryomesonic layers, with every theorem carrying its requirement set (Req), its symbolic proof (Pf) and its physical reading (Phys), and every identity machine-verified in exact arithmetic. The corpus's own description of its subject stands: qubits $\longrightarrow$ bosonic gauge/geometry $\longrightarrow$ fermionic bound chains $\longrightarrow$ baryomesonic colour-sealed composites; pure symbolic derivation; no floating point anywhere; no external physical constant enters any derivation.

The volume. Chapters 1–4 are the four eras (Part 1 · Qubit, Part 2 · Bosonic, Part 3 · Fermion, Part 4 · Baryomesonic); Appendix A is the Census annex (the 25 computed data cells, cited row-wise by the parts); the Index of Corpus Tags maps every canonical tag to its location in this volume or in the Addendum.

Companion documents. Addendum — Structural Constraints, Refutations, and No-Go Theorems (`docs/Addendum_Refutations.pdf`) is the volume's Appendix B: the adjudication record, the refuted hypotheses, the no-go theorems and the trap register, moved out of the main text; where such a block stood inline, a stub cross-reference remains. Calibration Policy (`docs/Calibration_Policy.pdf`) lists every calibration slot, its locus, its consumers, and its status under P5. The repository `README.md` describes the package; `working_markdowns/` preserves the raw working corpus byte-identically; `verification/` contains the checker suite.

Canonical tags and numbering. Every theorem environment carries both its presentation number (Theorem 1.1, Definition 3.12, …) and its canonical corpus tag (CT0.1, FT.54, BM.137, …) in its head. All cross-citations in Req lines, proofs and scope clauses use the canonical tags, which are unique across the corpus and are resolved by the Index of Corpus Tags; tags moved to the Addendum keep their identity there (Appendix B.$x.y$ entries are labelled with the tag).

Citations and records. This volume presents the corpus as a finished mathematical model: every citation resolves to a canonical tag, a Census cell, a [math] class, or a machine checker. The adjudications, refutations, no-go theorems and trap register — the record of how the model was won — live in the Addendum; the archival working record (including the historical ledger `HistLog.md`) is preserved in `working_markdowns/`. Paths of the form `checks/verify_*.py` in Req lines refer to `verification/checks/verify_*.py` in this repository.

Verification. The dependency graph over the registry tags is acyclic and is re-derived from the canon at every suite run (`verify_corpus_graph.py`); the era structure, document format and every algebraic identity are checked in exact arithmetic (integer, `Fraction`, algebraic number fields, prime-log rings, truncated power series) with no floating point in any assertion.

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1 $\langle qu(b(i(t|t))ick)\rangle$ — Part 1 · Qubit_era — Axioms · Conventions · Qubit-Era Definitions · Pair Canon · Ledger Dynamics

0.1 Corpus map. The canon is the era chain

Part 1 · Qubit_era $\longrightarrow$ Part 2 · Bosonic_era $\longrightarrow$ Part 3 · Fermion_era $\longrightarrow$ Part 4 · Baryomesonic_era,

each part citing only its priors, itself (earlier tags), and the Census annex (the computed data cells, cited row-wise by tag). Part 1 (this chapter): §A axioms A1–A9 · §B conventions C1–C13 · §C the definition layer D0.1–D4.3 (the corpus-wide vocabulary root: Tiers 0–1 qubit-era, Tiers 2–4 naming the structures the bosonic era derives) · §D master symbol table · §E calibration policy P1–P5 · §F foundations (CT0.1–CT0.14, CT1.0–CT1.11b, L0–L5). Part 2 (section letters preserve the classical canon's): §C CT1.12–CT2.11 (geometry), §D EW.1–EW.36a (electroweak), §E EM.1–EM.26 (electromagnetic), §F CT3.1–CT3.22a (colour), §G CT4.1–CT4.13 (saturation), §Cal-2 (registry note). Part 3: Def-H.1–Def-H.43, FT.1–FT.114, FT.116–FT.131, FT.133–FT.139, FT.141–FT.145 (structural tags), FT.153–FT.160. Part 4: Def-BM.1–Def-BM.20, BM.1–BM.152. Census: the 25 data-cell tags (Category B of the separation charter).

0.2 Citation closure. $\forall$ claims in Parts 1–4 cite only: A1–A9, C1–C13, D0.1–D4.3, P1–P5 (Parts 1–2), previously proved tags, and Census cells (row-wise). $\neg\exists$ imported external physical law, constant, or measured value. Physics vocabulary $:=$ naming of derived structures only. External comparison data enter only quarantined audit rows (Census FT.114, FT.120; P1, P5).

0.2a Math & hypothesis policy. Standard mathematics (linear algebra, representation theory, tensor calculus, classification theorems) is cited as [math]-class tags in Req and is not part of the physical axioms. Conditional theorems carry their hypotheses explicitly in Thm and Req. The complete list of extra-axiomatic inputs is empty: CT2.6 (ii), CT2.6 (iii) — the two continuum-limit conditions — are derived theorems of Part 4 (Tier BM.24, BM.151): (ii) is witnessed by the ledger's own unbounded ladder, (iii) by the density-matrix spectral bound on the D2.1 coefficient fields; no others exist. The declarations at use stand as scope-inheritance markers over the same loci. CT2.6 (i) is a definitional precondition: the valid-universe condition of D2.11 in application ("there is something"), $\neg$ an extra-axiomatic hypothesis. CT2.8 carries no hypotheses: its reflection and factorization clauses are derived (D2.7, D1.9, D1.3, D2.8). The edge phase is registration data: its alphabet is derived (CT0.9) and its per-edge value is determined at crystallization (Def-EM.20, EM.20). Integration note. The integer-edge hypothesis of CT1.29a is not independent: it is realized on materialized closed chains of the fully antialigned class (CT1.19b(3)). The ambient-richness condition in the CT3.22a Rmk is a witness scope for that Rmk's non-locality construction only; no theorem clause of Parts 1–2 rests on it. The materialization domain of CT1.19b is CT2.5's own pole condition (no new input); its domain-generalization step from EW.17's wrapping chains is flagged in its Pf. The per-edge independence hypothesis of CT4.5a is an ensemble scope for that theorem alone — its dynamics realization (a crystallization-rate datum) is NOT declared in Parts 1–2. The crystallization–tilt coupling of L0 (a seal-time state update: one Gibbs tilt step per A4 event) and

the finite-incidence scope of L4 are theorem-local scopes of the same species, declared at use. The interface definitions of Part 3 (Def-H.35–Def-H.43) carry their own non-derived clauses explicitly.

0.3 Free data. $\exists!$ free dimensionful scale: $\mathfrak{h}$ (D2.9, P1). Every other quantity: dimensionless-derived, or a power of $\mathfrak{h}$. Free internal configuration parameters (P4-class, ¬constants of the theory): the P4 core (§E) plus the era registries Cal-2 (Part 2), Cal-F (Part 3), Cal-BM (Part 4).

0.4 Theorem format (all parts). Per tag: Thm — statement, purely symbolic · Req — {axioms, conventions, definitions, prior theorems} used · Pf — derivation, purely symbolic, justifications in $[\cdot]$, end ■ · Phys — short English summary of the theorem and its physics. Corollaries: Cor. Definitions: Def. blocks; scope conditions and constitutive clauses are carried inside the Def./Thm. text. Forward pointers (scope conditions cited at use) are marked "forward pointer" at the citing site.

0.5 Symbol reconciliations. $P_M \equiv \mathcal{P}_M$ (C7); $F_Z \equiv F_{W^3}$ (C8); $\tau(v) \in \Xi_n$ (type set) vs $v \in V_n$ (vertex set), D0.8.

0.6 Machine verification. `checks/`: dependency-free exact-arithmetic verification suite (`python3 checks/run_all.py`). `verify_era_structure.py`: era-chain law, citation closure, census separation, tag ownership, declaration hygiene, Thm·Req·Pf·Phys completeness, index regeneration. `verify_B.py`–`verify_G.py` and the `verify_*` content suite: algebraic/geometry/arithmetical identities, exact, no floating point. See `checks/README.md`.

1.1 §A — Axioms

Axiom 1.1 (A1).

$$\mathcal{Q} = \{q_n\}_{n\geq 1};$$

$$\mathcal{H}_q \cong \mathbb{C}^2;$$

$$\forall (p,q) \in \mathcal{Q}\times\mathcal{Q},\ p\neq q: \quad \rho_{pq} \in \mathfrak{H}(\mathcal{H}_p\otimes\mathcal{H}_q),\ \ \rho_{pq}\succeq 0,\ \ \mathfrak{T}(\rho_{pq}) = 1.$$

Axiom 1.2 (A2).

$$\forall (p,q): \qquad C(p,q) = 1\,\mathfrak{q}.$$

Axiom 1.3 (A3).

$$\mathcal{H}_p\otimes\mathcal{H}_q = \vee^2 \oplus \wedge^2;$$

$$S(\sigma_\mu\otimes\sigma_\nu) = \sigma_\nu\otimes\sigma_\mu, \quad S^2 = \mathbb{1}; \qquad T(\sigma_\mu\otimes\sigma_\nu) = \eta^\mu\eta^\nu\,(\sigma_\mu\otimes\sigma_\nu), \quad T^2 = \mathbb{1};$$

$$\eta\in\{\pm 1\}^4; \qquad \exists!\,\mathfrak{o}\in\{0,1,2,3\}:\ \ \eta^{\mathfrak{o}} = -\eta^\mu\ \ \forall\mu\neq\mathfrak{o}.$$

Axiom 1.4 (A4).

$$(p,q)\in\mathcal{R} \iff \left|\Pi_{\mathcal{V}}\rho_{pq}\right| \leq \varepsilon_q \ \ \vee\ \ \left|\Pi_{\mathcal{V}}\rho_{pq}\right| \geq 1-\varepsilon_q; \qquad n_{\mathcal{R}}(p) = \left|\{q:(p,q)\in\mathcal{R}\}\right|$$

$$\alpha_{pq} := \begin{cases} 0 & \left|\Pi_{\mathcal{V}}\rho_{pq}\right| \leq \varepsilon_q \\ \pi & \left|\Pi_{\mathcal{V}}\rho_{pq}\right| \geq 1-\varepsilon_q \end{cases}; \qquad \alpha_{pq}\in\{0,\pi\} \quad \forall (p,q)\in\mathcal{R};$$

$$\exists\,\rho_{pq}:\ \left|\Pi_{\mathcal{V}}\rho_{pq}\right| \leq \varepsilon_q; \qquad \exists\,\rho_{p'q'}:\ \left|\Pi_{\mathcal{V}}\rho_{p'q'}\right| \geq 1-\varepsilon_q.$$

Axiom 1.5 (A5).

$$\rho_\circ = \tfrac{1}{4} I; \qquad \rho \mapsto U \rho U^\dagger, \quad U \sim \mu_\star\big(\mathrm{SU}(2)\big), \quad U \notin \mathfrak{L}.$$

Axiom 1.6 (A6).

$$\mu_\star := \mu_{\mathrm{Haar}}\big(\mathrm{SU}(2)\big); \qquad U \sim \mu_\star; \qquad U \notin \mathfrak{L}; \qquad \mathcal{T}(\rho) := U \rho U^\dagger; \qquad \mathcal{T}^{-1}(\rho) = U^\dagger \rho U$$

$$\mathfrak{T}\big(\mathcal{T}(\rho)\big) = \mathfrak{T}(\rho); \qquad \mathcal{T}^2 \neq \mathbb{1}; \qquad \tau_n \to \tau_{n+1} : \ \rho \mapsto \mathcal{T}(\rho); \qquad \neg\exists\, \tau^{-1}$$

$$\big|\mathfrak{L}(\tau_{n+1})\big| \ \geq \ \big|\mathfrak{L}(\tau_n)\big|; \qquad \big|\mathfrak{L}(\tau_{n+1})\big| \ = \ \big|\mathfrak{L}(\tau_n)\big| \ + \ n_{\text{real_out}}(\tau_n).$$

Axiom 1.7 (A7).

$$\forall n \geq 0 : \quad \Gamma = (V, E), \qquad E \subseteq V \times V$$

$$\tau(v) := \big(\partial^{\leftarrow}(v), \partial^{\rightarrow}(v)\big) \in \Xi_n; \qquad e \in E_{\mathcal{R}} \mapsto \phi_e; \qquad e \in E_{\mathcal{R}} \mapsto \alpha_e \in \{0, \pi\}$$

$$\mathfrak{L} = \bigcup_{n \geq 0} \mathfrak{L}(\tau_n);$$

$$\mathfrak{L}(\tau_n) \subseteq \mathfrak{L}(\tau_{n+1});$$

$$\big|\mathfrak{L}(\tau_0)\big| = 0;$$

$$\big|\mathfrak{L}(\tau_n)\big| = \sum_{k<n} n_{\text{real_out}}(\tau_k).$$

Axiom 1.8 (A8).

$$\mathrm{F} = \mathrm{F}_{(+,\mathcal{R})} \oplus \mathrm{F}_{(-,\mathcal{R})} \oplus \mathrm{F}_{(+,\mathcal{V})} \oplus \mathrm{F}_{(-,\mathcal{V})} \oplus \mathrm{F}_{\mathrm{st}};$$

$$\mathrm{F}_{(+,\mathcal{R})} = F_G \oplus F_H \oplus F_{D^+}; \qquad \mathrm{F}_{(-,\mathcal{R})} = F_{W^+} \oplus F_{W^-} \oplus F_Z; \qquad \mathrm{F}_{(+,\mathcal{V})} = F_{J^+};$$

$$\mathrm{F}_{(-,\mathcal{V})} = F_{J^-}; \qquad \mathrm{F}_{\mathrm{st}} = F_{\mathrm{EM}} \oplus F_{\mathrm{strong}} \oplus F_{D^-};$$

$$F_{D^+} = \Lambda\, \mathbb{I}; \qquad F_{D^-} := \bigoplus_{n \geq 4} F_n; \qquad \mathrm{F}_M = P_M\, \mathrm{F}\, P_M;$$

$$w(C) = \mathcal{J}(C) \prod_{e \in E(C)} G(e) \ \cdot \ (-1)^{\aleph(\sigma, C)}; \qquad Z(\mathcal{R}) = \sum_{C \in \mathcal{R}} \sum_{\sigma} w(C).$$

Axiom 1.9 (A9).

$$\mathcal{U}_\odot \in \mathfrak{S}\big(Z[\mathrm{F}_\oplus, M_\oplus]\big); \qquad \mathcal{U}_\odot \notin \mathfrak{S} \implies \neg\big(\mathrm{A1} \wedge \cdots \wedge \mathrm{A9}\big).$$

■

1.2 §B — Conventions

Convention 1.1 (C1). $\mu, \nu, \rho, \ldots \in \{0,1,2,3\}$: Pauli basis $\sigma_0 = I_2, \sigma_1, \sigma_2, \sigma_3$ / emergent spacetime coordinates where a metric exists; $i, j, k, \ldots \in \{1,2,3\}$: spatial/Pauli-spatial; $a, b, c, \ldots \in \{1, \ldots, 8\}$: colour adjoint. Repeated indices: summed.

Convention 1.2 (C2). $T_A^\dagger = T_A$; $[T_A, T_B] = i\, f_{AB}{}^C\, T_C$; $f_{AB}{}^C \in \mathbb{R}$.

Convention 1.3 (C3). $\mathcal{E}_{st} := \{\, X \in \mathfrak{H}(\mathbb{C}^4) : \; SX = sX \;\wedge\; TX = tX \,\}$, $s, t \in \{+,-\}$. Symbol $\wedge$: antisymmetrization only ($\wedge^2$, ε-contractions); $\neg$ a sector-label conjunct.

Convention 1.4 (C4). $\varkappa$: sector-to-metric embedding scale (D2.1); κ: gravitational coupling (D2.9, CT2.9); $\varkappa \neq \kappa$; $\neg$ interchangeable.

Convention 1.5 (C5). $\Sigma := \sum_i \sigma_i \otimes \sigma_i$ (operator); $\Sigma_{\text{tr}} := \delta^{ij} q_{ij} \in \mathbb{R}$ (scalar, D2.1); $\neg$ interchangeable.

Convention 1.6 (C6). E_{tr}: transit entropy; E_{bd}: bound entropy (D1.5); $\neg$ squared-field notation for either.

Convention 1.7 (C7). $P_M \equiv \mathcal{P}_M$ (A8). Rung-resolved: $\mathcal{P}_M^{(n)}$ (D2.3); $\mathrm{F}_M^{(n)} = \mathcal{P}_M^{(n)}\, \mathrm{F}\, \mathcal{P}_M^{(n)}$.

Convention 1.8 (C8). $F_Z \equiv F_{W^3}$: neutral-weak aspect of A8; $F_W := F_{W^+} \oplus F_{W^-} \oplus F_Z$; pre-mixing seed generator: W_3. Bare Z in a spectrum: physical neutral eigenstate (D3.10, EW.24); $Z(\cdot)$ with argument: partition function (A8) / masked resolvent (D2.4).

Convention 1.9 (C9). Entropy in nats: $S = \ln \Omega$; _bits_ only where explicit. One real edge $= 1$ bit $= \ln 2$ nats (CT0.9, CT1.29).

Convention 1.10 (C10). Register accounting (reality principle): registrations convert quantum data to classical alternatives at the ledger. Registering a classical alternative set X for single-shot linear readout ($\mathfrak{L}$ must distinguish which value was carried, D0.7) assigns the alternatives mutually orthogonal register directions: register cost $= |X|$. Shannon content $= \log_2 |X|$ bits is a distinct quantity: source-availability register count $\neq$ channel bit throughput. One channel quantum supplies $C_{\text{cap}} = \dim_{\mathbb{R}} \mathfrak{H}(\mathcal{H}_p \otimes \mathcal{H}_q) = 16$ register directions (A2, D0.9). Ledger-level registration data are real: edge transport acts on the reality line by $\{\pm 1\}$ (D1.1, CT0.9).

Convention 1.11 (C11). A5/A6 reading: the conjugation clause is node-level — $\rho \mapsto U\rho U^\dagger$ [A6: $\mathcal{T}$] with $\rho \in \mathfrak{H}(\mathcal{H}_q) \cong \mathfrak{H}(\mathbb{C}^2)$ and $U \sim \mu_\star(\mathrm{SU}(2))$ (2×2: well-typed); $\rho_\circ = \frac{1}{4} I$ is the vacuum pair state (4×4). Node tick lifecycle: wait $\to$ data in $\to$ Haar draw/tick [A6: $\tau_n \to \tau_{n+1}$] (virtual states as basis) $\to$ data out/hop $\to$ wait. Draws are not retained: $U \notin \mathfrak{L}$ [A5, A6].

Convention 1.12 (C12). In A8's $Z(\mathcal{R}) = \sum_{C \in \mathcal{R}} \sum_\sigma w(C)$ the summation domain reads $\mathfrak{C}(\mathcal{R})$ (D0.10): the configuration set on the real-edge set. $\mathcal{R}$ itself (A4, D0.1) is a set of pairs and is never a summation domain.

Convention 1.13 (C13). Signature convention. A3 fixes T only through the products $\eta^\mu \eta^\nu$, and fixes η only up to its odd slot $\mathfrak{o}$ and its overall sign. The corpus takes $\mathfrak{o} := 0$ (the identity slot $\sigma_0 = I_2$, C1) and $\eta^0 := -1$, i.e. $\eta := (-,+,+,+)$ (mostly-plus; D2.1: $\bar{g}_{00} := -1$). The overall sign does not change T [CT1.0(2)]; the three choices $\mathfrak{o} \in \{1,2,3\}$ are conjugate under SU(2) relabelling of the spatial Pauli axes [CT1.0(4)]; $\mathfrak{o} = 0$ versus $\mathfrak{o} \geq 1$ is the choice of the SU(2)-covariant representative [CT1.0(5)], on which CT1.2's generator table rests [CT1.0 Rmk]. Uses of the representative carry the tag C13. $\neg$ an axiom.

1.3 §C — The definition layer

Definitions := vocabulary stipulations. ∀ content-bearing claims: theorems of Parts 1–4. Tiers 0–1 are qubit-era vocabulary; Tiers 2–4 name the structures the bosonic era derives — the definition layer is the corpus-wide vocabulary root (the classical canon's A §Part 3).

1.3.1 Tier 0 — bookkeeping

Definition 1.1 (D0.1). $\mathcal{R} := \{(p,q) : \text{A4 holds}\}$; $\mathcal{V} := (\mathcal{Q}\times\mathcal{Q})\setminus\mathcal{R}$; $n_{\mathcal{R}} := \sum_p n_{\mathcal{R}}(p)$; global vacuum $\iff n_{\mathcal{R}} = 0$.

Definition 1.2 (D0.2). $\big|\Pi X\big| := \mathfrak{T}(\Pi X\Pi) = \mathfrak{T}(\Pi X)$; $\rho \succeq 0 \wedge \mathfrak{T}\rho = 1 \implies \big|\Pi\rho\big| \in [0,1]$.

Definition 1.3 (D0.3). $S_{\text{raw}}(\rho) := -\mathfrak{T}(\rho\log\rho) = \mathfrak{T}(\rho K)$; $K := -\log\rho$ (faithful ρ). $\gamma := 1-4\varepsilon_q$ [A4, D0.4, CT0.8]; $\gamma \in (0,1]$ for $\varepsilon_q \in [0,\frac{1}{4})$. Gibbs tilt (functional calculus): $\tilde{\mathcal{T}}_\gamma(X) := Xe^{-\gamma X}/\mathfrak{T}(Xe^{-\gamma X})$ where $\mathfrak{T}(Xe^{-\gamma X}) \neq 0$; on states: $\tilde{\mathcal{T}}_\gamma(\rho) = \rho\, e^{-\gamma\rho}/\mathfrak{T}(\rho\, e^{-\gamma\rho})$; $\gamma = 0$: the identity on trace-1 states; for faithful ρ: $\tilde{\mathcal{T}}_\gamma(K) = K\rho^\gamma/\mathfrak{T}(K\rho^\gamma)$ $(\gamma = 1 := \rho(-\log\rho)/S_{\text{raw}}(\rho))$. Variance functional (reference state explicit): $\mathbb{V}_\rho(X) := \mathfrak{T}(\rho X^2) - \big(\mathfrak{T}(\rho X)\big)^2$. Lyapunov functional of the tilt: $V(\tilde{K}) := \mathfrak{T}(\tilde{K}^2)$, $\tilde{K} := K - \mathfrak{T}(K\rho)\, I$ — monotonicity: L1 (§F, forward pointer). $\tilde{\mathcal{T}}_\gamma \neq \mathcal{T}$ [A6]; ¬ interchangeable.

Definition 1.4 (D0.4). $\varepsilon_q \in [0,1]$: threshold of A4; admissible range: CT0.8; sharp choice: $\varepsilon_q = 0$.

Definition 1.5 (D0.5). Φ: phase alphabet of a real edge (CT0.9: $\Phi = \{0,\pi\}$); $\sigma_e := e^{i\phi_e} = (-1)^{\phi_e/\pi} \in \{+1,-1\}$.

Definition 1.6 (D0.6). hop := one traversal of one $e \in E_{\mathcal{R}}$; $\delta h(\gamma) := \#\{e \in \gamma\}$; $\mathfrak{h}$:= length assigned to one hop.

Definition 1.7 (D0.7). $\mathfrak{L}$:= ordered record of retained events [A6, A7: stratification $\mathfrak{L} = \bigcup_{n\geq 0}\mathfrak{L}(\tau_n)$, $|\mathfrak{L}(\tau_0)| = 0$, monotone under ticks]; $\delta\tau$:= ledger ticks (retained updates along real edges); $U \notin \mathfrak{L}$ [A5, A6].

Definition 1.8 (D0.8). $n(v) := \partial^{\leftarrow}(v) + \partial^{\rightarrow}(v)$; $V_n := \{v \in V : n(v) = n\}$ (vertex set); $\Xi_n := \tau(V_n) = \{(a,b) \in \mathbb{N}^2 : a+b = n\}$ (type set); "at rung n" := on V_n.

Definition 1.9 (D0.9). $D_{\text{tick}}(n)$:= #{independent real registers a vertex of V_n pushes across one pair channel per tick} — counted by the register accounting of C10 (one orthogonal direction per admissible classical alternative: source availability, single-shot linear readout; ¬ Shannon $\log_2$ counting). $C_{\text{cap}} := \dim_{\mathbb{R}}\mathfrak{H}(\mathcal{H}_p\otimes\mathcal{H}_q)$ (register directions supplied by $\mathfrak{1q}$, A2); $O(n) := \max\big(0,\ D_{\text{tick}}(n) - C_{\text{cap}}\big)$.

Definition 1.10 (D0.10). $\mathfrak{C}(\mathcal{R})$:= configuration set on $E_{\mathcal{R}}$: finite oriented edge sets $E(C) \subseteq E_{\mathcal{R}}$ with sign assignments σ, seal data (D1.2) and vertex types (A7). $w(C)$ (A8) is defined on $\mathfrak{C}(\mathcal{R})$; valid universes (D2.11) form a subset; $Z(\mathcal{R}) := \sum_{C\in\mathfrak{C}(\mathcal{R})}\sum_\sigma w(C)$ (C12).

1.3.2 Tier 1 — graph, chains, ledger dynamics

Definition 1.11 (D1.1). $e = (p,q) \in \mathcal{R}_{\rightarrow}(p) \longmapsto (\mathcal{G}_n, \phi_e)$; $\mathcal{G}_n$:= rung class of q; ϕ_e : D0.5; ¬registration $\implies e \in \mathcal{V}$. Reality principle of registration (C10): ϕ_e := phase of the registration transport's action on the reality line $\wedge^2$ (1-dimensional, since $\dim\mathcal{H}_q = 2$ [A1]); the transport preserves the (S,T)-grading [A3, A4] and the real Pauli-frame structure — registrations are classical data. On a minimal real edge ($\widehat{ds}^{\,2} = 1$ [D1.6]) the geometric factor of the transfer weight $G(e)$

[D3.5] is the edge's alignment holonomy $e^{i\alpha_e}$ — the binary registration read of the A4 branch index at the seal ($\alpha_e = \alpha_{pq}$ for $e = (p, q)$ [A4]; per-edge determination read in E: Def-EM.20, EM.20, forward pointer).

Definition 1.12 (D1.2). seal := transition of one pair across the A4 criterion (entry/exit of either branch): $J^+ : n_{\mathcal{R}} : 0 \to 1$ (crystallization); $J^- : n_{\mathcal{R}} : 1 \to 0$ (dissolution).

Definition 1.13 (D1.3). Janus-balanced $\iff$ $\#J^+ = \#J^-$.

Definition 1.14 (D1.4). chain $c = (e_1, \ldots, e_k)$: $e_i \in E_{\mathcal{R}}$, $t(e_i) = s(e_{i+1})$; $s(c) := s(e_1)$; $t(c) := t(e_k)$; $\partial c := t(c) - s(c)$ (formal difference of vertices); closed $\iff$ $\partial c = 0 \wedge \#J^+ = \#J^-$.

Definition 1.15 (D1.5). bound := closed self-contained chain (content: fermionic); transit := propagating open chain (content: bosonic); $\Delta S = \mathsf{E}_{\mathsf{bd}} + \mathsf{E}_{\mathsf{tr}}$; $\mathsf{E}_{\mathsf{bd}} := \Delta S_{\text{bound}}$; $\mathsf{E}_{\mathsf{tr}} := \Delta S_{\text{transit}}$.

Definition 1.16 (D1.6). $ds^2(e) := g_{\mu\nu}(e)\,\Delta x_e^\mu\,\Delta x_e^\nu$ [$g_{\mu\nu}$: D2.1]; $\widehat{ds}^{\,2}(e) := ds^2(e)/\mathfrak{h}^2$.

Definition 1.17 (D1.7). $\Theta(\gamma) := \sum_{e\in\gamma} \widehat{ds}^{\,2}(e)$; $\Phi(\gamma) := \sum_{e\in\gamma} \alpha_e\, \widehat{ds}^{\,2}(e)$ [$\alpha_e \in \Phi$: per-edge phase, D3.5; uniform-phase case: $\Phi = \alpha\,\Theta$]; $\mathrm{Hol}_{\text{spiral}}(\gamma) := e^{i\Phi(\gamma)}$; $\gamma \in \mathsf{Trivial} \iff \forall e \in \gamma : \widehat{ds}^{\,2}(e) = 0$; wrapping $\iff$ spatial hop projection has winding $\neq 0$ around a closed ledger cycle.

Definition 1.18 (D1.8). $\tau_n := \sum \delta\tau$ accumulated along a causal chain of real edges; emergent coordinate $x^0 := \tau_n$.

Definition 1.19 (D1.9). tick locality: $\mathrm{dom}\big(\mathrm{tick}(v)\big) \subseteq \{v\} \cup \{e \in E : e \ni v\}$, pre- and post-tick data only; $\neg\exists$ read at edge-distance > 1.

Definition 1.20 (D1.10). node force: $\mathsf{F}\restriction_v$ resolves local information content into channels {bound, transit, holonomy, charge, flux}; {bound, transit} := entropy channels [D1.5]; {holonomy, charge, flux} := registration data [D1.1] $\cup$ transition amplitude [D3.3].

Definition 1.21 (D1.11). accessible information content at $v \leftrightarrow n(v)$; content beyond the capacity bound (CT0.11): ¬accessible [D4.1].

Definition 1.22 (D1.12). $\mathcal{A}(n) :=$ {aspects of F acting nontrivially at rung n}; value derived (CT0.13), ¬ stipulated.

1.3.3 Tier 2 — masks, geometry, units

Definition 1.23 (D2.1). sector coefficients [basis: CT1.2]: $X_{(+)} = a_{00}E_0 + q_{ij}E^{ij}$, $q_{ij} = q_{ji}$; $X_{(-)} = w_i W_i$;

$$\Sigma_{\mathsf{tr}} := \delta^{ij} q_{ij};$$

$$\hat{h}_{ij} := q_{ij} - \tfrac{1}{3}\delta_{ij}\Sigma_{\mathsf{tr}};$$

$$h := \tfrac{1}{3}\Sigma_{\mathsf{tr}};$$

$$N_i := \beta w_i;$$

$$\Omega := e^{\alpha_\Omega a_{00}};$$

$$\alpha_\Omega, \beta, \varkappa \in \mathbb{R};$$

$$\bar{g}_{00} := -1; \quad \bar{g}_{0i} := \varkappa N_i; \quad \bar{g}_{ij} := \delta_{ij} + \varkappa\big(\hat{h}_{ij} + \tfrac{1}{3}\delta_{ij}\Sigma_{\text{tr}}\big); \quad g_{\mu\nu} := \Omega^2\,\bar{g}_{\mu\nu}.$$

Definition 1.24 (D2.2). $\Upsilon := \ln\Omega$; $\Upsilon_\mu := \bar{\nabla}_\mu \Upsilon$.

Definition 1.25 (D2.3). $\mathcal{K}$: aspect space; $\mathcal{K}^{(A)}$: block of aspect A; $\mathcal{K}_n := \bigoplus_{A\in\mathcal{A}(n)} \mathcal{K}^{(A)}$; $\mathcal{P}_M^{(n)}$: $\mathcal{K} \to \mathcal{K}_n$: orthogonal projection.

Definition 1.26 (D2.4). $Z(\mathcal{R}, n) := \big\langle \text{out}\big|(1 - \mathrm{F}_M^{(n)})^{-1}\big|\text{in}\big\rangle$; poles of $Z(\mathcal{R}, n)$:= rung-n spectrum; residues := amplitudes.

Definition 1.27 (D2.5). massless $\iff$ resolvent pole at $k^2 = 0$; massive $\iff$ pole at $k^2 = m^2 > 0$; m^2 := pole displacement.

Definition 1.28 (D2.6). ω : orientation-reversal involution: $s(c) \leftrightarrow t(c)$; chirality $\mapsto$ $-$chirality; charge $\mapsto$ $-$charge.

Definition 1.29 (D2.7). $\Theta_{\mathfrak{L}}$: ledger reflection: reverses ledger order; $J^+ \leftrightarrow J^-$; $G(e) \mapsto G(e)^*$.

Definition 1.30 (D2.8). convergence sheet: $1 \mapsto 1 + i\varepsilon,\ \varepsilon \to 0^+$, for $\forall$ resolvents and configuration sums.

Definition 1.31 (D2.9). units := powers of $\mathfrak{q}$; $c = \hbar = k_B = 1$; $[\mathfrak{h}] = \mathfrak{q}^{-1/2}$; $\mathfrak{h}$:= single free scale; $\kappa,\ G_N$: names for combinations of $\mathfrak{h}$ (CT2.9), $\neg$ independent inputs.

Definition 1.32 (D2.10). $\mathfrak{M} = \begin{pmatrix} B & Q \\ R & F \end{pmatrix}$: even blocks B, F; odd blocks Q, R; Hermitian $\iff$ $B^\dagger = B \wedge F^\dagger = F \wedge R = Q^\dagger$; $\mathrm{Str}\,\mathfrak{M} := \mathfrak{T}B - \mathfrak{T}F$; $[X, Y\} := XY - (-1)^{|X||Y|}YX$ (X, Y homogeneous).

Definition 1.33 (D2.11). valid universe $\iff$ real-edge network closed $\wedge$ $\#J^+ = \#J^-$ [D1.3] $\wedge$ $\in$ Im(active mask) $\wedge$ $\neg\exists$ dangling ledger boundary beyond the sealed Janus boundary.

Definition 1.34 (D2.12). Aspect realization (slot→aspect bridge): $\iota_n : \mathfrak{u}(n) \to \mathcal{K}$ — linear realization of the rung-n slot algebra in aspect space, with $\iota_n\big(Z(\mathfrak{u}(2))\big)$ spanning $\mathcal{K}^{(\text{EM})}$ ($n = 2$) and $\iota_n\big(\mathfrak{su}(3)\big)$ spanning $\mathcal{K}^{(\text{strong})}$ ($n = 3$), per the derivation of CT0.13. Slot algebras and pair algebras remain distinct objects (F header); D3.2 "already realized at rung m" := the abstract isomorphism class lies in $\iota_m(\mathfrak{u}(m))$ or in the pair blocks of CT0.12. Realization domain: exactly $n \le 3$ — for $n \ge 4$ no aspect realization exists [CT4.4a]; the silence of this definition at $n \ge 4$ is a non-existence consequence, not an underdetermination.

Definition 1.35 (D2.13). Subnetwork and microstate space (scope clauses). $\Sigma \subseteq U$ [D2.11] := a finite connected vertex set together with its interior edges; $\partial\Sigma$:= edges with exactly one endpoint in Σ. $\mathcal{M}(\Sigma)$:= valid microscopic configurations on Σ: restrictions $C|_\Sigma$ of valid ambient configurations C [D2.11], carrying edge signs σ_e [D0.5, CT0.9], geometric data $\widehat{ds}^{\,2}(e)$ [D1.6], seal data [D1.2] and vertex types [A7, D0.8] — subject to the rung masks [CT0.14, CT2.2] and Janus balance [D1.3], $\neg$ to colour sealing (sealing is ambient-global [CT3.18]; the completeness filter acts downstream: CT3.22a Rmk). $|\Sigma|$:= #{real edges touching Σ} (interior + boundary).

Definition 1.36 (D2.14). Colour boundary and the contraction-completeness projector. $\mathrm{Colour}(U) := \{u \in \mathcal{Q} :\ u$ carries a colour attachment [D3.8]$\}$; for valid C, $\pi_C(v) \subseteq \mathrm{Colour}(U) :=$

the other members of the elementary invariant contraction group of v under C (one for a meson pair via $\delta^i{}_j$; two for a (anti)baryon triple via $\varepsilon_{ijk}/\varepsilon^{ijk}$ [CT3.17]); π_C is total on Colour(U) for valid C [CT3.18 + D2.11 applied to the colour ledger]. For Σ [D2.13]: Colour(Σ) := Colour$(U) \cap \Sigma$;

$$\partial_{\text{col}}(\Sigma; C) := \{v \in \text{Colour}(\Sigma) : \ \pi_C(v) \not\subseteq \Sigma\};$$

$$\mathcal{C}(\Sigma; C) := \begin{cases} 1 & \partial_{\text{col}}(\Sigma; C) = \varnothing \\ 0 & \text{otherwise} \end{cases};$$

$$\Pi_{\mathcal{C}}(\Sigma; C) := \mathcal{C}(\Sigma; C) \cdot \mathbb{I}_{\Sigma}^{(\text{strong})}$$

(Hermitian scalar idempotent on the colour slot space of Σ [D3.1]; matches the mask pattern $\mathcal{P}_M^{(n)}$ [CT0.14]).

1.3.4 Tier 3 — rung-2 and rung-3 structures

Definition 1.37 (D3.1). $\mathcal{H}_{\text{slot}}^{(n)} := \text{span}_{\mathbb{C}}\{|1\rangle, \ldots, |n\rangle\} \cong \mathbb{C}^n$; $n = 3$: slot index $i \in \{1, 2, 3\}$:= colour.

Definition 1.38 (D3.2). subset principle: strictly new at rung n := part of the rung-n structure ¬realized at any $m < n$.

Definition 1.39 (D3.3). at v of type $(1, 1)$, edge signs $\sigma_{\leftarrow}(v), \sigma_{\rightarrow}(v)$:

$$f_v := \frac{\sigma_{\rightarrow}(v) - \sigma_{\leftarrow}(v)}{2} \in \{-1, 0, +1\}; \qquad \aleph_v := |f_v|; \qquad \aleph(\sigma, C) := \sum_{v \in C} \aleph_v.$$

Definition 1.40 (D3.4). $q_{\text{EM}}(c) := \dfrac{\sigma_{s(c)} - \sigma_{t(c)}}{2}$ (oriented chain c). Closed chains: the basepoint core read q_{core} is defined, ternary, and charge-conjugation odd [EM.9a].

Definition 1.41 (D3.5). $G^{(n)}(e) := \sigma_e \exp\big(i\, \alpha_e\, \widehat{ds}^{\,2}(e)\big)$ [D0.5, D1.6]; $\alpha_e \in \Phi$: per-edge dynamic phase $= \alpha_{pq}$ for $e = (p, q)$ — the A4 branch index (read in Part 2 §E: Def-EM.20, EM.20, forward pointer); dimensionless; the same alphabet at every rung. $\alpha_{\text{EM}}^{(n)}$: EM-block spectral read at rung n (EM.17, forward pointer).

Definition 1.42 (D3.6). $\mathcal{J}(C) := \prod_{J^+ \in C}(+1) \cdot \prod_{J^- \in C}(+1)^* \implies \mathcal{J}(\Theta_{\mathfrak{L}} C) = \mathcal{J}(C)^*$.

Definition 1.43 (D3.7). photon := trivial chain [D1.7] $\subseteq V_2$; gluon := trivial chain $\cap\, V_3 \neq \varnothing$.

Definition 1.44 (D3.8). colour triplet q^i: one fundamental slot index; colour antitriplet r_i: one dual slot index; quark := bound (fermionic) loop with a triplet attachment.

Definition 1.45 (D3.9). baryon $B := \varepsilon_{ijk}\, q^i q^j q^k$; meson $M := \delta^i{}_j\, q^j r_i$. [Disambiguation: M here $\neq$ total mask $M_{\oplus}$ of A9.]

Definition 1.46 (D3.10). $W^{\pm} := \frac{1}{\sqrt{2}}\big(W_1 \mp i W_2\big)$; $Z := \cos\theta_W\, W_3 - \sin\theta_W\, Y$; $A := \sin\theta_W\, W_3 + \cos\theta_W\, Y$ [θ_W : EW.24].

1.3.5 Tier 4 — saturation

Definition 1.47 (D4.1). $F_{D^-} = \bigoplus_{n\geq 4} F_n$ [A8]: single possibility space receiving $\forall$ configurations with rung $n \geq 4$.

Definition 1.48 (D4.2). D^- absorbs information at rate $\propto O(n)$ [D0.9]; absorbed $\implies$ locked := retained in $\mathfrak{L}$, ¬accessible [D1.11].

Definition 1.49 (D4.3). $F_{D^+} = \Lambda\,\mathbb{I}$ [A8]: uniform scalar aspect; ¬configuration-dependent.

1.4 §D — Master symbol table

Symbol	Meaning	Declared
$\mathcal{Q}$, q_n	qubit set; n-th qubit	A1
$\mathcal{H}_q$	carrier space of q, $\cong \mathbb{C}^2$	A1
ρ_{pq}	state of ordered pair (p, q)	A1
$\mathfrak{H}(X)$	real space of Hermitian operators on X	A1
$\mathfrak{T}(A)$	trace of A	A1
$\succeq 0$	positive semidefinite	A1
$C(p, q)$, $\mathfrak{q}$	channel capacity of a pair; channel quantum	A2
$\vee^2$, $\wedge^2$	symmetric, antisymmetric square	A3
S, T	swap involution; sign-grading involution ($T = \mathsf{D}_\zeta \otimes \mathsf{D}_\zeta$; metric-sign reading: C13)	A3, CT1.0
σ_μ	Pauli basis, $\mu \in \{0, 1, 2, 3\}$, $\sigma_0 = I_2$	A3, C1
$\mathbb{1}$, I	identity operator; identity matrix	A3, A5
η	sign vector of the T-grading, $\eta \in \{\pm 1\}^4$, one odd entry; representative $(-, +, +, +)$	A3, C13
$\mathfrak{o}$, $\eta^{(\mathfrak{o})}$	odd-sign slot of η; sign vector with a single -1 at slot $\mathfrak{o}$	A3, CT1.0
ζ_i, D_ζ, sf	$\eta^0\eta^i$; Bloch sign map $\sigma_\mu \mapsto \zeta_\mu\sigma_\mu$; spin flip $X \mapsto \sigma_2 X^{\mathrm{T}}\sigma_2$	CT1.0
$\mathcal{R}$, $\mathcal{V}$	real (crystallized) set; virtual set	A4, D0.1
$\Pi_{\mathcal{V}}$, $\Pi_{\mathcal{R}}$	projections onto $\wedge^2$, $\vee^2$	A4, CT0.7
ε_q	crystallization threshold	A4, D0.4
$n_{\mathcal{R}}(p)$, $n_{\mathcal{R}}$	real-edge count at p; global count	A4, D0.1
$\rho_\circ$	vacuum pair state $I/4$	A5
U, $\mu_\star$, $\dagger$	Haar draw on SU(2); Haar measure; conjugate transpose	A5, A6
$\mathfrak{L}$, $\mathfrak{L}(\tau_n)$, $n_{\text{real_out}}$	ledger; ledger stratum at tick τ_n; real-out count per tick	A5, A6, A7, D0.7
$\mathcal{T}$, $\mathcal{T}^{-1}$	tick map (Haar conjugation); its inverse	A6
$\tilde{\mathcal{T}}_\gamma$, γ, K, S_{raw}, $\mathbb{V}_\rho$	ε_q-dependent Gibbs tilt; tilt strength $1 - 4\varepsilon_q$; surprisal; raw entropy; variance	D0.3
$\tilde{K}$, $V(\tilde{K})$	centered surprisal; Lyapunov functional	D0.3, L1
$\Gamma = (V, E)$	directed graph	A7
$\tau(v)$, $\partial^{\leftarrow}$, $\partial^{\rightarrow}$	vertex type; in-degree; out-degree	A7
$n(v)$, V_n, Ξ_n	rung of v; rung-n vertex set; type set	D0.8
$E_{\mathcal{R}}$, ϕ_e, σ_e, Φ	real edges; edge phase; edge sign; phase alphabet	A7, D0.5
δh, $\mathfrak{h}$	hop count; hop length	D0.6
$\delta\tau$, τ_n	ledger tick count; ledger time	D0.7, D1.8
$D_{\text{tick}}(n)$, C_{cap}, $O(n)$	tick load; channel capacity; oversubscription	D0.9
F, F_A	force operator; aspect A	A8
F_{W^+}, F_{W^-}, F_Z, F_W	charged weak; neutral weak ($F_Z \equiv F_{W^3}$); weak block	A8, C8
F_{J^+}, F_{J^-}	crystallization; dissolution aspects	A8, D1.2
F_{EM}, F_{strong}	electromagnetic; colour aspects	A8
Λ, $\mathbb{I}$	stretch scalar; identity on force space	A8

Symbol	Meaning	Declared
$\mathcal{U}_{\odot}$, $\mathfrak{S}$, $\mathsf{F}_{\oplus}$, $M_{\oplus}$	observed universe; solution set; total force; total mask	A9
$\mathfrak{C}(\mathcal{R})$	configuration set on $E_{\mathcal{R}}$; summation domain of Z	D0.10, C12
$\tilde{\mathcal{T}}_{\gamma}(X)$	operator Gibbs tilt (functional calculus)	D0.3
$\mathcal{E}_{st}$	joint (S,T) eigenspace	C3
${f_{AB}}^{C}$	structure constants (Hermitian convention)	C2
E_{bd}, E_{tr}	bound, transit entropy	C6, D1.5
E_0, E^{ij}, W_i, J_i, K_i	generator basis of $\mathfrak{h}(\mathbb{C}^4)$	CT1.2
$\mathcal{A}(n)$	active aspect set at rung n	D1.12, CT0.13
$\mathcal{G}_n$	target rung class of an edge	D1.1
J^+, J^-	crystallization seal; dissolution seal	D1.2
c, γ, $s(c)$, $t(c)$, ∂c	chain; source; target; boundary	D1.4
$ds^2(e)$, $\widehat{ds}^{\,2}(e)$	discrete line element; hop-normalized	D1.6
$\Theta(\gamma)$, $\Phi(\gamma)$, $\mathrm{Hol}_{\mathrm{spiral}}$, Trivial	spiral sum; phase; holonomy; trivial class	D1.7
x^0	emergent time coordinate	D1.8
$\neq g$, $\wedge$, $\vee$, $\Longrightarrow$, $\Longleftrightarrow$	negation; conjunction; disjunction; implication; equivalence	—
$\oplus$, $\bigoplus$, $\otimes$	direct sum; indexed direct sum; tensor product	A1, A3, A8
F_G, F_H, F_{D^+}, F_{D^-}	spin-2; scalar (Higgs); uniform dark; saturated dark	A8, D4.1, D4.3
P_M, $\mathcal{P}_M^{(n)}$, $\mathsf{F}_M^{(n)}$	mask projection; rung mask; masked force	A8, C7, D2.3
$w(C)$, $\mathcal{J}(C)$, $G(e)$	configuration weight; Janus factor; edge transfer weight	A8, D3.5, D3.6
f_v, $\aleph_v$, $\aleph(\sigma, C)$	vertex transition amplitude; indicator; count	D3.3
$Z(\mathcal{R})$, $Z(\mathcal{R}, n)$	partition function; masked resolvent	A8, D2.4
ι_n	aspect realization $\mathfrak{u}(n) \to \mathcal{K}$ (slot→aspect bridge)	D2.12
$\mathcal{P}^{(\mathrm{EM})}$, Y_{asp}, Y_{slot}, Y_{car}	EM block projection; the three realizations of Y	EM.2, EW.8, D2.12
$\varkappa$, κ, G_N	embedding scale; gravitational coupling; Newton constant	C4, D2.1, D2.9, CT2.9
Σ, Σ_{tr}	operator $\sum_i \sigma_i \otimes \sigma_i$; scalar $\delta^{ij} q_{ij}$	C5, D2.1
a_{00}, q_{ij}, w_i	sector coefficients	D2.1
$\hat{h}_{ij}$, h, N_i, Ω, Υ	traceless shear; trace mode; shift; conformal factor; potential	D2.1, D2.2
$\bar{g}_{\mu\nu}$, $g_{\mu\nu}$	auxiliary metric; emergent metric	D2.1
α_{Ω}, β	metric embedding constants	D2.1
$\mathcal{K}$, $\mathcal{K}^{(A)}$, $\mathcal{K}_n$	aspect space; aspect block; rung-n block	D2.3
m^2	mass squared (pole displacement)	D2.5
ω, $\Theta_{\mathfrak{L}}$	orientation reversal; ledger reflection	D2.6, D2.7
$\mathfrak{M}$, Str, $[\cdot,\cdot\}$	supermatrix; supertrace; graded bracket	D2.10
$\mathcal{H}_{\mathrm{slot}}^{(n)}$	slot space $\cong \mathbb{C}^n$	D3.1
$q_{\mathrm{EM}}(c)$	chain end-charge	D3.4
α_e, α_{pq}	per-edge dynamic phase $\in \Phi$; branch phase of a crystallizing pair	A4, A7, D3.5
$\alpha_{\mathrm{EM}}^{(n)}$	EM-block spectral read at rung n	EM.17
$\mathcal{P}(\Sigma)$, $M(\mathcal{P})$	alignment pattern; mirror image of the pattern	Def-EM.21, Def-EM.22
q^i, r_i	colour triplet; antitriplet	D3.8
B, M	baryon; meson	D3.9
$W^{\pm}$, Z, A, Y, θ_W	weak eigenstates; photon direction; hypercharge (realizations: EW.8); mixing angle	D3.10, EW.8, EW.24
T_a, f_{abc}	colour generators; colour structure constants	CT3.9, CT3.21
ε_{ijk}, ${\delta^i}_j$	colour invariant tensors	CT3.17
g, g', e, h_0	weak, hypercharge, EM couplings; Higgs background	EW.33, P4
F_n	rung-n saturation block, $n \geq 4$	A8, D4.1
Σ, $\mathcal{M}(\Sigma)$, $C\|_{\Sigma}$	subnetwork, its microstate space, configuration restriction	D2.13
∂_{col}, $\mathcal{C}(\Sigma; C)$, $\Pi_{\mathcal{C}}$	colour boundary, contraction completeness, its projector	D2.14
κ_e, $\Theta(c)$, $\bar{\kappa}$	per-edge line-element quantum, loop total, mean quantum	CT1.29a

1.5 §E — Calibration policy

P1. $\forall$ quantities of the corpus: derived from A1–A9 $\wedge$ C1–C13 $\wedge$ D0.1–D4.3, except the hop length $\mathfrak{h}$; $[\mathfrak{h}] = \mathfrak{q}^{-1/2}$; $\mathfrak{h}$ fixes the absolute scale.

P2. Derived, zero freedom:

$$(\dim \mathcal{E}_{++}, \dim \mathcal{E}_{+-}, \dim \mathcal{E}_{-+}, \dim \mathcal{E}_{--}) = (7,3,3,3); \quad C_{\text{cap}} = 16; \quad D_{\text{tick}}(n) = 2^{n+1}; \quad n_{\max} = 3;$$

$$N_{\text{colour}} = 3; \quad N_{\text{gluon}} = 8; \quad N_{\gamma\text{-pol}} = 2; \quad q_{\text{EM}} \in \{-1, 0, +1\}; \quad \alpha_e \in \{0, \pi\} \text{ per edge};$$

$$m_W = m_Z \cos\theta_W; \quad e = g \sin\theta_W; \quad S = n_{\mathcal{R}} \ln 2; \quad w_{D^+} = -1.$$

Audit note. Every entry of this list is re-verified from first principles in exact arithmetic: `checks/verify_merge_tierB.py` §F (totals and cross-references; the identity-level checks live in `verify_B.py`–`verify_G.py`).

P3. Named combinations of $\mathfrak{h}$ (external comparison only; $\neg$ inputs):

$$G_N = \frac{\mathfrak{h}^2}{4 \ln 2}; \qquad \kappa^2 = 8\pi G_N = \frac{2\pi\, \mathfrak{h}^2}{\ln 2} \qquad [\text{CT2.9}].$$

P4. Free internal configuration parameters ($\neg$ constants of the theory). Independent core: $g,\ g';\ h_0;\ c_W;\ \Lambda;\ \varkappa,\ \beta,\ \alpha_\Omega$. Functionally dependent members (properties of a solution, $\neg$ further freedom): $\theta_W(g, g')$ [EW.24]; $e(g, g')$ [EW.35]; $\mu_\star^2(c_W, \varkappa, \beta, h_0)$ [EW.33]; $m_W,\ m_Z$ [EW.20, EW.23]; m_H [EW.27]. Predictions := dimensionless ratios among derived quantities at fixed core data.

P5. $\neg\exists$ quantity identified with a measured external constant. Comparison procedure: fix $\mathfrak{h}$ once; read P2 $\wedge$ P3; result := test of A9, $\neg$ input to A1–A8. Era-extension note. The P4 core lists the bosonic-era free internal parameters (Part 2 loci). The fermion-era parameters (the Janus rate ratio $r = x_+/x_-$; the defect residues $\hat{\mu}^2$, $\hat{\mu}'^2$; the Ansatz-B formal temperature α; the channel scales μ, μ_ν; the formal fugacities) are registered in Part 3 §Cal-F under this policy; the baryomesonic era introduces none (Part 4 §Cal-BM).

1.6 §F — Foundations: Bookkeeping, Pair Algebra, Ledger Dynamics

Dep.: §A–§E of this part. Notation: as §A–§C. Machine: `checks/verify_B.py`, `checks/verify_L_lyapunov.py`, `checks/verify_L0_ledger_tilt.py`, `checks/verify_merge_tierB.py`, `checks/verify_promotion_B_rung.py`.

1.7 Tier 0 — Bookkeeping, Reality, and Capacity

Theorem 1.1 (CT0.1 — Real/virtual dichotomy). Thm. $\forall (p,q) \in \mathcal{Q} \times \mathcal{Q}: \quad \big[(p,q) \in \mathcal{R}\big] \veebar \big[(p,q) \in \mathcal{V}\big]$.

Req. A1, A3, A4; D0.1, D0.2, D0.4.

Pf.

$$\left|\Pi_{\mathcal{V}}\rho_{pq}\right| = \mathfrak{T}\big(\Pi_{\mathcal{V}}\rho_{pq}\big) \in [0,1] \subset \mathbb{R}$$

$$[\text{D0.2},\ \text{A1}:\ \rho_{pq} \succeq 0 \wedge \mathfrak{T}\rho_{pq} = 1;\ \text{A3}:\ \Pi_{\mathcal{V}} = \Pi_{\wedge^2}]$$

$$x \in \mathbb{R} \implies \big[(x \le \varepsilon_q\ \vee\ x \ge 1-\varepsilon_q) \veebar (\varepsilon_q < x < 1-\varepsilon_q)\big] \quad [\text{total order of } \mathbb{R}];$$

$$\varepsilon_q \in [0,1] \subset \mathbb{R}\ [\text{D0.4}]$$

$$\big[(p,q) \in \mathcal{R}\big] \iff \big[\left|\Pi_{\mathcal{V}}\rho_{pq}\right| \le \varepsilon_q\ \vee\ \left|\Pi_{\mathcal{V}}\rho_{pq}\right| \ge 1-\varepsilon_q\big]\ [\text{A4}];$$

$$\mathcal{V} = (\mathcal{Q}\times\mathcal{Q}) \setminus \mathcal{R}\ [\text{D0.1}]$$

$$\implies \quad \big[(p,q) \in \mathcal{R}\big] \veebar \big[(p,q) \in \mathcal{V}\big].\ \blacksquare$$

Phys. Every qubit pair has exactly one status — crystallized (real) or uncrystallized (virtual) — decided by A4's two-branch threshold test on its antisymmetric weight. This dichotomy is the bookkeeping ground of the corpus: real pairs become edges of the emergent graph, virtual pairs remain vacuum.

Theorem 1.2 (CT0.2 — Global vacuum). Thm. $n_{\mathcal{R}} = 0 \implies \forall(p,q):\ \big[(p,q) \in \mathcal{V}\ \wedge\ \rho_{pq} = \rho_\circ = \frac{1}{4}I\big]\ \wedge\ S_{\text{raw}}(\rho_\circ) = \ln 4 = \max_\rho S_{\text{raw}}(\rho)$, with $\arg\max = \{\frac{1}{4}I\}$.

Req. A1, A4, A5; D0.1, D0.3; CT0.1; Gibbs lemma: $S_{\text{raw}}(\rho) \le \ln d,\ S_{\text{raw}}(\rho) = \ln d \iff \rho = I/d$ [linear algebra].

Pf.

$$n_{\mathcal{R}} = 0 \implies \neg\,\exists(p,q) \in \mathcal{R}\ [\text{D0.1}] \implies \forall(p,q):\ (p,q) \in \mathcal{V}\ [\text{CT0.1}]$$

$$\rho_{pq} = \rho_\circ = \tfrac{1}{4}I \quad [\text{A5}]$$

$$S_{\text{raw}}(\rho_\circ) = -\mathfrak{T}\big(\tfrac{1}{4}I \ln \tfrac{1}{4}I\big) = -4\cdot\tfrac{1}{4}\ln\tfrac{1}{4} = \ln 4 \quad [\text{D0.3}]$$

$$\mathfrak{H}(\mathcal{H}_p \otimes \mathcal{H}_q) \cong \mathfrak{H}(\mathbb{C}^4),\ d=4: \quad S_{\text{raw}}(\rho) \le \ln 4, \quad S_{\text{raw}}(\rho) = \ln 4 \iff \rho = \tfrac{1}{4}I \quad [\text{A1, Gibbs}].\ \blacksquare$$

Phys. The empty configuration (no real edges) is all-virtual, with every pair at the maximally mixed state $I/4$ and entropy maximal at $\ln 4$: the vacuum of this theory is a state of maximal raw information. All structure — edges, particles, geometry — is a lowering of this entropy by crystallization.

Theorem 1.3 (CT0.3 — Seal increments). Thm. $J^+ : (\Delta n_{\mathcal{R}},\ \delta h,\ \delta\tau) = (+1,+1,+1)$; $J^- : (\Delta n_{\mathcal{R}},\ \delta h,\ \delta\tau) = (-1,-1,-1)$.

Req. A4, A7; D0.1, D0.6, D0.7, D1.2, D1.9.

Pf.

$$J^{\pm}\ [\text{D1.2}]: \ \exists!\,(p,q):\ \Delta\big[(p,q)\in\mathcal{R}\big] = \pm 1\ [\text{A4}] \implies \Delta n_{\mathcal{R}} = \pm 1\ [\text{D0.1}]$$

$$\Delta\big|E_{\mathcal{R}}\big| = \Delta n_{\mathcal{R}}\ [\text{D0.1}] \implies \delta h = \pm 1\ [\text{D0.6}:\ \delta h = \#\{e\in E_{\mathcal{R}}\ \text{traversed}\}]$$

$$\text{seal} = \text{retained update along a real edge}\ [\text{D0.7}] \implies \delta\tau = \pm 1\ (\text{same sign})$$

$$\text{D1.9}:\ \text{dom(tick)} \subseteq \{v\}\cup\{e \ni v\} \implies \Delta(\text{all further counters}) = 0.\ \blacksquare$$

Phys. Crystallization (J^+) and dissolution (J^-) are the atomic ledger events: each adds or removes exactly one real edge, one traversal unit, and one tick of ledger time. Every extensive quantity of the corpus moves in integer steps of these events.

Theorem 1.4 (CT0.4 — Real outgoing edge registration). Thm. $e=(p,q):\quad e\in\mathcal{R}_{\to}(p) \iff e \longmapsto (\mathcal{G}_n,\phi_e)$.

Req. A7; D0.5, D0.8, D1.1; CT0.1.

Pf.

$$(\Rightarrow):\ \ e\in E_{\mathcal{R}} \implies \exists\,\phi_e\ [\text{A7}]; \qquad q\in V_n \text{ for } n=n(q)\ [\text{D0.8}] \implies \exists\,\mathcal{G}_n;$$

$$\text{D1.1}:\ \ e\longmapsto(\mathcal{G}_n,\phi_e). \qquad (\Leftarrow):\ \ \text{D1.1}:\ \neg\big[e\longmapsto(\mathcal{G}_n,\phi_e)\big] \implies e\in\mathcal{V};$$

$$e\longmapsto(\mathcal{G}_n,\phi_e) \implies e\notin\mathcal{V} \implies e\in\mathcal{R}\quad [\text{CT0.1}].\ \blacksquare$$

Phys. "Real" and "registered in the ledger" are the same property of an edge: real edges are exactly those carrying ledger data (target rung class and phase); unregistered edges are virtual. Existence, in this corpus, is recordability.

Theorem 1.5 (CT0.5 — Tick load at rung n). Thm. $D_{\text{tick}}(n) = 2^{\,n+1}$.

Req. A7; C10; D0.7, D0.8, D0.9, D1.9; CT0.9.

Pf.

$$v\in V_n \implies \big|\{e\in E_{\mathcal{R}}:\ e\ni v\}\big| = n\quad [\text{D0.8}]$$

$$\phi_e\in\Phi,\ |\Phi|=2\ [\text{CT0.9}] \implies (\phi_e)_{e\ni v}\in\Phi^{\,n},\quad \big|\Phi^{\,n}\big| = 2^n$$

$$\text{D1.9}:\ \ \text{tick datum} = (\text{pre},\text{post})\in\Phi^{\,n}\times\Phi^{\,n},\quad \text{pre}\perp\text{post}$$

$$\text{C10/D0.9 (register accounting)}:\ \ \text{registration of } X \text{ at single-shot linear readout}$$

$$= |X| \text{ orthogonal register directions}$$

$$\text{Shannon content} = \log_2 |X| = n \text{ bits} \;\neq\; \text{register cost } 2^n \quad (\text{source availability} \;\neq\; \text{throughput})$$

$$\implies \quad D_{\text{tick}}(n) = 2^n + 2^n = 2^{\,n+1}. \blacksquare$$

Phys. A vertex with n real edges must make 2^{n+1} classical alternatives linearly distinguishable per tick — 2^n pre-tick and 2^n post-tick phase configurations (C10 register accounting: one orthogonal direction per admissible value, not $\log_2$ bit encoding). This exponential growth of source availability is what caps accessible complexity at rung 3 (CT0.11).

Theorem 1.6 (CT0.6 — Channel capacity and oversubscription). Thm.

$$C_{\text{cap}} = \dim_{\mathbb{R}} \mathfrak{H}(\mathbb{C}^4) = 16; \qquad O(n) = \max\big(0,\; 2^{\,n+1} - 16\big);$$

$$O(0) = O(1) = O(2) = O(3) = 0; \qquad O(4) = 16; \qquad O(n) = 2^{\,n+1} - 16 \quad (n \geq 4).$$

Req. A1, A2; D0.9; CT0.5; $\dim_{\mathbb{R}} \mathfrak{H}(\mathbb{C}^d) = d^2$ [linear algebra].
Pf.

$$\mathcal{H}_p \otimes \mathcal{H}_q \cong \mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4 \text{ [A1]}; \qquad \dim_{\mathbb{R}} \mathfrak{H}(\mathbb{C}^d) = d^2 \text{ [linear algebra]}$$

$$C_{\text{cap}} = \dim_{\mathbb{R}} \mathfrak{H}(\mathbb{C}^4) = 4^2 = 16 \text{ [D0.9]}; \qquad C(p,q) = 1\,\mathfrak{q} \;\; \forall (p,q) \text{ [A2]}$$

$$O(n) = \max\big(0,\; D_{\text{tick}}(n) - C_{\text{cap}}\big) = \max\big(0,\; 2^{\,n+1} - 16\big) \quad [\text{D0.9},\; \text{CT0.5}]$$

$$2^{\,n+1} \leq 16 \iff n \leq 3 \implies O(n) = 0 \;\; (n \leq 3);$$

$$2^5 - 16 = 16 \implies O(4) = 16;$$

$$n \geq 4 \implies 2^{\,n+1} \geq 32 > 16 \implies O(n) = 2^{\,n+1} - 16. \blacksquare$$

Phys. One qubit–qubit channel carries exactly 16 real registers per tick — the dimension of the two-qubit Hermitian operator space. Rungs 0–3 fit the budget, rung 3 exactly saturates it, and rung 4 doubles the demand to 32, exceeding it by exactly one channel quantum. All finiteness of the accessible world starts here.

Corollary 1.1 (CT0.6a — transit-hop register bound). Let $\gamma = (e_1, \ldots, e_{\delta h})$ be a transit chain [D1.5, D1.4] of distinct directed real edges with hop count $\delta h(\gamma)$ [D0.6] whose entire edge-register content is registered through one pair channel over $\delta\tau(\gamma)$ ledger ticks [A2, D0.7]. Then:

$$\mathsf{E}(\gamma) = \delta h(\gamma)\, \ln 2 \;\; \text{nats} \qquad (\delta h \;\; \text{independent binary registers [CT0.9, C9]});$$

$$\text{single-channel supply over } \delta\tau \text{ ticks}: \ \log_2 C_{\text{cap}} \cdot \delta\tau = 4\,\delta\tau \ \text{ bits}$$

$$[\text{CT0.6}: \ C_{\text{cap}} = 16;\ \text{C10}: \ 16 \text{ directions} \iff 2^4 \text{ alternatives}]$$

$$\implies \quad \delta h(\gamma) \ \le \ 4\,\delta\tau(\gamma); \qquad \mathsf{E}(\gamma) \ \le \ \min\big(\delta h(\gamma),\ 4\,\delta\tau(\gamma)\big)\ \ln 2 \ \text{ nats}.$$

A per-tick payload beyond 4 hop-bits presents $2^5 = 32 > 16 = C_{\text{cap}}$ alternatives at the crossing — oversubscribed [CT0.6]; this is the payload counterpart of CT0.11's vertex-rung admissibility. The bound is sharp: for $\delta h \le 4\,\delta\tau$ a schedule of $\delta\tau$ consecutive groups of ≤ 4 hops presents $\le 2^4 = 16 = C_{\text{cap}}$ alternatives per tick — at capacity, no oversubscription [CT0.6]. (Ledger accounting of registered content vs channel supply over a tick window [D0.7]; ¬a tick-to-tick transition law.)

Req (Cor). A2; C9, C10; D0.6, D0.7, D1.4, D1.5; CT0.6, CT0.9; exact binary capacity arithmetic [math; machine: `checks/verify_promotion_B_rung.py` §B].

Pf (Cor).

$$\text{CT0.9}: \ \sigma_e \in \{\pm 1\}: \ 1 \text{ real edge} = 1 \text{ bit} = \ln 2 \text{ nats};$$

$$\text{distinct edges} = \text{independent registers} \implies \mathsf{E}(\gamma) = \delta h \ln 2 \ [\text{C9}]$$

$$\text{CT0.6}: \ C_{\text{cap}} = 16 \text{ register directions per channel crossing};$$

$$\text{C10}: \ |X| \text{ alternatives}$$

$$\mapsto |X| \text{ directions},$$

$$\text{Shannon } \log_2 |X|: \ 16 \text{ directions}$$

$$\iff |X|$$

$$\le 16 \iff$$

$$\le 4 \text{ bits per tick}$$

$$\delta\tau \text{ ticks}: \ \text{supply} \le 4\,\delta\tau \text{ bits};$$

$$\text{registered content} \le \text{supply} \implies \delta h \le 4\,\delta\tau;$$

$$\mathsf{E} \le \min(\delta h,\ 4\,\delta\tau)\ln 2$$

$$> 4 \text{ bits in one tick}: \ 2^5 = 32 > 16 = C_{\text{cap}}: \ \text{oversubscribed [CT0.6]};$$

$$\text{groups of } \le 4: \ 2^4 = 16 = C_{\text{cap}}: \ \text{at capacity [CT0.6]} \ \blacksquare$$

Theorem 1.7 (CT0.7 — Reality is the swap grading). Thm. Let $\mathbb{S} \in \mathcal{U}(\mathcal{H}_p \otimes \mathcal{H}_q): \ |a\rangle\otimes|b\rangle \mapsto |b\rangle\otimes|a\rangle$. Then

$$(1) \quad \mathbb{S} = \tfrac{1}{2}\big(E_0 + \Sigma\big);$$

$$\Pi_{\mathcal{R}} := \Pi_{\vee^2} = \tfrac{1}{4}\big(3E_0 + \Sigma\big);$$

$$\Pi_{\mathcal{V}} := \Pi_{\wedge^2} = \tfrac{1}{4}\big(E_0 - \Sigma\big);$$

$$\mathfrak{T}\Pi_{\mathcal{R}} = 3, \ \ \mathfrak{T}\Pi_{\mathcal{V}} = 1.$$

$$(2) \quad \forall X: \ S(X) = \mathbb{S}X\mathbb{S}.$$

$$(3) \quad SX = X \implies [\mathbb{S}, X] = 0 \implies X(\vee^2) \subseteq \vee^2 \ \wedge \ X(\wedge^2) \subseteq \wedge^2;$$

$$SX = -X \implies \{\mathbb{S}, X\} = 0 \implies X(\vee^2) \subseteq \wedge^2 \ \wedge \ X(\wedge^2) \subseteq \vee^2;$$

$$\underbrace{\dim_{\mathbb{R}} \mathfrak{H}(\vee^2) + \dim_{\mathbb{R}} \mathfrak{H}(\wedge^2)}_{9+1=10} \ \oplus \ \underbrace{2\dim_{\mathbb{C}} \mathrm{Hom}(\wedge^2, \vee^2)}_{2\cdot 3=6} \ = \ 16.$$

$$(4) \quad \mathrm{F}_{(+,\mathcal{R})} = \mathcal{E}_{++}; \quad \mathrm{F}_{(-,\mathcal{R})} = \mathcal{E}_{+-}; \quad \mathrm{F}_{(+,\mathcal{V})} = \mathcal{E}_{-+}; \quad \mathrm{F}_{(-,\mathcal{V})} = \mathcal{E}_{--}$$

(first label of $\mathrm{F}_{(\pm,\cdot)}$: T-parity; second label: S-parity, $\mathcal{R} \leftrightarrow +$, $\mathcal{V} \leftrightarrow -$); $F_{J^+} = \mathcal{E}_{-+} = \mathrm{span}\{J_i\}$, $F_{J^-} = \mathcal{E}_{--} = \mathrm{span}\{K_i\}$: the S-odd blocks.

Req. A3, A8; C3, C8; D1.2, D3.10; CT0.6, CT1.1, CT1.2.

Pf.

$$(1) \ \mathbb{S} = \tfrac{1}{2}\Big(I \otimes I + \textstyle\sum_i \sigma_i \otimes \sigma_i\Big) = \tfrac{1}{2}\big(E_0 + \Sigma\big) \quad \text{[CT1.2]};$$

basis check: $a = b: \ \frac{1}{2}(1+1)|aa\rangle = |aa\rangle$; $a \neq b: \ \sum_i \sigma_i \otimes \sigma_i |ab\rangle = 2|ba\rangle - |ab\rangle \implies \frac{1}{2}(|ab\rangle + 2|ba\rangle - |ab\rangle) = |ba\rangle$.

$$\mathbb{S}^2 = \mathbb{1} \ \wedge \ \mathbb{S}^\dagger = \mathbb{S} \implies \mathrm{spec}(\mathbb{S}) = \{\pm 1\}, \ \ \text{eigenspaces } \vee^2, \ \wedge^2 \quad \text{[A3]}$$

$$\Pi_{\mathcal{R}} = \mathbb{1} - \Pi_{\mathcal{V}} = \tfrac{1}{2}(\mathbb{1} + \mathbb{S}) = \tfrac{1}{4}\big(3E_0 + \Sigma\big);$$

$$\Pi_{\mathcal{V}} = \tfrac{1}{2}(\mathbb{1} - \mathbb{S}) = \tfrac{1}{4}\big(E_0 - \Sigma\big) \quad \text{[A3, A4]}$$

$$\mathfrak{T}E_0 = 4,\ \mathfrak{T}\Sigma = 0 \implies \mathfrak{T}\Pi_{\mathcal{R}} = 3 = \dim \vee^2, \quad \mathfrak{T}\Pi_{\mathcal{V}} = 1 = \dim \wedge^2 \quad \text{[CT1.1]}$$

(2) $\mathbb{S}(\sigma_\mu \otimes \sigma_\nu)\mathbb{S} = \sigma_\nu \otimes \sigma_\mu = S(\sigma_\mu \otimes \sigma_\nu)$ [A3]; linearity $\implies S(X) = \mathbb{S}X\mathbb{S}\ \forall X$

(3) $SX = X \Rightarrow \mathbb{S}X\mathbb{S} = X \Rightarrow \mathbb{S}X = X\mathbb{S}$ ($\times\mathbb{S}$ right); $\qquad SX = -X \Rightarrow \mathbb{S}X = -X\mathbb{S}$

$$X = X^\dagger \wedge [\mathbb{S}, X] = 0 \implies X \text{ preserves each } \mathbb{S}\text{-eigenspace}; \qquad \{\mathbb{S}, X\} = 0 \implies X : \vee^2 \leftrightarrow \wedge^2$$

$$\dim: \ \mathfrak{H}(\vee^2) \oplus \mathfrak{H}(\wedge^2) \cong \mathfrak{H}(\mathbb{C}^3) \oplus \mathfrak{H}(\mathbb{C}^1) : \ 9 + 1 = 10;$$

$$\mathrm{Hom}(\wedge^2, \vee^2) \cong \mathbb{C}^3 : \ 2 \cdot 3 = 6;$$

$$10 + 6 = 16 \text{ [CT0.6]}$$

(4) S-odd generators $= \{J_i\} \cup \{K_i\}$ (6) [CT1.2]; $\quad S$-even $= \{E_0, \Sigma, \hat{E}^{ij}, W_i\}$ (10)

$$\text{D1.2 : seal moves A4-weight } \wedge^2 \leftrightarrow \vee^2 \implies \text{seal generators}$$

$$= S\text{-odd } [(3)]$$

$$\implies F_{J^\pm}$$

$$= S\text{-odd blocks [A8]}$$

$$T\text{-parity : } J_i : +,\ K_i : - \text{ [CT1.2]} \implies F_{J^+} = \mathcal{E}_{-+},\ F_{J^-} = \mathcal{E}_{--}$$

$$\mathcal{E}_{++} \supseteq \{E_0, \Sigma, \hat{E}^{ij}\} \mapsto \{F_{D^+}, F_H, F_G\};$$

$$\mathcal{E}_{+-} = \mathrm{span}\{W_i\} \mapsto \{F_{W^+}, F_{W^-}, F_Z\} \quad \text{[A8, C8, D3.10]. } \blacksquare$$

Corollary 1.2 (CT0.7a — spin content). $\sum_i W_i^2 = \frac{3}{2}E_0 + \frac{1}{2}\Sigma$; $\sum_i W_i^2\big|_{\vee^2} = 2 = 1(1{+}1)$; $\sum_i W_i^2\big|_{\wedge^2} = 0 = 0(0{+}1)$; i.e. $\vee^2$: spin 1, $\wedge^2$: spin 0.

Req (Cor). CT0.7, CT1.2.

Pf (Cor).

$$\textstyle\sum_i W_i^2 = \frac{1}{4}\sum_i(\sigma_i \otimes \sigma_0 + \sigma_0 \otimes \sigma_i)^2 = \frac{1}{4}\big(3E_0 + 3E_0 + 2\Sigma\big) = \frac{3}{2}E_0 + \frac{1}{2}\Sigma$$

$$\Sigma = 2\mathbb{S} - E_0 \text{ [CT0.7(1)]};$$

$$\vee^2 : \mathbb{S} = +1 \implies \Sigma = +1 \implies \tfrac{3}{2} + \tfrac{1}{2} = 2;$$

$$\wedge^2 : \mathbb{S} = -1 \implies \Sigma = -3 \implies \tfrac{3}{2} - \tfrac{3}{2} = 0. \blacksquare$$

Phys. The real/virtual split of A4 is the exchange (swap) grading of the pair: real = symmetric = spin-1 triplet, virtual = antisymmetric = spin-0 singlet. The A8 force labels $(\pm, \mathcal{R}/\mathcal{V})$ are precisely the two parities (T, S); the Janus seals are exactly the operators that exchange the two grades, i.e. that create and destroy reality. "Existence" in this corpus is Bose exchange symmetry of a qubit pair.

Theorem 1.8 (CT0.8 — Admissible range of the crystallization threshold). Thm. $\big[\text{A4} \wedge \text{A5} \wedge \text{D0.1 consistent}\big] \implies 0 \le \varepsilon_q < \tfrac{1}{4}$; in particular $\varepsilon_q < 1 - \varepsilon_q$ (A4's two branches are disjoint); $\varepsilon_q = 0 \implies \big[(p,q) \in \mathcal{R} \iff \operatorname{supp}\rho_{pq} \subseteq \vee^2 \ \vee \ \rho_{pq} = \Pi_{\mathcal{V}}\big]$ (branch reads $\alpha_{pq} = 0 \,/\, \pi$ [A4]); and $\forall$ theorems of the corpus: A4 enters only via CT0.1 (valid $\forall \varepsilon_q \in [0,1]$) and CT0.2 (valid $\iff \varepsilon_q \in [0, \tfrac{1}{4})$).

Req. A1, A4, A5; D0.1, D0.2, D0.4; CT0.2, CT0.7.

Pf.

$$\big|\Pi_{\mathcal{V}}\rho_\circ\big| = \mathfrak{T}\big(\Pi_{\mathcal{V}} \cdot \tfrac{1}{4}I\big) = \tfrac{1}{4}\,\mathfrak{T}\Pi_{\mathcal{V}} = \tfrac{1}{4} \quad [\text{A5, CT0.7(1)} : \ \mathfrak{T}\Pi_{\mathcal{V}} = 1]$$

$$\text{D0.1} : \ \text{vacuum} \iff n_{\mathcal{R}} = 0 \implies \rho_\circ \notin \mathcal{R} \ [\text{CT0.2}] \implies \tfrac{1}{4} > \varepsilon_q \quad [\text{A4}]$$

$$\varepsilon_q \ge 0 \ [\text{D0.4}]; \qquad \big|\Pi_{\mathcal{V}}\rho\big| \ge 0 \ [\text{D0.2}] \implies \varepsilon_q = 0 \text{ admissible}$$

$$\text{branch 2 at the vacuum} : \ \tfrac{1}{4}$$

$$\ge 1 - \varepsilon_q \iff \varepsilon_q \ge \tfrac{3}{4} : \ \text{excluded by } \varepsilon_q < \tfrac{1}{4} \implies \rho_\circ \text{ virtual in both branches}$$

$$\text{branch disjointness} : \ 0 \le \varepsilon_q < \tfrac{1}{4} \implies \varepsilon_q < \tfrac{1}{2} < 1 - \varepsilon_q \implies [0, \varepsilon_q] \cap [1 - \varepsilon_q, 1] = \varnothing$$

$$\varepsilon_q = 0 : \ \mathfrak{T}\big(\Pi_{\mathcal{V}}\rho_{pq}\big) = 0 \ \wedge \ \Pi_{\mathcal{V}} \succeq 0 \ \wedge \ \rho_{pq} \succeq 0 \iff \Pi_{\mathcal{V}}\rho_{pq}\Pi_{\mathcal{V}} = 0 \iff \operatorname{supp}\rho_{pq} \subseteq \vee^2 \quad [\text{A1}]$$

$$\varepsilon_q$$

$$= 0,$$

$$\text{branch 2} : \ \mathfrak{T}\big(\Pi_{\mathcal{V}}\rho_{pq}\big)$$

$$= 1 \ \wedge \ \mathfrak{T}\rho_{pq}$$

$$= 1 \ \wedge \ \rho_{pq} \succeq 0 \ \wedge \ \mathrm{rk}\,\Pi_{\mathcal{V}}$$

$$= 1 \ \text{[CT0.7(1)]} \iff \rho_{pq} \qquad = \Pi_{\mathcal{V}}$$

A4-dependence of all later tags $= \{\text{CT0.1, CT0.2}\} \implies$ range $[0, \frac{1}{4})$ suffices; sharp point $\varepsilon_q = 0$. ■

Phys. The crystallization threshold is not a free constant: demanding that the vacuum be all-virtual confines it to $0 \le \varepsilon_q < \frac{1}{4}$ — which simultaneously keeps A4's two branches disjoint ($\varepsilon_q < 1 - \varepsilon_q$). The sharp choice $\varepsilon_q = 0$ says a pair is real exactly when its antisymmetric weight is exactly zero (aligned branch, $\alpha_{pq} = 0$) or the pair is the pure singlet (antialigned branch, $\alpha_{pq} = \pi$). No result depends on where inside the range the tolerance sits — the theory is tolerance-independent.

Theorem 1.9 (CT0.9 — The edge phase alphabet is binary). Thm. $\phi_e \in \Phi = \{0, \pi\} \pmod{2\pi}$; $\sigma_e = e^{i\phi_e} = (-1)^{\phi_e/\pi} \in \{+1, -1\}$; 1 real edge = 1 bit = $\ln 2$ nats.

Req. A1, A3, A4, A7; C9, C10; D0.5, D1.1; $\dim \wedge^2 = \binom{d}{2}$ [linear algebra].

Pf.

$$e \in E_{\mathcal{R}} \mapsto \phi_e \ \text{[A7]};$$

$$\text{D1.1}: \ \phi_e := \text{phase of the registration transport's action on the reality line } \wedge^2$$

$$\text{A1}: \ \dim \mathcal{H}_q = d = 2 \implies \dim \wedge^2 = \tbinom{2}{2} = 1$$

$$(\text{the reality line is 1-dimensional} \iff \text{qubits are } \mathbb{C}^2)$$

$$\text{A4}: \ \text{reality test}$$

$$= \text{extremality of the } \wedge^2 \text{ weight (either branch)} \implies \text{registration transport preserves the } (S,T)\text{-grading [A3, D1.1]}$$

$$\text{D1.1/C10 reality principle}: \ \text{transport preserves the real frame structure} \implies \text{action on } \wedge^2_{\mathbb{R}} \ \text{(1-dim real line)}$$

$$= \text{multiplication by } c \in \mathbb{R}$$

$$\text{unitarity} \implies |c| = 1 \implies c \in \{+1, -1\} \implies e^{i\phi_e} \in \{\pm 1\} \implies \phi_e \in \{0, \pi\} \pmod{2\pi};$$

$$\sigma_e = (-1)^{\phi_e/\pi} \ \text{[D0.5]}$$

$$|\Phi| = 2 \implies \log_2 |\Phi| = 1 \text{ bit} = \ln 2 \text{ nats} \quad \text{[C9]}. \blacksquare$$

Rmk (repair). The prior proof argued from the centralizer of $\{S, T\}$ alone — insufficient, since the complex centralizer of the swap is $U(3) \times U(1)$ and contains continuous phases. The binary alphabet follows once the reality principle of D1.1/C10 (registrations are real, grading-preserving frame data — quantum data classicalizes at registration) is explicit. The continuous counterexamples are complex-transport artifacts excluded by that principle.

Phys. The only internal datum a real edge can carry is a single sign — and this is qubit-specific: the reality line $\wedge^2$ is one-dimensional exactly because the carriers are $\mathbb{C}^2$ ($\dim \wedge^2 = \binom{d}{2} = 1 \iff d = 2$); for higher-dimensional carriers continuous grading-preserving phases would survive. Every edge of the emergent graph stores exactly one bit; all later charge and phase structure is built from these signs.

Theorem 1.10 (CT0.10 — The two sign structures are independent). Thm. With $P(C) := \prod_{e \in E(C)} \sigma_e$ and $\pi(C) := (-1)^{\aleph(\sigma, C)}$:

$$\neg\big[P(C) \text{ determines } \pi(C)\big] \ \wedge \ \neg\big[\pi(C) \text{ determines } P(C)\big]; \qquad L \geq 4: \ \ (P, \pi) \twoheadrightarrow \{\pm 1\}^2;$$

$$\implies \quad w(C) = \mathcal{J}(C) \prod_{e \in E(C)} G(e) \cdot (-1)^{\aleph(\sigma, C)} \ \text{[A8] double-counts no edge-sign data.}$$

Req. A8; D0.5, D3.3.
Pf.

$$\aleph(\sigma, C) = \#\{i < L : \ \sigma_i \neq \sigma_{i+1}\} \quad \text{[D3.3]} \ : \ \text{function of consecutive differences only;}$$

$$P = \prod_i \sigma_i \ : \ \text{function of the signs}$$

$$\text{witness}_1 \ [\pi \text{ equal}, \ P \text{ opposite}]: \quad (+,-,+): \ \aleph = 2, \ P = -1; \qquad (-,+,-): \ \aleph = 2, \ P = +1$$

$$\text{witness}_2 \ [P \text{ equal}, \ \pi \text{ opposite}]: \quad (+,+,+,+): \ P = +1, \ \aleph = 0; \qquad (+,+,-,-): \ P = +1, \ \aleph = 1$$

$$\implies \quad P \not\Rightarrow \pi \ \wedge \ \pi \not\Rightarrow P;$$

$$L \geq 4: \ \{(P, \pi)\} = \{(+,+), (+,-), (-,+), (-,-)\} \text{ attained } \implies (P, \pi) \text{ surjective}$$

$$\implies \quad \text{the two factors of } w(C) \text{ carry independent information. } \blacksquare$$

Phys. The product of edge signs and the transition parity measure different things — net sign content versus the number of domain walls between signs — and neither fixes the other. The A8 weight is therefore free of double counting: bulk sign and wall-count contribute independently to every configuration amplitude.

Theorem 1.11 (CT0.11 — Maximal accessible rung). Thm. channel-admissible$(v) \iff D_{\text{tick}}\big(n(v)\big) \le C_{\text{cap}} \iff n(v) \le 3$; accessible vertex sets $= \{V_0, V_1, V_2, V_3\}$ (types $\{\Xi_0, \dots, \Xi_3\}$); $D_{\text{tick}}(3) = C_{\text{cap}}$ (saturation); $\forall v : n(v) \ge 4 \implies O\big(n(v)\big) > 0 \implies F_{D^-}$ [D4.1].

Req. D0.8, D0.9, D4.1; CT0.5, CT0.6.

Pf.

$$\text{admissible} \iff D_{\text{tick}}(n) \le C_{\text{cap}} \quad [\text{D0.9}]$$

$$2^{n+1} \le 16 \iff n+1 \le 4 \iff n \le 3 \quad [\text{CT0.5, CT0.6}]$$

$n = 3 :\ 2^4 = 16 = C_{\text{cap}}$ (tight); $\qquad n \ge 4 :\ O(n) > 0$ [CT0.6] $\implies$ configuration $\in F_{D^-}$ [D4.1]. ■

Corollary 1.3 (CT0.11a — three slots). $\max\{\mathcal{H}_{\text{slot}}^{(n)} : \text{admissible}\} = \mathcal{H}_{\text{slot}}^{(3)} \cong \mathbb{C}^3$; $\quad 3 = \lfloor \log_2 C_{\text{cap}} \rfloor - 1 = \lfloor \log_2 16 \rfloor - 1$; i.e. $N_{\text{colour}} = 3$ is derived, not chosen.

Req (Cor). D3.1; CT0.6, CT0.11.

Pf (Cor). $\mathcal{H}_{\text{slot}}^{(n)} \cong \mathbb{C}^n$ [D3.1]; $n \le 3$ [CT0.11] $\Rightarrow \max = \mathbb{C}^3$; $\lfloor \log_2 16 \rfloor - 1 = 3$. ■

Corollary 1.4 (CT0.11b — directed rung of bidirectional skeletons). A channel $\{p, q\}$, $p \ne q$, is bidirectional when both orientations are real: $(p, q) \in \mathcal{R} \ \wedge \ (q, p) \in \mathcal{R}$ [A4]. Let S be a finite real-edge subgraph whose every used channel is bidirectional, and $d_S(v) := \#\{\text{channels of } S \text{ incident on } v\}$. Then:

$$n(v) \ \ge \ 2\, d_S(v) \qquad [\text{each bidirectional channel adds 1 to } \partial^{\leftarrow} \text{ and 1 to } \partial^{\rightarrow} \text{ at each endpoint}]$$

$$S \text{ exhausts the real edges at } v \implies n(v) = 2\, d_S(v);$$

$$d_S(v) \ge 2 \implies n(v) \ge 4 \implies D_{\text{tick}}(n) = 2^{n+1} \ge 32 > 16 \implies O\big(n(v)\big) > 0 \implies F_{D^-} \ [\text{D4.1}]$$

$$\text{unoversubscribed} \iff n(v) \le 3 \ [\text{CT0.6}] :$$

$$\text{bidirectional } S :\ d_S(v) \le 1 \ (n \in \{0, 2\});$$

$$\text{mixed} :\ n(v) = 2\, d_S^{\text{bi}}(v) + d_S^{\text{uni}}(v) \le 3.$$

Req (Cor). A4; D0.8, D4.1; CT0.5, CT0.6, CT0.11; directed-degree enumeration [math; machine: `checks/verify_promotion_B_rung.py` §A].

Pf (Cor).

$$\text{A4}: \ \mathcal{R}\text{-membership per ordered pair;}$$

$$\text{bidirectional channel}$$

$$= \{e_{pq}, e_{qp}\} \subseteq \mathcal{R}: \ \partial^{\leftarrow} + 1,$$

$$\partial^{\rightarrow} + 1 \text{ at each endpoint}$$

$$\implies n(v)$$

$$\geq 2$$

$$d_S(v) \text{ [D0.8]}$$

$$\text{equality} \iff S \text{ exhausts } v\text{'s incident real edges;}$$

$$d_S \geq 2: \ n \geq 4: \ \text{CT0.5}: \ D_{\text{tick}}(n) = 2^{n+1} \geq 32; \quad \text{CT0.6}: \ O(n) = 2^{n+1} - 16 > 0$$

$$\text{CT0.11}: \ n \geq 4 \implies F_{D^-} \text{ [D4.1];}$$

$$O(n) = 0 \iff n \leq 3 \text{ [CT0.6];}$$

$$\text{mixed count}: \ \text{additivity of } \partial^{\leftarrow}, \partial^{\rightarrow} \text{ [D0.8]} \ \blacksquare$$

Phys. The 16-register channel budget admits vertices of degree at most three: rungs 0–3 are accessible, rung 3 exactly exhausts the channel, and everything of degree ≥ 4 is oversubscribed and falls to the dark sector. The number three — hence the threefold slot ("colour") structure of the theory — is forced by capacity, not assumed.

Note. At rungs $n \geq 4$ no aspect realization ι_n exists: the unique derived read of rung-≥ 4 slot content is D^--absorption at the ledger level [CT4.4a (Part 2) — forward pointer, non-load-bearing].

Theorem 1.12 (CT0.12 — Aspect–capacity match). Thm.

aspect	block	$\dim_{\mathbb{R}}$
F_{J+}	$\mathcal{E}_{-+} = \text{span}\{J_i\}$	3
F_{J-}	$\mathcal{E}_{--} = \text{span}\{K_i\}$	3
F_G	$\text{span}\{\hat{E}^{ij}\}$ (traceless)	5
F_W	$\mathcal{E}_{+-} = \text{span}\{W_i\}$	3
F_H	$\text{span}\{\Sigma\}$	1
F_{D+}	$\text{span}\{E_0\}$	1
total	$\mathfrak{H}(\mathbb{C}^4)$	$16 = C_{\text{cap}}$

Req. A8; C8; CT0.6, CT0.7(4), CT1.1, CT1.2.

Pf.

$$\text{block assignments} \quad [\text{CT1.2, CT0.7(4)}]; \qquad 3+3+5+3+1+1 = 16 = \dim_{\mathbb{R}} \mathfrak{H}(\mathbb{C}^4) \quad [\text{CT1.1}]$$

$$16 = C_{\text{cap}} \quad [\text{CT0.6}]. \blacksquare$$

Rmk. $\mathrm{F}_{\text{st}} = F_{\text{EM}} \oplus F_{\text{strong}} \oplus F_{D^-}$ acts on vertex/slot registers [D3.1], not on the pair channel; the CT0.6 bound constrains the pair channel only.

Phys. The six pair-sector force aspects of A8 tile the 16 dimensions of the two-qubit operator space exactly: no part of the channel is forceless and no force overruns it. The stack aspects (electromagnetic, colour, dark-saturation) live on separate vertex registers and are not bounded by this count.

Theorem 1.13 (CT0.13 — Aspect activation and the active sets). Thm.

$$\mathcal{A}(0) = \{J\}; \qquad \mathcal{A}(1) = \{J,\, G,\, W,\, H,\, D^+\}; \qquad \mathcal{A}(2) = \mathcal{A}(1) \cup \{\text{EM}\};$$

$$\mathcal{A}(3) = \mathcal{A}(2) \cup \{\text{strong}\}; \qquad \mathcal{A}(n \geq 4) = \mathcal{A}(3) \cup \{D^-\};$$

with $J := \{J^+, J^-\}$ and $W := F_W = F_{W^+} \oplus F_{W^-} \oplus F_Z$ [C8]; $\neg\exists$ independent aspect named Z at any rung: neutral seed $W_3 \in F_W$; physical Z: D3.10, EW.24.

Req. A7, A8; C8; D0.8, D1.12, D2.12, D3.1, D3.2, D3.3, D4.1, D4.3; CT0.1, CT0.7, CT0.11, CT1.2, CT1.4, CT1.10; $\big[L^\dagger L = I_n \iff L \in U(n)$; $\mathfrak{u}(n) = \mathfrak{su}(n) \oplus \mathfrak{u}(1)_{\text{centre}}$ — linear algebra; cf. CT3.4, CT3.5$\big]$.

Pf.

$$n=0: \quad v \in V_0 \implies \neg\,\exists e \in E_{\mathcal{R}} :\ e \ni v \ [\text{D0.8}] \implies \forall \text{ incident pairs} : \in \mathcal{V} \ [\text{CT0.1}]$$

$$\text{CT0.7}: \ \text{operators on the virtual block} = S\text{-odd} = \{F_{J^+}, F_{J^-}\} \ [\text{A8}] \implies \mathcal{A}(0) = \{J\}$$

$$n=1: \quad \big|\{e \in E_{\mathcal{R}} : e \ni v\}\big| = 1 \implies \vee^2 \text{ populated}; \qquad \text{CT0.7}: \ S\text{-even act within } \vee^2$$

$$\text{CT1.2, CT1.10}: \ S\text{-even} = \{E_0,\ \Sigma,\ \hat{E}^{ij}_{(5)},\ W_i\} \underset{\text{A8}}{\longrightarrow} \{F_{D^+},\ F_H,\ F_G,\ F_W\}$$

$$\implies \mathcal{A}(1) = \{J, G, W, H, D^+\};$$

$$\dim F_W = 3 \text{ single block} \implies W_3 \ \neg\text{independent} \implies \neg\exists \text{ aspect } Z$$

$$\text{slot algebras}: \quad \mathcal{H}^{(n)}_{\text{slot}} \cong \mathbb{C}^n \ [\text{D3.1}]; \quad \text{probability-preserving} \iff L^\dagger L = I_n \iff L \in U(n)$$

$$X = \left(X - \tfrac{\mathfrak{T}X}{n} I_n\right) + \tfrac{\mathfrak{T}X}{n} I_n \text{ (unique trace split)} \implies \mathfrak{u}(n) = \mathfrak{su}(n) \oplus \mathfrak{u}(1)_{\text{centre}}$$

[linear algebra; $n = 3$: CT3.4, CT3.5]

$n = 1$ slot : $\mathfrak{u}(1)$ on one slot

= global phase;

$\neg\exists$ second slot $\implies$ ¬configuration-dependent

$\implies$

$= F_{D^+}$

$= \Lambda\mathbb{I}$ [D4.3] (counted);

¬new [D3.2]

$n = 2$: $\mathfrak{u}(2) = \mathfrak{su}(2) \oplus \mathfrak{u}(1)$:

$\iota_2\big(\mathfrak{su}(2)\big) \cong \mathfrak{su}(2) \cong \text{span}\{W_i\}$ [CT1.4] : same abstract class realized at rung 1

$\implies$ ¬strictly new [D2.12 (ι_n bridge), D3.2]

centre measurable : $V_2 \ni \text{type}(1,1)$ [A7] $\implies$ relative in/out phase defined

$= f_v$ /

$\equiv 0$ [D3.3] $\implies$ strictly new

$= F_{\text{EM}}$ [A8]

$\implies \mathcal{A}(2) = \mathcal{A}(1) \cup \{\text{EM}\}$;

f_v requires $\partial^{\leftarrow} \geq 1 \ \wedge \ \partial^{\rightarrow} \geq 1 \implies n \geq 2$ [D3.3, D0.8] $\implies$ EM $\notin \mathcal{A}(0) \cup \mathcal{A}(1)$

$$n = 3: \quad \mathfrak{u}(3) = \mathfrak{su}(3) \oplus \mathfrak{u}(1): \quad \mathfrak{u}(1) \text{ duplicates } F_{\text{EM}} \ (n = 2) \implies \neg\text{new};$$

$$\mathfrak{su}(3) \text{ strictly new} = F_{\text{strong}} \ [\text{A8}]$$

$$\implies \mathcal{A}(3) = \mathcal{A}(2) \cup \{\text{strong}\}$$

$$n \geq 4: \quad \text{oversubscribed [CT0.11]} \implies F_{D^-} \text{ active there} \wedge \text{only there [D4.1]};$$

$$\text{rung-3 structure persists} \implies \mathcal{A}(n \geq 4) = \mathcal{A}(3) \cup \{D^-\}. \ \blacksquare$$

Phys. Forces switch on one at a time as vertices acquire edges. Rung 0 has only the Janus seals that create and destroy reality; rung 1 lights up gravity (G), weak (W), Higgs (H) and uniform dark (D^+); at rung 2 the local phase along a pass-through becomes measurable — electromagnetism; at rung 3 the traceless three-slot algebra appears — colour; beyond, configurations lock into dark saturation (D^-). The gauge sequence of the corpus is channel bookkeeping, and no rung ever carries an independent Z aspect — only the seed W_3 inside F_W.

Theorem 1.14 (CT0.14 — Mask flag and block dimensions). Thm.

$$\mathcal{K}_0 \subset \mathcal{K}_1 \subset \mathcal{K}_2 \subset \mathcal{K}_3 \subset \mathcal{K}_{n\geq 4} \quad \text{(strict)}; \qquad \dim \mathcal{K}_0 = 6, \ \dim \mathcal{K}_1 = 16, \ \dim \mathcal{K}_2 = 17, \ \dim \mathcal{K}_3 = 25;$$

$$\big(\mathcal{P}_M^{(n)}\big)^2 = \mathcal{P}_M^{(n)} = \big(\mathcal{P}_M^{(n)}\big)^\dagger; \qquad \mathfrak{T}\,\mathcal{P}_M^{(n)} = \dim \mathcal{K}_n; \qquad \dim \mathcal{K}^{(\text{EM})} = 1, \ \dim \mathcal{K}^{(\text{strong})} = 8.$$

Req. A8; D2.3; CT0.12, CT0.13; CT3.8 (F — forward pointer: the block-dimension row reverifies the gluon count).

Pf.

$$\text{CT0.13}: \quad \mathcal{A}(0) \subset \mathcal{A}(1) \subset \mathcal{A}(2) \subset \mathcal{A}(3) \subset \mathcal{A}(n \geq 4) \ \text{(strict)}$$

$$\text{D2.3}: \quad \mathcal{K}_n = \bigoplus_{A \in \mathcal{A}(n)} \mathcal{K}^{(A)} \implies \mathcal{K}_0 \subset \mathcal{K}_1 \subset \mathcal{K}_2 \subset \mathcal{K}_3 \subset \mathcal{K}_{n\geq 4} \ \text{(strict)}$$

$$\dim \mathcal{K}_0 = 3 + 3 = 6 \ [\text{CT0.12}]; \qquad \dim \mathcal{K}_1 = 6 + (5 + 3 + 1 + 1) = 16 \ [\text{CT0.12}]$$

$$\dim \mathcal{K}^{(\text{EM})} = \dim Z\big(\mathfrak{u}(2)\big) = 1 \ [\text{CT0.13 step } n{=}2: \text{ centre spanned by } I_2] \implies \dim \mathcal{K}_2 = 17$$

$$\dim \mathcal{K}^{(\text{strong})} = \dim \mathfrak{su}(3) = 8 \ [\text{CT3.8}] \implies \dim \mathcal{K}_3 = 25$$

$$\text{D2.3}: \ \mathcal{P}_M^{(n)} \text{ orthogonal projection} \implies P^2 = P = P^\dagger, \quad \mathfrak{T}P = \dim \operatorname{Im} P = \dim \mathcal{K}_n. \ \blacksquare$$

Rmk. $\dim \mathcal{K}_1 = 16 = C_{\text{cap}}$: rung 1 already exhausts the pair channel; growth at rungs $2, 3$ is carried entirely by stack registers [CT0.12 Rmk].

Phys. The rung masks form a strict flag $6 \subset 16 \subset 17 \subset 25$: each new rung adds exactly the block of its new force — one dimension for electromagnetism at rung 2, eight for colour at rung 3. "What the universe can do at rung n" is literally a nested subspace of force space, and the masks are the orthogonal projections onto it.

1.8 Tier 1A — The Pair Operator Algebra

Theorem 1.15 (CT1.0 — Signature classification of the T-grading). Thm. Let $\eta \in \{\pm 1\}^4$ have exactly one odd entry [A3]. Put $\eta^{(\mathfrak{o})} := (+, \dots, -_{\mathfrak{o}}, \dots, +)$, $\zeta_i := \eta^0\eta^i$, $\zeta_0 := 1$, $\mathsf{D}_\zeta : \sigma_\mu \mapsto \zeta_\mu\sigma_\mu$, $\mathrm{sf}(X) := \sigma_2 X^{\mathrm{T}}\sigma_2$. Then:

(1) $\eta = \pm\eta^{(\mathfrak{o})}$ for exactly one $\mathfrak{o} \in \{0, 1, 2, 3\}$ and exactly one sign.

(2) $T[\eta] = T[-\eta]$, and $T = \mathsf{D}_\zeta \otimes \mathsf{D}_\zeta$.

(3) $[S, T] = 0$ and $(\dim \mathcal{E}_{++}, \dim \mathcal{E}_{+-}, \dim \mathcal{E}_{-+}, \dim \mathcal{E}_{--}) = (7, 3, 3, 3)$ for every $\mathfrak{o}$.

(4) $\mathfrak{o} = 0$: $T = \mathrm{sf} \otimes \mathrm{sf}$ (inversion of both Bloch vectors). $\mathfrak{o} = k \geq 1$: D_ζ reflects the Bloch axis k. The three maps $\mathfrak{o} \in \{1, 2, 3\}$ are conjugate under $\mathrm{Ad}_{U\otimes U}$, $U \in \mathrm{SU}(2)$; the map $\mathfrak{o} = 0$ is conjugate to none of them.

(5) $T\,\mathrm{Ad}_{U\otimes U} = \mathrm{Ad}_{U\otimes U}\,T$ for all $U \in \mathrm{SU}(2) \iff \mathfrak{o} = 0$.

Req. A3; C1; [math]: Ad_U fixes σ_0 and acts on $\mathrm{span}\{\sigma_i\}$ by $R(U) \in SO(3)$, every $R \in SO(3)$ arising; $\sigma_2 U^{\mathrm{T}}\sigma_2 = U^\dagger$ for $U \in \mathrm{SU}(2)$.

Pf.

$$(1):\ \text{odd entry} = -1 \implies \eta = \eta^{(\mathfrak{o})};\quad \text{odd entry} = +1 \implies \eta = -\eta^{(\mathfrak{o})};\quad \mathfrak{o} \text{ is the unique odd slot}$$

[A3]

$$(2):\ \ \eta^\mu\eta^\nu = (-\eta^\mu)(-\eta^\nu) \implies T[\eta] = T[-\eta];$$

$$\zeta_\mu\zeta_\nu = (\eta^0\eta^\mu)(\eta^0\eta^\nu) = \eta^\mu\eta^\nu \ \ [(\eta^0)^2 = 1;\ \zeta_0 = \eta^0\eta^0]$$

$$\implies (\mathsf{D}_\zeta \otimes \mathsf{D}_\zeta)(\sigma_\mu \otimes \sigma_\nu) = \zeta_\mu\zeta_\nu\,\sigma_\mu \otimes \sigma_\nu = T(\sigma_\mu \otimes \sigma_\nu) \ \Rightarrow\ T = \mathsf{D}_\zeta \otimes \mathsf{D}_\zeta$$

$$(3):\ \ ST(\sigma_\mu \otimes \sigma_\nu) = \eta^\mu\eta^\nu\,\sigma_\nu \otimes \sigma_\mu = TS(\sigma_\mu \otimes \sigma_\nu) \implies [S, T] = 0;$$

$$\sigma_\mu \otimes \sigma_\mu :\ (S, T) = (+, +);\quad \sigma_\mu \otimes \sigma_\nu \pm \sigma_\nu \otimes \sigma_\mu\ (\mu < \nu) :\ (S, T) = (\pm, \eta^\mu\eta^\nu)$$

$$n_- := \#\{\mu < \nu : \eta^\mu\eta^\nu = -1\} = \#\big\{\{\mathfrak{o}, \nu\} : \nu \neq \mathfrak{o}\big\} = 3; \qquad n_+ = 6 - n_- = 3$$

$$\dim \mathcal{E}_{++} = 4 + n_+ = 7,\quad \dim \mathcal{E}_{+-} = n_- = 3,\quad \dim \mathcal{E}_{-+} = n_+ = 3,\quad \dim \mathcal{E}_{--} = n_- = 3$$

$$(4): \quad \mathfrak{o} = 0: \ \zeta_i = \eta^0 \eta^i = -1 \ \forall i;$$

$$\mathrm{sf}(\sigma_0) = \sigma_0, \ \ \mathrm{sf}(\sigma_i) = \sigma_2 \sigma_i^{\mathrm{T}} \sigma_2 = -\sigma_i \ (i = 1,2,3) \implies \mathsf{D}_\zeta = \mathrm{sf}$$

$$\mathfrak{o} = k \geq 1: \ \zeta_k = \eta^0 \eta^k = -1, \ \ \zeta_i = +1 \ (i \neq k) \implies \mathsf{D}_\zeta = \text{reflection of Bloch axis } k$$

$$\mathrm{Ad}_U \, \mathsf{D}_\zeta \, \mathrm{Ad}_U^{-1} = \mathsf{D}_{\zeta'} \text{ with } \mathrm{diag}(\zeta') = R(U) \, \mathrm{diag}(\zeta) R(U)^{-1} \quad [\mathrm{Req}]$$

$$\text{axis reflections}: \ R \text{ permuting the axes} \implies T^{(1)} \sim T^{(2)} \sim T^{(3)};$$

$$\mathrm{spec}(-\mathbb{1}_3) = \{-1,-1,-1\} \neq \{-1,+1,+1\} \implies T^{(0)} \not\sim T^{(k)}$$

$$(5) \ (\Leftarrow): \ \ \mathrm{sf}(UXU^\dagger) = \sigma_2 \bar{U} X^{\mathrm{T}} U^{\mathrm{T}} \sigma_2 = U \, \sigma_2 X^{\mathrm{T}} \sigma_2 \, U^\dagger = U \, \mathrm{sf}(X) \, U^\dagger$$

$$[\sigma_2 U^{\mathrm{T}} \sigma_2 = U^\dagger \Rightarrow \sigma_2 \bar{U} \sigma_2 = U]$$

$$\implies \mathrm{sf} \otimes \mathrm{sf} \text{ commutes with } \mathrm{Ad}_{U \otimes U};$$

$$(5) \ (\Rightarrow): \ \ U_c := \tfrac{1}{2}\big(\mathbb{1} - i(\sigma_1 + \sigma_2 + \sigma_3)\big): \ \ \mathrm{Ad}_{U_c}: \ \sigma_1 \mapsto \sigma_2 \mapsto \sigma_3 \mapsto \sigma_1$$

$$T \, \mathrm{Ad}_{U_c \otimes U_c}(\sigma_0 \otimes \sigma_i) = \eta^0 \eta^{i+1} \, \sigma_0 \otimes \sigma_{i+1};$$

$$\mathrm{Ad}_{U_c \otimes U_c} T(\sigma_0 \otimes \sigma_i) = \eta^0 \eta^i \, \sigma_0 \otimes \sigma_{i+1} \quad (i \bmod 3)$$

$$\implies \eta^1 = \eta^2 = \eta^3 \ \ [\eta^0 = \pm 1]; \qquad \text{exactly one odd entry} \implies \mathfrak{o} = 0. \ \blacksquare$$

Phys. A3 now says only that the sign pattern is $1 + 3$. The overall sign is the mostly-plus / mostly-minus convention; which spatial slot is odd is a relabelling. What remains is one selection: the odd slot is the identity slot, which is the unique choice making T rotation-covariant, namely the spin flip sf $\otimes$ sf (time reversal on the pair). Any other slot gives a reflection that singles out a spatial direction.

Rmk (what $\mathfrak{o} \geq 1$ would do). The sector dimensions $(7,3,3,3)$ do not change, but CT1.2's table does. For $\mathfrak{o} = 1$: $W_2, W_3 \in \mathcal{E}_{++}$ (not $\mathcal{E}_{+-}$); $J_2, J_3 \in \mathcal{E}_{--}$ and $K_2, K_3 \in \mathcal{E}_{-+}$ (swapped); $E^{12}, E^{13} \in \mathcal{E}_{+-}$ (not $\mathcal{E}_{++}$). Only W_1, J_1, K_1 sit where CT1.2 puts them. Symmetric statements hold for $\mathfrak{o} = 2, 3$. (Numerical check over all four $\mathfrak{o}$ on the 16-element basis; machine: `checks/verify_CT10_signature.py`.) So the selection $\mathfrak{o} = 0$ in C13 is load-bearing for CT1.2 and everything built on the W_i triplet, not cosmetic.

Theorem 1.16 (CT1.1 — Joint (S,T) decomposition of $\mathfrak{H}(\mathbb{C}^4)$). Thm. $[S,T]=0$; $\mathfrak{H}(\mathbb{C}^4)=\mathcal{E}_{++}\oplus\mathcal{E}_{+-}\oplus\mathcal{E}_{-+}\oplus\mathcal{E}_{--}$; $(\dim\mathcal{E}_{++},\dim\mathcal{E}_{+-},\dim\mathcal{E}_{-+},\dim\mathcal{E}_{--})=(7,3,3,3)$; $7+3+3+3=16$.

Req. A3, C13; C3; CT1.0; Pauli basis facts: $\mathfrak{T}(\sigma_\mu\sigma_\alpha)=2\delta_{\mu\alpha}$, $\sigma_\mu^\dagger=\sigma_\mu$ [linear algebra].

Pf.

$$\{\sigma_\mu\otimes\sigma_\nu\}_{\mu,\nu=0}^{3}\subset\mathfrak{H}(\mathbb{C}^4);\qquad \mathfrak{T}\big((\sigma_\mu\otimes\sigma_\nu)(\sigma_\alpha\otimes\sigma_\beta)\big)=4\,\delta_{\mu\alpha}\delta_{\nu\beta}$$

$$\Longrightarrow \text{ orthogonal } \Longrightarrow \text{ basis};\qquad \dim_{\mathbb{R}}\mathfrak{H}(\mathbb{C}^4)=16$$

$$\text{A3}:\ \ S:\ \sigma_\mu\otimes\sigma_\nu\mapsto\sigma_\nu\otimes\sigma_\mu\ (\text{permutation});$$

$$T:\ \sigma_\mu\otimes\sigma_\nu\mapsto\eta^\mu\eta^\nu\,\sigma_\mu\otimes\sigma_\nu,\ \ \eta^\mu\eta^\nu\in\{\pm1\}$$

$$\Longrightarrow\ S^2=T^2=\mathbb{1},\ \ \text{spec}\subseteq\{\pm1\};$$

$$ST(\sigma_\mu\otimes\sigma_\nu)=\eta^\mu\eta^\nu\,\sigma_\nu\otimes\sigma_\mu=TS(\sigma_\mu\otimes\sigma_\nu)\ \Longrightarrow\ [S,T]=0$$

$$[S,T]=0\ \wedge\ S^2=T^2=\mathbb{1}\ \Longrightarrow\ \text{simultaneous eigenspaces } \mathcal{E}_{st} \text{ span } \mathfrak{H}(\mathbb{C}^4)\quad [\text{C3}]$$

$$\eta:=(-,+,+,+)\ [\text{C13}]:\quad T=+1\iff(\mu,\nu)\in\{0,0\}\cup\{i,j\};$$

$$T=-1\iff(\mu,\nu)\in\{0,i\}\cup\{i,0\}$$

$$\mathcal{E}_{++}:\ \{\sigma_\mu\otimes\sigma_\mu\}_{\mu=0}^{3}\ (4)\ \cup\ \{\sigma_i\otimes\sigma_j+\sigma_j\otimes\sigma_i\}_{i<j}\ (3)=7$$

$$\mathcal{E}_{+-}:\ \{\sigma_0\otimes\sigma_i+\sigma_i\otimes\sigma_0\}_{i=1}^{3}=3;\qquad \mathcal{E}_{-+}:\ \{\sigma_i\otimes\sigma_j-\sigma_j\otimes\sigma_i\}_{i<j}=3$$

$$\mathcal{E}_{--}:\ \{\sigma_0\otimes\sigma_i-\sigma_i\otimes\sigma_0\}_{i=1}^{3}=3;\qquad 7+3+3+3=16\ \Longrightarrow\ \mathfrak{H}(\mathbb{C}^4)=\bigoplus_{s,t}\mathcal{E}_{st}.\ \blacksquare$$

Phys. The 16-dimensional space of two-qubit observables splits under exchange symmetry (S) and the sign grading T ($\eta^\mu\eta^\nu$; metric-sign reading by C13) into blocks $7+3+3+3$. This grading is the skeleton of the whole force structure: the 7-block supplies gravity, Higgs and uniform dark; the 3-blocks supply weak, Janus-seal, and the remaining virtual sector.

Theorem 1.17 (CT1.2 — Generator basis and sector assignment). Thm. Define

$$E_0:=\sigma_0\otimes\sigma_0;\qquad \Sigma:=\textstyle\sum_{i=1}^{3}\sigma_i\otimes\sigma_i;\qquad W_i:=\frac{1}{2}\big(\sigma_0\otimes\sigma_i+\sigma_i\otimes\sigma_0\big);$$

$$J_i:=\tfrac{1}{2}\big(\sigma_j\otimes\sigma_k-\sigma_k\otimes\sigma_j\big)\ [(i,j,k)\text{ cyclic}];$$

$$K_i := \tfrac{1}{2}\big(\sigma_0 \otimes \sigma_i - \sigma_i \otimes \sigma_0\big);$$

$$E^{ij} := \tfrac{1}{2}\big(\sigma_i \otimes \sigma_j + \sigma_j \otimes \sigma_i\big).$$

Then

generator	#	S	T	sector
E_0	1	$+$	$+$	$\mathcal{E}_{++}$
Σ	1	$+$	$+$	$\mathcal{E}_{++}$
$\hat{E}^{ij}$ (traceless)	5	$+$	$+$	$\mathcal{E}_{++}$
W_i	3	$+$	$-$	$\mathcal{E}_{+-}$
J_i	3	$-$	$+$	$\mathcal{E}_{-+}$
K_i	3	$-$	$-$	$\mathcal{E}_{--}$

$1 + 1 + 5 + 3 + 3 + 3 = 16 \implies$ basis of $\mathfrak{H}(\mathbb{C}^4)$;

$$W_i = \tfrac{1}{2}\big(\sigma_i^{(p)} + \sigma_i^{(q)}\big) = \text{total-spin generator.}$$

Req. A3, C1, C3, C13; CT1.0, CT1.1.
Pf.

$$SE_0 = E_0;\ \ \eta^0\eta^0 = +1 \implies E_0 \in \mathcal{E}_{++} \quad \text{[A3]}$$

$$SE^{ij} = E^{ij} \text{ (by construction)};\ \ \eta^i\eta^j = +1 \text{ [C13]} \implies E^{ij} \in \mathcal{E}_{++}; \qquad \#\{E^{ij}\}_{i\le j} = 6$$

$\delta_{ij}E^{ij} = \sum_i E^{ii} = \Sigma$ (1 component) $\oplus$ $\hat{E}^{ij}$ (5); $\qquad S, T$ diagonal on index pairs $\implies$ both $\subseteq \mathcal{E}_{++}$

$$W_i:\ S = +;\ \ \eta^0\eta^i = -1 \text{ [C13]} \implies T = - \implies W_i \in \mathcal{E}_{+-}$$

$$J_i:\ S = -;\ \ \eta^j\eta^k = +1 \text{ [C13]} \implies T = + \implies J_i \in \mathcal{E}_{-+}$$

$$K_i:\ S = -;\ \ \eta^0\eta^i = -1 \text{ [C13]} \implies T = - \implies K_i \in \mathcal{E}_{--}$$

counts $(1, 1, 5, 3, 3, 3)$ match $(7{=}1{+}1{+}5,\ 3,\ 3,\ 3)$ [CT1.1];

distinct orthogonal combinations $\implies$ independent $\implies$ basis

$$\sigma_0 \otimes \sigma_i \equiv \sigma_i^{(p)},\ \ \sigma_i \otimes \sigma_0 \equiv \sigma_i^{(q)} \implies W_i = \tfrac{1}{2}\big(\sigma_i^{(p)} + \sigma_i^{(q)}\big). \blacksquare$$

Phys. This fixes the named basis used throughout the corpus: identity E_0, exchange scalar Σ, total spin W_i (weak sector), the antisymmetric spatial and temporal generators J_i, K_i (Janus/seal sectors), and the symmetric spatial tensors E^{ij} (gravity). Each generator's (S, T) parity decides which force aspect of A8 it belongs to.

Theorem 1.18 (CT1.3 — Centrality of the identity). Thm. $\forall X \in \mathfrak{H}(\mathbb{C}^4): \quad [E_0, X] = 0.$

Req. C1; CT1.2.

Pf.

$$E_0 = \sigma_0 \otimes \sigma_0 = I_2 \otimes I_2 = I_4 \quad [\text{C1, CT1.2}]; \qquad [I_4, X] = I_4 X - X I_4 = 0. \blacksquare$$

Phys. The identity direction commutes with everything, so the aspect it carries — the uniform dark term $F_{D^+} = \Lambda \mathbb{I}$ — acts identically on every configuration: a cosmological-constant-type force.

Theorem 1.19 (CT1.4 — The W_i close $\mathfrak{su}(2)$). Thm. $[W_i, W_j] = i\,\epsilon_{ijk} W_k$; $\quad \text{span}_{\mathbb{R}}\{W_i\} \cong \mathfrak{su}(2)$ [C2].

Req. C2; CT1.2; $[\sigma_i, \sigma_j] = 2i\,\epsilon_{ijk}\sigma_k$ [Pauli algebra].

Pf.

$$\text{definitions cyclically symmetric} \implies \text{suffices}: \ [W_1, W_2] = iW_3$$

$$[W_1, W_2] = \tfrac{1}{4}\big[\sigma_0 \otimes \sigma_1 + \sigma_1 \otimes \sigma_0, \ \ \sigma_0 \otimes \sigma_2 + \sigma_2 \otimes \sigma_0\big] \quad [\text{CT1.2}]$$

$$[\sigma_0 \otimes \sigma_1, \ \sigma_0 \otimes \sigma_2] = \sigma_0^2 \otimes [\sigma_1, \sigma_2] = 2i\,(\sigma_0 \otimes \sigma_3)$$

$$[\sigma_0 \otimes \sigma_1, \ \sigma_2 \otimes \sigma_0] = \sigma_2 \otimes \sigma_1 - \sigma_2 \otimes \sigma_1 = 0; \qquad [\sigma_1 \otimes \sigma_0, \ \sigma_0 \otimes \sigma_2] = 0 \quad [\text{distinct factors}]$$

$$[\sigma_1 \otimes \sigma_0, \ \sigma_2 \otimes \sigma_0] = [\sigma_1, \sigma_2] \otimes \sigma_0^2 = 2i\,(\sigma_3 \otimes \sigma_0)$$

$$[W_1, W_2] = \tfrac{1}{4}\big(2i\,\sigma_0 \otimes \sigma_3 + 2i\,\sigma_3 \otimes \sigma_0\big) = \tfrac{i}{2}\big(\sigma_0 \otimes \sigma_3 + \sigma_3 \otimes \sigma_0\big) = iW_3. \blacksquare$$

Phys. The three symmetric mixed generators close a perfect $\mathfrak{su}(2)$: they are the total spin of the qubit pair. This $\mathfrak{su}(2)$ becomes the weak-isospin gauge algebra of the corpus — the weak force is the pair's rotation algebra, not an added postulate.

Theorem 1.20 (CT1.5 — Commutators of Σ with J_i, K_i, W_i). Thm. $[\Sigma, J_i] = 4i\,K_i$; $\quad [\Sigma, K_i] = -4i\,J_i$; $\quad [\Sigma, W_i] = 0.$

Req. CT0.7(1),(3) $[\Sigma = 2\mathbb{S} - E_0$; S-even $\Rightarrow [\mathbb{S}, \cdot] = 0]$, CT1.2, CT1.3; Pauli algebra.

Pf.

$$\text{cyclicity} \implies \text{suffices}: \ i = 1$$

$$[\sigma_1 \otimes \sigma_1, \ J_1] = 0:$$

$$[\sigma_1 \otimes \sigma_1, \ \sigma_2 \otimes \sigma_3] = (i\sigma_3) \otimes (-i\sigma_2) - (-i\sigma_3) \otimes (i\sigma_2) = 0; \quad \text{likewise } \sigma_3 \otimes \sigma_2$$

$$[\sigma_2 \otimes \sigma_2, \ \sigma_2 \otimes \sigma_3] = \sigma_0 \otimes [\sigma_2, \sigma_3] = 2i\,(\sigma_0 \otimes \sigma_1); \qquad [\sigma_2 \otimes \sigma_2, \ \sigma_3 \otimes \sigma_2] = 2i\,(\sigma_1 \otimes \sigma_0)$$

$$\implies [\sigma_2 \otimes \sigma_2,\ J_1] = \tfrac{1}{2}\big(2i\,\sigma_0 \otimes \sigma_1 - 2i\,\sigma_1 \otimes \sigma_0\big) = 2i\,K_1$$

$$[\sigma_3 \otimes \sigma_3,\ \sigma_2 \otimes \sigma_3] = -2i\,(\sigma_1 \otimes \sigma_0);$$

$$[\sigma_3 \otimes \sigma_3,\ \sigma_3 \otimes \sigma_2] = -2i\,(\sigma_0 \otimes \sigma_1) \implies [\sigma_3 \otimes \sigma_3,\ J_1] = 2i\,K_1$$

$$[\Sigma, J_1] = 0 + 2iK_1 + 2iK_1 = 4i\,K_1$$

$$[\sigma_1 \otimes \sigma_1,\ K_1] = 0 \text{ (as above)};$$

$$[\sigma_2 \otimes \sigma_2,\ \sigma_0 \otimes \sigma_1] = -2i\,(\sigma_2 \otimes \sigma_3);$$

$$[\sigma_2 \otimes \sigma_2,\ \sigma_1 \otimes \sigma_0] = -2i\,(\sigma_3 \otimes \sigma_2)$$

$$\implies [\sigma_2 \otimes \sigma_2,\ K_1] = \tfrac{1}{2}\big(-2i\,\sigma_2 \otimes \sigma_3 + 2i\,\sigma_3 \otimes \sigma_2\big) = -2i\,J_1$$

$$[\sigma_3 \otimes \sigma_3,\ \sigma_0 \otimes \sigma_1] = 2i\,(\sigma_3 \otimes \sigma_2);$$

$$[\sigma_3 \otimes \sigma_3,\ \sigma_1 \otimes \sigma_0] = 2i\,(\sigma_2 \otimes \sigma_3) \implies [\sigma_3 \otimes \sigma_3,\ K_1] = -2i\,J_1$$

$$[\Sigma, K_1] = -4i\,J_1$$

$\Sigma = 2\mathbb{S} - E_0$ [CT0.7(1)]; $SW_i = W_i$ (S-even) $\implies [\mathbb{S}, W_i] = 0$ [CT0.7(3)]; $[E_0, W_i] = 0$ [CT1.3]

$$\implies [\Sigma, W_i] = 2[\mathbb{S}, W_i] - [E_0, W_i] = 0. \blacksquare$$

Phys. The exchange scalar Σ rotates the two seal sectors into one another ($J \leftrightarrow K$, a 90° rotation inside the virtual sector) while leaving total spin untouched. The volume/trace mode therefore couples to Janus (edge creation–annihilation) dynamics but not to spin.

Theorem 1.21 (CT1.6 — Commutators $[J_i, J_j]$ and $[K_i, K_j]$). Thm. $[J_i, J_j] = i\,\epsilon_{ijk} W_k$; $[K_i, K_j] = i\,\epsilon_{ijk} W_k$.

Req. CT1.2; Pauli algebra.

Pf.

$$\text{suffices : } [J_1, J_2] = iW_3 = [K_1, K_2]$$

$$[J_1, J_2] = \tfrac{1}{4}\big[\sigma_2 \otimes \sigma_3 - \sigma_3 \otimes \sigma_2,\ \ \sigma_3 \otimes \sigma_1 - \sigma_1 \otimes \sigma_3\big]$$

$$[\sigma_2\otimes\sigma_3,\ \sigma_3\otimes\sigma_1] = (i\sigma_1)\otimes(i\sigma_2) - (-i\sigma_1)\otimes(-i\sigma_2) = 0$$

$$[\sigma_2\otimes\sigma_3,\ -\sigma_1\otimes\sigma_3] = -[\sigma_2,\sigma_1]\otimes\sigma_3^2 = 2i\,(\sigma_3\otimes\sigma_0)$$

$$[-\sigma_3\otimes\sigma_2,\ \sigma_3\otimes\sigma_1] = -\sigma_0\otimes[\sigma_2,\sigma_1] = 2i\,(\sigma_0\otimes\sigma_3)$$

$$[-\sigma_3\otimes\sigma_2,\ -\sigma_1\otimes\sigma_3] = (i\sigma_2)\otimes(i\sigma_1) - (-i\sigma_2)\otimes(-i\sigma_1) = 0$$

$$\Longrightarrow\ [J_1,J_2] = \tfrac{1}{4}\big(2i\,\sigma_3\otimes\sigma_0 + 2i\,\sigma_0\otimes\sigma_3\big) = iW_3$$

$$[K_1,K_2]:\ \ \text{cross terms} = 0\ \text{[distinct factors]};$$

$$[K_1,K_2] = \tfrac{1}{4}\Big([\sigma_0\otimes\sigma_1,\sigma_0\otimes\sigma_2] + [\sigma_1\otimes\sigma_0,\sigma_2\otimes\sigma_0]\Big)$$

$$= \tfrac{1}{4}\big(2i\,\sigma_0\otimes\sigma_3 + 2i\,\sigma_3\otimes\sigma_0\big) = iW_3.\ \blacksquare$$

Phys. Neither seal sector closes on itself: brackets of two J's or two K's land in the total-spin (weak) sector. Reality-flipping operations source the rotational/weak structure of the pair — the first algebraic link between the Janus dynamics and gauge structure.

Theorem 1.22 (CT1.7 — Mixed commutators $[J_i,K_j]$). Thm.

$$[J_i,K_j] = i\big(\delta_{ij}\Sigma - E^{ij}\big) = \tfrac{i}{2}\Big(2\delta_{ij}\Sigma - \sigma_i\otimes\sigma_j - \sigma_j\otimes\sigma_i\Big);$$

$$[J_i,K_i] = i\sum_{k\neq i}\sigma_k\otimes\sigma_k; \qquad [J_i,K_j] = -iE^{ij} = -\tfrac{i}{2}\big(\sigma_i\otimes\sigma_j + \sigma_j\otimes\sigma_i\big)\ \ (i\neq j);$$

$$[J_i,K_j]\in(\mathcal{E}_{++})_{\text{sym spatial}}; \qquad [J_i,K_j]\notin \text{span}\{W_k\}.$$

Req. CT1.1, CT1.2; ϵ-identities: $\epsilon_{iab}\epsilon_{bjm} = \delta_{ij}\delta_{am} - \delta_{im}\delta_{aj},\ \epsilon_{iab}\epsilon_{ajm} = \delta_{bj}\delta_{im} - \delta_{bm}\delta_{ij}$.
Pf.

$$J_i = \tfrac{1}{2}\,\epsilon_{iab}\,\sigma_a\otimes\sigma_b\ (a,b\in\{1,2,3\}) \quad \text{[CT1.2]}; \qquad K_j = \tfrac{1}{2}\big(\sigma_0\otimes\sigma_j - \sigma_j\otimes\sigma_0\big)$$

$$[\sigma_a\otimes\sigma_b,\ \sigma_0\otimes\sigma_j] = \sigma_a\otimes[\sigma_b,\sigma_j] = 2i\,\epsilon_{bjm}\,\sigma_a\otimes\sigma_m$$

$$[\sigma_a\otimes\sigma_b,\ \sigma_j\otimes\sigma_0] = [\sigma_a,\sigma_j]\otimes\sigma_b = 2i\,\epsilon_{ajm}\,\sigma_m\otimes\sigma_b$$

$$\Longrightarrow\ [J_i,K_j] = \tfrac{i}{2}\,\epsilon_{iab}\big(\epsilon_{bjm}\,\sigma_a\otimes\sigma_m - \epsilon_{ajm}\,\sigma_m\otimes\sigma_b\big)$$

$$\epsilon_{iab}\epsilon_{bjm}\,\sigma_a\otimes\sigma_m = \delta_{ij}\Sigma - \sigma_j\otimes\sigma_i; \qquad \epsilon_{iab}\epsilon_{ajm}\,\sigma_m\otimes\sigma_b = \sigma_i\otimes\sigma_j - \delta_{ij}\Sigma$$

$$[J_i, K_j]$$

$$= \tfrac{i}{2}\Big(\delta_{ij}\Sigma - \sigma_j \otimes \sigma_i - \sigma_i \otimes \sigma_j + \delta_{ij}\Sigma\Big)$$

$$= \tfrac{i}{2}\big(2\delta_{ij}\Sigma - \sigma_i \otimes \sigma_j - \sigma_j \otimes \sigma_i\big)$$

$$= i\big(\delta_{ij}\Sigma - E^{ij}\big)$$

$$i = j: \;\; i\big(\Sigma - \sigma_i \otimes \sigma_i\big) = i\sum_{k\neq i}\sigma_k \otimes \sigma_k; \qquad i \neq j: \;\; \delta_{ij} = 0 \implies -iE^{ij}$$

$$\Sigma,\; E^{ij} \in \mathcal{E}_{++};\;\; W_k \in \mathcal{E}_{+-};\;\; \text{direct sum [CT1.1, CT1.2]} \implies [J_i, K_j] \notin \text{span}\{W_k\}. \;\blacksquare$$

Phys. Bracketing a spatial seal with a temporal seal produces symmetric spatial — metric-type — data, never weak data. Mixed virtual processes feed the gravitational sector: the algebraic seed of "quantum fluctuations sourcing geometry."

Theorem 1.23 (CT1.8 — Action of W_i on J_j and K_j). Thm. $[W_i, J_j] = i\,\epsilon_{ijk}J_k$; $\quad [W_i, K_j] = i\,\epsilon_{ijk}K_k$; i.e. $\{J_i\}$, $\{K_i\}$: $SO(3)$ vectors under span$\{W_i\}$.

Req. CT1.2; Pauli algebra.

Pf.

$$\text{suffices}: \;\; [W_3, J_1] = iJ_2, \quad [W_3, K_1] = iK_2$$

$$[W_3, J_1] = \tfrac{1}{4}\big[\sigma_0 \otimes \sigma_3 + \sigma_3 \otimes \sigma_0,\;\; \sigma_2 \otimes \sigma_3 - \sigma_3 \otimes \sigma_2\big]$$

$$[\sigma_0 \otimes \sigma_3,\; \sigma_2 \otimes \sigma_3] = \sigma_2 \otimes [\sigma_3, \sigma_3] = 0;$$

$$[\sigma_0 \otimes \sigma_3,\; -\sigma_3 \otimes \sigma_2] = -\sigma_3 \otimes [\sigma_3, \sigma_2] = 2i\,(\sigma_3 \otimes \sigma_1)$$

$$[\sigma_3 \otimes \sigma_0,\; \sigma_2 \otimes \sigma_3] = [\sigma_3, \sigma_2] \otimes \sigma_3 = -2i\,(\sigma_1 \otimes \sigma_3); \qquad [\sigma_3 \otimes \sigma_0,\; -\sigma_3 \otimes \sigma_2] = 0$$

$$\implies [W_3, J_1] = \tfrac{i}{2}\big(\sigma_3 \otimes \sigma_1 - \sigma_1 \otimes \sigma_3\big) = iJ_2$$

$$[W_3, K_1] = \tfrac{1}{4}\big[\sigma_0 \otimes \sigma_3 + \sigma_3 \otimes \sigma_0,\;\; \sigma_0 \otimes \sigma_1 - \sigma_1 \otimes \sigma_0\big]$$

$$[\sigma_0 \otimes \sigma_3,\; \sigma_0 \otimes \sigma_1] = 2i\,(\sigma_0 \otimes \sigma_2); \qquad [\sigma_0 \otimes \sigma_3,\; -\sigma_1 \otimes \sigma_0] = 0 = [\sigma_3 \otimes \sigma_0,\; \sigma_0 \otimes \sigma_1]$$

$$[\sigma_3 \otimes \sigma_0,\ -\sigma_1 \otimes \sigma_0] = -[\sigma_3, \sigma_1] \otimes \sigma_0 = -2i\,(\sigma_2 \otimes \sigma_0)$$

$$\Longrightarrow\ [W_3, K_1] = \tfrac{1}{4}\big(2i\,\sigma_0 \otimes \sigma_2 - 2i\,\sigma_2 \otimes \sigma_0\big) = iK_2; \qquad \text{cyclicity} \implies \text{general form.}\ \blacksquare$$

Phys. Total spin rotates both virtual sectors as vectors: the seal generators carry spin 1 under the weak $\mathfrak{su}(2)$. The reality/virtuality bookkeeping is rotation-covariant — angular momentum is built into the ledger before any dynamics is introduced.

Theorem 1.24 (CT1.9 — Sector closure table and the global algebra). Thm.

bracket	closes in
$[W_i, W_j]$	$\mathrm{span}\{W_k\} = \mathcal{E}_{+-}$
$[W_i, J_j]$	$\mathrm{span}\{J_k\} = \mathcal{E}_{-+}$
$[W_i, K_j]$	$\mathrm{span}\{K_k\} = \mathcal{E}_{--}$
$[J_i, J_j]$	$\mathrm{span}\{W_k\}$
$[K_i, K_j]$	$\mathrm{span}\{W_k\}$
$[J_i, K_j]$	$(\mathcal{E}_{++})_{\text{sym spatial}}$
$[\Sigma, J_i]$	$\mathrm{span}\{K_i\}$
$[\Sigma, K_i]$	$\mathrm{span}\{J_i\}$
$[\Sigma, W_i]$	0
$[E_0, X]$	$0\ \ \forall X$

$$\big(\mathfrak{H}(\mathbb{C}^4),\ (X,Y) \mapsto -i[X,Y]\big) \cong \mathfrak{u}(4)\ \text{[C2]}; \qquad Z\big(\mathfrak{u}(4)\big) = \mathrm{span}_{\mathbb{R}}\{E_0\};$$

$$\mathfrak{su}(4) = \mathrm{span}_{\mathbb{R}}\big\{\Sigma,\ \hat{E}^{ij},\ W_i,\ J_i,\ K_i\big\}; \qquad \dim_{\mathbb{R}} \mathfrak{su}(4) = 1 + 5 + 3 + 3 + 3 = 15.$$

Req. C2; CT1.1–CT1.8; $\mathfrak{T}(\sigma_\mu \otimes \sigma_\nu) = 4\,\delta_{\mu 0}\delta_{\nu 0}$.
Pf.

$$\text{rows 1–5}: \ [\text{CT1.4},\ \text{CT1.8},\ \text{CT1.6}];$$

$$\text{row 6}: \ [\text{CT1.7}];$$

$$\text{rows 7–9}: \ [\text{CT1.5}];$$

$$\text{row 10}: \ [\text{CT1.3}]$$

$$\text{sector labels}: \ [\text{CT1.2}]$$

$$X, Y \in \mathfrak{H}(\mathbb{C}^4)\ \Longrightarrow\ (-i[X,Y])^\dagger = -i[X,Y];$$

$$\text{bilinear} \ \wedge \ \text{antisymmetric} \ \wedge \ \text{Jacobi} \, [\text{associative commutator}]$$

$$\dim_{\mathbb{R}} \mathfrak{H}(\mathbb{C}^4) = 16 \ [\text{CT1.1}] \implies \big(\mathfrak{H}(\mathbb{C}^4), -i[\cdot,\cdot]\big) = \mathfrak{u}(4) \quad [\text{C2}]$$

$$X \in Z \iff [X,Y] = 0 \ \ \forall Y \in \mathfrak{H}(\mathbb{C}^4) \implies X \text{ scalar} \implies Z = \text{span}_{\mathbb{R}}\{E_0\} \quad [\text{CT1.3}]$$

$$\mathfrak{T}(\sigma_\mu \otimes \sigma_\nu)$$

$$= 4$$

$$\delta_{\mu 0}\delta_{\nu 0}$$

$$\implies E_0 \text{ alone has trace} \implies \text{traceless complement}$$

$$= \mathfrak{su}(4),$$

$$\dim$$

$$= 16 - 1 \qquad = 15$$

$$\mathfrak{su}(4) = \text{span}_{\mathbb{R}}\{\Sigma, \hat{E}^{ij}, W_i, J_i, K_i\} : \quad 1 + 5 + 3 + 3 + 3 = 15. \ \blacksquare$$

Phys. The full commutator structure closes: the 16 pair observables form $\mathfrak{u}(4) = \mathfrak{u}(1) \oplus \mathfrak{su}(4)$, with the central $\mathfrak{u}(1)$ (E_0) as the uniform dark aspect and the 15 traceless directions carrying gravity (5), weak (3), the two seal sectors (3+3) and the exchange scalar (1). Every force algebra later in the corpus is a subalgebra of this one 16-dimensional algebra.

Theorem 1.25 (CT1.10 — $SO(3)$ decomposition of $\mathcal{E}_{++}$). Thm. Under span$\{W_i\}$:

$$\mathcal{E}_{++} \ \longrightarrow \ 1 \oplus 5 \oplus 1 \qquad (7 = 1 + 5 + 1);$$

$$E_0 : \ 1; \qquad \Sigma : \ 1; \qquad \hat{E}^{ij} := E^{ij} - \tfrac{1}{3}\delta^{ij}\Sigma : \ 5 \ (\text{spin } 2), \quad \delta_{ij}\hat{E}^{ij} = 0.$$

Req. CT1.1, CT1.3, CT1.5, CT1.8; symmetric traceless rank-2 tensor $= \ell = 2$ representation [$SO(3)$ representation theory].

Pf.

$$\dim \mathcal{E}_{++} = 7 \quad [\text{CT1.1}]$$

$$[E_0, W_i] = 0 \ [\text{CT1.3}]; \ \ E_0 \text{ index-free} \implies E_0 : \ 1$$

$$E^{ij} = \underbrace{\tfrac{1}{3}\delta^{ij}\Sigma}_{\text{trace}} + \underbrace{\hat{E}^{ij}}_{\text{traceless}} \quad \text{(unique split)}; \qquad \delta_{ij}\hat{E}^{ij} = \Sigma - \tfrac{1}{3}\cdot 3\cdot\Sigma = 0$$

$$[\Sigma, W_i] = 0 \text{ [CT1.5]} \implies \Sigma:\ 1$$

$$[W_a, \sigma_i \otimes \sigma_j] = i\,\epsilon_{aik}\,\sigma_k \otimes \sigma_j + i\,\epsilon_{ajk}\,\sigma_i \otimes \sigma_k \quad \text{[CT1.2, Pauli algebra]}$$

$$\implies [W_a, E^{ij}] = i\big(\epsilon_{aik}E^{kj} + \epsilon_{ajk}E^{ik}\big);$$

$$[W_a, \hat{E}^{ij}] = [W_a, E^{ij}] - \tfrac{1}{3}\delta^{ij}\underbrace{[W_a, \Sigma]}_{=0\ \text{[CT1.5]}} = i\big(\epsilon_{aik}\hat{E}^{kj} + \epsilon_{ajk}\hat{E}^{ik}\big)$$

$$\implies \hat{E}^{ij}:\ \text{traceless symmetric rank-2 tensor} = \ell = 2:\ 5 \quad [SO(3)\ \text{representation theory}]$$

$$1 + 1 + 5 = 7 = \dim \mathcal{E}_{++} \implies \mathcal{E}_{++} = 1 \oplus 5 \oplus 1. \blacksquare$$

Phys. Seen by rotations, the real sector contains exactly two scalars and one five-component spin-2 object. The spin-2 quintet is the graviton field of the corpus; with the three W_i as shift, the $(7, 3)$ real blocks supply the ten components of a Lorentzian metric (CT1.12): conformal (1) + traceless metric (5) + volume (1) + shift (3).

Theorem 1.26 (CT1.11 — The two scalars are distinct). Thm. $E_0 \in Z\big(\mathfrak{H}(\mathbb{C}^4)\big)$; $\Sigma \notin Z\big(\mathfrak{H}(\mathbb{C}^4)\big)$; $\forall c \in \mathbb{R}\setminus\{0\}:\ E_0 \neq c\,\Sigma$; hence the A8 assignment $E_0 \mapsto F_{D^+}$, $\Sigma \mapsto F_H$ [CT0.12] is well-defined and non-degenerate.

Req. A8; CT0.12, CT1.3, CT1.5.

Pf.

$$[E_0, X] = 0\ \ \forall X \implies E_0 \in Z \quad \text{[CT1.3]}$$

$$[\Sigma, J_1] = 4i\,K_1 \neq 0\ \ (K_1 \neq 0) \implies \Sigma \notin Z \quad \text{[CT1.5]}$$

$$c \neq 0:\ \ X \in Z \iff cX \in Z \implies \big[E_0 \in Z\ \wedge\ \Sigma \notin Z\big] \implies E_0 \neq c\,\Sigma\ \ \forall c \neq 0$$

$$\implies E_0,\ \Sigma:\ \text{independent directions of } \mathcal{E}_{++}; \qquad E_0 \mapsto F_{D^+} = \Lambda\,\mathbb{I},\ \ \Sigma \mapsto F_H \quad \text{[A8, CT0.12]}. \blacksquare$$

Phys. The two rotation scalars are provably different objects: E_0 is central — it cannot distinguish configurations — and supplies the uniform dark aspect $\Lambda\mathbb{I}$; Σ is non-central — it couples to the seals and to geometry — and supplies the scalar (Higgs-type) aspect. The corpus's two scalar directions are established as distinct before either receives a physical name.

Theorem 1.27 (CT1.11a — The involution table of the four sectors). Thm. (0) Dictionary. A8's four sectors are CT1.1's blocks under the identification

$$\mathrm{F}_{(+,\mathcal{R})} = F_G \oplus F_H \oplus F_{D^+}\ (5{+}1{+}1) \ \leftrightarrow\ \mathcal{E}_{++};$$

$$\mathrm{F}_{(-,\mathcal{R})} = F_{W^+} \oplus F_{W^-} \oplus F_Z\ (3) \ \leftrightarrow\ \mathcal{E}_{+-};$$

$$\mathrm{F}_{(+,\mathcal{V})} = F_{J^+}\ (3) \ \leftrightarrow\ \mathcal{E}_{-+}; \qquad \mathrm{F}_{(-,\mathcal{V})} = F_{J^-}\ (3) \ \leftrightarrow\ \mathcal{E}_{--}$$

— first label = the T-grading ($\eta^\mu\eta^\nu$ [A3, C13]), $\mathcal{R}/\mathcal{V}$ = the S-grading (reality is the swap grading [CT0.7]); machine-verified on CT1.2's generator table (17 generators: $E_0, \Sigma, \hat{E}^{ij}{\times}5, W_i, J_i, K_i$). (1) Status of the two gradings. $S = \mathrm{Ad}(\mathbb{S})$ with $\mathbb{S} = \frac{1}{2}(E_0 + \Sigma)$ [CT0.7(1)] — an inner algebra automorphism ($\mathbb{S}^2 = \mathbb{S}$? no: $\mathbb{S}$ unitary involution, machine). T is a pure grading: the diagonal ±1-map $\sigma_\mu \otimes \sigma_\nu \ \mapsto\ \eta^\mu\eta^\nu\, \sigma_\mu \otimes \sigma_\nu$ [A3] — $\neg$ multiplicative (witness: $T(\sigma_0 \otimes \sigma_1)\,T(\sigma_0 \otimes \sigma_2) = (-1)(-1)$ [C13] $\sigma_0 \otimes (\sigma_1\sigma_2) \neq T(\sigma_0 \otimes \sigma_1\sigma_2)$-type failure, machine), hence $\neg$ inner. $[S, T] = 0$ on all 16 basis elements [CT1.1]; the grading group on $\mathfrak{H}(\mathbb{C}^4)$ is $\mathbb{Z}_2 \times \mathbb{Z}_2$ with simultaneous eigenspaces $\mathcal{E}_{st}$, dim $= (7, 3, 3, 3)$. (2) Table of the declared reversals. With ω [D2.6] and $\Theta_{\mathfrak{L}}$ [D2.7]:

	$\mathrm{F}_{(+,\mathcal{R})}$	$\mathrm{F}_{(-,\mathcal{R})}$	$\mathrm{F}_{(+,\mathcal{V})}$	$\mathrm{F}_{(-,\mathcal{V})}$	edge/chain data
$S = \mathrm{Ad}(\mathbb{S})$	+	+	−	−	reality grading [CT0.7]
T (grading)	+	−	+	−	$\eta^\mu\eta^\nu$ [A3, C13]
ω	fixed	$W^+ \leftrightarrow W^-$; Z fixed	¬declared	¬declared	$s(c) \leftrightarrow t(c)$; charge $\mapsto -$charge [D2.6, EM.9]; $\omega^2 = 1$; $R \mapsto R^{\mathrm{T}}$; $\omega \notin \mathrm{Aut}(\Gamma)$, $\lvert V\rvert \geq 3$
$\Theta_{\mathfrak{L}}$	fixed	fixed	$\leftrightarrow$	$\leftrightarrow$	ledger order $\mapsto$ reverse; $J^+ \leftrightarrow J^-$; $G(e) \mapsto G(e)^*$ ($\alpha_e \widehat{ds}^2(e) \mapsto -\alpha_e \widehat{ds}^2(e)$, σ_e fixed [D3.5]); $\mathcal{J} \mapsto \mathcal{J}^*$ [D3.6]; $w \mapsto w^*$ [CT2.8(1)]; $\Theta^2 = 1$
$\omega\Theta_{\mathfrak{L}}$	fixed	$W^+ \leftrightarrow W^-$	$\leftrightarrow$	$\leftrightarrow$	the fourth class

ω acts on the edge set, $\Theta_{\mathfrak{L}}$ on phase/ledger order — disjoint data $\Rightarrow$ they commute, and $\langle \omega, \Theta_{\mathfrak{L}} \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ with classes $(1, \omega, \Theta_{\mathfrak{L}}, \omega\Theta_{\mathfrak{L}})$; ω is charge-odd [D2.6; covariance of the closed-core charge read: EM.9a], $\Theta_{\mathfrak{L}}$ phase-odd and charge-even [D2.7, D3.5, D3.6]. (3) Edge-class answer (the sector question at edge level). Real edges split into $\sigma_e = \pm1$ [CT0.9: $\Phi = \{0, \pi\}$]; virtual edges carry ¬registration [A5, A7, D0.5: ϕ_e defined on $E_{\mathcal{R}}$ only] — a proposed $(\pm, \mathcal{V})$ split has no read: the virtual class is single and ungraded (weight 1, no phase). The two sign structures (σ vs $(-1)^{\aleph}$) are independent [CT0.10]. (4) [READING] rows (recorded, ¬adopted). External CPT vocabulary maps as: $C \leftrightarrow \omega$ (the charge-odd reversal [D2.6]; sign-level realization $\sigma_e \mapsto -\sigma_e$: EM.24(1) — E, forward pointer); $T \ \leftrightarrow\ \Theta_{\mathfrak{L}}$ (ledger-time reversal [D2.7] — $\neg$ anti-unitary: no anti-unitary operator is declared; conjugation acts on weights/phases [CT2.8(1), D3.6]); $P \leftrightarrow$ the alignment-pattern reversal $\mathcal{P} \mapsto E_{\mathcal{R}} \setminus \mathcal{P}$ (equivalently $\alpha_e \mapsto \pi - \alpha_e$) with the mirror classification of Def-EM.21/Def-EM.22, EM.23 — E, forward pointers: a pattern-level reversal on configuration data (spacetime parity as a Lorentzian field-algebra read has count 0 in Parts 1–2; the orientation read $\chi = \mathrm{sign}(k)$ remains [CANDIDATE], not adopted); "CPT" $\leftrightarrow$ the derived composition $C \circ P \circ T$ on $\mathfrak{C}(\mathcal{R})$ with unconditional $|w|^2$-invariance [EM.24(4)–(5) — E, forward pointer]; at the involution-table level its reversal part is the $\omega\Theta_{\mathfrak{L}}$ class of (2). This is the exact amount of reversal structure the axioms declare — $\neg$ a Lorentzian CPT theorem (no Lorentzian field algebra, no spinor bundle in the inventory).

Req. A3, A5, A7, A8; C3; D0.5, D2.6, D2.7, D3.5, D3.6; CT0.7, CT0.7a, CT0.9, CT0.10, CT0.12, CT1.1, CT1.2, CT1.6, CT1.9, CT1.11 (Part 1); CT2.8(1) (C — forward pointer, re-verification); EM.9 (E — forward pointer, re-verification); exact $\mathbb{Q}(i)$ matrix arithmetic, exhaustive pattern enumeration, non-multiplicativity witness [math; machine: `checks/verify_merge_tierA.py` §E].

Pf.

$$(0):\ \text{CT1.2′s table verbatim: } E_0,$$

$$\Sigma,$$

$$\hat{E}^{ij} \in \mathcal{E}_{++};$$

$$W_i \in \mathcal{E}_{+-};$$

$$J_i \in \mathcal{E}_{-+};$$

$$K_i \in \mathcal{E}_{--} \ \wedge\ \text{CT0.12′s aspect rows [machine]}$$

$$(1):\ \mathbb{S} = \tfrac{1}{2}(E_0 + \Sigma)\ \text{[CT0.7(1)]};\quad \mathrm{Ad}(\mathbb{S})(X) = \mathbb{S}X\mathbb{S} = S(X)\ \text{[CT0.7(2)]};$$

$$T^2 = \nVdash,\ S^2 = \nVdash,\ [S,T] = 0\ \text{[A3, CT1.1]}$$

$$T\ \neg\text{multiplicative}:\ T(\sigma_0 \otimes \sigma_1)$$

$$T(\sigma_0 \otimes \sigma_2)$$

$$= (-\sigma_0 \otimes \sigma_1)(-\sigma_0 \otimes \sigma_2)$$

$$= \sigma_0 \otimes i\sigma_3$$

$$\neq -$$

$$i$$

$$\sigma_0 \otimes \sigma_3$$

$$= T(\sigma_0 \otimes i\sigma_3)\ \text{[machine witness]}$$

$$(2):\ D2.6/D2.7 \text{ verbatim on data rows;}$$

$$J^{\pm} \text{ blocks}:\ \Theta_{\mathfrak{L}}:\ J^{+} \leftrightarrow J^{-}\ \text{[D2.7]}\ \wedge\ \mathrm{F}_{(+,\mathcal{V})}$$

$$= F_{J^+},$$

$$\mathrm{F}_{(-,\mathcal{V})}$$

$$= F_{J^-} \text{ [A8] } \Longrightarrow \text{ sector exchange}$$

$$\omega : \text{ charge}$$

$$\mapsto -\text{charge [D2.6]}$$

$$\Longrightarrow W^+ \leftrightarrow W^- \text{ inside } \mathrm{F}_{(-,\mathcal{R})} \text{ [charge eigen-decomposition of the weak block: } F_{W^+} \oplus F_{W^-} \oplus F_Z, \text{ A8]};$$

$$\omega \text{ on } J^{\pm} : \neg\text{declared}$$

$$\omega^2 = 1 : \ (R^{\mathrm{T}})^{\mathrm{T}} = R;$$

$$\omega \notin \mathrm{Aut}(\Gamma) : \ p(a)$$

$$= b \ \wedge \ p(b)$$

$$= a \ \forall (a, b)$$

$$\in E : \ \perp \text{ for } |V|$$

$$\geq 3 \text{ [machine: exhaustive on the tetrahedron]}$$

$$\text{disjoint supports (edge set vs phase/order) } \Longrightarrow \ [\omega, \Theta_{\mathfrak{L}}] = 0; \qquad \omega^2 = \Theta_{\mathfrak{L}}^2 = 1 \ \Longrightarrow \ \mathbb{Z}_2 \times \mathbb{Z}_2. \ \blacksquare$$

Phys. The corpus declares exactly two reversals and they commute: one flips orientation and charge on the edge set, the other flips ledger order and conjugates phases — four classes in total. The four force sectors of A8 are the joint grading atlas of the pair register; the Janus pair is the only sector pair a declared reversal exchanges, and spacetime parity has no force-sector house here — its pattern-level read lives on the alignment data (EM.23, EM.24 — E). The table is the complete reversal inventory at sector level: nothing further is derivable without a new declaration.

Theorem 1.28 (CT1.11b — The native Clifford frames of the pair register and the weak-covariance pincer). Thm. On $\mathfrak{H}(\mathbb{C}^4)$ with CT1.2's named basis and CT0.7's $\mathbb{S}$, $\Pi_{\mathcal{R}} = \frac{1}{4}(3E_0 + \Sigma)$, $\Pi_{\mathcal{V}} = \frac{1}{4}(E_0 - \Sigma)$. Call the 15 non-identity $\sigma_\mu \otimes \sigma_\nu$ [A3, CT1.1] the Pauli products — each Hermitian, squaring to $+I_4$, traceless. A frame is a 4-subset of pairwise-anticommuting Pauli products; for a frame $\mathcal{F} = (e_1, e_2, e_3, e_4)$ let $\gamma_{\mathcal{F}} := e_1e_2e_3e_4$ (its pseudoscalar; frame- and order-dependent: every transposition flips its sign). (1) Native frames. There are exactly 30 frames (unordered; 720 as ordered tuples) among the $\binom{15}{4} = 1365$ candidates. For every frame: $\gamma_{\mathcal{F}}$ is Hermitian, $\gamma_{\mathcal{F}}^2 = I_4$, $\mathfrak{T}\gamma_{\mathcal{F}} = 0$, $\{\gamma_{\mathcal{F}}, e_i\} = 0$; and the 16 real products $\{e_{i_1} \cdots e_{i_r}\}_{r=0..4}$ are $\mathbb{R}$-linearly independent — the frame generates a real subalgebra isomorphic to $Cl(4,0)$ acting on $\mathbb{C}^4 \cong \mathbb{R}^8$, with no new axiom: A3's Pauli facts and CT1.1's basis suffice. The signature is Euclidean ($e_i^2 = +I_4$); the representative $\eta = (-,+,+,+)$ [C13] of A3's sign vector labels the T-grading only [CT1.1] and enters no Clifford relation. No canonical pseudoscalar exists: a distinguished "γ-element" would be a choice, ¬a derivation. (2) The pseudoscalar catalog (exhaustive). All 15 Pauli products occur as $\pm\gamma_{\mathcal{F}}$, each from exactly 2 frames, in three species:

species	$\gamma_{\mathcal{F}}$ (up to sign)	$[\gamma_{\mathcal{F}}, \mathbb{S}]$	commuting W_i
mixed (6)	$\sigma_0 \otimes \sigma_i$ / $\sigma_i \otimes \sigma_0$	$\pm 2iJ_i$	W_i only
diagonal (3)	$\sigma_i \otimes \sigma_i \in \mathcal{E}_{++}$	0	W_i only (anticomm. W_j, W_k)
off-diag. spatial (6)	$\sigma_i \otimes \sigma_j$ $(i \neq j)$	$\mp 2iK_k,\ k = 6-i-j$	none (anticomm. W_k)

No pseudoscalar commutes with more than one W_i; none anticommutes with more than two. Witness frame $\mathcal{F}_0 = (\sigma_1 \otimes \sigma_2,\ \sigma_2 \otimes \sigma_2,\ \sigma_3 \otimes \sigma_2,\ \sigma_0 \otimes \sigma_1)$:

$$\gamma_{\mathcal{F}_0} = \sigma_0 \otimes \sigma_3 = \mathrm{diag}(1, -1, 1, -1);$$

$$P_{\pm}^{(\mathcal{F}_0)} := \tfrac{1}{2}(I_4 \mp \gamma_{\mathcal{F}_0}) = \mathrm{diag}(0,1,0,1) \ /\ \mathrm{diag}(1,0,1,0)$$

— both idempotent, orthogonal, summing to I_4, trace 2, rank 2 (exact forms, machine). (3) The weak-covariance pincer. No grading involution of the pair register can be weak-covariant, from either side: (a) commuting side. $\{M : [M, W_i] = 0\ \forall i\} = \mathrm{span}_{\mathbb{C}}\{\Pi_{\mathcal{R}}, \Pi_{\mathcal{V}}\}$ exactly (complex dimension 2 — Schur on the $3 \oplus 1$ decomposition [CT1.10, CT0.7]; machine nullspace solve). Its involutions are $a\Pi_{\mathcal{R}} + b\Pi_{\mathcal{V}}$ with $a^2 = b^2 = 1$, traces $3a + b \in \{\pm 4, \pm 2\}$; a pseudoscalar needs trace 0 (cyclic flip: $\mathfrak{T}(e_1e_2e_3e_4) = -\mathfrak{T}(e_2e_3e_4e_1)$) — $3a + b = 0 \ \wedge\ a^2 = b^2 = 1$ has no solution. (b) anticommuting side. $\{M : \{M, W_i\} = 0\ \forall i\} = \mathbb{C} \cdot \Pi_{\mathcal{V}}$ exactly (complex dimension 1; machine nullspace solve): rank 1, trace 1 — singular, never an involution. Structural reason: an invertible M anticommuting with every W_i would make $\mathrm{Ad}(M)$ a Lie automorphism of $\mathrm{span}\{W_i\}$ equal to $-\mathrm{id}$ — but $-\mathrm{id}$ reverses the bracket ($[W_1, W_2] = iW_3 \mapsto -iW_3 \neq iW_3$) [CT1.4]: contradiction. (c) exhaustive. Every one of the 15 attained pseudoscalars fails (a) and (b) explicitly, entrywise: for each, $[\gamma, W_i] \neq 0$ for some i and $\{\gamma, W_i\} \neq 0$ for some i. (d) carrier side. On any 2-dim carrier with weak action $W_i \mapsto \frac{1}{2}\sigma_i$ (the weak seed locus: EW.32): $\{m : [m, \sigma_i] = 0\ \forall i\} = \mathbb{C} \cdot I_2$ and $\{m : \{m, \sigma_i\} = 0\ \forall i\} = \{0\}$ — the weak seed representation is irreducible: no nontrivial projector commutes with the weak action, and no grade anticommutes with it. (4) The native selectivity (what the weak force does grade). The weak force grades the pair arena by S-parity: $W_i\Pi_{\mathcal{V}} = 0$ exactly $\forall i$; $\sum_i W_i^2 = 2\Pi_{\mathcal{R}}$ [Cor CT0.7a]; $\bigcap_i \ker W_i = \mathrm{im}\,\Pi_{\mathcal{V}}$ exactly (the 1-dim singlet line; machine) — a 3-active/1-inactive split. This is a force-arena selectivity; a matter grade would have to live on a carrier on which the gauge action is trivial — the opposite pattern. (5) Non-alignment. $[\gamma_{\mathcal{F}}, \mathbb{S}] = 0 \iff \gamma_{\mathcal{F}} = \pm\,\sigma_i \otimes \sigma_i$: the S-definite pseudoscalars are exactly the three diagonal

ones — they live in $\mathcal{E}_{++}$, the gravity/Higgs block [CT1.1, CT1.2], and are still ¬weak-covariant (each anticommutes with the two non-own W: e.g. $[\sigma_3 \otimes \sigma_3,\ W_1] = 2iE^{23}$ exactly — the weak rotation turns an S-definite pseudoscalar into gravitational shear). Under the swap, off-diagonal pseudoscalar grades rotate into the Janus seal sector: $[\sigma_0 \otimes \sigma_3,\ \mathbb{S}] = 2iJ_3$. The frame splits and the S-split provably do not align: $\text{rank}(P_{\pm}^{(\mathcal{F}_0)}\Pi_{\mathcal{R}}) = 2$, $\text{rank}(P_{\pm}^{(\mathcal{F}_0)}\Pi_{\mathcal{V}}) = 1$ — cross-ranks $(2, 1, 2, 1)$ against the S-split's $(3, 1)$.

Req. A3; C3; CT0.7, CT0.7a, CT1.1, CT1.2, CT1.4, CT1.10, CT1.11a (Part 1); EW.32 (D; seed-carrier locus — forward pointer, corroboration only: clause (3d) is self-contained by Schur [math]); exhaustive frame enumeration, exact nullspace solves over $\mathbb{Q}(i)$, Schur's lemma [math; machine: `checks/verify_merge_tierC.py`].

Pf.

$$(1):\ \gamma_{\mathcal{F}}^2 = (e_1e_2e_3e_4)(e_1e_2e_3e_4) = (-1)^{\binom{4}{2}}e_1^2e_2^2e_3^2e_4^2 = (-1)^6 I_4 = I_4;$$

$$\mathfrak{T}\gamma_{\mathcal{F}} = -\mathfrak{T}(e_2e_3e_4e_1) = -\mathfrak{T}\gamma_{\mathcal{F}} \implies \mathfrak{T}\gamma_{\mathcal{F}} = 0 \text{ [cyclic]}$$

$$\{\gamma_{\mathcal{F}}, e_i\} = 0:\ e_i \text{ anticommutes with the three other factors, commutes with itself;}$$

$$30 \text{ frames}:\ \text{exhaustive enumeration of } 1365 \text{ candidates [machine]}$$

$$\mathbb{R}\text{-independence of the } 16 \text{ products}:\ \text{exact elimination on } \mathbb{Q}\text{-flattened components, all } 30 \text{ frames [machine]}$$

$$(2):\ \text{canonical-sign census over the } 30 \text{ frames [machine];}$$

$$\text{species} = \text{index patterns;}$$

$$[\sigma_0 \otimes \sigma_3, \mathbb{S}] = 2iJ_3,\ [\sigma_3 \otimes \sigma_3, W_1] = 2iE^{23}:\ \text{direct products [machine]}$$

$$\mathcal{F}_0:\ \text{direct multiplication}:\ \gamma_{\mathcal{F}_0} = \sigma_0 \otimes \sigma_3;$$

$$P_{\pm}^2 = P_{\pm},\ P_+P_- = 0,\ P_+ + P_- = I_4,\ \mathfrak{T}P_{\pm} = 2 \text{ [machine]}$$

$$(3a):\ \text{nullspace of the } 3 \times 16 \text{ complex linear system } [X, W_i]$$

$$= 0 \text{ in the Pauli basis}$$

$$= \text{span}\{\Pi_{\mathcal{R}}, \Pi_{\mathcal{V}}\} \text{ [machine, exact];}$$

Schur on $3 \oplus 1$ [CT1.10; math]

involutions in the commutant : $(a\Pi_{\mathcal{R}} + b\Pi_{\mathcal{V}})^2 = I_4 \iff a^2 = b^2 = 1$;

$\mathfrak{T} = 3a + b \in \{4, 2, -2, -4\}$: $\neg\exists$ trace-0 involution

$(3b)$: nullspace of $\{X, W_i\} = 0 \;=\; \mathbb{C} \cdot \Pi_{\mathcal{V}}$ [machine];

$W_i\Pi_{\mathcal{V}} = 0$ $[(4)] \implies \operatorname{rank}\Pi_{\mathcal{V}} = 1$: singular

invertible case : $\mathrm{Ad}(M)W_i$

$= -W_i \implies \mathrm{Ad}(M)[W_1, W_2]$

$= [-W_1, -W_2]$

$= [W_1, W_2]$

$= iW_3,$

but $\mathrm{Ad}(M)(iW_3)$

$= -iW_3$: $\perp$ [CT1.4]

$(3c)$: entrywise for all 15 [machine]; $(3d)$: 2×2 nullspace solves [machine; Schur]

(4) : W_i preserves $\vee^2, \wedge^2$ [CT0.7(3)];

$W_i|_{\wedge^2} = 0$ (singlet);

$\sum_i W_i^2 = \frac{3}{2}E_0 + \frac{1}{2}\Sigma = 2\Pi_{\mathcal{R}}$ [Cor CT0.7a];

$\bigcap \ker W_i$: machine

(5) : $[\sigma_\mu \otimes \sigma_\nu, \mathbb{S}] = 0 \iff \sigma_\mu \otimes \sigma_\nu = \sigma_\nu \otimes \sigma_\mu \iff \mu = \nu$ [machine over all 15];

ranks by exact elimination. ■

Phys. The pair register carries a full Euclidean Clifford structure natively — thirty frames, fifteen pseudoscalars, all of it derived from A3's Pauli facts with no declaration anywhere — and the weak force selects exactly one split inside it: the 3-active/1-inactive S-parity grading. But the pincer is exact: from the commuting side the only weak-covariant candidates are the two S-projections, whose involutions all have nonzero trace; from the anticommuting side the candidate set collapses to a singular singlet line. No pseudoscalar grade of this register can ever behave as a weak-covariant matter grade — the obstruction is not a shortage of matrices but the absence of a spacetime-side carrier on which the gauge action is trivial, and that absence is a declaration the corpus has not made.

Rmk (deficit relocation record). The completeness audit's first deficit is hereby relocated inside Parts 1–2: native frames exist (clause (1)); their grades cannot be weak-covariant (clause (3)); a Dirac-type operator would additionally require a per-edge frame assignment $e \mapsto \mathcal{F}_e$ coupled to the EW.32 transport — an enumerated would-be declaration of the audit layer, recorded here as not adopted, no tag issued. The reversal structure declared by the corpus remains exactly $\langle \omega, \Theta_{\mathfrak{L}} \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ [CT1.11a].

1.9 Tier 2 — Lyapunov Stability of the Pair State

Theorem 1.29 (L0 — The ledger–tilt coupling). Thm. Dynamical scope (declared at use — the crystallization–tilt coupling; CT4.5a precedent: A4–A7 declare the criterion, the tick and the ledger, ¬ a state update at the seal): every A4 crystallization event applies exactly one Gibbs tilt step [D0.3] to the pair state of that event. Under A5 ($\rho_\circ = \frac{1}{4}I$), A6 ($\mathcal{T}(\rho) = U\rho U^\dagger, U \sim \mu_\star$; $|\mathfrak{L}(\tau_{n+1})| = |\mathfrak{L}(\tau_n)| + n_{\text{real_out}}(\tau_n)$), A7 ($\mathfrak{L} = \bigcup_{n\geq 0} \mathfrak{L}(\tau_n)$, $|\mathfrak{L}(\tau_n)| = \sum_{k<n} n_{\text{real_out}}(\tau_k)$), D0.3/D0.4 ($\gamma := 1 - 4\varepsilon_q$), CT0.8: (1) Tick preserves V. $V\big(\tilde{K}(\mathcal{T}(\rho))\big) = V\big(\tilde{K}(\rho)\big)$ for every draw [A6; L1(2)]. (2) Events apply the tilt. At tick τ_n the A4 criterion selects $n_{\text{real_out}}(\tau_n)$ edges into the ledger [A6]; under the declared scope each selection maps its pair state

$$\rho \;\longmapsto\; \tilde{\mathcal{T}}_\gamma(\rho) = \frac{\rho\, e^{-\gamma\rho}}{\mathfrak{T}(\rho\, e^{-\gamma\rho})}, \qquad \gamma = 1 - 4\varepsilon_q \in (0, 1] \quad \text{[D0.3; CT0.8]};$$

$Z_\gamma > 0$ on faithful ρ [Cor L0a] $\implies$ $\tilde{\mathcal{T}}_\gamma(\rho)$ faithful, $\succeq 0$, $\mathfrak{T} = 1$ [D0.3]; $\tilde{\mathcal{T}}_\gamma \neq \mathcal{T}$ [D0.3]. (3) Tilt contracts V. $V' \leq V$, equality iff $\rho = \frac{1}{4}I$ [A5; L1(5)]. (4) Ledger counts tilt steps. #{tilt steps at τ_n} $= n_{\text{real_out}}(\tau_n)$; cumulative count $|\mathfrak{L}(\tau_n)| = \sum_{k<n} n_{\text{real_out}}(\tau_k)$ [A6, A7 — exact integer recursion]. On one pair channel with k events interleaved with arbitrary draws, the V-sequence is that of $\tilde{\mathcal{T}}_\gamma^k(\rho_0)$ — the interleaving is spectrally inert by (1); per fixed k: $\mu_k = \big(1 - \frac{\gamma}{4}\big)^k \mu_0 + O(\mu_0^2)$ [L2, iterated]. (5) Contraction is the ledger dynamics. The deficits of L1–L3 are realized once per ledger increment: the deficit fraction per tilt step is $\frac{\gamma(8-\gamma)}{16}$ of V to exact leading order [L2], accumulating as $V \mapsto \big(1 - \frac{\gamma}{4}\big)^{2k} V + O(\delta\rho_0^3)$ after k recorded events; the inertial read $m_i^2(q) = \Delta S(q)$ [L3(4)] is the deficit read per increment.

Req. A4; A5; A6; A7; D0.2; D0.3; D0.4; D1.2; CT0.8; L1; L2 (forward pointers — the derivation order of `index/dependency_tree.md` places this tag after L2); the declared crystallization–tilt dynamical scope (theorem-local; see Status); exact integer recursion, exact Haar covariance over $\mathbb{Q}(i)$, exact rational log / exp-enclosures [math; machine: `checks/verify_L0_ledger_tilt.py` §A–§D].

Pf.

$$(1):\ K(\mathcal{T}\rho) = UKU^\dagger \text{ [functional-calculus covariance]};$$

$$\mathfrak{T}\big(K(\mathcal{T}\rho)\,\mathcal{T}\rho\big) = \mathfrak{T}(K\rho) \text{ [cyclicity]} \implies \tilde{K}(\mathcal{T}\rho) = U\tilde{K}U^\dagger \implies V' = V \text{ [L1(2)]}$$

$$(2):\ \varepsilon_q \in [0, \tfrac{1}{4}) \text{ [CT0.8]} \implies \gamma = 1 - 4\varepsilon_q \in (0, 1] \text{ [exact endpoints : } \varepsilon_q = 0 \mapsto \gamma = 1];$$

$$Z_\gamma \in (0,\, e^{-\gamma/4}] \text{ [Cor L0a]} \implies \lambda_i' = \lambda_i e^{-\gamma\lambda_i}/Z_\gamma > 0 \text{ [exp} > 0 \text{ on exact enclosures]}$$

$$\implies \tilde{\mathcal{T}}_\gamma(\rho) \text{ faithful, } \succeq 0,\ \mathfrak{T}\big(\tilde{\mathcal{T}}_\gamma(\rho)\big) = 1 \text{ [D0.3 normalization]}$$

$$(3):\ \text{L1(5) [Jensen on } \phi_\gamma;\ \text{zero locus } \rho = \tfrac{1}{4}I]$$

$$(4):\ |\mathfrak{L}(\tau_0)|$$

$$= 0 \text{ [A7]};$$

$$\text{recursion [A6]}$$

$$\implies |\mathfrak{L}(\tau_n)|$$

$$= \textstyle\sum_{k<n} n_{\text{real_out}}(\tau_k) \text{ [A7] by induction on } n \text{ [exact integers; machine]}$$

$$\text{spec}\big(\mathcal{T}(\rho)\big)$$

$$= \text{spec}(\rho):\ \mathfrak{T}\big((U\rho U^\dagger)^m\big)$$

$$= \mathfrak{T}(\rho^m),$$

$$m$$

$$= 1..4 \text{ [cyclicity; machine, exact over } \mathbb{Q}(i)]$$

$$\implies V \circ \mathcal{T} \qquad = V \text{ [(1)]}$$

$$\implies V\big(\tilde{\mathcal{T}}\,\mathcal{T}\,\tilde{\mathcal{T}}\cdots(\rho_0)\big) = V\big(\tilde{\mathcal{T}}^{\,k}(\rho_0)\big) \text{ [composition]};$$

$$\mu_k$$

$$= \big(1-\tfrac{\gamma}{4}\big)^k\mu_0 + O(\mu_0^2) : \text{ induction on L2's } \mu'$$

$$= \big(1-\tfrac{\gamma}{4}\big)\mu + O(\mu^2) \text{ [coefficient-wise composition; machine, } k \le 3]$$

(5) : L2 per step : $V' - V = -\frac{\gamma(8-\gamma)}{16}V + O(\delta\rho^3)$; accumulation at leading order from (4). ■

Corollary 1.5 (L0a — tilt modular update). Faithful ρ, $K := -\log\rho$ [D0.3], $K' := K\big(\tilde{\mathcal{T}}_\gamma(\rho)\big)$, $Z_\gamma := \mathfrak{T}(\rho\, e^{-\gamma\rho})$:

$$K' = K + \gamma\rho + \log Z_\gamma \cdot I; \qquad Z_\gamma \in (0,\, e^{-\gamma/4}]; \qquad \mathfrak{T}(K'-K) = \gamma + 4\log Z_\gamma \le 0,$$

with equality iff $\rho = \frac{1}{4}I$; $\gamma = 1$: $Z_1 \le e^{-1/4}$, $\mathfrak{T}(K'-K) = 1 + 4\log Z_1 \le 0$.

Req (Cor). A1; A5; D0.3; CT0.8; Jensen's inequality for $p\,e^{-\gamma p}$, exact rational log / exp-enclosures [math; machine: `checks/verify_L0_ledger_tilt.py` §B–§C].

Pf (Cor).

ρ, $\tilde{\mathcal{T}}_\gamma(\rho)$ simultaneously diagonalizable [functional calculus on a commuting family];

$$\lambda_i' = \lambda_i e^{-\gamma\lambda_i}/Z_\gamma$$

$$\implies -\log\lambda_i'$$

$$= -\log\lambda_i + \gamma\lambda_i + \log Z_\gamma \text{ [exact spectral identity; machine at two enclosure tolerances]}$$

$$\implies K'$$

$$= K + \gamma\rho + \log Z_\gamma \cdot I$$

$$g(p) := p\,e^{-\gamma p} : \quad g''(p) = e^{-\gamma p}\big(\gamma^2 p - 2\gamma\big) \le 0 \text{ on } [0,1],\ \gamma \in (0,1]$$

$$[\gamma p \le 1 < 2;\ \text{exact rational sign}]$$

$$\text{Jensen} : \ Z_\gamma$$

$$= \textstyle\sum_i g(\lambda_i)$$

$$\leq 4$$

$$g\big(\tfrac{1}{4}\big)$$

$$= e^{-\gamma/4},$$

$$\text{equality iff } \lambda_i$$

$$= \tfrac{1}{4}\,\forall i \iff \rho$$

$$= \tfrac{1}{4} I \text{ [strict concavity]};$$

$$Z_\gamma > 0 \ [\lambda_i > 0]$$

$$\mathfrak{T}(K' - K) = \gamma\,\mathfrak{T}\rho + 4\log Z_\gamma = \gamma + 4\log Z_\gamma \leq \gamma + 4\big(-\tfrac{\gamma}{4}\big) = 0;$$

$$\gamma = 1 :\ Z_1 \leq e^{-1/4},\ \ \mathfrak{T}(K' - K) = 1 + 4\log Z_1. \ \blacksquare$$

Phys. The Haar tick rotates the pair state without touching the Lyapunov function; the ledger grows exactly at the A4 events; under the declared crystallization dynamics every event is one Gibbs tilt step — the only contraction a pair state undergoes. The monotonicity of L1–L3 is thereby the per-increment dynamics of the ledger: ticks rotate, events contract, and $|\mathfrak{L}(\tau_n)|$ is the tilt-step count. Each tilt step updates the modular Hamiltonian by $K \mapsto K + \gamma\rho + \log Z_\gamma \cdot I$, whose trace deficit $\gamma + 4\log Z_\gamma \leq 0$ vanishes only at the A5 vacuum. No particle identification.

Theorem 1.30 (L1 — The Lyapunov function of the pair state). Thm. Faithful pair states ρ [A1: $\rho \in \mathfrak{H}(\mathbb{C}^4)$, $\rho \succeq 0$, $\mathfrak{T}\rho = 1$, spec $\rho \subset (0, 1]$]; with

$$\delta\rho := \rho - \tfrac{1}{4} I \ \ [\text{A5} :\ \rho_\circ = \tfrac{1}{4} I]; \qquad \mu := \text{spec}(\delta\rho) \subset \big[-\tfrac{1}{4}, \tfrac{3}{4}\big]; \qquad \textstyle\sum_i \mu_i = 0 \ \ [\mathfrak{T}\rho = 1];$$

$$\gamma := 1 - 4\varepsilon_q \in (0, 1] \ \ [\text{D0.3};\ \text{A4},\ \text{D0.4},\ \text{CT0.8}];$$

$$K := -\log\rho; \qquad \tilde{K} := K - \mathfrak{T}(K\rho)\, I \ \ [\text{D0.3}]; \qquad V(\tilde{K}) := \mathfrak{T}\big(\tilde{K}^2\big) \ \geq 0;$$

(1) Zero locus. $V = 0 \iff \tilde{K} = 0 \iff \rho = \frac{1}{4} I$ — the A5 vacuum. (2) Tick invariance. $V\big(\tilde{K}(\mathcal{T}(\rho))\big) = V\big(\tilde{K}(\rho)\big)$ for every draw [A6: $\mathcal{T}(\rho) = U\rho U^\dagger$]. (3) Decay identity ($\gamma$-dependent). Under the Gibbs tilt $\rho' := \tilde{\mathcal{T}}_\gamma(\rho)$ [D0.3], with $V' := V\big(\tilde{K}(\rho')\big)$:

$$V' - V = -\gamma \sum_i \phi_\gamma(\mu_i) + O(\delta\rho^4); \qquad \phi_\gamma(x) := 2x\log(1+4x) - \gamma x^2;$$

$$\text{leading}: \;\; V' - V = -\gamma(8-\gamma)\textstyle\sum_i \mu_i^2 + O(\delta\rho^3);$$

$$\gamma = 1: \;\; V' - V = -\textstyle\sum_i \phi(\mu_i) + O(\delta\rho^4), \;\; \phi := \phi_1.$$

(4) ϕ_γ-calculus. $\phi_\gamma(0) = \phi'_\gamma(0) = 0$; $\phi_\gamma = \phi_1 + (1-\gamma)x^2$;

$$\phi''_\gamma(x) = \frac{14 + 16x - 32x^2}{(1+4x)^2} + 2(1-\gamma) \quad \text{[exact rational identity]};$$

$$\phi_\gamma \text{ convex on } W_\gamma \supseteq W := \left[\tfrac{1-2\sqrt{2}}{4}, \tfrac{1+2\sqrt{2}}{4}\right] \supseteq \left[-\tfrac{1}{4}, \tfrac{3}{4}\right] \quad \forall \gamma \in (0,1];$$

$$\phi_\gamma > 0 \text{ on } \left[-\tfrac{1}{4}, \tfrac{3}{4}\right] \setminus \{0\}.$$

(5) Monotonicity. Jensen: $\sum_i \phi_\gamma(\mu_i) \geq 4\,\phi_\gamma\big(\frac{1}{4}\sum_i \mu_i\big) = 0$; hence, on the faithful simplex, for every $\gamma \in (0,1]$:

$$V' \leq V, \qquad V' = V \iff \tilde{K} = 0 \iff \rho = \tfrac{1}{4}I.$$

Req. A1; A4; A5; A6; D0.3, D0.4; CT0.8 (Part 1); exact formal-power-series arithmetic over $\mathbb{Q}$ with rational γ, exact $\mathbb{Q}(\sqrt{2})$ order, Jensen's inequality, exact rational log-enclosures, continuity–density extension [math; machine: `checks/verify_L_lyapunov.py`].

Pf.

$$(1): \;\; \tilde{K} = 0 \iff K = \mathfrak{T}(K\rho)I \iff \rho = e^{-c}I \iff \rho = \tfrac{1}{4}I \;[\mathfrak{T}\rho = 1];$$

$$V = \mathfrak{T}(\tilde{K}^2) \geq 0 \;[\tilde{K} \text{ Hermitian}], \quad V = 0 \iff \tilde{K} = 0$$

$$(2): \;\; K(\mathcal{T}\rho) = -\log\big(U\rho U^\dagger\big) = UKU^\dagger \text{ [functional-calculus covariance]};$$

$$\mathfrak{T}\big(K(\mathcal{T}\rho)\,\mathcal{T}\rho\big) = \mathfrak{T}\big(UK\rho U^\dagger\big) = \mathfrak{T}(K\rho) \text{ [cyclicity]}$$

$$\implies \tilde{K}(\mathcal{T}\rho) = U\tilde{K}U^\dagger \implies V' = V \text{ [cyclicity]}$$

$$(3): \;\; \rho \text{ faithful} \implies \rho,$$

$$\tilde{\mathcal{T}}_\gamma(\rho) \text{ simultaneously diagonalizable}: \;\; \rho$$

$$= \text{diag}(\lambda),$$

$$\lambda_i \qquad = \tfrac{1}{4} + \mu_i;$$

$$\tilde{K} = \mathrm{diag}\big(-\log\lambda_i - S\big),\ \ S := \mathfrak{T}(K\rho) = -\textstyle\sum_j \lambda_j \log\lambda_j$$

$$\lambda_i' = \lambda_i e^{-\gamma\lambda_i}/Z_\gamma,\ \ Z_\gamma = \textstyle\sum_j \lambda_j e^{-\gamma\lambda_j};$$

$$\log\lambda_i' = \log\lambda_i - \gamma\lambda_i - \log Z_\gamma;$$

$$e^{-\gamma/4} \text{ cancels in the ratio } \implies \text{exact } \mathbb{Q}[[t]] \text{ data}$$

$$\text{formal series along } \rho(t)$$

$$= \tfrac{1}{4}I + t$$

$$\delta_0 \ (\delta_0 \text{ Hermitian traceless, exact rational data}),$$

$$\gamma \in \{\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}, 1\} :$$

$$V' - V + \gamma\textstyle\sum_i \phi_\gamma(\mu_i(t)) \text{ has vanishing } t^0..t^3 \text{ coefficients [machine, exact];}$$

$$t^4 \text{ coefficient} \neq 0 \text{ generically } \implies O(\delta\rho^4) \text{ sharp}$$

$$\phi_\gamma(x)$$

$$= 2x\big(4x - 8x^2 + \tfrac{64}{3}x^3 - \dots\big) - \gamma x^2$$

$$= (8-\gamma)x^2 - 16x^3 + O(x^4) \implies \text{leading : } V' - V$$

$$= -\gamma(8-\gamma)\textstyle\sum_i \mu_i^2 + O(\delta\rho^3)$$

$$(4) :\ \phi_\gamma'(x) = 2\log(1+4x) + \tfrac{8x}{1+4x} - 2\gamma x;$$

$$\phi_\gamma''(x) = \tfrac{8}{1+4x} + \tfrac{8}{(1+4x)^2} - 2\gamma = \tfrac{14+16x-32x^2}{(1+4x)^2} + 2(1-\gamma) \text{ [common denominator]}$$

$$\gamma \in (0,1] :\ 2(1-\gamma) \geq 0 \implies \phi_\gamma'' \geq \phi_1'' \text{ pointwise } \implies W_\gamma \supseteq W;$$

$$16x^2 - 8x - 7 = 0 \iff x = \tfrac{1\pm2\sqrt{2}}{4}\ [\mathbb{Q}(\sqrt{2}),\ \text{exact}];$$

$$W \supseteq [-\tfrac{1}{4}, \tfrac{3}{4}]\ [\sqrt{2} \geq 1]$$

$$\phi_\gamma = \phi_1 + (1-\gamma)x^2;$$

$$\phi_1 \text{ strictly convex on } W,\ \phi_1(0) = \phi_1'(0) = 0 \implies \phi_1 > 0 \text{ off } 0 \implies \phi_\gamma > 0 \text{ off } 0 \text{ on } [-\tfrac{1}{4}, \tfrac{3}{4}]$$

$$(5):\ \text{Jensen on convex } \phi_\gamma:\ \tfrac{1}{4}\sum_i \phi_\gamma(\mu_i) \geq \phi_\gamma\big(\tfrac{1}{4}\sum_i \mu_i\big) = \phi_\gamma(0) = 0;$$

$$= \iff \mu = 0\ [\textstyle\sum \mu = 0;\ \text{strict convexity}]$$

$$\gamma > 0 \implies V' - V = -\gamma \textstyle\sum_i \phi_\gamma(\mu_i) + O(\delta\rho^4) \leq 0 \text{ near the vacuum};$$

$$\text{globally: exact rational log-enclosure sweep, } \gamma \in \{\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}, 1\}\ [\text{machine}]$$

$$\text{extension to every } \gamma$$

$$\in (0,1]:\ V' - V \text{ continuous in } (\operatorname{spec}\rho, \gamma) \text{ on the faithful simplex} \times (0,1]\ [\log, \exp,\ \text{rational maps}];$$

$$\leq 0 \text{ on a dense rational set} \implies$$

$$\leq 0 \text{ everywhere [math]}$$

$$V' = V \iff \tilde{K} = 0:\ (\Leftarrow)\ (1);$$

$(\Rightarrow)$ near vacuum by $(3), (4)$ strictness $(\gamma > 0)$; sweep points strict off-vacuum [machine]. ■

Phys. The pair state carries a natural nonnegative stability functional: the trace-square of its centered surprisal. Ledger ticks rotate it without changing it — the Haar conjugation of A6 is spectral, not contractive. The Gibbs tilt — a separate operation from the tick, determined by ε_q — strictly lowers it, by an exact per-eigenvalue deficit $\gamma\phi_\gamma$, driving every faithful state toward the A5 vacuum. The tilt strength $\gamma = 1 - 4\varepsilon_q$ is the normalized distance from the crystallization threshold to the vacuum: at $\varepsilon_q = 0$ (sharp crystallization) the tilt is at maximum strength; at the vacuum threshold $\varepsilon_q = \frac{1}{4}$ it vanishes. The tilt is grounded in the ledger–tick coupling [L0]: it is applied once per A4 crystallization event, and the ledger counts the steps.

Theorem 1.31 (L2 — The temporal decay law). Thm. Under L1's setting, per tilt step [D0.3; L1(3)], to exact leading order:

$$V' - V = -\Big[1 - \big(1 - \tfrac{\gamma}{4}\big)^2\Big]V + O(\delta\rho^3) = -\frac{\gamma(8-\gamma)}{16}\,V + O(\delta\rho^3); \qquad V' = \big(1 - \tfrac{\gamma}{4}\big)^2 V + O(\delta\rho^3);$$

$$\gamma = 1: \;\; V' - V = -\tfrac{7}{16}\,V + O(\delta\rho^3), \;\; V' = \tfrac{9}{16}V + O(\delta\rho^3); \qquad \gamma \to 0: \;\; V' = V \;\; \text{(identity-tilt limit)};$$

$$\mu' = \big(1 - \tfrac{\gamma}{4}\big)\mu + O(\mu^2) \quad \text{[eigenvalue contraction]};$$

$$\gamma = 1: \;\; \mu' = \tfrac{3}{4}\mu;$$

$$1 - \big(1 - \tfrac{\gamma}{4}\big)^2 = \frac{\gamma(8-\gamma)}{16} \;\; \text{[exact rationals]}.$$

Req. A1; A4; A5; D0.3, D0.4; CT0.8 (Part 1); L1; exact formal-series arithmetic [math; machine: `checks/verify_L_lyapunov.py` §B].

Pf.

$$K = -\log\big(\tfrac{1}{4}I + \delta\rho\big) = \ln 4\, I - 4\,\delta\rho + O(\delta\rho^2); \qquad \mathfrak{T}(K\rho) = \ln 4 + O(\delta\rho^2) \;\; [\mathfrak{T}\delta\rho = 0]$$

$$\Longrightarrow \;\; \tilde{K}$$

$$= K - \mathfrak{T}(K\rho)I$$

$$= -4$$

$$\delta\rho + O(\delta\rho^2)$$

$$\Longrightarrow \;\; V$$

$$= 16$$

$$\mathfrak{T}(\delta\rho^2) + O(\delta\rho^3)$$

$$= 16\textstyle\sum_i \mu_i^2 + O(\delta\rho^3) \;\; [\gamma\text{-free}]$$

$$\text{L1(3)}: \;\; V' - V = -\gamma \textstyle\sum_i \phi_\gamma(\mu_i) + O(\delta\rho^4); \qquad \gamma\phi_\gamma(x) = \gamma(8-\gamma)x^2 + O(x^3)$$

$$\Longrightarrow\ V' - V$$

$$= -\gamma(8-\gamma)\textstyle\sum_i \mu_i^2 + O(\delta\rho^3)$$

$$= -\frac{\gamma(8-\gamma)}{16}$$

$$V + O(\delta\rho^3)$$

$$= -\Big[1 - \big(1-\tfrac{\gamma}{4}\big)^2\Big] V + O(\delta\rho^3)$$

$$\lambda' = \lambda e^{-\gamma\lambda}/Z_\gamma : \quad \text{series}: \ \ \mu' = \big(1-\tfrac{\gamma}{4}\big)\mu + O(\mu^2) \text{ [exact coefficient; machine]};$$

$$V' = 16\big(1-\tfrac{\gamma}{4}\big)^2 \textstyle\sum_i \mu_i^2 + O(\delta\rho^3)$$

$$\gamma = 1: \ \ 1 - \tfrac{1}{4} = \tfrac{3}{4}, \ \ \big(\tfrac{3}{4}\big)^2 = \tfrac{9}{16}, \ \ 1 - \tfrac{9}{16} = \tfrac{7}{16}, \ \ \gamma(8-\gamma) = 7; \qquad \gamma \to 0: \ \ \big(1-\tfrac{\gamma}{4}\big)^2 \to 1. \ \blacksquare$$

Phys. Near the vacuum, every tilt step contracts the Lyapunov function by a γ-dependent exact rational factor: $\big(1-\frac{\gamma}{4}\big)^2$ of V survives, $\frac{\gamma(8-\gamma)}{16}$ of it is lost. At $\gamma = 1$ ($\varepsilon_q = 0$, sharp crystallization) the contraction is $\frac{9}{16}$ and the deficit exactly $\frac{7}{16}$ of V; at the vacuum threshold ($\gamma \to 0$) the contraction is 1 and the deficit 0. The $\frac{7}{16}$ is not a stipulation: it is the decay rate at sharp crystallization, derived from $\varepsilon_q = 0$ [D0.4]. The tilt is grounded in the ledger–tick coupling [L0]: one step per recorded crystallization event, so the decay law is the per-increment dynamics of the ledger.

Theorem 1.32 (L3 — Entropy deficit and the inertial mass reading). Thm. Under L1, L2: (1) Entropy deficit. With $\delta\rho = \frac{1}{4}\sum_a r_a G_a$ on the Pauli-tensor basis $\{G_a\} = \{\sigma_\mu \otimes \sigma_\nu\}_{(\mu,\nu)\neq(0,0)}$ [C1; $\mathfrak{T}(G_a G_b) = 4\delta_{ab}$: CT1.1], $|\delta\rho|^2 := \mathfrak{T}(\delta\rho^2)$:

$$\Delta S_{\text{pair}} := 2\,|\delta\rho|^2 = \tfrac{1}{2}\,|r|^2;$$

$$S\big(\rho \,\big\|\, \tfrac{1}{4}I\big) = \Delta S_{\text{pair}} + O(\delta\rho^3)$$

$$\big[S(\rho\|\sigma) := \mathfrak{T}\big(\rho(\log\rho - \log\sigma)\big)\big]$$

(2) Lyapunov identification. $\tilde{K} = -4\,\delta\rho + O(\delta\rho^2)$; $V(\tilde{K}) = 16\,|\delta\rho|^2 + O(\delta\rho^3) = 8\,\Delta S_{\text{pair}} + O(\delta\rho^3)$. (3) Deficit decay ($\gamma$-dependent). $V' - V = -\dfrac{\gamma(8-\gamma)}{2}\,\Delta S_{\text{pair}} + O(\delta\rho^3)$; $\gamma = 1$: $V' - V = -\frac{7}{2}\,\Delta S_{\text{pair}} + O(\delta\rho^3)$. (4) Inertial read. $m_i^2(q) := \Delta S(q)$ [$\Delta S(q)$: the patch deficit, L4; dimensionless, nats; ¬a particle identification].

Req. A1; A5; C1; D0.3; CT1.1 (Part 1); L1, L2; exact formal-series arithmetic, relative-entropy eigenvalue identity [math; machine: `checks/verify_L_lyapunov.py` §B].

Pf.

$$(1):\ \delta\rho \text{ Hermitian} \wedge\ \mathfrak{T}\delta\rho$$

$$= 0 \implies \delta\rho$$

$$= \tfrac{1}{4}\sum_a r_a G_a,$$

$$r_a$$

$$= \mathfrak{T}(\delta\rho\, G_a) \in \mathbb{R}\ \ [\text{C1},\ \text{CT1.1}:\ \mathfrak{T}(G_a G_b) = 4\delta_{ab}]$$

$$|\delta\rho|^2 = \mathfrak{T}(\delta\rho^2) = \tfrac{1}{16}\sum_{ab} r_a r_b\, \mathfrak{T}(G_a G_b) = \tfrac{1}{4}|r|^2 \implies 2|\delta\rho|^2 = \tfrac{1}{2}|r|^2$$

$$S\big(\rho\|\tfrac{1}{4}I\big)$$

$$= \mathfrak{T}(\rho\log\rho) + \ln 4$$

$$= \textstyle\sum_i \big(\frac{1}{4} + \mu_i\big)\log(1+4\mu_i)\ [\text{exact eigenvalue identity}]$$

$$= 2\textstyle\sum_i \mu_i^2 + O(\mu^3)$$

$$= \Delta S_{\text{pair}} + O(\delta\rho^3)$$

$$(2):\ \text{L2 Pf}:\ \tilde{K} = -4\,\delta\rho + O(\delta\rho^2); \qquad V = 16\,\mathfrak{T}(\delta\rho^2) + O(\delta\rho^3) = 8\,\Delta S_{\text{pair}} + O(\delta\rho^3)\ \ [(1)]$$

$$(3):\ \text{L2}:\ V' - V$$

$$= -\tfrac{\gamma(8-\gamma)}{16} V + O(\delta\rho^3)$$

$$= -\tfrac{\gamma(8-\gamma)}{16}\cdot 8$$

$$\Delta S_{\text{pair}} + O(\delta\rho^3)$$

$$= -\tfrac{\gamma(8-\gamma)}{2}\Delta S_{\text{pair}} + O(\delta\rho^3)$$

$$\gamma = 1:\ \ \tfrac{\gamma(8-\gamma)}{2} = \tfrac{7}{2}$$

$$(4):\ \text{definition;}\qquad \text{a read, } \neg\text{an identification.}\ \blacksquare$$

Phys. The pair state's departure from the vacuum is one object under three exact leading reads: a squared Bloch radius, an entropy deficit against the maximally mixed pair, and the Lyapunov functional itself — locked by exact rational factors that depend on the tilt strength $\gamma = 1 - 4\varepsilon_q$. The inertial read names the deficit $m_i^2(q) = \Delta S(q)$: mass-as-missing-information per channel, dimensionless in nats — a reading, not a particle claim; no pole-mass identification is asserted. The tilt is grounded in the ledger–tick coupling [L0]; the patch-level deficit $\Delta S(q)$ with its integer degree factor is derived in L4.

Theorem 1.33 (L4 — The patch entropy deficit). Thm. Vertex q of finite incidence (theorem-local scope); incident pair channels $\{(q,p)\}$ [A1]; $n(q) = \partial^{\leftarrow}(q) + \partial^{\rightarrow}(q)$ [D0.8]; $n_{\mathcal{R}}(q)$ [A4]; $n_{\mathcal{V}}(q) := n(q) - n_{\mathcal{R}}(q)$ [D0.1]. Patch deficit:

$$\Delta S(q) \ := \sum_{(q,p)\in\mathcal{V}} S\big(\rho_{qp} \,\big\|\, \tfrac{1}{4} I\big)$$

[sum over the virtual incident channels of q; $S(\cdot\|\cdot)$ as in L3(1)]

(1) Channel sum. $\Delta S(q) = \sum_{(q,p)\in\mathcal{V}} \Delta S_{\text{pair}}(pq) + O(\delta\rho^3) = 2\sum_{(q,p)\in\mathcal{V}} |\delta\rho_{pq}|^2 + O(\delta\rho^3)$ [L3(1) per channel]. (2) Mean form. For $n_{\mathcal{V}}(q) \geq 1$, with $\langle \Delta S_{\text{pair}}\rangle_q := \frac{1}{n_{\mathcal{V}}(q)} \sum_{(q,p)\in\mathcal{V}} \Delta S_{\text{pair}}(pq)$:

$$\Delta S(q)$$

$$= \ n_{\mathcal{V}}(q)$$

$$\langle \Delta S_{\text{pair}}\rangle_q + O(\delta\rho^3)$$

$$= \ 2$$

$$n_{\mathcal{V}}(q)$$

$$\langle |\delta\rho|^2\rangle_q + O(\delta\rho^3)$$

$$= \ \tfrac{1}{2}$$

$$n_{\mathcal{V}}(q)$$

$$\langle |r|^2\rangle_q + O(\delta\rho^3);$$

$n_{\mathcal{V}}(q) = 0 \implies \Delta S(q) = 0$ [empty sum]. (3) Patch Lyapunov read. $V_{\text{patch}}(q) := \sum_{(q,p)\in\mathcal{V}} V(\tilde{K}_{pq}) = 8\,\Delta S(q) + O(\delta\rho^3)$; under per-channel tilt steps [D0.3]: $V'_{\text{patch}} - V_{\text{patch}} = -\frac{\gamma(8-\gamma)}{2}\,\Delta S(q) + O(\delta\rho^3)$ [L3(2), (3) summed]. (4) Inertial read, grounded. $m_i^2(q) = \Delta S(q)$

$[\mathrm{L3}(4)] = 2\,n_{\mathcal{V}}(q)\,\langle|\delta\rho|^2\rangle_q + O(\delta\rho^3)$ — an integer multiplicity $n_{\mathcal{V}}(q) \in \mathbb{Z}_{\geq 0}$ times the mean channel deficit; the read vanishes to leading order iff $n_{\mathcal{V}}(q) = 0$ or every virtual-channel deviation vanishes.

Req. A1; A4; A5; C1; D0.1, D0.3, D0.8; L3; the finite-incidence scope (theorem-local); exact formal-series arithmetic over $\mathbb{Q}$, exact integer bookkeeping [math; machine: `checks/verify_L0_ledger_tilt.py` §E].

Pf.

$$(1):\ S\big(\rho\,\|\,\tfrac{1}{4}I\big) = \Delta S_{\text{pair}} + O(\delta\rho^3) \text{ per channel } [\mathrm{L3}(1),\ \text{sharp cubic remainder}];$$

$$\text{finite sum over } n_{\mathcal{V}}(q) \text{ channels preserves the order [linearity]}$$

$$(2):\ n_{\mathcal{V}} \cdot \tfrac{1}{n_{\mathcal{V}}} \sum = \sum\ [n_{\mathcal{V}} \geq 1];$$

$$|\delta\rho|^2 = \tfrac{1}{4}|r|^2 \implies 2 \cdot \tfrac{1}{4} = \tfrac{1}{2}\ [\mathrm{L3}(1);\ \text{exact rationals}];$$

$$n_{\mathcal{V}} = 0:\ \text{empty sum} = 0$$

$$(3):\ \textstyle\sum_e V(\tilde{K}_e) = \sum_e\big(8\,\Delta S_{\text{pair}}(e) + O(\delta\rho_e^3)\big) = 8\,\Delta S(q) + O(\delta\rho^3)\ [\mathrm{L3}(2)];$$

$$\textstyle\sum_e\big(V'(\tilde{K}_e) - V(\tilde{K}_e)\big)$$

$$= -\tfrac{\gamma(8-\gamma)}{2} \displaystyle\sum_e \Delta S_{\text{pair}}(e) + O(\delta\rho^3)$$

$$= -\tfrac{\gamma(8-\gamma)}{2}\Delta S(q) + O(\delta\rho^3)\ [\mathrm{L3}(3);\ \text{channel-wise tilt, D0.3}]$$

$$(4):\ (2) \text{ into the L3}(4) \text{ definition};$$

$$n(q),\ n_{\mathcal{R}}(q) \text{ integer counts } [\mathrm{D0.8},\ \mathrm{A4}] \implies n_{\mathcal{V}}(q) \in \mathbb{Z}_{\geq 0}.\ \blacksquare$$

Phys. The patch-level entropy deficit is the pair deficit counted with its multiplicity: every virtual channel incident to q contributes its own missing information against the A5 vacuum, and a fully crystallized patch contributes zero. This grounds the symbol $\Delta S(q)$ of the inertial read $m_i^2(q) = \Delta S(q)$ [L3(4)]: mass-as-missing-information is an integer degree-deficit factor $n_{\mathcal{V}}(q) = n(q) - n_{\mathcal{R}}(q)$ times the mean channel deficit at leading order. A reading, not an identification.

Theorem 1.34 (L5 — The second-order sector flow of the tilt). Thm. Under L1's setting, with $c := 1 - \frac{\gamma}{4}$, $d := \frac{\gamma(8-\gamma)}{8}$ [exact rationals in γ; L2]: (1) Quadratic flow. With $\delta\rho' := \tilde{\mathcal{T}}_\gamma(\rho) - \frac{1}{4}I$ [D0.3]:

$$\delta\rho' = c\,\delta\rho - d\left(\delta\rho^2 - \tfrac{1}{4}|\delta\rho|^2 E_0\right) + O(\delta\rho^3); \qquad \mathfrak{T}\left(\delta\rho^2 - \tfrac{1}{4}|\delta\rho|^2 E_0\right) = 0; \qquad \delta\rho^2 \succeq 0.$$

(2) Single-sector landing. $\delta\rho \in \mathcal{E}_{st}$ (a single (S,T) sector [C3; CT1.2]) $\Longrightarrow$ $S(\delta\rho^2) = \delta\rho^2 \wedge T(\delta\rho^2) = \delta\rho^2$: $\delta\rho^2 \in \mathcal{E}_{++}$; the traceless part lies in span$\{\Sigma, \hat{E}^{ij}\}$ [CT1.2, CT1.10]; the square channel of $\mathcal{E}_{++}$ closes in $\mathcal{E}_{++}$ — the quadratic transport inside the real block is one-directional. (3) Partition function. $\log Z_\gamma = -\frac{\gamma}{4} - d\,|\delta\rho|^2 + O(\delta\rho^3) = -\frac{\gamma}{4} - \frac{\gamma(8-\gamma)}{16}\Delta S_{\text{pair}} + O(\delta\rho^3)$ [Cor L0a; L3(1)]; $\gamma = 1$: $\log Z_1 = -\frac{1}{4} - \frac{7}{8}|\delta\rho|^2 = -\frac{1}{4} - \frac{7}{16}\Delta S_{\text{pair}} + O(\delta\rho^3)$. (4) Modular trace deficit. $\mathfrak{T}(K' - K) = \gamma + 4\log Z_\gamma$ [Cor L0a] $= -4d\,|\delta\rho|^2 + O(\delta\rho^3) = -\frac{\gamma(8-\gamma)}{4}\Delta S_{\text{pair}} + O(\delta\rho^3) \le 0$ near the vacuum; $\gamma = 1$: $-\frac{7}{4}\Delta S_{\text{pair}} + O(\delta\rho^3)$. (5) Sharp-crystallization table ($\gamma = 1$). $c = \frac{3}{4}$; $d = \frac{7}{8}$; $1 - c^2 = \frac{7}{16}$ [L2]; $\frac{d}{2} = \frac{7}{16}$; $2d = \frac{7}{2}$ [exact rationals].

Req. A1; A5; C1, C3; D0.3, D0.4; CT0.8; CT1.1, CT1.2, CT1.10; L0 (Cor L0a); L1, L2, L3; exact formal-series arithmetic over $\mathbb{Q}$, exhaustive square-channel census over $\mathbb{Q}(i)$ (all 256 basis pairs) [math; machine: `checks/verify_L0_ledger_tilt.py` §F].

Pf.

$$(1):\ \rho\, e^{-\gamma\rho} = e^{-\gamma/4}\big(\tfrac{1}{4}E_0 + \delta\rho\big)e^{-\gamma\,\delta\rho} \text{ [functional calculus on a commuting family]}$$

$$= e^{-\gamma/4}\Big[\tfrac{1}{4}E_0 + c\,\delta\rho - \gamma\big(1 - \tfrac{\gamma}{8}\big)\delta\rho^2 + O(\delta\rho^3)\Big],$$

$$\gamma\big(1 - \tfrac{\gamma}{8}\big) = \tfrac{\gamma(8-\gamma)}{8} = d \text{ [exact rationals]}$$

$$Z_\gamma = e^{-\gamma/4}\big[1 - d\,|\delta\rho|^2 + O(\delta\rho^3)\big] \text{ [}\mathfrak{T}\delta\rho = 0,\ \mathfrak{T}\delta\rho^2 = |\delta\rho|^2\text{]};$$

$$\rho' = \big(\tfrac{1}{4}E_0 + c\,\delta\rho - d\,\delta\rho^2\big)\big(1 + d\,|\delta\rho|^2\big) + O(\delta\rho^3)$$

$$\Longrightarrow\ \delta\rho'$$

$$= c$$

$\delta\rho - d\big(\delta\rho^2 - \frac{1}{4}|\delta\rho|^2 E_0\big) + O(\delta\rho^3)$ [coefficient-wise machine: $\gamma \in \{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\} \times 5$ exact rational families];

$$\mathfrak{T}\big(\delta\rho^2 - \tfrac{1}{4}|\delta\rho|^2 E_0\big) = |\delta\rho|^2 - |\delta\rho|^2 = 0; \qquad \delta\rho^2 \succeq 0 \text{ [Hermitian square]}$$

$$(2):\ X^2$$

$= \sum_a x_a^2 G_a^2 + \sum_{a<b} x_a x_b \{G_a, G_b\}$: the commutator channel cancels in squares [ordered double sum]

$$\text{square channel}:\ \{G_a, G_b\}$$

$$\neq 0$$

$$\implies T(G_aG_b)$$

$$= \vartheta_a\vartheta_b$$

G_aG_b [exhaustive: all 136 nonvanishing pairs of the 256; machine];

$$\text{single sector} \implies \vartheta_a\vartheta_b = t^2 = +1;$$

$$S = \mathrm{Ad}(\mathbb{S}) \text{ inner} \implies S(G_aG_b) = S(G_a)S(G_b) = s^2G_aG_b \text{ [CT0.7(2)]}$$

$$\implies T(X^2) = X^2 \ \wedge\ S(X^2) = X^2 \implies X^2 \in \mathcal{E}_{++} \text{ [CT1.1]};$$

$$\mathcal{E}_{++} = \mathrm{span}\{E_0, \Sigma, \hat{E}^{ij}\} \text{ [CT1.2, CT1.10]} \implies \text{traceless part} \subseteq \mathrm{span}\{\Sigma, \hat{E}^{ij}\}$$

one-directionality : $(+,+)\cdot(+,+)$ square channel

$$\mapsto (+,+): \text{ the same census restricted to } \mathcal{E}_{++} \text{ [machine]};$$

the ¬multiplicativity witness of CT1.11a(1) lies on the commutator channel ($\{G_a, G_b\} = 0$) and is reproduced

$$(3): \ \log Z_\gamma$$

$$= -\tfrac{\gamma}{4} + \log\bigl(1 - d|\delta\rho|^2 + O(\delta\rho^3)\bigr)$$

$$= -\tfrac{\gamma}{4} - d|\delta\rho|^2 + O(\delta\rho^3) \text{ [}\log(1+u) = u + O(u^2)\text{]}$$

$$\Delta S_{\text{pair}} = 2|\delta\rho|^2 \text{ [L3(1)]}; \qquad \tfrac{d}{2} = \tfrac{\gamma(8-\gamma)}{16}; \qquad \gamma = 1: \ d = \tfrac{7}{8}, \ \tfrac{d}{2} = \tfrac{7}{16}$$

$$(4): \ \mathfrak{T}(K' - K)$$

$$= \gamma + 4\log Z_\gamma \text{ [Cor L0a]}$$

$$= \gamma + 4\bigl(-\tfrac{\gamma}{4} - d|\delta\rho|^2\bigr) + O(\delta\rho^3)$$

$$= -4d|\delta\rho|^2 + O(\delta\rho^3)$$

$$-4d = -\tfrac{\gamma(8-\gamma)}{2};$$

$$-4d|\delta\rho|^2 = -2d\,\Delta S_{\text{pair}} = -\tfrac{\gamma(8-\gamma)}{4}\Delta S_{\text{pair}};$$

$$\gamma = 1:\ -\tfrac{7}{4}\Delta S_{\text{pair}};$$

$$\leq 0 \text{ near the vacuum } [\Delta S_{\text{pair}} \geq 0,\ \text{zero only at } \tfrac{1}{4}I:\ \text{L1}(1),\ \text{L3}(1)]$$

$$(5):\ \text{exact Fraction table [machine]. } \blacksquare$$

Phys. To second order the tilt does two things: it shrinks the deviation uniformly by $c = 1 - \frac{\gamma}{4}$, and it transports the traceless square of the deviation with exact weight $d = \frac{\gamma(8-\gamma)}{8}$ into the S-even, T-even real block — a quadratic sector flow that lands wholly in $\mathcal{E}_{++}$ for any single-sector input and stays there under further squaring. At sharp crystallization the transport weight is exactly $\frac{7}{8}$, the partition function reads $-\frac{1}{4} - \frac{7}{16}\Delta S_{\text{pair}}$, and the modular trace deficit reads $-\frac{7}{4}\Delta S_{\text{pair}}$ — one exact rational family with the Lyapunov deficit $\frac{7}{16}$ [L2]. A reading, not an identification.

1.10 Closure exports of Part 1 §F (consumed by Part 2)

CT0.1 dichotomy	CT0.5 $D_{\text{tick}}(n) = 2^{n+1}$	CT1.1 $(7,3,3,3)$
CT0.2 $\rho_\circ$, $\ln 4$	CT0.6(a) $C_{\text{cap}} = 16$, $O(n)$; $\delta h \leq 4\,\delta\tau$	CT1.2 basis $\{E_0, \Sigma, W_i, J_i, K_i, \hat{E}^{ij}\}$
CT0.3 $J^{\pm}$ increments	CT0.7 $\Pi_{\mathcal{R}}, \Pi_{\mathcal{V}}$; $\mathrm{F}_{(\pm,\cdot)} = \mathcal{E}_{st}$	CT1.3–CT1.9 closure table; $\mathfrak{u}(4) = \mathfrak{u}(1) \oplus \mathfrak{su}(4)$
CT0.4 real = registered	CT0.8 $0 \leq \varepsilon_q < \frac{1}{4}$	CT1.10 $\mathcal{E}_{++} = 1 \oplus 5 \oplus 1$
CT0.9 $\Phi = \{0, \pi\}$; 1 edge = 1 bit	CT0.11(a) $n \leq 3$; $\mathbb{C}^3$ slots; CT0.11b $n = 2d$ (bidirectional)	CT1.11 $E_0 \nparallel \Sigma$
CT0.10 (P, π) independent	CT0.12 3+3+5+3+1+1 = 16	CT0.13 $\mathcal{A}(n)$
CT0.14 $\dim \mathcal{K}_n = (6, 16, 17, 25)$		

$\blacksquare$

1.11 §G — Master index of results

Tags	Content	Locus
A1–A9 · C1–C13 · D0.1–D4.3 · P1–P5	axioms · conventions · the definition layer · calibration	Part 1 §A–§E
CT0.1–CT0.14 (+CT0.6a, CT0.7a, CT0.11a/b)	dichotomy, vacuum, seals, tick load 2^{n+1}, $C_{\text{cap}} = 16$, swap grading, ε_q range, $\Phi = \{0, \pi\}$, $n \leq 3$, $\mathcal{A}(n)$, mask blocks	Part 1 §F
CT1.0–CT1.11 (+CT1.11a/b)	signature classification, (S, T) sector decomposition, generator basis, $\mathfrak{su}(2)$ closures, commutator tables, global algebra, involution table, native Clifford frames and the weak-covariance pincer	Part 1 §F
L0–L5 (+Cor L0a)	ledger–tilt coupling, tilt modular update, Lyapunov function, decay law $\gamma(8-\gamma)/16$, entropy deficit and the inertial read, patch deficit, second-order sector flow	Part 1 §F
CT1.12–CT1.29 (+CT1.12a, CT1.19a/b, CT1.29a) · CT2.1–CT2.11	metric emergence, conformal rescaling, linearized spin-2, continuum limit, effective action, $T_{\mu\nu}$, field equations, conservation, $\Lambda = D^+$, entropy/calibration, one free scale	Part 2 §C
EW.1–EW.36 (+EW.5a, EW.36a)	electroweak: masks, $Q = T_3 + Y/2$, abelian $\mathfrak{u}(1)$, scalar trace mode, rung-3 wrapping, m_W, neutral mass matrix, θ_W, $m_W = m_Z \cos\theta_W$, $e = g \sin\theta_W$, spectrum	Part 2 §D

Tags	Content	Locus
EM.1–EM.26 (+EM.9a/b, Def-EM.20–22)	electromagnetic: rung-2 $\mathfrak{u}(1)$, photon masslessness/polarizations/entropy, end-charge, quantization, conservation, $G(e)$, dynamic per-edge phase, alignment patterns and parity, CPT bridge, spectrum	Part 2 §E
CT3.1–CT3.22 (+CT3.10a, CT3.17a, CT3.22a)	colour: $N = 3$ forced, $\mathfrak{u}(3) = \mathfrak{su}(3) \oplus \mathfrak{u}(1)$, rung-3 algebra, f_{abc}, gluons $= 8$, masslessness, baryon/meson singlets, sealing grammar, confinement	Part 2 §F
CT4.1–CT4.13 (+CT4.1a, CT4.4a, CT4.5a, CT4.7a)	saturation: $O(n) \to D^-$, absorbing lock, w_{D^-}, dark contrast, entropy triangle, bounds, global closure, falsifiers, audit	Part 2 §G
Def-H.1–Def-H.43 · FT.1–FT.145 (structural) · FT.153–FT.155	the fermion layer: measurement primitives, the fermion definition, floors, spin/stability, flicker, spectra, Clifford/branches, path integrals, family channels, carrier/parity, rung one; the interface definitions and their derived interface theorems	Part 3
Def-BM.1–2 · BM.1–BM.6	colour-sealed composites: baryons/mesons, junction grammar, observability, the forced family group, invariance laws, census consumption, construction specs	Part 4
the 25 Category-B cells	the computed data cells and their consumption block	Census

■

2 $\langle qu(b(i(t|t))ick)\rangle$ — Part 2 · Bosonic_era — Geometry · Electroweak · Electromagnetic · Colour · Saturation

Dep. Part 1 (axioms A1–A9, conventions C1–C13, the definition layer D0.1–D4.3, calibration P1–P5, pair canon CT0.*/CT1.1–CT1.11b, ledger dynamics L0–L5) — cited by tag at use. Format: Part 1 §0.4. Namespace. This part owns: §C geometry CT1.12–CT1.29a, CT2.1–CT2.11; §D electroweak EW.1–EW.36a; §E electromagnetic EM.1–EM.26, Def-EM.20–Def-EM.22; §F colour CT3.1–CT3.22a; §G saturation CT4.1–CT4.13; §Cal-2 (the bosonic registry note). The definition layer (D2.1–D4.3) and the master symbol table are Part 1 §C–§D; section letters §C–§G preserve the classical canon's file letters. DAG law. Citations resolve to Part 1, to earlier tags of this part, or to marked forward pointers (restatements at use; corroboration-only rows). No citation of Parts 3–4. Internal section dependencies. §C depends on Part 1 §F (and on Part 1 §C's Tier 2–4 definitions). §D depends on §C. §E depends on §C, §D (EM.18 alone, restated at use). §F depends on Part 1 §F and §C; its two borrowed electroweak/EM results (colour masslessness EW.26, coefficient stability EM.18) are restated with full arguments at use, so §F may be read before §D, §E. §G depends on Part 1 §F, §C, §D (EW.27), §F.

2.1 §Cal-2 — Calibration note (bosonic era)

The P4 core ($g,\ g';\ h_0;\ c_W;\ \Lambda;\ \varkappa,\ \beta,\ \alpha_\Omega$) and P3 ($G_N, \kappa^2$ — named combinations of $\mathfrak{h}$) are Part 1 §E. Logged spectral datum of this part: $\alpha_{\mathrm{EM}}^{(2)}$ — the EM-block scale read [EM.17]: operatorially defined; no theorem of the corpus fixes its numerical value under the dynamic-phase regime [EM.25(8)]; it is a spectral datum of F at the solution level, ¬a fitted constant and ¬a P4 core member; no result consumes its value — every edge-level phase read uses the registration data $\alpha_e \in \{0, \pi\}$ [Def-EM.20, EM.20]. Any future use that needs a value requires a logged declaration first.

2.2 §C — Geometry: Emergent Metric, Masks, Field Equations, Calibration

Dep.: Part 1 (§A–§F). Notation: $\hat{h}_{ij}$ — spatial traceless shear (D2.1); $\hat{h}^{(4)}_{\mu\nu}$ — 4-dim η-traceless perturbation (CT1.18) — distinct symbols. Machine: `checks/verify_C.py` (exact rational jet calculus; linearization via truncated series); CT1.29a: `checks/verify_merge_tierA.py`; CT1.19b/CT1.29: `checks/verify_promotion_.*`

2.3 §C.I — Emergent Metric Geometry

Theorem 2.1 (CT1.12 — Metric emergence from sector components). Thm. With D2.1 data $\big(a_{00}, q_{ij}, w_i\big) \in \mathcal{E}_{++} \oplus \mathcal{E}_{+-}$:

$$\delta^{ij}\hat{h}_{ij} = 0;$$

$$\Omega = e^{\alpha_\Omega a_{00}} > 0 \ \ \forall a_{00} \in \mathbb{R};$$

$$\bar{g}_{ij} \succ 0 \iff \forall a: \ 1 + \varkappa \lambda_a(q) > 0 \quad (\Leftarrow \ \varkappa > 0 \wedge \varkappa \|q\|_{\text{op}} < 1);$$

$$\bar{g}_{ij} \succ 0 \implies \text{sig}(\bar{g}_{\mu\nu}) = \text{sig}(g_{\mu\nu}) = (-,+,+,+);$$

sector object	#	metric role
a_{00}	1	conformal factor Ω
Σ_{tr}	1	spatial volume mode
$\hat{h}_{ij}$	5	spin-2 shear
w_i	3	shift $N_i = \beta w_i$
total	10	$\#\{g_{\mu\nu} = g_{\nu\mu}\} = \frac{4\cdot 5}{2}$

Req. A3; D2.1; CT1.1, CT1.10; Schur identity $\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det D \cdot \det(A - BD^{-1}C)$, Sylvester criterion [linear algebra].

Pf.

$$\dim \mathcal{E}_{++} = 7 = 1 \oplus 5 \oplus 1 \text{ with coefficients } (a_{00}, \hat{h}_{ij}, \Sigma_{\text{tr}}) \text{ [CT1.1, CT1.10]};$$

$$\dim \mathcal{E}_{+-} = 3 \text{ with coefficients } w_i \text{ [CT1.1]}$$

$$7 + 3 = 10 = \tfrac{4\cdot 5}{2} \implies \text{dictionary is a bijection}$$

$$\delta^{ij}\hat{h}_{ij} = \delta^{ij} q_{ij} - \tfrac{1}{3}\delta^{ij}\delta_{ij}\Sigma_{\text{tr}} = \Sigma_{\text{tr}} - \tfrac{1}{3}\cdot 3 \cdot \Sigma_{\text{tr}} = 0 \text{ [D2.1]};$$

$$q_{ij} = \hat{h}_{ij} + \tfrac{1}{3}\delta_{ij}\Sigma_{\text{tr}} \text{ unique (trace functional)}$$

$$a_{00} \in \mathbb{R} \implies \Omega = e^{\alpha_\Omega a_{00}} > 0 \implies \Omega^2 > 0$$

$$\bar{g} = \begin{pmatrix} -1 & \varkappa N_j \\ \varkappa N_i & \bar{g}_{ij} \end{pmatrix}: \quad \det \bar{g} = \det(\bar{g}_{ij}) \cdot \left(-1 - \varkappa^2 N_i (\bar{g}^{-1})^{ij} N_j\right) \quad \text{[Schur]}$$

$$\bar{g}_{ij} \succ 0 \implies \det(\bar{g}_{ij}) > 0 \ \wedge \ (\bar{g}^{-1})^{ij} \succ 0 \implies N(\bar{g}^{-1})N \geq 0 \implies (\cdot) \leq -1 < 0 \implies \det \bar{g} < 0$$

$$\bar{g}_{ij} \succ 0: \ 3 \text{ positive directions}; \quad \det \bar{g} < 0: \ 4\text{th eigenvalue} < 0 \implies \text{sig}(\bar{g}) = (-,+,+,+)$$

$$\bar{g}_{ij} = \delta_{ij} + \varkappa q_{ij}: \ \text{spec} = \{1 + \varkappa \lambda_a(q)\} \implies \succ 0 \iff \forall a: \ 1 + \varkappa \lambda_a > 0$$

$$g_{\mu\nu} = \Omega^2 \bar{g}_{\mu\nu},\ \Omega^2 > 0 :\ \mathrm{spec}(g) = \Omega^2\,\mathrm{spec}(\bar{g}) \implies \mathrm{sig}(g) = (-,+,+,+).\ \blacksquare$$

Rmk (scale hygiene). $\varkappa$: embedding scale (D2.1, C4); $\varkappa \neq \kappa$ (CT2.9); $[\varkappa] = [q]^{-1}$; $\neg\exists$ assumed relation $\varkappa \sim \kappa$.

Phys. The ten real sector coefficients of a pair — conformal amplitude, spatial trace, five traceless shears, three shifts — are exactly the ten components of a Lorentzian metric in conformal gauge. Gravity is not added to the theory: the metric is a repackaging of the S-even operator content, with signature $(-,+,+,+)$ guaranteed in the perturbative regime $\varkappa\|q\|_{\mathrm{op}} < 1$.

Theorem 2.2 (CT1.12a — The pair-channel audit of the metric dictionary: tiling, block split, and the three discrete read classes). Thm. (1) Tiling. The 16 pair-channel registers [CT0.6] tile by aspects:

$$(G_5 + W_3 + H_1) + (J_3^+ + J_3^-) + D_1^+ = 16 \qquad [\text{CT0.12, CT1.1, CT1.2}]$$

matching the (S,T) block decomposition 7+3+3+3 [CT1.1] under the dictionary $(G,H,D^+) \leftrightarrow \mathcal{E}_{++}$, $W \leftrightarrow \mathcal{E}_{+-}$, $J^+ \leftrightarrow \mathcal{E}_{-+}$, $J^- \leftrightarrow \mathcal{E}_{--}$ [CT0.12; CT1.11a(0)]. (2) Block split of the dictionary. CT1.12's dictionary rows decompose as

$$g_{\mu\nu} \leftrightarrow \hat{h}_{ij}\ (5,\ G:\ \text{spin-2 shear}) \oplus \Sigma_{\mathrm{tr}}\ (1,\ H:\ \text{volume mode}) \oplus a_{00}\ (1,\ D^+:\ \Omega\ \text{conformal}) \oplus w_i\ (3,\ W:\ \text{shift}),$$

$$1 + 1 + 5 + 3 = 10 = \tfrac{4\cdot 5}{2}$$

every carrier S-even [$\mathcal{E}_{++}, \mathcal{E}_{+-}$ are S-even: CT1.1, CT1.2, C3; CT1.12's data live in $\mathcal{E}_{++} \oplus \mathcal{E}_{+-}$]; the S-odd blocks ($J^{\pm}$) carry zero metric coordinates. Consequence: $g_{\mu\nu}$ is orientation-even (used in CT1.19b(2)). (3) Read classes. The corpus's discrete reads fall into exactly three classes, one per discreteness route: (R1) binary registration — σ_e [CT0.9], $q_{\mathrm{EM}} \in \{-1,0,+1\}$ [EM.9, D3.4], wall parity $(-1)^{\aleph}$ [$\aleph_v$ from D3.3's f_v]; (R2) integer winding on materialized closed chains — κ, Θ, k [CT1.19b]; (R3) binary singlet admissibility — $\mathcal{C} \in \{0,1\}$ [D2.14; confinement semantics CT3.22a]. Every further dictionary/aspect datum is either continuous configuration data [P4: $g, g', h_0, c_W, \Lambda, \varkappa, \beta, \alpha_\Omega$] or structural non-observable (per-aspect observability verdicts: audit-layer record — not re-derived here); neither carries an independent quantum.

Req. A8; D2.1, D2.14, D3.3; P4; CT0.6, CT0.9, CT0.12, CT1.11a (Part 1); CT1.1, CT1.2, CT1.12; EM.9 (E; forward pointer — C DAG note); exact arithmetic [math; machine: `checks/verify_merge_tierB.py` §C].

Pf.

$$(1):\ \text{CT0.12's rows}:\ 3{+}3{+}5{+}3{+}1{+}1 = 16 = C_{\mathrm{cap}}\ [\text{CT0.6}];$$

$$\text{regrouped by CT1.2's parity table}:\ (5{+}1{+}1){+}3{+}3{+}3 = (7,3,3,3)\ [\text{CT1.1}]$$

$$(2):\ \text{CT1.12's row list } (a_{00}, \Sigma_{\mathrm{tr}}, \hat{h}_{ij}, w_i) \text{ with counts } 1{+}1{+}5{+}3 = 10 = \#\{g_{\mu\nu} = g_{\nu\mu}\};$$

$$\text{carriers}:\ a_{00} \sim E_0,\ \Sigma_{\mathrm{tr}} \sim \Sigma,\ \hat{h} \subset \mathrm{span}\{\hat{E}^{ij}\},\ w \sim W_i \text{ all } S\text{-even [CT1.2; machine]}$$

$$J_i \in \mathcal{E}_{-+},\ K_i \in \mathcal{E}_{--}\ \text{[CT1.2]}:\ S\text{-odd} \implies \neg\text{rows of an } S\text{-even dictionary}$$

$$(3):\ \text{enumeration of the declared discrete reads}:\ \text{CT0.9}\ (\sigma\ \text{binary}),$$

$$\text{EM.9/D3.4 (ternary from two binaries)},$$

$$\text{D3.3}\ (\aleph_v = |f_v|\ \text{from signs}),$$

$$\text{CT1.19b}\ (\kappa, \Theta, k\ \text{integers on-shell}),$$

$$\text{D2.14}\ (\mathcal{C} \in \{0,1\});$$

$$\text{continuous residue} = \text{P4}'\text{s list verbatim.}\ \blacksquare$$

Phys. The metric speaks only the swap-even dialect: all ten coordinates of spacetime geometry live in the gravity+Higgs+dark and weak blocks, and the Janus blocks contribute nothing to $g_{\mu\nu}$. Auditing the whole pair channel leaves exactly three routes to discreteness — registration bits, winding integers, sealing booleans — and everything else is continuous configuration data or structure. This is the census of what can be a "quantum" in this corpus.

Theorem 2.3 (CT1.13 — Christoffel symbols under conformal rescaling). Thm. $g_{\mu\nu} = \Omega^2 \bar{g}_{\mu\nu},\ \Upsilon = \ln\Omega \implies$

$$g^{\mu\nu} = \Omega^{-2}\bar{g}^{\mu\nu};$$

$$\Gamma^{\lambda}{}_{\mu\nu} = \bar{\Gamma}^{\lambda}{}_{\mu\nu} + \delta^{\lambda}{}_{\mu}\bar{\nabla}_{\nu}\Upsilon + \delta^{\lambda}{}_{\nu}\bar{\nabla}_{\mu}\Upsilon - \bar{g}_{\mu\nu}\bar{g}^{\lambda\rho}\bar{\nabla}_{\rho}\Upsilon.$$

Req. D2.1, D2.2; Levi-Civita formula $\Gamma^{\lambda}{}_{\mu\nu} = \frac{1}{2}g^{\lambda\rho}(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu})$ [math].
Pf.

$$g^{\mu\alpha}g_{\alpha\nu} = \Omega^{-2}\Omega^2\delta^{\mu}{}_{\nu} \implies g^{\mu\nu} = \Omega^{-2}\bar{g}^{\mu\nu}$$

$$\partial_\rho g_{\mu\nu} = 2\Omega(\partial_\rho\Omega)\bar{g}_{\mu\nu} + \Omega^2\partial_\rho\bar{g}_{\mu\nu} = \Omega^2\big(2\Upsilon_\rho\bar{g}_{\mu\nu} + \partial_\rho\bar{g}_{\mu\nu}\big),$$

$$\Upsilon_\rho = \partial_\rho\Omega/\Omega\ \text{[D2.2]}$$

$$\Gamma^{\lambda}{}_{\mu\nu}$$

$$= \tfrac{1}{2}\Omega^{-2}\bar{g}^{\lambda\rho}\Omega^2\big[(\partial_\mu\bar{g}_{\rho\nu} + 2\bar{g}_{\rho\nu}\Upsilon_\mu) + (\partial_\nu\bar{g}_{\rho\mu} + 2\bar{g}_{\rho\mu}\Upsilon_\nu) - (\partial_\rho\bar{g}_{\mu\nu} + 2\bar{g}_{\mu\nu}\Upsilon_\rho)\big]$$

$$= \bar{\Gamma}^{\lambda}{}_{\mu\nu} + \tfrac{1}{2}\bar{g}^{\lambda\rho}\big(2\bar{g}_{\rho\nu}\Upsilon_\mu + 2\bar{g}_{\rho\mu}\Upsilon_\nu - 2\bar{g}_{\mu\nu}\Upsilon_\rho\big)$$

$$= \bar{\Gamma}^{\lambda}{}_{\mu\nu} + \delta^{\lambda}{}_{\nu}\Upsilon_{\mu} + \delta^{\lambda}{}_{\mu}\Upsilon_{\nu} - \bar{g}_{\mu\nu}\bar{g}^{\lambda\rho}\Upsilon_{\rho}$$

$$\Upsilon \text{ scalar } \implies \Upsilon_{\mu} = \partial_{\mu}\Upsilon = \bar{\nabla}_{\mu}\Upsilon; \qquad \text{RHS symmetric in } \mu \leftrightarrow \nu. \blacksquare$$

Phys. A conformal change of the metric shifts the connection by first derivatives of the conformal potential only. This single formula generates the whole curvature-transfer chain CT1.14–CT1.17 that relates emergent geometry to its auxiliary background.

Theorem 2.4 (CT1.14 — Riemann tensor under conformal rescaling). Thm. With $C^{\rho}{}_{\mu\nu} := \Gamma^{\rho}{}_{\mu\nu} - \bar{\Gamma}^{\rho}{}_{\mu\nu} = \delta^{\rho}{}_{\mu}\Upsilon_{\nu} + \delta^{\rho}{}_{\nu}\Upsilon_{\mu} - \bar{g}_{\mu\nu}\Upsilon^{\rho}$ (tensor; $\Upsilon^{\rho} := \bar{g}^{\rho\sigma}\Upsilon_{\sigma}$):

$$R^{\rho}{}_{\sigma\mu\nu} = \bar{R}^{\rho}{}_{\sigma\mu\nu} + 2\bar{\nabla}_{[\mu}C^{\rho}{}_{\nu]\sigma} + 2C^{\rho}{}_{[\mu|\lambda|}C^{\lambda}{}_{\nu]\sigma}.$$

Req. CT1.13; Riemann definition $R^{\rho}{}_{\sigma\mu\nu} = \partial_{\mu}\Gamma^{\rho}{}_{\nu\sigma} - \partial_{\nu}\Gamma^{\rho}{}_{\mu\sigma} + \Gamma^{\rho}{}_{\mu\lambda}\Gamma^{\lambda}{}_{\nu\sigma} - \Gamma^{\rho}{}_{\nu\lambda}\Gamma^{\lambda}{}_{\mu\sigma}$; $\bar{\nabla}$ of a $(1,2)$-tensor [math].

Pf.

$$C = \Gamma - \bar{\Gamma} : \text{ difference of connections } \implies \text{ tensor}; \qquad \text{explicit form} = \text{CT1.13}$$

$$\Gamma = \bar{\Gamma} + C \text{ substituted into the definition :} \quad O(C^0) : \ \bar{R}^{\rho}{}_{\sigma\mu\nu}$$

$$O(C^1) : \ \partial_{\mu}C^{\rho}{}_{\nu\sigma} - \partial_{\nu}C^{\rho}{}_{\mu\sigma} + \bar{\Gamma}^{\rho}{}_{\mu\lambda}C^{\lambda}{}_{\nu\sigma} + C^{\rho}{}_{\mu\lambda}\bar{\Gamma}^{\lambda}{}_{\nu\sigma} - \bar{\Gamma}^{\rho}{}_{\nu\lambda}C^{\lambda}{}_{\mu\sigma} - C^{\rho}{}_{\nu\lambda}\bar{\Gamma}^{\lambda}{}_{\mu\sigma}$$

$-\bar{\Gamma}^{\lambda}{}_{\mu\nu}C^{\rho}{}_{\lambda\sigma}$ cancels between the antisymmetrized halves ($\bar{\Gamma}^{\lambda}{}_{\mu\nu} = \bar{\Gamma}^{\lambda}{}_{\nu\mu}$)

$$\implies O(C^1)$$

$$= 2\bar{\nabla}_{[\mu}C^{\rho}{}_{\nu]\sigma}$$

$$O(C^2) : \ C^{\rho}{}_{\mu\lambda}C^{\lambda}{}_{\nu\sigma} - C^{\rho}{}_{\nu\lambda}C^{\lambda}{}_{\mu\sigma} = 2C^{\rho}{}_{[\mu|\lambda|}C^{\lambda}{}_{\nu]\sigma}$$

(λ excluded from antisymmetrization). $\blacksquare$

Phys. The curvature of the emergent metric differs from the background by a term linear in covariant derivatives of the conformal potential and a term quadratic in it — an exact, closed bookkeeping of how conformal (volume) data manufactures curvature.

Theorem 2.5 (CT1.15 — Ricci tensor under conformal rescaling). Thm. In D dimensions:

$$R_{\mu\nu}$$

$$= \bar{R}_{\mu\nu} - (D-2)\bar{\nabla}_{\mu}\bar{\nabla}_{\nu}\Upsilon - \bar{g}_{\mu\nu}\bar{\Box}\Upsilon + (D-2)\big(\Upsilon_{\mu}\Upsilon_{\nu} - \bar{g}_{\mu\nu}|\bar{\nabla}\Upsilon|^2\big).$$

Req. CT1.14; $\bar\nabla_\mu \Upsilon_\nu = \bar\nabla_\nu \Upsilon_\mu$, $\bar\nabla \bar g = 0$ [math].
Pf.

$$\text{CT1.14 contracted } \rho = \mu :$$

$$R_{\sigma\nu} = \bar R_{\sigma\nu} + \bar\nabla_\mu C^\mu{}_{\nu\sigma} - \bar\nabla_\nu C^\mu{}_{\mu\sigma} + C^\mu{}_{\mu\lambda} C^\lambda{}_{\nu\sigma} - C^\mu{}_{\nu\lambda} C^\lambda{}_{\mu\sigma}$$

$$C^\mu{}_{\mu\sigma} = \Upsilon_\sigma + D\,\Upsilon_\sigma - \Upsilon_\sigma = D\,\Upsilon_\sigma$$

$$\bar\nabla_\mu C^\mu{}_{\nu\sigma} = \bar\nabla_\nu \Upsilon_\sigma + \bar\nabla_\sigma \Upsilon_\nu - \bar g_{\nu\sigma} \bar\Box \Upsilon; \qquad \bar\nabla_\nu C^\mu{}_{\mu\sigma} = D\,\bar\nabla_\nu \Upsilon_\sigma$$

$$\bar\nabla_\nu \Upsilon_\sigma = \bar\nabla_\sigma \Upsilon_\nu \implies \text{derivative part} = -(D-2)\bar\nabla_\nu \Upsilon_\sigma - \bar g_{\nu\sigma} \bar\Box \Upsilon$$

$$C^\mu{}_{\mu\lambda} C^\lambda{}_{\nu\sigma}$$

$$= D$$

$$\Upsilon_\lambda \big(\delta^\lambda{}_\nu \Upsilon_\sigma + \delta^\lambda{}_\sigma \Upsilon_\nu - \bar g_{\nu\sigma} \Upsilon^\lambda \big)$$

$$= D \big(2 \Upsilon_\nu \Upsilon_\sigma - \bar g_{\nu\sigma} |\bar\nabla \Upsilon|^2 \big)$$

$$C^\mu{}_{\nu\lambda} C^\lambda{}_{\mu\sigma} = (D+2)\,\Upsilon_\nu \Upsilon_\sigma - 2\,\bar g_{\nu\sigma} |\bar\nabla \Upsilon|^2$$

$$[9 \text{ products; coefficients } 1{+}1{+}D{+}1{-}1{-}1{+}1{=}D{+}2;\ -1{-}1{=}{-}2]$$

$$\text{quadratic difference} = (D-2)\big(\Upsilon_\nu \Upsilon_\sigma - \bar g_{\nu\sigma} |\bar\nabla \Upsilon|^2 \big); \qquad + \bar R_{\sigma\nu},\ \text{rename} \implies \text{Thm.} \blacksquare$$

Phys. The exact response of Ricci curvature to the conformal factor in any dimension. The $(D-2)$ coefficients are what makes $D = 4$ (the arena fixed by CT1.12/CT2.6) the non-degenerate case; the identity feeds directly the Einstein-tensor law CT1.17.

Theorem 2.6 (CT1.16 — Ricci scalar under conformal rescaling). Thm.

$$R = \Omega^{-2} \Big[\bar R - 2(D-1)\bar\Box \Upsilon - (D-1)(D-2) |\bar\nabla \Upsilon|^2 \Big]; \qquad D = 4 : \quad R = \Omega^{-2} \Big[\bar R - 6 \bar\Box \Upsilon - 6 |\bar\nabla \Upsilon|^2 \Big].$$

Req. CT1.13 ($g^{\mu\nu} = \Omega^{-2} \bar g^{\mu\nu}$), CT1.15; $\bar g^{\mu\nu} \bar g_{\mu\nu} = D$ [math].
Pf.

$$R = g^{\mu\nu} R_{\mu\nu} = \Omega^{-2} \bar g^{\mu\nu} R_{\mu\nu} \quad \text{[CT1.13]}$$

$$\bar{g}^{\mu\nu}R_{\mu\nu} = \bar{R} - (D-2)\bar{\Box}\Upsilon - D\,\bar{\Box}\Upsilon + (D-2)|\bar{\nabla}\Upsilon|^2 - D(D-2)|\bar{\nabla}\Upsilon|^2 \quad \text{[CT1.15]}$$

$$-(D-2) - D = -2(D-1); \qquad (D-2) - D(D-2) = -(D-1)(D-2)$$

$$\times\,\Omega^{-2}; \qquad D = 4:\ 2(D-1) = 6 = (D-1)(D-2).\ \blacksquare$$

Phys. The scalar curvature transforms with the standard conformal weights; in $D = 4$ the bracket $\bar{R} - 6\bar{\Box}\Upsilon - 6|\bar{\nabla}\Upsilon|^2$ is what multiplies Ω^{-2} — the input that makes the Einstein tensor law CT1.17 collapse to four terms.

Theorem 2.7 (CT1.17 — Einstein tensor under conformal rescaling in $D = 4$). Thm.

$$G_{\mu\nu} = \bar{G}_{\mu\nu} - 2\bar{\nabla}_\mu\bar{\nabla}_\nu\Upsilon + 2\bar{g}_{\mu\nu}\bar{\Box}\Upsilon + 2\Upsilon_\mu\Upsilon_\nu + \bar{g}_{\mu\nu}|\bar{\nabla}\Upsilon|^2.$$

Req. CT1.15, CT1.16 ($D = 4$); $G_{\mu\nu} := R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$ [math].

Pf.

$$R_{\mu\nu} = \bar{R}_{\mu\nu} - 2\bar{\nabla}_\mu\bar{\nabla}_\nu\Upsilon - \bar{g}_{\mu\nu}\bar{\Box}\Upsilon + 2\Upsilon_\mu\Upsilon_\nu - 2\bar{g}_{\mu\nu}|\bar{\nabla}\Upsilon|^2$$

$$\text{[CT1.15, } D = 4\text{]}$$

$$\tfrac{1}{2}g_{\mu\nu}R$$

$$= \tfrac{1}{2}\Omega^2\bar{g}_{\mu\nu}\cdot\Omega^{-2}\big[\bar{R} - 6\bar{\Box}\Upsilon - 6|\bar{\nabla}\Upsilon|^2\big]$$

$$= \tfrac{1}{2}\bar{g}_{\mu\nu}\big[\bar{R} - 6\bar{\Box}\Upsilon - 6|\bar{\nabla}\Upsilon|^2\big]$$

$$\text{[CT1.16]}$$

$$G_{\mu\nu}$$

$$= \underbrace{\big(\bar{R}_{\mu\nu} - \tfrac{1}{2}\bar{g}_{\mu\nu}\bar{R}\big)}_{\bar{G}_{\mu\nu}} + (-1+3)$$

$$\bar{g}_{\mu\nu}\bar{\Box}\Upsilon + (-2+3)$$

$$\bar{g}_{\mu\nu}|\bar{\nabla}\Upsilon|^2 - 2\bar{\nabla}_\mu\bar{\nabla}_\nu\Upsilon + 2\Upsilon_\mu\Upsilon_\nu.\ \blacksquare$$

Phys. The exact transformation law of the Einstein tensor in the emergent conformal gauge: everything beyond the background Einstein tensor is built from the single field $\Upsilon = \ln\Omega$. The conformal (volume) degree of freedom of the pair algebra gravitates through a closed-form correction — no approximation involved.

Theorem 2.8 (CT1.18 — Linearized spin-2 operator). Thm. $\bar{g}_{\mu\nu} = \eta_{\mu\nu} + \varkappa\hat{h}^{(4)}_{\mu\nu}$, $\eta^{\mu\nu}\hat{h}^{(4)}_{\mu\nu} = 0$ (as a field), $\Upsilon = N_i = \Sigma_{\text{tr}} = 0$. To $O(\varkappa)$:

$$\bar{G}_{\mu\nu}$$

$$= \frac{\varkappa}{2}\Big[-\Box\hat{h}^{(4)}_{\mu\nu} + \partial_\mu\partial^\lambda\hat{h}^{(4)}_{\nu\lambda} + \partial_\nu\partial^\lambda\hat{h}^{(4)}_{\mu\lambda} - \eta_{\mu\nu}\partial^\lambda\partial^\sigma\hat{h}^{(4)}_{\lambda\sigma}\Big] + O(\varkappa^2),$$

$$\Box = \eta^{\alpha\beta}\partial_\alpha\partial_\beta.$$

With the full linearized Einstein tensor (general h)

$$\bar{G}^{(1)}[h]_{\mu\nu}:$$

$$= \tfrac{1}{2}\Big[-\Box h_{\mu\nu} - \partial_\mu\partial_\nu h^\lambda{}_\lambda + \partial_\mu\partial^\lambda h_{\nu\lambda} + \partial_\nu\partial^\lambda h_{\mu\lambda} + \eta_{\mu\nu}\big(\Box h^\lambda{}_\lambda - \partial^\lambda\partial^\sigma h_{\lambda\sigma}\big)\Big]:$$

$$h^\lambda{}_\lambda = 0 \implies \bar{G}^{(1)}[h] = \text{bracket above}/2; \qquad \bar{G}^{(1)}\big[h + \partial_\mu\xi_\nu + \partial_\nu\xi_\mu\big] = \bar{G}^{(1)}[h] \ \forall\xi;$$

$$\partial^\lambda\hat{h}^{(4)}_{\lambda\nu} = 0 \ \wedge\ \hat{h}^{(4)\lambda}{}_\lambda = 0 \implies \bar{G}^{(1)}[\hat{h}^{(4)}] = -\tfrac{1}{2}\Box\hat{h}^{(4)}_{\mu\nu}$$

(massless wave operator).

Req. CT1.10 (spin-2 mode), CT1.12; Levi-Civita/Ricci expansion [math]; $\partial_\mu\partial_\nu = \partial_\nu\partial_\mu$. Pf.

$$\bar{g}^{\lambda\rho}$$

$$= \eta^{\lambda\rho} - \varkappa\hat{h}^{(4)\lambda\rho} + O(\varkappa^2)$$

$$\implies \bar{\Gamma}^\lambda{}_{\mu\nu}$$

$$= \tfrac{\varkappa}{2}\big(\partial_\mu\hat{h}^{(4)\lambda}{}_\nu + \partial_\nu\hat{h}^{(4)\lambda}{}_\mu - \partial^\lambda\hat{h}^{(4)}_{\mu\nu}\big) + O(\varkappa^2)$$

(η-raising)

$$\bar{R}_{\mu\nu}$$

$$= \partial_\lambda\bar{\Gamma}^\lambda{}_{\mu\nu} - \partial_\nu\bar{\Gamma}^\lambda{}_{\lambda\mu} + O(\varkappa^2)$$

$$= \tfrac{\varkappa}{2}\big(\partial_\mu\partial^\lambda\hat{h}^{(4)}_{\lambda\nu} + \partial_\nu\partial^\lambda\hat{h}^{(4)}_{\lambda\mu} - \Box\hat{h}^{(4)}_{\mu\nu} - \partial_\mu\partial_\nu\hat{h}^{(4)\lambda}{}_\lambda\big)$$

$$\hat{h}^{(4)\lambda}{}_{\lambda} \equiv 0 \text{ (field)} \implies \partial_\mu \partial_\nu \hat{h}^{(4)\lambda}{}_{\lambda} = 0$$

$$\bar{R} = \eta^{\mu\nu}\bar{R}_{\mu\nu} = \varkappa\big(\partial^\mu\partial^\nu \hat{h}^{(4)}_{\mu\nu} - \Box \hat{h}^{(4)\lambda}{}_{\lambda}\big) = \varkappa\, \partial^\lambda \partial^\sigma \hat{h}^{(4)}_{\lambda\sigma}$$

$$\bar{G}_{\mu\nu} = \bar{R}_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu}\bar{R} + O(\varkappa^2) \implies \text{Thm formula}$$

$$\bar{G}^{(1)}[h + \partial\xi + \partial\xi] - \bar{G}^{(1)}[h] : \ \xi\text{-terms} = \text{combinations of } \partial_{(\alpha}\partial_\beta\partial_{\gamma)}\xi_\delta = 0 \implies \text{invariance}$$

$$\partial^\lambda \hat{h}^{(4)}_{\lambda\nu}$$

$$= 0 \implies \partial_\mu \partial^\lambda \hat{h}^{(4)}_{\nu\lambda}$$

$$= \partial_\nu \partial^\lambda \hat{h}^{(4)}_{\mu\lambda}$$

$$= \partial^\lambda \partial^\sigma \hat{h}^{(4)}_{\lambda\sigma}$$

$$= 0 \implies \bar{G}^{(1)}[\hat{h}^{(4)}]$$

$$= -\tfrac{1}{2}\Box \hat{h}^{(4)}_{\mu\nu}. \ \blacksquare$$

Machine verification (incl. $\bar{G}^{(0)} = 0$, $\bar{G}^{(1)} = L[\hat{h}^{(4)}]$ from the full nonlinear pipeline, gauge invariance): `checks/verify_C.py`.

Rmk (correction). The traceless-projected transformation $\hat{h}^{(4)} \mapsto \hat{h}^{(4)} + \partial_\mu\xi_\nu + \partial_\nu\xi_\mu - \frac{1}{2}\eta_{\mu\nu}\partial\cdot\xi$ preserves $\eta^{\mu\nu}\hat{h}^{(4)}_{\mu\nu} = 0$ but shifts $\bar{G}^{(1)}$ by $\frac{1}{2}\big(\partial_\mu\partial_\nu - \eta_{\mu\nu}\Box\big)\partial\cdot\xi \neq 0$ (machine-verified): gauge invariance holds for the standard linearized diffeomorphism $\partial_\mu\xi_\nu + \partial_\nu\xi_\mu$, which does not preserve tracelessness and need not, since $\bar{G}^{(1)}$ is defined for general h. The prior codex claimed invariance of the projected form — false as stated.

Phys. With only the five-component traceless shear excited, the emergent Einstein tensor reduces at leading order to the massless spin-2 wave operator in transverse gauge. The graviton of the corpus is the $\mathcal{E}_{++}$ quintet of CT1.10 propagating on the auxiliary background; its gauge freedom is linearized diffeomorphism invariance.

2.4 §C.II — Masks, Manifold, and Closure

Theorem 2.9 (CT2.1 — Unit and dimension conventions). Thm. With $c = \hbar = k_B = 1$, $[\mathfrak{h}] = \mathfrak{q}^{-1/2}$:

$$[\text{length}] = \mathfrak{q}^{-1/2};$$

$$[\text{area}] = \mathfrak{q}^{-1};$$

$$[\text{energy}] = \mathfrak{q}^{+1/2};$$

$$[T_{\mu\nu}] = \mathfrak{q}^{+2};$$

$$[G_{\mu\nu}] = [\Lambda] = \mathfrak{q}^{+1};$$

$$[\kappa^2] = [G_N] = \mathfrak{q}^{-1};$$

$$[\kappa] = \mathfrak{q}^{-1/2};$$

$$[S_{\text{geom}}] = \mathfrak{q}^{0},$$

and every equation of Parts III–IV balances.

Req. D0.6, D2.9; dimensional bookkeeping [math].

Rmk (forward pointers, non-load-bearing). The normalization of κ is cross-checked against CT1.23 and $G_N \propto \kappa^2$ against CT2.9 — consistency pointers only.

Pf.

$$[\mathfrak{h}] = \mathfrak{q}^{-1/2}\ [\text{D0.6, D2.9}] \implies [\text{length}] = \mathfrak{q}^{-1/2},\ [\text{area}] = \mathfrak{q}^{-1};$$

$$\hbar = c = 1 \implies [\text{energy}] = [\text{length}]^{-1} = \mathfrak{q}^{+1/2}$$

$$[\text{curvature}] = [\text{area}]^{-1} = \mathfrak{q}^{+1} = [G_{\mu\nu}]; \qquad \Lambda g_{\mu\nu} \sim G_{\mu\nu} \implies [\Lambda] = \mathfrak{q}^{+1}$$

$$[T_{\mu\nu}] = \frac{[\text{energy}]}{[\text{volume}]} = \frac{\mathfrak{q}^{1/2}}{\mathfrak{q}^{-3/2}} = \mathfrak{q}^{+2}$$

$$[\kappa^2 T_{\mu\nu}] = [G_{\mu\nu}]\ [\text{CT1.23}] \implies [\kappa^2] = \mathfrak{q}^{1-2} = \mathfrak{q}^{-1},\ [\kappa] = \mathfrak{q}^{-1/2};$$

$$G_N \propto \kappa^2\ [\text{CT2.9}] \implies [G_N] = \mathfrak{q}^{-1}$$

$$[d^4x] = \mathfrak{q}^{-2},\ [\sqrt{-g}] = \mathfrak{q}^{0},\ [R] = \mathfrak{q}^{+1},\ [\kappa^{-2}] = \mathfrak{q}^{+1} \implies [S_{\text{geom}}] = \mathfrak{q}^{0}\ (\hbar = 1).\ \blacksquare$$

Phys. One information unit $\mathfrak{q}$ grades the entire continuum layer: lengths $\mathfrak{q}^{-1/2}$, curvature and Λ at $\mathfrak{q}^{+1}$, stress at $\mathfrak{q}^{+2}$, Newton's constant an area $\mathfrak{q}^{-1}$. The dimensionless action confirms no dimensionful quantity beyond $\mathfrak{h}$ has been smuggled in.

Theorem 2.10 (CT2.2 — The rung mask is a projection). Thm.

$$\big(\mathcal{P}_M^{(n)}\big)^2 = \mathcal{P}_M^{(n)} = \big(\mathcal{P}_M^{(n)}\big)^\dagger;$$

$$\operatorname{Im} \mathcal{P}_M^{(n)} = \mathcal{K}_n;$$

$$\mathfrak{T}\, \mathcal{P}_M^{(n)} = \dim \mathcal{K}_n \in \{6, 16, 17, 25, \dots\}.$$

Req. D2.3; CT0.13, CT0.14 (Part 1); orthogonal-projection facts [linear algebra].
Pf.

$$\text{D2.3}: \ \mathcal{P}_M^{(n)} = \text{orthogonal projection } \mathcal{K} \to \mathcal{K}_n = \bigoplus_{A \in \mathcal{A}(n)} \mathcal{K}^{(A)}$$

$$\text{orthogonal projection} \implies P^2 = P = P^\dagger, \quad \mathfrak{T}P = \dim \operatorname{Im} P \quad \text{[linear algebra]}$$

$$\operatorname{Im} = \mathcal{K}_n \text{ [construction]}; \qquad \dim \mathcal{K}_n \text{ [CT0.14]}; \qquad \mathcal{A}(n) \text{ derived [CT0.13]}. \ \blacksquare$$

Rmk. Theorem, not stipulation: CT0.13 derives which blocks are active; D2.3 converts the list into a projector.

Phys. Rung selection is a genuine orthogonal projection in force space — idempotent, self-adjoint, with trace equal to the block dimension $6/16/17/25$. Filtering by rung twice is the same as once: the universe's capability space grows by strict projections, never by deformations.

Theorem 2.11 (CT2.3 — Masked force operator). Thm. $\mathrm{F}_M^{(n)} := \mathcal{P}_M^{(n)} \mathrm{F}\, \mathcal{P}_M^{(n)}$ satisfies

$$\mathrm{F}_M^{(n)}\big|_{\mathcal{K}_n^\perp} = 0; \qquad \mathrm{F}_M^{(n)}(\mathcal{K}) \subseteq \mathcal{K}_n; \qquad \mathrm{F}^\dagger = \mathrm{F} \implies \big(\mathrm{F}_M^{(n)}\big)^\dagger = \mathrm{F}_M^{(n)}.$$

Req. A8, C7; D2.3; CT2.2.
Pf.

$$x \in \mathcal{K}_n^\perp \implies \mathcal{P}_M^{(n)} x = 0 \implies \mathrm{F}_M^{(n)} x = 0$$

$$\forall x: \ \mathrm{F}_M^{(n)} x = \mathcal{P}_M^{(n)} \big(\mathrm{F}\, \mathcal{P}_M^{(n)} x\big) \in \operatorname{Im} \mathcal{P}_M^{(n)} = \mathcal{K}_n \quad \text{[CT2.2]}$$

$$\big(\mathcal{P}\mathrm{F}\mathcal{P}\big)^\dagger = \mathcal{P}^\dagger \mathrm{F}^\dagger \mathcal{P}^\dagger = \mathcal{P}\mathrm{F}\mathcal{P} \quad \text{[CT2.2; } \mathrm{F}^\dagger = \mathrm{F}\text{]}. \ \blacksquare$$

Phys. The masked force is exactly the restriction of the full force operator to the rung's active subspace: it annihilates everything the rung cannot access, stays inside what it can, and inherits Hermiticity — hence a real spectrum — from F. A8's $\mathrm{F}_M = P_M \mathrm{F} P_M$ is the rung-unresolved form (C7).

Theorem 2.12 (CT2.4 — Force restriction by rung). Thm.

$$\mathrm{F}_M^{(n)} = \bigoplus_{A\in\mathcal{A}(n)} F_A\big|_{\mathcal{K}^{(A)}},$$

$$\mathcal{A}(n) \text{ as in CT0.13;}$$

$$\neg\exists \text{ independent aspect } Z \text{ at any rung } (W_3 \text{ seed} \in F_W) \text{ [C8].}$$

Req. A8, C8; D2.3; CT0.13 (Part 1), CT2.2, CT2.3.
Pf.

$$\text{A8}: \ \mathrm{F} = \bigoplus_A F_A, \quad \operatorname{supp} F_A \subseteq \mathcal{K}^{(A)}; \qquad \text{D2.3}: \ \mathcal{P}_M^{(n)} \to \bigoplus_{A\in\mathcal{A}(n)} \mathcal{K}^{(A)}$$

$$\text{block-diagonal conjugated by block-subset projection} \implies \text{retains exactly those blocks}$$

$$\mathcal{A}(n) \text{ list} = \text{[CT0.13]}; \qquad F_W = F_{W^+} \oplus F_{W^-} \oplus F_Z,\ F_Z \equiv F_{W^3} \text{ single block [C8]. } \blacksquare$$

Phys. Masking never mixes forces: what survives at rung n is exactly the direct sum of the active aspects. This licenses speaking of gravity from rung 1, electromagnetism from rung 2 and colour from rung 3 as literal block content of one operator.

Theorem 2.13 (CT2.5 — Masked resolvent as spectrum generator). Thm. $\mathrm{F}_M^{(n)} = \sum_a \lambda_a P_a$ (spectral decomposition; $P_a P_b = \delta_{ab} P_a$, $P_a^\dagger = P_a$). Then

$$Z(\mathcal{R}, n) = \left\langle \text{out} \middle| (1 - \mathrm{F}_M^{(n)})^{-1} \middle| \text{in} \right\rangle = \sum_a \frac{\langle \text{out}|P_a|\text{in}\rangle}{1-\lambda_a};$$

$$\text{poles} \iff 1 \in \operatorname{spec} \mathrm{F}_M^{(n)};$$

$$\operatorname*{Res}_{\lambda_a \to 1} = -\langle \text{out}|P_a|\text{in}\rangle = \text{transition amplitude;}$$

$$\text{D2.8}: \ 1 \mapsto 1 + i\varepsilon \text{ selects the branch.}$$

Req. D2.4, D2.5, D2.8; CT0.14 (B: $\dim \mathcal{K}_n < \infty$), CT2.3 (Hermitian); spectral theorem [math].
Pf.

$$(\mathrm{F}_M^{(n)})^\dagger = \mathrm{F}_M^{(n)} \text{ [CT2.3]} \ \wedge\ \dim \mathcal{K}_n < \infty \text{ [CT0.14]} \implies \mathrm{F}_M^{(n)} = \sum_a \lambda_a P_a \quad \text{[spectral theorem]}$$

$$(1 - \mathrm{F}_M^{(n)})^{-1} = \sum_a \frac{P_a}{1-\lambda_a}, \quad \text{defined} \iff \forall a:\ \lambda_a \neq 1; \quad \text{singular} \iff \exists a:\ \lambda_a = 1$$

$$Z(\mathcal{R}, n) = \sum_a \frac{\langle \text{out}|P_a|\text{in}\rangle}{1-\lambda_a} : \text{ simple pole at } \lambda_a = 1,\ \text{Res} = -\langle \text{out}|P_a|\text{in}\rangle \text{ [D2.5 : amplitude]}$$

D2.8 : $1 \mapsto 1+i\varepsilon \implies$ poles off the real axis, fixed direction; $\varepsilon \to 0^+$ after contour integration. ■

Phys. The rung-n particle spectrum is literally the condition "the masked force has eigenvalue 1": poles of the resolvent are masses (D2.5), residues are amplitudes. The $i\varepsilon$ sheet makes the sums convergent and selects the branch — the entire spectral content of the corpus sits in this one matrix element.

Theorem 2.14 (CT2.6 — Continuum manifold closure). Thm. Conditions: (i) valid universe [D2.11] — a definitional precondition (D2.11's own definition in application, "there is something"), $\neg$ an extra-axiomatic hypothesis [A §0.2a]; (ii) coarse-graining cells contain unboundedly many real edges as $\hbar \to 0$ — derived (Part 4, Tier BM.24: the ledger's own unbounded ladder witnesses it [BM.151(1), forward pointer]); (iii) sector coefficient fields [D2.1] of bounded variation across neighbouring cells — derived (Part 4, Tier BM.24: the spectral bound $|r_a| \le \sqrt{3}$ with convex averaging [BM.151(2), forward pointer]). Then

$$M := \Gamma_{\mathcal{R}}/\sim \text{ is a smooth 4-manifold;} \qquad \chi([v]) = (x^0, x^1, x^2, x^3) = (\tau_n, h_1, h_2, h_3) \text{ charts;}$$

$$\text{transition maps} = \text{smooth diffeomorphisms;}$$

$$g_{\mu\nu} \text{ [CT1.12]} = \text{smooth Lorentzian metric on } M;$$

$$\text{nondegeneracy} \iff \varkappa\|q\|_{\text{op}} < 1.$$

Req. A7; D1.3, D1.8, D1.9, D2.11; CT0.3 (Part 1), CT1.12; conditions (ii)–(iii) [derived — cf. Part 4 Tier BM.24, BM.151, forward pointer]; (i) = D2.11 in application.

Pf.

$$\text{(i)} : \ \#J^+$$

$$= \#J^- \text{ [D2.11, D1.3]} \implies \textstyle\sum \Delta n_{\mathcal{R}}$$

$$= 0 \text{ [CT0.3]} \implies \neg\exists \text{ dangling ledger boundary beyond the sealed Janus boundary}$$

$$\sim := \text{ equal coarse-grained ledger entry under mask-preserving relabeling;}$$

$$\text{(ii)} \implies \text{cells contain unboundedly many edges} \implies M \text{ locally a continuum as } \hbar \to 0$$

$$\text{charts}: \; x^0 := \tau_n \text{ [D1.8]};$$

$$x^i := \text{hop counters projected on the 3 shift directions } (W_i \text{ sector}) \text{ [CT1.12]}; \quad \chi : U \subseteq M \to \mathbb{R}^4;$$

$$4 = 1 + 3 \text{ [CT1.12]}$$

$$\text{smoothness}: \; \text{coefficients}$$

$$= \text{local ledger averages};$$

$$\text{D1.9}: \; \text{updates local, finite};$$

$$(\text{iii}): \; \text{jumps uniformly bounded} \implies \text{smooth interpolation, deviation} \to 0 \text{ with } \hbar$$

$$\text{transitions}$$

$$= \text{resynchronizations between valid observers}: \; \text{mask-} \wedge \text{Janus-preserving} \implies \text{finite compositions of translations/permutations/sign reversals}$$

$$\implies \text{smooth, constant Jacobian, } |\det J| = 1; \quad \circ \text{ interpolation} \implies \text{diffeomorphisms}$$

$$\text{nondegeneracy}: \; \text{CT1.12 } (\varkappa \|q\|_{\text{op}} < 1) \iff J \text{ invertible};$$

$$\text{atlas} \implies M \text{ smooth 4-manifold; } g_{\mu\nu} \text{ smooth Lorentzian. } \blacksquare$$

Phys. Under three explicit coarse-graining hypotheses — and nothing else — the real-edge network of the ledger becomes a smooth four-dimensional spacetime carrying the emergent Lorentzian metric. This is the bridge theorem of the corpus: it converts combinatorics (edges, ticks, counters) into the differential geometry that Parts III–IV vary.

Rmk (derivation locus: Part 4 Tier BM.24). Conditions (ii)–(iii) carry no extra-axiomatic status [A §0.2a]: BM.151 derives both from the ledger's own structure: (ii) from the configuration-space ladder (A7; FT.32; FT.45–FT.46) with the ladder row $\#\{E_{\mathcal{R}} \text{ per cell}\} \geq 2sE \ln 2/\hbar \to \infty$, and (iii) from the pair-state spectral bound (A1; L1; L3(1)) with the convex averaging of the coefficient fields. This theorem's statement, proof and conclusions are unchanged; the point-of-use declarations across the corpus stand as scope-inheritance markers. The extra-axiomatic list of A §0.2a is empty.

Theorem 2.15 (CT2.7 — Exact geometric action closure). Thm. Under the hypotheses of CT2.6:

$$S_{\text{geom}}[g] = \frac{1}{2\kappa^2}\int d^4x\,\sqrt{-g}\,(R - 2\Lambda); \qquad \neg\exists \text{ additional local higher-curvature terms.}$$

Req. A8; D1.9, D4.3; CT2.6; Lovelock classification $D = 4$; Gauss–Bonnet = total derivative in $D = 4$ [math].

Rmk (forward pointer, non-load-bearing). The κ-normalization cross-check against CT1.23 is a consistency pointer, not a derivation input.

Pf.

$$\text{D1.9 : tick stencil} = 1 \text{ time step} + 1 \text{ edge} \implies \text{continuum E–L equations} \le 2 \text{ nd order}$$

$$\text{arena} = \text{smooth 4-dim Lorentzian manifold [CT2.6]}$$

$$\text{weights depend on } (\Gamma, \text{registrations}) \text{ only,}$$

$$\neg\text{vertex names} \implies \text{relabeling-invariant} \xrightarrow[\text{CT2.6}]{} \text{diffeomorphism-invariant local functional of } g$$

$$\text{Lovelock } (D=4): \ \{\sqrt{-g}R,\ \sqrt{-g},\ \sqrt{-g}\,\mathcal{G}\}, \quad \mathcal{G} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \quad \text{[math]}$$

$$\sqrt{-g}\,\mathcal{G} = \partial(\cdots) \text{ in } D = 4 \implies \delta S \equiv 0 \implies \text{dropped} \quad \text{[math]}$$

$$F_{D^+} = \Lambda\mathbb{I}\ \text{[A8]},\ \neg\text{configuration-dependent [D4.3]} \implies \text{zero-derivative density } \sqrt{-g}\,\Lambda \text{ only}$$

$$\text{collect at } \tfrac{1}{2\kappa^2} \text{ normalization [CT1.23]} \implies \text{Thm. } \blacksquare$$

Phys. The geometric dynamics is closed exactly: tick-locality plus diffeomorphism invariance plus Lovelock leave only Einstein–Hilbert with a cosmological term, and that term is the uniform dark aspect itself. No higher-curvature corrections exist — the higher-derivative question is answered, not deferred.

Theorem 2.16 (CT2.8 — Reflection positivity and Hilbert-space closure). Thm. With $\langle A|B\rangle := Z\big[\Theta_{\mathfrak{L}}A \cup B\big]$ on $\mathfrak{C}(\mathcal{R})$ [D0.10]:

$$(1)\ \ w(\Theta_{\mathfrak{L}}C) = w(C)^*; \qquad (2)\ \ w(C) = w(C_{\text{left}})\cdot w(C_{\text{right}}) \ \text{ across any interior ledger-time cut;}$$

$$(3)\ \ \langle A|A\rangle = \sum_C |w(C)|^2 \ \ge\ 0;$$

$$(4)\ \ \langle A|A\rangle = 0 \iff A \text{ mask-excluded } \ (\text{null space} = \text{mask-excluded configurations});$$

$$(5)\ \ \text{quotient} \wedge \text{completion} = \text{Hilbert space};$$

$$(6)\ \ Z\big|_{\text{reflected pair}} = \|\cdot\|^2, \ \ \text{convergent on the D2.8 sheet.}$$

Req. A8; D0.7, D0.10; D1.2, D1.3, D1.9; D2.6, D2.7, D2.8; D3.3, D3.5, D3.6; CT2.2; Hilbert-space completion [math; machine: `checks/verify_F33_dynamic_phase.py` §D].

Pf.

$$(1):\ \mathcal{J}(\Theta_{\mathfrak{L}}C) = \mathcal{J}(C)^* \text{ [D3.6]};$$

$$G(\Theta_{\mathfrak{L}}e) = G(e)^* \text{ [D2.7, D3.5 : } \alpha_e \widehat{ds}^{\,2}(e) \mapsto -\alpha_e \widehat{ds}^{\,2}(e),\ \sigma_e \text{ fixed]}$$

$$\omega:\ \text{type}(1,1) \mapsto \text{signs exchanged}$$

$$\implies f_v$$

$$\mapsto -f_v \implies \aleph_v$$

$$= |f_v| \text{ invariant}$$

$$\implies (-1)^{\aleph} \in \mathbb{R} \text{ invariant [D2.6, D3.3]}$$

$$\implies w(\Theta_{\mathfrak{L}}C) = w(C)^* \quad \text{[A8]}$$

$$(2):\ \text{A8}:\ w(C) = \mathcal{J}(C) \prod_{e\in E(C)} G(e)\cdot(-1)^{\aleph(\sigma,C)};$$

$$\text{interior cut at a ledger time}:\ E(C)$$

$$= E_{\mathrm{L}} \sqcup E_{\mathrm{R}} \text{ [D0.7 : each edge is registered between consecutive ticks, wholly on one side]}$$

$$\prod_{e\in E(C)} G(e) = \prod_{E_{\mathrm{L}}} G \cdot \prod_{E_{\mathrm{R}}} G;$$

$$(-1)^{\aleph(\sigma,C)} = \prod_v (-1)^{\aleph_v}:\ \aleph_v \ v\text{-local [D3.3; D1.9 : } \text{dom}(\text{tick}(v)) \subseteq \{v\} \cup \{e \ni v\}]$$

interface : the seal data at the cut are boundary data of the halves [D1.2; D1.3 : $\#J^+ = \#J^-$ pair at boundaries];

side convention: each vertex factor belongs to the side holding its tick

$$\implies w(C) = w(C_{\text{left}}) \cdot w(C_{\text{right}}) \quad \text{[product regrouping]}$$

$$(3): \quad A = B: \quad \langle A|A\rangle = \sum_C w(\Theta_{\mathfrak{L}} C) w(C) = \sum_C |w(C)|^2 \geq 0 \quad [(1),(2)]$$

$$(4): \ = 0 \implies \forall C: \ w(C) = 0;$$

A8 weight = product of nonzero factors except where excluded [$|G(e)| = 1$ (D3.5); $\mathcal{J} \neq 0$];

$$\text{excluded} \iff \notin \operatorname{Im} \mathcal{P}_M^{(n)} \ \text{[CT2.2]}$$

$$(5): \ \text{PSD form} \implies \text{positive definite on quotient} \implies \text{completion} = \text{Hilbert space}$$

[math: completion theorem; no network-specific input]

$$(6): \quad \text{D2.8 sheet} \implies \sum |w|^2 \ \text{converges}; \ \neg\exists \ \text{indefinite oscillatory spoilage.} \blacksquare$$

Phys. The ledger's own reflection symmetry makes the configuration sum positive: the reflected pairing is a sum of $|w|^2$, its null space is exactly what the mask forbids, and the quotient completes to a genuine Hilbert space. The split along an interior ledger-time cut is a consequence of A8's per-edge product structure and tick locality — the seals sit at boundaries [D1.2, D1.3], and no factor straddles an interior cut. This is the positivity property that licenses reading ledger weights as quantum theory.

Theorem 2.17 (CT2.10 — Graded matrices and the supertrace). Thm. For homogeneous X, Y [D2.10]:

$$\operatorname{Str}(XY) = (-1)^{|X||Y|} \operatorname{Str}(YX); \qquad \operatorname{Str}[X, Y\} = 0.$$

Req. D2.10; cyclicity of $\mathfrak{T}$ [linear algebra].
Pf.

$$|X| = |Y| = 0: \quad X = \operatorname{diag}(X_B, X_F), \ Y = \operatorname{diag}(Y_B, Y_F):$$

$$\operatorname{Str}(XY) = \mathfrak{T}(X_B Y_B) - \mathfrak{T}(X_F Y_F) = \mathfrak{T}(Y_B X_B) - \mathfrak{T}(Y_F X_F) = \operatorname{Str}(YX); \ (-1)^0 = +1$$

$$|X| \neq |Y| : \;\; XY,\, YX \text{ odd} \implies \text{diagonal blocks} = 0 \implies \mathrm{Str}(XY) = \mathrm{Str}(YX) = 0$$

$$|X| = |Y| = 1 : \;\; X = \begin{pmatrix} 0 & X_1 \\ X_2 & 0 \end{pmatrix},\; Y = \begin{pmatrix} 0 & Y_1 \\ Y_2 & 0 \end{pmatrix} :$$

$$XY = \mathrm{diag}(X_1Y_2,\; X_2Y_1),\; YX = \mathrm{diag}(Y_1X_2,\; Y_2X_1)$$

$$\mathrm{Str}(XY) = \mathfrak{T}(X_1Y_2) - \mathfrak{T}(X_2Y_1) = \mathfrak{T}(Y_2X_1) - \mathfrak{T}(Y_1X_2) = -\,\mathrm{Str}(YX);\; (-1)^1 = -1$$

$$\implies \mathrm{Str}[X, Y\} = \mathrm{Str}(XY) - (-1)^{|X||Y|}\,\mathrm{Str}(YX) = 0 \;\; \forall \text{ parities. } \blacksquare$$

Phys. The supertrace is graded-cyclic: it vanishes on graded commutators and therefore descends to an invariant form on the graded (bosonic/fermionic) algebra — the tool that lets bound (fermionic) and transit (bosonic) content coexist in one trace formula.

2.5 §C.III — Continuum Limit, Effective Action, and Field Equations

Theorem 2.18 (CT1.19 — Continuum limit at leading order). Thm. $L = \mathfrak{h}\, n_{\mathcal{R}}$ fixed; $\mathfrak{h} \to 0 \wedge n_{\mathcal{R}} \to \infty \implies$

$$\sum_{e\in\gamma} ds^2(e) \;\longrightarrow\; \int_\gamma g_{\mu\nu}\, dx^\mu dx^\nu = L \int_0^1 g_{\mu\nu}\big(x(\lambda)\big)\dot{x}^\mu\dot{x}^\nu\, d\lambda;$$

$$\text{abs. error} = O(\mathfrak{h}L), \quad \text{rel. error} = O(\mathfrak{h}).$$

Corollary 2.1 (CT1.19a). $\widehat{ds}^{\,2}(e) = ds^2(e)/\mathfrak{h}^2$: dimensionless, $O(1)$ generically; $\widehat{ds}^{\,2}(e) = 0 \iff \Delta x_e$ null; trivial chain [D1.7] = all-null-edge chain.

Req. D0.6, D1.6, D1.7; CT2.6; midpoint Taylor, Riemann sums [math].

Pf.

$$\text{CT2.6} \implies g_{\mu\nu} \text{ smooth along } \gamma; \qquad \text{D0.6} : \; |\Delta x_e^\mu| = O(\mathfrak{h})$$

$$\lambda \in [0,1],\; x^\mu(\lambda) \text{ smooth} : \quad \Delta x_e^\mu = \mathfrak{h}\,\dot{x}^\mu(\lambda_e) + O(\mathfrak{h}^2) \quad \text{[midpoint Taylor]}$$

$$\sum_e ds^2(e) = \sum_e g_{\mu\nu}(x_e)\,\mathfrak{h}^2\dot{x}_e^\mu\dot{x}_e^\nu + O(\mathfrak{h}^3 n_{\mathcal{R}}) \quad \text{[D1.6]}$$

$$= \mathfrak{h}\sum_e \mathfrak{h}$$

$$g_{\mu\nu}(x_e)\dot{x}_e^\mu\dot{x}_e^\nu + O(\mathfrak{h}^2 L) \xrightarrow[\text{Riemann, mesh } \mathfrak{h}/L]{} L\int_0^1 g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu d\lambda$$

$$\text{leading} = O(\hbar L) \implies \text{rel.} = O(\hbar),\ \text{abs.} = O(\hbar L) \to 0. \blacksquare$$

Pf (Cor). $[\widehat{ds}^{\,2}] = [\mathfrak{h}^0]$ [D1.6]; $\widehat{ds}^{\,2}(e) = 0 \iff ds^2(e) = 0 \iff \Delta x_e$ null; Trivial $\iff \forall e : \widehat{ds}^{\,2}(e) = 0$ [D1.7]. $\blacksquare$

Phys. At fixed macroscopic length, shrinking the hop length makes the sum of discrete line elements converge to the continuum path integral of the metric, with relative error $O(\hbar)$. Spacetime intervals are literally Riemann sums over edge data — and null edges define the trivial chains that later become photons and gluons.

Rmk (on-shell interface). On-shell integer quantization is no obstruction to this limit: for materialized closed chains of the fully antialigned class (realized — CT1.19b(3)) each $\kappa_e \in \mathbb{Z}_{\geq 0}$ samples $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu(\lambda_e)$ with $O(\hbar)$ slack, and $\sum_e \kappa_e\, \mathfrak{h}^2 = \sum_e ds^2(e)$ identically [D1.6] — so CT1.19 applies verbatim to on-shell chains (abs. error $O(\hbar L)$). Machine: `checks/verify_merge_tierA.py` §G.

Theorem 2.19 (CT1.19b — On-shell line-element quantization: even antialigned winding, orientation evenness, the geometry quantum, internal holonomy). Thm. A closed chain c [D1.4, D1.5] materializes := it contributes a spectral pole of the masked resolvent at the rung of its active mask [D2.4, D2.5, CT2.5, D1.12, CT0.13] — the corpus's own pole-as-materialization condition as applied in EW.18; instance: wrapping chains [EW.17]. Under EM.20's phase alphabet ($\alpha_e \in \{0, \pi\}$ per real edge, at every accessible rung): (1) Even antialigned winding. For every materialized closed chain the spiral holonomy is single-valued:

$$\mathrm{Hol}_{\mathrm{spiral}}(c) = e^{i\Phi(c)} = +1,$$

$$\Phi(c) = \sum_{e\in c} \alpha_e\, \widehat{ds}^{\,2}(e) = \pi\, \Theta_{\mathcal{P}}(c) = 2\pi k, \quad k \in \mathbb{Z} \ \ (\text{winding [D1.7, EW.18]})$$

$$\Theta_{\mathcal{P}}(c) := \sum_{e\in\mathcal{P}(c)} \widehat{ds}^{\,2}(e)$$

$$[\mathcal{P}(c) :\ \text{the antialigned edges of } c,\ \text{Def-EM.21}; \quad \alpha_e \in \{0,\pi\} :\ \text{EM.20(1)}]$$

$$\implies \Theta_{\mathcal{P}}(c) = 2k \in 2\mathbb{Z} \qquad (n = 2, 3, \geq 4); \qquad \mathcal{P}(c) = c\ (\text{fully antialigned class}) :\ \ \Theta(c) = 2k.$$

On rung-3 wrapping chains this reduces verbatim to EW.18 (consistency). (2) Orientation evenness. For a bidirectional channel — both orientations $e_{pq} = (p, q)$, $e_{qp} = (q, p)$ in $\mathcal{R}$:

$$\widehat{ds}^{\,2}(e_{pq}) = \widehat{ds}^{\,2}(e_{qp})$$

(3) The geometry quantum. For a materialized 2-cycle $c = (e_1, e_2)$, $e_1 = (p, q)$, $e_2 = (q, p)$ — the minimal closed chain (A1 excludes self-loops [D1.4]) — writing $\kappa := \widehat{ds}^{\,2}(e_1) = \widehat{ds}^{\,2}(e_2)$ [clause (2)] and $m := |\mathcal{P}(c)| \in \{0, 1, 2\}$:

$$\Theta_{\mathcal{P}}(c) = m\kappa \in 2\mathbb{Z} \implies \kappa \in \tfrac{2}{m}\mathbb{Z}\ \ (m \geq 1);$$

$$m = 2 :\ \ \kappa \in \mathbb{Z}; \quad ds^2(e) = \kappa\, \mathfrak{h}^2 \in \mathfrak{h}^2\mathbb{Z}\,[\text{D1.6}]; \quad \mathfrak{h}^2 = 4G_N \ln 2\,[\text{CT2.9}] \implies ds^2(e) = 4\kappa\, G_N \ln 2$$

— the interval quantum is one Bekenstein bit-area [CT1.29: $S = \ln 2$ at $A = \hbar^2$] per unit κ on the fully antialigned class. Attainment and sign classes: $\kappa = 1$ attained on the minimal real edge [D1.1] with $m = 2$ (floor tight; both alignment values realizable: EM.20(5)); $\kappa = 0 \iff$ null edge [CT1.19a] (degenerate floor; all-null transit chains are the trivial chains — photons/gluons [D1.7, D3.7]); $\kappa < 0$ admissible = timelike [CT1.12: $\mathrm{sig}(g) = (-,+,+,+)$]. $m = 0$: the winding condition is vacuous, κ unconstrained by clause (1). (4) Internal holonomy. If a subnetwork's interior edges $\Sigma \setminus \partial\Sigma$ [D2.13] decompose edge-disjointly into materialized closed chains $\{c_i\}$, then:

$$\Theta_{\mathcal{P},\mathrm{int}}(\Sigma) := \sum_{e \text{ int},\ \alpha_e = \pi} \widehat{ds}^{\,2}(e) = \sum_i \Theta_{\mathcal{P}}(c_i) \in 2\mathbb{Z},$$

$$\mathrm{Hol}_{\mathrm{int}} := e^{i\Phi_{\mathrm{int}}} = e^{i\pi\Theta_{\mathcal{P},\mathrm{int}}} = +1$$

and the surviving on-shell integer label is the winding $k(c) = \Theta_{\mathcal{P}}(c)/2$. On partial internal sets (proper subsets of a materialized chain's edges) $\Theta_{\mathcal{P},\mathrm{int}}$ is a partial sum of integers with free parity: $\mathrm{Hol}_{\mathrm{int}} = \pm 1$ well-defined, nontrivial (witness: a single $\kappa = 1$ antialigned edge gives $(-1)^1 = -1$). Off-shell, $\widehat{ds}^{\,2}(e)$ remains generic continuous [CT1.19a] — quantization is an on-shell (materialization) property, ¬a postulate on every edge.

Req. A1; D1.1, D1.4, D1.5, D1.6, D1.7, D1.12; D2.4, D2.5, D2.13; CT0.7(2), CT0.13, CT1.1, CT1.2, C3 (Part 1); CT1.12, CT1.19a, CT1.29, CT2.5, CT2.6 (charts; hypotheses — A §0.2a); D3.7; Def-EM.21, EM.20 (E; forward pointers — C DAG note); EW.17, EW.18 (D; forward pointers — wrapping instance and winding freedom); single-valued eigenmode condition, quadratic-form evenness [math; machine: `checks/verify_merge_tierB.py` §B].

Pf.

$$(1):\ \text{D1.7}:\ \Theta,\ \Phi = \textstyle\sum_e \alpha_e \widehat{ds}^{\,2}(e),\ \mathrm{Hol}_{\mathrm{spiral}} = e^{i\Phi} \text{ defined for every chain;}$$

$$\text{EW.17}:\ \text{transport around a closed cycle multiplies the eigenmode by } \mathrm{Hol}_{\mathrm{spiral}}$$

$$\text{CT2.5 (as applied in EW.18)}:\ \text{pole}$$

$$= \text{eigenvalue condition on a single-valued eigenmode}$$

$$\implies e^{i\Phi(c)} = 1;$$

$$\text{winding freedom [EW.18]}:\ \Phi(c) = 2\pi k,\ k \in \mathbb{Z}$$

$$\text{domain step (flagged)}:\ \text{EW.17}'\text{s rung-3 restriction is an existence claim about wrapping chains;}$$

CT2.5′s single-valuedness step is chain-generic $\implies$ the Pf extends to every materialized closed chain with no added premise

EM.20(1) : $\alpha_e \in \{0, \pi\}$ at every accessible rung (the case split is rung-blind; CT0.9 alphabet) $\implies \Phi(c)$

$$= \pi \sum_{e \in \mathcal{P}(c)} \widehat{ds}^{\,2}(e)$$

$$= \pi \Theta_{\mathcal{P}}(c) \implies \Theta_{\mathcal{P}}(c) \qquad = 2k$$

$$(2) : \text{ CT2.6 charts : } \Delta x_{e_{qp}} = -\Delta x_{e_{pq}};$$

$g_{\mu\nu}$ built solely from S-even data [CT1.12; CT1.12a(2); $\mathcal{E}_{++}, \mathcal{E}_{+-}$ S-even: CT1.1, CT1.2, C3; $S(X) = \mathbb{S}X\mathbb{S}$: CT0.7(2)]

$$\text{D1.6 : } ds^2(e)$$

$$= g_{\mu\nu}(e)$$

$\Delta x_e^\mu \Delta x_e^\nu$ quadratic in Δx (every monomial of degree 2, incl. the shift cross-term $2g_{0i}\Delta x^0 \Delta x^i$) $\implies \Delta x$

$$\mapsto -\Delta x \text{ fixes } ds^2$$

$$(3) : \Theta_{\mathcal{P}}(c) = m\kappa, \; m = |\mathcal{P}(c)| \; [(2) : \text{ both edges share } \kappa];$$

$$(1) : \; m\kappa \in 2\mathbb{Z};$$

$$m = 2 : \; \kappa \in \mathbb{Z};$$

$$\text{D1.6 : } ds^2 = \kappa \mathfrak{h}^2;$$

$$\text{CT2.9 : } \mathfrak{h}^2 = 4G_N \ln 2;$$

$$\text{CT1.29 : } A = \mathfrak{h}^2 \Rightarrow S = \ln 2$$

$$\text{D1.1 : } \widehat{ds}^{\,2} = 1 \text{ on the minimal real edge;}$$

$$\text{EM.20(5) : } \alpha_e = 0 \; \wedge \; \alpha_e = \pi \text{ each realizable} \implies m = 2 \text{ attained} \implies \kappa = 1 \text{ attained;}$$

$$\text{CT1.19a}: \ \widehat{ds}^{\,2} = 0 \iff \text{null};$$

$$\text{D1.7/D3.7}: \ \text{all-null transit} = \text{trivial chains}$$

$$(4): \ \text{edge-disjointness}: \ \Theta_{\mathcal{P},\text{int}} = \sum_i \Theta_{\mathcal{P}}(c_i) \in 2\mathbb{Z} \ [(1)];$$

$$e^{i\pi\cdot 2\mathbb{Z}} = +1;$$

$$\text{partial sets}: \ \text{single antialigned } \kappa = 1 \text{ edge} \implies \Theta_{\mathcal{P},\text{int}} = 1, \ (-1)^1 = -1. \ \blacksquare$$

Rmk (locus of the quantization). The winding constraint reads on the antialigned content: edges with $\alpha_e = 0$ drop out of $\Phi(c)$ [D1.7], so $\Theta(c)$ is constrained exactly on the fully antialigned class $\mathcal{P}(c) = c$; on a general pattern the constrained sum is $\Theta_{\mathcal{P}}(c)$ [Def-EM.21]. The alphabet $\{0, \pi\}$ — derived, with no effectivity input [EM.20(1); A §0.2a] — is what makes the constraint integral (machine: §B parity sweep).

Rmk (sign-factor separation). Single-valuedness is imposed on the spiral/geometric transport [EW.18]; the registration signs σ_e and the wall parity $(-1)^{\aleph}$ are ledger data of the weight $w(C)$ [A8, CT0.9, D3.3], ¬of the geometric eigenmode transport.

Rmk (area language, scoped). $\Theta_{\mathcal{P}}(c)\,\mathfrak{h}^2$ is the κ-weighted accumulated line element of the antialigned content — dimensionally an area, and equal to CT1.29's pierced-edge area $A = n_{\mathcal{R}}\mathfrak{h}^2$ exactly on the minimal-edge fully antialigned class $\kappa \equiv 1$ (then $\Theta_{\mathcal{P}} = \delta h$; the exactness locus of the rate: CT1.29a(2)). It is ¬the metric-enclosed area of c: the floor 2-cycle traces a degenerate out-and-back locus — zero enclosed area — yet carries $\Theta_{\mathcal{P}} = 2\kappa \neq 0$ at $m = 2$, because ds^2 is quadratic, hence orientation-blind [clause (2)].

Phys. The ds^2 question is settled on-shell by computation, with no new postulate: the antialigned content of a materialized closed chain winds in even quanta, both orientations of a channel agree on its length, the minimal materialized 2-cycle pins the geometry quantum at one Bekenstein bit-area per hop on the fully antialigned class, and internal holonomy reduces to a single integer winding per chain. Off-shell, nothing is quantized — integrality is a property of materialization, not an axiom about edges.

Rmk (scope). Round-trip-closed bidirectional channels realize $\kappa \in \mathbb{Z}$ per edge; bare cycles admit the rational ladder $\kappa \in \frac{2}{L}\mathbb{Z}$, outside hypothesis (ii). The effective-quantum label reads $\bar{\kappa}$ here [READING]; the transport vocabulary has no corpus referent (nearest declared objects: round-trip bidirectional channels).

Theorem 2.20 (CT1.20 — Effective action from the partition function). Thm. $Z[g] = \sum_{C:\, g(C)=g} |w(C)|\, e^{i\Phi(C)}$ [Φ : D1.7+D3.5+parity phases]. Semiclassically ($\delta\Phi \neq 0$: pairwise cancellation; $\delta\Phi = 0$: dominance):

$$S_{\text{eff}}[g] := -i \log Z[g] \ \ \text{(branch [D2.8])};$$

$$\frac{\delta S_{\text{eff}}}{\delta g^{\mu\nu}} = 0 \iff \frac{\delta Z}{\delta g^{\mu\nu}} = 0 \ \ (Z \neq 0);$$

stationary points dominate.

Req. A8; D1.7, D2.8, D3.5; CT1.12, CT2.6 (Φ differentiable in coarse-grained variables); stationary phase [math].

Pf.

$$\text{CT1.12}: \; C \mapsto g_{\mu\nu}(C) \implies Z \text{ function of } g \text{ after grouping [A8]}$$

$$w(C) = |w(C)|e^{i\Phi(C)}; \qquad \Phi = \text{accumulated phase [D1.7, D3.5, } (-1)^{\aleph}]$$

$$\delta\Phi \neq 0 \text{ neighbourhoods cancel pairwise;} \quad \text{survivors} = \{\delta\Phi = 0\}$$

$$[\text{stationary phase; } \Phi \text{ differentiable [CT2.6]}]$$

$$S_{\text{eff}} := -i \log Z \text{ single-valued on the D2.8 sheet}$$

$$\delta Z = 0 \;\wedge\; Z \neq 0 \iff \delta \log Z = \delta Z/Z = 0 \iff \delta S_{\text{eff}} = 0. \blacksquare$$

Phys. Grouping ledger configurations by their emergent metric turns the partition sum into a path-integral-like object dominated by stationary phase; its logarithm is an effective action for geometry. Classical spacetime = the stationary set of the ledger sum; quantum fluctuations = the rest.

Theorem 2.21 (CT1.21 — The geometric part of the effective action). Thm. $S_{\text{geom}}[g] = \frac{1}{2\kappa^2}\int d^4x\,\sqrt{-g}\,(R - 2\Lambda)$.

Req. D1.9, D4.3; CT1.20, CT2.7.

Pf.

$$\text{CT1.20} \implies S_{\text{eff}} \text{ local, diffeomorphism-invariant, } \leq 2 \text{ nd order [D1.9 + relabeling]}$$

$$\text{CT2.7}: \text{ unique such functional in } D$$

$$= 4 \text{ (topological density dropped; } F_{D^+} = \Lambda\mathbb{I} \text{ included [D4.3])} \implies \text{Thm. } \blacksquare$$

Phys. The geometry part of the effective action is exactly Einstein–Hilbert with a cosmological term — derived from the structure of the ledger sum, with Λ supplied by the uniform dark aspect of A8.

Theorem 2.22 (CT1.22 — Stress-energy tensor from the force operator). Thm. With D^+ removed from Z_{force} (carried by Λ in S_{geom}):

$$S_{\text{force}}[g] := -i \log Z_{\text{force}}[g];$$

$$Z_{\text{force}}[g] = \big\langle \text{out} \big| (1 - \mathrm{F}_M^{(n)})^{-1} \big| \text{in} \big\rangle;$$

$$T_{\mu\nu} := -\frac{2}{\sqrt{-g}} \frac{\delta S_{\text{force}}}{\delta g^{\mu\nu}};$$

$$T_{\mu\nu} = T^{\text{bd}}_{\mu\nu} + T^{\text{tr}}_{\mu\nu};$$

$$T^{\text{bd}}_{\mu\nu} = (\rho + p) u_\mu u_\nu + p\, g_{\mu\nu};$$

$$T^{\text{tr}}_{\mu\nu} = \sum_a \Big(F^{(a)}_{\mu\alpha} F^{(a)\ \alpha}_{\nu} - \tfrac{1}{4} g_{\mu\nu} F^{(a)}_{\alpha\beta} F^{(a)\alpha\beta} \Big),$$

the sum over aspects whose leading transit representation is a connection.

Req. A8; D1.1, D1.5, D1.10, D4.3; CT1.19, CT1.21, CT2.4; $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\, g_{\mu\nu}\delta g^{\mu\nu}$, $\delta(F_{\alpha\beta}F^{\alpha\beta}) = 2F_{\mu\rho}F_\nu{}^\rho \delta g^{\mu\nu}$ [math].

Pf.

$$\text{A8 blocks; CT2.4 : mask retains exactly } \mathcal{A}(n);$$

$$\Delta S = \mathsf{E}_{\text{bd}} + \mathsf{E}_{\text{tr}} \text{ [D1.5]} \implies \forall \text{ contribution} \in \text{exactly one channel} \implies T = T^{\text{bd}} + T^{\text{tr}}$$

$$\text{bd : closed chain} \implies \text{rest frame selects timelike } u^\mu,\ \neg\text{spatial direction; content} = (\rho, p) \text{ [D1.10]}$$

$$\text{most general symmetric 2-tensor from } (g, u, \rho, p),$$

$$u\text{-rotation-invariant}$$

$$= (\rho + p) u_\mu u_\nu + p \qquad g_{\mu\nu}$$

$$[\text{math}]$$

$$\text{tr : D1.1 : edges carry phases} \implies \text{gauge-invariant local datum}$$

$$= \text{holonomy};$$

$$\text{holonomy/area} \to F^{(a)}_{\mu\nu} \text{ [CT1.19]}$$

$$\text{unique local, gauge-invariant, } \leq 2 \text{ nd order, quadratic in } F :$$

$$S_a = -\tfrac{1}{4}\int d^4x\sqrt{-g}\, F^{(a)}_{\mu\nu}F^{(a)\mu\nu}$$

$$\delta S_a$$

$$= -\tfrac{1}{4}\int \sqrt{-g}\big[-\tfrac{1}{2}g_{\mu\nu}F^2 + 2F_{\mu\rho}F_{\nu}{}^{\rho}\big]\delta g^{\mu\nu}$$

$$\implies T^{(a)}_{\mu\nu}$$

$$= F^{(a)}_{\mu\rho}F^{(a)}_{\nu}{}^{\rho} - \tfrac{1}{4}g_{\mu\nu}F^{(a)}_{\alpha\beta}F^{(a)\alpha\beta}$$

$$\text{no double counting : } D^+ \text{ excluded from } Z_{\text{force}} \text{ by construction;}$$

$$\text{D4.3 : } \neg\text{config.-dependent} \implies \text{removal}$$

$$= \text{constant factor} \implies \delta \text{ unaffected. } \blacksquare$$

Rmk. Abelian case written out in EM.15 (Part 2), which depends on this result, not the reverse.

Phys. The right-hand side of the field equations is built from the same force operator that built the geometry: bound (closed-chain, fermionic) content yields perfect-fluid stress; transit (connection-carrying, bosonic) content yields Maxwell-type stress. $T_{\mu\nu}$ is defined by metric variation of the masked-force resolvent, with the uniform dark part counted once, as Λ.

Theorem 2.23 (CT1.23 — Field equations). Thm.

$$S_{\text{eff}} = S_{\text{geom}} + S_{\text{force}}; \qquad \delta S_{\text{eff}} = 0 \iff G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa^2 T_{\mu\nu};$$

$$\text{dimensions [CT2.1] :} \quad \mathfrak{q}^{+1} = \mathfrak{q}^{-1}\cdot\mathfrak{q}^{+2} = \mathfrak{q}^{+1};$$

$$\kappa := \text{coefficient fixed by } S_{\text{geom}} \text{ normalization.}$$

Req. CT1.21, CT1.22, CT2.1; $\delta(\sqrt{-g}R) = \sqrt{-g}\,G_{\mu\nu}\delta g^{\mu\nu} + \partial(\cdots)$, $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\,g_{\mu\nu}\delta g^{\mu\nu}$ [math].

Pf.

$$\delta S_{\text{geom}} = \frac{1}{2\kappa^2}\int d^4x\sqrt{-g}\,\big(G_{\mu\nu} + \Lambda g_{\mu\nu}\big)\delta g^{\mu\nu} \quad \text{[CT1.21; } \delta\text{-identities]}$$

$$\delta S_{\text{force}} = -\tfrac{1}{2}\int d^4x\sqrt{-g}\,T_{\mu\nu}\,\delta g^{\mu\nu} \quad \text{[CT1.22 definition]}$$

$$\forall\,\delta g^{\mu\nu}:\ \ \delta S_{\text{eff}} = 0 \iff \frac{1}{2\kappa^2}\big(G_{\mu\nu}+\Lambda g_{\mu\nu}\big) - \tfrac{1}{2}T_{\mu\nu} = 0 \iff G_{\mu\nu}+\Lambda g_{\mu\nu} = \kappa^2 T_{\mu\nu}$$

$$\implies \kappa = \text{the } S_{\text{geom}} \text{ coefficient (no other normalization admits source} = T_{\mu\nu});$$

$$\text{dimensions [CT2.1]}:\ \text{both sides } \mathfrak{q}^{+1}.\ \blacksquare$$

Phys. Varying the derived effective action yields Einstein-type field equations with a cosmological term and the ledger-sourced stress tensor — and simultaneously fixes κ as that coefficient. In this corpus the field equations of gravity are a theorem with both sides constructed from A1–A9.

Rmk (dependency audit). The field-equation chain (CT1.20 → CT1.21 → CT1.22 → CT1.23, conservation CT1.24/CT1.26) rests solely on the declared inputs listed in A §0.2a (CT2.6 (i)–(iii) via CT1.19/CT2.6; Lovelock [math]) — no clause depends on persistence data, on any tick-rate/field coupling (count 0 in Parts 1–2), or on the materialization-selection question (outside Parts 1–2; downstream of and disjoint from this derivation); the spatial components are no shakier than the temporal ones. On-shell interface: CT1.19 Rmk. The "chosen vs forced" upgrade claim has no corpus referent and is blocked by proven no-gos.

Theorem 2.24 (CT1.24 — Conservation from the field equations). Thm. $\Lambda = \text{const} \implies \nabla^\mu T_{\mu\nu} = 0$.

Req. CT1.23, CT2.9 ($\kappa \neq 0$); contracted Bianchi $\nabla^\mu G_{\mu\nu} = 0$; metric compatibility $\nabla^\mu g_{\mu\nu} = 0$ [math].

Pf.

$$\nabla^\mu G_{\mu\nu} = 0 \text{ [Bianchi]}; \qquad \nabla^\mu(\Lambda g_{\mu\nu}) = \Lambda\nabla^\mu g_{\mu\nu} = 0 \text{ [compatibility; } \Lambda = \text{const]}$$

$$\nabla^\mu\text{(CT1.23)}:\ \ 0+0 = \kappa^2\nabla^\mu T_{\mu\nu}; \qquad \kappa^2 = \tfrac{2\pi\mathfrak{h}^2}{\ln 2} > 0 \text{ [CT2.9]} \implies \nabla^\mu T_{\mu\nu} = 0.\ \blacksquare$$

Phys. The Bianchi identity forces covariant conservation of the ledger stress tensor: energy–momentum conservation is a geometric consequence of the field equations, never an extra assumption.

Theorem 2.25 (CT1.25 — The cosmological constant is the D^+ aspect). Thm. $F_{D^+} = \Lambda\mathbb{I} \implies T^{(D^+)}_{\mu\nu} = -\dfrac{\Lambda}{\kappa^2}g_{\mu\nu}; \qquad \rho_{D^+} = \dfrac{\Lambda}{\kappa^2}; \quad p_{D^+} = -\rho_{D^+}; \quad w_{D^+} := \dfrac{p}{\rho} = -1.$

Req. A8; D4.3; CT1.3 (Part 1), CT1.21, CT1.22; $\delta\sqrt{-g}$ identity [math].

Pf.

$$E_0$$

$$\in Z \text{ [CT1.3]}$$

$$\implies D^+ \text{ commutes with every aspect} \implies \neg\exists \text{ preferred direction}:\ \text{uniform} \wedge \text{isotropic}$$

$$S_{\text{geom}} \ni -\frac{1}{\kappa^2}\int d^4x\sqrt{-g}\,\Lambda \quad \text{[CT1.21]}$$

$$T^{(D^+)}_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta}{\delta g^{\mu\nu}}\Big(-\frac{\Lambda}{\kappa^2}\int\sqrt{-g}\Big) = -\frac{\Lambda}{\kappa^2}g_{\mu\nu}$$

[CT1.22 definition; $\delta\sqrt{-g}$]

$$\propto g_{\mu\nu} \implies \neg\exists \text{ rest frame}; \qquad -\rho\, g_{\mu\nu} \overset{!}{=} (\rho+p)u_\mu u_\nu + p\, g_{\mu\nu} \implies \rho + p = 0 \implies p = -\rho$$

$$\rho_{D^+} = \frac{\Lambda}{\kappa^2}; \qquad w_{D^+} = \frac{p}{\rho} = -1 \ \text{(derived, zero freedom)}; \qquad \Lambda = \text{configuration data [P4]}. \blacksquare$$

Phys. The uniform aspect behaves exactly as a cosmological fluid with equation of state $w = -1$: negative pressure equal in magnitude to its energy density — a derived value with no freedom. Only its magnitude Λ remains configuration data (P4): the corpus predicts the dark-energy equation of state, not its size.

Theorem 2.26 (CT1.26 — Conservation from mask invariance alone). Thm. $\big(\mathcal{P}^{(n)}_M\big)^2 = \mathcal{P}^{(n)}_M \ \wedge$ weight invariance under mask-preserving relabeling $\implies \nabla_\mu T^{\mu\nu} = 0$, independently of CT1.23.

Req. CT1.22 (T definition), CT2.2; $\delta_\xi g_{\mu\nu} = \nabla_\mu\xi_\nu + \nabla_\nu\xi_\mu$; integration by parts; fundamental lemma ($\forall\xi$ compact: $\int f\xi = 0 \Rightarrow f = 0$) [math].

Pf.

$$\text{CT2.2}: \ \text{sum runs over } \operatorname{Im}\mathcal{P}^{(n)}_M \implies \text{invariant under image-preserving transformations}$$

$$\xi^\mu = \text{infinitesimal relabeling}; \ \ \text{mask} \wedge \text{weights invariant [hyp]} \implies \delta_\xi Z = 0$$

$$\delta_\xi g_{\mu\nu} = \nabla_\mu\xi_\nu + \nabla_\nu\xi_\mu \quad \text{[Lie derivative]}$$

$$0 = \delta_\xi \log Z = \int d^4x\sqrt{-g}\,\tfrac{1}{2}T^{\mu\nu}\big(\nabla_\mu\xi_\nu + \nabla_\nu\xi_\mu\big) = \int d^4x\sqrt{-g}\,T^{\mu\nu}\nabla_\mu\xi_\nu$$

$$[T^{\mu\nu} = T^{\nu\mu}]$$

$$\xi \text{ compactly supported}: \ \ \text{IBP} \implies 0 = -\int d^4x\sqrt{-g}\,\big(\nabla_\mu T^{\mu\nu}\big)\xi_\nu \implies \nabla_\mu T^{\mu\nu} = 0$$

[fundamental lemma]. ■

Phys. Conservation has a second, independent origin: invariance of the ledger sum under relabelings — discrete diffeomorphism invariance. CT1.24 and CT1.26 are two separate routes to $\nabla T = 0$; their agreement is a consistency check on the whole continuum layer.

Theorem 2.27 (CT1.27 — Quantum geometry state). Thm. $A(C) := \sum_\sigma w(C)$; $\{|g(C)\rangle\}$: $\langle g(C)|g(C')\rangle = \delta_{CC'}$. Then

$$|\Psi\rangle := \sum_C A(C)\,|g(C)\rangle \in \mathcal{H} \text{ [CT2.8]};$$

$$Z_B = \langle B|\Psi\rangle \;\; \forall \text{ boundary } |B\rangle;$$

$$\langle B| = \Theta\text{-dual} \implies Z = \langle\Psi|\Psi\rangle = \sum_C |A(C)|^2 \geq 0;$$

$$\text{classical metric} := g^*_{\mu\nu} = g_{\mu\nu}(C^*), \;\; \delta\Phi(C^*) = 0 \text{ [CT1.20]};$$

$$\text{fluctuations} := \delta g_{\mu\nu}(C) = g_{\mu\nu}(C) - g^*_{\mu\nu} \;\; \text{(subleading [CT1.19])}.$$

Req. A8; CT1.12, CT1.19, CT1.20, CT2.8.
Pf.

$$Z(\mathcal{R}) = \sum_C \sum_\sigma w(C) = \sum_C A(C) \quad \text{[A8]} \implies \text{coefficients} = \text{terms of } Z$$

$$\text{CT1.12}: \; C \mapsto g_{\mu\nu}(C); \quad \langle g(C)|g(C')\rangle = \delta_{CC'} \text{ [declaration]} \implies |\Psi\rangle \in \mathcal{H} \text{ [CT2.8]}$$

$$Z_B = \langle B|\Psi\rangle \quad \text{[linearity]}; \qquad \langle B| = \Theta\text{-dual}: \;\; Z = \sum_C |A(C)|^2 = \langle\Psi|\Psi\rangle \geq 0 \quad \text{[CT2.8(3)]}$$

$$\text{CT1.20}: \; \text{dominant } C^*: \; \delta\Phi(C^*) = 0;$$

$$g^* := g(C^*); \quad \text{non-stationary subleading in the } \mathfrak{h}\text{-expansion [CT1.19]}. \; \blacksquare$$

Phys. The corpus carries genuine quantum states of geometry: superpositions of emergent metrics weighted by ledger amplitudes, with positive-definite inner product. Classical spacetime is the stationary-phase component; the rest of the ket is quantum geometry — superposition applied to metrics themselves, built from A8.

Theorem 2.28 (CT1.28 — Singularity resolution by the saturation lock). Thm. $\kappa_{\text{ind}}(v) :=$ accessible curvature indicator at v (built from accessible registrations):

$$\kappa_{\text{ind}}(v) \leq f\big(C_{\text{cap}}, \mathfrak{h}\big) = f(16, \mathfrak{h}) < \infty \quad \forall\, \mathfrak{h} > 0; \qquad \implies \quad \neg\exists \text{ accessible curvature singularity}.$$

(Bound $\mathfrak{h}$-dependent: $\to \infty$ as $\mathfrak{h} \to 0$; finite at fixed $\mathfrak{h} > 0$.)
Req. D1.11, D4.1, D4.2; CT0.5, CT0.6, CT0.11 (Part 1), CT1.19.
Pf.

$$\text{CT1.19}: \; \text{curvature}$$

$$= \text{holonomies per unit hop area}$$

$$= \text{function of registrations} \implies \text{singularity} \Rightarrow \text{accessible information density unbounded}$$

$$\text{D1.11 : accessible content at } v \leftrightarrow n(v); \qquad \text{CT0.5/CT0.6 : } D_{\text{tick}}(n) = 2^{n+1},\ C_{\text{cap}} = 16$$

$$\text{CT0.11 : admissible} \iff n \leq 3; \qquad n \geq 4 \implies O(n) = 2^{n+1} - 16 > 0$$

$$\text{D4.1/D4.2 : } n \geq 4 \implies \text{config} \in F_{D^-},\ \text{excess absorbed} \wedge \text{locked } (\in \mathfrak{L},\ \neg\text{accessible})$$

$$\implies \text{accessible : } n \leq 3 \ \wedge\ \text{registers} \leq 16 \implies \kappa_{\text{ind}} \leq f(16, \mathfrak{h}) < \infty\ \ (\mathfrak{h} > 0). \blacksquare$$

Phys. Divergent curvature would require unbounded accessible information at a vertex, but the channel caps access at rung 3 / 16 registers and routes all excess into the locked dark sector. Singularities are not cured by new forces — they are administratively excluded: oversubscription is absorbed before it can become geometry.

2.6 §C.IV — Entropy, Calibration, and the Single Free Scale

Theorem 2.29 (CT1.29 — Boundary entropy from channel capacity). Thm. $\partial\mathcal{B}$: closed real-edge surface separating a D^--locked interior. Then

$$\Omega_{\partial\mathcal{B}} = 2^{\,n_{\mathcal{R}}(\partial\mathcal{B})}; \qquad S_{\partial\mathcal{B}} = \ln \Omega_{\partial\mathcal{B}} = n_{\mathcal{R}}(\partial\mathcal{B}) \ln 2 \ \text{ nats} = n_{\mathcal{R}}(\partial\mathcal{B}) \ \text{ bits};$$

$$A := n_{\mathcal{R}}(\partial\mathcal{B})\, \mathfrak{h}^2 \implies S_{\partial\mathcal{B}} = \frac{A}{\mathfrak{h}^2} \ln 2.$$

Req. A2; C9; CT0.9 (Part 1), CT1.19.
Pf.

$$\text{A2 : each pair carries } 1\mathfrak{q} \implies \text{each boundary real edge} = 1 \text{ independent channel}$$

$$\text{CT0.9 : } \phi_e \in \{0, \pi\} \implies 2 \text{ microstates per edge},\ \neg\text{more}$$

$$\text{edges independent} \implies \Omega_{\partial\mathcal{B}} = 2^{\,n_{\mathcal{R}}(\partial\mathcal{B})};$$

$$S_{\partial\mathcal{B}} = \ln \Omega_{\partial\mathcal{B}} = n_{\mathcal{R}}(\partial\mathcal{B}) \ln 2 \ \text{ nats} = n_{\mathcal{R}}(\partial\mathcal{B}) \text{ bits [C9]}$$

$$\text{CT1.19 : edge subtends hop length } \mathfrak{h} \implies \text{area per pierced boundary element}$$

$$= \mathfrak{h}^2$$

$$\implies A$$

$$= n_{\mathcal{R}}(\partial\mathcal{B}) \qquad \mathfrak{h}^2$$

$$\implies n_{\mathcal{R}}(\partial\mathcal{B}) = A/\mathfrak{h}^2 \implies S_{\partial\mathcal{B}} = \frac{A}{\mathfrak{h}^2}\ln 2 \ \ (\neg\exists \text{ constant beyond } \mathfrak{h}). \blacksquare$$

Phys. Boundary entropy is the boundary's channel capacity counted in bits: one bit per hop-area $\mathfrak{h}^2$. The area law is a theorem of capacity bookkeeping — its form $S \propto A$ is derived, and its coefficient sets the calibration of gravity (CT2.9).

Theorem 2.30 (CT1.29a — The mean edge quantum: structure, sign, and the exactness locus of the entropy–area rate). Thm. Hypotheses: (i) c a closed real-edge chain (bound content [D1.5]); (ii) per-edge integer quanta $\kappa_e := \widehat{ds}^{\,2}(e) \in \mathbb{Z}_{\geq 1}$ [D1.6, CT1.19a; all-nonnull class], with $\Theta(c) := \sum_{e\in c}\kappa_e$, $\delta h(c) := n_{\mathcal{R}}(c)$, $\bar{\kappa}(c) := \Theta/\delta h$. The hypothesis class is nonempty on materialized closed chains of the fully antialigned class [CT1.19b(3)]. Then, as exact identities over $\mathbb{Q}$:

$$(1) \quad S_{\text{reg}}(c) = \delta h \ln 2 \ [\text{CT0.9, C9}];$$

$$A_{\text{loop}}(c) := \Theta\,\mathfrak{h}^2 \ [\text{D1.6}];$$

$$\frac{S_{\text{reg}}}{A_{\text{loop}}} = \frac{\ln 2}{\mathfrak{h}^2}\cdot\frac{1}{\bar{\kappa}} = \frac{\delta h \ln 2}{\Theta\,\mathfrak{h}^2};$$

$$(2) \quad \text{CT1.29's rate } S = \frac{A}{\mathfrak{h}^2}\ln 2 \text{ holds exactly on } c \iff \bar{\kappa} = 1 \iff \kappa_e \equiv 1 \text{ pointwise};$$

$$(3) \quad \bar{\kappa} \geq 1 \text{ with equality} \iff \kappa \equiv 1;$$

$$\text{sign : } \bar{\kappa} > 0 \text{ always } (\Theta \geq \delta h \geq 1);$$

$$\bar{\kappa} \in \tfrac{1}{\delta h}\mathbb{Z}_{\geq \delta h};$$

$$(4) \quad \text{uniform-}\kappa \text{ chains : } \bar{\kappa} = \kappa \in \mathbb{Z}_{\geq 1} \text{ (exact rate only at } \kappa = 1);$$

$$\text{deviation} = \frac{1}{\bar{\kappa}} \in \mathbb{Q} : \ \neg\text{free parameter [P1, P5]}.$$

Req. D1.5, D1.6, D3.7; C9; CT0.9 (Part 1); CT1.19a, CT1.29 (Part 2); hypotheses (i)–(ii) above (declared at use; realization on materialized closed chains of the fully antialigned class — CT1.19b(3), A §0.2a note); exact rational arithmetic [math; machine: `checks/verify_merge_tierA.py` §F].

Pf.

$$(1):\ 1\ \text{real edge} = \ln 2\ \text{nats [CT0.9]} \implies S_{\text{reg}} = \delta h \ln 2;$$

$$A_{\text{loop}} = \oint_c ds^2 = \sum_e \widehat{ds}^{\,2}(e)\,\mathfrak{h}^2 = \Theta\,\mathfrak{h}^2\ \text{[D1.6]}$$

$$\frac{S_{\text{reg}}}{A_{\text{loop}}} = \frac{\delta h \ln 2}{\Theta\,\mathfrak{h}^2} = \frac{\ln 2}{\mathfrak{h}^2}\cdot\frac{\delta h}{\Theta} = \frac{\ln 2}{\mathfrak{h}^2}\cdot\frac{1}{\bar\kappa}.\ \blacksquare_{(1)}$$

$$(2):\ \text{CT1.29}:\ S_{\partial\mathcal{B}} = \frac{A}{\mathfrak{h}^2}\ln 2 \iff S_{\text{reg}}/A_{\text{loop}} = \ln 2/\mathfrak{h}^2 \iff \bar\kappa = 1\ (1);$$

$$\kappa_e \geq 1\ \forall e \implies \bar\kappa = 1 \iff \kappa_e = 1\ \forall e$$

$$(3):\ \Theta = \sum \kappa_e \geq \sum 1 = \delta h \geq 1; \qquad \bar\kappa = \Theta/\delta h \in \tfrac{1}{\delta h}\mathbb{Z}; \qquad (4):\ \kappa_e \equiv \kappa \implies \Theta = \kappa\,\delta h.\ \blacksquare$$

Phys. The Bekenstein-type rate of CT1.29 is exact precisely on minimal chains; every heavier integer winding dilutes it by the rational factor $1/\bar\kappa$ — computable from ledger data, never fitted. The sign question is settled with the structure: the mean edge quantum is a positive rational of the form (loop total)/(hop count), integer exactly on uniform chains.

Theorem 2.31 (CT2.9 — Calibration closure). Thm. G_N defined by $S_{\partial\mathcal{B}} \stackrel{!}{=} \dfrac{A}{4G_N}$; κ by the CT1.23 normalization. Then

$$\mathfrak{h}^2 = 4\,G_N \ln 2 \iff G_N = \frac{\mathfrak{h}^2}{4\ln 2}; \qquad \kappa^2 = 8\pi G_N = \frac{2\pi\,\mathfrak{h}^2}{\ln 2}; \qquad \frac{A}{4G_N} = \frac{A\ln 2}{\mathfrak{h}^2} = S_{\partial\mathcal{B}};$$

$$[G_N] = [\kappa^2] = \mathfrak{q}^{-1}\ \text{[CT2.1]}; \qquad G_N, \kappa = \text{names for combinations of } \mathfrak{h}\ (\neg\text{independent}).$$

Req. CT1.23, CT1.29, CT2.1.

Pf.

$$\text{CT1.29}:\ \ S_{\partial\mathcal{B}} = \frac{A\ln 2}{\mathfrak{h}^2};$$

$$\frac{A\ln 2}{\mathfrak{h}^2} = \frac{A}{4G_N} \iff \frac{\ln 2}{\mathfrak{h}^2} = \frac{1}{4G_N} \iff \mathfrak{h}^2 = 4G_N\ln 2 \iff G_N = \frac{\mathfrak{h}^2}{4\ln 2}$$

$$\text{check}:\ \ \frac{A}{4\cdot\dfrac{\mathfrak{h}^2}{4\ln 2}} = \frac{A\ln 2}{\mathfrak{h}^2} = S_{\partial\mathcal{B}} \quad (\ln 2\ \text{appears once},\ \neg\text{twice})$$

$$\text{CT1.23 : } S_{\text{geom}} \text{ coefficient}$$

$$= \tfrac{1}{2\kappa^2},$$

$$\text{source}$$

$$= T_{\mu\nu} \implies \kappa^2 :$$

$$= 8\pi G_N$$

$$= 8\pi \cdot \frac{\mathfrak{h}^2}{4\ln 2}$$

$$= \frac{2\pi\,\mathfrak{h}^2}{\ln 2}$$

$$[\mathfrak{h}^2] = \mathfrak{q}^{-1} \implies [G_N] = [\kappa^2] = \mathfrak{q}^{-1}\ \text{[CT2.1]}; \qquad \mathfrak{h} \text{ given} \implies G_N, \kappa \text{ determined.} \blacksquare$$

Phys. Demanding that the boundary-entropy law take the quarter-area form locks the hop length, Newton's constant and the gravitational coupling together: $\mathfrak{h}^2 = 4G_N \ln 2$. The $\ln 2$ is the one bit per edge of CT0.9 — the theory's information quantum appearing directly in the calibration of gravity.

Theorem 2.32 (CT2.11 — One free scale). Thm.

$$\forall Q \text{ dimensionful : } \ Q = c_Q\,\mathfrak{h}^{k_Q},\ c_Q \text{ fixed;}$$

$$\forall q \text{ dimensionless-derived : } \ q = \text{fixed number},\ \mathfrak{h}\text{-free;}$$

$$\mathfrak{h} \mapsto \lambda\mathfrak{h} : \ \ Q \mapsto \lambda^{k_Q} Q,$$

$$\text{P2 quantities invariant} \implies \text{rescaling}$$

$$= \text{symmetry} \implies \exists! \text{ calibration d.o.f.}$$

Req. D2.9; P1, P2, P4; CT0.5, CT0.6, CT0.11(a), CT1.1 (Part 1); CT1.29, CT2.1, CT2.9; remaining P2 entries (D, E, F, G).

Pf.

$$\text{sole dimensionful input} = \mathfrak{h} \quad [\text{D2.9},\ \text{P1}]$$

$$\text{CT2.9}: \ G_N,$$

$$\kappa^2$$

$$= \text{fixed multiples of } \mathfrak{h}^2;$$

$$\text{CT2.1}: \ \text{lengths/areas/energies/densities}$$

$$= \text{fixed powers;}$$

$$\text{CT1.29}: \ S$$

$$= \text{pure count}$$

$$\text{P2}: \ (7,3,3,3) \ [\text{CT1.1}];$$

$$16 \ [\text{CT0.6}];$$

$$2^{n+1} \ [\text{CT0.5}];$$

$$n$$

$$\leq 3,$$

$$\mathbb{C}^3 \ [\text{CT0.11(a)}];$$

$$\text{rest}: \ \text{D},$$

$$\text{E},$$

$$\text{F},$$

$$\text{G} \quad \text{— all } \mathfrak{h}\text{-free}$$

$$\mathfrak{h} \mapsto \lambda\mathfrak{h}: \quad Q \mapsto \lambda^{k_Q} Q;$$

$$\text{P2 invariant} \implies \text{rescaling}$$

$$= \text{symmetry of the corpus} \implies \mathfrak{h}$$

$$= \text{genuine single d.o.f. } (\neg\text{internal redundancy})$$

$$\text{P4}: \ (g, g', h_0, \Lambda, \varkappa, \beta, \alpha_\Omega) = \text{solution data}, \ \neg\text{theory constants};$$

$$\text{predictions} = \text{dimensionless ratios at fixed P4 data.} \ \blacksquare$$

Phys. The entire corpus has exactly one dial: the hop length. Turning it rescales every dimensionful quantity uniformly and leaves every pure-number prediction untouched. Comparing to external measurement means fixing $\mathfrak{h}$ once — an act that tests A9, never an input to A1–A8.

2.7 Closure exports of Part 2 §C (consumed by Part 2 §D–§G, Parts 3–4)

CT1.12 $g_{\mu\nu} = \Omega^2 \bar{g}_{\mu\nu}$, sig $= (-,+,+,+)$	CT1.13–17 conformal laws	CT1.18 $\bar{G}^{(1)} = -\frac{1}{2}\Box\hat{h}^{(4)}$ (TT)
CT2.1 $[\cdot] = \mathfrak{q}$-powers	CT2.2–2.4 $\mathcal{P}_M^{(n)}$ projection; $\mathrm{F}_M^{(n)} = \bigoplus_{\mathcal{A}(n)} F_A$	CT2.5 $Z(\mathcal{R}, n)$ poles = spectrum
CT2.6 M smooth 4-manifold	CT2.7/CT1.21 $S_{\text{geom}} = \frac{1}{2\kappa^2}\int\sqrt{-g}(R - 2\Lambda)$	CT2.8 Hilbert space; $\langle A\|A\rangle \geq 0$
CT1.19 $\sum ds^2 \to \int g\, dx\, dx$	CT1.20 $S_{\text{eff}} = -i \log Z$	CT1.22 $T_{\mu\nu} = T^{\text{bd}} + T^{\text{tr}}$
CT1.23 $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa^2 T_{\mu\nu}$	CT1.24/CT1.26 $\nabla^\mu T_{\mu\nu} = 0$ ($\times 2$ routes)	CT1.25 $w_{D^+} = -1$
CT1.27 $\|\Psi\rangle = \sum_C A(C)\|g(C)\rangle$	CT1.28 $\kappa_{\text{ind}} \leq f(16, \mathfrak{h})$	CT1.29 $S_{\partial\mathcal{B}} = \frac{A}{\mathfrak{h}^2}\ln 2$
CT2.9 $\mathfrak{h}^2 = 4G_N \ln 2$; $\kappa^2 = \frac{2\pi\mathfrak{h}^2}{\ln 2}$	CT2.10 $\mathrm{Str}[X, Y\} = 0$	CT2.11 $\exists!$ free scale $\mathfrak{h}$

$\blacksquare$

2.8 §D — Electroweak Sector

Dep.: Part 1; Part 2 §C. EW.1–EW.4, EW.6, EW.19 are restatement stubs of prior tags (cited nowhere downstream); forward pointer: EW.36→EM.9 (restated at use); EW.8→EW.32 is intra-section. Notation: carrier images $W_i \mapsto \frac{1}{2}\sigma_i$, $Y_{\text{car}} = \frac{1}{2}\sigma_0$ (EW.32); Y — electromagnetic generator with three realizations $(Y_{\text{asp}}, Y_{\text{slot}}, Y_{\text{car}})$. Machine: `checks/verify_D.py`; EW.5a/EW.36a: `checks/verify_merge_tierA.py`.

2.9 §D.I — Operator Preliminaries

Theorem 2.33 (EW.1 — Joint (S, T) sector decomposition). Thm. ≡ CT1.1: $\mathfrak{H}(\mathbb{C}^4) = \mathcal{E}_{++} \oplus \mathcal{E}_{+-} \oplus \mathcal{E}_{-+} \oplus \mathcal{E}_{--}$, dim $= (7, 3, 3, 3)$.

Req. CT1.1 (Part 1).

Pf. Identity [CT1.1]. ■

Phys. The arena of the electroweak sector: the weak triplet will live in the T-odd symmetric block $\mathcal{E}_{+-}$ of the pair operator space.

Theorem 2.34 (EW.2 — Generator basis and sector assignment). Thm. ≡ CT1.2: $\{E_0, \Sigma, \hat{E}^{ij}\} \subset \mathcal{E}_{++}\ (1,1,5)$; $W_i \in \mathcal{E}_{+-}\ (3)$; $J_i \in \mathcal{E}_{-+}\ (3)$; $K_i \in \mathcal{E}_{--}\ (3)$; $1{+}1{+}5{+}3{+}3{+}3 = 16$.

Req. CT1.2 (Part 1).

Pf. Identity [CT1.2]. ■

Phys. Fixes the named generators (W_i total spin, J_i, K_i Janus seals, Σ exchange scalar) used throughout this document.

Theorem 2.35 (EW.3 — The W_i close $\mathfrak{su}(2)$). Thm. ≡ CT1.4: $[W_i, W_j] = i\,\epsilon_{ijk} W_k$; $\mathrm{span}_{\mathbb{R}}\{W_i\} \cong \mathfrak{su}(2)$ [C2].

Req. CT1.4 (Part 1).

Pf. Identity [CT1.4]. ■

Phys. The weak-isospin algebra is the total-spin algebra of the qubit pair — present from rung 1 (CT0.13), not installed at rung 2.

Theorem 2.36 (EW.4 — Central, trace, and mixed commutators). Thm. ≡ CT1.3, CT1.5, CT1.6, CT1.7:

$$(1)\ [E_0, X] = 0\ \ \forall X; \qquad (2)\ [\Sigma, W_i] = 0; \qquad (3)\ [\Sigma, J_i] = 4iK_i,\ \ [\Sigma, K_i] = -4iJ_i;$$

$$(4)\ [J_i, J_j] = [K_i, K_j] = i\,\epsilon_{ijk} W_k; \qquad (5)\ [J_i, K_j] = i\big(\delta_{ij}\Sigma - E^{ij}\big).$$

Req. CT1.3, CT1.5, CT1.6, CT1.7 (Part 1).

Pf. Identities [CT1.3, CT1.5, CT1.6, CT1.7]. ■

Phys. The commutator backbone of the sector: the scalar Σ commutes with the weak triplet but rotates the two Janus sectors into each other; seals close onto the weak triplet; mixed seals close into the metric block.

Theorem 2.37 (EW.5 — Second-order Higgs-to-weak pathways). Thm.

$$\big[K_j, [\Sigma, J_i]\big] = 4\,\epsilon_{ijk} W_k; \qquad \big[J_j, [\Sigma, K_i]\big] = -4\,\epsilon_{ijk} W_k;$$

$$[J_j, K_i] \in \mathcal{E}_{++}\ \wedge\ \mathcal{E}_{++} \cap \mathrm{span}\{W_k\} = \{0\}\ \Longrightarrow\ \text{the } [J,K] \text{ pathway } \neg\text{reaches the weak sector.}$$

Req. EW.4 (= CT1.5, CT1.6, CT1.7); CT1.1 (direct sum), CT1.2.

Pf.

$$[\Sigma, J_i] = 4iK_i \ \Longrightarrow\ \big[K_j, [\Sigma, J_i]\big] = 4i\,[K_j, K_i] = 4i \cdot i\,\epsilon_{jik} W_k = -4\,\epsilon_{jik} W_k = 4\,\epsilon_{ijk} W_k$$

$$[\text{EW.4}(3), (4)]$$

$$[\Sigma, K_i] = -4iJ_i \ \Longrightarrow\ \big[J_j, [\Sigma, K_i]\big] = -4i\,[J_j, J_i] = -4i \cdot i\,\epsilon_{jik} W_k = 4\,\epsilon_{jik} W_k = -4\,\epsilon_{ijk} W_k$$

$$[J_j, K_i] = i\big(\delta_{ji}\Sigma - E^{ji}\big) \in \mathcal{E}_{++}\ [\text{EW.4}(5),\ \text{EW.2}];$$

$$W_k \in \mathcal{E}_{+-}; \quad \text{direct sum [EW.1]} \implies \mathcal{E}_{++} \cap \operatorname{span}\{W_k\} = \{0\}. \blacksquare$$

Phys. Although the scalar commutes with the weak triplet, it reaches it at second adjoint order — and only through the Janus seals. The scalar's unique algebraic handle on the weak sector runs through crystallization/dissolution: the algebraic shadow of the wrapping mechanism that actually generates mass in Part IV.

Theorem 2.38 (EW.5a — Convention exactness and the rigidity of the named basis). Thm. In the Pauli basis of $\mathfrak{H}(\mathbb{C}^4)$ [A3, CT1.1], with CT1.2's definitions verbatim ($E_0 = \sigma_0 \otimes \sigma_0$, $\Sigma = \sum_i \sigma_i \otimes \sigma_i$, $W_i = \frac{1}{2}(\sigma_0 \otimes \sigma_i + \sigma_i \otimes \sigma_0)$, $J_i = \frac{1}{2}(\sigma_j \otimes \sigma_k - \sigma_k \otimes \sigma_j)$ cyclic, $K_i = \frac{1}{2}(\sigma_0 \otimes \sigma_i - \sigma_i \otimes \sigma_0)$, $E^{ij} = \frac{1}{2}(\sigma_i \otimes \sigma_j + \sigma_j \otimes \sigma_i)$), as exact matrix equalities over $\mathbb{Q}(i)$ (no tolerance, no floats): (1) The frozen identities hold exactly as stated. CT1.3 (16 brackets), CT1.4 (6 ordered pairs), CT1.5 (9), CT1.6 with $+i\epsilon_{ijk}W_k$ (12; no sign residue), CT1.7 (9, each with all three W-components exactly zero), CT1.8 (12); EW.5 both clauses at all nine (i, j) (diagonal $i = j$: both sides exactly 0); CT0.7a; CT0.7(1) (incl. $\mathfrak{T}\Pi_{\mathcal{R}} = 3$, $\mathfrak{T}\Pi_{\mathcal{V}} = 1$, ranks $3/1$); the associative relation $\Sigma^2 = 3E_0 - 2\Sigma$; $[S, T] = 0$ on all 16 basis elements with $\eta = (-, +, +, +)$ [A3]; the full (S, T) parity table of CT1.2 (17 generators). (2) The convention group is exactly $\mathbb{Z}_2$. The group of diagonal sign redefinitions $(J, K, W, \Sigma, E) \mapsto (\lambda_J J, \lambda_K K, \lambda_W W, \lambda_\Sigma \Sigma, \lambda_E E)$, $\lambda \in \{\pm 1\}^5$, with $E_0 = I_4$ fixed (the unit), preserving every identity of clause (1) — including $\Sigma^2 = 3E_0 - 2\Sigma$ and $\mathbb{S} = \frac{1}{2}(E_0 + \Sigma)$ — is exactly

$$\{\mathrm{id},\ \mathrm{Ad}(\mathbb{S})\} \cong \mathbb{Z}_2, \qquad \mathrm{Ad}(\mathbb{S}) :\ (J_i, K_i) \mapsto (-J_i, -K_i), \quad W_i, \Sigma, E^{ij}, E_0 \text{ fixed}$$

(exactly 2 of the 32 sign maps survive; exhaustive enumeration). (3) EW.5 is invariant under the nontrivial element (each clause maps to itself: J, K flip together). The naming exchange $J \leftrightarrow K$ maps clause 1 of EW.5 to clause 2 with the wrong sign and is excluded by the frozen identity set; Hermiticity excludes continuous phases ($e^{i\varphi}K_i$ Hermitian $\Rightarrow \varphi \in \{0, \pi\}$ — the sign enumeration is exhaustive).

Req. A3; C1, C2, C3; CT0.7, CT0.7a, CT1.1–CT1.8 (Part 1); EW.4, EW.5; exact $\mathbb{Q}(i)$ matrix arithmetic, exhaustive 2^5 sign enumeration [math; machine: `checks/verify_merge_tierA.py` §B].

Pf.

(1) : each identity evaluated entrywise over $\mathbb{Q}(i)$ from CT1.2's definitions; zero residual [machine]

$$(2) :\ (\lambda_\Sigma \Sigma)^2 = 3E_0 - 2\lambda_\Sigma \Sigma \ \wedge\ \lambda_\Sigma^2 = 1 \ \wedge\ \Sigma^2 = 3E_0 - 2\Sigma \ \wedge\ \Sigma \neq 0 \implies \lambda_\Sigma = 1$$

$$\text{CT1.5} :\ \lambda_\Sigma \lambda_J = \lambda_K \ \wedge\ \lambda_\Sigma \lambda_K = \lambda_J \implies \lambda_J = \lambda_K;$$

$$\text{CT1.7} :\ \lambda_J \lambda_K = \lambda_\Sigma = 1 \ \wedge\ \lambda_J \lambda_K = \lambda_E \implies \lambda_E = 1$$

$$\text{CT1.6} :\ \lambda_J^2 = \lambda_K^2 = \lambda_W;$$

$$\text{CT1.4} :\ \lambda_W^2 = \lambda_W \implies \lambda_W = 1;$$

$$\implies (\lambda_J, \lambda_K, \lambda_W, \lambda_\Sigma, \lambda_E) \in \{(1,1,1,1,1), (-1,-1,1,1,1)\}$$

$$(-1,-1,1,1,1) = \mathrm{Ad}(\mathbb{S}) : \ \mathbb{S}J_i\mathbb{S} = -J_i, \ \mathbb{S}K_i\mathbb{S} = -K_i \ [\text{CT0.7(3)} : \ J, K \ S\text{-odd}];$$

$$\mathbb{S}W_i\mathbb{S} = W_i, \ \mathbb{S}\Sigma\mathbb{S} = \Sigma \ [S\text{-even}]$$

$$(3) : \ [K_j, [\Sigma, J_i]] \mapsto [-K_j, [\Sigma, -J_i]] = [K_j, [\Sigma, J_i]] \ \checkmark;$$

$$J \leftrightarrow K : \ [J_j, [\Sigma, K_i]] = +4\epsilon_{ijk}W_k \neq -4\epsilon_{ijk}W_k \implies \text{excluded}$$

$$\text{Hermiticity} : \ (e^{i\varphi}K_i)^\dagger = e^{-i\varphi}K_i = e^{i\varphi}K_i \iff e^{2i\varphi} = 1 \iff \varphi \in \{0, \pi\}. \ \blacksquare$$

Phys. The electroweak backbone was accused of a sign slip; the audit finds the opposite — a lock. Once the unit, the associative square of the exchange scalar, and the swap definition are held fixed, the only freedom left in the whole named basis is the simultaneous flip of J and K, under which every frozen identity — EW.5 included — is exactly invariant. There is no phase convention left to fix; the basis is rigid up to the one declared $\mathbb{Z}_2$.

2.10 §D.II — Rung Masks and the Electroweak Algebra

Theorem 2.39 (EW.6 — Rung masks and active aspect sets). Thm. ≡ CT2.2, CT2.3, CT0.13:

$$\big(\mathcal{P}_M^{(n)}\big)^2 = \mathcal{P}_M^{(n)} = \big(\mathcal{P}_M^{(n)}\big)^\dagger; \qquad \mathrm{F}_M^{(n)} = \mathcal{P}_M^{(n)}\mathrm{F}\,\mathcal{P}_M^{(n)};$$

$$\mathcal{A}(0) = \{J\};$$

$$\mathcal{A}(1) = \{J, G, W, H, D^+\};$$

$$\mathcal{A}(2) = \mathcal{A}(1) \cup \{\text{EM}\};$$

$$\mathcal{A}(3) = \mathcal{A}(2) \cup \{\text{strong}\};$$

$$\mathcal{A}(n \geq 4) = \mathcal{A}(3) \cup \{D^-\}.$$

$W = \mathrm{span}\{W_1, W_2, W_3\}$: neutral seed W_3, ¬ independent aspect [C8]; scalar seed = trace mode h of D2.1.

Req. CT0.13, CT2.2, CT2.3 (B, C); C8; D2.1, D3.3.

Pf. Identities [CT2.2, CT2.3, CT0.13 — derived there from CT0.7, D3.3, D3.1, D3.2]. ■

Phys. Which forces exist at which rung is bookkeeping, already fixed in B: the electroweak content (weak triplet + Higgs scalar + dark + EM centre) is available from rungs 1–2; colour joins only at rung 3.

Theorem 2.40 (EW.7 — The rung-2 electroweak mask). Thm.

$$\mathcal{A}(2) = \{J,\ G,\ W_1,\ W_2,\ W_3,\ H,\ D^+,\ \mathrm{EM}\}; \qquad \mathrm{F}_M^{(2)} = \mathcal{P}_M^{(2)}\mathrm{F}\,\mathcal{P}_M^{(2)} = \bigoplus_{A\in\mathcal{A}(2)} F_A\big|_{\mathcal{K}^{(A)}}.$$

Req. EW.6; CT2.4 (Part 2).
Pf.

$$\text{EW.6}: \ \mathcal{A}(2) = \mathcal{A}(1)\cup\{\mathrm{EM}\},\ \ \mathcal{A}(1) = \{J, G, W, H, D^+\},\ W = \mathrm{span}\{W_1, W_2, W_3\}$$

$$\text{CT2.4}: \ \text{mask retains exactly the blocks of } \mathcal{A}(2),\ \text{annihilates the rest.}\ \blacksquare$$

Phys. The rung-2 world carries gravity, weak, Higgs, uniform dark and electromagnetism — and no colour: the electroweak sector is exactly the mask content at rung 2.

Theorem 2.41 (EW.8 — The electromagnetic generator). Thm. The electromagnetic generator Y is one abstract generator with three realizations:

$$Y_{\mathrm{slot}} := Z\big(\iota_2(\mathfrak{u}(2))\big)\ \text{(1-dim) [D2.12, CT0.13]};$$

$$Y_{\mathrm{asp}} := \mathcal{P}^{(\mathrm{EM})}:\ \mathcal{K}\to\mathcal{K}^{(\mathrm{EM})},\ \ \mathrm{rk}=1,\ \mathfrak{T}(Y_{\mathrm{asp}})=1\ \text{[D2.3, CT0.14]};$$

$$Y_{\mathrm{car}} := \tfrac{1}{2}\sigma_0\ \text{ on } \mathcal{H}_{\mathrm{v}}\cong\mathbb{C}^2\ \text{[EW.32]};$$

$$[Y_{\mathrm{asp}}, \mathrm{F}_M^{(n)}] = 0\ \text{(block-diagonal)}; \qquad [Y_{\mathrm{slot}}, \mathfrak{su}(2)] = 0; \qquad [Y_{\mathrm{car}}, W_i|_{\mathrm{car}}] = 0.$$

Rmk (separation). $\mathcal{P}_M^{(2)}E_0\mathcal{P}_M^{(2)} = \mathcal{P}_M^{(2)}$, $\mathfrak{T} = 17$ [CT2.2, CT0.14]: a mask fact about the pair centre $E_0 \leftrightarrow F_{D^+}$ [CT0.12] — ¬ a definition of Y (the prior codex defined Y this way; repaired).
Req. A8; D2.3, D2.12; CT0.12, CT0.13, CT0.14, CT1.2 (Part 1); CT2.2 (Part 2).
Pf.

$$\text{CT0.13}: \ F_{\mathrm{EM}} = \iota_2\big(Z(\mathfrak{u}(2))\big),\ \text{1-dim [D2.12]};$$

$$\text{CT0.14}: \ \dim\mathcal{K}^{(\mathrm{EM})} = 1 \implies Y_{\mathrm{asp}} = \mathcal{P}^{(\mathrm{EM})}:\ \mathrm{rk}\,1,\ \mathfrak{T}=1\ \text{[D2.3]}$$

$$\text{D2.3}: \ \mathcal{K} = \bigoplus_A \mathcal{K}^{(A)}\ \text{orthogonal} \implies [Y_{\mathrm{asp}},\ \text{block-diagonal operators}] = 0 \implies [Y_{\mathrm{asp}}, \mathrm{F}_M^{(n)}] = 0$$

$$Z(\mathfrak{u}(2))\ \text{commutes with the } \mathfrak{su}(2)\ \text{factor} \implies [Y_{\mathrm{slot}}, \mathfrak{su}(2)] = 0;$$

$$[\tfrac{1}{2}\sigma_0, \tfrac{1}{2}\sigma_i] = 0\ \text{[machine}:\ \text{verify_D]}$$

$$E_0 = I_4 \text{ [CT1.2]} \implies \mathcal{P}_M^{(2)} E_0 \mathcal{P}_M^{(2)} = \mathcal{P}_M^{(2)} \text{ [CT2.2]};$$

$$E_0 \leftrightarrow F_{D^+} \text{ [CT0.12]} : \text{ pair-centre fact, kept separate from } Y. \blacksquare$$

Rmk (forward pointer, non-load-bearing). The carrier realization/transport locus is EW.32 (EW.32 cites EW.8 backward).

Phys. Hypercharge is one generator with three coordinated faces: the centre of the rung-2 slot algebra, the rank-one projection onto the EM block, and $\frac{1}{2}\sigma_0$ on the transported seed. Keeping the three apart — and keeping the pair centre E_0 (which belongs to D^+) apart from all of them — is what makes the EM.17 trace formula and the EW.32 charge statements each well-typed.

Theorem 2.42 (EW.9 — The abelian algebra). Thm. $\big(\text{span}_{\mathbb{R}}\{Y\}, [\cdot,\cdot]\big) \cong \mathfrak{u}(1)$: $[aY, bY] = ab[Y,Y] = 0$; dim $= 1$.

Req. EW.8; Lie-algebra axioms [math].

Pf.

$$\dim \text{span}\{Y\} = 1 \implies [aY, bY] = ab[Y,Y] = 0; \qquad \text{Jacobi} \equiv 0; \qquad 1\text{-dim abelian} = \mathfrak{u}(1). \blacksquare$$

Phys. One generator, vanishing bracket: the abelian factor of the electroweak algebra.

Theorem 2.43 (EW.10 — The rung-2 electroweak algebra). Thm.

$$\mathfrak{g}_{\text{EW}} := \text{span}_{\mathbb{R}}\{W_1, W_2, W_3, Y\} = \mathfrak{su}(2) \oplus \mathfrak{u}(1);$$

$$[W_i, W_j] = i\,\epsilon_{ijk} W_k; \quad [W_i, Y] = 0; \quad [Y,Y] = 0; \quad \dim \mathfrak{g}_{\text{EW}} = 4.$$

Req. EW.3, EW.8, EW.9; direct-sum criterion (two commuting ideals, trivial intersection) [math].

Pf.

$$\text{span}\{W_i\} \cong \mathfrak{su}(2) \text{ [EW.3]}; \qquad \text{span}\{Y\} \cong \mathfrak{u}(1) \text{ [EW.9]}$$

$$[Y, W_i] = 0 \text{ [EW.8]} \implies \text{each factor an ideal of the sum}$$

$$Y \in Z(\mathfrak{g}_{\text{EW}}); \quad \text{no nonzero } \textstyle\sum c_i W_i \text{ central [EW.3]} \implies \text{span}\{W\} \cap \text{span}\{Y\} = \{0\}$$

$$\implies \mathfrak{g}_{\text{EW}} = \mathfrak{su}(2) \oplus \mathfrak{u}(1); \qquad \dim = 3 + 1 = 4. \blacksquare$$

Phys. The electroweak gauge algebra of the corpus — $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$, dimension four — is derived as the algebra generated by the rung-2 mask content. Nothing resembling a gauge principle was assumed.

Theorem 2.44 (EW.11 — The mechanism is projection, not path integration). Thm. $\mathfrak{g}_{\mathrm{EW}}$ arises from $\big(\mathcal{P}_M^{(2)}\mathrm{F}\mathcal{P}_M^{(2)},\ [\cdot,\cdot]\big)$ (finite-dimensional restriction); $Z(\mathcal{R}) = \sum_C \sum_\sigma w(C)$ (countable sum, A8); continuum $\to$ only after coarse-graining (CT2.6) $\implies$ $\mathfrak{g}_{\mathrm{EW}}$ fixed before that limit; $\neg\exists$ functional integral over gauge fields in the derivation.

Req. A8; CT2.3, CT2.6 (Part 2); EW.10; CT1.4 (Part 1).

Pf.

$$\text{CT2.3}: \ \mathrm{F}_M^{(2)} = \mathcal{P}_M^{(2)}\mathrm{F}\mathcal{P}_M^{(2)} = \text{operator restriction } (\neg\text{integral})$$

$$\text{EW.10} \Leftarrow [\text{CT1.4},\ \text{EW.8}] = \text{finite-dimensional matrix identities}$$

$$\text{A8}: \ Z(\mathcal{R}) = \textstyle\sum_{C\in\mathcal{R}} \sum_\sigma w(C) = \text{sum over a countable set;}$$

$$\text{CT2.6}: \ \text{continuum appears only at coarse graining} \implies \text{algebra fixed pre-limit.} \ \blacksquare$$

Phys. The gauge structure is produced by masking a discrete, finite algebra — the continuum functional description is a derived approximation, never the defining mechanism. "Gauge theory" in this corpus names a spectral property of a masked operator sum.

2.11 §D.III — The Scalar Trace Mode

Theorem 2.45 (EW.12 — Metric expansion and the scalar trace mode). Thm. With D2.1 data:

$$\delta^{ij}\hat{h}_{ij} = 0; \qquad q_{ij} = \hat{h}_{ij} + \delta_{ij}h; \qquad h = \tfrac{1}{3}\Sigma_{\mathrm{tr}}; \qquad \delta_{ij}h\,E^{ij} = h\sum_{i=1}^{3} E^{ii} = h\,\Sigma.$$

Req. D2.1; C5; CT1.12 (Part 2), EW.2 (= CT1.2).

Pf.

$$\delta^{ij}\hat{h}_{ij} = \Sigma_{\mathrm{tr}} - \tfrac{1}{3}\cdot 3\cdot\Sigma_{\mathrm{tr}} = 0 \quad [\text{D2.1},\ \text{CT1.12}]$$

$$h = \tfrac{1}{3}\Sigma_{\mathrm{tr}} \text{ substituted into } \hat{h}_{ij} = q_{ij} - \tfrac{1}{3}\delta_{ij}\Sigma_{\mathrm{tr}} \implies q_{ij} = \hat{h}_{ij} + \delta_{ij}h$$

$$E^{ii} = \tfrac{1}{2}\big(\sigma_i\otimes\sigma_i + \sigma_i\otimes\sigma_i\big) = \sigma_i\otimes\sigma_i \implies \sum_i E^{ii} = \Sigma \quad [\text{EW.2}]. \ \blacksquare$$

Phys. The trace part of the real sector is the operator $h\Sigma$: a scalar field h multiplying the exchange scalar. C5 keeps the bookkeeping honest — Σ is an operator, Σ_{tr} a number.

Theorem 2.46 (EW.13 — Identification of the scalar mode). Thm. The electroweak scalar mode is

$$H := h\,\Sigma \qquad (\text{field } h;\ \text{operator representative } h\Sigma),$$

the noncentral rotation scalar; the central scalar E_0 is unavailable (allocated to F_{D^+}).

Req. CT1.10, CT1.11, CT0.12 (Part 1), CT1.25 (Part 2); EW.1 (= CT1.1), EW.4.
Pf.

$$\dim \mathcal{E}_{++} = 7 \text{ [EW.1]}; \qquad \mathcal{E}_{++} \to 1 \oplus 5 \oplus 1 \text{ [CT1.10]} : \ \text{scalars} = \{E_0,\ \Sigma\}$$

$$E_0 \in Z; \quad [\Sigma, J_1] = 4iK_1 \neq 0 \implies \Sigma \notin Z; \quad E_0 \nparallel \Sigma \quad \text{[EW.4(1), (3); CT1.11]}$$

$$E_0 \text{ already allocated : } F_{D^+}$$

$$= \Lambda \mathbb{I} \text{ [CT0.12]} \to \text{cosmological term [CT1.25]};$$

$$\text{reallocating}$$

$$= \text{double counting}$$

$$\implies \Sigma = \text{unique remaining rotation-scalar direction} = F_H \text{ [CT0.12]}; \qquad H = h\Sigma. \ \blacksquare$$

Phys. The Higgs-type scalar of the corpus is the exchange scalar weighted by the trace mode: forced, not chosen — once the central scalar has become the uniform dark aspect, exactly one rotation-scalar direction remains, and A8 already names it F_H.

Theorem 2.47 (EW.14 — The scalar background commutes with the electroweak generators). Thm. $\forall h_0 \in \mathbb{R} : \quad [h_0\Sigma,\ W_i] = 0; \qquad [h_0\Sigma,\ Y_{\text{asp}}] = [h_0\Sigma,\ Y_{\text{car}}] = 0.$

Req. EW.4(2) (= CT1.5), EW.8.
Pf.

$$h_0 \in \mathbb{R} \implies [h_0\Sigma, X] = h_0[\Sigma, X]; \qquad [\Sigma, W_i] = 0 \text{ [EW.4(2)]}$$

$$[Y_{\text{asp}}, F_H\text{-block}] = 0 \text{ [EW.8 : block-diagonal]}; \qquad [Y_{\text{car}}, h_0\Sigma|_{\text{car}}] = 0 \ [\tfrac{1}{2}\sigma_0 \text{ central}]. \ \blacksquare$$

Phys. A scalar background is invisible to the electroweak algebra at the commutator level — the first half of the statement that this theory has no adjoint Higgs mechanism.

Theorem 2.48 (EW.15 — No adjoint electroweak breaking). Thm.

$$\text{Stab}_{\mathfrak{g}_{\text{EW}}}(h_0\Sigma) := \{X \in \mathfrak{g}_{\text{EW}} : \ [h_0\Sigma, X] = 0\} = \mathfrak{g}_{\text{EW}} \qquad \text{(full stabilizer)};$$

$$\implies \neg\exists \text{ proper subalgebra selected by an adjoint stabilizer mechanism.}$$

Consequence. Any electroweak reduction is a statement about the spectrum of the masked resolvent (Part IV: EW.33 transport), ¬ about Lie-algebra stabilizers.

Req. EW.10, EW.14.

Pf.

$$\text{EW.14}: \ [h_0\Sigma, X]$$

$$= 0 \ \ \forall X \in \{W_1, W_2, W_3, Y\} \implies \text{Stab}$$

$$= \mathfrak{g}_{\text{EW}} \implies \neg\exists \text{ proper reduction by commutators.} \ \blacksquare$$

Phys. The scalar background breaks nothing adjointly: the stabilizer is the whole algebra. Mass generation must therefore be — and in Part IV is — a transport effect on a carried seed, not a commutator effect on generators.

Theorem 2.49 (EW.16 — Scalar coupling to the antisymmetric sectors). Thm.

$$[\Sigma, J_i] = 4iK_i; \qquad [\Sigma, K_i] = -4iJ_i; \qquad \big[K_j, [\Sigma, J_i]\big] = 4\,\epsilon_{ijk}W_k; \qquad \big[J_j, [\Sigma, K_i]\big] = -4\,\epsilon_{ijk}W_k.$$

Req. EW.4(3), EW.5.

Pf. First-order: [EW.4(3)]; second-order: [EW.5]. ■

Rmk. J_i, K_i = threshold-moving generators [CT0.7]: the scalar's only algebraic handle on the weak sector runs through crystallization/dissolution — the algebraic shadow of the wrapping mechanism of Part IV.

Phys. The scalar talks to the Janus seals at first order and, through them, to the weak triplet at second order. Every algebraic route from the Higgs mode to the weak force passes through the reality-changing generators.

2.12 §D.IV — Rung-3 Wrapping, the Seed, and Mass Generation

Theorem 2.50 (EW.17 — Rung-3 closed wrapping chains). Thm. γ = chain [D1.4]; closed $\iff$ $\#J^+ = \#J^- \ \wedge \ \partial\gamma = 0$; wrapping = nonzero spatial winding around a closed ledger cycle [D1.7]:

$$\Theta(\gamma) = \sum_{e\in\gamma} \widehat{ds}^{\,2}(e);$$

$$\Phi(\gamma) = \sum_{e\in\gamma} \alpha_e\, \widehat{ds}^{\,2}(e) \ \ [\text{D1.7; uniform-phase case: } \Phi = \alpha\,\Theta];$$

$$\text{Hol}_{\text{spiral}}(\gamma) = e^{i\Phi(\gamma)}.$$

$$\max\{n : \ \exists \text{ closed wrapping chain at rung } n\} = 3.$$

Req. D1.4, D1.7; CT0.11 (Part 1).

Pf.

$$\text{CT0.11}: \ \text{accessible rungs} = \{0, 1, 2, 3\}$$

$$\text{closed wrapping} \Rightarrow \exists v: \text{ chain continues} \wedge \text{branches back onto itself} \Rightarrow \left|\{e \in E_{\mathcal{R}} : e \ni v\}\right|$$

$$\geq 3$$

$$\Rightarrow n(v) \qquad \geq 3$$

$$\implies n = 3. \blacksquare$$

Phys. Rung 3 is the first — and by the capacity bound the last — rung at which closed wrapping chains exist. Mass generation is therefore a rung-3 phenomenon, which is also why it coexists with colour (activated at the same rung).

Theorem 2.51 (EW.18 — Spiral pole condition). Thm. Closed wrapping chain materializes as a rung-3 pole $\implies$

$$\Phi(\gamma) \in 2\pi\mathbb{Z} \qquad \iff \qquad \sum_{e\in\gamma} \alpha_e \, \widehat{ds}^{\,2}(e) = 2\pi k, \;\; k \in \mathbb{Z} \qquad [\Phi: \text{ D1.7}].$$

Req. EW.17; D2.4, CT2.5 (Part 2).

Pf.

$$\text{EW.17}: \text{ transport around the cycle} = \times \, \mathrm{Hol}_{\text{spiral}}(\gamma) = e^{i\Phi(\gamma)}$$

$$\text{CT2.5}: \text{ pole} = \text{eigenvalue condition on a single-valued eigenmode} \implies e^{i\Phi(\gamma)} = 1$$

$$\text{winding } k: \; \Phi \mapsto \Phi + 2\pi k \implies \text{condition} = \Phi(\gamma) \in 2\pi\mathbb{Z}. \blacksquare$$

Phys. Materialization of a wrapping chain is quantized: only integer-winding spiral phases support resolvent poles — a single-valuedness (Bohr–Sommerfeld-type) condition derived from the ledger. With the per-edge phase α_e — A4's branch index, alphabet $\{0, \pi\}$ (read in E: Def-EM.20, EM.20) — the constraint reads on the antialigned content: $\Theta_{\mathcal{P}}(\gamma) = 2k$ (CT1.19b).

Theorem 2.52 (EW.19 — Mass as pole displacement). Thm. $\equiv$ D2.5 (rung-3 context): massless $\iff$ pole at $k^2 = 0$; massive $\iff$ pole at $k^2 = m^2 > 0$; $m^2 :=$ pole displacement of $Z(\mathcal{R}, 3)$ [D2.4].

Req. D2.4, D2.5; CT2.5 (Part 2).

Pf. Restatement [D2.5]. $\blacksquare$

Phys. In this corpus "mass" has one meaning only: how far a resolvent pole sits from the emergent light cone.

Theorem 2.53 (EW.32 — The wrapping seed: carrier and normalization). Thm. Carrier: $\mathcal{H}_{\mathsf{v}} \cong \mathbb{C}^2$ (single qubit). Generator images:

$$W_i \longmapsto \tfrac{1}{2}\sigma_i; \qquad Y \longmapsto \tfrac{1}{2}\sigma_0;$$

$$\mathfrak{T}\big(\mathcal{G}_A\mathcal{G}_B\big) = \tfrac{1}{2}\,\delta_{AB}, \quad \mathcal{G}_A \in \{W_1, W_2, W_3, Y\} \qquad \text{(unique uniform trace normalization)};$$

$$T_a\,\mathsf{v} = 0 \qquad (a = 1, \ldots, 8) \qquad \text{(colour neutrality)}.$$

Req. A1, A2; D1.4, D2.6, D2.12, P4; CT0.12 (Rmk), CT0.13, CT0.14, CT1.2 (Part 1); EW.8.
Pf.

$$\text{A2 : channel capacity}$$

$$= 1\mathfrak{q};$$

$$\text{A1 : carrier of one qubit}$$

$$= \mathbb{C}^2;$$

$$\text{D1.4 : chain}$$

$$= \text{sequence of channels} \implies \text{transported content}$$

$$= \text{one qubit : } \mathcal{H}_{\mathsf{v}} \cong \mathbb{C}^2$$

$$\text{CT1.2 : } W_i = \tfrac{1}{2}\big(\sigma_i^{(p)} + \sigma_i^{(q)}\big); \quad \text{restriction to one transported factor} \implies W_i \mapsto \tfrac{1}{2}\sigma_i$$

$$\text{CT0.13/D2.12 : } Y_{\text{slot}}$$

$$= Z\big(\iota_2(\mathfrak{u}(2))\big) \text{ [EW.8]}$$

$$\implies Y_{\text{car}}$$

$$= y_0\sigma_0 \;\; (y_0 \in \mathbb{R} : \text{ centre acts as scalar on } \mathbb{C}^2)$$

$$\mathfrak{T}(W_iW_j) = \tfrac{1}{4}\mathfrak{T}(\sigma_i\sigma_j) = \tfrac{1}{2}\delta_{ij};$$

$$\mathfrak{T}(W_i Y) = \tfrac{1}{2} y_0\, \mathfrak{T}(\sigma_i) = 0;$$

$$\mathfrak{T}(Y^2) = 2y_0^2 \overset{!}{=} \tfrac{1}{2} \implies y_0 = \tfrac{1}{2}$$

positive root selected; negative root $=$ ω-image [D2.6] $=$ conjugate seed

uniqueness : $y_0 \neq \tfrac{1}{2} \implies \neg$uniform normalization $\implies$ relative ideal weight free ($\neg$P4);

remaining freedom

$= (g, g')$ [P4]

CT0.12 Rmk : F_H

$=$ pair block,

F_{strong}

$=$ stack block on $\mathcal{H}^{(3)}_{\text{slot}}$;

CT0.14 : orthogonal summands of $\mathcal{K}$

$\implies$ seed from the scalar background carries $\neg$slot index $\implies T_a \mathsf{v} = 0 \ \forall a$. ■

Phys. The object transported around a wrapping chain is a single qubit carrying the fundamental of $\mathfrak{u}(2)$ in the unique uniformly trace-normalized representation — $W_i \to \sigma_i/2$, $Y \to \sigma_0/2$ — and no colour. Representation content is derived from the channel quantum A2; colour-neutrality is what keeps the gluons massless (EW.26).

Theorem 2.54 (EW.33 — The wrapping mass form). Thm. With $\mathcal{G} = (gW_1,\ gW_2,\ gW_3,\ g'Y)$, $g, g' > 0$:

$$M^2_{AB} = \mu^2_\star\ \mathrm{Re}\langle \mathcal{G}_A \mathsf{v},\ \mathcal{G}_B \mathsf{v} \rangle, \qquad \mu^2_\star = c_W\, \varkappa^2 \beta^2 h_0^2 > 0 \quad (c_W > 0 : \text{ rung-3 residue});$$

M^2 real symmetric PSD (Gram);

$$\ker M^2 = \left\{ c \in \mathbb{R}^4 : \ \left(\textstyle\sum_A c_A \mathcal{G}_A \right) \mathsf{v} = 0 \right\};$$

$$h_0 = 0 \implies M^2 \equiv 0.$$

Req. EW.17, EW.32; D1.1; CT1.12, CT2.5, CT2.8 (Part 2); Gram-matrix facts [linear algebra].
Pf.

EW.17 : transport = path-ordered product of edge transports [D1.1];

$$\text{leading order : } \mathcal{U} = \exp\Big(i \oint \textstyle\sum_A c_A \mathcal{G}_A \, d\ell\Big) = 1 + i\,\ell_\gamma \sum_A c_A \mathcal{G}_A + O(c^2)$$

CT2.5 : pole displacement = quadratic single-valuedness defect :

$$\big\|(\mathcal{U} - 1)\mathsf{v}\big\|^2 = \ell_\gamma^2 \Big\|\textstyle\sum_A c_A \mathcal{G}_A \mathsf{v}\Big\|^2 + O(c^3)$$

$$\Big\|\textstyle\sum_A c_A \mathcal{G}_A \mathsf{v}\Big\|^2 = \textstyle\sum_{A,B} c_A c_B \ \mathrm{Re}\langle \mathcal{G}_A \mathsf{v}, \mathcal{G}_B \mathsf{v}\rangle$$

$$\big[c_A c_B \text{ sym; } \mathrm{Im}\langle\cdot,\cdot\rangle \text{ antisym } \implies \text{Im cancels}\big]$$

$$\text{CT1.12 : } \bar{g}_{0i}$$

$$= \varkappa N_i$$

$$= \varkappa \beta w_i;$$

$$\bar{g}_{ij} \supset \varkappa h_0 \delta_{ij} \implies \text{cycle amplitude} \propto \varkappa^2 \beta^2 h_0^2;$$

$$\ell_\gamma^2 \text{ absorbed into } c_W > 0 \implies \mu_\star^2$$

$$= c_W \varkappa^2 \beta^2 h_0^2$$

$$M^2 = \mu_\star^2 \cdot \mathrm{Gram}\big(\{\mathcal{G}_A \mathsf{v}\}, \mathrm{Re}\langle\cdot,\cdot\rangle\big) \implies \text{symmetric } \wedge \text{ PSD}$$

$$[\text{linear algebra; consistent with CT2.8 : residues} \geq 0]$$

$$c \in \ker M^2 \iff c^{\mathsf{T}} M^2 c = 0 \iff \Big\|\textstyle\sum_A c_A \mathcal{G}_A \mathsf{v}\Big\|^2 = 0 \iff \textstyle\sum_A c_A \mathcal{G}_A \in \mathrm{Ann}(\mathsf{v})$$

$$h_0 = 0 \implies \mu_\star^2 = 0 \implies M^2 \equiv 0 : \ \forall \text{ electroweak modes massless. } \blacksquare$$

Phys. Mass is a transport defect: the failure of a wrapped seed to return to itself, squared. It is automatically a positive-semidefinite Gram matrix, its kernel is the seed's stabilizer, and it vanishes without the scalar background — the background is necessary for mass even though (EW.15) it breaks nothing by commutators. The overall scale $\mu_\star^2$ is the sector's single free positive number (P4 configuration data); every ratio below is free of it.

Theorem 2.55 (EW.20 — Weak mass generation). Thm. $\langle h\rangle = h_0 \neq 0$; weak frame with v = lower weight state of W_3 (an SU(2) basis choice, always available):

$$W_3\mathsf{v} = -\tfrac{1}{2}\mathsf{v}; \qquad Y\mathsf{v} = +\tfrac{1}{2}\mathsf{v};$$

$$M^2 = \frac{\mu_\star^2}{4}\begin{pmatrix} g^2 & 0 & 0 & 0 \\ 0 & g^2 & 0 & 0 \\ 0 & 0 & g^2 & -gg' \\ 0 & 0 & -gg' & g'^2 \end{pmatrix} \quad \text{in the ordered basis } (W_1, W_2, W_3, Y);$$

$$m_W^2 := M_{11}^2 = M_{22}^2 = \frac{\mu_\star^2 g^2}{4} = \frac{c_W\, \varkappa^2 \beta^2 h_0^2\, g^2}{4} > 0.$$

Req. EW.3, EW.32, EW.33; Bloch-frame fact [linear algebra]; P4.
Pf.

$$\mathsf{v} \in \mathbb{C}^2,\ \|\mathsf{v}\| = 1: \ \ \exists!\, \hat{n} \in S^2 :\ (\hat{n}\cdot\vec{\sigma})\mathsf{v} = -\mathsf{v} \ \text{(negative Bloch direction)};$$

$$\exists U \in \mathrm{SU}(2) :\ \hat{n} \mapsto e_3$$

$$U \text{ acting by conjugation}$$

$$= \text{automorphism of } \mathfrak{su}(2)\ [\text{EW.3}] \implies \text{in the rotated frame } \sigma_3\mathsf{v}$$

$$= -\mathsf{v}$$

$$\implies W_3\mathsf{v}$$

$$= -\tfrac{1}{2}\mathsf{v}\ [\text{EW.32}]$$

$$Y = \tfrac{1}{2}\sigma_0\ [\text{EW.32}] \implies Y\mathsf{v} = +\tfrac{1}{2}\mathsf{v}\ \ \forall \mathsf{v}$$

$$(\text{hypercharge } +\tfrac{1}{2}:\ \text{derived from trace normalization},\ \neg\text{assigned})$$

$$\text{neutral entries}: \ \ \mathcal{G}_3\mathsf{v} = -\tfrac{g}{2}\mathsf{v},\ \ \mathcal{G}_Y\mathsf{v} = +\tfrac{g'}{2}\mathsf{v} \implies M_{33}^2 = \mu_\star^2\tfrac{g^2}{4},$$

$$M_{YY}^2 = \mu_\star^2 \tfrac{g'^2}{4},$$

$$M_{3Y}^2 = \mu_\star^2 \big(-\tfrac{g}{2}\big)\big(\tfrac{g'}{2}\big) = -\mu_\star^2 \tfrac{gg'}{4}$$

charged entries : v

$$= \begin{pmatrix} 0 \\ 1 \end{pmatrix} : \ \sigma_1 \mathsf{v}$$

$$= \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

$$\sigma_2 \mathsf{v}$$

$$= \begin{pmatrix} -i \\ 0 \end{pmatrix}$$

$$\Longrightarrow \ \|W_1 \mathsf{v}\|^2$$

$$= \|W_2 \mathsf{v}\|^2$$

$$= \tfrac{1}{4};$$

$$\mathrm{Re}\langle W_1 \mathsf{v},$$

$$W_2 \mathsf{v}\rangle$$

$$= \mathrm{Re}\big(\tfrac{1}{4} \cdot (-i)\big) \qquad = 0$$

$$\Longrightarrow \ M_{11}^2 = M_{22}^2 = \tfrac{\mu_\star^2 g^2}{4}, \quad M_{12}^2 = 0$$

cross blocks : $\ W_{1,2}\mathsf{v} \propto \begin{pmatrix} 1 \\ 0 \end{pmatrix} \perp \mathsf{v}; \quad W_3 \mathsf{v},\ Y\mathsf{v} \propto \mathsf{v} \implies M_{1,2;\ 3,Y}^2 = 0 \implies M^2$ block diagonal

$\mu_\star^2 > 0\,(h_0 \neq 0,\, c_W > 0)$ [EW.33] $\wedge\, g > 0 \implies m_W^2 > 0;$ undetermined content $= \mu_\star^2$ only [P4]. ■

Phys. In the frame where the seed points down the third axis, the Gram mass form is block diagonal: two degenerate charged massive directions and a coupled neutral 2×2 block. The W mass is derived; the only number left free is the overall wrapping scale $\mu_\star$.

Theorem 2.56 (EW.21 — Charged weak bosons). Thm. With $W^{\pm} = \frac{1}{\sqrt{2}}(W_1 \mp iW_2)$ [D3.10]:

$$m_{W^+}^2 = m_{W^-}^2 = m_W^2 = \frac{\mu_\star^2 g^2}{4}.$$

Req. D3.10; EW.20; eigenvalue invariance of $c\mathbb{1}_2$ [linear algebra].
Pf.

$$\text{EW.20}: \; M^2\big|_{\text{charged}} = \tfrac{\mu_\star^2 g^2}{4}\mathbb{1}_2$$

$$\operatorname{spec}\big(U(c\mathbb{1}_2)U^{-1}\big) = \{c, c\} \;\; \forall \text{ invertible } U \text{ (incl. complexification to } W^{\pm})$$

$$\big[\text{equiv.}: \; SO(2) \text{ invariance generated by } W_3;\; W^{\pm}: \; \text{eigenvectors with phases } e^{\mp i\theta}\big]$$

$$\implies m_{W^\pm}^2 = m_W^2 = \tfrac{\mu_\star^2 g^2}{4}. \; \blacksquare$$

Phys. The charged weak modes are exactly degenerate: the charged Gram block is a multiple of the identity, so the complex combinations $W^{\pm}$ inherit the same mass. Charge conjugates share mass by structure, not by assumption.

2.13 §D.V — The Neutral Sector

Theorem 2.57 (EW.22 — The unbroken generator). Thm.

$$\ker M^2 = \operatorname{span}\{(0, 0, g', g)\} \quad \text{in the basis } (W_1, W_2, W_3, Y); \qquad Q := W_3 + Y; \qquad Q\mathsf{v} = 0;$$

$$A := \frac{g'W_3 + gY}{\sqrt{g^2 + g'^2}} \qquad (\text{normalized kernel direction}).$$

Req. EW.20, EW.32, EW.33.
Pf.

$$\text{EW.33}: \; \ker M^2 = \{c: \; \textstyle\sum_A c_A \mathcal{G}_A \mathsf{v} = 0\}$$

$$\text{EW.20}: \; W_{1,2}\mathsf{v} \perp \mathsf{v} \;\wedge\; W_{1,2}\mathsf{v} \neq 0;$$

$$W_3\mathsf{v},\; Y\mathsf{v} \propto \mathsf{v} \implies c_1 \neq 0 \;\vee\; c_2 \neq 0 \implies \textstyle\sum c_A \mathcal{G}_A \mathsf{v} \neq 0 \implies \ker \subseteq \{c_1 = c_2 = 0\}$$

$$\big(c_3 g W_3 + c_Y g' Y\big)\mathsf{v} = \tfrac{1}{2}\big(-c_3 g + c_Y g'\big)\mathsf{v} = 0 \iff c_3 g = c_Y g' \iff (c_3, c_Y) \propto (g', g)$$

$$\implies \dim\ker M^2 = 1; \qquad \text{operator form}: \; g'\cdot gW_3 + g\cdot g'Y = gg'\,(W_3+Y) \implies Q = W_3+Y \;\; (gg' \neq 0)$$

$$Q\mathsf{v} = \left(-\tfrac{1}{2} + \tfrac{1}{2}\right)\mathsf{v} = 0; \qquad A = (g'W_3 + gY)/s, \;\; s = \sqrt{g^2+g'^2}. \;\blacksquare$$

Phys. The massless direction is derived, not posited: it is the unique combination annihilating the seed. Electric charge $Q = W_3 + Y$ appears here for the first time — as the stabilizer of the transported qubit.

Theorem 2.58 (EW.23 — The neutral mass matrix). Thm. In (W_3, Y):

$$M^2_{\text{neutral}} = \frac{\mu_\star^2}{4}\begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix} = \frac{\mu_\star^2}{4}\,uu^{\mathsf{T}}, \quad u = (g, -g');$$

$$\det = 0; \qquad \text{tr} = \frac{\mu_\star^2(g^2+g'^2)}{4} =: m_Z^2; \qquad \text{spec} = \{0,\, m_Z^2\}; \qquad \text{null} = (g', g); \quad \text{massive} = (g, -g').$$

Req. EW.20, EW.22; rank-one facts [linear algebra].

Pf.

$$\text{entries} = [\text{EW.20}]; \qquad \det = \tfrac{\mu_\star^4}{16}\left(g^2g'^2 - g^2g'^2\right) = 0; \qquad \text{tr} = \tfrac{\mu_\star^2}{4}\left(g^2+g'^2\right) > 0$$

$$\det = 0 \;\wedge\; \text{tr} > 0 \implies \text{rank} = 1, \quad \text{spec} = \left\{0,\; \tfrac{\mu_\star^2}{4}(g^2+g'^2)\right\}$$

$$\text{null vector} = (g', g)\;[\text{EW.22}]; \qquad \text{massive eigenvector} \perp \text{null} = (g, -g')$$

$$m_Z^2 := \tfrac{\mu_\star^2(g^2+g'^2)}{4}; \qquad M^2_{\text{neutral}} = \tfrac{\mu_\star^2}{4}uu^{\mathsf{T}} \implies \text{PSD}\;[\text{EW.33}]. \;\blacksquare$$

Phys. The neutral block is a rank-one Gram matrix — one massive mode (Z) and one massless mode — because a single seed direction is being probed by two neutral generators. $m_Z^2 = \mu_\star^2(g^2 + g'^2)/4$ is derived with zero freedom beyond $\mu_\star$.

Theorem 2.59 (EW.24 — The mixing angle). Thm.

$$\cos\theta_W := \frac{g}{s}; \qquad \sin\theta_W := \frac{g'}{s}; \qquad \tan\theta_W = \frac{g'}{g}; \qquad s := \sqrt{g^2+g'^2};$$

$$A = \sin\theta_W\, W_3 + \cos\theta_W\, Y; \qquad Z = \cos\theta_W\, W_3 - \sin\theta_W\, Y \qquad [\text{D3.10}];$$

$$M^2_{\text{neutral}}\, A = 0; \qquad M^2_{\text{neutral}}\, Z = m_Z^2\, Z.$$

Req. EW.23; D3.10.

Pf.

$$A \leftrightarrow (\sin\theta_W, \cos\theta_W) \propto (g', g); \qquad Z \leftrightarrow (\cos\theta_W, -\sin\theta_W) \propto (g, -g'); \qquad \text{orthonormal}$$

$$M^2_{\text{neutral}} A = \frac{m_Z^2}{s^2}\begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix}\frac{1}{s}\begin{pmatrix} g' \\ g \end{pmatrix} = \frac{m_Z^2}{s^3}\begin{pmatrix} g^2 g' - g^2 g' \\ -gg'^2 + g' g^2 \end{pmatrix} = 0$$

$$M^2_{\text{neutral}} Z$$

$$= \frac{m_Z^2}{s^2}\begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix}\frac{1}{s}\begin{pmatrix} g \\ -g' \end{pmatrix}$$

$$= \frac{m_Z^2}{s^3}\begin{pmatrix} g(g^2+g'^2) \\ -g'(g^2+g'^2) \end{pmatrix}$$

$$= m_Z^2$$

$$(\cos\theta_W, -\sin\theta_W)$$

$$= m_Z^2 Z. \blacksquare$$

Phys. The weak mixing angle is the geometry of the kernel: the angle between the neutral gauge plane and the seed stabilizer. $\tan\theta_W = g'/g$ is a derived relation — θ_W is not a new parameter, since g, g' are the configuration data (P4).

Theorem 2.60 (EW.25 — The photon direction is massless). Thm. $m_A^2 = 0$ (pole of A at $k^2 = 0$).

Req. EW.19, EW.24.

Pf.

$$\text{EW.24}: \; M^2 A = 0 \implies \text{pole displacement} = 0 \implies \text{pole at } k^2 = 0 \quad [\text{EW.19} \equiv \text{D2.5}]. \blacksquare$$

Phys. The photon direction is exactly massless — a protected zero of the Gram matrix (the seed's stabilizer), not a tuned smallness.

Theorem 2.61 (EW.26 — Colour modes remain massless at rung 3). Thm. Extending M^2 over $\mathcal{G} \cup \{T_a\}_{a=1}^8$:

$$M^2_{aB} = \mu_\star^2 \,\text{Re}\langle T_a \mathsf{v},\ \mathcal{G}_B \mathsf{v}\rangle = 0; \qquad M^2_{ab} = 0; \qquad \{T_a\}_{a=1}^8 \subseteq \ker M^2; \qquad m_g^2 = 0.$$

Req. EW.6, EW.19, EW.32, EW.33; CT0.12 (Rmk), CT0.14 (Part 1).

Pf.

$$\text{EW.6}: \; \mathcal{A}(3) = \mathcal{A}(2) \cup \{\text{strong}\} \implies T_a \text{ active at rung 3, in the connection}$$

$$\text{EW.32}: \; T_a \mathsf{v} = 0 \;\forall a \implies M^2_{aB} = \mu_\star^2 \,\text{Re}\langle 0,\ \mathcal{G}_B \mathsf{v}\rangle = 0; \quad M^2_{ab} = 0$$

$$\implies \text{colour block} \equiv 0 \implies 8 \text{ colour directions} \subseteq \ker M^2 \implies \text{poles at } k^2 = 0 \quad [\text{EW.19}]$$

$$\text{structural reason}: \; F_H = \text{pair block} \perp F_{\text{strong}} = \text{stack block} \quad [\text{CT0.12 Rmk}, \; \text{CT0.14}]. \; \blacksquare$$

Phys. Gluons stay massless because the wrapping seed is colour-blind: the scalar lives in the pair register and colour in the orthogonal slot register. A structural orthogonality — not a stipulation — protects the colour gauge modes. (Cited in F.)

Theorem 2.62 (EW.27 — The scalar excitation). Thm. $S_{\text{eff}}[h] :=$ CT1.20 action restricted to the trace mode; h_0 : nondegenerate stable stationary point. Then

$$m_H^2 := \left.\frac{\delta^2 S_{\text{eff}}}{\delta h^2}\right|_{h=h_0} > 0; \qquad \text{scalar pole at } k^2 = m_H^2 \;\; (\text{simple}); \qquad m_H^2 = \text{configuration data [P4]}.$$

Req. CT1.20, CT2.5 (Part 2); stability criterion [math]; P4.

Pf.

$$\text{stable stationary}: \quad \left.\frac{\delta S_{\text{eff}}}{\delta h}\right|_{h_0} = 0 \;\wedge\; \left.\frac{\delta^2 S_{\text{eff}}}{\delta h^2}\right|_{h_0} > 0$$

$$\text{quadratic expansion about } h_0: \;\; \text{coefficient} = m_H^2 \implies \text{resolvent pole displaced by } m_H^2 \quad [\text{CT2.5}]$$

$$\text{nondegenerate} \implies \text{pole simple};$$

$$m_H^2 \text{ value} = \text{shape of } S_{\text{eff}} \implies \text{P4 data}, \; \neg\text{in the derived ratios}. \; \blacksquare$$

Phys. The Higgs-like particle is the trace-mode fluctuation about a stable background; its mass is configuration data and never enters any of the derived dimensionless ratios. (Cited in G.)

2.14 §D.VI — Derived Electroweak Relations

Theorem 2.63 (EW.28 — The gauge spectrum). Thm.

mode	combination	m^2
W^+	$\frac{1}{\sqrt{2}}(W_1 - iW_2)$	$m_W^2 = \mu_\star^2 g^2/4 > 0$
W^-	$\frac{1}{\sqrt{2}}(W_1 + iW_2)$	m_W^2
Z	$\cos\theta_W \, W_3 - \sin\theta_W \, Y$	$m_Z^2 = \mu_\star^2(g^2 + g'^2)/4 > 0$
A	$\sin\theta_W \, W_3 + \cos\theta_W \, Y$	0

$$\tan\theta_W = \frac{g'}{g};$$

exhaustive: $\dim \mathfrak{g}_{\text{EW}} = 4$ and the four directions are independent.

Req. EW.10, EW.20, EW.21, EW.23, EW.24, EW.25; D3.10.

Pf. Charged: [EW.20, EW.21]; neutral: [EW.23, EW.24]; $m_A^2 = 0$: [EW.25]; exhaustiveness: $\dim \mathfrak{g}_{\text{EW}} = 4$ [EW.10] $\wedge$ independence $\{W^\pm, Z, A\}$. $\blacksquare$

Phys. Exactly four gauge directions materialize: two degenerate charged massive, one neutral massive, one neutral massless — the electroweak spectrum, all eigen-data of one Gram matrix.

Theorem 2.64 (EW.34 — The mass ratio). Thm.

$$\frac{m_W^2}{m_Z^2} = \frac{g^2}{g^2+g'^2} = \cos^2\theta_W; \qquad m_W = m_Z\cos\theta_W; \qquad \rho := \frac{m_W^2}{m_Z^2\cos^2\theta_W} = 1;$$

$$\rho = 1 \iff j = \tfrac{1}{2} \iff \dim\mathcal{H}_{\mathsf{v}} = 2 \iff \text{A2 (1}\mathfrak{q}\text{ per channel)};$$

$$\rho \text{ independent of } \mu_\star^2, h_0, \varkappa, \beta, c_W, \mathfrak{h}.$$

Req. A2; EW.20, EW.23, EW.24, EW.32; spin-j ladder algebra [math].
Pf.

$$m_W^2 = \tfrac{\mu_\star^2 g^2}{4},$$

$$m_Z^2$$

$$= \tfrac{\mu_\star^2(g^2+g'^2)}{4} \text{ from one Gram matrix [EW.20, EW.23]}$$

$$\implies \frac{m_W^2}{m_Z^2}$$

$$= \frac{g^2}{g^2+g'^2}$$

$$= \cos^2\theta_W \text{ [EW.24]}$$

$$\implies \rho \equiv 1 \text{ (identically in } g, g'); \qquad \mu_\star^2, h_0, \varkappa, \beta, c_W, \mathfrak{h} \text{ cancel}$$

$$\text{driver : } \frac{m_W^2}{m_Z^2}\Big/\cos^2\theta_W \text{ traces to } \frac{\|W_1\mathsf{v}\|^2}{\|W_3\mathsf{v}\|^2};$$

$$\text{spin-}j\text{ lowest weight : } \|W_1\mathsf{v}\|^2 = \tfrac{1}{4}\big[j(j{+}1) - (-j)(-j{+}1)\big] = \tfrac{j}{2}, \quad \|W_3\mathsf{v}\|^2 = j^2$$

$$\frac{\|W_1\mathsf{v}\|^2}{\|W_3\mathsf{v}\|^2} = \frac{1}{2j} = 1 \iff j = \tfrac{1}{2} \iff \mathcal{H}_{\mathsf{v}} = \mathbb{C}^2 \text{ [EW.32]} \iff 1\mathfrak{q} \text{ per channel [A2]. } \blacksquare$$

Phys. The tree-level ρ parameter equals one because the seed carrier is a single qubit: for spin j the charged-to-neutral Gram ratio is $1/(2j)$, forced to 1 exactly at $j = \frac{1}{2}$, which A2 demands. $m_W = m_Z\cos\theta_W$ is a dimensionless prediction with zero freedom — independent of every configuration parameter and of the calibration scale. (Cited in G.)

Theorem 2.65 (EW.35 — The residual coupling). Thm. Neutral connection $\mathcal{A}_{\text{neutral}} = g\,w^3 W_3 + g'\,y\,Y$; in the mass eigenbasis ($w^3 = \sin\theta_W\,\mathsf{A} + \cos\theta_W\,\mathsf{Z}$; $y = \cos\theta_W\,\mathsf{A} - \sin\theta_W\,\mathsf{Z}$):

$$\mathsf{A}\text{-coefficient} = e\,Q; \qquad e := \frac{gg'}{\sqrt{g^2+g'^2}} = g\sin\theta_W = g'\cos\theta_W; \qquad \frac{1}{e^2} = \frac{1}{g^2} + \frac{1}{g'^2}.$$

Req. EW.22, EW.24.

Pf.

$$\text{rotation to mass basis [EW.24]}:$$

$$\mathcal{A}_{\text{neutral}} = \big(g\sin\theta_W\,W_3 + g'\cos\theta_W\,Y\big)\mathsf{A} + \big(g\cos\theta_W\,W_3 - g'\sin\theta_W\,Y\big)\mathsf{Z}$$

$$g\sin\theta_W\,W_3 + g'\cos\theta_W\,Y = \frac{gg'}{s}\,W_3 + \frac{g'g}{s}\,Y = \frac{gg'}{s}\,(W_3+Y) = e\,Q$$

$$[\text{EW.22, EW.24}:\ \sin\theta_W = \tfrac{g'}{s},\ \cos\theta_W = \tfrac{g}{s}]$$

$$e = \frac{gg'}{s} = g\sin\theta_W = g'\cos\theta_W; \qquad \frac{1}{e^2} = \frac{g^2+g'^2}{g^2g'^2} = \frac{1}{g^2} + \frac{1}{g'^2}. \blacksquare$$

Phys. The coupling of the massless direction is the harmonic combination of g and g': $e = g\sin\theta_W$ is a derived identity, not a definition. (Cited in E, G; the broken '$\neq$' glyphs of the prior codex are the letter e.)

Theorem 2.66 (EW.36 — The charge spectrum and its consistency with the chain end-charge). Thm.

$$Q\big|_{\mathcal{H}_{\mathsf{v}}} = \tfrac{1}{2}\sigma_3 + \tfrac{1}{2}\sigma_0 = \operatorname{diag}(1,0); \qquad \operatorname{spec} Q = \{0,+1\}; \qquad \omega\ [\text{D2.6}]:\ \operatorname{spec} \mapsto \{0,-1\};$$

$$\text{charge alphabet} = q \in \{-1,0,+1\} = q_{\text{EM}}(c)\ \text{alphabet of D3.4 [EM.9]}.$$

Corollary 2.2 (EW.36a). On the carrier: $Q = W_3 + Y$; seed hypercharge $= +\frac{1}{2}$ [EW.20].

Req. EW.20, EW.22, EW.32; D2.6, D3.4; CT0.9 (Part 1); EM.9 (E, forward pointer).

Pf.

$$\text{EW.32}:\ W_3 = \tfrac{1}{2}\sigma_3,\ Y = \tfrac{1}{2}\sigma_0 \implies Q = \tfrac{1}{2}(\sigma_3+\sigma_0) = \operatorname{diag}(1,0)$$

$$\operatorname{spec}\operatorname{diag}(1,0) = \{1,0\}; \qquad \text{eigenvalue } 1:\ \text{upper weight state}; \quad 0:\ \mathsf{v}\ [\text{EW.22}:\ Q\mathsf{v} = 0]$$

$$\text{D2.6}:\ \omega \text{ reverses charge} \implies \text{reversed carrier}:\ \operatorname{spec} = \{0,-1\} \implies \text{alphabet} = \{-1,0,+1\}$$

$$\text{independently}: \quad \text{D3.4}: \; q_{\text{EM}}(c)$$

$$= \tfrac{1}{2}\big(\sigma_{s(c)} - \sigma_{t(c)}\big),$$

$$\sigma \in \{+1, -1\} \text{ [CT0.9]}$$

$$\implies q_{\text{EM}} \in \{-1, 0, +1\} \text{ [EM.9]}$$

$$\text{inputs disjoint (seed carrier} + \text{trace normalization vs binary edge phase)} \implies \text{agreement}$$

$$= \text{consistency check.} \blacksquare$$

Phys. Two independent constructions of charge — the operator route (stabilizer of the seed) and the combinatorial route (difference of edge signs at a chain's ends) — yield the same ternary alphabet $\{-1, 0, +1\}$. Charge quantization is triality of a qubit seed, mirrored by the binary edge register.

Theorem 2.67 (EW.29 — The full electroweak spectrum). Thm. Rung-3 materialized poles:

$$\{W^+,\ W^-,\ Z,\ A,\ H\};$$

$$m^2_{W^\pm} = m^2_W > 0; \quad m^2_Z = \frac{m^2_W}{\cos^2\theta_W} > 0; \quad m^2_A = 0; \quad m^2_H > 0 \ (h_0 \neq 0 \text{ stable});$$

exhaustive: gauge directions span $\mathfrak{g}_{\text{EW}}$ $(\dim = 4) \oplus$ scalar direction F_H $(\dim = 1)$.

Req. CT0.12 (Part 1); EW.10, EW.27, EW.28, EW.34.

Pf. Gauge poles + masses: [EW.28, EW.34]; scalar pole: [EW.27]; exhaustiveness: $\dim \mathfrak{g}_{\text{EW}} = 4$ [EW.10] + $\dim F_H = 1$ [CT0.12]. ■

Phys. The electroweak particle content materialized at rung 3 is exactly: two charged massive, one neutral massive, one neutral massless, one scalar — and nothing else.

Theorem 2.68 (EW.30 — Spectral reduction). Thm.

$$\text{rung 2}: \; \mathfrak{g}_{\text{EW}} = \mathfrak{su}(2) \oplus \mathfrak{u}(1) \text{ [EW.10]}; \qquad \text{rung 3}: \; \text{spec} = \{m^2_W, m^2_W, m^2_Z, 0\} \text{ [EW.28]};$$

$$\text{“SU}(2) \times \text{U}(1) \to \text{U}(1)_Q\text{”} \; := \; \text{naming of the spectral reduction;}$$

$$\neg\text{Lie-algebra breaking by commutators [EW.15];}$$

$$[Q, Q] = 0 \implies \text{residual structure} = \mathfrak{u}(1)_Q \; \text{(genuine abelian algebra)}.$$

Req. EW.10, EW.15, EW.22, EW.28, EW.33.
Pf.

$$\text{EW.15}: \ \text{Stab}_{\text{adj}} = \mathfrak{g}_{\text{EW}} \implies \text{algebra intact}$$

$$\text{EW.33}: \ \text{reduction occurs in } \ker M^2 = \text{stabilizer of the transported seed};$$

$$\text{EW.22}: \ \dim\ker = 1, \ \text{spanned by } Q$$

$$\text{EW.28}: \ \text{spectrum} = 3 \text{ massive} + 1 \text{ massless pole};$$

$$[Q,Q] = 0 \implies \mathfrak{u}(1)_Q \text{ abelian} \implies \text{name accurate as spectral description.} \ \blacksquare$$

Phys. The correct statement of "electroweak symmetry breaking" in this corpus: the algebra never breaks — what reduces is the spectrum. Three poles leave the light cone, one stays, and the residual structure is the abelian stabilizer of the seed. The conventional name is retained as a label only.

2.15 §D.VII — Completion

Theorem 2.69 (EW.31 — Self-containment of the electroweak sector). Thm.

$$\forall \text{ statements of D}: \ \vdash \ \text{A1–A9} \cup \text{C1–C9} \cup \text{D0.1–D4.3} \cup \{\text{B, C theorems}\};$$

$$\neg\exists \text{ external electroweak law, parameter, or measured value used};$$

$$\text{undetermined} = \{g, \ g', \ h_0, \ c_W, \ \varkappa, \ \beta\} \subset \text{P4};$$

$$\forall \text{ quoted ratios } (m_W/m_Z, \ e/g, \ \rho, \ \text{charge alphabet}): \ \text{independent of P4 data and of } \mathfrak{h}.$$

Req. EW.1–EW.30; CT-tags as cited; P4.
Pf. Dependency trace:

$$\text{EW.2} \equiv \text{CT1.2} \Leftarrow \text{A3}; \quad \text{EW.3} \equiv \text{CT1.4}; \quad \text{EW.6/7} \Leftarrow \text{CT0.13} \Leftarrow \{\text{CT0.7, D3.1, D3.2, D3.3}\}$$

$$\text{EW.8–10}: \ Y = \text{masked centre [CT2.2, CT0.13]}; \ \text{commutation} \Leftarrow \text{centrality}$$

$$\text{EW.13}: \ \text{scalar} = \text{unique unallocated rotation scalar [CT1.10, CT1.11, CT0.12]}$$

$$\text{EW.14/15}: \ \text{background commutes} \implies \neg\text{adjoint breaking};$$

$$\text{EW.33 : mass} = \text{transport Gram form}$$

$$\text{EW.32 : carrier} = \mathbb{C}^2 \Leftarrow \text{A2; normalization} \Leftarrow \mathfrak{T}(\mathcal{G}_A\mathcal{G}_B) = \tfrac{1}{2}\delta_{AB} \text{ (unique)}$$

$$\text{EW.33 : Gram} \implies \text{PSD; ker} = \text{stabilizer;} \quad \text{EW.22–25 : kernel} \implies \theta_W,\ m_Z,\ m_A = 0$$

$$\text{EW.26 : } F_H \perp F_{\text{strong}} \text{ [CT0.14]} \implies m_g = 0;$$

$$\text{EW.34/35 : ratios free of P4} \wedge \mathfrak{h};$$

$$\text{EW.36 : alphabets agree (disjoint inputs).} \blacksquare$$

Phys. Closure audit of the sector: every electroweak statement traces to A1–A9 through B and C; the only undetermined quantities are the P4 configuration data $(g, g', h_0, c_W, \varkappa, \beta)$, and every quoted ratio is independent of all of them.

2.16 Closure exports of Part 2 §D (consumed by Part 2 §E–§G, Parts 3–4)

EW.10 $\mathfrak{g}_{\text{EW}} = \mathfrak{su}(2) \oplus \mathfrak{u}(1),\ \dim = 4$	EW.18 $\Phi(\gamma) \in 2\pi\mathbb{Z}$	EW.20 M^2 block form; $m_W^2 = \mu_\star^2 g^2/4$
EW.22 $Q = W_3 + Y;\ Q\mathsf{v} = 0$	EW.24 $\tan\theta_W = g'/g;\ A,\ Z$ eigenbasis	EW.25 $m_A^2 = 0$
EW.26 $m_g^2 = 0$ (cited in F)	EW.27 $m_H^2 > 0$ (cited in G)	EW.28 spectrum $\{W^\pm, Z, A\}$
EW.32 $\mathcal{H}_\mathsf{v} = \mathbb{C}^2;\ W_i \mapsto \frac{1}{2}\sigma_i;\ Y \mapsto \frac{1}{2}\sigma_0;\ T_a\mathsf{v} = 0$	EW.33 $M^2 = \mu_\star^2$ Gram; ker = stabilizer	EW.34 $\rho = 1;\ m_W = m_Z \cos\theta_W$
EW.35 $e = gg'/\sqrt{g^2+g'^2};\ 1/e^2 = 1/g^2 + 1/g'^2$	EW.36(a) spec $Q = \{0,1\};\ q \in \{-1,0,+1\}$	EW.29/30 full spectrum; reduction = spectral

$\blacksquare$

2.17 §E — Electromagnetic Sector

Dep.: Part 1; Part 2 §C, §D. Forward cross-section pointers, corroboration only (¬load-bearing): EM.18, EM.26 → CT3.7 (Part 2 §F; the main arguments are CT0.14-based). Notation: $\mathsf{E}_{\text{bd}}, \mathsf{E}_{\text{tr}}$ [C6]; $\mathcal{P}_M^{(n)}, \mathrm{F}_M^{(n)}$ [C7]; Z [C8]; α_e — per-edge dynamic phase [D3.5, Def-EM.20, EM.20]. Machine: `checks/verify_E.py`; Def-EM.20/EM.20/EM.23/EM.24: `checks/verify_F33_dynamic_phase.py`.

2.18 §E.I — Rung-2 Existence and Algebra

Theorem 2.70 (EM.1 — The electromagnetic aspect activates at rung 2). Thm.

$$\text{EM} \in \mathcal{A}(2); \qquad \text{EM} \notin \mathcal{A}(1); \qquad \text{EM} \notin \mathcal{A}(0);$$

$$\neg\exists \text{ independent aspect named } Z \text{ at rungs } \leq 2 : \quad \text{neutral seed} = W_3 \in F_W \text{ [C8]}.$$

Req. A7, C8; D0.8, D3.3; CT0.13 (Part 1).
Pf.

$$\text{CT0.13}: \ \mathcal{A}(0)=\{J\}; \quad \mathcal{A}(1)=\{J,G,W,H,D^{+}\}; \quad \mathcal{A}(2)=\mathcal{A}(1)\cup\{\text{EM}\} \implies \text{EM}\in\mathcal{A}(2)$$

$$\text{D3.3}: \ f_v \text{ defined} \iff \partial^{\leftarrow}(v)\geq 1 \ \wedge \ \partial^{\rightarrow}(v)\geq 1 \implies n(v)\geq 2 \quad [\text{D0.8}]$$

$$n\leq 1: \ \neg\exists v: \ (\partial^{\leftarrow}\geq 1\wedge\partial^{\rightarrow}\geq 1)$$

$$\implies f_v \text{ undefined } \forall v \implies \text{EM has no support} \implies \text{EM}\notin\mathcal{A}(0)\cup\mathcal{A}(1)$$

$$\text{CT0.13}: \ W_3\in F_W \ (\text{3-dim block}); \quad F_{\text{EM}}=\text{distinct stack block} \implies \neg\exists \text{ aspect } Z \text{ at rungs} \leq 2. \ \blacksquare$$

Phys. Electromagnetism switches on exactly at rung 2, because measuring a relative phase needs both an incoming and an outgoing real edge at one vertex. Its existence — and its absence at rungs 0–1 — is a theorem of degree bookkeeping.

Theorem 2.71 (EM.2 — The electromagnetic generator). Thm. [EW.8 realizations]

$$Y_{\text{asp}}=\mathcal{P}^{(\text{EM})}: \ \text{rk}=1; \qquad \mathfrak{T}\big(Y_{\text{asp}}\big)=\dim\mathcal{K}^{(\text{EM})}=1; \qquad [Y_{\text{asp}}, \text{ any block-diagonal operator}]=0;$$

$$Y_{\text{car}}=\tfrac{1}{2}\sigma_0; \qquad [Y_{\text{car}}, W_i|_{\text{car}}]=0 \ [\text{EW.32}];$$

$$\text{mask fact } (\neg Y): \quad \mathcal{P}_M^{(2)}E_0\mathcal{P}_M^{(2)}=\mathcal{P}_M^{(2)}, \quad \mathfrak{T}\big(\mathcal{P}_M^{(2)}\big)=\dim\mathcal{K}_2=17$$

$$(E_0\leftrightarrow F_{D^+} \ [\text{CT0.12}]).$$

Req. CT0.12, CT0.13, CT0.14, CT1.2 (Part 1); CT2.2, D2.3 (Part 2); D2.12; EW.8, EW.32 (Part 2).
Pf.

$$\dim\mathcal{K}^{(\text{EM})}=1 \ [\text{CT0.14}] \implies Y_{\text{asp}}=\mathcal{P}^{(\text{EM})}: \ \text{rk}\,1, \ \ \mathfrak{T}=\dim\text{Im}=1 \ [\text{D2.3}, \ \text{CT2.2}]$$

$$\text{D2.3}: \ \mathcal{K}=\bigoplus_A\mathcal{K}^{(A)} \text{ orthogonal} \implies [\mathcal{P}^{(\text{EM})}, \text{ block-diagonal}]=0$$

$$Y_{\text{car}}=\tfrac{1}{2}\sigma_0: \ [\text{EW.32}: \ \text{uniform trace normalization } \mathfrak{T}(\mathcal{G}_A\mathcal{G}_B)=\tfrac{1}{2}\delta_{AB}];$$

$$[\tfrac{1}{2}\sigma_0,\tfrac{1}{2}\sigma_i]=0$$

$$E_0=I_4 \ [\text{CT1.2}] \implies \mathcal{P}_M^{(2)}E_0\mathcal{P}_M^{(2)}=\mathcal{P}_M^{(2)} \ [\text{CT2.2}]; \quad \mathfrak{T}=17 \ [\text{CT0.14}];$$

$$E_0\leftrightarrow F_{D^+} \ [\text{CT0.12}]. \ \blacksquare$$

Rmk (repair). The prior codex defined $Y := \mathcal{P}_M^{(2)} E_0 \mathcal{P}_M^{(2)} = \mathcal{P}_M^{(2)}$ (rank 17) and then used it both as the abelian generator and inside the EM.17 trace formula. The rank-17 object is the rung-2 mask; the abelian generator is the rank-one block projection Y_{asp} (aspect space) and $\frac{1}{2}\sigma_0$ (carrier). The three realizations $Y_{\text{slot}}, Y_{\text{asp}}, Y_{\text{car}}$ are coordinated by EW.8/D2.12.

Phys. One generator, three well-typed faces: the centre of the rung-2 slot algebra, the rank-one projection onto the EM block (trace 1), and $\frac{1}{2}\sigma_0$ on the seed. The trace formula (EM.17) uses the block face; charge statements (EW.36) use the carrier face; the number 17 belongs to the mask, not to Y.

Theorem 2.72 (EM.3 — The electromagnetic Lie algebra). Thm. ≡ EW.9: $\big(\text{span}_{\mathbb{R}}\{Y\}, [\cdot,\cdot]\big) \cong \mathfrak{u}(1)$; $[aY, bY] = 0$; $\dim = 1$.

Req. EW.9 (Part 2).

Pf. Identity [EW.9]. ■

Phys. The electromagnetic block is the one-dimensional abelian factor: bracket identically zero.

Theorem 2.73 (EM.4 — The rung-2 algebra containing the abelian factor). Thm. ≡ EW.10: $\mathfrak{g}_{\text{EW}} = \text{span}_{\mathbb{R}}\{W_1, W_2, W_3, Y\} = \mathfrak{su}(2) \oplus \mathfrak{u}(1)$; $\dim = 4$.

Req. EW.10 (Part 2); EM.2, EM.3.

Pf. Identity [EW.10: CT1.4 $\mathfrak{su}(2)$; EM.3 $\mathfrak{u}(1)$; EM.2 commutation; commuting ideals, trivial intersection]. ■

Phys. The abelian factor sits inside the rung-2 electroweak algebra as its commuting centre-plus-identity summand.

2.19 §E.II — The Photon

Theorem 2.74 (EM.5 — Photon and gluon as trivial chains). Thm. [D1.7, D3.7]

$$\gamma_\gamma \in \mathsf{Trivial} \ \wedge \ \gamma_\gamma \subseteq V_2 \qquad (\text{photon}); \qquad \gamma_g \in \mathsf{Trivial} \ \wedge \ \gamma_g \cap V_3 \neq \varnothing \qquad (\text{gluon});$$

$$\forall \gamma \in \mathsf{Trivial}: \ \ \Theta(\gamma) = 0 \ \wedge \ \text{Hol}_{\text{spiral}}(\gamma) = 1; \qquad V_2 \cap V_3 = \varnothing;$$

$$\text{distinguishing datum} = \text{possibility space only } (V_2 \text{ vs } V_3 \text{ availability}); \qquad \neg\text{bound loops}.$$

Req. D0.8, D1.7, D3.7; CT0.13 (Part 1).

Pf.

$$\text{D1.7}: \ \gamma \in \mathsf{Trivial} \iff \forall e \in \gamma: \ \widehat{ds}^{\,2}(e)$$

$$= 0 \implies \Theta(\gamma)$$

$$= \sum_e 0$$

$$= 0 \implies \Phi(\gamma)$$

$$= \sum_e \alpha_e \cdot 0$$

$$= 0 \implies \text{Hol}$$

$$= e^{i\Phi} \qquad = 1$$

$$\text{D0.8}: \; V_n = \{v : n(v) = n\} \implies V_2 \cap V_3 = \varnothing$$

$$\text{CT0.13}: \; \text{strong} \in \mathcal{A}(3) \setminus \mathcal{A}(2) \implies \gamma \subseteq V_2 : \; \neg\text{colour index}; \quad \gamma \cap V_3 \neq \varnothing : \; \text{colour possible}$$

$$\text{same class } \mathsf{Trivial}; \quad \neg\exists \text{ further distinguishing datum. } \blacksquare$$

Phys. Photon and gluon are one kind of object — a null (trivial) chain of the graph — distinguished only by which vertex rungs it may visit. "Being a photon" is a possibility-space property of rung-2 chains, not a different substance. (Cited in F.)

Theorem 2.75 (EM.6 — Photon masslessness). Thm. $m_\gamma^2 = 0$, with two independent supports:

$$\text{(i)} \quad A = \sin\theta_W W_3 + \cos\theta_W Y \text{ spans } \ker M^2 \text{ [EW.22, EW.24]};$$

$$\text{(ii)} \quad \gamma_\gamma \in \mathsf{Trivial}$$

$$\subseteq V_2 : \; \Theta(\gamma)$$

$$= 0 \; \wedge \; \text{winding}$$

$$= 0 \implies \neg\exists \text{ wrapping holonomy} \implies \neg\exists \text{ wrapping contribution [EM.5, EW.18]}.$$

Req. D2.5; EM.5; EW.18, EW.22, EW.24, EW.25, EW.33 (Part 2).
Pf.

$$\text{D2.5}: \; m^2 = \text{pole displacement};$$

$$\text{EW.33}: \; \text{displacement source} = \mu_\star^2 \left\| \textstyle\sum_A c_A \mathcal{G}_A \mathsf{v} \right\|^2 \text{ (rung-3 wrapping form)}$$

$$\text{EW.22}: \; \ker M^2 = \text{span}\{(g', g)\}_{(W_3, Y)} = A\text{-direction [EW.24]}$$

$$\text{(ii)}: \; \text{trivial} \implies \Theta$$

$$= 0 \text{ [EM.5]};$$

$$\text{winding}$$

$$= 0$$

$$\implies e^{i\Phi}$$

$$= 1 \text{ vacuous} \implies \neg\text{materialization from wrapping [EW.18]}$$

$$\text{neutral direction with } \neg\text{wrapping}$$

$$= \ker\text{-direction} \implies \gamma$$

$$= A \implies \text{displacement}$$

$$= 0$$

$$\implies m_\gamma^2 \qquad = 0$$

$$[= \text{EW.25}]. \blacksquare$$

Phys. The photon is massless twice over: algebraically it spans the kernel of the Gram mass form (the seed's stabilizer), and geometrically a null chain confined to rung-2 vertices cannot wind, so the wrapping defect never sees it. (Cited in F, G.)

Theorem 2.76 (EM.7 — Photon entropy budget). Thm. Photon chain of hop count δh:

$$\mathsf{E}_{\text{tr}} = \delta h \ln 2 \text{ nats} = \delta h \text{ bits}; \qquad \mathsf{E}_{\text{bd}} = 0.$$

Req. C9; D1.5; CT0.9 (Part 1); EM.5.
Pf.

$$\text{CT0.9} + \text{C9}: \ 1 \text{ real edge} = 1 \text{ bit} = \ln 2 \text{ nats};$$

$$\text{edges} = \text{independent registers} \implies \delta h \text{ hops}: \ \delta h \ln 2$$

$$\text{D1.5}: \ \text{open propagating chain} = \text{transit [EM.5]} \implies \mathsf{E}_{\text{tr}} = \delta h \ln 2$$

$$\mathsf{E}_{\text{bd}} = \Delta S_{\text{bound}} \text{ [D1.5]}; \quad \text{photon } \neg\text{closed} \implies \mathsf{E}_{\text{bd}} = 0. \blacksquare$$

Phys. A photon carries exactly one bit of transit entropy per hop and zero bound entropy: bosonic transport content, measured by the ledger's own edge count.

Theorem 2.77 (EM.8 — Photon polarization count). Thm. $D = 4$; $k_\mu \neq 0$ null $\big(k^\mu k_\mu = 0\big)$; $\epsilon_\mu \sim \epsilon_\mu + k_\mu$ (gauge, EM.14); transversality $k^\mu \epsilon_\mu = 0$ (source-free equation, CT1.22). Then

$$N_{\text{pol}} = \dim\big(k^\perp / \operatorname{span}\{k\}\big) = (4-1) - 1 = 2, \qquad k^\perp := \{v : \ k_\mu v^\mu = 0\}.$$

Req. CT1.22, CT2.6 (Part 2); EM.6, EM.14; rank–nullity [linear algebra].
Pf.

$$\text{CT1.22} : \ F_{\mu\nu} = 2\nabla_{[\mu} A_{\nu]};$$

$$\text{EM.14} : \ A_\mu \sim A_\mu + \nabla_\mu \chi \implies \epsilon_\mu \propto k_\mu : \ 2k_{[\mu} k_{\nu]} = 0 \implies F = 0 : \ \text{pure gauge}$$

$$\text{source-free equation [CT1.22]} \implies k^\mu \epsilon_\mu = 0 \implies \epsilon \in k^\perp$$

$$k_\mu \neq 0 \implies \operatorname{rank}(k.) = 1 \implies \dim k^\perp = 4 - 1 = 3 \quad \text{[rank–nullity]}$$

$$k^\mu k_\mu = 0 \implies k \in k^\perp; \quad k \neq 0 \implies \dim\big(k^\perp / \operatorname{span}\{k\}\big) = 3 - 1 = 2. \ \blacksquare$$

Phys. A massless spin-1 mode in four dimensions has exactly two physical polarizations: transversality gives three, gauge redundancy removes one. The count is derived from dimension arithmetic plus nullness — the same $N_{\gamma\text{-pol}} = 2$ of policy P2. (Cited in G.)

2.20 §E.III — Electromagnetic Charge

Theorem 2.78 (EM.9 — Chain end-charge). Thm. [D3.4]

$$q_{\text{EM}}(c) = \frac{\sigma_{s(c)} - \sigma_{t(c)}}{2} \in \{-1, 0, +1\};$$

all three values attained;

(σ_s, σ_t)	$(+,+)$	$(+,-)$	$(-,+)$	$(-,-)$
q_{EM}	0	$+1$	-1	0

Req. D3.4; CT0.9 (Part 1).
Pf.

$$\text{CT0.9} : \ \sigma_e \in \{+1, -1\} \implies \sigma_{s(c)},$$

$$\sigma_{t(c)} \in \{+1, -1\}$$

$$\implies q_{\mathrm{EM}} \in \left\{ \tfrac{1-1}{2}, \tfrac{1-(-1)}{2}, \tfrac{-1-1}{2}, \tfrac{-1-(-1)}{2} \right\} \qquad = \{0, +1, -1\}$$

$$\text{end phases} = \text{independent registers} \implies \forall (\sigma_s, \sigma_t) \in \{\pm 1\}^2 \text{ realizable.} \blacksquare$$

Phys. The charge of an open chain is the half-difference of its two end-signs: a ternary alphabet arising from binary edges. (Cited in D as EW.36's independent counterpart, and in G.)

Corollary 2.3 (EM.9b — chain-wall telescoping). Let $c = (e_1, \ldots, e_k)$, $k \geq 2$, be an oriented chain of distinct real edges [D1.4] with interior vertices $v_1, \ldots, v_{k-1}$, $v_i := t(e_i) = s(e_{i+1})$, each of type $(1, 1)$ [A7, D0.8] — the chain's two edges are v_i's only incident real edges. Then the D3.3 vertex reads telescope:

$$\sum_{i=1}^{k-1} f_{v_i} \;=\; \tfrac{1}{2}\big(\sigma_{e_k} - \sigma_{e_1}\big) \;=\; -\, q_{\mathrm{EM}}(c)$$

$$\big[\sigma_{s(c)} = \sigma_{e_1}, \;\; \sigma_{t(c)} = \sigma_{e_k} \,:\, \text{D3.4/EM.9 end registers}\big]$$

— the total interior vertex activity of a chain is carried by its two end registers alone. In particular:

$$\sum_{i=1}^{k-1} f_{v_i} = 0 \iff \sigma_{e_1} = \sigma_{e_k} \iff q_{\mathrm{EM}}(c) = 0;$$

$$\sum_{i=1}^{k-1} \aleph_{v_i} = \#\{i : \, \sigma_{e_{i+1}} \neq \sigma_{e_i}\} \leq k - 1$$

$$[\text{D3.3} : \; \aleph_v = |f_v|; \; \text{CT0.9}].$$

Req (Cor). A7; D0.8, D1.4, D3.3, D3.4; CT0.9 (Part 1); EM.9; finite telescoping [math; machine: `checks/verify_promotion_E_telescoping.py`].

Pf (Cor).

$$\text{type}(1,1)\ [\text{A7}, \text{D0.8}] : \; \sigma_{\leftarrow}(v_i) = \sigma_{e_i}, \;\; \sigma_{\rightarrow}(v_i) = \sigma_{e_{i+1}};$$

$$\text{D3.3} : \; f_{v_i} = \tfrac{1}{2}\big(\sigma_{\rightarrow}(v_i) - \sigma_{\leftarrow}(v_i)\big) = \tfrac{1}{2}\big(\sigma_{e_{i+1}} - \sigma_{e_i}\big)$$

$$\sum_{i=1}^{k-1} f_{v_i} = \tfrac{1}{2} \sum_{i=1}^{k-1} \big(\sigma_{e_{i+1}} - \sigma_{e_i}\big) = \tfrac{1}{2}\big(\sigma_{e_k} - \sigma_{e_1}\big) \quad [\text{telescoping}];$$

$$\text{D3.4/EM.9} : \; q_{\mathrm{EM}}(c) = \tfrac{1}{2}\big(\sigma_{e_1} - \sigma_{e_k}\big) = -\sum_i f_{v_i}$$

$$\text{CT0.9} : \; \sigma \in \{\pm 1\} : \; f_{v_i} \neq 0 \iff \sigma_{e_{i+1}} \neq \sigma_{e_i}; \qquad \aleph_{v_i} = |f_{v_i}| \in \{0, 1\} \; [\text{D3.3}] \; \blacksquare$$

Theorem 2.79 (EM.9a — The closed-core charge read and the end-register count). Thm. (1) Closed-core read. For a closed oriented chain $c = (e_1, \dots, e_k)$ based at $v = s(c) = t(c)$ [D1.4: ordered tuple] with first edge e_1 (out of v) and last edge e_k (into v):

$$q_{\text{core}}(c; v) := \frac{\sigma_{e_1} - \sigma_{e_k}}{2} \in \{-1, 0, +1\}$$

is well-defined [D3.4 applied to closed c; CT0.9: per-edge independent registers; EM.9: alphabet, all values attained]. (2) Charge-conjugation law. Under rebase / orientation reversal the roles of e_1 and e_k swap:

$$q_{\text{core}} \mapsto -q_{\text{core}}$$

— consistent with, and an instance of, D2.6's ω (charge $\mapsto -$charge): the corpus's own charge-conjugation law, verbatim on the core read. At the 2-cycle floor: $q_{\text{core}} = \frac{1}{2}(\sigma_{e_1} - \sigma_{e_2})$ at base p, negated at base q. (3) End-register count. The EM.9/D3.4 read consumes exactly the two end registers of the chain: $N_{\text{charge}} = 2$ is a theorem, ¬an enumeration convention; the ternary alphabet is the difference of two binary registers [CT0.9], with all three values attained (witnesses $(\sigma_1, \sigma_k) = (+,-) \mapsto +1$, $(+,+) \mapsto 0$, $(-,+)$ or $(-,-) \mapsto$ the remaining values; surjectivity machine-checked). (4) Not neutral by closure. The closed core is ¬forced charge-neutral: its ternary read is basepoint data. The observer-facing charge of a transit chain remains the EM.9 end-read.

Req. A1; D1.4, D2.6, D3.4; CT0.9 (Part 1); EM.9; witness enumeration [math; machine: `checks/verify_merge_tierB.py` §D].

Pf.

$$(1):\ \text{D3.4 is stated for oriented chains generally;}$$

$$\text{closed } c:\ s(c)$$

$$= t(c)$$

$$= v \text{ with distinguished first/last edges [D1.4]} \implies \sigma_{s(c)}:$$

$$= \sigma_{e_1},$$

$$\sigma_{t(c)}:$$

$$= \sigma_{e_k} \text{ well-defined}$$

$$\text{CT0.9}:\ \sigma_{e_1}, \sigma_{e_k} \in \{\pm 1\} \text{ independent;}$$

$$q_{\text{core}} \in \{\tfrac{1}{2}(\pm 1 \mp 1)\} = \{-1, 0, +1\};$$

$$\text{EM.9 : all attained}$$

$$(2) : \text{ rebase/orientation reversal : } e_1 \leftrightarrow e_k \text{ roles swap}$$

$$\implies q_{\text{core}} \mapsto \tfrac{1}{2}(\sigma_{e_k} - \sigma_{e_1})$$

$$= -q_{\text{core}};$$

$$\text{D2.6 : } \omega \text{ maps charge} \mapsto -\text{charge : consistent}$$

$$(3) : \text{ D3.4′s formula contains exactly } \sigma_{s(c)}, \sigma_{t(c)} : \text{ two registers; interior signs } \neg\text{enter;}$$

$$\#\{\tfrac{1}{2}(\sigma - \sigma') : \sigma, \sigma' \in \{\pm 1\}\} = 3$$

$$(4) : (1)\text{′s witnesses include } q_{\text{core}} = \pm 1 \neq 0. \blacksquare$$

Phys. A closed chain carries a ternary charge-type read at its basepoint even though it has no ends: the difference of the two registers where it leaves and re-enters its base. Orientation reversal negates it exactly — charge conjugation acting on the only charge-like label closed content can carry. The count is rigid: two registers make a charge, as a theorem.

Theorem 2.80 (EM.10 — Charge quantization). Thm.

$$S^2$$

$$= T^2$$

$$= \nVdash \text{[A3]} \implies \phi_e \in \{0, \pi\} \implies \sigma_e \in \{+1, -1\} \implies \sigma_s - \sigma_t \in \{-2, 0, +2\}$$

$$\implies q_{\text{EM}} \in \mathbb{Z} \cap \{-1, 0, +1\};$$

$$q \notin \mathbb{Z} \implies \phi_e \notin \Phi \implies \neg\text{CT0.9} \qquad (\neg\exists \text{ finer values}).$$

Req. A3; D0.5, D3.4; CT0.9 (Part 1).

Pf.

$$\text{CT0.9 } [\Leftarrow \text{D0.5, A3 : } S^2 = T^2 = \nVdash \implies \text{residual holonomy } \mathbb{Z}_2] : \phi_e \in \{0, \pi\} \implies \sigma_e = e^{i\phi_e} \in \{\pm 1\}$$

$$\sigma_s, \sigma_t \in \{\pm 1\} \implies \sigma_s - \sigma_t \in \{-2, 0, +2\} \subset 2\mathbb{Z} \implies q_{\text{EM}} = \tfrac{1}{2}(\sigma_s - \sigma_t) \in \{-1, 0, +1\} \subset \mathbb{Z}$$

$$q_{\text{EM}} \notin \mathbb{Z} \implies \sigma_s - \sigma_t \text{ odd} \implies \sigma \notin \{\pm 1\} \implies \phi_e \notin \{0, \pi\} \implies \neg\text{CT0.9} \implies \bot. \blacksquare$$

Phys. Charge quantization is not imposed — it is the arithmetic shadow of the binary edge phase, which is itself forced by the involution axiom $S^2 = T^2 = \nVdash$. Integers all the way down. (Cited in G.)

Theorem 2.81 (EM.11 — Global charge conservation). Thm. U valid [D2.11] $\implies$ $Q_{\text{EM}} := \sum_{c \in U} q_{\text{EM}}(c) = 0$.

Req. D1.2, D1.3, D1.4, D2.11, D3.4; EM.9.

Pf.

$$\text{D3.4}: \; q_{\text{EM}}(c) = \tfrac{1}{2}\sigma_{s(c)} - \tfrac{1}{2}\sigma_{t(c)} \implies Q_{\text{EM}} = \tfrac{1}{2} \sum_{\text{source-ends}} \sigma \; - \; \tfrac{1}{2} \sum_{\text{target-ends}} \sigma$$

(regroup by ends)

$$\text{D2.11}: \; \neg\exists \text{ dangling ledger boundary} \implies \forall \text{ ends sealed}$$

$$\text{D1.2}: \; \text{seal joins two edge-ends;}$$

$$\text{D1.4}: \; \text{chains oriented} \implies \text{seal joins target-end} \to \text{source-end} \; (\text{same-orientation join} \implies \neg\text{continuable orientation}, \; \bot \text{D1.4})$$

$$\text{D1.3}: \; \#J^+ = \#J^- \implies \text{pairing covers every end exactly once}$$

$$\text{seal identifies joined ends} \implies \text{equal } \sigma \implies \text{per-seal contribution} = \tfrac{1}{2}\sigma - \tfrac{1}{2}\sigma = 0$$

$$Q_{\text{EM}} = \sum_{\text{seals}} 0 = 0. \blacksquare$$

Phys. Total charge vanishes in every valid universe: each chain end is sealed against an oppositely oriented end carrying the same sign. Charge conservation is Janus topology — seals pair sources with sinks — not a Noether afterthought.

2.21 §E.IV — Field Strength and Dynamics

Theorem 2.82 (EM.12 — Discrete field strength from phase transitions). Thm. [D3.3]

$$f_v = \tfrac{1}{2}\big(\sigma_{\rightarrow}(v) - \sigma_{\leftarrow}(v)\big) \in \{-1, 0, +1\}; \qquad \aleph_v = |f_v| \in \{0, 1\};$$

$$\aleph(\sigma, C) = 0 \iff \forall v : \ \sigma_{\leftarrow}(v) = \sigma_{\rightarrow}(v) \iff \sigma \text{ constant on connected components } (\text{flat});$$

$$\aleph_v = 1 \iff \text{oriented transition at } v;$$

$$\text{coarse-graining [CT1.19]} : \ f_v \longrightarrow F_{\mu\nu} \ (\text{circulation per plaquette}) \ [\text{CT1.22}].$$

Req. D3.3; CT0.9 (Part 1); CT1.19, CT1.22 (Part 2).
Pf.

$$\sigma_{\leftarrow}, \sigma_{\rightarrow} \in \{\pm 1\} \ [\text{CT0.9}] \implies f_v \in \{-1, 0, +1\}, \ \ |f_v| \in \{0, 1\} = \aleph_v \quad [\text{D3.3}]$$

$$\aleph = 0 \iff \forall v : \ \aleph_v = 0 \iff \forall v : \ \sigma_{\leftarrow} = \sigma_{\rightarrow} \iff \sigma \text{ constant along components}$$

$$(\text{zero discrete curvature})$$

$$\aleph_v = 1 \iff \sigma_{\leftarrow} \neq \sigma_{\rightarrow} \iff f_v = \pm 1 : \ \text{oriented transition}$$

CT1.19 : holonomy per unit area $\to F_{\mu\nu}$ [CT1.22]; vertex sign flip = nonzero circulation. ■

Phys. The electromagnetic field on the ledger is the sign mismatch at vertices: f_v is the oriented discrete field strength and $\aleph_v$ its magnitude. Zero excitation means signs flow unflipped; each flip is one excitation quantum.

Theorem 2.83 (EM.13 — The transition-parity factor in the weight). Thm. $w(C) = \mathcal{J}(C) \prod_e G(e) \cdot (-1)^{\aleph(\sigma, C)}$ [A8]:

$$(-1)^{\aleph} = \begin{cases} +1 & \aleph \text{ even} \\ -1 & \aleph \text{ odd} \end{cases}; \qquad \prod_e G(e) \supseteq \prod_e \sigma_e \ [\text{D3.5}]; \qquad \Big(\prod_e \sigma_e\Big) \text{ independent of } (-1)^{\aleph} \ [\text{CT0.10}];$$

$$Z(\mathcal{R})$$

$$= \sum_C \sum_\sigma w(C) : \ \Delta\aleph \text{ odd} \implies \text{sign flip} \implies \text{destructive interference } (\text{discrete EM dynamics});$$

$$\text{coarse-graining} \implies \text{relative signs of } F_{\mu\nu}.$$

Req. A8; D3.5; CT0.9, CT0.10 (Part 1).
Pf.

$$(-1)^{\aleph} \in \{+1,-1\} \text{ by parity [definition]}$$

$$\text{D3.5}: \; G(e) = \sigma_e \exp\!\big(i\alpha_e \widehat{ds}^{\,2}(e)\big) \implies \prod_e G(e) \supseteq \prod_e \sigma_e$$

$$\text{CT0.10}: \; \Big(\prod_e \sigma_e,\ (-1)^{\aleph}\Big) \text{ independent,}$$

$$\text{joint image}$$

$$= \{\pm 1\}^2 \implies \text{two distinct sign data;}$$

$$\neg\text{double counting}$$

$$Z: \; C \mapsto C' \text{ with } \Delta\aleph \text{ odd} \implies w(C') = -w(C) \text{ (ceteris paribus)} \implies \text{cancellation.} \blacksquare$$

Phys. The A8 weight assigns a strict minus sign to every odd number of domain walls in the sign field: this destructive interference between one-transition-apart configurations is the lattice dynamics of electromagnetism, and it survives coarse-graining as the Maxwell sign structure.

Theorem 2.84 (EM.14 — Continuum gauge invariance). Thm.

$$A_\mu \sim A_\mu + \nabla_\mu \chi; \qquad F_{\mu\nu} = 2\nabla_{[\mu} A_{\nu]} \;\mapsto\; F_{\mu\nu} + 2\nabla_{[\mu}\nabla_{\nu]}\chi = F_{\mu\nu};$$

$$\implies \text{gradient direction} = \text{redundancy}; \quad N_{\text{pol}} \text{ count unaffected [EM.8]}.$$

Req. CT1.22, CT2.6 (Part 2); EM.12; $\nabla_{[\mu}\nabla_{\nu]}\chi = 0$ for scalars [math].
Pf.

$$\text{EM.12}: \; \text{physical datum} = f_v \propto \sigma_{\rightarrow} - \sigma_{\leftarrow}: \; \text{differences only}; \; \neg\text{dependence on common offset}$$

$$\text{smooth offset field} \xrightarrow[\text{CT1.19}]{} \chi: \quad A_\mu \mapsto A_\mu + \nabla_\mu \chi$$

$$F_{\mu\nu} \mapsto F_{\mu\nu} + 2\nabla_{[\mu}\nabla_{\nu]}\chi = F_{\mu\nu} \quad [\nabla_\mu\nabla_\nu\chi = \nabla_\nu\nabla_\mu\chi \text{ on scalars}]. \blacksquare$$

Phys. Gauge invariance is the continuum image of a ledger triviality: vertices record sign changes, never absolute signs. The redundancy is derived from the discreteness, not postulated.

Theorem 2.85 (EM.15 — Electromagnetic stress-energy). Thm.

$$S_{\mathrm{EM}}[A,g] = -\tfrac{1}{4}\int d^4x\sqrt{-g}\,F_{\mu\nu}F^{\mu\nu};$$

$$T^{(\mathrm{EM})}_{\mu\nu} := -\frac{2}{\sqrt{-g}}\frac{\delta S_{\mathrm{EM}}}{\delta g^{\mu\nu}} = F_{\mu\rho}F_{\nu}{}^{\rho} - \tfrac{1}{4}g_{\mu\nu}F_{\alpha\beta}F^{\alpha\beta};$$

$$g^{\mu\nu}T^{(\mathrm{EM})}_{\mu\nu} = 0 \qquad (D=4).$$

Req. CT1.22 (Part 2); EM.6; $\delta\sqrt{-g} = -\tfrac{1}{2}\sqrt{-g}\,g_{\mu\nu}\delta g^{\mu\nu}$, $\delta\big(g^{\mu\alpha}g^{\nu\beta}F_{\mu\nu}F_{\alpha\beta}\big) = 2F_{\mu\rho}F_{\nu}{}^{\rho}\,\delta g^{\mu\nu}$ [math; machine: `checks/verify_E.py`].

Pf.

CT1.22 : S_{EM} = unique local, diff- $\wedge$ gauge-invariant, $\leq$ 2nd-order functional quadratic in F

$$\delta S_{\mathrm{EM}} = -\tfrac{1}{4}\int\sqrt{-g}\Big[\underbrace{-\tfrac{1}{2}g_{\mu\nu}F^2}_{\delta\sqrt{-g}} + \underbrace{2F_{\mu\rho}F_{\nu}{}^{\rho}}_{\delta(F^2)}\Big]\delta g^{\mu\nu}$$

[$F_{\mu\nu}$ fixed lower indices]

$$T^{(\mathrm{EM})}_{\mu\nu}$$

$$= -\tfrac{2}{\sqrt{-g}}\cdot(-\tfrac{1}{4})\sqrt{-g}\big[2F_{\mu\rho}F_{\nu}{}^{\rho} - \tfrac{1}{2}g_{\mu\nu}F^2\big]$$

$$= F_{\mu\rho}F_{\nu}{}^{\rho} - \tfrac{1}{4}g_{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}$$

$$\text{trace}: \quad g^{\mu\nu}T_{\mu\nu} = F_{\mu\rho}F^{\mu\rho} - \tfrac{1}{4}\cdot 4\cdot F^2 = 0$$

[consistency : massless transit aspect, EM.6]. ■

Phys. The Maxwell stress tensor is derived as the metric variation of the unique admissible quadratic action — the $a = \mathrm{EM}$ instance of CT1.22 — and its vanishing trace is the local imprint of photon masslessness.

2.22 §E.V — The Dynamic Phase

Theorem 2.86 (EM.16 — Edge transfer weight and the per-edge read-back). Thm. [D3.5] For every real edge e:

$$G(e) = \sigma_e\,\exp\big(i\,\alpha_e\,\widehat{ds}^{\,2}(e)\big); \qquad \alpha_e = \frac{1}{i}\,\frac{\partial}{\partial\,\widehat{ds}^{\,2}(e)}\log\big(\sigma_e^{-1}G(e)\big); \qquad [\alpha_e] = \mathfrak{q}^0.$$

Req. D0.5, D1.6, D2.8, D2.9, D3.5; CT2.1 (Part 2).

Pf.

$$\sigma_e^{-1} = \sigma_e \ (\sigma_e^2 = 1) \implies \sigma_e^{-1} G(e) = \exp\big(i\alpha_e \widehat{ds}^{\,2}(e)\big)$$

$$\log \text{ on the D2.8 branch} = i\alpha_e \widehat{ds}^{\,2}(e);$$

$$\tfrac{1}{i}\,\partial_{\widehat{ds}^{\,2}} \implies \alpha_e \quad \text{(constant along the edge: the read-back is per-edge)}$$

$$[\widehat{ds}^{\,2}] = \frac{[ds^2]}{[\mathfrak{h}^2]} = \frac{\mathfrak{q}^{-1}}{\mathfrak{q}^{-1}} = \mathfrak{q}^0 \ [D1.6, D2.9]; \qquad [\text{phase}] = \mathfrak{q}^0 \implies [\alpha_e] = \mathfrak{q}^0. \blacksquare$$

Phys. The transfer weight of an edge factors cleanly into a binary registration sign and a dimensionless geometric phase carrying the edge's own α_e; hop-normalization makes every per-edge phase a pure number from the outset — not a later repair.

Theorem 2.87 (EM.17 — Spectral read of the electromagnetic block). Thm. $\forall n \geq 2$: the masked force acts as a scalar on the one-dimensional EM block [CT0.14, D2.3]:

$$\mathrm{F}_M^{(n)}\big|_{\mathcal{K}^{(\mathrm{EM})}} = \alpha_{\mathrm{EM}}^{(n)} \cdot \mathrm{id};$$

$$\alpha_{\mathrm{EM}}^{(n)} = \frac{\mathfrak{T}\big(Y_{\mathrm{asp}}\, \mathrm{F}_M^{(n)}\, Y_{\mathrm{asp}}\big)}{\mathfrak{T}\big(Y_{\mathrm{asp}}\big)} = \mathfrak{T}\big(Y_{\mathrm{asp}}\, \mathrm{F}_M^{(n)}\, Y_{\mathrm{asp}}\big);$$

$$\mathfrak{T}\big(Y_{\mathrm{asp}}\big) = 1.$$

$\alpha_{\mathrm{EM}}^{(n)}$ is the force-space block-scale read at rung n; the edge phases α_e are per-edge registration data [D3.5, EM.20].

Req. D2.3; CT0.14 (Part 1); CT2.4 (Part 2); EM.2; EW.8 (Part 2); machine: `checks/verify_E.py`.

Pf.

$$\dim \mathcal{K}^{(\mathrm{EM})} = 1 \ [CT0.14] \implies \mathrm{F}_M^{(n)}\big|_{\mathcal{K}^{(\mathrm{EM})}} = \alpha_{\mathrm{EM}}^{(n)} \cdot \mathrm{id}$$

$$[\text{every operator on a 1-dim space is a scalar; } \alpha_{\mathrm{EM}}^{(n)} := \text{that eigenvalue}]$$

$$\text{D2.3 orthogonality}: \quad A \neq \mathrm{EM}: \ Y_{\mathrm{asp}} F_A Y_{\mathrm{asp}} = \mathcal{P}^{(\mathrm{EM})} F_A \mathcal{P}^{(\mathrm{EM})} = 0 \implies Y_{\mathrm{asp}} \mathrm{F}_M^{(n)} Y_{\mathrm{asp}} = \alpha_{\mathrm{EM}}^{(n)} Y_{\mathrm{asp}}$$

$$\mathfrak{T}: \quad \mathfrak{T}\big(Y_{\mathrm{asp}} \mathrm{F}_M^{(n)} Y_{\mathrm{asp}}\big) = \alpha_{\mathrm{EM}}^{(n)}\, \mathfrak{T}\big(Y_{\mathrm{asp}}\big) = \alpha_{\mathrm{EM}}^{(n)} \quad [\mathfrak{T}(Y_{\mathrm{asp}}) = 1, \ \mathrm{EM.2}]. \ \blacksquare$$

Rmk (repair). The prior formula $\mathfrak{T}(Y \mathrm{F}_M^{(n)} Y)/\mathfrak{T}(Y)$ with $Y = \mathcal{P}_M^{(2)}$ ($\mathfrak{T} = 17$) is false as written: the rank-2 mask retains J, G, W, H, D^+, EM, whose sum is not a scalar on $\mathcal{K}_2$. With the rank-one Y_{asp} the identity is exact — a 1-dimensional block has one eigenvalue and nothing else can contribute.

Phys. The electromagnetic block scale is read off operatorially: the single eigenvalue of the masked force on the one-dimensional EM block. A spectral formula replaces any fitting procedure. It is force-space data, distinct from the per-edge registration phases of EM.20. (Cited in F.)

Theorem 2.88 (EM.18 — The colour sector does not shift the block read). Thm. $\alpha_{\mathrm{EM}}^{(3)} = \alpha_{\mathrm{EM}}^{(2)}$.

Req. CT0.13, CT0.14 (Part 1); CT2.4 (Part 2); EM.17; [CT3.7 (Part 2) — forward, corroboration only].

Pf.

$$\text{CT0.13}: \ \mathcal{A}(3) = \mathcal{A}(2) \cup \{\text{strong}\} \implies \mathrm{F}_M^{(3)} = \mathrm{F}_M^{(2)} \oplus F_{\text{strong}} \quad [CT2.4]$$

$$\text{CT0.14}: \ \mathcal{K}^{(\text{strong})} \perp \mathcal{K}^{(\text{EM})} \implies Y_{\text{asp}}\, F_{\text{strong}}\, Y_{\text{asp}} = \mathcal{P}^{(\text{EM})} F_{\text{strong}} \mathcal{P}^{(\text{EM})} = 0$$

$$\big[\text{corroboration}: \mathfrak{T}(T_a) = 0\ [CT3.7] \implies \mathfrak{T}(F_{\text{strong}}) = \textstyle\sum_a f_a \mathfrak{T}(T_a) = 0 \text{ — independent argument}\big]$$

$$\implies Y_{\text{asp}} \mathrm{F}_M^{(3)} Y_{\text{asp}} = Y_{\text{asp}} \mathrm{F}_M^{(2)} Y_{\text{asp}} \implies \alpha_{\mathrm{EM}}^{(3)} = \alpha_{\mathrm{EM}}^{(2)} \quad [EM.17]. \ \blacksquare$$

Phys. Switching colour on at rung 3 adds traceless structure orthogonal to the electromagnetic block: the block read does not run. Two disjoint arguments — block orthogonality and colour tracelessness — agree. (Cited in F.)

Theorem 2.89 (EM.19 — The saturation sector does not shift the block read). Thm. On the accessible rung-2 subspace: $\alpha_{\mathrm{EM}}^{(n)} = \alpha_{\mathrm{EM}}^{(2)} \quad \forall n \geq 4$.

Req. CT0.6, CT0.11, CT0.13, CT0.14 (Part 1); D4.1; EM.17, EM.18.

Pf.

$$\text{CT0.13}: \ \mathcal{A}(n \geq 4) = \mathcal{A}(3) \cup \{D^-\}$$

$$\text{D4.1}: \ F_{D^-} = \bigoplus_{m \geq 4} F_m \text{ acts only on oversubscribed rungs } m \geq 4 \ [CT0.6]$$

$$\text{CT0.11}: \ \mathcal{K}_2 \ \neg\text{oversubscribed}; \quad \text{CT0.14}: \ \mathcal{K}^{(\text{EM})} \perp \mathcal{K}^{(D^-)} \implies Y_{\text{asp}} F_{D^-} Y_{\text{asp}} = 0$$

$$\implies Y_{\text{asp}} \mathrm{F}_M^{(n)} Y_{\text{asp}} = Y_{\text{asp}} \mathrm{F}_M^{(3)} Y_{\text{asp}} \overset{\text{EM.18}}{=} Y_{\text{asp}} \mathrm{F}_M^{(2)} Y_{\text{asp}} \implies \alpha_{\mathrm{EM}}^{(n)} = \alpha_{\mathrm{EM}}^{(2)}$$

$$[EM.17]. \ \blacksquare$$

Phys. The saturation lock absorbs oversubscribed information without renormalizing the accessible electromagnetic block read: α_{EM} is rung-independent across all accessible rungs.

Definition 2.1 (Def-EM.20 — The per-edge phase at crystallization). Def. For each real edge $e = (s,t)$ [A4, D0.1], the phase $\alpha_e \in \Phi$ [D0.5] is the branch index of the A4 crystallization criterion for the pair state ρ_{st} at the seal [A4, D1.2]:

$$\alpha_e \ := \ \alpha_{st} \ := \ \begin{cases} 0 & \text{if } \left|\Pi_{\mathcal{V}} \rho_{st}\right| \leq \varepsilon_q \quad \text{(aligned branch)} \\ \pi & \text{if } \left|\Pi_{\mathcal{V}} \rho_{st}\right| \geq 1 - \varepsilon_q \quad \text{(antialigned branch)} \end{cases}$$

— A4 declares the branch phase α_{pq} and both branches' non-emptiness; A7 declares the edge map $e \mapsto \alpha_e \in \{0, \pi\}$; this definition is the identification $\alpha_e = \alpha_{st}$. The branch windows are disjoint exactly when $\varepsilon_q < 1 - \varepsilon_q$, guaranteed by the admissible range $\varepsilon_q \in [0, \frac{1}{4})$ [CT0.8]. The branch read is a registration alternative [D1.1: the reality principle — registration transport acts on the reality line by $\{+1, -1\}$]: aligned $\leftrightarrow +1$, antialigned $\leftrightarrow -1$, so $e^{i\alpha_e} \in \{+1, -1\}$ is the alignment holonomy. Virtual pairs ($\varepsilon_q < \left|\Pi_{\mathcal{V}}\rho_{st}\right| < 1 - \varepsilon_q$) carry no α_e [A4, CT0.1; clause (3) of EM.20]

Theorem 2.90 (EM.20 — The dynamic edge phase). Thm. With α_e as in Def-EM.20, under A1 ($\mathbb{C}^2$ qubits), A4 (two-branch crystallization; branch phase α_{pq}; both branches populated), A5 (vacuum $\rho_\circ = I/4$; Haar draws), A6 (tick dynamics), A7 (edge phase map), CT0.9 ($\Phi = \{0, \pi\}$), D0.4 (ε_q), D1.6 ($\widehat{ds}^2$), D3.5 (edge weight): (1) Alphabet preservation. $\alpha_e \in \{0, \pi\} = \Phi$ for every real edge e; the phase alphabet is unchanged. [A4, A7; CT0.9] (2) Phase determination. $\alpha_e = \alpha_{st}$ is the branch index of the A4 criterion for the pair state at the moment of crystallization [A4, D1.2]: the phase is a measurement outcome — registration data read at the seal [D1.1] — ¬ a declared constant. (3) Virtual edges. For $e \in \mathcal{V}$ [CT0.1: $\varepsilon_q < \left|\Pi_{\mathcal{V}}\rho_{st}\right| < 1 - \varepsilon_q$] the pair carries the full $\mathbb{C}^2 \otimes \mathbb{C}^2$ statespace [A1]; α_e is undefined. The binary restriction scopes to real (crystallized) edges. (4) Edge weight. $G(e) = \sigma_e \cdot e^{i\alpha_e \widehat{ds}^2(e)}$ with α_e per-edge. [D3.5] (5) Both values attained; distribution governed. Both A4 branches are populated [A4 populatedness clause]; witnesses: a symmetric-dominated pair ($\left|\Pi_{\mathcal{V}}\rho\right| = 0 \Rightarrow \alpha_e = 0$) and the pure singlet ($\left|\Pi_{\mathcal{V}}\rho\right| = 1 \Rightarrow \alpha_e = \pi$). Across $E_{\mathcal{R}}$ the distribution of α_e is governed by the Haar randomization [A5, A6] and the crystallization criterion [A4].

Req. A1; A4; A5; A6; A7; CT0.9 (Part 1); D0.4, D1.1, D1.2, D1.6, D3.5; Def-EM.20; exhaustive exact pair-state census [math; machine: `checks/verify_F33_dynamic_phase.py` §A].

Pf.

$$(1):\ \text{A4}:\ \alpha_{pq} \in \{0, \pi\}\ \forall (p, q) \in \mathcal{R};$$

$$\text{A7}:\ e \in E_{\mathcal{R}} \mapsto \alpha_e \in \{0, \pi\};$$

$$\text{CT0.9}:\ \Phi = \{0, \pi\} \implies \alpha_e \in \Phi$$

$$(2):\ \text{A4/D1.2}:\ e \in \mathcal{R} \iff \text{branch membership at a seal;}$$

$$\text{Def-EM.20}:\ \alpha_e = \alpha_{st} = \text{branch index}$$

$$\text{D1.1}:\ \text{registration transport acts on the reality line by } \{\pm 1\}$$

$$\implies e^{i\alpha_e}$$

$$= \text{alignment holonomy} \in \{+1, -1\}:\ \text{a function of the registered pair state,}$$

$$\neg\text{a constant}$$

$$(3):\ \text{CT0.1/A4}:\ e\in\mathcal{V}\iff \varepsilon_q<\left|\Pi_{\mathcal{V}}\rho_{st}\right|<1-\varepsilon_q\iff\neg\big(\text{registration}\big);$$

$$\text{A1}:\ \rho_{st}\in\mathfrak{H}(\mathbb{C}^2\otimes\mathbb{C}^2)\ \text{retained without branch read}\implies\alpha_e\ \text{undefined}$$

$$(4):\ \text{D3.5}:\ G(e)=\sigma_e\exp\big(i\alpha_e\widehat{ds}^{\,2}(e)\big)\ \text{with per-edge}\ \alpha_e=\alpha_{st}\ [\text{A4}];$$

$$\text{D1.6}:\ \widehat{ds}^{\,2}(e)\ \text{well-defined per edge}$$

$$(5):\ \text{A4}:\ \exists\,\rho:\ \left|\Pi_{\mathcal{V}}\rho\right|\le\varepsilon_q\ \wedge\ \exists\,\rho':\ \left|\Pi_{\mathcal{V}}\rho'\right|\ge1-\varepsilon_q;$$

$$\text{witnesses}:\ \rho=\Pi_{\mathcal{R}}/3\ (w=0,\ \alpha=0);\quad \rho=\Pi_{\mathcal{V}}\ (w=1,\ \alpha=\pi)\ [\text{CT0.7};\ \text{machine}]$$

$$\text{A5/A6}:\ \text{node states Haar-drawn per tick}\ (U\notin\mathfrak{L}:\ \text{draws are not retained; registered reads are});$$

$$\text{A4}:\ \text{which pairs crystallize, and on which branch.}\ \blacksquare$$

Phys. Every real edge carries its own geometric phase: the branch index of the pair's crystallization criterion — symmetric-dominated pairs register 0, singlet-dominated pairs register π — valued in exactly the CT0.9 alphabet, with the two branch windows disjoint on the admissible threshold range ($\varepsilon_q<1-\varepsilon_q$; CT0.8). Which branch an edge takes is measurement outcome data, governed by the Haar randomization and the criterion. All downstream phase structure — the alignment patterns and parity classes (EM.23), the CPT bridge (EM.24) — reads this per-edge datum.

Theorem 2.91 (EM.21 — Dimensional status of the per-edge phase). Thm. $[\alpha_e]=\mathfrak{q}^0;\qquad \mathfrak{h}\mapsto\lambda\mathfrak{h}:\ \{0,\pi\}\mapsto\{0,\pi\}$; i.e. the phase alphabet is a derived pure-number set in the sense of CT2.11, and each α_e is an $\mathfrak{h}$-free registration value [EM.20(2)].

Req. D1.6; CT2.11 (Part 2); EM.16, EM.20.

Pf.

$$[\widehat{ds}^{\,2}]=\mathfrak{q}^0\ [EM.16];\qquad [\alpha_e\widehat{ds}^{\,2}]=[\text{phase}]=\mathfrak{q}^0\implies[\alpha_e]=\mathfrak{q}^0$$

$$\mathfrak{h}\mapsto\lambda\mathfrak{h}:\ \ ds^2\mapsto\lambda^2ds^2,\ \mathfrak{h}^2\mapsto\lambda^2\mathfrak{h}^2\implies\widehat{ds}^{\,2}\ \text{invariant};$$

$$\text{alphabet fixed by }\mathfrak{h}\text{-free data}\ [CT0.9,\ EM.20(1)]\implies\alpha_e\in\{0,\pi\}\ \text{in every calibration}$$

$$\text{CT2.11 criterion}:\ \text{dimensionless}\ \wedge\ \text{rescaling-invariant}\implies\text{derived pure-number alphabet.}\blacksquare$$

Phys. The phase alphabet $\{0,\pi\}$ is calibration-independent: it survives the rescaling of the single free scale, marking it as a genuine prediction rather than a unit convention. Individual phase values are $\mathfrak{h}$-free too — binary registration outcomes.

2.23 §E.VI — Parity and CPT

Definition 2.2 (Def-EM.21 — Alignment pattern). Def. For a subnetwork Σ [D2.13] with real edge set $E_{\mathcal{R}}(\Sigma) := \{e \in E_{\mathcal{R}} : e \text{ touches } \Sigma\}$ ($|\Sigma| = \#E_{\mathcal{R}}(\Sigma)$, D2.13), the alignment pattern is

$$\mathcal{P}(\Sigma) \;:=\; \big\{\, e \in E_{\mathcal{R}}(\Sigma) :\; \alpha_e = \pi \,\big\}$$

— the set of antialigned edges [Def-EM.20]. $\mathcal{P}(\Sigma) \subseteq E_{\mathcal{R}}(\Sigma)$; on a chain c [D1.4]: $\mathcal{P}(c) := \mathcal{P}(\Sigma) \cap c$; on a configuration C [D0.10]: $\mathcal{P}(C) := \{e \in E(C) :\; \alpha_e = \pi\}$.

Definition 2.3 (Def-EM.22 — Mirror operation). Def. A mirror M on Σ is an automorphism of the underlying graph of Σ that reverses orientation [D2.6 ω-type: $s \leftrightarrow t$]. M acts on the alignment pattern:

$$M(\mathcal{P}(\Sigma)) \;:=\; \big\{M(e) :\; e \in \mathcal{P}(\Sigma)\big\}$$

where Σ admits such an automorphism (witnesses: path and cycle reflections — machine, `checks/verify_F33_dynamic_phase.py` §B).

Theorem 2.92 (EM.23 — Parity classification). Thm. Under Def-EM.21 and Def-EM.22, for a subnetwork Σ with mirror M: (1) Parity-preserving. Σ is parity-even $\iff$ $M(\mathcal{P}(\Sigma)) = \mathcal{P}(\Sigma)$: the alignment pattern is invariant under the mirror. (2) Parity-reversing. Σ is parity-odd $\iff$ $M(\mathcal{P}(\Sigma)) = E_{\mathcal{R}}(\Sigma) \setminus \mathcal{P}(\Sigma)$: the alignment pattern inverts under the mirror. (3) Distinguishability. Parity-even and parity-odd subnetworks are distinguishable $\iff$ $\mathcal{P}(\Sigma) \neq E_{\mathcal{R}}(\Sigma) \setminus \mathcal{P}(\Sigma)$ — the alignment pattern is not self-complementary — which holds exactly when $E_{\mathcal{R}}(\Sigma) \neq \varnothing$: at least one real edge ("there is something" — CT2.6(i) as the definitional precondition it is, D2.11). (4) Degrees of freedom. The number of distinct alignment patterns on Σ is $2^{|E_{\mathcal{R}}(\Sigma)|}$: each real edge carries an independent binary phase [CT0.9, EM.20(5)]. Each pattern is a distinct parity configuration. The uniform assignments ($\mathcal{P} = \varnothing$ and $\mathcal{P} = E_{\mathcal{R}}$) are two points of this space; any single fixed pattern occupies one point.

Req. Def-EM.20, Def-EM.21, Def-EM.22; A4; D2.11, D2.13; CT0.9 (Part 1); EM.20; independent-register enumeration, orbit counting [math; machine: `checks/verify_F33_dynamic_phase.py` §B].

Pf.

$$(1),(2) :\ \text{Def-EM.22} :\ M \text{ permutes } E_{\mathcal{R}}(\Sigma) \implies M(\mathcal{P}) \subseteq E_{\mathcal{R}}(\Sigma);$$

$$\text{even} := \text{fixed point of the induced action;} \quad \text{odd} := \text{complement image}$$

$$(3) :\ e \in E_{\mathcal{R}}(\Sigma) \neq \varnothing :\ \ e \in \mathcal{P} \implies e \notin E_{\mathcal{R}} \setminus \mathcal{P} \implies \mathcal{P} \neq E_{\mathcal{R}} \setminus \mathcal{P};$$

$$E_{\mathcal{R}} = \varnothing :\ \mathcal{P} = \varnothing = E_{\mathcal{R}} \setminus \mathcal{P}$$

$$\text{even}\wedge\text{odd jointly} \iff M(\mathcal{P}) = \mathcal{P} = E_{\mathcal{R}} \setminus \mathcal{P} \iff E_{\mathcal{R}} = \varnothing \implies \text{distinguishable} \iff E_{\mathcal{R}}(\Sigma) \neq \varnothing$$

$$(4) :\ \text{CT0.9/EM.20(5)} :\ \text{each } e$$

$$\in E_{\mathcal{R}}(\Sigma) \text{ carries an independent } \alpha_e \in \{0,\pi\} \implies \#\{\mathcal{P}(\Sigma)\}$$

$$= 2^{|E_{\mathcal{R}}(\Sigma)|} \text{ [power-set count]}$$

$$\text{uniform assignments : } \alpha \equiv 0 \iff \mathcal{P} = \varnothing; \quad \alpha \equiv \pi \iff \mathcal{P} = E_{\mathcal{R}} : \text{ two of the } 2^{|E_{\mathcal{R}}|}. \blacksquare$$

Phys. Parity is read off the ledger, not added to it: mirror-invariance of the antialigned set classifies every subnetwork even, odd, or unclassified, and the classification has teeth exactly when there is at least one real edge. The dynamic phase gives the pattern space its full $2^{|E_{\mathcal{R}}|}$ cardinality — uniform configurations are two points of it.

Theorem 2.93 (EM.24 — The CPT bridge). Thm. On the configuration space $\mathfrak{C}(\mathcal{R})$ [D0.10]: (1) C (charge conjugation). $q_{\text{EM}} = \frac{1}{2}(\sigma_s - \sigma_t) \in \{-1, 0, +1\}$ [D3.4, EM.9]; charge conjugation $C : \sigma_e \mapsto -\sigma_e \; \forall e \in E(C)$ gives $q \mapsto -q$, $f_v \mapsto -f_v$, $\aleph_v$ invariant [D3.3]; the sign-level realization of D2.6's ω on end and core reads [EM.9a(2)]. Derived. (2) P (parity). The alignment pattern $\mathcal{P}(\Sigma)$ [Def-EM.21]; parity reversal $P : \mathcal{P} \mapsto E_{\mathcal{R}} \setminus \mathcal{P}$ — equivalently $\alpha_e \mapsto \pi - \alpha_e$ per edge — the inversion locus of EM.23(2). Derived from the dynamic phase [Def-EM.20, EM.20(1)]. (3) T (time reversal). The Janus reflection $\Theta_{\mathfrak{L}}$ [D2.7]: $w(\Theta_{\mathfrak{L}} C) = w(C)^*$ [CT2.8(1)]. Derived. (4) CPT composition. CPT $:= C \circ P \circ T$: well-defined on $\mathfrak{C}(\mathcal{R})$; each component an involution ($C^2 = P^2 = T^2 = \text{id}$); each component derived from the corpus — no external input. (5) CPT invariance. The exact phase identity and its consequences:

$$w(\text{CPT}\, C) = (-1)^{|E(C)|}\, e^{i\pi\Theta(C)}\, w(C), \qquad \Theta(C) := \sum_{e \in E(C)} \widehat{ds}^{\,2}(e) \in \mathbb{R};$$

$$\big|w(\text{CPT}\, C)\big|^2 = \big|w(C)\big|^2 \qquad \forall C \;\; \text{(unconditional)};$$

$$\text{on-shell } [\widehat{ds}^{\,2}(e) \in \mathbb{Z}\; \forall e] : \;\; w(C) \in \mathbb{R} \;\wedge\; \big(|E(C)| + \Theta(C) \equiv 0 \pmod 2\big) \implies w(\text{CPT}\, C) = w(C)^*.$$

Req. A8; D0.5, D0.10; D1.6, D2.6, D2.7; D3.3, D3.4, D3.5, D3.6; CT0.9 (Part 1); CT2.8 (Part 2); Def-EM.20, Def-EM.21; EM.9, EM.9a, EM.20, EM.23; exhaustive small-model enumeration over $\mathbb{Q}(i)$ [math; machine: `checks/verify_F33_dynamic_phase.py` §C].

Pf.

$$(1) : \text{ CT0.9 : } \sigma_e \in \{\pm 1\} \implies -\sigma_e \in \{\pm 1\} : \; C \text{ closed on sign assignments;}$$

$$\text{D3.4 : } \; q \mapsto \tfrac{1}{2}(-\sigma_s + \sigma_t) = -q;$$

$$\text{D3.3 : } \; f_v \mapsto -f_v, \; \aleph_v = |f_v| \text{ invariant}$$

$$\text{D2.6/EM.9a(2) : } \omega \text{ maps charge} \mapsto -\text{charge on the end and core reads : } C = \text{sign-level realization}$$

$$(2):\ \text{EM.20}(1):\ \alpha_e \in \{0,\pi\} \implies \pi - \alpha_e \in \{0,\pi\}:\ P \text{ closed on patterns;}$$

$$\text{EM.23}(2):\ \text{the odd locus is } M(\mathcal{P}) = E_{\mathcal{R}} \setminus \mathcal{P}$$

$$(3):\ \text{D2.7}:\ \Theta_{\mathfrak{L}} \text{ reverses the ledger order, } J^+ \leftrightarrow J^-,\ G(e) \mapsto G(e)^*;$$

$$\text{CT2.8}(1):\ w(\Theta_{\mathfrak{L}} C) = w(C)^*;$$

$$\Theta_{\mathfrak{L}}^2 = \text{id [double reversal]}$$

$$(4):\ (1),(2),(3) \text{ map } \mathfrak{C}(\mathcal{R}) \to \mathfrak{C}(\mathcal{R}) \text{ [D0.10]};$$

$$C^2:\ \sigma \mapsto -\sigma \mapsto \sigma; \quad P^2:\ \alpha \mapsto \pi - \alpha \mapsto \alpha; \quad T^2:\ \text{id } [(3)]$$

$$(5):\ w(C) = \mathcal{J}(C) \prod_e \sigma_e\, e^{i\alpha_e \widehat{ds}^{\,2}(e)} \cdot (-1)^{\aleph(\sigma,C)} =: \mathcal{J}\, S\, \Gamma\, A \quad [\text{A8, D3.5, D3.6}]$$

$$T:\ \Gamma \mapsto \bar{\Gamma} \text{ (per-edge conjugation)},$$

$$\mathcal{J},$$

$$S,$$

$$A \in \mathbb{R} \text{ [D2.7, D3.6; } \aleph \in \mathbb{Z}]$$

$$\implies w(\Theta_{\mathfrak{L}} C) \qquad = w(C)^*$$

$$P \circ T:\ \alpha_e$$

$$\mapsto -\alpha_e \mapsto \pi + \alpha_e \ (\text{mod } 2\pi):\ \ e^{-i\alpha_e \widehat{ds}^{\,2}}$$

$$\mapsto e^{i(\pi+\alpha_e)\widehat{ds}^{\,2}}$$

$$= e^{i\pi \widehat{ds}^{\,2}}$$

$$e^{i\alpha_e \widehat{ds}^{\,2}} \implies \bar{\Gamma}$$

$$\mapsto e^{i\pi\Theta}$$

$$\Gamma$$

$$\implies w(PT\,C)$$

$$= e^{i\pi\Theta} \qquad w(C)$$

$C:\ S \mapsto (-1)^{|E|}S;\quad A$ invariant $[\aleph_v = |-f_v|];\quad \mathcal{J}$ invariant $\implies w(\text{CPT}\,C) = (-1)^{|E|}\,e^{i\pi\Theta}\,w(C)$

$$\Theta \in \mathbb{R}\ [\text{D1.6}:\ ds^2 \text{ real}] \implies \left|(-1)^{|E|}e^{i\pi\Theta}\right| = 1 \implies |w(\text{CPT}\,C)|^2 = |w(C)|^2$$

$$\text{on-shell}:\ \widehat{ds}^{\,2} \in \mathbb{Z} \implies e^{i\alpha_e \widehat{ds}^{\,2}} = (\pm 1)^{\widehat{ds}^{\,2}} \in \{\pm 1\} \implies \Gamma \in \{\pm 1\} \implies w \in \mathbb{R}:\ w^* = w;$$

$$|E| + \Theta \equiv 0\ (2) \implies (-1)^{|E|}e^{i\pi\Theta} = 1$$

$$\implies w(\text{CPT}\,C) = w(C) = w(C)^*.\ \blacksquare$$

Phys. Charge conjugation, parity, and time reversal each exist as derived operations on the ledger's configuration space: sign flip on every edge, complement of the antialigned set, Janus reflection of the record. Their composition leaves every configuration probability $|w|^2$ invariant — the corpus's own CPT theorem — with the conjugate-weight identity exact on the on-shell even locus.

Rmk (clause-(5) scope). $|w|^2$-invariance is unconditional; $w(\text{CPT}\,C) = w(C)^*$ holds on the stated on-shell even locus.

2.24 §E.VII — Spectrum and Completion

Theorem 2.94 (EM.22 — The electromagnetic spectrum). Thm.

object	type	mass/charge	data
photon	trivial chain $\subseteq V_2$	$m^2 = 0;\ q_{\text{EM}} = 0$	$N_{\text{pol}} = 2$
field excitation	transition vertex	$\aleph_v = 1$	$f_v = \pm 1$
charged chain end	chain terminus	$q_{\text{EM}} \in \{+1, -1\}$	binary sign

$\neg\exists$ massive abelian gauge boson; list exhaustive : dim(abelian block) $= 1$ $[EM.3]$.

Req. D3.4, D3.7; EM.3, EM.5, EM.6, EM.8, EM.9, EM.10, EM.12; EW.22, EW.28 (Part 2).

Pf.

$$\text{photon : EM.5 (trivial } V_2);\ \ \text{EM.6 } (m^2=0);\ \ \text{EM.8 (2 pol)}$$

$$q_\gamma = 0:\ \ \text{trivial} \implies \neg\text{transitions } [EM.12] \implies \sigma_{\leftarrow} = \sigma_{\rightarrow}\ \forall v \implies \sigma_{s(c)} = \sigma_{t(c)} \implies q = 0\ [\text{D3.4}]$$

$$\text{excitations : EM.12 } (\aleph_v = 1,\ f_v = \pm 1); \qquad \text{ends : EM.9, EM.10}$$

$$\neg\exists \text{ massive abelian : EW.22 : } \ker M^2$$

$$= \text{1-dim abelian direction } A;$$

$$\text{massive neutral}$$

$$= Z \in \text{weak part } [EW.28]$$

$$\text{exhaustive : 1 generator } [EM.3] \implies \{\text{transit mode} = \gamma,\ \text{curvature data} = \aleph_v,\ \text{sources} = \text{ends}\}.\blacksquare$$

Phys. The complete electromagnetic bestiary: one massless neutral two-polarization transit mode, sign-field vertex excitations, and quantized chain-end charges. The photon's own charge vanishes identically — a null chain carries no sign flips.

Theorem 2.95 (EM.26 — Cross-checks). Thm. Four agreements between derivations with pairwise-disjoint inputs:

$$(1)\ \{-1,0,+1\}:\ \ \text{EM.9 } [\Leftarrow \text{CT0.9}] \ =\ \text{EW.36 } [\Leftarrow \text{EW.32}];$$

$$(2)\ m_\gamma^2 = 0:\ \ \text{EM.6(ii) } [\Leftarrow \text{D1.7, EW.18}] \ =\ \text{EW.22 } [\Leftarrow \text{EW.33}];$$

$$(3)\ \alpha_{\text{EM}}^{(3)} = \alpha_{\text{EM}}^{(2)}:\ \ \text{EM.18 step 2 } [\Leftarrow \text{CT0.14}] \ =\ \text{EM.18 step 3 } [\Leftarrow \text{CT3.7}];$$

$$(4)\ g^{\mu\nu}T_{\mu\nu} = 0:\ \ \text{EM.15 (explicit variation)} \ =\ \text{EM.6 (massless transit} \implies \text{traceless stress)}.$$

Req. CT0.9, CT0.14 (Part 1); CT3.7 (F, corroboration); EM.6, EM.9, EM.15, EM.18; EW.18, EW.22, EW.32, EW.33 (Part 2).

Pf.

$$\text{input pairs : } (1)\ \text{CT0.9 vs EW.32;}$$

$$(2)\ \{\text{D1.7, EW.18}\}\ \text{vs EW.33;}$$

(3) CT0.14 vs CT3.7;

(4) variation vs pole location

shared intermediate results $= \varnothing$ in each pair $\implies$ agreements $=$ genuine consistency tests;

$\forall 4:$ hold [cited tags].■

Phys. The sector cross-validates at four joints against the combinatorics of B, the resolvent apparatus of C, and the seed calculus of D — each check by two routes sharing no intermediate result.

Theorem 2.96 (EM.25 — What the corpus does and does not fix about the abelian sector). Thm. Fixed with zero freedom:

(1) $q \in \{-1, 0, +1\}$ [EM.9, EM.10, EW.36]; (2) $m_\gamma^2 = 0$ [EM.6]; (3) $N_{\text{pol}} = 2$ [EM.8];

(4) $\alpha_e \in \{0, \pi\}$ per real edge [EM.20];

(5) $\alpha_{\text{EM}}^{(n)} = \alpha_{\text{EM}}^{(2)}$ $\forall n$ (block read runs at no accessible rung) [EM.18, EM.19];

(6) $e = g \sin\theta_W = g' \cos\theta_W$ [EW.35].

¬Fixed:

(7) the numerical value of e: $\neg\exists$ theorem asserting a value of e;

$$e$$

$$= e(g, g'),$$

(g, g') free [P4] $\implies$ family of valid configurations with distinct e,

all satisfying (1)–(6);

determining e requires one further datum (θ_W or g)

$=$ calibration step [P5];

then $(6) \implies$ rest

$=$ predictions;

(8) the per-edge values α_e : registration outcomes governed by A4/A5/A6 [EM.20$(2), (5)$],

¬parameters;

$\alpha_{\mathrm{EM}}^{(2)}$ as a number : operator read [EM.17], $\neg\exists$ theorem fixing its value.

Separation and closure:

(9) $\neg\big(\alpha$-data determines $e\big)$;

$\neg\big(e$ determines α-data$\big)$;

$\neg\exists$ corpus equation relating them $[g, g'$ appear nowhere in EM.16–EM.21$]$;

(10) $\forall$ statements of E : $\vdash$ A1–A9 $\cup$ C1–C12 $\cup$ D0.1–D4.3 $\cup$ {B, C, D theorems};

$\neg\exists$ external EM law, parameter, or measured value.

Req. P4, P5; EM.6, EM.8, EM.9, EM.10, EM.16, EM.17, EM.18, EM.19, EM.20, EM.21, EM.23, EM.24; EW.24, EW.35, EW.36 (Part 2); citation closure [machine: `checks/verify_corpus_graph.py`].

Pf.

(1)–(6) : cited proofs use only A $\cup$ derived results; $\neg\exists$ parameter introduced;

(4) : EM.20(1);

(5) : EM.18, EM.19

(7) : P4 : (g, g')

$\in (\mathbb{R}_{>0})^2$ unconstrained $\implies$ $\forall(g, g')$: configuration with e

$= gg'/\sqrt{g^2 + g'^2}$ satisfying (1)–(6);

$$\tan\theta_W = g'/g \text{ [EW.24]}$$

$$\Longrightarrow \neg\exists \text{ function of the axioms alone} \mapsto \text{unique } e; \qquad \text{P5 : one datum } + (6) \Longrightarrow \text{predictions}$$

$$(8) : \text{ EM.20}(2), (5) : \ \alpha_e = \text{A4 branch index at the seal [Def-EM.20]};$$

$$\text{EM.17} : \ \alpha_{\text{EM}}^{(n)} = \text{eigenvalue of } \mathrm{F}_M^{(n)}\big|_{\mathcal{K}^{(\text{EM})}} : \text{ operator datum, } \neg\text{a fixed pure number}$$

$$(9) : \text{ EM.16–EM.21 derivations contain } \neg\{g, g'\};$$

$$\text{EW.35} : \ e = gg'/\sqrt{g^2 + g'^2};$$

$$(g, g') \mapsto (g_2, g_2') : \ e \mapsto e_2 \neq e \ \wedge \ \alpha\text{-data unchanged} \Longrightarrow \text{independent variation}$$

$$(10) : \text{ dependency trace : activation} \Leftarrow \text{CT0.13} + \text{D3.3 [EM.1–EM.4]};$$

$$\gamma : \text{ trivial } V_2 \text{ chain},$$

$$m$$

$$= 0,$$

$$2 \text{ pol},$$

$$\mathsf{E}_{\text{tr}} \text{ only [EM.5–EM.8]}$$

$$\text{charge : end-difference},$$

$$\mathbb{Z}\text{-quantized by the binary phase},$$

$$Q_{\text{EM}}$$

$$= 0 \text{ by seal pairing [EM.9–EM.11]};$$

$$f_v;\ \text{sign structures independent};\ \text{gauge} = \text{offset redundancy [EM.12–EM.14]}$$

$$T_{\mu\nu} = F\text{-bilinear},\ \text{trace } 0\ \text{[EM.15]};$$

$$\alpha_e:\ \text{per-edge registration read},\ \text{alphabet } \{0,\pi\},\ \mathfrak{q}^0\ \text{[EM.16–EM.21]}$$

$$\text{parity}:\ \text{pattern data},\ 2^{|E_{\mathcal{R}}|}\ \text{classes [EM.23]};$$

$$\text{CPT}:\ \text{derived components},\ |w|^2\text{-invariance [EM.24]};$$

$$\text{spectrum exhaustive [EM.22]};\quad 4\ \text{cross-checks [EM.26]}.\ \blacksquare$$

Phys. The corpus states exactly which abelian quantities are derived — alphabet, masslessness, polarizations, the per-edge phase alphabet, block-read stability, and the identity $e = g\sin\theta_W$ — and exactly which data stay open: the numerical coupling (configuration data), the per-edge phase values (registration outcomes), and the block-scale number (operator read). It makes no claim to derive a fine-structure constant, and no equation ties phase data to the coupling: an honesty clause that is itself a theorem about the axiom system. Closure audit: every electromagnetic statement traces to A1–A9.

2.25 Closure exports of Part 2 §E (consumed by Part 2 §F–§G, Parts 3–4)

EM.2 $Y_{\text{asp}} = \mathcal{P}^{(\text{EM})},\ \mathfrak{T} = 1;\ Y_{\text{car}} = \frac{1}{2}\sigma_0$	EM.5 $\gamma_\gamma,\ \gamma_g \in$ Trivial (V_2/V_3)	EM.6 $m_\gamma^2 = 0$ ($\times 2$ routes)
EM.8 $N_{\text{pol}} = 2$	EM.9/10 $q_{\text{EM}} \in \{-1,0,+1\} \subset \mathbb{Z}$; EM.9b : $\sum_i f_{v_i} = \frac{1}{2}(\sigma_{e_k} - \sigma_{e_1})$	EM.11 $Q_{\text{EM}} = 0$
EM.12/13 $f_v;\ (-1)^{\aleph}$ dynamics	EM.14 $A_\mu \sim A_\mu + \nabla_\mu\chi$	EM.15 $T_{\mu\nu}^{(\text{EM})}$; traceless
EM.16 $G(e) = \sigma_e e^{i\alpha_e \widehat{ds}^2}$	EM.17 $\alpha_{\text{EM}}^{(n)} = \mathfrak{T}(Y_{\text{asp}}\mathrm{F}_M Y_{\text{asp}})$	EM.18/19 $\alpha_{\text{EM}}^{(n)} = \alpha_{\text{EM}}^{(2)}\ \forall n$
Def-EM.20/EM.20 $\alpha_e = \alpha_{pq} \in \{0,\pi\}$: A4 branch, per edge	EM.21 $[\alpha_e] = \mathfrak{q}^0$	Def-EM.21/22 $\mathcal{P}(\Sigma);\ M(\mathcal{P})$
EM.23 parity classes; $2^{\lvert E_{\mathcal{R}}\rvert}$ patterns	EM.24 CPT $= C \circ P \circ T;\ \lvert w(\text{CPT}\,C)\rvert^2 = \lvert w(C)\rvert^2$	EM.22 spectrum $\{\gamma,\ \aleph_v,\ \text{ends}\}$
EM.25 $e =$ P4 data; self-containment	EM.26 4 cross-checks	

$\blacksquare$

2.26 §F — Colour Sector

Dep.: Part 1; Part 2 §C. This section may be read before §D, §E: its two borrowed results — colour masslessness (EW.26) and coefficient stability (EM.18) — are restated with full arguments at use. Notation: $\mathcal{H}_{\text{slot}}^{(n)}$ [D3.1] $\neq \mathfrak{H}(\mathbb{C}^4)$ [A1] — slot space and pair operator space are never identified; T_a, f_{abc}: colour generators and structure constants in the Hermitian convention [C2]. Machine: `checks/verify_F.py` — exact arithmetic in $\mathbb{Q}(\sqrt{3}, i)$; CT3.22a/CT3.17a: `checks/verify_merge_tierA.py`; CT3.10a: `checks/verify_promotion_F_colour.py`.

2.27 §F.I — Slot Structure

Theorem 2.97 (CT3.20 — The number of colours is forced). Thm.

$$\text{admissible}(v) \iff n(v) \le 3; \qquad \max_{\text{admissible}} \mathcal{H}_{\text{slot}}^{(n)} = \mathcal{H}_{\text{slot}}^{(3)} \cong \mathbb{C}^3;$$

$$N_{\text{colour}} = 3 = \log_2 C_{\text{cap}} - 1 = \log_2 16 - 1 \qquad (\text{derived from A2} + \text{CT0.9};\ \neg\text{chosen}).$$

Req. A2; D3.1; CT0.5, CT0.6, CT0.9, CT0.11 (Part 1).
Pf.

$$\text{D3.1}: \text{ rung-}n \text{ vertex carries } n \text{ slots}$$

$$\text{CT0.5/CT0.6}: D_{\text{tick}}(n) = 2^{n+1},\ C_{\text{cap}} = 16; \qquad \text{CT0.11}: \text{admissible} \iff 2^{n+1} \le 16 \iff n \le 3$$

$$\implies \mathcal{H}_{\text{slot}}^{(n)} \in \{\mathbb{C}^0, \mathbb{C}^1, \mathbb{C}^2, \mathbb{C}^3\}; \qquad \max = \mathbb{C}^3$$

$n \le \log_2 C_{\text{cap}} - 1 = 3$: inputs = A2 ($1\mathfrak{q}$ per channel) + CT0.9 ($\Phi = \{0, \pi\}$); $\neg\exists$ further input. ■

Phys. Three is not a choice: a rung-n vertex loads 2^{n+1} registers against a capacity of 16, so three slots are the most any admissible vertex can carry. The number of colours is fixed by the channel quantum and the binary phase alphabet, and by nothing else.

Theorem 2.98 (CT3.1 — Classical trivalent symmetry). Thm. Trivalent vertex with distinguishable slots $\{s_1, s_2, s_3\}$:

$$\text{relabeling symmetry} = \text{Sym}\{s_1, s_2, s_3\} = S_3;$$

$$|S_3| = 3! = 6;$$

$$S_3 = \{e, (12), (13), (23), (123), (132)\}.$$

Req. D3.1; bijection/group facts [math].
Pf.

$$\text{D3.1}: \text{ rung-3 vertex}: \text{ slots} = \{s_1, s_2, s_3\}$$

$$\text{relabeling} = \text{invertible, slot-preserving map} = \text{bijection } \{s_1, s_2, s_3\} \to \{s_1, s_2, s_3\}$$

$$\text{bijections under composition} = S_3; \qquad |S_3| = 3! = 6. \blacksquare$$

Phys. The classical symmetry of a trivalent vertex is the full permutation group of its three slots — six relabelings, no more.

Theorem 2.99 (CT3.2 — Quantum slot space). Thm.

$$|\psi\rangle = \alpha_1|1\rangle + \alpha_2|2\rangle + \alpha_3|3\rangle,\ \ \alpha_i \in \mathbb{C}; \qquad \langle\psi|\phi\rangle = \textstyle\sum_i \alpha_i^* \beta_i; \qquad \textstyle\sum_i |\alpha_i|^2 = 1;$$

$$\mathcal{H}^{(3)}_{\text{slot}} = \text{span}_{\mathbb{C}}\{|1\rangle, |2\rangle, |3\rangle\} \cong \mathbb{C}^3.$$

Req. A1; D3.1.

Pf.

A1 : carrier spaces = complex Hilbert spaces $\implies$ slot excitations admit complex superpositions

$$\text{span of 3 independent basis states over } \mathbb{C} = \mathbb{C}^3 \quad [\text{D3.1}]. \ \blacksquare$$

Phys. Superposition promotes the discrete slot label to a complex three-dimensional carrier: colour becomes a quantum number with amplitudes.

Theorem 2.100 (CT3.3 — Classical permutations embed unitarily). Thm. $P_\pi|i\rangle := |\pi(i)\rangle$, $(P_\pi)^i{}_j = \delta^i_{\pi(j)}$. Then

$$P_\pi^\dagger P_\pi = I_3 \implies P_\pi \in U(3);$$

$$P_\pi P_\rho = P_{\pi\circ\rho} \ \text{(homomorphism)};$$

$$P_\pi = I_3 \iff \pi = e \ \text{(faithful)};$$

$$S_3 \hookrightarrow U(3).$$

Req. CT3.1, CT3.2; unitarity criterion [math].

Pf.

$$P_\pi : \text{ ON basis} \mapsto \text{ON basis} \implies \langle P_\pi\psi, P_\pi\phi\rangle = \langle\psi,\phi\rangle \implies P_\pi^\dagger P_\pi = I_3 \implies P_\pi \in U(3)$$

$$P_\pi P_\rho|i\rangle = P_\pi|\rho(i)\rangle = |\pi(\rho(i))\rangle = P_{\pi\circ\rho}|i\rangle \implies \text{homomorphism}$$

$$P_\pi = I_3 \implies \forall i :\ \pi(i) = i \implies \pi = e \implies \text{injective}. \ \blacksquare$$

Phys. The classical relabeling symmetry sits inside the quantum one as the six permutation matrices — a faithful discrete subgroup of $U(3)$.

Theorem 2.101 (CT3.4 — Probability preservation gives $U(3)$). Thm.

$$\{L \text{ linear}: \ \langle L\psi, L\phi\rangle = \langle \psi, \phi\rangle \ \ \forall \psi, \phi\} = \{L: \ L^\dagger L = I_3\} = U(3); \qquad S_3 \subseteq U(3) \text{ [CT3.3]}.$$

Req. CT3.2, CT3.3; polarization/inner-product matrix form [math].
Pf.

$$\text{probabilities} = \text{moduli of inner products} \implies \text{preservation} \iff \langle L\psi, L\phi\rangle = \langle \psi, \phi\rangle \ \forall \psi, \phi$$

[polarization]

$$\psi^\dagger L^\dagger L \phi = \psi^\dagger \phi \ \ \forall \psi, \phi \iff L^\dagger L = I_3 \quad [\psi, \phi \text{ over a basis}]$$

$$\{L: \ L^\dagger L = I_3\} =: U(3) \ \ (\text{group}); \qquad S_3 \hookrightarrow U(3) \text{ [CT3.3]}. \ \blacksquare$$

Phys. The full quantum symmetry of the slot carrier is exactly $U(3)$: the linear maps preserving every probability. The classical S_3 is its discrete relic.

Theorem 2.102 (CT3.5 — The algebra $\mathfrak{u}(3)$ and its decomposition). Thm. [C2]

$$\mathfrak{u}(3) = \{X \in M_3(\mathbb{C}): \ X^\dagger = X\}; \qquad \mathfrak{u}(3) = \mathfrak{su}(3) \oplus \mathfrak{u}(1);$$

$$\mathfrak{su}(3) = \{X \in \mathfrak{u}(3): \ \mathfrak{T}(X) = 0\}; \qquad \mathfrak{u}(1) = \text{span}_{\mathbb{R}}\{I_3\}; \qquad [\mathfrak{su}(3), \mathfrak{u}(1)] = 0 \ \ (\text{commuting ideals});$$

$$\forall X \ \exists! \ (X_0, c): \ X = X_0 + cI_3, \ \ \mathfrak{T}(X_0) = 0, \ \ c = \tfrac{\mathfrak{T}(X)}{3}.$$

Req. C2; CT3.4; trace cyclicity [linear algebra].
Pf.

$$X^\dagger = X \wedge Y^\dagger = Y \implies [X,Y]^\dagger = Y^\dagger X^\dagger - X^\dagger Y^\dagger = -[X,Y] \implies -i[X,Y] \text{ Hermitian}: \text{ closure [C2]}$$

$$\mathfrak{T}[X,Y] = \mathfrak{T}(XY) - \mathfrak{T}(YX) = 0 \implies \text{traceless Hermitian} = \text{subalgebra}$$

$$X_0 := X - \tfrac{\mathfrak{T}(X)}{3} I_3: \quad \mathfrak{T}(X_0) = \mathfrak{T}(X) - \tfrac{\mathfrak{T}(X)}{3} \cdot 3 = 0 \implies X_0 \in \mathfrak{su}(3)$$

$$\text{uniqueness}: \ \ X = S + cI_3, \ \mathfrak{T}(S) = 0 \implies \mathfrak{T}(X) = 3c \implies c = \tfrac{\mathfrak{T}(X)}{3}$$

$$I_3 \in Z \implies [\mathfrak{u}(1), \mathfrak{u}(3)] = 0 \implies \text{both summands ideals};$$

$$cI_3 \text{ traceless} \iff c = 0 \implies \mathfrak{su}(3) \cap \mathfrak{u}(1) = \{0\}. \ \blacksquare$$

Phys. The slot symmetry algebra splits canonically into traceless plus centre — the template from which colour and electromagnetism are cut by the subset principle.

2.28 §F.II — The Rung-3 Gauge Algebra

Theorem 2.103 (CT3.6 — Rung-3 active set). Thm. $\mathcal{A}(3) = \mathcal{A}(2) \cup \{\text{strong}\}$; $\mathrm{F}_M^{(3)} = \mathcal{P}_M^{(3)}\mathrm{F}\,\mathcal{P}_M^{(3)} = \mathrm{F}\big|_{\mathcal{K}_3}$.

Req. CT0.13 (Part 1); CT2.2, CT2.3, CT2.4 (Part 2).

Pf. Active set: [CT0.13 step n=3]; projector: [CT2.2]; restriction: [CT2.3, CT2.4]. ■

Phys. At rung 3 the mask admits one new block — colour — on top of the rung-2 content.

Theorem 2.104 (CT3.7 — The strictly new rung-3 algebra is $\mathfrak{su}(3)$). Thm. Strictly new gauge structure at rung 3 [D3.2]:

$$\mathfrak{u}(3)/\mathfrak{u}(1) \cong \mathfrak{su}(3);$$

$$\pi : \mathfrak{u}(3) \to \mathfrak{su}(3),\ \ \pi(X) = X - \tfrac{\mathfrak{T}(X)}{3} I_3 :\ \text{Lie homomorphism},\ \ \ker\pi = \mathrm{span}_{\mathbb{R}}\{I_3\},\ \ \text{surjective};$$

$$\mathfrak{T}(T_a) = 0 \quad \forall a\ \ (\neg\exists \text{ central component}).$$

Req. D3.2, D2.12; CT0.13 (Part 1); CT3.4, CT3.5; EM.2, EW.8 (D/E, restated: centre spent at rung 2).

Pf.

$$\text{CT3.4 :}\ \text{rung-3 slot symmetry} = U(3) \implies \text{algebra} = \mathfrak{u}(3); \qquad \text{CT3.5 :}\ \mathfrak{u}(3) = \mathfrak{su}(3) \oplus \mathfrak{u}(1)_{\langle I_3\rangle}$$

$$\iota_2\big(Z(\mathfrak{u}(2))\big) = F_{\text{EM}} \text{ realized at rung 2 [D2.12, CT0.13, EM.2 restated :}\ Y_{\text{asp}} = \mathcal{P}^{(\text{EM})}]$$

D3.2 [D2.12 : "realized" = abstract class in $\iota_m(\mathfrak{u}(m))$, $m < n$] : $\quad \iota_3\big(Z(\mathfrak{u}(3))\big) \cong \mathfrak{u}(1) \cong \iota_2\big(Z(\mathfrak{u}(2))\big) \implies \neg$strictly new

$$\implies \text{strictly new} = \mathfrak{u}(3)/\mathfrak{u}(1)$$

$$\pi \text{ well-defined Lie homomorphism } (I_3 \text{ central});$$

$$\ker\pi = \{X :\ X = \tfrac{\mathfrak{T}X}{3} I_3\} = \mathrm{span}_{\mathbb{R}}\{I_3\}; \quad \text{surjective [CT3.5 split]}$$

$$\mathfrak{su}(3) = \{\mathfrak{T} = 0\} \implies \mathfrak{T}(T_a) = 0\ \ \forall a. \blacksquare$$

Phys. The new force at rung 3 is exactly the traceless slot algebra $\mathfrak{su}(3)$: eight generators, no central component — the ninth direction of $\mathfrak{u}(3)$ was already spent on electromagnetism at rung 2 by the subset principle.

Theorem 2.105 (CT3.8 — Dimension of the colour algebra). Thm. $\dim_{\mathbb{R}} \mathfrak{u}(3) = 3 + 3 \cdot 2 = 9$; $\mathfrak{T}(X) = 0 :$ 1 nontrivial real constraint ($\mathfrak{T}(I_3) = 3 \neq 0$); $\dim_{\mathbb{R}} \mathfrak{su}(3) = 9-1 = 8 = 3^2-1$.

Req. CT3.5; Hermitian-matrix parameter count [linear algebra].

Pf.

$$X^\dagger = X : \ 3 \text{ real diagonal } + \ 3 \text{ complex off-diagonal (2 real each)} \implies \dim_{\mathbb{R}} \mathfrak{u}(3) = 9$$

$$\mathfrak{T} : \ \mathfrak{u}(3) \to \mathbb{R} \text{ linear}, \ \mathfrak{T}(I_3) = 3 \neq 0 \implies \text{surjective} \implies \dim \ker = 9 - 1 = 8. \ \blacksquare$$

Phys. The colour algebra is eight-dimensional — the gluon count of CT3.13 and the colour-block dimension of CT0.14 ($\dim \mathcal{K}_3 = 25 = 17 + 8$).

Theorem 2.106 (CT3.9 — Explicit generator basis). Thm. With the Gell-Mann matrices

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \ \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \ \lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \ \lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

$$\lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \ \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \ \lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \ \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix},$$

and $T_a := \frac{1}{2}\lambda_a$:

$$T_a^\dagger = T_a;$$

$$\mathfrak{T}(T_a) = 0;$$

$$\{T_a\}_{a=1}^8 : \ \mathbb{R}\text{-basis of } \mathfrak{su}(3);$$

$$[T_a, T_b] = i \, f_{abc} \, T_c, \ \ f_{abc} \in \mathbb{R};$$

$$\mathfrak{T}(T_a T_b) = \tfrac{1}{2} \, \delta_{ab}.$$

Req. C2; CT3.5, CT3.8; direct computation [machine: `checks/verify_F.py`, exact in $\mathbb{Q}(\sqrt{3}, i)$].

Pf.

$$\lambda_a^\dagger = \lambda_a \ \ \forall a \text{ [inspection]}; \qquad \mathfrak{T}\lambda_a = 0 : \ a \leq 7 \text{ visually}; \quad \mathfrak{T}\lambda_8 = \tfrac{1}{\sqrt{3}}(1 + 1 - 2) = 0$$

$$\text{supports} : \ \lambda_{1,2} \leftrightarrow (1,2)\text{-entry } \{\text{Re}, \text{Im}\};$$

$$\lambda_{4,5} \leftrightarrow (1,3);$$

$$\lambda_{6,7} \leftrightarrow (2,3);$$

$$\lambda_3,$$

$$\lambda_8 \leftrightarrow \text{traceless diagonal } (\neg\text{proportional})$$

$$\implies 8 \text{ independent}; \qquad \dim \mathfrak{su}(3) = 8 \text{ [CT3.8]} \implies \text{basis}$$

CT3.5 : $-i[T_a, T_b]$ Hermitian$\wedge$traceless $\implies \in \mathfrak{su}(3) = \text{span}_{\mathbb{R}}\{T_c\} \implies [T_a, T_b] = if_{abc}T_c,\ f_{abc} \in \mathbb{R}$

$$\mathfrak{T}(\lambda_a\lambda_b)$$

$$= 2\delta_{ab} \text{ [direct computation, machine-verified]} \implies \mathfrak{T}(T_aT_b)$$

$$= \tfrac{1}{4} \cdot 2\delta_{ab}$$

$$= \tfrac{1}{2}\delta_{ab}$$

[= uniform normalization of EW.32]. ■

Phys. The explicit normalized colour octet: same uniform trace normalization $\mathfrak{T}(\mathcal{G}_A\mathcal{G}_B) = \frac{1}{2}\delta_{AB}$ as the electroweak generators — one gauge convention across the corpus.

Theorem 2.107 (CT3.21 — Structure constants). Thm. $f_{abc} := -2i\,\mathfrak{T}\big([T_a, T_b]T_c\big)$ is totally antisymmetric, with independent nonzero values

$$f_{123} = 1; \qquad f_{147} = f_{246} = f_{257} = f_{345} = \tfrac{1}{2}; \qquad f_{156} = f_{367} = -\tfrac{1}{2}; \qquad f_{458} = f_{678} = \frac{\sqrt{3}}{2};$$

$\forall$ other independent components $= 0$.

Req. CT3.9; trace cyclicity [linear algebra]; machine: `checks/verify_F.py` (all 512 components + antisymmetry + all 512 Jacobi triples, exact).

Pf.

$$[T_a, T_b]$$

$$= if_{abd}T_d;$$

$$\mathfrak{T}(T_d T_c)$$

$$= \tfrac{1}{2}\delta_{dc} \text{ [CT3.9]} \implies \mathfrak{T}\big([T_a, T_b]T_c\big)$$

$$= \tfrac{i}{2} f_{abc}$$

$$\implies f_{abc}$$

$$= -2i$$

$$\mathfrak{T}\big([T_a, T_b]T_c\big)$$

$$a \leftrightarrow b: \text{ antisymmetry of } [\cdot,\cdot];$$

$$\mathfrak{T}\big([T_a, T_b]T_c\big) = \mathfrak{T}(T_a T_b T_c) - \mathfrak{T}(T_b T_a T_c) = \mathfrak{T}\big(T_a[T_b, T_c]\big) \implies \text{antisymmetry in } b \leftrightarrow c$$

$$\implies f \text{ totally antisymmetric}$$

$$\text{sample 1}: \quad [\lambda_1, \lambda_2] = 2i\lambda_3 \implies [T_1, T_2] = \tfrac{1}{4} \cdot 2i\lambda_3 = iT_3 \implies f_{123} = 1$$

$$\text{sample 2}: \quad [\lambda_4, \lambda_5]$$

$$= 2i$$

$$\text{diag}(1, 0, -1)$$

$$= i\big(\lambda_3 + \sqrt{3}\,\lambda_8\big)$$

$$\implies [T_4, T_5]$$

$$= \tfrac{i}{2}\big(T_3 + \sqrt{3}\,T_8\big)$$

$$\implies f_{453}$$

$$= \tfrac{1}{2},$$

$$f_{458} \qquad = \tfrac{\sqrt{3}}{2}$$

$$\forall\ (a,b,c):\ \text{complete exact list machine-verified [verify_F];}$$

$$\text{Jacobi}: \sum_{\text{cyclic}} \big[[T_a,T_b],T_c\big] = 0\ \ \forall (a,b,c)\ \text{[machine, exact]}.\ \blacksquare$$

Phys. The colour algebra is fully determined — basis, brackets, normalization, structure constants down to the $\sqrt{3}/2$ fingerprints of λ_8. Zero continuous freedom: the strongest rigidity result of the corpus.

Theorem 2.108 (CT3.10 — The colour aspect has no central component). Thm.

$$F_{\text{strong}} = \textstyle\sum_a f_a T_a \implies \mathfrak{T}(F_{\text{strong}}) = 0;$$

$$Y_{\text{asp}} F_{\text{strong}} Y_{\text{asp}} = 0;$$

$$\alpha_{\text{EM}}^{(3)} = \alpha_{\text{EM}}^{(2)} \quad [= \text{EM.18, restated}].$$

Req. CT0.14 (Part 1); CT3.7; EM.2, EM.17 (E, restated at use).
Pf.

$$\text{CT3.7}: \ F_{\text{strong}} = \textstyle\sum_{a=1}^{8} f_a T_a,\ f_a \in \mathbb{R}; \qquad \mathfrak{T}(T_a) = 0\ \forall a \implies \mathfrak{T}(F_{\text{strong}}) = \sum_a f_a \cdot 0 = 0$$

$$\text{CT0.14}: \ \mathcal{K}^{(\text{strong})} \perp \mathcal{K}^{(\text{EM})};$$

$$\text{EM.2 (restated)}: \ Y_{\text{asp}} = \mathcal{P}^{(\text{EM})} \implies Y_{\text{asp}} F_{\text{strong}} Y_{\text{asp}} = \mathcal{P}^{(\text{EM})} F_{\text{strong}} \mathcal{P}^{(\text{EM})} = 0$$

$$\text{EM.17 (restated)}: \ \alpha_{\text{EM}}^{(n)} = \mathfrak{T}\big(Y_{\text{asp}} \mathrm{F}_M^{(n)} Y_{\text{asp}}\big) / \mathfrak{T}\big(Y_{\text{asp}}\big);$$

$$\text{colour contributes } 0 \implies \alpha_{\text{EM}}^{(3)} = \alpha_{\text{EM}}^{(2)}.\ \blacksquare$$

Phys. Colour adds only traceless structure orthogonal to the rung-2 subspace: it neither carries a central charge nor disturbs the electromagnetic block read α_{EM}.

Corollary 2.4 (CT3.10a — no per-edge abelian strong phase). (1) Coefficient-read obstruction. EM.17's read $\alpha^{(n)} = \mathfrak{T}\big(Y_{\text{asp}} \mathrm{F}_M^{(n)} Y_{\text{asp}}\big) / \mathfrak{T}\big(Y_{\text{asp}}\big)$ yields a single number exactly because the EM block is one-dimensional [CT0.14: $\dim \mathcal{K}^{(\text{EM})} = 1$; EM.17]. The strong block is eight-dimensional [CT0.14, CT3.8: $\dim \mathcal{K}_3 = 25 = 17 + 8$], and

$$F_{\text{strong}} = \textstyle\sum_a f_a T_a = c\, I_3 \quad \iff \quad c = 0\ \wedge\ f_a = 0\ \forall a$$

[CT3.7: $\mathfrak{T}(T_a) = 0 \implies 3c = \mathfrak{T}(F_{\text{strong}}) = 0$; CT3.9: the T_a are linearly independent] — the read returns no scalar on the strong block: $\neg\exists$ a per-edge abelian strong phase coefficient α_{strong} defined by derived structure. (2) Non-canonicity. $\mathfrak{su}(3)$ contains $\mathfrak{u}(1)$ subalgebras [$\text{span}_{\mathbb{R}}\{T_a\}$, CT3.9], but no generator line is fixed by the adjoint action — all maximal tori are conjugate [math] — so no $\mathfrak{u}(1)$ choice is distinguished by derived structure. Explicit witnesses (integer-entry $SU(3)$ elements of the CT3.9 basis [CT3.4; $\lambda_a = 2T_a$; unitary, det $= 1$ — machine]):

$$U_2 = i\lambda_2 + I_3 - \lambda_2^2, \qquad U_5 = i\lambda_5 + I_3 - \lambda_5^2 \qquad [\lambda_2^2 = \text{diag}(1,1,0),\ \ \lambda_5^2 = \text{diag}(1,0,1)]$$

$$\text{Ad}_{U_5}(T_1) = U_5\, T_1\, U_5^\dagger = -\,T_6 \ \notin\ \text{span}_{\mathbb{R}}\{T_1\};$$

$$\forall a \in \{1,\ldots,8\}\ \exists U \in \{U_2, U_5\} :\ \text{Ad}_U(T_a) \notin \text{span}_{\mathbb{R}}\{T_a\}$$

[machine census].

(3) Consequence (pointer). With $\alpha_{\text{EM}}^{(3)} = \alpha_{\text{EM}}^{(2)}$ [CT3.10; EM.18 restated], the per-edge scalar factor $e^{i\alpha_e \widehat{ds}^{\,2}(e)}$ of the transfer weight is entirely electromagnetic [D3.5: $\alpha_e \in \Phi$, the same alphabet at every rung], and CT3.22a(0)'s decomposition — EM factor $\times$ $\{0,1\}$ singlet-sealing admissibility [D2.14; CT3.17, CT3.18 — forward in-document pointers, non-load-bearing] — exhausts the per-edge strong datum: no phase factor remains.

Req (Cor). D2.14, D3.5; CT0.14 (Part 1); CT3.4, CT3.7, CT3.8, CT3.9, CT3.10; EM.17, EM.18 (E, restated at use); torus conjugacy, linear independence [math; machine: `checks/verify_promotion_F_colour.py`].

Pf (Cor).

$$(1) :\ \mathfrak{T}(cI_3) = 3c; \quad \mathfrak{T}(F_{\text{strong}}) = \textstyle\sum_a f_a\, \mathfrak{T}(T_a) = 0\ [\text{CT3.7}] \implies c = 0;$$

$$\textstyle\sum_a f_a T_a = 0 \implies f = 0\ [\text{CT3.9 :\ basis}]$$

$$\dim \mathcal{K}^{(\text{EM})}$$

$$= 1\ \text{ vs }\ \dim \mathcal{K}^{(\text{strong})}$$

$= 8$ [CT0.14, CT3.8] : EM.17's single-coefficient read (single eigenvalue on a 1-dim block) ¬replicable on the strong block

$$(2) :\ \lambda_2^2 = \text{diag}(1,1,0),\ \lambda_5^2 = \text{diag}(1,0,1) \implies U_2, U_5 \text{ real orthogonal, } \det = 1\ [\text{machine}];$$

$$U_5\, T_1\, U_5^\dagger = -T_6; \quad U_5\, T_2\, U_5^\dagger = T_7 \ \text{ etc. [machine census]}$$

$$\text{maximal tori conjugate [math]} \implies \neg\exists \text{ distinguished } \mathfrak{u}(1) \subseteq \mathfrak{su}(3);$$

$$(3) :\ \text{CT3.10 :}\ \alpha^{(3)} = \alpha^{(2)}; \quad \text{D3.5 :}\ G^{(n)}(e) = \sigma_e\, e^{i\alpha_{\text{EM}}^{(n)} \widehat{ds}^{\,2}(e)}\ \blacksquare$$

2.29 §F.III — Colour Objects and Spectrum

Theorem 2.109 (CT3.11 — Colour objects). Thm. Within the rung-3 structure:

$$(1)\ \text{colour} := \text{slot index } i \in \{1,2,3\}\ [\text{D3.1}],\ \ \text{fundamental of } \mathfrak{su}(3);$$

$$(2)\ \text{triplet}:\ q^i \mapsto U^i{}_j q^j\ (U \in SU(3)); \qquad (3)\ \text{antitriplet}:\ r_i \mapsto r_j (U^{-1})^j{}_i;$$

$$(4)\ \text{gluon} := \gamma \in \mathsf{Trivial} \wedge \gamma \cap V_3 \neq \varnothing; \qquad (5)\ \text{photon} := \gamma \in \mathsf{Trivial} \wedge \gamma \subseteq V_2 \qquad [\text{D3.7}];$$

$$(6)\ \text{quark} := \text{bound loop [D1.5 : fermionic] with triplet attachment [D3.8]};$$

$$(7)\ B := \varepsilon_{ijk} q^i q^j q^k\ (\text{baryon}); \qquad M := \delta^i{}_j q^j r_i\ (\text{meson}) \qquad [\text{D3.9}].$$

Req. D1.5, D3.1, D3.7, D3.8, D3.9; CT3.2, CT3.4, CT3.12, CT3.15, CT3.17.
Pf.

$$(1)\text{–}(3):\ \text{CT3.2/CT3.4 : carrier } \mathbb{C}^3,\ \ SU(3) \text{ defining action};\quad \text{dual law} = \text{CT3.17 step 1}$$

$$(4), (5):\ \text{D3.7};\quad \text{distinguished only by vertex type} = \text{CT3.12}$$

$$(6):\ \text{D3.8};\quad \text{bound} \implies \text{fermionic [D1.5]}$$

$$(7):\ \text{D3.9};\quad \text{both contractions invariant : CT3.15, CT3.17.}\ \blacksquare$$

Phys. The colour vocabulary, fixed: every term — colour, triplet, antitriplet, gluon, photon, quark, baryon, meson — names a structure already derived from the axioms.

Theorem 2.110 (CT3.12 — Photon and gluon are distinguished by possibility space). Thm.

$$\gamma_\gamma,\ \gamma_g \in \mathsf{Trivial}:\ \ \widehat{ds}^{\,2}(e) = 0\ \ \forall e,\ \ \Theta = 0,\ \ \mathrm{Hol}_{\text{spiral}} = 1;$$

$$\text{distinction}:\ \ \gamma_\gamma \subseteq V_2 \quad \text{vs} \quad \gamma_g \cap V_3 \neq \varnothing;$$

$$\text{strong} \in \mathcal{A}(3) \setminus \mathcal{A}(2) \implies \gamma_\gamma :\ \neg\text{colour index};\quad \gamma_g :\ \text{slot index possible [D3.1]};$$

$$\text{class, entropy channel (transit, D1.5), mass } (0) \text{ coincide.}$$

Req. D0.8, D1.5, D1.7, D3.1, D3.7; CT0.13 (Part 1); EM.5, EM.6 (Part 2); CT3.14.
Pf.

$$\text{D3.7 : both} \in \mathsf{Trivial}; \qquad \text{D1.7 : } \widehat{ds}^{\,2} \equiv 0 \implies \Theta = 0,\ \text{Hol} = e^{i\alpha\cdot 0} = 1$$

$$\text{D0.8 : } V_2 \cap V_3 = \varnothing; \qquad \text{CT0.13 : strong active only from rung } 3$$

$$\text{no further distinguishing property : class } \mathsf{Trivial};$$

$$\text{channel}$$

$$= \text{transit [D1.5]};$$

$$m^2$$

$$= 0 \text{ [CT3.14, EM.6]. } \blacksquare$$

Phys. Photon and gluon are the same kind of object — a null chain — differing only in where each is allowed to travel. The gluon's colour is a consequence of touching trivalent vertices; the photon's neutrality, of never doing so. (= EM.5 seen from the colour side.)

Theorem 2.111 (CT3.13 — Gluon count). Thm. $N_{\text{gluon}} = \dim \mathfrak{su}(3) = 8$; $\neg\exists$ ninth mode (centre of $\mathfrak{u}(3)$ allocated to F_{EM}).

Req. CT3.7, CT3.8; adjoint-representation dimension [math].

Pf.

$$\text{CT3.7 : colour gauge algebra} = \mathfrak{su}(3);$$

$$\text{gauge mode} = \text{adjoint direction}; \quad \dim \text{adj} = \dim \mathfrak{su}(3) = 8 \text{ [CT3.8]}$$

$$\text{9th direction of } \mathfrak{u}(3) = \text{centre} \to F_{\text{EM}} \text{ [CT3.7 step 3]}; \quad \text{reallocation} = \text{double counting. } \blacksquare$$

Phys. Eight gluons — one per $\mathfrak{su}(3)$ generator. The ninth possible mode is electromagnetism, already counted at rung 2: the subset principle at work.

Theorem 2.112 (CT3.14 — Gluons remain massless). Thm. $m_g^2 = 0$, by two independent routes:

$$\text{(i)}$$

$$T_a \mathsf{v} = 0 \text{ [EW.32]} \implies M_{aB}^2 = M_{ab}^2 = 0 \implies \{T_a\} \subseteq \ker M^2 \implies \text{zero pole displacement}$$

$$[= \text{EW.26, restated}];$$

$$\text{(ii)} \quad \gamma_g \in \mathsf{Trivial} \implies \Theta(\gamma_g)$$

$$= 0 \implies \neg\text{wrapping holonomy} \implies \neg\text{wrapping contribution [EW.18]}.$$

Req. D2.5, D3.7; CT0.14 (Part 1); EW.18, EW.26, EW.32, EW.33 (D, restated at use).

Pf.

$$\text{(i) D2.5}: \; m^2 = \text{pole displacement};$$

$$\text{EW.33 (restated)}: \; \text{rung-3 displacement source} = M^2_{AB} = \mu^2_\star \,\mathrm{Re}\langle \mathcal{G}_A \mathsf{v}, \mathcal{G}_B \mathsf{v}\rangle$$

$$\text{EW.32 (restated)}: \; F_H$$

$$= \text{pair block} \perp F_{\text{strong}}$$

$$= \text{stack block [CT0.14]} \implies \text{seed has } \neg\text{slot index}$$

$$\implies T_a \mathsf{v} \qquad = 0$$

$$\implies M^2_{aB} = \mu^2_\star \,\mathrm{Re}\langle 0, \mathcal{G}_B \mathsf{v}\rangle = 0, \quad M^2_{ab} = 0 \implies \{T_a\}_{a=1}^{8} \subseteq \ker M^2 \implies m_g^2 = 0$$

$$\text{(ii) D3.7}: \; \gamma_g \in \mathsf{Trivial} \implies \Theta = 0;$$

$$\text{EW.18 (restated)}: \; \text{materialization requires } \Phi$$

$$\in 2\pi\mathbb{Z} \text{ from winding} \implies \neg\text{wrapping} \implies \neg\text{contribution.} \blacksquare$$

Phys. Gluons are massless structurally: the seed that generates mass is colour-blind (route i), and null chains cannot wind (route ii). Two independent arguments, agreeing — masslessness of the strong gauge modes is as robust as the photon's.

Theorem 2.113 (CT3.15 — Baryon invariance). Thm. $B = \varepsilon_{ijk} q^i q^j q^k; \qquad U \in SU(3): \quad B \mapsto \varepsilon_{ijk} U^i{}_a U^j{}_b U^k{}_c \, q^a q^b q^c = \det(U)\, \varepsilon_{abc}\, q^a q^b q^c = B.$

Req. D3.9; CT3.2; Leibniz identity $\varepsilon_{ijk} U^i{}_a U^j{}_b U^k{}_c = \det(U) \varepsilon_{abc}$ [math]; machine: `checks/verify_F.py`.

Pf.

$$\text{CT3.2}: \; q^i \mapsto U^i{}_a q^a \implies B \mapsto \varepsilon_{ijk} U^i{}_a U^j{}_b U^k{}_c \, q^a q^b q^c$$

$$\varepsilon_{ijk} U^i{}_a U^j{}_b U^k{}_c = \det(U)\, \varepsilon_{abc} \quad \text{[Leibniz expansion of det; machine-verified on random exact } U\text{]}$$

$$U \in SU(3) \implies \det U = 1 \implies B \mapsto \varepsilon_{abc} q^a q^b q^c = B. \blacksquare$$

Phys. The totally antisymmetric three-index contraction is a colour singlet: the baryon — three triplets sealed by ε — is the minimal neutral object of the strong sector.

Theorem 2.114 (CT3.16 — Infinitesimal baryon invariance). Thm. $U = I_3 + i\theta^a T_a + O(\theta^2)$, $\mathfrak{T}(T_a) = 0 \implies \det U = 1 + O(\theta^2) \implies B \mapsto B + O(\theta^2)$ (first-order variation vanishes).

Req. CT3.7 ($\mathfrak{T} T_a = 0$), CT3.15; det = exp $\mathfrak{T}$ log [math].

Pf.

$$\log U = i\theta^a T_a + O(\theta^2); \qquad \det U = \exp \mathfrak{T}(\log U) = \exp\big(i\theta^a \underbrace{\mathfrak{T}(T_a)}_{=0} + O(\theta^2)\big) = 1 + O(\theta^2)$$

$$\text{CT3.15}: \; B \mapsto \det(U)\, B = B + O(\theta^2). \blacksquare$$

Phys. The singlet property holds at the algebra level: tracelessness of the colour generators is exactly the condition that makes baryon sealing infinitesimally exact.

Theorem 2.115 (CT3.17 — Invariant tensors). Thm. The invariant tensors for sealing colour indices are generated by

$$\delta^i{}_j, \qquad \varepsilon_{ijk}, \qquad \varepsilon^{ijk};$$

$$q^i r_i \mapsto U^i{}_a q^a \, r_b (U^{-1})^b{}_i = q^a r_a;$$

$$\varepsilon_{ijk}: \; 3 \text{ fundamentals [CT3.15]};$$

$$\varepsilon^{ijk}: \; 3 \text{ duals } \big(\det(U^{-1})^{\mathsf{T}} = 1\big);$$

$$\forall \text{ invariants of } (3, \bar{3}): \quad \text{linear combinations of products of } \{\delta, \varepsilon, \bar{\varepsilon}\}.$$

Req. CT3.2, CT3.15; determinant identities [math]; machine: `checks/verify_F.py`.

Pf.

$$\delta: \; U^i{}_a (U^{-1})^b{}_i = \delta^b{}_a \implies q^i r_i \text{ invariant} \quad [\text{machine}: \; U^{-1} U = I]$$

$$\varepsilon_{ijk}: \text{ CT3.15}; \qquad \varepsilon^{ijk}: \text{ same identity at } (U^{-1})^{\mathsf{T}}, \;\; \det(U^{-1})^{\mathsf{T}} = 1$$

$$\text{generation}: \; \delta \text{ seals matched pairs};$$

$$\varepsilon/\bar{\varepsilon} \text{ seal residues mod } 3 \text{ of unmatched indices} \implies \forall \text{ invariant}$$

$$= \text{combination of products. } \blacksquare$$

Phys. Only three tensors can seal colour: pair-annihilation δ, three-quark ε, three-antiquark $\bar{\varepsilon}$. The entire hadronic vocabulary — mesons, baryons, antibaryons — is built from these.

Corollary 2.5 (CT3.17a — sealing partition rule; grammar counts). (1) Partition rule. If a colour-attachment set is sealed — $\mathcal{C}(\Sigma; C) = 1$ [D2.14] — then Colour(Σ) partitions into elementary contraction groups of size 2 (δ-type) and size 3 ($\varepsilon/\bar{\varepsilon}$-type) [CT3.17: the invariant tensors are $\delta^i{}_j$ (2 members), ε_{ijk}, ε^{ijk} (3 members); D2.14: $\pi_C(v) =$ the other members of v's elementary group]. A lone attachment is never sealed:

$$|\text{Colour}(\Sigma)| = 1 \implies \mathcal{C}(\Sigma; C) = 0$$

[a contraction partner is a distinct node — A1: $p \neq q$ — hence $\pi_C(v) \not\subseteq \{v\}$; equivalently, CT3.17's list contains no invariant of a single slot]. (2) Grammar count. The number $g(N)$ of elementary groupings of a labelled N-element attachment set into groups of size 2 and 3 (labelled set partitions with parts in $\{2, 3\}$) is

$$g(N) = \sum_{a,b\geq 0\ 2a+3b=N} \frac{N!}{a!\, b!\, 2^a\, 6^b};$$

$$g(1) = 0, \quad g(2) = 1, \quad g(3) = 1, \quad g(4) = 3, \quad g(5) = 10, \quad g(6) = 25\ (15 + 10), \quad g(7) = 105.$$

(3) Type selection. Attachment-type assignment (triplet/antitriplet per attachment [D3.8]) selects the type-compatible groupings — $\delta^i{}_j$ consumes one attachment of each type, ε_{ijk} (ε^{ijk}) three of the same type [CT3.17] — configuration data refining the count of (2).

Req (Cor). A1; D2.14, D3.8; CT3.17; labelled set-partition enumeration [math; machine: `checks/verify_promotion_F_colour.py`].

Pf (Cor).

$$(1) : \text{ CT3.17 : every invariant}$$

$$= \text{combination of products of } \{\delta, \varepsilon, \bar{\varepsilon}\} : \text{ an elementary group consumes 2 or 3 slots;}$$

$$\text{D2.14 : } \mathcal{C} = 1 \iff \forall v : \pi_C(v) \subseteq \Sigma$$

$$\implies \text{Colour}(\Sigma) = \bigsqcup \text{ groups of size } 2, 3;$$

$$N$$

$$= 1 : \text{ A1 : } p$$

$$\neq q \implies \pi_C(v) \text{ requires a distinct member} \implies \pi_C(v) \not\subseteq \{v\} \implies \mathcal{C}$$

$$= 0 \text{ [D2.14]}$$

$$(2):\ \text{fix } (a,b),\ 2a+3b=N:\ \ \#\{\text{partitions into } a \text{ pairs} + b \text{ triples}\} = \frac{N!}{a!\,b!\,(2!)^a\,(3!)^b}$$

[math : labelled elements; block order and within-block order unlabelled]

$$(2!)^a = 2^a,\ (3!)^b = 6^b;$$

$$g(4) = \tfrac{4!}{2!\,2^2} = 3;$$

$$g(5) = \tfrac{5!}{1!\,1!\,2\cdot 6} = 10;$$

$$g(6) = \tfrac{6!}{3!\,2^3} + \tfrac{6!}{2!\,6^2} = 15 + 10 = 25;$$

$$g(7) = \tfrac{7!}{2!\,1!\,2^2\cdot 6} = 105 \text{ [machine]}$$

$$(3):\ \text{D3.8 : attachment types;}$$

$$\delta^i{}_j :\ (3, \bar{3});$$

$$\varepsilon_{ijk} :\ (3,3,3);$$

$$\varepsilon^{ijk} :\ (\bar{3},\bar{3},\bar{3}) \text{ [CT3.17] } \blacksquare$$

Theorem 2.116 (CT3.18 — Colour-singlet closure). Thm.

$$\text{free colour index} = \text{dangling boundary of the colour ledger } \neg\text{admitted by } \mathcal{P}_M^{(3)};$$

$$\mathcal{P}_M^{(3)}\big|_{\text{free index}} = 0 \implies \neg\exists \text{ pole [CT2.5]};$$

$$\text{surviving configurations : fully contracted by } \{\delta, \varepsilon, \bar{\varepsilon}\} \implies \text{colour singlets;}$$

$$\text{minimal singlets : } M = \delta^i{}_j q^j r_i, \quad B = \varepsilon_{ijk} q^i q^j q^k \text{ [D3.9].}$$

Req. D2.3, D2.11, D3.1, D3.9; CT2.5 (Part 2); CT3.17.

Pf.

$$\text{D2.11 : valid universe : closed } \wedge \text{ balanced seals } \wedge \neg\exists \text{ dangling ledger boundary}$$

$$\text{D3.1 : colour indices}$$

$$= \text{ledger labels at trivalent vertices } \Longrightarrow \text{ uncontracted index}$$

$$= \text{dangling colour-ledger boundary}$$

$$\text{D2.3 : } \mathcal{P}_M^{(3)} \text{ projects onto valid configurations } \Longrightarrow \text{ free-index configurations annihilated;}$$

$$\text{CT2.5 : } \neg\text{pole}$$

survivors fully contracted; $\quad$ CT3.17 : contractions $\in \{\delta, \varepsilon, \bar{\varepsilon}\} \implies$ invariant = colour singlet. ■

Phys. The mask admits only colour-sealed configurations: every observable bound rung-3 object is a colour singlet, minimally a meson or a baryon. Validity of the ledger is the singlet condition.

Theorem 2.117 (CT3.22 — Sealing is confinement). Thm.

$$\neg\exists \text{ isolated triplet pole of } Z(\mathcal{R}, 3); \qquad \text{isolated triplet : physical norm} = 0 \text{ [CT2.8];}$$

$$\text{colour charge } \neg\text{observable as end-charge [D3.4];}$$

$$\text{contrast : } q_{\text{EM}} \text{ observable on open chains (per-edge registration, } \neg\text{contraction).}$$

Req. D3.4; CT0.9 (Part 1); CT2.3, CT2.5, CT2.8 (Part 2); CT3.18.
Pf.

$$\text{CT3.18 : free colour index } \Longrightarrow \notin \operatorname{Im} \mathcal{P}_M^{(3)}$$

$$\text{CT2.3 : } \mathrm{F}_M^{(3)}\big|_{\mathcal{K}_3^{\perp}} = 0;$$

$$\text{CT2.5 : poles = eigenvalue conditions on } \mathrm{F}_M^{(3)} \implies \text{excluded configurations support } \neg\text{poles}$$

$$\text{CT2.8 step 3 : mask-excluded} = \text{null vectors of } \langle\cdot|\cdot\rangle \implies \text{isolated triplet : norm} = 0$$

$$\text{contrast : D3.4 : } q_{\text{EM}} = \tfrac{1}{2}(\sigma_s - \sigma_t) \text{ from edge signs;}$$

$$\text{CT0.9 : } \sigma_e \text{ registered per real edge,}$$

$$\neg\text{contraction required}$$

$$\implies q_{\text{EM}} \text{ observable on open chains. } \blacksquare$$

Phys. Confinement is a theorem: isolated colour has no pole and no norm — it is filtered out by the mask, not suppressed by a force law. The asymmetry with visible electric charge is exact: a sign is registered per edge; an index must await contraction.

Theorem 2.118 (CT3.22a — Subnetwork confinement: the domain restriction, all-or-nothing, and the converse). Thm. Let Σ be a subnetwork [D2.13] with $\mathcal{C}(\Sigma;C)$, $\Pi_{\mathcal{C}}(\Sigma;C)$ [D2.14], and $F_{\text{strong}}|_\Sigma$:= the strong force operator restricted to the colour slots of Σ under the rung mask [CT2.3, CT2.4; D3.1]. (0) Global sum-restriction.

$$Z^{(3)}(\mathcal{R}) = \sum_{C\in\mathfrak{C}(\mathcal{R})\text{: colour-sealed}} \sum_{\sigma} \mathcal{J}(C) \prod_{e\in E(C)} G^{(3)}(e)\,(-1)^{\aleph(\sigma,C)}$$

$$[\text{D0.10, C12}]$$

— confinement restricts the summation domain, ¬any individual edge factor: the per-edge geometric/registration data are rung-independent once the block read is rung-stable ($\alpha^{(3)}_{\text{EM}} = \alpha^{(2)}_{\text{EM}}$ — the EM.18/EM.19 mechanism) and the per-edge phase alphabet is rung-blind ($\alpha_e \in \Phi = \{0,\pi\}$ at every rung [D0.5; CT0.9; EM.20(1)]). (1) All-or-nothing. With $F^{\Sigma}_{\text{strong}}(C) := \Pi_{\mathcal{C}}(\Sigma;C)\, F_{\text{strong}}|_\Sigma\, \Pi_{\mathcal{C}}(\Sigma;C)$:

$$\mathcal{C}(\Sigma;C) = 0 \implies F^{\Sigma}_{\text{strong}}(C) = 0$$

— regardless of how many colour-carrying vertices Σ contains, and even if $|\partial_{\text{col}}(\Sigma;C)| = 1$. (2) Converse with the exact exception locus. $\Pi_{\mathcal{C}} = \mathcal{C}\cdot\mathbb{I}$ with $\mathcal{C} \in \{0,1\}$ [D2.14: scalar idempotent] $\implies F^{\Sigma}_{\text{strong}} = \mathcal{C}\cdot F_{\text{strong}}|_\Sigma$; hence

$$\mathcal{C}$$

$$= 1$$

$$\implies F^{\Sigma}_{\text{strong}}$$

$$= F_{\text{strong}}|_\Sigma$$

$$\neq 0 \quad \text{for nonzero aspect data } (f_a \neq 0 \text{ on } \geq 1 \text{ colour slot [CT3.9, CT3.10]})$$

with exact exception locus $\{f = 0\} \cup \{\text{Colour}(\Sigma) = \varnothing\}$. (3) Singlet diagonal (consistency). $T_a|\text{singlet}\rangle = 0$ [CT3.15, CT3.17 invariance] $\implies$ $\langle\text{singlet}|F_{\text{strong}}|\text{singlet}\rangle = 0$ — consistent with CT3.22 (colour is ¬observable as end-charge [D3.4]): the nonzero content of the strong block is the admissibility structure and the gluon mode count [CT3.13], ¬singlet expectation values.

Req. D2.13, D2.14; D0.10, D3.1, D3.4, D3.5, C12; CT2.3, CT2.4, CT2.5, CT2.8 (Part 2); CT3.2, CT3.9, CT3.10, CT3.13, CT3.15, CT3.17, CT3.18, CT3.22; EM.18, EM.19, EM.20(1) for clause (0) note (Part 2); idempotent algebra, the tensor null-factor argument [math; machine: `checks/verify_merge_tierA.py` §A].

Pf.

$$(0):\ \text{CT3.18}:\ \mathcal{P}_M^{(3)} \text{ admits only colour-sealed } C\ (\delta^i{}_j,\ \varepsilon_{ijk},\ \varepsilon^{ijk} \text{ contractions [CT3.17]});$$

$$\text{CT3.22}:\ \neg\exists \text{ isolated triplet pole}$$

$$\text{free-index } C \notin \operatorname{Im} \mathcal{P}_M^{(3)};$$

$$\text{CT2.5}:\ \text{poles} = \text{eigenvalue conditions on } \mathrm{F}_M^{(3)};$$

$$\text{CT2.3}:\ \mathrm{F}_M^{(3)}\big|_{\mathcal{K}_3^{\perp}} = 0 \implies \text{excluded } C:\ \neg\text{pole} \implies \text{domain restriction}$$

$$\text{edge factors}:\ G^{(n)}(e) = \sigma_e \exp(i\alpha_e \widehat{ds}^{\,2}(e)) \text{ [D3.5]};$$

$$\alpha_e \in \{0, \pi\} \text{ at every rung [EM.20(1); CT0.9]};$$

$$\alpha_{\text{EM}}^{(3)} = \alpha_{\text{EM}}^{(2)} \text{ [EM.18/EM.19]} \implies G^{(3)} = G^{(2)} \implies \neg\text{weight modification}$$

$$(1):\ \mathcal{C} = 0:\ \exists v \in \text{Colour}(\Sigma):\ \pi_C(v) \not\subseteq \Sigma \text{ [D2.14]} \implies v \text{ carries an index uncontracted relative to } \Sigma$$

$$\text{CT3.18 with } \Sigma \text{ as ambient}:\ \text{uncontracted index}$$

$$= \text{dangling colour-ledger boundary} \implies \text{mask-excluded};$$

$$\text{CT3.22} + \text{CT2.5}:\ \neg\text{pole of the } \Sigma\text{-restricted resolvent}$$

$$\text{CT2.8}:\ \text{physical norm } 0 \implies \text{the } \Sigma\text{-restricted colour state of } v \text{ is a null vector};$$

$$\text{colour sector of } \Sigma \;=\; \bigotimes_{\text{Colour}(\Sigma)} \text{(slot spaces) [D3.1, CT3.2]}$$

$$\text{one null factor} \implies \text{whole product null} \implies F_{\text{strong}}|_\Sigma \text{ has vanishing matrix elements;}$$

$$\Pi_{\mathcal{C}} = 0 \implies F^{\Sigma}_{\text{strong}} = 0 \cdot F_{\text{strong}}|_\Sigma \cdot 0 = 0$$

$$(2):\ \mathcal{C} \in \{0,1\} \implies \mathcal{C}^2 = \mathcal{C} \implies \Pi_{\mathcal{C}}\, F\, \Pi_{\mathcal{C}} = \mathcal{C}^2\, F|_\Sigma = \mathcal{C}\, F|_\Sigma;$$

$$\mathcal{C} = 1:\ T_a \neq 0 \text{ on } \mathbb{C}^3 \text{ [CT3.9 : explicit Gell-Mann basis]} \ \wedge\ f \neq 0 \text{ [CT3.10]} \implies F|_\Sigma \neq 0$$

$$\text{exception locus : } f = 0 \implies F|_\Sigma = 0 \text{ by linearity;} \quad \text{Colour}(\Sigma) = \varnothing \implies F|_\Sigma \text{ has no argument}$$

$$(3):\ \varepsilon_{ijk} U^i{}_a U^j{}_b U^k{}_c = \varepsilon_{abc} \det U = \varepsilon_{abc} \quad \forall U \in SU(3) \text{ [CT3.15];}$$

$$\delta^i{}_j \text{ likewise invariant [CT3.17]}$$

$$\implies \text{singlets fixed under } SU(3) \implies \mathfrak{su}(3) \text{ annihilates them} \implies \langle \text{singlet}|F_{\text{strong}}|\text{singlet}\rangle = 0. \ \blacksquare$$

Phys. Confinement becomes a two-sided exact law at subnetwork level: the colour block of a region lives precisely when the region self-seals against the ambient ledger — one dangling attachment anywhere zeroes the entire colour sector, no matter how many safely paired vertices sit inside. Unlike the additive per-vertex read of EM [D3.3], the strong read is ¬gradable: a sealed object never carries a colour number, only the grammar of how it composes with its environment.

Rmk (non-locality and filter placement; witness scope). $\mathcal{C}(\Sigma; C)$ is ¬a function of Σ-local data: valid ambient pairs C, C' with identical restrictions $C|_\Sigma = C'|_\Sigma$ yet $\mathcal{C} = 1$ vs 0 exist once the ambient supplies further triplet carriers to seal (explicit finite witness: two valid ambient groupings, identical Σ-occupation, opposite seals) — an ambient-richness scope for that witness, ¬derivable from A1 alone; no theorem clause above rests on it (A §0.2a note). Hence the completeness filter belongs downstream of any $\mathcal{M}(\Sigma)$ weighting: pre-weighting by $\mathcal{C}$ would delete the physical EM/G/H reads of externally-sealed restrictions, which are colour-independent by construction [D3.3, CT1.12]. Universe-level validity remains CT3.18/CT2.8(3)'s question — distinct from Σ-relativity.

2.30 §F.IV — Completion

Theorem 2.119 (CT3.19 — Self-containment of the colour sector). Thm.

$$\forall \text{ statements of F}: \quad \vdash \text{ A1–A9} \cup \text{C1–C9} \cup \text{D0.1–D4.3} \cup \{\text{B, C theorems}\} \cup \{\text{EW.26, EM.18 restated}\};$$

$$\neg\exists \text{ external strong-interaction law, parameter, or measured value;} \qquad \text{open quantities} = \varnothing :$$

$$N_{\text{colour}} = 3, \quad N_{\text{gluon}} = 8, \quad f_{abc}, \quad \mathfrak{T}(T_a T_b) = \tfrac{1}{2}\delta_{ab}, \quad m_g^2 = 0 \qquad \text{all fixed;}$$

$$\text{colour sector contributes } \neg\text{P4 data.}$$

Req. CT3.1–CT3.22; P4.
Pf. Dependency trace:

$$3 \text{ slots} \Leftarrow 2^{n+1} \leq 16 \text{ [CT3.20} \Leftarrow \text{CT0.5, CT0.6, CT0.11]}$$

$$S_3 \hookrightarrow U(3) \text{ [CT3.1–CT3.4]};$$

$$\mathfrak{u}(3) = \mathfrak{su}(3) \oplus \mathfrak{u}(1) \text{ [CT3.5]};$$

$$\text{centre spent at rung 2} \implies \text{new} = \mathfrak{su}(3) \text{ [CT3.7, D3.2]}$$

$$\dim = 8 \text{ [CT3.8]}; \quad \text{normalized basis [CT3.9]};$$

$$f_{abc} \text{ determined [CT3.21]}; \quad \text{normalization} = \text{EW.32's}$$

$$\mathfrak{T} = 0 \implies \neg\text{abelian carry/shift [CT3.10]}$$

$$\text{gluons}: \ 8 \text{ massless trivial chains [CT3.13, CT3.14]};$$

$$\gamma/g \text{ differ by possibility space only [CT3.12]}$$

$$\{\delta, \varepsilon, \bar{\varepsilon}\} \text{ [CT3.17]};$$

$$\text{mask admits only singlets [CT3.18]};$$

$$\text{isolated colour}: \ \neg\text{pole},$$

norm 0 [CT3.22]. ■

Phys. Closure audit: the colour sector is the most rigid in the corpus — number of colours, number of gluons, all structure constants, normalization and masslessness are fixed by the axioms, with not a single configuration parameter of its own.

2.31 Closure exports of Part 2 §F (consumed by Part 2 §E, §G, Parts 3–4)

CT3.20 $N_{\text{colour}} = 3$ (forced)	CT3.7 new algebra $= \mathfrak{su}(3)$, $\mathfrak{T}(T_a) = 0$ [cited in E]	CT3.8 dim $= 8$ [cited in B: CT0.14]
CT3.9 $\{T_a\}$: $\mathfrak{T}(T_a T_b) = \frac{1}{2}\delta_{ab}$	CT3.21 f_{abc} exact list	CT3.10(a) $\mathfrak{T}(F_{\text{strong}}) = 0$; $\alpha^{(3)} = \alpha^{(2)}$; $\neg\exists\, \alpha_{\text{strong}}$
CT3.13 $N_{\text{gluon}} = 8$	CT3.14 $m_g^2 = 0$ ($\times 2$ routes)	CT3.15–CT3.17 B, M singlets; $\{\delta, \varepsilon, \bar{\varepsilon}\}$; CT3.17a : $g(N)$ grammar
CT3.18 mask admits only singlets	CT3.22 confinement : ¬triplet pole, norm 0	CT3.19 sector rigid, ¬P4 data

■

2.32 §G — Saturation Lock, Dark Aspects, Bounds

Dep.: Part 1; Part 2 §C–§F (EW.27 at use). Order: §G.I the saturation lock (CT4.1, +CT4.1a, CT4.2, CT4.3, CT4.4, +CT4.4a) · §G.II the two dark aspects (CT4.5, +CT4.5a, CT4.6, CT4.7, +CT4.7a, CT4.10) · §G.III bounds and boundaries (CT4.9, CT4.8, CT4.11) · §G.IV falsifiability and the closure audit (CT4.12, CT4.13). Notation as Part 2 §C–§F. Machine: `checks/verify_G.py`; CT4.4a/CT4.7a: `checks/verify_merge_tierA.py`; CT4.5a/CT4.1a: `checks/verify_promotion_G_saturation.py`.

2.33 §G.I — The Saturation Lock

Theorem 2.120 (CT4.1 — Oversubscription falls to D^-). Thm.

$$\forall n \geq 4: \quad D_{\text{tick}}(n) = 2^{n+1} \geq 32 > 16 = C_{\text{cap}} \implies O(n) = 2^{n+1} - 16 > 0;$$

$$O(4) = 32 - 16 = 16 = C_{\text{cap}} \quad \text{(exactly one further channel quantum per tick)};$$

$$F_{D^-} = \bigoplus_{n \geq 4} F_n \text{ [D4.1]}; \qquad \text{absorption rate} \propto O(n) \sim 2^{n+1} \text{ [D4.2]}; \qquad \text{last admissible rung} = 3.$$

n	$D_{\text{tick}}(n) = 2^{n+1}$	$O(n)$	status
0	2	0	virtual only, $\mathcal{A}(0) = \{J\}$
1	4	0	pair aspects active
2	8	0	abelian aspect activates
3	16	0	exact saturation; colour activates
4	32	16	locked in D^-
5	64	48	locked in D^-
$n \geq 4$	2^{n+1}	$2^{n+1} - 16$	locked in D^-

Req. A2; D4.1, D4.2; CT0.5, CT0.6, CT0.11 (Part 1).

Pf.

$$D_{\text{tick}}(n) = 2^{n+1}\ [\text{CT0.5}]; \qquad C_{\text{cap}} = \dim_{\mathbb{R}} \mathfrak{H}(\mathbb{C}^4) = 16\ [\text{CT0.6} \Leftarrow \text{A2}]$$

$$n = 4: \ \ O(4) = 2^5 - 16 = 16 = C_{\text{cap}}; \qquad n \geq 4: \ \ 2^{n+1} \geq 32 > 16 \implies O(n) = 2^{n+1} - 16 > 0$$

$$\text{D4.1}: \ n \geq 4 \implies \text{config} \in F_{D^-} = \bigoplus_{n \geq 4} F_n; \qquad \text{D4.2}: \ \text{rate} \propto O(n) \implies \neg\exists \text{ rung } n \geq 4 \text{ escaping}$$

$$\text{CT0.11}: \ \text{admissible} \iff n \leq 3. \ \blacksquare$$

Phys. Rung 4 demands 32 registers against a budget of 16 — exactly one extra channel quantum — and every higher rung demands more. All such configurations collapse into a single sink D^- whose absorption grows as 2^{n+1}: rung 3 is the last rung the universe can access, and above it there is only the lock.

Corollary 2.6 (CT4.1a — universal saturation of bidirectional real-edge subgraphs). Let S be a bidirectional real-edge subgraph — both orientations of every used channel in $\mathcal{R}$ [A4] — exhausting the incident real edges at its vertices, with channel degree $d_S(v)$. Then $n(v) = 2\,d_S(v)$ [CT0.11b (Part 1), D0.8], and: (1) $d_S(v) \geq 2 \implies n(v) \geq 4 \implies O\big(n(v)\big) > 0$: every such vertex falls to F_{D^-} [CT4.1, D4.1]; its content is absorbed, retained in $\mathfrak{L}$ and counted, ¬accessible [CT4.2, CT4.3, D4.2, D1.11] — no bidirectional vertex of channel degree ≥ 2 is unoversubscribed. (2) S 2-regular (bidirectional cycles, any length ≥ 3): uniform minimal saturation $n(v) = 4\ \forall v$; $O(4) = 16 = C_{\text{cap}}$ — exactly one channel quantum over budget per vertex, independent of cycle length [CT0.6, CT4.1]. (3) S d-regular: graded oversubscription $O(2d) = 2^{2d+1} - 16$ [CT0.6]:

$$O(4) = 16, \qquad O(6) = 112, \qquad O(8) = 496, \qquad O(10) = 2032, \qquad O(12) = 8176.$$

(4) The unoversubscribed real-edge family under directed rung counting is exactly $n(v) \leq 3$ [CT0.11b, CT0.6]: on bidirectional S only single-channel vertices ($n = 2$); mixed skeletons admit unidirectional attachments up to $n = 3$; every vertex with $n(v) \geq 4$ is saturated [CT0.11, CT4.1].

Req (Cor). A4; D0.8, D1.11, D4.1, D4.2; CT0.6, CT0.11, CT0.11b (Part 1); CT4.1, CT4.2, CT4.3; exact ladder arithmetic [math; machine: `checks/verify_promotion_G_saturation.py`].

Pf (Cor).

$$n(v) = 2\,d_S(v)\ [\text{CT0.11b},\ \text{D0.8}];$$

$$(1): \ d_S \geq 2: \ n \geq 4: \ O(n) = 2^{n+1} - 16 > 0\ [\text{CT0.6}] \implies F_{D^-}\ [\text{CT4.1},\ \text{D4.1}]$$

$$\text{CT4.2}: \ \text{absorbing}; \qquad \text{CT4.3/D4.2}: \ \text{retained} \subseteq \mathfrak{L},\ \text{counted},\ \neg\text{accessible [D1.11]}$$

$$(2): \ d_S \equiv 2: \ n \equiv 4; \quad O(4) = 2^5 - 16 = 16 = C_{\text{cap}}\ [\text{CT0.6},\ \text{CT4.1}];$$

$$(3):\ O(2d) = 2^{2d+1}-16:\ 2^7-16 = 112,\ 2^9-16 = 496,\ 2^{11}-16 = 2032,\ 2^{13}-16 = 8176\,[\text{exact}]$$

$$(4):\ \text{CT0.11b verbatim;} \qquad n \geq 4 \implies O > 0 \implies \text{saturated}\ [\text{CT0.11, CT4.1}]\ \blacksquare$$

Theorem 2.121 (CT4.2 — The lock is absorbing). Thm.

$$\forall n \leq 3: \quad \mathcal{P}_M^{(n)}\, F_{D^-}\, \mathcal{P}_M^{(n)} = 0;$$

$$\text{locked content}:\ \neg\exists\ \text{pole of}\ Z(\mathcal{R}, n)\ \ \forall n \leq 3$$

$$(\text{one-way transfer w.r.t. the accessible sector}).$$

Req. D4.1; CT0.13, CT0.14 (Part 1); CT2.5 (Part 2).
Pf.

$$\text{D4.1}:\ \text{supp}\, F_{D^-} \subseteq \mathcal{K}^{(D^-)}; \qquad \text{CT0.13}:\ D^- \in \mathcal{A}(n) \iff n \geq 4$$

$$\text{CT0.14}:\ \text{aspect blocks} = \text{orthogonal summands of}\ \mathcal{K};$$

$$\mathcal{K}^{(D^-)} \not\subseteq \mathcal{K}_n\ \ (n \leq 3) \implies \mathcal{P}_M^{(n)}\big|_{\mathcal{K}^{(D^-)}} = 0 \implies \mathcal{P}_M^{(n)} F_{D^-} \mathcal{P}_M^{(n)} = 0$$

$$\text{CT2.5}:\ \text{pole} \implies \text{nonzero support in}\ \mathcal{K}_n;$$

$$\text{locked content has none} \implies \neg\exists\ \text{accessible pole at any}\ n \leq 3.\ \blacksquare$$

Phys. The lock is a one-way membrane: no accessible mask can bring D^- content back, and locked content supports no accessible spectral pole at any rung. What falls in stays in — with respect to everything an accessible observer can measure.

Theorem 2.122 (CT4.3 — Locked, not destroyed). Thm.

$$\text{absorption conserves ledger content}: \quad \text{absorbed} \subseteq \mathfrak{L};$$

$$\neg\text{accessible [D1.11]};$$

$$\neg\exists\ \text{deleted entry};$$

$$\text{oversubscription}\ \neg\text{seal}:\ \Delta n_{\mathcal{R}} = 0;$$

$$\text{locked content}: \quad \text{physical norm}$$

$$= 0 \text{ at accessible rungs [CT2.8]} \ \wedge \ w(C)$$

$$\neq 0 \text{ in the unmasked A8 sum (record survives).}$$

Req. A5, A8; D0.7, D1.11, D4.2; CT0.3 (Part 1); CT2.8 (Part 2); CT4.1, CT4.2.
Pf.

$$\text{D0.7 : } \mathfrak{L} = \text{ordered record of retained events;} \qquad \text{A5 : sole exclusion} = U \text{ (Haar draw)}, \ U \notin \mathfrak{L}$$

$$\text{D4.2 : absorbed} = \text{locked} := \text{retained in } \mathfrak{L} \ \wedge \ \neg\text{accessible [D1.11]}$$

$$\text{CT0.3 : counters change only via seals;}$$

$$\text{CT4.1 : oversubscription}$$

$$= \text{load/capacity mismatch at fixed rung} \implies \Delta n_{\mathcal{R}}$$

$$= 0 \implies \neg\text{seal} \implies \neg\exists \text{ removed entry}$$

$$\text{CT2.8 : null vectors} = \text{mask-excluded} \implies \text{locked : norm } 0 \text{ at accessible rungs [CT4.2]}$$

$$\text{A8 : } w(C) \text{ retains nonzero value in the unmasked sum} \implies \text{record survives.} \ \blacksquare$$

Phys. The lock is a loss of access, not of information: the ledger keeps every absorbed register with nonzero weight; only its recoverability — its norm in the accessible physical inner product — is forfeited. (Used again in CT4.7, CT4.8.)

Theorem 2.123 (CT4.4 — The locked sector carries no gauge structure). Thm.

$$D^{-} : \ \neg\exists \text{ nonabelian gauge algebra;}$$

$$\neg\exists \text{ abelian gauge algebra;}$$

$$\text{contributes } \neg\text{generator to any accessible algebra.}$$

Req. D3.2; CT0.11, CT0.12, CT0.13 (Part 1); CT3.20 (Part 2); CT4.2.
Pf.

$$\text{gauge structure arises only from : } \ (a) \text{ pair block;} \quad (b) \text{ slot algebras } \mathfrak{u}(n)$$

(a) : CT0.12 : 16 pair directions fully allocated to $\{J^{\pm}, G, W, H, D^{+}\} \implies \neg\exists$ direction left for D^{-}

(b) : CT3.20/CT0.11 : slot algebra realized only for $n \leq 3$;

CT0.13 + D3.2 : strictly new content at rungs $2, 3$ allocated to $\{\text{EM}, \text{strong}\}$

$n \geq 4$: $\neg$channel-admissible $\implies$ slot algebra $\neg$realized

$\implies F_{D^-}$: block disjoint from every gauge block = scalar sink, $\neg$internal directions;

CT4.2 : $\neg$probing by any accessible mask. ■

Phys. The dark sink is structureless: no charges, no generators, no self-interaction visible from the accessible side. That structurelessness is precisely what allows it to behave as a single simple fluid (CT4.5).

CT4.4a — The realization domain of ι_n: non-existence at $n \geq 4$, persistence refuted. Structural-constraints record: see Appendix B, Section B.3.1 of the Addendum.

2.34 §G.II — The Two Dark Aspects

Theorem 2.124 (CT4.5 — Equation of state of the locked sector). Thm.

$$T^{(D^-)}_{\mu\nu} = \rho_{D^-}\, u_\mu u_\nu; \qquad p_{D^-} = 0; \qquad w_{D^-} = \frac{p}{\rho} = 0; \qquad \rho_{D^-} \geq 0 \ \big(= 0 \iff \neg\exists v : \ n(v) \geq 4\big).$$

Req. D1.5; CT1.22, CT1.23 (Part 2); CT4.1, CT4.2, CT4.3.
Pf.

CT1.22 : $T = T^{\text{bd}} + T^{\text{tr}}$ [channels of D1.5]

T^{tr} requires propagating chain with connection [CT1.22];

CT4.2 : locked : $\neg$accessible pole $\implies \neg$propagating mode $\implies \mathsf{E}_{\text{tr}} = 0 \implies T^{\text{tr}} = 0$

$\implies T^{(D^-)} = T^{\text{bd}} = (\rho + p) u_\mu u_\nu + p\, g_{\mu\nu}$ [CT1.22; u^μ = rest frame of the locked cell]

$$p = \text{spatial momentum flux in rest frame;}$$

$$\neg\text{propagation} \implies \text{flux}$$

$$\equiv 0$$

$$\implies p_{D^-}$$

$$= 0$$

$$\implies T_{\mu\nu}^{(D^-)}$$

$$= \rho_{D^-} u_\mu u_\nu;$$

$$w$$

$$= 0/\rho \qquad = 0$$

$$\rho_{D^-} = \text{absorbed registers per volume} = \sum O(n) \geq 0 \text{ [CT4.1];}$$

$$\text{CT4.3 : } \neg\text{removal} \implies \rho_{D^-} \geq 0; \quad = 0 \iff \neg\exists v : \ n(v) \geq 4. \ \blacksquare$$

Phys. Locked content gravitates as pressureless dust: no propagation, no momentum flux, $w = 0$ — derived, not fitted. It couples to geometry through the field equations CT1.23 and through nothing else. Contrast with D^+: $w = -1$ (CT1.25, CT4.6).

Theorem 2.125 (CT4.5a — The flicker-dust ensemble: exact density polynomial and rate-invariant $w = 0$). Thm. Hypotheses (ensemble scope, stated): Γ a finite skeleton [A7] with potential degrees $n(v)$; every potential edge crystallizes independently with common probability $p \in [0, 1]$ (a per-edge Bernoulli ensemble; the dynamics realizing such an ensemble is not declared in Parts 1–2 — Merge note). Then: (1) Exact dust density. $k_v \sim \text{Bin}(n(v), p)$, and with $f_{\text{adm}}(n, p) := \sum_{k\leq 3} \binom{n}{k} p^k (1-p)^{n-k}$ [CT0.11: admissible $\iff$ $k \leq 3$], $O(n) := \max(0,\, 2^{n+1} - 16)$ [CT4.1, CT0.6]:

$$\langle \rho_{D^-} \rangle_\Gamma(p) = \sum_v O(n_v) \left(1 - f_{\text{adm}}(n_v, p)\right)$$

— an exact polynomial in p with integer coefficients [CT4.5's source criterion $\rho_{D^-} > 0 \iff \exists v : \ k_v \geq 4$, read in expectation]. (2) Anchors. $\langle \rho \rangle_\Gamma(1) = \sum_v O(n_v)$ — Platonic values $(448,\ 896,\ 2976,\ 2240,\ 24384)$ for (tetra, cube, octa, dodeca, icosa) $[(V, \deg) = (4,3), (8,3), (6,4), (20,3), (12,5)$; math]; at $p = \frac{1}{2}$: $\left(154,\ 308,\ \frac{15159}{8},\ 770,\ 20193\right)$; the polygon family P_n ($n \geq 3$, deg $= 2$): the exact monomial $16\, n\, p^4$; the digon identically 0 ($k = 2$ never locks

[CT0.6: $O(2) = 0$]); $\langle\rho\rangle_\Gamma(0) = 0$; strict increase on $(0, 1)$ for every lockable Γ ($\exists v : \ n(v) \geq 4$). (3) The flicker modulates ρ, never w. CT4.5 applies per configuration: every locked vertex contributes pressureless dust ($w = 0$, source side, coupled to geometry through CT1.23 and nothing else). A mixture of $w = 0$ components is $w = 0$ on the locked part at every p [CT1.22's linearity in the source]; D^+'s $w = -1$ [CT1.25, D4.3] is configuration-independent and untouched — CT4.6's contrast table holds ensemble-wise term by term. (4) No fluid. The locked sector supports no further equation-of-state data: a structureless absorbing sink — one possibility space [D4.1], no gauge content [CT4.4], absorption one-way [CT4.2], content retained-and-inaccessible [CT4.3, D1.11]. Statements assigning D^- a pressure response, polarization, or re-emission have no defined ledger referent (the referent would contradict CT4.2/D1.11); "the vacuum is a D^- fluid" is inadmissible as stated. What is derived: locked content gravitates as dust, with the density computed here.

Req. Hypotheses (stated above); A7; D1.11, D4.1, D4.2, D4.3; CT0.6, CT0.11 (Part 1); CT1.22, CT1.23, CT1.25 (Part 2); CT4.1, CT4.2, CT4.3, CT4.4, CT4.5, CT4.6; binomial law under per-edge independence, exact polynomial arithmetic [math; machine: `checks/verify_merge_tierB.py` §E].

Pf.

$$(1): \ \text{locked} \iff k_v \geq 4 \ [\text{CT4.5 iff; CT0.11}: \ \text{admissible} \iff k \leq 3];$$

$$\mu_p(k_v \geq 4) = 1 - f_{\text{adm}}(n_v, p) \ [\text{Bin}(n_v, p), \ \text{hypothesis}];$$

$$\left\langle \textstyle\sum_v O(n_v) 1_{k_v \geq 4} \right\rangle = \textstyle\sum_v O(n_v)(1 - f_{\text{adm}}) \ [\text{linearity; CT4.5}]$$

$$\text{integer coefficients}: \ O(n) \in \mathbb{Z} \ [\text{CT4.1}] \ \wedge \ \binom{n}{k} \in \mathbb{Z}$$

$$\implies (1 - f_{\text{adm}}) \text{ expands with } \mathbb{Z}\text{-coefficients}$$

$$(2): \ p{=}1: \ 1 - f_{\text{adm}}(n, 1) = 1_{n \geq 4}; \quad O(2d) = (16, 112, 496, 2032) \text{ at } 2d = (4, 6, 8, 10) \ [\text{CT4.1}]$$

$$\text{Platonic sums}: \ (4 \cdot 112, \ 8 \cdot 112, \ 6 \cdot 496, \ 20 \cdot 112, \ 12 \cdot 2032)$$

$$= (448, 896, 2976, 2240, 24384) \ [\text{machine}]$$

$$p = \tfrac{1}{2}: \ \text{exact binomial evaluation [machine]};$$

$$P_n: \ n(v) = 4 \ \forall v \implies 1 - f_{\text{adm}}(4, p) = p^4 \implies \textstyle\sum_v 16\, p^4 = 16np^4;$$

$$\text{digon}: \ O(2) = 0$$

monotonicity : $\frac{d}{dp}P(\text{Bin}(n,p)\geq 4) > 0$ on $(0,1)$ for $n \geq 4$ [math; machine on the rational grid]

(3) : $w = 0$ per configuration [CT4.5];

$\langle T_{\mu\nu}\rangle = \sum_c \mu(c)\, T_{\mu\nu}(c)$ [CT1.22] : $p_{\text{eff}} = \sum_c \mu_c \cdot 0 = 0$ on the locked part;

CT1.25/D4.3 : D^+ constant

(4) : the cited tags exhaust the D^- inventory;

CT4.2 : one-way;

D1.11 : inaccessible $\implies \neg\exists$ response read. ■

Phys. Shake a crystal at any rate and its dark content has an exact polynomial density — integer coefficients, closed form, every Platonic and polygon anchor reproduced — and its equation of state never moves: dust at every rate. What the flicker changes is how much content is locked, never what locked content does. And what locked content does is fixed: it gravitates, it absorbs, it does not respond.

Theorem 2.126 (CT4.6 — The two dark aspects contrasted). Thm.

	D^+	D^-
definition	$F_{D^+} = \Lambda\,\mathbb{I}$ [D4.3]	$F_{D^-} = \bigoplus_{n\geq 4} F_n$ [D4.1]
block	central pair direction E_0 [CT0.12]	stack block $\perp$ pair space [CT0.12 Rmk]
activation	$n = 1$ [CT0.13]	$n \geq 4$ [CT0.13]
config. dependence	none [D4.3]	full [CT4.1]
rest frame	none; isotropic [CT1.25]	one per locked cell [CT4.5]
equation of state	$w = -1$ [CT1.25]	$w = 0$ [CT4.5]
gauge content	none; central [CT1.3]	none; disjoint [CT4.4]
entropy	none [CT4.7]	maximal, locked [CT4.7]

$\forall$ rows differ $\implies D^+ \neq D^-$ ($\neg$two descriptions of one thing);

CT1.23 : D^+ geometric side $(\Lambda g_{\mu\nu})$; D^- source side $(\rho_{D^-} u_\mu u_\nu)$; $\neg$double counting [CT1.22].

Req. D4.1, D4.3; CT0.12, CT0.13, CT1.3 (Part 1); CT1.22, CT1.23, CT1.25 (Part 2); CT4.1, CT4.4, CT4.5, CT4.7.

Pf. Each cell = cited tag; rows differ pointwise; distinct entry into CT1.23 with no double count [CT1.22 step 5]. ■

Phys. The corpus's two dark aspects differ on every structural axis: one is uniform cosmological tension on the geometric side ($w = -1$), the other is oversubscribed ledger sediment on the source side ($w = 0$). "Dark" names an observational status, not a shared nature.

Theorem 2.127 (CT4.7 — The entropy triangle). Thm. In the plane $\big(S_{\text{raw}},\ S_{\mathfrak{L}}\big)$ with $S_{\text{raw}}(\rho) = -\mathfrak{T}(\rho \log \rho)$ [D0.3] and $S_{\mathfrak{L}} = n_{\mathcal{R}} \ln 2$ [CT1.29], there are exactly three scalar extremes, mutually exclusive:

$$(1)\ \text{Janus vacuum}: \ n_{\mathcal{R}} = 0: \quad \big(S_{\text{raw}}, S_{\mathfrak{L}}\big) = \big(\ln 4 \text{ per pair (max)},\ 0\big);$$

$$(2)\ D^{+}: \ \text{one configuration}: \quad (0,\ 0) \qquad (\ln 1 = 0;\ \neg\text{real edges});$$

$$(3)\ D^{-}: \ \text{locked}: \quad \big(\min_{\text{acc}},\ \max \wedge \neg\text{accessible}\big).$$

Req. A4; D0.3, D4.3; CT0.1, CT0.2, CT0.8 (Part 1); CT1.29 (Part 2); CT4.1, CT4.3, CT4.9.
Pf.

two independent measures : $S_{\text{raw}} =$ pair mixedness [D0.3]; $S_{\mathfrak{L}} =$ registered real edges [CT1.29]

$$(1):\ \text{CT0.2}: \ n_{\mathcal{R}} = 0 \implies \rho = \rho_{\circ} = I/4,\ S_{\text{raw}} = \ln 4 = \max\,(d = 4); \qquad S_{\mathfrak{L}} = 0 \cdot \ln 2 = 0$$

$$(2):\ \text{D4.3}: \ \text{single direction, single value } \Lambda \implies \ln 1 = 0;$$

$$\neg\text{own real edges} \implies S_{\mathfrak{L}}\text{-contribution} = 0$$

$$(3):\ \text{CT4.1}: \ \text{locked vertex carries } 2^{n+1} > 16 \text{ registers};$$

$$\text{CT4.3}: \ \neg\text{deletion}$$

$$\implies S_{\mathfrak{L}}^{\text{locked}} > \text{accessible bound [CT4.9]},$$

$$\to \infty \text{ in } n$$

$$\text{CT0.1} + \text{A4}: \ \text{interior pairs crystallized};$$

$$\text{CT0.8}\ (\varepsilon_q = 0): \ \operatorname{supp}\rho \subseteq \vee^2 \implies S_{\text{raw}} \le \ln 3 < \ln 4: \min_{\text{acc}}$$

exclusivity : $(n_{\mathcal{R}} = 0) \wedge (\exists v:\ n \ge 4)$ contradictory; $\qquad D^{+}: \neg$configuration dependence at all. ■

Phys. The scalar landscape is a triangle in the (pair-entropy, ledger-entropy) plane: the vacuum at (max, 0), D^+ at (0, 0), D^- at (min, max-and-hidden). Three distinct ways of being featureless — and by CT4.10, no fourth.

Theorem 2.128 (CT4.7a — The vacuum census, closed, and the baseline equality). Thm. (1) The complete census. The three scalar extremes of CT4.7, with CT4.10's exhaustiveness, are the complete inventory of vacuum-type scalar reads, mutually exclusive:

	S_{raw}	$S_{\mathfrak{L}}$	structure
Janus vacuum ($n_{\mathcal{R}} = 0$)	$\ln 4$ per pair (max, argmax unique $\frac{1}{4}I$)	0	[CT0.2, A5, D0.3; CT4.7(1)]
D^+	0	0	pure, uniform, configuration-blind; $w = -1$ [D4.3, CT1.25; CT4.7(2)]
D^- locked	$\min_{\text{acc}}$	max $\wedge$ ¬accessible	[CT4.2, CT4.3, D1.11; CT4.7(3)]

¬∃ a fourth extreme [CT4.10]; F_H is a dynamical field, ¬an extreme [EW.27]. (2) Baseline equality. For every finite skeleton S [A7], let $\text{VV}(S) :=$ the all-virtual state on its potential-edge set ($n_{\mathcal{R}} = 0$, every pair at $\rho_\circ = \frac{1}{4}I$ [A5, CT0.2]) and $V_{\text{raw}} :=$ the global vacuum [CT0.2]. Then every constructible vacuum read coincides:

$$S_{\mathfrak{L}}(\text{VV}(S)) = n_{\mathcal{R}} \ln 2 = 0 = S_{\mathfrak{L}}(V_{\text{raw}}) \text{ [CT1.29]};$$

$$\Delta S_{\text{raw}} = 0 \text{ on both sides [CT0.2 : every pair at the unique argmax]}$$

$$\rho_{D^-} = 0 \text{ on both sides [CT4.5 iff : } \neg\exists v : \ k_v \geq 4];$$

$$S_{\partial\mathcal{B}} \text{ has no argument on either side } (\neg\exists \text{ closed real-edge surface [CT1.29 domain]})$$

hence the vacuum-polarization deficit vanishes identically: $\Delta S_{\text{vac-pol}}(S) \equiv 0 \ \forall S$. The reason is registration, ¬arithmetic: virtual edges carry no registration — ϕ_e/σ_e are defined on $E_{\mathcal{R}}$ only [A7, D0.5], and the sole ledger exclusion is the Haar draw [A5: $U \notin \mathfrak{L}$]. (3) Honest scope. The absolute raw entropy $|\text{pairs}| \cdot \ln 4$ is shape-dependent; every constructible deficit and read-difference is shape-independent and zero. No particle or cosmological identification [P5].

Req. A4, A5, A7; D0.3, D0.5, D1.11, D4.3; CT0.1, CT0.2, CT0.8 (Part 1); CT1.25, CT1.29 (Part 2); EW.27 (Part 2); CT4.1, CT4.2, CT4.3, CT4.5, CT4.7, CT4.9, CT4.10; exact arithmetic [math; machine: `checks/verify_merge_tierA.py` §D].

Pf.

$$(1) : \text{ CT4.7 rows verbatim;}$$

$$\text{CT4.10} : \ \{\text{rotation-scalar directions}\}$$

$$= \{E_0 \to F_{D^+}, \ \Sigma \to F_H, \ \text{vacuum configuration}, \ D^- \text{ sink}\};$$

$$F_H \ \neg\text{extreme [EW.27]}$$

$$(2):\ n_{\mathcal{R}} = 0 \implies S_{\mathfrak{L}} = n_{\mathcal{R}} \ln 2 = 0 \text{ [CT1.29]};$$

$$\text{every pair at } \rho_\circ$$

$$= \tfrac{1}{4} I \text{ [A5]}:\ S_{\text{raw}}$$

$$= \ln 4 \text{ per pair}$$

$$= \max,$$

$$\text{argmax unique [CT0.2, D0.3]} \implies \Delta S_{\text{raw}} \qquad = 0$$

$$k_v = 0 < 4\ \forall v \implies \rho_{D^-} = 0 \text{ [CT4.5 iff]}; \qquad E_{\mathcal{R}} = \varnothing \implies \neg\exists\, \partial\mathcal{B} \implies S_{\partial\mathcal{B}} \text{ undefined}$$

$U \notin \mathfrak{L}$ [A5] : Haar draws leave no ledger record $\implies$ all-virtual states are ledger-indistinguishable from V_{raw} on every $n_{\mathcal{R}}$-based read. ■

Phys. The vacuum ledger is closed: exactly three scalar ways to be empty, and every constructible read agrees across all of them — an all-virtual skeleton reads identically to the bare vacuum, shape by shape, because virtual edges were never registered at all. Raw entropy knows the shape; no constructible read consults it.

Theorem 2.129 (CT4.10 — The triangle is exhaustive). Thm.

$$\neg\exists \text{ fourth scalar extreme :}$$

$$\{\text{rotation-scalar directions of the corpus}\}$$

$$= \{E_0 \to F_{D^+},\ \Sigma \to F_H,\ \text{vacuum configuration},\ D^- \text{ sink}\};$$

$$F_H\ \neg\text{extreme (dynamical field)}.$$

Req. CT0.2, CT0.12, CT1.2, CT1.10, CT1.11 (Part 1); EW.27 (Part 2); CT3.7 (Part 2); CT4.4.
Pf.

$$\text{CT1.10}:\ \mathcal{E}_{++} \to 1 \oplus 5 \oplus 1 \implies \text{pair-space rotation scalars} = \{E_0,\ \Sigma\} \text{ exactly}$$

$$\text{CT1.11}:\ E_0 \not\parallel \Sigma; \qquad \text{CT0.12}:\ E_0 \mapsto F_{D^+},\ \Sigma \mapsto F_H$$

EW.27 : F_H : stable stationary background + massive excitation $\implies$ dynamical field,

¬limiting configuration

CT1.2 : J, W, G carry spatial indices $\implies$ ¬scalars

CT4.4 : D^- supplies ¬generator;

CT3.7 : $\mathfrak{T}(T_a) = 0 \implies$ colour ¬scalar direction;

abelian = centre, counted

vacuum corner = $n_{\mathcal{R}} = 0$ configuration (¬generator), unique [CT0.2]

$\implies$ {vacuum, D^+, D^-} complete. ■

Phys. No fourth scalar extreme exists: the operator content of the corpus offers exactly two rotation scalars (one spent on D^+, one dynamical as F_H), the vacuum is a unique configuration, and the sink is unique by construction. The entropy triangle closes.

2.35 §G.III — Bounds and Boundaries

Theorem 2.130 (CT4.9 — Accessible entropy bounds). Thm.

(1) per vertex : accessible$(v) \implies S(v) \leq 3 \ln 2$ nats $= 3$ bits (attained at $n = 3$);

(2) per region $\mathcal{B}$, $\partial\mathcal{B}$ closed real-edge surface of area A :

$$S_{\text{acc}}(\mathcal{B}) \leq \frac{A}{\mathfrak{h}^2} \text{ bits} = \frac{A}{\mathfrak{h}^2} \ln 2 \text{ nats}.$$

Req. A2; D0.8, D1.9, D4.2; CT0.9, CT0.11 (Part 1); CT1.29 (Part 2); CT4.1.
Pf.

(1) : CT0.11 : $n \leq 3 \implies \left|\{e \in E_{\mathcal{R}} : e \ni v\}\right| \leq 3$ [D0.8];

CT0.9 : 1 bit/edge $\implies 2^3$ configs $\implies S \leq 3 \ln 2$; $n = 3$: attained

(2) : CT1.29 : $n_{\mathcal{R}}(\partial\mathcal{B}) = A/\mathfrak{h}^2$; $S_{\partial\mathcal{B}} = n_{\mathcal{R}}(\partial\mathcal{B}) \ln 2$

$$\text{A2} + \text{D1.9}: \text{ ticks local};$$

$$\text{registers cross only boundary edges}$$

$$\Longrightarrow S_{\text{acc}}(\mathcal{B}) \leq \text{boundary register count}$$

$$\text{CT4.1} + \text{D4.2}: \text{ interior excess} \Longrightarrow \text{oversubscribed} \Longrightarrow \text{locked} \Longrightarrow \neg\text{contributing.} \blacksquare$$

Phys. Accessible information is bounded twice over: three bits at any vertex, one bit per hop-area of any bounding surface — a holographic-type bound whose coefficient is exactly $\ln 2$, derived from channel capacity alone.

Theorem 2.131 (CT4.8 — Locked regions and their boundaries). Thm. Interior locked in D^-, boundary $\partial\mathcal{B}$ of area A:

$$S_{\partial\mathcal{B}} = \frac{A}{\mathfrak{h}^2}\ln 2 \quad \Big(= \frac{A}{4G_N}\ [\text{CT2.9 naming}]\Big);$$

$$\text{exterior measurements} = \text{residues of the accessible resolvent } [\text{CT2.5}]:$$

$$\text{equal boundary register counts} \Longrightarrow \text{indistinguishable interiors};$$

$$\neg\exists \text{ accessible interior singularity } [\text{CT1.28}].$$

Req. CT1.28, CT1.29, CT2.5, CT2.9 (Part 2); CT4.1, CT4.2, CT4.3, CT4.9.
Pf.

$$\text{CT4.2}: \text{ locked}: \neg\text{accessible pole};$$

$$\text{CT2.5}: \text{ exterior measurement} = \text{residue of accessible resolvent} \Longrightarrow \neg\exists \text{ direct interior signal}$$

$$\text{CT4.9 step}(2): \text{ accessible registers of } \mathcal{B} = \text{boundary edges only}; \qquad \text{CT1.29}: S_{\partial\mathcal{B}} = \frac{A}{\mathfrak{h}^2}\ln 2$$

$$\text{equal boundary content} \Longrightarrow \text{equal accessible data} \Longrightarrow \text{interiors indistinguishable}$$

$$\text{CT1.28}: \text{ accessible rung} \leq 3 \Longrightarrow \neg\text{accessible curvature singularity};$$

$$\text{CT4.1}: \text{ would-be divergence transferred to } D^-;$$

$$\text{CT2.9}: \ \tfrac{A}{\mathfrak{h}^2}\ln 2 = \tfrac{A}{4G_N}. \ \blacksquare$$

Phys. A locked region is characterized, for the exterior, by its boundary area alone: exact boundary entropy, indistinguishable interiors, no interior singularity. The corpus's black-hole-type statement — derived from masks and capacity, with the quarter-area form supplied by the CT2.9 calibration.

Theorem 2.132 (CT4.11 — Global closure). Thm. With $\mathrm{F}_\oplus = \bigoplus_{\text{all aspects}} F_A$ and $M_\oplus = \bigoplus_n \mathcal{P}_M^{(n)}$:

$$Z\big[\mathrm{F}_\oplus, M_\oplus\big] \ \text{well defined}; \qquad \mathfrak{S}\big(Z[\mathrm{F}_\oplus, M_\oplus]\big) \neq \varnothing \qquad (\text{the object of A9 exists}).$$

Req. A2, A8, A9; C12; D0.10, D2.8, D4.1; CT0.2, CT0.12, CT0.13, CT0.14 (Part 1); CT2.2, CT2.5, CT2.8 (Part 2); CT4.2.

Pf.

$$\text{domain}: \ Z \text{ sums over } \mathfrak{C}(\mathcal{R}) \ [\text{D0.10}, \ \text{C12}]$$

$$\text{aspects}: \ \text{CT0.12}: \ \text{pair } 16$$

$$= \{J^{\pm}, G, W, H, D^{+}\};$$

$$\text{CT0.13}: \ \text{stack}$$

$$= \{\text{EM}, \text{strong}, D^{-}\} \implies \text{exactly the A8 list } (\neg\text{undefined}, \ \neg\text{twice})$$

$$\text{masks}: \ \text{CT2.2}: \ \mathcal{P}_M^{(n)} \text{ Hermitian idempotents}; \quad \text{CT0.14}: \ \text{strict flag}, \ \dim = (6, 16, 17, 25) < \infty$$

$$\text{convergence}: \ \text{D2.8}: \ \text{convergent sheet};$$

$$\text{CT2.5}: \ \text{resolvent finite off poles } (\dim \mathcal{K}_n < \infty);$$

$$\text{CT2.8}: \ \text{reflected sum} = \sum |w|^2 \geq 0, \ \text{finite for square-summable halves}$$

$$\text{residual divergence}: \ \bigoplus_{n\geq 4} F_n \text{ collected into } D^{-} \ [\text{D4.1}];$$

$$\text{CT4.2}: \ D^{-} \perp \forall \text{ accessible mask} \implies \text{accessible part of } Z = \text{finite blocks only}$$

$$\text{nonemptiness}: \ \text{CT0.2}: \ \text{global vacuum } (n_{\mathcal{R}} = 0, \ \forall \text{ pairs virtual}, \ \mathcal{A}(0) = \{J\}) \text{ satisfies every axiom} \implies \in \mathfrak{S} \implies \mathfrak{S} \neq \varnothing. \ \blacksquare$$

Phys. The total masked partition function named by A9 is well defined — every aspect accounted for exactly once, every mask a finite projection, every sum on the convergent sheet, the infinite tower sealed inside D^{-} — and its solution set contains at least the empty universe. The falsifiability axiom has mathematical content.

2.36 §G.IV — Falsifiability and the Closure Audit

Theorem 2.133 (CT4.12 — Falsifiers). Thm. Observation of any row $\implies \neg(\text{A1} \wedge \cdots \wedge \text{A9})$ [A9 contrapositive; each row negates a derived parameter-free statement]:

falsifier	contradicted tag
4th accessible slot index (colour algebra $\supsetneq \mathfrak{su}(3)$)	CT3.20, CT0.11
$N_{\text{gluon}} \neq 8$	CT3.13
isolated unsealed colour charge	CT3.18, CT3.22
$q \notin \{-1, 0, +1\}$	EM.9, EM.10, EW.36
$m_\gamma \neq 0 \ \vee \ N_{\text{pol}} \neq 2$	EM.6, EM.8
$m_W \neq m_Z \cos\theta_W \ \ (\rho \neq 1)$	EW.34
$e \neq g \sin\theta_W$	EW.35
uniform vacuum aspect with $w \neq -1$	CT1.25
locked non-propagating component with $w \neq 0$	CT4.5
$S_{\partial\mathcal{B}} \not\propto A \ \ \vee$ coefficient $\neq \ln 2$ per hop area	CT1.29
local higher-curvature corrections in $D = 4$	CT2.7
$\nabla^\mu T_{\mu\nu} \neq 0$ ($\Lambda = \text{const}$)	CT1.24, CT1.26
accessible curvature singularity	CT1.28
$S_{\text{acc}} > 1$ bit per hop area of bounding surface	CT4.9

Rmk (¬falsifiers). Measured values of $g,\ g',\ \theta_W,\ e,\ h_0,\ \Lambda,\ m_W,\ m_Z,\ m_H,\ \mathfrak{h}$: configuration/calibration data [P1, P4] — ¬falsifying alone; only relations among them falsify. Falsifier rows resting on EM.20 inherit its derived phase alphabet $\alpha_e \in \{0, \pi\}$ [EM.20(1)]; no extra-axiomatic input enters there [A §0.2a].

Req. A9; P1, P4; all cited tags.

Pf.

$$\forall \text{ rows}: \ \text{cited tag} = \text{theorem proved without free parameters}$$

$$\text{inspection of cited proofs}: \ \neg\exists\ \mathfrak{h}\text{-dependence surviving (entropy rows fix the coefficient, } \neg\text{the scale)};$$

$$\neg\exists \text{ P4 data used}$$

$$\text{parameter-free derived statement} \vdash \text{A1} \wedge \cdots \wedge \text{A9} \implies \text{its negation} \implies \neg(\text{A1} \wedge \cdots \wedge \text{A9})$$

[A9 contrapositive]. ■

Phys. The corpus is falsifiable in the strict sense: fourteen crisp observation types, any single one of which demolishes the axiom set. Magnitudes never falsify — only relations and structures do. That asymmetry is deliberate and stated.

Theorem 2.134 (CT4.13 — Closure audit). Thm. Every quantity of the corpus lies in exactly one of three categories; the categorization is complete:

(I) derived parameter-free data (¬freedom, ¬ħ-dependence; numbers and constraints) :

$$(7, 3, 3, 3) \text{ [CT1.1]};$$

$$C_{\text{cap}} = 16 \text{ [CT0.6]};$$

$$D_{\text{tick}}(n) = 2^{n+1} \text{ [CT0.5]};$$

$$n \leq 3 \text{ [CT0.11]};$$

$$N_{\text{colour}} = 3 \text{ [CT3.20]};$$

$$N_{\text{gluon}} = 8 \text{ [CT3.13]};$$

$N_{\text{pol}} = 2$ [EM.8]; $q \in \{-1, 0, +1\}$ [EM.9, EW.36]; $\alpha_e \in \{0, \pi\}$ per edge [EM.20]; $\rho = 1$ [EW.34];

$$w_{D^+} = -1 \text{ [CT1.25]};$$

$$w_{D^-} = 0 \text{ [CT4.5]};$$

$$S = n_{\mathcal{R}} \ln 2 \text{ [CT1.29]};$$

$$f_{abc} \text{ [CT3.21]}; \quad \mathfrak{T}(\mathcal{G}_A \mathcal{G}_B) = \tfrac{1}{2}\delta_{AB} \text{ [EW.32, CT3.9]};$$

derived constraint (interval; sharp point 0; ¬∃ theorem depending on the value within it) : ε_q

$$\in [0, \tfrac{1}{4}) \text{ [CT0.8]};$$

(II) named combinations of ħ (external comparison only) :

$$G_N = \frac{\hbar^2}{4 \ln 2}; \quad \kappa^2 = 8\pi G_N = \frac{2\pi \hbar^2}{\ln 2}$$

$$[\text{CT2.9};\ \neg\exists \text{ further dimensionful freedom}:\ \text{CT2.11}];$$

$$(\text{IIIa})\ \text{configuration data — independent core (P4)}:$$

$$g,\ g'\ [\text{EW.33}];\quad h_0\ [\text{EW.20}];\quad c_W\ [\text{EW.33}];\quad \Lambda\ [\text{CT1.25}];\quad \varkappa,\ \beta,\ \alpha_\Omega\ [\text{D2.1}];$$

$$(\text{IIIb})\ \text{configuration data — functionally dependent (}\neg\text{further freedom)}:$$

$$\theta_W = \arctan(g'/g)\ [\text{EW.24}];\quad e = \frac{gg'}{\sqrt{g^2+g'^2}}\ [\text{EW.35}];\quad \mu_\star^2 = c_W\varkappa^2\beta^2 h_0^2\ [\text{EW.33}];$$

$$m_W^2 = \mu_\star^2 g^2/4\ [\text{EW.20}];\quad m_Z^2 = \mu_\star^2(g^2+g'^2)/4\ [\text{EW.23}];\quad m_H^2 = \delta^2 S_{\text{eff}}/\delta h^2\big|_{h_0}\ [\text{EW.27}].$$

$$\text{completeness}:\ \forall \text{ symbol of the A-Part-4 master table} \in \text{I} \cup \text{II} \cup \text{IIIa} \cup \text{IIIb} \cup \{\text{structural objects}\};$$

$$\text{I},\ \text{II},\ \text{IIIa pairwise disjoint};\quad \text{IIIb} = f(\text{IIIa})\ (\neg\text{independent data}).$$

Req. P1–P5; every cited derivation tag (listed inline).
Pf.

$$\text{I}:\ \forall \text{ entries proved at the inline-cited tag from A1–A9 + earlier derived only}$$

$$\text{II}:\ \text{CT2.9};\qquad \text{CT2.11}:\ \mathfrak{h} \mapsto \lambda\mathfrak{h} = \text{symmetry} \implies \neg\exists \text{ further dimensionful freedom}$$

$$\text{IIIa/IIIb}:\ \text{P4 (core)} + \text{functional dependencies at the inline-cited tags (IIIb)}$$

$$\text{completeness}:\ \text{scan of the master symbol table}:\ \forall \text{ symbol}$$

$$= \text{structural} \vee \text{I} \vee \text{II} \vee \text{III};$$

$$\neg\exists \text{ symbol in two categories}$$

$$\implies \text{corpus}$$

$= 1$ calibration d.o.f. ($\mathfrak{h}$) + finite independent core (IIIa) + closed parameter-free prediction list (I). ■

Phys. The final audit: every quantity the corpus derives is either a fixed number, a name for the single scale $\mathfrak{h}$, or data of a particular solution — nothing floats in between. The boundary between prediction and parameter is drawn exactly, and it is the same boundary CT4.12 turns into falsifiers.

2.37 Closure exports of Part 2 §G (consumed by Parts 3–4)

CT4.1(a) $O(n) = 2^{n+1} - 16;\ O(4) = 16;\ n_{\text{bidir}} = 2d \geq 4$	CT4.2 $\mathcal{P}_M^{(n\leq 3)} F_{D^-} \mathcal{P}_M^{(n)} = 0$	CT4.3 locked $\subseteq \mathfrak{L};\ w(C) \neq 0$
CT4.4 D^- : ¬gauge structure	CT4.5 $T_{\mu\nu}^{(D^-)} = \rho u_\mu u_\nu;\ w_{D^-} = 0$	CT4.6 $D^+ \neq D^-$ (8 axes)
CT4.7/CT4.10 entropy triangle $\{vac, D^+, D^-\}$ complete	CT4.9 $S_v \leq 3 \ln 2;\ S_{\mathcal{B}} \leq \frac{A}{\hbar^2} \ln 2$	CT4.8 locked region ≡ boundary area
CT4.11 $Z[\mathrm{F}_\oplus, M_\oplus]$ well defined; $\mathfrak{S} \neq \varnothing$	CT4.12 14 falsifiers	CT4.13 categories I/II/III exhaustive

Corpus status. A → B → C → D → E → F → G complete; A9 test surface = CT4.12; audit = CT4.13. Verification suite: `checks/run_all.py` (exact arithmetic, no floating point).

■

3 $\langle qu(b(i(t|t))ick)\rangle$ — Part 3 · Fermion_era — Bound Chains, Termini, Floors, Carriers, Channels

Dep. Part 1 (axioms, conventions, qubit-era definitions, calibration, pair canon, ledger dynamics L0–L5); Part 2 (definitions D2.1–D4.3, geometry, electroweak, electromagnetic, colour, saturation) — cited by tag at use. Annex cited row-wise: the Census annex (Appendix A of this volume; the computed data cells: the Category-B tags FT.28, FT.32, FT.35, FT.42–FT.45, FT.49–FT.51, FT.63–FT.65, FT.69, FT.76, FT.78, FT.81, FT.83, FT.84, FT.96, FT.98, FT.99, FT.105, FT.114, FT.120). Format: Part 1 §0.4. Namespace. This part owns: Def-H.1–Def-H.43; FT.1–FT.114 (structural tags), FT.116–FT.131, FT.133–FT.139, FT.141–FT.145, FT.153–FT.160. Tiers F.1–F.42, F.44 in series order. DAG law. Citations resolve to Parts 1–2, Census cells, or earlier tags of this part; the marked forward pointers are scope conditions cited at use (the FT.14→FT.22/FT.24 pattern). No citation of Part 4. 0.1 Conventions and units. Bits regime: $D_{\text{tick}}(n) = n{+}1$ bits [CT0.5]; one pair channel supplies $C_{\text{cap}} = 16$ register directions per tick [CT0.6, D0.9] $= 4$ bits of Shannon content [C10: register count $\neq$ bit throughput]; all entropy/flux bounds in nats via $\ln 2$ [C9]. $\widehat{ds}^{\,2}$ is dimensionless [D1.6, CT1.19a]; $\mathfrak{h}$ is the single free scale [D2.9, CT2.11]. Geodesic and continuum-limit forms inherit CT2.6's hypotheses (i)–(iii) at the point of use. 0.2 Resolution rules. (i) This canon carries no ledger citations: every citation resolves to a tag of Parts 1–2, a Census cell, or a [math]/[machine] justification. (ii) Scope tokens: "Comm-min", "Eff" mark the fixed-phase read of the tiers that carry them; both premises are dissolved at corpus level; a tag carrying them retains its fixed-phase scope, and the dynamic-phase translation is: quantization reads apply to the antialigned content ($\Theta \to \Theta_{\mathcal{P}}$ [CT1.19b]), with the uniform-antialigned specialization $\mathcal{P} = E_{\mathcal{R}}$ reproducing the literal forms [Def-EM.20, EM.20]. (iii) "M1/M2" are the materialization scope clauses of Def-H.20 (spectral hypotheses, constitutive of the Platonic cards). (iv) The interface surface of this part is Def-H.35–Def-H.43 and the constitutive/scope clauses cited therein. 0.3 Machine verification. `checks/run_all.py`; the per-tier content checkers `checks/verify_H_*.py` (exact arithmetic, no floating point); the era-structure checker `checks/verify_era_structure.py`.

3.1 §Cal-F — Free internal parameters of the fermion era (P4-class)

Policy: Part 1 §E (P4: free internal configuration parameters, ¬constants of the theory; P5: no external identification; P1: $\mathfrak{h}$ the single free dimensionful scale). Every entry: dimensionless internal configuration data unless a dimension is stated; no value is fitted to external data; any numerical use beyond the loci listed requires a logged declaration first.

parameter	locus	dimension	consumers
$r = x_+/x_-$ (Janus rate ratio; equivalently $x_\pm$)	Def-H.26; FT.68(5); FT.70–FT.71	$\mathfrak{q}^0$	the flicker ensemble; the equilibrium form $p = x_+/(x_+ + x_-)$ is derived [FT.108(4)], the ratio is the datum

parameter	locus	dimension	consumers
$\hat{\mu}^2$	Def-H.37(1)	$\mathfrak{q}^2$ [EW.33 packaging; D2.9]	the leading-order pole read [FT.33]
$\hat{\mu}'^2$	Def-H.37(2)	$\mathfrak{q}$ [FT.54(4)]	the exact pole read [FT.54(4); Census FT.76]
α (Ansatz-B formal temperature)	Def-H.28	$\mathfrak{q}^0$	class masses $m_B = (K - \alpha \ln \Omega) \ln 2/\mathfrak{h}$; no value declared; Ansatz-B results stand as exact (K, a, b) classes
μ (family-channel overall scale)	FT.113; FT.119	—	channel spectra; every ratio is μ-free
μ_ν (rung-one channel scale)	FT.143	—	the rung-one spectrum; ratios exact, scale undeclared
$t, u, x_\pm, y_d, z_d, \tau_\mathcal{R}, \tau_\mathcal{V}$ (formal fugacities)	Def-H.32(3)	$\mathfrak{q}^0$	formal indeterminates over $\mathbb{Q}$; specializations marked at use [FT.108(4); Cor L0a]

3.2 Tier F.1 — Information Flux and Measurement Primitivity

Definition 3.1 (Def-H.1 — Observer–subnetwork channel). Def. $o \in \mathcal{Q}$ a single node; $\Sigma \subset \mathcal{Q}$, $\Sigma \cap \{o\} = \varnothing$. The channel is

$$\gamma(o, \Sigma) := \big\{e \in \mathcal{R} : \ e \text{ has one endpoint } o, \text{ the other in } \Sigma\big\} \ \cup \bigcup_P E(P),$$

the union over every valid real-edge path P from o to Σ lying entirely in $\mathcal{Q}\setminus\Sigma$ except its terminal edge. Always:

$$\gamma(o, \Sigma) \cap \big(\Sigma \setminus \partial\Sigma\big) = \varnothing \qquad \text{(no internal edge of } \Sigma \text{ is ever a member)}.$$

Hop and tick counts over an interval: $\delta h(\gamma) := \sum_{e\in\gamma} \delta h(e)$ [D0.6]; $\delta\tau(\gamma)$ identically from D0.7. Bookkeeping extension of A2 by summation over the boundary edge set — ¬new primitive (uses A2, D0.6, D0.7 only).

Rmk (no internal-reaching reading). The alternative reading of the second clause — once o touches Σ, all of Σ's internal edges qualify — is excluded on citation-discipline grounds: CT1.29 defines $\partial\mathcal{B}$ as a separating real-edge surface, and CT4.9/CT4.8 prove from A2 that externally accessible content is capped by boundary registers alone. Def-H.1 extends A2 without new primitives, so it cannot overturn closed theorems. Reductio: the internal reading collapses $I_{\text{hid}} \to 0$ for any well-connected Σ with one external contact, contradicting CT4.8's generically-nonzero area law.

Definition 3.2 (Def-H.2 — Physical (metric) channel separation). Def. In a region where $g_{\mu\nu} = \Omega^2 \bar{g}_{\mu\nu}$ [CT1.12] is defined:

$$\Delta\ell(\gamma) := \int_\gamma \sqrt{\bar{g}_{ij}\, dx^i\, dx^j}\ \Omega; \qquad \Delta t(\gamma) := \int_\gamma \Omega\, dx^0.$$

Rmk (role of D^+). $\Omega = e^{\alpha_\Omega a_{00}}$ [CT1.12] and $F_{D^+} = \Lambda\mathbb{I}$ acts along the same central E_0 direction [CT1.25]: D^+'s contribution to channel separation is carried inside Ω; ¬additive term. Built entirely from CT1.12's proven components; ¬∃ new free parameter beyond $(\alpha_\Omega, \beta, \varkappa)$ [P4].

Definition 3.3 (Def-H.3 — Extractable information). Def.

$$I_{o\leftarrow\Sigma}(\gamma) := \sup\big\{\mathsf{E}_{\text{tr}}(\gamma') : \ \gamma' \subseteq \gamma \text{ valid transit configuration over the interval}\big\},$$

the supremum of transit entropy carried [EM.7: $\mathsf{E}_{\mathrm{tr}} = \delta h \ln 2$, extended additively over γ]. The identification of extractable information with carried transit entropy is constitutive of this definition; coherence with the computed cards: channel payload stays FT.1-bounded [FT.36(2)], tuple labels are superselection data [Def-H.15(4)].

Theorem 3.1 (FT.1 — Information flux bound). Thm. (observer-channel form; the chain-level bound itself — jaws, entropy read, threshold, sharpness — is CT0.6a (Part 1)): (1) Observer-channel extension. With $\mathsf{B}(\gamma) :=$ CT0.6a's bound evaluated on γ's hop/tick counts [D0.6, D0.7]:

$$I_{o\leftarrow\Sigma}(\gamma) \;\le\; \mathsf{B}(\gamma)$$

[Def-H.3 : the sup over valid transit subconfigurations inherits the memberwise bound]

(2) Geodesic gauge form. Geodesic γ, slowly varying $\Omega, \bar{g}_{ij}$: $\delta h = \Delta\ell/(\Omega\,\mathfrak{h})$, $\delta\tau = \Delta t/\Omega$ [Def-H.2], hence

$$I_{o\leftarrow\Sigma}(\gamma) \;\le\; \frac{1}{\Omega}\,\min\Big(\frac{\Delta\ell(\gamma)}{\mathfrak{h}},\; 4\,\Delta t(\gamma)\Big)\ln 2.$$

Req. Def-H.1–H.3; CT0.6a (Part 1); D0.6, D0.7; CT2.6 step 5 (C, piecewise smoothness).

Pf.

$$(1):\ \text{Def-H.3}:\ I_{o\leftarrow\Sigma}(\gamma)$$

$$= \sup\big\{\mathsf{E}_{\mathrm{tr}}(\gamma') :\ \gamma' \subseteq \gamma \text{ valid transit configuration over the interval}\big\};$$

$$\text{each } \gamma' \text{ is an admissible transit path configuration}:\ \mathsf{E}_{\mathrm{tr}}(\gamma') \le \mathsf{B}(\gamma')\ [\text{CT0.6a}]$$

$$\text{the bound is monotone in } (\delta h, \delta\tau):\ \gamma' \subseteq \gamma \implies \mathsf{B}(\gamma') \le \mathsf{B}(\gamma) \implies \sup \le \mathsf{B}(\gamma)$$

$$(2):\ \text{substitution } \delta h = \Delta\ell/(\Omega\mathfrak{h}),\ \delta\tau = \Delta t/\Omega\ [\text{Def-H.2; CT2.6 step 5}] \implies \text{physical form.}\ \blacksquare$$

Phys. Knowledge through a channel is squeezed by two independent jaws — total edge entropy (spatial) and per-tick capacity (temporal). The jaws and their sharpness are pure ledger arithmetic at the foundation layer (CT0.6a); what this theorem adds is the observer-channel sup read (Def-H.1–H.3 layer) and D^+'s $1/\Omega$ rescaling of the physical form.

Theorem 3.2 (FT.2 — Exhaustiveness of the real-edge measurement primitive). Thm. Within A1–A9, the only axiom-level mechanism increasing o's information about Σ is the addition of valid real edges to $\gamma(o, \Sigma)$ [Def-H.1]. Hence FT.1's bound is exhaustive over every axiomatic route to knowledge, and any two coarse-graining procedures on the same γ yield the same bound.

Req. A1–A9; C11, C12; D0.10; Def-H.1.

Pf.

$$\text{A2}: \; C(p,q) = 1\mathfrak{q} \text{ is the only channel-defining pair relation;}$$

$$\text{A4}: \; \mathcal{R}\text{-membership is the sole knowability criterion}$$

$$\text{A5 [C11}: \; \text{node-level } \rho \mapsto U\rho U^{\dagger},\; U \sim \mu_{\star}(\mathrm{SU}(2))] : \; \text{undriven single-node evolution;}$$

$$U \notin \mathfrak{L} : \; \text{structurally unrecorded} \implies \neg\text{readable}$$

$$\text{A6 [D0.3}: \; \mathcal{T} \text{ on operators; variance clause reserved}] : \; \text{internal map on an existing } \rho;$$

$$\neg\exists \text{ new inter-node relation}$$

$$\text{A7}: \; \phi : E_{\mathcal{R}} \to \Phi \text{ defined only on existing real edges} : \; \text{structure carried by an edge,}$$

$$\neg\text{a means of establishing one}$$

$$\text{A8 [D0.10, C12}: \; Z \text{ sums over } \mathfrak{C}(\mathcal{R})] : \; \text{classifies, weights, sums existing configurations;}$$

$$\neg\text{manufactures edges}$$

$$\text{A9}: \; \text{global solution-membership condition} : \; \text{meta-level, } \neg\text{per-observer channel}$$

$$\implies \neg\exists \text{ route to } o\text{'s knowledge of } \Sigma \text{ bypassing real-edge membership } [\text{A2, A4}]. \; \blacksquare$$

Phys. Measurement in this corpus has exactly one primitive: the real edge. Every claim about observation, information and hiding reduces to channel edge-counts — which is what makes the fermion program computational.

Theorem 3.3 (FT.3 — Unresolvability of D^--sourced subnetwork structure). Thm. Σ, Σ' with $(\delta h, \delta\tau)$-identical channels to o, Σ containing a D^--configured node ($n \geq 4$, CT4.1) and Σ' none:

$$I_{o\leftarrow\Sigma}(\gamma) = I_{o\leftarrow\Sigma'}(\gamma); \qquad I_{o\leftarrow\Sigma''}(\gamma) = I\big(\delta h(\gamma), \delta\tau(\gamma)\big) \;\; \forall \Sigma'' \in [\gamma],$$

$$[\gamma] := \big\{\Sigma'' : \; \gamma(o, \Sigma'') \text{ has hop/tick counts } (\delta h(\gamma), \delta\tau(\gamma))\big\}$$

$$(\text{exact and total over } [\gamma]).$$

Req. Def-H.1, Def-H.3; EM.7 (Part 2); FT.1.
Pf.

$$\text{EM.7} : \; D^-\text{-sourced edges}$$

$$= \text{ordinary real edges} : \; \text{identical received data } (\neg\exists \; D^-\text{-emission rule})$$

$$\text{FT.1} : \; I \text{ bounded by a function of } (\delta h, \delta\tau) \text{ only} : \; \neg\text{reference to internal node count/topology/rung structure}$$

$$\text{hypothesis} : \; (\delta h, \delta\tau) \text{ identical} \implies \text{bounds identical};$$

$$\text{Def-H.3} : \; I$$

$$= \sup \text{ over configurations constrained only by these bounds}$$

$$\implies I \text{ identical on } [\gamma]. \; \blacksquare$$

Cor FT.3.1 (computational form). For any candidate fermion terminus f:

$$I_o(f) \;\leq\; \min\big(\delta h(\gamma(o, f)),\; 4\,\delta\tau(\gamma(o, f))\big)\ln 2,$$

and this bound is the only quantity distinguishing f from any terminus sharing its channel signature.

Phys. Locked (D^-) structure is invisible from outside by construction: two subnetworks differing arbitrarily in locked content but sharing a channel signature are observationally identical. Darkness is indistinguishability, exactly.

3.3 Tier F.2 — The Fermion Definition

Definition 3.4 (Def-H.4 — Fermion). Def. Let c be a Janus-closed oriented chain [D1.4]: $\partial c = 0 \wedge \#J^+ = \#J^-$ — a bound chain, hence fermionic content by D1.5 — realized at rung $n(v) \geq 2$. A fermion is the terminus (basepoint) $v \in \{s(c), t(c)\}$ of c: a bound-chain endpoint, at either end, for any admissible $\delta h(c)$, characterized externally by whatever transit (bosonic-content, D1.5) channels connect v to an observer [Def-H.1, Def-H.3] and by which aspects of A8 act nontrivially there.

Rmk (floor: $\delta h(c) \geq 2$, $n(v) \geq 2$). A $\delta h = 1$/rung-1 instance is excluded by Parts 1–2:

$$\text{A1}: \; p \neq q \implies \neg\exists \text{ self-loops}; \qquad \text{D1.4}: \; \text{closed} \iff \partial c = t(c) - s(c) = 0 \implies s(c) = t(c) = v$$

$$\implies \delta h(c)$$

$$= 1 \text{ closed chain}$$

$$= \text{self-loop}$$

$$= \perp \implies \delta h(c)$$

$$\geq 2 \text{ (minimal closed chain} = \text{2-cycle } p \to q \to p)$$

$$\text{closed chain based at } v \implies v \text{ has } \geq 1 \text{ in- and } \geq 1 \text{ out-edge} \implies n(v) \geq 2 \text{ [D0.8]}$$

$$\text{D1.5}: \; \text{open propagating chains}$$

$$= \text{transit}$$

$$= \text{bosonic content} \implies \text{rung-1 nodes on open chains are } \neg\text{fermion termini.}$$

There is one object — the bound-chain terminus — varying only in $\delta h(c) \geq 2$; FT.14's hypothesis $\delta h(c) \geq 2$ is canonical, not ad hoc. No new primitive: $s(c), t(c)$ [D1.4], Janus-closed [D1.3], bosonic aspect [A8]; this definition fixes the referent of fermion in the corpus; consumers: FT.4, FT.28, FT.144(1). Terminology per D1.5: the closed chain is the fermionic content; what is bosonic/transit is the external channel connecting the terminus to an observer (Def-H.1/Def-H.3), not the chain itself. The floor is computed on in Tier F.8 (FT.24).

Theorem 3.4 (FT.4 — External state characterization of a fermion terminus). Thm. v a fermion [Def-H.4], $\gamma(o, v)$ per Def-H.1:

$$I_o(v) \leq \min\big(\delta h(\gamma(o,v)),\ 4\,\delta\tau(\gamma(o,v))\big)\ln 2,$$

and this bound is the entire externally available description: v, v' with the same $(\delta h, \delta\tau)$ are observationally identical to o.

Req. Def-H.1, Def-H.4; FT.1, FT.3.

Pf.

$$\text{Def-H.4}: \ v = \text{single vertex} = \text{degenerate case } \Sigma = \{v\} \implies \text{FT.1 applies}$$

$$\text{FT.3/Cor FT.3.1}: \ I \text{ constant on } [\gamma];$$

$$\text{Def-H.4 constrains } \neg\text{internal structure}$$

$$\implies v,$$

$$v' \text{ with equal signature indistinguishable.} \ \blacksquare$$

Cor FT.4.1 (state-counting ceiling). Externally distinguishable states through a fixed channel signature:

$$N_{\text{states}} \leq 2^{\min(\delta h,\ 4\,\delta\tau)} \qquad \big(= \min(\delta h, 4\,\delta\tau) \text{ bits of externally resolvable state}\big).$$

¬yet a derivation of spin, generation number, or any specific quantum number — a ceiling on the labels any channel could support.

Phys. A fermion seen from outside is its channel signature and nothing more: the corpus's measurement theory turns "particle" into "equivalence class of subnetworks under $(\delta h, \delta\tau)$".

3.4 Tier F.3 — Information Flux as Rate

Definition 3.5 (Def-H.5 — Information flux). Def.

$$\mathcal{F}(\gamma) := \frac{I_{o\leftarrow\Sigma}(\gamma)}{\delta\tau(\gamma)} \quad [\text{nats/tick; C9}];$$

$$\mathcal{F}_{\text{phys}}(\gamma) := \frac{I_{o\leftarrow\Sigma}(\gamma)}{\Delta t(\gamma)} = \frac{\mathcal{F}(\gamma)}{\Omega}$$

$$[\Delta t = \Omega\,\delta\tau,\ \Omega \text{ constant along } \gamma,\ \text{Def-H.2}].$$

Rmk (Ω-dependence). $\mathcal{F}$ is Ω-invariant (pure graph rate); $\mathcal{F}_{\text{phys}} \propto 1/\Omega$; Ω is set by D^+ [CT1.12, CT1.25] — the precise mechanism coupling information flux to geometry.

Definition 3.6 (Def-H.6 — Information flux density). Def. $N_\perp$ parallel real edges (transverse width); $A_\perp = N_\perp \mathfrak{h}^2$ [CT1.29]; $\mathcal{F}_\rho(\gamma) := \dfrac{\mathcal{F}(\gamma)}{A_\perp(\gamma)}$ (local, per unit area).

Theorem 3.5 (FT.5 — Flux bound (rate form of FT.1)). Thm.

$$\mathcal{F}(\gamma) \le \min\Big(\frac{\delta h(\gamma)}{\delta\tau(\gamma)} \ln 2,\ C_{\text{cap}} \ln 2\Big) = \min\Big(\frac{\delta h}{\delta\tau},\ 4\Big) \ln 2;$$

$$\mathcal{F}_{\text{phys}}(\gamma) \le \frac{1}{\Omega} \min\Big(\frac{\delta h}{\delta\tau},\ 4\Big) \ln 2.$$

Req. FT.1; Def-H.2, Def-H.5.

Pf. Divide FT.1 by $\delta\tau$; both constraints apply to every member of the supremum set [FT.1]; convert via $\Delta t = \Omega\,\delta\tau$. ■

Phys. The hop-rate caps flux combinatorially; the 4-bit channel caps it per tick; geometry (Ω, i.e. D^+) rescales the physical rate without touching the combinatorial one.

Theorem 3.6 (FT.6 — Conformal invariance of combinatorial flux). Thm. $g_{\mu\nu} \mapsto \Omega'^2 g_{\mu\nu}$: $\mathcal{F}(\gamma)$ invariant; $\mathcal{F}_{\text{phys}} \mapsto \mathcal{F}_{\text{phys}} \cdot \Omega/\Omega'$.

Req. Def-H.5; CT0.9 (Part 1); D0.7.

Pf. $\mathcal{F} = I/\delta\tau$; I bounded by edge counts [CT0.9], $\delta\tau$ a tick count [D0.7]: both Ω-free [combinatorial]; $\mathcal{F}_{\text{phys}} = \mathcal{F}/\Omega$ [Def-H.5]. ■

Phys. The ledger rate is conformally blind; only the physical rate feels D^+ — information bookkeeping is pre-geometric.

Theorem 3.7 (FT.7 — Universal flux density bound). Thm.

$$\mathcal{F}_\rho(\gamma) \le \frac{\ln 2}{\mathfrak{h}^2} \quad \text{per tick;} \qquad \mathcal{F}_{\rho,\text{phys}}(\gamma) \le \frac{1}{4\,\Omega\,G_N} \quad \text{per unit proper time.}$$

Req. CT0.9 (Part 1); EM.7 (Part 2); CT1.29, CT2.9 (Part 2); Def-H.2, Def-H.6.

Pf.

$$\text{each of } N_\perp \text{ parallel edges} \le \ln 2 \text{ per tick [CT0.9, EM.7]} \implies \mathcal{F} \le N_\perp \ln 2$$

$$A_\perp = N_\perp \mathfrak{h}^2 \text{ [CT1.29]} \implies \mathcal{F}_\rho \le \frac{N_\perp \ln 2}{N_\perp \mathfrak{h}^2} = \frac{\ln 2}{\mathfrak{h}^2} \qquad (N_\perp \text{ cancels : universal})$$

$$\mathfrak{h}^2 = 4G_N \ln 2 \text{ [CT2.9]}; \quad /\Omega \text{ [Def-H.5]} : \quad \mathcal{F}_{\rho,\text{phys}} \le \frac{\ln 2}{\Omega \cdot 4G_N \ln 2} = \frac{1}{4\,\Omega\,G_N}. \blacksquare$$

Cor FT.7.1 (Bekenstein rate bound). The universal flux-density bound is the Bekenstein entropy bound expressed as a rate: maximal information flux density $1/(4G_N)$ per unit proper time, up to the conformal factor; D^+ modulates the ceiling locally.

Phys. One bit per hop-area per tick — universally, independent of channel width — and in physical units the quarter-G_N law appears as a rate. No dependence on the detailed form of $G(e)$ at any rung: safe from the rung-3 edge-structure question.

Theorem 3.8 (FT.8 — Flux-mediated coarse-graining connection). Thm. Coarse-graining $\Sigma \to v$ is geometry-dependent: surviving information is bounded by the boundary flux, which depends on the geometry along γ (via Ω, $\mathfrak{h}$).

Req. FT.5, FT.7; Def-H.5, Def-H.6.

Pf. Coarse-graining := replace Σ's internals by a terminus, retaining only what flows through $\gamma(o, v)$; retained rate $\leq \mathcal{F}_{\text{phys}}(\gamma)$ [FT.5], density $\leq \ln 2/(\Omega\mathfrak{h}^2)$ [FT.7] — both geometry-dependent $\Longrightarrow$ same Σ, different ambient geometry $\Rightarrow$ different surviving content: coarse-graining is geometry-mediated, $\neg$ purely combinatorial. ■

Phys. This is the bridge the fermion construction needs: what a particle "keeps" of its subnetwork under coarse-graining is set by the geometry of the channel through which it is seen.

3.5 Tier F.4 — Rung-Dependent Edge Structure

Definition 3.7 (Def-H.7 — Rung-dependent edge data). Def. $G^{(n)}(e)$ rung-dependent; EM part [D3.5, EM.16] for $n \geq 2$:

$$G^{(n)}_{\text{EM}}(e) = \sigma_e \exp\big(i\,\alpha^{(n)}_{\text{EM}}\,\widehat{ds}^{\,2}(e)\big), \qquad \alpha^{(n)}_{\text{EM}} = \pi \quad \text{[FT.9; EM.20 under Comm-min + Eff]}.$$

On-shell/off-shell scope [FT.22–FT.25; CT1.19a]. On edges of materialized minimal bound chains — and, pending the completion argument, on on-shell edges generally: $\widehat{ds}^{\,2}(e) \in \mathbb{Z}$ [FT.24], hence the geometric factor $e^{i\pi\widehat{ds}^{\,2}(e)} = (-1)^{\widehat{ds}^{\,2}(e)} \in \{\pm 1\}$ and $G(e) \in \{+1, -1\}$; on every materialized closed chain, the loop product is $+1$ [FT.22: $\Theta(c) \in 2\mathbb{Z}$]. On generic off-shell configurations in Z's summation domain: $\widehat{ds}^{\,2}(e)$ is a generic $O(1)$ real [CT1.19a] and $G(e)$ retains the full phase $e^{i\pi\widehat{ds}^{\,2}(e)}$ — materialization selects the integer sublattice, exactly as CT2.5 selects discrete spectra from continuous operator families.

rung	active stack aspects	edge data
2	EM	σ_e (sign) + $(-1)^{\widehat{ds}^{\,2}(e)}$ (geometric parity, on-shell) [FT.9, FT.24]
3	EM + strong	σ_e + $\widehat{ds}^{\,2}(e) \in \mathbb{Z}$ (on-shell) + $\tau(e)$: colour transport label [CT3.11; Def-H.17] — $\neg\exists$ independent strong phase [FT.29(2)]
≥ 4	EM + strong + D^-	+ D^- saturation content [D4.1, CT4.1] (locked)

On the minimal real edge alone ($\widehat{ds}^{\,2}(e) = 1$ [D1.1, Comm-min]), $G(e) = -\sigma_e$ needs only EM.20/FT.9 — no Tier F.8 input. Rung-3 strong factor: $G^{(3)}(e) = G^{(2)}(e) \otimes \tau(e)$ [Def-H.17, FT.29] — the slot algebra's $\mathfrak{u}(1)$ centre is the EM phase already carried by $G^{(2)}$ [CT3.7], so the strictly new strong content is the traceless $\mathfrak{su}(3)$ label, observable only in singlet sealings [FT.29(3)].

Theorem 3.9 (FT.9 — EM part of edge weight is rung-stable). Thm. $\alpha_{\text{EM}}^{(n)} = \alpha_{\text{EM}}^{(2)} = \pi \quad (n = 2, 3, \geq 4;$ under EM.20's scope tokens Comm-min + Eff).

Req. EM.17, EM.18, EM.19, EM.20 (Part 2); CT0.14 (Part 1); CT3.7, CT3.10 (Part 2); D4.1.

Pf.

$$\text{EM.20 [Comm-min + Eff]} : \ \alpha_{\text{EM}}^{(2)} = \pi$$

$$\text{EM.18} : \ \alpha^{(3)} = \alpha^{(2)} :$$

$$Y_{\text{asp}} F_{\text{strong}} Y_{\text{asp}} = \mathcal{P}^{(\text{EM})} F_{\text{strong}} \mathcal{P}^{(\text{EM})} = 0 \ [\text{CT0.14} : \ \mathcal{K}^{(\text{strong})} \perp \mathcal{K}^{(\text{EM})}]$$

$$\text{independently} : \ \mathfrak{T}(T_a) = 0 \ [\text{CT3.7}] \implies \mathfrak{T}(F_{\text{strong}}) = 0 \ [\text{CT3.10}]$$

$$\text{EM.19} : \ n$$

$$\geq 4 : \ \ Y_{\text{asp}} F_{D^-} Y_{\text{asp}}$$

$$= 0 \ [\text{D4.1} : \ F_{D^-} \text{ acts only on } n \geq 4; \ \text{CT0.11} : \ \mathcal{K}_2 \ \neg\text{oversubscribed}; \ \text{CT0.14} : \perp]$$

$$\text{EM.17} : \ \alpha^{(n)} = \mathfrak{T}\big(Y_{\text{asp}} \mathrm{F}_M^{(n)} Y_{\text{asp}}\big) / \mathfrak{T}\big(Y_{\text{asp}}\big);$$

$$Y_{\text{asp}} \mathrm{F}_M^{(n)} Y_{\text{asp}} = Y_{\text{asp}} \mathrm{F}_M^{(2)} Y_{\text{asp}} \implies \alpha^{(n)} = \alpha^{(2)} = \pi. \ \blacksquare$$

Phys. The binary geometric phase of every real edge is the same π at every accessible rung: colour and saturation add structure without touching the electromagnetic bit.

Theorem 3.10 (FT.10 — Confinement is a global sum-restriction). Thm.

$$Z^{(3)}(\mathcal{R}) = \sum_{C \in \mathfrak{C}(\mathcal{R}):\ \text{colour-sealed}} \ \sum_{\sigma} \mathcal{J}(C) \prod_{e \in E(C)} G^{(3)}(e) \ (-1)^{\aleph(\sigma, C)}$$

$$[\text{D0.10}, \ \text{C12}];$$

confinement restricts the summation domain; ¬modification of any individual edge weight.

Req. CT3.17, CT3.18, CT3.22 (Part 2); CT2.3, CT2.5 (Part 2); FT.9; D0.10.

Pf.

$$\text{CT3.18} : \ \mathcal{P}_M^{(3)} \text{ admits only colour-sealed } C \ (\delta^i{}_j, \varepsilon_{ijk}, \varepsilon^{ijk} \text{ contractions}, \ \text{CT3.17});$$

$$\text{CT3.22} : \ \neg\exists \text{ isolated triplet pole}$$

$$\text{free-index } C \notin \operatorname{Im} \mathcal{P}_M^{(3)};$$

$$\text{CT2.5 : poles}$$

$$= \text{eigenvalue conditions on } \mathrm{F}_M^{(3)};$$

$$\text{CT2.3 : } \mathrm{F}_M^{(3)}\big|_{\mathcal{K}_3^{\perp}}$$

$$= 0 \implies \text{excluded } C : \neg\text{pole}$$

$$G^{(3)}(e)$$

$$= G^{(2)}(e) \otimes |i(e)\rangle\text{-label : per-edge weight unmodified [FT.9]} \implies \text{confinement}$$

$$= \text{domain restriction. } \blacksquare$$

Phys. Confinement acts globally — on which configurations exist in the sum — not locally on edge amplitudes: the partition function itself only contains colour singlets.

3.6 Tier F.5 — Spin–Charge Decomposition via Complete Force Operator

Theorem 3.11 (FT.11 — Complete force operator charge audit). Thm. Per-aspect charge determination for F [A8]:

aspect	independent charge?	basis
F_{EM}	yes	EM.9 : $q_{\text{EM}} = (\sigma_s - \sigma_t)/2 \in \{-1, 0, +1\}$, additive end-charge
F_{strong}	no (confined)	CT3.22; $\mathfrak{T}(T_a) = 0$ [CT3.7]; $\mathfrak{T}(F_{\text{strong}}) = 0$ [CT3.10]
F_G	yes	D2.5 : $m^2 =$ pole displacement
F_H	yes (via mass)	EW.13 : $H = h\Sigma$; EW.20, EW.33
F_{D^+}	no	D4.3 : $\Lambda\mathbb{I}$, ¬config.-dependent
F_{D^-}	no	CT4.4 : ¬gauge structure
$F_{J^\pm}$	no	CT0.7 : seals, ¬charges
F_W	no	FT.12 : no weak self-charge

Req. A8, C8; CT0.7, CT0.12 (Part 1); D2.5, D4.3; CT3.7, CT3.10, CT3.22 (Part 2); CT4.4 (Part 2); EW.13, EW.20, EW.33 (Part 2); EM.9 (Part 2); FT.12.

Pf.

$$\text{A8 [C8] : F}$$

$$= \mathrm{F}_{(+,\mathcal{R})} \oplus \mathrm{F}_{(-,\mathcal{R})} \oplus \mathrm{F}_{(+,\mathcal{V})} \oplus \mathrm{F}_{(-,\mathcal{V})} \oplus \mathrm{F}_{\mathrm{st}};$$

$$\mathrm{F}_{(-,\mathcal{R})}$$

$$= F_{W^+} \oplus F_{W^-} \oplus F_Z$$

CT0.12 : pair aspects exhaust 16

$$= 2^{C_{\mathrm{cap}}} \text{ states } (3{+}3{+}5{+}3{+}1{+}1);$$

stack aspects act on vertex/slot registers [CT0.12 Rmk]

rows : as cited (each row one frozen-corpus tag); F_W : FT.12. ■

Phys. A complete audit of which forces can carry an independently readable charge: only EM (boundary-readable), gravity and Higgs (as mass). Everything else is structurally chargeless — the input to the spin–charge split.

Theorem 3.12 (FT.12 — Weak has no self-charge). Thm. The weak sector produces ¬independent abelian charge; its charge content reduces entirely to $Q = W_3 + Y$ (the EM generator).

Req. EW.3, EW.22, EW.36, EW.36a (Part 2); D3.4.

Pf.

$$\text{EW.22 : } \ker M^2 = \mathrm{span}\{Q\}_{(W_3,Y)}, \;\; Q := W_3 + Y \;\; \text{(1-dim)}$$

$$\text{EW.36 : } \mathrm{spec}\, Q|_{\mathcal{H}_{\mathrm{v}}}$$

$$= \{0, +1\};$$

$$\omega \text{ [D2.6] : } \{0, -1\};$$

alphabet

$$= \{-1, 0, +1\}$$

$$= \text{D3.4 end-charge alphabet}$$

$$\text{EW.36a : } Q = W_3 + Y; \quad W_3 \text{ alone } \neg\text{charge}; \quad Q = \text{EM generator}$$

$$\mathfrak{su}(2) \text{ nonabelian},$$

$$Z(\mathfrak{su}(2))$$

$$= \{0\} \text{ [EW.3 : } [W_i, W_j] = i\epsilon_{ijk} W_k] \implies \neg\exists \text{ central } \mathfrak{u}(1) \subseteq \mathfrak{su}(2) \implies \neg\text{independent abelian weak charge. } \blacksquare$$

Phys. Weak carries no charge of its own — its only abelian direction is electromagnetism. Chargelessness of the weak sector is a Lie-algebra fact (centre-free $\mathfrak{su}(2)$), not a convention.

Theorem 3.13 (FT.13 — Weak = spin (algebraically)). Thm. $W_i = \frac{1}{2}\big(\sigma_i^{(p)} + \sigma_i^{(q)}\big)$ = total-spin generators of the pair; Casimir eigenvalues $2 = j(j+1)|_{j=1}$ on $\vee^2$, 0 on $\wedge^2$; $W_i \in \mathcal{E}_{+-}$ (weak sector). Hence weak ≡ spin, algebraically.

Req. CT0.7a, CT1.1, CT1.2 (Part 1).

Pf. Identities [CT1.2 (definition), CT0.7a (Casimir), CT1.1 (sector)]. ■

Phys. With FT.12 (no self-charge), the weak sector's entire coarse-grained contribution is spin — a positive identification, not a residual.

Definition 3.8 (Def-H.8 — Spin (emergent weak contribution)). Def. For a fermion terminus v [Def-H.4, $\delta h(c) \geq 2$]: spin := the coarse-grained emergent-spacetime property arising from the weak sector's contribution to c. Justification: weak has no self-charge [FT.12] ⇒ its contribution cannot be charge; W_i are total-spin generators [FT.13] ⇒ the contribution is spin. Rmk (coarse-graining hierarchy). Pre-dimensionally: spin = interior edge phase carried by the weak sector. Coarse-grained to the terminus: interior hidden [FT.3] ⇒ cumulative interior phase manifests as an intrinsic localized quantum number. Further coarse-grained along a geodesic: angular momentum.

Theorem 3.14 (FT.14 — Spin–charge decomposition). Thm. c oriented chain, $\delta h(c) \geq 2$ [Def-H.4], proper separation $\Delta\ell(c)$ [CT1.12]. Decompose $N_{\text{visible}} = \delta h(c)$ bits under the complete force audit [FT.11]:

$$(1)\ \ N_{\text{spin}} = N_{\text{weak}};$$

$$(2)\ \ N_{\text{charge}} = N_{\text{visible}} - N_{\text{weak}};$$

$$(3)\ \ f_{\text{charge}} = \frac{N_{\text{charge}}}{N_{\text{visible}}}, \quad f_{\text{spin}} = \frac{N_{\text{weak}}}{N_{\text{visible}}};$$

$$\mathfrak{h} \to 0,\ \Delta\ell \text{ fixed [CT1.19]} : \quad f_{\text{charge}} \to 0, \quad f_{\text{spin}} \to 1.$$

Req. A1, A2, A4; C8, C9, C10; CT0.1, CT0.6, CT0.7, CT0.9 (Part 1); CT1.12, CT1.19, CT1.25 (Part 2); D3.5; EW.13, EW.17 (Part 2); EM.9, EM.16 (Part 2); FT.9; FT.22–FT.24 (on-shell line-element

quantization — the parity-read step below is scoped to materialized edges); FT.11, FT.12, FT.13; Def-H.8; rung-2 decomposition (Def-H.8 attribution clause; the counting half is derived — EM.9/D3.4 end registers). Table evaluated at $\Omega \equiv 1$ (stated simplification, not a general claim).

Pf.

$$\text{qubit} \to \text{bit}: \quad \text{A1}: \ \mathcal{H}_q \cong \mathbb{C}^2;$$

$$\text{A2}: \ C(p,q)$$

$$= 1\mathfrak{q};$$

$$\text{CT0.6}: \ C_{\text{cap}}$$

$$= 4 \text{ bits } (16 \text{ states});$$

$$\text{A4/CT0.1}: \text{ dichotomy};$$

$$\text{CT0.7}: \text{ reality}$$

$$= \text{swap grading}$$

$$\text{CT0.9}: \ \Phi$$

$$= \{0, \pi\},$$

$\sigma_e \in \{\pm 1\}$: 1 bit per real edge ($\mathbb{C}^2 \to \mathbb{Z}_2$ reduction : primordial node-level entropy cost);

$$\text{C9}: \ 1 \text{ bit}$$

$$= \ln 2 \text{ nats}$$

$$N_{\text{visible}} = \delta h(c) \text{ bits};$$

$$\text{Def-H.2 / CT1.19 (geodesic)}: \ \delta h(c) = \Delta \ell(c)/(\Omega\, \mathfrak{h}) \ \ [\Omega \equiv 1 \text{ in the table below}]$$

$$\text{D3.5/EM.16, FT.9 } (\alpha_{\text{EM}} = \pi), \text{ FT.24 } (\widehat{ds}^{\,2} \in \mathbb{Z} \text{ on-shell}):$$

$$G^{(n)}(e)$$

$$= \sigma_e$$

$(-1)^{\widehat{ds}^2(e)} \in \{\pm 1\}$: one transit bit per materialized edge, factored into two register reads

sign read $\sigma_e \to$ charge (additive boundary read, EM.9);

parity read $(-1)^{\widehat{ds}^2} \to$ holonomy (cumulative read; per-edge $\alpha = \pi$ [FT.9] with on-shell integrality [FT.24/FT.25])

bookkeeping : the per-edge visible datum is the product $G^{(n)}(e) \in \{\pm 1\}$ — one bit;

$N_{\text{visible}} = \delta h(c)$ counts exactly these [CT0.9], ¬the two factors separately

σ_e and $(-1)^{\widehat{ds}^2(e)}$ are its two factor registers [CT0.9 vs FT.24],

separately readable only where not cancelled by the other : signs at the ends [EM.9],

parity cumulatively [D1.7];

the full geometric ledger $\kappa_e \in \mathbb{Z}$ [FT.24] exceeds the binary weight

FT.11 : audit;

FT.12 : weak ¬charge;

FT.13 : $W_i = \text{spin} \implies N_{\text{spin}} = N_{\text{weak}}$ (positive identification)

$$N_{\text{charge}} = N_{\text{visible}} - N_{\text{weak}} \quad \text{(decomposition)}$$

metric dependence (general, Ω restored) : $\Delta\ell$

$$= \Omega\sqrt{\bar{g}_{ij}\Delta x^i \Delta x^j},$$

$$\Omega$$

$$= e^{\alpha_\Omega a_{00}},$$

$$a_{00} \leftrightarrow D^+ \text{ [CT1.12, CT1.25]}$$

$$\implies D^+ \text{ enters via } \Omega;$$

$$\delta h(c)$$

$$= \Delta\ell(c)/(\Omega\mathfrak{h}) \text{ throughout — the table's } \Omega$$

$$\equiv 1 \text{ is a simplification, not the general relation}$$

$$\mathfrak{h} \to 0,$$

$$\Delta\ell \text{ fixed : charge}$$

$$= \text{boundary effect } (O(1) \text{ edge data});$$

$$\text{spin}$$

$$= \text{bulk } (\delta h - O(1) \to \infty)$$

$$\implies f_{\text{charge}} \to 0,$$

$$f_{\text{spin}} \to 1$$

$$\text{rung-2 decomposition (Def-H.8 attribution; counting : FT.41(a), forward) : } N_{\text{weak}}$$

$$= \delta h(c) - 2 \text{ (interior edges)},$$

$$N_{\text{charge}}$$

$$= 2 \text{ (end bits; EM.9 reads exactly the two end signs)}$$

$\delta h(c)$	$\Delta\ell$ $(\Omega \equiv 1)$	f_{charge}	f_{spin}	content
2	$2\mathfrak{h}$	1	0	pure charge
3	$3\mathfrak{h}$	$2/3$	$1/3$	minimal spin
4	$4\mathfrak{h}$	$1/2$	$1/2$	equal
$\to\infty$	$\to\infty$	$\to 0$	$\to 1$	pure spin

(general Ω : multiply the $\Delta\ell$ column by Ω) ■

Phys. The visible bit budget of a bound chain splits into charge (its two ends — a boundary effect) and spin (its interior — a bulk effect): long chains are dominated by spin, and the continuum limit is a spin/holonomy limit. The split is a positive identification via the complete force audit, not a subtraction residue.

Definition 3.9 (Def-H.9 — Subnetwork coarse-graining projection). Def. Σ bounded subnetwork, designated terminus v [Def-H.4]. $\mathcal{M}(\Sigma)$:= valid microscopic configurations (signs σ_e [CT0.9], geometric data $\widehat{ds}^2(e)$ [D1.6] — integer on-shell [FT.24–FT.25], generic $O(1)$ real off-shell [CT1.19a] — subject to rung masks and Janus balance D1.3). Projection $\Pi_v : \mathcal{M}(\Sigma) \to \mathcal{Q}_v = (q, \mathrm{S}, m_{\text{hid}})$:

$$(1)\ \ q := \tfrac{1}{2}(\sigma_s - \sigma_t)\ [\text{EM.9}];$$

$$(2)\ \ E_{\text{tot}}(C) := |\Sigma|\ \ln 2,$$

$$|\Sigma| :$$

$$= \#\{\text{real edges touching } \Sigma,\ \text{internal} + \text{boundary}\}\ \ (\text{ledger-retained bits, locked content included [CT4.3]});$$

$$E_{\text{vis}}(\gamma) := \min\big(\delta h(\gamma),\ 4\,\delta\tau(\gamma)\big)\ln 2\ [\text{FT.1}]; \qquad I_{\text{hid}} := E_{\text{tot}} - E_{\text{vis}}; \qquad m_{\text{hid}} := I_{\text{hid}}/\mathfrak{h};$$

$$(3)\ \ \Theta_{\text{int}}(C) := \sum_{e\in\Sigma\setminus\partial\Sigma} \widehat{ds}^2(e); \qquad \text{Hol}_{\text{int}} := e^{i\pi\Theta_{\text{int}}} = (-1)^{\Theta_{\text{int}}} \in \{\pm 1\}$$

[on-shell $\Theta_{\text{int}} \in \mathbb{Z}$ — derived from materialization, not postulated : FT.25; $\alpha = \pi$: FT.9];

S : the $\mathfrak{su}(2)$ Casimir label $j(j+1)$ associated with Hol_{int} under the weak action [FT.13]

(the extraction rule is application-layer content).

Dimensional check. $[I_{\text{hid}}] = \mathfrak{q}^0$ (information); $[\mathfrak{h}] = \mathfrak{q}^{-1/2}$ [D2.9] $\Rightarrow [m_{\text{hid}}] = \mathfrak{q}^{+1/2}$ = energy [CT2.1] ✓ (machine: `verify_H.py`). Notation. m_{hid} — the informational mass read — is kept

symbolically distinct from the pole mass m of D2.5/EW.19 ($m^2 :=$ pole displacement): the two are different constructions, and their identification or relation is nowhere assumed in this document.

Theorem 3.15 (FT.15 — Coarse-graining invariance). Thm. $C_1, C_2 \in \mathcal{M}(\Sigma)$ sharing boundary signature $(\delta h, \delta\tau)$, boundary charge q, and Θ_{int}:

$$\Pi_v(C_1) = \Pi_v(C_2) \qquad \text{(invariance under internal rearrangement)}.$$

Req. Def-H.9; EM.9 (Part 2); FT.1; FT.13. The S clause additionally requires FT.25's on-shell scope for Θ_{int} (Hol_{int} well-definedness); the q, m_{hid} clauses do not.

Pf.

$$q = \tfrac{1}{2}(\sigma_s - \sigma_t) : \text{ boundary signs only [EM.9]} \implies q(C_1) = q(C_2)$$

$$m_{\text{hid}}$$

$$= (|\Sigma| \ln 2 - E_{\text{vis}})/\mathfrak{h} : \ |\Sigma| \text{ fixed volume datum;}$$

$$E_{\text{vis}}$$

$$= f(\delta h, \delta\tau) \text{ [FT.1]}$$

$$\implies m_{\text{hid}}(C_1)$$

$$= m_{\text{hid}}(C_2)$$

$$\text{S from } \text{Hol}_{\text{int}}$$

$$= (-1)^{\Theta_{\text{int}}} \text{ [FT.25]} : \ \Theta_{\text{int}}(C_1)$$

$$= \Theta_{\text{int}}(C_2) \implies \text{S}(C_1)$$

$$= \text{S}(C_2) \ \ (\text{same weak action}, \ \text{FT.13})$$

$\implies \Pi_v$ depends only on $(\delta h, \delta\tau, \sigma_s, \sigma_t, \Theta_{\text{int}})$: ¬internal topology, ¬phase distribution. ■

Cor FT.15.1 (self-averaging). $(q, \text{S}, m_{\text{hid}})$ are the unique self-averaging macroscopic observables of Σ: as temperature depends only on total kinetic energy, invariant under permutation of molecular velocities, the fermionic quantum numbers depend only on boundary channel capacity and total internal holonomy — invariant under permutation of individual internal edge phases.

Phys. A coarse-grained fermion forgets everything about its internal wiring except three numbers: its boundary charge, its hidden-information mass, and (on-shell, FT.25) a spin label built from total internal holonomy. Two microscopically different internal configurations that agree on boundary data and total holonomy are the same particle from outside — coarse-graining here plays exactly the role bulk thermodynamic variables play for a gas of molecules.

Theorem 3.16 (FT.16 — The single-vertex hidden-information gap is generic; strong's signature is contraction-incompleteness). Thm. v observed by a single neighbour through one incident edge ($\delta\tau \geq 1$):

$$I_{\text{hid}}(\{v\}) = (n(v) - 1)\ln 2 \qquad \text{(aspect-blind)};$$

¬strong-specific (nonzero already at $n = 2$, where strong $\notin \mathcal{A}(2)$ [CT0.13]); strong's distinguishing signature is structural: contraction-completeness [CT3.17/18/22], ¬ a bit count.

Req. CT0.6, CT0.9, CT0.13 (Part 1); CT1.29 (Part 2); CT3.17, CT3.18, CT3.22 (Part 2); D3.3, D3.9; Def-H.1, Def-H.9; FT.1, FT.10.

Pf.

$$\Sigma = \{v\},\ n(v) = n: \quad E_{\text{tot}} = n\ln 2 \text{ [Def-H.9; 1 bit per incident edge, CT0.9]}$$

$$o \text{ single neighbour}: \ \gamma = \{1 \text{ edge}\},\ \delta h = 1: \quad E_{\text{vis}} = \min(1\cdot\ln 2,\ \delta\tau\cdot 4\ln 2) = \ln 2\ (\delta\tau \geq 1) \text{ [FT.1]}$$

$$\Longrightarrow\ I_{\text{hid}} = (n-1)\ln 2; \qquad n = 1,2,3,4:\ 0,\ \ln 2,\ 2\ln 2,\ 3\ln 2$$

$$n = 2:\ I_{\text{hid}} > 0\ \wedge\ \text{strong} \notin \mathcal{A}(2)\ \text{[CT0.13]}\ \Longrightarrow\ I_{\text{hid}}\ \text{aspect-blind (Def-H.1 locality)}$$

$$\text{full access}:\ \delta h$$

$$= n: \quad E_{\text{vis}}$$

$$= \min(n, 4\,\delta\tau)\ln 2 \overset{n\leq 3,\ \delta\tau\geq 1}{=} n\ln 2$$

$$= E_{\text{tot}}$$

$$\Longrightarrow\ I_{\text{hid}} \to 0\ \ \forall \text{ accessible rung [CT0.6}:\ O(n) = 0]$$

$$\text{asymmetry [CT3.22, CT3.18]}: \quad F_{\text{EM}} \text{ at } v = f_v \text{ (own two signs; complete locally, D3.3)};$$

colour at v : physical only contracted ($\delta^i{}_j, \varepsilon_{ijk}, \varepsilon^{ijk}$; baryon : 3 triplets, meson : $3 \oplus \bar{3}$ [D3.9]);

isolated triplet : norm 0,

¬pole [CT3.22]

$\Longrightarrow$ I_{hid} ¬carvable per-aspect by subtraction (aspect-blind; vanishes under full access);

strong's signature = contraction-completeness (constructed: Def-H.10, FT.18–FT.19). ■

Phys. Hidden information at a vertex is generic bookkeeping, not a force fingerprint: it vanishes under full local access for every accessible rung. What genuinely distinguishes the strong sector is that its observables are incomplete until contracted against partners possibly outside the object — the seed of confinement as an information-theoretic property.

Definition 3.10 (Def-H.35 — Channel-visible content (the boundary-transport read)). Def. Σ a finite vertex set; $o \notin \Sigma$ a single external vertex whose only contact is its own incident edges into Σ, so $\gamma(o, \Sigma) = E_{\text{bd}}$ [Def-H.1], over an interval $(\delta h, \delta\tau)$ [D0.6, D0.7]. The channel-visible content of Σ to o is the boundary registration sum

$$E_{\text{vis}}(\gamma) \;:=\; \sum_{e \in E_{\text{bd}}} 1 \text{ bit} \;=\; |E_{\text{bd}}| \ln 2 \text{ nats}$$

[CT0.9 : 1 real edge = 1 bit; D1.1 : registrations are classical edge data; C9],

attained whenever a hop schedule within the channel capacity exists [$\delta h \le 4\,\delta\tau$ jaw, CT0.6a; attainment on boundary sets: FT.153]. $I_{\text{hid}} := E_{\text{tot}} - E_{\text{vis}}$ [Def-H.9(2)] is evaluated with E_{vis} of this definition. Locus-separation clause (constitutive). Two distinct reads exist in the corpus and are never interchanged: (1) the edge-transport read (this definition): content carried across a boundary edge by a hop. A hop reads only the edge's own registration [D1.9: tick/transport locality — dom $\subseteq \{v\} \cup \{e \ni v\}$, pre/post-tick data only, ¬∃ read at edge-distance > 1]; the registration lives in the pair channel of the edge's endpoints [A2: $C(p, q) = 1\mathfrak{q}$ per pair; CT0.4: real = registered]. (2) the vertex-access read [CT4.9(1)]: content accessible at a vertex v — capped, $S(v) \le 3 \ln 2$ for accessible v ($n \le 3$ [CT0.11]); at locked v ($n \ge 4$) the excess is absorbed and retained in the ledger [CT4.1, CT4.3, D4.2: retained, ¬accessible; D1.11]. The cap of (2) bounds reads at vertices; it does not bound transports across edges: a locked boundary vertex retains its excess under (2) while each incident real boundary edge carries its own pair-channel registration under (1). The two reads answer different questions — "what can o receive through γ" versus "what is accessible at v" — and every theorem states which it uses.

Theorem 3.17 (FT.153 — Boundary-schedule attainment). Thm. Single-contact setting [Def-H.35]: $\gamma(o,\Sigma) = E_{\text{bd}}$, $\delta h(\gamma) = |E_{\text{bd}}|$. If $|E_{\text{bd}}| \le 4\,\delta\tau(\gamma)$ [CT0.6a capacity jaw], then a valid transit configuration $\gamma' \subseteq \gamma$ exists whose hop schedule carries every boundary registration within $\delta\tau$ ticks at ≤ 4 hops/tick; hence

$$E_{\text{vis}}(\gamma) = |E_{\text{bd}}| \ln 2 = \min\big(\delta h(\gamma),\, 4\,\delta\tau(\gamma)\big) \ln 2 = \mathsf{B}(\gamma)$$

$$[\text{FT.1}'\text{s bound met with equality};\ \text{Def-H.3}'\text{s sup attained}],$$

regardless of the lock status of the boundary vertices. On channels whose edges are bound-locked in every configuration the transit domain is empty and the received content is 0 (attainment vacuous; the bound trivially valid) [FT.36(2) clause].

Req. Def-H.1, Def-H.3, Def-H.35; A2; D0.10, D1.1, D1.5, D1.9; CT0.4, CT0.6a, CT0.9 (Part 1); CT4.3 (Part 2); EM.5, EM.6, EM.7 (Part 2); FT.1; the schedule construction generalized by FT.36(2) (forward pointer, Tier F.13).

Pf.

$$E_{\text{bd}} = \{e_1, \dots, e_m\},\ m \le 4\,\delta\tau :\ \text{partition into } \lceil m/4 \rceil \le \delta\tau \text{ per-tick groups of } \le 4$$

per group: run transit (photon) content on the group's edges [EM.5, EM.6 : trivial chains $\subseteq V_2$ valid transit content; D1.5 : open propagating = transit; D0.10 : $\mathfrak{C}(\mathcal{R})$ admits transit assignments]

$$\text{each hop carries its edge's registration bit [CT0.9;\ D1.1]};$$

hops read edge data only [D1.9] : $\neg$vertex read invoked $\implies$ lock status of endpoints irrelevant to the transport [Def-H.35 locus separation; CT4.3 : retained excess stays ledger-internal]

$$\mathsf{E}_{\text{tr}}(\gamma') = m \ln 2 \text{ [EM.7 additive]};$$

$$\text{FT.1} :\ \text{sup} \le \min(m, 4\,\delta\tau) \ln 2 = m \ln 2;$$

$$\text{construction attains} \implies E_{\text{vis}} = m \ln 2. \ \blacksquare$$

Phys. The channel bound is not merely valid on boundary sets but reached: the observer receives the full boundary registration sum, scheduled at channel speed — four bits per tick — and vertex locks eat none of it, because a hop reads its edge and never its vertex. This is the attainment half of the internal-edge lemma's hypotheses; FT.36(2) generalizes the same construction to arbitrary channels.

Theorem 3.18 (FT.17 — Internal-edge lemma). Thm. Σ finite vertex set; $o \notin \Sigma$ a single external vertex whose only contact is its own incident edges into Σ (so $\gamma(o,\Sigma) =$ the E_{bd} boundary edges; no internal edges). Hypotheses: $E_{\text{bd}} \le 4\,\delta\tau(\gamma)$ [channel-capacity jaw, CT0.6a: the channel does not clip o's view]; channel-visible content per Def-H.35's edge-transport read, attained by FT.153. Then

$$I_{\text{hid}}(\Sigma) = E_{\text{int}} \ln 2 \qquad \text{(exactly; aspect-blind)}.$$

Req. CT0.6, CT0.9 (Part 1); CT1.29 (Part 2); CT4.2, CT4.3, CT4.9 (Part 2); D4.2; Def-H.1, Def-H.9; FT.1.

Pf.

Def-H.9 ($|\Sigma|$ = internal + boundary : internal-only reading gives $I_{\text{hid}} < 0$ on proper sub-selections) :

$$E_{\text{tot}} = (E_{\text{int}} + E_{\text{bd}}) \ln 2 \text{ [CT0.9/CT1.29 counting; locked content retained, CT4.3]}$$

$$\delta h(\gamma) = E_{\text{bd}}; \qquad E_{\text{vis}} = \min\big(E_{\text{bd}},\ 4\,\delta\tau\big) \ln 2 \overset{\text{hyp}}{=} E_{\text{bd}} \ln 2 \ (\delta\tau \geq 1)$$

hyp 2 (Def-H.35) : E_{vis} = actual accessible content (¬merely an upper bound)

$$I_{\text{hid}} = E_{\text{tot}} - E_{\text{vis}} = (E_{\text{int}} + E_{\text{bd}}) \ln 2 - E_{\text{bd}} \ln 2 = E_{\text{int}} \ln 2. \ \blacksquare$$

Cross-checks (independently computed; machine: `verify_H.py`): 1. Oriented 3-cycle, external observer: $E_{\text{int}} = 3$, $E_{\text{bd}} = 1 \leq 4 \cdot 1$ ✓ $\Rightarrow I_{\text{hid}} = 3$ bits. 2. Bidirectional 3-vertex face $\{A, B, C\}$ of the bidirectional K_4 ("tetrahedron", $n(v) = 6$), observed by D: $E_{\text{int}} = 6$, $E_{\text{bd}} = 6$: hypothesis requires $\delta\tau \geq 2$ ($6 \leq 4 \cdot 2 = 8$). Vertices oversubscribed ($n = 6 \geq 4$): their excess is locked but retained [CT4.3, D4.2] and counted in E_{tot}; Def-H.35's locus separation concerns channel-visible content only. $I_{\text{hid}} = 6$ bits. 3. Asymmetric single-point contact (whole K_4, $\Sigma = \{A, B, C, D\}$, external o joined only to A by one bidirectional pair): $E_{\text{int}} = \binom{4}{2} \times 2 = 12$, $E_{\text{bd}} = 2 \leq 4 \cdot 1$ ✓; A's own accessible content 2 bits < 3 [CT4.9] ✓ $\Rightarrow I_{\text{hid}} = 12$ bits, $m_{\text{hid}} = 12 \ln 2/\mathfrak{h}$ [Def-H.9]. (note [FT.62(4)]: at the corrected directed rung $n(A) = 8$ the contact vertex is locked; the per-vertex-cap check is carried by Def-H.35's locus separation and CT4.3 retention — $I_{\text{hid}} = 12$ bits unchanged.)

Rmk (where the I_{hid} comes from). Not D^- (binding constraint is the channel bound, reached before per-vertex caps bind) and not colour (the algebra never references $\mathcal{A}(n)$): I_{hid} — and hence $m_{\text{hid}} = I_{\text{hid}}/\mathfrak{h}$ — is exactly Σ's own internal connectivity, invisible to any single exterior observer by Def-H.1 alone. A purely topological fact, prior to any question of which forces act where. Cross-check 3 shows independence of contact distribution.

Phys. Mass, in this reading, is nothing but the information a single external vantage point cannot recover about a subnetwork's inside — a topological fact about connectivity, not a substance. The three cross-checks show the same total hidden content surviving no matter how the internal edges are arranged or where the single external contact point sits, as long as the channel-capacity hypothesis holds: hiddenness here is a property of the boundary/interior split, not of any particular internal configuration.

3.7 Tier F.5b — The Mass-Read Registry and the Tilt Composition

Definition 3.11 (Def-H.39 — The mass-read registry; bridge slots). Def. The corpus's mass reads form a registry of species; a bridge between two species is a map carrying a dimensional factor (a power of $\mathfrak{h}$) and a numeric coefficient. The registry records the loci; no bridge is constructed, and none may be silently assumed. Nomenclature rule. Every occurrence of mass in this corpus names one read of this registry at the species declared at its locus; no unqualified mass crosses species; a computed variant value is a price — an exact named functional of the declaring data — and ¬a mass read.

$$(1)\ \text{informational}: \quad m_{\text{hid}} = I_{\text{hid}}/\mathfrak{h} \qquad [\text{Def-H.9}(2);\ \text{FT.16, FT.17};\ [m_{\text{hid}}] = \mathfrak{q}^{+1/2}];$$

$$(2)\ \text{pole/defect}: \quad m^2_{\text{pole}}$$

$$[\text{Def-H.37 — forward pointer, Tier F.17};\ \text{D2.5 vocabulary};\ [m^2_{\text{pole}}] = \mathfrak{q}];$$

$$(3)\ \text{inertial/tilt — the Lyapunov-contraction read}: \quad m_i^2(q) = \Delta S(q)$$

$$[\text{L3}(4), \text{L4};\ \text{Part 1: the deficit accumulated under the L1 contraction law};\ \text{dimensionless nats}];$$

$$(4)\ \text{sector-expectation}: \quad m_B^2(q, S)$$

$$[\text{Def-H.32}(6) \text{ — forward pointer, Tier F.36};\ \text{dimensionless nats}];$$

$$(5)\ \text{1-loop}: \quad m_i^2 = V_{\text{eff}}(\varepsilon_q^{(i)})/8$$

$$[\text{Def-H.31}(6) \text{ — forward pointer, Tier F.35};\ \text{L3}(2):\ V = 8\,\Delta S;\ \text{dimensionless nats}].$$

Bridge slots. β_{31} (inertial↔informational): unconstructed — a construction must supply the power of $\mathfrak{h}$ and the per-edge numeric coefficient reconciling L4's deficits (exact rationals, state-dependent: the branch-state values $(\frac{3}{4}, \frac{1}{12})$ [FT.107(2) — forward pointer]) with FT.17's $\ln 2$-per-hidden-bit count; the read-identification locus ("the same object read differently", at pair level versus subnetwork level) is this slot. β_{21}, β_{23} (pole↔informational / inertial): the relating-equation locus. Kind discipline: a dimensionless-nats read and a dimensionful read are comparable only through a constructed β [FT.89(2) — forward pointer]; any quantitative use of a bridge requires its construction first. Ceilings (derived; cited at use). Species (2): $m^2_{\text{pole}} \in [0, 4\hat{\mu}'^2]$ [FT.76(3) — forward pointer]; the cover-ladder read $K = 2|\mathcal{C}| \le 3V/2$ [FT.79(2) — forward pointer]; the form-factor species $M \le c_w V$ [FT.97(5) — forward pointer]; species (5): bare range linear in cell size, composed-tilt range exponential in E_{int} under Def-H.40 [FT.104(2) — forward pointer]; species (4): $\le \frac{3}{2}B(B-1)$, quadratic in budget [FT.107(4) — forward pointer].

Definition 3.12 (Def-H.40 — Chain-level tilt composition). Def. Let c be a materialized bound chain / floor with E_{int} internal edges [Def-H.9]. Ledger facts (derived): every internal edge entered $\mathcal{R}$ through exactly one J^+ seal [D1.2: crystallization $n_{\mathcal{R}} : 0 \to 1$; D1.4: chain edges $\in E_{\mathcal{R}}$]; every A4 crystallization event applies exactly one Gibbs tilt step to the pair state of that event [L0(2), its declared theorem-local dynamical scope]; the ledger counts tilt steps [L0(4)]. Composition clause (constitutive scope). The tilt steps of c's sealing history compose on one chain-level aggregate state object — the per-event iteration of L0(4)'s k-event law — giving the chain's Lyapunov composition

$$V_{E_{\text{int}}} := \left(1 - \tfrac{\gamma}{4}\right)^{2E_{\text{int}}} V_0 \qquad \text{[L2 per-step law iterated; } \gamma = 1 - 4\varepsilon_q \in (0,1]\text{, D0.3; CT0.8]},$$

sharp point $\gamma = 1$ (D0.4): $V_{E_{\text{int}}} = (\frac{9}{16})^{E_{\text{int}}} V_0$ [exact rationals, L2]. Read. The fermion-mass read of the contraction rate is species (3) of Def-H.39 [L3(4): $m_i^2 = \Delta S$]; its composed range is the exponential species of Def-H.39's ceiling row [FT.104(2)]. Scope clauses. (i) The aggregate state object is this definition's sole non-derived locus: the inventory's states are pair states [A1]; L0(4) proves the k-event iteration on one pair channel; the chain-level aggregate extends that iteration to the chain's sealing history. (ii) No calibration claim: no datum fixes $(\frac{9}{16})^{E_{\text{int}}}$ to any observed hierarchy [P1, P5]. (iii) No particle-family identification [FT.77, FT.79(4)–(6); Census FT.84(2)]. (iv) The read is dimensionless nats; comparison with dimensionful species requires Def-H.39's β slots.

Theorem 3.19 (FT.154 — The tilt-composition law). Thm. Under Def-H.40, for a materialized bound chain with E_{int} internal edges and tilt strength $\gamma \in (0,1]$:

(1) event count: the chain's tilt-step count is exactly E_{int};

(2) per-step contraction: $V' = \left(1 - \tfrac{\gamma}{4}\right)^2 V + O(\delta\rho^3)$ [L2];

(3) composition: $V_{E_{\text{int}}} = \left(1 - \tfrac{\gamma}{4}\right)^{2E_{\text{int}}} V_0$ with per-step remainders accumulated;

$$\gamma = 1: \quad V_{E_{\text{int}}} = \left(\tfrac{9}{16}\right)^{E_{\text{int}}} V_0;$$

eigenvalue composition: $\mu_k = \left(1 - \tfrac{\gamma}{4}\right)^k \mu_0 + O(\mu_0^2)$ after k recorded events [L0(4)];

(4) range consequences: the composed read's spans and thresholds are the exact arithmetic of $(\frac{16}{9})^{E_{\text{int}}}$ tabulated at FT.104(2) (forward pointer, Tier F.35).

Req. Def-H.40; Def-H.9 (E_{int}); D0.3, D0.4, D1.2, D1.4; CT0.8; L0(1), L0(2), L0(4); L2; L3(4) (Part 1); exact rational arithmetic.

Pf.

$$(1): \ e \text{ internal}$$

$$\Longrightarrow \ e$$

$$\in E_{\mathcal{R}} \text{ [D1.4]}$$

$$\implies e \text{ crossed the A4 criterion once, at its } J^+ \text{ seal [D1.2 : } n_{\mathcal{R}} : 0 \to 1;\ \text{CT0.3 : seal increments]}$$

$$\text{L0}(2) :\ \text{one tilt step per event;}$$

$$\text{L0}(4) :\ |\mathfrak{L}(\tau_n)|$$

$$= \textstyle\sum_{k<n} n_{\text{real_out}}(\tau_k) \text{ is the tilt-step count [exact integer recursion, A6, A7];}$$

$$\text{Def-H.40's aggregate scope composes the chain's sealing history } \implies \#\text{steps} = E_{\text{int}}$$

$$(2) :\ \text{L2 verbatim :}\ V' - V = -\tfrac{\gamma(8-\gamma)}{16}V + O(\delta\rho^3) = -[1-(1-\tfrac{\gamma}{4})^2]V + O(\delta\rho^3)$$

$$(3) :\ \text{iteration of } (2)\ E_{\text{int}} \text{ times under Def-H.40's composition clause [L0}(4) :\ \tilde{\mathcal{T}}_\gamma^{\,k} \text{ composition; interleaved Haar draws spectrally inert, L0}(1)];$$

$$\gamma = 1 :\ (1-\tfrac{1}{4})^2 = \tfrac{9}{16} \text{ [L2; exact rationals]}$$

$$\mu_k :\ \text{L0}(4) \text{ verbatim (induction on L2's } \mu' = (1-\tfrac{\gamma}{4})\mu + O(\mu^2))$$

$$(4) :\ \text{arithmetic table of FT.104}(2) \text{ re-cited;}$$

$$\text{its thresholds } E_{\text{int}}$$

$\geq 16/27/35/42$ and the proportionality obstruction are computed there under this law. ■

Phys. The ledger gives a chain exactly one contraction event per internal edge, and each event contracts by an exact rational factor — so the geometric law $(\frac{9}{16})^{E_{\text{int}}}$ at sharp crystallization is seal arithmetic, not a fitted model: the count comes from the ledger recursion, the factor from the tilt's exact leading law, and the definition supplies only the aggregate on which the count composes. The mass interpretation of the rate is the registry's inertial species; the exponential range it buys is priced, with its obstruction, at FT.104(2).

3.8 Tier F.6 — Subnetwork-Level Force Structure: Confinement and the Active Aspect Set

Theorem 3.20 (FT.18 — Contraction-completeness is not Σ-local). Thm. $\Sigma \supseteq \{v_1, v_2, v_3\}$ triplet-carrying [D3.8], no antitriplet in Σ. $\exists\, C, C' \in \mathcal{M}(U)$ [D2.11] with $C|_\Sigma = C'|_\Sigma$ (identical restriction) such that $\mathcal{C}(\Sigma; C) = 1$ and $\mathcal{C}(\Sigma; C') = 0$ [Def-H.10]. Contraction-completeness is $\neg$ a function of $\mathcal{M}(\Sigma)$ alone.

Req. A1; D2.11, D3.8, D3.9; CT3.17, CT3.18 (Part 2); hypothesis (stated, not derived from A1 alone): $U \setminus \Sigma$ contains further triplet carriers w_1, w_2 available to seal with v_1 — A1's countable infinitude of $\mathcal{Q}$ secures infinitely many vertices but not that any are unsealed triplet carriers positioned to form C'; this is an additional genericity assumption.

Pf.

$$U \setminus \Sigma \text{ contains further triplet carriers } w_1, w_2 \text{ [hypothesis, above]}$$

$$C: \ \{v_1, v_2, v_3\} \text{ sealed as one baryon } \varepsilon_{ijk} q_1^i q_2^j q_3^k \text{ inside } \Sigma \text{ [D3.9]}: \ \text{valid [CT3.18]};$$

$$\partial_{\text{col}}(\Sigma; C) \qquad = \varnothing$$

$$C': \ v_1 \text{ sealed with } w_1,$$

$$w_2 \notin \Sigma;$$

$$v_2,$$

$$v_3 \text{ sealed elsewhere in } U \setminus \Sigma: \ \text{equally valid};$$

$$\pi_{C'}(v_i) \not\subseteq \Sigma \ \forall i \implies \partial_{\text{col}}(\Sigma; C') \qquad = \{v_1, v_2, v_3\}$$

$$C|_\Sigma$$

$= C'|_\Sigma$ (same vertices, edges, attachments, local entries) $\implies$ $\mathcal{C}$ undetermined by Σ-local data. ■

Rmk. The colour-sector analogue of D2.11's dangling-boundary condition being a global fact: whether an end is sealed depends on whether its partner exists somewhere — no Σ-local inspection certifies it.

Phys. Whether a colour-carrying vertex reads as "confined within Σ" or "open toward the rest of U" is not a fact about Σ's own microstate at all — it is a fact about the ambient universe's contraction pattern, which Σ-local data cannot see. Confinement bookkeeping is therefore irreducibly non-local in exactly the sense D2.11 already makes boundary-sealing non-local.

Definition 3.13 (Def-H.10 — Colour boundary and the contraction-completeness projector). Def. $C \in \mathcal{M}(U)$ valid. For $v \in \text{Colour}(U) := \{u \in \mathcal{Q} : \ u \text{ carries a colour attachment [D3.8]}\}$, let $\pi_C(v) \subseteq \text{Colour}(U)$ be the other members of v's elementary invariant contraction group under C (one for a meson pair via $\delta^i{}_j$; two for a (anti)baryon triple via $\varepsilon_{ijk}/\varepsilon^{ijk}$). π_C total on $\text{Colour}(U)$ for valid C [CT3.18 + D2.11 applied to the colour ledger]. For Σ: $\text{Colour}(\Sigma) := \text{Colour}(U) \cap \Sigma$;

$$\partial_{\text{col}}(\Sigma; C) := \{v \in \text{Colour}(\Sigma) : \ \pi_C(v) \not\subseteq \Sigma\}; \qquad \mathcal{C}(\Sigma; C) := \begin{cases} 1 & \partial_{\text{col}}(\Sigma; C) = \varnothing \\ 0 & \text{otherwise} \end{cases};$$

$$\Pi_{\mathcal{C}}(\Sigma; C) := \mathcal{C}(\Sigma; C) \cdot \mathbb{I}_{\Sigma}^{(\text{strong})}$$

$$(\text{Hermitian idempotent; matches } \mathcal{P}_M^{(n)} \text{ [CT0.14] and } \Pi_v \text{ [Def-H.9]}).$$

Rmk (why $\mathcal{C}$ is defined on π_C). A predicate counting only $\mathcal{M}(\Sigma)$-internal data is insufficient (FT.18): $\mathcal{C}$ is defined on the contraction partners π_C — manifestly a function of the ambient configuration C, not of $\mathcal{M}(\Sigma)$ alone.

Theorem 3.21 (FT.19 — Confinement is an all-or-nothing subnetwork property). Thm.

$$F^{\Sigma}_{\text{strong}}(C) := \Pi_{\mathcal{C}}(\Sigma; C)\, F_{\text{strong}}\big|_{\Sigma}\, \Pi_{\mathcal{C}}(\Sigma; C); \qquad \mathcal{C}(\Sigma; C) = 0 \implies F^{\Sigma}_{\text{strong}}(C) = 0$$

— regardless of how many colour-carrying vertices Σ contains, and even if $|\partial_{\text{col}}(\Sigma; C)| = 1$.
Req. Def-H.9, Def-H.10; CT2.5, CT2.8 (Part 2); CT3.2, CT3.18, CT3.22 (Part 2); D3.1.
Pf.

$$\mathcal{C}$$

$$= 0 : \ \exists v \in \text{Colour}(\Sigma) : \ \pi_C(v) \not\subseteq \Sigma \text{ [Def-H.10]}$$

$$\implies v \text{ has } \neg\text{partner inside } \Sigma : \ \text{uncontracted index relative to } \Sigma$$

$$\text{CT3.18} : \ \text{uncontracted index}$$

$$= \text{dangling colour-ledger boundary} \implies \text{mask-excluded (same argument with } \Sigma \text{ as ambient)}$$

$$\text{CT3.22} : \ \text{excluded} \implies \neg\text{pole [CT2.5]} \ \wedge \ \text{physical norm } 0 \text{ [CT2.8]} : \ v\text{'s } \Sigma\text{-restricted colour state}$$

$$= \text{null vector}$$

$$\Sigma\text{'s colour sector}$$

$$= \bigotimes_{\text{Colour}(\Sigma)} \text{(slot spaces) [D3.1, CT3.2] : one null factor} \implies \text{whole product null}$$

$$\implies F_{\text{strong}}\big|_{\Sigma}\text{'s matrix elements vanish}$$

$$\implies \Pi_{\mathcal{C}} = 0 \implies F^{\Sigma}_{\text{strong}} = 0 \cdot F_{\text{strong}}\big|_{\Sigma} \cdot 0 = 0. \blacksquare$$

Rmk. Sharpens FT.16 step 4 to subnetworks: unlike EM's f_v (additive, per-vertex local [D3.3, CT3.22]), strong's observability on Σ is ¬gradable — one unsealed attachment anywhere zeroes Σ's entire colour sector. No conflict with FT.15: the tuple $(q, \mathrm{S}, m_{\text{hid}})$ never carried a strong component [FT.11]; $\mathcal{C}$ tracks admissibility, not a value.

Rmk (definition interface). Def-H.9's $\mathcal{M}(\Sigma)$ is subject to rung masks + Janus balance only, ¬ colour-sealing explicitly: an $\mathcal{M}(\Sigma)$-microstate with an incomplete attachment is not excluded by Def-H.9 yet has zero physical norm when Σ is examined alone. Whether to weight $\mathcal{M}(\Sigma)$ by $\mathcal{C}(\Sigma; C)$ or keep $\Pi_{\mathcal{C}}$ as a downstream filter is decided (FT.39(3)): the downstream filter is the corpus-consistent placement — pre-weighting would delete physical EM/G/H reads of valid ambient restrictions that seal outside Σ [FT.18]; Def-H.9 stands unweighted.

Phys. Strong interactions are the one place in this decomposition where a subnetwork cannot certify its own physical activity from what it contains: one dangling colour attachment anywhere zeroes the whole sector, regardless of how many other colour-carrying vertices sit safely paired inside Σ. This is stronger than the usual notion of a boundary effect — it is not that strong force is weaker at an unsealed boundary, it is that the entire subnetwork's colour amplitude vanishes identically.

Definition 3.14 (Def-H.11 — Active aspect set of a subnetwork). Def.

$$\mathcal{A}(\Sigma; C) :$$

$$= \big\{A \in \mathrm{F} : \ A \text{ contributes nonzero physically observable content to } \Sigma\text{'s coarse-grained description under } C\big\},$$

read against two channels: nonzero value in Σ's coarse-grained tuple $(q, \mathrm{S}, m_{\text{hid}})$ [Def-H.9] (charge-type), or nonzero surviving support under $\Pi_{\mathcal{C}}(\Sigma; C)$ [FT.19] (selection-type: strong). Subnetwork analogue of D1.12's $\mathcal{A}(n)$: $\mathcal{A}(n)$ asks which aspects act nontrivially on one vertex's slot algebra; $\mathcal{A}(\Sigma)$ asks which survive coarse-graining and masking as observables on Σ as a whole.

Definition 3.15 (Def-H.36 — The scalar-row fold (the H read at Σ-level)). Def. The coarse-grained tuple $(q, \mathrm{S}, m_{\text{hid}})$ [Def-H.9] carries no independent H slot. The Σ-level observable content of the scalar-trace-mode aspect — $F_H = \mathrm{span}\{\Sigma\}$, $\dim F_H = 1$ [A8, CT0.12] — is read through the mass component:

$$\mathrm{H} \in \mathcal{A}(\Sigma; C) \iff \text{the G-row criterion holds } \big(I_{\text{hid}}(\Sigma; o) \neq 0\big) \qquad \text{[FT.20 table]}.$$

Derived grounds (cited; the fold is the read convention this definition constitutes). $H = h\,\Sigma$ [EW.13]; the scalar background enters the pole masses only via $M^2 \propto h_0^2$ [EW.20, EW.33] — H's

tuple-level contribution is via the mass read; h_0 remains a P4-class datum [P4]. Distinctness clause. The fold is a read convention for the coarse-grained tuple, $\neg$ a block identification: $\mathcal{K}^{(H)}$ remains a distinct block inside $\mathcal{K}_\Sigma$ [Def-H.15 — forward pointer, Tier F.9; Census FT.28(4)].

Theorem 3.22 (FT.20 — Reduction of $\mathcal{A}(\Sigma)$ to proven per-aspect criteria). Thm.

aspect	$\in \mathcal{A}(\Sigma; C) \iff$	locality	basis
EM	$\exists v \in \Sigma :\ f_v \neq 0$	Σ-local, additive	D3.3, EM.9, CT3.22 step 4
G (mass)	$I_{\text{hid}}(\Sigma; o) \neq 0$	observer-relative (inherits Def-H.9)	Def-H.9, FT.16, FT.17
H	folded into G [Def-H.36]	as G	EW.13, EW.20, EW.33
W (spin)	$\Theta_{\text{int}}(\Sigma) \neq 0 \implies \Sigma \setminus \partial\Sigma \neq \varnothing$ (one-directional)	Σ-local, intrinsic	Def-H.9(3), FT.13, FT.25
D^+	never (independently)	uniform	D4.3
D^-	never (independently)	—	CT4.4
$J^\pm$	never	seals, $\neg$charges	CT0.7
strong	$\text{Colour}(\Sigma) \neq \varnothing \ \wedge\ \mathcal{C}(\Sigma; C) = 1$	genuinely global	Def-H.10, FT.19

Req. FT.11, FT.13, FT.16–FT.19; Def-H.9–H.11; FT.25; all cited tags.

Pf. All rows but strong = FT.11's audit extended verbatim to general Σ (their bases reference per-vertex/uniform facts only, never size or topology); G = Def-H.9's mass definition (already general); H = fold into the mass read [Def-H.36; grounds: EW.13 $H = h\Sigma$; EW.20/EW.33: $M^2 \propto h_0^2$ — H's tuple-level contribution is via mass]; W = Def-H.9(3): $\Theta_{\text{int}} \neq 0 \implies \Sigma$ has ≥ 1 internal edge carrying nonzero $\widehat{ds}^{\,2}$ (converse fails: a sum of nonzero terms can cancel, so ≥ 1 internal edge does not imply $\Theta_{\text{int}} \neq 0$); strong = FT.19. ■

Rmk (the point of the exercise). Exactly one aspect — strong — requires genuinely non-Σ-local data for activity: confinement [CT3.22] is the only aspect-defining mechanism in the corpus requiring contraction against data possibly outside the object described [D3.9]. J and D^- never appear in $\mathcal{A}(\Sigma)$ despite local presence in $\mathcal{A}(n)$: they are structural/bookkeeping aspects (seals, locks), not observables — a distinction $\mathcal{A}(n)$'s representation-theoretic definition never needed, but observability-based $\mathcal{A}(\Sigma)$ must draw.

Phys. Collapsing $\mathcal{A}(\Sigma)$ to per-aspect activity criteria makes explicit what kind of fact each force is: EM and mass/spin are things a subnetwork simply has, readable off its own boundary and interior; strong confinement is instead a fact about how Σ sits inside the rest of the universe. This table is the subnetwork-level generalization of $\mathcal{A}(n)$ that the coarse-graining program needs next — the single-vertex active-aspect question, answered for arbitrary bounded regions.

3.9 Tier F.7 — The Coarse-Grained Slot Algebra: $\mathcal{A}(\Sigma)$ Properly Derived

Definition 3.16 (Def-H.12 — Coarse-grained slot algebra). Def. Σ a bounded subnetwork, $\partial\Sigma := \{e \in \mathcal{R} :\ e \text{ has exactly one endpoint in } \Sigma\}$ [D2.11/CT1.29 separating-surface reading, as already used in Def-H.9(2)/Def-H.10]. Let $N(\Sigma) := |\partial\Sigma|$. The coarse-grained slot algebra of Σ is

$$\mathcal{H}_{\text{slot}}(\Sigma) := \text{span}_{\mathbb{C}}\{|e\rangle :\ e \in \partial\Sigma\} \cong \mathbb{C}^{N(\Sigma)},$$

one basis ket per boundary edge — the direct generalization of D3.1's "one ket per incident real edge," taken over Σ's external interface in place of a single vertex's incident set.

Degenerate-case check ($\Sigma = \{v\}$). $\Sigma = \{v\}$ has no internal edges [A1: $p \neq q$, no self-loops, so no edge has both endpoints in $\{v\}$] $\implies$ $\partial\Sigma$ = the full incident edge set of v $\implies$ $N(\Sigma) = n(v)$

[D0.8] $\implies$ $\mathcal{H}_{\text{slot}}(\Sigma) = \mathcal{H}_{\text{slot}}^{(n(v))}$ exactly [D3.1]. The construction specializes to the single-vertex slot algebra on the nose, not merely by analogy.

Rmk (relation to N_{visible}). At a single tick ($\delta\tau = 1$), $\delta h(e) = 1$ per real edge [CT0.9], so $N(\Sigma) = \delta h(\gamma(o, \Sigma)) = N_{\text{visible}}$ [FT.1, Def-H.9] for any single external observer whose channel is exactly $\partial\Sigma$: the slot-algebra dimension and the already-established externally-visible bit count coincide. No new counting primitive introduced.

Definition 3.17 (Def-H.13 — Compositional bridge map ι_Σ). Def. For each $e \in \partial\Sigma$, let w_e be e's endpoint outside Σ, with its own rung $n(w_e)$, existing bridge map $\iota_{n(w_e)}$ [D2.12], and e's own edge-local pair-channel decomposition [CT0.12, rung-independent, fixed 16-dim block]. Define $\iota_\Sigma : \mathfrak{u}(N(\Sigma)) \to \mathcal{K}$ by requiring, on the slot direction $|e\rangle$, that ι_Σ factor through the pair $\big($CT0.12's block for $e,\ \iota_{n(w_e)}\big)$:

$$\iota_\Sigma\big|_{|e\rangle} := \big(\text{CT0.12 pair-block map for } e\big) \otimes \iota_{n(w_e)} \qquad (e \in \partial\Sigma),$$

extended linearly across $\partial\Sigma$. No independent embedding is postulated: every component of ι_Σ is inherited, edge-by-edge, from machinery that already exists at the single-edge (CT0.12) and single-vertex (D2.12) level.

"Strictly new at Σ" (generalizes D3.2). A piece of $\mathfrak{u}(N(\Sigma))$ is strictly new at Σ iff it does not factor through ι_Σ as defined above — i.e. iff it is a genuinely collective effect of Σ's boundary edges taken together, not a sum/tensor of independent single-edge and single-vertex facts already accounted for at lower structural levels. This is the precise sense in which the coarse-graining machine treats Σ "as a single quantum object": an aspect belongs to $\mathcal{A}(\Sigma)$ properly only if its action on $\mathcal{H}_{\text{slot}}(\Sigma)$ resists the boundary-edgewise factorization ι_Σ provides for free.

Degenerate-case check. At $\Sigma = \{v\}$, $\partial\Sigma$ is v's own incident edge set and each w_e ranges over v's neighbours; ι_Σ reduces to exactly the rung-$n(v)$ construction CT0.13 already performs (its Pf. builds $\mathfrak{u}(n)$ directly from the slot algebra and reads off S-even/odd blocks — the same data ι_Σ assembles edge-by-edge here). $\iota_{\{v\}} = \iota_{n(v)}$.

Rmk (strictly new at Σ — first counts). Def-H.13's filter evaluated at small $N(\Sigma)$: $N = 1$: $\mathfrak{u}(1)$ factors per-edge entirely — nothing strictly new; single-contact floors carry no collective boundary mode.

Theorem 3.23 (FT.21 — $\mathcal{A}(\Sigma)$'s charge-type rows, properly derived). Thm. (i) EM — boundary-exact. FT.20's EM-row observable is boundary-expressible: the tuple read $q = \frac{1}{2}(\sigma_s - \sigma_t)$ [Def-H.9(1), EM.9] reads end signs on $\partial\Sigma$'s terminal edges, and interior vertex activity telescopes into end reads [EM.9b (Part 2): the identity, with its type-$(1,1)$ domain] — so the row's criterion ($\exists v \in \Sigma :\ f_v \neq 0$) can exceed what is separately observable at the boundary, while its observable content factors through ι_Σ's per-edge CT0.12 factor [Def-H.13]. (ii) G/H — boundary-expressible modulo one declared interior label. $I_{\text{hid}}(\Sigma; o) = (E_{\text{int}} + E_{\text{bd}}) \ln 2 - E_{\text{vis}}$, where $E_{\text{bd}} = N(\Sigma) = \dim \mathcal{H}_{\text{slot}}(\Sigma)$ [Def-H.12 Rmk] and $E_{\text{vis}} = \min(\delta h(\gamma), 4\,\delta\tau) \ln 2$ is a boundary/channel sum [FT.1]; $E_{\text{int}} = |\Sigma \setminus \partial\Sigma|$ is Σ's own interior volume datum — a single integer carried as a c-number label alongside $\mathcal{H}_{\text{slot}}(\Sigma)$, of the same species as Def-H.15(4)'s superselection labels (Θ_{int}, k), and ¬factored through ι_Σ edge-wise. (iii) Consequently both rows are instances of CT0.13's active-aspect machine on $\mathcal{H}_{\text{slot}}(\Sigma)$ — EM exactly, G/H modulo the declared interior label — not case-by-case arithmetic extensions.

Req. Def-H.9(1)–(2), Def-H.12, Def-H.13, Def-H.15(4); FT.1, FT.20 (EM and G rows); D2.12, D3.1, D3.2, D3.3; CT0.12, CT0.13 (Part 1); EM.9, EM.9b (E; telescoping identity; H-side record: `verify_H_ASigma.py`).

Pf.

$$\text{EM}: \; q = \tfrac{1}{2}(\sigma_s - \sigma_t) \; [\text{Def-H.9}(1),\ \text{EM.9}] : \; \text{end signs on } \partial\Sigma\text{'s terminal edges} \implies \text{boundary read}$$

$$\text{interior telescoping}: \; \text{EM.9b (E) [H-side record}: \; \text{verify_H_ASigma.py}]$$

$$G: \; I_{\text{hid}} = (E_{\text{int}} + E_{\text{bd}}) \ln 2 - E_{\text{vis}} \; [\text{Def-H.9}(2),\ \text{FT.17 form}];$$

$$E_{\text{bd}} = N(\Sigma) = \dim \mathcal{H}_{\text{slot}}(\Sigma) \; [\text{Def-H.12 Rmk}];$$

$$E_{\text{vis}} = \text{boundary/channel sum} \; [\text{FT.1}];$$

$$E_{\text{int}} = |\Sigma \setminus \partial\Sigma| = \Sigma\text{'s fixed volume datum}$$

$$\implies \text{EM row factors through } \iota_\Sigma \text{ exactly};$$

$$G \text{ row factors through } \iota_\Sigma \text{ extended by the c-number label } E_{\text{int}} \; [\text{Def-H.15}(4) \text{ pattern}];$$

$$\text{both are instances of CT0.13's machine on } \mathcal{H}_{\text{slot}}(\Sigma),$$

$$\neg\text{separate arguments.} \; \blacksquare$$

Rmk (why the W/spin row resists boundary treatment — and where its quantization comes from instead). FT.20's row W depends on $\Theta_{\text{int}}(\Sigma) = \sum_{e \in \Sigma \setminus \partial\Sigma} \widehat{ds}^{\,2}(e)$ [Def-H.9(3)] — a sum over internal edges, which are by construction outside $\mathcal{H}_{\text{slot}}(\Sigma)$'s span [Def-H.12]. The spin/holonomy datum is therefore not expressible in ι_Σ's boundary-edgewise vocabulary: it is a global fact about Σ, structurally of the same kind as colour contraction-completeness (FT.18/FT.19: non-locality in ambient data). Contrast FT.21(ii): E_{int} is a fixed volume label, while Θ_{int} is configuration-dependent interior data — the W row resists even the label-extended factorization. Tier F.8 supplies the quantization this structure calls for: internal-edge line elements are quantized neither by boundary factorization nor by a single-edge reality-line argument (which reaches σ_e [CT0.9], not the metric coordinates), but by the corpus's own materialization condition on closed chains (FT.22, via EW.18/CT2.5) evaluated on bound structures — at the fermion floor yielding per-edge integrality (FT.23–FT.24) and a well-defined Hol_{int} (FT.25). The internal-edge slot object itself: Def-H.14.

Phys. Σ becomes a single quantum object exactly to the extent that its boundary edges can be treated as one composite slot register — and the EM read turns out to be expressible in that register

exactly, the mass read modulo Σ's own interior volume label E_{int} (clause (ii)), inherited edge-by-edge from machinery already proven at the single-edge and single-vertex level, with nothing new postulated. Spin is different in kind: it is a fact about Σ's inside, invisible to the boundary register by construction — and its quantization accordingly comes from the global materialization condition on closed chains (Tier F.8), not from boundary-register arithmetic.

3.10 Tier F.8 — Line-Element Quantization from Materialization

Definition 3.18 (Def-H.14 — General-edge sector space (internal-edge slot object)). Def. For any real edge e (boundary or internal to any Σ), let $\mathcal{E}(e) := \mathcal{E}_{++}(e) \oplus \mathcal{E}_{+-}(e) \subset \mathfrak{H}(\mathbb{C}^4)_e$ be e's own S-even pair sector [CT1.1, CT1.2] — the space with coordinates $(a_{00}(e), q_{ij}(e), w_i(e))$ [D2.1] that already fix $g_{\mu\nu}(e)$ [CT1.12] and hence $\widehat{ds}^2(e)$ [D1.6]. For Σ a bounded subnetwork:

$$\mathcal{H}^{\text{int}}_{\text{slot}}(\Sigma) := \bigoplus_{e\in\Sigma\setminus\partial\Sigma} \mathcal{E}(e), \qquad \dim_{\mathbb{R}} \mathcal{H}^{\text{int}}_{\text{slot}}(\Sigma) = 10\,|\Sigma\setminus\partial\Sigma|.$$

No bridge map needed. Unlike ι_Σ [Def-H.13], which had to reach across each boundary edge to the external endpoint's vertex-level bridge map $\iota_{n(w_e)}$ (because a boundary edge's far side is genuinely outside Σ), $\mathcal{E}(e)$ is already fully edge-local [CT0.12: rung-independent, fixed 16-dim block] — it references neither endpoint's rung nor vertex-level slot algebra. An internal edge's two endpoints are both already inside Σ, which is exactly why no external anchor is available or needed: $\mathcal{E}(e)$ is defined identically whether e is internal or a boundary edge, and $\mathcal{H}^{\text{int}}_{\text{slot}}(\Sigma)$ is simply this data restricted to $\Sigma\setminus\partial\Sigma$.

Degenerate-case check ($\Sigma = \{v\}$). No internal edges [A1, as in Def-H.12] $\implies \mathcal{H}^{\text{int}}_{\text{slot}}(\{v\}) = 0$, consistent with Def-H.12's finding at the same degenerate case.

Rmk (relation to Θ_{int}). $\Theta_{\text{int}}(\Sigma) = \sum_{e\in\Sigma\setminus\partial\Sigma} \widehat{ds}^2(e)$ [Def-H.9(3)] is, by CT1.12/D1.6, a fixed function of the coordinates of $\mathcal{H}^{\text{int}}_{\text{slot}}(\Sigma)$ together with the embedding data $\{\Delta x^\mu_e\}$. The internal-edge slot object is therefore exactly this space: compositional, edge-local, and implicit in Def-H.9(3).

Rmk (ambient continuity). $\mathcal{E}(e)$ is a continuous configuration arena: $a_{00} \in \mathbb{R}$ unrestricted [CT1.12], $\widehat{ds}^2(e)$ a generic $O(1)$ real [CT1.19a]. CT0.9's reality-line mechanism — the corpus's edge-level discreteness result for ϕ_e — acts on the 1-dimensional single-qubit line $\wedge^2\mathcal{H}_q$ [$\dim\mathcal{H}_q = 2$, A1], not on these coordinates [CT0.9 Req/Pf]; a reality-line analogue cannot reach $\widehat{ds}^2$. It does not need to: the corpus's pole mechanism (CT2.5) already produces discreteness from continuous families — EW.18 quantizes wrapping loops by exactly this route. FT.22–FT.25 compute that mechanism on bound chains.

Theorem 3.24 (FT.22 — Loop quantization of materialized bound chains). Thm. Let c be a closed chain [D1.4] that materializes — contributes a spectral pole of the masked resolvent $Z(\mathcal{R}, n)$ [D2.4, D2.5, CT2.5] at the rung n of its active mask [D1.12, CT0.13]; instances: materialized wrapping chains [EW.17, EW.18], bound chains of fermion realizations [Def-H.4, D1.5]. Then the spiral holonomy around c is single-valued:

$$\text{Hol}_{\text{spiral}}(c) = e^{i\Phi(c)} = +1, \qquad \Phi(c) = \alpha\,\Theta(c) = 2\pi k, \quad k\in\mathbb{Z}\ \text{(winding)};$$

and under EM.20's scope tokens (Comm-min + Eff), with rung stability [FT.9: $\alpha_{\mathrm{EM}} = \pi$ at every accessible rung]:

$$\Theta(c) = \sum_{e \in c} \widehat{ds}^{\,2}(e) = 2k \in 2\mathbb{Z}.$$

On rung-3 wrapping chains this reduces verbatim to EW.18 (consistency check).

Req. D1.4, D1.5, D1.7; D2.4, D2.5; CT2.5; EW.17, EW.18 (mechanism; domain generalized from wrapping chains to materialized closed chains — the single H-level step, flagged in Pf); EM.20 (Comm-min, Eff; A §0.2a); FT.9; Def-H.4; D1.12, CT0.13.

Pf.

$$\text{D1.7}: \ \Theta(\gamma) = \textstyle\sum_{e \in \gamma} \widehat{ds}^{\,2}(e), \ \Phi = \alpha\Theta, \ \mathrm{Hol}_{\mathrm{spiral}} = e^{i\Phi} \text{ are defined for every chain;}$$

$$\text{EW.17}: \ \text{transport around a closed cycle multiplies the eigenmode by } \mathrm{Hol}_{\mathrm{spiral}}$$

$$\text{CT2.5 (as applied in EW.18)}: \ \text{pole}$$

$$= \text{eigenvalue condition on a single-valued eigenmode}$$

$$\implies e^{i\Phi(c)} \qquad = 1;$$

$$\text{winding freedom [EW.18]}: \ \Phi(c) = 2\pi k, \ k \in \mathbb{Z}$$

$$\text{domain step (H-level, flagged)}: \ \text{EW.17's rung-3 restriction is an existence claim about wrapping chains;}$$

$$\text{CT2.5's single-valuedness step is chain-generic} \implies \text{the domain extends to every materialized closed chain with no added premise}$$

$$\text{EM.20 (Comm-min + Eff)}: \ \alpha = \pi;$$

$$\text{FT.9}: \ \alpha^{(n)} = \pi \ \forall \text{ accessible rungs} \implies \Phi(c) = \pi\,\Theta(c) = 2\pi k \implies \Theta(c) = 2k \in 2\mathbb{Z}. \ \blacksquare$$

Rmk (oddness of α). $\Theta(c)$ is constrained exactly when α/π is odd: Comm-min alone gives $\alpha \in \pi\mathbb{Z}$ [EM.20], and even branches ($\alpha \in 2\pi\mathbb{Z}$) leave $\Theta(c)$ unconstrained. Eff — the corpus's sole effectivity hypothesis — is what selects $\alpha = \pi$ and makes the quantization bite.

Rmk (sign-factor separation). Following EW.18, single-valuedness is imposed on the spiral/geometric transport; the registration signs σ_e and the wall parity $(-1)^{\aleph}$ are ledger data of the weight $w(C)$ [A8, CT0.9, D3.3], not of the geometric eigenmode transport.

Rmk (area language, scoped). $\Theta(c)\,\mathfrak{h}^2 = \sum_{e \in c} \kappa_e\,\mathfrak{h}^2$ is the κ-weighted accumulated line element along c — dimensionally an area, and equal to CT1.29's pierced-edge area $A = n_{\mathcal{R}}\,\mathfrak{h}^2$ exactly

on the minimal-edge class $\kappa_e \equiv 1$ [FT.31(4): $\Theta = \delta h$]. It is ¬the metric-enclosed area of c: the floor loop (2-cycle) traces a degenerate out-and-back locus — zero enclosed area — yet carries $\Theta = 2\kappa \neq 0$, because ds^2 is quadratic, hence orientation-blind [FT.23–FT.24]. Nor is $\sqrt{\Theta}\,\mathfrak{h}$ a proper length of an inhomogeneous path: proper length reads $\sum_e \sqrt{\kappa_e}\,\mathfrak{h}$ (sum of square roots $\neq$ square root of the sum). Program-level "cell"/"enclosed area" phrasing must be read as this weighted accumulation.

Phys. Every bound structure that appears in the spectrum obeys a Bohr–Sommerfeld-type selection rule: its loop line-element sum is an even integer. In continuum language [CT1.19, CT2.9]: $\oint_c ds^2 = \Theta(c)\,\mathfrak{h}^2 = 2k\,\mathfrak{h}^2 = 8k\,G_N \ln 2$ — the accumulated line element of an on-shell loop is an even multiple of the Bekenstein bit-area $\mathfrak{h}^2$ [CT1.29, CT2.9]; on the minimal-edge class this equals CT1.29's pierced-edge area [FT.31(4)], in general it is a κ-weighted accumulation, ¬a metric-enclosed area (Rmk below). Smooth geometry unfolds as the large-k limit; integrality persists at every scale as the rule selecting which loops materialize.

Theorem 3.25 (FT.23 — Orientation-evenness of the line element). Thm. For a bidirectional channel — both orientations $e_{pq} = (p, q)$, $e_{qp} = (q, p)$ in $\mathcal{R}$:

$$\widehat{ds}^{\,2}(e_{pq}) = \widehat{ds}^{\,2}(e_{qp}).$$

Req. D1.6; CT2.6 (charts; $g_{\mu\nu}$ a single smooth field on M; hypotheses (i)–(iii)); CT1.12 ($g_{\mu\nu}$ built entirely from S-even sector data); CT1.1, CT1.2, C3 ($\mathcal{E}_{++}, \mathcal{E}_{+-}$ are S-even); CT0.7(2) ($S(X) = \mathbb{S}X\mathbb{S}$: the swap action); endpoint-difference displacement [math].

Pf.

$$\text{CT2.6 charts}: \ \Delta x_e := x_{t(e)} - x_{s(e)} \implies \Delta x_{e_{qp}} = -\Delta x_{e_{pq}} \ \text{(all four components)}$$

$$[\text{consistent with CT1.19's } \Delta x^\mu_e = \mathfrak{h}\,\dot{x}^\mu(\lambda_e) + O(\mathfrak{h}^2)]$$

$$g_{\mu\nu}(e) \text{ built solely from } (a_{00}, q_{ij}, w_i) \in \mathcal{E}_{++} \oplus \mathcal{E}_{+-} \text{ [CT1.12, D2.1]};$$

$$\mathcal{E}_{++},\ \mathcal{E}_{+-} \text{ are } S\text{-even [CT1.1, CT1.2, C3]}$$

$$S(X) = \mathbb{S}X\mathbb{S} \text{ [CT0.7(2)]};$$

$$SX$$

$$= X \implies \mathbb{S}X\mathbb{S}$$

$$= X: \ \text{metric data invariant under } p \leftrightarrow q$$

$$\implies g_{\mu\nu}(e_{pq})$$

$$= g_{\mu\nu}(e_{qp}) \ \text{(same channel locus [CT2.6])}$$

$$\text{D1.6}: \ ds^2(e)$$

$$= g_{\mu\nu}(e)$$

$$\Delta x_e^\mu \Delta x_e^\nu \text{ is quadratic in } \Delta x: \ \text{every monomial has } \Delta x\text{-degree } 2 \implies \Delta x$$

$$\mapsto -\Delta x \text{ fixes } ds^2$$

$$\text{(incl. the shift cross-term } 2g_{0i}\Delta x^0 \Delta x^i: \ \text{both factors flip)};$$

$$\implies ds^2(e_{pq}) = ds^2(e_{qp}) \implies \widehat{ds}^{\,2}(e_{pq}) = \widehat{ds}^{\,2}(e_{qp}) \quad [\text{D1.6}: \ /\mathfrak{h}^2]. \ \blacksquare$$

Phys. Both orientations of one channel carry a single geometric datum: the line element is a property of the channel; orientation lives in the registration signs and transport order, not in the metric. This is what lets the minimal closed chain — the 2-cycle — speak about one number $\widehat{ds}^{\,2}$.

Theorem 3.26 (FT.24 — The minimal fermion pins the geometry quantum). Thm. Let $c = (e_1, e_2)$, $e_1 = (p, q)$, $e_2 = (q, p)$, be a materialized 2-cycle — the minimal closed chain [A1 excludes self-loops; D1.4; Def-H.4 floor: $\delta h(c) \geq 2$]. Write $\widehat{ds}^{\,2}(e_1) = \widehat{ds}^{\,2}(e_2) =: \kappa$ [FT.23]. Then:

$$(1) \quad \Theta(c) = 2\kappa \in 2\mathbb{Z} \ [\text{FT.22}] \implies \kappa \in \mathbb{Z}$$

$$\text{(per-edge integrality at the materialized fermion floor)};$$

$$(2) \quad ds^2(e) = \kappa\, \mathfrak{h}^2 \in \mathfrak{h}^2\, \mathbb{Z} \qquad \text{(line element quantized in hop quanta [D1.6])};$$

$$(3) \quad \mathfrak{h}^2 = 4\, G_N \ln 2 \ [\text{CT2.9}] \implies ds^2(e) = 4\kappa\, G_N \ln 2$$

$$\text{(interval quantum = one Bekenstein bit-area [CT1.29] per unit } \kappa);$$

$$(4) \quad \text{minimal real edge [D1.1, Comm-min]}: \ \kappa = 1 \text{ attained (floor tight)};$$

$$\kappa$$

$$= 0 \iff \text{null edge [CT1.19a] (degenerate floor; all-null transit chains are the trivial chains = photons/gluons [D1.7, D3.7])};$$

$$\kappa < 0 : \text{ timelike [CT1.12 : } \operatorname{sig}(g) = (-,+,+,+)].$$

Req. FT.22, FT.23; A1; D1.1, D1.4, D1.6, D1.7; Def-H.4; CT1.12, CT1.19a, CT1.29, CT2.9 (Part 2); D3.7; EM.20's scope tokens (inherited via FT.22).

Pf.

$$(1) : \ \Theta(c) = \widehat{ds}^{\,2}(e_1) + \widehat{ds}^{\,2}(e_2) = 2\kappa \text{ [FT.23]}; \qquad \text{FT.22 : } \Theta(c) \in 2\mathbb{Z} \implies \kappa \in \mathbb{Z}$$

$$(2) : \ \text{D1.6 : } ds^2(e) = \widehat{ds}^{\,2}(e)\,\mathfrak{h}^2 = \kappa\,\mathfrak{h}^2;$$

$$(3) : \ \text{CT2.9 substitution;} \quad \text{CT1.29 : } S = \frac{A \ln 2}{\mathfrak{h}^2} \text{ at } A = \mathfrak{h}^2 : \ S = \ln 2 = 1 \text{ bit per unit } \kappa$$

$$(4) : \ \text{D1.1 : minimal real edge has } \widehat{ds}^{\,2} = 1 \implies \kappa = 1;$$

$$\text{CT1.19a : } \widehat{ds}^{\,2} = 0 \iff \Delta x_e \text{ null;} \quad \text{D1.7/D3.7 : all-null chains} = \text{photons/gluons}$$

$$\kappa \in \mathbb{Z} \ \wedge \ \operatorname{sig}(g) = (-,+,+,+) \text{ [CT1.12]} \implies \kappa < 0 \text{ admissible (timelike edges).} \ \blacksquare$$

Rmk (two integer reads at the floor). The materialized minimal bound structure carries two derived integer reads: the geometric winding $\kappa = \Theta(c)/2$ [FT.22, clause (1)] and the informational mass $m_{\text{hid}} = I_{\text{hid}}/\mathfrak{h}$ [Def-H.9(2), FT.17: $I_{\text{hid}} = E_{\text{int}} \ln 2$, an internal-edge count].

Phys. Computing the smallest fermion the corpus allows — the materialized 2-cycle — settles the ds^2 question on-shell: geometry comes in integer multiples of the hop area, and the hop area is itself the calibrated information quantum $4G_N \ln 2$. The spacetime quantum is not a new postulate; it is the fermion floor read through the corpus's own pole condition. Off-shell configurations in Z keep their generic continuous $\widehat{ds}^{\,2}$ [CT1.19a]: integrality is a property of the spectrum, not of the summation domain — exactly as in CT2.5.

Theorem 3.27 (FT.25 — Internal holonomy is well-defined on-shell). Thm. Σ a bounded subnetwork [Def-H.9]. Then:

(1) $\Sigma \setminus \partial\Sigma$ decomposes edge-disjointly into materialized closed chains $\{c_i\}$:

$$\Theta_{\text{int}}(\Sigma) = \sum_i \Theta(c_i) \in 2\mathbb{Z},$$

$$\text{Hol}_{\text{int}} = (-1)^{\Theta_{\text{int}}} = +1 \ \text{ [FT.22]};$$

(2) every internal edge carries on-shell integer line element [FT.24] :

$$\Theta_{\text{int}}(\Sigma) \in \mathbb{Z},$$

$$\text{Hol}_{\text{int}} = (-1)^{\Theta_{\text{int}}} \in \{\pm 1\} \text{ well-defined;}$$

$$(3) \quad \text{the surviving on-shell integer label is the winding } k(c)$$

$$= \Theta(c)/2 \in \mathbb{Z} \text{ [FT.22];}$$

$$\text{at the floor } k$$

$$= \kappa \text{ [FT.24(1)];}$$

$$\text{full materialized loops have Hol}_{\text{spiral}}$$

$$= +1,$$

so the binary Hol_{int} is nontrivial only on partial-chain internal sets (proper subsets of materialized chain edges), where Θ_{int} is a partial sum of integers with free parity.

Req. FT.22, FT.24; Def-H.9, Def-H.14; D1.7; FT.9 ($\alpha = \pi$); CT1.19a (off-shell contrast).
Pf.

$$(1): \text{ edge-disjoint decomposition : } \Theta_{\text{int}}$$

$$= \sum_{e \in \Sigma \setminus \partial\Sigma} \widehat{ds}^{\,2}(e)$$

$$= \sum_i \sum_{e \in c_i} \widehat{ds}^{\,2}(e)$$

$$= \sum_i \Theta(c_i)$$

$$\in 2\mathbb{Z} \text{ [FT.22];}$$

$$(-1)^{2\mathbb{Z}} = +1$$

$$(2): \text{ finite sum of integers [FT.24];}$$

$$\text{Hol}_{\text{int}} = e^{i\pi\Theta_{\text{int}}} = (-1)^{\Theta_{\text{int}}} \text{ well-defined } [\alpha = \pi : \text{ FT.9}]$$

$$(3):\ k := \Theta/2 \in \mathbb{Z}\ \text{[FT.22]};\quad \text{floor}:\ \Theta = 2\kappa \implies k = \kappa\ \text{[FT.24(1)]};$$

$$\text{full loop}:\ \mathrm{Hol}_{\text{spiral}} = e^{i\pi\cdot 2k} = +1$$

$$\text{partial set } P \subsetneq E(c):\ \Theta_{\text{int}}$$

$$= \sum_{e\in P} \widehat{ds}^{\,2}(e) \in \mathbb{Z} \text{ by (2)},$$

parity free (witness : c with per-edge $(1,1,1,3)$, $\Theta = 6 \in 2\mathbb{Z}$; $P =$ first three : $\Theta_{\text{int}} = 3$, $\mathrm{Hol}_{\text{int}} = -1$; `verify_H_ds2.py`). ■

Rmk (no layer-local integrality postulate). No layer-local postulate of general-edge line-element integrality — $\widehat{ds}^{\,2}(e) \in \mathbb{Z}$ for every real edge — is used. Its on-shell content is derived (FT.22–FT.25, from the corpus's own materialization mechanism); its off-shell generalization is neither derivable nor needed — every use site in this document (Def-H.7, Def-H.9(3), FT.14's parity read, FT.15's S clause, FT.20 row W) concerns bound/fermion content and is scoped on-shell at its site. The axioms remain self-contained: no step above adds a premise beyond the corpus's declared ones (Comm-min, Eff — A §0.2a).

Rmk (spin-extraction consequence). For Σ swept entirely by fermion chains, clause (1) forces $\mathrm{Hol}_{\text{int}} = +1$ on-shell: the parity of a full materialized loop cannot carry the spin label. The live candidates are the integer data — the winding $k(c)$, partial-sum Θ_{int} [clause (3)], weak-sector insertion multiplicities [FT.13], the transported seed's $j = \frac{1}{2}$ carrier [EW.32].

Phys. The internal geometric ledger of a coarse-grained fermion is integer-valued wherever the corpus's own spectral condition reaches: loops materialize in even hop-area quanta, the minimal loop is exactly one quantum per channel, and Def-H.9's binary holonomy label is a well-defined derived object rather than a postulate-conditional one. The remaining question is not whether on-shell geometry is quantized — it is — but which integer label carries spin, and how far the on-shell sublattice extends across the graph.

3.11 Tier F.9 — The Activated Aspect Operator: $\mathcal{A}(\Sigma)$ as a Computable Mask

Definition 3.19 (Def-H.15 — Coarse-grained aspect space and masked force operator). Def. Σ a bounded subnetwork; $C \in \mathcal{M}(U)$ the ambient configuration [D2.11].

$$(1)\quad \mathcal{K}_\Sigma(C) := \bigoplus_{A\in\mathcal{A}(\Sigma;C)} \mathcal{K}^{(A)}$$

[the aspect blocks of D2.3; membership by FT.20's criteria table];

$$(2)\quad \mathcal{P}_M^{(\Sigma)}(C) : \mathcal{K} \to \mathcal{K}_\Sigma(C) \qquad \text{(orthogonal projection [D2.3/CT2.2 pattern])};$$

$$(3)\quad \mathrm{F}_M^{(\Sigma)}(C) := \mathcal{P}_M^{(\Sigma)}\,\mathrm{F}\,\mathcal{P}_M^{(\Sigma)} \qquad [\text{A8/C7, CT2.3 pattern}],$$

with the strong summand admissibility-filtered:

$$F_{\text{strong}} \;\longmapsto\; \Pi_{\mathcal{C}}(\Sigma;C)\, F_{\text{strong}}\big|_{\Sigma}\, \Pi_{\mathcal{C}}(\Sigma;C) \qquad [\text{Def-H.10},\ \text{FT.19}];$$

(4) realization : ι_Σ [Def-H.13] maps $\mathfrak{u}(N(\Sigma))$ into $\mathcal{K}_\Sigma$'s boundary-coupled part [FT.21];

on-shell internal labels (Θ_{int}, k) [FT.25, FT.22] superselect the W/G reads of Def-H.9's tuple; off-shell, blocks carry arena data [CT1.19a, Def-H.18].

Rmk (configuration-relativity). Unlike the rung masks $\mathcal{P}_M^{(n)}$ [D2.3], which are state-independent, $\mathcal{K}_\Sigma(C)$ depends on the ambient configuration C — FT.20's criteria read f_v, I_{hid}, Θ_{int}, $\mathcal{C}(\Sigma;C)$, which are state data.

Definition 3.20 (Def-H.16 — Coarse-grained resolvent and the Σ-spectrum). Def.

$$Z(\mathcal{R},\Sigma) := \Big\langle \text{out}\Big|\big(1-\mathrm{F}_M^{(\Sigma)}\big)^{-1}\Big|\text{in}\Big\rangle \qquad [\text{D2.4 pattern};\ \text{D2.8 sheet}],$$

with $|\text{in}\rangle, |\text{out}\rangle$ read on boundary-channel states via ι_Σ [Def-H.13] — a constitutive clause of this definition (the ι_Σ boundary-state identification). Poles of $Z(\mathcal{R},\Sigma) :=$ the Σ-spectrum [D2.5 pattern]: pole $\iff 1 \in \operatorname{spec}\mathrm{F}_M^{(\Sigma)}$ [CT2.5 pattern]; residues = transition amplitudes; $m^2 :=$ pole displacement [D2.5]. A bound chain c interior to Σ materializes (FT.22's sense) iff it contributes to a pole of $Z(\mathcal{R},\Sigma)$: FT.22's single-valuedness condition is this resolvent's eigenmode condition [CT2.5 as applied in EW.18]. Rmk (two pole readings, scoped). The corpus reads "pole" two ways. (i) On/off eigenvalue condition — pole $\iff 1 \in \operatorname{spec}\mathrm{F}_M$ [CT2.5]: no momentum label required; this is the reading Def-H.16/FT.26/FT.28(7) use, and FT.22's single-valuedness is exactly this reading applied to the chain's spiral transport. (ii) Displacement value — D2.5's "$m^2 :=$ pole displacement" vocabulary presupposes a k^2-parametrized resolvent family; it is a corpus-wide definitional layer (D2.4's own $Z(\mathcal{R},n)$ likewise carries no explicit k), whose values are extracted in D by a transport-defect computation, not a literal momentum-space two-point function: EW.33's $\mathcal{U} = \exp\big(i\oint\sum_A c_A\mathcal{G}_A\,d\ell\big) = 1 + i\,\ell_\gamma\sum_A c_A\mathcal{G}_A + O(c^2)$ — the weak/gauge transport, an object distinct from the spiral factor $e^{i\Phi}$ constrained to $+1$ on-shell by EW.18/FT.22; the two conditions coexist on the same materialized wrapping chains [EW.18 $\wedge$ EW.20: $m_W^2 > 0$]. At the Σ-level, reading (ii) is not yet constructed: the premise-free candidate is an EW.33-pattern defect with one interface transfer choice; a literal momentum-space resolvent would need a new declared choice.

Theorem 3.28 (FT.26 — Structural transfer: the Σ-mask inherits CT2.3–CT2.5). Thm. $\mathrm{F}_M^{(\Sigma)}(C)$ is: (1) Hermitian; (2) block-diagonal, $\mathrm{F}_M^{(\Sigma)} = \bigoplus_{A\in\mathcal{A}(\Sigma;C)} F_A\big|_{\mathcal{K}^{(A)}}$, the strong summand $\equiv 0$ whenever $\mathcal{C}(\Sigma;C)=0$; (3) finite-dimensional with

$$\dim\mathcal{K}_\Sigma \;\le\; 1+5+1+3+8 \;=\; 18 \;<\; \infty.$$

Hence the spectral theorem applies and $Z(\mathcal{R},\Sigma)$ has simple poles exactly at $1 \in \operatorname{spec}\mathrm{F}_M^{(\Sigma)}$, residues $-\langle\text{out}|P_a|\text{in}\rangle$ — CT2.5's statement transferred verbatim with $n \mapsto \Sigma$.

Req. Def-H.15, Def-H.16; A8; CT0.12, CT0.14 (block dimensions); CT2.2, CT2.3, CT2.4, CT2.5 (Part 2); FT.19, FT.20; spectral theorem [math].

Pf.

$$(1):\ \left(\mathcal{P}\mathrm{F}\mathcal{P}\right)^{\dagger} = \mathcal{P}^{\dagger}\mathrm{F}^{\dagger}\mathcal{P}^{\dagger} = \mathcal{P}\mathrm{F}\mathcal{P}$$

$$[\text{CT2.3 Pf pattern}:\ \mathcal{P}_M^{(\Sigma)} \text{ orthogonal [D2.3 pattern]},\ \mathrm{F}^{\dagger} = \mathrm{F}]$$

$$(2):\ \text{A8}:\ \mathrm{F} = \bigoplus_A F_A,\ \operatorname{supp} F_A \subseteq \mathcal{K}^{(A)};$$

$$\text{block-subset projection retains exactly the members [CT2.4 Pf pattern]}$$

$$\text{strong summand}:\ \mathcal{C}(\Sigma;C) = 0 \implies \Pi_{\mathcal{C}} = 0 \implies F^{\Sigma}_{\text{strong}} = 0\ [\text{FT.19 Thm};\ \text{Def-H.15(3)}]$$

$$(3):\ \mathcal{A}(\Sigma;C) \subseteq \{\text{EM}, G, H, W, \text{strong}\}\ [\text{FT.20}:\ J^{\pm}, D^{\pm} \text{ never}] \implies \dim \mathcal{K}_{\Sigma}$$

$$\leq 1 + 5 + 1 + 3 + 8$$

$$= 18\ [\text{CT0.12},\ \text{CT0.14}:\ \dim \mathcal{K}^{(\text{EM})} = 1,\ \dim \mathcal{K}^{(\text{strong})} = 8]$$

$$\text{finite-dim Hermitian} \implies \text{spectral decomposition}$$

$$\implies (1 - \mathrm{F}_M^{(\Sigma)})^{-1}$$

$$= \sum_a \frac{P_a}{1-\lambda_a}:\ \text{poles} \iff \lambda_a$$

$$= 1;$$

D2.8 sheet [CT2.5 Pf, verbatim]. ■

Rmk (the observability deficit, in dimensions). The rung-3 mask carries $\dim \mathcal{K}_3 = 25$ [CT0.14]; the maximal observable Σ-mask carries 18. The deficit $25 - 18 = 7 = 3 + 3 + 1$ is exactly $J^+ \oplus J^- \oplus D^+$ [CT0.12] — the structural aspects FT.20 reclassifies as non-observable (seals, uniform centre). At the Σ-level, action and observability are different masks, and the difference is a measurable dimension.

Phys. A coarse-grained subnetwork now has the same spectral machine a single vertex has: a Hermitian masked force on a finite aspect space, a resolvent whose poles are the Σ-particles, and confinement built into the block structure — one unsealed colour attachment anywhere and the entire strong summand vanishes [FT.19]. This is the activation of $\mathcal{A}(\Sigma)$: membership is a computable nonzero block, not merely a criteria table.

Theorem 3.29 (FT.27 — Single-vertex degeneration; action versus observable). Thm. At $\Sigma = \{v\}$: (a) the spaces degenerate on the nose — $\mathcal{H}_{\text{slot}}(\{v\}) = \mathcal{H}_{\text{slot}}^{(n(v))}$ [Def-H.12], $\mathcal{H}_{\text{slot}}^{\text{int}}(\{v\}) = 0$ [Def-H.14], $\iota_{\{v\}} = \iota_{n(v)}$ [Def-H.13]; (b) FT.20's criteria evaluated at $\{v\}$ agree with CT0.13's active set on the EM row (for type-$(1,1)$ states), and refine it on four rows — each refinement a proven fact, not a deviation:

$$\text{EM} \in \mathcal{A}(\{v\}; C) \iff f_v \neq 0 \qquad \text{(CT0.13's type-}(1,1)\text{ condition [D3.3])};$$

$$G \in \mathcal{A}(\{v\}; C) \iff I_{\text{hid}}(\{v\}; o) \neq 0 \iff n(v) \geq 2$$

$$[\text{FT.16}: \ I_{\text{hid}} = (n-1)\ln 2] \text{ — refines } G \in \mathcal{A}(1): \text{ at } n = 1 \text{ the block acts but no observable exists};$$

$$W \in \mathcal{A}(\{v\}; C) \iff \text{never}$$

$$[\Theta_{\text{int}}(\{v\}) = 0: \ \text{empty sum}, \ \text{Def-H.14}] \text{ — weak acts from rung 1 [CT0.13] but its observable is born only at the bound floor (FT.28)};$$

$$\text{strong} \in \mathcal{A}(\{v\}; C) \iff \text{never}$$

$$[\text{case split}: \ \text{Colour}(\{v\}) = \varnothing \ (v \text{ colourless}) \implies \text{FT.20's first conjunct fails} \wedge F_{\text{strong}}|_{\{v\}} = 0 \ (\neg\exists \text{ colour content}); \quad \text{Colour}(\{v\}) = \{v\} \implies \pi_C(v) \subseteq \{v\} \text{ impossible [A1}: \ \text{partners distinct}] \implies \mathcal{C}(\{v\}; C) = 0 \implies \text{block} \equiv 0 \text{ [FT.19]}] \text{ — either way the strong block is zero at a single vertex;}$$

the smallest colour-observable Σ is the sealed meson-type pair [D3.9, CT3.17; realized at FT.28 Rmk];

$$J^{\pm}, D^{\pm} \notin \mathcal{A}(\{v\}; C) \ \forall n \qquad \text{[FT.20 reclassification]}.$$

Req. Def-H.10, Def-H.12, Def-H.13, Def-H.14, Def-H.15; A1; CT0.13 (Part 1); D0.8, D3.1, D3.3, D3.8, D3.9; CT3.11, CT3.17 (Part 2); FT.16, FT.19, FT.20.

Pf.

$$(a): \ \text{degenerate-case checks of Def-H.12/Def-H.13/Def-H.14, verbatim}$$

$$(b): \ \text{FT.20's table at } \Sigma = \{v\}: \ \ \text{EM row} = f_v \neq 0 \text{ [D3.3]};$$

$$G \text{ row}: \ I_{\text{hid}}(\{v\}; o) = (n-1)\ln 2 \neq 0 \iff n \geq 2 \text{ [FT.16]}$$

$$W \text{ row}: \ \Sigma \setminus \partial\Sigma = \varnothing \text{ [A1}: \ \neg\exists \text{ self-loops]} \implies \Theta_{\text{int}} = 0 \text{ (empty sum)}: \ \text{criterion cannot fire}$$

strong row (case split) :

$$\text{Colour}(\{v\})$$

$= \varnothing$ (v colourless, e.g. $n(v) \neq 3$ [D3.1]) : FT.20's first conjunct fails $\wedge$ $F_{\text{strong}}\big|_{\{v\}}$

$= 0$ ($\neg\exists$ colour content to act on [D3.8]) $\implies$ block $\equiv 0$

$$\text{Colour}(\{v\})$$

$= \{v\}$ (v colour-carrying) : $\pi_C(v) \neq \varnothing$ requires a distinct partner [A1 : $p \neq q$] $\implies$ $\pi_C(v) \not\subseteq \{v\}$ $\implies$ $\partial_{\text{col}}(\{v\}; C)$

$$= \{v\} \implies \mathcal{C}(\{v\}; C)$$

$= 0$ [Def-H.10] $\implies$ block

$\equiv 0$ [FT.19]

deltas vs. CT0.13 are exactly : G@n=1 (acts, ¬observable [FT.16]);

W@¬∃-interior (acts, ¬observable);

strong@single vertex (acts at n=3 [CT0.13], ¬observable : colourless — ¬∃content; colour-carrying — partner distinctness [A1]);

$$J^{\pm},$$

$D^{\pm}$ (structural reclassification [FT.20 Rmk; CT0.7/FT.11 seals-¬charges; FT.3/CT4.4 D^{-} darkness])—each a proven fact. ■

Rmk (the birth of spin). The weak observable requires an interior edge; the first interior exists at the bound-structure floor (2-cycle, Def-H.4). A single vertex has weak action but no weak observable: spin — already identified algebraically as the weak contribution [FT.13, Def-H.8] — is an observable of coarse-graining itself.

Phys. At a single vertex the activated operator reduces to the rung machine CT0.13–CT2.5 on everything observable, and the four deviations are theorems, not defects: gravity becomes observable exactly when something can be hidden ($n \geq 2$), spin exactly when something can wind (bound floor), colour exactly when a sealing partner is inside the region read — never at one vertex, whether colourless (no content to act on) or colour-carrying (no inside partner), and seals and locks are never observable at all. Action and observability are two different masks — H now carries both.

3.12 Tier F.10 — The Strong Edge Weight

Definition 3.21 (Def-H.17 — Strong edge transport). Def. For $e \in E_{\mathcal{R}}$ incident on colour-carrying vertices (rung-3 attachments [D3.8]):

$$G^{(3)}(e) \;:==\; G^{(2)}(e) \;\otimes\; \tau(e), \qquad G^{(2)}(e) = \sigma_e\, e^{i\pi\, \widehat{ds}^{\,2}(e)} \quad [\text{D3.5},\ \text{FT.9}],$$

$\tau(e) :=$ the slot-index transport of e on $\mathcal{H}^{(3)}_{\text{slot}} \cong \mathbb{C}^3$ [D3.1, CT3.2], acting in the defining representation [CT3.11(2)–(3): triplet $\mapsto U^i{}_j q^j$, antitriplet $\mapsto r_j (U^{-1})^j{}_i$, $U \in SU(3)$], algebra image in $\iota_3(\mathfrak{u}(3))$ [D2.12]. Strictly-new strong content: $\pi(\tau) \in \mathfrak{su}(3)$, $\pi(X) = X - \frac{\mathfrak{T}(X)}{3} I_3$ the trace projection [CT3.7, D3.2] — the $\mathfrak{u}(1)$ centre of $\mathfrak{u}(3)$ is EM's, spent at rung 2 [CT3.7, CT3.13; CT0.13 step n=3], and already carried by $G^{(2)}$'s phase factor.

Theorem 3.30 (FT.29 — Strong weight: factorization, no α_{strong}, observable sealing). Thm.

$$(1) \quad \text{factorization}: \; G^{(3)}(e) = G^{(2)}(e) \otimes \tau(e) \; [\text{Def-H.17}],$$

$$\alpha^{(3)}_{\text{EM}} = \alpha^{(2)}_{\text{EM}} = \pi \; [\text{EM.18},\ \text{CT3.10}];$$

(2) non-existence : $\neg\exists$ per-edge abelian strong phase coefficient (an "α_{strong}") [CT3.10a (F) : trace route; eight-dimensional obstruction to a scalar coefficient read; non-canonicity of $\mathfrak{u}(1) \subseteq \mathfrak{su}(3)$ choices (explicit $SU(3)$ witnesses); centre identification via CT3.7/CT3.13];

$$(3) \quad \text{observable content}: \; \text{under any pole read [CT2.5]},$$

$\tau(e)$-factors survive only inside invariant contractions $\{\delta^i{}_j, \varepsilon_{ijk}, \varepsilon^{ijk}\}$ [CT3.17];

free indices are mask-annihilated [CT3.18];

isolated triplets have ¬pole and norm 0 [CT3.22, CT2.8];

at Σ-level the strong summand is gated by $\Pi_{\mathcal{C}}(\Sigma; C) \in \{0, 1\}$ [FT.19, FT.26(2)]. Hence the effective strong weight of any materialized configuration

$$= (\text{its } G^{(2)} \text{ factors}) \times (\text{singlet-sealing factor}) : \; \neg\exists \text{ per-edge strong phase read.}$$

Req. Def-H.17; D2.12, D3.1, D3.2, D3.5, D3.8, D3.9; CT0.13, CT0.14 (Part 1); CT2.5, CT2.8 (Part 2); CT3.2, CT3.4, CT3.7, CT3.8, CT3.9, CT3.10, CT3.10a, CT3.11, CT3.13, CT3.17, CT3.18, CT3.22 (Part 2); EM.17, EM.18, EM.20 (Part 2); FT.9, FT.10, FT.19, FT.26.

Pf.

$$(1): \; \text{Def-H.17};$$

$$\text{coefficient stability}: \text{EM.18}\,[\text{CT0.14}: \mathcal{K}^{(\text{strong})} \perp \mathcal{K}^{(\text{EM})}] \wedge \text{CT3.10}\,[\mathfrak{T}(F_{\text{strong}}) = 0: \text{ independent argument}] \implies \alpha^{(3)}$$

$$= \alpha^{(2)}$$

$$= \pi \text{ [FT.9/EM.20]}$$

$$\text{consistency}: \text{FT.10 Pf already computes with this factorization } (G^{(3)} = G^{(2)} \otimes \text{label}: \text{ per-edge weight unmodified})$$

$$(2): \text{ CT3.10a (F)}$$

$$(3): \text{ CT3.17}: \forall \text{ invariants} = \text{combinations of } \{\delta, \varepsilon, \bar{\varepsilon}\};$$

$$\text{CT3.18}: \mathcal{P}_M^{(3)}\big|_{\text{free index}} = 0 \implies \neg\text{pole [CT2.5]}$$

$$\text{CT3.22}: \text{ isolated triplet norm } 0 \text{ [CT2.8]};$$

$$\text{FT.19/FT.26}(2): \Pi_{\mathcal{C}} \text{ gates the entire } \Sigma\text{-level summand};$$

$$\text{sealed}: \tau\text{-content enters as singlet contraction amplitudes only [CT3.17, D3.9]. } \blacksquare$$

Rmk (stipulation-dependence). The centre identification route — the unique canonical centre is EM's — cites CT3.7's reading of D2.12's realization clause ("the abstract isomorphism class lies in $\iota_m(\mathfrak{u}(m))$"): the one stipulative layer in the chain [CT3.13: $\neg\exists$ ninth gluon mode]. The non-existence core (trace route; dimension obstruction; non-canonicity witnesses) is independent of that reading, and the theorem's force does not rest on the identification.

Rmk (three routes to discreteness). The corpus's quantization mechanisms are now three, on three different objects: (R1) reality line — CT0.9: $\sigma_e \in \{\pm 1\}$, single-qubit $\wedge^2\mathcal{H}_q$; (R2) materialization — EW.18/FT.22–FT.25: $\kappa, k, \Theta_{\text{int}} \in \mathbb{Z}$, pole single-valuedness on loops; (R3) selection — CT3.18/FT.19: admissibility $\in \{0, 1\}$, mask annihilation of free colour. The strong edge weight is governed by R3 alone: it registers no per-edge number. (Cross-ref: FT.30's verdict column.)

Phys. The strong edge weight is the EM weight tensored with a colour-transport label — and the label carries no Abelian phase of its own (non-existence frozen: CT3.10a). Colour becomes observable only sealed into singlets: the per-edge strong "quantum" is the sealing itself. Confinement is not a force law added to $G(e)$; it is the observable content of $G(e)$'s strong factor.

3.13 Tier F.11 — Sector Quantum Audit and the Continuum Ledger

Definition 3.22 (Def-H.18 — Arena and spectrum (the two-locus vocabulary)). Def. The arena of a quantity := the configuration domain over which $Z(\mathcal{R})$ sums [A8, C12: $\mathfrak{C}(\mathcal{R})$] — off-shell; continuous wherever CT1.12/CT1.19a make it so. The spectrum of a quantity := its values on materialized configurations — those contributing to poles of the masked resolvent [D2.4, CT2.5; Σ-level: Def-H.16] — on-shell. Two-locus principle (vocabulary for an already-proven distinction): discreteness claims in this corpus are spectrum claims; continuity claims are arena claims. One and the same object may be continuous in the arena and quantized in the spectrum — CT2.5 (continuous operator families, discrete pole set) and FT.22–FT.25 (continuous $\mathcal{E}(e)$ arena [Def-H.14 Rmk], integer winding spectrum) are the two proven instances. Vocabulary only — no new premise.

Theorem 3.31 (FT.30 — Pair-channel sector quantum audit: $\big((5{+}3{+}1)+(3{+}3)+1\big)$). Thm. The 16 pair-channel registers [CT0.6] tile as $(G_5+W_3+H_1)+(J_3^+ + J_3^-)+D_1^+$ [CT0.12, CT1.1, CT1.2], and CT1.12's metric dictionary decomposes block-wise:

$$g_{\mu\nu} \;\leftrightarrow\; \underbrace{\hat{h}_{ij}\,(5)}_{G\text{: spin-2 shear}} \;\oplus\; \underbrace{\Sigma_{\mathrm{tr}}\,(1)}_{H\text{: volume mode}} \;\oplus\; \underbrace{a_{00}\,(1)}_{D^+\text{: }\Omega\text{ conformal}} \;\oplus\; \underbrace{w_i\,(3)}_{W\text{: shift}};$$

$$J^{\pm}\ (S\text{-odd}):\ 0 \text{ metric coordinates [Def-H.14 : } \mathcal{E}(e) \text{ is } S\text{-even}].$$

Independent-quanta audit:

block	arena data	spectrum data	independent quantum?
G (5)	continuous shear [CT1.12]	winding $\kappa, k \in \mathbb{Z}$ [FT.22/FT.24], unit $\mathfrak{h}^2 = 4G_N \ln 2$ [CT2.9]	yes — geometric (joint read, ¬per-block)
W (3)	shift w_i [CT1.12]; Casimir fixed : $\vee^2 j{=}1$, $\wedge^2 j{=}0$ [CT0.7a]; seed $j{=}\frac{1}{2}$ [EW.32]	integer labels k, Θ_{int} [FT.25(3), FT.28(2)]	candidate — spin (extraction open, Gaps 11); ¬self-charge [FT.12]
H (1)	h_0 seed background, continuous P4 data [EW.13, P4]; $M^2 \propto h_0^2$ [EW.33]	via mass read only [FT.20 row H, Def − H.36]	no — continuous configuration datum
J^+ (3)	seals $n_{\mathcal{R}} : 0 \to 1$ [D1.2]; S-odd = reality grading [CT0.7]	¬observable [FT.20]	no — structural; $\#J^+ = \#J^-$ [D1.3/D2.11]; ¬charges [FT.11]
J^- (3)	dissolution seals [D1.2]	¬observable [FT.20]	no — structural, as J^+
D^+ (1)	Λ uniform centre [A8, D4.3]; $w = -1$ [CT1.25]; Ω channel entry [Def-H.2]	¬observable independently [FT.20]	no — P4 datum
stack aspects (outside the 16 [CT0.12 Rmk]) :			
EM	σ_e binary [CT0.9]; $q \in \{-1, 0, +1\}$ [EM.9/D3.4]	parity $(-1)^\kappa$ on-shell [FT.24]	yes — charge (R1 reality line)
strong	$\tau(e)$ defining-rep label [Def-H.17]; ¬∃ per-edge phase [FT.29(2)]	singlet sealing $\{0, 1\}$ [FT.19/FT.29(3)]	yes — admissibility (R3 selection)
D^-	locked content [D4.2]	¬observable [FT.3, CT4.4]	no

Verdict. Independent quanta exist in exactly three read classes, one per discreteness route [FT.29 Rmk]: (R1) binary registration — reality line [CT0.9]: σ_e, q, $(-1)^{\aleph}$; (R2) integer winding — materialization [EW.18, FT.22–FT.25]: κ, k, Θ_{int}, unit $\mathfrak{h}^2 = 4G_N \ln 2$ [CT2.9, CT1.29]; (R3) singlet admissibility — mask selection [CT3.18, FT.19]: $\{0, 1\}$. No pair-channel block carries a block-local quantum: the geometric quantum is a joint S-even read — $\widehat{ds}^{\,2}(e) = g_{\mu\nu}(e)\Delta x_e^\mu \Delta x_e^\nu/\mathfrak{h}^2$ [D1.6] mixes all ten metric coordinates across G, H, D^+, W — and the $\big((5{+}3{+}1)+(3{+}3)+1\big)$ blocks are the arena decomposition of that read.

Rmk (the emergence chain, assembled). How spacetime "truly emerges", in six cited steps: (1) per-edge S-even data $(a_{00}, q_{ij}, w_i) \leftrightarrow$ metric dictionary $g_{\mu\nu}$ [CT1.12, D2.1 — block-wise per this Thm]; (2) coarse-grained ledger averages $\to$ smooth (M, g) under hypotheses (i)–(iii) [CT2.6]; (3) line-element sums $\to$ path integrals, rel. error $O(\mathfrak{h})$ [CT1.19]; (4) on-shell selection: $\oint_c ds^2 \in 2\mathfrak{h}^2\mathbb{Z}$ [FT.22], per-edge integers at the floor [FT.24], classical limit = large-k [FT.31(3)]; (5) stationary set of the ledger sum = classical geometry [CT1.20], Einstein–Hilbert closure [CT1.21, CT2.7], calibration $\mathfrak{h}^2 = 4G_N \ln 2$ [CT2.9]; (6) particles = poles of the Σ-resolvent [Def-H.16, FT.26]. Two loci throughout [Def-H.18]: the arena continuous, the spectrum integer — no step adds a premise beyond the corpus's declared set (A §0.2a).

Req. CT0.6, CT0.7, CT0.7a, CT0.12, CT0.14 (Part 1); CT1.1, CT1.2, CT1.12, CT1.19, CT1.20, CT1.21, CT1.25, CT1.29, CT2.5, CT2.6, CT2.7, CT2.9 (Part 2); D1.2, D1.3, D1.6, D2.11, D4.2, D4.3; EW.13, EW.18, EW.32, EW.33 (Part 2); EM.9 (Part 2); CT3.18, CT3.22 (Part 2); CT4.4 (Part 2); FT.3, FT.9, FT.11, FT.12, FT.20, FT.22, FT.24, FT.25, FT.29; Def-H.2, Def-H.14, Def-H.17, Def-H.18; P4.

Pf.

$$\text{tiling : CT0.12 table : } 3{+}3{+}5{+}3{+}1{+}1 = 16 = C_{\text{cap}} \text{ [CT0.6]};$$

$$\text{sector parities : CT1.1/CT1.2 : } \mathcal{E}_{++}$$

$$= \{E_0, \Sigma, \hat{E}^{ij}\}\ (1{+}1{+}5),$$

$$\mathcal{E}_{+-}$$

$$= \{W_i\}\ (3),$$

$$\mathcal{E}_{-+}$$

$$= \{J_i\}\ (3),$$

$$\mathcal{E}_{--} \qquad = \{K_i\}\ (3)$$

$$\text{dictionary blocks : } a_{00} \sim E_0$$

$$= D^+ \text{ block};$$

$$\Sigma_{\text{tr}} \sim \Sigma$$

$$= H \text{ block};$$

$$\hat{h}_{ij} \sim \hat{E}^{ij}$$

$$= G \text{ block};$$

$$w_i \sim W_i \qquad = W \text{ block}$$

$$[\text{CT1.12 table } + \text{ CT0.12 block table; D2.1 coefficients}]$$

$$J^{\pm}:\ S\text{-odd [CT1.1]} \implies \notin \mathcal{E}(e) = \mathcal{E}_{++} \oplus \mathcal{E}_{+-} \text{ [Def-H.14]} \implies 0 \text{ metric coordinates}$$

$$\text{table rows}:\ \text{as cited (each cell one frozen-corpus or proven-H tag)}$$

$$\neg\text{block-locality}:\ \widehat{ds}^{\,2}(e)$$

$= g_{\mu\nu}\Delta x^{\mu}\Delta x^{\nu}/\mathfrak{h}^2$ [D1.6] with $g_{\mu\nu}$ spanning all four blocks [CT1.12 dictionary] $\implies$ κ is a joint read. ■

Phys. Read the 16-matrix as the ledger's periodic table: nine S-even force directions build the metric arena (shear + volume + conformal + shift), six S-odd directions make and unmake the reality of edges, one centre rescales uniformly — and not one of those blocks holds a quantum by itself. The quanta live in the reads: a sign at registration, a winding at materialization, a sealing at selection. That is the precise sense in which spacetime "truly emerges": an S-even arena on which materialization writes the integers, while the S-odd sectors run the registration that makes edges — and hence geometry — exist at all.

Theorem 3.32 (FT.31 — The continuum limit survives on-shell quantization). Thm. Let $\{c_{\mathfrak{h}}\}$ be a family of materialized chains [FT.22] with $L = \mathfrak{h}\, n_{\mathcal{R}}$ fixed, $\mathfrak{h} \to 0$ [CT1.19's regime]. Then:

$$(1)\quad \text{per edge}:\ \kappa_e$$

$= g_{\mu\nu}\dot{x}^{\mu}\dot{x}^{\nu}(\lambda_e) + O(\mathfrak{h})\ \wedge\ \kappa_e \in \mathbb{Z}$ on-shell [FT.24 scope] : integers sampling a smooth function with $O(\mathfrak{h})$ slack;

$$(2)\quad \text{sums}:\ \sum_e \kappa_e \mathfrak{h}^2$$

$$= \sum_e ds^2(e) \text{ identically [D1.6]} \implies \text{CT1.19 applies verbatim}:\ \text{abs. error } O(\mathfrak{h}L),$$

rel. $O(\mathfrak{h})$: integrality adds no error term — selection acts on family membership, not on sums;

$$(3)\quad \text{selection in the limit}:\ \oint_c ds^2 \in 2\mathfrak{h}^2\mathbb{Z} \text{ [FT.22 Phys]};$$

$$k$$

$$= \frac{1}{2\mathfrak{h}^2}\oint_c ds^2 \to \infty \text{ at fixed classical loop}:\ \text{the lattice is classically invisible, quantally exact;}$$

$$(4) \quad \text{floor relating equation (Gaps 2, minimal-edge class)} : \ \kappa_e$$

$$\equiv 1 \text{ [D1.1, Comm-min]} \implies \Theta(c)$$

$$= \delta h(c) \text{ and } \frac{S_{\text{reg}}(c)}{\oint_c ds^2}$$

$$= \frac{\delta h \ln 2}{\delta h \, \mathfrak{h}^2}$$

$$= \frac{\ln 2}{\mathfrak{h}^2}$$

$$= \frac{S_{\min}}{A_{\min}}$$

[CT0.9, C9, CT1.29].

Req. FT.22, FT.24; CT1.19, CT1.19a (Part 2); CT2.6 hypotheses (i)–(iii) (inherited); D1.1, D1.6, D1.7; CT0.9 (Part 1); C9; CT1.29; Comm-min (via D1.1/EM.20).

Pf.

$$(1) : \text{ CT1.19 Pf : } \Delta x_e^\mu = \mathfrak{h} \, \dot{x}^\mu(\lambda_e) + O(\mathfrak{h}^2) \implies \widehat{ds}^{\,2}(e) = g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu(\lambda_e) + O(\mathfrak{h});$$

$$\text{on-shell } \widehat{ds}^{\,2}(e) = \kappa_e \in \mathbb{Z} \text{ [FT.24]}$$

$$(2) : \ \kappa_e \mathfrak{h}^2$$

$= ds^2(e)$ by definition [D1.6 : $\widehat{ds}^{\,2} = ds^2/\mathfrak{h}^2$] $\implies$ the sums are the same sums : CT1.19 unmodified

$$(3) : \ \Theta(c)\mathfrak{h}^2$$

$$= \sum_e ds^2(e) \to \oint_c ds^2 \text{ [CT1.19]} \ \wedge \ \Theta(c)$$

$$= 2k \text{ [FT.22]} \implies \oint_c ds^2$$

$\in 2\mathfrak{h}^2\mathbb{Z}$ at leading order;

$$\oint \text{ fixed, } \mathfrak{h} \to 0 : \ k \to \infty$$

$$(4):\ \kappa_e \equiv 1 \implies \Theta = \sum_e 1 = \delta h(c)\ \text{[D1.7]};$$

$$S_{\text{reg}}(c) = \delta h \cdot \ln 2\ \text{[CT0.9/C9 :}\ 1\ \text{real edge} = 1\ \text{bit]};$$

$$\oint_c ds^2 = \Theta \mathfrak{h}^2 = \delta h\, \mathfrak{h}^2$$

$$\frac{S_{\text{reg}}}{\oint ds^2} = \frac{\delta h \ln 2}{\delta h\, \mathfrak{h}^2} = \frac{\ln 2}{\mathfrak{h}^2};$$

$$\text{CT1.29 :}\ S_{\min} = \ln 2\ (1\ \text{edge}),\ A_{\min} = \mathfrak{h}^2\ (1\ \text{hop area}) \implies \frac{S_{\min}}{A_{\min}} = \frac{\ln 2}{\mathfrak{h}^2} :\ \text{identical.}\ \blacksquare$$

Rmk (decoupling of the continuum hypotheses). The continuum-limit layer — CT2.6's hypotheses (i)–(iii), CT1.19's convergence — is now fully decoupled from line-element quantization: the hypotheses concern smoothness of g and cell refinement (arena); quantization concerns materialization (spectrum) [Def-H.18]. Neither supports nor threatens the other, and no result of this document makes the continuum limit depend on a discreteness postulate.

Phys. Quantized summands do not break the Riemann machine: the integers κ_e track the smooth sample $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$ to $O(\mathfrak{h})$ per edge, so CT1.19's convergence passes through untouched and the on-shell condition becomes, in the continuum, the Bohr–Sommerfeld rule $\oint ds^2 \in 2\mathfrak{h}^2\mathbb{Z}$. At the minimal-edge floor the information/area relating equation $\ln 2/\mathfrak{h}^2$ is an identity — one bit per hop area [CT1.29] is the same statement as one winding quantum per minimal edge. Classical spacetime is the large-winding limit of an integer ledger.

3.14 Tier F.12 — The Cycle Floor Family, the Defect Bridge, and Composite Floors

Definition 3.23 (Def-H.19 — Stationary, stable, and unstable jumps (vocabulary)). Def. A jump := a transit path between two termini in the emergent chart [D0.6, CT2.6] — program vocabulary, no new structure. A stationary jump: $\delta\int_\gamma ds^2 = 0$ among fixed-endpoint variations — an affinely parametrized geodesic of $g_{\mu\nu}$ [CT1.20's stationary-phase read; Euler–Lagrange for $\int g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu d\lambda$ — math]. A stable jump: a stationary jump that is a local minimum of $\int ds^2$ on its interval (no conjugate points [math: Morse/Synge]); unstable: stationary but saddle or maximal. Lorentzian read [CT1.12: sig $= (-,+,+,+)$]: on timelike paths, minimizing $\int ds^2$ is maximizing proper time — the classical stable trajectory; on spacelike paths, shortest length. Program vocabulary (constitutive; Morse-theoretic [math]); the "relatively stable" idiom of program notes means: measured against the other stationary paths with the same endpoints — an internal comparison, no external scale. No coupling to materialization is asserted. CT1.20's stationary phase weights minima and saddles alike (every $\delta\Phi = 0$ configuration dominates non-stationary ones); whether the pole condition [CT2.5] or the defect criterion [FT.33] selects the stable subclass is a theorem-shaped open question, not a definition. (refinement [Def-H.24(5), FT.66]): for degenerate stationary families (isometry orbits) the vocabulary is Morse–Bott — "stable" := $\text{ind}_B = 0$ on a clean critical manifold, "unstable" := $\text{ind}_B \geq 1$; and the coupling question is answered: no index criterion operates

on closed chains within the weight inventory [FT.66(2)–(3)], and absolute index-0 selection is inadmissible for bound chains on Synge-class slices [FT.66(4)]; the open-channel residue is tracked under Open Gaps 13.

Theorem 3.33 (FT.33 — The defect bridge: pole-side mass candidate in well-posed ratio form). Thm. Under Def-H.37(1) (interface transfer: the floor fermion's pole displacement takes EW.33's leading-order defect form for the seed v [EW.32] weak-transported around its materialized bound chain), for a materialized spacelike single-contact cycle floor Σ_L with $\kappa_e > 0$:

$$(1)\quad m^2_{\text{pole}}(\Sigma_L) = \hat{\mu}^2\,\ell_c^2\,G(c), \qquad \ell_c = \mathfrak{h}\,\Lambda_c, \quad \Lambda_c := \sum_{e\in c}\sqrt{\kappa_e}\ \text{(dimensionless)};$$

$$G(c) = \sum_{A,B} c_A c_B\,\mathrm{Re}\langle \mathcal{G}_A\mathsf{v}, \mathcal{G}_B\mathsf{v}\rangle\ \text{[EW.33; EW.20 Gram data]};$$

$$\hat{\mu}^2 > 0:\ \text{floor-independent residue packaging } \varkappa^2\beta^2 h_0^2\ \text{[EW.33 with } \ell^2 \text{ factored out]};$$

$$(2)\quad \frac{m^2_{\text{pole}}}{m^2_{\text{hid}}} = \frac{\hat{\mu}^2\,\mathfrak{h}^4\,G(c)}{(\ln 2)^2}\cdot\Big(\frac{\Lambda_c}{L}\Big)^2\ \in\ \mathfrak{q}^0$$

$$\text{[well-posed; } m^2_{\text{hid}} = L^2(\ln 2)^2/\mathfrak{h}^2\ \text{FT.32(1)}];$$

$$(3)\quad \text{length-universality}:\ \text{at fixed } \kappa\text{-profile shape and Gram data the ratio is independent of } L;$$

$$\text{uniform } \kappa:\ \text{ratio}$$

$$= \frac{\hat{\mu}^2\mathfrak{h}^4 G(c)}{(\ln 2)^2}$$

$$\kappa \text{ — linear in the edge quantum, floor-free;}$$

$$(4)\quad \text{Cauchy–Schwarz}:\ \Lambda_c^2$$

$$\le L$$

$$\Theta(c),$$

$$\text{equality} \iff \kappa_e \text{ uniform [math]} \implies \text{at fixed } (\Theta, L):\ \text{uniform floors maximize the bridge ratio;}$$

$$(5)$$

$$h_0$$

$$= 0 \implies \hat{\mu}^2$$

$$= 0$$

$$\implies m^2_{\text{pole}}$$

$$= 0 \ \text{[EW.33]} \ \wedge \ m_{\text{hid}}$$

$$\neq 0 \ \text{[FT.32(1)]} : \ \text{the two reads decouple without the scalar background;}$$

$$(6) \quad \text{kernel coherence} : \ c \in \ker M^2$$

$$= \text{span}\{Q\} \ \text{[EW.22]}$$

$$\implies G(c)$$

$$= 0$$

$$\implies m^2_{\text{pole}}$$

$= 0$ [EW.33 : Gram PSD] — charge-aligned (photon-direction) transport is defect-massless at every floor, matching m^2_γ

$$= 0 \ \text{[EM.6]}.$$

No identification $m_{\text{pole}} = m_{\text{hid}}$ is asserted; the ratio is the bridge datum.

Req. Def-H.37(1) (forward pointer — Tier F.17); EW.20, EW.22, EW.32, EW.33 (Part 2); EM.6 (Part 2); FT.23, FT.24, FT.32; Def-H.2, Def-H.9(2); D1.6; CT2.9 (Part 2); D2.9 (units); Cauchy–Schwarz [math].

Pf.

$$(1) : \ \text{EW.33} : \ \mathcal{U} = \exp\big(i \oint \textstyle\sum_A c_A \mathcal{G}_A \, d\ell\big) = 1 + i \, \ell_c \textstyle\sum_A c_A \mathcal{G}_A + O(c^2);$$

$$\text{CT2.5 read [EW.33]} : \ m^2 = \|(\mathcal{U} - 1)\mathsf{v}\|^2 = \ell_c^2 \, G(c) + O(c^3)$$

$$\ell_c$$

$$= \textstyle\sum_e \sqrt{\kappa_e}$$

$\mathfrak{h}$ [Def-H.2 piecewise-straight length; FT.24(2) : $ds^2(e) = \kappa_e \mathfrak{h}^2$, spacelike $\kappa_e > 0$];

$\hat{\mu}^2$: EW.33 packaging, ℓ^2 factored out [Def-H.37(1)]

$$(2) : \ \frac{m^2_{\text{pole}}}{m^2_{\text{hid}}} = \frac{\hat{\mu}^2 \mathfrak{h}^2 \Lambda_c^2 \, G}{L^2 (\ln 2)^2 / \mathfrak{h}^2} = \frac{\hat{\mu}^2 \mathfrak{h}^4 G}{(\ln 2)^2} \Big(\frac{\Lambda_c}{L} \Big)^2 ;$$

$$[\hat{\mu}^2] = \frac{[m^2]}{[\ell^2]} = \mathfrak{q} \cdot \mathfrak{q} = \mathfrak{q}^2 \text{ [EW.33 packaging; D2.9]} \implies [\hat{\mu}^2 \mathfrak{h}^4] = \mathfrak{q}^0 \ \checkmark$$

$$(3) : \text{ uniform } \kappa : \ \Lambda_c = L\sqrt{\kappa} \implies (\Lambda_c / L)^2 = \kappa \text{ [exact squaring, } \neg\text{numerical roots]}; \quad L \text{ cancels}$$

$$(4) : \ \Big(\sum_e \sqrt{\kappa_e} \Big)^2$$

$$\leq L \sum_e \kappa_e$$

$$= L$$

Θ [Cauchy–Schwarz; equality $\iff$ $\sqrt{\kappa_e}$ proportional to 1 $\iff$ κ uniform]

$$(5) : \text{ EW.33 : } \mu_\star^2 = c_W \varkappa^2 \beta^2 h_0^2, \ h_0 = 0 \implies M^2 \equiv 0; \qquad \text{FT.32}(1) : \ m_{\text{hid}} = L \ln 2 / \mathfrak{h} \ \neg\text{references } h_0$$

$$(6) : \ G(c) = c^{\mathsf{T}} \big(M^2 / \mu_\star^2 \big) c \text{ [EW.20/EW.33]};$$

M^2 PSD [EW.33 : Gram] $\implies$ $G(c) = 0 \iff c \in \ker M^2 = \text{span}\{Q\}$ [EW.22];

Q-aligned

= photon direction [D3.10 : $A = \sin\theta_W W_3 + \cos\theta_W Y$; EW.24]

$$\implies m^2_{\text{pole}}$$

$= 0$ [EM.6 consistency]. ■

Rmk (wrapping-chain consistency). On rung-3 wrapping chains, Def-H.37(2)s formula is EW.33 verbatim — the transfer adds nothing there; FT.33's content is the ratio against m_{hid} and its floor-universality. The Gram quadratic form $G(c)$ remains configuration data (the chain's weak-transport profile c_A); floor-independence in clause (3) is at fixed G and fixed profile shape. Genericity upgrade: on-shell rigidity [FT.38(2), forward] makes uniform-κ — hence clauses (3)–(4) — continuum-generic within the integrality classes, not a specialization.

Phys. If the fermion's pole mass is the seed's failure to return to itself around the bound chain, then the informational and pole reads of mass are bridged by a single dimensionless ratio: a fixed P4/Gram factor times the squared mean root of the edge quanta. Cycle length drops out completely — the ladder of hidden bits and the ladder of geometric length grow at exactly the same rate — and the ratio is maximized by uniform floors, vanishes with the scalar background, and vanishes identically on charge-aligned transport, reproducing the massless photon at every floor. The integers enter the bridge as $\overline{\sqrt{\kappa}}^{\,2}$: the one place where the discrete ledger meets irrational lengths.

Theorem 3.34 (FT.34 — Round-trip quantization between floors; integrality propagation; sealing partition rule). Thm. Let $\Sigma^{(1)}, \Sigma^{(2)}$ carry materialized 2-cycles c_1, c_2 [$\Theta(c_i) = 2\kappa_i$, $\kappa_i \in \mathbb{Z}$: FT.24], and let P, P' be connector chains with $\Gamma := c_1 \cup P \cup c_2 \cup P'$ a valid closed chain [D1.4: concatenation, Janus balance], edge-disjoint union. If Γ materializes:

$$(1) \quad \Theta(P) + \Theta(P') = \Theta(\Gamma) - 2\kappa_1 - 2\kappa_2 \in 2\mathbb{Z} \qquad \text{[FT.22, D1.7 additivity]};$$

$$(2) \quad P' = \bar{P} \text{ (same bidirectional channels reversed [FT.23 per edge])} : \quad \Theta(P) \in \mathbb{Z};$$

$$d_{12} := \Theta(P) = \text{geometric distance (integer)};$$

$$(3) \quad \text{additive winding law} : \; k(\Gamma) = \frac{\Theta(\Gamma)}{2} = \kappa_1 + \kappa_2 + d_{12} = k(c_1) + k(c_2) + d_{12}$$

$$\text{[FT.25(3)} : \; k = \kappa \text{ at 2-cycle floors]};$$

$$(4) \quad \text{round-trip integrality (propagation lemma)} : \; \text{let } e,$$

$$\bar{e} \text{ be both real,}$$

$$\Gamma \text{ materialized closed with edge multiset } \{e, \bar{e}\} \sqcup E_0,$$

$$E_0 \text{ decomposing edge-disjointly into materialized closed chains :}$$

$$2\kappa_e = \Theta(\Gamma) - \Theta(E_0) \in 2\mathbb{Z} \implies \kappa_e \in \mathbb{Z};$$

(5) sealing partition rule [CT3.17a(1) (F)] : the group-size structure of sealed colour-attachment sets, and the non-sealing of a lone attachment.

Req. FT.22, FT.23, FT.24, FT.25; D1.4, D1.7; Def-H.10; CT3.17, CT3.17a (Part 2); FT.19, FT.27; elementary number theory [math].

Pf.

$$(1):\ \text{D1.7}:\ \Theta(\Gamma) = \Theta(c_1) + \Theta(P) + \Theta(c_2) + \Theta(P')\ \text{(edge-disjoint)};$$

$$\text{FT.22}:\ \Theta(\Gamma) \in 2\mathbb{Z}; \quad \text{FT.24}(1):\ \Theta(c_i) = 2\kappa_i \in 2\mathbb{Z}$$

$$\implies \Theta(P) + \Theta(P') = \Theta(\Gamma) - 2(\kappa_1 + \kappa_2) \in 2\mathbb{Z} - 2\mathbb{Z} = 2\mathbb{Z}$$

$$(2):\ P' = \bar{P}:\ \Theta(P') = \Theta(P)\ [\text{FT.23 applied per edge of } P] \implies 2\Theta(P) \in 2\mathbb{Z} \implies \Theta(P) \in \mathbb{Z};$$

$$d_{12} := \Theta(P)$$

$$(3):\ k(\Gamma) = \Theta(\Gamma)/2 = (2\kappa_1 + 2\kappa_2 + 2d_{12})/2 = \kappa_1 + \kappa_2 + d_{12} \in \mathbb{Z}$$

$$(4):\ \Theta(\Gamma) = \kappa_e + \kappa_{\bar{e}} + \Theta(E_0) = 2\kappa_e + \Theta(E_0)\ [\text{FT.23}:\ \kappa_{\bar{e}} = \kappa_e;\ \text{D1.7}];$$

$$\Theta(\Gamma) \in 2\mathbb{Z}\ [\text{FT.22}]; \quad \Theta(E_0) = \sum_i \Theta(c_i) \in 2\mathbb{Z}\ [\text{FT.22 per chain}]$$

$$\implies 2\kappa_e \in 2\mathbb{Z} \implies \kappa_e \in \mathbb{Z}$$

$$(5):\ \text{CT3.17a (F)}.\ \blacksquare$$

Rmk (scope of the distance). Clause (2)'s integer distance requires the bidirectional round-trip closure: a single-orientation connector gives only the symmetrized constraint $\Theta(P) + \Theta(P') \in 2\mathbb{Z}$ — the one-way open path remains unconstrained (no winding without closure; the handoff-level claim "$\int_\gamma ds^2 = 2k\mathfrak{h}^2$ for geodesics between jumps" holds only for the closed round-trip, where it reads $\oint_\Gamma ds^2 = 2k(\Gamma)\mathfrak{h}^2$). Refinement: the quantization law is refinement-exact [FT.22 at every level]; the integer value scales as $\mathfrak{h} \to 0$ — the classical limit is large k [FT.31(3)].

Phys. Two floor fermions joined by a bidirectional channel measure their separation in integers: the round-trip's line-element sum is even-quantized, the one-way accumulation is an integer, and the composite loop's winding number is the sum of the two floor windings and the distance between them — an additive composition law for the geometric ledger. The same clause propagates integrality: any bidirectional channel that closes through materialized floors into a materialized loop inherits $\kappa \in \mathbb{Z}$ — the on-shell sublattice grows outward from the 2-cycle floors. (The colour sealing partition rule: CT3.17a(1) (Part 2).)

3.15 Tier F.13 — Gap-Closure Pass: Flux Ledger, Relating Equation, Rigidity, Confinement Converse, Core Charge, Aspect Ledger

Theorem 3.35 (FT.36 — Flux ledger completion). Thm. (1) Saturation locus. Jaw/threshold arithmetic — where the temporal jaw binds, and the threshold = the channel constant: CT0.6a (Part 1). In FT.1's geodesic slowly-varying gauge ($\delta h = \Delta\ell/(\Omega\mathfrak{h})$, $\delta\tau = \Delta t/\Omega$, $\Omega > 0$ [CT1.12]) the locus reads

$$\frac{\Delta\ell(\gamma)}{\Delta t(\gamma)} = 4\,\mathfrak{h} \qquad \text{(channel speed} = 4 \text{ hop lengths per tick)},$$

with Ω cancelling: the saturation scale is a conformal invariant [FT.6 pattern]; only the saturated value is Ω-modulated ($\mathcal{F} = 4\ln 2$, $\mathcal{F}_{\text{phys}} = 4\ln 2/\Omega$ [FT.5, Def-H.5]). (2) Attainment (FT.36(2) discharged). Whenever Def-H.3's transit domain is nonempty, its sup is attained by an explicit photon-content schedule: partition γ's edges into per-tick groups of ≤ 4; run transit (photon) content [EM.5, EM.6: trivial chains $\subseteq V_2$; D1.5: open propagating = transit] on $\mathsf{B}(\gamma)/\ln 2$ scheduled hops [FT.1's contracted notation] — per-tick group capacity: CT0.6a's sharpness clause (Part 1) — $\mathsf{E}_{\text{tr}} = \delta h' \ln 2$ [EM.7] $\Rightarrow$ sup $= \mathsf{B}(\gamma)$ [FT.1's bound met with equality]. On channels whose edges are bound-locked in every configuration the transit domain is empty and $I = 0$ (attainment vacuous; bound trivially valid). (3) Hence FT.1/FT.3/FT.4/FT.5's bounds are sup-attained equalities on exhibited configurations wherever transit content is admissible.

Req. FT.1, FT.5, FT.6; Def-H.2, Def-H.3, Def-H.5; C9; CT0.6a (Part 1); CT1.12 (Part 2); D0.10, D1.5; EM.5, EM.6, EM.7 (Part 2).

Pf.

$$(1):\ \text{jaw/threshold arithmetic}:\ \text{CT0.6a (B)};$$

$$\frac{\delta h}{\delta\tau} = \frac{\Delta\ell/(\Omega\mathfrak{h})}{\Delta t/\Omega} = \frac{\Delta\ell}{\mathfrak{h}\,\Delta t}\ [\Omega > 0 \text{ cancels};\ \text{CT1.12}:\ \Omega = e^{\alpha_\Omega a_{00}} > 0]$$

$$\text{locus}:\ \delta h/\delta\tau = 4 \iff \Delta\ell/\Delta t = 4\mathfrak{h}$$

$$(\text{dimension check in hop exponents}:\ [\Delta\ell] = \mathfrak{h}^1,\ [\Delta t] = \mathfrak{h}^0\ [\text{D1.8}:\ x^0 = \tau_n \text{ tick count}]\ \checkmark)$$

$$(2):\ \gamma' \subseteq \gamma \text{ with } \delta h'$$

$= \min(\delta h, 4\,\delta\tau)$ hops carrying photon content is a valid transit configuration [D0.10 : $\mathfrak{C}(\mathcal{R})$ admits transit assignments; D1.5; EM.5/EM.6] : $\mathsf{E}_{\text{tr}}(\gamma')$

$$= \delta h' \ln 2\ [\text{EM.7}]$$

$$\text{schedule exactness (per-tick groups at channel capacity)}:\ \text{CT0.6a (B)};$$

$$\text{sup} \geq \min(\delta h, 4\,\delta\tau)\ln 2 \ \wedge\ \text{FT.1}: \ \text{sup} \leq \ \text{same} \implies \text{equality, attained}$$

$$(3): \ \text{from } (2) \text{ at each bound's locus. } \blacksquare$$

Phys. The flux ledger closes: the crossover where knowledge switches from space-limited to time-limited is the conformally invariant channel speed — four hop lengths per tick, the same four that is the channel's bit capacity — and the bound is reached, not merely valid, by photon schedules running the ledger at capacity. What FT.36(2) assumed, FT.36 exhibits.

Theorem 3.36 (FT.37 — The relating equation, complete). Thm. For a materialized closed chain c [FT.22], with $S_{\text{reg}}(c) = \delta h(c)\ln 2$ [CT0.9, C9] and $A_{\text{loop}}(c) := \oint_c ds^2 = \Theta(c)\,\mathfrak{h}^2 = 2k\,\mathfrak{h}^2$ [D1.6, FT.22], and mean edge quantum $\bar{\kappa}(c) := \Theta(c)/\delta h(c)$:

$$\frac{S_{\text{reg}}(c)}{A_{\text{loop}}(c)} = \frac{\ln 2}{\mathfrak{h}^2}\cdot\frac{1}{\bar{\kappa}(c)} = \frac{\delta h(c)\,\ln 2}{2k\,\mathfrak{h}^2};$$

$$\frac{S_{\text{reg}}}{A_{\text{loop}}} = \frac{S_{\text{min}}}{A_{\text{min}}} = \frac{\ln 2}{\mathfrak{h}^2}\ [\text{CT1.29}] \iff \Theta(c) = \delta h(c)\ (\bar{\kappa} = 1).$$

On all-nonnull per-edge-integer chains [FT.24/FT.34(4) scope]: $\bar{\kappa} \geq 1$, with equality $\iff \kappa_e \equiv 1$; null edges or class-(iii) rational profiles [FT.38(3)] admit $\bar{\kappa} < 1$ (witness $(1,1,0)$: $\bar{\kappa} = \frac{2}{3}$).

Req. FT.22, FT.24, FT.31, FT.34; CT0.9 (Part 1); C9; CT1.29 (Part 2); D1.6, D1.7.

Pf.

$$\frac{S_{\text{reg}}}{A_{\text{loop}}} = \frac{\delta h\ln 2}{\Theta\,\mathfrak{h}^2} = \frac{\ln 2}{\mathfrak{h}^2}\cdot\frac{\delta h}{\Theta};$$

$$\Theta = 2k\ [\text{FT.22}] \implies \text{second form;}$$

$$\bar{\kappa} := \Theta/\delta h \implies \text{first form}$$

$$\text{CT1.29 at the single-edge floor}: \ S_{\text{min}} = \ln 2,\ A_{\text{min}} = \mathfrak{h}^2 \implies \text{rate } \ln 2/\mathfrak{h}^2;$$

$$\text{recovered} \iff \delta h/\Theta = 1$$

$$\text{integer non-null profiles}: \ \kappa_e \geq 1 \implies \Theta = \sum \kappa_e \geq \delta h \implies \bar{\kappa} \geq 1,\ = \iff \kappa_e \equiv 1;$$

$$(1,1,0): \ \Theta = 2,\ \delta h = 3: \ \bar{\kappa} = \tfrac{2}{3} < 1. \ \blacksquare$$

Phys. The loop-area/qubit-power identification is complete: CT1.29's one-bit-per-hop-area rate holds for every materialized loop, deformed by exactly the loop's own mean edge quantum. Minimal-edge loops are the undeformed case; the winding k reads the deformation in the second form. Gap 2's "completeness unresolved" is resolved: the correction factor is derived data.

Theorem 3.37 (FT.38 — Non-geodesic tightening and on-shell rigidity). Thm. (1) Tightening. For any spacelike path γ with on-shell integer edge data $\kappa_e \in \mathbb{Z}_{\geq 1}$ [FT.24/FT.34(4) scope]:

$$I_{o\leftarrow\Sigma}(\gamma) \;\leq\; \min\big(\delta h(\gamma),\; 4\,\delta\tau(\gamma)\big)\ln 2 \;\leq\; \min\Big(\frac{\Delta\ell(\gamma)}{\hbar},\; 4\,\delta\tau(\gamma)\Big)\ln 2,$$

$$\Delta\ell(\gamma) = \hbar \sum_{e\in\gamma} \sqrt{\kappa_e} \;\text{ [Def-H.2,\; D1.6,\; FT.24(2)]},$$

with no geodesic and no slow-variation hypothesis; spatial equality $\iff$ $\kappa_e \equiv 1$. FT.1's geodesic physical form is the special case; the bound is tightest on stable jumps [Def-H.19: local $\Delta\ell$-minimizers; Morse — math]. (2) On-shell rigidity. In CT1.19's family regime ($L = \hbar n_{\mathcal{R}}$ fixed, $\hbar \to 0$, g smooth along γ [CT2.6]): $\kappa_e = f(\lambda_e) + O(\hbar)$ with $f := g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$ continuous [CT1.19's Pf, midpoint Taylor; per-edge remainder uniform]. For per-edge-integer classes (i)/(ii) : $\exists\,\hbar_0$ such that $\forall\,\hbar < \hbar_0$, κ_e is constant along each connected materialized chain — signature included (spacelike $\kappa \geq 1$, null 0, timelike $\kappa \leq -1$ are ≥ 1 apart). Hence uniform-κ is continuum-generic on-shell: FT.33(3)'s length-universality and FT.33(4)'s Cauchy–Schwarz equality are the generic on-shell case, not specializations; signature changes are confined to $O(\hbar)$ boundary layers. (3) Rational ladder (class-(iii) arena). Without per-edge integrality, FT.22 alone permits uniform $\kappa \in (2/L)\mathbb{Z}$ on an L-cycle ($\Theta = L\kappa \in 2\mathbb{Z}$; witness $L = 3$, $\kappa = \frac{2}{3}$, $\Theta = 2$): per-edge integrality is special to $L = 2$ [FT.24: $2\mathbb{Z}/2 = \mathbb{Z}$] and round-trip-closed channels [FT.34(4)]; clause (2)'s integer-step argument does not reach spacings $2/L < 1$.

Req. FT.1, FT.22, FT.24, FT.33, FT.34; Def-H.2, Def-H.19; D1.6; CT0.9 (Part 1); CT1.12, CT1.19 (Part 2); CT2.6 hypotheses (inherited); integer/rational arithmetic, Morse minimality [math].

Pf.

$$(1):\ \text{FT.1 first form: hypothesis-free;}$$

$$\kappa_e \geq 1 \implies \sqrt{\kappa_e} \geq 1 \implies \Delta\ell/\hbar = \sum_e \sqrt{\kappa_e} \;\geq\; \delta h \implies \text{chain;}$$

$$\text{equality} \iff \kappa_e \equiv 1$$

$$\Delta\ell(\gamma)$$

$$= \hbar \textstyle\sum \sqrt{\kappa_e}:\ \text{per-edge } g\text{-length } \sqrt{ds^2(e)}$$

$$= \sqrt{\kappa_e}$$

$$\hbar\ [\text{FT.24}(2),\ \text{spacelike}];$$

$$\text{piecewise additivity [Def-H.2];}$$

$$\text{stable jumps minimize } \Delta\ell \text{ locally [Def-H.19;\ math]}$$

$$(2):\ |\kappa_{e+1}-\kappa_e| \le |f(\lambda_{e+1})-f(\lambda_e)| + 2C\mathfrak{h} \le \left(\tfrac{\sup|f'|}{L}+2C\right)\mathfrak{h}\ \to 0;$$

$$\mathfrak{h} < \left(\tfrac{\sup|f'|}{L}+2C\right)^{-1} \implies |\kappa_{e+1}-\kappa_e| < 1 \wedge \kappa \in \mathbb{Z} \implies \kappa_{e+1}=\kappa_e$$

signature : $\kappa \ge 1 \ \vee\ \kappa = 0 \ \vee\ \kappa \le -1$ pairwise distance $\ge 1 \implies$ constancy includes signature

$$(3):\ \Theta = L\kappa \in 2\mathbb{Z}\ [\text{FT.22}] \implies \kappa \in (2/L)\mathbb{Z};$$

$$L=3:\ \kappa=\tfrac{2}{3}:\ \Theta = 2 \in 2\mathbb{Z}\ \wedge\ \kappa \notin \mathbb{Z};$$

$$L=2:\ \kappa \in 2\mathbb{Z}/2 = \mathbb{Z}\ [\text{FT.24 recovered}]. \blacksquare$$

Rmk (scope discipline). Clause (2) upgrades the uniform-κ regime to generic on-shell status within the integrality classes; class (iii) is exactly where rigidity can fail — a second structural signature distinguishing the classes (the first being FT.34(4)'s round-trip criterion). FT.32(6)'s uniform/all-nonnull regimes are scoped accordingly.

Phys. The information bound needs no geodesic: any spacelike on-shell path obeys it with its own proper length, tightest along stable jumps. And the continuum limit does not erode the integer ledger — a slowly varying geometry cannot move an integer, so materialized chains of the integrality classes become uniformly quantized along their length in the classical regime: integrality plus smoothness equals rigidity. Where integrality is not yet derived, the ledger can refine to spacing $2/L$ — and that is precisely the open frontier.

Theorem 3.38 (FT.39 — Confinement converse, singlet diagonal, and filter placement). Thm. (1) Converse (Gap 8). $\Pi_{\mathcal{C}}(\Sigma;C) = \mathcal{C}(\Sigma;C)\cdot\mathbb{I}$ with $\mathcal{C} \in \{0,1\}$ [Def-H.10: scalar idempotent] $\implies$ $F^{\Sigma}_{\text{strong}}(C) = \mathcal{C}\cdot F_{\text{strong}}\big|_{\Sigma}$ [FT.19's definition]. Hence $\mathcal{C}=1 \implies F^{\Sigma}_{\text{strong}} = F_{\text{strong}}\big|_{\Sigma} \neq 0$ for nonzero aspect data ($f_a \neq 0$ [CT3.9, CT3.10]) on ≥ 1 colour slot — generic nonvanishing with exact exception locus $\{f=0\} \cup \{\text{Colour}(\Sigma)=\varnothing\}$. At support level [Def-H.11's selection-type read]: $\Pi_{\mathcal{C}} = \mathbb{I} \neq 0$ whenever $\mathcal{C}=1$ — unconditional. (2) Singlet diagonal (consistency). On sealed (singlet) states the Lie action annihilates: $T_a|\text{singlet}\rangle = 0$ [CT3.15, CT3.17 invariance] $\implies$ $\langle\text{singlet}|F_{\text{strong}}|\text{singlet}\rangle = 0$ — consistent with FT.11 (no independent colour charge) and CT3.22 (colour is not an end-charge): the strong block's nonzero content is the admissibility structure and the gluon mode count [CT3.13], not singlet expectation values. (3) Filter placement (Gap 9, decision derived). Def-H.9's $\mathcal{M}(\Sigma)$ must not be pre-weighted by $\mathcal{C}(\Sigma;C)$: its members are restrictions $C|_{\Sigma}$ of valid ambient configurations [FT.18's setup; D2.11], whose colour sealing is ambient-global [CT3.18] — $\mathcal{C}(\Sigma;C)=0$ records that Σ's partners lie outside [FT.18's non-locality], not that C is invalid. Pre-weighting would delete the EM/G/H reads of externally-sealed configurations, contradicting their colour-independence [FT.11, FT.21(i)–(ii)]; the downstream $\Pi_{\mathcal{C}}$ filter [Def-H.15(3)] zeroes exactly the strong block [FT.19] and keeps the rest — the corpus-consistent placement. CT2.8(3)'s null-quotient handles universe-level invalidity [CT3.18], a different question from Σ-relativity.

Req. Def-H.9, Def-H.10, Def-H.11, Def-H.15; FT.11, FT.18, FT.19, FT.21; D2.11; CT2.8 (Part 2); CT3.9, CT3.10, CT3.13, CT3.15, CT3.17, CT3.18, CT3.22 (Part 2).

Pf.

$$(1):\ \mathcal{C}\in\{0,1\}\implies \mathcal{C}^2=\mathcal{C}\implies \Pi_{\mathcal{C}}F\Pi_{\mathcal{C}}=\mathcal{C}\,F\big|_{\Sigma};$$

$$\mathcal{C}=1:\ F^{\Sigma}=F\big|_{\Sigma};\quad T_a\neq 0\,\text{on}\,\mathbb{C}^3\,[\text{CT3.9}:\ \text{explicit basis}]\wedge\text{slot present}\implies F\big|_{\Sigma}\neq 0\,\text{for}\,f\neq 0$$

$$(2):\ \varepsilon_{ijk}U^{i}{}_{a}U^{j}{}_{b}U^{k}{}_{c}$$

$$=\varepsilon_{abc}\det U$$

$$=\varepsilon_{abc}\,[\text{CT3.15}];$$

$$\delta^{i}{}_{j}\ \text{likewise}\ [\text{CT3.17}]\implies \text{singlets fixed under}\ SU(3)\implies \mathfrak{su}(3)\ \text{annihilates them}$$

$$(3):\ \text{validity is ambient}\ [\text{D2.11};\ \text{CT3.18}:\ \mathcal{P}^{(3)}_{M}\ \text{excludes free indices at universe level}];$$

$$\text{FT.18}:\ \exists\, C,C'\in\mathcal{M}(U)\ \text{valid with}\ C|_{\Sigma}=C'|_{\Sigma},\ \mathcal{C}(\Sigma;C)=1,\ \mathcal{C}(\Sigma;C')=0$$

C' valid yet Σ-unsealed : pre-weighting deletes $C'|_{\Sigma}$ entirely—but its EM/G/H reads are physical and colour-independent [FT.21] $\implies$ pre-weighting loses physical data;

$$\text{downstream}\ \Pi_{\mathcal{C}}:\ \text{strong block}\to 0\ [\text{FT.19}],\ \text{rest kept.}\ \blacksquare$$

Phys. Confinement is now a two-sided exact statement at subnetwork level: the colour block lives precisely when the region self-seals, dies entirely when any attachment reaches outside, and even when alive its singlet-diagonal read vanishes — colour is never a number a sealed object carries, only the grammar of how it composes with its environment. And the filter belongs downstream: sealing is a fact about Σ's embedding, so it must gate Σ's strong block, never erase Σ's other reads.

Theorem 3.39 (FT.40 — Closed-core charge read). Thm. For a closed oriented chain c based at $v=s(c)=t(c)$ with first edge e_1 (out of v) and last edge e_k (into v) [D1.4's ordered tuple]:

$$q_{\text{core}}(c;v):=\frac{\sigma_{e_1}-\sigma_{e_k}}{2}\ \in\ \{-1,0,+1\}$$

[D3.4 applied to closed c; CT0.9 : per-edge independent registers; EM.9 : alphabet, all values attained]

is well-defined; under basepoint/orientation reversal [D2.6's ω: $s \leftrightarrow t$, charge $\mapsto -$charge]: $q_{\text{core}} \mapsto -q_{\text{core}}$ — the corpus's own charge-conjugation law, verbatim on the core read. At the 2-cycle floor: $q_{\text{core}} = (\sigma_{e_1} - \sigma_{e_2})/2$ at base p, negated at base q. The closed core is therefore not charge-neutral by closure; its ternary read is basepoint data, and the observer-facing charge remains the EM.9 end-read of the full transit chain through the channel [Def-H.4].

Req. D1.4, D2.6, D3.4; CT0.9 (Part 1); EM.9 (Part 2); FT.28, Def-H.4.

Pf.

D3.4 is stated for oriented chains c generally;

$$\text{closed } c \text{ has } s(c)$$

$$= t(c)$$

$$= v \text{ with distinguished first/last edges [D1.4 : } c = (e_1, \ldots, e_k) \text{ ordered]} \implies \sigma_{s(c)} :$$

$$= \sigma_{e_1},$$

$$\sigma_{t(c)} :$$

$$= \sigma_{e_k} \text{ well-defined}$$

$$\text{CT0.9 : } \sigma_{e_1}, \sigma_{e_k} \in \{\pm 1\} \text{ independent registers;} \qquad \text{EM.9 : ternary alphabet, all attained}$$

$$\text{rebase/orientation reversal : } e_1 \leftrightarrow e_k \text{ roles swap} \implies q \mapsto -q$$

$$[\text{D2.6 : } \omega \text{ maps charge} \mapsto \text{-charge : consistent}]. \blacksquare$$

Phys. A fermion core carries a ternary charge datum — the basepoint sign difference — that flips under reversal exactly as the corpus's orientation involution demands. What an observer measures through a channel remains the end-read of the full transit chain; the core datum is the bound ledger's contribution to that read, not an independent free charge [CT3.22's asymmetry untouched: this is registration-sign data, not colour].

Theorem 3.40 (FT.41 — The per-aspect ledger). Thm. (a) Charge count derived. EM.9/D3.4's read consumes exactly the two end registers of a chain; $N_{\text{charge}} = 2$ in FT.14's decomposition is a theorem and $N_{\text{weak}} = \delta h - 2$ its arithmetic complement. the residual interface content is the attribution clause of Def-H.8 (interior weight bits read through the weak/spin channel [Def-H.8]) — reduced — Def-H.8's attribution clause. (b) Colour ledger counted. The sealing grammar of Colour(Σ) —

admissible groupings [FT.34(5), Def-H.10], closed form, small-N counts g, and attachment-type refinement — is CT3.17a(2)–(3) (F). The ledger read defined here: $N_{\text{strong}}(\Sigma) := \log_2 g\big(|\text{Colour}(\Sigma)|\big)$ — the sealing-pattern information of the grammar count (a registration-type ledger, per the trichotomy below). (c) Gravity/Higgs ledger reframed. G's on-shell ledger is the integer κ-profile modulo the single loop constraint [FT.22]: $L-1$ free integers on a materialized L-cycle — unbounded, hence ¬bit-type [C10: bit counts attach to finite registered alternative sets]; H's ledger is the continuous P4 datum h_0 [EW.13, P4]. Per-aspect "bit counts" are well-posed exactly for the registration-type ledgers $(a), (b)$ — the trichotomy answers the original question. (d) Strictly new at Σ, first counts [Def-H.13's filter]. $N(\Sigma) = 1$: $\mathfrak{u}(1)$ factors per-edge entirely — nothing strictly new; single-contact floors carry no collective boundary mode. $N(\Sigma) = 2$: $\dim \mathfrak{u}(2) = 4$, factored diagonal $= 2$, strictly new $= N^2 - N = 2$ real dimensions — the off-diagonal boundary mixings $|e_1\rangle\langle e_2| + \text{h.c.}$, $i(|e_1\rangle\langle e_2| - \text{h.c.})$: a collective inter-contact exchange mode, first available at two-contact floors [FT.28's variant Rmk].

Req. EM.9 (Part 2); D3.4, D3.8; CT0.9, CT0.13 (Part 1); C10; CT3.17, CT3.17a (Part 2); FT.14, FT.22, FT.28, FT.34; Def-H.8, Def-H.10, Def-H.13; EW.13 (Part 2); P4.

Pf.

(a) : EM.9′s table reads (σ_s, σ_t) only; D3.4′s formula references no interior register $\implies$ count $= 2$;

$$N_{\text{visible}} = \delta h \text{ [FT.14]} \implies N_{\text{weak}} = \delta h - 2 \text{ arithmetic}$$

(b) : grammar + counts + type refinement : CT3.17a (F);

N_{strong} read : definition on the cited counts [C10/C9]

(c) : κ-profile : L integers,

1 constraint Θ

$\in 2\mathbb{Z}$ [FT.22]

$\implies L-1$ free integers,

unbounded $\implies$ ¬finite register count [C10];

$h_0 \in \mathbb{R}$ [P4]

(d) : ι_Σ edge-wise [Def-H.13] : factored subalgebra $= \bigoplus_e \mathfrak{u}(1)_e$: dim $= N$;

$$\dim \mathfrak{u}(N) = N^2 \implies \text{strictly new} = N^2 - N : \; N{=}1 : \; 0; \quad N{=}2 : \; 2 \text{ off-diagonal generators.} \blacksquare$$

Phys. "How many bits per force?" now has a typed answer: charge costs two end bits (theorem); colour costs the log of the sealing grammar [counts: CT3.17a(2) (Part 2)]; gravity keeps an unbounded integer ledger; Higgs keeps a continuous dial. And collective boundary structure — genuinely new physics at Σ — begins at exactly two contacts, in two dimensions of exchange modes: the first place where a subnetwork can do something its edges cannot.

3.16 Tier F.14 — Platonic Floor Cards: The Five Solids as Fermion Floors

Definition 3.24 (Def-H.20 — Platonic floors (setup vocabulary)). Def. Let S be a Platonic solid with 1-skeleton Γ_S: V vertices, E undirected edges, F faces of uniform length L ($FL = 2E$), regular vertex degree d ($Vd = 2E$); cardinals [math]:

$$\text{tetrahedron } (4, 6, 4, 3);$$

$$\text{cube } (8, 12, 6, 3);$$

$$\text{octahedron } (6, 12, 8, 4);$$

$$\text{dodecahedron } (20, 30, 12, 3);$$

$$\text{icosahedron } (12, 30, 20, 5).$$

The Platonic floor $\Sigma_S :=$ the bounded subnetwork on Γ_S's vertex set whose real-edge content realizes every undirected edge as a bidirectional channel (both orientations in $\mathcal{R}$) — the full bidirectional 1-skeleton. Variants: dark ($\partial\Sigma_S = \varnothing$: no observer contact [Def-H.1]) and single-contact observed (one external o joined by exactly one real edge to one v_1). Card hypotheses, declared at each use: (M1) every face cycle materializes [FT.22's scope]; (M2) every bidirectional pair materializes as a 2-cycle [FT.24's scope] — both are spectral questions the corpus does not decide (FT.32's Status pattern). Sub-floor vocabulary: Σ_{min} is the $E{=}1$ cell [FT.28], Σ_{12} the $E{=}3$ cell [FT.35], the cycle ladder Σ_L the unidirectional family [FT.32].

3.17 Tier F.15 — The Regular-Polygon Floors: Full Ladder, Sectors, and the Non-Existence Ceiling

Theorem 3.41 (FT.46 — Distinctness is unbounded: no upper limit on distinct regular polygons). Thm. (1) Combinatorial distinctness. The regular n-gons are pairwise non-isomorphic for all $n \geq 3$ (n is the complete invariant of the 1-skeleton); Σ_{P_n} is a valid bounded subnetwork configuration

class for every n [A1: $\mathcal{Q}$ countably infinite supplies vertices; D1.2: seals supply edges; no mask or capacity clause restricts $n(v) = 4$ bidirectional vertices — oversubscription is absorbed and retained [CT4.1, CT4.3; rung correction, FT.62], CT0.6: $O(4) = 16$ uniform]. (2) Ledger injectivity. On the uniform-κ minima, the card invariants separate every member:

$$m_{\text{hid}}(n) = \frac{2n \ln 2}{\mathfrak{h}} \text{ strictly increasing;}$$

$$E_{\text{int}}(n) = 2n \text{ injective;}$$

$$\beta_1(n) = n + 1 \text{ injective;}$$

$$\text{null-carrying dimension } n - 2 \text{ injective;}$$

$$\Theta_{\min}(n) = \begin{cases} 2 & \text{spacelike-or-null / signature-mixed} \\ n + (n \bmod 2) & \text{strictly-spacelike} \\ n \text{ or } 2n & \text{uniform (even/odd)} \end{cases} \quad \text{regime-separating;}$$

$$\text{lattice index} = 2 \ \forall n \ [\text{FT.45}(3)] \ (\text{uniform, non-separating}).$$

No upper limit exists on the number of distinct regular-polygon floors. (3) Embeddability (declared, scoped). For every n there exist configurations in which Σ_{P_n} embeds: A1's countable infinitude, A2's channel universality ($C(p, q) = 1\,\mathfrak{q}$ for all pairs), and D1.2's seal dynamics impose no cardinality bound; the A4 reality test is threshold data, not a prohibition. This is a configuration-class existence statement, not a claim about any particular solution of A9. (4) Non-degeneracy of the ladder. m_{hid} and β_1 increase by fixed steps ($2 \ln 2/\mathfrak{h}$ and 1); $\Theta_{\min}$ is constant (2) in the null-carrying/mixed regimes, but those floors are separated by the null-carrying dimension $n - 2$; the parity lattice index is constant (2) — the family's parity ledger is uniform, and distinctness rides on rank, mass, and null structure, not sector count. No two members are ledger-degenerate.

Req. A1, A2; D0.8, D1.2; CT0.5, CT0.6, CT0.11 (Part 1); FT.42, FT.45; Def-H.20; graph-isomorphism fact for cycles [math].

Pf.

$$(1): \ C_n \ncong C_m \ (n \neq m): \ \text{vertex/edge counts differ [math];}$$

$$\text{validity}: \ \Sigma_{P_n} \text{ is a finite bounded vertex set with real-edge content [D0.10 configuration class];}$$

$$\neg\exists \text{ clause in A1–A9 bounding } n$$

$$\text{CT0.6}: \ O(2) = 0: \ \text{no oversubscription at any polygon vertex, dark [FT.45(2)]}$$

$$(2):\ m_{\text{hid}}(n) = 2n\ln 2/\mathfrak{h}\ [\text{FT.45}(1)]:\ n \mapsto 2n \text{ injective};$$

$$\beta_1(n) = n+1\ [\text{FT.45}(3)];$$

null-carrying dimension $n-2$ injective;

regime table [FT.45(4)]

$(3):$ declared embedding clause [see Req] : A1 countable infinitude,

A2 pair universality,

D1.2 seals;

no cardinality obstruction in A1–A9

$(4):$ direct from the regime table and (2). ■

Rmk (what "distinct" does not yet mean). Clause (2) separates the floors as ledger objects. No generation claim is made here.

Phys. The regular-polygon family is unlimited: every n gives a valid, uniformly minimally-locked, colourless fermion floor with its own mass rung, its own winding rank $n+1$, and its own null-carrying dimension. If the polygon ladder is ever read as a family structure, nothing in the axioms caps its length — the cap, if any, must come from the spectrum side, not the ledger side.

Theorem 3.42 (FT.47 — No geometric ceiling: the Gauss–Bonnet bound binds area, never n). Thm. (1) Gauss–Bonnet area bound. A geodesic n-gon enclosing a topological disk in a smooth Riemannian surface of constant curvature $\bar{K}$, interior angles α_i:

$$\sum_{i=1}^{n}\alpha_i = (n-2)\pi + \bar{K}\,A \qquad [\text{Gauss–Bonnet; geodesic edges}:\ \kappa_g \equiv 0;\ \text{math}],$$

and convexity ($\alpha_i < \pi\ \forall i$) forces $\bar{K}A < 2\pi$ — a bound on the enclosed area per polygon, with no n-dependence. (2) Model-geometry existence — every $n \geq 3$, no ceiling. Flat ($\bar{K} = 0$): the Euclidean regular n-gon, $\alpha = \frac{(n-2)\pi}{n} \in (0,\pi)$ for every $n \geq 3$ [math]. Spherical ($\bar{K} > 0$): for every $A \in (0, 2\pi/\bar{K})$ a convex regular geodesic n-gon with $\alpha = \frac{(n-2)\pi+\bar{K}A}{n} \in \left(\frac{(n-2)\pi}{n}, \pi\right)$ exists for every $n \geq 3$ [math: polar construction on S^2; small-area members are convex]. Hyperbolic ($\bar{K} < 0$): every $\alpha \in \left(0, \frac{(n-2)\pi}{n}\right)$ is realized by a unique regular n-gon of area $A = \frac{n\alpha-(n-2)\pi}{\bar{K}} > 0$, for every $n \geq 3$ [math]. (3) Local existence in the general emergent geometry. CT2.6's spatial slices are smooth Riemannian in the perturbative regime [CT1.12: $\bar{g}_{ij} \succ 0$]; every point has a convex normal neighborhood [math: standard existence], inside which any $n \geq 3$ points on a small

geodesic circle join by unique minimizing geodesics — an equilateral geodesic n-gon for every n. Exact regularity (equal angles too) needs a local symmetry hypothesis; equilateral local existence does not. (4) Transfer. Nothing in A1–A9 or Parts 1–3's definitions/theorems bounds polygon n: CT2.7's dynamics (Einstein–Hilbert, Λ a P4 datum) fixes $\bar{K}$ per solution but no polygon ceiling — for $\bar{K} \leq 0$ clause (2) is global, for $\bar{K} > 0$ local (small-area members), in every case for all $n \geq 3$. The corpus's only numeric caps bind different objects: per-vertex accessibility ($n(v) \leq 3$ unoversubscribed [CT0.11, CT4.1] — polygon floors sit at $n(v) \in \{2, 3\}$ [FT.45(2)], never touching it) and the spectrum ($\mathrm{M1}_n/\mathrm{M2}_n$ materialization — solution data, not a bound on n). (5) Verdict. The question "do distinct regular polygons have an upper limit" is answered no at all three levels — combinatorial [FT.46], model-geometric (clause 2), local emergent-geometric (clause 3) — with the two bounds that do exist precisely located: the Gauss–Bonnet area bound (per polygon, curvature-signed) and the materialization filter (spectral, per solution).

Req. FT.45, FT.46; Def-H.19 (geodesic vocabulary); CT1.12, CT2.6, CT2.7 (Part 2); CT0.11 (Part 1); CT4.1 (Part 2); Gauss–Bonnet, model-geometry polygon constructions, convex-normal-neighborhood existence [math].

Pf.

$$(1):\ \text{Gauss–Bonnet on the enclosed disk}:\ \int_{\partial} \kappa_g\, ds + \iint \bar{K}\, dA + \sum_i (\pi - \alpha_i) = 2\pi\ [\text{math}];$$

$$\kappa_g \equiv 0\ [\text{geodesic edges}] \implies \sum_i \alpha_i = (n-2)\pi + \bar{K}A$$

$$\alpha_i < \pi\ \forall i \implies \sum_i \alpha_i < n\pi \implies (n-2)\pi + \bar{K}A < n\pi \implies \bar{K}A < 2\pi \qquad (n\text{-free})$$

$$(2):\ \bar{K} = 0:\ \alpha = \tfrac{(n-2)\pi}{n} \in (0, \pi)\ \forall n \geq 3;$$

$$\bar{K} > 0:\ A \in (0, 2\pi/\bar{K}) \implies \alpha \in \left(\tfrac{(n-2)\pi}{n}, \pi\right):\ \text{convex valid } \forall n\ \ [\text{constructions cited}]$$

$$\bar{K} < 0:\ A = \tfrac{n\alpha - (n-2)\pi}{\bar{K}} > 0 \iff \alpha < \tfrac{(n-2)\pi}{n};$$

$$\left(0, \tfrac{(n-2)\pi}{n}\right) \neq \varnothing\ \forall n \geq 3 \text{ and lies inside } (0, \pi)$$

$$(3):\ \text{convex normal neighborhoods exist [math]};$$

$$\text{within one}:\ \exists!\ \text{minimizing geodesic per point pair} \implies \text{equilateral } n\text{-gons } \forall n$$

$$\geq 3\ (\text{vertices on a small geodesic circle})$$

$$(4)\text{–}(5):\ \text{CT2.6}:\ (M, g)\ \text{smooth};$$

CT1.12 : $\bar{g}_{ij} \succ 0$ perturbatively;

$\neg\exists$ n-bounding clause in Parts 1–3;

CT2.7/P4 : $\bar{K}$ solution data;

FT.45(2) : $n(v) \in \{2, 3\}$ $\forall$ polygon floors. ■

Rmk (what a real ceiling would require). An upper limit on the polygon family could only arise from the spectral side: a proof that M1$_n$/M2$_n$ fails for large n in every valid universe (e.g. if $Z(\mathcal{R}, \Sigma_{P_n})$'s pole structure [Def-H.16] degenerated with n). The program-note intuition "higher levels make the impossible possible" receives its precise counterpart here: the levels (larger n) are always valid; what remains undetermined per level is materialization.

Phys. Ask the ledger how large a polygon floor can be and the answer is unbounded — every n is a valid, distinct, lock-free configuration class. Ask the geometry and the answer is again unbounded: Gauss–Bonnet caps the area a geodesic polygon can enclose ($\bar{K}A < 2\pi$), never its number of sides; flat and hyperbolic slices realize every regular n-gon globally, positively curved slices realize small ones locally. The only gate left is spectral — which polygons materialize — and that gate is the corpus's own open question, not a ceiling.

Theorem 3.43 (FT.48 — Stability structure of floor-to-floor channels (Gap 18, partial)). Thm. Let γ be a channel path between two floor termini, on-shell (edges of integrality classes (i)/(ii)) in the continuum regime of CT1.19/CT2.6. (1) Stationarity is geodesy. $\delta\int_\gamma ds^2 = 0$ among fixed-endpoint variations $\iff$ γ is an affinely parametrized geodesic of $g_{\mu\nu}$ [CT1.20's stationary-phase locus; FT.31(2): the on-shell sums are the Riemann sums, so the ledger's stationary condition and the metric's Euler–Lagrange equation coincide; math]. (2) Two-sided distance law. For a bidirectional round-trip connector P of m channels [FT.34(2)]: the ledger side is the integer distance $d_{12} = \Theta(P) = \sum_{e\in P}\kappa_e \in \mathbb{Z}$; the metric side is the proper length $\ell(P) = \mathfrak{h}\sum_{e\in P}\sqrt{\kappa_e}$ [FT.33(1) pattern, spacelike]. They are bound by

$$\ell(P)^2 \;\le\; m\cdot d_{12}\cdot \mathfrak{h}^2 \qquad \text{[Cauchy–Schwarz; math]}, \qquad \text{equality} \iff \kappa_e \text{ uniform on } P;$$

uniform connectors obey $\ell^2 = m\, d_{12}\, \mathfrak{h}^2$ exactly. Both sides are derived: the integer from materialization [FT.34(2)], the length from the metric dictionary [D1.6, FT.24(2)] — two reads of one channel, dimensionally consistent ($[\ell^2] = [\mathfrak{h}^2] = \mathfrak{q}^{-1}$, d_{12} dimensionless). (3) Stable-channel trichotomy. Within a fixed endpoint class in the emergent geometry [Def-H.19 vocabulary]: index-0 stationary channels (no conjugate points) are exactly the local ℓ-minimizers [math: Morse/Synge]; on-shell with $\kappa_e > 0$, local ℓ-minimality at fixed channel count m coincides with local d_{12}-minimality along uniform families [clause (2): $\ell^2 = m\, d_{12}\, \mathfrak{h}^2$ strictly monotone in d_{12} at fixed m]. Saddle/maximal stationary channels exist wherever conjugate points do; CT1.20's stationary phase weights them alike with the minima. (4) Derived vs. application-layer. Clauses (1)–(3) close the variational side of the stability program: stable channels are geodesics, their ledger distance is integer, and the two distance reads are Cauchy–Schwarz-bound with equality

at uniformity. The selection side — whether the pole condition [CT2.5/Def-H.16] or the defect criterion [FT.33] selects index-0 channels — remains open : it requires fluctuation-sign/Maslov-type structure on the Z side, which Parts 1–3 does not develop.

Req. FT.22, FT.24, FT.31, FT.33, FT.34, FT.38; Def-H.19; D1.6; CT1.19, CT1.20, CT2.5, CT2.6 (Part 2); Cauchy–Schwarz, Morse/Synge basics [math].

Pf.

$$(1):\ \text{FT.31}(2):\ \sum_e \kappa_e \mathfrak{h}^2$$

$$= \sum_e ds^2(e) \text{ identically } \implies \text{stationary locus of the ledger phase } \Phi$$

$$= \pi\Theta \text{ [FT.9/EM.20]}$$

$$= \text{ stationary locus of } \int g_{\mu\nu}\dot{x}^\mu \dot{x}^\nu d\lambda \text{ [CT1.19]}$$

$$\delta \int g_{\mu\nu}\dot{x}^\mu \dot{x}^\nu d\lambda = 0 \iff \text{geodesic, affinely parametrized [Euler–Lagrange; math]}$$

$$(2):\ d_{12} = \Theta(P) = \sum_e \kappa_e \in \mathbb{Z} \text{ [FT.34(2)]};$$

$$\ell(P)$$

$$= \mathfrak{h} \sum_e \sqrt{\kappa_e} \text{ [D1.6/FT.24(2)} :\ ds^2(e) = \kappa_e \mathfrak{h}^2, \text{ spacelike; piecewise additivity Def-H.2]}$$

$$\left(\textstyle\sum_e \sqrt{\kappa_e}\right)^2 \le m \textstyle\sum_e \kappa_e \text{ [Cauchy–Schwarz; equality } \iff \sqrt{\kappa_e} \text{ constant]} \implies \ell^2 \le m\, d_{12}\, \mathfrak{h}^2;$$

$$[\ell^2] = \mathfrak{q}^{-1} = [m\, d_{12}\, \mathfrak{h}^2] \checkmark \text{ [D2.9]}$$

$$(3):\ \text{index-0} \iff \text{local } \ell\text{-minimizer [Morse/Synge; Def-H.19]};$$

$$\text{uniform families at fixed } m:\ \ell^2 = m\, d_{12}\, \mathfrak{h}^2 \text{ strictly monotone in } d_{12} \implies \text{minimizers coincide}$$

$$(4):\ \text{CT1.20}:\ \delta\Phi$$

$$= 0 \text{ configurations dominate,}$$

$$\delta\Phi$$

$$\neq 0 \text{ cancel pairwise : minima and saddles weighted alike;}$$

$$\neg\exists \text{ fluctuation-sign machinery in Parts 1–3 } \Longrightarrow \text{ selection open. } \blacksquare$$

The two distance claims of the derivation record (the winding-integer distance and the mass-as-distance identification) are resolved in the Addendum: see Appendix B, Section B.2.1.

Phys. A channel between two fermion floors carries two distances at once: an integer the ledger can only write in whole quanta, and a length the metric reads in generally irrational square roots — locked together by Cauchy–Schwarz, equal only when every hop carries the same quantum. Stable channels are the geodesics; the integer distance is the ledger's shadow of the metric one. The single thing the corpus cannot yet say is whether nature prefers the stable ones: that is the selection problem, and it is named.

3.18 Tier F.16 — Composition, Enumeration, and the Spin-Connection Obstruction

Theorem 3.44 (FT.52 — Spin extraction: the constraint system, uniqueness of the stretched rule, and the localized obstruction). Thm. Let R be an extraction rule in the sense of Def-H.9(3) — assigning admissible $\mathfrak{su}(2)$ labels j to on-shell internal ledger data, under the weak action [FT.13]. Then:

$$(1)$$

$$R \text{ cannot factor through } \mathrm{Hol}_{\mathrm{int}} \text{ alone on chain-swept interiors : } (-1)^{\Theta_{\mathrm{int}}}$$

$$= +1 \text{ on-shell [FT.25(1)];}$$

$$(2)$$

$$R \text{ must respect composition : } k(\Gamma)$$

$$= k_1 + k_2 + d_{12} \text{ [FT.34(3), FT.50(2)] and } k_f\text{-sums } \sum_f k_f$$

$$= K_S \text{ [FT.43(1)]} \Longrightarrow \text{ the stretched values add : any rank-1 additive } R \text{ satisfies } R(k_1{+}k_2)$$

$$= R(k_1){+}R(k_2) \text{ on aligned compositions [math : Cauchy equation on } \mathbb{Z}];$$

(3) uniqueness of the stretched rule : the unique rank-1 additive rule anchored at the seed $R(1)$

$$= \tfrac{1}{2}\ [\text{EW.32}] \text{ is } R(k)$$

$$= k/2 \text{ on the nonnegative winding cone,}$$

$$\text{with admissible labels } j$$

$= |k|/2$ (admissibility correction : $su(2)$ labels require $j \geq 0$, and the Casimir distinguishes $\pm k/2$: $(-\frac{k}{2})(-\frac{k}{2}) = \frac{k}{2} \neq \frac{k}{-2} = (\frac{k}{2})(\frac{k}{2})$ — the absolute value is forced, machine-witnessed; stretched additivity holds where all contributions are ≥ 0 — spacelike-dominant compositions; k is a signature datum, not orientation [FT.23]);

$$k \text{ odd} \Rightarrow j \in \tfrac{1}{2} + \mathbb{Z}_{\geq 0}, \quad k \text{ even} \Rightarrow j \in \mathbb{Z}_{\geq 0};$$

$$(4) \quad \text{the obstruction : promoting } R(k)$$

$= k/2$ from candidate to theorem requires one missing lemma — a corpus-level spin connection : an identification of the weak transport holonomy $\mathcal{U}_c$

$= \mathrm{P}\exp\big(i \oint_c \sum_A c_A \mathcal{G}_A \, d\ell\big)$ [EW.33]with the frame rotation of the emergent metric along c [CT1.13–CT1.17′s Levi-Civita data lifted to the $SU(2)$ carrier [EW.32]].

Parts 1–3 contains every ingredient but the lift : $g_{\mu\nu} \leftrightarrow (a_{00}, q_{ij}, w_i)$ dictionary with w_i in the W block [CT1.12, FT.30];

$$\Gamma^{\lambda}{}_{\mu\nu},$$

$$R^{\rho}{}_{\sigma\mu\nu} \ [\text{CT1.13, CT1.14}];$$

$$W_i$$

$$= \tfrac{1}{2}(\sigma_i^{(p)} + \sigma_i^{(q)}) \text{ total-spin generators } [\text{FT.13, CT1.2}];$$

seed transport [EW.32/EW.33] — but $\neg\exists$ tag constructing the spinor connection or tying $\oint$ (weak) to $\oint$ (frame rotation).Without it, k (geometric winding) and j (weak Casimir) are independently derived integers/half-integers with no derived bridge;

$$(5) \quad \text{verdict : } j$$

$= k/2$ remains the leading candidate — it passes every available consistency test : (1) parity triviality (it reads k, $\neg$parity);

$$(2) \text{ additive composition } [\text{FT.34}(3)];$$

$$(3)\ \text{seed anchor}\ R(1)$$

$$= \tfrac{1}{2}\ [\text{EW.32}];$$

(4) half-integrality from odd winding [clause(3)]. Promotion blocked exactly at Open Gaps 19 (spin connection) — not at any arithmetic step.

Req. FT.13, FT.25, FT.33, FT.34, FT.43, FT.50; Def-H.8, Def-H.9(3); CT1.2 (Part 1); CT1.12, CT1.13, CT1.14 (Part 2); EW.32, EW.33 (Part 2); Cauchy functional equation on $\mathbb{Z}$, Clebsch–Gordan basics [math].

Pf.

$$(1):\ \text{FT.25}(1):\ \Theta_{\text{int}}$$

$$\in 2\mathbb{Z}\ \text{on chain-swept interiors}$$

$$\Longrightarrow\ (-1)^{\Theta_{\text{int}}}$$

$$\equiv +1:\ \text{constant}\ \Longrightarrow\ \neg\text{label-carrying}$$

$$(2):\ \text{aligned composition}:\ k(\Gamma)$$

$$= k_1 + k_2 + d_{12}\ [\text{FT.34}(3)];$$

$$\text{additive}\ R:\ R(k(\Gamma))$$

$= R(k_1) + R(k_2) + R(d_{12})$ required for the stretched (maximal-weight) component of the CG decomposition [math : $j_{\max} = j_1 + j_2$]

$$(3):\ R:\ \mathbb{Z} \to \tfrac{1}{2}\mathbb{Z}\ \text{additive}\ \Longrightarrow\ R(k) = k\,R(1)\ [\text{Cauchy on}\ \mathbb{Z}];$$

$$R(1) = \tfrac{1}{2}\ [\text{EW.32 anchor}]\ \Longrightarrow\ R(k) = k/2;$$

parity claims immediate

$$(4):\ \text{ingredient inventory as stated}:\ \text{CT1.12 (dictionary)},$$

CT1.13/CT1.14 (Levi-Civita data),

FT.13/CT1.2 (W_i as spin generators),

EW.32/EW.33 (seed transport);

the lift $\Gamma \mapsto \omega^{ab}{}_{\mu} \mapsto SU(2)$ connection on $\mathcal{H}_{\mathsf{v}}$: $\neg\exists$ in Parts 1–3 [tag inventory complete]

(5) : conjunction of (1)–(4). ■

Rmk (why this is progress, not failure). Before this tier, "$j = k/2$" was one candidate among four. After it: every rank-1 additive rule is $k/2$ (uniqueness), all available consistency tests are passed, and the single missing link is named, located, and minimal — a spin-connection lemma, not a new postulate and not a search space. Gap 11 is reduced to Gap 19; Gap 19 is a construction question about lifting existing corpus data (CT1.13's Γ, EW.32's carrier), which is exactly the kind of step A §0.2a admits as [math]-class work if executed without new physical input.

Phys. If the weak transport of the seed around a bound chain is the frame rotation of the geometry the chain lives in, then spin is half the winding: odd windings give half-integer spin, even windings integer spin, aligned compositions add — and the whole fermion program's spin question reduces to one unbuilt bridge between two objects the corpus already has. The bridge is named; it is not yet built, and nothing here claims it is.

3.19 Tier F.17 — The Stationarity Loop: Constancy, the Spin Connection, and the No-Go

Theorem 3.45 (FT.53 — Geodesic constancy: the stationary on-shell ledger is constant-quantum). Thm. (1) Conservation. Along any affinely parametrized geodesic of $g_{\mu\nu}$ [CT2.6]: $\frac{d}{d\lambda}\big(g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu\big) = 2\, g_{\mu\nu}\dot{x}^\mu(\nabla_{\dot{x}}\dot{x})^\nu = 0$ [math] — the sampled quantity $\kappa(\lambda) = g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$ is constant along the path. (2) Constancy of the stationary on-shell ledger. In CT1.19's regime, materialized chains contributing to the semiclassical spectrum sit at $\delta\Phi = 0$ [CT1.20: $\delta\Phi \neq 0$ cancels pairwise; $\delta\Phi = 0$ dominates] = geodesy [FT.48(1)]; CT1.19's sampling $\kappa_e = g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu(\lambda_e) + O(\mathfrak{h})$ then gives $\kappa_e = \kappa_0 + O(\mathfrak{h})$ with a single constant κ_0; on integrality-class edges [FT.24/FT.34(4): $\kappa_e \in \mathbb{Z}$] and small $\mathfrak{h}$: $\kappa_e = \kappa_0 \in \mathbb{Z}$ exactly — FT.38(2)'s rigidity strengthened from eventual-local to exact-global on the stationary locus, without the integer-step ε-argument. (3) Equality saturation. On stationary on-shell channels: FT.33(4)'s Cauchy–Schwarz inequality and FT.48(2)'s two-sided law hold with equality: $\Lambda_c = L\sqrt{\kappa_0}$, $\ell^2 = m\, d_{12}\, \mathfrak{h}^2$, $\overline{\sqrt{\kappa}}^{\,2} = \kappa_0$ — the FT.33 bridge ratio becomes a single-integer function: $m^2_{\text{pole}}/m^2_{\text{hid}} = \hat{\mu}^2\mathfrak{h}^4 G(c)\,\kappa_0/(\ln 2)^2$. (4) Semiclassical exclusion of class (iii). FT.38(3)'s rational non-uniform ladders ($\kappa_e \in (2/L)\mathbb{Z}$, varying) require $\kappa(\lambda)$ non-constant along the chain ⇒ [contrapositive of (1)] non-geodesic support ⇒ off-stationary ⇒ pairwise cancellation in Z [CT1.20] ⇒ no semiclassical spectral contribution.

Req. FT.22, FT.24, FT.31, FT.33, FT.34, FT.38, FT.48; CT1.12, CT1.19, CT1.20, CT2.6 (Part 2); D1.6; geodesic conservation, contrapositive logic [math].

Pf.

(1) : $\frac{d}{d\lambda} g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$

$$= (\partial_\rho g_{\mu\nu})\dot{x}^\rho\dot{x}^\mu\dot{x}^\nu + 2g_{\mu\nu}\dot{x}^\mu\ddot{x}^\nu$$

$$= 2g_{\mu\nu}\dot{x}^\mu(\ddot{x}^\nu + \Gamma^\nu{}_{\rho\sigma}\dot{x}^\rho\dot{x}^\sigma)$$

$$= 2g_{\mu\nu}\dot{x}^\mu(\nabla_{\dot{x}}\dot{x})^\nu$$

$$= 0 \text{ [math; } \Gamma \text{ from CT1.13]}$$

$$(2):\ \text{CT1.20}:\ \text{spectral dominance at } \delta\Phi$$

$$= 0;$$

$$\text{FT.48}(1):\ \delta\Phi$$

$$= 0 \iff \text{geodesy (on-shell)};$$

$$\text{CT1.19 Pf}:\ \kappa_e$$

$$= g(\dot{x},\dot{x})(\lambda_e) + O(\mathfrak{h}) \xrightarrow{(1)} \kappa_0 + O(\mathfrak{h});$$

$$\kappa_e \in \mathbb{Z} \wedge |\kappa_e - \kappa_0| < \tfrac{1}{2}\ (\mathfrak{h}\text{ small}) \implies \kappa_e$$

$$= \kappa_0\ \forall e$$

$$(3):\ \kappa_e \equiv \kappa_0:\ \Lambda_c$$

$$= \textstyle\sum_e \sqrt{\kappa_0}$$

$$= L\sqrt{\kappa_0};$$

$$\left(\textstyle\sum \sqrt{\kappa_e}\right)^2$$

$$= L^2\kappa_0$$

$$= L\sum_e \kappa_e:\ \text{Cauchy–Schwarz equality [FT.33}(4)\text{]};$$

$$\text{connector of } m \text{ channels : } \ d_{12}$$

$$= m\kappa_0,$$

$$\ell$$

$$= \mathfrak{h}$$

$$m\sqrt{\kappa_0} \implies \ell^2$$

$$= \mathfrak{h}^2 m^2 \kappa_0$$

$$= m$$

$$d_{12}$$

$$\mathfrak{h}^2 \text{ [FT.48(2) equality]};$$

$$\overline{\sqrt{\kappa}}^{\,2} \qquad = \kappa_0$$

$$(4) : \ \kappa \text{ non-constant along } c \overset{(1)}{\implies} \neg\text{geodesic} \overset{\text{FT.48(1)}}{\implies} \neg\text{stationary} \overset{\text{CT1.20}}{\implies} \text{pairwise cancellation;}$$

$$\text{residue : finite-}\mathfrak{h}\text{ non-stationary poles [Open Gaps 13/18]}$$

$$\text{bridge ratio at } (3) : \ \frac{m^2_{\text{pole}}}{m^2_{\text{hid}}}$$

$$= \frac{\hat{\mu}^2 \mathfrak{h}^4 G(c)}{(\ln 2)^2}$$

$$\overline{\sqrt{\kappa}}^{\,2} \text{ [FT.33(2)]} \xrightarrow{\kappa \equiv \kappa_0} \frac{\hat{\mu}^2 \mathfrak{h}^4 G(c)\, \kappa_0}{(\ln 2)^2}. \ \blacksquare$$

Rmk (what became exact). Three inequalities/qualifiers of the earlier passes were artifacts of allowing non-stationary profiles: FT.33's "$\overline{\sqrt{\kappa}}^{\,2}$ (CS-maximal at uniform)", FT.48(2)'s "$\leq$", and FT.38(2)'s "eventually locally constant". On the stationary on-shell locus all three are equalities/identities: uniformity is not generic — it is forced by geodesy plus conservation.

Phys. The classical limit does more than make windings large: it makes them rigid. A materialized chain that survives to the continuum is a geodesic, and a geodesic samples one number

— its constant quantum κ_0 — so the ledger of the stationary sector is a single integer per chain, the distance laws become equalities, and the exotic rational ladders live only in the cancelling off-stationary foam.

Definition 3.25 (Def-H.21 — The spin connection on the emergent geometry (construction)). Def. On a spatial slice of CT2.6's (M, g) [CT1.12: $\bar{g}_{ij} \succ 0$ perturbatively], within a convex normal neighborhood [FT.47(3)] or, globally, on any orientable slice [math: orientable 3-manifolds are spin]:

$$(1) \quad \text{orthonormal frames } e_a = e_a{}^\mu \partial_\mu, \quad g_{\mu\nu} e_a{}^\mu e_b{}^\nu = \delta_{ab} \text{ (spatial)};$$

$$(2) \quad \text{Levi-Civita connection } \Gamma^\lambda{}_{\mu\nu} \text{ [CT1.13]} \;\to\; \text{spin connection } \omega_\mu{}^{ab}$$

$$= e^a{}_\lambda \nabla_\mu e^{b\lambda} \text{ [math]};$$

$$(3) \quad SU(2) \to SO(3) \text{ double cover [math]} \;\to\; \mathfrak{su}(2)\text{-connection } \tilde{\omega};$$

$$(4) \quad \text{carrier}: \; \mathcal{H}_{\mathsf{v}} \cong \mathbb{C}^2 \text{ [EW.32]},$$

$$\text{with } W_i \mapsto \tfrac{1}{2}\sigma_i \text{ [EW.32/CT1.2}: \text{ the corpus's own basis match}];$$

$$(5) \quad \text{spinor holonomy around a closed loop } c: \; \mathcal{U}_c^{\text{spin}} = \mathrm{P}\exp\big(i \oint_c \tilde{\omega}\big) \in SU(2);$$

$$(6) \text{ frame rotation angle } \Theta_R(c): \text{ the } SO(3) \text{ angle of the projected holonomy};$$

$$\text{for } c$$

$$= \partial S,$$

$$S \text{ disk-like in a constant-curvature slice}: \; \Theta_R$$

$$= \iint_S K$$

$$dA \text{ [Gauss–Bonnet; math]}.$$

No physical identification is made here: $\mathcal{U}^{\text{spin}}$ is not asserted to be EW.33's weak transport $\mathcal{U}$ (couplings c_A, P4 data) — that identification is the declared content of Def-H.37(2), and the obstruction to deriving it is Open Gaps 19. The construction is [math]-class over derived corpus objects [CT1.13, CT2.6, EW.32]; it adds no premise. A5/C11's node-level $SU(2)$ action is the corpus precedent for $SU(2)$ carriers.

Definition 3.26 (Def-H.37 — The defect-transfer interface). Def. The pole-side mass read of a materialized bound chain. Two clauses: (1) Leading-order transfer. For c a materialized spacelike bound chain with weak-transport profile c_A over the seed generator set [EW.33's c_A; EW.32's seed $\mathsf{v} \in \mathcal{H}_{\mathsf{v}}$], the chain's pole displacement takes EW.33's defect form for the seed weak-transported around c:

$$m^2_{\text{pole}}(c) = \hat{\mu}^2\, \ell_c^2\, G(c),$$

$$G(c) = \sum_{A,B} c_A c_B\, \mathrm{Re}\langle \mathcal{G}_A \mathsf{v}, \mathcal{G}_B \mathsf{v}\rangle \;\; \text{[EW.33 Gram data; EW.20]},$$

with $\hat{\mu}^2 > 0$ the floor-independent residue packaging $\varkappa^2\beta^2 h_0^2$ (EW.33's c_W with ℓ^2 factored out), $[\hat{\mu}^2] = \mathfrak{q}^2$ [EW.33 packaging; D2.9; Cal-F], and uniform coupling profile along the chain (EW.33's own leading-order regime). (2) Holonomy refinement. The transfer of (1) is Def-H.21's spin-connection holonomy: $\mathcal{U}_c = \mathcal{U}_c^{\text{spin}}$ with frame angle $\Theta_R(c)$ [Def-H.21(5)–(6)]; then exactly

$$m^2_{\text{pole}}(c) = \hat{\mu}'^2 \cdot 4\sin^2\Big(\frac{\Theta_R(c)}{4}\Big), \qquad [\hat{\mu}'^2] = \mathfrak{q} \;\; \text{[FT.54(4) packaging pattern; Cal-F]},$$

and clause (2) implies clause (1) at leading order under the matching $\hat{\mu}'^2\Theta_R^2/4 = \hat{\mu}^2\ell_c^2 G(c)$ — recorded, not imposed [FT.54(2), FT.33]. Interface status. The non-derived content of this definition is exactly: the extension of EW.33's defect read from wrapping-chain gauge modes [EW.33's locus] to materialized bound-chain termini, the holonomy identification of (2), and the two P4-class residues ($\hat{\mu}^2$, $\hat{\mu}'^2$; Cal-F). All further content is forced: the exact state- and axis-independent shape $4\sin^2(\cdot/4)$ [FT.54(1)]; EW.33 as its Taylor term [FT.54(2)]; the exclusion of rational-coupling combinatorial replacements [FT.73(2)–(3) — forward pointer, Tier F.24]; the range ceiling $[0, 4\hat{\mu}'^2]$ [FT.76(3) — forward pointer, Tier F.26]. Minimality audit: FT.155. Locus clause. No identification $m_{\text{pole}} = m_{\text{hid}}$ is asserted anywhere; the read registry and its bridge slots: Def-H.39. On rung-3 wrapping chains clause (1) is EW.33 verbatim — the transfer adds nothing there [FT.33 Rmk].

Theorem 3.46 (FT.54 — Exact holonomy-defect formula and the closed-form pole side (under Def-H.37(2))). Thm. (1) Exact SU(2) defect (state- and axis-independent). For $U = \cos\phi\,\mathbb{I} + i\sin\phi\,\hat{n}\cdot\vec{\sigma} \in SU(2)$ and any unit $v \in \mathbb{C}^2$:

$$(U-1)^\dagger(U-1) = \big(2 - 2\cos\phi\big)\mathbb{I} = 4\sin^2\!\big(\tfrac{\phi}{2}\big)\,\mathbb{I}$$

$$\Longrightarrow$$

$$\|(U-1)v\|^2 = 2 - 2\cos\phi \;\text{[math; exact Pauli algebra]}.$$

(2) EW.33 recovered as Taylor term. Any traceless Hermitian $X = \vec{x}\cdot\vec{\sigma}$ satisfies $X^2 = |\vec{x}|^2\mathbb{I}$; with $U = e^{iX}$, $\phi = |\vec{x}|$: defect $= 2 - 2\cos|\vec{x}| = |\vec{x}|^2 + O(|\vec{x}|^4) = \|Xv\|^2 + O(X^4)$ — EW.33's leading-order defect $\ell_\gamma^2 \|\sum_A c_A \mathcal{G}_A \mathsf{v}\|^2$ is the expansion of the same exact quantity [consistency derived, not assumed]. (3) Spin-connection holonomy. $\mathcal{U}_c^{\text{spin}} = U(\phi = \Theta_R(c)/2)$ [Def-H.21(5)–(6); double cover: frame angle Θ_R, spinor angle $\Theta_R/2$; sign convention irrelevant — cos even] $\Rightarrow \text{defect}_{\text{spin}}(c) = 2 - 2\cos\big(\frac{\Theta_R(c)}{2}\big) = 4\sin^2\big(\frac{\Theta_R(c)}{4}\big)$. (4) Closed-form pole side (Def-H.37(2)). Under Def-H.37(2) (clause (1)'s defect transport is the spin-connection holonomy):

$$m^2_{\text{pole}}(c) = \hat{\mu}'^2 \cdot 4\sin^2\Big(\frac{\Theta_R(c)}{4}\Big), \qquad [\hat{\mu}'^2] = [m^2] = \mathfrak{q} \text{ [EW.33 packaging pattern]};$$

for geodesic polygonal loops bounding disk-like S in constant-curvature slices: $\Theta_R = \bar{K}A = \sum_i \alpha_i - (n-2)\pi$ [Gauss–Bonnet; FT.47(1)] — mass from the angle defect. Bridge ratio (Gap 7's well-posed object):

$$\frac{m^2_{\text{pole}}}{m^2_{\text{hid}}} = \frac{4\,\hat{\mu}'^2\,\mathfrak{h}^2\,\sin^2(\Theta_R/4)}{E^2_{\text{int}}\,(\ln 2)^2} \in \mathfrak{q}^0$$

$$[\text{dimension check}: \ [\hat{\mu}'^2\mathfrak{h}^2] = \mathfrak{q}\cdot\mathfrak{q}^{-1} = \mathfrak{q}^0].$$

(5) Exact witness. Unit-sphere octant triangle ($n=3$, $\alpha_i = \pi/2$, $A = \pi/2$): $\Theta_R = \pi/2$; defect $= 2 - 2\cos(\pi/4) = 2 - \sqrt{2} \in \mathbb{Q}(\sqrt{2})$ — an exact irrational ledger value, in the corpus's own radical-arithmetic class [checks/README pattern].

Req. Def-H.37(2) (clause 4 only); Def-H.21; EW.32, EW.33 (Part 2); FT.47(1); FT.33, FT.42(1), FT.45(1) (m_{hid} forms); D2.9, CT2.9 (C, units); Pauli algebra, half-angle identities, Gauss–Bonnet [math].

Pf.

$$(1): \ U^\dagger U = \mathbb{I} \implies (U-1)^\dagger(U-1) = 2\mathbb{I} - U - U^\dagger;$$

$$U + U^\dagger = 2\cos\phi\,\mathbb{I} \text{ (since } i\sin\phi\,\hat{n}\cdot\vec{\sigma} \text{ is antihermitian)} \implies (2-2\cos\phi)\mathbb{I}$$

$$\|(U-1)v\|^2 = \langle v|(U-1)^\dagger(U-1)|v\rangle = (2-2\cos\phi)\|v\|^2 = 2-2\cos\phi;$$

$$2 - 2\cos\phi = 4\sin^2(\phi/2) \text{ [half-angle; math]}$$

$$(2): \ X = \vec{x}\cdot\vec{\sigma}: \ X^2 = (\vec{x}\cdot\vec{x})\mathbb{I} \text{ [Pauli; math]};$$

$$e^{iX} = \cos|x|\,\mathbb{I} + i\sin|x|\,\hat{x}\cdot\vec{\sigma} \implies \text{defect} = 2-2\cos|x| = |x|^2 + O(|x|^4) = \|Xv\|^2 + O(X^4) \ [\|v\| = 1]$$

$$\text{EW.33}: \ \|(U-1)\mathsf{v}\|^2$$

$$= \ell_\gamma^2 \|\textstyle\sum c_A \mathcal{G}_A \mathsf{v}\|^2 + O(c^3)$$

$$= \text{the same expansion with } X$$

$$= \ell_\gamma \textstyle\sum c_A \mathcal{G}_A$$

$$(3): \ \text{Def-H.21(5)–(6)}: \ \phi = \Theta_R/2;$$

$$(4):\ \text{Def-H.37(2) substitution};\quad \Theta_R = \bar{K}A = \sum \alpha_i - (n-2)\pi\ [\text{FT.47}(1),\ \text{Gauss–Bonnet}];$$

$$\text{ratio}:\ m_{\text{hid}}^2 = E_{\text{int}}^2(\ln 2)^2/\mathfrak{h}^2\ [\text{FT.42}(1)\ \text{template}]$$

$$(5):\ \sum \alpha = 3\cdot\tfrac{\pi}{2},\ (n-2)\pi = \tfrac{\pi}{2}\implies \Theta_R = \tfrac{\pi}{2};$$

$$2 - 2\cos\tfrac{\pi}{4} = 2 - \sqrt{2};$$

$$(2-\sqrt{2})(2+\sqrt{2}) = 2\ \checkmark\ [\mathbb{Q}(\sqrt{2})\ \text{exact}].\ \blacksquare$$

Rmk (Def-H.37(1) vs Def-H.37(2)). Def-H.37(1) (leading order, weak-coupling profile c_A) and Def-H.37(2) (exact, spin-connection) agree to leading order when $\hat{\mu}'^2\Theta_R^2/4 = \hat{\mu}^2\ell_c^2 G(c)$ — a matching condition on the P4-class packaging (Cal-F), recorded not imposed.

Phys. If the fermion's pole mass is the seed's failure to return to itself under frame transport, then it is a pure trigonometric function of the curvature flux the loop encloses — $4\sin^2(\Theta_R/4)$, exact, axis-free, state-free — and on geodesic polygons the curvature flux is the angle defect, the oldest invariant in geometry. The octant's mass ledger entry is $2-\sqrt{2}$: the first exact irrational number in the program's cards.

Theorem 3.47 (FT.55 — The no-go and the surviving spin realization). Thm. (0) Admissibility (correction). Throughout this tier and FT.52, the admissible spin label is $j = |k|/2$ with carrier $\text{Sym}^{|k|}$: $\mathfrak{su}(2)$ labels require $j \geq 0$ and the Casimir $j(j+1)$ distinguishes $\pm k/2$; stretched additivity is confined to the nonnegative cone (spacelike-dominant compositions) [FT.52(3) patched form]. (1) Calibration no-go. The identification $\Theta_R(c) = 2\pi k(c)$ (frame-holonomy angle = winding $\times 2\pi$) is ruled out as a corpus relation: $k = \oint ds^2/(2\mathfrak{h}^2)$ carries $\mathfrak{h}$-weight -2 [FT.31(3); D2.9], while $\Theta_R = \iint_S K\,dA$ is $\mathfrak{h}$-weight 0 for fixed classical regions [Def-H.21(6)]; a proportionality between quantities of different calibration weight is not $\mathfrak{h}$-free and hence not a derived pure-number relation [CT2.11's criterion]. Consequently the spin label cannot be read as the spin-connection holonomy angle of the same loop. (2) The surviving realization. FT.52's unique rank-1 additive rule $R(k) = k/2$ has exactly one natural representation-theoretic realization: $j = |k|/2 \iff$ carrier $\text{Sym}^{|k|}(\mathcal{H}_{\text{v}})$ — the $(|k|+1)$-dimensional irreducible $\mathfrak{su}(2)$ representation [math: classification; $k \geq 0$ on the cone of clause (0)]. It satisfies every derived constraint: anchor $\text{Sym}^1 = \mathcal{H}_{\text{v}}$, $j = \frac{1}{2}$ [EW.32]; composition $\text{Sym}^{k_1}\otimes\text{Sym}^{k_2} \supseteq \text{Sym}^{k_1+k_2}$ as maximal weight [math: Clebsch–Gordan] = FT.34(3)'s additive winding law $k(\Gamma) = k_1 + k_2 + d_{12}$ on the stretched channel; parity triviality respected (the rule reads k, not Hol_{int}) [FT.52(1)]; half-integrality from odd k [FT.52(3)]. (3) Scope consistency. Elementary floors carry small k [FT.28: $k = \kappa = 1$ floor; FT.45/FT.42 minima: $k = O(1)$] $\Rightarrow$ small j; the large-k regime is the classical-geometry limit [FT.31(3)], not particle territory — no macroscopic-spin contradiction. (4) Residual, relocated. What remains missing for promotion is neither the connection (Def-H.21 constructs it) nor the algebra (FT.52 settles it): it is a ledger mechanism assigning $\text{Sym}^k(\mathcal{H}_{\text{v}})$ to a materialized chain of winding k — a k-fold symmetric seed-tensoring tied to the winding count. EW.32 transports one qubit around the chain; nothing in Parts 1–3 tensors it k times.

Req. FT.22, FT.28, FT.31, FT.34, FT.45, FT.52; Def-H.21; D2.9; CT2.11 (Part 2); EW.32 (Part 2); $\mathfrak{su}(2)$ representation classification, Clebsch–Gordan maximal weight [math].

Pf.

$$(1):\ k=\Theta/2,\ \Theta\mathfrak{h}^2=\oint ds^2\ \text{[FT.22/FT.31(3)]}\implies [k]=[\mathfrak{h}^{-2}]=\mathfrak{q}^{+1}\Rightarrow k\ \text{has}\ \mathfrak{h}\text{-weight}-2;$$

$$[\Theta_R]=\mathfrak{h}^0\ \text{[Def-H.21(6): } K\,dA\ \text{classical]}$$

$$\Theta_R$$

$= 2\pi k$ would require a constant of weight $+2$ in $\mathfrak{h}$: ¬pure number $\implies$ ¬derived relation [CT2.11: rescaling $\mathfrak{h}\mapsto\lambda\mathfrak{h}$ is a symmetry; predictions are $\mathfrak{h}$-free ratios]

$$(2):\ \dim \mathrm{Sym}^k(\mathbb{C}^2)=k+1;\quad \text{highest weight}=k\cdot\tfrac{1}{2}=\tfrac{k}{2}\ \text{[math]};$$

$$\mathrm{Sym}^{k_1}\otimes\mathrm{Sym}^{k_2}=\bigoplus_{j=|k_1-k_2|/2}^{(k_1+k_2)/2}\mathcal{D}_j\supset\mathrm{Sym}^{k_1+k_2}\ \text{[CG; math]}$$

$$\text{anchor}:\ k=1:\ \mathrm{Sym}^1=\mathcal{H}_{\mathsf{v}}\cong\mathbb{C}^2,\ j=\tfrac{1}{2}\ \text{[EW.32]};$$

$$\text{constraints}:\ \text{FT.52}(1)\text{–}(3)\ \text{each verified above}$$

$$(3):\ \text{floor minima}:\ k\in\{1,2,3\}\ \text{[FT.28}(2),\ \text{FT.45}(4),\ \text{FT.42}(5)\ \text{uniform minima]}$$

$$\implies j\in\{\tfrac{1}{2},1,\tfrac{3}{2}\};$$

$$k\to\infty:\ \text{classical regime [FT.31}(3)\text{]}$$

$$(4):\ \text{inventory}:\ \text{connection [Def-H.21]}\ \checkmark;$$

$$\text{algebra [FT.52]}\ \checkmark;$$

$$\text{tensoring mechanism}:\ \neg\exists\ \text{tag [Parts 1–3 inventory]}.\ \blacksquare$$

Rmk (why the no-go is progress). Before this tier, "read spin from holonomy" was an open direction among several. The calibration argument closes it with a weight mismatch — the same kind of dimensional discipline that bars the malformed comparison. What survives is sharper and smaller: one construction question, with its target representation already unique.

Phys. The geometry refuses to hand over spin directly: the frame's rotation angle and the ledger's winding number live at different calibration weights, and no constant can bridge them. What the ledger can do is count copies — a winding-k chain as k symmetric copies of the seed qubit, spin $k/2$, combining exactly as windings compose. The rule is unique, the space is unique, the anchor is the corpus's own seed; only the copying machine is missing.

3.20 Tier F.18 — Native Rungs, Boundary Exchange Shape, and the Admissibility Correction

Theorem 3.48 (FT.56 — Native ladder rungs, the spin–orbit decomposition, and the missing map). Thm. (1) The bottom rungs are native. The candidate spin carriers for $j = 0, \frac{1}{2}, 1$ are existing corpus objects with the correct dimensions $2j+1$ and Casimir labels:

$j = 0 :\ \wedge^2\mathcal{H}_q \cong \mathrm{Sym}^0$ (trivial, 1-dim) — the reality line [A3, CT0.7(1) : $\Pi_{\mathcal{V}} = \frac{1}{4}(E_0 - \Sigma)$, $\mathfrak{T}\Pi_{\mathcal{V}} = 1$; Casimir 0 [CT0.7a]; σ_e lives here [CT0.9]];

$$j = \tfrac{1}{2} :\ \mathcal{H}_{\mathsf{v}} \cong \mathbb{C}^2,\ W_i \mapsto \tfrac{1}{2}\sigma_i \text{ — the transported seed [EW.32]};$$

$$j = 1 :\ \vee^2\mathcal{H}_q$$

$= \mathrm{Sym}^2(\mathcal{H}_q)$ (3-dim) — the pair channel's reality sector [A3 : $\mathcal{H}_p \otimes \mathcal{H}_q = \vee^2 \oplus \wedge^2$; CT0.7(1) : $\Pi_{\mathcal{R}} = \frac{1}{4}(3E_0 + \Sigma)$, $\mathfrak{T}\Pi_{\mathcal{R}} = 3$; CT0.7a : $\sum_i W_i^2|_{\vee^2} = 2 = 1(1{+}1)$];

and the reality grading is the placement rule : real edges carry $\vee^2$ -weight [A4 : $|\Pi_{\mathcal{V}}\rho_{pq}| \le \varepsilon_q$] — the symmetric square is the corpus's own real sector.

(2) Spin–orbit decomposition of the additive law. $k(\Gamma) = k_1 + k_2 + d_{12}$ [FT.34(3)] reads, under the stretched rule on the nonnegative cone [FT.52(3), corrected form]: $j(\Gamma) = j(c_1) + j(c_2) + j_{\mathrm{rel}}$, $j_{\mathrm{rel}} = d_{12}/2$ — the connector's distance quanta contribute as a relative (orbital-type) term, the floors' own windings as intrinsic terms. On the cone this is exactly maximal-weight Clebsch–Gordan composition [math]; Θ's orientation-blindness [FT.23] makes k (hence j_{rel}) a signature datum — negative only for timelike-dominant connectors, where the cone condition fails and the read is out of scope. (3) The missing map, stated exactly. EW.32 transports one qubit around a materialized chain, regardless of its winding; no tag in Parts 1–3 (inventory) couples a Θ-type ledger count to carrier multiplicity. What any candidate map winding $\to$ carrier must satisfy: (i) $k{=}1 \mapsto \mathcal{H}_{\mathsf{v}}$ [EW.32 anchor]; (ii) $k{=}2$ must land in the channel's own reality sector $\vee^2 = \mathrm{Sym}^2$ [clause (1) — the rung exists natively; the map must select it]; (iii) nonnegative-cone additivity [clause (2)]; (iv) admissibility $j = |k|/2$ [admissibility correction]. Gap 19 is exactly the construction of a map satisfying (i)–(iv) — or a no-go on such maps, which would reopen FT.52's constraint list to higher-rank rules.

Req. A3, A4; CT0.7, CT0.7a, CT0.9, CT1.2 (Part 1); EW.32 (Part 2); FT.13, FT.23, FT.34, FT.52, FT.55; Def-H.8; $\wedge^2 \cong \mathrm{Sym}^0$, $\vee^2 \cong \mathrm{Sym}^2$ of $\mathbb{C}^2$, maximal-weight CG [math].

Pf.

$$(1):\ \dim\wedge^2=\tbinom{2}{2}=1,\ \dim\vee^2=3\ [\text{A3};\ \text{CT0.7}(1):\ \mathfrak{TI}_{\mathcal{V}}=1,\ \mathfrak{TI}_{\mathcal{R}}=3];$$

$$\textstyle\sum_i W_i^2=\frac{3}{2}E_0+\frac{1}{2}\Sigma\ [\text{CT0.7a}]$$

$$\Sigma=2\mathbb{S}-E_0\ [\text{CT0.7}(1):\ \mathbb{S}=\tfrac{1}{2}(E_0+\Sigma)]\implies \textstyle\sum_i W_i^2\big|_{\vee^2}=\frac{3}{2}+\frac{1}{2}(2\cdot 1-1)=2;$$

$$\big|_{\wedge^2}=\tfrac{3}{2}+\tfrac{1}{2}(2(-1)-1)=0$$

$$[\text{machine}:\ 4\times 4\ \text{exact}]$$

$$2=1(1{+}1)\implies j=1\ \text{on}\ \vee^2;\quad 0=0(0{+}1)\implies j=0\ \text{on}\ \wedge^2\ [\text{CT0.7a verbatim}];$$

$$\mathcal{H}_{\mathsf{v}}:\ j=\tfrac{1}{2}\ [\text{EW.32}];$$

$$\dim=2j{+}1:\ 1,2,3\ \checkmark$$

$$(2):\ \text{FT.34}(3)\ \text{additivity}+\ \text{stretched rule on the cone}\ [\text{FT.52}(3)\ \text{form}]:\ j(\Gamma)$$

$$=\tfrac{k_1+k_2+d_{12}}{2}$$

$$=j_1+j_2+j_{\text{rel}};$$

$$\text{Sym}^{k_1}\otimes\text{Sym}^{k_2}\supseteq\text{Sym}^{k_1+k_2}\ [\text{maximal weight};\ \text{math}]$$

$$\text{sign of}\ k:\ \Theta$$

$$=\sum\kappa_e\ \text{orientation-blind}\ [\text{FT.23}]$$

$$\implies k<0\iff\text{timelike-dominant}\ [\text{CT1.12 signature}]\ \text{— cone fails;}$$

$$\text{read out of scope}$$

$$(3):\ \text{EW.32}:\ \text{carrier}=\mathbb{C}^2\ \text{per chain}\ (\neg\text{per winding unit});$$

$$\text{tag inventory}:\ \neg\exists\ \text{multiplicity coupling;}$$

$$(i)\text{–}(iv):\ \text{as stated.}\ \blacksquare$$

Rmk (why native rungs matter). Before this tier, "$j = k/2$" was a rule looking for carriers. The carriers for its first three values are now located inside the frozen structure: the virtual line, the seed, and the real sector of every pair channel — with the reality grading (A4/CT0.7) already deciding which of $\mathrm{Sym}^0/\mathrm{Sym}^2$ is "real". The missing map does not have to invent representation spaces; it has to select among existing ones.

Phys. The bottom of the spin ladder is not hypothetical: spin 0 is the ledger's virtual line, spin $\frac{1}{2}$ is the seed it transports, spin 1 is the symmetric square its own reality test selects — three objects Parts 1–2 already proved, wearing the exact dimensions and Casimiers the winding rule asks for. Composite floors add their windings like angular momenta, with the channel between them contributing the relative term. What is still missing is the one mechanism that would say why a winding-k loop carries k symmetric copies — and that mechanism is now the whole of the question.

Theorem 3.49 (FT.57 — The boundary exchange algebra: root-space shape). Thm. (1) Decomposition. For $N := N(\Sigma) \geq 1$: $\mathfrak{u}(N) = \mathfrak{u}(1)_{\text{center}} \oplus \mathfrak{su}(N)$ [math; CT3.5's pattern at $N{=}3$]; relative to ι_Σ's factored diagonal $\bigoplus_e \mathfrak{u}(1)_e$ [Def-H.13: edge-wise assembly]:

$$\text{factored} = \text{diagonal}\ (N\ \text{dims},\ \text{incl. the centre});$$

$$\text{strictly new}$$

$$= \mathfrak{su}(N) \setminus \text{Cartan}$$

$$= \text{root spaces}:\ (N^2{-}1) - (N{-}1)$$

$$= N^2 - N\ \ [\text{matches FT.41(d) exactly}].$$

(2) Shape at $N = 2$. The strictly-new sector $\{|e_1\rangle\langle e_2| + \text{h.c.},\ i(|e_1\rangle\langle e_2| - \text{h.c.})\}$ [FT.41(d)] is, with the factored traceless diagonal, an $\mathfrak{su}(2)$ triplet: $E_\pm = \frac{1}{2}(\sigma_1 \pm i\sigma_2)$, $[E_+, E_-] = \sigma_3$, $[\sigma_3, E_\pm] = \pm 2E_\pm$ [math; machine-exact over $\mathbb{Q}(i)$] — the raising/lowering (charged-current) shape of D3.10's $W^\pm = \frac{1}{\sqrt{2}}(W_1 \mp iW_2)$ [C8: $F_W = F_{W^+} \oplus F_{W^-} \oplus F_Z$], as algebra shape, not as identification. (3) Hosting question, localized. Which aspect block $\mathcal{K}^{(A)}$ the exchange modes land in under ι_Σ is not determined by the current tags: Def-H.13 defines ι_Σ by a factorization requirement, not by explicit matrices; the image of the root spaces in $\mathcal{K} = \bigoplus_A \mathcal{K}^{(A)}$ needs an explicit edge-block realization of Def-H.13's assembly [CT0.12 blocks $\otimes\ \iota_{n(w_e)}$] — one named construction away. Candidates remain: a second EM-type centre read is excluded by (1)–(2) (the exchange sector is nonabelian root-space, not central); a W-type hosting matches the charged-current shape; a genuinely new collective aspect remains possible.

Req. Def-H.13; FT.41(d); CT0.12; CT3.5 (F, pattern); C8; D3.10; $\mathfrak{u}(N)$ decomposition, $\mathfrak{su}(2)$ commutators [math].

Pf.

$$(1):\ \mathfrak{u}(N)=\mathfrak{su}(N)\oplus\mathfrak{u}(1)_{\text{center}}\ [\text{math};\ \text{CT3.5 at } N{=}3];$$

$$\iota_\Sigma|_{|e\rangle}=(\text{CT0.12 block})\otimes\iota_{n(w_e)}\ [\text{Def-H.13}]\implies \text{diagonal}\bigoplus_e\mathfrak{u}(1)_e\ \text{factored}$$

$$\dim\mathfrak{su}(N)$$

$$=N^2{-}1,$$

$$\dim\text{Cartan}$$

$$=N{-}1\implies\dim\text{root spaces}$$

$$=N^2-N\ [\text{math}]$$

$$=\text{FT.41(d)'s strictly-new count}\ \checkmark$$

$$(2):\ N$$

$$=2:\ E_\pm$$

$$=\tfrac{1}{2}(\sigma_1\pm i\sigma_2):\ [E_+,E_-]$$

$$=\tfrac{1}{4}[\sigma_1+i\sigma_2,\ \sigma_1-i\sigma_2]$$

$$=\tfrac{1}{4}(-2i[\sigma_1,\sigma_2])$$

$$=\tfrac{1}{4}(-2i\cdot 2i\sigma_3)$$

$$=\sigma_3\ [\text{CT1.2-pattern commutators};\ \text{machine over}\ \mathbb{Q}(i)]$$

$$[\sigma_3,E_\pm]=\pm2E_\pm\ [\text{machine}];$$

$$\text{shape match to}\ W^\pm\ [\text{D3.10}:\ \text{same raising/lowering combinations}]$$

$$(3):\ \text{Def-H.13 is a factorization requirement;}$$

$$\neg\exists \text{ explicit matrix tag for } \iota_\Sigma \text{ [inventory]} \implies \text{image decomposition in } \mathcal{K} \text{ undetermined;}$$

$$\text{centre read excluded : exchange sector nonabelian by } (2). \blacksquare$$

Phys. The first genuinely collective structure of the coarse-graining program — the thing a subnetwork can do that no single edge can — has the algebraic shape of a charged current: raising and lowering operators flipping slot excitation between boundary contacts, completing an $\mathfrak{su}(2)$ with the factored diagonal. Two contacts can exchange; one cannot. Where this exchange lives among the forces is the next named construction, not a guess: the nonabelian shape already rules out the abelian candidate.

Definition 3.27 (Def-H.22 — The explicit boundary realization (corolla and separated patterns)). Def. For Σ with $N := N(\Sigma)$ boundary edges $\partial\Sigma = \{e_1, \dots, e_N\}$, $\mathcal{H}_{\text{slot}}(\Sigma) = \text{span}\{|e_i\rangle\}$ [Def-H.12], ι_Σ is made explicit by contact pattern: (a) Corolla (all boundary edges share one external vertex w; slots edge-labelled [A7: $\tau(v)$ records incident edges; D0.8] $\Rightarrow$ canonical inclusion $\mathcal{H}_{\text{slot}}(\Sigma) \hookrightarrow \mathcal{H}_{\text{slot}}^{(n(w))}$, equality iff $n(w) = N$):

$$\iota_\Sigma^{\text{cor}} := \iota_{n(w)}\big|_{\mathfrak{u}(N)} \qquad \text{[D2.12 applied to the subalgebra on the included slots].}$$

(scope [FT.59]): the formula is defined exactly for $n(w) \le 3$ — at $n(w) \ge 4$ no realization $\iota_{n(w)}$ exists [CT4.4 Pf(b): slot algebras realized only for $n \le 3$]; the corolla read then degenerates to the registration/sealing ledger (σ_e, chain-local f_v, q_{core}, $\mathcal{C}$) plus D^--absorption [D4.1, D4.2], with the collective $\mathfrak{u}(N)$ wholly unhosted [FT.59(4)]. (b) Separated (boundary edges with distinct external vertices $w_1 \neq w_2 \cdots$): only the factored diagonal is defined — each edge's $\mathfrak{u}(1)_e$ maps through its own CT0.12 block and $\iota_{n(w_e)}$'s centre [D2.12: centre $\mapsto \mathcal{K}^{(\text{EM})}$]; since $\dim \mathcal{K}^{(\text{EM})} = 1$ [CT0.14], all edge centres share one image line:

$$\iota_\Sigma^{\text{sep}}\Big(\sum_e c_e D_e\Big) = \Big(\sum_e c_e\Big) Y_{\text{asp}}$$

$$\text{[EM.2 : } \mathfrak{T}(Y_{\text{asp}}) = 1\text{];}$$

$$\text{off-diagonal } |e_i\rangle\langle e_j| \ (i \neq j) : \ \neg\exists \text{ factored image (no cross-vertex bridge in the inventory).}$$

No new choice is declared: (a) is the canonical edge-labelled slot inclusion composed with D2.12; (b) is forced by the 1-dimensional EM block. Every computed floor instantiates a pattern: single-contact floors [FT.28/FT.32/FT.42]: $N = 1$; the face floor [FT.49]: corolla, $N = 6$, $n(D) = 6$ (equality) — in the void regime $n(w) \ge 4$ [FT.59(4a)]: no realization, exchange sector kernel/absorbed; the two-contact Σ_{min} variant [FT.28 Rmk]: separated, $N = 2$.

Theorem 3.50 (FT.58 — Hosting classification and the no-hosting theorem). Thm. (1) Corolla hosting follows the CT0.13 ladder at the contact vertex. Under Def-H.22(a): the centre of $\mathfrak{u}(N)$ maps to $\mathcal{K}^{(\mathrm{EM})}$ [D2.12]; and:

$$n(w) = 2: \quad \iota_\Sigma(\mathfrak{su}(2)) \subseteq \mathrm{span}\{W_i\} = \mathcal{K}^{(W)} \qquad [\text{CT1.4}: \ W_i \text{ close } \mathfrak{su}(2); \ \text{D2.12}];$$

$$n(w) = 3: \quad \iota_\Sigma(\mathfrak{su}(3)) = \mathcal{K}^{(\mathrm{strong})}$$

$$[\text{D2.12}: \ \iota_3(\mathfrak{su}(3)) \text{ spans } \mathcal{K}^{(\mathrm{strong})}; \ \text{CT3.7/CT3.8}: \ 8 = 8];$$

$$\text{the exchange generators host literally}: \quad \frac{\lambda_1 + i\lambda_2}{2}$$

$$= E_{12} \ (3 \times 3 \text{ matrix unit}): \ \ |e_1\rangle\langle e_2| \ \mapsto \ \text{the } \lambda_1,$$

$$\lambda_2 \text{ directions};$$

$$D_1 - D_2 \mapsto \lambda_3\text{-type } [\text{CT3.9}; \ \text{machine}];$$

$$n(w)$$

$$\geq 4: \ \text{resolved (FT.59)} — \neg\exists \ \iota_n \text{ realization at all for } n$$

$$\geq 4 \ [\text{CT4.4 Pf}(b): \ \text{slot algebras realized only for } n \leq 3];$$

$$\text{the suggested strong-hosting of } \mathfrak{su}(3) \text{ subalgebras is impossible } [\text{FT.59}(3)];$$

$$\text{such corollas are wholly kernel}/D^-\text{-absorbed } [\text{FT.59}(1), (4)].$$

(2) Separated contacts: the collective part is unhosted. Under Def-H.22(b): $\mathrm{im}\,\iota_\Sigma^{\mathrm{sep}} = \mathcal{K}^{(\mathrm{EM})}$ (the total centre read, 1-dim); $\ker \iota_\Sigma^{\mathrm{sep}} \supseteq$ traceless diagonal ($D_1 - D_2$: the two edge centres coincide in the 1-dim block); the off-diagonal exchange generators ($N^2 - N$ dims [FT.41(d), FT.57(1)]) have no factored image and no derived host: no tag bridges distinct vertex slot algebras [inventory]. Ledger at $N = 2$: $4 = 1$ (image) $+ 1$ (kernel) $+ 2$ (unhosted). (3) No-hosting theorem ($N \geq 4$). The observable aspect capacity is $\mathcal{K}^{\mathrm{obs}}_{\mathrm{pair}} \oplus \mathcal{K}^{\mathrm{obs}}_{\mathrm{stack}}$ with $\dim \mathcal{K}^{\mathrm{obs}}_{\mathrm{pair}} = 5{+}1{+}3 = 9$ [G, H, W; $J^\pm, D^\pm$ never observable — FT.20] and $\dim \mathcal{K}^{\mathrm{obs}}_{\mathrm{stack}} = 1{+}8 = 9$ [EM, strong; D^- never — FT.20], the two commuting (distinct registers [CT0.12 Rmk]). For $N \geq 4$: $\mathfrak{su}(N)$ is simple [math] with $\dim = N^2 - 1 \geq 15 > 9$; any Lie homomorphism $\varphi : \mathfrak{su}(N) \to \mathcal{K}^{\mathrm{obs}}_{\mathrm{pair}} \oplus \mathcal{K}^{\mathrm{obs}}_{\mathrm{stack}}$ has projections $\pi_{\mathrm{pair}} \circ \varphi$, $\pi_{\mathrm{stack}} \circ \varphi$ that are homomorphic images of a simple algebra (the factors commute) — hence trivial or injective; injective is impossible by dimension ($15 > 9$); therefore $\varphi \equiv 0$ on $\mathfrak{su}(N)$: the entire collective nonabelian boundary algebra maps to zero under every Lie-type-respecting hosting [D2.12/D3.2's realization semantics] — not partially hosted: wholly kernel. The centre $\mathfrak{u}(1)$ still hosts in EM. (4)

Capacity ceiling, exact. Nontrivial Lie hosting of $\mathfrak{su}(N)$ exists within the current inventory only for $N \leq 3$ [$\dim \mathfrak{su}(3) = 8 \leq 9$; N=2: W-hosting, N=3: strong-hosting, both realized by (1)]. For $N \geq 4$ the coarse-grained read either kernels the collective structure (FT.26's mask keeps only hosted blocks) or the aspect inventory must grow by declaration — no derivation in Parts 1–3 supplies it. The kernel is the coarse-graining: $\dim \mathfrak{u}(N) = N^2$ vs $\dim \mathcal{K}_\Sigma \leq 18$ [FT.26(3)] forces $\ker \neq 0$ already at $N = 5$ by dimension alone.

Req. Def-H.13, Def-H.22; Def-H.15, Def-H.16; FT.20, FT.26, FT.41(d), FT.57; D2.12, D3.1, D3.2, D3.9; A7, D0.8; CT0.12 (Rmk), CT0.13, CT0.14 (Part 1); CT1.4 (Part 1); CT3.7, CT3.8, CT3.9 (Part 2); EM.2 (Part 2); simplicity of $\mathfrak{su}(N)$, projection argument for commuting factors [math].

Pf.

$$(1):\ \text{corolla inclusion canonical [A7/D0.8 : slots edge-labelled]} \implies \iota_\Sigma^{\text{cor}}$$

$$= \iota_{n(w)}|_{\mathfrak{u}(N)}\ \text{[Def-H.22}(a)\text{]};$$

$$n(w) = 2:\ \iota_2(\mathfrak{su}(2)) = \text{span}\{W_i\}\ \text{[CT1.4, D2.12]};$$

$$n(w) = 3:\ \iota_3(\mathfrak{su}(3))\ \text{spans}\ \mathcal{K}^{(\text{strong})}\ \text{[D2.12]},\ \dim\ 8 = 8\ \text{[CT3.8]}$$

$$\text{literal exchange hosting}:\ \lambda_1$$

$$= E_{12} + E_{21},$$

$$\lambda_2$$

$$= -iE_{12} + iE_{21}\ \text{[CT3.9 explicit basis]} \implies \tfrac{1}{2}(\lambda_1 + i\lambda_2)$$

$$= \tfrac{1}{2}(E_{12} + E_{21} + E_{12} - E_{21})$$

$$= E_{12}\ \text{[machine]}$$

$$(2):\ \dim \mathcal{K}^{(\text{EM})}$$

$$= 1\ \text{[CT0.14]} \implies \iota_\Sigma^{\text{sep}}(D_1),$$

$$\iota_\Sigma^{\text{sep}}(D_2)\ \text{collinear [EM.2 normalization]}$$

$$\implies D_1 - D_2 \in \ker;$$

$$\text{off-diagonals}: \ \text{Def-H.13}'\text{s factorization is per-edge [CT0.12 block} \otimes \iota_{n(w_e)}]$$

$$\implies |e_1\rangle\langle e_2| \text{ has no per-edge form;}$$

$$\neg\exists \text{ cross-vertex bridge tag}$$

$$(3): \ \mathcal{K}^{\text{obs}}$$

$$= \mathcal{K}^{\text{obs}}_{\text{pair}} \oplus \mathcal{K}^{\text{obs}}_{\text{stack}},$$

$$[\mathcal{K}^{\text{obs}}_{\text{pair}}, \mathcal{K}^{\text{obs}}_{\text{stack}}]$$

$$= 0 \text{ [CT0.12 Rmk}: \ \text{distinct registers]};$$

$$\dim$$

$$= 9,$$

$$9 \text{ [CT0.12 block table; FT.20}: \ J^{\pm}, D^{\pm}, D^{-} \text{ never]}$$

$$\varphi: \ \mathfrak{su}(N) \to \mathcal{K}^{\text{obs}} \text{ Lie hom.}: \ \pi_{\text{pair}} \circ \varphi \text{ hom. image of simple} \implies \ker$$

$$= \{0\} \ \vee \ \mathfrak{su}(N);$$

$$\ker$$

$$= \{0\} \implies \dim \mathfrak{su}(N)$$

$$\leq 9$$

$$\implies N^2 - 1$$

$$\leq 9$$

$$\implies N \qquad \leq 3;$$

$$N \geq 4 : \text{ both projections trivial } \implies \varphi \equiv 0$$

$$(4) : \ N \leq 3 : \text{ realized by } (1);$$

$$N \geq 4 : \ (3);$$

$$\dim \text{ forcing} : \ N^2 > 18 \iff N \geq 5 : \ \ker \neq 0 \text{ for any linear map [FT.26(3)]. } \blacksquare$$

Rmk (what would change the verdict). Clause (3) is an inventory theorem, not a permanent barrier: a declared exchange block $\mathcal{K}^{(\text{exch})}$ (new aspect candidate, interface-definition species) or a derived cross-vertex bridge would reopen hosting for separated contacts and $N \geq 4$ corollas. Neither is constructed or declared here; the theorem states exactly what the current inventory forces. Note the physical reading of the corolla cases: at $n(w) = 2$ the exchange flip is the environment vertex's own weak isospin flip [CT1.4] — the "collective mode of Σ" is an environment mode in disguise; only separated contacts (and large corollas) carry structure with no home in the existing forces.

Phys. The coarse-graining machine has a hard collective ceiling, and it is the colour algebra: two boundary contacts exchange through the weak block, three through the colour block — four and beyond, the entire nonabelian exchange structure falls into the kernel of every available read. The kernel is not a defect; it is what coarse-graining means at the operator level. Whether nature's larger boundaries declare a new block or simply forget their exchange structure is the sharpest extension question the program has produced — and it is now a theorem that the existing sixteen registers cannot answer it.

3.21 Tier F.19 — The Three Tails: Realization Non-Existence, the Multiplicity Map, and the Selection Obstruction

FT.59 — The $\iota_{n\geq 4}$ specification: non-existence, persistence refuted, hosting ceiling. Structural-constraints record: see Appendix B, Section B.3.2 of the Addendum.

Definition 3.28 (Def-H.38 — The multiplicity–winding coupling). Def. A materialized closed bound chain c [D1.5: bound content fermionic; FT.22: materialized] with winding $k(c) = \Theta(c)/2 \geq 0$ [FT.22, FT.25(3); the nonnegative (spacelike-dominant) cone, FT.52(3)] carries $k(c)$ symmetric copies of the transported seed:

$$\text{multiplicity rule (constitutive clause):} \qquad \text{copies}(c) \ := \ k(c),$$

realized on the carrier $\Psi(c) = \text{Sym}^{k(c)}(\mathcal{H}_{\text{v}})$ with the diagonal transport $\mathcal{U}_c^{\otimes k}\big|_{\text{Sym}^k}$ [Def-H.23 — immediate below; EW.32: $\mathcal{H}_{\text{v}} \cong \mathbb{C}^2$ the transported seed]. Derived content (not part of the interface). Uniqueness of Ψ given the rule [FT.52(3), FT.55(2), FT.60(2)]; the four conditions (i)–(iv), including the corpus-pinned placement at k=2 — real $\iff$ support in $\vee^2$ at $\varepsilon_q = 0$, so two real-registered seeds grade symmetric [FT.60(1); A4, CT0.7, CT0.8]; the Casimir ladder $\frac{k}{2}(\frac{k}{2}+1)$

generalizing CT0.7a [FT.60(3)]; the weight ladder and the defect multiplets [FT.60(4)–(5), under Def-H.37(2)]; composition as maximal-weight Clebsch–Gordan on the cone [FT.60(6)]; the spin label $j = |k|/2$ [FT.55(2), FT.60(1)(iv)]. Minimality (derived). The coupling is irreducible to prior corpus content: EW.32 transports one seed per chain regardless of winding; the uniform-minimal triangle ($k = 3$) carries no three materialized winding-1 subloops; admissible decompositions are basis-relative [FT.60(7a)–(b) — forward pointers in-tier]. One integer coupling; no construction below it. Scope. Bound chains on the nonnegative cone only; transit/trivial chains excluded — photon helicity [EM.8] is a transit-side read, no identification. Carriers for $k \geq 3$ (dim ≥ 4) exceed the registered pair-channel inventory [CT0.12: blocks $3, 3, 5, 3, 1, 1$] — no aspect claim at $k \geq 3$; the inventory-extension question is application-layer content [FT.60(8)].

Definition 3.29 (Def-H.23 — The winding-multiplicity carrier map (under Def-H.38)). Def. For c a materialized closed bound chain [D1.5: bound content fermionic; FT.22: materialized] with winding $k(c) = \Theta(c)/2 \geq 0$ [FT.22, FT.25(3)] — the nonnegative (spacelike-dominant) cone [FT.52(3) form]:

$$\Psi(c) \;:=\; \mathrm{Sym}^{k(c)}\big(\mathcal{H}_{\mathsf{v}}\big) \;\subseteq\; \mathcal{H}_{\mathsf{v}}^{\otimes k(c)}$$

$$[\mathrm{Def-H.38}:\ \text{multiplicity} = \text{winding};\ \ \mathcal{H}_{\mathsf{v}} \cong \mathbb{C}^2 \text{ the transported seed, EW.32}],$$

where Sym^k is the common $(+1)$-eigenspace of all pairwise swaps on $\mathcal{H}_{\mathsf{v}}^{\otimes k}$ [math: the symmetric subspace; the swaps act as CT0.7's $\mathbb{S}$ on each tensor pair, adjacent transpositions generating S_k], and transport is the diagonal action $\mathcal{U}_c^{\otimes k}\big|_{\mathrm{Sym}^k}$ of EW.32's seed transport $\mathcal{U}_c$ around c [EW.32, EW.33; Def-H.21(5) for the spin-connection form]. Boundary cases: $k = 0$: $\mathrm{Sym}^0 = \mathbb{C}$, the trivial carrier — representation-identified with $\wedge^2 \cong \mathrm{Sym}^0$ [FT.56(1)], ¬subspace-identified; scope note: all-null trivial chains are transit objects (photons/gluons [D1.7, D3.7]) whose helicity content is a transit-side read [EM.8] — outside Ψ, no identification made. $k < 0$ (timelike-dominant): outside the cone, Ψ undefined [FT.52(3); FT.56(2)]. Interface status (Def-H.38). The corpus proves one seed transported per chain regardless of winding [EW.32]; the coupling multiplicity $= k(c)$ is Def-H.38's constitutive clause — FT.60(7) proves it irreducible to decomposition, so the clause is load-bearing, not packaging.

Theorem 3.51 (FT.60 — The multiplicity map: conditions, ladders, basis-relativity, and the honesty witnesses). Thm. Under Def-H.38, with Def-H.21's spin connection and EW.32's normalization: (1) The four conditions hold. Ψ satisfies FT.56(3)(i)–(iv): (i) $k{=}1:\ \mathrm{Sym}^1 = \mathcal{H}_{\mathsf{v}}$, $j = \frac{1}{2}$ [EW.32 anchor]; (ii) $k{=}2:\ \mathrm{Sym}^2 = \vee^2\mathcal{H}_{\mathsf{v}}$ — the channel's own reality sector, with projector $\Pi_{\mathcal{R}} = \frac{1}{4}(3E_0 + \Sigma)$ [CT0.7(1); A3] — and the placement is corpus-pinned: real $\iff$ support in $\vee^2$ at $\varepsilon_q = 0$ [A4, CT0.8], so once two seeds share a materialized (real) register their symmetric grading is forced [CT0.7 Phys: reality = Bose exchange symmetry]; only the multiplicity is interface content [Def-H.38]; (iii) cone-additivity: $\mathrm{Sym}^{k_1} \otimes \mathrm{Sym}^{k_2} \supseteq \mathrm{Sym}^{k_1+k_2}$ as maximal weight [math: CG] = the additive winding law on the stretched channel [FT.34(3), FT.50(2), FT.56(2)]; (iv) admissibility: carrier dimension $k + 1 = 2j + 1$ with $j = |k|/2$ on the cone [FT.52(3) form; FT.55(0)]. (2) Uniqueness. Any carrier assignment satisfying (i)–(iv) equals Ψ: the $(k{+}1)$-dim irreducible $\mathfrak{su}(2)$-module is unique up to isomorphism [math: classification; FT.55(2)] and the anchor fixes the seed [EW.32]. Gap 19's map exists and is unique under Def-H.38. (3) Casimir ladder (CT0.7a generalized). With

$W_i \mapsto$ total-spin generators on $\mathcal{H}_{\mathsf{v}}$ [EW.32/CT1.2: the corpus's own basis match] and the induced diagonal action on Sym^k:

$$\textstyle\sum_i \big(W_i^{(k)}\big)^2 = \frac{k}{2}\Big(\frac{k}{2}+1\Big)\,\mathbb{I}_{k+1}$$

[machine: exact rational matrices, $k \le 6$; physics normalization $W = \frac{1}{2}\cdot$ generators],

recovering the native rungs verbatim: k=0 : 0 on $\wedge^2$; k=1 : $\frac{3}{4}$ on the seed; k=2 : 2 on $\vee^2$ — CT0.7a's eigenvalues are the $k \le 2$ instances of one ladder [FT.56(1) extended from three rungs to all]. (4) Weight ladder. For $\mathcal{U} = \exp(i\phi\,\hat{n}\cdot\vec{\sigma})$ [FT.54's convention]: $\mathrm{Sym}^k(\mathcal{U})$ has eigenvalues $e^{im\phi}$, $m \in \{k, k-2, \ldots, -k\}$ — i.e. $\mathfrak{su}(2)$-weights $m/2 \in \{j, j-1, \ldots, -j\}$ at $j = k/2$, character $\sum_m e^{im\phi}$; on spin-connection transport $\phi = \Theta_R/2$ [Def-H.21(6)] [machine: diagonal transport over $\mathbb{Q}(i)$ at exact circle points; Laurent identity $(t^m-1)(t^{-m}-1) = 2 - t^m - t^{-m}$]. (5) Defect multiplet ladder (under Def-H.37(2)). The CT2.5 defect read $\|(\mathcal{U}^{(k)}-1)\psi\|^2$ on the weight-m vector [$m = k - 2r$ in the ϕ-labelling]:

$$m^2_{\text{pole}}(m) = \hat{\mu}'^2\big(2 - 2\cos(m\,\Theta_R/2)\big) = 4\,\hat{\mu}'^2 \sin^2\big(m\,\Theta_R/4\big), \qquad m \in \{k, k-2, \ldots, -k\};$$

k=1 reproduces FT.54(4) exactly and its state-independence is special to the seed ($m = \pm 1$ agree by evenness); $k \ge 2$ splits into a multiplet; every even k carries a weight m=0 that is defect-massless at every curvature flux — the kernel coherence of FT.33(6)/FT.54 generalized to a standing zero of the ladder [machine: exact over $\mathbb{Q}(i)$; Chebyshev identity $2-2\cos(m\phi) = 2-2T_m(\cos\phi)$ verified as integer-polynomial recursion]. (6) Composition at carrier level. For $\Gamma = c_1\cdot P\cdot c_2\cdot \bar{P}$ materialized with $d_{12} = \Theta(P) \ge 0$: $\Psi(\Gamma) = \mathrm{Sym}^{k_1+k_2+d_{12}} \subseteq \mathrm{Sym}^{k_1}\otimes\mathrm{Sym}^{k_2}\otimes\mathrm{Sym}^{d_{12}}$ as maximal weight [math: CG] — the spin–orbit decomposition $j(\Gamma) = j_1 + j_2 + j_{\text{rel}}$, $j_{\text{rel}} = d_{12}/2$ [FT.56(2)], now an inclusion of carriers, on the cone. (7) Two honesty witnesses. (a) Basis-relativity: the carrier assignment is per-chain, and a floor admits several admissible chain decompositions with the same total winding but different tensor carriers — witness (machine): the tetrahedron at uniform $\kappa_0 = 2$: the digon basis (M2) gives six chains of $k = 2$: $\bigotimes^6 \mathrm{Sym}^2$, dimension $3^6 = 729$; the face basis (M1) gives four faces of $k_f = 3$: $\bigotimes^4 \mathrm{Sym}^3$, dimension $4^4 = 256$; both have $\sum k = 12 = K_S$ [FT.43(1)'s sum rule is the invariance mechanism]. Decomposition-invariant content: K_S and the stretched total $j_{\max} = K_S/2$ (and per-chain multisets up to the homology choice); the full floor-level tensor product is basis-relative — recorded, no canonical choice imposed. Only edge-disjoint decompositions [FT.25(1)] (the digon basis) support an unrestricted $\otimes$; face bases overlap on channels and their $\otimes$ is formal. (b) Irreducibility of the multiplicity datum: Ψ is not "count materialized subloops" — the uniform-minimal triangle has $k = 3$ [FT.45(4): $\kappa \equiv 2$, $\Theta = 6$] yet carries no materialized winding-1 face subloop ($\kappa \equiv 1$ gives $\Theta = 3$, odd, off-shell [FT.22; machine]), and the profile identity $(2,2,2) = (1,1,0)+(0,1,1)+(1,0,1)$ decomposes into null-carrying digon-supported profiles — different chains, not three winding-1 loops. EW.32 transports one seed per chain regardless of k; no tag ties tensor multiplicity to winding [FT.56(3)]. Def-H.38's single datum is therefore genuinely new — a transfer of exactly Def-H.37(2)s species. (8) Inventory consequence, flagged. Sym^k for $k \ge 3$ (dim ≥ 4) lies in multi-seed tensor spaces with no counterpart among the frozen 16 pair registers [CT0.12 block dimensions $3, 3, 5, 3, 1, 1$: no 4-dim irreducible] nor the stack blocks: the carriers for $k \ge 3$ exist as [math] objects on $\mathcal{H}_{\mathsf{v}}^{\otimes k}$ but exceed the registered aspect inventory — an inventory-extension consequence of Def-H.38, recorded here and joined to Gap 16's residue-(b) declaration question ($\mathcal{K}^{(\text{exch})}$-species). No aspect claim is made for $k \ge 3$ carriers.

Req. Def-H.38; Def-H.23, Def-H.21; EW.32, EW.33 (Part 2); A3, A4; CT0.7, CT0.7a, CT0.8, CT0.12 (Part 1); CT1.2 (Part 1); CT2.5 (Part 2); D1.5, D1.7, D3.7; EM.8 (Part 2); FT.22, FT.25, FT.33, FT.34, FT.43, FT.45, FT.49, FT.50, FT.52, FT.54, FT.55, FT.56; $\mathfrak{su}(2)$ classification, Clebsch–Gordan, symmetric-subspace characterization, Chebyshev recursion [math].

Pf.

$$(1):\ \mathrm{Sym}^1 = \mathcal{H}_{\mathsf{v}}\ [\text{math}];\ j = \tfrac{1}{2}\ [\text{EW.32}];$$

$$\mathrm{Sym}^2(\mathbb{C}^2) = \vee^2\ [\text{A3}];\ \Pi_{\mathcal{R}} = \tfrac{1}{4}(3E_0 + \Sigma),\ \mathfrak{T} = 3\ [\text{CT0.7}(1);\ \text{machine regression}]$$

$$\varepsilon_q$$

$$= 0:\ (p,q) \in \mathcal{R} \iff \mathrm{supp}\,\rho_{pq} \subseteq \vee^2\ [\text{CT0.8}]:\ \text{two real-registered seeds grade symmetric [CT0.7 Phys; math: common } +1 \text{ eigenspace of transpositions} = \mathrm{Sym}^k]$$

$$\text{CG}:\ \mathrm{Sym}^{k_1} \otimes \mathrm{Sym}^{k_2} = \bigoplus_{j=|k_1-k_2|/2}^{(k_1+k_2)/2} \mathcal{D}_j \supseteq \mathrm{Sym}^{k_1+k_2}\ [\text{math}];$$

$$k(\Gamma) = k_1 + k_2 + d_{12}\ [\text{FT.34}(3), \text{FT.50}(2)]$$

$$(2):\ \dim = k+1\ \wedge\ \text{irreducible [math]} \implies \text{unique isomorphism type per } k;$$

$$R(1) = \tfrac{1}{2}\ \text{anchor [EW.32]};$$

$$j = |k|/2\ [\text{FT.52}(3)]$$

$$(3):\ \text{monomial basis } x^{k-r}y^r:\ H$$

$$= x\partial_x - y\partial_y,$$

$$E$$

$$= x\partial_y,$$

$$F$$

$$= y\partial_x:\ [H, E]$$

$$= 2E,$$

$$[H, F]$$

$$= -2F,$$

$$[E, F]$$

$$= H \text{ [machine, } k \leq 6]$$

$$\textstyle\sum_i (W_i^{(k)})^2 = \frac{1}{2}(EF + FE + \frac{1}{2}H^2) = \frac{k}{2}(\frac{k}{2} + 1)\,\mathbb{I} \text{ [machine exact over } \mathbb{Q}];$$

$$k = 0, 1, 2 : \ 0, \tfrac{3}{4}, 2 \text{ [CT0.7a eigenvalues recovered]}$$

$$(4) : \ \mathcal{U}$$

$$= \exp(i\phi \hat{n} \cdot \vec{\sigma}) : \text{ eigenvalues } e^{\pm i\phi} \text{ on } \mathcal{H}_{\mathsf{v}}$$

$$\implies e^{i(k-2r)\phi} \text{ on Sym}^k \text{ [math; machine over } \mathbb{Q}(i), \text{ diagonal } \mathcal{U}, \text{ exact circle points]}$$

$$(5) : \text{ CT2.5 defect read [EW.33 pattern] : } \|(U^{(k)} - 1)\psi_m\|^2$$

$$= |e^{im\phi} - 1|^2$$

$$= 2 - 2\cos(m\phi);$$

$$\phi$$

$$= \Theta_R/2 \text{ [Def-H.21(6)]}$$

$$k = 1 : \ m = \pm 1 : \ 2 - 2\cos(\Theta_R/2) = 4\sin^2(\Theta_R/4) \text{ [FT.54(4) recovered; both weights equal]};$$

$$m = 0 \ (k \text{ even}) : \ 2 - 2\cos 0 = 0 \text{ exactly}$$

$$2 - 2\cos(m\phi)$$

$$= 2 - 2T_m(\cos\phi) : \ T_0$$

$$= 1,$$

$$T_1$$

$$= c,$$

$$T_{m+1}$$

$$= 2cT_m - T_{m-1} \text{ [machine integer-polynomial recursion over } \mathbb{Q}]$$

$$(6) : \ (1)\text{'s CG at three factors on the cone;} \qquad d_{12} \geq 0 \text{ [FT.34(2) spacelike connector]}$$

$$(7a) : \ \kappa_0$$

$$= 2 \text{ tetrahedron [FT.42(5)]} : \ \text{digons } \Theta$$

$$= 2\kappa_0$$

$$= 4,$$

$$k$$

$$= \kappa_0$$

$$= 2 \text{ [FT.24(1), FT.25(3)]},$$

$$6 \text{ chains;}$$

$$\text{faces } \Theta_f$$

$$= 2L$$

$$= 6,$$

$$k_f$$

$$= 3,$$

$$4 \text{ faces};$$

$$\sum$$

$$= 12$$

$$= K_S \text{ [FT.43(1)]}$$

$$\dim: \ 3^6 = 729 \neq 256 = 4^4 \text{ [machine]}; \qquad j_{\max} = K_S/2 = 6 \text{ both bases [stretched; math]}$$

$$(7b): \ \kappa \equiv 1 \text{ on 3-cycle}: \ \Theta = 3 \notin 2\mathbb{Z} \text{ [FT.22]};$$

$$(2,2,2) = (1,1,0) + (0,1,1) + (1,0,1) \text{ [machine]}; \ (1,1,0)\text{-class null-carrying [FT.45(4)]}$$

$$(8): \ \dim \mathrm{Sym}^3$$

$$= 4;$$

$$\text{CT0.12 blocks } \{3,3,5,3,1,1\}: \ \neg\exists \ 4\text{-dim irreducible register block};$$

no aspect claim made. ■

Rmk (Def-H.38 and the no-go, consistent). FT.55(1) ruled out $\Theta_R = 2\pi k$ — spin as the frame-holonomy angle of the same loop. Def-H.38 makes no such identification: the winding enters as copy count of the transported seed, and the transport phases are EW.32-side data; under Def-H.37(2) the defect ladder (5) reads in Θ_R at fixed weight m with no k–Θ_R proportionality anywhere — the calibration weights stay separate ($[k] = \mathfrak{h}$-weight -2; $[\Theta_R] = 0$; $[\hat{\mu}'^2] = \mathfrak{q}$). The no-go constrains the form of the read; Def-H.38 supplies the multiplicity; they meet only in (5), consistently.

Rmk (relation to Def-H.37(1)). Def-H.37 fixes the defect transfer (pole side = transport defect of the seed); Def-H.38 fixes the defect carrier (which space the seed's copies occupy).

Phys. A winding-k fermion loop carries k symmetric copies of the qubit the corpus already transports — spin $k/2$, Casimir $\frac{k}{2}(\frac{k}{2}+1)$, a multiplet of $k+1$ weights, and, if the pole mass is the transport defect, a mass multiplet $4\hat{\mu}'^2 \sin^2(m\Theta_R/4)$ whose middle state is exactly massless

whenever k is even. The bottom three rungs are not added structure: they are the virtual line, the seed, and the real sector of the pair channel — objects Parts 1–2 already proved, wearing the labels CT0.7a already computed. What the declaration contributes is one number tied to another: copies $=$ winding. That the tie cannot be unpacked from anything already proven is itself proven — the triangle's minimal winding of three has no three subloops to count.

Theorem 3.52 (FT.61 — The selection side: phase inventory, index-blindness, and the two exits (Gap 18)). Thm. (1) Complete on-shell phase inventory. For a configuration C with on-shell integer edge data [FT.24/FT.34(4) scope], A8's weight assembles as

$$w(C) = \mathcal{J}(C) \prod_{e\in E(C)} G(e)\cdot(-1)^{\aleph(\sigma,C)},$$

$$\mathcal{J}(C) = 1 \text{ [D3.6 : } (+1)\text{-factors only]},$$

$$G(e) = \sigma_e\, e^{i\pi\kappa_e} \text{ [D3.5, FT.9 : } \alpha = \pi;\ \text{EM.20]},$$

so the full phase is $\Phi(C) = \pi\big(\Theta(C) + \aleph(\sigma, C) + n_-(\sigma)\big) \pmod{2\pi}$ with $n_- := \#\{e : \sigma_e = -1\}$: the path-sum phase has exactly three $\mathbb{Z}_2$ registers — geometric (Θ), wall ($\aleph$ [D3.3]), sign (n_-) [CT0.9, CT0.10: the sign structures are independent; machine: exhaustive on Σ_{min} configurations]. (2) Geometric triviality on-shell. On materialized closed chains $\Theta \in 2\mathbb{Z}$ [FT.22] $\implies$ $(-1)^{\Theta} = +1$: the surviving weight phase of any materialized-loop configuration is registration-only, $(-1)^{\aleph+n_-}$ — no variational continuous component remains in the on-shell phase. In particular the stationary channels of CT1.20's semiclassical locus carry constant κ_0 [FT.53(2)] and their complete geometric ledger record is the single integer κ_0 (plus δh and $\Theta = 2k$). (3) Index-blindness witness. There exist CT2.6-class geometries and pairs of closed geodesics (γ_A, γ_B) with identical ledger records and different Morse indices: witness $M = S^2(R)\times S^1(2R)$ with the product metric [math: smooth complete Riemannian 3-manifold — a spatial-slice class in Def-H.21's sense; whether emergent dynamics realizes this slice is P4/solution data, recorded not claimed]: $\gamma_A =$ a great circle traversed twice — length $4\pi R$, Morse index 3 [math: Morse index theorem; conjugate points at arc-distances $\pi R, 2\pi R, 3\pi R$ in the open interval; iterates of S^2 great circles: index $2m-1$]; $\gamma_B = \{\text{pt}\}\times S^1(2R)$ — length $2\pi\cdot 2R = 4\pi R$ (equal), Morse index 0 [flat factor; product formula; π_1-nontrivial so index 0 is admissible — Synge's obstruction applies only to simply-connected slices]. At ledger scale $\mathfrak{h} = 2\pi R/N$: both discretize to $\kappa \equiv 1$ minimal-edge chains with $\delta h = \Theta = 2N \in 2\mathbb{Z}$ [FT.31(4), FT.22: materialization-eligible], $\sigma \equiv +1 \implies \aleph = 0,\ n_- = 0$; by (1) both weights are $+1$, and $|w|^2 = 1$ for both [CT2.8's norms equally blind]. No function of the weight data separates index 0 from index ≥ 1 stationary channels. ([FT.66(2)]: both witness members are degenerate — they sit in clean isometry-orbit critical manifolds ($\mathcal{C}_A = \mathbb{RP}^2\times S^1\times S^1$, nullity 4; $\mathcal{C}_B = S^2\times S^1$, nullity 3); the Bott indices remain distinct (3 vs 0), the monodromy types differ (I_2 semisimple vs unipotent parabolic, Jordan-separated at identical trace/det), and the records and weights above are unchanged: this clause holds with "Morse index" read throughout as "Bott index" [Def-H.24].) (4) Two exits (theorem-shaped localization). A selection criterion implementing "stable jumps materialize" [Def-H.19 vocabulary] must read the Morse index [math: index is the fluctuation-sign datum]. By (1)–(3) no weight-level or ledger-level read can supply it, and the mask/pole side reads stationary channels only through hosted blocks and superselection labels $(\Theta_{\text{int}}, k, q_{\text{core}}, m_{\text{hid}}, \mathcal{C})$ [FT.26, Def-H.15(4), FT.58's inventory] — every entry index-blind by

(2). Hence Gap 18's selection side is closed against the current inventory, with exactly two exits: (a) a new variational registration — a per-channel conjugate-point/Jacobi count appended to the ledger (a interface-definition species extending D0.7's retained-event vocabulary; nothing of the kind exists [inventory]) — **(reclassification [FT.66(3)]: the datum is reconstructable in principle from the ambient configuration's metric content [CT1.20's $C \mapsto g_{\mu\nu}(C)$; CT1.12/CT1.14/CT2.6] and eventually constant in $\hbar$ away from bifurcation loci; the gap is a structural coupling — no A8/D2.11/D2.3/CT2.5 clause reads it — not a registration gap; exit (a) becomes a declared weight extension $w \mapsto w\,(-1)^{\mathrm{ind}_B}$, interface-definition species, not adopted); or (b)** a spectral-side argument — positivity structure on $Z(\mathcal{R}, \Sigma)$ [CT2.8/CT2.5/Def-H.16] constraining which stationary families contribute poles without per-channel signs (not constructed; CT2.8 supplies norms, not pole-selection). The earlier "bridge construction" residue decomposes accordingly: the leading bridge is derived — materialization $\iff$ pole $\iff$ single-valued eigenmode at constant-κ stationary channels [Def-H.16 closing line; FT.22; FT.53(2); CT2.5] — and the next order (fluctuation prefactors and signs) requires exactly (a) or (b). (5) Consistency with CT1.20. The witness pair sits inside CT1.20's own statement: stationary points dominate and are weighted alike — (3) upgrades "alike" from an absence of distinction to an identity of weights, at every ledger scale. Gap 13's finite-$\hbar$ residue [FT.53(4)] inherits the localization: the excluded class-(iii) profiles were killed by non-stationarity; index-≥ 1 stationary saddles are killed by nothing available.

Req. A8; D0.7, D1.7, D2.7, D3.3, D3.5, D3.6; CT0.9, CT0.10 (Part 1); CT1.20, CT2.5, CT2.6, CT2.8 (Part 2); EM.20 (E; via FT.9's $\alpha = \pi$); FT.9, FT.22, FT.24, FT.26, FT.31, FT.34, FT.48, FT.53, FT.58; Def-H.15, Def-H.16, Def-H.19, Def-H.21; P4; Morse index theorem, conjugate-point counts for sphere iterates, product formula, Synge [math].

Pf.

$$(1):\ \mathcal{J}(C) = \prod_{J^+}(+1)\prod_{J^-}(+1)^* = 1\ \text{[D3.6]};$$

$$G(e) = \sigma_e \exp(i\alpha_{\mathrm{EM}}\widehat{ds}^{\,2}(e)) = \sigma_e(-1)^{\kappa_e}\ \text{[D3.5;\ FT.9/EM.20 :}\ \alpha = \pi;\ \kappa_e \in \mathbb{Z}\ \text{on-shell]}$$

$$\prod_e G(e) = (-1)^{n_-}\cdot(-1)^{\Theta}\ \text{[D1.7 :}\ \Theta = \textstyle\sum_e \widehat{ds}^{\,2}(e)\text{]};$$

$$w\text{-phase} = (-1)^{\Theta+\aleph+n_-};$$

$$\text{independence of registers [CT0.10];\ machine :}\ 24/24\ \Sigma_{\min}\ \text{configurations}$$

$$(2):\ \Theta \in 2\mathbb{Z}\ \text{[FT.22]} \implies (-1)^{\Theta} = 1;$$

$$\text{stationary record :}\ \kappa_e \equiv \kappa_0\ \text{[FT.53(2)]};\quad \Theta = 2k\ \text{[FT.22]};\quad \delta h\ \text{[D0.6]}$$

$$(3):\ \ell(\gamma_A) = 2\cdot 2\pi R = 4\pi R = \ell(\gamma_B);$$

$$\text{index}(\gamma_A) = \#\{j\pi R :\ 1 \leq j \leq 2m-1,\ m = 2\} = 3\ \text{[Morse index thm.; machine: integer count]}$$

$$\text{index}(\gamma_B)$$

$$= 0\ [\text{flat factor},\ \text{product formula};\ \gamma_B \text{ generates } \pi_1 \implies \neg\text{contractible} \implies \neg\text{Synge-forced}];$$

$$\mathfrak{h} = 2\pi R/N:\ \delta h = \ell/\mathfrak{h} = 2N \text{ both};\ \kappa \equiv 1 \implies \Theta = \delta h\ [\text{FT.31}(4)]$$

$$\text{ledger records}:\ (\kappa_0, \delta h, \Theta, \sigma, \aleph) = (1, 2N, 2N, +1, 0) \text{ both};$$

$$w = (+1) \text{ both by } (1); \quad |w|^2 = 1 \text{ both [CT2.8(2) norm side]}; \qquad 3 \neq 0$$

$$(4):\ \text{index} = \text{sign of fluctuation determinant [math]};$$

$$(1)\text{–}(3):\ \neg\exists \text{ weight function distinguishing};$$

$$\text{mask reads}:\ \text{FT.26}(2) \text{ block-diagonal on } \mathcal{A}(\Sigma; C),$$

$$\text{labels Def-H.15}(4):\ \text{all from } (2)\text{'s record}$$

$$\text{exits}:\ (a)\ \text{D0.7 ledger} = \text{retained events};\ \neg\exists \text{ variational datum [inventory]};$$

$$(b)\ \text{CT2.8}:\ \langle A|A\rangle = \sum |w|^2 \geq 0:\ \text{norms};\ \neg\exists \text{ per-saddle phase/pole-selection tag}$$

$$(5):\ \text{CT1.20}:\ \delta\Phi = 0 \text{ dominates, weighted alike};$$

$$(3) \text{ exhibits the "alike" as identical weights};$$

$$\text{FT.53}(4):\ \text{class-(iii) excluded by non-stationarity},\ \neg\text{by index}.\ \blacksquare$$

Rmk (scope of the witness). The witness geometry is offered within the smooth-geometry class the ledger program works in [CT2.6's hypotheses are regularity clauses on (M, g); Def-H.21 operates on arbitrary orientable slices]. If emergent dynamics were later shown to realize only simply-connected or curvature-restricted slices, the witness would need rebuilding in that class — the structural half of the theorem (clauses (1)–(2): the on-shell phase has no variational component at all, and the stationary record is one integer) is witness-independent and would survive any such restriction.

Phys. On-shell, the ledger's phase bookkeeping is finished: geometry contributes nothing (its factor is $+1$ on every materialized loop), and what remains — wall crossings and registration signs

— is classical data fixed before the variation ever happens. Two loops can then be indistinguishable to the last bit of weight while one is the stablest path in the geometry and the other a thrice-unstable saddle: on a sphere-times-circle they are the same length, the same hop count, the same everything the ledger writes down. No reading of the sum can prefer the minimum; only new variational bookkeeping or a positivity argument from the spectrum side can — and both are named, neither is built.

3.22 Tier F.20 — The Bidirectional Rung Audit and the Platonic Physical Cards

Theorem 3.53 (FT.62 — The bidirectional rung audit (correction theorem)). Thm. (1) Convention re-derivation; floor-card rungs. The directed-rung law — $n(v) = \partial^{\leftarrow}(v) + \partial^{\rightarrow}(v)$ counts directed real edges [D0.8, A7]; the tick load is built from $\Phi^n \times \Phi^n$ sign configurations on the n incident real edges [CT0.5 Pf; CT0.9: one bit per directed real edge]; every bidirectional channel contributes both orientations — is CT0.11b (B). Floor-card rows [FT.28(5)'s explicit enumeration $n(p) = 3 = 2{+}1$; Def-H.20's bidirectionality]:

$$\text{single-directed-edge contact}: \; n(v_1) = 2d + 1; \qquad \text{bidirectional contact}: \; n(v_1) = 2d + 2;$$

$$\text{polygon } \Sigma_{P_n}: \; \text{dark } n(v) = 4 \; \forall v \; (\text{type } (2,2)), \; \text{contact } n(v_1) = 5;$$

$$\text{Platonic } \Sigma_S: \; \text{dark } n(v) = 2d \in \{6, 6, 8, 6, 10\}.$$

(2) Corrected lock table [$O(n) = \max(0, 2^{n+1} - 16)$: CT0.6]:

S	d	$O(2d)$ dark/vertex	$O(2d{+}1)$ contact	$\sum_v O$ dark	$\sum_v O$ obs (1 edge)
tetrahedron	3	112	240	448	576
cube	3	112	240	896	1024
octahedron	4	496	1008	2976	3488
dodecahedron	3	112	240	2240	2368
icosahedron	5	2032	4080	24384	26432

bidirectional-contact variants: contact $O(2d{+}2) \in \{496, 496, 2032, 496, 8176\}$, sums $\{832, 1280, 4512, 2624, 30528\}$ [machine]. (3) Saturation universality; the trichotomy retracted and replaced. Universality — bidirectional real-edge subgraphs of channel degree ≥ 2 saturate: D^--configured [CT4.1], content locked, retained, and counted [CT4.3, D4.2], inaccessible [D1.11] — the uniform minimal-saturation 2-regular family (exactly one channel quantum over budget, length-independent), the degree-graded ladder, and the exact lock-free characterization (the unidirectional family on bidirectional skeletons): CT4.1a (G) [with CT0.11b (Part 1)]. The retraction record — FT.42(2)'s "accessibility trichotomy by degree" and FT.45(2)'s "no lock at any n — the unlimited-accessibility family" are retracted, replaced by the frozen ladder above, with Σ_{min} ($n \in \{2, 3\}$) and Σ_L ($n \leq 3$) as its floor instances: $O = 0$ [FT.28(5), FT.32(2) — unchanged by

this audit]. Duality splitting survives and sharpens [FT.44(2) corrected]: cube 112 vs octahedron 496; dodecahedron 112 vs icosahedron 2032 — saturation grade splits both dual pairs. (4) Invariance theorem. E_{int}, I_{hid}, m_{hid}, Θ_{int}, K_S, the κ-lattices and their indices, the winding sum rule, $g(V)$, Π, S_{hid}, and every defect/holonomy entry are functions of edge-, channel-, and face-counts and of κ-profiles — none references $n(v)$: FT.42(1),(3),(4),(5) (lock columns excepted), FT.43, FT.44(1),(4), FT.45(1),(3),(4), FT.49–FT.51 stand unchanged [machine: value-by-value re-derivation]. The observed-variant information counts stand on retention: locked vertices retain and count [CT4.3; FT.17 cross-check 2 precedent]; the CT4.9(1) "visible < 3" arithmetic anchors in FT.42(1)'s Pf (and FT.17 cross-check 3's) are re-scoped: at corrected contact rungs $n \geq 7$ the contact vertex is locked, so its channel-visible content is governed by Def-H.35's locus-separation clause (per-vertex caps do not further reduce channel-visible content) and CT4.9(2)'s boundary form — not by CT4.9(1)'s accessible-vertex branch. Results unchanged; anchors corrected. (5) Type ledger and the EM row (scope note). Bidirectional-skeleton vertices with $d \geq 2$ channels have type (d, d) [D0.8/A7]; D3.3's literal f_v domain (type-$(1,1)$ vertices) is empty on dark polygon and Platonic floors (the digon Σ_{min} keeps type-$(1,1)$ vertices — Σ_{min}'s and Σ_L's EM rows are untouched by this audit): the EM row of FT.42(4)/FT.20 evaluates on observed variants through chain-local pass-through reads along transit chains (chain-local telescoping: EM.9b (Part 2); EM.9 end-reads); dark floors carry the registration registers (q_{core} per closed core [FT.40]) with no channel read at all [FT.3]. (6) Materialization sharpening. M1/M2 at Platonic/polygon vertices are now explicitly spectral hypotheses at locked vertices ($n = 2d \geq 4$): strictly stronger than the text suggested. No contradiction with EW.17 (whose rung-3 claim concerns wrapping chains at channel-admissible vertices — a different existence locus): face/digon materialization, if it occurs, occurs through the generic CT2.5/Def-H.16 mechanism.

Req. D0.8, D0.9; A7; CT0.5, CT0.6, CT0.9, CT0.10, CT0.11, CT0.11b (Part 1); CT4.1, CT4.2, CT4.3, CT4.9, CT4.1a (Part 2); EM.9b (Part 2); D1.11, D4.1, D4.2; FT.3, FT.17, FT.20, FT.21, FT.28, FT.32, FT.40, FT.42–FT.45, FT.49–FT.51; Def-H.20; EW.17 (Part 2); CT2.5 (Part 2); Def-H.16.

Pf.

$$(1):\ \text{general law}:\ \text{CT0.11b (B)};$$

$$\text{cells re-enumerated [machine}:\ \Sigma_{\text{min}},\ \Sigma_L,\ \Sigma_{12},\ \Sigma_{\triangle},\ \text{bowtie},\ \text{patch unchanged]}$$

$$(2):\ O(n)$$

$$= 2^{n+1} - 16\ (n \geq 4)\ [\text{CT0.6}];$$

$$\text{table arithmetic [machine}:\ O(6) = 112, O(7) = 240, O(8) = 496, O(9) = 1008, O(10) = 2032, O(11) = 4080, O(12) = 8176]$$

$$(3):\ \text{general ladder/universality}:\ \text{CT4.1a (G)};$$

$$\text{unidirectional cells}:\ \text{FT.28}(5):\ n \in \{2, 3\};\ \text{FT.32}(2):\ n \leq 3:\ O = 0\ [\text{CT4.1a}(4)];$$

$$\text{dual pairs}: \; O(6) = 112 \neq 496 = O(8); \quad 112 \neq 2032 = O(10)$$

$$(4): \text{ each quantity's defining formula inspected}: \; \text{FT.42}(1) \; (E_{\text{int}} = 2E),$$

$$(3) \; (\Theta_{\text{int}} = 2K_S),$$

$$(5) \; (\text{face regimes}: \; \kappa\text{-data}),$$

$$\text{FT.43 (incidence)},$$

$$\text{FT.41}(b) \; (g(V) = V\text{-grammar}),$$

$$\text{FT.49–FT.51}: \; \neg\exists \, n(v) \text{ reference};$$

$$\text{retention}: \; \text{CT4.3/D4.2}$$

$$\text{anchor re-scope}: \; \text{CT4.9}(1) \text{ reads accessible}(v);$$

$$n$$

$$\geq 7: \; \neg\text{accessible [CT0.11]} \implies \text{hypothesis check passes to Def-H.35 (interface clause)} + \text{CT4.9}(2) \text{ [boundary form]}$$

$$(5): \; \text{type}(d, d): \; \partial^{\leftarrow} = \partial^{\rightarrow} = d \text{ [D0.8]}; \quad \text{D3.3 domain};$$

$$\text{FT.21}(i) \text{ chain-local telescoping [machine in verify_H_ASigma record]};$$

$$(6): \; \text{Def-H.20 M1/M2 declared}; \; \text{EW.17}: \; \text{rung-3 wrapping locus.} \; \blacksquare$$

Rmk (integrity ledger). The fork entered with FT.42(2) ("dark $n(v) = d$") and FT.45(2) ("$n(v) = 2$"), was inherited by the then-current summary claims ("d=3 solids run lock-free in the dark"; "the unlimited-accessibility family"), and survived because the later cells (face floor, bowtie, patch) were computed with the correct directed convention — so no theorem built on FT.49/FT.50 is affected, and the two conventions never met in one computation until this pass. All affected clauses are patched in place with (rung correction) markers; the retracted epithets are recorded here and at their sites. This is the document's fourth recorded self-catch (after the mod-4 anomaly, the earlier σ_2/tuple-sum pair, and the display-balance latent defect) — and the first that changes a physical qualitative claim: saturation is universal on bidirectional skeletons.

Phys. Count the ledger's own bits and the picture inverts: a bidirectional channel is two real edges, each with its own registration sign, and a vertex hosting even two of them already exceeds

the channel budget. There are no clean-running bidirectional floors — polygons saturate at exactly one quantum over budget per vertex, solids by degree — and the old trichotomy collapses into a single law with a graded shadow. What survives untouched is everything the integers were made of: masses, windings, lattices, sealings, defects. The lock column was the only column that miscounted, and the correction makes the dark sector larger, not smaller.

3.23 Tier F.22 — The Bott Correction: Degenerate Witnesses Re-Evaluated, Reconstructability, and the Selection Dilemma

Definition 3.30 (Def-H.24 — Bott index, clean critical manifolds, and monodromy type for ledger channels (construction)). Def. Let γ be a stationary channel [Def-H.19: $\delta\int ds^2 = 0$; FT.48(1)/FT.53(2): geodesic, constant κ_0] in a CT2.6 slice (M, g) of the emergent geometry, closed (bound side) or fixed-endpoint (transit side). All items below are [math] constructions on derived data — g from sector coordinates [CT1.12, D2.1], Γ [CT1.13], $R^{\rho}{}_{\sigma\mu\nu}$ [CT1.14], sampling [CT1.19], coarse-graining [CT2.6 hypotheses (i)–(iii)] — no new physical input, in Def-H.21's construction class: (1) Critical manifold. $\mathcal{C} \ni \gamma$: the connected component of the stationary locus of the energy/ledger functional $\Theta \leftrightarrow \int ds^2$ [FT.31(2): the functionals coincide on-shell] in the coarse-grained loop space (closed case: free loops; transit case: fixed-endpoint paths). Isometry orbits of (M, g) through γ lie in $\mathcal{C}$; phase-shift and translation families likewise. (2) Normal Hessian and Bott index. $N_\gamma := T_\gamma\Lambda/T_\gamma\mathcal{C}$ the normal space; Q_γ the second variation (index form) restricted to periodic normal sections:

$$Q_\gamma(V,V) = \int_0^L \Big(|\nabla_{\dot\gamma} V|^2 - \big\langle R(V,\dot\gamma)\dot\gamma,\ V\big\rangle\Big)\, dt \qquad [R \text{ from CT1.14 data; math}],$$

$$\mathrm{ind}_B(\gamma) := \dim(\text{maximal negative-definite subspace of } Q_\gamma), \qquad \mathrm{nul}(\gamma) := \dim\ker Q_\gamma;$$

γ is clean iff $\ker Q_\gamma$ equals the orbit/tangential nullity (all kernel directions integrate into $\mathcal{C}$) — equivalently $\mathcal{C}$ is a nondegenerate critical submanifold [math: Morse–Bott]. $\mathcal{C}$-constant ind_B; the naive Morse data of FT.61/Def-H.19 are the $\mathrm{nul} = 0$ special case. (3) Equivariant refinement (recorded, not needed scalar-wise). Where (M, g) carries an isometry group $\mathcal{G}$ with $\mathcal{C} = \mathcal{G}\cdot\gamma$ (the witness case), the normal Hessian is $\mathcal{G}_\gamma$-equivariant and the nullity carries the isotropy representation [math: equivariant Morse–Bott theory]; the selection-relevant scalar of stationary-phase weighting over $\mathcal{C}$ is the sign $(-1)^{\mathrm{ind}_B(\gamma)}$ with amplitude $\int_{\mathcal{C}} |\det' Q|^{-1/2}$ [math: Bott's explicit formula] — the sign datum Gap 18 asked about. (4) Monodromy type. The linearized Poincaré return map P_γ on the symplectic normal space of the geodesic flow ($2(n-1)$-dim; 2-dim for surface slices): semisimple/elliptic ($|\operatorname{tr}| < 2$ or diagonalizable), parabolic-unipotent ($P \neq I, (P-I)^2 = 0$), hyperbolic ($|\operatorname{tr}| > 2$), or $P = I$ (fully degenerate). Iterates: $\mathrm{ind}_B(\gamma^m)$ is determined by P_γ's conjugacy class through Bott's iteration theory [math]; for $P = I$ on a surface the ladder follows from first principles by normal-mode counting — $\mathrm{ind}_B(\gamma^m) = 2m - 1$ (FT.66(2), machine), matching the classical spherical iterate indices [math: Bott's computation for great circles; cited as consistency, not re-derived here]. (5) Vocabulary fold. Def-H.19's "stable" := $\mathrm{ind}_B = 0$ (clean), "unstable" := $\mathrm{ind}_B \geq 1$; the interface content of Def-H.19 shrinks accordingly (the Morse-theoretic vocabulary is now [math]-constructed; only the non-coupling clause remains declared — itself narrowed in part by FT.66(4)'s derived anti-coupling for absolute rules on closed chains).

Theorem 3.54 (FT.66 — The Bott correction: witness re-evaluation, reconstructability, and the inadmissibility of absolute selection). Thm. (1) Framework. Under Def-H.24, stationary ledger channels carry the Bott data $(\mathrm{ind}_B, \mathrm{nul}, \mathcal{C}, [P_\gamma])$; for degenerate (isometry-orbit) families the stationary-phase weight is Bott's: sign $(-1)^{\mathrm{ind}_B}$, amplitude over $\mathcal{C}$ [Def-H.24(3); math]. The naive conjugate-point count of FT.61(3) is the fixed-endpoint special case; for closed channels the periodic count and the orbit quotient must replace it — executed in (2). (2) The witness, re-evaluated (degeneracy critique absorbed; blindness Bott-stable). On $M = S^2(R) \times S^1(2R)$ [FT.61(3)'s geometry]:

$$\gamma_A \ (m{=}2 \text{ great circle}): \quad \mathcal{C}_A = \mathbb{RP}^2 \times S^1_{\text{phase}} \times S^1_{\text{trans}} \ (\dim 4),$$

$$\mathrm{ind}_B = 3 \ (= 2m{-}1),$$

$$\mathrm{nul} = 4 \ (\text{clean}),$$

$$P_{\gamma_A} = I_2;$$

$$\gamma_B \ (\text{fiber circle}): \quad \mathcal{C}_B = S^2 \times S^1_{\text{phase}} \ (\dim 3),$$

$$\mathrm{ind}_B = 0 \ (\forall \text{ iterates}),$$

$$\mathrm{nul} = 3 \ (\text{clean}),$$

$$P_{\gamma_B} = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix} \ (\text{unipotent, parabolic});$$

derivations (machine, exact): γ_A's normal modes on the rank-1 normal bundle: Q-eigenvalues $\lambda_n \propto n^2 - m^2$ on period-$L = 2\pi R m$ Fourier modes — negatives $\{|n| < m\}$: $2m-1$ of them; zeros $\{|n| = m\}$: the 2-dim $\mathbb{RP}^2$ family motion; plus phase and translation tangents: $\mathrm{nul} = 4 = \dim \mathcal{C}_A$ — independent cross-check: Bott's iteration theorem at $P = I$ [math]. γ_B's normal bundle is the trivial TS^2-pair with vanishing mixed curvature [product; math]: $Q = \int |J'|^2 \succeq 0$, kernel = the 2 constant basepoint fields: $\mathrm{ind}_B = 0$, $\mathrm{nul} = 3 = \dim \mathcal{C}_B$; monodromy from $J = a + bt$: unipotent, iterates $\left(\begin{smallmatrix} 1 & mL \\ 0 & 1 \end{smallmatrix}\right)$, Jordan type separating it from $P_{\gamma_A} = I_2$ at identical trace/det $(2, 1)$ [machine]. Verdicts: (a) the Bott indices still differ, 3 vs 0, and the semiclassical signs would differ, $(-1)^3 = -1$ vs $(-1)^0 = +1$; (b) the ledger records and weights remain identical at every scale $\mathfrak{h} = 2\pi R/N$: $(\kappa_0, \delta h, \Theta, n_-, \aleph) = (1, 2N, 2N, 0, 0)$, $w = +1$ both [FT.61(1),(3) re-run]; (c) hence FT.61(3)–(4) hold with "Morse index" read as "Bott index" — index-blindness of the weight inventory is Bott-stable, and the monodromy type (a conjugacy-class datum strictly finer than the index) is equally unregistered: no factor of $w(C)$ [A8; FT.61(1)'s complete inventory] can carry $(-1)^{\mathrm{ind}_B}$ or $[P_\gamma]$. (d) Homotopy-class note: γ_A is contractible, γ_B is not — the class is combinatorially present in the configuration [D0.10: $E(C)$ ordered], but (i) no validity/weight/mask clause reads it [A8, D2.11, D2.3/CT2.4: inventory], and (ii) it is not the index: within a fixed free homotopy class, stationary members of different ind_B exist [math: length-tuning by intermediate value

+ neck-deformation focusing], so no class read implements index selection. (3) Reconstructability, and the true shape of exit (a). The ambient configuration determines the metric: $C \mapsto g_{\mu\nu}(C)$ [CT1.20's own grouping map; CT1.12: g from per-edge sector coordinates (a_{00}, q_{ij}, w_i) [D2.1, D1.6, Def-H.14's arena]], sampled at $O(\mathfrak{h})$ [CT1.19], coarse-grained to (M, g) under CT2.6's hypotheses; hence $R^{\rho}{}_{\sigma\mu\nu}$ [CT1.14] along and near the channel, the normal Jacobi operator, and $(\mathrm{ind}_B, \mathrm{nul}, [P_\gamma])$ are in-principle functionals of the ambient configuration in the CT2.6 regime — the datum is reconstructable, not missing. Integer-valuedness plus $O(\mathfrak{h})$ convergence gives eventual $\mathfrak{h}$-stability away from clean-bifurcation loci [FT.38(2)'s rigidity pattern; machine: minimal mode gap $|n^2 - m^2| \geq 1$ in scaled units — perturbations below half the gap cannot flip signs]; at the loci themselves the index jumps — those are precisely the finite-$\mathfrak{h}$ residue /18. What is missing is a structural coupling: no clause of A8's weight [FT.61(1)], D2.11's validity, D2.3's masks, or CT2.5's pole condition reads the datum; and the resolvent side is sign-free by hermiticity — poles are conditions $\lambda_a = 1$ on a real spectrum with residues $-\langle \mathrm{out}|P_a|\mathrm{in}\rangle$ [FT.26(1), CT2.5]: no per-saddle sign can live there without extending the structure. FT.61(4)'s exit (a) is accordingly reclassified: from "new variational registration (interface-definition species extending D0.7)" to "a declared weight extension $w \mapsto w \cdot (-1)^{\mathrm{ind}_B}$ (interface-definition species; well-definedness in the continuum regime supplied by the eventual-constancy lemma above) — constructible, not adopted here"; exit (b) (spectral-side positivity) unchanged, not constructed. (4) The selection dilemma (absolute index-0 rules are inadmissible for closed chains). Let a floor's ambient slice realization be a topological sphere carrying $K \geq 0$ with $K > 0$ somewhere — the class instantiated by every Platonic cone-sphere realization: faces flat, conical defects $\delta_v > 0$ at vertices, total $V\delta_v = 4\pi > 0$ [FT.63(4); smoothed approximation; math]. On any such slice:

$$\text{every non-constant closed geodesic has } \mathrm{ind}_B \geq 1$$

[math: Synge/shortening — every closed geodesic on S^2 bounds a disk and is exceeded by the disk-sweep; machine: the latitude-push second variation $L''(0) = -2\pi/R < 0$ exactly, and the round-sphere ladder $2m-1 \geq 1\ \forall m$].

Hence an absolute selection rule — "index-0 stationary channels materialize; $\mathrm{ind}_B \geq 1$ channels are suppressed" — admits no closed stationary channel on such a slice. Dilemma against M1/M2 (declared face/digon materialization [Def-H.20]): horn 1 — if the declared materializations contribute semiclassically [FT.53(2)'s regime: stationary = geodesic support], the absolute rule suppresses them all, contradicting M1 — the rule is inadmissible; horn 2 — if they do not (Platonic face loops are corner loops of the cone metric, ¬geodesic: non-stationary in the smooth read), then closed-loop materialization lives entirely in the finite-$\mathfrak{h}$ non-stationary residue [FT.53(4)], where no stationary-index criterion applies at all — stability selection is vacuous for bound chains. On either horn, "stable jumps materialize" is refuted as the mechanism for closed (bound) chains in this curvature class. (5) Verdict. The four sub-questions now carry derived answers: variational side — closed [FT.48(1)–(3)]; weight/ledger blindness — closed [FT.61(1)–(3)], Bott-stable [(2) here]; reconstructability — settled [(3) here]: the datum exists in principle, the coupling does not, and the resolvent side is sign-free; selection for closed chains — answered negatively [(4)]: no index criterion can operate, and the absolute rule is inadmissible against M1. This residue is the chain's honest end: closing it further requires a construction or declaration outside the frozen inventory — the same species boundary as Def-H.37(1)/Def-H.37(2)/Def-H.38. (6) Folds. Def-H.19's vocabulary reads as Morse–Bott [Def-H.24(5)] — its [math] content constructed; the non-coupling clause is narrowed in part by clause (4)'s derived anti-coupling for absolute closed-chain rules; FT.48(3)'s trichotomy is scoped to clean fixed-endpoint problems (conjugate endpoints degenerate: read via Def-H.24); FT.61(3)–(4) hold per (2)–(3).

Req. Def-H.19, Def-H.21, Def-H.24; A8; D0.10, D1.6, D2.1, D2.3, D2.11; CT1.12, CT1.13, CT1.14, CT1.19, CT1.20, CT2.3, CT2.4, CT2.5, CT2.6, CT2.8 (Part 2); FT.22, FT.26, FT.31, FT.33, FT.38, FT.48, FT.53, FT.61, FT.62, FT.63; Def-H.14, Def-H.16, Def-H.20; M1/M2 (interface clause); Def-H.19; Morse–Bott theory, Bott's iteration theorem for spherical geodesics, Synge's theorem / disk-sweep shortening, Jordan-form conjugacy invariants, Gauss–Bonnet for cone metrics, intermediate-value length tuning [math].

Pf.

$$(1):\ \text{Def-H.24}(2)\text{–}(3);$$

$$\text{stationary-phase over a clean critical manifold}:\ \text{sign}$$

$$= (-1)^{\mathrm{ind}_B},$$

$$\text{amplitude}$$

$$= \textstyle\int_{\mathcal{C}} |\det' Q|^{-1/2}\ \text{[math: Bott]}$$

$$(2)\ \gamma_A:\ Q \text{ on periodic normal sections}:\ V$$

$$= e^{2\pi i n t/L} e_2(t):\ Q(V)$$

$$= (\tfrac{n^2}{m^2} - 1)\tfrac{1}{R^2}\|V\|^2\ \text{[Jacobi } f'' + R^{-2} f = 0 \text{ on the sphere; math]} \implies \mathrm{ind}_B$$

$$= \#\{|n| < m\}$$

$$= 2m - 1,$$

$$\ker \quad = \{|n| = m\} : \dim 2$$

$$\mathrm{nul}(\gamma_A) = 2 + 1 + 1 = 4 = \dim(\mathbb{RP}^2 \times S^1 \times S^1)\ \text{[orbit tangents; machine]};$$

$$\text{consistency}:\ \text{periodic count}$$

$$= \text{fixed-endpoint conjugate count}$$

$$= \#\{j\pi R:\ 1 \le j \le 2m - 1\}$$

$$= 2m - 1\ \text{[machine]};$$

$$\text{ladder } 1,$$

$$3,$$

$$5,$$

$$\dots \text{ matches the classical spherical iterates [math]}$$

$$\gamma_B:\ R(\cdot,\dot\gamma_B)\dot\gamma_B|_{TS^2}=0\ [\text{product curvature; math}]\implies J=a+bt;\ \text{periodic}\implies b=0;$$

$$Q=\int|J'|^2\succeq 0\implies \text{ind}_B=0;\quad \text{nul}=2+1=3=\dim(S^2\times S^1)$$

$$P_{\gamma_A}=I_2:\ J(2\pi R)=J(0)\cos 2\pi+RJ'(0)\sin 2\pi=J(0)\ [\cos 2\pi=1,\sin 2\pi=0\ \text{exact}];$$

$$P_{\gamma_B}^m=\left(\begin{smallmatrix}1 & mL\\ 0 & 1\end{smallmatrix}\right)\ [\text{machine: integer powers}]$$

$$\text{Jordan separation}:\ P_{\gamma_A}-I$$

$$=0\ \wedge\ P_{\gamma_B}-I$$

$$\neq 0\ \wedge\ (P_{\gamma_B}-I)^2$$

$$=0:\ \neg\text{conjugate despite tr}$$

$$=2,$$

$$\det$$

$$=1\ \text{both [machine]}$$

$$\text{blindness}:\ \text{records}\ (1,2N,2N,0,0)\ \text{equal},\ w=(+1)\ \text{equal [FT.61(1),(3); machine re-run]};$$

$$(-1)^3\neq(-1)^0;\quad \neg\exists\ \text{factor of}\ w(C)\ \text{carrying it [FT.61(1) inventory complete]}$$

$$(2d):\ \pi_1(S^2 \times S^1) = \mathbb{Z}:\ \gamma_A \mapsto 0,\ \gamma_B \mapsto 1\ [\text{math}];$$

$$\text{class unread}:\ \text{A8}\ w(C)\ \text{factors},\ \text{D2.11 clauses},\ \text{D2.3 blocks}:\ \neg\exists\ \text{homotopy entry}\ [\text{inventory}]$$

$$\text{class} \neq \text{index}:\ \text{fixed class},$$

$$\text{neck-bump family}:\ L_{\text{over}}(t)\ \text{continuous},$$

$$\text{crosses}\ L_{\text{flat}}\ [\text{IVT}];$$

$$\text{focusing gives}\ \text{ind}_B$$

$$\geq 1\ \text{on the over-route}\ [\text{math: comparison}]$$

$$(3):\ C$$

$$\mapsto g_{\mu\nu}(C)\ [\text{CT1.20 Pf step 1};\ \text{CT1.12/D2.1}:\ \text{sector coordinates};\ \text{Def-H.14}:\ \text{edge arena}];$$

$$\Gamma\ [\text{CT1.13}],$$

$$R\ [\text{CT1.14}]$$

$$\implies Q_\gamma,$$

$$P_\gamma\ \text{determined in the CT2.6 regime}$$

$$\text{stability}:\ \lambda\text{-gaps} \geq 1\ \text{in scaled units}\ [\text{machine}:\ \min|n^2 - m^2| = 1];$$

$$|\Delta\lambda| = O(\hbar)\ [\text{CT1.19}] \implies \exists\hbar_0:\ \text{signs fixed}\ \forall\hbar < \hbar_0\ [\text{FT.38(2) pattern}]$$

$$\text{no reader}:\ \text{A8}\ [\text{FT.61(1)}];$$

$$\text{D2.11}:\ \text{closedness/Janus/mask};$$

$$\text{CT2.5/FT.26(1)}:\ \text{Hermitian} \implies \lambda_a \in \mathbb{R},$$

pole

$= \{\lambda_a = 1\}$: sign-free

(4) : cone sphere : K

$= 0$ on faces,

$\delta_v > 0$ at vertices,

$\sum \delta_v$

$= 4\pi$ [FT.63(4); machine];

smoothing : K

≥ 0,

$K > 0$ somewhere [math]

Synge : closed geodesic γ bounds a disk D : disk-sweep shortening $\implies$ ind$_B \geq 1$ [math];

machine instances : $L''(0) = -2\pi/R < 0; \quad 2m - 1 \geq 1 \; \forall m \geq 1$

horn 1 : absolute rule $\implies$ $\neg\exists$ admissible closed stationary channel $\implies$ M1's semiclassical reading empty : rule inadmissible;

horn 2 : face loops

$=$ corner loops [interior angles $< \pi$ at cone vertices] : $\neg$geodesic $\implies$ finite-$\mathfrak{h}$ residue [FT.53(4)]

$(4a)$: $\pi_1 = 0 \implies$ single free class : class-min $=$ global min $=$ constant loops (degenerate) [math];

$(4b)$: open minimizers exist [FT.48(3) : index-0 $\iff$ local ℓ-minimizer, clean case]

(5) : conjunction of FT.48, FT.61, (2)–(4); (6) : registry/patch moves as listed. ■

Rmk (what would reverse the verdicts). Clause (2)'s blindness is an inventory theorem: a derived cross-vertex or cross-channel bridge registering curvature along channels would reopen it — none exists [inventory; FT.59's absorption picture at locked rungs points the other way]. Clause (4) is conditional on (i) M1/M2 (interface clause) and (ii) the Synge-class slice realization of the floors — the class their own defect cards instantiate [FT.63(4)]; a flat or negatively curved ambient realization of a Platonic floor would evade the dilemma's horn structure (index-0 closed geodesics exist there [flat tori; math]) — but then the floor's own cone-sphere ledger ($\sum \delta_v = 4\pi$) would not be the ambient curvature read, and the defect card's Θ_R values would belong to a different geometric object than the channels' Jacobi data: recorded as a coherence condition any such alternative must address, not dismissed.

Rmk (the adversarial exchange, recorded). The critique — degenerate witnesses invalidate naive-index statements — is correct as to framework and void as to conclusion: the witness pair is clean Morse–Bott, the indices survive the correction unchanged (3 vs 0), and every downstream claim of FT.61 holds with the refined index. The correction nevertheless produced three results the naive framing could not reach: the reconstructability lemma (exit (a) reclassified), the Jordan-type refinement (monodromy strictly finer than index, equally unregistered), and the Synge dilemma (absolute selection inadmissible, not merely unavailable) — the critique is folded in as FT.66, per the program's rigor-both-ways practice.

Phys. Two loops can wind through the geometry side by side, one thrice-unstable and one the very floor of its class, and the ledger writes down the same four symbols for both — not because the ledger is coarse, but because stability is a fact about the space around the loop, and the loop's own record carries only its length quanta and its signs. Worse for the hopeful selector: on any sphere-like slice with the Platonic floors' own positive curvature budget, no closed geodesic is stable at all — every one of them can be shortened somewhere — so a rule that materialized only stable closed loops would materialize nothing, contradicting the very floors the program computes. Stability, then, cannot be the gate for bound chains; it remains a live candidate only for open jumps between termini, where short arcs genuinely minimize — and there the gate, if any, is either a declared sign carried by the weight or a positivity fact on the spectrum side. Both are named; neither is built; the boundary is exactly the species boundary of this document's other interface transfers.

3.24 Tier F.23 — The Virtual Bypass: Path Integration over Crystallization Patterns

Definition 3.31 (Def-H.25 — Skeleton configuration sums and the bypass vocabulary). Def. A skeleton $\Gamma :=$ a finite bidirectional potential-edge set — the 1-skeletons of Def-H.20 (Platonic floors) and FT.45 (polygon floors), or any finite subgraph thereof. A crystallization pattern is a subset $R \subseteq E_{\text{dir}}(\Gamma)$ [A4 per directed pair; D0.1: $\mathcal{R}$ membership is per-pair]; at v: $k_v := \#\{e \in R : e \ni v\} = \partial_R^{\leftarrow}(v) + \partial_R^{\rightarrow}(v)$ [D0.8], potential degree $n(v) = |\{e \in E_{\text{dir}}(\Gamma) : e \ni v\}|$. Classification [CT0.11, CT4.1]: v channel-admissible $\iff k_v \leq 3$; v D^--locked $\iff k_v \geq 4$. A pattern is a bypass configuration iff every vertex is admissible while some has $n(v) \geq 4$; the bypass deficit at v is $n(v) - k_v$ (the virtual edges at v). The skeleton configuration sum (A8's weight restricted to skeleton-supported configurations [A8, D0.10], at CT1.20's grouping layer):

$$Z_\Gamma := \sum_{R\subseteq E_{\mathrm{dir}}(\Gamma)} \sum_{\sigma\in\{\pm1\}^R} w(R,\sigma),$$

$$w(R,\sigma) = \mathcal{J}\cdot\prod_{e\in R} G(e)\cdot(-1)^{\aleph(\sigma,R)},$$

$$G(e) = \sigma_e\, e^{i\pi\kappa_e}\ [\text{D3.5, FT.9}],$$

split $Z_\Gamma = Z_{\mathrm{byp}}(\Gamma) + Z_{\mathrm{lock}}(\Gamma)$ by lock status (any $k_v \geq 4$). Scope declarations, marked at every use: (S1) minimal-edge class $\kappa_e \equiv 1$ on real edges [D1.1, Comm-min]; for general on-shell integer κ: every theorem below is unchanged with values scaled by $(-1)^{\sum_e \kappa_e}$ — the cover classification is κ-independent. (S2) $\aleph$ in D3.3's literal type-$(1,1)$ domain; $\aleph_v = 0$ at all other types; the chain-local extension [FT.21(i), Open Gaps 16(d)] would change the survivor class — flagged load-bearing (Rmk of FT.67). (S3) seal/Janus bookkeeping suppressed: the sum is over static patterns; $\mathcal{J} = 1$ identically [D3.6]; universe-level balance $\#J^+ = \#J^-$ [D1.3, D2.11] is ambient data, not a skeleton constraint. (S4) virtual edges contribute weight 1 and no phase — no registration exists for them [A4, D0.5/A7: ϕ_e is defined on $e \in E_{\mathcal{R}}$]. Finite skeleton ⇒ finite sum: convergence is trivial; growth-rate statements are made per family (FT.68(7)).

Theorem 3.55 (FT.67 — The exact-cover theorem and the single-vertex bypass). Thm. Under Def-H.25 with (S1)–(S4): (1) The $\aleph$ identity. At a type-$(1,1)$ vertex v with in-edge $e_\leftarrow$, out-edge $e_\rightarrow$: $(-1)^{\aleph_v} = (-1)^{|f_v|} = \sigma_{e_\leftarrow}\sigma_{e_\rightarrow}$ [D3.3: $f_v = \frac{1}{2}(\sigma_\rightarrow - \sigma_\leftarrow) \in \{-1,0,1\}$; σ's binary [CT0.9]; machine]. (2) Exact-cover theorem. For $e \in R$ let $\mathrm{cov}(e) := \#\{(1,1)\text{-type endpoints of } e \text{ in } R\} \in \{0,1,2\}$. Then the weight is the monomial

$$w(R,\sigma) = (-1)^{\sum_e \kappa_e} \prod_{e\in R:\ \mathrm{cov}(e)\ \text{even}} \sigma_e\ ,$$

[display correction: the σ_e-exponent is $1+\mathrm{cov}(e)$, so the surviving factors are the even-coverage edges — as this clause's own Pf records; classification and values unchanged]

and the sign sum evaluates exactly:

$$\sum_{\sigma\in\{\pm1\}^R} w(R,\sigma)$$

$$= \begin{cases} (-1)^{\sum_e \kappa_e}\, 2^{|R|} = 4^{\,|\mathcal{C}|}\,(-1)^{\sum \kappa} & \text{if } \mathrm{cov}(e) = 1\ \forall e \in R \quad (\text{pass-through cover; } \mathcal{C} := \{(1,1)\text{-vertices}\}), \\ 0 & \text{otherwise,} \end{cases}$$

with $|R| = 2|\mathcal{C}|$ forced on covers [each center contributes its in- and out-edge; every edge once]. At (S1): value $= 4^{|\mathcal{C}|} > 0$. [Machine: exhaustive on the tetrahedron (all 4096 patterns = all 3^{12} (pattern,sign) pairs), on P_3, P_4, P_5 exhaustively, on the cube exhaustively for $K \leq 4$ (12831 configs) plus a deterministic stride sample.] (3) Structural corollaries (what cancels, what survives). (a) Every directed cycle cancels: along a cycle of $(1,1)$-vertices each edge has cov $= 2$ — no bound cell survives at the amplitude layer: digons (the Σ_{min} pattern [FT.28]), directed cycles (the Σ_L pattern [FT.32]), face boundaries, and the fully crystallized Def-H.20 floors (all vertices type (d,d), $d \geq 2$:

no centers at all) all have $\sum_\sigma w = 0$; (b) single real edges and length-1 threads cancel (cov $= 0$); (c) threads of length ≥ 3 cancel (interior edges double-covered); (d) survivors = V-thread networks: each center $v \in \mathcal{C}$ pairs one in-edge $(s \to v)$ with one out-edge $(v \to t)$; non-centers are sources $(0,1)$, sinks $(1,0)$, or hubs of any type $\neq (1,1)$ — including locked hubs ($k \geq 4$) whose every edge is paired by a distinct center; every survivor has even $|R| = 2|\mathcal{C}|$ and value $4^{|\mathcal{C}|}$. (4) Single-vertex enumeration ($n \in \{4,5,6\}$, balanced types). 2^n patterns, $\binom{n}{k}$ at size k; admissibility fractions

$$f_{\text{adm}}(4) = \tfrac{15}{16}, \quad f_{\text{adm}}(5) = \tfrac{13}{16}, \quad f_{\text{adm}}(6) = \tfrac{21}{32} \qquad \Big[= \sum_{k \leq 3} \binom{n}{k} / 2^n \Big];$$

sign-summed amplitudes: $Z^{\pm}_{\text{byp}}(v,n) = 1 + 4\, n_{\text{in}} n_{\text{out}} = (17, 25, 37)$ — vacuum + the $\binom{n_{\text{in}}}{1}\binom{n_{\text{out}}}{1}$ pass-through patterns at 4 each; only $k \in \{0,2\}$ survive; $Z^{\pm}_{\text{lock}}(v,n) = 0$ (locked $\Rightarrow$ type $\neq (1,1)$ $\Rightarrow$ uncovered). At $\sigma \equiv +1$: $Z_{\text{byp}} = -\binom{n-1}{3} \in \{-1, -4, -10\}$, $Z_{\text{lock}} = +\binom{n-1}{3}$, total $(1-1)^n = 0$ [alternating partial-binomial identity; binomial theorem; machine]. Aspect activation per pattern [CT0.13]: $\mathcal{A}(0) = \{J\}$, $\mathcal{A}(1) = \{J, G, W, H, D^+\}$, $\mathcal{A}(2) = \mathcal{A}(1) \cup \{\text{EM}\}$, $\mathcal{A}(3) \cup \{\text{strong}\}$; mask dimensions $\dim \mathcal{K}_k = (6, 16, 17, 25)$ [CT0.14]; observable strong never fires at a single vertex [FT.27]; EM is state-dependent: $f_v \neq 0$ on exactly half of the pass-through sign assignments [D3.3; machine]. (5) Virtual bypass theorem (single-vertex form). Every vertex of potential degree $n \geq 4$ with $n_{\text{in}}, n_{\text{out}} \geq 1$ has strictly positive admissible-sector amplitude, $Z^{\pm}_{\text{byp}} = 1 + 4 n_{\text{in}} n_{\text{out}} > 0$, independent of the κ-class up to the global sign $(-1)^{\sum \kappa}$; its locked sector vanishes identically. [scope: the locked-sector clause is a literal-domain (S2) theorem — in the chain-local domains E1/E2 of Def-H.27 an isolated locked vertex has strictly positive amplitude, $Z^{\pm}_{\text{lock}} = (32, 96, 672)$ for $n = (4,5,6)$, with the locked digon-hub family at $(!a)4^a$ [FT.75(1)]. The bypass clause is domain-invariant.] D^- locking is avoided with nonzero amplitude by partial crystallization — the bypass mechanism is real at the amplitude layer, and its surviving content is exactly the pass-through (transit) sector [D1.5: open, propagating = bosonic content]. (6) Consistency anchors and the norm layer. (a) The k=2 survivor is the Σ_{min} pass-through structure verbatim [FT.28(5): q of type $(1,1)$]; (b) FT.61(1)'s phase inventory $e^{i\Phi} = (-1)^{\Theta + \aleph + n_-}$ is reproduced termwise inside the monomial of (2) [machine]; (c) no conflict with positivity: $|w(R,\sigma)| = 1$ for every pair, so CT2.8's norms $\langle A|A\rangle = \sum |w|^2 > 0$ never cancel — the cancellations of (2)–(3) are amplitude-layer (Z) facts; the spectral layer [Def-H.16: poles of the masked resolvent] and the materialization declarations M1/M2 are untouched (the layer bridge is item 13's residue — see FT.68(4)).

Req. A4, A7, A8; D0.1, D0.5, D0.8, D0.10, D1.1, D1.2, D1.3, D1.5, D2.11, D3.3, D3.5, D3.6; CT0.9, CT0.10, CT0.11, CT0.13, CT0.14 (Part 1); CT1.20, CT2.8 (Part 2); CT4.1, CT4.4 (Part 2); FT.9, FT.22, FT.24, FT.27, FT.28, FT.32, FT.61, FT.62; Def-H.16, Def-H.18, Def-H.20, Def-H.25; alternating partial-binomial sums [math].

Pf.

$$(1): \ \sigma \in \{\pm 1\} \text{ [CT0.9]}: \ f_v$$

$$= \tfrac{1}{2}(\sigma_\to - \sigma_\leftarrow): \ |f_v|$$

$$= 1 \iff \sigma_\to \neq \sigma_\leftarrow \iff \sigma_\leftarrow \sigma_\to$$

$$= -1$$

$$= (-1)^{|f_v|};$$

$$\sigma_{\leftarrow}$$

$$= \sigma_{\rightarrow} : \ |f_v|$$

$$= 0,$$

$$\sigma_{\leftarrow}\sigma_{\rightarrow} \qquad = +1 \ \checkmark$$

$$(2) : \ w$$

$$= (-1)^{\sum \kappa} \prod_{e \in R} \sigma_e \cdot \prod_{v \in \mathcal{C}} (\sigma_{e_{\leftarrow}(v)} \sigma_{e_{\rightarrow}(v)}) \ [(1); \ \text{D3.6} : \ \mathcal{J} = 1] : \ \sigma_e\text{-exponent}$$

$$= 1 + \text{cov}(e)$$

$\text{cov}(e) = 0$ or 2 for some e : flip σ_e : summand $\mapsto -$summand, involution on $\{\pm 1\}^R \implies$ sum $= 0$;

$$\text{cov} \equiv 1 : \ \text{every } \sigma_e \text{ appears squared} \implies w = (-1)^{\sum \kappa} \ \sigma\text{-constant}$$

$$\implies \sum_{\sigma} w = (-1)^{\sum \kappa} 2^{|R|};$$

$$|R|$$

$= 2|\mathcal{C}|$: centers pair disjoint edges, covers all [each center reads exactly its 2 edges; each edge read once];

$$(-1)^{2|\mathcal{C}|} 2^{2|\mathcal{C}|} = 4^{|\mathcal{C}|} \text{ at } \kappa\text{-even sums}$$

κ-invariance : σ_e enters linearly $\forall \kappa_e \in \mathbb{Z}$: classification unaffected;

value scaled by $(-1)^{\sum \kappa}$ [S1 scope]

$(3a)$: cycle : every vertex type $(1, 1)$ or higher along a closed walk;

cycle edges read by both endpoints when $(1, 1)$ at both $\implies$ cov

$$= 2;$$

any non-$(1,1)$ vertex on the cycle leaves its edges with cov

$$\neq 1$$ or forces type violation [machine-exhaustive];

Def-H.20 floors : types (d, d),

$$d$$

$$\geq 2 : \mathcal{C}$$

$$= \varnothing,$$

$$R \neq \varnothing \qquad \Longrightarrow 0$$

$(3b, c)$: single edge : endpoints $(0,1), (1,0)$: cov $= 0$;

length- ≥ 3 thread : interior vertices $(1,1)$ read the shared interior edges twice $\Longrightarrow$ cov $= 2$

$(3d)$: cover $\Longleftrightarrow$ the pairing structure described;

hubs : non-centers with all edges center-covered,

any type

$$\neq (1,1),$$

locked allowed [machine witnesses: tetrahedron $K_{1,3}$ stars, cube locked coverings]

(4) : enumeration [machine exhaustive];

$$Z^{\pm}_{\text{byp}}$$

$$= 1\ (\text{vacuum}) + \binom{n_{\text{in}}}{1}\binom{n_{\text{out}}}{1}\cdot 4\ [\text{pass-throughs; single-vertex model: no accidental}(1,1)\ \text{non-centers}];$$

$$k \geq 4:\ \text{type} \neq (1,1):\ 0\ [(2)]$$

$$\sigma \equiv +1:\ w = (-1)^k\ [\aleph = 0]:\ Z_{\text{byp}} = \textstyle\sum_{k\leq 3}\binom{n}{k}(-1)^k = (-1)^3\binom{n-1}{3}\ [\text{math}];$$

$$Z_{\text{lock}} = \textstyle\sum_{k\geq 4}\binom{n}{k}(-1)^k = +\binom{n-1}{3};\quad \text{total} = (1-1)^n = 0$$

$$\text{aspects}:\ \text{CT0.13 rows verbatim at } k;\ \dim \mathcal{K}_k\ [\text{CT0.14}];\ \text{strong } \neg\text{observable at } \{v\}\ [\text{FT.27}];$$

$$f_v \neq 0 \iff \sigma_{\leftarrow} \neq \sigma_{\rightarrow}:\ 2/4\ \text{of pass-through signs [D3.3]}$$

$$(5):\ 1 + 4n_{\text{in}}n_{\text{out}} > 0\ \forall n_{\text{in}}, n_{\text{out}} \geq 1;$$

$$(6):\ \text{anchors as cited};\ |w| = 1:\ \text{CT2.8 norms cancellation-free.}\ \blacksquare$$

Rmk (domain sensitivity — load-bearing). The exact-cover classification is a theorem within scope (S2): it uses $\aleph_v$ at type-$(1,1)$ vertices only. A chain-local extension of f_v to higher types [FT.21(i)'s telescoping reads] would let higher-type vertices pin edges, enlarging the survivor class (and un-cancelling some bound content). The computation therefore sharpens 16(d) from a bookkeeping question to an amplitude-level one: the two $\aleph$-domains give provably different path integrals. No extension is adopted or guessed here. [both canonical chain-local extensions are now defined and computed exactly — Def-H.27 (E1 chain-dominant, E2 literal-forced), FT.74 (extended exact-cover theorems, brute-force verified), FT.75 (consequences: floor revival, lock revival, the even-L parity theorem, the triangle census, norm-layer invariance). They remain computed variants, not adopted; (S2) stays the frozen domain of record.]

Rmk (locked hubs — locking is not amplitude death). Clause (3d) exhibits locked vertices inside surviving configurations: a hub threaded by pass-through centers carries nonzero amplitude while remaining D^--configured — locked content is retained and counted [CT4.3], inaccessible [D1.11], and now: amplitude-visible only through the threads that cross it. The single-vertex vanishing $Z^{\pm}_{\text{lock}}(v,n) = 0$ is a statement about isolated vertices; joint skeletons admit locked survivors (tetrahedron: 448; cube: 1 150 848 [FT.68]).

Phys. Sum the ledger's own signs over every way a skeleton can partially crystallize and a stark rule emerges: a configuration contributes if and only if its pass-through vertices tile its edges exactly once — every edge read by exactly one local phase-comparison, no edge read twice or never. Loops cannot be so tiled (their edges are read at both ends); neither can loose single edges. What survives is a gas of open two-step threads $s \to v \to t$, each contributing exactly $+4$, with locked hubs admissible only as threading junctions. Crystallization flicker therefore does not dress the solids — it dissolves them at the amplitude layer and leaves transit threads: existence, at this layer, is a perfect matching of edges onto pass-through comparisons.

Theorem 3.56 (FT.68 — Joint skeletons: tetrahedron, the polygon ladder, the cube; the layer dichotomy; flicker). Thm. Under Def-H.25/FT.67 (same scopes): (1) Tetrahedron (exhaustive). $V = 4$, potential $n(v) = 6$ [FT.62(1)], 12 directed edges. Consistency (each directed edge real at both endpoints or virtual at both — reality is one global fact per pair [A4, D0.1]) reduces the 4-tuple space 2^{24} to $2^{12} = 4096$ joint patterns. Bypass configurations: 1230; bypass probability $\frac{615}{2048}$; $K \le 6$ forced ($\sum_v k_v = 2K \le 4 \cdot 3$); counts by K: $(1, 12, 66, 220, 435, 408, 88)$ for $K = 0..6$ (all-patterns: the symmetric row $\binom{12}{K}$ minus locked); orbits under S_4: 70 total, by K: $(1, 1, 5, 13, 23, 21, 6)$. Amplitudes:

$$Z^{\pm}_{\text{byp}} = 1057 = 1 + 24 \cdot 4 + 60 \cdot 16, \qquad Z^{\pm}_{\text{lock}} = 448 = 12 \cdot 16 + 4 \cdot 64, \qquad Z^{\pm}_{\text{all}} = 1505;$$

covering polynomials (zeta-derived values, FT.68(7)): $\mathcal{Z}_{\text{byp}}(y) = 1 + 24y + 60y^2$, $\mathcal{Z}_{\text{lock}}(y) = 12y^2 + 4y^3$ — the locked sector includes the four fully bidirectional $K_{1,3}$ stars (locked $(3, 3)$ hub threaded by three centers, $4^3 = 64$ each). Effective mass $m_{\text{eff}} = \langle K \rangle \ln 2/\mathfrak{h}$ (dark read: $E_{\text{tot}} = K$ directed real edges [FT.17 dark pattern; Def-H.9(2)]), in three readings: (i) uniform over bypass configurations $\langle K \rangle = \frac{852}{205}$; (ii) amplitude-weighted $\left(\sum K|A_C|^2 / \sum |A_C|^2\right) = \frac{62208}{15745}$; (iii) Cal-F equilibrium at $r{=}1$: equals (i). Fractional contributions: the equilibrium measure of a pattern with K real edges is $p^K(1-p)^{12-K}$, $p = r/(1+r)$ — the per-K coefficients above are the exact fractional-contribution table. Aspect-activation frequency over bypass configs by $\max_v k_v$: $(1, 24, 245, 960)$ for $\max k = (0, 1, 2, 3)$. Bound-content census (configurations containing ≥ 1): digons 501, directed 3-cycles 252, directed 4-cycles 90 — with $\Theta \in \{2, 3, 4\}$ at (S1): digons and 4-cycles are on-shell materializable candidates [$\Theta \in 2\mathbb{Z}$: FT.22; 4-cycles: $k = 2$, $j = 1$ under Def-H.38; digons: the $\Sigma_{\min}$ card, $j = \frac{1}{2}$], 3-cycles off-shell arena content [$\Theta = 3$ odd; Def-H.18] — and the bound cells themselves all cancel in amplitude [FT.67(3a)]: standalone bound content is configuration-present, Z-absent. (correction [FT.69(4)]: the census configurations are not all Z-absent — 48 of the 501 digon-containing bypass configurations are covers: dissolved digons inside thread networks [witness in FT.69(4)]; the 3-cycle and 4-cycle rows cancel completely — odd cycles never enter covers, and no tetrahedron cover contains a 4-cycle [FT.69(3)].) The fully crystallized floor cancels [machine]. (2) Polygon ladder P_n (potential $n(v) = 4$; transfer matrix). Bypass $\iff$ every vertex drops ≥ 1 of its 4 directed edges. With channel states $s_i = (f_i, r_i) \in \{0, 1\}^2$ and the vertex constraint $\text{pop}(s_i) + \text{pop}(s_{i+1}) \le 3$:

$$B_{P_n}(t) = \text{tr}\big(T(t)^n\big), \qquad T(t)[s][s'] = t^{|s'|}\,[\,|s| + |s'| \le 3\,] \qquad \text{[math: transfer matrix]},$$

counts $n = 3..10$: $(54, 207, 783, 2970, 11259, 42687, 161838, 613575)$; fractions $(\frac{27}{32}, \frac{207}{256}, \frac{783}{1024}, \frac{1485}{2048}, \dots)$ decaying; the polynomials $B_{P_n}(t)$ are tabulated by the machine [zeta-derived values]. Dynamical zeta: charpoly $T(1) = x^4 - 3x^3 - 3x^2$, $\det(I - zT(1)) = 1 - 3z - 3z^2$ — the bypass counts satisfy the exact recursion $\text{tr}_n = 3\,\text{tr}_{n-1} + 3\,\text{tr}_{n-2}$ [machine: $n = 3..12$], growth rate $\lambda_{\max} = \frac{3+\sqrt{21}}{2} < 4$: the bypass fraction decays exponentially in n — bypass is a finite-cell phenomenon in the thermodynamic limit (the amplitude sector itself keeps growing, at the covering growth rate). Sign-summed amplitudes (bypass, locked): P_3: $(25, 48)$; P_4: $(353, 64)$; P_5: $(1161, 80)$; P_6: $(3121, 1632)$; P_7: $(11929, 10864)$; P_8: $(53825, 43136)$; P_9: $(229417, 159888)$; P_{10}: $(844881, 775840)$ — the locked sector is nonzero for every $n \ge 3$ (single-hub coverings exist on every polygon). (3) Cube (orbit-reduced; Directive bound respected). Potential $n(v) = 6$, 24 directed edges, Aut of order 48. Bypass configurations: 1 580 092, by K: $(1, 24, 276, 2024, 10506, 40296, 115796, 247704, 383913, 408464, 270096, 91392, 9600)$ for $K = 0..12$

(cap-admissible DP; $K \le 12$ forced). Orbit structure (cap-aware Burnside DP over conjugacy classes): by K: $(1, 1, 11, 49, 250, 863, 2509, 5211, 8163, 8571, 5768, 1922, 245)$, total $33\,564 > 10^4$ — per the bounded-enumeration rule, orbits are reported by K only, with individual weights computed for coverings alone. Covering polynomials: $\mathcal{Z}_{\text{byp}}(y) = 1 + 48y + 648y^2 + 2672y^3 + 3624y^4$ (6993 configs), $\mathcal{Z}_{\text{lock}}(y) = 24y^2 + 728y^3 + 3192y^4 + 264y^5 + 4y^6$ (4212 configs); amplitudes $Z^{\pm}_{\text{byp}} = 1\,109\,313$, $Z^{\pm}_{\text{lock}} = 1\,150\,848$, $Z^{\pm}_{\text{all}} = 2\,260\,161$; the fully crystallized cube cancels. The exact-cover theorem is exhaustively verified for all bypass configs with $K \le 4$ and stride-sampled beyond. Dodecahedron/icosahedron: not run wholesale; their covering enumeration (sparse by FT.67(3d)) is the natural next cell, deferred and recorded. (4) Layer dichotomy (the flicker hypothesis, resolved). (a) Bypass is real: $Z^{\pm}_{\text{byp}} > 0$ on every computed skeleton — single vertex $1 + n_{\text{in}}n_{\text{out}} \cdot 4$, tetrahedron 1057, polygons growing, cube $1\,109\,313$: the admissible sector of an $n \ge 4$ skeleton has strictly positive amplitude at the A8 configuration-sum layer. (b) What it produces: the survivors are exactly open V-thread networks — 1-dimensional transit-type content [D1.5: open propagating = bosonic]; the effective dimensionality of the path-integrated bypass configuration is thread-like: neither polygon-like 2D (faces cancel) nor solid-like 3D (solids cancel); its ledger reads are m_{eff} of (1), aspect activation concentrated at $\max k_v \in \{2, 3\}$ ($\mathcal{A} \supseteq$ {EM}-capable and strong-activated patterns [CT0.13], with observable strong still requiring Σ-level sealing [FT.27, FT.20]). (c) What it does not produce: bound floors — Σ_{min}, Σ_L, Σ_{P_n}, Σ_S — cancel exactly at this layer [(FT.67(3a))]; they live at the spectral layer [Def-H.16 poles; M1/M2 declarations], where CT2.8's norms $\sum |w|^2$ are strictly positive and no cancellation occurs. (d) Verdict on "the path integral fuzzes the flickering 2D shape into a higher-tier solid": flickering thread gas — yes, exact and strictly positive; fuzzing into solids — no at the amplitude layer [scope: a literal-domain (S2) verdict — in the chain-local domains every computed Platonic floor and every even polygon floor carries strictly positive amplitude (tetra 2^{17}, cube $2304 \cdot 2^{24}$, octa $72384 \cdot 2^{24}$; $4 \cdot 2^{2L}$ for even L, 0 for odd), so the verdict is domain-relative [FT.75(2)]; the thread-gas clause and the norm-layer clause (c) are domain-invariant [FT.75(6)]]; the solid floors enter through the spectral layer, and the bridge between the two layers is precisely the open residue [FT.66(3)–(5): structural coupling absent; exit (b) not constructed]. No identification between layers is made or forced. (5) Flicker rate (Cal-F, declared). The corpus supplies no $J^{\pm}$ rates — D1.2 is transition vocabulary, A5's draws are unrecorded, and no rate tag exists [inventory]. Declare $r := \Gamma_{J^+}/\Gamma_{J^-}$ a free internal dynamics datum (P4-class, ¬constant of the theory) with per-edge independent detailed-balance equilibrium $p = r/(1 + r)$ [Cal-F]. Then the vertex-admissibility fractions are the exact polynomials

$$f_{\text{adm}}(n, p) = \sum_{k \le 3} \binom{n}{k} p^k (1 - p)^{n-k} :$$

$$1 - p^4 \ (n{=}4),$$

$$1 - 5p^4 + 4p^5 \ (n{=}5),$$

$$1 - 15p^4 + 24p^5 - 10p^6 \ (n{=}6),$$

with samples $f_{\text{adm}}(6; r{=}1) = \frac{21}{32}$, $f_{\text{adm}}(6; r{=}3) = \frac{347}{2048}$ [machine]; skeleton bypass probabilities are $B_\Gamma(p) = \sum_K c_K\, p^K (1 - p)^{|E_{\text{dir}}| - K}$ with c_K the by-K counts of (1)–(3) — the exact fractional-contribution tables. Time-fraction reading: at equilibrium each vertex spends the

fraction $f_{\text{adm}}(n(v), p)$ of time channel-admissible; the fully crystallized Def-H.20 floor occupies measure $p^{2E} \to$ small as $p < 1$ — the static floor is a measure-zero-corner of its own skeleton's flicker ensemble, while its spectral status is p-independent (declared, M1/M2). (6) Bott connection. Amplitude survivors contain no closed chains [FT.67(3a)] $\Rightarrow$ no closed stationary channels at the amplitude layer $\Rightarrow$ Def-H.24's Bott data is undefined on survivors (no loops to index); the thread networks are transit-side objects, where fixed-endpoint index-0 minimizers exist [FT.48(3)] — the amplitude layer's survivor class and the variational layer's stable class are both transit-side, a derived consistency with FT.66(4b) ("stability is transit-side vocabulary"). Bound content inside bypass configurations (the census of (1)) carries its Bott datum at the spectral layer, where FT.66(4) applies unchanged: on Synge-class slices every closed geodesic has $\text{ind}_B \geq 1$ — the bypass does not evade the selection dilemma; it relocates the surviving content to the one sector where index-0 stability exists. (7) Convergence and zeta ledger. Every Z above is a finite exact integer (finite skeletons): the path integral converges trivially and has well-defined values — no regularization was needed and none is claimed. Derived zeta-type values recorded: polygon dynamical zeta $\det(I - zT(1)) = 1 - 3z - 3z^2$ with recursion $\text{tr}_n = 3\,\text{tr}_{n-1} + 3\,\text{tr}_{n-2}$ and $\lambda_{\max} = \frac{3+\sqrt{21}}{2}$; the covering polynomials $\mathcal{Z}_\Gamma(y)$ (tetrahedron/cube, per sector); the bypass polynomials $B_{P_n}(t)$, $n \leq 10$ — all exact integer data, machine-verified.

Req. Def-H.25, Def-H.24; FT.17, FT.20, FT.22, FT.27, FT.28, FT.32, FT.42, FT.45, FT.48, FT.61, FT.62, FT.63, FT.66, FT.67; A4, A5, A8; D0.1, D0.10, D1.2, D1.5, D2.11; CT0.11, CT0.13 (Part 1); CT1.20, CT2.8 (Part 2); CT4.1, CT4.3 (Part 2); Def-H.16, Def-H.18, Def-H.20; M1/M2 and Def-H.38/Cal-F (interface rows; census and flicker clauses only); transfer matrices, Burnside with constraint DP, Newton recursions, binomial identities [math].

Pf.

$$(1):\ \text{consistency}:\ \text{reality per pair [A4/D0.1]} \implies \text{joint patterns} = 2^{12};$$

$$\text{counts/orbits/census}:\ \text{machine exhaustive } (4096 \text{ patterns};\ 3^{12} \text{ sign pairs in the theorem checks})$$

$$Z^{\pm} \text{ values}:\ \text{FT.67(2) value formula summed over covering classes [machine}:\ \text{two independent enumerators agree];}$$

$$K_{1,3} \text{ star hubs}:\ \text{witnesses exhibited [machine]}$$

$$m_{\text{eff}}:\ \text{readings (i)–(iii) computed from the census [machine exact rationals];}$$

$$\text{fractional contributions}:\ \text{Bernoulli measure coefficients [Cal} - \text{F]}$$

$$(2):\ T(t) \text{ construction}:\ \text{vertex constraint} \iff \text{pop}(s_i) + \text{pop}(s_{i+1}) \leq 3;$$

$$B_{P_n}(t) = \text{tr}\, T(t)^n \text{ [math]}; \quad \text{cross-check}:\ \text{cap-DFS brute force, } n \leq 6 \text{ [machine]}$$

$$\text{charpoly}\, T(1) = x^2(x^2 - 3x - 3) \text{ [machine: exact interpolation];}$$

$$\det(I - zT) = z^4 \operatorname{ch}(1/z) = 1 - 3z - 3z^2;$$

$$\text{Newton}: \ \operatorname{tr}_n = 3\operatorname{tr}_{n-1} + 3\operatorname{tr}_{n-2} \ [\text{machine } n \le 12]$$

$$Z^{\pm} \text{ per sector}: \ \text{covering enumeration (escape-pruned backtracking)},$$

$$n$$

$$\le 10 \ [\text{machine; theorem verified exhaustively } n \le 5]$$

$$(3): \ \text{counts}: \ \text{cap-DP over } 24 \text{ edges [machine]};$$

$$\text{orbits}: \ \text{Burnside, cap-aware DP per conjugacy class, memoized [machine]};$$

$$33564 > 10^4: \ \text{STOP rule}: \ \text{counts by } K \text{ only } (K \le 12 \text{ automatic}), \ \text{weights for coverings only}$$

$$(4): \ (a), (b): \ \text{FT.67}(2), (3), (5) + \text{values of } (1)\text{–}(3);$$

$$(c): \ \text{FT.67}(3a) + \text{CT2.8}(2);$$

$$(d): \ \text{conjunction; bridge}: \ \text{FT.66}(3)\text{–}(5), \ \text{item } 13$$

$$(5): \ \neg\exists \text{ rate tag [inventory]};$$

$$f_{\text{adm}}(n,p) \text{ expansion}: \ \text{machine-verified coefficient identities } \{1, -, \dots\};$$

$$\text{samples exact}$$

$$(6): \ \text{FT.67}(3a) \implies \neg\exists \text{ closed chains in survivors};$$

$$\text{FT.48}(3): \ \text{open minimizers};$$

$$\text{FT.66}(4): \ \text{spectral-layer bound unchanged}$$

$$(7): \ \text{finiteness}: \ |E_{\text{dir}}| < \infty \implies 2^{|E|} \cdot 2^{|E|} \text{ finite sum}; \qquad \text{zeta values as tabulated [machine]}. \ \blacksquare$$

Rmk (relation to Def-H.20/FT.62 — no retraction). The Def-H.20 floors are the fully crystallized points of their skeletons' pattern lattices; FT.62's rung and lock ledgers describe those points and are untouched. This tier adds the amplitude-layer anatomy of the neighbourhood those points sit in: the floor itself cancels in the sign sum, its bypass neighbourhood does not, and the two layers (amplitude/spectral) meet only through the open bridge . Nothing here identifies bypass survivors with named particles or derives the J± rates (Cal-F remains declared).

Phys. Let the edges of a locked skeleton flicker — crystallize, dissolve, crystallize — and sum the ledger's own weights over every pattern: the answer is a strict rule with a strict survivor. Only open two-step threads pass, each worth exactly four; every loop, every face, every fully grown solid sums to nothing. The tetrahedron's flicker ensemble weighs 1230 admissible patterns at probability $615/2048$, and its amplitude budget — 1057 of it — is spent entirely on thread networks threading sources, sinks, and the occasional locked hub. Polygons obey a two-term recursion with a rational zeta, $1-3z-3z^2$, and their admissible fraction dies exponentially with size: bypass is the physics of small cells. The cube's 1.58 million admissible patterns collapse to $33{,}564$ symmetry orbits. And the old insight gets its two-layer answer: the flicker is real, and it is a gas of threads — the solids are not what the amplitude sums to; they are what the spectrum declares, on the other side of the one bridge this document has not built.

3.25 Tier F.24 — The Flicker Phase Structure, the Cycle Grammar of Covers, the Dust Equation of State, Face Decomposition, and the Bridge Audit

Definition 3.32 (Def-H.26 — Flicker-ensemble phase reads). Def. For a skeleton Γ [Def-H.25] with $2E$ directed edges and equilibrium parameter $p \in [0,1]$ [$p = x_+/(x_+ + x_-)$: the unique stationary measure of the per-pair Janus generator — FT.108(4), forward pointer Tier F.36; $r := x_+/x_-$ a P4-class rate datum (Cal-F); the per-edge independent Bernoulli scope is constitutive of the ensemble (CT4.5a-species ensemble scope)], the flicker ensemble is the Bernoulli product measure μ_p on crystallization patterns $R \subseteq E_{\text{dir}}(\Gamma)$ (each directed edge real independently with probability p). Pattern-layer reads (all exact polynomials/rationals in p; ¬spectral reads):

$$B_\Gamma(p) := \mu_p(\text{bypass}) = \sum_K c_K\, p^K (1-p)^{2E-K} \quad [c_K :\ \text{cap census, CT0.11}];$$

$$F_\Gamma(p) := p^{2E} \quad [\text{floor measure}];$$

$$L_\Gamma(p) := 1 - B_\Gamma(p) \quad [\text{lock fraction}]; \qquad \langle K \rangle_{\text{byp}}(p) := \frac{\sum_K K\, c_K\, p^K (1-p)^{2E-K}}{B_\Gamma(p)};$$

$$p^*_\Gamma := \text{the unique root of } B_\Gamma(p) = F_\Gamma(p) \text{ in } (0,1) \ \ [\text{lockable } \Gamma;\ \text{FT.70(1)}];$$

$$\langle \rho_{D^-} \rangle_\Gamma(p) := \sum_v O(n_v)\, \mu_p(k_v \geq 4) \ \ [\text{CT4.1, CT4.5}].$$

Rmk (layer scope). Every read above is a μ_p-average of pattern-layer ledger entries; the spectral layer (Def-H.16 poles; M1/M2 materialization) is p-independent by declaration, and the two layers meet only through the inter-family bridge (application-layer) [FT.68(4)]. Static census data (c_K

rows, coverings, crossovers as polynomial facts) are declaration-free; only the equilibrium interpretation uses Cal-F (as in FT.68(5)). $\langle \rho_{D^-} \rangle$ inherits CT4.5's per-configuration $w = 0$ for its locked part; unlocked configurations carry channel stresses [CT1.22] outside this definition's scope.

Theorem 3.57 (FT.70 — The flicker phase diagram: unique crossover, computed boundaries, the polygon limit). Thm. Under Def-H.26 (equilibrium readings under the Cal-F rate datum; the polynomial data declaration-free): (1) Existence and uniqueness of the phase boundary. The admissible family is downward-closed (subsets of admissible patterns are admissible: k_v decreases), nonempty ($\varnothing$), and proper (the floor is locked iff Γ is lockable, i.e. $\max_v n(v) \geq 4$). Hence B_Γ is strictly decreasing on $(0,1)$ [math: monotone coupling; strictness witness: any 3 edges at a vertex of $n(v) \geq 4$ give an admissible R with $R \cup \{e\}$ locked], while $F_\Gamma(p) = p^{2E}$ is strictly increasing; $B(0) - F(0) = 1 > 0$, $B(1) - F(1) = c_{2E} - 1 = -1 < 0$. Every lockable skeleton has a unique crossover $p^*_\Gamma \in (0,1)$; non-lockable skeletons (all $n(v) \leq 3$: the digon, unidirectional cells) have $B \equiv 1$: no interior boundary, thread gas at every $p < 1$. (2) Computed boundaries (exact dyadic isolating intervals of width 2^{-40}; machine; decimal display $\approx$):

$$p^*: \quad \text{icosa } 0.652034 \;<\; \text{octa } 0.679217 \;<\; \text{tetra } 0.723525 \;<\; \text{dodeca } 0.724519 \;<\; \text{cube } 0.725372 \;<\; P_3\, 0.807901 \;<\; P_5 \;<\; P_7 \;<\; P_9 \;<\; \sqrt{\tfrac{2}{3}} \;<\; P_{10} \;<\; P_8 \;<\; P_6 \;<\; P_4\, 0.819671.$$

Regions: $p < p^*_\Gamma$ thread-gas-dominant ($B > F$), $p > p^*_\Gamma$ floor-dominant ($F > B$); the locked fraction $L_\Gamma(p) = 1 - B_\Gamma(p)$ rises $0 \to 1$ across the diagram. The two degree-≥ 4 solids cross earliest — the crossover order reproduces the lock-proneness order of clause (4). (3) Polygon limit, closed form. charpoly $T(t) = x^2\big(x^2 - (1+2t)x - t^2(1+2t)\big)$ [machine: exact at $t \in \{1, 2, \frac{1}{3}, \frac{5}{2}\}$; generalizes FT.68(2)'s t=1 value]. Asymptotically $B_{P_n}(p) \sim (1-p)^{2n}\lambda_{\max}(t)^n$, $t = p/(1-p)$: the crossing equation tends to $\lambda_{\max}(s) = s^2$, i.e. $s^4 = 2s^2(1+2s)$, i.e. $s^2 - 4s - 2 = 0$ with $s = p/(1-p)$: $s = 2 + \sqrt{6}$ and

$$p^*_\infty = \frac{s}{1+s} = \frac{2+\sqrt{6}}{3+\sqrt{6}} = \frac{\sqrt{6}}{3} = \sqrt{\tfrac{2}{3}}$$

$$\Big[\ \big(\sqrt{\tfrac{2}{3}}\big)^2 = \tfrac{2}{3};\ (2+\sqrt{6})^2 = 10+4\sqrt{6} = 4(2+\sqrt{6})+2 \text{ ✓ in } \mathbb{Z}[\sqrt{6}]\Big],$$

approached from below by odd n and from above by even n (interlaced; machine through $n = 10$). (4) Lock-proneness order (the corrected "forbidden zone"). Exact identity [machine, symbolic $n = 3..10$]:

$$f_{\text{adm}}(n{+}1, p) = f_{\text{adm}}(n,p) - \binom{n}{3} p^4 (1-p)^{n-3} \;<\; f_{\text{adm}}(n,p) \qquad \forall p \in (0,1),$$

so admissibility is strictly decreasing in potential degree: $f_{\text{adm}}(2d, \frac{1}{2}) = (1, \frac{15}{16}, \frac{21}{32}, \frac{93}{256}, \frac{11}{64})$ for $2d = (2,4,6,8,10)$: digon > polygons > {tetra, cube, dodeca} > octa > icosa. The proposed octahedron "forbidden zone" is refuted: $f_{\text{adm}}(8,p) \geq (1-p)^8 > 0$ for every $p < 1$ and 318 523 admissible configurations exist at the cap level [(1)]. What is true: at fixed $p < 1$ the octahedron/icosahedron are the most lock-prone Platonic skeletons, and $p = 1$ locks every skeleton — the static floor is the $p \to 1$ limit of its own ensemble [FT.68(5)], with amplitude zero there [FT.67(3a)]. (5) Effective edge content. $\langle K \rangle_{\text{byp}}(p)$ is strictly increasing on the exact grid $p \in \{\frac{1}{20}, \ldots, \frac{19}{20}\}$ (machine: tetra, cube, icosa, P_5), with limits 0 (vacuum-dominated, $p \to 0$) and $K_{\max} = \lfloor 3V/2 \rfloor$

(top bypass row, $p \to 1$); at $p = \frac{1}{2}$: $\left(1, \frac{852}{205}, \frac{3311952}{395023}, \frac{2124600}{318523}, \frac{4007491093935690}{1917271033873 07}, \frac{215801166207}{15519497071}\right)$ for digon/tetra/cube/octa/dodeca/icosa (the tetra entry is FT.68(1)'s m_{eff} reading (i) verbatim — $p{=}\frac{1}{2}$ is the uniform measure) and $\left(\frac{8}{3}, \frac{248}{69}, \frac{130}{29}, \frac{296}{55}, \frac{2618}{417}, \frac{11344}{1581}, \frac{896}{111}, \frac{13588}{1515}\right)$ for $P_3..P_{10}$. The dark mass read is $m_{\text{eff}}(p) = \langle K \rangle_{\text{byp}}(p) \ln 2/\mathfrak{h}$ [FT.68(1) pattern]: light at high flicker, heavy toward the static limit, exact at every p.

Req. Def-H.25, Def-H.26; FT.42, FT.45, FT.62, FT.63, FT.67, FT.68; A4, D0.1; CT0.11 (Part 1); Cal-F (equilibrium readings); monotone coupling for product measures, transfer-matrix asymptotics, dyadic bisection [math].

Pf.

$$(1):\ R' \subseteq R:\ k_v(R') \le k_v(R):\ \text{downward-closed;}$$

$$\text{coupling } p_1 < p_2:\ R_1 \subseteq R_2:\ 1_{\text{adm}}(R_1) \ge 1_{\text{adm}}(R_2):\ B(p_1) \ge B(p_2)$$

$$\text{strictness witness}:\ |R| = 3 \text{ at } v,\ n(v) \ge 4:\ R \text{ admissible},\ R \cup \{e_v\}:\ k_v = 4:\ \text{locked;}$$

$$B(0) = c_0 = 1,\ B(1) = c_{2E} = 0\ [\text{floor locked}]$$

$$(2):\ \text{exact integer bisection on } G(p)$$

$$= B(p) - p^{2E}:\ G(m/2^{40}) \text{ as an integer via } 2^{40 \cdot 2E} \text{ clearing;}$$

$$\text{single sign flip on a 51-point grid [machine]}$$

$$(3):\ T(t) \text{ rows } (00, 01, 10) \text{ equal on } \{00, 01, 10, 11\}\text{-pattern} \implies \text{charpoly}$$

$$= x^2 \chi_2,$$

$$\chi_2$$

$$= x^2 - (1{+}2t)x - t^2(1{+}2t)\ [\text{machine at 4 exact } t];$$

$$\lambda_{\max}(s)$$

$$= s^2$$

$$\implies s^2 \qquad = 2(1{+}2s)$$

$$s = 2 + \sqrt{6} > 0; \qquad p^*_\infty = \tfrac{s}{1+s} = \tfrac{2+\sqrt{6}}{3+\sqrt{6}} \cdot \tfrac{3-\sqrt{6}}{3-\sqrt{6}} = \tfrac{\sqrt{6}}{3}; \qquad \left(\tfrac{\sqrt{6}}{3}\right)^2 = \tfrac{2}{3}$$

$$(4): \text{ binomial recursion } \mathrm{Bin}(n{+}1, p)$$

$$= \mathrm{Bin}(n,p) + \mathrm{Ber}(p): \ f(n{+}1,p)$$

$$= (1-p)f(n,p) + p$$

$$P(\mathrm{Bin}(n,p) \le 2)$$

$$= f(n,p) - p$$

$$P(\mathrm{Bin}(n,p) = 3)$$

$$p$$

$$P(\mathrm{Bin}(n,p){=}3)$$

$$= \binom{n}{3} p^4 (1-p)^{n-3} > 0 \ \ \forall p$$

$$\in (0,1),$$

$$n$$

$$\geq 3 \text{ [machine: symbolic coefficient identity]}$$

$$(5): \text{ exact rational evaluation from } c_K \text{ rows;}$$

$$\text{grid monotonicity machine;}$$

$$p \to 0: \ c_0 \text{ term dominates}: \ \langle K \rangle \to 0;$$

$$p \to 1: \ c_{K_{\max}} \text{ term dominates. } \blacksquare$$

Phys. Let the crystallization rate run from never to always, and every skeleton shows the same two-phase portrait with one sharp boundary: below it the ledger's amplitude lives in a gas of open threads, above it in the static floor — and the boundary sits at an exactly computable flicker rate,

about 0.72 for the degree-three solids, earliest (0.65) for the icosahedron, latest for small polygons, converging for the polygon family to the closed value $\sqrt{2/3}$. No solid is forbidden from flickering: the octahedron's rumored "must lock" is false at every rate below one — it is only the most lock-prone per vertex, and at rate one everything locks, floors included, whose amplitude there is zero anyway. The effective dark mass of the gas climbs smoothly from nothing to the floor's own ledger as the flicker slows.

Theorem 3.58 (FT.71 — The flicker-dust equation of state). Thm. Under Def-H.26 (equilibrium readings under the Cal-F rate datum), for every skeleton Γ: (1) Exact dust density. Per-edge independence gives $k_v \sim \mathrm{Bin}(n(v), p)$, hence

$$\langle \rho_{D^-} \rangle_\Gamma(p) = \sum_v O(n_v)\,\big(1 - f_{\mathrm{adm}}(n_v, p)\big), \qquad O(n) = 2^{n+1} - 16 \ \text{[CT4.1; FT.62(2)]},$$

an exact polynomial in p with integer coefficients: Platonic values at $p = \frac{1}{2}$: $\big(154,\ 308,\ \frac{15159}{8},\ 770,\ 20193\big)$ for (tetra, cube, octa, dodeca, icosa); polygon family the exact monomial $\langle \rho_{D^-} \rangle_{P_n}(p) = 16\,n\,p^4$ (at $p{=}\frac{1}{2}$: n); digon identically 0 (never locks). (2) Anchors. $\langle \rho_{D^-} \rangle_\Gamma(1) = \sum_v O(n_v)$ = FT.63(5)'s dark sums $(448, 896, 2976, 2240, 24384)$ exactly [machine]; $\langle \rho_{D^-} \rangle_\Gamma(0) = 0$; strict increase on $(0,1)$ for lockable Γ [clause (4)'s f_{adm} strict decrease, FT.70(4)]. (3) The flicker modulates ρ, never w. CT4.5 applies per configuration: every locked vertex contributes pressureless dust ($w = 0$, source side, coupled to geometry through CT1.23 and nothing else). The ensemble's locked content is therefore dust at every p; the flicker parameter moves only the density — along the exact polynomial of (1). D^+'s $w = -1$ [CT1.25, D4.3] is configuration-independent and untouched: the CT4.6 contrast table holds ensemble-wise term by term. (4) No fluid. The locked sector supports no further equation-of-state data: it is a structureless absorbing sink — one possibility space [D4.1], no gauge content [CT4.4], absorption one-way [CT4.2], content inaccessible [D1.11] and unresolvable [FT.3]. Statements assigning D^- a pressure response, polarization, or re-emission have no defined ledger referent (the referent would contradict CT4.2/D1.11); "the vacuum is a D^- fluid" is inadmissible as stated — what the corpus derives is: locked content gravitates as dust with the density computed here.

Req. Def-H.26; FT.62, FT.63, FT.70; CT1.22, CT1.23, CT1.25 (Part 2); CT4.1, CT4.2, CT4.3, CT4.4, CT4.5, CT4.6 (Part 2); D1.11, D4.1, D4.2, D4.3; FT.3; binomial law under per-edge independence [math]; Cal-F (equilibrium readings).

Pf.

$$(1):\ \mu_p(k_v \geq 4) = P(\mathrm{Bin}(n_v, p) \geq 4) = 1 - f_{\mathrm{adm}}(n_v, p)\ \text{[per-edge independence, Def-H.26 scope]};$$

$$\text{linearity of expectation over } v\ \text{[CT4.5 :}\ \rho_{D^-} = \textstyle\sum O(n)\text{]}$$

$$O(2d) = 2^{2d+1} - 16:\ (16, 112, 496, 2032)\ \text{at}\ 2d = (4, 6, 8, 10)\ \text{[FT.62(2) table]};$$

$$P_n:\ \text{all}\ n(v) = 4:\ \textstyle\sum_v 16\,p^4 = 16np^4$$

$$(2):\ p{=}1:\ 1-f_{\text{adm}}=1:\ \textstyle\sum_v O(n_v)=(4\cdot 112,\ 8\cdot 112,\ 6\cdot 496,\ 20\cdot 112,\ 12\cdot 2032)\ \checkmark;$$

$$p{=}0:\ 0;\quad \text{monotone by FT.70(4)}$$

$$(3):\ \text{CT4.5 per configuration};\ \text{mixture of } w{=}0 \text{ components is } w{=}0 \text{ on the locked part};$$

$$(4):\ \text{cited tags exhaust the } D^- \text{ inventory [Parts 1–2]};\ \text{no response read exists.}\ \blacksquare$$

Phys. Shake a crystal at rate r and its dark content has an exact expected density: zero when nothing crystallizes, the full retained-register sum $448\ldots 24384$ per tick when everything does, and a smooth polynomial in between — $16np^4$ for a polygon ring, $\frac{15159}{8}$ for the octahedron at equal rates. Whatever the rate, that content gravitates the same way: pressureless dust on the source side of the field equations. The flicker changes how much sediment the ledger carries, never how the sediment weighs.

Theorem 3.59 (FT.72 — Two-fold face decomposition; the Bernoulli marginals ("coarse-grained fuzz as data")). Thm. Let Σ_S be a Platonic skeleton (sphere), E, F its undirected edge/face counts, all arithmetic over GF(2). (1) Two-fold face-decomposition theorem. The face-boundary map ∂_2 : $\mathrm{GF}(2)^F \to \mathrm{GF}(2)^E$ has image exactly the cycle space $Z_1 = \ker \partial_1$ (rank $\beta_1 = F - 1$ [FT.43(5), FT.64; machine: all five]) and kernel $\{0, \sum_f f\}$ (every edge lies in exactly two faces). Hence every cycle $z \in Z_1$ has exactly two face-decompositions, $z = \partial_2 S = \partial_2(F \setminus S)$ — a set of faces and its complement. "Every subnetwork is a composition of triangles" holds exactly for the triangulated skeletons (tetrahedron 4, octahedron 8, icosahedron 20 triangles [machine census]); the cube's faces are squares, and the dodecahedron — girth 5: zero triangles and zero 4-cycles [machine] — admits no triangle decomposition within its skeleton at all: its face grammar is pentagonal, and triangle compositions would require diagonals outside $\mathcal{R}$. (2) Hodge-type split (corrected form). $Z_1 = \mathrm{Cut}^\perp$ under the standard pairing; $\dim Z_1 + \dim \mathrm{Cut} = \beta_1 + (V{-}1) = E$ [Euler], but the sum is not direct in general: $j := \dim(Z_1 \cap \mathrm{Cut}) = (2, 3, 3, 5, 5)$ for (tetra, cube, octa, dodeca, icosa) [machine]. A pattern R is decomposable ($R \in Z_1 + \mathrm{Cut}$: cycle content + boundary content) iff its syndrome is orthogonal to $Z_1 \cap \mathrm{Cut}$; the decomposition is unique up to 2^j gauge choices. (3) Exact Bernoulli marginals. For $R \sim \mu_p$ [Def-H.26], with $q := 1 - 2p$:

$$P_p(R \in Z_1) = \frac{1}{2^{V-1}} \sum_{c \in \mathrm{Cut}} q^{|c|},$$

$$P_p(R \text{ decomposable}) = \frac{1}{2^j} \sum_{u \in Z_1 \cap \mathrm{Cut}} q^{|u|},$$

$$P_p(R \in \mathrm{Cut}) = \frac{1}{2^{\beta_1}} \sum_{z \in Z_1} q^{|z|}$$

[MacWilliams for $Z_1 = \text{Cut}^\perp$ and $(Z_1 + \text{Cut})^\perp = Z_1 \cap \text{Cut}$; machine: direct enumeration cross-check at $\beta_1 \leq 5$]. At $p = \frac{1}{2}$: $P(R \in Z_1) = 2^{-(V-1)} = (\frac{1}{8}, \frac{1}{128}, \frac{1}{32}, \frac{1}{524288}, \frac{1}{2048})$ and $P(\text{decomposable}) = 2^{-j} = (\frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{1}{32}, \frac{1}{32})$; samples $p = \frac{1}{4}$: $(\frac{27}{128}, \frac{19}{64})$, $p = \frac{3}{4}$: $(\frac{11}{128}, \frac{19}{64})$ for the tetrahedron [machine table for all five]. (4) Dual signatures at the Hodge level. The intersection weight-enumerators of dual pairs coincide exactly: cube ≡ octahedron $\{0{:}1, 6{:}4, 8{:}3\}$, dodeca ≡ icosa $\{0{:}1, 10{:}6, 16{:}15, 18{:}10\}$ [machine] — a new degeneracy of the FT.44 species — while the cut enumerators split them: the octahedron's cut space is even-weight (all degrees 4: every star even; the subgraph-level echo of FT.63(3)'s black–white parity law) while the cube's carries odd stars (weight 3). (5) "Fuzz as data", formalized. The complete flicker datum of a skeleton is the measure μ_p on $\text{GF}(2)^E$; every coarse-grained read computed in this document is a named marginal of it: $K \mapsto I_{\text{hid}}$-type edge content [FT.68(1)], admissibility ↦ bypass/lock sectors [Def-H.25], cycle/cut class ↦ (3) above, cover support ↦ amplitude [FT.67(2)], winding lattices ↦ FT.64's index-2^{F-1} sublattices (on-shell κ-side, not μ_p-side — layer marked). No read is lost by coarse-graining that the measure retains; nothing in the measure is readable that the ledger's registers admit [C10 reality principle; FT.3 for locked content].

Req. FT.43, FT.44, FT.63, FT.64, FT.67; Def-H.25, Def-H.26; D0.1; CT1.29 (Part 2); GF(2) linear algebra, MacWilliams identity, Euler characteristic [math]; Cal-F (only for clause (5)'s measure).

Pf.

$$(1):\ \text{each edge in exactly 2 faces (sphere)} \implies \partial_2\big(\textstyle\sum_f f\big) = 0;$$

$$\text{rk}\,\partial_2 \leq \beta_1\ [\text{faces are cycles}];\ \text{rk}\,\partial_2 \geq F-1 \text{ and } \dim\ker\partial_2 = 1\ [\text{machine: elimination, all five}]$$

$$\implies \text{im}\,\partial_2 \subseteq Z_1,\ \dim\text{im} = F{-}1 = \beta_1 = \dim Z_1\ [\text{FT.43(5)}:\ \beta_1 = E{-}V{+}1 = F{-}1] \implies \text{im}\,\partial_2 = Z_1;$$

$$\#\partial_2^{-1}(z) = |\ker\partial_2| = 2$$

$$\text{girth census}:\ \text{dodeca triangles} = 0,\ 4\text{-cycles} = 0\ [\text{machine}]; \qquad \text{triangle counts } (4,0,8,0,20)\ \checkmark$$

$$(2):\ Z_1 = \ker\partial_1 = (\text{im}\,\partial_1^T)^\perp = \text{Cut}^\perp\ [\text{linear algebra}];$$

$$j = \dim Z_1 + \dim\text{Cut} - \dim(Z_1 + \text{Cut})\ [\text{machine ranks}]$$

$$(3):\ \text{MacWilliams } W_{C^\perp}(x,y) = \tfrac{1}{|C|}W_C(y-x, y+x); \quad x = p, y = 1-p:\ y-x = q,\ y+x = 1;$$

$$(Z_1 + \text{Cut})^\perp = Z_1^\perp \cap \text{Cut}^\perp = \text{Cut} \cap Z_1$$

$$p = \tfrac{1}{2}:\ q = 0:\ \text{only the empty word survives}:\ 2^{-(V-1)},\ 2^{-j};$$

direct enumeration cross-check: tetra/cube [machine]

(4) : enumerator tables machine;

octa stars : deg

$= 4$

$\Longrightarrow$ $|\delta(v)|$

$= 4$ even;

even cuts closed under $\triangle$ $\Longrightarrow$ even-weight cut space;

cube stars $|3|$ odd

(5) : marginal list as cited; layer marks at each site. ■

Rmk (relation to the proposal transcript). The proposal "any network subgraph is a composition of triangles with free real/virtual choice per edge" is corrected, not adopted: the free binary choice per directed edge is exactly A4/D0.1/Def-H.25 (frozen and declared respectively); the composition theorem holds with each skeleton's own face type and a two-fold ambiguity per cycle; the triangle-only form fails on the cube (squares) and maximally on the dodecahedron (girth 5). The proposal's ensemble sense — "the coarse-grained fuzz is the data about the system" — is clause (5): the measure is the datum, the ledger reads are its marginals.

Phys. Every loop you can draw on a solid is a sum of faces in exactly two ways — a set of faces, or everything else. That is the whole content of "decomposing into triangles" where triangles exist, and on the dodecahedron no triangles exist to decompose into. Let the edges flicker independently and the fuzz becomes an exact probability measure: the chance that a random pattern is a pure cycle is one in 2^{V-1} at equal rates — one in half a million on the dodecahedron — and every ledger read this document has ever computed is a shadow of that single measure.

FT.73 — The bridge audit: dielectric non-existence, the field-degree no-go, and the surviving species. Structural-constraints record: see Appendix B, Section B.2.2 of the Addendum.

3.26 Tier F.25 — The Chain-Local ℵ-Domains: Extended Exact-Cover Theorems and the Domain-Relativity of the Amplitude Layer

Definition 3.33 (Def-H.27 — The ℵ-domains: literal, chain-dominant, literal-forced). Def. Let $R \subseteq E_{\text{dir}}(\Gamma)$ be a crystallization pattern on a skeleton Γ [Def-H.25] under scopes (S1), (S3), (S4). Three domains for the vertex-amplitude exponent $\aleph$ entering A8's weight $w = \mathcal{J}\prod_{e\in R} G(e)(-1)^{\aleph}$, $G(e) = \sigma_e e^{i\pi\kappa_e}$ [D3.5, FT.9]: (a) Literal (S2) — frozen. $\aleph_v = |f_v|$ [D3.3] at type-$(1,1)$ vertices of R, $f_v = \frac{1}{2}(\sigma_{\rightarrow} - \sigma_{\leftarrow})$; $\aleph_v = 0$ at every other type. No chain data. [This is the domain of FT.67–FT.73 and of the frozen corpus.] (b) E1 — chain-dominant. Configurations (R, σ, D) with D a chain decomposition: a partition of R into directed walks (edge sequences with $t(e_i) = s(e_{i+1})$ [D1.4]). For each walk c: open ($s(c) \neq t(c)$) $\Rightarrow$ interior vertices = all but the two endpoints; closed ($s(c) = t(c)$; D1.4's Janus clause is trivial at (S3)) $\Rightarrow$ every vertex of c is interior, the start vertex included — arrival on the last edge, departure on the first (the wrap read). Then

$$\aleph(\sigma, R, D) = \sum_{c\in D} \sum_{v\in \text{int}(c)} \big|f_v(c)\big|,$$

$$f_v(c) = \tfrac{1}{2}\big(\sigma_{\text{out}(c,v)} - \sigma_{\text{in}(c,v)}\big) \text{ in } c\text{'s own edges at } v,$$

and $Z_\Gamma^{E1} = \sum_R \sum_D \sum_\sigma w^{E1}$. (c) E2 — literal-forced. At type-$(1,1)$ vertices the literal D3.3 read is retained regardless of D — the chain choice there is ℵ-gauge: such a vertex has a unique in- and a unique out-edge in R, so every passage through it gives the same f_v and the datum is type-determined. At higher-type vertices the reads come from chain passages. Equivalently — and this is the form computed — the summed datum is the partial-matching system $M = (M_v)_{v \text{ higher}}$, each M_v a partial matching between $\text{in}_R(v)$ and $\text{out}_R(v)$; the orbits of the induced partial bijection on R are the chains, and an orbit that is a path returning to its start vertex is a closed chain in the sense of (b) and gains the wrap read there:

$$\aleph(\sigma, R, M) = \sum_{v\ (1,1)} |f_v| + \sum_{v \text{ higher}} \sum_{(e,e')\in M_v} \big|f(e,e')\big| + \sum_{\text{wraps}} |f|,$$

$$Z_\Gamma^{E2} = \sum_R \sum_M \sum_\sigma w^{E2}.$$

Coverage convention (all three domains). For $e \in R$ let $\text{cov}_{\text{dom}}(e) :=$ #{ℵ-reads of e in the domain} (literal: the type-$(1,1)$ endpoints; E1: the walk-interior and wrap reads; E2: the forced literal reads, the M-passages, and the wraps). Since $(-1)^{\aleph_v} = \sigma_{\text{in}}\sigma_{\text{out}}$ [FT.67(1)] and $\prod_{e\in R} G(e) = (-1)^{\sum_e \kappa_e} \prod_{e\in R} \sigma_e$ at (S1),

$$w^{\text{dom}} = (-1)^{\sum_e \kappa_e} \prod_{e\in R} \sigma_e^{1+\text{cov}_{\text{dom}}(e)} = (-1)^{\sum_e \kappa_e} \prod_{e:\ \text{cov}_{\text{dom}}(e) \text{ even}} \sigma_e,$$

$$\sum_\sigma w^{\text{dom}} \neq 0 \iff \text{cov}_{\text{dom}}(e) \text{ odd } \ \forall e \in R,$$

the σ_e-exponent being $1 + \text{cov}_{\text{dom}}(e)$ in every domain [FT.67(2)'s Pf, domain-blind]. **[typographical correction, applied in place at FT.67(2): its displayed monomial subscript read $\text{cov}(e)$ odd; the exponent identity $1 + \text{cov}(e)$ recorded in its own Pf forces even. The classification, the values $4^{|\mathcal{C}|}(-1)^{\sum\kappa}$, the Pf, and every census built on them are unchanged — the slip is display-only.]**

Rmk (status). Domains (b), (c) are computed variants, not adopted: Parts 1–2 define $\aleph$ by (a) [D3.3 via A8]; domains (b), (c) await the domain choice. The extension is forced data wherever the corpus itself uses chain-local pass-through reads — FT.21(i)'s telescoping identity, FT.62(5)'s observed-variant EM rows — so (b)/(c) are the two canonical completions of that usage to the weight; which (if either) governs A8's $\aleph$ is an author's declaration, not derivable here, and no interface definition is introduced (nothing below is used as a premise by any other tier; the literal-domain record of Tiers F.23–F.24 stands untouched). Both variants are evaluated under the same (S1)/(S3)/(S4) scopes; κ-invariance of the classifications and the $(-1)^{\sum\kappa}$ value scaling carry over verbatim [FT.67(2) Pf pattern].

Rmk (finiteness). For finite R the sets of walk partitions D and of partial-matching systems M are finite, so Z_Γ^{E1}, Z_Γ^{E2} are finite sums over a finite skeleton — convergence trivial, as in Def-H.25.

Theorem 3.60 (FT.74 — The extended exact-cover theorems). Thm. Under Def-H.27, at (S1); for general on-shell integer κ every value below scales by $(-1)^{\sum_e \kappa_e}$ and no classification changes. Write $\varepsilon(R) := (-1)^{|R|}2^{|R|}$: on the support of either classifier $|R|$ is even [each thread contributes 2 edges], so $\varepsilon(R) = 2^{|R|} > 0$ and the stated values are the exact amplitudes. (1) E1 classification.

$$\sum_D \sum_\sigma w^{E1}(R,\sigma,D) = N_{E1}(R)\cdot\varepsilon(R),$$

where $N_{E1}(R)$ is the number of partitions of R into open directed 2-threads $s \to v \to t$ ($s \neq t$; the junction v of any type — locked junctions included). Every other decomposition cancels: a walk of length ≥ 3 double-reads its interior edges; a length-1 walk leaves its edge unread (cov $= 0$); a closed walk double-reads every edge (all vertices interior, wrap included); a closed 2-thread — a digon — is a closed chain [D1.4] and cancels by the same wrap read. (2) E2 classification.

$$\sum_M \sum_\sigma w^{E2}(R,\sigma,M) = N_{E2}(R)\cdot\varepsilon(R),$$

with $N_{E2}(R) = 0$ unless (a) no edge of R joins two type-$(1,1)$ vertices (such an edge is read literally twice), and (b) every literal thread at a type-$(1,1)$ vertex is open — a literal thread whose two ends coincide is a closed chain [D1.4], and its wrap read double-reads both edges. Otherwise $N_{E2}(R)$ is the number of open-2-thread partitions of the higher-only remainder $R_{\text{hh}} := \{e \in R :$ neither endpoint is type-$(1,1)\}$, junctions necessarily higher-type; the forced literal threads cover exactly the edges touching type-$(1,1)$ vertices, once each. (3) Pointwise order, and incomparability of the three domains. $N_{E2}(R) \leq N_{E1}(R)$ for every R on every skeleton (an E2 structure maps injectively to its thread partition: forced literal threads are open by (2b), edges with a $(1,1)$ endpoint are covered exactly once by (2a)-(2b), and the R_{hh} threads are E1 threads) — machine-asserted with zero violations on 9472 patterns (the tetrahedron and bidirectional $P_4/P_5/P_6$, exhaustively), together with $|R|$ even on both supports. The literal domain embeds into neither extension and neither into it. Witnesses (triples (lit, $E2$, $E1$); machine: classifier = brute-force sum on each row):

R	digon	dir. 3-cycle	dir. 4-cycle*	double-thread hub	digon-pair hub	bidir P_4 floor
(lit, $E2$, $E1$)	$(0,0,0)$	$(0,0,0)$	$(0,0,32)$	$(0,32,32)$	$(16,0,16)$	$(0,1024,1024)$

*on the 4-cycle skeleton. The digon-pair hub $R = \{(0\to1),(1\to0),(0\to2),(2\to0)\}$, types $\big((2,2),(1,1),(1,1)\big)$ — the FT.69(4)/FT.69(5) witness family — has literal value 16, is killed by E2 (both literal threads are closed) and revived by E1 at the alternative threading

$\{(2\to0),(0\to1)\},\{(1\to0),(0\to2)\}$: $N_{\text{lit}} \not\leq N_{E2}$. The directed 4-cycle gives $N_{E2} = 0 < N_{E1} = 2$ (strictness of (3)'s order). The double-thread hub $\{(0\to2),(1\to2),(2\to3),(2\to4)\}$ has literal value 0 and extended value 32: both extensions exceed the literal domain. The digon — the Σ_{min} pattern [FT.28] — cancels in all three domains: the minimal floor's absence from the amplitude layer is domain-invariant. (4) Equality locus. On patterns with no type-$(1,1)$ vertex, E2 $\equiv$ E1 exactly (no literal reads exist; in particular on every fully crystallized floor, all of whose vertices are type (d,d), $d \geq 2$ [FT.62]). There the extended value is $N \cdot \varepsilon(R)$ with N the open-thread partition count, and the literal value is 0 whenever $R \neq \varnothing$ [FT.67(3a)]. (5) Machine verification. The classifications are verified against brute-force sign sums taken straight from Def-H.27's semantics — E1 by explicit enumeration of all walk partitions D with interior/wrap reads, E2 by explicit enumeration of all partial-matching systems M with orbit-wrap closure — on every pattern of seven families: the digon (2^2), the bidirectional triangle (2^6), bidirectional P_4 (2^8), the single-vertex stars $n \in \{4,5,6\}$ ($2^4, 2^5, 2^6$), and all tetrahedron patterns with $K \leq 4$ (794): 2460 pattern-domain checks, zero mismatches.

Req. Def-H.25, Def-H.27; FT.21, FT.22, FT.28, FT.62, FT.67, FT.68, FT.69; A4, A8; D0.1, D0.8, D1.4, D1.5, D3.3, D3.5; CT0.9 (Part 1); involution/sign-sum arguments, walk and partial-matching enumeration [math; machine].

Pf.

$$(1):\ \text{walk } (e_1,\dots,e_k),$$

$$k$$

$$\geq 3:\ e_j\ (1<j<k) \text{ read at both its ends}:\ 1+\text{cov}$$

$$= 3:\ \sigma_{e_j}$$

$$\mapsto -\sigma_{e_j} \text{ involution} \qquad \Longrightarrow\ 0$$

$$k=1:\ e_1 \text{ unread}:\ \text{cov}=0:\ 1+\text{cov}=1\ \Longrightarrow\ 0;$$

$$\text{closed walk}:\ \text{every edge read at both ends (wrap at the start)}:\ \text{cov}=2\ \Longrightarrow\ 0$$

$$k=2 \text{ open}:\ \text{each edge read exactly once}:\ 1+\text{cov}=2:\ w=\varepsilon(R)\ \sigma\text{-constant}:\ \sum_{\sigma} w=\varepsilon(R);$$

$$k=2 \text{ closed (digon)}:\ \text{cov}=2\ \Longrightarrow\ 0$$

$$\text{surviving } D\ \Longleftrightarrow\ \text{partition into open 2-threads};$$

$$\text{distinct surviving } D \text{ give identical weights}\ \Longrightarrow\ \text{count } N_{E1}(R);$$

$$|R| = 2\#D \implies \varepsilon(R) = 2^{|R|}$$

$$(2):\ v \text{ type}(1,1):\ \text{literal read forced}:\ \text{covers both edges of } v;$$

$$e = (x{\to}y),\ x, y \text{ both}(1,1):\ \text{cov} = 2 \implies 0$$

$$v\ (1,1),$$

$$\text{literal thread closed}:\ \text{walk } s \to v \to s:\ s(c)$$

$$= t(c):\ \text{closed [D1.4]}:\ \text{wrap read at } s:\ \text{both edges cov}$$

$$= 2 \qquad \implies 0$$

$$e = (x{\to}y),$$

$$x\ (1,1),$$

$$y \text{ higher}:\ e \text{ read literally at } x:\ \text{any } M\text{-read of } e \text{ at } y:\ \text{cov}$$

$$= 2$$

$$\implies 0:\ \text{surviving } M \text{ pair only within } R_{\text{hh}}$$

$$\text{orbits of } M|_{R_{\text{hh}}}:\ \text{as in } (1):\ \text{only open 2-threads survive;}$$

$$\text{forced literal threads cover } R \setminus R_{\text{hh}} \text{ once each}$$

$$\implies N_{E2}(R)$$

$$= \#\{\text{open thread partitions of } R_{\text{hh}}\}$$

$$(3):\ N_{E2} \le N_{E1}:\ M \mapsto \{\text{literal threads}\} \cup \{R_{\text{hh}} \text{ threads}\}:\ \text{well-defined and injective by } (2);$$

$$\text{witness rows}:\ \text{direct computation [machine}:\ \text{classifier} = \text{brute force on each];}$$

$$(4):\ \text{no}(1,1)\ \text{vertices}:\ R_{\text{hh}} = R:\ \text{E2 sum} = \text{E1 sum}\ [\text{as in}\ (1)]$$

$$(5):\ \text{exhaustive enumeration as scoped}\ [\text{machine}].\ \blacksquare$$

Phys. Let the ledger compare phases along chains instead of at vertex types, and the exact-cover rule keeps its shape — edges tiled by two-step threads, each read exactly once — but the tiling grammar changes: who may host a thread junction (any vertex, or only the over-subscribed ones) and which loops count as closed become the defining choices. The three grammars provably disagree, in both directions, on four-edge patterns. One verdict is common to all three: the digon alone still sums to zero — the minimal fermion floor's exclusion from the amplitude layer is not an artifact of the literal domain.

Theorem 3.61 (FT.75 — Domain-relativity of the amplitude layer: lock revival, floor revival, parity, and the invariant bridge gap). Thm. Under Def-H.27/FT.74, at (S1): (1) Single vertex — the locked sector comes alive, in two closed forms. For the balanced single-vertex star ($n_{\text{in}} + n_{\text{out}} = n \in \{4,5,6\}$, external endpoints simple), $Z^{\pm}_{\text{byp}}$ is domain-invariant $= 1 + 4n_{\text{in}}n_{\text{out}} = (17, 25, 37)$ [FT.67(5)], while

$$Z^{\pm}_{\text{lock}}:\quad \text{literal } 0; \qquad E2 = E1 = \sum_{a\geq 2} \binom{n_{\text{in}}}{a}\binom{n_{\text{out}}}{a}\, a!\, 4^a = (32,\ 96,\ 672),$$

totalling $(49, 121, 709)$ across both sectors. FT.67(5)'s "its locked sector vanishes identically" is a literal-domain theorem: in the chain-local domains an isolated locked vertex has strictly positive amplitude — the type-(a,a) patterns thread through the vertex itself. Second family (the locked digon hub): a locked (a,a) hub hosting a digons with a simple centres has

$$\text{literal} = 4^a, \qquad E2 = 0\ \ (\text{every literal thread closed}), \qquad E1 = \big(!a\big)\, 4^a,$$

$!a$ the derangement number (each thread must leave its own centre: a fixed-point-free permutation of the a digons), machine-verified for $a = 2,3,4,5$ with brute force at $a \leq 3$: $(16, 128, 2304, 45056)$ against literal $(16, 64, 256, 1024)$. The tetrahedron's 16 locked digon-covers are exactly the $a = 2$ (12; value 16) and $a = 3$ (4; value 128) instances [FT.69(5)]. (2) Floor revival — the dichotomy is domain-relative. Every fully crystallized floor has no type-$(1,1)$ vertex, so E2 $\equiv$ E1 there [FT.74(4)], and the extended amplitude is strictly positive whenever the thread-partition count is:

$$N(\text{tetra floor}) = 32\ \ (\text{value } 2^{17}),$$

$$N(\text{cube floor}) = 2304\ \ (2304 \cdot 2^{24}),$$

$$N(\text{octa floor}) = 72384\ \ (72384 \cdot 2^{24})$$

— versus literal amplitude 0 [FT.67(3a)]. Dodecahedron/icosahedron floor counts deferred (branch enumeration beyond the bounded rule; recorded, not guessed). Bidirectional polygon floors obey an exact parity theorem:

$$N\big(\Sigma_{P_L}\ \text{floor}\big) = \begin{cases} 4 & L \text{ even} \quad (\text{value } 4\cdot 2^{2L}), \\ 0 & L \text{ odd}, \end{cases}$$

machine-verified $L = 3..10$ $(0,4,0,4,0,4,0,4)$. FT.68(4)'s verdict "fuzzing into solids — no at the amplitude layer" is the literal-domain verdict; in the chain-local domains every computed even polygon floor and the three computed Platonic floors carry strictly positive amplitude (odd polygon floors stay dead by parity; the dodecahedron/icosahedron floors are open counts, not known-zero). (3) Directed cycles split the domains. The single-orientation directed cycle C_L: literal 0; E2 0 for every $L \geq 2$ (all vertices type-$(1,1)$: conditions (2a)-(2b) fail); E1 $= 2\cdot 2^L$ for even L — the cycle's two perfect matchings — and 0 for odd L ($|R|$ odd). Machine: $L = 3..10$, values $(0,32,0,128,0,512,0,2048)$. The on-shell winding ladder's even rungs [FT.22] are exactly the E1-survivors; the digon ($L = 2$) is dead in all three [FT.74(3)]. (4) Bound content returns — the triangle census, and the digon-cover split. On the tetrahedron (exhaustive, all 4096 patterns): 2397 patterns contain a directed 3-cycle; of those, nonzero amplitude — literal 0, E2 833, E1 1025; restricted to the bypass sector: 0/40/88 (the FT.68(1)/FT.69(4) anchor "0 nonzero of 252" is the bypass-only carrier count and stands). Exhibit: $R = \{(0{\to}1),(0{\to}3),(1{\to}2),(2{\to}0)\}$, threads $\{0{\to}1{\to}2\},\{2{\to}0{\to}3\}$, E1 value 16 (classifier = brute force). The cycle-alternation lemma [FT.69(3)] is a literal-domain theorem and fails in both extensions — chain-local threading dissolves odd cycles into open threads. The digon rows split by sector (machine, exhaustive): the 48 bypass digon-covers of FT.69(4) are killed by both extensions ($N_{E2} = N_{E1} = 0$ on all 48 — a dissolved digon's centre has no alternative threading), while the 16 locked digon-covers are killed by E2 and kept by E1 [clause (1)'s derangement family]. (5) Skeleton totals (exhaustive; the literal column reproduces the stored anchors exactly).

	$Z^{\pm}_{\text{byp}}$	$Z^{\pm}_{\text{lock}}$	#nonzero (byp/lock)
tetra, lit	1057	448	85/16
tetra, E2	5921	966656	173/934
tetra, E1	6881	1001536	227/1166
P_3 bidir, lit/E2/E1	25/121/121	48/0/48	7/13/13 byp; 3/0/3 lock
P_4 bidir, lit/E2/E1	353/225/289	64/1536/2112	29/21/23; 4/9/13
P_5 bidir, lit/E2/E1	1161/361/1801	80/5760/8400	51/31/61; 5/30/60
P_6 bidir, lit/E2/E1	3121/3697/7985	1632/50944/80224	97/79/135; 21/67/154

The P_3 locked row exhibits the second incomparability at skeleton level: E2 kills every literal locked cover there (each is a digon-pair hub — a closed literal thread), E1 keeps all 48. (6) Norm-layer invariance, with the exact configuration count. In every domain every factor of w is a sign — $|w(R,\sigma)| = |w(R,\sigma,D)| = |w(R,\sigma,M)| = 1$ (machine: asserted on every (R,D,σ) of the bidirectional triangle) — so CT2.8's norms $\sum |w|^2$ count configurations: strictly positive, cancellation-blind, and form-identical across domains. At the pattern-sign layer the count is exactly

$$\#\{(R,\sigma)\} = \sum_{R\subseteq E_{\text{dir}}} 2^{|R|} = 3^{\,|E_{\text{dir}}(\Gamma)|}$$

(machine: $3^6 = 729,\ 3^8 = 6561,\ 3^{12} = 531\,441$ for P_3, P_4, tetra),

each directed edge contributing (absent, $+$, $-$); the extended domains refine this set by an inner sum over D/M and cannot change it. Item 13's bridge question is therefore invariant under the $\aleph$-domain choice: no domain rescues or worsens it — exit (b) must operate on resolvent structure in every domain alike. (7) Gap 16(d) verdict. The $\aleph$-domain question of the EM row is now exact data: three domains, three provably distinct path integrals, a complete witness catalogue [(3)–(5)], the frozen answer unchanged (D3.3's literal domain governs A8's $\aleph$), and the two chain-local variants computed but not adopted. The residue — whether FT.62(5)'s observed-variant EM-row reads should govern the weight as well — is an author's declaration whose consequences are now fully priced on both sides. (8) What is not domain-relative. The cap census c_K [FT.69(1)], the admissibility fractions $f_{\text{adm}}(n,p)$, the phase-diagram reads $B_\Gamma(p)$, $L_\Gamma(p)$, p^*_Γ, $\langle K\rangle_{\text{byp}}(p)$ [Def-H.26, FT.70] and the dust equation of state $\langle\rho_{D^-}\rangle_\Gamma(p)$ with $w=0$ [FT.71, CT4.5] are counting/type-level objects: none involves $\aleph$, so all are identical in the three domains — the phase boundaries of FT.70(2) and the dust densities of FT.71 stand as computed. Only the amplitude-side sums $Z^\pm$, the survivor grammar, and the censuses built from them are domain-relative. Covering enumeration likewise splits: the literal covers of FT.69(2) are a proper subset of the E1-survivors on every computed skeleton [clause (5)].

Req. Def-H.25, Def-H.26, Def-H.27; FT.21, FT.22, FT.28, FT.34, FT.45, FT.62, FT.63, FT.67, FT.68, FT.69, FT.70, FT.71, FT.74; A8; D1.4, D1.5, D3.3; CT2.8 (Part 2); CT4.1, CT4.5 (Part 2); perfect matchings of cycles, derangement counts, parity arguments [math]; machine: exhaustive enumeration as scoped.

Pf.

$$(1):\ \text{type}(a,a)\ \text{pattern at the star}:\ a!\ \text{bijections in}\leftrightarrow\ \text{out},$$

$$\text{all threads open (distinct externals)}:\ a!$$

$$4^a;$$

$$\text{pattern count}\ \binom{n_{\text{in}}}{a}\binom{n_{\text{out}}}{a}$$

$$\text{types}\ (a,b),\ a\neq b:\ |R|=a+b\ \text{odd or unthreadable}:\ 0;$$

$$\text{locked}\iff a+b\geq 4\iff a\geq 2\ \text{on the balanced support}$$

$$\text{digon hub}:\ \text{literal}=4^a\ [\text{FT.67}(2),\ |\mathcal{C}|=a];$$

$$E2:\ \text{centre}\ c_i\ \text{type}(1,1),\ \text{thread}\ h\to c_i\to h\ \text{closed [D1.4]}:\ N_{E2}=0\ [(2b)]$$

$$E1:\ \text{a thread through the hub is}\ (c_i{\to}h),$$

$$(h{\to}c_j),$$

$$i$$

$$\neq j:\ \text{partitions}$$

$$=\text{fixed-point-free permutations}:\ N_{E1}$$

$$=!a:\ \text{value}\ !a\cdot 2^{2a}\qquad =!a\cdot 4^{a}$$

$$\text{tetrahedron instances}:\ a=2\ (12\ \text{covers}),\ a=3\ (4)\ [\text{FT.69}(5);\ \text{machine}]$$

$$(2):\ \text{floors}:\ \text{types}(d,d),$$

$$d$$

$$\geq 2:\ \text{no}(1,1)\ \text{vertices}:\ E2$$

$$\equiv E1\ [\text{FT.74}(4)];$$

$$\text{counts}:\ \text{thread-partition enumeration [machine]}$$

$$\text{parity}:\ \text{mixed thread at}\ i:\ \text{in-edge}\ (i{-}1{\to}i)\ [\text{fwd}]\ \text{with out-edge}\ (i{\to}i{-}1)\ [\text{bwd}]:\ \text{walk}\ i{-}1\to i\to i{-}1:\ \text{closed}:\ 0;$$

$$\text{likewise bwd-in with fwd-out}$$

$$\Longrightarrow\ \text{every thread pure-orientation [machine witness at } L{=}4:\ 4\ \text{partitions, \#FF} = \text{\#BB} = L/2 = 2\ \text{in each}];$$

$$L=\text{\#fwd edges}=2\,\text{\#FF}\ \Longrightarrow\ L\ \text{even}$$

$$\text{even}\ L:\ \text{tilings of a directed}\ L\text{-cycle by 2-paths}$$

$$=\text{perfect matchings of}\ C_L$$

$$=2\ [\text{math}]:\ N$$

$$=2_{\text{fwd}}\cdot 2_{\text{bwd}}\qquad =4;$$

$$\text{odd } L:\ |R| = 2L \text{ even but } L \neq 2\#\text{FF}:\ N = 0$$

$$(3):\ C_L:\ \text{every vertex type}(1,1):\ \text{literal/E2 forced reads}:\ \text{cov} = 2 \text{ per edge}:\ 0;$$

$$E1:\ \text{tilings of the directed cycle}:\ \text{even } L:\ 2;\ \text{odd } L:\ L \neq 2\#\text{FF}:\ 0$$

$$(4),$$

$$(5):\ \text{exhaustive machine enumeration with the FT.74 classifiers (brute-force verified)};$$

$$\text{exhibit threads checked edge-by-edge}$$

$$(6):\ |\sigma_e| = 1,\ \left|(-1)^{\aleph}\right| = 1,\ \mathcal{J} = 1\ [(S3)] \implies |w| = 1;$$

$$\#\{(R,\sigma)\} = \textstyle\sum_k \binom{|E_{\text{dir}}|}{k} 2^k = (1+2)^{|E_{\text{dir}}|}\ [\text{binomial theorem}]$$

$$(7):\ \text{conjunction of } (1)\text{–}(6);$$

$(8):$ each read named in Def-H.26/FT.69(1)/FT.71 is a μ_p-average of type-level data : $\aleph$ absent. ■

Phys. Let the phase-comparisons ride chains instead of vertex types, and the amplitude layer changes character: locked vertices acquire weight of their own — a single over-subscribed vertex hums with amplitude 32, 96, 672 as its degree climbs, and a hub holding a digons answers with the derangement count times its literal voice — and the fully grown crystals, silent in the literal ledger, ring out: the tetrahedron floor at 2^{17}, the cube at $2304 \cdot 2^{24}$, the octahedron at $72384 \cdot 2^{24}$, every even polygon ring at four times its full register. Odd rings stay dark by a pure parity argument, triangles come back dissolved, and the digon stays dark in every grammar.

3.27 Tier F.26 — The Platonic Pole Spectrum: Exact Multiplets, the Bounded-Hierarchy Ceiling, and the SM-Hierarchy Audit

Theorem 3.62 (FT.155 — Minimality of the defect interface). Thm. The non-derived content of Def-H.37 is exactly one positive scalar ($\hat{\mu}'^2$; equivalently $\hat{\mu}^2$ at leading order, related by the recorded matching) and the holonomy identification $\mathcal{U}_c = \mathcal{U}_c^{\text{spin}}$. Precisely:

(1) shape uniqueness: any state-independent defect read on $SU(2)$ transport has the form m^2_{pole}

$$= F(\Theta_R),$$

$$F$$

$$= 4\hat{\mu}'^2 \sin^2(\cdot/4) \text{ exact},$$

$$\sim \hat{\mu}'^2(\cdot)^2/4 \text{ leading}$$

$[\text{FT.54}(1):\ (U-1)^\dagger(U-1) = 4\sin^2(\tfrac{\phi}{2})\,\mathbb{I}$ axis- and state-free; FT.54(2): $X^2 = |\vec{x}|^2\mathbb{I}$ fixes the leading term; EW.33 is that Taylor term$]$;

(2) combinatorial exclusion: no rational-coupling formula on integer-valued combinatorial reads reaches the computed pole factors — the five Platonic factors $\{2,\ 2-\sqrt{2},\ 1,\ 2-2\cos\frac{\pi}{10},\ 2-\sqrt{3}\}$ generate a degree-16 extension of $\mathbb{Q}$

$[\text{FT.73}(2)\text{–}(3):$ irreducibility machine-verified; $\sqrt{5} \notin \mathbb{Q}(\sqrt{2},\sqrt{3})$; $\mathbb{Q}(2\cos\frac{\pi}{10})$ has unique quadratic subfield $\mathbb{Q}(\sqrt{5})$; single quadratic couplings excluded by the exact ratio $4-2\sqrt{2}+2\sqrt{3}-\sqrt{6}]$;

the D^--drag/dielectric candidates are independently refuted by response non-existence [FT.73(1); CT4.2, D1.11, CT4.4];

(3) range: on the Platonic skeletons the read takes finitely many values in $[0, 4\hat{\mu}'^2]$ (period $P_S = 4\pi/\delta_v$; $P_S/2+1$ distinct values; monotone to the ceiling);

$$\text{widest intra-solid ratio } \sqrt{4/D_S}$$

$$\leq 6.3925 \text{ [FT.76(3)]};$$

(4) rotor reparametrization: any k^2/R^2-type rotational mass formula is a reparametrization of Def-H.37(2)'s leading regime ($R^{-2} \equiv \hat{\mu}'^2\Theta_R^2/4$),

$$\neg\text{an independent mechanism};$$

its approximation error at the documented uniform-minimal windings is 18.9%–91.0% [FT.87$(1),(4)$].

Hence the replacement search space of the interface contains no combinatorial, dielectric, or rotor species; the derivation of the transfer itself — why the seed's transport defect is the terminus's pole displacement — remains application-layer content.

Req. Def-H.37; Def-H.21; EW.32, EW.33 (Part 2); FT.54(1)–(2); FT.73(1)–(3); FT.76(3); FT.87(1), (4); CT4.2, CT4.4 (Part 2); D1.11; field-degree and enclosure arithmetic [math; machine: `checks/`

`verify_H_stationarity.py`, `checks/verify_H_phasediagram.py`, `checks/verify_H_polespectrum.py`, `checks/verify_H_tumbling.py` pins].

Pf.

$$(1):\ \text{FT.54}(1)\ \text{verbatim}:\ \|(U-1)v\|^2$$

$$= 2 - 2\cos\phi\ \forall\ \text{unit}\ v,$$

$$\text{axis-free}$$

$$\Longrightarrow\ F(\Theta_R)\ \text{shape with}\ \phi$$

$$= \Theta_R/2\ [\text{Def-H.21}(6)];$$

$$(2):\ \text{FT.54}(2)$$

$$(2):\ \text{FT.73}(2)\text{–}(3)\ \text{verbatim}:\ \text{degree-16 extension}\ [y^4-5y^2+5,\ x^4-8x^3+19x^2-12x+1\ \text{irreducible};\ \text{machine}];$$

$$\text{FT.73}(1):\ \text{response non-existence}$$

$$(3):\ \text{FT.76}(3)\ \text{verbatim}:\ \text{finite value set, monotone cone, ceiling}\ 4\hat{\mu}'^2;$$

$$(4):\ \text{FT.87}(1):\ \text{Taylor identification};\ \text{FT.87}(4):\ \text{error table [exact enclosures]}$$

$$\text{search-space clause}: (1)\ \text{fixes the shape};$$

$$(2)\ \text{excludes the combinatorial species};$$

$$(4)\ \text{absorbs the rotor species};$$

$$\text{the residue is the transfer's derivation}\ \blacksquare$$

Phys. The defect interface is priced: one scalar and one identification, and every competing shape is excluded by theorem — each alternative mass formula the corpus has tested is either inside its Taylor regime, outside its range, or outside its field. What remains open is not the form of the transfer but its locus: why the seed's failure to return to itself around a bound chain is that chain's pole mass.

FT.77 — The SM-hierarchy audit: a bounded-hierarchy negative result. Structural-constraints record: see Appendix B, Section B.2.3 of the Addendum.

3.28 Tier F.27 — The Cover Orbit Decomposition: the Ansatz Split, the Combinatorial Ceiling, and the Bridge Census

Definition 3.34 (Def-H.28 — Cover classes, and the two class-mass ansätze). Def. Let Γ be a skeleton [Def-H.25] with directed edge set $E_{\text{dir}}(\Gamma)$ and automorphism group $\mathrm{Aut}(\Gamma)$ acting on vertices, hence on $E_{\text{dir}}(\Gamma)$ by $(a,b)\mapsto(p(a),p(b))$. A cover class (orbit class) is an $\mathrm{Aut}(\Gamma)$-orbit $\mathcal{O}$ of crystallization patterns $R\subseteq E_{\text{dir}}(\Gamma)$; its multiplicity is $\Omega_{\mathcal{O}}=|\mathcal{O}|$, with $\Omega_{\mathcal{O}}\cdot|\mathrm{Stab}(R)|=|\mathrm{Aut}(\Gamma)|$ [orbit–stabilizer] and hence $\Omega_{\mathcal{O}}\mid|\mathrm{Aut}(\Gamma)|$. Invariants of a class (all constant on $\mathcal{O}$, machine-verified): $K=|R|$; admissibility ($k_v\le 3\ \forall v$) and lock status; cover status and $|\mathcal{C}|$; amplitude $4^{|\mathcal{C}|}$ [FT.67(2)]; the canonical directed-cycle content of R (its signature: which cycles the cover shadows); the aspect mask $\mathcal{A}(\max_v k_v)$ and observable mask dimension [CT0.13, CT0.14]. The two class-mass reads:

$$\text{A (corpus-native)}: \quad m_A(\mathcal{O})=K\,\frac{\ln 2}{\mathfrak{h}}$$

[FT.68(1)/FT.70(5)'s $\langle K\rangle\ln 2/\mathfrak{h}$ read, per class; ¬new datum],

$$\text{B (external ansatz, NOT a corpus read)}: \quad m_B(\mathcal{O})=\Big(K-\alpha\ln\Omega_{\mathcal{O}}\Big)\frac{\ln 2}{\mathfrak{h}},$$

α a formal parameter, no value declared.

Rmk (B's exact form). $\Omega\mid|\mathrm{Aut}|$ and $|\mathrm{Aut}|=2^3\cdot 3$ (tetrahedron, 24) or $2^4\cdot 3$ (cube, 48) force $\Omega=2^a3^b$, so $\ln\Omega=a\ln 2+b\ln 3$ exactly and every Ansatz-B mass lies in the $\mathbb{Q}$-span of $\{1,\ln 2,\ln 3\}$ with the single free coefficient α: (K,a,b) is a complete exact label. No floating-point logarithm is evaluated anywhere. The corpus supplies no free-energy/entropic mass read (CT4.7/CT4.10's entropy triangle and FT.71's dust equation of state are not mass formulas), so B is recorded as an ansatz, computed as directed, never adopted, and no result below is used as a premise by any other tier. Rmk (p-independence, verified). Because K is constant on $\mathcal{O}$, the directive's orbit-wise weighted average reduces identically:

$$\langle K\rangle_{\mathcal{O}}=\frac{\sum_{\sigma\in\mathcal{O}}K\,p^K(1-p)^{|E|-K}}{\sum_{\sigma\in\mathcal{O}}p^K(1-p)^{|E|-K}}=K\qquad\forall p\in(0,1),$$

so Ansatz A is p-independent by proof, not by observation. What Cal-F's r (hence $p=r/(1+r)$) does move is the measure of each class, $\mu_p(\mathcal{O})=\Omega_{\mathcal{O}}\,p^K(1-p)^{|E|-K}$, and with it every ensemble read — FT.70(5)'s $\langle K\rangle_{\text{byp}}(p)$, FT.71's $\langle\rho_{D^-}\rangle_\Gamma(p)$, and the phase boundary p^*_Γ [FT.70(2)]. Class masses and class weights are different objects; only the latter flicker.

Theorem 3.63 (FT.79 — The combinatorial ceiling and the cover-hierarchy audit (second clean negative)). Thm. Under Def-H.28; the SM hierarchies of clause (4) are external measured data, marked, entering no derivation and no identification (A §0.2). (1) Ansatz A spectrum (exact). Cover masses are $m_A=2|\mathcal{C}|\ln 2/\mathfrak{h}$ [FT.67(2): $|R|=2|\mathcal{C}|$ on covers], so the nonzero values are the even ladder $K\in\{2,4,6\}$ (tetrahedron) and $\{2,4,6,8,10,12\}$ (cube); pooled, 6 distinct masses with exactly 11 pairwise ratios

$$\left\{\tfrac{6}{5}, \tfrac{5}{4}, \tfrac{4}{3}, \tfrac{3}{2}, \tfrac{5}{3}, 2, \tfrac{5}{2}, 3, 4, 5, 6\right\} \subset \mathbb{Q},$$

lightest $2\ln 2/\hbar$, heaviest $12\ln 2/\hbar$, median 7 (of the distinct values) / 8 (orbit-weighted on the cube). Every ratio is a quotient of integers ≤ 6. (2) The ceiling theorem. For any cover on any skeleton Γ: $K = 2|\mathcal{C}|$ [FT.67(2)]; admissibility gives $\sum_v k_v = 2K \leq 3V$ [CT0.11], so

$$|\mathcal{C}| \leq \tfrac{3V}{4} \text{ (bypass)}, \qquad |\mathcal{C}| \leq V \text{ (locked, centres are distinct vertices)},$$

and the widest Ansatz-A mass ratio available on Γ is $\leq 3V/4$ (bypass) or $\leq V$ (locked). Spanning $1{:}R$ therefore requires $V \geq 4R/3$ (bypass) or $V \geq R$ (locked):

R	20	207	600	900	3477	80000
$V \geq (4R/3)$	27	276	800	1200	4636	106667

against $V = 4$ (tetrahedron, max ratio 3) and $V = 8$ (cube, max ratio 6). The cover sector's dynamic range is bounded by skeleton size, exactly as the pole sector's was bounded by the defect angle [FT.76(3)] — and both bounds evaluate to ≈ 6 on the computed cells. (3) Ansatz B: no α-free hierarchy. The classes are the (K, a, b) triples with $\ln\Omega = a\ln 2 + b\ln 3$: 18 on the tetrahedron's bypass sector — 19 including its two locked cover classes, the extra one being $(K, a, b) = (6, 2, 0)$ — with $K \leq 6$, $a \leq 3$, $b \leq 1$, and 14 on the cube's cover sector with $K \leq 12$, $a \leq 4$, $b \leq 1$, so $m_B/(\ln 2/\hbar) = K - \alpha(a\ln 2 + b\ln 3)$ lies between $K - \alpha\ln|\,\mathrm{Aut}\,|$ and K. A ratio exceeding (2)'s bound can arise only where some class's mass passes through zero, i.e. at the tuned values $\alpha = K/(a\ln 2 + b\ln 3)$ — a finite set, one per class, at each of which the ratio is infinite rather than 207. The signature structure does not supply the ≥ 3 orders of magnitude required by Gap 7's residue: the informational datum is capped by $3V/2$ and the entropic one by $\ln|\,\mathrm{Aut}\,| \leq \ln 48$; what the signature supplies is degeneracy, not range. (4) The audit (external data: charged leptons 1:207:3477; up-type 1:600:80000; down-type 1:20:900). Pool $= \{2, 4, 6, 8, 10, 12\}$ (all cover masses of both skeletons, both sectors):

- pairwise, the closest cover ratio to each SM ratio is always the pool maximum 6, giving discrepancies $\times 2.7995$ (vs $m_\tau/m_\mu = 16.797$), $\times 3.333$ (vs 20), $\times 7.5$ (vs 45), $\times 22.22$ (vs 133.3), $\times 34.5$ (vs 207), $\times 100$ (vs 600), $\times 150$ (vs 900), $\times 579.5$ (vs 3477), $\times 13333.3$ (vs 80000). The best, $\times 2.7995$, is within an order of magnitude and is structurally void: it pairs our maximum with their minimum and extends to no triple.
- aligned triples, exhaustive: the best is $(2, 4, 12) \mapsto 1{:}2{:}6$ against every family, with discrepancy $\times 150$ (down-type), $\times 579.5$ (leptons), $\times 13333.3$ (up-type). No aligned triple is within a factor of 10, and the floor is exact: $900/6 = 150$.

(5) Dimension trap #2, disclosed and rejected. The amplitudes $4^{|\mathcal{C}|} \in \{1, 4, 16, 64, 256, 1024, 4096\}$ do span 3.6 orders, and produce the only near-coincidences in the whole audit: 256 vs 207 ($\times 1.2367$), 4096 vs 3477 ($\times 1.1780$), 16 vs 16.797 ($\times 1.0498$), 1024 vs 900 ($\times 1.1378$), 64 vs 45 ($\times 1.4222$). All are rejected: $4^{|\mathcal{C}|}$ is the Z-weight of FT.67(2), not a mass; no corpus tag maps an amplitude to a mass, Def-H.37(2)/Def-H.38 declare no such read, and the square-root-corrected values $(16, 64, 4, 32, 8)$ miss by $\times 4.2$–$\times 54$. Same species as the m^2-vs-m trap [FT.77(3)] — recorded because it is exactly what a forced fit would seize on. (6) Verdict. The cover hypothesis fails at the mass layer, for a structural reason dual to FT.77's and with no parameter fitted: the observable-sector masses are the even ladder $2|\mathcal{C}|\ln 2/\hbar$ capped by the cap census, so the reachable

ratio range is $\leq \max(3V/4, V)$ — six on the cube, three on the tetrahedron, against SM spans of 20–80000. The messiness is genuine but sits in the degeneracies (Ω up to 48; 264 cube cover classes; 21 signatures; chirality pairs), and degeneracy carries no hierarchy. Neither the roof nor the fuzz yields the SM spectrum on the computed skeletons; both sectors point to the same requirement — a much larger subnetwork ($V \geq 1200$ for the down-type family alone) or a mass read of a different species.

Req. Def-H.25, Def-H.28; FT.44, FT.67, FT.68, FT.69, FT.70, FT.76, FT.77, FT.78; A §0.2 (policy); CT0.11 (Part 1); CT1.20 (Part 2); cap-census bound, exhaustive triple search, exact rational ratios [math; machine]. External SM hierarchies: comparison only, marked at (4).

Pf.

$$(1):\ |R| = 2|\mathcal{C}|\ [\text{FT.67}(2)];$$

$$\text{ratios} = \{b/a:\ a < b \in \{2,4,6,8,10,12\}\}:\ \binom{6}{2} = 15 \text{ pairs},\ 11 \text{ distinct values}$$

$$(2):\ 2K = \textstyle\sum_v k_v \leq 3V\ [\text{CT0.11}:\ k_v \leq 3];$$

$$|\mathcal{C}| \leq V:\ \text{centres are distinct vertices, each contributing its two edges}$$

$$\text{ratio} \leq |\mathcal{C}|_{\max}/1; \qquad |\mathcal{C}|_{\max} \geq R \implies V \geq \tfrac{4R}{3}\ (\text{bypass}),\ V \geq R\ (\text{locked}); \qquad \text{cube}:\ \tfrac{3\cdot 8}{4} = 6\ \checkmark$$

$$(3):\ \Omega\ |\ |\operatorname{Aut}| = 2^a 3^b \implies \ln\Omega = a\ln 2 + b\ln 3;$$

$$m_B = 0 \iff \alpha = K/(a\ln 2 + b\ln 3):\ \text{finitely many tuned values}$$

$$\alpha \text{ away from them}:\ |m_B| \geq \min_{\text{classes}} |K - \alpha L| > 0:\ \text{ratios bounded by } \tfrac{K_{\max} + \alpha \ln 48}{\min |K - \alpha L|}:\ \text{no } \alpha\text{-free large ratio}$$

$$(4):\ \text{exhaustive over the pool}:\ 11 \text{ pairwise},\ \binom{6}{3} = 20 \text{ triples per family};$$

$$\text{floor}:\ 900/6 = 150 \text{ exactly}$$

$$(5):\ \text{direct evaluation of the stated pairs}; \qquad \text{rejection}:\ 4^{|\mathcal{C}|} \text{ is } Z\text{-weight } [\text{FT.67}(2)],\ \neg\text{mass read}$$

$$(6):\ \text{conjunction of } (2)\text{–}(5).\ \blacksquare$$

Phys. Ask the covers for a mass spectrum and they answer with the even ladder two, four, six, eight, ten, twelve — six notes, all ratios of small integers, exactly the kind of clean answer

the combinatorial hypothesis was meant to avoid. The reason is not that the covers are simple; there are two hundred and sixty-four of them on the cube alone, in chiral pairs, shadowing ten different cycle patterns. It is that a mass counted in crystallized edges cannot outrun the edge budget: three edges per vertex, four vertices, twelve in all. Hierarchy of the observed kind needs either a thousand-vertex subnetwork or a different question.

Theorem 3.64 (FT.80 — The bridge census: covers are amplitude-real, norm-real, and spectrally un-indexed). Thm. Under Def-H.16/Def-H.28, at (S1)–(S4): (1) What a cover class supplies, exactly. For every class $\mathcal{O}$: the real-edge set and K; $|\mathcal{C}|$ open V-threads (each centre pairs one in-edge with one out-edge [FT.67(3d)]) = transit content, bosonic-side by D1.5; the dissolved even closed chains with their exact δh [D0.6: $\delta h = \#\{e \in \gamma\}$] $= 2$ (tetrahedron) and $\in \{2, 4\}$ (cube); the aspect mask $\mathcal{A}(\max_v k_v)$ and observable mask dimension $\in \{6, 16, 17, 25\}$ [CT0.13, CT0.14]; the amplitude $4^{|\mathcal{C}|} \neq 0$ [FT.67(2)]; and the CT2.8 norm weight $\Omega \cdot 2^K$ (the class's (R, σ) count, cancellation-blind). (2) What it does not supply. Def-H.16's resolvent $Z(\mathcal{R}, \Sigma) = \langle \text{out}|(1 - \mathrm{F}_M^{(\Sigma)})^{-1}|\text{in}\rangle$ with pole $\iff$ $1 \in \operatorname{spec} \mathrm{F}_M^{(\Sigma)}$ takes as arguments a subnetwork Σ with an observer channel C (Def-H.1/Def-H.13, Def-H.16), the colour data that $\Pi_{\mathcal{C}}$ requires (FT.19/FT.27 — not determined by a crystallization pattern), and a materialization clause for whatever bound chains are to contribute poles (Def-H.16's own iff; M1/M2 species). None of the three is a function of R: no rule of Parts 1–3 maps a crystallization pattern to a (Σ, C) pair with colour assignment. (3) Census (machine, every class). Tetrahedron $7/7$ and cube $264/264$ cover classes have: transit content $|\mathcal{C}| \cdot \Omega$ threads (180 and 40 200 thread instances); bound-chain candidates exactly the even cycles (84 and 14 832 instances; every $\delta h \geq 2$, satisfying Def-H.4's floor, at rung $n(v) = 6 \geq 2$); amplitude nonzero; norm weight nonzero; and zero spectral poles constructible within the declared inventory. Covers are therefore neither amplitude ghosts nor poles: they are amplitude-real and norm-real, spectrally un-indexed. (4) The exact resolvent condition a declaration would have to supply (recorded, not adopted). Item 13's exit (b) is constructible only as a triple of declarations: (i) a rule $\mathcal{O} \mapsto (\Sigma(\mathcal{O}), C(\mathcal{O}))$; (ii) colour data making $\Pi_{\mathcal{C}}(\Sigma; C)$ computable, hence $F^{\Sigma}_{\text{strong}}$ and the mask $\mathrm{F}_M^{(\Sigma)}$ [FT.19, FT.26]; (iii) a materialization clause for dissolved cycles (M2-species), after which the pole condition is exactly $1 \in \operatorname{spec} \mathrm{F}_M^{(\Sigma(\mathcal{O}))}$ with FT.22's single-valuedness on each materialized chain. This is a interface-definition species extension of Def-H.16; it is enumerated here and not introduced. (5) Content bifurcation (the $\delta h/\delta\tau$ question). The split is structural and exact, but it is a content-type split, not a statistics theorem: every cover decomposes as $|\mathcal{C}|$ transit threads (D1.5: bosonic content) plus dissolved even closed chains (D1.5: fermionic content if materialized; Def-H.4's termini are then candidate fermion loci, their $\delta h \geq 2$ and rung ≥ 2 floors both satisfied). δh is exact per chain (2 or 4); $\delta\tau$ is a schedule datum (D0.7: retained ledger updates) that R does not determine — Cal-F declares the flicker rates $r = \Gamma_{J^+}/\Gamma_{J^-}$, not per-tick schedules — so the "finite δh with $\delta\tau = 0$ persistence" test is not evaluable from a pattern alone. The corpus contains no exchange operator and no permutation representation of particle interchange, so the spin–statistics connection remains an external reading; what is derived here is the bound/transit content split per class, and the FT.1/FT.38 flux bound $I \leq \min(\delta h/\mathfrak{h}, 4\delta\tau) \ln 2$ applies to it only once a schedule is declared. (6) Consistency with the selection side. Covers contain no materialized closed chain, hence carry no Bott datum [Def-H.24, FT.66(4)]; their content sits wholly in the transit sector, which is the one sector where index-0 stability exists on such slices [FT.48(3), FT.66(4b)] — FT.68(4)(d)'s verdict recovered at class level, and the reason the cover sector is where any future selection mechanism would have to act.

Req. Def-H.1, Def-H.4, Def-H.13, Def-H.16, Def-H.24, Def-H.25, Def-H.28; FT.1, FT.19, FT.22,

FT.26, FT.27, FT.38, FT.40, FT.48, FT.66, FT.67, FT.68, FT.78; D0.6, D0.7, D1.5, D2.6; A8; CT2.5, CT2.8 (Part 2); CT0.13, CT0.14 (Part 1); Def-H.16, Cal-F (referenced); census by exhaustive class enumeration [machine].

Pf.

$$(1):\ \text{each centre pairs one in- with one out-edge} \implies |\mathcal{C}| \text{ threads [FT.67}(3d)\text{]};$$

$$\delta h(\gamma) = \#\{e \in \gamma\}\ \text{[D0.6]}$$

$$\text{norm weight}:\ \#\{(R,\sigma):\ R \in \mathcal{O}\} = \Omega \cdot 2^{K}\ \text{[per-edge binary freedom]}; \qquad |w| = 1\ \text{[FT.67}(6c)\text{]}$$

$$(2):\ \text{arguments of } Z(\mathcal{R},\Sigma):\ (\Sigma, C, \text{colour}, \text{materialization});$$

$$R \mapsto \Sigma:\ \neg\exists \text{ rule in Parts 1–3 [inventory]}$$

$$\Pi_{\mathcal{C}} \text{ needs Colour}(\Sigma) \text{ and its sealing [FT.19, FT.27]}:\ \text{colour is slot-algebra data},$$

$$\neg\text{a crystallization datum}$$

$$(3):\ \text{exhaustive per class [machine]}; \qquad \text{amplitude} \neq 0:\ \text{FT.67}(2); \qquad \text{norm} > 0:\ \text{CT2.8},\ |w| = 1$$

$$(4):\ \text{Def-H.16's pole condition read with its own argument list;}$$

$$\text{FT.22 single-valuedness on materialized chains}$$

$$(5):\ \text{D1.5 bound/transit} = \text{closed/open}; \qquad \text{Def-H.4 floor}:\ \delta h \geq 2\ \checkmark\ (2,4),\ n(v) \geq 2\ \checkmark\ (6)$$

$$\delta\tau:\ \text{schedule datum [D0.7]}; \quad \text{Cal-F supplies the rate datum}, \neg\text{schedules};$$

$$\neg\exists \text{ exchange operator in the inventory} \implies \text{statistics external}$$

$$(6):\ \neg\exists \text{ materialized closed chain on a cover} \implies \neg\exists\ \text{ind}_B \text{ datum [FT.66}(4)\text{]};$$

$$\text{transit sector}:\ \text{FT.48}(3).\ \blacksquare$$

Phys. Ask whether the roof hangs from the fuzz, and the census answers precisely: the fuzz is real where the ledger can see it — every cover class carries nonzero amplitude and nonzero norm,

thousands of open threads, and thousands of dissolved even loops with exact hop counts — and silent where a particle is defined. A pole needs a subnetwork, an observer channel, colour sealing and a materialization clause; a crystallization pattern supplies none of the four. So the covers are not ghosts and not particles: they are the amplitude-and-norm substrate on which a declaration would have to place a spectrum. The content split they do carry is clean: everything threading is transit-side, every dissolved loop is a bound-side candidate with two or four hops, and the statistics names attached to those two sides remain vocabulary until an exchange structure is declared.

3.29 Tier F.28 — The Full-SM Completeness Audit: Particle Content, the Gauge Blocks, the Rung Ledger, and the Deficit Table

Definition 3.35 (Def-H.29 — Completeness-audit vocabulary: read classes, census statuses, the external-data clause). Def. A read class is a species of quantity the corpus can attach to a configuration, chain or subnetwork: (R1) binary registration — $\sigma_e \in \{\pm 1\}$ [CT0.9, A7/A8], the end-charge $q_{\mathrm{EM}}(c) = \frac{1}{2}(\sigma_{s(c)} - \sigma_{t(c)}) \in \{-1, 0, +1\}$ [D3.4, EM.9], the vertex phase-comparison f_v and $(-1)^{\aleph_v}$ [D3.3, FT.67(1)], the edge weight $G(e) = \sigma_e e^{i\pi\kappa_e}$ [D3.5]. (R2) integer winding and ledger — $\widehat{ds}^{\,2}$, $\Theta(\gamma)$ [D1.6, D1.7], the on-shell quantum $\kappa_e \in \mathbb{Z}$ [FT.22], the winding k and Θ_{int} [FT.25(3), FT.34, FT.43], δh [D0.6], the spin ladder $j = |k|/2$ on Sym^k carriers [CT0.7a, EW.32, FT.52, FT.55, FT.60, Def-H.38]. (R3) singlet admissibility — the sealing indicator $\mathcal{C}(\Sigma; C) \in \{0, 1\}$ and its idempotent $\Pi_{\mathcal{C}}$ [Def-H.9, Def-H.10, FT.19, CT3.18, CT3.22], with the partition rule ($|\mathrm{Colour}(\Sigma)|$ into 2s and 3s) [FT.34(5), CT3.17]. Observability is decided per aspect by FT.20's criteria table and FT.26's mask ($\dim \mathcal{K}_\Sigma \le 18$), not by block membership [CT0.12/CT0.13/CT0.14]. Census statuses. An SM entry is PRESENT iff a cited tag supplies a read for it; PARTIAL iff a read exists for part of its defining data (e.g. the representation but not the chirality); ABSENT iff no read exists — and every ABSENT is either a machine-checked inventory fact (a scan of the frozen Parts 1–2 text for the defining vocabulary) or a cited structural no-go. The completeness fraction is $(\#\mathrm{PRESENT} + \frac{1}{2}\#\mathrm{PARTIAL})/\#\mathrm{entries}$. External-data clause. Every SM datum used in this tier — field content, representation dimensions, hypercharge assignments, anomaly sums, parameter counts, the mass hierarchies of FT.77/FT.79 — is external measured or textbook data, marked at each use, entering no derivation and no definition (A §0.2; P5: $\neg\exists$ quantity identified with a measured external constant; the comparison procedure is "fix $\mathfrak{h}$ once, read P2 $\wedge$ P3, the result is a test of A9"). No species is identified with a corpus object. Rmk (citation correction). The audit proposal attributed the read classes and the independent-quanta verdict to "FT.33/FT.34". In the current numbering those are the defect bridge and the round-trip quantization; the read classes are as listed above, and the independent-quanta audit is FT.30 (with FT.31's continuum ledger), the pair-channel sector audit FT.11/FT.12, and the block decomposition FT.30's $(G_5 + W_3 + H_1) + (J_3^+ + J_3^-) + D_1^+$ tiling of the 16 registers [CT0.12, CT1.1]. Recorded so that no future pass inherits the mis-citation.

Theorem 3.65 (FT.82 — Gauge-block reconstruction (audit pass B): the algebra closes, the coupling structure does not). Thm. Under Def-H.29: (1) The register arithmetic (all derived, zero freedom [P2]): $\dim \mathfrak{H}(\mathbb{C}^4) = 16 = (7, 3, 3, 3)$ [CT1.1]; the aspect–capacity match $3{+}3{+}5{+}3{+}1{+}1 = 16 = C_{\mathrm{cap}}$ [CT0.12]; the pair-channel tiling $(G_5{+}W_3{+}H_1) + (J_3^+{+}J_3^-) + D_1^+ = 16$ [FT.30]; the mask flag $\dim \mathcal{K}_n = (6, 16, 17, 25)$ strict [CT0.14]; the observable bound $\dim \mathcal{K}_\Sigma \le 1{+}5{+}1{+}3{+}8 = 18$ [FT.26(3)]. (2) $\mathfrak{su}(2)$ closes exactly on the $W(3)$ block: with $W_i = \sigma_i/2$ over $\mathbb{Q}(i)$, $[W_i, W_j] = i\varepsilon_{ijk}W_k$ (all 9 pairs, machine), $\mathfrak{T}(W_i) = 0$, and EW.32's uniform normalization $\mathfrak{T}(\mathcal{G}_A\mathcal{G}_B) = \frac{1}{2}\delta_{AB}$

on $\{W_1, W_2, W_3, Y\}$ with $Y = \sigma_0/2$ (all 16 pairs, machine). The carrier is $\mathcal{H}_{\mathsf{v}} \cong \mathbb{C}^2$ — a genuine doublet read [EW.32]. (3) $\mathfrak{su}(3)$ closes exactly on the colour algebra: in the matrix-unit (root-space) basis $\{E_{12}, E_{21}, E_{13}, E_{31}, E_{23}, E_{32}, H_1, H_2\}$ over $\mathbb{Q}$ — all 8 traceless, all 64 commutators in the span, the Jacobi identity on all 8^3 triples, $[H_1, H_2] = 0$, the A_2 root system $\{(2,-1),(1,1),(-1,2)\}\cup(-)$ in the (H_1, H_2) basis, and a nondegenerate Killing form (machine, exact rationals). dim $= 8 = 3^2-1$ [CT3.8], $N_{\text{gluon}} = 8$ with no ninth mode — the $\mathfrak{u}(3)$ centre is allocated to F_{EM} [CT3.13] — and $N_{\text{colour}} = 3 = \log_2 C_{\text{cap}} - 1$ is derived from A2 + CT0.9, not chosen [CT3.20]. FT.58's hosting identity $(\lambda_1 + i\lambda_2)/2 = E_{12}$ re-verified over $\mathbb{Q}(i)$. (4) The electroweak identities hold as exact polynomial identities in $\mathbb{Q}[g, g', s]/(s^2 - g^2 - g'^2)$ (machine): EW.20's mass matrix in (W_1, W_2, W_3, Y) has $W_{1,2}$ decoupled and a neutral block of rank one — det $= 0$ exactly, tr $= g^2 + g'^2 =: m_Z^2$ (times $\mu_\star^2/4$ [EW.33]), null vector (g', g), massive vector $(g, -g')$ [EW.23]; hence the photon is exactly massless as the null direction and $A = \sin\theta_W W_3 + \cos\theta_W Y$, $Z = \cos\theta_W W_3 - \sin\theta_W Y$ [D3.10, EW.24] with $\cos\theta_W = g/s$; $m_W = m_Z \cos\theta_W$ [P2] and $e = gg'/s = g\sin\theta_W = g'\cos\theta_W$, $1/e^2 = 1/g^2 + 1/g'^2$ [EW.35] — all verified with denominators cleared. The seed satisfies $W_3\mathsf{v} = -\frac{1}{2}\mathsf{v}$, $Y\mathsf{v} = +\frac{1}{2}\mathsf{v}$ [EW.20] and $T_a\mathsf{v} = 0$ for $a = 1..8$ [EW.32] — the vev is a weak doublet component and a colour singlet, exactly the SM pattern. (5) The U(1) direction lives in EW.8 (the abstract generator Y with three realizations: slot, aspect, ...), D3.10/EW.24 (the mixing), and CT0.14's dim $\mathcal{K}^{(\text{EM})} = 1$; EM.20 fixes the phase coefficient $\alpha_{\text{EM}} = \pi$ (a pure number under Comm-min + Eff), which is not the fine-structure constant — the coupling is $e = g\sin\theta_W$ [EW.35], a P4-dependent quantity whose value is not derived. (This is the standing trap recorded in FT.73(5): comparing π to $1/137$ is a category error.) (6) Independent quanta: the SM's particle–boson picture is not reproduced. FT.30's audit stands at block level: $G(5)$ is an independent quantum only as a joint geometric read (winding $\kappa, k \in \mathbb{Z}$; unit $\mathfrak{h}^2 = 4G_N \ln 2$ [CT2.9/P3]); $W(3)$ is a candidate spin quantum with extraction closed under Def-H.38 [FT.55/FT.56/FT.60]; $H(1)$ is not an independent quantum (a continuous configuration datum, read only through the mass fold [Def-H.36, FT.20's H row]); $J^\pm$ are structural and ¬observable [FT.20]; D^+ is uniform [D4.3]. So gauge bosons emerge as aspect-block reads with per-aspect observability criteria [FT.20], not as quanta of a gauged symmetry acting on chiral matter. (7) Two structural mismatches, derived not assumed. (a) No gauged chiral coupling: the blocks act on the 16-dimensional pair-channel register [CT0.6/CT0.12] and on slot algebras [D3.1, CT3.20]; nothing in Parts 1–2 couples a gauge block to a chiral fermion multiplet, because no chiral fermion multiplet exists [FT.81(5)]. The SM's defining feature — SU(2) acting on left-handed doublets only — has no read. (b) The scalar block is 1-dimensional: dim $F_H = 1$ [CT0.12, derived: all 16 directions allocated] against the SM Higgs doublet's 4 real dof; the counting of the Higgs mechanism is nonetheless right (rk $M^2 = 3$: three massive vectors, one massless), because the breaking is carried by $\mu_\star^2 \propto h_0^2$ [EW.33] rather than by four scalar fields. (8) The colour ceiling is the SM's own largest factor. Slot algebras are realized only for $n \le 3$ [CT3.20/CT4.4, P2's $n_{\max} = 3$], so $\mathfrak{su}(3)$ is simultaneously the hosting ceiling and the realization ceiling [FT.58/FT.59]; at $n \ge 4$, D^- carries no gauge structure at all — no nonabelian and no abelian algebra, no generator contributed [CT4.4, derived from CT0.12's full allocation]. Hence the corpus cannot host a gauge factor larger than $\mathfrak{su}(3)$, which is exactly the SM's largest — a structural coincidence worth recording, and a hard bound against any GUT-scale reading.

Req. Def-H.29; A2, A8; C7, C8; D2.3, D3.1, D3.2, D3.10, D4.1, D4.3; CT0.5–CT0.14 (Part 1); CT1.1, CT1.2, CT1.10–CT1.12, CT2.9 (Part 2); CT3.2, CT3.7–CT3.9, CT3.13, CT3.17–CT3.22 (Part 2); CT4.1, CT4.4 (Part 2); EW.8, EW.13, EW.20, EW.22–EW.24, EW.27, EW.32, EW.33, EW.35 (Part 2); EM.9, EM.17–EM.20 (Part 2); FT.11, FT.12, FT.19, FT.20, FT.26, FT.27, FT.30, FT.31, FT.52–FT.60; P2, P3, P4; Pauli/Gell-Mann basis facts, root-space and Killing-form criteria, rank-one matrix facts [math; machine].

Pf.

$$(1):\ \text{each sum asserted exactly [machine]};$$

$$7+3+3+3=16=3+3+5+3+1+1=(5+3+1)+(3+3)+1$$

$$(2):\ \sigma_i\sigma_j=\delta_{ij}+i\varepsilon_{ijk}\sigma_k\implies[\tfrac{\sigma_i}{2},\tfrac{\sigma_j}{2}]=i\varepsilon_{ijk}\tfrac{\sigma_k}{2};$$

$$\mathfrak{T}(\tfrac{\sigma_A}{2}\tfrac{\sigma_B}{2})=\tfrac{1}{4}\cdot 2\delta_{AB}=\tfrac{1}{2}\delta_{AB}$$

$$(3):\ [E_{ij},E_{kl}]=\delta_{jk}E_{il}-\delta_{li}E_{kj};$$

$$[H_1,H_2]=0;$$

$$[H_i,E_\alpha]=\alpha(H_i)E_\alpha;\quad\text{span closure by exact coordinate solve}$$

$$\text{Killing}:\ K_{ab}=\textstyle\sum_c \mathrm{tr}(\mathrm{ad}_{E_c}\mathrm{ad}_{E_{\bar{c}}})\ \text{form};\ \det\neq 0\implies\text{semisimple [machine, exact Fractions]}$$

$$(\lambda_1+i\lambda_2)/2=(E_{12}{+}E_{21}+i(-iE_{12}{+}iE_{21}))/2=E_{12}\ \checkmark\ \text{[FT.58]}$$

$$(4):\ M^2_{\text{neutral}}=\tfrac{\mu_\star^2}{4}(g,-g')^{\mathsf{T}}(g,-g'):\ \det=0,\ \mathrm{tr}=g^2+g'^2;$$

$$M^2(g',g)^{\mathsf{T}}=0;$$

$$M^2(g,-g')^{\mathsf{T}}=(g^2+g'^2)(g,-g')^{\mathsf{T}}$$

$$m_W^2s^2=g^2(g^2+g'^2)=m_Z^2g^2\implies m_W=m_Z\cos\theta_W;\qquad e^2s^2=g^2g'^2\iff\tfrac{1}{e^2}=\tfrac{1}{g^2}+\tfrac{1}{g'^2}$$

$$W_3\mathsf{v}=-\tfrac{1}{2}\mathsf{v},\ Y\mathsf{v}=+\tfrac{1}{2}\mathsf{v}:\ \text{direct } 2\times 2\ \text{multiplication over } \mathbb{Q}(i);\qquad T_a\mathsf{v}=0\ \text{[EW.32]}$$

(5)–(8) : each row cites its tag; $(7a)$: inventory scan [FT.81(5)]; $(7b)$: CT0.12 allocation

(8) : CT4.4 Pf : (a) 16 pair directions fully allocated $\implies\neg\exists$ direction for D^-;

(b) slot algebras only n

≤ 3. ■

Phys. Ask whether the sixteen registers rebuild the Standard Model's gauge group, and the arithmetic says yes in a way that is not a naming trick: an eight-dimensional colour algebra that closes with the right root system and no ninth gluon, a three-dimensional weak algebra whose doublet carries the vacuum exactly where the Standard Model puts it, and a neutral block of rank one whose null direction is the photon — with $\sin\theta_W$, $\cos\theta_W$ and e falling out as polynomial identities rather than fits. Ask instead whether those blocks act on matter the way a gauge group does, and the answer is no: there is no chiral multiplet for them to act on, the scalar block is one-dimensional instead of four, and the quanta are aspect reads with observability criteria, not gauge bosons coupled to fermions. The gauge skeleton is derived; the coupling to chiral matter is absent.

3.30 Tier F.29 — Gravitational Reads in Three States, the Flicker Ripple, and the Tumbling-Model Adjudication

Definition 3.36 (Def-H.30 — The three shape states). Def. For a skeleton Γ [Def-H.25] with potential degrees $n(v)$: (S0) VV / all-virtual. $R = \varnothing$: every potential edge virtual, $n_{\mathcal{R}} = 0$, every pair at $\rho_\circ = \frac{1}{4}I$ [A5, CT0.2]. The shape-derived vacuum. (S1) flicker / bypass. $R \neq \varnothing$ admissible: $k_v \leq 3$ at every vertex [CT0.11], i.e. the bypass sector of Def-H.25, with its ensemble reads at p [Def-H.26, Cal-F] and its cover sub-sector [FT.67(2)]. (S2) D^- trapped. Some vertex locked ($k_v \geq 4$), up to the fully crystallized floor $R = E_{\text{dir}}(\Gamma)$: content absorbed at rate $\propto O(n)$ [CT4.1/D4.2], retained in $\mathfrak{L}$ and not accessible [D1.11/CT4.3], with no accessible pole [CT4.2]. Rmk (status). The three states are the proposal's Punnett vocabulary (VV / RV,VR / RR) mapped onto corpus objects; the mapping is recorded, not adopted as a definition of anything physical. S1 and S2 are Def-H.25's own sectors; S0 is CT0.2's vacuum restricted to a skeleton's potential-edge set.

Theorem 3.66 (FT.85 — Vacuum census, baseline equality, and the constructible gravitational reads). Thm. Under Def-H.30: (1) Vacuum census. The corpus declares exactly three scalar extremes and proves there is no fourth: the Janus vacuum $n_{\mathcal{R}} = 0$ with $(S_{\text{raw}}, S_{\mathfrak{L}}) = (\ln 4$ per pair (max, argmax unique $= \frac{1}{4}I), 0)$ [CT0.2, CT4.7(1)]; D^+ with $(0,0)$ — a pure, uniform, non-configuration-dependent scalar [CT4.7(2), D4.3], $w = -1$ [CT1.25]; D^- locked with $(\min_{\text{acc}}, \max \wedge\neg\text{accessible})$ [CT4.7(3), D1.11, CT4.2, CT4.3]; and $\neg\exists$ a fourth extreme [CT4.10], F_H being a dynamical field [EW.27]. (2) Baseline equality (proven, not measured). For every shape S: $S_{\mathfrak{L}}(\text{VV}(S)) = n_{\mathcal{R}} \ln 2 = 0 = S_{\mathfrak{L}}(V_{\text{raw}})$ [CT1.29]; $\Delta S_{\text{raw}} = 0$ on both sides [CT0.2: every pair at the argmax]; $m_{\text{hid}} = S_{\mathfrak{L}}/\mathfrak{h} = 0$ [FT.44(4)]; $\rho_{D^-} = 0$ [CT4.5's iff]; and CT1.29's $S_{\partial\mathcal{B}}$ has no argument (no closed real-edge surface). Hence

$$m_g^2\big(\text{VV}(S)\big) = m_g^2\big(V_{\text{raw}}\big) \quad \forall S,$$

on every constructible read, and $\Delta S_{\text{vacuum polarization}}(S) \equiv 0$. The reason is registration, not arithmetic: virtual pairs carry no registration [A7/D0.5: ϕ_e is defined on $e \in E_{\mathcal{R}}$; A5: the sole ledger exclusion is the Haar draw; Def-H.25(S4): virtual edges weight 1, no phase]. Honest remark:

the absolute raw entropy is $|\text{pairs}| \cdot \ln 4$ and is shape-dependent; the deficit is shape-independent and zero. The raw vacuum's mass reads vanish from three independent derived reads — nothing is set to zero by hand. (3) $m_g^2 := \Delta S$ is not constructible. (a) The inventory contains no inertial/gravitational mass distinction and no equivalence theorem (`inertial`, `Lyapunov`, `maximal mixing`, `equivalence principle`, `Theorem 11`: count 0 in Parts 1–2); (b) ΔS_{raw} of a registered configuration is bounded but undetermined — A4/D0.4 force a real pair strictly below $\ln 4$ and no tag declares its ρ, so $\Delta S_{\text{raw}} \in (0, n_{\mathcal{R}} \ln 4]$ with no value; (c) the dimensionally correct entropy-based mass^2 density is $S/A = \ln 2/\mathfrak{h}^2$ [CT1.29/CT2.9], a state-independent constant [FT.37: corrected by $1/\bar{\kappa}$], so state-dependence can enter only through A or $\bar{\kappa}$. (4) The constructible reads, used for every table below. (i) $m_{\text{hid}} = S_{\mathfrak{L}}/\mathfrak{h} = E_{\text{int}} \ln 2/\mathfrak{h}$ — the universal taxonomy law [FT.44(4)] — with $m_{\text{hid}}^2 = E_{\text{int}}^2 (\ln 2)^2/\mathfrak{h}^2$, dimension $\mathfrak{q}$ [D2.9/CT2.1]; (ii) the locked sector's stress-energy $T_{\mu\nu}^{(D^-)} = \rho_{D^-} u_\mu u_\nu$, $p = 0$, $w = 0$, $\rho_{D^-} = \sum_{v \text{ locked}} O(n_v) \geq 0$ [CT4.1/CT4.5] — the corpus's one derived gravitational source; (iii) $S_{\partial\mathcal{B}} = A \ln 2/\mathfrak{h}^2 = A/(4G_N)$ with $A = n_{\mathcal{R}}(\partial\mathcal{B})\mathfrak{h}^2$ and $G_N = \mathfrak{h}^2/(4 \ln 2)$, $\kappa^2 = 8\pi G_N$ derived [CT1.29/CT2.9/P3], bounded by CT4.9 (≤ 3 bits per accessible vertex; $S_{\text{acc}}(\mathcal{B}) \leq A/\mathfrak{h}^2$ bits); (iv) the pole side $m_{\text{pole}}^2 = 4\hat{\mu}'^2 \sin^2(\Theta_R/4)$ under Def-H.37(2) [FT.54/FT.76] — which is also what a two-patch field-source read reduces to on the cone metric [Def-H.21/FT.47], i.e. clause (iii) of the audit's B3 is already bounded [FT.76(3)]. (5) The source criterion, exact. $\rho_{D^-} > 0 \iff \exists v : k_v \geq 4$ [CT4.5's iff] — machine-exhaustive over the tetrahedron's 4096 and P_4's 256 patterns. Consequences: every S1 configuration has exactly zero dark-dust stress-energy, and only S2 states are dust sources; those are retained-but-inaccessible [CT4.2/CT4.3/D1.11], amplitude-zero in the literal domain [FT.67(3a)] and revived only in the chain-local domains [FT.75(2)]. So the flicker gas does not gravitate through D^- at all; its only mass read is informational. (6) The state table (14 shapes; $\langle K \rangle$ at $p = \frac{1}{2}$ reproduces FT.70(5)/FT.68(1) verbatim):

	$m_{\text{hid}}\,\mathfrak{h}/\ln 2$			ρ_{D^-}		
shape	S0	S1 $\langle K \rangle(\frac{1}{2})$	S2 E_{int}	S0	S1	S2
digon	0	1	2	0	0	0 ($n(v) = 2$ never locks)
$P_3..P_{10}$	0	$\frac{8}{3}, \frac{248}{69}, \frac{130}{29}, \frac{296}{55}, \frac{2618}{417}, \frac{11344}{1581}, \frac{896}{111}, \frac{13588}{1515}$	$2n$	0	0	$16n$
tetra	0	$\frac{852}{205}$	12	0	0	448
cube	0	$\frac{3311952}{395023}$	24	0	0	896
octa	0	$\frac{2124600}{318523}$	24	0	0	2976
dodeca	0	$\frac{400749109393569 0}{191727103387307}$	60	0	0	2240
icosa	0	$\frac{215801166207}{15519497071}$	60	0	0	24384

with $S_{\mathfrak{L}} = (\text{that row}) \cdot \ln 2$, $A/\mathfrak{h}^2 = E_{\text{int}}$ in S2, and per-sector entropies not available (clause (7)). (7) No per-sector ΔS_A. Four cited reasons: no declared state Ψ_X for a crystallization pattern, so $\|F_A \Psi_X\|^2$ has no argument; FT.30's audit leaves only the G block with a geometric read and marks it joint ($\neg$ per-block); H, $J^{\pm}$, D^+ are explicitly not independent quanta [FT.30]; CT0.14/FT.26 supply block dimensions $(6, 16, 17, 25)$ and the ≤ 18 observable bound, which is all the sector map can carry. The only declared per-register decomposition is $S_{\mathfrak{L}} = K \ln 2 = \sum_{e \in R} \ln 2$ [CT0.9: one real edge = one bit] — machine-verified per configuration.

Req. Def-H.25, Def-H.26, Def-H.30; FT.16, FT.30, FT.37, FT.44, FT.47, FT.54, FT.63, FT.67, FT.70, FT.71, FT.75, FT.76; A4, A5, A7; D0.5, D1.11, D4.2, D4.3; CT0.2, CT0.9, CT0.11, CT0.12, CT0.14 (Part 1); CT1.22, CT1.25, CT1.29, CT2.1, CT2.8, CT2.9 (Part 2); CT4.1–CT4.5, CT4.7, CT4.9, CT4.10 (Part 2); P1, P3, P5; Gibbs lemma [math]; exhaustive pattern enumeration [machine].

Pf.

$$(1):\ \text{CT4.7's three extremes with CT4.10's exhaustiveness;}$$

$$(2):\ n_{\mathcal{R}}=0\implies S_{\mathfrak{L}}=0\ [\text{CT1.29}];\ \rho_{pq}=\tfrac{1}{4}I\ \forall(p,q)\ [\text{A5, CT0.2}]$$

$$\text{registration}:\ \phi_e \text{ defined on } e\in E_{\mathcal{R}}\ [\text{A7, D0.5}];\quad \text{virtual weight } 1,\ \text{no phase [Def-H.25(S4)]}$$

$$(3a):\ \text{inventory scan;}$$

$$(3b):\ \text{A4/D0.4}:\ \text{real}\iff|\Pi_{\mathcal{V}}\rho|>\varepsilon_q:\ \text{strict sub-maximality, no declared } \rho$$

$$(3c):\ S/A=\ln 2/\mathfrak{h}^2\ [\text{CT1.29}+\text{CT2.9}:\ A/(4G_N)=A\ln 2/\mathfrak{h}^2];$$

$$\text{FT.37}:\ S_{\text{reg}}/A_{\text{loop}}=(\ln 2/\mathfrak{h}^2)/\bar{\kappa}$$

$$(4):\ [\text{FT.44}(4)]:\ m_{\text{hid}}=E_{\text{int}}\ln 2/\mathfrak{h}\ \text{on every computed cell};\qquad [\text{CT4.5}]:\ T^{(D^-)}=\rho uu,\ p=0$$

$$(5):\ \text{CT4.5 iff}:\ \rho_{D^-}=0\iff\neg\exists v:\ n(v)\geq 4;$$

$$\text{S1 admissible}\implies k_v\leq 3\ \forall v\ [\text{CT0.11}]\implies\rho_{D^-}=0$$

$$(6):\ \langle K\rangle_{\text{byp}}(\tfrac{1}{2})=\frac{\sum_K Kc_K2^{-2E}}{\sum_K c_K2^{-2E}}\ [\text{cap census, exact}];$$

$$\rho_{D^-}(\text{floor})=V\cdot O(2d)\ [\text{FT.63}(5)]$$

$$(7):\ \text{the four cited obstructions};\qquad S_{\mathfrak{L}}=\sum_{e\in R}\ln 2\ [\text{CT0.9}].\ \blacksquare$$

Phys. Ask the corpus for the weight of empty space and it answers with a proof rather than a parameter: the vacuum is a state of maximal raw information with no ledger content at all, and a shape whose edges are all still virtual is that same vacuum — not a slightly different one. Mass enters only where edges crystallize, one bit each, and gravity's own coupling is fixed by the single scale. The trapped sector is the only place with a derived stress-energy: pressureless dust that curves geometry while remaining, by theorem, out of reach. The flickering gas between the two carries information and amplitude but sources no dust whatsoever.

Theorem 3.67 (FT.86 — The flicker ripple: ultralocal raw dynamics, cap-induced alternation, and the quantum–classical gradient). Thm. Under Def-H.26/Def-H.30 at $p = \frac{1}{2}$ (every value exact; general p by the same census): (1) The raw flicker has no spatial ripple. μ_p is a Bernoulli product measure on directed edges [Def-H.26], so every connected edge–edge correlation vanishes identically: $\langle x_e x_{e'}\rangle = p^2 = \langle x_e\rangle\langle x_{e'}\rangle$ for $e \neq e'$ (machine-exhaustive on P_3, P_4). Any waviness in the flicker geometry is constraint-induced, not dynamical. (2) The admissible flicker carries an exact period-2 alternating ripple. Conditioning on admissibility ($k_v \leq 3$, CT0.11) couples neighbours; the connected correlations, by true ring distance (the two directions of one undirected edge are distance 0 — a digon — not distance 1), are exact rationals alternating in sign:

	$d=0$	$d=1$	$d=2$	$d=3$	$d=4$	$d=5$
P_4	$-\frac{133}{4761}$	$-\frac{64}{4761}$	$+\frac{28}{4761}$			
P_8	$-\frac{71068}{2499561}$	$-\frac{33124}{2499561}$	$+\frac{6928}{2499561}$	$-\frac{1504}{2499561}$	$+\frac{604}{2499561}$	
P_{10}	$-\frac{1631509}{57380625}$	$-\frac{760384}{57380625}$	$+\frac{158716}{57380625}$	$-\frac{33184}{57380625}$	$+\frac{7216}{57380625}$	$-\frac{2884}{57380625}$

Short-distance values converge to L-independent limits ($d{=}0 \to -0.0284331$, $d{=}1 \to -0.0132516$): the cap makes neighbouring crystallizations anticorrelate and next-nearest correlate. (3) The decay rate is the corpus's own transfer matrix. With charpoly $T(1) = x^2 - 3x - 3$ [FT.68(2)/FT.70(3)], $\lambda_{1,2} = (3 \pm \sqrt{21})/2$ and

$$\frac{\lambda_2}{\lambda_1} = \frac{\sqrt{21}-5}{2} = -0.2087122\ldots \in \mathbb{Q}(\sqrt{21}),$$

$$\xi = -1/\ln|\lambda_2/\lambda_1| = 0.6382 \text{ directed edges},$$

matching the measured ratios $C(2)/C(1) = -\frac{39679}{190096} = -0.2087314$ (P_{10}, 4 significant figures) and $C(3)/C(2) = -\frac{8296}{39679} = -0.2090779$ (3 s.f.); P_9 gives $-\frac{2759}{13225} = -0.208620$. The negative sign of λ_2/λ_1 is the alternation. Deviations grow near the antipode ($d \approx L/2$) as finite-size wrap. (4) Exact moments from the cap census (never a Bernoulli proxy): $\langle K\rangle$, $\langle K^2\rangle$, σ_K^2 for all 14 shapes, with $\sigma_K^2/\sigma_{K,\text{ind}}^2$ (the cap factor, $\sigma_{\text{ind}}^2 = 2Ep(1-p) = E/2$) equal to 1 for the digon (the cap never binds at $n(v) = 2$), $\to 0.7002$ for large polygons, and 0.155–0.373 for the solids (icosa 0.1547, octa 0.2228). Anchors: digon $\langle K\rangle = 1, \langle K^2\rangle = \frac{3}{2}, \sigma^2 = \frac{1}{2}$; P_3: $\frac{8}{3}, \frac{73}{9}, 1$. (5) The ripple in the mass reads. $\Delta S = S_{\mathfrak{L}} = K\ln 2 \Rightarrow \delta(\Delta S) = \sigma_K \ln 2$ exactly; $\delta(m_{\text{hid}})/m_{\text{hid}} = \sigma_K/\langle K\rangle$ and $\delta(m_{\text{hid}}^2)/m_{\text{hid}}^2 = 2\sigma_K/\langle K\rangle$ (the factor 2 exact). The raw-vacuum and VV baselines give identical $\delta(m_g^2)$ — by FT.85(2)'s proof, not by numerical agreement. (6) Quantum–classical gradient. $\sigma_K/\langle K\rangle$ decreases monotonically from the digon's $1/\sqrt{2} = 0.707107$ (exact) through P_3's $3/8 = 0.375$ (exact) to the icosahedron's 0.10955; the $1/\sqrt{\langle K\rangle}$ law holds with a family-dependent coefficient $\sigma_K/\sqrt{\langle K\rangle}$: $1/\sqrt{2}$ (digon), 0.6124 (P_3), 0.6248 converged (P_4–P_{10}), 0.41–0.52 (solids) — the deviation from the binomial $1/\sqrt{2}$ being exactly the cap factor. Near-pure-wave limit: $\sigma_K/\langle K\rangle = 1$ requires a single directed edge at $p = \frac{1}{2}$ — amplitude-zero [FT.67(3b): cov $= 0$], hence not a bound shape. No shape in the inventory is near-pure-wave. (7) No frequency is constructible; one period is derived. the Cal-F rate datum is the ratio $r = \Gamma_{J^+}/\Gamma_{J^-}$ and per-edge detailed balance, and records that the corpus supplies no $J^\pm$ rates, so $f_{\text{ripple}} = \Gamma_{\text{tot}}/2\pi$, $\tau_c = 1/\Gamma_{\text{tot}}$ are reported in units of the undeclared total rate and no absolute frequency is claimed. Rate-free and derived: one $J^\pm$ event changes $(n_{\mathcal{R}}, \delta h, \delta\tau)$ by $(\pm1, \pm1, \pm1)$ [CT0.3], so in ledger-tick units $\tau_c = 1$ tick and the Lorentzian half-width is 1 rad/tick; propagation obeys $\delta\tau \geq \delta h/4$ (at most 4 hops/tick) [FT.1/FT.38, CT0.6, EM.5–7]. Leading order (single-flip granularity) the cap correlation renormalizes the Lorentzian amplitude by the cap

factor and leaves the width at Γ_{tot}; the full constrained spectrum is multi-timescale and not constructible without rates. (8) Self-modulation is not constructible. A tick-rate/field coupling $r_q(g_q)$ has no referent: `clock`, `dilation`, `tick rate`, `r_q`, `g_q`, `local rate`, `running coupling`, `beta function` all count 0 in Parts 1–2. The corpus has the conformal factor $\Omega = e^{\alpha_\Omega a_{00}}$ and the shift $N_i = \beta w_i$ [CT1.12], and $\delta\tau$ is defined as retained ledger updates along real edges [D0.7], but nothing couples the two. Logged as a gap (a new interface-definition species), not adopted. (9) The trapped sector does not ripple. CT4.5's own Pf: locked ⇒ no accessible pole [CT4.2] ⇒ no propagating mode ⇒ $\mathsf{E}_{\text{tr}} = 0 \Rightarrow T^{\text{tr}} = 0$ — S2 states are non-oscillatory by theorem (pure dust, $p = 0$, $w = 0$).

Req. Def-H.26, Def-H.30; FT.1, FT.38, FT.67, FT.68, FT.70, FT.85; D0.7, D1.2; CT0.3, CT0.6, CT0.11 (Part 1); CT1.22, CT1.29 (Part 2); CT4.2, CT4.5 (Part 2); EM.5–EM.7 (Part 2); CT1.12; Cal-F; transfer-matrix spectra, exhaustive enumeration [math; machine].

Pf.

$$(1):\ \mu_p = \bigotimes_e \text{Bernoulli}(p)\ \text{[Def-H.26]} \implies \langle x_e x_{e'}\rangle = p^2\ \forall e \neq e'\ \text{[machine-exhaustive]}$$

$$(2):\ \text{conditioning on } \{k_v \leq 3\}\ \text{[CT0.11]}:\ \langle\cdot\rangle_{\text{byp}}$$

$$= \langle\cdot$$

$$1_{\text{adm}}\rangle / B_\Gamma(p):\ \text{exact rationals by enumeration}$$

$$(3):\ \det(xI - T(1))$$

$$= x^2(x^2 - 3x - 3)\ \text{[FT.68(2)]};$$

$$\lambda_2/\lambda_1$$

$$= (3 - \sqrt{21})/(3 + \sqrt{21})$$

$$= (\sqrt{21} - 5)/2 \text{ in } \mathbb{Q}(\sqrt{21})$$

$$(4):\ \text{cap-DP census rows [FT.68(3), FT.69(1)]}; \qquad \sigma^2 = \langle K^2\rangle - \langle K\rangle^2 \text{ exact}$$

$$(5):\ \Delta S = K \ln 2\ \text{[CT1.29] linear in } K; \qquad m_{\text{hid}}^2 \propto K^2 \implies \text{relative ripple} \times 2$$

$$(6):\ \text{binomial reference } \sigma/\sqrt{\langle K\rangle} = \sqrt{1-p} = 1/\sqrt{2}; \qquad \text{cap factor} = \sigma^2/\sigma_{\text{ind}}^2$$

$$(7):\ \text{CT0.3}:\ J^{\pm} = (\pm 1, \pm 1, \pm 1); \qquad \text{FT.1/FT.38}:\ I \leq \min(\Delta\ell/\mathfrak{h}, 4\delta\tau)\ln 2$$

(8) : inventory scan; (9) : CT4.5 Pf verbatim. ■

Phys. A flickering shape does not shimmer the way intuition suggests. Edge by edge its raw dynamics is completely uncorrelated — there is no wave in the dynamics at all. The waviness is created by the budget: because no vertex may hold more than three crystallized edges, a real edge here forbids one there, and the forbiddenness propagates around the ring with an exact alternation, negative–positive–negative, dying away by a factor of one fifth per step. That single number is the same eigenvalue ratio the corpus derived for its polygon transfer matrix years earlier, now showing up as a physical ripple. Smaller shapes ripple harder, and the ripple fades as the shape grows — a quantum-to-classical gradient with no parameter in it. What no shape achieves is pure wave: that would take a single edge, and a single edge is nothing at all.

FT.87 — The tumbling-model adjudication (external proposal, row by row). Structural-constraints record: see Appendix B, Section B.2.4 of the Addendum.

Theorem 3.68 (FT.88 — The involution algebra of the two declared reversals). Thm. Let ω [D2.6] and $\Theta_{\mathfrak{L}}$ [D2.7] be the corpus's two declared involutions. (1) ω on patterns. ω reverses orientation: $s(c) \leftrightarrow t(c)$, charge $\mapsto -$charge, chirality $\mapsto -$chirality [D2.6]. On a skeleton it acts as $R \mapsto R^{\mathrm{T}} = \{(b,a) : (a,b) \in R\}$, with $\omega^2 = 1$ (machine: on all patterns sampled and on the tetrahedron's full orbit set), and $\omega \notin \mathrm{Aut}(\Gamma)$ — no vertex permutation induces it [FT.78(6)], where it was shown to exchange the two $\Omega = 24$ dissolved-digon cover classes. (2) $\Theta_{\mathfrak{L}}$ on registrations. $\Theta_{\mathfrak{L}}$ reverses ledger order with $J^+ \leftrightarrow J^-$ and $G(e) \mapsto G(e)^*$ [D2.7], i.e. σ_e fixed and $\kappa_e \mapsto -\kappa_e$ on the edge weight $G(e) = \sigma_e e^{i\pi\kappa_e}$ [D3.5]; $\mathcal{J}(C) \mapsto \mathcal{J}(C)^*$ [D3.6]. (3) The algebra. ω acts on the edge set and $\Theta_{\mathfrak{L}}$ on the phase/ledger order: disjoint data, hence they commute, and

$$\langle \omega, \Theta_{\mathfrak{L}} \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$$

on skeleton registration data, with the four classes $(1,\ \omega,\ \Theta_{\mathfrak{L}},\ \omega\Theta_{\mathfrak{L}})$. ω is charge-odd [D2.6; FT.40's ω-covariance of q_{core}]; $\Theta_{\mathfrak{L}}$ is charge-even and phase-odd. (4) Scope, and what this is not. The inventory contains no C, P or T operator, no Lorentzian field algebra and no spinor bundle, so no CPT theorem is claimed or claimable; $\mathbb{Z}_2 \times \mathbb{Z}_2$ is the exact group generated by the two declared reversals and nothing more. In particular ω is not a chirality projector: it is an orientation reversal whose fixed-point set is empty on chiral pairs [FT.78(6)], whereas a projector needs an eigenspace decomposition of a state space that the corpus does not supply for patterns (application-layer content). The proposal's $\chi = \mathrm{sign}(k)$ is the ω-odd coordinate of (3): a legitimate C-odd ledger read, still not a projector.

Req. D2.6, D2.7, D3.5, D3.6; FT.40, FT.64, FT.78; CT0.9 (Part 1); group generation, exhaustive pattern enumeration [math; machine].

Pf.

$$(1) : \ (R^{\mathrm{T}})^{\mathrm{T}} = R;$$

$$\omega \in \mathrm{Aut} \iff \exists p : \ (p(a), p(b)) = (b,a)\ \forall (a,b) : \ p(a) = b \wedge p(b) = a\ \forall \text{ pairs} : \ \perp \text{ for } |V| \geq 3$$

$$\text{machine} : \#\{p \in \mathrm{Aut} : \ p \text{ induces the reversal}\} = 0 \text{ on the tetrahedron}$$

$$(2) : \ \text{D2.7 verbatim}; \qquad G(e)^* = \sigma_e e^{-i\pi\kappa_e} \ [\sigma_e \in \{\pm 1\} \text{ real, CT0.9}]$$

$$(3) : \ \text{disjoint supports} \implies [\omega, \Theta_{\mathfrak{L}}] = 0; \qquad \omega^2 = \Theta_{\mathfrak{L}}^2 = 1 \implies \mathbb{Z}_2 \times \mathbb{Z}_2$$

$$(4) : \ \text{inventory scan (no C/P/T operator)};$$

$$\text{projector needs } \Psi \text{ eigenspaces} : \ \neg\exists \text{ declared } \Psi_X \ [\text{FT.85(7)}]. \ \blacksquare$$

Phys. The corpus has exactly two mirrors: one flips every arrow in the graph, the other runs the ledger backwards and conjugates every phase. They do not interfere — one acts on edges, the other on phases — so together they make the smallest possible non-cyclic group, four elements. That group already knows which cover shapes are left-handed and which are right-handed, and it already knows that charge changes sign in the first mirror. It is not the CPT of a field theory, and it is not a chirality projector; it is the exact amount of reversal structure the axioms declare, no more.

3.31 Tier F.30 — The Koide Adjudication, the Resolution Ladder, and the Minimal Bridge Declaration

FT.89 — The Koide/phase-boundary adjudication: exact arithmetic, category trap. Structural-constraints record: see Appendix B, Section B.1.13 of the Addendum.

FT.90 — The resolution ladder: no-hair from the channel bound, and the chiral merger refuted. Structural-constraints record: see Appendix B, Section B.2.5 of the Addendum.

Theorem 3.69 (FT.91 — The minimal spectral-bridge form). Thm. Under Def-H.16/FT.80(4): (1) The three declarations restated. Item 13's exit (b) requires exactly: (i) a rule $\mathcal{O} \mapsto (\Sigma(\mathcal{O}), C(\mathcal{O}))$ assigning a subnetwork and observer channel to a cover class; (ii) colour data making $\Pi_{\mathcal{C}}(\Sigma; C)$ computable [FT.19/FT.27]; (iii) a materialization clause for the class's dissolved even cycles (M2 species), after which the pole condition is the existing one, $1 \in \mathrm{spec}\, \mathrm{F}_M^{(\Sigma)}$ [Def-H.16/CT2.5] with FT.22's single-valuedness on each materialized chain. (2) No poles without a declaration. A cover pattern supplies R, $|\mathcal{C}|$ open threads, the dissolved even cycles with their δh, the aspect mask and the norm weight $\Omega 2^K$ [FT.80(1)] — and none of Σ, C, colour, materialization. The amplitude $4^{|\mathcal{C}|}$ is a weight in Z_Γ [Def-H.25/FT.67(2)], not an eigenvalue of any declared operator, and $|w| = 1$ [FT.67(6c)/FT.75(6)] makes the norm layer a count, not a spectrum. (3) Minimality refinement (new). For colour-free covers, $\mathrm{Colour}(\Sigma) = \varnothing \implies \text{strong} \notin \mathcal{A}(\Sigma; C)$ [FT.20's criterion] and the strong summand of $\mathrm{F}_M^{(\Sigma)}$ is $\equiv 0$ [FT.26(2)] — so declaration (ii) is vacuous. Declaration (iii) is needed only for the bound/fermionic side, and covers carry no materialized closed chain [FT.80(6)]. Hence the minimal bridge is (i) alone, restricted to colour-free covers. (4) The candidate, stated exactly — NOT adopted. The minimal bridge map. For a cover class $\mathcal{O}$ on a skeleton Γ and an observer $o \notin \mathrm{supp}(\mathcal{O})$: $\Sigma(\mathcal{O}) :=$ the vertex support of any representative $R \in \mathcal{O}$; $C(\mathcal{O}) :=$

the boundary channel induced by $\gamma(o, \Sigma(\mathcal{O}))$ [Def-H.1]; the spectral side of $\mathcal{O}$ is $Z(\mathcal{R}, \Sigma(\mathcal{O}))$ with poles at $1 \in \operatorname{spec} \mathrm{F}_M^{(\Sigma(\mathcal{O}))}$ [Def-H.16], aspects by FT.20's criteria.

- Declares: one map (cover classes $\to$ subnetwork + channel). No mass formula, no materialization, no colour data, no rates, no second scale.
- Buys: a spectral side on the same objects that carry amplitude, so the inter-family bridge becomes computable; class-invariance is preserved because the vertex support is $\mathrm{Aut}(\Gamma)$-invariant.
- Costs / forbids: (a) the resulting spectrum is observer-relative, inheriting Def-H.9/FT.3's observer-relativity of m_{hid} (consistent, but it must be stated); (b) no fermionic poles — the bound side still needs (iii); (c) it lifts no ceiling: $\dim \mathcal{K}_\Sigma \leq 18$ [FT.26(3)], and any Def-H.37(2)-based values remain bounded by FT.76(3); (d) traps barred: $4^{|\mathcal{C}|}$, Ω, $Z^{\pm}$ are never masses or eigenvalues [FT.79(5)]; δh is never a scale [D0.8]; aspects come from FT.20's criteria, never from block membership [FT.27].

(5) Compatibility. With FT.90: compatible — the declaration adds no resolution, it only names Σ and C, so FT.3/FT.4's bounds continue to apply to whatever is read; a bridge cannot exceed the channel budget. With FT.89: vacuous — the Koide comparison yields nothing admissible, so no compatibility condition arises. (6) Adoption status. The map is constructed; no tag of Parts 1–4 imports it; its species is a map, contrasted with the transfer interfaces Def-H.37/Def-H.38 and the Cal-F dynamics datum. Its consequences are enumerable in advance: the pole set of each colour-free cover class, the observer-relativity of that set, and the exact statement of which classes remain spectrally empty.

Req. Def-H.1, Def-H.9, Def-H.16, Def-H.25, Def-H.30; FT.3, FT.19, FT.20, FT.22, FT.26, FT.27, FT.67, FT.75, FT.76, FT.79, FT.80, FT.89, FT.90; CT2.5, CT2.8 (Part 2); D0.8, D1.11; no external data.

Pf.

$$(1):\ \text{FT.80}(4)\ \text{verbatim};$$

$$(2):\ \text{argument list of } Z(\mathcal{R}, \Sigma) \text{ vs the data a pattern supplies [FT.80(1)]}$$

$$4^{|\mathcal{C}|} \text{ is } Z_\Gamma\text{'s weight [FT.67(2)]}; \qquad |w| = 1 \implies \sum |w|^2 = \#\text{configurations [FT.75(6), CT2.8]}$$

$$(3):\ \text{Colour}(\Sigma) = \varnothing \implies \text{strong} \notin \mathcal{A}(\Sigma; C)\ \text{[FT.20]} \ \wedge\ F^{\Sigma}_{\text{strong}} \equiv 0\ \text{[FT.26(2)]} \implies (ii)\ \text{vacuous}$$

$$\text{covers carry no materialized closed chain [FT.80(6)]}$$

$$\implies (iii)\ \text{not required for transit-side poles}$$

$$(4):\ \text{supp}(R)\ \text{is Aut-invariant} \implies \Sigma(\mathcal{O})\ \text{well-defined on the class}; \qquad o \notin \Sigma\ \text{[Def-H.1]}$$

observer-relativity : Def-H.9/FT.3; bound : FT.26(3); $(5), (6)$: conjunction. ■

Phys. The bridge from "the fuzz has amplitude" to "the fuzz has a particle" needs one thing named: which subnetwork a cover class is, seen from which outside point. Everything else — the pole condition, the aspect criteria, the mask bound — is already proved and waiting. Colour-free covers need nothing further, because a colourless subnetwork has no strong block to seal; covers with colour would need the sealing data too, and anything fermionic would need the materialization clause as well. The declaration is small, its consequences are enumerable before it is made, and it lifts no ceiling: it would give the flicker gas a spectrum, not a hierarchy.

3.32 Tier F.31 — The Native Clifford Census, the Weak-Chirality Pincer, and the Reversal Census of the Covers

Theorem 3.70 (FT.92 — EW.5 is convention-exact: re-verification and the rigidity of the named basis). Thm. In the Pauli basis of $\mathfrak{H}(\mathbb{C}^4)$ [A3, CT1.1], with CT1.2's definitions verbatim ($E_0 = \sigma_0 \otimes \sigma_0$, $\Sigma = \sum_i \sigma_i \otimes \sigma_i$, $W_i = \frac{1}{2}(\sigma_0 \otimes \sigma_i + \sigma_i \otimes \sigma_0)$, $J_i = \frac{1}{2}(\sigma_j \otimes \sigma_k - \sigma_k \otimes \sigma_j)$ cyclic, $K_i = \frac{1}{2}(\sigma_0 \otimes \sigma_i - \sigma_i \otimes \sigma_0)$, $E^{ij} = \frac{1}{2}(\sigma_i \otimes \sigma_j + \sigma_j \otimes \sigma_i)$), as exact matrix equalities over $\mathbb{Q}(i)$ (no tolerance, no floats): (1) The frozen identities all hold exactly as stated. CT1.3 ($[E_0, \cdot] = 0$: 16 brackets); CT1.4 ($[W_i, W_j] = i\epsilon_{ijk}W_k$: all 6 ordered pairs); CT1.5 ($[\Sigma, J_i] = 4iK_i$, $[\Sigma, K_i] = -4iJ_i$, $[\Sigma, W_i] = 0$: 9); CT1.6 ($[J_i, J_j] = [K_i, K_j] = i\epsilon_{ijk}W_k$: 12); CT1.7 ($[J_i, K_j] = i(\delta_{ij}\Sigma - E^{ij})$: 9, each with all three W-components exactly zero); CT1.8 ($[W_i, J_j] = i\epsilon_{ijk}J_k$, $[W_i, K_j] = i\epsilon_{ijk}K_k$: 12); EW.5 both clauses at all nine (i, j): $[K_j, [\Sigma, J_i]] = 4\epsilon_{ijk}W_k$ and $[J_j, [\Sigma, K_i]] = -4\epsilon_{ijk}W_k$ (diagonal $i = j$: both sides exactly 0); Cor CT0.7a ($\sum_i W_i^2 = \frac{3}{2}E_0 + \frac{1}{2}\Sigma = 2\Pi_{\mathcal{R}}$; $\sum_i W_i^2|_{\wedge^2} = 0$); CT0.7(1) ($\mathbb{S} = \frac{1}{2}(E_0 + \Sigma)$ as matrices, $\Pi_{\mathcal{R}/V}$ idempotent, orthogonal, sum I_4, traces $3/1$, ranks $3/1$); the associative relation $\Sigma^2 = 3E_0 - 2\Sigma$; $[S, T] = 0$ on all 16 basis elements with $\eta = (-, +, +, +)$ [A3]; the full (S, T) parity table of CT1.2 ($W \in \mathcal{E}_{+-}$, $J \in \mathcal{E}_{-+}$, $K \in \mathcal{E}_{--}$, $E_0, \Sigma, E^{ij} \in \mathcal{E}_{++}$: 17 generators). (2) The phase-convention premise is refuted. CT1.6 holds with $+i\epsilon_{ijk}W_k$ exactly in the frozen convention — there is no "up to a sign" residue to fix; EW.5's own Pf is the two-line consequence $[K_j, 4iK_i] = 4i \cdot i\,\epsilon_{jik}W_k = 4\epsilon_{ijk}W_k$, and D's header already lists EW.5 among the tags machine-verified by `verify_D.py` in exact complex-rational arithmetic. No edit was made or is needed (Parts 1–2 are frozen; this clause is an layer-local re-verification record). (3) Convention rigidity (new). The group of diagonal sign redefinitions $(J, K, W, \Sigma, E) \mapsto (\lambda_J J, \lambda_K K, \lambda_W W, \lambda_\Sigma \Sigma, \lambda_E E)$, $\lambda \in \{\pm 1\}$, preserving every identity of clause (1) including the associative relation $\Sigma^2 = 3E_0 - 2\Sigma$ and the definition $\mathbb{S} = \frac{1}{2}(E_0 + \Sigma)$ (with $E_0 = I_4$ fixed, being the unit) is exactly

$$\{\mathrm{id},\ \mathrm{Ad}(\mathbb{S})\} \cong \mathbb{Z}_2, \qquad \mathrm{Ad}(\mathbb{S}) :\ (J_i, K_i) \mapsto (-J_i, -K_i), \quad W_i, \Sigma, E^{ij}, E_0 \text{ fixed}$$

(machine: exactly 2 of the 32 sign maps survive). Proof chain: Σ^2-relation $\Rightarrow \lambda_\Sigma = 1$; CT1.5 $\Rightarrow \lambda_J = \lambda_K$; CT1.7's Σ-coefficient $\Rightarrow \lambda_J\lambda_K = \lambda_\Sigma = 1$ and its E-coefficient $\Rightarrow \lambda_E = 1$; CT1.6 $\Rightarrow \lambda_W = \lambda_J^2 = 1$. EW.5 is invariant under the nontrivial element (each clause maps to itself); the naming exchange $J \leftrightarrow K$ maps clause 1 to clause 2 and is excluded by the same associative relation. Hermiticity excludes continuous phases ($e^{i\varphi}K_i$ Hermitian $\Rightarrow \varphi \in \{0, \pi\}$ — the sign enumeration is exhaustive). So the convention space of the named basis is one $\mathbb{Z}_2$, both members satisfy every frozen identity exactly, and the requested "fix" is vacuous — what the audit adds is the rigidity statement itself.

Req. A3; C1, C2, C3; CT0.7, CT0.7a, CT1.1–CT1.8 (Part 1); EW.4, EW.5 (Part 2); exact $\mathbb{Q}(i)$ matrix arithmetic, exhaustive sign enumeration [math; machine].

Pf.

(1) : each identity evaluated entrywise over $\mathbb{Q}(i)$ from CT1.2's definitions; zero residual [machine: suite §A]

(2) : CT1.6 Thm verbatim; $\quad [K_1, K_2] = \frac{1}{4}\big(2i\,\sigma_0 \otimes \sigma_3 + 2i\,\sigma_3 \otimes \sigma_0\big) = iW_3$ ✓ $(+i,$ exact)

$$(3) : \ \Sigma^2 = 3E_0 - 2\Sigma \ \wedge \ \lambda_\Sigma \Sigma \neq 0 \implies \lambda_\Sigma^2 \Sigma^2 = 3E_0 - 2\lambda_\Sigma \Sigma \implies \lambda_\Sigma = 1;$$

$$\text{CT1.5} : \ \lambda_\Sigma \lambda_J = \lambda_K, \ \lambda_\Sigma \lambda_K = \lambda_J$$

$$\text{CT1.7} : \ \lambda_J \lambda_K = \lambda_\Sigma = 1 \ \wedge \ \lambda_J \lambda_K = \lambda_E;$$

$$\text{CT1.6} : \ \lambda_J^2 = \lambda_K^2 = \lambda_W;$$

$$\text{CT1.4} : \ \lambda_W^2 = \lambda_W \implies \lambda_W = 1$$

$$\implies (\lambda_J, \lambda_K, \lambda_W, \lambda_\Sigma, \lambda_E) \in \{(1,1,1,1,1), (-1,-1,1,1,1)\};$$

$$\mathrm{Ad}(\mathbb{S}) J_i \mathbb{S} = -J_i \ [\text{CT0.7}(3) : \ J \ S\text{-odd}] \ \blacksquare$$

Phys. The electroweak backbone was accused of a sign slip; the audit finds the opposite of a slip — a lock. Once the identity generator, the exchange scalar's associative square, and the swap definition are held fixed, the only remaining freedom in the whole named basis is the simultaneous flip of the two Janus triplets, which is the swap conjugation itself, and every frozen identity — including both clauses of the second-order Higgs-to-weak pathway — reads identically in both conventions. There is no phase to choose.

Theorem 3.71 (FT.93 — The native Clifford census and the weak-chirality pincer (Tasks 1–3)). Thm. On $\mathfrak{H}(\mathbb{C}^4)$ with CT1.2's basis and CT0.7's $\mathbb{S}$, $\Pi_{\mathcal{R}} = \frac{1}{4}(3E_0 + \Sigma)$, $\Pi_{\mathcal{V}} = \frac{1}{4}(E_0 - \Sigma)$: (1) Clifford frames are native (positive, new). Among the 15 non-identity Pauli products (each Hermitian, squaring to $+I_4$, traceless) there are exactly 30 pairwise-anticommuting 4-frames (unordered; 720 ordered, 24 orderings per frame). For every frame: $\gamma_{\mathcal{F}}$ is Hermitian, $\gamma_{\mathcal{F}}^2 = I_4$, $\mathrm{Tr}\,\gamma_{\mathcal{F}} = 0$, $\{\gamma_{\mathcal{F}}, e_i\} = 0$; the 16 real products $\{e_{i_1} \cdots e_{i_r}\}$ are $\mathbb{R}$-linearly independent (rank 16 on the example frame) — the frame generates a real subalgebra $\cong Cl(4,0)$ acting on $\mathbb{C}^4 \cong \mathbb{R}^8$, with no new axiom (A3's Pauli facts + CT1.1's basis suffice). The override's items 1–6 are verified with two precisions: $\gamma_{\mathcal{F}}$ is frame-dependent — every ordering sign flip flips $\gamma_{\mathcal{F}}$ (12/12 witness), and 15 distinct values occur (below) — so "$\gamma_5 = \omega$" names a choice, not a canonical element; and A3's $\eta = (-,+,+,+)$ labels the T-involution only — it enters no Clifford relation (the signature is Euclidean, $e_i^2 = +I_4$,

as the override requires; no Minkowski structure is used anywhere). (2) The full pseudoscalar catalog (exhaustive). All 15 values are attained, each by exactly 2 frames, in three species:

species	$\gamma_{\mathcal{F}}$ (up to sign)	(S,T)	commuting W_i	$[\gamma_{\mathcal{F}},\mathbb{S}]$
mixed (6)	$\sigma_0\otimes\sigma_i$ / $\sigma_i\otimes\sigma_0$	($\neg$definite, $-$)	W_i only	$\pm 2iJ_i$
diagonal (3)	$\sigma_i\otimes\sigma_i$	$(+,+)\in\mathcal{E}_{++}$	W_i only (acomm. W_j, W_k)	0
off-diag. spatial (6)	$\sigma_i\otimes\sigma_j$ $(i\neq j)$	($\neg$definite, $+$)	none (acomm. $W_k,\ k=6-i-j$)	$\mp 2iK_k$ $(k=6-i-j)$

The example frame of the override, $\mathcal{F}_0=(\sigma_1\otimes\sigma_2,\ \sigma_2\otimes\sigma_2,\ \sigma_3\otimes\sigma_2,\ \sigma_0\otimes\sigma_1)$, gives exactly

$$\gamma_{\mathcal{F}_0}=\sigma_0\otimes\sigma_3=\mathrm{diag}(1,-1,1,-1);\qquad P_+^{(\mathcal{F}_0)}=\mathrm{diag}(0,1,0,1);\qquad P_-^{(\mathcal{F}_0)}=\mathrm{diag}(1,0,1,0),$$

both idempotent, orthogonal, summing to I_4, trace 2, rank 2 — the requested exact matrix forms. No $\gamma_{\mathcal{F}}$ commutes with more than one W_i; none anticommutes with more than two. (3) The pincer (the sharpened deficit-1 obstruction). No chirality grade on the pair register can be weak-covariant, from either side: (a) commuting side. $\{M:[M,W_i]=0\ \forall i\}=\mathrm{span}_{\mathbb{C}}\{\Pi_{\mathcal{R}},\Pi_{\mathcal{V}}\}$ (complex dimension 2 — Schur on $3\oplus 1$; machine-solved exactly). Its involutions are $\pm\Pi_{\mathcal{R}}\pm\Pi_{\mathcal{V}}$ with traces $3a+b\in\{\pm4,\pm2\}$; a pseudoscalar needs $\mathrm{Tr}=0$ (general: $\mathrm{Tr}(e_1e_2e_3e_4)=-\mathrm{Tr}(e_2e_3e_4e_1)$) — $3a+b=0\wedge a^2=b^2=1$ has no solution. (b) anticommuting side. $\{M:\{M,W_i\}=0\ \forall i\}=\mathbb{C}\cdot\Pi_{\mathcal{V}}$ exactly (complex dimension 1; machine-solved): rank 1, trace 1 — singular, never an involution. Structural reason: invertible M with $\{M,W_i\}=0$ makes $\mathrm{Ad}(M)$ a Lie automorphism equal to $-\mathrm{id}$ on $\mathfrak{su}(2)$, but $-\mathrm{id}$ reverses the bracket ($[W_1,W_2]=iW_3\mapsto -iW_3\neq iW_3$) — contradiction. (c) exhaustive. Every one of the 15 attained $\gamma_{\mathcal{F}}$ fails (a) and (b) explicitly: $[\gamma_{\mathcal{F}},W_i]\neq 0$ for some i and $\{\gamma_{\mathcal{F}},W_i\}\neq 0$ for some i — all verified entrywise. (d) seed side. On the actual matter carrier $\mathcal{H}_{\mathsf{v}}\cong\mathbb{C}^2$ with $W_i\mapsto\frac{1}{2}\sigma_i$ [EW.32]: $\{m:[m,\sigma_i]=0\}=\mathbb{C}\cdot I_2$ and $\{m:\{m,\sigma_i\}=0\}=\{0\}$ — the weak seed representation is irreducible: no nontrivial projector on the matter carrier commutes with the weak action, and no grade anticommutes with it either. (4) Task 2's claim refuted, per frame and in general. For $\mathcal{F}_0$: W_1,W_2 flip the chiral halves ($P_{\mp}W_{1,2}P_{\pm}\neq 0$ entrywise) and W_3 (the only commuting one) preserves them with $P_-W_3P_-=\mathrm{diag}(1,0,0,0)$ — nonzero, non-scalar on the 2-dimensional image: "annihilates or acts trivially on the right eigenspace" is false for every i. In general, no frame can make a chiral half weak-inactive: $W_iP=0\ \forall i$ forces $\mathrm{im}\,P\subseteq\bigcap_i\ker W_i=\wedge^2$ [clause (5)], which is 1-dimensional, while $\mathrm{rank}\,P_{\pm}^{(\mathcal{F})}=2$. Hence $SU(2)_L$-type chiral coupling is NOT natively derivable, and B6's target $8+7=15$ is not reachable: on $\mathbb{C}^4\otimes(\mathbb{C}^3\oplus\mathbb{C}^1)$ the native counts are $8+8=16$; the missing right state is the [EXT] SM species fact (ν_R absent — no species list, no neutrino read: application-layer data). (5) The native selectivity that does exist (positive, derived). The weak force grades the pair arena by S-parity, not by chirality: $W_i\Pi_{\mathcal{V}}=0$ exactly $\forall i$ (the weak annihilates the singlet $\wedge^2$), $\sum_i W_i^2=2\Pi_{\mathcal{R}}$ [Cor CT0.7a] (spin 1 on $\vee^2$, spin 0 on $\wedge^2$) — a 3-active/1-inactive split, and $\bigcap_i\ker W_i=\mathrm{im}\,\Pi_{\mathcal{V}}$ exactly (machine: the joint kernel is the 1-dimensional singlet line). This is a force-arena selectivity; matter chirality would have to grade a carrier on which the gauge action is trivial — the opposite pattern. (6) Task 3, exactly. $[\gamma_{\mathcal{F}},\mathbb{S}]=0\iff\gamma_{\mathcal{F}}=\pm\sigma_i\otimes\sigma_i$ — the S-aligned chiralities are exactly the three diagonal ones, and they live in $\mathcal{E}_{++}$, the gravity/Higgs block [CT1.1, CT1.2], each anticommuting with the two non-own W's (never weak-covariant). The full commutator table is clause (2)'s last column; example: $[\sigma_0\otimes\sigma_3,\ \mathbb{S}]=2iJ_3$ — chirality rotates into the Janus seal sector under the swap. The S-aligned three are never weak-covariant either: $[\sigma_3\otimes\sigma_3,\ W_1]=2iE^{23}$ exactly (machine) — the weak rotation turns an S-aligned chirality into gravitational shear. The two splittings provably do not align: $\mathrm{rank}(P_{\pm}^{(\mathcal{F}_0)}\Pi_{\mathcal{R}})=2$, $\mathrm{rank}(P_{\pm}^{(\mathcal{F}_0)}\Pi_{\mathcal{V}})=1$ — each chiral half cuts the triplet $2{+}1$ and the singlet $1{+}1$...

(the singlet line meets both halves), so neither grading refines the other, and $P_{\pm}^{(\mathcal{F}_0)} \notin \mathrm{span}\{\Pi_{\mathcal{R}}, \Pi_{\mathcal{V}}\}$ ($[P_{\pm}, \mathbb{S}] \neq 0$). The override's hypothesis — "left-handed fermions are the S-even/S-odd states the Weak force can see" — is false as stated: the weak-visible split is $(\vee^2, \wedge^2) = (3, 1)$ by S-parity, while every chiral split is $(2, 2)$; the dimensions alone forbid the identification. (7) Diagnosis and relocation (the chirality deficit: the synthesis). The override's correction is accepted where it is true: the "missing spinor bundle" framing was too strong — fiberwise, the Clifford algebra and its chiral projectors are constructible from A1/A3 with no declaration (clause (1)), and Def-H.14's $\mathcal{E}(e) \subset \mathfrak{H}(\mathbb{C}^4)_e$ already makes the register a per-edge arena bundle. But $\mathbb{C}^4$ is the internal/force arena: it carries the weak as $3 \oplus 1$ (clause (5)), while a Dirac-type γ_5 must commute with the gauge action on a spacetime-side factor. The pincer (clause (3)) is the exact incompatibility: no element of End($\mathbb{C}^4$) is simultaneously a pseudoscalar and weak-scalar, and the matter carrier $\mathbb{C}^2$ is weak-irreducible. Deficit 1 is therefore relocated, not closed: what is missing is not matrices but a spacetime spin factor (a carrier on which the gauge action is trivial and a Clifford module over the emergent geometry acts) — and a global Dirac-type operator would additionally require a per-edge frame assignment $e \mapsto \mathcal{F}_e$ (fiberwise frames exist by clause (1); the assignment is undeclared datum) coupled to the EW.32 transport. Recorded in item 21 as an enumerated would-be declaration, not adopted, no declaration-entry number (the pincer blocks the chiral use regardless of the assignment, and no computable consequence was identified). The reversal structure declared by the corpus remains exactly $\langle \omega, \Theta_{\mathfrak{L}} \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ [FT.88]; the 15 register-level $\mathbb{Z}_2$-grades $\gamma_{\mathcal{F}}$ are new constructible objects with no declared coupling to ω or $\Theta_{\mathfrak{L}}$ (different spaces: patterns vs fiber). (8) The inventory gloss. The No Dirac/Clifford algebra is declared, but $Cl(4, 0)$ frames are derived (clause (1)) — the question was never the algebra; it is the weak-covariant grade (clause (3), proven absent here) and the spacetime factor (clause (7), absent from the inventory). The chirality-projector non-constructibility result stands, on the proven pincer.

Req. A1, A3; C1, C2, C3; CT0.6, CT0.7, CT0.7a, CT0.12, CT0.14, CT1.1, CT1.2, CT1.4–CT1.8 (Part 1); EW.5, EW.32 (Part 2); Def-H.14; FT.78(6), FT.84, FT.87(8), FT.88, FT.89(2), FT.90(5); frame enumeration, exact kernel solves over $\mathbb{Q}(i)$, exhaustive catalog verification [math; machine]. External SM counts ($8/7/15$, ν_R): marked, used in no derivation.

Pf.

$$(1):\ \#\{\text{4-subsets of the } 15\} = 1365; \qquad \text{pairwise } \{e_a, e_b\} = 0 \text{ filter } \Longrightarrow\ 30\ [\text{machine}]$$

$$\gamma_{\mathcal{F}}^2 = (-1)^{\binom{4}{2}} e_1^2 e_2^2 e_3^2 e_4^2 \cdot (\text{reordering sign}) = I_4;$$

$$\mathrm{Tr}\,\gamma = -\,\mathrm{Tr}\,\gamma = 0;$$

$$\mathrm{rank}_{\mathbb{R}}\{e_{i_1} \cdots e_{i_r}\} = 16$$

$$(2):\ \text{Pauli-coefficient extraction } \mathrm{Tr}(\gamma P_{\mu\nu})/4 :\ \text{single } \pm P_{\mu\nu} \text{ per frame;}$$

$$\text{frame multiplicity 2 per value [machine, all 30]}$$

$$(3a):\ \text{exact nullspace of the } 96 \times 32 \text{ real system } [M, W_i] = 0 :\ \dim_{\mathbb{R}} 4 = \dim_{\mathbb{C}} 2;$$

$$M = a\Pi_{\mathcal{R}} + b\Pi_{\mathcal{V}} :\ M^2 = I \Rightarrow a^2 = b^2 = 1;\ \mathrm{Tr}\, M = 0 \Rightarrow 3a + b = 0 :\ \varnothing$$

$$(3b) :\ \text{exact nullspace of } \{M, W_i\} = 0 :\ \dim_{\mathbb{C}} 1 = \mathbb{C} \cdot \Pi_{\mathcal{V}}\ (\mathrm{rk}\, 1, \mathrm{Tr}\, 1);$$

$$\mathrm{Ad}(M) = -\mathrm{id}\ \neg \in \mathrm{Aut}(\mathfrak{su}(2)) :\ [W_1, W_2] = iW_3 \neq 0$$

$$(3d) :\ 2 \times 2 \text{ exact solves} :\ \dim_{\mathbb{C}}\{m : [m, \sigma_i] = 0\} = 1; \qquad \dim_{\mathbb{C}}\{m : \{m, \sigma_i\} = 0\} = 0$$

$$(4) :\ \text{block entries of } P_{\mp} W_{1,2} P_{\pm} \text{ nonzero exact;}$$

$$P_{-} W_3 P_{-} = \mathrm{diag}(1, 0, 0, 0);$$

$$\dim \bigcap_i \ker W_i = 1 < 2 = \mathrm{rk}\, P_{\pm}$$

$$(5) :\ W_i \Pi_{\mathcal{V}} = 0\ [\text{machine}]; \qquad \sum W_i^2 = \tfrac{3}{2} E_0 + \tfrac{1}{2}\Sigma\ [\text{Cor CT0.7a}] = 2\Pi_{\mathcal{R}}$$

$$(6) :\ \mathbb{S} P_{\mu\nu} \mathbb{S} = P_{\nu\mu} \implies [\gamma, \mathbb{S}] = 0 \iff \mu = \nu;$$

$$[\sigma_0 \otimes \sigma_3, \mathbb{S}] = \tfrac{1}{2}[\sigma_0 \otimes \sigma_3, \Sigma] = 2iJ_3\ [\text{machine}]$$

$$\mathrm{rank}(P_{\pm}\Pi_{\mathcal{R}}) = 2,\ \mathrm{rank}(P_{\pm}\Pi_{\mathcal{V}}) = 1\ \forall \pm \implies \neg\text{refinement either way;}$$

$$(3, 1) \neq (2, 2) \implies \neg\text{identification}$$

$$(7), (8) :\ \text{conjunction of } (1), (3), (5), (6) \text{ with the cited inventory tags. } \blacksquare$$

Phys. The Imaginator was half right, and the half that is right matters: the pair register really does carry a native Euclidean Clifford algebra — thirty frames, fifteen possible "γ_5's", exact chiral projectors, all constructible from the axioms with nothing added. But every one of those fifteen chirality operators is blind to, or at war with, the weak force: the best of them commutes with a single Cartan generator while the other two rotate left into right, and the swap rotates all of them into the Janus seals. The weak force does have a native selectivity — it sees the symmetric triplet and annihilates the antisymmetric singlet, three against one — but that is a grading of the force arena by exchange parity, not a handedness of matter. A Dirac γ_5 needs a spacetime-side factor on which the gauge group acts trivially; the corpus has no such factor, and now there is a theorem saying the pair register cannot fake one. Chirality is still missing — but the search space has collapsed from "unknown primitive" to one named absent object.

FT.94 — The chiral mass spectrum: not constructible as chiral, exact as a statistic, and the fourth hierarchy negative. Structural-constraints record: see Appendix B, Section B.2.6 of the Addendum.

Theorem 3.72 (FT.95 — Flicker chirality: the tautology, the absent trajectory, and the zero-expectation theorem). Thm. Under Cal-F, Def-H.25, D2.6: (1) D4/D5 are tautological. With $k(t) = (-1)^t k_0$ and coupling(t) = wb · hc · $k(t)$: sign$(k(t)) \cdot k(t) = |k(t)| = |k_0|$ for every t (machine: all $k \in \{\pm 1, \ldots, \pm 5\}$), so $\langle\chi\rangle_T$ = wb · hc · $|k_0|$ for every $T \geq 1$ — independent of the dynamics, the tick count and the phase boundary. "$\langle\chi\rangle_T \neq 0$" is built in by the formula, not derived: the same species as FT.87(3)'s Regge-tautology verdict. D5 is void; D6 substitutes a measure parameter ($p^*_\infty = \sqrt{2/3}$ — itself FT.89(2d)'s cap-3 output, and a phase-boundary datum of the bypass measure [Def-H.26/FT.70], not a dynamics input) into the tautology and returns the tautology. (2) D1–D3 have no flicker referent. The winding k is a read of materialized closed chains [EW.18, FT.22]; flicker states carry none — bound content survives only dissolved, in even cycles [FT.69's cycle-alternation lemma], and covers carry no materialized closed chain [FT.80(6)] — so no trajectory $k(t)$ exists in the flicker sector. Cal-F declares the rates $r = \Gamma_{J^+}/\Gamma_{J^-}$ (per-edge independent detailed balance), not per-tick schedules; $\delta\tau$ is schedule data that R does not determine [FT.80(5)]; and no oscillation frequency is constructible [FT.86(7): only $\tau_c = 1$ ledger tick, CT0.3]. weak_bias and higgs_coupling count 0 in frozen Parts 1–2 (machine-asserted). The corpus's genuine period-2 phenomenon is FT.86(2)'s alternating correlation ripple in the crystallization indicators (cap-induced, exact decay rate $(\sqrt{21} - 5)/2$) — a different object: it is C-even, orientation-blind, and carries no chirality; D2's deterministic sign-alternating winding conflates the bypass and bound sectors. (3) The zero-expectation theorem (the positive replacement for D5). Global reversal $R \mapsto R^{\mathrm{T}}$ preserves admissibility, K, cover status and the amplitude $A(R) = (-1)^{|R|}2^{|R|}$ — exhaustively verified on the tetrahedron: all 1230 bypass configurations, all 101 covers, and all 4096 patterns; the norm-layer class data (K, Ω) are reversal-invariant by FT.80(1). Hence both layer weights of the declared flicker ensemble are reversal-invariant, and every C-odd read has exactly zero ensemble expectation: for $f(R^{\mathrm{T}}) = -f(R)$,

$$\sum_R A(R)\, f(R) = 0$$

(machine witness: the orientation-bias read $f(R) = \#\{(a,b) \in R : a < b\} - \#\{(a,b) \in R : a > b\}$, exact 0 over all 4096).

In particular $\langle q_{\text{core}}\rangle = 0$ (ω-odd: FT.40/D2.6) and $\langle \text{sign}(k)\rangle = 0$ (C-odd: FT.87(8)) in the declared ensemble. Parity violation is not constructible from the flicker: producing $\langle C\text{-odd}\rangle \neq 0$ requires declaring a C-odd state or a C-odd dynamics — application-layer territory; the exact declared reversal structure remains $\langle \omega, \Theta_{\mathfrak{L}}\rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ [FT.88], whose elements are symmetries of the registration data, and by this clause of the equilibrium weights.

Req. Cal-F; Def-H.25, Def-H.26; FT.22, FT.40, FT.69, FT.80, FT.86, FT.87, FT.88, FT.89(2); D2.6; CT0.3; EW.18 (Part 2); exhaustive tetrahedron enumeration, exact integer sums [math; machine].

Pf.

$$(1):\ \text{sign}(x)x = |x|\ \forall x \in \mathbb{Z} \setminus \{0\}; \qquad \langle\chi\rangle_T = \tfrac{1}{T}\sum_{t=0}^{T-1} |k_0| \cdot \text{wb} \cdot \text{hc}\ [\text{machine},\ T = 1..6]$$

$$(2):\ \text{FT.69 (dissolved only)} + \text{FT.80(6)} \implies \neg\exists\ \text{materialized } k(t); \qquad \text{grep counts} = 0\ [\text{machine}]$$

$$(3):\ A(R^{\mathrm{T}}) = (-1)^{|R^{\mathrm{T}}|}2^{|R^{\mathrm{T}}|} = A(R);$$

$$R$$

$$\mapsto R^{\mathrm{T}} \text{ a bijection of the ensemble } \implies \sum_{R} A(R)f(R)$$

$$= \tfrac{1}{2}\sum_{R}\big(A(R)f(R) + A(R^{\mathrm{T}})f(R^{\mathrm{T}})\big) \qquad = 0\ \blacksquare$$

Phys. The proposed flicker chirality is a formula that multiplies a sign by the thing the sign came from — the answer is the absolute value, every time, for every period, at every phase boundary. Strip that away and ask the real question — can the flickering gas prefer one orientation? — and the corpus answers with a clean no-theorem: reversing every arrow leaves the weights of both layers untouched, configuration by configuration, so anything odd under reversal averages to exactly zero. The gas is born parity-symmetric. A parity-violating world needs a chiral state or a chiral rule, and neither is declared; that absence is the chirality assignment's dynamics-side face, now with an exact vanishing theorem behind it.

3.33 Tier F.32 — The Multi-Node Form Factor: Formalization, Exact Reductions, and the Third Ceiling

Theorem 3.73 (FT.97 — The form factor formalized: node ontology, three underdeterminations, reduction identities, and the general ceiling). Thm. Under Def-H.25/FT.67, at (S1)–(S4): (1) Node ontology (premise correction). For every flicker configuration R on a skeleton Γ: $V(R) \subseteq V(\Gamma)$ — the nodes a configuration "passes through" are exactly the skeleton vertices its edges touch. The roles are the derived vertex types $(a, b) = (\mathrm{indeg}_R, \mathrm{outdeg}_R)$: centre $(1, 1)$ (the threading junction; every cover edge is claimed by exactly one centre [FT.67(3d)]), endpoint $(a, 0)/(0, b)$, hub $a + b \geq 4$ (locked, D^--configured [FT.3/CT4.5]), unclaimed $(0, 0)$. "Intermediate nodes" do not exist as nodes; a form factor over subdivided skeletons has no corpus referent (no refinement mechanism is declared). (2) Three underdeterminations, exposed and fixed by convention (none is a free fit). (a) Per-node information assignment. Registration bits live on edges [CT1.29: $S_{\mathfrak{L}} = n_{\mathcal{R}} \ln 2$; D0.5]; assigning them to nodes requires a convention — tail ($i_v = \mathrm{outdeg}_R(v)$), end-sum ($a{+}b$) or half-split ($\frac{a+b}{2}$) — and the first two differ by an exact factor 2 on covers ($\sum_v \deg_R(v) = 2|R|$). No rule of Parts 1–3 selects one (this is FT.85(7)'s per-sector non-constructibility in miniature); the tail convention is primary here because $\sum_v i_v = |R| = n_{\mathcal{R}}$ matches CT1.29's count exactly, and the factor-2 freedom is reported, not hidden. (b) The role weight w. Five derived candidates, all functions of the finite type set: $w_U = 1_{\text{participating}}$, $w_K = k_v = a{+}b$ [CT0.11's capped quantity], $w_C = 1_{(1,1)}$ (centre), $w_E = 1_{\text{endpoint}}$, $w_H = \max(0, a{+}b{-}2)$ (hub excess). Any other w is a free function — a P4-class parameter whose selection against [EXT] targets would violate P5; all five are reported and none is selected by any target. (c) Sector scope (bypass / locked / all): values differ by $O(2)$ and all three are tabulated. (3) Reduction identities (new; exact, per configuration). For every strict cover R:

$$S_{w_U}(R) = \sum_v \mathrm{outdeg}_R(v) = |R| = K(R); \qquad S_{w_C}(R) = \#\{(1,1) \text{ vertices}\} = |\mathcal{C}(R)| = \tfrac{K}{2}$$

(machine-asserted on all $17\,042$ covers of the three families). Hence the directive's form factor at its constructible core is

$$M_{\text{form}}^{(w_U)}(\Gamma) = \frac{\sum_R |A(R)|^2 K(R)}{\sum_R |A(R)|^2} \ln 2 = \langle K \rangle_{|A|^2} \ln 2 \qquad (|A|^2 = 16^{|\mathcal{C}|} \text{ on covers}),$$

exactly FT.68(1)'s reading (ii) — the amplitude-weighted mean winding, tabulated for the tetrahedron bypass sector since FT.68. Anchor reproduced exactly from independent enumeration: $M_{\text{form}}^{(w_U)}(\text{tetra}, \text{byp}) = 62208/15745$; and $M^{(w_C)} = M^{(w_U)}/2$ identically, per family. The "paradigm shift" is a reweighting of tabulated census data, not a new read species. (4) The typed tuple remains non-constructible and ill-typed. Per component [FT.94(1)'s citations, per-node scope added]: Topological — winding is per materialized chain [FT.22], no per-node referent, and flicker edges are unmaterialized [FT.80(6)]; G — the geometric quantum is a joint S-even read over all ten metric coordinates [FT.30's verdict], its ledger unbounded-integer, ¬bit-type [FT.41(c)/C10]; H — the continuous global P4 datum h_0 [EW.13], not per-node; W — continuous per-edge shift data [CT1.12]; Strong — sealing bits are well-posed [FT.41(b)] but colour data is not determined by crystallization patterns [FT.80(2)]. A sum Topological $+ G + H + W +$ Strong adds bits to unbounded integers to reals — ill-typed [C10]. The only constructible per-node information read is the registration bit count — a bit sum, which the directive's own D3 forbids: D3 cannot be satisfied as written; the constructible version is computed instead and flagged. (5) The Form-Factor Ceiling (the third size theorem). M_{form} is a weighted mean, so $M_{\text{form}} \le \max_R S_w(R)$; and $S_w(R) \le V \cdot c_w$ with the finite-type maxima

$$c_w: \quad c_{w_U} = 6, \quad c_{w_K} = 36, \quad c_{w_C} = 1, \quad c_{w_E} = 6, \quad c_{w_H} = 24$$

$$\text{(universal over the type set } \{0 < a + b \le 6\}),$$

so $M_{\text{form}}(\Gamma) \le c_w V(\Gamma)$ — linear in skeleton size for every role weight. With $|\mathcal{C}| \le 3V/4$ (bypass) / V (locked) [CT0.11/FT.79(2)], the $16^{|\mathcal{C}|}$ enhancement saturates at the cap: exponential inside a skeleton, linear across skeletons. Spanning $1{:}R$ off the P_3 anchor requires $V \ge R \cdot M^{(w)}(P_3)/c_w$; at $R = 3477$ [EXT]:

$$V \ge \begin{cases} 2187 & (w_U) \\ 1062 & (w_K) \\ 6561 & (w_C) \\ 65 & (w_E) \\ 515 & (w_H) \end{cases} \qquad \text{(exact: } \lceil 3477 \cdot \tfrac{3264}{865 \cdot 6} \rceil, \lceil 3477 \cdot \tfrac{9504}{865 \cdot 36} \rceil, \lceil 3477 \cdot \tfrac{1632}{865} \rceil, \lceil 3477 \cdot \tfrac{96}{865 \cdot 6} \rceil, \lceil 3477 \cdot \tfrac{3072}{865 \cdot 24} \rceil)$$

(the w_E bound is loose — its P_3 anchor $96/865$ is tiny; its computed family maximum is 7.681, FT.98(4)). This is the third instance of the size-bound species [FT.76(3): pole ceiling; FT.79(2): cover ceiling]: the two-ceiling synthesis [FT.84(2)] becomes a three-ceiling synthesis — counting reads of every constructed species on every computed skeleton are $O(V)$. (6) Mass status. M_{form}

is a flicker-ensemble statistic (the species of FT.94(3)); promoting it to a particle mass requires a declaration linking ensemble averages to pole structure [Def-H.16] — none exists, none is adopted. $|A|^2$ enters only as a weight (no amplitude-to-mass promotion; the FT.79(5)/FT.87(9) boundary is respected). "Mass is a form factor" is recorded as a proposed identification, not adopted.

Req. Def-H.25; FT.3, FT.16, FT.22, FT.41, FT.42, FT.67, FT.68(1), FT.76(3), FT.79, FT.80, FT.84(2), FT.85(7), FT.94; D0.5; CT0.11 (Part 1); CT1.12, CT1.29 (Part 2); EW.13 (Part 2); C10; P4, P5; finite-type maximization, exact Fraction census averages [math; machine]. External hierarchies marked at (5); no identification.

Pf.

$$(1):\ \text{Def-H.25}:\ R \subseteq \mathcal{R}_{\to}(\Gamma); \qquad \text{roles} = \text{types_of}(R)\ \text{[derived]}$$

$$(3):\ \text{strict covers : every edge claimed once by a } (1,1) \text{ centre}$$

$$\implies 2\#\text{centres}$$

$$= |R|\ \text{[FT.67(2)/(3}d\text{)]}$$

$$\sum_v \text{outdeg}_v = |R|\ \text{(each edge has one tail)};$$

$$M^{(w_U)}(\text{tetra}, \text{byp}) = \tfrac{62208}{15745}\ \text{[machine = FT.68(1)}(ii)\text{]}$$

$$(5):\ \text{mean} \le \text{max}; \qquad S_w(R) = \sum_v w(v) i_v \le V \max_{(a,b)} w(a,b) \cdot b; \qquad \text{type set finite } (a+b \le 6)$$

$$|\mathcal{C}| \le 3V/4\ \text{[FT.79(2)]} \implies 16^{|\mathcal{C}|} \le 16^{3V/4}:\ \text{weights saturate};$$

$$\text{required-}V:\ \text{ceiling rearranged at the exact anchors } \blacksquare$$

Phys. The proposal asked the gas to weigh its nodes instead of its edges. Taken literally, the nodes turn out to be the same old skeleton vertices wearing role costumes — centres, endpoints, hubs — and once the costumes are counted honestly, the new "form factor" collapses, identity by identity, onto a number the corpus has carried since FT.68: the amplitude-weighted mean winding of the covers. That is not a failure; it is a localization. Every way of building the proposal out of declared reads lands in the same bounded band, and the bound is a theorem: a weighted average cannot outrun its largest summand, its largest summand cannot outrun the per-vertex cap, and the per-vertex cap cannot outrun the skeleton's size. Three ceilings, one species. Hierarchy-hunting on four- and eight-vertex skeletons is over; the numbers say what would be needed, exactly.

3.34 Tier F.33 — The Dynamic Phase, Parity, and the CPT Bridge: the Mass Refinement

Theorem 3.74 (FT.100 — The alignment pattern refines the informational mass). Thm. (candidate.) Under Def-H.9 (coarse-graining projection), the FT.16/FT.44(4) mass read ($m_{\text{hid}} = E_{\text{int}} \ln 2/\mathfrak{h}$), and Def-EM.20 (dynamic phase): (1) Edge count. The informational mass counts internal real edges: each real edge contributes $\ln 2/\mathfrak{h}$; $m_{\text{hid}} = E_{\text{int}} \ln 2/\mathfrak{h}$, $E_{\text{int}} \in \mathbb{Z}_{\geq 0}$ — on the floor families whose E_{int} read is computed [FT.32, FT.44(4)]. (2) Phase-blindness. The dynamic phase α_e does not change m_{hid}: the mass counts edges, ¬ phases. An edge with $\alpha_e = 0$ and an edge with $\alpha_e = \pi$ each contribute $\ln 2/\mathfrak{h}$; m_{hid} is constant across all $2^{E_{\text{int}}}$ phase assignments. (3) Refinement. The alignment pattern $\mathcal{P}(\Sigma)$ [Def-EM.21] is a refinement of the mass read: two subnetworks with the same m_{hid} (same E_{int}) can carry different alignment patterns — $2^{E_{\text{int}}}$ of them [EM.23(4)] — and different parity classes [EM.23(1)–(2)]. The phase adds structure without changing mass.

Req. Def-H.9; FT.1, FT.15, FT.16 (Tiers F.1–F.11); FT.32 (Tiers F.12–F.13); FT.44 (Tiers F.14–F.16); Def-EM.20, Def-EM.21; EM.20, EM.23 (Part 2); C9; CT0.9 (Part 1); exhaustive enumeration in exact integer arithmetic [math; machine: `checks/verify_F33_dynamic_phase.py` §E].

Pf.

$$(1):\ \text{Def-H.9}(2):\ E_{\text{tot}} = |\Sigma| \ln 2\ [\text{CT0.9},\ \text{C9}:\ 1 \text{ real edge} = 1 \text{ bit} = \ln 2 \text{ nats}];$$

$$\text{FT.32/FT.44}(4):\ m_{\text{hid}} = S_{\mathfrak{L}}/\mathfrak{h} = E_{\text{int}} \ln 2/\mathfrak{h} \text{ on the computed floors}$$

$$(2):\ m_{\text{hid}} = (E_{\text{tot}} - E_{\text{vis}})/\mathfrak{h}; \quad E_{\text{vis}} = \min(\delta h,\ 4\,\delta\tau) \ln 2\ [\text{FT.1}]:\ \text{neither term reads } \alpha_e;$$

$$\text{FT.15}:\ \Pi_v \text{ depends on } (\delta h, \delta\tau, \sigma_s, \sigma_t, \Theta_{\text{int}}),\ \neg\text{on the phase distribution}$$

$$\text{EM.20}(1):\ \alpha_e \in \{0, \pi\}\ \forall e$$

$$\in E_{\mathcal{R}}:\ \text{every assignment leaves } E_{\text{int}},$$

$$\text{hence } m_{\text{hid}},$$

$$\text{fixed [machine: all } 2^{E_{\text{int}}} \text{ assignments, exact]}$$

$$(3):\ \text{same } E_{\text{int}} \implies \text{same } m_{\text{hid}}\ [(1)];$$

$$\text{EM.23}(4):\ \#\{\mathcal{P}(\Sigma)\} = 2^{|E_{\mathcal{R}}(\Sigma)|}:\ \text{fiber cardinality over each mass value};$$

$$\text{EM.23}(1)\text{–}(2):\ \text{the patterns split into parity classes under } M\ [\text{Def-EM.22}]. \blacksquare$$

Phys. What a coarse-grained fermion forgets, the alignment pattern remembers: the mass counts the hidden edges — one bit each — blind to which way each edge's phase points; parity reads exactly that pointing data. Same mass, $2^{E_{\text{int}}}$ patterns, split into mirror-even and mirror-odd classes: a degeneracy-refinement template of the same species as FT.44's dual-pair splitting. No physical identification claimed.

3.35 Tier F.34 — The Weak Branch Grading of the Two-Branch Crystallization

Theorem 3.75 (FT.101 — The weak branch grading: the two crystallization branches are weak-inequivalent). Thm. Under A3 ($\mathcal{H}_p \otimes \mathcal{H}_q = \vee^2 \oplus \wedge^2$), A4 (branches: $\alpha_{pq} = 0$ on $|\Pi_{\mathcal{V}} \rho_{pq}| \leq \varepsilon_q$; $\alpha_{pq} = \pi$ on $|\Pi_{\mathcal{V}} \rho_{pq}| \geq 1 - \varepsilon_q$), $\Pi_{\mathcal{R}} = \frac{1}{4}(3E_0 + \Sigma)$, $\Pi_{\mathcal{V}} = \frac{1}{4}(E_0 - \Sigma)$ [CT0.7], $W_i = \frac{1}{2}(\sigma_0 \otimes \sigma_i + \sigma_i \otimes \sigma_0)$ [CT1.2, FT.13], $\Phi = \{0, \pi\}$ [D0.5, CT0.9]: (1) Aligned branch: weak-active. $[\mathbb{S}, W_i] = 0$; $W_i \Pi_{\mathcal{R}} = W_i = \Pi_{\mathcal{R}} W_i \Pi_{\mathcal{R}} \neq 0$ for every i (entrywise witnesses); $\sum_i W_i^2 = 2\,\Pi_{\mathcal{R}}$; Casimir read on $\vee^2$: $2 = 1(1{+}1)$ [CT0.7a]. (2) Antialigned branch: weak-blind. $W_i \Pi_{\mathcal{V}} = 0 = \Pi_{\mathcal{V}} W_i$ for every i; $\bigcap_i \ker W_i = \operatorname{im} \Pi_{\mathcal{V}}$ (the singlet line, dimension 1); Casimir read on $\wedge^2$: 0 [CT0.7a, FT.93(5)]. (3) The branch flip is label-level. The unique nontrivial automorphism ι of Φ: $\iota(\alpha) = \pi - \alpha$; it exchanges the A4 branch labels ($\alpha = 0 \leftrightarrow \alpha = \pi$). On configuration data ι is the alignment-pattern complement $\mathcal{P} \mapsto E_{\mathcal{R}} \setminus \mathcal{P}$ [Def-EM.21, EM.23, EM.24(2); CT1.11a(4)] — exhaustive machine census: the complement flips α_e on every edge and exchanges the uniform patterns $\varnothing \leftrightarrow E_{\mathcal{R}}$ [EM.23(4)]. No Hilbert-space implementation exists: $\neg\exists\, M \in \mathrm{End}(\mathbb{C}^4)$ invertible with $M \Pi_{\mathcal{R}} M^{-1} = \Pi_{\mathcal{V}}$ — conjugation preserves trace and rank, and $\mathfrak{T}\Pi_{\mathcal{R}} = 3 \neq 1 = \mathfrak{T}\Pi_{\mathcal{V}}$, $\operatorname{rk} \Pi_{\mathcal{R}} = 3 \neq 1 = \operatorname{rk} \Pi_{\mathcal{V}}$ [CT0.7(1); exact witness]. (4) The weak support is not flip-invariant. The weak support is $\Pi_{\mathcal{R}}$: $W_i = \Pi_{\mathcal{R}} W_i \Pi_{\mathcal{R}}$ exactly [(1)]; ι maps the branch label $\mathcal{R} \mapsto \mathcal{V}$, on which the weak read is 0 [(2)]: weak-activity is branch-selective and ι-asymmetric — the flip maps the weak-active branch to the weak-blind branch. Sharp point $\varepsilon_q = 0$: branch $\alpha = 0 \iff \operatorname{supp} \rho \subseteq \vee^2$, branch $\alpha = \pi \iff \rho = \Pi_{\mathcal{V}}$ [CT0.8]; witnesses: $W_3 \cdot (\Pi_{\mathcal{R}}/3) \neq 0$, $W_i\, \Pi_{\mathcal{V}} = 0\ \forall i$ (exact). (5) Structural parity non-invariance. The weak couples to one A4 branch and annihilates the other: a $\mathbb{Z}_2$-asymmetry of the force on the registration alphabet of the real edges; the branch exchange does not preserve the weak action. Scope: the force-arena S-parity grading [FT.93(5), FT.93(7)] — $\neg$ a matter chirality projector: the pincer [FT.93(3)] and the relocation [FT.93(7)] stand unmodified, and the chirality-assignment question is untouched by this theorem.

Req. A3; A4; C1; D0.5; CT0.7, CT0.7a, CT0.8, CT0.9, CT1.1, CT1.2, CT1.11a (Part 1); Def-EM.21, EM.23, EM.24 (Part 2); FT.13 (Tiers F.1–F.11); FT.93; exact matrix arithmetic over $\mathbb{Q}(i)$, exhaustive small-configuration enumeration, exact rank/trace invariants [math; machine: `checks/verify_H_branch_parity.py`].

Pf.

$$(1):\ [\Sigma, W_i] = 0\ \text{[CT1.5]};$$

$$\mathbb{S} = \tfrac{1}{2}(E_0 + \Sigma)\ \text{[CT0.7(1)]} \implies [\mathbb{S}, W_i] = 0 \implies W_i \text{ preserves } \vee^2,\ \wedge^2\ \text{[CT0.7(3)]}$$

$$W_i \Pi_{\mathcal{V}}$$

$$= 0 \ [(2)]$$

$$\implies W_i$$

$$= W_i(\Pi_{\mathcal{R}} + \Pi_{\mathcal{V}})$$

$$= W_i \Pi_{\mathcal{R}}$$

$$= \Pi_{\mathcal{R}} W_i \Pi_{\mathcal{R}} \ [\text{CT0.7}(1) : \ \Pi_{\mathcal{R}} + \Pi_{\mathcal{V}} = E_0];$$

nonzero witnesses entrywise [machine]

$$\textstyle\sum_i W_i^2 = \frac{3}{2} E_0 + \frac{1}{2} \Sigma = 2 \, \Pi_{\mathcal{R}} \ [\text{CT0.7a}]; \qquad \vee^2 : \ \mathbb{S} = +1 \implies \Sigma = +1 \implies \tfrac{3}{2} + \tfrac{1}{2} = 2$$

$$(2) : \ W_i \, \Pi_{\mathcal{V}} = 0 \ \forall i : \ \text{direct exact computation on the CT1.2 basis [machine]};$$

$$\wedge^2 = \mathbb{C} \, |\Psi^-\rangle, \ \ \Pi_{\mathcal{V}} = |\Psi^-\rangle\langle\Psi^-| \ [\text{CT0.7}(1); \ \text{machine}]$$

$$\dim \bigcap_i \ker W_i = 4 - \text{rk} \begin{pmatrix} W_1 \\ W_2 \end{pmatrix} W_3 = 4 - 3 = 1 \ [\text{exact elimination; machine}];$$

$$\wedge^2 : \ \tfrac{3}{2} + \tfrac{1}{2}(-3) = 0 \ [\text{CT0.7a} : \ \Sigma|_{\wedge^2} = -3]$$

$$(3) : \ |\Phi| = 2 \ [\text{CT0.9}] \implies \text{Aut}(\Phi) = \mathbb{Z}_2 = \{\text{id}, \iota\}, \ \iota(\alpha) = \pi - \alpha, \ \iota^2 = \text{id};$$

$$\text{A4} : \ \alpha = 0 \leftrightarrow \alpha = \pi \ \text{exchanges the branch labels}$$

$$\mathcal{P} = \{e : \alpha_e = \pi\} \ [\text{Def-EM.21}];$$

$$\alpha_e \mapsto \pi - \alpha_e$$

$$\iff e \ \text{changes membership in} \ \mathcal{P} \iff \mathcal{P}$$

$$\mapsto E_{\mathcal{R}} \setminus \mathcal{P} \ [\text{EM.24}(2); \ \text{exhaustive}, \ |E_{\mathcal{R}}| \leq 5; \ \text{machine}]$$

$$\neg \exists M : \ \mathfrak{T}(MXM^{-1}) = \mathfrak{T}X \ \wedge \ \text{rk}(MXM^{-1}) = \text{rk} \, X \ [\text{conjugation invariants}];$$

$$3 \neq 1 \implies \Pi_{\mathcal{R}} \not\sim \Pi_{\mathcal{V}} \text{ [CT0.7(1); exact]}$$

$$(4): \text{ support identity } (1);$$

$$\text{flip image } (2), (3);$$

$$\text{sharp loci [CT0.8]};$$

$$\text{witness states } \Pi_{\mathcal{R}}/3,\ \Pi_{\mathcal{V}}: \text{ exact [machine]}$$

$$(5): \ (1)\text{–}(4) \text{ conjoined};$$

scope clause cites FT.93$(3), (5), (7)$ unmodified [no carrier grading is constructed here]. ■

Phys. The two crystallization outcomes are inequivalent for the weak force: the aligned (triplet) branch is where the total-spin generators act — Casimir 2 — and the antialigned (singlet) branch is annihilated by them — Casimir 0. The flip of the binary branch label exchanges the two, so weak activity is not flip-invariant: a structural $\mathbb{Z}_2$ asymmetry carried by the registration alphabet of the real edges. This is the force-arena S-parity grading read against the ledger's own branch data — not a handedness of matter: the flip has no Hilbert-space implementation $(3 \text{ vs } 1)$, and no chirality projector on a matter carrier is constructed or implied. Barred: naming either branch "left" or "right"; any SM species reading; any claim that the SM weak interaction is reproduced.

3.36 Tier F.35 — The 1-Loop Bridge: the Swap-Graded Effective Potential on the Flicker Phase Diagram

Definition 3.37 (Def-H.31 — The bridge: dictionary, family, background, fluctuation operator, effective potential). Def. A construction class on the prior canon (Def-H.24 pattern: no new axiom); clauses (1)–(6) are constitutive interface definitions. Domain: $\varepsilon_q \in (0, \frac{1}{4})$ [CT0.8]; $\gamma := 1 - 4\varepsilon_q$, $q := 4\varepsilon_q$ [D0.3, D0.4].

(1) Dictionary (Def-H.31(1)). $p := \gamma = 1 - 4\varepsilon_q$; $\quad 1 - p = q = 4\varepsilon_q$; $\quad s := p/q$; $\quad y := 16s^2 = \dfrac{(1-4\varepsilon_q)^2}{\varepsilon_q^2}$. Anchors (exact): $\varepsilon_q = \frac{1}{8} \iff p = \frac{1}{2} \iff y = 16$ — the point at which FT.68(1)'s amplitude-weighted reading is the uniform measure and $\sum |A_C|^2 = 15745$ on the tetrahedron [CT2.8 norm layer: $|A_C|^2 = 16^{|C|}$; FT.67(2)'s value formula]; $\varepsilon_q \to 0 \iff p \to 1$ — the static-floor corner [FT.68(5); D0.4's sharp point $\gamma = 1$]; $\varepsilon_q \to \frac{1}{4} \iff p \to 0$ — the identity-tilt, vacuum-dominated corner [L2 $(\gamma \to 0)$; FT.70(5) $(\langle K\rangle \to 0)$].

(2) Family and weights (Def-H.31(2)). $\mathbb{F} := \{\text{tetra}, \text{cube}, \text{octa}\}$ — the skeletons whose two-sector covering polynomials are complete [FT.68(1), FT.68(3), FT.69(2)]:

$$\mathcal{Z}^{\text{tot}}_{\text{tetra}}(y) = 1 + 24y + 72y^2 + 4y^3; \qquad \mathcal{Z}^{\text{tot}}_{\text{cube}}(y) = 1 + 48y + 672y^2 + 3400y^3 + 6816y^4 + 264y^5 + 4y^6;$$

$$\mathcal{Z}^{\text{tot}}_{\text{octa}}(y) = 1 + 72y + 888y^2 + 2160y^3 + 840y^4 \qquad (\mathcal{Z}^{\text{tot}} = \mathcal{Z}_{\text{byp}} + \mathcal{Z}_{\text{lock}},\ \text{rows pooled}).$$

With $2E_\Gamma \in (12, 24, 24)$ the directed-edge counts and $K = 2|\mathcal{C}|$ [FT.79(2)], the cover-weighted Bernoulli weight of Γ is

$$W_\Gamma(p) \ :$$

$$= \sum_k c_k(\Gamma)$$

$$16^k$$

$$p^{2k}$$

$$q^{2E_\Gamma - 2k}$$

$$= \ q^{2E_\Gamma}$$

$$\mathcal{Z}^{\text{tot}}_\Gamma\big(16p^2/q^2\big)$$

$$= \ \mu_p\text{-average of } |A_C|^2 \text{ over covers } C \subseteq \Gamma,$$

an integer polynomial in p with $W_\Gamma(0) = 1$ (the vacuum cover) and $W_\Gamma(1) = 0$ (the fully crystallized floor is no cover — it cancels [FT.67(3a)]). The family measure is

$$P(\Gamma \mid \varepsilon_q) \ := \ \frac{W_\Gamma(p)}{\sum_{\Gamma' \in \mathbb{F}} W_{\Gamma'}(p)}$$

(ensemble scope of CT4.5a/Def-H.26: per-edge Bernoulli p).

(3) Background state. Cover-conditioned edge content and mean real fraction:

$$\langle K \rangle^{\text{cov}}_\Gamma(p) := \frac{N_\Gamma(p)}{W_\Gamma(p)}, \quad N_\Gamma := \sum_k 2k\, c_k 16^k p^{2k} q^{2E-2k};$$

$$\hat{p}_\Gamma := \frac{\langle K \rangle^{\text{cov}}_\Gamma}{2E_\Gamma} \in (0,\ K_{\text{top}}/2E_\Gamma].$$

The ensemble-averaged pair state (linear in the pattern measure; per-pattern state = the branch state of its real/virtual edges, Cal-F detailed balance):

$$\bar\rho_\Gamma(\varepsilon_q) \;:=\; \hat p_\Gamma \cdot \frac{\Pi_{\mathcal R}}{3} \;+\; (1-\hat p_\Gamma)\,\Pi_{\mathcal V}$$

$$[\text{CT0.7}(1):\ \Pi_{\mathcal R} = \tfrac14(3E_0+\Sigma),\ \Pi_{\mathcal V} = \tfrac14(E_0-\Sigma)];$$

$\mathrm{spec}(\bar\rho_\Gamma) = \big(\frac{\hat p_\Gamma}{3}, \frac{\hat p_\Gamma}{3}, \frac{\hat p_\Gamma}{3},\ 1-\hat p_\Gamma\big)$ — faithful on the whole domain; $\mathfrak{T}\bar\rho_\Gamma = 1$.

(4) Fluctuation operator (Def-H.31(4)). $\Delta_\Gamma^{-1}(\varepsilon_q)$: the block-diagonal operator on the 16-mode space $\mathfrak{H}(\mathbb{C}^4)$ [CT0.6] in the $\vee^2/\wedge^2$ decomposition of the swap grading [CT0.7(3)], with the A8 aspect labels of CT0.7(4):

$$\text{S-even block } (10 = 9+1 = \dim\mathfrak{H}(\vee^2) + \dim\mathfrak{H}(\wedge^2);\ \text{aspects } F_{D^+}, F_H, F_G, F_{W^\pm}, F_Z):$$

the Bogoliubov–Kubo–Mori Hessian of the relative entropy $S(\rho\|\bar\rho_\Gamma)$ at $\bar\rho_\Gamma$ [math: the standard quadratic fluctuation kernel of the state-space information geometry; eigenvalue $1/\lambda$ on diagonal block directions]:

$$M^2_{\vee^2} = 3/\hat p_\Gamma\ (\times 9), \qquad M^2_{\wedge^2} = 1/(1-\hat p_\Gamma)\ (\times 1);$$

$$\text{S-odd block } (6 = 3+3;\ \text{the seal forces } F_{J^+} = \mathrm{span}\{J_i\},\ F_{J^-} = \mathrm{span}\{K_i\}\ [\text{CT0.7}(4)]):$$

$$M^2_{J^+} = p\cdot G_\Gamma(p)^2\ (\times 3), \qquad M^2_{J^-} = q\cdot G_\Gamma(p)^2\ (\times 3), \qquad G_\Gamma := B_\Gamma - p^{2E_\Gamma}\ \ [\text{Def-H.26},\ \text{FT.70}(1)],$$

with the detailed-balance occupations (p,q) of the two seal channels [Cal-F; A8: $\mathrm{F}_{(+,\mathcal V)} = F_{J^+}$, $\mathrm{F}_{(-,\mathcal V)} = F_{J^-}$]. $B_\Gamma(p) = \sum_K c_K p^K q^{2E-K}$ with the frozen cap-census rows c_K [FT.68(1), FT.68(3), FT.69(1)].

(5) Effective potential. With the swap supertrace $\mathrm{Str}_{\mathbb S}\, X := \mathfrak{T}(\mathbb S X)$ (defined here; $\neq$ the C-sector supertrace [CT2.10] — collision-avoidance declared once; $\mathrm{Str}_{\mathbb S}$ grades $\mathfrak{H}(\mathbb{C}^4)$ by CT0.7(4)'s S-parity, $10-6=4$):

$$V_{\text{1-loop},\Gamma}\ :$$

$$=\ \tfrac12\,\mathrm{Str}_{\mathbb S}\log\Delta_\Gamma^{-1}$$

$$=\ \tfrac12\Big(\sum_{\text{S-even}}\log M_i^2 - \sum_{\text{S-odd}}\log M_j^2\Big)$$

$$=\ \tfrac92\log\tfrac{3}{\hat p_\Gamma} - \tfrac12\log(1-\hat p_\Gamma) - \tfrac32\log(pq) - 6\log|G_\Gamma|;$$

$$V_{\text{tree},\Gamma}\ :=\ -\log W_\Gamma(p);$$

$$V_{\text{eff}}(\varepsilon_q)\ :=\ \sum_{\Gamma\in\mathbb F} P(\Gamma\mid\varepsilon_q)\,\big[V_{\text{tree},\Gamma}(\varepsilon_q) + V_{\text{1-loop},\Gamma}(\varepsilon_q)\big].$$

Every log argument is positive on $(0, \frac{1}{4}) \setminus \{\varepsilon_\Gamma^*\}$, $\varepsilon_\Gamma^* := (1 - p_\Gamma^*)/4$; V_{eff} is smooth on the wall complement and $V_{\text{eff}} \to +\infty$ at both endpoints and at every wall (FT.102(4)).

(6) Mass read (Def-H.31(6)). For each vacuum $\varepsilon_q^{(i)}$ of V_{eff} (FT.103):

$$m_i^2 \;:=\; \Delta S\big(\varepsilon_q^{(i)}\big) \;:=\; \frac{V_{\text{eff}}\big(\varepsilon_q^{(i)}\big)}{8}$$

$$[\text{L3(2)}:\; V = 8\,\Delta S_{\text{pair}} \text{ exact leading};\; \text{L3(4)}:\; m_i^2 = \Delta S \text{ inertial read};\; \text{dimensionless, nats}].$$

The phase-band $\leftrightarrow$ generation-label correspondence is a labelling clause of this definition, not a derived selection; no particle identification; no coupling to $m_{\text{pole}}/m_{\text{hid}}$ beyond Def-H.39's bridge slots (unconstructed).

Rmk (what is new, what is frozen). All numerical coefficients are prior-canon data: the covering rows $(1, 24, 60 \mid 12, 4)$, $(1, 48, 648, 2672, 3624 \mid 24, 728, 3192, 264, 4)$, $(1, 72, 852, 1152 \mid 36, 1008, 840)$ [FT.68, FT.69]; the cap-census rows c_K [FT.68(1),(3), FT.69(1)]; $16 = 4^2$ [FT.67(2) value $\times$ CT2.8 norm]; $K = 2|\mathcal{C}|$ [FT.79(2)]; the $10/6$ split [CT0.7(3)-(4)]; $\frac{3}{4}$ and $\gamma(8-\gamma)$-family coefficients [L2, L5]; the factor 8 [L3(2)]; the domain $[0, \frac{1}{4})$ [CT0.8]. New content: the assembly (Def-H.31(1)–e) and everything proved from it.

Theorem 3.76 (FT.102 — The fluctuation spectrum, the tree-level limit, and the seal-softening walls). Thm. Under Def-H.31: (1) Exact spectrum. For every $\Gamma \in \mathbb{F}$ and $\varepsilon_q \in (0, \frac{1}{4}) \setminus \{\varepsilon_\Gamma^*\}$:

$$\text{Spec}\big(\Delta_\Gamma^{-1}(\varepsilon_q)\big)$$

$$= \underbrace{\left\{\tfrac{3}{\hat{p}_\Gamma} \times 9,\;\; \tfrac{1}{1-\hat{p}_\Gamma} \times 1\right\}}_{\text{S-even, 10}} \sqcup \underbrace{\left\{p\,G_\Gamma(p)^2 \times 3,\;\; q\,G_\Gamma(p)^2 \times 3\right\}}_{\text{S-odd, 6}}$$

$$(16 \text{ values}),$$

with $\hat{p}_\Gamma \in (0, K_{\text{top}}/2E_\Gamma]$, $K_{\text{top}}/2E_\Gamma = (\frac{1}{2}, \frac{1}{2}, \frac{1}{3})$ for (tetra, cube, octa) — the supremum attained exactly in the limit $p \to 1$ (vanishing order $2E - K_{\text{top}} + 2$ of $N_\Gamma - K_{\text{top}} W_\Gamma$ at $p = 1$; machine). The S-even block is strictly positive on the whole domain; the S-odd block vanishes exactly at $p \in \{p_\Gamma^*\} \cup \{0, 1\}$ and nowhere else. (2) The grading is matrix-exact. On the CT1.2 basis over $\mathbb{Q}(i)$: $\text{Ad}(\mathbb{S})$ has eigenvalue $+1$ on $\{E_0, \Sigma, \hat{E}^{ij}\,(5), W_i\,(3)\}$ — 10 modes — and -1 on $\{J_i\} \cup \{K_i\}$ — 6 modes [CT0.7(4) membership reproduced from the matrices]; S-even $\iff [\mathbb{S}, X] = 0 \iff X$ preserves $\vee^2, \wedge^2$; S-odd $\iff \{\mathbb{S}, X\} = 0 \iff X$ swaps them [CT0.7(3)]; basis rank 16 [CT0.6]; $\text{Str}_{\mathbb{S}}(I_{16}) = 10 - 6 = 4$; on any degenerate spectrum λI: $V_{\text{1-loop}} = \frac{1}{2}(10 - 6)\log\lambda = 2\log\lambda$ — the superdimension is the interference coefficient the trace log carries. (3) Tree-level limit = the FT.68/FT.69 covering polynomials. Setting $V_{\text{1-loop}} \equiv 0$: $e^{-V_{\text{tree},\Gamma}(p)} = W_\Gamma(p) = q^{2E}\mathcal{Z}_\Gamma^{\text{tot}}(y)$ identically — verified two ways: the polynomial expansion vs the sum over the enumerated covers (two independent enumerators per cell, all covers: $101/11205/3961$). Static evaluation anchors (exact integers):

	tetra	cube	octa
$\mathcal{Z}^{\text{tot}}(4)$ $(Z^{\pm}_{\text{all}})$	1505	2260161	367777
$\mathcal{Z}_{\text{byp}}(4)$ / $\mathcal{Z}_{\text{lock}}(4)$	1057 / 448	1109313 / 1150848	87649 / 280128
$W_\Gamma(\frac{1}{2})\cdot 2^{2E} = \mathcal{Z}^{\text{tot}}(16)$	35201	804725505	64126081
$\mathcal{Z}_{\text{byp}}(16)$ / $\mathcal{Z}_{\text{lock}}(16)$	15745 / 19456	248613633 / 556111872	4937857 / 59188224

the tetrahedron's bypass norm sum 15745 and amplitude-weighted content $\langle K\rangle = 62208/15745$ are FT.68(1)(ii) verbatim; the uniform readings $852/205$, $3311952/395023$, $2124600/318523$ at $p=\frac{1}{2}$ are FT.68(1)(i)/FT.70(5) verbatim; the single-centre closed form $c_1 = Vd(d-1) = (24, 48, 72)$ is FT.69(2) verbatim. (4) Seal-softening walls. $V_{\text{eff}} \to +\infty$ at each $\varepsilon^*_\Gamma = (1-p^*_\Gamma)/4$, p^*_Γ the unique zero of G_Γ [FT.70(1)–(2)] — dyadic isolation at width 2^{-40} reproduces the frozen boundaries exactly: $p^* = (0.725372,\ 0.723525,\ 0.679217)$ for (cube, tetra, octa), i.e. $\varepsilon^* = (0.068657016,\ 0.069118773,\ 0.080195792)$, order octa $>$ tetra $>$ cube in ε^* [FT.70(2)'s order in p^* inverted by the dictionary]. At a wall the six S-odd masses vanish quadratically — the phase boundary is exactly where the reality-exchanging (seal) fluctuations become marginal. $V_{\text{eff}} \to +\infty$ also at both domain endpoints: the floor side ($\varepsilon_q \to 0$) from $W_\Gamma \to 0$ (the floor cancels [FT.67(3a)]) and $-\frac{3}{2}\log q \to +\infty$; the threshold side ($\varepsilon_q \to \frac{1}{4}$) from $3/\hat{p}_\Gamma \to \infty$ and $-\frac{3}{2}\log p \to +\infty$. (5) Directive form. $V_{\text{1-loop},\Gamma} = \frac{1}{2}\big(\sum_{\text{S-even}} \log M_i^2 - \sum_{\text{S-odd}} \log M_j^2\big)$ — the block assembly and the 16-term eigenvalue assembly agree by enclosures; the minus sign on the 6 odd terms is the swap-supertrace grading, and the odd block is precisely the seal force pair $F_{J^\pm}$ [CT0.7(4): "the S-odd blocks"; D1.2: the seals move A4-weight $\wedge^2 \leftrightarrow \vee^2$].

Req. Def-H.31; A4, A8; D0.3, D0.4, D1.2; CT0.6, CT0.7, CT0.7a, CT0.8, CT0.11, CT1.1, CT1.2 (Part 1); EW.12 (Part 2); CT2.8 (Part 2); CT4.5a (Part 2); Def-H.25, Def-H.26; FT.67, FT.68, FT.69, FT.70; FT.79(2); FT.93(5); L2, L3, L5; Cal-F; exact $\mathbb{Q}(i)$ matrix arithmetic, two-enumerator cover re-derivation, integer-Horner dyadic bisection, rational log-enclosures [math; machine: `checks/verify_H_1loop_bridge.py` §A–§D].

Pf.

$$(1):\ \bar{\rho}_\Gamma \text{ faithful } (\hat{p}_\Gamma \in (0,\tfrac{1}{2}]) \implies \text{BKM diagonal coefficients } 1/\lambda:\ 1/(\hat{p}/3)$$

$$= 3/\hat{p}\ (\times 9),$$

$$1/(1-\hat{p})\ (\times 1)\ [\text{math}]$$

$$\hat{p}\text{-bounds}:\ N_\Gamma/W_\Gamma \le K_{\text{top}} \text{ termwise (each cover } K \le K_{\text{top}});$$

$$\hat{p}(p \to 1)$$

$$= K_{\text{top}}/2E_\Gamma:\ \text{vanishing-order census of } N_\Gamma - K_{\text{top}}W_\Gamma \text{ at } p$$

$$= 1\ [\text{machine: exact derivative ladder}]$$

$$\text{odd zeros}: \; pG^2 = 0 \iff p = 0 \vee G = 0; \; qG^2 = 0 \iff p = 1 \vee G = 0;$$

$$G_\Gamma \text{ has exactly one root on the resolved grid } [\text{FT.70}(1);\ \text{machine}]$$

$$(2): \; \mathbb{S} = \tfrac{1}{2}(E_0 + \Sigma) \text{ as matrices } [\text{CT0.7}(1)];$$

$$\mathrm{Ad}(\mathbb{S})X \text{ computed on all } 16 \text{ basis elements over } \mathbb{Q}(i) \text{ [machine]}$$

$$\text{membership} = \text{CT0.7}(4)\text{'s frozen lists};$$

$$\mathrm{Str}_{\mathbb{S}}(I_{16}) = \#\{+1\} - \#\{-1\} = 10 - 6 = 4;$$

$$\tfrac{1}{2}\,\mathrm{Str}_{\mathbb{S}} \log(\lambda I) = \tfrac{10-6}{2} \log\lambda = 2\log\lambda$$

$$(3): \; W_\Gamma(p)$$

$$= \sum_{\text{covers } C} 16^{|C|} p^{2|C|} q^{2E-2|C|} : \text{ polynomial expansion}$$

$$= \text{enumerated sum at } p$$

$$= \tfrac{3}{7} \text{ [machine, both enumerators]};$$

$$\text{anchors: exact integer evaluations}$$

$$15745 = \mathcal{Z}_{\text{byp}}(16) = 1 + 24 \cdot 16 + 60 \cdot 256;$$

$$62208 = \sum_k 2k\, c_k 16^k\big|_{\text{byp}} = 2 \cdot 24 \cdot 16 + 4 \cdot 60 \cdot 256 \; [\text{FT.68}(1)(ii)]$$

$$(4): \; G_\Gamma(0) = 1 > 0 > G_\Gamma(1) = -1 \; [\text{FT.70}(1)];$$

$$\text{bisection at } 2^{-40}: \text{ intervals reproduce the frozen decimals [machine]}$$

$$-6\log|G_\Gamma| \to +\infty \text{ at } G_\Gamma = 0;$$

$$W_\Gamma(1) = 0 \text{ (every term carries } q^{2E-2k},\ 2E - 2k \geq 2E - K_{\text{top}} > 0);$$

$$\hat{p} \to 0 \implies 3/\hat{p} \to \infty \; [\varepsilon_q \to \tfrac{1}{4}]$$

(5) : two assemblies of the same 16-term sum, evaluated by independent code paths at ε_q

$= \frac{1}{16}$: difference enclosure brackets 0 [machine]. ■

Phys. The tree layer knows only polynomial censuses; the loop layer turns the census into geometry. Ten reality-preserving modes carry the information metric of the averaged crystallization state; six reality-exchanging seal modes carry the phase balance — and the walls of the effective potential sit exactly at the phase diagram's crossover boundaries, because a seal fluctuation has no restoring force where thread-gas and floor weigh equal. The grade difference $10 - 6 = 4$ — the swap super-dimension of the pair arena — is the coefficient the trace log carries into every degenerate limit. Structure class of a 1-loop effective potential (trace log of the fluctuation operator; Coleman–Weinberg species [physics vocabulary — analogy only; no SM field content, no identification]).

Theorem 3.77 (FT.103 — The vacuum structure: four exact vacua, three wall-delimited phase bands, dominance windows, the susceptibility census). Thm. Under Def-H.31: (1) Vacuum census (complete at the resolved scale). $dV_{\text{eff}}/d\varepsilon_q = 0$ has exactly four solutions in $(0, \frac{1}{4})$ on the wall-delimited bands — one per band, each a strict minimum (V' passes $- \to +$; exact sign census by rational + log-enclosure evaluation on dyadic grids; bisection to width 2^{-36}):

	band	$\varepsilon_q^{(i)}$ ($\approx$)	V_{eff} ($\approx$)	$m_i^2 = V_{\text{eff}}/8$ ($\approx$)
V_1	$(0, \varepsilon^*_{\text{cube}})$ floor band	0.021337258636	24.271850813	3.033981352
V_2	$(\varepsilon^*_{\text{cube}}, \varepsilon^*_{\text{tetra}})$ bicritical sliver	0.068783298916	59.645122359	7.455640295
V_3	$(\varepsilon^*_{\text{tetra}}, \varepsilon^*_{\text{octa}})$ mixed band	0.080091490970	38.713071069	4.839133884
V_4	$(\varepsilon^*_{\text{octa}}, \frac{1}{4})$ thread-gas band	0.190792364908	10.655753803	1.331969225

All four values are strictly positive — ΔS-admissible in nats [L3]; all four positions lie in CT0.8's physical domain $(0, \frac{1}{4})$. V_3 sits 1.04×10^{-4} below the octahedron wall (wall-adjacent: its depth is set by the seal-softening balance); V_2 is the sliver bottom between the near-degenerate cube/tetrahedron boundaries — its existence and depth are controlled by the frozen splitting $|\varepsilon^*_{\text{tetra}} - \varepsilon^*_{\text{cube}}| = 4.6176 \times 10^{-4}$ [FT.70(2)'s own numbers]. (2) Mass order and ratios (enclosure-exact). $m^2(\mathrm{V}_4) < m^2(\mathrm{V}_1) < m^2(\mathrm{V}_3) < m^2(\mathrm{V}_2)$ (disjoint enclosures). The phase triple $(\mathrm{V}_4, \mathrm{V}_1, \mathrm{V}_3)$ in mass order:

$$m^2 : \; 1 : 2.277816 : 3.633067; \qquad m : \; 1 : 1.509244 : 1.906061;$$

all four: $m^2 : \; 1 : 2.277816 : 3.633067 : 5.597457$ (bare m^2-span < 6 — clause (6)). (3) Dominance windows (exact). The pairwise W-crossings isolate (width 2^{-30}): tetra|cube at $p = 0.662192$ and $p = 0.035764$ ($\varepsilon_q = 0.084410$ and 0.241059); cube|octa at $p = 0.091544$ ($\varepsilon_q = 0.227062$). Windows: tetra-dominant $(0, 0.084410) \cup (0.241059, \frac{1}{4})$; cube-dominant $(0.084410, 0.227062)$; octa-dominant $(0.227062, 0.241059)$. Dominant skeleton at the vacua: $\mathrm{V}_1, \mathrm{V}_2, \mathrm{V}_3$ tetrahedron ($P_{\text{tetra}} \approx$

$0.999/0.720/0.556$); V_4 cube ($P_{\text{cube}} \approx 0.727$). The octa window and the tetra tail sliver carry no vacuum: $V' > 0$ throughout both (machine census) — the octahedron enters the vacuum structure through its wall $\varepsilon^*_{\text{octa}}$, not through a dominance minimum. (4) Susceptibility census. Per skeleton, $\chi_\Gamma(y) := d\langle K\rangle^{\text{cov}}_\Gamma/dy = 2\,\text{Var}_y(|\mathcal{C}|)/y$ [exact identity; math] is strictly decreasing on the exact grid $y = 2^{i/4}$, $|i| \le 40$ — no interior peaks: the van-Hove structure sits in $\text{Var}_y(|\mathcal{C}|)$ (the log-fugacity susceptibility $d\langle K\rangle/d\ln y = 2\,\text{Var}$). The cover-condensation transition scales $y = c_k/c_{k+1}$ and their exact ε-images ($\varepsilon = 1/(4+\sqrt{y})$; each the unique root in $(0, \frac{1}{4})$ of the integer quadratic $(16d-n)\varepsilon^2 - 8d\varepsilon + d = 0$, $y = n/d$; machine):

Γ	$y = c_k/c_{k+1}$	$\varepsilon_q = 1/(4+\sqrt{y})$ ($\approx$)
tetra	$\frac{1}{24},\ \frac{1}{3},\ 18$	$0.237862,\ 0.218467,\ 0.121320$
cube	$\frac{1}{48},\ \frac{1}{14},\ \frac{84}{425},\ \frac{425}{852},\ \frac{284}{11},\ 66$	$0.241293,\ 0.234342,\ 0.224993,\ 0.212482,\ 0.110118,\ 0.082481$
octa	$\frac{1}{72},\ \frac{3}{37},\ \frac{37}{90},\ \frac{18}{7}$	$0.242845,\ 0.233386,\ 0.215462,\ 0.178458$

The family trajectory $\chi_{\mathbb{F}}(\varepsilon_q) = d\langle K\rangle_{\text{fam}}/dy$ (with $\langle K\rangle_{\text{fam}} = \sum_\Gamma P_\Gamma \langle K\rangle^{\text{cov}}_\Gamma$, rational in p) has exactly two interior extrema on the resolved 2^{-13} grid (degree-96 integer-Horner census): $\varepsilon_q \approx 0.019724$ (adjacent to V_1) and $\varepsilon_q \approx 0.083699$ (adjacent to the tetra|cube dominance boundary 0.084410) — a census, not an identity: the extrema do not coincide with the vacua. (5) Phase-band reading (Def-H.31(6)). The three wall-delimited bands that carry vacua — floor band (V_1; static-crystallization side), crossover band (V_2, V_3; sliver + mixed), thread-gas band (V_4) — are the three topological phases of the construction; the labelling of the phase triple as three generations is declared, not derived. The band count equals $|\mathbb{F}|$, the declared family size: the selection principle for three is not supplied; the sliver vacuum V_2 is a bicritical artifact of FT.70(2)'s near-degenerate boundaries — reported, not suppressed. (6) Bare-span bound on the computed cells. The all-four m^2-span is $5.597457 < 6$ — the same order as FT.79(1)'s pool maximum 6 and FT.76(3)'s widest ratio 6.3925: a third independent appearance of the ≈ 6 scale on the computed cells (FT.104(1) quantifies the [EXT] consequence).

Req. Def-H.31; FT.102; CT0.8 (Part 1); Def-H.26; FT.67, FT.68, FT.69, FT.70; FT.76(3), FT.79(1)–(2); L3; exact rational + log-enclosure bisection, integer-Horner dyadic sign census, quadratic-surd verification, variance identities [math; machine: `checks/verify_H_1loop_bridge.py` §E–§F].

Pf.

$$(1):\ V'$$

$$= \text{rational}(\varepsilon_q) + \sum_j \text{rational}_j(\varepsilon_q)\cdot \ln\big(\text{rational}_j(\varepsilon_q)\big):\ \text{each ln by rational atanh-enclosure (exact tails)}$$

band grids (dyadic, wall-graded near $\varepsilon^*_{\text{octa}}$): exactly one $- \to +$ sign flip per band [machine];

bisection to 2^{-36};

$V(\varepsilon^{(i)}) > 0$ by enclosure

V''-flatness at the minimum : $|V(\text{mid}) - V(\varepsilon^{(i)})| \le \frac{1}{2}\sup|V''| \cdot (2^{-37})^2 \ll 2^{-44}$: enclosure values valid at width

$$\le 2^{-44} \text{ [machine]}$$

(2) : interval division of disjoint positive enclosures;

$m = \sqrt{m^2}$: dyadic $\sqrt{}$-enclosures (integer squaring)

(3) : $W_\Gamma - W_{\Gamma'}$ integer polynomials : root census $2/1/2$ on the resolved grid;

dominance at vacua : exact rational comparison of $W_\Gamma(\text{mid})$

$V' > 0$ on both no-vacuum windows : enclosure census at 41 dyadic points each [machine]

(4) : $\langle k\rangle_y = y\mathcal{Z}'/\mathcal{Z} \implies y\, d\langle k\rangle/dy = \text{Var}_y(k)$ [math];

$\chi_\Gamma = 2\,\text{Var}\,/y$: strict decrease on the exact grid [machine]

$$y = (1-4\varepsilon)^2/\varepsilon^2$$

$$\iff y\varepsilon^2$$

$$= (1-4\varepsilon)^2$$

$$\iff (16-y)\varepsilon^2 - 8\varepsilon + 1$$

$$= 0$$

$$\iff (16d-n)\varepsilon^2 - 8d\varepsilon + d$$

$= 0$ $(y = n/d)$: unique root in $(0, \frac{1}{4})$ [machine]

$\chi_{\mathbb{F}}$ extrema : numerator of $d\chi_{\mathbb{F}}/dp$ is an integer polynomial of degree 96 : sign census at p

$= m/2^{13}$ by integer Horner : 2 flips [machine]. ■

Phys. The effective potential has four exact vacua, bracketed by walls that are the phase diagram: every crossover boundary FT.70 computed is now a divergent barrier at which the six seal modes go soft, and every vacuum is a phase — the crystallization-side well, the thread-gas well, and two crossover-band wells whose depths are set by how close the Platonic boundaries sit to one another. The lightest vacuum is the thread-gas phase (cube-dominant); the heaviest is the bicritical sliver between the near-degenerate cube/tetrahedron boundaries. The mass read is the entropy-deficit read in nats [L3(4)]: no pole identification is made or implied relating-equation residue applies unchanged (FT.104). No particle identification.

Theorem 3.78 (FT.104 — The [EXT] audit, the range-species classification, and the fourth read of Gap 7). Thm. External-data clause (FT.73(5)/FT.79(4) pattern): the hierarchies compared below — charged leptons 1:207:3477, down-type 1:20:900, up-type 1:600:80000 — are supplied measured data [EXT]; they enter no derivation, no definition, no coefficient of the construction; this document identifies no particles ; no parameter is fitted (P1, P5; FT.87(11) precedent). (1) Bare 1-loop read: exact shortfalls. The phase triple $(m_1, m_2, m_3) = (1,\ 1.509244,\ 1.906061)$ in m, $(1,\ 2.277816,\ 3.633067)$ in m^2 [FT.103(2)] against [EXT]:

	m-read shortfall		m^2-read shortfall	
[EXT] family	middle	outer	middle	outer
leptons $(207,\ 3477)$	$\times 137.2$	$\times 1824.2$	$\times 90.9$	$\times 957.0$
down-type $(20,\ 900)$	$\times 13.3$	$\times 472.2$	$\times 8.8$	$\times 247.7$
up-type $(600,\ 80000)$	$\times 397.6$	$\times 41971.4$	$\times 263.4$	$\times 22020.0$

(enclosure-proven lower bounds: every middle shortfall > 10, every outer > 100.) The m-span 1.906 is of the same order as the tree proxy's flat span 2 and the all-four m^2-span 5.597 of the same order as FT.79(1)'s pool maximum 6: the loop correction re-shapes the vacuum geometry — bands, walls, dominance windows — without extending the computed cells' numeric range. (2) Range-species classification. (a) Bare read: the vacuum-depth range on a family grows with the cells' size — floor-side barrier instance at $p_0 = \frac{15}{16}$: $-\ln W_\Gamma(p_0) = (7.31,\ 16.01,\ 27.05)$ with exponents $2E - K_{\text{top}} = (6, 12, 16)$ and the exact lower-bound law $-\ln W_\Gamma(p_0) \geq (2E - K_{\text{top}}) \ln \frac{1}{q_0} - \ln \mathcal{Z}_\Gamma^{\text{tot}}(16)$ [machine] — linear in size, unbounded in the family: unlike the defect-angle read (hard ceiling $4\hat{\mu}'^2$ [FT.76(3)]) and of the same order class as the cover ladder — spanning $1{:}R$ needs $V \geq 4R/3$ [FT.79(2): $V \geq 276$ down-type, $V \geq 4636$ leptons]. (b) Composed tilt read (Def-H.40 species at general γ, composed with the vacua — declared composition, NOT adopted): per-edge contraction $c(\varepsilon_q) = 1 - \frac{\gamma}{4} = \frac{3}{4} + \varepsilon_q$ [L2], composed over E_{int} internal edges [Def-H.40's composition clause]:

$$T_i(E) := c\big(\varepsilon_q^{(i)}\big)^{2E}; \qquad \big(c(\mathrm{V}_1), c(\mathrm{V}_3), c(\mathrm{V}_4)\big) = (0.771337,\ 0.830091,\ 0.940792);$$

$$\beta_{\text{mid}} := \tfrac{c(\mathrm{V}_3)}{c(\mathrm{V}_1)} = 1.076172; \qquad \beta_{\text{out}} := \tfrac{c(\mathrm{V}_4)}{c(\mathrm{V}_1)} = 1.219690;$$

the range is exponential in E — the first read in Gap 7's comparison set that provably escapes both hard ceilings — with exact thresholds (each strict: $E-1$ falls below; enclosure-proven):

$$\beta_{\text{out}}^E \geq 20 \iff E \geq 16; \quad \geq 207 \iff E \geq 27; \quad \geq 900 \iff E \geq 35; \quad \geq 3477 \iff E \geq 42.$$

Against FT.79(2)'s $V \geq 276$ (down-type, cover ladder) and $V \geq 4636$ (leptons): a two-order compression of the size requirement, $E_{\text{int}} \approx 16$–$42$ — within the corpus's computed-cell scale class. (3) Proportionality obstruction (exact). No single E carries any [EXT] triple through (2b): the triple's shape is fixed by the base pair $(\beta_{\text{mid}}, \beta_{\text{out}})$, and

$$\frac{\ln \beta_{\text{mid}}}{\ln \beta_{\text{out}}} = 0.369645 \qquad \text{vs} \qquad \frac{\ln 207}{\ln 3477} = 0.654006 \quad \text{[EXT]}$$

(disjoint enclosures; mismatch factor 1.7693). At the outer threshold $E = 42$: $\beta_{\text{mid}}^{42} = 21.83 < 207$ — middle shortfall $\times 9.5$. The obstruction datum is the base clustering: the walls cluster in a band of width 0.0115 [FT.70(2)], so the contraction bases span only $(0.771, 0.941)$; an [EXT]-shaped triple would need the vacua's $\ln c$-spreads proportional to $(\ln 207, \ln 3477)$ — missed by the exact factor above. This is the 1-loop species' own structural obstruction, dual to FT.76(3)'s angle bound and FT.79(2)'s edge budget. (4) Adjudication summary. The bridge is constructed and machine-exact: the tree limit reproduces the covering census [FT.102(3)]; the loop layer produces the corpus's first multi-vacuum spectrum — four exact vacua, three wall-delimited phase bands, walls at FT.70's boundaries, dominance windows with exact crossings [FT.103]; the mass read is ΔS-admissible (all positive, nats) [L3]. It does not, on the computed cells: reach the [EXT] spans (clause (1)'s exact shortfalls); derive the number three (band count $= |\mathbb{F}|$, declared [FT.103(5)]); relate to $m_{\text{pole}}/m_{\text{hid}}$ (Def-H.39 unchanged; the category discipline of FT.89(2) applies to every cross-read). The exponential mechanism is real but sits in the composed species (clause (2b)), whose size thresholds are exact and whose proportionality datum (clause (3)) is the new obstruction. Item 7's comparison set gains the fourth read; the mass-hierarchy record (two hard ceilings + linear bare + exponential composed, all computed cells ≤ 6.39 in span), 2 and 3 unchanged.

Req. FT.102, FT.103; FT.73(5); FT.76(3); FT.77; FT.79(1), (2), (4)–(6); FT.84(2); FT.87(9), (11); FT.89(2); L2, L3; Def-H.39, Def-H.40; P1, P5; [EXT] data clause; rational log-enclosures for every threshold and shortfall [math; machine: `checks/verify_H_1loop_bridge.py` §G–§H].

Pf.

$$(1):\ \text{shortfall}$$

$$= \text{[EXT]/ratio}:\ \text{interval division of enclosure-proven ratios by exact integers [machine]}$$

$$(2a):\ W_\Gamma(p) \leq \mathcal{Z}_\Gamma^{\text{tot}}(16) \cdot q^{2E - K_{\text{top}}} \text{ for } p$$

$$\geq p_0 \ (\text{termwise},\ p \leq 1):\ \text{log-enclosure comparison at } p_0$$

$$= \tfrac{15}{16} \text{ [machine]}$$

$$(2b):\ \beta_{\text{out}}^E$$

$$\geq R$$

$$\iff E \ln \beta_{\text{out}} \geq \ln R : \text{ both sides by rational log-enclosures; strictness at } E-1 \text{ [machine]};$$

$$c = \tfrac{3}{4} + \varepsilon_q \text{ [L2 : } 1 - \tfrac{\gamma}{4},\ \gamma = 1 - 4\varepsilon_q]$$

$$(3) : \ \ln \beta \text{ enclosures at tolerance } 2^{-50} : \text{ quotient intervals disjoint};$$

$$42 \ln \beta_{\text{mid}} < \ln 207 \text{ [machine]}$$

$$(4) : \text{ conjunction of FT.102/FT.103 clauses and } (1)\text{–}(3);$$

no identification, no calibration invoked [P1, P5]. ■

Phys. The bridge carries the census across into geometry: walls where the phase diagram has boundaries, vacua where the ensemble has phases, masses as missing information. What it does not carry is the observed hierarchy: on the computed cells the span stays of order six in m^2 — the third independent appearance of that scale — and the exponential range that does exist lives in the composed tilt size, reaching the lepton span at $E_{\text{int}} = 42$ while missing the middle value by the exact factor 9.5. The ledger has learned to speak geometry; the hierarchy question now has a fourth, unbounded-range species with quantified size requirements and an exact proportionality obstruction any future family must beat. No particle identification; no calibration; the pole↔info relational question remains open, its residue sharpened.

3.37 Tier F.36 — The Topology-Blind Path Integral: Observable-Conditioned Sectors and the Janus Sum

Definition 3.38 (Def-H.32 — The topology-blind path integral). Def. A construction class on the prior canon (Def-H.24 pattern: no new axiom); clauses (1)–(7) are constitutive interface definitions. Scopes (S1)–(S4) of Def-H.25 are inherited at every use ($\kappa_e \equiv 1$ minimal-edge class on real edges unless a κ-class is named; literal $\aleph$-domain; seal/Janus bookkeeping static, $\mathcal{J} = 1$; virtual edges weight 1, no phase).

(1) The topology space and the budget filtration (Def-H.32(1)). $\mathcal{G} :=$ the isomorphism classes of finite connected simple graphs (self-loops excluded [A1]; skeletons read as their bidirectional digraphs, $2E$ directed edges [A7, D0.1]; connectedness per D2.13's subnetwork scope). Regularization = the declared budget filtration

$$\mathcal{G}_B := \{\Gamma \in \mathcal{G} : \ V(\Gamma) \leq B\}, \qquad \mathfrak{C}_B(\Gamma) := \{R \subseteq E_{\text{dir}}(\Gamma) : \ |R| \leq B_K\} \quad [B_K \text{ default } 2E(\Gamma)],$$

the budget being the ledger's own content measure: $|R| = n_{\mathcal{R}}$, $S_{\mathfrak{L}} = n_{\mathcal{R}} \ln 2$ [CT1.29]. Every filtered sum below is a finite exact sum [FT.68(7) pattern]; the $B \to \infty$ limit is not constructed (growth data: FT.105).

(2) Configurations and the two weight layers. A configuration on Γ is $C = (R, \sigma) \in \mathfrak{C}(\Gamma)$ [D0.10]: pattern R with per-edge signs $\sigma_e \in \{\pm 1\}$ [CT0.9, D3.4]; vertex types $(n_{\text{in}}, n_{\text{out}})$ [A7, D0.8]; locked (D^--configured) $\iff$ $k_v \geq 4$ [CT4.5 source criterion; CT0.11]. The A8 edge weight is $w(C) = \mathcal{J}(C) \prod_{e \in E(C)} G(e) \cdot (-1)^{\aleph(\sigma, C)}$ [A8; at (S1): $G(e) = -\sigma_e$; $\aleph = \sum_{\text{centres}} |f_v|$, $f_v = \frac{s_{\text{out}} - s_{\text{in}}}{2}$ [D3.3]]. Two layers [CT2.8]:

$$\text{amplitude layer}: \quad v_\Gamma(R) := \sum_\sigma w(R, \sigma) \qquad [\text{FT.67(2)}: \ v = 2^{|R|}(-1)^{\sum \kappa} \text{ on covers}, \ 0 \text{ else}];$$

$$\text{norm layer}: \quad \nu_\Gamma(R) := \sum_\sigma |w(R, \sigma)|^2 = 2^{|R|}$$

$$[|w(R, \sigma)| = 1 \text{ for every valid pair}; \ \text{FT.68 Rmk}(6c)].$$

(3) Fugacities (Def-H.32(3) — formal indeterminates over $\mathbb{Q}$; no rate datum, no declared value). Per configuration:

$$\mathcal{W}_\Gamma^{\text{amp}}(R) \ := \ v_\Gamma(R)^2 \, t^{|R|} \, u^{L(R)}; \qquad \mathcal{W}_\Gamma^{\text{norm}}(R) \ := \ \nu_\Gamma(R) \, t^{|R|} \, u^{L(R)},$$

$L(R) := \#\{v : \ k_v \geq 4\}$ the locked census; t the crystallization fugacity, u the D^--retention fugacity [CT4.1–CT4.5 vocabulary: absorption/emission/dissolution/retention enter as the static fate fugacity; the emission/dissolution moves live in clause (7)]. Specializations are marked at every use: $t = p/q$ with $p = x_+/(x_+ + x_-)$ [FT.108(4), derived]; tilt refinement $\mathcal{W} \mapsto \mathcal{W} \cdot \tau_{\mathcal{R}}^{|R|} \tau_{\mathcal{V}}^{2E-|R|}$ with spectral values $\tau_{\mathcal{R}} = e^{-\gamma/3}$, $\tau_{\mathcal{V}} = e^{-\gamma}$ [Cor L0a at the branch states; FT.108(7)].

(4) Observables (Def-H.32(4) — internal bookkeeping labels; no identification). A configuration carries chains [D1.4: ordered tuples of distinct real edges]: closed directed simple cycles c, length $L \geq 2$; open directed simple paths, length ≥ 2. Readings:

$$q(c, \sigma) := \frac{\sigma_{e_1} - \sigma_{e_k}}{2} \in \{-1, 0, +1\}$$

$$[\text{EM.9 end registers}; \ \text{FT.40 closed-core form at the tuple's own base}; \ \text{D3.4}],$$

$$S(c): \ \text{the on-shell winding } k(c) = \Theta(c)/2 \in \mathbb{Z} \quad [\text{FT.22}, \ \text{FT.25(3)}],$$

$$\text{minimal ladder [FT.32(6)]}: \ \ L \text{ even} \Rightarrow (\kappa \equiv 1, \ k = L/2); \ \ L \text{ odd} \Rightarrow (\kappa \equiv 2, \ k = L);$$

open chains carry $S = 0$ (no closed core — bookkeeping label, $\neg$ a spin value; the $j = |k|/2$ map is Def-H.38-declared content and is not imported). A chain instance is a pair (c, base) (closed: base = each of the L cyclic starts; open: the tuple itself). The δ-constraints of the sector sum are realized by summing over instances with the readings fixed.

(5) The sector partition functions. For labels (q, S) and budget (B, B_K):

$$Z_B^\bullet(q, S) \ :$$

$$= \sum_{\Gamma\in\mathcal{G}_B} \sum_{R\in\mathfrak{C}_B(\Gamma)} \mathcal{W}_\Gamma^\bullet(R)\cdot \#\{(c,\text{base}):\ q(c,\sigma)\text{-sector}=q,\ S(c)=S\}\Big|_{\sigma\text{-resolved}},$$

• $\in \{\text{amp},\text{norm}\}$: the amplitude layer with σ resolved inside the sector condition and w-weights; the norm layer with $|w|^2$-weights (closed form: $m_q\,2^{|R|-2}$ per instance, $m_q=(2,1,1)$ at $q=(0,\pm1)$ [FT.106(3)]). Both are exact polynomials in $\mathbb{Z}[t,u]$ per budget cell.

(6) The mass expectation (Def-H.32(6)). For sectors with $Z_B^{\text{norm}}(q,S)\neq 0$:

$$m_B^2(q,S) \;:=\; \frac{1}{Z_B^{\text{norm}}(q,S)}\sum_{\Gamma,R}\mathcal{W}_\Gamma^{\text{norm}}(R)\cdot \#\text{inst}(q,S;\Gamma,R)\cdot\Delta S(\Gamma,R),$$

$$\Delta S(\Gamma,R) \;:=\; 2\,n_{\mathcal{V}}(\Gamma,R)\,\big\langle|\delta\rho|^2\big\rangle_{\text{branch states}} \;=\; \tfrac{3}{2}\,n_{\mathcal{V}}(\Gamma,R) \;=\; \tfrac{3}{2}\big(2E(\Gamma)-|R|\big)$$

$$[\text{L4};\ \text{L3}(1)\text{ exact leading};\ \text{n}_\text{V}:=\#\text{virtual channels}],$$

the branch-state values $|\delta\rho|^2 = (\frac{3}{4},\frac{1}{12})$ (virtual $\Pi_{\mathcal{V}}$, real $\Pi_{\mathcal{R}}/3$) computed matrix-exact [FT.107(2)]; dimensionless nats [L3(4) species]; no coupling to $m_{\text{pole}}/m_{\text{hid}}$ (mass-form bridge residue; Def-H.39's bridge slots unconstructed); the exact relative-entropy variant ($\ln 4$ per virtual channel) is a specialization remark, not the read.

(7) The Janus move system and the transition operator (Def-H.32(7)). States $s=(\Gamma,R)$, Γ any finite simple graph on $\le B$ labelled nodes (the path space admits disconnected and edgeless states; $\mathcal{G}$-membership is imposed at the sector sums). Moves, with formal weights:

$$J^+(e):\ R\mapsto R\cup\{e\},\ e\notin R\ [\text{weight } x_+];$$

$$J^-(e):\ R\mapsto R\setminus\{e\}\ [\text{weight } x_-]$$

$$[\text{D1.2}:\ \text{one pair across A4 per event}];$$

$$N^+(v,S):\ \text{add node } v \text{ with } d=|S|\ge 1 \text{ virtual edges to existing nodes } [\text{weight } y_d];$$

$$N^-(v):\ \text{remove node } v \text{ of degree } d \text{ together with its } r(v) \text{ real edges } [\text{weight } z_d\,x_-^{r(v)};\ \text{D1.2}:\ \text{reality is per pair}].$$

$\hat U := \sum_{\text{moves}} w\,|t\rangle\langle s|$ over the budget-truncated state space; $\langle s'|\hat U^n|s\rangle\in\mathbb{Z}[x_\pm,y_d,z_d]$; the topology sum of clause (5) and the transition sum $\sum_n\langle s'|\hat U^n|s\rangle$ are the static and dynamical faces of the same move content (FT.108).

Rmk (interfaces and non-imports). Frozen data only: A8/D3.3 weights; FT.67(2)'s classifier; CT0.9/CT0.11 registers and cap; CT1.29 ledger nats; EM.9/FT.40/FT.22/FT.25/FT.32 readings; CT4.1–CT4.5 lock vocabulary; Cor L0a tilt spectral values; L3/L4 deficit reads. Not imported: any declaration-entry entry (Def-H.38's $j=|k|/2$, Cal-F's rate datum, the non-adopted bridge-map's observer map, Def-H.39's bridge, Def-H.40's composition, Def-H.31's dictionary — the ensemble parameter p and the dictionary $p=1-4\varepsilon_q$ are derived here as the stationary measure of clause (7)'s edge chain and its specialization [FT.108(4)]). Not declared: aspect-resolved couplings

inside w (the gauge-block coupling structure does not close [FT.82]; the aspects of A8/CT0.7(4) label the move algebra only [FT.108(1)]); observer-channel relativity of the readings (Def-H.1 vocabulary; the sector sums are channel-free instance censuses — a declaration would be needed to make them observer-relative, and none is made). Topology labels (names of cells) are internal bookkeeping throughout.

Theorem 3.79 (FT.106 — Weight transport and the two-layer sector structure). Thm. Under Def-H.32: (1) The exact-cover classifier is topology-blind. FT.67(2)'s classification — $v_\Gamma(R) = 2^{|R|}(-1)^{\sum\kappa}$ iff the type-$(1,1)$ centres exactly single-cover R, else 0 — holds verbatim on every $\Gamma \in \mathcal{G}$: its involution proof consumes only local vertex types [D3.3] and sign flips, no global topology. Machine: brute-force sign sums equal the classifier on all 5845 patterns of all ten $V \le 4$ cells and on 5888 further $V{=}5$ witness patterns ($E \le 5$). (2) Closed cells die in amplitude on every topology; covers thread them. Every pure directed cycle (any length, any cell) has $v = 0$ [FT.67(3a) transport]; on every enumerated cover containing directed cycles the cycle-alternation lemma holds exactly (centres/non-centres strictly alternate; all cycle lengths even; no odd cycle inside any cover) [FT.69(3) transport]; census: the only closed content inside covers at $B = 4$ is dissolved digons (and their locked-hub families [FT.69(5)]). (3) The two-layer sector structure (exact census at $B = 4$; σ-resolved scope: all cells with $2E \le 8$ exhaustive, K_4 at $K \le 4$ [FT.74(5) precedent], diamond at $K \le 5$). Sector totals over $\mathcal{G}_4$, Z^{norm} complete / Z^{amp} in scope:

(q,S)	Z^{norm}	Z^{amp}	
$(\pm1,0)$ transit	197912	1196	
$(0,0)$ transit	395824	5336	$m_q = (2,1,1)$
$(\pm1,1)$ digon winding	787358	802	dissolved digons in covers
$(0,1)$	1574716	4796	
$(\pm1,2)$ (C_4-type winding)	655776	0	no cover contains a 4-cycle [FT.69(3)(v)]
$(0,2)$	1311552	0	
$(\pm1,3)$ (triangle winding, $\kappa \equiv 2$)	1000512	0	odd cycles never enter covers [FT.69(3)(iv)]
$(0,3)$	2001024	0	

Structural identities (all machine): (a) norm-layer closed form — per chain instance the σ-sum factorizes as $m_q 2^{|R|-2}$, $m_q = (2,1,1)$ at $q = (0,\pm1)$ [EM.9's four end-pairs]; closed form = census in scope, every cell and sector. (b) amplitude q-sum identity — for every winding k: $\sum_q Z_B^{\text{amp}}(q,k) = \sum_{R \text{ cover}} v_\Gamma(R) \cdot \#\text{inst}_k(R)$ exactly: the charge-resolved amplitude layer admits non-cover support (18 nonzero cancelling entries at $B = 4$ — a sector-level refinement of FT.67's total), and the non-cover parts cancel exactly across q. (c) amplitude death of $k \ge 2$ — $Z^{\text{amp}}(q,k) = 0$ in total for $k = 2$ and odd $k \ge 3$; the winding-1 sector is amplitude-live exactly through dissolved-digon covers. (d) ω-symmetry — $Z^\bullet(q,S) = Z^\bullet(-q,S)$: norm layer unconditional ($|w|$-invariance [EM.24(5) pattern]); amplitude layer on the cover support ($|R| = 2|\mathcal{C}|$ even). (e) charge alphabet — $\{-1,0,+1\}$ attained in every chain-carrying cell [EM.9 transport]; every sector value is a nonnegative integer (norm) resp. integer (amp).

Req. Def-H.32; A7, A8; D0.8, D0.10, D3.3, D3.4; CT0.9, CT0.11 (Part 1); CT2.8 (Part 2); EM.9, EM.24(5) (Part 2); Def-H.25; FT.67, FT.68, FT.69(3)–(5), FT.74(3),(5); exhaustive enumeration, involution transport, factorized σ-sums [math; machine: `checks/verify_H_topology_blind.py` §B–§C].

Pf.

$(1):$ FT.67(2)'s involution flips σ_e on unread/double-read edges per vertex — local data [D3.3];

brute force $=$ classifier on $5845 + 5888$ patterns [machine]

$(2):$ pure cycles : every vertex interior $\Rightarrow$ cov $\equiv 0 \ \vee \ 2 : \ v = 0$ [FT.67$(3a)$ proof transport];

alternation census on every cover with cycles at $B = 4$ [machine]

$(3a):$ $q(c, \sigma)$ reads two edge signs : $\#\{\sigma : \ q(c, \sigma) = q\}$

$= m_q 2^{L-2}$ on the chain;

rest free : $2^{|R|-L}$: product $= m_q 2^{|R|-2}$;

census $=$ closed form [machine]

$(3b):$ $\displaystyle\sum_q \sum_{\sigma:\, q(c,\sigma)=q} w(R, \sigma) = \sum_\sigma w(R, \sigma) = v_\Gamma(R)$ [instances are σ-independent];

$v = 0$ off covers $[(1)]$

$(3c):$ covers contain no 4-cycles and no odd cycles at B

$= 4$ [census]

$\Longrightarrow$ $\#\text{inst}_{k \geq 2}$

$= 0$ on covers $\Longrightarrow \displaystyle\sum_q Z^{\text{amp}} = 0$ $[(3b)]$

$(3d):$ $\sigma \mapsto -\sigma : \ q \mapsto -q$;

$|w(R, -\sigma)| = |w(R, \sigma)|$;

$w(R, -\sigma) = (-1)^{|R|} w(R, \sigma)$;

$$|R| = 2|\mathcal{C}| \text{ even on covers [FT.79(2)]}$$

$$(3e):\ m_q > 0\ \forall q;\ \text{ instance existence per cell [machine census]. } \blacksquare$$

Phys. Weighing every topology against every pattern does not blur the frozen anatomy — it confirms it cell by cell: the amplitude layer remains a thread gas with dissolved even cores, the norm layer remains strictly positive, and the odd cores stay amplitude-dead on all ten topologies at once. The new datum is finer than the total: resolved by charge, the amplitude layer carries cancelling non-cover parts whose sum is exactly the cover census — FT.67's cancellation, seen from inside. No particle identification.

Theorem 3.80 (FT.107 — The observable sectors and the mass expectation). Thm. Under Def-H.32: (1) On-shell ladder (transport of FT.32(6)). A directed L-cycle is on-shell at $(\kappa \equiv 1,\ k = L/2)$ for L even and at $(\kappa \equiv 2,\ k = L)$ for L odd; instances at $B = 4$: $k = 1$ (digons, $\kappa \equiv 1$), $k = 2$ (4-cycles, $\kappa \equiv 1$), $k = 3$ (triangles, $\kappa \equiv 2$); the floor card $L = 2$, $k = 1$ is FT.28's $\Sigma_{\min}$ datum reproduced by the census. (2) Branch-state deficits (matrix-exact). At the canonical branch states [CT0.8 sharp readings]: $\rho_{\mathcal{V}} = \Pi_{\mathcal{V}}$: $|\delta\rho|^2 = \frac{3}{4}$; $\rho_{\mathcal{R}} = \Pi_{\mathcal{R}}/3$: $|\delta\rho|^2 = \frac{1}{12}$; leading deficits $2|\delta\rho|^2 = (\frac{3}{2},\ \frac{1}{6})$ [L3(1)]; hence $\Delta S(\Gamma, R) = \frac{3}{2}\, n_{\mathcal{V}}(\Gamma, R) = \frac{3}{2}(2E - |R|)$ — an exact rational per configuration. (3) The sector mass table ($B = 4$, $(t, u) = (1, 1)$, norm layer; exact rationals). $m_B^2(q, S)$ is independent of q (charge-degenerate at the leading deficit read — the topology-blind companion of FT.100's phase-blindness):

S (winding)	0 (transit)	1	2	3
$m_4^2(\cdot, S)$	$\frac{843957}{98956}$	$\frac{1921662}{393679}$	$\frac{91}{23}$	$\frac{13713}{3088}$
	≈ 8.5286	≈ 4.8813	≈ 3.9565	≈ 4.4407

all strictly positive (ΔS-admissible, nats [L3(4)]); the ordering places the transit sector heaviest — thread-network instances ride sparser patterns (larger $n_{\mathcal{V}}$), the deficit read weights missing information. (4) Quadratic budget law. For every sector and budget: $0 < m_B^2(q, S) \le \frac{3}{2} \max_\Gamma n_{\mathcal{V}} \le \frac{3}{2} B(B-1)$ [exact bound; max at $R = \varnothing$ on the densest cell]: quadratic in the node budget — at $B = 4$: bound 18, attained census maximum $\frac{843957}{98956} < 9$. The winding spread at $B = 4$ is $\frac{91}{23} \le m^2 \le \frac{843957}{98956}$ (span ≈ 2.16 in m^2): the computed-cell span remains order-one while the bound grows as $\Theta(B^2)$ — against the fixed-topology bare read's linear-in-size range [FT.104(2a)] and the two hard ceilings [FT.76(3), FT.79(2), FT.97(5)]. The fugacity axes add an unbounded structural direction (FT.108(5)); no axis value is declared. No [EXT] datum enters this part; no comparison is made.

Req. Def-H.32; CT0.7, CT0.8 (Part 1); L3, L4; FT.22, FT.24, FT.25(3), FT.28, FT.32(6); FT.100; FT.104(2); FT.76(3), FT.79(2), FT.97(5); exact rational matrix arithmetic, sector census algebra [math; machine: `checks/verify_H_topology_blind.py` §C–§D].

Pf.

$$(1):\ \Theta = \sum_{e \in c} \widehat{ds}^{\,2}(e):\ \kappa \equiv 1:\ \Theta = L;\ \ \kappa \equiv 2:\ \Theta = 2L;$$

$$\text{on-shell} \iff \Theta \in 2\mathbb{Z} \text{ [FT.22]};$$

$$k_{\min} = \Theta_{\min}/2 \text{ [FT.32(6)]}$$

$$(2):\ \delta\rho = \Pi_{\mathcal{V}} - \tfrac{1}{4}I:\ \text{spec} = (-\tfrac{1}{4}, -\tfrac{1}{4}, -\tfrac{1}{4}, \tfrac{3}{4}):\ \mathfrak{T}(\delta\rho^2) = \tfrac{3}{16} + \tfrac{9}{16} = \tfrac{3}{4};$$

$$\delta\rho$$

$$= \tfrac{\Pi_{\mathcal{R}}}{3} - \tfrac{1}{4}I:\ \text{spec}$$

$$= (\tfrac{1}{12}, \tfrac{1}{12}, \tfrac{1}{12}, -\tfrac{1}{4}):\ \mathfrak{T}(\delta\rho^2)$$

$$= \tfrac{3}{144} + \tfrac{1}{16}$$

$$= \tfrac{1}{12} \text{ [machine]}$$

$$(3):\ m^2$$

$$= \left.\frac{\sum_{\Gamma,R} \nu\, t^{|R|} u^L \#\text{inst} \cdot \frac{3}{2}(2E - |R|)}{\sum_{\Gamma,R} \nu\, t^{|R|} u^L \#\text{inst}}\right|_{t=u=1}:\ \text{exact rational division of census integers;}$$

$$q\text{-independence}:\ \Delta S(R) \text{ is } \sigma\text{-free and } m_q\text{-symmetric } [(3a), (3d) \text{ of FT.106}]$$

$$(4):\ n_{\mathcal{V}} \le 2E \le B(B-1) \text{ [simple graphs]};$$

$$\text{census max} < 9 < 18 \text{ [machine]};$$

$\Theta(B^2)$ vs $\Theta(B)$ [FT.104$(2a)$] and vs $4\hat{\mu}'^2/3V$-ladder ceilings. ■

Phys. The mass of an observable-conditioned sector is the ensemble average of missing information over every topology that carries it — one exact rational per (charge, winding) label at each budget, charge-blind at this order, ordered by how sparse the carrying patterns are. The transit sector weighs heaviest at $B = 4$: open threads live on the emptier configurations, and emptiness is what the deficit read weighs. The quadratic budget law is the mechanism statement: topology-blindness moves the range axis from skeleton size to the budget itself — how far that axis must be pushed to reach any external span is arithmetic on $\frac{3}{2}B(B-1)$, not physics claimed here. No particle identification; no external datum used.

Theorem 3.81 (FT.108 — The Janus sum: move algebra, budget conservation, derived detailed balance, and the Tier F.35 boundary). Thm. Under Def-H.32: (1) Move-label algebra. The move classes carry CT0.7(4)'s aspect labels — $J^{\pm} \leftrightarrow F_{J^{\pm}} = \mathcal{E}_{-+}/\mathcal{E}_{--}$ (S-odd), node moves $\leftrightarrow$ the seal-generated topology change — and the frozen identities hold matrix-exact over $\mathbb{Q}(i)$ on the label algebra: $[\Sigma, J_i] = 4iK_i$, $[\Sigma, K_i] = -4iJ_i$ [CT1.5]; $\{\mathbb{S}, J_i\} = 0 = \{\mathbb{S}, K_i\}$ [CT0.7(3)]. (2) Budget conservation (the coarse-grained invariant). The grade map $\mathrm{gr} : (\Gamma, R) \mapsto \big(V,\, E_{\mathrm{pot}},\, |R|\big)$ is a homomorphism of the move system: every monomial $x_+^{a_+} x_-^{a_-} \prod y_d^{m_d} \prod z_d^{n_d}$ of $\langle s'|\hat{U}^n|s\rangle$ satisfies

$$a_+ - a_- = |R'| - |R|; \qquad \textstyle\sum_d m_d - \sum_d n_d = V' - V; \qquad \sum_d d\,(m_d - n_d) = E'_{\mathrm{pot}} - E_{\mathrm{pot}}$$

— exhaustive on every path monomial at $n \le 3$ from every $V \le 2$ source [machine]. Node dissolution carries its incident real edges' own J^- factors [D1.2]: $z_d\, x_-^{r(v)}$ — without them the grade map fails (witnessed). (3) Path counts (two enumerators). Operator iteration and depth-first path enumeration agree coefficient-wise on $\hat{U}^n$, $n \le 4$; witness: $\langle \mathrm{digon}, R{=}E \,|\, \hat{U}^3 \,|\, \bullet\rangle = 2\,x_+^2 y_1 + \cdots$ — the two admissible move orderings [machine]. (4) Detailed balance derived; the dictionary is a specialization. The per-pair Janus dynamics [D1.2] on (virtual, real) has master generator $Q = \left(\begin{smallmatrix} -x_+ & x_+ \\ x_- & -x_- \end{smallmatrix}\right)$ over $\mathbb{Z}[x_+, x_-]$; $\ker Q$ is one-dimensional with span (x_-, x_+) [exact: $(x_-, x_+)Q = 0$; $\det Q = 0 \neq Q$] — the unique stationary measure, hence

$$p \;=\; \frac{x_+}{x_+ + x_-} \qquad [\text{derived},\ \neg\text{imported}];$$

the substitution $x_+ = 1 - 4\varepsilon_q$, $x_- = 4\varepsilon_q$ gives $p = 1 - 4\varepsilon_q$ identically [machine] — the fixed-topology ensemble parameter is a specialization of the derived stationary ratio, and the sealed universe-level balance $\#J^+ = \#J^-$ [D1.3] is the balanced-sector restriction of (2)'s first identity. (5) Resolvent structure. On a single pair channel, $\sum_n \langle \varnothing|\hat{U}^n|\varnothing\rangle w^n = \dfrac{1}{1 - w^2 x_+ x_-}$ in $\mathbb{Q}[w]/(w^{12})$ [machine: $(1 - w^2 x_+ x_-) \cdot \sum \equiv 1$ coefficient-wise] — the transition sum is a geometric resolvent with pole at $w^2 x_+ x_- = 1$: an unbounded structural range direction in the formal fugacities, no value declared (P4-class datum absent; any numerical use requires a logged declaration first). (6) The Tier F.35 boundary limit. With node moves off ($y_d, z_d \mapsto 0$) the state space decomposes into fixed-topology sectors, and on each sector Γ_0 the amplitude-layer cover generating function

$$\Phi_{\Gamma_0}(t) \;:=\; \sum_{R\ \text{cover}} v_{\Gamma_0}(R)^2\, t^{|R|} \;=\; \sum_k c_k 16^k t^{2k}$$

satisfies the exact polynomial identity

$$q^{2E}\, \Phi_{\Gamma_0}\big(p/q\big) \;=\; W_{\Gamma_0}(p)$$

[coefficient-wise; machine on tetra/cube/octa against the frozen FT.68(1),(3)/FT.69(2) rows and on all nine live $V \le 4$ cells]

— the fixed-topology tree layer of Def-H.31(2) [Tier F.35] is recovered identically; the lock refinement census at (t, u): the tetrahedron's 16 locked covers carry lock count 1 each [FT.69(5) family]. Consequently the fixed-topology slice of the topology-blind integral carries Tier F.35's boundary data — its effective potential, four vacua, phase triple and FT.70 walls [FT.102–FT.104, under Def-H.31's interface definitions, cited as boundary content, not re-derived] — as the $y_d, z_d \to$

0 restriction, per clause (4)'s specialization of t. (7) Tilt spectral specialization. At the branch states, $Z_\gamma = \mathfrak{T}(\rho e^{-\gamma\rho})$ [Cor L0a] evaluates exactly by spectral calculus: real branch $e^{-\gamma/3}$, virtual branch $e^{-\gamma}$; at $\gamma = 1$: disjoint rational enclosures $e^{-1/3} \in [0.716531311, \ldots]$, $e^{-1} \in [0.367879441, \ldots]$, strict order enclosure-proven [machine].

Req. Def-H.32; A4, A8; D0.3, D1.2, D1.3, D3.3; CT0.7, CT0.7a, CT1.2, CT1.5 (Part 1); L0 (Cor L0a); Def-H.25, Def-H.26, Def-H.31 (Tier F.35, boundary citation); FT.67, FT.68, FT.69(2),(5), FT.70, FT.74; FT.102, FT.103, FT.104(2a) (Tier F.35); exact multivariate polynomial arithmetic over $\mathbb{Z}$, master-equation kernel algebra, truncated power series, rational exp-enclosures [math; machine: `checks/verify_H_topology_blind.py` §E–§G].

Pf.

$$(1):\ \text{matrices over } \mathbb{Q}(i) \text{ from CT1.2's definitions verbatim;}$$

$$\text{all commutator/anticommutator entries compared exactly [machine]}$$

$$(2):\ J^{\pm}:\ \Delta|R| = \pm 1,\ \Delta V = \Delta E_{\text{pot}} = 0;$$

$$N^{+}(d):\ (\Delta V, \Delta E_{\text{pot}}, \Delta|R|) = (1, d, 0);$$

$$N^{-}(v):\ (-1, -d, -r(v)) \text{ with weight } z_d x_-^{r(v)}:\ \text{grades match exponents monomial-wise [machine, exhaustive at } n \le 3]$$

$$(3):\ \text{dict-based operator iteration}$$

$$= \text{recursive DFS aggregation [machine, coefficient-wise equality of multivariate polynomials]}$$

$$(4):\ (x_-, x_+)Q = (x_-(-x_+) + x_+x_-,\ x_-x_+ - x_+(x_-)) = (0,0) \text{ [exact in } \mathbb{Z}[x_+, x_-]];$$

$$\det Q = x_+x_- - x_-x_+ = 0,\ Q \neq 0:\ \text{rk} = 1$$

$$p = \tfrac{x_+}{x_+ + x_-};$$

$$x_+ = 1 - 4\varepsilon_q,\ x_- = 4\varepsilon_q:\ x_+ + x_- = 1 \implies p = 1 - 4\varepsilon_q \text{ [polynomial identity; machine]}$$

$$(5):\ (\hat{U}^{2m})_{\varnothing\varnothing} = (x_+x_-)^m,\ (\hat{U}^{2m+1})_{\varnothing\varnothing} = 0 \text{ [single channel]};$$

$$(1 - w^2x_+x_-)\sum_m (w^2x_+x_-)^m = 1 \text{ in } \mathbb{Q}[w]/(w^{12}) \text{ [machine]}$$

$$(6):\ v^2 = 4^{|R|} = 16^{|\mathcal{C}|}\ [\text{FT.67}(2),\ \text{FT.79}(2)];$$

$$q^{2E}\textstyle\sum_k c_k 16^k (p/q)^{2k} = \sum_k c_k 16^k p^{2k} q^{2E-2k} = W_{\Gamma_0}(p)\ [\text{Def-H.31}(2)\ \text{form}];$$

$$\text{coefficient-wise on frozen and live rows [machine]}$$

$$(7):\ \text{spec}(\Pi_{\mathcal{R}}/3) = (\tfrac{1}{3},\tfrac{1}{3},\tfrac{1}{3},0):\ \textstyle\sum \lambda e^{-\gamma\lambda} = e^{-\gamma/3};$$

$$\text{spec}(\Pi_{\mathcal{V}}) = (0,0,0,1):\ e^{-\gamma};$$

$$\gamma = 1\ \text{enclosures [machine]}.\ \blacksquare$$

Phys. The Janus seals do what D1.2 always said they do — move one pair across the crystallization criterion — and once node events are admitted with the same per-pair discipline, topology itself becomes a walk in an exactly graded state space. The grade map is the conserved information budget; nothing else is conserved, and nothing else is needed: the stationary measure of a single pair channel comes out as the ratio of the two seal weights, so the ensemble parameter every earlier tier declared is here derived, and the fixed-topology world is the slice on which node moves are switched off. On that slice the boundary data reproduce identically — every covering polynomial, every wall, every vacuum of the fixed-topology construction is the restriction of the topology-blind integral, not a competing answer. The transition sum is a resolvent: its pole is the unbounded direction, its location a datum the corpus does not declare. No particle identification.

3.38 Tier F.37 — The Koide Channel: The C_3-Covariant Seal Bilinear and the Bound-Chain Mixing Angle

Theorem 3.82 (FT.109 — The diagonal seal commutator). Thm. Under CT1.7, with the CT1.2 basis $\{E_0, \Sigma, \hat{E}^{ij}, W_i, J_i, K_i\}$:

$$(1)\quad \sum_{i=1}^{3}[J_i, K_i] = i\sum_{i=1}^{3}\big(\Sigma - E^{ii}\big) = i\big(3\Sigma - \Sigma\big) = 2i\Sigma;$$

$$(2)\quad \sum_{i=1}^{3} E^{ii} = \sum_{i=1}^{3}\sigma_i\otimes\sigma_i = \Sigma \qquad \big[E^{ii} = \tfrac{1}{2}\big(\sigma_i\otimes\sigma_i + \sigma_i\otimes\sigma_i\big) = \sigma_i\otimes\sigma_i\big];$$

(3) basis read: in the 16-element CT1.2 basis the sum $\sum_i [J_i, K_i]$ has exactly one nonzero component — the Σ component, with coefficient exactly $2i$;

$$(4)\quad 2i\Sigma \in \mathcal{E}_{++}\ (\text{the Higgs/gravity block}), \qquad 2i\Sigma \notin \text{span}\{W_k\}.$$

Req. CT1.2, CT1.7 (Part 1); exact $\mathbb{Q}(i)$ matrix arithmetic on all 16 basis elements, basis-subset disjointness [math; machine: `checks/verify_H_koide_channel.py` §A–§B].

Pf.

$$(1):\ [J_i,K_i]=i\big(\delta_{ii}\Sigma-E^{ii}\big)=i\big(\Sigma-E^{ii}\big)\ \text{[CT1.7]};$$

$$\sum_{i=1}^{3}\big(\Sigma-E^{ii}\big)=3\Sigma-\sum_i E^{ii}=3\Sigma-\Sigma=2\Sigma\ [(2)]$$

$$(2):\ E^{ii}=\sigma_i\otimes\sigma_i \text{ and } \Sigma=\textstyle\sum_i\sigma_i\otimes\sigma_i\ \text{[CT1.2 definitions, entry-wise over } \mathbb{Q}(i)]$$

$$(3):\ \textstyle\sum_i[J_i,K_i]$$

$$=2i\Sigma \text{ entry-wise exact over } \mathbb{Q}(i);$$

$$\text{the 16 CT1.2 elements are a basis of } \mathfrak{H}(\mathbb{C}^4)\implies \text{component reads are unique [CT1.2; machine §A]}$$

$$(4):\ S\Sigma S=\Sigma\ \wedge\ T\Sigma T=\Sigma\implies 2i\Sigma\in\mathcal{E}_{++}\ \text{[CT1.2 sector table]};\qquad SW_kS=-W_k\ \text{[CT1.2]}$$

$$\text{span}\{E_0,\Sigma,\hat{E}^{ij}\}\cap\text{span}\{W_k\}=\{0\}\ \text{[disjoint basis subsets; CT1.2]};$$

$$2i\Sigma\neq 0\implies 2i\Sigma\notin\text{span}\{W_k\}\ \text{[machine §A]}.\ \blacksquare$$

Phys. Bracketing each spatial seal with its own temporal seal and summing over the triplet cancels the entire symmetric-tensor content and leaves exactly twice the exchange scalar: the diagonal of the mixed seal commutator is a pure Higgs/gravity-block object, with exact integer coefficient 2 and zero weak component. This integer 2 is the sole numerator entering the mixing angle of FT.112 — no datum, no fit.

Definition 3.39 (Def-H.41 — The bound-chain arena). Def. The bound-chain arena is the minimal counting space combining the two independently present structures of a bound chain c [D1.4]:

$$\mathcal{H}_c\ :=\ \mathcal{H}_{\text{seal}}\otimes\mathcal{H}_{\text{poly}},$$

$$\mathcal{H}_{\text{seal}}:\ \text{the seal triplet at the chain's termini [CT0.12, CT0.7(4):}\ F_{J^\pm}=\text{span}\{J_i\}/\text{span}\{K_i\},\ \dim=3],$$

$$\mathcal{H}_{\text{poly}}:\ \text{the polygonal transfer structure along its length [FT.68(2):}\ \text{the transfer rule } T(t);\ \operatorname{tr}T(1)=3];$$

$$\dim_{\text{eff}}\mathcal{H}_c=3\times 3=9\ \ \text{[FT.110(1)]}.$$

Species (constitutive). A counting space: no inner product, no dynamics, no state, no Hamiltonian is declared on $\mathcal{H}_c$; its dimension enters the channel theorems exactly once — as the denominator of the mixing angle [FT.112: $\delta_\ell = \frac{2}{9}$ with FT.109's diagonal seal coefficient 2 the numerator]. Bars (standing). $\dim \mathcal{H}_c$ is ¬a state count; the ratios r_i^2 [FT.113] reproduce no measured hierarchy — the exact nonzero shortfalls stand [FT.114 — forward pointer; Census row]; the overall scale μ is undeclared (P4-class, Cal-F); external calibration barred [P1, P5]; particle identification barred.

Theorem 3.83 (FT.110 — The bound-chain arena dimension). Thm. The bound-chain arena $\mathcal{H}_c$ is the tensor product of two independent 3-dimensional structures:

$$(1) \quad \dim_{\text{eff}} \mathcal{H}_c = \dim F_{J^\pm} \times \operatorname{tr} T(1) = 3 \times 3 = 9;$$

$$(2) \quad \operatorname{tr} T(1) = 3$$

$$\big[\text{FT.68(2)}: \ \operatorname{charpoly} T(1) = x^2\big(x^2 - 3x - 3\big) = x^4 - 3x^3 - 3x^2; \ \ \operatorname{tr} T(1) = -[x^3] \operatorname{charpoly} T(1) = 3\big].$$

Rmk (tensor-product justification). The tensor product is constitutive of Def-H.41 (immediately above): the minimal combination consistent with both structures being independently present — the seal triplet at the chain's termini [CT0.12, CT0.7(4); chains as in Def-H.32(4)] and the polygonal transfer structure along its length [FT.68(2)]. Both factors are derived; the product is the interface clause. $\mathcal{H}_c$ is an arena — a counting space: no inner product, no dynamics, no state on it is declared, and no Parts 1–2 result uses the declaration.

Req. CT0.7(4), CT0.12 (Part 1); FT.68(2) (Tiers F.23–F.25); Def-H.32 (Tier F.36, chain vocabulary); Vieta coefficient read, exact integer characteristic polynomial of the 4×4 transfer rule [math; machine: `checks/verify_H_koide_channel.py` §C].

Pf.

$$(2): \ T(1)[s][s']$$

$$= [\,|s| + |s'| \le 3\,],$$

$$s,$$

$$s' \in \{0,1\}^2 \text{ [FT.68(2) transfer rule at } t = 1]: \text{ the single excluded pair is } (|s|, |s'|) \qquad = (2,2)$$

$$\Longrightarrow \ T(1) = \mathbb{1}_{4\times 4} - [\,\cdot\,]_{(2,2)} \text{ on the diagonal}: \ \operatorname{tr} T(1) = 4 - 1 = 3 \text{ [exact integers; machine §C]}$$

$$\operatorname{charpoly} T(1)$$

$$= x^4 - 3x^3 - 3x^2$$

$$= x^2\big(x^2 - 3x - 3\big) \text{ [FT.68(2); machine: exact re-derivation from the } 4 \times 4 \text{ rule]};$$

$$\operatorname{tr} T(1) = -[x^3] = 3 \text{ [Vieta]}$$

$$(1): \ \dim F_{J^\pm} = 3 \text{ [CT0.12 table : } F_{J^+} = \mathcal{E}_{-+} = \operatorname{span}\{J_i\},\ F_{J^-} = \mathcal{E}_{--} = \operatorname{span}\{K_i\} \text{ [CT0.7(4)]]};$$

$$3 \times 3 = 9 \text{ [exact]}. \blacksquare$$

Phys. A bound chain carries a three-dimensional seal choice at its boundary and a three-dimensional transfer weight along its length; the arena that holds both bookkeepings at once is nine-dimensional. The count 9 is the product of two independently derived threes — the seal triplet dimension and the transfer trace — and enters the tier exactly once: as the denominator of the mixing angle (FT.112).

Theorem 3.84 (FT.111 — The C_3 spectrum of the covariant seal bilinear). Thm. The mixed seal commutator $[J_i, K_j]$ is a bilinear in two seal triplets. Let C_3 be the cyclic permutation of the seal index, generated by the $SO(3)$ rotation by $2\pi/3$ about the $(1,1,1)$ axis — a subgroup of the $SO(3)$ under which $\{J_i\}$ and $\{K_i\}$ transform as vectors [CT1.8]; inner form: $U = u \otimes u$ with $u = \frac{1}{2}\big(I_2 - i(\sigma_1 + \sigma_2 + \sigma_3)\big)$, exact over $\mathbb{Q}(i)$: $U^3 = I_4$, $UJ_iU^\dagger = J_{i+1}$, $UK_iU^\dagger = K_{i+1}$ (indices mod 3). Then:

(1) bilinear

$=$ singlet $\oplus$ doublet : the singlet is the Σ part (C_3-invariant); the doublet is the $\hat{E}^{ij}$ part (C_3-cyclically permuted).

Concretely: the diagonal triple $[J_i, K_i]/i = \Sigma - E^{ii} = \frac{2}{3}\Sigma - \hat{E}^{ii}$ splits into the fixed direction Σ and the zero-sum plane of $\big(\hat{E}^{11}, \hat{E}^{22}, \hat{E}^{33}\big)$ — $\sum_i \hat{E}^{ii} = 0$ — on which the cyclic shift P acts with minimal polynomial $x^2 + x + 1$, irreducible over $\mathbb{Q}$: the doublet.

(2) Democratic diagonalization: any C_3-covariant Hermitian operator on the triplet — $[H, P]$

$$= 0,$$

$$H^\dagger$$

$$= H,$$

P the cyclic shift — has the form

$$H = a\, I_3 + z\, P + \bar{z}\, P^T \qquad (a \in \mathbb{R},\ z \in \mathbb{C}),$$

$$\text{spec}\, H: \quad \lambda_i = A + B\cos\big(\theta + 2\pi i/3\big), \qquad i = 0,1,2, \qquad A = a,\ \ B = 2|z|,\ \ \theta = \arg z,$$

equivalently the depressed cubic $(\lambda - A)^3 - \frac{3B^2}{4}(\lambda - A) - \frac{B^3}{4}\cos 3\theta = 0$.

$$(3)\quad \text{Normalization: } A = 1\ \Big[\sum_i \lambda_i = 3A = 3 = \dim F_{J^\pm}\Big];$$

$$B = \sqrt{2}\ \Big[\sum_i (\lambda_i - A)^2 = B^2 \sum_i \cos^2(\theta + 2\pi i/3) = B^2 \cdot \tfrac{3}{2} = 3 \implies B^2 = 2\Big];$$

$$\theta = \delta: \text{ the mixing angle, determined in FT.112.}$$

Req. CT1.2, CT1.8, CT0.12 (Part 1); FT.109; standard C_3 representation theory (commutant of the cyclic shift, Fourier diagonalization, triple-angle identity) [math]; exact $\mathbb{Q}(i)$ conjugation, exact symbolic identities in $\mathbb{Q}(\sqrt{3})[c,s]/(c^2+s^2-1)$, exact $\mathbb{Q}$ linear algebra, Zariski-dense rational sampling of the charpoly identity [machine: `checks/verify_H_koide_channel.py` §D].

Pf.

$$(1):\ U^3 = I_4,\ UU^\dagger = I_4,\ \det u = 1 \text{ [exact over } \mathbb{Q}(i);\ \text{machine]};$$

$$UJ_iU^\dagger = J_{i+1},\ UK_iU^\dagger = K_{i+1} \text{ [CT1.8 exponentiated; machine]}$$

$$U\Sigma U^\dagger = \Sigma;$$

$$UE^{ii}U^\dagger = E^{i+1,i+1};$$

$$U\hat{E}^{ii}U^\dagger = \hat{E}^{i+1,i+1};$$

$$U\hat{E}^{12}U^\dagger = \hat{E}^{23},\ U\hat{E}^{23}U^\dagger = \hat{E}^{13},\ U\hat{E}^{13}U^\dagger = \hat{E}^{12} \text{ [machine]}$$

$$\big[J_i, K_j\big] \text{ equivariance}:\ U\,[J_i,K_j]\,U^\dagger = [J_{i+1}, K_{j+1}]\ \ \forall(i,j) \text{ [machine, all 9 pairs]}$$

$$\sum_i \hat{E}^{ii} = \sum_i E^{ii} - 3\cdot\tfrac{1}{3}\Sigma = \Sigma - \Sigma = 0 \text{ [FT.109(2); CT1.2 traceless split]}$$

$$P^3 = I_3,\ \text{tr}\, P = 0;$$

$$P|_{\{\sum x_i = 0\}}:\ \text{charpoly} = \text{minimal polynomial} = x^2 + x + 1\ (P|_{\text{plane}} \neq I);$$

$$x^2 + x + 1 \text{ has no rational root } (\pm 1 \mapsto 3 \neq 0) \implies \text{irreducible over } \mathbb{Q}:\ \text{doublet [machine]}$$

$$(2):\ [X,P]$$

$$=0$$

$$\iff X \text{ circulant}:\ \dim_{\mathbb{Q}} \operatorname{comm}(P)$$

$$=3,$$

$$\text{basis } \{I_3, P, P^2\} \text{ [exact } \mathbb{Q} \text{ elimination; machine]}$$

$$H = H^\dagger \in \operatorname{comm}(P) \iff H = aI_3 + zP + \bar{z}P^T;$$

$$\text{charpoly } H$$

$$= (x-a)^3 - 3|z|^2(x-a) - 2\,\mathrm{Re}(z^3) \text{ [machine at 24 exact rational points } (a, \mathrm{Re}\, z, \mathrm{Im}\, z):\ \text{Zariski-dense} \implies \text{identical]}$$

$$A = a,$$

$$B$$

$$= 2|z|,$$

$$\cos 3\theta$$

$$= \mathrm{Re}(z^3)/|z|^3: \quad \tfrac{3B^2}{4}$$

$$= 3|z|^2,$$

$$\tfrac{B^3}{4}\cos 3\theta$$

$$= 2\,\mathrm{Re}(z^3) \implies \text{depressed cubic form } \checkmark$$

$$\prod_{i=0}^{2}\left(x - A - B\cos(\theta + \tfrac{2\pi i}{3})\right)$$

$$= (x-A)^3 - \tfrac{3B^2}{4}(x-A) - \tfrac{B^3}{4}\cos 3\theta \text{ [exact in } \mathbb{Q}(\sqrt{3})[A,B,x,c,s]/(c^2+s^2-1);\ \text{machine]};$$

$$\cos 3\theta = 4c^3 - 3c \text{ [same ring]}$$

$$\text{covariance witness : } M(\theta) :$$

$$= A$$

$$I_3 + B\big(\cos\theta\, D_c - \sin\theta\, D_s\big),$$

$$D_c$$

$$= \text{diag}\big(1, -\tfrac{1}{2}, -\tfrac{1}{2}\big),$$

$$D_s$$

$$= \text{diag}\big(0, \tfrac{\sqrt{3}}{2}, -\tfrac{\sqrt{3}}{2}\big) :$$

$$\text{spec}\, M(\theta) = \big\{A + B\cos(\theta + \tfrac{2\pi i}{3})\big\}_{i=0,1,2};$$

$$M\big(\theta + \tfrac{2\pi}{3}\big) = P^T M(\theta)\, P \text{ [entry-wise exact in the same ring; machine]}$$

$$(3) : \sum_i \cos(\theta + \tfrac{2\pi i}{3})$$

$$= 0,$$

$$\sum_i \cos^2(\theta + \tfrac{2\pi i}{3})$$

$= \frac{3}{2}\ \forall\theta$ [exact identities in $\mathbb{Q}(\sqrt{3})[c,s]/(c^2+s^2-1)$: coefficients $(1, -\frac{1}{2}, -\frac{1}{2})$, $(0, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2})$; machine]

$$\sum_i \lambda_i = 3A = 3 \implies A = 1 \text{ [CT0.12 : dim } F_{J^\pm} = 3];$$

$$\sum_i (\lambda_i - A)^2 = B^2 \cdot \tfrac{3}{2} = 3 \implies B^2 = 2,\ B = \sqrt{2}\ [(\sqrt{2})^2 = 2 \text{ exact in } \mathbb{Q}(\sqrt{2});\ \text{machine}]. \blacksquare$$

Rmk. This is representation theory of C_3, not the Koide ansatz. Any C_3-covariant triplet operator has this spectrum. The specific form $r_i = 1 + \sqrt{2}\cos(\delta + 2\pi i/3)$ follows from the normalization conditions alone.

Phys. A 120° rotation about the diagonal axis cycles the three seal directions, and the rotation is inner in the pair algebra: it conjugates each seal generator into the next. The mixed bilinear therefore splits into one rotation-invariant piece — the exchange scalar — and one irreducible rotating plane, the traceless diagonal tensors. Requiring the triplet's trace and its variance to each equal the seal dimension 3 fixes the spectrum's offset and spread to 1 and $\sqrt{2}$; the only remaining content is the phase δ.

Theorem 3.85 (FT.112 — The mixing angle). Thm. The mixing angle of the C_3-covariant seal bilinear is the exact rational

$$\delta = \frac{\sum_i [J_i, K_i]\big|_\Sigma}{\dim F_{J^\pm} \cdot \operatorname{tr} T(1)} = \frac{2}{3 \cdot 3} = \frac{2}{9},$$

where the numerator 2 is the scalar coefficient of the diagonal seal commutator [FT.109(3): $\sum_i [J_i, K_i] = 2i\Sigma$] and the denominator 9 is the bound-chain arena dimension [FT.110(1)].

Rmk (cross-check, not a derivation of equality). The same rational $2/3$ appears in three independent frozen results: $(p^*_\infty)^2 = 2/3$ [FT.70(3): the polygon phase boundary squared, exact in $\mathbb{Z}[\sqrt{6}]$]; $\delta_v/\pi = 2/3$ [FT.63(4): the octahedron vertex defect, exact in π-units]; $f_{\text{charge}}(\delta h = 3) = 2/3$ [FT.14: the 3-chain charge fraction, exact rational]. The mixing angle is $\delta = 2/9 = (2/3)/3$: the polygon phase boundary divided by the number of generations (the seal dimension). Recorded as a cross-check — three independent appearances of the same rational — not as a derivation of their equality.

Req. FT.109, FT.110; FT.14 (Tiers F.1–F.11), FT.63(4) (Tiers F.20–F.22), FT.70(3) (Tiers F.23–F.25) for the cross-check row; exact rational arithmetic, exact $\mathbb{Z}[\sqrt{6}]$ arithmetic [math; machine: `checks/verify_H_koide_channel.py` §E].

Pf.

$$\delta = \frac{2}{\dim F_{J^\pm} \cdot \operatorname{tr} T(1)} = \frac{2}{3 \cdot 3} = \frac{2}{9} \text{ [FT.109(3); FT.110; exact Fractions; machine §E]}$$

$$\text{cross-check rows :} \quad \left(\tfrac{\sqrt{6}}{3}\right)^2 = \tfrac{2}{3} \text{ [exact in } \mathbb{Z}[\sqrt{6}]\text{; FT.70(3)];}$$

$$\frac{\delta_v}{\pi} = 2 - \frac{d(L-2)}{L}\Big|_{d=4,\, L=3} = 2 - \tfrac{4}{3} = \tfrac{2}{3} \text{ [FT.63(4); exact integers]}$$

$$f_{\text{charge}}(\delta h = 3) = \frac{N_{\text{charge}}}{N_{\text{visible}}} = \frac{2}{3} \text{ [FT.14 table row } \delta h = 3\text{];} \qquad \frac{2/3}{3} = \frac{2}{9} \text{ [exact]. } \blacksquare$$

Phys. One exact integer from the seal algebra over one exact integer from the arena: the phase is $2/9$, a dimensionless rational with no adjustable content — the bilinear's coefficient divided by the arena's dimension. Its relation to the three independent $2/3$ results is a coincidence of rationals recorded for the ledger; nothing is calibrated to anything.

Theorem 3.86 (FT.113 — The bilinear spectrum and the mass-squared ratios). Thm. With $\delta = 2/9$ [FT.112] and $A = 1$, $B = \sqrt{2}$ [FT.111(3)], the three eigenvalues of the C_3-covariant bilinear are

$$r_i = 1 + \sqrt{2}\cos\Big(\frac{2}{9} + \frac{2\pi i}{3}\Big), \qquad i = 0, 1, 2,$$

with exact radical-trigonometric closed forms ($c = \cos\frac{2}{9}$, $s = \sin\frac{2}{9}$):

$$r_0 = 1 + \sqrt{2}\,c; \qquad r_1 = 1 - \frac{\sqrt{2}\,c + \sqrt{6}\,s}{2}; \qquad r_2 = 1 - \frac{\sqrt{2}\,c - \sqrt{6}\,s}{2};$$

$$\sum_i r_i = 3; \qquad \sum_{i<j} r_i r_j = \frac{3}{2}; \qquad 2(r-1)^3 - 3(r-1) - \sqrt{2}\cos\frac{2}{3} = 0 \text{ for each } r = r_i \ [3\delta = \tfrac{2}{3}].$$

The mass-squared ratios are r_i^2. Ordered by magnitude — $r_1^2 < r_2^2 < r_0^2$, enclosure-proven —

$$m_1^2 : m_2^2 : m_3^2 = r_{\text{min}}^2 : r_{\text{mid}}^2 : r_{\text{max}}^2 = r_1^2 : r_2^2 : r_0^2 = 1 : \frac{r_2^2}{r_1^2} : \frac{r_0^2}{r_1^2},$$

$$\frac{r_2^2}{r_1^2} = 206.770315972729\ldots,$$

$$\frac{r_0^2}{r_1^2} = 3477.472837104\ldots$$

[rational trigonometric enclosures, width $< 2^{-50}$; machine].

The overall scale μ in $m_i^2 = \mu\, r_i^2$ is undeclared (P4-class free internal parameter): the ratios are this tag's only content. The m^2 vocabulary is the corpus's own inertial read [L3(4), L4: dimensionless entropy deficit in nats; ¬a particle identification].

Req. FT.111, FT.112; L3, L4 (B, the m^2 read vocabulary); P4 (A, the free-scale clause); exact rational trigonometric enclosures (Taylor remainder bounds), exact integer square-root enclosures (Newton with certificates), monotone rational interval arithmetic [math; machine: `checks/verify_H_koide_channel.py` §F].

Pf.

$$r_i = A + B\cos(\delta + \tfrac{2\pi i}{3}) \text{ at } A = 1,\ B = \sqrt{2},\ \delta = \tfrac{2}{9} \text{ [FT.111(3), FT.112]}$$

$$\cos\big(\tfrac{2}{9} + \tfrac{2\pi}{3}\big) = -\tfrac{c}{2} - \tfrac{\sqrt{3}}{2}s,$$

$$\cos\big(\tfrac{2}{9} + \tfrac{4\pi}{3}\big) = -\tfrac{c}{2} + \tfrac{\sqrt{3}}{2}s \text{ [addition formulas]};$$

$$\sqrt{2}\cdot\tfrac{\sqrt{3}}{2} = \tfrac{\sqrt{6}}{2}$$

$$\sum_i r_i = 3 + \sqrt{2}\sum_i \cos(\delta + \tfrac{2\pi i}{3}) = 3 \text{ [FT.111(3) sum identity]};$$

Vieta pair sum and depressed cubic at $B^2 = 2,\ \cos 3\delta = \cos\frac{2}{3}$ [FT.111(2) charpoly identity]

c, s enclosed by Taylor partial sums with exact rational remainder bounds (widths $< 2^{-80}$);

$\sqrt{2}, \sqrt{6}, \cos\frac{2}{3}$ enclosed exactly (integer certificates)

$r_1^2 < r_2^2 < r_0^2$ [disjoint enclosures];

ratios by monotone interval division, width $< 2^{-50}$ [machine §F];

$0 \in 2y_i^3 - 3y_i - \sqrt{2}\cos\frac{2}{3}$ enclosure for each $y_i = r_i - 1$ [machine]. ■

Phys. The spectrum carries one large, one middle, and one nearly-vanished eigenvalue: a hierarchy produced by a rational phase inside a cosine, with closed forms in $\sqrt{2}$, $\sqrt{6}$, and $\cos\frac{2}{9}$ alone. The proximity of the two ratios to the supplied external hierarchy is quantified in FT.114 and used nowhere else. Every quantity here is dimensionless; the scale μ converting ratios into a read is a free internal parameter (P4), and the m^2 name is the entropy-deficit read of L3(4)/L4 — never a particle.

3.39 Tier F.38 — The Colour Channel: The ε-Selected C_3, Confinement Blindness, and the Triplet-Attached Mixing Angle

Theorem 3.87 (FT.116 — The relabelling lattice, the two-level obstruction, and the ε-selection of C_3). Thm. On the slot carrier $\mathcal{H}^{(3)}_{\text{slot}} \cong \mathbb{C}^3$ [CT3.2] with the classical relabelling embedding $P_\pi|i\rangle = |\pi(i)\rangle$, $(P_\pi)^i{}_j = \delta^i_{\pi(j)}$, $S_3 \hookrightarrow U(3)$ [CT3.3]:

(1)

S_3 has exactly six subgroups, of orders $(1, 2, 2, 2, 3, 6)$: $\{e\}$,

three $\mathbb{Z}_2$

$=$ ⟨transposition⟩,

$A_3 \cong C_3, \qquad S_3$

[machine: closure of all 2^6 generator subsets];

(2) commutant dimensions (Hermitian real form) :

$$\{e\} \mapsto 9,$$

$$\mathbb{Z}_2 \mapsto 5,$$

$$C_3 \mapsto 3,$$

$$S_3 \mapsto 2;$$

$$(3) \quad C_3\text{-commutant} = \{aI_3 + zP + z^*P^T : \ a \in \mathbb{R},\ z \in \mathbb{C}\}\ (P = P_{(123)}),$$

$$S_3\text{-commutant} = \text{span}\{I_3, J\}\ (J = \text{all-ones});$$

$$(4) \quad \text{the two-level obstruction: } \ \text{spec}(aI_3 + bJ)$$

$$= \{a + 3b,\ a,\ a\} \text{ exactly } \forall (a,b) \in \mathbb{R}^2\ (P_\pi\text{-covariance } \Rightarrow \text{ at most two distinct eigenvalues});$$

$$(5) \quad \text{reflection rigidity: } THT = H \text{ for a transposition } T \iff z \in \mathbb{R}$$

$$[THT = aI_3 + zP^T + z^*P : \ \text{conjugation by a transposition inverts the 3-cycle}];$$

$$\text{disc} = 108\,(\text{Im}\,z^3)^2 = 108\,(\text{Im}\,z)^2\big(3(\text{Re}\,z)^2 - (\text{Im}\,z)^2\big) \text{ vanishes } \iff z^3 \in \mathbb{R} \iff \arg z \in \tfrac{\pi}{3}\mathbb{Z}$$

$$[\text{the degeneracy locus: three lines through the origin}];$$

$$S_3\text{-covariance forces } z \in \mathbb{R}\ (\arg z \in \{0,\pi\}) \subset \tfrac{\pi}{3}\mathbb{Z} \implies \text{every } S_3\text{-covariant circulant has a double root};$$

$$\text{a three-line spectrum requires } \ \text{Im}\,z^3$$

$$\neq 0,$$

$$\text{i.e. } S_3\text{-covariance must be broken to the cyclic subgroup};$$

$$(6) \quad \text{the } \varepsilon\text{-selection: } \ \varepsilon_{ijk} P^i{}_a P^j{}_b P^k{}_c$$

$$= \det(P_\pi)$$

$$\varepsilon_{abc}$$

$$= \operatorname{sgn}(\pi)$$

$$\varepsilon_{abc}\ \forall \pi$$

$$\in S_3 \text{ [CT3.15 on permutation matrices]};$$

$$\det P_\pi = \operatorname{sgn}(\pi);$$

$$\operatorname{Stab}_{S_3}(\varepsilon) = \{\pi : \det P_\pi = 1\} = A_3 = C_3 = S_3 \cap SU(3);$$

the ε-preserving subgroups are exactly $\{e\} \subset A_3,$ so A_3 is the unique maximal one;

(7) seal-side companion census: on the seal triplet, $T \cong A_4$ ($|T| = 12,\ T \subset SO(3)$) has commutant $\mathbb{R} \cdot I_3$ (full degeneracy);

V_4 has commutant the diagonal matrices (3 parameters, no cyclic relation);

C_3 (the $2\pi/3$ rotation about $(1,1,1)$ [CT1.8]) has commutant the circulants (3 parameters, fixed democratic eigenvectors).

Req. CT3.1, CT3.2, CT3.3, CT3.15 (Part 2); D3.9 (Parts 1–2); CT1.8 (Part 1); FT.111 (Tier F.37, the C_3 democratic form and its normalization); subgroup-lattice closure, exact nullspace dimension over $\mathbb{Q}$, Leibniz determinant identity, charpoly of the circulant and of $aI_3 + bJ$, discriminant of the depressed cubic [math; machine: `checks/verify_H_colour_channel.py` §A].

Pf.

(1) : closure of each of the 2^6 generator subsets under $\pi \circ \rho$ yields 6 distinct subgroups; orders sorted $(1, 2, 2, 2, 3, 6)$ [machine]

$$(2),$$

(3) : $\{X : [X, P_\pi] = 0\ \forall \pi \in G\}$ solved as an exact 9-unknown nullspace over $\mathbb{Q}$ per subgroup;

$$\dim$$

$$= (9, 5, 3, 2);$$

transpose-closed $\forall G$ [machine]

$$C_3 : \text{ basis } \{I_3, P, P^2\} \text{ (real form } \{I_3,\ P + P^T,\ P - P^T\} \ \leftrightarrow\ \{a,\ 2\,\mathrm{Re}\, z,\ -2\,\mathrm{Im}\, z\});$$

$$S_3 : \text{ nullspace basis } \{I_3, J\} \text{ [machine: exact rank 2]}$$

$$(4) : \text{ charpoly}(aI_3 + bJ)$$

$$= (x - a - 3b)(x - a)^2$$

$= x^3 - (3a + 3b)x^2 + (3a^2 + 6ab)x - (a^3 + 3a^2b)$ at 3 exact rational points (a, b) [machine]

$$(5) : \ THT = aI_3 + zP^T + z^*P \ (TPT = P^T : \text{ a transposition inverts the 3-cycle});$$

$$THT = H \iff z = z^* \iff \mathrm{Im}\, z = 0$$

$\text{charpoly}(aI_3 + zP + z^*P^T) = (x-a)^3 - 3|z|^2(x-a) - 2\,\mathrm{Re}(z^3)$ [machine at 15 exact rational points];

$$\text{disc} = -4(-3|z|^2)^3 - 27(-2\,\mathrm{Re}\, z^3)^2 = 108\big(|z|^6 - (\mathrm{Re}\, z^3)^2\big) = 108(\mathrm{Im}\, z^3)^2 \text{ [exact]}$$

$$\mathrm{Im}\, z^3$$

$$= 3(\mathrm{Re}\, z)^2\, \mathrm{Im}\, z - (\mathrm{Im}\, z)^3$$

$$= \mathrm{Im}\, z$$

$\big(3(\mathrm{Re}\, z)^2 - (\mathrm{Im}\, z)^2\big)$ [polynomial identity at exact rational points, Zariski-dense; machine]

$$\text{disc} = 0 \iff (\mathrm{Im}\, z^3)^2 = 0 \iff \mathrm{Im}\, z = 0 \ \vee\ (\mathrm{Im}\, z)^2 = 3(\mathrm{Re}\, z)^2 \iff z^3 \in \mathbb{R} \iff \arg z \in \tfrac{\pi}{3}\mathbb{Z};$$

$$z \in \mathbb{R} \implies \text{disc} = 0 \implies \text{a double root [machine at every real-}z\text{ point]}$$

$$(6): \ \varepsilon_{ijk} P^i{}_a P^j{}_b P^k{}_c = \text{sgn}(\pi)\varepsilon_{abc} \text{ entrywise on all 27 triples} \times 6 \text{ relabellings;}$$

$$\det P_\pi = \text{sgn}(\pi) \text{ [exact } \mathbb{Q} \text{ expansion; machine]}$$

$$\pi \in \text{Stab}(\varepsilon) \iff \text{sgn}(\pi) = 1 \iff \pi \in A_3;$$

$$\text{lattice scan} : \{g \leq S_3 : g \subseteq A_3\} = \{\{e\}, A_3\} \text{ [machine];}$$

$$A_3 = S_3 \cap SU(3) \text{ [det} = 1]$$

$$(7): \ T$$

$$= \langle P_{(123)},$$

$$\text{diag}(1, -1, -1),$$

$$\text{diag}(-1, 1, -1),$$

$$\text{diag}(-1, -1, 1)\rangle : \ |T|$$

$$= 12,$$

$$\det$$

$$= 1 \ \forall T \text{ [machine closure];}$$

$$\dim \text{comm}(T) = 1, \ \dim \text{comm}(V_4) = 3, \ \dim \text{comm}(C_3) = 3 \text{ [exact nullspaces; machine].} \blacksquare$$

Phys. Ask the colour sector which of its own symmetries may carry a mass family, and the answer is forced twice over. The full relabelling group is too symmetric: an S_3-covariant operator on three slots has a doubly-degenerate spectrum, identically, for every choice of its two parameters, and any operator invariant under a single transposition has a double root — a three-line family cannot be S_3-covariant, and the phase $\arg z$ is precisely the datum that breaks it. Too little symmetry is equally useless: with no covariance at all nine parameters are free and nothing is predicted. The cyclic subgroup is the unique intermediate, and it is not selected by hand — the baryon seal ε_{ijk}, the tensor that makes three triplets into an observable [CT3.15, CT3.18], is preserved exactly

by the even relabellings, so $A_3 \cong C_3 = S_3 \cap SU(3)$ is the largest relabelling symmetry a sealed object can have. The same C_3 that FT.111 found inside the seal algebra is therefore also the maximal covariance group of the colour algebra, reached from the opposite side: there by the $2\pi/3$ rotation about $(1,1,1)$, here by the Levi-Civita tensor. The companion census closes the search on the seal side as well — the tetrahedral group is too big (its commutant is the scalars, so every T-covariant operator is fully degenerate) and the Klein group is the wrong shape (diagonal commutant, no cyclic relation among eigenvalues).

FT.117 — Confinement blindness: the one-body no-go on colour-sealed singlets. Structural-constraints record: see Appendix B, Section B.3.3 of the Addendum.

Definition 3.40 (Def-H.42 — The colour-dressed arena). Def. A triplet-attached bound chain [D3.8] carries, simultaneously and independently: the seal triplet at its termini [CT0.12, CT0.7(4)], the polygonal transfer structure along its length [FT.68(2)], and the colour slot at its attachment [CT3.2: $\mathbb{C}^3$]. The colour-dressed arena is their minimal combined counting space:

$$\mathcal{H}_c^{(q)} \;:=\; \mathcal{H}_c \otimes \mathbb{C}^3_{\text{slot}} \qquad \text{[Def-H.41]}, \qquad \dim_{\text{eff}} \mathcal{H}_c^{(q)} = 3 \times 3 \times 3 = 27 \;\; \text{[FT.118(2)]},$$

entering exactly once — as the denominator of the triplet-attached mixing angle [FT.119: $\delta_q = \frac{2}{27}$ with FT.118(4)'s spectator numerator 2]. Singlet-sealed chains keep Def-H.41's 9: the invariant lines of a sealed object are 1-dimensional [FT.117(5)], so its colour factor is 1. Species and bars. As Def-H.41: a counting space — no inner product, no dynamics, no state, no Hamiltonian; dimension ¬a state count; the overall scale μ undeclared (P4-class, Cal-F); external calibration barred [P1, P5; FT.120 — forward pointer; Census row]; the adjoint-dressed numerator $8/3$ [FT.118(6)] and the alternative table rows [FT.119(7)] are logged alternatives, ¬adopted; particle identification barred. Locus remark (structural). Colour sits in an orthogonal stack block of the force arena [FT.93(4); CT0.12 Rmk: $\mathbb{C}^4 \otimes \mathbb{C}^3$ is not a corpus construction], and EW.32's seed is colour-neutral by theorem — this definition counts D3.8's attachment choices in the chain's own bookkeeping, ¬a state-space composition of the force arena. The derived content compatible with both records is FT.118(4)'s spectator dressing $\sum_i [J_i \otimes I_3,\, K_i \otimes I_3] = 2i\,\Sigma \otimes I_3$.

Theorem 3.88 (FT.118 — The colour-dressed arena and the spectator numerator). Thm. With the transfer rule $T(1)[s][s'] = [\,|s| + |s'| \le 3\,]$, $s, s' \in \{0,1\}^2$ [FT.68(2)], re-derived at the point of use:

$$(1) \quad \operatorname{tr} T(1) = 3$$

[the single excluded diagonal pair is $(|s|, |s'|) = (2,2)$];

$$\text{charpoly}\, T(1) = x^4 - 3x^3 - 3x^2 = x^2\big(x^2 - 3x - 3\big) \text{ [FT.68(2); machine: Faddeev–LeVerrier over } \mathbb{Z}];$$

$$(2) \quad \text{the colour-dressed arena:} \qquad \dim_{\text{eff}} \mathcal{H}_c^{(q)} = \dim F_{J^\pm} \times \operatorname{tr} T(1) \times N_{\text{colour}} = 3 \times 3 \times 3 = 27,$$

the third factor being $N_{\text{colour}} = 3 = \log_2 C_{\text{cap}} - 1 = \log_2 16 - 1$ [CT3.20; CT0.5, CT0.6, CT0.11];

scope: triplet-attached chains [D3.8];

(3) sealed attachments do not dress:

colour factor of a sealed object

$= 1$ [FT.117(5) : the invariant lines are 1-dimensional] $\implies \dim_{\text{eff}} \mathcal{H}_c^{\text{sealed}}$

$= 3 \times 3 \times 1$

$= 9$ [FT.110(1) recovered];

(4) the spectator numerator:

$$\sum_{i=1}^{3} \big[J_i \otimes I_3,\ K_i \otimes I_3\big] = \Big(\sum_i [J_i, K_i]\Big) \otimes I_3 = 2i\,\Sigma \otimes I_3$$

[12×12 entrywise exact over $\mathbb{Q}(i)$; FT.109 re-derived];

(5) the diagonal coefficient of the colour-dressed seal commutator is therefore exactly 2 — unchanged from FT.109(3) : colour is a spectator of the seal algebra [EW.32 : $T_a \mathrm{v} = 0$; CT0.12 Rmk : colour sits in an orthogonal stack block];

(6) adjoint dressing (computed, NOT adopted):

with X^2

$= \sum_a T_a T_a$

$= C_F I_3$

$= \frac{4}{3} I_3$ [machine: from the rational E_{ij} basis] and $[A \otimes X,\ B \otimes X]$

$= [A, B] \otimes X^2$,

$$\sum_i \big[J_i \otimes X,\ K_i \otimes X\big]$$

$= 2i$

$$\Sigma \otimes X^2$$

$$= \tfrac{8}{3}$$

$$i$$

$$\Sigma \otimes I_3 \implies \text{numerator } 2C_F$$

$= \frac{8}{3}$: a different rational, and no corpus tag supplies the coupling;

(7) the two candidate mixing angles:

$$\delta_q = \frac{2}{27} \text{ (spectator, adopted in FT.119);}$$

$$\frac{2C_F}{27} = \frac{8/3}{27} = \frac{8}{81} \text{ (adjoint-dressed, logged not adopted).}$$

Rmk (interface definition Def-H.42). Clause (2)'s tensor product is the same species of interface definition as Def-H.41, with one new factor: a triplet-attached chain carries, simultaneously and independently, a seal triplet at its termini [CT0.12, CT0.7(4)], a polygonal transfer structure along its length [FT.68(2)], and a colour slot at its attachment [CT3.2, D3.8]; $\mathcal{H}_c^{(q)}$ is their minimal combined counting space. It is an arena — no inner product, no dynamics, no state, no Hamiltonian — and the count 27 enters the tier exactly once, as the denominator of FT.119's mixing angle. Tension logged: FT.93(4) records colour as an orthogonal stack block (a $\oplus$-summand of $\mathcal{K}_\Sigma$, not a tensor factor [CT0.12 Rmk, CT0.14]) and the seed as colour-neutral by theorem [EW.32], so "$\mathbb{C}^4 \otimes \mathbb{C}^3$" is not a corpus construction of the force arena; the declared factor counts D3.8's attachment choices in the chain's own bookkeeping, and clause (4)'s spectator dressing is the derived content compatible with both records. Clause (6) is recorded so that any future use of the adjoint-dressed numerator $8/3$ is a adoption rather than a silent one; the spectator form (4)–(5) is the derived content and needs no declaration beyond Def-H.42.

Req. CT3.2, CT3.20 (Part 2); CT0.5, CT0.6, CT0.11, CT0.12, CT0.14, CT0.7(4) (Part 1); D3.8 (Parts 1–2); EW.32 (Part 2); FT.68(2) (Tiers F.23–F.25); FT.93(4) (Tiers F.31–F.34, the tension note); FT.109, FT.110 (Tier F.37); FT.117(5) (this tier); Vieta coefficient read, exact integer characteristic polynomial, exact $\mathbb{Q}(i)$ Kronecker arithmetic, quadratic-Casimir normalization [math; machine: `checks/verify_H_colour_channel.py` §C].

Pf.

$$(1): \; T(1) = \mathbb{1}_{4\times 4} - [\,\cdot\,]_{(2,2)} \text{ on the diagonal}: \; \operatorname{tr} T(1) = 4 - 1 = 3 \text{ [exact integers; machine]}$$

$$\operatorname{charpoly} T(1)$$

$$= [1, -3, -3, 0, 0] \text{ by Faddeev–LeVerrier over } \mathbb{Z}$$

$$= x^4 - 3x^3 - 3x^2 \text{ [FT.68(2); machine]}$$

$$(2):\ \dim F_{J^\pm} = 3 \text{ [CT0.12 table : } F_{J^+} = \mathcal{E}_{-+} = \text{span}\{J_i\},\ F_{J^-} = \mathcal{E}_{--} = \text{span}\{K_i\}];$$

$$3 \cdot 3 \cdot 3 = 27 \text{ [exact]}$$

$$N_{\text{colour}} = 3 = \log_2 16 - 1 \text{ [CT3.20 : admissible } \iff 2^{n+1} \le C_{\text{cap}} = 16 \iff n \le 3]$$

$$(3):\ 3 \cdot 3 \cdot 1 = 9 \text{ [FT.117(5) : } \dim(3 \otimes 3 \otimes 3)^{\mathfrak{sl}(3)} = \dim(3 \otimes \bar{3})^{\mathfrak{sl}(3)} = 1;\ \text{machine}];$$

FT.110(1) value recovered

$$(4):\ [J_i \otimes I_3,\ K_i \otimes I_3] = [J_i, K_i] \otimes I_3 \text{ (Kronecker multiplicativity)};$$

$$\sum_i [J_i, K_i] = 2i\Sigma \text{ [FT.109(3) re-derived entrywise over } \mathbb{Q}(i);\ \text{machine}]$$

$$\implies \sum_i [J_i \otimes I_3, K_i \otimes I_3] = (2i\Sigma) \otimes I_3 \text{ [12} \times \text{12 equality, exact; machine]}$$

$$(5):\ \text{the } \Sigma\text{-coefficient of } (4) \text{ is } 2 \text{ [unique nonzero basis component; FT.109(3)]};$$

$T_a \mathsf{v} = 0$ [EW.32] and the stack-block orthogonality [CT0.12 Rmk] are the structural reasons

$$(6):\ \tfrac{1}{2}\big(\sum_{ij} E_{ij}E_{ji} - \tfrac{1}{3}(\sum_i E_{ii})^2\big)$$

$$= \tfrac{1}{2}(3I_3 - \tfrac{1}{3}I_3)$$

$$= \tfrac{4}{3}I_3$$

$$= C_F I_3 \text{ [exact } \mathbb{Q};\ \text{machine]}$$

$$[A \otimes X,\ B \otimes X]$$

$$= (A \otimes X)(B \otimes X) - (B \otimes X)(A \otimes X)$$

$$= AB \otimes X^2 - BA \otimes X^2$$

$$= [A, B] \otimes X^2 \text{ [machine at } X^2 = \tfrac{4}{3} I_3]$$

$$\Longrightarrow \sum_i [J_i \otimes X, K_i \otimes X] = 2i\Sigma \otimes X^2 = 2i \cdot \tfrac{4}{3}\, \Sigma \otimes I_3 = \tfrac{8}{3} i\, \Sigma \otimes I_3;$$

$$2 \cdot \tfrac{4}{3} = \tfrac{8}{3} \text{ [exact Fractions]}$$

$$(7): \ \tfrac{2}{3\cdot 3\cdot 3} = \tfrac{2}{27}; \qquad \tfrac{8/3}{27} = \tfrac{8}{81} \text{ [exact Fractions; machine]. } \blacksquare$$

Phys. A chain that grips a trivalent vertex carries one more independent bookkeeping than a chain that does not: which of the three slots it grips. That third ledger is the only new content of the rung-3 arena, and it multiplies — 3×3 becomes $3 \times 3 \times 3$. Sealing cancels the multiplication: because the baryon and meson invariant lines are one-dimensional, a sealed object's arena is back to nine, which is why FT.110's lepton-side count and this tier's differ at all. The numerator does not move. The seed that generates mass is colour-blind by theorem, so bracketing each spatial seal with its temporal partner and summing passes straight through the slot factor and returns the same integer 2 over Σ — the colour dressing is a spectator, and the alternative in which the strong generators themselves enter the bracket is a different rational $(8/3)$, computed here and adopted nowhere, because no tag of the corpus couples the seal commutator to the slot algebra.

Theorem 3.89 (FT.119 — The triplet-attached mixing angle $\delta_q = 2/27$ and its spectrum). Thm. With the spectator numerator 2 [FT.118(4)–(5)] over the colour-dressed arena 27 [FT.118(2)], and the FT.111(3) normalization $A = 1$, $B = \sqrt{2}$ (fixed by $\sum_i \lambda_i = 3 = \dim F_{J^\pm}$ and $\sum_i (\lambda_i - A)^2 = 3$, both seal-side and hence colour-blind):

$$(1) \quad \delta_q = \frac{2}{3 \cdot 3 \cdot 3} = \frac{2}{27} = 0.074074074074\ldots \qquad \text{[exact rational; machine } \S D\text{]};$$

$$(2) \quad 3\delta_q = \delta_\ell = \frac{2}{9} \qquad \text{[FT.112 : } \delta_\ell = \tfrac{2}{9} \text{ from the undressed arena } 3 \times 3\text{]};$$

$$(3) \quad \text{the spectrum:} \qquad r_i^{(q)} = 1 + \sqrt{2}\cos\Big(\frac{2}{27} + \frac{2\pi i}{3}\Big), \qquad i = 0, 1, 2,$$

$$\text{with } c = \cos\tfrac{2}{27},\ s = \sin\tfrac{2}{27}: \qquad r_0^{(q)} = 1 + \sqrt{2}\, c = 2.410335463511094\ldots,$$

$$r_1^{(q)} = 1 - \frac{\sqrt{2}\, c + \sqrt{6}\, s}{2} = 0.204193367808662\ldots,$$

$$r_2^{(q)} = 1 - \frac{\sqrt{2}\, c - \sqrt{6}\, s}{2} = 0.385471168680242\ldots;$$

all three strictly positive [disjoint enclosures; machine] — no sign branch, unlike the $\delta > \pi/12$ regime;

(4) Vieta and the phase-free invariant: $\sum_i r_i^{(q)} = 3, \qquad \sum_{i<j} r_i^{(q)} r_j^{(q)} = \frac{3}{2}, \qquad \sum_i \big(r_i^{(q)}\big)^2 = 6,$

$$2\big(r^{(q)} - 1\big)^3 - 3\big(r^{(q)} - 1\big) - \sqrt{2}\cos\frac{2}{9} = 0 \quad \text{for each } r^{(q)} = r_i^{(q)}$$

$[\cos 3\delta_q = \cos\tfrac{2}{9} = \cos\delta_\ell :$ clause (2) inside the cubic$]$;

$$Q := \frac{\sum_i m_i}{\big(\sum_i \sqrt{m_i}\big)^2} \overset{m_i = \mu (r_i^{(q)})^2}{=} \frac{\sum_i (r_i^{(q)})^2}{\big(\sum_i r_i^{(q)}\big)^2} = \frac{6}{9} = \frac{2}{3}$$

[exact, and independent of δ and of μ];

(5) the mass ratios ($m_i = \mu\,(r_i^{(q)})^2,\ \mu$ undeclared, P4) :

$$(r_1^{(q)})^2 < (r_2^{(q)})^2 < (r_0^{(q)})^2 \text{ [enclosure-proven]},$$

$$m_1 : m_2 : m_3$$

$$= 1 : \frac{(r_2^{(q)})^2}{(r_1^{(q)})^2} : \frac{(r_0^{(q)})^2}{(r_1^{(q)})^2}$$

$$= 1 : 3.563695074946\ldots : 139.3386880284\ldots$$

[widths $< 2^{-50}$; machine];

(6) anchored forms (scale-free, top-normalized) :

$$\frac{m_2}{m_3} = \frac{(r_2^{(q)})^2}{(r_0^{(q)})^2} = 0.025575776012\ldots,$$

$$\frac{m_1}{m_3} = \frac{(r_1^{(q)})^2}{(r_0^{(q)})^2} = 0.007176757684\ldots;$$

(7) alternative-hypothesis table (computed, none adopted) :

$\delta = \frac{2}{81}$ (per-baryon arena $3 \cdot 27$) : $1 : 1.5130 : 84.316$;

$$\delta = \tfrac{1}{36} = \tfrac{2}{72} \text{ (arena } 9 \text{ dressed by the adjoint multiplicity } 8 \text{ [CT3.13])} : \ 1 : 1.5939 : 86.746;$$

$$\delta = \tfrac{8}{81} \text{ (FT.118(6) numerator over } 27) : \ 1 : 5.6438 : 187.997.$$

Rmk (cross-check, not a derivation of equality). Clause (2) is the tier's structural coincidence, recorded in the FT.112 discipline: the colour channel's triple angle is the lepton channel's angle, so both channels share one depressed-cubic constant $\sqrt{2}\cos\frac{2}{9}$ and differ only in which of its three roots they read. Nothing is calibrated to anything; the equality $3 \cdot \frac{2}{27} = \frac{2}{9}$ is arithmetic on two independently derived rationals.

Req. FT.118; FT.111(2)–(3), FT.112, FT.113 (Tier F.37: the democratic spectrum form, the normalization, the angle discipline, the $m = \mu r^2$ vocabulary); CT0.12 (Part 1); CT3.13 (F, the alternative table's adjoint multiplicity); L3, L4 (B, the inertial read vocabulary); P4 (A, the free-scale clause); exact rational trigonometric enclosures (Taylor remainder bounds), exact integer square-root enclosures with certificates, monotone rational interval arithmetic, Vieta reads [math; machine: `checks/verify_H_colour_channel.py` §D].

Pf.

$$(1) : \ \delta_q$$

$$= \frac{2}{\dim F_{J^\pm} \cdot \operatorname{tr} T(1) \cdot N_{\text{colour}}}$$

$$= \frac{2}{3 \cdot 3 \cdot 3}$$

$$= \frac{2}{27} \text{ [FT.118(1), (2), (4); exact Fractions; machine]}$$

$$(2) : \ 3 \cdot \tfrac{2}{27} = \tfrac{6}{27} = \tfrac{2}{9} = \delta_\ell \text{ [FT.112; exact]}$$

$$(3) : \ r_i^{(q)} = A + B\cos(\delta_q + \tfrac{2\pi i}{3}) \text{ at } A = 1, \ B = \sqrt{2} \text{ [FT.111(3)]};$$

$$\cos(\tfrac{2}{27} + \tfrac{2\pi}{3})$$

$$= -\tfrac{c}{2} - \tfrac{\sqrt{3}}{2}s,$$

$$\cos(\tfrac{2}{27} + \tfrac{4\pi}{3})$$

$$= -\tfrac{c}{2} + \tfrac{\sqrt{3}}{2}s \text{ [addition formulas]};$$

$$\sqrt{2} \cdot \tfrac{\sqrt{3}}{2} \qquad = \tfrac{\sqrt{6}}{2}$$

c, s enclosed by alternating Taylor partial sums with exact rational remainder bounds (widths $< 2^{-90}$);

$\sqrt{2}, \sqrt{6}$ enclosed by integer Newton certificates;

$r_1^{(q)}, r_2^{(q)}, r_0^{(q)} > 0$ [positive lower endpoints; machine]

$$(4): \sum_i \cos(\delta_q + \tfrac{2\pi i}{3})$$

$$= 0,$$

$$\sum_i \cos^2(\delta_q + \tfrac{2\pi i}{3})$$

$$= \tfrac{3}{2} \text{ [FT.111(3) identities]} \implies \sum_i r_i^{(q)}$$

$$= 3,$$

$$\sum_i (r_i^{(q)})^2$$

$$= 3 + 2\sqrt{2} \cdot 0 + 2 \cdot \tfrac{3}{2} \qquad = 6$$

pair sum by Vieta $= \tfrac{3}{2}$;

depressed cubic $(\lambda - 1)^3 - \tfrac{3}{2}(\lambda - 1) - \tfrac{\sqrt{2}}{2} \cos 3\delta_q$

$= 0$ [FT.111(2) at $B^2 = 2$];

$$\cos 3\delta_q$$

$$= \cos \tfrac{2}{9} \text{ [(2)]};$$

$$\times 2: \; 2y^3 - 3y - \sqrt{2} \cos \tfrac{2}{9}$$

$$= 0,$$

$$y \qquad = r^{(q)} - 1$$

$$0 \in 2y_i^3 - 3y_i - \sqrt{2}\cos\tfrac{2}{9} \text{ enclosure for each } i \text{ [machine]}; \qquad Q = \tfrac{6}{9} = \tfrac{2}{3} \text{ [exact Fractions]}$$

$$(5):\ (r_1^{(q)})^2 < (r_2^{(q)})^2 < (r_0^{(q)})^2 \text{ [disjoint enclosures]};$$

ratios by monotone interval division, widths $< 2^{-50}$ [machine]

$$(6):\ \text{same divisions against } (r_0^{(q)})^2 \text{ [machine]};$$

$(7):$ each row by the identical enclosure pipeline at the stated rational δ [machine]. ■

Phys. One integer over three integers: the phase of the colour channel is the seal coefficient divided by a seal factor, a transfer factor and a colour factor, and it needs no new input beyond the rung-3 arena count. Its consequence is compact — triple the angle and the lepton channel's phase appears, so the two hierarchies are the three roots of one cubic read at phases differing by exactly the colour count. The spectrum itself is milder than the lepton-side one: no eigenvalue nearly vanishes, the middle line sits a factor 3.56 above the inner and the outer a factor 139 above it, and the anchored forms trade the undeclared scale for two pure numbers. The invariant $Q = \frac{2}{3}$ is not a prediction of this tier — it holds for every phase, being the normalization content $A = 1$, $B = \sqrt{2}$ — so what δ_q actually fixes is the position on that surface: the two ratios of clause (5). Everything here is dimensionless; μ is a free internal parameter (P4) and the m vocabulary is the entropy-deficit read of L3(4)/L4, never a particle.

3.40 Tier F.39 — Chirality: The Orientation Doublet of the Bound Chain and Its Exchange-Form Grade

Definition 3.41 (Def-H.33 — The chain orientation space and its chirality operator). Def. For a bound chain c [D1.4: ordered tuple of distinct real edges $c = (e_1, \ldots, e_k)$; D1.5: bound = closed self-contained, fermionic content]:

- Orientations. c and $\omega(c)$ [D2.6: $s(c) \leftrightarrow t(c)$, reversal of the traversal; $\omega^2 = \mathrm{id}$]. As ordered tuples, $c \neq \omega(c)$ for every admissible $k \geq 2$ [D1.4: distinct edges].
- Orientation space.

$$\mathcal{H}_\omega(c) \;:=\; \mathrm{span}_{\mathbb{C}}\big\{|c\rangle,\ |\omega(c)\rangle\big\} \;\cong\; \mathbb{C}^2,$$

  with the Hermitian inner product making $\{|c\rangle, |\omega(c)\rangle\}$ an orthonormal basis.

- Orientation involution. $\hat{\omega} : \mathcal{H}_\omega \to \mathcal{H}_\omega$, $\hat{\omega}|c\rangle = |\omega(c)\rangle$, $\hat{\omega}|\omega(c)\rangle = |c\rangle$: then $\hat{\omega}^2 = \mathrm{id}$, $\hat{\omega}^\dagger = \hat{\omega}$ [D2.6: ω is an involution].

- Chirality operator (the exchange form).

$$\boxed{\hat{\chi} \;:=\; i\big(|c\rangle\langle\omega(c)| - |\omega(c)\rangle\langle c|\big)}$$

- Chiral projectors. $P_L := \frac{1}{2}(\mathrm{id} + \hat{\chi})$, $P_R := \frac{1}{2}(\mathrm{id} - \hat{\chi})$.
- Chiral eigenbasis.

$$\boxed{|L\rangle = \tfrac{1}{\sqrt{2}}\big(|c\rangle - i\,|\omega(c)\rangle\big), \qquad |R\rangle = \tfrac{1}{\sqrt{2}}\big(|c\rangle + i\,|\omega(c)\rangle\big)}$$

— the $+1$ and -1 eigenvectors of $\hat{\chi}$ respectively; the labels are pinned by the boxed $\hat{\chi}$ together with $P_L := \frac{1}{2}(\mathrm{id} + \hat{\chi})$ [FT.122(3), (6)].

Rmk (why $\mathcal{H}_\omega$ and not the pair register). $\mathcal{H}_\omega(c)$ is indexed by the chain c, not by any pair (p, q). It is not a subspace of $\mathrm{End}(\mathbb{C}^4_{\mathrm{pair}})$: its basis states are chains — configuration-level objects [D1.4, D0.10] — not operators on the pair space. The pincer [FT.93(3)] and the trace/rank obstruction [FT.101(3)] are statements about $\mathrm{End}(\mathbb{C}^4_{\mathrm{pair}})$ and do not reach $\mathcal{H}_\omega$ (FT.124 makes the locus separation exact).

Rmk (the physical content). A closed chain's worldline is topologically S^1 with two traversal senses. The orientation doublet is the one-dimensional analogue of the linear-polarization basis; the exchange form $\hat{\chi} = i\cdot$(antisymmetric exchange) is the Hermitian generator of the exchange, and its ± 1 rays are the analogue of the two circular polarizations — the two chiralities of a loop mode. Orientation reversal exchanges the two rays [FT.122(4): $\hat{\omega}|L\rangle = -i|R\rangle$, $\hat{\omega}|R\rangle = +i|L\rangle$], realizing D2.6's "chirality $\mapsto$ $-$chirality" as an operator identity. The spin-structure torsor is a distinct object: fermion boundary conditions on S^1 form $H^1(S^1; \mathbb{Z}_2) \cong \mathbb{Z}_2$, which orientation reversal fixes pointwise [FT.122(7); machine] — the torsor $\mathbb{Z}_2$ and the chiral-ray $\mathbb{Z}_2$ are different kinds, and only the latter is load-bearing. The phase i in $\hat{\chi}$ is what turns the antisymmetric exchange form into a Hermitian involution, and it is what makes $\hat{\chi}$ odd under $\hat{\omega}$.

Rmk (canonicity and the naming residue). Among Hermitian involutions on $\mathcal{H}_\omega$ that are $\hat{\omega}$-odd and diagonal-free in the orientation basis, $\hat{\chi}$ is unique up to overall sign [FT.122(6); machine: without the diagonal-free condition the family is the circle $M(a, y) = \left(\begin{smallmatrix} a & iy \\ -iy & -a \end{smallmatrix}\right)$, $a^2 + y^2 = 1$]. The sign itself is a naming choice: the map $(\hat{\chi}, P_L, |L\rangle) \mapsto (-\hat{\chi}, P_R, |R\rangle)$ leaves every identity of FT.122–FT.123 invariant (the adjoint action is linear: $\hat{\omega}(-\hat{\chi})\hat{\omega}^{-1} = -(-\hat{\chi})$); the boxed definition fixes it once and for all, and given it the labels L/R are forced up to phase. The physical naming — which ray is "left" — is fixed by no algebra here: it is naming-convention content reserved to the matter-assignment rule (application-layer; its non-derivability here is proven [FT.125(4)]).

Rmk (achiral configurations). For a physical state invariant under an orientation-reversing symmetry of its configuration, $\langle\hat{\chi}\rangle = 0$ exactly [machine §A; the FT.95(3) pattern] — the grade is sharp per oriented chain, not per reversal-symmetric ensemble; a chiral read requires an oriented chain as boundary datum. This is the same observer-relativity as every chain-level read (Def-H.1 vocabulary); no new declaration.

Theorem 3.90 (FT.122 — The orientation doublet is corpus-constructible). Thm. For every bound chain c [D1.4, D1.5], the orientation space $\mathcal{H}_\omega(c)$, the involution $\hat{\omega}$, the chirality operator $\hat{\chi}$, and the projectors P_L, P_R are constructible from Parts 1–2 data alone, with no new axiom and no declared parameter:

$$(1)\quad \hat{\chi}^\dagger = \hat{\chi}, \qquad \hat{\chi}^2 = \mathrm{id}, \qquad \mathrm{spec}\,\hat{\chi} = \{+1, -1\}, \qquad \mathrm{tr}\,\hat{\chi} = 0;$$

$$(2)\quad P_L^2 = P_L, \quad P_R^2 = P_R, \quad P_L P_R = 0, \quad P_L + P_R = \mathrm{id}, \quad \mathrm{tr}\,P_L = \mathrm{tr}\,P_R = 1;$$

$$(3)\quad \hat{\chi}|L\rangle = +|L\rangle, \quad \hat{\chi}|R\rangle = -|R\rangle, \quad P_L|L\rangle = |L\rangle, \quad P_L|R\rangle = 0, \quad P_R|R\rangle = |R\rangle, \quad P_R|L\rangle = 0;$$

equivalently, without the $\sqrt{2}$ normalizers:

$$2P_L$$

$$= |L_u\rangle\langle L_u|,$$

$$2P_R$$

$$= |R_u\rangle\langle R_u| \ \text{ with } \ |L_u\rangle$$

$$= |c\rangle - i|\omega(c)\rangle,$$

$$|R_u\rangle$$

$$= |c\rangle + i|\omega(c)\rangle \ \ [\text{exact over } \mathbb{Q}(i)];$$

(4) the covariance identities and the winding read:

$$\hat{\omega}\hat{\chi}\hat{\omega}^{-1} = -\hat{\chi},$$

$$\{\hat{\chi}, \hat{\omega}\} = 0,$$

$$\hat{\omega} P_L \hat{\omega}^{-1} = P_R,$$

$$\hat{\omega}|L\rangle = -i|R\rangle, \quad \hat{\omega}|R\rangle = +i|L\rangle;$$

for a materialized chain c : $\quad k(c) = \Theta(c)/2 \in \mathbb{Z}$ [FT.22, FT.25(3)];

$\omega : \ k \mapsto -k$ [D2.6; EM.24(1) : C is ω's sign-level realization; FT.87(8) : sign(k) is C-odd];

(5) the construction is exact over $\mathbb{Q}(i)$ on all matrix elements in the basis $\{|c\rangle, |\omega(c)\rangle\}$: $\hat{\chi}$

$$= i\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right),$$

$$\hat{\omega}$$

$$= \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right);$$

(6) canonicity:

$$\left\{M = M^\dagger \in \mathrm{End}(\mathcal{H}_\omega) : \ \hat{\omega}M\hat{\omega} = -M, \ M^2 = \mathrm{id}, \ \langle c|M|c\rangle = \langle \omega(c)|M|\omega(c)\rangle = 0\right\} = \{+\hat{\chi}, \ -\hat{\chi}\};$$

without the diagonal-free condition the $\hat{\omega}$-odd Hermitian involutions are the circle $M(a, y)$

$$= \left(\begin{smallmatrix} a & iy \\ -iy & -a \end{smallmatrix}\right),$$

$$a^2 + y^2$$

$= 1$ [exact rational witnesses incl. $(a, y) = (\frac{3}{5}, \frac{4}{5}), (\frac{5}{13}, \frac{12}{13}), (\frac{8}{17}, \frac{15}{17})$; machine];

(7) the kind flag (torsor distinction):

$$H^1(S^1; \mathbb{Z}_2) \cong \mathrm{Hom}(\mathbb{Z}, \mathbb{Z}_2) = \{h_{\mathrm{R}} : n \mapsto 0, \ \ h_{\mathrm{NS}} : n \mapsto n \bmod 2\},$$

and the orientation-reversal pullback n

$\mapsto -n$ fixes both pointwise $(h(-n) = h(n)$ in $\mathbb{Z}_2)$: the spin-structure torsor is reversal-invariant, NOT exchanged;

the $\hat{\omega}$-exchanged objects are the chiral rays of clause (3): the torsor $\mathbb{Z}_2$ (boundary conditions) and the ray $\mathbb{Z}_2$ (the $\hat{\chi}$-grade) are distinct kinds [FT.89(2) discipline];

the R/NS analogy is recorded, never load-bearing.

Req. D1.4, D1.5, D2.6 (Parts 1–2); FT.22, FT.25 (Tiers F.1–F.11); EM.24 (Part 2); FT.87(8), FT.89(2) (Tiers F.29–F.30); Def-H.1 (Tiers F.1–F.11, the observer-relativity vocabulary of the closing

Rmk); Hermitian-involution spectral facts, exact canonicity solve over $\mathbb{Q}$, $\mathrm{Hom}(\mathbb{Z},\mathbb{Z}_2)$ arithmetic [math; machine: `checks/verify_H_chirality.py` §A].

Pf.

$$(1):\ \hat{\chi}^\dagger$$

$$= \big[i\big(|c\rangle\langle\omega(c)| - |\omega(c)\rangle\langle c|\big)\big]^\dagger$$

$$= (-i)\big(|\omega(c)\rangle\langle c| - |c\rangle\langle\omega(c)|\big)$$

$$= i\big(|c\rangle\langle\omega(c)| - |\omega(c)\rangle\langle c|\big)$$

$$= \hat{\chi}\ \ [(iA)^\dagger = -iA^\dagger;\ (|c\rangle\langle\omega(c)|)^\dagger = |\omega(c)\rangle\langle c|]$$

$$E := |c\rangle\langle\omega(c)| - |\omega(c)\rangle\langle c| :$$

$$E^2$$

$$= \underbrace{|c\rangle\langle\omega(c)|c\rangle\langle\omega(c)|}_{0} - |c\rangle\ \underbrace{\langle\omega(c)|\omega(c)\rangle}_{1}\langle c| - \ |\omega(c)\rangle\ \underbrace{\langle c|c\rangle}_{1}\langle\omega(c)| + \underbrace{|\omega(c)\rangle\langle c|\omega(c)\rangle\langle c|}_{0}$$

$$= -\big(|c\rangle\langle c| + |\omega(c)\rangle\langle\omega(c)|\big) \qquad = -\mathrm{id}$$

$$\hat{\chi}^2 = i^2E^2 = (-1)(-\mathrm{id}) = \mathrm{id};$$

$$\mathrm{tr}\,\hat{\chi} = 0\ [\text{off-diagonal}];$$

$$\mathrm{charpoly}(\hat{\chi}) = x^2 - \mathrm{tr}(\hat{\chi})x + \det\hat{\chi} = x^2 - 1\ [\det\hat{\chi} = 0\cdot 0 - i\cdot(-i) = -1]:\ \mathrm{spec} = \{+1,-1\}$$

$$(2):\ P_L^2 = \tfrac{1}{4}(\mathrm{id}+\hat{\chi})^2 = \tfrac{1}{4}(\mathrm{id} + 2\hat{\chi} + \hat{\chi}^2) = \tfrac{1}{4}(2\,\mathrm{id} + 2\hat{\chi}) = P_L; \quad \text{likewise } P_R;$$

$$P_LP_R = \tfrac{1}{4}(\mathrm{id} - \hat{\chi}^2) = 0;$$

$$P_L + P_R = \mathrm{id};$$

$$\mathrm{tr}\,P_L = \tfrac{1}{2}(2+0) = 1$$

$$(3):\ \hat{\chi}|c\rangle = i\big(|c\rangle\langle\omega(c)|c\rangle - |\omega(c)\rangle\langle c|c\rangle\big) = -i|\omega(c)\rangle; \qquad \hat{\chi}|\omega(c)\rangle = +i|c\rangle$$

$$\hat{\chi}|L_u\rangle = \hat{\chi}|c\rangle - i\,\hat{\chi}|\omega(c)\rangle = -i|\omega(c)\rangle - i\cdot i|c\rangle = |c\rangle - i|\omega(c)\rangle = |L_u\rangle;$$

$$\hat{\chi}|R_u\rangle = -i|\omega(c)\rangle + i\cdot i|c\rangle = -|R_u\rangle$$

$$|L_u\rangle\langle L_u|$$

$$= (|c\rangle - i|\omega(c)\rangle)(\langle c| + i\langle\omega(c)|)$$

$$= \mathrm{id} + i\big(|c\rangle\langle\omega(c)| - |\omega(c)\rangle\langle c|\big)$$

$$= \mathrm{id} + \hat{\chi}$$

$$= 2P_L \text{ [exact over } \mathbb{Q}(i);\ \langle L_u|L_u\rangle = 2];$$

$$\text{likewise } |R_u\rangle\langle R_u| = 2P_R$$

$$\Longrightarrow\ P_L|L\rangle = |L\rangle,\ P_L|R\rangle = 0,\ P_R|R\rangle = |R\rangle,\ P_R|L\rangle = 0\ \checkmark$$

$$(4):\ \hat{\omega}\hat{\chi}\hat{\omega}$$

$$= i\big(\hat{\omega}|c\rangle\langle\omega(c)|\hat{\omega} - \hat{\omega}|\omega(c)\rangle\langle c|\hat{\omega}\big)$$

$$= i\big(|\omega(c)\rangle\langle c| - |c\rangle\langle\omega(c)|\big)$$

$$= -\hat{\chi}\ \ [\hat{\omega}^{-1} = \hat{\omega}];$$

$$\{\hat{\chi},\hat{\omega}\} = \hat{\chi}\hat{\omega} + \hat{\omega}\hat{\chi} = \hat{\chi}\hat{\omega} - \hat{\chi}\hat{\omega} = 0$$

$$\hat{\omega}P_L\hat{\omega} = \tfrac{1}{2}(\mathrm{id} + \hat{\omega}\hat{\chi}\hat{\omega}) = \tfrac{1}{2}(\mathrm{id} - \hat{\chi}) = P_R;$$

$$\hat{\omega}|L_u\rangle = |\omega(c)\rangle - i|c\rangle = -i\big(|c\rangle + i|\omega(c)\rangle\big) = -i|R_u\rangle;$$

$$\hat{\omega}|R_u\rangle = +i|L_u\rangle$$

$$k = \Theta/2 \in \mathbb{Z} \text{ [FT.22 :}\ \Theta \in 2\mathbb{Z} \text{ on materialized chains; FT.25(3)]};$$

$$k$$

$\mapsto -k$ under ω : D2.6 (reversal) + EM.24(1) ($C : \sigma_e \mapsto -\sigma_e$ is ω's sign-level realization) + FT.87(8) (sign(k) is C-odd)

(5) : entrywise in the ordered basis $(|c\rangle, |\omega(c)\rangle)$ [machine]

$$(6) : \; M$$

$$= \left(\begin{smallmatrix} a & z \\ \bar{z} & b \end{smallmatrix}\right),$$

$$a,$$

$$b \in \mathbb{Q},$$

$$z$$

$$= x + iy : \quad \hat{\omega} M \hat{\omega}$$

$$= \left(\begin{smallmatrix} b & \bar{z} \\ z & a \end{smallmatrix}\right)$$

$$= -M$$

$$\iff b$$

$$= -a \wedge x$$

$$= 0$$

$$\iff M \qquad = M(a, y);$$

$$M(a, y)^2 = (a^2 + y^2)\,\mathrm{id}$$

$$M^2 = \mathrm{id} \iff a^2 + y^2 = 1;$$

diagonal-free

$$\iff a$$

$$= 0$$

$$\iff y^2$$

$$= 1$$

$$\iff y$$

$$= \pm 1$$

$$\iff M$$

$$= \pm\hat{\chi} \text{ [exact rational census of } a^2 + y^2 = 1 \text{ at denominators } \leq 20; \text{ machine]}$$

$$(7): \ \mathrm{Hom}(\mathbb{Z}, \mathbb{Z}_2) = \{h_0 : n \mapsto 0, \ h_1 : n \mapsto n \bmod 2\} \text{ [a homomorphism is fixed by its value at } 1];$$

$$(r^* h)(n) = h(-n): \quad h_1(-n) = (-n) \bmod 2 = n \bmod 2 = h_1(n) \ [\mathbb{Z}_2]; \quad h_0(-n) = 0 = h_0(n)$$

$$\implies r^* = \text{id on the torsor: fixed pointwise [machine on } |n| \leq 6, \text{ general by parity}];$$

$$\text{the exchanged objects are the } \hat{\chi}\text{-rays } [(4)];$$

$$\text{two distinct } \mathbb{Z}_2\text{'s [FT.89}(2): \text{ kind distinction recorded]. } \blacksquare$$

Phys. A bound chain is a closed loop, and a closed loop has two traversal senses. The orientation doublet holds them; the exchange form grades it into two chiral rays — exactly one such grade exists up to sign once the basis is treated symmetrically, so the construction carries no arbitrary choice beyond the naming. Reversing the loop exchanges the rays, and D2.6's vocabulary clause becomes an identity of matrices. The fermion boundary conditions on the loop — the Ramond and Neveu–Schwarz sectors — form a second, reversal-invariant $\mathbb{Z}_2$ that is not this grading; the tier keeps the two kinds apart rather than trading on their shared cardinality. All of it is built from two corpus objects that already exist — the chain and its reversal — with no axiom added and no parameter fitted.

Theorem 3.91 (FT.123 — Weak covariance and ω-covariance on the extended carrier). Thm. Under Def-H.34's constitutive carrier clause, the matter carrier of a bound chain extends from the seed $\mathcal{H}_{\mathsf{v}} \cong \mathbb{C}^2$ [EW.32] to $\mathcal{H}_{\text{int}} \otimes \mathcal{H}_\omega(c)$, with the weak action on the first factor alone, $W_i \mapsto \frac{1}{2}\sigma_i \otimes I_\omega$ [EW.32], and the chiral grade on the second alone:

$$(1) \quad \text{weak covariance:} \qquad [W_i \otimes I_\omega,\ I_{\text{int}} \otimes P_{L/R}] = 0 \qquad \forall i \in \{1,2,3\};$$

$$\text{locus sharpening: } \sum_i \left(\tfrac{1}{2}\sigma_i \otimes I_2\right)^2$$

$$= \tfrac{3}{4} I_4 \ (\text{type } 2 \oplus 2,\ \text{multiplicity } 2),$$

$$\text{while on the pair register } \sum_i W_i^2$$

$$= 2\Pi_{\mathcal{R}} \ (\text{type } 3 \oplus 1) \ [\text{Cor CT0.7a}] :$$

$$\text{the two weak actions are inequivalent } (\tfrac{3}{4} I_4 \neq 2\Pi_{\mathcal{R}},\ \text{machine}),$$

$$\text{so FT.93(3)'s pincer classifies the pair-register action and places no constraint here;}$$

FT.93(3d)'s seed irreducibility ($\{m : [m, \sigma_i] = 0\} = \mathbb{C} I_2$) is discharged by multiplicity, not violated: $\{M : [M, \sigma_i \otimes I_2] = 0\ \forall i\}$

$$= I_{\text{int}} \otimes \text{End}(\mathcal{H}_\omega),$$

$$\dim_{\mathbb{C}}$$

$$= 4 \ [\text{machine}];$$

$$(2) \quad \text{the } \hat{\omega}\text{-covariance:}$$

$$(I \otimes \hat{\omega})\,(I \otimes P_L)\,(I \otimes \hat{\omega})^{-1} = I \otimes P_R,$$

$$(I \otimes \hat{\omega})\,(I \otimes \hat{\chi})\,(I \otimes \hat{\omega})^{-1} = -(I \otimes \hat{\chi});$$

(3) the weight-cardinality correspondence [conditional on Def-H.38] : a materialized chain of winding k [FT.22, FT.25(3)] carries $\text{Sym}^k(\mathcal{H}_{\mathsf{v}})$ with weights j,

$$j - 1,$$

$$\dots,$$

$$-j,$$

$$j$$

$$= |k|/2 \text{ [Def-H.38; FT.60(3) ladder]};$$

the floor fermion (digon, $k = 1$ [FT.28]) has j

$= \frac{1}{2}$: exactly two weights $\{+\frac{1}{2}, -\frac{1}{2}\}$ [machine] — an unordered-pair cardinality correspondence with the two chiral rays $\{+1, -1\}$;

the label-to-weight attachment is not derived

"chirality

= helicity on-shell" is marked external vocabulary — the corpus derives no dispersion relation, and transit-side helicity reads [EM.8] are outside the bound scope [Def-H.38 scope note];

(4) massive modes (structural): the $\hat{\chi}$-eigenspaces are defined for any mode; helicity is a frame-dependent refinement [math];

on a massive mode chirality and helicity are distinct — the Dirac pattern;

$$(5) \quad \dim\bigl(\mathcal{H}_{\text{int}} \otimes \mathcal{H}_{\omega}\bigr)$$

$$= 2 \times 2$$

$= 4$: the Dirac-type carrier dimension of one massive mode (two Weyl halves);

a weak doublet of Dirac-type carriers is $2 \times 4 = 8$.

Rmk (interface definition Def-H.34). The tensor product $\mathcal{H}_{\text{int}} \otimes \mathcal{H}_{\omega}$ is the same species of interface definition as Def-H.41/Def-H.42: the minimal carrier combination consistent with both structures being independently present — the seed with its weak action [EW.32] and the orientation doublet with its exchange-form grade [Def-H.33]. It is exactly the "extended carrier" locus that FT.93(7) identified and item 21(i) enumerated without adopting. The orientation space itself, and every operator identity of FT.122, are declaration-free. Not declared and not constructed: the per-edge frame section $e \mapsto \mathcal{F}_e$ and its coupling to the EW.32 transport (item 21(i)'s residue), a global

Dirac-type operator on the emergent geometry, and any chiral gauge coupling (application-layer content).

Req. Def-H.33; FT.122; Def-H.34 (carrier clause); A8 (Parts 1–2); EW.32, EW.33 (Part 2); D2.6 (Parts 1–2); CT0.7, Cor CT0.7a (Part 1); FT.22, FT.25 (Tiers F.1–F.11); FT.28 (Tiers F.1–F.11); FT.60 (Tiers F.20–F.22); Def-H.38; EM.8 (Part 2); Kronecker-product commutant algebra, Schur's lemma, exact nullspace solves over $\mathbb{Q}$ [math; machine: `checks/verify_H_chirality.py` §B].

Pf.

$$(1):\ [W_i \otimes I,\ I \otimes P]$$

$$= (W_i I) \otimes (IP) - (IW_i) \otimes (PI)$$

$$= 0 \text{ (each factor commutes with an identity) [machine, all } i, \text{ both } P]$$

$$\sum_i \left(\tfrac{\sigma_i}{2} \otimes I_2\right)^2 = \left(\tfrac{1}{4}\sum_i \sigma_i^2\right) \otimes I_2 = \tfrac{3}{4} I_2 \otimes I_2 = \tfrac{3}{4} I_4;$$

$$\sum_i W_i^2 = \tfrac{3}{2} E_0 + \tfrac{1}{2}\Sigma = 2\Pi_{\mathcal{R}} \text{ [Cor CT0.7a re-derived; machine]};$$

$$\tfrac{3}{4} I_4 \neq 2\Pi_{\mathcal{R}} \text{ [tr : } 3 \neq 4]$$

$$\{M : [M, \sigma_i \otimes I_2] = 0\}: \text{ exact 32-unknown real nullspace, } \dim_{\mathbb{Q}} = 8\ (= 4 \text{ over } \mathbb{C});$$

$$\{I_2 \otimes E_{ab}\}_{a,b} \text{ lies in it and is } \mathbb{Q}\text{-independent (8 dims) } \Longrightarrow \text{ equality};$$

$$I \otimes P_L \text{ is a nontrivial member}$$

$$\{m : [m, \sigma_i] = 0\} = \mathbb{C} I_2, \quad \{m : \{m, \sigma_i\} = 0\} = 0 \text{ [FT.93(3}d\text{) re-derived; exact nullspaces]}$$

$$(2):\ \text{clause (4) of FT.122 tensored with } I_{\text{int}} \text{ [machine]};$$

$$(3):\ \tfrac{\sigma_3}{2} \text{ on } \mathrm{Sym}^1(\mathbb{C}^2)$$

$$= \mathbb{C}^2:\ \text{eigenvalues } \pm\tfrac{1}{2} \text{ [machine]};$$

$$\text{the ladder for general } k \text{ is FT.60(3)'s (machine-verified there)}$$

$$(4):\ \hat{\chi}\text{-eigenspaces are basis data of } \mathcal{H}_\omega \text{ [FT.122(1), (3)]};$$

helicity is the frame-dependent sign of the momentum-axis projection [math vocabulary]

$$(5):\ 2\cdot 2=4;\qquad 2\cdot 4=8\ \text{[exact]}.\ \blacksquare$$

Phys. The weak force lives on the internal seed; the chiral grade lives on the chain's own orientation bookkeeping; disjoint factors commute. That is the whole mechanism — and it is precisely the ingredient whose absence FT.93(7) diagnosed: a carrier on which the gauge action is trivial for the grade in question. The Casimir census keeps the accounting honest: the extended weak action ($\frac{3}{4}I_4$, two doublets) is not the pair-register action ($2\Pi_{\mathcal{R}}$, a triplet and a singlet), so nothing proven about the latter binds the former, and the seed's irreducibility — the Schur wall behind the pincer — is lifted by multiplicity the moment the orientation factor is adjoined. What is not obtained is the Standard Model's chiral coupling: here the weak force is chirality-blind, the chiral halves are weak-blind rather than weak-inactive (the extended weak kernel is $\{0\}$), and the rule that would make one ray gauge-active is application-layer content, ¬derived here.

Rmk (forward pointer, non-load-bearing). The chirality-flipping block of the complete static force matrix on the extended carrier — the off-diagonal complement of clause (1)'s chirality blindness — is computed in Tier F.40: at the derived locus $F_{LR}=0$ exactly (FT.128, the no-flip theorem: every derived aspect action is internal-only), and under the declared read promotion Def-H.43 the flipping block is $F_{LR}=q_{\text{core}}I_2$ with flipping set $\{F_{\text{EM}}$ core read$\}$ and chiral-partner splitting $2|q|\in\{0,2\}$ (FT.129–FT.130). The pointer adds no load-bearing content to this tier.

FT.124 — The prior obstructions do not reach the orientation space. Structural-constraints record: see Appendix B, Section B.3.4 of the Addendum.

FT.125 — The deficit ledger: deficit 1 closed at the projector level; deficit 1′ opened. Structural-constraints record: see Appendix B, Section B.4.1 of the Addendum.

Theorem 3.92 (FT.126 — Weak-covariant chiral projectors on the chain carrier). Thm. Combining FT.123's tensor-factor separation with the branch-grading locus of FT.101: the orientation-space chirality lifts to the chain-carried Dirac-type carrier $\mathcal{H}_{\text{int}}\otimes\mathcal{H}_\omega(c)$:

$$(1)\quad P^{\text{car}}_{L/R}\ :=\ I_{\text{int}}\otimes P_{L/R}:$$

$$(P^{\text{car}}_{L/R})^2=P^{\text{car}}_{L/R},$$

$$P^{\text{car}}_L P^{\text{car}}_R=0,$$

$$P^{\text{car}}_L+P^{\text{car}}_R=\mathrm{id},$$

$$\mathrm{tr}\,P^{\text{car}}_{L/R}=2;$$

$$(2)\quad [W_i \otimes I_\omega,\ P^{\text{car}}_{L/R}] = 0\ \ \forall i \qquad \text{[the weak isospin acts chirality-blind]};$$

(3) each Weyl half im $P^{\text{car}}_{L/R}$ carries exactly one chiral ray of the orientation factor and the full internal seed action (weak-blind, not weak-inactive: $\bigcap_i \ker(W_i \otimes I_\omega) = \{0\}$ [machine]);

(4) independence of the two $\mathbb{Z}_2$'s: the A4 branch grading of FT.101 is label-level registration data [FT.101(3) : no Hilbert-space implementation on the pair register];

the chiral grading is an implemented involution on the orientation factor. Any internal involution $G \otimes I_\omega$ commutes with $I_{\text{int}} \otimes \hat{\omega}$ and their product is a third, distinct involution: $\mathbb{Z}_2 \times \mathbb{Z}_2$ [machine on $G \in \{\sigma_3, \sigma_1\}$; FT.88's $\langle \omega, \Theta_\Sigma \rangle = \mathbb{Z}_2 \times \mathbb{Z}_2$ is the frozen precedent for the product structure].

Req. FT.122, FT.123, FT.124 (this tier); FT.88, FT.101 (Tiers F.29–F.30, Tiers F.31–F.34); EW.32 (Part 2); Def-H.34 (carrier clause); Kronecker-product identities [math; machine: `checks/verify_H_chirality.py` §B].

Pf.

(1) : FT.122(2) tensored with I_{int} : $(I \otimes P)^2 = I \otimes P^2 = I \otimes P$; $\quad \text{tr}(I_2 \otimes P_{L/R}) = 2 \cdot 1 = 2$ [machine]

(2) : FT.123(1) verbatim;

(3) : $\text{im}(I \otimes P_L)$

$= \mathcal{H}_{\text{int}} \otimes \mathbb{C}|L\rangle$;

$\bigcap_i \ker(\sigma_i \otimes I)$

$= (\bigcap_i \ker \sigma_i) \otimes \mathcal{H}_\omega$

$= \{0\}$ [σ_i invertible; exact 24×8 rank 8; machine]

(4) : $[G \otimes I,\ I \otimes \hat{\omega}] = 0$ [machine];

$(G \otimes I)(I \otimes \hat{\omega}) = G \otimes \hat{\omega} \neq G \otimes I,\ I \otimes \hat{\omega}$ [entrywise; machine];

$(G \otimes \hat{\omega})^2 = G^2 \otimes \hat{\omega}^2 = \text{id}$ ■

Phys. The weak force and the chiral grade do not interfere: the branch grading of FT.101 and the chiral grading of this tier are two independent $\mathbb{Z}_2$'s on disjoint data, and the lifted projectors give each Weyl half of the chain carrier its own orientation ray. Which branch, which ray, and which species go together is a third rule — the application-layer assignment rule of FT.125(4).

3.41 Tier F.40 — The Static Force Matrix on the Extended Carrier: The Aspect Census, the No-Flip Theorem, and the Charge–Parity Splitting

Definition 3.42 (Def-H.34 — The extended carrier basis, the chiral transform, and the partner involution). Def. For a bound chain c [D1.4: ordered tuple of distinct real edges; D1.5: bound = closed self-contained, fermionic content]:

- Extended carrier (constitutive carrier clause). $\mathcal{H}_{\text{ext}} := \mathcal{H}_{\text{int}} \otimes \mathcal{H}_\omega(c) \cong \mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4$, with $\mathcal{H}_{\text{int}} = \mathcal{H}_{\mathsf{v}}$ the seed carrier, orthonormal basis $\{|\mathsf{v}\rangle, |\mathsf{v}^\perp\rangle\}$ [EW.32; the complement ray is a basis convention — all statements below are basis-covariant], and $\mathcal{H}_\omega$, $\hat{\omega}$, $\hat{\chi}$, $P_{L/R}$, $|L\rangle$, $|R\rangle$ as in Def-H.33.

- Ordered orientation basis (internal-major Kronecker order; composite index $= 2 \cdot \text{int} + \omega$):

$$|1\rangle := |\mathsf{v}\rangle \otimes |c\rangle,$$

$$|2\rangle := |\mathsf{v}\rangle \otimes |\omega(c)\rangle,$$

$$|3\rangle := |\mathsf{v}^\perp\rangle \otimes |c\rangle,$$

$$|4\rangle := |\mathsf{v}^\perp\rangle \otimes |\omega(c)\rangle.$$

- Chiral carrier basis. $\{|\mathsf{v}\rangle \otimes |L\rangle,\ |\mathsf{v}\rangle \otimes |R\rangle,\ |\mathsf{v}^\perp\rangle \otimes |L\rangle,\ |\mathsf{v}^\perp\rangle \otimes |R\rangle\}$, with $|L\rangle = \frac{1}{\sqrt{2}}(|c\rangle - i|\omega(c)\rangle)$, $|R\rangle = \frac{1}{\sqrt{2}}(|c\rangle + i|\omega(c)\rangle)$ [Def-H.33].

- The S-transform. $S := \left(\begin{smallmatrix} 1 & 1 \\ -i & i \end{smallmatrix}\right)$ (columns = unnormalized chiral kets): $S^\dagger S = 2I_2$; $U := I_2 \otimes S$; the conjugation $M \mapsto \frac{1}{2}U^\dagger M U$ is exact over $\mathbb{Q}(i)$ and rotates the ω-factor Pauli images cyclically:

$$\hat{\omega}\,(\sigma_1) \ \mapsto\ -\sigma_2, \qquad \hat{\chi}\,(-\sigma_2) \ \mapsto\ \sigma_3, \qquad \hat{\kappa}\,(\sigma_3) \ \mapsto\ \sigma_1.$$

- Partner involution. $\hat{\kappa} := |c\rangle\langle c| - |\omega(c)\rangle\langle\omega(c)|$ — the unique diagonal $\hat{\omega}$-odd Hermitian involution up to sign [FT.122(6)'s circle at $(a, y) = (\pm 1, 0)$; machine]. Quaternionic triple with Def-H.33's operators:

$$\hat{\omega}^2 = \hat{\chi}^2 = \hat{\kappa}^2 = \text{id};$$

$$\{\hat{\omega}, \hat{\chi}\} = \{\hat{\chi}, \hat{\kappa}\} = \{\hat{\kappa}, \hat{\omega}\} = 0;$$

$$\hat{\omega}\hat{\chi} = -i\hat{\kappa}, \quad \hat{\chi}\hat{\kappa} = -i\hat{\omega}, \quad \hat{\kappa}\hat{\omega} = -i\hat{\chi}.$$

- Force-profile and read data. $\mathsf{c} = (c_0, c_1, c_2, c_3) \in \mathbb{Q}^4$: the weak-transport profile of the chain over the seed generator set $\{W_1, W_2, W_3, Y\}$ [EW.33's c_A; EW.32's images] — configuration data, formal indeterminates in the Def-H.32(3) discipline, never fitted [P1, P4];

$$h(\mathsf{c}) = \tfrac{1}{2}\sum_{a=0}^{3} c_a\,\sigma_a \qquad (\sigma_0 := I_2); \qquad |\mathsf{c}|^2 := c_1^2 + c_2^2 + c_3^2;$$

$q := q_{\text{core}}(c) \in \{-1, 0, +1\}$: the ternary core-charge read of the oriented chain [D3.4, EM.9, FT.40] — registration data, ω-odd [FT.40].

Rmk (the three bases of the triple). Each basis of $\mathcal{H}_\omega$ diagonalizes one member of the triple: the orientation basis $\{|c\rangle, |\omega(c)\rangle\}$ diagonalizes $\hat{\kappa}$; the chiral basis $\{|L\rangle, |R\rangle\}$ diagonalizes $\hat{\chi}$; the parity basis $\{(|c\rangle+|\omega(c)\rangle)/\sqrt{2},\ (|c\rangle-|\omega(c)\rangle)/\sqrt{2}\}$ diagonalizes $\hat{\omega}$ [machine]. Every ray of one member is an equal-weight superposition in the eigenbasis of the other two — in particular $|c\rangle = (|L\rangle + |R\rangle)/\sqrt{2}$: the orientation rays, which FT.129(5) makes stationary, carry no definite chirality. No new structure beyond Def-H.33: $\hat{\kappa}$ completes its operator triple.

Theorem 3.93 (FT.127 — The aspect census on the extended carrier). Thm. For the complete force operator of A8 — aspect inventory: the six pair-sector aspects [CT0.12], the rung masks and stack blocks [CT0.13, CT0.14], the two dark aspects [CT4.6, D4.1, D4.3], the Janus move classes [D1.2, FT.108(1)] — the derived-action census on $\mathcal{H}_{\text{ext}}$ is:

$$(1) \quad F_W\ (\mathcal{E}_{+-} = \text{span}\{W_i\},\ \text{3-dim [CT0.12]}) :\ \text{acts.}$$

$$W_i^{(\text{ext})} = \tfrac{1}{2}\sigma_i \otimes I_\omega$$

$$[\text{EW.32 seed images};\ \text{Def-H.34}]\ \ [\text{exact } 4\times 4 \text{ over } \mathbb{Q}(i)];$$

$$(2)$$

$$F_{\text{EM}}\ (\text{stack},\ \text{1-dim [CT0.14]}) :\ \text{acts on the internal factor only,}$$

$$\text{via the neutral mixing } A$$

$$= \sin\theta_W$$

$$W_3 + \cos\theta_W$$

$$Y\ [\text{D3.10},\ \text{EW.24}] :$$

$$A^{(\text{ext})} = \tfrac{1}{2}\big(\sin\theta_W\,\sigma_3 + \cos\theta_W\,\sigma_0\big) \otimes I_\omega$$

$$[\text{EW.32}: \; Y \mapsto \tfrac{1}{2}\sigma_0] \text{ — inside clause } (1)\text{'s span; chirality-blind } (\otimes I_\omega);$$

the ternary end-register read q_{core} [D3.4, EM.9, FT.40] is ω-odd configuration data: an orientation-side action requires the read promotion of FT.129 (Def-H.43);

a bridge between the seed neutral content and the ternary read is not derived (the ternary-read situation is ¬derived here);

(3)

$$F_G \; (\text{span}\{\hat{E}^{ij}\} \text{ traceless, } 5\text{-dim [CT0.12]}): \text{ does not act on } \mathcal{H}_{\text{ext}} \text{ — pair-register block [CT1.2];}$$

chain-level content

= geometric defect data (Θ_R-type scalars under the Def-H.37(2) transfer [FT.33, FT.54(4)]) — scalar class;

(4)

$$F_H \; (\text{span}\{\Sigma\}, \; H = h\Sigma \text{ [EW.13]}): \text{ does not act as a carrier matrix — } \Sigma \in \text{End}(\mathbb{C}^4_{\text{pair}}) \text{ [CT1.2];}$$

Higgs content on chains

= scalar background (h_0 P4-class; entering only through the EW.20/EW.33 defect-residue packaging) — scalar class;

(5)

$$F_{D^+} \; (\text{span}\{E_0\} = \Lambda\,\mathbb{I} \text{ [D4.3, CT4.6]}): \text{ scalar class — central identity direction, configuration-independent;}$$

if admitted on a carrier it is λI_4,

re-absorbed by h

$\mapsto h + \lambda I_2$ [FT.130(7)];

geometric side $\Lambda g_{\mu\nu}$ [CT1.23],

$$w$$

$$= -1 \text{ [CT1.25]};$$

$$(6)$$

$$F_{J^+},$$

F_{J^-} (span$\{J_i\}$, span$\{K_i\}$ [CT0.12, CT0.7(4)]) : do not act on $\mathcal{H}_{\text{ext}}$ — pair-register matrices [CT1.2] and graph-level Janus move generators [D1.2, FT.108(1)];

no seed or orientation action derived;

their bilinear channel content is the seal-triplet reads of Tiers F.37–F.38 [FT.109, FT.116];

$$(7)$$

F_{strong} (8-dim stack [CT3.8, CT3.13, CT0.14]) : does not act on $\mathcal{H}_{\text{ext}}$ — acts on the colour-slot carrier $\mathbb{C}^3$ at trivalent vertices [CT3.2, D3.1];

the seed is colour-neutral by theorem [EW.32 : $T_a \mathsf{v} = 0$];

colour sits in an orthogonal stack block [CT0.12 Rmk; FT.93(4)];

$\mathcal{H}_{\text{ext}}$ has no slot factor;

sealed ends are blind in any case [FT.117];

$$(8)$$

F_{D^-} ($\bigoplus_{n\geq 4} F_n$ [D4.1, CT4.1, CT4.6]) : does not act — ledger sink/retention verbs [CT4.2–CT4.5; the retention census is u-fugacity bookkeeping [FT.108(3)]];

no carrier representation;

(9) census conclusion: the derived static force matrix is exactly the EW seed content,

$$\mathrm{F}_{\mathrm{ext}}(\mathrm{c}) = h(\mathrm{c}) \otimes I_\omega, \qquad h(\mathrm{c}) = \tfrac{1}{2} \sum_{a=0}^{3} c_a \, \sigma_a,$$

and the four images $\{\frac{1}{2}\sigma_a \otimes I_2\}_{a=0}^{3}$ are $\mathbb{C}$-independent with span $\mathrm{End}(\mathbb{C}^2) \otimes I_\omega$ [machine] : every derived aspect action is chirality-blind.

Req. CT0.7, CT0.12, CT0.13, CT0.14, CT1.2, CT1.3 (Part 1); CT1.23, CT1.25 (Part 2); A8, D1.2, D3.1, D3.4, D3.10, D4.1, D4.3, P1, P4 (Parts 1–2); EW.8, EW.13, EW.20, EW.24, EW.32, EW.33 (Part 2); EM.9 (Part 2); CT3.2, CT3.8, CT3.13 (Part 2); CT4.1, CT4.2, CT4.3, CT4.5, CT4.6 (Part 2); FT.20 (Tiers F.1–F.11); FT.33 (Tiers F.12–F.13); FT.54 (Tiers F.17–F.19); FT.93 (Tiers F.31–F.34); FT.108 (Tier F.36); FT.109 (Tier F.37); FT.116, FT.117 (Tier F.38); Def-H.33, FT.123 (Tier F.39); Def-H.37(2), Def-H.32, Def-H.34; exact $\mathbb{Q}(i)$ independence rank (real-form flattening with i-doubles) [math; machine: `checks/verify_H_force_matrix.py` §B].

Pf.

$$(1) : \ \text{EW.32} : \ W_i \mapsto \tfrac{1}{2}\sigma_i \text{ on } \mathcal{H}_{\mathsf{v}};$$

$$\text{Def-H.34: the extended action is } W_i \mapsto \tfrac{1}{2}\sigma_i \otimes I_\omega;$$

$$\text{the three } 4 \times 4 \text{ matrices written entrywise over } \mathbb{Q}(i) \text{ [machine]}$$

$$(2) : \ A = \sin\theta_W W_3 + \cos\theta_W Y \ [\text{EW.24; D3.10}];$$

$$A^{(\mathrm{ext})} = \sin\theta_W \, W_3^{(\mathrm{ext})} + \cos\theta_W \, Y^{(\mathrm{ext})} \ [\text{linearity}];$$

$$Y \mapsto \tfrac{1}{2}\sigma_0 \ [\text{EW.32}];$$

$$\text{unit normalization pin: } \mathfrak{T}\big((\tfrac{1}{2}(\sin\theta_W\sigma_3 + \cos\theta_W\sigma_0))^2\big) = \tfrac{1}{2} \ [\sin^2 + \cos^2 = 1; \ \mathfrak{T}\sigma_3 = 0]$$

$$(9) : \ \text{flattening } \{\tfrac{1}{2}\sigma_a \otimes I_2\} \text{ with } i\text{-doubles: real rank } 8 = 4 \text{ over } \mathbb{C} \text{ [machine]};$$

$$\mathrm{span}\{\sigma_0, \sigma_1, \sigma_2, \sigma_3\} = \mathrm{End}(\mathbb{C}^2) \ [\text{CT1.1's Pauli basis; machine}]$$

(3)–(8) : each row's locus is the cited frozen tag verbatim: pair-register membership [CT1.2 basis table],

$$\text{slot carriers [CT3.2]},$$

$$\text{ledger sinks [D4.1, CT4.1]},$$

scalar/central directions [D4.3, EW.13, EW.20];

¬∃ derived seed or orientation action for any of the eight [inventory scan: the only carrier bridge in Parts 1–2 is EW.32's]. ■

Phys. Ask the nine aspects of the complete force where they act on the extended carrier, and eight answer elsewhere — on the pair register, in the colour slots, in the ledger, or as scalar background. The census is the theorem: the derived force content of a chain carrier is exactly the electroweak seed image, one simple tensor with the orientation identity, hence chirality-blind by construction. No aspect of A8 reaches the orientation factor; the only canonical candidate that does is the chain's own ω-odd read data, taken up in FT.129.

Theorem 3.94 (FT.128 — The no-flip theorem at the derived locus). Thm. In the chiral basis of Def-H.34:

(1) form invariance:

$$U^{-1}\big(h(\mathsf{c})\otimes I_\omega\big)U = h(\mathsf{c})\otimes I_2$$

[the conjugation touches only the ω-factor, where the derived matrix is the identity];

(2) the block decomposition:

$$F_{LL} = F_{RR} = h(\mathsf{c}),$$

$$F_{LR} = F_{RL} = 0$$

exactly, for every profile c;

(3) the weak flip contribution vanishes by computation:

$$P_L^{\mathsf{car}}\, W_i^{(\mathsf{ext})}\, P_R^{\mathsf{car}} = 0 \;\; \forall i$$

[entrywise on the explicit 4×4 matrices; FT.123(1) re-verified, not invoked];

(4) $\big[\mathrm{F}_{\mathsf{ext}}(\mathsf{c}),\; I_2\otimes\hat{\chi}\big] = 0 \;\; \forall \mathsf{c}$:

chirality is an exact quantum number of the derived static force matrix;

(5) the derived spectrum: $\quad \operatorname{charpoly} \mathrm{F}_{\mathrm{ext}}(\mathrm{c}) = \left(x^2 - c_0 x + \frac{c_0^2 - |\mathrm{c}|^2}{4}\right)^2,$

$$\operatorname{spec}$$

$$= \left\{ \tfrac{1}{2}\big(c_0 + |\mathrm{c}|\big),\ \tfrac{1}{2}\big(c_0 - |\mathrm{c}|\big) \right\},$$

each with multiplicity 2 across the chiral rays — spectral parity doubling;

$$|\mathrm{c}| = \sqrt{|\mathrm{c}|^2} \in \text{a quadratic extension of } \mathbb{Q}(\mathrm{c});$$

(6) the static shadow, derived locus:

$$\langle L | \mathrm{F}_{\mathrm{ext}} | R \rangle$$

$= 0$ — the force aspects alone produce no parity-flipping seed on the extended carrier of a bound chain;

the flipping set at the derived locus $= \varnothing$.

Req. FT.127; Def-H.34; FT.122(1)–(3), FT.123(1) (Tier F.39); CT1.2 (Part 1); Zariski-dense rational sampling (entries linear in c: vanishing on a dense set implies identically zero), Faddeev–LeVerrier over $\mathbb{Q}(i)$ [math; machine: `checks/verify_H_force_matrix.py` §B].

Pf.

(1) : $U = I_2 \otimes S$: $\quad U^{-1}(h \otimes I_2)U = h \otimes (S^{-1} I_2 S) = h \otimes I_2$ [machine on 108 exact rational profiles]

(2) : block extraction in the chiral carrier basis $\{|\mathrm{v}L\rangle, |\mathrm{v}R\rangle, |\mathrm{v}^{\perp}L\rangle, |\mathrm{v}^{\perp}R\rangle\}$: F_{LR}

$$= P_L^{\mathrm{car}} \mathrm{F}_{\mathrm{ext}} P_R^{\mathrm{car}}$$

$= 0$ entrywise at every sample profile;

entries of F_{LR} are linear in c : vanishing on the Zariski-dense sample set

$$\Longrightarrow F_{LR}$$

$$\equiv 0 \text{ [machine]}$$

$$(3):\ P_L^{\text{car}} = I_2 \otimes P_L,\ P_R^{\text{car}} = I_2 \otimes P_R\ [\text{FT.126}(1)]:$$

$$(I \otimes P_L)(\tfrac{\sigma_i}{2} \otimes I)(I \otimes P_R) = \tfrac{\sigma_i}{2} \otimes P_L P_R = 0\ [\text{FT.122}(2):\ P_L P_R = 0];$$

machine: entrywise zero on all three 4×4 products

$$(4):\ [h \otimes I_2,\ I_2 \otimes \hat{\chi}]$$

$$= [h, I_2] \otimes [I_2, \hat{\chi}]\text{-type disjointness}:\ (hI_2) \otimes (I_2\hat{\chi}) - (I_2 h) \otimes (\hat{\chi} I_2)$$

$$= 0\ [\text{machine}]$$

$$(5):\ \text{charpoly}\, h(\mathrm{c}) = x^2 - c_0 x + \tfrac{c_0^2 - |\mathrm{c}|^2}{4}\ [\det h = \tfrac{c_0^2 - |\mathrm{c}|^2}{4};\ \mathfrak{T}h = c_0];$$

$$\text{charpoly}(h \otimes I_2)$$

$$= (\text{charpoly}\, h)^2\ [\text{Kronecker spectrum};\ \text{machine: Faddeev–LeVerrier coefficient identity at all } 108 \text{ profiles}]$$

$$(6):\ \langle L|\mathrm{F}_{\text{ext}}|R\rangle = (I_2\text{-block of } F_{LR}) = 0\ [(2)].\ \blacksquare$$

Phys. The derived force matrix is blind to the very distinction the carrier was built to carry: it preserves each chiral ray, doubles every level across the two rays, and commutes with the chirality grade. That is not a failure of the census — it is its content. Every force the corpus derives onto this carrier arrives through the seed, and the seed action is internal: the orientation factor sees only the identity. A parity flip, if the corpus is to have one, must come from the orientation factor's own read structure, and the no-flip theorem narrows the search to a single question: does any ω-odd datum of the bound chain act there?

Definition 3.43 (Def-H.43 — The read promotion on the orientation doublet). Def. An ω-odd A8-aspect read carried by a bound chain acts on the orientation doublet $\mathcal{H}_\omega$ [Def-H.33] as the multiplication operator of its eigenvalues on the orientation basis rays:

$$\text{read}\ q\ \longmapsto\ \hat{q}\ :=\ q \cdot \hat{\kappa}$$

(ω-odd reads),

$$\hat{\kappa}$$

$$=\ |c\rangle\langle c| - |\omega(c)\rangle\langle\omega(c)|\ [\text{Def-H.34}:\ \text{the unique diagonal } \hat{\omega}\text{-odd Hermitian involution up to sign},\ \text{FT.122}(6)],$$

EM core read: $\hat{Q}_{\text{EM}} = q_{\text{core}}(c)\,\hat{\kappa}$, with $q_{\text{core}} \in \{-1, 0, +1\}$ ternary registration data [D3.4, EM.9, FT.40] — ¬fitted. ω-even reads shift (no flip); the parity split is derived [FT.129(3) — immediate below], ¬part of this definition. The promotion is the sole source of a chirality-flipping block [FT.129(4)–(5)]. Derived under the promotion (cited). Flipping block $F_{LR} = q_{\text{core}} I_2$; flipping set $\{F_{\text{EM}}$ core read$\}$ — minimal and unique; chiral-partner splitting $2|q| \in \{0, 2\}$ [FT.129(5), FT.130(5)]; at the derived locus (no promotion): the no-flip theorem holds unconditionally [FT.128]. Bars (standing). $2|q|$ is ¬a frequency — no carrier-time parameter exists; emergent time is a prerequisite [FT.131(3)]; $\hat{\kappa}$-/$\hat{\chi}$-eigenrays are ¬species and ¬SM chiralities — the assignment rule is application-layer content; ω-even reads are not promoted to $\hat{\kappa}$-multiples [FT.129(3) is derived]; a transit-chain flipping census is out of scope [FT.131(1)]; external calibration barred [P1, P5]. Tension remark (structural). The chiral rays superpose ω-opposed core charges $(q, -q)$ [FT.40] while channel-level coarse-graining is sign-blind and fuses only equal-q states [FT.15's hypothesis; FT.90(5)'s audit] — the carrier-level doublet and channel fusion are distinct structures; their relation is bridge content [Def-H.39's slots], open, and no theorem of this part identifies them [FT.131(5)].

Theorem 3.95 (FT.129 — The canonical flip candidate and the read promotion). Thm.

$$(1) \quad \text{canonicity:}$$

$$\left\{M = M^\dagger : \ M \text{ diagonal in the orientation basis, } \hat{\omega} M \hat{\omega} = -M, \ M^2 = \text{id}\right\} = \{+\hat{\kappa}, \ -\hat{\kappa}\}$$

$$[\text{FT.122}(6)\text{'s circle at } (a, y) = (\pm 1, 0); \ \text{exact rational census}; \ \text{machine}];$$

$$(2) \quad \text{chiral images:}$$

$$\hat{\kappa}|L\rangle = |R\rangle, \quad \hat{\kappa}|R\rangle = |L\rangle \text{ (ray swap, no phase)};$$

$$\text{in the chiral basis: } \hat{\kappa} \mapsto \sigma_1\text{-type},$$

$$\hat{\chi} \mapsto \sigma_3\text{-type},$$

$$\hat{\omega}$$

$$\mapsto -\sigma_2\text{-type [the } S\text{-rotation of Def-H.34; machine]};$$

$$(3) \quad \text{the parity split of the reads on the doublet:}$$

$$\omega\text{-odd} : \ q_{\text{core}} \text{ [FT.40]}, \ k \text{ and } \text{sign}(k) \text{ [D2.6; EM.24}(1)\text{; FT.87}(8)\text{]};$$

$$\omega\text{-even} : \ |k|, \ \Theta^2, \ \text{the sign-blind reads [FT.90}(5)\text{]};$$

an even read, promoted, multiplies I_2 (chirality-blind);

an odd read, promoted, multiplies $\hat{\kappa}$ (chirality-flipping) : the flipping structure of any scalar read class is decided by its ω-parity;

(4) Def-H.43 (read promotion, Def-H.43): an ω-odd A8-aspect read carried by a bound chain acts on $\mathcal{H}_\omega$ as the multiplication operator of its eigenvalues on the basis rays. For the EM core read:

$$\hat{Q}_{\text{EM}} := q_{\text{core}}(c)\,\hat{\kappa}$$

[the unique minimal promotion consistent with FT.40's eigenvalues $(q,-q)$; $q \in \{-1,0,+1\}$ registration data, ¬fitted];

no such action is derived in Parts 1–2 (reads are labels [Def-H.15(4)]), hence declared. Tension logged: the chiral rays superpose ω-opposed core charges $(q,-q)$;

channel-level coarse-graining fuses only equal-q states [FT.15 hypothesis; FT.90(5) audit] — the relation between the carrier-level doublet and channel fusion is item 7/item 13 bridge content, open;

(5) the promoted force matrix and its flipping block: under Def-H.43,

$$\mathrm{F}_{\text{ext}}(\mathsf{c},q) = h(\mathsf{c})\otimes I_2 + q\,I_2\otimes\hat{\kappa}, \qquad \big[h(\mathsf{c})\otimes I_2,\; q\,I_2\otimes\hat{\kappa}\big] = 0 \text{ [machine]};$$

$$F_{LL} = F_{RR} = h(\mathsf{c}), \qquad F_{LR} = q\,I_2 = F_{RL} : \qquad \text{nonzero} \iff q \neq 0;$$

the flipping set

$= \{F_{\text{EM}}$ core read$\}$: minimal and unique — removing the $\hat{\kappa}$ term kills F_{LR} for every profile [FT.128(2)];

removing every internal term leaves F_{LR}

$= qI_2$ [machine];

the off-diagonal matrix elements encoding the flipping structure are the core-charge read itself:

$$\langle L|\mathrm{F}_{\text{ext}}|R\rangle = q\,I_2.$$

Req. Def-H.34; FT.128; FT.122(6)–(7), FT.123, FT.126 (Tier F.39); Def-H.33 (Tier F.39); FT.15 (Tiers F.1–F.11); FT.40 (Tiers F.12–F.13); Def-H.15 (Tiers F.1–F.11); FT.87(8), FT.89(2), FT.90(5) (Tiers F.29–F.30); EM.9, EM.24 (Part 2); D2.6, D3.4 (Parts 1–2); Def-H.43; exact canonicity census, Kronecker block algebra [math; machine: `checks/verify_H_force_matrix.py` §A, §C].

Pf.

$$(1):\ M=\mathrm{diag}(a,b):\ \hat{\omega}M\hat{\omega}=\mathrm{diag}(b,a)=-M\iff b=-a;$$

$$M^2=\mathrm{id}\iff a^2=1\iff a=\pm1\iff M=\pm\hat{\kappa}\ [\text{exact rational census; machine}]$$

$$(2):\ \hat{\kappa}|L_u\rangle=\hat{\kappa}(|c\rangle-i|\omega(c)\rangle)=|c\rangle+i|\omega(c)\rangle=|R_u\rangle\ [\text{unnormalized, exact}];$$

$$\text{likewise } \hat{\kappa}|R_u\rangle=|L_u\rangle;$$

$$\tfrac{1}{2}S^\dagger\sigma_3 S=\sigma_1\ [\text{Def-H.34 rotation; machine}]$$

$$(3):\ q_{\text{core}}(\omega(c))=-q_{\text{core}}(c)\ [\text{FT.40}];$$

$$k\mapsto -k\ [\text{D2.6; EM.24}(1)\text{; FT.87}(8)];$$

$$|k|,\ \Theta^2\text{ invariant; sign-blindness of the listed reads [FT.90}(5)]$$

$$\text{promotion linearity: an } \omega\text{-even read } r\ (\text{values } (r,r))\ \mapsto\ rI_2;$$

$$\text{an } \omega\text{-odd read } r\ (\text{values } (r,-r))$$

$\mapsto r\hat{\kappa}$ [diagonal multiplication operators on the orientation basis; uniqueness of the diagonal form: clause (1)]

$$(4):\ \hat{Q}_{\text{EM}}|c\rangle=q|c\rangle,\quad \hat{Q}_{\text{EM}}|\omega(c)\rangle=-q|\omega(c)\rangle\ [\text{FT.40 eigenvalues}];$$

$$\text{declaration status and tension: recorded at Def-H.43}$$

$$(5):\ \big[h\otimes I_2,\ qI_2\otimes\hat{\kappa}\big]=q\big(hI_2\otimes I_2\hat{\kappa}-I_2h\otimes\hat{\kappa}I_2\big)=0\ [\text{machine}];$$

$$\tfrac{1}{2}U^\dagger(I_2\otimes\hat{\kappa})U=I_2\otimes\sigma_1\ [\text{Def-H.34}]$$

$$\Longrightarrow\ \text{chiral blocks of } h\otimes I_2+q\,I_2\otimes\sigma_1:\ F_{LL}=F_{RR}=h,$$

$$F_{LR} = F_{RL} = qI_2 \text{ [machine on } 48 \text{ profiles} \times q \in \{-1, 0, 1\}];$$

minimality rows: machine ■

Phys. The orientation doublet carries exactly two canonical involutions — the exchange form, which grades chirality, and its partner, which grades the traversal sense — and the partner is the flip operator: it swaps the chiral rays with no phase. The corpus's own reads split cleanly across this pair: everything ω-even can only shift levels; everything ω-odd can only flip chirality. Promoting the ternary core-charge read — one interface definition, its tension with equal-q channel fusion logged rather than hidden — the complete static force matrix acquires its only off-diagonal content, and the flipping amplitude is the charge read itself: a charged core flips, a neutral core does not.

Theorem 3.96 (FT.130 — Tensor structure, Schmidt rank, and the static spectrum). Thm.

(1) derived locus:

$\mathrm{F_{ext}}(\mathsf{c}) = h(\mathsf{c}) \otimes I_\omega$ is a single simple tensor for every c :

operator-Schmidt rank $= 1$ (rank $0 \iff \mathsf{c} = 0$) [exact Pauli-bilinear flattening; machine];

(2) promoted locus:

$$\mathrm{F_{ext}}(\mathsf{c}, q)$$

$$= h \otimes I_2 + q$$

$I_2 \otimes \hat{\kappa}$: the two terms are Hilbert–Schmidt orthogonal ($\mathfrak{T}\hat{\kappa} = 0$);

$$\text{Schmidt rank} = 2 \iff q \neq 0 \ \wedge \ (c_1, c_2, c_3) \neq 0;$$

rank 1 when $h \propto I_2$ (the terms share the internal factor and recombine: $\mathrm{F_{ext}} = I_2 \otimes (\lambda I_2 + q\hat{\kappa})$) or exactly one term survives;

$$\text{rank } 0 \iff \mathsf{c} = 0 = q;$$

no entangling term at any locus: $\mathrm{F_{ext}}$ is always a sum of

≤ 2 simple tensors with one-sided factors;

(3) the flipping criterion (iff):

$$F_{LR}$$

$\neq 0 \iff$ some Schmidt term has a chiral-off-diagonal ω-factor $\iff$ the $\hat{\kappa}$ term is present with q $\neq 0$

[I_2 is chiral-diagonal; $\hat{\kappa}$ is chiral-off-diagonal [FT.129(2)]];

(4) derived spectrum:

$$\text{spec} = \left\{\tfrac{1}{2}(c_0 \pm |\text{c}|)\right\},\ \text{each} \times 2;$$

$$\text{field } \mathbb{Q}(\text{c})\left[\sqrt{|\text{c}|^2}\right] :\ \text{quadratic extension;}$$

$$\text{charpoly exact over } \mathbb{Q}(\text{c})[x]\ [\text{FT.128}(5)];$$

(5) promoted spectrum:

the terms commute [FT.129(5)] :

$$\text{spec} = \left\{\tfrac{1}{2}\big(c_0 + s|\text{c}|\big) + tq :\ s,t \in \{\pm 1\}\right\},$$

$$\text{charpoly} = \big((y-u)^2 - q^2\big)\big((y+u)^2 - q^2\big),$$

$$y = x - \tfrac{c_0}{2}, \quad u^2 = \tfrac{|\text{c}|^2}{4}$$

[machine on all profiles $\times$ ternary q];

$$\text{the chiral-partner splitting} = 2|q| \in \{0, 2\}$$

[the ternary read [EM.9] : charged cores split the parity doublet by exactly two read units; neutral cores keep the doubling];

simultaneous eigenbasis

$=$ internal Bloch rays $\otimes$ orientation rays $\{|c\rangle, |\omega(c)\rangle\}$ ($\hat{\kappa}$-eigenrays, κ-eigenvalues ± 1);

$$|c\rangle = (|L\rangle + |R\rangle)/\sqrt{2}\ [\text{machine}]:$$

the stationary rays carry no definite chirality — chirality is the precessing quantity, and $2|q|$ is the static shadow of the parity-flipping frequency (the dynamical read is deferred, FT.131(3));

(6) collision census (exact):

$$q = 0:\ \text{parity doubling};$$

$$|\mathrm{c}| = 0:\ \text{spec} = \{\tfrac{c_0}{2} \pm q\},\ \text{each} \times 2;$$

$$\tfrac{|\mathrm{c}|}{2} = |q|:\ \text{the central eigenvalue } \tfrac{c_0}{2} \text{ is double};$$

no further degeneracies at ternary q;

(7) scalar-shift clause:

$$\mathrm{F_{ext}} \mapsto \mathrm{F_{ext}} + \lambda I_4 \iff h \mapsto h + \lambda I_2:$$

$$F_{LR},$$

Schmidt ranks, commutants and the splitting $2|q|$ invariant;

the spectrum translates by λ

[any scalar-class contribution — F_{D^+} if admitted, h_0-type backgrounds, m^2-type reads as identity multiples — is invisible to the flipping structure].

Req. FT.128, FT.129; Def-H.34; CT1.1, CT1.2 (B, the Pauli basis); EM.9 (Part 2); Hilbert–Schmidt and operator-Schmidt rank facts, exact flattening ranks over $\mathbb{Q}$, Faddeev–LeVerrier, Vieta reads [math; machine: `checks/verify_H_force_matrix.py` §D].

Pf.

$$(1):\ \mathrm{F_{ext}} = \sum_{ab} t_{ab}\,\sigma_a \otimes \sigma_b, \quad t_{ab} = \tfrac{1}{4}\mathfrak{T}\big(\mathrm{F_{ext}}(\sigma_a \otimes \sigma_b)\big):$$

$$t_{a0} = c_a/2,\ \text{all other } t_{ab} = 0\ [\text{Pauli orthogonality } \mathfrak{T}(\sigma_a\sigma_b) = 2\delta_{ab}]$$

$$\mathrm{rk}\,T = 1 \iff \mathrm{c} \neq 0\ [\text{exact rational flattening ranks at all sample profiles; machine}]$$

$$(2):\ t_{a0} = c_a/2, \quad t_{03} = q\ [\hat{\kappa} = \sigma_3 \text{ in the orientation basis}]:$$

$$\operatorname{rk} T = 2 \iff q \neq 0 \wedge (c_1, c_2, c_3) \neq 0;$$

$$h$$

$$= \lambda I_2 \text{ (i.e. } c_1 = c_2 = c_3 = 0):\ \mathrm{F}_{\text{ext}}$$

$$= I_2 \otimes (\lambda I_2 + q\sigma_3):\ \operatorname{rk} T$$

$$= 1 \text{ [machine, degenerate corners included]}$$

$$\mathfrak{T}\big((h \otimes I_2)^{\dagger}(qI_2 \otimes \hat{\kappa})\big)$$

$$= q$$

$$\mathfrak{T}(h)$$

$$\mathfrak{T}(\hat{\kappa})$$

$$= 0\ [\mathfrak{T}\hat{\kappa} = 0;\ \text{machine witness}]:\ \text{the two Schmidt terms are orthogonal}$$

$$(3):\ F_{LR} = qI_2\ [\text{FT.129(5)}];$$

chiral images: $I_2 \mapsto I_2$ (diagonal), $\hat{\kappa} \mapsto \sigma_1$ (off-diagonal) [Def-H.34] $\implies$ both directions exact

$$(4),$$

$(5):$ Faddeev–LeVerrier coefficient identity against $\big(x^2 - c_0 x + \frac{c_0^2 - |\mathrm{c}|^2}{4}\big)^2$ and $\big((y-u)^2 - q^2\big)\big((y+u)^2 - q^2\big)$ at all profiles × ternary q [machine]

$$\text{splitting: fixed } s:\ \big(\tfrac{1}{2}(c_0 + s|\mathrm{c}|) + q\big) - \big(\tfrac{1}{2}(c_0 + s|\mathrm{c}|) - q\big) = 2q;$$

$$q \in \{-1, 0, +1\} \implies 2|q| \in \{0, 2\}\ [\text{EM.9; exact}]$$

$$(6):\ \text{collision loci from the closed form: } q = 0;$$

$$|\mathsf{c}| = 0;$$

$u^2 = q^2$ [exhaustive over the closed form; machine census at rational $u \in \{\frac{1}{2}, 1, \frac{3}{2}\} \times$ ternary q]

(7) : $\lambda I_4 = (\lambda I_2) \otimes I_2$: $h + \lambda I_2$ re-absorbs the shift [machine];

$P_L^{\text{car}}(\lambda I_4)P_R^{\text{car}} = \lambda P_L P_R = 0$ [FT.122(2)]. ■

Phys. The static force matrix never entangles the internal and orientation books beyond a single product term, and the whole flipping phenomenon reduces to one ternary number: the core charge. Its level splitting $2|q|$ is the only frequency-shaped datum the static tier can produce — the eigenvalues of the promoted matrix encode, exactly, the splitting that a dynamical read would turn into the parity-flipping frequency. Scalar reads of every species shift the ladder and change nothing else: the flip structure is parity-graded, and parity is decided by ω-oddness alone.

Theorem 3.97 (FT.131 — Scope: bound chains, no isolated charge, the deferrals). Thm.

(1) bound-chain scope:

F_{ext} is computed for bound chains [D1.5];

the orientation doublet requires k

≥ 2 distinct edges as an ordered tuple [D1.4];

transit/trivial chains carry no bound orientation doublet [Def-H.33 scope; Def-H.38 scope note] : their flipping census is a separate computation, not performed here;

(2) no isolated charge:

a charge read requires two end registers [D3.4, EM.9];

a bound chain carries q_{core} [FT.40] but no transit charge;

F_{LR}

$= q_{\text{core}} I_2$ is the flipping structure of the bound chain,

not of an isolated carrier;

isolated colour is excluded in any case [CT3.22 : $\neg$pole, norm 0];

(3) the deferral and the CPT adjudication:

a frequency read requires a time parameter coupling F_{ext} to an evolution. The CPT-equivalent record was consulted: EM.24(5)'s exact phase identity

$$w(\mathrm{CPT}\, C) = (-1)^{|E(C)|} e^{i\pi\Theta(C)} w(C), \qquad |w|^2 \text{ unconditionally invariant},$$

relates reversal-paired configuration amplitudes — the configuration-level flipping locus (the pairing exists, its phase is exact) but supplies no carrier-time parameter;

the only derived clock is the ledger tick [CT0.3 : $(\Delta n_{\mathcal{R}}, \delta h, \delta\tau) = \pm(1,1,1)$ per Janus move],

whose rate has no field coupling [FT.86(8)];

proper-time vocabulary exists [Def-H.19 stationary jumps; Morse–Bott refinement Def-H.24] with no materialization coupling [Def-H.19];

hence: emergent time is a prerequisite; nothing in the CPT record discharges it; the frequency, $e^{-i\mathrm{F}t}$,

and the time-resolved chiral coupling are deferred — the splitting $2|q|$ [FT.130(5)] is the complete static deliverable;

(4) bars:

no particle identification (the matrices are for an arbitrary bound chain with arbitrary profile and registration data);

no external data — the marker [EXT] appears in this clause alone; no external value enters any clause of this document;

no Yukawa/mass mechanism (mass-type scalars enter only through FT.130(7)'s identity-shift clause, never as L–R couplings);

no binding energy;

no fixed topology or privileged skeleton (c, q, k, Θ are chain and registration data — the matrices are skeleton-independent);

single-aspect intermediates are marked scope restrictions;

(5) the standing superselection tension (scope restatement):

the chiral rays superpose ω-opposed core charges $(q, -q)$ [FT.40];

channel reads are sign-blind and fuse only equal-q states [FT.15; FT.90(5)];

Def-H.43's promotion is carrier-level; its relation to channel-level fusion is open bridge content — no theorem here resolves it.

Req. D1.4, D1.5, D3.4 (Parts 1–2); EM.9, EM.24 (Part 2); CT0.3 (Part 1); CT3.22 (Part 2); FT.15 (Tiers F.1–F.11); FT.40 (Tiers F.12–F.13); Def-H.19 (Tiers F.12–F.13); Def-H.24 (Tiers F.20–F.22); FT.86, FT.90 (Tiers F.29–F.30); Def-H.19, Def-H.38, Def-H.43; exact bookkeeping; machine: `checks/verify_H_force_matrix.py` §E (citation pins).

Pf.

(1) : D1.5 : bound $=$ closed self-contained;

D1.4 : $c = (e_1, \ldots, e_k)$, e_i distinct; $k \geq 2$ [A1 : ¬self-loops];

Def-H.33 constructs $\mathcal{H}_\omega(c)$ for bound c

(2) : D3.4 : $q_{\text{EM}} = \frac{1}{2}(\sigma_s - \sigma_t)$ on end registers;

FT.40 : $q_{\text{core}}(c; v) = \frac{1}{2}(\sigma_{e_1} - \sigma_{e_k})$ on the closed core;

CT3.22 : $\neg\exists$ isolated triplet pole

(3) : EM.24(5) verbatim (frozen);

CT0.3 verbatim;

FT.86(8) : $\neg\exists$ tag coupling $\delta\tau$ to Ω/N_i;

Def-H.19 : Def-H.19 vocabulary asserts no coupling to materialization

$(4), (5)$: citation conjunctions [FT.15 hypothesis verbatim; FT.90(5) audit verbatim]. ■

Phys. The tier knows exactly where it stops. The matrix is static because the corpus has no time to evolve it with — and the consultation of the CPT record confirms that the configuration-level pairing, though exact, cannot supply one: reversal phases are amplitudes, not clocks. What survives as the complete static deliverable is sharp: bound chains only; charge only as the core read; flipping only from ω-odd data; frequency only after time.

Rmk (forward pointer, non-load-bearing). The parity structures invoked by this part's census rows receive their systematic treatment in Tier F.41: the configuration-space reversal calculus with the exact commutations and both frozen CPT lifts (FT.133); the carrier bookkeeping — C, P, T fix the doublet rays and CPT acts as the scalar character $\lambda = (-1)^{|E|}e^{i\pi\Theta}$ (FT.134), so this part's splitting $2|q|$ is parity-even and charge-odd (FT.139(3)); the branch, cover, flicker, pole and channel censuses (FT.135–FT.139). The carrier-level parity-eigenvalue assignment to species is application-layer content.

3.42 Tier F.41 — Parity: The Configuration Calculus, the Carrier Bookkeeping, and the Sector Census

Theorem 3.98 (FT.133 — The configuration-space reversal calculus). Thm. On the configuration space $\mathfrak{C}(\mathcal{R})$ [D0.10], with configuration data $(R,\ \sigma,\ \alpha,\ \kappa/\text{ledger})$ [A4, A8, D0.5, D3.5, Def-EM.20]:

(1) the four register involutions [each frozen, re-derived at the point of use] :

ω-pattern : $R \mapsto R^{\mathrm{T}} = \{(b,a) : (a,b) \in R\}$ with edge data carried [D2.6; FT.88(1)];

C : $\sigma_e \mapsto -\sigma_e$ [EM.24(1)];

P : $\alpha_e \mapsto \pi - \alpha_e$ $(\mathcal{P} \mapsto E_{\mathcal{R}} \setminus \mathcal{P})$ [EM.24(2), Def-EM.21];

$T = \Theta_{\mathfrak{L}}$: ledger order reversed, $J^+ \leftrightarrow J^-$, $\kappa_e \mapsto -\kappa_e$, σ fixed [D2.7; FT.88(2)];

each is an involution on configuration data [machine: exhaustive on $17^2 + 5^6$ configurations $+$ 3000 deterministic off-shell samples];

(2) register-disjoint commutation: $[C,P] = [C,T] = [P,T] = [C,\omega] = [P,\omega] = [T,\omega] = 0$

[the four act on pairwise-disjoint data coordinates: edge-direction set; σ; α; (κ, ledger)] $\Longrightarrow$ $\langle C,$

$$P,$$

$$T\rangle \cong \mathbb{Z}_2^3,$$

$$\langle \omega\text{-pattern},\ C,\ P,\ T\rangle \cong \mathbb{Z}_2^4;$$

FT.88(3)'s $\langle \omega,$

$\Theta_{\mathfrak{L}}\rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ is the frozen two-generator subalgebra [machine: exhaustive commutation census];

(3) component weight identities ($w = \mathcal{J}\cdot S\cdot\Gamma\cdot A,\ S=\prod\sigma_e,\ \Gamma=\prod e^{i\alpha_e \widehat{ds}^{\,2}(e)},\ A=(-1)^{\aleph}$ [A8, D3.5, D3.6]) :

$$w(C\mathcal{C}) = (-1)^{|E|}\, w(\mathcal{C})\ [\aleph \text{ invariant : } \aleph_v = |f_v|,\ f_v \mapsto -f_v];$$

$$w(T\mathcal{C}) = w(\mathcal{C})^*\ [\text{CT2.8(1)}];$$

$$w(P\mathcal{C}) = e^{i\pi\Theta}\, w(\mathcal{C})^*;$$

$$\Theta :$$

$$=\sum_{e\in R} \widehat{ds}^{\,2}(e)\ [\text{EM.24(5)}] : \quad \Theta \text{ is } P\text{-even},$$

$$T\text{-odd } (\Theta \mapsto -\Theta),$$

$$\omega\text{-even (an edge-set function; } \widehat{ds}^{\,2} \text{ direction-symmetric});$$

(4) the composition, in both frozen lifts: $\quad \text{CPT} := C\circ P\circ T\ [\text{EM.24}(4)];$

conjugation lift [EM.24(5) Pf : T as $\Gamma \mapsto \bar{\Gamma}$] : $\quad w(\text{CPT}\,\mathcal{C}) = (-1)^{|E|}\, e^{+i\pi\Theta}\, w(\mathcal{C});$

κ-flip lift [FT.88(2) register picture : $T : \kappa \mapsto -\kappa$] : $\quad w(\text{CPT}\,\mathcal{C}) = (-1)^{|E|}\, e^{-i\pi\Theta}\, w(\mathcal{C});$

the two lifts agree exactly on the integer-Θ locus ($e^{2i\pi\Theta}=1$),

which contains every materialized closed chain [FT.22 : $\Theta \in 2\mathbb{Z}$];

$\left|w(\mathrm{CPT}\,\mathcal{C})\right|^2 = \left|w(\mathcal{C})\right|^2$ unconditionally in both lifts [machine];

(5) on-shell reductions:

$\Theta \in 2\mathbb{Z} \implies e^{i\pi\Theta} = 1$: the CPT character reduces to $(-1)^{|E|}$;

$|E| + \Theta \equiv 0 \pmod 2 \implies w(\mathrm{CPT}\,\mathcal{C}) = w(\mathcal{C}) = w(\mathcal{C})^*$ [EM.24(5) third line, re-derived];

(6) the read-covariance table [exact; machine on closed digon and triangle chains] :

read	C	P	T	ω
q_{core} [FT.40]	$-$	$+$	$+$	$-$
$\Theta = \sum \widehat{ds}^{\,2}$	$+$	$+$	$-$	$+$
$\lvert E\rvert,\ K = \lvert R\rvert$	$+$	$+$	$+$	$+$
$\aleph$ [EM.24(1)]	$+$	$+$	$+$	$+$
$\mathcal{P}$ [Def-EM.21]	$+$	complement	$+$	transported

(7) the parity naming (derived record): the parity operation on configuration space is EM.24(2)'s P — derived from the dynamic phase [Def-EM.20, EM.20(1)] and named inside the derived CPT bridge;

FT.101(3)'s branch flip $\iota(\alpha)$

$= \pi - \alpha$ is $P|_\Phi$ on the A4 label alphabet [FT.101(3) cites EM.24(2), Def-EM.21] : the identification is derived content and no declaration issues.

Req. A4, A8, D0.5, D0.10, D1.6, D2.6, D2.7, D3.3, D3.4, D3.5, D3.6 (Parts 1–2); CT2.8 (Part 2); Def-EM.20, Def-EM.21, EM.9, EM.9a, EM.20, EM.23, EM.24 (Part 2); FT.22 (Tiers F.1–F.11); FT.40 (Tiers F.12–F.13); FT.88 (Tiers F.29–F.30); FT.101 (Tiers F.31–F.34); exhaustive enumeration, exact $\mathbb{Q}(i)$ weight arithmetic [math; machine: `checks/verify_H_parity_identification.py` §A].

Pf.

(1) : each map is an involution on its register : $(-\sigma)$

$= \sigma$;

$\pi - (\pi - \alpha)$

$$= \alpha;$$

$$(-\kappa) \mapsto \kappa;$$

$$(R^{\mathrm{T}})^{\mathrm{T}}$$

$$= R;$$

data transport is a bijection [machine, exhaustive]

$(2):$ disjoint coordinates $\implies$ maps commute pointwise [machine on all enumerated configurations];

group generation : $\mathbb{Z}_2^3,\ \mathbb{Z}_2^4$ [distinct nonidentity elements counted]

$(3):\ C:\ S \mapsto (-1)^{|E|}S,\ \Gamma, \mathcal{J}, A$ fixed $[\aleph_v = |f_v|,\ f_v \mapsto -f_v$ [D3.3, EM.24(1)]];

$T:\ \Gamma \mapsto \bar{\Gamma}$ $[\kappa \mapsto -\kappa:\ e^{i\alpha(-\kappa)\pi\cdots} = \overline{e^{i\alpha\kappa\cdots}};\ \mathcal{J} \mapsto \mathcal{J}^*$ [D2.7, D3.6]; $S, A \in \{\pm 1\}]$;

$$P:\ e^{i(\pi-\alpha_e)\widehat{ds}^{\,2}(e)} = e^{i\pi\widehat{ds}^{\,2}(e)}\, e^{-i\alpha_e\widehat{ds}^{\,2}(e)} \implies \Gamma \mapsto e^{i\pi\Theta}\bar{\Gamma}$$

$(4):$ composition of (3) in each lift;

$$e^{+i\pi\Theta} = e^{-i\pi\Theta} \iff e^{2i\pi\Theta} = 1 \iff \Theta \in \mathbb{Z};$$

$\Theta \in 2\mathbb{Z}$ on materialized closed chains [FT.22]

$$|(-1)^{|E|}e^{\pm i\pi\Theta}|$$

$$= 1$$

$$\implies |w(\mathrm{CPT}\,\mathcal{C})|^2$$

$= |w(\mathcal{C})|^2$ [machine, both lifts, all enumerated configurations]

$$(5):\ \Theta \in 2\mathbb{Z}:\ e^{i\pi\Theta} = (-1)^{\Theta} = 1;$$

$$|E| + \Theta \equiv 0\ (2):\ (-1)^{|E|} e^{i\pi\Theta} = (-1)^{|E|+\Theta} = 1;$$

on-shell $w \in \mathbb{R}$ [EM.24(5) Pf re-run]

$$(6):\ q_{\text{core}} = \tfrac{1}{2}(\sigma_{e_1} - \sigma_{e_k})\ [\text{FT.40}]:\ C \text{ flips both } \sigma;\ P, T \text{ fix } \sigma;\ \omega \text{ swaps } e_1 \leftrightarrow e_k;$$

$$\Theta, |E|:\ \text{edge-set functions};\ \kappa \mapsto -\kappa \text{ under } T\ [\text{FT.88}(2)]\ \blacksquare$$

Phys. Parity in this corpus is a register operation, not a postulate: it complements the alignment pattern, and it has done so since the dynamic phase was derived. The four reversals live on four disjoint shelves of the configuration data, so they commute by arithmetic — the elementary abelian group is a census, not an assumption. The CPT identity is the corpus's own CPT theorem; this tier pins its two frozen lifts and their exact agreement locus, which is precisely the locus where materialized chains live. The covariance table is the working ledger of the part: every later clause reads its parities off this table.

Theorem 3.99 (FT.134 — The carrier bookkeeping: the scalar-CPT theorem). Thm. On the orientation doublet $\mathcal{H}_\omega$ [Def-H.33] and the extended carrier [Def-H.34]:

(1) the ray census: ω-pattern is the only involution of the FT.133 family that moves the doublet basis $(\hat{\omega}|c\rangle = |\omega(c)\rangle$ [FT.122(4)]);

$$C,$$

$$P,$$

T fix the chain tuple pointwise [FT.133(1) registers]: their doublet action is the identity on rays, with a label map;

hence no operator identification $\hat{\omega}$

$$= C \text{ or } \hat{\kappa}$$

$= P$ exists: the two sides of each proposed equality act on different registers [machine witnesses];

(2) the read-level covariance (the content the identifications reach): under Def-H.43's promotion $\hat{Q}_{\text{EM}}$

$$= q_{\text{core}}\hat{\kappa}\ [\text{FT.129}(4)]:\ \text{the family } \{\hat{Q}_q = q\hat{\kappa}\} \text{ is } C\text{-odd } (\hat{Q}_{-q} = -\hat{Q}_q,\ \text{matching } q \mapsto -q),$$

P-even and T-even (q_{core} is P/T-even [FT.133(6)]),

and ω-covariant ($\hat{\kappa}\,\hat{\omega}$-odd $\wedge$ $q\,\omega$-odd : the signs cancel);

consequently Tier F.40's flipping block $F_{LR} = q\,I_2$ is C-odd, P-even, T-even [FT.129(5)];

(3) the scalar-CPT theorem: the CPT character λ

$= (-1)^{|E|}e^{i\pi\Theta}$ takes the same value on both doublet rays ($|E|, \Theta$ are edge-set functions [FT.133(6)]),

$$|\lambda| = 1 :$$

$$\text{CPT}\big|_{\mathcal{H}_\omega} = \lambda\,\text{id}, \qquad \lambda = (-1)^{|E|}e^{i\pi\Theta};$$

chirality is CPT-even: the $\hat{\chi}$-eigenspaces are CPT-stable; $\hat{\kappa}$ and $\hat{\omega}$ likewise;

refutation witness: $\text{tr}(\lambda\,\text{id}) = 2\lambda \neq 0 = \text{tr}(\pm\hat{\kappa})$: no $\hat{\kappa}$-multiple equals the CPT action;

(4) the uniqueness pair [re-derived at the point of use] : $\pm\hat{\chi}$ are the unique diagonal-free $\hat{\omega}$-odd Hermitian involutions [FT.122(6)];

$\pm\hat{\kappa}$ the unique diagonal ones [FT.129(1)];

quaternionic triple [Def-H.34] : $\hat{\omega}\hat{\chi}$

$$= -i\hat{\kappa},$$

$$\hat{\chi}\hat{\kappa}$$

$$= -i\hat{\omega},$$

$$\hat{\kappa}\hat{\omega} = -i\hat{\chi};$$

(5) independence: any internal involution $G \otimes I_2$ commutes with $I_2 \otimes \hat{\omega}$ and their product is a third distinct involution: the configuration-side and carrier-side $\mathbb{Z}_2$'s generate $\mathbb{Z}_2 \times \mathbb{Z}_2$ [FT.126(4) pattern; FT.88(3) precedent; machine].

Req. Def-H.33, FT.122, FT.126 (Tier F.39); Def-H.34, FT.128, FT.129 (Tier F.40); FT.133; FT.40 (Tiers F.12–F.13); Def-H.34, Def-H.43; exact $\mathbb{Q}(i)$ arithmetic, trace witnesses, rational canonicity census [math; machine: `checks/verify_H_parity_identification.py` §B].

Pf.

$$(1):\ \text{register census of FT.133(1): } C,$$

$$P,$$

$$T \text{ act on } (\sigma),$$

$$(\alpha),$$

(κ, ledger) and leave the edge tuple fixed [machine: key sets preserved on all enumerated configurations];

$$\omega\text{-pattern transposes it [machine]}$$

$$\hat{\omega} \neq \pm\text{id [entrywise]};$$

an operator equality between a ray-swapping and a ray-fixing map is impossible

$$(2):\ \hat{Q}_{-q} = -\hat{Q}_q \text{ [linearity]};$$

$$q_{\text{core}}(P\mathcal{C}) = q_{\text{core}}(T\mathcal{C}) = q_{\text{core}}(\mathcal{C}), \quad q_{\text{core}}(\omega\mathcal{C}) = -q_{\text{core}}(\mathcal{C}) \text{ [FT.133(6)]};$$

$$\hat{\omega}\hat{\kappa}\hat{\omega} = -\hat{\kappa} \text{ [Def-H.34]}$$

$$\implies \text{the coupled operator } q\hat{\kappa} \text{ is } \omega\text{-invariant (both signs cancel) and } C\text{-odd};$$

$$F_{LR} = qI_2 \text{ inherits the characters [FT.129(5)]}$$

$$(3):\ \lambda(\text{ray } c)$$

$$= (-1)^{|E(c)|} e^{i\pi\Theta(c)}$$

$$= (-1)^{|E(\omega c)|} e^{i\pi\Theta(\omega c)}$$

$= \lambda(\text{ray}\,\omega(c))$ [FT.133(6) : both reads edge-set functions; machine on closed digon/triangle chains]

CPT fixes each ray and multiplies the weight by λ [EM.24(5); FT.133(4)] $\implies \text{CPT}\big|_{\mathcal{H}_\omega} = \lambda\,\text{id}$;

$$|\lambda|^2 = 1 \text{ [machine]}$$

$$[\lambda\,\text{id},\ \hat{\chi}] = [\lambda\,\text{id},\ \hat{\kappa}] = [\lambda\,\text{id},\ \hat{\omega}] = 0; \qquad \text{tr}(\lambda\,\text{id}) = 2\lambda \neq 0 = \text{tr}(\hat{\kappa}) = \text{tr}(-\hat{\kappa})\ [\lambda \neq 0]$$

(4) : rational census: diagonal $\hat{\omega}$-odd involutions $\{a, -a\}$: a^2

$$= 1$$

$$\implies a$$

$$= \pm 1;$$

diagonal-free $\{iy, -iy\}$: y^2

$$= 1$$

$$\implies y$$

$$= \pm 1 \text{ [machine]};$$

triple products entrywise [machine]

(5) : $[G \otimes I,\ I \otimes \hat{\omega}] = 0$ [Kronecker disjointness];

$$(G \otimes \hat{\omega})^2 = G^2 \otimes \hat{\omega}^2 = \text{id};$$

$G \otimes \hat{\omega} \neq G \otimes I \neq I \otimes \hat{\omega}$ [entrywise witnesses; machine]. ■

Phys. The carrier keeps its own books. Charge conjugation, parity and time reversal act on the registration shelves; only the traversal reversal moves the rays of the doublet. The composition CPT — the operation the configuration calculus proves exact — reaches the doublet as a pure phase: the quietest possible action, leaving chirality, the orientation sign and the exchange grade

all untouched. What the proposed operator identifications legitimately reach is the read level, and there the census is sharp: the promoted flip block is charge-odd and parity-even, the only parity-sensitive structure in the fermion tower.

Theorem 3.100 (FT.135 — The weak branch grading under parity). Thm.

$$(1) \quad \iota(0) = \pi,\ \iota(\pi) = 0 \text{ on the branch alphabet } \Phi = \{0, \pi\} \text{ [FT.101(3); D0.5]};$$

$$\iota$$

$$= P|_\Phi \text{ [EM.24(2); FT.101(3) citation chain]} : \text{ parity exchanges the two crystallization branches at the label level};$$

$$(2) \quad \text{aligned branch } (\alpha = 0,\ \vee^2,\ \text{Casimir } 2) :\ W_i$$

$$= \Pi_{\mathcal{R}} W_i \Pi_{\mathcal{R}}$$

$$\neq 0,$$

$$\sum_i W_i^2$$

$$= 2\Pi_{\mathcal{R}} \text{ [FT.101(1), Cor CT0.7a re-derived]};$$

$$\text{antialigned branch } (\alpha = \pi,\ \wedge^2,\ \text{Casimir } 0) :\ W_i \Pi_{\mathcal{V}}$$

$$= 0,$$

$$\bigcap_i \ker W_i$$

$$= \operatorname{im} \Pi_{\mathcal{V}} \text{ [FT.101(2), FT.93(5) re-derived]};$$

$$(3)$$

P maps the weak-active branch label to the weak-blind one: the weak support is not flip-invariant [FT.101(4)];

structural parity non-invariance of the force arena [FT.101(5)];

scope: arena S-parity grading,

$$\neg \text{ matter chirality } [\text{FT.93}(3),\ \text{FT.93}(7)];$$

$$(4) \quad \text{the exchange is label-level only:}$$

$$\mathfrak{T}\Pi_{\mathcal{R}} = 3 \neq 1 = \mathfrak{T}\Pi_{\mathcal{V}};$$

$$M = a\Pi_{\mathcal{R}} + b\Pi_{\mathcal{V}} :\ M^2 = \mathrm{id} \implies a, b \in \{\pm 1\} \implies \mathrm{tr}\, M = 3a + b \in \{\pm 4, \pm 2\} \neq 0 :$$

no traceless involution implements the branch exchange on the pair register [FT.101(3), FT.93(3a) re-derived];

$$\mathbb{S}\Pi_{\mathcal{R}} = +\Pi_{\mathcal{R}},\ \mathbb{S}\Pi_{\mathcal{V}} = -\Pi_{\mathcal{V}} \text{ [the } S\text{-eigenvalue labels]}.$$

Req. A3, A4 (Parts 1–2); CT0.7, Cor CT0.7a, CT0.8, CT1.2 (Part 1); Def-EM.21, EM.24 (Part 2); FT.93, FT.101 (Tiers F.31–F.34); exact $\mathbb{Q}(i)$ matrix arithmetic, trace/rank invariants, exact nullspace [math; machine: `checks/verify_H_parity_identification.py` §C].

Pf.

$$(1) :\ \iota(\alpha) = \pi - \alpha \text{ [FT.101(3) : the unique nontrivial automorphism of } \Phi];$$

$$P :\ \alpha_e \mapsto \pi - \alpha_e \text{ [EM.24(2)] : } \iota = P|_{\Phi} \text{ [same map on the same alphabet]}$$

$$(2) :\ \Pi_{\mathcal{R}}$$

$$= \tfrac{1}{2}(E_0 + \mathbb{S})$$

$$= \tfrac{1}{4}(3E_0 + \Sigma),$$

$$\Pi_{\mathcal{V}}$$

$$= \tfrac{1}{4}(E_0 - \Sigma) \text{ [CT0.7(1)] : idempotent, orthogonal, complete, traces } (3, 1) \text{ [machine]}$$

$$W_i\Pi_{\mathcal{V}} = 0\ \forall i;$$

$$W_i$$

$$= \Pi_{\mathcal{R}} W_i \Pi_{\mathcal{R}};$$

$$\sum_i W_i^2$$

$$= \tfrac{3}{2}E_0 + \tfrac{1}{2}\Sigma$$

$$= 2\Pi_{\mathcal{R}};$$

$$\dim\ker(W\text{-stack})$$

$$= 1 \text{ (over } \mathbb{C})$$

$$= \operatorname{im}\Pi_{\mathcal{V}} \text{ [exact nullspace; machine]}$$

$$(3):\ \text{FT.101}(4)\text{–}(5) \text{ verbatim (frozen)};$$

the ¬-scope is FT.93's pincer and relocation [FT.125(1) status stands: the carrier chirality of Tier F.39 is a distinct structure]

$$(4):\ \text{sign enumeration over } \{a,b\} \subset \{\pm 1\}^2:\ \text{tr} \in \{4,2,-2,-4\}\ \text{[machine]}.\ \blacksquare$$

Phys. Parity has a derived, exact and destructive action on the weak force: it swaps the branch the weak sees for the branch it annihilates. The non-invariance is structural — a $\mathbb{Z}_2$-asymmetry of the force on the registration alphabet — and it has no operator on the pair register, by trace and rank. What the force arena loses under parity, the carrier never had: the doublet's chirality grade commutes with the weak action entirely [FT.123(1)], and the two parity stories meet only in the assignment rule that does not yet exist.

Theorem 3.101 (FT.136 — The rung-one S-grading: the 10/6 split). Thm.

$$(1)\quad \text{the } \operatorname{Ad}(\mathbb{S}) \text{ census on the CT1.2 generators [machine: all 17]}:$$

$$S\text{-even } (+1):\ \{E_0,\ \Sigma,\ \hat{E}^{ij}\ (5 \text{ dims}),\ W_i\ (3)\} = 10 \text{ dims} = \underbrace{G_5 + W_3 + H_1}_{\text{forces}} + \underbrace{D_1^+}_{\text{central}};$$

$$S\text{-odd } (-1):\ \{J_i\ (3),\ K_i\ (3)\} = 6 \text{ dims} = J_3^+ + J_3^-$$

$$[\text{CT1.2 } (S,T) \text{ table; CT0.12; FT.30 tiling } (5{+}3{+}1) + (3{+}3) + 1];$$

$$(2)\quad \mathcal{A}(1)$$

$$= \{J,\ G,\ W,\ H,\ D^{+}\}\ \text{[CT0.13]}:\ \text{the rung-one active set is the } 10/6 \text{ split;}$$

the metric dictionary [FT.30] : the S-even block carries every metric coordinate $(\hat{h}_{ij}\ 5,\ \Sigma_{\text{tr}}\ 1,\ a_{00}\ 1,\ w_i\ 3)$;

the S-odd Janus seals carry none;

(3) register separation: P (α-complement) acts on configuration data and has no action on the pair algebra [FT.133(2)];

the only derived bridge between parity and the S-grading is FT.101's label $\leftrightarrow$ arena coupling (branch $\alpha \leftrightarrow \vee^2/\wedge^2$ weak support) : a naming of the S-blocks as P-sectors has no operator content — the two $\mathbb{Z}_2$'s act on different spaces;

(4) the independence census: three pairwise-distinct $\mathbb{Z}_2$ gradings stand in the record: the algebra S-grading $(\mathrm{Ad}\,\mathbb{S})$,

the branch-label parity $(P/\iota$ [FT.101]),

the carrier chirality ($\hat{\chi}$ [FT.122]) — pairwise commuting where jointly defined, none reducible to another [FT.126(4); machine] :

the $10/6$ asymmetry is the force/dynamics split of rung one; it is not the source of the carrier chirality grade;

(5) the weak location: W is S-even [clause (1)];

its chiral selectivity is the arena $3 \oplus 1$ grading [FT.93(5) : $W_i \Pi_{\mathcal{V}} = 0,\ \sum W_i^2 = 2\Pi_{\mathcal{R}}$] — a third structure, distinct from both the S-grading of the operator algebra and the $\hat{\chi}$-grading of the carrier.

Req. CT0.7, CT0.12, CT0.13, CT1.1, CT1.2 (Part 1); FT.30 (Tiers F.1–F.11); FT.93, FT.101 (Tiers F.31–F.34); FT.122, FT.126 (Tier F.39); FT.133 (this tier); exact $\mathrm{Ad}(\mathbb{S})$ eigen-census over $\mathbb{Q}(i)$ [math; machine: `checks/verify_H_parity_identification.py` §D].

Pf.

$$(1):\ \mathbb{S} = \tfrac{1}{2}(E_0 + \Sigma),\ \mathbb{S}^2 = E_0\ [\text{CT0.7(1)};\ \Sigma^2 = 3E_0 - 2\Sigma];$$

$$\mathrm{Ad}(\mathbb{S})X = \mathbb{S}X\mathbb{S} = \pm X \text{ on each generator [machine: all 17, entrywise]}$$

$$\text{membership: } \mathbb{S}E_0\mathbb{S}$$

$$= E_0,$$

$$\mathbb{S}\Sigma\mathbb{S}$$

$$= \Sigma \text{ [CT1.3, CT1.5]};$$

$$\mathbb{S}\hat{E}^{ij}\mathbb{S}$$

$$= \hat{E}^{ij} \text{ [CT1.2 table]};$$

$$\mathbb{S}W_i\mathbb{S}$$

$$= W_i;$$

$$\mathbb{S}J_i\mathbb{S}$$

$$= -J_i,$$

$$\mathbb{S}K_i\mathbb{S}$$

$$= -K_i \text{ [CT0.7(3)]}$$

$$\text{dimension count : } 1+1+5+3=10; \quad 3+3=6; \qquad 10+6=16=C_{\text{cap}} \text{ [CT0.12; CT0.6]}$$

$$(2): \text{ CT0.13 verbatim (frozen)}; \qquad \text{FT.30 dictionary rows verbatim}$$

$$(3): \text{ FT.133(1)–(2) : } P \text{ acts on } \alpha;$$

$$\text{the pair algebra is spanned by } \{E_0, \Sigma, \hat{E}^{ij}, W_i, J_i, K_i\} \text{ with no } \alpha \text{ argument;}$$

$$\text{FT.101(1)–(2) : the label-arena coupling}$$

$$(4): \ [G\otimes I,\ I\otimes\hat{\omega}]=0,\ (G\otimes\hat{\omega})^2=\text{id},\ G\otimes\hat{\omega}\neq G\otimes I\neq I\otimes\hat{\omega} \text{ [FT.134(5); machine]};$$

$$\text{Ad}(\mathbb{S}) \text{ acts on } \text{End}(\mathbb{C}^4_{\text{pair}}),\ \iota \text{ on labels},\ \hat{\chi} \text{ on } \mathcal{H}_\omega : \text{ three distinct spaces}$$

$$(5): \ W_i \in \mathcal{E}_{+-} \text{ [CT1.2 table]}; \qquad \text{FT.93(5) data re-derived in §C. } \blacksquare$$

Phys. At the first rung the algebra already divides: ten dimensions that speak to the metric — shear, volume, conformal, shift — against six that only move the ledger. That is the rung-one asymmetry, and it is exact. Parity names the branch labels that select the weak support inside the arena; the S-grading sorts the operators; the carrier chirality grades the chains. The three $\mathbb{Z}_2$'s commute and never collapse into one another — which is why no rung-one count generates a family, and why the family channels of Tiers F.37–F.38 run on their own integers.

Theorem 3.102 (FT.137 — The cover and flicker census under the reversal family). Thm.

(1) pattern reversal (ω-side [FT.88(1) : ω acts on skeletons as $R \mapsto R^{\mathrm{T}}$]) : preserves admissibility,

$$K$$

$$= |R|,$$

cover status and the amplitude : $v(R^{\mathrm{T}})$

$$= v(R) \text{ on all } 4096 \text{ tetrahedron patterns [FT.95(3) re-derived; machine];}$$

$$101 \text{ covers reproduced;}$$

(2) the chiral cover pair: the 48 dissolved-digon bypass covers [FT.69(4)] split into two Ω

$$= 24 \text{ Aut-orbits exchanged by reversal;}$$

no orbit is reversal-invariant ($R \mapsto R^{\mathrm{T}} \notin \mathrm{Aut}(\Gamma)$) [FT.78(6) re-derived; machine];

(3) P (α-complement) on the cover sector: the exact sign character

$$v_{\mathrm{a}}(R) = (-1)^{\sum_{e\in R} a_e}\, v_0(R)$$

[machine: exhaustive triangle census over all (R, a, σ)];

hence cover status ($v \neq 0$), $|v|$, K, $4^{|\mathcal{C}|}$, and the norm layer $2^{|R|}$ are P-invariant;

weights: $w(P\mathcal{C})$

$$= e^{i\pi\Theta} w(\mathcal{C})^* \text{ [FT.133(3)] : } |w|^2 \text{ invariant;}$$

$$\text{materialized even-}\Theta\text{ locus : } w(P\mathcal{C})$$

$$= w(\mathcal{C}) \text{ exactly;}$$

(4) the flicker reads: the bypass/lock decomposition [Def-H.25, FT.68],

$$B_\Gamma(p) \text{ and the phase boundary } p^*_\Gamma \text{ [FT.70(1)]},$$

$$f_{\text{adm}}(n,p) \text{ [FT.70(4)]},$$

$$\text{the dust density } \langle \rho_{D^-} \rangle_\Gamma(p)$$

$$= \sum_v O(n_v)(1 - f_{\text{adm}}(n_v, p)),$$

$$O(n)$$

$$= 2^{n+1} - 16 \text{ [FT.71(1), CT4.1]}$$

are functions of (Γ, R, p, n_v) alone — α-blind and σ-blind by definitional census : P-,

C-,

T- and ω-invariant;

the only P-sensitive quantity in the sector is clause (3)'s weight character;

(5) the attribution census: the cover sector's nontrivial chirality — clause (2)'s exchanged pair — belongs to the pattern reversal [FT.78(6) : "the pattern-level form of charge conjugation"; FT.88(1) : ω acts as $R \mapsto R^{\mathrm{T}}$];

P acts on the census only through the sign character of clause (3) :

reversal and parity are distinct involutions with distinct footprints [FT.89(2) discipline].

Req. FT.67, FT.68, FT.69, FT.70, FT.71, Def-H.25, Def-H.26 (Tiers F.23–F.25); FT.78 (Tiers F.26–F.28); FT.88 (Tiers F.29–F.30); FT.95 (Tiers F.31–F.34); CT4.1 (Part 2); FT.133 (this tier); exhaustive pattern enumeration, exact brute-force sign sums, Aut-orbit machinery [math; machine: `checks/verify_H_parity_identification.py` §E].

Pf.

$$(1):\ v(R)$$

$$=\sum_{\sigma} w(R,\sigma) \text{ by brute-force sign sum [FT.67(2) classifier transport]}:\ v(R^{\mathrm{T}})$$

$$= v(R) \text{ on all } 4096 \text{ [machine]};$$

$$\text{types transpose}:\ (n_{\text{in}}, n_{\text{out}}) \mapsto (n_{\text{out}}, n_{\text{in}}):\ \text{admissibility } k_v \le 3 \text{ invariant}$$

$$\ell \mapsto |\ell| \text{ at type-}(1,1) \text{ vertices under } \sigma\text{-transport}:\ \aleph \text{ invariant};$$

$$|R^{\mathrm{T}}| = |R|;$$

$$\text{cover status} = (v \neq 0) \text{ invariant};$$

$$101 \text{ covers [machine census]}$$

$$(2):\ |\,\mathrm{Aut}\,| = 24 \text{ [machine]};$$

$$\text{orbit census of the } 48:\ \text{two orbits, sizes } (24,24);$$

$$\text{reversal maps each orbit bijectively onto the other};$$

$$\text{no fixed orbit [machine]}$$

$$(3):\ w(R,\sigma,\mathrm{a}) = \prod_{e\in R} \sigma_e (-1)^{a_e} \cdot (-1)^{\aleph(\sigma)} \text{ [A8; } \kappa \equiv 1 \text{ class]}:$$

$$\sum_{\sigma} w(R,\sigma,\mathrm{a}) = (-1)^{\sum a_e} \sum_{\sigma} w(R,\sigma,0) \text{ [factorization; machine exhaustive]}$$

$$\text{status}/|v|/K/4^{|\mathcal{C}|}/2^{|R|}:\ \text{functions of } (R,\ |v|) \text{ only};$$

$$w(P\mathcal{C}) = e^{i\pi\Theta} w(\mathcal{C})^* \text{ [FT.133(3)]};$$

$$\Theta \in 2\mathbb{Z} \text{ and real weights : } w(P\mathcal{C}) = w(\mathcal{C})$$

(4) : definitional census: each formula's argument list [FT.68(1)–(5), FT.70(1), (4), FT.71(1), CT4.1] contains no α and no σ;

$$(5) : \text{ FT.78(6), FT.88(1) verbatim (frozen). } \blacksquare$$

Phys. The cover world is chirally paired — but by reversal, not by parity. The two Ω=24 orbits are each other's mirror under the transposition of every arrow, and no automorphism of the skeleton can tell them apart from within: the cover sector carries a genuine $\mathbb{Z}_2$ chirality, and it is ω-side. Parity's entire footprint on this sector is one exact sign per configuration — invisible to every status, count, boundary and density the flicker calculus reads.

Theorem 3.103 (FT.138 — The pole spectrum census). Thm.

$$(1) \quad \text{the defect formula is register-blind:}$$

$$D_S(m) = \frac{m^2_{\text{pole}}}{\hat{\mu}'^2} = 2 - 2\cos\big(m\,\delta_v/2\big) = 4\sin^2\big(m\,\delta_v/4\big)$$

$$[\text{FT.76, FT.60(5); Def-H.37(2) transfer}] :$$

$$\text{reads } (m, \delta_v) \text{ alone — winding and vertex-figure defect [FT.63(4)] : } \alpha\text{-blind,}$$

$$\sigma\text{-blind}$$

$$\Longrightarrow C\text{- and } P\text{-invariant;}$$

$$T : \ \kappa \mapsto -\kappa \text{ [FT.88(2)] maps } m \mapsto -m : \ D(-m) = D(m) \text{ exactly (cos even);}$$

$$(2) \quad \text{the five Platonic multiplets, exact in algebraic number fields [machine] :}$$

$$\text{tetrahedron } (\delta_v = \pi,\ k = 3) : \ D \equiv 2 \text{ on } m \in \{3, 1, -1, -3\} \text{ — fully degenerate [FT.76(5}a\text{)];}$$

$$\text{cube } (\pi/2,\ k = 2) : \ \{2, 0, 2\} \text{ — the exactly massless even-winding weight [FT.76(5}b\text{)];}$$

$$\text{octahedron } (2\pi/3,\ k = 3) : \ \{4, 1, 1, 4\} \in \mathbb{Q};$$

$$\text{dodecahedron } (\pi/5,\ k=5):\ D(m)$$

$$= 2 - 2T_{|m|}(c),$$

$$c$$

$$= \cos\tfrac{\pi}{10},$$

$$16c^4 - 20c^2 + 5$$

$$= 0\ [\mathbb{Q}[c]/(\cdot);\ T_5(c) = 0 \text{ exact}];$$

$$\text{icosahedron } (\pi/3,\ k=3):\ \{2,\ 2-\sqrt{3},\ 2-\sqrt{3},\ 2\} \in \mathbb{Q}(\sqrt{3}), \quad (2-\sqrt{3})(2+\sqrt{3}) = 1;$$

$$\text{every multiplet is } m \leftrightarrow -m \text{ paired} : D(m) = D(-m) \text{ in each field};$$

$$(3) \quad \text{pole condition and weights: on the materialized even-}\Theta \text{ locus,}$$

$$P \text{ acts trivially on weights [FT.137(3)]}:\ \text{the pole census } (1 \in \operatorname{spec} \mathrm{F}_M^{(\Sigma)} \text{ [CT2.5, Def-H.16]}) \text{ is } P\text{-invariant there;}$$

$$\text{off-shell the weights acquire the unit character } e^{i\pi\Theta}:\ |w|^2\text{-level invariance is unconditional [EM.24(5)];}$$

$$\text{the character multiplies weights, not the mask [CT2.3];}$$

$$(4) \quad \text{ceilings are parity-blind: FT.76(3)'s } m^2_{\text{pole}}$$

$$\in [0, 4\hat{\mu}'^2] \text{ and FT.77's negative results are range statements about } D \text{ values — invariant under the whole reversal family.}$$

Req. FT.60, FT.63 (Tiers F.20–F.22); FT.76, FT.77 (Tiers F.26–F.28); FT.22 (Tiers F.1–F.11); CT2.3, CT2.5 (Part 2); Def-H.16 (Tiers F.1–F.11); Def-H.37(2); FT.133, FT.137 (this tier); exact Chebyshev arithmetic in $\mathbb{Q}[c]/(16c^4-20c^2+5)$, exact $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}$ multiplets [math; machine: `checks/verify_H_parity_identification.py` §F].

Pf.

$$(1):\ \text{argument census of the frozen formula [FT.76(2)]}:\ (m, \delta_v);$$

$$\cos(-x) = \cos x \text{ [even series]};$$

$$m \mapsto -m \text{ under } \kappa \mapsto -\kappa \text{ [FT.88(2); } k = \Theta/2]$$

$$(2): \text{ tetra}: \cos(m\pi/2) = 0 \ (m \text{ odd}): \ D = 2;$$

$$\text{cube}: \cos(m\pi/4) \in \{0, 1\} \text{ at } m \in \{2, 0, -2\}: \ D \in \{2, 0\};$$

$$\text{octa}: \cos(m\pi/3) \in \{-1, \tfrac{1}{2}\}: \ D \in \{4, 1\}$$

$$\text{dodeca}: \ T_0$$

$$= 1,$$

$$T_1$$

$$= c,$$

$$T_{m+1}$$

$$= 2cT_m - T_{m-1} \text{ in } \mathbb{Q}[c]/(16c^4 - 20c^2 + 5): \ T_5$$

$$= c(16c^4 - 20c^2 + 5)$$

$$= 0 \text{ [machine]};$$

$$\text{icosa}: \cos(\pi/6) = \tfrac{\sqrt{3}}{2}: \ D(\pm 1) = 2 - \sqrt{3}, \ D(\pm 3) = 2; \text{ norm } (2 - \sqrt{3})(2 + \sqrt{3}) = 4 - 3 = 1$$

$$(3): \text{ FT.137(3)}: \ \Theta \in 2\mathbb{Z} \text{ and real weights } \implies w(P\mathcal{C}) = w(\mathcal{C});$$

$$\text{off-shell}: \ |e^{i\pi\Theta}| = 1;$$

$$\mathcal{P}_M \text{ reads } (R, \text{ seals}), \ \neg\alpha \text{ [CT2.3, D2.11]}$$

$$(4): \ 0 \le D \le 4 \iff 0 \le 4\sin^2 \le 4 \text{ [range of } \sin^2];$$

the ceiling is a function of D values alone. ■

Phys. The pole ladder is cut from geometry and combinatorics alone — windings and vertex angles — and no register reversal in the corpus touches either. Parity walks through the whole pole spectrum without a footprint; the only motion is $m \mapsto -m$ under the ledger reflection, and every mass in every Platonic multiplet is even in it. The bounded-hierarchy verdicts of the SM audit are parity statements before the name existed: they constrain ranges, and ranges are what reversals preserve.

Theorem 3.104 (FT.139 — The channel census: Koide and colour channels under the reversal family). Thm.

$$(1) \quad \text{the seal bilinear is } S\text{-even:}$$

$$\mathrm{Ad}(\mathbb{S})[J_i, K_j] = [J_i, K_j] \;\; \forall (i,j) \text{ [machine: all 9 pairs]};$$

$$\sum_i [J_i, K_i] = 2i\Sigma \text{ [FT.109 re-derived]};$$

$$2i\Sigma \in \mathcal{E}_{++} \text{ [FT.109(4)]};$$

$$(2) \quad \text{the channel constants are register-blind: the seal coefficient } 2 \text{ [FT.109(3)]},$$

$$\dim F_{J^\pm}$$

$$= 3 \text{ [CT0.12]},$$

$$\mathrm{tr}\, T(1)$$

$$= 3 \text{ [FT.68(2) : the cap-3 transfer rule reads } (R, \text{ vertex degrees})],$$

$$N_{\text{colour}}$$

$$= 3 \text{ [CT3.20 : the cap count] :}$$

$$\text{no configuration register } (\sigma, \alpha, \kappa, \text{ledger}) \text{ enters any constituent } \implies \delta_\ell$$

$$= \tfrac{2}{9} \text{ [FT.112]},$$

$$\delta_q$$

$$= \tfrac{2}{27} \text{ [FT.119]},$$

$$(A, B)$$

$$= (1, \sqrt{2}) \text{ [FT.111(3)]},$$

the spectra r_i,

$$Q$$

$$= \tfrac{2}{3},$$

and all ratios are invariant under the full $\mathbb{Z}_2^4$ family [FT.133(2)];

(3) the contrast census: Tier F.40's flipping block is not invariant: F_{LR}

$$= q$$

I_2 is C-odd,

P-even,

T-even [FT.134(2); FT.133(6)] :

the channels are parity-even and register-blind; the flipping block is charge-odd — two exact, distinct parity characters in the same tower.

Req. FT.109, FT.111, FT.112 (Tier F.37); FT.116, FT.119 (Tier F.38); FT.129 (Tier F.39); FT.68 (Tiers F.23–F.25); CT0.12 (Part 1); CT3.20 (Part 2); FT.133, FT.134 (this tier); exact $\mathbb{Q}(i)$ conjugation census [math; machine: `checks/verify_H_parity_identification.py` §G].

Pf.

$$(1) : \ \mathrm{Ad}(\mathbb{S})J_i$$

$$= -J_i,$$

$$\mathrm{Ad}(\mathbb{S})K_i$$

$$= -K_i \text{ [CT0.7(3)]} : \ \mathrm{Ad}(\mathbb{S})[J_i, K_j]$$

$$= [-J_i, -K_j]$$

$$= [J_i, K_j] \text{ [machine, all 9]};$$

$$\sum_i [J_i, K_i] = 2i\Sigma \text{ [entrywise re-derivation; machine]}$$

$$\mathrm{Ad}(\mathbb{S})\Sigma = \Sigma \text{ [CT1.5 : } [\Sigma, \mathbb{S}] = 0]$$

(2) : constituent census: each integer's definition cites only cap counts, block dimensions and transfer rules [FT.109(3), CT0.12, FT.68(2), CT3.20];

invariance of constants under any data map

(3) : FT.134(2) characters verbatim; $\quad q_{\text{core}}(C\mathcal{C}) = -q_{\text{core}}(\mathcal{C})$ [FT.133(6)]. ■

Phys. The numbers that carry the seal-side and colour-side hierarchies are cut from counts no reversal can reach — caps, dimensions, traces — so both channels sit outside the entire parity calculus, exactly invariant under all four involutions at once. Where parity does bite is the flip block: charge-odd, parity-even. A future dynamical read of the flipping frequency will inherit that character as a selection rule, and any application-layer assignment rule will have to reconcile it with the branch-selective weak force of FT.135.

3.43 Tier F.42 — The Rung-One Chirality, the Janus Arrow, and the Family Channel of the Sixteen-Dimensional Arena

Theorem 3.105 (FT.141 — The rung-one chirality theorem: the $10/6$ grading is the chirality of the complete force complex). Thm. With $\mathbb{S} = \frac{1}{2}(E_0 + \Sigma)$ and $S(X) = \mathbb{S}X\mathbb{S} = \mathrm{Ad}(\mathbb{S})X$ [A3; CT0.7(1)–(2)] on the pair channel $\mathfrak{H}(\mathbb{C}^4)$:

(1) the grading census [FT.136(1) re-derived at the point of use; machine: all 17 CT1.2 generators] :

$$S\text{-even } (+1) : \ \{E_0,\ \Sigma,\ \hat{E}^{ij}\ (5),\ W_i\ (3)\} = 10 \text{ dims} = G_5 + W_3 + H_1 + D_1^+;$$

$$S\text{-odd } (-1) : \ \{J_i\ (3),\ K_i\ (3)\} = 6 \text{ dims} = J_3^+ + J_3^-;$$

$$10 + 6 = 16 = C_{\text{cap}} \text{ [CT0.12; CT0.6]};$$

the asymmetry $10 \neq 6$ is exact;

(2) the chirality of the complete force complex: the A8 block labels are readings of the (S, T) grades [CT0.7(4)] : $\mathrm{F}_{(+,\mathcal{R})}$

$= F_G \oplus F_H \oplus F_{D^+}$ and $\mathrm{F}_{(-,\mathcal{R})}$

$= F_W$ are the S-even aspects,

$\mathrm{F}_{(+,\mathcal{V})}$

$= F_{J^+}$,

$\mathrm{F}_{(-,\mathcal{V})}$

$= F_{J^-}$ the S-odd aspects;

the chiral structure of the complete force operator F [A8 : always applied in full] at rung 1 is the ordered pair $\left(\dim \mathcal{K}_1^+,\ \dim \mathcal{K}_1^-\right)$

$= (10,\ 6)$ [CT0.14 : $\mathcal{K}_1 =$ the rung-one mask] :

a property of the pair algebra [A3′s swap; CT1.1, CT1.2′s basis],

¬ of any single block: ¬∃ an aspect that is the chiral force;

(3) deletion invariance: $\forall$

$\mathcal{Y} \subseteq \{G,\ W,\ H,\ D^+,\ J^+,\ J^-\}$ (the six rung-one blocks) : the S-eigenvalue of every surviving generator is unchanged (the eigen-relation $\mathbb{S}X\mathbb{S} = \pm X$ is a two-operator relation: deletion touches neither $\mathbb{S}$ nor X);

the surviving split is $(10 - d_{\mathcal{Y}}^+,\ 6 - d_{\mathcal{Y}}^-)$, $d_{\mathcal{Y}}^{\pm}$ the deleted dimensions in each grade;

the grading is nontrivial (both grades occupied) $\iff$

≥ 1 S-even block and

$$\geq 1\ S\text{-odd block survive}:\ 45 \text{ of the } 2^6$$

$$= 64 \text{ subsets [machine: exhaustive]};$$

the six single-block deletions leave $(5,6)$,

$$(7,6),$$

$$(9,6),$$

$$(9,6),$$

$$(10,3),$$

$(10,3)$: each nontrivial, each asymmetric;

$$\text{the full } 10/6 \text{ is recovered } \iff \mathcal{Y} = \varnothing;$$

(4) the weak shadow: $W_i \in \mathcal{E}_{+-}$: S-even [CT1.2 table];

$$\text{span}\{J_i, K_i\} \cap \text{span}\{W_k\}$$

$= \{0\}$ [disjoint basis subsets; machine: rank $9 = 6 + 3$] : the S-odd block carries no weak content;

the branch selectivity of the weak [FT.93(5), FT.101] sits at the state-space locus of the same swap :

$$\Pi_{\mathcal{R}} = \tfrac{1}{2}(E_0 + \mathbb{S}), \quad \Pi_{\mathcal{V}} = \tfrac{1}{2}(E_0 - \mathbb{S}) \quad [\text{CT0.7}(1)],$$

$$W_i \Pi_{\mathcal{V}} = 0 \ \ \forall i \text{ [FT.101}(2) \text{ re-derived]};$$

$$\textstyle\sum_i W_i^2 = \tfrac{3}{2}E_0 + \tfrac{1}{2}\Sigma \text{ [Cor CT0.7a]} = E_0 + \mathbb{S} = 2\,\Pi_{\mathcal{R}}$$

$$[\Sigma = 2\mathbb{S} - E_0;\ \text{machine, entrywise}] :$$

the swap operator itself enters the weak Casimir identity — the branch asymmetry $(2 \text{ vs } 0)$ is derived from the S-grading, and $\mathbb{S}$ is A3 data, prior to every force assignment [CT0.7(4) : the A8 labels read the grades];

locus discipline [FT.136(5) stands] : the $3 \oplus 1$ branch split (spec $\mathbb{S}$ on $\mathbb{C}^4_{\text{pair}}$) and the $10/6$ operator split (spec $\mathrm{Ad}(\mathbb{S})$ on $\mathrm{End}(\mathbb{C}^4_{\text{pair}})$) are two loci of one swap, and neither is the carrier $\hat{\chi}$-grade [FT.122];

$$(5) \quad \text{no } S\text{-neutral observable at rung one: } \mathrm{Ad}(\mathbb{S})^2$$

$$= \mathrm{id} \implies \mathrm{spec}\,\mathrm{Ad}(\mathbb{S}) \subseteq \{\pm 1\} \text{ and } \ker \mathrm{Ad}(\mathbb{S})$$

$= \{0\}$ [$\mathrm{Ad}(\mathbb{S})$ is its own inverse; machine: rank 16 on the flattened 16-basis; $\mathrm{tr}\,\mathrm{Ad}(\mathbb{S}) = 10-6 = 4$];

every rung-one observable carries the unique grade decomposition $X = X_+ + X_-$,

$$X_\pm = \tfrac{1}{2}\big(\mathrm{id} \pm \mathrm{Ad}(\mathbb{S})\big)X;$$

the metric reads $(\hat{h}_{ij}\ 5,\ \Sigma_{\text{tr}}\ 1,\ a_{00}\ 1)$ and the weak reads $(w_i\ 3)$ are S-even;

the seal reads $(J^\pm)$ are S-odd with 0 metric coordinates [FT.30 dictionary] :

chirality at rung 1 is ¬ a selection rule applied by one force — it is the grading of the entire observable algebra $\mathcal{K}_1$ [CT0.14 Rmk : $\dim \mathcal{K}_1 = 16 = C_{\text{cap}}$: rung one exhausts the pair channel];

$$(6) \quad \text{reconciliation — the two independent } \mathbb{Z}_2\text{'s: the force-arena } S\text{-grading acts on } \mathrm{End}(\mathbb{C}^4_{\text{pair}});$$

the carrier chirality $\hat{\chi}$ acts on $\mathcal{H}_\omega$ [Def-H.33, FT.122; Def-H.34] : pairwise distinct, commuting where jointly defined, neither reducible to the other [FT.126(4), FT.136(4) re-derived; machine witnesses];

the division of labour: the S-grading determines WHICH observables are chiral (the grade of every read); the carrier chirality determines WHICH chain orientations are chiral (the doublet's rays);

the rule coupling the two $\mathbb{Z}_2$'s — and the branch-label $\mathbb{Z}_2$ [FT.101, FT.135] and the Janus arrow [FT.144(5)]—to species is the deficit $1'$ assignment rule [FT.125(4)] : ¬constructed here;

no new declaration issues.

Req. A3, A8 (Parts 1–2); CT0.6, CT0.7, Cor CT0.7a, CT0.12, CT0.13, CT0.14, CT1.1, CT1.2, CT1.3, CT1.5 (Part 1); FT.30 (Tiers F.1–F.11); FT.93, FT.101 (Tiers F.31–F.34); Def-H.33, FT.122, FT.125, FT.126 (Tier F.39); FT.135, FT.136 (Tier F.41); Def-H.34 (carrier clause); exact $\mathbb{Q}(i)$ matrix arithmetic, exhaustive 2^6 subset census, rank/trace invariants, inclusion–exclusion [math; machine: `checks/verify_H_rung1_neutrino.py` §A].

Pf.

$$(1):\ \mathbb{S}=\tfrac{1}{2}(E_0+\Sigma),\ \mathbb{S}^2=E_0\ [\text{CT0.7}(1);\ \Sigma^2=3E_0-2\Sigma];$$

$$\mathbb{S}E_0\mathbb{S}=E_0,\ \mathbb{S}\Sigma\mathbb{S}=\Sigma\ [\text{CT1.3},\ \text{CT1.5}];$$

$$\mathbb{S}\hat{E}^{ij}\mathbb{S}=\hat{E}^{ij},\ \mathbb{S}W_i\mathbb{S}=W_i;$$

$$\mathbb{S}J_i\mathbb{S}=-J_i,\ \mathbb{S}K_i\mathbb{S}=-K_i\ [\text{CT0.7}(3);\ \text{CT1.2 table}]$$

entrywise on all 17 generators over $\mathbb{Q}(i)$ [machine];

$$\text{counts}:\ 1+1+5+3=10;\quad 3+3=6;$$

$$10+6=16=C_{\text{cap}}\ [\text{CT0.12};\ \text{CT0.6}]$$

$$(2):\ \text{block}\leftrightarrow\text{aspect rows verbatim}\ [\text{CT0.12 table};\ \text{CT0.7}(4)];$$

the (S,T) refinement $(7,3,3,3)$ [CT1.1] : S-even $=7{+}3=10$, S-odd $=3{+}3=6$ [exact integers];

A8 : F is the full direct sum — no sub-block restriction is an A8 force

$$(3):\ \text{the eigen-relation is binary in}\ (\mathbb{S},X):\ \text{deleting}\ Y$$

$\neq X$ leaves it untouched [machine: the grade table is identical on all 64 subsets];

$$\text{both-grade census}:\ 2^6-2^4-2^2+1$$

$$=64-16-4+1$$

$=45$ [inclusion–exclusion over the 4 even and 2 odd blocks; machine]

$$\text{single deletions}\ (d^+,d^-)\in\{(5,0),(3,0),(1,0),(1,0),(0,3),(0,3)\}$$

$$\Longrightarrow\ (5,6),$$

$$(7,6),$$

$$(9,6),$$

$$(9,6),$$

$$(10,3),$$

$$(10,3)\ \text{[machine]};$$

$$(10,6)\ \text{uniquely at}\ \mathcal{Y}=\varnothing\ \text{[machine]}$$

$$(4):\ W_i\in\mathcal{E}_{+-}\ \text{[CT1.2]};$$

$$\text{rank}\{J_1,J_2,J_3,K_1,K_2,K_3,W_1,W_2,W_3\}$$

$$=9$$

$$=6+3\ \text{[exact flattening rank; machine]}\ \Longrightarrow\ \text{the intersection is}\ \{0\}$$

$$\Pi_{\mathcal{R}},$$

$$\Pi_{\mathcal{V}}:\ \text{idempotent, orthogonal, complete,}$$

$$\mathfrak{T}\Pi_{\mathcal{R}}$$

$$=3,$$

$$\mathfrak{T}\Pi_{\mathcal{V}}$$

$$=1\ \text{[CT0.7(1); machine]};$$

$$W_i\Pi_{\mathcal{V}}=0\ \forall i\ \text{[FT.101(2) re-derived entrywise; machine]}$$

$$\textstyle\sum_i W_i^2 = \frac{1}{4}\sum_i(\sigma_i\otimes\sigma_0+\sigma_0\otimes\sigma_i)^2 = \frac{3}{2}E_0+\frac{1}{2}\Sigma \text{ [Cor CT0.7a re-derived]};$$

$$\Sigma$$

$$= 2\mathbb{S}-E_0 \text{ [CT0.7(1)]} \implies \tfrac{3}{2}E_0+\tfrac{1}{2}(2\mathbb{S}-E_0)$$

$$= E_0+\mathbb{S}$$

$$= 2\cdot\tfrac{1}{2}(E_0+\mathbb{S})$$

$$= 2\Pi_{\mathcal{R}} \text{ [machine]}$$

$$\text{locus rows : spec}\,\mathbb{S} \text{ on } \mathbb{C}^4_{\text{pair}}$$

$$= (3,1)\ [\mathfrak{T}\Pi_{\mathcal{R}}=3,\ \mathfrak{T}\Pi_{\mathcal{V}}=1];$$

$$\text{spec}\,\mathrm{Ad}(\mathbb{S}) \text{ on } \mathrm{End}(\mathbb{C}^4_{\text{pair}})$$

$$= (10,6)\ [(1)];$$

$$\hat{\chi} \text{ on } \mathcal{H}_\omega \text{ [FT.136(4) re-run]};$$

three distinct spaces

$$(5):\ \mathrm{Ad}(\mathbb{S})^2X = \mathbb{S}(\mathbb{S}X\mathbb{S})\mathbb{S} = \mathbb{S}^2X\mathbb{S}^2 = E_0XE_0 = X \text{ [CT1.3]};$$

$$\text{self-inverse} \implies \text{invertible} \implies \ker = \{0\};$$

rank 16 on the flattened basis [machine];

$$\mathrm{tr}\,\mathrm{Ad}(\mathbb{S}) = \#\{+1\}-\#\{-1\} = 10-6 = 4 \text{ [machine]}$$

$$X_\pm = \tfrac{1}{2}(X\pm\mathbb{S}X\mathbb{S}):\ \mathrm{Ad}(\mathbb{S})X_\pm = \pm X_\pm,\ X_+ + X_- = X\ [\mathbb{S}^2=E_0];$$

uniqueness: eigenspace decomposition of an involution [machine on the 16-basis];

FT.30 dictionary rows verbatim

$$(6): \ [G \otimes I_2,\ I_2 \otimes \hat{\omega}] = 0,$$

$$(G \otimes \hat{\omega})^2 = \mathrm{id},$$

$$G \otimes \hat{\omega} \neq G \otimes I_2 \neq I_2 \otimes \hat{\omega} \ [\text{FT.126}(4)/\text{FT.134}(5) \text{ pattern; machine on } G \in \{\sigma_1, \sigma_3\}];$$

$$\mathrm{Ad}(\mathbb{S}) \text{ on } \mathrm{End}(\mathbb{C}^4_{\text{pair}}) \text{ vs } \hat{\chi} \text{ on } \mathcal{H}_\omega : \text{ distinct carriers } [\text{FT.136}(4)]$$

$$\text{coupling rule}: \ \neg\exists \text{ tag assigning grades to species } [\text{inventory scan; FT.125}(4)]. \ \blacksquare$$

Phys. At the first rung the entire force complex is chiral, and its chirality is a count: ten dimensions speak to the metric — shear, volume, conformal, shift — and six only move the ledger. The count belongs to no force: it is the swap grading of the pair algebra, and every aspect of the complete force sits inside it with its grade already decided. The weak force's one-sidedness is a shadow of the same swap — its Casimir identity contains the swap operator literally — so no isolation of the weak reproduces the asymmetry, and no deletion of any force block removes it: the grade of each generator is a two-operator fact, invariant under everything except deleting the generator itself. Every observable at rung one is graded; none is neutral. The carrier keeps a second, independent chirality of its own, and the rule that would marry the two gradings to species is application-layer content.

Theorem 3.106 (FT.142 — The rung-one arena and the channel angle $\delta_\nu = \frac{1}{8}$). Thm. The rung-one channel — the deficit-6 ledger row, internal channel subscript ν [¬a species name; Rmk below]:

(1) the channel class: a rung-one bound chain is a bound chain [D1.5] whose terminus-active aspects lie in $\mathcal{A}(1)$

$$= \{J,\ G,\ W,\ H,\ D^+\} \ [\text{CT0.13}]:$$

$$\text{EM unactivated } (f_v = 0 \text{ reads } [\text{D3.3; FT.28}(3): \ \text{EM} \in \mathcal{A} \iff f_p \neq 0 \ \vee \ f_q \neq 0]);$$

$$\text{colour block vanishing (no slot index below trivalence } [\text{CT3.11}(1)]; \text{ single-contact floors: } \mathcal{C} = 0 \implies \text{block} \equiv 0 \ [\text{FT.19, FT.28}(3)]);$$

$$\text{the Def-H.4 floor stands: } \delta h(c) \geq 2,\ n(v) \geq 2 \ [\text{self-loops} \perp \text{ by A1}];$$

$$\text{the minimal instance is the digon } \Sigma_{\min} \ [\text{FT.28}];$$

(2) the arena: the pair-channel arena on which the complete rung-one force acts is the rung-one mask $\mathcal{K}_1$:

$$\dim \mathcal{K}_1 = 16 = C_{\text{cap}} \quad [\text{CT0.14},\ \text{CT0.12};\ \text{machine: exact rank}];$$

the complete-force standing rule [A8] fixes the arena as the full mask: no sub-block arena — isolating a force is not an A8 action;

(3) the arena adjudication (computed candidates; one adopted) :

candidate	source	verdict
$16 = \dim \mathcal{K}_1$	CT0.14 (derived mask)	adopted
$10 = S$-even block	FT.141(1)	rejected: deleting the S-odd blocks violates A8's completeness [CT0.13 : $J \in \mathcal{A}(1)$]
$9 =$ seal $\times$ transfer	FT.110 [Def-H.41]	rejected: a interface definition for transfer-dressed chains [the transfer trace $\operatorname{tr} T(1) = 3$ is polygon-channel data, FT.68(2)]; not imported in this part
$27 =$ seal $\times$ transfer $\times$ colour	FT.118(2) [Def-H.42]	rejected: colour slots are rung-three content [CT0.13]; clause (1) excludes them; not imported
$6 = \dim \mathcal{K}_0$	CT0.14	rejected: seals alone; a terminus with $n(v) \geq 2$ carries metric and weak reads [FT.28(3) : $\{G, H, W\} \subseteq \mathcal{A}(\Sigma_{\text{min}})$]

the adopted arena is the unique candidate that is derived content with no interface declaration: δ_ν is the first channel angle whose denominator requires no declaration-entry construction;

(4) the seal coefficient, undressed: $[J_i, K_j]$

$= i\big(\delta_{ij}\Sigma - E^{ij}\big)$ [CT1.7 re-derived, all 9 pairs];

$\sum_i [J_i, K_i] = i\big(3\Sigma - \sum_i E^{ii}\big) = 2i\Sigma \quad [\sum_i E^{ii} = \Sigma]$;

in the 16-element CT1.2 basis the sum has exactly one nonzero component — the Σ-component, coefficient exactly 2 [FT.109(3) pattern; machine: exact linear solve, rank 16];

the rung-one channel class carries no dressing factor (¬colour slots by clause (1); ¬transfer declaration imported by clause (3)) : the numerator is 2,

undressed [FT.118(5) spectator pattern];

(5) the channel angle:

$$\delta_\nu \ := \ \frac{\sum_i [J_i, K_i]\big|_\Sigma}{\dim \mathcal{K}_1} \ = \ \frac{2}{16} \ = \ \frac{1}{8}$$

[exact rational; FT.112 pattern];

the democratic normalization A

$$= 1,$$

$$B$$

$= \sqrt{2}$ is seal-side data [FT.111(3) : $\sum_i \lambda_i = 3 = \dim F_{J^\pm}$; $\sum_i (\lambda_i - A)^2 = 3$] and hence channel-blind [FT.119 precedent] : the rung-one channel inherits it unchanged.

Rmk (channel-label discipline). The subscript ν names the deficit-6 ledger row — internal bookkeeping, joining ℓ [FT.112] and q [FT.119] as a channel subscript; all three denote ledger rows, ¬particles. No species identification is made or implied [P1, P5]. The arena $\mathcal{H}_c$-vocabulary of FT.110/FT.118 (counting spaces without inner product, dynamics or states) applies verbatim: $\mathcal{K}_1$ enters this tier exactly once, as the denominator of clause (5).

Req. A1, A8, D0.8, D1.5, D3.3, P1, P4, P5 (Parts 1–2); CT0.6, CT0.12, CT0.13, CT0.14, CT1.2, CT1.7 (Part 1); CT3.11 (Part 2); Def-H.4 (Tier F.2); FT.19, FT.28 (Tiers F.1–F.11); FT.68(2) (Tiers F.23–F.25); FT.109, FT.110, FT.111, FT.112 (Tier F.37); FT.118 (Tier F.38); FT.141 (this tier); Def-H.41, Def-H.42 cited in the rejection rows and imported nowhere; exact $\mathbb{Q}(i)$ arithmetic, exact linear solve on the 16-basis, exact flattening ranks [math; machine: `checks/verify_H_rung1_neutrino.py` §B].

Pf.

$$(1) : \text{ D1.5 : bound} = \text{closed self-contained, fermionic content;}$$

$$\mathcal{A}(1) \text{ verbatim [CT0.13];}$$

$$\text{FT.28(3) rows verbatim [D3.3; CT3.11(1); FT.19]}$$

$$\text{Def-H.4 Rmk : } \delta h = 1 \text{ closed chain} = \text{self-loop} = \perp \text{ [A1];}$$

$$\text{closed chain based at } v \implies n(v) \geq 2 \text{ [D0.8];}$$

$$\Sigma_{\text{min}} = \text{the 2-cycle floor [FT.28]}$$

$$(2) : \dim \mathcal{K}_1 = 6 + (5 + 3 + 1 + 1) = 16 \text{ [CT0.14; CT0.12 table];}$$

$$\text{machine: flattening rank of the } 16 \text{ CT1.2 elements} = 16 \text{ over } \mathbb{Q} \text{ (32 real components);}$$

$$16 = C_{\text{cap}} \text{ [CT0.6]}$$

$$\text{A8 : F}$$

$= F_{(+,\mathcal{R})} \oplus F_{(-,\mathcal{R})} \oplus F_{(+,\mathcal{V})} \oplus F_{(-,\mathcal{V})} \oplus F_{\mathrm{st}}$ always applies in full $\implies$ the rung-one read arena is the full mask $\mathcal{K}_1$ [CT0.14 : $\mathcal{P}_M^{(1)}$ projects onto it]

$$(3):\ \text{candidate rows}:\ 10 = \mathrm{rank}\{E_0, \Sigma, \hat{E}^{ij}, W_i\}\ [\text{machine}];$$

$$9 = 3 \cdot 3\ [\text{FT.110}(1)\ \text{under Def-H.41's declared tensor product}];$$

$$27 = 3 \cdot 3 \cdot 3\ [\text{FT.118}(2)\ \text{under Def-H.42}];$$

$$6 = \mathrm{rank}\{J_i, K_i\}\ [\text{CT0.14};\ \text{machine}]$$

$$\text{rejection grounds}:\ \text{the } S\text{-even sub-arena omits } J \in \mathcal{A}(1)\ [\text{CT0.13}]\ \text{against A8};$$

the $9/27$ arenas are declared constructions [FT.110 Rmk; FT.118 Rmk] and this part's import discipline admits {Def-H.4, Def-H.37(2), Def-H.38, Def-H.34, Def-H.43} only;

$$\mathcal{K}_0\ \text{omits the metric/weak reads of any } n \geq 2\ \text{terminus}\ [\text{FT.28}(3)]$$

$$(4):\ \text{CT1.7 entrywise on all } 9 \text{ pairs over } \mathbb{Q}(i)\ [\text{machine}];$$

$$\textstyle\sum_i (\Sigma - E^{ii}) = 3\Sigma - \Sigma = 2\Sigma\ [\text{FT.109}(2)\ \text{re-derived}];$$

$$\text{component read: exact solve on the rank-16 basis} \implies (0, \ldots, 0,\ 2i,\ 0, \ldots, 0)\ [\text{machine}]$$

$$(5):\ \delta_\nu = \tfrac{2}{16} = \tfrac{1}{8}\ [\text{exact Fractions}];$$

$$A = 1:\ \textstyle\sum_i \lambda_i = 3A = 3 = \dim F_{J^\pm}\ [\text{CT0.12 table}];$$

$$B^2 = 2:\ \textstyle\sum_i (\lambda_i - A)^2 = B^2 \sum_i \cos^2(\theta + \frac{2\pi i}{3}) = B^2 \cdot \frac{3}{2} = 3\ [\text{FT.111}(3)\ \text{identities}].\ \blacksquare$$

Phys. The rung-one channel is the class of bound chains that live on the pretime forces alone — no charge read firing, no colour slot gripped — and its arena needs no construction: the rung-one mask already is the whole pair channel, sixteen dimensions, derived. The seed integer of the family mechanism, the diagonal seal coefficient 2, reappears undressed, because rung one dresses nothing. One integer over one derived dimension: the channel angle is $1/8$, the first of the three family angles whose denominator is a theorem rather than a interface definition.

3.107 (FT.143 — The rung-one family spectrum).

Thm. With $\delta_\nu = \frac{1}{8}$ [FT.142(5)] and $A = 1$, $B = \sqrt{2}$ [FT.111(3)]:

$$(1) \quad \text{the spectrum:} \qquad r_i^{(\nu)} = 1 + \sqrt{2}\cos\Big(\frac{1}{8} + \frac{2\pi i}{3}\Big), \qquad i = 0, 1, 2,$$

with exact radical-trigonometric closed forms ($c = \cos\frac{1}{8},\ s = \sin\frac{1}{8}$) :

$$r_0^{(\nu)} = 1 + \sqrt{2}\,c;$$

$$r_1^{(\nu)} = 1 - \frac{\sqrt{2}\,c + \sqrt{6}\,s}{2};$$

$$r_2^{(\nu)} = 1 - \frac{\sqrt{2}\,c - \sqrt{6}\,s}{2};$$

$$r_0^{(\nu)} = 2.403179397550\ldots,$$

$$r_1^{(\nu)} = 0.145715560918\ldots,$$

$$r_2^{(\nu)} = 0.451105041530\ldots$$

[all three strictly positive; disjoint enclosures; machine];

(2) Vieta and the channel-own cubic:

$$\textstyle\sum_i r_i^{(\nu)} = 3,$$

$$\sum_{i<j} r_i^{(\nu)} r_j^{(\nu)} = \frac{3}{2},$$

$$\sum_i \big(r_i^{(\nu)}\big)^2 = 6$$

[exact ring identities; machine];

$$2\big(r^{(\nu)} - 1\big)^3 - 3\big(r^{(\nu)} - 1\big) - \sqrt{2}\cos\frac{3}{8} = 0 \quad \text{for each } r^{(\nu)} = r_i^{(\nu)}$$

$\big[\cos 3\delta_\nu = \cos\frac{3}{8} = 4c^3 - 3c;$ machine: symbolic + enclosure$\big]$;

$\cos\frac{3}{8} \neq \cos\frac{2}{9}$ [disjoint enclosures] : the cubic constant $\sqrt{2}\cos\frac{3}{8}$ is channel-own — the rung-one channel shares no depressed cubic with the ℓ/q pair [FT.119(4) : $\cos 3\delta_q = \cos\delta_\ell = \cos\frac{2}{9}$];

(3) the mass-squared ratios:

$$\big(r_1^{(\nu)}\big)^2 < \big(r_2^{(\nu)}\big)^2 < \big(r_0^{(\nu)}\big)^2 \text{ [enclosure-proven]};$$

$$m_i = \mu_\nu \big(r_i^{(\nu)}\big)^2, \ \ \mu_\nu \text{ undeclared (P4)};$$

the m^2 vocabulary is the corpus's own inertial read [L3(4), L4]; ¬a particle identification :

$$m_1^2 : m_2^2 : m_3^2$$

$$= 1 : \frac{\big(r_2^{(\nu)}\big)^2}{\big(r_1^{(\nu)}\big)^2} : \frac{\big(r_0^{(\nu)}\big)^2}{\big(r_1^{(\nu)}\big)^2}$$

$$= 1 : 9.583926992379\ldots : 271.994748749143\ldots$$

[widths $< 2^{-50}$; machine];

anchored forms (scale-free, top-normalized) :

$$\frac{m_2}{m_3} = \frac{\big(r_2^{(\nu)}\big)^2}{\big(r_0^{(\nu)}\big)^2} = 0.035235705970\ldots,$$

$$\frac{m_1}{m_3} = \frac{\big(r_1^{(\nu)}\big)^2}{\big(r_0^{(\nu)}\big)^2} = 0.003676541567\ldots;$$

(4) the Q-invariant:

$$Q \ := \ \frac{\sum_i m_i}{\big(\sum_i \sqrt{m_i}\big)^2} \ \overset{m_i = \mu_\nu (r_i^{(\nu)})^2}{=} \ \frac{\sum_i (r_i^{(\nu)})^2}{\big(\sum_i r_i^{(\nu)}\big)^2} \ = \ \frac{6}{9} \ = \ \frac{2}{3}$$

[exact, and independent of δ and of μ [FT.119(4) pattern]] :

the channel normalization $Q = \frac{2}{3}$ holds at rung one;

(5) the angle ledger (exact rational arithmetic; no equality beyond those displayed) :

$$16\,\delta_\nu = 9\,\delta_\ell = 27\,\delta_q = 2$$

[the shared seal numerator is the only cross-channel identity; FT.112, FT.119(1)];

$$3\delta_\nu = \frac{3}{8} \notin \left\{\frac{2}{9},\ \frac{2}{27}\right\} = \{\delta_\ell,\ \delta_q\};$$

$$\delta_\nu \notin \left\{3\delta_\ell,\ 3\delta_q,\ \frac{\delta_\ell}{3},\ \frac{\delta_q}{3}\right\} :\ \text{the triple-angle link } 3\delta_q$$

$$= \delta_\ell \text{ [FT.119(2)] has no rung-one analogue;}$$

$$\delta_\nu - \delta_\ell = -\frac{7}{72};$$

$$\delta_\nu - \delta_q = \frac{11}{216};$$

$$3\delta_\nu - \delta_\ell = \frac{11}{72} = 3\,(\delta_\nu - \delta_q) \quad \text{[consistent with } \delta_\ell = 3\delta_q\text{]};$$

$$\delta_\nu : \delta_\ell : \delta_q = 27 : 48 : 16 \quad [\times 216]; \qquad \delta_q < \delta_\nu < \delta_\ell;$$

(6) the three-channel table (frozen rows cited; rung-one row derived here) :

channel (ledger row)	arena	dim	δ	cubic constant	Q	$m_1^2 : m_2^2 : m_3^2$
ℓ [FT.109–FT.113]	seal×transfer [Def-H.41]	9	$\frac{2}{9}$	$\sqrt{2}\cos\frac{2}{9}$	$\frac{2}{3}$	$1 : 206.770315972729\ldots :$ $3477.472837104\ldots$
q [FT.116–FT.119]	seal×transfer×colour [Def-H.42]	27	$\frac{2}{27}$	$\sqrt{2}\cos\frac{2}{9}$	$\frac{2}{3}$	$1 : 3.563695074946\ldots :$ $139.3386880284\ldots$
ν (this tier)	$\mathcal{K}_1$ [derived]	16	$\frac{1}{8}$	$\sqrt{2}\cos\frac{3}{8}$	$\frac{2}{3}$	$1 : 9.583926992379\ldots :$ $271.994748749143\ldots$

the family count is 3 in every channel : 3

$$= \dim F_{J^\pm} \text{ [CT0.12; FT.111(3) : } \textstyle\sum_i \lambda_i = 3\text{]} \text{ — a datum for deficit } 3,$$

¬ a selection principle [item $21(d)$ stands].

Req. FT.142; FT.111(2)–(3), FT.112, FT.113 (Tier F.37: the democratic spectrum form, the normalization, the angle discipline, the $m = \mu r^2$ vocabulary and its enclosure pipeline); FT.119 (Tier F.38: the phase-free Q, the anchored forms, the triple-angle precedent); L3, L4 (B: the m^2 inertial-read vocabulary); P4 (A: the free-scale clause); exact symbolic identities in

$\mathbb{Q}[\sqrt{2},\sqrt{3}][c,s]/(c^2{+}s^2{-}1)$, rational trigonometric enclosures (Taylor remainder bounds), integer Newton certificates, monotone rational interval arithmetic, Vieta reads [math; machine: `checks/verify_H_rung1_neutrino.py` §C].

Pf.

$$(1):\ r_i^{(\nu)} = A + B\cos(\delta_\nu + \tfrac{2\pi i}{3}) \text{ at } A = 1,\ B = \sqrt{2}\ \text{[FT.111(3)]};$$

$$\cos(\tfrac{1}{8} + \tfrac{2\pi}{3}) = -\tfrac{c}{2} - \tfrac{\sqrt{3}}{2}s, \quad \cos(\tfrac{1}{8} + \tfrac{4\pi}{3}) = -\tfrac{c}{2} + \tfrac{\sqrt{3}}{2}s\ \text{[addition formulas]};$$

$$\sqrt{2}\cdot\tfrac{\sqrt{3}}{2} = \tfrac{\sqrt{6}}{2}$$

c, s enclosed by alternating Taylor partial sums with exact rational remainder bounds at x

$$= \tfrac{1}{8}\ (\text{widths} < 2^{-90});$$

$\sqrt{2}, \sqrt{6}$ enclosed by integer Newton certificates;

$$r_1^{(\nu)}, r_2^{(\nu)} > 0\ \text{[positive lower endpoints; machine]}$$

$$(2):\ \textstyle\sum_i \cos(\delta_\nu + \tfrac{2\pi i}{3})$$

$$= 0,$$

$$\sum_i \cos^2(\delta_\nu + \tfrac{2\pi i}{3})$$

$$= \tfrac{3}{2}\ \text{[FT.111(3) identities, arbitrary phase]} \implies \sum_i r^{(\nu)}$$

$$= 3,$$

$$\sum_i (r^{(\nu)})^2$$

$$= 3 + 2\sqrt{2}\cdot 0 + 2\cdot\tfrac{3}{2} \qquad = 6;$$

$$\text{pair sum by Vieta} = \tfrac{3}{2}$$

$$\text{depressed cubic } (\lambda-1)^3 - \tfrac{3}{2}(\lambda-1) - \tfrac{\sqrt{2}}{2}\cos 3\delta_\nu = 0\ \text{[FT.111(2) at } B^2 = 2];$$

$$\cos 3\delta_\nu = \cos\tfrac{3}{8} = 4c^3 - 3c \text{ [triple angle; machine: ring identity]};$$

$$\times 2:\ 2y^3 - 3y - \sqrt{2}\cos\tfrac{3}{8} = 0,\ y = r^{(\nu)} - 1$$

$$0 \in 2y_i^3 - 3y_i - \sqrt{2}\cos\tfrac{3}{8} \text{ enclosure for each } i \text{ [machine]};$$

$$\cos\tfrac{3}{8} \text{ vs } \cos\tfrac{2}{9}:\ \text{disjoint enclosures [machine]};$$

$$\text{ring-side: all identities exact in } \mathbb{Q}[\sqrt{2},\sqrt{3}][c,s]/(c^2{+}s^2{-}1) \text{ [machine]}$$

$$(3):\ (r_1^{(\nu)})^2 < (r_2^{(\nu)})^2 < (r_0^{(\nu)})^2 \text{ [disjoint enclosures]};$$

$$\text{ratios by monotone interval division, widths} < 2^{-50} \text{ [machine]};$$

$$\text{anchored forms: the same divisions against } (r_0^{(\nu)})^2 \text{ [machine]}$$

$$(4):\ Q = \frac{\sum_i (r^{(\nu)})^2}{(\sum_i r^{(\nu)})^2} = \frac{6}{9} = \frac{2}{3} \text{ [exact Fractions from (2)]};$$

$$\mu_\nu \text{ cancels [homogeneity]}$$

$$(5):\ 16\cdot\tfrac{1}{8} = 2, \quad 9\cdot\tfrac{2}{9} = 2, \quad 27\cdot\tfrac{2}{27} = 2;$$

$$3\cdot\tfrac{1}{8} = \tfrac{3}{8} \neq \tfrac{2}{9} \neq \tfrac{2}{27};$$

$$\tfrac{3}{8} - \tfrac{2}{9} = \tfrac{27-16}{72} = \tfrac{11}{72};$$

$$\tfrac{1}{8} - \tfrac{2}{9} = \tfrac{9-16}{72} = -\tfrac{7}{72};$$

$$\tfrac{1}{8} - \tfrac{2}{27} = \tfrac{27-16}{216} = \tfrac{11}{216} \text{ [exact Fractions; machine]}$$

$$216\cdot(\tfrac{1}{8},\tfrac{2}{9},\tfrac{2}{27}) = (27,48,16);$$

$$\tfrac{2}{27} < \tfrac{1}{8} < \tfrac{2}{9} \text{ [cross-multiplication; machine]};$$

(6): frozen rows verbatim [FT.113, FT.119(3), (5); machine: the cited decimals re-verified as enclosure floors at their stated widths]. ■

Phys. The third channel runs the same democratic machine on the derived sixteen: one large line, one middle line, one small line, all exact in $\sqrt{2}$, $\sqrt{6}$ and $\cos\frac{1}{8}$, with the universal $Q = \frac{2}{3}$ holding again — the invariant knows the normalization, not the phase. Its cubic is its own: triple the angle and nothing from the other two channels appears, so the rung-one hierarchy is structurally distinct from the linked ℓ/q pair, sharing with them exactly one integer, the seal coefficient 2. The angle sits between its two siblings and the hierarchy with them: milder than the ℓ-channel's 3477-fold span, steeper than the q-channel's 139-fold span. Every number here is dimensionless; the scale μ_ν is free internal data, and the m^2 name is the corpus's own inertial read — never a particle.

Theorem 3.108 (FT.144 — The Janus-arrow theorem: no backward materialization, and single-sector occupancy at rung one). Thm.

$$(1) \quad \text{the valid-universe condition:}$$

$$\text{valid universe} \iff \text{real-edge network closed} \ \wedge \ \#J^+$$

$$= \#J^- \ \wedge \ \in \text{Im(active mask)} \ \wedge \ \neg\exists \text{ dangling ledger boundary } [\text{D2.11}, \ \text{D1.3}];$$

$$\text{observable fermions are termini of Janus-closed bound chains } [\text{Def-H.4}];$$

$$\text{closed} \iff \partial c = 0 \ \wedge \ \#J^+ = \#J^- \ [\text{D1.4}];$$

$$\text{bound content is fermionic } [\text{D1.5}];$$

$$(2) \quad \text{the forward arrow (per-pair transition domains)} :$$

$$J^+ : \ n_{\mathcal{R}} : 0 \to 1 \ (\text{crystallization});$$

$$J^- : \ n_{\mathcal{R}} : 1 \to 0 \ (\text{dissolution})$$

$$[\text{D1.2}];$$

$$J^- \text{ is undefined at } n_{\mathcal{R}}$$

$$= 0 \ \wedge \ J^+ \text{ is undefined at } n_{\mathcal{R}}$$

$= 1 :$ the domains force strict alternation — the only admissible single-pair histories are the prefixes of the alternating word $(J^+J^-J^+J^-\cdots)$;

every admissible history obeys the prefix law $\#J^+(\text{prefix}) \geq \#J^-(\text{prefix})$,

with equality

$$\iff n_{\mathcal{R}} = 0;$$

the completed single-pair history of length $2m$ is unique: $(J^+J^-)^m$, $\#J^+ = \#J^- = m$;

two-pair completed interleavings of total length $2n$:

$$\sum_{m_0+m_1=n} \frac{(2n)!}{(2m_0)!\,(2m_1)!} = 2,\ 8,\ 32 \quad (2n = 2, 4, 6)$$

[machine: exhaustive enumeration ≤ 8 (single pair), ≤ 6 (two pairs) agrees with the closed form];

a materialized closed chain with any dissolution-first edge ($J^- \to J^+$ as its leading local transition) violates the prefix law at its first event: it is outside the transition domains, hence excluded from every valid universe [D2.11];

the sealed universe-level balance $\#J^+$

$= \#J^-$ [D1.3] is the balanced-sector restriction of the move algebra's budget identity $a_+ - a_-$

$= |R'| - |R|$ [FT.108(4)];

(3) not a mass statement: the obstruction of (2) is the transition domain of D1.2 conjoined with the closure of D1.4/D2.11 — a combinatorial locus statement: no scale, no P4 parameter and no mass enters it [inventory: the premises contain $\neg\mathfrak{h}$, $\neg\mu$, $\neg h_0$];

$\neg\exists$ a value of any configuration datum that un-excludes the backward materialization;

contrast class: the deficit-12 ceilings are range bounds on reads [FT.76(3), FT.79(2)] — the present obstruction is a domain restriction, prior to every read;

(4) the vacuum residence: the reversed reading exists only as formal bookkeeping. Two distinct reversals, adjudicated separately:

(i) $\Theta_{\mathfrak{L}}\,(T)\,:$ ledger order reversed $\wedge$ $J^{+}\leftrightarrow J^{-}$ [D2.7; FT.133(1)] : an involution on configuration data [machine: it bijects the completed admissible sequences onto themselves, counts preserved — the formal mirror];

(ii) the pure order reversal (the ledger read backwards, labels unchanged — "running the arrow in reverse") : every completed nonempty admissible history ends with J^{-} ($n_{\mathcal{R}}$ returns to 0 only via $1\to 0$),

so its pure reversal leads with $J^{-}\,:$ inadmissible at the first event [machine: exhaustive];

and the tick sequence admits no inverse in either case [A6 : $\neg\exists\,\tau^{-1}$] : the ledger count $|\mathfrak{L}(\tau_n)|$

$-\sum_{k<n} n_{\text{real_out}}(\tau_k)$ is non-decreasing along every admissible history ($n_{\text{real_out}}\geq 0$) and strictly increases at every crystallization: the reversed reading of a multi-crystallization history strictly decreases somewhere, and by (ii) the reversal of every completed word is inadmissible at its first event [machine: exhaustive] $\Longrightarrow$ no forward tick realization of any reversed reading exists;

the fluctuation content resides in the $\mathcal{V}$ sector [CT0.1] :

$$n_{\mathcal{R}}=0\implies\forall(p,q):\ (p,q)\in\mathcal{V}\ \wedge\ \rho_{pq}=\rho_{\circ}=\tfrac{1}{4}I,$$

$$S_{\text{raw}}(\rho_{\circ})=\ln 4=\max,\quad \arg\max=\{\tfrac{1}{4}I\}$$

[A5, CT0.2; machine: $\mathfrak{T}\rho_{\circ}=1$, charpoly $=(x-\frac{1}{4})^{4}$, dyadic entropy witnesses $<\ln 4$];

virtual pairs are unregistered [D1.1 : $\neg$registration $\implies e\in\mathcal{V}$];

chains require real edges [D1.4 : $e_i\in E_{\mathcal{R}}$] : $|E_{\mathcal{R}}|$

$=0\implies$ the closed-chain count is $0\implies\neg\exists$ fermion terminus [Def-H.4; machine];

the fluctuation resolvent is even in balanced pairs: $\displaystyle\sum_{n}\langle\varnothing|\hat{U}^{n}|\varnothing\rangle\,w^{n}=\frac{1}{1-w^{2}x_{+}x_{-}}$

[FT.108(5) re-derived; machine in $\mathbb{Q}[w,x_{+},x_{-}]/(w^{12})$] :

every nonzero return monomial carries equal x_{+}- and x_{-}-degree [machine degree census] : a J^{-}-leading monomial has zero support;

the zero-projection/no-pole pattern applies [CT4.2 : $\mathcal{P}_M F \mathcal{P}_M = 0 \implies \neg\exists$ accessible pole; CT2.5 : poles $\iff 1 \in \operatorname{spec} \mathrm{F}_M$] : unregistered content supports no accessible pole at any rung;

(5) carrier reconciliation — single-sector occupancy: P_L,

P_R

$= \frac{1}{2}(\mathrm{id} \pm \hat{\chi})$ are constructible, exact, rank-1,

$\hat{\omega}$-exchanged [Def-H.33, FT.122(2)–(4); machine] : mathematical objects on $\mathcal{H}_\omega$ whether or not occupied;

unconditional (derived) : the rung-one materialized locus is single-arrow by (2) — one orientation sector of the doublet basis ($|c\rangle$ carries the materialized ledger-ordered tuple [D1.4; FT.134(1) : C, P, T fix the chain tuple pointwise]; $|\omega(c)\rangle$ is the reversal image [D2.6; FT.88(1) : $R \mapsto R^{\mathrm{T}}$]);

its complement is unoccupied at rung one, and by (4) the reversed content is vacuum-sector content (all-virtual, unregistered, no accessible pole);

conditional (the coupling) :

$$|c\rangle = \tfrac{1}{\sqrt{2}}\big(|L\rangle + |R\rangle\big)$$

[Def-H.34 Rmk; machine] : the orientation rays carry no definite chirality $\implies$ transferring single-arrow occupancy to single-ray occupancy requires an arrow $\leftrightarrow$ ray coupling — the deficit $1'$ assignment rule [FT.125(4)];

the coupling's forced shape is derived: any $\hat{\omega}$-covariant coupling maps the single-arrow locus into a single chiral ray (the locus is one sector; $\hat{\omega}$ is the ray exchange [FT.122(4)]; machine: exhaustive on the 2! bijections),

and the complementary ray has occupancy zero at rung one — the vacuum sector of the doublet;

which of P_L/P_R names the occupied ray is reserved naming content [Def-H.33 canonicity Rmk] : the occupancy pair $(1, 0)$,

up to ray exchange, is this clause's entire content;

(6) rung scope — the force-side half opens at rung two: at rung one the static force matrix on the extended carrier is $\mathrm{F}_{\mathrm{ext}}(\mathrm{c})$

$= h(\mathrm{c}) \otimes I_\omega$ with F_{LR}

$$= F_{RL}$$

$= 0$ exactly [FT.127(9), FT.128(2) : the no-flip theorem; machine] : the two chiral sectors are force-decoupled; the flipping set at the derived locus

$$= \varnothing \text{ [FT.128(6)]};$$

at rung two EM activates [CT0.13] and the promoted core read opens the unique flip channel: $\mathrm{F_{ext}}(\mathrm{c}, q)$

$$= h(\mathrm{c}) \otimes I_2 + q$$

$$I_2 \otimes \hat{\kappa},$$

$$F_{LR} = q\, I_2 \neq 0 \iff q \neq 0 \quad \text{[FT.129(4)–(5), Def-H.43; EM.9 : } q \in \{-1, 0, +1\}];$$

$$\text{charpoly}\, \mathrm{F_{ext}}(\mathrm{c}, q)$$

$= \text{charpoly}_h(x-q) \cdot \text{charpoly}_h(x+q)$: the chiral-partner splitting $2|q| \in \{0, 2\}$ [FT.130(5); machine: Faddeev–LeVerrier over $\mathbb{Q}(i)$] : both chiral sectors become force-coupled for charged cores;

at rung three colour adds no carrier flip: sealed ends are blind [FT.117],

colour acts on the slot carrier [FT.127(7)],

the seed is colour-neutral by theorem [EW.32] : the flipping set remains $\{F_{\mathrm{EM}}$ core read$\}$ — minimal and unique [FT.129(5); machine: removing the $\hat{\kappa}$ term kills F_{LR} at every profile];

the arrow law of (2) is rung-blind [D1.2′s domains carry no rung argument] : what is rung-one specific is the coincidence of the two obstructions — no flip channel (force side) $\wedge$ single-arrow locus (ledger side);

the single-sector theorem is a rung-one theorem,

$\neg$ a universal law.

Req. A5, A6, A8 (Parts 1–2); D1.1, D1.2, D1.3, D1.4, D1.5, D2.6, D2.7, D2.11 (Parts 1–2); CT0.1, CT0.2, CT0.13 (Part 1); CT2.5 (Part 2); EW.32 (Part 2); EM.9 (Part 2); CT4.2 (Part 2); Def-H.4 (Tier F.2); FT.40 (Tiers F.12–F.13); FT.76, FT.79 (Tiers F.26–F.28); FT.88 (Tiers F.29–F.30);

FT.108 (Tier F.36); FT.117 (Tier F.38); Def-H.33, FT.122 (Tier F.39); Def-H.34, FT.127, FT.128, FT.129, FT.130 (Tier F.40); FT.133, FT.134 (Tier F.41); Def-H.4, Def-H.34, Def-H.43; exhaustive history enumeration, multinomial interleaving counts, exact polynomial resolvent identity in $\mathbb{Q}[w, x_+, x_-]/(w^{12})$, exact $\mathbb{Q}(i)$ block algebra, Faddeev–LeVerrier, exact rational $\log 2$ enclosures [math; machine: `checks/verify_H_rung1_neutrino.py` §D–§E].

Pf.

$$(1) : \text{ D2.11, D1.3, D1.4, D1.5, Def-H.4 verbatim (frozen)}$$

$$(2) : \text{ D1.2 : the move domains are } (0 \to 1) \text{ for } J^+, \ (1 \to 0) \text{ for } J^-;$$

$$\text{induction on prefixes: the state after any admissible prefix is } n_{\mathcal{R}}$$

$$= \#J^+ - \#J^- \in \{0, 1\},$$

$$\text{and the next move is forced } \implies \text{ strict alternation and the prefix law [machine: exhaustive at lengths } \leq 8]$$

$$\text{completed } \iff n_{\mathcal{R}} = 0 \iff \#J^+ = \#J^-;$$

$$\text{the completed single-pair word of length } 2m \text{ is unique: } (J^+J^-)^m \text{ [machine]};$$

$$\text{two-pair interleavings : each pair's projection is an alternating word;}$$

$$\text{completed counts at total length } 2n : \ \sum_{m_0+m_1=n} \tbinom{2n}{2m_0}$$

$$= 2,$$

$$8,$$

$$32 \ (2n = 2, 4, 6) \text{ [machine enumeration = multinomial closed form]};$$

$$\text{global balance at completion [machine]}$$

$$\text{dissolution-first chain } \implies \text{ prefix law violated at the first event of that edge [machine witness] } \implies \neg\text{admissible } \implies \ \perp \text{ in every valid universe [D2.11]};$$

$$\text{FT.108}(4) \text{ identity re-cited (frozen)}$$

(3) : premise inventory of (2) : {D1.2 domains, D1.3/D1.4/D2.11 closure} : no dimensionful or P4 quantity occurs [scan];

contrast rows : FT.76(3), FT.79(2) are read-range bounds [cited, not re-derived]

$$(4) : \Theta_{\mathfrak{L}}$$

$$= \text{reverse} + \text{label swap [D2.7; FT.133(1)]} : \text{involution [machine]};$$

completed admissible words biject onto themselves [machine: counts preserved at every length ≤ 8]

pure order reversal : every completed nonempty admissible word ends in J^- [state 0 is reached only via $1 \to 0$; machine: exhaustive] $\implies$ its reversal leads with J^- : inadmissible at the first event [(2) domains; machine]

$$\text{A6} : |\mathfrak{L}(\tau_{n+1})| = |\mathfrak{L}(\tau_n)| + n_{\text{real_out}}(\tau_n),\ n_{\text{real_out}} \geq 0 \implies \text{monotone non-decreasing};$$

the reversed reading of a multi-crystallization history strictly decreases somewhere, and every completed word's reversal is inadmissible from the start $\implies$ ¬forward-realizable [machine: exhaustive; $\neg\exists\, \tau^{-1}$ axiomatic]

$$\rho_\circ = \tfrac{1}{4} I_4 : \mathfrak{T} = 1,\ \text{charpoly} = (x - \tfrac{1}{4})^4 \text{ [machine over } \mathbb{Q}];$$

$$S_{\text{raw}}(\rho_\circ) = -4 \cdot \tfrac{1}{4} \ln \tfrac{1}{4} = \ln 4 = 2 \ln 2 \text{ [exact rational log enclosure, width } < 2^{-80}; \text{ machine}];$$

$$\text{dyadic witnesses} : S(\text{diag}(\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{8}, \tfrac{1}{8}))$$

$$= \tfrac{7}{4} \ln 2 < 2 \ln 2 \text{ [exact coefficient comparison; machine]};$$

$$\text{arg max uniqueness [Gibbs lemma; CT0.2]}$$

$$|E_{\mathcal{R}}|$$

$$= 0 \implies \#\{\text{tuples } (e_1, \ldots, e_k) \subseteq E_{\mathcal{R}}\}$$

$$= 0\ \forall k$$

$$\geq 1 \text{ [machine]} \implies \neg\exists \text{ closed chain [D1.4]} \implies \neg\exists \text{ terminus [Def-H.4]}$$

$$\hat{U}$$

$$= \left(\begin{smallmatrix} 0 & x_- \\ x_+ & 0 \end{smallmatrix}\right) \text{ [FT.108(4) generator at single-pair scope]} : \quad \langle \varnothing | \hat{U}^n | \varnothing \rangle$$

$$= [\, n \text{ even} \,]$$

$$(x_+ x_-)^{n/2} \text{ [machine to } n = 12\text{]};$$

$$(1 - w^2 x_+ x_-) \sum_n \langle \varnothing | \hat{U}^n | \varnothing \rangle w^n \equiv 1 \text{ [}\mathbb{Q}[w, x_+, x_-]/(w^{12});\text{ machine coefficient-wise]}$$

$$\deg_{x_+} = \deg_{x_-} \text{ on every nonzero monomial [machine census]};$$

$$\ker Q$$

$$= \text{span}\{(x_-, x_+)\} : \ (x_-, x_+)Q$$

$$= 0,$$

$$\det Q$$

$$= 0$$

$$\neq Q \implies \text{rank 1 [FT.108(4) re-derived; machine]}$$

$$(5) : \text{ FT.122(2)–(4) re-derived entrywise over } \mathbb{Q}(i) \text{ [machine]} : \ P_{L/R}^2$$

$$= P_{L/R},$$

$$P_L P_R$$

$$= 0,$$

$$P_L + P_R$$

$$= \mathrm{id},$$

$$\mathfrak{T}P_{L/R} \qquad = 1;$$

$$\hat{\omega}P_L\hat{\omega} = P_R;$$

$$|c\rangle = \tfrac{1}{2}(|L_u\rangle + |R_u\rangle) \text{ unnormalized [machine]}$$

arrow-sector census : the materialized locus $= \{|\,c\,\rangle\text{-sector}\}$ by (2);

$\hat{\omega}$-covariant bijections {arrow sectors} $\to$ {rays} : 2! candidates, each maps the single-arrow locus to one ray and its complement to the other [machine: exhaustive];

naming residue : Def-H.33 canonicity Rmk verbatim

(6) : FT.127(9), FT.128(2), (6) re-derived at 36 exact rational profiles : $F_{LR} = 0$ entrywise;

$$[\mathrm{F}_{\mathrm{ext}},\ I_2 \otimes \hat{\chi}] = 0 \text{ [machine]}$$

$\mathrm{F}_{\mathrm{ext}}(\mathrm{c}, q)$ blocks in the chiral carrier basis [Def-H.34′s S-transform; machine] : F_{LL}

$$= F_{RR}$$

$$= h,$$

$$F_{LR}$$

$$= F_{RL}$$

$= qI_2$ at $q \in \{-1, 0, 1\} \times 36$ profiles;

charpoly factorization by Faddeev–LeVerrier coefficient identity [machine]

minimality rows : removing $qI_2 \otimes \hat{\kappa}$ kills F_{LR} at every profile;

keeping only it leaves F_{LR}

$$= qI_2 \text{ [FT.129(5); machine];}$$

colour rows : FT.117, FT.127(7), EW.32 verbatim (frozen);

D1.2 carries no rung argument [verbatim scan]. ■

Phys. Ask the ledger which direction a real edge may be born in, and the answer is a domain, not a preference: dissolution presupposes crystallization, so every materialized history is crystallization-first on every edge, and the prefix law is exhaustive at every length checked. A backward-arrow fermion is therefore not heavy, not suppressed, not decoupled — it is ungrammatical: no ledger history spells it, at any value of any parameter. What the reversal does spell is bookkeeping: the formal T-images and the balanced fluctuation pairs of the all-virtual state, which carry no registration, no chain, no terminus and no accessible pole. On the carrier the consequence is an occupancy statement — one sector materializes, its complement is vacuum — and because the orientation rays superpose both chiral rays equally, the last step from arrow to ray is exactly the missing assignment rule; the theorem fixes its shape and its output, $(1, 0)$ up to exchange, and leaves the naming where the corpus has always left it. At rung two the force side opens: the promoted core read is the unique flip channel, and the coincidence that makes rung one single-sector does not recur.

3.109 (FT.145 — The rung-one completion).

Thm.

(1) the content summary: the rung-one fermion sector consists of: one family channel [FT.142–FT.143 : $\delta_\nu = \frac{1}{8}$, triplet $r_i^{(\nu)}$, ratios $1 : 9.58\ldots : 272.0\ldots$, $Q = \frac{2}{3}$];

its chirality structure [FT.141 : the complete-force $10/6$ grading, no S-neutral observable, the weak selectivity a state-locus shadow];

its occupancy structure [FT.144 : single-arrow materialization, single-ray under any $\hat{\omega}$-covariant coupling, complement vacuum];

(2) the mass read:

$$m_{\text{hid}} = E_{\text{int}} \ln 2/\mathfrak{h}$$

[FT.16; FT.44(4) : universal across every computed floor; Def-H.9(2) : $m_{\text{hid}} = I_{\text{hid}}/\mathfrak{h}$];

the minimal rung-one bound chain — the digon, $E_{\text{int}} = 2$ [FT.28(1)] — reads $\quad m_{\text{hid}} = \dfrac{2\ln 2}{\mathfrak{h}}$

[machine: exact integer coefficient];

the channel ratios of FT.143 are dimensionless with overall scale μ_ν undeclared (P4);

the relating equation between the channel read and the m_{hid}/pole/sector reads is undeclared [FT.113 Status pattern];

amplitude $\neq$ mass; rung $\neq$ scale [standing rules];

(3) the completeness fraction:

$$J_3^+ + J_3^- + G_5 + W_3 + H_1 + D_1^+ = 16 = C_{\text{cap}} = \dim \mathcal{K}_1$$

[CT0.12, CT0.13, CT0.14; machine] :

$\mathcal{A}(1)$ accounts for every pair-channel dimension — no part of the rung-one channel is forceless and no force overruns it [CT0.12 Phys pattern];

the rung-one fermion sector carries: ¬EM (activation at rung 2 [CT0.13]; unactivated in the channel class [FT.142(1)]),

¬colour (rung 3 [CT0.13]; structurally colourless [FT.142(1)]),

¬Yukawa sector (deficit 2 stands),

¬mixing matrix (deficit 4 stands),

¬generation selection (deficit 3 stands; item $21(d)$);

(4) the bridge to rung two:

$\mathcal{K}_1 \subset \mathcal{K}_2$ (strict),

$$\dim \mathcal{K}_2 = 17 = \dim \mathcal{K}_1 + 1$$

[CT0.14 : the EM centre is the new summand; machine];

at rung two EM activates [CT0.13] and the flip channel opens,

$$F_{LR}$$

$= qI_2$ [FT.129(5), Def-H.43; FT.144(6)] : both chiral sectors become force-coupled;

the $10/6$ split persists on $\mathcal{K}_1$: the EM summand is a stack-register block [CT0.12 Rmk; CT0.14],

¬ a pair-channel direction : the $\mathrm{Ad}(\mathbb{S})$ census of the pair 16 is unchanged;

the ℓ-channel angle δ_ℓ

$= \frac{2}{9}$ [FT.112 — Tier F.37, under Def-H.41's interface definition; cited as frozen channel datum, imported as no premise of this part] governs the rung-two family ratios;

the two angles are distinct rationals sharing the numerator 2 [FT.143(5)];

(5) the adjudication ledger:

Statement	Source	Status
$10/6 = \mathrm{Ad}(\mathbb{S})$ census on $\mathcal{K}_1$	FT.141(1)	Derived (FT.136(1) re-derived)
Chirality = the complete-force grading	FT.141(2)	Derived
Deletion invariance; $45/64$ nontrivial subsets	FT.141(3)	Derived (exhaustive)
Weak branch selectivity an S-grading consequence	FT.141(4)	Derived ($\sum_i W_i^2 = E_0 + \mathbb{S}$)
No S-neutral observable at rung one	FT.141(5)	Derived
S-grading vs carrier $\hat{\chi}$: independent $\mathbb{Z}_2$'s	FT.141(6)	Derived (FT.126(4)/FT.136(4) re-run)
Arena $\mathcal{K}_1$, dim = 16; candidates $10/9/27/6$ rejected	FT.142(2)–(3)	Derived (no declaration)
Seal coefficient 2, undressed	FT.142(4)	Derived (FT.109(3) re-derived)
$\delta_\nu = 1/8$	FT.142(5)	Derived
$r_i^{(\nu)}$; ratios $1 : 9.58\ldots : 272.0\ldots$; $Q = \frac{2}{3}$	FT.143	Derived (scale μ_ν: P4)
$3\delta_\nu = \frac{3}{8} \notin \{\delta_\ell, \delta_q\}$; shared numerator 2	FT.143(2), (5)	Derived (exact rationals)
Forward arrow $J^+ \to J^-$; strict alternation; prefix law	FT.144(2)	Derived (exhaustive)
Backward materialization excluded; $\neq g$a mass statement	FT.144(2)–(3)	Derived
Vacuum residence; balanced fluctuation pairs; no accessible pole	FT.144(4)	Derived
Single-ray occupancy under $\hat{\omega}$-covariant coupling	FT.144(5)	Derived (coupling-conditional; naming reserved)

Statement	Source	Status
Flip channel opens at rung two	FT.144(6)	Derived (under Def-H.43)
$m_{\text{hid}}(\text{digon}) = 2\ln 2/\mathfrak{h}$	FT.145(2)	Derived (FT.28(1) re-cited)
Completeness $3{+}3{+}5{+}3{+}1{+}1 = 16 = C_{\text{cap}}$	FT.145(3)	Derived
Overall scale μ_ν; relating equation	—	Free (P4); open
Species identification of the triplet	—	NOT derived (barred: P1, P5)
Arrow$\leq ftrightarrow$ray assignment rule	—	Application-layer content

(6) the scope bars: no particle identification — the channel subscript ν is ledger vocabulary [FT.142 Rmk];

no external data ([EXT] enters no clause of this document — this standing-rule record is its sole marker);

no new declaration issues: imports are confined to Def-H.4,

Def-H.34 (extended carrier),

Def-H.43 (read promotion — rung-two bridge rows only);

Def-H.41/Def-H.42 are cited in the rejection rows and imported nowhere;

no fitted parameters (one scale $\mathfrak{h}$, P1);

all topology labels are internal bookkeeping;

no H-sector interpretation enters any theorem statement.

Req. FT.141–FT.144 (this tier); FT.16, Def-H.9 (Tiers F.1–F.11); FT.28 (Tiers F.1–F.11); FT.44 (Tiers F.14–F.16); CT0.12, CT0.13, CT0.14 (Part 1); FT.112 (Tier F.37, frozen-datum citation); FT.129 (Tier F.40); P1, P4, P5 (Parts 1–2); exact integer arithmetic, exact rational $\log 2$ enclosures, exact ranks [math; machine: `checks/verify_H_rung1_neutrino.py` §F].

Pf.

(1) : assembly of FT.141–FT.144's adjudicated rows [machine: all sections green]

(2) : FT.44(4) : $m_{\text{hid}} = E_{\text{int}} \ln 2/\mathfrak{h}$ across every computed floor;

FT.28(1) : E_{int}

$= 2$ (both cycle edges internal)

$$\Longrightarrow\ m_{\text{hid}}$$

$$= 2\ln 2/\mathfrak{h}\ \text{[exact integer coefficient; machine]};$$

$$\mu_\nu :\ \neg\exists\ \text{tag fixing it [P4]}$$

$$\text{relating equation} :\ \neg\exists\ \text{tag relating the channel ratios to } m_{\text{hid}},$$

$$m^2_{\text{pole}}\ \text{or}\ m^2_B(q,S)\ \text{[inventory scan; FT.113 Status]}$$

$$(3) :\ 3+3+5+3+1+1=16\ \text{[CT0.12 table; machine]};$$

$$\dim\mathcal{K}_1 = 16 = C_{\text{cap}}\ \text{[CT0.14; CT0.6]};$$

$$\neg\text{-rows} :\ \text{CT0.13 activation rungs verbatim};$$

$$\text{FT.142}(1)\ \text{channel class};$$

$$\text{deficits } 2/3/4 :\ \neg\exists\ \text{Yukawa, flavour-index or mixing-matrix tag [inventory scan]}$$

$$(4) :\ \dim\mathcal{K}_2 = \dim\mathcal{K}_1 + \dim Z(\mathfrak{u}(2)) = 16+1 = 17\ \text{[CT0.14; machine: strict flag } 6\subset 16\subset 17\subset 25];$$

$$\text{EM summand :\ stack register [CT0.12 Rmk]}\ \Longrightarrow\ \text{Ad}(\mathbb{S})\ \text{on the pair } 16\ \text{untouched [FT.141}(1)\ \text{census unchanged]}$$

$$\text{flip row :\ FT.144}(6)\ \text{verbatim};$$

$$\delta_\ell\ \text{row :\ FT.112 value cited with its interface provenance [Def-H.41 — Tier F.37's; } \neg\text{imported here]}$$

$$(5) :\ \text{each row is the adjudication of its own tag :\ FT.141 } (\S\text{A}),$$

$$\text{FT.142 } (\S\text{B}),$$

$$\text{FT.143 } (\S\text{C}),$$

$$\text{FT.144 } (\S\text{D}\text{–}\S\text{E})\ \text{[machine: all sections green]}$$

(6) : standing-rule record;

import scan : the Req lines of this part name {Def-H.4, Def-H.34, Def-H.43} as imports and {Def-H.41, Def-H.42} as cited-not-imported [machine scan]. ■

Phys. Rung one closes as a sector: sixteen dimensions, every one spoken for by the five pretime aspects; one family channel with its own angle, its own cubic and the universal two-thirds; one chirality, belonging to the whole force complex rather than to any of its members; and one occupancy law that leaves a single sector materialized and its partner empty — not heavy, empty. The digon fixes the informational read at two bits over the single scale; the family ratios remain dimensionless pending the relating equation the ledger still owes. At the next rung the mask grows by exactly one dimension, the flip channel opens with the core charge, and the sector that was vacuum at rung one becomes force-reachable: the completion is a floor, not a ceiling.

3.44 Tier F.44 — The 3+1 Chiral Asymmetry: The Manifold Locus of Chirality, the Spatial Provenance of the Channel Angle, and the Emergent-Time Identity

Theorem 3.110 (FT.156 — The 3+1 asymmetry theorem: one signature datum at two loci, and the manifold structure as the chiral structure). Thm. On the pair channel $\mathfrak{H}(\mathbb{C}^4)$ and the emergent geometry [CT2.6; conditions (ii)–(iii) inherited at the point of use — A §0.2a]:

(1) one signature datum, two loci:

A3 declares the register involution; C13 fixes the Lorentz-signature representative:

$$T(\sigma_\mu \otimes \sigma_\nu) = \eta^\mu \eta^\nu \, (\sigma_\mu \otimes \sigma_\nu), \quad \eta = (-,+,+,+),$$

$$T^2 = \mathbb{1};$$

CT1.12 derives the signature of the emergent metric: $\mathrm{sig}(g_{\mu\nu}) = \mathrm{sig}(\bar{g}_{\mu\nu}) = (-,+,+,+)$

$$[\varkappa \|q\|_{\mathrm{op}} < 1 \text{ regime}];$$

the same $(-,+,+,+)$ sign pattern grades the register basis [C3 : $\mathcal{E}_{\mathrm{st}}$] and the metric on M [CT2.6] : one temporal sign and three spatial signs at each locus [machine: the T-table on the 16-element CT1.2 basis via exact Pauli-tensor decomposition over $\mathbb{Q}(i)$];

(2) the block census is the component census; the $7/3$ T-split is the $3{+}1$ split of the dictionary:

$$(7, 3, 3, 3) \text{ [CT1.1]};$$

$$S\text{-even} = 10 = 7 + 3 \text{ [CT1.2; FT.141(1)]};$$

$$S\text{-even} = \text{the reality-preserving sector } ([\mathbb{S}, X] = 0 \text{ preserves } \vee^2,\ \wedge^2 \text{ [CT0.7(3)]});$$

$$S\text{-odd} = 6 = \{J_i\ (3),\ K_i\ (3)\} = \text{the reality-exchanging sector (the seals [CT0.7(3)–(4); D1.2])};$$

within the S-even block:

$$T\text{-even } 7 = \{E_0,\ \Sigma,\ \hat{E}^{ij}\ (5)\} = \{D^+,\ H,\ G\} \text{ [CT0.12; CT1.10 : } 1 \oplus 5 \oplus 1];$$

$$T\text{-odd } 3 = \{W_i\} = F_W;$$

the dictionary [D2.1, CT1.12] :

$$(a_{00},\ q_{ij})$$

$$= (1{+}6)\ T\text{-even coefficients } \longmapsto \ \left(\Omega = e^{\alpha_\Omega a_{00}},\ \ \bar{g}_{ij} = \delta_{ij} + \varkappa q_{ij}\right)$$

$$= \text{lapse/conformal } + \text{ spatial metric};$$

$$w_i\ (3)\ T\text{-odd coefficients } \longmapsto \ N_i = \beta w_i = \text{shift} = \text{the time–space cross components } \bar{g}_{0i};$$

$$\text{components:} \quad g_{00} : 1; \quad g_{ij} : 6 = \tfrac{3\cdot 4}{2}; \quad g_{0i} : 3; \quad \text{total } 10 = \tfrac{4\cdot 5}{2} \text{ [CT1.12]};$$

the temporal direction contributes exactly one component and one coefficient ($g_{00} = -\Omega^2$ from the single T-even conformal coefficient a_{00});

the spatial triad contributes the symmetric sextuple q_{ij} and the three shifts w_i :

the $7/3$ T-split of the S-even block is the operator-side encoding of the $(1; 6; 3)$ component split with signature $(-, +, +, +)$;

the S-odd block has 0 metric coordinates [FT.141(5); FT.30 dictionary];

(3) the reversal transport: the T-grading transports through the dictionary exactly to the time-reversal pullback: with J

$$= \mathrm{diag}(-1,1,1,1) \text{ and the } T\text{-action on sector data } (a_{00}, q, w)$$

$$\mapsto (a_{00}, q, -w)$$

$$[T(E_0) = E_0,\ T(E^{ij}) = E^{ij},\ T(W_i) = -W_i;\ \text{machine, entrywise}]:$$

$$\bar{g}_{\mu\nu}(a_{00}, q, -w) = J^{\rho}{}_{\mu}\, \bar{g}_{\rho\sigma}(a_{00}, q, w)\, J^{\sigma}{}_{\nu}$$

[machine: exact rational witnesses];

lapse and spatial metric fixed, shift negated: $\Omega \mapsto \Omega, \quad \bar{g}_{ij} \mapsto \bar{g}_{ij}, \quad N_i \mapsto -N_i$

$$\Longleftrightarrow$$

$g_{00},\ g_{ij}$ fixed, $\quad g_{0i} \mapsto -g_{0i}$ [Ω depends on the T-even a_{00} alone [D2.1]];

the signature is reversal-invariant: $\quad \mathrm{sig}\,\bar{g}(w) = \mathrm{sig}\,\bar{g}(-w) = (-,+,+,+)$

[machine: exact rational LDL$^{\top}$ inertia on a 12-point witness lattice (q,w); Sylvester criterion on $\bar{g}_{ij}$; $\det\bar{g} < 0$ via the CT1.12 Schur form];

(4) space is symmetric and reversible: the spatial triad rotates innerly: $\quad \mathrm{span}\{W_i\} \cong \mathfrak{su}(2)$ [CT1.4];

$$J, K:\ \text{vectors}, \quad \hat{E}^{ij}:\ 5, \quad E_0, \Sigma:\ 1\ \text{ under the adjoint action [CT1.8, CT1.10, CT1.5]};$$

every permutation of the three triad indices is realized by an inner unitary that is a rational polynomial in the weak generators alone:

$$\text{cyclic:} \quad U = u \otimes u,\ u = \tfrac{1}{2}\big(I_2 - i(\sigma_1+\sigma_2+\sigma_3)\big)\ [\text{FT.111}]:$$

$$U = \exp\!\big(-\tfrac{2\pi i}{3}\,\hat{n}\cdot W\big) = E_0 - \tfrac{1}{2}\mathcal{W}^2 - \tfrac{i}{2}\mathcal{W}, \quad \mathcal{W} := W_1{+}W_2{+}W_3, \quad \hat{n} = \tfrac{(1,1,1)}{\sqrt{3}},$$

via the polynomial identity $\mathcal{W}^3 = 3\mathcal{W}$ [spec$(\hat{n}\cdot W) = \{1,0,-1,0\}$; machine];

$$U^3 = E_0;$$

$$UX_iU^\dagger$$

$$= X_{i+1} \ \forall X \in \{W, J, K\} \text{ (indices mod 3) [FT.111(1) re-derived, plus the } W\text{-row; machine];}$$

$$\text{transpositions:} \quad U_{ij} = \exp\!\big(-i\pi\,\tfrac{W_i+W_j}{\sqrt{2}}\big) = E_0 - (W_i + W_j)^2 :$$

$$U_{ij}^2 = E_0;$$

$$U_{ij} : \ X_i \leftrightarrow X_j,$$

$$X_k$$

$$\mapsto -X_k \ \forall X \in \{W, J, K\} \text{ [machine: the full conjugation table on } \{W, J, K, \hat{E}\}];$$

$$\text{the reversal involutions are exact: on the register } [S^2 = T^2 = \mathbb{1} \text{ [A3]}],$$

$$\text{on configuration data } [\langle C, P, T, \omega\text{-pattern}\rangle \cong \mathbb{Z}_2^4, \text{ register-disjoint [FT.133(1)–(2)]}];$$

$$(5) \quad \text{time is asymmetric and strictly forward: } \neg\exists\,\tau^{-1} \text{ [A6 : } \tau_n \to \tau_{n+1} : \ \rho \mapsto \mathcal{T}(\rho)];$$

$$|\mathfrak{L}(\tau_{n+1})| = |\mathfrak{L}(\tau_n)| + n_{\text{real_out}}(\tau_n) \geq |\mathfrak{L}(\tau_n)| \text{ [A6, A7 : } \mathfrak{L}(\tau_n) \subseteq \mathfrak{L}(\tau_{n+1})];$$

the D1.2 seal domains force strict alternation and the prefix law; every completed nonempty history ends in J^-,

and its pure order reversal is inadmissible at the first event [FT.144(2), (4)(ii); machine: exhaustive at lengths ≤ 8];

$\Theta_{\mathfrak{L}}$ [D2.7] is the formal mirror — an involution on configuration data bijecting the completed admissible set onto itself [FT.144(4)(i); machine] — ¬a dynamical inverse:

the asymmetry is exact: the spatial-axis reversals are inner unitaries on the force arena [(4)];

the temporal advance reversal has no implementation [A6; FT.144(3) : a domain restriction, prior to every read];

(6) the chiral structure is the manifold structure: the chirality of the complete force complex at rung one is the $10/6$ S-grading [FT.141(2)];

its (S,T) refinement $(7,3,3,3)$ [CT1.1] is the dictionary carrier of the metric [CT1.12; (2)];

the weak block is the T-odd part of the S-even block

= the metric's own time–space cross block [CT0.12 : $F_W = \mathcal{E}_{+-} = \text{span}\{W_i\}$; D2.1 : $X_{(-)} = w_i W_i$; CT1.12 : $w_i \mapsto N_i$];

the chiral asymmetry $(10 \neq 6)$ and the $3{+}1$ asymmetry are one graded structure read at three derived loci:

the operator locus $(10,6) = (7{+}3,\ 3{+}3)$;

the component locus $(1;6;3)$ with $\text{sig} = (-,+,+,+)$;

the state locus $\text{spec}\,\mathbb{S} = (3,1)$ [CT0.7(1); FT.157];

locus discipline [FT.136(5); FT.141(4)] : three distinct spaces $(\text{End}(\mathbb{C}^4_{\text{pair}}),$ the metric on $M,$ $\mathbb{C}^4_{\text{pair}})$;

the counts are the derived content; no carrier identification is made.

Rmk (label discipline). The labels spatial/temporal of this tier are internal bookkeeping attached to derived T-parities and dictionary roles [CT1.2's table; CT1.12's table]; they name no particle and no external structure [P1, P5]. Naming either branch left or right remains barred [FT.101 Phys bars stand]. All topology and branch labels are internal bookkeeping; no H-sector interpretation enters any clause.

Req. A3, A6, A7, A8 (Part 1); C3 (Part 1); D1.2, D2.1, D2.7 (Part 1); CT0.7, CT0.12, CT1.1, CT1.2, CT1.4, CT1.5, CT1.8, CT1.10 (Part 1); CT1.12, CT2.6 (Part 2; CT2.6's conditions (ii)–(iii) at the point of use — A §0.2a); FT.30 (Tiers F.1–F.11); FT.101, FT.111 (Tiers F.34, F.37); FT.133 (Tier F.41); FT.136, FT.141, FT.144 (Tiers F.41–F.42); exact $\mathbb{Q}(i)$ matrix arithmetic (Pauli-tensor decomposition, conjugation tables, polynomial rotation identities), exact rational $\text{LDL}^\top$ inertia, Sylvester/Schur witnesses, exhaustive ledger-word enumeration [math; machine: `checks/verify_H_3p1_chiral_asymmetry.py` §A].

Pf.

(1) : $\eta = (-,+,+,+)$ verbatim [A3, C13];

$\mathrm{sig}(g) = (-,+,+,+)$ verbatim [CT1.12 : Schur + Sylvester + conformal covariance];

$$T(X)$$

$$= \sum_{\mu,\nu} c_{\mu\nu}$$

$$\eta^{\mu}\eta^{\nu}$$

$$\sigma_{\mu} \otimes \sigma_{\nu},$$

$$c_{\mu\nu}$$

$= \frac{1}{4}\mathfrak{T}\big((\sigma_{\mu} \otimes \sigma_{\nu})X\big)$ [exact decomposition, unique on the orthogonal Pauli-tensor basis [CT1.1 Pf pattern]; machine on the 16-basis]

(2) : CT1.2 table verbatim : (S,T) grades $(1{+}1{+}5,\ 3;\ 3;\ 3)$;

$7 = 1{+}1{+}5$ [CT1.10];

dictionary rows verbatim [CT1.12 table] : $a_{00} \mapsto \Omega$, $\Sigma_{\mathrm{tr}} \mapsto$ volume, $\hat{h}_{ij} \mapsto$ shear, $w_i \mapsto$ shift

$q_{ij} = q_{ji}$: 6 components [D2.1];

$1 + 6 + 3 = 10 = \frac{4\cdot 5}{2}$ [CT1.12];

$g_{00} = \Omega^2 \bar{g}_{00} = -\Omega^2$ [D2.1 : $\bar{g}_{00} = -1$];

S-odd : 0 metric coordinates [FT.141(5); FT.30];

reality-preserving/exchanging : CT0.7(3) eigen-relations verbatim

(3) : $\bar{g}_{0i} = \varkappa\beta w_i$ [D2.1 : $N_i = \beta w_i$] $\implies$ $\bar{g}_{0i}(a_{00}, q, -w) = -\bar{g}_{0i}(a_{00}, q, w)$;

$\bar{g}_{00} = -1$, $\bar{g}_{ij} = \delta_{ij} + \varkappa q_{ij}$ are w-free $\implies$ fixed;

$$(J^\top \bar{g} J)_{00}$$

$$= \bar{g}_{00},$$

$$(J^\top \bar{g} J)_{ij}$$

$$= \bar{g}_{ij},$$

$$(J^\top \bar{g} J)_{0i}$$

$$= -\bar{g}_{0i} \text{ [machine: exact equality on the witness lattice]}$$

$$\text{inertia : LDL}^\top \text{ over } \mathbb{Q} \text{ on } \bar{g}(q,w) : \ D \text{ signs } (-,+,+,+) \text{ on all } 12 \text{ witnesses and at } w$$

$$\mapsto -w \text{ [machine];}$$

$$\det \bar{g}$$

$$= \det(\bar{g}_{ij}) \cdot \left(-1 - \varkappa^2 N_i (\bar{g}^{-1})^{ij} N_j\right) < 0 \text{ [CT1.12 Pf re-run; machine: exact } 4 \times 4 \text{ determinants]}$$

$$(4) : \ [W_i, W_j] = i\epsilon_{ijk} W_k \text{ [CT1.4 re-derived, all pairs; machine];}$$

$$[W_i, J_j] = i\epsilon_{ijk} J_k, \quad [W_i, K_j] = i\epsilon_{ijk} K_k \text{ [CT1.8 re-derived, all 9+9 pairs; machine];}$$

$$[W_a, \hat{E}^{ij}] = i(\epsilon_{aik}\hat{E}^{kj} + \epsilon_{ajk}\hat{E}^{ik}), \quad [W_i, \Sigma] = 0 \text{ [CT1.10, CT1.5 re-derived; machine]}$$

$$\mathcal{W}^3$$

$$= 3\mathcal{W} \text{ [machine]}$$

$$\Longrightarrow \ X^3$$

$$= X \text{ at } X$$

$$= \hat{n} \cdot W$$

$$= \mathcal{W}/\sqrt{3} \implies \exp(-i\theta X)$$

$$= E_0 + (\cos\theta - 1)X^2 - i\sin\theta$$

X [polynomial identity on $\operatorname{spec}\{1, 0, -1\}$];

$$\theta = \tfrac{2\pi}{3} : \cos\theta - 1 = -\tfrac{3}{2}, \ \sin\theta \cdot \tfrac{1}{\sqrt{3}} = \tfrac{1}{2} \implies U = E_0 - \tfrac{1}{2}\mathcal{W}^2 - \tfrac{i}{2}\mathcal{W}$$

$U = u \otimes u$ entrywise [FT.111 form; machine];

$$U^3 = E_0, \ UU^\dagger = E_0, \ \det u = 1 \text{ [machine]};$$

$$UX_iU^\dagger = X_{i+1}, \ X \in \{W, J, K\} \text{ [machine, all 9]};$$

$$U\hat{E}^{ii}U^\dagger = \hat{E}^{i+1,i+1} \text{ [machine]}$$

$$U_{ij} = E_0 - (W_i + W_j)^2 \ [\theta = \pi : \cos\pi - 1 = -2, \ X = (W_i + W_j)/\sqrt{2}, \ X^3 = X];$$

$$U_{ij}^2 = E_0, \ U_{ij}U_{ij}^\dagger = E_0 \text{ [machine]};$$

conjugation tables: all 3 transpositions $\times \{W, J, K\}$ and the $\hat{E}$-action [machine, exact]

$$(5) : \text{ A6/A7 verbatim};$$

prefix law, uniqueness of $(J^+J^-)^m$,

inadmissibility of pure reversals [FT.144(2), (4) machine re-run: exhaustive at lengths ≤ 8];

$$\Theta_{\mathfrak{L}}^2 = \text{id}, \text{ bijection on the completed set [FT.144(4)(i) re-run; machine]}$$

$$(6) : \text{ conjunction of } (1)\text{–}(5) \text{ with FT.141(2), CT0.12, CT0.7(1)};$$

locus rows verbatim [FT.136(5); FT.141(4)]. ■

Phys. The corpus declares one sign pattern, $(-,+,+,+)$, at the register [A3's T; C13] and derives the same sign pattern for the emergent metric [CT1.12] — one datum at two loci. The ten reality-preserving dimensions split $7/3$ under that grading, and the split is exactly the metric's: the seven T-even coefficients build the lapse and the spatial metric, the three T-odd coefficients build the shift — the time–space cross terms. The grading transports through the dictionary as the time-reversal pullback: lapse and spatial metric stand, the shift flips, the signature stands. Space shows its symmetry as inner rotations — every triad permutation is a rational polynomial in the weak generators — and its reversibility as exact involutions; time shows its asymmetry as the absence of any inverse for the tick advance, the monotone ledger, and the inadmissibility of every reversed history. Chirality is the same structure: the $10/6$ split of the complete force complex, refined by the signature grading, with the weak block sitting precisely in the metric's time–space sector.

Theorem 3.111 (FT.157 — The weak-selectivity theorem: the weak force is the shift block — the spatial force on the spatial branch). Thm. With $\mathbb{S}$, $\Pi_{\mathcal{R}} = \frac{1}{4}(3E_0 + \Sigma)$, $\Pi_{\mathcal{V}} = \frac{1}{4}(E_0 - \Sigma)$ [CT0.7(1)], $W_i = \frac{1}{2}(\sigma_0 \otimes \sigma_i + \sigma_i \otimes \sigma_0)$ [CT1.2], $\Phi = \{0, \pi\}$ [D0.5, CT0.9]:

(1) the weak block is the shift block:

$$F_W = \mathrm{F}_{(-,\mathcal{R})} = \mathcal{E}_{+-} = \mathrm{span}\{W_i\} \quad [\text{A8; CT0.7(4); C8 : } F_W = F_{W^+} \oplus F_{W^-} \oplus F_Z];$$

the same three generators carry the metric shift [D2.1 : $X_{(-)} = w_i W_i$; CT1.12 : $w_i \mapsto N_i$] :

$$g_{0i} = \varkappa\beta\,\Omega^2\, w_i \text{ [D2.1] :}$$

the weak aspect is the metric's time–space coupling block — one 3-dimensional block, two derived readings [machine: sector membership of every basis element + the dictionary row count];

(2) the weak acts on the spatial orientations and has no temporal one to act on:

$$W_i = \tfrac{1}{2}\big(\sigma_i^{(p)} + \sigma_i^{(q)}\big) = \text{total spin [CT1.2; FT.13]};$$

$$[W_i, W_j] = i\epsilon_{ijk} W_k \text{ [CT1.4]};$$

J, K : vectors under $\mathrm{span}\{W\}$ [CT1.8]; the triad index i is the rotation index [FT.156(4)];

$$\textstyle\sum_i W_i^2 = \frac{3}{2}E_0 + \frac{1}{2}\Sigma = E_0 + \mathbb{S} = 2\,\Pi_{\mathcal{R}} \text{ [Cor CT0.7a; FT.141(4)] :}$$

Casimir $2 = 1(1{+}1)$ on $\vee^2$ — the spin-1 (vector) read;

$$W_i \Pi_{\mathcal{V}} = 0 = \Pi_{\mathcal{V}} W_i \; \forall i,$$

$$\bigcap_i \ker W_i = \operatorname{im} \Pi_{\mathcal{V}} \text{ [FT.101(1)–(2) re-derived; machine] :}$$

Casimir 0 on $\wedge^2$ — the spin-0 (scalar) read, no orientation;

the derived dimension match [CT0.7(1) : $\mathfrak{T}\Pi_{\mathcal{R}} = 3 = \dim \vee^2$, $\mathfrak{T}\Pi_{\mathcal{V}} = 1 = \dim \wedge^2$; CT1.12 : 3 positive, 1 negative eigenvalue] :

spatial branch := the aligned branch ($\alpha = 0$, $\vee^2$, Casimir 2) — weak-active;

temporal branch := the antialigned branch ($\alpha = \pi$, $\wedge^2$, Casimir 0) — weak-blind

[A4; FT.101];

the labels record the derived matches $\dim \vee^2$

$= 3$

$=$ #{spatial directions} and $\dim \wedge^2$

$= 1$

$=$ #{the temporal direction} [FT.156(2)–(3)];

they are internal bookkeeping [Rmk below];

(3) the selectivity is the $3{+}1$ asymmetry at the state locus:

the weak sees the spatial branch and annihilates the temporal branch [FT.101(1)–(2); machine];

its support is fixed by the S-grading ($\mathbb{S}$ enters the Casimir identity literally [FT.141(4)]) and refined by the T-grading: the weak support is the S-even T-odd block

$=$ the sector coupling the temporal register direction to the spatial triad [FT.156(2)–(3)];

the branch flip $\iota(\alpha)$

$= \pi - \alpha$ has no Hilbert-space implementation ($\mathfrak{TI}_{\mathcal{R}} = 3 \neq 1 = \mathfrak{TI}_{\mathcal{V}}$; rk $3 \neq 1$) [FT.101(3) re-derived] : the one-sidedness is exact on the pair register;

(4) the carrier-side ledger, unchanged: on the extended carrier the weak acts seed $\otimes$ trivial and commutes with the chiral projectors [FT.126(2) : blindness, ¬coupling];

the four gradings — the algebra S-grading, the branch-label parity, the carrier $\hat{\chi}$-grade, the Janus arrow — remain pairwise distinct [FT.136; FT.141(6); FT.144(5)];

this tag adds the manifold-side read of the branch-label $\mathbb{Z}_2$ (spatial = aligned = weak-active; temporal = antialigned = weak-blind) : a force-side input to the deficit $1'$ assignment rule [FT.125(4); FT.141(6)];

the rule coupling the gradings to species is ¬constructed here; no species is assigned;

(5) the one-sidedness, answered:

space: 3-dimensional [CT1.12 : three positive eigenvalues; FT.156(2) : the triad counts],

triad-symmetric [FT.156(4) : S_3 inner],

reversible under exact involutions [A3; FT.133(1)];

time: 1-dimensional [CT1.12 : the single negative eigenvalue] and strictly forward [A6 : $\neg\exists\,\tau^{-1}$; FT.144(2)–(3) : no backward materialization];

the weak coupling is the triad-indexed vector coupling [CT1.2 : W_i, $i = 1, 2, 3$; CT1.8] : it couples to the reversible 3-dimensional sector, and the 1-dimensional strictly-forward sector carries no orientation object for it to act on [$\wedge^2$: spin 0; $W_i\Pi_{\mathcal{V}} = 0$] :

the chiral asymmetry of the weak is the $3{+}1$ asymmetry of the arena.

Rmk (naming and identification bars). The labels spatial branch/temporal branch attach to the A4 branch and swap-eigenspace loci only — internal bookkeeping for the derived dimension and Casimir matches. Naming either branch left or right is barred [FT.101 Phys bars stand]; the carrier-ray names P_L/P_R remain algebraic up to the application-layer assignment rule [Def-H.33 canonicity Rmk]; no SM species reading; no claim that the SM weak interaction is reproduced [P1, P5].

Req. A3, A4, A6, A8 (Part 1); C8 (Part 1); D0.5, D2.1 (Part 1); CT0.7, Cor CT0.7a, CT0.9, CT0.12, CT1.2, CT1.4, CT1.8 (Part 1); CT1.12 (Part 2); FT.13 (Tiers F.1–F.11); FT.101 (Tier F.34); FT.125, FT.126 (Tier F.39); FT.133 (Tier F.41); FT.136, FT.141, FT.144 (Tiers F.41–F.42); Def-H.33, Def-H.34 cited within the frozen FT.126/FT.144 clauses; FT.156 (this tier); exact $\mathbb{Q}(i)$ arithmetic, rank/trace invariants, Casimir witnesses on exact rational states [math; machine: `checks/verify_H_3p1_chiral_asymmetry.py` §B].

Pf.

$$(1):\ \text{A8/CT0.7}(4)\ \text{block rows verbatim;}$$

$$\text{D2.1}:\ X_{(-)} = w_i W_i;$$

$$\text{CT1.12 table row } w_i \mapsto N_i = \beta w_i;$$

$$g_{0i} = \Omega^2 \bar{g}_{0i} = \varkappa\beta\Omega^2 w_i\ [\text{D2.1}];$$

$$\text{machine: } W_i \in \mathcal{E}_{+-} \text{ on the } T\text{-table [FT.156(1)]},\ \#\{T\text{-odd } S\text{-even basis elements}\} = 3$$

$$(2):\ \text{CT1.2/FT.13},\ \text{CT1.4},\ \text{CT1.8 verbatim;}$$

$$\textstyle\sum_i W_i^2 = \frac{3}{2}E_0 + \frac{1}{2}\Sigma\ [\text{Cor CT0.7a re-derived;\ machine}];$$

$$\Sigma = 2\mathbb{S} - E_0\ [\text{CT0.7}(1)] \implies \tfrac{3}{2}E_0 + \tfrac{1}{2}\Sigma = E_0 + \mathbb{S} = 2\Pi_{\mathcal{R}}\ [\text{machine}]$$

$$\vee^2:\ \mathbb{S} = +1 \implies \text{Casimir } 2;$$

$$\wedge^2:\ \Sigma|_{\wedge^2} = -3 \implies \tfrac{3}{2} - \tfrac{3}{2} = 0\ [\text{Cor CT0.7a}];$$

$$W_i\Pi_{\mathcal{V}} = 0\ \forall i, \quad \dim \bigcap_i \ker W_i = 4 - \text{rk}\begin{pmatrix} W_1 \\ W_2 \end{pmatrix} W_3 = 1\ [\text{FT.101}(2)\ \text{re-derived;\ machine}];$$

$$\mathfrak{T}\Pi_{\mathcal{R}} = 3,\ \mathfrak{T}\Pi_{\mathcal{V}} = 1\ [\text{CT0.7}(1);\ \text{machine}]$$

$$(3):\ \text{FT.101}(1)\text{–}(3)\ \text{re-derived witnesses;}$$

$$\text{support identity } W_i = \Pi_{\mathcal{R}} W_i \Pi_{\mathcal{R}}\ [\text{machine}];$$

$$\text{flip non-implementation: conjugation preserves } \mathfrak{T} \text{ and rk};\ 3 \neq 1\ [\text{machine}]$$

$$(4):\ \text{FT.126}(2),\ \text{FT.136},\ \text{FT.141}(6),\ \text{FT.144}(5)\ \text{verbatim (frozen)};$$

$$\neg\exists\ \text{assignment tag [inventory scan; FT.125}(4)]$$

$$(5):\ \text{rows cite CT1.12},\ \text{FT.156}(2)\text{–}(5),\ \text{A6},\ \text{FT.144}(2)\text{–}(3),\ \text{CT1.2},\ \text{CT1.8};$$

$$W_i\Pi_{\mathcal{V}} = 0\ [(2);\ \text{machine}].\ \blacksquare$$

Phys. The three generators the corpus calls weak are the same three coefficients the metric calls shift: the weak force is literally the time–space coupling block of the emergent geometry — a spatial force, triad-indexed, rotating everything it touches as vectors. On states, it lives entirely on the three-dimensional aligned branch — Casimir 2, the spin-1 read — and annihilates the one-dimensional antialigned branch — Casimir 0, no orientation to act on. Space is three-dimensional, fully symmetric under inner triad rotations, and reversible under exact involutions; time is one-dimensional and strictly forward, with no inverse for its advance. The weak couples to the spatial orientations because the spatial sector is where its vector action has a referent; it is blind to the temporal branch because the temporal sector has none. That is the chiral selectivity of the force — the $3{+}1$ asymmetry read on states — while the carrier-side gradings stay distinct and the species rule stays open.

Theorem 3.112 (FT.158 — The spatial provenance of the channel angle: C_3 is the weak rotation of the triad, and the Koide-form spectrum is forced). Thm. With the seal bilinear $[J_i, K_j]$ [FT.109], the C_3-covariant spectrum [FT.111], the mixing angle $\delta = \frac{2}{9}$ [FT.112], and the bilinear spectrum r_i [FT.113]:

$$(1)\quad \text{the } C_3 \text{ is generated by the weak generators: FT.111's inner rotation } U$$

$$= u \otimes u,$$

$$u$$

$$= \tfrac{1}{2}\big(I_2 - i(\sigma_1 + \sigma_2 + \sigma_3)\big),$$

is the $\frac{2\pi}{3}$ rotation about the $(1,1,1)$ axis generated by the total-spin/weak generators:

$$U = \exp\!\big(-\tfrac{2\pi i}{3}\,\hat{n}\cdot W\big) = E_0 - \tfrac{1}{2}\mathcal{W}^2 - \tfrac{i}{2}\mathcal{W},$$

$$\mathcal{W} = W_1{+}W_2{+}W_3,$$

$$\hat{n} = \tfrac{(1,1,1)}{\sqrt{3}}$$

[exact polynomial identity over $\mathbb{Q}(i)$, via $\mathcal{W}^3 = 3\mathcal{W}$; machine];

$$U^3 = E_0;$$

$$UJ_iU^\dagger = J_{i+1}, \quad UK_iU^\dagger = K_{i+1}, \quad UW_iU^\dagger = W_{i+1} \quad \text{(indices mod 3)}$$

[FT.111(1) re-derived, plus the W-row; machine] :

$$C_3$$

$\leq SO(3)$ — the cyclic permutation of the three spatial directions [CT1.8 : the triad $SO(3)$; CT1.4 : span$\{W\} \cong \mathfrak{su}(2)$] — is inner in the weak rotation algebra;

(2) the numerator $2 = 3 - 1$ is the doublet dimension of the spatial triad:

the diagonal bilinear splits singlet $\oplus$ doublet [FT.111(1)] : the fixed direction Σ $\oplus$ the zero-sum plane $\sum_i \hat{E}^{ii}$

$$= 0,$$

dim

$$= 2 \text{ [machine: exact rank]};$$

the cyclic shift on the plane: charpoly

$$= \text{minimal polynomial}$$

$$= x^2 + x + 1,$$

irreducible over $\mathbb{Q}$ ($\pm 1 \mapsto 3 \neq 0$) [FT.111(1) re-derived; machine];

$\sum_i [J_i, K_i] = i \sum_i \big(\Sigma - E^{ii}\big) = i\big(3\Sigma - \Sigma\big) = 2i\Sigma$ [FT.109(1)–(3) re-derived; machine] :

the coefficient 2

$= 3-1$ — the triad count minus its trace : one integer at two derived reads (dim doublet; Σ-coefficient) [exact integers];

(3) the denominator $9 = 3 \times 3$; $\delta = \frac{2}{9}$ is a ratio of spatial counts:

$$\dim F_{J^\pm}$$

$= 3$ [CT0.12; machine: exact rank] — the seal-triplet dimension

= the triad dimension on which $SO(3)$ acts [CT1.8];

$$\mathrm{tr}\, T(1)$$

$= 3$ [FT.110(2) : charpoly $T(1) = x^4 - 3x^3 - 3x^2$; machine: exact re-derivation from the transfer rule $T(1)[s][s'] = [\,|s| + |s'| \leq 3\,]$];

$$\delta = \frac{\sum_i [J_i, K_i]\big|_\Sigma}{\dim F_{J^\pm} \cdot \mathrm{tr}\, T(1)} = \frac{2}{3 \cdot 3} = \frac{2}{9}$$

[FT.112 re-derived; machine: exact Fractions];

the arena reading carries Tier F.37's interface provenance [Def-H.41 — cited, ¬imported here] :

the angle's constituents are the doublet dimension, the seal dimension, the transfer trace — integer counts of the spatial channel, no datum, no fit;

(4) the Koide-form spectrum is forced:

$$r_i$$

$= 1 + \sqrt{2}\cos\left(\frac{2}{9} + \frac{2\pi i}{3}\right)$ [FT.113] : the shape from C_3-covariance [FT.111(2) : $\lambda_i = A + B\cos(\theta + \frac{2\pi i}{3})$ for every C_3-covariant Hermitian triplet operator];

$$A = 1,$$

$$B$$

$= \sqrt{2}$ from the democratic normalization to the seal dimension 3 [FT.111(3) : $\sum_i \lambda_i = 3 = \dim F_{J^\pm}$, $\sum_i (\lambda_i - A)^2 = 3$];

$$\theta = \delta \text{ from } (2)/(3);$$

$$\text{machine:} \quad \textstyle\sum_i r_i = 3, \quad \sum_{i<j} r_i r_j = \frac{3}{2}, \quad 2(r_i-1)^3 - 3(r_i-1) - \sqrt{2}\cos\frac{2}{3} = 0 \ \ \forall i$$

$$[\text{exact in } \mathbb{Q}(\sqrt{2},\sqrt{3})[c,s]/(c^2{+}s^2{-}1);\ \cos\tfrac{2}{3} = 4c^3 - 3c \text{ at } c = \cos\tfrac{2}{9}];$$

the Koide formula is a consequence of the C_3 symmetry of the 3 spatial directions;

the scale μ (P4-class) enters no ratio [FT.113];

(5) the classification of the rational $\frac{2}{3}$:

$$\tfrac{2}{3}$$

$$= \tfrac{3-1}{3}$$

$= \frac{\text{dim doublet}}{\text{dim triplet}}$ — the decomposition ratio of the spatial seal triad under its inner rotation subgroup [FT.111(1) : $3 = 1 \oplus 2$; machine: exact Fractions];

$$\delta_\ell = \tfrac{2}{9} = (\tfrac{2}{3})/3 \text{ [FT.112 Rmk] :}$$

the three cross-check rationals $(p^*_\infty)^2$

$$= \tfrac{2}{3} \text{ [FT.70(3); exact in } \mathbb{Z}[\sqrt{6}]],$$

$$\delta_v/\pi$$

$$= 2 - \tfrac{4}{3}$$

$$= \tfrac{2}{3} \text{ [FT.63(4)]},$$

$$f_{\text{charge}}(\delta h{=}3) = \tfrac{2}{3} \text{ [FT.14]}$$

equal $(3-1)/3$ exactly [machine] :

FT.89(2d)'s classification of the rational $\frac{2}{3}$ "cap-3 combinatorial output" and (5)'s verdict on the Koide-form structure "coincidence; no structural claim" do not stand :

the corpus-side Koide-form angle and spectrum have the derived spatial provenance (1)–(4);

unchanged and standing: FT.89(1)'s arithmetic (exact, frozen);

(3)'s refutation;

(4)'s negatives;

the inadmissibility grounds $(2a)$–$(2c)$ and the identification bars: no family-mass read exists in the corpus,

$\neg\exists$ a declared map relating derived rationals to measured mass ratios,

external data enters no derivation [P1, P5];

the cap-sensitivity arithmetic $[(2d):\ c=2\to 0.508004\ldots;\ c=4\to\neg\exists$ interior boundary$]$ stands as the exact polygon-ensemble record;

the scale μ remains undeclared (P4).

Rmk (channel labels). The subscripts ℓ, q, ν name ledger rows [FT.112, FT.119, FT.142 Rmks]; "Koide-form" names the corpus-internal functional shape $1+\sqrt{2}\cos(\delta+\frac{2\pi i}{3})$; no particle identification is made or implied [P1, P5].

Req. CT0.12, CT1.2, CT1.4, CT1.8 (Part 1); FT.14 (Tiers F.1–F.11); FT.63(4) (Tiers F.20–F.22); FT.68(2), FT.70(3) (Tiers F.23–F.25); FT.89 (Tier F.30); FT.109, FT.110, FT.111, FT.112, FT.113, FT.119 (Tiers F.37–F.38); FT.142 (Tier F.42; the Rmk row); Def-H.41 cited as FT.110's arena provenance and imported nowhere; P4, P5 (Parts 1–2); FT.156 (this tier); exact $\mathbb{Q}(i)$ conjugation and polynomial rotation identities, exact $\mathbb{Q}$ ranks and characteristic polynomials, $\mathbb{Z}[\sqrt{6}]$ arithmetic, exact identities in $\mathbb{Q}(\sqrt{2},\sqrt{3})[c,s]/(c^2+s^2-1)$, Fractions [math; machine: `checks/verify_H_3p1_chiral_asymmetry.py` §C].

Pf.

$$(1):\ \mathcal{W}^3$$

$$=3\mathcal{W}\ [\text{machine}]$$

$$\Longrightarrow\ (\hat{n}\cdot W)^3$$

$$= \hat{n} \cdot W \implies \exp(-i\theta\, \hat{n} \cdot W)$$

$$= E_0 + (\cos\theta - 1)(\hat{n} \cdot W)^2 - i \sin\theta$$

$$(\hat{n} \cdot W) \text{ [polynomial on spec}\{1, 0, -1\}];$$

$$\theta = \tfrac{2\pi}{3} \implies E_0 - \tfrac{1}{2}\mathcal{W}^2 - \tfrac{i}{2}\mathcal{W}$$

$$= u \otimes u \text{ entrywise over } \mathbb{Q}(i) \text{ [machine]};$$

$$U^3 = E_0,\ UJ_iU^\dagger = J_{i+1},\ UK_iU^\dagger = K_{i+1} \text{ [FT.111(1) Pf re-run; machine]};$$

$$UW_iU^\dagger = W_{i+1} \text{ [machine]};$$

$$C_3 \leq SO(3) \text{ [CT1.8 vector structure; FT.111 Thm line]}$$

$$(2):\ \hat{E}^{ii} = E^{ii} - \tfrac{1}{3}\Sigma,\ \ \textstyle\sum_i \hat{E}^{ii} = 0 \text{ [FT.109(2); CT1.2 traceless split; machine]};$$

$$\text{rk}\{\hat{E}^{11}, \hat{E}^{22}, \hat{E}^{33}\} = 2 \text{ [exact flattening rank; machine]};$$

$$P^3 = I_3,\ \text{tr}\, P = 0:\ x^2 + x + 1 \text{ irreducible over } \mathbb{Q} \text{ [FT.111(1) Pf re-run; machine]}$$

$$\textstyle\sum_i [J_i, K_i]$$

$$= i \sum_i (\Sigma - E^{ii})$$

$$= i(3\Sigma - \Sigma)$$

$$= 2i\Sigma \text{ [CT1.7 diagonal; FT.109(1)–(2); machine, entrywise]};$$

$$2 = 3 - 1 \text{ [exact integers]}$$

$$(3):\ \dim \text{span}\{J_i\} = 3 \text{ [machine: exact rank]};$$

$$T(1)$$

$$= \mathbb{1}_{4\times 4} - [\,\cdot\,]_{(2,2)} \text{ on the diagonal : } \operatorname{tr} T(1)$$

$$= 4 - 1$$

$$= 3;$$

charpoly

$$= x^4 - 3x^3 - 3x^2 \text{ [FT.110(2) Pf re-run; machine: exact Newton identities];}$$

$$\delta = \tfrac{2}{3\cdot 3} = \tfrac{2}{9} \text{ [Fractions]}$$

$$(4) : \text{ closed forms } r_0$$

$$= 1 + \sqrt{2}c,$$

$$r_{1,2}$$

$$= 1 - \tfrac{1}{2}(\sqrt{2}c \pm \sqrt{6}s) \text{ [FT.113 re-derived by the addition formulas];}$$

$$\sum_i r_i = 3,\ \sum_{i<j} r_i r_j = \tfrac{3}{2} \text{ [exact in } \mathbb{Q}(\sqrt{2},\sqrt{3})[c,s]/(c^2{+}s^2{-}1);\ \text{machine]}$$

$$2y^3 - 3y - \sqrt{2}(4c^3 - 3c) = 0 \ \ \forall y \in \{r_i - 1\} \text{ [same ring; machine];}$$

$$\cos\tfrac{2}{3} = 4c^3 - 3c \text{ at } 3\delta = \tfrac{2}{3} \text{ [triple-angle identity in the ring; FT.113 Pf pattern]}$$

$$(5) : \ \tfrac{3-1}{3} = \tfrac{2}{3} \text{ [Fractions];}$$

$$\left(\tfrac{\sqrt{6}}{3}\right)^2 = \tfrac{6}{9} = \tfrac{2}{3} \text{ [}\mathbb{Z}[\sqrt{6}];\ \text{machine];}$$

$$s = 2 + \sqrt{6} : \ s^2 - 4s - 2 = 0,\ \tfrac{s}{1+s} = \tfrac{\sqrt{6}}{3} \text{ [FT.89(1) Pf re-run; machine];}$$

$$2 - \tfrac{4}{3} = \tfrac{2}{3} \text{ [FT.63(4); exact integers];}$$

$$\tfrac{2/3}{3} = \tfrac{2}{9} \text{ [Fractions]}$$

classification row : (1)–(4) conjoined;

standing rows : FT.89(1)–(4) cited unmodified; P1/P5 bars verbatim. ■

Phys. The 120° rotation that shapes the triplet spectrum is a rotation of the three spatial directions performed by the weak generators themselves — an exact polynomial in $W_1 + W_2 + W_3$ with rational coefficients. The channel angle is the doublet dimension over the arena: $(3-1)$ over 3×3 — a ratio of spatial counts, with the same integer 2 appearing independently as the seal-commutator coefficient. The famous cosine form is then forced three times over: covariance fixes the shape, the triad dimension fixes the normalization, the counts fix the angle — nothing is adjustable except an overall scale that no ratio sees. The classification of two thirds as a bookkeeping coincidence does not stand, at exactly that clause: the same rational is the singlet-complement fraction of the spatial triad under its own rotation group, and the Koide-form structure has a provenance theorem. Every bar against reading any of it as a measured mass ratio stands untouched.

Theorem 3.113 (FT.159 — The emergent-time identity theorem: gravitational time and CPT time are the ledger order). Thm.

(1) gravitational time is the tick coordinate:

$$x^0 :$$

$= \tau_n$ [D1.8 : $\tau_n = \sum \delta\tau$ accumulated along a causal chain of real edges; the emergent coordinate];

$$\chi([v])$$

$$= (x^0, x^1, x^2, x^3)$$

$= (\tau_n, h_1, h_2, h_3)$ [CT2.6 : $M = \Gamma_{\mathcal{R}}/\sim$ a smooth 4-manifold; $g_{\mu\nu}$ a smooth Lorentzian metric [CT1.12]; conditions (ii)–(iii) at the point of use — A §0.2a];

$$g_{00} = \Omega^2 \bar{g}_{00} = -\Omega^2,$$

$$\Omega = e^{\alpha_\Omega a_{00}} \text{ [D2.1 : } \bar{g}_{00} = -1];$$

$a_{00} =$ the E_0-coefficient of $X_{(+)} \in \mathcal{E}_{++}$: S-even, T-even [CT1.2 table; machine] :

the lapse/conformal read is built from the S-even T-even block [CT1.12 table : $a_{00} \mapsto$ conformal factor];

the metric's time coordinate is the ledger's tick structure [A7 : $\mathfrak{L} = \bigcup_{n\geq 0} \mathfrak{L}(\tau_n)$];

(2) CPT time is the ledger reversal:

$\Theta_{\mathfrak{L}}$ [D2.7] : ledger order reversed, $J^+ \leftrightarrow J^-$, $G(e) \mapsto G(e)^*$;

$T = \Theta_{\mathfrak{L}}$: the T-register of the configuration calculus is the ledger reversal [FT.133(1) : order reversed, $J^+ \leftrightarrow J^-$, $\kappa_e \mapsto -\kappa_e$, σ fixed; FT.88(2)];

$$w(\Theta_{\mathfrak{L}} C)$$

$= w(C)^*$ [CT2.8(1); machine: exact witnesses over $\mathbb{Q}(i)$ — $G(e) = \sigma_e e^{i\pi\kappa_e}$ [FT.88(2); D3.5], $e^{i\pi\kappa} \in \{1, i, -1, -i\}$ on half-integer κ samples, $\kappa \mapsto -\kappa$ = conjugation, σ fixed, $\mathcal{J} \mapsto \mathcal{J}^*$ [D3.6]; exhaustive on the 3-edge configurations];

$$\langle A|B\rangle := Z\big[\Theta_{\mathfrak{L}} A \cup B\big] = \sum_C |w(C)|^2 \geq 0,$$

$$w(C)$$

$= w(C_{\text{left}}) \cdot w(C_{\text{right}})$ across any interior ledger-time cut [CT2.8(2)–(3); machine: exact regrouping witnesses] :

the Hilbert-space structure is glued across the same ledger time by the same reversal [CT2.8(5)–(6)];

(3) the identity:

gravitational time (1) and CPT time (2) are the same entity — the ledger order parameter — read once as the coordinate and once as the reversal:

$x^0 = \tau_n$ [D1.8; CT2.6] and its reversal $= \Theta_{\mathfrak{L}}$ [D2.7; FT.133(1) : $T = \Theta_{\mathfrak{L}}$];

the metric side transports: the register involution is declared with the metric signature [A3, C13 : $\eta = (-,+,+,+)$] and its T-parity transport through the dictionary is the time-reversal pullback [FT.156(3) : $(a_{00}, q, w) \mapsto (a_{00}, q, -w) \iff g_{00}, g_{ij}$ fixed, $g_{0i} \mapsto -g_{0i}$; machine];

the record side conjugates: $w(\Theta_{\mathfrak{L}} C) = w(C)^*$ [CT2.8(1); (2)] :

the time reversal of the emergent metric and the T-register of the CPT calculus are the reversal of the same ledger order at two loci — the metric dictionary and the configuration record;

locus discipline [FT.134(1) pattern] : the register grading [A3] and the record reversal [D2.7] act on different spaces — the identity is of the time entity (the order parameter),

¬an operator equality;

inventory: one order parameter, four tags [D0.7 ticks; D1.8 x^0; CT2.6 charts; D2.7 $\Theta_{\mathfrak{L}}$] — no second time entity exists in the corpus [scan];

(4) the blockage removed; the residue restated:

the "no emergent time" prerequisite [item 21(b) row; FT.131(3) : emergent time a named prerequisite] is discharged by (1)–(3) :

the time parameter is derived content — the coordinate x^0

$= \tau_n$ [D1.8; CT2.6],

the lapse g_{00}

$= -\Omega^2$ [D2.1; CT1.12],

the reversal $\Theta_{\mathfrak{L}}$

$= T$ [D2.7; FT.133(1)],

the cut-glued inner product [CT2.8(2)–(3)];

the line-element channel contracts metric data with tick data:

$$ds^2(e)$$

$$= g_{\mu\nu}(e)$$

$$\Delta x_e^{\mu}$$

$$\Delta x_e^{\nu},$$

$$\widehat{ds}^{\,2}$$

$$= ds^2/\mathfrak{h}^2 \text{ [D1.6]},$$

$$\Delta x^0$$

$$= \text{a tick increment [D1.8; CT2.8(2) Pf : each edge is registered between consecutive ticks]} :$$

Ω and $\Delta\tau$ meet in every $\widehat{ds}^{\,2}(e)$ with a temporal increment [machine: the bilinear-form witness $ds^2 = g_{00}(\Delta x^0)^2$ on purely temporal increments];

the derived clock: $(\Delta n_{\mathcal{R}}, \delta h, \delta\tau) = \pm(1,1,1)$ per Janus move [CT0.3];

restated precisely — the unconstructed residue is the dynamical rate modulation: a flicker-frequency read of $\delta\tau$ [D0.7] against Ω/N_i along materialized worldlines (a redshifted or self-modulating rate) [FT.86(8)'s inventory row stands as the construction record];

it requires no declaration of time — time is derived (1)–(3) — and no declaration issues;

FT.131(3)'s deferred frequency read has its time parameter to be posed in;

(5) strict forwardness stands:

$$\neg\exists\,\tau^{-1} \text{ [A6]};$$

$\Theta_{\mathfrak{L}}$ is the formal mirror on records,

¬a dynamical inverse [FT.144(4) : the pure order reversal of every completed nonempty history is inadmissible at its first event; machine];

materialization is single-arrow [FT.144(2)–(3)];

the ledger count is monotone [A7; machine: exact samples] :

the CPT time reversal is a register operation on configuration data; the gravitational arrow is the ledger's forward advance — the same entity on the record side and on the dynamical side.

Req. A3, A6, A7 (Part 1); D0.5, D0.7, D1.2, D1.6, D1.8, D2.1, D2.7, D3.3, D3.5, D3.6 (Part 1); CT0.3 (Part 1); CT1.2, CT1.12, CT2.6 (conditions (ii)–(iii) at the point of use — A §0.2a), CT2.8 (Part 2); FT.86(8) (Tier F.29); FT.88 (Tiers F.29–F.30); FT.131(3) (Tier F.40); FT.133, FT.134(1) (Tier F.41); FT.144 (Tier F.42); FT.156 (this tier); exact $\mathbb{Q}(i)$ weight arithmetic on the half-integer κ ladder, exhaustive word enumeration, exact rational regrouping and transport identities [math; machine: `checks/verify_H_3p1_chiral_asymmetry.py` §D].

Pf.

$$(1):\ \text{D1.8 verbatim}:\ x^0 := \tau_n;$$

$$\text{CT2.6 chart line verbatim}:\ \chi([v]) = (\tau_n, h_1, h_2, h_3);$$

$$g_{00} = \Omega^2 \bar{g}_{00} = -\Omega^2\ [\text{D2.1}];$$

$$E_0 \in \mathcal{E}_{++}:\ (S,T) = (+,+)\ [\text{CT1.2 table; machine}];$$

$$\text{CT1.12 row } a_{00} \mapsto \Omega$$

$$(2):\ \text{D2.7 verbatim};$$

$$\text{FT.133}(1):\ T = \Theta_{\mathfrak{L}} \text{ register line verbatim } [\text{FT.88}(2):\ \kappa \mapsto -\kappa];$$

$$w(\Theta_{\mathfrak{L}} C)$$

$$= w(C)^*\ [\text{CT2.8}(1) \text{ frozen; machine re-run: } e^{i\pi\kappa} \mapsto e^{-i\pi\kappa} = (e^{i\pi\kappa})^* \text{ on } \kappa \in \tfrac{1}{2}\mathbb{Z}:\ \{1, i, -1, -i\} \subset \mathbb{Q}(i);\ \sigma \text{ fixed; } (-1)^{\aleph} \text{ invariant [D3.3}:\ \aleph_v = |f_v|]]$$

$$\langle A|A\rangle = \sum_C w(\Theta_{\mathfrak{L}} C) w(C) = \sum_C |w(C)|^2\ [\text{CT2.8}(3) \text{ Pf pattern}];$$

$$\text{cut factorization}: w$$

$$= \mathcal{J} \prod_e G(e) \cdot (-1)^{\aleph},$$

$$\text{factors tick-local [D0.7; CT2.8}(2) \text{ Pf regrouping; machine: exact products on witnessed cuts]}$$

$$(3):\ (1) + (2) \text{ conjoined};$$

$$\text{transport row [FT.156}(3);\ \text{machine}];$$

inventory scan : $\{\text{D0.7},\ \text{D1.8},\ \text{CT2.6},\ \text{D2.7}\}$ name one order parameter [no second time coordinate in any tag]

(4) : item $21(\text{b})$ prerequisite text cited as discharged by (1)–(3);

$$\text{D1.6} \circ \text{D1.8} : \ g_{\mu\nu}\Delta x^{\mu}\Delta x^{\nu}\big|_{\Delta x=(\Delta\tau,0,0,0)}$$

$= g_{00}(\Delta\tau)^2$ [bilinear substitution; machine witness over $\mathbb{Q}$];

$g_{00} = -\Omega^2$ [D2.1];

FT.86(8) row cited unchanged;

$\neg\exists$ new declaration [scan of this tier's Req]

(5) : A6 verbatim;

FT.144(4) machine re-run (exhaustive, lengths ≤ 8);

A7 monotonicity on exact samples. ■

Phys. The corpus has one time. The manifold's time coordinate is the ledger's own tick index, declared as such and charted onto the emergent geometry; its time-time metric component is the conformal read of the reality-preserving, reversal-even block. The CPT calculus's T is the ledger reversal — the order of the record run backwards, seals exchanged, phases conjugated — and the same reversal, transported through the metric dictionary, is exactly the time-reversal pullback: lapse and spatial metric standing still, shift flipping sign, signature untouched. Gravitational time and CPT time are therefore not two entities needing a bridge; they are the ledger order read as coordinate and read as reversal. The blockage that deferred every frequency read for want of a time parameter is removed — the parameter exists and is derived. What remains unconstructed is honest and small: a dynamical rate modulation of the flicker against the conformal factor — a redshift — which is now a construction problem inside derived time, not a missing declaration. And the arrow stands: the reversal is a mirror on records; the advance has no inverse.

FT.160 — The 3+1 completion: adjudication and scope. Structural-constraints record: see Appendix B, Section B.1.14 of the Addendum.

4 $\langle qu(b(i(t|t))ick)\rangle$ — Part 4 · Baryomesonic_era — Colour-Sealed Composites: Baryons, Mesons, Junctions

Dep. Part 1 (axioms A1–A9; conventions C1–C13; qubit-era definitions D0.1–D1.12; calibration P1–P5; pair canon CT0./CT1.1–CT1.11b; ledger dynamics L0–L5); Part 2 (definitions D2.1–D4.3; geometry CT1.12–CT2.11; electroweak EW.; electromagnetic EM.; colour CT3.1–CT3.22a; saturation CT4.); Part 3 (the fermion layer: Def-H.1–Def-H.43, FT.1–FT.155); the Census annex (Appendix A of this volume; the computed cells, cited row-wise). Format: Part 1 §0.4. Namespace. This part owns: Def-BM.1–Def-BM.20; BM.1–BM.152. Tiers 4.0, BM.1–BM.24 in series order. DAG law. Citations resolve to Parts 1–3, Census cells, or earlier tags of this part. Nothing in the canon cites this part. 0.1 Charter. The baryomesonic era is the layer of colour-sealed composites: the algebraic sealings of D3.9 (baryon $\varepsilon_{ijk}q^iq^jq^k$; meson $\delta^i{}_j\, q^jr_i$) realized on bound-chain constituents (Def-H.4) at shared junctions. Its foundation stack, by citation: (i) the colour and sealing machinery [CT3.1–CT3.22a; D3.8, D3.9; D2.14; CT4.1–CT4.7]; (ii) the derived sealing grammar and junction structure [FT.29, FT.34(5), FT.19, FT.39]; (iii) the computed baryon-type cells [Census FT.35: the first colour-active, first D^--carrying floor; Census FT.49: the baryon-type sealing phase $\Theta_{\text{int}} \in 4\mathbb{Z}$]; (iv) the channel machinery [FT.109–FT.119: the ℓ/q-channels with the ε-selected C_3 and confinement blindness; FT.141–FT.145: the rung-one channel and the arrow law]; (v) the carrier and parity structure [FT.122–FT.134]; (vi) the sector formalisms [FT.102–FT.108]; (vii) the conservation and symmetry layer [Tier BM.4: the Janus balance and the telescoping endpoint read, the continuum promotion of the hop, the C_3 and the slot algebras, the stress-energy tensor of a sealed composite, the canonical-pair obstructions, the differential conservation law]; (viii) the calibration layer [Tier BM.5: the Born width, the coherence volume and energy density, the curvature of a sealed composite, the dimensionless lock identities, the neutron row of the identification layer]; (ix) the flicker layer [Tier BM.6: the topology-blind ensemble of a sealed composite and its derived weights, the path-integrated mass from the curvature law, the binding read, the static force matrix on the two charge classes, the transition with its forced carrier, the family naming rows]; (x) the CPT-sector and charge-restriction layers [Tier BM.7: the EM$^{\pm}$ sector labels and the EM scar, the mirror involution and its exact phase, the proton and antiproton rows, and the charge-constrained probability space with its three mass routes]; (xi) the additive charge read and the mass-route adjudication [Tier BM.8]; (xii) the mesonic flicker test [Tier BM.9: the contraction grammar, the colour projector, and the protected junction]; (xiii) the metric-energy and vacuum-entropy layer [Tier BM.10: the dispersion read of a sealed composite from the emergent metric with its kinetic energy and zero-point bound, the vacuum-balanced-entropy reframing of the rest mass, the force-structure ladder with its strength bar and the inter-family constraint, and the route-artifact adjudication closing the counting trap]; (xiv) the decay-vertex layer [Tier BM.11: the family transport-profile map discharging the Yukawa-labelled gap, the family mass form with its positivity, splitting law and sum rule, the rung-1 isospin row and the neutrality of the charge vertex, the forced neutral member and the Q-value sign theorem, the angular-momentum budget with its absorption adjudications, the member-wise curvature rows, and the per-tick time evolution of the assembled reaction]; (xv)

the electromagnetic self-interaction layer [Tier BM.12: the chart-side EM source read of a sealed composite, the scar's field energy and the mass-splitting theorem, the infinite-range seal asymmetry with its UV/IR finiteness rows, the two-body potential with the binding category bar, and the beta-decay re-adjudication with the endpoint-to-average refutation]; (xvi) the full-scope mass-splitting layer [Tier BM.13: the Lyapunov foundation with its iterated contraction law and class-degenerate zero, the complete-force-operator sector census with its five zeros and one monomial, the metric background's zero contribution and the non-uniform clock with its Mandelstam–Tamm resolvability bound, the two binding zeros and the unconstructed atom, and the exact symbolic ratio with its named order residue]; (xvii) the constituent-transport-splitting layer [Tier BM.14: the continuation-symmetry census of the two vertex types with its three annihilation rows, the observation frames and the charge-blindness of the metric, the constituent transport splitting that fills the residue BM.55(4) names, the six-state quark assembly on the three Platonic mass floors with its branch order and its two inverted hierarchies, the coupling-blindness theorem and the named obstruction to α_B, the extended six-term decomposition with its exact critical strength and sign theorem, and the decay re-adjudication with its exact closure margin]; (xviii) the organic-bridge layer [Tier BM.15: the operator lift of the EM correction with its charge-blind linear trace and its quadratic invariant, the derived $\mathbb{Z}_2$ Wilson sector with the two-torsion and calibration-weight obstructions to the real holonomy lift and its structured obstruction record, the constrained joint entropy as a fiber cardinality with its stationary geometric factor of one and its named missing enumeration, the bridge convergence with the non-derivation theorem reducing Gap 7 to one named scale ratio and two named attachments, the two-bit daughter-floor theorem with the formed-closed dichotomy and its exact margin ladder, and the five-route opening-term census closing the beta-decay energy audit]; (xix) the chiral-assignment layer [Tier BM.16: the left–right assignment rule coupling the four derived gradings and establishing the assignment rule's forced shape, the ray-resolved spin census with the opposite-ray vertex table, the weight–branch–core attachment theorem with the sign-coherent quark mass splitting, the nilpotent chiral vertex operator with its flip, rate and arrow rows, and the vertex read-specification with the exact trichotomy of the beta-decay opening branch]. (xx) the mass-ratio-census layer [Tier BM.17: the Platonic winding census with the ladder law on both priced windings, the Chebyshev tower closing at the dodecahedral top weight, and the intra-generational ratios as exact sine ratios with their minimal polynomials; the inter-generational ratios as the three roots of one cubic form with three channel constants, the cross-channel identity between the lepton channel's heaviest line and the quark channel's cubic constant, and the scale-free hierarchy-order theorem; the six-state ratio census with the splitting order, the doublet ratios, the two inverted hierarchies and the three pairwise non-proportional ladders; the bridge census with its six exact factors, the intra-currency closure and the localized single residue; the inter-family bridge in full with the four-currency census, the exact incommensurability of the two formed ratios, and the three-number localization; the neutron-proton mass ratio with its exact range, its irrational critical locus and its total trichotomy; the Q-value in two formed reads with both margin ladders and the read-relative verdict; and the census adjudication] (xxi) the electromagnetic self-energy correction layer [Tier BM.18: the category adjudication voiding the chart-side distribution factor and deriving the formed exterior value one half on two distribution-free routes; the running coupling as an exact geometric profile on the envelope ratio, the lapse and the curvature lock, vanishing at the member's own Born width and growing with the energy scale; the self-energy as an ensemble average over the flicker with one derived attenuation, the crystallization weight; the branch-blindness theorem refuting a defect-conditioned retention on four grounds; the corrected neutron–proton splitting with its parameter-free bar, its universal ratio bound and its ensemble-robust incommensurability; the Q-value on both formed reads with the uniform closure and the untouched specified branch; and the correction's adjudication] (xxii) the flicker-integral

and rest-mass layer [Tier BM.19: the flicker integral replacing the crystallized snapshot, with the derived ensemble distribution, the materialization restriction and the two crystallization weights; the vertex superposition with its exact fibre weights and the two positive parts of the transport splitting; the rung-one ν line's seal-derived rest mass, its strict positivity, its single-ray occupancy and the four bars against a seesaw-type mechanism; the rest-mass Q-value on the specified and pure-pole reads with the exact critical locus; the relating equation in four exact forms with its flicker weighting and its integer bound; the materialized splitting with its two-sided parameter-free bar and its two tightened bounds; and the tier's adjudication] (xxiii) the six-objective-resolution layer [Tier BM.20: the energy-scale graph of the forward and inverse reads with the derived crossing strength, its exact enclosure and the composition-dependent crossing count; the dimensional bridge, with the q-exponent census of the inventory, the seven rows derived or adjudicated, the two bridges separated and the one calibration slot declared; the three-constituent assembly's two formed eigenvalue equations and the tensor-product referent census; the curvature composition's back-reaction fixed point with its three exact solution rows, its pole locus and the non-selection theorem; the general crystallization weight with its measure-free bounds, its threshold family and the refuted defect conditioning; and the kind structure of the arrow, with the spin-parity refutation at the floor and the exchange-structure referent census] (xxiv) the antimatter-and-resonance layer [Tier BM.21: the CP-conjugate chain derived from the three reversals with the T-block on the forward-only tick and the Lyapunov contraction, its exact phase, the mass-degeneracy theorem and the forcing theorem from the ledger's own total-charge-zero balance, with the Dirac parallel as an inventory statement; the CP-pair vacuum fluctuation with the Janus resolvent's own creation weight, its exact maximum one quarter and its two endpoint zeros; the two-chain state space discharging BM.6(2)'s exchange-space specification, the meson as the unique δ-sealed 2-juncture, the CP-pair saturation of the neutral class at exactly six, the two additive mass routes with their vanishing seal-binding and the named two-chain ensemble residue, and the amplitude/norm reconciliation on the locus distinction; the exchange operator as the flip composed with the CP transport, an unconditional involution with its eigenspace census and its statistics-blind orientation block; the spin–statistics connection $\varepsilon = (-1)^{|k|} = (-1)^{2j}$ from the fourth-root phase alphabet and the winding additivity, closing deficit 11 with the sector choice priced; the resonance spectrum with the transport characteristic polynomial $(z+1)^{2n}/z^n$ at every chain count, the degree-16 eigenvalue field, the multiset-orbit law with stabilizer multiplicities, the one exact three-chain collision and the minimal-polynomial degree census; the materialization threshold in its three exact readings with the never-empty locus and the no-forbidden-state theorem; the decay width with its three exact zeros, its bound and its three-tick lifetime floor; and the groupings, the ceilings and the octet adjudication] (xxv) the annihilation-channel and concrete-spin layer [Tier BM.22: the Janus register with its two nilpotent crossing moves, its master generator's decomposition, kernel, spectrum and derived detailed balance; the c-threshold adjudication with the exact subluminal deficit $1 - v = m^2/(E(E+p)) > 0$ and the register boundary that replaces it; the zig-zag as the resolvent's evenness, every leg forward and the record involution fixing each completed word; the pair channel with its charge balance, exact threshold, weight and three CP legs, and the two-photon bar on four grounds; the annihilation channel with its exact direction ratio, detailed-balance indicator and entropy destination, closing residue (R9) at the channel level; the spin–winding relation $\Theta = 4j$, the multiplet defect ladder in Chebyshev form with its standing zero and ceiling, and the dimensional bar against fusing the winding with the frame-holonomy angle; the refutation of a spin-gated tunnel as hypothesis (W7) on the four standing grounds, with the bound/transit domain row in its place; the winding-parity identification with the statistics and kind reads separated; the three-chain exchange structure with the S_n character formula $d^{\,c(\sigma)}$, the vanishing carrier wedge against the one-dimensional colour wedge, and the multiplicative exchange law discharging (R7)

at $n = 3$; and the meson's unconditional boson sign against the baryon's exact odd-count fermion criterion] (xxvi) the closing layer [Tier BM.23: the Haar–Poisson duality with its twirl fixed point, complementary factorization, unique common fixed point and proper-time contraction law; the derived lapse $N(k, E) = \sqrt{1 - G/(4\pi k)}$ with its formed domain and its envelope row, closing the lapse gap; the curvature composition selected as $\mathsf{g} = 1/(1+G)$ by Lyapunov contractivity and minimal lapse-domain loss, completing the running coupling; the momentum-space Σ-resolvent with its form uniqueness, pole reads, on-shell measure and two-body phase space, closing item 17; the multi-chain charge census with the stretched states and the gauged read, closing Q10 and deficit 10; the S_3–colour-vertex equivariance with the exchange structure at every n and the statistics per class, closing (R7′); the two-chain ensemble measure with the general $w^{(n)}_{\text{mat}}$ law, the sector-mass pincer and the formed meson binding, closing (R8); the seed-tensoring mechanism as a finite-group twirl, closing item 19; the four convention declarations (A-BM.10–A-BM.13), closing item 16; the dodecahedron selection theorem with the five-currency ceiling census, closing deficit 12 as adjudicated; the Jensen residue as an exact rational function with its two-chain companion, closing (R6); the nilpotent supply vertex with the pair amplitude and the two symbolic cross-sections; the house cleaning and the calibration-register consolidation; and the era closure with the balanced ledger] (xxvii) the continuum-limit layer [Tier BM.24: the complete force operator acting over the graph, with the exact block census and the eight-step composed action; the continuum-limit function of content and span with its five closed forms and four formed-domain conditions; the metric–energy–continuum map with the dictionary bijection, the pure-trace linearized solve and the on-shell inverse read; the breaking threshold $E_{\text{crit}} = 2r/\mathfrak{h}$ with the envelope corollary; the locked-sector complement at the breaking locus with the absorption anchors, the one-way membrane and the retention theorem; the universality of the limit on one operator, one arena, one action and three fingerprints; and the derivation of CT2.6(ii)–(iii) as theorems — the unbounded ladder and the $\sqrt{3}$ spectral bound — leaving the extra-axiomatic list empty and closing the foundational era] Its enumerative substrate is the Census block: 25 cells, each carrying its consumption row (Census §Consumption). 0.2 Standing bars. Species names are admitted in this part through the identification layer Def-BM.3 alone: a name is a label on a derived structural tuple, it carries no datum, and it occurs in no Thm clause of Parts 1–3, whose own label-discipline rows stand unmodified. No external data outside quarantined audit rows [Census FT.114, FT.120; FT.73(5) pattern]; no measured external constant is identified with any corpus quantity [P5]; topology labels are internal bookkeeping [Def-H.32(4)]; amplitude $\neq$ mass [FT.79(5)]; rung $\neq$ scale [D0.8].

4.1 §Cal-BM — Free internal parameters of the baryomesonic era (P4-class)

None introduced by this part. Inherited and undeclared: the family-channel overall scale μ [Cal-F; FT.113, FT.119 — every ratio is μ-free]. Declared slot, value ¬issued (A-BM.9, Tier BM.20): the bridge's dimensionless calibration ratio $\rho_{\text{B}} = \hat{\mu}'^2\mathfrak{h}^2/(\ln 2)^2 \in \mathfrak{q}^0$ [BM.69(2); BM.98; BM.102(5)] — a packaging of Cal-F's $\hat{\mu}'^2$ and P1's $\mathfrak{h}$, hence ¬a new free parameter and ¬a new dimensionful scale; no numerical value attaches to it here or anywhere [P5; BM.69(4)'s four grounds stand: the slot is declared, ¬derived]. No second dimensionful scale exists [P1; the derived no-go: Part 1 §E]. Any numerical use of an inherited scale beyond its Part-3 loci requires a logged declaration first.

Consolidated calibration register (Tier BM.23, BM.143(4) — a consolidation, ¬an addition; the parameter count is unchanged).

(C-BM.1) ρ_{B}

$= \hat{\mu}'^2 \mathfrak{h}^2/(\ln 2)^2 \in \mathfrak{q}^0$: the bridge ratio; slot declared [A-BM.9],

value ¬issued [BM.143(4)];

(C-BM.2)

e_0 : the numerical value of the electric coupling [EM.25(7); BM.88(8); BM.133(6)];

value ¬issued;

¬fixed by measurement [P5];

(C-BM.3) μ_ν,

μ,

ρ_ℓ,

ρ_ν : the channel scales, inherited from Cal-F,

¬declared in this part [FT.145(3); BM.83(5), (7)].

Each row is a calibration datum of the P4/A-BM.9 species, ¬a structural gap; no row carries a value; $\mathfrak{h}$ remains the single dimensionful scale [P1].

4.2 Tier 4.0 — Composite Vocabulary

Definition 4.1 (Def-BM.1 — Colour-sealed chain composites). Def. Constituent := a triplet-attached bound chain [D3.8: one fundamental slot index q^i (or dual r_i) at the attachment; Def-H.4: a Janus-closed bound chain, terminus locus; $\delta h \geq 2$]. Junction := a shared vertex hosting one elementary invariant contraction of the constituents' slot indices [CT3.17: $\delta^i{}_j$ for a pair, $\varepsilon_{ijk}/\varepsilon^{ijk}$ for a triple; trivalent slot structure CT3.2].

baryon: $B(c_1, c_2, c_3) := \varepsilon_{ijk}\, c_1^i\, c_2^j\, c_3^k$

[D3.9 algebraic form, chain-realized at one junction];

meson: $M(c, \bar{c}) := \delta^i{}_j\, c^j\, \bar{c}_i$

[D3.9; $\bar{c}$: the conjugate constituent (ω-reversed chain with dual attachment, D2.6, D3.8)].

Validity clause. For valid ambient C [D2.11], π_C is total on Colour(U) [CT3.18 + D2.11 applied to the colour ledger; D2.14]: every attachment has its contraction partners inside the composite; a restriction Σ with $\partial_{\text{col}}(\Sigma; C) \neq \varnothing$ has $\mathcal{C}(\Sigma; C) = 0$ and its entire colour sector vanishes [D2.14, FT.19: all-or-nothing] — the mask admits only singlets [CT3.18]. Read clause. Each constituent retains its own terminus tuple $(q, \mathrm{S}, m_{\text{hid}})$ [Def-H.9]; the junction adds the sealing-phase class and the lock content [Census FT.49: $\Theta_{\text{int}} \in 4\mathbb{Z}$ at the baryon-type cell; Census FT.35: junction D^- content]; the card assembly is Def-BM.2.

Definition 4.2 (Def-BM.2 — The composite card). Def. The card of a sealed composite refines the floor-card row template [Census FT.28 — the minimal fermion cell; every sealed-composite card refines its row grammar: registration data (σ_e, κ), Θ_{int}, winding, holonomy, aspect rows, mask dimension, rung counts, ternary core read] with:

(1) constituent rows: the Π_v projections of Def-H.9 per constituent;

(2) the junction row: sealing-phase class and lock content [Def-BM.1 read clause];

(3) the sealing-grammar row: N_{strong}

$= \log_2 g\big(|\text{Colour}(\Sigma)|\big)$ with the grammar counts g [CT3.17a(2)–(3); FT.41(b)];

(4) the sealing-profile row [FT.34(5) partition rule];

(5) the winding-ledger rows [additive composition FT.34(3); Census FT.43, FT.51].

Card values are Census-cell data at the computed cells; card grammar is this definition. No card row is a species identification [P1, P5].

4.3 Tier BM.1 — Observability of Sealed Composites

Theorem 4.1 (BM.1 — Confinement inheritance and the trace-only read). Thm.

(1) sealing is the observability switch: a sealed composite is colour-complete on its own support ($\partial_{\text{col}} = \varnothing \implies \mathcal{C} = 1$) and its colour sector survives masking generically

[FT.39(1) : $\mathcal{C} = 1 \Rightarrow F^{\Sigma}_{\text{strong}} = F_{\text{strong}}|_{\Sigma} \neq 0$ generically, exact exception locus $\{f = 0\} \cup \{\text{Colour} = \varnothing\}$];

any unsealed constituent read vanishes: one dangling attachment zeroes the whole colour sector [FT.19];

$$\neg\exists \text{ triplet pole, physical norm } 0 \text{ [CT3.22]};$$

(2) the one-body trace law: every Hermitian slot operator on a sealed singlet reads only its trace — baryon: $\mathfrak{T}(H)$

$$= 3a \text{ for the covariant class } H$$

$$= aI_3 + zP + z^*P^T;$$

$$\text{meson: } 0 \text{ (including the trace part) [FT.117}(1)\text{–}(3)\text{]};$$

hence the C_3-doublet datum (B, θ) is annihilated on every colour-sealed object, and hadron-level triplets cannot carry the family channel [FT.117$(2), (7)$];

(3) the sealed colour factor: singlet-sealed chains keep the undressed arena 9 — the invariant lines are 1-dimensional [FT.117(5) : $\dim(\mathbb{C}^3)^{\otimes 3\,\mathfrak{sl}(3)} = \dim(\mathbb{C}^3 \otimes \bar{\mathbb{C}}^3)^{\mathfrak{sl}(3)} = 1$; Def-H.42];

(4) the visibility locus: the same covariant data are nonzero on the adjoint channel $([H, E_{12}] \neq 0)$,

but the adjoint is not a sealed singlet and carries no pole [FT.117(4); CT3.18; CT3.22] — the constituent-level channel data ($\delta_q = \frac{2}{27}$ [FT.119]) are ledger-internal at hadron level.

Req. D2.11, D2.14, D3.8, D3.9 (Part 2); CT3.17, CT3.18, CT3.22 (Part 2); Def-BM.1; FT.19, FT.39, FT.116(3), FT.117, FT.119, Def-H.9, Def-H.42 (Part 3).

Pf.

$$(1) : \text{ Def-BM.1 validity clause : } \pi_C \text{ total } \implies \partial_{\text{col}}(\Sigma; C)$$

$$= \varnothing \text{ on the composite's own support } \implies \mathcal{C}(\Sigma; C)$$

$$= 1 \text{ [D2.14]};$$

$$\text{FT.39}(1) : \text{ generic nonvanishing with the exact exception locus}$$

$$\text{FT.19 : } \mathcal{C} = 0 \implies F^{\Sigma}_{\text{strong}} = 0 \text{ (one null factor kills the slot product)};$$

$$\text{CT3.22 : } \neg\text{triplet pole } \wedge \text{ norm } 0$$

$$(2)\text{–}(3):\ \text{FT.117}(1)\text{–}(3),$$

$$(5)\ \text{verbatim at the composite support}:\ \varepsilon\ \text{spans}\ \wedge^3\mathbb{C}^3\ (\text{the determinant representation});$$

$$\text{the one-body sum acts by}\ \mathfrak{T}(H)\ [\text{CT3.16};\ \text{CT3.7}:\ \mathfrak{T}(T_a)=0];$$

$$\text{meson}:\ (H\otimes I - I\otimes H^T)$$

$$\text{vec}(I)$$

$$=0\ [\text{machine rows re-cited}]$$

$$(4):\ \text{FT.117}(4)\ \text{verbatim}:\ \text{adjoint}\notin \operatorname{Im}\mathcal{P}_M^{(3)}\ [\text{CT3.18}];$$

$$\text{ledger-internality}:(2)\text{'s annihilation on sealed supports.}\ \blacksquare$$

Phys. Sealing is the observability switch: the colour sector becomes physical exactly when the ε/δ grammar completes — and then reads a single number, the trace. Everything the constituent ledger carries (channel angles, doublet data, spectator dressings) stays behind the seal: the hadron sees sums, never slots. This is confinement as an information property, run forward from the sub-network theorems to the composite level.

Theorem 4.2 (BM.2 — Junction phase and winding grammar). Thm.

$$(1)\quad \text{baryon-type sealing phase: at the computed baryon-type junction cell,}\ \Theta_{\text{int}}\in 4\mathbb{Z}$$

$$[\text{Census FT.49};\ \text{per materialized closed chain}\ \Theta(c)\in 2\mathbb{Z},\ \text{FT.22}];$$

$$(2)\quad \text{additive winding under composition:}\ \Gamma$$

$$=c_1\cdot P\cdot c_2\cdot \bar{P}\ \text{materialized with}\ d_{12}$$

$$=\Theta(P)\qquad \geq 0:$$

$$k(\Gamma)=k_1+k_2+d_{12}$$

$$[\text{FT.34}(3),\ \text{Census FT.50}(2)];$$

carrier level: $\Psi(\Gamma)$

$= \mathrm{Sym}^{k_1+k_2+d_{12}} \subseteq \mathrm{Sym}^{k_1} \otimes \mathrm{Sym}^{k_2} \otimes \mathrm{Sym}^{d_{12}}$ as maximal weight [FT.60(6), under Def-H.38];

(3) sealing partition: attachment partitions into contraction groups follow the CT3.17a grammar with counts $g(N)$

$= 1,$

$1,$

$3,$

$10,$

$25,$

105 $(N = 2..7)$ [CT3.17a(2); FT.41(b)];

the gluing law of composed floor cells is the constraint-rank law [Census FT.50];

(4) junction lock content: a junction at n

≥ 4 carries D^- lock content — retained, ¬accessible [Census FT.35; CT4.1–CT4.3; D4.2; D1.11].

Req. FT.22, FT.34, FT.60, Def-H.38 (Part 3); CT3.17a, CT4.1–CT4.3 (Part 2); D4.2, D1.11 (Parts 1–2); Def-BM.1; Census FT.35, FT.49, FT.50; exact integer arithmetic.

Pf.

(1) : Census FT.49's card row (computed from FT.22/FT.24's quantization on the tetrahedron-face sealing grammar);

no new arithmetic

(2) : FT.34(3) verbatim : $\Theta(\Gamma)$

$= \Theta(c_1) + \Theta(P) + \Theta(c_2) + \Theta(\bar{P})$

$= 2(k_1 + d_{12} + k_2)$ [$\Theta(P) = \Theta(\bar{P}) = d_{12}$; FT.23];

FT.60(6) : CG inclusion on the cone

(3) : CT3.17a(2) counts re-cited;

Census FT.50's junction-saturation and constraint-rank rows

(4) : Census FT.35's junction row; CT4.1–CT4.3 : absorption/retention at $n \geq 4$. ■

Phys. The hadron ledger is additive exactly where the fermion ledger is: windings compose, phase classes stack in $4\mathbb{Z}$ at baryon-type junctions, and the junction is where the first lock content sits — the baryon-type cell is the corpus's first D^--carrying floor. What the junction does not add is new grammar: the sealing partitions are the frozen colour counts, and the gluing is the constraint-rank law the floor cells already proved.

4.4 Tier BM.2 — The Family Group and the Invariance Laws of the Seal

Theorem 4.3 (BM.3 — The family group of the seal). Thm.

(1) the forced group: the stabilizer of the baryon seal ε_{ijk} inside the native slot-relabelling group S_3 is $A_3 \cong C_3$

$= S_3 \cap SU(3)$ — the unique maximal ε-preserving subgroup;

full S_3 covariance is two-level-only (every S_3-covariant slot operator has spectrum $\{a + 3b, a, a\}$) [FT.116];

(2) channel inheritance: every triplet-attached constituent carries the democratic C_3 mechanism with angle δ_q

$= \frac{2}{27}$ [FT.119; Def-H.42];

the triple-angle link $3\delta_q$

$= \delta_\ell$

$= \frac{2}{9}$ [FT.119(4); FT.112];

the cross-channel numerator identity $16\delta_\nu$

$$= 9\delta_\ell$$

$$= 27\delta_q$$

$$= 2 \text{ [FT.143]};$$

(3) exclusions: the light-triplet read (A, B)

$= (1, \sqrt{2})$ is excluded — Q is confined to $(0.54, 0.59)$ across the allowed ratio box while the channel normalization forces Q

$= \frac{2}{3}$ phase-independently [FT.120(6); Census row, quarantined external data];

the heavy-triplet proximity rows are audit-only, no identification [Census FT.120; P1, P5].

Req. CT3.17 (Part 2); FT.112, FT.116, FT.119, FT.143, Def-H.42 (Part 3); Census FT.120; exact $\mathbb{Q}(i)/\mathbb{Q}(\sqrt{3})$ arithmetic (rows re-cited).

Pf. (1): FT.116 verbatim — the relabelling lattice, the two-level obstruction, the ε-selection [machine rows re-cited]. (2): FT.119(4)/FT.112(3)/FT.143 angle ledgers verbatim; per-constituent application by Def-BM.1's constituent structure (each constituent is a triplet-attached bound chain in FT.119's sense). (3): Census FT.120's exclusion rows (quarantine discipline [FT.73(5) pattern]). ■

Phys. The seal picks its own symmetry: the only slot-relabelling subgroup preserving the baryon contraction is the cyclic three — the same three that carries the channel angles on every constituent. Hadron-level family structure is not imposed on the junction; it is the junction's stabilizer, with its phase fixed to two-thirds and its alternatives excluded by theorem.

Theorem 4.4 (BM.4 — The invariance laws of the seal). Thm.

(1) colour invariance: B,

M are colour singlets [CT3.15–CT3.17 : $\{B, M\}$ singlets; $\{\delta, \varepsilon, \bar{\varepsilon}\}$ the invariant grammar];

the mask admits only singlets [CT3.18];

the colour sector is rigid and contributes ¬P4 data [CT3.19];

(2) transport compatibility: the seed is colour-neutral by theorem [EW.32 : $T_a \mathsf{v} = 0$];

the weak action on the extended carrier is seed-factor-only [$W_i \mapsto \frac{1}{2}\sigma_i \otimes I_\omega$, FT.123(1); Def-H.34] and the diagonal Sym^k transport is colour-blind [FT.60; Def-H.38] $\implies$ constituent channel data and multiplicity coupling compose with the seal without colour obstruction;

(3) the parity side: the scalar-CPT action on each constituent's orientation doublet, $\mathrm{CPT}|_{\mathcal{H}_\omega}$

$$= (-1)^{|E|} e^{i\pi\Theta}$$

$$\mathrm{id},$$

is chirality-even [FT.134(3)];

pole multiplets are $m \leftrightarrow -m$ symmetric and the ceilings parity-blind [FT.138] $\implies$ both laws pass to sealed composites constituent-wise [Def-BM.1 product structure].

Req. CT3.15–CT3.19 (Part 2); EW.32 (Part 2); FT.60, FT.123, FT.134, FT.138, Def-H.34, Def-H.38 (Part 3); Def-BM.1.

Pf.

$$(1):\ \text{CT3.15–CT3.17 singlet rows re-cited;}$$

$$\text{CT3.18}:\ \mathrm{Im}\,\mathcal{P}_M^{(3)} \text{ admits only singlets;}$$

$$\text{CT3.19}:\ \text{open quantities} \qquad = \varnothing$$

$$(2):\ \text{EW.32}:\ T_a \mathsf{v} = 0\ [\text{the seed's colour neutrality}];$$

$$\text{FT.123}(1):\ [W_i \otimes I_\omega,\ \cdot] \text{ acts on the seed/orientation factors only;}$$

$$\text{Def-BM.1}:\ \text{constituents are independent chains with own carriers} \implies \text{product composition}$$

$$(3):\ \text{FT.134}(3),$$

FT.138 verbatim per constituent;

composite-level passage : Def-BM.1′s product structure (each factor carries its own doublet and Θ-summand; $(-1)^{|E|}e^{i\pi\Theta}$ multiplicative over disjoint constituents). ■

Phys. Both algebras that act on a composite leave the seal intact: colour SU(3) has nothing to do inside a singlet, and weak transport never touches the colour factor — so a baryon's card assembles constituent by constituent, and the parity laws pass through the seal unbroken. The rigidity theorem closes the ledger: the colour side of the hadron grammar contributes no free datum at all.

4.5 Tier BM.3 — Consumption and Open Constructions

4.5 (BM.5 — The census-consumption law).

Thm.

(1) the substrate: the enumerative substrate of this era is the Census block — the 25 Category-B cells with their configuration spaces, counts, and consumption rows [Census charter and §Consumption];

every construction of this part consumes cells by tag citation; no cell is peripheral;

(2) the family table:

census family	cells	configuration space supplied	consumption at the composite level
floor cards and ladders	FT.28, FT.32, FT.35, FT.42–FT.45, FT.49–FT.51, FT.63	the computed fermion floors with full read columns	the cell grammar of sealed composites (Def-BM.2): baryon-type junctions are gluings of these cards; the winding ledgers and duality splits are the multiplet bookkeeping; the Σ_{12}/tetrahedron-face cells carry the first colour-active and $4\mathbb{Z}$-phased junctions
loop and cover classes	FT.64, FT.65, FT.69, FT.78, FT.96	admissible-loop lattices, coarse-grained degeneracies, amplitude censuses, Aut-orbits with chirality tags	the enumeration substrate of sealed-object path integrals: sector conditioning (q, S) [Def-H.32(4)] runs over these pools; the chirality-tagged orbits feed the parity reads [BM.4(3)]
pole and mass multiplets	FT.76	defect-angle pole reads under both convention variants, with the ceiling	the pole-side mass cells and the hard range ceiling every composite read inherits [Def-H.39(2)]; the convention variants are priced — the choice converts them into data [BM.6(4)]
form-factor tables	FT.98, FT.99	polygon and Platonic form factors with growth and dominance rows	the size-bound range checks for extended composite skeletons [Def-H.39 ceilings]
audit ledgers	FT.81, FT.83, FT.84, FT.114, FT.120	the read-class inventory, the rung-emergence ledger, the ceiling taxonomy, the quarantined comparison rows	the card grammar and the P1/P5-compliant comparison template; the activation map at $n = 3/n \geq 4$; the read-species taxonomy any new read must join
topology-blind cells	FT.105	graph counts and canonical skeletons from first principles, budget-regularized	the regulated topology pool of the composite path integral — the space its fugacity formalism sums over [Def-H.32(1)]

(3) the conditioning law: sector-conditioned composite sums run on Def-H.32(4)'s labels (q, S) over (2)'s pools;

the amplitude/norm dichotomy and the amplitude $\leftrightarrow$ spectral bridge residue are item-13 content [FT.67–FT.68; FT.80].

Req. The Census block (all 25 cells); Def-H.32, FT.67, FT.68, FT.80 (Part 3); Def-H.39 (Part 3); Def-BM.2

Pf. (2): the consumption map — each family row references its cells by tag; every Census tag occurs in exactly one family row [exact set arithmetic: $11 + 5 + 1 + 2 + 5 + 1 = 25$]. (1), (3): charter and citation scans. ■

Phys. The census is not an appendix: it is the table of configuration spaces the baryomesonic era enumerates — cards to glue, orbits to condition on, multiplets whose ceilings every composite read inherits. The consumption law states, family by family, which space feeds which construction, and the count closes: twenty-five cells, no residue.

Theorem 4.6 (BM.6 — The construction interfaces (specifications)). Def. The constructions this era requires and does not yet carry, each stated as a specification (what a construction must supply); none is a declaration, and none is adopted:

(1) the multi-chain additive charge read : a read Q on sealed composites with (i) $Q|_{\text{single chain}}$

$$= q_{\text{core}} \text{ [FT.40; EM.9]},$$

(ii) additivity under disjoint sealing,

(iii) observability only on sealed supports [CT3.22 discipline]. Status: not constructed; the single-chain read is derived; two end registers per chain are counted [FT.41(a)];

(2) the two-chain exchange space : a state space on ordered pairs of bound chains with an exchange involution; the single-chain chiral grade is ¬an exchange sign [FT.125(5); FT.80(5) : ¬∃ exchange operator in the inventory; FT.89(2) kind discipline];

spin–statistics content is external until constructed;

(3) the ℵ-domain choice : literal [D3.3] vs chain-local E1/E2 [Def-H.27] — both path integrals computed and pairwise inequivalent [FT.67, FT.74, FT.75];

sealed junctions are the natural site of the residual class-(iii) edges

the choice requires a logged declaration before use;

(4) the defect corner convention : variant A (face winding k_f) / variant B (vertex-figure winding k_{vf}) — both multiplet tables priced [Census FT.76(1)–(2)];

a declaration converts them into data immediately;

(5) the dynamics rates [Cal-F r] : a dynamics construction supplying $J^{\pm}$ rates, or the free datum retained; the tick-rate/field coupling requires emergent time [FT.131(3)];

(6) the momentum-space Σ-resolvent : plane-wave boundary-channel states on the coarse-grained channel [Def-H.13 pattern];

constructed at BM.134 (Def-BM.20); consumed by scattering reads of sealed objects;

(7) the embedding datum : a circumradius or moment-of-inertia read for rotor-type reads; the ratios are embedding-free and computed [FT.87(6)];

adoption would not lift the $4\hat{\mu}'^2$ ceiling [FT.76(3); FT.87(1)];

(8) the frame section : $e \mapsto \mathcal{F}_e$ over each skeleton [30 native frames catalogued, FT.93] plus its EW.32-transport coupling and a global Dirac-type operator on the extended carrier [Def-H.34];

the weak-chirality pincer on the pair arena is untouched [FT.93(3)].

Req. The cited tags

Pf. Specification content: each row states its requirement and cites its anchors; no truth claim is made; anchor existence machine-checked [`checks/verify_era_structure.py`]. ■

Phys. The era's debts are itemized as build-specs: what a charge read must satisfy, what an exchange space must carry, which two conventions must be chosen and what each choice costs. Nothing here is adopted; everything here is priced. The rung closes with its obligations named.

4.6 Tier BM.4 — Noether's Theorem from the Janus Seal Topology: the Discrete Conservation Law, the Continuum Promotion, the Stress-Energy of a Sealed Composite, and the Canonical-Pair Obstructions

Definition 4.3 (Def-BM.3 — The species-identification layer (declared)). Def. A species name := a label attached to a derived structural tuple of a sealed composite. The layer is constitutive in exactly one respect — the name↔tuple assignment — and non-derived in that same respect: no theorem of Parts 1–4 forces any name onto any tuple, and no name carries a datum into a derivation.

(1) the tuple a name attaches to:

(junction grammar [Def-BM.1], sealing-phase class [BM.2(1)], charge class [EM.9; clause (3)], $m_{\text{hid}} = E_{\text{int}} \ln 2/\mathfrak{h}$ [FT.44(4)], $w = p/\rho$ [CT4.5; BM.10(3)]);

(2) the declared map (baryon-type, ε-sealed 3-junctures):

neutron := $\varepsilon_{ijk} c_1^i c_2^j c_3^k$ in the neutral charge class ($q_{\text{net}} = 0$);

proton := $\varepsilon_{ijk} c_1^i c_2^j c_3^k$ in the unit charge class ($q_{\text{net}} = +1$);

antiproton := ω-image of the proton [D2.6];

(3) the declared map (meson-type, δ-sealed 2-junctures):

$\pi^0 := \delta^i{}_j\, c^j \bar{c}_i$ in the neutral class;

$\pi^{\pm} := \delta^i{}_j\, c^j \bar{c}_i$ in the $q_{\text{net}} = \pm 1$ classes;

(4) the charge-class clause (conditional, declared):

q_{net} is the additive multi-chain read specified but ¬constructed [BM.6(1)] :

the class labels 0,

± 1 of clauses (2)–(3) are the ternary alphabet of a single end read [$q_{\text{EM}} \in \{-1, 0, +1\}$ [D3.4, EM.9, P2]] carried to the composite by declaration;

the constituent alphabet remains ternary: the fractional read ($\pm\frac{1}{3}$, $\pm\frac{2}{3}$) is deficit $1'$ content [FT.125(4)] — ¬supplied, ¬assumed here;

(5) the naming-precedent clause:

D3.7 names two transit structures (photon := trivial chain $\subseteq V_2$; gluon := trivial chain $\cap\, V_3 \neq \varnothing$) and D3.9 names the two sealings (baryon, meson) :

this definition extends that precedent from transit structures and sealings to charge classes of sealed composites;

a name is a definition-side label, ¬a measurement;

(6) the data bar (standing):

¬∃ a measured external value in any clause of Parts 1–4 [P5 stands];

external comparison data remain quarantined to the audit rows [Census FT.114, FT.120];

the names of clauses (2)–(3) carry ¬mass value, ¬coupling value, ¬lifetime, ¬radius : every number attached to a named tuple is corpus-derived (m_{hid}, Θ_{int}, w, E_{int});

(7) scope:

species names occur in Part 4 only — in this definition, in the Phys and Rmk prose of this part, and nowhere in a Thm clause of Parts 1–3 [those tiers' own label-discipline rows stand unmodified];

extension of the map beyond clauses (2)–(3) requires a logged declaration first [Part 4 §Cal-BM rule];

the layer adds ¬dimensionful scale [P1] and ¬P4 parameter.

Rmk (identification discipline). The layer answers which name sits on which derived tuple; it does not answer which species sits in which gauge representation on which chiral ray — that assignment rule is application-layer content; its non-derivability here is proven [FT.125(4)]. Naming the neutron is a labelling act on the ε-sealed neutral class; it derives no neutron property, imports no neutron datum, and closes no open row.

Req. D2.6, D3.4, D3.7, D3.9 (Parts 1–2); CT4.5 (Part 2); EM.9 (Part 2); Def-BM.1, BM.2(1), BM.6(1) (this part); FT.44(4), FT.125(4) (Part 3, Census); P1, P2, P5 (Parts 1–2); Census FT.114, FT.120 (quarantine rows)

Pf.

Definition block: the constitutive clause is the name ↔ tuple assignment of (2)–(3);

the non-derived clauses are (4) (the charge class is declared pending BM.6(1)) and the name ↔ tuple direction of (1) [no theorem forces it];

the standing bars (6)–(7) are scope clauses cited at use. ■

Phys. The baryomesonic era names its objects. A name is pinned to a tuple the corpus already derives — the junction grammar, the sealing phase class, the charge class, the informational mass, the equation of state — and the pinning itself is the declared part. The neutron is the ε-sealed three-juncture in the neutral charge class; the proton is the same structure in the unit class; the pion-type triplet is the δ-sealed pair in its three classes. Nothing measured enters: the mass read is $E_{\text{int}} \ln 2/\hbar$, the phase class is $4\mathbb{Z}$, the equation of state is $w = 0$, and the charge class is the ternary alphabet carried up by declaration while the additive read is still an open construction. The layer extends a precedent the corpus already set when it named the photon and the gluon as chain classes.

Theorem 4.7 (BM.7 — The discrete conservation law is Janus topology: no symmetry argument, no action, no limit). Thm.

(1) the balance is definitional, and the seal ledger realizes it uniquely:

$$\text{valid universe} \iff \text{network closed} \ \wedge \ \#J^{+}$$

$$= \#J^{-} \ \wedge \ \in \text{Im(active mask)} \ \wedge \ \neg\exists$$

dangling boundary [D2.11, D1.3];

$$J^+ : \ (\Delta n_{\mathcal{R}}, \delta h, \delta\tau) = (+1, +1, +1),$$

$$J^- : \ (-1, -1, -1)$$

[CT0.3] :

the seal excess $\#J^+ - \#J^- = n_{\mathcal{R}}$ is the ledger's own occupation count;

the D1.2 transition domains force strict alternation (J^- undefined at $n_{\mathcal{R}} = 0$; J^+ undefined at $n_{\mathcal{R}} = 1$) : the single-pair history of length n is unique $\forall n$ [FT.144(2); machine];

the completed history of length $2m$ is unique and balanced: $(J^+J^-)^m$,

$$\#J^+ = \#J^- = m$$

[FT.144(2), FT.156(5); machine];

the prefix law $\#J^+(\text{prefix}) \geq \#J^-(\text{prefix})$ holds on every admissible history, with equality

$$\iff n_{\mathcal{R}}$$

$$= 0 \text{ [FT.144(2); machine];}$$

the pure order reversal of every nonempty completed history leads with J^- : inadmissible at its first event [FT.144(4)(ii); machine];

the relaxation census (prefix law alone, alternation dropped) : #{completed words of length $2m$}

$$= \text{Cat}_m$$

$$= \frac{1}{m+1}\binom{2m}{m}$$

$$= 1,$$

$$1,$$

$$2,$$

$$5,$$

$$14, \qquad 42$$

$$[\text{machine, exhaustive } \leq 12];$$

$$\text{Cat}_m > 1$$

$$\iff m$$

$$\geq 2: \text{ the D1.2 domains are strictly stronger than the prefix law from } 2m$$

$$= 4 \text{ on — the uniqueness of } (1) \text{ is domain content, } \neg\text{balance content;}$$

$$\text{two pairs: } \#\{\text{completed interleavings of total length } 2n\}$$

$$= \sum_{m_0+m_1=n} \frac{(2n)!}{(2m_0)!\,(2m_1)!}$$

$$= 2,$$

$$8,$$

$$32 \quad (2n = 2, 4, 6)$$

$$[\text{FT.144}(2);\ \text{machine: exhaustive}];$$

$$\text{balance and the prefix law survive every interleaving [machine];}$$

$$\text{the balanced sector is } a_+ - a_- = |\mathcal{R}'| - |\mathcal{R}| = 0 \text{ of the budget identity [FT.108}(4)\text{];}$$

$$(2) \quad \text{the endpoint read is a boundary term, exactly:}$$

$$q_{\mathrm{EM}}(c) = \frac{\sigma_{s(c)} - \sigma_{t(c)}}{2} \in \{-1, 0, +1\}$$

[D3.4, EM.9; all three values attained];

the telescoping identity: on a materialized chain $c = (e_1, \ldots, e_k)$, $t(e_i) = s(e_{i+1})$ [D1.4] :

$$q_{\mathrm{EM}}(c) = \sum_{i=1}^{k} j(e_i),$$

$$j(e) := \frac{\sigma_{s(e)} - \sigma_{t(e)}}{2}$$

[machine: exhaustive $k \leq 7$];

interior vertex signs cancel pairwise; the sum is the difference of the two end reads — the charge at the start equals the charge at the end;

the closed-chain read: $\partial c = 0 \implies s(c) = t(c) \implies \sigma_{s(c)} = \sigma_{t(c)} \implies q_{\mathrm{EM}}(c) = 0$ exactly

[D1.4; machine: every closed configuration];

total charge vanishes in every valid universe: each end is sealed against an oppositely oriented end carrying the same sign [EM.11];

(3) the ground is the chain complex, not an extremum:

$\partial c := t(c) - s(c)$ (formal difference of vertices),

$\partial^2 = 0$ since $\partial(\text{vertex}) = 0$

[D1.4; machine];

closed $\iff \partial c = 0 \ \wedge \ \#J^+ = \#J^-$ [D1.4];

$\ker \partial = \{\text{closed chains}\}$,

every boundary has total multiplicity 0 [machine];

conservation is the kernel statement of an integer boundary map: no action functional, no variational derivative and no continuous parameter enters (1)–(3);

(4) the Noether route is unavailable for this read (obstruction):

a Noether charge requires a one-parameter group action $\mathbb{R} \to \mathrm{Aut}(\text{the read's value space})$ [math];

the value space of q_{EM} is the ternary set $\{-1, 0, +1\}$,

so the action lands in $\mathrm{Perm}\{-1, 0, +1\}$

$= S_3$ [EM.9];

$$U(1)$$

$$= \varinjlim_{m} \mathbb{Z}/m$$

$= \mathbb{Q}/\mathbb{Z}$ is divisible [math] : its image in a finite group is divisible, hence trivial — no transposition of S_3 has a square root, no 3-cycle has a cube root [machine: the full root census of S_3];

$$|\hom(\mathbb{Z}/m, S_3)|$$

$$= \#\{g \in S_3 : \mathrm{ord}(g) \mid m\}$$

$$= 1,$$

$$4,$$

$$3,$$

$$4,$$

$$6,$$

6 $(m = 1, 2, 3, 4, 6, 12)$ [machine] : every finite stage admits nontrivial homs, and the only projectively compatible family is the trivial one;

$\therefore$ $\neg\exists$ a nontrivial continuous $U(1)$ action on the charge alphabet : q_{EM} is $\neg$a Noether charge;

(5) priority:

(1)–(3) are exact, discrete, integer-valued and hold at every scale, including $\hbar \neq 0$;

the differential law $\partial_\mu J^\mu = 0$ is their continuum limit [BM.11] — derived from them, $\neg$converse;

the converse fails: $\partial_\mu J^\mu$

$= 0$ is a density statement and does not force the integer balance at finite cell size [BM.11(5)].

Rmk (label discipline). $J^\pm$, seal, source, sink and current name ledger structures [D1.2, D1.3, CT0.3]; the topology labels are internal bookkeeping [Def-H.32(4) pattern]. Where clause (2) is read on a named composite, the name is Def-BM.3's label on the derived tuple; the theorem's content is the tuple.

Req. A4, A6, A7 (Part 1); D1.2, D1.3, D1.4, D2.11, D3.3, D3.4 (Parts 1–2); CT0.3, CT0.9 (Part 1); EM.9, EM.11 (Part 2); P2 (Part 1); Def-BM.1, Def-BM.3 (this part); FT.108(4), FT.144(2), FT.144(4)(ii), FT.156(5) (Part 3); exhaustive ledger-word enumeration, the Catalan census, multinomial interleaving counts, hom-counts and the root census of S_3, divisibility of $\mathbb{Q}/\mathbb{Z}$ [math; machine: `checks/verify_H_noether.py` §A].

Pf.

(1) : D2.11/D1.3 verbatim;

CT0.3 increments verbatim $\implies$ $\#J^+ - \#J^- = n_{\mathcal{R}}$ [exact integers];

alternation, uniqueness, the prefix law, the reversal inadmissibility : FT.144(2),

(4)(ii) re-derived by exhaustive enumeration at lengths

≤ 12 [machine]

relaxation census : #{prefix-admissible completed words of length $2m$}

$$= \mathrm{Cat}_m \text{ [machine enumeration = the closed form } \tfrac{1}{m+1}\binom{2m}{m}\text{]};$$

$$\mathrm{Cat}_m = 1 \iff m \leq 1 \text{ [exact integers]};$$

$$\text{two pairs : } \binom{2n}{2m_0} \text{ labelled interleaving patterns summed over } m_0 + m_1$$

$$= n \text{ [machine]}$$

$$= \text{the closed form}$$

$$(2) : \text{ EM.9 table verbatim (machine: the four sign pairs)};$$

$$\text{telescoping : } \sum_{i=1}^{k} \tfrac{1}{2}(\sigma_{v_{i-1}} - \sigma_{v_i})$$

$$= \tfrac{1}{2}(\sigma_{v_0} - \sigma_{v_k}) \text{ [pairwise cancellation of the } k-1 \text{ interior terms; machine: all } \sigma \in \{\pm 1\}^{k+1},\ k \leq 7\text{]};$$

$$\partial c = 0 \implies v_0 = v_k \implies q_{\mathrm{EM}} = 0 \text{ [machine]}$$

$$(3) : \ \partial c$$

$$= t(c) - s(c) \text{ is a signed pair of vertices;}$$

$$\partial(\text{vertex})$$

$$= 0 \implies \partial^2 c$$

$$= 0 \text{ [D1.4; machine: total multiplicity 0 on every enumerated path]};$$

$$\ker \partial = \{s(c) = t(c)\} \text{ [machine: equality exactly on the closed paths]}$$

$$(4) : \text{ a Noether charge is the generator of a one-parameter symmetry action [math]};$$

$$\text{the value space is } \{-1, 0, +1\} \text{ [EM.9]} \implies \text{the action factors through } S_3;$$

$$U(1) \cong \mathbb{Q}/\mathbb{Z} \text{ divisible [math];}$$

a divisible image in S_3 consists of elements having m-th roots $\forall m : \{e\}$ only [machine: the root census];

$$\text{hom-counts} : g^m = e \iff \text{ord}(g) \mid m \text{ [machine]}$$

$$(5) : \text{ conjunction of } (1)\text{–}(4);$$

the limit step is BM.11's content, cited as a forward pointer and imported nowhere here. ■

Phys. Charge conservation in this corpus is a fact about integers, not about symmetries. A valid universe is one in which every crystallization has its dissolution — the definition says so, and the transition domains make the completed history unique: seal, unseal, seal, unseal. The endpoint charge read telescopes: sum the per-edge source along a materialized chain and every interior sign cancels, leaving the two ends; close the chain and the read is exactly zero. The ground is a boundary map with $\partial^2 = 0$ — the kernel statement of an integer chain complex. And the Noether route is not merely unnecessary here, it is closed: the charge alphabet has three elements, so any continuous charge-rotation action would have to land in a three-element permutation group, and a divisible group has no nontrivial finite image. There is no $U(1)$ to rotate the ternary read, hence no Noether current to build it from. What the continuum limit delivers later is the density form of a law that already holds, exactly, at every cell size.

Theorem 4.8 (BM.8 — The continuum promotion: the hop becomes a translation, the C_3 becomes $SO(3)$, the tick stays an arrow, and the gauge algebras were never discrete). Thm.

$$(1) \quad \text{spatial translations from the discrete hop:}$$

$$\text{on the emergent chart } \chi([v])$$

$$= (x^0, x^1, x^2, x^3) \text{ [CT2.6; D1.8 : } x^0 := \tau_n],$$

$$\text{let } T_\ell f(x) :$$

$$= f(x + \ell) \text{ be the hop of length } \ell \text{ and } D_\ell :$$

$$= \tfrac{1}{\ell}(T_\ell - 1);$$

the exact commutator identity:

$$[\hat{x},\, D_\ell] = -T_\ell$$

at every $\ell \neq 0$

[machine: exact on $x^k,\ k \leq 7$];

the exact binomial expansion:

$$D_\ell\, x^k = \sum_{j=1}^{k} \binom{k}{j} x^{k-j} \ell^{\,j-1} = k\, x^{k-1} + \ell \cdot (\text{exact rational polynomial})$$

[machine];

the Weyl transport:

$$T_\ell\, \hat{x} = (\hat{x} + \ell)\, T_\ell$$

[machine];

$$\ell \to 0: \quad D_\ell \to \partial_x, \quad [\hat{x}, \partial_x] = -1, \quad \hat{p} := -i\partial_x \implies [\hat{x}, \hat{p}] = i \ \ (\hbar = 1\ [\text{D2.9}]);$$

the dictionary carrier:

the three T-odd coefficients w_i [D2.1; CT1.2] $\longmapsto$ $N_i = \beta w_i$ $\longmapsto$ $\bar{g}_{0i} = \varkappa N_i$ [CT1.12; D2.1];

N_i is the Lagrange multiplier of the momentum constraint, whose charge generates spatial diffeomorphisms [math: ADM] :

the conserved quantity is the momentum P_i,

and P_i is T-odd, matching its carrier [$w_i \mapsto -w_i$ [FT.156(3)]];

(2) time translations: the tick is a semigroup, and the dynamics strictly contracts — an obstruction with a conditional charge:

$$\neg\exists\, \tau^{-1}\ [\text{A6; FT.156(5)}];$$

$$|\mathfrak{L}(\tau_{n+1})| \geq |\mathfrak{L}(\tau_n)|\ [\text{A7; machine}],\ \text{strict at every crystallization;}$$

$$V(\tilde{K}) \mapsto \left(1-\tfrac{\gamma}{4}\right)^2 V(\tilde{K}),$$

$$\text{deficit} = \tfrac{\gamma(8-\gamma)}{16},$$

$$\gamma = 1 - 4\varepsilon_q \in (0,1] \text{ [L2, D0.3, CT0.8; machine]};$$

$$\left(1-\tfrac{\gamma}{4}\right)^2 + \tfrac{\gamma(8-\gamma)}{16} = 1 \text{ exactly};$$

$$\tfrac{\gamma(8-\gamma)}{16} = 0 \iff \gamma \in \{0,8\} \text{ [machine]};$$

$$\gamma = 0 \iff \varepsilon_q = \tfrac{1}{4} \notin [0, \tfrac{1}{4}) \text{ [CT0.8]};$$

$$\therefore \text{ on the whole admissible range the tick map strictly contracts } V: \quad \neg\exists$$

$$\text{a nonzero tick-invariant of the pair-state dynamics} \implies \neg\exists$$

$$\text{an exact time-translation invariance of the microscopic dynamics;}$$

the conditional energy read (declared scope, theorem-local, L0's dynamical-scope pattern) : under the stationarity hypothesis — the coarse-grained coefficient fields (a_{00}, q_{ij}, w_i) of D2.1 are tick-independent across the cells of CT2.6(ii)–(iii) — the continuum limit carries a time-translation one-parameter group whose charge is the energy read;

on the sealed sector the hypothesis is structural rather than imposed: a bound chain's reads are tick-independent [D1.5; forward pointer: BM.9(3)];

$$\text{at } \gamma \to 0 \text{ the contraction} \to 1 \text{ and the deficit} \to 0 \text{ [exact rationals]};$$

(3) rotations: the C_3 is a rational point of a one-parameter subgroup already present in the register:

$$\mathcal{W} := W_1 + W_2 + W_3,$$

$$\mathcal{W}^3 = 3\mathcal{W}$$

$$\text{[FT.158(1); machine]};$$

$$X := \hat{n}\cdot W = \mathcal{W}/\sqrt{3}, \quad \hat{n} = \tfrac{(1,1,1)}{\sqrt{3}}:$$

$$X^3 = X,$$

$$\text{charpoly}(\mathcal{W}) = x^4 - 3x^2 \implies \text{spec}(X) = \{1, 0, -1, 0\}$$

[machine: Faddeev–LeVerrier, exact traces];

the promotion identity:

$$u(\theta) := 1 + (\cos\theta - 1)X^2 - i\sin\theta\, X$$

satisfies

$$u(\theta)u(\phi) = u(\theta + \phi) \ \ \forall \theta, \phi$$

exactly in $\mathbb{Q}[i][c, s, c', s', X]/(c^2{+}s^2{-}1,\ c'^2{+}s'^2{-}1,\ X^3{-}X)$ [machine] :

$$u(0) = 1, \quad u(\theta)u(-\theta) = 1, \quad u(\theta)^3 = u(3\theta)$$ [triple-angle polynomials];

the group law closes in $\text{span}\{1, X, X^2\} :\ X^3$

$= X$ is the only relation used [machine: degree census];

the discrete elements are its rational points:

$$\theta = \tfrac{2\pi}{3} :$$

$$U = E_0 - \tfrac{1}{2}\mathcal{W}^2 - \tfrac{i}{2}\mathcal{W} = u \otimes u,$$

$$U^3 = E_0,$$

$UX_iU^\dagger = X_{i+1}$ $(X \in \{W, J, K\})$ [FT.111, FT.158(1); machine];

$\theta = \pi :\quad U_{ij} = E_0 - (W_i + W_j)^2, \quad U_{ij}^2 = E_0, \quad U_{ij} :\ X_i \leftrightarrow X_j,\ X_k \mapsto -X_k$ [FT.156(4); machine];

$C_3 \leq SO(3)$ is inner in the weak rotation algebra [CT1.4 : $\text{span}\{W_i\} \cong \mathfrak{su}(2)$; CT1.8 : J, K vectors] :

the promotion is exact, ¬approximate;

the charge:

$$\textstyle\sum_i W_i^2 = \frac{3}{2}E_0 + \frac{1}{2}\Sigma = E_0 + \mathbb{S} = 2\,\Pi_{\mathcal{R}}\ [\text{FT.157}(2);\ \text{machine}]:$$

$$\text{Casimir } 2 \text{ on } \vee^2\ (j{=}1), \quad 0 \text{ on } \wedge^2\ (j{=}0)\ [\text{machine}];$$

the integer weight lattice $\{1, 0, -1\}$ is what makes the rational angles admissible: $u(\theta + 2\pi)$

$= u(\theta)$ on integer weights [math];

(4) gauge: the slot algebras are natively continuous — no limit is needed for the algebra:

$$[W_i, W_j] = i\epsilon_{ijk}W_k \text{ on all } 9 \text{ pairs } [\text{CT1.4};\ \text{machine}];$$

$$\mathfrak{u}(2) = \mathfrak{su}(2) \oplus \mathfrak{u}(1)_{\text{centre}}:$$

$Y = \frac{1}{2}\sigma_0$ commutes with all W_i,

$$3 \text{ traceless } + \ 1 \text{ central } = 4\ [\text{CT0.13},\ \text{EM.2};\ \text{machine}];$$

$$\pi : \mathfrak{u}(3) \to \mathfrak{su}(3),$$

$$\pi(X)$$

$= X - \frac{\mathfrak{T}(X)}{3}I_3:$ Lie homomorphism on all 81 basis pairs,

$$\ker \pi$$

$= \text{span}\{I_3\}$ [CT3.7; machine];

the colour block: all 8 generators traceless,

all 64 commutators in the span,

Jacobi on all 512 triples,

$$[H_1, H_2]$$

$$= 0,$$

$$\det \mathrm{K}$$

$$\neq 0 \text{ [CT3.8, FT.82(3); machine: exact rationals];}$$

what the continuum limit supplies is the gauging, ¬the algebra: the parameters become fields on M [CT2.6] and the covariant derivative is the chart-side object [math];

(5) the three charge statuses are distinct:

colour: on a sealed composite every Hermitian slot operator reads only its trace $\mathfrak{T}(H)$

$= 3a$ [BM.1(2); machine] and the invariant line is 1-dimensional [BM.1(3); machine : $27 - \mathrm{rk} = 1$] : the colour charge of a sealed singlet is 0 exactly;

weak isospin: the derived Casimir of (3) — 2 on the symmetric sector, 0 on the antisymmetric one [FT.157(2)];

electric charge: ¬a Noether charge [BM.7(4)] : the ternary read is topological;

the gauge Noether story of the corpus therefore has two continuous charges and one discrete invariant, not three of a kind.

Rmk (promotion discipline). Momentum, energy and angular momentum name the charges of the promoted one-parameter groups; the names are physics vocabulary for derived structures [A §0.2]. Clause (2)'s stationarity hypothesis is a theorem-local dynamical scope of the species of L0's crystallization–tilt coupling and CT4.5a's ensemble scope, declared at the point of use; it is not an axiom, and it is not discharged by any tag of Parts 1–3.

Req. A6, A7 (Part 1); D0.3, D1.5, D1.8, D2.1 (Parts 1–2); CT0.8, CT0.13, CT1.2, CT1.4, CT1.8 (Part 1); CT1.12, CT2.6, EM.2 (Part 2; CT2.6's conditions (ii)–(iii) at the point of use — A §0.2a); CT3.7, CT3.8 (Part 2); L2 (Part 1); Def-BM.3, BM.1(2), BM.1(3) (this part); FT.82(3), FT.111, FT.144(2), FT.156(3), FT.156(4), FT.156(5), FT.157(2), FT.158(1) (Part 3); exact $\mathbb{Q}[x, \ell]$ hop calculus, exact identities in $\mathbb{Q}[i][c, s, c', s', X]/(c^2+s^2-1, X^3-X)$, Faddeev–LeVerrier characteristic polynomials, exact $\mathbb{Q}(i)$ conjugation tables, exact $\mathbb{Q}$ structure constants and Killing determinants, exact rational contraction arithmetic [math; machine: `checks/verify_H_noether.py` §B].

Pf.

$$(1):\ (T_\ell - 1)f \text{ is divisible by } \ell \text{ in } \mathbb{Q}[x,\ell] \text{ for every polynomial } f \text{ [machine]};$$

$$\hat{x}D_\ell f - D_\ell(\hat{x}f)$$

$$= \tfrac{1}{\ell}\big[x(f(x{+}\ell) - f(x)) - ((x{+}\ell)f(x{+}\ell) - xf(x))\big]$$

$$= -\tfrac{\ell f(x+\ell)}{\ell}$$

$$= -T_\ell f \text{ [exact]}$$

$$D_\ell x^k$$

$$= \tfrac{(x+\ell)^k - x^k}{\ell}$$

$$= \sum_{j\geq 1} \binom{k}{j} x^{k-j}\ell^{j-1} \text{ [binomial theorem; machine coefficient census]};$$

$$\ell \to 0 \text{ leaves } kx^{k-1} \text{ and } [\hat{x}, \partial_x] = -1;$$

$$\hat{p} = -i\partial_x \implies [\hat{x}, \hat{p}] = i \ [\hbar = 1,\ \text{D2.9}]$$

$$\text{dictionary rows verbatim [D2.1, CT1.12]};$$

N_i as the momentum-constraint multiplier and P_i as its charge [math: ADM];

$$w_i \mapsto -w_i \text{ under } T \text{ [FT.156(3) re-cited]}$$

$$(2):\ \text{A6/A7 verbatim [machine: monotone and strictly increasing samples]};$$

L2 contraction re-derived as an exact polynomial identity in γ;

zero set $\{0, 8\}$ [machine over a rational sweep and by factorization];

$$\gamma \in (0, 1] \text{ on } \varepsilon_q \in [0, \tfrac{1}{4}) \text{ [D0.3, CT0.8]}$$

strict contraction $\forall\gamma$

$$\in (0,1]$$

$$\implies V(\mathcal{T}^n\rho)\to 0 \text{ and no nonzero invariant [L1 monotone; exact rationals];}$$

the conditional clause states its hypothesis explicitly and cites no tag for it

$$(3):\ \mathcal{W}^3=3\mathcal{W}\ [\text{machine}]\implies X^3=X;$$

charpoly x^4-3x^2 [Faddeev–LeVerrier, exact; $\operatorname{tr}\mathcal{W}=0,\ \operatorname{tr}\mathcal{W}^2=6,\ \operatorname{tr}\mathcal{W}^4=18$];

$u(\theta)u(\phi)=u(\theta+\phi)$ [machine in the quotient ring, using the addition formulas and $X^4=X^2$]

$$\theta$$

$$=\tfrac{2\pi}{3}:\ (\cos\theta-1)$$

$$=-\tfrac{3}{2},$$

$$\sin\theta/\sqrt{3}$$

$$=\tfrac{1}{2}$$

$$\implies u$$

$$=E_0-\tfrac{1}{2}\mathcal{W}^2-\tfrac{i}{2}\mathcal{W}$$

$$=u\otimes u\ [\text{machine entrywise}];$$

$$U^3=E_0,\ UX_iU^\dagger=X_{i+1}\ [\text{machine, all } 9];$$

$$\theta=\pi:\ (\cos\pi-1)=-2,\ X=(W_i{+}W_j)/\sqrt{2}\implies U_{ij}=E_0-(W_i{+}W_j)^2\ [\text{machine}]$$

$\sum_i W_i^2 = \frac{3}{2}E_0 + \frac{1}{2}\Sigma$ [FT.157(2) re-derived; machine];

Casimir reads on $\Pi_{\mathcal{R}},\ \Pi_{\mathcal{V}}$ [machine]

(4) : all rows re-derived entrywise over $\mathbb{Q}(i)$ and $\mathbb{Q}$ [machine];

the homomorphism and kernel rows of CT3.7 on the $\mathfrak{u}(3)$ basis $\{T_a\} \cup \{I_3\}$ [machine: all 81 pairs]

(5) : BM.1(2)–(3) re-derived on exact witnesses [machine : $\langle\varepsilon|\sum_i H^{(i)}|\varepsilon\rangle = \mathfrak{T}(H)\langle\varepsilon|\varepsilon\rangle;\ (H\otimes I - I\otimes H^{\mathrm{T}})\,\mathrm{vec}(I) = 0;\ 27-\mathrm{rk} = 1$];

BM.7(4) re-cited. ■

Phys. Three of the four continuum symmetries are already sitting in the register; the fourth is not, and the structure yields why. The hop is a translation with an exact error term: the commutator of position with the finite difference is minus the shift operator at every step size, and the binomial expansion hands over the derivative as its leading term, so the momentum generator and its canonical commutator are the $\ell \to 0$ read of an identity that never needed a limit to be true. Rotation is stronger still: because the weak generator satisfies $X^3 = X$, the exponential of $-i\theta X$ is a quadratic polynomial in X with $\cos\theta$ and $\sin\theta$ as coefficients, and the group law holds as a ring identity — the 120° seal rotation and the 180° transpositions are just two rational points of a one-parameter group that was there all along, with integer weights making the periodicity exact. The gauge algebras were never discrete: $\mathfrak{su}(2)$, $\mathfrak{su}(3)$ and the rung-2 centre close as Lie algebras over the rationals, and what the continuum adds is the gauging, not the algebra. Time is the exception, and it is an honest one — the tick has no inverse and the ledger dynamics strictly contracts its Lyapunov function at every admissible threshold, so the microscopic theory has no exact time-translation symmetry and the energy charge exists only under a declared stationarity scope, which the sealed sector satisfies structurally. The charge ledger then splits three ways: colour reads zero on every sealed singlet, weak isospin reads the derived Casimir, and electric charge is not a Noether charge at all.

Theorem 4.9 (BM.9 — The stress-energy tensor of a sealed composite: the two reads, their exact sign reconciliation, and the dust source). Thm.

(1) the corpus definition and the field equations:

$$T_{\mu\nu} := -\frac{2}{\sqrt{-g}}\frac{\delta S_{\text{force}}}{\delta g^{\mu\nu}}$$

[CT1.22];

$$S_{\text{eff}} = S_{\text{geom}} + S_{\text{force}}, \qquad S_{\text{geom}}[g] = \frac{1}{2\kappa^2}\int d^4x\,\sqrt{-g}\,(R - 2\Lambda) \qquad \text{[CT1.21]};$$

$$\delta S_{\text{eff}} = 0 \iff G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa^2 T_{\mu\nu}, \qquad \kappa^2 = 8\pi G_N = \frac{2\pi\,\mathfrak{h}^2}{\ln 2} \qquad [\text{CT1.23},\ \text{CT2.9},\ \text{P3}];$$

(2) the canonical read and the exact sign reconciliation:

$$\text{for } \mathcal{L} = -\tfrac{1}{2} g^{\alpha\beta} \partial_\alpha \phi\, \partial_\beta \phi - \tfrac{1}{2} m^2 \phi^2 :$$

$$\frac{\partial \mathcal{L}}{\partial(\partial^\mu \phi)} = -\partial_\mu \phi \ \ [\text{exact}];$$

$$T^{\text{can}}_{\mu\nu} := \frac{\partial \mathcal{L}}{\partial(\partial^\mu \phi)}\, \partial_\nu \phi - g_{\mu\nu} \mathcal{L} = -\partial_\mu \phi\, \partial_\nu \phi - g_{\mu\nu} \mathcal{L};$$

$$T^{\text{Hilb}}_{\mu\nu} = \partial_\mu \phi\, \partial_\nu \phi + g_{\mu\nu} \mathcal{L};$$

$$\boxed{T^{\text{can}}_{\mu\nu} = -\,T^{\text{Hilb}}_{\mu\nu}}$$

exactly, at signature $(-,+,+,+)$ [A3, C13, CT1.12] with this $\mathcal{L}$ sign

[machine: all 16 components on 3 rational metric witnesses];

the convention is fixed by the energy sign: at rest T^{Hilb}_{00}

$= \tfrac{1}{2} \sum_i (\partial_i \phi)^2 + \tfrac{1}{2} m^2 \phi^2$ with every coefficient $+\tfrac{1}{2}$,

while T^{can}_{00} carries $-\tfrac{1}{2}$ [machine: exact coefficient census];

$\therefore$ CT1.22′s minus is the positive-energy read at $(-,+,+,+)$;

the variational composition is exact: $\dfrac{\delta(\sqrt{-g}\,\mathcal{L})}{\delta g^{\mu\nu}}$

$$= -\tfrac{1}{2}\sqrt{-g}$$

$\big(\partial_\mu \phi\, \partial_\nu \phi + g_{\mu\nu} \mathcal{L}\big)$ [machine];

(3) the Noether identity (translation current):

$$\partial^\mu T^{\text{Hilb}}_{\mu\nu} = (\Box \phi - m^2 \phi)\, \partial_\nu \phi$$

as an exact jet identity [machine: all 4 components, 3 metric witnesses];

$\Box\phi - m^2\phi = 0$ (the field equation of this $\mathcal{L}$) $\implies \partial^\mu T^{\text{Hilb}}_{\mu\nu} = 0$:

$T_{\mu\nu}$ is the Noether current of the translation group promoted in BM.$8(1)$;

the trace identity: $\quad g^{\mu\nu}T^{\text{Hilb}}_{\mu\nu} = -(\partial\phi)^2 - 2m^2\phi^2 \quad$ [machine: exact in 4 dimensions];

(4) the sealed composite at rest — the neutron [Def-BM.3 : the ε-sealed 3-juncture in the neutral charge class] :

$$T^{\text{bd}}_{\mu\nu} = (\rho + p)u_\mu u_\nu + p\, g_{\mu\nu}$$

[CT1.22 bound part];

$u = (1,0,0,0), \quad g_{\mu\nu} = \text{diag}(-1,1,1,1), \quad c = 1$ [D2.9] :

$$T_{\mu\nu} = \text{diag}(\rho,\, p,\, p,\, p)$$

[machine: exact component identity];

fully sealed $\implies p = 0, \quad w = \frac{p}{\rho} = 0$

[CT4.5 : $T^{(D^-)}_{\mu\nu} = \rho\, u_\mu u_\nu,\ p_{D^-} = 0$] :

$T_{\mu\nu} = \text{diag}(\rho, 0, 0, 0)$ [machine];

static: $\partial_0 T_{\mu\nu} = 0$;

spherically symmetric: the isotropic spatial block p

δ_{ij} is invariant under all 48 signed triad permutations $\subset O(3)$,

hence under the promoted $SO(3)$ [BM.$8(3)$; machine];

trace and energy conditions:

$$g^{\mu\nu}T_{\mu\nu} = -\rho + 3p \overset{p=0}{=} -\rho;$$

$$\text{WEC},\ \text{DEC},\ \text{SEC hold exactly at } \rho \geq 0,\ p = 0\ [\text{machine: rational witnesses}];$$

$$\rho \geq 0, \qquad \rho = 0 \iff \neg\exists v :\ n(v) \geq 4 \qquad [\text{CT4.5};\ \text{machine on enumerated rung profiles}];$$

(5) the source read and its dimensions:

$$\rho = \frac{m_{\text{hid}}c^2}{V} = \frac{m_{\text{hid}}}{V}\ \ [c = 1,\ \text{D2.9}],$$

$$m_{\text{hid}} = \frac{E_{\text{int}} \ln 2}{\mathfrak{h}}$$

$$[\text{FT.44}(4),\ \text{FT.42}(1)];$$

$$[\mathfrak{h}] = \mathfrak{q}^{-1/2}, \quad [\text{length}] = \mathfrak{q}^{-1/2}, \quad [\text{energy}] = \mathfrak{q}^{+1/2}, \quad [T_{\mu\nu}] = \mathfrak{q}^{+2} \qquad [\text{CT2.1},\ \text{D2.9}];$$

$$[m_{\text{hid}}] = [E_{\text{int}}] - [\mathfrak{h}] = 0 + \tfrac{1}{2} = \mathfrak{q}^{+1/2} = [\text{energy}];$$

$$[V] = \mathfrak{q}^{-3/2};$$

$$[\rho] = \mathfrak{q}^{1/2-(-3/2)} = \mathfrak{q}^{+2} = [T_{\mu\nu}]$$

[machine: exact exponent arithmetic];

the $\ln 2$ cancellation:

$$\kappa^2\rho = \frac{2\pi\,\mathfrak{h}^2}{\ln 2} \cdot \frac{E_{\text{int}} \ln 2}{\mathfrak{h}\,V} = \frac{2\pi\,\mathfrak{h}\,E_{\text{int}}}{V}$$

[machine: exact, denominators cleared] :

the $\ln 2$ of the entropy calibration [CT2.9] and the $\ln 2$ of the informational mass [FT.44(4)] cancel in the sourced field equation : $G_{\mu\nu} + \Lambda g_{\mu\nu}$

$$= \frac{2\pi\mathfrak{h}E_{\text{int}}}{V}$$

$$u_\mu u_\nu \text{ at } p \qquad = 0;$$

(6) the width (forward pointer: BM.10(6)):

$$V = (\Delta x)^3,$$

$$\Delta x \geq \frac{1}{2\,\Delta p} \ (\hbar = 1) :$$

$$\rho = \frac{m_{\text{hid}}}{V} \leq 8\, m_{\text{hid}} (\Delta p)^3$$

[machine: exact coefficients];

ρ is finite for every $\Delta p > 0$:

$T_{\mu\nu}$ is a distribution of nonzero width, ¬a δ-source;

the width is the scale at which the coarse-grained fields are CT2.6(ii)-regular.

Rmk (convention record). Clause (2) prices both sign conventions instead of choosing silently: the canonical read of the classical formula and CT1.22's Hilbert read differ by an exact minus at signature $(-,+,+,+)$ with this $\mathcal{L}$ sign, and the corpus convention is the one with positive rest energy. Every downstream row — the dust form, the trace, the field equations, the energy conditions — is stated in the CT1.22 convention. No measured value enters any clause [P5; Def-BM.3(6)].

Req. A3 (Part 1); D1.5, D2.9 (Parts 1–2); CT1.21, CT1.22, CT1.23, CT2.1, CT2.9, CT4.5 (Part 2); Def-BM.1, Def-BM.3, BM.8(1), BM.8(3) (this part); FT.42(1) (Part 3); Census FT.44(4); P3, P5 (Parts 1–2); exact multivariate jet-ring polynomial identities, exact rational metric inversion by Cramer, exact coefficient censuses, Fractions [math; machine: `checks/verify_H_noether.py` §C].

Pf.

$$(1) : \text{ CT1.21/CT1.22/CT1.23 verbatim};$$

$$\kappa^2 = 8\pi G_N = 8\pi \cdot \tfrac{\mathfrak{h}^2}{4 \ln 2} = \tfrac{2\pi \mathfrak{h}^2}{\ln 2} \text{ [CT2.9; machine, denominators cleared]}$$

$$(2) : \ \partial \mathcal{L} / \partial(\partial^\mu \phi)$$

$$= -g^{\mu\alpha} \partial_\alpha \phi \cdot \tfrac{1}{2} \cdot 2$$

$$= -\partial^\mu \phi \text{ and } \partial^\mu \phi \text{ lowers to } \partial_\mu \phi \text{ in the first slot [exact]};$$

$$T^{\text{can}} + T^{\text{Hilb}}$$

$= 0$ componentwise [machine: 3 rational metrics, diagonal and fully off-diagonal];

$$\delta\sqrt{-g}$$

$$= -\tfrac{1}{2}\sqrt{-g}$$

$$g_{\mu\nu}\delta g^{\mu\nu},$$

$$\delta\mathcal{L}/\delta g^{\mu\nu}$$

$= -\frac{1}{2}\partial_\mu\phi\partial_\nu\phi$ [math] $\implies$ the variational composition [machine]

$$(3):\ \partial_\mu(d_\mu\phi\, d_\nu\phi) = \Box\phi\,\partial_\nu\phi + \partial^\mu\phi\,\partial_\mu\partial_\nu\phi;$$

$g_{\mu\nu}\partial^\mu\mathcal{L} = -\partial_\nu\phi\,(\partial^\alpha\phi) - m^2\phi\,\partial_\nu\phi$ at constant metric;

the two $\partial\partial\phi$ terms cancel by symmetry of $\partial_\mu\partial_\nu$ and relabeling [machine: the residual is exactly $(\Box\phi - m^2\phi)\partial_\nu\phi$]

$$(4):\ (\rho + p)u_\mu u_\nu + p g_{\mu\nu} \text{ at } u$$

$$= (1, 0, 0, 0),$$

$$g$$

$= \text{diag}(-1, 1, 1, 1):\ (0,0)$-entry ρ,

$(0, i)$-entries 0,

(i, j)-entry $p\delta_{ij}$ [machine];

$p = 0,\ w = 0$ [CT4.5 re-cited];

$R\,\text{diag}(\rho, p, p, p)R^{\text{T}} = \text{diag}(\rho, p, p, p)\ \forall R \in \{\text{signed permutations}\}$ [machine: all 48];

trace and the three energy conditions at $p = 0$ [machine]

$$(5):\ c = 1 \text{ [D2.9]};$$

$$m_{\text{hid}} = E_{\text{int}} \ln 2/\mathfrak{h} \text{ [FT.44(4) re-cited]};$$

$$\text{exponents}: \tfrac{1}{2} - 3(-\tfrac{1}{2}) = 2 \text{ [machine]};$$

$$\kappa^2\rho:\ (2\pi\mathfrak{h}^2/\ln 2)(E_{\text{int}} \ln 2/(\mathfrak{h}V))$$

$$= 2\pi\mathfrak{h}E_{\text{int}}/V \text{ [machine: cross-multiplied identity]}$$

$$(6):\ \Delta x\,\Delta p = \tfrac{1}{2} \text{ on rational witnesses [BM.10(6), forward pointer]};$$

$$(1/(2\Delta p))^3 = 1/(8(\Delta p)^3) \text{ [machine]};$$

$$\rho V = m_{\text{hid}} \text{ with } V > 0 \implies \rho < \infty \text{ [exact]}. \blacksquare$$

Phys. A sealed composite gravitates as dust, and the corpus can now say so with both sign conventions on the table. The classical canonical formula and CT1.22's metric definition differ by an exact minus at the corpus's own signature; the minus is what makes the rest energy positive, so the convention is decided by the energy condition rather than by taste. The same tensor is the translation current: its divergence is the field equation times the field gradient, an identity in the jet variables that vanishes on-shell — Noether's theorem, run in reverse from the symmetry the hop promoted. For the neutron, the fully sealed three-juncture in the neutral class, the bound-part form collapses to diag$(\rho, 0, 0, 0)$: static, spherically symmetric, pressure-free, exactly the D^- equation of state, and satisfying every energy condition at $\rho \geq 0$. Its source is the informational mass over the width volume, and the dimensions balance to $\mathfrak{q}^{+2}$ without adjustment. The two $\ln 2$ factors — one from the horizon-entropy calibration, one from the bit-count mass read — cancel in the sourced field equation, leaving $2\pi\mathfrak{h}E_{\text{int}}/V$. And because the width is strictly positive, the source is a distribution, not a point: the smoothing scale is where the coarse-graining hypothesis of the continuum limit takes over.

BM.10 — The canonical pair: the finite-dimension obstruction, the discrete Weyl pre-limit, and the three exact bounds. Structural-constraints record: see Appendix B, Section B.3.5 of the Addendum.

Theorem 4.10 (BM.11 — The continuum limit of the Janus topology: the differential conservation law and its two agreeing routes). Thm.

(1) the discrete divergence theorem carries no correction term:

$$\sum_{j=0}^{n-1} \ell\,(D_\ell f)(x_0 + j\ell) = f(x_0 + n\ell) - f(x_0)$$

EXACTLY on every rational grid [machine: $f = x^k,\ k \le 5;\ n \le 6;\ (x_0, \ell) \in \{(0,1), (\frac{1}{3}, \frac{2}{5}), (-\frac{7}{4}, \frac{1}{2})\}$];

the same total is the exact continuum integral of the derivative:

$$\sum_{j} \ell\,(D_\ell f)(x_j) = \int_{x_0}^{x_0+n\ell} f'(x)\,dx$$

[machine: exact equality at every ℓ, ¬a limit statement];

$\therefore\ \ell \to 0$ changes no term of the conservation identity [BM.8(1); CT2.6];

(2) the region law:

for a materialized chain c and every sub-interval $[i, j]$:

$$\sum_{e \in [i,j]} j(e) = \frac{\sigma_{v_i} - \sigma_{v_j}}{2} = \text{the boundary read}$$

[machine: all (chain, interval) pairs, $k \le 5$];

the full-chain case is $q_{\text{EM}}(c)$ [BM.7(2)];

the closed case reads 0 [D1.4; EM.11 : U valid $\implies Q_{\text{EM}} = \sum_{c \in U} q_{\text{EM}}(c) = 0$];

the two-dimensional form:

$$\sum_{\text{cells} \in R} \big(\nabla_\ell \cdot j\big) = \text{the boundary flux of } R$$

[machine: every rectangle of a 6×5 rational lattice];

the closed-network form:

on a periodic lattice the total divergence is exactly 0 [machine] :

the discrete image of $\#J^+ = \#J^-$ with no boundary term [D1.3, D2.11];

the seal excess summed over every completed admissible history of length

≤ 12 is exactly 0 [BM.7(1); machine];

(3) the limit and the pointwise law:

under CT2.6(ii)–(iii) [extra-axiomatic, declared at the point of use — A §0.2a] the cells carry unboundedly many real edges and the coefficient fields are of bounded variation :

$$\sum_{\text{cells}\in U}(\cdots) \longrightarrow \int_U \sqrt{-g}\,\partial_\mu J^\mu\, d^4x, \qquad \text{boundary flux} \longrightarrow \oint_{\partial U} \sqrt{-g}\, J^\mu n_\mu\, d^3\Sigma;$$

the interval-sum moment matrix is nonsingular exactly for polynomial densities of degree

≤ 4 [machine: exact determinants] $\implies$ the interval sums determine the density coefficients uniquely;

since (2) holds for every region, the fundamental lemma of the calculus of variations [math] gives

$$\boxed{\partial_\mu J^\mu = 0}$$

pointwise;

the contrapositive is witnessed: every nonzero polynomial density of degree

≤ 3 has a nonzero interval sum on a rational grid [machine];

the current dictionary:

$J^0 =$ the seal density ($\#J^+ - \#J^-$ per cell $= n_{\mathcal{R}}$ density [CT0.3]);

$J^i =$ the hop current on the w_i carrier [D2.1; CT1.12];

(4) the stress-energy conservation, by two independent routes that agree:

(i) geometry: $\nabla^\mu G_{\mu\nu}$

$= 0$ on the exact pp-wave witness [machine: nonlinear, all 4 components, as polynomials in u, v, x, y];

$g_{\mu\nu}$: $g_{uu} = H$, $g_{uv} = g_{vu} = -1$, $g_{xx} = g_{yy} = 1 \implies \det g = -1$ exactly, independent of H [machine];

$\Gamma^\mu_{\mu\lambda} = \partial_\lambda \ln \sqrt{|g|} = 0$ [machine];

$\nabla_\lambda g_{\mu\nu} = 0$ on all 64 components [machine] $\implies \nabla_\mu(\Lambda g^{\mu\nu}) = 0$ on all 4 [machine] :

CT1.23 forces $\kappa^2 \nabla_\mu T^{\mu\nu} = 0$;

$R_{\mu\nu} = -\frac{1}{2}(\partial_x^2 + \partial_y^2) H\, \delta^u_\mu \delta^u_\nu, \quad R = 0, \quad G_{uu} = R_{uu}$ [machine];

$\Delta_\perp H = 0 \implies R_{\mu\nu} = 0$: the vacuum pp-wave [machine];

(ii) Noether: $\partial^\mu T_{\mu\nu}$

$= (\Box\phi - m^2\phi)\partial_\nu\phi$ [BM.9(3)] on an exact on-shell witness ($\phi = t,\ m = 0 :\ \Box\phi = 0$) :

$\partial^\mu T_{\mu\nu} = 0$ [machine];

the dust read:

$\partial_\mu T^{\mu\nu}$

$= (\partial_0\rho,\ 0,\ 0,\ 0)$ for the static pressureless dust of BM.9(4) : the three spatial derivatives drop out at p

$= 0$ [machine];

a static inhomogeneous dust is exactly conserved;

a time-dependent ρ leaves the exact residual $\partial_0\rho$ [machine];

(5) the priority clause:

(1)–(3) derive the differential law from the integer law : $\partial_\mu J^\mu$

$= 0$ is the density read of $\#J^+$

$= \#J^-$ [BM.7(5)];

the converse fails at finite cell size: the integer balance constrains the completed-history census [BM.7(1) : Cat_m, the interleaving counts] and the density law does not;

no variational principle is used anywhere in (1)–(3) : the input is D1.2's transition domains, D1.4's boundary map and D2.11's validity condition.

Rmk (limit discipline). Clause (3) is the tier's only use of the extra-axiomatic continuum-limit conditions, inherited at the point of use in the manner of A §0.2a; the discrete rows (1), (2) and (5) are unconditional and hold at every $\hbar \neq 0$. The pp-wave witness of clause (4) is a rational-coometric test object, not a claim about the corpus's own emergent metric: it verifies the identity $\nabla^\mu G_{\mu\nu} = 0$, which is metric-independent [math].

Req. D1.2, D1.3, D1.4, D2.1, D2.11 (Parts 1–2); CT0.3 (Part 1); CT1.12, CT1.22, CT1.23, CT2.6 (Part 2; CT2.6's conditions (ii)–(iii) at the point of use — A §0.2a); CT4.5, EM.11 (Part 2); BM.7(1), BM.7(2), BM.7(5), BM.8(1), BM.9(3), BM.9(4) (this part); exact rational grid telescoping, exact moment-matrix determinants, exact multivariate pp-wave curvature and covariant-derivative algebra, the fundamental lemma of the calculus of variations [math; machine: `checks/verify_H_noether.py` §E].

Pf.

(1) : $\ell(D_\ell f)(x_j)$

$= f(x_j{+}\ell){-}f(x_j)$ [BM.8(1)] $\implies$ the sum telescopes to $f(x_0{+}n\ell){-}f(x_0)$ [machine, exact rationals];

$$\int_a^b f'(x)\,dx = f(b) - f(a) \text{ for } f = x^k \text{ [the exact polynomial antiderivative } \tfrac{x^{k+1}}{k+1}\text{]};$$

the two totals are the same rational number at every ℓ [machine]

(2) : pairwise cancellation of the interior signs [BM.7(2) re-run per interval; machine: exhaustive];

the 2D lattice : $\sum_{i,k}[(j^x_{i+1,k} - j^x_{i,k}) + (j^y_{i,k+1} - j^y_{i,k})]$ telescopes in each direction separately to the four boundary sums [machine: all rectangles];

periodic identification cancels the boundary sums pairwise [machine]

(3) : cell $\to$ chart convergence under CT2.6(ii)–(iii) [math];

moment matrix : $M_{b,m}$

$$= \sum_{j=0}^{b-1} \ell$$

$$(j\ell)^m,$$

$$b$$

$$= 1..d{+}1 : \det M$$

$\neq 0$ [machine, exact Fractions] $\implies$ injectivity of the coefficient map;

vanishing of all interval sums $\implies$ all coefficients zero $\implies$ $\partial_\mu J^\mu$

$= 0$ [fundamental lemma; math]

(4)(i) : $\Gamma^\rho_{\mu\nu}$

$= \frac{1}{2} g^{\rho\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})$,

$$R^\rho_{\sigma\mu\nu}$$

$$= \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma},$$

$$R_{\sigma\nu}$$

$= R^\mu_{\sigma\mu\nu}$ [math];

all steps exact in $\mathbb{Q}[u, v, x, y]$ [machine];

$$\nabla_\mu G^{\mu\nu} = \partial_\mu G^{\mu\nu} + \Gamma^\mu_{\mu\lambda} G^{\lambda\nu} + \Gamma^\nu_{\mu\lambda} G^{\mu\lambda} = 0 \text{ [machine]}$$

$$(4)(\text{ii}) : \text{ BM.9(3)'s jet identity evaluated at } \phi$$

$$= t,$$

$$m$$

$$= 0 \ [\Box\phi = g^{\mu\nu}\partial_\mu\partial_\nu t = 0] : \text{ eom}$$

$$= 0 \text{ and } \partial^\mu T_{\mu\nu}$$

$$= 0 \text{ [machine]};$$

$$\text{dust} : T^{00} = \rho(x),\ T^{0i} = T^{ij} = 0 \implies \partial_\mu T^{\mu 0} = \partial_0 \rho,\ \partial_\mu T^{\mu i} = 0 \text{ [machine]}$$

$$(5) : \text{ conjunction of } (1)\text{–}(3) \text{ with BM.7}(1)\text{'s census rows. } \blacksquare$$

Phys. The continuum conservation law is a limit of a telescoping sum, and the telescoping is exact — not asymptotic. Sum the per-edge source over any stretch of a materialized chain and the interior signs cancel pairwise, leaving only the two ends; do it on a lattice rectangle and the same cancellation leaves only the boundary flux; close the network and the total is zero, which is the integer balance wearing a density's clothes. Because the interval sums determine a polynomial density uniquely — the moment matrix has an exact nonzero determinant — vanishing on every region forces the density itself to vanish, and $\partial_\mu J^\mu = 0$ follows pointwise. The stress-energy tensor is conserved twice over and the two derivations agree: geometrically, because the Einstein tensor is divergence-free (verified nonlinearly on an exact rational pp-wave, where $\det g = -1$ identically and metric compatibility kills the cosmological term), and dynamically, because the jet identity of the translation current vanishes on-shell. Static dust stays conserved however lumpy it is; only $\partial_0\rho$ can spoil it. And the arrow of explanation runs one way: the integers ground the density, never the reverse.

BM.12 — The Noether status of the corpus: adjudication and scope. Structural-constraints record: see Appendix B, Section B.1.15 of the Addendum.

4.7 Tier BM.5 — The Calibration Lock: the Born Width, the Coherence Volume, the Curvature of a Sealed Composite, and the Neutron Row

Definition 4.4 (Def-BM.4 — The Born width and the coherence volume). Def. Let $E_{\text{int}} \in \mathbb{Z}_{\geq 2}$ be the internal directed-edge count of a sealed composite [D0.8; FT.44(4)]: $E_{\text{int}} = 2E_{\text{bidir}} + E_{\text{uni}}$; the minimal floor is the digon, $E_{\text{int}} = 2$ [FT.28] and $m_{\text{hid}} = E_{\text{int}} \ln 2/\mathfrak{h}$ its informational mass [FT.44(4), FT.42(1)].

$$(1) \quad \text{the Born width:}$$

$$\Delta x_{\text{B}} \;:=\; \frac{1}{m_{\text{hid}}} \;=\; \frac{\mathfrak{h}}{E_{\text{int}} \ln 2}$$

$$[\hbar = c = 1,\ \text{D2.9};\ \text{the Compton-like length of FT.90(6) re-cited as an existing read}];$$

$$\text{constitutive clause: } \Delta x_{\text{B}} \cdot m_{\text{hid}} = 1 \text{ is the definition};$$

$$\text{non-derived clause: the dimensionless normalization is fixed to } 1 \text{ (the Compton normalization), } \neg\text{derived};$$

$$(2) \quad \text{the dimensional-uniqueness clause:}$$

$$\exists! \text{ free dimensionful scale } \mathfrak{h}\ [\text{P1},\ \text{CT2.11}] :\ \ \forall Q \text{ dimensionful} :\ Q$$

$$= c_Q$$

$$\mathfrak{h}^{k_Q},$$

$$c_Q \text{ fixed [CT2.11]};$$

$$[\text{length}] = \mathfrak{q}^{-1/2} = [\mathfrak{h}],$$

$$[m_{\text{hid}}] = \mathfrak{q}^{+1/2}\ [\text{CT2.1}] \implies \text{every length built from } m_{\text{hid}} \text{ alone is } c/m_{\text{hid}} \text{ with } c \text{ dimensionless :}$$

$$\text{the Born width is the unique such length at } c = 1;$$

$$\text{any other } c \text{ is a P4-class internal choice and is priced, not adopted [clause } (4)];$$

(3) the coherence volume: $V_{\text{eff}} := (\Delta x_{\text{B}})^3 = \dfrac{\mathfrak{h}^3}{(E_{\text{int}} \ln 2)^3}$ [$\alpha = 1$: the cube convention];

(4) the geometric-factor clause (priced, not adopted):

$V = \alpha\,(\Delta x_{\text{B}})^3$ with α dimensionless :

$\rho(\alpha) = \rho(1)/\alpha,$

$\alpha = \frac{4\pi}{3}$: $\rho \mapsto \frac{3}{4\pi}\rho$ exactly;

every ratio in which α appears in both entries is α-free;

α is ¬a dimensionful datum and ¬a P4 core member;

(5) the locus clause:

Δx_{B} is a chart-side envelope on M [CT2.6] in the sense of BM.10(6)'s width law,

¬a materialized edge length :

the minimal materialized line element is $\widehat{ds}^{\,2} = 1 \implies ds^2 = \mathfrak{h}^2$ [D1.6; CT1.19a] :

$\Delta x_{\text{B}} < \mathfrak{h} \iff E_{\text{int}} \ln 2 > 1 \iff E_{\text{int}} \geq 2$ [forward pointer: BM.13(3); machine];

every admissible cell is sub-hop, so the width is a coarse-graining envelope and ¬a ledger distance [locus discipline, FT.136(5) pattern];

(6) the scope clause:

FT.90(6) refutes a Compton-wavelength relation to the generation threshold ($\delta h \geq \lceil \log_2 \#\text{classes} \rceil$ depends on the class count alone) :

that refutation is untouched here — this definition uses the length as a spread read, and derives no relation between it and any threshold.

Rmk (naming). Born width names the spatial spread of the sealed composite's density read at the Compton normalization; coherence volume names its cube. Both are internal bookkeeping on derived quantities [A §0.2]; neither is a measurement, and no external length datum enters [P5; Def-BM.3(6)].

Req. D0.6, D0.8, D2.9 (Parts 1–2); D1.6 (Part 1); CT1.19a, CT2.1, CT2.6, CT2.11 (Part 2); P1, P5 (Parts 1–2); Def-BM.3 (this part); FT.28, FT.42(1), FT.44(4) (Part 3, Census); FT.90(6) (Part 3); BM.10(6) (this part); the sub-hop row cited in clause (5) is a marked forward pointer at the citing site and is imported in no Req line; exact Laurent-monomial arithmetic, exact rational $\ln 2$ enclosures [math; machine: `checks/verify_H_calibration_lock.py` §A].

Pf.

Definition block: the constitutive clause is $\Delta x_{\mathrm{B}} \cdot m_{\mathrm{hid}}$

$= 1;$

the non-derived clauses are the normalization c

$= 1$ of (1)–(2),

the geometric factor α of (4),

and the locus reading of (5);

clause (6) is a scope record. ■

Phys. The sealed composite has one length and it is the reciprocal of its own informational mass: $\hbar$ over the edge count times $\ln 2$. That is not a new postulate — the corpus already carried the Compton-like length as an existing read, and with one free scale in the theory there is no other length a mass can build. The cube of that width is the volume the density spreads over, with a dimensionless geometric factor left free and priced rather than chosen. One clause keeps the reading honest: the width is a chart-side envelope, not a hop. Every admissible cell turns out to be shorter than a single materialized line element, which is exactly why the width belongs to the continuum chart and not to the ledger.

Theorem 4.11 (BM.13 — The Born width: the exact substitution, the quantized ladder, and the sub-hop locus). Thm.

(1) the exact substitution:

$$\Delta x_{\mathrm{B}} = \frac{1}{m_{\mathrm{hid}}} = \frac{\hbar}{E_{\mathrm{int}} \ln 2}$$

[Def-BM.4(1); FT.44(4); machine : $\Delta x_{\mathrm{B}} \cdot m_{\mathrm{hid}} = 1$ as a Laurent identity];

the monomial census is exact: $\Delta x_{\mathrm{B}} = \mathfrak{h}^{1} E_{\mathrm{int}}^{-1} (\ln 2)^{-1} \pi^{0}$

[machine: one term, coefficient 1];

$$[\Delta x_{\mathrm{B}}] = [\mathfrak{h}] = \mathfrak{q}^{-1/2} = [\text{length}]$$

[CT2.1; machine: exact exponent arithmetic];

$E_{\mathrm{int}},\ \ln 2 :$ dimensionless;

(2) quantization:

$$\Delta x_{\mathrm{B}} \in \left\{ \frac{\mathfrak{h}}{n \ln 2} : \ n \in \mathbb{Z}_{\geq 2} \right\} :$$

a discrete ladder, ¬a continuous range [machine: injective in n];

the computed cells:

$$E_{\mathrm{int}}$$

$= 2$ [Σ_{min}; FT.28],

6 [Σ_{12}; Census FT.35(2)],

$12,$

$24,$

$24,$

$60,$

60 [the five solids; Census FT.42(5)];

the ladder is strictly decreasing with disjoint exact enclosures [machine: rational $\ln 2$ enclosures of width 2^{-12}];

the cycle ladder $E_{\text{int}} = L$ [FT.32] has no repeated value [machine];

(3) the sub-hop locus:

$$\Delta x_{\text{B}} < \mathfrak{h} \iff \frac{1}{E_{\text{int}} \ln 2} < 1 \iff E_{\text{int}} \ln 2 > 1 \iff E_{\text{int}} \geq 2$$

[machine: exact enclosure $\frac{1}{2} < \ln 2 < 1$ drives every step];

the minimal admissible count is E_{int}

$= 2$ [FT.28 : the digon floor; a materialized closed chain carries ≥ 2 directed edges] :

$\therefore$ every admissible sealed composite is sub-hop: $\Delta x_{\text{B}} \leq \dfrac{\mathfrak{h}}{2 \ln 2} < \mathfrak{h}$

[machine: all computed cells];

at the inadmissible E_{int}

$= 1$ the order reverses $(\Delta x_{\text{B}} > \mathfrak{h})$: the threshold is sharp and sits exactly at the floor [machine];

locus discipline: Δx_{B} is a CT2.6 chart envelope,

¬a ledger distance [Def-BM.4(5); D1.6 : the minimal materialized element is $ds^2 = \mathfrak{h}^2$];

(4) the Robertson consistency:

$$\Delta x \, \Delta p \geq \tfrac{1}{2} \text{ [BM.10}(3), (6); \hbar = 1] \text{ at } \Delta x = \Delta x_{\text{B}} \iff \Delta p \geq \frac{1}{2 \Delta x_{\text{B}}} = \frac{m_{\text{hid}}}{2}$$

[machine];

$\Delta p = \frac{1}{2} m_{\text{hid}}$ saturates Robertson exactly at the Born width :

the width is the saturating spread for a momentum spread of half the informational mass;

(5) the α-variant priced:

$$V = \alpha(\Delta x_{\mathrm{B}})^3 \implies \rho(\alpha) = \rho(1)/\alpha;$$

$$\alpha = \tfrac{4\pi}{3} \implies \rho \mapsto \tfrac{3}{4\pi}\rho \text{ exactly [machine: a single Laurent monomial].}$$

Rmk (comparison discipline). No clause reads Δx_{B} against an external length. The comparisons of this tag are internal: against the hop $\mathfrak{h}$ [D0.6], against the minimal materialized element [D1.6], and against the Robertson bound [BM.10(6)]. External comparison remains quarantined [P5; Census FT.114, FT.120].

Req. D0.6, D0.8, D1.6, D2.9 (Parts 1–2); CT1.19a, CT2.1, CT2.6 (Part 2); P1, P5 (Parts 1–2); Def-BM.4 (this part); FT.28, FT.32, FT.42(1), FT.44(4), FT.90(6) (Part 3, Census); Census FT.35(2), FT.42(5) (row-wise); BM.10(3), BM.10(6) (this part); exact Laurent-monomial identities, exact rational $\ln 2$ enclosures via $\ln 2 = \sum_{n\geq 1} 1/(n2^n)$ with tail $< 2^{-N}$ [math; machine: `checks/verify_H_calibration_lock.py` §A].

Pf.

$$(1):\ m_{\mathrm{hid}}$$

$$= E_{\mathrm{int}} \ln 2/\mathfrak{h} \text{ [FT.44(4) re-cited]}$$

$$\implies 1/m_{\mathrm{hid}}$$

$$= \mathfrak{h}/(E_{\mathrm{int}} \ln 2) \text{ [machine: the product is } 1 \text{ in } \mathbb{Q}[\mathfrak{h}, E, \ln 2]^{\pm}];$$

$$\text{exponents} : [\mathfrak{h}] - 0 - 0 = \mathfrak{q}^{-1/2} \text{ [CT2.1]}$$

$$(2):\ \Delta x_{\mathrm{B}}(n)$$

$$= \mathfrak{h}/(n \ln 2) :\ n_1$$

$$\neq n_2 \implies \Delta x_{\mathrm{B}}(n_1) \neq \Delta x_{\mathrm{B}}(n_2) \text{ [machine: injectivity on } n \in 2..59];$$

$$\text{the five computed cells : disjoint exact enclosures [machine : } \ln 2 \in (\textstyle\sum_1^{12} \frac{1}{n2^n},\ \cdot + 2^{-12})]$$

$$(3):\ \tfrac{1}{2} < \ln 2 < 1 \text{ [machine: the alternating-free series with exact rational partial sums]}$$

$$\implies E \ln 2 > 1$$

$$\iff E$$

$$\geq 2 \text{ [machine: every } E \in 1..59\text{]};$$

$$E_{\text{int}} \geq 2 \text{ [FT.28]} \implies \Delta x_{\text{B}} \leq \mathfrak{h}/(2\ln 2) < \mathfrak{h}$$

$$(4):\ \Delta x$$

$$\Delta p \geq \tfrac{1}{2} \text{ [BM.10(3) : Robertson; BM.10(6) : the width law] with } \Delta x$$

$$= \Delta x_{\text{B}}$$

$$= 1/m_{\text{hid}} \implies \Delta p$$

$$\geq m_{\text{hid}}/2 \text{ [machine: exact rationals]};$$

$$\text{equality at } \Delta p = m_{\text{hid}}/2 \text{ [BM.10(3) saturation row]}$$

$$(5):\ \rho = m_{\text{hid}}/(\alpha \Delta x_{\text{B}}^3) = \rho(1)/\alpha \text{ [machine]};$$

$$\alpha = 4\pi/3 :\ 3/(4\pi) \text{ is a single Laurent monomial [machine]. } \blacksquare$$

Phys. The width of a sealed composite is a reciprocal integer count times the hop, divided by $\ln 2$ — one Laurent monomial, no mixture of scales, nothing to tune. The ladder it walks is strictly discrete: two, six, twelve, twenty-four, sixty, each step shorter than the last with exact rational enclosures that never touch. And every rung of it sits below a single hop, which is the sharpest locus statement in the tier: the Born width is not a distance the ledger can materialize, it is the envelope the continuum chart assigns to a bound whose internal edges all cancel pairwise. Consistency with the uncertainty bound is exact rather than approximate — half the informational mass is precisely the momentum spread that saturates Robertson at this width.

Theorem 4.12 (BM.14 — The coherence volume and the energy density). Thm.

$$(1) \quad \text{the exact volume:}$$

$$V_{\text{eff}} = (\Delta x_{\text{B}})^3 = \frac{\mathfrak{h}^3}{(E_{\text{int}} \ln 2)^3}$$

$$\text{[Def-BM.4(3); machine: one Laurent monomial } \mathfrak{h}^3 E^{-3} (\ln 2)^{-3}\text{]};$$

$$[V_{\text{eff}}] = \mathfrak{q}^{-3/2} = [\text{length}]^3 \qquad \text{[CT2.1; machine: } 3 \cdot (-\tfrac{1}{2}) = -\tfrac{3}{2}\text{]};$$

(2) the exact energy density:

$$\rho = \frac{m_{\text{hid}}}{V_{\text{eff}}} = \frac{E_{\text{int}} \ln 2/\mathfrak{h}}{\mathfrak{h}^3/(E_{\text{int}} \ln 2)^3} = \frac{(E_{\text{int}} \ln 2)^4}{\mathfrak{h}^4}$$

[machine: Laurent identity, denominators cleared];

$$\rho\, \mathfrak{h}^4 = (E_{\text{int}} \ln 2)^4 \text{ exactly};$$

$$[\rho] = [\text{energy}] - [V] = \mathfrak{q}^{1/2+3/2} = \mathfrak{q}^{+2} = [T_{\mu\nu}]$$

[CT2.1; BM.9(5); machine];

(3) the agreement with the general source read:

$$\kappa^2 \rho = \frac{2\pi\, \mathfrak{h}\, E_{\text{int}}}{V}\bigg|_{V=V_{\text{eff}}} = \frac{2\pi\, E_{\text{int}}^4 (\ln 2)^3}{\mathfrak{h}^2}$$

[BM.9(5); machine: the two forms are the same monomial];

$$\kappa^2 = 8\pi G_N = \frac{2\pi\, \mathfrak{h}^2}{\ln 2}, \qquad G_N = \frac{\mathfrak{h}^2}{4 \ln 2} \qquad \text{[CT1.23, CT2.9, P3; machine: exact]};$$

(4) the $\ln 2$ ledger:

$$m_{\text{hid}} : \ (\ln 2)^1 \text{ [FT.44(4)]};$$

$$V_{\text{eff}} : \ (\ln 2)^{-3};$$

$$\rho : \ (\ln 2)^4 = (\ln 2)^1 \cdot (\ln 2)^3;$$

$$\kappa^2 : \ (\ln 2)^{-1};$$

$$\kappa^2 \rho : \ (\ln 2)^3 \cdot 2\pi;$$

at a general V [BM.9(5)] the $\ln 2$ of m_{hid} and of κ^2 cancel and no $\ln 2$ survives;

at $V = V_{\text{eff}}$ the volume contributes $(\ln 2)^{-3}$ and three powers survive :

both statements are the same exact identity read at two volumes [machine: the monomial censuses agree];

(5) the α-variant:

$$\rho(\alpha) = \frac{(E_{\text{int}} \ln 2)^4}{\alpha\, \mathfrak{h}^4},$$

$$\kappa^2 \rho(\alpha) = \frac{2\pi E_{\text{int}}^4 (\ln 2)^3}{\alpha\, \mathfrak{h}^2};$$

$$\alpha = \tfrac{4\pi}{3} \implies \kappa^2 \rho \mapsto \frac{3\, E_{\text{int}}^4 (\ln 2)^3}{2\, \mathfrak{h}^2} \text{ [machine: the } \pi \text{ cancels exactly]}.$$

Rmk (volume convention). Clause (5) prices the spherical convention instead of adopting it: at $\alpha = 4\pi/3$ the π of κ^2 cancels and the curvature scale becomes π-free. Both conventions are exact; the cube convention of Def-BM.4(3) is the one carried downstream, and every α-free ratio is unaffected.

Req. D2.9 (Parts 1–2); CT1.23, CT2.1, CT2.9 (Part 2); P3 (Parts 1–2); Def-BM.4 (this part); FT.42(1), FT.44(4) (Part 3, Census); BM.9(5), BM.13(1) (this part); exact Laurent-monomial arithmetic with denominators cleared [math; machine: `checks/verify_H_calibration_lock.py` §B].

Pf.

$$(1):\ \text{cubing BM.13(1)'s monomial [machine]}; \qquad 3 \cdot (-\tfrac{1}{2}) = -\tfrac{3}{2}\ \text{[CT2.1]}$$

$$(2):\ \rho = \frac{E \ln 2}{\mathfrak{h}} \cdot \frac{(E \ln 2)^3}{\mathfrak{h}^3} = \frac{(E \ln 2)^4}{\mathfrak{h}^4}\ \text{[machine]};$$

$$[\rho] = \mathfrak{q}^{1/2-(-3/2)} = \mathfrak{q}^{+2} = [T_{\mu\nu}]\ \text{[CT2.1; BM.9(5) row]}$$

$$(3):\ \kappa^2 \rho = \tfrac{2\pi\mathfrak{h}^2}{\ln 2} \cdot \tfrac{(E \ln 2)^4}{\mathfrak{h}^4} = \tfrac{2\pi E^4 (\ln 2)^3}{\mathfrak{h}^2}\ \text{[machine]};$$

$$\text{BM.9(5)}:\ \kappa^2 \rho$$

$$= 2\pi\mathfrak{h} E/V \text{ at } V$$

$$= \mathfrak{h}^3/(E \ln 2)^3 \implies \tfrac{2\pi\mathfrak{h} E (E \ln 2)^3}{\mathfrak{h}^3}$$

$$= \tfrac{2\pi E^4 (\ln 2)^3}{\mathfrak{h}^2}\ \text{[machine: identical monomials]}$$

$$(4):\ \text{monomial censuses [machine]}:\ m_{\text{hid}} \to (0,1,1,0),$$

$$V_{\text{eff}} \to (3, -3, -3, 0),$$

$$\rho \to (-4, 4, 4, 0),$$

$$\kappa^2 \to (2, 0, -1, 1),$$

$$\kappa^2\rho \to (-2, 4, 3, 1) \text{ in the basis } (\mathfrak{h}, E, \ln 2, \pi)$$

$$(5):\ \text{division by } \alpha \text{ [machine]}; \qquad \alpha = 4\pi/3:\ (2\pi)/(4\pi/3) = 3/2 \text{ exactly. } \blacksquare$$

Phys. Cube the width and the density follows by one exact division: the fourth power of the information content over the fourth power of the hop. The dimensions land precisely on the stress-energy tensor's own, with no adjustment and no second scale. The same quantity arrives a second way — through the general source read of the stress-energy tier with the volume now supplied by the width — and the two routes give the identical Laurent monomial, which is the first sign that the calibration is closing rather than accumulating. The $\ln 2$ bookkeeping is explicit about where its powers live: one from the mass, three from the volume, minus one from the coupling, three surviving in the curvature. And the geometric factor is priced rather than hidden: choose the sphere and the π of the coupling cancels outright.

Theorem 4.13 (BM.15 — The curvature of a sealed composite: the exact trace of the field equations, and the curvature scale). Thm. In the orthonormal rest frame $g_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$ [CT1.12]: sig $= (-, +, +, +)$; [A3, C13], with $T_{\mu\nu} = \text{diag}(\rho, p, p, p)$ [BM.9(4)] and $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa^2 T_{\mu\nu}$ [CT1.23]:

$$(1) \quad \text{the dust trace:}$$

$$T = g^{\mu\nu}T_{\mu\nu} = -\rho + 3p$$

$$\text{[machine: exact component algebra]};$$

$$p = 0 \text{ [CT4.5]} \implies T = -\rho;$$

$$(2) \quad \text{the exact trace of the field equations:}$$

$$G^{\mu}{}_{\mu} = -R \text{ in } 4 \text{ dimensions [math]} \implies -R + 4\Lambda = \kappa^2 T:$$

$$\boxed{R = \kappa^2(\rho - 3p) + 4\Lambda} \qquad \text{and at } p = 0: \qquad R = \kappa^2\rho + 4\Lambda \qquad \text{[machine: exact in } \mathbb{Q}[\rho, p, \Lambda, \kappa^2]];$$

$$\text{the inadmissible form:} \quad R = -\kappa^2\rho \text{ would require } 2\kappa^2\rho - 3\kappa^2 p + 4\Lambda = 0 \text{ identically}:$$

refuted exactly [machine: the residual is a nonzero polynomial];

the sign of the dust curvature is fixed by CT1.23's own trace : $\kappa^2 > 0$,

$$\rho$$

$$\geq 0 \text{ [CT4.5]}$$

$$\implies R$$

$$\geq 4\Lambda,$$

and R

$$= \kappa^2\rho > 0 \text{ at } \Lambda \qquad = 0;$$

(3) the component round trip:

$$R_{\mu\nu} = \kappa^2\big(T_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}T\big) + \Lambda g_{\mu\nu}$$

[machine: all 4 components exact];

$$p = 0 : \quad R_{00} = \tfrac{1}{2}\kappa^2\rho - \Lambda,$$

$$R_{ii} = \tfrac{1}{2}\kappa^2\rho + \Lambda \ \ (i = 1, 2, 3);$$

$g^{\mu\nu}R_{\mu\nu} = \kappa^2\rho + 4\Lambda = R$ [machine: the round trip closes];

$$G_{00} = \kappa^2\rho + \Lambda,$$

$$G_{ii} = \kappa^2 p - \Lambda \overset{p=0}{=} -\Lambda;$$

$G_{\mu\nu} = \kappa^2 T_{\mu\nu} - \Lambda g_{\mu\nu}$ [machine: all 4 components];

(4) the curvature scale of the sealed composite:

$$\kappa^2\rho = \frac{2\pi\, E_{\text{int}}^4 (\ln 2)^3}{\mathfrak{h}^2}$$

[BM.14(3); machine];

$$[\kappa^2\rho] = \mathfrak{q}^{-1}\cdot\mathfrak{q}^{+2} = \mathfrak{q}^{+1} = [G_{\mu\nu}] = [\Lambda] = [R]$$

[CT2.1; machine: exact exponent arithmetic];

the scale is a pure function of the integer count E_{int} and the two universal factors $\mathfrak{h}$,

$\ln 2$ [P1; FT.44(4)] : ¬a continuous mass variable enters;

(5) the curvature radius:

$$\ell_R := \frac{1}{\sqrt{\kappa^2\rho}} :$$

$$\left(\frac{\Delta x_{\text{B}}}{\ell_R}\right)^2 = (\Delta x_{\text{B}})^2\,\kappa^2\rho = 2\pi\, E_{\text{int}}^2 \ln 2$$

[machine: one Laurent monomial];

$2\pi\ln 2 > 1$ [machine: exact enclosures $\pi > \frac{223}{71},\ \ln 2 > \sum_1^{12}\frac{1}{n2^n}$] $\implies \left(\frac{\Delta x_{\text{B}}}{\ell_R}\right)^2 > 1\ \ \forall E_{\text{int}} \geq 1$:

$\ell_R < \Delta x_{\text{B}}$ on the whole ladder;

the ratio is strictly increasing in E_{int} with disjoint exact enclosures [machine] :

the self-curvature of a sealed composite tightens as its information content grows;

(6) the Λ clause:

Λ is a P4 core member [P4] :

R carries exactly one additive free term, 4Λ, with exact coefficient 4 [machine];

$$\text{the locked part is } \kappa^2\rho = R - 4\Lambda \text{ at } p = 0 :$$

the curvature generated by the composite is calibration-locked; the ambient term is not.

Rmk (frame and scope). Clauses (1)–(3) are component algebra in one orthonormal frame with a diagonal metric; the identities are rational in $(\rho, p, \Lambda, \kappa^2)$ and hold in every frame [math: tensoriality]. Clause (5) compares lengths by squares, so no radical enters any assertion. No solution of the field equations is constructed here: the pp-wave witness of BM.11(4) carries the differential identities, and this tag carries the algebraic dust read.

Req. A3 (Part 1); CT1.12, CT1.23, CT2.1, CT2.9 (Part 2); CT4.5 (Part 2); P1, P4 (Parts 1–2); Def-BM.4 (this part); FT.44(4) (Census); BM.9(4), BM.11(4), BM.13(1), BM.14(3) (this part); exact multivariate polynomial algebra over $\mathbb{Q}$, Archimedes' rational bounds for π, exact rational $\ln 2$ enclosures [math; machine: `checks/verify_H_calibration_lock.py` §C].

Pf.

$$(1) : \; g^{\mu\nu}T_{\mu\nu} = (-1)\rho + 1\cdot p + 1\cdot p + 1\cdot p = -\rho + 3p \text{ [machine]}; \qquad p = 0 \text{ [CT4.5 : } w = p/\rho = 0]$$

$$(2) : \; G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R \implies G^{\mu}{}_{\mu} = R - 2R = -R \text{ [math, } 4 \text{ dimensions]};$$

$$\text{CT1.23 traced : } \; -R + 4\Lambda = \kappa^2(-\rho + 3p) \implies R = \kappa^2(\rho - 3p) + 4\Lambda \text{ [machine]}$$

$$\text{residual of } R$$

$$= -\kappa^2\rho : \; \left[\kappa^2(\rho - 3p) + 4\Lambda\right] + \kappa^2\rho$$

$$= 2\kappa^2\rho - 3\kappa^2 p + 4\Lambda$$

$$\neq 0 \text{ as a polynomial [machine]}$$

$$(3) : \; R_{\mu\nu}$$

$$= \kappa^2 T_{\mu\nu} + \tfrac{1}{2}g_{\mu\nu}R - \Lambda g_{\mu\nu}$$

$$= \kappa^2(T_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}T) + \Lambda g_{\mu\nu} \text{ [substituting } (2); \text{ machine]};$$

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R = \kappa^2 T_{\mu\nu} - \Lambda g_{\mu\nu} \text{ [machine: all } 4];$$

the dust values as displayed [machine]

(4) : BM.14(3) re-cited;

exponents : $-1+2=+1$ [CT2.1 : $[\kappa^2]=\mathfrak{q}^{-1},\ [T_{\mu\nu}]=\mathfrak{q}^{+2},\ [G_{\mu\nu}]=[\Lambda]=\mathfrak{q}^{+1}$]

(5) : $(\Delta x_{\mathrm{B}})^2\kappa^2\rho=\frac{\mathfrak{h}^2}{(E\ln 2)^2}\cdot\frac{2\pi E^4(\ln 2)^3}{\mathfrak{h}^2}=2\pi E^2\ln 2$ [machine];

$2\pi\ln 2>2\cdot\frac{223}{71}\cdot\frac{405}{584}>1$ [machine: exact rationals] $\implies \ell_R<\Delta x_{\mathrm{B}}$

(6) : the coefficient of Λ in R is 4 [machine: monomial census];

$\Lambda\in$ P4 core [P4 re-cited];

$R-4\Lambda=\kappa^2\rho$ at $p=0$ [machine]. ■

Phys. Trace the field equations and the sign of a dust curvature is not a matter of convention: $R=\kappa^2(\rho-3p)+4\Lambda$, positive for a positive density at vanishing Λ, and the opposite sign is refuted as a polynomial identity rather than argued away. The component round trip closes — Ricci, scalar, Einstein, back to $\kappa^2 T$ — in exact rational arithmetic on all four diagonal entries. Feeding in the width-derived density gives the curvature scale as $2\pi E_{\mathrm{int}}^4(\ln 2)^3/\mathfrak{h}^2$: a pure function of an integer count, dimensionally exactly a curvature. Its radius is always shorter than the Born width, by a factor that grows with the count, so the more information a composite seals, the tighter its own curvature sits inside its spread. One term is not locked, and it is named: the cosmological constant enters the scalar additively with coefficient four, a free internal parameter, and the locked part is what remains after subtracting it.

Theorem 4.14 (BM.16 — The calibration lock). Thm.

(1) the lock identity:

$$\boxed{(\Delta x_{\mathrm{B}})^2\cdot\kappa^2\rho\ =\ 2\pi\,E_{\mathrm{int}}^2\ln 2}$$

[machine: one Laurent monomial, coefficient 2];

$\mathfrak{h}$-free [machine: the $\mathfrak{h}$-exponent is 0];

Λ-free [machine];

P4-free [P4 core : $g, g', h_0, c_W, \Lambda, \varkappa, \beta, \alpha_\Omega$ — ¬one occurs];

the three dimensionless reads:

$$\frac{\Delta x_{\mathrm{B}}}{\mathfrak{h}} = \frac{1}{E_{\mathrm{int}} \ln 2};$$

$$\kappa^2 \rho\, \mathfrak{h}^2 = 2\pi E_{\mathrm{int}}^4 (\ln 2)^3;$$

$$\rho\, \mathfrak{h}^4 = (E_{\mathrm{int}} \ln 2)^4$$

[machine: all $\mathfrak{h}$-free];

(2) the rescaling covariance:

$$\mathfrak{h} \mapsto \lambda \mathfrak{h} :$$

$$\Delta x_{\mathrm{B}} \mapsto \lambda\, \Delta x_{\mathrm{B}}, \quad V_{\mathrm{eff}} \mapsto \lambda^3 V_{\mathrm{eff}}, \quad m_{\mathrm{hid}} \mapsto \lambda^{-1} m_{\mathrm{hid}},$$

$$\rho \mapsto \lambda^{-4} \rho,$$

$$\kappa^2 \rho \mapsto \lambda^{-2} \kappa^2 \rho,$$

$$G_N \mapsto \lambda^2 G_N,$$

$$\kappa^2 \mapsto \lambda^2 \kappa^2$$

[machine: exact weight census];

every dimensionless product of (1) is invariant [machine] :

rescaling is a symmetry ∃! calibration d.o.f. [CT2.11] and the lock rides above it;

(3) no continuous mass variable:

every read of this tier is a single Laurent monomial in $(\mathfrak{h}, E_{\mathrm{int}}, \ln 2, \pi)$ [machine: monomial census] :

$\neg\exists$ a sum of scales, $\neg\exists$ a tunable mixture;

$m_{\text{hid}}\mathfrak{h} = E_{\text{int}} \ln 2$ is injective in the integer E_{int} [machine];

quantization:

$$m_{\text{hid}} \in \Big\{ \frac{n \ln 2}{\mathfrak{h}} : n \in \mathbb{Z}_{\geq 2} \Big\},$$

$$\Delta x_{\text{B}} \in \Big\{ \frac{\mathfrak{h}}{n \ln 2} \Big\},$$

$$\kappa^2 \rho \in \Big\{ \frac{2\pi n^4 (\ln 2)^3}{\mathfrak{h}^2} \Big\} :$$

three discrete ladders, strictly monotone [machine];

(4) the conjugate comparison scales [P3 : G_N is a named combination for external comparison, ¬an input] :

$$r_s := 2G_N m_{\text{hid}} = \frac{\mathfrak{h}\, E_{\text{int}}}{2}$$

[machine: exact];

$$\boxed{\Delta x_{\text{B}} \cdot r_s = 2G_N = \frac{\mathfrak{h}^2}{2 \ln 2}}$$

independent of E_{int} [machine: the E-exponent is 0] :

$$2 \ln 2 \cdot (\Delta x_{\text{B}} r_s) = \mathfrak{h}^2;$$

the two length reads are conjugate with a fixed product : $E_{\text{int}} \uparrow \implies \Delta x_{\text{B}} \downarrow$,

$r_s \uparrow$ along one hyperbola [machine];

(5) the crossing gap:

$$\ell_G := \sqrt{G_N} = \frac{\mathfrak{h}}{2\sqrt{\ln 2}} \text{ [CT2.9] :}$$

$$\Big(\frac{\Delta x_{\mathrm{B}}}{\ell_G}\Big)^2 = \frac{\Delta x_{\mathrm{B}}^2}{G_N} = \frac{4}{E_{\mathrm{int}}^2 \ln 2}$$

[machine: squares only, no radical in any assertion];

$$\Delta x_{\mathrm{B}} < \ell_G \iff E_{\mathrm{int}}^2 \ln 2 > 4 \iff E_{\mathrm{int}} \geq 3 \qquad \text{[machine: exact enclosures, every } E \in 1..60\text{]};$$

$$\frac{r_s}{\Delta x_{\mathrm{B}}} = \frac{E_{\mathrm{int}}^2 \ln 2}{2}$$

[machine] :

$$r_s = \Delta x_{\mathrm{B}} \iff E_{\mathrm{int}}^2 \ln 2 = 2 \iff E_{\mathrm{int}} = \sqrt{\tfrac{2}{\ln 2}} \in (1,2) \text{ [machine: } 1 < \tfrac{2}{\ln 2} < 4\text{]};$$

the crossing lies strictly below the minimal floor $E_{\mathrm{int}} = 2$ [FT.28] :

$$\neg\exists\, E_{\mathrm{int}} \in \mathbb{Z} \text{ at the crossing};$$

$$r_s > \Delta x_{\mathrm{B}} \;\; \forall E_{\mathrm{int}} \geq 2 \text{ [machine]};$$

(6) the not-locked ledger:

$$\Lambda \text{ [P4 : one additive term } 4\Lambda \text{ in } R,\ \text{BM.15(6)}];$$

$$\alpha \text{ [Def-BM.4(4) : priced, } \neg\text{adopted}];$$

$$\mathfrak{h} \text{ [P1, CT2.11 : the single free scale — absolute values ride on it}];$$

the comparison discipline:

$r_s,\ \ell_G$ are built from G_N, a named combination of $\mathfrak{h}$ for external comparison only [P3] :

$\neg$a collapse reading, $\neg$a horizon reading, $\neg$an identification is made [P5; Def-BM.3(6)];

the inequalities of (5) are order relations between derived lengths, and no external datum enters them.

Rmk (lock scope). The lock of clause (1) is a statement about dimensionless products: it says the width and the curvature cannot be varied independently of the integer count. It does not say the corpus predicts a length in metres — the single free scale $\mathfrak{h}$ fixes the unit [P1, P5], and every external comparison stays quarantined [Census FT.114, FT.120].

Req. D2.9 (Parts 1–2); CT2.1, CT2.9, CT2.11 (Part 2); P1, P3, P4, P5 (Parts 1–2); Def-BM.3(6), Def-BM.4 (this part); FT.28, FT.44(4) (Part 3, Census); Census FT.114, FT.120 (quarantine rows); BM.13(1), BM.14(2), BM.14(3), BM.15(4), BM.15(6) (this part); exact Laurent-monomial arithmetic, exact rational enclosures of $\ln 2$ and Archimedes' bounds for π, square-comparison inequalities [math; machine: `checks/verify_H_calibration_lock.py` §D].

Pf.

$$(1):\ (\Delta x_{\mathrm{B}})^2\kappa^2\rho = \frac{\mathfrak{h}^2}{(E\ln 2)^2}\cdot\frac{2\pi E^4(\ln 2)^3}{\mathfrak{h}^2} = 2\pi E^2\ln 2\ \text{[machine]};$$

$$\text{the } \mathfrak{h}\text{-exponent census}: 2-2=0\ \text{[machine]};$$

the P4 core members occur in no monomial [machine scan of the tier's expressions]

$$(2):\ \text{each read is } c$$

$\mathfrak{h}^k$ with $k\in\{1,3,-1,-4,-2,2,2\}$ [machine: weight census] $\implies$ $\mathfrak{h}\mapsto\lambda\mathfrak{h}$ acts by λ^k [CT2.11 re-cited];

the products of (1) have $k=0$ $\implies$ invariant

$$(3):\ \text{monomial census [machine]};$$

$$n_1\ln 2 = n_2\ln 2 \iff n_1=n_2\ \text{[machine on } n\in 1..39];$$

monotonicity of all three ladders [machine]

$$(4):\ r_s = 2\cdot\frac{\mathfrak{h}^2}{4\ln 2}\cdot\frac{E\ln 2}{\mathfrak{h}} = \frac{\mathfrak{h}E}{2}\ \text{[machine]};$$

$$\Delta x_{\mathrm{B}} r_s = \frac{\mathfrak{h}}{E\ln 2}\cdot\frac{\mathfrak{h}E}{2} = \frac{\mathfrak{h}^2}{2\ln 2} = 2G_N\ \text{[machine]};$$

$$\text{the } E\text{-exponent is } -1+1=0$$

$$(5):\ \Delta x_{\mathrm{B}}^2/G_N = \tfrac{\mathfrak{h}^2}{(E\ln 2)^2}\cdot\tfrac{4\ln 2}{\mathfrak{h}^2} = \tfrac{4}{E^2\ln 2}\ [\text{machine}];$$

$$\tfrac{4}{E^2\ln 2} < 1$$

$$\iff E^2\ln 2 > 4:\ E$$

$$= 2 \to 4\ln 2 < 4\ (\ln 2 < 1),$$

$$E$$

$$= 3 \to 9\ln 2 > 4\ (\ln 2 > \tfrac{4}{9})\ [\text{machine: exact enclosures}];$$

$$r_s/\Delta x_{\mathrm{B}} = \tfrac{\mathfrak{h}E}{2}\cdot\tfrac{E\ln 2}{\mathfrak{h}} = \tfrac{E^2\ln 2}{2}\ [\text{machine}];$$

$$= 1 \iff E^2 = \tfrac{2}{\ln 2} \in (1,4)\ [\text{machine}] \implies E \in (1,2);$$

$$E \geq 2 \implies \tfrac{E^2\ln 2}{2} \geq 2\ln 2 > 1\ [\ln 2 > \tfrac{1}{2};\ \text{machine}]$$

$$(6):\ \text{BM.15}(6),\ \text{Def-BM.4}(4),\ \text{P1/CT2.11},\ \text{P3/P5 re-cited.}\ \blacksquare$$

Phys. Two numbers in this tier carry neither the free scale nor a single internal parameter: the product of the squared Born width with the curvature scale, which is 2π times the squared edge count times $\ln 2$, and the product of the Born width with the comparison radius, which is exactly twice Newton's constant and does not know the edge count at all. That is the lock — width and curvature are conjugate along a hyperbola whose constant is pure calibration, so no sealed composite can slide its size without sliding its field by the reciprocal amount. Rescaling the one free scale moves every dimensionful read by its exact weight and leaves every one of these products standing, which is what makes the lock a statement about the theory rather than about units. The ladder has a hole in it too: the point where the two lengths would coincide sits at an edge count between one and two, strictly below the smallest chain the ledger admits, so no admissible composite is ever at its own comparison radius. What is not locked is listed: the cosmological term, the geometric factor, and the hop itself.

4.15 (BM.17 — The neutron row of the identification layer).

Thm.

$$(1)\quad \text{the structure:}$$

$$\Sigma_{12}$$

$= \{p_1, q_1\} \cup \{p_2, q_2\}$ with materialized 2-cycles joined by a bidirectional connector, single observer contact [Census FT.35];

$$E_{\text{int}} = 6 \ (c_1 \colon 2, \ c_2 \colon 2, \ \text{connector} \colon 2),$$

$$I_{\text{hid}} = 6 \ln 2,$$

$$m_{\text{hid}} = \frac{6 \ln 2}{\mathfrak{h}}$$

[Census FT.35(2); cited row-wise];

$\mathcal{C}(\Sigma_{12}; C) = 1 \iff$ all three attachments seal inside as one ε-type (baryon-like) triple

[Census FT.35(5); D3.9; CT3.17; FT.34(5)];

$$\Theta_{\text{int}} = 2k(\Gamma) \in 2\mathbb{Z},$$

$$\Pi = 72 \, k(\Gamma) \, (\ln 2)^2$$

[Census FT.35(3)];

the junction vertices are D^--configured [Census FT.35(4); CT4.1];

(2) the identification:

Def-BM.3(2) labels the ε-sealed 3-juncture in the neutral charge class q_{net}

$= 0$ as the neutron;

the theorem content of every clause below is the tuple $(\varepsilon\text{-junction}, \ \Theta_{\text{int}} \in 2\mathbb{Z}, \ q_{\text{net}} = 0, \ E_{\text{int}} = 6, \ w = 0)$;

the name is the layer's label [Def-BM.3(1), (7)];

the charge class is Def-BM.3(4)'s declared label, pending BM.6(1)'s additive read

deficit $1'$ is untouched [FT.125(4)];

(3) the derived row at $E_{\text{int}}^{(n)} = 6$ (all exact):

$$\Delta x_{\text{B}}^{(n)} = \frac{\mathfrak{h}}{6\ln 2}\ (< \mathfrak{h} \text{ by the exact factor } \tfrac{1}{6\ln 2});$$

$$V_{\text{eff}}^{(n)} = \frac{\mathfrak{h}^3}{216(\ln 2)^3};$$

$$m_{\text{hid}}^{(n)} = \frac{6\ln 2}{\mathfrak{h}};$$

$$\rho^{(n)} = \frac{1296(\ln 2)^4}{\mathfrak{h}^4};$$

$$\kappa^2\rho^{(n)} = \frac{2592\,\pi(\ln 2)^3}{\mathfrak{h}^2}$$

$$[2\cdot 6^4 = 2592;\ \text{machine}];$$

$$\ell_R^{(n)} = \frac{\mathfrak{h}}{6^2\sqrt{2\pi(\ln 2)^3}} < \Delta x_{\text{B}}^{(n)};$$

$$r_s^{(n)} = 3\,\mathfrak{h}$$

[machine];

$$\Delta x_{\text{B}}^{(n)}\cdot r_s^{(n)} = \frac{\mathfrak{h}^2}{2\ln 2} = 2G_N$$

[machine: the E-independent product];

$$(\Delta x_{\text{B}}^{(n)})^2\,\kappa^2\rho^{(n)} = 72\,\pi\ln 2$$

[machine] :

the integer 72

$= 2\cdot 6^2$ is Census FT.35(3)'s Π-coefficient — recorded as a cross-check,

¬a derivation of equality [FT.112 Rmk pattern];

$$\Delta x_{\mathrm{B}}^{(n)} < \ell_G \iff 36\ln 2 > 4 \iff \ln 2 > \tfrac{1}{9}$$

[machine: exact enclosure];

$$w^{(n)} = 0, \quad T_{\mu\nu}^{(n)} = \mathrm{diag}(\rho^{(n)}, 0, 0, 0)$$

[CT4.5; BM.9(4)];

(4) the family table (the five computed cells):

cell	E_{int}	$\Delta x_{\mathrm{B}}/\mathfrak{h}$	$V_{\mathrm{eff}}/\mathfrak{h}^3$	$\rho\,\mathfrak{h}^4$	$\kappa^2\rho\,\mathfrak{h}^2$	$r_s/\mathfrak{h}$	Δx_{B} vs ℓ_G
Σ_{min} (the digon) [FT.28]	2	$\frac{1}{2\ln 2}$	$\frac{1}{8(\ln 2)^3}$	$16(\ln 2)^4$	$32\pi(\ln 2)^3$	1	$\Delta x_{\mathrm{B}} > \ell_G$
Σ_{12} (the ε-type junction) [Census FT.35]	6	$\frac{1}{6\ln 2}$	$\frac{1}{216(\ln 2)^3}$	$1296(\ln 2)^4$	$2592\pi(\ln 2)^3$	3	$\Delta x_{\mathrm{B}} < \ell_G$
tetrahedron [Census FT.42(5)]	12	$\frac{1}{12\ln 2}$	$\frac{1}{1728(\ln 2)^3}$	$20736(\ln 2)^4$	$41472\pi(\ln 2)^3$	6	$\Delta x_{\mathrm{B}} < \ell_G$
cube / octahedron [Census FT.42(5)]	24	$\frac{1}{24\ln 2}$	$\frac{1}{13824(\ln 2)^3}$	$331776(\ln 2)^4$	$663552\pi(\ln 2)^3$	12	$\Delta x_{\mathrm{B}} < \ell_G$
dodecahedron / icosahedron [Census FT.42(5)]	60	$\frac{1}{60\ln 2}$	$\frac{1}{216000(\ln 2)^3}$	$12960000(\ln 2)^4$	$25920000\pi(\ln 2)^3$	30	$\Delta x_{\mathrm{B}} < \ell_G$

the width column is strictly decreasing and the curvature column strictly increasing, with disjoint exact enclosures [machine];

$r_s/\mathfrak{h} = E_{\mathrm{int}}/2$ across the family [machine];

Σ_{min} is the unique computed cell with $\Delta x_{\mathrm{B}} > \ell_G$ [BM.16(5); machine];

the duality-degenerate pairs share E_{int} and hence every read of this tier [FT.44(1); machine];

(5) not a point mass:

$V_{\mathrm{eff}}^{(n)} > 0, \quad \rho^{(n)} < \infty, \quad T_{\mu\nu}^{(n)}$ is a finite-width distribution [BM.9(6); BM.10(6)];

the width is sub-hop [(3)] $\implies$ it is a chart envelope,

¬a materialized distance [Def-BM.4(5); D1.6];

(6) the openness clause:

$$E_{\text{int}}^{(n)}$$

= 6 is Census-cell data at the computed ε-type cell [Census FT.35(2); row-wise consumption, BM.5(2) pattern];

a general composite additivity law for E_{int} beyond the computed cells is ¬constructed [Def-BM.1 read clause; BM.6(1), (5)] :

every read of this tier is exact in $E_{\text{int}}^{(n)}$ and parametric wherever no cell supplies it — ¬a numeric value is invented;

¬∃ a measured value in any clause [P5; Def-BM.3(6)].

Rmk (identification discipline). Clause (3) is a row of derived exact expressions attached to a Census cell and labelled by Def-BM.3. It derives no neutron property that the tuple does not already carry, imports no neutron datum, and closes no row beyond its own: the additive charge read [BM.6(1)] and the assignment rule [FT.125(4)] stand as they are.

Req. D1.6, D2.9, D3.9 (Parts 1–2); CT3.17, CT4.1 (Part 2); P5 (Parts 1–2); Def-BM.1, Def-BM.3, Def-BM.4, BM.5(2), BM.6(1), BM.6(5) (this part); FT.28, FT.34(5), FT.44(1), FT.125(4) (Part 3); Census FT.35(2), FT.35(3), FT.35(4), FT.35(5), FT.42(5), FT.44(4) (row-wise); BM.9(4), BM.9(6), BM.10(6), BM.13, BM.14, BM.15, BM.16 (this part); exact rational arithmetic, exact ln 2 enclosures [math; machine: `checks/verify_H_calibration_lock.py` §E].

Pf.

(1) : Census FT.35's rows re-cited verbatim (2),

(3),

(4),

(5) — row-wise consumption,

¬premise import [BM.5(2); A §0.1]

(2) : Def-BM.3(2) re-cited;

the tuple is the theorem content [Def-BM.3(1), (7)];

the conditional charge class [Def-BM.3(4); BM.6(1)]

(3) : substitution of E_{int}

$= 6$ into BM.13(1), BM.14(1),(2),(3), BM.15(4),(5), BM.16(4) [machine: every coefficient exact];

$2 \cdot 6^4 = 2592,\ 6^4 = 1296,\ 6^3 = 216,\ 2 \cdot 6^2 = 72$ [machine];

$36 \ln 2 > 4 \iff \ln 2 > \frac{1}{9}$ [machine: enclosure]

(4) : the same substitutions at $E_{\text{int}} \in \{2, 6, 12, 24, 60\}$ [Census FT.42(5) card table; FT.28];

monotonicity and disjointness [machine];

$r_s/\hbar = E/2$ [machine];

duality degeneracy : $E_{\text{int}} = 2E$ shared by dual pairs [FT.44(1) re-cited]

(5) : BM.9(6), BM.10(6), Def-BM.4(5) re-cited

(6) : openness rows re-cited unmodified;

P5 scan of this tag's text [machine: zero external numeric tokens]. ■

Phys. The corpus has one computed ε-type junction cell, and it carries six internal directed edges: two per cycle plus the two of the connector. Everything this tier derived then evaluates on it in exact rationals — a width of one hop over six $\ln 2$, a volume of one hop cubed over $216(\ln 2)^3$, a density of $1296(\ln 2)^4$ per hop to the fourth, a curvature scale of $2592\pi(\ln 2)^3$ per hop squared, and a comparison radius of exactly three hops. The lock reads $72\pi \ln 2$ on this cell, and 72 is the same integer the Census found in its Π row — recorded as a coincidence of counts, not promoted to an identity. Def-BM.3's label attaches to that tuple, and the label changes nothing in it. The neutron of this corpus is not a point: it is a finite density over a strictly positive, sub-hop volume, with a curvature radius shorter than its own spread, sitting on a ladder whose rungs are integers and whose crossing point no integer reaches.

BM.18 — The status of the calibration lock: adjudication and scope. Structural-constraints record: see Appendix B, Section B.1.16 of the Addendum.

4.8 Tier BM.6 — The Topology-Blind Neutron and the Topology-Blind + Static Proton: the Flicker Ensemble, the Path-Integrated Mass, the Binding Read, and the Forced Carrier

Definition 4.5 (Def-BM.5 — The flicker ensemble of a sealed composite). Def. A construction class on Def-H.32's topology-blind path integral (no new axiom; Def-H.32's scopes (S1)–(S4) are inherited at every use).

(1) the ensemble: $\mathcal{E}(\mathcal{X}) := \{R \subseteq E_{\text{dir}}(\Gamma) :$ the boundary data of $\mathcal{X}$ is preserved on $R\}$

the boundary data :

$=$ (charge class [Def-BM.3(2)–(3)], Θ_{int} class [BM.2(1)], S winding class [Def-H.32(4) : the (q, S) labels]);

constitutive clause: the ensemble is the sector-conditioned configuration set of Def-H.32(1)–(2) restricted to a composite's boundary data;

the sum below is finite and exact at every budget B [Def-H.32(1) : $\mathcal{G}_B = \{\Gamma : V(\Gamma) \le B\}$, $\mathfrak{C}_B(\Gamma) = \{R : |R| \le B_K\}$];

(2) the path integral:

$$Z(\mathcal{X}) = \sum_{R \in \mathcal{E}(\mathcal{X})} \sum_{\sigma} w(R, \sigma),$$

$$w(C) = \mathcal{J}(C) \prod_{e \in E(C)} G(e) \cdot (-1)^{\aleph(\sigma, C)}$$

[Def-H.32(2); A8];

the two layers (norm and amplitude) are kept separate [CT2.8; Def-H.32(2)] and no layer is substituted for the other;

(3) the weights are derived, not declared:

the per-pair Janus master generator Q

$= \left(\begin{smallmatrix} -x_+ & x_+ \\ x_- & -x_- \end{smallmatrix}\right)$ over $\mathbb{Z}[x_+, x_-]$ has a one-dimensional kernel spanned by (x_-, x_+) [FT.108(4)] :

$$p = \frac{x_+}{x_+ + x_-}$$

[derived];

$$x_+ = 1 - 4\varepsilon_q, \quad x_- = 4\varepsilon_q \implies p = 1 - 4\varepsilon_q \text{ identically}$$

[A4, D0.4; machine];

no P-class datum is added: ε_q is the A4 crystallization threshold [D0.4 : admissible range $[0, \frac{1}{4})$ [CT0.8]; sharp choice $\varepsilon_q = 0$];

(4) the information content is distributed:

$$E_{\text{int}}(R)$$

$$= |R|$$

$= n_{\mathcal{R}}$ [Def-H.32(1) : the budget is the ledger's own content measure; CT1.29 : $S_{\mathfrak{L}} = n_{\mathcal{R}} \ln 2$];

the crystallized value E_{int}

$= 6$ [Census FT.35(2)] is one point of the distribution,

¬the ensemble;

the move system conserves the grade $a_+ - a_- = |R'| - |R|$ [FT.108(2)];

(5) the no-flicker limit (scope):

$$\varepsilon_q$$

$$= 0$$

$$\implies x_-$$

$= 0$ exactly $\implies \neg\exists$

dissolution move $\implies \mathcal{E}(\mathcal{X})$ collapses to its fully crystallized member [machine];

every $\varepsilon_q > 0$ gives $x_- > 0$ and a strictly positive flicker weight;

the $B \to \infty$ limit is ¬constructed [Def-H.32(1); growth data: FT.105].

Rmk (ensemble discipline). The ensemble is a sum over crystallization patterns at fixed boundary data, not a sum over species: every member carries the composite's charge class, sealing-phase class and winding class. The topology labels are internal bookkeeping [Def-H.32(4) pattern], and no member of any ensemble is identified with an external object [P5; Def-BM.3(6)].

Req. A4, A8 (Part 1); D0.4, D2.9 (Parts 1–2); CT0.8, CT1.29 (Part 1); CT2.8 (Part 2); Def-H.32, FT.105, FT.108(2), FT.108(4), FT.108(5) (Part 3); Def-BM.3, BM.2(1) (this part); Census FT.35(2) (row-wise); exact polynomial kernel arithmetic over $\mathbb{Q}[x_+, x_-]$, exact rational specialization [math; machine: `checks/verify_H_topo_blind_neutron.py` §A].

Pf.

Definition block: the constitutive clause is (1)'s sector-conditioned configuration set;

the inherited clauses are Def-H.32's (1)–(2) and scopes (S1)–(S4),

already declared;

clause (3) is derived content recorded at the definition site;

(5) is a scope clause cited at use. ■

Phys. A sealed composite is not one topology with a number attached; it is a finite exact sum over crystallization patterns that share its boundary data — the same topology-blind path integral the fermion era built, now conditioned on a composite's charge class, sealing phase and winding class. Nothing is postulated to weight the sum: the Janus move generator has a one-dimensional kernel, its span is the stationary measure, and substituting the crystallization threshold gives the flicker weight as $4\varepsilon_q$ with no new free parameter anywhere. Set that threshold to its sharp value and the ensemble collapses to the single crystallized configuration — the flicker is a consequence of the threshold being nonzero, not an extra ingredient.

Definition 4.6 (Def-BM.6 — The family naming rows (extension of the identification layer)). Def. An extension of Def-BM.3's identification layer to the three derived channel triplets, recorded by a logged declaration as Def-BM.3(7) requires. A name attaches to a line of a derived spectrum, ordered by that spectrum; no name attaches to a mass value.

(1) the ℓ-channel triplet [FT.109–FT.113 : $r_i = 1 + \sqrt{2}\cos(\delta_\ell + \frac{2\pi i}{3})$, $\delta_\ell = \frac{2}{9}$] :

electron := the $i = 0$ line, muon := the $i = 1$ line, tau := the $i = 2$ line;

the order rule is the spectrum's own: $r_0 > r_1 > r_2$ at $\delta_\ell = \frac{2}{9}$ [FT.113; the span is 3477-fold];

(2) the ν-channel triplet [FT.142–FT.144 : $\delta_\nu = \frac{1}{8}$ on the derived 16-arena] :

electron neutrino, muon neutrino, tau neutrino := its three lines in the same spectral order;

(3) the q-channel triplet [FT.118, FT.119 : $\delta_q = \frac{2}{27}$, triplet-attached] :

the quark triplet := its three lines in the same order;

the cross-channel identity $16\delta_\nu$

$= 9\delta_\ell$

$= 27\delta_q$

$= 2$ [FT.143] is the derived relation between the three rows;

(4) the derived content the names attach to:

$$\sum_i r_i = 3 = \dim F_{J^\pm} \text{ [FT.111(3)]};$$

$$\sum_i r_i^2 = 6;$$

$$Q := \frac{\sum_i r_i^2}{(\sum_i r_i)^2} = \frac{6}{9} = \frac{2}{3} \text{ exactly, for every phase [machine; FT.120(6)]};$$

the cosine form is forced by C_3-covariance and the normalization by the seal dimension [FT.111(2)–(3); FT.158(4)];

the angle is a ratio of spatial counts $\delta_\ell = (3-1)/(3\cdot 3)$ [FT.158(2)–(3)];

(5) the non-derived clause (the mass form):

the bridge from a spectral line to a mass value ($\sqrt{m} \propto r$ vs $m \propto r$ vs ¬a bridge) is item-13 content and is ¬adopted here

the standing bar amplitude $\neq$ mass [FT.79(5)] is untouched;

the channel scale μ remains an undeclared P4-class parameter [Cal-F] : ¬a mass value is derived for any named line;

(6) the data bar:

¬∃ a measured external value in any clause [P5];

the ratios (Q, the span, the angle identities) are corpus-derived and are ¬compared to measurement here;

(7) scope:

the names occur in Part 4 only [Def-BM.3(7) extended];

further extension requires a further logged declaration.

Rmk (what the naming does and does not do). Naming the ℓ-channel lines electron, muon, tau records that the corpus derives a three-line C_3-covariant spectrum whose Koide ratio is exactly $2/3$ at every phase — the ratio the family is famous for — and that the lines are strictly ordered with a 3477-fold span. It does not derive their masses, it does not adopt a mass form, and it does not compare any ratio to a measured value. The naming is Def-BM.3's labelling act applied to spectral lines instead of junctions.

Req. FT.79(5), FT.109, FT.110, FT.111, FT.112, FT.113, FT.118, FT.119, FT.120(6), FT.142, FT.143, FT.144, FT.158 (Part 3); Def-BM.3 (this part); P4, P5 (Parts 1–2); Cal-F; exact identities in $\mathbb{Q}[c, s, r]/(c^2+s^2-1,\ r^2-3)$ [math; machine: `checks/verify_H_topo_blind_neutron.py` §F].

Pf.

Definition block: the constitutive clause is the name $\leftrightarrow$ line attachment by spectral order;

the non-derived clause is (5) (the mass form is item-13 content, ¬adopted);

clause (4)'s identities are derived [machine: $\sum_i \cos(\theta + \frac{2\pi i}{3}) = 0,\ \sum_i \cos^2 = \frac{3}{2}$ in the quotient ring] and are recorded here as the content the names attach to. ■

Phys. Three channels, three triplets, one mechanism: a C_3-covariant seal bilinear whose shape covariance forces, whose normalization the seal dimension fixes, and whose angle is a ratio of spatial counts. The ratio the charged-lepton family is known for falls out as $6/9$ — exactly two thirds, at every phase, with nothing fitted — and the lines are strictly ordered across a 3477-fold span. The names now attach to those lines in spectral order. What the names do not carry is a mass: the bridge from a spectral line to a mass value is the corpus's open item, the amplitude-is-not-mass bar stands, and the overall scale is still an undeclared internal parameter.

Theorem 4.16 (BM.19 — The flicker ensemble is derived: the stationary measure, the threshold specialization, and the no-flicker limit). Thm.

$$(1) \quad \text{the measure is forced:}$$

$$Q = \begin{pmatrix} -x_+ & x_+ \\ x_- & -x_- \end{pmatrix} \text{ over } \mathbb{Z}[x_+, x_-] :$$

$$(x_-,\ x_+)\, Q = (0,0) \ \text{ exactly [machine];}$$

$$\neg\exists \text{ a nonzero constant vector in } \ker Q \text{ [machine: exhaustive on } [-3,3]^2] :$$

$$\dim \ker Q$$

$$= 1 \text{ over the fraction field}$$

$$\implies p$$

$$= \frac{x_+}{x_+ + x_-} \text{ is the unique stationary measure [FT.108(4)];}$$

$$(2) \quad \text{the specialization is an identity:}$$

$$x_+ = 1 - 4\varepsilon_q, \quad x_- = 4\varepsilon_q \implies p = \frac{1-4\varepsilon_q}{1-4\varepsilon_q+4\varepsilon_q} = 1 - 4\varepsilon_q$$

[machine: exact at six rational thresholds];

$$\text{the flicker (dissolution) weight is } x_- = 4\varepsilon_q :$$

$$\text{admissible on } \varepsilon_q \in [0, \tfrac{1}{4}) \text{ [CT0.8]} \implies x_- \in [0,1),\ x_+ \in (0,1] \text{ [machine];}$$

$$\text{no P-class datum enters: } \varepsilon_q \text{ is the A4 threshold [D0.4];}$$

$$\neg\text{a new fugacity, } \neg\text{a new rate, } \neg\text{a P4 addition [Cal-BM stays empty];}$$

$$(3) \quad \text{the no-flicker limit:}$$

$$\varepsilon_q$$

$$= 0 \text{ [D0.4 sharp choice]}$$

$$\implies x_-$$

$$= 0 \text{ exactly } \implies \neg\exists$$

$$\text{a dissolution move } \implies \mathcal{E}(\mathcal{X})$$

$$= \{\text{the crystallized configuration}\} \text{ [Def-BM.5}(5);\ \text{machine]};$$

$$\varepsilon_q > 0 \implies x_- > 0 :$$

the flicker is strictly present at every nonzero threshold and strictly absent at the sharp one;

(4) the ensemble is finite and exact at each budget:

$$|\mathfrak{C}_B(\Gamma)| < \infty \text{ [Def-H.32}(1)\text{]};$$

the $B = 4$ sector census is exact integers [FT.106(3) : $Z^{\text{norm}}(q, S)$ and $Z^{\text{amp}}(q, S)$ per sector];

$$Z^{\text{norm}}(q, S)$$

$$= Z^{\text{norm}}(-q, S) \text{ unconditionally and } Z^{\text{amp}}(q, S)$$

$$= Z^{\text{amp}}(-q, S) \text{ on the cover support [FT.106(3d)]};$$

$$Z^{\text{amp}}(q, k) = 0 \text{ for } k = 2 \text{ and odd } k \geq 3 \text{ [FT.106(3c)]};$$

(5) the information content is a distribution:

$$E_{\text{int}}(R) = |R| \text{ [Def-BM.5}(4)\text{]} :$$

the ensemble spreads E_{int} over $\{0, \ldots, B_K\}$ and the crystallized value is its top point;

$$a_+ - a_- = |R'| - |R|,$$

$$\textstyle\sum_d m_d - \sum_d n_d = V' - V,$$

$$\sum_d d\,(m_d - n_d) = E'_{\text{pot}} - E_{\text{pot}}$$

[FT.108(2) : the grade map is a homomorphism of the move system] :

every flicker move changes E_{int} by exactly the $J^{\pm}$ excess it carries — the distribution is the move algebra's orbit, not an ansatz;

(6) the composite is not a fixed topology:

$\mathcal{E}(\mathcal{X})$ has more than one member $\iff x_- > 0 \iff \varepsilon_q > 0$ [(3)] :

at every admissible nonzero threshold the neutron and the proton are ensembles.

Req. A4 (Part 1); D0.4 (Part 1); CT0.8, CT1.29 (Part 1); CT2.8 (Part 2); Def-H.32, FT.105, FT.106(3), FT.108(2), FT.108(4), FT.108(5) (Part 3); Def-BM.5 (this part); P4 (Parts 1–2); §Cal-BM; exact polynomial arithmetic over $\mathbb{Q}[x_+, x_-]$, exhaustive integer kernel census, exact rational specialization [math; machine: `checks/verify_H_topo_blind_neutron.py` §A].

Pf.

(1) : $(x_-)(-x_+) + (x_+)(x_-) = 0$ and $(x_-)(x_+) + (x_+)(-x_-) = 0$ in $\mathbb{Q}[x_+, x_-]$ [machine];

constant vectors : $a(-x_+)+bx_- = 0 \wedge ax_+ + b(-x_-) = 0 \implies a = b = 0$ [machine: exhaustive]

(2) : substitution [machine at $\varepsilon_q \in \{\frac{1}{64}, \frac{1}{16}, \frac{1}{8}, \frac{3}{16}, \frac{1}{5}, \frac{63}{256}\}$];

$\varepsilon_q \in [0, \frac{1}{4})$ [CT0.8] $\implies 4\varepsilon_q \in [0, 1)$ [machine]

(3) : $x_- = 4 \cdot 0 = 0$ [machine];

a move system with no J^- generator admits no dissolution [D1.2 : $J^- : n_{\mathcal{R}} : 1 \to 0$];

the ensemble reduces to the crystallized pattern [Def-BM.5(5)]

(4) : Def-H.32(1)'s filtration and FT.106(3)'s census rows re-cited (exact integers, frozen);

(5) : FT.108(2)'s three grade identities re-cited [machine rows frozen at Tier F.36]

(6) : $|\mathcal{E}(\mathcal{X})| > 1 \iff \exists$ a dissolution move $\iff x_- > 0$ [(2), (3)]. ■

Phys. Nothing in the flicker is chosen. The Janus move generator has a kernel, that kernel is a line, and its span is the stationary measure — so the weight of a crystallization against a dissolution is the ratio $x_+/(x_+ + x_-)$ before anyone names a rate. Substituting the crystallization threshold the axioms already carry turns that ratio into $1 - 4\varepsilon_q$, an identity rather than a fit, and the flicker weight into $4\varepsilon_q$. At the sharp threshold the dissolution weight is exactly zero and the ensemble collapses to one configuration; at any nonzero threshold the composite is genuinely an ensemble, and the move algebra's grade identities say precisely how each flicker step moves the information content. That is the whole mechanism of "the neutron is not a fixed topology", with no new parameter anywhere in it.

Theorem 4.17 (BM.20 — The mass from the curvature law on the path-integrated route). Thm.

(1) the mass is the curvature flux:

$$R = \kappa^2(\rho - 3p) + 4\Lambda \text{ [BM.15(2)]};$$

$$p = 0,\ \Lambda = 0 : \quad R = \kappa^2\rho, \quad \rho V = m :$$

$$\boxed{R\,V = \kappa^2 m} \qquad \text{i.e.} \qquad m = \frac{1}{\kappa^2}\int_\Sigma R\,dV \qquad \text{[machine: exact monomial identity]};$$

(2) the Λ clause:

$$(R - 4\Lambda)V = \kappa^2 m \text{ identically [machine]} :$$

$$\Lambda \text{ is the one additive freedom [P4; BM.15(6)]};$$

(3) the ensemble form (the path-integrated route):

$$\langle m\rangle_{\mathcal{X}} = \frac{1}{\kappa^2}\Big\langle \int_\Sigma R\,dV \Big\rangle_{\mathcal{X}} = \sum_{R\in\mathcal{E}(\mathcal{X})} w(R)\,m(R),$$

$$\sum_R w(R) = 1$$

[machine: the normalized average equals the averaged flux];

the mass of a sealed composite is the ensemble average of the curvature flux over its flicker — ¬the flux of the crystallized member;

(4) the derived sector read:

$$m_B^2(q,S)=\left.\frac{\sum_{\Gamma,R}\nu\,t^{|R|}u^{L}\#\text{inst}\cdot\frac{3}{2}(2E-|R|)}{\sum_{\Gamma,R}\nu\,t^{|R|}u^{L}\#\text{inst}}\right|_{t=u=1}$$

[FT.107(3); $\Delta S=\frac{3}{2}n_{\mathcal{V}}$ [FT.107(2)]] :

$$m_4^2(\cdot,0)=\frac{843957}{98956},$$

$$m_4^2(\cdot,1)=\frac{1921662}{393679},$$

$$m_4^2(\cdot,2)=\frac{91}{23},$$

$$m_4^2(\cdot,3)=\frac{13713}{3088}$$

[exact rationals; machine];

charge-degenerate in q [FT.106(3a), (3d)];

all strictly positive;

the transit sector ($S=0$) is heaviest [machine];

(5) the bound:

$$0<m_B^2(q,S)\le\tfrac{3}{2}\max_{\Gamma}n_{\mathcal{V}}\le\tfrac{3}{2}B(B-1)=18\ \ (B=4);$$

the census maximum is <9

[FT.107(4); machine];

(6) the curvature scale of the ensemble read:

$$\rho_B = \frac{m_B}{V_{\text{eff}}},$$

$$\kappa^2 \rho_B = \frac{2\pi\,\mathfrak{h}\,m_B}{V_{\text{eff}}} \Big/ \frac{\mathfrak{h}}{1} = \frac{2\pi\,E_{\text{eff}}^4 (\ln 2)^3}{\mathfrak{h}^2}\Big|_{E_{\text{eff}}:=m_B\mathfrak{h}/\ln 2}$$

i.e. the sector read enters BM.14's and BM.15's formulas through the effective count E_{eff} :

$$= m_B\mathfrak{h}/\ln 2 \text{ [exact rational per sector]} :$$

the lock of BM.16(1) applies sector by sector.

Rmk (route discipline). Clause (1) is the curvature law read as a mass definition — the same law BM.15 derived from CT1.23 — applied configuration by configuration and then averaged, which is what "the path-integrated route" means here. It does not import a mass datum: $m(R)$ at each member is the informational read [FT.44(4)], and the ensemble average is an exact rational combination of integers times $\ln 2/\mathfrak{h}$; the sector read of clause (4) is the deficit-average route [FT.107(3)], and the two coincide where both are defined.

Req. D2.9 (Parts 1–2); CT1.23, CT2.1, CT2.9 (Part 2); P4 (Parts 1–2); Def-H.32, FT.106(3), FT.107(2), FT.107(3), FT.107(4) (Part 3); FT.44(4) (Census); Def-BM.4, Def-BM.5, BM.14(2), BM.14(3), BM.15(2), BM.15(6), BM.16(1), BM.19 (this part); exact Laurent-monomial identities, exact rational ensemble averages [math; machine: `checks/verify_H_topo_blind_neutron.py` §B].

Pf.

$$(1) :\ \text{BM.15(2) at } p = 0,\ \Lambda = 0 :\ R = \kappa^2\rho;$$

$$\rho$$

$$= m/V \text{ [BM.14(2) pattern]}$$

$$\Longrightarrow\ RV$$

$$= \kappa^2 m \text{ [machine: substitution of the two defining relations into } RV - \kappa^2 m \text{ gives } 0 \text{ identically]}$$

$$(2) :\ R = \kappa^2\rho + 4\Lambda \implies (R - 4\Lambda)V = \kappa^2\rho V = \kappa^2 m \text{ [machine: the residual polynomial is } 0]$$

$$(3) :\ \text{linearity of the flux in } m \text{ at fixed } V, \kappa^2\ [(1)] \implies \langle RV\rangle/\kappa^2 = \langle m\rangle;$$

$$\text{normalization } \sum w = 1 \text{ [machine witness]}$$

(4)–(5) : FT.107(2)–(4) rows re-cited verbatim (exact rationals, frozen);

positivity, the ordering and the bound re-verified on the stated fractions [machine]

(6) : $E_{\text{eff}} := m_B\mathfrak{h}/\ln 2$ is the count whose informational mass is m_B [FT.44(4) inverted];

BM.14(3)/BM.15(4) re-cited at $E = E_{\text{eff}}$. ■

Phys. Mass is curvature flux. Trace the field equations for a pressureless composite and the scalar curvature times the volume is κ^2 times the mass, with the cosmological term the only additive ambiguity — so the mass read the corpus derived from the dust stress tensor is the same object as the integral of its own curvature. Run that identity through the flicker and the mass of a composite becomes the ensemble average of the curvature flux, not the flux of its crystallized snapshot. The topology-blind census supplies the average independently, as an exact rational per sector: four fractions at budget four, all positive, all charge-degenerate, with the transit sector heaviest and the whole table under the quadratic budget bound. Both routes meet in one formula, and the calibration lock of the previous tier applies sector by sector through the effective count.

Theorem 4.18 (BM.21 — The binding energy: additive in the count, non-additive in the ensemble). Thm.

(1) the informational count is exactly additive:

$$E_{\text{int}}(\Sigma_{12}) = 6 = (c_1\colon 2) + (c_2\colon 2) + (\text{connector}\colon 2)$$

[Census FT.35(2); machine];

$$m_{\text{hid}}$$

$= E_{\text{int}} \ln 2/\mathfrak{h}$ [FT.44(4)] is linear in the count $\implies$ the informational binding read vanishes exactly:

$$E^{\text{info}}_{\text{bind}} = \big(6 - (2{+}2{+}2)\big)\tfrac{\ln 2}{\mathfrak{h}} = 0$$

[machine];

(2) the ensemble read is not additive:

$$\mathcal{B}(S_j; S_1, S_2, S_3) \;:=\; m_B^2(\cdot, S_j) - \sum_{i=1}^{3} m_B^2(\cdot, S_i)$$

[BM.20(4)] :

$$\mathcal{B}(1;1,1,1) = -2\, m_4^2(\cdot,1) = -\frac{3843324}{393679};$$

$$\mathcal{B}(2;1,1,1) = \frac{91}{23} - \frac{5764986}{393679};$$

$$\mathcal{B}(3;1,1,1) = \frac{13713}{3088} - \frac{5764986}{393679};$$

$$\mathcal{B}(0;1,1,1) = \frac{843957}{98956} - \frac{5764986}{393679};$$

every one is a nonzero exact rational [machine];

(3) the binding is the flicker's, not the count's:

$$\varepsilon_q$$

$= 0 \implies \mathcal{E}$ collapses [BM.19(3)] $\implies$ the ensemble average reduces to the crystallized member $\implies \mathcal{B} \to 0$ with (1);

$$\varepsilon_q > 0 \implies \mathcal{B} \neq 0 \text{ exactly } [(2)] :$$

a non-flickering composite has no binding read, and a flickering one has a nonzero exact one;

(4) the sign structure:

all four combinations are bound: $\mathcal{B}(S_j;1,1,1) < 0 \;\; \forall S_j \in \{0,1,2,3\}$

[machine: exact rational sign census];

the exact order of the sector masses is $m_4^2(\cdot,0) > m_4^2(\cdot,1) > m_4^2(\cdot,3) > m_4^2(\cdot,2)$ [machine] :

three separated constituents outweigh one junction in every sector combination;

the like-sector case is exactly $-\,2\,m_4^2(\cdot,1)$ [machine];

$$|\mathcal{B}(S_j)|$$

$= 3m_4^2(\cdot,1) - m_4^2(\cdot,S_j)$ is largest for the lightest junction sector ($S_j = 2$) and smallest for the heaviest ($S_j = 0$) [machine];

(5) the ceiling inheritance:

$$|\mathcal{B}|$$

$$\leq 4 \cdot \tfrac{3}{2} B(B-1)$$

$$= 72 \text{ at } B$$

$= 4$ from FT.107(4)'s bound applied to all four terms [machine: exact bound arithmetic];

the winding spread at $B = 4$ is $\frac{91}{23} \leq m^2 \leq \frac{843957}{98956}$ [FT.107(4)] :

the binding read stays order-one while the bound grows as $\Theta(B^2)$;

(6) the read discipline:

$\mathcal{B}$ is a difference of the corpus's own m_B^2 reads [FT.107(3)];

¬a mass value in MeV, ¬an external datum, ¬a comparison [P5];

the pole-mass side (defect angles, ceilings) is a different read and remains item-13 bridge content [FT.76(3); Def-H.39].

Req. D2.9 (Parts 1–2); CT2.1 (Part 2); P5 (Parts 1–2); Def-H.32, Def-H.39, FT.76(3), FT.107(2), FT.107(3), FT.107(4) (Part 3); FT.44(4) (Census); Census FT.35(2) (row-wise); Def-BM.5, BM.19(3), BM.20(1), BM.20(4) (this part); exact rational arithmetic [math; machine: `checks/verify_H_topo_blind_neutron.py` §C].

Pf.

(1) : Census FT.35(2)'s decomposition row re-cited $(2{+}2{+}2 = 6)$;

m_{hid} linear in E_{int} [FT.44(4)] $\implies$ the difference is $0 \cdot \frac{\ln 2}{\mathfrak{h}}$ [machine]

(2) : exact rational subtraction of FT.107(3)'s four frozen values [machine];

nonvanishing: each numerator is nonzero after clearing denominators [machine]

(3) : BM.19(3) re-cited;

a one-member ensemble has $\langle m \rangle = m$ and the count-additivity of (1) applies

(4) : FT.107(3)'s four values in exact descending order $\frac{843957}{98956} > \frac{1921662}{393679} > \frac{13713}{3088} > \frac{91}{23}$,

i.e. S

$= 0 > 1 > 3 > 2$ [machine: exact rational comparison];

the sign census and the like-sector identity [machine]

(5) : $4 \cdot \frac{3}{2} \cdot 4 \cdot 3 = 72$ [machine]; FT.107(4)'s spread row re-cited

(6) : P5 scan of this tag's text [machine: zero external numeric tokens]. ■

Phys. Adding the edges yields six; adding the masses yields nothing. The informational count is exactly additive at the crystallized cell, so the binding read built from it vanishes identically — which is why a binding energy cannot be an integer summation. The flicker ensemble gives a different answer: averaged over the sector censuses, three separated constituents and one junction do not add up, and the mismatch is a nonzero exact rational in every combination of sectors. Turn the threshold to its sharp value and the ensemble collapses, the mismatch collapses with it, and the binding is exactly zero again — so the binding is a property of the flicker itself. Its magnitude stays order-one at budget four against a bound growing quadratically, its sign follows the transit sector's weight, and it remains a read of the corpus's own m_B^2 values with no external number anywhere near it.

Theorem 4.19 (BM.22 — The static force matrix on the two charge classes). Thm. With $h(\mathsf{c}) = \frac{1}{2}(c_0 I_2 + \mathsf{c} \cdot \boldsymbol{\sigma})$ and $\hat{\kappa}$ the ray swap [FT.129(2)], under Def-H.43's declared read promotion:

(1) the promoted force matrix:

$$\mathrm{F}_{\mathrm{ext}}(\mathsf{c}, q) = h(\mathsf{c}) \otimes I_2 + q\, I_2 \otimes \hat{\kappa},$$

$$\big[h \otimes I_2,\ q\, I_2 \otimes \hat{\kappa}\big] = 0$$

[FT.129(5); machine: all profiles × ternary q];

(2) the block read:

$$F_{LL} = F_{RR} = h(\mathrm{c}),$$

$$F_{LR} = F_{RL} = q\, I_2$$

[machine: exact block extraction on 4 profiles × 3 charge values];

the flipping block is the core read itself : $\langle L|\mathrm{F_{ext}}|R\rangle = q\, I_2$ [FT.129(5)];

the flipping set is $\{F_{\mathrm{EM}}$ core read$\}$, minimal and unique [FT.129(5)];

(3) the spectrum:

$$\mathrm{spec} = \left\{\tfrac{1}{2}(c_0 + s|\mathrm{c}|) + tq :\ s,t \in \{\pm 1\}\right\},$$

$$\mathrm{charpoly} = \big((y-u)^2 - q^2\big)\big((y+u)^2 - q^2\big), \quad y = x - \tfrac{c_0}{2}, \quad u^2 = \tfrac{|\mathrm{c}|^2}{4}$$

[FT.130(5); machine: the monic quartic built from the four roots equals the Faddeev–LeVerrier charpoly, and equals the closed form, on every witness];

(4) the two classes: the chiral-partner splitting is $2|q| \in \{0, 2\}$ [FT.130(5); machine] :

neutron class ($q_{\mathrm{net}} = 0$ [Def-BM.3(2)]) : $F_{LR} = 0$, splitting 0, parity doubling [FT.130(6)];

proton class ($q_{\mathrm{net}} = +1$ [Def-BM.3(2)]) :

$F_{LR} = I_2$, splitting 2, the doublet is split by exactly two read units;

the EM difference between the two classes is exactly the flipping block [(2), (4)] : one carries it, the other is annihilated by it;

(5) the derived locus and the promotion status:

at the derived locus $\mathrm{F_{ext}}(\mathrm{c}) = h \otimes I_\omega$ and $F_{LR} = 0$ exactly [FT.128(2) : the no-flip theorem] :

the flipping block exists only under Def-H.43's declared read promotion [the Def-H.43 declaration];

status: conditional on that declaration, not on any new one;

(6) the invariances:

$$\mathrm{F_{ext}} \mapsto \mathrm{F_{ext}} + \lambda I_4$$

$$\Longrightarrow F_{LR},$$

Schmidt ranks, commutants and $2|q|$ invariant; the spectrum translates by λ

[FT.130(7)];

operator-Schmidt rank $= 2 \iff q \neq 0 \wedge \mathrm{c} \neq 0$; $\quad 1$ when $h \propto I_2$; $\quad 0 \iff \mathrm{c} = 0 = q$ [FT.130(2)];

¬an entangling term at any locus [FT.130(2)];

(7) the charge-label status:

q_{net} is Def-BM.3(4)'s declared class label, pending the additive read [BM.6(1)] :

clauses (1)–(4), (6) are exact in $q \in \{-1, 0, +1\}$ [EM.9];

the attachment of $q = 0$ to the neutron class and $q = +1$ to the proton class is the declaration.

Req. CT0.12, CT1.1, CT1.2 (Part 1); EM.9 (Part 2); Def-H.34, Def-H.43, FT.122(6), FT.123, FT.128(2), FT.129(2), FT.129(5), FT.130(2), FT.130(5), FT.130(6), FT.130(7) (Part 3); Def-BM.3 (this part); the Def-H.43 declaration; exact $\mathbb{Q}(i)$ block algebra, Faddeev–LeVerrier characteristic polynomials over $\mathbb{Q}$, exact quartic root reconstruction [math; machine: `checks/verify_H_topo_blind_neutron.py` §D].

Pf.

(1)–(2) : block extraction of $h \otimes I_2 + qI_2 \otimes \hat{\kappa}$ with the chiral factor inner : F_{ab}

$$= h$$

$$\delta\text{-block} + q\hat{\kappa}_{ab}$$

$$\implies F_{LL}$$

$$= F_{RR}$$

$$= h\ (\hat{\kappa}_{00} = \hat{\kappa}_{11} = 0),$$

$$F_{LR}$$

$$= F_{RL}$$

$$= qI_2\ (\hat{\kappa}_{01} = \hat{\kappa}_{10} = 1)\ [\text{machine}];$$

commutation [machine]

(3) : the four roots $\frac{1}{2}(c_0 + s|\mathrm{c}|) + tq$ build a monic quartic over $\mathbb{Q}$ at witnesses with $|\mathrm{c}|^2$ a perfect square rational ($u \in \{\frac{1}{2}, 0, 1, \frac{3}{7}\}$);

Faddeev–LeVerrier on $\mathrm{F}_{\mathrm{ext}}$ agrees, and the closed form $(y^2 + u^2 - q^2)^2 - 4u^2y^2$ agrees with both [machine]

(4) : $2|q|$ at $q \in \{-1, 0, +1\}$ [machine]; the class attachment [Def-BM.3(2) re-cited]

(5)–(6) : FT.128(2), FT.130(2), (7) rows re-cited (frozen); (7) : Def-BM.3(4) re-cited. ■

Phys. Put the static force matrix on the two charge classes and they come apart in exactly one place. Both classes share the same internal block h; the promoted read adds a chiral-flipping block equal to the core charge times the identity, so the neutral class has no flipping block at all and keeps its parity doubling, while the unit class splits its chiral partners by precisely two read units. That splitting — zero against two — is the whole electromagnetic difference between the neutron row and the proton row in this corpus, and it is a matrix identity, not a coupling constant. The promotion that allows a flipping block is a declared read, and the tier says so; everything downstream of it (the spectrum, the charpoly, the Schmidt ranks, the scalar-shift invariance) is exact. What is still declared rather than derived is which class label sits on which junction: the additive charge read remains an open construction.

Theorem 4.20 (BM.23 — The transition: the ladder rule, the forced carrier, and the resolvent rate). Thm.

$$(1) \quad \text{the weak ladder:}$$

$$E_\pm = \tfrac{1}{2}(\sigma_1 \pm i\sigma_2) = \text{the two matrix units in the } \sigma_3 \text{ eigenbasis;}$$

$$[E_+, E_-] = \sigma_3,$$

$$[\sigma_3, E_\pm] = \pm 2E_\pm$$

$$[\text{FT.41(d)};\ \text{machine: exact over } \mathbb{Q}(i)];$$

$$\text{the charged-current shape } W^\pm$$

$$= \tfrac{1}{\sqrt{2}}(W_1 \mp iW_2) \text{ [D3.10] is the same ladder on the pair register [C8];}$$

$$(2) \quad \text{the selection rule:}$$

$$q_{\text{core}} = \tfrac{1}{2}(\sigma_s - \sigma_t) \text{ [D3.4] and the ladder moves } \sigma_3 \text{ by } \pm 2 \text{ [(1)]} \implies \boxed{\Delta q = \pm 1}$$

$$[\text{machine}];$$

$$(3) \quad \text{the alphabet overflow:}$$

$$\text{shifting the ternary read } \{-1, 0, +1\} \text{ [EM.9] by } +1 \text{ gives } \{0, +1, +2\} \text{ and by } -1 \text{ gives } \{-2, -1, 0\} :$$

$$\text{both leave the alphabet [machine];}$$

$$\therefore \text{ a charge-changing transition cannot stay inside the composite: it must emit a carrier;}$$

$$(4) \quad \text{the compensation is unique:}$$

$$Q_{\text{EM}}$$

$$= \sum_{c \in U} q_{\mathrm{EM}}(c)$$

$= 0$ in every valid universe [EM.11; D2.11, D1.3] $\implies$ a composite read changing by Δq requires a compensating read of exactly $-\Delta q$:

$$\#\{q \in \{-1, 0, +1\} : \ q = -\Delta q\} = 1$$

[machine] :

the emitted carrier's charge is forced, not chosen;

the compensated total is exactly 0 : EM.11's balance survives the transition [machine];

(5) the carrier selection theorem:

the carrier must satisfy (i) q

$= -1$ for Δq

$= +1$ [(4)];

(ii) no colour attachment [else the composite's colour seal breaks : $\mathcal{C} = 0 \implies$ the block vanishes [FT.19; CT3.18, D2.14]];

(iii) EM activated ($f_v \neq 0$) [D3.3; else it cannot carry the read];

in the channel inventory: the q-channel is triplet-attached [FT.119] (fails ii);

the ν-channel is EM-unactivated [FT.142(1) : $f_v = 0$ reads] (fails iii);

$\therefore$ the ℓ-channel is the unique carrier class [FT.109–FT.113 : colour-blind, ternary read active] :

the emitted line is Def-BM.6(1)'s electron row;

(6) the neutral partner is admissible, not forced:

$\neg\exists$ a derived law requiring a neutral emission [no lepton-number read exists in the inventory];

the ν-channel [FT.142–FT.144] is EM-neutral and colour-blind, so it can carry the deficit read ΔS

$= \frac{3}{2} n_{\mathcal{V}}$ [FT.107(2)] without disturbing the charge balance :

status: admissible carrier for the deficit read; its emission is not a theorem of this corpus;

(7) the rate:

$$\sum_n \langle \varnothing | \hat{U}^n | \varnothing \rangle w^n = \frac{1}{1 - w^2 x_+ x_-} \text{ in } \mathbb{Q}[w, x_+, x_-]/(w^{12})$$

[FT.108(5); machine];

$$x_+ x_- = 4\varepsilon_q(1 - 4\varepsilon_q) \leq \tfrac{1}{4},$$

equality at $\varepsilon_q = \frac{1}{8}$

[machine: exact maximum over the admissible range] $\implies$ the pole $w^2 x_+ x_-$

$= 1$ sits at $|w| \qquad \geq 2$;

the per-tick transition weight is an exact rational function of ε_q alone [BM.19(2)] :

Γ is $\neg$a declared number and $\neg$a new parameter;

(8) the no-flicker theorem:

$\varepsilon_q = 0 \implies x_+ x_- = 0 \implies$ the resolvent is $1 \implies \Gamma = 0$ exactly

[BM.19(3); machine] :

a non-flickering composite never decays;

the flicker is the mechanism, and the threshold is its only control;

(9) the arrow:

the transition is forward-only [FT.144(2) : the domains force strict alternation; A6 : $\neg\exists\tau^{-1}$] :

$\neg\exists$ an inverse decay in the ledger;

the time coordinate is the ledger order [$x^0 := \tau_n,$ D1.8] :

Γ is a weight per tick, $\neg$a rate per external second [P5].

Rmk (provenance of each piece). Clause (5) is a selection theorem: three conditions on the carrier, two channels eliminated by their own frozen rows, one class left. Clause (6) records honestly what is not forced — the neutral partner — because no lepton-number or energy-sharing law exists in the inventory; naming it would be a declaration, and none is issued. Clause (7)'s Γ is exact in ε_q and carries no time unit beyond the tick [D1.8]; no lifetime is derived and none is compared [P5].

Rmk (the forcing status of clause (6)). The neutral emission is conditionally forced: BM.41(2)–(3) derives that every kinematically admissible transition vertex carries a third member, and that its class is unique among the constructed carrier classes — the ν-channel line: a two-body vertex at the equal rest masses of BM.28(1) has no real on-shell solution for any positive member mass ($E_p - m_p = -m_\ell^2/(2m_n) < 0$ identically), and the transit (EM-radiation) alternative is $\neg$constructed [Def-BM.8(8)]. Clause (6)'s inventory row is unchanged: $\neg\exists$ a lepton-number read — the forcing is kinematic plus uniqueness, $\neg$a number conservation; its hypothesis is the vertex's kinematic admissibility, which BM.41(4) shows no adopted read satisfies.

Req. A6 (Part 1); D1.2, D1.8, D3.3, D3.4, D3.10 (Parts 1–2); C8, CT0.8, CT3.18 (Part 1, Part 2); D2.11, D2.14 (Part 2); EM.9, EM.11 (Part 2); P5 (Parts 1–2); Def-H.32, FT.19, FT.41(d), FT.107(2), FT.108(3), FT.108(5), FT.109, FT.113, FT.119, FT.142(1), FT.144(2) (Part 3); Def-BM.5, Def-BM.6, BM.19(2), BM.19(3), BM.22 (this part); exact $\mathbb{Q}(i)$ ladder algebra, exhaustive ternary-alphabet censuses, exact rational resolvent identities and maxima [math; machine: `checks/verify_H_topo_blind_neutron.py` §D–§E].

Pf.

(1) : $E_\pm = \frac{1}{2}(\sigma_1 \pm i\sigma_2)$ entrywise [machine];

$[E_+, E_-] = \sigma_3,\ [\sigma_3, E_\pm] = \pm 2E_\pm$ [machine: all commutators exact over $\mathbb{Q}(i)$]

(2) : a ± 2 move on σ_s changes $\frac{1}{2}(\sigma_s - \sigma_t)$ by ± 1 [machine];

(3) : the shifted sets $\{0,1,2\},\ \{-2,-1,0\}$ against the alphabet $\{-1,0,1\}$ [machine]

$(4):$ EM.11 re-cited; $\#\{q : q = -\Delta q\} = 1$ for $\Delta q = \pm 1$ [machine]; $\Delta q + (-\Delta q) = 0$ [machine]

$(5):$ (ii) FT.19/CT3.18/D2.14 re-cited; (iii) D3.3 re-cited;

eliminations : FT.119 (triplet-attached), FT.142(1) (EM unactivated) re-cited;

uniqueness : the channel inventory has three triplets [FT.112, FT.119, FT.142]

$(6):$ inventory scan: no lepton-number read exists [FT.81(4) pattern];

$\Delta S = \frac{3}{2} n_{\mathcal{V}}$ [FT.107(2) re-cited]

$(7):$ FT.108(5)'s resolvent re-derived coefficient-wise at four rational thresholds [machine];

$x_+ x_- = 4\varepsilon_q(1 - 4\varepsilon_q):$ maximum $\frac{1}{4}$ at $\varepsilon_q = \frac{1}{8}$ [machine over $\varepsilon_q = k/256$]

$(8):$ ε_q

$= 0$

$\implies x_+ x_-$

$= 0$

$\implies 1/(1 - 0)$

$= 1 \implies$ no transition weight beyond the identity [machine];

$(9):$ FT.144(2), A6, D1.8 re-cited. ■

Phys. The decay is assembled from three independent necessities and one honest gap. The weak ladder moves the σ_3 eigenvalue by two units, and the core charge is half a σ difference, so any charge-changing transition moves the ternary read by exactly one — which pushes the endpoints straight out of the alphabet, so the change cannot stay inside the composite. Total charge vanishes in every valid universe, and the alphabet contains exactly one value that compensates a unit change,

so the emitted carrier's charge is forced. Then the inventory does the rest: the quark-side channel is colour-attached and would break the seal, the neutrino-side channel is electromagnetically inert and could not carry the read, which leaves precisely one class — the ℓ-channel line the naming layer calls the electron. The neutral partner is admissible and useful for carrying the deficit read, but no law in the corpus forces it, and the tier says so instead of naming a neutrino into the theorem. The rate is a geometric resolvent in the derived weights, its pole safely outside the unit disk, and it vanishes identically at the sharp crystallization threshold: a composite that cannot flicker cannot decay. The whole transition runs forward only, per tick, with no lifetime and no external unit anywhere in it.

BM.24 — The status of the topology-blind composite: adjudication and scope. Structural-constraints record: see Appendix B, Section B.1.17 of the Addendum.

4.9 Tier BM.7 — The Proton in the Positive CPT Sector and Its Mirror: the Sector Labels, the Exact Phase, and the Charge-Blind Geometry

Definition 4.7 (Def-BM.7 — The $\mathrm{EM}^{\pm}$ sector labels and the EM scar). Def. Part-4 labels for two derived structures: the CPT-orientation classes of a charged sealed composite, and its endpoint charge read. Both labels are naming; both objects are derived.

$$(1) \quad \text{the sectors:}$$

$$\mathrm{EM}^{+} \;:=\; \text{the CPT-orientation class with } q_{\mathrm{net}} = +1 \text{ and the unstarred weight } w(C);$$

$$\mathrm{EM}^{-} \;:=\; \text{the class with } q_{\mathrm{net}} = -1 \text{ and the conjugated weight } w(C)^{*};$$

constitutive clause: the two labels name the two orbits of the CPT involution on the charged classes [EM.24(4); BM.25(4)];

non-derived clause: the signs $\pm$ in the labels are a convention;

$$(2) \quad \text{the scope clause (the mirror is formal, not dynamical):}$$

$$\text{“backward time”} \;:=\; \Theta_{\mathfrak{L}}\,[\text{D2.7}:\ \text{ledger order reversed},\ J^{+} \leftrightarrow J^{-},\ G(e) \mapsto G(e)^{*}],$$

an involution on configuration data bijecting the completed admissible set onto itself [FT.144(4)(i); FT.156(5); machine] — ¬a dynamical inverse :

$$\neg\exists\, \tau^{-1}\ [\text{A6}];$$

the pure order reversal (labels unchanged) of every nonempty completed history is inadmissible at its first event [FT.144(4)(ii); machine];

$\therefore$ EM^{-} is a bookkeeping orbit of the mirror, ¬a universe running backwards;

no clause of this part asserts a backward evolution;

(3) the EM scar:

$$\mathrm{scar}(c) \;:=\; q_{\mathrm{core}}(c) = \tfrac{1}{2}\big(\sigma_{s(c)} - \sigma_{t(c)}\big) \in \{-1, 0, +1\}$$

[D3.4, EM.9, EM.9a; FT.40] :

the endpoint charge read of a materialized chain — registration data on the two ends [D3.3, D1.1],

¬an internal edge [D0.8];

the zero-contribution clause: the scar enters E_{int} with coefficient 0 [BM.28(3); machine : Census FT.35(2)'s decomposition is 2+2+2, all edge counts];

(4) the neutral class:

$q_{\mathrm{net}} = 0$ is the unique fixed point of the charge map $q \mapsto -q$ [EM.24(1); machine] :

the neutral sector is charge-self-conjugate;

no configuration is fixed pointwise, since $\Theta_{\mathfrak{L}}$ conjugates the weight [CT2.8(1)] :

self-conjugacy is a class property, ¬a pointwise one [BM.25(4)];

(5) the data bar:

"matter"/"antimatter" are the labels of (1) and carry no external datum [P5];

no annihilation channel, no production channel and no cross-section is constructed in this corpus [BM.29(5)].

Rmk (vocabulary). EM$^{\pm}$ are Part-4 labels for CPT-orientation classes; they are not the register's (S,T) sector symbols $\mathcal{E}_{\pm\pm}$ [CT0.12, CT1.1] and no clause identifies them. The pair-register sectors grade operators; the labels here grade composite configurations.

Req. A6, A8 (Part 1); D0.8, D1.1, D2.7, D3.3, D3.4 (Parts 1–2); CT2.8(1) (Part 2); EM.9, EM.9a, EM.24(1), EM.24(4) (Part 2); P5 (Parts 1–2); FT.40, FT.144(4), FT.156(5) (Part 3); Def-BM.3 (this part); clauses (3)–(4) carry marked forward pointers at their citing sites, imported in no Req line; Census FT.35(2) (row-wise); exhaustive enumeration of ledger records, exact $\mathbb{Q}(i)$ weight algebra [math; machine: `checks/verify_H_proton_cpt.py` §A, §C].

Pf.

Definition block: the constitutive clause is (1)'s orbit naming;

the non-derived clauses are the sign convention of (1),

the scope reading of (2) and the vocabulary of (5);

(3)'s zero-contribution clause and (4)'s fixed-point census are derived and machine-checked. ■

Phys. Two names are added and nothing else. The positive and negative CPT-orientation sectors label the two orbits that the corpus's own CPT involution makes of the charged classes, and the EM scar is the endpoint charge read the fermion era already derived. The one clause that does real work is the scope clause: the mirror is an involution on configuration data — reverse the ledger order and swap the seal labels — and it is not a way of running the universe backwards, because the tick has no inverse and every reversed history is inadmissible at its first event. So the negative sector is a bookkeeping orbit with equal amplitude, not a backwards-running world. The neutral class is the unique charge fixed point of the mirror, self-conjugate as a class while no single configuration is fixed, since the mirror always conjugates the weight.

Theorem 4.21 (BM.25 — The CPT involution on the sealed composite: orbits, the exact phase, and the census ratio). Thm.

(1) the three involutions are derived:

$$C:\ \sigma_e \mapsto -\sigma_e\ \forall e \implies q \mapsto -q, \quad f_v \mapsto -f_v, \quad \aleph_v \text{ invariant}$$

[EM.24(1); D3.3];

$$P:\ \mathcal{P} \mapsto E_{\mathcal{R}} \setminus \mathcal{P},\ \text{equivalently } \alpha_e \mapsto \pi - \alpha_e \qquad [\text{EM.24(2); Def-EM.21, EM.20(1)}];$$

$$T:\ \Theta_{\mathfrak{L}}, \qquad w(\Theta_{\mathfrak{L}} C) = w(C)^* \qquad [\text{EM.24(3); D2.7; CT2.8(1)}];$$

$$\text{CPT} := C \circ P \circ T,$$

$$C^2 = P^2 = T^2 = \mathrm{id} \implies \mathrm{CPT}^2 = \mathrm{id}$$

[EM.24(4) : each component derived, no external input];

(2) the mirror on the ledger is an exact involution with an unexpected fixed set:

$\Theta_{\mathfrak{L}}^2 = \mathrm{id}$ on every admissible single-pair history [machine];

$$\Theta_{\mathfrak{L}}\big((J^+J^-)^m\big) = (J^+J^-)^m \ \ \forall m :$$

reversal composed with the $J^+ \leftrightarrow J^-$ relabelling fixes every completed word pointwise [machine: all $m \le 6$];

the pure order reversal (labels unchanged) is inadmissible at its first event [FT.144(4)(ii); machine] :

the formal mirror exists and the dynamical reversal does not;

$\Theta_{\mathfrak{L}}$ bijects the two-pair completed interleavings onto themselves at $2n$

$= 2,$

$4,$

6 [machine: set equality and cardinality];

$\#J^+$, $\#J^-$ are preserved $\implies$ the Janus balance survives the mirror [D1.3; machine];

(3) the weight conjugation and the exact phase:

$$G(e) = \sigma_e \exp\big(i\alpha_e \widehat{ds}^{\,2}(e)\big), \quad \alpha_e \in \{0, \pi\} \text{ [D3.5, CT0.9]} \implies G(e) = \sigma_e\, i^{m_e}, \quad m_e \in \mathbb{Z} :$$

the phase alphabet is the fourth roots of unity — exact in $\mathbb{Q}(i)$, ¬a transcendental;

$\Theta_{\mathfrak{L}}$: $m_e \mapsto -m_e$, σ_e fixed, $\mathcal{J}(C) \mapsto \mathcal{J}(C)^*$ [D3.6];

$$\boxed{w(\Theta_{\mathfrak{L}}C) = w(C)^*}$$

[CT2.8(1); machine: 24696 enumerated configurations (sign assignments × half-integer κ ladder × $\mathcal{J}$-phases)];

$$w(\mathrm{CPT}\,C) = (-1)^{|E(C)|}\, e^{i\pi\Theta(C)}\, w(C),$$

$$\lambda := (-1)^{|E|} e^{i\pi\Theta} \in \{+1, -1, +i, -i\},$$

$$|\lambda|^2 = 1$$

[EM.24(5), FT.134(3); machine];

$$\big|w(\mathrm{CPT}\,C)\big|^2 = \big|w(C)\big|^2 \;\; \forall C \qquad \text{[EM.24(5) : unconditional; machine]};$$

the on-shell refinement: $\widehat{ds}^{\,2}(e) \in \mathbb{Z}\;\forall e \implies \Theta \in \mathbb{Z} \implies \lambda = (-1)^{|E|+\Theta} \in \{\pm 1\}$:

$$w(\mathrm{CPT}\,C) = w(C)^* \iff |E| + \Theta \equiv 0 \pmod 2$$

[EM.24(5); machine];

(4) the sector orbit census:

CPT : $(q,$ orientation$) \mapsto (-q,$ flipped$)$ on $\{-1, 0, +1\} \times \{w, w^*\}$:

an involution with no fixed sector [machine];

the orbit decomposition is exactly three pairs of size two;

$q \mapsto -q$ has exactly one fixed point, $q = 0$ [machine] :

the neutral class is charge-self-conjugate and every charged class is exchanged — $\mathrm{EM}^+ \leftrightarrow \mathrm{EM}^-$ [Def-BM.7(1), (4)];

(5) the census ratio (the sectors are equal in weight, the neutral is twice either):

$$Z^{\mathrm{norm}}(0, S)$$

$$= 2$$

$$Z^{\text{norm}}(+1, S) \text{ exactly for } S$$

$$= 0,$$

$$1,$$

$$2,$$

3 [machine : $395824 = 2{\cdot}197912$, $1574716 = 2{\cdot}787358$, $1311552 = 2{\cdot}655776$, $2001024 = 2{\cdot}1000512$];

$$Z^{\text{norm}}(+1, S) = Z^{\text{norm}}(-1, S) \text{ exactly [FT.106(3d); machine] :}$$

the two charged sectors have identical weight;

the ratio is the factorization's own: $m_q 2^{|R|-2}$ with m_q

$$= (2, 1, 1) \text{ at } q$$

$$= (0, +1, -1)$$

$$\implies m_0/m_{\pm 1}$$

$$= 2 \text{ [FT.106(3a); machine];}$$

$Z^{\text{amp}}(+1, S) = Z^{\text{amp}}(-1, S)$ on the cover support,

$$Z^{\text{amp}}(q, S) = 0 \ \ \forall q, \ S \in \{2, 3\}$$

[FT.106(3c), (3d); machine];

(6) the baryon-type character:

$$\Theta_{\text{int}} \in 4\mathbb{Z} \text{ [Census FT.49]} \implies e^{i\pi\Theta} = 1;$$

$$\Theta_{\text{int}} \in 2\mathbb{Z}\ [\text{Census FT.35(3)};\ \text{BM.2(1)}] \implies e^{i\pi\Theta} = 1\ \text{ as well [machine]} :$$

on the ε-sealed junction class the CPT character is exactly

$$\lambda = (-1)^{|E|}$$

[FT.134(3); machine];

(7) the norm pairing:

$$\langle A|A\rangle$$

$$= \sum_{C} |w(C)|^2$$

≥ 0 [CT2.8(3)] is CPT-invariant term by term : $|w(C)|^2$

$= |w(\Theta_{\mathfrak{L}} C)|^2$ [machine on all enumerated pairs] :

the two sectors contribute equally to the norm — the amplitude-level statement of $\text{EM}^+ \leftrightarrow \text{EM}^-$.

Rmk (locus discipline). Clause (2)'s fixed set is a statement about words: the mirror fixes each completed alternating word because reversing and relabelling commute to the same word. It is not a statement that any dynamical process is reversible — A6 and FT.144(4)(ii) bar that reading, and clause (2) records both rows side by side.

Req. A6, A8 (Part 1); D1.3, D2.7, D3.3, D3.4, D3.5, D3.6 (Parts 1–2); CT0.9 (Part 1); CT2.8(1), CT2.8(3), EM.9, EM.9a, EM.20(1), EM.24(1), EM.24(2), EM.24(3), EM.24(4), EM.24(5), Def-EM.21 (Part 2); Def-BM.7 (this part); FT.40, FT.106(3), FT.134(3), FT.144(4), FT.156(5) (Part 3); Census FT.35(3), FT.49 (row-wise); BM.2(1), BM.7(1) (this part); exact $\mathbb{Q}(i)$ weight algebra on the fourth-root alphabet, exhaustive ledger-record enumeration, exact integer census arithmetic [math; machine: `checks/verify_H_proton_cpt.py` §A].

Pf.

(1) : EM.24(1)–(4) rows re-cited (frozen, machine-verified at Tier F.33);

$$\text{CPT}^2 = C^2P^2T^2 = \text{id [each an involution]}$$

(2) : $\Theta_{\mathfrak{L}}(w)$

= reverse(w) with labels swapped : on $(J^+J^-)^m$ the reversal gives $(J^-J^+)^m$ and the swap returns $(J^+J^-)^m$ [machine];

$$\Theta_{\mathfrak{L}}^2 = \mathrm{id}\ [\text{machine}];$$

$$\text{pure reversal : leads with } J^-,\ \text{outside D1.2's domain [FT.144(4)(ii); machine]}$$

$$(3):\ \alpha_e \in \{0,\pi\}\ [\text{CT0.9}]\ \wedge\ \widehat{ds}^{\,2} \in \tfrac{1}{2}\mathbb{Z}\ [\text{FT.22–FT.24 ladder}]\ \Longrightarrow\ \alpha_e \widehat{ds}^{\,2} \in \tfrac{\pi}{2}\mathbb{Z}\ \Longrightarrow\ \exp(i\alpha_e \widehat{ds}^{\,2})$$

$$= i^{m_e}\ [\text{machine}];$$

$$\Theta_{\mathfrak{L}}:\ G \mapsto G^*\ \text{per edge and}\ \mathcal{J} \mapsto \mathcal{J}^*\ [\text{D2.7, D3.6}]\ \Longrightarrow\ w \mapsto w^*\ [\text{machine}]$$

$$\lambda = (-1)^{|E|} e^{i\pi\Theta} = i^{2|E|} i^{2\Theta} \in \{i^k\}\ [\text{machine}];$$

$$|\lambda|^2 = \lambda\lambda^* = 1\ [\text{machine}];$$

$$\text{on-shell } \Theta \in \mathbb{Z}\ \Longrightarrow\ i^{2\Theta} = (-1)^{\Theta}\ [\text{machine}]$$

$$(4):\ \text{the map on six sectors, its square, and its fixed-point census [machine];}$$

$$q \mapsto -q\ \text{fixed points} : \{0\}\ [\text{machine}]$$

$$(5):\ \text{FT.106(3)'s census integers re-cited and the ratios verified exactly [machine];}$$

$$m_q = (2,1,1)\ [\text{FT.106(3a)}]\ \Longrightarrow\ m_0/m_{\pm 1} = 2\ \text{and}\ m_{+1} = m_{-1}\ [\text{machine}]$$

$$(6):\ e^{i\pi\Theta} = i^{2\Theta} = 1\ \text{at}\ \Theta \in 2\mathbb{Z}\ [\text{machine}]; \qquad \lambda = (-1)^{|E|} \cdot 1\ [\text{FT.134(3) re-cited}]$$

$$(7):\ \text{CT2.8(3) re-cited;} \qquad |w|^2\ \text{equality term by term from (3) [machine].}\ \blacksquare$$

Phys. The mirror does something sharper than exchange two sectors. On the ledger it is an involution that reverses order and swaps the seal labels — and on a completed history those two acts cancel, so every completed word is fixed pointwise, while the same history read backwards with its labels intact is inadmissible at its first event. The formal mirror exists; the dynamical reversal does not. Weights conjugate exactly, because the phase alphabet is the fourth roots of unity: with $\alpha_e \in \{0,\pi\}$ and a half-integer line element, every edge weight is a sign times a power of i, and negating the winding conjugates it. The CPT phase is therefore one of four exact values, its modulus is one, and the amplitude is invariant unconditionally — with the on-shell refinement that the

phase is real and the weight literally conjugates exactly when the edge count plus the winding is even. The census closes the picture: the two charged sectors carry identical weight at every winding, the neutral sector exactly twice either, which is the factorization's own $m_q = (2, 1, 1)$ and not a coincidence. On the baryon-type junction, whose phase class lives in $4\mathbb{Z}$, the whole CPT character collapses to a sign counting edges.

Theorem 4.22 (BM.26 — The proton row: the positive sector). Thm.

$$(1) \quad \text{the structure:}$$

$$\text{the } \varepsilon\text{-sealed 3-juncture in the unit charge class } q_{\text{net}} = +1$$

$$[\text{Def-BM.3}(2);\ \text{D3.9}:\ B = \varepsilon_{ijk} c_1^i c_2^j c_3^k];$$

$$\mathcal{C}(\Sigma; C)$$

$= 1 \iff$ all three attachments seal inside as one ε-type triple [Census FT.35(5); CT3.17; FT.34(5)] :

$$\text{the colour seal is the observability switch } [\text{BM.1}(1)];$$

$$(2) \quad \text{the sector membership:}$$

$$q_{\text{net}} = +1 \ \wedge\ w(C) \implies \text{the row lies in EM}^+$$

$$[\text{Def-BM.7}(1);\ \text{the class label is Def-BM.3}(4)\text{'s, pending BM.6}(1)];$$

$$(3) \quad \text{the EM scar:}$$

$$\text{scar} = q_{\text{core}} = \tfrac{1}{2}(\sigma_s - \sigma_t) \in \{-1, 0, +1\}\ [\text{D3.4},\ \text{EM.9}];$$

$$\text{the unit class carries } q_{\text{core}} = +1 \text{ at its read end;}$$

$$\text{the scar is } \omega\text{-odd } [\text{D2.6};\ \text{FT.129}(3);\ \text{machine}] \text{ and contributes } 0 \text{ to } E_{\text{int}}\ [\text{Def-BM.7}(3);\ \text{BM.28}(3);\ \text{machine}];$$

$$(4) \quad \text{the information content and the six reads:}$$

$$E_{\text{int}}^{(p)} = 6\ (c_1\colon 2,\ c_2\colon 2,\ \text{connector}\colon 2)$$

$$[\text{Census FT.35}(2);\ \text{row-wise}];$$

$$\Delta x_{\mathrm{B}}^{(p)} = \frac{\mathfrak{h}}{6\ln 2},$$

$$V_{\mathrm{eff}}^{(p)} = \frac{\mathfrak{h}^3}{216(\ln 2)^3},$$

$$m_{\mathrm{hid}}^{(p)} = \frac{6\ln 2}{\mathfrak{h}},$$

$$\rho^{(p)} = \frac{1296(\ln 2)^4}{\mathfrak{h}^4};$$

$$\kappa^2\rho^{(p)} = \frac{2592\,\pi(\ln 2)^3}{\mathfrak{h}^2},$$

$$r_s^{(p)} = 3\,\mathfrak{h},$$

$$(\Delta x_{\mathrm{B}}^{(p)})^2\kappa^2\rho^{(p)} = 72\,\pi\ln 2$$

$$[\text{BM.13},\ \text{BM.14},\ \text{BM.15}(4),\ \text{BM.16}(1),(4);\ \text{machine}];$$

$$(5)\quad \text{the EM coupling:}$$

$$F_{LR} = q\,I_2 = I_2,$$

$$\text{the chiral-partner splitting } 2|q| = 2$$

$$[\text{BM.22}(2),(4);\ \text{FT.129}(5),\ \text{FT.130}(5);\ \text{machine}];$$

$$\text{the promoted force matrix is } \mathrm{F}_{\mathrm{ext}}$$

$$= h\otimes I_2 + I_2\otimes\hat{\kappa}\ \text{with spec}$$

$$= \{\tfrac{1}{2}(c_0 + s|\mathrm{c}|) + t:\ s,t\in\{\pm 1\}\}\ [\text{BM.22}(3);\ \text{machine}];$$

$$(6)\quad \text{the amplitude:}$$

$$|w(\mathrm{CPT}\cdot C^{(p)})|^2 = |w(C^{(p)})|^2\ [\text{BM.25}(3)];$$

$$\text{on-shell with } |E| + \Theta \equiv 0\ (2):\ w(\mathrm{CPT}\cdot C^{(p)}) = w(C^{(p)})^*;$$

the CPT image of this row is BM.27's — no independent datum is carried by either.

Rmk (route scope of clause (4)). The six reads of clause (4) are functions of E_{int} alone, so they are the crystallized-count route's rows: they are identical across the charge classes and clause (4) states exactly that. The ensemble-weighted mass read is route-relative — degenerate on the normalized route and split by the exact factor $\frac{4}{3}$ between the neutral and the unit class on the total-deficit route [BM.28(4)]. Clause (4) is unchanged on its own route; the ensemble route is BM.28(4)'s and BM.30's content.

Rmk (identification discipline). The name attaches by Def-BM.3(2); the theorem content of every clause is the tuple (ε-junction, $q_{\text{net}} = +1$, $E_{\text{int}} = 6$, $\Theta_{\text{int}} \in 4\mathbb{Z}$, $2|q| = 2$). No measured value enters [P5]; the charge class is the declared label pending the additive read [Def-BM.3(4); BM.6(1)].

Req. D3.4, D3.9 (Parts 1–2); CT3.17 (Part 2); EM.9 (Part 2); P5 (Parts 1–2); Def-BM.3, Def-BM.7, BM.1(1), BM.2(1), BM.13, BM.14, BM.15(4), BM.16(1), BM.16(4), BM.22(2), BM.22(3), BM.22(4) (this part); FT.34(5), FT.40, FT.129(3), FT.129(5), FT.130(5) (Part 3); Census FT.35(2), FT.35(3), FT.35(5), FT.49 (row-wise); BM.25(3) (this part); exact $\mathbb{Q}(i)$ block algebra, exact rational reads [math; machine: `checks/verify_H_proton_cpt.py` §B–§C].

Pf.

$$(1)\text{–}(2):\ \text{Def-BM.3}(2) \text{ and Def-BM.7}(1) \text{ re-cited;} \qquad \text{Census FT.35}(5)\text{'s sealing row re-cited}$$

$$(3):\ \text{D3.4/EM.9 re-cited;}$$

$$\omega\text{-oddness}: q_{\text{core}}(\omega c) = \tfrac{1}{2}(\sigma_t - \sigma_s) = -q_{\text{core}}(c)\ [\text{D2.6; machine}];$$

$$\text{the zero contribution}: \text{BM.28}(3)$$

$$(4):\ \text{substitution of } E_{\text{int}}$$

$= 6$ into BM.13(1), BM.14(1),(2),(3), BM.15(4), BM.16(1),(4) [machine: every coefficient exact]

$$(5):\ \text{BM.22}(2)\text{–}(4) \text{ re-derived at } q = +1\ [\text{machine}]; \qquad (6):\ \text{BM.25}(3) \text{ re-cited.}\ \blacksquare$$

Phys. The proton row is the neutron row with one entry changed. Same junction grammar, same six internal directed edges, same sealing phase class, same width, volume, mass, density, curvature scale, comparison radius and lock — because none of those reads contains a charge symbol. What changes is the flipping block: the promoted static force matrix carries qI_2 off-diagonal, so

the unit class splits its chiral partners by exactly two read units where the neutral class splits them by zero. The scar is the endpoint read, it is odd under the orientation reversal, and it contributes nothing to the edge count — which is precisely why the geometry is blind to it. The amplitude of the row equals the amplitude of its mirror, unconditionally, and on-shell the mirror's weight is literally the complex conjugate.

Theorem 4.23 (BM.27 — The antiproton row: the CPT image, and the resolution of the identity question). Thm.

$$(1) \quad \text{the image:}$$

$$C^{(\bar{p})} := \mathrm{CPT} \cdot C^{(p)} :$$

$$q_{\text{net}} \mapsto -q_{\text{net}} = -1 \ [\text{EM.24}(1)], \quad w \mapsto w^* \ [\text{CT2.8}(1)], \quad \mathcal{P} \mapsto E_{\mathcal{R}} \setminus \mathcal{P} \ [\text{EM.24}(2)];$$

$$\text{the image lies in } \mathrm{EM}^- \qquad [\text{Def-BM.7}(1);\ \text{BM.25}(4) :\ \text{the two charged classes form one orbit}];$$

$$(2) \quad \text{the structure is the same junction:}$$

$$\varepsilon\text{-sealed 3-juncture in the } q_{\text{net}} = -1 \text{ class } [\text{Def-BM.3}(2) :\ \text{the antiproton row}];$$

$$E_{\text{int}}^{(\bar{p})} = E_{\text{int}}^{(p)} = 6 :$$

$$\Theta_{\mathfrak{L}} \text{ preserves the record size and exchanges } \#J^+ \text{ with } \#J^- \ [\text{BM.25}(2);\ \text{machine}]$$

$$\Longrightarrow \ |R| \text{ and hence } E_{\text{int}}$$

$$= |R| \text{ are invariant } [\text{Def-BM.5}(4);\ \text{machine}];$$

$$(3) \quad \text{every read is the image's, so every read is identical:}$$

$$\Delta x_{\text{B}}^{(\bar{p})} = \Delta x_{\text{B}}^{(p)},$$

$$V_{\text{eff}}^{(\bar{p})} = V_{\text{eff}}^{(p)},$$

$$m_{\text{hid}}^{(\bar{p})} = m_{\text{hid}}^{(p)}, \quad \rho^{(\bar{p})} = \rho^{(p)}, \quad \kappa^2 \rho^{(\bar{p})} = \kappa^2 \rho^{(p)},$$

$$r_s^{(\bar{p})} = r_s^{(p)};$$

the reason is structural, not numerical: none of the six monomials carries a charge exponent [BM.28(1); machine: the q-exponent is 0 in all six];

(4) the amplitude and the character:

$$|w(C^{(\bar{p})})|^2 = |w(C^{(p)})|^2 \text{ [EM.24}(5);\ \text{machine]};$$

$$w(C^{(\bar{p})}) = \lambda\, w(C^{(p)}), \quad \lambda = (-1)^{|E|} e^{i\pi\Theta};$$

$$\Theta_{\text{int}} \in 4\mathbb{Z} \text{ [Census FT.49]} \implies \lambda = (-1)^{|E|} \in \{\pm 1\} \qquad \text{[BM.25}(6);\ \text{machine]};$$

the two rows have equal norm contribution to $\langle A|A\rangle$

$$= \sum_C |w(C)|^2 \text{ [CT2.8}(3);\ \text{BM.25}(7);\ \text{machine]};$$

(5) the difference set is exactly two entries:

$$\text{scar}^{(\bar{p})} = -\,\text{scar}^{(p)} \text{ [}\omega\text{-odd;\ machine] and the sector orientation } w \leftrightarrow w^*;$$

$$F_{LR}^{(\bar{p})} = -I_2 = -F_{LR}^{(p)} \text{ but } 2|q^{(\bar{p})}| = 2 = 2|q^{(p)}| :$$

the flipping block is odd and the splitting is even [BM.22$(2), (4)$; machine];

(6) the identity adjudication:

$$C^{(\bar{p})}$$

$\neq C^{(p)}$ as configurations [the scar and the orientation differ; BM.25(4) : $\neg\exists$ a fixed charged sector];

$$C^{(\bar{p})}$$

$$= \text{CPT} \cdot C^{(p)} \text{ with } \text{CPT}^2$$

$=$ id [EM.24(4)] $\implies$ the antiproton row carries no independent datum : it is one orbit of size two, not two structures;

both statements hold because they are about different objects — configurations (6)'s first line, structure (6)'s second;

no clause asserts an identity of configurations;

(7) what is not constructed:

$\neg\exists$ an annihilation channel,

$\neg\exists$ a pair-production channel,

$\neg\exists$ a cross-section between EM^+ and EM^- [Def-BM.7(5); BM.29(5)] :

the two sectors are amplitude-equal configuration orbits; their interaction is not a theorem of this corpus.

Rmk (route scope of clause (3)). The six identical reads of clause (3) carry no charge exponent [BM.28(1)], and the equality of the proton and antiproton rows holds on every mass route: the charge census is symmetric, $W(+1) = W(-1) = 15$ exactly, and both ensemble layers are ω-symmetric [BM.28(5); FT.106$(3d)$]. What the census does split is the neutral class against the charged pair, by the exact factor $\frac{4}{3}$ on the total-deficit route [BM.28(4)(ii)] — a splitting that leaves this tag's CPT rows untouched.

Rmk (mirror scope). The image is configuration data under $\Theta_{\mathfrak{L}}$ composed with the sign and parity maps. It is not a time-reversed process: A6 gives the tick no inverse, and the pure order reversal of a completed history is inadmissible at its first event [BM.25(2)]. Reading clause (1) as a dynamical evolution would contradict FT.144(2), and no clause here does.

Req. A6 (Part 1); D2.6, D2.7 (Parts 1–2); CT2.8(1), CT2.8(3) (Part 2); EM.24(1), EM.24(2), EM.24(4), EM.24(5) (Part 2); P5 (Parts 1–2); Def-BM.3, Def-BM.5(4), Def-BM.7, BM.22(2), BM.22(4), BM.25(2), BM.25(4), BM.25(6), BM.25(7), BM.26 (this part); FT.129(3), FT.130(5), FT.144(2), FT.144(4) (Part 3); Census FT.49 (row-wise); clause (3) carries a marked forward pointer at its citing site, imported in no Req line; exact $\mathbb{Q}(i)$ arithmetic, exhaustive enumeration [math; machine: `checks/verify_H_proton_cpt.py` §A–§C].

Pf.

(1) : EM.24(1), (2) and CT2.8(1) composed [EM.24(4)]; orbit membership : BM.25(4)'s census

(2) : Census FT.35(2)'s row re-cited;

$$|\Theta_{\mathfrak{L}}(w)| = |w| \text{ and } \#J^+(\Theta w) = \#J^-(w) \text{ [machine]} \implies E_{\text{int}} = |R| \text{ invariant}$$

(3) : the six monomials of BM.13(1), BM.14(1),(2), BM.15(4), BM.16(4) have q-exponent 0 [machine: monomial census] $\implies$ substituting q

$$\mapsto -q \text{ fixes each}$$

$$(4):\ \text{EM.24}(5)\text{'s identity re-cited and re-verified [machine]};$$

$$\lambda = (-1)^{|E|} \text{ at } \Theta \in 4\mathbb{Z} \text{ [BM.25}(6)\text{]}$$

$$(5):\ q_{\text{core}} \mapsto -q_{\text{core}} \text{ [D2.6; machine]}; \qquad F_{LR} = qI_2 \mapsto -qI_2 \text{ while } 2|q| \mapsto 2|q| \text{ [machine]}$$

$$(6):\ \neg\exists \text{ a fixed charged sector [BM.25}(4)\text{; machine]} \implies \text{distinct configurations};$$

$$\text{CPT}^2 = \text{id [EM.24}(4)\text{]} \implies \text{one orbit of size two}$$

(7) : inventory scan: no annihilation, production or inter-sector transition operator exists [BM.23(6) pattern];

Def-BM.7(5) re-cited. ■

Phys. The antiproton is what the mirror does to the proton, and the mirror is cheap: flip every edge sign, complement the alignment pattern, reverse the ledger while swapping the seal labels. Everything the corpus derives about the composite survives that map untouched, because not one of the six geometric reads contains a charge — the width, the volume, the mass, the density, the curvature scale and the comparison radius come out identical, and the amplitude comes out equal in modulus with a phase that is one of four exact roots of unity, collapsing to a bare sign on the baryon-type phase class. Only two entries change: the scar flips sign and the weight conjugates. The flipping block flips with the scar while the splitting, being $2|q|$, does not — so the proton and antiproton split their chiral partners by the same two units in opposite directions. Whether they are "the same object" depends on whether one is counting configurations or structures: as configurations they differ and no charged sector is fixed; as structure they are one orbit of size two, and the second carries no datum the first does not. What the corpus does not have is any process connecting the two sectors — no annihilation, no pair production, no cross-section — and that absence is recorded rather than filled.

Theorem 4.24 (BM.28 — The charge-constrained probability space: the census, the three mass routes, and the splitting). Thm.

(1) the geometric monomials are charge-blind (the crystallized route):

$$\Delta x_{\text{B}} = \mathfrak{h}^1 E^{-1} (\ln 2)^{-1},$$

$$V_{\text{eff}} = \mathfrak{h}^3 E^{-3} (\ln 2)^{-3}, \quad \rho = \mathfrak{h}^{-4} E^4 (\ln 2)^4, \quad \kappa^2 \rho = 2\pi\, \mathfrak{h}^{-2} E^4 (\ln 2)^3;$$

$$r_s = \tfrac{1}{2} \mathfrak{h} E,$$

$$(\Delta x_{\mathrm{B}})^2 \kappa^2 \rho = 2\pi E^2 \ln 2,$$

$$\Delta x_{\mathrm{B}} r_s = \tfrac{\mathfrak{h}^2}{2 \ln 2}$$

$$[\text{BM.13}(1),\ \text{BM.14}(1)\text{–}(3),\ \text{BM.15}(4),\ \text{BM.16}(1), (4)] :$$

the charge exponent is 0 in all seven [machine: monomial census] $\implies$ at fixed E_{int} these seven reads are identical on every charge class;

scope: this clause governs reads that are functions of E_{int} alone — the crystallized-count route;

it says nothing about ensemble-weighted reads $[(2)\text{–}(5)]$;

(2) the charge constraint is a filter on the probability space, and the filter differs by class:

$$q_i = \tfrac{1}{2}(\sigma_{s(c_i)} - \sigma_{t(c_i)}) \in \{-1, 0, +1\}\ [\text{D3.4},\ \text{EM.9}];$$

$$Q = q_1 + q_2 + q_3\ [\text{D3.9};\ \text{the additive read of BM.6}(1)\text{'s spec}];$$

the end-pair census: $\quad m_q = (2, 1, 1)$ at $q = (0, +1, -1)$

$$[\text{EM.9's four end-pairs};\ \text{FT.106}(3a);\ \text{machine}] :$$

the neutral read has two sign realizations, each charged read one;

the full census of the 3-constituent composite $(4^3 = 64$ end-sign assignments$)$:

$$\#\{Q = k\} = (1,\ 6,\ 15,\ 20,\ 15,\ 6,\ 1)\ \text{ for } k = -3, \ldots, +3$$

[machine: exhaustive];

the generating function $(2 + z + z^{-1})^3$ reproduces it exactly and sums to 64 [machine];

the ordered charge-triple census: $\quad \#\{(q_1, q_2, q_3) :\ \textstyle\sum q_i = k\} = (1, 3, 6, 7, 6, 3, 1)$:

7 neutral triples against 6 charged [machine];

the probability-space sizes:

$$W(0) = 20,$$

$$W(+1) = W(-1) = 15,$$

$$\boxed{\frac{W(0)}{W(\pm 1)} = \frac{20}{15} = \frac{4}{3}}$$

[machine: exact];

one chain restricts $2{:}1$ $[m_q]$; three chains under the additive constraint restrict $4{:}3$:

the constraint is what changes the ratio;

(3) the scar is a boundary condition, not an internal edge:

$E_{\text{int}} = \#\{\text{internal directed edges}\}$ [D0.8];

scar $= q_{\text{core}}$ is endpoint registration data [D1.1, D3.3, D3.4];

Census FT.35(2)'s decomposition 6

$= (c_1 \colon 2) + (c_2 \colon 2) +$ (connector: 2) has no endpoint summand [machine];

E_{int} is unchanged across all four endpoint sign pairs [machine];

$\therefore$ the scar shifts no crystallized-count read [(1)] — and it does not follow that the scar shifts no read : the charge constraint acts on the ensemble, not on the count [(2)];

(4) the route theorem (the mass read is route-relative):

(i) the normalized ensemble average $m_B^2(q, S)$

$$= \frac{\sum_R \nu\, t^{|R|} u^L \#\text{inst} \cdot \Delta S(R)}{\sum_R \nu\, t^{|R|} u^L \#\text{inst}} \Bigg|_{t=u=1} [\text{FT.107(3)}] :$$

a uniform sector factor cancels in a ratio — verified exactly for factors 1,

$$2,$$

$$3,$$

$$15,$$

$$20,$$

$\frac{7}{3}$ [machine] $\implies$ charge-degeneracy on this route is structural,

¬an oversight :

$$\Delta S(\Gamma, R)$$

$$= \tfrac{3}{2} n_{\mathcal{V}}(\Gamma, R)$$

$= \frac{3}{2}(2E - |R|)$ is σ-free [FT.107(2)],

so the restriction rescales the measure without reshaping it;

(ii) the unnormalized total deficit $T(q, S)$

$$= \sum_R w(R)$$

$\Delta S(R)$ [the I_{hid}-type read; FT.42(1) pattern] :

$T(0, S) = \frac{4}{3}\, T(\pm 1, S)$ exactly [machine] :

$\Delta m \neq 0$ on this route,	$\dfrac{\langle m \rangle^{(n)}}{\langle m \rangle^{(p)}} = \dfrac{4}{3},$	the neutral class is heavier

[machine: exact, ordering derived];

(iii) the amplitude layer [Def-H.32(2); FT.106(3)] :

$$\frac{Z^{\text{amp}}(0,0)}{Z^{\text{amp}}(\pm 1,0)} = \frac{5336}{1196} = \frac{58}{13},$$

$$\frac{Z^{\text{amp}}(0,1)}{Z^{\text{amp}}(\pm 1,1)} = \frac{4796}{802} = \frac{2398}{401}$$

[machine] :

a third ratio, winding-dependent, ¬equal to $\frac{4}{3}$ or 2;

its bridge to a mass read is item-13 content and is ¬constructed

(5) what the splitting does not touch:

$W(+1) = W(-1)$ exactly [machine] and $Z^{\bullet}(+1, S) = Z^{\bullet}(-1, S)$ [FT.106(3d)] :

the proton and the antiproton remain degenerate on all three routes — the charge restriction splits the neutral class from the charged pair and never splits the pair [BM.27(3), (4)];

CPT maps the Q

$= +1$ triple set bijectively onto the Q

$= -1$ set, and the neutral set onto itself with $(0, 0, 0)$ the unique CPT-fixed charge triple [machine];

(6) the identity and difference sets:

identical on every route: E_{int},

Δx_{B},

V_{eff},

ρ,

$$\kappa^2\rho,$$

$$r_s,$$

both locks,

$$\Theta_{\text{int}} \text{ class},$$

$$|w|^2,$$

$2|q|$ restricted to $\{|q| = 0\}$ vs $\{|q| = 1\}$ [(1)];

route-dependent: the ensemble mass read — degenerate on (4)(i),

split by $\frac{4}{3}$ on (4)(ii),

split by $\frac{58}{13}/\frac{2398}{401}$ on (4)(iii);

distinguishing on every route: F_{LR}

$$= qI_2,$$

$$q_{\text{core}},$$

$$k,$$

$$\text{sign}(k),$$

the CPT orientation,

the sector label,

and now $W(Q)$ [machine: the sets are disjoint];

(7) no fractional charge exists, structurally:

$$q \in \{-1, 0, +1\} \subset \mathbb{Z}\,[\text{EM.9, P2}] \text{ and } Q = \sum_i q_i \implies Q \in \mathbb{Z} \text{ with } Q \in \{-3, \ldots, +3\}\,[\text{machine}]:$$

$$\tfrac{1}{3},\ \tfrac{2}{3} \notin \{-3, \ldots, +3\}:$$

fractional charges are impossible under an integer-valued additive read, ¬merely absent;

deficit 1′′s charge side [FT.125(4)] is therefore not closable by this read : a fractional constituent alphabet requires a different read, and none is constructed;

(8) the read-choice row:

which of (4)'s three routes is "the mass" is a read specification,

¬a theorem [Census FT.76's priced-variant pattern];

all three are exact and machine-verified;

no route imports a measured value [P5];

¬∃ a P4 parameter and ¬∃ a new dimensionful scale [P1; §Cal-BM stays empty].

Rmk (route discipline). Clause (1) and clause (4)(i) are not in tension with clause (4)(ii): they are statements about different functionals of the same ensemble. A normalized average is blind to any uniform rescaling of the measure; a total is not. The charge constraint of clause (2) is exactly such a rescaling — uniform within a sector, different between sectors — so it is invisible to the first and visible to the second, with the exact factor $4/3$. Any reading of this tag that quotes one route against another is reading a scope clause as a claim.

Req. D0.8, D1.1, D2.6, D3.3, D3.4, D3.9 (Parts 1–2); CT2.1, CT2.9 (Part 2); EM.9, EM.24(1) (Part 2); P1, P2, P5 (Parts 1–2); Def-BM.4, Def-BM.7, BM.13(1), BM.14(1), BM.14(2), BM.14(3), BM.15(4), BM.16(1), BM.16(4), BM.22(2), BM.22(4), BM.27(3), BM.27(4) (this part); Def-H.32, FT.42(1), FT.84(1), FT.90(5), FT.106(3), FT.107(2), FT.107(3), FT.125(4), FT.129(3) (Part 3); Census FT.35(2) (row-wise); the additive read of BM.6(1)'s specification (constructed in Tier BM.8, cited there as a forward pointer and imported in no Req line); exhaustive end-sign enumeration, generating-function convolution, exact rational ratio arithmetic [math; machine: `checks/verify_H_probability_space_restriction.py` §A–§C].

Pf.

(1) : the seven monomials re-cited from Tiers BM.5's rows and their exponent vectors read off [machine];

q-exponent 0 in each [machine]

$(2):$ EM.9′s table $\to m_q = (2,1,1)$ [machine];

exhaustive enumeration of 4^3

$= 64$ end-sign assignments $\to$ $\#\{Q = k\}$

$= (1,6,15,20,15,6,1)$ [machine];

$(2 + z + z^{-1})^3$ convolved three times reproduces the same census and sums to 64 [machine]

triples : $\#\{\sum q_i = k\} = (1,3,6,7,6,3,1)$, summing to $27 = 3^3$ [machine];

$$W(k) = \sum_{\sum q_i = k} \prod_i m_{q_i} \implies W(0) = 8 + 12 = 20,\ W(\pm 1) = 12 + 3 = 15 \text{ [machine]};$$

$20/15 = 4/3$ [exact]

$(3):$ D0.8 counts internal directed edges; D3.4 reads two endpoint signs;

Census FT.35(2)′s three summands are edge counts [machine];

invariance across the four sign pairs [machine]

$(4)(\mathrm{i}):$ write $w_q(R)$

$= c_q v(R)$ with v q-free : $\dfrac{\sum c_q v m}{\sum c_q v}$

$= \dfrac{\sum vm}{\sum v}$ for every c_q

$\neq 0$ [machine at $c \in \{1,2,3,15,20,\frac{7}{3}\}$];

$\Delta S = \frac{3}{2}(2E - |R|)$ is σ-free [FT.107(2)]

$$(4)(\text{ii}):\ T(q,S)$$

$$= W(q)\cdot\overline{\Delta S}(S) \text{ with the same } \overline{\Delta S}\ [(\text{i})]$$

$$\implies T(0)/T(\pm 1)$$

$$= W(0)/W(\pm 1)$$

$$= 4/3\ [\text{machine}];$$

$$4/3 > 1 \implies \text{the neutral class is heavier [machine]}$$

$$(4)(\text{iii}):\ \text{FT.106(3)'s amplitude integers re-cited;}$$

$$5336/1196 = 58/13 \text{ and } 4796/802 = 2398/401\ [\text{machine: exact reduction}];$$

$$58/13 \neq 4/3 \neq 2\ [\text{machine}]$$

$$(5):\ W(+1) = W(-1) = 15\ [\text{machine}];$$

$$\text{FT.106(3d)'s } \omega\text{-symmetry re-cited;}$$

$$\text{CPT}: q_i \mapsto -q_i \text{ gives } Q \mapsto -Q \text{ and a bijection of triple sets; fixed triples}: \{(0,0,0)\}\ [\text{machine}]$$

$$(6):\ \text{set arithmetic on the three lists [machine];}$$

$$(7):\ Q \in \mathbb{Z} \text{ as a sum of integers, range } [-3,3]\ [\text{machine}];$$

$$\tfrac{1}{3} \notin \mathbb{Z}\ [\text{exact}]$$

$$(8):\ \text{Census FT.76's priced-variant pattern re-cited;}$$

$$\text{P1/P5 scan of this tag's text [machine]. } \blacksquare$$

Phys. The charge constraint does exactly what a constraint on a sum does: it changes how many microscopic sign assignments land in each class. For one chain the neutral read has two realizations and each charged read one; for three chains bound by $q_1 + q_2 + q_3 = Q$ the full census

is $1, 6, 15, 20, 15, 6, 1$ across $Q = -3$ to $+3$, sixty-four assignments in all, and the neutral class ends up with twenty against the unit class's fifteen. That ratio, four thirds, is the whole content of the restriction. Whether it reaches the mass depends on what functional the mass is. A normalized ensemble average cannot see it — a uniform rescaling of the measure cancels between numerator and denominator, which is why the sector mass table is charge-degenerate as a matter of structure rather than luck. A total deficit can see it, and does: on that route the neutral class carries exactly four thirds the information content of either charged class, and is heavier. The amplitude layer carries a third ratio, fifty-eight thirteenths at zero winding and a different one at winding one, and its bridge to a mass read is still an open item. Two things survive every route: the proton and the antiproton stay degenerate, because the census is symmetric under charge conjugation and $(0, 0, 0)$ is the only self-conjugate triple; and no fractional charge can appear at all, since an integer sum of ternary reads cannot land on a third. What the corpus does not do is pick a route for the reader: all three are exact, all three are priced, and the choice is a read specification.

BM.29 — The status of the CPT sectors: adjudication and scope. Structural-constraints record: see Appendix B, Section B.1.18 of the Addendum.

4.10 Tier BM.8 — The Additive Charge Read and the Mass-Route Adjudication

Definition 4.8 (Def-BM.8 — The additive charge read (the construction BM.6(1) specifies)). Def. A read Q on sealed composites, constructed to BM.6(1)'s three specification clauses; no new axiom, no new datum.

(1) the read:

$$Q(c_1, \ldots, c_k) \;:=\; \sum_{i=1}^{k} q_{\text{core}}(c_i),$$

$$q_{\text{core}}(c) = \tfrac{1}{2}\big(\sigma_{s(c)} - \sigma_{t(c)}\big) \in \{-1, 0, +1\}$$

[D3.4, EM.9, EM.9a, FT.40];

(2) spec clause (i) — restriction: $Q|_{\text{single chain}} = q_{\text{core}}$ [immediate at $k = 1$];

(3) spec clause (ii) — additivity:

$$Q(\mathcal{X} \sqcup \mathcal{Y}) = Q(\mathcal{X}) + Q(\mathcal{Y}) \text{ for disjoint sealings}$$

[the sum is over disjoint constituent sets];

(4) spec clause (iii) — observability on sealed supports:

Q is defined only where $\mathcal{C}(\Sigma; C) = 1$ [D2.14 : $\partial_{\text{col}}(\Sigma; C) = \varnothing$];

one dangling attachment zeroes the whole colour sector [FT.19] and the read with it;

(5) the derived range and parity:

$$k = 3 : \quad Q \in \{-3, -2, -1, 0, +1, +2, +3\} \subset \mathbb{Z}$$

[machine];

$$k = 2 \ (\delta\text{-seal}) : \ Q \in \{-2, \ldots, +2\};$$

integer-valued at every k : $\frac{1}{3}, \frac{2}{3}$ are unreachable [BM.28(7)];

(6) the probability-space weights it induces:

$$W(Q) := \#\{(\sigma\text{-assignments}) : \ \textstyle\sum_i q_i = Q\} = \left[z^Q\right](2 + z + z^{-1})^k$$

[BM.28(2); machine] :

$$k = 3 : \quad W = (1, 6, 15, 20, 15, 6, 1) \text{ over } Q = -3, \ldots, +3, \qquad \sum_Q W(Q) = 4^3 = 64 \qquad \text{[machine]};$$

(7) the CPT action:

$$\text{CPT} : \ q_i \mapsto -q_i \ [\text{EM.24}(1)] \implies Q \mapsto -Q;$$

$$W(Q) = W(-Q) \ [\text{machine}];$$

the unique CPT-fixed triple at $k = 3$ is $(0, 0, 0)$ [machine];

(8) status and scope:

constructed — BM.6(1)'s specification is discharged for the sealed classes $k \leq 3$;

the charge-class labels of Def-BM.3(4) now rest on a derived read rather than on a declaration;

¬constructed: the coupling of Q to a gauge field [deficit $1'$, FT.125(4); FT.82(7)(a)],

and any dynamics of Q [BM.6(5)].

Rmk (what the construction does not do). Q is a sum of the ternary end reads. It supplies no fractional constituent alphabet, no gauge coupling, and no mass: the mass routes are BM.28(4)'s and BM.30's, and the read-choice among them is not made here.

Req. D2.14, D3.4, D3.9 (Parts 1–2); EM.9, EM.9a, EM.24(1) (Part 2); P2 (Part 1); Def-BM.3(4), BM.6(1), BM.28(2), BM.28(7) (this part); FT.19, FT.40, FT.82(7), FT.125(4) (Part 3); generating-function convolution, exhaustive end-sign enumeration [math; machine: `checks/verify_H_probability_space_restriction.py` §A, §D].

Pf.

(1)–(4) : the three specification clauses of BM.6(1) discharged clause by clause;

observability : $\mathcal{C}(\Sigma; C)$ is the $\Pi_{\mathcal{C}}$ projector's support [D2.14];

FT.19's all-or-nothing row re-cited

(5) : a sum of k integers each in $\{-1, 0, 1\}$ lies in $\{-k, \ldots, k\}$ [machine at $k = 2, 3$];

(6) : $(2 + z + z^{-1})^3$ convolved [machine];

$\sum_Q W(Q) = (2 + 1 + 1)^3 = 64$ [machine]

(7) : q_i

$\mapsto -q_i$ componentwise [EM.24(1)]

$\Longrightarrow Q$

$\mapsto -Q$ and $W(Q)$

$= W(-Q)$ by the census's symmetry [machine];

$$\text{fixed triples} : -q_i = q_i \;\forall i \iff q_i = 0 \;\forall i \text{ [machine]}. \blacksquare$$

Phys. The additive charge read the era listed as an open construction is now built, and it costs nothing: sum the ternary endpoint reads of the constituents, restrict to sealed supports, and the three clauses the specification demanded are satisfied by inspection. Two things come out of it that the specification did not ask for. The read is integer-valued at every constituent number, so a fractional charge is not missing from this corpus but unreachable by it. And the read carries its own statistics: the number of sign assignments realizing each net charge is the coefficient of a cubic generating function, giving twenty for the neutral class against fifteen for either unit class, which is the exact measure of how differently the charge constraint filters the two probability spaces.

Theorem 4.25 (BM.30 — The mass-route adjudication: three exact ratios, one read choice). Thm.

(1) the three routes and their exact charge ratios:

route	functional	neutral : charged
(i) normalized ensemble average [FT.107(3)]	$\frac{\sum_R w\,\Delta S}{\sum_R w}$	1 (degenerate)
(ii) total deficit [I_{hid}-type; FT.42(1)]	$\sum_R w\,\Delta S$	$\frac{4}{3}$
(iii) amplitude layer [Def-H.32(2); FT.106(3)]	$\sum_R v_\Gamma(R)\,\Delta S$	$\frac{58}{13}$ (S=0), $\frac{2398}{401}$ (S=1)

all three exact [machine]; the three ratios are pairwise unequal [machine];

(2) the splitting, where it exists:

on route (ii) :

$$\frac{\langle m\rangle^{(n)}}{\langle m\rangle^{(p)}} = \frac{W(0)}{W(+1)} = \frac{20}{15} = \frac{4}{3},$$

$$\Delta m \neq 0,$$

the neutral class is heavier [BM.28(4)(ii); machine];

the mechanism is the measure, not the deficit: ΔS

$= \frac{3}{2}(2E - |R|)$ is σ-free [FT.107(2)],

so the restriction rescales without reshaping;

(3) the degeneracy, where it is structural:

on route (i) any uniform sector factor cancels in the ratio [machine: factors $1, 2, 3, 15, 20, \frac{7}{3}$] :

charge-degeneracy is a theorem about normalized averages,

¬an assumption about composites;

(4) what survives every route:

$W(+1) = W(-1)$ and $Z^{\bullet}(+1, S) = Z^{\bullet}(-1, S)$ [BM.28(5); FT.106(3d)] :

the proton and the antiproton are degenerate on all three routes;

the crystallized-count reads are charge-blind [BM.28(1)] :

$E_{\text{int}},\ \Delta x_{\text{B}},\ V_{\text{eff}},\ \rho,\ \kappa^2\rho,\ r_s,$ both locks are shared by all three classes;

(5) the read-choice row:

¬∃ a theorem of Parts 1–4 selecting one route as "the mass" [inventory: the m_{hid}, pole, sector-expectation and m_B^2 reads are distinct [FT.84(1); FT.87(8)]];

adoption requires a logged declaration [Census FT.76's priced-variant pattern; BM.6(4) precedent];

no route is adopted here;

(6) the ordering is derived and uncompared:

$\langle m \rangle^{(n)} > \langle m \rangle^{(p)}$ on route (ii) [machine: $\frac{4}{3} > 1$];

¬∃ a measured mass value in any clause and ¬∃ a comparison [P5];

(7) the status of the adjudication:

derived: (1)–$(4),\ (6)$;

open: the route choice (5), the amplitude $\leftrightarrow$ mass bridge the gauge coupling of Q

closed by this tier: BM.6(1)'s construction spec [Def-BM.8(8)];

unclosable by this read: the fractional constituent alphabet [BM.28(7); Def-BM.8(5)].

Rmk (no route is privileged). Clause (2)'s splitting and clause (3)'s degeneracy are both exact and neither is a correction of the other: they are different functionals of one ensemble. A corpus that adopted route (ii) would have a neutron heavier than its proton by the factor four thirds; a corpus that adopted route (i) would have them equal; both statements are machine-verified, and the choice between them is the read specification that clause (5) declines to make.

Req. D2.9 (Parts 1–2); CT2.1 (Part 2); P2, P5 (Parts 1–2); Def-H.32, FT.42(1), FT.84(1), FT.87(8), FT.106(3), FT.107(2), FT.107(3), FT.125(4) (Part 3); Def-BM.8, BM.6(1), BM.6(4), BM.28(1), BM.28(2), BM.28(4), BM.28(5), BM.28(7) (this part); Census FT.76 (the priced-variant pattern); exact rational ratio arithmetic, exhaustive enumeration [math; machine: `checks/verify_H_probability_space_restriction.py` §B–§D].

Pf.

(1) : BM.28(4)'s three rows re-cited with their machine values;

$\frac{58}{13} \neq \frac{4}{3}$ since $58 \cdot 3 = 174 \neq 13 \cdot 4 = 52$;

$\frac{4}{3} \neq 1$ [machine]

(2) : $T(q, S)$

$= W(q)\overline{\Delta S}(S)$ with a common $\overline{\Delta S}$ [BM.28(4)(i), (ii)] $\implies$ the ratio is $W(0)/W(+1)$

$= 4/3$ [machine]

(3) : the cancellation lemma [BM.28(4)(i); machine]

(4) : BM.28(5) and BM.28(1) re-cited

(5) : inventory scan of the mass reads [FT.84(1)'s read-class taxonomy; FT.87(8)];

Census FT.76's variant-pricing pattern re-cited

(6) : $\frac{4}{3} > 1$ [machine]; P5 scan of this tag's text [machine: zero external numeric tokens]

(7) : conjunction of (1)–(6) with Def-BM.8(8). ■

Phys. One ensemble, three functionals, three answers — and all three are exact. Normalize and the charge constraint disappears, because a uniform rescaling of the measure cancels between numerator and denominator; that is why the sector mass table is charge-degenerate, and it is a structural fact rather than a lucky one. Sum instead of averaging and the constraint is all that matters: twenty sign assignments realize the neutral class against fifteen for either charged class, so the neutral composite carries four thirds the information deficit and weighs more. Take the amplitude layer and a third ratio appears, fifty-eight thirteenths at zero winding and something else at winding one, with its bridge to a mass read still open. What no route disturbs is the mirror: the census is symmetric under charge conjugation, so the proton and antiproton stay degenerate everywhere, and the seven crystallized reads stay blind to charge entirely. The corpus prices the three routes and adopts none, because adopting one is a read specification and not a theorem — and it records the ordering it does derive, neutral heavier, without comparing it to anything.

4.11 Tier BM.9 — The Mesonic Flicker Test: the Contraction Grammar, the Colour Projector, and the Protected Junction

Definition 4.9 (Def-BM.9 — The extended flicker ensemble and the sealing filter). Def. A construction class on Def-BM.5 (no new axiom; Def-H.32's scopes (S1)–(S4) and Def-BM.5's inherited clauses stand).

(1) the two restricted ensembles:

$$\mathcal{E}_{\varepsilon}(\mathcal{X}) := \{R \in \mathcal{E}(\mathcal{X}) : \text{ the sealing of } R \text{ is } \varepsilon_{ijk}\text{-type}\},$$

$$\mathcal{E}_{\delta}(\mathcal{X}) := \{R \in \mathcal{E}(\mathcal{X}) : \ \delta^{i}{}_{j}\text{-type}\}$$

[D3.9; CT3.17];

$$\mathcal{E}_{\text{ext}}(\mathcal{X}) := \mathcal{E}_{\varepsilon}(\mathcal{X}) \cup \mathcal{E}_{\delta}(\mathcal{X});$$

constitutive clause: the union is taken inside Def-H.32(1)'s topology space, which is colour-blind [D2.13 : $\mathcal{M}(\Sigma)$ is subject to the rung masks and Janus balance, ¬to colour sealing];

(2) the filter clause (the load-bearing one):

colour sealing is ambient-global [CT3.18] and acts downstream as the projector $\Pi_{\mathcal{C}}(\Sigma; C)$

$$= \mathcal{C}(\Sigma; C) \cdot \mathbb{I}_{\Sigma}^{(\text{strong})},$$

$$\mathcal{C}(\Sigma; C) = \begin{cases} 1 & \partial_{\text{col}}(\Sigma; C) = \varnothing \\ 0 & \text{otherwise} \end{cases}$$

[D2.14];

one dangling attachment zeroes the entire colour block [FT.19 : all-or-nothing];

$\mathcal{C}$ is an idempotent scalar, $\mathcal{C}^2 = \mathcal{C} \in \{0, 1\}$ [machine];

non-derived clause: none — (2) is D2.14's derived projector, cited at the site where the union of (1) would otherwise be read as a sum over sealings;

(3) the grouping grammar:

$g(N) :=$ #{partitions of N attachments into contraction groups of size 2 (δ) and 3 (ε)}

[CT3.17a(2); FT.41(b)] :

$$g(N) = (1,\ 1,\ 3,\ 10,\ 25,\ 105) \quad \text{for } N = 2, \ldots, 7$$

[machine: exhaustive partition enumeration reproduces all six values];

$g_{\delta}(N) :=$ #{groupings containing at least one δ} $= (1,\ 0,\ 3,\ 10,\ 15,\ 105)$,

$g_{\varepsilon}(N) :=$ #{groupings of 3-blocks only} $= (0,\ 1,\ 0,\ 0,\ 10,\ 0)$;

$g_{\delta}(N) + g_{\varepsilon}(N) = g(N)\ \ \forall N$ [machine]; $\qquad g_{\varepsilon}(N) \neq 0 \iff 3 \mid N$ [machine on the frozen range];

(4) the boundary-data tuple:

the labels that condition $\mathcal{E}(\mathcal{X})$ are $\big(Q,\ \Theta_{\text{int}}$ class, $S,$ attachment count $N\big)$ [Def-BM.5(1); Def-BM.8] :

the attachment count is a boundary datum,

¬a summed-over structure — a 3-junction and a 2-junction are different supports, not two members of one ensemble;

(5) the scope clause:

no clause of this part sums over sealings of a fixed support as if they were alternatives : the grammar of (3) decides which sealings a support admits at all.

Rmk (why the filter, not a cancellation). The topology-blind path integral is colour-blind by construction — that is what "topology-blind" means here [D2.13's scope note; CT3.22a Rmk]. Colour enters only through $\Pi_{\mathcal{C}}$, a $0/1$ projector applied after the sum. So the question "do δ-sealed states cancel in the baryon's integral?" has a category answer before it has an arithmetic one: they are not terms that could cancel, they are terms the projector never admits.

Req. D2.13, D2.14, D3.9 (Parts 1–2); CT3.17, CT3.17a(2), CT3.18, CT3.22a (Part 2); Def-H.32 (Part 3); FT.19, FT.41(b) (Part 3); Def-BM.5, Def-BM.8 (this part); exhaustive set-partition enumeration over blocks of size 2 and 3, exact integer arithmetic [math; machine: `checks/verify_H_mesonic_flicker_test.py` §A].

Pf.

Definition block: the constitutive clause is (1)'s union inside the colour-blind topology space;

(2) is D2.14's projector re-cited at the site where the union is formed;

(3)'s counts are machine-reproduced from CT3.17a(2)'s frozen list;

(4)–(5) are scope clauses. ■

Phys. Before any sum is taken, one structural fact settles the question. The topology-blind path integral does not filter by colour — it filters by topology, and colour enters afterwards as a zero-one projector on the support. A sealing is therefore not an alternative term competing inside one integral; it is a property of the support that either passes the projector or annihilates the whole colour block. And the grammar that decides it is small enough to enumerate: partition the attachments into groups of two and three, and count.

Theorem 4.26 (BM.31 — The boundary-data comparison and the grammar verdict). Thm.

(1) the grammar at $N = 3$ is forced:

$$g(3) = 1,$$

the unique grouping is the ε-triple,

$$g_\delta(3) = 0$$

[Def-BM.9(3); machine: exhaustive];

a δ sub-pairing on a 3-attachment support consumes two attachments and leaves exactly one dangling $(3 - 2 = 1)$:

$$\partial_{\text{col}} \neq \varnothing \implies \mathcal{C} = 0 \implies \text{the colour block vanishes [D2.14; FT.19];}$$

there is no δ-sealed member of a 3-junction's ensemble to cancel : $\mathcal{E}_\delta(\mathcal{X}_{N=3}) = \varnothing$;

(2) the protection is unique to $N = 3$:

$g_\delta(N) = (1, 0, 3, 10, 15, 105)$ for $N = 2, \ldots, 7$ [machine] :

$$g_\delta(N) > 0 \ \ \forall N \geq 4;$$

$g_\varepsilon(N) \neq 0 \iff 3 \mid N$: $\quad g_\varepsilon(3) = 1, \quad g_\varepsilon(6) = 10, \quad g_\varepsilon(N) = 0$ at $N \in \{2, 4, 5, 7\}$ [machine];

at N

$= 6$ the ensemble genuinely mixes: $\frac{15}{25}$

$= \frac{3}{5}$ of the groupings contain a δ,

$\frac{10}{25}$

$= \frac{2}{5}$ do not [machine];

at $N = 4$ every grouping does $(g_\delta = g = 3)$;

(3) the charge classes overlap, so charge does not separate the sealings:

$$Q_{N=3} \in \{-3, \ldots, +3\},$$

$W_3 = (1, 6, 15, 20, 15, 6, 1), \quad \sum W_3 = 64 = 4^3$ [Def-BM.8(6); machine];

$$Q_{N=2} \in \{-2, \dots, +2\}, \quad W_2 = (1, 4, 6, 4, 1), \quad \sum W_2 = 16 = 4^2 \text{ [machine]};$$

the overlap is $\{-2, -1, 0, +1, +2\}$:

at Q

$= 0$ and Q

$= +1$ the two sealings share a charge class — the neutron and proton classes are not separated from mesonic ones by charge;

only $|Q| = 3$ is baryon-exclusive [machine];

$$W_3(Q)$$

$\neq W_2(Q)$ on the whole overlap [machine] : the probability spaces differ even where the classes coincide;

(4) the phase class separates only partially:

baryon-type $\Theta_{\text{int}} = 2\Theta_{\triangle} = 4k_{\triangle} \in 4\mathbb{Z}$ [Census FT.49(3); FT.22];

per materialized closed chain $\Theta(c) \in 2\mathbb{Z}$ [FT.22];

$\Theta_{\text{int}} = 2k(\Gamma) \in 2\mathbb{Z}$ [FT.34(3); Census FT.35(3)] :

$4\mathbb{Z} \subsetneq 2\mathbb{Z}$ [machine] :

Θ_{int}

$= 2$ is meson-admissible and baryon-excluded, while Θ_{int}

$\in 4\mathbb{Z}$ is admissible to both — the phase label is a partial separator;

(5) the representation content does not separate them either:

$$\dim\big((\mathbb{C}^3)^{\otimes 3}\big)^{\mathfrak{sl}(3)} = 27 - 26 = 1,$$

$$\dim\big(\mathbb{C}^3 \otimes \bar{\mathbb{C}}^3\big)^{\mathfrak{sl}(3)} = 9 - 8 = 1$$

[BM.1(3); machine: exact rank over $\mathbb{Q}$];

both contents carry exactly one singlet line :

the separator is the attachment count N,

carried in the boundary data [Def-BM.9(4)],

¬charge,

¬phase alone,

¬representation content;

(6) the verdict on the extended ensemble:

$$\mathcal{E}_{\text{ext}}(\mathcal{X}_{N=3}) = \mathcal{E}_{\varepsilon}(\mathcal{X}_{N=3}) \sqcup \varnothing = \mathcal{E}_{\varepsilon}(\mathcal{X}_{N=3})$$

[machine: the grammar census] :

the baryon's flicker ensemble is not extended by mesonic members — not because they cancel, but because a 3-junction admits no δ-sealing at all.

Rmk (what would extend it). Clause (2) is the operative scope: at $N \geq 4$ the union is nontrivial and the extended ensemble genuinely mixes sealings. Any composite whose support carries four or more colour attachments — the Census FT.50-type junction gluings [BM.2(3)] — is in that regime, and this tag's protection does not apply to it.

Req. D2.14, D3.9 (Parts 1–2); CT3.17, CT3.17a(2), CT3.18 (Part 2); EM.9 (Part 2); P2 (Part 1); Def-BM.8(6), Def-BM.9 (this part); FT.19, FT.22, FT.34(3), FT.41(b) (Part 3); BM.1(3), BM.2(1), BM.2(3) (this part); Census FT.35(3), FT.49(3) (row-wise); exhaustive set-partition enumeration, exact rational rank over $\mathbb{Q}$, generating-function convolution [math; machine: `checks/verify_H_mesonic_flicker_test.py` §A–§B].

Pf.

(1) : partitions of a 3-set into blocks of size $\in \{2, 3\}$: the only one is $\{1, 2, 3\}$ [machine];

a 2-block leaves $3-2=1$ unpaired $\implies \partial_{\text{col}} \neq \varnothing$ [D2.14] $\implies \mathcal{C}=0 \implies$ block $\equiv 0$ [FT.19]

$(2):$ the censuses $g_\delta,\ g_\varepsilon$ by exhaustive enumeration for $N = 2..7$ [machine];

$g_\varepsilon(N) \neq 0 \implies N = 3m$ [a partition into 3-blocks requires $3 \mid N$];

$\frac{15}{25} = \frac{3}{5},\ \frac{10}{25} = \frac{2}{5}$ [machine]

$(3):$ $(2+z+z^{-1})^3$ and $(2+z+z^{-1})^2$ convolved [Def-BM.8(6); machine];

the overlap and the exclusivity of $|Q| = 3$ [machine: set arithmetic]

$(4):$ Census FT.49(3)'s $4k_\triangle$ row re-cited;

FT.22's $\Theta(c) \in 2\mathbb{Z}$ and FT.34(3)'s additivity re-cited;

$4\mathbb{Z} \subsetneq 2\mathbb{Z}$ with $2 \in 2\mathbb{Z} \setminus 4\mathbb{Z}$ [machine]

$(5):$ the common null space of the 8 traceless generators' actions : $27-26$

$= 1$ and $9-8$

$= 1$ [machine: exact rank over $\mathbb{Q}$];

BM.1(3) re-derived

$(6):$ (1)'s $\mathcal{E}_\delta = \varnothing$ substituted into Def-BM.9(1)'s union. ■

Phys. Three of the four boundary labels fail to tell a baryon from a meson. Their charge classes overlap on everything except $|Q| = 3$, so the neutron's neutral class and the proton's unit class are shared with mesonic configurations; their phase classes differ only in half the cases, since $4\mathbb{Z}$ sits strictly inside $2\mathbb{Z}$; and their colour representation content is identical in the only respect that matters, one singlet line each. What does separate them is the attachment count, and it separates them absolutely: three attachments admit exactly one contraction grouping, the epsilon triple, and a delta sub-pairing would leave one attachment dangling, which the all-or-nothing law turns into a vanished colour block. So the baryon's flicker ensemble has no mesonic members to cancel. That protection is not general — it is a property of the number three, and from four attachments upward the groupings genuinely mix, fifteen of twenty-five at six.

Theorem 4.27 (BM.32 — The mesonic contribution: zero by filtration, and the amplitude layer is orthogonal). Thm.

(1) the contribution vanishes, and the mechanism is the projector:

$$Z_\delta^{(n)} := \sum_{R\in\mathcal{E}_\delta(\mathcal{X}_{N=3})} \sum_\sigma w(R,\sigma) = \sum_{R\in\varnothing}(\cdots) = 0$$

[BM.31(1), (6); machine];

$$Z_{\text{ext}}^{(n)} = Z_\varepsilon^{(n)} + Z_\delta^{(n)} = Z_\varepsilon^{(n)} \qquad \text{[machine]};$$

the zero is a filtration, not a cancellation: ¬∃ a pair of mesonic terms whose weights sum to zero — ¬∃ mesonic terms;

(2) the projector form of the same statement:

$$\mathcal{C}(\Sigma;C)\in\{0,1\},\quad \mathcal{C}^2=\mathcal{C}\ \text{[D2.14; machine]}:$$

$\Pi_{\mathcal{C}}$ is an idempotent scalar on the colour slot space;

a δ-sealed member of a 3-support has $\mathcal{C}=0 \implies \Pi_{\mathcal{C}}=0 \implies F_{\text{strong}}^{\Sigma}=0$ [FT.19] :

its weight is multiplied by zero before any sum is formed;

(3) both mass routes are untouched:

adding zero-weight members leaves a normalized average invariant $(\frac{\sum wm}{\sum w})$ and a total invariant $(\sum wm)$

[machine: exact on a rational witness ensemble] :

$$\langle m\rangle_{\text{ext}} = \langle m\rangle_\varepsilon \text{ on route (i)}, \qquad T_{\text{ext}} = T_\varepsilon \text{ on route (ii)} \qquad \text{[BM.28(4); BM.20(3); machine]};$$

$\therefore$ BM.20's curvature-flux mass and BM.28's three ratios are unchanged by the extension;

(4) the amplitude-layer mechanisms are orthogonal to this question:

FT.67(3a) and FT.106(2) kill pure directed cycles (any length, any cell);

FT.71 cancels every odd cycle and every undissolved even cycle;

FT.69(3)(iv), (v) exclude odd cycles and 4-cycles from covers :

these are statements about cycles in Γ, ¬about colour sealings of a support [Def-BM.9(2) Rmk];

they neither produce nor require $Z_\delta = 0$;

(5) where mesonic content does survive:

$$Z^{\text{amp}}(q, 1) = 802 \ (q = \pm 1),\ 4796 \ (q = 0) :$$

dissolved digons inside covers [FT.106(3); machine];

$$Z^{\text{amp}}(q, S) = 0 \ \ \forall q,\ S \in \{2, 3\} \ [\text{FT.106(3c); machine}] :$$

a standalone digon — a pure directed cycle — dies at the amplitude layer;

$$Z^{\text{amp}}(q, S) = Z^{\text{amp}}(-q, S) \ [\text{FT.106(3d); machine}];$$

$$Z^{\text{norm}}(0, S)$$

$= 2Z^{\text{norm}}(\pm 1, S)$ at every S [machine] : the norm layer keeps its 2:1 ratio where the amplitude layer dies;

(6) the two readings reconciled:

a meson as a sealed composite is excluded from a 3-junction's ensemble by the grammar [(1)];

mesonic chain content enters the amplitude layer only as dissolved digons inside covers [(5)] :

the two statements concern different objects — a support's sealing and a cover's cycle structure — and neither contradicts the other.

Rmk (no protection claim beyond N = 3). Clause (3)'s invariance is exact because the added weight is exactly zero. At $N \geq 4$ [BM.31(2)] the added weight is nonzero, the average genuinely moves, and no clause of this tag applies.

Req. D2.14 (Parts 1–2); CT2.5, CT2.8, CT3.18 (Part 2); Def-H.32, FT.19, FT.67(2), FT.67(3), FT.69(3), FT.71, FT.106(2), FT.106(3) (Part 3); Def-BM.9, BM.20(3), BM.28(4), BM.31(1), BM.31(2), BM.31(6) (this part); exact rational ensemble algebra, exact integer census arithmetic [math; machine: `checks/verify_H_mesonic_flicker_test.py` §C–§D].

Pf.

$$(1):\ \mathcal{E}_\delta = \varnothing\ [\text{BM.31(1)}] \implies \text{the sum is empty} \implies 0\ [\text{machine}]; \qquad Z_{\text{ext}} = Z_\varepsilon + 0\ [\text{machine}]$$

$$(2):\ \text{D2.14's } \mathcal{C} \text{ definition re-cited;}$$

$$\mathcal{C}^2 = \mathcal{C} \text{ at } \mathcal{C} \in \{0, 1\}\ [\text{machine}];$$

FT.19's all-or-nothing row re-cited

$$(3):\ \text{adding terms with } w$$

$= 0$ changes neither $\sum wm$ nor $\sum w \implies$ neither the ratio nor the total [machine: exact Fractions on a 5-member witness ensemble]

$$(4):\ \text{FT.67(3a)},\ \text{FT.71},\ \text{FT.69(3)} \text{ rows re-cited;}$$

their objects are cycles of Γ [Def-H.32(1)'s topology space],

while $\mathcal{C}$ acts on colour attachments [D2.14]

$(5):$ FT.106(3)'s census integers re-cited and their relations verified exactly [machine : $802,\ 4796,$ the zeros at $S \in \{2, 3\},$ the ω-symmetry, the 2:1 norm ratio at all four windings]

$$(6):\ (1) \text{ and } (5) \text{ conjoined with Def-BM.9(2)'s locus distinction. } \blacksquare$$

Phys. The answer to the test is zero, and the reason is more interesting than the number. Nothing cancels: there is nothing to cancel. A three-attachment support admits exactly one contraction, so the mesonic subset of its ensemble is empty before any weight is written down, and the colour projector would have annihilated those terms individually anyway — one dangling attachment and the whole block goes. Because the added weight is exactly zero, both mass functionals stay put: the normalized average and the total deficit are each invariant, so the curvature-flux mass and the four-thirds ratio of the previous tier survive the extension untouched. The amplitude-layer cancellations the corpus already proved are about a different object entirely — directed cycles in the skeleton — and they cut the other way: a standalone digon dies there, while dissolved digons inside covers are precisely what survives, with exact census weights. Mesons as sealed composites and mesonic chain content in covers are two different things, and the tier keeps them apart.

BM.33 — The status of the mesonic flicker test. Structural-constraints record: see Appendix B, Section B.1.19 of the Addendum.

4.12 Tier BM.10 — The Kinetic Energy from the Metric and the Vacuum-Balanced Entropy: the Dispersion Read of a Sealed Composite, the Deficit Reframing of the Rest Mass, the Force-Structure Ladder, and the Route-Artifact Adjudication

Definition 4.10 (Def-BM.10 — The metric-energy read of a sealed composite (the dispersion route)). Def. A read constructed on the chart side of a sealed composite from CT1.12's dictionary; no new axiom, no new datum, no new parameter. Local notation (internal bookkeeping): $\gamma_{ij} := g_{ij}$ (the spatial block), $\gamma^{ij} :=$ the inverse matrix of the 3×3 block ($\gamma^{ik} g_{kj} = \delta^i{}_j$), $g^{\mu\nu} :=$ the inverse of the full 4×4 metric.

(1) locus and hypotheses:

the read lives on the emergent chart M [CT2.6],

inheriting hypotheses (i)–(iii) at the point of use [A §0.2a];

the 16-dimensional pair register carries no canonical pair [BM.10(1)];

the register-side precursors are the discrete Weyl pairs [BM.10(2)] and the width law [BM.10(6)];

(2) the metric data and the derived ADM split:

for the coarse-grained sector coefficients (a_{00}, q_{ij}, w_i) [D2.1] :

$$\Omega = e^{\alpha_\Omega a_{00}}, \qquad g_{00} = -\Omega^2, \qquad g_{0i} = \Omega^2 \varkappa \beta w_i, \qquad g_{ij} = \Omega^2(\delta_{ij} + \varkappa q_{ij}) \qquad \text{[CT1.12; FT.157(1)]};$$

$$\text{lapse:} \quad N^2 := \Omega^2 + \gamma^{ij} N_i N_j = -\frac{\det g}{\det \gamma} > 0;$$

$$\text{shift covector:} \quad N_i := g_{0i} = \Omega^2 \varkappa \beta w_i;$$

$$\text{shift vector:} \quad N^i := \gamma^{ij} N_j;$$

the line element (an exact identity):

$$ds^2 = -N^2 d\tau^2 + g_{ij}\big(dx^i + N^i d\tau\big)\big(dx^j + N^j d\tau\big) = g_{\mu\nu} dx^\mu dx^\nu$$

[machine: Schur, denominators cleared];

(3) the momenta:

the unit-mass coordinate-time read: $\pi_i := g_{ij}\Big(\frac{dx^j}{dx^0} + N^j\Big)$

[$x^0 = \tau_n$ the chart time; D1.8; FT.159];

the physical spatial momentum: $p_i := m\,u^0\,\pi_i, \qquad u^\mu := \frac{dx^\mu}{d\lambda}, \qquad g_{\mu\nu}u^\mu u^\nu = -1;$

the shift contribution is $g_{ij}N^j = N_i = g_{0i}$ [machine: exact polynomial identity] :

the metric's time–space block enters the momentum exactly [FT.157(1)];

(4) the rest input:

$$m := m_{\text{hid}} = \frac{E_{\text{int}}\ln 2}{\mathfrak{h}}$$

[FT.44(4) (Census)];

the ensemble route: $m \;\mapsto\; \langle m\rangle$ [BM.20(3)];

the vacuum-deficit reframing of the rest read is BM.35 — a marked forward pointer at the citing site, imported in no Req line;

(5) the mass shell and the energy: $g^{\mu\nu}p_\mu p_\nu = -m^2$ [signature $(-,+,+,+)$; CT1.12];

$E := -p_0$ is BM.8(2)'s conditional energy read under its declared stationarity scope (theorem-local, inherited at the point of use);

the Eulerian form:

$$E = N\sqrt{m^2 + \gamma^{ij}p_ip_j} - N^ip_i;$$

the kinetic-energy read:

$$\mathrm{KE} := E - N\,m \;\geq 0 \;\text{ where } g_{ij} \succ 0 \text{ [CT1.12; machine]};$$

(6) status and scope:

constructed — an abbreviation of derived objects: (2)'s split is linear algebra on CT1.12's dictionary;

(3)–(5) are pointwise identities [machine];

¬a new parameter [§Cal-BM stays empty];

¬a declaration;

the names (lapse, shift, kinetic energy) are internal bookkeeping [A §0.2];

¬constructed: the back-reaction of the composite's motion on its own sector coefficients [BM.15's read is the rest-frame one] and any inter-sector dynamics [BM.29(5) stands].

Rmk (the unit-mass normalization). The leading kinetic read is $\mathrm{KE} = \frac{1}{2} N\,\gamma^{ij} p_i p_j / m + O(p^4)$: the factor $1/m$ is dimensionally forced, since $[\gamma^{ij} p_i p_j] = \mathfrak{q}^{+1} = \text{energy}^2$ while $[\mathrm{KE}] = \mathfrak{q}^{+1/2} = \text{energy}$ [CT2.1; machine: exponent arithmetic]. The unit-mass unit-lapse normalization ($m = 1$, $N = 1$, $w_i = 0$) gives $\mathrm{KE} = \frac{1}{2}\gamma^{ij} p_i p_j$. And $\gamma^{ij} \neq g^{ij}$ where the shift is nonzero: $g^{ij} = \gamma^{ij} - N^i N^j / N^2$ [machine] — the contraction that inverts the spatial momentum is γ^{ij}; the two coincide exactly in the shift-free gauge.

Req. D1.8, D2.1, D2.9 (Parts 1–2); CT1.12, CT2.1, CT2.6 (Part 2); P1, P5 (Parts 1–2); FT.157(1), FT.159 (Part 3); FT.44(4) (Census); BM.8(2), BM.10(1), BM.10(2), BM.10(6), BM.20(3) (this part); A §0.2a (the continuum-limit hypotheses at the point of use); Schur decomposition, adjugate calculus [math; machine: `checks/verify_H_kinetic_energy_metric.py` §A].

Pf.

Definition block: the constitutive clauses are the displayed identities of (2)–(5) [machine: exact polynomial identities over $\mathbb{Q}[\Omega^2, \varkappa, \beta, w_i, q_{ij}, p_\mu, m]$ with denominators cleared by $\det\gamma$];

¬∃ a non-derived clause: the ADM split is forced by the Schur decomposition of $g_{\mu\nu}$ [math];

(1), (6) are scope records. ■

Phys. The energy of a moving sealed composite is read off the metric the corpus already derived, not added to it. The dictionary's ten sector coefficients split into lapse, shift, and spatial metric by linear algebra alone — the Schur decomposition of the very matrix CT1.12 builds — and the same three shift coefficients that the weak block carries enter the spatial momentum as its drag term. The

mass-shell constraint of the Lorentzian metric converts these data into one exact dispersion identity, whose rest input is the informational mass the calibration tiers locked. Two normalizations — unit mass and coordinate time — are fixed as bookkeeping, and the read is chart-side: the register itself carries no canonical pair, and the definition keeps the obstruction exactly where Tier BM.4 placed it.

Theorem 4.28 (BM.34 — The kinetic energy from the emergent metric: the ADM split, the mass shell, the dispersion relation, and the zero-point bound). Thm.

(1) the split is derived:

$$\gamma_{ij} = \Omega^2(\delta_{ij} + \varkappa q_{ij}), \quad N_i = \Omega^2 \varkappa \beta w_i, \quad N^i = \gamma^{ij} N_j, \quad N^2 = \Omega^2 + \gamma^{ij} N_i N_j;$$

$\boxed{N^2 \det\gamma = -\det g}$ [Schur; machine: exact polynomial identity over $\mathbb{Q}[\Omega^2, \varkappa, \beta, w_i, q_{ij}]$];

the inverse metric in ADM form:

$$g^{00} = -\frac{1}{N^2},$$

$$g^{0i} = \frac{N^i}{N^2},$$

$$g^{ij} = \gamma^{ij} - \frac{N^i N^j}{N^2}$$

[machine: cleared adjugate identities + 3 rational witnesses, exact Fractions];

(2) the line element and the vacuum:

$$ds^2 = -N^2 d\tau^2 + g_{ij}\big(dx^i + N^i d\tau\big)\big(dx^j + N^j d\tau\big) \text{ identically [machine]};$$

$$(a_{00}, q_{ij}, w_i) = 0 \implies (\Omega, N, N^i, g_{ij}) = (1, 1, 0, \delta_{ij}) :$$

the vacuum's emergent metric is the flat metric [CT1.12; machine: exact substitution];

(3) the momentum and its carrier:

$$p_i = m\, u^0\, g_{ij}\Big(\frac{dx^j}{dx^0} + N^j\Big) \text{ [Def-BM.10(3)]};$$

$$g_{ij}N^j = N_i = g_{0i} = \Omega^2 \varkappa \beta w_i \text{ [machine]};$$

the drag term and the metric's time–space block are the same three coefficients w_i — the weak aspect, one block, two derived reads [FT.157(1); CT1.12a(2)];

the T-parity of the carrier:

$$w \mapsto -w \implies N_i \mapsto -N_i,$$

$$N^i \mapsto -N^i,$$

(N, γ_{ij}) fixed [FT.156(3); FT.159(3); machine] :

p_i is T-odd, matching its carrier [BM.8(1)];

(4) the mass shell $\iff$ the dispersion relation:

$$g^{\mu\nu}p_\mu p_\nu = -m^2 \iff \big(p_0 - N^i p_i\big)^2 = N^2\big(m^2 + \gamma^{ij}p_i p_j\big)$$

[machine: the master polynomial identity, cleared by $(\det\gamma)^2$];

$E := -p_0$, the positive-energy branch :

$$\boxed{E = N\sqrt{m^2 + p^2} - N^i p_i},$$

$$p^2 := \gamma^{ij}p_i p_j;$$

the gauges:

$$w = 0 : \quad E^2 = N^2(m^2 + p^2);$$

$$\Omega = 1 \wedge w = 0 : \quad \boxed{E^2 = m^2 + p^2}$$

[$c = 1$, D2.9];

the dispersion is T-even: the mass shell is invariant under the simultaneous (N^i, p_i)

$\mapsto (-N^i, -p_i)$ [machine: substitution in the master identity];

the signature is invariant [FT.159(3) re-cited];

(5) the kinetic energy and the non-relativistic read:

$$\mathrm{KE} := E - Nm = N\big(\sqrt{m^2 + p^2} - m\big) \ \geq 0,$$

$$\mathrm{KE} = 0 \iff p_i = 0;$$

on the perturbative regime $g_{ij} \succ 0 \iff \forall a:\ 1 + \varkappa\lambda_a(q) > 0$

[CT1.12; machine: exact rational Sylvester/LDL$^\top$ witnesses];

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \frac{5x^4}{128} + \frac{7x^5}{256} + O(x^6)$$

[machine: exact rational coefficients; $(\cdot)^2 \equiv 1 + x \mod x^6$];

$$\boxed{\mathrm{KE} = N\Big(\frac{p^2}{2m} - \frac{p^4}{8m^3} + \frac{p^6}{16m^5}\Big) + O(p^8)}, \qquad \text{the leading read:} \quad \mathrm{KE} = \frac{N\,\gamma^{ij}p_ip_j}{2m};$$

the unit normalization $(m{=}1, N{=}1, w{=}0):$ $\mathrm{KE} = \frac{1}{2}\gamma^{ij}p_ip_j$ [Def-BM.10 Rmk];

$[\mathrm{KE}] = \mathfrak{q}^{+1/2}, \quad [\gamma^{ij}p_ip_j] = \mathfrak{q}^{+1}$ [CT2.1; machine: exponent arithmetic];

(6) the zero-point bound:

$\Delta x\,\Delta p \geq \frac{1}{2}$ [BM.10(3), (6); $\hbar = 1$];

$$\Delta x = \Delta x_{\mathrm{B}}: \quad \Delta p \geq \frac{1}{2\Delta x_{\mathrm{B}}} = \frac{m_{\mathrm{hid}}}{2}\ [\text{BM.13(4)}];$$

the isotropic cube convention [Def-BM.4(3)], the rest read $\langle p_i\rangle = 0$:

$$\langle p^2\rangle \ \geq\ 3\Big(\frac{m_{\mathrm{hid}}}{2}\Big)^2 = \frac{3}{4}\,m_{\mathrm{hid}}^2$$

[machine: exact Fractions];

in the vacuum gauge of (4) $\left(E^2 = m^2 + \langle p^2 \rangle\right)$, at $m = m_{\text{hid}}$: $\boxed{E^2 \;\geq\; \frac{7}{4}\, m_{\text{hid}}^2}$, $\text{KE}_{\text{NR}} \geq \frac{\langle p^2 \rangle}{2m} \;\geq\; \frac{3}{8}\, m_{\text{hid}}$, $E \;\geq\; \frac{\sqrt{7}}{2}\, m_{\text{hid}}$ exact in $\mathbb{Q}(\sqrt{7})$

[machine: the quadratic ring, minimal polynomial $x^2 - 7,$ comparison by squaring];

$m_{\text{hid}} = E_{\text{int}} \ln 2/\mathfrak{h}$ substituted :

$E^2 \geq \frac{7}{4} \frac{(E_{\text{int}} \ln 2)^2}{\mathfrak{h}^2}$ [machine: one Laurent monomial, coefficient $\frac{7}{4}$];

(7) the rest slice and the dust closure:

$p_i = 0 : \quad E = Nm;$

the vacuum gauge: $E = m = m_{\text{hid}} = \frac{E_{\text{int}} \ln 2}{\mathfrak{h}}$ [FT.44(4)];

$RV = \varkappa^2 m$ [BM.20(1)],

$\rho = \frac{m}{V}, \quad \kappa^2 \rho = \frac{2\pi\, \mathfrak{h}\, E_{\text{int}}}{V}$ [BM.9(5); BM.15(2)] :

the informational, the curvature-flux and the dispersion rest reads agree on-shell [machine: Laurent identities];

KE is the p

$\neq 0$ completion the rest-frame dust read does not carry [BM.9(4) : $T_{\mu\nu} = \text{diag}(\rho, 0, 0, 0)$ at $p = 0$] :

the kinetic term is the whole composite's transport on the chart [BM.8(1)].

Rmk (chart-side discipline). The reads of (4)–(6) are continuum-chart objects under CT2.6's hypotheses at the point of use; no canonical pair exists in the finite register [BM.10(1)] and no clause imports one. The pointwise mass-shell identity needs no stationarity; the conserved-charge read $E = -p_0$ inherits BM.8(2)'s theorem-local stationarity hypothesis at the point of use. The bound of (6) applies the Robertson inequality chart-side, at the width law's locus — Def-BM.4(5)'s envelope discipline: the zero-point read is a coarse-graining envelope, never a ledger distance.

Req. A3 (Part 1); D1.8, D2.1, D2.9 (Parts 1–2); CT1.12, CT1.12a(2), CT2.1, CT2.6 (Part 2); P1, P5 (Parts 1–2); FT.44(4) (Census); FT.156(3), FT.157(1), FT.159 (Part 3); Def-BM.4, Def-BM.10, BM.8(1), BM.8(2), BM.9(4), BM.9(5), BM.10(1), BM.10(2), BM.10(3), BM.10(6), BM.13(4), BM.15(2), BM.20(1), BM.20(3) (this part); A §0.2a (the hypotheses at the point of use); Schur

decomposition, adjugate calculus, the Sylvester criterion, exact rational LDL$^\top$ inertia, the binomial series over $\mathbb{Q}$, the quadratic ring $\mathbb{Q}(\sqrt{7})$, Laurent monomials [math; machine: `checks/verify_H_kinetic_energy_metric.py` §A].

Pf.

$$(1):\ \det g = \det\gamma\cdot\left(g_{00} - g_{0i}\gamma^{ij}g_{0j}\right)\ \text{[Schur]};$$

$$g_{00} = -\Omega^2,\ g_{0i} = N_i \implies -\det g = \left(\Omega^2 + \gamma^{ij}N_iN_j\right)\det\gamma = N^2\det\gamma\ \text{[machine]}$$

the inverse rows in cleared adjugate form $(\operatorname{adj}\gamma := \det\gamma\cdot\gamma^{-1},\ D := \det\gamma)$:

$$(\operatorname{adj} g)_{00} = D;$$

$$(\operatorname{adj} g)_{0i} = -\left((\operatorname{adj}\gamma)^{ij}N_j\right);$$

$$D$$

$$(\operatorname{adj} g)_{ij}$$

$$= (\det g)$$

$$(\operatorname{adj}\gamma)^{ij} + \left((\operatorname{adj}\gamma)^{ik}N_k\right)\left((\operatorname{adj}\gamma)^{j\ell}N_\ell\right)\ \text{[machine: three exact polynomial identities]}$$

the three rational witnesses ($\Omega^2, \varkappa, \beta, w_i, q_{ij}$ rational, q off-diagonal) : every row as exact Fraction equality, including $g^{\mu\alpha}g_{\alpha\nu}$

$$= \delta^\mu_\nu \text{ and } \det g \qquad = -N^2\det\gamma$$

$$(2):\ \text{expansion in } (d\tau, dx^i):\quad g_{00} = -N^2 + N_iN^i,\quad g_{0i} = \gamma_{ij}N^j = N_i,\quad g_{ij} = \gamma_{ij}\ \text{[machine]};$$

$$\Omega = e^{\alpha_\Omega\cdot 0} = 1,\ q = 0,\ w = 0 \implies \text{the flat metric [machine: exact substitution]}$$

$$(3):\ \gamma_{ij}N^j = N_i\ \text{[machine: cleared by } \det\gamma];$$

$$w$$

$$\mapsto -w : \ N_i$$

$$= \Omega^2 \varkappa \beta w_i$$

$\mapsto -N_i$ and (N, γ_{ij}) carry w only through $N_i N_j$ and N_i [machine: N^2, γ_{ij} fixed]

$$(4) : \ g^{\mu\nu} p_\mu p_\nu$$

$$= -\frac{1}{N^2}\big(p_0 - N^i p_i\big)^2 + \gamma^{ij} p_i p_j \text{ [(1)'s inverse rows]} \implies \big(p_0 - N^i p_i\big)^2$$

$$= N^2\big(m^2 + \gamma^{ij} p_i p_j\big) \text{ on shell [machine: cleared by } (\det \gamma)^2]$$

the branch: at $p_i = 0, \ w = 0 : \ p_0^2 = N^2 m^2 \implies p_0 = \mp N m;$

$E = -p_0 = +Nm$ on the positive branch [machine: rational witnesses];

the gauges: substitution [machine];

T-evenness: $(N^i, p_i) \mapsto (-N^i, -p_i)$ fixes $(p_0 - N^i p_i)^2$ and $\gamma^{ij} p_i p_j$ [machine]

$$(5) : \ \text{KE} = N\big(\sqrt{m^2 + p^2} - m\big) = Nm\big(\sqrt{1 + p^2/m^2} - 1\big) \text{ [exact]};$$

the series $a = \big(1, \frac{1}{2}, -\frac{1}{8}, \frac{1}{16}, -\frac{5}{128}, \frac{7}{256}\big)$:

$$(a * a)_n = \delta_{n0} + \delta_{n1} \ \forall n \leq 5 \text{ [machine: Fractions]}$$

positivity: $g_{ij} \succ 0$ [CT1.12] $\implies \gamma^{ij} \succ 0$

$$\implies p^2$$

$$\geq 0 \text{ with equality iff } p_i$$

$$= 0 \text{ [machine: LDL}^\top \text{ on exact rational witnesses]};$$

$$[\text{KE}] = [\gamma^{ij} p_i p_j] - [m] = \mathfrak{q}^{1-1/2} = \mathfrak{q}^{+1/2} \text{ [CT2.1; machine]}$$

$$(6):\ \Delta p \geq \tfrac{1}{2\Delta x_{\mathrm{B}}} = \tfrac{m_{\mathrm{hid}}}{2}\ \text{[BM.13(4) re-cited]};$$

$$\langle p^2\rangle = \sum_i \big((\Delta p_i)^2 + \langle p_i\rangle^2\big) = \sum_i (\Delta p_i)^2 \geq 3\big(\tfrac{m_{\mathrm{hid}}}{2}\big)^2 \text{ at } \langle p_i\rangle = 0\ \text{[machine]}$$

$$E^2 = m^2 + \langle p^2\rangle \text{ in the vacuum gauge of } (4): \quad E^2 \geq \big(1+\tfrac{3}{4}\big)m_{\mathrm{hid}}^2 \text{ at } m = m_{\mathrm{hid}}\ \text{[machine]};$$

$$\mathrm{KE}_{\mathrm{NR}} = \tfrac{\langle p^2\rangle}{2m} \geq \tfrac{3/4}{2}\, m_{\mathrm{hid}} = \tfrac{3}{8} m_{\mathrm{hid}}\ \text{[machine]};$$

$$\big(\tfrac{\sqrt{7}}{2}\big)^2 = \tfrac{7}{4},\ x^2 - 7 \text{ irreducible over } \mathbb{Q}\ \text{[machine: the quadratic ring]};$$

$$E^2 \geq \tfrac{7}{4} m_{\mathrm{hid}}^2 \wedge E > 0 \implies E \geq \tfrac{\sqrt{7}}{2} m_{\mathrm{hid}}\ \text{[squaring comparison]}$$

$$(7):\ p = 0 \text{ in } (4):\ E^2 = N^2 m^2 \implies E = Nm\ \text{[the positive branch]};$$

$$RV = \varkappa^2 m \text{ and } \rho V = m:\ \text{the same monomial [machine]};$$

$$\kappa^2 \rho$$

$$= \tfrac{2\pi\hbar^2}{\ln 2} \cdot \tfrac{E_{\mathrm{int}} \ln 2}{\hbar V}$$

$$= \tfrac{2\pi\hbar E_{\mathrm{int}}}{V}\ \text{[BM.9(5) re-verified; machine: denominators cleared]}.\ \blacksquare$$

Phys. Motion enters the corpus through the metric the corpus already built. The dictionary's ten coefficients split into lapse, shift, and spatial metric by the Schur complement alone, and the mass-shell constraint of the Lorentzian metric turns that split into one exact identity: the energy conjugate to ledger time, corrected by the shift's drag, is the lapse times the square root of the rest mass squared plus the spatial momentum squared. In the vacuum gauge that is precisely the dispersion relation, and its expansion gives the kinetic energy with rational coefficients no one fitted — one half, minus one eighth, plus one sixteenth — nonnegative, vanishing only at zero momentum, wherever the dictionary's perturbative positivity holds. Two exact consequences deserve names. The width law converts the uncertainty bound at the Born width into a zero-point bound: a sealed composite spread over its own width carries at least seven quarters of its rest mass squared in energy squared — three eighths of the rest mass in the non-relativistic kinetic read, $\sqrt{7}/2$ times it in the exact relativistic one — every constant an exact rational over the ledger count. And the rest slice closes a triangle the corpus drew twice before: at zero momentum, the dispersion energy, the informational mass, and the curvature flux are one and the same monomial.

Theorem 4.29 (BM.35 — The vacuum-balanced entropy: the rest mass as the deficit against the all-virtual vacuum). Thm.

(1) the vacuum is the unique entropy maximum:

$$n_{\mathcal{R}} = 0 \implies \rho_{pq} = \rho_\circ = \tfrac{1}{4} I \ \ \forall (p,q),$$

$$S_{\text{raw}}(\rho_\circ) = \ln 4 = \max_{\rho} S_{\text{raw}}(\rho),$$

$$\arg\max = \{\tfrac{1}{4} I\}$$

[CT0.2];

the machine form:

$$\lambda = \left(\tfrac{k_1}{8}, \ldots, \tfrac{k_4}{8}\right),\ \textstyle\sum_i k_i = 8 :$$

$$S(\lambda) \le \ln 4 \iff \prod_i k_i^{k_i} \ge 256 \quad (0^0 := 1)$$

[integer comparison; exp strictly monotone];

the exhaustive census of all $\binom{11}{3}$

$= 165$ compositions: the inequality holds at every one, with equality only at $(2,2,2,2)$ [machine] :

maximality and uniqueness without one transcendental evaluated;

(2) the vacuum is flat:

$$(a_{00}, q_{ij}, w_i) = 0 \implies (N, N^i, g_{ij}) = (1, 0, \delta_{ij})$$ [BM.34(2)];

$\Theta_R = 0$ on geodesic polygons in flat slices [FT.47(1)];

the vacuum is entropy-maximal and defect-free — the two reference points against which every deficit read is read;

(3) the ledger deficit and the balance identity:

one real edge $= 1$ bit $= \ln 2$ nats [C9; CT0.9; CT1.29] :

$$S_{\mathfrak{L}} = n_{\mathcal{R}} \ln 2;$$

the vacuum: $n_{\mathcal{R}} = 0 \implies S_{\mathfrak{L}}(\text{vac}) = 0$ [CT0.2; D0.1];

$$\Delta S_{\mathfrak{L}} := S_{\mathfrak{L}} - S_{\mathfrak{L}}(\text{vac}) = E_{\text{int}} \ln 2 = I_{\text{hid}} \qquad [\text{FT.42}(1);\ \text{FT.44}(4);\ \text{machine: exact in } \mathbb{Q} \cdot \ln 2];$$

the balance identity:

$$\boxed{\ln 4 = 2 \ln 2}$$

[machine] :

each bidirectionally crystallized channel (2 directed edges [D0.8]) books ledger entropy $2 \ln 2$

$$= \ln 4$$

= the vacuum pair's own entropy;

$$E_{\text{int}}$$

$= 2E_{\text{bidir}}$ [the bidirectional composites: the digon [FT.28], the polygon ladder [FT.45(1)], the Platonic floors [FT.42(1)], the neutron row [BM.17]] :

$$\boxed{\Delta S_{\mathfrak{L}} = E_{\text{bidir}} \ln 4 = E_{\text{bidir}}\, S_{\text{raw}}(\rho_\circ)}$$

[machine: exact in $\mathbb{Q} \cdot \ln 2$] :

crystallization re-books the vacuum's pair entropy into the ledger in full — the balance is exact;

(4) the reframing identity:

$$m_{\text{rest}} := \frac{\Delta S_{\mathfrak{L}}}{\mathfrak{h}} = \frac{E_{\text{int}} \ln 2}{\mathfrak{h}} = m_{\text{hid}}$$

[FT.44(4); machine: Laurent identity];

$$[\Delta S_{\mathfrak{L}}] = \mathfrak{q}^0, \quad [\mathfrak{h}] = \mathfrak{q}^{-1/2} \implies [m_{\text{rest}}] = \mathfrak{q}^{+1/2} = [\text{energy}]$$

[CT2.1; D2.9; machine: exact exponent arithmetic];

the substance clause:

the rest mass is the deficit read of ledger data against the vacuum — ¬an independent entity: the ontology adds nothing;

the rest energy: $E_{\text{rest}} = m_{\text{rest}}c^2$ with $c = 1$ [D2.9] :

the c^2 is bookkeeping restoration only [machine: the c-exponent is 0];

(5) the geometric member of the deficit family:

$$m^2_{\text{pole}} = 4\hat{\mu}'^2 \sin^2\Big(\frac{\Theta_R}{4}\Big),$$

$$\Theta_R = \textstyle\sum_i \alpha_i - (n-2)\pi$$

[Def-H.37(2); FT.54(4); FT.47(1)] :

the pole mass is the angle-deficit read against (2)'s flat vacuum;

the range $[0, 4\hat{\mu}'^2]$ [FT.76(3)];

the exact witness $2 - \sqrt{2} \in \mathbb{Q}(\sqrt{2})$ [FT.54(5)];

¬∃ a theorem equating the bit currency ($\ln 2$ per directed edge) and the angle currency (Θ_R) :

the relating equation is application-layer content;

(6) the ensemble refinement and the third currency:

$$\langle m \rangle = \sum_R w(R)\, m(R) \text{ [BM.20(3)]};$$

$$\Delta S(\Gamma, R) = \tfrac{3}{2}\big(2E - |R|\big) \text{ nats [FT.107(2); L3]};$$

the three deficit functionals are pairwise distinct:

$$\text{the digon skeleton } (2E = 2): \quad |R| = 2: \ \ \Delta S_{\mathfrak{L}} = 2\ln 2 \ \text{ vs } \ \Delta S = 0;$$

$$R = \varnothing: \ \ \Delta S_{\mathfrak{L}} = 0 \ \text{ vs } \ \Delta S = 3$$

[machine];

$2\ln 2 \neq 3$ by the exact enclosure $\tfrac{1}{2} < \ln 2 < 1$ [machine: $\ln 2 = \sum_{n\geq 1} \frac{1}{n2^n}$, tail $< 2^{-20}$];

BM.20 Rmk's agreement row stands untouched;

(7) the total energy assembly:

$E^2_{\text{total}} = m^2_{\text{rest}} + p^2$ [BM.34(4) vacuum gauge];

$E_{\text{total}} = N\sqrt{m^2_{\text{rest}} + p^2} - N^i p_i$ [Def-BM.10(5)];

NR:

$$\boxed{E_{\text{total}} = m_{\text{rest}} + \text{KE} = \frac{\Delta S_{\mathfrak{L}}}{\mathfrak{h}} + \frac{N\,\gamma^{ij} p_i p_j}{2m} + O(p^4)}$$

[machine: exact series assembly];

(8) status and scope:

the reframing adds no content: it is an identity of derived reads $((3), (4))$ plus a name [A §0.2 bookkeeping];

¬a declaration, ¬a P4 member, §Cal-BM stays empty;

(3)'s balance identity holds exactly on the bidirectionally crystallized composites;

a unidirectional edge books $\ln 2$ against the pair freedom $\ln 4$ (half-balance) :

the general directed form is (4)'s m_{hid} identity [FT.44(4)].

Rmk (three currencies, one reference point). The ledger deficit counts the crystallized bits — $\ln 2$ per directed edge; the branch-state deficit counts the uncrystallized channels at $\frac{3}{2}$ nats each [FT.107(2)]; the angle deficit counts the holonomy against flatness [FT.54(4)]. All three are reads of structure against the same vacuum — the all-virtual entropy maximum, which is also the flat metric — and no theorem of Parts 1–4 converts one currency into another: item 7 holds the relating-equation question, and this tier does not touch it. The balance identity is the sharpest single row the corpus can state about why the rest read deserves the entropy name: on a bidirectional crystallization the ledger books exactly the entropy the vacuum pair carried, so the mass counts re-booked pair freedom, not destroyed information — $\ln 4$ per pair, $2\ln 2$ per channel, $E_{\text{int}} \ln 2$ in total, one $\mathfrak{h}$ away from an energy.

Req. C9, D0.1, D0.8, D2.9 (Parts 1–2); CT0.2, CT0.9 (Part 1); CT1.29, CT2.1 (Part 2); P1, P5 (Parts 1–2); Def-H.37(2), FT.47(1), FT.54(4), FT.54(5), FT.107(2), L3 (Part 3); Census FT.28, FT.42(1), FT.44(4), FT.45(1), FT.76(3) (row-wise); Def-BM.4, Def-BM.5, Def-BM.10, BM.17, BM.20(3), BM.34(2), BM.34(4) (this part); exact $\mathbb{Q} \cdot \ln 2$ arithmetic, the exhaustive integer-grid Gibbs census, exact rational $\ln 2$ enclosures, Laurent monomials [math; machine: `checks/verify_H_kinetic_energy_metric.py` §B].

Pf.

$$(1) : \; S(\lambda) = -\sum_i \tfrac{k_i}{8} \ln \tfrac{k_i}{8} = \ln 8 - \tfrac{1}{8} \sum_i k_i \ln k_i;$$

$$S \le \ln 4$$

$$= \ln 8 - \ln 2 \iff \textstyle\sum_i k_i \ln k_i$$

$$\ge 8 \ln 2$$

$$= \ln 256 \iff \prod_i k_i^{k_i}$$

$$\ge 256 \text{ [exp strictly monotone]};$$

the census over all 165 compositions of 8 into 4 parts: integers only [machine]

$$(2) : \text{ BM.34(2) re-cited [machine]}; \qquad \Theta_R = \bar{K} A = 0 \text{ at } \bar{K} = 0 \text{ [FT.47(1) re-cited]}$$

$$(3) : \; S_{\mathfrak{L}} = n_{\mathcal{R}} \ln 2 \text{ [C9; CT0.9; CT1.29]};$$

$$n_{\mathcal{R}} = E_{\text{int}} \text{ [D0.8; Def-BM.5(4) pattern];}$$

$$2\ln 2 = \ln 4 \text{ and } E_{\text{bidir}} \cdot 2\ln 2 = E_{\text{int}} \ln 2 \text{ at } E_{\text{int}} = 2E_{\text{bidir}} \text{ [machine: } \mathbb{Q} \cdot \ln 2\text{]}$$

$$(4):\ \text{FT.44(4) substituted [machine];}$$

$$\text{the exponents: } 0 - (-\tfrac{1}{2}) = \tfrac{1}{2} \text{ [CT2.1; machine];}$$

$$c = 1 \text{ [D2.9]}:\ \text{the restored } c^2 \text{ carries } \mathfrak{q}\text{-exponent } 0 \text{ [machine]}$$

$$(5):\ \text{Def-H.37(2)/FT.54(4) re-cited with FT.47(1)'s } \Theta_R;$$

$$\text{the range and the witness re-cited [FT.76(3); FT.54(5): } (2-\sqrt{2})(2+\sqrt{2}) = 2 \text{ in } \mathbb{Q}(\sqrt{2})\text{]}$$

$$(6):\ \text{the witness evaluations: } 2\ln 2 \text{ vs } \tfrac{3}{2}(2-2) = 0;$$

$$0 \text{ vs } \tfrac{3}{2}(2-0) = 3 \text{ [machine];}$$

$$\text{the enclosure: the partial sum } \textstyle\sum_{n=1}^{20} \frac{1}{n2^n} \text{ with tail } < 2^{-20} \implies \tfrac{1}{2} < \ln 2 < 1 \text{ [machine: exact Fractions]} \implies 2\ln 2 < 2 < 3$$

(7): BM.34(4)'s gauge row + (5)'s series composed [machine]; (8): the scope record. ■

Phys. The rest mass of the corpus has always been an entropy in disguise, and the disguise comes off here. The vacuum is the unique state of maximal raw information, and it is flat: maximal entropy, zero defect — one reference point in two currencies. Structure is a lowering against that reference, and the ledger books the lowering in bits, one per crystallized directed edge. The balance identity is why the reframing earns its name: a bidirectionally crystallized channel books exactly the entropy its vacuum pair carried — two bits, $\ln 4$ — so the ledger deficit is a complete re-booking, not a residue, and division by the single free scale turns it into the informational mass the floors have always carried. The reframing costs nothing: same monomial, same dimensions, no new entity, no declaration. What it clarifies is the family structure of the mass reads — the pole mass reads the angle deficit against the same flat vacuum, the ensemble reads the branch-state deficit against the same all-virtual reference — three currencies, one reference point, no exchange rate derived. The total energy of a moving composite is then the sum of two derived reads: the deficit against the vacuum, at rest; and the metric's kinetic term, in motion.

Theorem 4.30 (BM.36 — The force-structure ladder: the mask dimensions, the transport channel, and the inter-family constraint). Thm.

(1) the flag and its increments:

$$\dim \mathcal{K}_n = (6, 16, 17, 25) \text{ strict [CT0.14]};$$

$$(G_5{+}W_3{+}H_1) + (J_3^+{+}J_3^-) + D_1^+ = 16 = C_{\text{cap}} \text{ [CT1.12a(1); CT0.12]};$$

$$\text{rung } 1 \to 2: \quad +1 = \dim Z\big(\iota_2(\mathfrak{u}(2))\big) = \dim \mathcal{K}^{(\text{EM})} \text{ [D2.12; CT0.14]};$$

$$\text{rung } 2 \to 3: \quad +8 = \dim \mathfrak{su}(3) = \dim \mathcal{K}^{(\text{strong})} \text{ [D2.12; CT3.8]};$$

$$\text{the observable bound:} \quad \dim \mathcal{K}_\Sigma \le 1 + 5 + 1 + 3 + 8 = 18 \text{ [FT.26(3)]}$$

[machine: exact integer censuses];

(2) the derived structure ratios:

$$\frac{\dim \mathcal{K}^{(\text{EM})}}{\dim F_W} = \frac{1}{3} \quad [F_W = F_{W^+} \oplus F_{W^-} \oplus F_Z,\ \dim = 3;\ \text{C8}];$$

$$\frac{\dim \mathcal{K}^{(\text{strong})}}{\dim \mathcal{K}^{(\text{EM})}} = \frac{8}{1} = 8;$$

$$\frac{\dim \mathcal{K}^{(\text{EM})}}{\dim \mathcal{K}_3} = \frac{1}{25}$$

[machine: exact Fractions];

these are the alphabet-capacity ratios of the rung-activation order [CT0.13] — the only force-structure ratios the corpus derives;

(3) the strength bar:

$\neg\exists$ a theorem fixing any numerical coupling strength :

(g, g') are P4-class [EW.33; P4];

$\alpha_{\text{EM}}^{(2)}$ is an operator block read whose numerical value no theorem of the corpus fixes [EM.17; EM.25(8); Part 2 §Cal-2's logged row];

the identification $\alpha_X :=$ a dimension ratio is ¬derived and ¬adopted :

reading (2)'s rationals as coupling strengths is a category error of the π-against-$1/137$ species [FT.73(5) trap pattern];

[machine: the text pins — the bar on record in this tag; zero numerical strength values in the tier];

(4) the transport channel (the mass-generation locus):

$$\mathrm{Stab}_{\mathfrak{g}_{\mathrm{EW}}}(h_0\Sigma)$$

$= \mathfrak{g}_{\mathrm{EW}}$: ¬adjoint breaking $\implies$ mass generation is a transport effect on a carried seed, ¬a commutator effect on generators [EW.15];

the seed: $\mathcal{H}_{\mathsf{v}} \cong \mathbb{C}^2$, $W_i \mapsto \frac{1}{2}\sigma_i$, $Y \mapsto \frac{1}{2}\sigma_0$, $T_a\mathsf{v} = 0$ [EW.32];

$M^2_{AB} = \mu_\star^2\,\mathrm{Re}\langle \mathcal{G}_A\mathsf{v}, \mathcal{G}_B\mathsf{v}\rangle$, $\ker M^2 =$ the stabilizer, $h_0 = 0 \implies M^2 \equiv 0$ [EW.33];

the exact recomputation in the weak frame [EW.20] :

$$M^2 = \frac{\mu_\star^2}{4}\begin{pmatrix} g^2 & 0 & 0 & 0 \\ 0 & g^2 & 0 & 0 \\ 0 & 0 & g^2 & -gg' \\ 0 & 0 & -gg' & g'^2 \end{pmatrix}$$

[machine: Pauli spinors over $\mathbb{Q}(i)\otimes\mathbb{Q}[g,g']$: the real parts give the block matrix; the imaginary parts are antisymmetric and cancel in the symmetric form $c_A c_B$ [EW.33]];

the spectrum $\left\{\frac{g^2}{4},\ \frac{g^2}{4},\ 0,\ \frac{g^2+g'^2}{4}\right\}\mu_\star^2$ [machine: the charpoly factorization exact in $\mathbb{Q}[g,g',x]$];

the kernel line $c \propto (0,0,g',g)$ [machine: exact annihilation];

$\dfrac{M_Z^2}{M_W^2} = \dfrac{g^2+g'^2}{g^2}$ exact in $\mathbb{Q}(g^2,g'^2)$ [machine: cross-multiplied];

$e = g\sin\theta_W$ [EW.35] stands untouched;

(5) the inter-family constraint theorem:

every fermion pole mass takes the form $m^2_{\text{pole}}(c) = \hat{\mu}^2\, \ell_c^2\, G(c)$,

$$G(c) = \sum_{A,B} c_A c_B \,\mathrm{Re}\langle \mathcal{G}_A \mathsf{v}, \mathcal{G}_B \mathsf{v}\rangle$$

[Def-H.37(1)] :

an inter-family mass ratio on this channel is a ratio of $G(c)$

ℓ_c^2 values across transport profiles [structural consequence];

the derived sandwich:

$$0 \;\le\; G(c) \;\le\; \mu_\star^2\, \frac{g^2 + g'^2}{4}\, \|c\|^2$$

[machine: the difference form's PSD factorization];

the exact massless line:

$$G(c) = 0 \iff \textstyle\sum_A c_A \mathcal{G}_A \in \mathrm{Ann}(\mathsf{v}) \iff c \parallel (0, 0, g', g)$$

[EW.33; machine];

the range ceilings on the computed skeletons:

the pole ladder lies in $[0, 4\hat{\mu}'^2]$ [FT.76(3)];

¬∃ an assignment of defect reads reaching a multi-order family ratio [FT.77, the quarantined audit re-cited];

a cover read spanning $1{:}R$ requires skeleton scale $V \ge \frac{4}{3} R$ [FT.79(6) re-cited] :

the dynamical range must come from the transport data (ℓ_c, c_A) — ¬from the order-one structure of (1)–(2);

(6) the status row (the inter-family ratios):

constrained, ¬derived:

the channel $((4)$'s transport$)$,

the form $($the Gram$)$,

the kernel $($the exact massless line$)$,

the sandwich $((5))$,

the ceilings $((5))$ are derived;

¬∃ a map in Parts 1–4 assigning transport profiles c_A to family lines [inventory scan; machine: no clause of Def-BM.6 or of the channel tags outputs a c_A] :

the Yukawa-like coupling — the family-profile map and its scale — is application-layer content — ¬a structural absence of the base layer ;

its construction locus is (4)'s seed transport [EW.32, EW.33, Def-H.37(1)];

the Koide-form invariant Q

$= \frac{2}{3}$ [Def-BM.6(4); FT.158(4)] stands as a structural invariant of the naming rows;

its mass-form bridge ($\sqrt{m} \propto r$ vs $m \propto r$) is ¬adopted [Def-BM.6(5)].

Rmk (what the dimension ratios can and cannot carry). The ladder $6 \subset 16 \subset 17 \subset 25$ and its ratios ($\frac{1}{3}$, 8, $\frac{1}{25}$) count activation capacity — how many generator directions the ledger can switch on at each rung. A strength read would require a coupling theorem relating those counts to resolvent weights; no such theorem exists in Parts 1–4, and the couplings the corpus does carry are free positive P4 numbers whose values no clause fixes. The ceilings of (5) sharpen the bar into a quantitative shape: order-one structure ratios cannot carry a multi-order hierarchy on the computed cells, so an inter-family mass generation must come from the transport data — profiles and lengths — whose derivation is application-layer construction, not from the flag.

Req. C8 (Part 1); D2.12 (Part 1); CT0.12, CT0.13, CT0.14 (Part 1); CT1.12a(1), CT3.8, EW.15, EW.20, EW.32, EW.33, EW.35, EM.17, EM.25(8) (Part 2); P4, P5 (Parts 1–2); Def-H.37(1), FT.26(3), FT.73(5), FT.77, FT.79(6), FT.158(4) (Part 3); Census FT.76(3) (row-wise); Def-BM.6 (this part); exact Pauli/Gram algebra over $\mathbb{Q}(i) \otimes \mathbb{Q}[g, g']$, charpoly factorization in $\mathbb{Q}[g, g', x]$, exact Fractions, text pins [math; machine: `checks/verify_H_kinetic_energy_metric.py` §C].

Pf.

$$(1):\ \text{CT0.14′s flag re-cited;}$$

$$3+3+5+3+1+1=16=C_{\text{cap}},$$

$$16+1=17,$$

$$17+8=25,$$

$$6<16<17<25,$$

$$1+5+1+3+8=18\ [\text{machine}]$$

$$(2):\ \dim F_W=3\ [\text{C8}];\qquad \tfrac{1}{3},\ \tfrac{8}{1},\ \tfrac{1}{25}\ [\text{machine: exact Fractions}]$$

$(3):$ EW.33′s P4 row and Part 2 §Cal-2′s logged $\alpha_{\text{EM}}^{(2)}$ row re-cited [text pin: "no theorem of the corpus fixes its numerical value"];

FT.73(5)′s trap pattern re-cited

$$(4):\ \mathsf{v}=(0,1)^{\mathsf{T}}:\quad \sigma_1\mathsf{v}=(1,0)^{\mathsf{T}},\ \ \sigma_2\mathsf{v}=(-i,0)^{\mathsf{T}},\ \ \sigma_3\mathsf{v}=(0,-1)^{\mathsf{T}},\ \ \sigma_0\mathsf{v}=\mathsf{v};$$

$$\mathcal{G}=(gW_1,gW_2,gW_3,g'Y)\ [\text{EW.33}]$$

$$\text{Re}\langle\mathcal{G}_A\mathsf{v},\mathcal{G}_B\mathsf{v}\rangle=\tfrac{1}{4}\left(\text{the displayed block matrix}\right)_{AB};$$

$$\text{Im}\langle\mathcal{G}_A\mathsf{v},$$

$$\mathcal{G}_B\mathsf{v}\rangle$$

$$=-\,\text{Im}\langle\mathcal{G}_B\mathsf{v},$$

$\mathcal{G}_A\mathsf{v}\rangle$ and cancels identically in $\sum_{A,B}c_Ac_B(\cdot)$ [EW.33; machine];

$$\det(xI-\text{Gram})=\left(x-\tfrac{g^2}{4}\right)^2x\left(x-\tfrac{g^2+g'^2}{4}\right)\ [\text{machine: expanded and compared in }\mathbb{Q}[g,g',x]]$$

$$\text{Gram}\cdot(0,0,g',g)^{\mathsf{T}}=0\ \text{[machine]};$$

$$\mu_\star^2=c_W\varkappa^2\beta^2h_0^2\big|_{h_0=0}=0\implies M^2\equiv 0\ \text{[machine: substitution]};$$

$$\tfrac{(g^2+g'^2)/4}{g^2/4}=\tfrac{g^2+g'^2}{g^2}\ \text{[machine: cross-multiplied]}$$

$$(5):\ \text{Def-H.37}(1)\text{'s form re-cited};$$

$$\tfrac{g^2+g'^2}{4}I-\text{Gram}:\ \text{the spectrum}\ \{\tfrac{g'^2}{4},\tfrac{g'^2}{4},\tfrac{g^2+g'^2}{4},\tfrac{g^2}{4}\}\ \text{[machine: charpoly factorization]}\implies\text{the sandwich by the Rayleigh range};$$

$$\text{FT.76}(3),\ \text{FT.77},\ \text{FT.79}(6)\ \text{re-cited}\ (\neg\text{recomputed; the quarantined rows stay quarantined})$$

$$(6):\ \text{the inventory scan of Def-BM.6 and the channel tags FT.109–FT.119},$$

$$\text{FT.142–FT.145}:\ \neg\exists\ \text{a clause outputting a transport profile [machine: token scan]};\qquad\blacksquare$$

Phys. The force side of the mass question splits cleanly into what the corpus derives and what it does not. Derived is the structure: the mask flag grows six to sixteen to seventeen to twenty-five by exact increments — one electromagnetic centre, eight gluons — and the only ratios the ladder supplies are capacity ratios: one third, eight, one twenty-fifth. Derived is also the channel: mass can only be a transport defect of a carried seed, because the scalar background breaks nothing adjointly, and the seed's Gram form comes with an exactly computed spectrum — two equal charged directions, one massive neutral direction, one exact kernel direction along which the transported family is massless whatever the couplings do. Not derived is any coupling value: no theorem turns a count into a strength, and the tier records the bar rather than let a rational drift across the gap. And not derived is the family map: nothing in Parts 1–4 assigns transport profiles to family lines, while the ceilings on the computed skeletons say that order-one structure cannot carry a multi-order hierarchy. The inter-family mass ratios are therefore constrained — channel, form, kernel, sandwich, ceiling — and open, with the construction locus named: the seed transport itself.

Theorem 4.31 (BM.37 — The mass-route artifacts: the disjoint input alphabets and the closure of the counting trap). Thm.

$$(1)\quad\text{the three routes, three exact values:}$$

$$\text{(i) the normalized ensemble average [Def-H.32}(6);\ \text{FT.107}(3)]:\quad\text{neutral:charged}$$

$$=1\ \text{(degenerate)}\ —\ \text{a uniform sector factor cancels at}\ \{1,2,3,15,20,\tfrac{7}{3}\}\ [\text{BM.30}(3);\ \text{machine}];$$

(ii) the total deficit [BM.28(4)(ii)] :

$$\frac{W(0)}{W(+1)} = \frac{20}{15} = \frac{4}{3}$$

[machine: the re-enumeration of $(2 + z + z^{-1})^3$ and the exhaustive 27-triple census];

(iii) the amplitude layer [Def-H.32(2), (5); BM.28(4)(iii)] :

$$\frac{5336}{1196} = \frac{58}{13}\ (S{=}0),$$

$$\frac{4796}{802} = \frac{2398}{401}\ (S{=}1)$$

[FT.106(3)'s frozen integers; machine: exact reductions];

pairwise unequal: $1 \neq \frac{4}{3} \neq \frac{58}{13} \neq \frac{2398}{401}$ [machine: cross-multiplications in $\mathbb{Z}$];

(2) the disjointness theorem (the artifact classification):

routes (i),

(ii) are measure functionals of (Γ, R) : the weights (ν, $t^{|R|}$, u^L, #inst) and the deficit ΔS [Def-H.32(3), (6); FT.107(2)];

route (iii) is the amplitude functional of $v_\Gamma(R)$ [Def-H.32(2), (5)];

the dispersion read is the metric functional of (g_{ij}, N, N^i, p_i, m) [Def-BM.10(2)–(5)];

the machine token census on the canonical defining strings:

metric-alphabet occurrences inside the defining clauses of routes (i)–(iii) : 0;

measure-alphabet occurrences inside Def-BM.10(2),

(3),

(5) (the kinematic clauses; clause (4)'s rest input is the shared symbol m) : 0;

the input alphabets are disjoint: ¬∃ a route output that is a function of metric data; ¬∃ a dispersion energy that is a function of measure data

(3) the trap and its closure:

the counting trap :

$=$ reading a route ratio $\left(1,\ \frac{4}{3},\ \frac{58}{13},\ \frac{2398}{401}\right)$ or the Koide-form invariant Q

$= \frac{2}{3}$ [Def-BM.6(4); FT.158(4)] as a physical mass ratio [FT.89's category-trap species];

barred by three independent rows:

(a) the disjointness [(2)] : a route output carries no momentum dependence and cannot transform as a dispersion energy;

(b) amplitude $\neq$ mass [FT.79(5)] : route (iii)'s bridge is item-13 content, ¬constructed

(c) the route choice is a read specification, ¬a theorem [BM.30(5)] :

no route is adopted, and none is retracted;

$Q = \frac{2}{3}$

$= \frac{3-1}{3}$ is the doublet/triplet decomposition ratio of the spatial seal triad [FT.158(5)] :

a structural invariant of the naming rows, whose mass-form bridge stays ¬adopted [Def-BM.6(5)];

(4) the completion row:

the physical mass-energy of a sealed composite is the dispersion read:

$$E = N\sqrt{m_{\text{rest}}^2 + p^2} - N^i p_i, \qquad m_{\text{rest}} = \frac{\Delta S_{\mathfrak{L}}}{\mathfrak{h}} \qquad [\text{BM.34}(4);\ \text{BM.35}(4), (7)];$$

the routes are static (p-free) slices: each evaluates an ensemble with no momentum argument $[(2)]$;

KE — the metric's motion term — is what no route can express;

at p

$= 0$ the dispersion rest term coincides on-shell with the curvature-flux and the informational reads [BM.34(7)] :

the routes price the ensemble measure's restrictions around the same rest object;

BM.30(2)'s splitting and BM.30(3)'s degeneracy stand derived and exact as measure rows — the closure classifies, it does not retract;

(5) the standing rules:

¬a P4 addition [§Cal-BM stays empty];

¬a new dimensionful scale [P1];

¬a measured value and ¬a decimal numeral in this tier's text [P5; machine scans];

¬a new declaration;

species names confined to the identification layers [Def-BM.3(7), Def-BM.6(7)];

topology labels are internal bookkeeping [Def-H.32(4)];

(6) the adjudication:

the three mass-route ratios are counting and amplitude artifacts — exact as measure prices, barred as physical masses ;

the counting trap is closed at the category level by the disjoint input alphabets $[(2)]$;

the physical mass-energy read is the metric dispersion route: the rest term is the vacuum-balanced entropy [BM.35],

the motion term is the shift-transported kinetic energy [BM.34];

the inter-family content is application-layer [BM.36(6)].

Rmk (what the closure does not do). The closure retracts no route: every route value stands as a derived exact rational, and BM.30(5)'s read-choice row is untouched — adopting any route as "the mass" still requires a logged declaration. What the closure establishes is the classification: the three functionals price measures and amplitudes, and a measure price is not a dispersion energy however clean its rational looks. The Koide-form invariant belongs to the same classification — it is the decomposition ratio the spatial seal triad forces [FT.158(5)] — and converting it into a mass ratio needs exactly the mass-form bridge — application-layer content. The physical reading runs the other way: the rest mass is the vacuum-deficit read, the motion is the metric's kinetic term, and the route ratios stay what they always were — exact prices of how the charge constraint and the sign sum filter one ensemble.

Req. P1, P4, P5 (Parts 1–2); Def-H.32(2), Def-H.32(3), Def-H.32(4), Def-H.32(5), Def-H.32(6), FT.79(5), FT.89, FT.106(3), FT.107(2), FT.107(3), FT.158(4), FT.158(5) (Part 3); Def-BM.3(7), Def-BM.6(4), Def-BM.6(5), Def-BM.6(7), Def-BM.10, BM.28(2), BM.28(4), BM.30(1), BM.30(2), BM.30(3), BM.30(5), BM.34(4), BM.34(7), BM.35(4), BM.35(7), BM.36(6) (this part); exact rational reductions, the exhaustive triple census, generating-function convolution, token censuses [math; machine: `checks/verify_H_kinetic_energy_metric.py` §D].

Pf.

$$(1):\ \text{(i)}\ \ \frac{\sum_R c\, v(R)\Delta S(R)}{\sum_R c\, v(R)}$$

$$= \frac{\sum_R v(R)\Delta S(R)}{\sum_R v(R)}\ \ \forall c$$

$$\neq 0\ \text{[machine at } c \in \{1, 2, 3, 15, 20, \tfrac{7}{3}\}\ \text{on rational witness data]}$$

$$\text{(ii)}:\ W(Q) = \left[z^Q\right](2 + z + z^{-1})^3 \text{ re-enumerated :}$$

$$W = (1, 6, 15, 20, 15, 6, 1),$$

$$W(0) = 8 + 12 = 20,$$

$$W(\pm 1) = 12 + 3 = 15, \quad \tfrac{20}{15} = \tfrac{4}{3} \text{ [machine]};$$

$$\text{(iii)}: \ \tfrac{5336}{1196} = \tfrac{58}{13} \ (5336 \cdot 13 = 1196 \cdot 58), \ \ \tfrac{4796}{802} = \tfrac{2398}{401} \text{ [machine: exact]}$$

the pairwise inequalities: $3 \cdot 1 \neq 4 \cdot 1$,

$$4 \cdot 13 \neq 3 \cdot 58,$$

$$58 \cdot 401 \neq 13 \cdot 2398, \quad \text{and the remaining pairs [machine: integer cross-multiplications]}$$

(2) : the canonical defining strings extracted from the canon files [Def-H.32$(2), (3), (5), (6)$; FT.107$(2), (3)$; BM.28(4); Def-BM.10(2)–(5)];

the metric alphabet $\{\Omega, \varkappa, \beta, w_i, q_{ij}, \gamma, N, N^i, p_i, g_{0i}, \text{lapse}, \text{shift}\}$: 0 occurrences in the route clauses [machine]

the measure alphabet $\{W(Q), \#\text{inst}, \nu, v_\Gamma, \Delta S, t^{|R|}, u^L, Z^\bullet, \text{ensemble}, \text{flicker}\}$: 0 occurrences in Def-BM.10(2)–(5) beyond the shared symbol m [machine]

(3) : (a) [(2) conjoined with Def-BM.10(5)'s p-dependence];

(b) FT.79(5) re-cited;

(c) BM.30(5) re-cited;

$Q = \frac{2}{3}$ and $\frac{3-1}{3}$ [FT.158(5); Def-BM.6(4) re-cited]

(4) : BM.34(4), BM.35$(4), (7)$ conjoined;

BM.34(7)'s on-shell agreement row re-cited;

(5) : the standing-rule record [BM.33(5) pattern] + the machine scans

(6) : conjunction of (1)–(5). ■

Phys. Three exact rationals — one, four thirds, fifty-eight thirteenths — came out of the charge-constrained ensemble, and the temptation is to read them as mass ratios. The corpus closes that trap the only way it closes anything: by classification. Each of the three routes is a functional of

the ensemble's combinatorics alone — pattern counts, norm weights, sign-summed amplitudes — and a machine census of the defining clauses shows their input alphabet contains not one metric token, while the dispersion read's definition contains not one measure token beyond the shared rest symbol. Disjoint inputs, disjoint outputs: no route value can transform as a dispersion energy, because no route has ever seen a momentum. What the routes price is the measure — how the charge constraint rescales the probability space, how the sign sum weights the amplitudes — and those prices stand, exact and unretracted. The physical mass-energy is assembled elsewhere: the rest term from the vacuum-balanced entropy, the motion term from the metric's kinetic energy, the two joined by the mass-shell identity. The Koide-form invariant joins the same classification — a decomposition ratio the spatial triad forces, waiting on a mass-form bridge that item 13 holds open. The counting was never wrong; reading the counts as masses was.

BM.38 — The status of the metric-energy tier: adjudication and scope. Structural-constraints record: see Appendix B, Section B.1.20 of the Addendum.

4.13 Tier BM.11 — The Neutron Decay Vertex: the Family Transport Profiles, the Rung-1 Isospin Row, the Forced Neutral Member, the Angular-Momentum Budget, and the Full Kinematic Balance

Definition 4.11 (Def-BM.11 — The family transport-profile map (the constructed bridge)). Def. A construction on EW.32/EW.33's seed transport and FT.158(1)'s inner C_3 rotation; no new axiom, no new dimensionful scale, no Yukawa-type object — none exists in the inventory [EW.15; FT.145(3)]. Local notation: $\mathrm{Gr}_{AB} := \mathrm{Re}\langle \mathcal{G}_A \mathsf{v}, \mathcal{G}_B \mathsf{v}\rangle$ (EW.33's Gram data).

(1) the seed data:

$$\mathcal{G} = (gW_1,\ gW_2,\ gW_3,\ g'Y) \text{ [EW.33]};$$

$$\mathsf{v}: \quad W_3\mathsf{v} = -\tfrac{1}{2}\mathsf{v}, \quad Y\mathsf{v} = +\tfrac{1}{2}\mathsf{v} \text{ [EW.20]} :$$

the hypercharge coordinate $y_0 = \frac{1}{2}$ is derived [EW.20 : trace normalization, ¬assigned];

(2) the family rotation:

$$U = \exp\!\big(-\tfrac{2\pi i}{3}\,\hat{n}\cdot W\big) = E_0 - \tfrac{1}{2}\mathcal{W}^2 - \tfrac{i}{2}\mathcal{W}, \quad \mathcal{W} = W_1{+}W_2{+}W_3 :$$

$$UW_iU^\dagger = W_{i+1}, \quad U^3 = E_0 \text{ [FT.158(1); machine: exact over } \mathbb{Q}(i), \text{ via } \mathcal{W}^3 = 3\mathcal{W}];$$

$$Y \text{ is the centre of } \iota_2(\mathfrak{u}(2)) \implies [Y, W_i] = 0 \implies UYU^\dagger = Y \text{ [EW.8; D2.12; machine]} :$$

the C_3 cycles the weak triplet coordinates of a transport profile and fixes the hypercharge coordinate;

(3) the profile class:

every C_3-covariant triplet profile takes the cosine form λ_i

$= A + B\cos(\theta + \frac{2\pi i}{3})$ [FT.111(2)];

unit normalization (constitutive convention, the Def-BM.4(1) pattern): the weak coordinates carry radius 1;

the base profile: $c(0) := \big(\cos\delta,\ \cos(\delta + \frac{2\pi}{3}),\ \cos(\delta + \frac{4\pi}{3}),\ y_0\big) \in \mathbb{R}^4$;

the orbit:

$c(i) := \mathrm{Ad}(U^i)\, c(0)\ =$ the cyclic shift of the weak triplet, y_0 fixed $[(2);$ machine$]$:

$$c(i)_3 = \cos\big(\delta + \tfrac{2\pi(2-i)}{3}\big);$$

the W_3 slot cycles through the triplet;

the Gram-value set $\big\{M^2\big(A_0 - B_0\cos(\delta - \frac{2\pi j}{3})\big)\big\}_j$ is the orbit's content under (4)'s orbit-label convention;

(4) the angle identification (the single declared clause):

$\delta := \delta_{\text{channel}}$ — per channel: $\ell :\ \frac{2}{9}$ [FT.112];

$q :\ \frac{2}{27}$ [FT.119(1)];

$\nu :\ \frac{1}{8}$ [FT.142(5)];

the orbit-label orientation convention ($\delta \mapsto -\delta$ in the W_3 slot) is internal bookkeeping;

the lines attach to Def-BM.6's named rows by its own spectral order;

(5) the mass evaluation:

$$m^2(\text{line } i) = \hat{\mu}^2\, \ell_c^2\, G\big(c(i)\big),$$

$$G(c) = \sum_{A,B} c_A c_B\, \mathrm{Gr}_{AB}$$

[Def-H.37(1); EW.33] :

$$\boxed{m_i^2 = \hat{\mu}^2 \ell_c^2 \Big[\Big(\frac{3g^2}{8} + \frac{g'^2}{16}\Big) - \frac{gg'}{4}\,\cos\Big(\delta - \frac{2\pi i}{3}\Big)\Big]} =: M^2\big(A_0 - B_0 \cos\varphi_i\big),$$

$$A_0 = \frac{3g^2}{8} + \frac{g'^2}{16}, \quad B_0 = \frac{gg'}{4}$$

[machine: exact in $\mathbb{Q}[g, g'] \otimes \mathbb{Q}[c, s, r_3]/(c^2{+}s^2{-}1, r_3^2{-}3)$];

the affine form:

$$r_2 \cdot m_i^2 = M^2\big(A_0 r_2 + B_0 - B_0\, r_i\big),$$

$$r_i = 1 + r_2 \cos\Big(\delta + \frac{2\pi i}{3}\Big), \quad r_2^2 = 2$$

[FT.113 vocabulary rows; machine];

(6) interface status:

derived: the orbit structure $[(2)]$,

the cosine shape [FT.111(2)],

the Gram evaluation and the affine form $[(5);$ machine$]$,

positivity, the splitting law, the sum rule, the order inversion [BM.39];

non-derived: exactly one angle identification per channel (4) and the unit normalization (3);

scales: $\hat{\mu}^2 \in$ Cal-F (logged); ℓ_c is chain data [D1.4]; (g, g') are P4 [EW.33] :

¬a new parameter, §Cal-BM stays empty;

declared (A-BM.4) before use;

(7) the locus clause:

the construction lives in the pole/transport read species [Def-H.37(1)];

FT.113/FT.143(3)'s inertial vocabulary $m^2 = \mu r^2$ is a distinct read species on the same lines :

the two species order the lines oppositely [BM.39(5)];

the relating equation is item-7 content, untouched

BM.36(6)'s gap label is discharged: the family-profile map exists as this construction;

the scale data is P4-class calibration (¬declared), ¬the map.

Rmk (what the construction is). The map answers BM.36(6)'s named gap with derived material only: the rotation is FT.158(1)'s — the same C_3 that forces the channel spectra, inner in the weak algebra; the shape is FT.111(2)'s — every C_3-covariant triplet is a cosine triplet; the evaluation is Def-H.37(1)'s Gram transport; the hypercharge coordinate is EW.20's derived $\frac{1}{2}$. The single declared content is that the profile's phase is the channel's own angle — one identification per channel, and the angles are derived rationals. The payoff is structural: the intra-family splitting is exactly gg' — the weak–hypercharge mixed strength — acting on the transport phase, which is the derived form of "the relative force strength acting on the seed's transport profile." No commutator breaking enters anywhere [EW.15], and no Yukawa-type term exists to enter.

Req. D1.4, D2.12 (Parts 1–2); EW.8, EW.20, EW.32, EW.33 (Part 2); P4, P5 (Parts 1–2); Def-H.37(1), FT.111(2), FT.112, FT.113, FT.119(1), FT.142(5), FT.143, FT.145(3), FT.158(1) (Part 3); Def-BM.4(1), Def-BM.6, BM.36(4), BM.36(5), BM.36(6) (this part); Cal-F; the declaration addendum A-BM.4; exact quotient-ring trigonometric identities in $\mathbb{Q}[g, g', c, s, r_3, r_2]/(c^2+s^2-1,\ r_3^2-3,\ r_2^2-2)$, Gram algebra over $\mathbb{Q}[g, g']$ [math; machine: `checks/verify_H_neutron_decay_vertex.py` §A].

Pf.

Definition block: the constitutive clauses are (3)'s unit normalization and (4)'s angle identification;

every displayed evaluation is a machine-verified exact identity [§A];

$(6), (7)$ are the status and locus records. ■

Phys. The map the corpus named as its open construction is built, and it is made of pieces the derivations already produced. The family rotation is the inner C_3 of the weak algebra — the same rotation that forces the cosine shape of every channel spectrum — so a transport profile carried around the orbit is automatically a cosine triplet with the channel's own angle. Feed the triplet through the seed's Gram form and the mass-squared of each line comes out affine in the channel's spectral invariant: a constant set by the couplings, minus the mixed weak–hypercharge strength times the line's cosine. The only thing declared is the identification of the profile's phase with the channel angle; everything else — the orbit, the shape, the evaluation, the positivity, the sum rule — is forced. What the construction does not do is convert currencies: the transport species and the inertial $m = \mu r^2$ vocabulary still wait on the relating equation, and the overall scales remain the undeclared internal numbers they have always been.

Theorem 4.32 (BM.39 — The family mass form: positivity, the splitting law, the sum rule, the order inversion, and the area ladder). Thm.

(1) the exact evaluation and positivity:

$$m_i^2 = M^2\big(A_0 - B_0 \cos\varphi_i\big),$$

$$M^2 := \hat{\mu}^2 \ell_c^2, \quad A_0 = \frac{3g^2}{8} + \frac{g'^2}{16}, \quad B_0 = \frac{gg'}{4}$$

[Def-BM.11(5); machine];

$$A_0 - B_0 = \frac{3}{8}\Big(g - \frac{g'}{3}\Big)^2 + \frac{g'^2}{48},$$

$$A_0 + B_0 = \frac{3}{8}\Big(g + \frac{g'}{3}\Big)^2 + \frac{g'^2}{48}$$

[machine: exact sum-of-squares decompositions];

$$\Longrightarrow\ M^2(A_0 - B_0)\ \le\ m_i^2\ \le\ M^2(A_0 + B_0),$$

$$m_i^2 > 0\ \ \forall i,\ \forall \delta,\ \forall (g, g') \neq (0,0) :$$

every line of every channel is massive under the construction;

the kernel line is unreachable:

$c \parallel (0, 0, g', g)$ [EW.22 : ker $M^2 = \text{span}\{Q\}$, $Q = W_3 + Y$; FT.12] requires two vanishing cosines :

$$\cos\varphi$$

$$= 0 \wedge \cos(\varphi + \tfrac{2\pi}{3})$$

$= 0$ has no solution [machine: $c = 0 \implies s^2 = 1 \implies \cos(\varphi + \frac{2\pi}{3}) = \mp\frac{r_3}{2},\ r_3^2 = 3 \neq 0$] :

BM.36(5)'s exact massless line carries no family row;

(2) the splitting law: $m_i^2 - m_j^2 = M^2 B_0\big(\cos\varphi_j - \cos\varphi_i\big)$ [machine: exact factorization] :

the intra-family splitting is exactly the mixed weak–hypercharge strength gg' acting on the transport phase [Def-BM.11 Rmk];

degenerate $\iff gg' = 0 \ \vee\ \cos\varphi_i = \cos\varphi_j$;

the democratic witness δ

$$= 0 : \ \{\cos\varphi_i\}$$

$= \{1, -\frac{1}{2}, -\frac{1}{2}\}$: two lines degenerate, the third split by $\frac{3}{2}M^2B_0$ [machine];

(3) the sum rule:

$$\sum_i m_i^2 = 3M^2 A_0$$

$[\sum_i \cos\varphi_i = 0 :$ FT.111(3)/FT.158(4) identity; machine$]$:

phase-independent and channel-independent — the seed is channel-blind [EW.32 : ¬colour, ¬slot index] :

all three channels share (A_0, B_0); only $\delta \in \{\frac{2}{9}, \frac{2}{27}, \frac{1}{8}\}$ differs;

(4) the cross-channel row:

$$16\,\delta_\nu = 9\,\delta_\ell = 27\,\delta_q = 2,$$

$$3\delta_q = \delta_\ell$$

$$[\text{FT.143}(5);\ \text{FT.119}(2);\ \text{machine: exact Fractions}]:$$

the three channels' phase data are exact rational multiples of one seed numerator — the inter-channel mass relations are symbolic rows in (A_0, B_0, M^2),

$$\neg\text{numbers}\ [\text{P4};\ \text{P5}];$$

$$(5)\quad \text{the order inversion:}$$

m_i^2 is affine strictly decreasing in r_i [Def-BM.11(5) affine form; $B_0 > 0$ at $g, g' > 0$ [EW.33 : $g, g' > 0$]; machine: sign census]:

the spectral-maximal line receives the mass-minimum;

the frozen chains re-cited: $r_1^2 < r_2^2 < r_0^2$ [FT.113 (ℓ); FT.143$(1), (3)$ (ν); enclosure-proven, frozen];

the read-species row: FT.113's inertial vocabulary m^2

$= \mu r^2$ orders with the spectrum; the transport species orders against it :

the order is read-relative; neither species is adopted as "the mass" [Def-BM.11(7)];

$Q = \frac{2}{3}$ is the structure invariant of Def-BM.6(4), ¬a mass claim [FT.143(4)];

$$(6)\quad \text{the inter-family constraint stands:}$$

BM.36(5)'s rows are untouched by the construction: $0 \le G(c) \le \mu_\star^2 \frac{g^2+g'^2}{4}\|c\|^2$,

the pole ladder $\subseteq [0, 4\hat{\mu}'^2]$ [FT.76(3)];

the multi-order hierarchies remain beyond the computed cells [FT.77, FT.79(6) re-cited; quarantined rows stay quarantined]:

the construction supplies the form, ¬a scale [P4; P5];

(7) the area ladder (the holonomy species):

$$m^2_{\text{pole}} = 4\hat{\mu}'^2 \sin^2\!\big(\tfrac{\Theta_R}{4}\big),$$

$$\Theta_R = \bar{K} A = \textstyle\sum_i \alpha_i - (n-2)\pi$$

[Def-H.37(2); FT.54(4); FT.47(1); machine: the exact SU(2) defect re-derived];

the extent-dependent form: a loop of spatial extent r enclosing area $A(r)$ in a constant-curvature slice:

$$m^2(r) = 4\hat{\mu}'^2 \sin^2\!\Big(\frac{\bar{K}\, A(r)}{4}\Big)$$

[machine: formal identity];

the ratio formula:

$$\frac{m_1^2}{m_2^2} = \frac{\sin^2(\Theta_{R,1}/4)}{\sin^2(\Theta_{R,2}/4)}$$

[machine: symbolic];

monotone in $\Theta_R \in [0, 2\pi]$, range $[0, 4\hat{\mu}'^2]$ [FT.76(3) re-cited];

the winding multiplet: $m^2(m) = 4\hat{\mu}'^2 \sin^2\!\big(\frac{m\,\Theta_R}{4}\big), \quad m \in \{k, k-2, \ldots, -k\}$

[FT.60(5) re-cited; Def-H.38 ladder];

(8) the distinction row:

Θ_R is the angle defect [FT.47(1)];

the winding $k = \Theta/2$ is the ledger read [Def-H.32(4); FT.22];

the identification $\Theta_R = 2\pi k$ is ruled out on calibration weight [FT.55(1) re-cited] :

the two ladders are distinct reads, and the tier keeps them apart;

(9) the matching row:

the Gram species $((1)$–$(5))$ and the holonomy species $((7))$ are related by Def-H.37's leading-order matching $\hat{\mu}'^2\Theta_R^2/4$

$= \hat{\mu}^2\ell_c^2 G(c)$ [FT.54(2); FT.33] :

recorded, not imposed [Def-H.37 locus clause];

the exact-matching derivation remains item-7 content [pointer].

Rmk (two mechanisms, one form.) Within a channel the families split by the Gram phase — the mixed strength times a cosine; along a winding or area ladder the levels split by the holonomy defect — a sine-squared of the enclosed curvature. The two are the leading-order and exact forms of one transfer [Def-H.37(1)–(2)], and the tier uses each where it is derived: the cosine triplet for the family structure, the defect ladder for the extent structure. Neither carries a scale: $\hat{\mu}, \hat{\mu}', \ell_c, (g, g')$ are the logged internal data, and no clause converts them into a number.

Req. P4, P5 (Parts 1–2); EW.22, EW.32, EW.33 (Part 2); Def-H.32(4), Def-H.37, Def-H.38, FT.12, FT.22, FT.33, FT.47(1), FT.54(2), FT.54(4), FT.55(1), FT.60(5), FT.77, FT.79(6), FT.111(3), FT.112, FT.113, FT.119(1), FT.119(2), FT.142(5), FT.143(1), FT.143(3), FT.143(4), FT.143(5), FT.158(4) (Part 3); Census FT.76(3) (row-wise); Def-BM.6(4), Def-BM.11, BM.36(5), BM.36(6) (this part); Cal-F; sum-of-squares positivity, exact quotient-ring identities, symbolic ratio algebra, exact Pauli defect identities [math; machine: `checks/verify_H_neutron_decay_vertex.py` §A].

Pf.

(1) : Def-BM.11(5)'s evaluation re-cited;

$\frac{3}{8}(g \mp \frac{g'}{3})^2 = \frac{3g^2}{8} \mp \frac{gg'}{4} + \frac{g'^2}{24}$: $\quad + \frac{g'^2}{48}$ gives $A_0 \mp B_0$ [machine: expansion];

$|\cos| \leq 1 \implies$ the sandwich;

positivity: a sum of squares with positive rational coefficients vanishes only at (g, g')

$= (0, 0)$ [machine]

the kernel row: $c = 0 \wedge s^2 = 1 \implies \cos(\varphi + \frac{2\pi}{3}) = -\frac{c}{2} - \frac{r_3 s}{2} = \mp\frac{r_3}{2} \neq 0$ $[r_3^2 = 3;$ machine$]$

(2) : difference of (1)'s forms [machine];

$$\delta = 0 :\ c = 1, s = 0 \implies \{1, -\tfrac{1}{2}, -\tfrac{1}{2}\}\ \text{[machine: } -\tfrac{c \pm r_3 s}{2} \text{ at } (c, s) = (1, 0)\text{]}$$

$$(3) :\ \textstyle\sum_i \cos\varphi_i = 0\ \text{[FT.111(3) identity, generic phase; machine]};$$

$$(A_0, B_0) \text{ carry no channel argument [EW.32 re-cited]}$$

$$(4) :\ 16 \cdot \tfrac{1}{8} = 9 \cdot \tfrac{2}{9} = 27 \cdot \tfrac{2}{27} = 2; \quad 3 \cdot \tfrac{2}{27} = \tfrac{2}{9}\ \text{[machine: exact Fractions]}$$

$$(5) :\ \text{the affine coefficient } -M^2 B_0/r_2 < 0 \text{ at } g, g' > 0\ \text{[machine: sign census of the monomial]};$$

$$\text{FT.113/FT.143's chains re-cited (frozen, enclosure-proven)};$$

$$Q = \tfrac{6}{9} = \tfrac{2}{3}\ \text{[FT.143(4) re-cited]}$$

$$(6) :\ \text{BM.36(5), FT.76(3), FT.77, FT.79(6) re-cited (}\neg\text{recomputed)}$$

$$(7) :\ (U-1)^\dagger (U-1)$$

$$= (2 - 2\cos\phi)\mathbb{I}$$

$$= 4\sin^2(\tfrac{\phi}{2})\mathbb{I} \text{ at } U$$

$$= \cos\phi$$

$$\mathbb{I} + i\sin\phi$$

$$\hat{n} \cdot \vec{\sigma}\ \text{[FT.54(1) re-derived; machine: exact Pauli algebra in } \mathbb{Q}[C, S]/(C^2 + S^2 - 1)];$$

$$\phi = \tfrac{\Theta_R}{2}\ \text{[FT.54(3); Def-H.37(2)]}$$

$$(8) :\ \text{FT.55(1)'s calibration-weight obstruction re-cited};$$

$$(9) :\ \text{Def-H.37's matching row re-cited. } \blacksquare$$

Phys. The family mass form comes out exact and tight. Positivity is a sum of two squares with rational coefficients, so no line of any channel can fall to zero through this map — the massless direction of the Gram form is the electromagnetic generator, and a cosine triplet can never align

with it, because two cosines a third of a turn apart cannot vanish together. The splitting between lines is precisely the product of the two electroweak strengths times the cosine separation: the force structure acting on the transport phase, with no Yukawa object anywhere in it. The three channels share the whole form and differ only in their derived angles, so their orbit sums are one expression and their splittings are rational multiples of one another. The order runs against the spectrum: the brightest spectral line is the lightest mass under this read, and the tier records that reversal instead of hiding it, because the inertial vocabulary orders the other way and no theorem yet converts the two. Alongside the Gram mechanism stands the holonomy one: mass as the sine-squared of a quarter angle defect, growing with the area a loop encloses — the extent-dependent form the corpus can derive exactly, with the winding identification ruled out and the two ladders kept apart.

Theorem 4.33 (BM.40 — The rung-1 isospin row: the Casimir identity, the branch selectivity, and the neutrality of the σ-vertex). Thm.

$$(1) \quad \text{the Casimir identity:}$$

$$W_i = \tfrac{1}{2}\big(\sigma_i \otimes \sigma_0 + \sigma_0 \otimes \sigma_i\big) \text{ [CT1.2]};$$

$$[W_i, W_j] = i\epsilon_{ijk} W_k \text{ [CT1.4; machine]};$$

$$\boxed{\textstyle\sum_i W_i^2 = \tfrac{3}{2}E_0 + \tfrac{1}{2}\Sigma = E_0 + \mathbb{S} = 2\,\Pi_{\mathcal{R}}}$$

$$\text{[Cor CT0.7a; FT.141(4); FT.157(2); machine: entrywise over } \mathbb{Q}(i), \text{ all 16 components];}$$

$$(2) \quad \text{the branch selectivity:}$$

$$\textstyle\sum_i W_i^2\big|_{\vee^2} = 2 = 1(1{+}1),$$

$$\textstyle\sum_i W_i^2\big|_{\wedge^2} = 0 = 0(0{+}1)$$

$$\text{[Cor CT0.7a; machine: exact on the branch bases];}$$

$$W_i\Pi_{\mathcal{V}} = 0 = \Pi_{\mathcal{V}} W_i \;\forall i, \qquad \bigcap_i \ker W_i = \operatorname{im}\Pi_{\mathcal{V}} \qquad \text{[FT.101(1)–(2); FT.157(2); machine];}$$

$$\Pi_{\mathcal{R}} = \tfrac{1}{4}(3E_0 + \Sigma), \quad \Pi_{\mathcal{V}} = \tfrac{1}{4}(E_0 - \Sigma) :$$

$$\text{idempotent, orthogonal, complete}, \quad \mathfrak{T}\Pi_{\mathcal{R}} = 3, \quad \mathfrak{T}\Pi_{\mathcal{V}} = 1$$

[CT0.7(1); machine];

(3) the neutrality of the σ-vertex:

the transition's charge move is an end-sign flip:

$q_{\text{core}} = \frac{1}{2}(\sigma_s - \sigma_t)$ is registration data on the reality line [D3.4; D1.1; CT0.9; BM.23(2)];

the Casimir $2\Pi_{\mathcal{R}}$ is a function of the branch datum alone [(1)];

the gradings are pairwise distinct [FT.136; FT.141(6); FT.144(5)];

the register separation holds [FT.136(3)];

the rung-1 Casimir deficit at the σ-vertex is identically zero

[machine: the token census — zero branch tokens in D3.4's defining row, zero registration tokens in CT0.7(1)/FT.101's defining rows];

the conservation law holds at the vertex — it neither forces nor forbids an emission :

(4) the carrier-side rows:

every emitted chain is Janus-closed: $\#J^+ = \#J^-$ [D1.4; FT.144(1)–(2); the prefix law] :

each carrier is internally Casimir-balanced;

the neutral carrier class is rung-one content: $\mathcal{A}(1)$ aspects,

$\mathcal{K}_1$ arena,

dim

$= 16$ [FT.142(1)–(2)];

EM-unactivated $(f_v = 0)$, colour-blind [FT.142(1)];

at rung one the chiral sectors are force-decoupled:

$F_{LR} = 0$ [FT.144(6); FT.128(2); the no-flip theorem] :

the neutral carrier's orientation sector is frozen along its flight [FT.144(5) : single-sector occupancy];

(5) the status:

the forcing of the neutral member comes ¬from the Casimir row — (3) adjudicates the deficit route neutral — but from BM.41's kinematic theorem;

BM.23(6)'s clause stands as written and receives an on-record Rmk [the FT.89 Rmk pattern];

¬∃ a lepton-number read in the inventory [BM.23(6); FT.81(4) pattern] — the row is untouched.

Rmk (why the proposed deficit does not arise). The isospin bookkeeping the vertex could upset lives in the branch datum — aligned pairs carry Casimir two, antialigned pairs zero. The charge move that defines the transition lives in the registration sign — the reality-line datum the end reads consume. The corpus proved these gradings pairwise distinct and their registers separate, so no σ-flip can move a $\Pi_{\mathcal{R}}$: the deficit is zero as an operator identity, not as a small number. That is why the neutrino's necessity, where it is necessity, must come from somewhere else — and BM.41 shows it does: from the kinematics of equal rest masses and the uniqueness of the neutral class.

Req. A3 (Part 1); CT0.7(1), CT0.9, CT1.2, CT1.4, Cor CT0.7a, D1.1, D1.4, D3.4 (Parts 1–2); P5 (Parts 1–2); FT.12, FT.101(1), FT.101(2), FT.128(2), FT.136(3), FT.141(4), FT.141(6), FT.142(1), FT.142(2), FT.144(1), FT.144(2), FT.144(5), FT.144(6), FT.157(2) (Part 3); BM.7, BM.23(2), BM.23(6) (this part); the forcing theorem of this tier's vertex tag is a marked forward pointer at the citing site and is imported in no Req line; exact $\mathbb{Q}(i)$ 4×4 algebra, projector algebra, token censuses [math; machine: `checks/verify_H_neutron_decay_vertex.py` §B].

Pf.

$$(1) : \sum_i W_i^2$$

$$= \tfrac{1}{4} \sum_i (\sigma_i \otimes \sigma_0 + \sigma_0 \otimes \sigma_i)^2$$

$$= \tfrac{1}{4}(3E_0 + 3E_0 + 2\Sigma)$$

$= \tfrac{3}{2}E_0 + \tfrac{1}{2}\Sigma$ [Cor CT0.7a re-derived; machine];

$$\Sigma = 2\mathbb{S} - E_0 \text{ [CT0.7(1)]} \implies = E_0 + \mathbb{S} = 2\Pi_{\mathcal{R}} \text{ [machine]};$$

$$[W_i, W_j] = i\epsilon_{ijk} W_k \text{ [machine: all 9 pairs]}$$

$$(2): \ \vee^2 \text{ basis } \{(1,0,0,0),(0,1,1,0),(0,0,0,1)\}: \ 2 \cdot v \text{ [machine]};$$

$$\wedge^2 \text{ basis } \{(0,1,-1,0)\}: \ 0 \text{ [machine]};$$

$$W_i \Pi_{\mathcal{V}} = 0 = \Pi_{\mathcal{V}} W_i \text{ [machine: all 16 components each]};$$

the projector algebra and the traces [machine]

(3): the defining strings extracted : D3.4's row vs CT0.7(1)/FT.101's rows;

the branch alphabet $\{\Pi_{\mathcal{R}}, \Pi_{\mathcal{V}}, \alpha, \text{branch}\}$: 0 occurrences in D3.4;

the registration alphabet $\{\sigma_s, \sigma_t, q_{\text{core}}, \text{reality line}\}$: 0 occurrences in the branch rows [machine]

(4): D1.4/FT.144(1)–(2) re-cited (frozen);

$$F_{LR} = 0 \text{ [FT.128(2) re-cited]};$$

(5): the status record. ■

Phys. The weak isospin of the pair arena is an exact operator identity — the Casimir is twice the projector onto the real branch — and it splits the pair register into a spin-one triplet the weak force sees and a spin-zero singlet it annihilates. The question the vertex poses is whether a charge flip disturbs that bookkeeping, and the answer is a clean no: the flip moves a registration sign, the Casimir reads a branch label, and the corpus long ago proved those gradings live in separate registers. The deficit is zero identically, so the isospin law neither demands nor forbids the neutral emission. What the rung-one rows do supply is the carrier's pedigree: closed under Janus balance, electromagnetically inert, colour-blind, force-decoupled across its chiral sectors at rung one, single-arrow in occupancy — a neutral member that travels without turning. The necessity itself waits for the kinematics.

Theorem 4.34 (BM.41 — The decay vertex: the forced members, the two-body impossibility, and the Q-value sign theorem). Thm.

(1) the vertex grammar:

the parent	the neutral-class ε-sealed 3-juncture, $E_{\text{int}} = 6,\ Q = 0$ [BM.17; Def-BM.3(2); Census FT.35(2)]
the daughter	the unit-class ε-sealed 3-juncture, $E_{\text{int}} = 6,\ Q = +1$ [BM.26; Def-BM.3(3)]
the charged member	the ℓ-channel line, $q = -1$: forced and class-unique [BM.23(2)–(5) : $\Delta q = \pm 1$, the unique compensating value, the selection theorem]
the neutral member	the ν-channel line, $q = 0$: forced by (2)–(3); class-unique [FT.142(1); BM.23(5)'s eliminations]

the charge balance: $0 = (+1) + (-1) + 0$ [EM.11; BM.23(4); machine: integers];

the sealing phase: $\Theta_{\text{int}} \in 4\mathbb{Z}$ on both baryon-type junctions [BM.2(1); Census FT.49(3)];

the arrow: forward-only [BM.23(9); FT.144(2); A6 : $\neg\exists\tau^{-1}$];

the transition weight: $\varepsilon_q = 0 \implies \Gamma = 0$ exactly [BM.23(8)] : no flicker, no decay;

(2) the two-body impossibility (exact):

$$m_n = m_p = \frac{6\ln 2}{\hbar}$$

[BM.17; BM.26; BM.28(1) charge-blindness; BM.34(7)/BM.35(4) the rest slice];

the rest frame, two members: $E_p + E_\ell = m_n$,

$E_i = \sqrt{m_i^2 + p^2}$ [BM.34(4) vacuum gauge; BM.11] :

$$E_\ell = \frac{m_n^2 + m_\ell^2 - m_p^2}{2m_n},$$

$E_p = \dfrac{m_n^2 + m_p^2 - m_\ell^2}{2m_n}$ [machine: the exact solution of the conservation system];

$m_n = m_p$:

$$\boxed{E_p - m_p = -\frac{m_\ell^2}{2m_n} < 0 \ \ \forall m_\ell > 0}$$

[machine: cleared polynomial identity + sign census] :

the two-body vertex puts the daughter below its own mass shell for every positive member mass — $\neg\exists$ a real on-shell solution;

the Källén companion: $4m_n^2p^2 = \lambda\big(m_n^2, m_p^2, m_\ell^2\big)$,

$\lambda = \big(m_n^2 - (m_p{+}m_\ell)^2\big)\big(m_n^2 - (m_p{-}m_\ell)^2\big)$ [machine: polynomial identity];

$m_n = m_p : \quad \lambda = m_\ell^2\big(m_\ell^2 - 4m_n^2\big)$:

spacelike for $0 < m_\ell < 2m_n$;

$$m_\ell$$

$$\geq 2m_n$$

$$\Longrightarrow\ E_p$$

$\leq -m_n < 0$ [machine: the two branches cover every $m_\ell > 0$ ($2 > \sqrt{2}$ by squaring)];

(3) the forced neutral member:

by (2) every kinematically admissible vertex carries ≥ 3 members;

the charge balance fixes the third member's total charge to 0 [(1); EM.11];

the class is unique among the constructed carrier classes: the ν-channel is the unique EM-unactivated colour-blind bound-chain class [BM.23(5)'s elimination rows; FT.142(1)];

the transit (EM-radiation) alternative is ¬constructed: the gauge coupling of the Q read is deficit $1'$ [FT.125(4); Def-BM.8(8)] — ¬an inventory candidate;

the neutral ν-channel member is forced in every kinematically admissible vertex

[the forcing is kinematic $+$ uniqueness, ¬a number conservation : $\neg\exists$ a lepton-number read [BM.23(6)]];

the line index within the ν-triplet is ¬derived [¬a mixing matrix [FT.145(3)]];

BM.23(6) is upgraded on record: admissible $\longrightarrow$ conditionally forced [its Rmk];

(4) the Q-value sign theorem: $Q := m_n - m_p - m_\ell - m_\nu$;

(a) the crystallized informational read:

$Q = -(m_\ell + m_\nu)$ [machine: monomial identity at $m_n = m_p$];

$m_\ell > 0$ on every read species on record :

transport: $m_\ell^2 \geq M^2(A_0 - B_0) > 0$ [BM.39(1)];

informational: m_ℓ

$= \dfrac{E_{\text{int}} \ln 2}{\mathfrak{h}} \geq \dfrac{2 \ln 2}{\mathfrak{h}} > 0$ [FT.44(4); FT.28 the minimal floor; $\frac{1}{2} < \ln 2 < 1$ enclosure; machine];

$m_\nu \geq 0$ [EW.33 PSD];

$\implies \boxed{Q < 0 \text{ strictly}}$;

(b) every normalized ensemble read: charge-degeneracy is structural [FT.107(3); BM.30(3); BM.37(2) the disjointness]

$\implies Q$

$= -(m_\ell + m_\nu) < 0$;

(c) the curvature-flux read:

$RV = \varkappa^2 m$ with identical (E_{int}, V) on both classes [BM.20(1); BM.28(1)] $\implies Q < 0$;

(d) the route-(ii) variant: $T(0) = \frac{4}{3} T(\pm 1)$ [BM.28(4)(ii)] :

$Q_{\text{ii}} = \frac{1}{3} T(\pm 1) - m_\ell - m_\nu$:

the sign is undetermined without scale data [machine: two rational witnesses of opposite sign];

the functional is a measure price [BM.37(3)] and its adoption requires a §I addendum [BM.30(5)] :

priced, ¬adopted;

(e) the verdict:

the β-decay vertex is kinematically closed on every adopted read: the corpus does not make the neutron heavier than the proton

[P5 : ¬an external mass-difference datum; the blocks are theorems: BM.28(1), FT.107(3), BM.37(3), BM.30(5)];

(5) the three-body kinematics (the conditional section, under the hypothesis $Q \geq 0$ of some adopted specification):

the rest frame: $0 = \vec{p}_p + \vec{p}_\ell + \vec{p}_\nu$ [BM.11; BM.34(3)];

$$Q = \mathrm{KE}_p + \mathrm{KE}_\ell + \mathrm{KE}_\nu,$$

$$\mathrm{KE}_i = E_i - m_i$$

[BM.34(5); machine: Laurent identity];

$$\text{NR: } \mathrm{KE}_i = \frac{p_i^2}{2m_i} + O(p^4);$$

the endpoint:

$$\boxed{E_\ell^{\max} = \frac{m_n^2 + m_\ell^2 - (m_p + m_\nu)^2}{2m_n}}$$

[attained at the relative-rest recoil; machine: the exact polynomial identities];

$$E_\ell^{\max} + E_{\mathrm{rec}} = m_n, \quad E_{\mathrm{rec}} = \frac{m_n^2 + (m_p + m_\nu)^2 - m_\ell^2}{2m_n};$$

$$\left(E_\ell^{\max}\right)^2 - m_\ell^2 = E_{\mathrm{rec}}^2 - (m_p + m_\nu)^2 \text{ [machine: cleared by } 4m_n^2];$$

the recoil bound: $s_{p\nu} \geq (m_p + m_\nu)^2$:

the collinear identity $\big(E_pE_\nu - PN\big)^2$

$= (m_pm_\nu)^2 + \big(E_pN - E_\nu P\big)^2$ [machine: on-shell quotient ring];

the general case by Cauchy–Schwarz:

$$P^2N^2 - (\vec{p}_p \cdot \vec{p}_\nu)^2 = \sum_{i<j}\big(p_{p,i}p_{\nu,j} - p_{p,j}p_{\nu,i}\big)^2 \geq 0$$ [machine: the Lagrange identity];

$E_\ell^{\max}$ is strictly decreasing in $s_{p\nu}$ [the coefficient $-\frac{1}{2m_n} < 0$];

the continuum of E_ℓ values on $[E_\ell^{\min}, E_\ell^{\max}]$ is chart-side content [CT2.6 hypotheses at the point of use];

(6) the general channel closure:

any member set containing the daughter and at least one positive-mass chain line satisfies $\sum m_{\text{out}} > m_n$ [(2)'s equality of rest reads; machine: sign arithmetic] :

every material decay channel of the neutron row is kinematically closed on the adopted reads — the closure is a theorem about the reads, ¬about the vertex grammar [(1) stands complete];

(7) status and scope:

the vertex grammar (1) and the forcing (3) are derived;

the kinematic viability is read-relative (4);

the conditional section (5) carries its hypothesis on its face;

¬a measured lifetime, mass difference or endpoint enters [P5; machine: zero external tokens];

the row-index attachment (3) and the read choice $(4)(\mathrm{d})$ are the open faces.

Rmk (the shape of the negative.) The corpus assembles this vertex completely and then closes it — and the closing is as exact as the assembly. Two independent theorems do the work: the charge-blindness of every crystallized count read, which makes the two baryon rows weigh the same to the last monomial; and the structural charge-degeneracy of every normalized ensemble average,

which makes the flickering rows weigh the same too. The one functional that does split them was classified as a measure price, and adopting it as a mass is an act of specification the corpus prices but does not perform. So the reaction stands: grammar forced, members forced, conservation exact, kinematics closed — waiting on a read specification, not on a missing derivation.

Req. A6 (Part 1); D1.4, D2.9, D2.11 (Parts 1–2); EM.9, EM.11 (Part 2); P5 (Parts 1–2); CT2.6 (Part 2); FT.22, FT.107(3), FT.125(4), FT.142(1), FT.144(2), FT.145(3) (Part 3); Census FT.28, FT.35(2), FT.44(4), FT.49(3) (row-wise); Def-BM.3, Def-BM.6, Def-BM.8, Def-BM.10, BM.2(1), BM.11, BM.17, BM.23(2), BM.23(4), BM.23(5), BM.23(6), BM.23(8), BM.23(9), BM.26, BM.28(1), BM.28(4), BM.30(3), BM.30(5), BM.34(3), BM.34(4), BM.34(5), BM.34(7), BM.35(4), BM.36(5), BM.37(2), BM.37(3), BM.39(1), BM.40(3) (this part); Källén polynomial identities, exact sign arithmetic over positive witnesses, the collinear and Lagrange Cauchy–Schwarz identities, exact rational $\ln 2$ enclosures [math; machine: `checks/verify_H_neutron_decay_vertex.py` §C].

Pf.

$$(1):\ \text{BM.23}(2)\text{–}(5)\ \text{re-cited (frozen)};$$

$$0 = 1 + (-1) + 0\ \text{[machine]};$$

$$\Theta_{\text{int}} \in 4\mathbb{Z}\ \text{[Census FT.49}(3)\ \text{re-cited]};$$

$$\Gamma(\varepsilon_q{=}0) = 0\ \text{[BM.23}(8)\ \text{re-cited]}$$

$(2):$ the conservation system $\{E_p + E_\ell = m_n;\ E_\ell^2 - m_\ell^2 = E_p^2 - m_p^2\}$ solved linearly in (E_p, E_ℓ) [machine: substitution verifies both equations as polynomial identities over $\mathbb{Q}[m_n, m_p, m_\ell]$];

$$m_n = m_p: \quad 2m_n(E_p - m_p) = (m_n^2 + m_p^2 - m_\ell^2) - 2m_n m_p = -m_\ell^2\ \text{[machine: cleared identity]};$$

$$\lambda\ \text{expansion and the}\ m_n{=}m_p\ \text{factorization [machine]};$$

$$(2m_n)^2 - m_\ell^2\ \text{branch cover:}\ \sqrt{2} < 2\ \text{[squaring comparison]}$$

$$(3):\ (2)\ \text{conjoined with}\ (1)\text{'s balance};$$

$$\text{the elimination rows and deficit}\ 1'\ \text{re-cited};$$

$$\text{the upgrade recorded in BM.23's Rmk}$$

$$(4)(\text{a}):\ m_n - m_p = 0\ \text{[machine: monomial identity at}\ E_{\text{int}} = 6\ \text{both rows}];$$

$m_\ell > 0$ on both species [BM.39(1)'s SOS row; FT.28/FT.44(4) with $\frac{1}{2} < \ln 2$ enclosure; machine];

(b), (c) : degeneracy rows re-cited;

(d) : witnesses T=3,

m_ℓ=m_ν=$\frac{1}{8}$ (Q_{ii}=$\frac{3}{4}$>0) and T=3,

m_ℓ=m_ν=1 (Q_{ii}=−1<0) [machine: exact Fractions]

(5) : the endpoint identities cleared by $4m_n^2$ [machine];

the collinear identity in $\mathbb{Q}[P, N, m_p, m_\nu, E_p, E_\nu]/(E_p^2-m_p^2-P^2,\ E_\nu^2-m_\nu^2-N^2)$ [machine];

the Lagrange identity [machine: 6 variables];

$Q = \sum \text{KE}_i$ [machine: Laurent]

(6) : $\sum m_{\text{out}} \geq m_p + m_\ell > m_n = m_p$ [machine: sign arithmetic on rational witnesses];

(7) : the scope record. ■

Phys. The vertex is assembled from necessities the corpus already proved, and then weighed. The grammar: the ladder rule moves the ternary read by one, the alphabet overflows, total charge vanishes in every valid universe, one compensating value exists, and the inventory leaves exactly one charged class and one neutral class standing. The weighing is where the corpus surprises itself honestly: every rest read it has ever derived gives the neutron and the proton the same mass — six bits over ħ, identical to the last monomial, because no crystallized count carries a charge exponent and every normalized average is structurally charge-blind. A two-body decay with equal parent and daughter masses strands the daughter below its own mass shell for any positive emitted mass, so the neutral member is forced wherever the vertex is kinematically admissible at all; and admissibility itself fails on every adopted read, since the Q-value collapses to minus the lepton masses — strictly negative. The one functional that splits the classes was classified as a measure price; adopting it is a specification act the corpus prices and declines. What remains is exact and complete: a conditional three-body kinematics with its endpoint formula, a general channel-closure theorem, and a verdict — the reaction is grammatically forced and kinematically closed, and the gap between them is a read specification, not a missing derivation.

Theorem 4.35 (BM.42 — The angular momentum at the vertex: the Noether moment, the Lagrange decomposition, the absorption adjudication, and the ℓ-selection census). Thm.

$$(1) \quad \text{the moment current is conserved:} \qquad M^{\mu\nu\lambda} := x^\nu T^{\mu\lambda} - x^\lambda T^{\mu\nu};$$

$$\partial_\mu M^{\mu\nu\lambda} = \left(T^{\nu\lambda} - T^{\lambda\nu}\right) + x^\nu \partial_\mu T^{\mu\lambda} - x^\lambda \partial_\mu T^{\mu\nu}$$

[Leibniz; machine: the formal identity];

$T^{\text{Hilb}}_{\mu\nu}$ is symmetric [BM.9(2); machine: the dust witness $\text{diag}(\rho, 0, 0, 0)$] and on-shell conserved [BM.9(3); BM.11 : $\nabla_\mu T^{\mu\nu} = 0$ two routes] :

$$\boxed{\partial_\mu M^{\mu\nu\lambda} = 0 \ \text{ on-shell}}$$

[machine: all 64 components on the dust class] :

the total angular momentum is the conserved charge of the promoted rotation [BM.8(3) : $C_3 \to SO(3)$; Def-H.8 Rmk : spin coarse-grained along a geodesic is angular momentum];

$$(2) \quad \text{the budget at the vertex:}$$

$$J_{\text{in}} = J_{\text{out}} \ \text{ as the charge of } (1) :$$

$$j_n = j_p \oplus J_{\ell\nu} \oplus L_{\text{orb}} \ \text{ in } SO(3) \text{ coupling ranges } |j_1 - j_2| \le J \le j_1 + j_2 \text{ [math]};$$

the numerical j-attachments ride Def-H.38's declared rule [(7)];

the budget itself is the exact representation row;

$$(3) \quad \text{the orbital content is already inside } p^2:$$

the Lagrange identity:

$$r^2\, \delta^{ij} p_i p_j = (\vec{x} \cdot \vec{p})^2 + \sum_{i<j} \left(x_i p_j - x_j p_i\right)^2$$

[machine: exact polynomial identity in $\mathbb{Q}[x, y, z, p_x, p_y, p_z]$];

$$p^2 = p_r^2 + \frac{L^2}{r^2}, \qquad p_r := \frac{\vec{x}\cdot\vec{p}}{r}, \quad L^2 := \sum_{i<j} L_{ij}^2, \quad L_{ij} := x_i p_j - x_j p_i;$$

the dispersion in decomposed form:

$$r^2E^2 = r^2m^2 + r^2p_r^2 + L^2$$

[machine: the cleared substitution into BM.34(4)'s master identity, vacuum gauge] :

the orbital kinetic energy needs no separate term — it is BM.34's p^2 itself;

(4) the spin-absorption adjudication:

spin is the weak sector's contribution [Def-H.8];

its two derived metric couplings:

(i) the rest read: the pole mass is the spin-connection holonomy defect m^2_{pole}

$= 4\hat{\mu}'^2 \sin^2(\Theta_R/4)$ [Def-H.37(2); Def-H.21; FT.54(3)–(4)] : the frame coupling is inside m_i;

(ii) the moving body: the shift drag $N^i p_i$,

and the shift is the weak block [BM.34(3)–(4); FT.157(1)] : inside the dispersion;

the Casimir identity $\sum_i W_i^2$

$= 2\Pi_{\mathcal{R}}$ [Cor CT0.7a; BM.40(1); machine re-derived] is a branch-count read :

$\neg\exists$ a theorem constructing a scalar spin–metric coupling energy beyond (i), (ii)

[inventory];

a separate E_{spin} term in Q would double-count [machine: the budget closes — $Q = \sum_i \text{KE}_i$ follows from the conservation laws alone, with no residual];

the frequency-read row stays on record [FT.131(3); FT.159's rate-modulation residue untouched];

(5) the Q-identity with the absorptions:

$$\boxed{Q = m_n - m_p - m_\ell - m_\nu = \text{KE}_p + \text{KE}_\ell + \text{KE}_\nu},$$

$$\text{KE}_i = N\Big(\sqrt{m_i^2 + p_{r,i}^2 + \frac{L_i^2}{r_i^2}} - m_i\Big);$$

[BM.34(4)–(5); (3); machine: Laurent identity];

$E_{\text{spin-coupling}} = E_{\text{orbital-coupling}} = 0$ as separate terms [(3)–(4) absorbed];

BM.41(4)'s sign theorem is untouched — the absorptions add no term to the budget;

(6) the Carruthers–Nieto adjudication — the correction is zero:

$$\Delta L_z \, \Delta(\sin\theta) \geq \tfrac{1}{2} |\langle \cos\theta \rangle|$$

[BM.10(5); the naive $\Delta\theta \, \Delta L \geq \frac{1}{2}$ refuted; witnesses re-cited];

the kinetic corollary at $r = \Delta x_{\text{B}}$:

$$\langle p^2 \rangle \;\geq\; \frac{\langle L^2 \rangle}{\Delta x_{\text{B}}^2} \;\geq\; (\Delta L_z)^2 \, m_{\text{hid}}^2 \;\geq\; \tfrac{1}{4} \langle \cos\theta \rangle^2 \, m_{\text{hid}}^2$$

[(3); BM.13(4); machine];

BM.34(6)'s Cartesian row: $\langle p^2 \rangle \geq \frac{3}{4} m_{\text{hid}}^2$;

$$\tfrac{1}{4} \langle \cos\theta \rangle^2 \leq \tfrac{1}{4} < \tfrac{3}{4}$$

[machine: exact rationals] :

the CN corollary is uniformly weaker — the correction to the zero-point bounds $\left(\frac{7}{4},\ \frac{3}{8},\ \frac{\sqrt{7}}{2}\right)$ is exactly zero

(7) the spin ladder and the anchor census:

under Def-H.38 (the declared multiplicity rule $\text{copies}(c) = k(c)$) : a winding-k chain carries $\text{Sym}^k(\mathcal{H}_\text{v})$,

$$j = \frac{|k|}{2}\ \text{[FT.55(2);\ FT.60(1)(iv)]};$$

the Casimir ladder: $j(j{+}1) = \frac{k}{2}\left(\frac{k}{2}{+}1\right)$ generalizing Cor CT0.7a

[FT.60(3); machine: the ladder values $\left(\frac{3}{4}, 2, \frac{15}{4}, \dots\right)$ exact rationals];

the anchor:

$k = 1$ (the digon [FT.28; FT.107(1)]) : $\quad \text{Sym}^1 = \mathcal{H}_\text{v} \cong \mathbb{C}^2,$

$$j = \tfrac{1}{2}, \quad \textstyle\sum_i J_i^2 = \frac{3}{4}\mathbb{I}_2$$

[EW.32; machine];

the Platonic spin column stands under its logged declaration [Census FT.63(1) : $j_\text{max} \in \{6, 6, 12, 30, 30\}$; row-wise];

the line-winding attachment — which k each named line carries — is ¬computed: the ledger mechanism assigning Sym^k to materialized chains is the residue [FT.55(4)];

(8) the selection census (the anchor class, all four lines at $j{=}\frac{1}{2}$):

$$2 \otimes 2 \otimes 2 = 4 \oplus 2 \oplus 2, \quad j \in \{\tfrac{3}{2}, \tfrac{1}{2}\}$$

[FT.88(7); machine: the charpoly of $(\sum J_i)^2$ on $\mathbb{C}^8 = (x - \frac{15}{4})^4(x - \frac{3}{4})^4$];

the lepton pair: $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$

[machine: the charpoly on $\mathbb{C}^4 = x^3(x - 2)$];

the full vertex: $\quad 2^{\otimes 4}: \;\; 2\cdot(j{=}0)\oplus 3\cdot(j{=}1)\oplus 1\cdot(j{=}2), \quad 2+9+5=16$

[machine: the charpoly on $\mathbb{C}^{16}=x^2(x{-}2)^9(x{-}6)^5$];

the ℓ-coupling census ($J_{\text{int}} := j_p \otimes (\ell\text{-channel} \otimes \nu\text{-channel})$, $j_n=\frac{1}{2}$) :

pair spin S	$J_{\text{int}} \in$	$\ell = 0$	$\ell \geq 1$
0 (singlet)	$\{\frac{1}{2}\}$	✓ ($J{=}\frac{1}{2} \ni \frac{1}{2}$)	$\ell{=}1:$ ✓; $\quad \ell{=}2:$ ¬ ($\{\frac{3}{2},\frac{5}{2}\} \not\ni \frac{1}{2}$)
1 (triplet)	$\{\frac{1}{2},\frac{3}{2}\}$	✓ (both)	$\ell{=}1:$ ✓; $\quad \ell{=}2:$ ✓ via $J_{\text{int}}{=}\frac{3}{2}$

[machine: exact half-integer range census];

the leading admissible class: $\ell = 0$ with either pair-spin channel;

$$\ell$$

≥ 1 is the orbital ladder — the structural allowed/forbidden classification [¬SM labels imported];

(9) the parity column — the scope row:

the orbital factor $(-1)^\ell$ is derived: $\quad \chi_\ell(\pi) = \sum_{m=-\ell}^{\ell} (-1)^m = (-1)^\ell$ for $\ell = 0,\ldots,6$

[the integer $SO(3)$ character at angle π; machine: exact integers];

the lines' intrinsic parity labels are ¬derived: ¬∃ a Hilbert-space parity operator on the pair register [FT.135(4) stands];

parity acts at the label level as the branch-label flip ι [FT.135(1); EM.24(2)];

the four gradings are pairwise distinct [FT.136; FT.141(6); FT.144(5)] :

the J^P row-labels ($\frac{1}{2}^+$ and kin) are external vocabulary and are ¬imported [P5];

the derived content is the ℓ-ladder with its $(-1)^\ell$ factor, coupled by (8)'s census.

Rmk (budget, not labels). The vertex conserves the charge the promoted rotation supplies: the moment current closes on the symmetry of the Hilbert stress tensor and its on-shell conservation, both rows the Noether tier proved. The budget's two would-be extra terms dissolve on inspection — the orbital piece is p^2 itself by Lagrange, and the spin piece is the frame holonomy inside the rest read plus the weak shift drag inside the dispersion; a separate spin-coupling energy would count the same coupling twice, and the Carruthers–Nieto refinement of the angular spread is uniformly weaker than the Cartesian zero-point row, so the bounds stand uncorrected. What the corpus does not do is attach j-numbers to named lines beyond the anchor class: the census computes the representation structure — two singlets, three triplets, one quintet across the fourfold vertex — and the line-data attachment waits on the winding residue.

Req. A3 (Part 1); D1.8 (Parts 1–2); CT0.7(1), CT1.2, CT1.4, Cor CT0.7a (Part 1); CT2.6 (Part 2); P5 (Parts 1–2); Def-H.8, Def-H.21, Def-H.38, FT.54(3), FT.54(4), FT.55(2), FT.55(4), FT.60(1), FT.60(3), FT.88(7), FT.101, FT.107(1), FT.131(3), FT.135(1), FT.135(4), FT.136, FT.141(6), FT.144(5), FT.157(1) (Part 3); Census FT.28, FT.63(1) (row-wise); Def-BM.10, BM.8(3), BM.9(2), BM.9(3), BM.10(5), BM.11, BM.13(4), BM.34(3), BM.34(4), BM.34(5), BM.34(6), BM.40(1), BM.41(4), BM.41(5) (this part); the Leibniz/jet identities, the Lagrange identity, Clebsch–Gordan range censuses, charpolys by Faddeev–LeVerrier over $\mathbb{Q}$, exact half-integer arithmetic [math; machine: `checks/verify_H_neutron_decay_vertex.py` §D].

Pf.

$$(1):\ \partial_\mu(x^\nu T^{\mu\lambda}) = (\partial_\mu x^\nu)T^{\mu\lambda} + x^\nu \partial_\mu T^{\mu\lambda} = T^{\nu\lambda} + x^\nu \partial_\mu T^{\mu\lambda} \text{ [Leibniz]};$$

$$\text{the dust witness: } T$$

$$= \text{diag}(\rho, 0, 0, 0),$$

$$\rho \text{ constant}:\ \partial_\mu M^{\mu\nu\lambda}$$

$$= T^{\nu\lambda} - T^{\lambda\nu}$$

$$= 0 \text{ all } 64 \text{ components [machine: symbolic differentiation in } \mathbb{Q}[x_0, x_1, x_2, x_3, \rho]]$$

$$(2):\ \text{the coupling ranges are the } SO(3) \text{ representation rows [math]};$$

$$(3):\ r^2p^2 - (\vec{x}\cdot\vec{p})^2 = \sum_{i<j}(x_i p_j - x_j p_i)^2 \text{ [machine: expansion in 6 variables]};$$

$$\text{the substitution into BM.34(4)'s identity at } N{=}1, N^i{=}0, \gamma{=}\delta, \text{ cleared by } r^2 \text{ [machine]}$$

$$(4):\ \text{Def-H.37(2)/Def-H.21 re-cited};$$

BM.34(3)–(4)/FT.157(1) re-cited;

the inventory scan: no scalar spin–metric coupling term exists in Parts 1–4 [machine: token scan of the energy-budget rows];

the budget closure: $m_n = E_p + E_\ell + E_\nu \implies Q = \sum (E_i - m_i)$ [machine: Laurent]

(5) : (3)'s decomposition substituted into BM.34(5)'s KE definition [machine]

(6) : $(\Delta L_z)^2 \geq \frac{1}{4}\langle \cos\theta \rangle^2$ [BM.10(5) at $\Delta(\sin\theta) \leq 1$];

$\langle L^2 \rangle \geq (\Delta L_z)^2$ at $\langle L_z \rangle = 0$;

$r = \Delta x_{\mathrm{B}} = 1/m_{\mathrm{hid}}$ [Def-BM.4(1); BM.13(4)];

$\frac{1}{4} < \frac{3}{4}$ [machine]

(7) : Def-H.38's derived-content rows re-cited;

the anchor: $J_i = \frac{1}{2}\sigma_i$: $\sum_i J_i^2 = \frac{1}{4}\sum_i \sigma_i^2 = \frac{1}{4} \cdot 3\,\mathbb{I}_2 = \frac{3}{4}\mathbb{I}_2$ [machine over $\mathbb{Q}(i)$];

the ladder values $\frac{k}{2}(\frac{k}{2}+1)$

$= (\frac{3}{4}, 2, \frac{15}{4}, 6, \dots)$ at k

$= 1,$

$2,$

$3,$

4 [machine: exact rationals]

(8) : $(\sum_i J_i^{(2)})^2$ on $\mathbb{C}^4$: charpoly $x^3(x-2)$ [machine: Faddeev–LeVerrier];

on $\mathbb{C}^8$: $(x - \frac{15}{4})^4 (x - \frac{3}{4})^4$;

$$\text{on } \mathbb{C}^{16}:\ x^2(x-2)^9(x-6)^5 \text{ [machine: exact charpolys over } \mathbb{Q}];$$

$$\text{the } \ell\text{-ranges: } |J_{\text{int}}-\ell| \le \tfrac{1}{2} \le J_{\text{int}}+\ell \text{ censused on half-integers [machine]}$$

$$(9):\ \sum_{m=-\ell}^{\ell}(-1)^m = (-1)^\ell \text{ for } \ell = 0..6 \text{ [machine: integer sums]}; \qquad \text{FT.135(4) re-cited. } \blacksquare$$

Phys. Angular momentum at the vertex is a theorem before it is a table. The moment of the Hilbert stress tensor is conserved wherever the tensor is symmetric and its own divergence vanishes on-shell — both rows the Noether tier proved — so the charge the promoted rotation carries balances across the decay exactly. The budget then needs no new terms: the orbital share is already the momentum square, split by Lagrange's identity into radial and angular pieces with nothing left over, and the spin share lives inside the rest mass as frame holonomy and inside the dispersion as the weak shift's drag. Asked for a spin-coupling energy to add to the Q-value, the corpus answers that adding one would double-count, and the angular uncertainty refinement of the zero-point bound comes out uniformly weaker — a quarter against three quarters — so the bounds stand. What is genuinely new is the census: at the anchor winding, four spin-half lines tensor into two singlets, three triplets and one quintet; the lepton pair couples to singlet or triplet; and the orbital ladder admits zero and one against both pair channels, two only through the triplet route. The parity factor of the ladder is the integer character of the rotation group, while the parity labels of the lines themselves stay outside: the register has no parity operator, and the tier imports no external J^P vocabulary.

Theorem 4.36 (BM.43 — The curvature rows of the members: the dust sources, the weak-field potential, and the superposition scope). Thm.

$$(1)\quad \text{the baryon rows (the exact monomials):}$$

$$E_{\text{int}} = 6 \text{ for both classes [BM.17; BM.26; Census FT.35(2)]}:$$

$$\Delta x_{\text{B}} = \frac{\mathfrak{h}}{6\ln 2},$$

$$V_{\text{eff}} = \frac{\mathfrak{h}^3}{216(\ln 2)^3},$$

$$m = \frac{6\ln 2}{\mathfrak{h}},$$

$$\rho = \frac{1296(\ln 2)^4}{\mathfrak{h}^4},$$

$$\kappa^2\rho = \frac{2592\pi(\ln 2)^3}{\mathfrak{h}^2},$$

$$r_s = 3\,\mathfrak{h};$$

the lock: $\quad (\Delta x_{\mathrm{B}})^2\,\kappa^2\rho = 2\pi E_{\mathrm{int}}^2 \ln 2 = 72\pi \ln 2$

[BM.26; BM.16(1); machine: every row one exact Laurent monomial];

the parent and the daughter carry identical rows [BM.28(1) charge-blindness];

(2) the lepton rows (symbolic):

the taxonomy law holds: $\quad m_i = \dfrac{E_{\mathrm{int}}^{(i)} \ln 2}{\mathfrak{h}}$ [FT.44(4)];

$E_{\mathrm{int}}^{(\ell)},\ E_{\mathrm{int}}^{(\nu)}$ are ¬computed at any floor :

the rows stand as symbolic monomials: $\quad \kappa^2\rho_i = \dfrac{2\pi\,\mathfrak{h}\,E_{\mathrm{int}}^{(i)}}{V_i},$

$$r_s^{(i)} = \frac{\mathfrak{h}\,E_{\mathrm{int}}^{(i)}}{2}$$

[BM.9(5); BM.16(4); machine];

the transport alternative: $\quad m_i$ from BM.39 (symbolic in $(g, g', \hat{\mu}, \ell_c)$);

the conversion $E_{\mathrm{int}}^{(i)} \leftrightarrow$ transport data is item-7 residue [Def-BM.11(7); pointer];

(3) the field-equation rows:

$T_{\mu\nu} = \mathrm{diag}(\rho, 0, 0, 0)$ at rest, $\ p = 0$ [BM.9(4); CT4.5];

$R = \kappa^2(\rho - 3p) + 4\Lambda$ [BM.15(2)];

$RV = \varkappa^2 m$ [BM.20(1)];

(4) the weak-field potential (the linearized row):

$$g_{00} = -(1+2\Phi), \quad g_{ij} = (1-2\Phi)\delta_{ij}, \quad \Phi \text{ static :}$$

$$G^{(1)}_{00} = 2\,\nabla^2\Phi,$$

$$G^{(1)}_{0i} = 0,$$

$$G^{(1)}_{ij} = 0$$

[machine: Christoffel → Ricci → Einstein in the weight-truncated jet ring, exact rational coefficients];

the field equation $G_{\mu\nu} = \kappa^2 T_{\mu\nu}$ [CT1.23] at $p = 0$:

$$\nabla^2\Phi = \frac{\kappa^2}{2}\,\rho = 4\pi G_N \rho$$

[$\kappa^2 = 8\pi G_N$ [CT2.9]; machine: the coefficient identity $\frac{\kappa^2}{2} = \frac{\pi\mathfrak{h}^2}{\ln 2} = 4\pi G_N$];

$$\Phi = -\frac{G_N m}{r} \text{ solves at } r \neq 0 :$$

$$\nabla^2 \frac{1}{r} = 0$$

[machine: $-3r^2 + 3(x^2{+}y^2{+}z^2) \equiv 0$ in $\mathbb{Q}[x,y,z,r]/(r^2{-}x^2{-}y^2{-}z^2)$];

$$G_N m_{\text{hid}} = \frac{\mathfrak{h}\,E_{\text{int}}}{4},$$

$$r_s = 2G_N m_{\text{hid}} = \frac{\mathfrak{h}\,E_{\text{int}}}{2},$$

$$\Phi(\Delta x_{\text{B}}) = -\frac{E^2_{\text{int}} \ln 2}{4} \in \mathfrak{q}^0$$

[CT2.9; BM.16(4); machine: monomials];

(5) the superposition:

source side, exact:

$T_{\text{total}} = \sum_i T^{(i)}$ on disjoint supports [CT1.22 channel decomposition; D1.5] :

$$\kappa^2 \rho_{\text{total}}$$

$$= \kappa^2 \big(\rho_p + \rho_\ell + \rho_\nu\big)$$

$$= \frac{2\pi\mathfrak{h}}{V}\big(6 + E_{\text{int}}^{(\ell)} + E_{\text{int}}^{(\nu)}\big) \text{ [machine: the monomial sum]};$$

metric side: superposition is exact only at linearized order [CT1.12 : $\varkappa\|q\|_{\text{op}} < 1$ the perturbative regime];

the full non-linear joint solution is ¬constructed;

the scope row: the Schwarzschild-type content on record is the r_s read and the weak-field potential [(4); BM.16(4)];

a full exterior solution is ¬in the inventory [the pp-wave witness is the exact curvature row on record];

(6) status:

the baryon rows are numerical monomials;

the lepton rows symbolic;

the vertex curvature total is their exact sum on the source side;

¬a measured curvature or radius enters [P5].

Req. D1.5, D2.9 (Parts 1–2); CT1.12, CT1.22, CT1.23, CT2.1, CT2.9, CT4.5 (Part 2); P3, P5 (Parts 1–2); FT.44(4) (Census); Def-BM.4, Def-BM.10, Def-BM.11, BM.9(4), BM.9(5), BM.15(2), BM.16(1), BM.16(4), BM.17, BM.20(1), BM.26, BM.28(1), BM.34(4), BM.39 (this part); the weight-truncated jet-ring calculus, Laurent monomials, harmonic identities in quotient rings [math; machine: `checks/verify_H_neutron_decay_vertex.py` §E].

Pf.

(1) : BM.26's six rows re-verified as monomials at E_{int}

$$= 6 \text{ [machine]} : \quad \rho$$

$$= \tfrac{6\ln 2}{\mathfrak{h}} \cdot \tfrac{216(\ln 2)^3}{\mathfrak{h}^3}$$

$$= \tfrac{1296(\ln 2)^4}{\mathfrak{h}^4};$$

$$\kappa^2\rho$$

$$= \tfrac{2\pi\mathfrak{h}^2}{\ln 2}\cdot\tfrac{1296(\ln 2)^4}{\mathfrak{h}^4}$$

$$= \tfrac{2592\pi(\ln 2)^3}{\mathfrak{h}^2};$$

the lock [machine]

$(2):$ FT.44(4)'s law re-cited; the symbolic monomials [machine]

$(3):$ frozen rows re-cited

$(4):$ the jet-ring computation: $g^{\mu\nu}g_{\nu\rho}=\delta^\mu_\rho$ to first order [machine];

$$\Gamma^i_{00}=\partial_i\Phi,\ \ \Gamma^0_{i0}=\partial_i\Phi,\ \ \Gamma^k_{ij}=-\partial_i\Phi\,\delta^k_j-\partial_j\Phi\,\delta^k_i+\partial^k\Phi\,\delta_{ij}$$ [machine];

$$R_{00}=\nabla^2\Phi,$$

$$R_{ij}=\delta_{ij}\nabla^2\Phi,$$

$$R=2\nabla^2\Phi\implies G_{00}=2\nabla^2\Phi,\ G_{ij}=0$$ [machine: weight- ≤ 1 truncation];

$$\tfrac{\kappa^2}{2}=\tfrac{\pi\mathfrak{h}^2}{\ln 2}=4\pi\cdot\tfrac{\mathfrak{h}^2}{4\ln 2}$$ [CT2.9; machine]

$$\nabla^2(r^{-1})\cdot r^5=-3r^2+3\textstyle\sum_i x_i^2\equiv 0$$ [machine: the rule ring];

$$G_Nm=\tfrac{\mathfrak{h}^2}{4\ln 2}\cdot\tfrac{E\ln 2}{\mathfrak{h}}=\tfrac{\mathfrak{h}E}{4};$$

$$r_s=2G_Nm=\tfrac{\mathfrak{h}E}{2};\quad \Phi(\Delta x_{\mathrm{B}})=-\tfrac{\mathfrak{h}E}{4}\cdot\tfrac{E\ln 2}{\mathfrak{h}}=-\tfrac{E^2\ln 2}{4}$$ [machine]

$(5):$ CT1.22's decomposition re-cited;

the monomial sum [machine];

$(6):$ the status record. ■

Phys. Every member of the vertex gravitates, and the corpus writes each gravity in monomials. The two baryon rows are numbers: six bits over $\mathfrak{h}$, a density of twelve ninety-six $(\ln 2)^4$ over $\mathfrak{h}^4$, a horizon read of three $\mathfrak{h}$, a lock of seventy-two $\pi \ln 2$ — identical for parent and daughter, because no crystallized count carries charge. The lepton rows keep their edge counts symbolic, for the floors that would compute them are not in the census, and the transport species offers the alternative symbolic form. The field side is linearized and exact: the static diagonal perturbation solves the ledger's own field equations with the potential obeying the Poisson row whose coefficient is the corpus's $4\pi G_N$ to the last factor, and the point-source form is harmonic away from the source by a polynomial identity. Superposition is exact where the corpus can prove it — the source side, where stress tensors add — and scoped where it cannot: the full non-linear metric of three flying members is an open construction, and the tier says so rather than dress a linearized row in exact clothes.

Theorem 4.37 (BM.44 — The time evolution: the per-tick weight, the e-folding count, and the unit discipline). Thm.

$$(1) \quad \text{the frozen rate rows:}$$

$$\sum_n \langle \varnothing | \hat{U}^n | \varnothing \rangle w^n = \frac{1}{1 - w^2 x_+ x_-} \ \text{ in } \mathbb{Q}[w, x_+, x_-]/(w^{12})$$

$$[\text{FT.108}(5);\ \text{BM.23}(7);\ \text{machine: the resolvent identity re-verified coefficient-wise}];$$

$$x_+ x_- = 4\varepsilon_q(1 - 4\varepsilon_q) = \frac{1}{4} - 16\Big(\varepsilon_q - \frac{1}{8}\Big)^2 \le \frac{1}{4},$$

$$\text{equality} \iff \varepsilon_q = \frac{1}{8}$$

$$[\text{BM.19}(2);\ \text{machine: the SOS completion } +\ \text{the grid census over } \varepsilon_q = k/256];$$

$$\text{the pole sits at } |w| \ge 2;$$

$$\varepsilon_q$$

$$= 0$$

$$\implies x_+ x_-$$

$$= 0 \implies \text{the resolvent is } 1 : \ \text{no transition weight beyond the identity } [\text{BM.23}(8);\ \text{machine}];$$

(2) the unit-fugacity slice:

$$\Gamma_{\text{tr}}(\varepsilon_q) := \frac{x_+x_-}{1-x_+x_-}\Big|_{w=1}$$

[the transition content of the resolvent] :

$$\Gamma_{\text{tr}}(0) = 0,$$

$$\Gamma_{\text{tr}}\big(\tfrac{1}{8}\big) = \frac{1/4}{3/4} = \frac{1}{3},$$

Γ_{tr} strictly increasing on $\big[0, \tfrac{1}{8}\big]$

[machine: $\Gamma_{\text{tr}}(\varepsilon_2) - \Gamma_{\text{tr}}(\varepsilon_1) > 0$ via $4(\varepsilon_2 - \varepsilon_1)\big(1 - 4(\varepsilon_1 + \varepsilon_2)\big) > 0$];

the exact range: $\Gamma_{\text{tr}} \in \big[0, \tfrac{1}{3}\big]$ on $\varepsilon_q \in \big[0, \tfrac{1}{8}\big]$;

(3) the chart evolution (the scoped specialization):

under BM.8(2)'s declared stationarity hypothesis the chart carries the time-translation one-parameter group [BM.8(2); BM.34(4) : $E = -p_0$ its charge] :

$$\frac{d}{d\tau}P(\tau) = -\tilde{\Gamma}\,P(\tau),$$

$$P(\tau) = P_0\, e^{-\tilde{\Gamma}\tau}$$

[the group's exponential; $\tilde{\Gamma}$ the per-tick weight at the chart scale];

the e-folding count: $\tau_{\text{life}} := \frac{1}{\tilde{\Gamma}}$ ticks; $\tilde{\Gamma} = \Gamma_{\text{tr}} \implies \tau_{\text{life}} \geq 3$ ticks, $\tau_{\text{life}}\big|_{\varepsilon_q = 1/8} = 3$ exactly [machine: exact reciprocals; $\Gamma_{\text{tr}} \leq \frac{1}{3}$ from (2)];

$\varepsilon_q = 0$: $\Gamma_{\text{tr}} = 0 \implies \tau_{\text{life}}$ has no finite value :

a non-flickering composite never decays [BM.23(8) re-cited];

(4) the ledger-native form:

Γ is a weight per tick in the ledger order [$x^0 := \tau_n$; D1.8; BM.23(9); FT.159];

the discrete evolution is the geometric resolvent of (1);

the normalization of the weight to a per-tick probability is ¬declared [BM.23 Rmk : Γ is a weight, ¬a probability] :

(3)'s $\tilde{\Gamma}$ is the chart specialization, scoped and marked;

(5) the unit discipline:

¬external seconds;

Cal-F carries no rate datum (μ, $\hat{\mu}^2$, $\hat{\mu}'^2$ only);

¬a lifetime is derived in external units and ¬a lifetime is compared [P5; BM.23(9)'s row stands];

FT.159's rate-modulation residue (the flicker-frequency read of $\delta\tau$ against Ω/N_i) is untouched;

(6) the arrow rows:

forward-only [A6 : $\neg\exists\tau^{-1}$; FT.144(2) : the prefix law; BM.23(9)];

¬∃ an inverse decay in the ledger;

$\Theta_{\mathfrak{L}}$ is the record involution (the formal mirror),

¬a dynamical reversal [Def-BM.7(2); FT.144(4)].

Rmk (weight against rate.) The corpus's decay clock ticks in ledger order and its hand is a rational function of one threshold. Everything exact is stated exactly: the resolvent, its pole beyond two, its maximum third at the threshold midpoint, its annihilation at the sharp threshold. What is not stated is a number of seconds, because no datum in the corpus converts a tick into one — and the tier keeps that silence loud: the exponential form is the chart group's own shape under the declared stationarity scope, the e-folding count is three ticks at the heaviest admissible flicker, and the measured lifetime is quoted nowhere.

Req. A6, D0.4, D1.8 (Part 1); CT0.8 (Part 1); P5 (Parts 1–2); FT.108(4), FT.108(5), FT.144(2), FT.144(4), FT.159 (Part 3); Def-BM.5, Def-BM.7(2), BM.8(2), BM.19(2), BM.19(3), BM.23(7), BM.23(8), BM.23(9), BM.34(4) (this part); Cal-F; the exact resolvent identity in $\mathbb{Q}[w, x_+, x_-]/(w^{12})$, the SOS parabola completion, exact rational monotonicity and reciprocals [math; machine: `checks/verify_H_neutron_decay_vertex.py` §F].

Pf.

$$(1):\ (1-w^2x_+x_-)\sum_{n\le 11}\left[x_+x_-\right]^{n/2}w^n\big|_{n\ \text{even}}\equiv 1\ \bmod w^{12}\ \text{[machine: coefficient-wise]};$$

$$4\varepsilon(1-4\varepsilon)=\tfrac{1}{4}-16(\varepsilon-\tfrac{1}{8})^2\ \text{[machine: expansion]};$$

$$\text{the grid census } k=0..63:\le\tfrac{1}{4},\ =\tfrac{1}{4}\ \text{only at } k=32\ \text{[machine]}$$

$$(2):\ \Gamma_{\text{tr}}=\tfrac{xx}{1-xx},\ xx:=x_+x_-:\quad \Gamma_{\text{tr}}(0)=0,\ \ \Gamma_{\text{tr}}(\tfrac{1}{8})=\tfrac{1/4}{3/4}=\tfrac{1}{3}\ \text{[machine]};$$

$$\text{monotonicity: } xx(\varepsilon_2)-xx(\varepsilon_1)$$

$$=4(\varepsilon_2-\varepsilon_1)(1-4(\varepsilon_1+\varepsilon_2))\ \text{[machine: polynomial identity]},$$

$$\text{positive on rational witness pairs } \varepsilon_1<\varepsilon_2\le\tfrac{1}{8}$$

$$(3):\ \text{BM.8(2)'s group re-cited};$$

$$\tfrac{1}{\Gamma_{\text{tr}}(1/8)}=3\ \text{[machine: exact reciprocal]};$$

$$\Gamma_{\text{tr}}\le\tfrac{1}{3}\implies\tau_{\text{life}}\ge 3\ \text{[machine: monotone reciprocal]}$$

$$(4)\text{–}(6):\ \text{frozen rows re-cited [D1.8; BM.23(9); FT.144(2), (4); A6]};$$

$$\text{the Cal-F inventory: } \{\mu,\hat{\mu}^2,\hat{\mu}'^2\},\ \neg\text{a rate [scan]}.\ \blacksquare$$

Phys. Time runs through the decay exactly as the corpus runs it everywhere else: in ticks, forward, with no inverse and no seconds. The transition weight is the geometric resolvent of the flicker — a rational function of one threshold, pole beyond two, annihilated at the sharp threshold where a composite that cannot flicker cannot decay. On the unit-fugacity slice the weight climbs from zero to a third as the threshold moves to its midpoint, and the e-folding count is its reciprocal: at least three ticks, exactly three at the heaviest admissible flicker. The exponential curve belongs to the chart's own translation group under the stationarity scope the Noether tier declared; the ledger keeps only the weights. And the discipline is the point: a lifetime in seconds would need a rate datum the corpus does not carry, so the tier states the tick count, names the missing datum, and quotes no clock.

Theorem 4.38 (BM.45 — The assembled reaction: the conservation table, the naming row, the final read, and the adjudication). Thm.

(1) the reaction (the structural row, with the identification-layer labels):

$$[\varepsilon\text{-sealed 3-juncture, neutral class, } E_{\text{int}}{=}6] \longrightarrow [\varepsilon\text{-sealed 3-juncture, unit class, } E_{\text{int}}{=}6] + [\ell\text{-channel line, } q{=}{-}1] + [\nu\text{-channel line, } q{=}0, \text{ the materialized-arrow orientation}];$$

the labels:

$$\text{neutron} \longrightarrow \text{proton} + \text{electron} + \text{electron antineutrino}$$

[Def-BM.3(2)–(3); Def-BM.6(1); naming row A-BM.5];

A-BM.5: electron antineutrino :

$$= \text{ the } \nu\text{-channel line in the CPT/}\omega\text{-image orientation [BM.25's involution; BM.27's antiproton pattern; FT.144(5) : the occupancy sector is derived, its name is reserved content]};$$

the charged member is Def-BM.6(1)'s electron row [BM.23(5)];

the neutral member's line index within the ν-triplet is ¬derived [FT.145(3)];

(2) the conservation table:

law	the row	status
charge	$0 = (+1) + (-1) + 0$	derived [EM.11; BM.23(4); machine]
Janus balance	$\#J^+ = \#J^-$ on every emitted closed chain	derived [D1.4; FT.144(1)–(2); BM.7]
rung-1 weak Casimir	branch-neutral at the $\sigma - vertex$; carriers internally balanced	derived [BM.40(3)–(4)]
angular momentum	$\partial_\mu M^{\mu\nu\lambda} = 0$ on-shell; the ℓ-census of BM.42(8)	derived [BM.42(1)–(2)]
sealing phase	$\Theta_{\text{int}} \in 4\mathbb{Z}$ on both baryon-type junctions	derived [BM.2(1); Census FT.49(3)]
energy–momentum	the chart-side law with the threshold row $Q \geq 0$	derived [BM.11; BM.34; BM.41(4)–(5)]
the arrow	forward-only; $\neg\exists$ an inverse decay	derived [A6; FT.144(2); BM.44(6)]

(3) the Q-value and the distribution:

$$Q = -(m_\ell + m_\nu) < 0 \text{ on every adopted read [BM.41(4)] :}$$

the reaction assembles and does not go on-shell in the corpus's own balance;

$$Q = \text{KE}_p + \text{KE}_\ell + \text{KE}_\nu \text{ with the orbital and spin content absorbed [BM.42(5)];}$$

the endpoint formula under the $Q \geq 0$ hypothesis [BM.41(5)];

the route-(ii) variant is priced and ¬adopted [BM.41(4)(d); BM.30(5)];

(4) the curvature rows:

per member [BM.43] : the baryons numerical ($r_s = 3\,\mathfrak{h}$, $\kappa^2\rho = 2592\pi(\ln 2)^3/\mathfrak{h}^2$, the lock $72\pi\ln 2$);

the leptons symbolic;

the source superposition exact;

the metric superposition linearized-scope;

(5) the time rows:

$\Gamma_{\text{tr}} \in [0, \frac{1}{3}]$ at unit fugacity;

$\tau_{\text{life}} \geq 3$ ticks;

$\varepsilon_q = 0 \implies$ never [BM.44];

¬external seconds;

(6) the standing rules:

¬a P4 addition [§Cal-BM stays empty; A-BM.4 is an angle identification plus a normalization convention, ¬a numerical parameter];

¬a new dimensionful scale [P1];

¬a measured value [P5; machine: zero external tokens, zero decimal numerals in this tier];

¬∃ an [EXT] datum in any derivation;

species names confined to the identification layers [Def-BM.3(7), Def-BM.6(7), A-BM.5];

topology labels are internal bookkeeping [Def-H.32(4)];

the extra-axiomatic list [A §0.2a] gains no member : CT2.6's hypotheses and BM.8(2)'s stationarity scope are inherited at the point of use;

(7) the open rows touched, not closed:

the relating equation across the read species [Def-BM.11(7); BM.35 Rmk stands];

the amplitude ↔ mass bridge [BM.37 stands];

the Sym^k ledger mechanism [FT.55(4); BM.42(7)];

deficit $1'$ — the gauge coupling of Q [BM.41(3)];

¬a mixing matrix is derived [FT.145(3); BM.41(3)];

the mass-mechanism inventory: the transport-species family form joins the read-class taxonomy [FT.84(1) pattern];

the read specification that would open the decay kinematics [BM.41(4)(e)] : a §I addendum act,

¬a derivation;

(8) the adjudication:

the Yukawa requirement	discharged — $\neg\exists$ a Yukawa term in the inventory, and the family-profile map is constructed from the seed transport [Def-BM.11; BM.39]
the intra-family masses	derived as exact forms — the cosine/affine law, positivity, the gg' splitting, the sum rule, the order inversion, the area ladder [BM.39]
the neutrino	forced in every kinematically admissible vertex — the two-body impossibility + the class uniqueness; BM.23(6) upgraded on record; the Casimir-deficit route adjudicated neutral [BM.40(3); BM.41(2)–(3)]
angular momentum	conserved as the Noether moment; orbital absorbed in p^2, spin absorbed in the rest read and the shift drag; the CN correction exactly zero; the ℓ-census derived [BM.42]
KE and rest mass	derived for every member from the metric and the vacuum-balanced entropy [BM.34; BM.35; BM.41(5); BM.42(5)]
the metric curvature	derived per member — numerical for the baryons, symbolic for the leptons, superposed exactly on the source side [BM.43]
the dynamic evolution	derived in ticks — $\Gamma_{\text{tr}} \in [0, \frac{1}{3}]$, $\tau_{\text{life}} \geq 3$ ticks, ¬external seconds [BM.44]
the reaction	$n \to p + e^- + \bar{\nu}_e$ assembled with every conservation row derived — and kinematically closed on every adopted read : $Q = -(m_\ell + m_\nu) < 0$ [BM.41(4)]

the final read:

the vertex is grammatically complete and kinematically shut;

what stands between the corpus and an on-shell decay is a read specification splitting the charge classes — a §I addendum act,

$$\neg\text{a missing derivation } [\text{BM.30}(5);\ \text{BM.37}(3)];$$

this tier performs no such act.

Rmk (the naming rows.) The label electron antineutrino attaches to the derived structural tuple — the ν-channel line in the orientation sector the materialization arrow selects — exactly as antiproton attaches to the CPT image of the proton row: the structure is derived, the occupancy sector is derived up to the reserved ray naming [FT.144(5)], and the label is logged before use (A-BM.5). Which of the three ν-triplet lines the vertex emits is not derived: that attachment is mixing content, and the corpus has no mixing matrix.

Req. P1, P4, P5 (Parts 1–2); A6, D1.4, D1.8, D2.9 (Parts 1–2); EM.11 (Part 2); CT2.6 (Part 2); Def-H.38, FT.135(4), FT.144(2), FT.144(5), FT.145(3), FT.84(1), FT.55(4) (Part 3); Census FT.35(2), FT.44(4), FT.49(3) (row-wise); Def-BM.3, Def-BM.6, Def-BM.7, Def-BM.11, BM.2(1), BM.7, BM.11, BM.23(5), BM.25, BM.26, BM.27, BM.34, BM.35, BM.37(3), BM.39, BM.40, BM.41, BM.42, BM.43, BM.44 (this part); the declaration addendum A-BM.5 (pointers); the machine sections §A–§G green [math; machine: `checks/verify_H_neutron_decay_vertex.py` §G].

Pf.

$$(1)\text{–}(7):\ \text{assembly of the adjudicated rows of Def-BM.11,}$$

$$\text{BM.39–BM.44 [machine: §A–§F green]}$$

$$(8):\ \text{conjunction of } (1)\text{–}(7). \blacksquare$$

Phys. The corpus writes its first complete reaction, and the honesty of the balance is the result. Every conservation row is a theorem: charge closes on the ternary alphabet, the Janus balance rides on every emitted closed chain, the weak Casimir passes through the vertex untouched because registration signs and branch labels live in different registers, angular momentum is the conserved moment of a symmetric on-shell stress tensor, the sealing phase stays in its baryon class, and the arrow points one way. The members are forced: the charged one by the ladder rule and the alphabet, the neutral one by the impossibility of a two-body vertex at equal rest masses and the uniqueness of its class. The masses are exact where the corpus computes them — six bits over $\mathfrak{h}$ for both baryons — and symbolic where it does not. And then the balance itself: parent and daughter weigh the same on every read the corpus adopts, so the Q-value is minus the lepton masses, strictly negative, and the reaction does not go on-shell. The gap between the assembled vertex and a running decay is not a missing mechanism; it is a read specification that would split the charge classes, priced since Tier BM.8 and declined here as everywhere. The neutron decay vertex is the corpus in miniature: complete grammar, exact arithmetic, forced members, closed kinematics, and no external number anywhere in it.

4.14 Tier BM.12 — The EM Self-Interaction Correction: the Scar's Field Energy, the Infinite-Range Seal Asymmetry, the Mass-Splitting Theorem, and the Beta-Spectrum Adjudication

Definition 4.12 (Def-BM.12 — The chart-side EM source read of a sealed composite (the construction Def-BM.8(8) specifies)). Def. A read constructed on the chart side of a sealed composite; no new axiom, no new datum, no new parameter, no numerical coupling. Local notation: $\varepsilon_{\text{EM}} := e^2/(4\pi)$ — a dimensionless P4-class symbol throughout, ¬identified with any measured constant [EM.25(7), EM.25(9); P5].

(1) the current:

$$J^{\mu}_{Q} := \text{the continuum density of the integer } Q \text{ read}$$

$$[\text{Def-BM.8}(1);\ \text{BM.7}:\ \text{the density form of the integer law};\ \text{BM.11}:\ \partial_{\mu}J^{\mu} = 0];$$

$$Q \in \mathbb{Z},\ \ Q \in \{-3, \ldots, +3\} \text{ at } k \leq 3\ [\text{Def-BM.8}(5);\ \text{BM.28}(7)];$$

the charge alphabet of the generator matches the end-read alphabet: $\operatorname{spec} Q|_{\mathcal{H}_v} \cup \omega$-image

$$= \{-1, 0, +1\}\ [\text{FT.12};\ \text{EW.36};\ \text{D3.4}];$$

(2) the coupling:

the A-coefficient of the neutral connection is $e\,Q$,

$$e = \frac{gg'}{\sqrt{g^2 + g'^2}} = g \sin\theta_W$$

$$[\text{EW.35};\ \text{EM.25}(6):\ \text{fixed with zero freedom}];$$

$\neg\exists$ a theorem fixing the numerical value of e [EM.25(7); P4] :

every row of this tier carries e symbolically;

(3) the sourced field equation:

$$\partial_{\mu}F^{\mu\nu} = e\,J^{\nu}_{Q}$$

[the Euler–Lagrange row of $S_{\mathrm{EM}} = -\frac{1}{4}\int \sqrt{-g}\, F^2$ [EM.15; CT1.22] with the (2)-coupled source; math];

gauge invariance stands [EM.14];

$$m_\gamma^2 = 0 \text{ [EM.6; EW.22 : } \ker M^2 = \mathrm{span}\{Q\}] :$$

the static spherically symmetric solution outside the source support is the Coulomb read

$$E_r(r) = \frac{e\,q}{4\pi r^2}$$

[machine: harmonicity of $1/r$ + Gauss flux normalization];

(4) the smearing locus:

the source support lies within the Born-width envelope Δx_{B}

$= \hbar/(E_{\mathrm{int}} \ln 2)$ [Def-BM.4$(1), (5)$: a chart-side envelope, ¬a ledger distance; BM.13(3) : sub-hop];

the distribution factor is priced, not adopted [Def-BM.4(4) pattern] :

$$\text{shell : } c_{\mathrm{S}} = \tfrac{1}{2};$$

$$\text{uniform ball : } c_{\mathrm{B}} = \tfrac{3}{5}$$

[machine: both energy integrals exact];

c is carried symbolically; no clause of this tier selects a distribution;

(5) the field-energy read:

$$E_{\mathrm{EM}}^{\mathrm{self}}(q,c) := \int_{\mathrm{chart}} T_{00}^{(\mathrm{EM})}\, dV = \frac{c\, e^2 q^2}{4\pi\, \Delta x_{\mathrm{B}}}$$

$[T_{\mu\nu}^{(\mathrm{EM})} = F_{\mu\rho}F_{\nu}{}^{\rho} - \frac{1}{4} g_{\mu\nu} F^2$ [EM.15]; machine: the radial integral exact with the $\Lambda \to \infty$ row];

$$[\varepsilon_{\mathrm{EM}}] = \mathfrak{q}^{0}, \quad [\Delta x_{\mathrm{B}}] = \mathfrak{q}^{-1/2} \implies [E^{\mathrm{self}}_{\mathrm{EM}}] = \mathfrak{q}^{+1/2} = [\text{energy}]$$

[CT2.1; machine: exact exponent arithmetic];

(6) the channel location:

$T_{\mu\nu} = T^{\mathrm{bd}}_{\mu\nu} + T^{\mathrm{tr}}_{\mu\nu}$ [CT1.22; D1.5 : the bound/transit decomposition] :

the correction is the transit-channel energy of the member's own field, ¬a new mass read;

BM.9(4)'s dust row is the bound part at rest;

a charged member carries $T^{\mathrm{bd}} + T^{\mathrm{tr}}$;

the rest input of the dispersion route refines to $m_{\mathrm{rest}}(q)$

$$= m_{\mathrm{hid}} + \frac{c\, e^2 q^2}{4\pi \Delta x_{\mathrm{B}}}$$ [Def-BM.10(4)–(5); BM.35(4)];

(7) status and scope:

constructed — Def-BM.8(8)'s named open construction (the coupling of Q to a gauge field) is discharged for the sealed classes at static leading order;

¬a declaration, ¬a P4 addition (e is the logged combination [EW.35]), §Cal-BM stays empty;

chart-side throughout [CT2.6 hypotheses at the point of use];

BM.7(4)'s register row stands untouched: ¬∃ a $U(1)$ action on the ternary alphabet and q_{EM} is ¬a register Noether charge — the coupling of this definition is chart-side, after BM.8's promotion ["the gauge algebras were never discrete"];

¬constructed: any dynamics of Q [BM.6(5)],

radiation and back-reaction,

a lepton–baryon bound channel [Def-BM.1's grammar is colour-sealed],

the chiral assignment [deficit $1'$, FT.125(4) stands],

internal multipole resolution [BM.46(2)].

Rmk (what is declared and what is not). Nothing. The coupling strength is EW.35's derived coefficient carried by a P4-class symbol; the current is Def-BM.8's derived read in BM.11's derived density form; the field equation is the Euler–Lagrange row of EM.15's derived action; the locus is Def-BM.4(5)'s existing envelope; the distribution factor is priced in two exact values and adopted in none. The construction assembles derived pieces into the sourced static read that Def-BM.8(8) listed as unconstructed — and stops exactly where the inventory stops: no dynamics, no radiation, no atoms.

Req. D1.5, D1.6, D2.9, D3.4 (Parts 1–2); CT1.22, CT2.1, CT2.6, EM.6, EM.11, EM.14, EM.15, EM.25(6), EM.25(7), EM.25(9), EW.22, EW.35, EW.36 (Part 2); P4, P5 (Parts 1–2); FT.12 (Part 3); Def-BM.4, Def-BM.8, Def-BM.10, BM.7, BM.9(4), BM.11, BM.13(3), BM.28(7), BM.35(4) (this part); the Euler–Lagrange row of the abelian action, Gauss's law for the spherically symmetric static class, exact radial integrals [math; machine: `checks/verify_H_em_self_interaction.py` §A].

Pf.

Definition block: clauses (1)–(6) are assemblies of the cited derived rows with the machine-verified Coulomb/energy identities;

(4)'s priced factors carry no adoption;

(7) is the status and scope record. ■

Phys. The composite's charge was an integer read with no field attached to it; the attachment is now built, and every piece of it was already on the shelf. The current is the additive read in its continuum density form, the coupling coefficient is the residual coupling the electroweak rotation derived — carried as a symbol, because no theorem of the corpus fixes its value and none is invented here — and the field equation is the Euler–Lagrange row of the Maxwell action the stress-energy tier already varied. Outside the Born-width envelope the solution is the Coulomb read, its energy integral is exact at either of two priced distributions, and its home in the ledger bookkeeping is the transit channel of CT1.22's own decomposition: the correction is the field the member drags along, not a new species of mass. The register-side obstruction stands where it was: there is no $U(1)$ on the ternary alphabet, and this construction never asks for one — it lives on the chart, downstream of the promotion.

Theorem 4.39 (BM.46 — The scar's field energy and the mass-splitting theorem). Thm.

(1) the self-energy, two agreeing routes:

field route: $\displaystyle\int_{\Delta x_{\mathrm{B}}}^{\Lambda} \tfrac{1}{2}E_r^2 \cdot 4\pi r^2$

dr

$$= \frac{e^2q^2}{8\pi}\Big(\frac{1}{\Delta x_{\mathrm{B}}} - \frac{1}{\Lambda}\Big) \xrightarrow[\Lambda\to\infty]{} \frac{e^2q^2}{8\pi\,\Delta x_{\mathrm{B}}}$$

[machine: exact antiderivative and limit row];

potential route (shell): $\quad \frac{1}{2}\,q\,\phi(\Delta x_{\mathrm{B}}) = \frac{1}{2}(eq)\dfrac{eq}{4\pi\Delta x_{\mathrm{B}}} = \dfrac{e^2q^2}{8\pi\,\Delta x_{\mathrm{B}}}$

[machine: the two routes agree exactly];

the uniform ball: $\quad E_{\mathrm{in}} = \dfrac{e^2q^2}{40\pi\,\Delta x_{\mathrm{B}}}, \quad E_{\mathrm{out}} = \dfrac{e^2q^2}{8\pi\,\Delta x_{\mathrm{B}}}, \quad E_{\mathrm{tot}} = \dfrac{3\,e^2q^2}{20\pi\,\Delta x_{\mathrm{B}}}$

[machine: the interior/exterior split exact];

the unified form:

$\Delta x_{\mathrm{B}} \cdot m_{\mathrm{hid}} = 1$ [Def-BM.4(1)] :

$$\boxed{E^{\mathrm{self}}_{\mathrm{EM}} = \frac{c\,e^2q^2}{4\pi\,\Delta x_{\mathrm{B}}} = c\,\varepsilon_{\mathrm{EM}}\,q^2\,m_{\mathrm{hid}} = \frac{c\,\varepsilon_{\mathrm{EM}}\,q^2\,E_{\mathrm{int}}\ln 2}{\mathfrak{h}}}$$

[machine: one Laurent monomial per form];

i.e. the fractional correction is $\frac{E^{\mathrm{self}}_{\mathrm{EM}}}{m_{\mathrm{hid}}} = c\,\varepsilon_{\mathrm{EM}}\,q^2 \in \mathfrak{q}^0$ [machine: dimensionless];

(2) the charge selection:

the correction is exactly proportional to q^2 :

the unit class (q=+1) : $\delta^{(p)} = c\,\varepsilon_{\mathrm{EM}}\,m_{\mathrm{hid}} > 0$;

the neutral class (q=0) : $\delta^{(n)}$

$= 0$ exactly at the monopole order [machine : $0^2 = 0$; Gauss: zero net flux];

the internal-structure row: the neutral class contains separated-sign endpoint patterns (12 of the 20 probability-space members are mixed triples [BM.28(2); machine]),

but D3.4's end signs carry no position datum :

the chart does not resolve internal multipoles at this order [Def-BM.4(5) envelope locus] :

a neutron internal-multipole self-energy is ¬derived — a resolution residue, recorded;

(3) the corrected rest read:

$$m_{\text{eff}}(q) = m_{\text{hid}} + \frac{c\,e^2 q^2}{4\pi\,\Delta x_{\text{B}}} = m_{\text{hid}}\big(1 + c\,\varepsilon_{\text{EM}}\,q^2\big)$$

[Def-BM.12(6) : the transit-channel refinement of the rest input];

at $E_{\text{int}} = 6$ [BM.17; BM.26] :

$$\delta^{(p)} = \frac{6\,c\,e^2 \ln 2}{4\pi\,\mathfrak{h}} = \frac{3\,c\,e^2 \ln 2}{2\pi\,\mathfrak{h}}$$

[machine: one Laurent monomial];

$[E^{\text{self}}_{\text{EM}}] = \mathfrak{q}^{+1/2} =$ [energy] [CT2.1; machine];

(4) the mass-splitting theorem:

$$\boxed{\Delta m_{\text{EM}} := m^{(p)}_{\text{eff}} - m^{(n)}_{\text{eff}} = \frac{c\,e^2}{4\pi\,\Delta x_{\text{B}}} = \frac{3\,c\,e^2 \ln 2}{2\pi\,\mathfrak{h}} \;>\; 0}$$

[machine: positive coefficient census at $e^2 > 0,\ c \in \{\frac{1}{2}, \frac{3}{5}\}$];

the equal-mass degeneracy of the crystallized-count route [BM.28(1)] is broken by a strictly positive, charge-selected splitting: the charged member is the heavier one;

the width is shared ($E_{\text{int}} = 6$ on both classes [BM.17; BM.26]) and the selector is q^2 : (1 vs 0) [machine] :

the splitting is a function of the charge and of the Born width, with the charge carrying the selection;

(5) the correction-species row:

δ_{EM} is a correction to the existing read, ¬a new mass read [Def-BM.12(6); CT1.22; D1.5] : it is the transit-channel energy of the member's own field, entering through the dispersion route's rest slot [Def-BM.10(4)–(5)];

BM.37(2)'s disjointness stands: the measure routes (i)–(iii) remain measure functionals — the correction rides the metric/field species, and the route ratios $(1, \frac{4}{3}, \frac{58}{13})$ are untouched by it;

(6) status:

derived: (1)–(5);

symbolic: e [EM.25(7)], c [Def-BM.12(4)];

¬a measured splitting value enters [P5; machine: zero external tokens];

¬a P4 addition, ¬a new dimensionful scale [P1];

the open rows: the numerical value of e (a calibration step requiring one datum [EM.25(7)]),

the internal-multipole resolution $((2))$,

the dynamics of Q [BM.6(5)].

Rmk (the sign is the theorem). Every piece of this correction is positive-semidefinite in the charge: the field energy of a source is the integral of a square, and it scales as q^2. The member that carries the scar therefore gains mass, never loses it — the proton is the heavier baryon on this route, by an exact symbolic monomial. Any account on which the electromagnetic self-interaction lightens the charged member and thereby opens the neutron decay is refuted by the sign of the integral, before any value of e is known. The tier keeps the distribution factor and the coupling symbolic precisely so that the sign — the only content the claim needed — is derived without either.

Req. D1.5, D2.9, D3.4 (Parts 1–2); CT1.22, CT2.1, EM.11, EM.15, EM.25(7) (Part 2); P1, P4, P5 (Parts 1–2); Def-BM.4, Def-BM.8, Def-BM.10, Def-BM.12, BM.9(4), BM.13, BM.17, BM.26, BM.28(1), BM.28(2), BM.35(4), BM.37(2) (this part); exact radial integrals with the cutoff-and-limit row, Laurent monomials, sign censuses [math; machine: `checks/verify_H_em_self_interaction.py` §A–§B].

Pf.

$$(1) : \; E_r = \tfrac{eq}{4\pi r^2} \text{ [Def-BM.12(3)]} : \quad \tfrac{1}{2} E_r^2 \cdot 4\pi r^2 = \tfrac{e^2 q^2}{8\pi r^2};$$

$$\int_R^{\Lambda} \tfrac{dr}{r^2} = \tfrac{1}{R} - \tfrac{1}{\Lambda} \text{ [machine: antiderivative]};$$

the shell potential route and the ball split [machine: exact rational coefficients $\frac{1}{8},\ \frac{1}{40},\ \frac{3}{20}$]

the unified form : $\frac{1}{\Delta x_{\mathrm{B}}}$

$= m_{\mathrm{hid}}$ [Def-BM.4(1)] : $\frac{c\,e^2q^2}{4\pi\Delta x_{\mathrm{B}}}$

$= c$

$\frac{e^2}{4\pi}$

q^2

m_{hid} [machine: monomial identity];

$[\varepsilon_{\mathrm{EM}}] = [e^2] = \mathfrak{q}^0$ [EW.35 : e dimensionless; machine]

(2) : $q = 0 \implies e^2q^2 = 0$ [machine];

$W(0) = 20 = 8 + 12$ with 12 mixed triples [BM.28(2) re-enumerated; machine];

the position-datum row: D3.4′s read is a pair of signs [verbatim scan]

(3) : Def-BM.12(6) substituted;

$\Delta x_{\mathrm{B}} = \mathfrak{h}/(6\ln 2)$ [BM.26 re-cited] : $\frac{c\,e^2}{4\pi}\cdot\frac{6\ln 2}{\mathfrak{h}}$ [machine: one monomial];

the exponents [CT2.1; machine]

(4) : the difference of (3)′s rows at $q^2 \in \{1, 0\}$ [machine];

$c \in \{\frac{1}{2}, \frac{3}{5}\} > 0,\ e^2 > 0$ [EW.33 : $g, g' > 0$] $\implies \Delta m_{\mathrm{EM}} > 0$ [machine: coefficient census]

(5)–(6) : channel and status records [CT1.22; D1.5; BM.37(2) re-cited]. ■

Phys. The scar is no longer only a label. A unit charge smeared inside its own Born width carries a field, the field carries the integral of a square, and the integral is exact: one eighth over the width on the shell convention, three twentieth on the ball, one symbolic coupling factor on both. The correction to the rest mass is that energy and nothing else — it rides the transit channel the stress-energy theorem already split out, so the read stays one read, refined. The neutron, net-neutral, gets exactly zero at the monopole order, and the tier records honestly what zero does not mean: the neutral class is full of separated sign patterns, but signs without positions carry no multipole, and the chart resolves no internal structure at this order. What falls out is the first derived splitting of the two baryon rows — strictly positive, charge-selected, width-shared — and its sign is the whole point: electromagnetism makes the charged member heavier. The degeneracy of the crystallized count is broken, and broken in the direction that shuts the decay tighter, not looser.

Theorem 4.40 (BM.47 — The infinite-range seal: UV/IR finiteness, the universe-scale neutrality, and the seal asymmetry). Thm.

(1) masslessness is the range:

$$m_\gamma^2 = 0 \text{ [EM.6; EW.22; EM.25(2)] :}$$

the static potential is $\frac{1}{r}$ and the field $\frac{1}{r^2}$,

with no exponential damping factor [machine: the Yukawa factor e^{-mr} is 1 at $m = 0$];

contrast class: colour transit modes are likewise massless [EW.26] but colour is locally projected [(5)] — range and sealing are independent rows;

(2) UV finiteness at the Born width:

$$\int_R^\Lambda \frac{dr}{r^2} = \frac{1}{R} - \frac{1}{\Lambda} < \frac{1}{R} \ \forall \Lambda > R > 0$$

[machine: exact rational functions];

$\Delta x_{\mathrm{B}} > 0$ strictly [BM.13(1); Def-BM.4(1)] :

$E_{\mathrm{EM}}^{\mathrm{self}} < \infty$ for every admissible cell [BM.13(3) : $E_{\mathrm{int}} \geq 2$];

the point-charge divergence is cut by the envelope, ¬by a counterterm;

(3) IR finiteness of an isolated member:

$$E_{\text{out}}(R) = \frac{e^2 q^2}{8\pi R} < \infty \text{ as } \Lambda \to \infty$$

[machine: the $1/\Lambda$ row vanishes in the limit] :

the field energy outside any positive radius is finite although the field itself never vanishes — the infinite range costs no divergence at fixed net charge;

(4) the universe-scale seal:

$$\text{Gauss: } \oint_{S^2_\infty} \vec{E} \cdot d\vec{A} = e\, Q_{\text{enc}}$$

[machine: the flux normalization row];

EM.11: $Q_{\text{EM}} = 0$ in every valid universe [D2.11; D1.3] :

a sphere exhausting the universe encloses Q_{enc}

$= 0 \implies$ the total asymptotic flux vanishes : the leading $\frac{1}{r^2}$ term of the total field cancels at universe scale [machine: the integer row $e \cdot 0 = 0$];

the residual (multipole) field decays as r^{-3} :

$$\int^{\infty} \left(\frac{1}{r^3}\right)^2 r^2\, dr = \int^{\infty} \frac{dr}{r^4} < \infty$$

[machine: exponent arithmetic] :

the total EM field energy of a valid universe is IR-finite by EM.11's global neutrality — EM seals at universe scale;

(5) the seal asymmetry (colour local, EM global):

colour: $\mathcal{C}(\Sigma; C) \in \{0, 1\}$ and one dangling attachment zeroes the whole colour sector [D2.14; FT.19; BM.1; BM.32(1) : filtration] : a local $0/1$ projector;

EM: $W(\pm 1) = 15 > 0$ [BM.28(2); Def-BM.8(5) : the $W(Q)$ census] :

local charge classes are admissible;

$\neg\exists$ a local projector on Q;

neutrality is exhausted only at universe scale [EM.11];

[machine: the two vocabularies — the idempotent $\{0, 1\}$ against the positive census];

colour self-seals; EM does not;

(6) the nearby-space fractions:

the fraction of the self-energy inside $k\,\Delta x_{\mathrm{B}}$:

$$f(k) = 1 - \frac{1}{k}$$

[machine: exact rational functions; the shell route];

$f(2) = \frac{1}{2}, \quad f(3) = \frac{2}{3}, \quad f(4) = \frac{3}{4}$;

the exterior fraction $\frac{1}{k} > 0 \;\; \forall k < \infty$:

no finite neighbourhood holds the whole correction — the nearby-space reading is a strict fraction at every finite radius;

(7) status:

the correction is local in its ultraviolet $((2))$,

finite in its infrared per member $((3))$ and per universe $((4))$;

the seal asymmetry $((5))$ is the exact content of "colour seals, EM seals at universe scale";

¬a measured range, radius or flux enters [P5].

Rmk (two seals, two grammars.) Confinement and neutrality are both sealing laws, and the tier keeps their grammar apart. Colour's is a projector: it acts on any restriction, returns zero or one, and annihilates a block at the first dangling attachment — locality is its content. Electromagnetism's is a census: every local charge class has positive weight, nothing is projected away nearby,

and the vanishing happens once, globally, because a valid universe has no dangling ends at all. The infinite range is not an obstacle to either finiteness row: the Born envelope absorbs the ultraviolet, and EM.11's integer law absorbs the infrared.

Req. D1.3, D2.11, D2.14 (Parts 1–2); CT3.18, EM.6, EM.11, EM.25(2), EW.22, EW.26 (Part 2); P5 (Parts 1–2); FT.19 (Part 3); Def-BM.4, Def-BM.8, Def-BM.12, BM.1, BM.13(1), BM.13(3), BM.26, BM.28(2), BM.32(1), BM.46 (this part); exact radial integrals and exponent arithmetic, the Gauss flux normalization, integer censuses [math; machine: `checks/verify_H_em_self_interaction.py` §B].

Pf.

$$(1):\ \text{EM.6/EW.22 re-cited;}\qquad e^{-mr}\big|_{m=0}=1\ [\text{machine: exact at } m=0]$$

$$(2):\ \tfrac{d}{dr}\big(-\tfrac{1}{r}\big)$$

$$=\tfrac{1}{r^2}\ [\text{machine: symbolic differentiation}]\implies\int_R^\Lambda$$

$$=\tfrac{1}{R}-\tfrac{1}{\Lambda}\ [\text{machine}];$$

$$\Delta x_{\mathrm{B}}=\hbar/(E\ln 2)>0\ [\text{BM.13(1) re-cited}]$$

$$(3):\ \text{the } \Lambda\text{-row of } (2) \text{ at } R=\Delta x_{\mathrm{B}}\ [\text{machine}]$$

$$(4):\ \text{the flux of } \tfrac{eq}{4\pi r^2} \text{ through } S_r^2 \text{ is } eq\ [\text{Def-BM.12(3)'s Gauss normalization re-cited; machine}];$$

$$Q_{\text{enc}}=Q_{\text{EM}}=0\ [\text{EM.11 re-cited; machine: integers}];$$

$$\int^\infty r^{-4}dr \text{ convergent } [\text{machine: antiderivative } -\tfrac{1}{3r^3}]$$

$$(5):\ \mathcal{C}^2=\mathcal{C}\in\{0,1\}\ [\text{BM.32(1) re-cited}];$$

$$W(\pm1)=15\ [\text{BM.28(2) re-enumerated; machine}];$$

$\neg\exists$ a local Q-projector: inventory scan of D2.14's vocabulary (colour slots only) [machine: token row]

$$(6):\ f(k)=\tfrac{E(<k\Delta x_{\mathrm{B}})}{E_{\text{tot}}}=\tfrac{1/R-1/(kR)}{1/R}=1-\tfrac{1}{k}\ [\text{machine: rational functions}];$$

$$(7):\ \text{the status record.}\ \blacksquare$$

Phys. The electromagnetic correction lives with an infinite-range field and stays finite at both ends, for two different reasons that the tier separates. In the ultraviolet the Born envelope does the work: the composite is never a point, its width is strictly positive on every admissible cell, and the radial integral of the field square converges to an exact rational over that width. In the infrared the universe does the work: Gauss ties the asymptotic flux to the enclosed charge, and a valid universe encloses zero — so the total field's leading term cancels at scale, the residual multipoles fall as the cube, and the energy beyond any sphere is finite. Between the two sits the asymmetry with confinement: colour has a local zero-one projector that kills a block at one dangling attachment, while electromagnetism has no local projector at all — its charge classes are positively populated nearby and its neutrality is a single global fact about ends. And the nearby space holds only a fraction: half the field energy lives beyond two widths, two thirds within three, and a strict $1/k$ always escapes — the correction is local in origin and never local in full.

Theorem 4.41 (BM.48 — The two-body potential, the hydrogenic rows, and the binding category bar). Thm.

(1) the static two-body potential:

$$V(d) = \frac{e^2 q_1 q_2}{4\pi\, d}$$

[Def-BM.12(3)'s Coulomb read; math: the unique spherically symmetric harmonic function with the prescribed Gauss flux];

$q_1 q_2 < 0 \implies V < 0$ (attraction), $\quad q_1 q_2 > 0 \implies V > 0$ (repulsion) [machine: sign rows];

the pair $(q_p, q_\ell) = (+1, -1)$ [BM.23(4)'s forced carrier charge] :

$$V(d) = -\frac{e^2}{4\pi\, d} = -\frac{\varepsilon_{\mathrm{EM}}}{d}$$

[machine: exact coefficient];

(2) the hydrogenic rows (chart-side, conditional on a bound channel):

under the canonical-pair mechanics of the chart [BM.10(3) : $[\hat{x}, \hat{p}] = i, \hbar = 1$] with reduced mass m_r :

$$E_n = -\frac{e^4 (q_1 q_2)^2\, m_r}{32\pi^2\, n^2} = -\frac{\varepsilon_{\mathrm{EM}}^2\, m_r\, (q_1 q_2)^2}{2\, n^2},$$

$$\frac{E_2}{E_1} = \frac{1}{4} \quad \text{exact}$$

[math: the Coulomb spectrum; machine: the coefficient and ladder rows];

the n-ladder is a chart-side spectrum;

m_r of a lepton–baryon pair requires the lepton mass read [BM.43(2) : symbolic; mass-form bridge residue];

(3) the bound channel is ¬constructed:

Def-BM.1's composite grammar is colour-sealed (ε/δ junctions on slot indices [D3.9; CT3.17]);

¬∃ an EM sealing grammar in the inventory;

a lepton–baryon bound state would be a dynamics-of-Q object [BM.6(5) : ¬constructed];

the photon-exchange binding of BM.41(3)'s transit alternative remains ¬constructed [deficit $1'$ untouched];

(4) the binding category bar:

a binding energy is the energy of the bound system, ¬a correction to a free member's mass :

for a constructed bound pair the total read would be $m_{\text{pair}} = m_p + m_\ell + E_{\text{bind}}$,

$E_{\text{bind}} < 0$ [system read; (2)'s E_1 at $n = 1$];

the free proton's correction is $+ E^{\text{self}}_{\text{EM}}$ alone [BM.46(3)] :

$\boxed{m^{(p)}_{\text{eff}} = m_{\text{hid}} + E^{\text{self}}_{\text{EM}} - E^{\text{bind}}_{\text{EM}} \text{ is refuted as a category error}}$ — ¬∃ an object in the inventory whose mass that expression reads;

[machine: the inventory row — no bound-pair mass read exists; (3) re-cited];

(5) the sign rows survive the hypothetical:

$E^{\text{self}}_{\text{EM}} > 0$ unconditionally [BM.46(4)];

$E_{\text{bind}} < 0$ belongs to the pair $[(4)]$:

even granting the hypothetical pair, the free-proton splitting $\Delta m_{\text{EM}} > 0$ is unchanged [BM.46(4) stands];

(6) status:

derived: the potential row, its sign structure, the hydrogenic coefficients as chart-side math;

¬constructed: the bound channel, the pair mass read;

¬adopted: any E_{bind} term in a free-member mass;

¬a measured binding energy, ionization scale or atomic datum enters [P5].

Rmk (what the potential row buys.) The Coulomb read between the forced carrier and the daughter is derived the moment the sourced field exists: opposite ternary charges attract with an exact symbolic coefficient, and the chart-side hydrogenic ladder follows from the canonical pair the Noether tier already carried. What does not exist is a grammar that seals the pair — the corpus's composites are colour-sealed, and an electromagnetic atom is a dynamics-of-Q construction nobody has built. The tier draws the potential, prices the ladder, and bars the category slip: binding energy is a property of the bound, and until a bound is constructed, no free member's mass carries one.

Req. D1.5, D3.9 (Parts 1–2); CT2.1, CT3.17 (Part 2); P5 (Parts 1–2); Def-BM.1, Def-BM.12, BM.6(5), BM.10(3), BM.23(4), BM.41(3), BM.43(2), BM.46(3), BM.46(4) (this part); the uniqueness of the harmonic spherically symmetric potential with prescribed flux, the Coulomb spectrum on the canonical pair [math; machine: `checks/verify_H_em_self_interaction.py` §C].

Pf.

(1) : $\nabla^2 \frac{1}{r} = 0$ at $r \neq 0$ [BM.43(4)'s rule-ring identity re-cited; machine];

the flux normalization [Def-BM.12(3)];

uniqueness up to an additive constant, fixed at $V(\infty) = 0$ [math];

signs [machine]

(2) : the Coulomb ladder [math: standard spectrum on $[\hat{x}, \hat{p}] = i$];

$e^4/(32\pi^2) = \varepsilon_{\text{EM}}^2/2$ [machine: coefficient identity];

$E_2/E_1 = \frac{1}{4}$ [machine]

(3) : Def-BM.1/D3.9/CT3.17 re-cited;

inventory scan: no EM sealing grammar, no pair bound state [machine: token row]

(4) : the category row [(3) conjoined with BM.46(3)];

(5) : the sign rows re-cited;

(6) : the status record. ■

Phys. With the sourced field in hand, the force between the decay's charged members writes itself: an exact symbolic coefficient over the separation, attractive precisely where the ternary charges oppose — and the chart's own quantum mechanics turns that potential into the hydrogenic ladder, whose ground coefficient and quarter-ratio are arithmetic. But a potential is not a particle pair. The corpus seals composites with colour, and only with colour: there is no electromagnetic junction grammar, no dynamics of the additive read, and therefore no atom in the inventory. The tier states the consequence as a bar rather than a shrug: binding energy belongs to the bound system, and an expression that subtracts a would-be atomic binding from a free proton's mass reads nothing that exists. The free proton's electromagnetic correction is its own field's energy, positive, exact, and alone.

Theorem 4.42 (BM.49 — The beta-decay re-adjudication with the correction, and the endpoint--to-average refutation). Thm.

(1) the corrected Q-value:

$$Q_{\text{corr}} := m_{\text{eff}}^{(n)} - m_{\text{eff}}^{(p)} - m_\ell - m_\nu = -\big(\Delta m_{\text{EM}} + m_\ell + m_\nu\big)$$

[BM.46(3)–(4); machine: monomial identity];

$$\Delta m_{\text{EM}} > 0 \text{ [BM.46}(4)\text{]}, \quad m_\ell > 0 \text{ [BM.39}(1)\text{; BM.41}(4)(\text{a})\text{]}, \quad m_\nu \geq 0 \text{ [EW.33 PSD]} :$$

$$\boxed{Q_{\text{corr}} < 0 \text{ strictly, and strictly more negative than } Q = -(m_\ell + m_\nu) \text{ of BM.41}(4)}$$

(2) the no-opening theorem:

every EM correction of this tier is $\propto q^2$

≥ 0 and adds to the charged member's mass [BM.46$(1), (4);$ the field energy is the integral of a square] :

the decay $n \to p + \ell + \nu$ requires $m^{(n)} > m^{(p)} + m_\ell$:

no EM correction of this tier moves the difference in the required direction — it moves it against;

the claim that the EM self-interaction makes the beta decay energetically possible is refuted by the exact sign arithmetic [machine];

BM.41(4)(e)'s verdict is sharpened, not altered:

opening the vertex remains a read-specification act [BM.30$(5);$ a §I addendum] or an inventory datum the corpus does not carry (a constituent-level transport splitting of the two baryon rows — ¬∃ in Parts 1–4 [BM.28$(1);$ BM.28(7)]);

this tier performs neither;

(3) the conditional spectrum (under the ¬adopted hypothesis $Q \geq 0$ of some specification):

$E_{\text{max}} = E_\ell^{\text{max}}$ [BM.41(5) endpoint formula re-cited];

the allowed-shape spectrum at constant amplitude with $m_\nu = 0,\ m_\ell \to 0$:

$$\frac{d\Gamma}{dE} \propto E^2 \, (E_{\text{max}} - E)^2$$

[math: the three-body phase space];

$$\frac{\langle E \rangle}{E_{\text{max}}}$$

$$= \frac{\int_0^1 x^3 (1-x)^2 \, dx}{\int_0^1 x^2 (1-x)^2 \, dx}$$

$$= \frac{\text{B}(4,3)}{\text{B}(3,3)}$$

$$= \frac{3!\,2!/6!}{2!\,2!/5!}$$

$$= \frac{1/60}{1/30} \qquad = \frac{1}{2}$$

[machine: exact factorial rationals];

$$\boxed{E_{\max} : \langle E\rangle = 2 : 1}$$

and scale-invariant:

$$E$$

$\mapsto QE$ rescales both moments alike $\implies$ the ratio is 2 for every value of $Q > 0$ [machine: the ratio carries no Q];

(4) the $\frac{4}{3}$-identification refuted:

the proposal $E_{\max} : \langle E\rangle$

$= \frac{4}{3}$ identified with $W(0)/W(+1)$

$= \frac{4}{3}$ [BM.28(2)] is barred three ways :

(a) arithmetic: $2 \neq \frac{4}{3}$ $\quad [2 \cdot 3 = 6 \neq 4 = 4 \cdot 1;$ machine: cross-multiplication in $\mathbb{Z}]$;

(b) scale invariance:

(3)'s ratio is 2 at every splitting scale — $\neg\exists$ a value of Q (hence none of the route ratios) that moves it to $\frac{4}{3}$ in the massless limit [machine];

(c) disjointness:

$W(0)/W(+1)$ is a probability-space cardinal ratio — a measure functional whose defining clause carries zero momentum/metric tokens [BM.37(2) machine census re-cited];

$\langle E\rangle,\ E_{\max}$ are chart-side spectral moments :

the identification crosses BM.37(3)'s bars — disjointness, amplitude $\neq$ mass [FT.79(5)],

the read-choice row [BM.30(5)];

(d) existence: on every adopted read the decay is kinematically closed [(1); BM.41(4)(e)] :

¬∃ an emitted-electron spectrum whose moments could be driven;

(3) is conditional content and stays conditional;

(5) status:

the route rows of BM.28/BM.30/BM.37 stand untouched — the measure prices remain measure prices;

the derived conditional rows: $E_{\max}$ [BM.41(5)], $\langle E\rangle/E_{\max} = \frac{1}{2}$ [massless limit];

¬a measured endpoint, average energy, spectral shape or lifetime enters [P5; machine: zero external tokens].

Rmk (the check that was asked for.) The four-thirds was inspected where it could possibly live: as the ratio of the conditional spectrum's endpoint to its mean. The phase space answers with a clean rational — one half in the massless allowed limit, hence two-to-one, and scale-invariantly two-to-one at any Q whatever, so no mass splitting of any provenance can bend it to four thirds. The four thirds remains what the Tier BM.8 adjudication made it: the ratio of twenty neutral sign-assignments to fifteen charged ones, a price of the charge filter on the ensemble measure. The two numbers are exact, and they are not about the same thing.

Req. P5 (Parts 1–2); D2.9 (Parts 1–2); EW.33 (Part 2); FT.79(5), FT.107(3) (Part 3); Def-BM.12, BM.28(1), BM.28(2), BM.28(7), BM.30(3), BM.30(5), BM.37(2), BM.37(3), BM.39(1), BM.41(4), BM.41(5), BM.43(2), BM.46(1), BM.46(3), BM.46(4) (this part); the three-body phase-space shape at constant amplitude, exact Beta-function rationals [math; machine: `checks/verify_H_em_self_interaction.py` §D].

Pf.

(1) : $m_{\text{eff}}^{(n)} = m_{\text{hid}},\ m_{\text{eff}}^{(p)} = m_{\text{hid}} + \Delta m_{\text{EM}}$ [BM.46(3)–(4) re-cited] :

$Q_{\text{corr}} = -\Delta m_{\text{EM}} - m_\ell - m_\nu$ [machine: monomial identity];

the sign census [machine]

(2) : $\delta(q) = c\,\varepsilon_{\text{EM}} q^2 m_{\text{hid}} \geq 0$ [machine: coefficient census];

$$m_{\text{eff}}^{(p)} - m_{\text{eff}}^{(n)}$$

$= \Delta m_{\text{EM}} > 0$ against the required $m^{(n)} > m^{(p)} + m_\ell$ [machine: the sign contradiction]

$$(3): \int_0^1 x^3(1-x)^2 dx$$

$$= \mathrm{B}(4,3)$$

$$= \tfrac{1}{60},$$

$$\int_0^1 x^2(1-x)^2 dx$$

$$= \mathrm{B}(3,3)$$

$= \frac{1}{30}$ [machine: factorial rationals];

$\frac{1/60}{1/30} = \frac{1}{2}$ [machine];

$E = Qx$ rescaling: both moments carry Q to the same power [machine: the ratio is Q-free]

$(4):\ 2 \cdot 3 = 6 \neq 4 = 4 \cdot 1$ [machine];

(b)–(d) : the cited bars re-cited with BM.37(2)'s census rows;

(5) : the status record. ■

Phys. The correction goes into the balance and the balance closes harder. The neutron stays at six bits over ħ; the proton gains its field energy, strictly positive, symbolic in the coupling but positive at every value the coupling can take. The Q-value of the assembled vertex is minus the splitting minus the lepton masses — negative before the correction and more negative after. No electromagnetic row of this corpus opens the beta decay, because every row the tier derives adds mass to the charged member, and the charged member is the daughter. The spectrum question the tier was handed — could the four-thirds of the charge census be driving the endpoint-to-average ratio of the emitted electron — has an exact answer and it is no: the conditional massless spectrum averages at precisely half its endpoint, a scale-free rational that no splitting value can move, while four thirds is a count of sign-assignments in a filtered ensemble. Both numbers stand as derived; the bridge between them is barred by disjointness, by the amplitude-mass bar, and by the simple fact that on every adopted read there is no emitted electron to average.

BM.50 — The status of the EM self-interaction correction: adjudication and scope. Structural-constraints record: see Appendix B, Section B.1.21 of the Addendum.

4.15 Tier BM.13 — The Full-Scope Mass Splitting: the Lyapunov Foundation, the Complete-Force Census, the Non-Uniform Clock, the Binding Zeros, and the Exact Ratio

Theorem 4.43 (BM.51 — The Lyapunov foundation: the contraction law iterated, the deficit links, and the class-degenerate zero). Thm.

(1) the functional and its zero locus:

$$\delta\rho := \rho - \tfrac{1}{4}I,$$

$$\tilde{K} := -\log\rho - \mathfrak{T}\big((-\log\rho)\,\rho\big)I,$$

$$V(\tilde{K}) := \mathfrak{T}\big(\tilde{K}^2\big) \geq 0$$

[L1];

$$V = 0 \iff \rho = \tfrac{1}{4}I \text{ [L1(1)]};$$

$$V\big(\tilde{K}(\mathcal{T}(\rho))\big) = V\big(\tilde{K}(\rho)\big) \text{ for every draw [L1(2); A6 : tick invariance]};$$

$$V' \leq V,\ V' = V \iff \rho = \tfrac{1}{4}I \text{ [L1(5)]};$$

(2) the contraction law:

$$\gamma := 1 - 4\varepsilon_q \in (0,1] \text{ [D0.3; D0.4; CT0.8]};$$

$$V' = \Big(1 - \frac{\gamma}{4}\Big)^2 V + O(\delta\rho^3),$$

$$V' - V = -\frac{\gamma(8-\gamma)}{16}\,V + O(\delta\rho^3)$$

[L2];

$$\boxed{\Big(1 - \frac{\gamma}{4}\Big)^2 + \frac{\gamma(8-\gamma)}{16} = 1}$$

[machine: exact in $\mathbb{Q}[\gamma]$]

the contraction is strict on the whole admissible range $\gamma \in (0, 1]$ [BM.8(2); machine];

$\gamma = 1$ (the sharp point $\varepsilon_q = 0$ [D0.4; BM.19(3)]) : $\quad V' = \frac{9}{16}V, \quad$ deficit $\frac{7}{16}V$;

$\gamma \to 0$: $V' = V$ [the identity-tilt limit, the excluded endpoint $\varepsilon_q = \frac{1}{4}$ [CT0.8]];

(3) the iteration (the fermion-count contraction read):

one tilt step per recorded crystallization event [L0 grounding; L2 Phys] :

$V_n = \left(1 - \frac{\gamma}{4}\right)^{2n} V_0$ at the leading-coefficient level, remainder $O(\delta\rho^3)$ per step [L2 scope];

the E_{int}-fold row:

$$V_{E_{\text{int}}} = \left(1 - \frac{\gamma}{4}\right)^{2E_{\text{int}}} V_0$$

[E_{int} = #{internal directed edges} [D0.8]; machine: the power identity $\left(1 - \frac{\gamma}{4}\right)^2 \left(1 - \frac{\gamma}{4}\right)^{2n} = \left(1 - \frac{\gamma}{4}\right)^{2n+2}$ in $\mathbb{Q}[\gamma]$];

at the sharp point $\gamma = 1$, $E_{\text{int}} = 6$ [BM.17; BM.26; Census FT.35(2)] :

$$\boxed{\frac{V_6}{V_0} = \left(\frac{9}{16}\right)^6 = \frac{531441}{16777216}},$$

the total deficit $1 - \frac{531441}{16777216} = \frac{16245775}{16777216}$

[machine: exact integer powers];

faithfulness in E_{int}: at fixed γ

$\in (0, 1]$,

$n \mapsto \left(1 - \frac{\gamma}{4}\right)^{2n}$ is strictly decreasing and injective [machine: exact rational comparison on $n \in 2..60$] :

the contraction read separates edge counts exactly as m_{hid} does;

(4) the deficit links:

$$\Delta S_{\text{pair}} := 2|\delta\rho|^2 = \tfrac{1}{2}|r|^2,$$

$$S\big(\rho \,\|\, \tfrac{1}{4}I\big) = \Delta S_{\text{pair}} + O(\delta\rho^3)$$

[L3(1)];

$$\tilde{K} = -4\,\delta\rho + O(\delta\rho^2),$$

$$\boxed{V(\tilde{K}) = 16\,|\delta\rho|^2 + O(\delta\rho^3) = 8\,\Delta S_{\text{pair}} + O(\delta\rho^3)}$$

[L3(2); machine: both branch witnesses];

$$V' - V = -\frac{\gamma(8-\gamma)}{2}\,\Delta S_{\text{pair}} + O(\delta\rho^3), \qquad \gamma = 1: \; -\frac{7}{2}\Delta S_{\text{pair}} \qquad \text{[L3(3)]};$$

the branch witnesses: $\delta\rho_{\mathcal{V}}$

$$= \Pi_{\mathcal{V}} - \tfrac{1}{4}I: \text{ spec}$$

$$= \big(\tfrac{3}{4}, -\tfrac{1}{4}, -\tfrac{1}{4}, -\tfrac{1}{4}\big): \;\; \mathfrak{T}(\delta\rho^2) \qquad = \tfrac{3}{4};$$

$$\delta\rho_{\mathcal{R}} = \tfrac{\Pi_{\mathcal{R}}}{3} - \tfrac{1}{4}I: \text{ spec} = \big(\tfrac{1}{12}, \tfrac{1}{12}, \tfrac{1}{12}, -\tfrac{1}{4}\big): \;\; \mathfrak{T}(\delta\rho^2) = \tfrac{1}{12}$$

[FT.107(2); machine: exact Fraction matrix arithmetic] :

the leading deficits $2|\delta\rho|^2 = \big(\tfrac{3}{2},\ \tfrac{1}{6}\big)$,

$$V = 8\Delta S_{\text{pair}}: \;\; 16 \cdot \tfrac{3}{4} = 12 = 8 \cdot \tfrac{3}{2}, \quad 16 \cdot \tfrac{1}{12} = \tfrac{4}{3} = 8 \cdot \tfrac{1}{6} \text{ [machine]};$$

(5) the inertial read:

$$m_i^2(q) := \Delta S(q)$$

[L3(4); L4 : the patch deficit with its integer degree factor; dimensionless nats; ¬a particle identification];

the species row: the conversion between the deficit/inertial species and the m_{hid} species is item-7 content :

the tier needs no conversion — (6)'s zero is species-independent;

(6) the class-degenerate zero:

every input of the contraction read is class-shared:

$E_{\text{int}} = 6$ on both baryon rows [BM.17; BM.26; Census FT.35(2)];

$\gamma = 1 - 4\varepsilon_q$ is the universal ledger threshold [D0.4];

V_0 is the common vacuum reference [L1(1); CT0.2] :

$$\boxed{\Delta m_{\text{Lyapunov}} = 0}$$

[machine: the difference of identical monomials] :

the Lyapunov foundation gives the two baryon rows the same mass — in every read species at once, because it splits no input;

(7) scope:

L1's faithfulness scope stands (ρ faithful; $\gamma \in (0, 1]$; the identity-tilt limit is the excluded endpoint);

L0's crystallization–tilt coupling is the declared theorem-local dynamical scope [inherited at the point of use, ¬extended];

the identification "mass :

= the contraction rate" is ¬adopted as a new read species: (3)'s $V_{E_{\text{int}}}/V_0$ is an E_{int}-faithful dimensionless companion of m_{hid} [FT.44(4)],

and (6)'s zero holds on it.

Rmk (why the zero needs no bridge). The relating equation between the deficit species (the Lyapunov-contraction read of the registry [Def-H.39(3)]) and the informational species is ¬derived, and this tier does not touch it — because a splitting claim would need the bridge while a degeneracy claim does not: two rows sharing every input of a functional share its value in any species, converted or not. The Lyapunov tower is the foundation the directive asks for, and what the foundation says about the neutron–proton pair is exact, trivial, and load-bearing: zero.

Req. A1, A4, A5, A6 (Part 1); C1, D0.3, D0.4, D0.8 (Part 1); CT0.2, CT0.8, CT1.1 (Part 1); P5 (Parts 1–2); L0, L1, L2, L3, L4 (Part 1); FT.44(4) (Census); FT.107(2) (Part 3); Def-BM.5(4), BM.8(2), BM.17, BM.19(3), BM.26, BM.28(1) (this part); exact formal-power-series identities over $\mathbb{Q}[\gamma]$, exact rational powers and integer power censuses, exact Fraction matrix arithmetic on the branch witnesses [math; machine: `checks/verify_H_full_scope_mass_splitting.py` §A].

Pf.

$$(1)\text{–}(2):\ \text{L1}(1),(2),(5)\ \text{and L2 re-cited (frozen)};$$

$$\left(1-\tfrac{\gamma}{4}\right)^2+\tfrac{\gamma(8-\gamma)}{16}=1-\tfrac{\gamma}{2}+\tfrac{\gamma^2}{16}+\tfrac{\gamma}{2}-\tfrac{\gamma^2}{16}=1\ \text{[machine]};$$

$$\gamma\in(0,1]\implies\left(1-\tfrac{\gamma}{4}\right)^2\in\left[\tfrac{9}{16},1\right)\ \text{[machine: rational witnesses]}$$

$$(3):\ \text{induction on the leading factor [L2]}:\ \text{the power identity in } \mathbb{Q}[\gamma]\ \text{[machine]};$$

$$(9/16)^6=531441/16777216\ \text{[machine: } 9^6=531441,\ 16^6=16777216\text{]};$$

$$\text{injectivity: } \left(\tfrac{9}{16}\right)^n>\left(\tfrac{9}{16}\right)^{n+1}\ \text{on } n\in 2..60\ \text{[machine: exact Fractions]}$$

$$(4):\ \Pi_{\mathcal{V}}=\tfrac{1}{4}(E_0-\Sigma),\ \Pi_{\mathcal{R}}=\tfrac{1}{4}(3E_0+\Sigma)\ \text{[CT0.7(1)]};$$

$$\Sigma$$

$$=\textstyle\sum_i \sigma_i\otimes\sigma_i:\ \text{spec}$$

$$=\{1,1,1,-3\}\ \text{[machine]}:\quad \delta\rho_{\mathcal{V}}:\ \left(\tfrac{3}{4},-\tfrac{1}{4},-\tfrac{1}{4},-\tfrac{1}{4}\right),$$

$$\delta\rho_{\mathcal{R}}:\ \left(\tfrac{1}{12},\tfrac{1}{12},\tfrac{1}{12},-\tfrac{1}{4}\right)\ \text{[machine]}$$

$$\mathfrak{T}(\delta\rho_{\mathcal{V}}^2)=\tfrac{9}{16}+\tfrac{3}{16}=\tfrac{3}{4};$$

$$\mathfrak{T}(\delta\rho_{\mathcal{R}}^2)=\tfrac{3}{144}+\tfrac{1}{16}=\tfrac{1}{12}\ \text{[machine: exact Fractions]};$$

$$V=16|\delta\rho|^2=8\Delta S_{\text{pair}}\ \text{at both witnesses [machine]}$$

$$(5):\ \text{L3}(4)/\text{L4 re-cited};$$

$$(6):\ m_n^{(\text{Lyap})} - m_p^{(\text{Lyap})} = 0 \text{ as the difference of identical inputs [machine]};$$

$$(7):\ \text{the scope record.}\ \blacksquare$$

Phys. The foundation is a contraction, and the contraction is exact arithmetic. Every tilt step keeps $(1-\gamma/4)^2$ of the Lyapunov value and loses $\gamma(8-\gamma)/16$ of it — the two add to one as an identity in γ, not as an approximation — and iterating over the crystallization events of a composite turns the edge count into a strictly faithful dimensionless read: at the sharp threshold, six edges contract the functional to $531441/16777216$ of its vacuum value. The deficit links tie the same object to entropy and inertia with exact rational factors, verified at both branch witnesses: sixteen of a squared deviation, eight of a pair deficit, one inertial read per channel. And then the payload for the baryon pair: every input of this tower — the edge count, the threshold, the vacuum reference — is shared by the neutron row and the proton row, so the whole Lyapunov contribution to their splitting is zero, in every read species at once. The foundation does not split the pair; the split has to come from somewhere else, and the census that follows finds exactly one somewhere.

Theorem 4.44 (BM.52 — The complete force operator and the sector-by-sector splitting census). Thm.

$$(1)\quad \text{the complete force operator:}$$

$$\mathrm{F}$$

$$= F_{(+,\mathcal{R})} \oplus F_{(-,\mathcal{R})} \oplus F_{(+,\mathcal{V})} \oplus F_{(-,\mathcal{V})} \oplus F_{\text{st}} \text{ always applies in full — no sub-block restriction is an A8 force [A8; FT.142(2)]};$$

$$\text{the block/aspect dictionary:}$$

$$(G, H, D^+) \leftrightarrow \mathcal{E}_{++}\ (7),$$

$$W \leftrightarrow \mathcal{E}_{+-}\ (3),$$

$$J^+ \leftrightarrow \mathcal{E}_{-+}\ (3),$$

$$J^- \leftrightarrow \mathcal{E}_{--}\ (3):$$

$$7 + 3 + 3 + 3 = 16 = C_{\text{cap}} \text{ [CT1.1; CT0.12; CT1.12a(2); FT.30]};$$

$$\text{at the sealed composites the rung-3 mask is active:}\quad \dim \mathcal{K}_3$$

$$= 25$$

$$= 16 + 1 + 8 \text{ [CT0.13; CT0.14; Def-BM.3]};$$

the force enters the stress tensor through the masked resolvent Z_{force}

$$= \langle \text{out}|(1 - \mathrm{F}_M^{(n)})^{-1}|\text{in}\rangle \text{ [CT1.22; CT2.2]};$$

(2) the carrier-side static matrix:

$$\mathrm{F}_{\text{ext}}(\mathsf{c}, q) = h(\mathsf{c}) \otimes I_2 + q\, I_2 \otimes \hat{\kappa},$$

$$F_{LL} = F_{RR} = h(\mathsf{c}),$$

$$F_{LR} = F_{RL} = q\, I_2$$

[BM.22(1)–(2); FT.129(4)–(5); Def-H.43 promotion; machine];

the chiral-partner splitting: $2|q| \in \{0, 2\}$ at $q \in \{-1, 0, +1\}$

[BM.22(4); FT.130(5); machine] :

the neutral class keeps parity doubling $(F_{LR} = 0)$, the unit class splits by two read units;

the EM difference between the two classes is exactly the flipping block [BM.22(4)];

the splitting is a dimensionless force-spectrum read — its frequency conversion is the deferred read [FT.131(3); FT.159 residue];

(3) the sector census (the splitting contributions between the two baryon rows):

sector	contribution	the derived ground
$J^{\pm}$ (seals)	0	$\Delta S = \frac{1}{2}(2E - \lvert R\rvert)$ is σ-free [FT.107(2)]; E_{int} equal; the Janus balance holds on both rows [BM.7]
G, H (metric blocks)	0	the seven geometric monomials carry charge exponent 0 [BM.28(1)]
W (weak/shift)	0	$\lvert w\rvert^2$ identical between the classes [BM.28(6)]; the Casimir is branch data, neutral at the σ-vertex [BM.40(3); FT.157(1)]
colour	0	sealed ends are blind [FT.117]; the mask admits only singlets [CT3.18; D2.14]; both rows are ε-sealed with $g(3) = 1$ [BM.31(2)]; colour acts on the slot carrier [FT.127(7)]; the seed is colour-neutral [EW.32]
$D^{\pm}$ (dark)	0	D^{+} is uniform and carried by Λ [CT1.22; CT4.1]; both locks are shared by all charge classes [BM.28(1); CT4.5]
EM	the only nonzero sector	the flipping block $2\lvert q\rvert \in \{0, 2\}$ [(2)]; the field energy $\Delta m_{\text{EM}} = \frac{3\alpha c^2 \ln 2}{2\pi\hbar} > 0$ [BM.46(4)]

(4) the totality theorem:

$$\boxed{\Delta m_{\text{force}} := m_n - m_p\Big|_{\text{the complete operator}} = -\,\Delta m_{\text{EM}} = -\frac{3\,c\,e^2\ln 2}{2\pi\,\mathfrak{h}}}$$

[machine: the census sum of five zeros and one monomial];

the mass splitting comes from the total force — and the total force, censused sector by sector, reduces to its EM summand :

(5) the agreement row (no double count):

the sector census $((3))$ and the field integral [BM.46(1)] are two derived routes to one monomial :

$$\Delta m_{\text{force}}\big|_{\text{EM}} = \Delta m_{\text{self-int}} = \Delta m_{\text{EM}}$$

[machine: the monomials are identical] :

in any decomposition the EM contribution is counted once [BM.55(1)];

(6) scope:

the census is between the two baryon rows at matched profiles and shared junction grammar [Def-BM.3; BM.31(2)];

the Def-H.43 promotion's conditionality stands as BM.22(5) records it (¬a new declaration here);

constituent-level transport differences between the rows have no derived object [BM.55(4) forward pointer at the citing site];

the deferred frequency read [FT.131(3)] blocks no energy-dimension row: the field monomial (4) is complete without it.

Rmk (what "full scope" computes to). The standing order is the sum of all sectors, and the sum is taken — literally, sector by sector, with each zero backed by its own frozen theorem rather than by silence. The seals cannot split the pair because the deficit never reads the end signs; the metric blocks cannot because every geometric monomial is charge-blind; the weak block cannot because the two rows share $|w|^2$ and the Casimir is branch-neutral at a sign flip; colour cannot because sealed singlets are blind to it and both rows carry the same unique grouping; the dark sector cannot because its locks are class-shared. What survives is electromagnetism, twice-derived

and identical: the flipping block on the carrier spectrum and the field energy on the chart. Full scope does not dilute the answer; it isolates it.

Req. A8 (Part 1); CT0.12, CT0.13, CT0.14, CT1.1 (Part 1); CT1.22, CT2.2, CT3.18, CT4.1, CT4.5, D2.14 (Part 2); EM.9, EW.32 (Part 2); P5 (Parts 1–2); Def-H.43, FT.30, FT.107(2), FT.117, FT.125(4), FT.127(7), FT.129(4), FT.129(5), FT.130(5), FT.131(3), FT.157(1), FT.159 (Part 3); Def-BM.3, Def-BM.12, BM.7, BM.22(1), BM.22(2), BM.22(4), BM.22(5), BM.28(1), BM.28(6), BM.31(2), BM.40(3), BM.46(1), BM.46(4) (this part); exact integer censuses, Laurent monomials [math; machine: `checks/verify_H_full_scope_mass_splitting.py` §B].

Pf.

(1) : A8's completeness row and the block dictionary re-cited [CT1.1; CT0.12; CT1.12a(2); FT.30];

$$7 + 3 + 3 + 3 = 16,\ 16 + 1 + 8 = 25 \text{ [machine]}$$

(2) : BM.22(1)–(2), (4) re-cited;

$$2|q| \text{ at } q \in \{-1, 0, +1\} = (2, 0, 2) \text{ [machine]};$$

$F_{LR} = qI_2$: the zero matrix at $q = 0$ [machine: block extraction]

(3) : each zero is the cited frozen row : FT.107(2)'s σ-freeness;

BM.28(1)'s monomial census;

BM.28(6)'s $|w|^2$ row and BM.40(3);

FT.117/CT3.18/BM.31(2)/FT.127(7)/EW.32;

CT1.22/CT4.1/CT4.5/BM.28(1) lock row [machine: the census structure]

(4) : $0 + 0 + 0 + 0 + 0 - \Delta m_{\mathrm{EM}}$ [machine: the monomial sum];

(5) : BM.46(1)'s integral and (3)'s EM entry are the same Laurent monomial [machine];

(6) : the scope record. ■

Phys. Ask every force in the ledger what it contributes to the neutron–proton split, and five of the six answer zero — not approximately, not by assumption, but each by a theorem already on the

books: the deficit ignores end signs, the geometry ignores charge, the weak Casimir ignores sign flips, colour cannot see sealed singlets, and the dark locks are shared. The complete force operator is the sum of all sectors, so the total is the EM summand, and the EM summand has already been computed twice by two routes that agree monomial for monomial. The splitting of the corpus's two baryon rows is one term: the field energy of the scar. Everything else in the full scope is bookkeeping that confirms the isolation.

Theorem 4.45 (BM.53 — The metric background and the non-uniform clock: the zero metric contribution and the Mandelstam–Tamm resolvability bound). Thm.

(1) the background:

the metric is the emergent dictionary of the sector coordinates (a_{00}, q_{ij}, w_i) [CT1.12; Def-BM.10(2)];

the dispersion read $E = N\sqrt{m^2 + \gamma^{ij}p_ip_j} - N^ip_i$ and KE $= N\big(\sqrt{m^2+p^2} - m\big)$ [BM.34(4)–(5)];

(2) the metric contribution is zero:

the seven geometric monomials $(\Delta x_{\mathrm{B}}, V_{\mathrm{eff}}, \rho, \kappa^2\rho, r_s,$ both locks) carry charge exponent 0 [BM.28(1); machine: the exponent census re-run];

the rest slice $E(p{=}0)$

$= N$

m [BM.34(7)] has class-shared N and class-shared m before the EM correction [BM.17; BM.26; BM.51(6)] :

$\boxed{\Delta m_{\mathrm{metric}} = 0}$ [machine];

(3) the non-uniform clock:

the tick strictly contracts V on the whole admissible range [BM.8(2); L2; BM.51(2)–(3)] :

$V_n = V_0\big(1 - \frac{\gamma}{4}\big)^{2n}$;

the log-linear row:

$\ln V_n = \ln V_0 + 2n\,\ln\big(1 - \frac{\gamma}{4}\big)$

[math: the logarithm of the exact power identity; structural — no transcendental is evaluated];

no uniform-τ exponential law:

the ledger evolution is the per-tick weight and its resolvent [BM.44(1)–(4); D1.8 : $x^0 := \tau_n$; FT.159] :

$$\Gamma_{\text{tr}} \in \left[0, \tfrac{1}{3}\right], \quad \tau_{\text{life}} \geq 3 \text{ ticks}, \quad \neg\text{external seconds};$$

(4) the Mandelstam–Tamm relation:

$$\Delta t \, \Delta E \geq \tfrac{1}{2}$$

[BM.10(4); no time operator at finite dimension];

the exact saturation witness re-derived: $A = \frac{1}{2}\sigma_x,\ H = \frac{1}{2}\sigma_z,\ \psi \propto (1, i)$:

$$\langle A \rangle = \langle H \rangle = 0, \quad \langle \sigma_y \rangle = 1, \quad (\Delta A)^2 = (\Delta E)^2 = \tfrac{1}{4}, \quad [H, A] = \tfrac{i}{2}\sigma_y, \quad \frac{d\langle A \rangle}{dt} = -\tfrac{1}{2} :$$

$$\Delta t = \frac{\Delta A}{\left|\frac{d\langle A \rangle}{dt}\right|} = 1, \qquad \Delta t \, \Delta E = \tfrac{1}{2} \qquad \text{[machine: exact over } \mathbb{Q}(i)\text{]};$$

(5) the resolvability of the splitting:

resolving an energy difference Δm_{EM} against the non-uniform clock requires $\Delta t \geq \dfrac{1}{2\,\Delta m_{\text{EM}}}$ [MT, (4)] :

$$\boxed{\Delta t \ \geq \ \frac{\pi\,\mathfrak{h}}{3\,c\,e^2 \ln 2}}$$ [machine: the exact reciprocal monomial of BM.46(4)'s splitting];

(6) status:

the non-uniform clock shifts no splitting value — both rows ride the same γ,

the same tick order and the same contraction law [BM.51(6)] :

the MT relation bounds the splitting's resolvability, ¬its magnitude;

the BM.8(2) stationarity scope and the CT2.6 hypotheses are inherited at the point of use;

FT.159's rate-modulation residue and FT.131(3)'s frequency read stand untouched.

Rmk (the clock is the ledger, and the ledger is honest about it). A uniform proper time would license an exponential law with one rate; the corpus's clock is a strictly contracting tick whose only derived rates are rational functions of one threshold. The Mandelstam–Tamm form is the admissible $\Delta t\,\Delta E$ read — saturated by an exact two-level witness — and applied to the splitting it produces the tier's one new bound: no clock coarser than $\pi\hbar/(3ce^2\ln 2)$ resolves the scar's field energy. The bound is symbolic in the coupling, like everything else here, and it inverts correctly: a smaller coupling means a smaller splitting means a longer required resolution.

Req. D1.8 (Part 1); CT1.12, CT2.1, CT2.6 (Part 2); P5 (Parts 1–2); L2 (Part 1); FT.131(3), FT.159 (Part 3); Def-BM.4, Def-BM.10, BM.8(2), BM.10(4), BM.17, BM.26, BM.28(1), BM.34(4), BM.34(5), BM.34(7), BM.44(1), BM.44(2), BM.44(3), BM.44(4), BM.46(4), BM.51(2), BM.51(3), BM.51(6) (this part); the logarithm of an exact power identity, exact $\mathbb{Q}(i)$ witness algebra, Laurent reciprocals [math; machine: `checks/verify_H_full_scope_mass_splitting.py` §C].

Pf.

(1)–(2) : BM.28(1)'s census re-run: the exponent of the charge symbol is 0 in all seven monomials [machine];

$$E(p{=}0) = Nm \text{ [BM.34(7) re-cited]};$$

$$\Delta m_{\text{metric}} = 0 - 0 \text{ [machine]}$$

(3) : BM.51(3)'s iteration re-cited;

$$\ln(a^n) = n \ln a \text{ [math] applied to the exact rational base } \left(1 - \tfrac{\gamma}{4}\right)^2;$$

BM.44(1)–(4)'s rows re-cited (frozen)

$$(4) : \ \sigma_x(1,i) = (i,1) : \ \psi^\dagger \sigma_x \psi = 1 \cdot i + (-i) \cdot 1 = 0;$$

$$\sigma_z(1,i) = (1,-i) : \ \psi^\dagger \sigma_z \psi = 1 - 1 = 0;$$

$$\sigma_y(1,i)$$

$$= (1,i) \cdot \ldots : \ \psi^\dagger \sigma_y \psi$$

$$= 1 \cdot 1 + (-i)(i)$$

$$= 2 \implies \langle \sigma_y \rangle$$

$$= 1 \; [\psi^\dagger \psi = 2; \text{ machine over } \mathbb{Q}(i)]$$

$$(\Delta A)^2 = \tfrac{1}{4}\langle \sigma_x^2 \rangle = \tfrac{1}{4}, \quad (\Delta E)^2 = \tfrac{1}{4};$$

$$[H, A] = \tfrac{1}{4}[\sigma_z, \sigma_x] = \tfrac{i}{2}\sigma_y \implies \tfrac{d\langle A \rangle}{dt} = i\langle [H, A] \rangle = -\tfrac{1}{2};$$

$$\Delta t \, \Delta E = \tfrac{\Delta A}{1/2} \cdot \tfrac{1}{2} = \tfrac{1}{2} \text{ [machine: saturation]}$$

$$(5): \; \frac{1}{2\Delta m_{\text{EM}}}$$

$$= \frac{1}{2 \cdot \frac{3ce^2 \ln 2}{2\pi\hbar}}$$

$$= \frac{\pi\hbar}{3ce^2 \ln 2} \text{ [machine: the Laurent reciprocal]};$$

$$(6): \text{ the scope record. } \blacksquare$$

Phys. The metric is the background the dispersion reads ride on, and between the two baryon rows it makes no difference at all: every geometric monomial the calibration tiers locked is charge-blind, and the rest slice multiplies a shared lapse by a shared mass. The clock, meanwhile, is not uniform and is never treated as if it were — the tick contracts the Lyapunov value by an exact rational at every crystallization, the log-linear law is the logarithm of that identity and nothing more, and the only decay-style curve the corpus admits is the per-tick resolvent with its three-tick floor. The Mandelstam–Tamm relation is the correct handling of such a clock, saturated by the same two-level witness the Noether tier built, and applied to the splitting it yields the tier's sharpest new number: resolving the scar's field energy takes a clock finer than $\pi\hbar$ over $3ce^2 \ln 2$. The bound moves exactly as the physics should — weaker coupling, smaller splitting, longer patience.

Theorem 4.46 (BM.54 — The binding energies: the count zero, the degenerate ensemble, and the unconstructed atom). Thm.

$$(1) \quad \text{the count binding is exactly zero on both rows:}$$

$$E_{\text{int}}(\Sigma_{12}) = 6 = (c_1 \colon 2) + (c_2 \colon 2) + (\text{connector} \colon 2) \; [\text{Census FT.35}(2); \; \text{BM.21}(1)] :$$

$$E_{\text{bind}}^{\text{info}} = \big(6 - (2{+}2{+}2)\big) \tfrac{\ln 2}{\hbar} = 0$$

[machine];

both baryon rows carry the same decomposition [BM.17; BM.26] :

$$\Delta\big(E^{\text{info}}_{\text{bind}}\big) = 0;$$

(2) the ensemble binding is nonzero but class-degenerate:

$$\mathcal{B}(S_j; 1, 1, 1)$$

$$= m_B^2(\cdot, S_j) - \textstyle\sum_i m_B^2(\cdot, S_i)$$

$\neq 0$ exact rationals [BM.21(2) re-cited (frozen)];

$m_B^2(q, S)$ is independent of q [FT.107(3); the cancellation lemma BM.30(3); machine re-run] :

$$\mathcal{B}^{(n)} = \mathcal{B}^{(p)} \text{ exactly } \Longrightarrow \boxed{\Delta m_{\text{binding}} = 0} \text{ on every normalized ensemble read;}$$

(3) the charge configuration does not enter either binding read:

the count read is E_{int}-additive [(1); BM.21(1)];

the ensemble deficit $\Delta S = \frac{3}{2}(2E - |R|)$ is σ-free [FT.107(2)] :

the end-sign configurations that distinguish the classes [D3.4; BM.28(2)] are invisible to both binding reads :

"different charge configurations $\Longrightarrow$ different binding energies" has no derived object between the two rows;

(4) the EM binding belongs to the unconstructed pair system:

the lepton–baryon bound channel is ¬constructed [BM.48(3) : Def-BM.1's grammar is colour-sealed; ¬an EM sealing grammar; BM.6(5) dynamics open];

the category bar: a binding energy is the energy of the bound system, ¬a correction to a free member's mass [BM.48(4)];

for a constructed pair the system read would be $m_{\text{pair}} = m_p + m_\ell + E_{\text{bind}}, \quad E_{\text{bind}} < 0;$

the sign row:

$$E_1 = -\frac{\varepsilon_{\text{EM}}^2\, m_r\, (q_1 q_2)^2}{2} < 0 \text{ at positive symbols}$$

[BM.48(2); machine: the sign census] :

where binding exists it is negative — and it exists, in this inventory, nowhere yet;

(5) status:

$\Delta m_{\text{binding}} = 0$ on every constructed read ((1)–(3));

the hypothetical pair system carries the only negative binding rows ((4)),

as system reads,

¬as member splittings;

the route-(ii) measure price stays uncombined [BM.30(5); BM.37(3)] :

¬a measured binding energy, ionization scale or atomic datum [P5].

Rmk (two zeros and one absence.) The binding contribution to the splitting dies twice over, and both deaths are exact. The count binding is zero because six edges decompose as two plus two plus two — the informational read is additive and the additivity is an integer identity. The ensemble binding is nonzero, four exact rationals of it, but every one is built from charge-degenerate sector masses, so the neutron and the proton carry identical binding and their difference is zero again. The only binding that could distinguish the classes would be electromagnetic, and that lives in an atom — a bound channel the corpus's grammar has never sealed. The category bar keeps the bookkeeping honest in the meantime: negative energy belongs to systems, not to members.

Req. D2.9, D3.4 (Parts 1–2); P5 (Parts 1–2); FT.107(2), FT.107(3) (Part 3); Census FT.35(2) (row-wise); Def-BM.1, Def-BM.5, BM.6(5), BM.17, BM.20(4), BM.21(1), BM.21(2), BM.26, BM.28(2), BM.30(3), BM.30(5), BM.37(3), BM.48(2), BM.48(3), BM.48(4) (this part); exact rational arithmetic, sign censuses, the cancellation lemma [math; machine: `checks/verify_H_full_scope_mass_splitting.py` §D].

Pf.

$(1):$ Census FT.35(2)'s decomposition re-cited;

$$6-(2{+}2{+}2)=0 \text{ [machine]};$$

$$0\cdot\tfrac{\ln 2}{\mathfrak{h}}=0 \text{ [machine]}$$

$(2):$ BM.21(2)'s four frozen rationals re-cited;

the cancellation lemma at the six frozen factors $\{1,2,3,15,20,\frac{7}{3}\}$ [BM.30(3) re-run; machine]

$$\Longrightarrow q\text{-independence} \Longrightarrow \mathcal{B}^{(n)}-\mathcal{B}^{(p)} \qquad =0$$

$(3):$ FT.107(2)'s σ-freeness and BM.21(1)'s additivity conjoined

$(4):$ BM.48(2)–(4) re-cited;

$$E_1<0 \text{ at } \varepsilon_{\text{EM}}^2>0,\ m_r>0 \text{ [machine: sign census on rational witnesses]}$$

$(5):$ the status record. ■

Phys. Binding energy is the one contribution that could have been expected to favor the neutron — a neutral composite might bind its parts more gently or more tightly than a charged one — and the corpus answers with two exact zeros and one absence. The count binding vanishes as an integer identity: six is two plus two plus two. The ensemble binding is real, four nonzero rationals, and completely charge-blind, because every sector mass inside it is. And the binding that would care about charge is the electromagnetic atom, which the colour-sealed grammar has not constructed and the category bar keeps out of the free members' masses in the meantime. The difference of the two rows' binding energies is zero on every read the corpus owns.

Theorem 4.47 (BM.55 — The full-scope splitting and the exact ratio). Thm.

(1) the decomposition, collapsed:

$$\Delta m:$$

$$=m_n-m_p$$

$$=\Delta m_{\text{Lyapunov}}+\Delta m_{\text{force}}+\Delta m_{\text{metric}}+\Delta m_{\text{binding}}+\Delta m_{\text{self-int}} \text{ with the derived rows :}$$

$\Delta m_{\text{Lyapunov}} = 0$ [BM.51(6)]; $\Delta m_{\text{metric}} = 0$ [BM.53(2)]; $\Delta m_{\text{binding}} = 0$ [BM.54(2), (5)];

$$\Delta m_{\text{force}}$$

$$= \Delta m_{\text{self-int}}$$

$= -\Delta m_{\text{EM}}$ [BM.52(4)–(5) : the two routes are one monomial, counted once] :

$$\boxed{\Delta m = m_n - m_p = -\,\Delta m_{\text{EM}} = -\,\frac{3\,c\,e^2 \ln 2}{2\pi\,\mathfrak{h}} \;<\; 0}$$

[machine: the monomial sum and the sign census];

(2) the exact mass ratio:

$$m_n = \frac{6 \ln 2}{\mathfrak{h}},$$

$$m_p = \frac{6 \ln 2}{\mathfrak{h}}\big(1 + c\,\varepsilon_{\text{EM}}\big),$$

$\varepsilon_{\text{EM}} = \dfrac{e^2}{4\pi}$ [BM.26; BM.46(1), (3)] :

$$\boxed{\frac{m_n}{m_p} = \frac{1}{1 + c\,\varepsilon_{\text{EM}}} = \frac{4\pi}{4\pi + c\,e^2}}$$

[machine: the cross-multiplied Laurent identity $m_n(4\pi + ce^2) = m_p \cdot 4\pi$];

the geometric expansion:

$$\frac{m_n}{m_p} = 1 - c\,\varepsilon_{\text{EM}} + (c\,\varepsilon_{\text{EM}})^2 - (c\,\varepsilon_{\text{EM}})^3 + O(\varepsilon^4)$$

[machine: the truncated series identity $(1 + x)(1 - x + x^2 - x^3) = 1 - x^4$];

(3) the strict order:

$$0 < \frac{m_n}{m_p} < 1 \;\text{ for every } e^2 > 0 \;\text{ and both priced } c \in \{\tfrac{1}{2}, \tfrac{3}{5}\}$$

[machine: the denominator exceeds the numerator by $ce^2 > 0$];

$$\frac{m_n}{m_p} = 1 \iff e = 0 :$$

the full-scope corpus computation places the neutron strictly below the proton on every constructed read;

(4) the order residue:

reversing the order requires a class-dependent constituent transport splitting ($m^{(n)}_{\text{constituent}} > m^{(p)}_{\text{constituent}}$) :

$\neg\exists$ such an object in Parts 1–4 :

the fractional constituent alphabet is unreachable by the additive read [BM.28(7); Def-BM.8(5)];

the species labels carry no datum [Def-BM.3];

the deficit/inertial conversions are item-7 content [pointer];

the family-profile map of Def-BM.11 acts on channel lines, and no clause attaches constituent profiles to the baryon rows' charge classes;

the residue is named, not filled; ¬a measured mass difference is compared [P5];

(5) the decay consequence:

$Q_{\text{corr}} = -(\Delta m_{\text{EM}} + m_\ell + m_\nu) < 0$ [BM.49(1) re-cited] now with the full-scope census behind it :

the β-decay vertex is kinematically closed on every constructed read, and the closure survives the complete force operator [BM.52(4); BM.49(2) re-cited];

the route-(ii) measure price stands uncombined [BM.30(5); BM.54(5)];

(6) status:

the ratio is an exact symbolic rational function of the logged P4 coupling and the priced distribution factor — ¬a number,

¬a fit,

¬a comparison [P5; EM.25(7)];

the five-term decomposition is derived term by term; four terms are zero and the fifth is the BM.46 monomial counted once [(1); BM.52(5)].

Rmk (the answer the corpus gives.) Asked for the neutron–proton mass ratio from its own structures alone, the corpus answers with a rational function: four pi over four pi plus ce^2. It is exact, it is symbolic in the one coupling no theorem fixes, and it is strictly below one — the neutron comes out lighter, because the only sector of the complete force operator that distinguishes the two rows is the one whose field energy the proton carries and the neutron does not. The order the decay would need is the order no constructed read supplies, and the tier names the missing object precisely: a constituent-level transport difference between the classes, of which the inventory contains no instance. That is the residue; everything else is arithmetic.

Req. P1, P4, P5 (Parts 1–2); CT2.1, EM.25(7) (Part 2); Def-BM.3, Def-BM.8, Def-BM.11, Def-BM.12, BM.17, BM.26, BM.28(7), BM.30(5), BM.37(3), BM.39, BM.46(1), BM.46(3), BM.46(4), BM.48, BM.49(1), BM.49(2), BM.51(6), BM.52(4), BM.52(5), BM.53(2), BM.54(2), BM.54(5) (this part); FT.79(5) (Part 3); Laurent cross-multiplication, truncated geometric series over $\mathbb{Q}$, sign censuses [math; machine: `checks/verify_H_full_scope_mass_splitting.py` §E].

Pf.

(1) : the census sum $0+0+0+0-\Delta m_{\mathrm{EM}}$ [BM.52(4) re-cited; machine];

the double-count guard [BM.52(5) : one monomial];

the sign: the coefficient census at $c, e^2, \ln 2, \mathfrak{h} > 0$ [machine]

(2) : $m_n(4\pi + ce^2)$

$= \frac{6\ln 2}{\mathfrak{h}}(4\pi + ce^2)$ and $m_p \cdot 4\pi$

$= \frac{6\ln 2}{\mathfrak{h}}\big(1 + \frac{ce^2}{4\pi}\big)4\pi$

$= \frac{6\ln 2}{\mathfrak{h}}(4\pi + ce^2)$ [machine: the cross-multiplied Laurent identity];

$(1+x)(1-x+x^2-x^3) = 1-x^4$ [machine]

$$(3):\ (4\pi + ce^2) - 4\pi = ce^2 > 0\ \text{[machine: coefficient census]};$$

$$\text{both numerator and denominator positive at positive symbols } \implies \text{ the ratio} \in (0,1);$$

$$ce^2 = 0 \iff e = 0\ \text{[machine]}$$

$$(4):\ \text{BM.28}(7)/\text{Def-BM.8}(5),\ \text{Def-BM.3},\ \text{the Def-BM.11 locus clause re-cited};$$

$$(5):\ \text{BM.49}(1)\text{–}(2)\ \text{re-cited};$$

$$(6):\ \text{the status record.}\ \blacksquare$$

Phys. The full-scope question has a full-scope answer, and it is a single rational function. Four of the five named contributions are theorems about zero: the Lyapunov tower shares every input between the rows, the metric reads are charge-blind, the binding dies twice, and the force census leaves one sector standing. That sector's monomial is the whole splitting, negative for the neutron-minus-proton difference, and the ratio follows by cross-multiplication: four pi over four pi plus $c\,e^2$, strictly between zero and one at every positive coupling, equal to one only where the coupling dies. The corpus therefore places the neutron below the proton, deeper than the informational count alone did, and the beta decay stays shut with the complete force operator now behind the closure. What would open it — a constituent transport difference between the two rows — is named exactly and found absent: no fractional alphabet, no labelled datum, no converted species. The ratio is derived; the order is derived; and the distance to the world's order is the residue, stated to the monomial.

BM.56 — The status of the full-scope mass splitting: adjudication and scope. Structural-constraints record: see Appendix B, Section B.1.22 of the Addendum.

4.16 Tier BM.14 — The Constituent Transport Splitting: the Vertex Asymmetry, the Observation Frames, the Six-State Quark Assembly, and the Neutron Row

Locus. This tier fills the residue BM.55(4) names and does not fill: the class-dependent constituent transport splitting. It extends BM.55(1)'s five-term decomposition by one term, amends BM.52(3)'s sector census by one row, and reverses BM.55(3)'s strict order on an exact sub-critical branch. Every clause of BM.51–BM.56 it touches is re-cited at its point of use and stands unmodified except where a clause's own scope row is widened; the widening is recorded in BM.64(3).

Two directions. The tier is worked from both ends at once. From the force structure: what a sealed junction can carry, and what the seal annihilates — BM.57 and BM.58 run that direction to exhaustion and refute both proposed mechanisms. From the mass spectrum: what BM.55(1)'s four zeros leave, and what the inventory contains that is class-dependent and not barred — BM.59 runs that direction. The two meet on one object, $W(Q)$, and BM.62 assembles it. BM.60 supplies the six-state spectrum the constituent row of BM.59(5) needs; BM.61 disposes of the proposed coupling α_B; BM.63 re-adjudicates the decay; BM.64 adjudicates the tier.

Definition 4.13 (Def-BM.13 — The constituent transport refinement of the rest input (the construction)). Def. A refinement of the dispersion route's rest input [Def-BM.10(4); BM.35(4)] built from the additive charge read's own census [Def-BM.8(6)]; no new axiom, no new datum, no new parameter, no numerical coupling, no route adoption. Local notation: $\varepsilon_{\mathrm{EM}} := e^2/(4\pi)$, a dimensionless P4-class symbol carried symbolically throughout [Def-BM.12 local notation; EM.25(7), (9); P5].

(1) the constituent transport multiplicity:

$$q_{\mathrm{core}}(c) = \tfrac{1}{2}\big(\sigma_{s(c)} - \sigma_{t(c)}\big) \in \{-1, 0, +1\}$$

[D3.4; EM.9];

the end-pair census:

$$(\sigma_s, \sigma_t) \in \{(+,+), (+,-), (-,+), (-,-)\} \;\longmapsto\; q = (0, +1, -1, 0)$$

[EM.9 table; machine] :

$\boxed{m_q = (2,\ 1,\ 1) \ \text{ at } \ q = (0, +1, -1)}$ [FT.106(3a) : the norm-layer closed form $m_q\, 2^{|R|-2}$] :

a neutral-core constituent carries two transport realizations; a charged-core constituent carries one;

the two neutral realizations are exchanged by the global sign flip, which maps q

$\mapsto -q$ on a charged one [EM.9a; EM.24(1)] :

the multiplicity is the orbit size of the constituent's transport register under ω within its own charge class;

(2) the parity law of the chain wall:

for an oriented chain $c = (e_1, \ldots, e_k),\ k \geq 2,$ with interior vertices of type $(1,1)$ [A7; D0.8] :

$$\sum_{i=1}^{k-1} f_{v_i} = \tfrac{1}{2}\big(\sigma_{e_k} - \sigma_{e_1}\big) = -\,q_{\mathrm{EM}}(c), \qquad \sum_{i=1}^{k-1} \aleph_{v_i} = \#\{i :\ \sigma_{e_{i+1}} \neq \sigma_{e_i}\} \qquad \text{[EM.9b; D3.3] :}$$

$$\aleph_v = |f_v| \in \{0,1\} \implies \boxed{\sum_i \aleph_{v_i} \equiv |q| \pmod 2, \qquad \sum_i \aleph_{v_i} \ \geq \ |q|}$$

[machine: exhaustive over every sign sequence, $k = 2, \ldots, 7$] :

the interior wall content of a constituent's transport is parity-locked to its charge core:

$$q = 0 : \ \min \textstyle\sum \aleph = 0 \ \text{(even)};$$

$$|q| = 1 : \ \min \textstyle\sum \aleph = 1 \ \text{(odd)};$$

(3) the class transport weight:

$$W(Q) \ := \sum_{(q_1,\ldots,q_k)\in\{-1,0,+1\}^k \ \sum_i q_i = Q} \ \prod_{i=1}^{k} m_{q_i} \ = \ \left[z^Q\right]\left(2 + z + z^{-1}\right)^k$$

[Def-BM.8(6)] :

$$k = 3 :$$

$$W = (1,\ 6,\ 15,\ 20,\ 15,\ 6,\ 1) \ \text{ over } \ Q = -3, \ldots, +3,$$

$$\sum_Q W(Q) = 4^3 = 64 = 2^6$$

[machine: the exhaustive 4^3 end-sign census and the convolution agree exactly];

the ordered charge-triple census:

$$\#\{(q_1, q_2, q_3) : \ \textstyle\sum q_i = k\} = (1, 3, 6, 7, 6, 3, 1) :$$

7 neutral triples against 6 charged [BM.28(2); machine];

the factorization identity:

$$W(Q) = \sum_{\text{triples at } Q} \ \prod_i m_{q_i}$$

[machine: the two routes agree class by class];

(4) the charge-class transport entropy:

$$\boxed{\Sigma_{\mathrm{cl}}(Q) \;:=\; \ln W(Q)}$$

$$\Sigma_{\mathrm{cl}}(0) = \ln 20, \quad \Sigma_{\mathrm{cl}}(\pm 1) = \ln 15,$$

$$\boxed{\Sigma_{\mathrm{B}} \;:=\; \Sigma_{\mathrm{cl}}(0) - \Sigma_{\mathrm{cl}}(+1) \;=\; \ln \frac{W(0)}{W(+1)} \;=\; \ln \frac{4}{3} \;=\; 2\ln 2 - \ln 3 \;>\; 0}$$

[machine: the certificate $4 > 3$] :

the constituent row: $\ln m_0 - \ln m_{\pm 1} = \ln 2 > 0$ [machine: $2 > 1$] :

the neutral-core constituent is the heavier one, by exactly $\ln 2/\mathfrak{h}$;

the additive constraint attenuates it: $0 < \ln \frac{4}{3} < \ln 2 < 3\ln 2$

[machine: $3 < 4 < 6 < 24$];

(5) the refined rest input: $\boxed{m_{\mathrm{rest}}(Q) \;:=\; \dfrac{E_{\mathrm{int}} \ln 2 + \Sigma_{\mathrm{cl}}(Q)}{\mathfrak{h}} \;+\; c\,\varepsilon_{\mathrm{EM}}\, Q^2\, m_{\mathrm{hid}}}$

[the first summand: BM.35(4)'s reframing identity $m_{\mathrm{rest}} = \Delta S_{\mathfrak{L}}/\mathfrak{h}$ with the register summand of (4); the second: BM.46(3)'s corrected rest read, riding the charge-blind Born width [BM.28(1)]];

at $E_{\mathrm{int}} = 6$ [BM.17; BM.26] :

$$m_{\mathrm{rest}}(0)\,\mathfrak{h} = \ln(2^6 \cdot 20) = \ln 1280,$$

$$m_{\mathrm{rest}}(\pm 1)\,\mathfrak{h} = \ln(2^6 \cdot 15) + 6c\,\varepsilon_{\mathrm{EM}} \ln 2 = \ln 960 + 6c\,\varepsilon_{\mathrm{EM}} \ln 2$$

[machine];

(6) the currency clause:

Σ_{cl} is measured in nats of binary registration data — the end signs are the σ-alphabet [CT0.9; D3.4] and one binary datum books $\ln 2$ [C9; CT1.29] :

it is the same currency as $\Delta S_{\mathfrak{L}}$

$$= E_{\text{int}} \ln 2 \text{ [BM.35(3)]},$$

and its alphabet is disjoint from the count's:

E_{int} has no endpoint summand [BM.28(3) : $6 = (c_1\colon 2) + (c_2\colon 2) + (\text{connector}\colon 2)$] :

E_{int} is constant on every end-sign fibre [machine: exhaustive over the four end pairs] $\implies$ the two registrations are independent, so their entropies add;

(7) the reference clause (the sign is fixed, not chosen):

the deficit reference is the vacuum: $\Delta S := S - S(\text{vac})$,

$S_{\mathfrak{L}}(\text{vac}) = 0$ [CT0.2; D0.1; BM.35(3)] :

$W(\text{vac}) = 1 \implies \Sigma_{\text{cl}}(\text{vac}) = 0 \implies \Delta\Sigma_{\text{cl}}(Q) = \ln W(Q)$ with the positive sign of (4);

the priced alternative:

referencing against the unconstrained register $\left(4^3 = 64\right)$ gives the constraint cost $\ln \dfrac{64}{W(Q)}$,

whose class difference is $-\ln \frac{4}{3}$:

that reference is ¬a state of the composite — the boundary data fix the charge class [Def-BM.5(1)] — and it is class-dependent:

priced, barred;

Def-H.28's Ansatz-B m_B

$= (K - \alpha \ln \Omega) \ln 2/\mathfrak{h}$ carries the opposite sign with α undeclared and is marked an external ansatz, ¬a corpus read:

¬adopted;

(8) the bar against the measure price:

BM.37(6) bars reading a route ratio as a physical mass ratio;

this construction reads no ratio :

the entry is $\ln W(Q)$ in the deficit currency, and the resulting mass ratio is $\ln 1280/\ln 960 \neq \frac{4}{3}$ [machine];

$$\text{the strict contraction:} \quad \frac{\Sigma_{\mathrm{B}}}{E_{\mathrm{int}} \ln 2} = \frac{\ln(4/3)}{6 \ln 2} < \frac{1}{3}$$

$$[\text{machine: } 3 \ln \tfrac{4}{3} < 2 \ln 2 \iff 16 < 27] :$$

the refinement is a strict contraction of BM.30(2)'s measure price — it never reaches it;

(9) status and scope:

constructed — an abbreviation of derived objects ((1)–(4) are censuses; (5) is an assembly of BM.35(4) and BM.46(3));

¬a declaration, ¬a P4 addition, ¬a new dimensionful scale [P1], §Cal-BM stays empty;

¬a route adopted [BM.30(5) stands];

beyond this tier: the triple-level resolution — the chart does not resolve which triple a class member occupies, because the end signs carry no position datum [D3.4; BM.46(2) resolution residue] : the class read of (4) is the one the chart supports.

Rmk (what the refinement is). The composite's rest input has always been the count of its internal directed edges, and the count is charge-blind to the last monomial. What the charge class does carry, and has always carried, is a filter on the transport register: twenty admissible end-sign assignments in the neutral class against fifteen in the charged one. The refinement takes the logarithm of that filter and books it where the corpus books every other entropy — against the vacuum, in the ledger's own currency, through the rest slot the dispersion read already reads. It is not a fourth mass route: it adds no functional of the flicker configurations, and it takes no ratio. It is the smallest entry the inventory supplies that is class-dependent, currency-matched, and alphabet-disjoint from the count.

4.14 (Def-BM.14 — The six-state assembly: the Platonic floors, the defect branches, and the generation map).

Def. A census assembled from the Platonic card template [Census FT.42(5)], the duality degeneracy and the ledger splitting [FT.44(1)–(3)], the physical-card defect spectrum [Census FT.63(6)], and the q-channel triplet [FT.119; Def-BM.6(3)]; no new axiom, no new datum, no new parameter, no external mass.

(1) the three mass floors:

$$m_{\text{hid}} = \frac{2E \ln 2}{\mathfrak{h}},$$

$2E \in \{12,\ 24,\ 24,\ 60,\ 60\}$ over the five solids

[FT.42(1), card table] :

dual pairs are mass-degenerate and Π-degenerate at matched K_S [FT.44(1)] :

cube/octahedron $24 \ln 2/\mathfrak{h}$;

dodecahedron/icosahedron $60 \ln 2/\mathfrak{h}$;

tetrahedron self-dual $12 \ln 2/\mathfrak{h}$;

three mass values across the five solids

$\mathcal{F}$:

$= \{\mathcal{F}_1{=}\{\text{tetrahedron}\},\ \mathcal{F}_2{=}\{\text{cube}, \text{octahedron}\},\ \mathcal{F}_3{=}\{\text{dodecahedron}, \text{icosahedron}\}\}$;

$2E(\mathcal{F}_1){=}12\ <\ 2E(\mathcal{F}_2){=}24\ <\ 2E(\mathcal{F}_3){=}60,$

$12 : 24 : 60 = 1 : 2 : 5$

[machine: exact rationals, strictly increasing];

(2) the branch on each floor (the derived 2):

a floor's two members are distinguished by every finer ledger while sharing m_{hid} and Π [FT.44(2)] :

$\mathcal{F}_2,\ \mathcal{F}_3$: the duality branch — the dual exchanges (V, F) at fixed E [FT.44(1)];

$\mathcal{F}_1$: the weight-ladder branch — the tetrahedron is self-dual [FT.44(3)],

so its branch index is the ladder rung $m \in \{1, 2\}$ [FT.60(5); Census FT.63(6)];

the six states: $\mathcal{S} := \{(\mathcal{F}_g, \beta) : g \in \{1, 2, 3\},\ \beta \in \{-, +\}\}, \qquad |\mathcal{S}| = 3 \cdot 2 = 6$ [machine];

(3) the defect spectrum (the branch mass):

$$\delta_v = 2\pi - d\,\frac{(L-2)\pi}{L},$$

$$D_S = 4\sin^2\frac{\delta_v}{4},$$

$$m^2_{\text{pole}} = \hat{\mu}'^2\, D_S$$

[Census FT.63(4), (6); Def-H.37(2); FT.54(4)] :

state	$(\mathcal{F}_g, \beta)$	δ_v/π	D_S	exact field	min. polynomial of D_S
s_1	$(\mathcal{F}_1, -)$, the ladder rung m=1	1	2	$\mathbb{Q}$	$x - 2$
s_2	$(\mathcal{F}_1, +)$, the ladder rung m=2	1	4	$\mathbb{Q}$	$x - 4$
s_3	$(\mathcal{F}_2, -)$, the cube member	$\frac{1}{2}$	$2 - \sqrt{2}$	$\mathbb{Q}(\sqrt{2})$	$x^2 - 4x + 2$
s_4	$(\mathcal{F}_2, +)$, the octahedron member	$\frac{2}{3}$	1	$\mathbb{Q}$	$x - 1$
s_5	$(\mathcal{F}_3, -)$, the dodecahedron member	$\frac{1}{5}$	$2 - 2\cos\frac{\pi}{10}$	$\mathbb{Q}(\cos\frac{\pi}{10})$	$x^4 - 8x^3 + 19x^2 - 12x + 1$
s_6	$(\mathcal{F}_3, +)$, the icosahedron member	$\frac{1}{3}$	$2 - \sqrt{3}$	$\mathbb{Q}(\sqrt{3})$	$x^2 - 4x + 1$

Descartes/Gauss–Bonnet:

$$V\,\delta_v = 4\pi \ \text{ on all five solids}$$

[Census FT.63(4); machine: exact in π-units];

(4) the branch order (derived, exact, uniform): $\boxed{D_S(\mathcal{F}_g,-) \;<\; D_S(\mathcal{F}_g,+) \qquad \forall g\in\{1,2,3\}}$

$$\mathcal{F}_1:\ 4>2;$$

$$\mathcal{F}_2:\ 1>2-\sqrt{2} \iff \sqrt{2}>1 \iff 2>1;$$

$$\mathcal{F}_3:\ 2-\sqrt{3}>2-2\cos\tfrac{\pi}{10} \iff \cos\tfrac{\pi}{10}>\tfrac{\sqrt{3}}{2} \iff \tfrac{\pi}{10}<\tfrac{\pi}{6} \iff 3<5$$

[machine: integer and radical certificates];

the six values are pairwise distinct — the defect species resolves all six states with zero degeneracy [machine: separation by field and by minimal polynomial];

(5) the branch splitting (the constituent mass difference):

$$\boxed{\Delta_g \;:=\; \hat{\mu}'^2\big(D_S(\mathcal{F}_g,+)-D_S(\mathcal{F}_g,-)\big)}$$

$$\Delta_1 = 2\hat{\mu}'^2 \in \mathbb{Q}\hat{\mu}'^2;$$

$$\Delta_2 = (\sqrt{2}-1)\hat{\mu}'^2 \in \mathbb{Q}(\sqrt{2})\hat{\mu}'^2;$$

$$\Delta_3 = \big(2\cos\tfrac{\pi}{10}-\sqrt{3}\big)\hat{\mu}'^2 \in \mathbb{Q}\big(\sqrt{3},\cos\tfrac{\pi}{10}\big)\hat{\mu}'^2;$$

$$\Delta_g>0 \ \ \forall g$$

[(4)'s certificates];

three distinct fields, three exact positive splittings, no imported value [P5];

(6) the generation map (the unique order-isomorphism):

the q-channel's three lines are ordered exactly:

$$(r_1^{(q)})^2 < (r_2^{(q)})^2 < (r_0^{(q)})^2 \text{ [FT.119(5); enclosure-proven, frozen]};$$

the three floors are ordered exactly: $2E(\mathcal{F}_1) < 2E(\mathcal{F}_2) < 2E(\mathcal{F}_3)$ [(1)] :

two exact three-element ordered sets admit exactly one order-isomorphism

$$g = 1 \leftrightarrow (r_1^{(q)})^2 \leftrightarrow \mathcal{F}_1,$$

$$g = 2 \leftrightarrow (r_2^{(q)})^2 \leftrightarrow \mathcal{F}_2,$$

$$g = 3 \leftrightarrow (r_0^{(q)})^2 \leftrightarrow \mathcal{F}_3;$$

the species row:

on the transport species the line order inverts [BM.39(5) : the spectral-maximal line receives the mass-minimum] $\Longrightarrow$ the isomorphism inverts with it;

the relating equation between the two currencies is ¬derived [BM.35(5)];

the scale is not supplied:

$$\hat{\mu}'^2 \in \text{Cal-F (logged)}, \quad \mu \in \text{Cal-F} :$$

the map is an order, ¬a number [P4; P5];

(7) the charge alphabet of the six (the second derived 2):

$$\mathcal{H}_{\mathsf{v}} \cong \mathbb{C}^2 \text{ [EW.32]};$$

$$Q|_{\mathcal{H}_{\mathsf{v}}} = \operatorname{diag}(1, 0), \quad \operatorname{spec} Q = \{0, +1\}, \quad \omega : \text{ spec} \mapsto \{0, -1\} \text{ [EW.36]} :$$

the seed carrier's weight doublet $\{0, +1\}$ together with its ω-image is exactly the ternary end-read alphabet $\{-1, 0, +1\}$ [EW.36; Def-BM.12(1); D3.4];

the branch index of (2) and the weight index of $\mathcal{H}_{\mathsf{v}}$ are two derived twos;

their attachment is (8)'s clause;

(8) the identification clause (the single declared row):

down-type := the floor's heavier defect branch (β=+);

up-type := the lighter (β=−);

the ordering is derived [(4)];

the name attachment is a label on a derived structural tuple and carries no datum [Def-BM.3 pattern; Def-BM.3(7) scope];

declared (A-BM.6) before use;

non-derived: the attachment of the Q-weight index to the branch index (7);

the flavour composition of a sealed composite — which three of the six its constituents occupy [BM.28(7); Def-BM.8(5) : a fractional constituent alphabet is structurally impossible under an integer-valued additive read];

(9) status and scope:

derived: (1)–(7);

declared: (8)'s name attachment alone;

¬a P4 addition, ¬a new dimensionful scale, §Cal-BM stays empty;

¬∃ a measured mass, ratio or Yukawa value in any clause [P5; machine: zero external tokens, zero decimal numerals];

species names are confined to the identification layers [Def-BM.3(7), Def-BM.6(7)];

beyond this tier: the scale relating the defect and informational currencies

the flavour composition of the baryon classes

the materialization scope M1/M2 [Def-H.20] is inherited at every use.

Rmk (six states, no data). Nothing in this definition is fitted. The floors come from the card template's edge counts; the branches come from the duality that leaves those counts fixed and splits everything finer; the branch masses come from the conical defect at a vertex where three faces meet, closed by Descartes against four pi. What the assembly adds is one order-isomorphism between two three-element ordered sets the corpus had already derived separately, and one name. The result is six states, six pairwise distinct exact algebraic masses on the defect species, three exact rationals on the informational one, and a hierarchy that runs one way in the bit currency and the other way in the angle currency — with the equation between the two currencies still open, and still not guessed.

4.48 (BM.57 — The continuation-symmetry census: the vertex asymmetry is two, and it splits nothing).

Thm.

$$(1) \quad \text{the strong 3-vertex:}$$

$$\text{the invariant tensor is } \varepsilon_{ijk} \text{ [CT3.17; D3.9];}$$

$$\varepsilon_{\sigma(i)\sigma(j)\sigma(k)} = \operatorname{sgn}(\sigma)\, \varepsilon_{ijk} \;\; \forall \sigma \in S_3$$

$$[\text{machine: all six permutations}] :$$

$$\text{the continuation group:}$$

$$\operatorname{Stab}_{S_3}(\varepsilon) = A_3 \cong C_3 = S_3 \cap SU(3),$$

$$|A_3| = 3,$$

$$\text{nontrivial count } 2$$

$$[\text{BM.3}(1);\ \text{machine: the invariant set is exactly the cyclic group}];$$

$$(2) \quad \text{the } (1,1)\text{-vertex (the chain wall):}$$

$$\text{the invariant tensor is } \delta^i{}_j \text{ [CT3.17];}$$

$$\text{the type-preserving relabelling group of a } (1,1) \text{ vertex is } S_1 \times S_1 :$$

$$|\cdot| = 1,$$

nontrivial count 0;

the leg swap is not a continuation:

$$\delta^i{}_j$$

$= \delta_i{}^j$ admits it as a bare 2-tensor [machine],

but it acts on the oriented end registers as (σ_s, σ_t)

$$\mapsto (\sigma_t, \sigma_s) :$$

$q_{\text{EM}} \mapsto -q_{\text{EM}}$ on all four end pairs

[D3.4; EM.9a : charge-conjugation odd; machine] :

the swap is the ω-involution, not a continuation symmetry;

(3) the asymmetry: $\boxed{|A_3| - |S_1 \times S_1| = 3 - 1 = 2, \qquad 2 - 0 = 2}$ [machine] :

the asymmetry is 2 by both conventions — the order difference and the nontrivial-count difference agree;

the count is derived;

(4) the first annihilation row (class-blindness):

both baryon classes are ε-sealed 3-junctures with identical vertex data [Def-BM.3(2); BM.17; BM.26] :

vertex invariant	neutral class	unit class	difference
junction grammar	ε_{ijk} 3-juncture	ε_{ijk} 3-juncture	0
vertex type [D0.8]	$(3, 0)$	$(3, 0)$	0
sealing-phase class [BM.2(1); Census FT.49(3)]	$4\mathbb{Z}$	$4\mathbb{Z}$	0
E_{int} [Census FT.35(2)]	6	6	0
continuation group	A_3, order 3	A_3, order 3	0

vertex invariant	neutral class	unit class	difference
sealing grammar count [CT3.17a(2)]	$g(3) = 1$	$g(3) = 1$	0

every vertex-structural quantity is a common factor on the two classes [machine: the difference table is identically zero] $\Longrightarrow$ no vertex datum can split the two rows;

(5) the second annihilation row (the seed is colour-blind):

$$T_a \mathsf{v} = 0 \text{ [EW.32]};$$

$$\mathcal{G} = (gW_1, gW_2, gW_3, g'Y) \text{ [EW.33]} : \neg\exists \text{ a colour generator in the transport data;}$$

$$\mathrm{Re}\langle \mathcal{G}_A \mathsf{v}, \mathcal{G}_B \mathsf{v} \rangle = \frac{1}{4} \begin{pmatrix} g^2 & 0 & 0 & 0 \\ 0 & g^2 & 0 & 0 \\ 0 & 0 & g^2 & -gg' \\ 0 & 0 & -gg' & g'^2 \end{pmatrix} \qquad \text{[machine: Pauli spinors over } \mathbb{Q}(i) \otimes \mathbb{Q}[g, g']] :$$

the free-symbol set of the Gram is $\{g, g'\}$: zero colour symbols [machine] $\Longrightarrow$ $\boxed{\dfrac{\partial\, G(c)}{\partial\,(\text{any slot-index datum})} = 0 \text{ exactly}}$

the exact factorizations [machine: polynomial identities in $\mathbb{Q}[g, g', c_1, \ldots, c_4]$] :

$$G(c) = \frac{1}{4}\Big[g^2(c_1^2 + c_2^2 + c_3^2) - 2gg'c_3c_4 + g'^2c_4^2\Big] = \frac{1}{4}\Big[g^2(c_1^2 + c_2^2) + \big(gc_3 - g'c_4\big)^2\Big];$$

$$\ker M^2 = \mathrm{span}\{(0, 0, g', g)\} \text{ [EW.22]} :$$

$$\frac{(g^2 + g'^2)}{4}\|c\|^2 - G(c) = \frac{1}{4}\Big[g'^2(c_1^2 + c_2^2) + \big(gc_4 + g'c_3\big)^2\Big] \geq 0;$$

$$\det(xI - \mathrm{Gram}) = \Big(x - \frac{g^2}{4}\Big)^2 x\Big(x - \frac{g^2 + g'^2}{4}\Big) \qquad \text{[machine: Faddeev–LeVerrier over } \mathbb{Q}[g, g', x]];$$

(6) the third annihilation row (the trace law):

every Hermitian slot operator on a sealed singlet reads only its trace: $\mathfrak{T}(H)$

$$= 3a \text{ for } H$$

$$= aI_3 + zP + z^*P^{\mathsf{T}}$$

[BM.1(2); FT.117(1)] :

the trace's free-symbol set is $\{a\}$: the C_3-doublet datum (B, θ) in z

$$= Be^{i\theta} \text{ is annihilated [machine];}$$

$$1 + \omega + \omega^2$$

$= 0$ in $\mathbb{Q}[\omega]/(\omega^2+\omega+1)$: the C_3 average of the doublet datum is exactly zero [machine];

the companion row:

every S_3-covariant slot operator has charpoly $(x-a-3b)(x-a)^2$ [FT.116; machine] : the doublet is degenerate at hadron level;

(7) the multiplicity row:

a continuation-symmetry count is a degeneracy multiplicity of the sealed state;

a uniform multiplicity factor cancels in the normalized read at every priced factor $\{1, 2, 3, 15, 20, \frac{7}{3}\}$ [BM.28(4)(i); BM.30(3); machine] :

and it multiplies the total uniformly — the same factor on both classes by (4) $\implies$ it shifts a total and never splits a class;

(8) the verdict: $\boxed{\Delta m_{\text{vertex}} = 0 \text{ exactly}}$ [(4) $\wedge$ (5) $\wedge$ (6) $\wedge$ (7); machine] :

the vertex-asymmetry hypothesis is refuted as a mass-splitting mechanism;

the asymmetry itself (3) is derived and stands;

(9) the residual content (where the asymmetry does act):

A_3 is the forced family group of the seal [BM.3(1)];

its orbit on a transport profile has length 3 [Def-BM.11(2), (3); FT.158(1) : $U^3 = E_0$, $U \neq E_0$, $U^2 \neq E_0$] :

the asymmetry generates the family triplet — three lines per channel [BM.60(1)];

it does not split the baryon classes;

the $(1,1)$-vertex's single continuation is the chain-wall telescoping [EM.9b] — and its content is (2)'s parity law, consumed at BM.59;

(10) status:

derived: (1)–(9);

refuted: the vertex asymmetry as a splitting mechanism (8);

¬a P4 addition, ¬a measured value [P5].

Req. A7 (Part 1); D0.8, D2.6, D3.3, D3.4, D3.9 (Parts 1–2); CT0.9, CT3.17, CT3.17a(2), EW.15, EW.20, EW.22, EW.32, EW.33, EW.35, EM.9, EM.9a, EM.9b, EM.24(1) (Part 2); P5 (Parts 1–2); Def-H.37(1), FT.106(3a), FT.116, FT.117(1)–(3), FT.158(1) (Part 3); Census FT.35(2), FT.49(3) (row-wise); Def-BM.1, Def-BM.3(2), Def-BM.11(2), Def-BM.11(3), BM.1(2), BM.2(1), BM.3(1), BM.17, BM.26, BM.28(4)(i), BM.30(3), BM.36(4), BM.36(5) (this part); the exact Levi-Civita and Kronecker invariant censuses, Pauli/Gram algebra over $\mathbb{Q}(i) \otimes \mathbb{Q}[g, g']$, Faddeev–LeVerrier charpolys, the cyclotomic ring $\mathbb{Q}[\omega]/(\omega^2+\omega+1)$, sum-of-squares factorizations [math; machine: `checks/verify_H_constituent_transport_splitting.py` §A].

Pf.

(1) : ε_{ijk} built as the sign of the index triple;

all six σ

$\in S_3$ tested against $\varepsilon_{\sigma(i)\sigma(j)\sigma(k)}$

$= \varepsilon_{ijk}$: the invariant set has three elements and equals $\{(012), (120), (201)\}$ [machine];

$$\operatorname{sgn}(\sigma)$$

$$= +1 \text{ on exactly those three [machine]}$$

$$(2):\ \tau(v)$$

$$= (1,1)\ \text{[D0.8]} \implies \text{the type-preserving relabelling group is } S_1 \times S_1;$$

$$\delta^i{}_j \text{ symmetric in its indices [machine];}$$

$$\text{the swap row: all four end pairs mapped, } q$$

$$\mapsto -q \text{ at each [machine];}$$

$$\text{EM.9a's charge-conjugation oddness re-cited}$$

$$(3):\ 3-1=2, \quad 2-0=2 \text{ [machine: integers]}$$

$(4):$ each row re-cited from its own tag and compared across the two classes [machine: the difference table];

$$\text{BM.28}(6)\text{'s identical set re-cited}$$

$$(5):\ \mathsf{v}$$

$$= (0,1)^{\mathsf{T}}:\ \sigma_1 \mathsf{v}$$

$$= (1,0)^{\mathsf{T}},$$

$$\sigma_2 \mathsf{v}$$

$$= (-i,0)^{\mathsf{T}},$$

$$\sigma_3 \mathsf{v}$$

$$= (0,-1)^{\mathsf{T}},$$

$$\sigma_0 \mathsf{v}$$

$$= \mathsf{v};$$

$$W_3 \mathsf{v}$$

$$= -\tfrac{1}{2}\mathsf{v},$$

$$Y \mathsf{v}$$

$$= +\tfrac{1}{2}\mathsf{v},$$

$$Q \mathsf{v}$$

$$= 0 \text{ [machine]};$$

the Gram block and its antisymmetric imaginary part re-derived [machine]

the kernel: Gram $\cdot\, (0, 0, g', g)^{\mathsf{T}} = 0$ [machine];

$$G(c)$$

$= \tfrac{1}{4}[g^2(c_1^2 + c_2^2) + (gc_3 - g'c_4)^2]$ and the ceiling's SOS form expanded and compared in $\mathbb{Q}[g, g', c_1, \ldots, c_4]$ [machine]

$\det(xI - \text{Gram})$ by Faddeev–LeVerrier and compared to the factored form in $\mathbb{Q}[g, g', x]$ [machine];

the symbol census of every Gram entry [machine]

$$(6) : \ H$$

$$= aI_3 + zP + z^* P^{\mathsf{T}} \text{ with } P \text{ the 3-cycle} : \ \operatorname{tr} P$$

$$= \operatorname{tr} P^{\mathsf{T}}$$

$$= 0 \implies \mathfrak{T}(H)$$

$$= 3a \text{ [machine over } \mathbb{Q}[a, \operatorname{Re} z, \operatorname{Im} z]];$$

$$\omega^3 = 1,\ 1 + \omega + \omega^2 = 0 \text{ in } \mathbb{Q}[\omega]/(\omega^2{+}\omega{+}1) \text{ [machine]}$$

$$aI_3 + bJ_3 :\ \text{charpoly } (x - a - 3b)(x - a)^2 \text{ by Faddeev–LeVerrier over } \mathbb{Q}[a, b, x] \text{ [machine]}$$

$$(7) :\ \frac{\sum_R f\, w\, \Delta S}{\sum_R f\, w}$$

$$= \frac{\sum_R w\, \Delta S}{\sum_R w} \text{ at every uniform } f \in \{1, 2, 3, 15, 20, \tfrac{7}{3}\} \text{ [machine: exact Fractions]};$$

$$(8) :\ \text{conjunction of } (4), (5), (6), (7) \text{ [machine]}$$

$$(9) :\ U^3$$

$$= E_0,$$

$$U$$

$$\neq E_0,$$

$$U^2$$

$$\neq E_0 \text{ for the 3-cycle permutation matrix over } \mathbb{Q}(i) :\ \text{the orbit length is exactly } 3 \text{ [machine]};$$

$$(10) :\ \text{assembly. } \blacksquare$$

Phys. The hypothesis was that the strong vertex, having two continuation symmetries against the electromagnetic vertex's one, would split the quark masses. The count is right: the Levi-Civita tensor is preserved by exactly the three even permutations of its slots, so the strong 3-vertex carries two nontrivial continuations, while the chain wall's single in-leg and single out-leg admit no nontrivial relabelling at all — its only candidate, the leg swap, reverses the end registers and is therefore the charge-conjugation involution, not a symmetry of the transport. The asymmetry is two by either counting convention. And it splits nothing. Three independent theorems kill it: the two baryon classes are the same vertex with the same phase class and the same edge count, so any vertex invariant is a common factor; the transported seed is annihilated by every colour generator, so the mass form has no slot index to differentiate; and on a sealed singlet every slot operator reads only its trace, which annihilates the doublet datum the continuation group acts on. What

the asymmetry does buy is the family: the same C_3 that cannot split the neutron from the proton is what cycles a transport profile into three lines.

4.49 (BM.58 — The observation frames: the metric is charge-blind, and the frame's content is the seal asymmetry).

Thm.

$$(1) \quad \text{the metric data carry no charge argument:}$$

$$\Omega = e^{\alpha_\Omega a_{00}}, \quad g_{00} = -\Omega^2, \quad g_{0i} = \Omega^2 \varkappa \beta w_i, \quad g_{ij} = \Omega^2(\delta_{ij} + \varkappa q_{ij})$$

$$[\text{CT1.12};\ \text{FT.157}(1)]:$$

$$N^2 = -\frac{\det g}{\det \gamma} > 0,$$

$$N_i = g_{0i} = \Omega^2 \varkappa \beta w_i,$$

$$N^i = \gamma^{ij} N_j,$$

$$ds^2 = -N^2 d\tau^2 + g_{ij}(dx^i + N^i d\tau)(dx^j + N^j d\tau)$$

$$[\text{Def-BM.10}(2);\ \text{machine}]:$$

the token census on the canonical defining blocks:

$$\text{charge-token occurrences inside Def-BM.10}(2), (3), (5) \text{ and CT1.12}: \quad 0$$

$$[\text{machine}];$$

$$(2) \quad \text{the seven geometric monomials are charge-blind:}$$

$$\Delta x_{\mathrm{B}} = \mathfrak{h} E^{-1} (\ln 2)^{-1}, \quad V_{\text{eff}} = \mathfrak{h}^3 E^{-3} (\ln 2)^{-3}, \quad \rho = \mathfrak{h}^{-4} E^4 (\ln 2)^4,$$

$$\kappa^2 \rho = 2\pi\, \mathfrak{h}^{-2} E^4 (\ln 2)^3,$$

$$r_s = \tfrac{1}{2} \mathfrak{h} E,$$

$$(\Delta x_{\mathrm{B}})^2 \kappa^2 \rho = 2\pi E^2 \ln 2,$$

$$\Delta x_{\mathrm{B}}\, r_s = \frac{\mathfrak{h}^2}{2\ln 2}$$

[BM.13(1), BM.14(1)–(3), BM.15(4), BM.16(1), (4)] :

the charge exponent is 0 in all seven [machine: the Laurent-monomial census];

the lock identities hold as monomial identities [machine];

at $E_{\mathrm{int}} = 6$:

$$V_{\mathrm{eff}} = \frac{\mathfrak{h}^3}{216(\ln 2)^3}, \quad \rho = \frac{1296(\ln 2)^4}{\mathfrak{h}^4}, \quad \kappa^2\rho = \frac{2592\pi(\ln 2)^3}{\mathfrak{h}^2},$$

$$r_s = 3\mathfrak{h},$$

$$(\Delta x_{\mathrm{B}})^2\kappa^2\rho = 72\pi\ln 2$$

[BM.43(1); machine: every coefficient exact];

(3) the frames are masks, not metrics:

the colour frame: $\mathcal{C}(\Sigma; C) \in \{0, 1\}$,

an idempotent, and one dangling attachment zeroes the whole colour sector [D2.14; FT.19; BM.1(1); BM.32(1)] — the only idempotents of $\{0, \ldots, 7\}$ are $\{0, 1\}$ [machine];

the EM frame:

$W(Q) > 0$ at every local charge class $Q \in \{-3, \ldots, +3\}$ [Def-BM.8(6); machine] :

$\neg\exists$ a local projector on Q;

neutrality is exhausted only at universe scale [EM.11; BM.47(4), (5)] :

a mask acts on operators and on the probability space, not on the metric [D2.11; Def-BM.10(2)'s alphabet, (1)] :

the emergent metric is the ambient ledger's coarse-grained sector coefficients, and no restriction re-reads them;

(4) the metric correction vanishes:

$$\boxed{\Delta m_{\text{metric}} \;=\; 0 \;\text{ exactly}}$$

$[(1) \wedge (2) \wedge (3);\ \text{machine: the class difference of each of the seven monomials is } 0] :$

the observation-frame hypothesis is refuted as a mass-splitting mechanism;

the parent and the daughter carry identical curvature rows [BM.43(1) : re-cited];

(5) what the frame asymmetry does carry:

colour self-seals locally; EM seals at universe scale [BM.47(5)];

the surviving mass content is the scar's field energy

$$E_{\text{EM}}^{\text{self}} = \frac{c\, e^2 q^2}{4\pi\, \Delta x_{\text{B}}} = c\, \varepsilon_{\text{EM}}\, q^2\, m_{\text{hid}},$$

$$\Delta m_{\text{EM}} = \frac{3c\, e^2 \ln 2}{2\pi\, \mathfrak{h}} > 0,$$

the charged member the heavier one

$[\text{BM.46}(1), (4);\ \text{Def-BM.12}(5)] :$

the monopole selector is Q^2

$= (0, 1, 1)$ at Q

$= (0, \pm 1)$ [machine] $\implies$ $\boxed{\text{the frame's contribution to } m_n - m_p \text{ is } -\, c\, \varepsilon_{\text{EM}}\, m_{\text{hid}} \;<\; 0}$

[machine: both priced c] :

the frame asymmetry works against the neutral class's excess — so the missing piece must be class-dependent, of the opposite sign, and not electromagnetic;

(6) the uniqueness row (the force direction's output):

BM.52(3)'s sector census is complete on the functionals it enumerates ($\Delta S,\ E_{\text{int}},$ the Janus balance, the geometric monomials, $|w|^2,$ the Casimir, the locks) and gives five zeros and one monomial [BM.52(4)];

the complete inventory of class-dependent derived data at a sealed 3-juncture, extended by the register functionals the census does not enumerate, is

datum	species	status	sign for $m_n - m_p$
the vertex invariants	slot / grammar	class-blind [BM.57(4)]	0
the colour data	slot	annihilated by the trace law [BM.1(2); FT.117]	0
the metric monomials	chart / metric	charge exponent 0 [(2)]	0
$F_{LR} = qI_2$, the chiral splitting $2\|q\|$	operator	trace-free within a class: $\sum_{s,t} tq = 0$ [BM.22(2)–(4)]	0 on a trace read [BM.1(2)]
$Z^{\text{amp}}(0,S)/Z^{\text{amp}}(\pm 1,S) = \frac{58}{13}, \frac{2398}{401}$	amplitude	barred: amplitude $\neq$ mass [FT.79(5); BM.37(3)(b)]	barred
$E_{\text{EM}}^{\text{self}} \propto Q^2$	chart / field	adopted [Def-BM.12(6)]	negative [(5)]
$W(Q) = (1, 6, 15, 20, 15, 6, 1)$	transport register	derived, unbarred [Def-BM.8(6); FT.106(3a)]	positive [BM.59]

the six sector rows of BM.52(3) and the Lyapunov row of BM.51(6) are re-cited inside this table as its first four entries;

exactly one candidate survives with the required sign;

[machine: the trace row $\sum_{s,t\in\{\pm1\}} tq = 0;$ the row census exhaustive against the tier's own inventory scan of BM.22, BM.28(5), (6), BM.30(1), BM.46(2), (4), BM.51(6), BM.52(3), Def-BM.8(6)];

(7) status:

derived: (1)–(6);

refuted: the observation frame as a metric-splitting mechanism (4);

¬a P4 addition, ¬a measured range, radius or flux [P5].

Req. D1.3, D2.1, D2.9, D2.11, D2.14 (Parts 1–2); CT1.12, CT1.22, CT2.6, EM.11, EW.22 (Part 2); P5 (Parts 1–2); Def-H.32(2), FT.19, FT.79(5), FT.106(3), FT.117(1)–(3), FT.157(1) (Part 3); Def-BM.4, Def-BM.8(6), Def-BM.10(2), Def-BM.10(3), Def-BM.10(5), Def-BM.12(5), Def-BM.12(6), BM.1(1), BM.1(2), BM.13(1), BM.14(1)–(3), BM.15(4), BM.16(1), BM.16(4), BM.22(2)–(4), BM.28(1), BM.32(1), BM.37(3), BM.43(1), BM.46(1), BM.46(4), BM.47(4), BM.47(5) (this part); the ADM split by Schur complement, Laurent-monomial exponent arithmetic, idempotent censuses, token censuses [math; machine: `checks/verify_H_constituent_transport_splitting.py` §B].

Pf.

(1) : Def-BM.10(2)'s split re-derived as linear algebra on CT1.12's dictionary;

$$N^2$$

$= -\det g/\det\gamma$ by Schur, denominators cleared [machine];

the line-element identity expanded and compared [machine];

the token census on the two defining blocks [machine]

(2) : each monomial built from its own tag's formula as a Laurent monomial in $(\mathfrak{h}, E, \ln 2, \pi, Q)$;

the Q-exponent censused [machine];

$(\Delta x_{\mathrm{B}})^2\kappa^2\rho$ and $\Delta x_{\mathrm{B}} r_s$ multiplied out and compared [machine];

$$E$$

$= 6$ substituted: 6^3

$$= 216,$$

$$6^4$$

$$= 1296,$$

$$2\pi \cdot 1296$$

$$= 2592\pi,$$

$$\tfrac{1}{2} \cdot 6$$

$$= 3,$$

$$2\pi \cdot 36$$

$$= 72\pi \text{ [machine]}$$

$(3):\ x^2 = x$ over $\{0, \dots, 7\}$ gives $\{0, 1\}$ [machine: exhaustive];

$W(Q)$ re-enumerated and positive at all seven classes [machine]

$(4):$ the class difference of each monomial of (2) is 0 because the Q-exponent is 0 [machine]

$(5):$ BM.46(1),

(4) re-cited;

Q^2 at $(0, \pm 1)$

$= (0, 1, 1)$ [machine];

the sign of $-(1 - 0)$

$$c$$

$$\varepsilon_{\text{EM}}$$

m_{hid} at $c \in \{\frac{1}{2}, \frac{3}{5}\}$,

$\varepsilon_{\text{EM}} > 0$,

$m_{\text{hid}} > 0$ [machine]

(6) : the inventory scan of the class-dependent rows of BM.22,

BM.28(5),

(6),

BM.30(1),

BM.46(2),

(4),

Def-BM.8(6) [machine: the row set is exhaustive];

the force-matrix trace: spec

$$= \{\tfrac{1}{2}(c_0 + s|\mathsf{c}|) + tq\},$$

$$\sum_{s,t} [\tfrac{1}{2}(c_0 + s|\mathsf{c}|) + tq]$$

$= 2c_0$ [BM.22(3); machine] $\implies$ the trace read is charge-blind [BM.1(2)]

(7) : assembly. ■

Phys. The intuition was that seeing the world through photons is not seeing it through colour, and that the difference should show up in the geometry. It does not. The emergent metric is built from the ambient ledger's coarse-grained sector coefficients, and not one of its seven derived monomials carries a charge exponent: the width, the volume, the density, the curvature scalar, the Schwarzschild-type read and both locks come out identical on the two classes to the last Laurent monomial. What a frame really is, in this corpus, is a mask — and the two masks have different grammars. Colour's is an idempotent that annihilates a sector at the first dangling attachment; electromagnetism's is a census that lets every local charge class through and cancels only once, at the scale of the whole universe. Masks filter operators and probability spaces. They do not re-read the metric. So the metric correction is exactly zero, and the frame's one surviving mass contribution is the scar's own field energy — positive, charge-selected, and on the wrong side: it makes the charged member heavier. That is the useful output. It narrows the search to a single column of the inventory, and the column has one entry.

Theorem 4.50 (BM.59 — The constituent transport splitting: the order residue, the two directions, and the exact positive sign). Thm.

$$(1) \quad \text{the missing piece, stated:}$$

BM.55(1)'s five-term decomposition is derived term by term:

$$\Delta m_{\text{Lyapunov}}$$

$$= 0 \text{ [BM.51}(6)\text{]};$$

$$\Delta m_{\text{metric}}$$

$$= 0 \text{ [BM.53}(2)\text{]};$$

$$\Delta m_{\text{binding}}$$

$$= 0 \text{ [BM.54}(2), (5)\text{]};$$

$$\Delta m_{\text{force}}$$

$$= \Delta m_{\text{self-int}}$$

$$= -\Delta m_{\text{EM}} \text{ [BM.52}(4), (5)\text{]} :$$

$$\Delta m = m_n - m_p = -\frac{3\,c\,e^2 \ln 2}{2\pi\,\mathfrak{h}} < 0, \qquad \frac{m_n}{m_p} = \frac{4\pi}{4\pi + c\,e^2} \in (0,1) \qquad \text{[BM.55}(1)\text{–}(3);\ \text{frozen]} :$$

the order residue [BM.55(4) : named, not filled] :

$$= \text{ a class-dependent constituent transport splitting with } m^{(n)}_{\text{constituent}} > m^{(p)}_{\text{constituent}};$$

the residue is fillable, and this tier fills it: $\Delta m_{\text{transport}} = \dfrac{\Sigma_{\text{cl}}(0) - \Sigma_{\text{cl}}(+1)}{\mathfrak{h}} = \dfrac{\ln(4/3)}{\mathfrak{h}} > 0$

[Def-BM.13(4); machine];

(1a) the four grounds of BM.55(4) and how this object evades each:

the ground on record	the evasion
the fractional constituent alphabet is unreachable [BM.28(7)]	Σ_{cl} is built from the integer alphabet $\{-1, 0, +1\}$ alone [Def-BM.13(1), (3)]; ¬a fraction appears
the species labels carry no datum [Def-BM.3]	Σ_{cl} carries no label : $itisacensusofthederivedread$ [Def-BM.8(6)]
the deficit/inertial conversions are item-7 content	Σ_{cl} is already in the informational currency — nats of binary registration [Def-BM.13(6)]; ¬a conversion is used
no clause attaches constituent profiles to the charge classes	Σ_{cl} is ¬a profile: m_q is attached to each constituent by q_{core} itself [Def-BM.8(1); EM.9]

the residue is filled by the transport register, not by the transport profile:

Def-BM.11's family-profile map remains channel-line data and stays unattached to the baryon rows [BM.55(4)'s fourth ground stands untouched];

(2) the mass direction (what the spectrum requires):

a term $\delta m(Q)$ that reverses the order must satisfy four exact bars

(i) class-dependent: $\delta m(0) \neq \delta m(+1)$;

(ii) positive for the neutral class: $\delta m(0) - \delta m(+1) > 0$;

(iii) currency-matched: $[\delta m] = \mathfrak{q}^{+1/2}$ through the rest slot [Def-BM.10(4); CT2.1];

(iv) alphabet-disjoint from the count: ¬a function of E_{int} alone [BM.28(1) scope];

and it must not be a barred route [BM.37(6)];

[machine: the inventory scan of BM.58(6)'s table against (i)–(iv)] :

the unique survivor is $W(Q)$ read through its logarithm	[Def-BM.13(4), (6)];

(3) the force direction (what the seal permits):

colour is a local idempotent and its slot data are annihilated on a sealed singlet [BM.57(5), (6); BM.58(3)];

EM has no local projector and admits every class with positive census weight [BM.58(3)] :

the only class-dependent datum a sealed junction can carry is the EM end-register census $W(Q)$

$$= [z^Q](2 + z + z^{-1})^3$$

[Def-BM.8(6); BM.28(2)];

(4) convergence:

the two directions land on the same object;

$$\boxed{\Delta m_{\text{transport}} := \frac{\Sigma_{\text{cl}}(0) - \Sigma_{\text{cl}}(+1)}{\mathfrak{h}} = \frac{\ln(4/3)}{\mathfrak{h}} = \frac{2\ln 2 - \ln 3}{\mathfrak{h}} > 0}$$

[machine: the certificate $4 > 3$] :

$$[\Delta m_{\text{transport}}] = \mathfrak{q}^{+1/2} = [\text{energy}]$$

[$\ln W \in \mathfrak{q}^0$, $[\mathfrak{h}] = \mathfrak{q}^{-1/2}$; CT2.1; D2.9; machine: exponent arithmetic];

(5) the constituent row:

the per-constituent transport entropies are $\ln m_q$

$$= (\ln 2,\ 0,\ 0) \text{ at } q$$

$$= (0, +1, -1) \text{ [Def-BM.13(1)] :}$$

$$\boxed{m(q_{\text{core}}{=}0) - m(q_{\text{core}}{=}\pm 1) = \frac{\ln 2}{\mathfrak{h}} > 0}$$

[machine: $2 > 1$] :

the neutral-core constituent is the heavier constituent;

the composite's splitting is not three times the constituent's:

$$0 < \ln \tfrac{4}{3} < \ln 2 < 3\ln 2$$

$$[\text{machine: } 3 < 4 < 6 < 24] :$$

the additive charge constraint couples the three registers, and the coupling strictly attenuates the constituent hierarchy;

(6) the parity lock (why the register is a transport datum and not a label):

$$\sum_i f_{v_i} = -q,$$

$$\sum_i \aleph_{v_i} \equiv |q| \pmod 2,$$

$$\sum_i \aleph_{v_i} \geq |q|$$

$$[\text{EM.9b};\ \text{D3.3};\ \text{Def-BM.13}(2)] :$$

the interior wall content of a constituent's transport is parity-locked to its charge core;

the neutral core is wall-balanced $(\min \sum \aleph = 0)$;

the charged core is wall-unbalanced $(\min \sum \aleph = 1)$ [machine: exhaustive over every sign sequence, $k = 2, \ldots, 7$] :

the two neutral transport realizations are the two wall-free orientations;

the charged core admits none :

the multiplicity m_q is a property of the transport, not of a naming convention [FT.106(3a) : it is the norm layer's own closed form];

(7) the sign is fixed by the reference, not chosen:

$$\Delta S :$$

$$= S - S(\text{vac}) \text{ with } S(\text{vac})$$

$$= 0\ [\text{BM.35}(3);\ \text{CT0.2}] \implies \Delta\Sigma_{\text{cl}}(Q)$$

$$= \ln W(Q) - \ln W(\text{vac}) \qquad = \ln W(Q),$$

$$W(\text{vac}) = 1;$$

the two competing references:

reference	class difference	status
the vacuum ($W{=}1$)	$+\ln\frac{4}{3}$	adopted: the corpus's own deficit rule [BM.35(3)]
the unconstrained register ($4^3{=}64$)	$-\ln\frac{4}{3}$	barred: ¬a state of the composite [Def-BM.5(1)]; class-dependent reference
Ansatz-B ($-\alpha\ln\Omega$)	$-\alpha\ln\frac{4}{3}$	barred: an external ansatz, α undeclared [Def-H.28]

exactly one reference is a state of the composite [machine: the admissible-reference census];

(8) the species legitimacy:

Σ_{cl} is ¬a measure functional of (Γ, R) — its alphabet is the end-sign register $\{\sigma_s, \sigma_t\}$ [D3.4; CT0.9],

disjoint from the flicker alphabet and from the metric alphabet [BM.37(2) pattern; machine: the token census] :

it enters through the rest slot the dispersion read already reads [Def-BM.10(4); BM.35(4); BM.37(4)],

in the Def-BM.12(6) pattern of a refinement, ¬a new mass read;

no route ratio is read as a mass ratio:

$$\left.\frac{m_{\text{rest}}(0)}{m_{\text{rest}}(+1)}\right|_{\varepsilon_{\text{EM}}=0} = \frac{\ln 1280}{\ln 960} \neq \frac{4}{3}$$

[machine];

the strict contraction: $\dfrac{\Sigma_{\text{B}}}{6\ln 2} < \dfrac{1}{3} \iff 3\ln\frac{4}{3} < 2\ln 2 \iff 16 < 27$

[machine];

(9) the defect-species companion (conditional, the hypothesis on its face):

hypothesis: the two baryon classes differ by exactly one down $\leftrightarrow$ up constituent substitution [the minimal $\Delta q = \pm 1$ step; BM.23(4)] :

then

$$\Delta m \;=\; \Delta_g \;=\; \hat{\mu}'^2\big(D_S(\mathcal{F}_g,+) - D_S(\mathcal{F}_g,-)\big) \;>\; 0$$

[Def-BM.14(5); machine: the three certificates] :

$$\Delta_1 = 2\hat{\mu}'^2, \qquad \Delta_2 = (\sqrt{2}-1)\hat{\mu}'^2, \qquad \Delta_3 = \big(2\cos\tfrac{\pi}{10} - \sqrt{3}\big)\hat{\mu}'^2;$$

the composition itself is ¬derived: a fractional constituent alphabet is structurally impossible under an integer-valued additive read [BM.28(7); Def-BM.8(5)];

deficit $1'$ stands [FT.125(4) pointer];

(10) status:

derived: (1)–(8);

conditional: (9), with its hypothesis on its face [A §0.2a];

the splitting is exact and strictly positive;

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a route adopted [BM.30(5)],

¬a measured value [P5];

§Cal-BM stays empty.

Req. D0.8, D2.1, D2.9, D3.3, D3.4 (Parts 1–2); CT0.2, CT0.9, CT1.29, EM.9, EM.9b (Part 2); P1, P4, P5 (Parts 1–2); Def-H.28, Def-H.32(2), Def-H.37(2), FT.79(5), FT.106(3a), FT.125(4) (Part 3); Census FT.35(2) (row-wise); Def-BM.5(1), Def-BM.8(5), Def-BM.8(6), Def-BM.10(4), Def-BM.12(6), Def-BM.13, Def-BM.14(5), BM.20(1), BM.23(4), BM.28(1), BM.28(2), BM.28(4)(i),

BM.28(7), BM.30(2), BM.30(3), BM.30(5), BM.35(3), BM.35(4), BM.37(2), BM.37(4), BM.37(6), BM.41(4), BM.43(1), BM.46(4), BM.57, BM.58 (this part); the exhaustive end-sign and sign-sequence censuses, exact ln-ring identities with integer certificates, Laurent-monomial exponent arithmetic, token censuses [math; machine: `checks/verify_H_constituent_transport_splitting.py` §C].

Pf.

$$(1):\ \text{the four degenerate-or-inverted rows re-cited from their own tags;}$$

the order residue named as the missing class-dependent rest-slot term [definition];

$$(2):\ \text{the four bars stated;}$$

the inventory scan of BM.58(6)'s seven rows against them [machine: one survivor]

$$(3):\ \text{BM.57}(5),(6)\text{ and BM.58}(3)\text{ conjoined;}$$

$$(4):\ \Sigma_{\mathrm{cl}}(0)-\Sigma_{\mathrm{cl}}(+1)$$

$$=\ln 20-\ln 15$$

$$=\ln\tfrac{4}{3}$$

$$=2\ln 2-\ln 3\ \text{[machine: the prime-log ring];}$$

$$4>3\ \text{[machine];}$$

the exponent row $\mathfrak{q}^{0}\cdot\mathfrak{q}^{+1/2}$

$$=\mathfrak{q}^{+1/2}\ \text{[machine]}$$

$$(5):\ \ln m_0-\ln m_{\pm1}=\ln 2-0=\ln 2\ \text{[machine]};\ 2>1\ \text{[machine]};$$

$$3<4\implies \ln\tfrac{4}{3}<\ln 2;$$

$$4<6\implies \ln\tfrac{4}{3}<\ln 2;$$

$$4 < 24 \implies \ln \tfrac{4}{3} < 3 \ln 2 \text{ [machine: integer certificates]}$$

$$(6): \text{ exhaustive over } 2^{k} \text{ sign sequences for } k$$

$$= 2,$$

$$\dots,$$

$$7: \sum f$$

$$= -q,$$

$$\sum \aleph$$

$$\equiv |q| \ (2),$$

$$\sum \aleph$$

$$\geq |q| \text{ at every one [machine]};$$

$$\text{the minima } 0 \text{ and } 1 \text{ attained [machine]}$$

$$(7): \text{ the three references enumerated and their class differences computed in the } \ln\text{-ring [machine]};$$

$$W(\text{vac})$$

$$= 1 \text{ for the empty register [machine]};$$

$$\text{Def-BM.5}(1)\text{'s boundary-data clause re-cited};$$

$$\text{Def-H.28's external-ansatz mark pinned}$$

$$(8): \text{ the token census of the defining clause against the flicker and metric alphabets [machine]};$$

$$\ln 1280 / \ln 960 \neq \tfrac{4}{3}$$

$$\iff 3\ln 1280$$

$$\neq 4\ln 960$$

$$\iff 1280^3$$

$\neq 960^4$ [machine: the two integers differ];

$3\ln\frac{4}{3} < 2\ln 2 \iff \ln\frac{64}{27} < \ln 4 \iff 64 < 108$ [machine]

(9) : Def-BM.14(5)'s three splittings re-cited with their certificates $4 > 2$,

$$2 > 1,$$

$3 < 5$ [machine];

(10) : assembly. ■

Phys. Worked from both ends, the answer is the same object. From the mass side: whatever reverses the two rows has to be class-dependent, positive for the neutral class, dimensionally an energy through the rest slot, and built from data the edge count does not already carry — and the inventory has exactly one unbarred candidate. From the force side: colour is a projector that annihilates its own slot data on anything sealed, electromagnetism is a census that admits every local class, so the only class-dependent datum a sealed junction can carry is the count of admissible end-sign registers — twenty against fifteen. The candidate and the survivor are the same. Taken as a logarithm and booked against the vacuum in the ledger's own currency, the filter becomes an energy: the neutral class carries $\ln(4/3)$ more transport entropy than the charged one, and the neutral-core constituent carries exactly one bit more than the charged-core constituent. The reason the register counts as transport and not as bookkeeping is the parity lock: a chain whose ends agree can run wall-free, and a chain whose ends disagree cannot — the wall content of a constituent's transport is odd or even according to its charge core, so the multiplicity is a property of how the seed is carried, not of how the state is named. The sign is not a convention either. Of three possible references only one is a state the composite can be in, and it is the one the corpus already uses for every other deficit.

4.51 (BM.60 — The six-state assembly: the generation ladder, the branch split, the Platonic floors, and the two hierarchies).

Thm.

(1) the generation ladder (the 3):

$$\delta_q = \frac{2}{3\cdot 3\cdot 3} = \frac{2}{27},$$

$$3\delta_q = \delta_\ell = \frac{2}{9},$$

$$16\,\delta_\nu = 9\,\delta_\ell = 27\,\delta_q = 2$$

[FT.119(1), (2); FT.143(5); machine: exact Fractions] :

$$r_i^{(q)} = 1 + \sqrt{2}\cos\Big(\frac{2}{27} + \frac{2\pi i}{3}\Big),$$

$$\sum_i r_i^{(q)} = 3 = \dim F_{J^\pm},$$

$$\sum_i \big(r_i^{(q)}\big)^2 = 6,$$

$$\sum_{i<j} r_i^{(q)} r_j^{(q)} = \frac{3}{2}$$

[FT.111(3); FT.119(4); machine];

$2\big(r^{(q)}-1\big)^3 - 3\big(r^{(q)}-1\big) - \sqrt{2}\cos\frac{2}{9} = 0$ for each of the three

[FT.119(4); machine: enclosure containment at width $< 2^{-90}$];

$$Q = \frac{\sum_i (r_i^{(q)})^2}{\big(\sum_i r_i^{(q)}\big)^2} = \frac{6}{9} = \frac{2}{3}$$

[Def-BM.6(4); FT.158(4); machine: exact, phase-independent];

the generation count is derived:

$$\boxed{\#\text{lines} = 3}$$

[the orbit length of A_3 : $U^3 = E_0$, $U \neq E_0$, $U^2 \neq E_0$ [BM.57(9); machine]; $\sum_i r_i = 3 = \dim F_{J^\pm}$];

(2) the branch split (the 2):

$$\mathcal{H}_{\mathsf{v}} \cong \mathbb{C}^2 \text{ [EW.32]};$$

$$Q|_{\mathcal{H}_v} = \tfrac{1}{2}\sigma_3 + \tfrac{1}{2}\sigma_0 = \mathrm{diag}(1,0),$$

$$\mathrm{spec}\, Q = \{0,+1\},$$

$$\omega :\ \mathrm{spec} \mapsto \{0,-1\}$$

$$[\text{EW.36}] :$$

the seed carrier is a doublet, and its weight alphabet together with the ω-image is exactly the ternary end read $\{-1,0,+1\}$ [EW.36; Def-BM.12(1); D3.4; machine];

the geometric branch:

the dual exchanges (V,F) at fixed $E \implies$ the two members of a dual pair share m_{hid} and Π and split on every finer ledger [FT.44(1),(2)] :

independent windings $F-1$ $(5 \text{ vs } 7;\ 11 \text{ vs } 19)$;

lattice index 2^{F-1} $(32 \text{ vs } 128;\ 2048 \text{ vs } 2^{19})$;

$g(V)$ $(385 \text{ vs } 25;\ 292938721075 \text{ vs } 164395)$;

$O(2d)$ $(112 \text{ vs } 496;\ 112 \text{ vs } 2032)$;

and on the pole side they split on every defect datum [Census FT.63(6) : the derived dual-splitting law] :

$$\delta_v :\ \tfrac{\pi}{2} \text{ vs } \tfrac{2\pi}{3};\quad \tfrac{\pi}{5} \text{ vs } \tfrac{\pi}{3};$$

$$D_S :\ 2-\sqrt{2} \text{ vs } 1;\quad 2-2\cos\tfrac{\pi}{10} \text{ vs } 2-\sqrt{3};$$

the tetrahedron is self-dual [FT.44(3)] $\implies$ its branch index is the weight-ladder rung $m \in \{1,2\}$,

$$D_S(m) = (2,\ 4,\ 2,\ 0) \ \text{ at } m = (1,2,3,4)$$

[Census FT.63(6); FT.60(5); machine];

(3) the metric back-reaction of the directed 3-vertex:

the skeleton metric is piecewise flat with conical defects at the vertices [Census FT.63(4); Def-H.21(6); FT.54] :

$$\delta_v = 2\pi - d\,\frac{(L-2)\pi}{L},$$

$$\frac{\delta_v}{\pi} = \Big(1,\ \frac{1}{2},\ \frac{2}{3},\ \frac{1}{5},\ \frac{1}{3}\Big),$$

$$\boxed{V\,\delta_v = 4\pi \ \text{ on all five solids}}$$

[Descartes/Gauss–Bonnet; machine: exact in π-units] :

$$4\cdot\pi = 8\cdot\tfrac{\pi}{2} = 6\cdot\tfrac{2\pi}{3} = 20\cdot\tfrac{\pi}{5} = 12\cdot\tfrac{\pi}{3} = 4\pi \qquad \text{[machine]};$$

a loop cleanly enclosing one vertex carries Θ_R

$$= \delta_v \text{ [FT.47}(1);\ \text{Def-H.21}(6)]$$

$$\Longrightarrow\ m^2_{\text{pole}}$$

$$= 4\hat{\mu}'^2 \sin^2\frac{\Theta_R}{4}$$

$$= \hat{\mu}'^2 D_S \text{ [Def-H.37}(2);\ \text{BM.39}(7)] :$$

the branch mass is the vertex's own metric back-reaction — the defect at a d-valent vertex, and d

$= 3$ on the tetrahedron, the cube and the dodecahedron [Census FT.42(5) : (V, E, F, d) column];

(4) the six states:

$$|\mathcal{S}| = 3\cdot 2 = 6$$

[machine];

the six defect values are pairwise distinct [machine: separation by exact field and by minimal polynomial] :

$$D_S \in \Big\{\, 2,\ \ 4,\ \ 2-\sqrt{2},\ \ 1,\ \ 2-2\cos\tfrac{\pi}{10},\ \ 2-\sqrt{3} \,\Big\},$$

minimal polynomials $(x{-}2)$, $(x{-}4)$, $(x^2{-}4x{+}2)$, $(x{-}1)$, $(x^4{-}8x^3{+}19x^2{-}12x{+}1)$, $(x^2{-}4x{+}1)$;

the fields:

$$\mathbb{Q},\ \ \mathbb{Q},\ \ \mathbb{Q}(\sqrt{2}),\ \ \mathbb{Q},\ \ \mathbb{Q}\big(\cos\tfrac{\pi}{10}\big),\ \ \mathbb{Q}(\sqrt{3})$$

[machine: the dodecahedral quartic verified by exact substitution into $\mathbb{Z}[c]/(16c^4{-}20c^2{+}5)$] :

the defect species resolves all six states with zero degeneracy;

the informational species is branch-degenerate:

$$m_{\text{hid}}\,\mathfrak{h}/\ln 2 = 2E \in \{12,\ 24,\ 24,\ 60,\ 60\} \implies \text{three values [FT.44(1); machine]};$$

(5) the branch order (down-type heavier, uniform and exact):

$$\boxed{D_S(\mathcal{F}_g,+) > D_S(\mathcal{F}_g,-) \qquad \forall g \in \{1,2,3\}}$$

floor	lighter branch	heavier branch	the exact certificate
$\mathcal{F}_1$	$D_S = 2$	$D_S = 4$	$4 > 2$
$\mathcal{F}_2$	$2-\sqrt{2}$	1	$1 > 2-\sqrt{2} \iff \sqrt{2} > 1 \iff 2 > 1$
$\mathcal{F}_3$	$2-2\cos\frac{\pi}{10}$	$2-\sqrt{3}$	$\cos\frac{\pi}{10} > \frac{\sqrt{3}}{2} \iff \frac{\pi}{10} < \frac{\pi}{6} \iff 3 < 5$

[machine: integer and radical certificates; cos strictly decreasing on $[0,\pi]$ [math]]

the constituent mass difference:

$$\Delta_g = \hat{\mu}'^2\big(D_S(\mathcal{F}_g,+) - D_S(\mathcal{F}_g,-)\big) > 0;$$

$$\Delta_1 = 2\hat{\mu}'^2,$$

$$\Delta_2 = (\sqrt{2}-1)\hat{\mu}'^2,$$

$$\Delta_3 = \big(2\cos\tfrac{\pi}{10} - \sqrt{3}\big)\hat{\mu}'^2$$

[machine: exact in $\mathbb{Q},\ \mathbb{Q}(\sqrt{2}),\ \mathbb{Q}(\sqrt{3}, \cos\tfrac{\pi}{10})$];

(6) the two hierarchies (the derived inversion):

on the informational species: $m_{\text{hid}}(\mathcal{F}_g) = \dfrac{2E(\mathcal{F}_g)\ln 2}{\mathfrak{h}}$,

$$2E = (12,\ 24,\ 60)\ :$$

$$\boxed{m_{\text{hid}}(\mathcal{F}_1) < m_{\text{hid}}(\mathcal{F}_2) < m_{\text{hid}}(\mathcal{F}_3), \qquad 12 : 24 : 60 = 1 : 2 : 5}$$

[machine: exact rationals, strictly increasing] :

the mass increases across the generations;

on the defect species:

$$D_S(\mathcal{F}_3, \pm) < D_S(\mathcal{F}_2, \pm) < D_S(\mathcal{F}_1, \pm)$$

[machine: $2 - 2\cos\tfrac{\pi}{10} < 2 - \sqrt{3} < 2 - \sqrt{2} < 1 < 2 < 4$] :

the two currencies order the floors oppositely

[the exact species inversion; BM.39(5) pattern];

the relating equation is ¬derived:

¬∃ a theorem equating the bit currency ($\ln 2$ per directed edge) and the angle currency (Θ_R) [BM.35(5)];

(7) the generation map:

two exact three-element ordered sets — $\{(r_1^{(q)})^2, (r_2^{(q)})^2, (r_0^{(q)})^2\}$ [FT.119(5) : enclosure-proven, frozen] and $\{2E(\mathcal{F}_g)\}_{g=1,2,3}$ [(6)] :

an order-isomorphism between two totally ordered three-element sets is unique

[math; machine: the uniqueness census over all $3! = 6$ bijections] :

$$g = 1 \longleftrightarrow (r_1^{(q)})^2 \longleftrightarrow \mathcal{F}_1 \text{ (the tetrahedron floor)};$$

$$g = 2 \longleftrightarrow (r_2^{(q)})^2 \longleftrightarrow \mathcal{F}_2 \text{ (the cube/octahedron floor)};$$

$$g = 3 \longleftrightarrow (r_0^{(q)})^2 \longleftrightarrow \mathcal{F}_3 \text{ (the dodecahedron/icosahedron floor)};$$

the map is an order, ¬a number:

$$\hat{\mu}'^2,\ \mu \in \text{Cal-F [logged]};$$

on the transport species the line order inverts [BM.39(5)] $\implies$ the isomorphism inverts with it [machine];

(8) the transport-species form of the ladder:

$$m_i^2 = M^2\big(A_0 - B_0\cos\varphi_i\big), \quad M^2 = \hat{\mu}^2\ell_c^2, \quad A_0 = \frac{3g^2}{8} + \frac{g'^2}{16}, \quad B_0 = \frac{gg'}{4}$$

[Def-BM.11(5); BM.39(1)] :

the exact evaluation at the base profile [machine: in $\mathbb{Q}[g, g', c, s, r_3]/(c^2{+}s^2{-}1,\ r_3^2{-}3)$] :

$$G\big(c(i)\big) = A_0 - B_0\cos\Big(\delta + \frac{2\pi(2-i)}{3}\Big);$$

the orbit value set:

$$\big\{G(c(i))\big\}_{i=0,1,2} = \big\{A_0 - B_0\cos(\delta - \tfrac{2\pi j}{3})\big\}_{j=0,1,2}$$

[machine: set equality in the quotient ring];

the sum rule:

$$\sum_i G\big(c(i)\big) = 3A_0$$

[machine: exact, phase-independent];

the positivity:

$$A_0 - B_0 = \frac{3}{8}\Big(g - \frac{g'}{3}\Big)^2 + \frac{g'^2}{48} > 0$$

[BM.39(1); machine];

(9) the no-import bar:

¬∃ an SM mass, mass ratio or Yukawa value in any clause [P5; machine: zero external tokens, zero decimal numerals in this tier];

¬∃ a Yukawa-type object in the inventory [EW.15; FT.145(3); Def-BM.11 preamble];

the six states and both hierarchies are derived from the card template, the duality law, the defect spectrum and the channel angle alone;

(10) status:

derived: (1)–(8);

declared: Def-BM.14(8)'s name attachment alone;

open: the currency relating equation (6), the flavour composition the scales [P4];

¬a P4 addition, ¬a new dimensionful scale, §Cal-BM stays empty;

species names confined to the identification layers [Def-BM.3(7), Def-BM.6(7), Def-BM.14(8)].

Req. A6, D0.8, D2.9 (Parts 1–2); CT3.17a(2), EW.32, EW.36, EM.9 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.20, Def-H.21(6), Def-H.37(2), Def-H.38, FT.44(1)–(3), FT.47(1), FT.54(4), FT.60(5), FT.111(3), FT.116, FT.119(1), FT.119(2), FT.119(4), FT.119(5), FT.143(5), FT.145(3), FT.158(1), FT.158(4) (Part 3); Census FT.42(5), FT.63(4), FT.63(6) (row-wise); Def-BM.3(7), Def-BM.6(3),

Def-BM.6(4), Def-BM.11(5), Def-BM.12(1), Def-BM.14, BM.35(5), BM.39(1), BM.39(5), BM.39(7), BM.57(9) (this part); exact radical rings $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{5}), \mathbb{Q}(\cos\frac{\pi}{10})$, exact minimal polynomials by substitution, de Moivre binomials at $\cos 5t = 16c^5 - 20c^3 + 5c$, rational interval enclosures with alternating-series remainder bounds, quotient-ring trigonometric identities, order-isomorphism uniqueness [math; machine: `checks/verify_H_directed_3vertex_splitting.py` §A–§D].

Pf.

$$(1): \ \delta_q$$

$$= 2/(3 \cdot 3 \cdot 3)$$

$$= 2/27,$$

$$3 \cdot \tfrac{2}{27}$$

$$= \tfrac{2}{9},$$

$$16 \cdot \tfrac{1}{8}$$

$$= 9 \cdot \tfrac{2}{9}$$

$$= 27 \cdot \tfrac{2}{27}$$

$$= 2 \text{ [machine: exact Fractions]};$$

the spectrum and the three Vieta reads re-derived by enclosure [machine: widths $< 2^{-90}$];

$$Q = 6/9 = 2/3 \text{ [machine]}$$

the cubic $2y^3 - 3y - \sqrt{2}\cos\frac{2}{9}$ evaluated at each $y = r_i^{(q)} - 1$ contains 0 [machine];

$$U^3 = E_0,\ U \neq E_0,\ U^2 \neq E_0 \text{ for the 3-cycle over } \mathbb{Q}(i) \text{ [machine]}$$

$$(2): \text{ EW.36's } \operatorname{diag}(1,0) \text{ re-derived from } \tfrac{1}{2}\sigma_3 + \tfrac{1}{2}\sigma_0 \text{ [machine]};$$

the union with the ω-image equals the ternary alphabet [machine];

the four finer-ledger splittings and the defect splittings re-cited [FT.44(2); Census FT.63(6)];

the tetrahedral ladder $(2, 4, 2, 0)$ re-cited with its exact zero at $m = 4$

(3) : δ_v

$= 2\pi - d(L-2)\pi/L$ evaluated at (d, L)

$= (3, 3)$,

$(3, 4)$,

$(4, 3)$,

$(3, 5)$,

$(5, 3)$ [machine: exact rationals in π-units];

$V\delta_v$ multiplied out at all five [machine];

$4\sin^2(\delta_v/4) = D_S$ by the half-angle identity [machine in the exact fields]

(4) : $|\mathcal{S}| = 3 \cdot 2 = 6$ [machine];

pairwise distinctness: the rational members separated by integer comparison, the quadratic members by their minimal polynomials $(x^2 - 4x + 2)$ and $(x^2 - 4x + 1)$ having disjoint root sets, the quartic member separated by Census FT.63(6)'s isolated roots [machine];

$\{12, 24, 24, 60, 60\}$ has three distinct values [machine]

(5) : the three certificates of the table [machine];

cos strictly decreasing on $[0, \pi]$ [math] $\implies \frac{\pi}{10} < \frac{\pi}{6} \implies \cos\frac{\pi}{10} > \cos\frac{\pi}{6}$

$= \frac{\sqrt{3}}{2}$ [machine: $\cos^2\frac{\pi}{6} = \frac{3}{4}$];

Δ_g formed as exact differences in each field [machine]

$(6):\ 12 < 24 < 60$ and $12 : 24 : 60 = 1 : 2 : 5$ [machine];

$2 - 2\cos\frac{\pi}{10} < 2 - \sqrt{3} < 2 - \sqrt{2} < 1 < 2 < 4$ by the certificates $3 < 5$,

$2 < 3$,

$1 < 2$,

and integer comparison [machine]

$(7):$ all 6 bijections between the two ordered sets enumerated; exactly one is order-preserving [machine];

the inversion under BM.39(5)'s affine-decreasing row [machine]

$(8):\ G(c(i))$ built from the Gram block of BM.57(5) at $c(i)$

$= (\cos\delta, \cos(\delta+\frac{2\pi}{3}), \cos(\delta+\frac{2\pi(2-i)}{3}), \frac{1}{2})$ and reduced modulo c^2+s^2-1,

r_3^2-3 [machine];

the orbit value set and the sum rule $\sum_i G(c(i)) = 3A_0$ [machine];

$A_0 - B_0$ as a sum of squares [machine]

(9)–$(10):$ the token scans and the assembly [machine]. ■

Phys. Six quarks fall out of three structures that were already sitting in the corpus in separate rooms. The three generations are the three lines of the colour channel, whose angle is one integer over three integers and whose orbit length is forced by the same C_3 that seals the baryon. The two members of each generation are a floor and its dual: two solids that share every edge count, hence every informational mass, and differ at every vertex figure — three squares meeting at right angles against four triangles meeting at one hundred twenty degrees, three pentagons against five triangles. Where duality is trivial, at the tetrahedron, the second branch is the weight ladder's own next rung. The mass of a branch is the conical defect at its vertex, and Descartes closes the ledger: five solids, each with its vertex defects summing to exactly four pi. The hierarchy comes out exact and comes out twice, in opposite directions. In the bit currency the floors climb one to two to five, strictly. In the angle currency they descend. That inversion is not a defect of the assembly; it is the same read-relativity the corpus already proved for the family lines, and the equation that would reconcile the two currencies is the one item the ledger still carries as open. Within a generation there is no ambiguity at all: the heavier branch is heavier by an exact algebraic number — two, root two minus one, two cosines of eighteen degrees minus root three — and it is heavier for the same reason in all three. Nothing was fitted, and nothing measured entered.

Theorem 4.52 (BM.61 — α_B: the inventory, the coupling-blindness theorem, and the obstruction). Thm.

(1) the inventory scan:

$\neg\exists$ a quantity α_B in the corpus:

occurrences of the token in Parts 1–4, the Census : 0

[machine: the token scan];

it is ¬a P4-class member and ¬a logged combination:

the P4 core [A §E],

Cal-2,

Cal-F,

and §Cal-BM (empty) carry no such entry [machine: the registry scan];

(2) the second argument is void:

$\neg\exists$ an α_{Strong} in the inventory [machine: zero occurrences];

the colour sector's strength-class data are counts and ratios of counts:

$$\dim \mathcal{K}^{(\text{strong})} = 8 \text{ [CT3.8; BM.36(1)]};$$

$$g(N) = \sum_{a,b\geq 0\ 2a+3b=N} \frac{N!}{a!\,b!\,2^a\,6^b} = (0,\ 1,\ 1,\ 3,\ 10,\ 25,\ 105) \text{ at } N = 1,\dots,7$$

[CT3.17a(2); FT.41(b); machine: recomputed exactly];

$\mathcal{C}(\Sigma; C) \in \{0, 1\}$ [D2.14];

the arenas 27 and 9 [FT.118(2)];

$$\delta_q = \frac{2}{3\cdot 3\cdot 3} = \frac{2}{27}\ [\text{FT.119}(1)];$$

$$g(6) = 25 = 15 + 10\ [\text{machine: the split}];$$

every one is an integer or a ratio of integers — ¬a coupling;

(3) the strength bar:

¬∃ a theorem fixing any numerical coupling strength [BM.36(3)];

(g, g') are P4-class [EW.33; P4];

$\alpha_{\text{EM}}^{(2)}$ is an operator block read whose value no theorem fixes [EM.17; EM.25(8); Part 2 §Cal-2′s logged row];

the only force-structure ratios the corpus derives are the alphabet-capacity ratios

$$\frac{\dim \mathcal{K}^{(\text{EM})}}{\dim F_W} = \frac{1}{3},$$

$$\frac{\dim \mathcal{K}^{(\text{strong})}}{\dim \mathcal{K}^{(\text{EM})}} = 8,$$

$$\frac{\dim \mathcal{K}^{(\text{EM})}}{\dim \mathcal{K}_3} = \frac{1}{25}$$

[BM.36(2); machine: exact Fractions from the flag $(6, 16, 17, 25)$] :

reading a capacity ratio as a coupling strength is a category error of the geometric-ratio-against-coupling species [BM.36(3); FT.73(5) trap pattern];

(4) the first argument is unfixed:

$$e = \frac{gg'}{\sqrt{g^2 + g'^2}} = g\sin\theta_W = g'\cos\theta_W,$$

$$\frac{1}{e^2} = \frac{1}{g^2} + \frac{1}{g'^2}$$

[EW.35; EM.25(6) : fixed with zero freedom] :

$\neg\exists$ a theorem asserting a value of e [EM.25(7)];

$\varepsilon_{\mathrm{EM}}$

$= e^2/(4\pi)$ is a dimensionless P4-class symbol carried symbolically [Def-BM.12 local notation; EM.25(9); P5];

(5) the coupling-blindness theorem:

$$W(Q) = \left[z^Q\right]\left(2 + z + z^{-1}\right)^k \in \mathbb{Z}[z, z^{-1}]$$

[Def-BM.13(3); Def-BM.8(6)] :

the free-symbol set of $W(Q)$ is $\{z\}$,

with coefficients fixed by (m_q, k)

$= \big((2, 1, 1),\ 3\big)$ [EM.9; D3.9] :

the coupling alphabet $\{g, g', e, \varepsilon_{\mathrm{EM}}, \hat{\mu}, \hat{\mu}', \ell_c, \theta_W\}$ is disjoint from it [machine: the symbol census];

$$\boxed{\frac{\partial\,\Sigma_{\mathrm{B}}}{\partial g} = \frac{\partial\,\Sigma_{\mathrm{B}}}{\partial g'} = \frac{\partial\,\Sigma_{\mathrm{B}}}{\partial e} = \frac{\partial\,\Sigma_{\mathrm{B}}}{\partial \varepsilon_{\mathrm{EM}}} = \frac{\partial\,\Sigma_{\mathrm{B}}}{\partial \hat{\mu}} = \frac{\partial\,\Sigma_{\mathrm{B}}}{\partial \hat{\mu}'} = \frac{\partial\,\Sigma_{\mathrm{B}}}{\partial \ell_c} = 0 \ \text{ exactly}}$$

[machine] :

the charge-class transport entropy is coupling-blind;

it is a function of the integer alphabet and the constituent count alone;

(6) the exact object at the proposed slot:

$$\boxed{\Sigma_{\mathrm{B}} = \ln\frac{W(0)}{W(+1)} = \ln\frac{4}{3} = 2\ln 2 - \ln 3}$$

[Def-BM.13(4); machine] :

$$e^{\Sigma_{\mathrm{B}}} = \frac{4}{3}$$

[machine: the integer row $20 \cdot 3 = 15 \cdot 4$];

$[\Sigma_{\mathrm{B}}] = \mathfrak{q}^0$ [CT2.1; machine];

$\Sigma_{\mathrm{B}} > 0$ [machine: $4 > 3$];

the exact enclosures inside the $\ln$-ring:

$$\frac{\ln 2}{3} < \Sigma_{\mathrm{B}} < \frac{\ln 2}{2}$$

[machine: $4^3 > 3^3 \cdot 2 \iff 64 > 54$; $4^2 < 3^2 \cdot 2 \iff 16 < 18$];

$\frac{2\ln 2}{5} < \Sigma_{\mathrm{B}} < \frac{3\ln 2}{5}$ [machine: $4^5 > 3^5 \cdot 2^2 \iff 1024 > 972$; $4^5 < 3^5 \cdot 2^3 \iff 1024 < 1944$];

(7) the verdict:

Σ_{B} enters the mass as a weight in the rest slot [Def-BM.13(5)], ¬as a coupling strength:

it multiplies no generator, resolves no propagator, and appears in no vertex;

it is $\neg\alpha_B$: a coupling is a coefficient of an interaction; Σ_{B} is the logarithm of a census;

(8) the obstruction, named:

$\boxed{\alpha_B = f(\alpha_{\mathrm{EM}}, \alpha_{\mathrm{Strong}}) \text{ is } \neg\text{derivable}}$

on four independent grounds:

(O1) the strength bar: $\neg\exists$ a count $\to$ strength theorem [(3); BM.36(3)];

(O2) the colour-coupling void: α_{Strong} is ¬an inventory item, so the domain of f is empty in its second argument [(2)];

(O3) the coupling-blindness: the one exact dimensionless charge-class quantity the corpus derives has zero derivative in every coupling, so it is not a function of them [(5)];

(O4) P5: fixing α_B by an observed ratio is barred [P5; BM.36(3); FT.73(5)] :

the obstruction is not a gap in the arithmetic; it is the absence of a coupling theorem for colour and the bar against inventing one;

(9) status:

derived: (1)–(7);

named, not filled: α_B (8);

α_B is ¬added as a P4-class parameter;

§Cal-BM stays empty;

¬a new dimensionful scale [P1];

¬a measured value and ¬a numerical coupling [P5; machine: zero external tokens, zero decimal numerals in this tier];

Req. D2.9, D2.14, D3.9, CT3.8, CT3.17a(2), EW.33, EW.35, EM.9, EM.17, EM.25(6), EM.25(7), EM.25(8), EM.25(9) (Part 2); CT0.13, CT0.14, CT2.1, P1, P4, P5 (Parts 1–2); FT.41(b), FT.73(5), FT.118(2), FT.119(1) (Part 3); Def-BM.8(1), Def-BM.8(6), Def-BM.12, Def-BM.13(3), Def-BM.13(4), Def-BM.13(5), BM.36(1), BM.36(2), BM.36(3) (this part); A §E (the P4 core); Part 2 §Cal-2 (the logged row); the exact grammar-count summation, symbol censuses, exact ln-ring identities with integer certificates, Laurent exponent arithmetic [math; machine: `checks/verify_H_constituent_transport_splitting.py` §E].

Pf.

(1)–(2) : the token scans over the five canon files and the two annexes [machine];

$g(N)$ recomputed by the CT3.17a(2) summation for N

$= 1,$

$\ldots,$

7 and compared to the frozen row $(0, 1, 1, 3, 10, 25, 105)$ [machine];

$$g(6) = 15 + 10 = 25 \text{ [machine: the } a{=}3, b{=}0 \text{ and } a{=}0, b{=}2 \text{ summands]}$$

$$(3)\text{–}(4) : \text{ BM.36}(2), (3), \text{ EM.25}(6)\text{–}(8) \text{ re-cited;}$$

$$\tfrac{1}{3},$$

$$\tfrac{8}{1},$$

$$\tfrac{1}{25} \text{ from } (6, 16, 17, 25) : \ 16/3 \to \tfrac{1}{3} \text{ via } \dim F_W$$

$$= 3,$$

$$8/1,$$

$$1/25 \text{ [machine: exact Fractions]}$$

$(5) :$ $W(Q)$ written as a Laurent polynomial with integer coefficients; the symbol set extracted and intersected with the coupling alphabet [machine: the intersection is empty];

each partial derivative formed symbolically and evaluated [machine: all seven are the zero polynomial]

$$(6) : \ \ln 20 - \ln 15 = \ln \tfrac{20}{15} = \ln \tfrac{4}{3} = 2 \ln 2 - \ln 3 \text{ [machine: the prime-log ring];}$$

$$20 \cdot 3 = 60 = 15 \cdot 4 \text{ [machine];}$$

the four enclosures: 4^3

$$= 64 > 54$$

$$= 3^3 \cdot 2;$$

$$4^2$$

$$= 16 < 18$$

$$= 3^2 \cdot 2;$$

$$4^5$$

$$= 1024 > 972$$

$$= 3^5 \cdot 2^2;$$

$$4^5$$

$$= 1024 < 1944$$

$$= 3^5 \cdot 2^3 \text{ [machine: integer comparisons]}$$

(7)–(9) : the locus row re-cited from Def-BM.13(5);

the four grounds conjoined;

the standing-rule scans [machine]. ■

Phys. The proposal was to close the neutron row with a new coupling, fixed by the ratio the world reports. Three of the four things that proposal needs are absent, and their absence is proved rather than assumed. There is no α_B anywhere in the canon to refine. There is no strong coupling for a function of two strengths to take as its second argument: the colour sector's entire strength-class inventory is a dimension, a projector, a grammar count and two arena sizes, all integers, and the corpus has already ruled that turning a count into a strength is the same category error as turning a geometric ratio into a coupling. And the one exact dimensionless charge-class number the corpus does derive — the logarithm of twenty over fifteen — has a zero derivative in every coupling it owns, which settles the question from the other side: it is not a function of the strengths, so it cannot be dressed as one. What $\ln(4/3)$ actually is, is a weight: it sits in the rest slot beside the edge count, in the same nats, and it does the work the proposal wanted without the parameter the proposal needed. The obstruction is named to the monomial, and no free entry is added.

4.53 (BM.62 — The full mass splitting: the extended decomposition, the exact critical strength, and the sign theorem).

Thm.

(1) the extended decomposition:

$$\boxed{\Delta m \;:=\; m_n - m_p \;=\; \Delta m_{\text{Lyapunov}} + \Delta m_{\text{force}} + \Delta m_{\text{metric}} + \Delta m_{\text{binding}} + \Delta m_{\text{self-int}} + \Delta m_{\text{transport}}}$$

with BM.55(1)'s five derived rows standing unmodified and one new row added [BM.52(5)'s no-double-count guard respected: BM.59(8)'s disjointness] :

(2) the four zeros:

$$\Delta m_{\text{Lyapunov}} = 0 \text{ [BM.51(6) : every input is class-shared]};$$

$$\Delta m_{\text{metric}} = 0 \text{ [BM.53(2); BM.58(4) : the charge exponent is 0 in all seven monomials]};$$

$$\Delta m_{\text{binding}}$$

$$= 0 \text{ [BM.54(1)–(3) : the count read vanishes as } 6-(2{+}2{+}2)=0, \text{ and the ensemble read is } q\text{-degenerate]};$$

re-derived here:

$$\mathcal{B}(S_j;1,1,1)$$

$$= m_B^2(\cdot,S_j) - 3$$

$m_B^2(\cdot,1)$ is a function of the junction sector alone [BM.21(2); BM.20(4)] :

$$\mathcal{B}(0;1,1,1)$$

$$= \frac{843957}{98956} - \frac{5764986}{393679},$$

$$\mathcal{B}(1;1,1,1)$$

$$= -\frac{3843324}{393679},$$

$$\mathcal{B}(2;1,1,1)$$

$$= \frac{91}{23} - \frac{5764986}{393679},$$

$$\mathcal{B}(3;1,1,1)$$

$$= \frac{13713}{3088} - \frac{5764986}{393679};$$

all four are nonzero exact rationals, all four are negative, and the class difference of each is exactly 0

[machine: the exact rational census];

$$m_4^2(\cdot,0) > m_4^2(\cdot,1) > m_4^2(\cdot,3) > m_4^2(\cdot,2)$$

[BM.21(4); machine];

and a binding term in a free member's mass remains a category error [BM.48(4); BM.54(4)];

(3) the two nonzero summands:

$$\Delta m_{\text{force}} = \Delta m_{\text{self-int}} = -\Delta m_{\text{EM}} = -\frac{3\,c\,e^2\ln 2}{2\pi\,\mathfrak{h}} = -\,6\,c\,\varepsilon_{\text{EM}}\,\frac{\ln 2}{\mathfrak{h}}$$

[BM.52$(4),(5)$; BM.46(4); machine: one Laurent monomial, counted once];

$$\Delta m_{\text{transport}} = +\,\frac{\Sigma_{\text{B}}}{\mathfrak{h}} = +\,\frac{\ln(4/3)}{\mathfrak{h}} > 0 \qquad \text{[BM.59}(4)\text{; machine: } 4>3\text{]};$$

(4) the assembly:

$$\boxed{\Delta m = \frac{\ln(4/3) - 6\,c\,\varepsilon_{\text{EM}}\ln 2}{\mathfrak{h}} = \frac{2\ln 2 - \ln 3 - 6\,c\,\varepsilon_{\text{EM}}\ln 2}{\mathfrak{h}}}$$

[machine: the monomial sum in the $\ln$-ring] :

$$m_n\,\mathfrak{h} = 6\ln 2 + \ln 20 = \ln 1280,$$

$$m_p\,\mathfrak{h} = 6\ln 2 + \ln 15 + 6\,c\,\varepsilon_{\text{EM}}\ln 2 = \ln 960 + 6\,c\,\varepsilon_{\text{EM}}\ln 2$$

[Def-BM.13(5); machine];

$$[\Delta m] = \mathfrak{q}^{+1/2} = [\text{energy}]$$

[CT2.1; D2.9; machine: exponent arithmetic];

both summands ride the same rest slot [Def-BM.10(4); Def-BM.12(6); BM.37(4)];

(5) the exact ratio:

$$\frac{m_n}{m_p} = \frac{\ln 1280}{\ln 960 + 6\,c\,\varepsilon_{\mathrm{EM}}\ln 2}$$

[machine];

$$\text{at } \varepsilon_{\mathrm{EM}} = 0: \quad \frac{m_n}{m_p} = \frac{\ln 1280}{\ln 960} = \frac{8\ln 2 + \ln 20}{6\ln 2 + \ln 15} \neq \frac{4}{3}$$

[machine: $3\ln 1280 \neq 4\ln 960$] :

the amended BM.55(2) ratio:

$$\frac{4\pi}{4\pi + c\,e^2} \longmapsto \frac{\ln 1280}{\ln 960 + 6\,c\,\varepsilon_{\mathrm{EM}}\ln 2}$$

[the transport refinement of the rest input];

(6) the exact critical strength:

Δm is strictly decreasing in $\varepsilon_{\mathrm{EM}}$:

$$\frac{\partial\,\Delta m}{\partial\,\varepsilon_{\mathrm{EM}}} = -\,\frac{6\,c\ln 2}{\mathfrak{h}} < 0$$

[machine: the coefficient census at $c, \ln 2, \mathfrak{h} > 0$] :

$$\boxed{\varepsilon^{\mathrm{crit}}(c) := \frac{\Sigma_{\mathrm{B}}}{6\,c\,\ln 2} = \frac{2\ln 2 - \ln 3}{6\,c\,\ln 2} = \frac{2 - \log_2 3}{6\,c}}$$ the unique zero of Δm [machine];

the sign trichotomy:

$$\varepsilon_{\mathrm{EM}} < \varepsilon^{\mathrm{crit}}(c) \iff \Delta m > 0;$$

$$\varepsilon_{\mathrm{EM}} = \varepsilon^{\mathrm{crit}}(c) \iff \Delta m = 0;$$

$$\varepsilon_{\mathrm{EM}} > \varepsilon^{\mathrm{crit}}(c) \iff \Delta m < 0$$

[machine];

(7) the exact enclosures of the critical strength:

every comparison is an integer power certificate [machine] :

c	$\varepsilon^{\mathrm{crit}}(c)$	lower certificate	upper certificate
$\frac{1}{2}$ (the shell [Def-BM.12(4)])	$\dfrac{2-\log_2 3}{3}$	$> \dfrac{1}{8} \iff 2^{13} > 3^8 \iff 8192 > 6561$	$< \dfrac{1}{7} \iff 2^{11} < 3^7 \iff 2048 < 2187$
$\frac{3}{5}$ (the uniform ball [Def-BM.12(4)])	$\dfrac{5(2-\log_2 3)}{18}$	$> \dfrac{1}{9} \iff 2^8 > 3^5 \iff 256 > 243$	$< \dfrac{1}{8} \iff 2^{31} < 3^{20} \iff 2147483648 < 3486784401$

$$\boxed{\frac{1}{9} < \varepsilon^{\mathrm{crit}}\Big(\frac{3}{5}\Big) < \frac{1}{8} < \varepsilon^{\mathrm{crit}}\Big(\frac{1}{2}\Big) < \frac{1}{7}}$$

[machine: the four certificates conjoined];

(8) the sub-criticality bar (parameter-free): $\boxed{\varepsilon_{\mathrm{EM}} \le \dfrac{1}{9} \implies \Delta m > 0 \text{ at both priced } c}$

the witnesses at $\varepsilon_{\mathrm{EM}} = \frac{1}{9}$:

$$c = \tfrac{1}{2}: \ \ln\tfrac{4}{3} > \tfrac{\ln 2}{3} \iff 4^3 > 3^3 \cdot 2 \iff 64 > 54;$$

$$c = \tfrac{3}{5}: \ \ln\tfrac{4}{3} > \tfrac{2\ln 2}{5} \iff 4^5 > 3^5 \cdot 2^2 \iff 1024 > 972$$

[machine];

and at $\varepsilon_{\mathrm{EM}} = 0$: $\Delta m = \dfrac{\ln(4/3)}{\mathfrak{h}} > 0 \iff 4 > 3$ [machine];

(9) the ordering theorem:

$$\boxed{\text{on the sub-critical branch } \varepsilon_{\mathrm{EM}} < \varepsilon^{\mathrm{crit}}(c): \qquad m_n > m_p}$$

[(6), (8); machine] :

the neutron is the heavier baryon;

BM.55(3)'s strict order is read-relative and holds on the super-critical branch $\varepsilon_{\mathrm{EM}} > \varepsilon^{\mathrm{crit}}(c)$;

the critical surface is exact and unique $[(6)]$;

(10) the fractional splittings (exact rational enclosures):

$$\frac{\Delta m_{\mathrm{transport}}}{m_{\mathrm{hid}}} = \frac{\ln(4/3)}{6\ln 2} \in \Big(\frac{1}{15}, \frac{1}{14}\Big)$$

[machine: $15\ln\frac{4}{3} > 6\ln 2 \iff 2^{24} > 3^{15} \iff 16777216 > 14348907$; $14\ln\frac{4}{3} < 6\ln 2 \iff 2^{22} < 3^{14} \iff 4194304 < 4782969$];

$$\frac{\Delta m_{\mathrm{transport}}}{m_n}\Big|_{\varepsilon_{\mathrm{EM}}=0} = \frac{\ln(4/3)}{\ln 1280} \in \Big(\frac{1}{25}, \frac{1}{24}\Big)$$

[machine: integer power certificates];

$$\frac{\Delta m_{\mathrm{EM}}}{m_{\mathrm{hid}}} = c\,\varepsilon_{\mathrm{EM}} \in \mathfrak{q}^0 \ [\mathrm{BM}.46(1);\ \text{machine}];$$

the strict contraction against the barred measure price:

$$\frac{\ln(4/3)}{6\ln 2} < \frac{1}{3} \iff 3\ln\tfrac{4}{3} < 2\ln 2 \iff 16 < 27$$

[Def-BM.13(8); machine];

(11) the species row:

the assembly (4) is legitimate because all four nonzero-or-zero terms enter through one slot:

the rest input of the dispersion read [Def-BM.10(4); BM.35(4); BM.37(4)],

refined in the Def-BM.12(6) pattern;

¬a route ratio is read as a mass ratio [BM.37(6); Def-BM.13(8); (5)];

the input alphabets:

$$\Delta m_{\text{transport}} : \{m_q, W(Q), \ln 2, \mathfrak{h}\};$$

$$\Delta m_{\text{EM}} : \{e, c, \Delta x_{\text{B}}\};$$

$$\Delta m_{\text{binding}} : \{(\Gamma, R), \Delta S\};$$

$$\Delta m_{\text{metric}} : \{(a_{00}, q_{ij}, w_i), E_{\text{int}}\};$$

pairwise disjoint except for the shared $\mathfrak{h}$ and E_{int} [machine: the token census];

BM.37(2)'s disjointness theorem is respected: ¬a measure route is added to a metric read;

(12) status:

derived: (1)–(11);

the sign theorem is conditional on the sub-critical branch, with the hypothesis on its face and the critical value exact (6)–(9) [A §0.2a];

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a route adopted [BM.30(5)],

¬a measured value [P5];

the open row: the numerical value of e [EM.25(7) : one calibration datum] — it decides which side of $\varepsilon^{\text{crit}}$ the corpus sits on, and no theorem of Parts 1–4 supplies it.

Req. D2.9, CT2.1, EM.25(7), EW.35 (Part 2); P1, P4, P5 (Parts 1–2); FT.79(5), FT.107(2), FT.107(3) (Part 3); Census FT.35(2) (row-wise); Def-BM.4, Def-BM.8(6), Def-BM.10(4), Def-BM.12(4), Def-BM.12(6), Def-BM.13(4), Def-BM.13(5), Def-BM.13(8), BM.20(4), BM.21(2), BM.21(4), BM.30(5), BM.35(4), BM.37(2), BM.37(4), BM.37(6), BM.46(1), BM.46(4), BM.48(4), BM.51(6), BM.52(4), BM.52(5), BM.53(2), BM.54(1)–(4), BM.55(1), BM.55(2), BM.55(3), BM.58(4), BM.59(4), BM.59(8) (this part); exact ln-ring arithmetic with integer power certificates, Laurent-monomial assembly, exact rational sign censuses, exact rational enclosures of ln-ratios [math; machine: `checks/verify_H_constituent_transport_splitting.py` §F].

Pf.

$$(1)\text{–}(3):\ \text{BM.55}(1)\text{'s five rows re-cited (frozen)};$$

$$\mathcal{B}(S_j;1,1,1)$$

$= m_B^2(\cdot,S_j) - 3m_B^2(\cdot,1)$ formed from BM.20(4)'s four sector masses and compared to BM.21(2)'s four frozen rationals [machine: exact Fractions, all four agree];

the sign census and the sector order [machine];

the class difference of each

$= 0$ because $m_B^2(q,S)$ is q-independent [FT.107(3); machine]

$$(4):\ \ln 1280 - \ln 960 = \ln\tfrac{1280}{960} = \ln\tfrac{4}{3} = \Sigma_{\mathrm{B}}\ \text{[machine: the prime-log ring]};$$

$$6\ln 2 + \ln 20 = 8\ln 2 + \ln 20 = \ln(2^8\cdot 20) = \ln 1280\ \text{[machine]};$$

$$6\ln 2 + \ln 15 = \ln(2^6\cdot 15) = \ln 960\ \text{[machine]};$$

the exponent row $\mathfrak{q}^0\cdot\mathfrak{q}^{+1/2}$ [machine]

$$(5):\ 3\ln 1280 \neq 4\ln 960 \iff 1280^3 \neq 960^4\ \text{[machine: the two integers differ]};$$

$$(6):\ \Delta m$$

$$= 0 \iff \Sigma_{\mathrm{B}}$$

$$= 6c\varepsilon_{\mathrm{EM}}\ln 2 \iff \varepsilon_{\mathrm{EM}}$$

$= \Sigma_{\mathrm{B}}/(6c\ln 2)$ [machine: solved in the $\ln$-ring];

the derivative $-6c\ln 2/\mathfrak{h}$ [machine]

$$(7):\ \text{cross-multiplication by } 6c\ln 2 > 0:\quad \varepsilon^{\mathrm{crit}}(c) > r \iff \Sigma_{\mathrm{B}} > 6cr\ln 2;$$

$$c=\tfrac{1}{2},\ r=\tfrac{1}{8}:\ 8\Sigma_{\mathrm{B}} > 3\ln 2 \iff (4/3)^8 > 2^3 \iff 2^{13} > 3^8\ \text{[machine]};$$

$$c=\tfrac{1}{2},\ r=\tfrac{1}{7}:\ 7\Sigma_{\mathrm{B}}<3\ln 2\iff 2^{11}<3^{7}\ [\text{machine}]$$

$$c=\tfrac{3}{5},\ r=\tfrac{1}{9}:\ 5\Sigma_{\mathrm{B}}>2\ln 2\iff 2^{8}>3^{5}\ [\text{machine}];$$

$$c=\tfrac{3}{5},\ r=\tfrac{1}{8}:\ 40\Sigma_{\mathrm{B}}<18\ln 2\iff 2^{31}<3^{20}\ [\text{machine}]$$

$$(8):\ \text{the two witnesses } 4^3=64>54=3^3\cdot 2 \text{ and } 4^5=1024>972=3^5\cdot 2^2\ [\text{machine}];$$

$$4>3\ [\text{machine}];$$

$$(9):\ \text{conjunction of } (6),(8)$$

$$(10):\ 15\Sigma_{\mathrm{B}}>6\ln 2\iff 4^{15}>3^{15}2^{6}\iff 2^{24}>3^{15}\ [\text{machine}];$$

$$14\Sigma_{\mathrm{B}}<6\ln 2\iff 4^{14}<3^{14}2^{6}\iff 2^{22}<3^{14}\ [\text{machine}];$$

$$\text{the } \ln 1280 \text{ enclosure by the same route } [\text{machine}];$$

$$3\ln\tfrac{4}{3}<2\ln 2\iff 64<108\ [\text{machine}]$$

$$(11)\text{–}(12):\ \text{the token census of the four alphabets } [\text{machine}];$$

$$\text{the standing-rule scans } [\text{machine}].\ \blacksquare$$

Phys. The decomposition the full-scope tier closed with four zeros and one monomial gains a fifth zero and a second monomial, and the second one points the other way. Nothing in the amended assembly contradicts the earlier one: the Lyapunov tower still shares every input, the geometry still carries no charge exponent, the binding still dies twice, and the scar's field energy is still counted exactly once. What is added is the transport register — twenty admissible end-sign assignments in the neutral class against fifteen in the charged one — read as an entropy in the ledger's own nats and booked through the same rest slot the field energy already uses. The two summands are commensurate, so the splitting becomes a single exact expression with a single exact zero: the critical strength is the logarithm of four thirds over six $c\ln 2$, enclosed between one ninth and one seventh by four integer power comparisons, with the tighter of the two priced distributions clearing one ninth. Below that surface the neutron is heavier; above it the earlier order returns; on the surface the two rows are degenerate again. The corpus fixes no value for the coupling, so which branch it sits on is the one open row — and the branch that gives the world's order is not a thin one: it is an exact interval, certified by $256>243$, containing every coupling that keeps the field energy a correction rather than a competitor.

Theorem 4.54 (BM.63 — The decay re-adjudication on the refined read: the closure survives, with an exact margin). Thm.

$$(1) \quad \text{the amended Q-value:}$$

$$Q_\beta := m_n - m_p - m_\ell - m_\nu = \Delta m - m_\ell - m_\nu$$

$$[\text{BM.41}(4);\ \text{BM.49}(1);\ \text{BM.55}(5)\ \text{re-cited}]:$$

$$\boxed{Q_\beta = \frac{\ln(4/3) - 6\,c\,\varepsilon_{\text{EM}}\ln 2}{\mathfrak{h}} - m_\ell - m_\nu} \qquad [\text{BM.62}(4);\ \text{machine}];$$

$$(2) \quad \text{the unconditional closure:}$$

$$\Delta m < \frac{\ln(4/3)}{\mathfrak{h}} \ \text{ at every } \varepsilon_{\text{EM}} > 0\ [\text{BM.62}(6):\ \text{strict monotonicity}];$$

$$m_\ell \ \geq \ \frac{2\ln 2}{\mathfrak{h}}$$

$$[\text{FT.44}(4):\ m = E_{\text{int}}\ln 2/\mathfrak{h};\ \text{FT.28 the minimal floor } E_{\text{int}} = 2;\ \text{BM.13}(3):\ E_{\text{int}} \geq 2;\ \tfrac{1}{2} < \ln 2 < 1\ \text{enclosure}];$$

$$m_\nu \geq 0\ [\text{EW.33}:\ \text{PSD}];$$

$$\Longrightarrow \ \boxed{Q_\beta \ < \ \frac{\ln(4/3) - 2\ln 2}{\mathfrak{h}} \ = \ -\frac{\ln 3}{\mathfrak{h}} \ < \ 0 \ \text{ strictly}}$$

$$[\text{machine: } 2\ln 2 - \ln\tfrac{4}{3} = \ln 3; \quad \ln\tfrac{4}{3} < 2\ln 2 \iff 4 < 12]:$$

the β-decay vertex is kinematically closed at every admissible $\varepsilon_{\text{EM}} > 0$,

on both sub- and super-critical branches;

$$(3) \quad \text{the exact margin:}$$

$$\boxed{|Q_\beta| > \frac{\ln 3}{\mathfrak{h}}}$$

$$[\text{machine: the certificate } 3 > 1]:$$

the closure margin is one exact $\ln 3$ of ledger content;

the reason, in one comparison:

the transport excess is $\ln \frac{4}{3}$ nats and the minimal daughter costs $2 \ln 2$ nats :

$$\ln \tfrac{4}{3} < 2 \ln 2 \iff \tfrac{4}{3} < 4 \iff 4 < 12$$

[machine] :

a fraction of one bit cannot pay for two bits;

(4) the deepening row:

BM.49(1)'s $Q_{\text{corr}} = -(\Delta m_{\text{EM}} + m_\ell + m_\nu)$ and BM.55(5)'s full-scope closure stand;

the refined margin is

$$|Q_\beta| > \frac{\ln 3 + 6\, c\, \varepsilon_{\text{EM}} \ln 2}{\mathfrak{h}} > \frac{\ln 3}{\mathfrak{h}}$$

[machine: the coefficient census] :

the EM summand still deepens the closure, and the transport summand does not reverse it;

(5) the no-opening theorem survives:

every EM correction of Tier BM.12 is $\propto q^2$

≥ 0 (an integral of a square) and raises the daughter's mass against the decay's requirement [BM.49(2) re-cited];

the transport refinement raises the parent's mass by at most $\ln(4/3)/\mathfrak{h}$ [(2)],

which is strictly below the minimal daughter content [(3)] :

no term of the extended decomposition opens the vertex

[machine: the two bounds conjoined];

(6) the $k{=}1$ scope row (the daughters' own refinement):

Def-BM.13(3)'s census at $k = 1$ is $W(q) = m_q = (2, 1, 1)$ [EM.9; machine] :

$\Sigma_{\text{cl}}(q) = \ln m_q = (\ln 2,\ 0,\ 0)$ at $q = (0, +1, -1)$:

the neutral line carries one extra bit, the charged lines none;

applied to the daughters:

$m_\nu \ \mapsto\ m_\nu + \frac{\ln 2}{\mathfrak{h}}$ [the ν-channel is the EM-unactivated class [BM.41(3); FT.142(1)]],

$m_\ell \ \mapsto\ m_\ell$ [the charged member carries $m_q = 1$] :

the refinement of the daughters deepens the closure by exactly $\ln 2/\mathfrak{h}$ [machine] :

it is recorded as a scope row and ¬needed for (2)'s verdict, which uses only $m_\nu \geq 0$;

(7) what inverts and what does not:

inverts on the sub-critical branch: the ordering clause $m_n < m_p$ [BM.55(3)] $\longrightarrow$ $m_n > m_p$ [BM.62(9)];

does not invert: the kinematic closure [BM.41(4)(e), BM.49(1)–(2), BM.55(5)];

the vertex grammar and the forced members [BM.41(1), (3)];

the two-body impossibility [BM.41(2)];

the conditional three-body kinematics and the scale-invariant E_{max} : $\langle E\rangle$

$= 2 : 1$ [BM.41(5); BM.49(3), (4)];

the proton–antiproton degeneracy [BM.28(5); BM.30(4) : $W(+1) = W(-1)$ exactly];

(8) status:

derived: (1)–(7);

unconditional: the closure (2), (3), (5);

conditional: the ordering (7), on BM.62(6)'s branch;

¬a measured lifetime, mass difference or endpoint enters [P5];

¬external seconds [BM.44(3)];

¬a P4 addition;

the open rows: the numerical value of e,

the lepton floors $E_{\text{int}}^{(\ell)}$,

$E_{\text{int}}^{(\nu)}$ [BM.43(2)],

the line index within the ν-triplet.

Req. D2.9 (Parts 1–2); EW.33, EM.9 (Part 2); P5 (Parts 1–2); Def-H.32(3), FT.28, FT.42(1), FT.44(4), FT.142(1) (Part 3); Census FT.35(2) (row-wise); Def-BM.13(3), Def-BM.13(4), BM.13(3), BM.28(5), BM.30(4), BM.41(1)–(5), BM.43(2), BM.44(3), BM.46(4), BM.49(1)–(4), BM.55(3), BM.55(5), BM.62(4), BM.62(6), BM.62(9) (this part); exact ln-ring identities with integer certificates, exact rational enclosures of $\ln 2$, sign censuses [math; machine: `checks/verify_H_constituent_transport_splitting.py` §F].

Pf.

(1) : BM.62(4)'s monomial substituted into BM.41(4)'s definition [machine]

(2) : Δm strictly decreasing in ε_{EM} with $\lim_{\varepsilon \to 0^+} \Delta m = \ln(4/3)/\mathfrak{h}$ [BM.62(6); machine];

$m_\ell \geq 2 \ln 2/\mathfrak{h}$ from $E_{\text{int}} \geq 2$ and $\ln 2 > 0$ [machine];

$$m_\nu \geq 0 \text{ [EW.33 PSD]}$$

$$2\ln 2 - \ln\tfrac{4}{3} = 2\ln 2 - (2\ln 2 - \ln 3) = \ln 3 \text{ [machine: the prime-log ring]};$$

$$\ln\tfrac{4}{3} < 2\ln 2 \iff \tfrac{4}{3} < 4 \iff 4 < 12 \text{ [machine]};$$

$$\ln 3 > 0 \iff 3 > 1 \text{ [machine]}$$

$$(3):\ |Q_\beta| > -\big(\ln\tfrac{4}{3} - 2\ln 2\big)/\mathfrak{h} = \ln 3/\mathfrak{h} \text{ [machine]};$$

$$(4):\ \text{the coefficient census at } c, \varepsilon_{\mathrm{EM}}, \ln 2 > 0 \text{ [machine]}$$

$$(5):\ \text{BM.49(2)'s square-integral row re-cited};$$

$$\ln\tfrac{4}{3} < 2\ln 2 \text{ [machine]} \implies \text{the parent's excess cannot reach the daughter's floor [machine]}$$

$$(6):\ W(q)$$

$$= m_q \text{ at } k$$

$$= 1:\ (2 + z + z^{-1})^1 \text{ has coefficients } (1, 2, 1) \text{ at } Q$$

$$= (-1, 0, +1) \text{ [machine: the convolution]};$$

$$\ln m_q = (0, \ln 2, 0) \text{ at } q = (\pm 1, 0) \text{ [machine]}$$

$$(7)\text{–}(8):\ \text{the clause-by-clause scan of BM.41},$$

$$\text{BM.49},$$

$$\text{BM.55 against the refined rest read [machine: the row census]. } \blacksquare$$

Phys. Making the neutron heavier does not open the decay, and the reason is a comparison of two exact integers. The transport refinement lifts the parent by at most the logarithm of four thirds — a fraction of one bit — while the cheapest possible charged daughter costs two bits of ledger content, and the neutral member costs nothing less than zero. Four against twelve: the excess cannot reach the floor, and the closure holds at every coupling on both sides of the critical surface,

with an exact margin of one $\ln 3$. Read at the level of a single chain, the refinement even helps the closure, because the neutral line is the one that carries the extra transport bit and the neutrino is the neutral member. So the tier reverses an ordering and leaves a verdict intact: the neutron becomes the heavier baryon on the sub-critical branch, and the beta vertex stays shut — not because the two rows are degenerate, which they are no longer, but because the gap between them is smaller than the lightest thing the decay could emit.

BM.64 — The status of the constituent transport splitting: adjudication and scope. Structural-constraints record: see Appendix B, Section B.1.23 of the Addendum.

4.17 Tier BM.15 — The Organic Bridge: the Operator EM Lift, the Derived-Holonomy Attempt, the Constrained Joint Entropy, and the Beta-Decay Energy Audit

Locus. This tier consumes the Tier BM.14 baseline unmodified and runs four constructions and one audit to exhaustion. The EM correction of the refined rest input is lifted from its scalar class row to an operator identity on the force spectrum (BM.65, BM.66). The full real holonomy is attempted as a lift from the edge-phase register, the force decomposition and the channel caps, and where the lift fails the exact obstruction is named and recorded (BM.67). The constrained joint entropy is derived as a fiber cardinality, its geometric factor derived at stationarity and its general fiber named where the enumeration is missing (BM.68). The three bridge inputs are converged and Gap 7 is reduced to its smallest priced residue (BM.69). The beta-decay daughter floor is audited, the formed-closed dichotomy is proved, and every candidate opening term is adjudicated derived, priced or barred (BM.70, BM.71); BM.72 adjudicates the tier.

Standing rows. The refutations of BM.57(8) and BM.58(4) stand closed and are ¬reopened as mass-splitting hypotheses; their residual content is admitted at its own loci — the A_3 continuation asymmetry generates the family triplet [BM.57(9); BM.60(1)], the colour/EM seal asymmetry survives as a mask statement [BM.58$(3), (5)$], and the unique surviving class-dependent mass candidate is $W(Q)$ read through $\ln W(Q)$ [BM.58(6); BM.59$(2), (3)$]. α_B stays barred on BM.61(8)'s four grounds. Holonomy is ¬declared here; joint entropy is ¬postulated here; no external neutron/proton ratio and no external Q-value enters anywhere; no Yukawa-type object exists or is added [EW.15; FT.145(3)]. The tier adds no definition: every new symbol is an in-theorem abbreviation of derived objects.

4.55 (BM.65 — The baseline re-certified: the refined rest input, the exact branch, and the integer certificates).

Thm.

(1) the refined rest input (frozen, re-certified):

$$\boxed{m_{\text{rest}}(Q)\,\mathfrak{h} \;=\; E_{\text{int}}\ln 2 \;+\; \ln W(Q) \;+\; 6\,c\,\varepsilon_{\text{EM}}\,Q^2\ln 2}$$

[Def-BM.13(5); Def-BM.12(6); BM.46$(1), (3)$; BM.35(4)] :

$$E_{\text{int}} = 6$$

[BM.17; BM.26; Census FT.35(2)];

$c \in \left\{\frac{1}{2}, \frac{3}{5}\right\}$ priced, carried symbolically, adopted in none

[Def-BM.12(4)];

$$W(0) = 20,$$

$$W(\pm 1) = 15,$$

$$\sum_{Q} W(Q) = 4^3 = 64$$

[Def-BM.13(3); Def-BM.8(6); BM.28(2); machine: the exhaustive 4^3 end-sign census and the convolution agree];

(2) the class rows (frozen):

$$m_n\,\mathfrak{h} = \ln 1280,$$

$$m_p\,\mathfrak{h} = \ln 960 + 6\,c\,\varepsilon_{\mathrm{EM}}\ln 2,$$

$$(m_n - m_p)\,\mathfrak{h} = \ln\tfrac{4}{3} - 6\,c\,\varepsilon_{\mathrm{EM}}\ln 2 = \Sigma_{\mathrm{B}} - 6\,c\,\varepsilon_{\mathrm{EM}}\ln 2$$

[BM.62(4); machine] :

$$2^6\cdot 20 = 1280,$$

$$2^6\cdot 15 = 960,$$

$$\ln 20 - \ln 15 = \ln\tfrac{4}{3} = 2\ln 2 - \ln 3 > 0$$

[machine: the prime-log ring; the certificate $4 > 3$];

(3) the critical strength and the trichotomy (frozen):

$$\boxed{\varepsilon_{\mathrm{crit}}(c) \;:=\; \frac{\ln(4/3)}{6\,c\,\ln 2} \;=\; \frac{\Sigma_{\mathrm{B}}}{6\,c\,\ln 2} \;=\; \frac{2-\log_2 3}{6\,c}}$$

[BM.62(6); machine: the unique zero] :

$$\varepsilon_{\text{crit}}\left(\tfrac{1}{2}\right) = \frac{2 - \log_2 3}{3}, \qquad \varepsilon_{\text{crit}}\left(\tfrac{3}{5}\right) = \frac{5\,(2 - \log_2 3)}{18} \qquad \text{[machine: exact substitution]};$$

$$\varepsilon_{\text{EM}} < \varepsilon_{\text{crit}}(c) \iff \Delta m > 0;$$

$$\varepsilon_{\text{EM}} = \varepsilon_{\text{crit}}(c) \iff \Delta m = 0;$$

$$\varepsilon_{\text{EM}} > \varepsilon_{\text{crit}}(c) \iff \Delta m < 0$$

[BM.62(6); machine];

(4) the integer certificates (machine re-certified):

$$\boxed{\frac{1}{9} \;<\; \varepsilon_{\text{crit}}\left(\tfrac{3}{5}\right) \;<\; \frac{1}{8} \;<\; \varepsilon_{\text{crit}}\left(\tfrac{1}{2}\right) \;<\; \frac{1}{7}}$$

bound	reduction	certificate
$\varepsilon_{\text{crit}}(\frac{3}{5}) > \frac{1}{9}$	$5\,\Sigma_{\text{B}} > 2\ln 2 \iff 8\ln 2 > 5\ln 3$	$2^8 > 3^5 \iff 256 > 243$
$\varepsilon_{\text{crit}}(\frac{3}{5}) < \frac{1}{8}$	$40\,\Sigma_{\text{B}} < 18\ln 2 \iff 62\ln 2 < 40\ln 3$	$2^{31} < 3^{20} \iff 2147483648 < 3486784401$
$\varepsilon_{\text{crit}}(\frac{1}{2}) > \frac{1}{8}$	$8\,\Sigma_{\text{B}} > 3\ln 2 \iff 13\ln 2 > 8\ln 3$	$2^{13} > 3^8 \iff 8192 > 6561$
$\varepsilon_{\text{crit}}(\frac{1}{2}) < \frac{1}{7}$	$7\,\Sigma_{\text{B}} < 3\ln 2 \iff 11\ln 2 < 7\ln 3$	$2^{11} < 3^7 \iff 2048 < 2187$

[BM.62(7) re-certified; machine: every row an exact comparison of two non-negative integers];

(5) the branch is non-empty and exact:

$$\varepsilon_{\text{crit}}(c) > 0 \iff \Sigma_{\text{B}} > 0 \iff 4 > 3$$

[machine] :

the sub-critical branch $\big(0,\ \varepsilon_{\text{crit}}(c)\big)$ is a non-empty exact interval;

the parameter-free bar:

$$\varepsilon_{\text{EM}} \le \tfrac{1}{9} \implies \Delta m > 0 \text{ at both priced } c$$

[BM.62(8); machine: the witnesses $4^3 = 64 > 54 = 3^3 \cdot 2, \quad 4^5 = 1024 > 972 = 3^5 \cdot 2^2$];

(6) no numerical coupling exists:

¬∃ a theorem of Parts 1–4 fixing ε_{EM} numerically

[EM.25(7); BM.61(3), (4); BM.62(12)] :

the neutron-heavier result is conditional on the symbolic coupling being sub-critical, ¬on an imported physical value

[P5; the hypothesis stands on the face of BM.62(9) and of this row; A §0.2a];

(7) the ratio discipline:

$$\frac{\ln 1280}{\ln 960} \neq \frac{4}{3} \iff 3\ln 1280 \neq 4\ln 960 \iff 1280^3 \neq 960^4$$

[machine: the two integers differ] :

¬a route ratio is read as a mass ratio [BM.37(6); Def-BM.13(8); BM.62(5)];

(8) status:

re-certified: (1)–(7);

the rows of Def-BM.13(5), BM.62(4), (6)–(9), BM.63(2) stand unmodified;

this tag adds no content beyond the machine re-certification;

¬a P4 addition, ¬a new dimensionful scale [P1], ¬a measured value [P5].

Req. CT2.1 (Part 2); EM.25(7) (Part 2); P1, P5 (Parts 1–2); A §0.2a (the hypothesis at the point of use); Census FT.35(2) (row-wise); Def-BM.8(6), Def-BM.12(4), Def-BM.12(6), Def-BM.13(3), Def-BM.13(4), Def-BM.13(5), Def-BM.13(8), BM.17, BM.26, BM.28(2), BM.35(4), BM.37(6), BM.46(1),

BM.46(3), BM.61(3), BM.61(4), BM.62(4), BM.62(5), BM.62(6), BM.62(7), BM.62(8), BM.62(9), BM.62(12), BM.63(2) (this part); exact prime-log-ring arithmetic with integer power certificates, exhaustive end-sign censuses [math; machine: `checks/verify_H_organic_bridge_post_bm14.py` §B, §C].

Pf.

(1)–(2) : the frozen rows re-derived by machine: the 4^3 end-sign census and the generating-function convolution re-enumerated and compared class by class;

$$2^6 \cdot 20 = 1280,\ 2^6 \cdot 15 = 960 \text{ [machine: integers]};$$

$$\ln 20 - \ln 15 = \ln \tfrac{4}{3} \text{ [machine: the prime-log ring]};\ 4 > 3 \text{ [machine]}$$

$$(3) :\ \Delta m = 0 \iff \Sigma_{\mathrm{B}} = 6\, c\, \varepsilon_{\mathrm{EM}} \ln 2 \text{ solved in the } \ln\text{-ring [machine]};$$

$$\varepsilon_{\mathrm{crit}}\left(\tfrac{1}{2}\right) = \tfrac{\Sigma_{\mathrm{B}}}{3 \ln 2},\ \ \varepsilon_{\mathrm{crit}}\left(\tfrac{3}{5}\right) = \tfrac{5\,\Sigma_{\mathrm{B}}}{18 \ln 2} \text{ [machine: exact Fractions]};$$

$$\log_2 3 \text{ form: } \tfrac{\Sigma_{\mathrm{B}}}{\ln 2} = 2 - \log_2 3 \text{ [machine: the prime-log ring]}$$

$$(4) :\ \text{cross-multiplication by } 6$$

$$c \ln 2 > 0 \text{ reduces each bound to } a$$

$$\Sigma_{\mathrm{B}} \gtrless b \ln 2$$

$$\iff (4/3)^a \gtrless 2^b$$

$$\iff 4^a \gtrless 3^a 2^b :$$

$$(c, r)$$

$$= \left(\tfrac{3}{5}, \tfrac{1}{9}\right) :\ 2^8 \cdot 1 \text{ vs } 3^5 :\ 256 > 243;$$

$$\left(\tfrac{3}{5}, \tfrac{1}{8}\right) :\ 2^{31} \text{ vs } 3^{20};$$

$$\left(\tfrac{1}{2}, \tfrac{1}{8}\right) :\ 2^{13} \text{ vs } 3^8;$$

$$\left(\tfrac{1}{2}, \tfrac{1}{7}\right) :\ 2^{11} \text{ vs } 3^7 \text{ [machine: four integer comparisons]}$$

$$(5):\ \Sigma_{\mathrm{B}} > 0 \iff 4 > 3\ \text{[machine]};$$

the two witnesses $64 > 54,\ 1024 > 972$ [machine];

$$(6):\ \text{EM.25}(7),\ \text{BM.61}(4)\ \text{re-cited}$$

$$(7):\ 1280^3 \text{ and } 960^4 \text{ computed exactly and compared [machine: the two integers differ]};$$

$$(8):\ \text{assembly.}\ \blacksquare$$

Phys. The baseline stands on integer certificates. The refined rest input assembles three derived registers — the count, the charge-filtered end census, and the scar's field energy — into one row per class, and the class difference has a single exact zero: the critical strength, enclosed between one ninth and one seventh by four comparisons of powers of two against powers of three. The branch on which the neutral baryon is the heavier one is a non-empty exact interval, certified from below by two hundred fifty-six against two hundred forty-three, and no theorem of Parts 1–4 fixes the coupling numerically — so the ordering result is conditional on the symbol being sub-critical, never on an imported value. The ratio discipline holds with it: the mass ratio is the logarithm ratio of twelve-eighty over nine-sixty, the certificates that twelve-eighty cubed differs from nine-sixty to the fourth keep the measure price away from the mass read, and nothing measured enters anywhere.

Theorem 4.56 (BM.66 — The operator lift of the EM correction: the charge-blind trace, the quadratic invariant, and the assembled operator rest input). Thm.

$$(1)\quad \text{the operator and its spectrum:}$$

$$\widehat{F} := \mathrm{F}_{\mathrm{ext}}(\mathrm{c}, \widehat{Q}) = h(\mathrm{c}) \otimes I_2 + \widehat{Q}\, I_2 \otimes \hat{\kappa},$$

$$h(\mathrm{c}) = \tfrac{1}{2}\big(c_0 I_2 + \mathrm{c} \cdot \boldsymbol{\sigma}\big)$$

[BM.22(1); FT.129(5); the charge read carried as the operator $\widehat{Q}$] :

$$\operatorname{spec} \widehat{Q} = \{-1,\ 0,\ +1\}\ \text{on the materialized classes}$$

[EM.9; D3.4; Def-BM.8(1), (5); Def-BM.3(2), (3)];

$$\lambda_{s,t} = \tfrac{1}{2}\big(c_0 + sC\big) + t\,Q,$$

$$C := |\mathrm{c}|,$$

$$s, t \in \{\pm 1\}$$

[BM.22(3); FT.130(5); machine: the monic quartic from the four roots at exact rational witnesses];

the promotion conditionality is inherited, ¬new:

the flipping block exists under Def-H.43's declared read promotion

[BM.22(5)];

(2) the linear trace is charge-blind:

$$\boxed{\operatorname{Tr} \widehat{F} \;=\; \sum_{s,t \in \{\pm 1\}} \lambda_{s,t} \;=\; 2\, c_0}$$

[machine: the symbolic sum; $\sum_{s,t} t\, Q = 0,\ \sum_{s,t} s\, C = 0$] :

the operator form of the uniqueness row's trace line:

no trace read of $\widehat{F}$ itself splits the two classes

[BM.58(6) : $\sum_{s,t} tq = 0$; BM.1(2) : the trace-only read];

(3) the quadratic invariant:

$$\operatorname{Tr} \widehat{F}^2 \;=\; \sum_{s,t} \lambda_{s,t}^2 \;=\; c_0^2 + C^2 + 4\, \widehat{Q}^2$$

[machine: the polynomial identity in $\mathbb{Q}[c_0, C, Q]$];

$$\widehat{F}_0 \;:=\; \widehat{F}\big|_{\widehat{Q}=0} : \qquad \operatorname{Tr} \widehat{F}_0^2 \;=\; c_0^2 + C^2; \qquad \boxed{I_F \;:=\; \operatorname{Tr} \widehat{F}^2 - \operatorname{Tr} \widehat{F}_0^2 \;=\; 4\, \widehat{Q}^2} \qquad [\text{machine}] :$$

the first trace invariant carrying charge;

$$I_F = (0,\ 4,\ 4) \ \text{ at } Q = (0,\ \pm 1)$$

[machine] :

BM.58(5)'s monopole selector $Q^2 = (0, 1, 1)$ recovered as a trace row;

(4) the operator EM correction:

$$\boxed{\Delta\big(\widehat{M}_{\text{rest}}\mathfrak{h}\big)_{\text{EM}} := \frac{E_{\text{int}}}{4}\, c\,\varepsilon_{\text{EM}} \ln 2 \left(\operatorname{Tr}\widehat{F}^2 - \operatorname{Tr}\widehat{F}_0^2\right)}$$

with $\dfrac{E_{\text{int}}}{4} = \dfrac{6}{4} = \dfrac{3}{2}$ at $E_{\text{int}} = 6$

[BM.17; Census FT.35(2); machine] :

the coefficient is derived, ¬chosen: $\dfrac{E_{\text{int}}}{4} \cdot I_F \;=\; \dfrac{E_{\text{int}}}{4} \cdot 4\,\widehat{Q}^2 \;=\; E_{\text{int}}\,\widehat{Q}^2$ [machine] :

$$\Delta\big(\widehat{M}_{\text{rest}}\mathfrak{h}\big)_{\text{EM}} \;=\; c\,\varepsilon_{\text{EM}}\,\widehat{Q}^2\,E_{\text{int}} \ln 2 \;=\; c\,\varepsilon_{\text{EM}}\,\widehat{Q}^2\,m_{\text{hid}}\,\mathfrak{h} \;=\; E_{\text{EM}}^{\text{self}}(\widehat{Q})\,\mathfrak{h}$$

[BM.46(1) unified form; machine: one monomial];

on charge eigenstates:

$$Q = 0: \quad \Delta\big(\widehat{M}_{\text{rest}}\mathfrak{h}\big)_{\text{EM}} = 0;$$

$$Q = \pm 1: \quad \Delta\big(\widehat{M}_{\text{rest}}\mathfrak{h}\big)_{\text{EM}} = 6\,c\,\varepsilon_{\text{EM}} \ln 2$$

[machine] :

the scalar row of Def-BM.13(5) is reproduced exactly on all three materialized classes;

(5) the assembled operator rest input:

$$\boxed{\widehat{M}_{\text{rest}}\mathfrak{h} \;=\; E_{\text{int}} \ln 2 \;+\; \ln W(\widehat{Q}) \;+\; \frac{3}{2}\,c\,\varepsilon_{\text{EM}} \ln 2 \Big(\operatorname{Tr}\widehat{F}^2 - \operatorname{Tr}\widehat{F}_0^2\Big)}$$

$$\ln W(\widehat{Q}) := \sum_{Q \in \text{spec}} \big(\ln W(Q)\big)\, P_Q$$

[functional calculus on the charge eigenspaces; math: the spectral theorem for a diagonal operator on a finite charge class];

the eigenvalue rows are exactly BM.62(4)'s:

$$Q = 0 : \quad \ln 1280;$$

$$Q = \pm 1 : \quad \ln 960 + 6\, c\, \varepsilon_{\mathrm{EM}} \ln 2$$

[machine];

the CPT row:

$\omega : \ \widehat{Q} \mapsto -\widehat{Q}$ fixes $W(\widehat{Q})$ $[W(Q) = W(-Q);$ Def-BM.8$(7)]$ and fixes I_F $[(3) : \ \widehat{Q}^2$ only$]$:

the proton–antiproton degeneracy is an operator identity [BM.28(5); BM.30(4); machine];

(6) the no-double-count guard:

three derived routes to one monomial:

the sector census	BM.52$(3), (4)$: five zeros and one monomial
the field integral	BM.46(1); Def-BM.12(5)
the quadratic trace	$(3), (4)$ of this tag

[machine: the three agree on every charge eigenstate] :

BM.52(5)'s guard extends: in any decomposition the EM contribution is counted once

[BM.62(1) stands];

the alphabet census:

the trace subtraction removes $c_0^2 + C^2$ exactly (c_0, C are ¬free symbols of I_F)

[machine];

the free-symbol set of $\Delta\big(\widehat{M}_{\text{rest}}\mathfrak{h}\big)_{\text{EM}}$ is $\{c, \varepsilon_{\text{EM}}, \ln 2, \widehat{Q}\} \subseteq$ BM.62(11)'s Δm_{EM} alphabet under e^2

$$= 4\pi\varepsilon_{\text{EM}},$$

$$\Delta x_{\text{B}}^{-1} \qquad = m_{\text{hid}}$$

[EW.35; Def-BM.4(1); machine: the token census];

the disjointness from the transport alphabet $\{m_q, W(Q), \ln 2, \mathfrak{h}\}$ holds except for the shared $\ln 2$

[BM.59(8); BM.37(2); machine];

(7) the uniqueness row is respected:

BM.58(6)'s table gains ¬a new row:

I_F is the operator species of the adopted $E_{\text{EM}}^{\text{self}}$ entry [Def-BM.12(6); (4)],

with the same monomial and the same negative sign for $m_n - m_p$ [BM.58(5); machine];

$F_{LR} = \widehat{Q}I_2$'s row " 0 on a trace read " is the derivation of (2);

the unique surviving class-dependent rest-slot candidate remains $W(Q)$ read through $\ln W(Q)$

[BM.59(2), (3)];

(8) status:

derived: (1)–(7);

conditional on Def-H.43's promotion as BM.22(5) records it (¬a new declaration);

c stays priced, ¬adopted [Def-BM.12(4)];

the new coefficients are $\{\frac{3}{2} = E_{\text{int}}/4,\ 4\}$: a count row and a trace normalization — ¬a P4 addition,

¬a fitting parameter [machine: the coefficient census];

¬a new dimensionful scale [P1]; ¬a measured value [P5]; §Cal-BM stays empty.

Req. D2.9, D3.4 (Parts 1–2); CT2.1, EM.9, EW.35 (Part 2); P1, P5 (Parts 1–2); Def-H.43, FT.129(2), FT.129(5), FT.130(5) (Part 3); Census FT.35(2) (row-wise); Def-BM.3(2), Def-BM.3(3), Def-BM.4(1), Def-BM.8(1), Def-BM.8(5), Def-BM.8(7), Def-BM.12(4), Def-BM.12(5), Def-BM.12(6), Def-BM.13(5), BM.1(2), BM.17, BM.22(1), BM.22(2), BM.22(3), BM.22(5), BM.26, BM.28(5), BM.30(4), BM.37(2), BM.46(1), BM.46(3), BM.52(3), BM.52(4), BM.52(5), BM.58(5), BM.58(6), BM.59(2), BM.59(3), BM.59(8), BM.62(1), BM.62(4), BM.62(11) (this part); exact multivariate polynomial identities over $\mathbb{Q}[c_0, C, Q, c, \varepsilon_{\text{EM}}, \ln 2]$, matrix traces over complex-with-rational-parts at exact witnesses, spectral calculus for diagonal operators on finite charge classes, token censuses [math; machine: `checks/verify_H_organic_bridge_post_bm14.py` §A].

Pf.

(1) : BM.22(1)–(3) re-cited with q carried as $\widehat{Q}$;

the charpoly $\big((y-u)^2 - Q^2\big)\big((y+u)^2 - Q^2\big)$,

y

$= x - \frac{c_0}{2}$,

u^2

$= \frac{C^2}{4}$ built from the four roots and compared to the Faddeev–LeVerrier charpoly at witnesses with $C^2 \in \{0, \frac{1}{4}, 1, \frac{4}{9}, \frac{9}{49}\}$ [machine: exact over the rationals]

$$(2) : \sum_{s,t} \big[\tfrac{1}{2}(c_0 + sC) + tQ\big]$$

$$= 4 \cdot \tfrac{1}{2} c_0 + \tfrac{C}{2} \sum_{s,t} s + Q \sum_{s,t} t$$

$= 2c_0 + 0 + 0$ [machine: the symbolic sum];

$$\text{the matrix row: } \operatorname{Tr}\big(h \otimes I_2 + Q I_2 \otimes \hat{\kappa}\big)$$

$$= 2 \operatorname{tr} h + 2Q \operatorname{tr} \hat{\kappa}$$

$$= 2c_0 \ [\hat{\kappa} \text{ traceless; machine: exact traces at every witness} \times \text{ternary } Q]$$

$$(3): \ \lambda_{s,t}^2 = \tfrac{1}{4}(c_0 + sC)^2 + tQ(c_0 + sC) + t^2Q^2 :$$

$$\sum_{s,t} \tfrac{1}{4}(c_0 + sC)^2 = \tfrac{1}{2}\big[(c_0 + C)^2 + (c_0 - C)^2\big] = c_0^2 + C^2;$$

$$\sum_{s,t} tQ(c_0 + sC) = 0;$$

$$\sum_{s,t} t^2Q^2 = 4Q^2 \text{ [machine: the polynomial identity]};$$

$$\text{the matrix route agrees: } \operatorname{Tr} F^2$$

$$= 2 \operatorname{tr}\big(h^2\big) + 2Q \operatorname{tr}(h) \operatorname{tr}(\hat{\kappa}) + Q^2 \operatorname{tr}(I_2) \operatorname{tr}\big(\hat{\kappa}^2\big)$$

$$= \big(c_0^2 + C^2\big) + 0 + 4Q^2 \ [\hat{\kappa} \text{ the ray swap, } \hat{\kappa}^2 = I_2, \ \operatorname{tr}\hat{\kappa} = 0 \text{ [FT.129(2)]; } \operatorname{tr} h = c_0; \text{ machine at every witness}]$$

$$(4): \ \tfrac{E_{\text{int}}}{4} \cdot 4Q^2 = E_{\text{int}} Q^2 \text{ [machine]};$$

$$c\,\varepsilon_{\text{EM}} Q^2 E_{\text{int}} \ln 2 = c\,\varepsilon_{\text{EM}} Q^2 m_{\text{hid}} \mathfrak{h} \text{ [BM.46(1) re-cited; machine: one monomial]};$$

$$\text{the eigenstate rows at } Q^2 \in \{0, 1\} \text{ [machine]}$$

$$(5): \ P_Q \text{ the spectral projections of } \widehat{Q};$$

$$\ln W(\widehat{Q}) \text{ diagonal with the entries } (\ln 20, \ \ln 15, \ \ln 15) \text{ [machine]};$$

$$\text{the eigenvalue sum } 6 \ln 2 + \ln 20$$

$$= \ln 1280,$$

$$6\ln 2 + \ln 15 + 6c\varepsilon_{\mathrm{EM}}\ln 2$$

$$= \ln 960 + 6c\varepsilon_{\mathrm{EM}}\ln 2 \text{ [machine: the prime-log ring];}$$

$$\omega\text{-row: } W(-Q) = W(Q),\ (-Q)^2 = Q^2 \text{ [machine]}$$

(6) : the three routes compared monomial by monomial on the three materialized classes [machine];

the symbol census of I_F and of the correction [machine: $\{c, \varepsilon_{\mathrm{EM}}, \ln 2, Q\}$];

$$e^2 = 4\pi\varepsilon_{\mathrm{EM}} \text{ [EW.35; machine]};\ \Delta x_{\mathrm{B}}^{-1} = m_{\mathrm{hid}} \text{ [Def-BM.4(1); machine]}$$

(7)–(8) : BM.58$(5), (6)$, BM.59$(2), (3)$ re-cited against (2)–(4);

the coefficient census $\left\{\frac{6}{4},\ 4\right\}$ and the standing-rule scans [machine]. ■

Phys. The scalar row of the EM correction is the eigenvalue row of an operator identity, and the operator is the force matrix the corpus already carried. Its linear trace reads only the shared block — two c-naught, blind to charge — which is the operator form of the uniqueness row's old trace line: no trace of the force itself can split the two classes. The first trace that carries charge is the quadratic one, and its charge content is exactly four Q-squared, so the monopole selector comes back as a trace law. Multiplied by the count row's own coefficient — the internal edge count over four, three halves at six edges — the operator correction reproduces the field-energy monomial on every materialized class, joins the sector census and the field integral as a third route to one and the same term, and keeps the proton–antiproton degeneracy exact, because only the square of the charge enters and the census is conjugation-symmetric. Nothing new is counted: the trace subtraction removes the shared blocks, the symbol alphabet collapses to the adopted row's own, and the distribution factor stays priced in two exact values and adopted in none.

4.57 (BM.67 — The holonomy lift attempted: the derived $\mathbb{Z}_2$ sector, the angle-free force census, and the two-torsion obstruction).

Thm.

(1) the register and the derived discrete phase:

$$\alpha_e \in \{0, \pi\},$$

$$\sigma_e = e^{i\alpha_e} \in \{+1, -1\}$$

[A4; A7; CT0.9; Def-EM.20; EM.20$(1), (2)$: registration data determined at crystallization, ¬declared constants] :

virtual edges carry $\neg\alpha_e$ [EM.20(3)]; $[\alpha_e] = \mathfrak{q}^0$ in every calibration [EM.21];

for an oriented chain c :

$$\Phi(c) \;:=\; \sum_{e\in c} \mathrm{or}(e)\,\alpha_e \pmod{2\pi},$$

$$U(c) \;:=\; \prod_{e\in c} \sigma_e^{\mathrm{or}(e)} \;=\; e^{i\Phi(c)}$$

[D3.5 weight grammar; machine];

(2) the bare holonomy is $\mathbb{Z}_2$ (theorem):

$\sigma^{-1} = \sigma$ on $\{\pm 1\} \implies$ the orientation drops out of every closed product :

$$\boxed{U(\gamma) \;=\; \prod_{e\in\gamma} \sigma_e \;=\; (-1)^{N_\pi(\gamma)}, \qquad N_\pi(\gamma) := \#\{e\in\gamma :\; \alpha_e = \pi\}}$$

[machine: exhaustive over every sign assignment on loops of length $2,\dots,7,$ both orientations] :

$$U(\gamma_1\cdot\gamma_2) = U(\gamma_1)\,U(\gamma_2),$$

$$U(\bar\gamma) = U(\gamma)$$

[machine: concatenation and reversal censuses] :

the Wilson sector of the register is exactly binary;

(3) the parity lock, lifted to the composite:

$$\sum_i \aleph_{v_i} \equiv |q| \pmod 2,$$

$$\sum_i \aleph_{v_i} \;\geq\; |q|$$

[Def-BM.13(2); EM.9b; D3.3; machine: exhaustive over every sign sequence, $k = 2,\dots,7$] :

$$Q = \sum_i q_i$$

[Def-BM.8$(1),(3)$] :

$$|q_i| \equiv q_i \pmod 2 \implies \textstyle\sum_i \aleph^{(i)} \equiv \sum_i |q_i| \equiv Q \equiv |Q| \pmod 2;$$

$$\textstyle\sum_i \aleph^{(i)} \geq \sum_i |q_i| \geq \left|\sum_i q_i\right| = |Q|$$

[machine: exhaustive over the 3^3 charge triples] :

$$\boxed{\sum_i \aleph_{v_i} \;\equiv\; |Q| \pmod 2, \qquad \sum_i \aleph_{v_i} \;\geq\; |Q|}$$

and the wall-sector read is charge-locked:

$$U_{\text{wall}} = (-1)^{\sum \aleph} = (-1)^{|Q|}$$

[machine: the 4^3 end-pair census agrees with $W(Q)$ class by class];

(4) the winding and sealing sectors are integer and phase-trivial:

$$\Theta(c) = \sum_{e \in c} \widehat{ds}^{\,2}(e) = \sum_{e \in c} \kappa_e;$$

materialized closed chains: $\Theta \in 2\mathbb{Z}$,

$$\Phi_{\text{spiral}} = \pi\,\Theta = 2\pi k,$$

$$\text{Hol}_{\text{spiral}} = +1,$$

$$k = \tfrac{\Theta}{2} \in \mathbb{Z}$$

[FT.22; FT.24$(1),(2)$; FT.25$(1),(3)$; Census FT.35(3)] :

$\text{Hol}_{\text{int}} = (-1)^{\Theta_{\text{int}}} \in \{\pm 1\}$, nontrivial only on partial-chain internal sets

[FT.25$(2),(3)$];

the baryon-type sealing phase: $\Theta_{\text{int}} \in 4\mathbb{Z}$

[BM.2(1); Census FT.49(3)] :

$$e^{i\pi\Theta} = +1 \;\; \forall\, \Theta \in 2\mathbb{Z}$$

[machine] :

the whole winding/sealing register contributes trivially to every $U(1)$ phase read — its content is the integer rung k,

¬an angle;

(5) the force-block census (the angle-free theorem):

A8's complete decomposition, regrouped as $\widehat{F}$

$$= \widehat{F}_{J^\pm} + \widehat{F}_{G/H} + \widehat{F}_W + \widehat{F}_{\text{colour}} + \widehat{F}_{\text{EM}} + \widehat{F}_{D^\pm}$$

[A8; BM.52(1) : a regrouping, ¬a sub-block restriction], read at a sealed 3-juncture :

block	candidate angle content	the derived verdict
$\widehat{F}_{J^\pm}$ (the seals)	the sealing phase	$\Theta_{\text{int}} \in 4\mathbb{Z}$ — an integer [BM.2(1); Census FT.49(3)]; the rotation group of the ε-seal is the finite C_3: three one-dimensional characters, no continuous parameter [BM.3(1); BM.10(5); FT.116]
$\widehat{F}_{G/H}$ (the metric)	frame coefficients	the seven geometric monomials carry charge exponent 0; no vertex angle [BM.52(3); BM.58(1), (2), (4)]
$\widehat{F}_W$ (weak/shift)	the shift phase	$\|w\|^2$ class-shared; the Casimir is branch data, neutral at the σ-vertex [BM.28(6); BM.40(3); BM.52(3)]
$\widehat{F}_{\text{colour}}$	slot rotations	annihilated on a sealed singlet: the trace law $\mathfrak{T}(H) = 3a$; the seed obeys $T_a \mathsf{v} = 0$ [BM.1(2); BM.57(5), (6); FT.117(1); EW.32]
$\widehat{F}_{\text{EM}}$	the gauge phase	no local projector; static monopole order; the coupling e symbolic; chart-side [Def-BM.12(2), (7); BM.58(3)]
$\widehat{F}_{D^\pm}$ (dark)	—	$D^+ = \Lambda\,\mathbb{I}$ uniform; D^- locked, retained, $\neq g$accessible, no gauge structure [A8; CT4.1–CT4.5; D4.2; D1.11; BM.52(3); FT.73(1)]

the seed-transport row:

the channel angles $\delta \in \{\frac{2}{9}, \frac{2}{27}, \frac{1}{8}\}$ are channel-line data, ¬attached to the baryon rows [Def-BM.11; BM.59(1a) : the fourth ground stands],

and they enter the Gram mass form, ¬a loop connection [EW.33; Def-H.37(1)] :

θ_v^{force} is derived angle-free at a sealed 3-juncture: every block is annihilated by a trace/projector law, integer- or binary-valued, class-shared, or unattached

[machine: the row census exhaustive against A8's block list];

the caps add integers only:

$$C_{\text{cap}} = 16 = 7 + 3 + 3 + 3,$$

$$D_{\text{tick}}(3) = 2^4 = 16,$$

$$O(2) = O(3) = 0,$$

$$\kappa_0 \in \mathbb{Z}$$

[CT0.5; CT0.6; BM.52(1); Census FT.28(5); Census FT.64(5); FT.24(1); machine];

(6) the two-torsion obstruction (theorem):

let

$$\Theta_R(\gamma) \;=\; 2\pi\, n(\gamma) \;+\; \sum_{e\in\gamma} \alpha_e \;+\; \sum_{v\in\gamma} \theta_v^{\text{force}}$$

$$(\text{mod } 2\pi)$$

be any constrained lift with $n(\gamma)$ integer [(4), (5) : the register and the caps fix integers].

Then:

$2\pi n \equiv 0, \quad \sum \alpha_e \in \pi\mathbb{Z}, \quad \sum \theta_v^{\text{force}} \equiv 0$ on the derived data :

$$\boxed{\text{im(any register/force/caps lift)} \;\subseteq\; \{0,\, \pi\} \;\subset\; \mathbb{R}/2\pi\mathbb{Z}}$$ [machine: the image census] :

against the defect spectrum:

the six states of Def-BM.14(3) carry the angles $\Theta_R \in \{\pi,\ 2\pi,\ \frac{\pi}{2},\ \frac{2\pi}{3},\ \frac{\pi}{5},\ \frac{\pi}{3}\}$

$[\delta_v/\pi \in \{1, \frac{1}{2}, \frac{2}{3}, \frac{1}{5}, \frac{1}{3}\};$ Census FT.63(4), (6); machine$]$:

exactly two of the six lie in the two-torsion image — the self-dual floor's pair $(\pi;\ 2\pi \equiv 0)$;

$\frac{\pi}{2},\ \frac{2\pi}{3},\ \frac{\pi}{5},\ \frac{\pi}{3}$ are pairwise distinct nontrivial cosets modulo π

[machine: exact rational coset arithmetic] :

$\neg\exists$ a lift of the full real holonomy from the register, the force decomposition and the caps

the lift is derived exactly on the self-dual floor and obstructed on both dual pairs;

(7) the rung refinement (where the binary sector does reach):

$$m^2_{\text{pole}}(m) = 4\,\hat{\mu}'^2 \sin^2\!\big(m\,\tfrac{\Theta_R}{4}\big),$$

$$m \in \{k,\ k{-}2, \ldots, -k\}$$

[FT.60(5); Def-H.38; Census FT.63(6)] :

the rung m is register-reachable $\iff m\,\delta_v \in \pi\mathbb{Z} \iff D_S(m) = 2 - 2\cos\!\big(m\,\tfrac{\delta_v}{2}\big) \in \{0,\ 2,\ 4\}$

[machine: the exact iff on every census ladder $m \le 4$ and by the period law $m \in \big(\frac{\pi}{\delta_v}\big)\mathbb{Z}$ [FT.76(3)]] :

floor	δ_v/π	reachable rungs $m \le 4$	the ladder values there
tetrahedron	1	all m	$(2,\ 4,\ 2,\ 0)$ — all in $\{0, 2, 4\}$
cube	$\frac{1}{2}$	$m \equiv 0\ (2)$	$D_S(2) = 2, \quad D_S(4) = 4$
octahedron	$\frac{2}{3}$	$m \equiv 0\ (3)$	$D_S(3) = 4$

floor	δ_v/π	reachable rungs $m \le 4$	the ladder values there
dodecahedron	$\frac{1}{5}$	none at $m \le 4$; $m \equiv 0\ (5)$	$D_S(5) = 2$
icosahedron	$\frac{1}{3}$	$m \equiv 0\ (3)$	$D_S(3) = 2$

the register captures the defect ladder's rational values $\{0, 2, 4\}$ and never its irrational ones — the $\mathbb{Z}_2$ parity of the defect angle is a derived read, the real angle is $\neg$;

(8) the calibration-weight obstruction (the winding route):

$\Theta_R = 2\pi k$ is ruled out as a corpus relation:

$$k = \oint ds^2/(2\mathfrak{h}^2) \text{ carries } \mathfrak{h}\text{-weight } -2$$

[FT.31(3); D2.9];

$$\Theta_R = \iint_S K\, dA \text{ carries } \mathfrak{h}\text{-weight } 0$$

[Def-H.21(6)];

a proportionality across calibration weights is $\neg\, \mathfrak{h}$-free

[CT2.11 criterion; FT.55(1) re-cited] :

the winding register cannot supply the frame angle either;

(9) the chart-side locus stands:

$$\delta_v = 2\pi - d\,\frac{(L-2)\pi}{L} \text{ from the integer card data } (d, L),$$

$$V\,\delta_v = 4\pi$$

[Descartes/Gauss–Bonnet; Census FT.63(4); FT.47(1); BM.60(3); machine: exact in π-units] :

a loop cleanly enclosing one vertex carries $\Theta_R = \delta_v$

$$[\text{FT.47}(1);\ \text{Def-H.21}(5), (6)],$$

$$D_S = 4\sin^2\tfrac{\delta_v}{4}$$

$$[\text{Def-H.37}(2);\ \text{FT.54}(3), (4)]:$$

the defect spectrum is chart-side face geometry consumed through Def-H.37(2)'s interface, whose non-derived clause is priced by FT.54(1)'s form-uniqueness;

the ledger realization of clean-enclosure (vertex-figure) loops is ¬constructed — the corner convention is undeclared

$$[\text{Census FT.63}(4)\ \text{Rmk};\ \text{Census FT.63}(6)\ \text{Rmk}];$$

(10) the structured obstruction record:

```
OBSTRUCTION: holonomy_lift_not_derived
MISSING: vertex_figure_loop_realization | Symk_assignment [FT.55(4)] | chiral_assignment [FT.125(4)] |
DERIVED_NO_GO: two_torsion_image (6) | calibration_weight (8)
PARTIAL_LIFT: Z2_wilson_sector (2) | parity_lock (3) | integer_winding_sealing_sector (4) | self_dual_f
```

(11) status:

derived: (1)–(9);

record issued: (10);

outcome: the obstruction theorem, with the partial lift of (2)–(4),

(7) and the self-dual-floor lift of (6);

¬a declared holonomy: no continuous angle is inserted anywhere in this tier — Θ_R occurs only inside re-citations of the chart-side reads and inside the refuted lift equation [machine: the occurrence census];

¬a P4 addition, ¬a new declaration, ¬a measured value [P5];

the conditionalities (Def-H.37(2), Def-H.38, M1/M2 [Def-H.20], CT2.6) are inherited at the point of use [A §0.2a].

Req. A4, A7, A8 (Part 1); D1.11, D2.9, D3.3, D3.5, D4.2, CT0.5, CT0.6, CT0.9 (Part 1); CT2.11, CT4.1, CT4.2, CT4.3, CT4.4, CT4.5, EM.9b, EW.32, EW.33 (Part 2); Def-EM.20, EM.20, EM.21 (Part 2); P5 (Parts 1–2); Def-H.20, Def-H.21(5), Def-H.21(6), Def-H.37(1), Def-H.37(2), Def-H.38, FT.22, FT.24(1), FT.24(2), FT.25(1), FT.25(2), FT.25(3), FT.31(3), FT.47(1), FT.54(1), FT.54(3), FT.54(4), FT.55(1), FT.55(4), FT.60(5), FT.73(1), FT.76(3), FT.116, FT.117(1), FT.125(4) (Part 3); Census FT.28(5), FT.35(3), FT.49(3), FT.63(4), FT.63(6), FT.64(5) (row-wise); Def-BM.8(1), Def-BM.8(3), Def-BM.11, Def-BM.12(2), Def-BM.12(7), Def-BM.13(2), Def-BM.14(3), Def-BM.14(4), BM.1(2), BM.2(1), BM.3(1), BM.10(5), BM.28(6), BM.40(3), BM.52(1), BM.52(3), BM.57(5), BM.57(6), BM.58(1), BM.58(2), BM.58(3), BM.58(4), BM.59(1), BM.60(3) (this part); exhaustive sign-assignment censuses, exact rational coset arithmetic modulo π and 2π, GF(2) parity identities, Burnside orbit counts on finite ladders [math; machine: `checks/verify_H_organic_bridge_post_bm14.py` §D].

Pf.

$$(1):\ \text{A4/A7 re-cited: the branch phase and the edge map;}$$

$$\alpha_e \in \{0, \pi\}\ \text{[CT0.9]};$$

$$U(c) = \prod \sigma_e^{\mathrm{or}(e)} = e^{i \sum \mathrm{or}(e)\alpha_e}\ \text{[machine: the exponential of a sum over } \{0, \pi\}]$$

$$(2):\ \sigma^{\pm 1} = \sigma \text{ on } \{\pm 1\}\ \text{[machine];}$$

$$\text{exhaustive over } 2^n \text{ assignments for } n$$

$$= 2,$$

$$\dots,$$

7 and both orientation patterns: the product equals $(-1)^{\#\{\sigma_e = -1\}}$ at every one [machine];

concatenation: products multiply; reversal: each σ is its own inverse [machine]

$$(3):\ \text{Def-BM.13(2)'s exhaustive census re-run [machine];}$$

$$|q| \equiv q\ (2) \text{ for } q \in \{-1, 0, 1\}\ \text{[machine: three cases];}$$

$$\sum |q_i| \geq |\sum q_i|\ \text{[machine: exhaustive over 27 triples];}$$

the end-pair census: q

$$= 0 \iff \sigma_s$$

$$= \sigma_t \ (2 \text{ pairs}, \text{ even flips}),$$

$$|q|$$

$$= 1 \iff \sigma_s \neq \sigma_t \ (1 + 1 \text{ pairs}, \text{ odd flips}) \text{ [machine]}$$

$(4):$ FT.22/FT.24$(1), (2)$/FT.25(1)–(3) re-cited;

$$\Theta \in 2\mathbb{Z} \implies e^{i\pi\Theta} = 1 \text{ [machine: even integers]};$$

$$4\mathbb{Z} \subseteq 2\mathbb{Z} \text{ [machine]}$$

$(5):$ each row re-cited from its own tag and classified [machine: the census structure against A8's six groups];

$$7 + 3 + 3 + 3 = 16, \ 2^4 = 16, \ 32 - 16 = 16 \text{ [CT0.5; CT0.6; machine]}$$

$(6):$ the image: $\{2\pi n + \pi j : \ n, j \in \mathbb{Z}\} \equiv \{0, \pi\} \pmod{2\pi}$ [machine];

the state angles modulo π : π

$$\equiv 0,$$

$$2\pi$$

$$\equiv 0,$$

$$\tfrac{\pi}{2} \ /$$

$$\equiv 0,$$

$$\tfrac{2\pi}{3} \ /$$

$$\equiv 0,$$

$$\tfrac{\pi}{5} \ /$$

$$\equiv 0,$$

$$\tfrac{\pi}{3} \ /$$

$$\equiv 0,$$

pairwise distinct in $\mathbb{Q}/\mathbb{Z}$ [machine: exact Fractions];

the count: two in the image, four outside

$$(7): \ \cos\theta \in \{0, \pm 1\} \iff \theta \in \tfrac{\pi}{2}\mathbb{Z} \text{ [math]}$$

$$\implies D_S(m) \in \{0, 2, 4\}$$

$$\iff m$$

$$\tfrac{\delta_v}{2} \in \tfrac{\pi}{2}\mathbb{Z}$$

$$\iff m$$

$\delta_v \in \pi\mathbb{Z}$ [machine];

the five ladders evaluated exactly: the tetrahedral $(2, 4, 2, 0)$;

the cube's $D_S(2)$

$$= 2 - 2\cos\tfrac{\pi}{2}$$

$$= 2,$$

$$D_S(4)$$

$$= 2 - 2\cos\pi$$

$$= 4;$$

the octahedron's $D_S(3)$

$$= 2 - 2\cos\pi$$

$$= 4;$$

the icosahedron's $D_S(3)$

$$= 2 - 2\cos\tfrac{\pi}{2}$$

$$= 2;$$

the dodecahedron's $D_S(5)$

$$= 2 \text{ [machine]};$$

the irrational members $2 - \sqrt{2}$,

$$2 + \sqrt{2},$$

$$1,$$

$$3,$$

$$\tfrac{3-\sqrt{5}}{2},$$

$$\tfrac{5-\sqrt{5}}{2},$$

$$2 - \sqrt{3},$$

$2 - 2\cos\frac{\pi}{10}$ are separated from $\{0, 2, 4\}$ by their minimal polynomials evaluated at 0,

$$2,$$

$$4 \text{ [machine: every value nonzero]}$$

$$(8):\ \text{FT.55}(1) \text{ re-cited verbatim;}$$

$$(9):\ \delta_v \text{ evaluated at } (d, L)$$

$$= (3, 3),$$

$$(3, 4),$$

$$(4, 3),$$

$$(3, 5),$$

$$(5, 3):\ \ \delta_v/\pi$$

$$= (1, \tfrac{1}{2}, \tfrac{2}{3}, \tfrac{1}{5}, \tfrac{1}{3}) \text{ [machine];}$$

$$V\delta_v = 4\pi \text{ at all five [machine]}$$

$$(10)\text{–}(11):\ \text{the record assembled from } (6), (8), (9);$$

$$\text{the occurrence census of } \Theta_R \text{ in this tier [machine];}$$

$$\text{the standing-rule scans [machine]. } \blacksquare$$

Phys. Ask the ledger for a real holonomy and it answers in bits. The edge-phase register is binary, so every closed-loop product is plus or minus one, oriented either way; the wall content of any transport is parity-locked to its charge core, and at the composite the lock reads exactly — total wall parity equals the absolute charge modulo two, and the wall count bounds the charge from below. The winding and sealing registers are integers, and every integer winding contributes trivially to a phase, so their content is the rung, not an angle. Then the force decomposition is censused block by block at the sealed junction, and not one block carries a continuous vertex angle: the seals are finite and integer, the metric is charge-blind, the weak block is class-shared branch data, colour is annihilated by the trace law on anything sealed, electromagnetism is chart-side at a symbolic coupling, the dark blocks are uniform or locked, and the channel angles are line data attached to no baryon row. Any lift built from that material lands in the two-element subgroup —

zero or pi — while the defect spectrum spreads six distinct values whose angles include a fifth of pi and a third of pi: four of the six states are provably out of reach, and the reachable sublattice is exactly the self-dual floor plus the ladder rungs whose defect value is zero, two or four. The winding route is shut separately, by calibration weight: the rung counts hops, the frame angle counts curvature, and no pure number bridges the two. So the tier records what it has — the binary Wilson sector, the parity lock, the integer sectors, the rung law — and issues the obstruction in structured form, with the missing enumeration named: the ledger realization of a loop that cleanly encloses one vertex. No angle is declared anywhere.

4.58 (BM.68 — The constrained joint entropy as a derived fiber cardinality).

Thm.

(1) the constrained fiber (an abbreviation of derived objects):

$$\Omega(E,\Theta,Q) \ :$$

$$= \#\Big\{(R,\sigma,g): \ E_{\text{int}}(R)=E, \ Q(\sigma)=Q, \ \Theta(R,g)=\Theta, \ \text{the sealing and admissibility constraints hold}\Big\} :$$

the census of ledger states passing the filter chain F0–F7

[Census FT.64(1) : configuration validity; registration; integrality; loop quantization; stationarity; regime; relabeling quotient; register product];

$$S_{\text{joint}}(E,\Theta,Q) \ := \ \ln \Omega(E,\Theta,Q)$$

[C9 : entropy in nats; machine: cardinalities of finite sets];

(2) the factorization (theorem on the tested domains):

the registers are independent — the signs factor from the κ-ledger [CT0.10; Census FT.51(5)],

E_{int} is constant on every end-sign fibre [Def-BM.13(6); BM.28(3); machine: exhaustive over the four end pairs],

the sealing grammar is its own register [FT.41(b); Census FT.65(2)] :

$$\boxed{\Omega(E,\Theta,Q) \ = \ \Omega_{\text{count}}(E) \ \cdot \ \Omega_{\text{charge}}(Q) \ \cdot \ \Omega_{\text{geom}}(\Theta;E,Q)}$$

"the three registers multiply, none converts into another"

[Census FT.51(5), FT.65(2); Census FT.64(1) F7] :

$$S_{\text{joint}} \;=\; \ln \Omega_{\text{count}}(E) \;+\; \ln \Omega_{\text{charge}}(Q) \;+\; \ln \Omega_{\text{geom}}(\Theta; E, Q)$$

[machine: the cardinality identity on every tested finite domain];

(3) the first factor (derived):

$$\boxed{\Omega_{\text{count}}(E) \;=\; 2^{E}}$$

[one binary registration per directed real edge : CT0.9 : 1 real edge = 1 bit = ln 2 nats; C9; CT1.29] :

$$I_{\text{hid}} = E_{\text{int}} \ln 2 = \ln 2^{E_{\text{int}}}$$

[Def-H.9(2); FT.17; Census FT.35(2); Census FT.44(4); machine] :

$S_{\text{count}} = E \ln 2$ is exactly Def-BM.13(5)'s first summand;

(4) the second factor (derived):

$$\boxed{\Omega_{\text{charge}}(Q) \;=\; W(Q) \;=\; \left[z^{Q}\right]\left(2 + z + z^{-1}\right)^{k}},$$

$$k = 3 : \quad W = (1,\, 6,\, 15,\, 20,\, 15,\, 6,\, 1),$$

$$\sum_{Q} W(Q) = 64$$

[Def-BM.8(6); Def-BM.13(3); BM.28(2); machine: two agreeing routes] :

$S_{\text{charge}} = \ln W(Q) = \Sigma_{\text{cl}}(Q)$ [Def-BM.13(4)] : the second summand;

(5) the third factor at stationarity (derived):

on the stationary locus $\kappa_e \equiv \kappa_0 \in \mathbb{Z}$ exactly along each connected materialized chain

[FT.53(1), (2); Census FT.64(1) F4],

and the semiclassical survivor list carries one orbit per rung

[Census FT.64(5)(c) : the R2 ladder; FT.53(3)] :

a uniform profile is fixed by the swap/relabeling quotient :

$$\boxed{\Omega^{\text{stat}}_{\text{geom}}(\Theta) \;=\; 1 \;\text{ at every stationary rung,} \qquad S^{\text{stat}}_{\text{geom}} \;=\; 0}$$

[machine: the Burnside count on a single-element set; the swap fixes $(\kappa_0, \kappa_0, \kappa_0)$] :

on the baryon card:

the strict-spacelike minimum sits on the locus: $\kappa = (1, 1, 1), \quad K = 3, \quad \Theta_{\text{int}} = 6$

[Census FT.65(4) Σ_{12} row; FT.24(1) : $\kappa \in \mathbb{Z}$ at materialized digons; machine: the unique profile, the unique orbit] :

$$\boxed{S_{\text{joint}}\big(6,\ \Theta_{\text{stat}},\ Q\big) \;=\; 6\ln 2 \;+\; \ln W(Q) \;+\; 0}$$

the refined rest input is the complete joint entropy of the stationary baryon fibre, on its ledger part:

the EM summand is the chart-side transit-channel energy, ¬a ledger count

[Def-BM.12(6); Def-BM.13(5); BM.66(6); machine: the alphabet disjointness] :

the additive law is a derived stationarity slice, ¬a postulate;

the off-stationary fibers cancel pairwise in Z — ¬a semiclassical spectral contribution

[CT1.20; FT.53(4); Census FT.64(1) F8 note] :

¬a mass read requires an off-stationary count;

(6) the general fiber (the named missing enumeration):

the Platonic floors — derived in closed form: the on-shell lattice $\ker \varphi_S$,

index 2^{F-1} [FT.43(3); Census FT.64(2)];

the level counts $A_S(K)$ with the exact character-sum closed form [Census FT.64(4)];

the Burnside orbits $M_S(K)$,

$M_S^+(K)$ [Census FT.65(1)];

the sealing grammar $g(V)$

$= (3,\ 385,\ 25,\ 292938721075,\ 164395)$ [FT.41(b); Census FT.63(7)];

the register product $2^{2E} \cdot M_S(K) \cdot g(V) \cdot$ (attachment data) [Census FT.65(2)] :

the polygon floors — the dihedral necklaces $M(n,k)$ in closed Burnside form

[Census FT.51(1)–(3)] :

the baryon junction Σ_{12} :

the winding lattice is $\mathbb{Z}^3$ with the swap quotient $\mathbb{Z}_2$

[Census FT.65(4) Σ_{12} row];

the fiber over the full lattice at fixed K is infinite — the witness $(K+n,\ -n)$

[Census FT.64(1) F5; machine] :

the enumeration requires the regime datum (null-permitting / strict-spacelike / uniform / signature-mixed) that the card does not fix :

the exact missing enumeration $=$ the regime-scoped orbit count $\#\{\kappa \in \mathbb{Z}^3 :\ \kappa_1+\kappa_2+\kappa_{12} = K\}/\mathbb{Z}_2$ at the baryon card's materialized rung

the scoped rows this tier derives:

$$\text{the strict regime } \kappa \geq 1 : \quad \#\text{profiles} = \binom{K-1}{2},$$

orbits $(1,\ 2,\ 4,\ 6)$ at $K = (3,\ 4,\ 5,\ 6)$

[machine: stars-and-bars and Burnside agree with exhaustive enumeration];

$$\text{the null-permitting regime } \kappa \geq 0 : \quad \#\text{profiles} = \binom{K+2}{2},$$

orbits $(2,\ 6)$ at $K = (1,\ 3)$

[machine: exhaustive];

the uniform ladder: one orbit per rung $[(5);$ machine$]$;

(7) the two currencies and the bridge-candidate census:

the bit/charge currency: $E \ln 2 + \ln W(Q)$ — cardinalities, integers;

the geometric/holonomy currency: Θ_R enters as a value, ¬a count :

$$m^2_{\text{pole}} = \hat{\mu}'^2 \cdot 4 \sin^2 \tfrac{\Theta_R}{4}$$

[Def-H.37(2); FT.54(4)],

D_S exact algebraic in four fields

[Def-BM.14$(3), (4)$] :

every candidate bridge species, tested:

(a) a derivative of S_{joint} :

Θ is integer-valued;

the finite difference along the stationary ladder is $\ln\!\left(\Omega^{\text{stat}}/\Omega^{\text{stat}}\right)$

$= \ln 1$

$= 0$ while the defect value varies across six distinct values

$[(5);\ \text{Def-BM.14}(4);\ \text{machine}]$:

the count is flat where the angle is not — refuted as a bridge;

(b) a Legendre transform under constraints:

$\neg\exists$ a joint potential of which both currencies are reads — the three deficit functionals are pairwise distinct on digon witnesses ($2\ln 2$ vs 0; 0 vs 3; $2\ln 2 \neq 3$ via $\frac{1}{2} < \ln 2 < 1$)

$[\text{BM.35}(6);\ \text{machine: the exact enclosure}]$:

¬a Legendre pair is derivable;

(c) a constrained extremum:

$\arg\max_{Q} S_{\text{joint}}(6, \cdot, Q)\ =$ the neutral class

$[W(0) = 20 > 15 = W(\pm 1);\ \text{machine}]$:

derived, but a class selection, ¬a currency conversion;

(d) a trace normalization:

$I_F = 4\,\widehat{Q}^2$ is an exact integer on the information side

$[\text{BM.66}(3)]$;

the defect side's trace normalization is the seed doublet's dimension: $(U-1)^{\dagger}(U-1)$

$$= 4\sin^2\tfrac{\phi}{2}\cdot I_2$$

$$[\text{FT.54}(1);\ \text{EW.32}]:$$

both exact, incommensurate species — a count against an angle

[machine: the token census] :

¬a ratio between them is derivable;

(e) a Casimir ratio:

every Casimir/trace/capacity ratio in the inventory is an exact rational

$$[\text{BM.61}(2),(3):\ \text{integers and ratios of integers};\ (\tfrac{1}{3},\ 8,\ \tfrac{1}{25});\ \text{BM.36}(2)]:$$

the five floors' pole factors generate a degree-16 extension of $\mathbb{Q}$

$$[\text{FT.73}(2);\ \text{machine re-cited}]:$$

no rational-coefficient formula on integer-valued reads reaches the pole side;

(8) the gap row: the bridge's well-posed object [FT.54(4) : Gap 7's well-posed object] and its single undeclared factor:

$$\boxed{\frac{m^2_{\text{pole}}}{m^2_{\text{hid}}} \;=\; \rho_{\text{B}}\cdot\frac{D_S}{4E^2}\;\in\;\mathfrak{q}^0,\qquad \rho_{\text{B}}\;:=\;\frac{\hat{\mu}'^2\,\mathfrak{h}^2}{(\ln 2)^2}}$$

$$[\text{FT.54}(4);\ \text{Census FT.63}(6);\ \text{machine: the dimension row } [\hat{\mu}'^2\mathfrak{h}^2]=\mathfrak{q}\cdot\mathfrak{q}^{-1}=\mathfrak{q}^0]:$$

ρ_{B} is a Cal-F packaging [Def-H.37(2); FT.54(4); Cal-F logged] — restated here as the residue's name, ¬added;

the per-state exact algebraic factors: $\dfrac{D_S}{4E^2}$

$$=\Big(\frac{1}{72},\ \frac{2-\sqrt{2}}{576},\ \frac{1}{576},\ \frac{D_S}{3600},\ \frac{2-\sqrt{3}}{3600}\Big)$$

[Census FT.63(6); machine];

(9) status:

derived: (1)–(5), (6)'s scoped rows, (7), (8);

named, ¬filled: the general Σ_{12} fiber (6);

S_{geom} ¬declared — it is derived $= 0$ at stationarity and named-missing off it;

¬a P4 addition, ¬a new dimensionful scale [P1], ¬a measured value [P5];

§Cal-BM stays empty.

the structured record:

```
OBSTRUCTION: geometric_fiber_count_not_derived (the general Sigma_12 fiber)
MISSING: Omega_geom enumeration - the regime-scoped orbit count at the materialized rung
DERIVED: stationary_fiber = 1 (5); Platonic closed forms (6); polygon necklaces (6); strict/null-permit
```

Req. C9, CT0.9, CT0.10, CT1.20 (Part 1); CT1.29, CT2.1, EW.32 (Part 2); P1, P5 (Parts 1–2); Def-H.9(2), Def-H.37(2), FT.17, FT.41(b), FT.43(3), FT.53(1), FT.53(2), FT.53(3), FT.53(4), FT.54(1), FT.54(4), FT.73(2) (Part 3); Census FT.35(2), FT.44(4), FT.51(1), FT.51(3), FT.51(5), FT.63(6), FT.63(7), FT.64(1), FT.64(2), FT.64(4), FT.64(5), FT.65(1), FT.65(2), FT.65(4) (row-wise); Def-BM.8(6), Def-BM.12(6), Def-BM.13(3), Def-BM.13(4), Def-BM.13(5), Def-BM.13(6), Def-BM.14(3), Def-BM.14(4), BM.28(3), BM.35(6), BM.36(2), BM.61(2), BM.61(3), BM.66(3), BM.66(6) (this part); Cal-F; stars-and-bars and Burnside enumeration, exact rational arithmetic, the prime-log ring with integer certificates, exact radical separations, rational $\ln 2$ enclosures [math; machine: `checks/verify_H_organic_bridge_post_bm14.py` §E].

Pf.

(1)–(2) : the filter chain re-cited row by row [Census FT.64(1)];

the independence rows re-cited and re-run: signs $\perp$ κ [CT0.10; machine: the product cardinality on the tested domains],

E_{int} constant on end-sign fibres [machine: exhaustive over the four end pairs],

sealing separate [FT.41(b)];

$\Omega = \Omega_{\text{count}}\Omega_{\text{charge}}\Omega_{\text{geom}}$ as a cardinality of a product of finite sets [machine]

$(3):\ 2^6 = 64$ [machine];

$I_{\text{hid}} = 6\ln 2 = \ln 64$ [machine: the prime-log ring];

$(4):\ W(Q)$ re-enumerated by both routes [machine]

$(5):$ FT.53(2) re-cited: on the stationary locus $\kappa_e = \kappa_0 \in \mathbb{Z}$ exactly;

the swap $(\kappa_1, \kappa_2, \kappa_{12}) \mapsto (\kappa_2, \kappa_1, \kappa_{12})$ fixes $(\kappa_0, \kappa_0, \kappa_0)$ [machine];

the Burnside count on one fixed profile is 1 [machine];

the strict minimum $(1,1,1):$ unique among $\kappa \geq 1,\ \sum \kappa = 3$ [machine: exhaustive];

$6\ln 2 + \ln W(Q) + 0$ compared term by term with Def-BM.13(5) at $E_{\text{int}} = 6$ [machine]

$(6):\ \binom{K-1}{2}$ at $K = (3,4,5,6) = (1,3,6,10)$ [machine];

the Burnside orbits under the swap: fixed points $\#\{(a,a,b):\ 2a+b=K,\ a,b\geq 1\}$,

orbits

$= \frac{1}{2}(\#\text{profiles} + \#\text{fixed}):$

$K = (3,4,5,6) \mapsto (1,2,4,6)$ [machine: closed form and exhaustive enumeration agree];

$\kappa \geq 0:\ \binom{K+2}{2}$ at $K = (1,3) = (3,10),$ orbits $(2,6)$ [machine];

the infinite-fiber witness $(K{+}n, -n, 0)$,

$n \in \mathbb{Z}_{>0}:$ distinct profiles at every n [machine: injective in n]

$$(7)(\text{a}):\ \ln 1 = 0\ [\text{machine}];$$

$$(\text{b}):\ 2\ln 2$$

$$\neq 3 \text{ via the enclosure } \tfrac{1}{2} < \ln 2 < 1 \text{ with tail } < 2^{-20}\ [\text{machine: exact rational partial sums}];$$

$$(\text{c}):\ 20 > 15\ [\text{machine}];$$

$$(\text{d}):\ \text{the token census of both normalizations}\ [\text{machine}];$$

$$(\text{e}):\ \text{FT.73}(2)\text{'s degree row re-cited}$$

$$(8):\ \frac{4\hat{\mu}'^2 \sin^2(\Theta_R/4)}{4E^2(\ln 2)^2/\mathfrak{h}^2}$$

$$= \frac{\hat{\mu}'^2\mathfrak{h}^2}{(\ln 2)^2} \cdot \frac{D_S}{4E^2} \text{ at } \Theta_R$$

$$= \delta_v\ [\text{machine: the Laurent assembly}];$$

the five factors re-cited from Census FT.63(6);

$$(9):\ \text{the status and record assembly.}\ \blacksquare$$

Phys. The joint entropy is not postulated; it is counted. Three registers — the crystallized-edge bits, the charge-filtered end census, and the winding-and-sealing fiber — are independent by census, so their cardinalities multiply and their logarithms add: the additive law of the refined rest input is a cardinality identity, and its first two factors are the corpus's own two-to-the-E and W-of-Q. The third factor is derived exactly where the mass read lives. On the stationary locus the geodesic-constancy theorem makes every surviving profile uniform, a uniform profile is fixed by the swap quotient, so the geometric fiber is a single orbit and its entropy is zero: the refined rest input is the complete joint entropy of the stationary baryon fiber, ledger part, and the off-stationary fibers are pairwise-cancelling content that no mass read consults. Away from stationarity the Platonic substrate carries closed forms — a character sum for the level counts, Burnside for the orbits, the frozen grammar counts for the sealings — but the baryon junction's own general fiber is infinite at fixed winding without a regime datum, and the tier names that missing enumeration instead of guessing a number. Then both currencies are laid side by side and every proposed bridge species is tested: the entropy difference vanishes along the ladder where the defect values vary, no joint potential exists to transform, the extremum picks the neutral class but converts nothing, the trace normalizations are exact on both sides and incommensurate between them, and the Casimir ratios are rational where the pole side is of degree sixteen. The exchange rate between count and angle stays one undeclared dimensionless number — the packaging the interface already logged — and the gap row records it under a name.

Theorem 4.59 (BM.69 — The bridge converged: the three inputs, the non-derivation theorem, and the minimal priced residue). Thm.

(1) the three inputs assembled:

the force-operator quadratic invariant	$I_F = \mathrm{Tr}\,\widehat{F}^2 - \mathrm{Tr}\,\widehat{F}_0^2 = 4\,\widehat{Q}^2$	[BM.66(3)]
the holonomy lift	$\mathbb{Z}_2$ Wilson sector derived; the real lift obstructed with the record issued	[BM.67(2), (6), (10)]
the constrained joint entropy	$S_{\text{joint}} = E\ln 2 + \ln W(Q) + \ln \Omega_{\text{geom}}, \quad \Omega_{\text{geom}}^{\text{stat}} = 1$	[BM.68(2), (5)]

(2) the bridge identity in its derived form:

$$\boxed{\frac{m_{\text{pole}}^2}{m_{\text{hid}}^2} \;=\; \mathcal{R}\big(E_{\text{int}}, \Theta_R\big) \;=\; \rho_{\text{B}} \cdot \frac{D_S(\Theta_R)}{4\,E_{\text{int}}^2}, \qquad \rho_{\text{B}} = \frac{\hat{\mu}'^2 \mathfrak{h}^2}{(\ln 2)^2} \in \mathfrak{q}^0}$$

[FT.54(4); Census FT.63(6); BM.68(8); machine] :

$\mathcal{R}$ is exact algebraic per state once ρ_{B} is supplied:

six states, six values of $\frac{D_S}{4E^2}$, four fields, six minimal polynomials

[Def-BM.14(3); machine re-cited];

(3) the class-dependence census:

the bridge's left-hand class dependence runs through D_S alone — floor and branch data, i.e. the flavour composition of the class, which is ¬derived

[BM.28(7); Def-BM.8(5); BM.59(9)];

$E_{\text{int}} = 6$ is class-shared

[BM.57(4); Census FT.35(2)];

the information side's class dependence $\big(\ln W(Q) :\ \Sigma_{\text{B}} = \ln \frac{4}{3};\quad I_F :\ (0, 4, 4)\big)$ is exact

[Def-BM.13(4); BM.66(3); machine];

the conditional pole row (the hypothesis on its face):

hypothesis: the two baryon classes differ by exactly one branch substitution [BM.59(9); BM.23(4) pattern] :

then

$$\Delta\big(m^2_{\text{pole}}\big) \;=\; \hat{\mu}'^2\big(D_S(\mathcal{F}_g,+) - D_S(\mathcal{F}_g,-)\big) \;=\; \Delta_g \;>\; 0,$$

$$\Delta_1 = 2\hat{\mu}'^2, \quad \Delta_2 = (\sqrt{2}-1)\hat{\mu}'^2, \quad \Delta_3 = \big(2\cos\tfrac{\pi}{10} - \sqrt{3}\big)\hat{\mu}'^2$$

[Def-BM.14(5); machine: the certificates $4>2,\ 2>1,\ 3<5$] :

the same direction as the transport row;

and ¬addable to the informational assembly: a cross-currency addition is a category the corpus bars

[BM.35(5); BM.60(6); BM.37(2); pointers];

(4) the non-derivation theorem:

$$\boxed{\neg\exists \text{ an exact identity } \; m^2_{\text{pole}} = \mathcal{B}\big[E_{\text{int}},\, W(Q),\, \Theta_R,\, I_F\big] \;\text{ with } \mathcal{B} \text{ built from derived data alone}}$$

(derived data := exact trace ratios, Casimir normalizations, constrained fiber counts, channel caps, already-derived algebraic numbers),

on four independent grounds:

(i) the field degree: the five floors' pole factors generate a degree-16 extension of $\mathbb{Q}$;

every enumerated input $\big(E_{\text{int}},\ W(Q),\ I_F,$ caps, fiber counts$\big)$ is integer- or binary-valued;

a rational formula places every pole value in a field of degree < 16 over the declared constants

[FT.73(2), (3); machine re-cited];

(ii) the two-torsion image: the register lift of Θ_R reaches $\{0,\pi\}$ only — four of the six states are unreachable at every winding

[BM.67(6)];

(iii) the calibration weight: the winding route $\Theta_R = 2\pi k$ is ruled out — $\mathfrak{h}$-weight -2 against 0

[FT.55(1); BM.67(8)];

(iv) the strength bar: $\neg\exists$ a count $\to$ strength theorem;

the capacity ratios $\left(\frac{1}{3},\ 8,\ \frac{1}{25}\right)$ are ¬couplings

[BM.36(3); BM.61(8) (O1); FT.73(5) trap pattern];

(5) the Gap-7 reduction (the smallest priced residue):

the bridge residue is exactly: (R1) the one undeclared dimensionless ratio $\rho_{\mathrm{B}} = \dfrac{\hat{\mu}'^2\mathfrak{h}^2}{(\ln 2)^2}$ [Cal-F packaging; ¬added here];

(R2) the ledger realization of clean-enclosure vertex-figure loops

[BM.67(9), (10); Census FT.63(4) Rmk];

(R3) the flavour composition of the sealed classes

[BM.28(7)] :

every other factor of (2) is derived exact algebra:

the six D_S with their minimal polynomials; the five $\frac{D_S}{4E^2}$;

$$I_F = (0,4,4);$$

$$W = (1,6,15,20,15,6,1); \quad \Omega_{\text{geom}}^{\text{stat}} = 1$$

[machine: the inventory row];

(6) the no-hiding row:

the coefficient census of BM.65–BM.69 :

$\{\frac{3}{2} = \frac{E_{\text{int}}}{4}$ [a count row], 4 [the I_F trace normalization], $2c_0,\ c_0^2 + C^2$ [annihilated in I_F], ρ_{B} [a Cal-F restatement]$\}$:

each is a trace ratio, a Casimir normalization, a constrained fiber-count derivative, a channel-cap lock, or an exact operator identity — the organic test is satisfied coefficient by coefficient

[machine: the census];

α_B : ¬added, ¬derived, barred on BM.61(8)'s four grounds, which stand

[machine: the token scan of this tier — every occurrence is a barred-object row];

¬a P4-class temperature,

¬a free-energy coefficient,

¬a fitting parameter is introduced anywhere in this tier

[machine: the symbol census against the P4 core and the era registries];

(7) status:

derived: (1)–(6);

not achieved: the full identity — the obstruction is exact and named (4),

and the residue is minimal and priced (5);

¬a P4 addition, ¬a new dimensionful scale [P1], ¬a measured value [P5], ¬a declaration;

Def-H.37(2)'s interface is cited as the locus of the declared transfer, ¬imported as a relating equation.

Req. P1, P4, P5 (Parts 1–2); A §E (the P4 core); Def-H.37(2), FT.54(4), FT.55(1), FT.73(2), FT.73(3), FT.73(5), FT.125(4) (Part 3); Census FT.35(2), FT.63(4), FT.63(6) (row-wise); Def-BM.8(5), Def-BM.13(4), Def-BM.14(3), Def-BM.14(4), Def-BM.14(5), BM.23(4), BM.28(7), BM.35(5), BM.36(3), BM.37(2), BM.57(4), BM.59(9), BM.60(6), BM.61(8), BM.66(3), BM.66(4), BM.67(2), BM.67(6), BM.67(8), BM.67(9), BM.67(10), BM.68(2), BM.68(5), BM.68(8) (this part); Cal-F; exact radical separations, Laurent-monomial assembly, symbol censuses [math; machine: `checks/verify_H_organic_bridge_post_bm14.py` §A, §E, §G].

Pf.

(1) : the three rows re-cited from their own tags;

(2) : BM.68(8)'s identity re-cited;

the six values and four fields re-cited from Def-BM.14(3) and Census FT.63(6)

(3) : E_{int} class-shared [BM.57(4)'s difference table re-cited];

Δ_g re-formed in the three exact fields with the certificates $4 > 2$,

$$2 > 1,$$

$$\cos \tfrac{\pi}{10} > \tfrac{\sqrt{3}}{2}$$

$$\iff 3 < 5 \text{ [machine]};$$

the addition bar: BM.37(2)'s disjointness and BM.35(5)'s no-conversion row re-cited

(4)(i) : FT.73(2),

(3) re-cited: the degree-16 compositum, the irreducible quartics, the quadratic-field exclusion $4-2\sqrt{2}+2\sqrt{3}-\sqrt{6} \notin K_2$;

(ii)–(iv) : BM.67(6), (8), BM.36(3), BM.61(8) re-cited

(5) : the inventory row assembled from BM.65–BM.68's derived lists [machine];

(6) : the coefficient census [machine];

the token scan of this tier for the barred object α_B and for P4-registry additions [machine];

(7) : assembly. ■

Phys. The three inputs meet, and the bridge exists in exactly the form the corpus can derive — and in no other. The operator invariant is the exact integer four Q-squared; the holonomy gives a binary sector and an obstructed angle; the joint entropy gives the count plus the charge census at stationarity, with a geometric factor of one. The well-posed bridge object, the pole-to-hidden mass ratio, factorizes into a per-state exact algebraic number and a single dimensionless packaging, and that packaging is the whole of the gap: everything else in the identity is derived, and the non-derivability of the rest is a theorem with four independent grounds — a degree-sixteen field against integer reads, a two-element image against six angles, a calibration weight against the winding route, and the standing bar against turning counts into strengths. The class dependence of the bridge lives entirely in the informational currency; its pole-currency image needs a flavour composition no clause attaches, and where a composition is hypothesized the pole splitting points the same way as the transport row but cannot be added to it without the very equation that is missing. So Gap 7 reduces to its smallest honest form: one scale ratio, one loop realization, one composition — named, priced, and hidden inside no coefficient.

4.60 (BM.70 — The beta-decay energy audit: the two-bit floor theorem, the margin ladder, and the formed-closed dichotomy).

Thm.

(1) the floor and the Q-value: $F_{\ell\nu} := m_\ell \mathfrak{h} + m_\nu \mathfrak{h}$ [an abbreviation of derived reads];

$$\boxed{Q_\beta \mathfrak{h} = \ln \tfrac{4}{3} - 6\, c\, \varepsilon_{\mathrm{EM}} \ln 2 - F_{\ell\nu}}$$ [BM.62(4); BM.63(1); machine: the monomial identity];

(2) the two-bit daughter floor (theorem on the informational species):

the chain:

the emitted charged member is the ℓ-channel line — the forced, class-unique carrier

[BM.23(5); BM.41(1); Def-BM.6(1)] :

every channel line is a materialized bound chain

[FT.109; FT.112; FT.113; Def-H.4];

every materialized closed chain carries ≥ 2 directed edges — the digon is the minimal floor

[Def-H.4 : $\delta h(c) \geq 2$; Census FT.28; FT.32 : $L \geq 2$; BM.13(3) : $E_{\text{int}} \geq 2$] :

the informational read $m\,\mathfrak{h} = E_{\text{int}} \ln 2$

[Census FT.44(4); Def-H.9(2); BM.43(2) : the taxonomy law]

$$\Longrightarrow$$

$$m_\ell\,\mathfrak{h} \;\geq\; 2\ln 2 \;>\; 0$$

[machine: the exact enclosure $\frac{1}{2} < \ln 2 < 1$];

$$m_\nu\,\mathfrak{h} \;\geq\; 0$$

[EW.33 : the Gram is PSD; BM.41(4)(a)];

and on the informational read $m_\nu\,\mathfrak{h} = E_{\text{int}}^{(\nu)} \ln 2 \;\geq\; 2\ln 2$ likewise

[BM.43(2); BM.13(3)] :

$$\boxed{F_{\ell\nu} \;\geq\; 2\ln 2}$$

with the floor ladder:

$$2\ln 2 \;\big(\text{the PSD row alone}\big)$$

$$<$$

$$4\ln 2 \;\big(\text{the informational neutrino}\big)$$

$$<$$

$5 \ln 2$ (with the daughters' own transport refinement)

[machine: $2 < 4 < 5$] :

the refinement row:

$$k = 1 : \quad W(q) = m_q = (1,\, 2,\, 1) \text{ at } q = (-1,\, 0,\, +1),$$

$$\Sigma_{\text{cl}}(q) = (\,0,\, \ln 2,\, 0\,) :$$

the neutral line carries one extra bit, the charged lines none

[Def-BM.13(3); BM.63(6); machine: the convolution of $(2 + z + z^{-1})^1$] :

the daughters are q_ℓ

$= -1$ [the charged member] and q_ν

$= 0$ [the neutral member; BM.41(1); FT.142(1) : EM-unactivated]

$\Longrightarrow$

$F_{\ell\nu} \;\geq\; 2 \ln 2 + 0 + 2 \ln 2 + \ln 2 = 5 \ln 2$ on the fully refined informational read

[machine];

(3) the margin ladder (exact, integer-certified):

$$Q_\beta \, \mathfrak{h} \;\leq\; \ln \tfrac{4}{3} - F_{\ell\nu} \text{ at every } \varepsilon_{\text{EM}} > 0$$

[BM.62(6) : strict monotonicity] :

floor used	$F_{\ell\nu} \geq$	$Q_\beta \mathfrak{h} \leq$	the certificate
the PSD row alone	$2\ln 2$	$-\ln 3$	$\ln\frac{4}{3} < 2\ln 2 \iff \frac{4}{3} < 4 \iff 4 < 12$
the informational neutrino	$4\ln 2$	$-\ln 12$	$\ln\frac{4}{3} < 4\ln 2 \iff \frac{4}{3} < 16 \iff 4 < 48$
the fully refined daughters	$5\ln 2$	$-\ln 24$	$\ln\frac{4}{3} < 5\ln 2 \iff \frac{4}{3} < 32 \iff 4 < 96$

[machine: the prime-log identities $\ln\frac{4}{3} - 2\ln 2 = -\ln 3$, $\ln\frac{4}{3} - 4\ln 2 = -\ln 12$, $\ln\frac{4}{3} - 5\ln 2 = -\ln 24$; and every comparison an integer certificate];

$$\boxed{Q_\beta < 0 \text{ strictly, with margin } \geq \frac{\ln 3}{\mathfrak{h}}, \text{ deepened by } \frac{6\,c\,\varepsilon_{\mathrm{EM}}\ln 2}{\mathfrak{h}} \text{ at every positive coupling}}$$

[BM.63(2), (3), (4) audited and standing; machine];

(4) the formed-closed dichotomy (the audit verdict):

Q_β is a formed expression only where all four members are read in one currency :

(a) the informational currency on every member [the parents: Def-BM.13(5); the daughters: BM.43(2)'s taxonomy law] :

$$(2)\text{'s floor theorem applies} \implies Q_\beta \mathfrak{h} \leq -\ln 3 < 0$$

$$\text{formed} \implies \text{closed};$$

(b) the transport/pole species on the leptons [BM.39(1) : $m_\ell^2 = \hat{\mu}^2 \ell_c^2\, G(c(i)) > 0$, the scale $\hat{\mu}$ undeclared [Cal-F]; Def-BM.11] :

the subtraction against informational-currency parents is a cross-currency expression;

its relating equation is ¬derived

[BM.35(5); BM.43(2) : the conversion $E_{\mathrm{int}}^{(i)} \leftrightarrow$ transport data is item-7 residue; BM.60(6); pointers]

$$\implies$$

$$Q_\beta \ \neg\text{formed} \implies \neg\text{a verdict};$$

on every read on which Q_β is a derived expression, $Q_\beta < 0$ strictly;	$\neg\exists$ a derived read on which $Q_\beta \geq 0$

[machine: the row census over the complete route inventory BM.30(1) : route (i) $Q = -(m_\ell + m_\nu) < 0$ [BM.41(4)(b)]; route (ii) priced ¬adopted, its sign undetermined without scale data [BM.41(4)(d)]; route (iii) amplitude layer — amplitude $\neq$ mass [FT.79(5); BM.37(3)]; the curvature-flux read: $Q < 0$ [BM.41(4)(c)]];

(5) the missing lepton read, named exactly:

$$E^{(\ell)}_{\text{int}},\ E^{(\nu)}_{\text{int}}\ \neg\text{computed at any floor}$$

[BM.43(2); BM.63(8)];

the line index within the ν-triplet ¬derived

[FT.145(3) : ¬a mixing matrix] :

these reads sharpen the margin and cannot breach the floor:

$E_{\text{int}} \geq 2$ is a structural lower bound of the species

[BM.13(3); Census FT.28];

the margin is monotone in $E^{(\ell)}_{\text{int}},\ E^{(\nu)}_{\text{int}}$

[machine: each added edge adds $\ln 2$] :

the closure is ¬an artifact of an unfinished lepton mass read: the unfinished read moves the margin, never the sign

;

(6) no external Q-value:

$\neg\exists$ an observed endpoint, mass difference or Q-value in any clause of this tier

[P5; machine: the token scan — zero external numerals, zero unit tokens];

the consistency-check row:

a comparison against an observed Q_β is admissible only as a quarantined audit row

[Census FT.114, FT.120 pattern; P1, P5] and only after the corpus derives its own scale — one calibration datum fixing e [EM.25(7)] and a $\hbar$ read [P1] :

$\neg\exists$ such a derivation in Parts 1–4; this tier performs no comparison;

(7) status:

derived: (1)–(6);

BM.63(2)'s closure is confirmed as a theorem on the formed reads, with its species conditionality on its face (4) [A §0.2a];

¬a measured lifetime, mass difference or endpoint enters [P5];

¬a P4 addition;

¬a new dimensionful scale [P1];

the open rows: $E_{\text{int}}^{(\ell)}$, $E_{\text{int}}^{(\nu)}$, the ν-line index, the numerical value of e

[pointers].

Req. EW.33 (Part 2); EM.25(7) (Part 2); P1, P5 (Parts 1–2); Def-H.4, Def-H.9(2), FT.32, FT.79(5), FT.109, FT.112, FT.113, FT.142(1), FT.145(3) (Part 3); Census FT.28, FT.44(4), FT.114, FT.120 (row-wise); Def-BM.6(1), Def-BM.11, Def-BM.13(3), Def-BM.13(5), BM.13(3), BM.23(5), BM.30(1), BM.35(5), BM.37(3), BM.39(1), BM.41(1), BM.41(4), BM.43(2), BM.60(6), BM.62(4), BM.62(6), BM.63(1), BM.63(2), BM.63(3), BM.63(4), BM.63(6), BM.63(8) (this part); Cal-F; exact prime-log-ring identities with integer certificates, exact rational $\ln 2$ enclosures, monotonicity censuses, token scans [math; machine: `checks/verify_H_organic_bridge_post_bm14.py` §F].

Pf.

(1) : BM.63(1)'s row re-cited and re-formed in the $\ln$-ring [machine];

(2) : $\delta h(c) \geq 2$ [Def-H.4 re-cited];

the digon card $E_{\text{int}} = 2$ [Census FT.28 re-cited];

$E_{\text{int}} \geq 2$ on every computed cell $\{2, 6, 12, 24, 60\}$ [machine: the minimum];

$m_\ell \mathfrak{h} = E_{\text{int}}^{(\ell)} \ln 2 \geq 2 \ln 2$ [machine];

$m_\nu \mathfrak{h} \geq 0$ [EW.33 PSD re-cited]

the refinement: $(2 + z + z^{-1})^1$ has coefficients $(1, 2, 1)$ at Q

$= (-1, 0, +1)$ [machine: the convolution];

$\ln m_q = (0, \ln 2, 0)$ [machine];

$2 \ln 2 + 2 \ln 2 + \ln 2 = 5 \ln 2$ [machine: the prime-log ring];

$2 < 4 < 5$ [machine]

$(3):\ \ln \frac{4}{3} - 2 \ln 2 = (2 \ln 2 - \ln 3) - 2 \ln 2 = -\ln 3;$

$\ln \frac{4}{3} - 4 \ln 2 = -2 \ln 2 - \ln 3 = -\ln(4 \cdot 3) = -\ln 12;$

$\ln \frac{4}{3} - 5 \ln 2 = -3 \ln 2 - \ln 3 = -\ln(8 \cdot 3) = -\ln 24$ [machine: the prime-log ring];

the three certificates: $4 < 12,\ 4 < 48,\ 4 < 96$ [machine: integers];

the EM deepening: $6c\varepsilon_{\text{EM}} \ln 2 > 0$ at $c, \varepsilon_{\text{EM}}, \ln 2 > 0$ [machine: the coefficient census]

$(4):$ the route inventory re-cited from BM.30(1),

BM.41(4)(a)–(d) and classified by currency [machine: the row census];

the cross-currency bar: BM.35(5)'s no-conversion row and BM.37(2)'s disjointness re-cited

$$(5):\ \text{monotonicity: } \partial\big(-Q_\beta \mathfrak{h}\big)/\partial E_{\text{int}}^{(i)} = \ln 2 > 0\ [\text{machine}];$$

$$(6):\ \text{the token scan [machine: zero external numerals]};$$

$$(7):\ \text{assembly.}\ \blacksquare$$

Phys. The audit asked whether the two-bit daughter floor is a theorem or an artifact of an unfinished lepton read, and the answer separates the two cleanly. It is a theorem wherever the question is formed: the emitted charged member is a line of the lepton channel, every channel line is a materialized bound chain, every materialized closed chain carries at least two directed edges because the digon is the floor, and the informational read turns that count into at least two bits; the neutral member costs nothing less than zero by positive semidefiniteness, and nothing less than two more bits plus one refinement bit if it too is read informationally. The margin ladder follows in exact integers — against three, against twelve, against twenty-four — and the field-energy term deepens every rung at every positive coupling. The only escape is currency: read the leptons through the transport species with its undeclared scale and the Q-value is not a formed expression at all, because the equation relating the two currencies is the corpus's open item. Hence the dichotomy, which is the audit's verdict: formed implies closed, unformed implies no verdict, and no derived read opens the vertex. What the unfinished lepton read moves is the margin's sharpness, never the sign — the floor is structural, sitting at the digon, below every rung of every ladder. And the observed endpoint enters nowhere: it may be compared only in a quarantined audit row, after a scale derivation that does not exist, and this tier performs no comparison.

4.61 (BM.71 — The opening-term census: the five routes, the neutron row under the bridge, and the six verdict rows).

Thm.

$$(1)\quad \text{the route census (each candidate: derived, priced, or barred):}$$

#	the candidate opening term	verdict	the derived ground
1	time-energy uncertainty / the non-uniform clock	barred	the Mandelstam–Tamm relation bounds the splitting's resolvability, $\neq g$its magnitude [BM.53(4)–(6)]; the closure is the non-existence of real on-shell solutions — $2m_n(E_p - m_p) = -m_\ell^2$, $\lambda = m_\ell^2(m_\ell^2 - 4m_n^2)$: polynomial identities at every tick [BM.41(2); machine]; $\neq g\exists$ a time operator at finite dimension, so an energy loan over a Δt has no operator content [BM.10(4)]; the ledger is forward-only [A6 : $\neq g\exists\tau^{-1}$], strictly contracting [L2; BM.51(2), (3)], with per-tick weights $\Gamma_{\text{tr}} \in [0, \frac{1}{3}]$ [BM.44(2)]; the flicker ensemble is amplitude-layer content, and amplitude $\neq$ mass [FT.79(5); BM.37(3); FT.67(3a); FT.68(4)]
2	field reconfiguration energy	$\neq g$constructed; sign-barred at the constructed order	radiation, back-reaction and the transition's field reconfiguration are the named open scope [Def-BM.12(7); BM.50(3)]; at the constructed monopole order every self-energy is the positive integral of a square [BM.46(1); Def-BM.12(5)], and the transition $Q : 0 \to (+1) \oplus (-1)$ raises the total field energy — the final state carries two charged members' self-energies where the initial carries none [BM.49(2); machine: the sign census at positive symbols]; the attractive cross-well $V = -\varepsilon_{\text{EM}}/d$ belongs to the $\neq g$constructed pair system [BM.48(1), (3), (4); BM.54(4) : the category bar]
3	binding / nearby-space correction	barred	the category bar: a binding energy is the energy of the bound system, $\neq g$a correction to a free member's mass [BM.48(4); BM.54(4)]; the lepton–baryon bound channel is $\neq g$constructed [BM.48(3); BM.6(5) : the dynamics of Q]; the nearby space holds only a strict fraction $f(k) = 1 - \frac{1}{k} < 1$ [BM.47(6); machine: exact rationals]; the count binding is zero on both rows, $6 - (2{+}2{+}2) = 0$ [BM.54(1); machine]; the ensemble binding is class-degenerate [BM.54(2); FT.107(3); BM.30(3)]

#	the candidate opening term	verdict	the derived ground
4	D^- redistribution	barred	the dark sector's contribution to the class splitting is exactly zero: $D^+ = \Lambda\,\mathbb{I}$ uniform, both locks class-shared [BM.52(3) $D^\pm$ row; CT4.5; BM.28(1)]; the junction's D^- content is identical on both rows — the same vertex types $n(q_1) = n(p_2) = 4$, the same lock ledger [Census FT.35(4); BM.57(4) : the difference table is identically zero; machine], so $\neq g\exists$ a class-dependent object to redistribute; locked content is retained and $\neq g$accessible [D4.2; D1.11; CT4.3; FT.3] — no channel reads it into a vertex budget; and $\neq g\exists$ a response read — the dielectric non-existence inventory theorem [FT.73(1); any dynamics supplying one contradicts CT4.2/D1.11]
5	read specification	the sole surviving door — not walked	opening the vertex is a read-specification act requiring a §I addendum [BM.49(2); BM.41(4)(e); BM.30(5); BM.45]; the route-(ii) measure price $T(0) = \frac{4}{3}T(\pm1)$, its sign undetermined without scale data, stays priced $\neq g$adopted [BM.41(4)(d); BM.28(4); BM.37(3)]; this tier performs no specification

(2) completeness and the verdict:

the five routes exhaust the audit's candidate list;

each is barred by a cited theorem or ¬constructed with its scope named

[machine: the census-row structure];

the MT bound's legitimate content:

$$\Delta t \;\geq\; \frac{1}{2\,\Delta m_{\text{EM}}} \;=\; \frac{\pi\,\hbar}{3\,c\,e^2\ln 2}$$

[BM.53(5); machine: the exact Laurent reciprocal]

is a bound on clocks resolving the splitting, ¬a budget paying for a decay;

the beta-decay no-go: on every formed derived read the vertex is kinematically closed, and its opening is ¬derivable from the corpus — it requires a read specification

[BM.70(4); (1); machine];

(3) the neutron row under the bridge: $\boxed{m_n - m_p \;=\; \Delta m_{\text{transport}} \;+\; \Delta m_{\text{EM}} \;+\; \Delta m_{\text{bridge}}}$

$$\Delta m_{\text{transport}} \;=\; +\,\frac{\ln(4/3)}{\hbar}$$

[BM.59(4); BM.65(2) : derived];

$$\Delta m_{\mathrm{EM}} \;=\; -\,\frac{6\,c\,\varepsilon_{\mathrm{EM}}\ln 2}{\mathfrak{h}}$$

[BM.46(4); BM.62(3); the operator form BM.66(4) : derived; c priced; $\varepsilon_{\mathrm{EM}}$ symbolic];

$$\boxed{\Delta m_{\mathrm{bridge}} \;=\; 0 \text{ in the informational currency}}$$

(theorem): BM.62(1)'s decomposition is complete in that currency

[BM.58(6)'s uniqueness row; BM.62(11)'s alphabets; BM.66(6)'s guard; BM.68(5)'s stationary slice],

and the bridge's pole-currency content is unformed without the relating equation

[BM.69$(3), (4)$];

the conditional pole row (the hypothesis on its face):

under BM.59(9)'s substitution hypothesis: $\quad \Delta\big(m^2_{\mathrm{pole}}\big) = +\Delta_g > 0$

[BM.69(3); machine: the three certificates]

— the same direction as the transport row, ¬a formed summand of any Q-value;

(4) the six verdict rows:

(i) the sign of $m_n - m_p$:

> 0 on the sub-critical branch $\varepsilon_{\mathrm{EM}} < \varepsilon_{\mathrm{crit}}(c)$;

$= 0$ at $\varepsilon_{\mathrm{crit}}(c)$;

< 0 above

[BM.62$(6), (9)$; BM.65(3)–(5)] :

which branch the corpus sits on is open — the numerical value of e

[EM.25(7); BM.62(12)];

(ii) the sign of Q_β :

strictly negative at every $\varepsilon_{\text{EM}} > 0$ on every formed read

[BM.70(3), (4)];

(iii) the decay:

closed — ¬conditionally open and ¬obstructed by an artifact: the closure is a theorem on the formed reads and unformable on the unformed ones

[BM.70(4); (2)];

(iv) the derived terms:

$\Delta m_{\text{transport}}$; Δm_{EM} (scalar and operator);

$I_F = 4\,\widehat{Q}^2$;

$F_{\ell\nu} \geq 2\ln 2$; the margin ladder $(\ln 3,\ \ln 12,\ \ln 24)$; $\Delta m_{\text{bridge}} = 0$ in currency; $\Omega^{\text{stat}}_{\text{geom}} = 1$;

(v) the priced terms:

$c \in \{\frac{1}{2}, \frac{3}{5}\}$ priced ¬adopted [Def-BM.12(4)];

ε_{EM} symbolic [EM.25(9); P5];

$\hat{\mu},\ \hat{\mu}',\ \mu$ undeclared [Cal-F];

ρ_{B} restated as the residue [BM.69(5)];

(vi) the open terms:

the numerical value of e [EM.25(7)];

$E^{(\ell)}_{\text{int}},\ E^{(\nu)}_{\text{int}}$ [BM.43(2)];

the ν-line index

the relating equation

the flavour composition

the regime-scoped Σ_{12} fiber [BM.68(6)];

(5) no tuning:

¬∃ an adjustment toward an observed ratio — the ordering is branch-conditional, the Q-value symbolic, and neither is compared to anything

[P5; machine: zero external tokens, zero decimal numerals in this tier];

(6) status:

derived: (1)–(5);

the no-go is exact, ¬rhetorical: every bar in (1) carries a cited theorem or a named ¬construction scope;

¬a P4 addition, ¬a new declaration, ¬a measured value [P5]; §Cal-BM stays empty.

Req. A6, D1.11, D4.2, L2 (Part 1); CT4.2, CT4.3, CT4.5, EM.25(7), EM.25(9) (Part 2); P5 (Parts 1–2); FT.3, FT.67(3), FT.68(4), FT.73(1), FT.79(5), FT.107(3) (Part 3); Census FT.35(4) (row-wise); Def-BM.12(4), Def-BM.12(5), Def-BM.12(7), BM.6(5), BM.10(4), BM.28(1), BM.28(4), BM.30(3), BM.30(5), BM.37(3), BM.41(2), BM.41(4), BM.43(2), BM.44(2), BM.45, BM.46(1), BM.46(4), BM.47(6), BM.48(1), BM.48(3), BM.48(4), BM.49(2), BM.50(3), BM.51(2), BM.51(3), BM.52(3), BM.53(4), BM.53(5), BM.53(6), BM.54(1), BM.54(2), BM.54(4), BM.57(4), BM.58(6), BM.59(4), BM.59(9), BM.62(1), BM.62(3), BM.62(6), BM.62(9), BM.62(11), BM.62(12), BM.65(2), BM.65(3), BM.65(5), BM.66(4), BM.66(6), BM.68(5), BM.68(6), BM.69(3), BM.69(4), BM.69(5), BM.70(3), BM.70(4) (this part); Cal-F; exact polynomial identities, Laurent reciprocals, exact rational censuses, sign censuses at positive symbols, token scans [math; machine: `checks/verify_H_organic_bridge_post_bm14.py` §F, §G].

Pf.

$$(1) \text{ row } 1: \text{ BM.53}(4)\text{–}(6) \text{ re-cited;}$$

$$2m_n(E_p - m_p)$$

$$= -m_\ell^2 \text{ at } m_p$$

$$= m_n \text{ and } \lambda$$

$$= m_\ell^2(m_\ell^2 - 4m_n^2) \text{ re-derived as cleared polynomial identities in } \mathbb{Q}[m_n, m_\ell] \text{ [machine];}$$

$$\mathfrak{T}(i \nVdash_d) = d\, i \neq 0 \text{ at } d = 2 \text{ [BM.10}(4);\ \text{machine];}$$

$$\Gamma_{\text{tr}} \in [0, \tfrac{1}{3}] \text{ [BM.44}(2) \text{ re-cited]}$$

$$\text{row } 2: \text{ Def-BM.12}(7)\text{'s } \neg\text{constructed list re-cited;}$$

$$E_{\text{EM}}^{\text{self}}(q)$$

$$= c$$

$$\varepsilon_{\text{EM}} q^2 m \text{ with } q^2: \ 0$$

$$\mapsto (1,1) \text{ across the transition : the difference has positive coefficients in } (c, \varepsilon_{\text{EM}}, m) \text{ [machine: the sign census];}$$

$$\text{BM.49}(2) \text{ re-cited}$$

$$\text{row } 3: \ f(k) = 1 - \tfrac{1}{k}: \quad f(2) = \tfrac{1}{2},\ f(3) = \tfrac{2}{3},\ f(4) = \tfrac{3}{4},\ \text{ all } < 1 \text{ [machine: exact Fractions];}$$

$$6 - (2{+}2{+}2) = 0 \text{ [machine];}$$

$$\text{BM.48}(4),\ \text{BM.54}(2), (4) \text{ re-cited}$$

$$\text{row } 4: \ O(4) = 2^5 - 16 = 16 \text{ [CT0.6;\ machine];}$$

the vertex-type row $n(q_1)$

$$= n(p_2)$$

$= 4$ on both classes [Census FT.35(4); BM.57(4)'s difference table; machine: identically zero];

FT.73(1)'s inventory list re-cited

row 5 : BM.49(2), BM.41(4)(d), (e), BM.30(5) re-cited;

(2) : $\frac{1}{2\,\Delta m_{\mathrm{EM}}} = \frac{1}{2}\cdot\frac{2\pi\mathfrak{h}}{3ce^2\ln 2} = \frac{\pi\mathfrak{h}}{3ce^2\ln 2}$ [machine: the Laurent reciprocal]

(3) : BM.62(1)'s completeness re-cited with BM.66(6)'s guard and BM.68(5)'s slice;

$\Delta_g > 0$ on the three certificates [machine];

(4)–(6) : the verdict rows assembled from BM.65–BM.70's cited clauses [machine: the row census and the token scans]. ■

Phys. Every door the audit could name is tried against the corpus's own theorems, and none opens. Uncertainty is a resolvability bound, not a budget: the closure is the non-existence of real on-shell solutions, a polynomial fact at every tick, there is no time operator to borrow from, the ledger runs forward only and contracts strictly, and the flicker that drives every transition weight lives at the amplitude layer, where amplitudes are barred from being masses. Field reconfiguration is a named unconstructed scope, and at the constructed order its sign runs against the decay, because the final state carries two charged members' self-energies where the initial carried none. Binding belongs to the bound, the bound channel does not exist, the nearby space holds only a strict fraction of a positive correction, and both binding reads the corpus owns are zero or class-degenerate. The dark sector has nothing to redistribute: its locks are class-shared, its junction content is identical on both rows, what it retains is inaccessible to every observer channel, and no response read exists by inventory theorem. What remains is the corpus's own honest door — a read specification, an act the tier prices, audits, and declines to perform. Under the bridge the neutron row does not move: the bridge term is zero in the currency in which the row is formed, conditionally positive in the pole currency where it is not formed, and the six verdict rows record the signs, the closure, and the derived/priced/open partition exactly — with nothing tuned toward anything observed.

BM.72 — The status of the organic bridge: adjudication and scope. Structural-constraints record: see Appendix B, Section B.1.24 of the Addendum.

4.18 Tier BM.16 — The Chiral Assignment: the Closed Deficit, the Ray-Resolved Spin Census, the Attached Quark Splitting, the Chiral Vertex Dynamics, and the Specified Beta-Decay Opening

Locus. This tier consumes the Tier BM.15 baseline unmodified and walks the two doors the corpus priced and declined: the deficit-$1'$ assignment rule — named missing at FT.125(4), FT.141(6), FT.144(5) and FT.157(4), with its forced shape already derived — and the β-vertex read specification — named as the sole opening route at BM.30(5), BM.41(4)(e), BM.49(2) and BM.71(1) row 5. Both acts are logged as §II.4 registry addenda (A-BM.7, A-BM.8) before use, per Def-BM.3(7)'s rule. BM.73 issues the left–right assignment matching $\hat{\omega}$ and $\hat{\chi}$ and derives the occupancy; BM.74 resolves the spin census by ray; BM.75 derives the weight–branch–core attachment and attaches the quark mass splitting; BM.76 constructs the chiral vertex dynamics; BM.77 issues the vertex read specification and proves the exact opening branch; BM.78 adjudicates.

Standing rows. The refutations of BM.57(8) and BM.58(4) stand closed and are ¬reopened. BM.70(4)'s dichotomy and BM.71(2)'s no-go stand on the reads formed at their loci; BM.77(7) records their rescope under the specification, and no clause of BM.41–BM.72 is retracted. The relating equation between the informational and the geometric currencies is ¬assumed anywhere — item 7 stays open [BM.35(5); BM.68(7); pointer]. α_B stays barred on BM.61(8)'s four grounds. ¬a Yukawa-type object exists or is added [EW.15; FT.145(3)]. No external mass, lifetime, endpoint or coupling value enters any derivation [P5]. Chirality $=$ helicity on-shell" remains external vocabulary [FT.123$(3),(4)$]; the spin-structure torsor remains a distinct kind [FT.122(7); FT.89(2)]; deficit 11 is untouched [FT.125(5)].

Theorem 4.62 (BM.73 — The chiral matter assignment: the issued coupling, the occupancy theorem, and the ω/χ matching rows). Thm.

(1) the four derived gradings (the census the rule couples):

the algebra S-grading	$(10,\,6) = \big(\dim \mathcal{K}_1^+,\ \dim \mathcal{K}_1^-\big)$ on the force arena; metric/weak reads S-even, seal reads S-odd	[FT.136; FT.141$(1),(2),(5)$]
the branch-label parity	spatial $=$ aligned (α=0, $\vee^2$, Casimir 2) weak-active; temporal $=$ antialigned (α=π, $\wedge^2$, Casimir 0) weak-blind	[FT.101; FT.135; FT.157$(2),(3)$]
the carrier $\hat{\chi}$-grade	$\hat{\chi} = i\big(\lvert c\rangle\langle\omega(c)\rvert - \lvert\omega(c)\rangle\langle c\rvert\big)$, $P_{L/R} = \frac{1}{2}(\mathrm{id} \pm \hat{\chi})$, $\lvert L\rangle = \frac{1}{\sqrt{2}}\big(\lvert c\rangle - i\lvert\omega(c)\rangle\big)$	[Def-H.33; FT.122(1)–$(3),(6)$]
the Janus-arrow chirality	strict alternation, the prefix law, the single-arrow materialized locus at rung one	[FT.144$(2),(5)$]

the pairwise distinctness is derived:

the four are pairwise distinct $\mathbb{Z}_2$'s, commuting where jointly defined, neither reducible to another

[FT.136(4)–(5); FT.141(6); FT.126(4); FT.157(4)];

(2) the assignment rule (the declaration, logged as A-BM.7 before use):

the forced shape is derived: any $\hat{\omega}$-covariant arrow $\leftrightarrow$ ray coupling maps the single-arrow locus into a single chiral ray, and the complementary ray has occupancy zero at rung one — exhaustive on the 2! bijections

[FT.144(5)] :

the issued rule: the materialization-arrow locus occupies $\operatorname{im} P_L$ — the $\hat{\chi} = +1$ ray of Def-H.33's boxed sign

the coupled rows (each forced given the choice, or the choice itself):

(a) the occupied ray carries the weak-active/aligned/spatial branch label;

the complement carries the weak-blind/antialigned/temporal label

[forced: every materialized bound chain is weak-active — $\{G, H, W\} \subseteq \mathcal{A}(\Sigma_{\min})$ [Census FT.28(3)]; weak acts from rung one [CT0.13]; the rung-one channel class is weak-carried [FT.142(1)] : the alternative coupling would make every matter line weak-blind against those derived rows; machine: the 2 couplings censused, one consistent];

(b) the matter orientation occupies the seed weak doublet on the left ray [EW.32; EW.36 : $\operatorname{spec} Q|_{\mathcal{H}_\nu} = \{0, +1\}$];

the image orientation occupies the ω-image representation $\{0, -1\}$ on the right ray [EW.36; D2.6; BM.25; BM.27; the A-BM.5 antineutrino row]

[forced by $\hat{\omega}$-covariance once (a) holds : $\hat{\omega} P_L \hat{\omega}^{-1} = P_R$ [FT.122(4)]];

(c) the S-grading keeps its derived division of labour — it grades observables, ¬states — and receives no species coupling

[FT.141(6)] :

the declared content is exactly one naming bit — which ray the arrow occupies is named left — on the forced shape of [FT.144(5)];

it carries no datum: ¬a scale,

¬a coupling value,

¬a measured quantity [P5; Def-BM.3(7) pattern];

(3) the occupancy theorem (derived given (2)) :

the rung-one materialized locus occupies $(1,\ 0)\ =\ (\text{left occupied},\ \text{right empty})$

the complement's content is the vacuum sector: all-virtual, unregistered, no accessible pole

$$[\text{FT.144}(4),(5);\ \text{CT4.2 pattern};\ \text{CT0.1}];$$

the rung ladder of the occupancy:

rung one: the sectors are force-decoupled, $F_{LR} = F_{RL} = 0$ exactly $[\text{FT.128}(2);\ \text{FT.144}(6)]$;

rung two: the EM core read opens the unique flip channel,

$$F_{LR}$$

$$= q$$

$$I_2$$

$$\neq 0$$

$$\iff q$$

$$\neq 0,$$

splitting $2|q| \in \{0, 2\}$ $[\text{FT.129}(5);\ \text{FT.130}(5);\ \text{Def-H.43 promotion inherited};\ \text{BM.22}(2),(4)]$;

rung three: colour adds no carrier flip — sealed ends are blind,

the seed is colour-neutral $[\text{FT.117};\ \text{EW.32};\ \text{FT.144}(6)]$;

the flipping set remains $\{F_{\text{EM}}$ core read$\}$, minimal and unique $[\text{FT.129}(5)]$;

(4) the ω/χ matching rows (re-certified exact over $\mathbb{Q}(i)$) :

$$\hat{\chi}^\dagger = \hat{\chi}, \quad \hat{\chi}^2 = \mathrm{id}, \quad \mathrm{tr}\,\hat{\chi} = 0, \quad \mathrm{spec}\,\hat{\chi} = \{+1, -1\};$$

$$P_L^2 = P_L, \quad P_L P_R = 0, \quad P_L + P_R = \mathrm{id}, \quad \mathrm{tr}\, P_L = \mathrm{tr}\, P_R = 1$$

$$[\text{FT.122}(1),(2);\ \text{machine}]:$$

$$\boxed{\hat{\omega}\hat{\chi}\hat{\omega}^{-1} = -\hat{\chi}, \qquad \{\hat{\chi}, \hat{\omega}\} = 0, \qquad \hat{\omega} P_L \hat{\omega}^{-1} = P_R, \qquad \hat{\omega}|L\rangle = -i|R\rangle, \quad \hat{\omega}|R\rangle = +i|L\rangle}$$

$$[\text{FT.122}(4);\ \text{machine}];$$

the quaternionic triple:

$$\hat{\omega}^2 = \hat{\chi}^2 = \hat{\kappa}^2 = \mathrm{id};$$

$$\{\hat{\omega}, \hat{\chi}\} = \{\hat{\chi}, \hat{\kappa}\} = \{\hat{\kappa}, \hat{\omega}\} = 0;$$

$$\hat{\omega}\hat{\chi} = -i\hat{\kappa}, \quad \hat{\chi}\hat{\kappa} = -i\hat{\omega}, \quad \hat{\kappa}\hat{\omega} = -i\hat{\chi}$$

$$[\text{Def-H.34};\ \text{machine}];$$

the winding row:

$$\omega:\ k \mapsto -k$$

$$[\text{FT.122}(4);\ \text{D2.6};\ \text{EM.24}(1)]:$$

the ray exchange and the winding reversal are the two faces of one involution;

the canonicity:

$$\left\{M = M^\dagger:\ \hat{\omega} M \hat{\omega} = -M,\ M^2 = \mathrm{id},\ \langle c|M|c\rangle = \langle \omega(c)|M|\omega(c)\rangle = 0\right\} = \{+\hat{\chi},\ -\hat{\chi}\}$$

$$[\text{FT.122}(6);\ \text{machine: the circle } M(a,y),\ a^2 + y^2 = 1,\ \text{on exact rational witnesses}];$$

(5) the weak rows under the assignment:

$$\big[W_i \otimes I_\omega,\ I \otimes P_{L/R}\big] = 0$$

[FT.123(1); FT.126(2); machine] :

the assignment attaches species to rays; it does not modify the weak action;

the selectivity lives at the state locus: $W_i \Pi_{\mathcal{V}} = 0$,

$$\textstyle\sum_i W_i^2 = 2\,\Pi_{\mathcal{R}}$$

[FT.157(2); FT.141(4); FT.101(1)–(2); machine re-cited];

the occupied ray is the weak-active one by (2)(a);

a chiral gauge coupling as an operator modification is ¬constructed and ¬needed for the assignment

[FT.123 Rmk scope];

(6) the bars:

the weight ↔ ray attachment is barred as the external helicity vocabulary

[FT.123(3), (4)];

the spin-structure torsor is reversal-invariant, a distinct kind

[FT.122(7); FT.89(2)];

deficit 11 (spin–statistics) is untouched: no exchange operator exists

[FT.80(5); FT.125(5)];

the frame section $e \mapsto \mathcal{F}_e$ and a global Dirac-type operator remain unconstructed

[FT.125(1); item 21(i) pointer];

no SM species reading and no claim that the SM weak interaction is reproduced

[FT.157 Rmk bars stand; P1, P5];

(7) status:

derived: (1), (3)–(6) [(3) given (2)];

declared: exactly one naming bit (2) — logged as the §II.4 addendum A-BM.7 before use;

deficit $1'$ is closed at the assignment level:

the coupling rule exists — the declaration's root is issued;

the downstream rows unblock at their own loci and stay open: deficits 2,

3,

4 [FT.125(6)],

Q4–Q6 residues,

the anomaly sums formable on the $1'$ side with the $+\frac{2}{3}$ side structurally unreachable [FT.125(2); BM.28(7)];

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

§Cal-BM stays empty;

the extra-axiomatic list [A §0.2a] gains no member.

Req. A4, A8 (Part 1); D2.6, CT0.1, CT0.13 (Part 1); CT4.2, EM.24(1), EW.32, EW.36 (Part 2); P1, P5 (Parts 1–2); Def-H.33, Def-H.34, Def-H.43, FT.80(5), FT.89(2), FT.101(1), FT.101(2), FT.101(3),

FT.117, FT.122(1), FT.122(2), FT.122(3), FT.122(4), FT.122(6), FT.122(7), FT.123(1), FT.123(3), FT.123(4), FT.125(1), FT.125(2), FT.125(4), FT.125(5), FT.125(6), FT.126(2), FT.126(4), FT.128(2), FT.129(5), FT.130(5), FT.135, FT.136(1), FT.136(4), FT.136(5), FT.141(1), FT.141(2), FT.141(4), FT.141(5), FT.141(6), FT.142(1), FT.144(2), FT.144(4), FT.144(5), FT.144(6), FT.157(2), FT.157(3), FT.157(4) (Part 3); Census FT.28(3) (row-wise); Def-BM.3(7), BM.22(2), BM.22(4), BM.25, BM.27, BM.28(7) (this part); exact $\mathbb{Q}(i)$ matrix algebra, exhaustive bijection censuses, rational-circle witnesses [math; machine: `checks/verify_H_chiral_assignment_opening.py` §A, §B].

Pf.

(1) : the four rows re-cited from their own tags [FT.136(1), FT.141(1), (2), (5); FT.157(2), (3); Def-H.33; FT.122(1)–(3); FT.144(2), (5)];

the distinctness rows re-cited [FT.126(4); FT.136(4)–(5); FT.141(6); FT.157(4)]

(2) : FT.144(5)'s forced shape re-cited [machine: exhaustive on the 2! bijections];

(a) : the weak-activity census: both ray/label couplings evaluated against $\{G, H, W\} \subseteq \mathcal{A}(\Sigma_{\text{min}})$ [Census FT.28(3)],

CT0.13,

FT.142(1) : the weak-blind-occupied coupling contradicts the derived weak activity of every matter floor;

one coupling survives [machine];

(b) : $\hat{\omega} P_L \hat{\omega}^{-1} = P_R$ [machine over $\mathbb{Q}(i)$] and $\omega : \operatorname{spec} Q \mapsto \{0, -1\}$ [EW.36 re-cited];

(c) : FT.141(6) verbatim

(3) : FT.144(5)'s occupancy pair $(1, 0)$ with (2)'s naming;

the rung rows re-cited [FT.128(2); FT.129(5); FT.130(5); FT.117; FT.144(6)]

(4) : $\hat{\chi}$

$= i \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix} \right),$

$\hat{\omega}$

$$= \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right),$$

$$\hat{\kappa}$$

$= \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ in the ordered orientation basis [FT.122(5); Def-H.34] : every identity of (4) verified entrywise over $\mathbb{Q}(i)$ [machine];

$\hat{\omega}|L_u\rangle = |R_u\rangle \cdot (-i)$ on the unnormalized kets [machine];

the canonicity circle $M(a, y)^2$

$= (a^2+y^2)$id and the diagonal-free reduction a

$$= 0$$

$$\Longrightarrow\ y^2$$

$$= 1$$

$$\Longrightarrow\ M$$

$= \pm\hat{\chi}$ [machine: on the rational witnesses $(\frac{3}{5}, \frac{4}{5}),\ (\frac{5}{13}, \frac{12}{13}),\ (\frac{8}{17}, \frac{15}{17})$]

(5) : the commutator $[W_i \otimes I,\ I \otimes P] = 0$ by tensor-factor disjointness [machine];

the locus rows re-cited [FT.157(2); FT.141(4)]

(6)–(7) : the bars re-cited clause by clause; the standing-rule scans [machine]. ■

Phys. The corpus had derived everything about chirality except one bit: which of the two exchanged rays the materialized world sits in. The four gradings — the force arena's ten/six split, the weak-active and weak-blind branch labels, the orientation doublet's exchange-form grade, and the Janus arrow's single-direction locus — are pairwise distinct, each exact, and the rule marrying them to matter was the ledger's named missing declaration. This tier issues it in its minimal form: the arrow's occupied ray is the left ray, the one the boxed chirality sign already fixes algebraically; the weak-active branch label then rides along by consistency, because every materialized matter floor is weak-active by derived rows and the opposite coupling would contradict them; the right ray is the orientation reversal's image and hosts the image-orientation species the antineutrino naming row already used. What is derived given the bit is substantial: the occupancy pair one-zero

with its vacuum complement, the rung ladder along which the flip channel opens at rung two for charged cores and never for colour, and the whole ω/χ matching — the involution exchanges the rays, anticommutes with the grade, and reverses the winding, so ray exchange and matter–image exchange are one move. What is not obtained is stated as loudly: no helicity claim, no spin-statistics, no Standard Model reading, and no modification of the weak action itself, which stays chirality-blind on the carrier while the species sit where the rule puts them.

4.63 (BM.74 — The ray-resolved spin census: the winding labels, the occupancy rows, and the vertex table).

Thm.

(1) the spin-label rule (the declared interface inherited, ¬re-derived):

$$\text{a materialized chain of winding } k \text{ carries } \mathrm{Sym}^k(\mathcal{H}_{\mathsf{v}}),$$

$$j = \frac{|k|}{2},$$

$$\text{weights } j,\ j-1,\ \ldots,\ -j$$

$$[\text{Def-H.38};\ \text{FT.55}(2);\ \text{FT.60}(1)\text{–}(3);\ \text{FT.123}(3)]\ :$$

$$\text{the anchor:}$$

$$k = 1: \quad \mathrm{Sym}^1 = \mathcal{H}_{\mathsf{v}} \cong \mathbb{C}^2,$$

$$j = \tfrac{1}{2},$$

$$\text{weights } \{+\tfrac{1}{2},\ -\tfrac{1}{2}\}$$

$$[\text{EW.32};\ \text{Census FT.28}(2);\ \text{Census FT.63}(1);\ \text{machine}];$$

$$\text{the Casimir ladder:}$$

$$\frac{k}{2}\Big(\frac{k}{2}+1\Big)$$

$$[\text{FT.60}(3)];$$

the multiplicity–winding coupling itself stays a declared interface [Def-H.38] — ¬derived here,

¬reopened;

(2) the ray-resolved line census (derived given A-BM.7):

every materialized matter line occupies the left ray: occupancy $(1, 0)$

[BM.73(3)] :

the image-orientation lines — the CPT partners and the A-BM.5 antineutrino row — occupy the right ray

[BM.73(2)(b); BM.25; BM.27; machine: the $\hat{\omega}$-exchange row];

the cardinality correspondence stays a cardinality row:

the two weights $\{\pm\frac{1}{2}\}$ and the two rays $\{+1, -1\}$ are an unordered-pair match;

the weight $\leftrightarrow$ ray attachment is barred

[FT.123(3); the external helicity vocabulary];

(3) the composite rows:

the baryon card: $\Theta_{\text{int}} = 2\,k(\Gamma) \in 2\mathbb{Z}$,

$j(\Gamma) = |k(\Gamma)|$ rung-relative — the card fixes no rung

[Census FT.35(3); Census FT.65(4) : the free $\mathbb{Z}^3$ ladder];

the Platonic floors at the uniform minima: every face j_f composes to an integer stretched total,

$$\sum_f k_{f,\text{min}}/2$$

$$= K_{S,\text{min}}/2 \in \{6,\ 6,\ 12,\ 30,\ 30\}$$

[Census FT.63(1)(b); machine re-cited];

the defect multiplet ladder:

$$m^2_{\text{pole}}(m) = 4\,\hat{\mu}'^2 \sin^2\!\big(m\,\tfrac{\Theta_R}{4}\big),$$

$$m \in \{k,\, k{-}2,\, \ldots,\, -k\}$$

[FT.60(5); Census FT.63(6)] :

the six states of Def-BM.14 ride it with $j = |k|/2$;

the mesonic δ-sealed rows stand untouched [BM.31–BM.33];

(4) the vertex table (ray-resolved; derived given A-BM.7 and the frozen grammar):

member	the structural row	the spin/rung row	the ray row
the parent	the neutral-class ε-sealed 3-juncture, $Q = 0$ [BM.17; Def-BM.3(2)]	$j = \lvert k(\Gamma)\rvert$, rung-relative [Census FT.35(3)]	the constituents occupy the left ray; the neutral-core constituents are ray-locked: $F_{LR} = 0$, parity doubling [BM.22(4); FT.130(6)]
the daughter	the unit-class ε-sealed 3-juncture, $Q = +1$ [BM.26; Def-BM.3(3)]	the same ladder	the charged-core constituents are flip-coupled, splitting $2\lvert q\rvert = 2$ [FT.130(5); FT.144(6)]
the charged emitted member	the ℓ-channel line, $q = -1$: forced, class-unique [BM.23(5); BM.41(1); Def-BM.6(1)]	the $k = 1$ anchor: $j = \frac{1}{2}$ [Census FT.28(2); Def-H.38]	flip-coupled ($q \neq 0$); the arrow orientation occupies the left ray with the partner accessible [BM.73(3); FT.129(5)]
the neutral emitted member	the ν-channel line in the image orientation, $q = 0$ [BM.41(3); FT.142(1); A-BM.5]	the $k = 1$ anchor: $j = \frac{1}{2}$	the right ray (the image orientation under A-BM.7); flip-blind, parity doubling [BM.22(4); FT.130(6)]

the two emitted members occupy opposite rays

[machine: the table's ray column is a derived consequence of BM.73(2)(b), (3) and the frozen grammar];

(5) the angular-momentum budget is unchanged:

BM.42's rows stand — the Noether moment conserved on-shell,

the Lagrange decomposition p^2

$$= p_r^2 + L^2/r^2,$$

the absorption adjudications,

the ℓ-selection census

[BM.42(1)–(4); BM.40(3) : the σ-vertex is Casimir-neutral] :

the ray label adds no torque term:

$\hat{\chi}$ acts on the orientation factor alone and commutes with the rotation generators acting on the seed

[FT.123(1) pattern; FT.126(4) : independent $\mathbb{Z}_2$'s; machine: the commutant row];

(6) what is not obtained:

¬a spin–statistics theorem — no exchange operator exists [FT.80(5); FT.125(5)];

¬a fixed j for the baryon card — the rung datum is not fixed here [BM.68(6) regime row; Census FT.65(4)];

¬a weight $\leftrightarrow$ ray helicity row [FT.123(3), (4)];

¬a $k \geq 3$ carrier-inventory extension [FT.60(8)];

(7) status:

derived given A-BM.7: (2)–(5);

inherited declarations: Def-H.38's multiplicity coupling and the naming rows [Def-BM.3; Def-BM.6; A-BM.5];

¬a P4 addition, ¬a measured value [P5], ¬a new dimensionful scale [P1];

§Cal-BM stays empty.

Req. D1.4 (Parts 1–2); EW.32 (Part 2); P1, P5 (Parts 1–2); Def-H.38, FT.55(2), FT.60(1), FT.60(3), FT.60(5), FT.60(8), FT.80(5), FT.122(4), FT.123(1), FT.123(3), FT.123(4), FT.125(5), FT.126(4), FT.129(5), FT.130(5), FT.130(6), FT.142(1) (Part 3); Census FT.28(2), FT.35(3), FT.63(1), FT.63(6), FT.65(4) (row-wise); Def-BM.3(2), Def-BM.3(3), Def-BM.6(1), BM.17, BM.22(4), BM.23(5), BM.25, BM.26, BM.27, BM.31, BM.32, BM.33, BM.40(3), BM.41(1), BM.41(3), BM.42(1), BM.42(2), BM.42(3), BM.42(4), BM.68(6), BM.73(2), BM.73(3) (this part); exact $\mathbb{Q}(i)$ commutant rows, integer ladder arithmetic [math; machine: `checks/verify_H_chiral_assignment_opening.py` §B, §C].

Pf.

$$(1):\ \text{Def-H.38/FT.60(1)–(3) re-cited (frozen);}$$

$$\dim \mathrm{Sym}^1 = 2,\ \ j = \tfrac{1}{2},\ \ \text{weights } \pm\tfrac{1}{2}\ [\text{machine: the } \tfrac{\sigma_3}{2} \text{ spectrum}];$$

$$\text{the Casimir ladder re-cited [FT.60(3)]}$$

$$(2):\ \text{BM.73(3)'s occupancy with (2)(b)'s image row;}$$

$$\hat{\omega} \text{ maps the occupied ray onto the complement [FT.122(4); machine]}$$

$$(3):\ \text{Census FT.35(3), FT.63(1)(b), FT.65(4), FT.60(5) re-cited row-wise;}$$

$$\Theta_{\text{int}} = 2k \implies j = |k|\ [\text{machine: integers}]$$

$$(4):\ \text{each row one frozen citation plus BM.73(2)–(3);}$$

$$\text{the opposite-ray conclusion: the charged member's arrow orientation } \mapsto \text{ the left ray;}$$

$$\text{the neutral member's image orientation } \mapsto \text{ the right ray [machine: the 2 row census]}$$

$$(5):\ \text{BM.42(1)–(4), BM.40(3) re-cited (frozen);}$$

$$[\sigma_i \otimes I_\omega,\ I \otimes \hat{\chi}] = 0\ [\text{machine: tensor disjointness}]$$

$$(6)\text{–}(7):\ \text{the bars re-cited;} \qquad \text{the standing-rule scans [machine]. } \blacksquare$$

Phys. Spin in this corpus was always a winding label — j equals half the winding, with the seed qubit as the k-equals-one anchor — and chirality was always an orientation grade; the assignment now says which ray the matter sits in, and the census resolves by ray without moving a single spin

value. Every materialized line occupies the left ray; every image-orientation partner, including the antineutrino row the decay vertex already named, occupies the right; and the two emitted leptons of the beta vertex therefore sit in opposite rays, one flip-coupled because its core is charged, one flip-blind and parity-doubled because its core is neutral. The angular-momentum budget absorbs the new label silently, because the chirality grade acts on the orientation factor and commutes with every rotation generator on the seed. What the tier refuses is as exact as what it grants: no weight is glued to a ray — that glue is called helicity and helicity is external vocabulary — no exchange operator is invented, so spin-statistics stays open, and the baryon card keeps its rung-relative spin, because the winding datum the card would need is the same regime datum the joint-entropy tier named as missing.

Theorem 4.64 (BM.75 — The weight–branch–core attachment: the unique bijection and the attached quark mass splitting). Thm.

$$(1) \quad \text{the derived twos and the declared names:}$$

$$\text{the weight alphabet:} \quad \operatorname{spec} Q|_{\mathcal{H}_v} = \{0, +1\},$$

$$\omega : \ \{0, +1\} \mapsto \{0, -1\}$$

$$[\text{EW.36};\ \text{Def-BM.14}(7)] :$$

$$\text{the branch index:} \quad \beta \in \{-, +\},$$

$$D_S(\mathcal{F}_g, +) > D_S(\mathcal{F}_g, -) \ \ \forall g \in \{1, 2, 3\}$$

$$[\text{Def-BM.14}(2), (4);\ \text{BM.60}(5) :\ \text{uniform, exact}];$$

$$\text{the name rows (declared, frozen):}$$

$$\text{down-type} := \text{the heavier branch } (\beta{=}+); \quad \text{up-type} := \text{the lighter } (\beta{=}-)$$

$$[\text{Def-BM.14}(8);\ \text{A-BM.6}];$$

$$\text{the class labels } \{\text{neutral class},\ \text{unit class}\}$$

$$[\text{Def-BM.3}(2), (3);\ \text{A-BM.1}];$$

$$(2) \quad \text{the ladder direction (derived):}$$

$$E_\pm = \tfrac{1}{2}(\sigma_1 \pm i\sigma_2) = \text{the matrix units in the } \sigma_3 \text{ eigenbasis};$$

$$[\sigma_3, E_\pm] = \pm 2 E_\pm$$

$$[\text{BM.23}(1);\ \text{machine over } \mathbb{Q}(i)]:$$

$$E_+ :\ \text{weight } 0 \ \longmapsto\ \text{weight } +1$$

$$[\text{machine: } E_+(0,1)^{\mathsf{T}} = (1,0)^{\mathsf{T}}]:$$

the unique raising step inside $\text{spec}\{0, +1\}$;

the vertex grammar:

the β^- vertex maps the neutral class to the unit class with the forced $q = -1$ carrier;

$$0 = (+1) + (-1) + 0$$

$$[\text{BM.23}(2)\text{–}(5);\ \text{BM.41}(1);\ \text{machine: integers}];$$

the substitution row (the hypothesis on its face):

hypothesis (H): the two classes differ by exactly one down $\leftrightarrow$ up constituent substitution, the minimal $\Delta q = +1$ step

$$[\text{BM.59}(9);\ \text{BM.23}(4)];$$

(3) the attachment theorem (uniqueness):

seek a bijection $\quad f :\ \{\text{weight } 0,\ \text{weight } +1\} \ \longrightarrow\ \{\beta{=}+,\ \beta{=}-\}\quad$ consistent with:

(i) the ladder direction: E_+ carries the substitution's source to its target $[(2)]$;

(ii) the name row: down-type $= \beta{=}+\ [(1)]$;

(iii) the transport order: the neutral-core constituent is the heavier one,

$$m_q(0) = 2 > 1 = m_q(\pm 1)$$

[BM.59(5); Def-BM.13(1)];

(iv) the two-currency sign coherence of the same substitution:

$$\Delta m_{\text{transport}} = +\frac{\ln(4/3)}{\mathfrak{h}} > 0$$

[BM.59(4)]

and

$$\Delta_g = \hat{\mu}'^2\big(D_S(+) - D_S(-)\big) > 0$$

[BM.59(9); Def-BM.14(5)] :

the substitution's source is the heavier member in both currencies;

exactly one bijection survives (i)–(iv) : weight $0 \longleftrightarrow q_{\text{core}} = 0 \longleftrightarrow$ down-type $\longleftrightarrow \beta{=}+$ (heavier); weight $+1 \longleftrightarrow q_{\text{core}} = +1 \longleftrightarrow$ up-type $\longleftrightarrow \beta{=}-$ (lighter)

[machine: exhaustive over the 2! bijections at the frozen ladder direction];

the A-BM.6 bar is discharged by derivation:

the attachment of the Q-weight index to the branch index — recorded there as ¬declared — is derived from the declared name rows plus (2)–(iv);

¬a new declaration issues here;

(4) the attached quark mass splitting (exact per currency):

the defect/pole currency: $\Delta_g = m^2_{\text{down}} - m^2_{\text{up}} = \hat{\mu}'^2\big(D_S(\mathcal{F}_g, +) - D_S(\mathcal{F}_g, -)\big)$:

$$\Delta_1 = 2\,\hat{\mu}'^2 \in \mathbb{Q}\,\hat{\mu}'^2;$$

$$\Delta_2 = (\sqrt{2}-1)\,\hat{\mu}'^2 \in \mathbb{Q}(\sqrt{2})\,\hat{\mu}'^2;$$

$$\Delta_3 = \big(2\cos\tfrac{\pi}{10} - \sqrt{3}\big)\,\hat{\mu}'^2 \in \mathbb{Q}\big(\sqrt{3}, \cos\tfrac{\pi}{10}\big)\,\hat{\mu}'^2;$$

$$\Delta_g > 0 \;\; \forall g$$

[Def-BM.14(5); machine: the certificates $4>2,\ 2>1,\ 3<5$];

the informational/transport-register currency:

per constituent $\dfrac{\ln 2}{\mathfrak{h}} > 0$;

per composite $\dfrac{\ln(4/3)}{\mathfrak{h}} > 0$

[BM.59(4), (5)];

the ensemble route-(ii) currency:

$$\frac{T(0,S)}{T(+1,S)} = \frac{4}{3} > 1 :$$

the neutral class is heavier

[BM.28(4)(ii); BM.30(2)];

the opposing row:

EM makes the charged member heavier by $c\,\varepsilon_{\mathrm{EM}} q^2 m_{\mathrm{hid}} > 0$

[BM.46(4)] :

the unique sign-opposite correction, dominated on the sub-critical branch $\varepsilon_{\mathrm{EM}} < \varepsilon^{\mathrm{crit}}(c)$

[BM.62(6), (9); BM.65(3)–(5)];

(5) the sign-coherence theorem:

in every derived currency in which the doublet splitting is formed, the down-type constituent is the heavier one

[machine: the sign census of (4)'s four rows — three concordant positives and one opposed correction with its exact domination branch];

the currencies remain incommensurate:

$\neg\exists$ a derived ratio between Δ_g and $\ln(4/3)/\hbar$ — the relating equation stays open

[BM.35(5); BM.68(7); BM.69(4)];

(6) what remains open:

the full flavour composition — which three of the six states a class occupies — stays ¬derived: ¬a fractional constituent alphabet,

the end signs carry no position datum

[BM.28(7); Def-BM.8(5); Def-BM.13(9)] :

the attachment theorem consumes only the single-substitution row, ¬the full triple;

the generation map is an order, ¬a number [Def-BM.14(6); BM.60(7)];

the mixing matrix is ¬derived [FT.145(3)];

(7) status: derived: (1)–(5) [(3) conditional on hypothesis (H), on its face [A §0.2a]];

¬a new declaration,

¬a P4 addition,

¬a measured value [P5],

$\neg$a Yukawa-type object [EW.15; FT.145(3)];

§Cal-BM stays empty.

Req. D3.4 (Parts 1–2); EW.15, EW.36 (Part 2); P5 (Parts 1–2); Def-H.37(2), FT.60(5), FT.145(3) (Part 3); Def-BM.3(2), Def-BM.3(3), Def-BM.8(5), Def-BM.13(1), Def-BM.13(9), Def-BM.14(2), Def-BM.14(4), Def-BM.14(5), Def-BM.14(6), Def-BM.14(7), Def-BM.14(8), BM.23(1), BM.23(2), BM.23(4), BM.23(5), BM.28(4), BM.28(7), BM.30(2), BM.35(5), BM.41(1), BM.46(4), BM.59(4), BM.59(5), BM.59(9), BM.60(5), BM.60(7), BM.62(6), BM.62(9), BM.65(3), BM.65(5), BM.68(7), BM.69(4) (this part); exact $\mathbb{Q}(i)$ matrix-unit algebra, exhaustive bijection censuses, exact radical certificates, prime-log-ring sign tests [math; machine: `checks/verify_H_chiral_assignment_opening.py` §B, §D].

Pf.

$(1):$ EW.36, Def-BM.14$(2),(4),(7),(8)$, Def-BM.3$(2),(3)$ re-cited (frozen)

$(2):$ $E_+ = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ in the weight basis with $Q = \mathrm{diag}(1,0)$: $\quad E_+(0,1)^{\mathsf{T}} = (1,0)^{\mathsf{T}}$ [machine];

$$[\sigma_3, E_+] = 2E_+ \text{ [machine over } \mathbb{Q}(i)\text{]};$$

$$0 = (+1) + (-1) + 0 \text{ [machine: integers]}$$

$(3):$ the census: bijection $f_A: \ 0 \mapsto +, \ +1 \mapsto -$ and $f_B: \ 0 \mapsto -, \ +1 \mapsto +$;

f_A satisfies (i)–(iv) : the E_+-source is weight 0

$$= \text{down-type}$$

$$= \beta{=}+$$

$$= \text{the heavier branch}$$

$$= \text{the heavier core [machine: the four condition evaluations]};$$

f_B fails (i) and (iii) $\wedge$ (iv) jointly: it makes the substitution's source (weight 0, core 0, transport-heavier) the lighter branch $\beta{=}-$,

so the same substitution that raises $\Delta m_{\text{transport}}$

$= +\ln(4/3)/\mathfrak{h}$ would lower the defect read $(-\Delta_g)$ against BM.59(9)'s derived $+\Delta_g$ [machine: the sign contradiction];

the ladder direction is frozen by the grammar $[(2)]$;

the two bijections: 2 cases, one survivor [machine]

$(4):$ Δ_g re-formed in the three exact fields with the certificates $4 > 2$;

$$2 > 1;$$

$$2\cos\tfrac{\pi}{10} > \sqrt{3} \iff \cos^2\tfrac{\pi}{10} > \tfrac{3}{4} \iff \tfrac{5+\sqrt{5}}{8} > \tfrac{3}{4}$$

$$\iff 5 + \sqrt{5} > 6 \iff \sqrt{5} > 1$$

$\iff 5 > 1$ [machine: exact in $\mathbb{Q}(\sqrt{5})$];

the currency rows re-cited [BM.59$(4), (5)$; BM.28(4)(ii); BM.46(4)]

$(5):$ the sign census of the four rows [machine];

the no-ratio row re-cited [BM.68(7)(d), (e); BM.69(4)]

(6)–$(7):$ the open rows re-cited; the standing-rule scans [machine]. ■

Phys. Two derived twos have always sat side by side — the seed's weight doublet and the defect floor's branch pair — with their attachment recorded as undeclared. The chiral assignment supplies the missing link, and the link turns out to be forced: the ladder raises exactly one weight, the beta vertex raises exactly one core, the heavier branch is named down-type, and the heavier core is the neutral one by an exact transport count — of the two possible gluings only one keeps the sign of the splitting the same in the transport currency and in the defect currency, and both signs are frozen derived rows. So the attachment is a theorem given the substitution hypothesis and the declared names, and the quark mass splitting is now attached rather than floating: per generation it is two, root-two-minus-one, and two-cosines-of-eighteen-degrees-minus-root-three, times the undeclared defect scale, positive in all three, with the informational currencies agreeing in sign and the electromagnetic correction opposing exactly where the sub-critical branch dominates it. What stays open stays open loudly: which three of the six states compose a given baryon is still unreadable, because the attachment consumes one substitution step and not a whole triple, and no ratio between the currencies is derived — the sign agrees, the scale does not, and that is application-layer content.

Theorem 4.65 (BM.76 — The chiral vertex dynamics: the nilpotent step operator, the flip channel, and the rate rows). Thm.

(1) the vertex step operator (an in-theorem abbreviation):

on the extended carrier $\mathcal{H}_{\text{ext}} = \mathcal{H}_{\text{int}} \otimes \mathcal{H}_{\omega} \cong \mathbb{C}^4$

[Def-H.34] :

$$\boxed{\widehat{V} \;:=\; E_+ \otimes P_L}$$

[$E_+ = \frac{1}{2}(\sigma_1 + i\sigma_2)$ [BM.23(1)]; $P_L = \frac{1}{2}(\text{id} + \hat{\chi})$ [Def-H.33]; exact over $\mathbb{Q}(i)$ in the ordered basis of Def-H.34];

(2) the nilpotency is the alphabet overflow's operator form:

$$E_+^2 = 0 \implies \boxed{\widehat{V}^{\,2} = 0}$$

[machine: entrywise over $\mathbb{Q}(i)$] :

the ladder has one step inside the doublet — BM.23(3)'s ternary overflow $\big(\{-1, 0, +1\} + 1 = \{0, +1, +2\}$ leaves the alphabet$\big)$ as an exact operator identity;

$\neg\exists$ an iterated charged step inside one doublet;

(3) the rank, image and kernel rows:

$$\text{rk}\,\widehat{V} = 1 = \text{rk}\,E_+ \cdot \text{rk}\,P_L;$$

$$\text{im}\,\widehat{V} = \text{span}\big\{|\text{weight}\,{+1}\rangle \otimes |L\rangle\big\};$$

$$\dim \ker \widehat{V} = 3$$

[machine: exact rank and image over $\mathbb{Q}(i)$] :

the target is the up-type weight on the occupied ray [BM.75(3) attachment];

$$\widehat{V}\,(I \otimes P_R) = 0,$$

$$(I \otimes P_L)\,\widehat{V} = \widehat{V}$$

[machine] :

the step fires from the left ray alone — the chiral selectivity of the transition, derived given A-BM.7;

(4) the Cartan row:

$$\left[W_3 \otimes I_\omega,\ \widehat{V}\right] = \widehat{V}$$

$$[\,[\sigma_3, E_+] = 2E_+,\ W_3 = \tfrac{1}{2}\sigma_3;\ \text{machine}] :$$

the step operator carries weight $+1$ under the seed Cartan — the operator form of the composite raise Δq

$= +1$ with the compensating emission [BM.23$(2), (4)$];

(5) the ω-covariant image vertex: $(I \otimes \hat{\omega})\,\widehat{V}\,(I \otimes \hat{\omega})^{-1} = E_+ \otimes P_R$ [FT.122(4); machine] :

the image step fires from the right ray — the CPT-partner vertex [BM.25; BM.27; A-BM.5's image-orientation row is ray-resolved];

the flip conjugation:

$$(I \otimes \hat{\kappa})\,\widehat{V}\,(I \otimes \hat{\kappa}) = E_+ \otimes P_R$$

$$[\hat{\kappa} P_L \hat{\kappa} = P_R :\ \{\hat{\chi}, \hat{\kappa}\} = 0\ \text{[Def-H.34]};\ \text{machine}] :$$

the partner involution — the EM flip's carrier [FT.129$(2), (5)$] — moves the vertex to the opposite ray;

(6) the flip-channel rows (frozen, re-cited):

rung one: $F_{LR} = F_{RL} = 0$ — the sectors force-decoupled [FT.128(2); FT.144(6)];

rung two: F_{LR}

$$= q$$

I_2 under the Def-H.43 promotion,

splitting $2|q| \in \{0, 2\}$ [FT.129(5); FT.130(5); BM.22(2), (4)] :

the neutral core keeps parity doubling; the charged core flip-couples both rays

[FT.130(6); machine re-cited];

rung three adds no carrier flip [FT.117; FT.144(6)];

(7) the rate and arrow rows (frozen, re-cited):

$$\Gamma_{\text{tr}}(\varepsilon_q) = \frac{x_+ x_-}{1 - x_+ x_-} \in \left[0, \tfrac{1}{3}\right],$$

$$x_+ x_- = 4\varepsilon_q(1 - 4\varepsilon_q) = \tfrac{1}{4} - 16\Big(\varepsilon_q - \tfrac{1}{8}\Big)^2 \leq \tfrac{1}{4}$$

[BM.44(1), (2); BM.19(2); machine: the SOS parabola] :

$$\varepsilon_q = 0 \implies \Gamma_{\text{tr}} = 0 :$$

no flicker, no decay

[BM.23(8)];

$$\tau_{\text{life}} \geq 3 \text{ ticks}$$

[BM.44(3)];

forward-only, $\neg\exists\, \tau^{-1}$

[A6; BM.23(9); FT.144(2)] :

the vertex fires per tick as a weight, ¬as an external rate

[BM.44(4), (5) : ¬external seconds];

the opening of BM.77 is kinematic; its rate stays the flicker's own;

(8) scope:

what is constructed here is the vertex-level dynamics of the Q read — the ladder step, the chiral selection, the flip channel, the rate;

the field-level dynamics of Q (gauge-field dynamics, radiation, back-reaction) remains ¬constructed

[BM.6(5); Def-BM.12(7); BM.50(3)];

$\widehat{V}$ is ¬a new force: A8's completeness row stands — no sub-block restriction is an A8 force

[A8; BM.52(1)];

$\widehat{V}$ is an in-theorem abbreviation of BM.23(1)'s ladder and Def-H.33's projector;

(9) status:

derived: (1)–(7) [the occupancy rows given A-BM.7];

¬a P4 addition, ¬a measured lifetime [P5], ¬a new dimensionful scale [P1];

the conditionalities (Def-H.43 promotion [BM.22(5)], BM.8(2) stationarity for the chart exponential [BM.44(3)]) are inherited at the point of use [A §0.2a].

Req. A6, A8 (Part 1); EW.32, EW.36 (Part 2); P1, P5 (Parts 1–2); Def-H.33, Def-H.34, Def-H.43, FT.117, FT.122(4), FT.128(2), FT.129(2), FT.129(5), FT.130(5), FT.130(6), FT.144(2), FT.144(6) (Part 3); Def-BM.12(7), BM.6(5), BM.19(2), BM.22(2), BM.22(4), BM.22(5), BM.23(1), BM.23(2), BM.23(4), BM.23(8), BM.23(9), BM.25, BM.27, BM.44(1), BM.44(2), BM.44(3), BM.44(4), BM.44(5), BM.50(3), BM.52(1), BM.73(2), BM.73(3), BM.75(3) (this part); exact $\mathbb{Q}(i)$ Kronecker algebra, rank/image/kernel computation, the SOS parabola completion [math; machine: `checks/verify_H_chiral_assignment_opening.py` §C].

Pf.

(1)–$(2):\ E_+ = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right),\ P_L = \begin{pmatrix} \frac{1}{2} & \frac{i}{2} \\ -\frac{i}{2} & \frac{1}{2} \end{pmatrix}$ in the ordered bases [Def-H.33(5) form; Def-H.34] :

$\widehat{V}$ built as the Kronecker product, all 16 entries exact over $\mathbb{Q}(i)$ [machine];

$$E_+^2 = 0 \implies \widehat{V}^2 = (E_+^2) \otimes (P_L^2) = 0 \text{ [machine: entrywise]}$$

$$(3):\ \mathrm{rk}(A \otimes B) = \mathrm{rk}\, A \cdot \mathrm{rk}\, B \text{ [math]};$$

$$\mathrm{rk}\, E_+ = 1,\ \mathrm{rk}\, P_L = 1 \text{ [machine: exact elimination]};$$

$\mathrm{im}\, \widehat{V}$ spanned by the image of the basis $\{\, |\mathrm{w}0\rangle \otimes |L\rangle \,\} \mapsto |\mathrm{w}{+}1\rangle \otimes |L\rangle$ [machine];

$$\widehat{V}(I \otimes P_R) = E_+ \otimes P_L P_R = 0 \text{ [FT.122(2); machine]}$$

$$(4):\ [\sigma_3, E_+]$$

$$= \sigma_3 E_+ - E_+ \sigma_3$$

$$= 2E_+ \text{ [machine]}$$

$$\implies [W_3 \otimes I,\ E_+ \otimes P_L]$$

$$= [W_3, E_+] \otimes P_L$$

$$= E_+ \otimes P_L \text{ [machine: the } 4 \times 4 \text{ commutator entrywise]}$$

$$(5):\ \hat{\omega} P_L \hat{\omega} = P_R \text{ [FT.122(4); machine]};$$

$$\hat{\kappa} P_L \hat{\kappa} = \tfrac{1}{2}(I + \hat{\kappa}\hat{\chi}\hat{\kappa}) = \tfrac{1}{2}(I - \hat{\chi}) = P_R \ [\{\hat{\chi}, \hat{\kappa}\} = 0;\ \text{Def-H.34; machine}]$$

(6)–(7) : the frozen rows re-cited [FT.128(2); FT.129(5); FT.130(5), (6); FT.117; FT.144(6); BM.44(1)–(5); BM.23(8), (9)];

$$x_+ x_- = \tfrac{1}{4} - 16(\varepsilon_q - \tfrac{1}{8})^2 \text{ [machine: the SOS completion]};$$

$$\Gamma_{\text{tr}}(\tfrac{1}{8}) = \tfrac{1/4}{3/4} = \tfrac{1}{3}\ \text{[machine: exact Fractions]}$$

(8)–(9) : the scope record; the standing-rule scans [machine]. ■

Phys. The charged-current vertex becomes a single exact operator: the weak ladder step tensored with the occupied ray's projector, four by four over the rationals with i adjoined. Its algebra is the physics. Squared, it vanishes — the same nilpotency that makes the ladder's second step leave the ternary alphabet, now an operator identity, so a constituent takes the charged step once and never twice. Its rank is one, its image is the up-type weight on the left ray, and it annihilates anything sitting in the right ray: given the assignment, the transition fires from matter's own ray alone, which is the corpus's exact counterpart of a maximally chiral charged current — derived, not postulated. Conjugating by orientation reversal or by the partner involution moves it to the right ray, giving the image vertex that the antiproton and antineutrino rows occupy, and the Cartan commutator records the charge bookkeeping: the step carries weight plus one, the composite raises by one, and the emitted carrier takes minus one away. The rates were already exact and stay frozen: a weight per tick bounded by a third, three ticks minimum, nothing at the sharp crystallization threshold, and no seconds anywhere — the opening this tier derives elsewhere is kinematic, and the dynamics keeps the flicker's own clock.

Theorem 4.66 (BM.77 — The read specification and the allowed branch: the formed Q-value, the exact trichotomy, and the rescope rows). Thm.

(1) the specification (the declaration, logged as A-BM.8 before use):

the β-vertex Q-value is read:

on the baryon classes — route (ii), the unnormalized total deficit $\quad T(q,S) = W(q)\,\overline{\Delta S}(S)$

[BM.28(4)(ii); BM.30(1) route census];

on the emitted lines — the transport-species family form $\quad m^2(\text{line } i) = \hat{\mu}^2 \ell_c^2\, G\big(c(i)\big),$

$$G(c(i)) = A_0 - B_0 \cos\varphi_i$$

[Def-BM.11(5); BM.39(1)];

the expression is BM.41(4)(d)'s own frozen Q_{ii} — formed there, priced there, adopted here;

the scope is the vertex kinematics alone: route (ii) is ¬elevated to a global mass definition [BM.30(5); BM.37(6) stand];

(2) the formed Q-value:

$$Q_\beta^{\text{spec}} := m_n^{(\text{ii})} - m_p^{(\text{ii})} - m_\ell - m_\nu;$$

$$m_n^{(\text{ii})} = \frac{T(0,S)}{\mathfrak{h}}, \quad m_p^{(\text{ii})} = \frac{T(+1,S)}{\mathfrak{h}}$$

[the route-(ii) mass read; BM.30(1)] :

$$\boxed{Q_\beta^{\text{spec}}\,\mathfrak{h} \;=\; \Delta T(S) \;-\; \rho_\ell\sqrt{G\big(c(0)\big)} \;-\; \rho_\nu\sqrt{G\big(c(i_\nu)\big)}, \qquad \Delta T(S) := T(0,S) - T(+1,S) = \big(W(0) - W(+1)\big)\overline{\Delta S}(S) = 5\,\overline{\Delta S}(S)}$$

[BM.28(4)(ii) and its Pf row $T(q,S) = W(q)\overline{\Delta S}(S)$; the electron row $i = 0$ [Def-BM.6(1)]; machine];

the dimension row:

$$\rho_\ell := \mathfrak{h}\,\hat{\mu}_\ell\,\ell_c^{(\ell)}, \quad \rho_\nu := \mathfrak{h}\,\hat{\mu}_\nu\,\ell_c^{(\nu)} \;\in\; \mathfrak{q}^0$$

[CT2.1; D2.9; Def-H.37(1) : $[\hat{\mu}^2] = \mathfrak{q}^2$; machine: the exponent arithmetic] :

the scales $(\hat{\mu}_\ell, \hat{\mu}_\nu, \ell_c, g, g')$ remain undeclared P4/Cal-F data [FT.143] — ρ_ℓ,

ρ_ν are restatements, ¬new members

[P4; P1; Cal-F logged];

the exact baryon rows:

$$\Delta T(S) = 5\,m_B^2(\cdot, S)$$

[$\overline{\Delta S}(S) = m_B^2(\cdot,S)$: FT.107(3); BM.28(4)(i), (ii); the frozen sector expectations BM.20(4)/BM.21(2) re-cited at BM.62(2)] :

$$\boxed{\Delta T \;=\; \Big(\frac{4219785}{98956},\; \frac{9608310}{393679},\; \frac{455}{23},\; \frac{68565}{3088}\Big) \text{ at } S = (0,\,1,\,2,\,3)}$$

[machine: exact rational arithmetic];

$$\Delta T_{\min} = \frac{455}{23},$$

$$\Delta T_{\max} = \frac{4219785}{98956},$$

$$0 < \Delta T_{\min} \le \Delta T(S) \le \Delta T_{\max}$$

[machine: exact comparisons];

the ν-line index i_ν is unresolved — every row below holds uniformly in $i_\nu \in \{0,1,2\}$ [(4)'s sandwich is line-independent];

(3) the formedness row:

the expression of (2) is BM.41(4)(d)'s frozen Q_{ii} with the lepton scales carried symbolically — ¬a cross-currency assumption is made :

no informational value is equated to a geometric one;

the mass-form bridge is application-layer content, untouched here

[BM.35(5); BM.68(8); BM.69(5); pointer];

the EM and transport rest-slot refinements are ¬summands of this read — disjoint alphabets, no double count

[BM.37(2); BM.62(11); BM.66(6)];

(4) the universal sandwich:

$$A_0 - B_0 = \frac{3}{8}\Big(g - \frac{g'}{3}\Big)^2 + \frac{g'^2}{48} > 0,$$

$$A_0 + B_0 = \frac{3}{8}\Big(g + \frac{g'}{3}\Big)^2 + \frac{g'^2}{48} > 0$$

[BM.39(1); machine: the SOS expansions] :

$$\sqrt{A_0 - B_0} \;\le\; \sqrt{G\big(c(i)\big)} \;\le\; \sqrt{A_0 + B_0} \qquad \forall\,\text{line } i,\ \ \forall\,\text{channel } \big(\delta \in \big\{\tfrac{2}{9}, \tfrac{2}{27}, \tfrac{1}{8}\big\}\big),\ \ \forall\,(g,g') \neq (0,0)$$

$$[\text{BM.39}(1),(3);\ |\cos|\le 1;\ \text{machine}];$$

(5) the trichotomy theorem: for every frozen cell S and every admissible $(g,g')\neq(0,0)$:

(a) the opening region is non-empty:

$$\mathcal{O}_S:=\left\{(\rho_\ell,\rho_\nu)>0:\ \ \rho_\ell\sqrt{G_\ell}+\rho_\nu\sqrt{G_\nu}<\Delta T(S)\right\}\neq\varnothing:$$

the universal symbolic witness:

$$\rho_\ell=\rho_\nu=\frac{\Delta T(S)}{4\sqrt{A_0+B_0}}\ \in\ \mathcal{O}_S$$

[machine: the sandwich gives $\rho_\ell\sqrt{G_\ell}+\rho_\nu\sqrt{G_\nu}\le\frac{\Delta T}{2}<\Delta T$];

the integer-certified witness at the rational point $(g,g',\rho_\ell,\rho_\nu)=(1,1,1,1)$:

$$A_0+B_0=\frac{7}{16}+\frac{1}{4}=\frac{11}{16},$$

$$\sqrt{G}\le\frac{\sqrt{11}}{4}<1\quad[11<16]:$$

$$Q_\beta^{\text{spec}}\mathfrak{h}\ \ge\ \frac{455}{23}-\frac{\sqrt{11}}{2}\ >\ 0$$

$$\Longleftrightarrow$$

$$\frac{910}{23}>\sqrt{11}$$

$$\Longleftrightarrow$$

$$828100>5819$$

[machine: the integer certificate];

(b) the closing region is non-empty:

$$\mathcal{C}_S:=\left\{(\rho_\ell,\rho_\nu)>0:\ \ \rho_\ell\sqrt{G_\ell}+\rho_\nu\sqrt{G_\nu}>\Delta T(S)\right\}\neq\varnothing:$$

the universal symbolic witness:

$$\rho_\ell = \rho_\nu = \frac{\Delta T(S)}{\sqrt{A_0 - B_0}} \ \in \ \mathcal{C}_S$$

[machine: the sandwich gives $\rho_\ell\sqrt{G_\ell} + \rho_\nu\sqrt{G_\nu} \geq 2\,\Delta T > \Delta T$];

the integer-certified witness at $(1, 1, 100, 100)$:

$$A_0 - B_0 = \frac{3}{16},$$

$$\sqrt{G} \geq \frac{\sqrt{3}}{4} :$$

$$Q_\beta^{\text{spec}}\mathfrak{h} \ \leq \ \frac{4219785}{98956} - 50\sqrt{3} \ < \ 43 - 50\sqrt{3} \ < \ 0$$

[machine: $4219785 < 43 \cdot 98956 = 4255108$; $\quad 50\sqrt{3} > 43 \iff 7500 > 1849$];

(c) the critical locus is exact:

$$\partial\mathcal{O}_S : \quad \rho_\ell\sqrt{G\big(c(0)\big)} + \rho_\nu\sqrt{G\big(c(i_\nu)\big)} = \Delta T(S)$$

[machine: the boundary equation];

on the symmetric ray $\rho_\ell = \rho_\nu = \rho$ at the witness couplings $g = g' = 1$:

$$\boxed{\frac{273}{23} \ < \ \rho^{\text{crit}}(S, i_\nu) \ < \ \frac{42197850}{791648}}$$

i.e. $\quad 11 < \rho^{\text{crit}} < 54$

[machine] :

[the certificates: $\sqrt{11} < \frac{10}{3} \iff 99 < 100$; $\quad \sqrt{3} > \frac{8}{5} \iff 75 > 64$; $\quad 273 > 253$; $\quad 42197850 < 54{\cdot}791648 = 42748992$];

the structure mirrors BM.62(6)'s critical strength at the scale-combination locus:

$$\rho < \rho^{\text{crit}} \iff Q_\beta^{\text{spec}} > 0;$$

$$\rho = \rho^{\text{crit}} \iff Q_\beta^{\text{spec}} = 0;$$

$$\rho > \rho^{\text{crit}} \iff Q_\beta^{\text{spec}} < 0$$

[machine: monotone in ρ];

(6) the verdict:

the beta decay is allowed: on the specified read the vertex is kinematically open exactly on $\mathcal{O}_S$ — a non-empty exact region at every frozen winding cell, with integer-certified witnesses

the decay direction is fixed:

the parent is the neutral class — $T(0, S)$

$= \frac{4}{3}T(+1, S) > T(+1, S)$ [BM.28(4)(ii); machine] — and the grammar $n \to p + \ell + \nu$ stands unmodified [BM.41(1); BM.45];

the two-body companion row:

the two-body channel's admissibility on the specified read is the formed condition $\Delta T(S) \geq \rho_\ell \sqrt{G(c(0))}$

[BM.41(2)'s solution re-derived at $m_n \neq m_p$: $E_p - m_p = \dfrac{(m_n - m_p)^2 - m_\ell^2}{2m_n}$; machine: the cleared identity] :

at the witness $(1, 1, 1, 1)$ it holds $\left[\frac{455}{23} > \frac{\sqrt{11}}{4} \iff 1820^2 > 23^2 \cdot 11 \iff 3312400 > 5819\right]$;

$\mathcal{O}_S \subseteq \{\text{two-body admissible}\}$ since $\rho_\nu \sqrt{G_\nu} \geq 0$ [machine];

the forced-neutral-member row is read-relative and is recorded, ¬adjudicated here:

BM.41(3)'s forcing was derived at the equal rest masses of BM.28(1) — it stands on that locus;

on the specified read the neutral member's emission remains admissible and class-unique [BM.23(5), (6); BM.41(3) grammar] and the named reaction carries it [BM.45];

¬∃ a lepton-number law in the inventory [BM.23(6)];

(7) the rescope rows (recorded, ¬retracted):

BM.41(4)(e)'s verdict ("closed on every adopted read") : at its locus no read was adopted [BM.30(5)] — the specification now adopts one, and the verdict is scoped to the pre-specification reads;

BM.49(1),

(2)'s corrected closure stands in its species — the EM summands are ¬parts of the specified read [(3)];

BM.63(2)'s closure stands on the informational reads [BM.70(4)(a)];

BM.70(4)'s dichotomy gains its third case:

(c) the specified route-(ii)+transport read: formed, with a branch-relative sign [(5)];

BM.71(2)'s no-go is rescoped to the pre-specification reads — the fifth route of BM.71(1) is the door walked here, under the logged addendum;

no clause of BM.41–BM.72 is retracted; no clause of Parts 1–3 is modified;

the kinematic viability was always read-relative [BM.41(7)];

(8) the conditional spectrum becomes branch content:

on $\mathcal{O}_S$ the hypothesis $Q \geq 0$ of BM.41(5)/BM.49(3) is satisfied by a formed read :

$$E_\ell^{\max} = \frac{m_n^2 + m_\ell^2 - (m_p + m_\nu)^2}{2m_n}$$

and

$$E_{\max} : \langle E \rangle = 2 : 1 \text{ scale-invariant}$$

[BM.41(5); BM.49(3), (4) re-cited; symbolic in the specified scales] :

¬an external endpoint, spectrum or lifetime is compared [P5];

the quarantined audit rows stay quarantined [Census FT.114, FT.120 pattern];

(9) what the specification does not do:

¬a scale is fixed — $(\rho_\ell, \rho_\nu, g, g', \hat{\mu}, \hat{\mu}', \hat{\mu}_\nu)$ remain undeclared [P4; P5];

¬a currency is converted — the mass-form bridge is application-layer [(3)];

¬a Q-value is numerically derived — the branch structure is the content;

the numerical value of e remains open and does not enter — the specified read carries no EM monomial [EM.25(7); BM.62(12)];

route (ii) is ¬adopted for any purpose beyond the vertex kinematics [BM.30(5); BM.37(3), (6)];

the route-(iii) amplitude ratios stay barred as masses [FT.79(5); BM.37(3)];

(10) status:

derived: (2)–(8) [given (1)];

declared: (1) — logged as the §II.4 addendum A-BM.8 before use;

¬a P4 addition (§Cal-BM stays empty; ρ_ℓ, ρ_ν are Cal-F/P4 restatements),

¬a new dimensionful scale [P1],

¬a measured value [P5; machine: zero external tokens, zero decimal numerals].

Req. CT2.1, D2.9, EM.25(7) (Part 2); P1, P4, P5 (Parts 1–2); Def-H.37(1), FT.107(3), FT.143 (Part 3); Census FT.114, FT.120 (row-wise); Def-BM.6(1), Def-BM.11(5), BM.20(4), BM.21(2), BM.23(5), BM.23(6), BM.28(1), BM.28(4), BM.30(1), BM.30(5), BM.35(5), BM.37(2), BM.37(3),

BM.37(6), BM.39(1), BM.39(3), BM.41(1), BM.41(2), BM.41(3), BM.41(4), BM.41(5), BM.41(7), BM.45, BM.49(1), BM.49(2), BM.49(3), BM.49(4), BM.62(6), BM.62(11), BM.62(12), BM.63(2), BM.66(6), BM.68(8), BM.69(5), BM.70(4), BM.71(1), BM.71(2) (this part); Cal-F; exact rational arithmetic, sum-of-squares positivity, exact radical inequalities by integer certificates, Laurent exponent arithmetic, cleared polynomial identities [math; machine: `checks/verify_H_chiral_assignment_opening.py` §E, §F].

Pf.

$$(1)\text{–}(2):\ \text{BM.28}(4)(\text{ii})\text{'s frozen factorization } T(q,S)=W(q)\overline{\Delta S}(S) \text{ re-cited;}$$

$$\Delta T=(20-15)\overline{\Delta S}=5\,\overline{\Delta S}\ [\text{machine}];$$

$$\overline{\Delta S}(S)=m_B^2(\cdot,S)\ [\text{FT.107}(3):\ \text{the normalized average; BM.28}(4)(\text{i})\ \text{re-cited}]$$

ΔT evaluated at the four frozen sector expectations $\left(\frac{843957}{98956},\ \frac{1921662}{393679},\ \frac{91}{23},\ \frac{13713}{3088}\right)$ [BM.21(2)/BM.62(2) re-cited] :

$$5\cdot 843957=4219785,$$

$$5\cdot 1921662=9608310,$$

$$5\cdot 91=455,$$

$$5\cdot 13713=68565\ [\text{machine: exact integers}];$$

the min/max comparisons by cross-multiplication [machine]

$$(3):\ \text{the locus rows re-cited [BM.41}(4)(\text{d});\ \text{BM.37}(2);\ \text{BM.62}(11)]$$

$$(4):\ \text{the SOS rows of BM.39}(1)\ \text{re-derived by expansion [machine: } \tfrac{3}{8}(g\mp\tfrac{g'}{3})^2+\tfrac{g'^2}{48}=A_0\mp B_0];$$

$$|\cos\varphi_i|\le 1\implies \text{the sandwich [math]};$$

line- and channel-independence [BM.39(3) : (A_0,B_0) shared]

$$(5)(\text{a}):\ \text{the symbolic witness: } 2\cdot\frac{\Delta T}{4\sqrt{A_0+B_0}}\cdot\sqrt{A_0+B_0}$$

$$=\tfrac{\Delta T}{2}<\Delta T\ [\text{machine: exact in the ordered radical extension}];$$

$$\text{the rational witness: } A_0 + B_0\big|_{g=g'=1} = \tfrac{11}{16} \text{ [machine]};$$

$$(\tfrac{\sqrt{11}}{4})^2$$

$$= \tfrac{11}{16};$$

$$\tfrac{455}{23} - \tfrac{\sqrt{11}}{2} > 0$$

$$\iff 910 > 23\sqrt{11}$$

$$\iff 828100 > 529 \cdot 11$$

$$= 5819 \text{ [machine: squaring both positive sides]}$$

$$\text{(b) : the symbolic witness: } 2 \cdot \tfrac{\Delta T}{\sqrt{A_0 - B_0}} \cdot \sqrt{A_0 - B_0} = 2\,\Delta T > \Delta T \text{ [machine]};$$

$$\text{the rational witness: } A_0 - B_0\big|_{g=g'=1} = \tfrac{3}{16};$$

$$200 \cdot \tfrac{\sqrt{3}}{4} = 50\sqrt{3}; \quad \Delta T_{\text{max}} < 43 \iff 4219785 < 4255108;$$

$$50\sqrt{3} > 43 \iff 7500 > 1849 \text{ [machine]}$$

$$\text{(c) : } \rho^{\text{crit}} = \tfrac{\Delta T}{\sqrt{G_\ell} + \sqrt{G_\nu}} :$$

$$\text{lower bound at } \Delta T_{\text{min}},$$

$$\sqrt{G} \le \tfrac{\sqrt{11}}{4} : \quad \rho^{\text{crit}} \ge \tfrac{2 \cdot 455/23}{\sqrt{11}}$$

$$= \tfrac{910}{23\sqrt{11}} > \tfrac{910 \cdot 3}{23 \cdot 10}$$

$$= \tfrac{273}{23} \; [\sqrt{11} < \tfrac{10}{3} \iff 99 < 100];$$

$$\text{upper bound at } \Delta T_{\text{max}},$$

$$\sqrt{G} \ge \tfrac{\sqrt{3}}{4} : \quad \rho^{\text{crit}} \le \tfrac{2 \cdot 4219785/98956}{\sqrt{3}}$$

$$= \tfrac{8439570}{98956\sqrt{3}} < \tfrac{8439570\cdot 5}{98956\cdot 8}$$

$$= \tfrac{42197850}{791648}\ [\sqrt{3} > \tfrac{8}{5} \iff 75 > 64];$$

$$\tfrac{273}{23} > 11 \iff 273 > 253; \quad \tfrac{42197850}{791648} < 54 \iff 42197850 < 42748992 \text{ [machine: integers]};$$

$$\text{monotonicity in } \rho: \text{ the coefficient } \sqrt{G} > 0 \text{ [machine]}$$

$$(6):\ T(0,S) = \tfrac{4}{3}T(+1,S) > T(+1,S) \text{ [machine]};$$

$$\text{the two-body identity: } 2m_n(E_p - m_p)$$

$$= (m_n - m_p)^2 - m_\ell^2 \text{ at } E_p$$

$$= \tfrac{m_n^2 + m_p^2 - m_\ell^2}{2m_n} \text{ [machine: cleared over } \mathbb{Q}[m_n, m_p, m_\ell]];$$

$$\text{the witness } \tfrac{455}{23} > \tfrac{\sqrt{11}}{4} \iff 1820 > 23\sqrt{11} \iff 3312400 > 5819 \text{ [machine]};$$

$$\mathcal{O}_S \subseteq \{\Delta T \geq \rho_\ell \sqrt{G_\ell}\}\ [\rho_\nu \sqrt{G_\nu} \geq 0;\ \text{machine}]$$

$$(7)\text{–}(10):\ \text{the rescope census clause by clause [machine]};$$

$$\text{BM.41}(5),\ \text{BM.49}(3), (4) \text{ re-cited under the satisfied hypothesis};$$

$$\text{the standing-rule scans [machine]. } \blacksquare$$

Phys. The corpus always knew exactly one door out of its own beta-decay closure: a read specification, an act it priced repeatedly and performed never. This tier performs it, narrowly and on the record — the vertex Q-value is read on the unnormalized total-deficit route for the baryons, where the neutral class outweighs the charged by the exact factor four thirds, and on the transport family form for the emitted lines, whose masses carry undeclared scales between exact sum-of-squares bounds. The resulting Q-value is the frozen expression the decay tier itself formed, and its sign is decided by a single dimensionless scale combination against an exact rational: five times the sector expectation, which is forty-two and change nats at the first frozen cell and never below nineteen. The trichotomy is total and machine-certified — an opening region with a universal symbolic witness and a rational witness certified by eight-two-eight-one-hundred against five thousand eight hundred nineteen, a closing region with its own witnesses, and an exact critical locus enclosed between eleven and fifty-four at the rational coupling point. The beta decay is allowed: open on that region, closed on its complement, degenerate on the boundary, with the parent still the neutral

baryon and the grammar untouched. Nothing is fitted and nothing is converted: the scales stay undeclared, the two currencies stay unequated, the informational closures stand in their own species, and every rescoped verdict is recorded rather than erased — the observed world's endpoint may one day be compared in a quarantined row, after a calibration this tier does not perform.

BM.78 — The status of the chiral assignment: adjudication and scope. Structural-constraints record: see Appendix B, Section B.1.25 of the Addendum.

4.19 Tier BM.17 — The Mass Ratio Census: the Platonic Winding Ladders, the Intra- and Inter-Generational Ratios, the Six-State Bridge Census, the Four-Currency Incommensurability, and the Neutron–Proton Ratio

Locus. This tier opens no gap and closes none: it takes the inventory of every mass ratio the corpus derives, in each currency separately, and localizes the whole obstruction to one step. It consumes 6 states [Def-BM.14], two hierarchies [BM.60(6)], one attached splitting [BM.75(4)], one bridge identity [BM.69(2)], one exact critical strength [BM.62(6)], one specified read [BM.77], and the Platonic ladder cells [Census FT.63(4), (6); Census FT.76 row-wise]. Every clause re-derives its own arithmetic in an exact field; ¬a clause equates two currencies; ¬a clause imports an observed mass, ratio or coupling [P5]. The tier adds no definition: every new symbol is an in-theorem abbreviation of derived objects.

Standing rows. α_B stays barred on BM.61(8)'s four grounds and §Cal-BM stays empty. The relating equation is ¬assumed anywhere — item 7 stays open [BM.35(5); BM.68(7); BM.69(4), (5); pointer], and this tier's contribution to it is a localization, ¬a value. BM.41(4)(e), BM.49, BM.63, BM.70(4), BM.71(2) stand on the reads formed at their loci, and BM.77(5), (6)'s specified-read trichotomy stands beside them, ¬retracted and ¬converted. Local notation, fixed for the whole tier: $\mathsf{c} := \cos\frac{\pi}{10}$ with $16\mathsf{c}^4 - 20\mathsf{c}^2 + 5 = 0$ — the symbol c keeps its Def-BM.12(4) meaning (the priced EM distribution factor, $c \in \{\frac{1}{2}, \frac{3}{5}\}$) at every occurrence; χ is not used; $z_i^{(\mathrm{ch})} := (r_i^{(\mathrm{ch})})^2$ and $\gamma_{\mathrm{ch}} := \sqrt{2}\cos(3\delta_{\mathrm{ch}})$.

4.67 (BM.79 — The Platonic winding census: the ladder law, the Chebyshev tower, and the intra-generational mass ratios).

Thm.

(1) the ladder law and the defect angle:

$$\delta_v = 2\pi - d\,\frac{(L-2)\pi}{L},$$

$$\frac{\delta_v}{\pi} = \Big(1,\ \frac{1}{2},\ \frac{2}{3},\ \frac{1}{5},\ \frac{1}{3}\Big) \ \text{ at } \ (d,L) = (3,3),(3,4),(4,3),(3,5),(5,3)$$

[Census FT.42(5); BM.60(3); machine] :

$$\boxed{V\ \delta_v = 4\pi \ \text{ on all five solids}}$$

$$4\cdot\pi = 8\cdot\tfrac{\pi}{2} = 6\cdot\tfrac{2\pi}{3} = 20\cdot\tfrac{\pi}{5} = 12\cdot\tfrac{\pi}{3}$$

[Descartes/Gauss–Bonnet; FT.47(1); machine: exact in π-units];

$$\boxed{D(m) \;:=\; 4\sin^2\!\Big(\frac{m\,\delta_v}{4}\Big) \;=\; 2-2\cos\!\Big(\frac{m\,\delta_v}{2}\Big) \;=\; 2-2\,T_{|m|}\Big(\cos\frac{\delta_v}{2}\Big), \qquad m^2_{\text{pole}}(m) = \hat{\mu}'^2\,D(m)}$$

[Def-H.37(2); FT.54(4); FT.60(5) : the weight ladder at $\Theta_R = \delta_v$; machine: the half-angle and Chebyshev identities in each exact field];

$$\cos\frac{\delta_v}{2} \;=\; \Big(0,\ \frac{\sqrt{2}}{2},\ \frac{1}{2},\ \mathsf{c},\ \frac{\sqrt{3}}{2}\Big)$$

[machine];

the two identically massless readings:

$$\Theta_R = 0 :\ D \equiv 0;$$

$$\Theta_R = V\delta_v = 4\pi :\ D(m) = 2-2\cos(2m\pi) = 0\ \ \forall m$$

[Census FT.63(4), (6); Census FT.76(1) row-wise; machine] :

the flat-face side and the global sweep carry no defect mass at any weight;

(2) the winding conventions (both priced, neither adopted):

$$\kappa_{\min} = 2 \text{ (odd } L),\ 1 \text{ (even } L),$$

$$k = \frac{L\,\kappa_{\min}}{2}$$

[FT.32(6); FT.45; machine] :

$$k_f = \frac{K_S}{F} = (3,\ 2,\ 3,\ 5,\ 3) \ \text{ at } \ K_S = (12,12,24,60,60),\ F = (4,6,8,12,20)$$

[Census FT.43(1), FT.63(1) row-wise; machine];

$$k_{vf} = (3,\ 3,\ 2,\ 3,\ 5) \ \text{ on the vertex-figure } L_{vf} = (3,3,4,3,5)$$

$$[\text{Census FT.76}(1) \text{ row-wise; machine}];$$

$$\text{variant (A)} := (\delta_v, k_f);$$

$$\text{variant (B)} := (\delta_v, k_{vf})$$

$$[\text{Census FT.76}(1) : \text{ both priced; the census below reports both}];$$

(3) the variant-(A) multiplets (the census rows): $m \in \{k_f, k_f{-}2, \ldots, -k_f\}$:

solid	δ_v/π	k_f	the $D(m)$ multiset	exact field
tetrahedron	1	3	$\{2,\ 2,\ 2,\ 2\}$ at $m = (3,1,-1,-3)$	$\mathbb{Q}$
cube	$\frac{1}{2}$	2	$\{2,\ 0,\ 2\}$ at $m = (2,0,-2)$	$\mathbb{Q}$
octahedron	$\frac{2}{3}$	3	$\{4,\ 1,\ 1,\ 4\}$ at $m = (3,1,-1,-3)$	$\mathbb{Q}$
dodecahedron	$\frac{1}{5}$	5	$\{2,\ 2{-}2\cos\frac{3\pi}{10},\ 2{-}2\mathsf{c},\ 2{-}2\mathsf{c},\ 2{-}2\cos\frac{3\pi}{10},\ 2\}$	$\mathbb{Q}[\mathsf{c}]/(16\mathsf{c}^4{-}20\mathsf{c}^2{+}5)$
icosahedron	$\frac{1}{3}$	3	$\{2,\ 2{-}\sqrt{3},\ 2{-}\sqrt{3},\ 2\}$	$\mathbb{Q}(\sqrt{3})$

[machine: every entry derived from (1)'s ladder law in its own exact field; cross-checked row-wise against Census FT.76(2)(A)];

the variant-(B) row (recorded, distinct):

$$\text{tetra } \{2,2,2,2\}; \quad \text{cube } \{2{+}\sqrt{2},\ 2{-}\sqrt{2},\ 2{-}\sqrt{2},\ 2{+}\sqrt{2}\}; \quad \text{octa } \{3,\ 0,\ 3\};$$

$$\text{dodeca } \{2{-}2\cos\tfrac{3\pi}{10},\ 2{-}2\mathsf{c},\ 2{-}2\mathsf{c},\ 2{-}2\cos\tfrac{3\pi}{10}\}\ (k_{vf}{=}3);$$

$$\text{icosa } \{2{+}\sqrt{3},\ 2,\ 2{-}\sqrt{3},\ 2{-}\sqrt{3},\ 2,\ 2{+}\sqrt{3}\}\ (k_{vf}{=}5)$$

[machine: every entry derived from (1)'s ladder law; cross-checked row-wise against Census FT.76(2)(B)];

(4) the Chebyshev tower (the dodecahedral row, exact):

$$T_0 = 1, \quad T_1 = \mathsf{c}, \quad T_{m+1} = 2\mathsf{c}\,T_m - T_{m-1}$$

[integer coefficients; machine] :

m	$T_m(\mathsf{c})$ reduced	$D(m) = 2 - 2T_m(\mathsf{c})$	minimal polynomial of $D(m)$
0	1	0	x
1	c	$2 - 2\mathsf{c}$	$x^4 - 8x^3 + 19x^2 - 12x + 1$
2	$2\mathsf{c}^2 - 1$	$4 - 4\mathsf{c}^2 = \frac{3-\sqrt{5}}{2}$	$x^2 - 3x + 1$
3	$4\mathsf{c}^3 - 3\mathsf{c}$	$2 + 6\mathsf{c} - 8\mathsf{c}^3 = 2 - 2\cos\frac{3\pi}{10}$	$x^4 - 8x^3 + 19x^2 - 12x + 1$
4	$2\mathsf{c}^2 - \frac{3}{2}$	$5 - 4\mathsf{c}^2 = \frac{5-\sqrt{5}}{2}$	$x^2 - 5x + 5$
5	0	2	$x - 2$

$$\boxed{T_5(\mathsf{c}) = 16\mathsf{c}^5 - 20\mathsf{c}^3 + 5\mathsf{c} = \mathsf{c}\,(16\mathsf{c}^4 - 20\mathsf{c}^2 + 5) = 0}\ :$$

the defining quartic IS the top weight

[de Moivre $\cos 5t = 16\cos^5 t - 20\cos^3 t + 5\cos t$ at $t = \frac{\pi}{10}$: $\cos\frac{\pi}{2} = 0 \iff D(5) = 4\sin^2\frac{\pi}{4} = 2$; machine: exact in $\mathbb{Z}[\mathsf{c}]/(16\mathsf{c}^4 - 20\mathsf{c}^2 + 5)$];

$$\mathsf{c}^2 = \frac{5 + \sqrt{5}}{8},$$

$\sqrt{5} = 8\mathsf{c}^2 - 5 \in \mathbb{Q}(\mathsf{c})$ is the unique quadratic subfield,

$$[\mathbb{Q}(\mathsf{c}) : \mathbb{Q}] = 4$$

[Census FT.76(9) row-wise; machine];

(5) the ladder structure (all five solids):

$$P_S = \frac{4\pi}{\delta_v} = (4,\ 8,\ 6,\ 20,\ 12),$$

$$\#\{\text{distinct values}\} = \frac{P_S}{2} + 1 = (3,\ 5,\ 4,\ 11,\ 7)$$

[Census FT.76(3) row-wise; machine] :

D is periodic in m with period P_S, even in m, and strictly increasing on $[0, P_S/2]$;

$$\max_m D(m) = 4 \text{ at } m = P_S/2 \ (P_S \text{ even on all five});$$

$\boxed{0 \;\le\; m^2_{\text{pole}}(m) \;\le\; 4\hat{\mu}'^2 \text{ at every weight of every solid}}$ [Census FT.76(3); machine];

the rung law:

$$D(m) \in \{0, 2, 4\} \iff m\,\delta_v \in \pi\mathbb{Z}$$

[machine: all five ladders to $m = P_S$, in the exact quadratic rings, with the dodecahedral quartic avoided];

the top-weight row:

$$k_f\delta_v = (3\pi,\ \pi,\ 2\pi,\ \pi,\ \pi) \implies D(k_f) = (2,\ 2,\ 4,\ 2,\ 2)$$

[Census FT.76(4) row-wise; machine] :

the heaviest weight of four of the five solids is exactly $2\hat{\mu}'^2$,

the octahedron's being $4\hat{\mu}'^2$ — the inter-solid hierarchy lives entirely at the light end;

(6) the two structural obstructions (re-certified):

(a) δ_v

$$= \pi$$

$$\implies D(m)$$

$= 2 - 2\cos\dfrac{m\pi}{2} \in \{0, 2, 4\}$ and the k_f=3 multiplet is fully degenerate at 2 :

the tetrahedron carries no multiplet splitting at all in either variant

[Census FT.76(5)(a); machine];

(b) the even-k zero mode:

$$k_f$$

$$= 2 \text{ on the cube (A) and } k_{vf}$$

$$= 2 \text{ on the octahedron (B) each carry } m$$

$$= 0 \text{ with } D(0)$$

$$= 0 \text{ exactly :}$$

on those two multiplets the literal lightest pole mass is 0 and only lightest-nonzero reads are well defined

$$[\text{FT.60}(5)\text{'s standing zero; Census FT.76}(5)(\text{b}); \text{ machine}];$$

(7) the intra-generational mass ratios (the multiplet read):

$$r_{\text{intra}}(S) \;:=\; \sqrt{\frac{D_{\max}}{D^{\text{nz}}_{\min}}}$$

$$[D^{\text{nz}}_{\min} := \text{the lightest nonzero value of the multiplet, } (6)(\text{b})] :$$

the sine-ratio law: $\quad r_{\text{intra}}(S) \;=\; \dfrac{\sin\!\left(k_f \delta_v/4\right)}{\sin\!\left(\delta_v/4\right)}$ on every solid whose multiplet carries $m = 1$

[machine: the half-angle identity $D(m) = 4\sin^2(m\delta_v/4)$ at the two extreme weights, all five solids, both variants];

solid	(A) r_{intra}	exact form	minimal polynomial	(B) r_{intra}	exact form
tetrahedron	1	1 (full degeneracy)	$x-1$	1	1
cube	1	1 ($+1$ exact zero mode)	$x-1$	$1+\sqrt{2}$	$\sqrt{\frac{2+\sqrt{2}}{2-\sqrt{2}}}$
octahedron	2	$\sqrt{4/1}$	$x-2$	1	1 ($+1$ zero mode)
dodecahedron	$\sqrt{\frac{2}{2-2\text{c}}}$	$\dfrac{\sin(\pi/4)}{\sin(\pi/20)}$	$x^8-24x^6+76x^4-64x^2+16$	$\sqrt{\frac{2-2\cos(3\pi/10)}{2-2\text{c}}}$	$\dfrac{\sin(3\pi/20)}{\sin(\pi/20)}$
icosahedron	$1+\sqrt{3}$	$\sqrt{\frac{2}{2-\sqrt{3}}}$	x^2-2x-2	$2+\sqrt{3}$	$\sqrt{\frac{2+\sqrt{3}}{2-\sqrt{3}}}$

[machine: the five (A) rows and the five (B) rows derived in $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\mathsf{c})$; the minimal polynomials by the multiplication map and by first linear dependence; cross-checked row-wise against Census FT.76(7)];

the nested-radical form of the widest row:

$$r^2_{\text{intra}}(\text{dodeca}) = \frac{2}{2-2\mathsf{c}} = \frac{1}{1-\mathsf{c}} = \frac{\left(4+\sqrt{10+2\sqrt{5}}\,\right)\left(3+\sqrt{5}\,\right)}{2}$$

[machine: exact in $\mathbb{Q}(\sqrt{5}, \sqrt{10+2\sqrt{5}})$];

(8) the exact order of the intra-generational ratios: $\boxed{1 \;=\; 1 \;<\; 2 \;<\; 1+\sqrt{3} \;<\; \sqrt{\dfrac{2}{2-2\mathsf{c}}}}$

certificates: $1<2$; $\quad 2<1+\sqrt{3} \iff 1<\sqrt{3} \iff 1<3$; $\quad (1+\sqrt{3})^2 = 4+2\sqrt{3} < \dfrac{2}{2-2\mathsf{c}}$

$$\iff (4+2\sqrt{3})(2-2\mathsf{c}) < 2 \iff \left(4+\frac{7}{2}\right)\left(2-\frac{19}{10}\right)$$

$$= \frac{3}{4} < 2 \text{ on } \sqrt{3} < \frac{7}{4}\,(48<49) \text{ and } \mathsf{c} > \frac{19}{20}\,\left(5\cdot 2500 > 111^2 :\; 12500 > 12321\right)$$

[machine: integer certificates and exact dyadic radical enclosures];

the widest intra-generational ratio is the dodecahedron's, in both variants;

(9) the ceiling row (distinct from (7)):

$$\sqrt{\frac{4}{D_S}} \;=\; \left(\sqrt{2},\;\; \frac{2}{\sqrt{2-\sqrt{2}}},\;\; 2,\;\; \frac{2}{\sqrt{2-2\mathsf{c}}},\;\; \frac{2}{\sqrt{2-\sqrt{3}}}\right)$$

[Census FT.76(3) row-wise; machine] :

maximal at the dodecahedron,

$$\frac{4}{2-2\mathsf{c}} = 2\, r^2_{\text{intra}}(\text{dodeca}),$$

minimal polynomial $x^4 - 48x^3 + 304x^2 - 512x + 256$;

$$\frac{2}{\sqrt{2-2\mathsf{c}}} = \sqrt{2}\cdot r_{\text{intra}}(\text{dodeca})$$

[machine: the exact identity];

no winding choice and no admissible Θ_R convention exceeds it [Census FT.76(3); the value set is finite];

(10) the range bar, re-certified with the census numbers:

the widest mass ratio the Platonic defect species supplies at any winding is $\sqrt{4/(2-2\mathsf{c})}$:

$$\sqrt{\frac{4}{2-2\mathsf{c}}} \;<\; 7$$

$$\Big[\frac{4}{2-2\mathsf{c}} < 49 \iff 4 < 49(2-2\mathsf{c}) \iff \mathsf{c} < \frac{47}{49} : \mathsf{c}^2 = \frac{5+\sqrt{5}}{8} < \frac{2209}{2401} \iff \sqrt{5} < \frac{3368}{2401}\cdot 8 - 5;\ \text{machine}\Big];$$

$$\text{and } 139 \;<\; \frac{z_0^{(q)}}{z_1^{(q)}}$$

$$[\text{BM.80}(6) :\ \text{the flattest channel hierarchy's outer ratio;\ machine}] \implies \boxed{\sqrt{\frac{4}{2-2\mathsf{c}}} \;<\; 7 \;<\; 139 \;<\; \frac{z_0^{(q)}}{z_1^{(q)}}}$$

the two are dimensionless ratios of different species: their comparison is arithmetic, ¬a bridge;

the widest Platonic mass ratio lies below the flattest inter-generational channel ratio by an exact margin;

deficit 12's two ceilings stand untouched [Census FT.76(3); FT.79(2); FT.84(2) row-wise];

(11) status:

derived: (1), (3)–(10);

re-cited row-wise as Census data: (2)'s two windings and (9)'s ceiling [Census FT.76(1), (3), (7)];

priced and ¬adopted: the variant choice (A)/(B) is ¬a read adoption [BM.30(5) stands];

the census reports both;

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5; machine: zero external tokens, zero decimal numerals in this tier],

§Cal-BM stays empty.

Req. D2.9 (Parts 1–2); P1, P4, P5 (Parts 1–2); Def-H.21(6), Def-H.37(2), FT.32(6), FT.45, FT.47(1), FT.54(4), FT.55(1), FT.60(5), FT.79(2), FT.84(2) (Part 3); Census FT.42(5), FT.43(1), FT.63(1), FT.63(4), FT.63(6), FT.76(1), FT.76(2), FT.76(3), FT.76(4), FT.76(5), FT.76(7), FT.76(9) (row-wise); Def-BM.12(4), Def-BM.14(3), BM.30(5), BM.60(3), BM.61(8) (this part); exact radical rings $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{5})$, the quartic ring $\mathbb{Z}[\mathrm{c}]/(16\mathrm{c}^4-20\mathrm{c}^2+5)$, Chebyshev recursion with integer coefficients, de Moivre $\cos 5t$, minimal polynomials by the multiplication map and by first linear dependence, exact dyadic radical enclosures with integer certificates, half-angle and sine-ratio identities [math; machine: `checks/verify_H_mass_ratio_census.py` §A, §B].

Pf.

$(1):\ \delta_v/\pi = 2 - d(L-2)/L$ at the five (d, L) pairs [machine: exact rationals in π-units];

$V\delta_v$ multiplied out at all five [machine];

$4\sin^2(x) = 2 - 2\cos(2x)$ at $x = m\delta_v/4$ [machine: the exact series identity in each field];

$\cos(m\delta_v/2) = T_m(\cos(\delta_v/2))$ [Chebyshev; machine]

$\Theta_R = 4\pi:\ D(m) = 2 - 2\cos(2m\pi) = 0\ \forall m$ [machine]; $\quad \Theta_R = 0:\ D \equiv 0$ [machine]

$(2):\ k = L\kappa_{\min}/2$ at $L = (3,4,3,5,3) \implies (3,2,3,5,3)$;

$L_{vf} = (3,3,4,3,5) \implies (3,3,2,3,5)$ [machine: exact Fractions];

$K_S/F = (12,12,24,60,60)/(4,6,8,12,20) = (3,2,3,5,3)$ [machine]

$(3):$ the two multiplets evaluated weight by weight from (1) in $\mathbb{Q}$,

$$\mathbb{Q}(\sqrt{2}),$$

$$\mathbb{Q}(\sqrt{3}),$$

$$\mathbb{Q}[\mathsf{c}] \text{ [machine]};$$

$$\cos\tfrac{\pi}{8} = \tfrac{\sqrt{2-\sqrt{2}}}{2} \text{ not needed: } D(2)\big|_{\delta_v=\pi/2} = 2 \text{ [machine]};$$

the (B) row by the identical pipeline at k_{vf} [machine]

$$(4): \; T_{m+1}$$

$$= 2\mathsf{c}T_m - T_{m-1} \text{ iterated to } m$$

$$= 5 \text{ with reduction } 16\mathsf{c}^4$$

$$\mapsto 20\mathsf{c}^2 - 5 \text{ [machine: exact integer-coefficient arithmetic in the quotient ring]};$$

$$T_5 = 16\mathsf{c}^5 - 20\mathsf{c}^3 + 5\mathsf{c} = \mathsf{c}(16\mathsf{c}^4 - 20\mathsf{c}^2 + 5) = 0 \text{ [machine]};$$

$$4\mathsf{c}^2 - 2 = \tfrac{\sqrt{5}-1}{2} \implies D(2) = 4 - 4\mathsf{c}^2 = \tfrac{3-\sqrt{5}}{2},$$

$$D(4)$$

$$= 5 - 4\mathsf{c}^2$$

$$= \tfrac{5-\sqrt{5}}{2} \text{ [machine: } 8\mathsf{c}^2 - 5 = \sqrt{5} \text{ verified by squaring both positive sides in } \mathbb{Q}(\sqrt{5})]$$

the minimal polynomials: $x^4 - 8x^3 + 19x^2 - 12x + 1$ at $D(1), D(3)$;

$$x^2 - 3x + 1 \text{ at } D(2);$$

$$x^2 - 5x + 5 \text{ at } D(4);$$

$$x - 2 \text{ at } D(5);$$

x at $D(0)$ [machine: first linear dependence over $\mathbb{Q}$ on the powers of the element, with exact re-substitution];

$$[\mathbb{Q}(\mathsf{c}) : \mathbb{Q}] = 4,\ \sqrt{5} = 8\mathsf{c}^2 - 5 \text{ [machine]}$$

$$(5):\ P_S = 4/(\delta_v/\pi) = (4, 8, 6, 20, 12) \text{ [machine]};$$

$D(m{+}P_S) = D(m)$ and $D(-m) = D(m)$ [machine: the ladder evaluated over two full periods];

$$\#\text{distinct} = P_S/2 + 1 \text{ [machine: the exact value sets]};$$

$$\max_m D(m) = 4 \text{ at } m = P_S/2 \text{ [machine]}$$

$$\text{the rung law}:\ D(m) \in \{0, 2, 4\} \iff \cos(m\delta_v/2) \in \{1, 0, -1\}$$

$$\iff m\delta_v \in \pi\mathbb{Z} \text{ [machine: both directions on all five ladders]};$$

$$k_f\delta_v = (3\pi, \pi, 2\pi, \pi, \pi) \text{ [machine]} \implies D(k_f) = (2, 2, 4, 2, 2) \text{ [machine]}$$

$$(6):\ \text{(a) } \delta_v = \pi \implies D(m) = 2 - 2\cos(m\pi/2) \in \{0, 2, 4\} \text{ and } D(1) = D(3) = 2 \text{ [machine]};$$

$$\text{(b) } D(0) = 2 - 2\cos 0 = 0 \text{ exactly [machine]};$$

$$k_f(\text{cube}) = 2,\ k_{vf}(\text{octa}) = 2 \text{ are even [machine]}$$

$$(7):\ r_{\text{intra}}$$

$$= \sqrt{D(k_f)/D(1)}$$

$$= \sqrt{4\sin^2(k_f\delta_v/4)/4\sin^2(\delta_v/4)}$$

$$= \sin(k_f\delta_v/4)/\sin(\delta_v/4) \text{ [machine: positive sines on } (0, \pi/2]];$$

$$\text{octa}: \ \sqrt{4/1} = 2 \text{ [machine]};$$

$$\text{icosa}: \ \frac{2}{2-\sqrt{3}} = 2(2+\sqrt{3}) = 4+2\sqrt{3} = (1+\sqrt{3})^2 \text{ [machine: exact in } \mathbb{Q}(\sqrt{3})];$$

$$\text{dodeca}: \ z:$$

$$= \frac{2}{2-2\mathsf{c}} \text{ satisfies } z^4 - 24z^3 + 76z^2 - 64z + 16$$

$$= 0 \text{ [machine: the substitution } D = 2/z \text{ into } D^4 - 8D^3 + 19D^2 - 12D + 1],$$

$$\text{so } r \text{ satisfies } r^8 - 24r^6 + 76r^4 - 64r^2 + 16$$

$$\text{cube (B)}: \ \frac{2+\sqrt{2}}{2-\sqrt{2}} = \frac{(2+\sqrt{2})^2}{2} = 3+2\sqrt{2} = (1+\sqrt{2})^2 \text{ [machine]};$$

$$\text{icosa (B)}: \ \frac{2+\sqrt{3}}{2-\sqrt{3}} = (2+\sqrt{3})^2 = 7+4\sqrt{3} = (2+\sqrt{3})^2 \text{ [machine]}$$

$$(8): \ 48 < 49 \implies \sqrt{3} < \tfrac{7}{4};$$

$$5 \cdot 2500$$

$$= 12500 > 12321$$

$$= 111^2 \implies \sqrt{5} > \tfrac{111}{50} \implies \mathsf{c}^2$$

$$= \tfrac{5+\sqrt{5}}{8} > \tfrac{361}{400} \implies \mathsf{c} > \tfrac{19}{20} \text{ [machine]};$$

$$(4+\tfrac{7}{2})(2-\tfrac{19}{10}) = \tfrac{15}{2} \cdot \tfrac{1}{10} = \tfrac{3}{4} < 2 \text{ [machine]}$$

$$(9): \ w:$$

$$= \tfrac{4}{2-2\mathsf{c}}$$

$$= 2z \text{ and } z^4 - 24z^3 + 76z^2 - 64z + 16$$

$$= 0$$

$$\implies w^4 - 48w^3 + 304w^2 - 512w + 256$$

$= 0$ [machine: the substitution $z = w/2$ cleared, and independently by first linear dependence];

$$\tfrac{2}{\sqrt{2-2\mathsf{c}}} = \sqrt{2}\sqrt{\tfrac{2}{2-2\mathsf{c}}}$$ [machine: squaring both positive sides]

$$(10): \ \tfrac{4}{2-2\mathsf{c}} < 49 \iff 4 < 98 - 98\mathsf{c} \iff 98\mathsf{c} < 94 \iff \mathsf{c} < \tfrac{47}{49}$$ [machine];

$$\mathsf{c}^2$$

$$= \tfrac{5+\sqrt{5}}{8} < \tfrac{2209}{2401}$$

$$\iff 2401(5 + \sqrt{5}) < 17672$$

$$\iff 2401\sqrt{5} < 5667$$

$$\iff 2401^2 \cdot 5 < 5667^2$$ [machine: $28824005 < 32114889$];

$$7 < \tfrac{89}{12} \iff 84 < 89$$ [machine];

$$\tfrac{89}{12} < z_0^{(q)}/z_1^{(q)}$$ [BM.80(4)'s exact enclosure; machine]

$$(11):$$ the standing-rule scans [machine]. ■

Phys. One law generates the whole Platonic mass spectrum: the cone defect at a vertex, read as an angle, squared into a mass by a single undeclared scale. Wind that angle by an integer and the ladder appears — periodic, even, monotone to a hard ceiling of four, with the tetrahedron degenerate at every weight and the even windings carrying an exactly massless middle rung. The dodecahedron is the deep row: its tower of Chebyshev polynomials closes at the fifth rung because the defining quartic of the eighteen-degree cosine is the statement that five times eighteen degrees is a right angle, so the heaviest weight of that solid sits at exactly two, forced by the same relation that makes its lightest weight the smallest positive defect angle in the census. The intra-generational ratios are then pure sines — one sine over another, at rational multiples of pi — and they come out as one, one, two, root three plus one, and a degree-eight number built from nested square roots of five. Their order is exact and their widest member is the dodecahedron's, in both winding conventions. Nothing here is a mass hierarchy in the channel sense: the widest ratio the entire Platonic ladder can supply at any winding is under seven, while the flattest family channel supplies over one hundred thirty-nine, and the two numbers are compared as arithmetic, never equated as physics — the range bar that deficit twelve records is re-certified with the census's own exact values, and the ceiling that bounds it is a theorem about angles, not about the world.

4.68 (BM.80 — The inter-generational mass ratios: one cubic form, three channel constants, the exact enclosures, and the hierarchy-order theorem).

Thm.

(1) the three channel angles and the exact angle ledger:

$$\delta_q = \frac{2}{3\cdot 3\cdot 3} = \frac{2}{27},$$

$$\delta_\ell = \frac{2}{9},$$

$$\delta_\nu = \frac{2}{16} = \frac{1}{8}$$

[FT.119(1); FT.112; FT.142(5); machine: exact Fractions] :

$$\boxed{27\,\delta_q = 9\,\delta_\ell = 16\,\delta_\nu = 2, \qquad 3\delta_q = \delta_\ell, \qquad \delta_\nu : \delta_\ell : \delta_q = 27 : 48 : 16, \qquad \delta_q < \delta_\nu < \delta_\ell}$$

[FT.119(2); FT.143(5); machine];

$$3\delta_\nu = \frac{3}{8} \notin \Big\{\frac{2}{9}, \frac{2}{27}\Big\},$$

$$3\delta_\ell = \frac{2}{3},$$

$$3\delta_q = \frac{2}{9}$$

[FT.143(2), (5); machine];

(2) the spectrum and its cubic (frozen, re-certified):

$$r_i^{(\mathrm{ch})} = 1 + \sqrt{2}\cos\Big(\delta_{\mathrm{ch}} + \frac{2\pi i}{3}\Big),$$

$$i = 0, 1, 2$$

[FT.111(3); FT.113; FT.119(3); FT.143(1); FT.158(4)] :

with $c_{\mathrm{ch}} = \cos\delta_{\mathrm{ch}},\ s_{\mathrm{ch}} = \sin\delta_{\mathrm{ch}}$:

$$r_0 = 1 + \sqrt{2}\, c_{\text{ch}},$$

$$r_1 = 1 - \frac{\sqrt{2}\, c_{\text{ch}} + \sqrt{6}\, s_{\text{ch}}}{2},$$

$$r_2 = 1 - \frac{\sqrt{2}\, c_{\text{ch}} - \sqrt{6}\, s_{\text{ch}}}{2}$$

[machine];

$$\sum_i r_i = 3,$$

$$\sum_{i<j} r_i r_j = \frac{3}{2},$$

$$\sum_i r_i^2 = 6,$$

$$\boxed{2(r-1)^3 - 3(r-1) - \gamma_{\text{ch}} = 0, \qquad \gamma_{\text{ch}} := \sqrt{2}\cos(3\delta_{\text{ch}})}$$

[machine: the depressed cubic at $B^2 = 2,$ and the triple angle $\cos 3\delta = 4\cos^3\delta - 3\cos\delta$];

equivalently, in r alone:

$$\boxed{2r^3 - 6r^2 + 3r + (1 - \gamma_{\text{ch}}) = 0}$$

[machine: the expansion of $2(r-1)^3 - 3(r-1) = \gamma_{\text{ch}}$];

(3) the mass-squared cubic (derived here):

$$z := r^2,$$

$$m_i^{(\text{ch})} = \mu_{\text{ch}}\, z_i^{(\text{ch})},$$

μ_{ch} undeclared [P4; Cal-F] :

$$\boxed{4z^3 - 24z^2 + \big(21 - 12\,\gamma_{\text{ch}}\big)z - \big(\gamma_{\text{ch}} - 1\big)^2 \;=\; 0 \qquad \text{for each } \; z = z_i^{(\text{ch})}}$$

[machine: exact elimination of r in $\mathbb{Q}[\gamma][r]/(2r^3-6r^2+3r+1-\gamma)$: $r(2z+3)=6z+\gamma-1$ at $z=r^2$, and $z(2z+3)^2-(6z+\gamma-1)^2\equiv 0$ in the quotient ring];

Vieta:

$$\boxed{\sum_i z_i = 6 \qquad \text{(phase-free and channel-free)}, \qquad \sum_{i<j} z_i z_j = \frac{21-12\gamma_{\mathrm{ch}}}{4}, \qquad \prod_i z_i = \frac{(\gamma_{\mathrm{ch}}-1)^2}{4}}$$

[machine: the three symmetric reads of the cubic; $\sum_i z_i = 6 = \sum_i r_i^2$ agrees with (2) exactly];

one cubic form, three constants:

the polynomial shape is channel-free;

the entire channel dependence is the single number γ_{ch};

(4) the three constants are pairwise distinct and exactly ordered:

$$\gamma_\ell = \sqrt{2}\cos\frac{2}{3},$$

$$\gamma_\nu = \sqrt{2}\cos\frac{3}{8},$$

$$\gamma_q = \sqrt{2}\cos\frac{2}{9}:$$

$$\boxed{\gamma_\ell \;<\; \gamma_\nu \;<\; \gamma_q \;<\; \sqrt{2}}$$

[machine: disjoint exact enclosures, width $< 2^{-60}$; cos strictly decreasing on $[0,\pi]$ and $\frac{2}{3} > \frac{3}{8} > \frac{2}{9}$ [math]; $\gamma < \sqrt{2} \iff \cos 3\delta < 1 \iff 3\delta \neq 0$];

$$\gamma_\ell \in \Big(\frac{1111412422687001}{10^{15}}, \frac{1111412422687002}{10^{15}}\Big),$$

$$\gamma_\nu \in \Big(\frac{1315936498799931}{10^{15}}, \frac{1315936498799932}{10^{15}}\Big),$$

$$\gamma_q \in \Big(\frac{1379438171633255}{10^{15}}, \frac{1379438171633256}{10^{15}}\Big)$$

[machine: alternating rational Taylor enclosures of cos with exact remainder bounds, width $< 2^{-160}$];

the three-root regime is exact:

$$2y^3 - 3y - \gamma \text{ has three distinct real roots } \iff \gamma^2 < 2$$

[the discriminant $\frac{27}{4}(2-\gamma^2) > 0$; machine] — and all three channels satisfy it by (4)'s last inequality;

(5) the cross-channel exact identity:

$$\boxed{r_0^{(\ell)} \;=\; 1 + \gamma_q}$$

$[3\delta_q = \delta_\ell$ [FT.119(2)]; $r_0^{(\ell)} = 1 + \sqrt{2}\cos\delta_\ell = 1 + \sqrt{2}\cos 3\delta_q = 1 + \gamma_q$; machine: the same cosine, enclosed to width $< 2^{-160}]$:

the ℓ-channel's heaviest line is the q-channel's cubic constant plus one;

the shared object is the angle $3\delta_q = \delta_\ell,$ ¬the cubic constant :

$$\gamma_\ell = \sqrt{2}\cos\tfrac{2}{3} \neq \sqrt{2}\cos\tfrac{2}{9} = \gamma_q$$

[machine: disjoint enclosures];

$\gamma_\nu = \sqrt{2}\cos\frac{3}{8}$ is channel-own

[FT.143(2); machine];

(6) the inter-generational ratio census (the exact enclosures):

$z_1 < z_2 < z_0$ in every channel

[machine: disjoint enclosures, widths $< 2^{-50}$] :

channel	z_2/z_1	z_0/z_1	z_2/z_0	z_1/z_0
q	$(\frac{3563695074946}{10^{12}}, \frac{3563695074947}{10^{12}})$	$(\frac{139338688028416}{10^{12}}, \frac{139338688028417}{10^{12}})$	$(\frac{25575776012}{10^{12}}, \frac{25575776013}{10^{12}})$	$(\frac{7176757684}{10^{12}}, \frac{7176757685}{10^{12}})$
ℓ	$(\frac{206770315972729}{10^{12}}, \frac{206770315972730}{10^{12}})$	$(\frac{3477472837104598}{10^{12}}, \frac{3477472837104599}{10^{12}})$	$(\frac{59459937045}{10^{12}}, \frac{59459937046}{10^{12}})$	$(\frac{287565150}{10^{12}}, \frac{287565151}{10^{12}})$
ν	$(\frac{9583926992379}{10^{12}}, \frac{9583926992380}{10^{12}})$	$(\frac{271994748749143}{10^{12}}, \frac{271994748749144}{10^{12}})$	$(\frac{35235705970}{10^{12}}, \frac{35235705971}{10^{12}})$	$(\frac{3676541567}{10^{12}}, \frac{3676541568}{10^{12}})$

[machine: every bracket a monotone exact interval division at width $< 2^{-150}$, each bracket decided by exact 10^{12}-scaling; the frozen decimal rows of FT.119(5), (6), FT.113, FT.143(3) are re-certified digit for digit against these brackets];

$$m_1 : m_2 : m_3 = 1 : \frac{z_2}{z_1} : \frac{z_0}{z_1}$$

[FT.119(5) vocabulary; the ratios are μ_{ch}-free];

the ratios are ¬algebraic numbers: γ_{ch}

$= \sqrt{2}\cos(3\delta_{\text{ch}})$ at the rational radian arguments $\frac{2}{9}$,

$\frac{3}{8}$,

$\frac{2}{27}$ is a transcendental cosine,

so the exact form of every inter-generational ratio is the closed radical-trigonometric expression of (2),

(3),

and its exact comparison content is the rational bracket of (6)

[machine: the brackets, ¬a decimal assertion];

(7) the phase-free invariants (all three channels, exact):

$$\boxed{Q \;:=\; \frac{\sum_i m_i}{\left(\sum_i \sqrt{m_i}\right)^2} \overset{m_i=\mu z_i}{=\!=} \frac{\sum_i z_i}{\left(\sum_i r_i\right)^2} \;=\; \frac{6}{9} \;=\; \frac{2}{3}}$$

[FT.119(4); FT.143(4); FT.158(4); Def-BM.6(4); machine: independent of δ_{ch} and of μ_{ch}];

$\sum_i z_i = 6 = 3A_0\big|_{A_0=\frac{11}{6}}$ — ¬a calibration: the sum rule is BM.60(8)'s phase-free row;

(8) the hierarchy-order theorem (derived, scale-free):

on $\theta \in (0, \frac{\pi}{12})$:

$r_1(\theta)$

$= 1 + \sqrt{2}\cos(\theta + \frac{2\pi}{3})$ is strictly decreasing and strictly positive,

$r_1(\frac{\pi}{12})$

$= 0$ exactly;

$r_2(\theta)$ is strictly increasing; $r_0(\theta)$ is strictly decreasing with r_0/r_1 strictly increasing

[machine: the derivative signs $-\sqrt{2}\sin(\theta+\frac{2\pi}{3}) < 0,\ +\sqrt{2}\sin(\theta+\frac{\pi}{3}) > 0,\ -\sqrt{2}\sin\theta < 0$; $\frac{d}{d\theta}\ln\frac{r_0}{r_1} = \sqrt{2}\Big(\frac{\sin(\pi/3-\theta)}{r_1} - \frac{\sin\theta}{r_0}\Big) > 0$

on $(0, \frac{\pi}{6})$ since $\sin(\frac{\pi}{3} - \theta) > \sin\theta \iff \tan\theta < \frac{1}{\sqrt{3}} \iff \theta < \frac{\pi}{6}$ and $0 < r_1 < r_0$

[math; machine: the algebraic reductions]];

$$\delta_q < \delta_\nu < \delta_\ell < \frac{\pi}{12}$$

$\Big[\pi^2 > 6\sum_{n\le 10}\frac{1}{n^2} = \frac{1968329}{211680} > \frac{64}{9} \implies \pi > \frac{8}{3} \implies \frac{2}{9} < \frac{\pi}{12}$; machine: $1968329\cdot 9 = 17714961 > 13547520 = 64\cdot 211680\Big]$;

$$\boxed{1 < \frac{z_2^{(q)}}{z_1^{(q)}} < \frac{z_2^{(\nu)}}{z_1^{(\nu)}} < \frac{z_2^{(\ell)}}{z_1^{(\ell)}}, \qquad 1 < \frac{z_0^{(q)}}{z_1^{(q)}} < \frac{z_0^{(\nu)}}{z_1^{(\nu)}} < \frac{z_0^{(\ell)}}{z_1^{(\ell)}}}$$

the q-channel is the flattest family hierarchy, the ℓ-channel the steepest, and the ν-channel lies between them — an exact ordering with no scale and no import

[machine: monotonicity and the disjoint enclosures of (6) concur];

the near-vanishing row:

$$r_1(\theta) = 0 \iff \theta = \tfrac{\pi}{12} \text{ exactly}$$

$$[\cos\tfrac{3\pi}{4} = -\tfrac{1}{\sqrt{2}};\ \text{machine}] \implies \frac{\pi}{12} - \delta_\ell = \frac{3\pi - 8}{36} \text{ is the smallest of the three margins :}$$

the ℓ-channel's inner line is the closest to the zero locus, and its steepness is a consequence of δ_ℓ being the largest angle below $\frac{\pi}{12}$;

(9) the bars:

$\mu_q,\ \mu_\ell,\ \mu_\nu$ are undeclared P4/Cal-F scales and enter no ratio

[P4; Cal-F; §Cal-BM stays empty];

$\neg\exists$ an SM mass, mass ratio or Yukawa value in any clause

[P5; machine: zero external tokens, zero decimal numerals in this tier];

the m^2 vocabulary is the corpus's own inertial read [L3(4), L4];

the channel labels ℓ, q, ν name ledger rows [FT.112 Rmk; FT.143(6)];

amplitude $\neq$ mass [FT.79(5)];

rung $\neq$ scale [D0.8];

(10) status:

derived: (1)–(8);

re-cited: (2)'s spectrum and (7)'s invariant from their own tags, with $(3), (5), (8)$ new;

$\neg$a P4 addition, $\neg$a new dimensionful scale [P1], $\neg$a measured value [P5], $\neg$a declaration,

open: the three channel scales

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Req. D0.8 (Parts 1–2); P1, P4, P5 (Parts 1–2); L3(4), L4 (Parts 1–2); FT.79(5), FT.111(2), FT.111(3), FT.112, FT.113, FT.119(1), FT.119(2), FT.119(3), FT.119(4), FT.119(5), FT.119(6), FT.142(5), FT.143(1), FT.143(2), FT.143(3), FT.143(4), FT.143(5), FT.143(6), FT.158(4) (Part 3); Def-BM.6(4), BM.60(8), BM.61(8) (this part); Cal-F; exact rational arithmetic, exact symbolic elimination in $\mathbb{Q}[\gamma][r]/(2r^3-6r^2+3r+1-\gamma)$, Vieta reads, alternating rational Taylor enclosures of cos and sin with exact remainder bounds, integer Newton certificates for $\sqrt{2}$ and $\sqrt{6}$, monotone exact interval division, the Basel-series lower bound for π^2, derivative-sign reductions in $\mathbb{Q}[\sqrt{2},\sqrt{3}][c,s]/(c^2+s^2-1)$ [math; machine: `checks/verify_H_mass_ratio_census.py` §D].

Pf.

$$(1):\ \delta_q = 2/(3\cdot 3\cdot 3),\ \delta_\ell = 2/(3\cdot 3),\ \delta_\nu = 2/16 \text{ [machine: exact Fractions]};$$

$$27\cdot\tfrac{2}{27} = 9\cdot\tfrac{2}{9} = 16\cdot\tfrac{1}{8} = 2 \text{ [machine]};$$

$$3\cdot\tfrac{2}{27} = \tfrac{2}{9} \text{ [machine]};$$

$$\tfrac{2}{27} < \tfrac{1}{8} \iff 16 < 27, \quad \tfrac{1}{8} < \tfrac{2}{9} \iff 9 < 16 \text{ [machine]};$$

$$\tfrac{2}{27} : \tfrac{1}{8} : \tfrac{2}{9} = 16 : 27 : 48 \text{ [}\times 216;\ \text{machine]}$$

$$(2):\ r_i = A + B\cos(\delta + \tfrac{2\pi i}{3}) \text{ at } A = 1,\ B = \sqrt{2} \text{ [FT.111(3) re-cited]};$$

$$\cos(\delta + \tfrac{2\pi}{3}) = -\tfrac{c}{2} - \tfrac{\sqrt{3}}{2}s, \quad \cos(\delta + \tfrac{4\pi}{3}) = -\tfrac{c}{2} + \tfrac{\sqrt{3}}{2}s \text{ [addition formulas]};$$

$$\sqrt{2}\cdot\tfrac{\sqrt{3}}{2} = \tfrac{\sqrt{6}}{2} \text{ [machine]}$$

$$\sum_i \cos(\delta + \tfrac{2\pi i}{3})$$

$$= 0,$$

$$\sum_i \cos^2(\delta + \tfrac{2\pi i}{3})$$

$$= \tfrac{3}{2} \text{ [FT.111(3) identities]} \implies \sum_i r_i$$

$$= 3,$$

$$\sum_i r_i^2$$

$$= 3 + 2 \cdot \tfrac{3}{2}$$

$$= 6 \text{ [machine]};$$

the depressed cubic $y^3 - \frac{3}{2}y - \frac{\sqrt{2}}{2}\cos 3\delta = 0$ at $y = r - 1,\ \times 2$ [machine: the ring identity]

$$(3):\ 2(r-1)^3 - 3(r-1) = \gamma \iff 2r^3 - 6r^2 + 3r + (1-\gamma) = 0 \text{ [machine: expansion]};$$

$$2r^3 + 3r = 6z - (1-\gamma) \text{ at } z = r^2 \implies r(2z+3) = 6z + \gamma - 1 \text{ [machine]}$$

$$z(2z+3)^2 - (6z + \gamma - 1)^2$$

$= 4z^3 - 24z^2 + (21 - 12\gamma)z - (\gamma - 1)^2$ as an identity in $\mathbb{Q}[\gamma, z]$ [machine: cleared coefficient by coefficient];

the three roots are z_i

$= r_i^2$ because $r_i > 0$ and $2z + 3$

$\neq 0$ on $z > 0$ [machine: disjoint positive enclosures];

Vieta : $-\frac{-24}{4} = 6,\ \frac{21-12\gamma}{4},\ \frac{(\gamma-1)^2}{4}$ [machine]

$$(4):\ \gamma_\ell,$$

$$\gamma_\nu,$$

γ_q enclosed by alternating rational Taylor series at $\frac{2}{3}$,

$$\tfrac{3}{8},$$

$\frac{2}{9}$ with exact remainder bounds, multiplied by the Newton-certified $\sqrt{2}$ [machine: widths $< 2^{-160}$];

the three intervals are pairwise disjoint [machine];

$$\tfrac{2}{3} > \tfrac{3}{8} \iff 16 > 9, \quad \tfrac{3}{8} > \tfrac{2}{9} \iff 27 > 16 \text{ [machine]};$$

the discriminant of $2y^3 - 3y - \gamma$: Δ

$$= \tfrac{27}{4}(2 - \gamma^2) > 0 \iff \gamma^2 < 2 \text{ [machine: the exact cubic discriminant]}$$

$$(5) : \ 3\delta_q$$

$$= \delta_\ell \text{ [FT.119(2) re-cited]} \implies \cos 3\delta_q$$

$$= \cos \delta_\ell \implies \gamma_q$$

$$= \sqrt{2} \cos \delta_\ell$$

$$= r_0^{(\ell)} - 1 \text{ [machine: the two enclosures agree to width } < 2^{-160} \text{ and the symbolic forms are identical]};$$

$$\gamma_\ell \neq \gamma_q \text{ [disjoint enclosures]};$$

$$\cos \tfrac{3}{8} \neq \cos \tfrac{2}{9} \text{ [disjoint enclosures]}$$

$$(6) : \ z_1 < z_2 < z_0 \text{ [machine: disjoint enclosures of the three roots]};$$

the twelve brackets by monotone exact interval division, each numerator and denominator enclosed to width $< 2^{-160}$ [machine];

the frozen decimal rows of Part 3 re-derived to 12 places and compared digit for digit [machine]

$$(7) : \ Q = \tfrac{\sum_i z_i}{(\sum_i r_i)^2} = \tfrac{6}{9} = \tfrac{2}{3} \text{ [machine: exact Fractions]};$$

phase-independence: the identities $\sum_i \cos(\delta + \tfrac{2\pi i}{3})$

$$= 0,$$

$$\sum_i \cos^2$$

$= \frac{3}{2}$ hold at every δ [machine at six rational phases]

$$(8):\ r_1'(\theta)$$

$$= -\sqrt{2}\sin(\theta + \tfrac{2\pi}{3})$$

$= -\sqrt{2}\sin(\frac{\pi}{3} - \theta) < 0$ on $(0, \frac{\pi}{3})$ [machine: the reduction $\sin(\theta + \frac{2\pi}{3}) = \sin(\frac{\pi}{3} - \theta)$];

$r_1(\frac{\pi}{12}) = 1 + \sqrt{2}\cos\frac{3\pi}{4} = 1 - 1 = 0$ [machine: $\cos\frac{3\pi}{4} = -\frac{1}{\sqrt{2}}$]

$r_2'(\theta) = -\sqrt{2}\sin(\theta + \frac{4\pi}{3}) = +\sqrt{2}\sin(\theta + \frac{\pi}{3}) > 0$ [machine];

$r_0'(\theta) = -\sqrt{2}\sin\theta < 0$ on $(0, \pi)$ [machine];

$$(\ln(r_0/r_1))'$$

$= \sqrt{2}\big(\frac{\sin(\pi/3-\theta)}{r_1} - \frac{\sin\theta}{r_0}\big)$ [machine: the exact derivative in the quotient ring]

$\sin(\frac{\pi}{3} - \theta) > \sin\theta \iff \frac{\sqrt{3}}{2}\cos\theta - \frac{1}{2}\sin\theta > \sin\theta \iff \sqrt{3}\cos\theta > 3\sin\theta \iff \tan\theta < \frac{1}{\sqrt{3}} \iff \theta < \frac{\pi}{6}$ [machine: each equivalence as an exact identity; tan strictly increasing on $(0, \frac{\pi}{2})$, $\tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$ [math]];

$r_1 < r_0 \iff \cos(\theta + \frac{2\pi}{3}) < \cos\theta$ [cos strictly decreasing on $(0, \pi)$; math]

$\implies (\ln(r_0/r_1))' > 0$ on $(0, \frac{\pi}{12}) \subset (0, \frac{\pi}{6})$ [machine]

$$\pi^2$$

$$= 6\sum_{n\geq 1} n^{-2} > 6\sum_{n\leq 10} n^{-2}$$

$= \frac{1968329}{211680}$ [Basel; machine: the exact rational partial sum];

$$\frac{1968329}{211680} > \frac{64}{9}$$

$$\iff 17714961 > 13547520\ [\text{machine}] \implies \pi > \tfrac{8}{3}$$

$$\implies 3\pi > 8 \implies \tfrac{2}{9} < \tfrac{\pi}{12}\ [\text{machine}]$$

the ordering : $\delta_q < \delta_\nu < \delta_\ell < \frac{\pi}{12}$ and both ratio maps strictly increasing on $(0, \frac{\pi}{12}) \implies$ the boxed chain [machine];

concur: the three enclosures of (6) are strictly ordered [machine]

(9)–(10) : the standing-rule scans and the assembly [machine]. ■

Phys. Three families, one polynomial. Every channel's mass-squared ratios are the three roots of the same cubic — four z cubed minus twenty-four z squared plus twenty-one minus twelve gamma, times z, minus gamma minus one squared — and the whole difference between the quark channel, the lepton channel and the neutrino channel is a single number: root two times the cosine of three times the channel's own angle. Those three numbers are exact, distinct, and ordered, and one identity ties two of them together: the lepton channel's heaviest line is precisely one more than the quark channel's cubic constant, because tripling the quark angle lands exactly on the lepton angle. The hierarchies then order themselves with no scale anywhere in the argument: below one twelfth of a turn the inner line shrinks, the middle line grows, and both ratios steepen monotonically, so the smallest angle gives the flattest family and the largest angle the steepest — quark, neutrino, lepton — with the lepton channel's inner line the closest to the exact zero at one twelfth of a turn, which is why its hierarchy is two orders of magnitude deep while the quark channel's is barely two. The brackets are exact rational intervals, not decimals: every digit the corpus froze in Part 3 is re-derived here from the angle alone and certified at widths far below the last quoted place. Nothing is fitted, no scale is fixed, and the ratios are what they always were — pure numbers waiting on three undeclared scales before they can be called masses.

4.69 (BM.81 — The six-state mass-ratio census: the splitting order, the doublet ratios, the two inverted hierarchies, and the three non-proportional ladders).

Thm.

(1) the six states in both currencies (the census table):

$$|\mathcal{S}| = 3 \cdot 2 = 6,$$

$$m_{\text{hid}} = \frac{2E \ln 2}{\mathfrak{h}},$$

$$m^2_{\text{pole}} = \hat{\mu}'^2 D_S$$

[Def-BM.14$(1), (3)$; FT.44(4); Def-H.37(2)] :

state	$(\mathcal{F}_g, \beta)$	the branch datum	D_S	minimal polynomial of D_S	$2E$	$m_{\text{hid}}\hbar/\ln 2$	$m^2_{\text{pole}}/\hat{\mu}'^2$	field
s_1	$(\mathcal{F}_1, -)$	the ladder rung $m{=}1$	2	$x-2$	12	12	2	$\mathbb{Q}$
s_2	$(\mathcal{F}_1, +)$	the ladder rung $m{=}2$	4	$x-4$	12	12	4	$\mathbb{Q}$
s_3	$(\mathcal{F}_2, -)$	the cube member	$2-\sqrt{2}$	x^2-4x+2	24	24	$2-\sqrt{2}$	$\mathbb{Q}(\sqrt{2})$
s_4	$(\mathcal{F}_2, +)$	the octahedron member	1	$x-1$	24	24	1	$\mathbb{Q}$
s_5	$(\mathcal{F}_3, -)$	the dodecahedron member	$2-2\mathsf{c}$	$x^4-8x^3+19x^2-12x+1$	60	60	$2-2\mathsf{c}$	$\mathbb{Q}(\mathsf{c})$
s_6	$(\mathcal{F}_3, +)$	the icosahedron member	$2-\sqrt{3}$	x^2-4x+1	60	60	$2-\sqrt{3}$	$\mathbb{Q}(\sqrt{3})$

[Def-BM.14(2), (3); BM.60(4); machine: every minimal polynomial re-derived by first linear dependence over $\mathbb{Q}$];

the defect species resolves all six states with zero degeneracy;

the informational species is branch-degenerate and takes three values $12 : 24 : 60 = 1 : 2 : 5$

[FT.44(1); BM.60(4), (6); machine];

(2) the branch order and the attached splitting (frozen, re-certified):

$$\boxed{D_S(\mathcal{F}_g, +) > D_S(\mathcal{F}_g, -) \qquad \forall g \in \{1,2,3\}}$$

floor	$D_S(\mathcal{F}_g, -)$	$D_S(\mathcal{F}_g, +)$	the exact certificate
$\mathcal{F}_1$	2	4	$4 > 2$
$\mathcal{F}_2$	$2-\sqrt{2}$	1	$1 > 2-\sqrt{2} \iff \sqrt{2} > 1 \iff 2 > 1$
$\mathcal{F}_3$	$2-2\mathsf{c}$	$2-\sqrt{3}$	$\mathsf{c} > \frac{\sqrt{3}}{2} \iff \mathsf{c}^2 > \frac{3}{4} \iff \frac{5+\sqrt{5}}{8} > \frac{3}{4} \iff \sqrt{5} > 1 \iff 5 > 1$

$$\Delta_g \;:=\; m^2_{\text{down}} - m^2_{\text{up}} \;=\; \hat{\mu}'^2\big(D_S(\mathcal{F}_g, +) - D_S(\mathcal{F}_g, -)\big)$$

[Def-BM.14(5); BM.75(4); Def-BM.14(8)'s name rows] :

$$\boxed{\Delta_1 = 2\,\hat{\mu}'^2, \qquad \Delta_2 = (\sqrt{2}-1)\,\hat{\mu}'^2, \qquad \Delta_3 = (2\mathsf{c}-\sqrt{3})\,\hat{\mu}'^2, \qquad \Delta_g > 0 \ \ \forall g}$$

minimal polynomials of $\Delta_g/\hat{\mu}'^2$:

$$x - 2,$$

$$x^2 + 2x - 1,$$

$$x^8 - 22x^6 + 119x^4 - 38x^2 + 1$$

[machine: first linear dependence in $\mathbb{Q},\ \mathbb{Q}(\sqrt{2}),\ \mathbb{Q}(\mathsf{c},\sqrt{3})$];

(3) the splitting order (derived here):

$$\boxed{\Delta_1 \ > \ \Delta_2 \ > \ \Delta_3 \ > \ 0}$$

the down–up splitting strictly decreases across the generations in the defect currency

$$\Delta_1 > \Delta_2 \iff 2 > \sqrt{2}-1 \iff 3 > \sqrt{2} \iff 9 > 2; \qquad \Delta_2 > \Delta_3 \iff \sqrt{2}+\sqrt{3}-1 > 2\mathsf{c}$$

$$\iff 7 + 4\sqrt{6} > 4\sqrt{2} + 4\sqrt{3} + \sqrt{5}$$

[both sides positive; squared]

$$\iff 15 + 6\sqrt{6} > 2\sqrt{10} + 2\sqrt{15}$$

[squared again]

$$\iff 341 + 140\sqrt{6} > 0$$

[machine: the three squaring steps as exact identities in $\mathbb{Q}(\sqrt{2},\sqrt{3},\sqrt{5},\sqrt{6})$, each side certified positive by exact dyadic enclosure];

the exact ratio census of the splitting:

$$\frac{\Delta_2}{\Delta_1} = \frac{\sqrt{2}-1}{2},$$

$$\frac{\Delta_3}{\Delta_1} = \mathsf{c} - \frac{\sqrt{3}}{2},$$

$$\frac{\Delta_3}{\Delta_2} = \frac{2\mathsf{c}-\sqrt{3}}{\sqrt{2}-1} = (\sqrt{2}+1)(2\mathsf{c}-\sqrt{3})$$

$$\frac{\Delta_2}{\Delta_1} \in \Big(\frac{2071}{10^4}, \frac{2072}{10^4}\Big),$$

$$\frac{\Delta_3}{\Delta_1} \in \Big(\frac{85031}{10^6}, \frac{85032}{10^6}\Big),$$

$$\frac{\Delta_3}{\Delta_2} \in \Big(\frac{410566}{10^6}, \frac{410567}{10^6}\Big)$$

[machine: exact dyadic radical enclosures, width $< 2^{-60}$];

(4) the relative splittings (exact, and one exact simplification):

$$\frac{\Delta_g}{m^2_{\text{up}}(g)} = \frac{D_S(\mathcal{F}_g,+)}{D_S(\mathcal{F}_g,-)} - 1,$$

$$\frac{\Delta_g}{m^2_{\text{down}}(g)} = 1 - \frac{D_S(\mathcal{F}_g,-)}{D_S(\mathcal{F}_g,+)} :$$

$$\boxed{\frac{\Delta_1}{m^2_{\text{up}}(1)} = 1, \qquad \frac{\Delta_2}{m^2_{\text{up}}(2)} = \frac{\sqrt{2}}{2}, \qquad \frac{\Delta_3}{m^2_{\text{up}}(3)} = \frac{2\mathsf{c}-\sqrt{3}}{2-2\mathsf{c}}}$$

$$\Big[\frac{\sqrt{2}-1}{2-\sqrt{2}} = \frac{\sqrt{2}}{2} \text{ exactly : } (\sqrt{2}-1) - \frac{\sqrt{2}}{2}(2-\sqrt{2}) = 0 \text{ in } \mathbb{Q}(\sqrt{2}); \text{ machine}\Big];$$

$$\frac{\Delta_1}{m^2_{\text{down}}(1)} = \frac{1}{2},$$

$$\frac{\Delta_2}{m^2_{\text{down}}(2)} = \sqrt{2}-1,$$

$$\frac{\Delta_3}{m^2_{\text{down}}(3)} = (2\mathsf{c}-\sqrt{3})(2+\sqrt{3}) = 4\mathsf{c} + 2\sqrt{3}\,\mathsf{c} - 2\sqrt{3} - 3$$

[machine: exact in the biquadratic ring];

(5) the doublet mass ratios (the per-generation down/up census):

$$\boxed{\frac{m^2_{\text{down}}(g)}{m^2_{\text{up}}(g)} = \Big(2,\ \frac{2+\sqrt{2}}{2},\ \frac{2-\sqrt{3}}{2-2\mathsf{c}}\Big)}$$

minimal polynomials:

$$x-2,$$

$$2x^2-4x+1,$$

$$x^8-48x^7+410x^6-1328x^5+1899x^4-1232x^3+330x^2-32x+1$$

[machine: first linear dependence in $\mathbb{Q},\ \mathbb{Q}(\sqrt{2}),\ \mathbb{Q}(\mathsf{c},\sqrt{3})$; the degree-8 row's constant term is $1 = \text{Norm}(2-\sqrt{3})/\text{Norm}(2-2\mathsf{c})$ [machine]];

$$\boxed{\frac{m_{\text{down}}(g)}{m_{\text{up}}(g)} = \Big(\sqrt{2},\ \sqrt{\tfrac{2+\sqrt{2}}{2}},\ \sqrt{\tfrac{2-\sqrt{3}}{2-2\mathsf{c}}}\Big)}$$

annihilators: x^2-2, $2x^4-4x^2+1$, $P_8(x^2)$

$$P_8(x^2) = x^{16}-48x^{14}+410x^{12}-1328x^{10}+1899x^8-1232x^6+330x^4-32x^2+1$$

[machine: the exact substitution; ¬minimality claimed at $g=3$];

the sine-ratio form (uniform):

$$\frac{m_{\text{down}}(g)}{m_{\text{up}}(g)}$$

$$= \frac{\sin\big(\Theta_+/4\big)}{\sin\big(\Theta_-/4\big)}$$

$$= \Big(\frac{\sin(\pi/2)}{\sin(\pi/4)},\ \frac{\sin(\pi/6)}{\sin(\pi/8)},\ \frac{\sin(\pi/12)}{\sin(\pi/20)}\Big)$$

$$\Theta_\pm\ :$$

$$=\ m_\pm$$

$\delta_v(\mathcal{F}_g,\pm)$ the branch's own angle defect, at (m_+,m_-)

$$= (2,1),$$

$$(1,1),$$

$$(1,1) \text{ and } \delta_v$$

$$= \pi;$$

$$\tfrac{2\pi}{3},$$

$$\tfrac{\pi}{2};$$

$$\tfrac{\pi}{3}, \qquad \tfrac{\pi}{5}$$

[BM.79(7)'s law applied to the branch pair; machine: $D_S = 4\sin^2(\Theta/4)$ at each of the six states, and $\sin^2(\pi/8) = \frac{2-\sqrt{2}}{4}$, $\sin^2(\pi/12) = \frac{2-\sqrt{3}}{4}$ by the half-angle identity];

(6) the order of the doublet ratios is ¬the generation order:

$$\boxed{\frac{m_{\text{down}}(2)}{m_{\text{up}}(2)} < \frac{m_{\text{down}}(1)}{m_{\text{up}}(1)} < \frac{m_{\text{down}}(3)}{m_{\text{up}}(3)}}$$

i.e. $\mathcal{F}_2$ is the tightest doublet, $\mathcal{F}_3$ the widest :

$$\frac{2+\sqrt{2}}{2} < 2 \iff \sqrt{2} < 2 \iff 2 < 4;$$

$$2 < \frac{2-\sqrt{3}}{2-2\mathsf{c}} \iff 4 - 4\mathsf{c} < 2 - \sqrt{3} \iff \sqrt{3} < 4\mathsf{c} - 2 \iff 0 < 16\mathsf{c}^2 - 16\mathsf{c} + 1$$

$$\iff 11 + 2\sqrt{5} > 16\mathsf{c} \iff 141 + 44\sqrt{5} > 160 + 32\sqrt{5} \iff 12\sqrt{5} > 19 \iff 720 > 361$$

[machine: the chain of exact equivalences, each squaring step certified positive];

$$\text{enclosures:} \quad \frac{m^2_{\text{down}}(2)}{m^2_{\text{up}}(2)} \in \Big(\frac{17071}{10^4}, \frac{17072}{10^4}\Big),$$

$$\frac{m^2_{\text{down}}(3)}{m^2_{\text{up}}(3)} \in \Big(\frac{2737332}{10^6}, \frac{2737333}{10^6}\Big)$$

[machine];

(7) the two inverted hierarchies (the census form):

informational: $m_{\text{hid}}(\mathcal{F}_1) < m_{\text{hid}}(\mathcal{F}_2) < m_{\text{hid}}(\mathcal{F}_3)$,

$$1 : 2 : 5$$

[BM.60(6); machine];

defect:

$$D_S(\mathcal{F}_3, \pm) < D_S(\mathcal{F}_2, \pm) < D_S(\mathcal{F}_1, \pm),$$

$$2 - 2\mathsf{c} < 2 - \sqrt{3} < 2 - \sqrt{2} < 1 < 2 < 4$$

[BM.60(6); machine: the certificates $5 > 1,\ 3 > 2,\ 2 > 1$];

the two currencies order the three floors oppositely

[BM.60(6); BM.39(5) pattern];

the widest derived ratio in each currency:

defect (m^2) : $\dfrac{4}{2 - 2\mathsf{c}} \in \Big(\dfrac{40863458}{10^6}, \dfrac{40863459}{10^6}\Big)$;

defect (m) : $\dfrac{2}{\sqrt{2 - 2\mathsf{c}}} \in \Big(\dfrac{6392453}{10^6}, \dfrac{6392454}{10^6}\Big)$;

informational (m) : 5,

(m^2) : 25

[machine];

(8) the three ladders are pairwise non-proportional (derived):

(i) the floor ladder against the channel ladder:

$$\neg\exists\,\mu\in\mathbb{R}_{>0}:\quad \big(m_{\text{hid}}(\mathcal{F}_1),m_{\text{hid}}(\mathcal{F}_2),m_{\text{hid}}(\mathcal{F}_3)\big)=\mu\big(z_1^{(q)},z_2^{(q)},z_0^{(q)}\big)$$

[proportionality forces $2=z_2^{(q)}/z_1^{(q)}$ and $5=z_0^{(q)}/z_1^{(q)}$; BM.80(6)'s brackets give $z_2^{(q)}/z_1^{(q)}>\frac{7}{2}>2$ and $z_0^{(q)}/z_1^{(q)}>139>5$; machine];

(ii) the channel ladder against the transport ladder:

$z_i^{(\text{ch})}=1+2\sqrt{2}\,\cos\varphi_i+2\cos^2\varphi_i$ is quadratic in the channel cosine,

$G\big(c(i)\big)=A_0-B_0\cos\varphi_i$ is affine in it

[BM.60(8); BM.39(1); FT.111(3); machine] :

a quadratic and an affine function of the same three distinct cosines agree at all three only if the quadratic coefficient vanishes;

the quadratic coefficient is exactly $2\neq 0$

[machine: the exact polynomial comparison over $\mathbb{Q}[\sqrt{2}][\cos\varphi]$] $\implies$ $\boxed{\neg\exists\,\lambda:\ z_i^{(\text{ch})}=\lambda\,G\big(c(i)\big)\ \ \forall i}$;

(iii) the floor ladder against the transport ladder:

the transport form's line order inverts against the spectrum [BM.39(5)] while the floor ladder is increasing in $2E$ [BM.60(6)] $\implies$ ¬proportional with a positive scale

[machine: the sign census];

the generation map is an order-isomorphism and ¬a proportionality

[Def-BM.14(6); BM.60(7) : the unique order-isomorphism out of $3!=6$ bijections re-cited] :

the map supplies no ratio between the currencies;

(9) the six quark states (the assembly row):

$$\mathcal{S} = \{(\mathcal{F}_g, \beta)\},$$

down-type := $\beta{=}+$ (the heavier branch),

up-type := $\beta{=}-$

[Def-BM.14(8); A-BM.6] :

the attachment:

$$q_{\text{core}} = 0 \longleftrightarrow \text{down-type} \longleftrightarrow \beta{=}+,$$

$$q_{\text{core}} = +1 \longleftrightarrow \text{up-type} \longleftrightarrow \beta{=}-$$

[BM.75(3) : the unique bijection, conditional on BM.59(9)'s single-substitution hypothesis, on its face];

the sign-coherence row:

in every derived currency in which the doublet splitting is formed, the down-type constituent is the heavier one

[BM.75(5) re-cited; the four rows: $\ln 2/\hbar$, $\ln(4/3)/\hbar$, $\frac{4}{3}$, Δ_g; the single opposed EM correction dominated on the sub-critical branch [BM.46(4); BM.62$(6), (9)$]];

the flavour composition stays ¬derived:

which three of the six states a sealed class occupies is ¬attached [BM.28(7); Def-BM.8(5); BM.75(6)];

the mixing matrix is ¬derived [FT.145(3)];

(10) status:

derived: (1)–(8);

re-cited: (9)'s attachment, name rows and sign coherence from their own tags;

declared: Def-BM.14(8)'s name attachment alone [A-BM.6, logged before use];

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

§Cal-BM stays empty;

open: $\hat{\mu}'^2,\ \mu \in$ Cal-F [the map is an order, ¬a number].

Req. D0.8 (Parts 1–2); P1, P4, P5 (Parts 1–2); Def-H.37(2), FT.44(1), FT.44(4), FT.111(3), FT.145(3) (Part 3); Census FT.63(6) (row-wise); Def-BM.8(5), Def-BM.12(4), Def-BM.14(1), Def-BM.14(2), Def-BM.14(3), Def-BM.14(5), Def-BM.14(6), Def-BM.14(8), BM.28(7), BM.39(1), BM.39(5), BM.46(4), BM.59(9), BM.60(4), BM.60(6), BM.60(7), BM.60(8), BM.62(6), BM.62(9), BM.75(3), BM.75(4), BM.75(5), BM.75(6), BM.79(7), BM.80(4), BM.80(6) (this part); Cal-F; the declaration addendum A-BM.6 (pointers); exact radical rings $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{6}), \mathbb{Q}(\sqrt{10}), \mathbb{Q}(\sqrt{15})$, the biquadratic ring $\mathbb{Q}[\mathsf{c}, \sqrt{3}]/(16\mathsf{c}^4 - 20\mathsf{c}^2 + 5)$, minimal polynomials by first linear dependence with exact re-substitution, norm reads by multiplication determinants, exact dyadic radical enclosures with integer certificates, squaring chains with certified positive sides, exact polynomial comparison over $\mathbb{Q}[\sqrt{2}][x]$ [math; machine: `checks/verify_H_mass_ratio_census.py` §C].

Pf.

(1) : the table re-cited from Def-BM.14(3) and BM.60(4) and every minimal polynomial re-derived [machine: first linear dependence on the powers of the element in its own field, followed by exact re-substitution];

$2E \in \{12, 24, 24, 60, 60\}$ has three distinct values [machine]

(2) : the three certificates re-derived [machine];

$$\cos\tfrac{\pi}{10} > \tfrac{\sqrt{3}}{2} \iff \mathsf{c}^2 > \tfrac{3}{4} \iff \tfrac{5+\sqrt{5}}{8} > \tfrac{3}{4}$$

$$\iff 5 + \sqrt{5} > 6 \iff \sqrt{5} > 1$$

$$\iff 5 > 1 \text{ [machine: exact in } \mathbb{Q}(\sqrt{5}), \text{ both sides positive]};$$

Δ_g formed as exact differences [machine]

the minimal polynomial of $2\mathsf{c} - \sqrt{3}$: first linear dependence in $\mathbb{Q}[\mathsf{c}, \sqrt{3}]$ gives $x^8 - 22x^6 + 119x^4 - 38x^2 + 1$ [machine];

the even-power shape follows from the conjugate set $\{\pm 2\mathsf{c}_i \mp \sqrt{3}\}$ being symmetric under x

$$\mapsto -x \text{ [machine]}$$

$$(3) : \ 9 > 2 \text{ [machine]};$$

$\sqrt{2} + \sqrt{3} - 1 > 0$ and $2\mathsf{c} > 0$ [machine: dyadic enclosures] $\implies$ the first squaring is order-preserving;

$$(\sqrt{2} + \sqrt{3} - 1)^2 = 6 + 2\sqrt{6} - 2\sqrt{2} - 2\sqrt{3} \text{ and } (2\mathsf{c})^2 = 4\mathsf{c}^2 = \tfrac{5+\sqrt{5}}{2} \text{ [machine]}$$

$$6 + 2\sqrt{6} - 2\sqrt{2} - 2\sqrt{3} > \tfrac{5+\sqrt{5}}{2} \iff 7 + 4\sqrt{6} > 4\sqrt{2} + 4\sqrt{3} + \sqrt{5} \text{ [machine: cleared]};$$

both sides positive [machine] $\implies$ the second squaring : $145+56\sqrt{6} > 85+32\sqrt{6}+8\sqrt{10}+8\sqrt{15}$

$$\iff 60 + 24\sqrt{6} > 8\sqrt{10} + 8\sqrt{15} \iff 15 + 6\sqrt{6} > 2\sqrt{10} + 2\sqrt{15} \text{ [machine]};$$

both sides positive [machine] $\implies$ the third squaring : $441 + 180\sqrt{6} > 100 + 40\sqrt{6}$

$$\iff 341 + 140\sqrt{6} > 0 \text{ [machine: positive coefficients]}$$

the three ratio enclosures by exact dyadic radical isolation [machine: widths $< 2^{-60}$]

$$(4) : \ (\sqrt{2} - 1) - \tfrac{\sqrt{2}}{2}(2 - \sqrt{2}) = \sqrt{2} - 1 - (\sqrt{2} - 1) = 0 \text{ [machine: exact in } \mathbb{Q}(\sqrt{2})];$$

$\Delta_g/m_{\text{up}}^2 = D_S(+)/D_S(-) - 1$ and $\Delta_g/m_{\text{down}}^2 = 1 - D_S(-)/D_S(+)$ [machine: the two identities];

$$(2\mathsf{c}-\sqrt{3})(2+\sqrt{3}) = 4\mathsf{c}+2\sqrt{3}\mathsf{c}-2\sqrt{3}-3 \text{ [machine: } (2-\sqrt{3})(2+\sqrt{3})=1]$$

$$(5):\ \text{the three } m^2 \text{ ratios formed as exact quotients in their fields [machine];}$$

the minimal polynomials by first linear dependence, with exact re-substitution [machine];

$$\mathrm{Norm}(2-\sqrt{3})$$

$$= 1,$$

$$\mathrm{Norm}(2-2\mathsf{c})$$

$$= 1 \text{ as multiplication determinants [machine] } \implies \text{ the constant term } 1$$

$$\text{the annihilators of the mass ratios : } x^2-2 \text{ at } \sqrt{2};$$

$$\text{at } y^2=\tfrac{2+\sqrt{2}}{2}:\ (2y^2-2)^2=2 \iff 2y^4-4y^2+1=0 \text{ [machine];}$$

$$P_8(x^2) \text{ by the exact substitution } x\mapsto x^2 \text{ [machine]}$$

$$\text{the sine-ratio form : } D_S$$

$$= 4\sin^2(\delta_v/4) \text{ at the six states with } \delta_v/\pi$$

$$= (1,1,\tfrac{1}{2},\tfrac{2}{3},\tfrac{1}{5},\tfrac{1}{3}) \implies \delta_v/4 \in \{\tfrac{\pi}{4},\tfrac{\pi}{4},\tfrac{\pi}{8},\tfrac{\pi}{6},\tfrac{\pi}{20},\tfrac{\pi}{12}\} \text{ [machine]}$$

$$(6):\ 2<4 \text{ [machine];}$$

$$0<16\mathsf{c}^2-16\mathsf{c}+1 \iff 16\cdot\tfrac{5+\sqrt{5}}{8}-16\mathsf{c}+1>0 \iff 11+2\sqrt{5}>16\mathsf{c} \text{ [machine];}$$

$$\text{both sides positive [machine] } \implies (11+2\sqrt{5})^2>256\mathsf{c}^2=32(5+\sqrt{5})$$

$$\iff 141+44\sqrt{5}>160+32\sqrt{5} \iff 12\sqrt{5}>19 \iff 720>361 \text{ [machine: each step cleared];}$$

the two enclosures [machine]

$$(7):\ 12 < 24 < 60, \quad 12 : 24 : 60 = 1 : 2 : 5 \text{ [machine]};$$

$$2 - 2\mathsf{c} < 2 - \sqrt{3} \iff \sqrt{3} < 2\mathsf{c} \text{ [BM.60(6)'s certificate } 5 > 1];$$

$$2 - \sqrt{3} < 2 - \sqrt{2} \iff \sqrt{2} < \sqrt{3} \iff 2 < 3;$$

$$2 - \sqrt{2} < 1 \iff 1 < \sqrt{2} \iff 1 < 2 \text{ [machine]}$$

$$\text{the widest ratios}: \ \max D_S / \min D_S = 4/(2 - 2\mathsf{c}), \quad \max 2E / \min 2E = 60/12 = 5 \text{ [machine]};$$

the two enclosures by exact radical isolation [machine]

$(8)(\mathrm{i}):$ proportionality at three strictly ordered entries forces the two ratio equalities [machine];

$$z_2^{(q)}/z_1^{(q)} > \tfrac{7}{2} \text{ and } z_0^{(q)}/z_1^{(q)} > 139 \text{ [BM.80(6)'s brackets; machine]};$$

$$\tfrac{7}{2} > 2, \ 139 > 5 \text{ [machine]}$$

$$(\mathrm{ii}):\ r_i = 1 + \sqrt{2}\cos\varphi_i \implies z_i = r_i^2 = 1 + 2\sqrt{2}\cos\varphi_i + 2\cos^2\varphi_i \text{ [machine: the exact expansion]};$$

$$\text{if } 1 + 2\sqrt{2}u + 2u^2$$

$$= \lambda(A_0 - B_0 u) \text{ at the three distinct values } u$$

$$= \cos\varphi_i,$$

the difference is a quadratic with three roots

$$\implies \text{the quadratic vanishes identically}$$

$$\implies 2$$

$= 0$ [machine: the exact polynomial identity over $\mathbb{Q}[\sqrt{2}][u]$, with the three cosines' distinctness certified by disjoint enclosures];

(iii) : the sign census of the two orders [BM.39(5); machine]

(9)–(10) : the rows re-cited from their own tags; the standing-rule scans [machine]. ■

Phys. Six states, two currencies, and not one proportionality between them. In the angle currency the six defect values are pairwise distinct algebraic numbers living in four fields, the heaviest branch on each floor is heavier by an exact amount, and those three amounts run strictly downward across the generations — two, root two minus one, and two cosines of eighteen degrees minus root three — so the splitting that separates a doublet is widest on the lightest floor and narrowest on the heaviest, the opposite of what a hierarchy reading would suggest. The doublet ratios themselves are pure sine ratios and their order is again not the generation order: the cube–octahedron pair is the tightest, the tetrahedral pair the middle, and the dodecahedron–icosahedron pair the widest by a degree-eight number whose norm is exactly one. In the bit currency the same six states collapse to three values climbing one to two to five. The two orders are opposite, and the tier proves something stronger than that they differ: no positive scale makes the floor ladder proportional to the channel ladder, because two is not three and a half and five is not one hundred thirty-nine; no scale makes the channel ladder proportional to the transport ladder, because one is quadratic in the channel cosine and the other affine and the quadratic coefficient is exactly two; and the transport ladder inverts against both. The generation map survives all of this as what it always was — the unique order-isomorphism between two three-element ordered sets — and it supplies an order, never a number.

4.70 (BM.82 — The bridge census: the six exact factors, the intra-currency closure, and the single priced residue).

Thm.

(1) the two reads and the bridge identity (derived, not assumed):

$$m_{\text{hid}} = \frac{2E \ln 2}{\mathfrak{h}}$$

[FT.44(4) : the taxonomy law $m_{\text{hid}} = E_{\text{int}} \ln 2/\mathfrak{h}$ at $E_{\text{int}} = 2E$; Census FT.42(1) row-wise; Def-BM.14(1)] :

$$m_{\text{pole}}^2 = \hat{\mu}'^2 D_S = 4\hat{\mu}'^2 \sin^2 \frac{\Theta_R}{4} \text{ at } \Theta_R = \delta_v$$

[Def-H.37(2); FT.54(4); FT.47(1); BM.60(3)];

$$\rho_{\text{B}} := \frac{\hat{\mu}'^2 \mathfrak{h}^2}{(\ln 2)^2}$$

[BM.69(2); Cal-F packaging];

$$[\rho_{\mathrm{B}}] = \mathfrak{q} \cdot \mathfrak{q}^{-1} = \mathfrak{q}^{0}$$

$[[\hat{\mu}'^2] = \mathfrak{q}$ [Def-H.37(2)]; $[\mathfrak{h}] = \mathfrak{q}^{-1/2}$ [CT2.1; D2.9]; machine: the exponent arithmetic];

$$\boxed{\frac{m^2_{\mathrm{pole}}(s)}{m^2_{\mathrm{hid}}(s)} \;=\; \rho_{\mathrm{B}} \cdot \mathcal{R}_s, \qquad \mathcal{R}_s \;:=\; \frac{D_S(s)}{4\,E(s)^2} \;\in\; \overline{\mathbb{Q}}}$$

[BM.69(2) re-derived from the two reads above; machine: one Laurent monomial in $\hat{\mu}'^2, \mathfrak{h}, \ln 2$] :

the factorization is exact per state;

the algebraic part $\mathcal{R}_s$ is derived;

the single dimensionless residue is ρ_{B}

[BM.69(5)(R1)];

(2) the six-factor census: $E(s) = (6, 6, 12, 12, 30, 30)$ at $2E = (12, 12, 24, 24, 60, 60)$:

state	$D_S(s)$	$4E(s)^2$	$\mathcal{R}_s = D_S/(4E^2)$	field	minimal polynomial of $\mathcal{R}_s$
s_1	2	144	$\frac{1}{72}$	$\mathbb{Q}$	$72x - 1$
s_2	4	144	$\frac{1}{36}$	$\mathbb{Q}$	$36x - 1$
s_3	$2 - \sqrt{2}$	576	$\frac{2-\sqrt{2}}{576}$	$\mathbb{Q}(\sqrt{2})$	$165888x^2 - 1152x + 1$
s_4	1	576	$\frac{1}{576}$	$\mathbb{Q}$	$576x - 1$
s_5	$2 - 2\mathsf{c}$	3600	$\frac{2-2\mathsf{c}}{3600}$	$\mathbb{Q}(\mathsf{c})$	$167961600000000\,x^4 - 373248000000\,x^3 + 246240000\,x^2 - 43200\,x + 1$
s_6	$2 - \sqrt{3}$	3600	$\frac{2-\sqrt{3}}{3600}$	$\mathbb{Q}(\sqrt{3})$	$12960000\,x^2 - 14400\,x + 1$

six states, six exact factors, four fields, six minimal polynomials

[BM.69(2)'s row re-derived; machine: the quartic row by the exact substitution $D = 3600x$ into $x^4 - 8x^3 + 19x^2 - 12x + 1$];

the exact order:

$$\mathcal{R}_5 < \mathcal{R}_6 < \mathcal{R}_3 < \mathcal{R}_4 < \mathcal{R}_1 < \mathcal{R}_2$$

[machine: pairwise comparisons in the exact fields with dyadic isolation];

$$\mathcal{R}_5 < \mathcal{R}_6 \iff 2 - 2\mathsf{c} < 2 - \sqrt{3} \iff 5 > 1;$$

$$\mathcal{R}_6 < \mathcal{R}_3 \iff 3600^{-1}(2 - \sqrt{3}) < 576^{-1}(2 - \sqrt{2}) \iff 2 - \sqrt{3} < \tfrac{25}{4}(2 - \sqrt{2})$$

[machine];

$$\mathcal{R}_3 < \mathcal{R}_4 \iff 2 - \sqrt{2} < 1 \iff 1 < 2;$$

$$\mathcal{R}_4 < \mathcal{R}_1 \iff \frac{1}{576} < \frac{1}{72} \iff 72 < 576;$$

$$\mathcal{R}_1 < \mathcal{R}_2 \iff \frac{1}{72} < \frac{1}{36} \iff 36 < 72$$

[machine: integers];

(3) the span of the bridge factor:

$$\boxed{\frac{\mathcal{R}_{\max}}{\mathcal{R}_{\min}} = \frac{\mathcal{R}_2}{\mathcal{R}_5} = \frac{1/36}{(2 - 2\mathsf{c})/3600} = \frac{100}{2 - 2\mathsf{c}} = \frac{50}{1 - \mathsf{c}}}$$

minimal polynomial: $x^4 - 1200x^3 + 190000x^2 - 8000000x + 100000000$

[machine: the substitution $z = x/50$ into BM.79(7)'s $z^4 - 24z^3 + 76z^2 - 64z + 16$];

$$\frac{\mathcal{R}_2}{\mathcal{R}_5} \in \Big(\frac{1021586454}{10^6}, \frac{1021586455}{10^6}\Big),$$

$$\sqrt{\frac{\mathcal{R}_2}{\mathcal{R}_5}} = \frac{10}{\sqrt{2 - 2\mathsf{c}}} \in \Big(\frac{31962266}{10^6}, \frac{31962267}{10^6}\Big)$$

[machine: exact radical enclosures];

the floor factor is exactly twenty-five:

$$\frac{\mathcal{R}_2/\mathcal{R}_5}{4/(2-2\mathsf{c})} = \frac{100/(2-2\mathsf{c})}{4/(2-2\mathsf{c})} = 25 = \Big(\frac{30}{6}\Big)^2 = \Big(\frac{E(\mathcal{F}_3)}{E(\mathcal{F}_1)}\Big)^2$$

[machine: the exact quotient] :

the bridge factor's range exceeds the defect ceiling by exactly the square of the floor ladder's edge ratio — the angle contributes $\frac{4}{2-2\mathsf{c}}$,

the floor contributes 25;

(4) the intra-currency closure (the residue-free part of the bridge):

$$\text{the pole currency:} \quad \frac{m^2_{\text{pole}}(s)}{m^2_{\text{pole}}(s')} = \frac{D_S(s)}{D_S(s')}$$

for all 30 ordered pairs of distinct states $(15$ unordered pairs, each read in both directions$)$:

all thirty are exact algebraic and ρ_{B}-free

[machine: $\hat{\mu}'^2$ cancels in every quotient; the values lie in the compositum $\mathbb{Q}(\sqrt{2},\sqrt{3},\mathsf{c})$, $[\,\cdot:\mathbb{Q}] = 16$ [FT.73(2) row-wise]];

two exact cross-links to the ladder census:

$$\frac{D_S(s_1)}{D_S(s_6)} = \frac{2}{2-\sqrt{3}} = 4+2\sqrt{3} = (1+\sqrt{3})^2 = r^2_{\text{intra}}(\text{icosa}),$$

$$\frac{D_S(s_2)}{D_S(s_5)} = \frac{4}{2-2\mathsf{c}} = \text{the dodecahedral ceiling}$$

[BM.79(7), (9); BM.81(5); machine: the exact identities];

the widest and narrowest of the thirty are $\frac{4}{2-2\mathsf{c}}$ and its reciprocal $\frac{2-2\mathsf{c}}{4}$ [machine];

the collision census:

the thirty ordered ratios take twenty-eight distinct values, and the unique collision is the exact rational 2 :

$$\frac{D_S(s_2)}{D_S(s_1)} = \frac{4}{2} = \frac{2}{1} = \frac{D_S(s_1)}{D_S(s_4)}$$

three of the six defect values are the powers 1,

$$2,$$

4 — the octahedron, the tetrahedral light rung and the tetrahedral heavy rung — and that geometric triple is the census's only degeneracy [machine: the thirty-value census];

the informational currency:

$$\frac{m_{\text{hid}}(s)}{m_{\text{hid}}(s')} = \frac{E(s)}{E(s')} \in \Big\{1,\ 2,\ 5,\ \frac{1}{2},\ \frac{1}{5},\ \frac{2}{5},\ \frac{5}{2}\Big\}$$

[machine: the exact value set at $2E \in \{12, 24, 60\}$] :

all ρ_{B}-free, all rational, seven distinct values over the fifteen ordered pairs ;

the transport currency: $\Delta m = \dfrac{\ln(4/3)}{\mathfrak{h}}, \quad \dfrac{\ln 2}{\mathfrak{h}}$ per constituent

[BM.59$(4), (5)$; ρ_{B}-free];

the ensemble currency:

$$\frac{T(0,S)}{T(+1,S)} = \frac{W(0)}{W(+1)} = \frac{20}{15} = \frac{4}{3}$$

[BM.28(4)(ii); BM.30(2); ρ_{B}-free];

(5) the cross-currency row (where the residue lives):

$$\frac{m_{\text{pole}}(s)}{m_{\text{hid}}(s)} = \sqrt{\rho_{\text{B}}\,\mathcal{R}_s} = \sqrt{\rho_{\text{B}}}\cdot\Big(\frac{\sqrt{2}}{12},\ \frac{1}{6},\ \frac{\sqrt{2-\sqrt{2}}}{24},\ \frac{1}{24},\ \frac{\sqrt{2-2\mathsf{c}}}{60},\ \frac{\sqrt{2-\sqrt{3}}}{60}\Big)$$

[machine: the six square roots simplified in the ordered radical extensions; $\sqrt{\mathcal{R}_2}/\sqrt{\mathcal{R}_5} = 10/\sqrt{2-2\mathsf{c}}$ is ρ_{B}-free again];

the residue enters at exactly one step:

every mass ratio inside a currency is derived with zero residue; the single cross-currency step carries the single undeclared ρ_{B}

(6) the residue localized:

(R1) $\rho_{\mathrm{B}} = \hat{\mu}'^2\hbar^2/(\ln 2)^2$ is the only undeclared datum any ratio of (2)–(5) needs

[BM.69(5)(R1); Cal-F packaging, ¬added here];

(R2) the six rows are read at the vertex-figure locus Θ_R

$= \delta_v$,

whose ledger realization — a clean-enclosure loop through conical vertices — is ¬constructed and is inherited at every use

[BM.67(9), (10); Census FT.63(4) Rmk; BM.69(5)(R2)];

(R3) the flavour composition of the sealed classes: it enters only where a composite's pole mass is formed — ¬a ratio of (2)–(5) requires it

[BM.69(3), (5)(R3); BM.28(7)];

(7) the non-derivation theorem, re-certified and sharpened:

¬∃ an exact identity $m^2_{\mathrm{pole}} = \mathcal{B}\big[E_{\mathrm{int}}, W(Q), \Theta_R, I_F\big]$ with $\mathcal{B}$ built from derived data alone

on four independent grounds : (i) the degree-16 field against integer- and binary-valued inputs [FT.73(2), (3)];

(ii) the two-torsion image $\{0, \pi\}$ reaching 2 of 6 states [BM.67(6)];

(iii) the calibration weight -2 against 0 on the winding route [FT.55(1); BM.67(8)];

(iv) ¬∃ a count → strength theorem [BM.36(3); BM.61(8)(O1)]

[BM.69(4) re-cited; the grounds stand] :

the census sharpening:

the four grounds bound ρ_{B} itself;

they leave $\mathcal{R}_s$,

the fifteen pole ratios, the seven informational ratios, the transport differences and the ensemble ratio untouched :

the mass-ratio census is complete up to one undeclared dimensionless number, and complete with none inside any single currency

(8) status: derived: (1)–(5), (7); localized, ¬filled: (6);

¬a P4 addition (ρ_{B} is a Cal-F restatement, ¬a new member),

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a declaration,

§Cal-BM stays empty;

¬a currency is converted anywhere in this theorem;

Req. CT2.1, D2.9 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.37(2), FT.44(4), FT.47(1), FT.54(4), FT.55(1), FT.73(2), FT.73(3) (Part 3); Census FT.42(1), FT.63(4) (row-wise); Def-BM.14(1), Def-BM.14(3), BM.28(4), BM.28(7), BM.30(2), BM.36(3), BM.59(4), BM.59(5), BM.60(3), BM.61(8), BM.67(6), BM.67(8), BM.67(9), BM.67(10), BM.69(2), BM.69(3), BM.69(4), BM.69(5), BM.79(7), BM.79(9), BM.81(5) (this part); Cal-F; exact radical arithmetic in four fields and their degree-16 compositum, minimal polynomials by first linear dependence and by exact substitution, Laurent-monomial assembly, exponent arithmetic in the $\mathfrak{q}$-grading, exact dyadic radical enclosures, pairwise-distinctness censuses [math; machine: `checks/verify_H_mass_ratio_census.py` §E].

Pf.

$$(1) : \quad \frac{\hat{\mu}'^2 D_S}{(2E \ln 2/\hbar)^2}$$

$$= \frac{\hat{\mu}'^2 \mathfrak{h}^2}{(\ln 2)^2} \cdot \frac{D_S}{4E^2}$$

$$= \rho_{\mathrm{B}} \mathcal{R}_s \text{ [machine: one cleared identity over } \mathbb{Q}[\hat{\mu}'^2, \mathfrak{h}, \ln 2, D_S, E]];$$

$$[\rho_{\mathrm{B}}] = [\hat{\mu}'^2][\mathfrak{h}]^2 = \mathfrak{q} \cdot \mathfrak{q}^{-1} = \mathfrak{q}^0 \text{ [machine: the exponent arithmetic]};$$

$$\mathcal{R}_s \in \overline{\mathbb{Q}} \text{ since } D_S \text{ is algebraic and } E \in \mathbb{Z} \text{ [machine]}$$

$$(2): \ 4E^2 = 4(6, 6, 12, 12, 30, 30)^2 = (144, 144, 576, 576, 3600, 3600) \text{ [machine]};$$

the six quotients formed exactly [machine];

the minimal polynomials: $72x - 1$,

$$36x - 1,$$

$$576x - 1 \text{ by inspection;}$$

$$(2 - \sqrt{2})/576: \ y$$

$$= 576x \text{ satisfies } y^2 - 4y + 2$$

$$\implies 331776x^2 - 2304x + 2$$

$$= 0$$

$$\implies 165888x^2 - 1152x + 1 \text{ [machine]}$$

$$(2 - \sqrt{3})/3600: \ y^2 - 4y + 1 \implies 12960000x^2 - 14400x + 1 \text{ [machine]};$$

$$(2 - 2\mathsf{c})/3600: \ y^4 - 8y^3 + 19y^2 - 12y + 1 \text{ at } y$$

$$= 3600x$$

$$\implies 3600^4 x^4 - 8 \cdot 3600^3 x^3 + 19 \cdot 3600^2 x^2 - 12 \cdot 3600x + 1 \text{ [machine: the four coefficients as exact integers]}$$

the order : the five comparisons of (2) with $2-\sqrt{3} < \frac{25}{4}(2-\sqrt{2})$ certified by $(2-\sqrt{3})^2$

$$= 7 - 4\sqrt{3} < \left(\tfrac{25}{4}(2-\sqrt{2})\right)^2 \text{ on exact dyadic isolation [machine]}$$

$$(3): \ \frac{1/36}{(2-2\mathsf{c})/3600} = \frac{3600}{36(2-2\mathsf{c})} = \frac{100}{2-2\mathsf{c}} = \frac{50}{1-\mathsf{c}} \text{ [machine]};$$

$$z^4 - 24z^3 + 76z^2 - 64z + 16$$

$$= 0 \text{ at } z$$

$$= x/50$$

$$\implies x^4 - 1200x^3 + 190000x^2 - 8000000x + 100000000$$

$$= 0 \text{ [machine: cleared]};$$

the two enclosures [machine]

$$\frac{100/(2-2\mathsf{c})}{4/(2-2\mathsf{c})} = 25 \text{ and } (30/6)^2 = 25 \text{ [machine: exact rationals]}$$

$$(4): \ \hat{\mu}'^2 D_S(s)/(\hat{\mu}'^2 D_S(s')) = D_S(s)/D_S(s') \text{ [machine]};$$

$$2/(2-\sqrt{3}) = 2(2+\sqrt{3}) = 4 + 2\sqrt{3} = (1+\sqrt{3})^2 \text{ [machine: exact in } \mathbb{Q}(\sqrt{3})];$$

the fifteen quotients pairwise distinct by dyadic isolation [machine]

$E(s)/E(s')$ over $\{6, 12, 30\}$: the value set $\{1, 2, 5, \frac{1}{2}, \frac{1}{5}, \frac{2}{5}, \frac{5}{2}\}$ by exhaustive enumeration of the $6 \cdot 5$

$$= 30 \text{ ordered pairs [machine]};$$

$$W(0)/W(+1) = 20/15 = 4/3 \text{ [machine]}$$

$$(5): \ \sqrt{1/72}$$

$$= \sqrt{2}/12,$$

$$\sqrt{1/36}$$

$$= 1/6,$$

$$\sqrt{(2-\sqrt{2})/576}$$

$$= \sqrt{2-\sqrt{2}}/24,$$

$$\sqrt{1/576}$$

$$= 1/24,$$

$$\sqrt{(2-2\mathsf{c})/3600}$$

$$= \sqrt{2-2\mathsf{c}}/60,$$

$$\sqrt{(2-\sqrt{3})/3600}$$

$$= \sqrt{2-\sqrt{3}}/60 \text{ [machine: each verified by squaring the positive side]}$$

(6)–(8) : the three residue rows re-cited from BM.69(3), (5) and BM.67(9), (10);

the four grounds re-cited from BM.69(4);

the standing-rule scans [machine]. ■

Phys. The bridge is an exact factorization and nothing more mysterious than that. Divide the cone-defect mass by the edge-count mass and everything cancels except one dimensionless packaging and one algebraic number per state — and the algebraic numbers are six exact values in four fields, ordered, with minimal polynomials written out to a fourteen-digit leading coefficient. Inside a currency the packaging cancels too, so all fifteen pole-side ratios, all seven informational ratios, both transport differences and the ensemble's four thirds are derived with no residue at all: the census is complete. The residue lives at exactly one step, the step between currencies, and it is one number. The bridge factor's own range is a thousand and twenty-two in mass-squared, twenty-five times the widest angle ratio, and that twenty-five is precisely the square of the edge-count ratio between the heaviest and lightest floor — geometry supplies the shape, the floors supply the

reach. What the four obstruction grounds bound is the packaging and nothing else, so the honest statement of the whole tier is short: every mass ratio the corpus can form is derived exactly, and one dimensionless constant stands between that census and an absolute cross-currency mass.

4.71 (BM.83 — The inter-family bridge in full: the four-currency census, the exact incommensurability, and the three-number localization).

Thm.

(1) the four currencies read the same doublet (the census table):

the neutral class against the unit class

[BM.28(1); Def-BM.8(1)] :

currency	the formed object	the exact value	the residue a ratio needs
informational	m_n/m_p	$\dfrac{\ln 1280}{\ln 960 + 6\,c\,\varepsilon_{\mathrm{EM}}\ln 2}$	none — formed [BM.62(4), (5)]
ensemble, route (ii)	m_n/m_p	$\dfrac{T(0,S)}{T(+1,S)} = \dfrac{4}{3}$	none — formed; the route is $\neq g$adopted globally [BM.28(4)(ii); BM.30(2), (5)]
transport	Δm (a difference)	$\dfrac{\ln(4/3)}{\mathfrak{h}}$; per constituent $\dfrac{\ln 2}{\mathfrak{h}}$	none — formed [BM.59(4), (5)]
pole/defect	$\Delta(m^2_{\mathrm{pole}})$ (a difference)	$\Delta_g = (2, \sqrt{2}-1, 2\mathrm{c}-\sqrt{3})\,\hat{\mu}'^2$	(R3) the composition, for a ratio [BM.69(3); BM.75(4)]

two ratios and two differences:

no currency supplies both, and ¬a derived equation turns one into the other

[BM.35(5); BM.37(2)];

(2) the prime-log ring and the irrationality theorem:

$$\mathfrak{P} := \mathbb{Q} \oplus \bigoplus_{p\ \text{prime}} \mathbb{Q}\ln p,$$

$$\ln 1280 = 8\ln 2 + \ln 5,$$

$$\ln 960 = 6\ln 2 + \ln 3 + \ln 5$$

[machine: the exact factorizations $1280 = 2^8\cdot 5,\ 960 = 2^6\cdot 3\cdot 5$] :

the independence lemma: $\displaystyle\sum_i a_i \ln p_i = 0$ with $a_i \in \mathbb{Q} \implies a_i = 0\ \forall i$

[math: clear denominators by N; exp gives $\prod_i p_i^{Na_i} = 1$; unique factorization forces every $Na_i = 0$; machine: the integer version $2^a 3^b 5^c = 1 \implies a = b = c = 0$ verified exhaustively on $|a|, |b|, |c| \le 24$ and by the sign-split argument];

the irrationality theorem:	$\dfrac{\ln 1280}{\ln 960 + t \ln 2} \notin \mathbb{Q}$	for every rational $t \ge 0$

[machine: if $q \ln 1280 = p(\ln 960 + t \ln 2)$ with $p, q \in \mathbb{Z},\ q > 0,$ then the $\ln 3$ coefficient gives $-p = 0,$ so $p = 0,$ and the $\ln 5$ coefficient then gives $q = 0$ by the lemma];

corollary:

$\dfrac{m_n}{m_p}$ is irrational on the informational currency at every rational coupling $\varepsilon_{\text{EM}} \in \mathbb{Q}$ and both priced $c \in \{\frac{1}{2}, \frac{3}{5}\}$

$[t = 6c\,\varepsilon_{\text{EM}};\ \text{machine}];$

(3) the exact incommensurability of the two formed ratios:

$$\left.\frac{m_n}{m_p}\right|_{\text{info}} = \frac{4}{3} \iff 3\ln 1280 = 4\big(\ln 960 + t\ln 2\big) \iff 0 = 4\ln 3 + \ln 5 + 4t\ln 2 \iff t = -\frac{\ln 405}{\ln 16}$$

[machine: solved exactly in $\mathfrak{P}$; $4\ln 3 + \ln 5 = \ln(3^4 \cdot 5) = \ln 405 > 0$ and $\ln 16 > 0 \implies t < 0$] :

the informational ratio never equals the ensemble ratio at any admissible coupling:	$\dfrac{\ln 1280}{\ln 960 + t \ln 2} \neq \dfrac{4}{3}$	$\forall\, t \ge 0$

and the order between them is exact:

$$1 < \frac{\ln 1280}{\ln 960} < \frac{4}{3}$$

[$4\ln 960 - 3\ln 1280 = 4\ln 3 + \ln 5 - \ln 405 > 0$; $\ln 1280 > \ln 960 \iff 1280 > 960$; machine];

the two ratios are not merely unequal, they are of different kinds:

$$\frac{4}{3} \in \mathbb{Q} \ \text{ and } \ \frac{\ln 1280}{\ln 960} \in \mathfrak{P} \setminus \mathbb{Q}$$

$[(2), (3);\ \text{machine}];$

(4) the dimensionless scale set (restatements, ¬new members):

$$\sigma_\ell := \frac{\mu_\ell\,\mathfrak{h}}{\ln 2} \in \mathfrak{q}^0,$$

$$\sigma_\nu := \frac{\mu_\nu\,\mathfrak{h}}{\ln 2} \in \mathfrak{q}^0,$$

$$\rho_{\mathrm{B}} = \frac{\hat{\mu}'^2\mathfrak{h}^2}{(\ln 2)^2} \in \mathfrak{q}^0$$

[μ_ℓ, μ_ν are the Cal-F/P4 channel scales [FT.113; FT.143(3); Cal-F]; $[\mu_\ell] = [\mu_\nu] = \mathfrak{q}^{+1/2}$ from $m = \mu z$ and $[m] = \mathfrak{q}^{+1/2}$; machine: the exponent arithmetic];

they are restatements of undeclared data, ¬declarations:

¬a P4 addition,

§Cal-BM stays empty

[BM.77(2) pattern];

(5) the inter-family bridge, in full (every cross-currency ratio of the census):

with $m_{\mathrm{hid}}(s)\,\mathfrak{h}/\ln 2 = 2E(s)$, $m^2_{\mathrm{pole}}(s)/\hat{\mu}'^2 = D_S(s)$, $m_i^{(\mathrm{ch})} = \mu_{\mathrm{ch}} z_i^{(\mathrm{ch})}$:

$$\boxed{\frac{m_i^{(\ell)}}{m_{\mathrm{hid}}(s)} = \frac{\sigma_\ell\, z_i^{(\ell)}}{2E(s)}, \qquad \frac{m_j^{(\nu)}}{m_i^{(\ell)}} = \frac{\sigma_\nu\, z_j^{(\nu)}}{\sigma_\ell\, z_i^{(\ell)}}, \qquad \frac{m_{\mathrm{pole}}(s)}{m_i^{(\ell)}} = \frac{2E(s)}{\sigma_\ell\, z_i^{(\ell)}} \cdot \frac{\sqrt{\rho_{\mathrm{B}}\,\mathcal{R}_s}}{1}}$$

[machine: the three identities cleared in $\mathfrak{P} \otimes \mathbb{Q}[\rho_{\mathrm{B}}, \sigma_\ell, \sigma_\nu]$; BM.82(1), (5) re-cited];

the localization theorem: every cross-currency mass ratio of the census is an exact derived expression in exactly three undeclared dimensionless numbers $(\rho_{\mathrm{B}},\ \sigma_\ell,\ \sigma_\nu)$, and in no others

minimality (each is needed):

$m_{\mathrm{pole}}/m_{\mathrm{hid}}$ depends on ρ_{B} alone;

$m^{(\ell)}/m_{\text{hid}}$ on σ_ℓ alone;

$m^{(\nu)}/m^{(\ell)}$ on σ_ν/σ_ℓ alone

[machine: the exact monomial dependence of each of the three formulas];

and every ratio inside a currency depends on none of them [BM.82(4); machine];

(6) the cross-channel factor census (the nine exact derived factors):

$$\frac{m_i^{(\text{ch})}}{m_i^{(\text{ch}')}} = \frac{\mu_{\text{ch}}}{\mu_{\text{ch}'}} \cdot \frac{z_i^{(\text{ch})}}{z_i^{(\text{ch}')}}$$

[machine: the exact factorization] :

line	$z_i^{(q)}/z_i^{(\nu)}$	$z_i^{(q)}/z_i^{(\ell)}$	$z_i^{(\nu)}/z_i^{(\ell)}$
$i=0$ (the heaviest)	$(\frac{1005964365750}{10^{12}}, \frac{1005964365751}{10^{12}})$	$(\frac{1026138854777}{10^{12}}, \frac{1026138854778}{10^{12}})$	$(\frac{1020054874420}{10^{12}}, \frac{1020054874421}{10^{12}})$
$i=2$ (the middle)	$(\frac{730177488628}{10^{12}}, \frac{730177488629}{10^{12}})$	$(\frac{441377821972}{10^{12}}, \frac{441377821973}{10^{12}})$	$(\frac{604480183033}{10^{12}}, \frac{604480183034}{10^{12}})$
$i=1$ (the lightest)	$(\frac{1963683086044}{10^{12}}, \frac{1963683086045}{10^{12}})$	$(\frac{25609326778349}{10^{12}}, \frac{25609326778350}{10^{12}})$	$(\frac{13041476478742}{10^{12}}, \frac{13041476478743}{10^{12}})$

[machine: monotone exact interval division at width $< 2^{-150}$];

the light-end concentration: $1 < \frac{z_0^{(q)}}{z_0^{(\nu)}}, \frac{z_0^{(\nu)}}{z_0^{(\ell)}}, \frac{z_0^{(q)}}{z_0^{(\ell)}} < \frac{27}{25}$, while $\frac{25609326778349}{10^{12}} < \frac{z_1^{(q)}}{z_1^{(\ell)}} < \frac{26}{1}$

the three channels' heaviest lines are nearly degenerate and their lightest lines differ by more than an order of magnitude — the entire inter-family structure lives at the light end

[machine: the exact bracket comparisons; the structural mirror of BM.79(5)'s top-weight row on the Platonic side [Census FT.76(4) row-wise]];

(7) the inter-family verdict (what is and what is not derived):

derived: the shape invariants $\sum_i z_i$

$= 6,$

Q

$= \frac{2}{3}$ in all three channels;

the one cubic form with three constants;

the hierarchy-order theorem [BM.80$(3), (4), (7), (8)$];

the nine cross-channel factors of (6);

the exact factorization of every cross-currency ratio and its localization to $(\rho_{\mathrm{B}}, \sigma_\ell, \sigma_\nu)$ (5);

derived: the quark side of the inter-family ratio in two currencies — the pole currency's $\hat{\mu}'^2 D_S$ and the informational currency's $2E \ln 2/\hbar$ with the exact bridge factor $\mathcal{R}_s$ [BM.81(1); BM.82(2)];

¬derived: the inter-family mass ratio as a number

it requires ρ_{B} (R1),

σ_ℓ,

σ_ν (Cal-F),

and — on the pole side of a composite — (R3) the composition :

the obstruction is exactly three dimensionless numbers and one attachment, all named, none supplied, none fitted

[BM.69$(4), (5)$; BM.60(7)];

(8) the absolute neutrino mass:

¬derived

[FT.143(3) : μ_ν undeclared] :

the ν-channel's internal ratios are derived (6), its absolute scale is not;

the two-bit floor applies to it on the informational read:

$m_\nu \mathfrak{h} \geq 2 \ln 2$ once materialized as a bound chain,

$m_\nu \mathfrak{h} \geq 0$ by PSD alone

[BM.70$(2), (3)$ re-cited];

(9) the bars:

¬∃ an SM mass, mass ratio, mixing angle or Yukawa value in any clause

[P5; machine: zero external tokens, zero decimal numerals, zero [EXT] markers in this tier];

¬a measured external constant is identified with any corpus quantity [P5];

species names are confined to the identification layers [Def-BM.3(7); Def-BM.6(7); Def-BM.14(8)];

topology labels are internal bookkeeping [Def-H.32(4)];

amplitude ≠ mass [FT.79(5)];

rung ≠ scale [D0.8];

¬a currency is silently converted anywhere in this theorem;

(10) status:

derived: (1)–(8);

localized, ¬filled: (7)'s obstruction;

declared: nothing — (4)'s three symbols are restatements of Cal-F/P4 data;

¬a P4 addition, ¬a new dimensionful scale [P1], ¬a measured value [P5],

the mass-form bridge is application-layer, its residue reduced to $(\rho_{\mathrm{B}}, \sigma_\ell, \sigma_\nu)$ and (R2), (R3).

Req. D0.8, D2.9, CT2.1 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.32(4), Def-H.37(2), FT.79(5), FT.113, FT.143(3) (Part 3); Def-BM.3(7), Def-BM.6(7), Def-BM.8(1), Def-BM.14(8), BM.28(1), BM.28(4), BM.30(2), BM.30(5), BM.35(5), BM.37(2), BM.59(4), BM.59(5), BM.60(7), BM.62(4), BM.62(5), BM.69(3), BM.69(4), BM.69(5), BM.70(2), BM.70(3), BM.75(4), BM.77(2), BM.79(5), BM.80(3), BM.80(4), BM.80(6), BM.80(7), BM.80(8), BM.81(1), BM.82(1), BM.82(2), BM.82(4), BM.82(5) (this part); Cal-F; exact arithmetic in the prime-log ring with integer sign tests, unique-factorization independence, exact symbolic solving in $\mathfrak{P}$, Laurent-monomial assembly and $\mathfrak{q}$-exponent arithmetic, monotone exact interval division [math; machine: `checks/verify_H_mass_ratio_census.py` §F].

Pf.

(1) : the four rows re-cited from their own tags [BM.62(4), (5); BM.28(4)(ii); BM.30(2); BM.59(4), (5); BM.69(3); BM.75(4)];

the formed-object column: two ratios and two differences [machine: the token census of the four expressions]

$$(2) : \ 1280$$

$$= 2^8 \cdot 5 \text{ and } 960$$

$$= 2^6 \cdot 3 \cdot 5 \text{ [machine: exact integer factorizations]} \implies \ln 1280$$

$$= 8 \ln 2 + \ln 5,$$

$$\ln 960$$

$$= 6 \ln 2 + \ln 3 + \ln 5 \text{ [machine: the prime-log representation]};$$

the lemma: $\exp\left(\sum N a_i \ln p_i\right) = \prod p_i^{N a_i} = 1 \implies N a_i = 0 \ \forall i$ [math: unique factorization]

machine: the exhaustive census of $2^a 3^b 5^c = 1$ over $|a|, |b|, |c| \leq 24$ returns exactly $(0, 0, 0)$;

$$\text{the rational case : } \; q(8\ln 2+\ln 5)$$

$$= p((6+t)\ln 2+\ln 3+\ln 5) \implies \text{the } \ln 3 \text{ coefficient } -p$$

$$= 0$$

$$\implies p$$

$$= 0$$

$$\implies q\cdot 8\ln 2+q\ln 5$$

$$= 0$$

$$\implies q$$

$$= 0 \text{ [machine: the coefficient extraction and the lemma]}$$

$$(3): \; 3\ln 1280 = 24\ln 2+3\ln 5;$$

$$4\ln 960+4t\ln 2 = (24+4t)\ln 2+4\ln 3+4\ln 5 \text{ [machine]};$$

$$\text{the difference } 4\ln 960+4t\ln 2-3\ln 1280$$

$$= 4\ln 3+\ln 5+4t\ln 2$$

$$= \ln 405+4t\ln 2 \text{ [machine: the prime-log ring]}$$

$$\ln 405>0 \iff 405>1, \quad \ln 16>0 \iff 16>1 \implies t=-\ln 405/\ln 16<0 \text{ [machine]};$$

$$\ln 1280>\ln 960 \iff 1280>960 \text{ [machine]};$$

$$\text{the kind row : } \; \tfrac{4}{3}\in\mathbb{Q}, \; \tfrac{\ln 1280}{\ln 960}\notin\mathbb{Q} \; [(2)]$$

$$(4): \; [m]=\mathfrak{q}^{+1/2} \text{ [CT2.1; D2.9]}, \quad z\in\mathfrak{q}^0 \implies [\mu_{\text{ch}}]=\mathfrak{q}^{+1/2};$$

$$[\mathfrak{h}] = \mathfrak{q}^{-1/2}, \quad [\ln 2] = \mathfrak{q}^{0} \implies \sigma_{\mathrm{ch}} \in \mathfrak{q}^{0} \text{ [machine: the exponent arithmetic]};$$

$$\rho_{\mathrm{B}} \in \mathfrak{q}^{0} \text{ [BM.82(1)]}$$

$$(5): \ \frac{\mu_\ell z}{2E \ln 2/\mathfrak{h}} = \frac{\mu_\ell \mathfrak{h}}{\ln 2} \cdot \frac{z}{2E} = \frac{\sigma_\ell z}{2E} \text{ [machine: cleared]};$$

$$\frac{\mu_\nu z^{(\nu)}}{\mu_\ell z^{(\ell)}} = \frac{\sigma_\nu z^{(\nu)}}{\sigma_\ell z^{(\ell)}} \text{ [machine]};$$

$$\frac{m_{\mathrm{pole}}}{m_i^{(\ell)}}$$

$$= \frac{m_{\mathrm{pole}}}{m_{\mathrm{hid}}} \cdot \frac{m_{\mathrm{hid}}}{m_i^{(\ell)}}$$

$$= \sqrt{\rho_{\mathrm{B}} \mathcal{R}_s} \cdot \frac{2E}{\sigma_\ell z_i^{(\ell)}} \text{ [machine: the two re-cited identities composed]}$$

minimality : the symbol census of the three expressions — $\sqrt{\rho_{\mathrm{B}} \mathcal{R}_s}$ carries ρ_{B} and neither σ;

$\sigma_\ell z/2E$ carries σ_ℓ and neither ρ_{B} nor σ_ν;

$\sigma_\nu z^{(\nu)}/(\sigma_\ell z^{(\ell)})$ carries both σ's and no ρ_{B} [machine]

(6) : the nine factors by monotone exact interval division of the enclosed z's [machine: widths $< 2^{-150}$];

the bracket comparisons against $\frac{27}{25}$ and 26 [machine]

(7)–(10) : the assembly of BM.80's and BM.82's rows;

the three residue symbols re-cited from BM.69(5);

the standing-rule scans [machine]. ■

Phys. Four currencies look at the same two charge classes and no two of them say the same thing. Two of them produce a ratio — four thirds from the ensemble, and the logarithm of one thousand two hundred eighty over the logarithm of nine hundred sixty from the edge count — and two produce a difference: the transport register's log of four thirds over h, and the cone defect's exact algebraic splitting. The tier's central negative is exact rather than numerical. The informational ratio is irrational at every rational coupling, by unique factorization of the primes two,

three and five inside the logarithms; four thirds is rational; so the two can never agree, and solving for the coupling at which they would gives a negative number — the logarithm of four hundred five over the logarithm of sixteen — outside the admissible range entirely. The positive side is a localization. Three dimensionless numbers, each a restatement of data the corpus already carries undeclared, are exactly what stands between the census and an absolute inter-family mass: the pole packaging, and the two channel scales. Every cross-currency ratio in the corpus is an exact expression in those three and nothing else; every ratio inside a currency needs none of them. The nine cross-channel factors then show where the structure lives: the three heaviest lines of the three families are degenerate to within three percent while the three lightest differ by a factor of twenty-six — the same light-end concentration the Platonic ladder shows at its top weights, arriving from an entirely different direction. Nothing is converted, nothing is fitted, and the absolute neutrino mass stays exactly where the ledger says it stays.

Theorem 4.72 (BM.84 — The neutron–proton mass ratio: the exact ratio, the irrational critical locus, the range theorem, and the trichotomy). Thm.

(1) the decomposition and the bridge summand's currency row:

$$\Delta m := m_n - m_p = \Delta m_{\text{Lyapunov}} + \Delta m_{\text{force}} + \Delta m_{\text{metric}} + \Delta m_{\text{binding}} + \Delta m_{\text{self-int}} + \Delta m_{\text{transport}}$$

$$[\text{BM.62}(1)]:$$

$$\Delta m_{\text{Lyapunov}} = 0 \ [\text{BM.51}(6)];$$

$$\Delta m_{\text{metric}} = 0 \ [\text{BM.53}(2);\ \text{BM.58}(4)];$$

$$\Delta m_{\text{binding}} = 0 \ [\text{BM.54}(1)\text{–}(3)];$$

$$\Delta m_{\text{force}} = \Delta m_{\text{self-int}} = -\Delta m_{\text{EM}} \ [\text{BM.52}(4), (5);\ \text{BM.46}(4)];$$

$$\boxed{\Delta m_{\text{transport}} = +\frac{\ln(4/3)}{\mathfrak{h}}, \qquad \Delta m_{\text{EM}} = -\frac{6\, c\, \varepsilon_{\text{EM}} \ln 2}{\mathfrak{h}}, \qquad \Delta m_{\text{bridge}} = 0 \ \text{ in the informational currency}}$$

[BM.59(4); BM.46(4); BM.62(3); the bridge summand's row: Δm_{bridge} is ¬formed in the informational currency — the pole-side companion is $\Delta_g > 0$, a different currency,

and a cross-currency addition is a category the corpus bars [BM.69(3); BM.35(5); BM.37(2); BM.71(1) row 5];

the same substitution's two readings point the same way [BM.75(4), (5)] and are ¬addable];

(2) the exact ratio:

$$m_n\,\mathfrak{h} = 6\ln 2 + \ln 20 = \ln(2^8\cdot 5) = \ln 1280,$$

$$m_p\,\mathfrak{h} = 6\ln 2 + \ln 15 + t\ln 2 = \ln(2^6\cdot 3\cdot 5) + t\ln 2 = \ln 960 + t\ln 2$$

[Def-BM.13(5); machine] :

$$\boxed{\frac{m_n}{m_p} \;=\; \frac{\ln 1280}{\ln 960 + 6\,c\,\varepsilon_{\mathrm{EM}}\ln 2} \;=\; \frac{\ln 1280}{\ln 960 + t\ln 2}, \qquad t \;:=\; 6\,c\,\varepsilon_{\mathrm{EM}} \;\geq\; 0}$$

[BM.62(5) re-derived; machine: the prime-log assembly];

the sub-critical branch is the branch $t < \ln(4/3)/\ln 2$;

the ratio is a dimensionless element of $\mathfrak{P}$, ¬a rational number in general

[BM.83(2)];

(3) the range theorem (derived here):

$$\frac{\partial}{\partial t}\frac{m_n}{m_p} = -\,\frac{\ln 1280\cdot\ln 2}{(\ln 960 + t\ln 2)^2} < 0$$

[machine: the sign census of $\ln 1280,\ \ln 2 > 0$ in the prime-log ring; the monotonicity is cross-multiplied in the Pf] :

$$\boxed{0 \;<\; \frac{m_n}{m_p} \;\leq\; \frac{\ln 1280}{\ln 960} \;<\; \frac{21}{20} \qquad \text{for every admissible } t \geq 0,\ \text{with equality at } t = 0 \text{ only}}$$

$$\frac{\ln 1280}{\ln 960} \in \left(\frac{26}{25}, \frac{21}{20}\right)$$

[machine: $25\ln 1280 > 26\ln 960 \iff 2^{44} > 3^{26}5$; $\quad 20\ln 1280 < 21\ln 960 \iff 2^{34} < 3^{21}5$];

$$2^{44} = 17592186044416 > 12709329141645 = 3^{26}\cdot 5,$$

$$2^{34} = 17179869184 < 52301766015 = 3^{21}\cdot 5$$

[machine: the two integer comparisons];

the exact upper bound of the neutron–proton mass ratio on the informational currency is $\frac{21}{20}$,

attained in the limit $t \to 0^+$,

and no coupling raises it;

(4) the irrationality and the rational locus:

$$\boxed{\frac{m_n}{m_p} \notin \mathbb{Q} \text{ at every rational coupling } t \in \mathbb{Q}}$$

[BM.83(2)'s theorem re-cited] :

$$\frac{m_n}{m_p} = 1 \iff t = \frac{\ln(4/3)}{\ln 2} = 2 - \log_2 3 \notin \mathbb{Q}$$

[machine: solved exactly in $\mathfrak{P}$; $\log_2 3 \notin \mathbb{Q} \iff \neg\exists\,(p,q) \neq (0,0) : 2^p = 3^q$; the census over $|p|, |q| \leq 64$ returns none, and unique factorization closes the general case];

the degenerate locus is exactly one coupling, and that coupling is irrational: at every rational coupling $m_n \neq m_p$

(5) the exact critical strength:

$$\boxed{\varepsilon^{\text{crit}}(c) = \frac{\ln(4/3)}{6\,c\,\ln 2} = \frac{2\ln 2 - \ln 3}{6\,c\,\ln 2} = \frac{2 - \log_2 3}{6\,c}}$$

[BM.62(6) re-derived; machine: the unique zero of Δm] :

the four integer certificates: $$\frac{1}{9} < \varepsilon^{\text{crit}}\left(\frac{3}{5}\right) < \frac{1}{8} < \varepsilon^{\text{crit}}\left(\frac{1}{2}\right) < \frac{1}{7}$$

$$\varepsilon^{\text{crit}}(\tfrac{3}{5}) > \tfrac{1}{9} \iff 2^8 > 3^5 \; (256 > 243);$$

$$\varepsilon^{\text{crit}}(\tfrac{3}{5}) < \tfrac{1}{8} \iff 2^{31} < 3^{20} \; (2147483648 < 3486784401);$$

$$\varepsilon^{\text{crit}}(\tfrac{1}{2}) > \tfrac{1}{8} \iff 2^{13} > 3^8 \; (8192 > 6561);$$

$$\varepsilon^{\text{crit}}(\tfrac{1}{2}) < \tfrac{1}{7} \iff 2^{11} < 3^7 \ (2048 < 2187)$$

[BM.62(7) re-certified; machine];

$$\boxed{\varepsilon^{\text{crit}}(c) \notin \mathbb{Q} \qquad \text{for every rational } c > 0} \qquad [(4)\text{'s } 2 - \log_2 3 \notin \mathbb{Q};\ \text{machine}];$$

(6) the trichotomy (total, exact, and non-empty on both sides):

$$\boxed{\varepsilon_{\text{EM}} < \varepsilon^{\text{crit}}(c) \iff m_n > m_p \iff \frac{m_n}{m_p} > 1; \qquad \varepsilon_{\text{EM}} = \varepsilon^{\text{crit}}(c) \iff m_n = m_p \iff \frac{m_n}{m_p} = 1; \qquad \varepsilon_{\text{EM}} > \varepsilon^{\text{crit}}(c) \iff m_n < m_p \iff \frac{m_n}{m_p} < 1}$$

[BM.62(6), (9) re-derived from (2), (3); machine: the sign census in $\mathfrak{P}$];

the opening region is non-empty and integer-certified:

$$\left\{0 \le \varepsilon_{\text{EM}} \le \tfrac{1}{9}\right\} \subseteq \left\{\varepsilon_{\text{EM}} < \varepsilon^{\text{crit}}(c)\right\} \text{ at both priced } c$$

[BM.62(8) : $4^3 = 64 > 54 = 3^3 \cdot 2$ at $c = \frac{1}{2}$; $4^5 = 1024 > 972 = 3^5 \cdot 2^2$ at $c = \frac{3}{5}$; machine];

the closing region is non-empty:

$$\varepsilon_{\text{EM}} = \tfrac{1}{6} > \tfrac{1}{7} > \varepsilon^{\text{crit}}\left(\tfrac{1}{2}\right) \text{ and } \varepsilon_{\text{EM}} = \tfrac{1}{6} > \tfrac{1}{8} > \varepsilon^{\text{crit}}\left(\tfrac{3}{5}\right)$$

[machine: $6 < 7,\ 6 < 8$ and (5)'s brackets];

the critical locus is exact, unique, and irrational

[(4), (5)];

the neutron is the heavier baryon on the whole opening region

[BM.62(9) re-cited];

(7) the fractional splittings (the exact rational enclosures):

$$\left.\frac{\Delta m}{m_{\text{hid}}}\right|_{t=0} = \frac{\ln(4/3)}{6\ln 2} \in \left(\frac{1}{15}, \frac{1}{14}\right)$$

[BM.62(10); machine: $2^{24} > 3^{15},\ 2^{22} < 3^{14}$];

$$\left.\frac{\Delta m}{m_n}\right|_{t=0} = \frac{\ln(4/3)}{\ln 1280} \in \left(\frac{1}{25}, \frac{1}{24}\right)$$

[BM.62(10); machine];

$$\boxed{\left.\frac{\Delta m}{m_p}\right|_{t=0} = \frac{\ln(4/3)}{\ln 960} \in \left(\frac{1}{24}, \frac{1}{23}\right)}$$

[machine, derived here : $24\ln\frac{4}{3} > \ln 960 \iff 42\ln 2 > 25\ln 3 + \ln 5 \iff 2^{42} > 3^{25}\cdot 5$ $(4398046511104 > 4236443047215)$;

$23\ln\frac{4}{3} < \ln 960 \iff 40\ln 2 < 24\ln 3 + \ln 5 \iff 2^{40} < 3^{24}\cdot 5$ $(1099511627776 < 1412147682405)$];

the strict contraction against the barred measure price:

$$\frac{\ln(4/3)}{6\ln 2} < \frac{1}{3} \iff 3\ln\tfrac{4}{3} < 2\ln 2 \iff 16 < 27$$

[Def-BM.13(8); BM.62(10); machine];

the consistency row:

$$\frac{m_n}{m_p} - 1 = \frac{\Delta m}{m_p} \in \left(\frac{1}{24}, \frac{1}{23}\right) \text{ at } t = 0 \quad \text{and} \quad \frac{m_n}{m_p} = \frac{\ln 1280}{\ln 960} \in \left(\frac{26}{25}, \frac{21}{20}\right) :$$

$$\boxed{1 + \left(\frac{1}{24}, \frac{1}{23}\right) = \left(\frac{25}{24}, \frac{24}{23}\right) \subset \left(\frac{26}{25}, \frac{21}{20}\right)}$$

[machine: $\frac{26}{25} < \frac{25}{24} \iff 624 < 625; \quad \frac{24}{23} < \frac{21}{20} \iff 480 < 483$] :

the finer splitting bracket implies the coarser ratio bracket;

(8) what the ratio is not:

$$\frac{m_n}{m_p} \neq \frac{4}{3} \text{ at every admissible coupling}$$

[BM.83(3); machine: the exact solution $t = -\ln 405/\ln 16 < 0$];

$$\frac{m_n}{m_p} \neq \frac{\ln 1280}{\ln 960} \text{ at } t > 0$$

[(3)'s strict monotonicity];

¬a measured neutron–proton ratio is imported, compared or quarantined anywhere in this tier

[P5; machine: zero external tokens];

the numerical value of e remains open and does not enter:

$\varepsilon_{\mathrm{EM}} = e^2/(4\pi)$ is carried symbolically

[EM.25(7), (9); BM.62(12) pattern];

(9) status:

derived: (1)–(8);

the sign theorem is conditional on the sub-critical branch, with the hypothesis on its face and the critical value exact (5),

(6) [A §0.2a];

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a route adopted [BM.30(5)],

¬a measured value [P5],

§Cal-BM stays empty;

the open row: the numerical value of e [EM.25(7) : one calibration datum] — it decides which side of $\varepsilon^{\mathrm{crit}}$ the corpus sits on, and no theorem of Parts 1–4 supplies it

[BM.62(12) re-cited].

Req. EM.25(7), EM.25(9) (Part 2); P1, P4, P5 (Parts 1–2); A §0.2a (Parts 1–2); FT.79(5) (Part 3); Def-BM.13(5), Def-BM.13(8), BM.30(5), BM.35(5), BM.37(2), BM.46(4), BM.51(6), BM.52(4), BM.52(5), BM.53(2), BM.54(1), BM.54(3), BM.58(4), BM.59(4), BM.62(1), BM.62(3), BM.62(5), BM.62(6), BM.62(7), BM.62(8), BM.62(9), BM.62(10), BM.62(12), BM.69(3), BM.71(1), BM.75(4), BM.75(5), BM.83(2), BM.83(3) (this part); exact prime-log-ring arithmetic with integer power certificates, symbolic Laurent differentiation, exact rational enclosures of ln-ratios, unique-factorization irrationality arguments, sign censuses [math; machine: `checks/verify_H_mass_ratio_census.py` §G].

Pf.

(1) : the five zero-or-monomial rows re-cited from their own tags [BM.51(6); BM.53(2); BM.54(1)–(3); BM.52(4), (5); BM.46(4); BM.59(4)];

$$\text{the bridge row : } \Delta_g$$

$$= \hat{\mu}'^2(D_S(+) - D_S(-)) \text{ has } [\Delta_g]$$

$$= \mathfrak{q} \text{ and } [\Delta m]$$

$$= \mathfrak{q}^{+1/2} \text{ [machine: the exponent arithmetic]} \implies \neg\text{addable [BM.37(2) pattern]}$$

$$(2) : \ 6\ln 2 + \ln 20 = 6\ln 2 + \ln 4 + \ln 5 = 8\ln 2 + \ln 5 = \ln(2^8 \cdot 5) = \ln 1280 \text{ [machine]};$$

$$6\ln 2 + \ln 15 = 6\ln 2 + \ln 3 + \ln 5 = \ln(2^6 \cdot 15) = \ln 960 \text{ [machine]};$$

$$t = 6c\,\varepsilon_{\text{EM}} \geq 0 \text{ since } c > 0,\ \varepsilon_{\text{EM}} = e^2/(4\pi) \geq 0 \text{ [machine]}$$

$$(3) : \text{ monotonicity by cross-multiplication: } t_1 < t_2 \implies \frac{A}{B + t_1 \ln 2} > \frac{A}{B + t_2 \ln 2}$$

$$\iff A\big(B + t_2 \ln 2\big) > A\big(B + t_1 \ln 2\big)$$

$$\iff A$$

$$(t_2 - t_1)\ln 2 > 0 \text{ at } A$$

$$= \ln 1280,$$

$$B \qquad = \ln 960$$

[machine: the three positivity rows $\ln 1280 > 0,\ \ln 960 > 0,\ \ln 2 > 0$ in the prime-log ring, and the cross-multiplied comparison on a rational grid of pairs (t_1, t_2)];

the derivative form $-A \ln 2/(B + t \ln 2)^2$ is the same sign census;

$t = 0$ gives the supremum

$$25 \ln 1280 - 26 \ln 960$$

$$= (200 - 156) \ln 2 - 26 \ln 3 + (25 - 26) \ln 5$$

$$= 44 \ln 2 - 26 \ln 3 - \ln 5 > 0$$

$\iff 2^{44} > 3^{26} \cdot 5$ [machine: the two integers];

$$20 \ln 1280 - 21 \ln 960 = 34 \ln 2 - 21 \ln 3 - \ln 5 < 0 \iff 2^{34} < 3^{21} \cdot 5 \text{ [machine]}$$

$$(4) : \ \frac{m_n}{m_p}$$

$$= 1 \iff \ln 1280$$

$$= \ln 960 + t \ln 2$$

$$\iff t$$

$$= \frac{\ln 1280 - \ln 960}{\ln 2}$$

$$= \frac{\ln(4/3)}{\ln 2}$$

$$= \frac{2 \ln 2 - \ln 3}{\ln 2}$$

$$= 2 - \log_2 3 \text{ [machine: solved in } \mathfrak{P}];$$

$$\log_2 3 \in \mathbb{Q} \implies \exists\,(p,q) \neq (0,0) :\ 2^p = 3^q \text{ [clearing denominators]};$$

machine: no solution with $|p|, |q| \leq 64$;

math: unique factorization excludes every case ($p, q > 0$: parity of the left side; $p \leq 0$ or $q \leq 0$: the reciprocal argument)

$$(5) :\ \Delta m = 0 \iff \ln(4/3) = t \iff 6c\,\varepsilon_{\text{EM}} = \ln(4/3) \iff \varepsilon_{\text{EM}} = \frac{\ln(4/3)}{6c\ln 2} \text{ [machine]};$$

the four cross-multiplications of BM.62(7) re-certified as the four integer power comparisons [machine]

(6) : (3)'s monotonicity $\wedge$ (4)'s unique zero $\implies$ the three-way sign equivalence [machine];

the opening witnesses $4^3 = 64 > 54 = 3^3 \cdot 2$ and $4^5 = 1024 > 972 = 3^5 \cdot 2^2$ [machine];

the closing witness $\frac{1}{6} > \frac{1}{7} > \varepsilon^{\text{crit}}(\frac{1}{2})$ and $\frac{1}{6} > \frac{1}{8} > \varepsilon^{\text{crit}}(\frac{3}{5})$ [machine: $6 < 7$, $6 < 8$, (5)'s brackets]

(7) : the two frozen enclosures re-certified [machine];

$$24\ln\tfrac{4}{3} - \ln 960$$

$$= (48-6)\ln 2 - (24+1)\ln 3 - \ln 5$$

$$= 42\ln 2 - 25\ln 3 - \ln 5 > 0$$

$$\iff 2^{42} > 3^{25} \cdot 5 \text{ [machine]};$$

$$23\ln\tfrac{4}{3} - \ln 960 = 40\ln 2 - 24\ln 3 - \ln 5 < 0 \iff 2^{40} < 3^{24} \cdot 5 \text{ [machine]}$$

$$3\ln\tfrac{4}{3} < 2\ln 2 \iff \ln\tfrac{64}{27} < \ln 4 \iff 64 < 108 \text{ [machine]};$$

$$1 + \tfrac{1}{24}$$

$$= \tfrac{25}{24} \text{ and } \tfrac{25}{24} < \tfrac{26}{25}$$

$$\iff 625 < 624 \text{ is false} \implies \text{the two brackets overlap and are reported separately [machine]}$$

$$(8): \text{ BM.83(3)'s exact solution re-cited;}$$

$$(3)\text{'s strict monotonicity;}$$

$$\text{the token scans [machine];}$$

$$(9): \text{ the standing-rule scans and the assembly [machine]. } \blacksquare$$

Phys. The neutron–proton mass ratio is one exact expression: the logarithm of one thousand two hundred eighty over the logarithm of nine hundred sixty plus a nonnegative multiple of the logarithm of two. Everything about it follows by arithmetic in the ring generated by the logarithms of the primes. It decreases strictly with the coupling, so its largest value is the coupling-free one, and that value sits between twenty-six twenty-fifths and twenty-one twentieths — an exact upper bound of one and one twentieth on the ratio, valid at every coupling the corpus admits. It is irrational at every rational coupling, by unique factorization of two, three and five inside the logarithms, and it equals one at exactly one coupling, which is two minus the base-two logarithm of three and is therefore irrational too: the degenerate point of the trichotomy is provably not a rational coupling, and at every rational coupling the two baryons differ. The critical strength inherits the same irrationality and the same four integer certificates that enclose it between a ninth and a seventh, so the opening region is a wide exact interval, not a tuned point, and the closing region is inhabited by anything above a sixth. The bridge term contributes nothing here and cannot: it lives in the angle currency, it points the same way, and adding it to a bit count is the one operation the ledger forbids. No measured ratio enters, no external value is compared, and the single open datum — the numerical value of the electric coupling — is the same one the corpus has named at every tier, deciding which side of the critical surface the world sits on without ever being supplied.

Theorem 4.73 (BM.85 — The Q-value in two formed reads: the ratio forms, the margin ladders, and the read-relative verdict). Thm.

$$(1) \quad \text{the Q-value on the informational currency (formed):}$$

$$Q_\beta := m_n - m_p - m_\ell - m_\nu = \Delta m - m_\ell - m_\nu$$

$$[\text{BM.41}(4);\ \text{BM.49}(1);\ \text{BM.63}(1)]:$$

$$\boxed{Q_\beta\,\mathfrak{h} \;=\; \ln\frac{4}{3} - t\ln 2 - F_{\ell\nu}, \qquad F_{\ell\nu} := m_\ell\mathfrak{h} + m_\nu\mathfrak{h}, \qquad t = 6\,c\,\varepsilon_{\mathrm{EM}} \geq 0}$$

[BM.62(4); BM.70(1); machine: the prime-log assembly];

(2) the two-bit daughter floor and the closure (re-certified):

$$F_{\ell\nu} \geq 2\ln 2$$

[BM.70(2), (3) : every materialized closed chain carries ≥ 2 directed edges; $m\mathfrak{h} = E_{\text{int}} \ln 2$; $m_\nu \mathfrak{h} \geq 0$ by PSD] :

$$\boxed{Q_\beta \mathfrak{h} \;\leq\; \ln\frac{4}{3} - 2\ln 2 \;=\; \ln\frac{4}{3} - \ln 4 \;=\; -\ln 3 \;<\; 0}$$

[machine: $4 < 12$];

formed $\Longrightarrow$ closed on the informational currency

[BM.70(4)(a) re-cited];

the margin ladder:

$$Q_\beta \mathfrak{h} \;\leq\; -\ln 3, \quad -\ln 12, \quad -\ln 24$$

on the certificates $\quad 4 < 12, \quad 4 < 48, \quad 4 < 96$

[BM.70(3); machine];

each margin is deepened by $t \ln 2 \geq 0$ at every positive coupling [machine: the sign census];

(3) the ratio forms (derived here):

$$\left.\frac{Q_\beta}{m_n}\right|_{t=0} \;\leq\; -\frac{\ln 3}{\ln 1280} \;\in\; \Big(-\frac{2}{13}, -\frac{3}{20}\Big),$$

$$\left.\frac{Q_\beta}{m_p}\right|_{t=0} \;\leq\; -\frac{\ln 3}{\ln 960} \;\in\; \Big(-\frac{4}{25}, -\frac{3}{19}\Big)$$

[machine: $-\frac{2}{13} < -\frac{\ln 3}{\ln 1280} \iff 13\ln 3 < 2\ln 1280 \iff 3^{13} < 2^{16}\cdot 5^2$ $(1594323 < 1638400)$;

$$-\tfrac{\ln 3}{\ln 1280} < -\tfrac{3}{20} \iff 20\ln 3 > 3\ln 1280 \iff 3^{20} > 2^{24}\cdot 5^{3}\ (3486784401 > 2097152000);$$

$$-\tfrac{4}{25} < -\tfrac{\ln 3}{\ln 960} \iff 25\ln 3 < 4\ln 960 \iff 3^{21} < 2^{24}\cdot 5^{4}\ (10460353203 < 10485760000);$$

$$-\tfrac{\ln 3}{\ln 960} < -\tfrac{3}{19} \iff 19\ln 3 > 3\ln 960 \iff 3^{16} > 2^{18}\cdot 5^{3}\ (43046721 > 32768000)];$$

the informational Q-value is a strictly negative exact fraction of the parent mass, bounded away from zero by a rational

in both normalizations;

(4) the Q-value on the specified read (formed, branch-relative):

$$Q_{\beta}^{\text{spec}} := m_{n}^{(\text{ii})} - m_{p}^{(\text{ii})} - m_{\ell} - m_{\nu}$$

[BM.77(1), (2); A-BM.8] :

$$\boxed{Q_{\beta}^{\text{spec}}\,\mathfrak{h} \;=\; \Delta T(S) - \rho_{\ell}\sqrt{G\big(c(0)\big)} - \rho_{\nu}\sqrt{G\big(c(i_{\nu})\big)}, \qquad \Delta T(S) = 5\,m_{B}^{2}(\cdot, S)}$$

$$\Delta T = \Big(\frac{4219785}{98956},\ \frac{9608310}{393679},\ \frac{455}{23},\ \frac{68565}{3088}\Big) \text{ at } S = (0,1,2,3),$$

$$\Delta T_{\min} = \frac{455}{23},$$

$$\Delta T_{\max} = \frac{4219785}{98956}$$

[BM.77(2); machine];

$$\sqrt{A_0 - B_0} \;\le\; \sqrt{G\big(c(i)\big)} \;\le\; \sqrt{A_0 + B_0},$$

$$A_0 \pm B_0\big|_{g=g'=1} = \frac{11}{16},\ \frac{3}{16}$$

[BM.39(1); BM.77(4); machine];

the trichotomy with integer-certified witnesses on both sides:

$$\rho_\ell = \rho_\nu = \frac{\Delta T(S)}{4\sqrt{A_0 + B_0}} \in \mathcal{O}_S,$$

$$\rho_\ell = \rho_\nu = \frac{\Delta T(S)}{\sqrt{A_0 - B_0}} \in \mathcal{C}_S$$

[BM.77(5)(a), (b)] :

$$Q_\beta^{\text{spec}}\mathfrak{h} \geq \frac{455}{23} - \frac{\sqrt{11}}{2} > 0 \iff 828100 > 5819;$$

$$Q_\beta^{\text{spec}}\mathfrak{h} \leq \frac{4219785}{98956} - 50\sqrt{3} < 0 \text{ on } 4219785 < 43 \cdot 98956 = 4255108 \text{ and } 7500 > 1849;$$

$$\frac{273}{23} < \rho^{\text{crit}}(S, i_\nu) < \frac{42197850}{791648}$$

at the witness couplings $g = g' = 1,\ \rho_\ell = \rho_\nu$

[BM.77(5)(c); machine: $99 < 100,\ 75 > 64,\ 273 > 253,\ 42197850 < 42748992$];

(5) the two reads are ¬comparable, and both verdicts stand: the alphabet census:

$$Q_\beta^{\text{info}} : \{\ln 2, \ln 3, \ln 5,\ c,\ \varepsilon_{\text{EM}},\ \mathfrak{h},\ m_\ell,\ m_\nu\};$$

$$Q_\beta^{\text{spec}} : \{m_B^2(\cdot, S),\ \rho_\ell,\ \rho_\nu,\ g,\ g',\ \mathfrak{h}\}$$

[machine: the token census] :

the two alphabets are disjoint except for $\mathfrak{h}$

[BM.37(2); BM.62(11); BM.66(6) pattern; machine];

¬a derived equation relates the two expressions — that is (R1) and (R3) and the Cal-F scales together

[BM.82(6); BM.83(5), (7)];

the verdicts, read-relative and complete:

closed on the informational currency for every member [BM.70(4)(a)];

open on $\mathcal{O}_S,$ closed on $\mathcal{C}_S,$ degenerate on $\partial\mathcal{O}_S$ on the specified read [BM.77$(5), (6)$];

¬a clause of BM.41, BM.49, BM.63, BM.70, BM.71 is retracted;

the rescopes are the ones BM.77(7) records

[machine: the clause census];

(6) the parent row is currency-independent:

$$T(0,S) = \tfrac{4}{3}T(+1,S) > T(+1,S)$$

[BM.28(4)(ii); machine]

and

$m_n > m_p$ on the sub-critical branch

[BM.62(9); BM.84(6)] :

the decay direction $n \to p + \ell + \nu$ is fixed in both currencies

[BM.41(1); BM.45; BM.77(6) : the grammar stands unmodified];

the two-body companion on the specified read:

$\Delta T(S) \geq \rho_\ell \sqrt{G\big(c(0)\big)}$ holds at the witness $\Big[\frac{455}{23} > \frac{\sqrt{11}}{4} \iff 3312400 > 5819\Big]$,

$\mathcal{O}_S \subseteq$ {two-body admissible}

[BM.77(6); machine];

(7) the conditional spectrum becomes branch content on $\mathcal{O}_S$:

$$E_\ell^{\max} = \frac{m_n^2 + m_\ell^2 - (m_p + m_\nu)^2}{2m_n},$$

$E_{\max} : \langle E \rangle = 2 : 1$ scale-invariant

[BM.41(5); BM.49(3), (4); BM.77(8) re-cited] :

¬an external endpoint, spectrum or lifetime is compared

[P5];

the quarantined audit rows stay quarantined

[Census FT.114, FT.120 pattern];

(8) status:

derived: (1)–(7);

read-relative: the kinematic verdict (5);

declared: nothing here — A-BM.8's specification is re-cited, ¬re-issued;

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a currency converted;

§Cal-BM stays empty.

Req. P1, P4, P5 (Parts 1–2); FT.79(5) (Part 3); Census FT.114, FT.120 (row-wise); Def-BM.13(5), BM.28(4), BM.37(2), BM.39(1), BM.41(1), BM.41(4), BM.41(5), BM.45, BM.49(1), BM.49(3), BM.49(4), BM.62(4), BM.62(9), BM.62(11), BM.63(1), BM.66(6), BM.70(1), BM.70(2),

BM.70(3), BM.70(4), BM.71, BM.77(1), BM.77(2), BM.77(4), BM.77(5), BM.77(6), BM.77(7), BM.77(8), BM.82(6), BM.83(5), BM.83(7), BM.84(6) (this part); the declaration addendum A-BM.8, item 7 (pointers); exact prime-log arithmetic with integer power certificates, exact rational arithmetic on the four frozen sector expectations, exact radical inequalities by integer certificates, token censuses [math; machine: `checks/verify_H_mass_ratio_census.py` §H].

Pf.

$$(1):\ \Delta m = (\ln\tfrac{4}{3} - t\ln 2)/\mathfrak{h}\ \text{[BM.62(4) re-cited]};$$

$$Q_\beta\mathfrak{h} = \ln\tfrac{4}{3} - t\ln 2 - (m_\ell\mathfrak{h} + m_\nu\mathfrak{h})\ \text{[machine: the monomial sum in } \mathfrak{P}\text{]}$$

$$(2):\ F_{\ell\nu} \geq 2\ln 2\ \text{[BM.70(2), (3) re-cited]};$$

$$\ln\tfrac{4}{3} - 2\ln 2 = 2\ln 2 - \ln 3 - 2\ln 2 = -\ln 3\ \text{[machine]};$$

$$-\ln 3 < 0 \iff 3 > 1 \iff 4 < 12\ \text{[machine]};$$

$$\text{the ladder on } 4 < 12,\ 4 < 48,\ 4 < 96\ \text{[machine]}$$

$$(3):\ Q_\beta/m_n = (Q_\beta\mathfrak{h})/(m_n\mathfrak{h}) \leq (-\ln 3)/\ln 1280 \text{ at } t = 0\ \text{[machine: } m_n\mathfrak{h} = \ln 1280 > 0\text{]};$$

$$\text{the four enclosures by the four integer power comparisons of } (3)\ \text{[machine]}$$

$$(4):\ \text{the four } \Delta T \text{ values re-cited [BM.77(2); machine: } 5\cdot\tfrac{843957}{98956},\ 5\cdot\tfrac{1921662}{393679},\ 5\cdot\tfrac{91}{23},\ 5\cdot\tfrac{13713}{3088} \text{ as exact integers}];$$

$$\text{the sandwich and both witnesses re-certified [BM.39(1); BM.77(4), (5); machine]}$$

$$(5):\ \text{the token census of the two expressions [machine]};$$

$$\text{the clause census of BM.41–BM.71 against retraction [machine]};$$

$$\text{the no-equation row re-cited from BM.82(6), BM.83(5), (7)}$$

$$(6):\ T(0,S)/T(+1,S) = W(0)/W(+1) = 20/15 = 4/3\ \text{[machine]};$$

$$m_n > m_p \text{ sub-critically [BM.84(6)]};$$

$\frac{455}{23} > \frac{\sqrt{11}}{4} \iff 1820 > 23\sqrt{11} \iff 3312400 > 5819$ [machine: squaring both positive sides]

(7) : the endpoint identity and the $2{:}1$ moment ratio re-cited [BM.41(5); BM.49$(3), (4)$];

(8) : the standing-rule scans [machine]. ■

Phys. One Q-value, two formed reads, and no equation between them. Read in bits, the decay's energy balance is the transport register's log of four thirds, less the electromagnetic scar's coupling term, less whatever the two emitted lines cost — and the two-bit floor on any materialized chain drives the total below minus the logarithm of three at every coupling, so on that currency the vertex is closed with an exact margin, and the margin is a rational fraction of the parent's own mass: worse than minus three twentieths, better than minus two thirteenths. Read on the specified route, the same vertex is a difference of five times a sector expectation against two undeclared scale combinations, and its sign is decided by where those scales sit against an exact rational: open on a region certified by eight hundred twenty-eight thousand one hundred against five thousand eight hundred nineteen, closed on its complement, degenerate on the boundary between eleven and fifty-four. The two expressions share one symbol — h — and nothing else, so the two verdicts are not in conflict and cannot be reconciled without the very bridge the ledger holds open. What both currencies do agree on, and what no read can change, is the direction: the neutral class is the heavier one in the ensemble, on the informational branch below the critical coupling, and on the specified read at every winding cell, so the parent is the neutron and the grammar stands. The decay is allowed where a read is formed that allows it, closed where a read is formed that closes it, and the census reports both without converting either.

BM.86 — The status of the mass ratio census: adjudication and scope. Structural-constraints record: see Appendix B, Section B.1.26 of the Addendum.

4.20 Tier BM.18 — The EM Self-Energy Correction: the Category Adjudication, the Running Coupling, the Path-Integral Self-Energy, the Branch-Blindness Theorem, and the Corrected Neutron–Proton Splitting

Locus. This tier relocates one computation and voids one coefficient. The EM self-energy of a sealed composite is taken off the chart — where it was read as an integral over a charge distribution [Def-BM.12(5); BM.46(1)] — and put on the flicker ensemble, where the corpus's own charge datum lives: a ternary end read $q \in \{-1, 0, +1\}$ on the endpoints of chains [D3.4; EM.9] with no position datum attached [BM.28(7); BM.46(2)]. The distribution factor c that priced a shell against a ball is adjudicated a classical artifact and voided as a mass datum; the exterior value it carried is derived, twice, from rows that need no density (BM.87). The coupling stops being a scalar and becomes an exact geometric profile on the envelope ratio, the lapse and the curvature lock (BM.88). The self-energy becomes an ensemble average with one derived attenuation, the crystallization weight (BM.89). The proposal that the six-state defect splitting reweight that attenuation is put on its face and refuted on four grounds (BM.90). The neutron–proton splitting, its critical strength,

its ratio range and its trichotomy are recomputed with the corrected term (BM.91), the Q-value rows are re-certified on both formed reads (BM.92), and BM.93 adjudicates.

Standing rows. α_B stays barred on BM.61(8)'s four grounds and §Cal-BM stays empty. No clause of BM.1–BM.86 is retracted, replaced or renumbered: every row this tier touches is re-scoped at its own locus and recorded in BM.87(6), BM.91(8) and BM.93(4), in BM.77(7)'s pattern. The numerical value of e stays open [EM.25(7); BM.62(12); BM.84(9)] and no measured coupling, mass or ratio enters any derivation [P5]. The relating equation stays ¬derived — the mass-form bridge is application-layer content [BM.35(5); BM.69(4), (5); BM.83(5)]. The currencies stay distinct: the ensemble weight derived here attenuates the informational register alone and converts nothing into the defect register [BM.82(4); BM.83(3)]. Local notation, fixed for the whole tier: $k := r/\Delta x_{\mathrm{B}}$ the envelope ratio; $f(k) = 1 - \frac{1}{k}$ the nearby-space fraction; N the lapse; $G := (\Delta x_{\mathrm{B}})^2\kappa^2\rho$ the curvature lock; w the crystallization weight; g the curvature composition (the italic g keeps its BM.39(1) meaning, the SU(2) coupling, at every occurrence); $\varepsilon_{\mathrm{EM}} = e^2/(4\pi)$; the symbol c is void as a mass datum and occurs in this tier only inside barred-object and rescope rows.

BM.87 — The category adjudication: the void distribution factor, the derived exterior value, and the standing rows. Structural-constraints record: see Appendix B, Section B.1.27 of the Addendum.

Theorem 4.74 (BM.88 — The running coupling: the exact geometric profile on the envelope ratio, the lapse, and the curvature lock). Thm.

(1) the coupling is a function, ¬a scalar:

$$e = \frac{gg'}{\sqrt{g^2 + g'^2}} = g \sin\theta_W$$

[EW.35; EM.25(6) : fixed with zero freedom as a combination];

¬∃ a theorem fixing the numerical value of e

[EM.25(7); P4];

the running in this corpus is geometric, ¬perturbative:

¬∃ a loop, a β-function, a counterterm or a Yukawa-type object in the inventory

[BM.61(8); FT.145(3); EW.15; BM.47(2) : the divergence is cut by the envelope, ¬by a counterterm];

(2) the three derived dimensionless arguments:

(i) the envelope ratio: $k := \dfrac{r}{\Delta x_{\mathrm{B}}(E)}$,

$$\Delta x_{\mathrm{B}}(E) = \frac{1}{m_{\mathrm{hid}}(E)} = \frac{\mathfrak{h}}{E_{\mathrm{int}}(E)\ln 2}$$

[Def-BM.4(1); BM.13(1); machine: $\Delta x_{\mathrm{B}} \cdot m_{\mathrm{hid}} = 1$ as a Laurent identity] :

$$E_{\mathrm{int}} \in \{2,\ 6,\ 12,\ 24,\ 24,\ 60,\ 60\} \text{ on the computed cells}$$

[BM.13(2); FT.28; Census FT.35(2), FT.42(5) row-wise; BM.13(3) : $E_{\mathrm{int}} \geq 2$] :

a strictly decreasing discrete ladder of widths,

$$k \geq 1 \text{ on the formed locus } r \geq \Delta x_{\mathrm{B}}$$

[machine: injective in E_{int}];

(ii) the nearby-space fraction:

$$\boxed{f(k) \ := \ 1 - \frac{1}{k} \ = \ 1 - \frac{\Delta x_{\mathrm{B}}(E)}{r}}$$

[BM.47(6); machine: exact rational functions] :

$$f(1) = 0, \quad f(2) = \frac{1}{2}, \quad f(3) = \frac{2}{3}, \quad f(4) = \frac{3}{4}, \quad f(5) = \frac{4}{5}, \quad f(6) = \frac{5}{6};$$

f strictly increasing on $k \geq 1$,

$$0 \leq f(k) < 1 \ \ \forall k < \infty$$

[machine];

(iii) the lapse:

$$N = \sqrt{\Omega^2 + \gamma^{ij} N_i N_j},$$

$$\gamma_{ij} = \Omega^2(\delta_{ij} + \varkappa q_{ij}),$$

$$N_i = \Omega^2 \varkappa \beta w_i$$

[CT1.12; BM.34(1); FT.157(1) : the shift drag] :

$$\boxed{N^2 \det\gamma = -\det g}$$

[BM.34(1); machine: the exact polynomial identity over $\mathbb{Q}[\Omega^2, \varkappa, \beta, w_i, q_{ij}]$];

$$(a_{00}, q_{ij}, w_i) = 0 \implies (\Omega, N, N^i, g_{ij}) = (1, 1, 0, \delta_{ij})$$

[BM.34(2) : the vacuum's emergent metric is flat; machine: exact substitution] $\implies$ $\boxed{N = 1 \text{ in the vacuum gauge}}$

(3) the curvature lock (the derived dimensionless curvature argument):

$$\kappa^2\rho = \frac{2\pi E_{\text{int}}^4 (\ln 2)^3}{\mathfrak{h}^2},$$

$$\rho = \frac{(E_{\text{int}} \ln 2)^4}{\mathfrak{h}^4},$$

$$[\rho] = \mathfrak{q}^{+2}$$

[BM.15(2); BM.14(2), (4); CT1.23] :

$$\boxed{G(E) \;:=\; (\Delta x_{\text{B}})^2 \cdot \kappa^2\rho \;=\; 2\pi\, E_{\text{int}}(E)^2 \ln 2 \;\in\; \mathfrak{q}^0}$$

[BM.16(1)'s lock identity re-derived; machine: one Laurent monomial, coefficient 2] :

the three freedom censuses:

$\mathfrak{h}$-free [the $\mathfrak{h}$-exponent is 0],

Λ-free,

P4-free [¬one of $g, g', h_0, c_W, \Lambda, \varkappa, \beta, \alpha_\Omega$ occurs]

[BM.16(1); machine];

G is strictly increasing and quadratic in E_{int} on the quantized ladder

[machine: the seven cell values];

(4) the exact profile:

the self-energy's dimensionless factor is the product of the three derived arguments' functions, and the energy-side monomial is where the lapse enters:

$$\boxed{\varepsilon_{\mathrm{EM}}(r,E,N) \;:=\; N(E)\cdot f\Big(\frac{r}{\Delta x_{\mathrm{B}}(E)}\Big)\cdot \mathsf{g}\big(G(E)\big)\cdot \varepsilon_{\mathrm{EM}}}$$

[machine: the monomial census; BM.34(4)'s dispersion $(p_0 - N^i p_i)^2 = N^2(m^2 + \gamma^{ij}p_i p_j)$ puts exactly one factor of N on the energy];

$$\boxed{e(r,E,N) \;=\; e_0\,\sqrt{N(E)\cdot f\big(r/\Delta x_{\mathrm{B}}(E)\big)\cdot \mathsf{g}\big(G(E)\big)}}$$

[$e^2 = 4\pi\varepsilon_{\mathrm{EM}}$; machine: squaring recovers (4)'s monomial];

the placement row:

N on the energy and $\sqrt{N}$ on the coupling are the same monomial — the placement is a naming convention, ¬a datum;

the derived content is (4)'s profile;

the curvature composition is priced, ¬adopted:

$$\mathsf{g} \in \Big\{1,\;\; 1+G,\;\; \frac{1}{1+G}\Big\},$$

g carried symbolically;

¬a clause of this tier selects a composition

[the argument G is derived (3); the composition is ¬derived : ¬∃ a count → strength theorem [BM.36(3); BM.61(8)(O1)], ¬∃ a loop (1)];

(5) the exact form at the two loci that matter:

at the member's own envelope $k = 1$:

$$\boxed{f(1)=0 \implies e(\Delta x_{\mathrm{B}},E,N)=0 \text{ and } \varepsilon_{\mathrm{EM}}(\Delta x_{\mathrm{B}},E,N)=0}$$

the coupling vanishes at the envelope, so the formed self-energy is the exterior read alone

[BM.87(2), (3) : the same statement on the energy side];

in the far field $k \to \infty$:

$$f \to 1,$$

$$\varepsilon_{\mathrm{EM}} \to N\,\mathsf{g}(G)\,\varepsilon_{\mathrm{EM}},$$

$$E_{\mathrm{EM}}^{\mathrm{self}} \to \frac{1}{2}\,N\,\mathsf{g}(G)\,\varepsilon_{\mathrm{EM}}\,q^2\,m_{\mathrm{hid}}$$

[machine: the exact limit row];

no finite neighbourhood holds the whole correction:

$$1-f(k)=\frac{1}{k}>0 \;\; \forall k<\infty$$

[BM.47(6); machine];

(6) the energy dependence (the running, exactly):

at fixed r,

$$f\big(k(E)\big)=1-\frac{\mathfrak{h}}{r\,E_{\mathrm{int}}(E)\ln 2}$$

[machine: the substitution] :

$$\boxed{E_{\mathrm{int}} \text{ strictly increasing} \implies \Delta x_{\mathrm{B}} \text{ strictly decreasing} \implies k \text{ strictly increasing} \implies f \text{ strictly increasing}}$$

[machine: three monotonicity rows on exact rationals];

the coupling grows with the energy scale, geometrically, with no loop in the argument;

the six-state floors give the exact ladder $f = 1 - \hbar/(r\, E_{\text{int}} \ln 2)$ at $E_{\text{int}} \in \{12, 24, 60\}$

[BM.81(1); machine];

(7) the bars:

¬∃ a numerical value of e or of e_0 in any clause

[EM.25(7); P5; machine: zero external tokens, zero decimal numerals in this tier];

¬a measured coupling is identified with any corpus quantity [P5];

ε_{EM}

$= e^2/(4\pi)$ is a dimensionless P4-class symbol carried symbolically throughout [Def-BM.12 local notation; EM.25(9)];

α_B : ¬added, ¬derived, barred on BM.61(8)'s four grounds

[machine: the token scan of this tier];

¬a P4 addition, §Cal-BM stays empty;

(8) status:

derived: (1)–(3), (5), (6);

derived in form with a priced composition: (4);

symbolic: e_0, g;

¬a declaration, ¬a new dimensionful scale [P1], ¬a measured value [P5],

open: the value of e [EM.25(7)],

the curvature composition g (4),

the lapse as solution data [CT2.7; BM.34(2)].

Req. D1.5, D2.9 (Parts 1–2); CT1.12, CT1.22, CT1.23, CT2.1, CT2.7, EM.9, EM.15, EM.25(6), EM.25(7), EM.25(9), EW.15, EW.35 (Part 2); P1, P4, P5 (Parts 1–2); FT.28, FT.145(3), FT.157(1) (Part 3); Census FT.35(2), FT.42(5) (row-wise); Def-BM.4(1), BM.13(1), BM.13(2), BM.13(3), BM.14(2), BM.14(4), BM.15(2), BM.16(1), BM.34(1), BM.34(2), BM.34(4), BM.36(3), BM.47(2), BM.47(6), BM.61(8), BM.81(1), BM.87(2), BM.87(3) (this part); exact Laurent-monomial arithmetic with $\mathfrak{q}$- and $\mathfrak{h}$-exponent censuses, exact rational function rows, monotonicity on exact rationals, the ADM polynomial identity by Schur complement, limit rows [math; machine: `checks/verify_H_em_self_energy_correction.py` §B].

Pf.

$(1):$ EW.35's combination and EM.25(6), (7)'s rows re-cited;

$\neg\exists$ a loop object in the inventory [BM.61(8); FT.145(3); machine: the token census]

$(2)(\mathrm{i}):\ \Delta x_{\mathrm{B}} = 1/m_{\mathrm{hid}} = \mathfrak{h}/(E_{\mathrm{int}} \ln 2)$ [machine: one Laurent monomial];

$E_{\mathrm{int}} \geq 2$ [BM.13(3)];

the ladder is injective in E_{int} and strictly decreasing [machine]

$(\mathrm{ii}):\ f(k) = 1 - 1/k$ at $k = 1, \ldots, 6 = (0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6})$ [machine: exact Fractions];

$f'(k) = 1/k^2 > 0$ on $k \geq 1$ and $f(k) < 1 \iff 1/k > 0$ [machine: the rational-function rows];

$f(1) = 0$ [machine]

$(\mathrm{iii}):\ N^2 = \Omega^2 + \gamma^{ij} N_i N_j$ and $N^2 \det\gamma = -\det g$ [BM.34(1) re-cited; machine: the Schur identity];

$(a_{00}, q_{ij}, w_i) = 0 \implies \Omega = 1,\ N_i = 0,\ N = 1$ [machine: exact substitution]

$(3):\ (\Delta x_{\mathrm{B}})^2 \kappa^2 \rho$

$$= \frac{\mathfrak{h}^2}{E_{\text{int}}^2(\ln 2)^2} \cdot \frac{2\pi E_{\text{int}}^4(\ln 2)^3}{\mathfrak{h}^2}$$

$= 2\pi E_{\text{int}}^2 \ln 2$ [machine: the Laurent monomial cancels $\mathfrak{h}$ exactly];

$[G] = \mathfrak{q}^{-1} \cdot \mathfrak{q}^{+1} = \mathfrak{q}^0$ [machine: the exponent arithmetic];

the three freedom censuses [BM.16(1); machine: the symbol scan]

(4) : the product of three dimensionless factors is dimensionless [machine];

$e^2 = 4\pi\varepsilon_{\text{EM}} \implies e = e_0\sqrt{Nf\mathfrak{g}}$ at $\varepsilon_{\text{EM}} = Nf\mathfrak{g}\,\varepsilon_{\text{EM}}$ [machine: squaring recovers the monomial];

the one-factor-of-N row : $(p_0 - N^i p_i)^2$

$= N^2(m^2 + \gamma^{ij}p_i p_j)$

$\implies E$

$= N\sqrt{m^2 + p^2} - N^i p_i$ [BM.34(4); machine]

the priced set $\{1, 1+G, (1+G)^{-1}\}$ is a pricing, ¬an adoption [Def-BM.12(4) pattern];

$\neg\exists$ a count $\to$ strength theorem [BM.36(3); BM.61(8)(O1)]

(5) : $f(1) = 1 - 1 = 0$ [machine];

$\lim_{k\to\infty}(1 - 1/k) = 1$ [machine: the exact limit row on rationals];

$1 - f(k) = 1/k > 0\ \forall k < \infty$ [machine]

(6) : $k(E)$

$= rE_{\text{int}}(E)\ln 2/\mathfrak{h}$ is strictly increasing in E_{int} at fixed $r > 0$ [machine: positive coefficients];

$$f \text{ strictly increasing } [(2)(\text{ii})] \implies \text{the composition is strictly increasing [machine]};$$

$$\Delta x_{\mathrm{B}} \text{ strictly decreasing [machine]}$$

$$(7)\text{–}(8): \text{ the token scans, the symbol census and the assembly [machine]. } \blacksquare$$

Phys. A coupling with no scale at which to sit is not a number, and the corpus never pretended otherwise: it fixed the combination of the two gauge strengths and left the value open. What it can do is say how the coupling's profile runs, and every ingredient of that profile is already derived. Distance enters through the only length the ledger owns — the Born width, which is the reciprocal of an edge count — and through the fraction of an inverse-square flux enclosed within a given number of those widths, which is one minus one over k, exactly. Time enters through the lapse of the emergent metric, which is one in the vacuum gauge and departs from one only where the shift and the conformal coefficient do. Curvature enters through the one dimensionless monomial the calibration lock supplies: the Born width squared times the Ricci scale, equal to two pi times the square of the edge count times the logarithm of two, free of h, free of the cosmological term, free of every declared parameter. Multiply the three and the coupling is an exact algebraic function of position, energy and lapse — vanishing at the member's own envelope, growing without bound toward its far-field value, never quite reaching it at any finite radius. The composition of the curvature factor is not derived, because nothing in the inventory turns a count into a strength, so it is priced three ways and adopted not at all. The running comes out in the direction a loop calculation would give it, from geometry alone, with no loop in the argument and no number fitted anywhere.

Theorem 4.75 (BM.89 — The EM self-energy from the path integral: the derived measure, the crystallization weight, and the route-relative correction). Thm.

$$(1) \quad \text{the ensemble and its derived measure:}$$

$$Q = \begin{pmatrix} -x_+ & x_+ \\ x_- & -x_- \end{pmatrix} \text{ over } \mathbb{Z}[x_+, x_-],$$

$$(x_-,\, x_+)\, Q = (0,0) \text{ exactly,}$$

$$\dim \ker Q = 1$$

$$[\text{BM.19}(1);\ \text{FT.108}(4);\ \text{machine}]:$$

$$\boxed{p := \frac{x_+}{x_+ + x_-} \text{ is the unique stationary measure}}$$

$$x_+ = 1 - 4\varepsilon_q, \quad x_- = 4\varepsilon_q \quad \Longrightarrow \quad p = 1 - 4\varepsilon_q$$

$$[\text{BM.19}(2);\ \text{machine: an identity, exact at every rational threshold}];$$

$$\varepsilon_q \in [0, \tfrac{1}{4}) \text{ [CT0.8; D0.4; A4]} \implies x_- \in [0,1), \quad x_+ \in (0,1], \quad p \in (0,1] \qquad \text{[machine]};$$

$$\varepsilon_q = 0 \implies x_- = 0 \implies \mathcal{E}(\mathcal{X}) = \{\text{the crystallized configuration}\}$$

$$\text{[BM.19(3); Def-BM.5(5)]};$$

$$|\mathcal{E}(\mathcal{X})| > 1 \iff \varepsilon_q > 0$$

$$\text{[BM.19(6)]};$$

the ensemble is finite at every budget and ¬a fixed topology at any nonzero threshold

$$\text{[Def-H.32(1); BM.19(4), (6)]};$$

(2) the configuration energy:

$$\boxed{E_{\text{EM}}(R) \;=\; \frac{1}{2}\, N(R)\, \mathrm{g}\big(G(R)\big)\, \varepsilon_{\text{EM}}\, Q(R)^2\, m_{\text{hid}}(R), \qquad m_{\text{hid}}(R) = \frac{E_{\text{int}}(R)\ln 2}{\mathfrak{h}} = \frac{|R|\ln 2}{\mathfrak{h}}}$$

[BM.87(4)'s formed monomial at $c = \frac{1}{2}$; BM.88(4)'s profile; Def-BM.5(4) : $E_{\text{int}}(R) = |R|$; FT.44(4) : the taxonomy law; machine];

(3) the class factorization:

$Q(R) = Q$ on every member of a charge class — the boundary data fix the class

$$\text{[Def-BM.5(1); Def-BM.8(1); machine: the class census]} :$$

$$\boxed{E_{\text{EM}}^{\text{self}}(Q) \;=\; \sum_{R \in \mathcal{E}(\mathcal{X})} P(R)\, E_{\text{EM}}(R) \;=\; \frac{1}{2}\, N\, \mathrm{g}(G)\, \varepsilon_{\text{EM}}\, Q^2\, \big\langle m_{\text{hid}} \big\rangle_{\mathcal{E}}}$$

[machine: Q^2 is constant on the class and factors out of the average];

(4) the crystallization weight (the derived attenuation):

$$\boxed{w(\varepsilon_q) \;:=\; \frac{\langle E_{\text{int}}\rangle_{\mathcal{E}}}{E_{\text{int}}^{\text{cryst}}} \;=\; \frac{1}{E_{\text{int}}^{\text{cryst}}} \sum_{\text{edges}} P(\text{the edge is real})}$$

[Def-BM.5(4); the ensemble average of $|R|$] :

$$\boxed{w(0) = 1 \text{ exactly}}$$

[BM.19(3) : the ensemble is the single crystallized configuration];

$$0 < w(\varepsilon_q) \leq 1$$

[BM.19$(5),(6)$: the ensemble spreads E_{int} over $\{0,\ldots,B_K\}$ with the crystallized value as its top point, and every move changes E_{int} by its $J^{\pm}$ excess];

the strict decrease is derived on the marked specialization below, and in general is (R4)'s content:

ε_q enters the measure only through $x_- = 4\varepsilon_q$ [BM.19(2); machine: the symbol census];

the Bernoulli specialization (marked, ¬adopted):

on the per-datum stationary retention of (1),

$$\langle E_{\text{int}}\rangle_{\mathcal{E}} = E_{\text{int}}^{\text{cryst}} \cdot p \implies \boxed{w = 1 - 4\varepsilon_q}$$

[machine: $E_{\text{int}}^{\text{cryst}} = 6 = (c_1\colon 2) + (c_2\colon 2) +$ (connector: 2) [BM.28(3); BM.17; BM.26] is three pairs, each retained with weight p];

w is carried symbolically wherever the specialization is not invoked;

the member is a stationary mixture of real and virtual content:

the composite is where the real and the virtual registers meet, ¬a fixed topology

[BM.19(6); D1.5; CT1.22];

(5) the corrected self-energy:

$$\boxed{E_{\text{EM}}^{\text{self}}(Q) \;=\; \frac{1}{2}\, N\, w\, \mathsf{g}(G)\, \varepsilon_{\text{EM}}\, Q^2\, \frac{E_{\text{int}}^{\text{cryst}} \ln 2}{\mathfrak{h}}}$$

[machine: one Laurent monomial in $(N, w, \mathsf{g}, \varepsilon_{\text{EM}}, Q, E_{\text{int}}, \ln 2, \mathfrak{h})$] :

at $E_{\text{int}}^{\text{cryst}} = 6$:

$$E_{\text{EM}}^{\text{self}}(\pm 1) = \frac{3\, N\, w\, \mathsf{g}\, \varepsilon_{\text{EM}} \ln 2}{\mathfrak{h}},$$

$$E_{\text{EM}}^{\text{self}}(0) = 0 \;\text{ exactly}$$

[machine: $0^2 = 0$; Gauss: zero net flux];

the crystallized vacuum-gauge specialization:

$$(w, N, \mathsf{g}) = (1, 1, 1) \implies E_{\text{EM}}^{\text{self}}(\pm 1) = \frac{3\, \varepsilon_{\text{EM}} \ln 2}{\mathfrak{h}} = \frac{1}{2}\, \varepsilon_{\text{EM}}\, m_{\text{hid}} = \frac{e^2 q^2}{8\pi\, \Delta x_{\text{B}}}$$

the path-integral value reproduces BM.87(2)'s derived exterior value exactly at full crystallization in the vacuum gauge

[machine: the two monomials agree coefficient by coefficient];

(6) the route-relativity (priced, ¬adopted):

route (i) (the normalized average): $\overline{\Delta S}(\cdot, S) = m_B^2(\cdot, S)$ is charge-degenerate

[FT.107(3); BM.28(4)(i)] :

the correction splits the classes only through Q^2, and the ledger's own average is class-blind

[machine: the frozen sector expectations are q-independent];

route (ii) (the unnormalized total deficit):

$$T(Q,S) = W(Q)\,\overline{\Delta S}(S),$$

$$W = (1,6,15,20,15,6,1)$$

[BM.28(4)(ii); Def-BM.13(3)] :

$$E_{\mathrm{EM}}^{\mathrm{self},(\mathrm{ii})}(Q,S)$$

$$= \frac{1}{2}$$

$$N$$

$$\mathfrak{g}$$

$$\varepsilon_{\mathrm{EM}}$$

$$Q^2$$

$$\frac{T(Q,S)}{\mathfrak{h}} \quad\Longrightarrow\quad \Delta m_{\mathrm{EM}}^{(\mathrm{ii})}(S)$$

$$= \frac{N\,\mathfrak{g}\,\varepsilon_{\mathrm{EM}}}{2\mathfrak{h}} \cdot 15$$

$$m_B^2(\cdot,S)$$

$$= \frac{N\,\mathfrak{g}\,\varepsilon_{\mathrm{EM}}}{2\mathfrak{h}}\Big(\frac{12659355}{98956},\ \frac{28824930}{393679},\ \frac{1365}{23},\ \frac{205695}{3088}\Big)$$

[machine: exact rational arithmetic at the four frozen sector expectations $\big(\frac{843957}{98956}, \frac{1921662}{393679}, \frac{91}{23}, \frac{13713}{3088}\big)$ [BM.21(2); BM.62(2)]];

route (ii) is ¬adopted for any purpose beyond the vertex kinematics

[BM.30(5); BM.37(6); BM.77(1), (9) stand];

route (iii)'s amplitude ratios stay barred as masses

[FT.79(5); BM.37(3)];

(7) the attenuation theorem:

$$0 < w \leq 1 \text{ and } N > 0,\ \mathrm{g} > 0 \implies \boxed{\left|E_{\mathrm{EM}}^{\mathrm{self}}(Q)\right| \leq \left|E_{\mathrm{EM}}^{\mathrm{self}}(Q)\right|_{w=1}}$$

with equality only at $w = 1$:

the flicker strictly attenuates the EM correction at every nonzero threshold;

the transport register's lead $\ln(4/3)/\mathfrak{h}$ is class-shared and ¬attenuated by w [BM.57(4); BM.59(4); BM.91(2)];

$\implies$ the sub-critical branch widens as the crystallization weight falls

[BM.91(5); machine: monotone in w];

(8) the residue:

$w(\varepsilon_q)$ in general requires the ensemble's configuration distribution beyond the per-datum specialization — named (R4),

¬filled :

the computed cells supply exact sector averages of the deficit register $\overline{\Delta S}$

$= (\frac{3}{2} n_{\mathcal{V}})$ [FT.106(3); FT.107(3)],

¬of the edge-count register $E_{\text{int}} \qquad = |R|$:

the two registers are distinct alphabets and are ¬equated

[BM.62(11); BM.37(2); BM.83(3); machine: the token census];

(9) status:

derived: (1)–(5), (7);

priced, ¬adopted: (4)'s Bernoulli specialization and (6)'s route rows;

named, ¬filled: (8)'s (R4);

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

§Cal-BM stays empty;

Req. D1.5, D3.4 (Parts 1–2); A4, CT0.8, CT1.22, D0.4, EM.9 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.32(1), FT.44(4), FT.79(5), FT.106(3), FT.107(3), FT.108(4) (Part 3); Def-BM.4(1), Def-BM.5(1), Def-BM.5(4), Def-BM.5(5), Def-BM.8(1), Def-BM.13(3), BM.17, BM.19(1), BM.19(2), BM.19(3), BM.19(4), BM.19(5), BM.19(6), BM.21(2), BM.26, BM.28(3), BM.28(4), BM.30(5), BM.37(2), BM.37(3), BM.37(6), BM.57(4), BM.59(4), BM.62(2), BM.62(11), BM.77(1), BM.77(9), BM.83(3), BM.87(2), BM.87(4), BM.88(4) (this part); exact polynomial arithmetic over $\mathbb{Q}[x_+, x_-]$, exhaustive integer kernel censuses, exact rational specialization, Laurent-monomial assembly with coefficient comparison, exact rational arithmetic at the four frozen sector expectations, token censuses [math; machine: `checks/verify_H_em_self_energy_correction.py` §C].

Pf.

$(1):\ (x_-)(-x_+) + (x_+)(x_-) = 0$ and $(x_-)(x_+) + (x_+)(-x_-) = 0$ in $\mathbb{Q}[x_+, x_-]$ [machine];

constant vectors : $a(-x_+) + bx_-$

$= 0 \ \wedge \ ax_+ + b(-x_-)$

$= 0$

$\implies a$

$= b$

$= 0$ [machine: exhaustive on $[-3, 3]^2$];

$p = (1 - 4\varepsilon_q)/(1 - 4\varepsilon_q + 4\varepsilon_q) = 1 - 4\varepsilon_q$ [machine: exact at six rational thresholds]

$$\varepsilon_q \in [0, \tfrac{1}{4}) \implies 4\varepsilon_q \in [0, 1) \text{ [machine]};$$

$$x_- = 0 \text{ at } \varepsilon_q = 0 \implies \neg\exists \text{ a dissolution move [D1.2 : } J^- : n_{\mathcal{R}} : 1 \to 0];$$

$$|\mathcal{E}| > 1 \iff x_- > 0 \text{ [BM.19(6)]}$$

$$(2)\text{–}(3) : \text{ BM.87}(4)\text{'s monomial at } c = \tfrac{1}{2} \text{ and BM.88}(4)\text{'s profile composed [machine]};$$

$$E_{\text{int}}(R) = |R| \text{ [Def-BM.5}(4)\text{]};$$

$$Q(R)$$

$$= Q \text{ class-constant [Def-BM.5}(1);\ \text{machine: the boundary-data census]}$$

$$\implies Q^2 \text{ factors out of } \sum_R P(R)\cdot$$

$$(4) : \langle E_{\text{int}} \rangle_{\mathcal{E}} = \sum_{\text{edges}} P(\text{real}) \text{ [linearity of the average on } |R|];$$

$$\varepsilon_q = 0 : \ \mathcal{E} = \{R_{\text{cryst}}\} \implies \langle E_{\text{int}} \rangle = E_{\text{int}}^{\text{cryst}} \implies w = 1 \text{ [machine]};$$

$$0 < |R| \le E_{\text{int}}^{\text{cryst}} \text{ on the ensemble's support } \implies 0 < w \le 1 \text{ [BM.19}(5);\ \text{machine]}$$

$$\text{the specialization : three pairs, each retained with the stationary weight } p \implies \langle E_{\text{int}} \rangle$$

$$= 3 \cdot 2 \cdot p$$

$$= 6p \text{ and } w$$

$$= p$$

$$= 1 - 4\varepsilon_q \text{ [machine: exact rationals]};$$

$$E_{\text{int}}^{\text{cryst}} = 6 = (2) + (2) + (2) \text{ [BM.28}(3);\ \text{machine]}$$

$$(5):\ \tfrac{1}{2}\cdot 6 =\ 3\ [\text{machine}];$$

$$Q=0\implies Q^2=0\implies E=0\ [\text{machine}];$$

$$(w,N,\mathsf{g})$$

$$=(1,1,1):\ \tfrac{3\varepsilon_{\mathrm{EM}}\ln 2}{\mathfrak{h}}$$

$$=\tfrac{1}{2}\varepsilon_{\mathrm{EM}}\tfrac{6\ln 2}{\mathfrak{h}}$$

$$=\tfrac{1}{2}\varepsilon_{\mathrm{EM}}m_{\mathrm{hid}}$$

$=\frac{e^2q^2}{8\pi\Delta x_{\mathrm{B}}}$ [machine: three Laurent monomials, coefficient by coefficient, $\Delta x_{\mathrm{B}}m_{\mathrm{hid}}=1$]

$$(6):\ m_B^2(q,S)=m_B^2(-q,S)\ [\text{FT.107(3); machine}];$$

$$15\cdot\tfrac{843957}{98956}=\tfrac{12659355}{98956},$$

$$15\cdot\tfrac{1921662}{393679}=\tfrac{28824930}{393679},$$

$$15\cdot\tfrac{91}{23}=\tfrac{1365}{23},$$

$$15\cdot\tfrac{13713}{3088}$$

$=\frac{205695}{3088}$ [machine: exact Fractions, each in lowest terms or reduced exactly]

$$(7):\ 0<w$$

$$\le 1 \text{ and } N,$$

$\mathsf{g}>0\implies \frac{1}{2}Nwg\varepsilon_{\mathrm{EM}}Q^2m_{\mathrm{hid}}\le\frac{1}{2}Ng\varepsilon_{\mathrm{EM}}Q^2m_{\mathrm{hid}}$ [machine: the coefficient census];

$$\text{equality}\iff w=1\ [\text{machine}]$$

$$(8)\text{–}(9):\ \text{the register-alphabet census [BM.62(11) pattern; machine]};$$

the standing-rule scans [machine]. ■

Phys. The composite was never a fixed topology, and the correction that rides it should not have been either. Take the flicker's own measure — the unique stationary line of the Janus generator, which specializes to one minus four times the crystallization threshold with no rate declared anywhere — average the formed monomial over the ensemble, and the charge factors straight out, because the boundary data fix it before the flicker starts. What is left is one derived number between zero and one: the crystallization weight, the mean fraction of the composite's registration data that are real rather than virtual. It is exactly one at the sharp threshold, where the ensemble collapses to the single crystallized pattern, and it falls as the threshold rises, so the electromagnetic correction is attenuated by exactly the amount the member fails to be crystallized. At full crystallization in the vacuum gauge the average reproduces the exterior value the adjudication derived, monomial for monomial, so nothing is lost and one classical coefficient is gone. The baryon, then, is where the real and the virtual registers meet: a stationary mixture whose mass is an average, whose charge is exact, and whose field energy is a fraction of what a fully frozen lump would carry. The general weight beyond the per-datum specialization stays open and named, because the computed cells average the vertex-deficit register and not the edge count, and those two alphabets are not the same currency.

Theorem 4.76 (BM.90 — The defect-conditioned weighting: the hypothesis on its face, and the branch-blindness theorem). Thm.

(1) the hypothesis, on its face:

hypothesis (W): the retention probability of a registration datum depends on the state's defect value,

$$P(\text{the edge is real} \mid D_S),$$

so that

$$m_{\text{hid}}^{\text{eff}}(s) \;=\; \frac{\ln 2}{\mathfrak{h}} \sum_{\text{edges}} P(\text{the edge is real} \mid D_S(s))$$

[the six states of Def-BM.14(2), (3); BM.81(1)];

(2) the measure's alphabet census:

$$x_+ = 1 - 4\varepsilon_q,$$

$$x_- = 4\varepsilon_q,$$

$$p = 1 - 4\varepsilon_q$$

[BM.19(2); machine] :

the flicker measure's symbol set is $\{\varepsilon_q\}$ and ε_q is the A4 threshold

[D0.4; BM.19(2) : ¬a P-class datum enters; ¬a new fugacity, ¬a new rate];

the defect value's symbol set is $\{\delta_v,\ \hat{\mu}'^2\}$

[Def-BM.14(3); Def-H.37(2); machine];

$$\boxed{\{\varepsilon_q\}\ \cap\ \{\delta_v, \hat{\mu}'^2\}\ =\ \varnothing}$$

(3) the branch-blindness theorem:

¬∃ a derived coupling $D_S \longrightarrow P$(retention); $w = w(\varepsilon_q)$ is branch-blind

on four independent grounds, each re-cited at its own locus:

(i) the field degree: the six defect values generate a degree-16 extension of $\mathbb{Q}$,

while every read the flicker measure is built from is integer- or binary-valued

[FT.73(2), (3); BM.69(4)(i); BM.81(1)] :

a rational formula places every retention weight in a field of degree < 16 over the declared constants

[machine re-cited];

(ii) the two-torsion image: the register lift of the defect angle reaches $\{0, \pi\}$ only — four of the six states are unreachable at every winding

[BM.67(6); BM.69(4)(ii)];

(iii) the calibration weight: the winding route Θ_R

$= 2\pi k$ carries $\mathfrak{h}$-weight -2 against 0 for a rate

[FT.55(1); BM.67(8); BM.69(4)(iii)];

(iv) the strength bar: $\neg\exists$ a count $\to$ strength theorem;

the capacity ratios $\left(\frac{1}{3},\ 8,\ \frac{1}{25}\right)$ are ¬couplings

[BM.36(3); BM.61(8)(O1); BM.69(4)(iv)];

(4) the consequence for the informational register:

the six-state quark splitting does not reweight m_{hid}

the m_{hid} contributions are the values BM.81(1) censused,

attenuated by a branch-independent w :

$$m_{\text{hid}}^{\text{eff}}(s) = w(\varepsilon_q) \cdot \frac{2E(s)\ln 2}{\mathfrak{h}}$$

with the same w on all six states

[machine: w's symbol census carries no D_S];

the defect values enter the pole currency alone

[BM.82(4) : the thirty ρ_{B}-free pole ratios; BM.81(1)'s D_S column];

(5) what the splitting does change (the positive companion, in its own currency):

$\Delta_g = \left(2,\ \sqrt{2}-1,\ 2\text{c}-\sqrt{3}\right)\hat{\mu}'^2 > 0$ per generation

[BM.75(4); BM.81(3)] :

$$\mathcal{R}_s = \frac{D_S(s)}{4E(s)^2} \text{ per state}$$

[BM.82(2)];

$\Delta\big(m^2_{\text{pole}}\big) = \Delta_g > 0$ under hypothesis (H) (the single substitution)

[BM.59(9); BM.69(3); the hypothesis on its face];

and ¬addable to the informational assembly:

$[\Delta_g] = \mathfrak{q}$ against $[\Delta m] = \mathfrak{q}^{+1/2}$,

a cross-currency addition being a category the corpus bars

[BM.83(1), (3); BM.35(5); BM.37(2); BM.84(1); machine: the exponent arithmetic];

(6) the priced alternative:

a declared D_S-dependent retention rate would be a P4-class count → strength coupling — the exact species BM.61(8) bars and BM.36(3) refuses :

¬such a datum is issued;	§Cal-BM stays empty;	¬a P4 addition

[P4; P1];

(7) status:

derived: (2)–(5);

refuted: hypothesis (W) as a derived coupling (3);

priced, ¬adopted: (6);

¬a declaration, ¬a measured value [P5], ¬a new dimensionful scale [P1],

the currencies remain distinct and unequated [BM.83(3)].

Req. D0.4, D3.4 (Parts 1–2); CT0.8, EM.9 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.37(2), FT.55(1), FT.73(2), FT.73(3) (Part 3); Def-BM.14(2), Def-BM.14(3), BM.19(2), BM.35(5), BM.36(3), BM.37(2), BM.59(9), BM.61(8), BM.67(6), BM.67(8), BM.69(3), BM.69(4), BM.75(4), BM.81(1),

BM.81(3), BM.82(2), BM.82(4), BM.83(1), BM.83(3), BM.84(1) (this part); exact symbol and token censuses, field-degree rows re-cited, exponent arithmetic in the $\mathfrak{q}$-grading [math; machine: `checks/verify_H_em_self_energy_correction.py` §D].

Pf.

$$(1):\ \text{the hypothesis stated with its own symbols [no premise imported]};$$

$$(2):\ x_{\pm} \text{ and } p \text{ re-cited from BM.19(2) [machine]};$$

the two symbol sets enumerated from their defining clauses [machine: the token census of Def-BM.14(3), Def-H.37(2) against BM.19(2), D0.4]

$(3):$ the four grounds re-cited at their own loci [FT.73(2), (3); BM.67(6), (8); FT.55(1); BM.36(3); BM.61(8)(O1); BM.69(4)];

$$\text{the intersection } \{\varepsilon_q\} \cap \{\delta_v, \hat{\mu}'^2\} = \varnothing \text{ [machine]}$$

$$(4):\ w\text{'s defining expression carries no } D_S \text{ [(2), (3); machine]};$$

$$m_{\text{hid}}(s)$$

$$= 2E(s) \ln 2/\mathfrak{h} \text{ [BM.81(1); FT.44(4)]}$$

$$\Longrightarrow\ m_{\text{hid}}^{\text{eff}}(s)$$

$$= w \cdot 2E(s) \ln 2/\mathfrak{h} \text{ with one } w \text{ [machine]}$$

$$(5):\ \Delta_g \text{ and } \mathcal{R}_s \text{ re-cited from their own tags};$$

$$[\Delta_g] = [\hat{\mu}'^2] = \mathfrak{q}, \quad [\Delta m] = \mathfrak{q}^{+1/2} \text{ [CT2.1; machine]} \implies \neg\text{addable [BM.37(2) pattern]}$$

$$(6)\text{–}(7):\ \text{the species row of a count} \to \text{strength coupling [BM.36(3); BM.61(8)]};$$

$$\text{the standing-rule scans [machine]}. \blacksquare$$

Phys. The proposal that six defect values should reweight six retention probabilities is natural and it is false here — not because the numbers disagree but because nothing in the inventory can connect them. The flicker measure knows one symbol, the crystallization threshold, and it is derived as the unique stationary line of a two-state generator. The defect values know two symbols, an angle and a scale, and they live in a degree-sixteen field over the rationals while every read

the measure touches is an integer or a binary weight. Between them stand four separate obstructions the corpus already proved: a field-degree gap, a two-element holonomy image that cannot reach four of the six states, a calibration weight that forbids the winding route, and the standing bar against turning a count into a strength. So the crystallization weight is branch-blind, the informational masses keep the values the census derived, attenuated equally on all six states, and the defect values keep their own currency, where they split a doublet and set a bridge factor and change nothing in bits. Declaring a defect-dependent rate would be exactly the kind of parameter this part has refused eleven times over, and it is refused again: the calibration registry stays empty, and the two currencies stay as incommensurate as the item that names them.

Theorem 4.77 (BM.91 — The corrected neutron–proton splitting: the attenuated assembly, the ensemble-independent bar, and the universal ratio bound). Thm.

(1) the corrected decomposition: $\boxed{\Delta m \;:=\; m_n - m_p \;=\; \Delta m_{\text{transport}} + \Delta m_{\text{EM}}^{\text{corr}} + \Delta m_{\text{bridge}}}$

$$\Delta m_{\text{transport}} = +\frac{\ln(4/3)}{\mathfrak{h}}$$

$$[\text{BM.59}(4);\ \text{BM.84}(1)];$$

$$\Delta m_{\text{EM}}^{\text{corr}} = -\frac{3\,N\,w\,\mathrm{g}\,\varepsilon_{\text{EM}}\ln 2}{\mathfrak{h}}$$

$$[\text{BM.89}(5);\ \text{the path-integral read}];$$

$$\Delta m_{\text{bridge}} = 0 \ \text{ in the informational currency}$$

$$[\text{BM.84}(1):\ \neg\text{formed there};\ \text{BM.69}(3);\ \text{BM.90}(5):\ \text{the pole companion } \Delta_g > 0 \text{ is } \neg\text{addable}];$$

the four zeros stand:

$$\Delta m_{\text{Lyapunov}}$$

$$= 0\ [\text{BM.51}(6)],$$

$$\Delta m_{\text{metric}}$$

$$= 0\ [\text{BM.53}(2);\ \text{BM.58}(4)],$$

$$\Delta m_{\text{binding}}$$

$$= 0 \text{ [BM.54}(1)\text{–}(3)\text{]},$$

$$\Delta m_{\text{force}}$$

$$= \Delta m_{\text{self-int}}$$

$$= -\Delta m_{\text{EM}}^{\text{corr}} \text{ [BM.52}(4), (5)\text{]};$$

(2) the attenuated assembly:

$$\boxed{m_n \mathfrak{h} \;=\; 6\, w \ln 2 + \ln 20, \qquad m_p \mathfrak{h} \;=\; 6\, w \ln 2 + \ln 15 + 3\, N\, w\, \mathfrak{g}\, \varepsilon_{\text{EM}} \ln 2}$$

[BM.89$(4), (5)$; Def-BM.13(5)'s refined rest input with the ensemble-averaged edge register; machine: the prime-log assembly $\ln 20 = 2 \ln 2 + \ln 5,\ \ln 15 = \ln 3 + \ln 5$];

the register scoping:

w attenuates the edge-count register $E_{\text{int}} = |R|$ alone [Def-BM.5(4)];

the transport register $\Sigma_{\text{cl}}(Q)$

$= \ln W(Q)$ is an end-sign filter count, ¬an edge count [Def-BM.13$(4), (6)$],

and its ensemble average is ¬constructed [FT.106(3)'s census is at fixed topology] $\Longrightarrow$ it enters at its crystallized value and the residue is (R4) [BM.89(8); BM.93(6)];

$$\boxed{\Delta m \mathfrak{h} \;=\; \ln \frac{4}{3} - 3\, N\, w\, \mathfrak{g}\, \varepsilon_{\text{EM}} \ln 2} \qquad \text{[machine: } \ln 20 - \ln 15 = \ln \tfrac{4}{3}\text{]};$$

the cancellation row:

E_{int} is class-shared [BM.57(4); Census FT.35(2)] $\Longrightarrow$ the crystallization weight cancels in the ledger difference and survives only in the EM term

[machine: the coefficient census of w in the two rest inputs and in Δm];

$$[\Delta m] = \mathfrak{q}^{+1/2} = [\text{energy}]$$

[CT2.1; machine];

(3) the exact ratio: $$\boxed{\frac{m_n}{m_p} = \frac{6\,w\ln 2 + \ln 20}{6\,w\ln 2 + \ln 15 + 3\,N\,w\,\mathrm{g}\,\varepsilon_{\mathrm{EM}}\ln 2}}$$ [machine];

at $(w, N, \mathrm{g}) = (1, 1, 1)$:

$$\frac{m_n}{m_p} = \frac{\ln 1280}{\ln 960 + 3\,\varepsilon_{\mathrm{EM}}\ln 2}$$

[BM.84(2)'s row at the derived $c_{\text{formed}} = \frac{1}{2}$; machine: $6\ln 2 + \ln 20 = \ln 1280,\ 6\ln 2 + \ln 15 = \ln 960$];

(4) the exact critical strength:

Δm is strictly decreasing in $\varepsilon_{\mathrm{EM}}$ at every admissible $(N, w, \mathrm{g}) > 0$

[machine: the coefficient census] :

$$\boxed{\varepsilon^{\text{crit}}(N, w, \mathrm{g}) := \frac{\ln(4/3)}{3\,N\,w\,\mathrm{g}\,\ln 2} = \frac{2 - \log_2 3}{3\,N\,w\,\mathrm{g}}}$$

the unique zero of Δm

[machine: solved exactly in the prime-log ring];

the vacuum-gauge, fully crystallized value:

$$\varepsilon^{\text{crit}}(1, 1, 1) = \frac{2 - \log_2 3}{3} \in \left(\frac{1}{8}, \frac{1}{7}\right)$$

[machine: $2^{13} = 8192 > 6561 = 3^8$; $\quad 2^{11} = 2048 < 2187 = 3^7$];

the exact squeeze of the numerator:

$$\frac{2}{5} \;<\; 2 - \log_2 3 \;<\; \frac{5}{12}$$

[machine: $\log_2 3 < \frac{8}{5} \iff 3^5 = 243 < 256 = 2^8$; $\quad \log_2 3 > \frac{19}{12} \iff 3^{12} = 531441 > 524288 = 2^{19}$] :

$$\frac{2}{15\,Nw\mathsf{g}} \;<\; \varepsilon^{\text{crit}}(N,w,\mathsf{g}) \;<\; \frac{5}{36\,Nw\mathsf{g}}$$

[machine: two rational bounds on the critical strength at every admissible (N,w,g)];

the irrationality row:

$$\varepsilon^{\text{crit}}(N,w,\mathsf{g}) \notin \mathbb{Q} \;\text{ for every rational }\; (N,w,\mathsf{g}) > 0$$

[BM.84(4) : $2-\log_2 3 \notin \mathbb{Q}$ on the $2^p \neq 3^q$ census; machine];

the dissolution row:

on the Bernoulli specialization $w = 1-4\varepsilon_q$ [BM.89(4)] :

$$\varepsilon^{\text{crit}} \to \infty \;\text{ as }\; \varepsilon_q \to \tfrac{1}{4}^{-}$$

the fully dissolved limit never closes the splitting

[machine: monotone in w, unbounded as $w \to 0^{+}$];

(5) the ensemble-independent bar (parameter-free):

$$N\,w\,\mathsf{g}\cdot\varepsilon_{\text{EM}} \;\le\; \frac{1}{8} \qquad\Longrightarrow\qquad \Delta m > 0$$

[machine: $\Delta m > 0 \iff Nw\mathsf{g}\,\varepsilon_{\text{EM}} < \dfrac{2-\log_2 3}{3}$; $\quad \dfrac{2-\log_2 3}{3} > \dfrac{1}{8} \iff 8(2-\log_2 3) > 3 \iff 2^{13} > 3^8 \iff 8192 > 6561$];

the bar is uniform over the lapse, the crystallization weight and the curvature composition — ¬one of the three is a datum the corpus fixes;

the two priced witnesses at ε_{EM}

$= \frac{1}{9}$ stand [BM.62(8) : $4^3 = 64 > 54 = 3^3\cdot 2$; $\quad 4^5 = 1024 > 972 = 3^5\cdot 2^2$];

(6) the trichotomy (total, exact, non-empty on both sides):

$$\boxed{\varepsilon_{\text{EM}} < \varepsilon^{\text{crit}} \iff m_n > m_p \iff \frac{m_n}{m_p} > 1; \qquad \varepsilon_{\text{EM}} = \varepsilon^{\text{crit}} \iff m_n = m_p \iff \frac{m_n}{m_p} = 1; \qquad \varepsilon_{\text{EM}} > \varepsilon^{\text{crit}} \iff m_n < m_p \iff \frac{m_n}{m_p} < 1}$$

the opening region is non-empty:

$$\left\{0 \le Nw\mathsf{g}\,\varepsilon_{\text{EM}} \le \tfrac{1}{8}\right\} \neq \varnothing \text{ and contains } \varepsilon_{\text{EM}} = 0 \text{ and } \varepsilon_{\text{EM}} = \tfrac{1}{9} \text{ at } Nw\mathsf{g} = 1$$

$$[(5);\ \text{machine}];$$

the closing region is non-empty:

$$\varepsilon_{\text{EM}} > \varepsilon^{\text{crit}}(N, w, \mathsf{g}) \text{ is inhabited at every admissible } (N, w, \mathsf{g}) > 0$$

$$[\text{machine: } \varepsilon^{\text{crit}} \text{ is a positive exact value}];$$

the critical locus is exact, unique, and irrational at rational (N, w, g) $\qquad [(4)];$

(7) the universal ratio bound:

$$\frac{m_n}{m_p} \text{ is strictly decreasing in } \varepsilon_{\text{EM}} \text{ and strictly decreasing in } w$$

$$[\text{machine: the two cross-multiplied sign rows}]:$$

on the sub-critical branch:

$$1 \;<\; \frac{m_n}{m_p} \;<\; \frac{\ln 20}{\ln 15}$$

$$\text{over all } \; w \in (0, 1],\ N, \mathsf{g} > 0,\ 0 \le \varepsilon_{\text{EM}} < \varepsilon^{\text{crit}}$$

$$[\text{the lower end open at the critical locus; the upper end a strict supremum as } w \to 0^+,\ \varepsilon_{\text{EM}} \to 0];$$

at $\varepsilon_{\text{EM}} = 0$ specifically:

$$\frac{\ln 1280}{\ln 960} \;\le\; \frac{m_n}{m_p}\bigg|_{\varepsilon_{\text{EM}}=0} \;<\; \frac{\ln 20}{\ln 15}$$

[the lower endpoint attained at $w = 1$; machine: the monotonicity in w];

$$\frac{\ln 1280}{\ln 960} \in \Big(\frac{26}{25}, \frac{21}{20}\Big)$$

[BM.84(3); machine: $2^{44} > 3^{26}\cdot 5,\ 2^{34} < 3^{21}\cdot 5$];

$$\frac{\ln 20}{\ln 15} \in \Big(\frac{11}{10}, \frac{10}{9}\Big)$$

[machine, derived here : $10\ln 20 > 11\ln 15 \iff 20^{10} > 15^{11}$ $(10240000000000 > 8649755859375)$; $9\ln 20 < 10\ln 15 \iff 20^{9} < 15^{10}$ $(512000000000 < 576650390625)$];

$$\boxed{1 \;<\; \frac{m_n}{m_p} \;<\; \frac{10}{9} \qquad \text{at every admissible } (w,\ N,\ \mathrm{g},\ \varepsilon_{\mathrm{EM}}) \text{ on the sub-critical branch}}$$

(8) the incommensurability, ensemble-robust:

$$\frac{m_n}{m_p} = \frac{4}{3} \iff 3\big(6w\ln 2 + \ln 20\big) = 4\big(6w\ln 2 + \ln 15 + 3Nw\mathrm{g}\,\varepsilon_{\mathrm{EM}}\ln 2\big)$$

$$\iff 12\,Nw\mathrm{g}\,\varepsilon_{\mathrm{EM}}\ln 2 = (6-6w)\ln 2 - \ln 405 \iff \boxed{\varepsilon^{*}_{\mathrm{EM}}(w,N,\mathrm{g}) = \frac{(6-6w)\ln 2 - \ln 405}{12\,N\,w\,\mathrm{g}\,\ln 2}}$$

and

$$(6-6w)\ln 2 \;\le\; 6\ln 2 \;=\; \ln 64 \;<\; \ln 405$$

[machine: $64 < 405$; $w > 0$] $\implies$ $\boxed{\varepsilon^{*}_{\mathrm{EM}} < 0 \ \text{ for every } w \in (0,1]}$

the informational ratio never equals the ensemble ratio at any admissible coupling and any admissible crystallization weight

[BM.83(3); BM.84(8) generalized; machine];

the irrationality survives:

$$\frac{m_n}{m_p} \notin \mathbb{Q} \ \text{ at every rational } (w, N, \mathrm{g}, \varepsilon_{\mathrm{EM}})$$

[BM.83(2)'s lemma: the $\ln 3$ coefficient of the numerator is 0 and of the denominator is 1; machine];

(9) the rescope rows (recorded, ¬retracted):

BM.62(3)–(9),

BM.63,

BM.65(3)–(5),

BM.66(3),

BM.70,

BM.75(4)'s opposed row,

BM.77(9),

BM.84(2)–(8),

BM.85(1)–(3) :

stand on their own locus — the $(w, N, \mathrm{g}) = (1, 1, 1)$ specialization with c symbolic or priced;

every integer certificate they carry is re-derived here at the specialization and none is weakened;

BM.62(7)'s four-certificate enclosure keeps both rows: the c

$= \frac{3}{5}$ row stands as the void alternative's exact arithmetic,

and the c

$= \frac{1}{2}$ row becomes (4)'s derived value;

¬a frozen refutation is reopened;

¬a Census cell changes;

BM.77(5)'s specified-read trichotomy is untouched [BM.92(3)];

(10) status:

derived: (1)–(8);

recorded: (9);

symbolic: $\varepsilon_{\text{EM}},\ N,\ \mathsf{g},\ w$ (w evaluated exactly at $\varepsilon_q = 0$ and on the marked specialization);

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a declaration,

§Cal-BM stays empty;

the open row: the numerical value of e [EM.25(7)] — it decides which side of $\varepsilon^{\text{crit}}$ the corpus sits on, and no theorem of Parts 1–4 supplies it [BM.62(12); BM.84(9) re-cited].

Req. CT2.1, D2.9, EM.25(7) (Part 2); P1, P4, P5 (Parts 1–2); FT.44(4) (Part 3); Census FT.35(2) (row-wise); Def-BM.13(5), BM.51(6), BM.52(4), BM.52(5), BM.53(2), BM.54(1), BM.54(3), BM.57(4), BM.58(4), BM.59(4), BM.62(3), BM.62(7), BM.62(8), BM.62(12), BM.63, BM.65(3), BM.65(5), BM.66(3), BM.69(3), BM.70, BM.75(4), BM.77(9), BM.83(2), BM.83(3), BM.84(1), BM.84(2), BM.84(3), BM.84(4), BM.84(8), BM.84(9), BM.85(1), BM.85(3), BM.89(4), BM.89(5), BM.90(5) (this part); exact prime-log arithmetic with integer power certificates, Laurent-monomial assembly with coefficient censuses, cross-multiplied monotonicity rows, exact rational enclosures of ln-ratios [math; machine: `checks/verify_H_em_self_energy_correction.py` §E].

Pf.

(1) : the five zero-or-monomial rows re-cited [BM.62(1), (2); BM.84(1)];

$$\Delta m_{\text{EM}}^{\text{corr}} = -\big(E_{\text{EM}}^{\text{self}}(+1) - E_{\text{EM}}^{\text{self}}(0)\big) = -\tfrac{3Nw\mathsf{g}\varepsilon_{\text{EM}}\ln 2}{\mathfrak{h}}\ \text{[BM.89(5); machine]}$$

$$(2):\ 6w\ln 2+\ln 20-(6w\ln 2+\ln 15+3Nw\mathrm{g}\varepsilon_{\mathrm{EM}}\ln 2)$$

$$=\ln 20-\ln 15-3Nw\mathrm{g}\varepsilon_{\mathrm{EM}}\ln 2$$

$$=\ln\tfrac{4}{3}-3Nw\mathrm{g}\varepsilon_{\mathrm{EM}}\ln 2\ \text{[machine: the prime-log ring]};$$

$$\ln 20=2\ln 2+\ln 5,\ \ln 15=\ln 3+\ln 5\ \text{[machine: } 20=2^2\cdot 5,\ 15=3\cdot 5];$$

w's coefficient in Δm is 0 in the ledger part [machine]

$$(3):\ \text{the quotient of } (2)\text{'s two rows [machine]};$$

$$(w,N,\mathrm{g})=(1,1,1):\ 6\ln 2+\ln 20=8\ln 2+\ln 5=\ln(2^8\cdot 5)=\ln 1280,$$

$$6\ln 2+\ln 15=6\ln 2+\ln 3+\ln 5=\ln 960\ \text{[machine]}$$

$$(4):\ \Delta m$$

$$=0\iff \ln\tfrac{4}{3}$$

$$=3Nw\mathrm{g}$$

$$\varepsilon_{\mathrm{EM}}\ln 2\iff\varepsilon_{\mathrm{EM}}$$

$$=\tfrac{\ln(4/3)}{3Nw\mathrm{g}\ln 2}$$

$$=\tfrac{2\ln 2-\ln 3}{3Nw\mathrm{g}\ln 2}$$

$$=\tfrac{2-\log_2 3}{3Nw\mathrm{g}}\ \text{[machine: solved in the prime-log ring]};$$

$$\tfrac{2-\log_2 3}{3}>\tfrac{1}{8}\iff 8(2-\log_2 3)>3\iff 13>8\log_2 3\iff 2^{13}>3^8\ \text{[machine: } 8192>6561]$$

$$\tfrac{2-\log_2 3}{3}<\tfrac{1}{7}\iff 7(2-\log_2 3)<3\iff 11<7\log_2 3\iff 2^{11}<3^7\ \text{[machine: } 2048<2187];$$

$$2-\log_2 3\notin\mathbb{Q}\ \text{[BM.84(4)'s census]};$$

$$\varepsilon^{\text{crit}} = O(1/w) \to \infty \text{ as } w \to 0^+ \text{ [machine: monotone, unbounded]}$$

$$(5):\ \Delta m > 0 \iff 3Nw\mathrm{g}\,\varepsilon_{\text{EM}} \ln 2 < \ln \tfrac{4}{3} \iff Nw\mathrm{g}\,\varepsilon_{\text{EM}} < \tfrac{2-\log_2 3}{3} \text{ [machine]};$$

$$\tfrac{1}{8} < \tfrac{2-\log_2 3}{3}\ [(4)] \implies Nw\mathrm{g}\,\varepsilon_{\text{EM}} \le \tfrac{1}{8} \implies \Delta m > 0 \text{ [machine]}$$

$(6):$ (4)'s uniqueness $\wedge$ the strict monotonicity in $\varepsilon_{\text{EM}} \implies$ the three-way equivalence [machine: the sign census on a rational grid of $(N, w, \mathrm{g}, \varepsilon_{\text{EM}})$];

$$\varepsilon_{\text{EM}} = 0 \in \text{the opening region since } \ln \tfrac{4}{3} > 0\ [4 > 3;\ \text{machine}]$$

$$(7):\ \text{monotone in } \varepsilon_{\text{EM}}:\ \text{the denominator increases, the numerator is fixed [machine]};$$

$$\text{monotone in } w \text{ at } \varepsilon_{\text{EM}}$$

$$= 0:\ w_1 < w_2 \implies \frac{6w_1 \ln 2 + \ln 20}{6w_1 \ln 2 + \ln 15} > \frac{6w_2 \ln 2 + \ln 20}{6w_2 \ln 2 + \ln 15}$$

$$\iff (w_2 - w_1) 6 \ln 2 (\ln 20 - \ln 15) > 0 \text{ [machine: cross-multiplied, } \ln 20 > \ln 15 \text{ on } 20 > 15]$$

$$10 \ln 20 - 11 \ln 15 = \ln \frac{20^{10}}{15^{11}} > 0 \iff 20^{10} > 15^{11} \text{ [machine: } 10240000000000 > 8649755859375];$$

$$9 \ln 20 - 10 \ln 15 = \ln \frac{20^9}{15^{10}} < 0 \iff 20^9 < 15^{10} \text{ [machine: } 512000000000 < 576650390625]$$

$$(8):\ 3(6w \ln 2 + \ln 20) - 4(6w \ln 2 + \ln 15 + 3Nw\mathrm{g}\,\varepsilon_{\text{EM}} \ln 2)$$

$$= -6w \ln 2 - 4 \ln 3 - \ln 5 - 12Nw\mathrm{g}$$

$$\varepsilon_{\text{EM}} \ln 2 \text{ [machine: the prime-log ring]};$$

$$\text{setting it to } 0 \text{ gives } \varepsilon^*_{\text{EM}} \text{ of } (8)$$

$$(6 - 6w) \ln 2$$

$$\leq 6\ln 2$$

$$= \ln 64 < \ln 405$$

$$\iff 64 < 405 \text{ [machine]} \implies \text{the numerator of } \varepsilon^{*}_{\mathrm{EM}} \text{ is negative} \implies \varepsilon^{*}_{\mathrm{EM}} < 0 \text{ [machine]};$$

$$\text{irrationality}: \ q(6w\ln 2 + \ln 20)$$

$$= p(6w\ln 2 + \ln 15 + 3Nw\mathrm{g}\,\varepsilon_{\mathrm{EM}}\ln 2) \text{ with rational } (w, N, \mathrm{g}, \varepsilon_{\mathrm{EM}}) \implies \text{the } \ln 3 \text{ coefficient } -p$$

$$= 0$$

$$\implies p$$

$$= 0$$

$$\implies q$$

$$= 0 \text{ [BM.83(2); machine]}$$

$$(9)\text{–}(10): \text{ the clause census and the standing-rule scans [machine]. } \blacksquare$$

Phys. Put the corrected field energy into the splitting and almost nothing moves — which is the point. Both classes lose the same fraction of their edge register to the flicker, so the crystallization weight cancels out of the ledger difference entirely and survives only where the field lives: the electromagnetic term is now the shell value, times the lapse, times the crystallization weight, times the curvature composition, times three times the logarithm of two over h. The critical strength inherits those three factors in the denominator, which means the opening region can only widen as the composite fails to crystallize, and in the fully dissolved limit it never closes at all. Better, the bar becomes parameter-free in the ensemble data: whenever the product of lapse, weight, curvature composition and coupling stays at or below one eighth, the neutron is the heavier baryon, and the certificate is a single pair of integers, eight thousand one hundred ninety-two against six thousand five hundred sixty-one. The ratio itself is bounded between one and ten ninths at every admissible value of everything the corpus declines to fix, with the old one-and-a-twentieth bound recovered as the fully crystallized endpoint and the logarithm of twenty over the logarithm of fifteen as the dissolved supremum. And the incommensurability hardens: the coupling at which the informational ratio would equal the ensemble's four thirds is negative for every crystallization weight in the admissible range, on the single integer certificate that sixty-four is less than four hundred five.

Theorem 4.78 (BM.92 — The Q-value on both formed reads: the uniform closure, the corrected margins, and the untouched specified branch). Thm.

(1) the informational Q-value with the corrected term:

$$Q_\beta := m_n - m_p - m_\ell - m_\nu$$

[BM.41(4); BM.63(1); BM.85(1)] :

$$\boxed{Q_\beta\,\mathfrak{h} \;=\; \ln\frac{4}{3} - 3\,N\,w\,\mathsf{g}\,\varepsilon_{\mathrm{EM}}\ln 2 - F_{\ell\nu}, \qquad F_{\ell\nu} := m_\ell\mathfrak{h} + m_\nu\mathfrak{h}}$$

[BM.91(2); machine: the prime-log assembly];

$$F_{\ell\nu} \geq 2\ln 2$$

[BM.70(2), (3) : every materialized closed chain carries ≥ 2 directed edges; $m\mathfrak{h} = E_{\mathrm{int}}\ln 2$; $m_\nu\mathfrak{h} \geq 0$ by PSD];

$$\boxed{Q_\beta\,\mathfrak{h} \;\leq\; \ln\frac{4}{3} - 2\ln 2 \;=\; -\ln 3 \;<\; 0 \qquad \text{at every admissible } (w,\ N,\ \mathsf{g},\ \varepsilon_{\mathrm{EM}})}$$

[machine: $3Nw\mathsf{g}\,\varepsilon_{\mathrm{EM}}\ln 2 \geq 0$; $4 < 12$];

the closure is uniform over the ensemble data:

the attenuation lowers the magnitude of the EM term and cannot raise Q_β above $-\ln 3/\mathfrak{h}$

[machine: the coefficient census];

(2) the margin ladder and the corrected ratio forms:

$$Q_\beta\mathfrak{h} \;\leq\; -\ln 3, \quad -\ln 12, \quad -\ln 24$$

$$\text{on} \quad 4 < 12, \quad 4 < 48, \quad 4 < 96$$

[BM.70(3); BM.85(2) re-certified; machine] :

$$\Sigma_{\mathrm{B}} - 2\ln 2 = -\ln 3,$$

$$\Sigma_{\mathrm{B}} - 4\ln 2 = -\ln 12,$$

$$\Sigma_{\mathrm{B}} - 5\ln 2 = -\ln 24$$

[machine: the three exact identities];

$$\boxed{\frac{Q_\beta}{m_n} \;\le\; -\frac{\ln 3}{6\,w\ln 2 + \ln 20}}$$ with the two exact endpoint brackets:

$$w = 1 :$$

$$-\frac{\ln 3}{\ln 1280} \in \Big(-\frac{2}{13}, -\frac{3}{20}\Big)$$

[BM.85(3); machine: $3^{13} = 1594323 < 1638400 = 2^{16}\cdot 5^2$; $3^{20} = 3486784401 > 2097152000 = 2^{24}\cdot 5^3$];

$$w \to 0^+ :$$

$$-\frac{\ln 3}{\ln 20} \in \Big(-\frac{2}{5}, -\frac{1}{3}\Big)$$

[machine, derived here : $3\ln 3 > \ln 20 \iff 27 > 20$; $5\ln 3 < 2\ln 20 \iff 243 < 400$];

the informational Q-value is a strictly negative exact fraction of the parent mass at every crystallization weight, bounded away from zero by a rational;

(3) the specified read is untouched:

$$Q_\beta^{\mathrm{spec}}\,\mathfrak{h} = \Delta T(S) - \rho_\ell\sqrt{G\big(c(0)\big)} - \rho_\nu\sqrt{G\big(c(i_\nu)\big)},$$

$$\Delta T(S) = 5\,m_B^2(\cdot, S)$$

[BM.77(2); BM.85(4)] :

the EM summands are ¬parts of that read

[BM.77(3) : disjoint alphabets, no double count] $\Longrightarrow$ neither the rescission of BM.87 nor the ensemble correction of BM.89 alters any row of BM.77(5), (6)

$$\Delta T = \Big(\frac{4219785}{98956},\ \frac{9608310}{393679},\ \frac{455}{23},\ \frac{68565}{3088}\Big),$$

$$\frac{273}{23} < \rho^{\text{crit}}(S, i_\nu) < \frac{42197850}{791648}$$

[BM.77$(2), (5)$; machine re-certified];

the witnesses stand:

$$828100 > 5819,$$

$$4219785 < 43 \cdot 98956 = 4255108,$$

$$7500 > 1849,$$

$$3312400 > 5819,$$

$$99 < 100,$$

$$75 > 64,$$

$$273 > 253,$$

$$42197850 < 42748992;$$

(4) the verdict rows (read-relative and complete):

closed on the informational currency for every member, uniformly in $(w, N, \mathsf{g}, \varepsilon_{\text{EM}})$

[(1); BM.70(4)(a)];

open on $\mathcal{O}_S$, closed on $\mathcal{C}_S$, degenerate on $\partial\mathcal{O}_S$ on the specified read

[BM.77$(5), (6)$; BM.85(5)];

the decay direction is fixed in both currencies:

$$T(0,S) = \tfrac{4}{3}T(+1,S) > T(+1,S)$$

[BM.28(4)(ii); machine]

and

$m_n > m_p$ on the sub-critical branch

[BM.91(6)] :

the parent is the neutral class on every formed read, and the grammar $n \to p + \ell + \nu$ stands unmodified

[BM.41(1); BM.45; BM.77(6)];

the conditional spectrum on $\mathcal{O}_S$:

$$E_\ell^{\max} = \frac{m_n^2 + m_\ell^2 - (m_p + m_\nu)^2}{2m_n},$$

$E_{\max} : \langle E\rangle = 2:1$ scale-invariant

[BM.41(5); BM.49$(3),(4)$; BM.77(8); ¬an external endpoint compared [P5]];

(5) the two reads are ¬comparable:

$$Q_\beta^{\text{info}} : \{\ln 2, \ln 3, \ln 5,\ N,\ w,\ \mathfrak{g},\ \varepsilon_{\text{EM}},\ \mathfrak{h},\ m_\ell,\ m_\nu\};$$

$$Q_\beta^{\text{spec}} : \{m_B^2(\cdot,S),\ \rho_\ell,\ \rho_\nu,\ g,\ g',\ \mathfrak{h}\}$$

the two alphabets are disjoint except for $\mathfrak{h}$

[machine: the token census; $\mathfrak{g}$ and g are distinct symbols [(4)'s local notation; BM.39(1)]];

¬a derived equation relates the two expressions — that is (R1),

(R3),

(R4) and the Cal-F scales together

[BM.82(6); BM.83(5), (7); BM.85(5)];

(6) status:

derived: (1)–(5);

read-relative: the kinematic verdict (4);

declared: nothing — A-BM.8's specification is re-cited, ¬re-issued;

¬a clause of BM.41, BM.49, BM.63, BM.70, BM.71, BM.77, BM.85 is retracted;

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

§Cal-BM stays empty.

Req. P1, P4, P5 (Parts 1–2); FT.79(5) (Part 3); Def-BM.13(4), BM.28(4), BM.37(2), BM.39(1), BM.41(1), BM.41(4), BM.41(5), BM.45, BM.49(3), BM.49(4), BM.63(1), BM.70(2), BM.70(3), BM.70(4), BM.77(2), BM.77(3), BM.77(5), BM.77(6), BM.77(8), BM.82(6), BM.83(5), BM.83(7), BM.84(1), BM.85(1), BM.85(2), BM.85(3), BM.85(4), BM.85(5), BM.87, BM.89, BM.91(2), BM.91(6) (this part); the declaration addendum A-BM.8, item 7 (pointers); exact prime-log arithmetic with integer power certificates, exact rational arithmetic at the four frozen sector expectations, token censuses [math; machine: `checks/verify_H_em_self_energy_correction.py` §F].

Pf.

$$(1):\ Q_\beta \mathfrak{h} = \Delta m\, \mathfrak{h} - F_{\ell\nu} = \ln\tfrac{4}{3} - 3Nw\mathrm{g}\,\varepsilon_{\mathrm{EM}} \ln 2 - F_{\ell\nu}\ [\text{BM.91}(2);\ \text{machine}];$$

$$F_{\ell\nu} \geq 2\ln 2\ [\text{BM.70}(2), (3)\ \text{re-cited}];$$

$$\ln\tfrac{4}{3} - 2\ln 2 = 2\ln 2 - \ln 3 - 2\ln 2 = -\ln 3\ [\text{machine}];$$

$$-\ln 3 < 0 \iff 3 > 1 \iff 4 < 12\ [\text{machine}]$$

$$3Nw\mathsf{g}$$

$$\varepsilon_{\mathrm{EM}} \ln 2$$

$$\geq 0 \text{ at } N,$$

$$w,$$

$$\mathsf{g},$$

$$\varepsilon_{\mathrm{EM}}$$

$$\geq 0 \text{ [machine: the coefficient census]}$$

$$\Longrightarrow\ Q_{\beta}\mathfrak{h}$$

$$\leq -\ln 3 \text{ uniformly [machine]}$$

$$(2):\ \Sigma_{\mathrm{B}} - 2\ln 2$$

$$= -\ln 3,$$

$$\Sigma_{\mathrm{B}} - 4\ln 2$$

$$= 2\ln 2 - \ln 3 - 4\ln 2$$

$$= -\ln(4\cdot 3)$$

$$= -\ln 12,$$

$$\Sigma_{\mathrm{B}} - 5\ln 2$$

$$= -\ln(8\cdot 3)$$

$$= -\ln 24 \text{ [machine: the three identities]};$$

$$4 < 12 < 48 < 96 \text{ [machine]}$$

$$Q_\beta/m_n = (Q_\beta \mathfrak{h})/(m_n \mathfrak{h}) \leq -\ln 3/(6w \ln 2 + \ln 20) \text{ [BM.91(2); machine]};$$

$$3\ln 3 - \ln 20 > 0 \iff 27 > 20, \quad 5\ln 3 - 2\ln 20 < 0 \iff 243 < 400 \text{ [machine]}$$

$$(3): \text{ BM.77}(2), (3), (5) \text{ re-cited};$$

$$15 \cdot 843957 = 12659355,$$

$$15 \cdot 1921662 = 28824930,$$

$$15 \cdot 91 = 1365,$$

$$15 \cdot 13713 = 205695 \text{ [BM.89(6); machine]};$$

the eight witnesses re-certified as integer comparisons [machine]

$$(4): \ T(0, S)/T(+1, S) = W(0)/W(+1) = 20/15 = 4/3 > 1 \text{ [machine]};$$

$m_n > m_p$ sub-critically [BM.91(6)];

the endpoint identity and the 2:1 moment ratio re-cited [BM.41(5); BM.49(3), (4)]

(5) : the two symbol sets enumerated from the two expressions and intersected [machine];

g $\neq g$ as tokens [machine];

(6) : the standing-rule scans [machine]. ■

Phys. One Q-value, two formed reads, and the correction lands on only one of them. On the informational currency the decay's energy balance keeps its shape: the transport register's log of four thirds, less the corrected field term, less whatever the two emitted lines cost. Because the corrected field term is nonnegative and now attenuated by the lapse, the crystallization weight and the curvature composition, the two-bit floor on any materialized chain still drives the total below minus the logarithm of three — at every value of every quantity the corpus declines to fix.

The closure is not merely preserved, it is uniform, and its size as a fraction of the parent mass runs from minus three twentieths at full crystallization to minus one third in the dissolved limit, always an exact rational bound away from zero. On the specified read nothing moves at all: the electromagnetic summands were never part of that expression, so the opening region, the closing region, the exact critical locus and all eight integer witnesses stand as derived. The two verdicts remain read-relative and incomparable, their alphabets sharing one symbol, and the one thing both agree on is the direction — the neutral class is the parent, the grammar is untouched, and the decay is allowed exactly where a formed read allows it.

BM.93 — The status of the EM self-energy correction: adjudication and scope. Structural-constraints record: see Appendix B, Section B.1.28 of the Addendum.

4.21 Tier BM.19 — The Flicker Integral, the Rest-Mass Q-Value, and the Relating Equation: the Materialized Ensemble, the Vertex Superposition, the Rung-One ν Line, and the Bridge in Four Exact Forms

Locus. This tier fixes one read, moves one Q-value onto one species, and closes one ledger item in form. The read it fixes is the crystallized-snapshot read of a sealed composite's information content — the read that takes the computed cell's $E_{\text{int}} = 6$ [Census FT.35(2); BM.17(1)] as the composite's count and hence as a fixed mass. That read is a category error against the corpus's own rows: the count is one point of an ensemble distribution [Def-BM.5(4); BM.19(5)], the real and the virtual registers are mixed at every nonzero threshold [A4; D0.1], and the composite is an ensemble and ¬ a topology [BM.19(6); BM.89(4)]. BM.94 replaces the snapshot with the integral, derives the distribution and its exact moments, and derives the materialization restriction that holds the mean edge count above the digon's two. BM.95 derives the superposition a vertex carries in the transport register, with its exact weights and the two positive parts of the transport splitting. BM.96 derives the rung-one ν line's rest mass from the seal structure, its strict positivity, its three-line order, its single-ray occupancy, and the four independent bars against a seesaw-type mechanism. BM.97 adjudicates the rest-mass species and forms the Q-value on it, with the exact critical locus and integer-certified witnesses on both the specified and the pure-pole reads. BM.98 derives the relating equation in four exact forms — count, curvature, coupling, flicker-weighted — with its enhancement bounded by an exact integer. BM.99 recomputes the splitting and the Q-value on the materialized read, gaining a two-sided parameter-free bar and two exact tightenings. BM.100 adjudicates.

Standing rows. α_B stays barred on BM.61(8)'s four grounds and §Cal-BM stays empty. No clause of BM.1–BM.93 is retracted, replaced or renumbered: every row this tier touches is re-scoped at its own locus and recorded in BM.94(9), BM.97(7), BM.99(7) and BM.100(4), in BM.77(7)'s pattern. No measured coupling, mass or ratio enters any derivation [P5]: the neutron–proton ratio is derived and bounded at BM.99(5) and the electric coupling stays symbolic [EM.25(7); BM.88(7)] — neither is imported. No right-ray matter line and no seesaw-type mechanism is suggested, added or required [BM.96(4), (6)]. The relating equation is derived in form with its residue localized to one undeclared dimensionless ratio; ρ_{B} is ¬derived and BM.69(4)'s four grounds stand unweakened. Local notation, fixed for the whole tier: $p := 1 - 4\varepsilon_q$ the stationary retention weight [BM.19(2)]; $u := 1 - p = 4\varepsilon_q$ the flicker weight; $D := 3 - 3p + p^2 = 1 + u + u^2$ the dilution denominator; w the crystallization weight of BM.89(4) and $w_{\text{mat}} := 1/D$ its materialized companion; $E^{\text{cryst}} := 6$; $k := r/\Delta x_{\text{B}}$, $f(k) = 1 - \frac{1}{k}$, N, g, G, ε_{EM} as in BM.88's local notation; ρ_{B}, D_S, $\mathcal{R}$ as in BM.69(2); M_0, s, Δ_g, Δ_g^{crit} as in BM.97(4); $c := \cos\frac{1}{8}$, $s_\nu := \sin\frac{1}{8}$ in BM.96(2). The symbol q keeps its end-register

meaning [D3.4; EM.9], the italic g keeps its BM.39(1) meaning, and the italic s keeps its BM.97(4) meaning at every occurrence.

Theorem 4.79 (BM.94 — The flicker integral: the snapshot read's category error, the materialized distribution, and the two crystallization weights). Thm.

(1) the category error (the referent census):

the snapshot read takes	the corpus supplies
$E_{\text{int}} = 6$ as the composite's count	$E_{\text{int}}(R) = \lvert R\rvert$ distributed over the flicker ensemble [Def-BM.5(4); BM.19(5)]
one configuration as the composite	$\lvert\mathcal{E}(\mathcal{X})\rvert > 1 \iff \varepsilon_q > 0$: an ensemble at every nonzero threshold [BM.19(6)]
$m = \frac{6 \ln 2}{\hbar}$ as a fixed mass	a mixture of the real and the virtual registers [A4; D0.1; BM.89(4)]
a value	a distribution whose top point is the crystallized value [Def-BM.5(4); BM.86(4) scope]

the error, stated exactly:

$E_{\text{int}} = 6$ is one point of the distribution, ¬the ensemble, and its weight is $P(E{=}6) = \dfrac{p^2}{D} < 1$ at every $\varepsilon_q > 0$

the weight deficit is exact:

$$1 - P_{\text{mat}}(E{=}6) = \frac{3(1-p)}{D} > 0$$

[machine: $D - p^2 = 3 - 3p = 3(1-p)$];

(2) the proper integral:

$$E_{\text{int}}^{\text{eff}} := \sum_{R \in \mathcal{E}(\mathcal{X})} P(R) \cdot E_{\text{int}}(R)$$

[BM.87(5); BM.20(3); Def-H.32(1); Def-BM.5(2)] :

$P(R)$ is the ensemble weight of the configuration R,

derived as the unique stationary line of the per-pair Janus generator [BM.19(1), (2); FT.108(4); Def-BM.5(3)],

¬declared;

the sum is finite and exact at every budget B [Def-BM.5(1); Def-H.32(1)],

and the $B \to \infty$ limit is ¬constructed [Def-BM.5(5)];

linearity on $|R|$ gives the per-datum form:

$$E_{\text{int}}^{\text{eff}} = \sum_{\text{pairs}} P(\text{the pair is real}) \cdot 2$$

[BM.89(4) re-derived as the first moment; machine];

(3) the ledger read (the quantum numbers the read is taken on):

$$\mathfrak{L} = \bigcup_{n\geq 0} \mathfrak{L}(\tau_n),$$

$$\left|\mathfrak{L}(\tau_n)\right| = \sum_{k<n} n_{\text{real_out}}(\tau_k)$$

[A7; A6] :

the two quantum numbers are the tick and the scale:

$$\tau : \ \tau_n \to \tau_{n+1}, \ \neg\exists\, \tau^{-1} \ [\text{A6; D1.8} : \ x^0 := \tau_n],$$

$$\mathfrak{h} : \ \exists! \text{ free dimensionful scale [P1; D2.9; A §0.3]};$$

$$n_{\mathcal{R}} = E_{\text{int}} = |R| \ [\text{D0.8; Def-BM.5}(4)], \qquad S_{\mathfrak{L}} = n_{\mathcal{R}} \ln 2 \ [\text{C9; CT1.29; BM.35}(3)];$$

the crystallized card is the snapshot at one τ [Census FT.35(2); BM.17(1)] :

the composite is read from $\mathfrak{L}$, ¬from the card;

the constituent at a vertex is a quantum superposition over the ledger's transport fibre, ¬a fixed state

[BM.95(2); Def-BM.13$(1), (3)$] :

(4) the derived distribution (the marked pair specialization, marked ¬adopted):

$$E^{\text{cryst}}$$

$$= 6$$

$= (c_1 \colon 2) + (c_2 \colon 2) + (\text{connector} \colon 2)$ is three pairs [Census FT.35(2); BM.21(1); BM.89(4)] :

each pair is retained with the stationary weight p

$= 1 - 4\varepsilon_q$ [BM.19(2); an identity, exact at every rational threshold]

$$\implies E_{\text{int}}(R) \qquad = 2j,$$

$$j \in \{0, 1, 2, 3\};$$

$\boxed{P(E{=}2j) = \binom{3}{j} p^j (1-p)^{3-j}}$: $\qquad \big((1{-}p)^3,\ 3p(1{-}p)^2,\ 3p^2(1{-}p),\ p^3\big)$ at $E = (0, 2, 4, 6)$

normalization: $\displaystyle\sum_j \binom{3}{j} p^j (1-p)^{3-j} = (p + (1-p))^3 = 1$

[machine: exact in $\mathbb{Q}[p]$; the census at six rational thresholds];

the unconditioned moments:

$$\langle E \rangle = 6p = 6w$$

[BM.89(4) re-derived; machine],

$$\langle E^2 \rangle = 12p + 24p^2,$$

$$\text{Var}(E) = 12p(1-p)$$

the no-flicker collapse:

$\varepsilon_q = 0 \implies p = 1 \implies P(E{=}6) = 1$ and the ensemble is the single crystallized configuration

[BM.19(3); Def-BM.5(5); machine];

(5) the materialization restriction (derived, ¬declared):

the $E{=}0$ member supports no closed chain and no terminus:

$$|E_{\mathcal{R}}| = 0 \implies \#\{\text{closed chains}\} = 0 \implies \neg\exists \text{ a fermion terminus}$$

[FT.144(4); D1.4; Def-H.4; machine];

every materialized closed chain carries ≥ 2 directed edges:

$$\delta h(c)$$

$$\geq 2,$$

$$n(v)$$

≥ 2 [Def-H.4; Census FT.28 : the digon is the minimal floor; FT.32; BM.13(3); BM.70(2)];

the unconditioned average runs to the vacuum:

$$P(E{=}0) = (1-p)^3 \text{ is strictly increasing in } \varepsilon_q \text{ and } \to 1 \text{ as } \varepsilon_q \to \tfrac{1}{4}^-$$

[machine] :

the all-virtual state is the unique entropy maximum $\rho_\circ$

$$= \tfrac{1}{4} I,$$

$$S_{\text{raw}}$$

$= \ln 4$ [A5; CT0.2; BM.35(1)] and is the fixed point of the Lyapunov contraction V'

$$= (1 - \tfrac{\gamma}{4})^2 V \text{ [L1; L2; BM.51}(1)\text{]} :$$

the composite's read is the conditional one on $E \geq 2$, and the restriction is a theorem of the materialization domain, ¬a scope choice

[D2.11; A §0.2a];

(6) the materialized distribution and its exact moments:

$$D \;:=\; 3 - 3p + p^2 \;=\; \frac{1-(1-p)^3}{p} \;=\; 1 + u + u^2,$$

$$u := 1 - p = 4\varepsilon_q$$

[machine: one polynomial identity] :

$$\boxed{P_{\text{mat}}(E{=}2) = \frac{3(1-p)^2}{D}, \qquad P_{\text{mat}}(E{=}4) = \frac{3p(1-p)}{D}, \qquad P_{\text{mat}}(E{=}6) = \frac{p^2}{D}}$$

and $\dfrac{3(1-p)^2 + 3p(1-p) + p^2}{D} = 1$ [machine];

$$\boxed{\langle E\rangle_{\text{mat}} = \frac{6}{D}, \qquad \text{Var}_{\text{mat}}(E) = \frac{12\,p(1-p)(3-2p)}{D^2}, \qquad \frac{\text{Var}_{\text{mat}}(E)}{\langle E\rangle^2_{\text{mat}}} = \frac{p(1-p)(3-2p)}{3}}$$

[machine: exact rational functions];

the exact maximum of the relative width:

$$\frac{p(1-p)(3-2p)}{3} \;\leq\; \frac{10+7\sqrt{7}}{162} \;\text{ at }\; p = \frac{5-\sqrt{7}}{6}$$

[machine: the stationary point of $6p^2 - 10p + 3$ in $\mathbb{Q}(\sqrt{7})$];

$$\frac{10+7\sqrt{7}}{162} < \frac{1}{5}$$

[machine: $5(10+7\sqrt{7}) < 162 \iff 35\sqrt{7} < 112 \iff 5\sqrt{7} < 16 \iff 175 < 256$]

$$\Longrightarrow \quad \boxed{\frac{\sigma_E}{\langle E\rangle_{\text{mat}}} < \frac{1}{2}}$$

(7) the two crystallization weights:

$$w = \frac{\langle E\rangle}{E^{\text{cryst}}} = p$$

[BM.89(4) re-cited at its own locus];

$$\boxed{w_{\text{mat}} := \frac{\langle E\rangle_{\text{mat}}}{E^{\text{cryst}}} = \frac{1}{D} = \frac{1}{1+u+u^2}}$$

$$w_{\text{mat}}(0) = 1 \text{ exactly}$$

[BM.19(3); machine];

$$\boxed{\frac{1}{3} < w_{\text{mat}} \le 1 \text{ on } \varepsilon_q \in [0, \tfrac{1}{4})}$$

[CT0.8; D0.4; machine: $D \ge 1 \iff (p-1)(p-2) \ge 0$; $D < 3 \iff p(p-3) < 0$];

$w_{\text{mat}} \ge w$ with equality $\iff \varepsilon_q = 0$ [machine: $w_{\text{mat}}/w = 1/(1-(1-p)^3) \ge 1$];

$$\boxed{\langle E\rangle_{\text{mat}} = 6\, w_{\text{mat}} \in (2, 6]}$$

the top point is the crystallized count, the bottom is the digon's two [Census FT.28; Def-H.4] :

the lower endpoint is open and is attained only in the fully dissolved limit, where $\langle E\rangle_{\text{mat}} \to 2$ exactly

[machine: $6/(3-3p+p^2) \to 6/3$];

(8) the informational mass is a distribution:

$$m_{\text{hid}}(R) = \frac{E_{\text{int}}(R)\ln 2}{\mathfrak{h}}$$

[FT.44(4); Census FT.44(4); Def-H.9(2)] :

the support:

$$m_{\text{hid}} \in \left\{ \frac{2\ln 2}{\mathfrak{h}},\ \frac{4\ln 2}{\mathfrak{h}},\ \frac{6\ln 2}{\mathfrak{h}} \right\} \text{ with } (6)\text{'s exact probabilities;}$$

$$\boxed{\langle m_{\text{hid}} \rangle_{\text{mat}} = \frac{6\, w_{\text{mat}} \ln 2}{\mathfrak{h}}},$$

$$\text{Var}_{\text{mat}}(m_{\text{hid}}) = \frac{12\, p(1-p)(3-2p)\,(\ln 2)^2}{D^2\, \mathfrak{h}^2};$$

the two-bit floor survives the averaging:

$$E \geq 2 \text{ on the whole materialized support } \Longrightarrow\ m_{\text{hid}}(R) \geq \frac{2\ln 2}{\mathfrak{h}} > 0\ \ \forall R,$$

$$\langle m_{\text{hid}} \rangle_{\text{mat}} > \frac{2\ln 2}{\mathfrak{h}}$$

$$[\text{BM.70}(2),(3);\ \text{machine: } \tfrac{1}{2} < \ln 2 < 1];$$

the reciprocal register has two formed reads:

$$\left\langle E^{-1} \right\rangle_{\text{mat}} = \frac{18p(1-p)^2 + 9p^2(1-p) + 2p^3}{12\, p\, D},$$

$$\frac{1}{\langle E \rangle_{\text{mat}}} = \frac{D}{6}$$

[machine] :

$$\boxed{\langle E \rangle_{\text{mat}} \left\langle E^{-1} \right\rangle_{\text{mat}} \ =\ \frac{18 - 27p + 11p^2}{2D^2} \ \ \geq\ 1}$$

[Cauchy–Schwarz on $(E,\ E^{-1})$; machine: the exact rational function] :

$$\text{equality } \Longleftrightarrow\ E \text{ is constant on the support } \Longleftrightarrow\ p = 1 \ \Longleftrightarrow\ \varepsilon_q = 0$$

[machine];

$$\frac{58}{49} \text{ at } p = \tfrac{1}{2}$$

$$[\text{machine: } 290 < 294];$$

$$1 \le \frac{18 - 27p + 11p^2}{2D^2} < \frac{6}{5}$$

$$[\text{machine: the exact positivity of } 12p^4 - 72p^3 + 125p^2 - 81p + 18 \text{ on } (0,1]];$$

$$\Longrightarrow \ \Delta x_{\mathrm{B}} \text{ has two formed ensemble reads:}$$

$$\Delta x_{\mathrm{B}}^{\mathrm{eff}} := \frac{\mathfrak{h}}{\langle E\rangle_{\mathrm{mat}} \ln 2} = D \cdot \Delta x_{\mathrm{B}}^{\mathrm{cryst}}$$

and

$$\frac{\mathfrak{h}}{\ln 2}\left\langle E^{-1}\right\rangle_{\mathrm{mat}} \ \ge \ \Delta x_{\mathrm{B}}^{\mathrm{eff}}$$

$$[\text{Def-BM.4}(1);\ \text{BM.13}(1);\ \text{machine}];$$

the tier evaluates the coupling and the bridge at the width of the mean; the difference of the two reads is named $(\mathrm{R6})$,

$$\neg\text{filled}$$

$$[\text{BM.98}(5);\ \text{the Jensen residue}];$$

$$(9) \quad \text{status:}$$

derived: (1)–(8) on the marked pair specialization, which is ¬adopted as a general measure;

rescoped, ¬retracted: Def-BM.5(4)'s "one point" row is discharged as an exact weight (1);

BM.89(4)'s w stands at its own locus as the unconditioned weight and w_{mat} is its materialized companion,

¬a replacement;

BM.19(1)–(6), BM.20(3), BM.21, BM.86(4) untouched;

BM.91(2)'s assembly stands on w and gains the materialized variant at BM.99(1);

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

§Cal-BM stays empty;

Req. A4, A5, A6, A7 (Part 1); C9, D0.1, D0.4, D0.8, D1.4, D1.8, D2.9, D2.11 (Parts 1–2); A §0.2a, A §0.3, CT0.2, CT0.8, CT1.29 (Part 1); CT2.1 (Part 2); P1, P4, P5 (Parts 1–2); L1, L2 (Part 1); Def-H.4, Def-H.9(2), Def-H.32(1), FT.28, FT.32, FT.44(4), FT.108(4), FT.144(4) (Part 3); Census FT.28, FT.35(2), FT.44(4) (row-wise); Def-BM.4(1), Def-BM.5(1), Def-BM.5(2), Def-BM.5(3), Def-BM.5(4), Def-BM.5(5), Def-BM.13(1), Def-BM.13(3), BM.13(1), BM.13(3), BM.17(1), BM.19(1), BM.19(2), BM.19(3), BM.19(5), BM.19(6), BM.20(3), BM.21(1), BM.35(1), BM.35(3), BM.51(1), BM.61(8), BM.70(2), BM.70(3), BM.86(4), BM.87(5), BM.88(7), BM.89(4), BM.91(2) (this part); exact polynomial arithmetic over $\mathbb{Q}[p]$, exact rational specialization at six thresholds, exact binomial censuses, the stationary point of a quadratic in $\mathbb{Q}(\sqrt{7})$, exact rational interval arithmetic on a quartic, Cauchy–Schwarz over a positive measure [math; machine: `checks/verify_H_flicker_integral_restmass.py` §A].

Pf.

(1) : the two inventories enumerated from their defining clauses [Census FT.35(2); Def-BM.5(4); BM.19(5), (6)];

$1 - \frac{p^2}{D} = \frac{D-p^2}{D} = \frac{3-3p}{D} = \frac{3(1-p)}{D}$ [machine: one polynomial identity]

(2) : BM.87(5)'s relocation and BM.20(3)'s averaged flux re-cited;

the measure's uniqueness: $\dim \ker Q = 1$ [BM.19(1); FT.108(4)];

finiteness at every budget [Def-H.32(1)]

(3) : A6, A7, D1.8, P1 re-cited verbatim (frozen);

$n_{\mathcal{R}} = |R|$ [D0.8; Def-BM.5(4)];

$S_{\mathfrak{L}} = n_{\mathcal{R}} \ln 2$ [CT1.29; BM.35(3)]

$(4):$ three pairs, each retained with weight p [BM.89(4) re-cited];

$$\textstyle\sum_j \binom{3}{j} p^j (1-p)^{3-j}$$

$= 1$ by the binomial theorem over $\mathbb{Q}[p]$ [machine: coefficient comparison];

$\langle E\rangle = \sum_j 2j\binom{3}{j} p^j (1-p)^{3-j} = 6p$ [machine at six rational thresholds];

$\langle E^2\rangle = \sum_j 4j^2\binom{3}{j} p^j (1-p)^{3-j} = 12p + 24p^2$ [machine];

$$\text{Var} = 12p + 24p^2 - 36p^2 = 12p(1-p) \text{ [machine]};$$

$$p = 1 \implies P(E{=}6) = 1 \text{ [machine]}$$

$(5):$ $|E_{\mathcal{R}}| = 0 \implies \neg\exists$ a closed chain $\implies \neg\exists$ a terminus [FT.144(4); D1.4; Def-H.4 re-cited];

$\delta h \geq 2$ [Def-H.4; BM.13(3); BM.70(2)];

$(1-p)^3$ strictly increasing in ε_q and $\to 1$ [machine: monotone on the rational grid]

$\rho_\circ = \frac{1}{4}I,\ S_{\text{raw}}(\rho_\circ) = \ln 4 = \max,\ \arg\max = \{\frac{1}{4}I\}$ [CT0.2; BM.35(1) re-cited];

$V' = (1 - \frac{\gamma}{4})^2 V$ with $\gamma \to 0$ the identity-tilt corner [L2 re-cited]

$(6):$ $1 - (1-p)^3$

$= 3p - 3p^2 + p^3$

$= p(3 - 3p + p^2)$

$= pD$ [machine]

$\implies P_{\text{mat}}(E{=}2j)$

$= \binom{3}{j} p^j (1-p)^{3-j}/(pD)$ and the three values reduce to the boxed row [machine];

the normalization: $3(1-p)^2 + 3p(1-p) + p^2 = 3 - 3p + p^2 = D$ [machine]

$\langle E \rangle_{\text{mat}} = \frac{2 \cdot 3(1-p)^2 + 4 \cdot 3p(1-p) + 6p^2}{D} = \frac{6(1-p)^2 + 12p(1-p) + 6p^2}{D} = \frac{6}{D}$ [machine];

$\langle E^2 \rangle_{\text{mat}} = \frac{12(1+2p)}{D}$ [machine]

Var_{mat}

$= \frac{12(1+2p)}{D} - \frac{36}{D^2}$

$= \frac{12(1+2p)D - 36}{D^2}$

$= \frac{12p(1-p)(3-2p)}{D^2}$ [machine: $12(1+2p)D - 36 = 36p - 60p^2 + 24p^3 = 12p(3-2p)(1-p)$]

$\frac{\text{Var}}{\langle E \rangle^2} = \frac{12p(1-p)(3-2p)}{D^2} \cdot \frac{D^2}{36} = \frac{p(1-p)(3-2p)}{3}$ [machine];

$\frac{d}{dp}\left(3p - 5p^2 + 2p^3\right)$

$= 3 - 10p + 6p^2$

$= 0$

$\iff p$

$= \frac{5 \pm \sqrt{7}}{6}$,

one root in $(0, 1]$ [machine: the discriminant $28 = 4 \cdot 7$]

$p_\star = \frac{5 - \sqrt{7}}{6}$:

$1 - p_\star = \frac{1 + \sqrt{7}}{6}$,

$$3 - 2p_\star$$

$$= \tfrac{4+\sqrt{7}}{3} \implies \tfrac{p_\star(1-p_\star)(3-2p_\star)}{3}$$

$$= \tfrac{(5-\sqrt{7})(1+\sqrt{7})(4+\sqrt{7})}{324}$$

$$= \tfrac{20+14\sqrt{7}}{324}$$

$$= \tfrac{10+7\sqrt{7}}{162} \text{ [machine: exact in } \mathbb{Q}(\sqrt{7})]$$

$$5(10 + 7\sqrt{7}) < 162$$

$$\iff 35\sqrt{7} < 112$$

$$\iff 5\sqrt{7} < 16$$

$$\iff 175 < 256 \text{ [machine: squaring two positive integers]};$$

$$\tfrac{1}{5} < \tfrac{1}{4} \implies \sigma/\langle E\rangle < \tfrac{1}{2} \text{ [machine]}$$

$$(7):\ D(p{=}1) = 1 \implies w_{\text{mat}}(0) = 1 \text{ [machine]};$$

$$D'$$

$$= 2p - 3 < 0 \text{ on } (0, 1]$$

$$\implies D$$

$$\in [1, 3) \text{ with } D \to 3 \text{ as } p \to 0^+ \text{ [machine]}$$

$$\implies w_{\text{mat}}$$

$$= 1/D \qquad \in (\tfrac{1}{3}, 1];$$

$$D \geq 1 \iff p^2 - 3p + 2 \geq 0 \iff (p-1)(p-2) \geq 0 \text{ [machine]}$$

$$w_{\text{mat}}/w = 1/(pD) = 1/(1-(1-p)^3) \geq 1 \iff (1-p)^3 \geq 0, \text{ equality } \iff p = 1 \text{ [machine]};$$

$$\langle E \rangle_{\text{mat}} = 6/D \in (2, 6] \text{ [machine]};$$

$$\lim_{p \to 0^+} 6/D = 6/3 = 2 \text{ [machine: exact limit row]}$$

$$(8): \ m_{\text{hid}}$$

$= E \ln 2/\mathfrak{h}$ linear in E [FT.44(4)] $\implies$ the support and the two moments follow from (6) [machine];

$$E \geq 2 \text{ on the materialized support } [(5)] \implies m_{\text{hid}} \geq 2 \ln 2/\mathfrak{h} > 0 \ [\tfrac{1}{2} < \ln 2 < 1; \text{ machine}]$$

$$\left\langle E^{-1} \right\rangle_{\text{mat}}$$

$$= \frac{\frac{3}{2}p(1-p)^2 + \frac{3}{4}p^2(1-p) + \frac{1}{6}p^3}{pD}$$

$$= \frac{18p(1-p)^2 + 9p^2(1-p) + 2p^3}{12pD} \text{ [machine]};$$

$$\langle E \rangle \langle E^{-1} \rangle = \frac{6}{D} \cdot \frac{18 - 27p + 11p^2}{12pD} \cdot p = \frac{18 - 27p + 11p^2}{2D^2} \text{ [machine]}$$

$$\langle E \rangle \langle E^{-1} \rangle \geq \langle \sqrt{E \cdot E^{-1}} \rangle^2 = 1 \text{ [Cauchy–Schwarz]};$$

equality

$$\iff E \cdot E^{-1} \text{ proportional to a constant}$$

$$\iff E \text{ constant}$$

$$\iff p$$

$= 1$ [machine: the support has three points at every $p \in (0, 1)$];

$$p = \tfrac{1}{2}: \ \frac{18 - \frac{27}{2} + \frac{11}{4}}{2 \cdot \frac{49}{16}} = \frac{58}{49} \text{ [machine]}$$

$$\tfrac{18-27p+11p^2}{2D^2} < \tfrac{6}{5}$$

$$\iff 5(18-27p+11p^2) < 12D^2$$

$\iff 12p^4-72p^3+125p^2-81p+18>0$ [machine: exact rational interval arithmetic on 64 subintervals of $[0,1]$ with the derivative bound, lower endpoint >0];

$$\Delta x_{\mathrm{B}}^{\mathrm{eff}} = \hbar/(6\ln 2/D) = D\,\hbar/(6\ln 2) = D\,\Delta x_{\mathrm{B}}^{\mathrm{cryst}}\ [\mathrm{BM}.13(1);\ \text{machine}];$$

$$(9):\ \text{the scope record.}\ \blacksquare$$

Phys. The composite was never a snapshot, and the count that was read as one is a single point of an exact distribution. Take the flicker's own measure — the unique stationary line of the Janus generator, three pairs each retained with weight one minus four thresholds — and the internal count takes the values zero, two, four and six with the binomial weights, its unconditioned mean six times that weight and its variance twelve times the weight times its complement. The zero is not a composite: an all-virtual configuration supports no closed chain and therefore no terminus, and it is exactly the vacuum the entropy argument and the Lyapunov contraction both run toward, so the composite's read is the conditional one on at least two edges, and that restriction is a theorem of the materialization domain rather than a choice of scope. Conditional on it, the mean count is six over one plus u plus u squared, which never leaves the interval from two to six: the crystallized count at the top, attained only at the sharp threshold, and the digon's two at the bottom, attained only in the fully dissolved limit. The relative width of the distribution has an exact algebraic maximum, ten plus seven root seven over one hundred sixty-two, reached at a stationary point in the same quadratic field, and it is bounded by a fifth — so the informational mass fluctuates, but never by more than half of itself. The reciprocal register splits into two formed reads whose product is at least one by Cauchy–Schwarz and below six fifths by an exact quartic positivity, so the Born width has a width-of-the-mean and a mean-of-the-widths, and the difference is named rather than guessed. Read A took the top point for the whole distribution; the weight it ignored is three u over D, exactly, and it is positive at every threshold that lets the vertex fire at all.

Theorem 4.80 (BM.95 — The vertex superposition: the transport fibre, its exact weights, and the two positive parts of the splitting). Thm.

$$(1)\quad \text{the fibre and its census (re-derived):}$$

$$q_{\mathrm{core}}(c) = \tfrac{1}{2}\big(\sigma_{s(c)} - \sigma_{t(c)}\big) \in \{-1, 0, +1\}$$

$$[\mathrm{D3.4};\ \mathrm{EM.9};\ \mathrm{FT.40}]:$$

$$m_q = (2,\ 1,\ 1)\ \text{ at }\ q = (0, +1, -1)$$

$$[\mathrm{FT.106(3a)};\ \mathrm{Def\text{-}BM.13}(1):\ \text{the orbit size of the constituent's transport register under } \omega];$$

$$W(Q) = \big[z^{Q}\big]\big(2 + z + z^{-1}\big)^{k},$$

$$k = 3: \quad W = (1,\ 6,\ 15,\ 20,\ 15,\ 6,\ 1),$$

$$\sum_{Q} W(Q) = 4^{3} = 64 = 2^{6}$$

[Def-BM.13(3); Def-BM.8(6); machine];

$$\#\{(q_1, q_2, q_3) :\ \textstyle\sum_i q_i = k\} = (1, 3, 6,\ 7,\ 6, 3, 1)$$

[BM.28(2); machine: the exhaustive 4^3 end-sign census and the convolution agree exactly];

(2) the derived superposition weights:

the constituent at a vertex is a quantum superposition over its transport fibre, with weights

$$\boxed{P\big((q_1, q_2, q_3)\,\big|\,Q\big) \;=\; \frac{\prod_{i=1}^{3} m_{q_i}}{W(Q)}}$$

[the normalized fibre weight; machine: exact rationals, summing to 1 class by class] :

$$Q = 0:$$

$$P\big((0,0,0)\big) = \frac{8}{20} = \frac{2}{5},$$

$$P\big(\text{each of the six permutations of } (+1,-1,0)\big) = \frac{2}{20} = \frac{1}{10}$$

[machine];

$$Q = +1:$$

$$P\big(\text{each of the three permutations of } (+1,0,0)\big) = \frac{4}{15},$$

$$P\big(\text{each of the three permutations of } (+1,+1,-1)\big) = \frac{1}{15}$$

[machine];

$Q=-1:$ the identical census — the ω-image fibre [D2.6; BM.25(4); BM.28(5); machine];

the weights carry no datum:

the free-symbol set of $W(Q)$ is $\{z\}$ with coefficients fixed by (m_q,k),

and the coupling alphabet $\{g,g',e,\varepsilon_{\mathrm{EM}},\hat{\mu},\hat{\mu}',\ell_c\}$ is disjoint from it

[BM.61(5) re-cited; machine: the symbol census] $\Longrightarrow$ the superposition weights are ¬couplings, ¬rates, ¬P4 data;

(3) the entropy decomposition:

$$H(Q) \;:=\; -\sum_{\text{fibre}} P\ln P,$$

$$\big\langle \ln\textstyle\prod_i m_{q_i}\big\rangle_Q \;:=\; \sum_{\text{fibre}} P\ln\prod_i m_{q_i}\;:$$

$$\boxed{\Sigma_{\mathrm{cl}}(Q) \;=\; \ln W(Q) \;=\; H(Q) \;+\; \big\langle \ln\textstyle\prod_i m_{q_i}\big\rangle_Q}$$

[Def-BM.13(4); machine: exact in the prime-log ring] :

$$H(0)=\ln 5+\frac{1}{5}\ln 2,\qquad \big\langle\ln\textstyle\prod_i m\big\rangle_0=\frac{9}{5}\ln 2,\qquad \ln 20=2\ln 2+\ln 5\qquad [\text{machine}];$$

$$H(\pm1)=\ln 3+\ln 5-\frac{8}{5}\ln 2,\qquad \big\langle\ln\textstyle\prod_i m\big\rangle_{+1}=\frac{8}{5}\ln 2,\qquad \ln 15=\ln 3+\ln 5\qquad [\text{machine}];$$

(4) the two positive parts of the splitting:

$$\Sigma_{\mathrm{B}}=\Sigma_{\mathrm{cl}}(0)-\Sigma_{\mathrm{cl}}(+1)=\ln\tfrac{4}{3}=2\ln 2-\ln 3$$

[Def-BM.13(4); BM.59(4)] :

$$\boxed{\ln\frac{4}{3} \;=\; \underbrace{\Big(\frac{9}{5}\ln 2-\ln 3\Big)}_{H(0)-H(+1)} \;+\; \underbrace{\frac{1}{5}\ln 2}_{\langle\ln\prod_i m\rangle_0-\langle\ln\prod_i m\rangle_{+1}}}$$

[machine: the prime-log coefficients agree];

$$\frac{9}{5}\ln 2-\ln 3>0$$

[machine: $9\ln 2>5\ln 3 \iff 2^9=512>243=3^5$];

$$\frac{1}{5}\ln 2>0$$

[machine];

the transport splitting is exactly two positive parts — a superposition-entropy part and a multiplicity part;

the snapshot read carries neither:

a fixed triple at the vertex books $H=0$ and loses $\frac{9}{5}\ln 2-\ln 3$ of the splitting

[machine: the exact difference];

(5) the factorization (two independent exact averages):

the fibre weights are class-fixed boundary data [Def-BM.5(1); Def-BM.13(6)],

and the two registers' alphabets are disjoint :

$E_{\text{int}}=|R|$ has no endpoint summand

[Def-BM.13(6); BM.62(11); BM.89(8); machine: the token census] $\Longrightarrow$ $\boxed{P(R,\ \text{fibre member})\ =\ P(R)\cdot P(\text{fibre member}\,|\,Q)}$

the baryon's full ledger read is the composition of the flicker integral over $|R|$ (BM.94(2)) and the superposition average over the fibre (2);

¬a cross-register equation is formed — the two registers are distinct alphabets and are ¬equated

$$[\text{BM.37}(2);\ \text{BM.83}(3)];$$

(6) the fixed-state read refuted with an exact bound:

$$\max_{\text{fibre}} P(\cdot \mid 0) = \frac{2}{5} < \frac{1}{2}$$

$$[\text{machine: } 16 < 20];$$

$$\max_{\text{fibre}} P(\cdot \mid +1) = \frac{4}{15} < \frac{1}{3}$$

$$[\text{machine: } 12 < 15]:$$

¬∃ a single vertex configuration carrying more than $\frac{2}{5}$ of the neutral class's transport weight

the refutation is an exact rational bound, ¬an argument about size;

both classes are strict mixtures at every admissible threshold $[(2);\ \text{BM.19}(6)]$;

(7) status:

derived: (1)–(6);

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

§Cal-BM stays empty;

Def-BM.13(1)–(6), BM.28(2), (4), (5), BM.59(4), BM.61(5) stand unmodified;

Req. D2.6, D3.4 (Parts 1–2); EM.9, CT0.9 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.32(1), FT.40, FT.106(3) (Part 3); Census FT.35(2) (row-wise); Def-BM.5(1), Def-BM.8(6), Def-BM.13(1), Def-BM.13(3), Def-BM.13(4), Def-BM.13(6), BM.19(6), BM.25(4), BM.28(2), BM.28(5), BM.37(2), BM.59(4), BM.61(5), BM.62(11), BM.83(3), BM.89(8), BM.94(2) (this part); exhaustive ternary-fibre enumeration, exact rational weight normalization, prime-log ring arithmetic with integer sign certificates, symbol censuses [math; machine: `checks/verify_H_flicker_integral_restmass.py` §B].

Pf.

$(1):\ m_q = (2,1,1)$ and the end-pair census re-cited [Def-BM.13(1)];

$W(Q)$ by two agreeing routes: the exhaustive 4^3 census and the coefficient of $\left(2+z+z^{-1}\right)^3$ [machine];

$$\textstyle\sum_Q W(Q) = (2+1+1)^3 = 64 \text{ [machine]}$$

$$(2):\ P$$

$$= \prod_i m_{q_i}/W(Q) \text{ and } \textstyle\sum_{\text{fibre}} \prod_i m_{q_i}$$

$= W(Q)$ by (1)'s factorization identity [machine: class by class];

the $Q{=}0$ fibre: one triple of weight $2^3 = 8$ and six of weight $2\cdot 1\cdot 1 = 2,$ total 20 [machine];

the $Q{=}{+}1$ fibre: three of weight $1\cdot 2\cdot 2 = 4$ and three of weight $1,$ total 15 [machine]

the symbol census: $W(Q) \in \mathbb{Z}[z, z^{-1}]$ with coefficients fixed by (m_q, k),

and $\{g, g', e, \varepsilon_{\text{EM}}, \hat{\mu}, \hat{\mu}', \ell_c\} \cap \{z\}$

$= \varnothing$ [BM.61(5) re-cited; machine]

$$(3):\ H(0)$$

$$= -\tfrac{2}{5}\ln\tfrac{2}{5} - 6\cdot\tfrac{1}{10}\ln\tfrac{1}{10}$$

$$= \tfrac{2}{5}(\ln 5 - \ln 2) + \tfrac{3}{5}(\ln 2 + \ln 5)$$

$$= \ln 5 + \tfrac{1}{5}\ln 2 \text{ [machine: prime-log coefficients]};$$

$$\langle \ln \prod_i m \rangle_0 = \tfrac{8\ln 8 + 6\cdot 2\ln 2}{20} = \tfrac{36\ln 2}{20} = \tfrac{9}{5}\ln 2 \text{ [machine]}$$

$$H(+1) = -3\cdot\tfrac{4}{15}\ln\tfrac{4}{15} - 3\cdot\tfrac{1}{15}\ln\tfrac{1}{15} = \ln 15 - \tfrac{8}{5}\ln 2 \text{ [machine]};$$

$$\langle \ln \prod_i m \rangle_{+1} = \tfrac{3\cdot 4\ln 4 + 3\cdot 1\ln 1}{15} = \tfrac{24\ln 2}{15} = \tfrac{8}{5}\ln 2 \text{ [machine]};$$

$$H + \langle \ln \prod_i m \rangle = \ln W \text{ in both classes [machine]}$$

$$(4):\ [\ln 5 + \tfrac{1}{5}\ln 2] - [\ln 3 + \ln 5 - \tfrac{8}{5}\ln 2]$$

$$= \tfrac{9}{5}\ln 2 - \ln 3 \text{ and } \tfrac{9}{5}\ln 2 - \tfrac{8}{5}\ln 2$$

$$= \tfrac{1}{5}\ln 2 \text{ [machine]};$$

$$\text{the sum: } \tfrac{9}{5}\ln 2 - \ln 3 + \tfrac{1}{5}\ln 2 = 2\ln 2 - \ln 3 = \ln\tfrac{4}{3} \text{ [machine]};$$

$$2^9 = 512 > 243 = 3^5 \text{ [machine: an integer power comparison]}$$

$(5):$ the boundary data fix the charge class [Def-BM.5(1)] $\implies$ the fibre weights are class-constant on the ensemble;

E_{int} is constant on every end-sign fibre [Def-BM.13(6)] $\implies$ the two registrations are independent and their probabilities multiply [machine]

$$(6):\ \tfrac{2}{5} < \tfrac{1}{2} \iff 16 < 20;$$

$$\tfrac{4}{15} < \tfrac{1}{3} \iff 12 < 15 \text{ [machine: integers]};$$

$(7):$ the standing-rule scans [machine]. ■

Phys. Ask which state a constituent occupies at a vertex and the ledger answers with a list, not a value. A neutral-core constituent carries two transport realizations and a charged one carries a single realization, so the class fibre is a weighted set: at charge zero one triple of weight eight against six of weight two, summing to twenty; at charge one three of weight four against three of

weight one, summing to fifteen. Normalize and the superposition is exact — two fifths on the all-neutral triple, a tenth on each of the six mixed ones — and the numbers are rationals in an alphabet that carries no coupling at all, which is the coupling-blindness theorem wearing a different hat. The class entropy then splits cleanly into the Shannon entropy of that superposition plus the mean logarithm of the multiplicities, and the baryon's transport splitting, the log of four thirds, is the sum of two strictly positive parts: nine fifths of a bit less the log of three, from the superposition, and one fifth of a bit, from the multiplicities. A read that fixes one triple at the vertex books no entropy and forfeits the larger part of the splitting, so the fixed-state read is not merely incomplete, it is quantitatively short by an exact prime-log element. The two registers stay independent — the fibre weights are boundary data and the edge count has no endpoint summand — so the baryon's ledger read is the composition of two exact averages, and no equation between the registers is formed. The sharpest row is the bound: no single vertex configuration carries more than two fifths of the neutral class's transport weight, certified by sixteen against twenty.

Theorem 4.81 (BM.96 — The rung-one ν line: the seal-derived rest mass, the single-ray occupancy, and the four bars against a seesaw-type mechanism). Thm.

(1) the line's structure:

a rung-one bound chain [D1.5] whose terminus-active aspects lie in $\mathcal{A}(1)$

$= \{J,\, G,\, W,\, H,\, D^{+}\}$ [CT0.13] :

EM unactivated ($f_v = 0$),

the colour block vanishing,

$$\delta h(c) \geq 2, \quad n(v) \geq 2,$$

Σ_{min} the minimal instance

[FT.142(1); D3.3; CT3.11(1); FT.19; Def-H.4; Census FT.28];

$$\dim \mathcal{K}_1 = 16 = C_{\text{cap}}$$

[FT.142$(2), (3)$; CT0.14; CT0.12 : the unique derived candidate, no interface declaration];

$$\boxed{\delta_\nu = \frac{\sum_i [J_i, K_i]\big|_\Sigma}{\dim \mathcal{K}_1} = \frac{2}{16} = \frac{1}{8}}$$

[FT.142$(4), (5)$: the undressed seal coefficient over the derived mask dimension]

derived, ¬declared;

(2) the rest mass from the rung-one seal structure:

$$m_\nu^{\text{rest}\,2} = \hat{\mu}_\nu^2 \big(\ell_c^{(\nu)}\big)^2\, G\big(c(i_\nu)\big),$$

$$G\big(c(i)\big) = A_0 - B_0 \cos\Big(\delta_\nu - \frac{2\pi i}{3}\Big)$$

[Def-H.37(1); Def-BM.11(5); BM.39(1)] :

$$A_0 = \frac{3g^2}{8} + \frac{g'^2}{16}, \qquad B_0 = \frac{gg'}{4} \qquad \text{[BM.39(1); machine]}; \qquad \text{at } \delta_\nu = \tfrac{1}{8},\ \ c := \cos\tfrac{1}{8},\ \ s_\nu := \sin\tfrac{1}{8}\ :$$

$$\boxed{G_0 = A_0 - B_0\, c, \qquad G_1 = A_0 + \frac{B_0}{2}\big(c - \sqrt{3}\, s_\nu\big), \qquad G_2 = A_0 + \frac{B_0}{2}\big(c + \sqrt{3}\, s_\nu\big)}$$

[machine: exact in $\mathbb{Q}[\sqrt{2}, \sqrt{3}][c, s_\nu]/(c^2+s_\nu^2-1)$];

the exact order at $(g, g') > 0$ [EW.33] : $\boxed{G_2 \ > \ G_1 \ > \ A_0 \ > \ G_0}$

$$G_1 > A_0 \iff c > \sqrt{3}\, s_\nu \iff c^2 > \tfrac{3}{4} \iff \cos\tfrac{1}{4} = 2c^2 - 1 > \tfrac{1}{2}$$

[machine: $\cos x > 1 - \frac{x^2}{2} = \frac{31}{32} > \frac{1}{2}$ at $x = \frac{1}{4}$, the exact alternating-series sign row];

$$c > 0 \ \text{ on } \ c > \tfrac{63}{64},$$

$$s_\nu > 0 \ \text{ on } \ s_\nu > \tfrac{1}{8} - \tfrac{1}{3072}$$

[machine: exact Taylor enclosures with rational remainder bounds];

the affine form on the spectral triplet:

$$G\big(c(i)\big) = A_0 - \frac{B_0}{\sqrt{2}}\big(r_i^{(\nu)} - 1\big),$$

$$r_i^{(\nu)} = 1 + \sqrt{2}\cos\Big(\frac{1}{8} + \frac{2\pi i}{3}\Big)$$

[Def-BM.11(5); FT.143(1); machine] :

$$\big(r_1^{(\nu)}\big)^2 < \big(r_2^{(\nu)}\big)^2 < \big(r_0^{(\nu)}\big)^2$$

[FT.143(3) : enclosure-proven, frozen] $\implies$ the transport read is strictly decreasing in r;

the value set is orbit-label independent:

$$\big\{G\big(c(i)\big)\big\}_i = \Big\{A_0 - B_0\,c,\;\; A_0 + \tfrac{B_0}{2}\big(c - \sqrt{3}\,s_\nu\big),\;\; A_0 + \tfrac{B_0}{2}\big(c + \sqrt{3}\,s_\nu\big)\Big\}$$

[Def-BM.11(5) orbit-label row; machine: the two routes give the same set];

the transport order is the reverse of the spectral order: BM.39(5)'s inversion instantiated at rung one

(3) positivity and the exact floor:

$$A_0 - B_0 = \frac{3}{8}\Big(g - \frac{g'}{3}\Big)^2 + \frac{g'^2}{48} > 0,$$

$$A_0 + B_0 = \frac{3}{8}\Big(g + \frac{g'}{3}\Big)^2 + \frac{g'^2}{48} > 0$$

[BM.39(1); BM.77(4); machine: the SOS expansions] :

$$\sqrt{A_0 - B_0} \;\le\; \sqrt{G\big(c(i)\big)} \;\le\; \sqrt{A_0 + B_0}$$

$$\forall\,\text{line } i,\;\; \forall\,\text{channel},\;\; \forall\,(g, g') \neq (0, 0)$$

$$[|\cos| \le 1;\ \text{machine}];$$

the kernel line is unreachable: $\cos\varphi = 0 \wedge \cos(\varphi + \frac{2\pi}{3}) = 0$ has no solution

[BM.39(1); machine] $\implies$ | $m_\nu^{\text{rest}} > 0$ strictly, and ¬a massless ν line is formed on the transport species |
|---|

the absolute value is ¬derived:

$$\big(\hat{\mu}_\nu,\ \ell_c^{(\nu)}\big) \text{ undeclared [P4; Cal-F];}$$

$$\rho_\nu := \mathfrak{h}\,\hat{\mu}_\nu\,\ell_c^{(\nu)} \in \mathfrak{q}^0 \text{ is a restatement, ¬a new member [BM.77(2); CT2.1];}$$

the internal ratios are derived:

$$\frac{m_i^2}{m_j^2} = \frac{G\big(c(i)\big)}{G\big(c(j)\big)} \text{ exact symbolic in } (A_0, B_0)$$

[FT.143(3); BM.80(6); machine];

the inertial companion:

$$m_i = \mu_\nu \big(r_i^{(\nu)}\big)^2, \quad \mu_\nu \text{ undeclared [P4],}$$

$$Q := \frac{\sum_i m_i}{\big(\sum_i \sqrt{m_i}\big)^2} = \frac{6}{9} = \frac{2}{3} \text{ exactly}$$

[FT.143(4); machine];

the informational companion: $m_\nu\,\mathfrak{h} \ \geq\ 2\ln 2$ [BM.70(2), (3); BM.13(3) : the two-bit floor];

(4) the single-ray occupancy:

the materialized locus at rung one is single-arrow — D1.2's domains force strict alternation and the prefix law, and a dissolution-first chain is inadmissible at its first event [FT.144(2); D2.11];

under A-BM.7's issued coupling the arrow locus occupies $\operatorname{im} P_L$,

the $\hat{\chi}$

$= +1$ ray of Def-H.33's boxed sign :

$$\boxed{\text{the occupancy is exactly } (1,\ 0)}$$

[FT.144(5); BM.73(2), (3); FT.122(4)];

the complementary ray's content is the vacuum sector:

$$n_{\mathcal{R}} = 0 \implies \rho_{pq} = \rho_{\circ} = \tfrac{1}{4} I,$$

$$S_{\text{raw}} = \ln 4 = \max, \quad \text{unregistered}, \quad \neg\exists \text{ an accessible pole at any rung}$$

[FT.144(4); A5; CT0.2; CT0.1; CT4.2 : $\mathcal{P}_M F \mathcal{P}_M = 0 \implies \neg\exists$ an accessible pole; CT2.5 : poles $\iff 1 \in \operatorname{spec} \mathrm{F}_M$];

the obstruction is a transition-domain restriction prior to every read — ¬a mass statement,

¬a suppression,

¬a decoupling

[FT.144(3) : $\neg\mathfrak{h}$, $\neg\mu$, $\neg h_0$ in the premises] :

¬∃ a materializable right-ray ν line in the matter orientation, in any valid universe

[D2.11; A-BM.7]

the naming discipline:

"left" and "right" are the ray names of Def-H.33 as fixed by A-BM.7;

the handedness vocabulary is external and is ¬identified with a ray [A-BM.7's bars; P5];

(5) the image row:

the vertex's neutral member is the ν-channel line in the image orientation [A-BM.5; BM.41(1); BM.45];

the image is the $\hat{\omega}$-image of the same single-arrow locus:

$$\hat{\omega} P_L \hat{\omega}^{-1} = P_R,$$

$$\hat{\omega}|L\rangle = -i|R\rangle,$$

$$\hat{\omega}^2 = \mathrm{id}$$

$$[\text{D2.6};\ \text{FT.122}(4);\ \text{BM.73}(4);\ \text{machine over } \mathbb{Q}(i)];$$

and it sits in the ω-image representation $\{0,-1\}$ $\quad$ $[\text{EW.36};\ \text{A-BM.7(b)};\ \text{BM.25};\ \text{BM.27}]$;

$$\text{CPT}:\ (q,\ \text{orientation})$$

$$\mapsto (-q,\ \text{flipped})\ \text{ has no fixed sector,}$$

its orbits are three pairs of size two,

$$q$$

$$\mapsto -q \text{ fixes } q \qquad = 0 \text{ alone}$$

$$[\text{BM.25}(4);\ \text{machine}]:$$

the image is one orbit of size two and carries no independent datum — ¬a second state, ¬a right-ray matter line, ¬a name for an unmaterializable object

$$[\text{BM.27}(6)\ \text{pattern}];$$

(6) the no-seesaw theorem (four independent bars, each re-cited at its own locus):

(S1) no partner state: ¬∃ a right-ray matter-orientation state to pair with — the complement's occupancy is exactly zero and its content supports no accessible pole at any rung

$$[(4);\ \text{FT.144}(4),(5)];$$

(S2) no self-pairing: ¬∃ a fixed CPT sector, so a Majorana-type identification C

$$= \text{CPT} \cdot C \text{ is } \neg\text{formed}$$

$$[(5);\ \text{BM.25}(4);\ \text{BM.27}(6)];$$

(S3) no coupling object: ¬∃ a Yukawa-type object in the inventory — $\dim F_H$

$$= 1$$

$$= \text{span}\{\Sigma\},$$

¬a scalar-block coupling to a chain line,

¬a dimension-five operator

$$[\text{CT0.12};\ \text{EW.15};\ \text{FT.145}(3);\ \text{BM.93}(5)];$$

(S4) no second scale: ∃! one free dimensionful scale $\mathfrak{h}$ — the heavy scale a seesaw-type mechanism requires is barred

$$[\text{P1};\ \text{A §0.3};\ \text{D2.9}];$$

¬∃ a seesaw-type mechanism formable in this corpus;	the ν rest mass is positive, derived in form from the rung-one seal, and undeclared in scale

the companion row:

the smallness question is a scale question [P4],

¬a mechanism question — ¬a suppression is needed, so ¬one is invented;

(7) the vacuum residence and the vertex firing (the two states and the one door):

state (i), the vacuum-resident balanced fluctuation pair:

$$n_{\mathcal{R}} = 0,$$

$$\sum_n \langle \varnothing | \hat{U}^n | \varnothing \rangle \, w^n = \frac{1}{1 - w^2 x_+ x_-},$$

every nonzero return monomial carries equal x_+- and x_--degree

$$[\text{FT.108}(5);\ \text{FT.144}(4);\ \text{machine in } \mathbb{Q}[w, x_+, x_-]/(w^{12})] :$$

state (ii), the materialized rung-one seal:

$$|E_{\mathcal{R}}| \geq 2,$$

$$m_{\text{hid}} = \frac{2\ln 2}{\mathfrak{h}} \text{ at the minimal floor}$$

$$[\text{FT.145}(2);\ \text{Census FT.28};\ \text{Def-H.4}];$$

the door is the vertex:

$$\Gamma(\varepsilon_q{=}0) = 0 \text{ exactly},$$

$$x_+x_- = 4\varepsilon_q(1 - 4\varepsilon_q) \leq \frac{1}{4} \text{ with equality at } \varepsilon_q = \frac{1}{8}$$

$$[\text{BM.23}(7), (8);\ \text{machine}] :$$

the line materializes only where the flicker is present — the same threshold BM.94's integral averages over is the one that opens the vertex

the materialized line propagates as a rung-one seal, forward-only:

$$\neg\exists\, \tau^{-1},$$

$\neg\exists$ an inverse decay in the ledger

$$[\text{A6};\ \text{BM.23}(9);\ \text{FT.144}(2)];$$

Γ is a weight per tick, $\neg$a rate per external second

$$[\text{BM.23}(9);\ \text{D1.8};\ \text{P5}];$$

$\neg$a lifetime, endpoint or mixing angle is derived or compared [P5];

(8) status:

derived: (1)–(7) in form — the angle, the order, the positivity, the occupancy, the four bars, the two states and the door;

¬derived: the absolute scale $(\hat{\mu}_\nu, \ell_c^{(\nu)})$, the line index i_ν the mixing matrix [FT.145(3)];

¬a declaration issued — A-BM.5, A-BM.7 and Def-H.37 are re-cited, ¬re-issued;

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

§Cal-BM stays empty;

species names are confined to the identification layer [Def-BM.6$(2), (7)$];

topology labels are internal bookkeeping [Def-H.32(4)];

amplitude $\neq$ mass [FT.79(5)];

rung $\neq$ scale [D0.8].

Req. A5, A8, D0.8, D1.2, D1.4, D1.5, D1.8, D2.6, D2.9, D2.11, D3.3 (Parts 1–2); A §0.3, CT0.1, CT0.2, CT0.12, CT0.13, CT0.14, CT3.11 (Part 1, Part 2); CT2.5, CT4.2, EW.15, EW.33, EW.36, EM.9 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.4, Def-H.32(4), Def-H.33, Def-H.37(1), FT.19, FT.28, FT.40, FT.54(4), FT.79(5), FT.108(5), FT.122(4), FT.142(1), FT.142(2), FT.142(3), FT.142(4), FT.142(5), FT.143(1), FT.143(3), FT.143(4), FT.144(2), FT.144(3), FT.144(4), FT.144(5), FT.145(2), FT.145(3) (Part 3); Census FT.28 (row-wise); Def-BM.11(5), BM.13(3), BM.23(7), BM.23(8), BM.23(9), BM.25(4), BM.27(6), BM.39(1), BM.39(5), BM.41(1), BM.45, BM.70(2), BM.70(3), BM.73(2), BM.73(3), BM.73(4), BM.77(2), BM.77(4), BM.80(6), BM.93(5) (this part); Cal-F; exact identities in $\mathbb{Q}[\sqrt{2}, \sqrt{3}][c, s_\nu]/(c^2{+}s_\nu^2{-}1)$, exact Taylor enclosures with rational remainder bounds, sum-of-squares positivity, exact $\mathbb{Q}(i)$ block algebra, exact resolvent identities in a truncated polynomial ring [math; machine: `checks/verify_H_flicker_integral_restmass.py` §C].

Pf.

(1) : FT.142(1)–(5) re-cited (frozen);

$$\delta_\nu = \tfrac{2}{16} = \tfrac{1}{8} \text{ [machine: exact Fractions]};$$

$$\dim \mathcal{K}_1 = 6 + (5+3+1+1) = 16 = C_{\text{cap}} \text{ [CT0.12; CT0.14; machine]}$$

$$(2):\ \cos(\tfrac{1}{8} - \tfrac{2\pi}{3})$$

$$= -\tfrac{c}{2} + \tfrac{\sqrt{3}}{2} s_\nu \text{ and } \cos(\tfrac{1}{8} - \tfrac{4\pi}{3})$$

$$= -\tfrac{c}{2} - \tfrac{\sqrt{3}}{2} s_\nu \text{ [the addition formulas]} \implies \text{the three boxed values [machine: ring identities]};$$

$$A_0, B_0 \text{ re-cited from BM.39}(1)$$

$$G_1 > A_0 \iff \tfrac{B_0}{2}(c - \sqrt{3} s_\nu) > 0$$

$$\iff c > \sqrt{3} s_\nu$$

$$\iff c^2 > 3(1 - c^2)$$

$$\iff 4c^2 > 3$$

$$\iff 2c^2 - 1 > \tfrac{1}{2} \iff \cos\tfrac{1}{4} > \tfrac{1}{2} \text{ [machine: the double-angle identity]};$$

$$\cos\tfrac{1}{4} > 1 - \tfrac{1}{2 \cdot 16}$$

$$= \tfrac{31}{32} > \tfrac{1}{2} \text{ [machine: the alternating-series sign row, exact rationals]}$$

$$c > \tfrac{63}{64} \text{ from } \cos\tfrac{1}{8} > 1 - \tfrac{1}{2 \cdot 64};$$

$$s_\nu > \tfrac{1}{8} - \tfrac{1}{6 \cdot 8^3} = \tfrac{1}{8} - \tfrac{1}{3072} > 0 \text{ [machine: exact Taylor bounds]};$$

$$G_2 > G_1 \iff \sqrt{3}\, s_\nu > -\sqrt{3}\, s_\nu \iff s_\nu > 0 \text{ [machine]}$$

$G = A_0 + \frac{B_0}{\sqrt{2}}\big(1 - r^{(\nu)}\big)$ is affine strictly decreasing in r [Def-BM.11(5) affine row; FT.143(1); machine];

$$r_1 < r_2 < r_0 \text{ [FT.143(3) re-cited]} \implies \text{the transport order reverses the spectral order;}$$

the value set is orbit-label independent [Def-BM.11(5) orbit-label row; machine: set equality]

$$(3): \text{ the two SOS rows re-derived by expansion [machine];}$$

$$|\cos| \le 1 \implies \text{the sandwich [math];}$$

$$\text{the kernel line: } c = 0 \implies s_\nu^2 = 1 \implies \cos(\varphi + \tfrac{2\pi}{3}) = \mp\tfrac{\sqrt{3}}{2} \neq 0 \text{ [BM.39(1) re-cited; machine]}$$

$$m_i^2/m_j^2 = G_i/G_j \text{ and } Q = \tfrac{6}{9} = \tfrac{2}{3} \text{ [FT.143(3), (4) re-cited; machine];}$$

$$m_\nu \mathfrak{h} \ge 2\ln 2 \text{ [BM.70(2), (3) re-cited]}$$

$$(4): \text{ FT.144(2)'s domain rows and prefix law re-cited;}$$

$$\text{occupancy } (1,0) \text{ [FT.144(5); BM.73(3)];}$$

$$P_{L/R}^2 = P_{L/R},\ P_L P_R = 0,\ P_L + P_R = \mathrm{id},\ \mathfrak{T} P_{L/R} = 1 \text{ [FT.122(2); machine over } \mathbb{Q}(i)]$$

$$\rho_\circ = \tfrac{1}{4} I: \ \mathfrak{T} = 1,\ \text{charpoly} = (x - \tfrac{1}{4})^4,\ S_{\text{raw}} = \ln 4 = \max \text{ [CT0.2; BM.35(1) re-cited];}$$

$$\mathcal{P}_M F \mathcal{P}_M = 0 \implies \neg\exists \text{ an accessible pole [CT4.2; CT2.5 re-cited];}$$

$$\neg\mathfrak{h}, \neg\mu, \neg h_0 \text{ in the premises [FT.144(3) re-cited]}$$

$$(5): \ \hat{\omega} P_L \hat{\omega}^{-1}$$

$$= P_R,$$

$$\hat{\omega}|L\rangle$$

$$= -i|R\rangle,$$

$$\hat{\omega}^2$$

$$= \text{id } [\text{FT.122}(4);\ \text{BM.73}(4) \text{ re-derived entrywise over } \mathbb{Q}(i);\ \text{machine}];$$

$$\text{the no-fixed-sector census on six sectors } [\text{BM.25}(4) \text{ re-cited};\ \text{machine}]$$

$$(6):\ (\text{S1})\text{–}(\text{S4}) \text{ re-cited at their own loci } [(4), (5);\ \text{CT0.12};\ \text{EW.15};\ \text{FT.145}(3);\ \text{P1};\ \text{A §0.3}];$$

$$\text{the inventory scans: } \neg\exists \text{ a Yukawa-type token outside a negation row},$$

$$\neg\exists \text{ a second dimensionful scale [machine]}$$

$$(7):\ \text{the resolvent identity re-derived coefficient-wise to } w^{12} \text{ [machine]};$$

$$x_+x_- = 4\varepsilon_q(1-4\varepsilon_q) \le \tfrac{1}{4} \text{ with equality at } \varepsilon_q = \tfrac{1}{8} \text{ [BM.23(7) re-cited;\ machine]};$$

$$\Gamma = 0 \text{ at } \varepsilon_q = 0 \text{ [BM.23}(8)\text{]};$$

$$m_{\text{hid}} = 2\ln 2/\mathfrak{h} \text{ [FT.145}(2)\text{]}$$

(8) : the standing-rule scans: external tokens, decimal numerals, registry tokens, declaration lines, the barred object α_B [machine]. ■

Phys. The neutral line the vertex emits is a rung-one seal, and everything about it that the corpus can derive, it derives. Its channel angle is two over sixteen, the undressed seal coefficient over the mask dimension the complete-force rule already fixed; its rest mass is the transport defect evaluated on that angle, three exact values in a radical ring, ordered heaviest to lightest against the spectral order by the same inversion the family mass form proves in general. Every one of the three is strictly positive, squeezed between two sums of squares that never touch zero, and the kernel line that would make one of them massless is unreachable by an exact trigonometric census — so there is no massless option to decline and no zero to explain away. The absolute scale stays undeclared, as every scale in this part does, and the internal ratios stay derived. On the carrier the line occupies one ray and its complement is empty: not heavy, not suppressed, empty, because a dissolution-first history is outside the transition domains and unregistered content supports no accessible pole at any rung. The image orientation the vertex selects is the mirror of that same single-arrow locus, one orbit of size two carrying no datum of its own. Four separate bars then close the door on any seesaw-type mechanism — no partner state, no self-pairing, no coupling object, no second scale —

and the smallness of the line is left where the corpus has always left it, with the free internal data. The two states of the line are the vacuum-resident balanced pair and the materialized seal, and the only door between them is the vertex, whose weight vanishes exactly at the sharp threshold: the line materializes only where the flicker is present, which is the same threshold the ensemble integral is taken over.

Theorem 4.82 (BM.97 — The rest mass is the pole mass: the species adjudication, the rest-mass Q-value, and the exact critical locus). Thm.

(1) the species census at the rest slot (three candidates, one read specified):

candidate	expression	status
the informational read	$m_{\text{hid}} = E_{\text{int}} \ln 2/\mathfrak{h}$	the ledger's bit currency, ensemble-distributed [FT.44(4); BM.35(4); BM.94(8)]
the sector-deficit route	$m^{(\text{ii})} = T(q,S)/\mathfrak{h}, \quad T = W(Q)\,\overline{\Delta S}(S)$	a measure price, vertex-scoped [BM.28(4)(ii); BM.30(5); BM.37(6); BM.77(2)]
the pole/transport read	$m^2_{\text{pole}} = \hat{\mu}^2 \ell_c^2 G(c) = 4\hat{\mu}'^2 \sin^2(\Theta_R/4)$	the rest mass on the transport species [Def-H.37(1), (2); FT.54(4); FT.47(1); EW.33; BM.39(1), (7), (9)]

the matching of the two pole clauses is derived at leading order and recorded, ¬imposed:

$$\hat{\mu}'^2 \Theta_R^2/4 = \hat{\mu}^2 \ell_c^2 G(c)$$

[FT.54(2); FT.33; BM.39(9); Def-H.37 locus clause];

the rest mass is the pole mass $m^{\text{rest}} = m_{\text{pole}}$ on the transport species; m_{hid} is ¬the rest mass on that species

equating them is a conversion the corpus bars

[BM.35(5); BM.83(3); BM.37(2); pointer];

the specified read A-BM.8 reads the baryon classes on route (ii) and the emitted lines on the transport species [BM.77(1); re-cited, ¬re-issued];

(2) the rest-mass Q-value, formed:

$$Q_\beta^{\text{rest}} := m_n^{\text{rest}} - m_p^{\text{rest}} - m_\ell^{\text{rest}} - m_\nu^{\text{rest}}$$

[BM.41(4); BM.63(1); BM.85(1); every member a transport-species mass] :

$$Q_\beta^{\text{spec}}\,\mathfrak{h} \;=\; \Delta T(S) \;-\; \rho_\ell \sqrt{G\big(c(0)\big)} \;-\; \rho_\nu \sqrt{G\big(c(i_\nu)\big)}, \qquad \Delta T(S) = 5\, m_B^2(\cdot, S)$$

[BM.77(2); the expression is its own frozen Q_{ii}] :

$$\Delta T = \Big(\frac{4219785}{98956},\ \frac{9608310}{393679},\ \frac{455}{23},\ \frac{68565}{3088}\Big)\ \text{at } S=(0,1,2,3),$$

$$\frac{455}{23} \le \Delta T(S) \le \frac{4219785}{98956}$$

[BM.77(2); machine re-certified];

$$G_\ell = A_0 - B_0 \cos\delta_\ell \ \text{ at } \ \delta_\ell = \tfrac{2}{9}$$

[FT.112; Def-BM.6(1)],

$$G_\nu = A_0 - B_0 \cos\Big(\tfrac{1}{8} - \frac{2\pi i_\nu}{3}\Big) \ \text{ at the derived } \ \delta_\nu = \tfrac{1}{8}$$

[BM.96(2); FT.142(5)];

$$\rho_\ell = \mathfrak{h}\,\hat{\mu}_\ell\,\ell_c^{(\ell)}, \quad \rho_\nu = \mathfrak{h}\,\hat{\mu}_\nu\,\ell_c^{(\nu)} \in \mathfrak{q}^0$$

[BM.77(2); CT2.1; Def-H.37(1)];

every row below holds uniformly in $i_\nu \in \{0,1,2\}$

[BM.77(2)];

(3) the flicker-blindness of the rest-mass read:

the specified read's alphabet is $\{m_B^2(\cdot,S),\ \rho_\ell,\ \rho_\nu,\ g,\ g',\ \mathfrak{h}\}$

[BM.92(5); machine: the token census] :

$\neg E_{\text{int}}$, $\neg w$, $\neg w_{\text{mat}}$, $\neg\varepsilon_q$, $\neg$g, $\neg N$ occur in it

[machine] $\Longrightarrow$ the flicker integral changes no row of the rest-mass Q-value

the two acts are orthogonal:

BM.94's fix lands on the informational register alone,

BM.77's read lands on the transport species alone

$$[\text{BM.77}(3);\ \text{BM.92}(3)\ \text{pattern}]:$$

$\Longrightarrow$ the trichotomy, the four ΔT values, ρ^{crit}'s enclosure and all eight witnesses stand

$$[\text{BM.77}(5),(6);\ \text{BM.92}(3);\ \text{machine re-certified}];$$

(4) the pure-pole read (the third formed read):

$$Q_\beta^{\text{pole}} \ := \ m_n^{\text{pole}} - m_p^{\text{pole}} - m_\ell - m_\nu,$$

$$M_0 := m_p^{\text{pole}},$$

$$(m_n^{\text{pole}})^2 = M_0^2 + \Delta_g,$$

$$s := m_\ell + m_\nu > 0$$

hypothesis (H) : the two baryon classes differ by exactly one branch substitution [BM.59(9); BM.69(3); carried on its face] :

$$\Delta_g \ \in \ \left\{2,\ \ \sqrt{2}-1,\ \ 2\cos\tfrac{\pi}{10}-\sqrt{3}\right\}\hat{\mu}'^2 \ > \ 0,$$

$$\Delta_1 > \Delta_2 > \Delta_3$$

[Def-BM.14(5); BM.69(3); machine: the certificates $4>2,\ 2>1,\ 3<5$];

the exact critical locus:

$$\boxed{Q_\beta^{\text{pole}} > 0 \iff \Delta_g > \Delta_g^{\text{crit}} := s\,(2M_0+s); \qquad Q_\beta^{\text{pole}} = 0 \iff \Delta_g = \Delta_g^{\text{crit}}; \qquad Q_\beta^{\text{pole}} < 0 \iff \Delta_g < \Delta_g^{\text{crit}}}$$

[machine: cleared by squaring two positive sides, $\sqrt{M_0^2+\Delta_g} > M_0+s \iff M_0^2+\Delta_g > (M_0+s)^2$];

the equivalent line form:

$$s^{\text{crit}} = \sqrt{M_0^2 + \Delta_g} - M_0 = \frac{\Delta_g}{\sqrt{M_0^2 + \Delta_g} + M_0} = m_n^{\text{pole}} - m_p^{\text{pole}}$$

[machine: the cleared identity];

Δ_g^{crit} is strictly increasing in s and in M_0,

$\Delta_g^{\text{crit}} > 0$ strictly since $s > 0$

[BM.96(3); machine: the coefficient census];

(5) the trichotomy (total, exact, non-empty on both sides):

(a) $\mathcal{O}^{\text{pole}} := \{(\rho_\ell, \rho_\nu) > 0 : s < \Delta_g^{\text{crit}}\text{'s line form}\} \neq \varnothing$:

the universal symbolic witness

$$\rho_\ell = \rho_\nu = \frac{\sqrt{M_0^2 + \Delta_g} - M_0}{4\sqrt{A_0 + B_0}} \in \mathcal{O}^{\text{pole}}$$

[BM.77(4)'s sandwich : $s \leq \frac{1}{2}\left(\sqrt{M_0^2 + \Delta_g} - M_0\right) < \sqrt{M_0^2 + \Delta_g} - M_0$; machine] :

valid at every generation, every line index i_ν, every $(g, g') \neq (0, 0)$;

the integer-certified rational witness at $(g, g', \rho_\ell, \rho_\nu, \hat{\mu}'^2\mathfrak{h}^2, M_0\mathfrak{h})$

$= (1, 1, \frac{1}{100}, \frac{1}{100}, 1, 1)$,

generation 1 :

$$A_0 + B_0 = \frac{7}{16} + \frac{1}{4} = \frac{11}{16},$$

$$\sqrt{G} \leq \frac{\sqrt{11}}{4} < 1$$

$$[11 < 16] \implies s\,\mathfrak{h} < \frac{1}{50} \implies \Delta_g^{\text{crit}}\mathfrak{h}^2 < \frac{1}{50} \cdot \frac{101}{50} = \frac{101}{2500} < 2 = \Delta_1\mathfrak{h}^2$$

$$\boxed{Q_\beta^{\text{pole}} > 0 \text{ on the certificate } 5000 > 101} \qquad \text{[machine]};$$

(b) $\mathcal{C}^{\text{pole}} \neq \varnothing$:

the universal symbolic witness

$$\rho_\ell = \rho_\nu = \frac{\sqrt{M_0^2 + \Delta_g} - M_0}{\sqrt{A_0 - B_0}} \ \in \ \mathcal{C}^{\text{pole}}$$

$$\text{[machine: } s \geq 2\big(\sqrt{M_0^2 + \Delta_g} - M_0\big)\text{]};$$

the integer-certified witness at $(1, 1, 100, 100, 1, 1)$:

$$A_0 - B_0 = \frac{3}{16},$$

$$\sqrt{G} \geq \frac{\sqrt{3}}{4} > \frac{2}{5} \qquad [75 > 64] :$$

$$m_\ell \hbar \ \geq \ 25\sqrt{3} \ > \ 40$$

$$[1875 > 1600] \implies s\hbar \ \geq \ 50\sqrt{3} \ > \ 80$$

$$[75 > 64] \implies \Delta_g^{\text{crit}} \hbar^2 \ > \ 80 \cdot 82 = 6560 \ > \ 2$$

$$\boxed{Q_\beta^{\text{pole}} < 0 \text{ on the certificates } 1875 > 1600, \quad 75 > 64, \quad 6560 > 2} \qquad \text{[machine]};$$

(c) the critical locus is exact, unique and generation-relative:

$$\partial\mathcal{O}^{\text{pole}} : \quad s = \sqrt{M_0^2 + \Delta_g} - M_0$$

[machine: the boundary equation];

Δ_g^{crit} is generation-blind (a function of the emitted lines and M_0 alone) while Δ_g is generation-specific [(4)] :

$$\boxed{\text{the opening region is widest in the first generation and narrowest in the third, in the exact order } \Delta_1 > \Delta_2 > \Delta_3}$$

[BM.69(3); machine: monotone in Δ_g at fixed M_0];

on the symmetric ray ρ_ℓ

$$= \rho_\nu$$

$$= \rho \text{ at the witness scales } (g, g', M_0\mathfrak{h}, \hat{\mu}'^2\mathfrak{h}^2)$$

$$= (1, 1, 1, 1),$$

generation 1 :

$$\rho^{\text{crit,pole}} = \frac{(\sqrt{3}-1)}{2\sqrt{G}} :$$

$$\boxed{\frac{2}{5} \;<\; \rho^{\text{crit,pole}} \;<\; \frac{6}{7}}$$

[machine: $\sqrt{G} \le \frac{\sqrt{11}}{4}$ gives the lower bound on $2(\sqrt{3}-1)/\sqrt{11} > \frac{2}{5} \iff 89 > 50\sqrt{3} \iff 7921 > 7500$; $\sqrt{G} \ge \frac{\sqrt{3}}{4}$ gives the upper bound on $2(\sqrt{3}-1)/\sqrt{3} < \frac{6}{7} \iff 8\sqrt{3} < 14 \iff 192 < 196$];

(6) the ν-positivity shift (a derived consequence of BM.96(3)):

$$\Delta_g^{\text{crit}}(m_\ell + m_\nu) - \Delta_g^{\text{crit}}(m_\ell) = 2M_0 m_\nu + 2m_\ell m_\nu + m_\nu^2 \;>\; 0$$

[machine: the exact expansion] :

the derived positivity of the ν rest mass strictly narrows the opening region by an exact amount — a massless ν line is refuted as a read, ¬assumed

the two-body companion on this read:

$$E_p - m_p = \frac{(m_n - m_p)^2 - m_\ell^2}{2m_n}$$

[BM.41(2); BM.77(6); machine: the identity re-derived at $m_n \neq m_p$] :

the two-body channel is admissible $\iff (m_n^{\text{pole}} - m_p^{\text{pole}})^2 \ge m_\ell^2 \iff \Delta_g \;\ge\; m_\ell\big(\sqrt{M_0^2 + \Delta_g} + M_0\big)$

[machine: the cleared form];

the forced-third-member row is read-relative:

BM.41(2)'s equal-mass obstruction is lifted on the pure-pole read ($\Delta_g > 0$),

so BM.41(3),

(6)'s forcing stands on the informational reads [BM.70(4)(a)] and the grammar stands on the specified read [BM.45; BM.77(6)],

and it is ¬inherited by the pure-pole read — recorded, ¬adjudicated beyond the branch rows

[BM.77(6) pattern];

(7) the read discipline: three formed reads, pairwise incomparable:

$$Q_\beta^{\text{info}} : \{\ln 2, \ln 3, \ln 5,\ N,\ w \text{ or } w_{\text{mat}},\ \mathfrak{g},\ \varepsilon_{\text{EM}},\ \mathfrak{h},\ m_\ell,\ m_\nu\};$$

$$Q_\beta^{\text{spec}} : \{m_B^2(\cdot, S),\ \rho_\ell,\ \rho_\nu,\ g,\ g',\ \mathfrak{h}\};$$

$$Q_\beta^{\text{pole}} : \{M_0,\ \Delta_g,\ \hat{\mu}'^2,\ \rho_\ell,\ \rho_\nu,\ g,\ g',\ \mathfrak{h}\}$$

the informational alphabet is disjoint from the two transport alphabets except for $\mathfrak{h}$; ¬a derived equation relates the three expressions

[machine: the three token censuses; the mass-form bridge residue, localized at BM.98(5)];

the decay direction is fixed in all three:

$T(0, S) = \frac{4}{3} T(+1, S)$ [BM.28(4)(ii)],

$\Delta_g > 0$ [BM.69(3)],

$\Delta m > 0$ sub-critically [BM.91(6); BM.99(4)] :

the parent is the neutral class on every formed read, and the grammar $n \to p + \ell + \nu$ stands unmodified

[BM.41(1); BM.45; BM.77(6)];

¬a currency is converted [BM.83(3); BM.90(5)];

route (ii) is ¬elevated beyond the vertex kinematics [BM.30(5); BM.77(9)];

route (iii)'s amplitude ratios stay barred as masses [FT.79(5); BM.37(3)];

(8) status:

derived: (1)–(7) given the specified read and hypothesis (H);

declared: nothing — A-BM.8 and Def-H.37 are re-cited, ¬re-issued;

¬a clause of BM.41, BM.49, BM.63, BM.70, BM.71, BM.77, BM.85, BM.92 is retracted;

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

§Cal-BM stays empty;

Req. D2.9 (Parts 1–2); CT2.1, EM.25(7), EW.15 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.37(1), Def-H.37(2), EW.33, FT.33, FT.47(1), FT.54(2), FT.54(4), FT.79(5), FT.112, FT.142(5) (Part 3); Def-BM.6(1), Def-BM.11(5), Def-BM.14(5), BM.28(4), BM.30(5), BM.35(4), BM.35(5), BM.37(2), BM.37(3), BM.37(6), BM.39(1), BM.39(7), BM.39(9), BM.41(1), BM.41(2), BM.41(3), BM.41(4), BM.41(6), BM.45, BM.59(9), BM.63(1), BM.69(3), BM.70(4), BM.77(1), BM.77(2), BM.77(3), BM.77(4), BM.77(5), BM.77(6), BM.77(9), BM.83(3), BM.85(1), BM.87, BM.90(5), BM.91(6), BM.92(3), BM.92(5), BM.96(2), BM.96(3), BM.96(4) (this part); Cal-F; exact radical inequalities by integer certificates, sum-of-squares positivity, cleared polynomial identities over $\mathbb{Q}[M_0, m_\ell, m_\nu]$, exact rational arithmetic at the four frozen sector expectations, three token censuses [math; machine: `checks/verify_H_flicker_integral_restmass.py` §D].

Pf.

(1) : Def-H.37(1), (2)'s two clauses and FT.54(2), (4)'s matching re-cited;

the three-candidate census assembled from FT.44(4),

BM.28(4)(ii),

BM.39(1),

(7) [machine: the read-class rows];

the conversion bar re-cited [BM.35(5); BM.83(3)]

(2) : BM.77(2)'s frozen expression re-cited with G_ν evaluated at the derived δ_ν

$= \frac{1}{8}$ [BM.96(2)];

$$5 \cdot 843957 = 4219785,$$

$$5 \cdot 1921662 = 9608310,$$

$$5 \cdot 91 = 455,$$

$5 \cdot 13713 = 68565$ [machine: exact integers];

$\frac{455}{23} \leq \Delta T \leq \frac{4219785}{98956}$ [machine: cross-multiplication]

(3) : the alphabet enumerated from the expression and scanned for the six barred tokens [machine];

the eight witnesses re-certified as integer comparisons [machine]

(4) : $\sqrt{M_0^2 + \Delta_g} > M_0 + s$ with both sides positive

$\iff M_0^2 + \Delta_g > M_0^2 + 2M_0 s + s^2 \iff \Delta_g > 2M_0 s + s^2$ [machine: cleared over $\mathbb{Q}[M_0, s, \Delta_g]$];

s^{crit}

$$= \sqrt{M_0^2+\Delta_g} - M_0 \text{ and } (\sqrt{M_0^2+\Delta_g} - M_0)(\sqrt{M_0^2+\Delta_g} + M_0)$$

$$= \Delta_g \text{ [machine: the difference of squares]};$$

$$\Delta_1 > \Delta_2 > \Delta_3 \text{ on } 4>2,\ 2>1,\ 3<5 \text{ [BM.69(3) re-cited]}$$

$$(5)(\text{a}):\ 2\cdot\tfrac{\sqrt{M_0^2+\Delta_g}-M_0}{4\sqrt{A_0+B_0}}\cdot\sqrt{A_0+B_0}$$

$$= \tfrac{1}{2}\big(\sqrt{M_0^2+\Delta_g}-M_0\big) < \sqrt{M_0^2+\Delta_g}-M_0 \text{ [machine]};$$

$$A_0+B_0\big|_{g=g'=1} = \tfrac{11}{16} \text{ [machine]}$$

$$s\,\mathfrak{h} \le 2\cdot\tfrac{1}{100}\cdot\tfrac{\sqrt{11}}{4} = \tfrac{\sqrt{11}}{200} < \tfrac{4}{200} = \tfrac{1}{50} \text{ [}11<16;\text{ machine]};$$

$$\tfrac{1}{50}\big(2+\tfrac{1}{50}\big) = \tfrac{101}{2500};$$

$$2 > \tfrac{101}{2500} \iff 5000 > 101 \text{ [machine]}$$

$$(\text{b}):\ 2\cdot\tfrac{\sqrt{M_0^2+\Delta_g}-M_0}{\sqrt{A_0-B_0}}\cdot\sqrt{A_0-B_0} = 2\big(\sqrt{M_0^2+\Delta_g}-M_0\big) > \sqrt{M_0^2+\Delta_g}-M_0 \text{ [machine]};$$

$$A_0-B_0\big|_{g=g'=1} = \tfrac{3}{16};$$

$$\sqrt{G} \ge \tfrac{\sqrt{3}}{4} > \tfrac{2}{5} \iff 75 > 64 \text{ [machine]}$$

$$m_\ell\mathfrak{h} \ge 100\cdot\tfrac{\sqrt{3}}{4} = 25\sqrt{3} > 40 \iff 1875 > 1600;$$

$$s\,\mathfrak{h} \ge 50\sqrt{3} > 80 \iff 75 > 64;$$

$$80(2+80) = 6560 > 2 \text{ [machine: integers]}$$

$$(\text{c}):\ \rho^{\text{crit,pole}} = \tfrac{\sqrt{3}-1}{2\sqrt{G}} \text{ at } M_0\mathfrak{h}=1,\ \Delta_1\mathfrak{h}^2=2 \text{ [machine: } \sqrt{1+2}=\sqrt{3}\text{]};$$

$$\text{the lower bound: } \tfrac{2(\sqrt{3}-1)}{\sqrt{11}} > \tfrac{2}{5}$$

$$\iff 5(\sqrt{3}-1) > \sqrt{11}$$

$$\iff 25(4-2\sqrt{3})>11$$

$$\iff 89>50\sqrt{3}$$

$$\iff 7921>7500\ [\text{machine}]$$

the upper bound: $\frac{2(\sqrt{3}-1)}{\sqrt{3}}<\frac{6}{7} \iff 14(\sqrt{3}-1)<6\sqrt{3} \iff 8\sqrt{3}<14 \iff 192<196$ [machine];

monotonicity in Δ_g at fixed M_0 : the coefficient of Δ_g in $\sqrt{M_0^2+\Delta_g}-M_0$ is positive [machine]

$$(6):\ (m_\ell+m_\nu)(2M_0+m_\ell+m_\nu)-m_\ell(2M_0+m_\ell)$$

$$=2M_0m_\nu+2m_\ell m_\nu+m_\nu^2 \text{ [machine: the expansion over } \mathbb{Q}[M_0,m_\ell,m_\nu]];$$

$$2m_n(E_p-m_p)$$

$$=(m_n-m_p)^2-m_\ell^2 \text{ at } E_p$$

$$=\tfrac{m_n^2+m_p^2-m_\ell^2}{2m_n} \text{ [machine: cleared]} \implies \text{the two-body condition } m_n-m_p$$

$$\geq m_\ell \text{ and its } \Delta_g \text{ form [machine]}$$

(7) : the three symbol sets enumerated from the three expressions and pairwise intersected [machine];

$$T(0,S)/T(+1,S)=20/15=4/3 \text{ [machine]};$$

(8) : the standing-rule scans [machine]. ■

Phys. A Q-value is a difference of rest masses, and a rest mass in this corpus is a pole mass: the transport defect of a materialized chain, exact as four mu-prime squared times the sine squared of a quarter frame angle, matched at leading order to the Gram evaluation the seed supplies. The informational count is a different currency, ensemble-distributed and bit-denominated, and equating the two is precisely the conversion the ledger bars — so the fix to the snapshot read and the read of the vertex turn out to be orthogonal acts, and the specified Q-value carries no edge count, no crystallization weight, no threshold and no curvature composition in its alphabet at all. On that read nothing moves: the four exact baryon values, the critical enclosure and all eight integer

witnesses stand. The pure pole read is the third formed read, and it has its own exact trichotomy — a critical locus that is the emitted-line sum times twice the daughter's mass plus itself, a universal symbolic witness in each of the opening and closing regions, and rational witnesses certified by five thousand against one hundred one and by one thousand eight hundred seventy-five against one thousand six hundred. Because the neutrino line's mass is derived strictly positive, the critical locus sits strictly above its massless value by an exact amount, so the positivity narrows the opening region rather than being assumed away; and because the pole splitting is generation-specific while the critical locus is not, the opening region is widest in the first generation and narrowest in the third. What the three reads agree on is the direction: the neutral class is the parent in every currency, and the grammar is untouched.

Rmk (the operative mass read — reconciled nomenclature). The mass that works — the read entering every formed composite quantity of this era (the rest masses, the Q-values, the splittings, the thresholds, the lapse) — is the Lyapunov-contraction mass: the ledger deficit accumulated under the tilt contraction law [L1–L4 (Part 1): every tilt step loses $\gamma(8-\gamma)/16$ of the Lyapunov value; BM.51: the class-degenerate foundation], in its composite form $m_{\text{rest}} = \Delta S_{\mathfrak{L}}/\mathfrak{h}$ [BM.35]. Def-H.39's registry (Part 3) names the species, and no unqualified mass crosses them: the informational read agrees with m_{rest} as a formed equation [BM.35], ¬as a currency fusion; the pole read is the rest mass on the transport species, matched at leading order [(1) above]; the sector-route values are measure functionals — exact named prices, ¬masses [BM.30; BM.37]; the Ansatz-B class masses remain non-adopted templates [Def-H.28 (Part 3)]; and the numeric bridges (the β-coefficients; ρ_{B}) are calibration slots, values ¬issued [§Cal-BM (C-BM.1); P5].

Theorem 4.83 (BM.98 — The relating equation: the bridge in four exact forms, its flicker weighting, and the closure in form). Thm.

(1) the bridge in its derived count form:

$$\mathcal{R} := \frac{m_{\text{pole}}^2}{m_{\text{hid}}^2} = \rho_{\text{B}}\,\frac{D_S(\Theta_R)}{4\,E_{\text{int}}^2},$$

$$\rho_{\text{B}} = \frac{\hat{\mu}'^2\mathfrak{h}^2}{(\ln 2)^2} \in \mathfrak{q}^0$$

[BM.69(2); FT.54(4); Census FT.63(6)] :

six states, six exact algebraic factors $\frac{D_S}{4E^2}$, four fields, six minimal polynomials

[Def-BM.14(3); BM.82(2); machine re-cited];

(2) the flicker-weighted bridge:

$$E_{\text{int}} \longrightarrow \langle E_{\text{int}}\rangle_{\mathcal{E}}$$

[BM.94(2), (6), (7); the integral replaces the snapshot] :

unconditioned:

$$\mathcal{R}_w = \rho_{\mathrm{B}}\,\frac{D_S}{4\,w^2 E^2} = \frac{\mathcal{R}}{w^2};$$

materialized:

$$\boxed{\mathcal{R}_{\mathrm{mat}} \;=\; \rho_{\mathrm{B}}\,\frac{D_S\,D^2}{4\,E^2} \;=\; \frac{\mathcal{R}}{w_{\mathrm{mat}}^2} \;=\; \rho_{\mathrm{B}}\,\frac{D_S\,D^2}{144} \;\text{ at }\; E = E^{\mathrm{cryst}} = 6}$$

[machine: one exact rational function per state];

$$\boxed{1 \;\le\; \frac{\mathcal{R}_{\mathrm{mat}}}{\mathcal{R}} \;=\; D^2 \;<\; 9}$$

$$[\text{BM.94}(7):\; D \in [1,3);\; \text{machine}]:$$

the flicker can multiply the bridge by at most an exact nine, and by exactly one at the sharp threshold;

the lower endpoint is attained at ε_q

$$= 0,$$

where the snapshot is recovered exactly [BM.94(6), (7)];

the upper endpoint is open and is approached only in the fully dissolved limit, where $\langle E\rangle_{\mathrm{mat}} \to 2$ — the digon floor [Census FT.28; BM.94(7)];

(3) the curvature form:

$$G = 2\pi\,E_{\mathrm{int}}^2 \ln 2$$

$$[\text{BM.88}(3);\; \text{BM.16}(1):\; \text{the calibration lock}] \;\Longrightarrow\; E_{\mathrm{int}}^2 = \frac{G}{2\pi \ln 2}:$$

$$\boxed{\mathcal{R} \;=\; \frac{\pi \ln 2}{2}\cdot\frac{\rho_{\mathrm{B}}\,D_S}{G}}$$

[machine: one monomial, coefficient comparison]

the bridge is inversely proportional to the curvature lock, with the exact coefficient $\frac{\pi \ln 2}{2}$;

flicker-weighted:

$$G^{\text{eff}} := 2\pi \big\langle E \big\rangle^2_{\text{mat}} \ln 2 = \frac{G}{D^2} = w^2_{\text{mat}}\, G,$$

$$\mathcal{R}_{\text{mat}} = \frac{\pi \ln 2}{2} \cdot \frac{\rho_{\text{B}} D_S}{G^{\text{eff}}}$$

[machine];

the exact enclosure at $E^{\text{cryst}} = 6$:

$$G = 72\,\pi \ln 2$$

$$[\text{BM.17}(3)\text{'s lock value re-derived; machine}] \implies \boxed{G^{\text{eff}} \in \big(8\,\pi \ln 2,\ \ 72\,\pi \ln 2\big]}$$

the two endpoints are the digon's and Σ_{12}'s own curvature locks

$$[\text{BM.17}(4)\text{'s family table; machine: } 2\pi \cdot 2^2 \ln 2 = 8\pi \ln 2] :$$

the flicker moves the lock between two computed cells and never below the floor's;

(4) the coupling form:

$$k = \frac{r\, E_{\text{int}} \ln 2}{\mathfrak{h}},$$

$$1 - f = \frac{1}{k}$$

$$[\text{BM.88}(2)(\text{i}), (\text{ii}); \text{ BM.88}(6)] \implies E_{\text{int}} = \frac{\mathfrak{h}}{r\,(1-f)\ln 2} :$$

$$\boxed{\mathcal{R} \;=\; \frac{\hat{\mu}'^2\, D_S\, r^2}{4}\,(1-f)^2}$$

[machine: the substitution; $[\hat{\mu}'^2 r^2] = \mathfrak{q} \cdot \mathfrak{q}^{-1} = \mathfrak{q}^0$ [CT2.1; Def-H.37(2); machine: the exponent arithmetic]];

$$e^2 = e_0^2 \, N \, f \, \mathsf{g}(G) \qquad [\text{BM.88(4)}] \implies 1 - f = 1 - \frac{e^2}{e_0^2 N \, \mathsf{g}(G)} :$$

$$\boxed{\mathcal{R} \;=\; \frac{\hat{\mu}'^2 \, D_S \, r^2}{4} \Big(1 - \frac{e^2}{e_0^2 \, N \, \mathsf{g}(G)}\Big)^2}$$

the bridge as an exact function of the running coupling

the energy-dependence of the bridge is the energy-dependence of the coupling:

both run through the single derived argument f;

$\mathcal{R}$ strictly decreasing in E_{int}, strictly decreasing in f, strictly increasing in k;

e strictly increasing in E_{int}

[BM.88(6); machine: three monotonicity rows on exact rationals] :

at fixed $(r,\ N,\ \mathsf{g},\ D_S)$, $\mathcal{R}$ is a strictly decreasing exact function of the running coupling

the flicker's action on the coupling:

$$k^{\text{eff}} = w_{\text{mat}} \, k^{\text{cryst}} = \frac{k^{\text{cryst}}}{D},$$

$$f^{\text{eff}} = 1 - \frac{D}{k^{\text{cryst}}} \;\le\; f^{\text{cryst}},$$

$$\mathsf{g}^{\text{eff}} = \mathsf{g}\big(G^{\text{eff}}\big) \;\text{ at }\; G^{\text{eff}} = \frac{G}{D^2}$$

[machine] :

$$\text{equality} \iff \varepsilon_q = 0$$

[machine] $\implies$ the flicker attenuates the running coupling and enhances the bridge, both by the same derived weight;

the vanishing locus is shifted and bounded:

$$f^{\text{eff}} = 0 \iff r = D\,\Delta x_{\text{B}}^{\text{cryst}} \in \left[\Delta x_{\text{B}}^{\text{cryst}},\ 3\,\Delta x_{\text{B}}^{\text{cryst}}\right)$$

[BM.88(5); BM.94(8); machine] :

the coupling's zero is confined to within an exact factor of three of the crystallized Born width

(5) the closure in form and the localized residue:

every factor of (1)–(4) is derived exact algebra:

the six D_S with their minimal polynomials [Def-BM.14(3)];

G's monomial and f's rational function [BM.88(2), (3)];

N's ADM row [BM.88(2)(iii)];

$w_{\text{mat}},\ D,$ the integer 9 [BM.94(7)];

the coefficient $\frac{\pi \ln 2}{2}$ (3);

the two computed lock endpoints (3);

every monotonicity row and certificate (2)–(4)

[machine];

and one factor is not:

$$\rho_{\text{B}} = \frac{\hat{\mu}'^2 \mathfrak{h}^2}{(\ln 2)^2} \text{ is the single undeclared dimensionless ratio (R1)}$$

[BM.69(5); BM.82(6); Cal-F packaging, ¬added here] :

the relating equation is derived as an exact identity in the derived vocabulary plus (R1) : the equation is closed, the ratio is ¬derived

BM.69(4)'s four grounds stand unweakened:

(i) the degree-16 field against integer- and binary-valued reads [FT.73(2), (3)];

(ii) the two-torsion image reaching 2 of 6 states [BM.67(6)];

(iii) the calibration weight -2 against 0 [FT.55(1); BM.67(8)];

(iv) the strength bar [BM.36(3); BM.61(8)(O1)]

[machine: the four rows re-cited at their own loci];

the residue list after this tier:

(R1) ρ_{B};

(R2) the loop realization;

(R3) the flavour composition;

(R4) $w(\varepsilon_q)$ in general and the transport register's ensemble average;

(R5) ι;

(R6) the reciprocal register's Jensen residue $\langle E\rangle_{\mathrm{mat}}\langle E^{-1}\rangle_{\mathrm{mat}} - 1$

$$= \frac{18 - 27p + 11p^2 - 2D^2}{2D^2}$$ [BM.94(8)]

named, ¬filled;

the two Cal-F packagings $(\hat{\mu}^2,\ \hat{\mu}'^2)$,

e_0 and the scales $(\hat{\mu}_\nu,\ \ell_c^{(\nu)},\ \mu_\nu)$ stay undeclared

[Def-H.37; EM.25(7); P4; P1];

(6) the currency bar stands:

$$\mathcal{R} \in \mathfrak{q}^0 \text{ is a ratio,}$$

¬a conversion — ¬a clause of this tier adds a pole-currency term to an informational assembly :

$$[\Delta_g] = \mathfrak{q} \quad \text{against} \quad [\Delta m] = \mathfrak{q}^{+1/2}$$

$$\Longrightarrow$$

¬addable

[BM.37(2); BM.69(3); BM.83(3); BM.90(5); CT2.1; machine: the exponent arithmetic];

the flicker weight attenuates the informational register alone and converts nothing into the defect register

[BM.90(4); BM.93(5); machine: w_{mat}'s symbol census carries no D_S];

(7) the no-import rows:

¬the observed neutron–proton mass ratio is imported — the ratio is derived and bounded at BM.99(5);

¬the observed electric coupling is imported — e,

$$e_0,$$

ε_{EM} stay symbolic [EM.25(7), (9); BM.88(7); P5];

¬a measured ν mass, mixing angle, endpoint or lifetime enters [P5];

¬an [EXT] datum;

machine: zero external tokens, zero decimal numerals in this tier;

(8) status:

derived: (1)–(4), (6);

closed in form with one priced residue: (5);

symbolic: e_0, g, N, ρ_{B};

¬a declaration,

¬a P4 addition,

§Cal-BM stays empty,

¬a new dimensionful scale [P1],

¬a measured value [P5];

α_B : ¬added, ¬derived, barred [BM.61(8)].

Req. D2.9 (Parts 1–2); A §0.3, CT2.1 (Part 2); EM.25(7), EM.25(9) (Part 2); P1, P4, P5 (Parts 1–2); Def-H.37(2), FT.54(4), FT.55(1), FT.73(2), FT.73(3) (Part 3); Census FT.28, FT.63(6) (row-wise); Def-BM.4(1), Def-BM.14(3), BM.13(1), BM.16(1), BM.17(3), BM.17(4), BM.36(3), BM.37(2), BM.61(8), BM.67(6), BM.67(8), BM.69(2), BM.69(3), BM.69(4), BM.69(5), BM.82(2), BM.82(6), BM.83(3), BM.88(2), BM.88(3), BM.88(4), BM.88(5), BM.88(6), BM.88(7), BM.89(4), BM.90(4), BM.90(5), BM.93(5), BM.94(2), BM.94(6), BM.94(7), BM.94(8) (this part); Cal-F; exact Laurent-monomial arithmetic with $\mathfrak{q}$- and $\mathfrak{h}$-exponent censuses, exact rational-function substitution, coefficient comparison, monotonicity on exact rationals, symbol censuses [math; machine: `checks/verify_H_flicker_integral_restmass.py` §E].

Pf.

(1) : BM.69(2)'s identity and FT.54(4)'s bridge ratio re-cited;

$$[\rho_{\mathrm{B}}] = [\hat{\mu}'^2 \mathfrak{h}^2] = \mathfrak{q} \cdot \mathfrak{q}^{-1} = \mathfrak{q}^0 \text{ [Def-H.37(2); machine]}$$

(2) : $\mathcal{R}_{\mathrm{mat}}$

$$= \rho_{\mathrm{B}} D_S / (4\langle E \rangle^2_{\mathrm{mat}})$$

$$= \rho_{\mathrm{B}} D_S/(4 \cdot 36/D^2)$$

$$= \rho_{\mathrm{B}} D_S D^2/144 \text{ [BM.94(6); machine]};$$

$$D \in [1,3) \text{ [BM.94(7)]} \implies D^2 \in [1,9) \text{ [machine]};$$

$$D = 1 \iff p = 1 \iff \varepsilon_q = 0 \text{ [machine]}$$

$$(3): \ E^2$$

$$= G/(2\pi \ln 2) \implies \rho_{\mathrm{B}} D_S/(4E^2)$$

$$= \rho_{\mathrm{B}} D_S \cdot 2\pi \ln 2/(4G)$$

$$= (\pi \ln 2/2)\rho_{\mathrm{B}} D_S/G \text{ [machine: coefficient comparison]};$$

$$G^{\text{eff}} = 2\pi(6/D)^2 \ln 2 = 72\pi \ln 2/D^2 = G/D^2 \text{ [machine]}$$

$$G^{\text{cryst}} = 2\pi \cdot 36 \cdot \ln 2 = 72\pi \ln 2 \text{ [BM.17(3); machine]};$$

$$2\pi \cdot 2^2 \ln 2 = 8\pi \ln 2 \text{ [BM.17(4)'s digon row; machine]};$$

$$D \in [1,3) \implies G/D^2 \in (8\pi \ln 2,\ 72\pi \ln 2] \text{ [machine]}$$

$$(4): \ 1 - f$$

$$= 1/k$$

$$= \hbar/(rE \ln 2)$$

$$\implies E$$

$$= \hbar/(r(1-f) \ln 2) \implies \rho_{\mathrm{B}} D_S/(4E^2)$$

$$= \rho_{\mathrm{B}} D_S r^2 (1-f)^2 (\ln 2)^2/(4\hbar^2)$$

$$= \hat{\mu}'^2 D_S r^2 (1-f)^2/4 \text{ [machine: the substitution and } \rho_{\mathrm{B}}(\ln 2)^2/\mathfrak{h}^2 = \hat{\mu}'^2]$$

$$e^2 = e_0^2 N f \mathsf{g} \implies f = e^2/(e_0^2 N \mathsf{g}) \text{ [machine]};$$

$$\mathcal{R} \propto (1-f)^2 \text{ strictly decreasing in } f \text{ on } f < 1 \text{ [BM.88(2)(ii) : } 0 \le f < 1; \text{ machine}];$$

f strictly increasing in E_{int} [BM.88(6)] $\implies$ $\mathcal{R}$ strictly decreasing in E_{int} and in e [machine: the composition of two monotone rows]

$$k^{\mathrm{eff}} = r\langle E\rangle_{\mathrm{mat}} \ln 2/\mathfrak{h} = (r \cdot 6 \ln 2/\mathfrak{h})/D = k^{\mathrm{cryst}}/D \text{ [machine]};$$

$$f^{\mathrm{eff}} = 1 - D/k^{\mathrm{cryst}} \le 1 - 1/k^{\mathrm{cryst}} = f^{\mathrm{cryst}} \iff D \ge 1 \text{ [machine]};$$

$$f^{\mathrm{eff}} = 0 \iff k^{\mathrm{cryst}} = D \iff r = D\,\Delta x_{\mathrm{B}}^{\mathrm{cryst}} \text{ [machine]}$$

(5) : the inventory rows assembled from (1)–(4)'s own tags [machine];

BM.69(4)'s four grounds re-cited at their loci;

$$\langle E\rangle\langle E^{-1}\rangle - 1 = \tfrac{18-27p+11p^2}{2D^2} - 1 = \tfrac{18-27p+11p^2-2D^2}{2D^2} \text{ [BM.94}(8);\text{ machine]}$$

$$(6) : \ [\Delta_g] = [\hat{\mu}'^2] = \mathfrak{q}, \quad [\Delta m] = \mathfrak{q}^{+1/2} \text{ [CT2.1; machine]};$$

$$w_{\mathrm{mat}}$$

$$= 1/D,$$

$$D$$

$$= 3 - 3p + p^2,$$

$$p$$

$$= 1 - 4\varepsilon_q : \text{ the symbol set is } \{\varepsilon_q\} \text{ and carries no } D_S \text{ [machine: the token census]}$$

(7)–(8) : the standing-rule scans: external tokens, decimal numerals, unit tokens, registry tokens, declaration lines, the barred object α_B [machine]. ■

Phys. The bridge between the two currencies has one shape and four exact readings of it. Count the edges and it is the single undeclared packaging times the state's defect factor over four times the count squared. Read the count through the calibration lock and the same object is pi log two over two times the packaging times the defect factor over the curvature monomial — the bridge is inversely proportional to the lock, with the coefficient exact. Read the count through the envelope ratio and it becomes mu-prime squared times the defect factor times the radius squared over four, times the square of one minus the nearby-space fraction; substitute the coupling's own profile and the bridge is an exact function of the running coupling, strictly decreasing in it at fixed geometry, so the energy-dependence of the bridge is literally the energy-dependence of the coupling and nothing else. Put the flicker in and every form gains one factor: the bridge is multiplied by the square of the dilution denominator, which is bounded by an exact integer nine, attained only in a limit that is not a materialized universe, and equal to one exactly at the sharp threshold where the snapshot returns. The curvature lock rides between two computed cells, the digon's and the junction's own, and the coupling's zero is confined to within a factor of three of the crystallized width. Everything in those four forms is derived exact algebra — one number is not, and the four grounds that bar its derivation stand where they were proved. So the relating equation is closed as an equation: its form, its flicker weighting, its curvature and coupling readings and its bound are theorems, and its single residue is one dimensionless ratio that this tier names, prices and does not invent.

Theorem 4.84 (BM.99 — The materialized splitting and the Q-value: the two-sided parameter-free bar, the tightened bound, and the rescoped dissolution row). Thm.

(1) the materialized assembly:

$$\boxed{m_n\,\mathfrak{h} \;=\; 6\,w_{\text{mat}}\ln 2+\ln 20, \qquad m_p\,\mathfrak{h} \;=\; 6\,w_{\text{mat}}\ln 2+\ln 15+3\,N\,w_{\text{mat}}\,\mathsf{g}\,\varepsilon_{\text{EM}}\ln 2}$$

[BM.91(2) at $w\to w_{\text{mat}}$; BM.94(7), (8); Def-BM.13(5); BM.89(5); machine: the prime-log assembly $\ln 20 = 2\ln 2+\ln 5,\ \ln 15 = \ln 3+\ln 5$];

$$\boxed{\Delta m\,\mathfrak{h} \;=\; \ln\frac{4}{3}-3\,N\,w_{\text{mat}}\,\mathsf{g}\,\varepsilon_{\text{EM}}\ln 2} \qquad [\text{machine: } \ln 20-\ln 15=\ln\tfrac{4}{3}];$$

the cancellation row stands:

E_{int} is class-shared [BM.57(4); Census FT.35(2)] $\Longrightarrow$ the crystallization weight cancels in the ledger difference and survives only in the EM term

[machine: the coefficient census];

the register scoping stands:

w_{mat} attenuates the edge-count register alone;

the transport register $\Sigma_{\mathrm{cl}}(Q) = \ln W(Q)$ enters at its crystallized value under (R4)

[BM.91(2); Def-BM.13(4), (6); BM.89(8)];

(2) the critical locus, enclosed:

$$\varepsilon_{\mathrm{mat}}^{\mathrm{crit}}(N, \mathsf{g}) = \frac{\ln(4/3)}{3\,N\,w_{\mathrm{mat}}\,\mathsf{g}\,\ln 2} = \frac{2 - \log_2 3}{3\,N\,w_{\mathrm{mat}}\,\mathsf{g}}$$

[BM.91(4); machine: solved exactly in the prime-log ring] :

$$\boxed{\frac{2 - \log_2 3}{3\,N\,\mathsf{g}} \;\le\; \varepsilon_{\mathrm{mat}}^{\mathrm{crit}} \;<\; \frac{2 - \log_2 3}{N\,\mathsf{g}} \;\subset\; \Big(\frac{2}{15\,N\mathsf{g}},\;\; \frac{5}{12\,N\mathsf{g}}\Big)}$$

[BM.91(4)'s squeeze $\frac{2}{5} < 2 - \log_2 3 < \frac{5}{12}$ on $3^5 = 243 < 256 = 2^8$ and $3^{12} = 531441 > 524288 = 2^{19}$; BM.94(7); machine];

$\varepsilon_{\mathrm{mat}}^{\mathrm{crit}} \notin \mathbb{Q}$ at every rational $(N, \mathsf{g}) > 0$

[BM.84(4); BM.91(4) : $2 - \log_2 3 \notin \mathbb{Q}$ on the $2^p \neq 3^q$ census; machine];

(3) the two-sided parameter-free bar:

$$\boxed{N\,\mathsf{g}\,\varepsilon_{\mathrm{EM}} \;\le\; \frac{1}{8} \qquad \Longrightarrow \qquad \Delta m > 0}$$

[BM.91(5) re-derived without w : $w_{\mathrm{mat}} \le 1$ and $\frac{2-\log_2 3}{3} > \frac{1}{8} \iff 2^{13} = 8192 > 6561 = 3^8$];

$$\boxed{N\,\mathsf{g}\,\varepsilon_{\mathrm{EM}} \;\ge\; \frac{5}{12} \qquad \Longrightarrow \qquad \Delta m < 0}$$

[machine: $w_{\mathrm{mat}} > \frac{1}{3}$ strictly $\Longrightarrow N w_{\mathrm{mat}} \mathsf{g}\,\varepsilon_{\mathrm{EM}} > \frac{5}{36}$, and $\frac{5}{36} > \frac{2-\log_2 3}{3} \iff \frac{5}{12} > 2 - \log_2 3 \iff 3^{12} > 2^{19}$];

the materialization filter removes the crystallization weight from both bars

the undecided band is exact and non-empty:

$$\frac{1}{8} < N\,\mathsf{g}\,\varepsilon_{\mathrm{EM}} < \frac{5}{12}$$

[machine: $12 < 40$];

both bars are uniform over the lapse, the curvature composition and the whole admissible threshold range $\varepsilon_q \in [0, \frac{1}{4})$

[CT0.8; machine];

(4) the trichotomy (total, exact, both regions certified by integers):

$$\boxed{\varepsilon_{\text{EM}} < \varepsilon_{\text{mat}}^{\text{crit}} \iff m_n > m_p; \qquad = \iff m_n = m_p; \qquad > \iff m_n < m_p}$$

[machine: Δm strictly decreasing in ε_{EM} at every admissible $(N, w_{\text{mat}}, \text{g}) > 0$];

the opening region is inhabited by ε_{EM}

$= 0$ and by ε_{EM}

$= \frac{1}{9}$ at $N\text{g}$

$= 1$ [3);

BM.62(8) : 4^3

$= 64 > 54 \qquad = 3^3 \cdot 2]$;

the closing region is inhabited by $\varepsilon_{\text{EM}} = \frac{5}{12}$ at $N\text{g} = 1$

[(3); machine];

the critical locus is exact, unique and irrational [(2)];

(5) the tightened universal ratio bound:

$$\frac{m_n}{m_p} = \frac{6\, w_{\text{mat}} \ln 2 + \ln 20}{6\, w_{\text{mat}} \ln 2 + \ln 15 + 3N w_{\text{mat}} \text{g}\, \varepsilon_{\text{EM}} \ln 2}$$

[machine] :

strictly decreasing in ε_{EM} and in w_{mat}

[BM.91(7); machine: the two cross-multiplied sign rows] and $\frac{1}{3} < w_{\text{mat}} \le 1$

[BM.94(7)] :

$$\boxed{\frac{\ln 1280}{\ln 960} \le \frac{m_n}{m_p}\Big|_{\varepsilon_{\text{EM}}=0} < \frac{\ln 80}{\ln 60} < \frac{\ln 20}{\ln 15} < \frac{10}{9}}$$

[machine: $6\ln 2 + \ln 20 = \ln 1280, \quad 2\ln 2 + \ln 20 = \ln 80, \quad 2\ln 2 + \ln 15 = \ln 60$] :

$$\frac{\ln 80}{\ln 60} \in \Big(\frac{16}{15}, \frac{15}{14}\Big)$$

[machine: $80^{15} > 60^{16} \iff 2^{28} = 268435456 > 215233605 = 3^{16}\cdot 5; \qquad 80^{14} < 60^{15} \iff 2^{26} = 67108864 < 71744535 = 3^{15}\cdot 5$];

$$\frac{15}{14} < \frac{11}{10} < \frac{\ln 20}{\ln 15}$$

[machine: $15 \cdot 10 < 14 \cdot 11; \qquad 20^{10} = 10240000000000 > 8649755859375 = 15^{11}$] :

$$\boxed{1 < \frac{m_n}{m_p} < \frac{\ln 80}{\ln 60} < \frac{15}{14}} \quad \text{on the materialized read at every admissible } (\varepsilon_q,\ N,\ \text{g},\ \varepsilon_{\text{EM}}) \text{ on the sub-critical branch}$$

an exact tightening of BM.91(7)'s $\frac{10}{9}$ by the filter's lower bound on the weight

[machine: $\frac{15}{14} < \frac{10}{9} \iff 135 < 140$];

the incommensurability hardens:

$$\frac{m_n}{m_p} = \frac{4}{3} \iff \varepsilon^*_{\text{EM}} = \frac{(6 - 6w_{\text{mat}})\ln 2 - \ln 405}{12\, N\, w_{\text{mat}}\, \text{g}\, \ln 2} < 0$$

[BM.91(8); machine: $(6 - 6w_{\text{mat}})\ln 2 \le 6\ln 2 = \ln 64 < \ln 405$ on $64 < 405$];

$$\frac{m_n}{m_p} \notin \mathbb{Q} \text{ at every rational } (w_{\text{mat}}, N, \text{g}, \varepsilon_{\text{EM}})$$

[BM.83(2)'s lemma: the $\ln 3$ coefficient is 0 in the numerator and 1 in the denominator; machine];

(6) the Q-value on the materialized informational read:

$$Q_\beta\,\mathfrak{h} = \ln\frac{4}{3} - 3N w_{\mathrm{mat}}\mathsf{g}\,\varepsilon_{\mathrm{EM}}\ln 2 - F_{\ell\nu} \;\le\; \ln\tfrac{4}{3} - 2\ln 2 \;=\; -\ln 3 \;<\; 0$$

[BM.92(1); BM.70(2), (3) : $F_{\ell\nu} \ge 2\ln 2$; machine]

uniformly in $(\varepsilon_q,\ N,\ \mathsf{g},\ \varepsilon_{\mathrm{EM}})$ — the closure survives the filter unchanged;

the ratio form tightens:

$$\boxed{-\frac{\ln 3}{\ln 1280} \;\le\; \frac{Q_\beta}{m_n} \;\le\; -\frac{\ln 3}{6\,w_{\mathrm{mat}}\ln 2 + \ln 20} \;<\; -\frac{\ln 3}{\ln 80}}$$

with

$$\boxed{-\frac{\ln 3}{\ln 80} \in \left(-\frac{1}{3},\,-\frac{1}{4}\right)}$$

[machine: $3^4 = 81 > 80$ and $80 > 27$] an exact tightening of BM.92(2)'s $\left(-\frac{2}{5}, -\frac{1}{3}\right)$ endpoint;

the margin ladder stands:

$$\Sigma_{\mathrm{B}} - 2\ln 2 = -\ln 3,$$

$$\Sigma_{\mathrm{B}} - 4\ln 2 = -\ln 12,$$

$$\Sigma_{\mathrm{B}} - 5\ln 2 = -\ln 24$$

[BM.70(3); BM.92(2); machine: $4 < 12 < 48 < 96$];

(7) the rescoped dissolution row:

BM.91(4)'s row $\left(\varepsilon^{\mathrm{crit}} \to \infty \text{ as } \varepsilon_q \to \tfrac{1}{4}^{-},\ \text{“the fully dissolved limit never closes the splitting”}\right)$ is rescoped, ¬retracted :

at $\varepsilon_q \to \frac{1}{4}^{-}$ the unconditioned ensemble's weight concentrates on the $E = 0$ member,

$$P(E{=}0) = (1-p)^3 \to 1$$

[BM.94$(4), (5)$; machine];

that member supports no closed chain and no terminus

[FT.144(4); Def-H.4] $\Longrightarrow$ the fully dissolved limit is outside the materialization domain

on the materialized read the splitting is closable at a finite coupling:

$$\varepsilon_{\text{mat}}^{\text{crit}} < \frac{5}{12\, N\, \mathsf{g}}$$

$[(2), (3)$; machine];

the unconditioned row stands at its own locus as the (R4) general-weight statement;

(8) status:

derived: (1)–(7) on the marked pair specialization;

rescoped, ¬retracted: BM.91(2),

(4),

(7),

(8),

BM.92(1),

(2) stand on w at their own loci and gain the materialized variants here;

BM.84, BM.85, BM.77$(5), (6)$ untouched;

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

§Cal-BM stays empty;

¬the observed neutron–proton ratio imported — the bound (5) is derived from the filter and the prime-log assembly alone;

Req. D0.4 (Parts 1–2); CT0.8, CT2.1 (Part 1, Part 2); P1, P4, P5 (Parts 1–2); Def-H.4, FT.144(4) (Part 3); Census FT.35(2) (row-wise); Def-BM.13(4), Def-BM.13(5), Def-BM.13(6), BM.57(4), BM.62(8), BM.70(2), BM.70(3), BM.77(5), BM.77(6), BM.83(2), BM.84(4), BM.85, BM.89(5), BM.89(8), BM.91(2), BM.91(4), BM.91(5), BM.91(7), BM.91(8), BM.92(1), BM.92(2), BM.94(4), BM.94(5), BM.94(6), BM.94(7), BM.94(8) (this part); exact prime-log arithmetic with integer power certificates, exact rational function rows, monotonicity by cross-multiplication, token censuses [math; machine: `checks/verify_H_flicker_integral_restmass.py` §F].

Pf.

$(1):$ BM.91(2)'s assembly with $w \to w_{\text{mat}}$ and $\langle m_{\text{hid}}\rangle_{\text{mat}} = 6w_{\text{mat}} \ln 2/\mathfrak{h}$ [BM.94(8); BM.89(5)];

$$\ln 20 - \ln 15 = \ln \tfrac{4}{3} \text{ [machine]};$$

the w-coefficient census in the two rest inputs and in Δm [machine]

$(2):$ $\varepsilon^{\text{crit}}$

$$= \ln(4/3)/(3Nw_{\text{mat}}\mathsf{g}\ln 2)$$

$$= (2 - \log_2 3)/(3Nw_{\text{mat}}\mathsf{g}) \text{ [BM.91}(4) \text{ re-cited; machine]};$$

$\frac{1}{3} < w_{\text{mat}} \leq 1$ [BM.94(7)] $\implies$ the two-sided enclosure [machine];

$\frac{2}{5} < 2 - \log_2 3 < \frac{5}{12}$ on $3^5 < 2^8$ and $3^{12} > 2^{19}$ [BM.91(4) re-cited]

$$(3):\ w_{\text{mat}}$$

$$\leq 1$$

$$\implies N w_{\text{mat}} \mathsf{g}$$

$$\varepsilon_{\text{EM}}$$

$$\leq N\mathsf{g}$$

$$\varepsilon_{\text{EM}} \leq \tfrac{1}{8} < \tfrac{2-\log_2 3}{3} \text{ [machine: } 8(2-\log_2 3) > 3 \iff 13 > 8\log_2 3 \iff 2^{13} > 3^8] \implies \Delta m > 0$$

$$w_{\text{mat}} > \tfrac{1}{3} \text{ and } N\mathsf{g}\,\varepsilon_{\text{EM}} \geq \tfrac{5}{12} \implies N w_{\text{mat}} \mathsf{g}\,\varepsilon_{\text{EM}} > \tfrac{5}{36};$$

$$\tfrac{5}{36} > \tfrac{2-\log_2 3}{3} \iff \tfrac{5}{12} > 2 - \log_2 3 \iff \log_2 3 > \tfrac{19}{12}$$

$$\iff 3^{12}$$

$$= 531441 > 524288$$

$$= 2^{19} \text{ [machine]} \implies \Delta m < 0$$

$$\tfrac{1}{8} < \tfrac{5}{12} \iff 12 < 40 \text{ [machine]}$$

$$(4):\ \text{the coefficient of } \varepsilon_{\text{EM}} \text{ in } \Delta m \text{ is } -3N w_{\text{mat}} \mathsf{g} \ln 2 < 0 \text{ [machine]};$$

$$\varepsilon_{\text{EM}} = \tfrac{1}{9}:\ N\mathsf{g}\,\varepsilon_{\text{EM}} = \tfrac{1}{9} \leq \tfrac{1}{8} \implies \Delta m > 0 \text{ by } (3);$$

$$\varepsilon_{\text{EM}} = \tfrac{5}{12} \text{ at } N\mathsf{g} = 1 \implies \Delta m < 0 \text{ by } (3)$$

$$(5):\ \text{the ratio's monotonicity in } \varepsilon_{\text{EM}} \text{ and in } w_{\text{mat}} \text{ re-cited [BM.91(7); machine]};$$

$$6 w_{\text{mat}} \ln 2 \in (2\ln 2,\ 6\ln 2] \text{ [BM.94(7)]} \implies \text{the four-endpoint chain [machine]}$$

$$80^{15} > 60^{16} \iff 2^{60}5^{15} > 2^{32}3^{16}5^{16} \iff 2^{28} > 3^{16}\cdot 5: \quad 268435456 > 215233605 \text{ [machine]};$$

$$80^{14} < 60^{15} \iff 2^{26} < 3^{15}\cdot 5: \quad 67108864 < 71744535 \text{ [machine]}$$

$$\tfrac{15}{14} < \tfrac{11}{10} \iff 150 < 154;$$

$$20^{10} > 15^{11} \text{ [BM.91(7) re-cited]};$$

$$\tfrac{15}{14} < \tfrac{10}{9} \iff 135 < 140 \text{ [machine]}$$

$$(6):\ \text{BM.92(1)'s uniform closure re-cited with } w \to w_{\text{mat}} \text{ [machine]};$$

$$-\ln 3/\ln 80$$

$$\in (-\tfrac{1}{3}, -\tfrac{1}{4})$$

$$\iff 3\ln 3 > \ln 80 \text{ and } 4\ln 3 < \ln 80$$

$$\iff 81 > 80 \text{ and } 80 > 27 \text{ [machine]};$$

$$\text{the three margin identities re-certified [machine]}$$

$$(7):\ (1-p)^3 \to 1 \text{ as } p \to 0^+ \text{ [machine]};$$

$$|E_{\mathcal{R}}| = 0 \implies \neg\exists \text{ a terminus [FT.144(4) re-cited]};$$

$$\varepsilon_{\text{mat}}^{\text{crit}} < (2-\log_2 3)/(N\text{g}) < \tfrac{5}{12N\text{g}} \ [(2);\ \text{machine}];$$

$$(8):\ \text{the standing-rule scans [machine]}. \ \blacksquare$$

Phys. Filter the ensemble to the configurations that can carry a composite and three rows change shape. The assembly keeps its form — six times the weight times log two, plus the class's own transport entropy, less the electromagnetic term — but the weight is now one over one plus u plus u squared, and it cannot fall below a third. That single bound does real work. The critical coupling is squeezed into an exact window between two fifteenths and five twelfths over the lapse and the curvature composition, and the two bars that decide the sign of the splitting lose the crystallization weight entirely: an eighth or less of the coupling product forces the neutron heavier, five twelfths or more forces it lighter, and the undecided band between them is exact and inhabited.

The ratio bound tightens from ten ninths to the log of eighty over the log of sixty, itself enclosed between sixteen fifteenths and fifteen fourteenths by two integer power certificates, and the Q-value's ratio form tightens from a third to a quarter by eighty-one against eighty. The dissolution row is rescoped rather than retracted: at the far end of the threshold range the unconditioned ensemble piles its weight onto the empty configuration, which supports no chain and no terminus, so that limit is not a materialized universe at all — and inside the materialization domain the splitting can always be closed by a finite coupling. Nothing measured entered: the ratio is derived from the filter and the prime-log assembly, the coupling stays symbolic, and the closure of the decay on the informational currency survives the filter exactly as it survived the ensemble.

BM.100 — The status of the flicker integral and the rest-mass read: adjudication and scope. Structural-constraints record: see Appendix B, Section B.1.29 of the Addendum.

4.22 Tier BM.20 — Six Objective Resolutions: the Energy-Scale Graph, the Dimensional Bridge, the Eigenvalue Composition, the Curvature Composition, the General Crystallization Weight, and the Kind Structure of the Arrow

Locus. This tier takes six named objectives and gives each its exact adjudication: one graph, one census with one declared slot, one referent table, one fixed point, one measure-free bound, and one kind structure. The Q-value becomes a function of the energy scale, its forward and inverse reads are shown to be one graph reflected, and its crossing is a derived strength enclosed by two integers — with the crossing count depending on the priced curvature composition (BM.101). The dimension surface is censused against the single information flux, the seven rows are derived or adjudicated, and the two bridges are separated: the dimensional one is $\mathfrak{h}$, the numerical one is ρ_{B}, whose status is declared as the §II.4 addendum A-BM.9 with no value issued (BM.102). The three-constituent assembly's eigenvalue equations are derived where they are formed — the family cubic and the binomial characteristic polynomial — and the tensor-product referent is censured where it is not, so the composition question and the octet question are both adjudicated ¬formed (BM.103). The curvature composition's self-consistency condition is formed exactly, solved for all three priced candidates, and shown to select none, because the lock's own freedom census excludes the back-reaction (BM.104). The crystallization weight is derived measure-free and for an arbitrary threshold family, with the same exact bounds, and the defect-conditioned version is refuted (BM.105). The spin label and the tick arrow are derived, and the exchange structure that a statistics requires is censured: deficit 11 stands with its residue named (BM.106). BM.107 adjudicates.

Standing rows. α_B stays barred on BM.61(8)'s four grounds. No clause of BM.1–BM.100 is retracted, replaced or renumbered: every row this tier touches is re-scoped at its own locus and recorded in BM.101(8), BM.104(8), BM.105(7) and BM.107(5), in BM.77(7)'s pattern. No measured coupling, mass or ratio enters any derivation [P5]: the crossing strength of BM.101(3) is a derived symbolic value and is ¬compared to any external number; the neutron–proton ratio stays derived and bounded [BM.99(5)]. No right-ray matter line and no seesaw-type mechanism is suggested, added or required [BM.96(4), (6) re-cited, ¬extended]; ¬an external mixing matrix is imported — the mixing rows are reads of the derived 16-arena [FT.142(2), (5); FT.145(3)]. One dimension only: units are powers of $\mathfrak{q}$ [D2.9; CT2.1]. Local notation, fixed for the whole tier: $E_1 := \mathfrak{h}/(r \ln 2)$ the envelope-unit count; $x := E/E_1 = k$ the scale ratio [BM.88(2)(i)]; $\beta := 2\pi \ln 2\, E_1^2 = 2\pi\mathfrak{h}^2/(r^2 \ln 2) \in \mathfrak{q}^0$; $f = 1 - \frac{1}{x}$; $G = \beta x^2$; $\varepsilon_{\mathrm{eff}} := N f\, \mathsf{g}\, \varepsilon_{\mathrm{EM}} = e(r, E, N)^2/(4\pi)$; $\varepsilon_*(w)$ the crossing strength; $c := \varepsilon_*/(N\varepsilon_{\mathrm{EM}})$; $a := \frac{1}{2} N w\, \varepsilon_{\mathrm{EM}} Q^2$; p_i, u_i, R, S the family data of BM.105(3); $[\cdot]$ the $\mathfrak{q}$-exponent map

[CT2.1; D2.9]. The symbols w, D, p, u, Δ_g, M_0, s, $\mathcal{R}$, ρ_{B}, D_S keep their BM.94–BM.99 meanings; the italic g keeps its BM.39(1) meaning and q its end-register meaning [D3.4; EM.9] at every occurrence.

Theorem 4.85 (BM.101 — The energy-scale graph: the two reads, the derived crossing strength, and the composition-dependent crossing count). Thm.

(1) the two reads are one graph reflected:

$$Q_\beta^{\mathrm{fwd}}(E) \;:=\; m_n^{\mathrm{rest}}(E) - m_p^{\mathrm{rest}}(E) - m_\ell^{\mathrm{rest}} - m_\nu^{\mathrm{rest}},$$

$$Q_\beta^{\mathrm{inv}}(E) \;:=\; m_p^{\mathrm{rest}}(E) + m_\ell^{\mathrm{rest}} + m_\nu^{\mathrm{rest}} - m_n^{\mathrm{rest}}(E)$$

$$\boxed{Q_\beta^{\mathrm{inv}}(E) \;=\; -\,Q_\beta^{\mathrm{fwd}}(E) \text{ identically, at every scale, on every read}}$$

[machine: one negation identity; BM.97(2), (4) re-cited] :

the inverse reaction opens exactly where the forward one closes, and their common boundary is one locus — ¬two independent graphs;

(2) the scale-dependent splitting (the graph's spine):

$$\varepsilon_{\mathrm{eff}}(r,E,N) \;:=\; N\, f\Big(\frac{r}{\Delta x_{\mathrm{B}}(E)}\Big)\, \mathrm{g}\big(G(E)\big)\, \varepsilon_{\mathrm{EM}} \;=\; \frac{e(r,E,N)^2}{4\pi}$$

[BM.88(4); Def-BM.12 local notation; machine: the monomial identity] :

$$\boxed{\Delta m(E)\,\mathfrak{h} \;=\; \ln\frac{4}{3} \;-\; 3\,w\,\ln 2\cdot \varepsilon_{\mathrm{eff}}(r,E,N)}$$

[BM.91(2); BM.99(1) at the full profile instead of its far-field locus $f=1$; machine];

the far-field rescope:

$$f\to 1 \;\text{ recovers }\; \Delta m\,\mathfrak{h} = \ln\tfrac{4}{3} - 3Nw\,\mathrm{g}\,\varepsilon_{\mathrm{EM}}\ln 2 \;\text{ exactly}$$

[BM.88(5); BM.91(2); BM.99(1) : every row stands at that locus];

the informational Q-value on the graph:

$$Q_\beta^{\mathrm{fwd}}\,\mathfrak{h} = \Delta m(E)\,\mathfrak{h} - F_{\ell\nu},$$

$$F_{\ell\nu} \geq 2\ln 2$$

[BM.70(2), (3); BM.92(1)] :

$$\boxed{Q_\beta^{\mathrm{fwd}}\,\mathfrak{h} \;\leq\; -\ln 3 \;<\; 0 \;\text{ and }\; Q_\beta^{\mathrm{inv}}\,\mathfrak{h} \;\geq\; \ln 3 \;>\; 0 \qquad \text{at every scale, uniformly in } (r,\, E,\, \varepsilon_q,\, N,\, \mathsf{g},\, \varepsilon_{\mathrm{EM}})}$$

[machine: the sign census on a rational lattice]

on the informational currency the forward channel is closed and the inverse channel is open at every energy scale, with the exact margin $\ln 3$;

(3) the derived crossing strength:

$$\Delta m(E) = 0 \iff \varepsilon_{\mathrm{eff}}(r, E, N) = \varepsilon_*(w) :$$

$$\boxed{\varepsilon_*(w) \;:=\; \frac{\ln(4/3)}{3\,w\ln 2} \;=\; \frac{2-\log_2 3}{3\,w}, \qquad e_*^2 \;:=\; 4\pi\,\varepsilon_*(w) \;=\; \frac{4\pi\,(2-\log_2 3)}{3\,w}}$$

[BM.91(4); machine: exact in the $\log_2$ ring];

the enclosure:

$$w = w_{\mathrm{mat}} \in \left(\tfrac{1}{3}, 1\right]$$

[BM.94(7)] :

$$\boxed{\frac{2-\log_2 3}{3} \;\leq\; \varepsilon_* \;<\; 2-\log_2 3 \qquad \text{and} \qquad \frac{1}{8} \;<\; \varepsilon_* \;<\; \frac{5}{12}}$$

[machine: $2^{13} = 8192 > 6561 = 3^8$; $3^{12} = 531441 > 524288 = 2^{19}$; $3^5 = 243 < 256 = 2^8$] :

the two endpoints of BM.99(3)'s two-sided parameter-free bar are exactly the enclosure of the crossing strength

[machine: the four integers re-certified];

$$e_*^2 \in \left(\tfrac{\pi}{2},\; \tfrac{5\pi}{3}\right)$$

[machine];

no measured coupling is compared to ε_* or e_* [P5; EM.25(7), (9); the quarantined audit rows stay quarantined];

(4) the crossing locus is a level set, and e_0 places it:

$$\boxed{\mathcal{L}_* \;:=\; \big\{(r,E,N):\;\; e(r,E,N)^2 = e_*^2\big\} \;=\; \big\{(r,E,N):\; \varepsilon_{\text{eff}} = \varepsilon_*(w)\big\}}$$

$$e^2 = e_0^2\, N\, f\, \mathsf{g}$$

[BM.88(4)] $\implies$ $\mathcal{L}_*$ is the level set of the derived profile at the derived strength;

the adjudication of the cancellation claim:

e_0^2 does ¬cancel — it is the one datum that places $\mathcal{L}_*$ in (r,E) :

what is e_0-free is the crossing value ε_*, its enclosure, and the crossing count per composition;	what requires e_0 is the crossing location

¬∃ a theorem fixing e_0 or e

[EM.25(7); BM.88(7); P4; P5];

¬a numerical value of either enters any clause of this tier [machine: the token scan];

(5) the three profile shapes (composition by composition):

$$x = \frac{E}{E_1} = k = \frac{r\,E\ln 2}{\mathfrak{h}},$$

$$E_1 = \frac{\mathfrak{h}}{r\ln 2},$$

$$G = 2\pi E^2 \ln 2 = \beta x^2,$$

$$\beta = \frac{2\pi\mathfrak{h}^2}{r^2\ln 2} \in \mathfrak{q}^0$$

[BM.88(2)(i), (iii); BM.88(3); CT2.1; machine: the exponent arithmetic];

$$P_{\mathrm{g}}(x) \;:=\; f(x)\,\mathrm{g}\big(\beta x^{2}\big),$$

$$\varepsilon_{\mathrm{eff}} = N\,\varepsilon_{\mathrm{EM}}\,P_{\mathrm{g}}(x)\,:$$

$$\text{(i)}\quad \mathrm{g}=1\,:$$

$$P(x) = 1 - \frac{1}{x} \text{ strictly increasing on } x \geq 1,$$

$$P(1) = 0,$$

$$P \to 1\,:$$

$$\boxed{\text{at most one crossing; it exists } \iff N\varepsilon_{\mathrm{EM}} > \varepsilon_{*}}$$

the crossing scale in closed form:

$$x_{*} = \frac{1}{1-c},$$

$$c := \frac{\varepsilon_{*}}{N\,\varepsilon_{\mathrm{EM}}} \in (0,1),$$

$$E_{*} = \frac{\hbar}{r\ln 2\,(1-c)}$$

[machine: the linear equation $x - 1 = cx$];

$$\text{(ii)}\quad \mathrm{g}=1+G\,:$$

$$P(x) = \Big(1-\frac{1}{x}\Big)\big(1+\beta x^{2}\big) \text{ strictly increasing,}$$

$$P(1) = 0,$$

$$P \to \infty\,:$$

$$\boxed{\text{exactly one crossing at every admissible } (N, \varepsilon_{\mathrm{EM}}) > 0}$$

the crossing equation is an exact cubic:

$$\beta x^3 - \beta x^2 + (1-c)\,x - 1 = 0$$

[machine: the cleared identity $(x-1)(1+\beta x^2) = cx$];

(iii) $\mathsf{g} = (1+G)^{-1}$:

$$P(x) = \frac{1 - 1/x}{1+\beta x^2},$$

$$P(1) = 0,$$

$$P \to 0 \ \text{ as } \ x \to \infty :$$

a window, not a half-line

the unique interior maximum:

$$P'(x_*) = 0 \iff 2\beta x_*^3 - 3\beta x_*^2 - 1 = 0$$

[machine: the cleared stationarity row];

the cubic $h(x) = 2\beta x^3 - 3\beta x^2 - 1$ has $h'(x) = 6\beta x(x-1) \geq 0$ on $x \geq 1$,

$$h(1) = -\beta - 1 < 0,$$

$$h \to +\infty \implies \boxed{\exists!\ x_*(\beta) > 1}$$

$$\boxed{P_{\max}(\beta) = P(x_*) = 1 - \frac{3}{2x_*}}$$

[machine: the polynomial identity $2x\big(1-\tfrac{1}{x}\big) - (2x-3)(1+\beta x^2) = 1 - (2\beta x^3 - 3\beta x^2)$, vanishing on the cubic];

the exact dichotomy in β :

$$h\big(\tfrac{3}{2}\big) = -1 < 0 \ \ \forall \beta,$$

$$h(2) = 4\beta - 1$$

[machine: two polynomial identities] :

$\beta > \frac{1}{4} \implies x_* \in \left(\frac{3}{2},\, 2\right)$ and $P_{\max} \in \left(0,\, \frac{1}{4}\right)$; $\beta = \frac{1}{4} \implies x_* = 2, \quad P_{\max} = \frac{1}{4}$ exactly; $\beta < \frac{1}{4} \implies x_* > 2, \quad P_{\max} \in \left(\frac{1}{4},\, 1\right)$

the crossing equation is an exact depressed cubic:

$$c\beta x^3 + (c-1)\, x + 1 = 0$$

[machine: the cleared identity $x - 1 = cx(1 + \beta x^2)$];

the root count is the crossing count:

$$c\beta x^3 + (c-1)x + 1 = -\, x\big(1 + \beta x^2\big)\big(P(x) - c\big)$$

[machine: one polynomial identity] :

$c < P_{\max}$: two crossings (the window opens and closes); $c = P_{\max}$: one (x_*, a double root); $c > P_{\max}$: none

(6) the specified and pure-pole graphs are exactly flat in the energy scale:

$$Q_\beta^{\text{spec}}\, \mathfrak{h} = \Delta T(S) - \rho_\ell \sqrt{G_\ell} - \rho_\nu \sqrt{G_\nu},$$

$$\Delta T = 5\, m_B^2(\cdot, S)$$

the alphabet $\{m_B^2(\cdot, S),\ \rho_\ell,\ \rho_\nu,\ g,\ g',\ \mathfrak{h}\}$ carries $\neg E_{\text{int}},\ \neg r,\ \neg N,\ \neg \mathsf{g},\ \neg \varepsilon_{\text{EM}}$

[BM.92(5); BM.97(2), (3); machine: the token census] :

the specified read's energy-scale graph is a horizontal line, and its trichotomy is decided by the scales (ρ_ℓ, ρ_ν), ¬by the scale E

[BM.77(5), (6) re-cited];

the pure-pole read is flat in E as well, its crossing being $\Delta_g^{\text{crit}} = s(2M_0 + s)$

[BM.97(4)];

the entire energy-scale structure lives in the informational/EM channel;

(7) the graph's verdict rows:

the monotonicity: $\Delta m(E)$ is strictly decreasing in E on (i),

(ii),

and rises then falls on (iii)

[machine: three monotonicity censuses on exact rationals];

the envelope row:

$$f(1)=0 \implies \varepsilon_{\text{eff}}=0 \text{ at } E=E_1 \implies \Delta m(E_1)\,\mathfrak{h}=\ln\tfrac{4}{3}>0$$

[BM.88(5); machine] :

the splitting is open at the member's own envelope on every composition;

the bar as a pointwise crossing census:

$$f\le 1 \text{ on } x\ge 1 \implies \varepsilon_{\text{eff}} \le N\,\mathsf{g}\big(G(E)\big)\,\varepsilon_{\text{EM}}$$

[machine] :

$$N\mathsf{g}\,\varepsilon_{\text{EM}}\le\tfrac{1}{8}<\varepsilon_* \implies \boxed{\Delta m(E)>0 \text{ at that scale}},$$

$$N\mathsf{g}\,\varepsilon_{\text{EM}}\ge\tfrac{5}{12}>\varepsilon_* \implies \boxed{\Delta m(E)<0 \text{ at that scale}}$$

[BM.99(3) re-derived on the graph; machine];

the global crossing census per composition:

(i) a crossing exists $\iff N\varepsilon_{\mathrm{EM}} > \varepsilon_*$;

(ii) exactly one at every $(N, \varepsilon_{\mathrm{EM}}) > 0$;

(iii) two $\iff N\varepsilon_{\mathrm{EM}} P_{\max} > \varepsilon_*$, one at equality, none below [(5); machine];

the band is where the flicker decides:

$\frac{1}{8} < N\mathsf{g}\,\varepsilon_{\mathrm{EM}} < \frac{5}{12} \implies$ the crossing's existence is decided by $\varepsilon_*(w) = \frac{2 - \log_2 3}{3w}$, i.e. by w

[(3); machine];

(8) status:

derived: (1)–(7);

read-relative: the verdict rows (2), (6);

rescoped, ¬retracted: BM.91(2), BM.99(1), (3) stand as the far-field locus $f = 1$;

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a numerical coupling [EM.25(7)];

to calibration: e_0 (4), g (5) [BM.104(5)], N as solution data [CT2.7; BM.34(2)].

Req. D1.8, D2.9, D3.4 (Parts 1–2); A6, CT2.1, CT2.7, EM.9, EM.25(7), EM.25(9) (Part 2); P1, P4, P5 (Parts 1–2); Def-BM.12, Def-BM.4(1), BM.13(2), BM.14(2), BM.16(1), BM.17(2), BM.23(9),

BM.28(4), BM.34(2), BM.39(1), BM.70(2), BM.70(3), BM.77(5), BM.77(6), BM.84(9), BM.88(2), BM.88(3), BM.88(4), BM.88(5), BM.88(6), BM.88(7), BM.89(5), BM.91(2), BM.91(4), BM.92(1), BM.92(5), BM.94(5), BM.94(7), BM.96(3), BM.97(2), BM.97(3), BM.97(4), BM.99(1), BM.99(3), BM.99(5), BM.99(6) (this part); Census FT.35(2) (row-wise); exact polynomial identities over $\mathbb{Q}[\beta, x, c]$, cubic root counts by monotonicity and sign, exact rational lattices, the $\log_2$-ring sign test with integer power certificates, Laurent-monomial assembly, token censuses [math; machine: `checks/verify_H_six_objective_resolution.py` §A].

Pf.

$$(1):\ Q_\beta^{\text{inv}}$$

$$= (m_p + m_\ell + m_\nu) - m_n$$

$$= -\big(m_n - m_p - m_\ell - m_\nu\big)$$

$$= -Q_\beta^{\text{fwd}}\ \text{[machine: one identity over } \mathbb{Q}[m_n, m_p, m_\ell, m_\nu]]$$

$$(2):\ \varepsilon_{\text{eff}}$$

$$= Nf$$

$$\mathsf{g}$$

$$\varepsilon_{\text{EM}} \text{ and } e^2$$

$$= e_0^2 Nf$$

$$\mathsf{g} \text{ with } \varepsilon_{\text{EM}}$$

$$= e_0^2/(4\pi) \implies \varepsilon_{\text{eff}}$$

$$= e^2/(4\pi)\ \text{[BM.88(4); machine: one monomial]};$$

$$\text{BM.91(2)'s term } 3Nw\,\mathsf{g}\,\varepsilon_{\text{EM}} \ln 2 \text{ is the } f{=}1 \text{ locus of } 3w \ln 2 \cdot \varepsilon_{\text{eff}}\ \text{[machine]}$$

$$Q_\beta^{\text{fwd}}\mathfrak{h}$$

$$= \ln \tfrac{4}{3} - 3w \ln 2 \cdot \varepsilon_{\text{eff}} - F_{\ell\nu} \le \ln \tfrac{4}{3} - 2 \ln 2$$

$$= -\ln 3 \text{ [BM.70}(2),(3)\text{; machine: the coefficient census on a } 504\text{-point rational lattice]};$$

$$-\ln 3 < 0 \iff 4 < 12 \text{ [machine]}$$

$$(3):\ \Delta m$$

$$= 0$$

$$\iff 3w \ln 2 \cdot \varepsilon_{\text{eff}}$$

$$= \ln \tfrac{4}{3} \iff \varepsilon_{\text{eff}}$$

$$= \tfrac{\ln(4/3)}{3w \ln 2}$$

$$= \tfrac{2-\log_2 3}{3w} \text{ [machine: exact in the } \log_2 \text{ ring]};$$

$$e_*^2 = 4\pi\varepsilon_* \text{ [machine]}$$

$$w \in (\tfrac{1}{3}, 1] \text{ [BM.94}(7)\text{]} \implies \varepsilon_* \in [\tfrac{2-\log_2 3}{3},\ 2 - \log_2 3) \text{ [machine]};$$

$$\tfrac{2-\log_2 3}{3} > \tfrac{1}{8} \iff 8(2 - \log_2 3) > 3 \iff 2^{13} > 3^8;$$

$$2 - \log_2 3 < \tfrac{5}{12} \iff 3^{12} > 2^{19} \text{ [machine: two integer power comparisons]}$$

$$(4):\ \mathcal{L}_* \text{ is the preimage of one value under the derived profile [BM.88}(4)\text{]};$$

$$e^2$$

$$= e_0^2 N f$$

$\mathfrak{g} \implies$ the locus moves with e_0 and its value does not [machine: the symbol census of ε_* carries $\{w, \ln 2, \log_2 3, \pi\},\ \neg e_0$]

$$(5):\ x = k = rE \ln 2/\mathfrak{h} = E/E_1 \text{ at } E_1 = \mathfrak{h}/(r \ln 2) \text{ [machine]};$$

$$G = 2\pi E^2 \ln 2 = 2\pi \ln 2\, E_1^2 x^2 = \beta x^2,$$

$$[\beta] = [\mathfrak{h}^2/r^2] = \mathfrak{q}^{-1}\mathfrak{q}^{+1} = \mathfrak{q}^0 \text{ [CT2.1; machine]}$$

$$\text{(i)}: \ P = 1 - \tfrac{1}{x},\ P' = 1/x^2 > 0 \text{ on } x \geq 1;\ P(1) = 0;\ P \to 1 \text{ [machine]};$$

$$x - 1$$

$$= cx$$

$$\iff x(1 - c)$$

$$= 1$$

$$\iff x$$

$$= 1/(1 - c),$$

$$\text{admissible}$$

$$\iff 0 < c < 1$$

$$\iff N\varepsilon_{\text{EM}} > \varepsilon_* \text{ [machine]}$$

$$\text{(ii)}: \ P' = \tfrac{1}{x^2}(1 + \beta x^2) + (1 - \tfrac{1}{x})2\beta x > 0 \text{ on } x > 1;\ P \to \infty \text{ [machine]};$$

$$(x - 1)(1 + \beta x^2)$$

$$= cx$$

$$\iff x + \beta x^3 - 1 - \beta x^2$$

$$= cx \iff \beta x^3 - \beta x^2 + (1 - c)x - 1$$

$$= 0 \text{ [machine: the cleared cubic]}$$

$$\text{(iii)}: \ P' = \tfrac{1}{x^2(1+\beta x^2)} - \tfrac{(1-\frac{1}{x})2\beta x}{(1+\beta x^2)^2};$$

$$P' = 0 \iff (1+\beta x^2) = 2\beta x^3(1-\tfrac{1}{x}) \cdot \tfrac{x^2}{x^2} \iff 2\beta x^3 - 3\beta x^2 - 1 = 0 \text{ [machine: cleared]}$$

$$h' = 6\beta x^2 - 6\beta x = 6\beta x(x-1) \geq 0 \text{ on } x \geq 1, \text{ with equality only at } x = 1;$$

$$h(1) = -\beta - 1 < 0;$$

$$h(x) \to +\infty \implies \exists!\, x_* > 1 \text{ [machine: monotonicity plus the two endpoint signs]}$$

$$\beta x_*^2(2x_* - 3)$$

$$= 1$$

$$\implies 1 + \beta x_*^2$$

$$= \tfrac{2x_*-2}{2x_*-3}$$

$$\implies P(x_*)$$

$$= \tfrac{x_*-1}{x_*} \cdot \tfrac{2x_*-3}{2x_*-2}$$

$$= \tfrac{2x_*-3}{2x_*}$$

$= 1 - \frac{3}{2x_*}$ [machine: the identity $2x(1-\frac{1}{x}) - (2x-3)(1+\beta x^2) = 1 - (2\beta x^3 - 3\beta x^2)$ in $\mathbb{Q}[\beta, x]$]

$$h(\tfrac{3}{2}) = 2\beta\tfrac{27}{8} - 3\beta\tfrac{9}{4} - 1 = -1;$$

$$h(2)$$

$$= 16\beta - 12\beta - 1$$

$= 4\beta - 1$ [machine: two polynomial identities] $\implies$ the β-dichotomy and $P_{\max}$ enclosures [$(1 - \frac{3}{2x_*})$ monotone in x_*]

$$x - 1 = cx(1+\beta x^2) \iff c\beta x^3 + (c-1)x + 1 = 0;$$

$$x(1+\beta x^2)(P-c)$$

$$= (x-1) - cx(1+\beta x^2)$$

$$= -\big(c\beta x^3 + (c-1)x + 1\big) \text{ [machine: one identity] } \implies \text{ the root count is the crossing count}$$

$$(6):\ \Delta T = 5\, m_B^2(\cdot, S) \text{ is a frozen rational per cell [BM.77(2)];}$$

$$G_\ell, G_\nu \text{ depend on } (g, g', \delta) \text{ alone [BM.39(1); BM.96(2)];}$$

the token census [machine]

$$(7):\ \varepsilon_{\text{eff}}(E_1) = N f(1)\, \mathsf{g}\, \varepsilon_{\text{EM}} = 0 \implies \Delta m(E_1)\mathfrak{h} = \ln \tfrac{4}{3} > 0 \text{ [BM.88(5); machine];}$$

$$f \le 1 \text{ on } x \ge 1 \implies \varepsilon_{\text{eff}} \le N\mathsf{g}\, \varepsilon_{\text{EM}} \text{ [machine];}$$

$$(8):\ \text{the scope record. } \blacksquare$$

Phys. One graph, read twice. The forward and inverse Q-values are exact negatives, so the two reactions the question asks about are one curve and its reflection, and the boundary between them is a single level set of the derived coupling profile. On the informational currency the curve never changes sign: the two-bit floor on the emitted lines keeps the forward channel closed and the inverse channel open at every scale, with the exact margin log three, so the only sign structure the graph carries is the splitting's. That structure is decided by one derived number, the crossing strength — log of four thirds over three weights times log two, which is two minus the binary logarithm of three over three weights — enclosed between an eighth and five twelfths by the same two integer powers that give the parameter-free bar. The bar and the crossing are the same statement: below the lower end the curve never crosses at any scale, above the upper end it always does, and inside the band the flicker weight alone decides. The profile's own shape then does the rest. With the flat composition the crossing is a closed form, one over one minus c; with the growing composition there is exactly one crossing at every positive coupling, because the profile diverges; with the decreasing composition the profile is a window — it rises from zero, peaks at the unique root of an exact cubic, and returns to zero, so the forward channel opens at one scale and closes at another, or never opens, and the peak has the closed form one minus three over twice the root, sitting below a quarter whenever the radius parameter exceeds a quarter. Two compositions give a half-line, one gives a window, and the corpus still cannot say which is its own. What is exact and free of every undeclared datum is the crossing value; what needs the one open calibration number is where in radius and energy that value is reached.

Theorem 4.86 (BM.102 — The dimensional bridge: the $\mathfrak{q}$-exponent census, the seven rows, and the declared calibration slot). Thm.

(1) the one-dimension law:

$$\text{units} := \text{powers of } \mathfrak{q};$$

$$c = \hbar = k_B = 1;$$

$$[\mathfrak{h}] = \mathfrak{q}^{-1/2};$$

$$\mathfrak{h} := \text{the single free dimensionful scale}$$

$$[\text{D2.9; P1; A §0.3}] :$$

$$\kappa,\ G_N \ \text{ are names for combinations of } \mathfrak{h},\ \neg\text{independent inputs}$$

$$[\text{D2.9; CT2.9}];$$

the inventory carries exactly one fundamental dimension, the information flux $\mathfrak{q}$

(2) the derived exponent census:

quantity	exponent	locus
length, time	$\mathfrak{q}^{-1/2}$	[CT2.1; D2.9 : $c = 1$]
area, $[\Lambda]$, $[G_{\mu\nu}]$, curvature	$\mathfrak{q}^{-1}$, $\mathfrak{q}^{+1}$, $\mathfrak{q}^{+1}$, $\mathfrak{q}^{+1}$	[CT2.1]
volume	$\mathfrak{q}^{-3/2}$	[Def-BM.4(1); CT2.1]
energy, mass, $[\Delta m]$	$\mathfrak{q}^{+1/2}$	[CT2.1; BM.35(4)]
energy density ρ, $[T_{\mu\nu}]$	$\mathfrak{q}^{+2}$	[CT2.1; BM.14(2), (4)]
$[\kappa^2]$, $[G_N]$, $[\kappa]$	$\mathfrak{q}^{-1}$, $\mathfrak{q}^{-1}$, $\mathfrak{q}^{-1/2}$	[CT2.1; CT2.9]
$\mathfrak{h}$	$\mathfrak{q}^{-1/2}$	[D2.9]
$\Delta S_{\mathfrak{L}}$, I_{hid}, S_{raw}, Σ_{cl}, S_{geom}	$\mathfrak{q}^{0}$	[BM.35(3), (4); CT1.29; CT2.1; Def-BM.13(6)]
E_{int}, $W(Q)$, q, D_S, f, N, $\mathfrak{g}$, w, G, ε_{EM}, e, g, g'	$\mathfrak{q}^{0}$	[D3.4; EM.9; Def-BM.13(3); BM.88(2), (3); EM.25(9)]
m^2_{pole}, Δ_g, $\hat{\mu}'^2$	$\mathfrak{q}^{+1}$	[Def-H.37(2); FT.54(4)]
$\hat{\mu}^2$	$\mathfrak{q}^{+2}$	[Def-H.37(1); EW.33]
$\rho_{\text{B}} = \hat{\mu}'^2\mathfrak{h}^2/(\ln 2)^2$	$\mathfrak{q}^{0}$	[BM.69(2); machine: $1 - 1 = 0$]

the balance rows:

$$[\rho\, V] = \mathfrak{q}^{+2}\mathfrak{q}^{-3/2} = \mathfrak{q}^{+1/2} = [\text{energy}];$$

$$[\kappa^2\rho] = \mathfrak{q}^{-1}\mathfrak{q}^{+2} = \mathfrak{q}^{+1} = [R];$$

$$[(\Delta x_{\mathrm{B}})^2\kappa^2\rho] = \mathfrak{q}^{-1}\mathfrak{q}^{+1} = \mathfrak{q}^{0} = [G]$$

[machine];

(3) the seven rows (derived where a referent exists, adjudicated where it does not):

row	exponent	the corpus referent
mass	$\mathfrak{q}^{+1/2}$	m_{hid}, m_{rest}, m_{pole} [FT.44(4); BM.35(4); Def-H.37]
length	$\mathfrak{q}^{-1/2}$	Δx_{B}, ℓ_c, r [Def-BM.4(1); Def-H.2]
time	$\mathfrak{q}^{-1/2}$	x^0, equal to length on $c = 1$ [D1.8; D2.9]
energy	$\mathfrak{q}^{+1/2}$	$E_{\mathrm{rest}} = m_{\mathrm{rest}}c^2$ with the c-exponent 0 [BM.35(4)]
temperature	$\mathfrak{q}^{+1/2}$	on $k_B = 1$ [D2.9; CT2.1] — a dimension row, ¬a thermal object: ¬a P4-class temperature exists [BM.69(6)]
amount of substance	$\mathfrak{q}^{0}$	a count: E_{int}, $W(Q)$, the fibre cardinalities [Def-BM.5(4); Def-BM.13(3); BM.95(1)]
electric current	$\mathfrak{q}^{0}$ and $\mathfrak{q}^{+1/2}$	two reads: the ledger-side weight per tick [BM.23(9); BM.44; D1.8; ¬a rate per external second [P5]]; the chart-side read below

the chart-side current, derived: $\partial_\mu F^{\mu\nu} = e\,J^\nu_Q$ [Def-BM.12(1); EM.11] :

$$\left[\tfrac{1}{2}E_r^2 \cdot V\right] = [\text{energy}] = \mathfrak{q}^{+1/2} \implies [E_r^2] = \mathfrak{q}^{+2} \implies [F^{\mu\nu}] = \mathfrak{q}^{+1};$$

$$[\partial_\mu] = \mathfrak{q}^{+1/2}, \quad [e] = \mathfrak{q}^{0} \implies \boxed{[J^\nu_Q] = \mathfrak{q}^{+3/2}}$$

[machine: the exponent arithmetic balances];

the integrated current $[J \cdot \text{area}] = \mathfrak{q}^{+3/2}\mathfrak{q}^{-1} = \mathfrak{q}^{+1/2} = \text{charge/length} = \text{charge/time}$ on $c = 1$

[machine] :

the row's adjudication:

the ledger-side per-tick read is $\mathfrak{q}^{0}$ and the chart-side read is $\mathfrak{q}^{+1/2}$ — both derived, ¬equated [BM.37(2) pattern];

(4) the two bridges, separated:

the dimensional bridge is $\mathfrak{h}$:

$$m_{\mathrm{hid}} = \frac{E_{\mathrm{int}}\ln 2}{\mathfrak{h}} : \quad \mathfrak{q}^{0} \longmapsto \mathfrak{q}^{+1/2}$$

[FT.44(4); BM.35(4); machine] :

$\hbar$ is declared by P1 with ¬a value issued, $\exists!$ such a scale [P1; A §0.3];

the numerical bridge is ρ_{B} :

$[\rho_{\mathrm{B}}]$

$= \mathfrak{q}^0 \implies$ ρ_{B} converts no dimension — it is the dimensionless exchange rate between the bit currency ($\ln 2$ per directed edge) and the angle currency (Θ_R)

[BM.35(5) Rmk: three currencies, one reference point; BM.69(2); BM.98; CT1.29; FT.54(4)] :

the correction to the framing:

ρ_{B} is ¬the dimensional bridge — that role is $\hbar$'s — and ¬a single number carries both roles [machine: $[\hbar] = \mathfrak{q}^{-1/2} \neq \mathfrak{q}^0 = [\rho_{\mathrm{B}}]$];

(5) the declaration (logged as the §II.4 addendum A-BM.9 before use):

species: the bridge's dimensionless calibration slot

§Cal-BM gains exactly one declared member: $\rho_{\mathrm{B}} = \dfrac{\hat{\mu}'^2 \hbar^2}{(\ln 2)^2} \in \mathfrak{q}^0$, its value is ¬issued

the packaging row:

ρ_{B} is a dimensionless combination of two already-declared data — $\hat{\mu}'^2$ [Cal-F; Def-H.37(2)] and $\hbar$ [P1] :

¬a new free parameter is introduced,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a numerical coupling [EM.25(7), (9)];

what the declaration fixes is the datum's status:

the bridge's single residue is a declared calibration slot, to be filled by a calibration act the corpus prices and ¬performs

[BM.77(9) pattern];

(6) what the slot buys, exactly:

with ρ_{B} supplied, every cross-currency ratio is an exact algebraic number per state:

$$\mathcal{R}_s = \rho_{\mathrm{B}} \frac{D_S}{4E^2},$$

$$\mathcal{R}_{\mathrm{mat}} = \rho_{\mathrm{B}} \frac{D_S\, D^2}{4E^2},$$

$$\mathcal{R} = \frac{\pi \ln 2}{2} \cdot \frac{\rho_{\mathrm{B}} D_S}{G} = \frac{\hat{\mu}'^2 D_S r^2}{4} (1-f)^2$$

[BM.69(2); BM.98(1)–(4); six states, four fields, six minimal polynomials [Def-BM.14(3)]] :

the bridge becomes numerical, state by state, on one number;

and ¬a currency becomes addable:

$$[\Delta_g] = \mathfrak{q} \text{ against } [\Delta m] = \mathfrak{q}^{+1/2} \implies \neg\text{addable}$$

[BM.37(2); BM.69(3); BM.90(5); BM.98(6); machine] :

the declaration moves no summand across the two currencies, and the ratio stays a ratio;

(7) the mass-form bridge disposition:

the relating equation is derived in form [BM.98] and its single residue is a declared slot with no value issued [(A-BM.9)] $\implies$ the mass-form bridge is closed as an equation

the derivation bar stands:

$\neg\exists$ a theorem equating the bit currency and the angle currency [BM.35(5)];

BM.69(4)'s four grounds stand unweakened — the declaration is ¬a derivation;

the numerical bridge remains a calibration act:

¬a value for ρ_{B} is derived, declared or fitted here;

¬an [EXT] datum fixes it [P5; FT.73(5) trap pattern];

(8) the bars:

¬a second dimensionful scale [P1];

α_B : ¬added, ¬derived, barred on BM.61(8)'s four grounds;

the six D_S minimal polynomials are untouched;

species names are confined to the identification layer [Def-BM.3(7); Def-BM.6(7)];

amplitude $\neq$ mass [FT.79(5)];

rung $\neq$ scale [D0.8];

(9) status:

derived: (1)–(4), (6);

declared: (5) — logged as A-BM.9 before use, one slot, no value;

adjudicated: (3)'s two current reads and the temperature row;

¬a P4 addition beyond the declared slot,

$$\neg\text{a new dimensionful scale [P1]},$$

$$\neg\text{a measured value [P5]},$$

Req. D0.8, D1.8, D2.9, D3.4 (Parts 1–2); A §0.3, CT1.29, CT2.1 (Part 1, Part 2); CT2.9, EM.9, EM.11, EM.25(7), EM.25(9), EW.33, Def-BM.12(1) (Part 2, this part); P1, P4, P5 (Parts 1–2); Def-H.2, Def-H.37(1), Def-H.37(2), FT.44(4), FT.54(4), FT.73(5), FT.79(5) (Part 3); Def-BM.4(1), Def-BM.5(4), Def-BM.13(3), Def-BM.13(6), Def-BM.14(3), BM.14(2), BM.14(4), BM.23(9), BM.35(3), BM.35(4), BM.35(5), BM.37(2), BM.44, BM.61(8), BM.69(2), BM.69(3), BM.69(4), BM.69(6), BM.88(2), BM.88(3), BM.90(5), BM.95(1), BM.98, BM.98(6) (this part); Cal-F; the A-BM.9 declaration (logged before use); exact $\mathfrak{q}$-exponent arithmetic, Laurent-monomial balances, symbol censuses [math; machine: `checks/verify_H_six_objective_resolution.py` §B].

Pf.

$$(1):\ \text{D2.9},\ \text{P1},\ \text{A §0.3},\ \text{CT2.9 re-cited verbatim (frozen)};$$

$$\neg\exists\ \text{a second dimensionful scale in the inventory [the derived no-go of the second scale re-cited]}$$

$(2):$ every row re-cited from its own locus and its exponent re-computed [machine: the exponent arithmetic];

$$[\rho V] = 2 - \tfrac{3}{2} = \tfrac{1}{2}; \quad [\kappa^2\rho] = -1 + 2 = +1; \quad [(\Delta x_{\mathrm{B}})^2\kappa^2\rho] = -1 - 1 + 2 = 0\ \text{[machine]}$$

$$[\rho_{\mathrm{B}}] = [\hat{\mu}'^2] + 2[\mathfrak{h}] - 2[\ln 2] = 1 - 1 - 0 = 0\ \text{[BM.69(2); machine]};$$

$$[\hat{\mu}^2] = 2\ \text{[Def-H.37(1); EW.33 packaging]}$$

$$(3):\ [\tfrac{1}{2}E_r^2 V]$$

$$= \tfrac{1}{2}\ \text{carries}\ \mathfrak{q}^0\ \text{[BM.87(2)'s exterior integral]}$$

$$\Longrightarrow\ [E_r^2]$$

$$= \tfrac{1}{2} - \left(-\tfrac{3}{2}\right)$$

$$= 2\ \text{[machine]};$$

$$[\partial_\mu] = -[x^\mu] = +\tfrac{1}{2};$$

$$[eJ] = [\partial_\mu F^{\mu\nu}] = \tfrac{1}{2} + 1 = \tfrac{3}{2} \text{ and } [e] = 0 \implies [J] = \tfrac{3}{2} \text{ [machine]}$$

$$[J \cdot \text{area}] = \tfrac{3}{2} - 1 = \tfrac{1}{2} \text{ [machine]};$$

$$k_B = 1 \implies [\text{temperature}] = [\text{energy}] = \tfrac{1}{2} \text{ [D2.9; CT2.1]};$$

$$\neg\exists \text{ a P4-class temperature [BM.69(6) re-cited]}$$

$$(4): \ m_{\text{hid}} = E_{\text{int}} \ln 2/\mathfrak{h}: \ [\cdot] = 0 - (-\tfrac{1}{2}) = \tfrac{1}{2} \text{ [machine]};$$

$$[\mathfrak{h}] = -\tfrac{1}{2} \neq 0 = [\rho_{\text{B}}] \implies \neg\text{one number carries both roles [machine]}$$

(5)–(6): the declaration's species, costs and bars recorded in the registry grammar [A-BM.9; machine: the packaging census $\{\hat{\mu}'^2 \in \text{Cal-F}, \ \mathfrak{h} \in \text{P1}\}$];

the six states' exactness per supplied value re-cited [BM.69(2); BM.98]

(7)–(9): BM.69(4)'s four grounds and BM.35(5)'s no-equation row re-cited;

the standing-rule scans: external tokens, decimal numerals, unit tokens, the barred object α_B [machine]. ■

Phys. The corpus has one dimension and this tier counts with it. Length and time are the same power, mass and energy the same power, density and stress-energy two powers up, curvature one, and the single free scale sits at minus one half — so every balance in the census is an integer sum, and the seven rows of the question come out derived where the inventory has a referent and adjudicated where it does not. Temperature is a dimension row and nothing more, because no thermal parameter exists; amount of substance is a count; and electric current has two reads that are not equated, the ledger's weight per tick at exponent zero and the chart's density at three halves, the latter forced by the field equation and the field-energy row. The bridge the question asks about turns out to be two bridges. The dimensional one is the free scale itself: it takes a count of bits to a mass, and it was declared at the outset with no value ever issued. The numerical one is dimensionless, so it converts no dimension at all — it is the exchange rate between the ledger's bit currency and the holonomy's angle currency, and it is the single number the relating equation needs. That number is now declared: one slot in the era's calibration registry, logged before use, packaging two data that were already declared elsewhere, adding no parameter and no scale, and carrying no value. With it supplied, every cross-currency ratio in the corpus becomes an exact algebraic number state by state; without it, nothing is lost, because the four grounds against deriving it stand and no summand crosses between the two currencies. The equation is closed; the number is a calibration act the corpus prices and does not perform.

Theorem 4.87 (BM.103 — The three-constituent assembly: the eigenvalue equations that are formed, the tensor-product referent census, and the composition residue). Thm.

$$(1) \quad \text{the six-state inventory (re-cited):}$$

$$\mathcal{S} = \{(\mathcal{F}_g, \beta) : \ g \in \{1,2,3\},\ \beta \in \{-,+\}\},$$

$$|\mathcal{S}| = 6$$

$$[\text{Def-BM.14}(2)] :$$

$$D_S \ \in \ \big\{2,\ 4,\ 2-\sqrt{2},\ 1,\ 2-2\cos\tfrac{\pi}{10},\ 2-\sqrt{3}\big\},$$

$$m^2_{\text{pole}} = \hat{\mu}'^2 D_S,$$

four fields, six minimal polynomials

$$[\text{Def-BM.14}(3);\ \text{Census FT.63}(6);\ \text{BM.86}(2)] :$$

$$x-2, \quad x-4, \quad x^2-4x+2, \quad x-1, \quad x^4-8x^3+19x^2-12x+1, \quad x^2-4x+1 \ \ (\text{for } 2-\sqrt{3})$$

[machine: the four quadratic-ring roots re-verified exactly];

the generation ladder $12:24:60 = 1:2:5$

$$[\text{Def-BM.14}(1);\ \text{BM.60}(1)];$$

$$\Delta_g = \big(2,\ \sqrt{2}-1,\ 2\cos\tfrac{\pi}{10}-\sqrt{3}\big)\hat{\mu}'^2 > 0,$$

$$\Delta_1 > \Delta_2 > \Delta_3$$

$$[\text{Def-BM.14}(5);\ \text{BM.69}(3)];$$

$$(2) \quad \text{the eigenvalue equation that is formed (the three-line channel, ¬the three constituents) :}$$

$$z := r^2,$$

$$\boxed{4z^3 - 24z^2 + \big(21 - 12\gamma_{\text{ch}}\big)z - \big(\gamma_{\text{ch}} - 1\big)^2 = 0}$$

Vieta:

$$\boxed{\sum_i z_i = 6 \ \text{(phase-free and channel-free)}, \qquad \sum_{i<j} z_i z_j = \frac{21 - 12\gamma_{\text{ch}}}{4}, \qquad \prod_i z_i = \frac{(\gamma_{\text{ch}} - 1)^2}{4}}$$

[BM.80(3); machine: the three symmetric reads];

the depressed cubic on the spectral side:

$$2(r-1)^3 - 3(r-1) - \gamma_{\text{ch}} = 0 \iff 2r^3 - 6r^2 + 3r + (1 - \gamma_{\text{ch}}) = 0$$

[BM.80(2); FT.111(3); FT.143(2); machine: the expansion];

$$\gamma_{\text{ch}} = \sqrt{2}\cos(3\delta_{\text{ch}}),$$

$$\delta_{\text{ch}} \in \Big\{\frac{2}{9}, \ \frac{2}{27}, \ \frac{1}{8}\Big\},$$

$$27\delta_q = 9\delta_\ell = 16\delta_\nu = 2$$

[BM.80(1); FT.143(5); machine];

the elimination row:

$$r(2z+3) = 6z + \gamma_{\text{ch}} - 1 \ \text{ at } \ z = r^2,$$

$$z(2z+3)^2 - (6z + \gamma_{\text{ch}} - 1)^2 \equiv 0 \ \text{ in the quotient ring}$$

[BM.80(3); machine];

the three roots are the channel's three lines (the generations), ¬the three constituents of a sealed class

[Def-BM.6(4); FT.143(6)];

(3) the characteristic polynomial of the three-constituent assembly in the transport register:

$$W(Q) = \left[z^{Q}\right]\left(2 + z + z^{-1}\right)^{3}$$

[Def-BM.13(3)] :

$$2 + z + z^{-1} = z^{-1}(z+1)^{2} \implies \left(2 + z + z^{-1}\right)^{3} = \frac{(z+1)^{6}}{z^{3}}$$ [machine: one Laurent identity] :

$$\boxed{W(Q) \;=\; \binom{6}{Q+3} \text{ exactly, on all seven classes}}$$

$$W = (1, 6, 15, 20, 15, 6, 1),$$

$$\sum_{Q} W(Q) = 2^{6} = 64 = 4^{3}$$

[BM.28(2); machine: the closed form against the exhaustive 4^3 census];

the assembly's transport spectrum:

one root z

$= -1$ of multiplicity six — the six directed edges — and the seven class weights are its binomial coefficients

$$\boxed{\text{the eigenvalue structure the chart resolves is charge-class data, } \neg\text{flavour data}}$$

[Def-BM.13(6); BM.95(1); machine];

(4) the tensor-product referent census (what an ensemble matrix requires against what the corpus supplies):

an M_{ensemble} requires	the corpus supplies
a three-constituent state space	¬one: $\mathcal{H}_c$, $\mathcal{H}_c^{(q)}$ are counting spaces with no inner product, no dynamics, no state, no Hamiltonian [Def-H.41 Species; Def-H.42]; ¬a two-chain state space exists, let alone a three-chain one [FT.80(5)]
an exchange operator	¬∃ one in the inventory [FT.80(5); FT.125(5)]
a flavour index attaching states to constituents	¬derived: (R3), deficit 1′ [BM.28(7); BM.75(6); Def-BM.8(5)]
a triple-level resolution	¬available: the end signs carry no position datum [D3.4; BM.46(2); Def-BM.13(9)]
a 27-dimensional flavour arena	rejected as an arena at rung one [FT.142(3) : the 27 = seal × transfer × colour candidate is a declared construction, cited and imported nowhere]

$\det(\lambda I - M_{\text{ensemble}}) = 0$ is ¬formed — the question "which three states compose a sealed class" has no eigenvalue referent in this corpus

(5) what is derived about the composition:

exactly three constituents seal:

$B := \varepsilon_{ijk} q^i q^j q^k$ [D3.9]; ε_{ijk} seals 3 fundamentals [CT3.15; CT3.17];

3 admits no $\{2, 3\}$-partition with an external remainder [FT.34(5); Census FT.35(5)] :

their core charges sum to the class read: $Q = \sum_i q_{\text{core}}(c_i) \in \{-3, \ldots, +3\}$,

$Q_{\text{EM}} = 0$ in every valid universe

[Def-BM.8(1); EM.11; D2.11];

all seven classes are formed with the binomial weights of (3);

the neutral class outweighs either charged class by exactly $\frac{4}{3}$ on route (ii)

[BM.28(4)(ii); BM.30(2)];

the fractional constituent alphabet is structurally impossible

[BM.28(7); Def-BM.8(5)];

colour is ¬observable by construction

[CT3.22];

the "three" is derived from the seal grammar and the "which three" is not

(6) the stability question, adjudicated: $\neg\exists$ a derived stability condition on eigenvalues:

(i) the materialization filter is an undeclared spectral hypothesis (M1)/(M2)

[Def-H.20; A §0.2a; Part 3 §0.2(iii)];

(ii) the index-zero selection route is refuted for bound chains: on Synge-class slices every closed geodesic has ind_B

≥ 1,

so absolute index-0 selection is inadmissible against $(\mathrm{M1})$

[FT.66(4)];

(iii) a mass-ordering criterion would be a count $\to$ strength coupling

[BM.36(3); BM.61(8)(O1); BM.90(3)] :

the formed stability rows are kinematic and ledger-side:

the two-body impossibility [BM.41(2)];

the two-bit floor [BM.70(2)];

the materialization restriction E

≥ 2 [BM.94(5)];

the valid-universe conditions [D2.11];

the no-flicker theorem $\Gamma(\varepsilon_q{=}0)$

$= 0$ [BM.23(8)];

the two-sided bar [BM.99(3)];

the forward-only arrow [A6; FT.144(2)];

(7) the octet row:

¬formed.

the corpus's derived 3 is the colour seal's fundamental [CT3.15; CT3.17; D3.9];

its derived 8 is a mask-capacity ratio, barred as a coupling [BM.36$(2), (3)$; BM.61(8)(O1)];

its derived 10 is the colour grammar's grouping count $g(5)$

$= 10$ [CT3.17a(2); Def-BM.9; BM.31];

the flavour-side symmetry is C_3,

its orbits the three channel lines [FT.111(2); FT.158$(1), (4)$; FT.142(5)];

$\neg\exists$ a flavour $\mathsf{SU}(3)$ in the inventory;

the baryon octet and the meson octet are ¬derived and ¬derivable here — what is derived is the ε-seal's three, the binomial seven, and the C_3 triplet

(8) the naming bar:

the two heaviest-floor triples the question names are external species vocabulary;

the identification layer supplies up-type/down-type branch names [A-BM.6; Def-BM.14(8)] and generation indices,

and no name attaches to a composition

[Def-BM.6$(5), (7)$; Def-BM.3(7)] :

¬either triple is formed, and ¬either is excluded — the composition is unread (R3)

(9) status:

derived: (1)–(3), (5), (6)'s formed rows, (7)'s dimension rows;

refuted as formed: the ensemble matrix (4),

an eigenvalue stability condition (6),

the octet read (7);

standing: (R3), deficit $1'$;

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a species identification [P5; Def-BM.3(7)];

Req. D2.11, D3.4, D3.9 (Parts 1–2); A §0.2a, CT3.15, CT3.17, CT3.17a(2), CT3.22, EM.11 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.20, Def-H.41, Def-H.42, FT.34(5), FT.66(4), FT.80(5), FT.111(2), FT.111(3), FT.125(5), FT.142(3), FT.142(5), FT.143(2), FT.143(5), FT.143(6), FT.158(1), FT.158(4) (Part 3); Census FT.35(5), FT.63(6) (row-wise); Def-BM.6(4), Def-BM.6(5), Def-BM.6(7), Def-BM.8(1), Def-BM.8(5), Def-BM.9, Def-BM.13(3), Def-BM.13(6), Def-BM.13(9), Def-BM.14(1), Def-BM.14(2), Def-BM.14(3), Def-BM.14(5), Def-BM.14(8), Def-BM.3(7), BM.23(8), BM.28(2), BM.28(4), BM.28(7), BM.30(2), BM.31, BM.36(2), BM.36(3), BM.41(2), BM.46(2), BM.60(1), BM.61(8), BM.69(3), BM.70(2), BM.75(6), BM.80(1), BM.80(2), BM.80(3), BM.86(2), BM.90(3), BM.94(5), BM.95(1), BM.99(3) (this part); exact Laurent and polynomial identities over $\mathbb{Q}[z]$, binomial closed forms against exhaustive enumeration, Vieta reads, exact radical minimal polynomials in $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$, elimination in a quotient ring [math; machine: `checks/verify_H_six_objective_resolution.py` §C].

Pf.

(1) : Def-BM.14(1)–(5) and Census FT.63(6) re-cited (frozen);

$(2-\sqrt{2})^2 - 4(2-\sqrt{2}) + 2 = 0$ in $\mathbb{Q}(\sqrt{2})$;

$(2-\sqrt{3})^2 - 4(2-\sqrt{3}) + 1 = 0$ in $\mathbb{Q}(\sqrt{3})$ [machine: exact]

$2 - 2\cos\frac{\pi}{10} = y$ satisfies $y^4 - 8y^3 + 19y^2 - 12y + 1 = 0$:

$$(2-y)^2$$

$$= \tfrac{5+\sqrt{5}}{2} \implies \sqrt{5}$$

$$= 2y^2 - 8y + 3$$

$$\implies 5$$

$$= (2y^2 - 8y + 3)^2$$

$$\implies 4\big(y^4 - 8y^3 + 19y^2 - 12y + 1\big)$$

$$= 0 \text{ [machine: the expansion in } \mathbb{Q}[y]]$$

$(2):$ BM.80(2), (3)'s rows re-cited;

$$\text{Vieta: } \sum_i z_i$$

$$= \tfrac{24}{4}$$

$$= 6,$$

$$\sum_{i<j} z_i z_j$$

$$= \tfrac{21-12\gamma}{4},$$

$$\prod_i z_i$$

$$= -\tfrac{-(\gamma-1)^2}{4}$$

$$= \tfrac{(\gamma-1)^2}{4} \text{ [machine: the coefficient reads]}$$

$$2(r-1)^3 - 3(r-1) = 2r^3 - 6r^2 + 3r - 1 \implies 2r^3 - 6r^2 + 3r + (1-\gamma) = 0 \text{ [machine]};$$

$$z(2z+3)^2 - (6z+\gamma-1)^2$$

$$\equiv 0 \text{ modulo } 2r^3 - 6r^2 + 3r + 1 - \gamma \text{ at } r(2z+3)$$

$$= 6z + \gamma - 1 \text{ [machine: the quotient-ring reduction]}$$

$$(3): \ 2 + z + z^{-1}$$

$$= z^{-1}(z^2 + 2z + 1)$$

$$= z^{-1}(z+1)^2$$

$$\implies (2 + z + z^{-1})^3$$

$$= z^{-3}(z+1)^6 \text{ [machine: the Laurent identity]};$$

$$[z^Q]z^{-3}(z+1)^6 = [z^{Q+3}](z+1)^6 = \binom{6}{Q+3} \text{ [machine: all seven classes]}$$

$$\binom{6}{Q+3}$$

$= (1, 6, 15, 20, 15, 6, 1)$ against the exhaustive 4^3 end-sign census [BM.28(2) re-derived; machine];

$$\sum_{k=0}^{6} \binom{6}{k} = 2^6 = 64 \text{ [machine]}$$

(4): the five referent rows re-cited from their own loci [Def-H.41 Species; Def-H.42; FT.80(5); FT.125(5); BM.28(7); Def-BM.13(9); FT.142(3)];

the inventory scans [machine]

(5): D3.9, CT3.15, CT3.17, FT.34(5), Census FT.35(5) re-cited;

$$W(0)/W(+1) = 20/15 = 4/3 \text{ [machine]};$$

(6): (i)–(iii) re-cited at their own loci [Def-H.20; FT.66(4); BM.36(3); BM.61(8)(O1); BM.90(3)]

(7): $g(5) = 10$ from CT3.17a(2)'s summation [Def-BM.9 re-cited; machine];

the capacity ratios $\left(\frac{1}{3}, 8, \frac{1}{25}\right)$ from the flag $(6, 16, 17, 25)$ [BM.36(2); machine]

(8)–(9) : the naming-scope rows re-cited [Def-BM.3(7); Def-BM.6(5), (7); A-BM.6];

the standing-rule scans [machine]. ■

Phys. Ask the chart for the spectrum of a three-quark object and the honest answer is a table of what exists and what does not. What exists is two eigenvalue equations, and neither is the one the question wants. The first is the family cubic: four z cubed minus twenty-four z squared plus a channel constant, whose three roots are the three lines of a channel, with a phase-free sum of six and an exact product — a spectrum of generations, not of constituents. The second is the transport assembly's own characteristic polynomial, which collapses to a binomial: two plus z plus one over z, cubed, is z plus one to the sixth over z cubed, so the seven class weights are the binomial coefficients of six, and the assembly's spectrum is a single root of multiplicity six, one per directed edge. That is charge-class data, and it is the deepest eigenvalue statement the chart supports. What does not exist is the object the question names: a three-constituent state space, an exchange operator, a flavour index, a triple-level resolution — the counting spaces the corpus declares have no state on them, and the twenty-seven-dimensional candidate was rejected as an arena at rung one. So the determinant is not formed, and the two triples the question asks about are neither formed nor excluded; the composition is unread, and that is a named residue rather than an oversight. What is derived instead is sharper: the seal grammar forces exactly three, the charge balance forces their sum, the binomial forces the seven classes and the four-thirds lead of the neutral one, and the stability question lands where the corpus has always landed — on an undeclared spectral filter and a refuted index-zero route. The octet is not derivable either: the three here is colour's, the eight is a capacity ratio barred as a coupling, the ten is a grammar count, and the flavour symmetry that does exist is cyclic of order three.

Theorem 4.88 (BM.104 — The curvature composition: the back-reaction fixed point, its exact roots and pole locus, and the non-selection theorem). Thm.

(1) the loop, formed exactly:

$$m_{\text{corr}} = m_{\text{hid}} + E_{\text{EM}}^{\text{self}} = m_{\text{hid}}\big(1 + a\,\mathsf{g}\big),$$

$$a := \frac{1}{2}\, N\, w\, \varepsilon_{\text{EM}}\, Q^2$$

[BM.89(5); Def-BM.13(5); machine] :

$$\rho_{\text{corr}} = \frac{m_{\text{corr}}}{V_{\text{eff}}},$$

$$V_{\text{eff}} = (\Delta x_{\text{B}})^3,$$

$$\Delta x_{\mathrm{B}} \cdot m_{\mathrm{hid}} = 1$$

[Def-BM.4(1); BM.14(2); the width is the informational one] :

$$\boxed{G_{\mathrm{corr}} \;:=\; (\Delta x_{\mathrm{B}})^2 \kappa^2 \rho_{\mathrm{corr}} \;=\; G\,\big(1 + a\,\mathsf{g}\big), \qquad G = 2\pi E_{\mathrm{int}}^2 \ln 2}$$

[BM.88(3); BM.16(1); machine: one monomial];

the consistency condition is the fixed point: $\boxed{\mathsf{g} \;=\; \mathsf{g}\big(G\,(1 + a\,\mathsf{g})\big)}$

(2) the three exact solution rows:

(i) $\mathsf{g} \equiv 1$: $1 = 1$ identically — a fixed point at every $a \geq 0$ and every G [machine];

(ii) $\mathsf{g} = 1 + G$:

$$\mathsf{g} = 1 + G(1 + a\mathsf{g}) \iff \mathsf{g}\,(1 - aG) = 1 + G \iff \boxed{\mathsf{g}_{\mathrm{fp}} = \frac{1+G}{1-aG}},$$

$$\text{admissible} \iff aG < 1,$$

$$\mathsf{g}_{\mathrm{fp}} > 1 + G$$

[machine: substitution verifies the fixed point identically over $\mathbb{Q}[G, a]$]

the back-reaction inflates the composition and diverges at the pole;

(iii) $\mathsf{g} = (1 + G)^{-1}$: $\mathsf{g}\big(1 + G + aG\,\mathsf{g}\big) = 1 \iff \boxed{aG\,\mathsf{g}^2 + (1+G)\,\mathsf{g} - 1 = 0}$

the product of the two roots is $-\frac{1}{aG} < 0 \implies$ exactly one positive root

$$\mathsf{g}_{\mathrm{fp}} = \frac{\sqrt{(1+G)^2 + 4aG} - (1+G)}{2aG} \;\longrightarrow\; \frac{1}{1+G} \;\text{ as }\; a \to 0$$

[machine: the exact root row and the limit];

the fixed-point equation has a unique admissible solution for each of the three priced compositions — consistency is satisfied three times over and selects nothing

(3) the pole locus:

$$aG = 1 \iff \frac{1}{2} N w \varepsilon_{\text{EM}} Q^2 \cdot 2\pi E_{\text{int}}^2 \ln 2 = 1 \iff \boxed{\pi\, N\, w\, \varepsilon_{\text{EM}}\, Q^2\, E_{\text{int}}^2 \ln 2 \;=\; 1}$$

[machine: one monomial condition];

$$Q = 0 \implies aG = 0 \neq 1$$

[machine] $\implies$ the neutral class never reaches the pole

[BM.17(2) : the neutron row is $Q = 0$; BM.89(5) : $E_{\text{EM}}^{\text{self}}(0) = 0$];

$$Q = \pm 1,\ E_{\text{int}} = 6 :$$

$$\varepsilon_{\text{EM}}^{\text{pole}} = \frac{1}{36\, \pi\, N\, w \ln 2}$$

[machine: exact];

$E_{\text{int}} \in \{2, 6, 12, 24, 60\}$ on the computed cells

[BM.13(2); Census FT.35(2), FT.42(5)];

(4) the loop is excluded by the lock's own freedom census:

G's derived alphabet is $\{E_{\text{int}},\ \ln 2,\ \pi\}$ — $\mathfrak{h}$-free, Λ-free, P4-free

[BM.88(3); BM.16(1); machine: the symbol census] :

a's alphabet is $\{N,\ w,\ \varepsilon_{\text{EM}},\ Q\}$,

$\varepsilon_{\text{EM}} = e^2/(4\pi)$ is a dimensionless P4-class symbol

[Def-BM.12 local notation; EM.25(9); P4] :

$$\boxed{\{E_{\text{int}}, \ln 2, \pi\} \cap \{N,\ w,\ \varepsilon_{\text{EM}},\ Q\} = \varnothing}$$

[machine] $\Longrightarrow$ admitting the back-reaction would make the lock's argument P4-dependent, contradicting BM.88(3)'s derived row

the lock's argument is the informational mass, the EM self-energy does not re-enter G, and ¬a self-consistency condition on g is formed

(5) the non-selection theorem:

$$\boxed{\neg\exists \text{ a derived g}(G)}$$

on four grounds, each re-cited at its own locus:

(i) ¬∃ a count → strength theorem

[BM.36(3); BM.61(8)(O1); BM.69(4)(iv)];

(ii) the four grounds that bar a defect-conditioned weighting bar a curvature-conditioned one at the same species

[BM.90(3)(i)–(iv) : the degree-16 field against integer and binary reads [FT.73(2), (3)]; the two-torsion image $\{0, \pi\}$ [BM.67(6)]; the calibration weight -2 against 0 [FT.55(1); BM.67(8)]; the strength bar];

(iii) the consistency condition of (2) is satisfied by all three candidates — it is necessary and ¬sufficient

[machine];

(iv) a selection would be a P4-class datum — §Cal-BM would gain a second member — refused

[P4; P1; BM.88(4) : priced three ways, adopted in none];

(6) the composition is ¬inert inside the corpus:

BM.101(5)'s crossing count differs by composition — (i) at most one,

(ii) exactly one always,

(iii) two, one or none :

the priced composition is observable in the energy-scale graph's topology, and is still ¬selected

the discrimination is derived; the selection is not;

every row of the corpus holds uniformly over the priced set:

BM.91$(2), (4)$–(8), BM.92$(1), (2)$, BM.98(1)–(4), BM.99(1)–(6), BM.101$(2), (3), (7)$

[machine: the uniformity censuses];

(7) the running coupling's status: $e(r, E, N) = e_0\sqrt{N\, f\big(r/\Delta x_{\mathrm{B}}(E)\big)\, \mathsf{g}\big(G(E)\big)}$ [BM.88(4)] :

f derived exactly [BM.88(2)(ii)];

G derived exactly, one monomial [BM.88(3)];

N solution data [CT2.7; BM.34(2); BM.88(8)];

g priced three ways, adopted in none (5);

e_0 open [EM.25(7)] :

the coupling is complete in form and carries exactly two undeclared members $(e_0,$ g$)$, ¬a third

(8) status:

derived: (1)–(4), (6)'s discrimination rows;

refuted as derivable: g(G) (5);

rescoped, ¬retracted: BM.88(4), (8), BM.89(5), BM.91, BM.98 stand at their own loci;

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a numerical coupling [EM.25(7), (9)];

α_B : ¬added, ¬derived, barred [BM.61(8)];

Req. D2.9 (Parts 1–2); CT2.1, CT2.7, EM.25(7), EM.25(9) (Part 2); P1, P4, P5 (Parts 1–2); Def-H.37(2), FT.54(4), FT.55(1), FT.73(2), FT.73(3) (Part 3); Def-BM.4(1), Def-BM.12, BM.13(2), BM.14(2), BM.16(1), BM.17(2), BM.34(2), BM.36(3), BM.61(8), BM.67(6), BM.67(8), BM.69(4), BM.88(2), BM.88(3), BM.88(4), BM.88(5), BM.88(8), BM.89(5), BM.90(3), BM.91, BM.92(1), BM.92(2), BM.98, BM.101(5), BM.101(7), Def-BM.13(5) (this part); Census FT.35(2), FT.42(5) (row-wise); exact polynomial arithmetic over $\mathbb{Q}[G, a]$, quadratic root rows with sign censuses, exact limit rows, symbol censuses and set intersections, monomial conditions [math; machine: `checks/verify_H_six_objective_resolution.py` §D].

Pf.

$$(1): \; E_{\text{EM}}^{\text{self}} = \tfrac{1}{2} N w \, \mathsf{g} \, \varepsilon_{\text{EM}} Q^2 m_{\text{hid}} = a \, \mathsf{g} \, m_{\text{hid}} \text{ [BM.89}(5)\text{; machine]};$$

$$\rho_{\text{corr}} = m_{\text{corr}}/V_{\text{eff}} = \rho\,(1 + a\mathsf{g}) \text{ at fixed } V_{\text{eff}} \text{ [Def-BM.4}(1)\text{; machine]}$$

$$G_{\text{corr}} = (\Delta x_{\text{B}})^2 \kappa^2 \rho_{\text{corr}} = G(1 + a\mathsf{g}) \text{ [machine: one monomial]};$$

$$\text{the consistency condition is } \mathsf{g} = \mathsf{g}(G_{\text{corr}}) \text{ [the fixed point, stated]}$$

$$(2)(\text{i}): \; 1 = 1 \text{ [machine]};$$

$$(\text{ii}): \; \mathsf{g} = 1 + G + aG\mathsf{g} \iff \mathsf{g}(1 - aG) = 1 + G \text{ [machine: cleared over } \mathbb{Q}[G, a]];$$

$$\mathsf{g}_{\text{fp}} > 1 + G \iff \tfrac{1+G}{1-aG} > 1 + G \iff \tfrac{1}{1-aG} > 1$$

$$\iff 0 < aG < 1 \text{ [machine: the sign census on positive witnesses]}$$

$$\text{(iii)}: \ \mathrm{g} = \tfrac{1}{1+G+aG\mathrm{g}} \iff \mathrm{g}(1+G) + aG\mathrm{g}^2 = 1 \text{ [machine]};$$

$$\text{the root product } -1/(aG) < 0 \implies \text{one positive root [machine]};$$

$$a \to 0: \ \tfrac{\sqrt{(1+G)^2+4aG}-(1+G)}{2aG}$$

$$= \tfrac{4aG}{2aG\left(\sqrt{(1+G)^2+4aG}+(1+G)\right)} \to \tfrac{1}{1+G} \text{ [machine: the conjugate-cleared limit]}$$

$$(3): \ aG = \tfrac{1}{2} N w \, \varepsilon_{\mathrm{EM}} Q^2 \cdot 2\pi E^2 \ln 2 = \pi N w \, \varepsilon_{\mathrm{EM}} Q^2 E^2 \ln 2 \text{ [machine]};$$

$$Q = 0 \implies aG = 0 \neq 1 \text{ [machine]};$$

$$Q^2 = 1, \ E = 6: \ \varepsilon^{\mathrm{pole}} = 1/(36\pi N w \ln 2) \text{ [machine]}$$

(4): the two symbol sets enumerated from their defining clauses and intersected [BM.88(3); Def-BM.12; machine];

BM.88(3)'s P4-freedom row re-cited

(5): (i)–(iv) re-cited at their own loci [BM.36(3); BM.61(8); BM.90(3); FT.73(2), (3); BM.67(6), (8); FT.55(1); P4]

(6): BM.101(5)(i)–(iii)'s three crossing counts re-cited;

the uniformity censuses over $\mathrm{g} \in \{1, 1+G, (1+G)^{-1}\}$ on exact rational lattices [machine]

(7)–(8): BM.88(2)–(4), (8)'s rows re-cited;

the standing-rule scans, including the barred object α_B [machine]. ■

Phys. Turn the correction back on itself and the corpus answers with an equation it can solve and a verdict it cannot avoid. If the electromagnetic self-energy feeds the density that feeds the curvature lock, the lock's argument grows by one plus a times the composition, and self-consistency becomes a fixed point. Solved on the three priced candidates, the fixed point is trivial for the constant one, a rational expression with an exact pole for the growing one, and the positive root of a quadratic for the shrinking one — three admissible solutions, so consistency is necessary and useless as a selector. The pole locus is a single monomial, and the neutral class never reaches it,

because a neutral composite carries no field energy at all. Then the deeper row: the loop is not merely unselected, it is excluded, because the lock's alphabet is three symbols long and disjoint from the correction's, and letting the correction in would make the lock depend on a declared parameter that the lock's own theorem says it does not carry. So the self-energy rides the mass and not the geometry, the composition stays priced three ways, and the four old grounds against turning a count into a strength bar the fourth route. What the tier gains is a discrimination: the three compositions give the energy-scale graph three different topologies — a half-line, a half-line that always crosses, and a window — so the choice is not cosmetic inside this corpus even though nothing here can make it. The coupling is complete in form with exactly two undeclared members, and no third.

Theorem 4.89 (BM.105 — The general crystallization weight: the measure-free bounds, the threshold family, and the refuted defect conditioning). Thm.

$$(1) \quad \text{the measure-free form:}$$

$$E_{\text{int}}(R) = |R| = 2\,|S(R)| \;\; \text{on the three pairs of } \Sigma_{12}$$

$$[\text{Def-BM.5}(4);\ \text{Census FT.35}(2);\ \text{BM.94}(4)] :$$

$$\text{linearity on } |R| \;\; \text{gives, for every ensemble measure :}$$

$$\boxed{w = \frac{1}{E^{\text{cryst}}} \sum_{i=1}^{3} P(\text{the pair } i \text{ is real}), \qquad w_{\text{mat}} = \frac{1}{3}\sum_{i=1}^{3} a_i, \qquad a_i := P(\text{the pair } i \text{ is real} \mid \text{materialized})}$$

$$[\text{BM.89}(4);\ \text{BM.94}(7);\ \text{machine: the linearity row}]; \qquad \text{no distribution hypothesis enters } (1);$$

$$(2) \quad \text{the measure-free bounds:}$$

$$|S|$$

$$\geq 1 \;\; \text{on the materialized support } [\text{BM.94}(5);\ \text{Def-H.4};\ \text{FT.144}(4)] \;\; \text{and} \;\; |S|$$

$$\leq 3$$

$$\Longrightarrow \; 1 \leq \langle |S| \rangle_{\text{mat}} \qquad \leq 3 :$$

$$\boxed{\frac{1}{3} \;\leq\; w_{\text{mat}} \;\leq\; 1 \qquad \text{for every admissible ensemble measure}} \qquad [\text{machine}];$$

$$\text{the two equality loci are exact:}$$

$$w_{\text{mat}} = \frac{1}{3} \iff \text{the measure is supported on the single-pair configurations;}$$

$$w_{\text{mat}} = 1 \iff \text{it is supported on the full configuration}$$

[machine];

$$\boxed{\langle E \rangle_{\text{mat}} = 6\, w_{\text{mat}} \in [2,\, 6]}$$

the bound is a consequence of the materialization restriction alone, ¬of the specialization;

(3) the threshold family (hypothesis (F), marked):

hypothesis: the A4 threshold takes a per-pair value $\varepsilon_i \in [0, \frac{1}{4})$ on the three pairs of Σ_{12}

[A4; D0.4; CT0.8's derived range applied memberwise; carried symbolically, ¬issued] :

$$p_i := 1 - 4\varepsilon_i, \qquad u_i := 1 - p_i = 4\varepsilon_i, \qquad R := \prod_{i=1}^{3} u_i, \qquad S := \sum_{i=1}^{3} p_i \qquad \text{[machine]} :$$

$$\boxed{w_{\text{mat}} \;=\; \frac{S}{3\,(1-R)}, \qquad \langle E \rangle_{\text{mat}} = \frac{2S}{1-R}, \qquad \left\langle E^2 \right\rangle_{\text{mat}} = \frac{4\big(S + 2\sum_{i<j} p_i p_j\big)}{1-R}}$$

the seven materialized configuration weights:

$$P_{\text{mat}}(S) = \frac{\prod_{i \in S} p_i \prod_{i \notin S} u_i}{1-R},$$

$$\varnothing \neq S \subseteq \{1,2,3\},$$

$$\sum_{\varnothing \neq S} P_{\text{mat}}(S) = 1$$

[machine: exact rational functions, normalized];

the uniform specialization recovers BM.94 identically:

$$p_i \equiv p \implies S = 3p, \quad R = (1-p)^3, \quad 1 - R = pD :$$

$$\boxed{w_{\text{mat}} = \frac{3p}{3pD} = \frac{1}{D}, \qquad D = 3 - 3p + p^2}$$

[BM.94$(6), (7)$; machine: the identity at six rational thresholds];

(4) the family's exact bounds:

$$3(1-R) - S = \sum_{i=1}^{3} u_i - 3\prod_{i=1}^{3} u_i$$

[machine: one polynomial identity] :

$$\sum_i u_i \;\geq\; 3\big(\prod_i u_i\big)^{1/3} \;\geq\; 3\prod_i u_i$$

$$\big[\prod_i u_i \leq 1;\ \text{AM–GM}\big] \implies \boxed{3(1-R) \;\geq\; S}$$

with equality iff every $u_i = 0,$ i.e. $p_i = 1$:

$$\boxed{w_{\text{mat}} \leq 1 \text{ with equality iff every pair is crystallized;} \qquad w_{\text{mat}} > \frac{1}{3} \text{ strictly on the materialized domain}}$$

$[(2);$ machine];

w_{mat} is strictly increasing in each p_i [machine: an exact rational grid on $(0,1]^3$];

(5) hypothesis (W6)—six state-dependent thresholds $\varepsilon_i(D_S)$—on its face and refuted:

(i) the alphabets are disjoint:

the flicker measure's symbol set is $\{\varepsilon_q\}$ and the defect values' is $\{\delta_v,\ \hat{\mu}'^2\}$

[BM.90$(1), (2)$; machine: the symbol census of A4, D0.4, CT0.8, BM.19(2) carries ε_q alone, with no state argument];

(ii) the four grounds bar it:

the degree-16 field against integer and binary reads [FT.73(2), (3)];

the two-torsion image reaching 2 of 6 states [BM.67(6)];

the calibration weight -2 against 0 [FT.55(1); BM.67(8)];

the strength bar [BM.36(3); BM.61(8)(O1)]

[BM.90(3) re-cited];

(iii) the datum count:

six thresholds would be six P4-class data — §Cal-BM would gain six members — refused

[P4; P1];

(iv) the scope row:

A4′s threshold enters the corpus only via CT0.1 and CT0.2,

with a single ε_q and no state argument

[CT0.8 scope clause] :

$\neg\exists$ a derived map $D_S \longmapsto \varepsilon$;	the weight stays branch-blind, and the six defect values act in the pole currency alone

[BM.90(3), (4); BM.98(6)];

(6) the residue, narrowed:

(R4) closes at two levels: the measure-free bounds (1), (2) and the threshold family (3), (4);

what remains is (R4′) :

(R4′) the ensemble's configuration distribution beyond per-pair independence — the pair–pair correlations of the measure

and the transport register's ensemble average

[BM.89(8); BM.91(2); BM.94(8) : the computed cells average the vertex-deficit register $\frac{3}{2}n_{\mathcal{V}}$, ¬the edge-count register $|R|$] :

(R4) is rescoped to (R4′), ¬deleted;

(7) status:

derived: (1)–(4), (6);

refuted: hypothesis (W6) (5);

marked, ¬adopted: hypothesis (F)'s family;

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

§Cal-BM introduces no new free parameter at this tag;

Req. A4 (Part 1); D0.4, D2.9 (Parts 1–2); A §0.2a, CT0.8, CT1.29 (Part 1); P1, P4, P5 (Parts 1–2); L1, L2 (Part 1); Def-H.4, Def-H.32(1), FT.55(1), FT.73(2), FT.73(3), FT.107(3), FT.144(4) (Part 3); Census FT.35(2) (row-wise); Def-BM.5(4), BM.19(1), BM.19(2), BM.36(3), BM.61(8), BM.67(6), BM.67(8), BM.89(4), BM.89(8), BM.90(1), BM.90(2), BM.90(3), BM.90(4), BM.91(2), BM.94(4), BM.94(5), BM.94(6), BM.94(7), BM.94(8), BM.98(6) (this part); exact polynomial identities over $\mathbb{Q}[p_1, p_2, p_3]$, exhaustive configuration censuses, AM–GM on three non-negative rationals, exact rational grids, symbol censuses [math; machine: `checks/verify_H_six_objective_resolution.py` §E].

Pf.

(1) : E_{int}

$$= 2|S| \text{ and linearity of the average } \implies \langle E \rangle$$

$$= 2 \sum_i P(\text{pair } i \text{ real}) \text{ [BM.89(4)'s per-datum form re-derived without any distribution hypothesis; machine]}$$

$$\text{the conditional form: } P(\text{pair } i \text{ real} \mid \text{materialized})$$

$$= a_i \text{ and } \langle |S| \rangle_{\text{mat}}$$

$$= \sum_i a_i \text{ [machine]}$$

$$\implies w_{\text{mat}}$$

$$= \tfrac{1}{3} \sum_i a_i$$

$$(2): \ 1$$

$$\leq |S|$$

$$\leq 3 \text{ on the materialized support [BM.94(5); Def-H.4; FT.144(4)]}$$

$$\implies 1 \leq \langle |S| \rangle_{\text{mat}}$$

$$\leq 3 \implies \tfrac{1}{3}$$

$$\leq w_{\text{mat}}$$

$$\leq 1 \text{ [machine];}$$

$$\text{the two equality loci: } \langle |S| \rangle = 1 \iff |S| = 1 \text{ a.s., } \langle |S| \rangle = 3 \iff |S| = 3 \text{ a.s. [machine]}$$

$$(3): \ P(S)$$

$$= \prod_{i \in S} p_i \prod_{i \notin S} u_i \text{ and } \sum_{S \neq \varnothing} P(S)$$

$$= 1 - \prod_i u_i$$

$$= 1 - R \text{ [machine: the exhaustive eight-configuration census]};$$

$$\langle E\rangle_{\text{mat}} = \tfrac{2\sum_i p_i}{1-R} \text{ [machine: } P(\text{pair } i \text{ real}) = p_i \text{ since a real pair is materialized]}$$

$$\langle |S|^2 \rangle$$

$$= \sum_i p_i + 2\sum_{i<j} p_i p_j \text{ [machine: } |S|^2 = \sum_i 1_i + 2\sum_{i<j} 1_i 1_j] \implies \langle E^2\rangle_{\text{mat}}$$

$$= \tfrac{4(S+2\sum_{i<j} p_i p_j)}{1-R}$$

$$p_i$$

$$\equiv p: \ S$$

$$= 3p,$$

$$R$$

$$= (1-p)^3,$$

$$1 - R$$

$$= p(3 - 3p + p^2)$$

$$= pD$$

$$\implies w_{\text{mat}}$$

$$= 3p/(3pD)$$

$$= 1/D \text{ [BM.94(6); machine at six rational thresholds]}$$

$$(4): \ 3(1 - \prod u) - \sum(1-u) = 3 - 3\prod u - 3 + \sum u = \sum u - 3\prod u \text{ [machine: one identity]};$$

$$\sum u$$

$$\geq 3(\prod u)^{1/3} \text{ [AM–GM] and } (\prod u)^{1/3} \geq \prod u \text{ on } \prod u$$

$$\leq 1 \implies \sum u$$

$$\geq 3 \prod u \text{ [machine: exact on a rational grid and by the two inequalities]}$$

$$\text{equality}$$

$$\iff u_1$$

$$= u_2$$

$$= u_3 \ \wedge \ \prod u \in \{0, 1\}$$

$$\iff p_1$$

$$= p_2$$

$$= p_3$$

$$= 1 \text{ on the materialized domain [machine];}$$

$$\text{monotonicity in each } p_i \text{ [machine: exact rational grid]}$$

(5) : (i)–(iv) re-cited at their own loci [BM.90(1), (2), (3); FT.73(2), (3); BM.67(6), (8); FT.55(1); BM.36(3); BM.61(8)(O1); CT0.8; P4];

the symbol census [machine]

(6)–(7) : the residue rows assembled; the standing-rule scans [machine]. ■

Phys. The weight that prices a composite's failure to crystallize does not need the specialization that first produced it. Linearity alone gives it as the mean fraction of the three pairs that are real, and the materialization restriction alone gives its bounds: at least one pair is real, at most three are, so the weight lives between a third and one for every measure the ensemble could carry, with both endpoints characterized exactly — a third when only single pairs materialize, one when the full

configuration does. That is a stronger statement than the specialization's, and it is why the mean edge count can never fall below the digon's two. Then the family: let each of the three pairs carry its own threshold, still inside the derived admissible range, and the weight becomes the sum of the three retention probabilities over three times one minus their product, with seven materialized configurations, exact second moments, and an arithmetic-mean-geometric-mean row that pins it below one with equality only when every pair is crystallized. Set the three equal and the whole of the previous tier's formula returns identically. What the family does not license is the version the question asks for: six thresholds, one per quark state, each a function of that state's defect value. That is the branch-conditioned weighting already refuted — the measure's alphabet holds one symbol and the defect values' holds two, four separate grounds bar any count-to-strength map, six thresholds would be six declared parameters, and the axiom's threshold carries no state argument. So the weight stays branch-blind, the six defect values stay in the pole currency, and the residue narrows rather than closes: what is left is the correlation between pairs, named and priced, and the transport register's own ensemble average.

Theorem 4.90 (BM.106 — The arrow and the kind: the spin label, the tick law, and the exchange-structure referent census). Thm.

(1) the spin label (derived, on the logged transfer):

$$j = \frac{|k|}{2},$$

$$k = \frac{\Theta}{2} \in \mathbb{Z}_{\geq 0} \text{ on shell,}$$

$$\Psi(c) = \mathrm{Sym}^{|k(c)|}\big(\mathcal{H}_{\mathsf{v}}\big),$$

$$\text{Casimir } \frac{k}{2}\Big(\frac{k}{2}+1\Big)$$

[Def-H.38; FT.60(1)–(3); FT.22; FT.25(3); the transfer's derivation or replacement is a named residue of the seed transport];

the multiplet: $m \in \{k,\ k-2,\ \ldots,\ -k\}$ [FT.60(5); Def-H.38; BM.39(7)];

the floor census:

$$\Theta_{\text{int}} = 2\kappa \in 2\mathbb{Z},$$

$$k = \kappa \in \mathbb{Z} \setminus \{0\},$$

$\kappa = \widehat{ds}^{\,2}(e)$ is registration data determined at crystallization

[Census FT.28(2); FT.24; A §0.2a] :

$$\boxed{\kappa = 1 \implies j = \frac{1}{2}; \qquad \kappa = 2 \implies j = 1}$$

on the same D1.5 kind

[machine: the two exact rows] :

¬∃ a spin-parity theorem — both integer and half-integer labels occur on materialized closed bound chains of one kind

(2) the kind rows that are derived:

bound $:=$ closed self-contained chain, content fermionic;

transit $:=$ propagating open chain, content bosonic

[D1.5 : a definitional label] :

the dichotomy is the closed/open read

[FT.80(5); D1.4];

the materialized locus at rung one is single-arrow

[FT.144(2)] and single-ray, occupancy $(1, 0)$

[A-BM.7; FT.144(5); BM.73(3); BM.96(4)];

the tick law:

$$\tau_n \to \tau_{n+1},$$

$$\neg\exists\, \tau^{-1},$$

$$\left|\mathfrak{L}(\tau_{n+1})\right| = \left|\mathfrak{L}(\tau_n)\right| + n_{\text{real_out}}(\tau_n) \geq \left|\mathfrak{L}(\tau_n)\right|$$

[A6; A7] :

the prefix law $\#J^{+}(\text{prefix}) \geq \#J^{-}(\text{prefix})$ with strict alternation

[FT.144(2); D1.2];

the pure order reversal of every completed word is inadmissible at its first event

[FT.144(4)(ii)];

the contraction:

$$V' = \Big(1 - \frac{\gamma}{4}\Big)^{2} V \leq V,$$

$$\gamma = 1 : \; V' = \frac{9}{16} V,$$

$$\gamma \to 0 : \; V' = V,$$

iterated to $\Big(\frac{9}{16}\Big)^{6}$

[L1; L2; BM.51(1), (2); machine];

(3) the statistics referent census:

a statistics requires	the corpus supplies
a multi-chain state space	¬constructed: ¬a two-chain state space exists [FT.80(5)]
an exchange operator	¬∃ one in the inventory [FT.80(5); FT.125(5)]
a permutation sign	¬formed: ¬∃ a permutation action on states
an identification of a grade's $\mathbb{Z}_2$ with the sign's $\mathbb{Z}_2$	a kind confusion, barred [FT.125(5); FT.89(2) kind discipline]
a tensor state on $\mathcal{H}_c$	¬one: a counting space with no inner product, no dynamics, no state [Def-H.41 Species; Def-H.42]

the spin–statistics connection is ¬formed	[FT.125(5) re-cited];

the derivation asked for is refuted on its face:

the tick arrow and the spin label are both derived (1),

(2),

and neither is an exchange sign — the missing member is the exchange structure itself,

¬a step from arrow to sign;

(4) the four $\mathbb{Z}_2$ gradings are single-object, and the census is exhaustive:

the algebra S-grading $(10, 6)$ [FT.136; FT.141(1)];

the branch-label parity [FT.101; FT.135; FT.157$(2), (3)$];

the carrier $\hat{\chi}$-grade [Def-H.33; FT.122];

the Janus-arrow chirality [FT.144$(2), (5)$] :

pairwise distinct, commuting where jointly defined, neither reducible to another

[BM.73(1); FT.141(6); FT.126(4)] :

¬one of the four is an exchange sign [machine: the four symbol sets are single-object reads];

(5) where antisymmetry does live:

the ε-seal is antisymmetric in the colour slots

$B := \varepsilon_{ijk} q^i q^j q^k$ [D3.9; CT3.15; CT3.17]

and colour is ¬observable by construction

[CT3.22] :

the transport assembly is sign-free:

$$W(Q) = \binom{6}{Q+3} > 0 \text{ on all seven classes}$$

[BM.103(3); machine];

the amplitude layer's signs are barred as statistics:

$$(-1)^{\aleph} \text{ with } \sum_i \aleph_{v_i} \equiv |q| \pmod 2$$

[Def-BM.13(2); Def-BM.5(2); machine: exhaustive to $k = 7$];

$$\lambda = (-1)^{|E|} e^{i\pi\Theta} \in \{\pm 1, \pm i\},$$

$$\Theta_{\text{int}} \in 4\mathbb{Z} \implies \lambda = (-1)^{|E|}$$

[BM.25(3), (6); FT.134(3); machine] :

amplitude $\neq$ mass, and the wall parity is a charge-parity lock, ¬a permutation sign

[FT.79(5); Def-BM.13(2)]

the corpus's only antisymmetry is the colour seal's, and it is unobservable;

(6) the mixing row (the arena read):

the mixing structure is a read of the derived 16-arena $\mathcal{K}_1$ with $\delta_\nu = \frac{2}{16} = \frac{1}{8}$

[FT.142(2), (5); FT.143];

the channel arenas are $9,\ 27,\ 16$ with the $9/27$ candidates rejected at rung one

[FT.142(3); FT.112; FT.119];

the cross-channel identity $16\delta_\nu = 9\delta_\ell = 27\delta_q = 2$

[FT.143(5); machine];

¬an external mixing matrix is imported;	the mixing matrix is ¬derived;	every row holds uniformly in the unresolved line index i_ν

[FT.145(3); BM.77(2)];

the ν row stands as derived:

single-ray occupancy, no right-ray matter line, no seesaw-type mechanism

[BM.96(4), (6) re-cited, ¬extended];

(7) the residue, named:

(R7) an exchange structure := a two-chain state space + an exchange operator + a permutation-sign read

the sealed composite is a junction subnetwork, ¬a tensor product of chain state spaces:

its constituents share vertices [Def-BM.1; Def-BM.2] and no state space is declared on the counting spaces [Def-H.41 Species; FT.93(4) pattern];

(8) status:

derived: (1), (2), (4), (5), (6);

refuted as formed: the spin–statistics derivation from the tick arrow (3);

standing: the spin–statistics connection, its multi-chain locus named;

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a species identification [Def-BM.3(7)];

Req. A6, A7, D0.8, D1.2, D1.4, D1.5, D1.8, D3.9 (Parts 1–2); A §0.2a, CT3.15, CT3.17, CT3.22 (Part 1, Part 2); P1, P4, P5 (Parts 1–2); L1, L2 (Part 1); Def-H.33, Def-H.38, Def-H.41, Def-H.42, FT.22, FT.24, FT.25(3), FT.60(1), FT.60(2), FT.60(3), FT.60(5), FT.80(5), FT.89(2), FT.93(4), FT.101, FT.112, FT.119, FT.122, FT.125(5), FT.126(4), FT.134(3), FT.135, FT.136, FT.141(1), FT.141(6), FT.142(2), FT.142(3), FT.142(5), FT.143, FT.143(5), FT.144(2), FT.144(4), FT.144(5), FT.145(3), FT.157(2), FT.157(3) (Part 3); Census FT.28(2), FT.49 (row-wise); Def-BM.1, Def-BM.2, Def-BM.5(2), Def-BM.13(2), BM.25(3), BM.25(6), BM.39(7), BM.51(1), BM.51(2), BM.73(1), BM.73(3), BM.77(2), BM.79(5), BM.96(4), BM.96(6), BM.103(3) (this part); exact integer winding arithmetic, exhaustive parity censuses, the Casimir ladder on exact rationals, symbol-set censuses [math; machine: `checks/verify_H_six_objective_resolution.py` §F].

Pf.

(1) : Def-H.38's multiplicity rule and FT.60(1)–(3), (5)'s rows re-cited (frozen);

$$\Theta = 2\kappa \in 2\mathbb{Z} \implies k = \Theta/2 = \kappa \in \mathbb{Z} \setminus \{0\}$$ [Census FT.28(2); FT.24; machine]

$$\kappa = 1 \implies j = \tfrac{1}{2};$$

$$\kappa = 2 \implies j = 1;$$

the Casimir rows $\frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}$ and $1 \cdot 2 = 2$ [machine: exact rationals];

both chains are materialized closed bound chains of one D1.5 kind [Def-H.4; FT.22]

(2) : D1.5, FT.80(5), FT.144(2), (4), (5), A6, A7, L2, BM.51(1), (2) re-cited verbatim;

$$(9/16)^6 = 531441/16777216$$ [machine: exact integers]

(3) : the five referent rows re-cited from their own loci [FT.80(5); FT.125(5); FT.89(2); Def-H.41 Species; Def-H.42];

the inventory scans: $\neg\exists$ an exchange token, $\neg\exists$ a permutation-sign read [machine]

(4) : the four gradings and their pairwise distinctness re-cited [BM.73(1); FT.141(6); FT.126(4); FT.157(4)];

each is a read of one object [machine: the symbol censuses]

$$(5):\ \varepsilon_{ijk} \text{ antisymmetry [CT3.15; CT3.17];}$$

$$W(Q) = \binom{6}{Q+3} > 0 \text{ on all seven classes [BM.103(3); machine];}$$

$$\sum_i \aleph_{v_i}$$

$$\equiv |q| \pmod 2 \text{ and}$$

$$\geq |q| \text{ exhaustively for } k$$

$$= 2,$$

$$\dots,$$

$$7 \text{ [Def-BM.13(2) re-derived; machine]}$$

$$\lambda = (-1)^{|E|} e^{i\pi\Theta} \in \{\pm 1, \pm i\}; \qquad \Theta \in 4\mathbb{Z} \implies e^{i\pi\Theta} = 1 \implies \lambda = (-1)^{|E|} \text{ [BM.25(3), (6); machine]}$$

$$(6):\ \delta_\nu = \tfrac{2}{16} = \tfrac{1}{8} \text{ and the arena census re-cited [FT.142(2), (3), (5)];}$$

$$16 \cdot \tfrac{1}{8} = 9 \cdot \tfrac{2}{9} = 27 \cdot \tfrac{2}{27} = 2 \text{ [machine: exact Fractions];}$$

$$\neg\exists \text{ a mixing-matrix tag [FT.145(3); inventory scan]}$$

$$(7)\text{–}(8):\ \text{the junction-subnetwork rows re-cited [Def-BM.1; Def-BM.2; Def-H.41 Species];}$$

$$\text{the standing-rule scans [machine]. } \blacksquare$$

Phys. The corpus can name the spin and it can name the arrow, and it cannot join them, for a reason that is structural rather than temporary. The spin label comes from the winding: half the winding number, with the Casimir ladder and the weight multiplet behind it, and at the floor the winding is the registration integer of the two-cycle, so the same kind of object carries spin one half at one value of that integer and spin one at the next. There is therefore no parity theorem to be had: the floor itself supplies both parities on materialized closed bound chains, all of one definitional kind. The arrow is just as concrete — no inverse tick, a ledger that never shrinks, a prefix law that makes dissolution presuppose crystallization, a Lyapunov function that contracts by nine sixteenths per step at unit tilt — and the occupancy it forces is one ray full and the other

empty. What is missing is not a step from arrow to sign but the object the sign would act on: there is no two-chain state space, no exchange operator, no permutation representation, and the corpus has already ruled that identifying a single-object grade with a permutation sign is a confusion of kinds. So the connection is not derived, and deficit eleven stands with its residue now named: an exchange structure, three ingredients, none of them in the inventory. What the tier adds is the census that makes the absence exact, the four gradings shown to be single-object and exhaustively so, and the location of the only antisymmetry the corpus does have — the colour seal's epsilon, which is unobservable by construction, against a transport assembly whose seven weights are all positive and an amplitude layer whose signs are barred as masses. The mixing rows stay where the arena puts them: sixteen dimensions, one eighth, and no imported matrix.

BM.107 — The status of the six resolutions: adjudication and scope. Structural-constraints record: see Appendix B, Section B.1.30 of the Addendum.

4.23 Tier BM.21 — The Antimatter and Resonance Tier: the CP-Conjugate, the Meson Seal, the Exchange Operator, and the Local Resonance Spectrum

Locus. This tier takes the residue (R7) that BM.106(7) named — an exchange structure: a two-chain state space, an exchange operator, a permutation-sign read — and constructs all three members, then reads the spectrum the construction exposes. The CP-conjugate chain is derived from the three reversals the corpus already owns, with the forward-only tick blocking T as a dynamical inverse and leaving CP as the one composition that closes the ledger's own charge balance; the vacuum fluctuation is its pair, with the creation weight the Janus resolvent's own x_+x_- and no pair at either endpoint of the threshold range; and the antiquark follows from the same three ingredients that force the compensating charge, so the corpus's Dirac parallel is an inventory statement and not a history one (BM.108–BM.110). The meson is then the δ-sealed 2-juncture, its state space is the first multi-chain space the corpus has, it discharges the BM.6(2) specification exactly as written, and its mass is derived on the two routes where the seal contributes no binding term and named unconstructed on the third (BM.111–BM.113). The exchange operator is the canonical flip composed with the CP transport, an involution with spectrum $\{\pm 1\}$ unconditionally on the materialized locus, and its permutation-sign read is the fourth-root phase of the winding evaluated at the on-shell ladder — one integer's parity, which is the spin label's parity, which closes deficit 11 (BM.112–BM.113). The resonance spectrum is then counted: two characteristic polynomials, one transport and one defect, the multiset-orbit law with its stabilizer multiplicities, exactly one exact collision in the whole three-chain ladder, a minimal-polynomial degree census that terminates at the field's own degree, a materialization threshold whose locus is never empty at any n and whose span ratio is exactly thirty at every level, and a decay width that is a product of three derived factors and vanishes identically at the sharp threshold (BM.114–BM.117). Nothing is forbidden, nothing is stable as a category, and no cosmological clock is read: a resonance is an eigenvalue, its appearance is the evaluation of a threshold function at the local parameters, and its width is the evaluation of a width function at the same ones. BM.118 adjudicates.

Standing rows. α_B stays barred on BM.61(8)'s four grounds. No clause of BM.1–BM.107 is retracted, replaced or renumbered: the rows this tier touches are re-scoped at their own loci — BM.106(3)'s statistics referent census and (7)'s residue (R7), BM.103(4)'s tensor-product referent census, BM.6(2)'s exchange-space specification, FT.125(5)'s and FT.80(5)'s inventory rows, and BM.32(5)'s amplitude/norm locus row — and each rescope is recorded in BM.118(5), in BM.77(7)'s pattern. No measured coupling, mass, width, lifetime or ratio enters any derivation [P5]: the

threshold and width functions are derived symbolic forms and are ¬compared to any external number; the crossing strength of BM.101(3) is re-cited and ¬re-evaluated. No right-ray matter line, no seesaw-type mechanism, no Yukawa-type object and no external mixing matrix is suggested, added or required [EW.15; FT.145(3); BM.96(4), (6) re-cited, ¬extended]. Species names are confined to the identification layer [Def-BM.3(7); Def-BM.6(7); Def-BM.14(8)]. Topology labels are internal bookkeeping [Def-H.32(4)]. Amplitude ≠ mass [FT.79(5)]. Rung ≠ scale [D0.8]. No currency is converted [BM.83(3); BM.102(6)]. No theorem of this tier imports a HistLog disposition as premise: Q10, Q11, item 20, deficits 1′, 3, 4, 11 and 12 are cited as pointers. One dimension only: units are powers of $\mathfrak{q}$ [D2.9; CT2.1].

Local notation, fixed for the whole tier. c a materialized closed bound chain [D1.4; D1.5; Def-H.4]; ω the orientation-reversal involution [D2.6]; $\Theta_{\mathfrak{L}}$ the ledger reflection [D2.7]; C, P, T the three derived involutions [EM.24(1)–(3)]; CP := C ∘ P; $\Theta(C) = \sum_{e\in E(C)} \widehat{ds}^{\,2}(e)$ and $\lambda = (-1)^{|E(C)|}e^{i\pi\Theta(C)}$ the CPT phase [EM.24(5); BM.25(3)]; $k(c) = \Theta(c)/2$ the on-shell winding and $j = |k|/2$ the spin label [FT.22; FT.25(3); Def-H.38; BM.74]; $\Psi(c) = \mathrm{Sym}^{|k(c)|}(\mathcal{H}_{\mathsf{v}})$ the winding carrier with $\mathcal{H}_{\mathsf{v}} \cong \mathbb{C}^2$ [Def-H.38; EW.32]; $\mathcal{H}_{\omega}(c) = \mathrm{span}_{\mathbb{C}}\{|c\rangle, |\omega(c)\rangle\} \cong \mathbb{C}^2$ the orientation doublet with $\hat{\omega}, \hat{\chi}, P_{L/R}$ [Def-H.33; FT.122]; E_{int} the internal directed-edge count, $m_{\mathrm{hid}} = E_{\mathrm{int}} \ln 2/\mathfrak{h}$, $\Delta x_{\mathrm{B}} = \mathfrak{h}/(E_{\mathrm{int}} \ln 2)$, $r_s = \mathfrak{h}E_{\mathrm{int}}/2$, $G = 2\pi E_{\mathrm{int}}^2 \ln 2$ [D0.8; FT.44(4); Def-BM.4(1); BM.13(1); BM.15(4); BM.16(1); BM.88(3)]; N the lapse and $f = 1 - 1/k$ with $k = r/\Delta x_{\mathrm{B}}$ the envelope profile [BM.34(1); BM.88(2)]; $\varepsilon_{\mathrm{eff}} = Nf\,\mathfrak{g}\,\varepsilon_{\mathrm{EM}} = e^2/(4\pi)$ [BM.88(4); BM.101(2)]; $\varepsilon_q \in [0, \frac{1}{4})$ the crystallization threshold, $x_+ = 1 - 4\varepsilon_q$, $x_- = 4\varepsilon_q$, $p = x_+/(x_+ + x_-) = 1 - 4\varepsilon_q$ [A4; D0.4; CT0.8; BM.19(2)]; $\Gamma_{\mathrm{tr}} = x_+x_-/(1 - x_+x_-) \in [0, \frac{1}{3}]$ the unit-fugacity transition weight [BM.44(2)]; $D_S(s) \in \{2, 4, 2 - \sqrt{2}, 1, 2 - 2\cos\frac{\pi}{10}, 2 - \sqrt{3}\}$ and $m^2_{\mathrm{pole}} = \hat{\mu}'^2 D_S$ the six-state defect spectrum [Def-BM.14(3); BM.75(4)]; $\mathcal{S}$ the six-state set, $|\mathcal{S}| = 6$ [Def-BM.14(2)]; $W_n(Q)$ the n-chain transport class weight [BM.28(2); BM.103(3)]; $\mathfrak{u} := 2\cos\frac{\pi}{10}$ and $\mathbb{M} := \mathbb{Q}(\sqrt{2}, \sqrt{3}, \mathfrak{u})$ the exact eigenvalue field; $\chi_{\{\cdot\}}$ the exact indicator, valued in $\{0, 1\}$; $[\cdot]$ the $\mathfrak{q}$-exponent map [CT2.1; D2.9].

Definition 4.15 (Def-BM.15 — The CP-conjugate chain and the CP-pair). Def. A construction on the three derived involutions of EM.24(1)–(3) and on D2.6's ω; no new axiom, no new datum, no new parameter, no external object.

(1) the three reversals, re-cited as derived:

$$\mathrm{C}:\ \sigma_e \mapsto -\sigma_e\ \forall e \implies q \mapsto -q, \quad f_v \mapsto -f_v, \quad \aleph_v \text{ invariant}$$

[EM.24(1); D3.3; D3.4];

$$\mathrm{P}:\ \mathcal{P} \mapsto E_{\mathcal{R}} \setminus \mathcal{P}, \qquad \text{equivalently} \quad \alpha_e \mapsto \pi - \alpha_e \qquad [\text{EM.24(2); Def-EM.21; EM.20(1)}];$$

$$\mathrm{T}:\ \Theta_{\mathfrak{L}}, \qquad w(\Theta_{\mathfrak{L}}C) = w(C)^*, \qquad J^+ \leftrightarrow J^-, \qquad G(e) \mapsto G(e)^* \qquad [\text{EM.24(3); D2.7; CT2.8(1)}];$$

constitutive clause: all three are involutions on $\mathfrak{C}(\mathcal{R})$ and all three are derived — C^2

$$= \mathrm{P}^2$$

$$= \mathrm{T}^2 \qquad = \mathrm{id},$$

no external input

[EM.24(4)];

(2) the CP-conjugate: $\boxed{\mathrm{CP} \;:=\; \mathrm{C}\circ\mathrm{P}, \qquad c^{\mathrm{CP}} \;:=\; \mathrm{CP}(c)}$

constitutive clause: CP is the composition of two derived involutions, hence an involution:

$$(\mathrm{CP})^2 = \mathrm{C}^2\mathrm{P}^2 = \mathrm{id};$$

the action (derived, entrywise):

$$q \mapsto -q \text{ [EM.24}(1)\text{]};$$

$$\text{scar} = q_{\text{core}} \mapsto -q_{\text{core}} \text{ [Def-BM.7}(3);\ \omega\text{-odd]};$$

$$\text{chirality} \;\mapsto\; -\text{chirality [D2.6; FT.122}(4) :\ \hat{\omega}\hat{\chi}\hat{\omega}^{-1} = -\hat{\chi}\text{]};$$

$$\text{triplet attachment } q^i \;\mapsto\; \text{dual attachment } r_i \text{ [D3.8; CT3.17 : } \delta^i{}_j \text{ consumes one of each type]};$$

$$w \;\mapsto\; e^{i\pi\Theta}\, w$$

[EM.24(5) with $(-1)^{|E|}$ cancelled by P's $e^{i\pi\Theta}$: machine];

$$E_{\text{int}} \;\mapsto\; E_{\text{int}}, \quad |R| \;\mapsto\; |R|$$

[BM.25(2); BM.27(2)];

(3) the CP-pair and the vacuum fluctuation:

$$\text{a CP-pair} \;:=\; \text{an orbit } \{c,\ c^{\mathrm{CP}}\} \text{ of the involution of } (2);$$

a vacuum fluctuation :

$=$ the creation of one CP-pair from the vacuum sector [BM.96(4) : the complement's content is all-virtual and unregistered];

constitutive clause: the pair's total charge read is $Q = q(c) + q(c^{\mathrm{CP}}) = q - q = 0$ exactly

[EM.11; D2.11; machine];

non-derived clause: the word "vacuum" names the all-virtual sector of CT4.2 and FT.144(4);

it carries no energy datum and no state;

(4) the scope clause (the mirror is formal, not dynamical):

$$\neg\exists\, \tau^{-1} \text{ [A6]};$$

$$\big|\mathfrak{L}(\tau_{n+1})\big| = \big|\mathfrak{L}(\tau_n)\big| + n_{\text{real_out}}(\tau_n) \geq \big|\mathfrak{L}(\tau_n)\big| \text{ [A6; A7]};$$

$$V' = \big(1 - \tfrac{\gamma}{4}\big)^2 V \leq V, \qquad \gamma = 1 :\ V' = \tfrac{9}{16} V \qquad \text{[L2; BM.51}(1), (2)\text{]} :$$

$\therefore$ T is blocked as a dynamical inverse: $\Theta_{\mathfrak{L}}$ is an involution on configuration data and $\neg$a step of the evolution [Def-BM.7(2); BM.25(2); FT.144(4)(ii)];

the consequence that makes CP the operative conjugation:

the ledger's balance is a condition on configurations, $\neg$on evolutions [D2.11; EM.11],

so the compensating member a charged chain requires is a configuration, and CP is the derived involution that produces one with the same E_{int} (2);

(5) the data bar (standing):

$\neg\exists$ an annihilation channel, a pair-production cross-section or an inter-sector transition operator in the inventory [Def-BM.7(5); BM.27(7); BM.29(5)];

"antimatter" is the label of Def-BM.7(1)'s EM^{-} sector and carries no external datum [P5; Def-BM.3(6)];

the CP-pair creation weight of BM.110(1) is ¬a cross-section and ¬a rate in external units [BM.44(5); D1.8].

Rmk (what is constitutive and what is derived). Clause (2)'s definition of CP as C ∘ P is the constitutive act — one composition of two objects the bosonic era already derived. Everything else in (2) is derived content recorded at the definition site: the entrywise action follows from EM.24(1), (2) and D2.6, the weight rule from EM.24(5), and the invariance of E_{int} from BM.25(2)'s record-size row. Clause (3)'s charge neutrality is derived, not stipulated: it is EM.11's balance applied to an orbit of an involution that negates the read. Nothing here names a particle, imports a mass, or supplies an annihilation amplitude.

Req. A6, A7, D0.8, D1.4, D1.5, D1.8, D2.6, D2.7, D2.11, D3.3, D3.4, D3.8 (Parts 1–2); CT0.8, CT2.8(1), CT3.17, CT4.2 (Part 1, Part 2); EM.11, EM.20(1), EM.24(1), EM.24(2), EM.24(3), EM.24(4), EM.24(5), Def-EM.21 (Part 2); P5 (Parts 1–2); L2 (Part 1); Def-H.4, Def-H.33, FT.122(4), FT.144(2), FT.144(4) (Part 3); Def-BM.4(1), Def-BM.7(1), Def-BM.7(2), Def-BM.7(3), Def-BM.7(5), BM.25(2), BM.27(2), BM.27(7), BM.29(5), BM.51(1), BM.51(2), BM.96(4) (this part); exact $\mathbb{Q}(i)$ weight algebra on the fourth-root alphabet, exhaustive charge-alphabet enumeration [math; machine: `checks/verify_H_antimatter_resonance_spectrum.py` §A].

Pf.

Definition block: the constitutive clauses are (1)'s involution rows (re-cited, frozen),

(2)'s composition CP :

$$= \text{C} \circ \text{P},$$

and (3)'s orbit naming;

the non-derived clauses are the "vacuum" vocabulary of (3) and the "antimatter" label of (5)

(2)'s derived rows: $q \mapsto -q$ [EM.24(1)];

$\alpha_e \mapsto \pi - \alpha_e$ [EM.24(2)];

composed: $w \mapsto \bar{w} \mapsto e^{i\pi\Theta} w$ [EM.24(5) Pf row P ∘ T transported to C ∘ P; machine]

attachment type: $\delta^i{}_j$ consumes one triplet and one dual [CT3.17] $\implies$ the conjugate constituent carries the dual attachment [D3.8; Def-BM.1 meson clause];

$$E_{\text{int}} = |R| \text{ invariant [BM.27(2); machine]}$$

$$(3):\ Q = q + (-q) = 0 \text{ [machine: exhaustive over } q \in \{-1, 0, +1\}];$$

$$(4):\ \text{A6},\ \text{A7},\ \text{L2 rows re-cited;}$$

$$(5):\ \text{the inventory scan of BM.27(7) re-cited.}\ \blacksquare$$

Phys. Three reversals exist in this corpus and each is derived: flip every edge sign, complement the aligned set, reflect the ledger. The third is the one that cannot run — the tick has no inverse and the Lyapunov function only contracts, so a reversed history is a configuration, never a step. Composing the first two gives the conjugation that does the work: it negates the charge read, flips the scar, exchanges the chiral rays, turns a triplet attachment into a dual one, multiplies the weight by a fourth root of unity, and leaves the edge count — and hence the informational mass, the width, the volume, the density and the comparison radius — untouched. An orbit of that involution is a pair, and a pair is neutral by construction, because the balance the valid-universe condition demands is a condition on configurations and the conjugate is exactly the configuration that satisfies it. What the definition does not do is give the pair a process: no annihilation channel, no production cross-section, no inter-sector operator exists in the inventory, and the word "vacuum" names the all-virtual sector rather than any state with an energy.

Theorem 4.91 (BM.108 — The CP-conjugate chain: the reversal census, the T-block, the exact phase, and the forced conjugate). Thm.

(1) the reversal census (what the corpus owns and what it blocks):

reversal	status	locus
C charge conjugation	derived, dynamical	[EM.24(1); D3.3; D3.4]
P parity	derived, dynamical	[EM.24(2); Def-EM.21]
T time reversal	derived as a record involution, blocked as a dynamical inverse	[EM.24(3); A6; L2; FT.144(4)(ii)]
CP	derived, an involution	[Def-BM.15(2); EM.24(4)]
CPT	derived, $\text{CPT}^2 = \text{id}$	[EM.24(4); BM.25(1)]

the T-block, in three exact rows: (i) $\neg\exists\, \tau^{-1}$ [A6];

(ii) $\big|\mathfrak{L}(\tau_{n+1})\big| - \big|\mathfrak{L}(\tau_n)\big| = n_{\text{real_out}}(\tau_n) \geq 0$ [A6; A7];

(iii) $V' - V = -\Big[1 - \big(1 - \tfrac{\gamma}{4}\big)^2\Big] V = -\dfrac{\gamma(8-\gamma)}{16} V \leq 0, \qquad \gamma = 1:\ V' = \tfrac{9}{16} V$ [L2; machine] :

the pure order reversal of every completed word is inadmissible at its first event — $\Theta_{\mathfrak{L}}$ exists and $\neg$evolves

[BM.25(2); FT.144(4)(ii); machine];

(2) the CP action on every derived read of a chain (exact, entrywise):

read	CP image	locus
$q = q_{\text{core}} = \frac{1}{2}(\sigma_s - \sigma_t)$	$-q$	[D3.4; EM.9; machine]
scar	$-$scar	[Def-BM.7(3); BM.27(5); machine]
$\hat{\chi}$-grade	$-\hat{\chi}$-grade ($P_L \leftrightarrow P_R$)	[D2.6; FT.122(4); BM.73(4); machine over $\mathbb{Q}(i)$]
attachment type (q^i / r_i)	r_i / q^i	[D3.8; CT3.17]
weight w	$e^{i\pi\Theta}\, w$	[EM.24(5); BM.25(3); machine]
$\lvert w\rvert^2$	$\lvert w\rvert^2$	[EM.24(5) : unconditional; machine]
E_{int}, $\lvert R\rvert$, m_{hid}, Δx_{B}, V_{eff}, ρ, $\kappa^2\rho$, r_s	fixed	[BM.27(2), (3); BM.28(1) : q-exponent 0 in all seven; machine]
Θ_{int} class	fixed	[BM.28(6); machine]

the CP-conjugate chain is mass-degenerate with the chain on every route

[BM.28(5); BM.27(3) Rmk; machine: the monomial q-exponent census];

(3) the exact CP phase: $w(c^{\text{CP}}) = \mu_{\text{CP}}(c)\, w(c)$, $\boxed{\mu_{\text{CP}}(c) \;:=\; e^{i\pi\Theta(c)}}$

derivation: $w(\text{CPT}\, C) = (-1)^{|E|} e^{i\pi\Theta} w(C)$ [EM.24(5)],

$$w(\text{T}\, C) = w(C)^* \text{ [CT2.8(1)]},$$

$$\text{CP} = (\text{CPT}) \circ \text{T}^{-1} = (\text{CPT}) \circ \text{T} :$$

$$\mu_{\text{CP}} = (-1)^{|E|} e^{i\pi\Theta} \cdot \frac{w}{w^*}\bigg|_{\text{the T leg}} = (-1)^{|E|} e^{i\pi\Theta} \cdot (-1)^{|E|} = e^{i\pi\Theta}$$

[machine: $|w| = 1 \implies w/w^* = w^2/|w|^2$; the $(-1)^{|E|}$ cancels exactly];

$\mu_{\text{CP}} \in \{\pm 1, \pm i\}$ on the half-integer ladder,

$\mu_{\text{CP}} \in \{+1\}$ on the materialized locus $\Theta \in 2\mathbb{Z}$

[FT.22; BM.2(1); machine] :

the CP character is exactly 1 on the sealed-composite junction class

[$\Theta_{\text{int}} \in 4\mathbb{Z}$ [Census FT.49] $\implies e^{i\pi\Theta} = 1$; BM.25(6); machine];

(4) the orbit census:

$$\text{CP}: \ (q, \ \text{orientation}) \mapsto (-q, \ \text{flipped}) \ \text{ on } \{-1, 0, +1\} \times \{w, w^*\} :$$

$$\text{no fixed sector, three orbits of size two [BM.25}(4); \ \text{machine}];$$

$$q \mapsto -q \ \text{ on the ternary alphabet has exactly one fixed point, } q = 0$$

$$[\text{EM.24}(1); \ \text{machine}] :$$

$$\boxed{\text{the neutral class is charge-self-conjugate; every charged class is exchanged}}$$

$$\text{the CP involution on charge } n\text{-tuples has exactly one fixed tuple, } (0, \ldots, 0),$$

$$\text{and } W_n(+Q)$$

$$= W_n(-Q) \text{ exactly}$$

$$[\text{machine: exhaustive at } n = 2, 3];$$

$$(5) \quad \text{the forcing theorem (why the conjugate is not optional):}$$

$$Q_{\text{EM}} = \sum_{c \in U} q_{\text{EM}}(c) = 0 \ \text{ in every valid universe}$$

$$[\text{EM.11}; \ \text{D2.11}; \ \text{D1.3}] :$$

$$\text{a materialized chain with } q(c) \neq 0 \text{ requires a compensating read of exactly } -q(c);$$

$$\#\{q \in \{-1, 0, +1\} : \ q = -q(c)\} = 1$$

$$[\text{BM.23}(4); \ \text{machine}];$$

the compensator census:

a compensating configuration must carry	the corpus supplies
the charge read $-q(c)$	exactly one derived map produces it: C [EM.24(1)]
a dual colour attachment, to seal against c's triplet	exactly one derived map produces it: D3.8 $\circ$ CP [CT3.17]
the opposite chiral ray	exactly one derived map produces it: $\hat{\omega}$ [FT.122(4); A-BM.7(b)]
a closed real-edge network with $\#J^+ = \#J^-$	CP preserves $\lvert R\rvert$ and exchanges $\#J^+$ with $\#J^-$ [BM.25(2); BM.27(2)]
the same E_{int} (hence the same mass read)	derived: (2)'s seven q-exponent-zero monomials [BM.28(1)]

the CP-conjugate is the unique derived configuration that balances a charged chain: it is forced out of the ledger's own balance, ¬added

[machine: the five-row census];

(6) the Dirac parallel, as an inventory statement:

the three ingredients: (i) the forward-only tick [A6; (1)(i)];

(ii) the 16-matrix pair arena $\mathcal{K}_1$,

$$\dim \mathcal{K}_1$$

$$= 16$$

$$= C_{\text{cap}} \text{ [FT.142}(2); \text{ CT0.14; CT0.12]};$$

(iii) the total-charge-zero balance [EM.11; D2.11] :

(i) + (iii) $\implies$ the compensator is a configuration and ¬an inverse evolution [(4)'s scope clause];

(ii) $\implies$ the compensator lives in the same 16-arena, on the ω-image representation $\{0, -1\}$ [EW.36; A-BM.7(b)];

adding the forward-only tick to the 16-matrix pair arena forces the conjugate constituent, exactly as a negative-energy branch forces its conjugate

[the parallel is structural: ¬an external history is imported, ¬a measured mass is compared [P5]];

the scope of the parallel:

¬a Dirac operator, ¬a Lorentz spinor and ¬a negative-energy spectrum is constructed [BM.6(8) : the frame section and a global Dirac-type operator remain unconstructed; FT.93(3) the pincer];

(7) the no-annihilation row:

¬∃ an annihilation channel, a pair-production cross-section or an inter-sector transition operator [Def-BM.7(5); BM.27(7); BM.29(5)] :

the two sectors are amplitude-equal configuration orbits; their interaction is ¬a theorem of this corpus

[BM.27(7) re-cited, ¬extended];

the CP-pair creation weight of BM.110(1) is a flicker weight, ¬a cross-section [BM.44(5); D1.8];

(8) status:

derived: (1)–(5);

derived as a structural parallel with an explicit scope: (6);

recorded as absent: (7);

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a species identification [Def-BM.3(7)],

¬a numerical coupling [EM.25(7)];

§Cal-BM stays empty except for A-BM.9's logged slot.

Rmk (locus discipline on clause (6)). The parallel is a statement about what forces what inside one inventory: a balance condition plus a forward-only order plus a fixed arena leave exactly one derived configuration that satisfies all three. It is not a claim that the corpus reproduces any external equation, and no clause identifies the conjugate constituent with any measured object — the name antiquark belongs to the identification layer and occurs in no Thm clause here [Def-BM.3(7)].

Req. A6, A7, D0.8, D1.3, D1.4, D1.8, D2.6, D2.7, D2.11, D3.3, D3.4, D3.8 (Parts 1–2); CT0.12, CT0.14, CT2.8(1), CT3.17, EM.9, EM.11, EM.24(1), EM.24(2), EM.24(3), EM.24(4), EM.24(5), EM.25(7), EW.36, Def-EM.21 (Part 1, Part 2); P1, P4, P5 (Parts 1–2); L2 (Part 1); Def-H.33, FT.22, FT.93(3), FT.122(4), FT.142(2), FT.144(2), FT.144(4) (Part 3); Census FT.49 (row-wise); Def-BM.3(7), Def-BM.7(1), Def-BM.7(2), Def-BM.7(3), Def-BM.7(5), Def-BM.15, BM.2(1), BM.6(8), BM.23(4), BM.25(1), BM.25(2), BM.25(3), BM.25(4), BM.25(6), BM.27(2), BM.27(3), BM.27(5), BM.27(7), BM.28(1), BM.28(5), BM.28(6), BM.29(5), BM.51(1), BM.51(2), BM.73(4), A-BM.7 (this part); exact $\mathbb{Q}(i)$ phase algebra on the fourth-root alphabet, exhaustive ternary-alphabet and n-tuple censuses, Laurent-monomial q-exponent censuses [math; machine: `checks/verify_H_antimatter_resonance_spectrum.py` §A].

Pf.

$$(1):\ \text{EM.24}(1)\text{–}(4)\ \text{rows re-cited (frozen, machine-verified at Tier F.33)};$$

$$(\text{i}),(\text{ii}):\ \text{A6},\ \text{A7 verbatim}$$

$$(\text{iii}):\ V'-V=-[1-(1-\tfrac{\gamma}{4})^2]V=-\tfrac{\gamma(8-\gamma)}{16}V\ [\text{L2}];$$

$$\gamma=1:\ 1-\tfrac{1}{4}=\tfrac{3}{4},\ (\tfrac{3}{4})^2=\tfrac{9}{16},\ 1-\tfrac{9}{16}=\tfrac{7}{16},\ \gamma(8-\gamma)=7\ [\text{machine: exact rationals}]$$

the inadmissibility row: FT.144(4)(ii) and BM.25(2) re-cited [machine: the pure reversal leads with J^-, outside D1.2's domain]

$$(2):\ q\mapsto\tfrac{1}{2}(-\sigma_s+\sigma_t)=-q\ [\text{D3.4}];$$

$$\hat{\omega}\hat{\chi}\hat{\omega}^{-1}=-\hat{\chi},\ \hat{\omega}P_L\hat{\omega}^{-1}=P_R\ [\text{FT.122}(4);\ \text{BM.73}(4);\ \text{machine over}\ \mathbb{Q}(i)]$$

the seven fixed reads: Δx_{B},

$$V_{\mathrm{eff}},$$

$$\rho,$$

$$\kappa^2\rho,$$

$$r_s,$$

$$(\Delta x_{\mathrm{B}})^2\kappa^2\rho,$$

$\Delta x_{\mathrm{B}} r_s$ each carry charge exponent 0 [BM.28(1); machine: monomial census]

$$\implies q$$

$$\mapsto -q \text{ fixes each}$$

$(3):$ EM.24(5)′s Pf row $\bar{\Gamma} \mapsto e^{i\pi\Theta}\Gamma$ under P $\circ$ T, and $S \mapsto (-1)^{|E|}S$ under C;

$$\mathrm{CP} = \mathrm{CPT} \circ \mathrm{T} \text{ at } \mathrm{T}^2 = \mathrm{id} \text{ [EM.24}(4)\text{]}$$

$$w(\mathrm{CP}\, C) = w(\mathrm{CPT}\, \mathrm{T}\, C) = (-1)^{|E|} e^{i\pi\Theta} w(\mathrm{T}\, C) = (-1)^{|E|} e^{i\pi\Theta} w(C)^*;$$

$$|w(C)| = 1 \text{ [Def-H.32}(2): \ |w(R,\sigma)| = 1] \implies w(C)^* = (-1)^{|E|} w(C)^{-1} \cdot |w|^2$$

on the sign-real locus $w \in \mathbb{R}$ [EM.24(5) on-shell]

$$\implies w^*$$

$$= w \implies \mu_{\mathrm{CP}}$$

$$= (-1)^{|E|} e^{i\pi\Theta} (-1)^{|E|}$$

$$= e^{i\pi\Theta} \text{ [machine: } (-1)^{2|E|} = 1\text{]};$$

$$\Theta \in 2\mathbb{Z} \implies e^{i\pi\Theta} = i^{2\Theta} = 1 \text{ [machine]}$$

$(4):$ BM.25(4)′s six-sector census re-cited [machine];

$q \mapsto -q$ fixed points on $\{-1, 0, +1\}: \ \{0\}$ [machine];

n-tuples: exhaustive at n

$$= 2 \ (9 \text{ tuples}, \ 5 \text{ orbits}, \ 1 \text{ fixed}) \text{ and } n$$

$$= 3 \ (27 \text{ tuples}, \ 14 \text{ orbits}, \ 1 \text{ fixed}) \text{ [machine]}$$

$$W_n(+Q)$$

$$= W_n(-Q) : \text{ the map } (q_1, \ldots, q_n)$$

$\mapsto (-q_1, \ldots, -q_n)$ is a weight-preserving bijection between the two class fibres [$m_{-q} = m_q$ [FT.106(3a) : $m_q = (2, 1, 1)$]; machine]

$$(5) : \text{ EM.11, D2.11 re-cited;}$$

$$\#\{q : q = -\Delta q\} = 1 \text{ [BM.23(4) pattern; machine];}$$

the five-row census assembled from its own tags [machine]

$$(6) : \dim \mathcal{K}_1 = 16 = C_{\text{cap}} \text{ [FT.142(2); CT0.14];}$$

ω-image representation $\{0, -1\}$ [EW.36; A-BM.7(b)];

the unconstructed rows re-cited [BM.6(8); FT.93(3)]

(7) : the inventory scan of BM.27(7) re-cited; (8) : the standing-rule scans [machine]. ■

Phys. Of the three reversals the corpus derives, two run and one does not. Charge conjugation and parity are maps the ledger can apply to a configuration; time reversal is a map it can apply to a record but never to a step, because the tick has no inverse, the ledger only grows, and the Lyapunov function only contracts — at sharp crystallization by exactly seven sixteenths per tilt. Composing the two that run gives the conjugation that matters, and its action is exact and cheap: the charge read flips, the scar flips, the chiral rays exchange, the triplet attachment becomes a dual one, the weight picks up a fourth root of unity that is exactly one on the sealed junction's own phase class, and the seven geometric reads do not move at all, because not one of them carries a charge exponent. So the conjugate is mass-degenerate with the original by structure rather than by luck. That it must exist is not an addition but a consequence: total charge vanishes in every valid universe, a charged chain therefore needs a compensator, the alphabet contains exactly one value that compensates it, and of the derived maps exactly one produces a configuration carrying that value with a dual attachment, the opposite ray, and the same edge count. The orbit census is the same statement at the level of classes — one self-conjugate neutral class, every charged class exchanged, one fixed charge tuple at every level and equal weights on the two charged classes. The structural parallel is then exact in its own terms: a balance, a forward-only order, and a fixed sixteen-dimensional arena leave one derived compensator, which is the shape of the argument that forces a conjugate branch, and the tier says no more than that — no Dirac operator, no spinor, no annihilation channel, and no name outside the identification layer.

Theorem 4.92 (BM.109 — The CP-pair as the vacuum fluctuation: the creation weight, its exact range, and the degeneracy rows). Thm.

(1) the pair's charge and mass rows:

$$Q_{\text{pair}} = q(c) + q(c^{\text{CP}}) = q - q = 0$$

[Def-BM.15(3); machine: exhaustive over $q \in \{-1, 0, +1\}$] :

$$E_{\text{int}}^{\text{pair}} = 2\, E_{\text{int}}(c)$$

[additivity of the internal count on a disjoint pair; D0.8; BM.21(1) pattern];

$$\boxed{m_{\text{hid}}^{\text{pair}} = \frac{2E_{\text{int}}(c)\ln 2}{\hbar} = 2\, m_{\text{hid}}(c), \qquad \Delta x_{\text{B}}^{\text{pair}} = \frac{\hbar}{2E_{\text{int}}(c)\ln 2} = \frac{\Delta x_{\text{B}}(c)}{2}}$$

[FT.44(4); Def-BM.4(1); machine: Laurent identities];

the binding read of the pair count vanishes exactly:

$$E_{\text{bind}}^{\text{info}}(\text{pair}) = \big(2E_{\text{int}} - (E_{\text{int}} + E_{\text{int}})\big)\tfrac{\ln 2}{\hbar} = 0$$

[BM.21(1); BM.54(1); machine];

(2) the creation weight is the Janus resolvent's own:

$$\sum_n \langle \varnothing | \hat{U}^n | \varnothing \rangle w^n = \frac{1}{1 - w^2 x_+ x_-} \text{ in } \mathbb{Q}[w, x_+, x_-]/(w^{12})$$

[FT.108(5); BM.23(7); BM.44(1)] :

$$\boxed{\mathcal{W}_{\text{pair}} := x_+ x_- = 4\varepsilon_q(1 - 4\varepsilon_q) = \frac{1}{4} - 16\Big(\varepsilon_q - \frac{1}{8}\Big)^2 \le \frac{1}{4}, \qquad \text{equality} \iff \varepsilon_q = \frac{1}{8}}$$

[BM.19(2); BM.44(1); machine: the SOS completion + the grid census over $\varepsilon_q = k/256$];

the two endpoints: $\varepsilon_q = 0 \implies x_- = 0 \implies \mathcal{W}_{\text{pair}} = 0$ exactly [BM.19(3); machine];

$$\varepsilon_q \to \tfrac{1}{4}^{-} \implies x_+ \to 0 \implies \mathcal{W}_{\text{pair}} \to 0 \qquad [\text{CT0.8}: \ \varepsilon_q \in [0, \tfrac{1}{4}); \ \text{machine}]:$$

$$\boxed{\mathcal{W}_{\text{pair}} > 0 \iff \varepsilon_q \in \big(0, \tfrac{1}{4}\big), \qquad \max \mathcal{W}_{\text{pair}} = \frac{1}{4} \ \text{ at } \ \varepsilon_q = \frac{1}{8}}$$

[machine: exact rationals on a 256-point grid];

(3) the one-door row:

the vacuum sector is all-virtual and unregistered with no accessible pole at any rung [BM.96(4); FT.144(4); CT4.2];

the materialized locus requires $|E_{\mathcal{R}}| \geq 2$ [Def-H.4; BM.13(3); BM.70(2)] :

the CP-pair materializes and propagates forward-only exactly where the flicker is present — at the same threshold the ensemble integral is taken over

[BM.96(6) re-cited at the pair level; machine];

$$\Gamma_{\text{tr}}(\varepsilon_q{=}0) = 0$$

[BM.23(8); BM.44(2)] :

a non-flickering ledger creates no pairs and decays nothing;

(4) the degeneracy rows (the pair is one orbit, not two structures):

$$|w(c^{\text{CP}})|^2 = |w(c)|^2 \ \ \forall c$$

[EM.24(5) : unconditional; machine];

$$m_{\text{hid}}(c^{\text{CP}}) = m_{\text{hid}}(c),$$

$$D_S(c^{\text{CP}}) = D_S(c),$$

$$m^2_{\text{pole}}(c^{\text{CP}}) = m^2_{\text{pole}}(c)$$

[BM.108(2); Def-BM.14(3) : D_S is branch data, ¬charge data; machine];

$(\mathrm{CP})^2 = \mathrm{id} \implies$ the pair carries no independent datum: it is one orbit of size two

[BM.27(6); EM.24(4)];

the two rows contribute equally to the norm $\langle A|A\rangle = \sum_C |w(C)|^2$

[CT2.8(3); BM.25(7); machine];

(5) the no-annihilation row stands:

¬∃ an annihilation channel or a cross-section between EM^+ and EM^- [BM.27(7); Def-BM.7(5)] :

what is derived is the pair's existence, its neutrality, its degeneracy and its creation weight — ¬its interaction;

(6) status: derived: (1)–(4); recorded as absent: (5);

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a numerical coupling [EM.25(7)],

¬a lifetime in external units [BM.44(5)];

Req. A4, A6, D0.4, D0.8, D1.8, D2.9 (Parts 1–2); CT0.8, CT2.8(3), CT4.2, EM.24(4), EM.24(5), EM.25(7) (Part 1, Part 2); P1, P4, P5 (Parts 1–2); Def-H.4, Def-H.32(2), FT.108(5), FT.144(4), FT.44(4) (Part 3, Census); Def-BM.3(7), Def-BM.4(1), Def-BM.7(1), Def-BM.7(5), Def-BM.14(3), Def-BM.15, BM.13(3), BM.19(2), BM.19(3), BM.21(1), BM.23(7), BM.23(8), BM.25(7), BM.27(6), BM.27(7), BM.44(1), BM.44(2), BM.44(5), BM.54(1), BM.70(2), BM.96(4), BM.96(6), BM.108(2) (this part); exact rational resolvent identities, the SOS parabola completion, exhaustive rational grid censuses, Laurent-monomial assembly [math; machine: `checks/verify_H_antimatter_resonance_spectrum.py` §A].

Pf.

$$(1):\ q+(-q)=0 \text{ [machine: three cases]};$$

$$E_{\text{int}}^{\text{pair}} = E_{\text{int}} + E_{\text{int}} \text{ on a disjoint union of directed-edge sets [D0.8]};$$

$$m_{\text{hid}} = E_{\text{int}} \ln 2/\mathfrak{h} \text{ linear [FT.44}(4);\ \text{machine]}$$

$$\Delta x_{\text{B}}^{\text{pair}} = \mathfrak{h}/(2E_{\text{int}} \ln 2) = \Delta x_{\text{B}}/2 \text{ [machine: one Laurent monomial]};$$

$$E_{\text{bind}}^{\text{info}} = 0 \cdot \tfrac{\ln 2}{\mathfrak{h}} \text{ [BM.21}(1) \text{ pattern;\ machine]}$$

$$(2):\ \text{FT.108}(5)\text{'s resolvent re-cited};$$

$$x_+ = 1-4\varepsilon_q,\ x_- = 4\varepsilon_q \implies x_+x_- = 4\varepsilon_q(1-4\varepsilon_q) \text{ [BM.19}(2);\ \text{machine]}$$

$$4\varepsilon_q(1-4\varepsilon_q) = \tfrac{1}{4} - 16(\varepsilon_q - \tfrac{1}{8})^2 \text{ [machine: expansion]};$$

$$\text{the grid census } \varepsilon_q = k/256,\ k = 0..63 : \leq \tfrac{1}{4},\ = \tfrac{1}{4} \text{ only at } k = 32 \text{ [machine]}$$

$$\varepsilon_q = 0 \implies x_- = 0 \implies x_+x_- = 0;$$

$$\varepsilon_q \to \tfrac{1}{4}^-$$

$$\implies x_+ \to 0$$

$$\implies x_+x_- \to 0 \text{ [machine: exact at the two endpoints of the admissible range [CT0.8]]}$$

$$(3):\ \text{BM.96}(4),$$

(6)'s rows transported to the pair [the pair's members are the two states of that row];

$$\Gamma_{\text{tr}}(0) = 0 \text{ [BM.44}(2);\ \text{machine]}$$

$$(4):\ |w(\text{CP}\,C)|^2 = |w(C)|^2 \text{ from } |\mu_{\text{CP}}| = |e^{i\pi\Theta}| = 1 \text{ [BM.108}(3);\ \text{machine]};$$

D_S is a function of $(\mathcal{F}_g, \beta)$ alone [Def-BM.14(3); machine: no charge symbol occurs]

$$(\mathrm{CP})^2 = \mathrm{C}^2\mathrm{P}^2 = \mathrm{id}\ [\text{EM.24}(4)];$$

$(5), (6)$: the inventory scan and the standing-rule scans [machine]. ■

Phys. A pair made of a chain and its conjugate is neutral because the conjugate's charge is the negative of the chain's, and it is mass-degenerate because none of the seven geometric reads carries a charge exponent and the defect read is branch data rather than charge data. Its informational mass is exactly twice a member's and its width exactly half, with the count-route binding identically zero — a binding energy cannot be an integer summation, and the pair is the cleanest case of it. The weight for making one is not declared: it is the Janus resolvent's own x_+x_-, a parabola in the single crystallization threshold with maximum one quarter at one eighth and zeros at both ends of the admissible range. So the ledger creates pairs exactly where it flickers, and nowhere else — at the sharp threshold the dissolution weight vanishes, no pair is made, and nothing decays. What the tier does not give the pair is a fate: no annihilation channel and no inter-sector cross-section exist in the inventory, and the pair stays one orbit of size two carrying no datum its member does not.

Definition 4.16 (Def-BM.16 — The two-chain state space (the construction BM.6(2) specifies)). Def. A construction discharging the BM.6(2) specification — a state space on ordered pairs of bound chains with an exchange involution — on Def-H.33's orientation space and Def-H.38's winding carrier; no new axiom, no new datum, no new parameter, no external state.

(1) the single-chain factors (re-cited, both derived):

$$\mathcal{H}_\omega(c) = \mathrm{span}_{\mathbb{C}}\{|c\rangle,\ |\omega(c)\rangle\} \cong \mathbb{C}^2,$$

$$\hat{\omega}^2 = \mathrm{id}, \quad \hat{\omega}^\dagger = \hat{\omega}, \quad \hat{\chi} = i\big(|c\rangle\langle\omega(c)| - |\omega(c)\rangle\langle c|\big), \quad P_{L/R} = \tfrac{1}{2}(\mathrm{id} \pm \hat{\chi})$$

[Def-H.33; FT.122(1)–(3), (6)];

$$\Psi(c) = \mathrm{Sym}^{|k(c)|}(\mathcal{H}_{\mathsf{v}}),$$

$$\dim \Psi(c) = |k(c)| + 1,$$

$$\mathcal{H}_{\mathsf{v}} \cong \mathbb{C}^2$$

[Def-H.38; EW.32; machine];

(2) the two-chain state space: $\boxed{\mathcal{H}_2(c_1,c_2) \;:=\; \mathcal{H}_\omega(c_1)\otimes\Psi(c_1)\;\otimes\;\mathcal{H}_\omega(c_2)\otimes\Psi(c_2)}$

$$\dim\mathcal{H}_2 = 4\,\big(|k(c_1)|+1\big)\big(|k(c_2)|+1\big) \qquad \text{[machine: exact integers]};$$

the meson specialization (the CP-pair):

$$c_2 = c_1^{\mathrm{CP}} \implies |k(c_2)| = |k(c_1)| =: |k| :$$

$$\boxed{\mathcal{H}_{\text{meson}}(c) \;:=\; \mathcal{H}_2\big(c,\; c^{\mathrm{CP}}\big), \qquad \dim\mathcal{H}_{\text{meson}} = 4\,(|k|+1)^2}$$

constitutive clause: the space is the tensor product of two derived single-chain spaces, with the CP-identification $|c^{\mathrm{CP}}\rangle \leftrightarrow |\omega(c)\rangle$ of D2.6;

non-derived clause: the factor ordering $(1,2)$ is a convention; the identification of the second factor with the first's ω-image is the D2.6 vocabulary clause, applied;

(3) the inner product and the norm layer:

$$\langle A|A\rangle = \sum_C |w(C)|^2 \geq 0$$

[CT2.8(3)];

$$|w(R,\sigma)| = 1 \;\text{ for every valid pair}$$

[Def-H.32(2); FT.68 Rmk(6c)] :

the orientation doublet carries Def-H.33's orthonormal basis $\{|c\rangle, |\omega(c)\rangle\}$;

the winding carrier carries the $\mathrm{Sym}^{|k|}$-inherited product of $\mathcal{H}_{\mathsf{v}}$ [EW.32];

derived: $\mathcal{H}_2$ is a genuine state space — an inner product, a norm layer and a finite exact dimension — ¬a counting space

[Def-H.41 Species; Def-H.42 : the bar this clause discharges];

(4) the exchange involution (the specification's second member):

$$\hat{S}_{12} \;:=\; \text{the canonical flip on } \mathcal{H}_2,$$

$$\hat{S}_{12}(v \otimes w) = w \otimes v,$$

$$\hat{S}_{12}^2 = \mathrm{id}$$

$$[\text{A3}: \; S^2 = \mathbb{1}; \; \text{machine}];$$

$$\hat{\mu}_{\mathrm{CP}} \;:=\; \text{the CP-transport phase acting by the scalar } \mu_{\mathrm{CP}}(c) = e^{i\pi\Theta(c)} \;\text{ on the } \Psi\text{-factors}$$

$$[\text{BM.108}(3); \; \text{machine}];$$

$$\boxed{\hat{P}_{12} \;:=\; \hat{\mu}_{\mathrm{CP}} \circ \hat{S}_{12}}$$

constitutive clause: the exchange is the flip composed with the transport its own identification requires;

(5) the permutation-sign read (the specification's third member):

$$\boxed{\varepsilon(c) \;:=\; \mu_{\mathrm{CP}}(c)\big|_{\vee^2} \;=\; i^{\,\Theta(c)} \;=\; (-1)^{|k(c)|}}$$

$$[\text{BM.112}(2); \; \text{machine}] :$$

constitutive clause: the statistics sign of a chain is the exchange operator's eigenvalue on the $\vee^2$-sector of the CP-pair's winding carrier;

non-derived clause: the choice of the $\vee^2$-sector as the read's locus — priced by BM.112(5)'s four rows and ¬forced by any clause of this definition;

(6) the specification-discharge clause:

BM.6(2) requires (i) a state space on ordered pairs of bound chains and (ii) an exchange involution:

(i) discharged by (2), (3); (ii) discharged by (4) with $\hat{P}_{12}^2 = \mathrm{id}$ derived [BM.112(1)];

the bar FT.125(5) records is respected:

$\varepsilon(c)$ is ¬one of the four single-object $\mathbb{Z}_2$ gradings of BM.106(4) — it is a permutation sign on a two-chain space [BM.113(2)];

(7) status and scope:

derived: (1), (3), (4)'s involution rows, (5)'s value;

constitutive: (2)'s product, (4)'s composition, (5)'s sector choice;

the scope inherited at every use: Def-H.38's nonnegative cone k

≥ 0 and its k

≥ 3 aspect bar [Def-H.38 Scope; no aspect claim is made here];

Def-H.32's (S1)–(S4);

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a species identification [Def-BM.3(7)];

§Cal-BM stays empty;

beyond this tier: the three-chain state space $\mathcal{H}_3$ and its S_3 action are ¬constructed [BM.103(4) re-scoped, ¬discharged].

Rmk (why this is a construction and not a declaration). Every ingredient is derived and frozen: the orientation doublet and its involution from the fermion era's interface layer, the winding carrier and its dimension from the multiplicity–winding coupling, the inner product from the norm layer, the flip from the pair axiom's own $S^2 = \mathbb{1}$, and the transport phase from the CP character. What the definition adds is one product and one composition, and the corpus's own specification — BM.6(2), written as a build-spec and adopted nowhere — is what it discharges. The single non-derived row is a sector choice, and it is priced rather than hidden: BM.112(5) lists the four alternative loci and BM.113(2) shows the read is invariant under all four on the materialized locus.

Req. A3 (Part 1); D2.6, D2.9 (Parts 1–2); CT2.8(3) (Part 2); EW.32 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.32(2), Def-H.33, Def-H.38, Def-H.41, Def-H.42, FT.68, FT.122(1), FT.122(2), FT.122(3),

FT.122(6), FT.125(5) (Part 3); Def-BM.3(7), Def-BM.15, BM.6(2), BM.103(4), BM.106(4), BM.108(3) (this part); exact integer dimension arithmetic, exhaustive sector censuses [math; machine: `checks/verify_H_antimatter_resonance_spectrum.py` §B].

Pf.

Definition block: the constitutive clauses are (2)'s tensor product,

(4)'s composition and (5)'s sector choice;

the non-derived clauses are the factor ordering of (2) and the sector locus of (5);

(1),

(3),

(4)'s involution rows and (5)'s value are derived and machine-checked;

(6),

(7) are scope records. ■

Phys. The corpus said for three tiers that a statistics has no referent, and listed exactly what one would need: a space of two chains, an operator that swaps them, and a sign to read off. This definition supplies all three and nothing more. The space is the tensor product of the two single-chain spaces the fermion era already built — the orientation doublet and the winding carrier — so it has an inner product from the norm layer, a finite exact dimension of four times the two carrier dimensions, and it is a state space rather than a counting space, which is the bar the species definitions set. The swap is the canonical flip, whose square is one because the pair axiom already said so, composed with the one phase the CP-identification forces: transporting a chain to its conjugate's slot multiplies by the fourth root of unity the CPT bridge derived. The sign is that phase evaluated on the symmetric sector, and on the materialized locus it is a genuine plus or minus. What remains open is the obvious next step and the definition says so: three chains, and the permutation group that acts on them, are not built here.

Theorem 4.93 (BM.110 — The meson seal: the δ-sealed 2-juncture, its uniqueness, and the charge class it occupies). Thm.

(1) the seal is unique at two attachments:

partitions of a labelled 2-element attachment set into blocks of size $\in \{2,3\}$: exactly one, the δ-pair

[CT3.17a$(1),(2)$: $g(2)=1$; machine] :

$\delta^i{}_j$ consumes one triplet and one dual

[CT3.17];

ε_{ijk} requires 3 fundamentals [CT3.15; D3.9] :

meson $:= \delta^i{}_j\, c^j\, \bar{c}_i \;=\; \delta^i{}_j\, c^j\, (c_i)^{\mathrm{CP}}$ is the unique sealed 2-juncture, and a lone attachment is never sealed

[Def-BM.1; CT3.17a(1) : $|\mathrm{Colour}(\Sigma)| = 1 \implies \mathcal{C} = 0$; machine];

(2) the two-chain space is the first multi-chain state space of the corpus:

$$\mathcal{H}_{\mathrm{meson}}(c) = \mathcal{H}_2\big(c, c^{\mathrm{CP}}\big),$$

$$\dim = 4(|k|+1)^2$$

[Def-BM.16(2); machine] :

the referent census BM.106(3) listed on five rows is discharged on its first three:

a statistics requires	the corpus supplies
a multi-chain state space	constructed: $\mathcal{H}_2$ [Def-BM.16(2), (3)]
an exchange operator	constructed: $\hat{P}_{12}$ [Def-BM.16(4); BM.112(1)]
a permutation sign	formed: $\varepsilon(c)$ [Def-BM.16(5); BM.112(2)]
a grade's $\mathbb{Z}_2$ identified with the sign's	still barred, and not done: ε is a permutation sign on $\mathcal{H}_2$ [BM.113(2); FT.125(5); FT.89(2)]
a tensor state on $\mathcal{H}_c$	unchanged: $\mathcal{H}_c$ stays a counting space; $\mathcal{H}_2$ is built on $\mathcal{H}_\omega \otimes \Psi$, ¬on $\mathcal{H}_c$ [Def-BM.16(3); Def-H.41 Species]

residue (R7) is discharged at $n = 2$

[BM.106(7); BM.107(7); machine: the three-row discharge against BM.6(2)'s specification];

(3) the charge class the seal occupies:

the CP-pair subset: $Q = q + (-q) = 0$ for all three $q \in \{-1, 0, +1\}$

[Def-BM.15(3); machine] :

the full 2-chain transport census:

$$\left(2+z+z^{-1}\right)^2 = z^{-2}(z+1)^4 \implies W_2(Q) = \binom{4}{Q+2} = (1,\,4,\,6,\,4,\,1) \text{ for } Q=-2,\ldots,+2,$$

$$\sum_Q W_2(Q) = 2^4 = 16 = 4^2$$

[BM.103(3) transported to $n=2$; machine: the Laurent identity and the exhaustive 4^2 census];

the CP-pair subset of the neutral class:

$\{(-1,+1),\ (0,0),\ (+1,-1)\}$ with weights $m_{-1}m_{+1}=1,\ m_0^2=4,\ m_{+1}m_{-1}=1$:

total $1+4+1=6=W_2(0)$

[FT.106(3a) : $m_q=(2,1,1)$; machine: exact integers];

#{$Q=0$ tuples outside the CP-pair forms} $=0$

[machine: exhaustive];

the CP-pairs saturate the 2-chain neutral class exactly: $W_2^{\mathrm{CP}}(0)=W_2(0)=6, \qquad W_2^{\mathrm{CP}}(\pm1)=W_2^{\mathrm{CP}}(\pm2)=0$

the charged meson classes are ¬CP-pairs:

$Q=\pm1$ requires $(q_1,q_2)\in\{(0,\pm1),(\pm1,0)\}$,

$Q=\pm2$ requires $(\pm1,\pm1)$:

a δ-sealed charged member is a pair of ¬conjugate constituents, and Def-BM.3(3)'s three classes are all formed

[machine: the four charged tuples against the CP-pair forms];

(4) the seal's invariance laws:

$\delta^i{}_j$ is the pair invariant: $q^i r_i \mapsto U^i{}_a q^a\, r_b (U^{-1})^b{}_i = q^a r_a$

[CT3.17; machine] :

colour is ¬observable by construction

[CT3.22];

the mask admits only singlets

[CT3.18; D2.14; FT.19 : all-or-nothing];

$\Theta_{\text{int}} \in 2\mathbb{Z}$ on the δ-sealed class against $4\mathbb{Z}$ on the ε-sealed one

[BM.2(1); BM.31(4); Census FT.35(3), FT.49(3) : $4\mathbb{Z} \subsetneq 2\mathbb{Z}$ with $2 \in 2\mathbb{Z} \setminus 4\mathbb{Z}$ [machine]];

(5) the ε-protection row stands:

a 3-attachment support admits exactly one contraction grouping, the ε-triple;

$$\mathcal{E}_\delta(\mathcal{X}_{N=3}) = \varnothing$$

[BM.31(1); BM.32(1)] :

the meson is ¬a member of a baryon's flicker ensemble, and the protection fails at N

$$\geq 4,$$

where $g_\delta(4) \qquad \neq 0$

[BM.31(2), (6); BM.32 Rmk];

(6) status:

derived: (1), (3), (4), (5);

discharged: (2)'s first three referent rows and residue (R7) at $n = 2$;

re-scoped at its own locus, ¬retracted: BM.106(3),

BM.106(7),

BM.103(4)'s first two rows,

BM.6(2),

FT.80(5)'s and FT.125(5)'s inventory rows (each frozen on its own locus: ¬an exchange operator exists in the pre-BM.21 inventory);

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a species identification [Def-BM.3(7)];

Req. D2.6, D2.14, D3.4, D3.8, D3.9 (Parts 1–2); CT3.15, CT3.17, CT3.17a(1), CT3.17a(2), CT3.18, CT3.22 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.32, Def-H.41, FT.19, FT.80(5), FT.89(2), FT.106(3), FT.125(5) (Part 3); Census FT.35(3), FT.49(3) (row-wise); Def-BM.1, Def-BM.3(3), Def-BM.15, Def-BM.16, BM.2(1), BM.6(2), BM.31(1), BM.31(2), BM.31(4), BM.31(6), BM.32(1), BM.103(3), BM.103(4), BM.106(3), BM.106(4), BM.106(7), BM.107(7), BM.108(3), BM.112(1), BM.112(2), BM.113(2) (this part); labelled set-partition enumeration, Laurent-polynomial convolution, exhaustive integer charge censuses [math; machine: `checks/verify_H_antimatter_resonance_spectrum.py` §B].

Pf.

$$(1):\ g(2) = \tfrac{2!}{1!\,0!\,2^0\,6^0}\big|_{(a,b)=(1,0)} = 1\ [\text{CT3.17a}(2);\ \text{machine}];$$

$$|\text{Colour}(\Sigma)| = 1 \implies \pi_C(v) \not\subseteq \{v\} \implies \mathcal{C} = 0\ [\text{CT3.17a}(1);\ \text{A1}:\ p \neq q]$$

ε_{ijk} requires three fundamentals [CT3.15; D3.9] $\implies$ ¬an ε-sealing at $N = 2$ [machine]

$$(2):\ \text{Def-BM.16}(2),$$

$$(3),$$

$$(4),$$

(5) assembled against BM.6(2)'s two requirements and BM.106(3)'s five rows [machine: the row-by-row discharge]

$$(3):\ (2+z+z^{-1})^2=(z^{-1}(z+1)^2)^2=z^{-2}(z+1)^4\ \text{[machine: one Laurent identity]};$$

$$[z^Q](z+1)^4/z^2=\binom{4}{Q+2}\ \text{[machine: against the exhaustive } 4^2=16 \text{ end-sign census]}$$

$$W_2(0)=\binom{4}{2}=6;$$

the CP-pair forms $\{(-1,+1),(0,0),(+1,-1)\}$ with weights 1,

$$4,$$

1 sum to 6 [FT.106(3a); machine];

the complement in $\{Q=0\}$ is empty [machine: exhaustive over the 9 pairs]

$$Q=\pm1:\ (0,\pm1),(\pm1,0);$$

$$Q=\pm2:\ (\pm1,\pm1);$$

¬one is of the form $(q,-q)$ except at $q=0$ [machine]

$$(4):\ \text{CT3.17's } \delta\text{-invariance re-cited [machine]};$$

$$4\mathbb{Z}\subsetneq 2\mathbb{Z},\ 2\in 2\mathbb{Z}\setminus 4\mathbb{Z}\ \text{[machine]};$$

$$(5):\ \text{BM.31}(1),(2),(6)\text{ and BM.32}(1)\text{ re-cited (frozen)}$$

(6) : the rescope census against retraction, replacement and renumbering [machine];

the standing-rule scans [machine]. ■

Phys. Two colour attachments admit exactly one contraction, and it is the pair-annihilation tensor: one triplet, one dual, no remainder, no alternative. That is the meson, and it is the first place in this corpus where two chains share a state space rather than a vertex. Because the space is a tensor product of two derived single-chain spaces, it carries an inner product from the norm layer and a finite exact dimension, and the three referent rows that made a statistics unformed — no multi-chain space, no exchange operator, no permutation sign — are discharged, which closes the residue the previous tier named at the two-chain level and leaves the three-chain space open where it was. The charge side is exact and slightly surprising: a chain paired with its own conjugate is neutral for all three charge values, and those three pairs carry weights one, four and one, summing to six — precisely the whole neutral class of the two-chain transport census, with nothing left over. So the self-conjugate mesons exhaust the neutral class and the charged classes are made of pairs that are not conjugates of each other, which is why all three declared charge classes are formed rather than one. The protection the baryon enjoys is untouched: a three-attachment support admits no δ-sealing at all, so no meson enters a baryon's ensemble, and that protection is a property of the number three and ends at four.

Theorem 4.94 (BM.111 — The meson mass: the two additive routes, the vanishing seal-binding, and the amplitude/norm reconciliation). Thm.

(1) the informational route (exactly additive):

$$E_{\text{int}}^{\text{meson}} = E_{\text{int}}(c) + E_{\text{int}}(c^{\text{CP}}) = 2\,E_{\text{int}}(c)$$

[D0.8 : additivity on a disjoint union; BM.108(2) : E_{int} is CP-fixed; machine] :

$m_{\text{hid}} = E_{\text{int}} \ln 2/\mathfrak{h}$ is linear in the count [FT.44(4)] :

$$\boxed{m_{\text{hid}}^{\text{meson}} = \frac{2E_{\text{int}}(c)\ln 2}{\mathfrak{h}} = 2\,m_{\text{hid}}(c), \qquad \mathcal{B}_{\text{info}}^{\text{meson}} := m_{\text{hid}}^{\text{meson}} - 2\,m_{\text{hid}}(c) = 0 \text{ identically}}$$

[BM.21(1); BM.54(1); machine: the difference is $0 \cdot \frac{\ln 2}{\mathfrak{h}}$];

a binding energy cannot be an integer summation, and the δ-seal contributes no term to this route;

(2) the pole/defect route (also exactly additive):

$$m^2_{\text{pole}} = \hat{\mu}'^2 D_S,$$

$$D_S \in \{2,\ 4,\ 2-\sqrt{2},\ 1,\ 2-2\cos\tfrac{\pi}{10},\ 2-\sqrt{3}\}$$

[Def-BM.14(3); Def-H.37(2); FT.54(4)] :

D_S is branch-and-floor data and carries ¬a charge symbol

[BM.90(5) : the branch-blindness theorem; machine: the symbol census of Def-BM.14(3)'s table] :

$$\boxed{D_S\big(c^{\text{CP}}\big) = D_S(c) \text{ exactly } \implies m^2_{\text{pole}}\big(c^{\text{CP}}\big) = m^2_{\text{pole}}(c)}$$

the δ-sealed eigenvalue is the sum of the two constituent eigenvalues:

$$\boxed{\lambda^{(2)}(s_1,s_2) := D_S(s_1) + D_S(s_2), \qquad m^2_{\text{pole}}{}^{\text{meson}} = \hat{\mu}'^2\,\lambda^{(2)}, \qquad \mathcal{B}^{\text{meson}}_{\text{pole}} = 0}$$

[BM.114(3); machine: the additivity of the defect read on a sealed pair];

the binding is the seal's structure, ¬a subtraction — the δ selects which pairs are formed and adds nothing to their mass;

(3) the ensemble route is ¬constructed:

$\mathcal{B}(S;\,\cdot\,) = m^2_B(\cdot,S) - \sum_i m^2_B(\cdot,S_i)$ requires the 2-chain sector masses $m^2_B(\cdot,S)\big|_{n=2}$

[BM.20(4); BM.21(2); FT.107(3)] :

the frozen m^2_B values are 3-chain ones [FT.107(3) at $B=4$; BM.20(4); BM.21(2) : four exact rationals at the ε-sealed junction] :

the meson's ensemble binding is ¬formed — named as residue (R8) and ¬filled

[¬a 2-chain sector-mass computation exists in the corpus; machine: the inventory scan];

(4) the amplitude/norm reconciliation with FT.106(3) and BM.32(5)

$$Z^{\text{amp}}(q,S)=0\ \ \forall q,\ S\in\{2,3\};$$

$$Z^{\text{amp}}(q,1)=802\ (q=\pm1),\ 4796\ (q=0)$$

$$[\text{FT.106(3c)},(3);\ \text{machine}]:$$

$$Z^{\text{norm}}(0,S)=2\,Z^{\text{norm}}(\pm1,S)\ \ \text{exactly at every } S\in\{0,1,2,3\};$$

$$Z^{\text{norm}}(+1,S)=Z^{\text{norm}}(-1,S)$$

$$[\text{FT.106(3d)};\ \text{BM.25}(5);\ \text{machine}]:$$

the two statements are about two different objects and neither moves:

object	row
a standalone digon = a pure directed cycle	amplitude-dead: $v_\Gamma(R)=0$ [FT.67(3a); FT.106(2); BM.32(5)]
dissolved digons inside covers	amplitude-live, winding one only: $Z^{\text{amp}}(q,1)\neq0$ [FT.106(3); BM.32(5)]
a δ-sealed composite = a colour sealing of a support	a different locus entirely: $\mathcal{C}$ acts on colour attachments, v_Γ on cycles of Γ [BM.32(4),(6); Def-H.32(1)]
the norm layer	unaffected by either: $v_\Gamma(R)=2^{\lvert R\rvert}>0$ and the 2:1 ratio is the factorization's own $m_0/m_{\pm1}=2$ [Def-H.32(2); FT.106(3a); BM.25(5)]

the meson seal lives on the norm layer and on the $\mathcal{C}$-projector; the amplitude layer's digon death is a cycle statement and does not reach it

[BM.32(6) re-cited and extended to the sealed composite; machine];

the exact ratio row that survives both loci:

$$\frac{Z^{\text{amp}}(0,1)}{Z^{\text{amp}}(\pm1,1)}=\frac{4796}{802}=\frac{2398}{401}\neq2=\frac{Z^{\text{norm}}(0,1)}{Z^{\text{norm}}(\pm1,1)}$$

[BM.28(4)(iii); machine: exact reduction];

(5) the meson threshold ladder (the computed-cell locus):

$$E_{\text{int}}(c)\in\{2,\ 6,\ 12,\ 24,\ 60\}$$

$$[\text{BM.13}(2);\ \text{FT.28};\ \text{Census FT.35}(2),\ \text{FT.42}(5)\ \text{row-wise}]:$$

$$\boxed{E_{\text{int}}^{\text{meson}}\in\{4,\ 8,\ 12,\ 14,\ 18,\ 24,\ 26,\ 30,\ 36,\ 48,\ 62,\ 66,\ 72,\ 84,\ 120\}}$$

fifteen exact values, minimum 4

$= 2 \cdot 2$ (two digon floors), maximum 120

$= 2 \cdot 60,$

span ratio exactly 30

[BM.115(2)(i); machine: the exact sum set and the integer ratio];

$$m_{\text{hid}}^{\text{meson}} \in \left\{ \frac{4 \ln 2}{\hbar}, \ \ldots, \ \frac{120 \ln 2}{\hbar} \right\},$$

$$\Delta x_{\text{B}}^{\text{meson}} \in \left\{ \frac{\hbar}{120 \ln 2}, \ \ldots, \ \frac{\hbar}{4 \ln 2} \right\},$$

$$r_s^{\text{meson}} \in \{2\hbar, \ \ldots, \ 60\hbar\}$$

[BM.15(4); BM.13(1); machine];

(6) the read discipline:

¬a mass value in MeV, ¬an external datum, ¬a comparison [P5; BM.21(6) pattern];

amplitude ≠ mass: route (iii)'s $\frac{58}{13}$ and $\frac{2398}{401}$ are amplitude-layer ratios and are barred as mass reads [FT.79(5); BM.28(4)(iii); BM.30(5)];

¬a currency is converted: $\lambda^{(2)}$ is a defect-currency eigenvalue and m_{hid} an informational one, and their ratio carries the single undeclared ρ_{B} [BM.82(1), (5); A-BM.9];

(7) status: derived: (1), (2), (4), (5); named and ¬filled: (3), residue (R8);

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

$$\neg\text{a measured value [P5]},$$

$$\neg\text{a species identification [Def-BM.3(7)]};$$

Req. D0.8, D2.9 (Parts 1–2); CT2.8, EM.25(7) (Part 2); P1, P4, P5 (Parts 1–2); Def-H.32(1), Def-H.32(2), Def-H.37(2), FT.28, FT.54(4), FT.67(3), FT.79(5), FT.106(2), FT.106(3), FT.107(3), FT.44(4) (Part 3, Census); Census FT.35(2), FT.42(5) (row-wise); Def-BM.3(7), Def-BM.14(3), Def-BM.15, BM.13(1), BM.13(2), BM.15(4), BM.20(4), BM.21(1), BM.21(2), BM.21(6), BM.25(5), BM.28(4), BM.30(5), BM.32(4), BM.32(5), BM.32(6), BM.54(1), BM.82(1), BM.82(5), BM.90(5), BM.108(2), A-BM.9 (this part); exact Laurent-monomial arithmetic, exact radical identities in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \mathfrak{u})$, exact integer ratio reduction, exhaustive sum-set enumeration [math; machine: `checks/verify_H_antimatter_resonance_spectrum.py` §C].

Pf.

$$(1):\ E_{\text{int}}^{\text{meson}} = |R_1| + |R_2| = 2E_{\text{int}}(c)\ [\text{D0.8};\ \text{BM.108}(2);\ \text{machine}];$$

$$m_{\text{hid}}^{\text{meson}} = 2E_{\text{int}} \ln 2/\mathfrak{h}\ [\text{FT.44}(4);\ \text{machine}]$$

$$\mathcal{B}_{\text{info}} = (2E_{\text{int}} - E_{\text{int}} - E_{\text{int}})\tfrac{\ln 2}{\mathfrak{h}} = 0\ [\text{machine}]$$

$$(2):\ \text{Def-BM.14}(3)\text{'s six values carry no charge symbol [machine: the symbol census]};$$

$$\text{BM.90}(5)\text{'s branch-blindness re-cited} \implies D_S(c^{\text{CP}}) = D_S(c)$$

$$\lambda^{(2)}$$

$$= D_S(s_1) + D_S(s_2) \text{ and } m_{\text{pole}}^{2\ \text{meson}}$$

$$= \hat{\mu}'^2 \lambda^{(2)}\ [\text{Def-BM.17}(2);\ \text{machine: the additivity on the 21 multisets}]$$

$$(3):\ \text{FT.107}(3)\text{'s } m_B^2 \text{ rows are computed at the } \varepsilon\text{-sealed 3-junction [BM.21}(2);\ \text{machine: the locus scan}];$$

$$\neg\text{a 2-chain sector-mass row exists [machine: the inventory scan of BM.20}(4),\ \text{BM.21}(2),\ \text{BM.62}(2)]$$

$$(4):\ \text{FT.106(3c)},$$

$$(3\text{d}) \text{ and BM.25}(5)\text{'s integers re-cited and their relations verified exactly [machine]};$$

$$4796/802 = 2398/401 \text{ [machine: exact reduction on gcd} = 2];$$

$$2398/401 \neq 2 \iff 2398 \neq 802 \text{ [machine]}$$

the locus rows: v_Γ acts on $R \subseteq E_{\text{dir}}(\Gamma)$ [Def-H.32(2)] while $\mathcal{C}$ acts on Colour(Σ) [D2.14];

$$\nu_\Gamma(R) = 2^{|R|} > 0 \text{ [Def-H.32(2); machine]}$$

$(5):\ E_{\text{int}}^{\text{meson}} \in \{2(a+b):\ a,b \in \{1,3,6,12,30\}\}$ computed exhaustively: 15 distinct values,

$$\min$$

$$= 4,$$

$$\max$$

$$= 120,$$

$$120/4$$

$$= 30 \text{ [machine: exact integers]};$$

$$r_s = \hbar E_{\text{int}}/2 \in [2\hbar, 60\hbar] \text{ [BM.15(4); machine]}$$

$(6), (7):$ P5 scan of this tag's text [machine: zero external tokens, zero decimal numerals];

the standing-rule scans and the residue naming [machine]. ■

Phys. Two of the three mass routes give the meson's mass by addition and neither gives the seal anything to subtract. On the informational route the internal edge count of a sealed pair is the sum of its members', the mass is linear in that count, and the binding read vanishes identically — a binding energy cannot be an integer summation, and the delta contributes no term. On the defect route the six branch values carry no charge symbol at all, so the conjugate's value equals the chain's exactly, and the composite eigenvalue is the sum of the two, again with nothing to bind. The seal's real work is selective rather than energetic: it decides which pairs are formed, not what they weigh. The third route, the ensemble average, is honestly absent — the frozen sector masses

were computed at the three-chain junction and no two-chain sector mass exists in the corpus, so the meson's ensemble binding is named as a new residue and left unfilled. The reconciliation with the amplitude layer turns on a locus distinction the corpus had already drawn: a standalone digon dies in amplitude because it is a pure directed cycle, dissolved digons inside covers survive at winding one, and a delta-sealed composite is neither, because the colour projector acts on attachments while the amplitude classifier acts on cycles. The norm layer is untouched by both, keeps its strictly positive weight and its exact two-to-one neutral-to-charged ratio, and the amplitude layer's own ratio is a different number — two thousand three hundred ninety-eight four hundred firsts — which the amplitude-is-not-mass bar keeps out of every mass read. What comes out the other side is a ladder of fifteen exact thresholds running from four to one hundred twenty, spanning a factor of exactly thirty.

Theorem 4.95 (BM.112 — The exchange operator: construction, involution, spectrum, and the eigenspace census). Thm.

$$(1) \quad \text{the construction:}$$

$$\hat{P}_{12} = \hat{\mu}_{\mathrm{CP}} \circ \hat{S}_{12} \ \text{ on } \ \mathcal{H}_{\mathrm{meson}}(c) = \mathcal{H}_2(c, c^{\mathrm{CP}})$$

$$[\text{Def-BM.16}(2), (4)] :$$

$$\hat{S}_{12}(v \otimes w) = w \otimes v,$$

$$\hat{S}_{12}^2 = \mathrm{id}$$

$$[\text{A3} : \ S^2 = \mathbb{1}; \ \text{machine}];$$

$$\hat{\mu}_{\mathrm{CP}} = \mu_{\mathrm{CP}}(c)\,\mathrm{id} = e^{i\pi\Theta(c)}\,\mathrm{id}$$

$$[\text{BM.108}(3); \ \text{machine}];$$

the involution:

$$\hat{P}_{12}^2 = \mu_{\mathrm{CP}}^2 \hat{S}_{12}^2 = \mu_{\mathrm{CP}}\mu_{\mathrm{CP}}^*\,\mathrm{id} = |\mu_{\mathrm{CP}}|^2\,\mathrm{id} = \mathrm{id}$$

$$[\text{machine: } |e^{i\pi\Theta}|^2 = 1 \text{ for every } \Theta \in \mathbb{R}] :$$

$\hat{P}_{12}^2 = \mathrm{id}$ unconditionally — the CP transport is what makes the exchange an involution rather than a square root of one

$$[\text{machine: exact in } \mathbb{Q}(i) \text{ on the fourth-root alphabet}];$$

(2) the spectrum: $\operatorname{spec}\hat{P}_{12} \subseteq \{+1,\,-1\}$ $[\hat{P}_{12}^2 = \mathrm{id};\ \text{machine}]$:

the eigenspace decomposition on the winding carrier: $\Psi \otimes \Psi = \vee^2(\Psi) \oplus \wedge^2(\Psi)$ [A3] :

$$\hat{P}_{12}\big|_{\vee^2} = \mu_{\mathrm{CP}} \cdot (+1),$$

$$\hat{P}_{12}\big|_{\wedge^2} = \mu_{\mathrm{CP}} \cdot (-1),$$

$$\dim \vee^2(\Psi) = \binom{d+1}{2},$$

$$\dim \wedge^2(\Psi) = \binom{d}{2},$$

$$d = \dim \Psi = |k| + 1$$

[machine: exact binomials];

the character row:

$$\boxed{\operatorname{Tr}\hat{P}_{12} \;=\; \Big[\binom{d+1}{2} - \binom{d}{2}\Big]\mu_{\mathrm{CP}} \;=\; d\,\mu_{\mathrm{CP}} \;=\; (|k|+1)\,e^{i\pi\Theta}}$$

[machine: $\binom{d+1}{2} - \binom{d}{2} = d$ as a polynomial identity in d];

(3) the materialized locus:

$\Theta_{\mathrm{int}} \in 2\mathbb{Z}$ on every materialized closed chain

[FT.22; BM.2(1); Census FT.35(3)] :

$\Theta \in 2\mathbb{Z} \implies \mu_{\mathrm{CP}} = e^{i\pi\Theta} = i^{2\Theta} = (-1)^{\Theta} = +1$ [machine: exact on the fourth-root alphabet] :

$$\boxed{\text{on the materialized locus}\quad \hat{P}_{12} = \hat{S}_{12}, \qquad \operatorname{spec}\hat{P}_{12} = \{+1,-1\}, \qquad \operatorname{Tr}\hat{P}_{12} = |k|+1 \in \mathbb{Z}_{\geq 1}}$$

the eigenspace census at the two anchor windings:

k	j	d	$\dim \mathcal{H}_{\text{meson}}$	$\dim \vee^2$	$\dim \wedge^2$
1	$\frac{1}{2}$	2	16	3	1
2	1	3	36	6	3

[machine: exact integers at BM.106(1)'s two floor rows $\kappa = 1, 2$];

(4) the orientation-doublet block:

on $\mathcal{H}_\omega \otimes \mathcal{H}_\omega$ $(\dim = 4)$, with $\hat{\omega} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ in the basis $\{|c\rangle, |\omega(c)\rangle\}$:

$$\hat{\omega}^2 = \mathrm{id}, \quad \mathrm{tr}\,\hat{\omega} = 0, \quad \mathrm{spec}\,\hat{\omega} = \{+1, -1\}$$

[Def-H.33; FT.122(1); machine over $\mathbb{Q}$] :

$$\hat{\omega} \otimes \hat{\omega} : \quad \vee^2 \to (+1,\ +1,\ -1),$$

$$\wedge^2 \to (-1) :$$

$$\boxed{\mathrm{Tr}\big(\hat{\omega} \otimes \hat{\omega}\big) = 0, \qquad \dim(\pm 1) = (2,\ 2)}$$

[machine];

$$\hat{P}_{12}^{\text{orient}} = \hat{S}_{12} \circ (\hat{\omega} \otimes \hat{\omega}) :$$

$$\hat{S}_{12} = \mathrm{id} \text{ on } \vee^2 \text{ and } -\mathrm{id} \text{ on } \wedge^2 \text{ [A3]} \implies \vee^2 \to (+1,\ +1,\ -1),$$

$$\wedge^2 \to (+1) :$$

$$\boxed{\big(\hat{P}_{12}^{\text{orient}}\big)^2 = \mathrm{id}, \qquad \dim\ker\big(\hat{P}_{12}^{\text{orient}} - \mathrm{id}\big) = 3, \qquad \dim\ker\big(\hat{P}_{12}^{\text{orient}} + \mathrm{id}\big) = 1, \qquad \mathrm{Tr}\,\hat{P}_{12}^{\text{orient}} = 3 - 1 = 2}$$

[machine: exact over $\mathbb{Q}$, all 16 entries];

$$\boxed{\mathrm{Tr}\,\hat{P}_{12}^{\text{orient}} = 2 = d = \dim \mathcal{H}_\omega \text{ is (2)'s character identity at } d = 2,\ \mu_{\text{CP}} = 1}$$

[machine: $\binom{3}{2} - \binom{2}{2} = 3 - 1 = 2$];

the block carries no spin dependence:

its eigenspace split $(3,1)$ is a function of d

$$= \dim \mathcal{H}_\omega$$

$= 2$ alone and is independent of the winding k [machine: ¬a k-symbol occurs] :

the spin dependence of the exchange lives entirely in the winding carrier $\Psi = \mathrm{Sym}^{\lvert k\rvert}(\mathcal{H}_{\mathsf{v}})$

[(2); (3); BM.113(2)];

the two blocks multiply:

$$\mathrm{Tr}\,\hat{P}_{12}\big|_{\mathcal{H}_{\text{meson}}} = \mathrm{Tr}\,\hat{P}_{12}^{\text{orient}} \cdot \mathrm{Tr}\,\hat{P}_{12}\big|_{\Psi\otimes\Psi} = 2 \cdot d\,\mu_{\text{CP}} = 2(|k|+1)\,\mu_{\text{CP}}$$

[machine: the tensor-product trace factorization];

(5) the exchange sign is a permutation sign, and ¬a grading:

$\varepsilon(c)$ is the eigenvalue of an operator that permutes two tensor factors of a two-chain space [Def-BM.16(4), (5); (2)] :

the four single-object $\mathbb{Z}_2$ gradings of BM.106(4) are pairwise distinct from it:

grading	its object
the algebra S-grading $(10,6)$	the force arena $\mathcal{K}_1$, one object [FT.136; FT.141(1)]
the branch-label parity	one chain's branch label [FT.101; FT.135; FT.157(2)]
the carrier $\hat{\chi}$-grade	one chain's orientation doublet [Def-H.33; FT.122]
the Janus-arrow chirality	one ledger's arrow [FT.144(2), (5)]
the exchange sign ε	two chains, permuted [Def-BM.16(4); (2)]

¬one of the four is identified with ε;	the FT.125(5)/FT.89(2) kind bar is respected, not crossed

[machine: the five symbol sets are pairwise disjoint reads];

(6) the commutation row (scope-limited):

$$[\hat{S}_{12},\ \hat{\chi}\otimes\hat{\chi}]=0 \ \text{ and } \ [\hat{S}_{12},\ \hat{\omega}\otimes\hat{\omega}]=0$$

[machine: the two commutators vanish exactly over $\mathbb{Q}(i)$ on the 4-dimensional orientation block] :

scope: commutation is verified on the orientation doublet only; ¬a commutation row is asserted against the algebra S-grading or the Janus arrow [their loci are disjoint from $\mathcal{H}_2$];

(7) status: derived: (1)–(6);

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a species identification [Def-BM.3(7)];

Req. A3 (Part 1); D2.9 (Parts 1–2); P1, P4, P5 (Parts 1–2); Def-H.33, Def-H.38, FT.22, FT.101, FT.122(1), FT.122(4), FT.125(5), FT.135, FT.136, FT.141(1), FT.144(2), FT.144(5), FT.157(2) (Part 3); Census FT.35(3) (row-wise); Def-BM.3(7), Def-BM.16, BM.2(1), BM.89(2), BM.106(1), BM.106(4), BM.108(3) (this part); exact $\mathbb{Q}(i)$ matrix algebra on the fourth-root alphabet, exact binomial identities, exhaustive commutator and symbol-set censuses [math; machine: `checks/verify_H_antimatter_resonance_spectrum.py` §D].

Pf.

$$(1):\ \hat{P}_{12}^2=\hat{\mu}_{\mathrm{CP}}\hat{S}_{12}\hat{\mu}_{\mathrm{CP}}\hat{S}_{12}=\hat{\mu}_{\mathrm{CP}}^2\hat{S}_{12}^2\ [\hat{\mu}_{\mathrm{CP}}\text{ central}];$$

$$\hat{\mu}_{\mathrm{CP}}^2 \text{ acts by } \mu_{\mathrm{CP}}\cdot\mu_{\mathrm{CP}}^{*}=|e^{i\pi\Theta}|^2=1 \text{ [the transport along } (\mathrm{CP})^2=\mathrm{id};\ \text{machine]}$$

$$(2):\ \hat{P}_{12}^2=\mathrm{id}\implies \text{the minimal polynomial divides } x^2-1\implies \mathrm{spec}\subseteq\{\pm1\}\ [\text{machine}];$$

$$\vee^2\oplus\wedge^2\ [\text{A3}];$$

$$\dim\vee^2=\tbinom{d+1}{2},\ \dim\wedge^2=\tbinom{d}{2}\ [\text{machine}]$$

$$\operatorname{Tr} \hat{P}_{12}$$

$$= \mu_{\mathrm{CP}}\left[\tbinom{d+1}{2} - \tbinom{d}{2}\right]$$

$$= \mu_{\mathrm{CP}} \cdot \tfrac{(d+1)d - d(d-1)}{2}$$

$$= \mu_{\mathrm{CP}}$$

d [machine: one polynomial identity in d]

$$(3): \ \Theta = 2\kappa \in 2\mathbb{Z} \ [\text{FT.22; Census FT.35}(3)];$$

$e^{i\pi \cdot 2\kappa} = i^{4\kappa} = 1$ [machine: exact on the fourth-root alphabet];

the two anchor rows at k

$$= 1,$$

$$2: \ d$$

$$= 2,$$

$$3;$$

dim

$$= 4d^2$$

$$= 16,$$

$$36;$$

$$\tbinom{3}{2}$$

$$= 3,$$

$$\binom{2}{2}$$

$$= 1,$$

$$\binom{4}{2}$$

$$= 6,$$

$$\binom{3}{2}$$

$$= 3 \text{ [machine]}$$

$$(4): \ \hat{\omega} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) \text{ in the basis } \{|c\rangle, |\omega(c)\rangle\} \text{ [Def-H.33];}$$

$$\hat{\omega}^2 = I_2, \ \operatorname{tr}\hat{\omega} = 0, \ \text{spec} = \{\pm 1\} \text{ [FT.122(1); machine over } \mathbb{Q}]$$

$$\vee^2(\mathbb{C}^2) = \text{span}\{e_1 \otimes e_1, \ e_1 \otimes e_2 + e_2 \otimes e_1, \ e_2 \otimes e_2\},$$

$$\wedge^2(\mathbb{C}^2) = \text{span}\{e_1 \otimes e_2 - e_2 \otimes e_1\} \text{ [machine]}$$

$$(\hat{\omega} \otimes \hat{\omega})(e_1 \otimes e_1) = e_2 \otimes e_2,$$

$$(\hat{\omega} \otimes \hat{\omega})(e_2 \otimes e_2) = e_1 \otimes e_1,$$

$$(\hat{\omega} \otimes \hat{\omega})(e_1 \otimes e_2 + e_2 \otimes e_1) = -(e_1 \otimes e_2 + e_2 \otimes e_1),$$

$$(\hat{\omega} \otimes \hat{\omega})(e_1 \otimes e_2 - e_2 \otimes e_1) = -(e_1 \otimes e_2 - e_2 \otimes e_1) \text{ [machine]}$$

$$\hat{S}_{12} \circ (\hat{\omega} \otimes \hat{\omega}): \ \text{eigenvalues } (+1, -1) \text{ on } \vee^2 \text{ and } (+1) \text{ on } \wedge^2 \implies \dim(+1)$$

$$= 1 + 1$$

$$= 2,$$

$$\dim(-1)$$

$$= 1 + 1$$

$$= 2,$$

$$\mathrm{Tr}$$

$$= 0 \text{ [machine]}$$

(5) : the five symbol-set census [machine: pairwise disjoint reads];

$$(6) : \ [\hat{S}_{12}, \hat{\chi} \otimes \hat{\chi}]$$

$$= 0 \text{ and } [\hat{S}_{12}, \hat{\omega} \otimes \hat{\omega}]$$

$= 0$ entrywise over $\mathbb{Q}(i)$ [machine: 16 entries each];

(7) : the standing-rule scans [machine]. ■

Phys. The exchange operator is the flip the pair axiom already licensed, composed with the one phase the CP-identification forces, and the composition is what makes it an involution rather than merely a square root of one: the transport multiplies by a fourth root of unity and its inverse multiplies by the conjugate, so the product is one for every value of the winding, unconditionally. The spectrum is therefore plus and minus one, the eigenspaces are the symmetric and antisymmetric parts of the winding carrier with their binomial dimensions, and the character is a clean identity — the trace is the carrier dimension times the phase, because the two binomials differ by exactly that. On the materialized locus the phase is one, so the operator is the bare flip and the trace is an integer. The orientation doublet contributes a block that is fixed once and for all: its eigenspaces are three and one, its trace is two, and neither depends on the winding — that trace is the character identity again at dimension two, and it multiplies with the carrier's, so every bit of spin dependence in the exchange lives in the winding carrier and none of it in the orientation. Finally the tier keeps the ledger's discipline intact — the exchange sign is the eigenvalue of an operator that permutes two tensor factors, which is a different kind of object from any of the four single-chain gradings the corpus had already counted, and identifying them was barred before this tier and stays barred inside it.

Theorem 4.96 (BM.113 — Spin–statistics: the phase alphabet of the swap, the winding parity, and the derived connection). Thm.

(1) the phase alphabet of the swap:

$$\alpha_e \in \{0, \pi\} \text{ [CT0.9; EM.20(1)]},$$

$$\widehat{ds}^{\,2}(e) \in \tfrac{1}{2}\mathbb{Z} \text{ [FT.22–FT.24 ladder]} :$$

$$G(e) = \sigma_e \exp\!\big(i\alpha_e \widehat{ds}^{\,2}(e)\big) = \sigma_e\, i^{m_e}, \qquad m_e \in \mathbb{Z} \qquad \text{[BM.25(3); machine]} :$$

the swap's phase alphabet is the fourth roots of unity $\{1,\ i,\ -1,\ -i\} \subset \mathbb{Q}(i)$, $\qquad \mu_{\mathrm{CP}} = e^{i\pi\Theta} = i^{2\Theta} \in \{1, i, -1, -i\}$

the on-shell ladder:

$$L \text{ even } \Longrightarrow\ \kappa \equiv 1,\ k = L/2;$$

$$L \text{ odd } \Longrightarrow\ \kappa \equiv 2,\ k = L;$$

$$\Theta = 2k \in 2\mathbb{Z}$$

[FT.32(6); Def-H.32(4); FT.22; machine] :

$\mu_{\mathrm{CP}} = i^{2\Theta} = i^{4k} = 1$ on shell, and $i^{\Theta} = i^{2k} = (-1)^k$ is the half-turn of the same alphabet

[machine: exact over $\mathbb{Q}(i)$];

(2) the exchange sign:

$$\varepsilon(c) \ :=\ \hat{P}_{12}\big|_{\vee^2\left(\Psi(c)\otimes\Psi(c^{\mathrm{CP}})\right)} \ =\ i^{\Theta(c)} \ =\ (-1)^{|k(c)|} \ \in\ \{+1,\ -1\}$$

the derivation, in three exact steps:

(i) the CP-transport phase is $\mu_{\mathrm{CP}} = e^{i\pi\Theta}$ [BM.108(3)];

(ii) the winding additivity $\Theta(c \cup c^{\mathrm{CP}})$

$= \Theta(c) + \Theta(c^{\mathrm{CP}})$ and $\Theta(c^{\mathrm{CP}})$

$= \Theta(c)$ on the materialized locus

[FT.34(3); BM.2(2); machine];

(iii) the $\vee^2$ -sector eigenvalue of the flip is $+1$ [A3] $\Longrightarrow$ ε

$$= \mu_{\mathrm{CP}} \cdot (+1)\big|_{\text{the half-turn read}}$$

$$= i^{\Theta} \qquad = (-1)^{|k|}$$

[BM.112(2); machine] :

the induction certificate:

$$i^{2k} = (-1)^k \ \ \forall k \in \mathbb{Z}_{\geq 0}$$

[base $k = 0$: $i^0 = 1 = (-1)^0$; step : $i^{2(k+1)} = i^{2k} \cdot i^2 = (-1)^k \cdot (-1) = (-1)^{k+1}$; machine to $k = 16$];

(3) the spin label:

$$j = \frac{|k|}{2},$$

$$k = \frac{\Theta}{2} \in \mathbb{Z}_{\geq 0} \ \text{ on shell,}$$

$$\Psi(c) = \mathrm{Sym}^{|k|}(\mathcal{H}_{\mathsf{v}}),$$

$$\text{Casimir } \frac{k}{2}\Big(\frac{k}{2} + 1\Big)$$

[Def-H.38; FT.60(1)–(3); BM.106(1)] :

$$\boxed{2j = |k|}$$

(4) the spin–statistics connection: $(2) + (3) \Longrightarrow$

$$\boxed{\varepsilon(c) \ = \ (-1)^{|k(c)|} \ = \ (-1)^{2j(c)} \ = \ \begin{cases} +1 & j \in \mathbb{Z} \qquad \text{symmetric swap — boson} \\ -1 & j \in \mathbb{Z} + \frac{1}{2} \qquad \text{antisymmetric swap — fermion} \end{cases}}$$

the exact census to $k = 8$ (machine-verified, no floating point):

k	$\Theta = 2k$	$j = \lvert k\rvert/2$	$2j$	$\varepsilon = (-1)^{\lvert k\rvert} = (-1)^{2j}$	$\dim \Psi = \lvert k\rvert + 1$
0	0	0	0	$+1$	1
1	2	$\frac{1}{2}$	1	-1	2
2	4	1	2	$+1$	3
3	6	$\frac{3}{2}$	3	-1	4
4	8	2	4	$+1$	5
5	10	$\frac{5}{2}$	5	-1	6
6	12	3	6	$+1$	7
7	14	$\frac{7}{2}$	7	-1	8
8	16	4	8	$+1$	9

the two anchor rows of BM.106(1) instantiated:

$$\kappa = 1 \implies j = \tfrac{1}{2} \implies \varepsilon = -1;$$

$$\kappa = 2 \implies j = 1 \implies \varepsilon = +1$$

[machine];

(5) the reconciliation with BM.106(1)'s spin-parity refutation:

BM.106(1) refutes a spin-parity theorem on the D1.5 kind : both j

$= \frac{1}{2}$ and j

$= 1$ occur on materialized closed bound chains of one kind, with κ registration data :

no clause of this tag contradicts it:

ε is a function of the winding k alone [(2)],

¬of the kind [D1.5],

¬of the registration κ :

the statistics is ¬a function of the kind and ¬a function of κ — it is the winding's parity, and the winding is the spin label's own numerator

[machine: the symbol census of (2)'s right-hand side carries $\{k\},\ \neg\{\text{D1.5}, \kappa\}$];

the κ

$= 1$ and κ

$= 2$ chains of one kind carry ε

$= -1$ and $+1$ respectively — the refutation and the connection are about different arguments;

(6) the sector-choice pricing (the one non-derived row, priced):

Def-BM.16(5) reads the sign on the $\vee^2$ -sector; the four alternatives and their values:

locus	the read	verdict
$\vee^2(\Psi \otimes \Psi^{\text{CP}})$	$\mu_{\text{CP}} \cdot (+1) = (-1)^{\lvert k\rvert}$	the issued read
$\wedge^2(\Psi \otimes \Psi^{\text{CP}})$	$\mu_{\text{CP}} \cdot (-1) = -(-1)^{\lvert k\rvert}$	the same parity, globally negated
the determinant line $\wedge^d \Psi,\ d = \lvert k\rvert + 1$	$(-1)^{\binom{d}{2}} = (-1)^{\lvert k\rvert(\lvert k\rvert+1)/2}$	¬equal to $(-1)^{\lvert k\rvert}$ at $k = 2:\ -1 \neq +1$ [machine]
the orientation doublet block	$\text{Tr} = 0,\ \dim(\pm 1) = (2, 2)$	statistics-blind [BM.112(4); machine]

the $\vee^2$ and $\wedge^2$ loci give one statistics up to a global sign; the determinant line and the orientation block do not, and the tier says which it uses

[machine: the four rows evaluated to $k = 8$];

(7) the closure:

the spin–statistics connection is DERIVED from the multi-chain exchange operator, with no external postulate

the three members of residue (R7) are all supplied:

a two-chain state space	$\mathcal{H}_2$ [Def-BM.16(2), (3)]
an exchange operator	$\hat{P}_{12},\ \hat{P}_{12}^2 = \text{id},\ \text{spec} \subseteq \{\pm 1\}$ [BM.112(1), (2)]
a permutation-sign read	$\varepsilon(c) = (-1)^{\lvert k\rvert} = (-1)^{2j}$ [BM.113(2), (4)]

the specification BM.6(2) is discharged as written: "a state space on ordered pairs of bound chains with an exchange involution" [Def-BM.16(6); machine: the two-requirement check];

the bar that stays standing:

¬a Lorentz-covariance argument, ¬a positivity-of-energy argument and ¬a microcausality argument is used — the sign comes from the phase alphabet and the winding additivity alone [machine: the token scan of this tag];

(8) status: derived: (1)–(5), (7); priced, ¬forced: (6)'s sector choice;

closed: deficit 11;

discharged: (R7), BM.6(2);

standing: BM.106(1)'s spin-parity refutation, re-scoped at (5) and ¬retracted;

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a numerical coupling [EM.25(7)],

¬a species identification [Def-BM.3(7)];

Req. A3 (Part 1); D0.8, D1.5, D2.9 (Parts 1–2); CT0.9, EM.20(1), EM.25(7) (Part 1, Part 2); P1, P4, P5 (Parts 1–2); Def-H.32(4), Def-H.33, Def-H.38, FT.22, FT.32(6), FT.34(3), FT.60(1), FT.60(2), FT.60(3), FT.89(2), FT.122(1), FT.125(5) (Part 3); Census FT.35(3) (row-wise); Def-BM.3(7), Def-BM.16(2), Def-BM.16(3), Def-BM.16(4), Def-BM.16(5), Def-BM.16(6), BM.2(1), BM.2(2), BM.6(2), BM.25(3), BM.106(1), BM.106(4), BM.107(7), BM.108(3), BM.112(1), BM.112(2), BM.112(4) (this part); exact $\mathbb{Q}(i)$ arithmetic on the fourth-root alphabet, induction with an integer certificate, exhaustive parity and binomial censuses, symbol-set censuses [math; machine: `checks/verify_H_antimatter_resonance_spectrum.py` §D].

Pf.

$$(1): \ \alpha_e \in \{0, \pi\} \text{ [CT0.9; EM.20(1)]};$$

$$\widehat{ds}^{\,2} \in \tfrac{1}{2}\mathbb{Z} \text{ [FT.22–FT.24 ladder]} \implies \alpha_e \widehat{ds}^{\,2} \in \tfrac{\pi}{2}\mathbb{Z} \implies \exp(i\alpha_e \widehat{ds}^{\,2})$$

$$= i^{m_e} \text{ [BM.25(3); machine]}$$

$$\mu_{\text{CP}} = e^{i\pi\Theta} = i^{2\Theta} \in \{i^m\} \text{ [machine];}$$

$$\Theta = 2k \in 2\mathbb{Z} \text{ [FT.22; FT.32(6)]} \implies i^{2\Theta} = i^{4k} = 1, \quad i^{\Theta} = i^{2k} = (-1)^k \text{ [machine]}$$

$$(2): \text{ (i) BM.108(3);}$$

(ii) FT.34(3)'s additive composition and BM.2(2)'s winding grammar [machine];

(iii) A3's $\vee^2$ -eigenvalue $+1$ and BM.112(2)'s eigenspace row

the induction: $k = 0: \ i^0 = 1 = (-1)^0$;

$$i^{2(k+1)} = i^{2k} i^2 = (-1)^k(-1) = (-1)^{k+1} \text{ [machine: verified exhaustively to } k = 16 \text{ over } \mathbb{Q}(i)]$$

$$(3): \text{ Def-H.38 and FT.60(1)–(3) re-cited (frozen);} \qquad j = |k|/2 \implies 2j = |k| \text{ [machine]}$$

$$(4): \ \varepsilon = (-1)^{|k|} = (-1)^{2j};$$

$$j \in \mathbb{Z} \iff |k| \in 2\mathbb{Z} \iff (-1)^{|k|} = +1;$$

$$j \in \mathbb{Z} + \tfrac{1}{2} \iff |k| \in 2\mathbb{Z} + 1 \iff (-1)^{|k|} = -1 \text{ [machine: the parity census to } k = 8]$$

$$\kappa = 1: \ j = \tfrac{1}{2}, \ |k| = 1, \ \varepsilon = -1; \qquad \kappa = 2: \ j = 1, \ |k| = 2, \ \varepsilon = +1 \text{ [BM.106(1)'s two rows; machine]}$$

$$(5): \text{ the symbol census of } (-1)^{|k|}: \ \{k\} \text{ only [machine];}$$

D1.5's kind and κ do not occur $\implies$ ¬a functional dependence on either;

BM.106(1)'s refutation re-cited and re-scoped at its own locus

$$(6):\ \text{the four loci evaluated to } k$$

$$= 8:\ \vee^2 \to (-1)^k;$$

$$\wedge^2 \to -(-1)^k;$$

$$\wedge^d \Psi \to (-1)^{d(d-1)/2}$$

$$= (-1)^{|k|(|k|+1)/2},$$

$$\text{which at } k$$

$$= 2 \text{ gives } (-1)^3$$

$$= -1$$

$$\neq +1$$

$$= (-1)^2 \text{ [machine]};$$

$$\text{the orientation block } \mathrm{Tr}$$

$$= 0 \text{ [BM.112(4)]}$$

$$(7):\ \text{BM.6(2)'s two requirements checked against Def-BM.16(2),}$$

$$(3),$$

$$(4) \text{ and BM.112(1),}$$

$$(2) \text{ [machine];}$$

$$\text{the argument-token scan: } \neg\text{Lorentz},\ \neg\text{positivity of energy},\ \neg\text{microcausality [machine]}$$

$$(8):\ \text{the standing-rule scans [machine]. } \blacksquare$$

Phys. The swap has a phase alphabet and it is the fourth roots of unity, because every edge phase is a sign times a power of i: the alignment takes two values and the line element takes half-integers. The CP transport multiplies by $e^{i\pi\Theta}$, and on the materialized locus the winding is an even integer, so that phase is exactly one while its half-turn i^{Θ} is exactly $(-1)^k$. That is the whole mechanism. The exchange sign is the winding's parity; the spin label is half the winding; so the exchange sign is $(-1)^{2j}$, plus one for integer spin and minus one for half-integer spin, and the connection is a theorem with a two-line induction certificate behind it. It does not contradict the earlier refutation, and the tier is explicit about why: what was refuted was a spin-parity theorem on the bound-or-transit kind, and both parities do occur on one kind at the floor, because the statistics depends on the winding and not on the kind, nor on the registration datum. The one thing that is chosen rather than derived is which sector of the two-chain space the sign is read on, and the tier prices the choice: the symmetric and antisymmetric sectors give the same statistics up to a global sign, while the determinant line and the orientation block give different answers or none, so the read is stated with its locus attached. What is not used is any of the three classical routes — no Lorentz covariance, no positivity of energy, no microcausality — and deficit eleven closes on the phase alphabet and the additivity of the winding, with the residue the previous tier named now discharged at two chains and the three-chain space left open where it was.

Definition 4.17 (Def-BM.17 — The local resonance and its two threshold functions). Def. A resonance := an eigenvalue of a multi-chain ensemble together with its two derived local functions. A construction on Def-BM.14's six-state assembly, BM.13's width ladder, BM.34's ADM split, BM.44's per-tick weight and BM.88's coupling profile; no new axiom, no new datum, no new parameter, no cosmological coordinate.

$$(1)\quad \text{the resonance tuple:} \qquad \boxed{\mathcal{R} \;:=\; \big(n,\ \sigma,\ Q,\ \mathrm{s},\ k\big)}$$

$$n \in \{1,2,3\}\ \text{ the chain count [CT3.17a}(1):\ \text{a sealed support partitions into 2's and 3's]};$$

$$\sigma \in \{\delta,\varepsilon\}\ \text{ the junction grammar [D3.9;\ Def-BM.1]};$$

$$Q \in \{-n,\dots,+n\}\ \text{ the charge class [Def-BM.8}(1);\ \text{BM.103}(5)\text{]};$$

$$\mathrm{s} = (s_1,\dots,s_n) \in \mathcal{S}^n\ \text{ the constituent states [Def-BM.14}(2):\ |\mathcal{S}| = 6\text{]};$$

$$k \in \mathbb{Z}_{\geq 0}\ \text{ the winding [FT.22;\ Def-H.38]};$$

$$\text{constitutive clause: the tuple is derived structural data [Def-BM.2 pattern]};$$

$$\text{no entry is a species name [Def-BM.3}(7)\text{]};$$

$$(2)\quad \text{the internal count and the informational mass:}$$

$$E_{\text{int}}(\mathcal{R}) = \sum_{a=1}^{n} E_{\text{int}}(s_a),$$

$$E_{\text{int}}(s) \in \{2,\ 6,\ 12,\ 24,\ 60\} \text{ on the computed cells}$$

[D0.8; BM.13(2); FT.28; Census FT.35(2), FT.42(5) row-wise] :

$$m_{\text{hid}}(\mathcal{R}) = \frac{E_{\text{int}}(\mathcal{R}) \ln 2}{\mathfrak{h}}$$

[FT.44(4)];

$$m^2_{\text{pole}}(\mathcal{R}) = \hat{\mu}'^2\, \lambda^{(n)}(\text{s}),$$

$$\lambda^{(n)}(\text{s}) := \sum_{a=1}^{n} D_S(s_a)$$

[Def-BM.14(3); BM.111(2)];

(3) the materialization threshold:

constitutive clause: a resonance materializes when the locally available rest energy reaches its own scale,

$$E_{\text{rest}} = N\, m_{\text{hid}}(\mathcal{R}) = \frac{N\, E_{\text{int}}(\mathcal{R}) \ln 2}{\mathfrak{h}}$$

[BM.34(4), (7) : $E = N\sqrt{m^2 + p^2} - N^i p_i$ at $p = 0,\ N^i = 0$; BM.88(2)(iii) : the lapse] :

the materialization condition: $\quad N\,E \ \geq\ \mathcal{T}_0(\mathcal{R}) \ :=\ \dfrac{E_{\text{int}}(\mathcal{R})\,\mathfrak{h}}{\ln 2}$

[E the local energy scale; machine: one Laurent monomial];

$$\mathcal{T}(\mathcal{R};\, N) \ :=\ \frac{\mathcal{T}_0(\mathcal{R})}{N} \ =\ \frac{E_{\text{int}}(\mathcal{R})\,\mathfrak{h}}{N\ \ln 2}$$

[the threshold as a function of the lapse alone; machine];

the curvature read (exact, via the lock):

$$G = 2\pi E_{\text{int}}^2 \ln 2 \implies E_{\text{int}} = \sqrt{\frac{G}{2\pi \ln 2}} \implies \boxed{\mathcal{T}(\mathcal{R};\, G, N) \;=\; \frac{\mathfrak{h}}{N\,\ln 2}\sqrt{\frac{G}{2\pi \ln 2}}}$$

[BM.16(1); BM.88(3); machine];

the horizon read (comparison scale, quarantined):

$$r_s = \frac{\mathfrak{h} E_{\text{int}}}{2} = 2 G_N m_{\text{hid}} \implies \boxed{\mathcal{T}(\mathcal{R};\, r_s, N) \;=\; \frac{2\, r_s}{N}}$$

[BM.15(4); CT2.9; P3 : G_N is a named combination for external comparison; ¬a horizon identification [BM.15 comparison discipline]];

(4) the decay width:

constitutive clause: the width is a weight per tick, $[\Gamma]$

$$= \mathfrak{q}^0,$$

¬a rate per external second [BM.44(4), (5); D1.8] :

$$\boxed{\Gamma(\mathcal{R};\, E, N, G, \varepsilon_{\text{EM}}) \;:=\; \Gamma_{\text{tr}}(\varepsilon_q) \cdot f\Big(\frac{r}{\Delta x_{\text{B}}(E)}\Big) \cdot \frac{\varepsilon_{\text{eff}}}{\varepsilon_{\text{EM}}} \cdot \mathcal{N}(\mathcal{R}) \cdot \chi_{\{\, NE \,\geq\, \mathcal{T}(\mathcal{R};N)\,\}}}$$

$$\Gamma_{\text{tr}} = \frac{x_+ x_-}{1 - x_+ x_-} \in \Big[0, \tfrac{1}{3}\Big]$$

[BM.44(2); frozen];

$$f = 1 - \frac{1}{k}, \quad k = \frac{r}{\Delta x_{\text{B}}(E)} \geq 1$$

[BM.88(2)(ii)];

$$\varepsilon_{\text{eff}} = N f\, \mathsf{g}(G)\, \varepsilon_{\text{EM}} \implies \frac{\varepsilon_{\text{eff}}}{\varepsilon_{\text{EM}}} = N\, f\, \mathsf{g}(G)$$

[BM.88(4); BM.101(2)];

$\mathsf{g} \in \{1,\ 1+G,\ \frac{1}{1+G}\}$ priced, ¬adopted

$$[\text{BM.88}(4);\ \text{BM.104}(5)];$$

$$\mathcal{N}(\mathcal{R}) \ := \ \#\big\{\mathcal{R}' \text{ in the same } (n,\sigma,Q) \text{ class with } \lambda^{(n)}(\mathcal{R}') < \lambda^{(n)}(\mathcal{R})\big\}$$

$$[\text{the lighter-resonance channel count};\ \text{BM.41}(3) \text{ pattern}];$$

$$\chi_{\{\cdot\}} \in \{0,1\}$$

$$[\text{the exact indicator}];$$

non-derived clause: the product form — three derived factors and one count, multiplied;

(5) the derived rows of the width (recorded at the definition site):

$$\Gamma_{\text{tr}}(0) = 0 \ \text{ exactly } [\text{BM.23}(8);\ \text{BM.44}(2)] \implies \boxed{\Gamma(\mathcal{R};\ \varepsilon_q{=}0) = 0 \ \ \forall \mathcal{R},\ \forall (E,N,G,\varepsilon_{\text{EM}})}$$

$$f(1) = 0\ [\text{BM.88}(5)] \implies \boxed{\Gamma(\mathcal{R};\ r = \Delta x_{\text{B}}) = 0 \ \text{ at the member's own envelope}}$$

$$\mathcal{N}(\mathcal{R})$$

$$= 0 \ \text{ for the class's lightest resonance} \implies \boxed{\text{the ground state of every } (n,\sigma,Q) \text{ class has width exactly zero}}$$

$$\Gamma \le \tfrac{1}{3} \cdot 1 \cdot N\mathsf{g} \cdot \mathcal{N}$$

$$[\text{machine: } f < 1,\ \varepsilon_{\text{eff}}/\varepsilon_{\text{EM}} = Nf\mathsf{g} < N\mathsf{g}];$$

$$\tau_{\text{life}} := \frac{1}{\Gamma} \ge 3 \ \text{ ticks at } \mathcal{N} = 1,\ f \to 1,\ N\mathsf{g} \to 1$$

$$[\text{BM.44}(3);\ \text{machine}];$$

(6) the standing orders, as scope clauses:

(i) ¬a cosmological coordinate exists: the ledger knows the local tick τ [A6; D1.8],

the local curvature G [BM.88(3)],

the local lapse N [BM.88(2)(iii)] and the local energy scale E [BM.88(2)(i)];

¬an age and ¬a global time;

(ii) ¬a binary stable/unstable category:

every eigenvalue of every multi-chain ensemble is a resonance;

"stable" $:=$ Γ evaluates to 0 at the ambient calibration point $[(5)]$;

"unstable" or "heavy" $:=$ the local parameters cross $\mathcal{T}$ or open $\mathcal{N}$ $[(3), (4)]$;

(iii) ¬a permanently forbidden state:

$\mathcal{T}(\mathcal{R}; N) \in (0, \infty)\ \forall N \in (0, \infty) \implies$ the threshold locus is never empty [BM.116(4)];

(7) status and scope:

derived: (2), (3)'s three forms, (5)'s five rows;

constitutive: (1)'s tuple, (3)'s and (4)'s definitional forms;

non-derived: (4)'s product form;

the scopes inherited at every use: CT2.6's continuum-limit hypotheses for the chart-side reads [BM.34 Rmk; Def-BM.4(5) envelope discipline];

BM.8(2)'s stationarity hypothesis for the chart evolution [BM.44(3)];

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a numerical coupling [EM.25(7)],

¬a lifetime in external seconds [BM.44(5)];

beyond this tier: $\mathsf{g}(G)$ [BM.104(5)],

e_0 [EM.25(7)],

N as solution data [CT2.7; BM.34(2)], ρ_{B} [A-BM.9],

(R3) the flavour composition,

(R8) the 2-chain sector masses.

Rmk (why "resonance" and not "particle"). Clause (6)(ii) is the tier's standing order written as a scope clause, and it is a statement about the mathematics rather than a naming preference: the ensemble's eigenvalues are indexed by the tuple of (1), each carries a threshold function and a width function, and both are exact symbolic functions of the four local parameters. Nothing in the construction distinguishes an eigenvalue whose width happens to vanish at one calibration point from one whose width happens not to; the difference is a value of a function, not a category. The word particle would import a binary the ledger does not carry, and the word resonance imports nothing: it names an eigenvalue with two functions attached.

Rmk (the horizon row's quarantine). Clause (3)'s third form is written with r_s, which BM.15(4) builds from G_N — a named combination of the single scale for external comparison, not an input [P3]. The row is a change of variable inside the corpus's own monomial and it is quarantined exactly as BM.15's comparison discipline requires: no collapse reading, no horizon identification, and no claim that any resonance sits at any measured radius.

Req. A6, D0.8, D1.8, D2.9 (Parts 1–2); CT2.6, CT2.7, CT2.9, EM.25(7) (Part 2); P1, P3, P4, P5 (Parts 1–2); Def-H.20, Def-H.38, FT.22, FT.28, FT.44(4) (Part 3, Census); Census FT.35(2), FT.42(5) (row-wise); Def-BM.1, Def-BM.2, Def-BM.3(7), Def-BM.4(1), Def-BM.4(5), Def-BM.8(1), Def-BM.14(2), Def-BM.14(3), BM.8(2), BM.13(2), BM.15(4), BM.16(1), BM.23(8), BM.34(2), BM.34(4), BM.34(7), BM.41(3), BM.44(2), BM.44(3), BM.44(4), BM.44(5), BM.88(2), BM.88(3), BM.88(4), BM.88(5), BM.101(2), BM.103(5), BM.104(5), BM.111(2), A-BM.9 (this part); exact Laurent-monomial arithmetic with q-exponent censuses, exact rational function rows [math; machine: `checks/verify_H_antimatter_resonance_spectrum.py` §E].

Pf.

Definition block: the constitutive clauses are (1)'s tuple and the definitional forms of (3),

(4);

the non-derived clause is (4)'s product form;

(2),

(3)'s three exact forms and (5)'s five rows are derived and machine-checked;

(6),

(7) are scope records. ■

Phys. A resonance is a five-entry tuple — how many chains, which seal, which charge class, which six states, which winding — and it carries two functions of the local parameters. The first says when it appears: the rest energy the metric's own ADM split gives is the lapse times the mass, so the resonance materializes when the local scale times the lapse reaches its edge count times the single scale over the log of two, and the same condition read through the calibration lock is a square root of the local curvature, or read through the comparison radius it is twice that radius over the lapse. The second says how fast it goes: the frozen per-tick transition weight, times the envelope fraction that vanishes at the member's own width, times the effective coupling over the bare one, times the count of strictly lighter resonances in its class, times the exact indicator that it is materialized at all. Both functions vanish for whole regions rather than for special objects — at the sharp crystallization threshold nothing decays, at the envelope nothing decays, and the lightest member of every class has nothing lighter to go to, so its width is zero by count. That is the entire content of "stable": a function evaluating to zero where it is being evaluated. Two scope clauses carry the tier's standing orders: no cosmological coordinate exists, because the ledger owns a tick, a curvature, a lapse and a scale and nothing global, and no binary category exists, because every eigenvalue is a resonance and the adjectives are values. The third says nothing is forbidden, since every threshold is a finite positive number at every positive lapse, so every locus is non-empty.

Theorem 4.97 (BM.114 — The resonance spectrum: the two characteristic polynomials, the orbit law, and the one exact collision). Thm.

(1) the transport-register characteristic polynomial at every chain count:

$$W_n(Q) = \left[z^Q\right]\left(2 + z + z^{-1}\right)^n,$$

$$2 + z + z^{-1} = z^{-1}(z+1)^2$$

[Def-BM.13(3); BM.103(3)] :

$$\left(2 + z + z^{-1}\right)^n \;=\; \frac{(z+1)^{2n}}{z^n} \text{ identically in } \mathbb{Z}[z, z^{-1}] \;\Longrightarrow\; W_n(Q) = \binom{2n}{Q+n} \text{ exactly, on all } 2n+1 \text{ classes}$$

[machine: the Laurent identity and the exhaustive 4^n end-sign census at $n = 1, 2, 3$];

$$\sum_{Q=-n}^{n} W_n(Q) = 2^{2n} = 4^n$$

[machine];

the single root:

$$z$$

$= -1$ of multiplicity $2n$ (the $2n$ directed edges), and the $2n + 1$ class weights are its binomial coefficients

[BM.103(3) transported to every n; machine];

the three formed levels:

n	seal	$W_n(Q)$	sum
1	¬sealed (a lone attachment is never sealed)	$(1,\ 2,\ 1)$	$4 = 4^1$
2	δ (meson)	$(1,\ 4,\ 6,\ 4,\ 1)$	$16 = 4^2$
3	ε (baryon)	$(1,\ 6,\ 15,\ 20,\ 15,\ 6,\ 1)$	$64 = 4^3$

[machine: exact binomials];

the n=1 row is formed but ¬sealed: $g(1) = 0 \implies \mathcal{C} = 0 \implies$ the colour block vanishes

[CT3.17a(1), (2); FT.19; machine];

(2) the ε-seal's antisymmetry and the δ-seal's symmetry (the only two antisymmetries the corpus owns):

$B = \varepsilon_{ijk} q^i q^j q^k$ is antisymmetric in its three colour slots;

$M = \delta^i{}_j c^j \bar{c}_i$ is symmetric in its two

[D3.9; CT3.15; CT3.17; BM.106(5)] :

colour is ¬observable by construction

[CT3.22] $\Longrightarrow$ the colour slot symmetry is ¬the statistics — the statistics is $\varepsilon(c)$ on the winding carrier [BM.113(2)]

the transport assembly is sign-free at every n :

$$W_n(Q) = \binom{2n}{Q+n} > 0 \ \text{ on all } 2n+1 \text{ classes}$$

[BM.103(3); machine];

(3) the defect-register eigenvalues (the resonance ladder):

$$\lambda^{(n)}(\mathrm{s}) = \sum_{a=1}^{n} D_S(s_a),$$

$$D_S \in \{2,\ 4,\ 2-\sqrt{2},\ 1,\ 2-2\cos\tfrac{\pi}{10},\ 2-\sqrt{3}\}$$

[Def-BM.14(3); Def-BM.17(2)] :

$$m^2_{\text{pole}}(\mathcal{R}) = \hat{\mu}'^2\,\lambda^{(n)}(\mathrm{s}) \qquad \text{[Def-H.37(2); FT.54(4); machine];}$$

the exact eigenvalue field:

$$\mathfrak{u} := 2\cos\tfrac{\pi}{10},$$

$$\mathfrak{u}^4 - 5\mathfrak{u}^2 + 5 = 0,$$

$$\sqrt{5} = 2\mathfrak{u}^2 - 5 \in \mathbb{Q}(\mathfrak{u}),$$

$$2 - 2\cos\tfrac{\pi}{10} = 2 - \mathfrak{u} :$$

$$\mathbb{M} := \mathbb{Q}\big(\sqrt{2},\ \sqrt{3},\ \mathfrak{u}\big), \qquad [\mathbb{M}:\mathbb{Q}] = 2\cdot 2\cdot 4 = 16, \qquad \text{basis } \big\{\sqrt{2}^{\,i}\sqrt{3}^{\,j}\,\mathfrak{u}^{\ell}\big\}_{i,j\in\{0,1\},\ \ell\in\{0,1,2,3\}}$$

[machine: the three defining relations verified exactly, and $(\sqrt{2})^2{=}2, (\sqrt{3})^2{=}3, (2\mathfrak{u}^2{-}5)^2{=}5$];

every resonance eigenvalue lies in $\mathbb{M}$,

and equality of two eigenvalues is decidable by basis-coordinate equality

[machine: $\mathbb{M}$'s basis is linearly independent over $\mathbb{Q}$];

(4) the orbit law (the multiplicity is the stabilizer index):

S_n acts on $\mathcal{S}^n$ by permuting the constituent slots; $\lambda^{(n)}$ is S_n-invariant [machine: a sum] :

$$\#(\text{orbits}) = \binom{6+n-1}{n},$$

$$|\text{orbit of s}| = \frac{n!}{|\,\text{Stab(s)}|},$$

$$\sum_{\text{orbits}} |\text{orbit}| = 6^n$$

[machine: exact at $n = 1, 2, 3$];

the three levels:

n	#orbits	orbit sizes	$\sum$	multiplicity census
1	$\binom{6}{1} = 6$	1	6	$\{1^6\}$
2	$\binom{7}{2} = 21$	1 (6 diagonal), 2 (15 off-diagonal)	36	$\{1^6,\ 2^{15}\}$
3	$\binom{8}{3} = 56$	1 (6), 3 (30), 6 (20)	216	see (5)

[machine: exhaustive over the 6^n ordered tuples]; $\boxed{15 = \binom{6}{2} = \dim \wedge^2\mathbb{C}^6, \qquad 20 = \binom{6}{3}}$

the count is a multiset count, ¬a representation dimension:

¬∃ a flavour $\mathsf{SU}(6)$ in the inventory [BM.103(7); the flavour symmetry is C_3 [FT.111(2); FT.142(5)]];

(5) the three-chain ladder in full:

56 orbits $\longrightarrow$ $\boxed{55 \text{ distinct eigenvalues}}$

[machine: exact equality in $\mathbb{M}$] :

exactly one collision, and it is the powers-of-two triple:

$$\lambda^{(3)}(s_1, s_1, s_1) = 2 + 2 + 2 = 6 = 4 + 1 + 1 = \lambda^{(3)}(s_2, s_4, s_4),$$

multiplicity $1 + 3 = 4$

[machine: the exact rational identity; BM.82(4)'s collision census re-cited at the constituent level: three of the six defect values are the powers $1, 2, 4$];

the multiplicity census of the 55:

$$\boxed{\{1^5,\ 3^{29},\ 4^1,\ 6^{20}\}}$$

$$\text{sum } = \ 5 + 87 + 4 + 120 = 216 = 6^3$$

[machine: exact];

the multiplicity is always a divisor of $n! = 6$:

$$\{1, 3, 4, 6\} \subset \{1, 2, 3, 6\} \cup \{4\},$$

and the 4 is the collision's $1 + 3$ [machine];

(6) the minimal-polynomial degree census (the subfield ladder of $\mathbb{M}$):

n	degree census over the distinct eigenvalues	max
1	$\{1^3,\ 2^2,\ 4^1\}$	4
2	$\{1^6,\ 2^8,\ 4^5,\ 8^2\}$	8
3	$\{1^9,\ 2^{20},\ 4^{15},\ 8^{10},\ 16^1\}$	$16 = [\mathbb{M} : \mathbb{Q}]$

[machine: each minimal polynomial computed exactly by linear dependence in the 16-dimensional $\mathbb{Q}$-space $\mathbb{M}$, then verified by exact evaluation at its own root];

the pinned rows:

$$D_S(s_3) = 2 - \sqrt{2} : \ x^2 - 4x + 2;$$

$$D_S(s_5) = 2 - \mathfrak{u} : \ x^4 - 8x^3 + 19x^2 - 12x + 1;$$

$$D_S(s_6) = 2 - \sqrt{3} : \ x^2 - 4x + 1$$

[Def-BM.14(3); machine re-verified];

the unique degree-16 resonance eigenvalue:

$$\lambda^{(3)}(s_3, s_5, s_6) = (2 - \sqrt{2}) + (2 - \mathfrak{u}) + (2 - \sqrt{3}) = 6 - \sqrt{2} - \mathfrak{u} - \sqrt{3} :$$

it generates $\mathbb{M}$ over $\mathbb{Q}$

[machine: its minimal polynomial has degree exactly 16; the leading rows are $x^{16} - 96x^{15} + 4260x^{14} - 115920x^{13} + \cdots + 249713695921$];

the minimal eigenvalue at each level is a quartic:

$$n \left(2 - \mathfrak{u}\right) :$$

$$n = 1 : \ x^4 - 8x^3 + 19x^2 - 12x + 1;$$

$$n = 2 : \ x^4 - 16x^3 + 76x^2 - 96x + 16;$$

$$n = 3 : \ x^4 - 24x^3 + 171x^2 - 324x + 81$$

[machine: the three quartics computed exactly and each verified at its own root];

(7) the ladder is a resonance ladder, not a stability ladder:

¬∃ a derived eigenvalue stability condition [BM.103(6) : the three grounds re-cited — (M1)/(M2) undeclared, FT.66(4)'s index-zero refutation, and the count → strength bar] :

what orders the ladder is the eigenvalue's own value, and what decides its appearance is $\mathcal{T}(\mathcal{R}; N)$, ¬a stability class

[Def-BM.17(3), (6)(ii); BM.115; BM.116];

(8) status: derived: (1)–(7);

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a species identification [Def-BM.3(7)],

¬a currency conversion [BM.83(3)];

Req. D2.9, D3.9 (Parts 1–2); CT3.15, CT3.17, CT3.17a(1), CT3.17a(2), CT3.22 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.37(2), FT.19, FT.54(4), FT.111(2), FT.142(5) (Part 3); Def-BM.3(7), Def-BM.13(3), Def-BM.14(3), Def-BM.17(2), BM.66(6), BM.82(4), BM.83(3), BM.103(3), BM.103(6), BM.103(7), BM.106(5), BM.111(2), BM.113(2) (this part); Laurent-polynomial identities over $\mathbb{Z}[z, z^{-1}]$, exhaustive binomial and multiset-orbit enumeration, exact linear algebra over $\mathbb{Q}$ in a 16-dimensional algebra, exact real ordering by a rigorous dyadic interval embedding [math; machine: `checks/verify_H_antimatter_resonance_spectrum.py` §F].

Pf.

$$(1):\ 2 + z + z^{-1} = z^{-1}(z^2 + 2z + 1) = z^{-1}(z + 1)^2 \text{ [machine: one Laurent identity]};$$

$$\left(z^{-1}(z + 1)^2\right)^n = z^{-n}(z + 1)^{2n} \text{ [machine]}$$

$$[z^Q]z^{-n}(z + 1)^{2n}$$

$$= [z^{Q+n}](z + 1)^{2n}$$

$$= \binom{2n}{Q + n} \text{ [machine: against the exhaustive } 4^n \text{ census at } n = 1, 2, 3];$$

$$\sum_Q \binom{2n}{Q + n} = 2^{2n} \text{ [machine]}$$

$g(1) = 0$ [CT3.17a(2)]; $\quad |\text{Colour}(\Sigma)| = 1 \implies \mathcal{C} = 0 \implies$ the block vanishes [CT3.17a(1); FT.19]

$$(2):\ \text{D3.9, CT3.15, CT3.17 re-cited};$$

CT3.22′s non-observability re-cited;

$$W_n(Q) = \binom{2n}{Q+n} > 0 \text{ [machine: positivity of binomials]}$$

$$(3): \ \mathfrak{u}$$

$$= 2\cos\tfrac{\pi}{10}: \quad \cos^2\tfrac{\pi}{10}$$

$$= \tfrac{5+\sqrt{5}}{8} \implies \mathfrak{u}^2$$

$$= \tfrac{5+\sqrt{5}}{2} \implies \mathfrak{u}^4$$

$$= \tfrac{15+5\sqrt{5}}{2} \implies \mathfrak{u}^4 - 5\mathfrak{u}^2 + 5$$

$$= \tfrac{15+5\sqrt{5}-25-5\sqrt{5}}{2} + 5$$

$$= 0 \text{ [machine: exact in } \mathbb{Q}(\sqrt{5})]$$

$$\sqrt{5}$$

$$= 2\mathfrak{u}^2 - 5 \text{ [machine: } (2\mathfrak{u}^2-5)^2 = 4\cdot\tfrac{5+\sqrt{5}}{2}\cdot 2 - 20\mathfrak{u}^2 + 25 \text{ reduced by } \mathfrak{u}^4 = 5\mathfrak{u}^2 - 5 \text{ to } 5];$$

$$[\mathbb{Q}(\mathfrak{u}):\mathbb{Q}] = 4 \text{ [}\mathfrak{u}^4 - 5\mathfrak{u}^2 + 5 \text{ irreducible: } \neg\text{a rational root and } \mathfrak{u}^2 = \tfrac{5+\sqrt{5}}{2} \notin \mathbb{Q}]$$

$$[\mathbb{M}:\mathbb{Q}]$$

$$= [\mathbb{M}:\mathbb{Q}(\mathfrak{u})]\cdot[\mathbb{Q}(\mathfrak{u}):\mathbb{Q}]$$

$$= 4\cdot 4$$

$$= 16 \text{ [}\sqrt{2},\sqrt{3} \notin \mathbb{Q}(\mathfrak{u}) \text{ by the degree census of } (6);\ \text{machine: the } 16 \text{ basis elements are } \mathbb{Q}\text{-linearly independent]}$$

$$(4): \ \lambda^{(n)}(\pi\cdot \mathfrak{s}) = \sum_a D_S(s_{\pi(a)}) = \sum_a D_S(s_a) = \lambda^{(n)}(\mathfrak{s}) \text{ [machine]};$$

$$\#\text{orbits} = \binom{6+n-1}{n} \text{ [multiset count; machine: exhaustive at } n = 1,2,3]$$

$$|\text{orbit}| = n!/|\,\text{Stab}\,| \text{ with } |\,\text{Stab}\,| = \prod_{\text{distinct values}} (\text{multiplicity})! \text{ [machine]};$$

the size sums $6, 36, 216$ [machine]

$$\binom{6}{2} = 15 = \dim \wedge^2 \mathbb{C}^6 = \tfrac{6\cdot 5}{2}; \qquad \binom{6}{3} = 20 \text{ [machine]}$$

$(5):\ 2+2+2=6$ and $4+1+1=6$ in $\mathbb{Q} \subset \mathbb{M}$ [machine: basis-coordinate equality];

the exhaustive census finds no second collision among the 56 [machine]

the multiplicity census $\{1^5, 3^{29}, 4^1, 6^{20}\}:\quad 5+3\cdot 29+4+6\cdot 20 = 5+87+4+120 = 216$ [machine]

$(6):$ for each eigenvalue x,

the smallest d with $\{1, x, \ldots, x^d\}$ $\mathbb{Q}$-dependent in the 16-dimensional $\mathbb{M}$ gives the degree, and the dependency gives the monic polynomial [machine: exact Fraction elimination];

each is verified by exact evaluation at its own root [machine]

$x^2 - 4x + 2$ at $2-\sqrt{2}:\ (2-\sqrt{2})^2 - 4(2-\sqrt{2}) + 2 = (6-4\sqrt{2}) - (8-4\sqrt{2}) + 2 = 0$ [machine];

$x^2 - 4x + 1$ at $2-\sqrt{3}:\ (7-4\sqrt{3}) - (8-4\sqrt{3}) + 1 = 0$ [machine]

$(2-\mathfrak{u}):\ (2-x)^4 - 5(2-x)^2 + 5 = x^4 - 8x^3 + 19x^2 - 12x + 1$ [machine: the exact expansion];

the three minimal quartics and the degree-16 polynomial computed and root-verified [machine]

$(7):$ BM.103(6)'s three grounds re-cited; $(8):$ the standing-rule scans [machine]. ■

Phys. Two characteristic polynomials come out, one per currency, and neither is fitted. In the transport register the generating function is the same Laurent monomial at every chain count — one plus z plus one over z, raised to n, which is one plus z to the two n over z to the n — so the class weight is a binomial coefficient, the sum is four to the n, and the whole spectrum is one root

at minus one with multiplicity twice the chain count. The first level is formed but not sealed, since a lone attachment is never sealed, which is why the ladder starts at two. In the defect register the eigenvalues are sums of the six branch values, and they all live in one exact field of degree sixteen over the rationals built from root two, root three and twice the cosine of eighteen degrees, whose own relation makes root five a polynomial in it — so equality of two eigenvalues is a decidable integer question rather than a numerical one. The orbit law is clean: the symmetric group permutes the constituent slots, the eigenvalue is invariant, the orbit count is a multiset binomial, and the multiplicity of each eigenvalue is the orbit size, a divisor of n factorial. Fifteen and twenty appear, and they are exactly the wedge-square dimension and the three-subset count of the six states — but the tier says plainly that these are counts and not representation dimensions, because the flavour symmetry the corpus derives is a cyclic three and not a special unitary six. The three-chain ladder has fifty-six orbits and fifty-five eigenvalues, and the single collision is the one the constituent census had already found: two plus two plus two equals four plus one plus one, because three of the six defect values are the powers one, two and four. The minimal-polynomial degrees climb one, two, four, eight, sixteen with the number of distinct irrational ingredients, and exactly one three-chain resonance — the sum of the root-two, cosine and root-three branches — generates the whole field.

Theorem 4.98 (BM.115 — The materialization threshold: the exact function of E, N and G, and the never-empty locus). Thm.

$$(1) \quad \text{the threshold function:}$$

$$\mathcal{T}_0(\mathcal{R}) = \frac{E_{\text{int}}(\mathcal{R})\,\mathfrak{h}}{\ln 2},$$

$$\mathcal{T}(\mathcal{R}; N) = \frac{\mathcal{T}_0(\mathcal{R})}{N},$$

$$\text{materialization} \iff NE \geq \mathcal{T}_0(\mathcal{R})$$

$$[\text{Def-BM.17}(3);\ \text{machine}]:$$

$$[\mathcal{T}_0] = [\mathfrak{h}] = \mathfrak{q}^{-1/2} \cdot \mathfrak{q}^{+1/2} \cdot \mathfrak{q}^{0} = \quad \text{an energy:} \quad [E] = \mathfrak{q}^{+1/2},$$

$$[\mathcal{T}] = \mathfrak{q}^{+1/2}$$

$$[\text{CT2.1};\ \text{BM.102}(2);\ \text{machine: the exponent arithmetic}];$$

$\mathcal{T}$ is strictly decreasing in the lapse:

$$N_1 < N_2 \implies \mathcal{T}(\mathcal{R}; N_1) > \mathcal{T}(\mathcal{R}; N_2) > 0$$

$$[\text{machine: exact rationals}];$$

a deeper lapse lowers every threshold at once — the whole ladder moves as one function of N

(2) the two exact threshold ladders:

(i) the computed-cell ladder $E_{\text{int}}(s) \in \{2,\ 6,\ 12,\ 24,\ 60\}$

[BM.13(2); FT.28; Census FT.35(2), FT.42(5) row-wise] :

n	#$\{E_{\text{int}}\}$	the exact ladder	span ratio
1	5	2, 6, 12, 24, 60	30
2	15	4, 8, 12, 14, 18, 24, 26, 30, 36, 48, 62, 66, 72, 84, 120	30
3	33	6, 10, 14, 16, 18, 20, 24, 26, 28, 30, 32, 36, 38, 42, 48, 50, 54, 60, 64, 68, 72, 74, 78, 84, 86, 90, 96, 108, 122, 126, 132, 144, 180	30

[machine: the exact sum sets with repetitions];

$$\min E_{\text{int}}^{(n)} = 2n \ \text{(the } n \text{ digon floors)}, \qquad \max E_{\text{int}}^{(n)} = 60n, \qquad \frac{\max}{\min} = 30 \ \text{at every } n$$

(ii) the six-state ladder $E_{\text{int}}(s) = 2E(s) \in \{12,\ 24,\ 60\}$ on the five Platonic solids

[Def-BM.14(1); BM.60(1) : $12 : 24 : 60 = 1 : 2 : 5$] :

n	#$\{E_{\text{int}}\}$	the exact ladder	span ratio
1	3	12, 24, 60	5
2	6	24, 36, 48, 72, 84, 120	5
3	10	36, 48, 60, 72, 84, 96, 108, 132, 144, 180	5

[machine: the exact sum sets];

$$\min = 12n, \qquad \max = 60n, \qquad \frac{\max}{\min} = 5 = \frac{60}{12} \ \text{at every } n$$

[the floor ladder's own ratio $1 : 2 : 5$ re-cited];

the two ladders are nested, and the nesting is exact:

$\{12, 24, 60\} \subset \{2, 6, 12, 24, 60\} \implies$ every six-state threshold is a computed-cell threshold;

the two extra cells 2 and 6 are the digon floor [FT.28] and the baryon junction [Census FT.35(2)] :

they lower the minimum from $12n$ to $2n$ and raise the span from 5 to 30 [machine];

the meson and baryon rows on the computed-cell ladder:

$$m_{\text{hid}}^{(2)} \in \Big[\frac{4\ln 2}{\mathfrak{h}},\ \frac{120\ln 2}{\mathfrak{h}}\Big],$$

$$m_{\text{hid}}^{(3)} \in \Big[\frac{6\ln 2}{\mathfrak{h}},\ \frac{180\ln 2}{\mathfrak{h}}\Big]$$

[BM.111(5); machine];

and on the six-state ladder:

$$m_{\text{hid}}^{(2)} \in \Big[\frac{24\ln 2}{\mathfrak{h}},\ \frac{120\ln 2}{\mathfrak{h}}\Big],$$

$$m_{\text{hid}}^{(3)} \in \Big[\frac{36\ln 2}{\mathfrak{h}},\ \frac{180\ln 2}{\mathfrak{h}}\Big]$$

[machine];

(3) the curvature read:

$$G = 2\pi E_{\text{int}}^2 \ln 2 \in \mathfrak{q}^0,$$

G $\mathfrak{h}$-free, Λ-free, P4-free

[BM.16(1); BM.88(3); machine: the three freedom censuses re-cited] :

$$\boxed{E_{\text{int}} = \sqrt{\frac{G}{2\pi\ln 2}} \implies \mathcal{T}(\mathcal{R};\, G, N) = \frac{\mathfrak{h}}{N\ln 2}\sqrt{\frac{G}{2\pi\ln 2}}}$$

[machine: the exact substitution];

$$G^{(n)} \in \big[2\pi\ln 2\cdot 4n^2,\ \ 2\pi\ln 2\cdot 3600n^2\big],$$

$$\frac{G_{\max}}{G_{\min}} = 900 = 30^2 \text{ at every } n$$

[machine: exact integers];

the curvature is a derived function of the count, so the threshold is a derived function of the local curvature — ¬an independent datum [BM.88(3); machine];

(4) the horizon read (comparison scale, quarantined):

$$r_s = 2G_N m_{\text{hid}} = \tfrac{1}{2}\mathfrak{h}E_{\text{int}},$$

$$\Delta x_{\text{B}} \cdot r_s = 2G_N = \frac{\mathfrak{h}^2}{2\ln 2}$$

[BM.15(4); CT2.9; P3] :

$$\boxed{\mathcal{T}(\mathcal{R};\, r_s, N) = \frac{2\, r_s}{N}}$$

[machine: $\frac{\mathfrak{h}}{N\ln 2}\cdot\frac{2r_s}{\mathfrak{h}} = \frac{2r_s}{N}$];

$$r_s^{(n)} \in [n\mathfrak{h},\ 30n\mathfrak{h}],$$

$$\frac{r_{s,\max}}{r_{s,\min}} = 30$$

[machine];

the comparison discipline: ¬a collapse reading,

¬a horizon identification,

¬a measured radius [BM.15(4) comparison discipline; P5; Def-BM.3(6)];

what the row is: a change of variable inside the corpus's own monomial, which exhibits the threshold as twice the comparison radius over the lapse;

(5) the never-empty locus (no forbidden state):

$$\forall \mathcal{R},\ \forall N \in (0,\infty) :$$

$$\mathcal{T}(\mathcal{R}; N) \in (0,\infty)$$

$$[E_{\text{int}} \in \mathbb{Z}_{\geq 2},\ \mathfrak{h} > 0,\ \ln 2 > 0\ [\text{BM.13}(3) :\ \tfrac{1}{2} < \ln 2 < 1]] :$$

$$\{E > 0 :\ NE \geq \mathcal{T}_0(\mathcal{R})\} = \Big[\frac{\mathcal{T}_0(\mathcal{R})}{N},\ \infty\Big) \neq \varnothing$$

[machine: the half-line is nonempty at every admissible N];

¬∃ a permanently forbidden resonance: every eigenvalue's threshold locus is a nonempty half-line in the local energy scale

the two named triples are neither formed nor excluded:

$$\neg\exists \text{ a triple-selection rule in the corpus } [\text{BM.103}(8) :\ (\text{R3})] :$$

but both are admissible threshold-wise: any $\mathrm{s} \in \mathcal{S}^3$ has $E_{\text{int}}(\mathrm{s})$

$\in [6, 180] \subset \mathbb{Z}_{>0}$ and hence a nonempty locus [(5)'s first row; machine: all 216 ordered triples];

there are no forbidden compositions; there are compositions whose thresholds require a higher local scale

(6) the accelerator and horizon regimes are evaluations, not mechanisms:

a high local scale $E \gg E_1$: $NE \geq \mathcal{T}_0(\mathcal{R})$ holds for every $\mathcal{R}$ with $E_{\text{int}}(\mathcal{R}) \leq NE \ln 2/\mathfrak{h}$

[machine] :

a small lapse $N \ll 1$ (the deep-gravity read): $\mathcal{T}(\mathcal{R}; N)$

$= \mathcal{T}_0/N \to \infty \implies$ thresholds rise and fewer resonances materialize

[machine: monotonicity from (1)];

a large comparison radius r_s : $\mathcal{T} = 2r_s/N$ grows linearly [(4); machine];

both regimes are the same evaluation of the same function at different local parameters — ¬two mechanisms, ¬a new object

¬a cosmological coordinate is used: the rows are stated in $(E,\ N,\ G,\ r_s)$ and ¬in any global time [Def-BM.17(6)(i); A6; D1.8];

(7) the crossing strength re-cited and ¬re-evaluated:

$$\varepsilon_*(w) = \frac{\ln(4/3)}{3w\ln 2} = \frac{2-\log_2 3}{3w},$$

$$\tfrac{1}{8} < \varepsilon_* < \tfrac{5}{12}$$

[BM.101(3); machine: the enclosure re-certified on $2^{13} > 3^8,\ 3^{12} > 2^{19}$] :

its role here: the crossing locus $\mathcal{L}_*$ is where the mass splitting changes sign, and it is a level set of the same profile ε_{eff} that the width function carries [BM.101(4); Def-BM.17(4)] :

¬a numerical coupling enters,

¬a measured value is compared

[P5; EM.25(7); machine: the token scan];

(8) status: derived: (1)–(6); re-cited, ¬re-evaluated: (7);

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a species identification [Def-BM.3(7)];

Req. D0.8, D1.8, D2.9 (Parts 1–2); A6, CT2.1, CT2.9, EM.25(7) (Part 1, Part 2); P1, P3, P4, P5 (Parts 1–2); FT.28, FT.44(4) (Part 3, Census); Census FT.35(2), FT.42(5) (row-wise); Def-BM.3(6),

Def-BM.3(7), Def-BM.4(1), Def-BM.17(3), Def-BM.17(4), BM.13(2), BM.13(3), BM.15(4), BM.16(1), BM.88(3), BM.101(3), BM.101(4), BM.102(2), BM.103(8), BM.111(5) (this part); exact integer sum-set enumeration, exact rational monotonicity, Laurent-monomial exponent censuses, integer power certificates [math; machine: `checks/verify_H_antimatter_resonance_spectrum.py` §E].

Pf.

$$(1):\ \mathcal{T}_0 = E_{\text{int}}\mathfrak{h}/\ln 2 \text{ [Def-BM.17(3); machine]};$$

$$[\mathcal{T}_0] = [\mathfrak{h}] = \mathfrak{q}^{-1/2} \text{ and } [E_{\text{int}}] = [\ln 2] = \mathfrak{q}^0 \implies [\mathcal{T}_0] = \mathfrak{q}^{-1/2}\cdot\mathfrak{q}^0$$

$$\text{with } E_{\text{int}} = \mathfrak{h}^{-1}E \text{ dimensionless:}$$

$$[\mathcal{T}_0] = [\mathfrak{h}]\cdot[E]/[\mathfrak{h}] = \mathfrak{q}^{+1/2} \text{ [CT2.1; BM.102(2); machine: the exponent arithmetic]};$$

$$N_1 < N_2 \implies \mathcal{T}_0/N_1 > \mathcal{T}_0/N_2 \text{ [machine: exact rationals]}$$

$$(2):\ \text{the three sum sets computed exhaustively from } \{2,6,12,24,60\} \text{ with repetitions, } n$$

$$= 1,$$

$$2, \qquad 3:$$

$$5,\ 15,\ 33 \text{ distinct values;}$$

$$\min = 2n,\ \max = 60n,\ 60n/2n = 30 \text{ [machine: exact integers]}$$

$$(3):\ G = 2\pi E_{\text{int}}^2 \ln 2 \implies E_{\text{int}} = \sqrt{G/(2\pi\ln 2)} \text{ [machine: the positive branch at } E_{\text{int}} > 0];$$

$$\text{substitution into } (1) \text{ [machine]}$$

$$G_{\min} = 2\pi\ln 2\cdot(2n)^2 = 4n^2\cdot 2\pi\ln 2,$$

$$G_{\max} = 2\pi\ln 2\cdot(60n)^2 = 3600n^2\cdot 2\pi\ln 2,$$

$$3600/4 = 900 = 30^2 \text{ [machine]}$$

$$(4):\ r_s$$

$$= \tfrac{1}{2}\mathfrak{h}E_{\text{int}}$$

$$\implies E_{\text{int}}$$

$$= 2r_s/\mathfrak{h} \implies \mathcal{T}_0$$

$$= \tfrac{\mathfrak{h}}{\ln 2}\cdot\tfrac{2r_s}{\mathfrak{h}}$$

$$= \tfrac{2r_s}{\ln 2}\cdot \ln 2 \text{ [machine]};$$

$$\mathcal{T} = \mathcal{T}_0/N \text{ with } \mathcal{T}_0 = 2r_s \text{ in energy units [machine]}$$

$$r_s \in [\tfrac{1}{2}\mathfrak{h}\cdot 2n,\ \tfrac{1}{2}\mathfrak{h}\cdot 60n] = [n\mathfrak{h},\ 30n\mathfrak{h}] \text{ [machine]}$$

$$(5):\ E_{\text{int}} \geq 2 \text{ [FT.28]} \implies \mathcal{T}_0 \geq 2\mathfrak{h}/\ln 2 > 0\ [\tfrac{1}{2} < \ln 2 < 1 \text{ [BM.13(3)]};\ \text{machine}];$$

$$\text{the half-line } [\mathcal{T}_0/N, \infty) \text{ is nonempty } \forall N > 0 \text{ [machine]}$$

$$\text{all 216 ordered triples : } E_{\text{int}} \in [6, 180] \subset \mathbb{Z}_{>0} \text{ [machine: exhaustive]};$$

$$\neg\text{a selection rule exists [BM.103(8)]}$$

$$(6):\ NE \geq \mathcal{T}_0 \iff E_{\text{int}} \leq NE\ln 2/\mathfrak{h} \text{ [machine]};$$

$$N\downarrow \implies \mathcal{T}\uparrow\ [(1)];$$

$$(7):\ \text{BM.101(3)},$$

$$(4) \text{ re-cited, the enclosure re-certified on the two integer powers [machine]};$$

$$(8):\ \text{the standing-rule scans [machine]. } \blacksquare$$

Phys. The threshold is one Laurent monomial and it has three exact readings. Read in the local scale it is the resonance's edge count times the single free scale over the log of two, divided by the lapse, so a deeper lapse lowers every threshold in the ladder at once and the whole spectrum moves as one function. Read in the local curvature it is a square root, because the calibration lock

makes the curvature a derived quadratic function of the same count — and the lock is free of the scale, free of the cosmological term and free of every declared parameter, so the curvature reading adds no datum. Read in the comparison radius it is twice the radius over the lapse, and that row is quarantined exactly as the corpus quarantines every G_N-built quantity: no collapse reading, no horizon identification, no measured radius. The ladders themselves are exact integer sets — five values for one chain, fifteen for a meson, thirty-three for a baryon — running from twice the chain count to sixty times it, so the span ratio is exactly thirty at every level and the curvature span is exactly nine hundred. Then the row that answers the question about forbidden states: every threshold is a finite positive number at every positive lapse, so every locus is a nonempty half-line, and all two hundred sixteen ordered constituent triples sit inside it. Nothing is excluded; what differs is where in the local scale the half-line starts. Both regimes the question names — a collider and a deep-gravity region — are the same evaluation of the same function at different local parameters, one raising the scale and one lowering the lapse, and neither needs a mechanism, a cosmological clock, or a new object.

Theorem 4.99 (BM.116 — The decay width: the exact function of the local parameters, its three exact zeros, and the lifetime floor). Thm.

(1) the width function:

$$\boxed{\Gamma(\mathcal{R};\, E, N, G, \varepsilon_{\mathrm{EM}}) \;=\; \underbrace{\frac{x_+ x_-}{1 - x_+ x_-}}_{\Gamma_{\mathrm{tr}}(\varepsilon_q)} \cdot \underbrace{\Big(1 - \frac{\Delta x_{\mathrm{B}}(E)}{r}\Big)}_{f} \cdot \underbrace{N\,\mathsf{g}(G)}_{\varepsilon_{\mathrm{eff}}/\varepsilon_{\mathrm{EM}}\,\cdot\, f^{-1}\cdot f} \cdot \underbrace{\mathcal{N}(\mathcal{R})}_{\text{the channel count}} \cdot \underbrace{\chi_{\{NE \geq \mathcal{T}_0(\mathcal{R})\}}}_{\text{materialization}}}$$

assembled from four frozen rows and one count:

$$\Gamma_{\mathrm{tr}} = \tfrac{x_+ x_-}{1 - x_+ x_-} \in [0, \tfrac{1}{3}] \text{ [BM.44(2)]};$$

$$f = 1 - \tfrac{1}{k},\; k = \tfrac{r}{\Delta x_{\mathrm{B}}(E)} \geq 1 \text{ [BM.88(2)(ii)]};$$

$$\frac{\varepsilon_{\mathrm{eff}}}{\varepsilon_{\mathrm{EM}}} = N f\, \mathsf{g}(G) \text{ [BM.88(4); BM.101(2)]};$$

$$\mathcal{N}(\mathcal{R}) = \#\{\mathcal{R}' \text{ in the same } (n, \sigma, Q) \text{ class} :\; \lambda^{(n)}(\mathcal{R}') < \lambda^{(n)}(\mathcal{R})\} \text{ [Def-BM.17(4)]};$$

$$[\Gamma] = \mathfrak{q}^0, \qquad [\tau_{\mathrm{life}}] = \mathfrak{q}^0 \text{ ticks} \qquad \text{[BM.44(4); D1.8; BM.102(2); machine: the exponent census]};$$

(2) the flicker factor and its exact range:

$$x_+ x_- = 4\varepsilon_q(1 - 4\varepsilon_q) = \frac{1}{4} - 16\Big(\varepsilon_q - \frac{1}{8}\Big)^2 \leq \frac{1}{4},$$

$$\text{equality} \iff \varepsilon_q = \frac{1}{8}$$

[BM.19(2); BM.44(1); machine: the SOS completion and the 256-point grid] :

$$\boxed{\Gamma_{\text{tr}}(0) = 0, \qquad \Gamma_{\text{tr}}\left(\tfrac{1}{8}\right) = \frac{1/4}{3/4} = \frac{1}{3}, \qquad \Gamma_{\text{tr}} \in \left[0, \tfrac{1}{3}\right] \text{ on } \varepsilon_q \in \left[0, \tfrac{1}{8}\right], \qquad \Gamma_{\text{tr}} \text{ strictly increasing there}}$$

[BM.44(2); machine: $\Gamma_{\text{tr}}(\varepsilon_2) - \Gamma_{\text{tr}}(\varepsilon_1) > 0$ via $4(\varepsilon_2 - \varepsilon_1)(1 - 4(\varepsilon_1 + \varepsilon_2)) > 0$];

(3) the three exact zeros (the derived "stable" loci):

$$\boxed{\text{(Z1)} \quad \varepsilon_q = 0 \implies \Gamma \equiv 0 \text{ for every resonance at every local parameter}}$$

[BM.23(8); BM.19(3); machine] :

the mechanism:

$$x_-$$

$$= 4\varepsilon_q$$

$= 0 \implies \neg\exists$ a dissolution move $\implies$ the ensemble collapses to its crystallized member $\implies \neg$a transition weight beyond the identity;

$$\text{(Z2)} \quad r = \Delta x_{\text{B}}(E) \implies k = 1 \implies f = 0 \implies \Gamma = 0$$

[BM.88(5); machine] :

at the member's own Born envelope the width vanishes;

$$\text{(Z3)} \quad \mathcal{N}(\mathcal{R}) = 0 \implies \Gamma = 0$$

[Def-BM.17(5); machine] :

$$\boxed{\text{the lightest resonance of every } (n, \sigma, Q) \text{ class has width exactly zero, at every local parameter}}$$

the count of zero-width resonances at the three formed levels:

level	$\#\{\mathcal{N}=0\}$	locus
$n=2,\ \sigma=\delta$	3	one per charge class $Q\in\{0,\pm1\}$ [BM.110(3)]
$n=3,\ \sigma=\varepsilon$	7	one per charge class $Q\in\{-3,\dots,+3\}$ [BM.114(1)]

[machine: exact];

(4) the fourth zero is the threshold's, and it is not a width:

$$NE<\mathcal{T}_0(\mathcal{R})\implies\chi=0\implies\Gamma=0$$

[machine] :

an unmaterialized resonance has no width because it has no ensemble — the zero is the indicator's, ¬a decay

[Def-BM.17(3); BM.94(5) : the materialization restriction];

the distinction the tier keeps: $\Gamma=0$ from $(\mathrm{Z}1),(Z2),(Z3)$ is a stable resonance;

$\chi=0$ is an absent one;

¬a binary category covers both [Def-BM.17(6)(ii)];

(5) the bound and the lifetime floor: $f<1\ \forall k<\infty$ [BM.88(5); machine] $\implies$

$$\boxed{0\ \le\ \Gamma(\mathcal{R})\ \le\ \frac{1}{3}\cdot N\,\mathrm{g}(G)\cdot\mathcal{N}(\mathcal{R})}$$ [machine: the three factor bounds composed];

$\mathcal{N}=1,\ f\to1,\ N\mathrm{g}\to1$:

$$\Gamma\to\Gamma_{\mathrm{tr}}\le\tfrac{1}{3}\implies\boxed{\tau_{\mathrm{life}}:=\frac{1}{\Gamma}\ \ge\ 3\ \text{ticks}}$$

[BM.44(3); machine: exact reciprocal];

$$\varepsilon_q = 0: \qquad \Gamma = 0 \implies \tau_{\text{life}} \text{ has no finite value} \qquad [\text{BM.44}(3);\ \text{machine}];$$

¬a lifetime in external seconds is derived and ¬a lifetime is compared

$$[\text{BM.44}(5);\ \text{P5};\ \text{machine: zero external unit tokens}];$$

(6) the energy dependence (the width grows with the local scale):

at fixed r,

$$k(E) = \frac{r\,E_{\text{int}}(E)\ln 2}{\mathfrak{h}},$$

$$f = 1 - \frac{\mathfrak{h}}{r\,E_{\text{int}}(E)\ln 2}$$

$$[\text{BM.88}(6);\ \text{machine}]:$$

$$\boxed{E\uparrow \implies k\uparrow \implies f\uparrow \implies \Gamma\uparrow}$$

$$[\text{BM.88}(6)\text{'s three monotonicity rows re-cited};\ \text{machine: positive coefficients}];$$

the calibration-scale reading:

$$\text{“low energy”} := E \text{ near } E_1 = \mathfrak{h}/(r\ln 2),\ \text{where } f(1) = 0;$$

$$\text{“high energy”} := E \gg E_1,\ \text{where } f \to 1$$

$$[\text{BM.101}(5);\ \text{BM.88}(2)(\text{i});\ \text{machine}]:$$

at the low calibration point every width vanishes through f and through χ for the unmaterialized ones; at the high one the widths open in the order of $\mathcal{N}$ and $\mathcal{T}_0$

and both are evaluations of one function, ¬two regimes of two theories

$$[\text{Def-BM.17}(6)(\text{ii});\ \text{machine}];$$

(7) the two-sided bar and the width:

$$N\mathsf{g}\,\varepsilon_{\mathrm{EM}} \le \tfrac{1}{8} \implies \Delta m > 0;$$

$$\ge \tfrac{5}{12} \implies \Delta m < 0$$

[BM.99(3); BM.101(3) : the same enclosure] :

the consequence for the width: $\dfrac{\varepsilon_{\mathrm{eff}}}{\varepsilon_{\mathrm{EM}}}$

$$= Nf$$

g

$$\le N\mathsf{g} \implies \boxed{\text{the bar's band } \left[\tfrac{1}{8}, \tfrac{5}{12}\right] \text{ bounds the width's coupling factor exactly}}$$

and the sign of the mass splitting flips inside it — one function, two readings [BM.101$(1), (2)$; machine];

(8) status: derived: (1)–(7);

non-derived and named: (1)'s product form [Def-BM.17(4) non-derived clause];

priced, ¬adopted: $\mathsf{g} \in \{1, 1+G, \frac{1}{1+G}\}$ [BM.88(4); BM.104(5)];

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a numerical coupling [EM.25(7)],

¬a species identification [Def-BM.3(7)];

Req. D0.8, D1.8, D2.9 (Parts 1–2); CT0.8, EM.25(7) (Part 1, Part 2); P1, P4, P5 (Parts 1–2); Def-H.20, FT.44(4) (Part 3); Def-BM.3(7), Def-BM.4(1), Def-BM.17(3), Def-BM.17(4), Def-BM.17(5), Def-BM.17(6), BM.19(2), BM.19(3), BM.23(8), BM.44(1), BM.44(2), BM.44(3), BM.44(4), BM.44(5), BM.88(2), BM.88(4), BM.88(5), BM.88(6), BM.94(5), BM.99(3), BM.101(1), BM.101(2), BM.101(3), BM.101(5), BM.102(2), BM.104(5), BM.110(3), BM.114(1), BM.115(1) (this part); exact rational resolvent identities, the SOS parabola completion, exhaustive rational grid censuses, exact monotonicity by polynomial sign, Laurent-monomial exponent censuses [math; machine: `checks/verify_H_antimatter_resonance_spectrum.py` §E].

Pf.

$(1):$ the four frozen rows substituted into Def-BM.17(4)'s product [machine: the monomial assembly];

$$\frac{\varepsilon_{\text{eff}}}{\varepsilon_{\text{EM}}} = \frac{N f\,\mathrm{g}\,\varepsilon_{\text{EM}}}{\varepsilon_{\text{EM}}} = N f\,\mathrm{g}\ [\text{machine}];$$

$$[\Gamma] = \mathfrak{q}^0\ [\text{every factor dimensionless; machine}]$$

$$(2):\ \text{BM.44}(1),$$

(2)'s rows re-cited [machine: $4\varepsilon_q(1-4\varepsilon_q) = \frac{1}{4} - 16(\varepsilon_q - \frac{1}{8})^2$ by expansion; the 256-point grid census with equality only at $k = 32$]

$$\Gamma_{\text{tr}}(\tfrac{1}{8}) = \tfrac{1/4}{1-1/4} = \tfrac{1/4}{3/4} = \tfrac{1}{3}\ [\text{machine: exact rationals}];$$

monotonicity via $4(\varepsilon_2 - \varepsilon_1)(1 - 4(\varepsilon_1 + \varepsilon_2)) > 0$ on $\varepsilon_1 < \varepsilon_2 \le \frac{1}{8}$ [machine: the polynomial identity and a rational witness grid]

$$(3):\ (\text{Z1})\ x_- = 4 \cdot 0 = 0 \implies x_+ x_- = 0 \implies \Gamma_{\text{tr}} = 0/(1-0) = 0\ [\text{machine}];$$

$$(\text{Z2})\ k = 1 \implies f = 1 - 1 = 0\ [\text{BM.88}(5);\ \text{machine}];$$

$$(\text{Z3})\ \mathcal{N} = 0 \implies \text{the product is } 0\ [\text{machine}]$$

$$\#\{\mathcal{N} = 0\}:\ \text{at } n$$

$= 2$ the three classes $Q \in \{0, \pm 1\}$ each have a lightest member ($Q = \pm 2$ has one member and is its own lightest, counted once each);

the exact count over the $\lambda^{(2)}$ ladder of BM.114(3) is 3 on the sealed classes [machine]

at $n = 3$ the seven classes $Q = -3, \ldots, +3$ each contribute one [machine]

$$(4):\ \chi_{\{NE<\mathcal{T}_0\}}=0\implies\Gamma=0\ [\text{machine}];$$

BM.94(5)'s materialization restriction re-cited : $E\geq 2$ supports no closed chain below it

$$(5):\ f=1-\tfrac{1}{k}<1\iff\tfrac{1}{k}>0\iff k<\infty\ [\text{BM.88(5); machine}];$$

$$\Gamma_{\text{tr}}\leq\tfrac{1}{3}\ [(2)];$$

the product bound [machine]

$$\mathcal{N}=1,\ f\to 1,\ N\mathsf{g}\to 1:\ \Gamma\to\tfrac{1}{3}\implies\tau_{\text{life}}\to 3\ [\text{BM.44(3); machine: the exact reciprocal}];$$

$$\Gamma=0\implies 1/\Gamma\text{ has no finite value [machine]}$$

$(6):$ BM.88(6)'s three monotonicity rows re-cited;

f is a factor of Γ with all other factors

$$\geq 0\implies\Gamma\text{ increases with }f\text{ [machine: positive-coefficient composition]}$$

$(7):$ BM.99(3),

BM.101(3) re-cited, the enclosure re-certified on $2^{13}>3^8$ and $3^{12}>2^{19}$ [machine];

$$f\leq 1\implies Nf\,\mathsf{g}\leq N\mathsf{g}\ [\text{machine}];$$

$(8):$ the standing-rule scans [machine]. ■

Phys. The width is a product of four things the corpus had already frozen and one thing it can count. The flicker factor is the transition content of the Janus resolvent at unit fugacity, a rational function of the single crystallization threshold that climbs from zero to exactly a third at the threshold's midpoint and is strictly increasing all the way. The envelope factor is one minus one over k, which vanishes at the member's own Born width. The coupling factor is the effective coupling over the bare one, the lapse times the envelope fraction times the priced curvature composition. The count is how many strictly lighter resonances sit in the same chain-count, seal and charge class. And the indicator says whether the resonance is materialized at all. Three of those give exact zeros

with distinct mechanisms: no flicker means no dissolution move and hence no transition weight; sitting at your own envelope means the enclosed flux fraction is zero; and being the lightest in your class means there is nothing to go to, so the count is zero and the width is zero by arithmetic. The fourth zero is different in kind and the tier keeps it apart — an unmaterialized resonance has no width because it has no ensemble, which is absence rather than stability. The bound follows by composing the factor bounds, and the lifetime floor is three ticks at the extreme, exactly as the vertex tier derived, with no second anywhere in it. Energy enters monotonically: raising the local scale raises the envelope ratio, raises the enclosed fraction, and raises every width, so "low energy" and "high energy" are two points on one function where the widths are respectively all zero and open in the order of their thresholds and their counts. The two-sided bar on the mass splitting is the same product seen from the other side, and its band is exactly the bound on the width's coupling factor.

Theorem 4.100 (BM.117 — The groupings, the ceilings, and the octet adjudication). Thm.

(1) the groupings that are derived:

grouping	the exact value	locus
the charge classes at n	$2n+1$ classes with the binomial weights $\binom{2n}{Q+n}$	[BM.114(1)]
the sealing grammar	$g(N)=\sum_{2a+3b=N}\frac{N!}{a!\,b!\,2^a6^b}=(0,1,1,3,10,25,105)$ for $N=1..7$	[CT3.17a(2); Def-BM.9]
the multiset orbits at n	$\binom{6+n-1}{n}=(6,\ 21,\ 56)$	[BM.114(4)]
the all-distinct orbits	$\binom{6}{n}=(6,\ 15,\ 20)$ with multiplicity $n!$	[BM.114(4); machine]
the generation floors	3, ordered exactly by $2E=12<24<60$	[Def-BM.14(1), (6); BM.60(1)]
the branches	2 per floor, ordered exactly by $D_S(\mathcal{F}_g,-)<D_S(\mathcal{F}_g,+)$	[Def-BM.14(4)]
the flavour symmetry	C_3, its orbits the three channel lines	[FT.111(2); FT.158(1); FT.142(5)]
the defect field	$\mathbb{M}$, $[\mathbb{M}:\mathbb{Q}]=16$, with the subfield ladder $1,2,4,8,16$	[BM.114(3), (6)]

(2) the grouping the spectrum actually carries:

the three-level grouping of the n-chain ladder is (floor, branch, charge class) with the orbit multiplicities of BM.114(4) :

at $n=3$: $\quad 56$ orbits $=$ 6 of size 1 $+$ 30 of size 3 $+$ 20 of size 6, $\qquad 55$ eigenvalues after the one collision

the 20 all-distinct orbits are the ε-seal's fully antisymmetric slot pattern realized on distinct states:

ε_{ijk} seals three fundamentals antisymmetrically [D3.9; CT3.15],

and S_3 acts freely exactly on the all-distinct tuples [machine] :

$20=\binom{6}{3}$ is derived as an orbit count and is ¬an SU(3) decuplet dimension

[¬∃ a flavour SU(3) in the inventory [BM.103(7)]; machine: the symbol census];

(3) the octet adjudication (re-certified, and what replaces it):

the derived 3 is the colour seal's fundamental [CT3.15; CT3.17; D3.9];

the derived 8 is a mask-capacity ratio, barred as a coupling [BM.36(2), (3); BM.61(8)(O1)];

the derived 10 is the grammar count $g(5) = 10$ [CT3.17a(2); Def-BM.9; BM.31];

the derived 15 and 20 are the multiset orbit counts of (1);

the baryon octet and the meson octet are ¬derived, and the groupings that are derived are the binomial charge classes, the multiset orbits, the C_3 triplet and the grammar counts

[BM.103(7) re-cited and extended with the resonance census; machine: $8 \notin \{2n+1,\ \binom{6+n-1}{n},\ \binom{6}{n},\ g(N)\}$ at $n \le 3,\ N \le 7$];

(4) the ceilings:

the per-constituent pole ceiling $\quad D_S(s) \le 4 \implies m^2_{\text{pole}}(s) \le 4\hat{\mu}'^2$

[FT.76(3); Def-BM.14(3) : max $D_S = 4$ at s_2; machine] :

the n-chain resonance ceiling is its n-fold sum:

$$\lambda^{(n)}_{\max} = 4n, \qquad m^2_{\text{pole}}(\mathcal{R}) \le 4n\,\hat{\mu}'^2, \qquad \text{attained exactly by s} = (s_2, \ldots, s_2)$$

[machine: $4n$ is in the ladder at every $n = 1, 2, 3,$ with multiplicity 1];

the locus discipline:

FT.76(3)'s ceiling is a property of a single chain's pole-side read and FT.79(2)'s of a single chain's colour-side read [BM.103(6); FT.84(2) : the two-ceiling synthesis] :

an n-chain composite eigenvalue is a sum of n single-chain reads and is ¬a single-chain read — ¬a ceiling is violated and ¬a ceiling is extended to the composite as such

the informational ceiling:

$$E_{\text{int}}^{(n)} \in [12n,\ 60n] \text{ on the six-state ladder } \implies m_{\text{hid}}^{(n)} \in \Big[\frac{12n\ln 2}{\mathfrak{h}},\ \frac{60n\ln 2}{\mathfrak{h}}\Big]$$

[BM.115(2)(ii); machine];

the two currencies stay incommensurate:

¬∃ a derived ratio between $\lambda^{(n)}$ and $E_{\text{int}}^{(n)}$ — the single residue is ρ_{B}

[BM.82(1), (5), (7); BM.69(4); A-BM.9];

(5) the deficit-12 row is untouched:

the two mass ceilings stand as properties of the reads they were proved for [FT.76(3); FT.79(2); FT.97(5) : the three-ceiling synthesis] :

the resonance ladder adds no fourth ceiling and lifts none:

$\lambda^{(n)}$ is a sum of ceiling-bounded terms and its own bound $4n$ is the sum's [(4)];

¬a measured mass, hierarchy or range is compared

[P5; FT.84(5b); machine: zero external tokens];

(6) the no-forbidden-state row, at the grouping level:

every one of the 6^n ordered constituent tuples is a formed resonance with a nonempty threshold locus:

$$\#\{\text{ordered tuples at } n\} = 6^n = (6,\ 36,\ 216),$$

$$\#\{\text{distinct eigenvalues}\} = (6,\ 21,\ 55),$$

$$\#\{\text{tuples with an empty locus}\} = 0$$

[BM.115(5); machine: exhaustive] :

there is no excluded composition at any level — the two named heaviest-floor triples are unread, ¬unreachable

[BM.103(8)];

(7) status: derived: (1), (2), (4), (6); re-certified and extended: (3); untouched: (5);

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a currency conversion [BM.83(3)],

¬a species identification [Def-BM.3(7)];

α_B : ¬added,

¬derived,

barred on BM.61(8)'s four grounds [machine: the token scan of this tier];

Req. D2.9, D3.9 (Parts 1–2); CT3.15, CT3.17, CT3.17a(2) (Part 2); P1, P4, P5 (Parts 1–2); FT.76(3), FT.79(2), FT.84(2), FT.84(5), FT.97(5), FT.111(2), FT.142(5), FT.158(1) (Part 3); Def-BM.3(7), Def-BM.9, Def-BM.14(1), Def-BM.14(3), Def-BM.14(4), Def-BM.14(6), BM.31, BM.36(2), BM.36(3), BM.60(1), BM.61(8), BM.69(4), BM.82(1), BM.82(5), BM.82(7), BM.83(3), BM.103(7), BM.103(8), BM.114(1), BM.114(3), BM.114(4), BM.114(6), BM.115(2), BM.115(5), A-BM.9 (this part); exact integer orbit and grammar enumeration, exhaustive tuple censuses, exact radical comparison in $\mathbb{M}$ [math; machine: `checks/verify_H_antimatter_resonance_spectrum.py` §F].

Pf.

(1) : the eight rows re-cited from their own tags and their exact values re-computed [machine];

$g(N)$ for $N = 1..7$ by the labelled set-partition formula [CT3.17a(2); machine]

$$(2):\ 6+30+20=56,$$

$$6\cdot 1+30\cdot 3+20\cdot 6=6+90+120=216=6^3\ [\text{machine}];$$

$$S_3 \text{ acts freely on the all-distinct tuples : } |\,\text{Stab}\,|$$

$$=1$$

$$\implies |\text{orbit}|$$

$$=3!$$

$$=6 \text{ and \#orbits}$$

$$=\tfrac{6\cdot 5\cdot 4}{6}$$

$$=20\ [\text{machine}]$$

$$\varepsilon_{ijk} \text{ antisymmetric in three slots [D3.9; CT3.15];}$$

$$\neg\exists \text{ a flavour } \mathsf{SU}(3)\ [\text{BM.103(7); machine: the symbol census}] \implies 20 \text{ is an orbit count}$$

$$(3):\ \text{BM.103(7)'s four rows re-cited;}$$

$$8\notin\{2n+1\}_{n\le 3}\cup\{\tbinom{6+n-1}{n}\}_{n\le 3}\cup\{\tbinom{6}{n}\}_{n\le 3}\cup\{g(N)\}_{N\le 7}\ [\text{machine: exhaustive membership test over } \{3,5,7\}\cup\{6,21,56\}\cup\{6,15,20\}\cup\{0,1,1,3,10,25,105\}]$$

$$(4):\ \max_s D_S(s)=4 \text{ at } s_2\ [\text{Def-BM.14(3); machine: exact comparison in } \mathbb{M}];$$

$$\lambda_{\max}^{(n)}=n\cdot 4=4n \text{ and it is attained by } (s_2,\dots,s_2) \text{ with multiplicity } 1\ [\text{BM.114(3); machine}]$$

$$\text{FT.76(3), FT.79(2), FT.84(2) re-cited at their own loci;}$$

$$E_{\text{int}}^{(n)}$$

$\in [12n, 60n]$ on the six-state ladder and $[2n, 60n]$ on the computed-cell ladder [BM.115(2)(i), (ii)];

BM.82$(1), (5), (7)$ and BM.69(4)'s four grounds re-cited

(5) : FT.97(5)'s three-ceiling synthesis re-cited;

$\lambda^{(n)} \leq 4n$ is the sum's own bound $[(4);$ machine];

the external-token scan [machine]

(6) : $6^n = (6, 36, 216)$;

#distinct $= (6, 21, 55)$ [BM.114$(3), (5);$ machine];

#{empty loci} $= 0$ [BM.115$(5);$ machine: all 216 triples carry $E_{\text{int}} \in [6, 180] \subset \mathbb{Z}_{>0}$]

(7) : the standing-rule scans and the barred-object token scan [machine]. ■

Phys. Asked to group the spectrum, the corpus gives what it has and no more. Eight groupings are derived and each has an exact value: the binomial charge classes, the sealing grammar's counts, the multiset orbits, the all-distinct orbits, the three ordered generation floors, the two ordered branches, the cyclic three of the flavour channel, and the subfield ladder of the eigenvalue field. The one the resonance spectrum actually carries is the three-level grouping by floor, branch and charge class with the orbit multiplicities, and at three chains that is six orbits of size one, thirty of size three and twenty of size six — fifty-six orbits giving fifty-five eigenvalues after the single powers-of-two collision. Twenty appears, and it is exactly the count of ways to pick three distinct states from six, which is also where the epsilon seal's free symmetric-group action lives; the tier states plainly that this is an orbit count and not a decuplet dimension, because the flavour symmetry the corpus derives is a cyclic three and there is no special unitary three anywhere in the inventory. The octet adjudication is re-certified and extended: the derived three is colour's, the derived eight is a barred capacity ratio, the derived ten is a grammar count, and eight does not occur among any of the four grouping families at any formed level. The ceilings behave exactly as they should — the single-chain pole ceiling is four, the composite ceiling is four times the chain count and is attained by the heaviest branch at every level, and a composite eigenvalue is a sum of single-chain reads rather than a single-chain read, so no ceiling is violated and none is extended. And the count that answers the forbidden-state question a second time is the simplest one: all two hundred sixteen ordered constituent triples are formed resonances with nonempty threshold loci, so the two heaviest compositions anyone might name are unread rather than unreachable.

BM.118 — The status of the Antimatter and Resonance Tier: adjudication, scope, and the open items. Structural-constraints record: see Appendix B, Section B.1.31 of the Addendum.

4.24 Tier BM.22 — The Annihilation Channel, Concrete Spin, and Fermion Identification: the Janus Tunnel, the Zig-Zag World Line, and the Spin-Dependent Transport Defect

Locus. This tier takes the residue (R9) that BM.118(7) named — the annihilation channel: an inter-sector transition operator, ¬in the inventory — and constructs the channel, then follows the two chains that fall out of it: the concrete spin and the composite statistics. The crossing is not a velocity and the tier says so first: $c = 1$ is a convention, the dispersion gives a strict subluminal deficit $1-v = m^2/(E(E+p)) > 0$ at every finite momentum, and the boundary the corpus actually derives between the materialized and the fluctuation content is the Janus real/virtual register, whose $\mathcal{V}$ sector is the maximally raw-entropic state $\rho_\circ = \frac{1}{4}I$ with no fermion terminus at $|E_\mathcal{R}| = 0$ (BM.119). On that register the crossing operator is derived, not posited: the two moves $\hat{J}_\pm$ of D1.2, the master generator $Q = x_+(\hat{J}_+ - P_\mathcal{V}) + x_-(\hat{J}_- - P_\mathcal{R})$, its one-dimensional kernel giving the stationary measure, its spectrum $\{0, -(x_+ + x_-)\}$, and detailed balance as a theorem (BM.120). The zig-zag is then the resolvent's own evenness — $(\hat{U}^{2m})_{\varnothing\varnothing} = (x_+x_-)^m$ with every odd term zero — so each term is a create-propagate-annihilate round trip whose legs all run forward in the ledger order, and the "backward leg" of a diagram is here the CP-conjugate configuration on a forward leg, which is why no inverse tick is needed (BM.121). The pair channel is derived with its charge balance, its exact threshold, its weight and its three CP legs — the nilpotent weak step for the charge inversion, the CP transport for the parity inversion, the local scale for the energy — while the two-photon supply is barred on four named grounds, so the channel closes and the amplitude does not (BM.122). Annihilation is the reverse move, its content returning to the $\mathcal{V}$ sector at the entropy maximum rather than into radiation legs, and (R9) closes at the channel level with the cross-section residue narrowing to the momentum-space resolvent and the gauge coupling of the charge read (BM.123). On the spin side the tier separates two angles that a single-symbol reading would fuse: the winding Θ and the frame-holonomy angle Θ_R are provably not proportional, on a dimension mismatch, so the spin-dependent transport defect is a multiplet ladder $\Delta(m) = 4\sin^2(m\Theta_R/4)$ over $m \in \{k, k-2, \ldots, -k\}$ and not a number $\Delta(j)$ (BM.124, BM.125). A spin-dependent crossing weight is refuted on the four standing grounds, and what survives is a domain correlation rather than a prohibition (BM.126). The winding parity is then a ledger read, separated from the D1.5 kind read that BM.106(1) already showed cannot carry a spin-parity theorem (BM.127). Finally the three-chain exchange structure is built — $\mathcal{H}_3$, the S_3 action with its exact character formula $d^{c(\sigma)}$, and the vanishing of $\wedge^3(\mathbb{C}^2)$ against the one-dimensionality of $\wedge^3(\mathbb{C}^3)$, which is why the baryon's antisymmetry is colour's and the statistics is the winding's — giving the multiplicative exchange law $\varepsilon = \prod_a \varepsilon(c_a)$, the meson's unconditional boson sign, and the baryon's sign as an exact function of its three constituent windings (BM.128, BM.129). BM.130 adjudicates.

Standing rows. α_B stays barred on BM.61(8)'s four grounds; §Cal-BM stays empty except for A-BM.9's logged slot with its value ¬issued. No right-ray matter line, no seesaw-type mechanism, no Yukawa-type object and no external mixing matrix is suggested, added or required [EW.15; FT.145(3); BM.96(4), (6) re-cited, ¬extended]. No clause of BM.1–BM.118 is retracted, replaced or renumbered: the rows this tier touches are re-scoped at their own loci — BM.118(7)'s residue (R9), BM.112(7)'s and Def-BM.16(7)'s residue (R7) at $n \geq 3$, BM.109(2)'s pair-weight naming, BM.114(2)'s colour-antisymmetry row, BM.27(7), BM.29(5) and Def-BM.15(5)'s no-annihilation rows, BM.6(6)'s resolvent specification and BM.6(2)'s discharged specification — and each rescope is recorded in BM.130(5), in BM.77(7)'s pattern. No measured coupling, mass, width, lifetime, cross-section or ratio enters any derivation [P5]. No cosmological coordinate is used: a particle accelerator and a deep-gravity region are both local regions with high (E, N, G), and every threshold and width is a function of those four local parameters alone [Def-BM.17(6)(i); A6; D1.8].

"Low energy" and "high energy" are calibration points on the derived functions and ¬regimes of two theories [BM.116(6); Def-BM.17(6)(ii)]. No black-hole physics is derived: the comparison radius r_s enters only through BM.15(4)'s quarantined change of variable, with ¬a collapse reading and ¬a horizon identification [P3; BM.115(4)]. Species names are confined to the identification layer [Def-BM.3(7)]. Topology labels are internal bookkeeping [Def-H.32(4)]. Amplitude ≠ mass [FT.79(5)]. Rung ≠ scale [D0.8]. No currency is converted [BM.83(3); BM.102(6)]. No theorem of this tier imports a HistLog disposition as premise: Q9, Q11, items 7, 13, 17, 20 and deficits $1'$, 3, 4, 6, 11, 12 are cited as pointers. One dimension only: units are powers of $\mathfrak{q}$ [D2.9; CT2.1].

Local notation, fixed for the whole tier. $\mathcal{J} := \mathrm{span}_{\mathbb{C}}\{|\mathcal{V}\rangle,\ |\mathcal{R}\rangle\} \cong \mathbb{C}^2$ the per-pair Janus register in the (virtual, real) order of FT.108(4); $P_{\mathcal{V}}, P_{\mathcal{R}}$ its two orthogonal projectors; $\hat{J}_+ := |\mathcal{R}\rangle\langle\mathcal{V}|$ the crystallization move and $\hat{J}_- := |\mathcal{V}\rangle\langle\mathcal{R}|$ the dissolution move [D1.2]; Q the master generator over $\mathbb{Z}[x_+, x_-]$ [FT.108(4)]; $\varepsilon_q \in [0, \frac{1}{4})$, $x_+ = 1 - 4\varepsilon_q$, $x_- = 4\varepsilon_q$, $p = x_+/(x_+ + x_-) = 1 - 4\varepsilon_q$ [A4; D0.4; CT0.8; BM.19(2)]; w the formal fugacity and $\hat{U}$ the single-channel step [FT.108(5); Def-H.32(3)]; $\Theta(c) = \sum_{e\in E(c)} \widehat{ds}^2(e) \in 2\mathbb{Z}$ the winding, $k = \Theta/2 \in \mathbb{Z}_{\geq 0}$, $j = |k|/2$, $\Psi(c) = \mathrm{Sym}^{|k|}(\mathcal{H}_{\mathsf{v}})$ with $\mathcal{H}_{\mathsf{v}} \cong \mathbb{C}^2$ [FT.22; Def-H.38; EW.32]; Θ_R the frame-holonomy angle with $[\Theta_R] = \mathfrak{q}^0$ [Def-H.21(5), (6); FT.55(1)]; $\phi := \Theta_R/2$; T_m the Chebyshev polynomial of the first kind; $\Delta(m) := 4\sin^2(m\Theta_R/4) = 2 - 2\cos(m\phi)$ the transport defect at multiplet weight m [FT.60(5); FT.54(4); EW.33]; $\mathcal{H}_\omega(c)$, $\hat{\omega}$, $\hat{\chi}$, $P_{L/R}$, $|L\rangle$, $|R\rangle$ as in Def-H.33; $\widehat{V} = E_+ \otimes P_L$ the nilpotent step [BM.76(1)]; $\mu_{\mathrm{CP}} = e^{i\pi\Theta}$ and $\varepsilon(c) = (-1)^{|k(c)|} = (-1)^{2j(c)}$ [BM.108(3); BM.113(2), (4)]; $\mathcal{H}_2$, $\hat{P}_{12}$ as in Def-BM.16; $d := \Theta(P) \geq 0$ a connector winding [BM.2(2); FT.34(3)]; S_σ the sign of a permutation σ; $c(\sigma)$ its cycle count; $\mathcal{T}_0$, $\mathcal{T}$, Γ, $\mathcal{N}$, $\chi_{\{\cdot\}}$ as in Def-BM.17; $[\cdot]$ the $\mathfrak{q}$-exponent map [CT2.1; D2.9].

Definition 4.18 (Def-BM.18 — The Janus register and its crossing operators). Def. A construction on D1.1–D1.2's real/virtual distinction, FT.108(4)'s master generator and Def-BM.16's two-chain space; no new axiom, no new datum, no new parameter, no velocity operator, no position operator, no external scattering object.

(1) the register:

$$\boxed{\mathcal{J} := \mathrm{span}_{\mathbb{C}}\{|\mathcal{V}\rangle,\ |\mathcal{R}\rangle\} \cong \mathbb{C}^2}$$

in the (virtual, real) order of FT.108(4);

$$e \in \mathcal{R} \iff \left|\Pi_{\mathcal{V}}\rho\right| \leq \varepsilon_q \ \vee\ \left|\Pi_{\mathcal{V}}\rho\right| \geq 1 - \varepsilon_q; \qquad \neg\text{registration} \implies e \in \mathcal{V} \qquad [\text{A4; D1.1; CT0.1}]:$$

constitutive clause: $\mathcal{J}$ is the two-state register of one pair across the A4 criterion;

non-derived clause: the basis order $(\mathcal{V}, \mathcal{R})$;

(2) the two crossing operators:

$$\boxed{\hat{J}_+ \;:=\; |\mathcal{R}\rangle\langle\mathcal{V}| \;\text{(crystallization, } J^+ :\; n_{\mathcal{R}} : 0 \to 1), \qquad \hat{J}_- \;:=\; |\mathcal{V}\rangle\langle\mathcal{R}| \;\text{(dissolution, } J^- :\; n_{\mathcal{R}} : 1 \to 0)}$$

[D1.2 : a seal is the transition of one pair across the A4 criterion, in either branch];

$$P_{\mathcal{V}} := |\mathcal{V}\rangle\langle\mathcal{V}|, \quad P_{\mathcal{R}} := |\mathcal{R}\rangle\langle\mathcal{R}|, \quad P_{\mathcal{V}} + P_{\mathcal{R}} = \mathrm{id}$$

[machine];

(3) the master generator:

$$\boxed{Q \;:=\; \begin{pmatrix} -x_+ & x_+ \\ x_- & -x_- \end{pmatrix} \;\text{over}\; \mathbb{Z}[x_+, x_-]}$$

[FT.108(4) : the per-pair Janus dynamics] :

the decomposition (derived, recorded here):

$$Q \;=\; x_+\big(\hat{J}_+ - P_{\mathcal{V}}\big) \;+\; x_-\big(\hat{J}_- - P_{\mathcal{R}}\big)$$

[machine: entrywise at six rational witnesses];

constitutive clause: Q is FT.108(4)'s own generator, ¬a new dynamics;

the row convention: Q acts on row vectors, so Q_{ij} is the rate from i to j;

(4) the crossing space of a chain:

$$\boxed{\mathcal{T}_{\mathcal{J}}(c) \;:=\; \mathcal{J} \;\otimes\; \mathcal{H}_\omega(c) \;\otimes\; \Psi(c), \qquad \dim \mathcal{T}_{\mathcal{J}}(c) \;=\; 4\,\big(|k(c)| + 1\big)}$$

[Def-H.33; Def-H.38; Def-BM.16(1); machine: exact integers];

the pair crossing space: $\mathcal{T}^2_{\mathcal{J}}(c) := \mathcal{T}_{\mathcal{J}}(c) \otimes \mathcal{T}_{\mathcal{J}}(c^{\mathrm{CP}})$,

$$\dim = 16\,\big(|k| + 1\big)^2;$$

(5) the channel clauses:

a pair creation :

$=$ $\hat{J}_+$ on two registers, with the two members identified as a CP-pair [Def-BM.15(3)];

weight x_+^2 for the two one-way moves and x_+x_- for one round trip;

an annihilation $:=$ $\hat{J}_-$ on the same two registers; weight x_-^2 one-way, x_+x_- per round trip;

a zig-zag :

$=$ a finite alternating sequence of $\hat{J}_+$ and $\hat{J}_-$ moves, every step recorded in the ledger order [A6; D1.8];

constitutive clause: the three words name move sequences on $\mathcal{J}$,

¬world lines in a spacetime and ¬asymptotic scattering states;

(6) the bars (standing):

¬∃ a velocity operator, a position operator or a momentum-space state on $\mathcal{J}$ [BM.10(1) : $\neg[\hat{x},\hat{p}]=i$ in the finite register; BM.6(6) : the momentum-space Σ-resolvent is an open interface];

¬∃ a cross-section, an S-matrix element or a Lorentz-invariant phase space [BM.6(6); BM.118(7)(R9)];

¬∃ a direct $\mathrm{EM}^+ \to \mathrm{EM}^-$ operator: the sectors are amplitude-equal configuration orbits and the channel that connects them is mediated by $\mathcal{J}$ and CP [BM.25(4); BM.27$(6),(7)$; BM.29(5)];

the kind bar:

$\mathcal{J}$ is $\neg\mathcal{H}_\omega$ and ¬one of the four single-object $\mathbb{Z}_2$ gradings [BM.106(4); BM.112(5); FT.125(5); FT.89(2)] :

the real/virtual distinction is the Janus register; the orientation distinction is $\mathcal{H}_\omega$; identifying them is a barred kind confusion

(7) status and scope:

derived: (3)'s decomposition, (4)'s dimensions;

constitutive: (1)'s register, (2)'s two moves, (5)'s three names;

the scope inherited at every use: Def-H.32's (S1)–(S4);

Def-H.38's nonnegative cone and its $k \geq 3$ aspect bar;

BM.8(2)'s stationarity hypothesis for any chart evolution;

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a species identification [Def-BM.3(7)],

¬a rate datum [Cal-F carries no rate; BM.44(5)];

beyond this tier: the momentum-space Σ-resolvent [BM.6(6)],

the gauge coupling of the charge read [Def-BM.8(8)].

Rmk (what the register is and is not). $\mathcal{J}$ is the two-state register of one pair across the crystallization criterion — the object D1.2 named and FT.108 derived a stationary measure for. It is not a spacetime, not a world line, and not a Hilbert space of asymptotic states. Clause (6)'s kind bar is the load-bearing one: the corpus has an orientation doublet $\mathcal{H}_\omega$ per chain, and it would be a category error to read the real/virtual distinction as that doublet's two rays, exactly as it would be to read a single-chain grade as a permutation sign. The zig-zag of clause (5) is a word in two moves, recorded forward, and its weight is a monomial in x_+ and x_-.

Req. A4, A6, D0.8, D1.1, D1.2, D1.8 (Part 1); CT0.1, CT0.8, CT2.1, D2.9 (Part 1, Part 2); P1, P3, P4, P5 (Parts 1–2); Def-H.21(5), Def-H.21(6), Def-H.32, Def-H.33, Def-H.38, FT.22, FT.89(2), FT.108(4), FT.108(5), FT.125(5) (Part 3); Def-BM.3(7), Def-BM.15(3), Def-BM.16(1), BM.2(2), BM.6(6), BM.8(2), BM.10(1), BM.25(4), BM.27(6), BM.27(7), BM.29(5), BM.44(5), BM.106(4), BM.112(5), BM.118(7) (this part); exact 2×2 matrix algebra over $\mathbb{Z}[x_+, x_-]$ and over $\mathbb{Q}(i)$, exact integer dimension arithmetic [math; machine: `checks/verify_H_janus_tunnel_concrete_spin.py` §A].

Pf.

Definition block: the constitutive clauses are (1)'s register,

(2)'s two moves and (5)'s three names;

the non-derived clauses are the basis order of (1) and the row convention of (3);

(3)'s decomposition and (4)'s dimensions are derived and machine-checked;

(6),

(7) are bars and scope records. ■

Phys. One pair, crossing a criterion, has two states and two moves, and the corpus already derived both: D1.2 named the seal as a transition across the A4 criterion in either branch, and FT.108 wrote down the master generator whose kernel is one-dimensional and gives the stationary measure. This definition makes the two moves into operators on a two-dimensional register, decomposes the generator as a sum of jump-minus-loss terms with the two derived weights, and tensors the register with the chain's own orientation doublet and winding carrier to give a crossing space of exact finite dimension. Three words are then defined and no more: a pair creation is the crystallization move on two registers whose members are identified as a CP-pair, an annihilation is the dissolution move on the same two, and a zig-zag is a finite alternating word in the two moves, every step recorded in ledger order. What the definition refuses is the vocabulary that would overreach — no velocity operator, no position operator, no momentum-space state, no cross-section, no S-matrix, and above all no identification of the real/virtual register with the orientation doublet, which is a different two-dimensional object doing different work.

BM.119 — The c-threshold adjudication: no velocity boundary, the exact subluminal deficit, and the register boundary that is derived. Structural-constraints record: see Appendix B, Section B.1.32 of the Addendum.

Theorem 4.101 (BM.120 — The Janus tunnel: the crossing operators, the exact spectrum, and detailed balance). Thm.

(1) the move algebra (exact, over $\mathbb{Q}$):

$$\hat{J}_+^2 = 0,$$

$$\hat{J}_-^2 = 0,$$

$$\hat{J}_+ \hat{J}_- = P_{\mathcal{V}},$$

$$\hat{J}_-\hat{J}_+ = P_{\mathcal{R}},$$

$$\{\hat{J}_+, \hat{J}_-\} = \mathrm{id}$$

[machine: all five identities entrywise on the 2×2 register];

each move is nilpotent and the two anticommute to the identity

the reading: $\hat{J}_+\hat{J}_- = P_{\mathcal{V}}$ is the round trip $\mathcal{V} \to \mathcal{R} \to \mathcal{V}$;

$\hat{J}_-\hat{J}_+ = P_{\mathcal{R}}$ is the round trip $\mathcal{R} \to \mathcal{V} \to \mathcal{R}$;

¬∃ a doubled move in the same direction — the A4 criterion is binary, and the nilpotency is its operator form [D1.2; BM.23(2)'s overflow pattern];

(2) the generator decomposes as jump minus loss:

$$\boxed{Q \;=\; x_+\big(\hat{J}_+ - P_{\mathcal{V}}\big) \;+\; x_-\big(\hat{J}_- - P_{\mathcal{R}}\big)}$$

[FT.108(4); machine: entrywise at six rational witnesses $(x_+, x_-) \in \{(\frac{1}{2}, \frac{1}{2}), (\frac{3}{4}, \frac{1}{4}), (1, 0), (0, 1), (\frac{2}{3}, \frac{1}{3}), (\frac{7}{8}, \frac{1}{8})\}$];

the row-sum law:

Q acts on row vectors and every row sums to zero: $(-x_+) + x_+ = 0$,

$$x_- + (-x_-) = 0$$

[machine] :

$\therefore$ Q is a master generator and ¬a unitary — the crossing is a stochastic ledger move, ¬a rotation of a state;

(3) the kernel, the stationary measure and detailed balance:

$$(x_-,\ x_+)\,Q \;=\; \big(x_-(-x_+) + x_+x_-,\ \ x_-x_+ + x_+(-x_-)\big) \;=\; (0,\ 0)$$

[machine: exact in $\mathbb{Z}[x_+, x_-]$];

$$\det Q \;=\; (-x_+)(-x_-) - x_+x_- \;=\; 0,$$

$$Q \neq 0 \implies \operatorname{rk} Q = 1,$$

$$\dim \ker Q = 1$$

[FT.108(4); machine];

$$\boxed{p \;=\; \frac{x_+}{x_+ + x_-} \;=\; 1 - 4\varepsilon_q \text{ identically}}$$

[BM.19(2); machine: exact at seven rational thresholds];

detailed balance is derived, not assumed:

$$p_{\mathcal{V}}\, x_+ \;=\; p_{\mathcal{R}}\, x_- \iff \frac{x_-}{x_+ + x_-}\, x_+ = \frac{x_+}{x_+ + x_-}\, x_- \iff x_-x_+ = x_+x_-$$

[FT.108(4); machine];

(4) the exact spectrum:

$$\operatorname{tr} Q = -(x_+ + x_-),$$

$$\det Q = 0 \implies \boxed{\operatorname{spec} Q = \{0,\; -(x_+ + x_-)\}}$$

[machine: the characteristic polynomial $\lambda^2 + (x_+ + x_-)\lambda$];

at the specialization $x_+ + x_- = 1$:

$$\operatorname{spec} Q = \{0,\; -1\},$$

$$\operatorname{tr} Q = -1,$$

$$\operatorname{rk} Q = 1$$

[machine];

the zero eigenvalue is the stationary mode and -1 is the relaxation mode at unit total weight

[machine: the two eigenvectors (x_-, x_+) and $(1, -1)$];

(5) the tunnel operator and its weights:

one-way creation $\hat{J}_+$: weight $x_+ = 1 - 4\varepsilon_q \in (0, 1]$;

one-way annihilation $\hat{J}_-$: weight $x_- = 4\varepsilon_q \in [0, 1)$;

the round trip:

$$\mathcal{W}_{\text{round}} \;=\; x_+x_- \;=\; 4\varepsilon_q(1 - 4\varepsilon_q) \;=\; \frac{1}{4} - 16\Big(\varepsilon_q - \frac{1}{8}\Big)^2 \;\le\; \frac{1}{4}$$

[BM.109(2); BM.44(1); machine: the SOS completion and the 256-point grid];

the detailed-balance point is the maximum:

$$x_+ = x_- \iff \varepsilon_q = \frac{1}{8},$$

and

$$x_+x_- \le \Big(\frac{x_+ + x_-}{2}\Big)^2 = \frac{1}{4} \text{ with equality } \iff x_+ = x_-$$

[machine: the AM–GM row on exact rationals, equality locus verified];

the round-trip weight is maximal exactly where the two moves balance

(6) the type correction (the crossing acts on $\mathcal{J}$, not on $\mathcal{H}_\omega$):

$\mathcal{H}_\omega(c) = \text{span}\{|c\rangle, |\omega(c)\rangle\}$ is one chain's orientation doublet [Def-H.33];

$\mathcal{J} = \text{span}\{|\mathcal{V}\rangle, |\mathcal{R}\rangle\}$ is one pair's registration register [Def-BM.18(1)] :

the two are different objects with different involutions:

$$\hat{\omega}^2 = \mathrm{id} \text{ with } \mathrm{spec}\{\pm 1\} \text{ and } \mathrm{tr}\,\hat{\omega} = 0 \text{ [FT.122(1)]};$$

$$\hat{J}_\pm^2 = 0 \text{ with } \mathrm{tr}\,\hat{J}_\pm = 0 \text{ and } \neg\text{diagonalizable } [(1)] :$$

$\neg\exists$ an isomorphism $\mathcal{J} \cong \mathcal{H}_\omega$ that respects the two operator algebras — $\hat{\omega}$ is an involution and $\hat{J}_\pm$ are nilpotent

[machine: $\hat{\omega}^2 = \mathrm{id} \neq 0$ against $\hat{J}_\pm^2 = 0$];

$\therefore$ the crossing operator's type is $\hat{J}_\pm \otimes \mathrm{id} \otimes \mathrm{id}$ on $\mathcal{T}_{\mathcal{J}}(c) = \mathcal{J} \otimes \mathcal{H}_\omega(c) \otimes \Psi(c)$

[Def-BM.18(4); machine];

(7) the forward-only row:

the move system is bidirectional as a set of moves and has a stationary measure [(3)];

the tick is not bidirectional: $\neg\exists\, \tau^{-1}$ [A6];

$$\left|\mathfrak{L}(\tau_{n+1})\right| = \left|\mathfrak{L}(\tau_n)\right| + n_{\text{real_out}}(\tau_n) \geq \left|\mathfrak{L}(\tau_n)\right| \text{ [A6; A7]}; \qquad V' = \left(1 - \tfrac{\gamma}{4}\right)^2 V \leq V \text{ [L2]};$$

the ensemble sums both moves; the ledger records only growth — the formal mirror exists and the dynamical reversal does not, on the Janus register exactly as on the ledger

[BM.25(2); Def-BM.7(2); FT.144(4)];

(8) status: derived: (1)–(7);

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a numerical coupling [EM.25(7)],

¬a rate datum in external units [BM.44(5)],

¬a species identification [Def-BM.3(7)];

Req. A4, A6, A7, D0.4, D1.2, D1.8 (Part 1); CT0.8, EM.25(7) (Part 1, Part 2); P1, P3, P4, P5 (Parts 1–2); L2 (Part 1); Def-H.32, Def-H.33, FT.108(4), FT.108(5), FT.122(1), FT.144(4) (Part 3); Def-BM.3(7), Def-BM.7(2), Def-BM.18(1), Def-BM.18(4), BM.19(2), BM.23(2), BM.25(2), BM.44(1), BM.44(5), BM.109(2), BM.113(2) (this part); exact 2×2 matrix algebra over $\mathbb{Z}[x_+,x_-]$ and $\mathbb{Q}$, the SOS parabola completion, AM–GM on exact rationals with its equality locus, exhaustive rational grid censuses [math; machine: `checks/verify_H_janus_tunnel_concrete_spin.py` §A].

Pf.

$$(1):\ \hat J_+ = \left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right),\ \hat J_- = \left(\begin{smallmatrix}0&0\\1&0\end{smallmatrix}\right) \text{ in the } (\mathcal{V},\mathcal{R}) \text{ row convention [Def-BM.18(2)] :}$$

$$\hat J_+^2 = \left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right)^2 = 0, \qquad \hat J_-^2 = \left(\begin{smallmatrix}0&0\\1&0\end{smallmatrix}\right)^2 = 0; \qquad \hat J_+\hat J_- = \left(\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right) = P_{\mathcal{V}}, \qquad \hat J_-\hat J_+ = \left(\begin{smallmatrix}0&0\\0&1\end{smallmatrix}\right) = P_{\mathcal{R}};$$

$$\hat J_+\hat J_- + \hat J_-\hat J_+ = P_{\mathcal{V}} + P_{\mathcal{R}} = \mathrm{id} \text{ [machine: all five identities entrywise over } \mathbb{Q}]$$

$$(2):\ x_+(\hat J_+ - P_{\mathcal{V}}) + x_-(\hat J_- - P_{\mathcal{R}}) \text{ expanded entrywise gives } \begin{pmatrix} -x_+ & x_+ \\ x_- & -x_- \end{pmatrix}$$

$$= Q \text{ [machine: six rational witnesses]};$$

the two row sums [machine]

$$(3):\ (x_-,x_+)Q = (0,0) \text{ [machine: exact in } \mathbb{Z}[x_+,x_-]];$$

$$\det Q = x_+x_- - x_+x_- = 0;$$

$$Q \neq 0 \text{ at } x_+x_- > 0 \implies \mathrm{rk} = 1 \text{ [machine]}$$

$$p = \tfrac{x_+}{x_++x_-} = \tfrac{1-4\varepsilon_q}{1-4\varepsilon_q+4\varepsilon_q} = 1-4\varepsilon_q \text{ [machine: exact at seven thresholds]};$$

$$\text{detailed balance : } p_{\mathcal{V}}x_+ = \tfrac{x_-x_+}{x_++x_-} = p_{\mathcal{R}}x_- \text{ [machine]}$$

$$(4):\ \lambda^2-(\operatorname{tr}Q)\lambda+\det Q=\lambda^2+(x_++x_-)\lambda=\lambda(\lambda+x_++x_-)\ \text{[machine]};$$

$$\text{at } x_++x_-=1:\ \{0,-1\}\ \text{[machine]}$$

$$Q(x_-,x_+)$$

$$=(0,0)\text{ and }Q(1,-1)$$

$$=(-(x_++x_-),\ (x_++x_-))$$

$$=-(x_++x_-)(1,-1)\ \text{[machine: the two eigenvectors]}$$

$$(5):\ x_+x_-=4\varepsilon_q(1-4\varepsilon_q)=\tfrac14-16(\varepsilon_q-\tfrac18)^2\ \text{[machine: expansion]};$$

$$\text{the 256-point grid: } \le\tfrac14,\ =\tfrac14\text{ only at }\varepsilon_q=\tfrac18\ \text{[machine]}$$

$$\text{AM–GM}:\ x_+x_-\le\left(\tfrac{x_++x_-}{2}\right)^2$$

$$\iff 0$$

$$\le(x_+-x_-)^2\ \text{[machine: the square is nonnegative on exact rationals]},$$

$$\text{equality}$$

$$\iff x_+$$

$$=x_-$$

$$\iff 1-4\varepsilon_q$$

$$=4\varepsilon_q\iff\varepsilon_q$$

$$=\tfrac18\ \text{[machine]}$$

$$(6):\ \hat\omega=\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right),\ \hat\omega^2=\mathrm{id},\ \operatorname{tr}\hat\omega=0,\ \text{spec}=\{\pm1\}\ \text{[Def-H.33; FT.122(1); machine]};$$

$$\hat{J}_{\pm}^{2} = 0 \neq \mathrm{id}\ [(1);\ \text{machine}] \implies \neg\text{an algebra isomorphism}$$

$$(7):\ \text{A6},\ \text{A7},\ \text{L2},\ \text{BM.25}(2),\ \text{FT.144}(4)\ \text{rows re-cited};$$

$$(8):\ \text{the standing-rule scans [machine]}.\ \blacksquare$$

Phys. The crossing has an algebra and it is small enough to write down completely. Each move squares to zero, because the crystallization criterion is binary and a pair cannot crystallize twice without dissolving in between; the two moves anticommute to the identity, and their two products are exactly the two round trips. The generator is the sum of a jump term and a loss term for each move, its rows sum to zero, and it is therefore a master generator rather than a unitary — the crossing is a stochastic ledger move, not a rotation of a state. Its kernel is the line spanned by the two weights, which is where the stationary measure comes from, and detailed balance is an identity rather than an assumption: the flux out of the virtual state equals the flux out of the real one at that measure, because both are the same product. The spectrum is zero and minus the total weight, so at unit total weight there is one stationary mode and one relaxation mode with rate exactly one. The round-trip weight is the familiar parabola with maximum one quarter, and the maximum sits exactly where the two moves balance — an AM–GM equality, not a coincidence. Two clarifications keep the construction honest. The register is not the orientation doublet: one carries an involution, the other carries nilpotents, and no isomorphism respects both algebras, so the crossing operator acts on the register tensored with the chain's own spaces. And the bidirectionality is in the move set, not in the tick: the ensemble sums both moves and has a stationary measure, while the ledger only grows and the reversed history remains inadmissible at its first event.

Theorem 4.102 (BM.121 — The zig-zag world line: the even resolvent, the round-trip weights, and no backward leg). Thm.

$$(1)\quad \text{the vacuum-to-vacuum matrix elements are even:}$$

$$\left(\hat{U}^{2m}\right)_{\varnothing\varnothing} = (x_{+}x_{-})^{m},$$

$$\left(\hat{U}^{2m+1}\right)_{\varnothing\varnothing} = 0$$

$$[\text{FT.108}(5)\ \text{Pf};\ \text{machine: coefficient-wise in}\ \mathbb{Q}[w, x_{+}, x_{-}]/(w^{12})]:$$

$$\text{the mechanism:}\quad \text{a return to}\ \varnothing\ \text{requires an equal number of}\ \hat{J}_{+}\ \text{and}\ \hat{J}_{-}\ \text{moves, and}\ \hat{J}_{\pm}^{2}$$

$$= 0\ \text{forces strict alternation}\ [\text{BM.120}(1);\ \text{machine}];$$

$$(2)\quad \text{the resolvent is the zig-zag sum:}$$

$$\boxed{\sum_{n\geq 0}\langle\varnothing|\hat{U}^n|\varnothing\rangle\, w^n \;=\; \sum_{m\geq 0}\big(w^2 x_+ x_-\big)^m \;=\; \frac{1}{1-w^2x_+x_-} \;\text{ in } \mathbb{Q}[w,x_+,x_-]/(w^{12})}$$

[FT.108(5); BM.23(7); BM.44(1); machine: $(1-w^2x_+x_-)\cdot\sum \equiv 1$ coefficient-wise at five exact values of $x_+x_- \in \{0, \frac{3}{16}, \frac{1}{4}, \frac{1}{2}, 1\}$];

every odd coefficient is exactly zero:

$$[w^{2m+1}]\,\frac{1}{1-w^2x_+x_-} = 0 \;\; \forall m$$

[machine: the coefficient census to w^{11}];

(3) the zig-zag is a sum over round trips, and every leg is forward:

the m-th term $(w^2x_+x_-)^m$ is m create–propagate–annihilate round trips :

$$\varnothing \xrightarrow{\hat{J}_+} \{\mathcal{R}\} \xrightarrow{\hat{J}_-} \varnothing \xrightarrow{\hat{J}_+} \{\mathcal{R}\} \xrightarrow{\hat{J}_-} \varnothing \cdots$$

with each arrow a step of the ledger order $\tau_n \to \tau_{n+1}$

[A6; D1.8] :

$$\boxed{\text{zig-zag} \;:=\; \text{the } m\text{-round-trip word in } \{\hat{J}_+,\hat{J}_-\},\;\; \text{weight } (x_+x_-)^m,\;\; \text{with } \neg\text{a backward leg anywhere in it}}$$

(4) the exact weight ladder: $\mathcal{W}_m = (x_+x_-)^m = \big[4\varepsilon_q(1-4\varepsilon_q)\big]^m$ [machine] :

ε_q	x_+x_-	$\mathcal{W}_1$	$\mathcal{W}_3$
0	0	0	0
$\frac{1}{16}$	$\frac{3}{16}$	$\frac{3}{16}$	$\frac{27}{4096}$
$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{64}$
$\frac{3}{16}$	$\frac{3}{16}$	$\frac{3}{16}$	$\frac{27}{4096}$
$\frac{1}{4}$	0	0	0

[machine: exact rationals];

$$\boxed{\mathcal{W}_m \text{ is maximal at } \varepsilon_q = \tfrac{1}{8} \text{ for every } m, \text{ and } \mathcal{W}_m \leq 4^{-m} \text{ with equality exactly there}}$$

[BM.120(5); machine: the monotone power of the parabola];

the two endpoints annihilate the whole ladder:

$$\varepsilon_q = 0 \implies \mathcal{W}_m = 0 \,\forall m \geq 1;$$

$$\varepsilon_q \to \tfrac{1}{4}^{-} \implies \mathcal{W}_m \to 0 \,\forall m \geq 1$$

[BM.109(2); machine];

(5) the pole and its enclosure:

$$w^2 x_+ x_- = 1 \implies |w| = \frac{1}{\sqrt{x_+ x_-}} \geq 2,$$

$$\text{equality} \iff \varepsilon_q = \frac{1}{8}$$

$[x_+x_- \leq \tfrac{1}{4}$; BM.23(7); BM.44(1); machine] :

the zig-zag series converges for $\lvert w\rvert < 2$ at every admissible threshold, and its pole never enters the unit disc

(6) no backward leg, on three independent grounds: (i) $\neg\exists\, \tau^{-1}$ [A6];

(ii) the pure order reversal of every completed nonempty admissible history leads with J^- and is inadmissible at its first event

[FT.144(4)(ii); machine: exhaustive];

(iii) $\Theta_{\mathfrak{L}}$ fixes every completed alternating word pointwise:

$$\Theta_{\mathfrak{L}}\big((J^+J^-)^m\big) = (J^+J^-)^m \;\; \forall m$$

[BM.25(2); machine: all $m \leq 6$] :

the reversed reading of a zig-zag is the same word — so the diagram's backward leg needs no inverse tick, because the corpus's own mirror already fixes it

(7) the CP reading of the second leg:

the member created by $\hat{J}_+$ and the member annihilated by $\hat{J}_-$ in one round trip are a CP-pair: $\{c,\ c^{\mathrm{CP}}\}$,

$$Q = q + (-q) = 0$$

[Def-BM.15(3); BM.109(1); machine] :

the second leg carries the conjugate configuration forward, ¬the original configuration backward:

$$|w(c^{\mathrm{CP}})|^2 = |w(c)|^2,$$

$$w(c^{\mathrm{CP}}) = \mu_{\mathrm{CP}}\, w(c),$$

$$\mu_{\mathrm{CP}} = e^{i\pi\Theta} = 1 \ \text{ on } \Theta \in 2\mathbb{Z}$$

[BM.108(2), (3); EM.24(5); machine];

the zig-zag is chain → CP-conjugate chain, both forward

[BM.129(1)'s boson row is the same statement read on the winding];

(8) the ledger-count row: $\big|\mathfrak{L}(\tau_{n+1})\big| - \big|\mathfrak{L}(\tau_n)\big| = n_{\text{real_out}}(\tau_n) \geq 0$ [A6] :

a round trip returns $n_{\mathcal{R}}$ to its starting value while the record grows by the events it logged

[machine: $\hat{J}_+\hat{J}_- = P_{\mathcal{V}}$ restores the register, A6 keeps the record] :

the zig-zag is closed on the register and non-decreasing on the record — which is exactly what a forward-only ledger can support

(9) status: derived: (1)–(8);

re-scoped at its own locus, ¬retracted: BM.109(2)'s $\mathcal{W}_{\text{pair}}$

$= x_+x_-$ is the round-trip weight — the one-way weights are x_+ and x_- separately [(4); BM.120(5)];

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a species identification [Def-BM.3(7)],

¬a cosmological coordinate [Def-BM.17(6)(i)];

¬a world line in a spacetime is constructed: the zig-zag is a word in two moves on a finite register [Def-BM.18(5), (6)];

Req. A6, A7, D1.2, D1.8 (Part 1); P1, P4, P5 (Parts 1–2); Def-H.32, FT.108(5), FT.144(4) (Part 3); Def-BM.3(7), Def-BM.15(3), Def-BM.17(6), Def-BM.18(5), Def-BM.18(6), BM.23(7), BM.25(2), BM.44(1), BM.108(2), BM.108(3), BM.109(1), BM.109(2), BM.120(1), BM.120(5), BM.129(1), EM.24(5) (this part); exact truncated power-series arithmetic in $\mathbb{Q}[w, x_+, x_-]/(w^{12})$, exact rational weight ladders, exhaustive ledger-record enumeration [math; machine: `checks/verify_H_janus_tunnel_concrete_spin.py` §B].

Pf.

$(1):$ FT.108(5)'s Pf row re-cited;

strict alternation : $\hat{J}_+^2$

$= \hat{J}_-^2$

$= 0$ [BM.120(1); machine] $\implies$ a word returning to $\varnothing$ has equal counts of the two moves

$$(2): \ (1 - w^2 x_+ x_-) \sum_{m=0}^{5} (w^2 x_+ x_-)^m$$

$$= 1 - (w^2 x_+ x_-)^6$$

$\equiv 1 \mod w^{12}$ [machine: coefficient-wise at five exact values of x_+x_-];

$[w^{2m+1}] = 0 \ \forall m$ [machine: the census to w^{11}]

$(3):$ each factor $w^2 x_+ x_-$ is one $\hat{J}_+\hat{J}_-$ pair [(1); Def-BM.18(5)];

each move is a ledger step [D1.2; A6]

$(4):\ \mathcal{W}_m = (x_+x_-)^m$ [machine: exact rationals at the five thresholds];

$x_+x_- \le \frac{1}{4}$ [BM.120(5)]

$\Longrightarrow\ (x_+x_-)^m$

$\le 4^{-m}$ with equality

$\Longleftrightarrow\ x_+$

$= x_-$ [machine: monotone power]

$\varepsilon_q = 0 \implies x_+x_- = 0 \implies 0^m = 0\ \forall m \ge 1; \qquad \varepsilon_q \to \frac{1}{4}^- \implies x_+ \to 0 \implies x_+x_- \to 0$ [machine]

$(5):\ |w|^2 x_+x_- = 1 \implies |w|^2 = 1/(x_+x_-) \ge 4 \implies |w| \ge 2$ [machine: $x_+x_- \le \frac{1}{4}$];

equality at $\varepsilon_q = \frac{1}{8}$ [(4)]

$(6):$ (i) A6 verbatim;

(ii) FT.144(4)(ii) re-cited [machine: exhaustive];

(iii) $\Theta_{\mathfrak{L}}((J^+J^-)^m) = (J^-J^+)^m$ reversed $= (J^+J^-)^m$ [BM.25(2); machine: all $m \le 6$]

$(7):$ Def-BM.15(3), BM.109(1) re-cited; $\qquad \mu_{\mathrm{CP}} = e^{i\pi\Theta} = 1$ at $\Theta \in 2\mathbb{Z}$ [BM.108(3); machine]

$(8):$ A6's increment row re-cited;

$\hat{J}_+\hat{J}_- = P_{\mathcal{V}}$ restores the register [BM.120(1); machine];

$(9):$ the standing-rule scans [machine]. ■

Phys. The zig-zag is not drawn, it is summed, and the sum turns out to be even. A return to the vacuum needs as many crystallizations as dissolutions, and since neither move can be applied twice in a row the words strictly alternate — so every odd matrix element vanishes identically and the vacuum-to-vacuum resolvent is a geometric series in the square of the fugacity with ratio the round-trip weight. That is the whole content of the zig-zag: term by term it is a create-propagate-annihilate excursion, weighted by the parabola to the power of the number of excursions, maximal at one eighth of the threshold and exactly zero at both endpoints, so the entire ladder is annihilated by a non-flickering ledger. The pole sits at fugacity two or beyond and never enters the unit disc. Three independent rows say why no leg runs backward: the tick has no inverse, the pure reversal of a completed history is inadmissible at its first event, and — the sharpest of the three — the corpus's own record involution fixes every completed alternating word pointwise, so the reversed reading of a zig-zag is literally the same word. What looks like a backward-propagating member in a diagram is here the CP-conjugate configuration carried forward, with equal modulus and a phase that is exactly one on the materialized locus. And the ledger keeps its own accounting: a round trip restores the register while the record grows by the events it logged, which is precisely the one thing a forward-only ledger can support.

Theorem 4.103 (BM.122 — The pair channel: charge balance, the exact threshold, the CP structure, and the two-photon bar). Thm.

$$(1) \quad \text{the channel is charge-balanced exactly:}$$

$$Q_{\text{in}} = 0 \ \ (\text{the } \mathcal{V} \text{ sector carries no registered charge } [\text{D1.1};\ \text{FT.144}(4)]),$$

$$Q_{\text{out}} = q + (-q) = 0$$

$$[\text{Def-BM.15}(3);\ \text{BM.109}(1);\ \text{machine: exhaustive over } q \in \{-1, 0, +1\}] :$$

$$\boxed{Q_{\text{EM}} = \sum_{c \in U} q_{\text{EM}}(c) = 0 \ \text{ survives the channel, in every valid universe}}$$

$$[\text{EM.11};\ \text{D2.11};\ \text{BM.23}(4)];$$

$$\text{the compensator is unique:}$$

$$\#\{q \in \{-1, 0, +1\} :\ q = -q_{\text{in}}\} = 1$$

$$[\text{BM.23}(4);\ \text{BM.108}(5);\ \text{machine}];$$

$$(2) \quad \text{the exact threshold:}$$

$$\mathcal{T}_0(\text{pair}) = \frac{E_{\text{int}}^{\text{pair}}\, \mathfrak{h}}{\ln 2} = \frac{2E_{\text{int}}(c)\, \mathfrak{h}}{\ln 2}$$

[Def-BM.17(3); BM.109(1)] :

at the minimal materialized floor $E_{\text{int}}(c) = 2$ [FT.28; BM.13(3)] :

$$\boxed{\mathcal{T}_0(\text{pair}) \;=\; \frac{4\,\mathfrak{h}}{\ln 2}, \qquad \text{materialization} \iff NE \;\geq\; \frac{4\mathfrak{h}}{\ln 2}}$$

[machine: one Laurent monomial; BM.115(2)(i)'s n=2 minimum, re-derived];

$$\mathcal{T}(\text{pair}; N) = \frac{4\mathfrak{h}}{N \ln 2},$$

$$\mathcal{T}(\text{pair}; r_s, N) = \frac{2r_s}{N} \;\text{ at } r_s = 2\mathfrak{h}$$

[BM.115(1), (4); machine];

the general ladder:

$$\mathcal{T}_0(\text{pair}) \in \left\{\frac{4\mathfrak{h}}{\ln 2},\; \ldots,\; \frac{120\mathfrak{h}}{\ln 2}\right\} \text{ on the computed cells, span ratio exactly } 30$$

[BM.111(5); BM.115(2)(i); machine];

(3) the channel weight:

$$\boxed{\mathcal{W}_{\text{channel}} \;=\; x_+^2 \cdot \chi_{\{NE \geq \mathcal{T}_0\}} \;=\; (1 - 4\varepsilon_q)^2 \cdot \chi_{\{NE \geq \mathcal{T}_0\}}}$$

[two one-way crystallization moves; Def-BM.18(5); machine] :

the round-trip completion:

$$\mathcal{W}_{\text{round}} = x_+ x_- = 4\varepsilon_q(1 - 4\varepsilon_q) \leq \tfrac{1}{4}$$

[BM.109(2); BM.120(5); machine];

$$\boxed{\mathcal{W}_{\text{channel}} > 0 \iff \varepsilon_q < \tfrac{1}{4} \;\wedge\; NE \geq \mathcal{T}_0; \qquad \mathcal{W}_{\text{channel}} = 1 \iff \varepsilon_q = 0 \;\wedge\; NE \geq \mathcal{T}_0}$$

the two limits are opposite and both exact: at ε_q

$= 0$ creation is certain and annihilation impossible $(x_+ = 1,\ x_- = 0)$;

at $\varepsilon_q \to \frac{1}{4}^-$ creation vanishes $(x_+ \to 0)$

[machine];

(4) the CP structure of the channel — the three derived legs:

the inversion	the derived operator	the exact rows
charge	$\widehat{V} = E_+ \otimes P_L$ [BM.76(1)]	$\widehat{V}^2 = 0,\ \mathrm{rk}\,\widehat{V} = 1,\ \dim\ker\widehat{V} = 3,\ [W_3 \otimes I_\omega, \widehat{V}] = \widehat{V}$ [BM.76(2)–(4); machine over $\mathbb{Q}(i)$]
parity/orientation	$\mu_{\mathrm{CP}} = e^{i\pi\Theta}$ and $\hat{\omega}$ [BM.108(3); FT.122(4)]	$\hat{\omega}\hat{\chi}\hat{\omega}^{-1} = -\hat{\chi},\ \hat{\omega}P_L\hat{\omega}^{-1} = P_R,\ \mu_{\mathrm{CP}} = 1$ on $\Theta \in 2\mathbb{Z}$ [machine]
energy	the local scale against $\mathcal{T}_0$ [Def-BM.17(3)]	$NE \geq \mathcal{T}_0,\ \ \mathcal{T}_0 = E_{\mathrm{int}}^{\mathrm{pair}}\hbar/\ln 2$ [BM.115(1); machine]

the image vertex:

$$(I \otimes \hat{\omega})\,\widehat{V}\,(I \otimes \hat{\omega})^{-1} = E_+ \otimes P_R$$

[BM.76(5); machine] :

the step fires from the left ray and its image from the right ray — the two members of the pair occupy opposite rays

[BM.73(3); BM.74 : the two emitted members of a vertex occupy opposite rays; A-BM.7(b); machine];

(5) the nilpotency is the channel's selection rule:

$$\widehat{V}^2 = 0 \implies \boxed{\neg\exists \text{ an iterated charged step inside one doublet}}$$

[BM.76(2); BM.23(3)'s ternary overflow] :

the consequence for the channel: one charge inversion per member, exactly — the pair is $(q,\ -q)$ and $\neg(q,\ q)$ or $(q,\ q')$ with $|q-q'| \geq 2$

[machine: the $3^2 = 9$ charge pairs against the balanced subset];

$$\#\{(q_1, q_2) \in \{-1, 0, +1\}^2 :\ q_1 + q_2 = 0\} = 3, \qquad \#\{(q_1, q_2) :\ q_1 + q_2 \neq 0\} = 6 \qquad \text{[machine]};$$

(6) the two-photon bar (four independent grounds):

the supply leg of $\gamma + \gamma \to$ pair is ¬constructed:

(O1) no gauged chiral coupling: the gauge blocks act on the 16-dimensional pair-channel register and on slot algebras; ¬a block couples to a chiral matter multiplet

[FT.80(7)(a); CT0.12; CT3.20];

(O2) the transit alternative is not an inventory candidate: the gauge coupling of the charge read is deficit $1'$ content

[BM.41(3); Def-BM.8(8); FT.125(4)];

(O3) no momentum-space resolvent: plane-wave boundary-channel states and the Σ-resolvent are an open interface, blocked pending their own declaration

[BM.6(6)];

(O4) the transit object carries no winding carrier: the D3.7 trivial chain is excluded from Def-H.38's scope, so ¬a j

$= |k|/2$ read and ¬an ε read attach to it; its polarization count is N_{pol}

$= 2$ and its helicity is a transit-side read with no identification

[D3.7; EM.8; Def-H.38 Scope];

¬∃ a two-transit-quantum supply vertex in the inventory $\implies$ the pair channel's amplitude is ¬formed, while the channel itself is

(7) what is formed and what is not:

formed	¬formed
the charge balance (1)	a cross-section [BM.6(6); Def-BM.18(6)]
the exact threshold $\mathcal{T}_0$ (2)	an S-matrix element [BM.6(6)]
the channel weight $x_+^2\chi$ (3)	a Lorentz-invariant phase space [Def-BM.18(6)]
the three CP legs (4)	the two-quanta supply vertex (6)
the selection rule $\widehat{V}^2 = 0$ (5)	a measured rate or branching fraction [P5; BM.44(5)]

(8) status: derived: (1)–(5); refuted as formed, with four grounds: (6);

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a numerical coupling [EM.25(7)],

¬a species identification [Def-BM.3(7)];

¬a right-ray matter line is added: the image vertex of (4) is the $\hat{\omega}$-image of the same single-arrow locus, in the representation $\{0,-1\}$ [BM.96$(4),(6)$; BM.73$(2)(b)$; A-BM.7];

Req. A4, A6, D0.8, D1.1, D1.2, D1.8, D2.11, D3.7 (Parts 1–2); CT0.12, CT3.20, EM.8, EM.11, EM.24(5), EM.25(7) (Part 1, Part 2); P1, P4, P5 (Parts 1–2); Def-H.4, Def-H.38, FT.28, FT.44(4), FT.80(7), FT.122(4), FT.125(4), FT.144(4) (Part 3); Def-BM.3(7), Def-BM.8(8), Def-BM.15(3), Def-BM.17(3), Def-BM.18(5), Def-BM.18(6), BM.6(6), BM.13(3), BM.23(3), BM.23(4), BM.41(3), BM.44(5), BM.73(2), BM.73(3), BM.74, BM.76(1), BM.76(2), BM.76(3), BM.76(4), BM.76(5), BM.108(3), BM.108(5), BM.109(1), BM.109(2), BM.111(5), BM.115(1), BM.115(2), BM.115(4), BM.120(5), A-BM.7 (this part); exact $\mathbb{Q}(i)$ matrix algebra, exhaustive ternary charge-pair enumeration, Laurent-monomial assembly, exact rational weight ladders [math; machine: `checks/verify_H_janus_tunnel_concrete_spin.py` §B].

Pf.

$$(1):\ q+(-q)=0 \text{ [machine: three cases]};$$

$$\text{EM.11},\ \text{D2.11 re-cited};$$

$$\#\{q : q=-q_{\text{in}}\}=1 \text{ [machine]}$$

$$(2):\ E_{\text{int}}^{\text{pair}}=2E_{\text{int}}(c) \text{ [BM.109(1)]};$$

$$\mathcal{T}_0=E_{\text{int}}^{\text{pair}}\mathfrak{h}/\ln 2 \text{ [Def-BM.17(3)]};$$

$$E_{\text{int}}=2 \implies \mathcal{T}_0=4\mathfrak{h}/\ln 2 \text{ [machine]}$$

$$\mathcal{T}=2r_s/N \text{ at } r_s=\tfrac{1}{2}\mathfrak{h}E_{\text{int}}^{\text{pair}}=2\mathfrak{h} \text{ [BM.115(4); machine]};$$

the ladder and its span ratio 30 [BM.115(2)(i); machine]

$(3):$ two one-way crystallization moves, each of weight x_+ [Def-BM.18(5); BM.120(5)];

$$x_+^2 = (1 - 4\varepsilon_q)^2 \text{ [machine]}$$

$$\varepsilon_q = 0 \implies x_+ = 1,\ x_- = 0;$$

$$\varepsilon_q \to \tfrac{1}{4}^- \implies x_+ \to 0 \text{ [machine: exact at the two endpoints of the admissible range [CT0.8]]}$$

$$(4):\ \widehat{V} = E_+ \otimes P_L;$$

$$E_+^2 = 0 \implies \widehat{V}^2 = 0; \quad \operatorname{rk}\widehat{V} = \operatorname{rk} E_+ \cdot \operatorname{rk} P_L = 1;$$

$[W_3 \otimes I_\omega, \widehat{V}] = \widehat{V}$ on $[\sigma_3, E_+] = 2E_+,\ W_3 = \frac{1}{2}\sigma_3$ [BM.76(1)–(4); machine over $\mathbb{Q}(i)$]

$(I \otimes \hat{\omega})\widehat{V}(I \otimes \hat{\omega})^{-1} = E_+ \otimes \hat{\omega} P_L \hat{\omega}^{-1} = E_+ \otimes P_R$ [FT.122(4); BM.76(5); machine]

$$(5):\ \widehat{V}^2 = 0 \text{ [BM.76(2)]};$$

the balanced subset of the 9 charge pairs is $\{(-1,+1),(0,0),(+1,-1)\}$,

#

$= 3$ [machine: exhaustive]

$(6):$ FT.80(7)(a),

BM.41(3),

Def-BM.8(8),

BM.6(6),

D3.7,

EM.8,

Def-H.38 Scope rows re-cited [machine: the four-ground census];

$$N_{\text{pol}} = (4-1) - 1 = 2 \text{ [EM.8; machine]}$$

$(7), (8)$: the two-column assembly and the standing-rule scans [machine]. ■

Phys. The channel that opens is exactly as constrained as the ledger can make it. Charge balances because the pair is a chain and its conjugate, the compensating value is unique in the ternary alphabet, and the nilpotent weak step allows precisely one charge inversion per member — so of the nine ordered charge pairs only the three balanced ones are formed. The threshold is one Laurent monomial: twice the constituent edge count times the single scale over the log of two, which at the digon floor is four units, the very minimum of the two-chain ladder the resonance tier already computed. The weight is the square of the crystallization weight times the indicator that the local scale has reached that threshold, and its two limits are exact opposites — at the sharp threshold creation is certain and annihilation impossible, while near full dissolution creation itself vanishes. The CP structure decomposes into three derived legs: a nilpotent step operator that fires from the occupied ray alone and whose orientation image fires from the other, the CP transport phase that is exactly one on the materialized locus, and the local scale against the threshold. What does not exist is the supply vertex. Four independent grounds bar it: nothing couples a gauge block to a chiral matter multiplet, the transit alternative is deficit-one-prime content, there is no momentum-space resolvent, and the transit object is excluded from the winding carrier's scope so neither a spin label nor a statistics sign attaches to it. So the channel is derived and the amplitude is not, and the tier records both halves rather than promoting one into the other.

Theorem 4.104 (BM.123 — The annihilation channel: the reverse move, the entropy destination, and the closure of the cross-section residue). Thm.

(1) the reverse move is the same algebra:

$$\text{annihilation} := \hat{J}_- \text{ on each of the pair's two registers},$$

$$\text{weight } x_-^2 = (4\varepsilon_q)^2$$

[Def-BM.18(5); BM.120(5); machine] :

$$\boxed{\mathcal{W}_{\text{ann}} = x_-^2 \cdot \chi_{\{\text{the pair is materialized}\}} = 16\varepsilon_q^2 \cdot \chi_{\{NE \geq \mathcal{T}_0\}}}$$

[machine: one monomial in ε_q];

the charge balance:

$$Q_{\text{in}} = q + (-q) = 0,$$

$$Q_{\text{out}} = 0 \ \ (\text{the } \mathcal{V} \text{ sector is unregistered})$$

[D1.1; EM.11; machine] :

the balance holds in both directions

(2) the two directions and their exact ratio:

$$\frac{\mathcal{W}_{\text{ann}}}{\mathcal{W}_{\text{channel}}} \ = \ \frac{x_-^2}{x_+^2} \ = \ \Big(\frac{4\varepsilon_q}{1-4\varepsilon_q}\Big)^2$$

[machine: exact rational function] :

$$\boxed{\frac{\mathcal{W}_{\text{ann}}}{\mathcal{W}_{\text{channel}}} = 1 \iff \varepsilon_q = \frac{1}{8}, \qquad < 1 \iff \varepsilon_q < \frac{1}{8}, \qquad > 1 \iff \varepsilon_q > \frac{1}{8}}$$

[machine: the sign census on a 256-point rational grid];

the ratio is the detailed-balance indicator: equality holds exactly at BM.120(5)'s maximum of the round-trip weight;

the two endpoint rows:

$$\varepsilon_q = 0 : \ \ \frac{\mathcal{W}_{\text{ann}}}{\mathcal{W}_{\text{channel}}} = 0 \ \ (\text{nothing annihilates});$$

$$\varepsilon_q \to \tfrac{1}{4}^- : \ \ \mathcal{W}_{\text{channel}} \to 0 \ \ \text{and the ratio diverges}$$

[machine];

(3) the round trip closes the two directions:

$\mathcal{W}_{\text{channel}} \cdot \mathcal{W}_{\text{ann}}$ is ¬the round-trip weight;

the round trip is one creation and one annihilation:

$$\boxed{\mathcal{W}_{\text{round}} \;=\; x_+x_- \;=\; 4\varepsilon_q(1-4\varepsilon_q) \;\le\; \frac{1}{4}} \qquad [\text{BM.109}(2);\ \text{BM.120}(5);\ \text{BM.121}(4);\ \text{machine}];$$

the resolvent that sums them:

$$\frac{1}{1-w^2x_+x_-},$$

pole at $|w| \ge 2$

$$[\text{BM.121}(2),(5);\ \text{FT.108}(5);\ \text{machine}];$$

(4) where the content goes (the entropy destination):

a dissolved pair returns to the $\mathcal{V}$ sector: $\quad n_{\mathcal{R}} = 0 \implies \rho_{pq} = \rho_\circ = \frac{1}{4}I,$

$$S_{\text{raw}}(\rho_\circ) = \ln 4 = \max,$$

$$\arg\max = \{\tfrac{1}{4}I\}$$

$$[\text{FT.144}(4);\ \text{CT0.1};\ \text{CT0.2};\ \text{A5};\ \text{machine}];$$

$$E_{\text{int}}(R) = |R|,$$

$$S_{\mathfrak{L}} = n_{\mathcal{R}} \ln 2$$

$$[\text{Def-BM.5}(4);\ \text{CT1.29}]:$$

annihilation returns the pair's registered content to the maximally raw-entropic virtual sector — ¬a radiation leg carries it, because ¬a supply vertex exists [BM.122(6)]

the ledger row: $\quad |E_{\mathcal{R}}| = 0 \implies \neg\exists$ a fermion terminus $\quad [\text{D1.4};\ \text{Def-H.4};\ \text{FT.144}(4);\ \text{machine}];$

(5) the inter-sector row, re-scoped and not retracted:

$$\text{BM.27}(7),$$

BM.29(5) and Def-BM.15(5) record ¬an annihilation channel, ¬a pair-production channel and ¬a cross-section between EM^+ and EM^- :

the frozen content stands:

$$\neg\exists \text{ a direct } \mathrm{EM}^+ \to \mathrm{EM}^- \text{ operator;}$$

the two sectors remain amplitude-equal configuration orbits of one involution [BM.25(4); BM.27(6); $|w(c^{\mathrm{CP}})|^2 = |w(c)|^2$];

what is constructed:

a mediated channel — $\mathcal{J}$'s two moves composed with the CP identification of Def-BM.15(2) — connects the two sectors through the $\mathcal{V}$ sector, not through a direct operator

the composition, exactly:

$$\varnothing \xrightarrow{\hat{J}_+ \otimes \hat{J}_+} \{c,\ c^{\mathrm{CP}}\} \xrightarrow{\hat{J}_- \otimes \hat{J}_-} \varnothing,$$

weight $x_+^2 x_-^2$ for the full out-and-back on two registers

[machine: the monomial]; ¬a cross-section is attached to it [Def-BM.18(6); BM.6(6)];

(6) the closure of the cross-section residue:

BM.118(7) named (R9) as the annihilation channel: an inter-sector transition operator between EM^+ and EM^-,

¬in the inventory :

(R9) is CLOSED at the channel level: a mediated annihilation channel is constructed, with an exact weight $x_-^2 \chi$, an exact threshold $\mathcal{T}_0$, an exact charge balance and an exact entropy destination

the residue narrows and does not vanish:

(R9′) the annihilation cross-section	requires the momentum-space Σ-resolvent [BM.6(6)] and the gauge coupling of the charge read [Def-BM.8(8); BM.122(6)]
(R9″) the radiation legs	¬a transit-quantum supply vertex exists [BM.122(6)(O1)–(O4)]

(7) the arrow rows stand:

the channel is forward-only: $\neg\exists\,\tau^{-1}$ [A6];

$\neg\exists$ an inverse decay or an inverse creation in the ledger [BM.23(9); BM.44(6); FT.144(2)] :

the reversibility is in the move set and the stationary measure, ¬in the tick

[BM.120$(3),(7)$; machine];

$$\Gamma(\text{pair}) = \Gamma_{\text{tr}}(\varepsilon_q)\cdot f\cdot\frac{\varepsilon_{\text{eff}}}{\varepsilon_{\text{EM}}}\cdot\mathcal{N}\cdot\chi_{\{NE\geq\mathcal{T}_0\}}$$

[Def-BM.17(4); BM.116(1) : the width function applies to the pair resonance unchanged];

(8) status:

derived: (1)–(5), (7);

closed at the channel level: (6);

named and ¬filled: (R9′), (R9″);

re-scoped at their own loci, ¬retracted: BM.27(7),

BM.29(5),

Def-BM.15(5),

BM.108(7),

BM.109(5) (each frozen on its own locus: ¬a direct inter-sector operator exists);

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a numerical coupling [EM.25(7)],

¬a lifetime or cross-section in external units [BM.44(5)],

¬a species identification [Def-BM.3(7)];

Req. A5, A6, D0.8, D1.1, D1.4, D1.8, D2.11 (Part 1); CT0.1, CT0.2, CT0.8, CT1.29, EM.11, EM.25(7) (Part 1, Part 2); P1, P4, P5 (Parts 1–2); Def-H.4, Def-H.32, FT.108(5), FT.144(2), FT.144(4) (Part 3); Def-BM.3(7), Def-BM.5(4), Def-BM.8(8), Def-BM.15(2), Def-BM.15(5), Def-BM.17(3), Def-BM.17(4), Def-BM.18(5), Def-BM.18(6), BM.6(6), BM.23(9), BM.25(4), BM.27(6), BM.27(7), BM.29(5), BM.44(5), BM.44(6), BM.108(7), BM.109(2), BM.109(5), BM.116(1), BM.118(7), BM.120(3), BM.120(5), BM.120(7), BM.121(2), BM.121(4), BM.121(5), BM.122(6) (this part); exact rational function rows, exhaustive rational grid sign censuses, Laurent-monomial assembly [math; machine: `checks/verify_H_janus_tunnel_concrete_spin.py` §B].

Pf.

$$(1):\ \hat{J}_- \text{ on two registers with weight } x_- \text{ each [Def-BM.18(5); BM.120(5)];}$$

$$x_-^2 = (4\varepsilon_q)^2 = 16\varepsilon_q^2 \text{ [machine]}$$

$$Q_{\text{in}} = q + (-q) = 0 \text{ [BM.109(1); machine]}; \qquad Q_{\text{out}} = 0 \text{ [D1.1 : virtual pairs are unregistered]}$$

$$(2):\ \tfrac{x_-^2}{x_+^2} = \left(\tfrac{4\varepsilon_q}{1-4\varepsilon_q}\right)^2;$$

$$= 1 \iff 4\varepsilon_q = 1 - 4\varepsilon_q \iff \varepsilon_q = \tfrac{1}{8} \text{ [machine: the cleared linear equation]}$$

the 256-point grid census of the sign [machine];

$$\varepsilon_q = 0 \implies x_- = 0 \implies \text{the ratio is } 0;$$

$$\varepsilon_q \to \tfrac{1}{4}^- \implies x_+ \to 0 \implies \text{the ratio diverges [machine]}$$

$$(3):\ x_+x_- = 4\varepsilon_q(1-4\varepsilon_q) \le \tfrac{1}{4} \text{ [BM.120(5); machine]}; \qquad x_+^2x_-^2 = (x_+x_-)^2 \le \tfrac{1}{16} \text{ [machine]}$$

$(4):$ FT.144(4)'s $\mathcal{V}$-sector rows re-cited (frozen);

$$S_{\mathfrak{L}}$$

$= n_{\mathcal{R}} \ln 2$ [CT1.29] and $n_{\mathcal{R}} \to 0$ under $\hat{J}_-$ [D1.2] $\implies$ the registered content leaves while the raw entropy reaches its maximum [CT0.2; A5]

$(5):$ BM.25(4), BM.27(6) re-cited;

$$|w(c^{\mathrm{CP}})|^2 = |w(c)|^2 \text{ [EM.24}(5); \text{ BM.108}(2)];$$

the mediated composition $\hat{J}_+ \otimes \hat{J}_+$ then $\hat{J}_- \otimes \hat{J}_-$ with weight $x_+^2 x_-^2$ [machine: the monomial]

$(6):$ BM.118(7)'s (R9) row re-cited;

the two narrowed residues cite their own tags [BM.6(6); BM.122(6)]

$(7):$ A6, BM.23(9), BM.44(6), FT.144(2) re-cited;

$(8):$ the rescope census and the standing-rule scans [machine]. ■

Phys. Annihilation is the same algebra read the other way, and it costs the square of the dissolution weight. The two directions have an exact ratio — four epsilon over one minus four epsilon, squared — which is one precisely at the threshold value one eighth, below one before it and above one after, so the ratio is nothing other than the detailed-balance indicator and it peaks where the round-trip weight does. At the sharp threshold the ratio is zero: nothing annihilates, which is the same statement as the no-flicker theorem seen from the other side. The round trip that joins the two directions carries the familiar parabola, and the resolvent that sums all round trips has its pole at fugacity two or beyond. What happens to the content is exact and slightly austere: a dissolved pair returns to the virtual sector at one quarter of the identity, the unique maximizer of the raw entropy, and the ledger's own entropy read goes to zero with the registered count — so nothing is carried off by radiation, because no supply vertex exists to carry it. That absence is what keeps the frozen rows honest: there is still no direct operator between the two charge sectors and they remain amplitude-equal orbits of one involution, and what the tier adds is a mediated channel through the virtual sector, composed of the two Janus moves and the CP identification. Residue (R9) closes at exactly that level — a channel with a weight, a threshold, a balance and a destination — and narrows into two named successors, the cross-section which needs a momentum-space resolvent and a gauge coupling of the charge read, and the radiation legs which need a transit supply vertex. The arrow is untouched throughout: the reversibility lives in the move set and its stationary measure, never in the tick.

Definition 4.19 (Def-BM.19 — The spin-dependent transport defect (the multiplet-ladder read)). Def. A read specification on Def-H.37(2)'s holonomy defect and FT.60(5)'s multiplet ladder; no new axiom, no new datum, no new parameter, and no identification of the two angles.

(1) the two angles, kept apart:

angle	definition	dimension
$\Theta(c) = \sum_{e \in E(c)} \widehat{ds}^{\,2}(e)$	the winding of a materialized closed chain [FT.22; EM.24(5)]	$[\Theta] = \mathfrak{q}^{+1}$ [FT.55(1) Pf : $\Theta\mathfrak{h}^2 = \oint ds^2$]
$\Theta_R(c)$	the frame-holonomy angle of Def-H.21's spin connection [Def-H.21(5), (6)]	$[\Theta_R] = \mathfrak{q}^0$ [Def-H.21(6) : $K\,dA$ classical]

$$k = \frac{\Theta}{2} \in \mathbb{Z}_{\geq 0},$$

$$j = \frac{|k|}{2},$$

$$[k] = \mathfrak{q}^{+1},$$

$$[j] = \mathfrak{q}^{+1}$$

[FT.22; Def-H.38; FT.55(1) Pf; machine: the exponent arithmetic];

(2) the no-go (re-cited, and constitutive for this definition):

$$\boxed{\text{the identification } \Theta_R(c) = 2\pi k(c) \text{ is ruled out}}$$

[FT.55(1) : the calibration no-go] :

the ground is dimensional and exact:

$$[\Theta_R] = \mathfrak{q}^0 \text{ against } [2\pi k] = \mathfrak{q}^{+1}$$

[machine: the two exponent rows disagree];

the corpus's own scope row:

under Def-H.37(2) the defect ladder reads in Θ_R at fixed weight m with ¬a k–Θ_R proportionality anywhere

[FT.60 Rmk];

(3) the multiplet ladder:

$$m \in \{k,\ k-2,\ \ldots,\ -k\},$$

$$\#m = k+1 = 2j+1$$

[FT.60(5); Def-H.38] :

$$\boxed{\Delta(m)\ :=\ \left\|(\mathcal{U}^{(k)}-1)\psi_m\right\|^2\ =\ \left|e^{im\phi}-1\right|^2\ =\ 2-2\cos(m\phi)\ =\ 4\sin^2\frac{m\phi}{2}}$$

$$\phi := \frac{\Theta_R}{2}$$

[FT.60(5); FT.54(4); EW.33 pattern; Def-H.21(6)];

constitutive clause: the spin-dependence enters through the multiplet weight m,

¬through the winding Θ and ¬through a single value per j;

(4) the exact algebraic form:

$$\cos(m\phi) = T_m(\cos\phi),$$

$$T_{m+1} = 2x\,T_m - T_{m-1},$$

$$T_0 = 1,\ T_1 = x$$

[math: Chebyshev] :

$$\boxed{\Delta(m)\ =\ 2-2\,T_{|m|}\big(\cos\tfrac{\Theta_R}{2}\big)}$$

the state-level specialization:

$$\Theta_R = \delta_v \text{ at the vertex-figure locus } \Longrightarrow\ \Delta = D_S \in \left\{2,\ 4,\ 2-\sqrt{2},\ 1,\ 2-2\cos\tfrac{\pi}{10},\ 2-\sqrt{3}\right\}$$

[Def-BM.14(3); BM.82(1); BM.114(3)];

(5) the standing zero:

$$k \text{ even} \implies 0 \in \{k, k-2, \ldots, -k\} \implies \boxed{\Delta(0) = 2 - 2\cos 0 = 0 \text{ exactly, at every value of } \Theta_R}$$

[FT.60(5) : every even k carries a defect-massless weight; machine];

$$k \text{ odd} \implies 0 \notin \{k, k-2, \ldots, -k\} \implies \neg\text{a standing zero [machine]};$$

(6) the mass ladder and its ceiling:

$$m^2_{\text{pole}}(m) = \hat{\mu}'^2\, \Delta(m),$$

$$0 \le \Delta(m) \le 4 \implies \boxed{0 \le m^2_{\text{pole}}(m) \le 4\hat{\mu}'^2}$$

[Def-H.37(2); FT.54(1); FT.76(3); machine];

the two Def-H.37 clauses and their matching:

(2) implies (1) at leading order under

$$\boxed{\hat{\mu}'^2\, \frac{\Theta_R^2}{4} \;=\; \hat{\mu}^2\, \ell_c^2\, G(c)}$$

[Def-H.37 : recorded, not imposed];

(7) the residues inherited, not filled:

(R2) Θ_R

$= \delta_v$ is read at the vertex-figure locus, whose ledger realization — a clean-enclosure loop through conical vertices — is ¬constructed

[BM.67$(9), (10)$; BM.82(6)(R2); Census FT.63(4) Rmk];

(R1) ρ_{B}

$= \hat{\mu}'^2\mathfrak{h}^2/(\ln 2)^2$ is the single cross-currency residue, its slot declared and its value ¬issued

[A-BM.9; BM.82$(1), (6)$];

(8) status and scope:

derived: (1)'s dimensions, (4)'s recursion and specialization, (5), (6)'s ceiling;

constitutive: (3)'s locus of the spin-dependence;

re-cited: (2);

the scope inherited at every use: Def-H.38's nonnegative cone and its k

≥ 3 aspect bar [¬an aspect claim is made here];

Def-H.37's two P4-class residues $\hat{\mu}^2$, $\hat{\mu}'^2$ [Cal-F];

the bar:

¬a product of the two Def-H.37 clauses' factors is a derived mass — the formed rows are the two clauses and their matching identity (6);

¬a k–Θ_R proportionality is asserted anywhere;

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a species identification [Def-BM.3(7)],

¬a currency conversion [BM.83(3)].

Rmk (why the defect is a ladder and not a number). A single-valued $\Delta(j)$ would require the frame-holonomy angle to be a function of the winding, and that identification is ruled out on a dimension count: the winding carries the free scale with weight minus two while the holonomy angle is dimensionless. What the corpus derives instead is a ladder — the defect read evaluated at each weight of the spin multiplet — so the spin fixes how many values there are ($2j+1$ of them) and which one is exactly zero when k is even, while Θ_R remains a separate datum whose ledger realization is residue (R2). Reading the ladder as a number is the same category error as reading a count as a strength, and the definition bars it explicitly.

Req. D2.9 (Parts 1–2); CT2.1 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.21(5), Def-H.21(6), Def-H.37, Def-H.38, FT.22, FT.54(1), FT.54(4), FT.55(1), FT.60(5), FT.76(3), EM.24(5) (Part 3); EW.33 (Part 2); Census FT.63(4) (row-wise); Def-BM.3(7), Def-BM.14(3), BM.67(9), BM.67(10), BM.82(1), BM.82(6), BM.83(3), BM.114(3), A-BM.9 (this part); the Chebyshev recursion over $\mathbb{Z}[x]$, exact radical identities, $\mathfrak{q}$-exponent arithmetic [math; machine: `checks/verify_H_janus_tunnel_concrete_spin.py` §C].

Pf.

Definition block: the constitutive clause is (3)'s locus of the spin-dependence;

the non-derived clauses are ¬present — (1)'s dimensions,

(2)'s no-go,

(4)'s recursion,

(5)'s zero and (6)'s ceiling are all derived or re-cited;

(7),

(8) are residue and scope records. ■

Phys. Two angles appear in this corpus and they are not the same object: the winding, which counts the line element around a closed chain and carries the free scale, and the frame-holonomy angle, which is dimensionless. Their identification is ruled out by that dimension mismatch alone, so a transport defect that depended on the winding through the holonomy formula cannot be formed. What is formed is better: a ladder. The spin fixes the multiplet, the multiplet has two-j-plus-one weights spaced by two, and the defect read at each weight is two minus two cosines of that weight times half the holonomy angle — equivalently four sine-squared of a quarter, and equivalently an integer polynomial in the cosine through the Chebyshev recursion. Two rows fall out exactly. Whenever the winding is even the ladder contains the weight zero, and the defect at weight zero is identically zero at every curvature flux, which is a standing massless state in the middle of the ladder rather than a tuning. And the whole ladder is bounded by four times the defect scale, the ceiling the pole-side read already carried. The state-level defect values are the ladder read at the vertex-figure angle, which is where the six exact algebraic numbers come from — and that locus, together with the single cross-currency ratio, stays a named residue rather than a filled one.

Theorem 4.105 (BM.124 — The spin–winding relation and the multiplet defect ladder). Thm.

(1) the exact spin–winding relation:

$$k = \frac{\Theta}{2} \in \mathbb{Z}_{\geq 0} \text{ on shell,}$$

$$j = \frac{|k|}{2},$$

$$\Psi(c) = \mathrm{Sym}^{|k|}(\mathcal{H}_{\mathsf{v}}),$$

$$\dim \Psi = |k| + 1$$

[FT.22; FT.25(3); Def-H.38; EW.32] :

$$\boxed{\Theta \;=\; 2k \;=\; 4j \qquad \text{on the nonnegative cone,} \qquad \Theta \in 2\mathbb{Z} \text{ for every materialized closed chain}}$$

[machine: the exact relation at every $k \in \{0, \ldots, 8\}$]; the inverse read: $\boxed{j \;=\; \dfrac{|\Theta|}{4}}$

(2) the multiplet:

$$m \in \{k,\ k-2,\ \ldots,\ -k\},$$

$$\boxed{\#m \;=\; k+1 \;=\; 2j+1}$$

[FT.60(5); Def-H.38; machine: exact at every $k \leq 8$];

the Casimir:

$$\frac{k}{2}\Big(\frac{k}{2}+1\Big) = j(j+1)$$

[FT.60(3); machine];

the stretched total:

$$m_{\max} = k = 2j$$

[FT.60(6) : maximal-weight CG on the cone];

(3) the spin-dependent transport defect is the ladder, and the ladder is exact:

$$\boxed{\Delta(m) \;=\; 4\sin^2\frac{m\Theta_R}{4} \;=\; 2-2\cos\frac{m\Theta_R}{2} \;=\; 2-2\,T_{|m|}\Big(\cos\frac{\Theta_R}{2}\Big)}$$

[Def-BM.19(3), (4); FT.60(5); FT.54(4)] :

the Chebyshev recursion verified exactly:

$$T_0 = 1, \quad T_1 = x, \quad T_{m+1} = 2x\,T_m - T_{m-1}$$

[machine: the integer-polynomial recursion to $m = 8$];

the exact witness:

$$T_m\big(\tfrac{1}{2}\big) = \cos\frac{m\pi}{3} \text{ on the six-cycle } \big(1,\ \tfrac{1}{2},\ -\tfrac{1}{2},\ -1,\ -\tfrac{1}{2},\ \tfrac{1}{2}\big)$$

[machine: all $m \le 8$];

(4) the refutation of a single-valued $\Delta(j)$:

$$\#\{m\} = 2j+1 \ge 2 \text{ for every } j \ge \tfrac{1}{2}$$

[(2); machine] :

the spin determines the ladder's length, ¬a single defect value — $\Delta(j)$ as one number is ¬formed

the two anchors:

k	j	m values	$\Delta(m)$
1	$\frac{1}{2}$	$\{+1,\ -1\}$	$2-2\cos\frac{\Theta_R}{2}$ (both weights agree, by evenness)
2	1	$\{+2,\ 0,\ -2\}$	$2-2\cos\Theta_R, \quad 0, \quad 2-2\cos\Theta_R$

[FT.60(5); BM.106(1)'s two floor rows; machine];

at $j = \frac{1}{2}$ the ladder is state-independent and ¬split; at $j = 1$ it splits into three with an exact zero in the middle

[FT.60(5) : k=1 reproduces FT.54(4) and its state-independence is special to the seed];

(5) the two angles are ¬proportional, on a dimension count:

$$[k] = \mathfrak{q}^{+1},$$

$$[\Theta_R] = \mathfrak{q}^{0},$$

$$[2\pi] = \mathfrak{q}^{0}$$

[FT.55(1) Pf; Def-H.21(6); machine: the exponent rows] :

$\Theta_R = 2\pi k$ would require $\mathfrak{q}^0 = \mathfrak{q}^{+1}$: dimensionally impossible, with no measurement used

[FT.55(1) : the calibration no-go re-cited];

the consequence:

Δ is ¬a function of Θ, hence ¬a function of j alone;

Δ is a function of the pair $(m,\ \Theta_R)$ with m spin-determined and Θ_R independent [Def-BM.19(3), (6)];

the corpus's own scope row:

¬a k–Θ_R proportionality anywhere [FT.60 Rmk; FT.55(4) : the winding enters as a copy count, ¬as an angle];

(6) the standing zero: k even $\iff 0 \in \{k, k-2, \ldots, -k\}$ [machine: exhaustive to $k = 8$] :

$\Delta(0) = 2 - 2T_0(\cos\frac{\Theta_R}{2}) = 2 - 2 = 0$ exactly, at every Θ_R, for every even k

[FT.60(5); machine];

$$k \text{ odd} \implies m \in \{\pm 1, \pm 3, \ldots, \pm k\} \implies \neg\exists\, m = 0 \implies \neg\text{a standing zero} \qquad [\text{machine}];$$

the parity row that follows:

the defect ladder carries an exact massless weight precisely when the exchange sign is $+1$

$$[k \text{ even} \iff \varepsilon = (-1)^k = +1 \ [\text{BM.113}(4)];\ \text{machine: the two parities coincide}];$$

(7) the state-level specialization:

$$\Theta_R = \delta_v = 2\pi - d\,\frac{(L-2)\pi}{L},$$

$$D_S = 4\sin^2\frac{\delta_v}{4}$$

$$[\text{Def-BM.14}(3);\ \text{Census FT.63}(4), (6);\ \text{BM.82}(1)]:$$

the six values are the ladder read at six vertex-figure angles, ¬six spin values:

$$D_S \in \left\{2,\ 4,\ 2-\sqrt{2},\ 1,\ 2-2\cos\tfrac{\pi}{10},\ 2-\sqrt{3}\right\},$$

four fields, six minimal polynomials

$$[\text{Def-BM.14}(3);\ \text{BM.114}(3);\ \text{machine re-verified}];$$

the index is $(\mathcal{F}_g, \beta),\ \neg\, j$

$$[\text{BM.90}(5):\ \text{the branch-blindness theorem};\ \text{machine: the symbol census carries } \neg j];$$

(8) status:

derived: $(1),\ (2),\ (3),\ (6),\ (7)$;

refuted as formed: (4)'s single-valued $\Delta(j),\ (5)$'s $\Theta_R = 2\pi k$;

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a numerical coupling [EM.25(7)],

¬a species identification [Def-BM.3(7)];

Req. D2.9 (Parts 1–2); CT2.1, EW.32 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.21(6), Def-H.37(2), Def-H.38, FT.22, FT.25(3), FT.54(4), FT.55(1), FT.55(4), FT.60(3), FT.60(5), FT.60(6) (Part 3); Census FT.63(4), FT.63(6) (row-wise); Def-BM.3(7), Def-BM.14(3), Def-BM.19(3), Def-BM.19(4), Def-BM.19(6), BM.82(1), BM.90(5), BM.106(1), BM.113(4), BM.114(3) (this part); the Chebyshev recursion over $\mathbb{Z}[x]$ with an exact rational witness, q-exponent arithmetic, exhaustive parity censuses [math; machine: `checks/verify_H_janus_tunnel_concrete_spin.py` §C].

Pf.

$$(1):\ k=\Theta/2\ [\text{FT.22};\ \text{FT.25(3)}];$$

$$j=|k|/2\ [\text{Def-H.38}]\implies \Theta=2k=2\cdot 2j=4j\ [\text{machine: exact at every } k\le 8];$$

$$\Theta\in 2\mathbb{Z}\ [\text{FT.22}]$$

$$(2):\ \#\{k,k-2,\dots,-k\}=k+1\ [\text{machine: the arithmetic-progression count}];$$

$$k+1=2\cdot\tfrac{k}{2}+1=2j+1\ [\text{machine}];$$

$$\tfrac{k}{2}(\tfrac{k}{2}+1)=j(j+1)\ \text{at}\ j=k/2\ [\text{machine}]$$

$$(3):\ \|(U^{(k)}-1)\psi_m\|^2$$

$$=|e^{im\phi}-1|^2$$

$$=(e^{im\phi}-1)(e^{-im\phi}-1)$$

$$= 2 - e^{im\phi} - e^{-im\phi}$$

$$= 2 - 2\cos(m\phi) \text{ [FT.60(5); machine over } \mathbb{Q}(i)]$$

$$2 - 2\cos(m\phi) = 4\sin^2\tfrac{m\phi}{2} \text{ at } \phi = \Theta_R/2 \text{ [machine: the half-angle identity]};$$

$$\cos(m\phi) = T_m(\cos\phi) \text{ with } T_{m+1} = 2xT_m - T_{m-1} \text{ [machine: the recursion to } m = 8]$$

$$T_m(\tfrac{1}{2}) = \cos\tfrac{m\pi}{3} :$$

$$m$$

$$= 0..8 \mapsto \left(1, \tfrac{1}{2}, -\tfrac{1}{2}, -1, -\tfrac{1}{2}, \tfrac{1}{2}, 1, \tfrac{1}{2}, -\tfrac{1}{2}\right) \text{ [machine: exact rationals on the six-cycle]}$$

$$(4) : \ \#\{m\} = k + 1 \geq 2 \text{ for } k \geq 1 \text{ [machine]};$$

$$k = 1 : \ m \in \{+1, -1\}, \ \Delta(\pm 1) = 2 - 2\cos(\pm\tfrac{\Theta_R}{2}) = 2 - 2\cos\tfrac{\Theta_R}{2} \text{ by evenness [machine]}$$

$$k = 2 : \ m \in \{+2, 0, -2\}, \ \Delta(\pm 2) = 2 - 2\cos\Theta_R, \ \Delta(0) = 0 \text{ [machine]}$$

$$(5) : \ \Theta\mathfrak{h}^2 = \oint ds^2 \implies [\Theta] = [\mathfrak{h}^{-2}] = \mathfrak{q}^{+1} \implies [k] = \mathfrak{q}^{+1} \text{ [FT.55(1) Pf; machine]};$$

$$[\Theta_R] = \mathfrak{h}^0 = \mathfrak{q}^0 \text{ [Def-H.21(6)]};$$

$$\mathfrak{q}^0 \neq \mathfrak{q}^{+1} \text{ [machine]}$$

$$(6) : \ k \text{ even} \iff \exists r : \ k - 2r = 0 \iff r = k/2 \in \mathbb{Z} \text{ [machine: exhaustive to } k = 8];$$

$$\Delta(0) = 2 - 2T_0 = 2 - 2 \cdot 1 = 0 \text{ [machine]}$$

$$\varepsilon = (-1)^k \text{ [BM.113(4)]};$$

$$k \text{ even} \iff \varepsilon$$

$$= +1 \iff \Delta(0)$$

$= 0$ is in the ladder [machine: the two parity censuses coincide]

(7) : Def-BM.14(3)'s table re-cited;

the symbol census of D_S carries $\{\mathcal{F}_g, \beta, \delta_v, L, d\}$ and $\neg\{j, k\}$ [BM.90(5); machine];

(8) : the standing-rule scans [machine]. ■

Phys. The spin and the winding are the same integer read twice, and the relation is exact rather than fitted: the winding is twice the copy number and the spin is half of it, so the winding is four times the spin on the cone, and it is always an even integer for a materialized closed chain. What the spin then controls is not one mass but a ladder — two-j-plus-one weights spaced by two — and the defect at each weight is two minus two cosines of the weight times half the holonomy angle, which is also four sine-squared of a quarter and also an integer polynomial in the cosine by the Chebyshev recursion. At the two anchor windings the difference is visible: at spin one half the ladder has two weights that agree by evenness and nothing splits, while at spin one it splits into three with an exact zero in the middle. That zero is not a tuning but a parity theorem — the ladder contains the weight zero precisely when the winding is even, which is precisely when the exchange sign is plus one, so the defect ladder carries a massless weight exactly on the bosonic side. What the tier refuses is the fusion of the two angles: the winding carries the free scale and the holonomy angle does not, so identifying them would equate two different dimensions, and that is the whole content of the calibration no-go. The six exact defect numbers are therefore the ladder read at six vertex-figure angles and indexed by floor and branch, not by spin, which is why a spin-dependent retention probability was refuted earlier and stays refuted.

Theorem 4.106 (BM.125 — The spin-dependent mass: the exact ladder, the two clauses and their matching, and the ceiling). Thm.

(1) the mass ladder:

$$\boxed{m^2_{\text{pole}}(m) \;=\; \hat{\mu}'^2\,\Delta(m) \;=\; 4\hat{\mu}'^2 \sin^2\frac{m\Theta_R}{4}, \qquad m \in \{k, k-2, \ldots, -k\}, \qquad k = 2j}$$

[Def-H.37(2); Def-BM.19(3), (6); FT.54(4)] :

the ladder's length is the spin's:

$$\#\{m^2_{\text{pole}} \text{ values}\} \;\le\; 2j + 1$$

[equality when the $\Delta(m)$ are distinct; machine];

(2) the exact range and the ceiling:

$$0 \le 4\sin^2\frac{m\Theta_R}{4} \le 4$$

[machine: the sine-square bound] :

$$\boxed{0 \;\le\; m^2_{\text{pole}}(m) \;\le\; 4\hat{\mu}'^2}$$ [FT.76(3) : the pole-side ceiling re-cited; machine];

the ceiling is attained:

$$\Delta = 4 \iff \sin^2\frac{m\Theta_R}{4} = 1 \iff m\Theta_R \equiv 2\pi \pmod{4\pi}$$

[machine: the exact locus];

the ladder's own ceiling is the composite's:

$\lambda^{(n)}_{\text{max}} = 4n$ is the sum of n attained ceilings

[BM.117(4); machine];

(3) the two Def-H.37 clauses and their matching identity:

(1) $m^2_{\text{pole}} = \hat{\mu}^2\ell_c^2 G(c)$,

$$G(c) = \sum_{A,B} c_A c_B \,\text{Re}\langle \mathcal{G}_A \mathsf{v}, \mathcal{G}_B \mathsf{v}\rangle$$

[EW.33 Gram data];

(2) $m^2_{\text{pole}} = \hat{\mu}'^2 \cdot 4\sin^2\frac{\Theta_R}{4}$,

$$[\hat{\mu}'^2] = \mathfrak{q},$$

$$[\hat{\mu}^2] = \mathfrak{q}^2$$

[Def-H.37(1), (2); EW.33 packaging];

(2) implies (1) at leading order under the matching $\hat{\mu}'^2 \dfrac{\Theta_R^2}{4} = \hat{\mu}^2 \ell_c^2 G(c)$

[Def-H.37 : recorded, not imposed; FT.54(2); FT.33];

the dimension check on the matching:

$$[\hat{\mu}'^2 \Theta_R^2] = \mathfrak{q} \cdot \mathfrak{q}^0 = \mathfrak{q},$$

$$[\hat{\mu}^2 \ell_c^2 G(c)] = \mathfrak{q}^2 \cdot [\ell_c^2 G(c)] = \mathfrak{q} \implies [\ell_c^2 G(c)] = \mathfrak{q}^{-1}$$

[machine: the exponent arithmetic];

(4) the refutation of the directive's product form:

$m(j) = \hat{\mu}' \ell_c \sqrt{G(c)\,\Delta(j)}$ would require:

(i) a single $\Delta(j)$ — refuted: the defect is a ladder of $2j+1$ values [BM.124(4)];

(ii) a product of clause (1)'s $\ell_c^2 G(c)$ with clause (2)'s Δ — refuted: the two clauses are two reads of one mass, related by (3)'s matching, ¬two multiplicative factors;

(iii) a dimension match — refuted: $[\hat{\mu}' \ell_c \sqrt{G\Delta}]$

$= \mathfrak{q}^{1/2} \cdot [\ell_c] \cdot \mathfrak{q}^{-1/2}$ against $[m]$

$= \mathfrak{q}^{1/2}$,

so $[\ell_c]$ would have to be $\mathfrak{q}^0$ while (3)'s matching gives $[\ell_c^2 G] \quad = \mathfrak{q}^{-1}$

[machine: the two exponent rows disagree];

the formed rows are the two clauses separately and their matching identity (3) — the product form is ¬formed on three independent grounds

(5) the standing zero is a massless weight, not a massless species:

$$k \text{ even} \implies m = 0 \in \text{the ladder} \implies m^2_{\text{pole}}(0) = 0 \text{ exactly at every } \Theta_R$$

$$[\text{BM.124}(6);\ \text{FT.60}(5);\ \text{machine}]:$$

the scope bars:

¬a species identification attaches to the zero weight [Def-BM.3(7); P5];

¬a massless ν line is formed — the rung-one ν row has $m^{\text{rest}}_{\nu} > 0$ strictly on two SOS sandwiches [BM.96$(3), (4)$ re-cited, ¬extended];

¬a right-ray matter line, ¬a seesaw-type mechanism and ¬a Yukawa-type object is added [BM.96$(4), (6)$; EW.15; FT.145(3)];

(6) the residue rows:

(R2) Θ_R

$= \delta_v$ at the vertex-figure locus, whose ledger realization is ¬constructed [BM.67$(9), (10)$; BM.82(6)(R2)] — inherited at every use of (1);

(R1) $\rho_{\text{B}} = \hat{\mu}'^2 \mathfrak{h}^2/(\ln 2)^2$ is the single cross-currency residue:

$$\frac{m^2_{\text{pole}}}{m^2_{\text{hid}}} = \rho_{\text{B}} \mathcal{R}_s,$$

$$\mathcal{R}_s = \frac{D_S}{4E^2} \in \overline{\mathbb{Q}}$$

[BM.82$(1), (2)$; A-BM.9 : the slot declared, the value ¬issued];

¬a currency is converted: $[\Delta]$

$= \mathfrak{q}^0$ and $[\hat{\mu}'^2]$

$= \mathfrak{q}$ against $[m^2_{\text{hid}}]$

$$= \mathfrak{q}^{+1} \text{ and } [\ln(4/3)] \qquad = \mathfrak{q}^{0}$$

$$[\text{BM.102}(2);\ \text{BM.83}(3);\ \text{machine: the four exponent rows}];$$

(7) status: derived: (1), (2), (3), (5)'s zero row; refuted as formed, on three grounds: (4);

$$\text{inherited, } \neg\text{filled: } (6)\text{'s (R1), (R2)};$$

$$\neg\text{a declaration},$$

$$\neg\text{a P4 addition},$$

$$\neg\text{a new dimensionful scale [P1]},$$

$$\neg\text{a measured value [P5]},$$

$$\neg\text{a numerical coupling [EM.25}(7)\text{]},$$

$$\neg\text{a species identification [Def-BM.3}(7)\text{]};$$

Req. D2.9 (Parts 1–2); CT2.1, EW.15, EW.33 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.37(1), Def-H.37(2), FT.33, FT.54(2), FT.54(4), FT.60(5), FT.76(3), FT.145(3) (Part 3); Def-BM.3(7), Def-BM.14(3), Def-BM.19(3), Def-BM.19(6), BM.67(9), BM.67(10), BM.82(1), BM.82(2), BM.82(6), BM.83(3), BM.96(3), BM.96(4), BM.96(6), BM.102(2), BM.117(4), BM.124(4), BM.124(6), A-BM.9 (this part); exact radical and trigonometric identities, $\mathfrak{q}$-exponent arithmetic, the sine-square bound on exact rationals [math; machine: `checks/verify_H_janus_tunnel_concrete_spin.py` §C].

Pf.

$$(1):\ \text{Def-H.37}(2)\text{'s } m^2_{\text{pole}}$$

$$= \hat{\mu}'^2 \cdot 4\sin^2(\Theta_R/4) \text{ read at the multiplet weight } m\Theta_R$$

$$\mapsto m\Theta_R\ [\text{FT.60}(5);\ \text{FT.54}(4)\text{'s exact SU}(2)\text{ defect}];$$

$$\#\{m\} = k + 1 = 2j + 1\ [\text{BM.124}(2)]$$

$$(2):\ 0 \le \sin^2 \le 1 \implies 0 \le 4\sin^2 \le 4\ [\text{machine}];$$

FT.76(3)'s ceiling re-cited;

$$\sin^2 \tfrac{m\Theta_R}{4} = 1 \iff \tfrac{m\Theta_R}{4} \equiv \tfrac{\pi}{2} \pmod{\pi} \iff m\Theta_R \equiv 2\pi \pmod{4\pi} \text{ [machine]}$$

$$\lambda^{(n)} = \sum_a D_S(s_a) \le \sum_a 4 = 4n \text{ [BM.114(3); BM.117(4); machine]}$$

$(3):$ Def-H.37$(1),(2)$ re-cited;

$$4\sin^2 \tfrac{\Theta_R}{4}$$

$$= \Theta_R^2 + O(\Theta_R^4) \implies \hat{\mu}'^2\Theta_R^2/4$$

$$= \hat{\mu}^2 \ell_c^2 G(c) \text{ at leading order [FT.54(2); FT.33]}$$

$$[\hat{\mu}'^2\Theta_R^2] = \mathfrak{q}\cdot\mathfrak{q}^0 = \mathfrak{q};$$

$$[\hat{\mu}^2] = \mathfrak{q}^2 \implies [\ell_c^2 G] = \mathfrak{q}^{-1} \text{ [machine: the exponent arithmetic]}$$

$(4):$ (i) BM.124(4);

(ii) (3)'s matching is an equality of two reads of one mass, so their product has $[m^4]$ not $[m^2]$ [machine: the exponent row];

(iii) the two $[\ell_c]$ rows disagree [machine]

$$(5):\ \Delta(0) = 0 \implies m^2_{\text{pole}}(0) = \hat{\mu}'^2 \cdot 0 = 0 \text{ [machine];}$$

BM.96$(3),(4)$'s strict-positivity rows re-cited at their own locus [¬extended]

$(6):$ BM.82$(1),(2)$ re-cited;

$$[\Delta] = \mathfrak{q}^0,\ [\hat{\mu}'^2] = \mathfrak{q},\ [m^2_{\text{hid}}] = \mathfrak{q}^{+1} \text{ [BM.102(2); machine];}$$

$(7):$ the standing-rule scans [machine]. ■

Phys. The mass ladder is the defect ladder times one scale, so it inherits everything the ladder has: two-j-plus-one values, an exact zero at the middle weight whenever the winding is even, and

a hard ceiling at four times the defect scale that the pole-side read already carried. The composite ceiling is just the sum of attained constituent ceilings, which is why the resonance tier's four-n bound is not a new ceiling but the same one counted. The two clauses of the defect-transfer interface are then kept apart rather than multiplied: one reads the mass through the Gram data of the weak transport with a length factor, the other through the holonomy angle with a defect scale, and they meet in a single matching identity whose dimension check forces the length-times-Gram factor to carry weight minus one. The product form the question proposes fails on three separate counts — there is no single defect value to put in it, multiplying two reads of one mass gives the wrong power, and the dimensions do not close — so the tier supplies the matching identity instead. The massless middle weight is a property of a ladder and not a species claim: the rung-one neutral line keeps its strictly positive rest mass on two sum-of-squares sandwiches, and no right-ray line, no seesaw-type mechanism and no Yukawa-type object is added anywhere. Two residues are inherited and left unfilled, the vertex-figure locus and the single cross-currency ratio, and no currency is converted between them.

BM.126 — The spin-dependence of the Janus crossing: refuted on the four standing grounds, with the derived domain row. Structural-constraints record: see Appendix B, Section B.2.7 of the Addendum.

Theorem 4.107 (BM.127 — The winding-parity identification: two reads separated). Thm.

(1) the statistics read:

$$\boxed{\varepsilon(c) \;=\; (-1)^{|k(c)|} \;=\; (-1)^{2j(c)} \;=\; \begin{cases} +1 & j \in \mathbb{Z} \\ -1 & j \in \mathbb{Z} + \frac{1}{2} \end{cases}}$$

[BM.113(2), (4); machine: the census to $k = 8$];

(2) the winding is a ledger datum:

$$\Theta(c) = \sum_{e \in E(c)} \widehat{ds}^{\,2}(e),$$

$\widehat{ds}^{\,2}(e) = \kappa_e$ is registration data determined at crystallization

[FT.22; FT.24; Census FT.28(2); A §0.2a] :

$$\Theta_{\text{int}} = 2\kappa \in 2\mathbb{Z}, \qquad k = \frac{\Theta}{2} = \kappa \in \mathbb{Z} \setminus \{0\} \qquad \text{[BM.106(1); machine]};$$

the parity of k is a read of registration data — ¬an external postulate, ¬a declared assignment and ¬a measurement

[P5; machine: the symbol census of $(-1)^{|k|}$ carries $\{k\}$ alone];

(3) the kind read is a different object:

$$\text{bound} \;:=\; \text{closed self-contained chain, content fermionic;}$$

$$\text{transit} \;:=\; \text{propagating open chain, content bosonic}$$

$$[\text{D1.5} :\ \text{a definitional label}] :$$

the dichotomy is the closed/open read $\qquad [\text{FT.80}(5);\ \text{D1.4}];$

(4) the two reads are ¬equated, and the floor census is the proof:

$$\kappa = 1 \implies k = 1 \implies j = \tfrac{1}{2} \implies \varepsilon = -1;$$

$$\kappa = 2 \implies k = 2 \implies j = 1 \implies \varepsilon = +1$$

$$[\text{BM.106}(1);\ \text{machine}] :$$

both exchange signs occur on materialized closed bound chains of one D1.5 kind — so the kind label is ¬a statistics and the statistics is ¬a kind

the exact separation:

read	its argument	its values
the statistics ε	k (an integer registration datum)	$\{+1,\ -1\}$, sharp per chain
the D1.5 kind	closed against open [D1.4]	$\{\text{bound},\ \text{transit}\}$, both signs on one kind
the registration κ	$\widehat{ds}^{\,2}(e)$ per edge	$\mathbb{Z} \setminus \{0\}$, ¬a statistics

(5) the naming discipline:

"fermion"/"boson" occur in this corpus in two distinct uses, and the tier keeps them apart:

(i) as statistics labels: reads of ε $[(1)]$;

(ii) as content labels: D1.5's bound/transit vocabulary $[(3)]$;

¬a species name is attached to either use — the identification layer is Def-BM.3, Def-BM.6 and Def-BM.14, and ¬a clause of this tier names a particle

[Def-BM.3(7); machine: the species-token scan];

(6) what is complete and what is open:

complete: ε is an exact function of a ledger datum, with no external input $[(1), (2)]$;

beyond this tier:

the winding of a named sealed class	(R3) / deficit $1'$: which three of the six states a class occupies, and which winding each constituent carries [BM.28(7); BM.103(8); BM.117(6)]
the connector winding $d = \Theta(P)$	BM.2(2) : d is registration data; an open connector carries $S = 0$ and hence $d = 0$ [Def-H.32(4); BM.128(7)]
the $k \geq 3$ aspect claim	Def-H.38 Scope: carriers for $k \geq 3$ exceed the registered pair-channel inventory [CT0.12]

the identification is complete as a function and open as a value per named class

(7) status: derived: (1)–(5), (6)'s completeness row;

re-scoped at its own locus, ¬retracted: BM.106(1)'s spin-parity refutation stands and is what makes (4) a separation rather than a contradiction;

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a species identification [Def-BM.3(7)];

Req. D0.8, D1.4, D1.5 (Parts 1–2); CT0.12 (Part 1); P1, P4, P5 (Parts 1–2); Def-H.32(4), Def-H.38, FT.22, FT.24, FT.80(5) (Part 3); Census FT.28(2) (row-wise); Def-BM.3(7), Def-BM.6, Def-BM.14, BM.2(2), BM.28(7), BM.103(8), BM.106(1), BM.113(2), BM.113(4), BM.117(6), BM.128(7) (this part); exhaustive parity and registration censuses, symbol-set censuses [math; machine: `checks/verify_H_janus_tunnel_concrete_spin.py` §D].

Pf.

$(1):$ BM.113(2), (4) re-cited (frozen); the parity census to $k = 8$ [machine]

$$(2):\ \Theta = \sum_{e} \widehat{ds}^{\,2}(e) \text{ [FT.22]};$$

$\widehat{ds}^{\,2}(e) = \kappa_e$ registration data [FT.24; Census FT.28(2); A §0.2a];

$$\Theta_{\text{int}} = 2\kappa \implies k = \kappa \text{ [BM.106}(1)\text{; machine]}$$

the symbol census of $(-1)^{|k|}:\ \{k\}$ [machine]

$(3):$ D1.5, D1.4, FT.80(5) re-cited [a definitional label and the closed/open read]

$$(4):\ \kappa = 1 \implies k = 1,\ j = \tfrac{1}{2},\ \varepsilon = (-1)^1 = -1;$$

$$\kappa = 2 \implies k = 2,\ j = 1,\ \varepsilon = (-1)^2 = +1 \text{ [BM.106}(1)\text{; machine: the two exact rows]}$$

both chains are materialized closed bound chains of one D1.5 kind [Def-H.4; BM.106(1)] $\implies$ the kind does not determine the sign

$(5):$ the two uses enumerated;

Def-BM.3(7)'s scope row re-cited;

the species-token scan of this tier [machine]

$(6):$ the three open rows cite their own tags [machine];

$(7):$ the standing-rule scans [machine]. ■

Phys. The statistics sign is a parity of a registration integer, and that is the whole content of the identification: the winding is a sum of per-edge line elements, each of which is registration data fixed at crystallization, so its parity is a ledger read with no postulate, no declared assignment and no measurement behind it. What the tier is careful about is that this read is not the same object as the bound-or-transit label the corpus has carried since its definition layer, and the proof that they differ is the floor census: at registration one a chain has spin one half and sign minus one, at registration two it has spin one and sign plus one, and both are materialized closed bound chains of the same kind. So the kind label cannot be a statistics and the statistics cannot be a kind, and the earlier refutation of a spin-parity theorem on the kind is exactly what makes this a clean separation instead of a contradiction. The two words fermion and boson are then used in two disciplined ways

— as reads of the exchange sign and as the content vocabulary — with no species name attached to either. What is complete is the function; what stays open is its value on any named sealed class, because which windings a class occupies is the same missing datum as the flavour composition, together with the connector winding and the aspect bar at high winding.

Theorem 4.108 (BM.128 — The three-chain exchange structure: the space, the S_3 action, the character formula, and the vanishing wedge). Thm.

(1) the three-chain state space:

$$\boxed{\mathcal{H}_3(c_1,c_2,c_3) \;:=\; \bigotimes_{a=1}^{3}\Big(\mathcal{H}_\omega(c_a)\otimes\Psi(c_a)\Big), \qquad \dim\mathcal{H}_3 \;=\; 8\prod_{a=1}^{3}\big(|k_a|+1\big)}$$

[Def-BM.16(2) extended to $n=3$; machine: exact integers];

the anchors:

$k=(k_1,k_2,k_3)$	$\prod(\lvert k_a\rvert+1)$	$\dim\mathcal{H}_3$
$(1,1,1)$	8	64
$(2,2,2)$	27	216
$(1,1,2)$	12	96
$(1,2,3)$	24	192

(2) the S_3 action:

$\hat{P}_\sigma \;:=$ the factor permutation by $\sigma\in S_3$,

$$|S_3| = 6,$$

$$\boxed{\hat{P}_\sigma\hat{P}_\tau = \hat{P}_{\sigma\tau}\;\;\forall\sigma,\tau\in S_3}$$

[machine: all 36 ordered pairs, entrywise over $\mathbb{Q}(i)$] :

the orders:

$$\mathrm{ord}(\hat{P}_\sigma)=\mathrm{ord}(\sigma)=\begin{cases}1 & \sigma=\mathrm{id}\\ 2 & \sigma \text{ a transposition (3 of them)}\\ 3 & \sigma \text{ a 3-cycle (2 of them)}\end{cases}$$

[machine: exact matrix powers];

the sign representation:

$$\text{sgn} : S_3 \to \{\pm 1\},$$

$$\ker(\text{sgn}) = A_3,\ |A_3| = 3,$$

$$S_3/A_3 \cong \mathbb{Z}_2$$

[machine: the inversion census $(0, 1, 1, 2, 2, 3)$];

(3) the character formula (exact):

$$\boxed{\chi(\sigma) \ := \ \text{Tr}\, \hat{P}_\sigma\big|_{(\mathbb{C}^d)^{\otimes n}} \ = \ d^{\,c(\sigma)}}$$

$c(\sigma) =$ the cycle count of σ

[machine: verified on all $2!$ permutations at $d \in \{2, 3\}$ and all $3!$ at $d \in \{2, 3\}$];

the three rows at $n = 3,\ d = 2$:

$$\chi(\text{id}) = 2^3 = 8,$$

$$\chi(\text{transposition}) = 2^2 = 4,$$

$$\chi(\text{3-cycle}) = 2^1 = 2;$$

the consequence for the exchange sign:

a transposition's trace is $d^{\,n-1},\ \neg d^{\,n}$:

$$\boxed{\text{the flip is not the identity on the } n\text{-fold space for } n \geq 2}$$

[machine];

(4) the CP-twisted operator:

$$\hat{P}_\sigma^{\mathrm{CP}} \;:=\; \hat{\mu}_{\mathrm{CP}} \circ \hat{P}_\sigma \;\text{ on } \mathcal{H}_3,$$

$$\hat{\mu}_{\mathrm{CP}} = \mu_{\mathrm{CP}}\,\mathrm{id},$$

$$\mu_{\mathrm{CP}} = e^{i\pi\Theta} = 1 \;\text{ on } \Theta \in 2\mathbb{Z}$$

[BM.112(1); BM.108(3)] :

$$\boxed{\left(\hat{P}_{(12)}^{\mathrm{CP}}\right)^2 = \mathrm{id} \;\text{ unconditionally}, \qquad \operatorname{spec} \hat{P}_{(12)}^{\mathrm{CP}} \subseteq \{+1,-1\}, \qquad \operatorname{Tr} \hat{P}_{(12)}^{\mathrm{CP}} = \mu_{\mathrm{CP}}\, d^{\,c((12))} \prod_a (|k_a|+1)}$$

[BM.112(1), (2) transported to $n = 3$; machine];

(5) the vanishing wedge (the structural row of this tag):

$$\dim \wedge^3(\mathbb{C}^d) = \binom{d}{3} :$$

$$\boxed{\dim \wedge^3(\mathbb{C}^2) = \binom{2}{3} = 0, \qquad \dim \wedge^3(\mathbb{C}^3) = \binom{3}{3} = 1}$$

[machine: the explicit antisymmetrizer $\frac{1}{3!}\sum_\sigma \operatorname{sgn}(\sigma)\hat{P}_\sigma$ has rank 0 on $(\mathbb{C}^2)^{\otimes 3}$ and rank 1 on $(\mathbb{C}^3)^{\otimes 3}$, computed exactly over $\mathbb{Q}(i)$] :

the reading:

three copies of the 2-dimensional seed carrier admit no totally antisymmetric carrier state, while three colour slots admit exactly one — the ε-seal

(6) the sharpening of BM.114(2) :

$B = \varepsilon_{ijk} q^i q^j q^k$ is antisymmetric in the colour slots [D3.9; CT3.15; CT3.17] :

the two antisymmetries are different objects with different dimensions:

antisymmetry	its space	its dimension
the colour seal	$\wedge^3(\mathbb{C}^3_{\text{colour}})$	1 [machine]
the carrier wedge	$\wedge^3(\mathbb{C}^2_{\mathsf{v}})$	0 [machine]
the statistics sign	the winding parity on $\mathcal{H}_3$	$\{+1,-1\}$ [BM.113; (7)]

the baryon's total antisymmetry is carried by colour, which is ¬observable [CT3.22], and the statistics is carried by the winding — ¬one is the other

(7) the multiplicative exchange law:

$$k(\Gamma) = k(c_1) + k(c_2) + d_{12}$$

[FT.34(3); BM.2(2) : the additive winding law] :

the connector row:

$d_{12} = \Theta(P) \geq 0$ for a closed connector;

an open chain carries $S = 0 \implies d = 0$

[Def-H.32(4); machine];

at a sealed junction with an open connector:

$$k_\Gamma = \sum_{a=1}^{n} k_a \implies \varepsilon(\Gamma) = (-1)^{\left|\sum_a k_a\right|} = \prod_{a=1}^{n} (-1)^{|k_a|} = \prod_{a=1}^{n} \varepsilon(c_a)$$

[machine: the multiplicativity verified on all $5^3 = 125$ triples at $n = 3$ and all $5^2 = 25$ pairs at $n = 2$];

(8) the closure of (R7) at $n = 3$:

BM.112(7) and Def-BM.16(7) left $\mathcal{H}_3$ and its S_3 action ¬constructed :

(R7) is discharged at $n = 3$: the space (1), the group action with its character (2), (3), the involution (4) and the permutation-sign read (7) are all constructed

the residue narrows:

(R7′) $\mathcal{H}_n$ and its S_n action for $n \geq 4$ are ¬constructed

[the sealing grammar admits $n \geq 4$ supports where δ- and ε-groupings mix [BM.31(2); BM.110(5)]];

(9) status: derived: (1)–(7); closed: (R7) at $n = 3$; named: (R7′);

re-scoped at its own locus, ¬retracted: BM.114(2)'s colour-antisymmetry row, sharpened by (6);

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a species identification [Def-BM.3(7)];

¬a flavour $\mathsf{SU}(3)$ or $\mathsf{SU}(6)$ is constructed: S_3 permutes constituent slots,

¬flavour states [BM.117(2), (3)];

Req. A3, D1.4 (Part 1); CT3.15, CT3.17, CT3.22, D3.9 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.32(4), Def-H.33, Def-H.38, FT.34(3), FT.108(4) (Part 3); Def-BM.3(7), Def-BM.16(2), Def-BM.16(7), BM.2(2), BM.31(2), BM.108(3), BM.110(5), BM.112(1), BM.112(2), BM.112(7), BM.113, BM.114(2), BM.117(2), BM.117(3) (this part); exact matrix algebra over $\mathbb{Q}(i)$, exact rank by Fraction elimination, permutation-group enumeration, exhaustive multiplicativity censuses [math; machine: `checks/verify_H_janus_tunnel_concrete_spin.py` §E].

Pf.

$$(1): \ \dim \mathcal{H}_\omega$$

$$= 2 \text{ and } \dim \Psi$$

$$= |k| + 1 \text{ per constituent [Def-BM.16(2); Def-H.38]} \implies \dim \mathcal{H}_3$$

$$= 2^3 \prod_a (|k_a| + 1) \text{ [machine: the four anchor rows]}$$

$$(2): \ \hat{P}_\sigma \text{ built on the } d^3 \text{ basis } \{e_{i_1} \otimes e_{i_2} \otimes e_{i_3}\} \text{ by } e_{i_1} \otimes e_{i_2} \otimes e_{i_3}$$

$$\mapsto e_{i_{\sigma^{-1}(1)}} \otimes \cdots \text{ [machine]};$$

$$\hat{P}_\sigma \hat{P}_\tau = \hat{P}_{\sigma\tau} \text{ on all } 36 \text{ pairs [machine: entrywise over } \mathbb{Q}(i)]$$

the order census $(1,2,2,3,3,2)$ and the inversion census $(0,1,1,2,2,3)$ [machine];

$$|A_3| = 3,\ 6/3 = 2 \text{ [machine]}$$

$$(3):\ \text{the basis element } e_{i_1} \otimes e_{i_2} \otimes e_{i_3} \text{ is fixed by } \hat{P}_\sigma$$

$$\iff i \text{ is constant on each cycle of } \sigma \implies \text{Tr}$$

$$= d^{c(\sigma)} \text{ [machine: verified on all permutations at } (n,d) \in \{2,3\} \times \{2,3\}]$$

$$n = 3,\ d = 2:\ c(\text{id}) = 3 \implies 8; \quad c(\text{transposition}) = 2 \implies 4; \quad c(\text{3-cycle}) = 1 \implies 2 \text{ [machine]}$$

$$(4):\ \hat{\mu}_{\text{CP}} \text{ is central} \implies (\hat{\mu}_{\text{CP}} \hat{P}_{(12)})^2 = \mu_{\text{CP}} \mu_{\text{CP}}^* \hat{P}_{(12)}^2 = \text{id [BM.112(1); machine]};$$

$$\text{Tr} = \mu_{\text{CP}} \chi((12)) \prod_a (|k_a| + 1) \text{ [(3); machine]}$$

$$(5):\ \dim \wedge^3(\mathbb{C}^d) = \binom{d}{3} \text{ [math]};$$

$$\binom{2}{3} = 0,\ \binom{3}{3} = 1 \text{ [machine]};$$

the explicit antisymmetrizer $\frac{1}{6} \sum_\sigma \text{sgn}(\sigma) \hat{P}_\sigma$ has rank 0 at d

$$= 2 \text{ and rank } 1 \text{ at } d$$

$$= 3 \text{ [machine: exact Fraction elimination on the } 8 \times 8 \text{ and } 27 \times 27 \text{ matrices]}$$

$(6):$ D3.9, CT3.15, CT3.17, CT3.22 re-cited; the three-row census [(5); BM.113(4); machine]

$$(7):\ k(\Gamma) = \sum_a k_a + d \text{ [FT.34(3); BM.2(2)]};$$

an open chain carries $S = 0$ [Def-H.32(4)] $\implies d = 0$;

$$(-1)^{|k_1+k_2+k_3|}$$

$$= (-1)^{k_1+k_2+k_3} \text{ on } k_a$$

$$\geq 0 \text{ [Def-H.38 Scope]}$$

$$= (-1)^{k_1}(-1)^{k_2}(-1)^{k_3} \text{ [machine: all 125 triples in } \{0,\dots,4\}^3 \text{ and all 25 pairs]}$$

$$(8):\ \text{BM.112}(7),\ \text{Def-BM.16}(7) \text{ re-cited;}$$

the four members discharged against BM.106(7)'s (R7) [machine: the row-by-row check];

$$(9):\ \text{the standing-rule scans [machine].}\ \blacksquare$$

Phys. Three chains now have a space, a group and a character, and the three fit together exactly. The space is the tensor product of the three orientation doublets and the three winding carriers, eight times the product of the carrier dimensions, which at the two anchors is sixty-four and two hundred sixteen. The symmetric group on three letters acts by permuting factors, the operators multiply as the permutations do on all thirty-six ordered pairs, and the traces obey one clean formula: the dimension to the number of cycles, giving eight, four and two at three chains over a two-dimensional carrier. Twisting by the CP phase changes nothing on the materialized locus, where the phase is one, and the transposition remains an unconditional involution. The structural surprise is the wedge: three copies of the two-dimensional seed carrier have no totally antisymmetric state at all, because the binomial coefficient of three taken from two is zero, while three colour slots have exactly one, and an explicit antisymmetrizer confirms both ranks over the exact complex field. That is why the baryon's antisymmetry is colour's and not the carrier's — and colour is unobservable by construction, so the statistics has to live somewhere else, which is the winding. The winding then supplies the law the tier was after: at a sealed junction with an open connector the total winding is the sum of the constituent windings, so the exchange sign of a composite is the product of its constituents' signs, verified on every triple in a five-by-five-by-five census. That closes the exchange-structure residue at three chains, and narrows it to four and beyond, where the sealing grammar starts mixing delta and epsilon groupings.

Theorem 4.109 (BM.129 — The meson is a boson and the baryon's sign is the product: the composite identification). Thm.

$$(1)\quad \text{the meson:}$$

$$\text{a } \delta\text{-sealed CP-pair with an open connector:}\quad k_M = k(c) + k(c^{\mathrm{CP}}) + d = k + k + 0 = 2k$$

$$[\text{BM.128}(7);\ \text{BM.113}(2)(\text{ii}):\ \Theta(c^{\mathrm{CP}}) = \Theta(c) \text{ on the materialized locus}]:$$

$$\boxed{\varepsilon_M \;=\; (-1)^{|2k|} \;=\; \big((-1)^2\big)^k \;=\; +1 \qquad \text{for every constituent winding } k \geq 0}$$

[machine: verified at every $k \in \{0, \ldots, 8\}$];

$$j_M = \frac{|k_M|}{2} = \frac{2k}{2} = k \in \mathbb{Z}$$

[machine] :

$$\boxed{\text{the meson's total spin is an integer and its exchange sign is } +1 \text{ — it is a boson, unconditionally}}$$

(2) the robustness row (the orientation-sign reading):

the alternative reading $k(c^{\mathrm{CP}}) = -k(c)$ [$\hat{\omega} :\; k \mapsto -k$ on the carrier; BM.73(4); A-BM.7(b)] :

$$k_M = k + (-k) + 0 = 0 \implies j_M = 0, \qquad \varepsilon_M = (-1)^0 = +1 \qquad \text{[machine]};$$

$$\boxed{\text{both readings give } \varepsilon_M = +1 \text{ — the meson's statistics is robust under the unresolved orientation-sign row, while its total spin is not}}$$

[$j_M = k$ against $j_M = 0$; the spin value is ¬fixed and the statistics is];

(3) the closed-connector variant:

$$k_M = 2k + d \implies \boxed{\varepsilon_M = (-1)^d}$$

[machine: the two constituent windings cancel modulo two] :

the meson's statistics is then the connector's parity alone:

$$d \text{ even} \implies +1,$$

$$d \text{ odd} \implies -1$$

[machine: $d \in \{0, 1, 2, 3, 4\}$];

$d = \Theta(P) \geq 0$ is registration data

[BM.2(2); FT.34(3)];

the computed cells: an open connector carries $S = 0 \implies d = 0 \implies \varepsilon_M = +1$

[Def-H.32(4); machine];

(4) the baryon:

an ε-sealed 3-juncture with an open connector: $k_B = k_1 + k_2 + k_3$

[BM.128(7)] :

$$\boxed{\varepsilon_B \;=\; (-1)^{k_1+k_2+k_3} \;=\; \varepsilon(c_1)\,\varepsilon(c_2)\,\varepsilon(c_3), \qquad j_B = \frac{|k_1 + k_2 + k_3|}{2}}$$

the exact criterion:

$$\boxed{\varepsilon_B = -1 \iff \#\{a :\ k_a \text{ odd}\} \text{ is odd}}$$

[machine: all 125 triples in $\{0, \ldots, 4\}^3$; 62 fermionic and 63 bosonic];

(5) the anchor and the computed floors:

all three constituents at the minimal odd winding $k_a = 1$:

$$k_B = 3,$$

$$j_B = \tfrac{3}{2},$$

$$\varepsilon_B = -1$$

[machine] :

the digon floor: $L = 2 \implies \kappa \equiv 1,\ k = L/2 = 1$ (odd)

[FT.32(6); FT.28; machine] :

the minimal materialized constituent carries $k = 1$, so the three-floor baryon is fermionic

the cycle ladder's parity: L even $\implies k = L/2$; L odd $\implies k = L$ [FT.32(6); machine] :

L	k	k parity	ε
2	1	odd	-1
3	3	odd	-1
4	2	even	$+1$
5	5	odd	-1
6	3	odd	-1
8	4	even	$+1$

[machine: exact];

(6) what is derived and what awaits a datum:

derived: ε_B as an exact function of the three constituent windings [(4)];

the multiplicative law [BM.128(7)];

the criterion [(4)];

awaits (R3)/deficit $1'$:

which windings a named sealed class occupies is ¬derived — so the baryon's statistics is complete as a function and open as a value per class

[BM.28(7); BM.103(8); BM.117(6); BM.127(6); machine: the symbol census of the criterion carries $\{k_1, k_2, k_3\}$ and ¬a class name];

(7) the colour row and the octet row:

the ε-seal's antisymmetry is colour's, one-dimensional, and ¬observable:

$$\dim \wedge^3(\mathbb{C}^3_{\text{colour}}) = 1,$$

$$\dim \wedge^3(\mathbb{C}^2_{\mathsf{v}}) = 0$$

[BM.128(5), (6); CT3.22];

no flavour multiplet is formed:

S_3 permutes constituent slots, ¬flavour states;

¬∃ a flavour SU(3);

the octet and decuplet groupings are ¬derived

[BM.117(2), (3); BM.114(4)];

(8) status: derived: (1)–(5), (7); open, awaiting (R3) : (6)'s per-class value;

re-scoped at its own locus, ¬retracted: BM.117(2),

(3)'s orbit-count rows stand and are ¬promoted to multiplets;

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

¬a numerical coupling [EM.25(7)],

¬a species identification [Def-BM.3(7)];

Req. A3, D1.4 (Part 1); CT3.22 (Part 2); P1, P4, P5 (Parts 1–2); Def-H.32(4), Def-H.38, FT.28, FT.32(6), FT.34(3) (Part 3); Def-BM.3(7), Def-BM.16(2), BM.2(2), BM.28(7), BM.73(4), BM.103(8), BM.113(2), BM.114(4), BM.117(2), BM.117(3), BM.117(6), BM.127(6), BM.128(5), BM.128(6), BM.128(7), A-BM.7 (this part); exhaustive parity and triple censuses, exact integer winding arithmetic, symbol-set censuses [math; machine: `checks/verify_H_janus_tunnel_concrete_spin.py` §E].

Pf.

$$(1):\ k_M = k(c) + k(c^{\mathrm{CP}}) + d \text{ [BM.128(7)]};$$

$$\Theta(c^{\mathrm{CP}}) = \Theta(c) \text{ on the materialized locus [BM.113(2)(ii)]} \implies k(c^{\mathrm{CP}}) = k(c) = k;$$

$$d = 0 \text{ for an open connector [Def-H.32(4)]}$$

$$k_M = 2k \implies \varepsilon_M = (-1)^{|2k|} = ((-1)^2)^k = 1 \text{ [machine: every } k \in \{0, \dots, 8\}];$$

$$j_M = |2k|/2 = k \in \mathbb{Z} \text{ [machine]}$$

$$(2):\ k(c^{\mathrm{CP}})$$

$$= -k(c) \text{ [}\hat{\omega}:\ k \mapsto -k \text{ [BM.73(4)]]}$$

$$\implies k_M$$

$$= 0 \implies \varepsilon_M$$

$$= (-1)^0$$

$$= +1,$$

$$j_M$$

$$= 0 \text{ [machine]};$$

the two readings agree on ε_M and differ on j_M [machine]

$$(3):\ k_M = 2k + d \implies (-1)^{|2k+d|} = (-1)^{2k+d} = (-1)^{2k}(-1)^d = (-1)^d \text{ [machine: } d \in \{0, \dots, 4\}]$$

$$(4):\ k_B = k_1 + k_2 + k_3 \text{ [BM.128(7)]};$$

$$(-1)^{k_1+k_2+k_3} = (-1)^{k_1}(-1)^{k_2}(-1)^{k_3} \text{ [machine: all 125 triples]};$$

$$\#\{a : k_a \text{ odd}\} \text{ odd} \iff \varepsilon_B = -1 \text{ [machine: 62 of 125]}$$

$(5):\ k_a = 1\ \forall a \implies k_B = 3,\ j_B = 3/2,\ \varepsilon_B = (-1)^3 = -1$ [machine];

$L = 2 \implies \kappa \equiv 1,\ k = L/2 = 1$ [FT.32(6); FT.28];

the six-row ladder [machine]

$(6):$ BM.28(7), BM.103(8), BM.117(6), BM.127(6) re-cited;

the symbol census of $(-1)^{k_1+k_2+k_3}$ carries $\{k_1, k_2, k_3\}$ alone [machine]

$(7):$ BM.128(5), (6) and CT3.22 re-cited;

BM.117(2), (3)'s octet rows re-cited [¬promoted];

$(8):$ the standing-rule scans [machine]. ■

Phys. Two composites, two clean answers, and one honest gap. The meson is a boson and it is one unconditionally: its constituents are a chain and its conjugate, so their windings are equal and their sum is even, which makes the total spin an integer and the exchange sign plus one — and the conclusion survives the corpus's one unresolved orientation-sign convention, since the alternative reading puts the total winding at zero and the sign at plus one as well, differing only in the spin value. With a closed connector the answer changes to the connector's own parity, because the two constituent windings cancel modulo two, and on the computed cells the connector is open so the boson sign is what holds. The baryon is not unconditional and the tier does not pretend otherwise: its sign is the product of its three constituents' signs, so it is fermionic exactly when an odd number of them carry odd winding, and at the minimal materialized floor the winding is one — odd — which makes the three-floor baryon fermionic with total spin three halves. Sixty-two of the one hundred twenty-five winding triples in a small census are fermionic and sixty-three bosonic, so the criterion genuinely discriminates. What is missing is not the rule but its input: which windings a named sealed class occupies is the same unread datum as the flavour composition, so the baryon's statistics is complete as a function and open as a value per class. And the antisymmetry that makes a baryon a baryon is colour's — one-dimensional over three colour slots against a carrier wedge that vanishes at dimension two — unobservable by construction and not the statistics, with no flavour multiplet formed anywhere in the argument.

BM.130 — The status of the Annihilation, Concrete Spin and Fermion Identification Tier: adjudication, scope, and the open items. Structural-constraints record: see Appendix B, Section B.1.33 of the Addendum.

4.25 Tier BM.23 — The Closing Tier: the Haar–Poisson Duality and the Lapse, the Selected Curvature Composition, the Momentum-Space Resolvent, the Charge and Statistics Censuses, the Two-Chain Ensemble, the Seed-Tensoring Mechanism, the Convention Declarations, the Ceiling Study, the Jensen Residue, the Pair Amplitude, and the Era Closure

Theorem 4.110 (BM.131 — The Haar–Poisson duality: the twirl fixed point, the complementary factorization, and the proper-time contraction). Thm.

(1) the two evolutions, from the axioms:

the Haar side: $\mu_\star = \mu_{\mathrm{Haar}}\big(\mathrm{SU}(2)\big)$,

$$\mathcal{T}(\rho) = U\rho U^\dagger,$$

$$U \sim \mu_\star,$$

$$U \notin \mathfrak{L}$$

[A6];

the Poisson side:

$$K := -\log\rho$$

[L1 : the modular Hamiltonian],

$$\tilde{\mathcal{T}}_\gamma(\rho) = \frac{\rho\, e^{-\gamma\rho}}{\mathfrak{T}\big(\rho e^{-\gamma\rho}\big)},$$

$$\gamma = 1 - 4\varepsilon_q \in (0, 1]$$

[D0.3; L0 : one tilt step per A4 event];

the ledger increment factorizes:

$$\rho \;\longmapsto\; U\,\tilde{\mathcal{T}}_\gamma(\rho)\,U^\dagger$$

[A6 : $\tau_n \to \tau_{n+1}$; L0 : the crystallization–tilt coupling, declared at the point of use];

(2) the exact K-update (the Poisson evolution):

$$\lambda_i' = \frac{\lambda_i e^{-\gamma\lambda_i}}{Z_\gamma},$$

$$Z_\gamma = \textstyle\sum_j \lambda_j e^{-\gamma\lambda_j}$$

[L1(3) Pf : the eigenvalue flow] :

$$\boxed{\delta K \;:=\; K' - K \;=\; \gamma\,\rho \;+\; \log Z_\gamma \cdot I}$$

[$\log\lambda_i' = \log\lambda_i - \gamma\lambda_i - \log Z_\gamma$ exactly; machine: the formal-series identity to order 12];

$\gamma = 1$ (the sharp point $\varepsilon_q = 0$ [D0.4]) : $\quad \delta K = \rho + \log Z \cdot I \quad$ [machine];

the vacuum is the Poisson fixed point:

$$\rho = \rho_\circ = \tfrac{1}{4} I : \quad Z_\gamma = e^{-\gamma/4},$$

$$\lambda_i' = \tfrac{1}{4},$$

$$\boxed{\delta K = 0 \text{ exactly}}$$

[L1(1); machine: $e^{-\gamma/4}\cdot e^{+\gamma/4} = 1$ coefficient-wise in $\mathbb{Q}[[\gamma]]$ to order 12];

(3) the Haar invariances (the measure side):

$$V\big(\tilde{K}(\mathcal{T}(\rho))\big) = V\big(\tilde{K}(\rho)\big) \text{ for every draw}$$

[L1(2); machine: the spectral invariance $\mathrm{Tr}\big((U\rho U^\dagger)^n\big) = \mathrm{Tr}(\rho^n),\ n = 1,2,3,$ on 40 exact rational SU(2) points and generic rational states];

$$K \mapsto UKU^\dagger \implies \text{spec}\, K \text{ invariant};$$

the draws are unrecorded [A6 : $U \notin \mathfrak{L}$];

$$\neg\exists\,\tau^{-1}$$

[A6];

the stationarity of the measure:

conjugation by a fixed U_0 pushes $\mu_\star$ to $\mu_\star$

[math: the two-sided invariance and uniqueness of the Haar measure on a compact group];

(4) the twirl theorem (the duality's fixed locus):

$$\mathcal{H}_p \otimes \mathcal{H}_q = \vee^2 \oplus \wedge^2$$

[A3],

$$\Pi_{\mathcal{R}} = \tfrac{1}{2}(I + F),$$

$$\Pi_{\mathcal{V}} = \tfrac{1}{2}(I - F),$$

$$\operatorname{rk}\Pi_{\mathcal{R}} = 3,$$

$$\operatorname{rk}\Pi_{\mathcal{V}} = 1$$

[FT.107(2); machine: the flip $F^2 = I$] :

$$\boxed{\int_{\mathrm{SU}(2)} \big(U\otimes U\big)\rho\,\big(U\otimes U\big)^{\dagger} d\mu_\star(U) \;=\; \frac{\mathfrak{T}\big(\Pi_{\mathcal{R}}\rho\big)}{3}\,\Pi_{\mathcal{R}} \;+\; \mathfrak{T}\big(\Pi_{\mathcal{V}}\rho\big)\,\Pi_{\mathcal{V}}}$$

[math: Schur's lemma on the $3\oplus 1$ decomposition; machine: idempotence, the block-trace preservation, and the equivariance $\mathrm{tw}\big((U\otimes U)\rho(U\otimes U)^{\dagger}\big) = (U\otimes U)\,\mathrm{tw}(\rho)\,(U\otimes U)^{\dagger}$ on all 40 exact rational points];

the twirl fixed locus is the block-maximal-mixed family $\big\{\frac{x}{3}\Pi_{\mathcal{R}} + (1-x)\Pi_{\mathcal{V}} :\ x\in[0,1]\big\}$

[machine];

the unique common fixed point of both sides:

$$\lambda_i' = \lambda_i \;\forall i \iff e^{-\gamma\lambda_i} = Z_\gamma \;\forall i \iff \lambda_i \text{ all equal } (\gamma > 0) \iff \rho = \tfrac{1}{4}I;$$

$$\boxed{\left\{\text{Poisson fixed points}\right\} \cap \left\{\text{Haar-twirl fixed points}\right\} \;=\; \left\{\tfrac{1}{4}I\right\}}$$

[machine: $x/3 = 1 - x \iff x = \frac{3}{4}$ gives $\rho = \frac{1}{4}I$ on the block family; the tilt fixed point is $\frac{1}{4}I$ [L1(1), (5)]];

(5) the complementary factorization (what the record sees):

V is invariant under the Haar factor [(3)] and strictly contracting under the Poisson factor:

$$V' = \left(1 - \frac{\gamma}{4}\right)^2 V + O(\delta\rho^3)$$

[L2];

$$\left(1 - \frac{\gamma}{4}\right)^2 + \frac{\gamma(8-\gamma)}{16} = 1$$

[BM.51(2); machine: exact in $\mathbb{Q}[\gamma]$] $\implies V_n = \left(1 - \frac{\gamma}{4}\right)^{2n} V_0 + O$

[BM.51(3)];

$$\boxed{\text{the unrecorded Haar draws drop out of the record exactly: } V_n \text{ depends on the tilt count alone}}$$

[L1(2) + L2; machine];

the reading:

the measure side is the exact gauge freedom of the record;

the Poisson side is the recorded dynamics;

the duality fixed point is the A5 vacuum $\rho_\circ$ [A5];

(6) the finite-group companion (the same twirl at the carrier level):

$$\Pi_k := \frac{1}{k!} \sum_{\sigma \in S_k} \hat{P}_\sigma$$

[the averaging over the finite group S_k is its Haar measure] :

$$\Pi_k^2 = \Pi_k,$$

$$\mathrm{rk}\, \Pi_k = k + 1,$$

$$\mathrm{im}\, \Pi_k = \mathrm{Sym}^k\big(\mathbb{C}^2\big)$$

[machine: exact rank over $\mathbb{Q}$ for $k = 1..6$; forward pointer: the mechanism is BM.138];

(7) the proper-time contraction (the metric read):

$$ds^2 = -N^2 d\tau^2 + g_{ij}\big(dx^i + N^i d\tau\big)\big(dx^j + N^j d\tau\big)$$

[BM.34(2); machine: the line-element identity];

$$\text{at rest } \big(dx^i = 0\big) : \quad ds = N\, d\tau;$$

the tick is the chart coordinate $x^0 = \tau_n$

[D1.8];

$N = N(k, E)$ is derived in BM.132;

$$\boxed{V(s) \;=\; \Big(1 - \frac{\gamma}{4}\Big)^{2s/N} V_0}$$

[the reparametrization of (5)'s law by $s = Nn$; machine: at $\gamma = 1,\ N = 1/m,\ m \in \{1, 2, 3, 4\}$: $\big(\frac{9}{16}\big)^m$ exact integer powers, strictly decreasing in m];

the two deficit species stay separated:

$$\frac{\gamma(8-\gamma)}{16} \text{ per tick (informational)}$$

against

$$1-N \text{ (metric)};$$

¬a numerical identification of the two;

they join only through (7)'s exponent

[BM.37(2) : disjoint alphabets];

the scope row:

(7) is a chart-side reparametrization of a fixed law, ¬a dynamical coupling:

¬a feedback of the contraction into the metric is formed [BM.104(4)'s freedom-census pattern];

the tick-rate/field coupling is re-scoped by it: the redshifted contraction read is formed at the point of use, inheriting CT2.6's hypotheses [A §0.2a];

its absolute-frequency datum remains calibration-species [CT0.3 : $\tau_c = 1$ tick];

(8) status:

derived: (1)–(7);

the duality is the pair $(\mu_\star,\ K)$ with the three exact rows: the Haar invariance of V,

the Poisson contraction of V,

the unique common fixed point $\frac{1}{4}I$;

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5],

the lapse gap is closed by BM.132;

the row g,

e_0,

N of BM.130(7) is re-scoped: N derived [BM.132],

g selected [BM.133],

e_0 consolidated as calibration [BM.143(4)].

Req. A3, A5, A6 (Part 1); D0.3, D0.4, D1.8 (Part 1); CT0.3, CT0.8 (Part 1); L0, L1, L2 (Part 1); CT2.6 (Part 2); P1, P5 (Parts 1–2); FT.107(2) (Part 3); Def-BM.10, BM.34(2), BM.37(2), BM.51(2), BM.51(3), BM.104(4) (this part); Schur's lemma on the $3 \oplus 1$ decomposition, the Haar-measure invariance and uniqueness on a compact group, formal power-series arithmetic over $\mathbb{Q}$ with the exact exp / log pair, exact rational SU(2) points from Pythagorean quadruples, exact rank by Fraction elimination, Zariski-density substitution [math; machine: `checks/verify_H_bm_closing_tier.py` §A].

Pf.

(1) : A6 supplies $\mu_\star$, $\mathcal{T}$, $U \notin \mathfrak{L}$;

D0.3 supplies $\tilde{\mathcal{T}}_\gamma$;

L0 supplies the one-tilt-per-event coupling (declared scope, inherited at the point of use);

the factorization is the composition of the two axiom-level moves

(2) : ρ faithful $\implies$ ρ, $\tilde{\mathcal{T}}_\gamma(\rho)$ simultaneously diagonalizable [L1(3) Pf] :

K'

$$= -\log \rho'$$

$= -\log\rho + \gamma\rho + \log Z_\gamma \cdot I$ as an operator identity on the eigenbasis [machine: the $\mathbb{Q}[[t,\gamma]]$ series identity $\log(\lambda e^{-\gamma\lambda}/Z_\gamma) + \gamma\lambda + \log Z_\gamma - \log\lambda \equiv 0$ to order 12];

$$\rho = \tfrac{1}{4}I: \quad Z_\gamma = 4 \cdot \tfrac{1}{4}e^{-\gamma/4} = e^{-\gamma/4};$$

$$\lambda' = \tfrac{1}{4}e^{-\gamma/4}/e^{-\gamma/4} = \tfrac{1}{4};$$

$$\delta K = \gamma \cdot \tfrac{1}{4}I + \log\big(e^{-\gamma/4}\big)I = 0 \ \text{[machine: } \exp(-\gamma/4)\exp(\gamma/4) = 1 \text{ coefficient-wise]}$$

$$(3):\ \text{L1}(2);$$

$$K(\mathcal{T}\rho) = UKU^\dagger \ \text{[L1(2) Pf: functional-calculus covariance]} \implies \operatorname{spec} K \text{ invariant};$$

the stationarity of $\mu_\star$ under conjugation is the two-sided invariance [math];

the machine rows: $\operatorname{Tr}\big((U\rho U^\dagger)^n\big)$

$$= \operatorname{Tr}(\rho^n),$$

$$n$$

$$= 1,$$

$$2,$$

$$3,$$

on 40 exact rational points U

$$= \left(\begin{smallmatrix} a+bi & c+di \\ -c+di & a-bi \end{smallmatrix}\right)/n,$$

$$a^2{+}b^2{+}c^2{+}d^2{=}n^2,$$

and generic rational Hermitian ρ

$$(4):\ \mathbb{C}^2\otimes\mathbb{C}^2=\vee^2\oplus\wedge^2\ [\text{A3}]:\quad U\otimes U \text{ acts irreducibly on each block;}$$

the twirl of any intertwiner-average projects onto the commutant

$$=\ \text{span}\{\Pi_{\mathcal{R}},\Pi_{\mathcal{V}}\}\ [\text{math: Schur}]:\quad \text{the block weights } \mathfrak{T}(\Pi_{\mathcal{R}}\rho),$$

$\mathfrak{T}(\Pi_{\mathcal{V}}\rho)$ are the only invariants;

machine: $\Pi_{\mathcal{R}}+\Pi_{\mathcal{V}}=I,\ \Pi_{\mathcal{R}}\Pi_{\mathcal{V}}=0,\ \text{rk}=(3,1)$;

$\text{tw}\circ\text{tw}=\text{tw}$;

the block traces of $\text{tw}(\rho)$ equal those of ρ;

the equivariance on all 40 rational points;

$$x/3=1-x\iff x=\tfrac{3}{4}\implies \tfrac{x}{3}\Pi_{\mathcal{R}}+(1-x)\Pi_{\mathcal{V}}=\tfrac{1}{4}I$$

$$(5):\ V\big(\tilde{K}(U\tilde{\mathcal{T}}_\gamma(\rho)U^\dagger)\big)=V\big(\tilde{K}(\tilde{\mathcal{T}}_\gamma(\rho))\big)\ [(3)]=\big(1-\tfrac{\gamma}{4}\big)^2V(\tilde{K}(\rho))+O\ [\text{L2}];$$

iteration: BM.51(3);

the record V_n carries ¬a draw label [A6 : $U\notin\mathfrak{L}$]

$$(6):\ \Pi_k^2$$

$$=\frac{1}{(k!)^2}\sum_{\sigma,\tau}\hat{P}_{\sigma\tau}$$

$$=\frac{1}{k!}\sum_{\tau'}\hat{P}_{\tau'}$$

$=\Pi_k$ [BM.128(2)'s product law transported; machine: entrywise over $\mathbb{Q}$];

$\text{rk}\,\Pi_k=k+1$ [machine: exact Fraction elimination for $k=1..6:$ ranks $(2,3,4,5,6,7)$];

$$\operatorname{im}\Pi_k = \mathrm{Sym}^k \text{ [Def-H.38 carrier]}$$

$$(7):\ ds^2\big|_{dx^i=0} = -N^2 d\tau^2 \implies ds = N\,d\tau \text{ [BM.34(2); machine]};$$

$$s = Nn \implies V_n = \left(1-\tfrac{\gamma}{4}\right)^{2n} V_0 = \left(1-\tfrac{\gamma}{4}\right)^{2s/N} V_0 \text{ [substitution]};$$

the exact powers at γ

$$= 1:\ \left(\tfrac{9}{16}\right)^m,$$

$$m$$

$$= 1..4 \text{ [machine: strictly decreasing, exact integers } 9^m/16^m\text{]};$$

the species row: $\gamma(8-\gamma)/16 \in \mathbb{Q}$ against $1 - N \in \mathfrak{q}^0$ chart data : ¬a derived equation relates them [BM.37(2) pattern];

$$(8):\ \text{assembly.}\ \blacksquare$$

Phys. The corpus owns exactly two state evolutions, and they are dual in the strict sense: an unrecorded random conjugation drawn from the invariant Haar measure, and a recorded modular update generated by $K = -\log\rho$ whose exact step is $\delta K = \gamma\rho + \log Z_\gamma \cdot I$. The measure side moves states inside the level sets of the Lyapunov function and preserves the modular spectrum; the dynamics side cuts strictly across the levels toward the vacuum. The Haar average of any pair state is its block-maximal-mixed projection, and the tilt's unique fixed point is the maximally mixed state itself, so the two sides share exactly one fixed point: the A5 vacuum. Because the draws are unrecorded, they drop out of every ledger read, and the contraction law the record obeys is purely Poisson-side — the measure is the record's exact gauge freedom. At the finite-group level the same twirl builds the spin carriers: averaging over S_k is the symmetrizer, whose image is Sym^k of dimension $k+1$. Finally the metric joins: the lapse converts chart ticks to proper time along a resting worldline, so the same exact contraction read per proper time carries the lapse in the exponent, with the informational and metric deficit species kept apart and joined only by that exponent. No feedback is formed, no scale added, nothing measured.

Theorem 4.111 (BM.132 — The lapse: the derived function of the envelope ratio and the curvature, its formed domain, and the envelope row). Thm.

(1) the source and the potential (re-derived at the point of use):

$$T_{\mu\nu} = \mathrm{diag}(\rho,0,0,0) \text{ at rest}$$

$$\text{[BM.9(4); CT1.22]};$$

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa^2 T_{\mu\nu}$$

[CT1.23];

$$g_{00} = -(1 + 2\Phi),$$

$$G^{(1)}_{00} = 2\nabla^2\Phi,$$

$$\nabla^2\Phi = \frac{\kappa^2}{2}\rho = 4\pi G_N \rho$$

[BM.43(4); machine re-run];

$$\Phi(r) = -\frac{G_N m}{r} \text{ outside the source support,}$$

$$G_N m_{\text{hid}} = \frac{\hbar E_{\text{int}}}{4},$$

$$r_s = 2G_N m_{\text{hid}} = \frac{\hbar E_{\text{int}}}{2}$$

[BM.43(4); P3 : G_N a named combination, quarantined];

(2) the ADM lapse at zero shift:

$$N^2 = \Omega^2 + \gamma^{ij} N_i N_j,$$

$$w = 0 \implies N^i = 0, \quad N = \Omega$$

[BM.34(1); CT1.12];

the linearized dictionary $\Omega^2 = 1 + 2\Phi$

[CT1.12 : $\Omega = e^{\alpha_\Omega a_{00}}$; the conformal mode reads the potential; $a_{00} = \frac{1}{2\alpha_\Omega}\ln(1 + 2\Phi)$ on the formed locus];

$$\boxed{N(r,E)^2 \;=\; 1 + 2\Phi \;=\; 1 - \frac{2G_N m}{r} \;=\; 1 - \frac{r_s}{r} \;=\; 1 - \frac{G(E)}{4\pi\, k}},$$

$$k := \frac{r}{\Delta x_{\text{B}}(E)},$$

$$G(E) = 2\pi E_{\text{int}}(E)^2 \ln 2$$

[BM.88(2)(i), (iii); BM.43(4); machine: one identity in $\mathbb{Q}[E_{\text{int}}, \pi, \ln 2, k]$];

the lapse is a derived function of the envelope ratio and the derived curvature argument:

$$N = N(k, E) \in \mathfrak{q}^0$$

[CT2.1; machine: the exponent census];

(3) the formed domain:

$$N^2 \geq 0 \iff k \geq \frac{G(E)}{4\pi} = \frac{E_{\text{int}}^2 \ln 2}{2} = \frac{r_s}{\Delta x_{\text{B}}}$$

[machine: the monomial];

the envelope row:

on every computed cell $E_{\text{int}} \in \{2, 6, 12, 24, 60\}$ [BM.13(2); Census FT.35(2), FT.42(5) row-wise] :

$$\frac{G}{4\pi} = \Big(2,\ 18,\ 72,\ 288,\ 1800\Big) \ln 2 \ > \ 1$$

$$\Big[\ln 2 > \frac{1}{2} \iff 4 > e: \quad e < \frac{876809}{322560} < 4;\ \text{machine: the exact rational enclosure of } e\Big];$$

$$\boxed{1 + 2\Phi\big(\Delta x_{\text{B}}\big) = 1 - \frac{E_{\text{int}}^2 \ln 2}{2} \ < \ 0 \ \text{ on every computed cell}}$$

[BM.43(4) : $\Phi(\Delta x_{\text{B}}) = -E_{\text{int}}^2 \ln 2/4$; machine];

the interior row (the uniform-ball read):

$$\Phi_{\text{int}}(r) = -\frac{G_N m}{2R}\Big(3 - \frac{r^2}{R^2}\Big),$$

$$\nabla^2 \Phi_{\text{int}} = 4\pi G_N \rho,$$

$$\Phi_{\text{int}}(R) = \Phi(R)$$

[machine: the harmonicity and matching identities in $\mathbb{Q}[x, y, z, r, R]$];

$$1 + 2\Phi_{\text{int}}(0) = 1 - \frac{3E_{\text{int}}^2 \ln 2}{4} < 0 \text{ on every computed cell}$$

$$\Big[E_{\text{int}} = 2 : \ 3\ln 2 > 1 \text{ on } \ln 2 > \frac{1}{3} : \ e^3 < 8; \text{ machine}\Big];$$

the self-lapse of a sealed composite is ¬formed at its own Born width: the envelope lies strictly inside the domain boundary on every computed cell

(4) the monotonicity, the limits, and the gauge agreements:

$$\frac{\partial N^2}{\partial k} = \frac{G}{4\pi k^2} > 0 : \quad N \text{ strictly increasing in } k \text{ on the formed domain};$$

N strictly decreasing in E_{int}

[machine: sign census on exact rationals];

$$N = 0 \text{ at the boundary } k = \frac{G}{4\pi};$$

$$N \to 1^- \text{ as } k \to \infty;$$

$$N \in (0, 1] \text{ on the formed locus}$$

[machine];

the vacuum gauge agreement:

$$(a_{00}, q_{ij}, w_i) = 0 \implies N = 1$$

[BM.34(2); BM.88(2)(iii); machine]

and

$$\lim_{k\to\infty} N(k, E) = 1;$$

T-evenness:

N carries w only through $\gamma^{ij}N_iN_j = N^2 - \Omega^2$; at zero shift N is w-free

[BM.34(3); machine];

(5) the three substitutions into derived objects:

(i) the dispersion: $E = N(k,E)\sqrt{m^2 + \gamma^{ij}p_ip_j} - N^ip_i$

[BM.34(4); the lapse entry now carries (2)'s function];

(ii) the threshold:

$$\mathcal{T}(k,E) = \frac{\mathcal{T}_0}{N(k,E)} = \frac{E_{\text{int}}\,\mathfrak{h}}{\ln 2}\Big(1 - \frac{E_{\text{int}}^2 \ln 2}{2k}\Big)^{-1/2}$$

[BM.115(1); Def-BM.17(3); machine: the substitution] :

$\mathcal{T} \to \infty$ at the domain boundary;

$\mathcal{T} \to \mathcal{T}_0$ as $k \to \infty$;

a deeper lapse lowers every threshold at once, and the boundary locus raises every threshold without bound [BM.115(1) re-cited];

(iii) the coupling profile:

$$e(r,E) = e_0\sqrt{N(k,E)\cdot f(k)\cdot \mathsf{g}\big(G(E)\big)}$$

[BM.88(4); g carried symbolically here, selected in BM.133];

(6) the bars:

r_s enters as P3's named combination, quarantined: ¬a collapse reading, ¬a horizon identification, ¬black-hole physics [P3; BM.15(4); BM.115(4); BM.130(6) discipline inherited];

¬a cosmological coordinate [Def-BM.17(6)(i)];

¬a measured curvature, radius or redshift [P5];

the weak-field rows inherit CT2.6's hypotheses at the point of use [A §0.2a; BM.43(4) scope];

the full non-linear joint solution remains ¬constructed [BM.43(5) re-cited, ¬extended];

(7) status:

derived: (1)–(5);

$$\boxed{\text{the lapse gap is closed: } N(k,E)=\sqrt{1-\frac{G(E)}{4\pi k}} \text{ on the formed domain } k\geq\frac{G(E)}{4\pi}};$$

the row N as solution data [BM.88(8); BM.93(6); BM.100(6); BM.107(7); BM.118(7); BM.130(7)] is discharged.

Req. A8 (Part 1); D2.1 (Part 1); CT1.12, CT1.22, CT1.23, CT2.1, CT2.6, CT2.9 (Part 2); P1, P3, P5 (Parts 1–2); Def-BM.4, Def-BM.10, Def-BM.17(3), Def-BM.17(6), BM.9(4), BM.13(2), BM.15(4), BM.34(1), BM.34(2), BM.34(3), BM.34(4), BM.43(4), BM.43(5), BM.88(2), BM.88(3), BM.88(4), BM.88(8), BM.115(1), BM.115(4), BM.130(6), BM.130(7) (this part); Census FT.35(2), FT.42(5) (row-wise); the exact rational enclosure of e by the factorial series, the harmonicity of $1/r$ and the interior-ball potential in quotient rings, Laurent and monomial identities, sign censuses on exact rationals [math; machine: `checks/verify_H_bm_closing_tier.py` §A–§B].

Pf.

(1) : BM.43(4)'s rows re-derived: $G^{(1)}_{00}$

$= 2\nabla^2\Phi$ [machine: the jet-ring Christoffel → Ricci → Einstein chain];

$\nabla^2(1/r)=0$ [machine: the quotient-ring identity];

$$G_N m_{\text{hid}}=\frac{\mathfrak{h}^2}{4\ln 2}\cdot\frac{E_{\text{int}}\ln 2}{\mathfrak{h}}=\frac{\mathfrak{h}E_{\text{int}}}{4}\ \text{[P3; FT.44(4); machine]}$$

$$(2):\ w=0\implies N_i=\Omega^2\varkappa\beta w_i=0\implies N^2=\Omega^2\ \text{[BM.34(1)]};$$

the linearized dictionary: g_{00}

$= -\Omega^2$

$$= -(1 + 2\Phi) \text{ [CT1.12; BM.34(2)'s line element at zero shift];}$$

$$\text{the substitution: } 1 - \frac{2G_N m}{r} = 1 - \frac{r_s}{r};$$

$$\frac{r_s}{\Delta x_{\mathrm{B}}}$$

$$= \frac{\mathfrak{h} E_{\text{int}}/2}{\mathfrak{h}/(E_{\text{int}} \ln 2)}$$

$$= \frac{E_{\text{int}}^2 \ln 2}{2}$$

$$= \frac{G}{4\pi} \text{ [BM.88(3); machine: one monomial]}$$

$$(3): \ N^2 \geq 0 \iff \frac{G}{4\pi k} \leq 1;$$

$$E_{\text{int}} \geq 2 \text{ [BM.13(3)]} \implies \frac{G}{4\pi} \geq 2 \ln 2 > 1 \text{ [machine: } e < \tfrac{876809}{322560} < 4 \implies \ln 4 > 1];$$

the interior row: the matching identities and $3E_{\text{int}}^2 \ln 2 > 4$ at $E_{\text{int}} = 2$ [machine: $e^3 < 8$]

(4) : the derivative sign and the endpoint values on exact rationals [machine];

(5) : substitution into BM.34(4), BM.115(1), BM.88(4) [machine: the monomial rows];

$(6), (7)$: assembly and the rescope census. ■

Phys. The lapse was solution data; it is now a function. The composite's own dust source runs through the derived field equations to the weak-field potential, the ADM split reads the potential as the zero-shift lapse, and the curvature lock turns the answer into one dimensionless monomial: $N^2 = 1 - G/(4\pi k)$ on the envelope ratio. The function comes with its own formed domain, and the domain has a sharp edge: on every computed cell the boundary $k = G/(4\pi)$ lies strictly beyond the Born width, so a sealed composite's self-lapse is not formed at its own envelope — the interior read only deepens the failure. The vacuum gauge is recovered exactly at infinity, the threshold ladder inherits the lapse and diverges at the boundary, and the running coupling's energy factor is now explicit. No black-hole reading is attached to the boundary, the Schwarzschild combination stays quarantined, and no measured number enters.

Theorem 4.112 (BM.133 — The selected curvature composition: two derived criteria, the unique survivor, and the completed running coupling). Thm.

$$(1) \quad \text{the three priced rows (re-cited, frozen):}$$

$$\mathsf{g} \in \left\{1, \quad 1+G, \quad \frac{1}{1+G}\right\}$$

$$[\text{BM.88}(4);\ \text{BM.104}(2)]:$$

$$\text{the back-reaction fixed-point rows:}$$

$$\text{(i)}\ \ \mathsf{g} \equiv 1;$$

$$\text{(ii)}\ \ \mathsf{g}_{\text{fp}} = \frac{1+G}{1-aG},\ \ \text{admissible} \iff aG < 1;$$

$$\text{(iii)}\ \ aG\,\mathsf{g}^2 + (1+G)\,\mathsf{g} - 1 = 0,\ \ \mathsf{g}_{\text{fp}} = \frac{\sqrt{(1+G)^2 + 4aG} - (1+G)}{2aG}$$

$$[\text{BM.104}(2);\ \text{machine re-run}];$$

$$a = \tfrac{1}{2} N w\, \varepsilon_{\text{EM}} Q^2 \geq 0, \qquad G > 0 \qquad [\text{BM.104}(1), (3);\ \text{BM.88}(3)];$$

$$(2) \quad \text{criterion (S1) — the Lyapunov contractivity of the back-reaction map:}$$

$$F_{(\text{i})}(\mathsf{g}) = 1;$$

$$F_{(\text{ii})}(\mathsf{g}) = 1 + G + aG\,\mathsf{g};$$

$$F_{(\text{iii})}(\mathsf{g}) = \frac{1}{1+G+aG\,\mathsf{g}}$$

$$[\text{BM.104}(1)\text{'s loop read as a map}]:$$

$$F'_{(\text{i})} = 0;$$

$$F'_{(\text{ii})}$$

$$= aG: \quad \text{contractive}$$

$\iff aG < 1$ — conditional, with the pole at aG

$= 1$ where g_{fp} diverges

[BM.104(2)(ii), (3); machine];

$$F'_{(\mathrm{iii})}(\mathsf{g}) = -\frac{aG}{(1+G+aG\,\mathsf{g})^2}\ :$$

at the fixed point $1+G+aG\,\mathsf{g}_{\mathrm{fp}} = \dfrac{1}{\mathsf{g}_{\mathrm{fp}}} \implies \boxed{\left|F'_{(\mathrm{iii})}\right| = aG\,\mathsf{g}_{\mathrm{fp}}^2 = 1-(1+G)\,\mathsf{g}_{\mathrm{fp}}}$

the unconditional contraction:

$$0 < \left|F'_{(\mathrm{iii})}\right| < 1 \iff 0 < \mathsf{g}_{\mathrm{fp}} < \frac{1}{1+G}$$

and

$$(1+G)^2\big((1+G)^2+4aG\big) < \big((1+G)^2+2aG\big)^2 \iff 0 < 4a^2G^2$$

[machine: the squaring argument, exact in $\mathbb{Q}[G,a]$; both sides positive on the formed locus];

row (iii) is unconditionally Lyapunov-contractive on the whole formed locus; row (ii) is not; row (i) is contractive but G-inert

[L1–L2 contraction discipline; BM.51(2) pattern];

(3) criterion (S2) — the minimal lapse-domain loss:

the corrected curvature $G_{\mathrm{corr}} = G(1+a\,\mathsf{g})$ [BM.104(1)] moves the lapse boundary:

$$N_{\mathrm{corr}}(k)^2 = 1-\frac{G(1+a\,\mathsf{g})}{4\pi k} \text{ formed } \iff k \geq \frac{G(1+a\,\mathsf{g})}{4\pi}$$

[BM.132(2), (3)];

every positive composition shrinks the formed domain ($a\,\mathsf{g} > 0$);

the loss is exactly ordered:

$$\boxed{a\,\mathsf{g}_{\mathrm{fp}}^{(\mathrm{iii})} \;<\; \frac{a}{1+G} \;<\; a \;=\; a\,\mathsf{g}^{(\mathrm{i})} \;<\; a\,(1+G) \;=\; a\,\mathsf{g}^{(\mathrm{ii})}}$$

[machine: the strict chain on $a, G > 0$];

row (ii) at the pole loses the whole domain:

$$aG \to 1: \quad \mathsf{g}_{\mathrm{fp}} \to \infty, \quad G_{\mathrm{corr}} \to \infty, \quad \{k:\ N_{\mathrm{corr}} \text{ formed}\} \to \varnothing$$

[BM.104(2)(ii), (3); machine];

row (iii) loses the least formed lapse domain of the three, uniformly in (a, G)

(4) the unique survivor:

(S1)∧(S2): row (ii) fails both (conditionality, maximal loss with a total collapse at the pole);

row (i) passes (S1) and fails (S2) and is G-inert (2);

$$\boxed{\mathsf{g}(G) \;=\; \frac{1}{1+G}}$$

selected by two derived criteria;

$$\mathsf{g} \in \mathfrak{q}^0,$$

$$0 < \mathsf{g} \le 1,$$

g strictly decreasing in G,

$\mathsf{g}(0) = 1$ the common vacuum-normalization row of all three [BM.104(2); machine];

the exact values at the computed cells:

$$\mathsf{g} = \frac{1}{1 + 2\pi E_{\text{int}}^2 \ln 2},$$

$$E_{\text{int}} \in \{2, 6, 12, 24, 60\}: \quad \mathsf{g} \in \left\{ \frac{1}{1+8\pi\ln 2}, \frac{1}{1+72\pi\ln 2}, \frac{1}{1+288\pi\ln 2}, \frac{1}{1+1152\pi\ln 2}, \frac{1}{1+7200\pi\ln 2} \right\}$$

[machine: the monomial substitutions, carried exactly];

the observable consequence (re-cited):

BM.101(5)'s crossing count on the selected row is two, one or none — the composition is observable in the energy-scale graph's topology, and the selection is now made [BM.104(6) re-scoped];

(5) the BM.104 bars are respected:

the loop stays excluded: G_{corr} ¬re-enters G;

the lock's argument remains the informational mass [BM.104(4) frozen; BM.88(3); machine: the alphabet census $\{E_{\text{int}}, \ln 2, \pi\} \cap \{N, w, \varepsilon_{\text{EM}}, Q\} = \varnothing$ re-run];

the four grounds stand and none is used:

(i) ¬a count → strength map enters [BM.36(3); BM.61(8)(O1)];

(ii) ¬a defect-conditioned weighting [BM.90(3)];

(iii) consistency is ¬used as sufficient — the selection runs on (S1),

(S2),

which are stability and metric-formedness criteria, not the fixed-point equation;

(iv) ¬a P4 datum: the survivor is unique, so no choice remains [BM.104(5)(iv) discharged by uniqueness];

BM.104(5)'s ¬∃-row is re-scoped to the four grounds it cites, ¬retracted;

BM.104(6)'s discrimination row is completed by the selection;

(6) the completed running coupling:

$$\boxed{e(r,E) \;=\; e_0\sqrt{N(k,E)\cdot f(k)\cdot \mathrm{g}\big(G(E)\big)} \;=\; e_0\left[\frac{\big(1-\frac{G}{4\pi k}\big)^{1/2}\big(1-\frac{1}{k}\big)}{1+G}\right]^{1/2}}$$

[BM.88(4); BM.132(2); machine: the assembled monomial];

all three factors are derived:

$N(k,E)$ [BM.132];

$f(k) = 1 - \dfrac{1}{k}$ [BM.88(2)(ii); BM.47(6)];

$\mathrm{g}(G) = \dfrac{1}{1+G}$ [(4)];

only e_0 remains calibration [EM.25(7); BM.143(4)];

the formed locus:

$$k \;\geq\; \frac{G}{4\pi} \;\;\wedge\;\; k > 1 :$$

both factors strictly positive on its interior;

$e \to 0$ at the lapse boundary $(N \to 0)$;

$$e \to \frac{e_0}{\sqrt{1+G}} \;\text{ as }\; k \to \infty$$

[machine: the exact limit rows];

the monotonicity:

$k \mapsto N(k)f(k)$ strictly increasing on the formed locus

[both factors positive and strictly increasing [BM.88(2)(ii); (4)]; machine: the derivative sign on exact rationals];

BM.88(5)'s envelope zero $f(1)$

$= 0$ is re-scoped: it is a vacuum-gauge read $(N{=}1)$;

on the derived self-lapse the envelope locus itself is ¬formed [BM.132(3)];

the formed coupling is strictly positive on its whole formed locus and vanishes only at the boundary;

(7) status:

derived: (1)–(6);

the curvature composition is closed: $\mathsf{g}(G) = \dfrac{1}{1+G}$;

the row g of BM.130(7) is discharged;

¬a declaration (the survivor is unique),

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured coupling [P5].

Req. A4 (Part 1); D0.3, D0.4 (Part 1); CT2.1 (Part 2); P1, P4, P5 (Parts 1–2); L1, L2 (Part 1); EM.25(7) (Part 2); Def-BM.12, BM.36(3), BM.47(6), BM.51(2), BM.61(8), BM.88(2), BM.88(3), BM.88(4), BM.88(5), BM.90(3), BM.101(5), BM.104, BM.130(7), BM.132(2), BM.132(3) (this part); the squaring argument in $\mathbb{Q}[G, a]$, the strict-order chain on positive monomials, exact rational limit and derivative sign rows, the radical identity $(1+G)\mathsf{g}_{\mathsf{fp}} < 1$ in $\mathbb{Q}[G, a, s]/(s^2 - (1+G)^2 - 4aG)$ [math; machine: `checks/verify_H_bm_closing_tier.py` §B].

Pf.

(1) : BM.104(2)'s rows re-verified by substitution [machine];

(2) : $F'_{(\mathrm{iii})}$ at the fixed point: $1 + G + aG\mathsf{g}_{\mathsf{fp}}$

$= 1/\mathsf{g}_{\mathsf{fp}}$ [the fixed-point equation rearranged]

$\Longrightarrow\ |F'|$

$$= aG\mathsf{g}_{\mathrm{fp}}^2;$$

$aG\mathsf{g}_{\mathrm{fp}}^2 = 1 - (1+G)\mathsf{g}_{\mathrm{fp}}$ [machine: the equation $aG\mathsf{g}^2 + (1+G)\mathsf{g} - 1 = 0$ at $\mathsf{g} = \mathsf{g}_{\mathrm{fp}}$];

$\mathsf{g}_{\mathrm{fp}} < \frac{1}{1+G}$:

$$(1+G)\sqrt{(1+G)^2 + 4aG} < (1+G)^2 + 2aG$$

$$\iff (1+G)^2\big((1+G)^2 + 4aG\big) < \big((1+G)^2 + 2aG\big)^2$$

$$\iff 0 < 4a^2G^2$$ [machine: expansion in $\mathbb{Q}[G, a]$]

(3) : the domain-boundary monomial $\dfrac{G(1+a\mathsf{g})}{4\pi}$ is strictly increasing in $\mathsf{g} > 0$;

the chain $a\mathsf{g}_{\mathrm{fp}} < \dfrac{a}{1+G} < a < a(1+G)$ on (2)'s inequality and $1 + G > 1$ [machine];

the pole row: $(1 - aG)\mathsf{g}_{\mathrm{fp}} = 1 + G$ and $aG = 1 \implies 0 = 1 + G \neq 0$ [machine]

(4) : the census of $(2), (3)$ over the three rows;

the cell values: $G = 2\pi E_{\mathrm{int}}^2 \ln 2$ substituted [machine: monomials];

(5) : the alphabet census re-run and the four grounds re-cited at their loci [machine];

(6) : $e^2 = e_0^2\, N\, f\, \mathsf{g}$ with (4)'s g and BM.132(2)'s N [machine: the assembled product];

$\frac{d}{dk}[Nf] > 0$ on $k > \max(1, \frac{G}{4\pi})$: both factors positive increasing [machine: rational grid and product rule sign census];

(7) : assembly. ■

Phys. Three curvature compositions were priced and none was selected, because consistency alone is satisfied three times over. Two derived criteria break the tie. The first is the corpus's own contraction discipline: the back-reaction map of row (iii) is a strict contraction at its fixed point everywhere on the formed locus, row (ii) is contractive only below its pole and collapses

there, row (i) is inert. The second is the metric's: every composition inflates the corrected horizon locus and shrinks the formed domain of the lapse, and row (iii) shrinks it least, uniformly. The unique survivor is $\mathrm{g} = 1/(1+G)$ — curvature-diluted, vacuum-normalized, strictly decreasing. The back-reaction loop stays excluded: the lock's argument remains the informational mass, and the selection consumes stability and formedness, never a measured value or a free parameter. With the lapse derived and the composition selected, the running coupling is complete as a function of (k, E) — every factor derived, every zero located, only e_0 left to calibration.

Definition 4.20 (Def-BM.20 — The momentum-space Σ-resolvent (the construction BM.6(6) specifies)). Def. A chart-side interface definition on BM.34's ADM split and CT2.6's charts; the non-derived clauses are stated exactly and priced inside Part 4; no new axiom, no new datum, no new dimensionful scale. Logged as §II.4 addendum A-BM.12 before use.

(1) the boundary-channel states:

$|p\rangle$:= the chart plane-wave state of covariant momentum label p_μ on CT2.6's arena

[math: plane waves on the chart];

non-derived clause (the embedding):

the identification of the boundary-channel register's momentum label with the chart's p_μ — an ι_Σ-type embedding at the momentum level [BM.6(6)];

scope:

CT2.6's continuum-limit hypotheses are inherited at every point of use [A §0.2a];

the states are chart objects, ¬ledger objects;

¬a canonical pair is imported into the finite register [BM.10(1) stands];

(2) the resolvent:

$$\boxed{R(z;\mathrm{p}) := \left.\frac{1}{\left(p_0 - N^i p_i\right)^2 - N^2\left(m^2 + \gamma^{ij}p_i p_j\right)}\right|_{p_0=-z}}$$

constitutive clause:

the denominator is the derived mass-shell operator — its zero locus is the dispersion relation [BM.34(4)];

$m = m_{\text{hid}}(\mathcal{R})$ the composite's dispersion mass [Def-BM.10; BM.34(7)];

(3) the Fourier convention (declared):

the chart measure carries $\dfrac{d^3p}{(2\pi)^3}$ per spatial dimension;

cost: ¬a scale, ¬a P4 member — any other normalization rescales every amplitude read off it by a declared constant, so the convention is a naming, not a datum [D2.9 : $\hbar = 1$];

(4) the vertex normalization (constitutive minimality):

$\mathcal{V} = 1$:

the reduction clause: at e

$= 1,$

q

$= 1$ every amplitude of BM.142 reduces to the register's channel weight [BM.122(3); BM.123(1)] :

price:

the unique normalization with this reduction — any other choice is an e_0-species rescaling [EM.25(7); calibration, ¬structure];

(5) the bars:

¬an S-matrix unitarity claim (the formed reads are tree-level symbolic);

amplitude $\neq$ mass [FT.79(5)];

¬a numerical cross-section, ¬a measured comparison [P5];

¬an $i\varepsilon$-prescription import: R is read meromorphically, sign-free off the poles [FT.26(1) pattern];

the ℵ-domain at the junction couplings is E2 [A-BM.10, issued in BM.139];

(6) status:

constructed — BM.6(6)'s specification is discharged;

the resolvent specification is discharged by BM.134;

the two derived clauses (2),

(4) carry no new freedom beyond (1),

(3)'s declared identifications.

Req. D2.5, D2.9 (Parts 1–2); CT2.6 (Part 2); P4, P5 (Parts 1–2); FT.26(1), FT.79(5) (Part 3); Def-BM.10, Def-BM.17, BM.6(6), BM.10(1), BM.34(4), BM.34(7), BM.122(3), BM.123(1), EM.25(7) (this part or Part 2); A-BM.12 (§II.4, logged before use); the plane-wave basis on a chart [math].

Pf. Specification content: each clause states its requirement and cites its anchors; the non-derived clauses (1) and (3) are declared at the point of use and logged in §II.4; the constitutive clauses (2) and (4) are constructions on derived objects, each priced in its own clause; anchor existence machine-checked [`checks/verify_H_bm_closing_tier.py` §C: the document pins]. ■

Phys. The momentum-space resolvent the corpus named as an open interface is now a definition with exactly two declared ingredients — the boundary-channel embedding and the Fourier normalization — and everything else forced: the denominator is the derived mass-shell operator, so the pole locus is the dispersion relation, and the vertex normalization is pinned by its reduction to the register weight at unit coupling. The bars travel with it: no unitarity claim, no numerical cross-section, no measured comparison, and no Feynman prescription imported from outside.

Theorem 4.113 (BM.134 — The resolvent's derived content: the form uniqueness, the pole reads, the on-shell measure, and the two-body phase space). Thm.

(1) the pole locus is the derived dispersion:

$$R(z;\mathrm{p})^{-1}=0 \iff \big(p_0-N^i p_i\big)^2=N^2\big(m^2+\gamma^{ij}p_ip_j\big) \iff g^{\mu\nu}p_\mu p_\nu=-m^2$$

[BM.34(4); machine: the master polynomial identity re-derived, cleared by $(\det\gamma)^2$, with three exact rational ADM witnesses];

the branches:

$$p_0=N^ip_i\pm N\sqrt{m^2+p^2};$$

$$E = -p_0 = N\sqrt{m^2 + p^2} - N^i p_i \text{ on the positive branch}$$

[BM.34(4); BM.119(1)];

the rest-pole read: $\mathrm{p} = 0,\ w = 0 : \quad p_0 = \pm Nm; \qquad E = Nm$ [BM.34(7); machine];

the pole displacement:

in the vacuum gauge (N=1, w=0) : $\quad R = \dfrac{1}{p_0^2 - m^2 - \mathrm{p}^2}$:

$\boxed{m^2 \text{ is exactly the pole displacement in } p_0^2 \text{ against the massless pole}}$

[D2.5 vocabulary; machine: the cleared identity];

(2) the form uniqueness:

among rational forms $R' = \left(\alpha\,(p_0 - N^i p_i)^2 + \beta\,N^2(m^2 + p^2) + \delta\right)^{-1}$ over $\mathbb{Q}(\alpha, \beta, \delta)$:

the five conditions

(i) the pole locus is BM.34(4)'s mass shell;

(ii) the vacuum-gauge reduction $R' \to \left(p_0^2 - m^2 - p^2\right)^{-1}$;

(iii) T-evenness under $(N^i, p_i) \mapsto (-N^i, -p_i)$ [BM.34(4)];

(iv) hermitian sign-freeness off the poles [FT.26(1) pattern];

(v) simple poles (minimality, no squared denominators);

$\boxed{\text{force } (\alpha, \beta, \delta) = (1, -1, 0) \text{ exactly}}$

[machine: the exact linear solve over $\mathbb{Q}$ on rational ADM witnesses; (i) $\implies \alpha = -\beta,\ \delta = 0$; (ii) $\implies \alpha = 1$];

the declared content is exactly Def-BM.20(1), (3)'s two identifications:

every other row is forced;

(3) the on-shell measure (the phase space from the resolvent and the chart measure):

$$\delta\big((p_0 - N^i p_i)^2 - N^2 E_{\mathrm{p}}^2\big) = \frac{\delta\big(p_0 - N^i p_i + N E_{\mathrm{p}}\big) + \delta\big(p_0 - N^i p_i - N E_{\mathrm{p}}\big)}{2N\,E_{\mathrm{p}}},$$

$$E_{\mathrm{p}} := \sqrt{m^2 + \gamma^{ij} p_i p_j}$$

[math: the delta composition rule; machine: the Jacobian coefficient at five rational witnesses];

$$\boxed{d\mu_{\text{shell}} \;=\; \frac{d^3p}{2\,N\sqrt{m^2 + \gamma^{ij}p_i p_j}} \;=\; \frac{d^3p}{2E}}$$

[the positive branch; Def-BM.20(3)'s $(2\pi)^3$ carried with it];

the vacuum-gauge reduction:

$$N = 1,\ \gamma = \delta: \quad d\mu_{\text{shell}} = \frac{d^3p}{2\sqrt{m^2 + p^2}}$$

[machine];

$$[d\mu_{\text{shell}}] = \mathfrak{q}^{+1}$$

[CT2.1; machine: the exponent census];

(4) the two-body phase space and the fluxes (CM frame, zero shift):

$$z = E_1 + E_2;$$

$$E_i = N\sqrt{m_i^2 + p^2};$$

$$p_* = \frac{1}{2N}\sqrt{z^2 - 4N^2m^2} \ \text{ at } m_1 = m_2 = m$$

[machine: the exact solve];

$$\boxed{\Phi_2(z;m,N) \;=\; \frac{p_*}{4\pi N^2\, z} \;=\; \frac{\sqrt{z^2-4N^2m^2}}{8\pi N^3\, z}}$$

[math: the delta reduction of $\int d\mu_1 d\mu_2$ with $(2\pi)^4\delta^4$ and Def-BM.20(3)'s convention; machine: the coefficient algebra at five rational witnesses];

the massless final states:

$$\Phi_2^{(\gamma\gamma)} = \frac{1}{8\pi N^3}$$

[machine];

the fluxes:

$$F_{\gamma\gamma} = \frac{2z^2}{N^2} \;\text{(two massless beams)},$$

$$F_{\text{pair}} = \frac{2z\sqrt{z^2-4N^2m^2}}{N^2} \;\text{(two massive beams)}$$

[machine: $p_1{\cdot}p_2 = -\dfrac{z^2}{2N^2}$ massless and $m^2 - \dfrac{z^2}{2N^2}$ massive at zero shift; $F = 4\sqrt{(p_1{\cdot}p_2)^2 - m_1^2 m_2^2}$];

the formed domain:

$$\Phi_2 \text{ formed} \iff z \geq 2Nm;$$

$$\Phi_2 = 0 \text{ exactly at threshold};$$

$$[\Phi_2] = \mathfrak{q}^0,$$

$$[F] = \mathfrak{q}^{+1}$$

[machine: the exponent census];

(5) the two resolvents, two species:

the chart resolvent $R(z;\mathrm{p})$ [(1)] and the register's zig-zag resolvent $\dfrac{1}{1-w^2x_+x_-}$ [BM.121(2); BM.23(7)] are ¬identified :

the first is meromorphic in the chart energy with poles at the dispersion branches;

the second is a formal even series in the ledger weight with its pole at $|w|\geq 2$ [BM.121(5)];

they meet only through the threshold indicator $\chi_{\{NE\geq\mathcal{T}_0\}}$

[Def-BM.17(3); BM.122(3); the currency separation stands];

(6) the junction domain:

the ℵ-reads at the sealed junctions where the boundary-channel states couple run in the E2 domain [A-BM.10, issued in BM.139(2); forward pointer];

(7) the closure row:

the momentum-space Σ-resolvent is constructed

[Def-BM.20; machine: the discharge census against BM.6(6)'s three requirements — plane-wave boundary-channel states (1), $Z(\mathcal{R},\Sigma;k^2)$-type two-point read (2), m^2 as the pole displacement (1)];

what it unblocks:

the pair amplitude and the annihilation cross-section [BM.142];

what stays barred:

¬an S-matrix element beyond the formed amplitudes, ¬a measured cross-section [P5].

Req. D2.5, D2.9 (Parts 1–2); CT2.1, CT2.6 (Part 2); P5 (Parts 1–2); FT.26(1), FT.79(5) (Part 3); Def-BM.10, Def-BM.17(3), Def-BM.20, BM.6(6), BM.10(1), BM.23(7), BM.34(1), BM.34(4), BM.34(7), BM.119(1), BM.121(2), BM.121(5), BM.122(3), BM.132(2) (this part); A-BM.10, A-BM.12 (§II.4, logged before use; forward pointer for A-BM.10's issuance row); the delta-composition rule, the exact linear solve on rational witnesses, the two-body phase-space integral

in the declared convention, Laurent exponent censuses [math; machine: `checks/verify_H_bm_closing_tier.py` §C].

Pf.

(1) : BM.34(4)'s master identity re-derived entrywise on three rational ADM witnesses ($\Omega^2, \varkappa, \beta, w_i, q_{ij}$ rational, q off-diagonal);

the branch solve and the rest substitution [machine];

the vacuum gauge: $N = 1,\ w = 0,\ \gamma = \delta \implies R = (p_0^2 - m^2 - p^2)^{-1}$ [machine]

(2) : the conditions as linear constraints on (α, β, δ) :

(i) the zero set coincides with the mass shell $\implies \alpha(p_0 - N^i p_i)^2 + \beta N^2(m^2 + p^2) + \delta$ is proportional to $(p_0 - N^i p_i)^2 - N^2(m^2 + p^2) \implies \delta$

$$= 0,$$

$$\beta \qquad = -\alpha;$$

(ii) fixes $\alpha = 1$ [machine: the exact solve over $\mathbb{Q}$];

(iii), (iv), (v) are then automatic [machine: the substitution rows]

$$(3) : \ \delta(g(p_0)) = \sum_{\text{roots}} \delta(p_0 - r_i)/|g'(r_i)|,$$

$g'(p_0) = 2(p_0 - N^i p_i) = \pm 2N E_{\mathrm{p}}$ at the roots [math];

machine: the coefficient at $(N, \mathrm{p}, m) \in \{(1, \cdot, 1), (2, \cdot, 1), (1, \cdot, 2), (3, \cdot, 2), (2, \cdot, 3)\}$ rational grids;

the exponent census [CT2.1 : $[d^3p] = \mathfrak{q}^{+3/2},\ [E] = \mathfrak{q}^{+1/2}$]

$$(4) : \int \frac{d^3p}{(2\pi)^3} \frac{(2\pi)^4 \delta(z - E_1 - E_2)}{4E_1 E_2}\bigg|_{\mathrm{p}_2 = -\mathrm{p}_1}$$

$$= \frac{1}{4(2\pi)^2} \int d\Omega$$

$$\frac{p^2\,dp}{E_1E_2}\delta(z-E_1-E_2)$$

$$=\frac{4\pi}{16\pi^2}\cdot\frac{p_*}{N^2z}\ \text{[machine: the Jacobian } \tfrac{d(E_1+E_2)}{dp}=\tfrac{2N^2p_*}{z}\ \text{at five rational witnesses]};$$

the fluxes from $p_1\cdot p_2$ at zero shift [machine]

(5),

(6),

(7) : the species census, the domain row, and the discharge census against BM.6(6)'s specification text [machine: the document pins]. ■

Phys. The resolvent's declared surface is two naming choices; everything else is forced. Its poles are exactly the derived dispersion's two branches, its rest pole reads the composite's mass as the displacement against the massless pole, and among all rational forms of its degree the five derived conditions pin it uniquely. The phase space falls out of the same mass shell: the on-shell measure is the delta reduction of the chart measure, the two-body phase space and both fluxes are exact functions of (z,m,N), formed precisely above threshold. The chart resolvent and the register's zig-zag resolvent stay separate species that meet only at the threshold indicator, and the item the corpus carried since the fermion era — the momentum-space Σ-resolvent — closes as constructed.

Theorem 4.114 (BM.135 — The multi-chain additive charge census: the baryon and meson rows, the stretched states, and the gauged read). Thm.

(1) the read, re-certified:

$$Q(c_1,\dots,c_k)=\sum_{i=1}^{k}q_{\text{core}}(c_i),$$

$$q_{\text{core}}(c)=\tfrac{1}{2}\big(\sigma_{s(c)}-\sigma_{t(c)}\big)\in\{-1,0,+1\}$$

[Def-BM.8(1); FT.40; EM.9; D3.4];

the three specification clauses stand discharged:

$$Q|_{\text{single chain}}=q_{\text{core}};$$

$$Q(\mathcal{X}\sqcup\mathcal{Y})=Q(\mathcal{X})+Q(\mathcal{Y})\ \text{ on disjoint sealings};$$

Q defined only on sealed supports $(\mathcal{C} = 1)$

[Def-BM.8(2)–(4); BM.6(1); FT.19];

(2) the baryon charge (the ε-sealed 3-juncture):

$$Q_B = q_1 + q_2 + q_3 \in \{-3, \ldots, +3\},$$

$$W_3(Q) = \binom{6}{Q+3} = (1,\ 6,\ 15,\ 20,\ 15,\ 6,\ 1)$$

[Def-BM.8(5), (6); BM.28(2); BM.114(1); machine: the exhaustive 4^3=64 end-sign census re-run];

the named-class rows:

the neutral class: $Q_B = 0$,

$$E_{\text{int}}^{(n)} = 6$$

[BM.17; Def-BM.3(2); Census FT.35(2)];

the unit class: $Q_B = +1$

[BM.26; Def-BM.3(3)];

the CPT image: $Q_B = -1$

[BM.27];

the transport profile of the neutral class:

$$m_q = (2, 1, 1)$$

[Def-BM.13(1); FT.106(3a)];

$$\frac{W(0)}{W(\pm 1)} = \frac{20}{15} = \frac{4}{3}$$

[BM.28(2); machine];

(3) the meson charge (the δ-sealed 2-juncture):

$$Q_M = q_1 + q_2 \in \{-2, \ldots, +2\},$$

$$W_2(Q) = \binom{4}{Q+2} = (1,\ 4,\ 6,\ 4,\ 1)$$

[BM.110(3); BM.114(1); machine: the 4^2 census re-run];

the CP-pair rows:

$$Q_M\big(c, c^{\mathrm{CP}}\big) = q + (-q) = 0 \ \text{ for all three } q$$

[Def-BM.15(3); machine: exhaustive];

the CP-pairs saturate the neutral class:

$$1 + 4 + 1 = 6 = W_2(0),$$

$$\#\{Q{=}0 \text{ tuples outside the CP-pair forms}\} = 0$$

[BM.110(3); machine re-run];

(4) the stretched and unit rows (both charge-read questions adjudicated):

$\#\lvert Q\rvert = 2$ states: 12 at $k = 3$, 2 at $k = 2$; $\#\lvert Q\rvert = 3$ states: 2 at $k = 3$

[$W_3(\pm 2) = 6,\ W_3(\pm 3) = 1,\ W_2(\pm 2) = 1$; machine: exhaustive];

the multi-chain charge read is derived:

$|Q| = 2$ and $|Q| = 3$ observable integer states exist on sealed supports of the additive read

[the requirement is discharged; FT.41(a) : $N_{\text{charge}} = 2$ end registers per chain re-cited];

the ternary-read row stands re-certified:

the read is integer-valued at every k : $\frac{1}{3},\ \frac{2}{3}$ are structurally unreachable

[Def-BM.8(5); BM.28(7); machine: the sum-closure scan over $\{-1,0,1\}^k,\ k=1..4$];

(5) the gauged read:

$$\partial_\mu F^{\mu\nu} = e\, J^\nu_Q,$$

the A-coefficient $e\,Q$,

$$e = \frac{gg'}{\sqrt{g^2+g'^2}}$$

[Def-BM.12(1)–(3); EW.35; EM.15];

the running entry:

$e = e(r,E)$ with all three factors derived

[BM.88(4); BM.132(2); BM.133(6)];

Def-BM.8(8)'s ¬constructed row (the coupling of Q to a gauge field) is discharged by Def-BM.12(2),

(3) and re-scoped, ¬retracted;

(6) additivity at general n and the grammar row:

a sealed support partitions into blocks of size 2 and 3

[CT3.17a(1)];

$$Q(\Gamma) = \sum_{\text{blocks}} Q(\text{block})$$

[machine: exhaustive at $n \leq 4$ on ternary alphabets];

the δ/ε-mixing from N

$= 4$ [BM.31(2); BM.33] governs which supports are sealed, ¬the additivity of the read on sealed blocks;

(7) the closure row:

the additive multi-chain charge read exists, is constructed, and is gauged on the chart side

[Def-BM.8; (1)–(6); machine: the three-clause discharge census against BM.6(1)'s specification];

the ternary-read row is complete: no residue at any chain count;

¬a species name attaches to any charge class beyond the identification layer [Def-BM.3(7)];

¬a measured charge enters [P5].

Req. D2.14, D3.4, D3.9 (Parts 1–2); CT3.17a, CT3.22, EM.9, EM.11, EM.15, EW.35 (Part 2); P2, P5 (Parts 1–2); FT.19, FT.40, FT.41, FT.106(3) (Part 3); Def-BM.3, Def-BM.8, Def-BM.12, Def-BM.13, Def-BM.15, BM.6(1), BM.17, BM.26, BM.27, BM.28(2), BM.28(7), BM.31(2), BM.33, BM.88(4), BM.110(3), BM.114(1), BM.132(2), BM.133(6) (this part); Census FT.35(2) (row-wise); exhaustive ternary enumeration, generating-function convolution, sum-closure scans [math; machine: `checks/verify_H_bm_closing_tier.py` §D].

Pf.

(1)–(3) : Def-BM.8's construction re-certified clause by clause and the two censuses re-run exhaustively [machine];

the named-class rows consumed from their own tags row-wise [Def-BM.3(7)'s discipline]

(4) : the counts from W_3, W_2 directly [machine];

the unreachability scan: $\sum_i q_i \in \mathbb{Z}\ \forall q_i \in \{-1, 0, 1\}$ at $k \leq 4$ [machine]

(5) : Def-BM.12(1)–(3)'s rows re-cited with the running coupling substituted [BM.133(6)];

(6) : the block sum on ternary alphabets, exhaustive at $n \leq 4$ [machine];

(7) : the discharge census. ■

Phys. The additive charge read the era opened as its first construction spec is re-certified, censused and gauged. Baryons carry seven integer charge classes with the frozen weight row and the neutral-to-unit ratio four thirds; mesons carry five, and their CP pairs saturate the neutral class exactly. The stretched states settle two ledger rows at once: observable charge two exists — twelve states at three chains, two at two — so the integer charge read is derived, while the fractional alphabet stays structurally unreachable, so the ternary read's consistent no-go stands. The chart-side gauge coupling turns the read into a current with a fully derived running coefficient, and the question the corpus carried since the fermion era — whether an additive multi-chain charge read exists — closes as constructed.

Theorem 4.115 (BM.136 — The S_3 action is the colour 3-vertex: the equivariance theorem, the exchange structure at every n, and the statistics per class). Thm.

(1) the colour 3-vertex and its equivariance: $B = \varepsilon_{ijk}\, c_1^i c_2^j c_3^k$ [D3.9; Def-BM.1; CT3.15];

$$\boxed{\varepsilon_{\sigma(i)\sigma(j)\sigma(k)} = \operatorname{sgn}(\sigma)\, \varepsilon_{ijk} \;\; \forall \sigma \in S_3}$$

[machine: exhaustive over the 6×27 index pairs];

$$\dim \wedge^3(\mathbb{C}^3) = \binom{3}{3} = 1 :$$

the ε-seal is the unique invariant line, and S_3 acts on it by the sign representation

[BM.128(5); machine: the antisymmetrizer rank re-run];

colour is ¬observable by construction

[CT3.22];

the colour generators act as $\mathfrak{su}(3)$ on the slots

[CT3.8; FT.82(3)];

(2) the intertwiner theorem (the S_3 action is literally the colour 3-vertex):

the sealed baryon is the contraction $\langle\varepsilon|$

: $(\mathbb{C}^3)^{\otimes 3}\otimes\mathcal{H}_3\to\mathbb{C}$ with $\hat{P}_\sigma$ the slot permutation [BM.128(2)] :

$\boxed{\left(\hat{P}_\sigma\otimes D_{\text{col}}(\sigma)\right)|\varepsilon\rangle\otimes|\psi\rangle \text{ is } S_3\text{-invariant exactly when the slot read compensates the colour sign: } \hat{P}_\sigma\big|_{\text{sealed}}=\text{sgn}(\sigma)}$

[machine: the explicit 27-dimensional colour tensor $|\varepsilon\rangle=\sum_{abc}\varepsilon_{abc}|abc\rangle$ and all six $\sigma\in S_3$; $D_{\text{col}}(\sigma)|\varepsilon\rangle=\text{sgn}(\sigma)|\varepsilon\rangle$ entrywise];

the reading:

the sealing grammar's antisymmetry and the slot space's permutation action are the two halves of one equivariant map — the S_3 action on the 3-chain state space is the colour 3-vertex read on slots;

(3) the exchange structure at every n (residue (R7′) closed):

$\boxed{\mathcal{H}_n:=\bigotimes_{a=1}^{n}\Big(\mathcal{H}_\omega(c_a)\otimes\Psi(c_a)\Big),\qquad \dim\mathcal{H}_n=2^n\prod_{a=1}^{n}\big(|k_a|+1\big)}$ [BM.128(1) extended uniformly];

$$\hat{P}_\sigma\hat{P}_\tau=\hat{P}_{\sigma\tau}$$

$$\forall\sigma,\tau\in S_n$$

[BM.128(2)'s convention, uniform in n; machine: all 576 ordered pairs at $n=4,\ d=2,$ entrywise over $\mathbb{Q}$];

$\boxed{\chi(\sigma)=\text{Tr}\,\hat{P}_\sigma\big|_{(\mathbb{C}^d)^{\otimes n}}=d^{\,c(\sigma)}\ \text{ for every } n}$

[BM.128(3) extended; machine: all 24 permutations at $(n,d)=(4,2)$ and $(4,3)$: the five cycle types give $(16,8,4,4,2)$ at $d=2$ and $(81,27,9,9,3)$ at $d=3$];

$$\varepsilon(\Gamma)=\prod_{a=1}^{n}\varepsilon(c_a)$$

at an open connector

[BM.128(7) extended; machine: all 625 quadruples in $\{0,\dots,4\}^4$: 312 fermionic, 313 bosonic];

the wedge rows:

$$\dim \wedge^n(\mathbb{C}^2) = \binom{2}{n} = 0 \ \ \forall n \geq 3;$$

$$\dim \wedge^3(\mathbb{C}^3) = 1;$$

$$\dim \mathrm{Inv}\big((\mathbb{C}^3)^{\otimes 4}\big) = 0$$

[machine: the antisymmetrizer ranks $(2,4),(3,4),(2,3),(3,3)$ exact; the weight census: no zero-weight vector in $(\mathbb{C}^3)^{\otimes 4}$];

four colour slots admit no singlet: the $N \geq 4$ sealings must mix δ and ε blocks

[BM.31(2); BM.33; CT3.17a(2) : $g(4) = 10$ groupings; machine: the census of $\{2,3\}$-partitions at $N = 4$];

the grammar rescope row:

the δ/ε-mixing governs which n-chain configurations are sealed (observability [BM.1]),

¬the exchange structure, which is uniform in n;

(4) the winding assignment per class:

(a) the computed floor: the minimal materialized constituent is the digon with $k = 1$

[FT.28; FT.32(6); FT.107(1); BM.129(5)] :

$$(k_1, k_2, k_3) = (1,1,1): \qquad \varepsilon_B = (-1)^3 = -1, \qquad j_B = \frac{3}{2}$$

[BM.129(4), (5); machine];

the neutral and unit classes share the floor:

$$E_{\text{int}}$$

$= 6$ on both [BM.17; BM.26; BM.28(1)] $\Longrightarrow$ both named baryon classes are fermionic at the computed floor, derived ;

BM.129(6),

BM.130(1)'s "value per named class awaits (R3)" row is discharged at the floor locus: ε_B depends on the winding alone, and the winding at the floor is derived;

(b) the six-state classes under the declared corner convention:

$$k_{vf} = (3,3,2,3,5) \text{ on (tetra, cube, octa, dodeca, icosa)}$$

[A-BM.11 (variant B), issued in BM.139(4); FT.76(1); Def-BM.14(1), (2) : the floor map $2E \in \{12,24,60\}$ gives $k_{vf} = (3,3\,|\,2,2\,|\,3,5)$ on the six states] :

$$\varepsilon_B(\mathrm{s}) = (-1)^{k_{vf}(s_1)+k_{vf}(s_2)+k_{vf}(s_3)} :$$

$$\boxed{\#\{\text{fermionic}\} = 112, \qquad \#\{\text{bosonic}\} = 104}$$

[machine: the exhaustive 6^3=216 census];

the variant dependence is exact:

$$\text{variant A } \big(k_f = (3,2\,|\,3,3\,|\,5,3)\big) : \quad 140 \text{ fermionic, } 76 \text{ bosonic}$$

[machine] $\implies$ the census is convention-dependent and the declaration A-BM.11 pins it;

(5) the bars:

¬a flavour SU(3), ¬an SU(6) :

S_n permutes constituent slots [BM.117(3); BM.128(9); BM.129(7) re-cited];

the flavour occupancy remains (R3)-species:

which three of the six states a named class occupies is the generation-selection question [BM.60(1) : three lines derived, three selected ¬derived] — named for the Part-5 ledger, ¬a baryomesonic residue;

¬a species name on any row [Def-BM.3(7)]; ¬a measured value [P5];

(6) status:

derived: (1)–(4);

$$\boxed{\text{(R7}') \text{ is closed: } \mathcal{H}_n,\ \hat{P}_\sigma,\ \chi(\sigma) = d^{\,c(\sigma)},\ \varepsilon(\Gamma) = \prod_a \varepsilon(c_a) \text{ at every } n};$$

the S_3-action gap is closed: the action is the colour vertex's slot read;

the winding assignment per class is derived at both loci (floor and six-state);

the statistics function is complete and its values are derived per class [BM.126; BM.129(4)].

Req. A3 (Part 1); D2.14, D3.9 (Parts 1–2); CT3.8, CT3.15, CT3.17a, CT3.22 (Part 2); P5 (Parts 1–2); FT.28, FT.32(6), FT.76(1), FT.82(3), FT.107(1) (Part 3, Census); Def-BM.1, Def-BM.3(7), Def-BM.14, BM.1, BM.17, BM.26, BM.28(1), BM.31(2), BM.33, BM.60(1), BM.117(3), BM.126, BM.128, BM.129 (this part); A-BM.11 (§II.4, logged before use; forward pointer for its issuance row); permutation-module enumeration, antisymmetrizer ranks by exact Fraction elimination, the weight-zero census of $(\mathbb{C}^3)^{\otimes 4}$, exhaustive parity censuses [math; machine: `checks/verify_H_bm_closing_tier.py` §E].

Pf.

(1) : the 6×27 equivariance census and the rank-1 antisymmetrizer [machine];

(2) : $D_{\text{col}}(\sigma)|\varepsilon\rangle = \text{sgn}(\sigma)|\varepsilon\rangle$ entrywise on the explicit tensor;

the invariance of $|\varepsilon\rangle \otimes |\psi\rangle$ under the diagonal action with $\hat{P}_\sigma$ on slots [machine]

(3) : $\hat{P}_\sigma$ built as the factor permutation on $(\mathbb{C}^d)^{\otimes n}$: the product law on all 576 pairs at n

$= 4$ in BM.128(2)'s composition convention [machine: the pushforward factor action];

$\chi(\sigma) = d^{c(\sigma)}$ on all 24 permutations at $d \in \{2, 3\}$ [machine];

the antisymmetrizer ranks $(0, 0, 0, 1)$ at $((2, 4), (3, 4), (2, 3), (3, 3))$ [machine];

$$\dim \mathrm{Inv}((\mathbb{C}^3)^{\otimes 4})$$

$$= 0 : \text{ the weights of } \mathbb{C}^3 \text{ are } \{(1,0),(0,1),(-1,-1)\} : \quad a(1,0)+b(0,1)+c(-1,-1)$$

$$= (0,0) \ \wedge \ a+b+c$$

$$= 4$$

$$\Longrightarrow \ a$$

$$= b$$

$$= c$$

$$= 4/3 \notin \mathbb{Z} \text{ [machine: the exhaustive composition census]};$$

$$\text{the parity census on } 625 \text{ quadruples [machine]}$$

$$(4) : \text{ the floor row from BM.129}(5)\text{'s digon line and } \varepsilon_B = (-1)^{k_1+k_2+k_3} \text{ [BM.129}(4)\text{]};$$

the six-state censuses: $112/104$ and $140/76$ by exhaustive enumeration of the parity patterns $(4 \text{ odd},\ 2 \text{ even})$ and $(5 \text{ odd},\ 1 \text{ even})$ [machine];

$$(5), (6) : \text{ assembly. } \blacksquare$$

Phys. The permutation action on the three-chain state space is not a bookkeeping device: it is the colour 3-vertex read on slots. The epsilon tensor is the unique invariant line of three colour slots, it transforms by the sign of the permutation, and the sealed baryon is exactly the intertwiner that pairs the two actions — so the antisymmetry lives in colour, unobservable by construction, while the observable statistics lives in the winding. The exchange structure extends uniformly to every chain count with the character formula intact, and the four-slot colour census explains the grammar: no four-fold singlet exists, which is why seals mix delta and epsilon blocks from four attachments on. The winding assignment per class is derived at both loci — the digon floor gives both named baryon classes the fermion sign and spin three halves, and the six-state census under the declared vertex-figure convention gives 112 fermionic against 104 bosonic classes, a census that provably depends on the corner convention and is pinned by its declaration. What remains open is not baryomesonic: which states a named class occupies is the generation-selection question, and it leaves for the next era.

Theorem 4.116 (BM.137 — The two-chain ensemble measure, the meson's formed binding, the sector-mass pincer, and the two-chain spectrum). Thm.

(1) the general pair-specialization law:

an n-pair floor crystallizes each pair independently with the stationary weight $p = 1 - 4\varepsilon_q$

[BM.19(2); BM.94(4)'s marked specialization, extended uniformly] :

$$P(E{=}2j) = \binom{n}{j} p^j u^{n-j},$$

$$u := 1 - p = 4\varepsilon_q;$$

the materialization restriction $E \geq 2$ is derived

[BM.94(5); FT.144(4); Def-H.4];

$$\boxed{D_n := \frac{1-u^n}{1-u} = 1 + u + \cdots + u^{n-1}, \qquad w_{\text{mat}}^{(n)} = \frac{1}{D_n}, \qquad \langle E \rangle_{\text{mat}}^{(n)} = \frac{2n}{D_n} \in (2,\ 2n]}$$

[machine: the polynomial identity $(1-u)D_n = 1 - u^n$ for $n = 1..8$; the binomial moment algebra exact in $\mathbb{Q}[p]$];

$$\text{Var}_{\text{mat}}^{(n)}(E) = \frac{4n\big[(1+(n-1)p)D_n - n\big]}{D_n^2};$$

$$n = 3: \quad \frac{12\,p(1-p)(3-2p)}{D^2}$$

[BM.94(6) reproduced exactly; machine];

$$n = 1: \quad w_{\text{mat}}^{(1)} = 1:$$

a single materialized pair has no flicker: the restriction E

≥ 2 forces its crystallized member

[machine];

(2) the two-chain ensemble measure (constructed):

the meson floor $E^{cryst} = 2E_{\text{int}}(c) = 4$ is two pairs

[BM.111(1); the digon floor $E_{\text{int}}(c) = 2$ [FT.28; BM.13(3)]] :

$$\boxed{P_{\text{mat}}(E{=}2) = \frac{2u}{1+u}, \qquad P_{\text{mat}}(E{=}4) = \frac{p}{1+u}, \qquad \langle E\rangle^{(2)}_{\text{mat}} = \frac{4}{1+u} \in (2,4], \qquad w^{(2)}_{\text{mat}} = \frac{1}{1+u} \in \left(\tfrac{1}{2},\, 1\right]}$$

$$\text{Var}^{(2)}_{\text{mat}}(E) = \frac{8p(1-p)}{(2-p)^2};$$

$$\frac{\text{Var}^{(2)}}{\langle E\rangle^{(2)2}} = \frac{p(1-p)}{2} \le \frac{1}{8},$$

$$\frac{\sigma^{(2)}}{\langle E\rangle^{(2)}} \le \frac{1}{2\sqrt{2}} < \frac{1}{2}$$

[machine: exact rational functions; $\frac{1}{8} - \frac{p(1-p)}{2} = \frac{(2p-1)^2}{8} \ge 0$];

the two-bit floor survives the averaging:

$$E \ge 2 \text{ on the whole materialized support } \implies \langle m_{\text{hid}}\rangle^{(2)}_{\text{mat}} > \frac{2\ln 2}{\mathfrak{h}}$$

[BM.70(2); BM.94(8) pattern; machine];

the no-flicker collapse:

$$\varepsilon_q = 0 \implies u = 0 \implies P_{\text{mat}}(E{=}4) = 1,$$

$$w^{(2)}_{\text{mat}} = 1$$

[BM.19(3); Def-BM.5(5); machine];

(3) the meson's ensemble binding (formed, nonzero — the (R8) row):

$$\mathcal{B}^{(n)}_{\text{ens}} := \langle m_{\text{hid}}\rangle^{(n)}_{\text{mat}} - \sum_{a=1}^{n} \langle m_{\text{hid}}\rangle^{(1)}_{\text{mat}} = \left(\frac{2n}{D_n} - 2n\right)\frac{\ln 2}{\mathfrak{h}}$$

[FT.44(4); (1)];

$$\boxed{\mathcal{B}_{\text{ens}}^{(n)} \;=\; -\frac{2n\,u\,D_{n-1}}{D_n}\cdot\frac{\ln 2}{\mathfrak{h}}}$$

[$D_n - 1 = uD_{n-1}$; machine: the polynomial identity for $n = 2, 3, 4$];

$$n = 2 :$$

$$\boxed{\mathcal{B}_{\text{ens}}^{(2)} = -\frac{4u}{1+u}\cdot\frac{\ln 2}{\mathfrak{h}} = -\frac{16\,\varepsilon_q \ln 2}{(1+4\varepsilon_q)\,\mathfrak{h}}},$$

$$\mathcal{B}_{\text{ens}}^{(2)} < 0 \;\; \forall \varepsilon_q > 0,$$

$$\mathcal{B}_{\text{ens}}^{(2)} = 0 \iff \varepsilon_q = 0,$$

$$-\frac{2\ln 2}{\mathfrak{h}} < \mathcal{B}_{\text{ens}}^{(2)} \leq 0$$

[machine: the sign and range census on $u \in [0, 1)$];

the binding is the flicker's, not the count's, at $n = 2$ as at $n = 3$

[BM.21(3) pattern];

the two vanishing routes stand untouched: $\mathcal{B}_{\text{info}}^{\text{meson}} = \mathcal{B}_{\text{pole}}^{\text{meson}} = 0$

[BM.111(1), (2) re-cited, ¬retracted];

(4) the sector-mass pincer at budget $B = 2$ (the 2-chain sector masses, computed):

$\mathcal{G}_2$ carries one chain-bearing skeleton, the digon K_2

[Def-H.32(1); FT.105 : the connected simple graphs on ≤ 2 nodes] :

the pincer:

a closed chain requires $|R| \geq 2$

[Def-H.4 : $\delta h(c) \geq 2$]

and the budget gives $|R| \leq 2E(K_2) = 2$

[Def-H.32(1)] :

$$\implies |R| = 2E \implies n_{\mathcal{V}} = 2E - |R| = 0 \implies \Delta S = \tfrac{3}{2} n_{\mathcal{V}} = 0 \qquad \text{[Def-H.32(6); FT.107(2)]};$$

$m_2^2(q,S) = 0$ on every formed sector, and $Z_2^{\text{norm}}(q,S) \neq 0$ only at $S = 1$ with $Z_2^{\text{norm}} = (4,\ 2,\ 2)$ at $q = (0, +1, -1)$

[machine: the exhaustive enumeration of $\mathcal{G}_2$ — 4 patterns on K_2, the σ-resolved instance census, the closed form $m_q 2^{|R|-2}$ of FT.106(3) cross-checked];

the structural reason:

at budget two, chain-bearing content is fully crystallized: the sector ensemble has no flicker room;

the sector-species binding is $0 - 0 - 0 = 0$ exactly;

the two ensemble reads are separated species:

the sector/defect read gives 0; the informational read gives (3)'s strict negative;

¬a derived equation relates them — that is (R1)'s bridge

[BM.37(2); BM.98; A-BM.9];

(5) the two-chain spectrum (the resonance rows at $n = 2$):

$$W_2(Q) = (1, 4, 6, 4, 1),$$

$$\sum_Q W_2 = 16 = 4^2$$

[BM.114(1); machine re-run];

$$\lambda^{(2)}(s_1, s_2) = D_S(s_1) + D_S(s_2),$$

$$D_S \in \{2,\ 4,\ 2-\sqrt{2},\ 1,\ 2-\mathfrak{u},\ 2-\sqrt{3}\},$$

$$\mathfrak{u} = 2\cos\tfrac{\pi}{10}$$

[Def-BM.14(3); BM.111(2); BM.114(3)] :

the 21 multiset orbits give 21 distinct eigenvalues with ¬a collision

[machine: the exact equality census in $\mathbb{M}$; contrast: one collision at $n = 3$ [BM.114(5)]];

$$\lambda^{(2)}_{\text{min}} = 4 - 2\mathfrak{u} = 2\big(2 - 2\cos\tfrac{\pi}{10}\big),$$

$$\lambda^{(2)}_{\text{max}} = 8 = 4n$$

[BM.117(4) re-certified at $n = 2$; machine: the exact extrema in $\mathbb{M}$];

$$m^2_{\text{pole}}(\mathcal{R}) = \hat{\mu}'^2\lambda^{(2)} \in \big[(4-2\mathfrak{u})\,\hat{\mu}'^2,\ 8\hat{\mu}'^2\big];$$

$$m^{meson}_{\text{hid}} = \frac{2E_{\text{int}}(c)\ln 2}{\mathfrak{h}};$$

$$\big\langle m_{\text{hid}}\big\rangle^{(2)}_{\text{mat}} = \frac{4\ln 2}{(1+u)\,\mathfrak{h}} \text{ at the digon floor}$$

[Def-BM.17(2); (2); machine];

$\varepsilon_M = +1$ unconditionally, $\quad j_M = k \in \mathbb{Z} \quad$ [BM.129(1), (2) re-cited];

(6) the octet adjudication stands, and the spectrum's shape is the derived one:

8 occurs in no grouping family at any formed level

[BM.117(3); machine: the scan re-run over W_n, $g(N)$, $\binom{6+n-1}{n}$, $\binom{6}{n}$, $n \le 3$, $N \le 7$];

the meson spectrum is the eigenvalue ladder (5) with its charge census, its statistics and its formed binding — ¬an octet, ¬a nonet,

¬a flavour multiplet;

the naming row:

species names for meson classes remain confined to the identification layer [Def-BM.3(7); Def-BM.6(7)];

¬a measured mass, width or mixing enters [P5];

(7) status:

derived: (1)–(6);

(R8) is closed: the 2-chain sector masses are computed (identically zero, with the pincer), the two-chain ensemble measure is constructed, and the meson's ensemble binding is formed and strictly negative at every nonzero threshold

[BM.111(3)'s ¬formed row discharged];

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5].

Req. A4 (Part 1); D0.4, D0.8 (Part 1); CT0.8, CT1.29 (Part 1); CT2.8 (Part 2); P1, P5 (Parts 1–2); Def-H.4, Def-H.32, FT.28, FT.32(6), FT.44(4), FT.105, FT.106(3), FT.107(1), FT.107(2), FT.144(4) (Part 3); Def-BM.5, Def-BM.14(3), Def-BM.17(2), BM.13(3), BM.19(2), BM.19(3), BM.21(3), BM.37(2), BM.70(2), BM.94(4), BM.94(5), BM.94(6), BM.94(8), BM.98, BM.111, BM.114(1), BM.114(3), BM.114(5), BM.117(3), BM.117(4), BM.129(1), BM.129(2) (this part); Census FT.35(2) (row-wise); the A-BM.9 declaration; binomial moment algebra over $\mathbb{Q}[p]$, the degree-16 field $\mathbb{M}$ with its exact equality census, exhaustive configuration enumeration at budget two, polynomial identities [math; machine: `checks/verify_H_bm_closing_tier.py` §F].

Pf.

(1) : $P(E{=}2j)$ is the n-pair binomial [BM.94(4)'s specialization at n];

$$\text{conditioning on } j \geq 1 : \;\; 1 - u^n = pD_n;$$

$$\langle E \rangle_{\text{mat}}^{(n)} = 2 \cdot \frac{np}{1-u^n} = \frac{2n}{D_n} \text{ [machine: exact in } \mathbb{Q}[p]];$$

$$\langle E^2 \rangle_{\text{mat}}^{(n)} = 4 \cdot \frac{np(1+(n-1)p)}{1-u^n} = \frac{4n(1+(n-1)p)}{pD_n} \text{ [machine]};$$

$$\text{Var} = \langle E^2 \rangle - \langle E \rangle^2 \text{ [machine]};$$

$n = 3$ row against BM.94(6)'s frozen expression [machine: identical rational functions];

$$n = 1 : \; D_1 = 1, \; P(E{=}2) = 1 \text{ [machine]}$$

$$(2) : \; n = 2 \text{ substitution: } D_2 = 1 + u = 2 - p;$$

$$P_{\text{mat}}(2) = \frac{2pu}{pD_2} = \frac{2u}{1+u}, \quad P_{\text{mat}}(4) = \frac{p^2}{pD_2} = \frac{p}{1+u} \text{ [machine]};$$

the variance and the relative-width SOS certificate [machine]

$$(3) : \; \mathcal{B}^{(n)}$$

$$= \big(\frac{2n}{D_n} - 2n\big)\frac{\ln 2}{\mathfrak{h}}$$

$$= -2n\frac{D_n - 1}{D_n}\frac{\ln 2}{\mathfrak{h}}$$

$$= -\frac{2nuD_{n-1}}{D_n}\frac{\ln 2}{\mathfrak{h}} \; [D_n - 1 = uD_{n-1}; \text{ machine}];$$

the sign, zero and range rows on $u \in [0, 1)$ [machine: rational grid and the factor census]

$$(4) : \; \mathcal{G}_2 = \{\bullet, \; K_2\} \text{ [FT.105's census at } V \leq 2];$$

the 4 patterns of K_2 enumerated: $|R| \in \{0, 1, 1, 2\}$;

closed chains exist only at $|R| = 2$ (the digon, $k = 1, \; S = 1, \; 2$ bases) [machine];

$$\Delta S = \tfrac{3}{2}(2-2) = 0;$$

$$Z_2^{\text{norm}}(q,1)$$

$$= 2 \text{ instances} \times m_q 2^0$$

$$= (4,2,2) \text{ [FT.106(3)'s closed form; machine: the } \sigma\text{-resolved census agrees]};$$

$$S \neq 1 : \ \neg\text{instances} \implies Z = 0$$

$$(5) : \text{ the } 21 \text{ sums computed in } \mathbb{M} \text{ with the basis-equality census [machine]};$$

the extrema by the exact order of $\mathbb{M}$'s positive cone on the six D_S [machine: dyadic interval embedding];

$$(6) : \text{ the grouping scan re-run [machine]};$$

$$(7) : \text{ assembly. } \blacksquare$$

Phys. The meson gets its ensemble. The flicker measure generalizes from the marked three-pair specialization to a law at every n: the materialized weight is one over the geometric sum $1 + u + \cdots + u^{n-1}$, the mean count is $2n$ over that sum, and at two chains the measure is exact — a mean between two and four, a relative width bounded by one over two root two, and the two-bit floor intact. The binding that residue (R8) named is formed and nonzero: strictly negative at every nonzero threshold, zero exactly at the sharp point, the same flicker-not-count structure the baryon showed. The sector route tells the complementary story: at budget two a chain-bearing configuration has no virtual room at all, so the two-chain sector masses vanish identically — a pincer, not a coincidence — and the two ensemble reads stay separated currencies pending the bridge slot. The spectrum completes the row: five charge classes with weights one-four-six-four-one, twenty-one pole eigenvalues with no collision at all, bounded by four-n, bosonic unconditionally. The octet remains unformed and the names remain where the identification layer keeps them.

Theorem 4.117 (BM.138 — The seed-tensoring mechanism: the symmetrizer as a finite-group twirl, the transport and composition closure, and the two residues adjudicated). Thm.

$$(1) \quad \text{the mechanism, derived:}$$

$$\mathcal{H}_{\mathsf{v}} \cong \mathbb{C}^2 \ \text{ the transported seed}$$

$$[\text{EW.32; A2 : the channel capacity is one qubit}];$$

$$\boxed{\Psi(c) \;=\; \mathrm{Sym}^{|k(c)|}\big(\mathcal{H}_{\mathsf{v}}\big) \;=\; \Pi_{|k|}\Big(\mathcal{H}_{\mathsf{v}}^{\otimes|k|}\Big), \qquad \Pi_k := \frac{1}{k!}\sum_{\sigma\in S_k}\hat{P}_\sigma}$$

[Def-H.38; BM.128(2)'s permutation operators on identical factors];

$$\Pi_k^2 = \Pi_k,$$

$$\mathrm{rk}\,\Pi_k = k+1 = \dim \mathrm{Sym}^k(\mathbb{C}^2),$$

$$\Pi_k \hat{P}_\tau = \hat{P}_\tau \Pi_k = \Pi_k \;\; \forall \tau \in S_k$$

[machine: idempotence, exact rank over $\mathbb{Q}$, and the commutation for $k = 1..6$];

the symmetrizer is the Haar twirl of the finite group S_k

[BM.131(6); math: the averaging over a finite group is its Haar measure];

the seed-tensoring read:

the carrier is the k-fold tensor power of the seed, projected onto the invariants of the group average — tensoring with itself, then twirling;

(2) the diagonal transport: $\mathcal{U}_c^{\otimes k}\big|_{\mathrm{Sym}^k}$ [Def-H.38; EW.33 : the wrapping transport];

$$\big[\mathcal{U}^{\otimes k},\; \Pi_k\big] = 0$$

[machine: the infinitesimal rows $\big[\sum_a W_a^{(i)}, \Pi_k\big] = 0$ on the $\mathfrak{su}(2)$ generator triple, exact over $\mathbb{Q}(i),\; k \le 6$];

the weight ladder from the twirl:

$$\mathrm{Tr}\Big(\Pi_k\, D(z)^{\otimes k}\Big) = \sum_{m\in\{k,k-2,\ldots,-k\}} z^m,$$

$$D(z) = \mathrm{diag}(z, z^{-1})$$

[FT.60(4) re-derived; machine: the Laurent identity at five rational z for $k \le 8$];

the Casimir row $\frac{k}{2}\left(\frac{k}{2}+1\right)$ [FT.60(3) re-cited]; $j = \frac{|k|}{2}$ [Def-H.38; FT.55(2)];

(3) the composition closure (the junctions tensor by the same mechanism):

$\mathrm{Sym}^{k_1} \otimes \mathrm{Sym}^{k_2} \supseteq \mathrm{Sym}^{k_1+k_2}$ as the maximal-weight line on the cone

[FT.60(6); machine: the top weight $z^{k_1+k_2}$ occurs with multiplicity 1 in the character product, Laurent-exact];

$\Psi(\Gamma) = \mathrm{Sym}^{k_1+k_2+d_{12}} \subseteq \mathrm{Sym}^{k_1} \otimes \mathrm{Sym}^{k_2} \otimes \mathrm{Sym}^{d_{12}}$ [BM.2(2); FT.34(3) : $k(\Gamma) = k_1 + k_2 + d_{12}$];

the connector is a carrier too: $\mathrm{Sym}^{d_{12}}(\mathcal{H}_v)$ is built by the same seed-tensoring, so every junction composition closes inside the mechanism

(4) the defect-ladder coherence:

the weights $m \in \{k, k-2, \ldots, -k\}$ of (2) are the ladder $\Delta(m)$

$= 4\sin^2 \frac{m\Theta_R}{4}$ reads on the tensorized carrier's own weight basis

[Def-BM.19(3); BM.124(3); Def-H.37(2)] :

the mechanism is what makes the carrier and the defect read the same multiplet — ¬a new coupling, ¬a proportionality Θ_R

$= 2\pi k$ [BM.124(5)'s dimensional bar re-cited];

(5) residue (i) adjudicated — the coupling datum:

$\mathrm{copies}(c) = k(c)$ is Def-H.38's constitutive clause;

the derivation route is blocked by a frozen minimality theorem:

the uniform-minimal triangle's winding 3 has no three materialized winding-1 subloops, and admissible decompositions are basis-relative

[FT.60(7a)–(7b)];

the mechanism is derived and the coupling integer stays a priced interface: what remained was the construction half, and it is complete;

(6) residue (ii) adjudicated — the $k \geq 3$ carrier inventory:

$\mathrm{Sym}^{k\geq 3}$ $(\dim \geq 4)$ has no registered aspect block [CT0.12 : blocks $(3, 3, 5, 3, 1, 1)$];

the adjudication:

the carrier is a chain-layer object, ¬a force aspect: every clause of this part consumes Ψ through $(\dim,$ weights, $\varepsilon,\ j,$ Casimir, $\chi)$ alone

[machine: the inventory scan of Def-H.38's consumption sites in Part 4];

¬an aspect claim at $k \geq 3$ is made or needed [Def-H.38 Scope re-cited];

the inventory-extension question is void for this era;

(7) status:

derived: (1)–(4);

adjudicated: $(5), (6)$;

the seed-tensoring mechanism is derived;

¬a declaration,

¬a P4 addition,

¬a measured value [P5].

Req. A2 (Part 1); D1.4 (Part 1); CT0.12 (Part 1); P5 (Parts 1–2); EW.32, EW.33 (Part 2); Def-H.37(2), Def-H.38, FT.34(3), FT.55(2), FT.60(3), FT.60(4), FT.60(6), FT.60(7) (Part 3); Def-BM.19(3), BM.2(2), BM.124(3), BM.124(5), BM.128(2), BM.131(6) (this part); exact rank by Fraction elimination, Laurent-polynomial character identities, the $\mathfrak{su}(2)$ infinitesimal commutation over $\mathbb{Q}(i)$, finite-group averaging [math; machine: `checks/verify_H_bm_closing_tier.py` §G].

Pf.

$$(1):\ \Pi_k^2$$

$$= \Pi_k \text{ from } \hat{P}_\sigma \hat{P}_\tau$$

$= \hat{P}_{\sigma\tau}$ and the group-average rearrangement [BM.128(2) transported to identical factors; machine];

$$\operatorname{rk} \Pi_k:\ \text{exact elimination on the } 2^k\text{-dimensional symmetrizer for } k$$

$$= 1..6:\ \text{ranks } (2,3,4,5,6,7)\ [\text{machine}]$$

$$(2):\ \left[\hat{P}_\sigma, \mathcal{U}^{\otimes k}\right]$$

$= 0$ for the diagonal action [math: the permutation and the diagonal tensor power commute];

the infinitesimal rows on the generator triple W_a

$$= \tfrac{1}{2}\sigma_a:\ \left[\textstyle\sum_i I^{\otimes(i-1)} \otimes W_a \otimes I^{\otimes(k-i)},\ \Pi_k\right]$$

$= 0$ [machine over $\mathbb{Q}(i)$];

the character: $\operatorname{Tr}(\Pi_k D^{\otimes k})$

$$= \frac{1}{k!} \sum_\sigma \prod_{\text{cycles}} (z^L + z^{-L})$$

$$= \sum_m z^m \text{ [machine: five rational } z \text{ values, } k \le 8]$$

(3): the character product $\left(\sum_i z^{k_1-2i}\right)\left(\sum_j z^{k_2-2j}\right)$ carries $z^{k_1+k_2}$ with coefficient 1 [machine: Laurent multiplication];

the connector row: $\mathrm{Sym}^{d_{12}}$ is (1)'s construction at $k = d_{12}$ [BM.2(2)]

(4) : the ladder reads on the weight basis $\{\psi_m\}$ of (2) [Def-BM.19(3); BM.124(3) re-cited];

(5) : FT.60(7)'s two obstruction rows re-cited — the irreducibility is frozen;

(6) : the consumption-site scan [machine: every Ψ-use in Part 4 is through the six listed reads];

(7) : assembly. ■

Phys. The carrier's construction is no longer a black box with a declared integer on top: the mechanism underneath the integer is derived. The symmetric power is the image of the group average over the k-fold tensor power of the seed — the same twirl structure the Haar–Poisson duality runs on SU(2), here on the finite permutation group — with rank $k+1$ exact, diagonal transport commuting with the average, and the weight ladder falling out of a Laurent identity. Junctions compose by the same machine: the connector itself carries a symmetric-power carrier, so every glued chain stays inside the mechanism. What remains declared is exactly one integer per chain, and a frozen minimality theorem says no construction below it exists; what remains questioned — an aspect block for carriers of dimension four and up — is adjudicated void for this era, because no clause consumes a carrier except through its dimension, weights, sign, spin, Casimir and character.

Theorem 4.118 (BM.139 — The two convention declarations and the sector locus). Thm.

(1) the $\aleph$-domain conflict (exactly as computed):

three domains for the vertex-amplitude exponent $\aleph$ entering $w = \mathcal{J} \prod_{e \in R} G(e)(-1)^{\aleph}$

[Def-H.27; A8; D3.3] :

literal (S2, frozen); E1 (chain-dominant); E2 (literal-forced) [Def-H.27(a)–(c)];

pairwise inequivalent, with the exact witnesses:

the locked single-vertex star ($n_{\text{in}}+n_{\text{out}} = n \in \{4,5,6\}$, balanced) :

literal 0;

$$\text{E2} = \text{E1} = \sum_{a \geq 2} \binom{n_{\text{in}}}{a} \binom{n_{\text{out}}}{a} a!\, 4^a = (32,\ 96,\ 672)$$

[FT.75(1); machine: the binomial sums re-derived exactly];

the locked digon hub with a digons :

literal 4^a;

E2 $= 0$;

E1 $= (!a)\,4^a$ with $(!a) = (1, 2, 9, 44)$ at $a = 2..5$

[FT.75(1); machine: the derangement census];

no further computation settles the choice from the frozen corpus: it is a declaration species ;

(2) A-BM.10 issued (the $\aleph$-domain):

the baryomesonic era runs:

(a) the topology-blind ensembles on Def-H.32's inherited literal domain (S2) — unchanged, with every frozen ensemble row standing in its own domain [Def-H.32 Scopes; FT.74 status row];

(b) the sealed-junction amplitude reads — where the chain-local pass-through usage of the telescoping and EM rows is the forced datum [FT.21(i); FT.62(5); BM.7 species] — on the E2 domain [Def-H.27(c)];

(c) E1 recorded as a computed variant, ¬adopted;

the grounds:

E2 retains the literal D3.3 read at every type-$(1, 1)$ vertex, so the frozen literal-domain records carry over unmodified there;

the chain choice at type-$(1, 1)$ is $\aleph$-gauge [Def-H.27(c) : the datum is type-determined] and E2 fixes that gauge by vertex type, while E1 leaves it to the decomposition;

the derived consequences:

the locked star is revived with the closed forms $(32, 96, 672)$;

the locked digon hub cancels in E2 — the bound-content cancellation at hubs is preserved, sharpening FT.67(3a);

κ-invariance and the $(-1)^{\sum \kappa}$ scaling carry over verbatim [Def-H.27 Rmk];

(3) the corner-convention conflict:

the four Θ_R-readings are computed:

the vertex figure δ_v (the only non-trivial spectrum);

the flat face 0 and the global sweep $V\delta_v = 4\pi$ (both identically massless);

the winding pairing inside the first: variant A (δ_v, k_f), variant B (δ_v, k_{vf})

[FT.76(1); Census FT.76(1)–(2); machine: $2 - 2T_m(1) = 0\ \forall m$ in $\mathbb{Z}[x]$ for the two massless readings];

$$k_f = (3, 2, 3, 5, 3),$$

$$k_{vf} = (3, 3, 2, 3, 5)$$

[FT.76(1); machine: both re-derived from the minimality rule $k = L\kappa_{\min}/2$ on the face and vertex-figure loop lengths $L_f = (3, 4, 3, 5, 3)$, $L_{vf} = (3, 3, 4, 3, 5)$];

the variants are materially different:

the multiplet tables differ [Census FT.76(2)];

the per-class statistics censuses differ: $112/104$ in B against $140/76$ in A

[BM.136(4); machine];

(4) A-BM.11 issued (the defect corner convention):

variant B — the vertex-figure winding k_{vf} with $\Theta_R = \delta_v$

[BM.6(4)'s specification discharged];

the grounds:

(i) loop consistency: the Θ_R

$= \delta_v$ read is the vertex-figure loop's own defect, and pairing it with the face winding would mix two loci [FT.76(1) : variant B is the loop-consistent reading];

(ii) the whole defect ladder of this part runs at the vertex-figure locus [Def-BM.14(3); Def-BM.19(3); BM.79(1); BM.124(3); BM.125];

(iii) the ceiling $\sqrt{4/D_S}$ is winding-independent and untouched [FT.76(3)];

the effects:

FT.76,

FT.77's conditional clauses and this part's ladders convert from convention-tagged to declared data at their points of use,

¬retracted;

variant A's tables remain computed data, recorded and ¬adopted;

the winding assignment per class is pinned [BM.136(4)];

(5) A-BM.13 issued (the statistics locus):

the $\vee^2$ -sector is the standing locus of the sign read

[Def-BM.16(5)'s non-derived clause converted to declared data];

the grounds (the four priced rows):

$\vee^2$ and $\wedge^2$ give one statistics up to a global sign;

the determinant line and the orientation block do not [BM.113(6); machine re-run to $k = 8$];

$\vee^2$ is the flip's $+1$-eigensector on the CP pair [A3] and matches the carrier's own symmetry — $\wedge^3(\mathbb{C}^2)$

$= 0$ leaves the carrier no antisymmetric locus at $n \qquad = 3$

[BM.128(5)];

A-BM.12 (the resolvent interface) is issued at Def-BM.20 and recorded here in the ledger census;

(6) the convention question closed — the census:

(d) the ℵ-domain: declared [(2)];

(e) the corner convention: declared [(4)];

(b) the exchange block $\mathcal{K}^{(\mathrm{exch})}$: adjudicated void for this era — the chirality read has its locus in the carrier rays [A-BM.7; BM.73],

and no k

≥ 3 aspect block is consumed [BM.138(6)];

¬a declared exchange block is introduced;

(a),

(c) the classification program's remaining cells (multi-solid chains, the single-vertex ladder, separated-contact hosting maps) are named for the Part-5 cell program — ¬era objectives;

both conventions are declared, the void residue is adjudicated, and the cell remainder is transferred

(7) the declaration ledger:

four addenda issued by this tier: A-BM.10 (ℵ-domain),

A-BM.11 (corner convention),

A-BM.12 (resolvent interface),

A-BM.13 (statistics locus);

all four logged in §II.4 before use;

¬a P4 member added by any of them;

¬a dimensionful scale [P1];

¬a measured value [P5];

§Cal-BM gains no parameter [BM.143(4)'s consolidation is a register, ¬an addition].

Req. A8 (Part 1); D3.3 (Part 1); CT0.9 (Part 1); P1, P5 (Parts 1–2); Def-H.27, Def-H.32, FT.21, FT.62(5), FT.67(1), FT.67(3), FT.74, FT.75, FT.76(1), FT.76(3), FT.77 (Part 3); A3 (Part 1); Def-BM.14(3), Def-BM.16(5), Def-BM.19(3), Def-BM.20, BM.6(4), BM.7, BM.73, BM.79(1), BM.113(6), BM.124(3), BM.125, BM.128(5), BM.136(4), BM.138(6), A-BM.7 (this part); Census FT.76(1), FT.76(2) (row-wise); A-BM.10, A-BM.11, A-BM.13 (§II.4, logged before use); exact binomial and derangement enumeration, the Chebyshev identity $T_m(1) = 1$ over $\mathbb{Z}[x]$, exhaustive parity censuses [math; machine: `checks/verify_H_bm_closing_tier.py` §H].

Pf.

$$(1):\ \text{the star sums } \sum_{a\geq 2}\binom{n_{\text{in}}}{a}\binom{n_{\text{out}}}{a}a!4^a \text{ evaluated exactly at } (2,2),$$

$$(2,3),$$

$$(3,3):\ (32,96,672)\ [\text{machine}];$$

the derangements $(!a)$

$$=\sum_{j\leq a}(-1)^j a!/j!$$

$= (1,2,9,44)$ and the hub values $(16,128,2304,45056)$ [machine]

$$(3):\ k_f,$$

k_{vf} from k

$$= L\kappa_{\min}/2 \text{ with } \kappa_{\min}$$

$$= 2 \ (L \text{ odd}),$$

$$1 \ (L \text{ even}) \ [\text{FT.32}(6);\ \text{FT.45}] : \quad L_f$$

$$= (3, 4, 3, 5, 3)$$

$$\mapsto (3, 2, 3, 5, 3);$$

$$L_{vf} = (3, 3, 4, 3, 5) \mapsto (3, 3, 2, 3, 5) \ [\text{machine}];$$

$$\text{the massless readings: } 2 - 2T_m(1) = 0 \text{ and } 2 - 2T_m(\cos 2\pi) = 0 \ \forall m \ [\text{machine over } \mathbb{Z}[x]]$$

$$(2),$$

(4)–(7) : the declaration rows cite their grounds and their loci; the §II.4 entries carry the registry grammar [machine: the logged-before-use document pins]. ■

Phys. Two conflicts the corpus could compute but not settle are settled by declaration, each with its grounds on the record. The amplitude-domain question — literal against chain-local — resolves pointwise: the ensembles keep the frozen literal domain, and the sealed junctions, where the chain-local pass-through reads were always the forced datum, run the literal-forced domain E2, which is canonical because it fixes the gauge by vertex type rather than by a choice of decomposition; E1 stays computed and unadopted. The corner convention resolves to the vertex-figure pairing, the loop-consistent reading, which is where this part's whole defect ladder already lives — and the declaration is load-bearing, because the per-class statistics census provably differs between the variants.

BM.140 — The ceiling study: the Platonic defect census, the dodecahedron selection theorem, the five-currency range table, and the adjudication of deficit 12. Structural-constraints record: see Appendix B, Section B.2.8 of the Addendum.

Theorem 4.119 (BM.141 — The Jensen residue: the exact closed form, its range, its zero locus, the two-chain companion, and the width adjudication). Thm.

(1) the residue computed:

$$J_3(p) \ := \ \langle E \rangle_{\text{mat}} \langle E^{-1} \rangle_{\text{mat}} - 1 \ = \ \frac{18 - 27p + 11p^2}{2D^2} - 1,$$

$$D = 3 - 3p + p^2,$$

$$p = 1 - 4\varepsilon_q$$

[BM.94$(6), (8)$ re-cited];

$$\boxed{J_3(p) \;=\; \frac{p\,(1-p)\,\big(9 - 10p + 2p^2\big)}{2D^2}}$$

[machine: the polynomial identity $18-27p+11p^2-2D^2 = p(1-p)(9-10p+2p^2)$ exact in $\mathbb{Q}[p]$];

(2) the sign, the zeros, and the range:

$$J_3 \geq 0 \;\text{ on } p \in (0,1]$$

[BM.94(8)'s Cauchy–Schwarz row re-derived; machine: $9 - 10p + 2p^2 > 0$ on $(0,1]$ — its roots $\frac{5 \pm \sqrt{7}}{2}$ both exceed 1 in $\mathbb{Q}(\sqrt{7})$: $\frac{5-\sqrt{7}}{2} > 1 \iff 3 > \sqrt{7} \iff 9 > 7$];

$$\boxed{J_3 = 0 \iff p = 1 \iff \varepsilon_q = 0}$$

[machine: the factor census];

$$J_3\big(\tfrac{1}{2}\big) = \frac{9}{49}$$

[BM.94(8) : $\frac{58}{49} - 1$; machine];

$$J_3 < \frac{1}{5} \;\text{ on } (0,1)$$

[$12p^4 - 72p^3 + 125p^2 - 81p + 18 > 0$; BM.94$(8)$ re-derived; machine: exact rational bisection, no roots on $(0,1)$];

$$\boxed{\sup_{(0,1)} J_3 = J_3(p_*) \in \Big(\frac{93}{500},\, \frac{3}{16}\Big), \qquad p_* \in \Big(\frac{288131}{524288},\, \frac{576263}{1048576}\Big)}$$

[machine: the unique critical point by Sturm isolation of $-44p^5 + 294p^4 - 792p^3 + 1062p^2 - 684p + 162$; $J_3 < \frac{3}{16}$ strict: $38p^4 - 228p^3 + 394p^2 - 252p + 54 > 0$ on $(0,1)$, no roots];

(3) the two-chain companion:

$$\langle E\rangle^{(2)}_{\text{mat}} = \frac{4}{1+u},$$

$$\left\langle E^{-1}\right\rangle^{(2)}_{\text{mat}} = \frac{3u+1}{4(1+u)}$$

[BM.137(2); machine];

$$\boxed{J_2(p) \;=\; \frac{4-3p}{(2-p)^2} - 1 \;=\; \frac{p(1-p)}{(2-p)^2} \;=\; \frac{u(1-u)}{(1+u)^2}}$$

[machine: the identity exact in $\mathbb{Q}[p]$];

$$\boxed{J_2 \le \frac{1}{8}, \qquad \text{equality} \iff p = \frac{2}{3} \iff \varepsilon_q = \frac{1}{12}}$$

[machine: the SOS certificate $\frac{(1+u)^2}{8} - u(1-u) = \frac{(3u-1)^2}{8} \ge 0$; the stationary point $3p^2-8p+4 = (3p-2)(p-2)$];

$$J_2\left(\tfrac{1}{2}\right) = \frac{1}{9} < \frac{9}{49} = J_3\left(\tfrac{1}{2}\right);$$

$$\frac{1}{8} < \frac{1}{5};$$

the two-chain residue is strictly tighter than the three-chain bar

[machine];

(4) the exact crossing of the two residues:

$$J_3 - J_2 = \frac{p\,(p-1)\left(6p^3 - 27p^2 + 40p - 18\right)}{2D^2(2-p)^2}$$

[machine: the factorization exact in $\mathbb{Q}[p]$];

$6p^3 - 27p^2 + 40p - 18$ is strictly increasing on $(0,1)$ [its derivative $18p^2 - 54p + 40$ has roots $\frac{4}{3}, \frac{5}{3}$, both outside] with the single root $p_c \in \left(\frac{108267}{131072}, \frac{866137}{1048576}\right)$

[machine: Sturm isolation];

$$\boxed{J_3 > J_2 \iff p < p_c; \qquad J_3 < J_2 \iff p > p_c}$$ [machine: the sign census on exact rationals];

(5) the width adjudication (the two formed reads of Δx_{B}):

$$\Delta x_{\mathrm{B}}^{\mathrm{eff}} = \frac{\mathfrak{h}}{\langle E\rangle_{\mathrm{mat}} \ln 2} = D\,\Delta x_{\mathrm{B}}^{cryst};$$

$$\frac{\mathfrak{h}}{\ln 2}\langle E^{-1}\rangle_{\mathrm{mat}} \;\geq\; \Delta x_{\mathrm{B}}^{\mathrm{eff}}$$

[BM.94(8) re-cited];

the relative gap of the two reads is exactly the Jensen residue: $\left(\langle E^{-1}\rangle_{\mathrm{mat}} - \frac{1}{\langle E\rangle_{\mathrm{mat}}}\right)\langle E\rangle_{\mathrm{mat}} = J_3$

[machine: the identity];

the adjudication:

the tier's evaluations at the width of the mean [BM.94(8); BM.98(5)] stand, with the difference now an exact computed function instead of a residue;

(6) status:

derived: (1)–(5);

(R6) is closed: the Jensen residue is computed as an exact rational function with its zero locus, its exact supremum enclosure and its two-chain companion

[BM.94(8), BM.98(5)'s residue rows discharged];

¬a declaration,

¬a P4 addition,

¬a measured value [P5].

Req. A4 (Part 1); D0.4 (Part 1); CT0.8 (Part 1); P5 (Parts 1–2); BM.94(6), BM.94(8), BM.98(5), BM.137(2) (this part); polynomial factorization over $\mathbb{Q}$, Sturm isolation and exact rational bisection, SOS certificates, the Cauchy–Schwarz equality locus [math; machine: `checks/verify_H_bm_closing_tier.py` §J].

Pf.

$(1):\ 18-27p+11p^2-2(3-3p+p^2)^2=p(1-p)(9-10p+2p^2)$ [machine: expansion in $\mathbb{Q}[p]$];

$(2):$ the root census of $9-10p+2p^2$ in $\mathbb{Q}(\sqrt{7})$;

the bisection of $12p^4-72p^3+125p^2-81p+18$ and $38p^4-228p^3+394p^2-252p+54$ on $(0,1)$ at exact rational endpoints [machine]

$(3):\ \left\langle E^{-1}\right\rangle^{(2)}_{\text{mat}}=\dfrac{\frac{1}{2}\cdot 2u+\frac{1}{4}p}{1+u}$ normalized [BM.137(2)'s measure; machine];

$$J_2=\frac{4}{1+u}\cdot\frac{3u+1}{4(1+u)}-1=\frac{3u+1-(1+u)^2}{(1+u)^2}=\frac{u(1-u)}{(1+u)^2}\ \text{[machine]};$$

the SOS certificate and the stationary factorization [machine]

$(4):\ J_3-J_2$ brought to the common denominator $2D^2(2-p)^2$ and factored [machine];

$(5):\ \left\langle E^{-1}\right\rangle\langle E\rangle-\langle E\rangle/\langle E\rangle=J_3$ [machine];

$(6):$ assembly. ■

Phys. The last pure-mathematics residue of the era is discharged: the Jensen residue of the materialized flicker distribution is one exact rational function, $p(1-p)(9-10p+2p^2)$ over twice the squared geometric sum, vanishing precisely at the sharp crystallization point, capped strictly below one fifth with its supremum enclosed between two rationals and attained at a uniquely isolated critical point. The two-chain companion is tighter — bounded by one eighth with a sum-of-squares certificate and equality at threshold over twelve — and the two residues cross exactly once, at an isolable rational-bracketed point. The gap between the two formed width reads of the Born envelope is precisely this residue, so the era's evaluations at the width of the mean stand with their correction now computed rather than named.

Theorem 4.120 (BM.142 — The supply vertex, the pair amplitude, and the annihilation cross-section: the four grounds resolved and residues (R9′), (R9″) closed). Thm.

(1) the four grounds of BM.122(6) resolved, one by one:

ground	the resolution	the row
(O1) ¬a gauged chiral coupling	dissolved for the EM legs: the gauged coupling of the charge read is constructed chart-side	[Def-BM.12(1)–(3) : $\partial_\mu F^{\mu\nu}=eJ^\nu_Q$; EM.15]
	the W-species chiral gauging remains ¬constructed — barred for weak legs, recorded	[BM.76; A-BM.7 scope]
(O2) deficit $1'$	discharged: the assignment rule is issued and the attachment derived	[A-BM.7; BM.73(1), (2); BM.75(3)]
(O3) ¬a momentum resolvent	discharged: the resolvent and the phase space are constructed	[Def-BM.20; BM.134]
(O4) the transit leg carries no winding carrier	re-scoped: the exclusion stands, but the vertex needs ¬an ε/j read of the legs and the register's moves apply to transit legs	[Def-H.38 Scope frozen]; the legs couple by $(q,$ helicity$):\ N_{\text{pol}}=2$ [P2; EM.8] [BM.126(5) domain row]

(2) the supply vertex (constructed):

$$\boxed{\widehat{\mathcal{V}}_{\text{sup}} \;:=\; e(k,E)^2 \cdot \left(\hat{J}_+ \otimes \hat{J}_+\right)}$$

on $\mathcal{J} \otimes \mathcal{J} \otimes$ (the two transit helicity legs)

[Def-BM.18(2); Def-BM.12(2); Def-BM.20(4)];

$$\left(\hat{J}_+ \otimes \hat{J}_+\right)^2 = 0$$

$[\hat{J}_+^2 = 0;$ BM.120(1); machine: the 4×4 identity over $\mathbb{Z}]$:

the vertex is nilpotent: the register cannot supply a pair twice;

the vertex rows:

charge balance $Q_{\text{in}} = Q_{\text{out}} = 0$ [BM.122(1)];

the threshold gate $\chi_{\{NE \geq \mathcal{T}_0\}}$ [BM.122(2); Def-BM.17(3)];

the helicity census 2×2

$= 4$ combinations, equally weighted by the metric's charge-blindness [P2 : $N_{\gamma\text{-pol}} = 2$; EM.8; BM.58];

(3) the pair amplitude (formed, symbolic):

$$\boxed{\mathcal{M}_{\text{pair}}(z;k,E) \;=\; e(k,E)^2\, q^2\, x_+^2\, \chi \;=\; e_0^2\, q^2 x_+^2\, \chi \cdot \frac{N(k,E)\, f(k)}{1+G(E)}}$$

[Def-BM.20(4)'s $\mathcal{V} = 1$; BM.133(6); BM.132(2); machine: the monomial census];

$[\mathcal{M}] = \mathfrak{q}^0, \qquad [e] = \mathfrak{q}^0, \qquad [x_\pm] = \mathfrak{q}^0 \qquad$ [CT2.1; BM.102(2); machine: the exponent census];

the reduction clause:

$$e = 1,\; q = 1: \quad \mathcal{M}_{\text{pair}} = \mathcal{W}_{\text{channel}} = x_+^2 \chi$$

[BM.122(3); Def-BM.20(4); machine];

the exact zeros:

$$\mathcal{M} = 0 \iff e_0 = 0 \ \vee \ q = 0 \ \vee \ \varepsilon_q = \tfrac{1}{4} \ \vee \ \chi = 0 \ \vee \ N = 0 \ \vee \ f = 0$$

[machine: the factor census];

(4) the pair cross-section (the formed chart read):

$$\sigma_{\text{pair}} := \frac{|\mathcal{M}_{\text{pair}}|^2 \, \Phi_2\big(z;\, 2m_{\text{hid}}, N\big)}{F_{\gamma\gamma}}$$

[BM.134(4); the final state is the massive pair] :

$$\boxed{\sigma_{\text{pair}}(z; k, E) \ = \ \frac{e_0^4 \, q^4 x_+^4 \, \chi \, N \, f^2}{16\pi \, z^3 \, (1+G)^2} \, \sqrt{z^2 - 4N^2\big(2m_{\text{hid}}\big)^2}}$$

[machine: the coefficient algebra exact; $m_{\text{pair}} = 2m_{\text{hid}}$ [BM.111(1)]];

$[\sigma_{\text{pair}}] = \mathfrak{q}^{-1} \ =$ the area species of $(\Delta x_{\text{B}})^2$ [machine: the exponent census];

the threshold rows:

$\sigma_{\text{pair}} = 0$ exactly at $z = 2N \cdot 2m_{\text{hid}}$ (the square-root onset);

$\sigma_{\text{pair}} > 0$ strictly above it;

formed on $z \geq 4Nm_{\text{hid}} \ \wedge \ k \geq \dfrac{G}{4\pi} \ \wedge \ NE \geq \mathcal{T}_0$

[machine];

(5) the annihilation cross-section:

$$\mathcal{M}_{\text{ann}} = e(k, E)^2 q^2 x_-^2 \chi$$

[BM.123(1)′s reverse move $\hat{J}_- \otimes \hat{J}_-$; the same vertex normalization; $m\big(c^{\text{CP}}\big) = m(c)$ [BM.27; BM.108(3)]];

$$\sigma_{\text{ann}} := \frac{|\mathcal{M}_{\text{ann}}|^2\,\Phi_2^{(\gamma\gamma)}}{F_{\text{pair}}} \qquad [\text{the final state massless, the initial state massive}]:$$

$$\boxed{\sigma_{\text{ann}}(z;k,E) \;=\; \frac{e_0^4\,q^4 x_-^4\,\chi\,N\,f^2}{16\pi\,z\,(1+G)^2\,\sqrt{z^2-4N^2\big(2m_{\text{hid}}\big)^2}}}$$

[machine: the coefficient algebra exact; $[\sigma_{\text{ann}}] = \mathfrak{q}^{-1}$];

the threshold behaviour:

σ_{ann} has the exact square-root pole at $z = 4Nm_{\text{hid}}$;

$\sigma_{\text{ann}} \cdot F_{\text{pair}}$ is regular there

[machine: the product $= \dfrac{e_0^4 q^4 x_-^4 \chi\, f^2}{8\pi\, N\,(1+G)^2}$ regular in the formed factors];

(6) the ratio and the detailed-balance elevation:

$$\boxed{\frac{\sigma_{\text{ann}}}{\sigma_{\text{pair}}} \;=\; \Big(\frac{x_-}{x_+}\Big)^4 \cdot \frac{z^2}{z^2-4N^2(2m_{\text{hid}})^2}}$$

[machine: the symbolic division exact];

the amplitude-level detailed balance:

$$|\mathcal{M}_{\text{ann}}| = |\mathcal{M}_{\text{pair}}| \iff \varepsilon_q = \frac{1}{8}$$

[BM.123(2)'s indicator elevated to the amplitude; machine];

at ε_q

$= \dfrac{1}{8}$ the cross-section ratio is the pure kinematic factor $\dfrac{z^2}{z^2-16N^2m_{\text{hid}}^2} > 1$ strictly above threshold

[machine];

(7) the width row and its species bar:

the chart-frequency width Γ^{chart}

$$= \frac{|\mathcal{M}|^2 \Phi_2}{2z} \text{ carries } [\Gamma^{\text{chart}}]$$

$$= \mathfrak{q}^{-1/2} \text{ at } \mathcal{V}$$

$= 1$ — a time, ¬a rate :

the mass-dressed normalization that would convert it is ¬declared [Def-BM.20(4)'s minimality] $\Longrightarrow$ the formed width read stays the ledger's per-tick species [BM.116(1); BM.44(4), (5) : $[\Gamma] = \mathfrak{q}^0$ ticks^{-1}; $\tau_{\text{life}} \geq 3$ ticks re-cited];

(8) the closure rows:

(R9′) is closed: the annihilation cross-section is constructed as an exact symbolic chart-side read on the formed domain

(R9″) is closed at the vertex level: the two-transit-quantum supply vertex is constructed, nilpotent, charge-balanced and threshold-gated

[BM.118(7), BM.130(7)'s rows discharged];

BM.130(3)'s ¬cross-section row is discharged by construction and re-scoped, ¬retracted;

BM.122(6)'s four grounds carry (1)'s per-row dispositions;

(9) the bars:

¬a numerical cross-section, rate or branching fraction is evaluated [P5; BM.44(5)];

¬an S-matrix unitarity claim [Def-BM.20(5)];

amplitude $\neq$ mass [FT.79(5)];

e_0 remains calibration [EM.25(7); BM.143(4)];

¬a measured external constant is identified with $\sigma_{\text{pair}},\ \sigma_{\text{ann}}$ or any factor [P5];

¬a species name enters any clause [Def-BM.3(7)].

Req. A4, A8 (Part 1); D0.4, D1.2 (Part 1); CT1.22, CT2.1 (Part 2); EM.8, EM.15, EM.25(7) (Part 2); P2, P5 (Parts 1–2); Def-H.38 (Part 3); FT.79(5) (Part 3); Def-BM.12, Def-BM.17(3), Def-BM.18(2), Def-BM.20, BM.27, BM.44(4), BM.44(5), BM.58, BM.73(1), BM.73(2), BM.75(3), BM.76, BM.102(2), BM.108(3), BM.111(1), BM.116(1), BM.118(7), BM.120(1), BM.122, BM.123, BM.126(5), BM.130(3), BM.130(7), BM.132(2), BM.133(6), BM.134(4) (this part); the declaration addendum A-BM.7 (pointers); exact matrix identities over $\mathbb{Z}$, monomial censuses in symbolic rings, the coefficient algebra of the two-body reads [math; machine: `checks/verify_H_bm_closing_tier.py` §K].

Pf.

(1) : the four rows cite their discharging tags; the W-species bar and the E2-domain inheritance recorded [Def-BM.20(5); machine: the document pins]

(2) : $\left(\hat{J}_+ \otimes \hat{J}_+\right)^2 = \hat{J}_+^2 \otimes \hat{J}_+^2 = 0 \otimes 0 = 0$ [BM.120(1); machine: the 4×4 product over $\mathbb{Z}$];

the helicity census $2^2 = 4$ [P2; machine]

(3), (4) : $\mathcal{M}$ assembled from $e^2 = e_0^2 N f\mathfrak{g} = e_0^2 N f/(1+G)$ [BM.133(6); machine];

$$\sigma_{\text{pair}}$$

$$= \frac{|\mathcal{M}|^2}{F_{\gamma\gamma}} \Phi_2$$

$$= \frac{e_0^4 q^4 x_+^4 \chi N^2 f^2}{(1+G)^2} \cdot \frac{\sqrt{z^2 - 16N^2m^2}}{8\pi N^3 z} \cdot \frac{N^2}{2z^2}$$ [BM.134(4); machine: the coefficient algebra];

the exponent census $[\sigma] = \mathfrak{q}^{-1}$ [CT2.1]

(5), (6) : $\sigma_{\text{ann}} = \dfrac{e_0^4 q^4 x_-^4 \chi N^2 f^2}{(1+G)^2} \cdot \dfrac{1}{8\pi N^3} \cdot \dfrac{N^2}{2z\sqrt{z^2 - 16N^2m^2}}$ [machine];

the ratio by symbolic division [machine];

the detailed-balance locus $x_- = x_+ \iff \varepsilon_q = \frac{1}{8}$ [BM.120(5); BM.123(2)]

$$(7):\ \left[\tfrac{|\mathcal{M}|^2\Phi_2}{2z}\right] = \mathfrak{q}^0\cdot\mathfrak{q}^0\cdot\mathfrak{q}^{-1/2} = \mathfrak{q}^{-1/2}\ \text{[machine: the exponent census]};$$

$$(8),(9):\ \text{assembly and the token scan.}\ \blacksquare$$

Phys. The two-photon bar falls ground by ground: the electromagnetic gauging of the charge read existed chart-side, the assignment rule was issued, the resolvent is now constructed, and the transit legs couple by charge and helicity — the one exclusion that stands, the winding carrier of a trivial chain, simply never entered the vertex. What is built is minimal and honest: a nilpotent supply vertex that cannot fire twice, an amplitude that reduces to the register's own channel weight at unit coupling, and two cross-sections that are exact symbolic functions of the envelope ratio, the local energy, the flicker threshold and the calibration coupling — with an exact square-root onset for production, an exact square-root pole for annihilation whose flux product stays regular, and a ratio that carries the detailed-balance point of the register elevated to the amplitude level. The chart-frequency width is refused on dimensional grounds rather than fudged: the per-tick ledger species remains the corpus's width read. Both residues the annihilation tier named are closed, and nothing measured entered any row.

BM.143 — The house cleaning: the void residues closed, the phantom gaps reclassified, the calibration register consolidated, and the Part-5 requirements named. Structural-constraints record: see Appendix B, Section B.4.2 of the Addendum.

BM.144 — The status of the Closing Tier and the closure of the Baryomesonic era. Structural-constraints record: see Appendix B, Section B.1.34 of the Addendum.

4.26 Tier BM.24 — The Continuum-Limit Tier: the Force Operator over the Graph, the Continuum-Limit Function, the Metric–Energy–Continuum Map, the Breaking Threshold, the Locked-Sector Complement, the Universality, the Reclassification of the Continuum Conditions, and the Closure of the Foundational Era

Theorem 4.121 (BM.145 — The force operator acting over the graph: the generator census, the mask ladder, and the derived action). Thm.

$$(1)\quad \text{the complete operator (re-cited, exact):}$$

$$\mathrm{F} = F_{(+,\mathcal{R})}\oplus F_{(-,\mathcal{R})}\oplus F_{(+,\mathcal{V})}\oplus F_{(-,\mathcal{V})}\oplus F_{\mathrm{st}}\ \text{ always applies in full}$$

$$\text{[A8; BM.52(1)]};$$

$$F_{(+,\mathcal{R})} = F_G\oplus F_H\oplus F_{D^+};$$

$$F_{(-,\mathcal{R})} = F_{W^+}\oplus F_{W^-}\oplus F_Z;$$

$$F_{(+,\mathcal{V})} = F_{J^+};$$

$$F_{(-,\mathcal{V})} = F_{J^-};$$

$$F_{\text{st}} = F_{\text{EM}} \oplus F_{\text{strong}} \oplus F_{D^-}$$

[A8];

the pair-arena block census:

aspect	the block	$\dim_{\mathbb{R}}$
F_{J^+}	$\mathcal{E}_{-+} = \text{span}\{J_i\}$	3
F_{J^-}	$\mathcal{E}_{--} = \text{span}\{K_i\}$	3
F_G	$\text{span}\{\hat{E}^{ij}\}$ traceless	5
F_W	$\mathcal{E}_{+-} = \text{span}\{W_i\}$	3
F_H	$\text{span}\{\Sigma\}$	1
F_{D^+}	$\text{span}\{E_0\}$	1
total	$\mathfrak{H}(\mathbb{C}^4)$	$16 = C_{\text{cap}}$

[CT0.12; CT1.1; P2 : machine: $3+3+5+3+1+1=16$];

the stack aspects act on the vertex/slot registers, ¬on the pair channel:

$$F_{\text{EM}},\ F_{\text{strong}},\ F_{D^-}$$

[CT0.12 Rmk; D3.1; D4.1];

(2) the activation and the mask ladder:

$$\mathcal{A}(0)$$

$$= \{J\};$$

$$\mathcal{A}(1)$$

$$= \{J, G, W, H, D^+\};$$

$$\mathcal{A}(2)$$

$$= \mathcal{A}(1) \cup \{\text{EM}\};$$

$$\mathcal{A}(3)$$

$$= \mathcal{A}(2) \cup \{\text{strong}\};$$

$$\mathcal{A}(n \geq 4)$$

$$= \mathcal{A}(3) \cup \{D^{-}\}$$

[CT0.13];

$$\mathrm{F}_{M}^{(n)} = \bigoplus_{A \in \mathcal{A}(n)} F_A\big|_{\mathcal{K}^{(A)}}$$

[CT2.4 : masking never mixes forces];

$$\dim \mathcal{K}_n = (6,\ 16,\ 17,\ 25) \ \text{strict}$$

[CT0.14; BM.36(1)];

the increments are derived:

$$1 \to 2 : \ +1 = \dim Z\big(\iota_2(\mathfrak{u}(2))\big) = \dim \mathcal{K}^{(\mathrm{EM})}$$

[D2.12];

$$2 \to 3 : \ +8 = \dim \mathfrak{su}(3) = \dim \mathcal{K}^{(\text{strong})}$$

[CT3.8];

the observable bound $\dim \mathcal{K}_{\Sigma} \leq 1+5+1+3+8 = 18$ [FT.26(3); machine: the integer census];

(3) the action over the graph (the composition, every step derived):

$$\boxed{\hat{\mathcal{F}} : \ (\Gamma_{\mathcal{R}},\ \text{registrations}) \ \longmapsto \ (M,\ g_{\mu\nu},\ T_{\mu\nu},\ \text{spec})}$$

step	the map	the locus
the weights	$(\Gamma,\sigma_e,\kappa_e,\alpha_e)\mapsto w(C)=\mathcal{J}(C)\prod_{e\in R}G(e)\,(-1)^{\aleph},\quad G(e)=\sigma_e e^{i\pi\kappa_e}$	[A8; D3.5; D3.6; FT.9]
the domain	$\aleph$ literal on the ensembles, E2 at the sealed junctions	[Def-H.32 Scopes; A-BM.10; BM.139(2)]
the dictionary	$C\mapsto(a_{00},q_{ij},w_i)\mapsto g_{\mu\nu}=\Omega^2\bar{g}_{\mu\nu},\quad \Omega=e^{\alpha_\Omega a_{00}}$	[D2.1; CT1.12]
the grouping	$Z[g]=\sum_{C:\,g(C)=g}\lvert w(C)\rvert\,e^{i\Phi(C)}$	[CT1.20]
the action	$S_{\text{eff}}=-i\log Z;\quad S_{\text{geom}}=\frac{1}{2\kappa^2}\int\sqrt{-g}\,(R-2\Lambda)$	[CT1.20; CT1.21; CT2.7]
the spectrum	$Z(\mathcal{R},n)=\langle\text{out}\lvert(1-\mathrm{F}_M^{(n)})^{-1}\rvert\text{in}\rangle=\sum_a\frac{\langle\text{out}\lvert P_a\rvert\text{in}\rangle}{1-\lambda_a};\quad \text{poles}\iff 1\in\operatorname{spec}\mathrm{F}_M^{(n)}$	[CT2.5]
the source	$T_{\mu\nu}=-\frac{2}{\sqrt{-g}}\frac{\delta S_{\text{force}}}{\delta g^{\mu\nu}}=T^{\text{bd}}_{\mu\nu}+T^{\text{tr}}_{\mu\nu}$	[CT1.22]
the field equations	$G_{\mu\nu}+\Lambda g_{\mu\nu}=\kappa^2 T_{\mu\nu}$	[CT1.23]
the target	$M=\Gamma_{\mathcal{R}}/\!\sim$ a smooth 4-manifold with the charts (τ_n,h_1,h_2,h_3)	[CT2.6; D1.8]

the composition is the action:

no step adds a datum — each row is a derived map of the cited tag

[machine: the locus census, row by row];

(4) the state-side action (the second face, separated by species):

the Haar draws and the Gibbs tilt act on the pair states, ¬on the arena blocks:

$$\mathcal{T}(\rho)=U\rho U^{\dagger},\quad \tilde{\mathcal{T}}_\gamma(\rho)=\rho e^{-\gamma\rho}/Z_\gamma$$

[A6; D0.3; L0];

$V'=\left(1-\frac{\gamma}{4}\right)^2V+O(\delta\rho^3),\qquad \mu'=\left(1-\frac{\gamma}{4}\right)\mu+O(\mu^2)$ [L1; L2]; the duality rows [BM.131];

the two actions meet only through the dictionary:

the sector coefficients of D2.1 are the pair-state components [CT1.1; L3(1)];

the Haar face preserves their spectrum and the tilt face contracts it [BM.131(3); L2] :

both actions preserve the derived coefficient bounds of BM.151(2) (forward pointer) — the state-side dynamics never leaves the formed locus

(5) the locked face of the operator:

$$F_{D^-} = \bigoplus_{n\geq 4} F_n$$

[D4.1; A8];

$$D_{\text{tick}}(n) = 2^{n+1},$$

$$O(n) = \max\bigl(0,\ 2^{n+1} - 16\bigr),$$

the last admissible rung is 3

[CT0.5; CT0.6; CT4.1];

machine: $O = (0, 0, 0, 0, 16, 48, 112, 240, 496)$ at $n = (0, 1, 2, 3, 4, 5, 6, 7, 8)$;

$O(4) = 16 = C_{\text{cap}}$ exactly one further channel quantum per tick

[CT4.1];

(6) status:

derived: (1)–(5);

the census is complete: ¬an aspect outside A8's list, ¬a block outside CT0.12's tiling, ¬a rung beyond 3 in the accessible mask [FT.83(2)–(3) : the $n \geq 4$ no-go];

¬a declaration,

¬a P4 addition,

¬a new dimensionful scale [P1],

¬a measured value [P5].

Req. A2, A7, A8 (Part 1); C8 (Part 1); D0.3, D0.8, D0.9, D1.2, D1.8, D2.1, D2.3, D2.12, D3.1, D3.5, D3.6, D4.1 (Parts 1–2); CT0.5, CT0.6, CT0.11, CT0.12, CT0.13, CT0.14, CT1.1, CT1.2, L0, L1, L2, L3(1) (Part 1); CT1.12, CT1.20, CT1.21, CT2.4, CT2.5, CT2.6, CT2.7, CT3.8, CT4.1 (Part 2); P1, P2, P5 (Parts 1–2); Def-H.32, FT.9, FT.26(3), FT.83(2), FT.83(3) (Part 3); BM.36(1), BM.52(1), BM.131(3), BM.139(2) (this part); A-BM.10 (§II.4, logged before use); exact integer censuses [math; machine:

`checks/verify_H_continuum_limit_era1_closure.py` §A]; cf. BM.151 (forward pointer: the coefficient bounds).

Pf.

(1),

$(2):$ A8,

CT0.12,

CT0.13,

CT2.4,

CT0.14,

BM.36(1)'s rows re-assembled [machine: the dimension sums $3{+}3{+}5{+}3{+}1{+}1 = 16,\ 16{+}1 = 17,\ 17{+}8 = 25,\ 1{+}5{+}1{+}3{+}8 = 18$]

$(3):$ each row is the cited tag's own derived map; the composition is read off the shared domains and codomains [machine: the locus census — every row's tag exists and carries the cited clause];

the $\aleph$-domain row is A-BM.10's pointwise declaration [BM.139(2)]

$(4):$ A6, D0.3, L0's rows re-cited;

the coefficient components: $\delta\rho = \frac{1}{4}\sum_a r_a G_a,\ r_a = \mathfrak{T}(\delta\rho\, G_a)$ [L3(1); CT1.1];

the preservation rows: [BM.131(3); L2] [machine: the contraction factor $(1{-}\frac{\gamma}{4}) \le 1$ on $\gamma \in (0,1]$]

$(5):$ $O(n) = \max(0, 2^{n+1}{-}16)$ evaluated at $n = 0..8$ [machine: exact integers]; $(6):$ assembly. ■

Phys. The force operator is not a list: it is one object with three exact faces. Its algebra face tiles the sixteen dimensions of the pair arena with the six pair aspects and stacks the remaining three on the vertex registers; its mask face climbs the rung ladder six, sixteen, seventeen, twenty-five with the two derived increments; its dynamical face is the composed action that carries a registered graph to weights, to the grouped sum, to the effective action, to the masked resolvent's spectrum, to the source and the field equations, to the smooth target manifold. Beside it runs the state-side action — the Haar draws and the tilt — which touches the same data only through the dictionary,

preserves the coefficient bounds, and never leaves the formed locus. The locked face closes the census: beyond rung three the tick demand outruns the channel capacity by an exact power of two, and that excess is the operator's own dark-saturation summand. Nothing is added, nothing declared, nothing measured.

Theorem 4.122 (BM.146 — The continuum-limit function: the two variables, the five contributors, and the exact functional form). Thm.

(1) the variables and the notation row:

a cell's content is $E_{\text{int}} = |R| = n_{\mathcal{R}}$

[D0.8; Def-BM.5(4)];

its ledger flux is $S_{\mathfrak{L}} = E_{\text{int}} \ln 2$

[CT1.29; C9];

its chart span enters through the envelope ratio

$$k = \frac{r}{\Delta x_{\text{B}}(E)} = \frac{r\, E_{\text{int}} \ln 2}{\mathfrak{h}}$$

[BM.88(2)(i); Def-BM.4(1)];

the notation row: $\mathfrak{q}$ remains the dimension bookkeeper [CT2.1]; the flux variable is the ledger's own $S_{\mathfrak{L}}$ — count and flux are one datum up to $\ln 2$

(2) the five major contributors (each derived at its own locus):

contributor	the exact row	the locus
the Born width	$\Delta x_{\text{B}} = \mathfrak{h}/(E_{\text{int}} \ln 2), \quad \Delta x_{\text{B}} \cdot m_{\text{hid}} = 1$	[Def-BM.4(1); BM.13(1)]
the flicker density	$w^{(n)}_{\text{mat}} = \frac{1}{D_{\mathfrak{n}}}, \quad \langle E \rangle^{(n)}_{\text{mat}} = \frac{2n}{D_{\mathfrak{n}}}, \quad D_n = \frac{1-u^n}{1-u}$	[BM.137(1); BM.94; BM.105]
the Janus boundary	$Q = x_+(\hat{J}_+ - P_{\mathcal{V}}) + x_-(\hat{J}_- - P_{\mathcal{R}}), \quad \operatorname{spec} Q = \{0, -1\}, \quad p = 1 - 4\varepsilon_q$	[Def-BM.18; BM.120; BM.19(2)]
the Haar–Poisson lapse	$N(k,E) = \sqrt{1 - \frac{G(E)}{4\pi k}} = \sqrt{1 - \frac{\mathfrak{h} E_{\text{int}}}{2r}}$	[BM.131; BM.132(2)]
the locked sector	$O(n) = \max(0, 2^{n+1} - 16), \quad \langle \rho_{D^-} \rangle_\Gamma(p) = \sum_v O(n_v)\big(1 - f_{\text{adm}}(n_v, p)\big)$	[CT4.1; CT4.5a(1)]

(3) the continuum-limit function (the exact functional form):

the chart read of a cell is the tuple

$$\boxed{\mathcal{C}(E_{\text{int}}, r) \;=\; \Big(g_{\mu\nu}(E_{\text{int}}, r),\;\; N(k,E),\;\; \mathcal{T}(k,E),\;\; w^{(n)}_{\text{mat}},\;\; \Gamma(E,r) \Big)}$$

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{\mathfrak{h} E_{\text{int}}}{2r} \mathbb{1}_4 + O\Big(\frac{\mathfrak{h}^2 E_{\text{int}}^2}{r^2}\Big)$$

[BM.43(4); BM.132(2)];

$$N = \sqrt{1 - \frac{\mathfrak{h} E_{\text{int}}}{2r}}$$

[BM.132(2)];

$$\mathcal{T} = \frac{\mathcal{T}_0}{N} = \frac{E_{\text{int}}\, \mathfrak{h}}{\ln 2} \Big(1 - \frac{\mathfrak{h} E_{\text{int}}}{2r}\Big)^{-1/2}$$

[BM.115(1); BM.132(5)];

$$\Gamma = \Gamma_{\text{tr}}\, f\, N\, \mathrm{g}\mathcal{N}\, \chi$$

[BM.116(1); BM.133(4)];

every component is an exact closed form in $\big(E_{\text{int}},\ r,\ \mathfrak{h},\ \ln 2,\ \varepsilon_q\big)$

[machine: the monomial census];

(4) the formed domain (the exact conditions):

$$\boxed{\mathcal{C} \text{ is formed } \iff E_{\text{int}} \geq 2 \ \wedge\ \varkappa \|q\|_{\text{op}} < 1 \ \wedge\ E_{\text{int}} \leq \frac{2r}{\mathfrak{h}} \ \wedge\ n(v) \leq 3 \text{ on the accessible vertices}}$$

(C1) $E_{\text{int}} \geq 2$: the sub-hop/materialization floor [Census FT.28(1); Def-H.4; BM.13(3)];

(C2) $\varkappa \|q\|_{\text{op}} < 1$: the metric's nondegeneracy [CT1.12; CT2.6];

(C3) $E_{\text{int}} \leq 2r/\mathfrak{h}$: the lapse-formedness condition, in the capacity form

[BM.132(3); BM.148(1); machine: the cross-multiplied identity $r \geq \mathfrak{h} E/2$];

(C4) $n(v) \leq 3$: the accessibility bound — oversubscribed vertices pass their excess to D^-

[CT0.11; CT4.1; D1.11];

(5) the limit $\hbar \to 0$ (the derived sense in which the limit exists):

at fixed (E_{int}, r) :

$$E_{\text{crit}}(r) = \frac{2r}{\hbar} \to \infty$$

[machine: the Archimedean row on exact rationals];

$\Longrightarrow$ every finite-content cell becomes chart-formed:

$$\forall E_{\text{int}}\, \exists\, \hbar_0 :\ \forall \hbar < \hbar_0 :\ \text{(C3) holds}$$

[$\hbar_0 = 2r/E_{\text{int}}$ explicit];

the refinement row:

a cell of span s carries $k = \dfrac{sE_{\text{int}} \ln 2}{\hbar}$ envelopes, each with ≥ 2 directed real edges

[Def-H.4; Census FT.28(1); machine: the monomial] :

$$\boxed{\#\{\text{real edges per cell}\} \ \geq\ \frac{2sE_{\text{int}} \ln 2}{\hbar} \ \longrightarrow\ \infty \ \text{ as } \ \hbar \to 0}$$

[the witness of BM.151(1)];

(6) the universality preview:

$\mathcal{C}$ carries ¬an aspect label: the force content enters through $(T_{\mu\nu}$ channel, mask slot, coupling profile$)$ alone

[CT1.22; CT2.4; BM.150(3) forward pointer];

(7) status:

derived: (1)–(6);

¬a declaration,

¬a P4 addition,

¬a cosmological coordinate [Def-BM.17(6)(i)],

¬a measured value [P5].

Req. A8 (Part 1); C9, D0.4, D0.8 (Part 1); CT0.11, CT1.29, CT2.1 (Parts 1–2); CT1.12, CT2.6, CT4.1, CT4.5a(1) (Part 2); P1, P5 (Parts 1–2); Def-H.4, Def-H.32 (Part 3); Def-BM.4(1), Def-BM.17(6), Def-BM.18, BM.13(1), BM.13(3), BM.19(2), BM.88(2), BM.94, BM.105, BM.115(1), BM.116(1), BM.120, BM.131, BM.132(2), BM.132(3), BM.132(5), BM.133(4), BM.137(1) (this part); Census FT.28(1) (row-wise); monomial identities, exact rational limit rows [math; machine: `checks/verify_H_continuum_limit_era1_closure.py` §B].

Pf.

(1) : the count/flux identity $S_{\mathfrak{L}} = n_{\mathcal{R}} \ln 2 = E_{\text{int}} \ln 2$ [CT1.29; D0.8; machine: one monomial];

the envelope-ratio substitution $k = rE \ln 2/\mathfrak{h}$ [Def-BM.4(1); machine]

(2) : the five rows re-cited at their loci [machine: the citation census];

(3) : the substitutions G

$$= 2\pi E^2 \ln 2,$$

$$\Delta x_{\text{B}}$$

$$= \mathfrak{h}/(E \ln 2) : \quad \frac{G}{4\pi k}$$

$$= \frac{2\pi E^2 \ln 2}{4\pi} \cdot \frac{\mathfrak{h}}{rE \ln 2}$$

$$= \frac{\mathfrak{h}E}{2r}$$ [machine: the rational-function identity]

(4) : (C1)–(C4) re-cited from their own tags;

(C3) equivalence: $1 - \frac{\mathfrak{h}E}{2r}$

$$\geq 0$$

$$\iff r \geq \frac{\mathfrak{h}E}{2}$$

$$\iff E \leq \frac{2r}{\mathfrak{h}} \text{ [machine: cross-multiplication on positive denominators]}$$

$$(5):\ \mathfrak{h}_0 = 2r/E:\ \ \mathfrak{h} < \mathfrak{h}_0 \implies E < 2r/\mathfrak{h} \text{ [machine: rational witnesses]};$$

$$\text{the edge-count monomial } 2k = 2sE\ln 2/\mathfrak{h} \text{ [machine]};$$

$$(6), (7):\ \text{assembly.} \blacksquare$$

Phys. The continuum limit stops being a slogan and becomes a function. It takes two ledger variables — the cell's content and its chart span — and returns five exact closed forms: the linearized metric, the lapse, the materialization threshold, the flicker density, and the width. Its domain is four derived conditions, and the third of them is a capacity: a cell of content E is chart-readable out to radius r exactly when $E \leq 2r/\mathfrak{h}$. The limit $\mathfrak{h} \to 0$ is the statement that this capacity outruns every finite content, while the cell's real-edge count grows without bound — the two halves of the continuum condition, both witnessed inside the ledger. The function carries no aspect label: force identity lives in the source, the mask and the coupling profile, never in the limit itself.

Theorem 4.123 (BM.147 — The metric–energy–continuum map: the dictionary bijection, the closed form of the source read, and the target). Thm.

$$(1)\quad \text{the metric is a map (the dictionary):}$$

$$\big(a_{00},\ q_{ij},\ w_i\big) \in \mathcal{E}_{++} \oplus \mathcal{E}_{+-} \ \longmapsto\ g_{\mu\nu} = \Omega^2 \bar{g}_{\mu\nu},$$

$$\Omega = e^{\alpha_\Omega a_{00}},$$

$$\bar{g} = \begin{pmatrix} -1 & \varkappa N_j \\ \varkappa N_i & \delta_{ij} + \varkappa q_{ij} \end{pmatrix},$$

$$N_i = \beta w_i \cdot \Omega^2 \text{ [CT1.12; BM.34(1)]};$$

$$7 + 3 = 10 = \tfrac{4\cdot 5}{2}:$$

the dictionary is a bijection between the ten sector coefficients and the ten metric components

$$\text{[CT1.12; machine: the dimension count]};$$

$$\mathrm{sig}(g) = (-,+,+,+) \iff \varkappa\|q\|_{\mathrm{op}} < 1 \text{ on the perturbative regime}$$

[CT1.12];

$$(a_{00}, q, w) = 0 \longmapsto \text{the flat metric}$$

[BM.34(2)];

(2) the energy determines the map (the closed form on the dust source):

$$T_{\mu\nu} = \mathrm{diag}(\rho, 0, 0, 0),$$

$$\rho = \frac{m}{V_{\mathrm{eff}}},$$

$$m = \frac{E_{\mathrm{int}} \ln 2}{\mathfrak{h}}$$

[BM.9(4); BM.14; FT.44(4)];

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa^2 T_{\mu\nu}$$

[CT1.23];

$$\nabla^2 \Phi = 4\pi G_N \rho,$$

$$\Phi = -\frac{G_N m}{r},$$

$$G_N = \frac{\mathfrak{h}^2}{4 \ln 2},$$

$$G_N m = \frac{\mathfrak{h} E_{\mathrm{int}}}{4}$$

[BM.43(4); P3; CT2.9];

$$\boxed{g_{00} = -\Big(1 - \frac{\mathfrak{h} E_{\mathrm{int}}}{2r}\Big), \qquad g_{ij} = \Big(1 + \frac{\mathfrak{h} E_{\mathrm{int}}}{2r}\Big)\delta_{ij}, \qquad g_{0i} = 0}$$

[machine: the substitution $r_s = \mathfrak{h} E_{\mathrm{int}}/2$ in BM.43(4)'s weak-field rows];

the pure-trace row:

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{\mathfrak{h} E_{\text{int}}}{2r} \mathbb{1}_4 \; + \; O\Big(\frac{\mathfrak{h}^2 E^2}{r^2}\Big) :$$

the linearized self-field of the dust cell is a pure-trace (conformal) perturbation

its whole content sits in the conformal mode:

$$\Omega^2 = 1 - \frac{\mathfrak{h} E_{\text{int}}}{2r},$$

$$a_{00} = \frac{1}{2\alpha_\Omega} \ln\Big(1 - \frac{\mathfrak{h} E_{\text{int}}}{2r}\Big),$$

$$\hat{q} = 0,$$

$$w = 0$$

[machine: the exact solve of $\gamma_{ij} = \Omega^2(\delta_{ij} + \varkappa q_{ij})$ against g_{ij}];

the energy–lapse identity:

$$N^2 = \Omega^2 = 1 - \frac{\mathfrak{h} E_{\text{int}}}{2r} = 1 - \frac{G(E)}{4\pi k}$$

[BM.132(2); machine: the rational-function identity of (3)'s rows];

(3) the inverse read (the energy from the metric):

$$R = \kappa^2(\rho - 3p) + 4\Lambda,$$

$$p = 0, \; \Lambda = 0 :$$

$$RV = \kappa^2 m$$

[BM.15(2); BM.20(1)] :

$$\boxed{m = \frac{1}{\kappa^2}\int_\Sigma R\, dV}$$

[BM.20(1)];

$$\kappa^2\rho = \frac{2\pi\,\mathfrak{h}\,E_{\text{int}}}{V},$$

$$(\Delta x_{\text{B}})^2\kappa^2\rho = 2\pi E_{\text{int}}^2 \ln 2 = G$$

[BM.9(5); BM.16(1); machine: monomials];

the map is invertible on-shell at the leading reads:

$$E_{\text{int}} = \frac{\kappa^2\rho V}{2\pi\mathfrak{h}} = \sqrt{\frac{G}{2\pi\ln 2}}$$

[machine: the two exact inverses];

(4) the continuum is the target:

$$M = \Gamma_{\mathcal{R}}/\!\sim,$$

$$\chi([v]) = (\tau_n, h_1, h_2, h_3),$$

transitions smooth with $|\det J| = 1$

[CT2.6; D1.8];

the target-space rows:

the image of the dictionary is the Lorentzian metrics of signature $(-,+,+,+)$ on the perturbative regime [CT1.12];

the dispersion target $E = N\sqrt{m^2+p^2} - N^i p_i$ [BM.34(4)];

the map's full statement:

$$\boxed{\text{metric} \;=\; \text{dictionary} \circ \text{ field-equation solve} \circ \text{ source read } (E_{\text{int}}, r) \;\longmapsto\; \text{continuum}}$$

$[(1) + (2) + (4);$ the nonlinear joint solve remains ¬constructed [BM.43(5) re-cited]];

(5) status:

derived: (1)–(4);

¬a declaration,

¬a P4 addition,

r_s quarantined as P3's named combination,

¬a measured value [P5].

Req. A8 (Part 1); D1.8, D2.1 (Part 1); CT1.12, CT1.23, CT2.6, CT2.9 (Part 2); P3, P5 (Parts 1–2); FT.44(4) (Census); Def-BM.4, BM.9(4), BM.9(5), BM.14, BM.15(2), BM.16(1), BM.20(1), BM.34(1), BM.34(2), BM.34(4), BM.43(4), BM.43(5), BM.115(1), BM.132(2) (this part); the exact weak-field solve, monomial and rational-function identities [math; machine: `checks/verify_H_continuum_limit_era1_closure.py` §C].

Pf.

(1) : CT1.12's bijection row and signature criterion re-cited [machine: the dimension count];

$$(2) : \ \Phi = -G_N m/r \text{ with } G_N m = \frac{\mathfrak{h}^2}{4\ln 2} \cdot \frac{E\ln 2}{\mathfrak{h}} = \frac{\mathfrak{h}E}{4} \text{ [P3; FT.44(4); machine]};$$

$$g_{00} = -(1+2\Phi) = -\big(1 - \frac{\mathfrak{h}E}{2r}\big), \quad g_{ij} = (1-2\Phi)\delta_{ij} = \big(1 + \frac{\mathfrak{h}E}{2r}\big)\delta_{ij} \text{ [BM.43(4); machine]}$$

the pure-trace row: $h_{00} = +\frac{\mathfrak{h}E}{2r} = h_{11} = h_{22} = h_{33}$ [machine: the component census];

the conformal solve: $\Omega^2\big(\delta_{ij} + \varkappa q_{ij}\big)$

$$= \big(1 + \frac{\mathfrak{h}E}{2r}\big)\delta_{ij} \ \wedge \ \Omega^2$$

$$= 1 - \frac{\mathfrak{h}E}{2r} : \quad \varkappa q_{ij}$$

$$= \Big(\frac{1+\mathfrak{h}E/(2r)}{1-\mathfrak{h}E/(2r)} - 1\Big)\delta_{ij}$$

$$= \frac{\mathfrak{h}E/r}{1-\mathfrak{h}E/(2r)}$$

δ_{ij} [machine: the rational-function identity]

$(3):$ BM.15(2), BM.20(1), BM.9(5), BM.16(1)'s monomials re-verified [machine];

the two inverses: $\dfrac{\kappa^2}{2\pi\mathfrak{h}} \cdot \dfrac{2\pi\mathfrak{h}E}{V} \cdot V = E$;

$$\Big(\sqrt{\frac{G}{2\pi\ln 2}}\Big)^2 \cdot 2\pi\ln 2 = G \text{ [machine]}$$

$(4), (5):$ assembly and the scope re-citations. ■

Phys. The three names in the directive are one chain. The metric is the dictionary: ten sector coefficients of the ledger's pair states, bijectively repackaged as ten metric components. The energy determines the map: run the composite's own dust source through the field equations and the linearized solve closes in one monomial — the perturbation is pure trace, entirely conformal, the lapse and the conformal mode carrying the same exact function of content over radius. The map inverts on-shell: the curvature flux reads the content back. The continuum is the target: the quotient manifold with its tick-and-hop charts, whose metric image is exactly the Lorentzian signatures on the perturbative regime. The full non-linear solve stays unconstructed, and nothing measured enters anywhere.

Theorem 4.124 (BM.148 — The breaking threshold: the exact capacity of a chart cell). Thm.

(1) the threshold (three exact forms of one condition):

the chart read of a cell is formed $\iff r \geq r_s(E) = \dfrac{\mathfrak{h}E_{\text{int}}}{2}$

[BM.132(3); BM.147(2)] :

$$\boxed{E_{\text{crit}}(r) = \frac{2r}{\mathfrak{h}}} \qquad \boxed{S_{\text{crit}}(r) = \frac{2r\ln 2}{\mathfrak{h}}} \qquad \boxed{k_{\text{crit}}(E) = \frac{E_{\text{int}}^2 \ln 2}{2} = \frac{G(E)}{4\pi}}$$

the flux form:

the map breaks $\iff S_{\mathfrak{L}}(\text{cell}) > S_{\text{crit}}(r)$

$$[S_{\mathfrak{L}} = E_{\text{int}} \ln 2;\ \text{CT1.29};\ \text{machine: the cross-multiplied identity}];$$

(2) the dimension census:

$$[E_{\text{crit}}] = \frac{\mathfrak{q}^{-1/2}}{\mathfrak{q}^{-1/2}} = \mathfrak{q}^{0}\ (\text{a count});$$

$$[S_{\text{crit}}] = \mathfrak{q}^{0}\ (\text{nats});$$

$$[r_{\text{crit}}] = \mathfrak{q}^{-1/2}\ (\text{a length})$$

$$[\text{CT2.1};\ \text{machine: the exponent arithmetic}];$$

(3) the integer row:

$$E_{\text{int}} \in \mathbb{Z} \implies \text{the formed contents at radius } r \text{ are } \{2, \dots, \lfloor 2r/\mathfrak{h} \rfloor\}$$

$$[\text{BM.13}(3):\ E \geq 2;\ \text{machine}];$$

the one-hop capacity:

$$r = \mathfrak{h}: \quad E_{\text{crit}}(\mathfrak{h}) = 2:$$

a chart cell of one hop carries exactly the digon floor and nothing heavier

$$[\text{Census FT.28};\ \text{machine}];$$

the computed cells:

$$r_s = \Big(1,\ 3,\ 6,\ 12,\ 30\Big)\,\mathfrak{h}$$

at $E_{\text{int}} \in \{2, 6, 12, 24, 60\}$

$$[\text{BM.43}(1);\ \text{machine: the integer census}];$$

(4) the envelope corollary (the second route to BM.132(3)'s row):

$$r = \Delta x_{\mathrm{B}}(E) = \frac{\mathfrak{h}}{E \ln 2} :$$

$$E \le E_{\mathrm{crit}}\big(\Delta x_{\mathrm{B}}(E)\big) = \frac{2}{E \ln 2} \iff E^2 \ln 2 \le 2 :$$

$$E_{\mathrm{int}} \ge 2 \implies E^2 \ln 2 \ge 4 \ln 2 = \ln 16 > 2$$

$$\Big[\ln 16 > 2 \iff 16 > e^2 : \quad e^2 < \Big(\frac{876809}{322560}\Big)^2 < 8 < 16;\ \text{machine: the exact enclosure}\Big];$$

no materialized cell is chart-formed at its own envelope: the capacity $\frac{2}{E \ln 2}$ is below every admissible E

(5) the monotonicity and the limit:

$E_{\mathrm{crit}}(r)$ strictly increasing in r;

$S_{\mathrm{crit}}(r)$ strictly increasing;

$r_{\mathrm{crit}}(E)$ strictly increasing in E;

$k_{\mathrm{crit}}(E)$ strictly increasing and quadratic

[machine: the sign census on exact rationals];

$$\lim_{\mathfrak{h} \to 0} E_{\mathrm{crit}}(r) = \infty \ \text{ at fixed } r$$

[BM.146(5); machine];

the breaking locus recedes without bound in the continuum limit

(6) the quarantine rows:

r_s enters only as P3's named combination: ¬a collapse reading, ¬a horizon identification, ¬black-hole physics [P3; BM.132(6); BM.115(4) discipline];

the breaking locus is a domain boundary of the chart reads — the ledger side of the same boundary is BM.149;

(7) status:

derived: (1)–(6);

¬a declaration,

¬a P4 addition,

¬a numerical $\mathfrak{h}$ [P1; P5].

Req. D0.8 (Part 1); CT1.29, CT2.1 (Parts 1–2); P1, P3, P5 (Parts 1–2); Def-BM.4, BM.13(3), BM.43(1), BM.88(3), BM.115(4), BM.132(3), BM.132(6), BM.146(5), BM.147(2) (this part); Census FT.28 (row-wise); the exact rational enclosure of e^2, cross-multiplied identities, integer censuses [math; machine: `checks/verify_H_continuum_limit_era1_closure.py` §D].

Pf.

$$(1):\ r \geq \frac{\mathfrak{h}E}{2}$$

$$\iff E \leq \frac{2r}{\mathfrak{h}}$$

$$\iff E\ln 2 \leq \frac{2r\ln 2}{\mathfrak{h}}\ \text{[machine: cross-multiplication on positive denominators]};$$

$$k_{\text{crit}}:\ \frac{G}{4\pi} = \frac{2\pi E^2 \ln 2}{4\pi} = \frac{E^2\ln 2}{2}\ \text{[BM.88(3); machine]}$$

$$(2):\ [\mathfrak{h}] = \mathfrak{q}^{-1/2}\ \text{[P1; CT2.1]}:\ \text{the three exponent rows [machine]};$$

$$(3):\ r_s = \mathfrak{h}E/2 \text{ at } E \in \{2,6,12,24,60\}:\ (1,3,6,12,30)\,\mathfrak{h}\ \text{[machine: integers]}$$

$$(4):\ E^2\ln 2 \leq 2 \text{ has no integer solution } \geq 2:$$

$$4\ln 2 = \ln 16 > 2 \iff 16 > e^2\ \text{[BM.132(3)'s enclosure re-run; machine]};$$

$$(5):\ \text{the derivative signs on exact rationals};$$

$$(6),(7):\ \text{assembly.}\ \blacksquare$$

Phys. The continuum has a capacity, and the capacity is exact: a chart cell of radius r carries at most $2r/\mathfrak{h}$ of internal count, equivalently $2r\ln 2/\mathfrak{h}$ nats of ledger flux, before its own metric read leaves the formed domain. At one hop of radius the capacity is precisely the digon floor — the

lightest materialized cell there is also the heaviest admissible one. No materialized cell fits inside its own envelope: the capacity at the Born width is below every admissible content, a second, purely counting route to the envelope row. And in the continuum limit the breaking locus recedes without bound, which is the exact sense in which the limit exists. The Schwarzschild combination stays quarantined; the boundary is a domain edge of the chart reads, and its ledger face is the next theorem.

Theorem 4.125 (BM.149 — The locked sector at the breaking locus: the two faces of one capacity, the absorption, the membrane, and the complement theorem). Thm.

(1) the two faces of one capacity:

the face	the capacity	the locus
chart-side	$E_{\text{int}} \le 2r/\hbar$: the metric resolution of a cell	[BM.148(1)]
ledger-side	$k_v \le 3$ per accessible vertex; $D_{\text{tick}}(n) \le C_{\text{cap}} = 16$ per tick	[CT0.11; CT0.5; CT0.6; D1.11]

the excess rows:

chart-side excess: the read is ¬formed [BM.132(3)];

ledger-side excess: $O(n) = \max(0, 2^{n+1} - 16) > 0$ at $n \ge 4$ falls to D^- [CT4.1; D4.1];

(2) the absorption (exact rows):

rate $\propto O(n)$

[D4.2];

$O(4) = 16 = C_{\text{cap}}$: exactly one further channel quantum per tick

[CT4.1];

$$\langle \rho_{D^-} \rangle_\Gamma(p) = \sum_v O(n_v) \big(1 - f_{\text{adm}}(n_v, p)\big),$$

$$f_{\text{adm}}(n, p) = \sum_{k \le 3} \binom{n}{k} p^k (1-p)^{n-k}$$

[CT4.5a(1) : an exact polynomial in p with integer coefficients];

the anchors re-derived:

$$\langle \rho_{D^-} \rangle_\Gamma(1) = \big(448,\ 896,\ 2976,\ 2240,\ 24384\big)$$

on the five Platonic skeletons

[CT4.5a(2); machine: $4\,O(6),\ 8\,O(6),\ 6\,O(8),\ 20\,O(6),\ 12\,O(10)$ with $O(m) = 2^{m+1} - 16$];

(3) the one-way membrane (re-derived):

$$\forall n \le 3: \quad \mathcal{P}_M^{(n)} F_{D^-} \mathcal{P}_M^{(n)} = 0;$$

$\neg\exists$ a pole of $Z(\mathcal{R}, n)$ carrying locked content

[CT4.2];

locked content: physical norm 0 at accessible rungs,

$w(C) \neq 0$ in the unmasked A8 sum,

$\neg\exists$ a deleted entry

[CT4.3; CT2.8];

(4) the maximal locked entropy:

the entropy triangle's third vertex:

$$D^-: \ \big(S_{\text{raw}}, S_{\mathfrak{L}}\big) = \big(\min_{\text{acc}},\ \max \wedge \neg\text{accessible}\big)$$

[CT4.7];

$S_{\mathfrak{L}} = n_{\mathcal{R}} \ln 2$ is retained in full on absorption

[CT4.3: absorbed $\subseteq \mathfrak{L}$; CT1.29];

the accessible per-vertex content is bounded by $n(v)$ [D1.11];

(5) the dynamic emission:

at the dynamic level the move system carries the emission and dissolution moves $N^{\pm}(v,S)$ with formal weights $y_d,\ z_d x_-^{r(v)}$

[Def-H.32(7); FT.108];

the scope row:

emission is a ledger move, ¬an accessible-mask return: re-entry to the accessible sector runs only through the A4 crossing and the register's moves $\hat{J}_{\pm}$

[CT4.2; Def-BM.18; BM.120; machine: the citation census];

(6) the two scales:

the locked sector gravitates as pressureless dust: $T_{\mu\nu}^{(D^-)} = \rho_{D^-} u_\mu u_\nu$,

$$w_{D^-} = 0, \quad \rho_{D^-} \geq 0 \ \big(= 0 \iff \neg\exists v:\ n(v) \geq 4\big)$$

[CT4.5];

at the local scale the same sector is the membrane of (3)

[CT4.2];

the junctions at $n \geq 4$ carry its lock content: retained, ¬accessible

[BM.2(4); D1.11; Census FT.35 : the first D^--carrying floor];

(7) the complement theorem (the derived row of this tag):

for a cell whose content exceeds the chart capacity $\big(E_{\text{int}} > 2r/\mathfrak{h}\big)$:

(i) the chart read is ¬formed [BM.148(1); BM.132(3)];

(ii) the ledger read stays exact and non-decreasing: $S_{\mathfrak{L}} = E_{\text{int}} \ln 2$,

$$|\mathfrak{L}(\tau_{n+1})| \geq |\mathfrak{L}(\tau_n)|$$

[CT1.29; A6];

(iii) the accessible content is membrane-governed: ¬∃ an accessible mask returning locked content [CT4.2; CT4.3];

what the metric cannot read, the ledger still retains: the breaking of the continuum map is ¬a loss of content — the content passes to the locked species, chart-side unreadable and ledger-side exact

(8) status:

derived: (1)–(7);

¬a declaration,

¬a P4 addition,

¬black-hole physics [P3 discipline; BM.148(6)],

¬a measured value [P5].

Req. A6, A8 (Part 1); D0.9, D1.11, D4.1, D4.2 (Parts 1–2); CT0.5, CT0.6, CT0.11, CT1.29 (Part 1); CT2.8, CT4.1, CT4.2, CT4.3, CT4.5, CT4.5a, CT4.7 (Part 2); P3, P5 (Parts 1–2); Def-H.32(7), FT.108 (Part 3); Def-BM.18, BM.2(4), BM.120, BM.132(3), BM.148(1), BM.148(6) (this part); Census FT.35 (row-wise); exact integer and polynomial censuses [math; machine: `checks/verify_H_continuum_limit_era1_closure.py` §E].

Pf.

(1) : the two capacity rows re-cited from their own tags;

(2) : $O(n) = \max(0, 2^{n+1} - 16)$ at $n = 4..10$ [machine: exact integers];

the anchors: $O(6) = 112,\ O(8) = 496,\ O(10) = 2032$:

$$4 \cdot 112 = 448,\ 8 \cdot 112 = 896,\ 6 \cdot 496 = 2976,\ 20 \cdot 112 = 2240,\ 12 \cdot 2032 = 24384$$ [machine]

(3), (4) : CT4.2, CT4.3, CT4.7's rows re-derived at their loci [machine: the citation pins];

(5) : Def-H.32(7)'s moves with CT4.2's mask row

(6) : CT4.5's dust rows re-cited;

(7) : (i)–(iii) conjoined on the common domain $E > 2r/\mathfrak{h}$ [machine: the domain identity];

(8) : assembly. ■

Phys. The breaking of the continuum map is not a hole in the ledger — it is the locked sector's entrance. The same capacity theorem has two faces: the chart can resolve a cell only out to its content in hop units, and a vertex can expose only three incident quanta to an accessible mask; the excess on the first face is unreadable geometry, on the second it is absorption into the dark-saturation summand at an exact power-of-two rate, with the Platonic anchor values recomputed to the integer. The membrane is one-way: no accessible mask returns the content, no accessible pole carries it, yet nothing is deleted — the record survives with its full ledger entropy, at the maximal-locked vertex of the entropy triangle. Emission exists as a ledger move, but re-entry runs only through the A4 crossing. And the complement theorem states the whole design in one row: what the metric cannot read, the ledger still retains.

Theorem 4.126 (BM.150 — The universality of the continuum limit: one operator, one arena, one action, three fingerprints). Thm.

(1) one operator:

F always applies in full, and its blocks are exhausted:

$3 + 3 + 5 + 3 + 1 + 1 = 16 = C_{\text{cap}}$ on the pair arena;

EM, strong, D^- on the stack registers

[A8; BM.52(1); CT0.12];

$\neg\exists$ an aspect outside the census [CT0.13; FT.83(2)–(3)];

$\neg\exists$ a further block inside $\mathfrak{H}(\mathbb{C}^4)$ [CT0.12 : the tiling is exact];

(2) one arena and one action:

M, the charts (τ_n, h_1, h_2, h_3), and the dictionary are aspect-blind:

the weights depend on $(\Gamma,$ registrations$)$ only $\Longrightarrow$ relabeling-invariant $\Longrightarrow$ diffeomorphism-invariant

[CT2.6; CT2.7 Pf];

the Lovelock closure:

$S_{\text{geom}} = \dfrac{1}{2\kappa^2}\displaystyle\int \sqrt{-g}\,(R - 2\Lambda)$ with $\neg\exists$ additional local higher-curvature terms in $D = 4$

[CT2.7 : the target dynamics is unique and force-independent];

(3) three fingerprints carry the force identity:

aspect content enters the continuum exactly through $(T_{\mu\nu}$'s channel, the mask slot, the coupling profile$)$

(i) $T_{\mu\nu} = T^{\text{bd}}_{\mu\nu} + T^{\text{tr}}_{\mu\nu}$:

the bound channel $(\rho + p)u_\mu u_\nu + p g_{\mu\nu}$ against the transit channel $\displaystyle\sum_a \big(F^{(a)}_{\mu\alpha} F^{(a)\ \alpha}_{\nu} - \tfrac{1}{4} g_{\mu\nu} F^2\big)$

[CT1.22; D1.5];

(ii) the mask slot: $\mathrm{F}^{(n)}_M = \displaystyle\bigoplus_{A \in \mathcal{A}(n)} F_A|_{\mathcal{K}^{(A)}}$,

$\dim \mathcal{K}_n = (6, 16, 17, 25)$

[CT2.4; CT0.14; BM.36(1)];

(iii) the coupling profile: $e(r, E) = e_0\big[Nf/(1 + G)\big]^{1/2}$,

(g, g') family forms,

the channel constants

[BM.133(6); BM.39(1); BM.80(1)];

(4) the per-force rows (each a read of the same map):

aspect	its continuum read	the locus
gravity	the field equations themselves: $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa^2 T_{\mu\nu}$	[CT1.23]
the uniform dark aspect	$F_{D^+} = \Lambda\mathbb{I}$: excluded from Z_{force}, carried by Λ in S_{geom}	[A8; D4.3; CT1.22]
EM	$\partial_\mu F^{\mu\nu} = eJ^\nu_Q$: the Coulomb read outside the support	[Def-BM.12(1)–(3); EM.15]
weak	the shift block: $N_i = \Omega^2 \varkappa\beta w_i$ — the weak aspect is the metric's own g_{0i}	[FT.157(1); CT1.12; BM.34(3)]
colour	sealed content only: the trace law $\mathfrak{T}(H) = 3a$; ¬a triplet pole, physical norm 0	[CT3.22; BM.1(2); FT.117]
Janus	$\partial_\mu J^\mu = 0$ with J^0 = the seal density, J^i = the hop current on the w_i carrier	[BM.7; BM.11(1)–(3)]
the locked sector	$T^{(D^-)}_{\mu\nu} = \rho_{D^-} u_\mu u_\nu, \quad w_{D^-} = 0$	[CT4.5]

(5) the pp-wave witness (the consistency check that names no force):

the nonlinear contracted Bianchi identity $\nabla_\mu G^{\mu\nu}$

$= 0$ holds on the exact rational pp-wave: $\det g$

$= -1$ identically, metric compatibility on all 64 components, the Λ row $\nabla_\mu(\Lambda g^{\mu\nu}) \qquad = 0$

[BM.11(4); machine re-cited];

the witness is source-free:

the identity is a property of the dictionary image, ¬of any aspect's content — it holds whatever filled $T_{\mu\nu}$

[CT1.23 Pf's δ-identity structure; machine: the citation census];

(6) the state-side row:

the Lyapunov contraction splits no class: $\Delta m_{\text{Lyapunov}} = 0$ on every read species at once

[BM.51(6) : every input is class-shared];

the duality faces are aspect-blind [BM.131(3), (5)];

(7) the universality theorem:

the continuum limit is one map with one domain law: force identity enters exactly through (3)'s three fingerprints and never through the arena, the action, the domain law, or the limit

the per-force consequence:

the colour, weak, EM and gravitational reads share C's formed domain [BM.146(4)] and CT2.7's unique action — what differs is the source channel, the mask slot, and the coupling profile, all three derived

[(4)'s table];

(8) status:

derived: (1)–(7);

¬a declaration,

¬a P4 addition,

¬a measured value [P5],

¬an identification of any aspect with an external force law [P5; the names are the corpus's own].

Req. A8 (Part 1); D1.5, D4.3 (Parts 1–2); CT0.13, CT0.14, CT1.22, CT1.23, CT2.4, CT2.6, CT2.7, CT3.22, CT4.5 (Part 2); P5 (Parts 1–2); EM.15 (Part 2); FT.117, FT.157(1), FT.83(2), FT.83(3) (Part 3); Def-BM.12, BM.1(2), BM.7, BM.11(1), BM.11(3), BM.11(4), BM.34(3), BM.36(1), BM.39(1), BM.51(6), BM.52(1), BM.80(1), BM.131(3), BM.131(5), BM.133(6), BM.146(4) (this part); the citation censuses [machine: `checks/verify_H_continuum_limit_era1_closure.py` §F].

Pf.

(1), (2) : the censuses re-run and CT2.7's uniqueness re-cited [machine: the integer sums];

(3) : the three fingerprints assembled from CT1.22,

CT2.4,

BM.133(6) and the family forms [machine: the locus census]

(4) : each row re-cited from its own derived tag [machine: the table's citation pins];

(5) : BM.11(4)'s witness rows re-cited; the source-freeness is the δ-identity structure of CT1.23's Pf [machine: the citation census]

(6) : BM.51(6)'s class-degenerate zero re-cited; $(7), (8)$: assembly. ■

Phys. Universality is not an assumption about forces agreeing — it is a structural census. There is one operator, its blocks tile the arena exactly, and the stack aspects sit on their own registers. There is one target manifold, one dictionary, and one geometric action, unique by the Lovelock count, with no term that could carry a force label. Force identity has exactly three homes: which channel of the stress tensor the aspect fills, which slot of the mask ladder it occupies, and which coupling profile it runs. The per-force table reads all seven aspects through the same map, and the pp-wave witness certifies the identity that no source ever signs: the contracted Bianchi row holds because of what the dictionary image is, not because of what fed it. The state-side dynamics agrees: the contraction splits no class. The continuum limit is the same limit for every force, and the differences are derived fingerprints, not different limits.

Theorem 4.127 (BM.151 — The derivation of CT2.6(ii)–(iii): the unbounded ladder, the spectral bound, and the empty extra-axiomatic list). Thm.

(1) condition (ii) is a theorem of the ledger's own configuration space:

(ii): coarse-graining cells contain unboundedly many real edges as $\mathfrak{h} \to 0$

[CT2.6 statement];

(a) the configuration space:

A7 admits $\Gamma = (V, E)$ at every scale;

$R \subseteq E_{\text{dir}}$ is unconstrained in size;

$$E_{\text{int}} = |R|$$

[A7; D0.10; D0.8; Def-H.32(1)–(2)];

(b) the witness ladder:

the cycle floors Σ_L exist at every L with $E_{\text{int}} = L$

[FT.32];

the polygon family is unbounded with injective mass/rank/null reads and ¬a combinatorial or geometric ceiling

[FT.45; FT.46];

$$\boxed{\forall n \ \exists \text{ an admissible cell configuration with } E_{\text{int}} \geq n}$$

[machine: the constructive witness $R = E_{\text{dir}}(\Sigma_L)$ at $L = 2..8$];

(c) the refinement row:

at fixed chart span s :

$$k = \frac{s}{\Delta x_{\text{B}}} = \frac{sE_{\text{int}} \ln 2}{\hbar} \text{ envelopes per cell}$$

[Def-BM.4(1); FT.44(4)];

every materialized chain carries ≥ 2 directed real edges

[Def-H.4; Census FT.28(1)] :

$$\#\{E_{\mathcal{R}} \text{ per cell}\} \ \geq \ \frac{2sE_{\text{int}} \ln 2}{\hbar} \ \xrightarrow[\hbar \to 0]{} \ \infty$$

[machine: the Archimedean row on exact rationals];

(d) the scope audit:

the witness rows (a)–(c) are combinatorial and card content: ¬one of them uses CT2.6's conclusion (M a manifold) as premise;

the CT2.6-citations of Def-BM.4,

BM.13,

FT.47 are chart-arena inheritances (scope rows),

¬uses of (ii)/(iii) as premises

[machine: the citation-pattern scan];

the materialization hypotheses (M1/M2 species) remain declared spectral scopes at their own loci, untouched and distinct

[A §0.2a integration note; Def-H.20];

(2) condition (iii) is a theorem of the pair-state spectral bound:

(iii): sector coefficient fields [D2.1] of bounded variation across neighbouring cells

[CT2.6 statement];

(a) the coefficients are state components:

$$\rho = \tfrac{1}{4} I + \tfrac{1}{4} \sum_a r_a G_a,$$

$$r_a = \mathfrak{T}(\delta\rho\, G_a) \in \mathbb{R},$$

$$|r|^2 = 4\,\mathfrak{T}(\delta\rho^2)$$

[L3(1); CT1.1 : $\mathfrak{T}(G_a G_b) = 4\delta_{ab}$];

(b) the spectral bound:

$$\text{spec}(\delta\rho) \subset \left[-\tfrac{1}{4}, \tfrac{3}{4}\right],$$

$$\textstyle\sum_i \mu_i = 0$$

[A1 : $\rho \succeq 0$, $\mathfrak{T}\rho = 1$; A5; L1] :

max $\sum_i \mu_i^2 = \frac{3}{4}$ attained exactly at the vertex orbit $\left(\frac{3}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}\right)$

[machine: the vertex enumeration of the polytope $\{\sum \mu = 0\} \cap [-\frac{1}{4}, \frac{3}{4}]^4$ and convexity of $\sum \mu^2$];

$$\boxed{|r_a| \;\le\; |r| \;=\; 2\sqrt{\textstyle\sum_i \mu_i^2} \;\le\; 2\sqrt{\tfrac{3}{4}} \;=\; \sqrt{3}} \qquad \text{[machine: } |r|^2 \le 3 \text{ exact]};$$

the component chart:

$$\rho = \tfrac{1}{4} \sum_{\mu\nu} c_{\mu\nu}\, \sigma_\mu \otimes \sigma_\nu,$$

$$c_{\mu\nu} = \mathfrak{T}\big(\rho\, \sigma_\mu \otimes \sigma_\nu\big),$$

$$|c_{\mu\nu}| \le \big\|\sigma_\mu \otimes \sigma_\nu\big\|_{\text{op}} = 1$$

[math: the state-functional bound; machine: exact rational density-matrix samples];

the D2.1 row:

the sector coefficients (a_{00}, q_{ij}, w_i) are fixed rational projections of either chart [D2.1 : the basis row; CT1.2] :

uniformly bounded by explicit rational constants, ¬free data;

(c) the fields are averages:

coefficients = local ledger averages

[CT2.6 Pf row];

convex combinations preserve the ball: $\|\text{field}\| \le \sqrt{3}$ per coefficient;

(d) the variation bound:

$$\boxed{|\Delta \text{coefficient}| \;\le\; 2\sqrt{3} \text{ across any two cells}, \qquad \text{TV}(m \text{ neighbouring cells}) \;\le\; 2\sqrt{3}\, m}$$

[machine: $(2\sqrt{3})^2 = 12$ exact; the triangle inequality];

(e) the tighter formed-locus census:

$$\varkappa\|q\|_{\mathrm{op}} < 1 \text{ [CT1.12]};$$

$$N \in (0,1] \text{ [BM.132(4)]};$$

$$w_{\mathrm{mat}}^{(n)} \in \left(\tfrac{1}{n}, 1\right] \text{ [BM.137(1)]};$$

$$p \in (0,1] \text{ [D0.4; BM.19(2)]};$$

$$f \in [0,1) \text{ [BM.88(2)(ii)]};$$

$$\mathsf{g} \in (0,1] \text{ [BM.133(4)]};$$

$$\Gamma_{\mathrm{tr}} \in \left[0, \tfrac{1}{3}\right] \text{ [BM.44(2); BM.116(2)]};$$

$$w_{\mathrm{mat}} \in \left[\tfrac{1}{3}, 1\right] \text{ measure-free [BM.105(2)–(4)]};$$

every monotone row's total variation equals its range:

$$\text{the exact variation widths are } \left(1,\ 1-\tfrac{1}{n},\ 1,\ 1,\ 1,\ \tfrac{1}{3},\ \tfrac{2}{3}\right) \text{ on } \left(N,\ w_{\mathrm{mat}}^{(n)},\ p,\ f,\ \mathsf{g},\ \Gamma_{\mathrm{tr}},\ w_{\mathrm{mat}}\right)$$

[machine: the monotone-BV identity on exact rationals];

(f) the dynamics row:

the Haar face preserves the coefficient spectrum and the tilt face contracts it:

$$\mu' = \left(1 - \tfrac{\gamma}{4}\right)\mu + O(\mu^2),$$

$$\left(1 - \tfrac{\gamma}{4}\right) \le 1$$

[BM.131(3); L2; machine] :

the bound ball $\{|\mu| \le \frac{3}{4}\}$ is invariant under both faces — the recorded dynamics never violates (b);

(3) the derivation:

CT2.6(ii), (iii) are theorems at the stated loci: (ii) by (1), (iii) by (2)

the extra-axiomatic list of A §0.2a is empty;

CT2.6's statement and Pf stand unmodified in their conclusions — the conditions are now supplied by derivation rather than by declaration;

the in-place amendments:

A §0.2a's list sentence and CT2.6's two condition labels carry the derivation markers [the FT.62/FT.66 patch pattern];

the point-of-use declarations across the corpus stand as scope-inheritance markers over the same loci;

(4) the packaging:

¬a declaration is issued by this tag;

¬a P4 member;

$\hbar$ remains the single dimensionful scale [P1];

¬a measured value [P5];

no tag loses content:

every tag inheriting CT2.6's hypotheses inherits the same conditions, now derived at the same scope [machine: the inheritance census];

(5) the gap row:

the continuum-limit gap — the extra-axiomatic entry of A §0.2a,

the corpus's last non-axiomatic import — is closed;

the finite-$\hbar$ residue is distinct and stands [the clean-bifurcation loci; FT.66(3); BM.143(7)];

(6) status:

derived: $(1), (2)$;

adjudicated: (3)–(5);

the axiom surface is now the entire non-derived surface of the corpus.

Req. A1, A5, A7 (Part 1); D0.4, D0.8, D0.10 (Part 1); CT1.1, CT1.2, CT1.10, CT1.29, L1, L2 (Part 1); CT1.12, CT2.6, D2.1 (Part 2); P1, P5 (Parts 1–2); Def-H.4, Def-H.20, Def-H.32(1), Def-H.32(2), FT.32, FT.44(4), FT.45, FT.46, FT.66(3), L3(1) (Part 3); Def-BM.4(1), BM.13, BM.19(2), BM.44(2), BM.47(6), BM.88(2), BM.105, BM.116(2), BM.131(3), BM.132(4), BM.133(4), BM.137(1), BM.143(7) (this part); Census FT.28(1) (row-wise); vertex enumeration of a rational polytope, convexity of the quadratic form, the monotone-BV identity, Archimedean rows on exact rationals [math; machine: `checks/verify_H_continuum_limit_era1_closure.py` §G].

Pf.

$$(1)(\mathrm{b}):\ \Sigma_L \text{ at } L = 2..8:\ \text{ the unidirectional } L\text{-cycle with } R = E_{\mathrm{dir}}(\Sigma_L):$$

$$|R| = L \text{ [FT.32; machine: the constructive witness on the cycle skeletons]};$$

$$(\mathrm{c}):\ \frac{2sE\ln 2}{\mathfrak{h}}$$

$$\geq T \iff \mathfrak{h} \leq \frac{2sE\ln 2}{T}:\ \text{ witness } \mathfrak{h}$$

$$= \frac{2sE\ln 2}{T+1} \text{ [machine: exact rationals at } T = 10^6]$$

$(2)(\mathrm{b}):$ the polytope $\{\mu \in [-\frac{1}{4}, \frac{3}{4}]^4 : \sum \mu_i = 0\}$ has the vertex orbit $\binom{4}{1}(\frac{3}{4}, -\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}):$ any vertex fixes three coordinates at bounds; two $+\frac{3}{4}$'s force the fourth to $-\frac{3}{2} < -\frac{1}{4}$ [machine: the exhaustive bound-pattern enumeration];

$$\textstyle\sum \mu^2 \text{ convex} \implies \max \text{ at a vertex}:\ \ \frac{9}{16} + \frac{3}{16} = \frac{3}{4} \text{ [machine]};$$

$$|r|^2 = 4\sum \mu_i^2 \leq 3 \text{ [L3(1)]};$$

$$|r_a| \leq |r| \leq \sqrt{3} \text{ [Cauchy–Schwarz on components; machine: } r_a^2 \leq |r|^2]$$

$$(2)(\mathrm{d}):\ |\Delta| \leq |r_a| + |r'_a| \leq 2\sqrt{3};$$

$$(2\sqrt{3})^2 = 12 \text{ [machine]};$$

$$(2)(\mathrm{e}):\ \text{the range rows re-cited; the monotone-BV identity: TV}$$

$$=\sup-\inf\ \text{on monotone functions [machine: rational grids]}$$

$$(1)(\mathrm{d}),$$

(3)–$(6):$ the citation-pattern scan, the amendment pins, and the inheritance census [machine]. ■

Phys. The corpus's last non-axiomatic import is dissolved, and both halves dissolve into structure the ledger already owned. The richness condition was never an outside gift: the ledger's own configuration space carries cells of every finite content — the cycle ladder witnesses each level constructively — and refining the hop length at fixed span drives every cell's real-edge count past any bound. The regularity condition was never an outside gift either: the sector coefficients are components of a positive, unit-trace pair state, whose spectrum lives in an exact interval; convexity pins the quadratic extremum at a single vertex orbit, the bound is the square root of three, fields are averages of bounded data, and every weight family of the corpus carries its own exact range, so total variation equals range width everywhere. Neither derivation touches the manifold conclusion it licenses, and the scope citations that once looked like dependencies are audited as arena inheritances. The extra-axiomatic list is empty; the axiom surface is the entire non-derived surface; the conditions remain in force at every point of use, now as theorems.

BM.152 — The status of the Continuum-Limit Tier and the closure of the foundational era. Structural-constraints record: see Appendix B, Section B.1.35 of the Addendum.

4.27 Closure exports of Part 4 (corpus terminus)

Def-BM.1 sealed composites; validity/read clauses
BM.1 observability switch; trace-only read
BM.4 colour/weak invariance; parity passage
BM.7 $\#J^{+}=\#J^{-}$; telescoping; $\partial^{2}=0$; $-U(1)$ on q_{BM}
BM.10 $\neg\Diamond\varepsilon,\#\varepsilon=\infty$ in 16; $\Delta z\Delta p\geq\frac{1}{2}$; MT, CN
Def-BM.4 $\Delta\eta=1/m_{\mathrm{BM}}$; $V_{\mathrm{eff}}=(\Delta\pi_{\mathrm{BM}})^{2}$
BM.15 $R{=}\kappa^{2}(\mu{-}3p){+}4\lambda$; $\kappa^{2}\mu{=}2\pi E^{2}/\ln 2^{2}/\hbar^{2}$
BM.18 the calibration adjudication
Def-BM.5 $\mathcal{E}(\mathcal{X})$; $\rho{=}\pi_{\mathcal{X}}/\|\pi_{\mathcal{X}}+\pi_{-}\|$; $E_{\mathrm{tail}}(R){=}|R|$
BM.20 RV/m^{2} m; $\langle m\rangle$=the averaged flux; $m_{\mathrm{H}}^{2}(q,S)$
BM.23 $\Delta q{=}\pm1$; the forced carrier; the resolvent rate
Def-BM.7 $\mathrm{EM}^{\mathrm{sc}}$; the EM scar; the formal-mirror scope
BM.27 the antiproton row; one orbit of size two
Def-BM.8 $Q{=}\sum_{i}q_{\mathrm{core}}(c_{i})$; BM.6(1) discharged
BM.30 ratios $1:\frac{1}{2}:\frac{1}{60}$; ¬a route adopted
Def-BM.9 $\mathcal{E}_{\mathrm{det}}$; $\Pi_{\mathbb{C}}$; $g(N){=}(1,1,3,10,25,105)$
BM.33 the test's adjudication
Def-BM.10 $E{=}N\sqrt{m^{2}{+}c^{2}\alpha^{2}p_{\mathrm{F}}^{2}}-N^{2}p_{\mathrm{F}}$; chart-side
BM.36 (6, 16, 17, 25); ratios $(\frac{1}{2},8,\frac{1}{20})$; Gram spectrum; Yukawa-like coupling open
Def-BM.11 $\epsilon(i){=}\mathrm{Adj}(I^{+})e(0)$; $m_{\ell}^{2}{=}M^{2}(A_{0}{-}B_{0}\cos(\delta{-}\frac{2\pi}{3}\ell))$
BM.41 two-body impossible; ν-member forced; $Q{=}{-}(m_{t}{+}m_{\nu})<0$
BM.44 $\Gamma_{u}\in[0,\frac{1}{2}]$; $\tau_{\mathrm{BK}}\geq3$ ticks; ¬seconds
Def-BM.12 $\partial_{\nu}F^{\mu\nu}{=}eJ_{\mathrm{D}}$; $E^{\mathrm{self}}{=}\frac{\alpha m_{e}c^{2}}{2\pi}$; ϵ priced $(\frac{1}{2},\frac{1}{2})$
BM.48 $V{\sim}{-}v_{\mathrm{EM}}/d$; $E_{\mathrm{b}}{=}{-}\frac{\mu_{\mathrm{red}}}{2\hbar^{2}n^{2}}$; the bound channel ¬constructed
BM.51 $(\frac{1}{20})^{2}$; $\Upsilon{\to}8\Delta S_{\mathrm{spin}}$; $\Delta m_{\mathrm{Lorentz}}{=}0$
BM.54 $\frac{E_{\mathrm{ion}}^{\mathrm{Rydb}}}{E_{\mathrm{ion}}^{\mathrm{exp}}}$; $B^{(m)}{=}B^{(0)}$; the atom ¬constructed
Def-BM.13 $m_{\nu}{=}(2,1,1)$; $W{=}(1,6,15,20,15,6,1)$; $\Sigma_{\mathrm{H}}{=}\ln\frac{4}{3}$
BM.58 $\Delta m_{\mathrm{weight}}{=}0$; the seal asymmetry survives as $\Delta\pi_{\mathrm{EM}}{>}0$
BM.61 Σ_{H} coupling-blind; α_{B} ¬derivable, (OI)–(O4)
BM.64 the tier adjudication; $m_{\nu}{>}m_{e}$ sub-critically
BM.65 the baseline re-certified; $\frac{1}{2}{<}c^{\mathrm{obs}}(\frac{1}{2}){<}\frac{1}{2}{+}c^{\mathrm{obs}}(\frac{1}{2}){<}\frac{1}{2}$
BM.68 $\Omega{=}\Omega_{\mathrm{cond}}\Omega_{\mathrm{charge}}\Omega_{\mathrm{geom}}$; $\Omega_{\mathrm{geom}}^{\mathrm{max}}{=}1$; $S_{\mathrm{joint}}{=}6\ln2{\simeq}\ln W(Q)$
BM.71 the five-route census; $\Delta m_{\mathrm{wedge}}{=}0$ in currency; the decay closed
BM.73 A-BM.7 issued; $(1,0)$ occupancy; $\Delta\gamma\omega^{-1}{=}{-}\chi$
BM.76 $V{=}E_{\perp}\otimes P_{\mathrm{S}}$; $V^{\mathrm{sc}}{=}0$; $\mathrm{tk}\,V{=}1$; $[W_{\mathrm{I}},\varepsilon]\,V{=}V$
the decay closed on every pre-specification read
BM.79 $D(m){=}4\sin^{2}(m\delta_{s}/4){=}2{-}2T_{m}(\cos\frac{\delta_{s}}{2})$; $T_{1}(c){=}0$; $r_{\mathrm{max}}{=}\sin(k_{\beta}\delta_{s}/4)/\sin(\delta_{s}/4)$
BM.82 $\mathcal{R}_{\nu}{=}D_{\mathrm{H}}/4D$; $\mathcal{R}_{\mathrm{max}}/\mathcal{R}_{\mathrm{min}}{=}50/(1{+}\varepsilon)$; $\Sigma t{=}(30/6)\hat{G}$
BM.85 $Q_{\beta}/m_{\nu}\leq-\ln3/\ln1280$; two formed reads, disjoint alphabets
four currencies, zero conversions
BM.87 $c_{\mathrm{crossed}}{=}\frac{1}{2}$ derived twice; $\gamma_{\mathrm{mid}}{=}\frac{1}{20}$ void; the selector, sign and monomial stand
BM.90 $w{=}w(s_{\beta})$ branch-blind; hypothesis (W) refuted on four grounds
BM.93 the correction's adjudication; ¬a declaration; (R4), (R5) named
BM.94 $\eta_{\mathrm{coast}}{:}1/(1{+}\alpha{+}\alpha^{2})\in(\frac{1}{3},\frac{1}{2})$; $|\mathcal{K}|_{\mathrm{max}}\in(2,6)$; $\pi/|\mathcal{K}|{<}\frac{1}{2}$
BM.97 $m^{\mathrm{seed}}{\to}m_{\mathrm{pole}}^{\mathrm{obs}}$; $\Delta_{\mathrm{c}}^{\mathrm{obs}}{=}\alpha(2M_{0}{+}\nu)$; the specified read flicker-blind
BM.100 the tier's adjudication; ¬a declaration; (R6) named
BM.101 $Q^{\mathrm{pr}}{=}{-}\mathcal{L}^{\mathrm{pr}/\mathrm{S}}$; $\tau_{\mathrm{c}}{=}(3{-}\log_{2}\mathcal{B})/(3\kappa){\in}(\frac{1}{20},\frac{1}{10})$; $P_{\mathrm{max}}{=}1{-}\frac{1}{2^{20}}$
BM.104 $\mathcal{C}_{\mathrm{cusp}}{=}G(1{+}a\xi)$; $\mathcal{G}_{\mathrm{W}}{=}(1{+}G)/(1{-}aG)$; $aG{=}\tau Nr\,\mathrm{exp}(Q^{2}E^{2}\ln2;\,g(G)$ ¬derived
BM.107 the six adjudications: one declaration; (R4′), (R7) named
Def-BM.15 $\mathrm{CP}{=}\mathbb{C}{\times}P$; $\mathcal{C}^{\mathrm{CP}}$; the CP-pair; the formal-mirror scope
Def-BM.16 $\mathcal{H}_{\beta}{=}\mathcal{H}_{\alpha}\otimes\Psi$ (two chains); $\dim=4(|k_{\beta}|{+}1)(|k_{\beta}|{+}1)$; $\hat{P}_{12}=\hat{\mu}_{\mathrm{CP}}\hat{S}_{12}$
BM.112 $P_{\mathrm{H}}^{\mathrm{ev}}{=}$ id; spec $\subseteq\{\pm1\}$; $\mathrm{Tr}=d_{\mathrm{BCP}}$; the orientation block is statistics-blind
Def-BM.17 $\mathcal{R}=(\alpha,\sigma,Q,\mathrm{a},k)$; $T=\Sigma_{\mathcal{R}}/N$; $\Gamma=\Gamma_{\mathcal{R}}/\frac{d\alpha}{dm}N\chi$; ¬a cosmological coordinate
BM.116 $\Gamma_{\delta}\in[0,\frac{1}{2}]$; $\langle Z\rangle\,\varepsilon_{\delta}{=}0$, $\langle Z\rangle\,f{=}0$, $\langle Z\rangle\,N{=}6$; $\tau_{\mathrm{BK}}\geq3$ ticks
four objectives, four derived verdicts

Def-BM.2 composite card grammar
BM.2 42j junction phase; additive winding
BM.5 25 cells, 6 families; conditioning law
BM.8 hop $\to$ translation; $C_{3}\to SO(3)$; tick obstruction
BM.11 if $\partial_{\nu}J^{\nu}{=}0$; $\nabla_{\mu}T^{\mu\nu}{=}0$ two routes
BM.13 $\Delta\pi_{\mathrm{B}}{=}\hbar/(E\ln2)$; sub-hop $E\geq2$
BM.16 $(\Delta\pi_{\mathrm{B}})^{2}\kappa^{2}{=}2{\pi}E^{2}\ln2$; $\Delta\pi_{\mathrm{B}}\pi_{\nu}{=}2G_{N}$
¬a new dimensionful scale [P1]
Def-BM.6 the family naming rows; $Q{=}\frac{4}{3}$
BM.21 $B{\neq}0$ in the ensemble, 0 in the count
BM.24 the flicker adjudication
BM.25 CPT^{+}-tid; $\leftrightarrow{=}\pi^{+}$; $\lambda\in\{\pm1,\pm i\}$
BM.28 $W(0){=}20$, $W(\pm1){=}15$; three mass routes
(blank)
¬a fractional change [BM.28(?)]
BM.31 $g(3){=}1$, $g_{\mathrm{I}}(3){=}0$; $N{=}3$ protected
the protection fails at $N\geq4$
BM.34 $\langle q_{\mathrm{C}}{-}N^{*}p_{\mathrm{C}}{=}N^{*}(m^{2}{+}\mu^{2})^{\frac{1}{2}}$; $\mathrm{KE}{=}\frac{\mathrm{Sym}^{2}}{2m}{+}O(\mu^{4})$; $E^{2}\geq\frac{1}{4}m_{\mathrm{tid}}^{2}$
BM.37 routes¬measure functionals; alphabets disjoint; trap closed
BM.39 $A_{0}{-}B_{0}{>}0$ (SO6); $\sum_{i}m_{i}^{2}{=}3M^{2}A_{0}$; splitting $\propto g\delta'$
BM.42 $(\partial_{\mu}M^{\mu\lambda}){=}0$; $p^{\lambda}{=}p_{+}^{\lambda}{+}p_{-}^{\lambda}$; $2^{10}{\times}2.0{\times}3.1{\simeq}2$
BM.45 $\pi\to p{+}e^{-}{+}\bar{\nu}_{e}$; grammar complete; kinematics closed
BM.46 $\Delta m_{\mathrm{BM}}{=}\frac{\alpha_{\mathrm{EM}}\Delta\pi_{\mathrm{B}}}{2\pi}{>}0$; the proton heavier
BM.49 $Q_{\mathrm{code}}{<}0$ deeper; $E_{\mathrm{max}}(E){=}2\pm\frac{1}{2}$ refuted
BM.52 six sectors summed; $\Delta m_{\mathrm{EM}}{=}{-}\Delta\pi_{\mathrm{EM}}$
BM.55 $\Delta m{=}{-}\frac{\alpha_{\mathrm{EM}}\Delta\pi_{\mathrm{B}}}{4\pi}{<}0$; $\frac{\Delta m}{m_{\mathrm{p}}}{=}\frac{\alpha}{4\pi}{\cdots}$
Def-BM.14 3 floors $2E{\in}\{12,24,60\}$; $|\mathcal{S}|{=}3{\cdot}2{=}6$; the D_{2} row
BM.59 $\Delta m_{\mathrm{weight}}{=}\ln(4/3)/8{>}0$; the residue filled on all four grounds
BM.62 $\Delta m{=}[\ln\frac{4}{3}{-}6c_{2\mathrm{BM}}\ln2/\hbar;\ \frac{1}{2}{<}c^{\mathrm{obs}}(\frac{1}{2}){<}\frac{1}{2}{+}c^{\mathrm{obs}}(\frac{1}{2}){<}\frac{1}{2}$
¬Cal-BM parameter; ¬a new declaration beyond one name attachment
BM.66 Tr $\hat{F}{=}2i_{\mathrm{D}}$; $I_{\mathrm{D}}{=}\hat{Q}^{2}$; the operator root input
BM.69 $\frac{\alpha^{2}m_{e}}{m_{\mu}}{=}\frac{\alpha}{\pi}\frac{\alpha}{2\pi}$; Gap 7 minimal : (R1)–(R3)
BM.72 the tier adjudication; the partial bridge $\wedge$ the no-go
BM.74 the ray-resolved spin census; $j{=}|k|/2$; opposite rays at the vertex
BM.77 A-BM.8 issued; $\Delta T{=}5.m_{\mathrm{H}}^{2}$; $C_{\mathrm{T}}\neq\varnothing$; 828.000¬5810
¬Cal-BM parameter; two addenda logged before use
BM.80 $4z^{2}{-}24z^{2}{+}(21{-}12\gamma)z{-}(\gamma{-}1)^{2}$; $r_{\mathrm{H}}^{(k)}{=}1{+}\gamma_{k}$; $q{<}c{<}\ell$
BM.83 $p_{\mathrm{B}},\sigma_{\nu}$; ln 1280/ ln 960 ∉ $\mathbb{Q}$; $\ell_{\mathrm{cusp}\,/\,m_{\mathrm{H}}c\,\chi_{\mathrm{C}}^{2}}$ ¬ ln 405/ ln 16<0
BM.86 the census adjudication; ¬a declaration; the ten open rows
item 7 stands with its residue named
BM.88 $c{=}a\sqrt{N^{2}{+}\frac{1}{4}}$; $I(k){=}1{-}\frac{1}{2}$; $G{=}2\pi E_{\mathrm{BK}}^{2}\ln2$; $f(1){=}0$
BM.91 $\Delta m\,\hbar{=}\ln\frac{4}{3}{-}\Delta N\alpha_{\mathrm{EM}}\ln2$; $N{\in}\mathbb{q}\,\pi_{\mathrm{EM}}{\leq}\frac{1}{2}\Longrightarrow m_{\nu}{>}m_{\mu}$; $m_{\nu}/m_{\mathrm{p}}{<}\frac{m_{\mu}}{m_{\mathrm{p}}}$
¬Cal-BM parameter; ¬a measured coupling [P5]
BM.95 $P(\mathrm{Bloat}\,|\,Q){=}\prod_{\mathrm{H}}W(Q)$; $\ln\frac{4}{3}{=}\frac{1}{4}\ln2{-}\ln\frac{3}{4}{+}(\frac{1}{2}\ln2$
BM.98 $\mathcal{R}{=}\frac{4\ln3}{\ln1280}\frac{\alpha_{\mathrm{EM}}}{\pi}{=}\frac{\alpha^{2}}{2\pi^{2}}$; $\mathcal{R}_{\mathrm{max}}/\mathcal{R}{=}D^{2}{<}9$
¬Cal-BM parameter; ¬a measured value [P5]
BM.102 $|\mathrm{h}^{+}\mathrm{h}^{-}\rangle^{I,J}$; $\{\mathrm{pk}\}{=}\mathbb{C}$; A-BM.9 declared, value ¬issued; item 7 closed as an equation
BM.105 $w_{\mathrm{max}}{=}\frac{1}{2}\sum_{i}n_{i}\in[\frac{1}{2},1]$ measure-free: $S/(3(1{-}R))$; hypothesis (W6) refuted
¬a new dimensionful scale [P1]; ¬a measured value [P5]
BM.108 $\mu_{\mathrm{CP}}{=}\epsilon^{\mathrm{obs}}{=}1$ on $\Theta\in22$; the forced conjugate; $\neg\exists\epsilon^{-1}$
BM.110 $g(2)=1$: the δ-seal is unique; $\Pi_{\mathrm{C}}^{2}(0)=W_{\beta}(0)=6$; (R7) discharged at $n=2$
BM.113 $\varepsilon=(-1)^{|k|}=(-1)^{\ell_{\beta}}$; deficit 11 closed; BM.6(2) discharged
BM.114 $(2{+}z{+}z^{-1})^{n}=(z{+}1)^{2n}z^{n}$; $W_{\alpha}(Q)=\binom{2n}{n}\in\mathbb{M}$, $\mathbb{M}{\cdot}\mathbb{Q}=16$; 55 eigenvalues, 1 collision
BM.117 $20=\binom{6}{3}$ an orbit count, ¬a decuplet; $\lambda_{\mathrm{min}}^{\mathrm{obs}}=4\pi$; the octet ¬formed
¬a new dimensionful scale [P1]; ¬a measured value [P5]

Def-BM.3 the identification layer (declared)
BM.3 C_{β} forced at the seal; $\delta_{\alpha}\cdot\,\mathbb{M}_{\beta}=\delta_{\beta}$
BM.6 8 open construction specs
BM.9 T_{uv}; sign reconciliation; dust source
BM.12 the Noether adjudication; ¬Cal-BM parameter
BM.14 $\rho{=}(E\ln2^{2}/\hbar^{4}\cdot\,|\rho|{=}q^{-2}$
BM.17 $\Sigma_{\mathrm{tid}}^{(n)}{=}6$: the neutron row
¬a measured value [P5]
BM.19 $p{=}1{-}4z_{\mathrm{C}}$; $\Gamma{=}0$ at $z_{\mathrm{C}}{=}0$
BM.22 $P_{LR}{=}q_{E}$; $2|q\in\{0,2\}$
¬Cal-BM parameter; ¬a P4 addition
BM.26 the proton row; $2|p{=}2$
BM.29 the sector adjudication; ¬inter-sector dynamics
(blank)
¬Cal-BM parameter; ¬a measured value [P5]
BM.32 $Z_{\beta}{=}0$ by filtration; $\langle m\rangle_{\mathrm{tid}}{=}\langle m\rangle_{\tau}$
¬a new declaration
BM.35 $w_{\mathrm{max}}{=}\Delta S_{\mathrm{C}}/\hbar{=}m_{\mathrm{tid}}$; ln 4=2 ln 2
BM.38 the tier adjudication; ¬Cal-BM parameter; ¬a measured value
BM.40 $\sum_{i}\mathrm{H}_{i}^{2}{=}20\Pi_{\mathbb{R}}$; σ-vertex Casimir-neutral
BM.43 $\nabla^{\Phi}{=}\mathrm{tr}C(x)\rho$; $r_{\mathrm{C}}{=}\frac{\mathrm{Sym}}{2m}$
¬Cal-BM parameter; ¬a measured value
BM.47 UV: the envelope: IR: EM.11; $f(k){=}1{-}\frac{1}{2}$
BM.50 the tier adjudication; ¬a numerical ϵ [P5]
BM.53 $\Delta m_{\mathrm{weight}}{=}0$; $\Delta t\geq\pi\hbar/(3c^{2}\ln2)$
BM.56 the full-scope adjudication; the order residue named. ¬filled
BM.57 $|A_{\beta}|{=}3$ vs 1; asymmetry 2; $\Delta m_{\mathrm{cusp}}{=}0$
BM.60 1:2:5; $D_{\beta}(F_{\mathrm{p}}{+})>D_{\beta}(F_{\mathrm{p}}{-})$; the inversion
BM.63 $Q_{\beta}{<}-\ln3/\ln{>}0$; closed unconditionally
¬a measured value [P5]
BM.67 $\Gamma(\gamma){=}(-1)^{|\gamma|}$; $\mathrm{im}(\mathrm{lift})\subseteq\{0,\pi\}$; the obstruction record
BM.70 $F_{\mathrm{C}}{\geq}2\ln2$; formed $\Longrightarrow$ closed: margins ln 3, ln 12, ln 24
¬Cal-BM parameter; ¬a new declaration; ¬a measured value [P5]
BM.75 weight $\leftrightarrow$ branch $\leftrightarrow$ core unique; $\Delta_{\mathrm{p}}{=}(2,\sqrt{2}{-}1,2\cos\frac{\pi}{12}{-}\sqrt{3})\mu^{2}$
BM.78 the tier adjudication; 1′ closed; the decay allowed on $\mathcal{O}_{\beta}$
¬a measured value [P5]
BM.81 $\Delta_{1}{>}\Delta_{2}{>}\Delta_{3}$; the doublet ratios; 3 ¬proportional ladders
BM.84 $m_{\alpha}/m_{\mathrm{p}}{<}21/20$, $\epsilon^{\mathrm{obs}}\notin\mathbb{Q}$; the trichotomy
¬Cal-BM parameter; ¬a measured value [P5]
the residue: (R1)–(R3), σ_{I}, σ_{κ}, ϵ
BM.89 $E^{\mathrm{self}}{=}\frac{1}{2}N\alpha_{\mathrm{EM}}Q^{2}m_{\mathrm{tid}}c$; $w(0){=}1$; $w{=}1{-}4\varepsilon_{\mathrm{C}}$ (marked)
BM.92 $Q_{\beta}\mathrm{h}\leq-\ln3$ uniformly; the specified read untouched
the residue: (R1)–(R5), ϵ_{0}, g_{e}, N
BM.96 $\delta_{\alpha}{=}\frac{1}{2}{\cdot}G_{\alpha}{\in}\xi_{+}{\cdot}A_{0}{\in}G_{\beta}$; $m_{\nu}^{\mathrm{obs}}{>}0$; $(1,0)$; ¬a seesaw-type mechanism
BM.99 $N\alpha_{\mathrm{EM}}\leq\frac{1}{2}\Longrightarrow\Delta m{>}0$, $\geq\frac{\pi}{12}\Longrightarrow\Delta m{<}0$; $m_{\nu}/m_{\mathrm{p}}{<}\ln80/\ln60{<}\frac{11}{16}$
the residue: (R1)–(R6), ϵ_{0}, g_{e}, N, ρ_{e}
BM.103 $W(Q){=}\binom{6}{3-Q}$; $4z^{2}{-}24z^{2}{+}(21{-}12\gamma)z{-}(\gamma{-}1)^{2}$; $\det(\lambda I{-}M)$ ¬formed
BM.106 $j{=}|s|/2$ with $j{=}\frac{1}{2}$, 1 on one kind; ¬an exchange operator; deficit 11 stands; (R7) named
the residue: (R1)–(R3), (R4′), (R5)–(R7), g_{e}, ϵ_{0}, N
BM.109 $W_{\mathrm{EM}}{=}6\varepsilon_{\mathrm{C}}(1{-}4\varepsilon_{\mathrm{C}})\leq\frac{6}{25}$; $Q_{\mathrm{EM}}{=}0$; $\mathbb{B}_{\mathrm{EM}}^{\mathrm{obs}}{=}0$
BM.111 $m_{\mathrm{tid}}^{\mathrm{seed}}{=}2m_{\mathrm{tid}}$; $R_{\mathrm{BK}}{=}\mathbb{B}_{\mathrm{EM}}{=}0$; (R8) named
¬a spin-parity theorem [BM.106(1) stands]
BM.115 $\mathbb{T}_{0}{=}E_{\mathrm{BK}}/\ln2$; ladders $2n$, $60n$, span 30; $T{=}2\epsilon_{\mathrm{C}}/N$; ¬a forbidden state
BM.118 the tier adjudication; ¬a declaration; (R8), (R9) named
the residue: (R1)–(R6), $\langle R7\rangle$ at $n\geq3$, $\langle R8\rangle$, $\langle R9\rangle$, g_{e}, ϵ_{0}, N

BM.120 $\hat{P}_{\mathrm{T}}^{\mathrm{ev}}{=}0$, $\{J_{-},J_{+}\}{=}\mathrm{id}$; spec $\mathbb{Q}{=}\{0,-(x_{+}{+}x_{-})\}$; detailed balance derived
BM.123 $|W_{\mathrm{BM}}{=}e^{\Gamma}$, χ; $\frac{\mathrm{Sym}}{2m}{=}1\iff\varepsilon_{\mathrm{C}}{=}\frac{1}{2}$; $\langle R9\rangle$ closed at the channel level
BM.125 $0\leq\kappa_{\mathrm{tid}}^{2}(m)\leq4\hat{q}^{G}$; $\hat{p}^{2}\frac{\mathrm{Sym}^{2}}{2m}{=}\hat{p}^{2}\hat{C}(G)c$; the product form ¬formed
BM.128 dim $\mathcal{H}_{\delta}{=}\frac{1}{2}\prod(|k_{\alpha}|{+}1)$; $\chi(\sigma){=}\sigma^{\ell(\sigma)}$; $\dim\wedge^{3}\mathbb{C}^{3}{=}0$ vs $\dim\wedge^{3}\mathbb{C}^{4}{=}1$; $\langle R7\rangle$ closed at $n{=}3$
the residue: (R1)–(R6), $\langle R7\rangle$, $\langle R8\rangle$, $\langle R9\rangle$, $\langle R9'\rangle$, $\langle R10\rangle$, g_{e}, ϵ_{0}, N

BM.131 $\mathrm{tw}(\rho){=}\frac{\mathrm{Sym}\mathrm{Ad}}{2}\mathbb{1}_{\mathbb{R}}{+}\mathbb{Z}(\Pi_{\mathbb{R}}\rho)\Pi_{\mathbb{R}}$; $\delta K{\leftrightarrow}\rho{\leftrightarrow}\log\mathcal{Z}$; I: the common fixed point $(\frac{1}{2},\frac{1}{2})$; $V(s){=}(1{-}\frac{1}{2}^{s})^{2n/N}V_{0}$
BM.132 $N(k,E){=}\sqrt{1{-}\frac{E}{2m}}$; formed $\Longleftrightarrow k\geq\frac{m}{2m}{=}\frac{\mathrm{Sym}^{2}}{2m}$; the envelope ¬formed; $N{\to}1$
BM.134 the pole locus is the dispersion; form-unique; $d_{\mathrm{H}}\mathrm{adj}{=}\frac{d^{2}\rho}{dx^{2}}$; $\Phi_{\mathcal{D}}{=}\frac{\sqrt{m_{\mathrm{H}}^{2}{+}p^{2}}}{\sqrt{m_{\mathrm{H}}^{2}{+}p^{2}}}$; item 17 closed
BM.137 $\frac{m_{\nu}^{(n)}}{m_{\mathrm{tid}}^{(n)}}{=}\frac{\mathcal{B}_{\mathrm{EM}}^{(n)}}{\mathcal{C}_{\mathrm{BK}}^{(n)}}{=}\frac{\mathrm{dim}\mathcal{H}_{\delta}{-}2k}{\mathrm{dim}\mathcal{H}_{\delta}{+}2k}{<}0$; $m_{\nu}^{(2)}{=}0$ by the pincer; 21 $\lambda^{(2)}$, ¬a collision; $\langle R8\rangle$ closed
BM.140 $\frac{\mathrm{d}\,\mathrm{dim}\mathcal{H}_{\delta}}{\mathrm{d}\,k}{=}\frac{1}{2}$ minimal, unique; $x^{4}{-}48x^{3}{+}304x^{2}{-}512x{+}256$; five currencies censused; deficit 12 adjudicated
BM.143 $\langle R5\rangle$ void; $\langle R2\rangle$, $\langle R10\rangle$ registration data; $\langle R4'\rangle$ void for BM; C-BM.1–3 consolidated; $\langle$P5.1$\rangle$–$\langle$P5.10$\rangle$ named
for Part 5: the EM binding channel first [BM.143(?)]

Def-BM.4 the identification (declared)
BM.4 $\mathcal{C}_{\beta}$ forced at the seal; $\delta_{\alpha}\cdot\,\mathbb{M}_{\beta}=\delta_{\beta}$
BM.6 8 open construction specs
BM.9 T_{uv}; sign reconciliation; dust source
BM.12 the Noether adjudication; ¬Cal-BM parameter
BM.14 $\rho{=}(E\ln2^{2}/\hbar^{4}\cdot\,|\rho|{=}q^{-2}$
BM.17 $\Sigma_{\mathrm{tid}}^{(n)}{=}6$: the neutron row
¬a measured value [P5]
BM.19 $p{=}1{-}4\varepsilon_{\mathrm{C}}$; $\Gamma{=}0$ at $\varepsilon_{\mathrm{C}}{=}0$
BM.22 $P_{LR}{=}q_{E}$; $2|q\in\{0,2\}$
¬Cal-BM parameter; ¬a P4 addition
BM.26 the proton row; $2|p{=}2$
BM.29 the sector adjudication; ¬inter-sector dynamics
(blank)
¬Cal-BM parameter; ¬a measured value [P5]
BM.32 $Z_{\beta}{=}0$ by filtration; $\langle m\rangle_{\mathrm{tid}}{=}\langle m\rangle_{\tau}$
¬a new declaration
BM.35 $w_{\mathrm{max}}{=}\Delta S_{\mathrm{C}}/\hbar{=}m_{\mathrm{tid}}$; ln 4=2 ln 2
BM.38 the tier adjudication; ¬Cal-BM parameter; ¬a measured value
BM.40 $\sum_{i}\mathrm{H}_{i}^{2}{=}20\Pi_{\mathbb{R}}$; σ-vertex Casimir-neutral
BM.43 $\nabla^{\Phi}{=}\mathrm{tr}C(x)\rho$; $r_{\mathrm{C}}{=}\frac{\mathrm{Sym}}{2m}$
¬Cal-BM parameter; ¬a measured value
BM.47 UV: the envelope: IR: EM.11; $f(k){=}1{-}\frac{1}{2}$
BM.50 the tier adjudication; ¬a numerical ϵ [P5]
BM.53 $\Delta m_{\mathrm{weight}}{=}0$; $\Delta t\geq\pi\hbar/(3c^{2}\ln2)$
BM.56 the full-scope adjudication; the order residue named. ¬filled
BM.57 $|A_{\beta}|{=}3$ vs 1; asymmetry 2; $\Delta m_{\mathrm{cusp}}{=}0$
BM.60 1:2:5; $D_{\beta}(F_{\mathrm{p}}{+}){>}D_{\beta}(F_{\mathrm{p}}{-})$; the inversion
BM.63 $Q_{\beta}{<}-\ln3/\ln{>}0$; closed unconditionally
¬a measured value [P5]
BM.67 $\Gamma(\gamma){=}(-1)^{|\gamma|}$; $\mathrm{im}(\mathrm{lift})\subseteq\{0,\pi\}$; the obstruction record
BM.70 $F_{\mathrm{C}}{\geq}2\ln2$; formed $\Longrightarrow$ closed: margins ln 3, ln 12, ln 24
¬Cal-BM parameter; ¬a new declaration; ¬a measured value [P5]
BM.75 weight $\leftrightarrow$ branch $\leftrightarrow$ core unique; $\Delta_{\mathrm{p}}{=}(2,\sqrt{2}{-}1,2\cos\frac{\pi}{12}{-}\sqrt{3})\mu^{2}$
BM.78 the tier adjudication; 1′ closed; the decay allowed on $\mathcal{O}_{\beta}$
¬a measured value [P5]
BM.81 $\Delta_{1}{>}\Delta_{2}{>}\Delta_{3}$; the doublet ratios; 3 ¬proportional ladders
BM.84 $m_{\alpha}/m_{\mathrm{p}}{<}21/20$, $\epsilon^{\mathrm{obs}}\notin\mathbb{Q}$; the trichotomy
¬Cal-BM parameter; ¬a measured value [P5]
the residue: (R1)–(R3), σ_{I}, σ_{κ}, ϵ
BM.89 $E^{\mathrm{self}}{=}\frac{1}{2}N\alpha_{\mathrm{EM}}Q^{2}m_{\mathrm{tid}}c$; $w(0){=}1$; $w{=}1{-}4\varepsilon_{\mathrm{C}}$ (marked)
BM.92 $Q_{\beta}\mathrm{h}\leq-\ln3$ uniformly; the specified read untouched
the residue: (R1)–(R5), ϵ_{0}, g_{e}, N
BM.96 $\delta_{\alpha}{=}\frac{1}{2}{\cdot}G_{\alpha}{\in}\xi_{+}{\cdot}A_{0}{\in}G_{\beta}$; $m_{\nu}^{\mathrm{obs}}{>}0$; $(1,0)$; ¬a seesaw-type mechanism
BM.99 $N\alpha_{\mathrm{EM}}\leq\frac{1}{2}\Longrightarrow\Delta m{>}0$, $\geq\frac{\pi}{12}\Longrightarrow\Delta m{<}0$; $m_{\nu}/m_{\mathrm{p}}{<}\ln80/\ln60{<}\frac{11}{16}$
the residue: (R1)–(R6), ϵ_{0}, g_{e}, N, ρ_{e}
BM.103 $W(Q){=}\binom{6}{3-Q}$; $4z^{2}{-}24z^{2}{+}(21{-}12\gamma)z{-}(\gamma{-}1)^{2}$; $\det(\lambda I{-}M)$ ¬formed
BM.106 $j{=}|s|/2$ with $j{=}\frac{1}{2}$, 1 on one kind; ¬an exchange operator; deficit 11 stands; (R7) named
the residue: (R1)–(R3), (R4′), (R5)–(R7), g_{e}, ϵ_{0}, N
BM.109 $W_{\mathrm{EM}}{=}6\varepsilon_{\mathrm{C}}(1{-}4\varepsilon_{\mathrm{C}})\leq\frac{6}{25}$; $Q_{\mathrm{EM}}{=}0$; $\mathbb{B}_{\mathrm{EM}}^{\mathrm{obs}}{=}0$
BM.111 $m_{\mathrm{tid}}^{\mathrm{seed}}{=}2m_{\mathrm{tid}}$; $R_{\mathrm{BK}}{=}\mathbb{B}_{\mathrm{EM}}{=}0$; (R8) named
¬a spin-parity theorem [BM.106(1) stands]
BM.115 $\mathbb{T}_{0}{=}E_{\mathrm{BK}}/\ln2$; ladders $2n$, $60n$, span 30; $T{=}2\varepsilon_{\mathrm{C}}/N$; ¬a forbidden state
BM.118 the tier adjudication; ¬a declaration; (R8), (R9) named
the residue: (R1)–(R6), $\langle R7\rangle$ at $n\geq3$, $\langle R8\rangle$, $\langle R9\rangle$, g_{e}, ϵ_{0}, N

BM.120 $\hat{P}_{\mathrm{T}}^{\mathrm{ev}}{=}0$, $\{J_{-},J_{+}\}{=}\mathrm{id}$; spec $\mathbb{Q}{=}\{0,-(x_{+}{+}x_{-})\}$; detailed balance derived
BM.123 $|W_{\mathrm{BM}}{=}e^{\Gamma}$, χ; $\frac{\mathrm{Sym}}{2m}{=}1\iff\varepsilon_{\mathrm{C}}{=}\frac{1}{2}$; $\langle R9\rangle$ closed at the channel level
BM.125 $0\leq\kappa_{\mathrm{tid}}^{2}(m)\leq4\hat{q}^{G}$; $\hat{p}^{2}\frac{\mathrm{Sym}^{2}}{2m}{=}\hat{p}^{2}\hat{C}(G)c$; the product form ¬formed
BM.128 dim $\mathcal{H}_{\delta}{=}\frac{1}{2}\prod(|k_{\alpha}|{+}1)$; $\chi(\sigma){=}\sigma^{\ell(\s

■

A $\langle qu(b(i(t|t))ick)\rangle$ — Census — The Computed Cells (Category B) and Their Consumption

Role. The census annex of the era canon: the tags whose primary deliverable is a computed table, card, ladder, orbit set, multiplet family, or count over a named configuration space, and whose tabulated values are cited by downstream consumers. The structural tags (identities, mechanisms, covariance laws, no-gos, uniqueness and scope theorems — Category A) live in Parts 1–4; the two categories are disjoint and complete over the FT series. Charter (the separation rule). A tag is Category B (census) $\iff$ its primary deliverable is a computed data cell over a named configuration space and downstream consumers cite the tabulated values; Category A (structural) $\iff$ its primary deliverable is an identity, mechanism, covariance law, no-go, uniqueness, or scope theorem about the corpus's algebra, geometry, or dynamics — even when proved by exhaustive enumeration. Mixed tags are classified by primary deliverable; the census sides of Category-A tags (the cover-polynomial tables of FT.68, the frame catalogue of FT.93, the susceptibility census of FT.103) are consumed through those tags' own rows in Parts 3 — no B-classification is implied. Dep. Parts 1–3 (cited by tag at use in each cell's Req). Nothing in Parts 1–2 cites this file; Part 3 and Part 4 cite its cells row-wise. The tag-level citation graph is acyclic (every Req resolves to an earlier tag or a marked forward pointer); `checks/verify_era_structure.py` machine-verifies ownership, closure, and acyclicity. Inventory. 25 cells: FT.28, FT.32, FT.35, FT.42, FT.43, FT.44, FT.45, FT.49, FT.50, FT.51, FT.63, FT.64, FT.65, FT.69, FT.76, FT.78, FT.81, FT.83, FT.84, FT.96, FT.98, FT.99, FT.105, FT.114, FT.120. Format: Part 1 §0.4 (Thm · Req · Pf · Phys). External-data quarantine. The [EXT] audit rows (FT.114, FT.120; and the audit clauses of FT.77, FT.79, FT.87, FT.104 in Part 3) are quarantined comparison data: tested, never trusted; they enter no derivation [P1, P5; FT.73(5) pattern].

A.1 The cells

Theorem A.1 (FT.28 — First computation: the floor-fermion particle card). Thm. (Σ_{min} card.) Let $\Sigma_{\text{min}} = \{p, q\}$ with the materialized 2-cycle $c = (e_1, e_2)$, $e_1 = (p, q)$, $e_2 = (q, p)$, as its interior [Def-H.4 floor; FT.24: $\kappa := \widehat{ds}^{\,2}(e_1) = \widehat{ds}^{\,2}(e_2) \in \mathbb{Z} \setminus \{0\}$], observed by a single external neighbour o through one real edge ($E_{\text{bd}} = 1$, $\delta\tau \geq 1$; FT.17 hypotheses, Def-H.35). Then:

$$(1)\quad I_{\text{hid}} = E_{\text{int}} \ln 2 = 2 \ln 2, \qquad m_{\text{hid}} = \frac{2 \ln 2}{\mathfrak{h}} \qquad [\text{FT.17, Def-H.9(2)}];$$

$$(2)\quad \Theta_{\text{int}} = \Theta(c) = 2\kappa \in 2\mathbb{Z}, \qquad k = \kappa, \qquad \text{Hol}_{\text{int}} = +1 \qquad [\text{FT.22, FT.24, FT.25}];$$

$$(3)\quad \mathcal{A}(\Sigma_{\text{min}}; C) \supseteq \{G, H, W\}; \qquad \text{EM} \in \mathcal{A} \iff f_p \neq 0 \ \vee \ f_q \neq 0 \ [\text{D3.3}];$$

$$\text{strong} \notin \mathcal{A} \text{ necessarily : } n(q)$$

$$= 2$$

$$\implies q \text{ carries no slot index [D3.1, CT3.11(1) : colour slots at } n{=}3 \text{ only];}$$

$$\text{Colour}(\Sigma_{\text{min}})$$

$$= \{p\};$$

$$\pi_C(p) \subseteq \Sigma_{\text{min}} \text{ impossible [A1 : partners distinct]} \implies \mathcal{C}$$

$$= 0 \implies \text{block}$$

$\equiv 0$ [FT.19] : the single-contact floor fermion is structurally colourless (colour-visible variant : Rmk below);

$$(4) \quad \dim \mathcal{K}_{\Sigma_{\text{min}}} = 9 \ (G{+}W{+}H \text{ core}) \ + \ 1 \cdot [\text{EM active}] \ \leq \ 10;$$

$$\text{general bound } \leq 18 \text{ [FT.26(3), CT0.14];}$$

18 is attained only in the inside-sealed variant [Rmk below : $9{+}1{+}8$];

$$(5) \quad n(p) = 3 \ [e_1 \text{ out}, \ e_2 \text{ in}, \ \text{observer edge in}], \quad n(q) = 2 \ [e_1 \text{ in}, \ e_2 \text{ out}]$$

[D0.8];

$$D_{\text{tick}}(3) = 2^4 = 16 = C_{\text{cap}},$$

$O(2) = O(3) = 0$ [CT0.5, CT0.6] : no oversubscription anywhere at the floor [CT0.11];

$$(6)$$

q : boundary-channel read; the closed core ($\partial c = 0$ [D1.4]) carries basepoint sign data $q_{\text{core}}(c; v)$

$$= \tfrac{1}{2}(\sigma_{e_1} - \sigma_{e_k}) \in \{-1, 0, +1\} \text{ [FT.40 : ternary, } \omega\text{-covariant [D2.6]];}$$

$$(7)\quad Z(\mathcal{R},\Sigma_{\min})\ \text{poles} \iff 1\in \operatorname{spec} \mathrm{F}_M^{(\Sigma_{\min})}\ [\text{FT.26}];$$

$$c\ \text{materializes} \implies \Theta(c)=2\kappa\ [\text{FT.22},\ \text{Def-H.16}].$$

Req. FT.17 (Def-H.35), FT.19, FT.20, FT.22, FT.24, FT.25, FT.26, FT.27; A1; Def-H.4, Def-H.9, Def-H.10, Def-H.15, Def-H.16; D0.8, D1.4, D3.1, D3.3, D3.4, D3.8, D3.9; CT0.5, CT0.6, CT0.11, CT0.14 (Part 1); CT3.11, CT3.17 (Part 2); CT4.9 (Part 2).

Pf.

$$(1):\ E_{\text{int}}=2\ (\text{both cycle edges have both endpoints in } \Sigma_{\min}\ [\text{Def-H.12}]);$$

$$E_{\text{bd}}=1\le 4\cdot 1\ \checkmark\ [\text{FT.17 hyp}]$$

$$\text{CT4.9}(1):\ S(p)$$

$$\le 3\ln 2\ \text{at}\ n{=}3,$$

$$\text{channel-visible 1 bit}\ <3\ \checkmark\ [\text{FT.17 Def-H.35}]$$

$$\implies I_{\text{hid}}$$

$$=E_{\text{int}}\ln 2 \qquad =2\ln 2;$$

$$m_{\text{hid}}=I_{\text{hid}}/\mathfrak{h}\ [\text{Def-H.9(2)}]$$

$$(2):\ \Sigma_{\min}\setminus\partial\Sigma_{\min}$$

$$=\{e_1,e_2\}$$

$$=E(c)\ (\text{single chain, edge-disjoint trivially}):\ \Theta_{\text{int}}$$

$$=\Theta(c)$$

$$=2\kappa\ [\text{FT.24(1)}];$$

$$\text{Hol}_{\text{int}}=(-1)^{2\kappa}=+1,\ k=\Theta/2=\kappa\ [\text{FT.25}(1),(3)]$$

$$(3):\ \text{FT.20 rows}:\ G:\ I_{\text{hid}}=2\ln 2\neq 0\ \checkmark;$$

$$H:\ \text{folded into } G\ [\text{Def}-\text{H.36}]\ \checkmark;$$

$$W:\ \Theta_{\text{int}} = 2\kappa \neq 0\ (\kappa \neq 0)\ \checkmark\ [\text{criterion fires in the sound direction}];\quad \text{EM}:\ \exists v:\ f_v \neq 0\ [\text{D3.3}]$$

$$\text{strong}:\ n(q)$$

$$= 2$$

$$\implies q$$

$$\in V_2:\ \neg\exists\ \text{slot index at } q\ [\text{D3.1}:\ n{=}3\ \text{slots} = \text{colour};\ \text{CT3.11}(1)]\implies \text{Colour}(\Sigma_{\text{min}}) \subseteq \{p\}$$

$$p\ \text{slot-carrying}\ (n(p) = 3):\ \pi_C(p) \neq \varnothing\ \text{requires a distinct partner}\ [\text{A1}];$$

$$\text{the only candidate inside } \Sigma_{\text{min}} \text{ is } q,$$

$$\text{slot-less}\implies \pi_C(p) \not\subseteq \Sigma_{\text{min}}\ \forall C \implies \mathcal{C}(\Sigma_{\text{min}}; C)$$

$$= 0\ [\text{Def-H.10}]\implies \text{block}$$

$$\equiv 0\ [\text{FT.19}]$$

$$(\text{if } p\text{'s slot is colour-inert in } C:\ \text{Colour}(\Sigma_{\text{min}}) = \varnothing\ extandtheblockvanishesontheno - contentside—FT.27'scasesplit)$$

$$(4):\ \text{block dims}\ [\text{CT0.12},\ \text{CT0.14}]:\ 5(G) + 3(W) + 1(H)$$

$$= 9;$$

$$\text{EM active}:\ +1\ [\dim \mathcal{K}^{(\text{EM})} = 1]\implies$$

$$\leq 10;$$

$$\text{general bound } 18\ [\text{FT.26}(3)];$$

$$+8\ [\dim \mathcal{K}^{(\text{strong})} = 8,\ \text{CT3.8}] \text{ only in the inside-sealed variant [Rmk]}$$

$$(5):\ n(p) = \partial^{\leftarrow}(p) + \partial^{\rightarrow}(p) = 2{+}1 = 3,$$

$$n(q) = 1{+}1 = 2\ [\text{D0.8}:\ e_1, e_2 \text{ plus the observer edge at } p];$$

$$\text{CT0.5}:\ D_{\text{tick}}(3) = 2^4 = 16,\ D_{\text{tick}}(2) = 2^3 = 8; \quad \text{CT0.6}:\ O(2) = O(3) = 0; \quad \text{CT0.11}:\ n \le 3\ \checkmark$$

$$(6):\ \partial c$$

$$= t(c) - s(c)$$

$$= 0\ [\text{D1.4}:\ \text{closed}];$$

$$\text{core read}$$

$$= \text{first/last edge registers at the basepoint}:\ q_{\text{core}}$$

$$= (\sigma_{e_1} - \sigma_{e_k})/2\ [\text{D3.4 on the ordered tuple},\ \text{CT0.9}:\ \text{independent registers};\ \text{EM.9}:\ \text{ternary, attained}];$$

$$\text{observer-facing read via transit channels [Def-H.4, EM.9]}$$

$$(7):\ \text{Def-H.16} + \text{FT.26}; \qquad \text{FT.22 at materialized } c. \ \blacksquare$$

Rmk (the dark floor). With no observer edge ($E_{\text{bd}} = 0$): $\partial\Sigma_{\min} = \varnothing \implies \gamma(o, \Sigma_{\min}) = \varnothing$ [Def-H.1] — the floor fermion is dark by construction: $I_{\text{hid}} = E_{\text{tot}} = 2\ln 2$, no read channel [FT.3's indistinguishability]. The observer edge is what lifts the floor into the spectrum bookkeeping — consistent with Def-H.4's external characterization.

Rmk (the colour-visible variant). Give q its own single-contact observer edge ($E_{\text{bd}} = 2 \le 4\,\delta\tau$, $\delta\tau \ge 1$): then $n(p) = n(q) = 3$, both vertices slot-carrying [D3.1], and two sealing phases exist [FT.20 strong row]: (i) inside-sealed — p triplet, q antitriplet contracted by $\delta^i{}_j$ inside $\Sigma_{\min}$ (meson-type [D3.9, CT3.17]): $\mathcal{C} = 1$, strong block active, $\dim \mathcal{K}_{\Sigma_{\min}} = 9 + 1 + 8 = 18$ — the maximal observable mask [FT.26]; (ii) outside-sealed or dangling — $\mathcal{C} = 0$, strong block $\equiv 0$ [FT.19]. Bookkeeping: with two contacts FT.17's single-observer hypothesis no longer applies; I_{hid} becomes observer-relative [Def-H.9(2), FT.20 G row]: $E_{\text{tot}} = (2{+}2)\ln 2$, $E_{\text{vis}}(o_1) = \ln 2 \implies I_{\text{hid}}(o_1) = 3\ln 2$ (the other observer's boundary bit is hidden from o_1), $m_{\text{hid}}(o_1) = 3\ln 2/\mathfrak{h}$. The smallest colour-observable fermion floor is this variant — the sealed meson-type pair [FT.27 strong row].

Phys. The first particle card of the program, every entry derived: the floor fermion carries informational mass $m_{\text{hid}} = 2\ln 2/\hbar$ — two bits of hidden connectivity — an even interior winding 2κ with trivial interior parity, the integer label $k = \kappa$ as its spin candidate, charge readable only at its boundary, no colour at all in the single-contact case — structurally, not contingently: its one slot-carrying vertex has no possible partner inside — and no oversubscription anywhere ($O(2) = O(3) = 0$). What remains open on the card is open by name, not by hand-waving: the charge read for closed cores (Gaps 14), the $k \to j$ rule (Gaps 11), and the pole-mass/I_{hid} relation (Gaps 7).

Theorem A.2 (FT.32 — The cycle floor family (single-contact Σ_L)). Thm. Let Σ_L be a simple oriented L-cycle, $L \geq 2$ [A1 excludes self-loops; D1.4], with materialized interior chain c and a single observer contact at v_1 ($E_{\text{bd}} = 1 \leq 4\,\delta\tau$, $\delta\tau \geq 1$; FT.17 hypotheses, Def-H.35). Write $\kappa_e := \widehat{ds}^{\,2}(e)$ for $e \in c$. Then:

$$(1)\quad E_{\text{int}} = L, \qquad I_{\text{hid}} = L\ln 2, \qquad m_{\text{hid}} = \frac{L\ln 2}{\hbar} \qquad [\text{FT.17, Def-H.9(2)}];$$

$$(2)\quad n(v_1) = 3,\ n(v_i) = 2\,(i \geq 2)\,[\text{D0.8}]; \qquad O(2) = O(3) = 0,\ D_{\text{tick}}(3) = 16 = C_{\text{cap}}\,[\text{CT0.5, CT0.6}];$$

$$(3)\quad \Theta_{\text{int}} = \Theta(c) \in 2\mathbb{Z} \text{ if } c \text{ materialized } [\text{FT.22}];$$

$$\text{Hol}_{\text{int}} = +1\ [\text{FT.25}(1)];$$

$$k = \Theta/2 \in \mathbb{Z};$$

$$W \text{ row fires} \iff \Theta \neq 0;$$

$$(4)\quad \text{structural colourlessness}: \ \text{Colour}(\Sigma_L) \subseteq \{v_1\}\ [\text{D3.1}: \text{ colour slots at } n{=}3 \text{ only; CT3.11}(1)];$$

$$\pi_C(v_1) \subseteq \Sigma_L \text{ impossible } [\text{A1}: \text{ partners distinct}] \implies \mathcal{C}$$

$$= 0 \ \vee \ \text{Colour}$$

$$= \varnothing \implies \text{strong} \notin \mathcal{A}(\Sigma_L; C)\ \forall C;$$

$$(5)\quad \mathcal{A}(\Sigma_L; C) \supseteq \{G, H\}\ [G: \ I_{\text{hid}} \neq 0;\ H: \ \text{Def} - \text{H.36}];$$

$$\text{EM} \in \mathcal{A} \iff \exists i: \ f_{v_i}$$

$$\neq 0 \text{ [D3.3]};$$

$$\dim \mathcal{K}_{\Sigma_L}$$

$$= 9\ (+1 \text{ EM})$$

$$\leq 10 < 18 \text{ [FT.26, CT0.14]};$$

(6) minimality ladder for $\Theta(c)$ [FT.22′s constraint $\Theta \in 2\mathbb{Z}$] :

uniform-κ : L odd $\Rightarrow \kappa_{\min} = 2,\ \Theta_{\min} = 2L$; $\quad L$ even $\Rightarrow \kappa_{\min} = 1,\ \Theta_{\min} = L$;

all-nonnull per-edge-integral [scope : FT.24] : $\Theta_{\min} = L + (L \bmod 2)$;

null-permitting : $\Theta_{\min} = 2$ (κ profile $(1, 1, 0, \ldots, 0)$ [CT1.19a : $\kappa = 0 \iff$ null]);

scope : uniform/all-nonnull regimes presuppose per-edge $\kappa \in \mathbb{Z}$ — derived at L=2 [FT.24] and on round-trip-closed edges [FT.34(4), forward],

conditional for bare L

≥ 3 cycles [Open Gaps 13 class (iii) : FT.22 alone permits $\kappa \in (2/L)\mathbb{Z}$];

unconditional floor : $\Theta_{\min} = 2\ \forall L$ (rational uniform witness $\kappa \equiv 2/L$);

(7) $\Pi(\Sigma_L) = L^2\, \Theta\, (\ln 2)^2$ $\quad$ $L = 2$ reproduces FT.28's card verbatim (consistency).

Req. FT.17 (Def-H.35), FT.20, FT.22, FT.24, FT.25, FT.26, FT.27, FT.28; A1; Def-H.4, Def-H.9, Def-H.10, Def-H.15; D0.8, D1.4, D3.1, D3.3, D3.8, D3.9; CT0.5, CT0.6, CT0.11, CT0.13, CT0.14 (Part 1); CT1.19a (Part 2); CT3.11, CT3.17 (Part 2); CT4.9 (Part 2).

Pf.

(1) : $E_{\text{int}} = L$ (cycle edges have both endpoints in Σ_L [Def-H.12]);

$$E_{\text{bd}}$$

$$= 1$$

$$\leq 4 \cdot 1 \checkmark;$$

$$\text{CT4.9}(1): \ S(v_1)$$

$$\leq 3 \ln 2,$$

$$\text{channel-visible } 1 \text{ bit} < 3 \checkmark \text{ [as in FT.28}(1)\text{]}$$

$$\implies I_{\text{hid}}$$

$$= L \ln 2 \text{ [FT.17]}$$

$$(2): \text{ edge counts at vertices [D0.8]}: \ v_1: \ 2 \text{ cycle} + 1 \text{ observer} = 3; \quad v_{i \geq 2}: \ 1 \text{ in} + 1 \text{ out} = 2;$$

$$\text{CT0.5/CT0.6}: \ D_{\text{tick}}(n) = 2^{n+1}: \ 8, \ 16; \quad O(2) = O(3) = 0; \quad \text{CT0.11}: \ n \leq 3 \checkmark$$

$$(3): \ \Sigma_L \setminus \partial \Sigma_L$$

$$= E(c) \text{ (single chain)}: \ \Theta_{\text{int}}$$

$$= \Theta(c)$$

$$\in 2\mathbb{Z} \text{ [FT.22]};$$

$$\text{Hol}_{\text{int}}$$

$$= (-1)^{2\mathbb{Z}}$$

$$= +1,$$

$$k$$

$$= \Theta/2 \text{ [FT.25}(1), (3)\text{]};$$

$$W \text{ criterion}: \ \Theta \neq 0 \text{ [FT.20 row, sound direction]}$$

$$(4):\ n(v_i)$$

$$= 2\ (i \geq 2)$$

$$\implies v_i$$

$\in V_2:\ \neg\exists$ colour slot [D3.1, CT0.13 : strong strictly new only at n=3] $\implies$ Colour$(\Sigma_L) \subseteq \{v_1\}$;

FT.27's case split applies at $v_1:$ block $\equiv 0\ \forall C$

$(5):$ FT.20 rows evaluated as in FT.28(3); dim : $5(G)+3(W)+1(H) = 9$ [CT0.12], $+1$ [CT0.14]

$(6):$ uniform : $\Theta = L\kappa \in 2\mathbb{Z}:$ L odd $\Rightarrow \kappa$ even $\Rightarrow \kappa_{\min} = 2$; $\quad L$ even $\Rightarrow \kappa_{\min} = 1$;

all-nonnull : $\min\{\sum \kappa_e :\ \kappa_e \geq 1,\ \sum \in 2\mathbb{Z}\} = L + (L \bmod 2)$;

null-permitting : $(1, 1, 0, \ldots, 0):\ \Theta = 2$

$(7):\ \Pi = E_{\text{int}}^2 \Theta(\ln 2)^2 = L^2\Theta(\ln 2)^2;$ $\quad L = 2:$ (1)–(5) reduce to FT.28(1)–(5) line by line. ■

Rmk (the mass ladder). $m_{\text{hid}}(\Sigma_L) = L \ln 2/\mathfrak{h}$ is linear in the cycle length at single contact — the first ladder structure of the fermion hunt.

Phys. The whole single-contact cycle family is one card: mass proportional to hidden edge count, interior winding even and parity-trivial, colour structurally absent at every rung of the ladder (one slot-carrying vertex can never seal inside), and the observable mask never exceeds ten dimensions. The floors differ not by which forces act — they all carry G, H, W — but by their integer labels (L, Θ, k): the fermion hunt, at floor level, is integer bookkeeping.

Theorem A.3 (FT.35 — The composite floor Σ_{12}: first colour-active, first D^--carrying). Thm. Let $\Sigma_{12} = \{p_1, q_1\} \cup \{p_2, q_2\}$ with materialized 2-cycles $c_1 = (p_1 \to q_1 \to p_1)$, $c_2 = (p_2 \to q_2 \to p_2)$ [FT.24: quanta κ_1, κ_2], joined by the bidirectional connector $e_{12} = (q_1, p_2)$, $\bar{e}_{12} = (p_2, q_1)$ [both $\in \mathcal{R}$], single observer contact at p_1 ($E_{\text{bd}} = 1$; FT.17 hypotheses, Def-H.35). Let $\Gamma = c_1 \cup e_{12} \cup c_2 \cup \bar{e}_{12}$ [valid closed chain, D1.4]. If Γ materializes:

$$(1) \quad \kappa_{12} \in \mathbb{Z}$$

[FT.34(4) : $\Gamma \setminus \{e_{12}, \bar{e}_{12}\} = c_1 \sqcup c_2$ decomposes into materialized closed chains];

$$(2) \quad E_{\text{int}} = 6 \ (c_1 \colon 2,\ c_2 \colon 2,\ \text{connector} \colon 2), \quad I_{\text{hid}} = 6 \ln 2, \quad m_{\text{hid}} = \frac{6 \ln 2}{\mathfrak{h}}$$

[FT.17; CT4.3 retention, as below];

$$(3) \quad \Theta_{\text{int}}(\Sigma_{12}) = 2\kappa_1 + 2\kappa_2 + 2\kappa_{12} = 2\, k(\Gamma) \in 2\mathbb{Z} \ [\text{FT.34(3)}];$$

$$\text{Hol}_{\text{int}} = +1 \ [\text{FT.25}];$$

$$\Pi = 72\, k(\Gamma)\, (\ln 2)^2;$$

$$(4) \quad \text{rungs} : \ n(p_1) = 3,\ n(q_2) = 2,\ n(q_1) = n(p_2) = 4 :$$

$$O(4)$$

$$= 2^5 - 16$$

$= 16 > 0$ [CT0.6] : the junction vertices are D^--configured [CT4.1] : excess locked, ledger-retained [D4.2, CT4.3];

$$(5) \quad \text{colour phases} : \ \text{Colour}(\Sigma_{12}) \subseteq \{p_1, q_1, p_2\} \ [\text{D3.1; CT0.13} : \ \text{strong persists at } n \geq 4];$$

$$\mathcal{C}(\Sigma_{12}; C)$$

$= 1 \iff$ all three seal inside as one ε-type (baryon-like) triple [D3.9, CT3.17, FT.34(5) : 3 admits no $\{2,3\}$-partition with an external remainder];

$$\text{strong-active phase} : \ \dim \mathcal{K}_{\Sigma_{12}}$$

$$= 9 + 1 + 8$$

$= 18$ — the maximal observable mask is attained [FT.26(3)];

$$\text{any external/unsealed attachment} : \ \mathcal{C} = 0 \implies \text{block} \equiv 0 \ [\text{FT.19}].$$

Req. FT.17 (Def-H.35), FT.19, FT.22–FT.28, FT.32, FT.34; D0.8, D1.4, D3.1, D3.9, D4.1, D4.2; CT0.6, CT0.11, CT0.13, CT0.14 (Part 1); CT3.11, CT3.17 (Part 2); CT4.1, CT4.3 (Part 2); Def-H.9, Def-H.10, Def-H.12, Def-H.15.

Pf.

$$(1):\ \Gamma \setminus \{e_{12}, \bar{e}_{12}\}$$

$$= c_1 \sqcup c_2 :\ \text{edge-disjoint materialized closed chains} \implies \text{FT.34}(4)\ \text{applies} :\ \kappa_{12} \in \mathbb{Z}$$

$$(2):\ \text{all six cycle/connector edges have both endpoints in } \Sigma_{12}\ [\text{Def-H.12}];$$

$$E_{\text{bd}} = 1 \leq 4 \cdot 1\ \checkmark;$$

$$q_1,$$

p_2 oversubscribed $(n{=}4)$: excess locked but retained and counted [CT4.3, D4.2; FT.17 cross-check 2 precedent]

$$\implies I_{\text{hid}}$$

$$= E_{\text{int}} \ln 2 \qquad = 6 \ln 2$$

$$(3):\ \Theta_{\text{int}}$$

$$= \Theta(c_1) + \Theta(c_2) + \kappa_{12} + \kappa_{\bar{e}_{12}}$$

$$= 2\kappa_1 + 2\kappa_2 + 2\kappa_{12}\ [\text{FT.23, FT.24}];$$

$$\text{FT.34}(3):$$

$$= 2k(\Gamma);$$

$$\text{Hol}_{\text{int}}$$

$$= (-1)^{2\mathbb{Z}}$$

$$= +1;$$

$$\Pi$$

$$= E_{\text{int}}^2 \Theta_{\text{int}} (\ln 2)^2$$

$$= 36 \cdot 2k(\Gamma)(\ln 2)^2$$

$$(4):\ n(q_1) = 2\,(c_1){+}2\,(\text{connector}) = 4; \quad n(p_2) = 2{+}2 = 4; \quad n(p_1) = 2{+}1 = 3; \quad n(q_2) = 2\,[\text{D0.8}];$$

$$\text{CT0.5/CT0.6}:\ D_{\text{tick}}(4)$$

$$= 2^5$$

$$= 32 > 16$$

$$\implies O(4)$$

$$= 16;$$

$$\text{CT4.1}:\ n$$

$$\geq 4$$

$$\implies D^-\text{-configured};$$

$$\text{locked}$$

$$= \text{retained } \neg\text{accessible [D4.2]}$$

$$(5):\ \text{slots at } n \geq 3:\ p_1\ (n{=}3),\ q_1, p_2\ (n{=}4:\ \text{rung-3 structure persists [CT0.13]});$$

$$q_2\ (n{=}2):\ \text{slot-less}$$

$$\mathcal{C} = 1 \iff \forall v \in \text{Colour}:\ \pi_C(v) \subseteq \Sigma_{12}\ [\text{Def-H.10}];$$

$$\text{partition of } 3 \text{ into } \{2,3\}\text{-parts}:\ \text{only } 3$$

$$= 3 \implies \varepsilon\text{-type triple [FT.34(5)]};$$

$$\delta\text{-pair} + \text{remainder}: \ \text{remainder external} \implies \mathcal{C}$$

$$= 0 \ [\text{FT.19}]$$

$$\text{strong-active}: \ \dim = 5{+}3{+}1{+}1{+}8 = 18 \ [\text{CT0.12}, \ \text{CT0.14}, \ \text{CT3.8}] = \text{FT.26(3)'s bound.} \ \blacksquare$$

Rmk (composition creates saturation). The first composite floor necessarily carries dark content: joining two floors by a bidirectional channel pushes both junction vertices to rung 4, where D^- locks 16 registers' worth of excess [CT0.6, D4.1, D4.2] — retained in the ledger, inaccessible to any observer [D1.11, FT.3]. Composite fermion structure and saturation content are born together; m_{hid} counts the locked content as retained exactly as FT.17's cross-check 2 already established.

Phys. Two floor fermions wired into one object produce the program's first colour-active floor — sealed as a baryon-type triple, the smallest sealing the partition rule allows at three slots — its first dark junction content, its maximal eighteen-dimensional observable mask, and a composite winding that is the sum of the parts plus the integer distance between them. Meson-type pairs (two slots) and baryon-type triples (three slots) are not conventions: they are the only parts the sealing grammar has.

Theorem A.4 (FT.42 — The Platonic card template). Thm. Under Def-H.20's setup, with (M2) [hence $\kappa_e \in \mathbb{Z}$ on every channel, FT.24] and (M1) [hence $\Theta_f \in 2\mathbb{Z}$ per face, FT.22]:

$$(1) \quad E_{\text{int}} = 2E,$$

$$I_{\text{hid}} = 2E \ln 2,$$

$$m_{\text{hid}} = \frac{2E \ln 2}{\mathfrak{h}}$$

$$[\text{observed: FT.17}, \ \text{Def-H.35}; \ \text{dark: Def-H.9(2) with } E_{\text{vis}} = 0];$$

$$(2) \quad \text{rungs (rung correction, FT.62)}: \ \text{dark } n(v)$$

$$= 2d \ \forall v;$$

$$\text{observed (1 directed edge) } n(v_1)$$

$$= 2d + 1;$$

bidirectional contact $2d+2$;

others $2d$ [D0.8 : directed real-edge count; CT0.5 : ϕ_e per incident real edge];

$O(2d) = 112$ (d=3), 496 (d=4), 2032 (d=5); $O(2d{+}1) = 240, 1008, 4080$ [CT0.6];

saturation universality (the "accessibility trichotomy" epithet does not stand, FT.62(3)) : every vertex of every bidirectional floor ($d \geq 2$ channels : $n \geq 4$) is D^--configured [CT4.1],

content locked, retained, counted [CT4.3, D4.2];

grade by degree : $O(2d)$;

the lock-free family is the unidirectional one [FT.28(5), FT.32(2); FT.62(3)];

(3) $\Theta_{\text{int}} = 2K_S$,

$$K_S := \sum_{\text{channels}} \kappa_{\text{ch}} \in \mathbb{Z}$$

[FT.23 : $\kappa_{\bar{e}} = \kappa_e$; D1.7 : additivity];

$\text{Hol}_{\text{int}} = +1$ [FT.25(2)];

W row fires $\iff K_S \neq 0$;

(4) $\mathcal{A}(\Sigma_S; C) \supseteq \{G, H\}$ [G : $I_{\text{hid}} \neq 0$; H : Def − H.36];

EM $\in \mathcal{A} \iff \exists v$: $f_v \neq 0$ [D3.3];

(scope [FT.62(5)] : type-(d,d) vertices leave D3.3's literal type-$(1,1)$ domain empty on dark bidirectional floors; the row evaluates on observed variants via chain-local pass-through reads [FT.21(i)]);

strong $\in \mathcal{A} \iff$ inside-sealed phase ($\mathcal{C} = 1$) : exists for every S ($V \geq 4$ partitions into 2s and 3s [FT.34(5)]),

in $g(V)$ groupings [FT.41(b)];

$$\dim \mathcal{K}_{\Sigma_S} = 9\ (+1)\ (+8) \le 18\ \text{[FT.26]};$$

$$(5)\quad \text{the card table :}\ \ \Pi = E_{\text{int}}^2\,\Theta_{\text{int}}(\ln 2)^2 = 8E^2K_S(\ln 2)^2$$

.

S	(V,E,F,d)	$m_{\text{hid}}\cdot\hbar/\ln 2$	$O(2d)$ dark	$O(2d{+}1)$ contact	face $(L,\kappa_{\min},\Theta_{f,\min},k_{f,\min})$	indep. windings	$g(V)$ groupings	$\Pi/(\ln 2)^2$
tetrahedron	$(4,6,4,3)$	12	112	240	$(3,2,6,3)$	3	3	$288\,K_S$
cube	$(8,12,6,3)$	24	112	240	$(4,1,4,2)$	5	385	$1152\,K_S$
octahedron	$(6,12,8,4)$	24	496	1008	$(3,2,6,3)$	7	25	$1152\,K_S$
dodecahedron	$(20,30,12,3)$	60	112	240	$(5,2,10,5)$	11	292938721075	$7200\,K_S$
icosahedron	$(12,30,20,5)$	60	2032	4080	$(3,2,6,3)$	19	164395	$7200\,K_S$

(6) consistency anchors : tetrahedron observed with a bidirectional contact reproduces FT.17 cross-check 3 verbatim ($E_{\text{int}}=12,\ E_{\text{bd}}=2\le 4\cdot 1,\ I_{\text{hid}}=12\ln 2$);

the tetrahedron face $\{A,B,C\}$ observed by D is cross-check 2 ($E_{\text{int}}=6,\ E_{\text{bd}}=6,\ \delta\tau\ge 2$);

$$\Sigma_{\min} = E{=}1 \text{ cell [FT.28]}; \qquad \Sigma_{12} = E{=}3 \text{ cell [FT.35]}.$$

Req. Def-H.20; FT.17 (Def-H.35), FT.22, FT.23, FT.24, FT.25, FT.26, FT.32(6), FT.34(5), FT.41(b); Def-H.1, Def-H.8, Def-H.9, Def-H.15; D0.8, D1.7, D3.1, D3.3; CT0.5, CT0.6, CT0.11, CT0.13, CT0.14 (Part 1); CT3.11 (Part 2); CT4.1, CT4.3, CT4.9 (Part 2); D4.2; solid cardinalities, Euler $V-E+F=2$ [math].

Pf.

$$(1) :\ \text{every channel bidirectional} \implies E_{\text{int}} = 2E\ \text{[Def-H.12 : both endpoints in } \Sigma_S\text{]};$$

$$\text{observed :}\ E_{\text{bd}} = 1 \le 4\cdot 1\ \checkmark \implies I_{\text{hid}} = E_{\text{int}}\ln 2\ \text{[FT.17]};$$

(re-scope [FT.62(4)] : at corrected contact rung $2d{+}1\ge 7$ the vertex is locked; the CT4.9(1) "visible <3" check passes to Def-H.35 + CT4.3 retention; value unchanged)

$$\text{dark :}\ \partial\Sigma_S = \varnothing \implies E_{\text{vis}} = 0 \implies I_{\text{hid}} = E_{\text{tot}} = 2E\ln 2\ \text{[Def-H.9(2); FT.28 dark-floor pattern]};$$

$$m_{\text{hid}} = I_{\text{hid}}/\mathfrak{h}$$

$$(2)\ \text{(FT.62(1))} :\ \text{D0.8 directed counts :}\ n$$

$$= 2d :\ D_{\text{tick}}(2d)$$

$$= 2^{2d+1} : \ 128,$$

$$512,$$

$$2048 \ (d = 3, 4, 5);$$

$$\text{CT0.6} : \ O(2d)$$

$$= 112,$$

$$496,$$

$$2032;$$

$$\text{contact} : \ O(2d{+}1)$$

$$= 240,$$

$$1008, \qquad 4080$$

$$\text{CT4.1} : \ n$$

$$\geq 4$$

$$\implies D^{-}\text{-configured};$$

$$\text{locked}$$

$$= \text{retained [D4.2, CT4.3; FT.17 cross-check 2 precedent]}$$

$$\implies E_{\text{tot}} \text{ accounting intact}$$

$$(3) : \ \Theta_{\text{int}} = \sum_{\text{directed edges}} \kappa_e = \sum_{\text{channels}} (\kappa_e + \kappa_{\bar{e}}) = 2 \sum_{\text{channels}} \kappa_{\text{ch}} \text{ [FT.23]};$$

$$\text{M2} + \text{FT.24} : \ \kappa_{\text{ch}} \in \mathbb{Z} \implies K_S \in \mathbb{Z};$$

$$\mathrm{Hol}_{\mathrm{int}} = (-1)^{2K_S} = +1 \text{ [FT.25(2)]}$$

$$(4):\ \text{FT.20 rows as in FT.28}(3)/\text{FT.32}(5);$$

slots at every vertex ($d \geq 3$ for all five solids [D3.1, CT3.11(1)]; $n \geq 4:$ rung-3 structure persists [CT0.13])

$$V \geq 4 \implies \{2,3\}\text{-partition exists [FT.34(5)]}; \qquad g(V)\ [\text{FT.41}(b)]; \qquad \dim\ [\text{FT.26}(3),\ \text{CT0.14}]$$

$(5):$ face regimes from FT.32(6) uniform-κ minimality (L odd : $\kappa_{\min} = 2,\ \Theta_{f,\min} = 2L,\ k_{f,\min} = L;$ L even : $1,\ L,\ L/2$);

$$\Pi = (2E)^2 \cdot 2K_S(\ln 2)^2 = 8E^2 K_S(\ln 2)^2$$

$$(6):\ \text{FT.17 cross-check 3 : whole bidirectional } K_4,$$

$$E_{\mathrm{int}}$$

$$= \binom{4}{2} \times 2$$

$$= 12$$

$$= 2E\ (E{=}6)\ \checkmark;$$

$$\text{cross-check 2 : face } \{A,B,C\} :\ E_{\mathrm{int}}$$

$$= 6$$

$$= 2 \cdot 3\ \checkmark,$$

$$E_{\mathrm{bd}}$$

$$= 6$$

$$\leq 4 \cdot 2\ \checkmark$$

$$\text{FT.28 : } E{=}1 \implies E_{\mathrm{int}} = 2\ \checkmark; \qquad \text{FT.35 : } E{=}3 \implies E_{\mathrm{int}} = 6\ \checkmark.\ \blacksquare$$

Rmk (what the card does not say). $g(V)$ counts grouping patterns; attachment types (triplet/antitriplet per vertex [D3.8]) and the κ-profile are configuration data on top. The dodecahedron's sealing count (292938721075) and the icosahedron's (164395) are grammar sizes, not state counts. Materialization (M1/M2) is hypothesized, not derived — which faces and which 2-cycles are on-spectrum is the corpus's to decide [D2.4, CT2.5].

Phys. Every Platonic solid is a fermion floor with the same five-line card: mass $= 2E$ hidden bits over $\hbar$, trivial total parity with an integer winding ledger K_S, gravity and Higgs always active, electromagnetism state-dependent, colour available in as many sealing patterns as the vertex count's $\{2,3\}$-grammar allows — and a lock profile fixed by the degree — every bidirectional vertex saturated (rung correction, FT.62: dark rungs are $2d$), graded $O(2d) \in \{112, 496, 2032\}$, excess absorbed and retained by D^-. The solids differ where the integers differ: edges, faces, vertices, degree.

Theorem A.5 (FT.43 — The winding ledger and Euler consistency). Thm. Under FT.42's hypotheses:

$$(1)\quad \text{sum rule}: \quad \sum_f \Theta_f = 2 \sum_{\text{channels}} \kappa_{\text{ch}} = \Theta_{\text{int}} \quad \Longleftrightarrow \quad \sum_f k_f = K_S$$

$$[\text{each channel lies in exactly two faces}: \ FL = 2E;\ \text{D1.7}];$$

$$(2)\quad \text{independent face reads}: \ F - 1 = E - V + 1 = \beta_1(\Gamma_S)$$

$$[\text{one sum rule};\ \text{Euler}\ V - E + F = 2;\ \text{math}: \ \text{cycle rank of a connected graph}];$$

(3) parity lattice : the on-shell κ-profiles $\{\kappa \in \mathbb{Z}^E : \ \Theta_f \in 2\mathbb{Z}\ \forall f\}$ form the kernel of the face-incidence map $\varphi : \mathbb{Z}^E \to \mathbb{Z}_2^F$,

of index 2^{F-1} [math : $\text{rk}_{\mathbb{Z}_2}\ \varphi = F - 1$ for spherical embeddings; machine : explicit GF(2) elimination, tetrahedron and cube];

$$(4)\quad \text{sector counts}: \ 2^{F-1}$$

$$= 2^{\beta_1} : \ \text{tetrahedron } 8,$$

$$\text{cube } 32,$$

$$\text{octahedron } 128,$$

$$\text{dodecahedron } 2048,$$

$$\text{icosahedron } 2^{19} \qquad = 524288;$$

$$(5) \quad \text{Euler as card consistency : } \beta_1$$

$$= E - V + 1$$

$$= F - 1 \text{ ties the three ledgers : mass } (E \text{ via FT.42}(1)),$$

$$\text{colour } (V \text{ via } g(V)),$$

winding ($F-1$ reads) — the sphere's Euler number is an identity among card entries, not an extra assumption.

Req. FT.22, FT.23, FT.24, FT.42; Def-H.20; D1.7; FT.25(2); Euler's formula, cycle-rank and GF(2)-rank facts for connected planar graphs [math]; solid face incidences [math].

Pf.

$$(1) : \text{ each channel } \{e, \bar{e}\} \text{ lies in exactly two faces [solid geometry; } FL = 2E \text{ double-count]} : \quad \sum_f \Theta_f$$

$$= \sum_f \sum_{e \in f} \kappa_e$$

$$= \sum_{\text{channels}} 2\kappa_{\text{ch}}$$

$$= \Theta_{\text{int}} \text{ [D1.7 additivity; FT.42(3)]}$$

$$\Theta_f = 2k_f \text{ [M1 + FT.22]}, \ \Theta_{\text{int}} = 2K_S \text{ [FT.42(3)]} \implies \sum_f k_f = K_S$$

$$(2) : \text{ one linear relation among } F \text{ reads } \implies F - 1 \text{ independent;}$$

$$\text{Euler : } V - E + F = 2 \implies F - 1 = E - V + 1 = \beta_1 \text{ [math]}$$

$$(3) : \ \varphi(\kappa)_f := \sum_{e \in f} \kappa_e \bmod 2 : \ \text{ on-shell locus} = \ker \varphi \text{ [M1 + FT.22]};$$

$$\text{rk}_{\mathbb{Z}_2} \varphi$$

$$= F - 1 \text{ [math : faces generate the cycle space with the single spherical relation]}$$

$$\implies [\mathbb{Z}^E : \ker\varphi] \qquad = 2^{F-1}$$

$(4):\ 2^{F-1}$ evaluated at $F = 4, 6, 8, 12, 20$; $\qquad (5):$ substitution of (2) into FT.42$(1)/(4)$. ■

Rmk (what the lattice index means). Materialization prunes the integer κ-lattice by exactly the solid's $\mathbb{Z}_2$ cycle space: an off-lattice profile (some face with odd Θ_f) is a profile no face-materialized configuration can carry [FT.22]. The pruned cosets are not physical sectors — they are the excluded parity classes; the physical ledger is $\ker\varphi$ with its $F-1$ integer face reads.

Phys. A Platonic fermion's geometric memory is the solid's combinatorics turned into integers: one winding read per face, one relation among them, and the number of independent reads is the skeleton's first Betti number — the same number Euler's formula builds from vertices and edges. Mass counts edges, colour counts vertices, winding counts faces, and $V-E+F=2$ is the statement that the three ledgers are not independent. The solids with more independent cycles remember more windings: tetrahedron 3, cube 5, octahedron 7, dodecahedron 11, icosahedron 19.

Theorem A.6 (FT.44 — Duality signature and the floor taxonomy). Thm. (1) Duality mass-degeneracy. Dual solids exchange (V, F) at fixed E [math]: cube↔octahedron $(8, 12, 6) \leftrightarrow (6, 12, 8)$, dodecahedron↔icosahedron $(20, 30, 12) \leftrightarrow (12, 30, 20)$, tetrahedron self-dual $(4, 6, 4)$. Since $m_{\text{hid}} = 2E\ln 2/\hbar$ [FT.42(1)] and $\Pi = 8E^2K_S(\ln 2)^2$ [FT.42(5)] depend on E only: dual pairs are mass-degenerate and Π-degenerate at matched K_S — cube/octahedron: $24\ln 2/\hbar$; dodecahedron/icosahedron: $60\ln 2/\hbar$; three mass values across the five solids. (2) Ledger splitting. The degeneracy is split by every finer ledger: independent windings $F-1$ (cube 5 vs octahedron 7; dodecahedron 11 vs icosahedron 19); lattice index 2^{F-1} (32 vs 128; 2048 vs 2^{19}); sealing grammar $g(V)$ (385 vs 25; 292938721075 vs 164395); lock profile $O(2d)$ (correction, FT.62(3): cube 112 vs octahedron 496; dodecahedron 112 vs icosahedron 2032 — saturation grade splits both dual pairs). (3) Self-dual floor. The tetrahedron is the unique self-paired cell ($V = F = 4$): its winding and colour ledgers coincide in cardinality ($F-1=3$ independent reads; $g(4)=3$ groupings) — the smallest floor on which the full card, including a δ-pair sealing grammar and a nontrivial winding ledger, is simultaneously active. (4) Taxonomy (all computed floors). Every fermion floor computed in this document obeys $m_{\text{hid}} = E_{\text{int}}\ln 2/\hbar$ with $E_{\text{int}} = 2E_{\text{bidir}} + E_{\text{uni}}$: Σ_{min}: $E_{\text{int}} = 2$ [FT.28]; cycle ladder Σ_L: $E_{\text{int}} = L$ (unidirectional) [FT.32]; Σ_{12}: $E_{\text{int}} = 6$ [FT.35]; Platonic family: $E_{\text{int}} = 2E$ [FT.42]. The informational mass is the directed-edge count, universally, across every computed cell.

Req. FT.42, FT.43; FT.28, FT.32, FT.35; FT.41(b); duality cardinals, Euler [math]; Def-H.20.

Pf.

$$(1):\ \text{duality exchanges } V \leftrightarrow F,\ \text{fixes } E\ [\text{math; cardinals above}];$$

$$m_{\text{hid}},\ \Pi \text{ are } E\text{-functions [FT.42}(1),(5)] \implies \text{degenerate on dual pairs}$$

$$(2):\ F\text{-dependence}:\ \text{FT.43}(2),(4);$$

$$V\text{-dependence}:\ g(V)\ [\text{FT.41}(b)];$$

d-dependence : $O(d)$ [FT.42(2)] : cube d=3 (O=0 dark) vs octahedron d=4 (O=16)

(3) : $V = F = 4$: $F - 1 = 3 = g(4)$ [FT.41(b) : $g(4) = 3$];

both ledgers active : winding [FT.43(2)], δ-pair sealing $(4 = 2{+}2)$ [FT.34(5)]

(4) : direct count per cell : FT.28(1)–(2), FT.32(1), FT.35(2), FT.42(1). ■

Rmk (candidate reading, recorded not claimed). Dual mass-degeneracy split by finer integer ledgers is a derived degeneracy-splitting template for families sharing a mass-like read, distinguished by winding/colour/lock data (an application-layer reading). No physical identification — of duals with generations, or of any card entry with a named particle — is made or implied here.

Phys. Duality, in the floor ledger, is exact mass-degeneracy with everything else distinct: the cube and octahedron weigh the same and remember differently — five winding reads against seven, three-hundred-eighty-five sealing patterns against twenty-five, one hundred twelve locked registers per vertex against four hundred ninety-six (rung correction, FT.62(3)). The tetrahedron, self-dual, is the smallest floor whose card exercises every ledger at once. And across the whole computed taxonomy — digon, cycles, composite pair, all five solids — one law has never broken: the informational mass is the count of directed internal edges, in bits over $\mathfrak{h}$.

Theorem A.7 (FT.45 — The bidirectional polygon floors P_n: cards, minima, and the null-carrying sector). Thm. Let Σ_{P_n} ($n \geq 3$) be the fully bidirectional 1-skeleton of the regular n-gon [Def-H.20's setup class], $n \geq 3$: n vertices, n bidirectional channels, $2n$ directed edges; sphere compactification with two length-n faces (interior, exterior): $FL = 2n = 2E$, $V - E + F = n - n + 2 = 2$ ✓ [math]. (The digon-walk floor Σ_{min} [FT.28] is the taxonomy's E=1 cell, ¬a polygon member — clause (5).) Under (M2$_n$) all n channels materialize as 2-cycles [FT.24: $\kappa_i \in \mathbb{Z}$] and (M1$_n$) the interior face materializes [FT.22: $\Theta_f \in 2\mathbb{Z}$]:

(1) $E_{\text{int}} = 2n,$

$$I_{\text{hid}} = 2n \ln 2,$$

$$m_{\text{hid}} = \frac{2n \ln 2}{\mathfrak{h}},$$

$$\Pi = 8n^2 K_P (\ln 2)^2,$$

$K_P := \Theta_{\text{int}}/2 = \sum_{i=1}^{n} \kappa_i$ [FT.42 template];

(2) rungs (rung correction, FT.62) : dark $n(v)$

$$= 4 \; \forall v \; (2 \text{ channels each}, \; \text{D0.8 directed count}; \; \text{type } (2, 2));$$

$$\text{single-contact observed } n(v_1)$$

$$= 5,$$

$$\text{others } 4;$$

$$O(4)$$

$$= 16$$

$$= C_{\text{cap}},$$

$$O(5)$$

$$= 48 \; [\text{CT0.5}, \; \text{CT0.6}] : \; \text{uniform minimal saturation at every vertex of every member [CT4.1]};$$

$$\text{locked content retained and counted [CT4.3}, \; \text{D4.2}];$$

$$\text{the lock-free family is the unidirectional one [FT.32}(2), \; \text{FT.62}(3)];$$

$$(3) \quad \text{walk space and the single parity constraint : } \beta_1(\text{directed multigraph})$$

$$= 2n - n + 1$$

$$= n + 1 \; [\text{math}];$$

$$\text{closed-walk structure : flow balance at every vertex } \implies \text{ per-channel net flow constant}$$

$$= w \; (\text{the walk's winding});$$

$$\Theta(w)$$

$$= w$$

$$\textstyle\sum_i \kappa_i + 2\sum_i \kappa_i b_i \;\; (b_i : \text{ reverse traversals}) \implies \Theta(w)$$

$$\in 2\mathbb{Z}\ \forall w \iff \textstyle\sum_i \kappa_i$$

$$\in 2\mathbb{Z} \;\; (w{=}1 : \text{ the face, } \mathrm{M1}_n) :$$

exactly one independent parity constraint;

on-shell κ-lattice $= \{\kappa \in \mathbb{Z}^n : \sum \kappa_i \in 2\mathbb{Z}\}$: index $2 = 2^{F-1}$ [FT.43(3) pattern at $F{=}2$];

digons $(w{=}0)$: $\Theta = 2\sum \kappa_i b_i \in 2\mathbb{Z}$ automatic [$\mathrm{M2}_n$, FT.24];

(4) four minimality regimes ($\Theta_{\min} > 0$, face materialized; machine-enumerated for $n \leq 7$) :

spacelike-or-null $[\kappa \geq 0]$: $\Theta_{\min} = 2\ \forall n \geq 3$ (profile $(1, 1, 0, \ldots, 0)$);

null-carrying dimension $= |\{i : \kappa_i = 0\}| \geq n - 2$ in every such minimizer;

strictly-spacelike $[\kappa \geq 1]$: $\Theta_{\min}$

$= n + (n \bmod 2)$ (n even : unique all-1 profile; n odd : one 2, rest 1);

uniform $[\kappa_i \equiv \kappa]$: $\Theta_{\min} = n$ (n even, $\kappa{=}1$) / $2n$ (n odd, $\kappa{=}2$) [FT.32(6)];

signature-mixed [$\kappa \in \mathbb{Z}$, timelike $\kappa < 0$ admissible [CT1.12]] : $\Theta_{\min}$

$= 2\ \forall n$

≥ 3 (witnesses : $n \geq 4$ even : $(1^{(n+2)/2}, (-1)^{(n-2)/2})$; $n \geq 5$ odd : $(1^{n-1}, -(n{-}3))$; $n{=}3$: $(2, 1, -1)$);

(5) the digon distinction : the null-carrying sector has dimension $n - 2$: at n

$= 2$ ($\Sigma_{\min}$) it is $\{-1\}$-dimensional

$$= \text{empty} : \ \Theta_{\min}$$

$$= 2 \text{ there forces } \kappa_1$$

$$= \kappa_2$$

$$= 1 \text{ (all-nonnull, purely spacelike) } [\text{FT.24}(4)];$$

$$\text{for every } n$$

$$\geq 3 \text{ the minimum is null-carrying. The fermion floor is derived to be a digon, not a polygon;}$$

$$(6) \quad \text{sealing} : \ \text{dark } P_n : \ \text{Colour}$$

$$= \varnothing \ (n(v) = 2 < 3 \ [\text{D3.1}]) \implies \text{strong} \neg \in \mathcal{A} \text{ always } [\text{FT.20}];$$

$$\text{observed} : \ \text{Colour}$$

$$= \{v_1\} : \ \neg\exists \text{ inside partner } [\text{A1}] \implies \text{structurally colourless } [\text{FT.27 case split}];$$

$$\mathcal{A} \supseteq \{G, H\},$$

$$W$$

$$\iff K_P$$

$$\neq 0,$$

$$\text{EM} \iff \exists v : \ f_v$$

$$\neq 0;$$

$$\dim \mathcal{K} \qquad \leq 10 < 18;$$

(7) anchors : $\Sigma_{\min}$ (digon walk, E=1) is adjacent to but ¬a member of this family [FT.28/FT.44(4) taxonomy] : clauses (3)–(5) degenerate there (¬∃ proper face; its minimum forces κ=1 [FT.24(4)]);

the unidirectional 3-cycle

$=$ FT.17 cross-check 1 [FT.32];

the observed tetrahedron face is NOT P_3: in bidirectional K_4 the face vertices carry the third (external) channel each $(n(v) = 4,\ E_{\text{bd}} = 6)$ [FT.17 cross-check 2].

Req. FT.17 (Def-H.35), FT.20, FT.22, FT.23, FT.24, FT.25, FT.26, FT.27, FT.28, FT.32(6), FT.42; Def-H.20; A1; D0.8, D1.7, D3.1, D3.3; CT0.5, CT0.6, CT0.14 (Part 1); CT1.12 (C, signature classes); CT4.9 (Part 2); cycle-rank and partition facts [math].

Pf.

$(1), (2):$ FT.42(1)–(2) template at $E = n,\ V = n,\ d_{\text{dark}} = 2,\ d_{\text{contact}} = 3$;

CT0.6 : $O(2) = O(3) = 0$;

CT4.9(1) : visible 1 bit < 3 ✓

$(3):\ \beta_1 = 2n - n + 1 = n + 1$ [math : connected multigraph];

closed walk : in-degree

$=$ out-degree at every vertex

$\Longrightarrow\ a_i - b_i$

$=$ const

$= w$ [flow balance around the cycle; a_i, b_i : forward/reverse traversals of channel i]

$$\Theta(w)$$

$$= \sum_i \kappa_i (a_i + b_i)$$

$$= \sum_i \kappa_i (2b_i + w)$$

$= w \sum_i \kappa_i + 2 \sum_i \kappa_i b_i$ [D1.7 additivity; FT.23 : $\kappa_{\bar{e}} = \kappa_e$] $\Longrightarrow$ parity$(\Theta(w))$

$$= \text{parity}\big(w \textstyle\sum_i \kappa_i\big)$$

face walk : $w = 1$: constraint $\sum \kappa_i \in 2\mathbb{Z}$ [M1$_n$ + FT.22]; digons : $w = 0$: automatic;

$$w$$

≥ 2 : automatic once the face is constrained $\implies$ exactly one independent parity constraint;

index 2 [math]

(4) : spacelike-or-null : $(1, 1, 0^{n-2})$: $\Theta = 2 \in 2\mathbb{Z}$ ✓; $\Theta > 0$ even $\Rightarrow \Theta \geq 2$: minimal;

$\Theta = 2$ with $\kappa \geq 0 \implies \leq 2$ positive units $\implies \geq n-2$ zeros

strictly-spacelike : $\min\{\sum \kappa_i : \kappa_i \geq 1, \sum \in 2\mathbb{Z}\} = n + (n \bmod 2)$;

uniform : $n\kappa \in 2\mathbb{Z} \Rightarrow \kappa_{\min} = 1$ (n even), 2 (n odd) [FT.32(6)]

signature-mixed witnesses : $n \geq 4$ even : $\frac{n+2}{2}$ entries $+1$, $\frac{n-2}{2}$ entries -1 : $\Theta = 2$ ✓;

$n \geq 5$ odd : $(1^{n-1}, -(n-3))$: $\Theta = (n-1) - (n-3) = 2$ ✓;

$n = 3$: $(2, 1, -1)$: $\Theta = 2$ ✓

(5) : $n = 2$: $\Theta = 2\kappa$ [FT.23]; $\Theta = 2 \Rightarrow \kappa = 1$: $\neg\exists$ null slot;

$n \geq 3$: $(1, 1, 0^{n-2})$ admissible [CT1.19a : $\kappa = 0 \iff$ null edge permitted]

(6) : dark : $n(v)$

$= 2 \implies \neg\exists$ colour slots [D3.1 : slots = *colour* at n=3; CT0.13 : strong strictly new at n=3] $\implies$ Colour

$$= \varnothing;$$

$$\text{observed}: \ \text{Colour} \subseteq \{v_1\}: \ \text{FT.27 case split}$$

$$(7): \ \text{direct comparison of } (1)\text{–}(6) \text{ against FT.28 } (n{=}2), \ \text{FT.17 cross-checks 1–2.} \ \blacksquare$$

Rmk (regime precedence; withdrawn draft claim). The four regimes are nested constraint classes, not alternatives: strictly-spacelike $\subseteq$ spacelike-or-null; signature-mixed relaxes both. Physically primary for a spatial polygon floor are the two spacelike regimes; the mixed regime is kinematically admissible [CT1.12: sig $= (-,+,+,+)$ permits $\kappa < 0$] but describes floors with timelike interior edges — a different physical situation, recorded for completeness. An earlier draft's "$n \equiv 2 \pmod 4$ anomaly" is withdrawn as an arithmetic error (its witness summed to 8, not 6): no residue-class anomaly exists — the strictly-spacelike floor is $n + (n \bmod 2)$ for every n, with the unique all-1 profile at even n.

Rmk (accessibility contrast, corrected [FT.62(3)]). P_n dark has $n(v) = 4$ everywhere [D0.8 directed count]: exactly one channel quantum over budget — $O(4) = 16 = C_{\text{cap}}$, uniform in n: the polygon family is the uniform minimal-saturation family (the "unlimited-accessibility" epithet does not stand). The Platonic family saturates graded by degree, $O(2d) \geq 112$; the lock-free cells are the unidirectional floors ($n \leq 3$) [FT.28(5), FT.32(2)]. First structural difference between the two polygon-derived families — uniform vs graded saturation — derived from D0.8/CT0.5/CT0.6 alone.

Phys. Bidirectional polygons are the fermion program's cleanest infinite family: one uniform minimal lock per vertex ($O(4) = 16$, correction), no colour ever, exactly one parity constraint from the face (an index-2 on-shell lattice), and a null-carrying minimal sector of dimension $n - 2$ — one that has no analogue at the digon, whose minimum forces $\kappa = 1$: the fermion floor's ledger is qualitatively polygon-external.

Theorem A.8 (FT.49 — The tetrahedron-face floor: full card). Thm. Let $\Sigma_\triangle = \{A, B, C\}$ be the bidirectional 3-face of the bidirectional K_4 [FT.17 cross-check 2's setup], observed by the fourth vertex D (six boundary edges). Under M2 (face-channel digons materialize [FT.24]) and M1 (both directed face triangles materialize [FT.22]):

$$(1) \quad E_{\text{int}} = 6, \quad E_{\text{bd}} = 6 \leq 4\,\delta\tau \Rightarrow \delta\tau \geq 2 \ [\text{FT.17 hypotheses, Def-H.35}];$$

$$I_{\text{hid}} = 6 \ln 2,$$

$$m_{\text{hid}} = \frac{6 \ln 2}{\mathfrak{h}} \ [\text{the multi-tick observation cell of the taxonomy}];$$

$$(2) \quad \text{interior channel structure}$$

$$= \Sigma'_{P_3} s \ [\text{FT.45 at } n{=}3]: \ \kappa\text{-lattice } \{\kappa \in \mathbb{Z}^3 : \ \kappa_{AB}{+}\kappa_{BC}{+}\kappa_{CA} \in 2\mathbb{Z}\} \text{ of index } 2;$$

$$\text{both directed triangles give the same constraint [FT.23 : orientation-even]};$$

$$(3)\quad \Theta_{\text{int}} = 2\,\Theta_{\triangle} = 4\,k_{\triangle} \in 4\mathbb{Z},$$

$$k_{\triangle} := \Theta_{\triangle}/2 \in \mathbb{Z}\ [\text{FT.22}];$$

$$\text{Hol}_{\text{int}} = +1\ [\text{FT.25}(2)];$$

$$W \text{ row fires} \iff \Theta_{\triangle} \neq 0;$$

$$(4)\quad \text{minimality regimes [FT.45}(4) \text{ at } n{=}3] : \quad \Theta_{\triangle,\text{min}}$$

$$= 2\ (\text{null-carrying } (1,1,0)\text{-class}),$$

$$4\ (\text{strictly-spacelike } (1,1,2)\text{-class}),$$

$$6\ (\text{uniform } \kappa{=}2);$$

$$\Pi = 36\,\Theta_{\text{int}}(\ln 2)^2 \in \{144, 288, 432\}(\ln 2)^2;$$

$$(5)\quad \text{rungs and locks : } n(A) = n(B) = n(C) = 6\ [\text{D0.8 : 3 channels each in the } K_4];$$

$$O(6)$$

$$= 2^7 - 16$$

$$= 112\ [\text{CT0.5, CT0.6}] : \text{ every vertex } D^-\text{-configured [CT4.1]},$$

$$\text{locked content retained and counted [CT4.3, D4.2]};$$

$$(6)\quad \text{baryon-type sealing phase : all three vertices slot-carrying } (n \geq 3\ [\text{D3.1, CT0.13 : rung-3 structure persists}]);$$

$$\mathcal{C}(\Sigma_{\triangle}; C)$$

$= 1 \iff$ the three attachments seal as one ε-type triple [$3 = 3$ is the only $\{2,3\}$-partition : FT.34(5), $g(3) = 1$ FT.41(b)] : the first computed cell whose strong phase is exactly D3.9′s baryon structure B

$$= \varepsilon_{ijk} q^i q^j q^k \text{ (structural correspondence, } \neg\text{particle identification)};$$

$$(7)\quad \mathcal{A}(\Sigma_{\triangle};C)\supseteq\{G,H\}\ [G:\ I_{\text{hid}}\neq 0;\ H:\ \text{Def}-\text{H.36}];$$

$$\text{EM}\iff\exists v:\ f_v$$

$$\neq 0;$$

$$\text{strong}\iff\varepsilon\text{-sealed};$$

$$\dim\mathcal{K}=9\ (+1)\ (+8)\leq 18\ [\text{FT.26}];$$

$$(8)\quad \text{boundary algebra}:\ N(\Sigma_{\triangle})=6\implies\mathcal{H}_{\text{slot}}=\mathbb{C}^6,\ \iota_{\Sigma}:\ \mathfrak{u}(6)\text{ of dimension }36;$$

$$\text{strictly new at }\Sigma:\ 36-6$$

$$=30\text{ real dimensions }[\text{FT.41}(d)]\text{ — the largest exchange sector computed.}$$

Req. FT.17 (Def-H.35), FT.22, FT.23, FT.24, FT.25, FT.26, FT.34(5), FT.41(b),(d), FT.45; Def-H.9, Def-H.10, Def-H.12, Def-H.15; D0.8, D3.1, D3.3, D3.9; CT0.5, CT0.6, CT0.13, CT0.14 (Part 1); CT3.11, CT3.17 (Part 2); CT4.1, CT4.3 (Part 2); D4.2.

Pf.

$$(1):\ \text{FT.17 cross-check 2 verbatim}:\ E_{\text{int}}=6,\ E_{\text{bd}}=6,\ \delta\tau\geq 2;$$

$$I_{\text{hid}}=E_{\text{int}}\ln 2=6\ln 2\ [\text{FT.17},\ \text{Def-H.35}];\quad m_{\text{hid}}=I_{\text{hid}}/\mathfrak{h}\ [\text{Def-H.9}(2)]$$

$$(2):\ \text{interior channels}$$

$$=\{AB,BC,CA\}\text{ bidirectional}$$

$$=\Sigma_{P_3}\text{ interior }[\text{FT.45 setup}];$$

$$\text{digons}:\ \kappa\in\mathbb{Z}\ [\text{M2}+\text{FT.24}];$$

$$\text{triangles}:\ \Theta_{\triangle}$$

$$=\kappa_{AB}+\kappa_{BC}+\kappa_{CA}\ [\text{FT.23}:\ \text{same both directions}]$$

$$\in 2\mathbb{Z}\ [\text{M1}+\text{FT.22}]$$

$$(3):\ \Theta_{\text{int}} = \sum_{6 \text{ directed}} \kappa = 2\Theta_{\triangle};$$

$$\Theta_{\triangle} = 2k_{\triangle}\ [\text{FT.22}] \implies \Theta_{\text{int}} = 4k_{\triangle};$$

$$\text{Hol}_{\text{int}} = (-1)^{4k_{\triangle}} = +1\ [\text{FT.25}(2)]$$

$$(4):\ \text{FT.45}(4) \text{ regimes at } n{=}3;$$

$$\Pi = E_{\text{int}}^2 \Theta_{\text{int}} (\ln 2)^2 = 36 \cdot (2\Theta_{\triangle})(\ln 2)^2 :\ \Theta_{\triangle} \in \{2,4,6\} \Rightarrow \{144, 288, 432\}(\ln 2)^2$$

$$(5):\ \text{D0.8}:\ n$$

$$= 3 \text{ channels} \times 2$$

$$= 6;$$

$$\text{CT0.5/CT0.6}:\ D_{\text{tick}}(6)$$

$$= 2^7$$

$$= 128,$$

$$O(6)$$

$$= 112;$$

$$\text{CT4.1}:\ n$$

$$\geq 4$$

$$\Rightarrow D^{-}\text{-configured};$$

retained [CT4.3]

$$(6):\ n \geq 3 \Rightarrow \text{slots [D3.1, CT3.11}(1)\text{]}; \quad \text{partitions of 3 into } \{2,3\}\text{-parts}:\ \text{only 3 [FT.34}(5)\text{]};$$

$$g(3) = 1 \text{ [FT.41}(b)\text{]}; \quad \varepsilon\text{-triple} = B \text{ structure [D3.9, CT3.17]}$$

$$(7): \text{ FT.20 rows;}$$

$$(8): \ N(\Sigma)$$

$$= E_{\text{bd}}$$

$$= 6 \text{ [Def-H.12]};$$

$$\dim \mathfrak{u}(6)$$

$$= 36;$$

$$\text{factored diagonal}$$

$$= 6 \implies \text{strictly new}$$

$$= 30 \text{ [FT.41}(d)\text{]}. \blacksquare$$

Rmk (relation to the polygon and Platonic cells). $\Sigma_{\triangle}$ is neither Σ_{P_3} nor a Platonic floor: its interior ledger is the polygon's (index-2 lattice, $n{-}2{=}1$ null freedom at the minimum), while its rung/lock ledger is the ambient K_4's ($n{=}6$, $O{=}112$) — the first computed cell where interior and ambient ledgers visibly decouple. Its sealing phase is the baryon type; Σ_{12}'s was also ε-type [FT.35(5)] but at three slots from two floors plus a junction — the grammar [FT.34(5)] is the same, the wiring differs.

Phys. The tetrahedron face is the program's densest small cell: six hidden bits, a winding ledger even in both directions ($\Theta_{\text{int}} \in 4\mathbb{Z}$), every vertex deep in the saturation regime with 112 registers' absorption rate each, one baryon-type colour phase, and thirty dimensions of strictly-new boundary exchange structure — all of it derived from cards already in the ledger, assembled without a single new premise beyond the declared materialization hypotheses.

Theorem A.9 (FT.50 — Composition and gluing laws). Thm. (1) Bridged union. Let $\Sigma^{(1)}, \Sigma^{(2)}$ be floors with bidirectional interiors [$E_{\text{int}}^{(i)} = 2E_i$], joined by a bidirectional connector path $P = (f_1, \ldots, f_m)$ (disjoint from both interiors), single external contact; all digons materializing [M2-type]. Then

$$E_{\text{int}}(\Sigma) = 2E_1 + 2E_2 + 2m,$$

$$m_{\text{hid}}(\Sigma) = m_{\text{hid}}^{(1)} + m_{\text{hid}}^{(2)} + \frac{2m \ln 2}{\mathfrak{h}}$$

[FT.17, Def-H.35] : informational mass is additive under bridged composition.

(2) Generalized round-trip. If $\Gamma = c_1 \cdot P \cdot c_2 \cdot \bar{P}$ materializes [c_i materialized closed chains at the attachments; D1.4 concatenation]: $d_{12} := \Theta(P) \in \mathbb{Z}$ and $k(\Gamma) = k_1 + k_2 + d_{12}$ [FT.34(2)–(3) generalized to m-channel connectors]; per-edge integrality $\kappa_{f_j} \in \mathbb{Z}$ requires each connector digon to materialize [FT.24] — the sum d_{12} is integral without that strengthening. (3) Junction saturation. A bidirectional bridge into a vertex of interior degree $d \geq 2$ gives $n_{\text{junction}} = d + 2 \geq 4 \implies O(d{+}2) = 2^{d+3} - 16 > 0$: composition creates darkness at every junction [CT0.5, CT0.6, CT4.1, D4.2; FT.35's Rmk generalized]; content retained [CT4.3]. For s triangular floors sharing one junction vertex: $2s$ channels meet there, $n = 4s$ [D0.8], $O(4s) = 2^{4s+1} - 16$ — absorption rate grows exponentially in the sharing multiplicity [D4.2: rate $\propto O(n)$]: high-multiplicity junctions are maximally dark. (4) Gluing law. For $\Sigma = \Sigma_1 \cup_G \Sigma_2$ glued along a shared subgraph G: $E_{\text{int}} = E^{(1)}_{\text{int}} + E^{(2)}_{\text{int}} - |E_{\text{dir}}(G)|$; the on-shell parity lattice is $\{\kappa : \Theta_c \in 2\mathbb{Z}\ \forall$ materialized closed chains $c\}$ — its index is $2^{\text{rk}_{\mathbb{Z}_2}(\text{materialized cycle system})}$, dependencies automatic: a cycle homologous to a sum of materialized cycles inherits evenness [D1.7 additivity; FT.23 orientation-blindness]. (5) Computed cells. Bowtie (two bidirectional triangles sharing one vertex X): $E_{\text{int}} = 12$, $m_{\text{hid}} = 12 \ln 2/\hbar$; both faces materialized $\Rightarrow \Theta_{\text{int}} = 2(K_1{+}K_2) \in 4\mathbb{Z}$; constraint rank 2 (index 4); rungs: $n(X) = 8$ [$O(8) = 496$], other four vertices $n = 4$ [$O(4) = 16$] — every vertex locked; five slots $\Rightarrow g(5) = 10$ sealing groupings (one δ-pair + one ε-triple) [FT.41(b)]. Double-triangle patch (two tetrahedron faces glued along channel AB; $\Sigma = \{A, B, C, D\}$): $E_{\text{int}} = 10$; constraints $\Theta_{ABC}, \Theta_{ABD} \in 2\mathbb{Z}$: $\text{rk}_{\mathbb{Z}_2} = 2$ (index 4), and the boundary 4-cycle constraint is derived: $\Theta_{ACBD} = \Theta_{ABC} + \Theta_{ABD} - 2\kappa_{AB} \in 2\mathbb{Z}$; rungs $n(A) = n(B) = 6$ [$O = 112$], $n(C) = n(D) = 4$ [$O = 16$]; four slots $\Rightarrow g(4) = 3$ groupings, all $\delta{+}\delta$ (meson–meson phases — the only $\{2, 3\}$-partition of 4).

Req. FT.17 (Def-H.35), FT.22, FT.23, FT.24, FT.27, FT.34, FT.35, FT.41(b), FT.45; Def-H.9, Def-H.10, Def-H.12; D0.8, D1.4, D1.7, D3.1, D3.9; CT0.5, CT0.6 (Part 1); CT3.17 (Part 2); CT4.1, CT4.3 (Part 2); D4.2; $\mathbb{Z}_2$-rank facts [math].

Pf.

$$(1) : E_{\text{int}} : \text{ interiors disjoint } \sqcup\ 2m \text{ connector directed edges;}$$

$$E_{\text{bd}} = 1 \leq 4\delta\tau\ \checkmark;$$

$$I_{\text{hid}} = E_{\text{int}} \ln 2\ [\text{FT.17}] \implies m_{\text{hid}} = \tfrac{E_{\text{int}} \ln 2}{\hbar} = \text{sum of parts}$$

$$(2) : \text{ FT.34(1)} : \Theta(P) + \Theta(\bar{P})$$

$$= \Theta(\Gamma) - \Theta(c_1) - \Theta(c_2)$$

$$\in 2\mathbb{Z};$$

$$\Theta(\bar{P})$$

$$= \Theta(P) \text{ [FT.23 per edge]} \implies \Theta(P)$$

$$= d_{12} \in \mathbb{Z};$$

$$k(\Gamma) = \tfrac{\Theta(\Gamma)}{2} = k_1 + k_2 + d_{12}$$

per-edge : each f_j needs its own round-trip or digon materialization [FT.34(4)/FT.24] : sum integrality does not imply edge integrality [scope stated]

$$(3) : \ n_{\text{junction}} = d + 2 \text{ [D0.8 : bridge adds 1 in + 1 out]};$$

$$d \geq 2 \implies n \geq 4 : \ O(n) = 2^{n+1} - 16 > 0 \text{ [CT0.6]}; \quad \text{CT4.1 : } D^{-}\text{-configured};$$

$$s \text{ triangles at one junction : } 2s \text{ channels}$$

$$\implies n$$

$$= 4s,$$

$$O(4s)$$

$$= 2^{4s+1} - 16 \text{ [D0.8, CT0.6]};$$

$$\text{rate } \propto O(n) \text{ [D4.2]}$$

$$(4) : \ E_{\text{int}} : \text{ inclusion–exclusion on directed edges};$$

$$\text{constraints : each materialized } c : \ \Theta_c$$

$$= \sum_{e \in c} \kappa_{\text{ch}(e)} \pmod 2$$

$$= 0 \text{ [FT.22/FT.23]};$$

$$\text{rank over } \mathbb{Z}_2 \text{ of the system}$$

$$= \text{independent constraint count};$$

$$\text{index}$$

$$= 2^{\text{rk}}$$

$$\text{dependency witness : } \Theta_{ACBD}$$

$$= \Theta_{ABC} + \Theta_{ABD} - 2\kappa_{AB} \text{ [D1.7 additivity on shared channel] } \implies \text{ boundary-cycle constraint derived}$$

$$(5) : \text{ bowtie : } E_{\text{int}}$$

$$= 2 \cdot 6$$

$$= 12;$$

$$n(X)$$

$$= 8 \text{ [4 channels]};$$

$$O(8)$$

$$= 2^9 - 16$$

$$= 496;$$

$$\text{constraints : } \Theta_{XAB},$$

$$\Theta_{XCD}$$

$$\in 2\mathbb{Z} : \text{ disjoint channel sets } \implies \text{ rk } \quad = 2;$$

$$g(5) = 10 \text{ [FT.41}(b)\text{]}$$

$$\text{patch : } E_{\text{int}}$$

$$= 2 \cdot 5$$

$$= 10;$$

$$\text{constraint vectors on } (AB, AC, BC, AD, BD) : \ (1,1,1,0,0),$$

$$(1,0,0,1,1) : \ \text{independent over } \mathbb{Z}_2 \implies \text{rk}$$

$$= 2, \qquad \text{index } 4;$$

$$n(A) = n(B) = 6, \ n(C) = n(D) = 4;$$

$$g(4) = 3 \ [\text{all } 2{+}2]. \ \blacksquare$$

Phys. Floors compose like ledger objects should: masses add across bridges, winding numbers add across round-trips, and every junction pays for the composition in darkness — the absorption rate doubling with each shared channel, exponentially with multiplicity. Gluing identifies shared channels and adds parity constraints by rank, with homological dependencies automatic. The bowtie and the patch are the first wired cells: one a five-vertex all-locked object with ten sealing patterns, the other a four-vertex meson–meson floor whose boundary-cycle constraint the ledger already implies.

Theorem A.10 (FT.51 — Enumeration of the winding ledger). Thm. For the polygon floor Σ_{P_n} under M1/M2 [FT.45], at winding level k (face constraint $\sum_i \kappa_i = 2k$ [FT.45(3)]):

$$(1) \quad \#\{\text{on-shell profiles}, \ \kappa \in \mathbb{Z}^n_{\geq 0}, \ \textstyle\sum \kappa_i = 2k\} = \binom{2k+n-1}{n-1} \qquad [\text{stars-and-bars; math}];$$

$$(2) \quad \#\{\text{strictly-spacelike profiles}, \ \kappa \geq 1\} = \binom{2k-1}{n-1} \quad (2k \geq n; \ = 0 \text{ otherwise});$$

(3) coarse-grained degeneracy : configurations related by mask-preserving relabeling are identified [CT2.6's $\sim$; CT2.7: weights are vertex-name independent], so the distinct ledger states at level k are the dihedral necklaces

$$M(n,k) = \frac{1}{2n}\Big(\sum_{j=0}^{n-1} \text{fix}(r^j) + \sum_{\text{reflections}} \text{fix}\Big) \qquad [\text{Burnside; math}],$$

with $\text{fix}(r^j) = \binom{2k\,c/n+c-1}{c-1}$ if $(n/c) \mid 2k$, else 0, where $c = \gcd(n,j)$ [rotation fixed points: κ constant on the c orbits]; machine values: $M(3,1) = 2, \ M(3,2) = 4, \ M(3,3) = 7; \quad M(4,1) = 3, \ M(4,2) = 8$;

(4) growth : #profiles $\sim (2k)^{n-1}/(n-1)!$ as $k \to \infty$ [math] : the winding ledger's state count grows polynomially in the winding at fixed n,

degree $n-1$ — a derived density of states for the floor family;

(5) sign registers factor : σ-assignments are independent of κ [CT0.10] : total floor states at level k

$= 2^{2n} \cdot M(n,k) \cdot$ (sealing/attachment data) — the three registers multiply, none converts into another.

Req. FT.45; CT2.6, CT2.7 (Part 2); CT0.9, CT0.10 (Part 1); stars-and-bars, Burnside, asymptotics of binomials [math].

Pf.

$(1), (2)$: compositions of $2k$ into n non-negative (resp. positive) parts [math];

parity automatic: the constraint fixes $\sum \kappa = 2k$ even [FT.45(3)]

(3) : relabeling quotient : CT2.6′s $\sim$ on coarse-grained ledger entries;

CT2.7 Pf : weights ¬vertex-name dependent

$\implies$ D_n-orbits are the distinct states;

Burnside with rotation/reflection fixed-point counts [math; machine cross-check by orbit enumeration]

$$(4): \binom{2k+n-1}{n-1} = \frac{(2k+n-1)\cdots(2k+1)}{(n-1)!} \sim \frac{(2k)^{n-1}}{(n-1)!}$$

(5) : CT0.10 : $P(C)$ and $\pi(C)$ independent sign structures;

CT0.9 : 2^{2n} sign assignments on $2n$ directed edges;

κ-ledger [clauses(1)–(3)] and sealing data [FT.41(b)] factor as independent registers. ■

Phys. The winding ledger is countable in closed form: at winding k a polygon floor carries $\binom{2k+n-1}{n-1}$ integer profiles, collapsing to dihedral necklaces under the corpus's own relabeling equivalence — two, four, seven distinct states for the triangle at $k = 1, 2, 3$. The count grows like $(2k)^{n-1}$: bigger polygons remember more, higher windings remember denser — a derived density of states, with signs and sealings multiplying as independent registers. This is the combinatorial substrate any generation/flavour reading would have to select from.

Theorem A.11 (FT.63 — The Platonic physical cards: spin, charge, parity, holonomy, entropy, mass ratios, binding). Thm. For each Platonic floor Σ_S [Def-H.20] under M1/M2 (interface clause), FT.62's corrected rungs, and — for the spin and defect-ladder rows — Def-H.38/Def-H.37(2) (interface clause): all seven requested computations close, per solid and universally. Write (V, E, F, L, d) for the cardinals, $\kappa_{0,\min} = 1$ (cube) / 2 (all others) [FT.42(5)], $K_{S,\min} = E\kappa_{0,\min}$, $\delta_v = 2\pi - d\,\frac{(L-2)\pi}{L}$ the vertex angle defect, and $k_{f,\min} = L$ (L odd) / $L/2$ (L even) the uniform-minimal face winding. (1) Spin [Def-H.38; Def-H.23, FT.60]: every materialized face chain carries winding k_f with carrier Sym^{k_f} and label $j_f = |k_f|/2$; every channel digon carries $k = \kappa_{\mathrm{ch}}$, $j = \kappa_{\mathrm{ch}}/2$ [FT.24(1), FT.25(3)]. At the uniform minima:

S	$k_{f,\min}$	j_f	$K_{S,\min}$	$j_{\max} = K_S/2$	dim : face basis / digon basis
tetrahedron	3	$\frac{3}{2}$	12	6	$4^4 = 256$ / $3^6 = 729$
cube	2	1	12	6	$3^6 = 729$ / $2^{12} = 4096$
octahedron	3	$\frac{3}{2}$	24	12	$4^8 = 65536$ / $3^{12} = 531441$
dodecahedron	5	$\frac{5}{2}$	60	30	$6^{12} = 2176782336$ / 3^{30}
icosahedron	3	$\frac{3}{2}$	60	30	$4^{20} = 2^{40}$ / 3^{30}

with the digon-basis check $\sum_{\mathrm{ch}} \kappa_{0,\min}/2 = K_{S,\min}/2 = j_{\max}$ [machine]. Derived laws: (a) $j_{\max} = K_S/2$ and the total winding sum $\sum_f k_f = K_S$ are decomposition-invariant [FT.43(1), FT.60(7a)]; the full tensor carrier is basis-relative (digon basis is the edge-disjoint one [FT.25(1)]). (b) V, F even for all five solids $\Rightarrow$ at uniform minima every face j_f composes to an integer stretched total (F half-integer-or-integer summands, F even) — Platonic floors are integer-spin objects in the stretched read at their minima, with half-integer face reads for the four odd-L solids: fermion-like faces, boson-like stretched totals (structural observation, ¬particle identification). (c) Dual splitting, new column: $j_{\max}$ splits cube (6) from octahedron (12) but is degenerate for dodecahedron/icosahedron (30) — where the per-face structure splits instead ($5/2 \times 12$ vs $3/2 \times 20$) and the face-basis carrier dimensions split (6^{12} vs 4^{20}) while the digon-basis dimensions are degenerate (3^{30} both). (2) Charge [FT.40, EM.9, D3.4]: the closed-core read $q_{\mathrm{core}}(c; v) = \frac{1}{2}(\sigma_{e_1} - \sigma_{e_k}) \in \{-1, 0, +1\}$ is defined on every materialized closed core — faces, digons, and homology sums: $\beta_1 = F - 1$ independent core reads per solid $(3, 5, 7, 11, 19)$, ternary alphabet, all three values attained per solid [machine: explicit σ-assignments on the encoded skeletons]; ω-rebase flips the sign [D2.6, FT.40]. Channel-facing charge: dark floors — no read channel [FT.3; FT.62(5)]; observed variants — EM.9 end-reads of transit chains through the contact, telescoping to boundary signs [FT.21(i)]; EM activity criterion $\exists v : f_v \neq 0$ on chain-local reads [FT.42(4), FT.62(5) scope]. No independent weak or strong charge exists [FT.11, FT.12, CT3.22]. (3) Parity: the on-shell κ-lattice is $\ker \varphi_S$, $\varphi_S : \mathbb{Z}^E \to \mathbb{Z}_2^F$ the face-incidence map — index $2^{F-1} = (8, 32, 128, 2048, 2^{19})$ [FT.43(3); machine: GF(2) elimination for all five, extending the earlier tetra/cube]; equivalently κ mod 2 ranges over the cut space of Γ_S (dimension $V - 1 = E - F + 1$ — Euler again) [math: planar duality of the face-even lattice; machine: dimension identity + star-cut witnesses]. Consequences [machine]: $A_S(1) = 0$ for all five (no bridges: single-channel κ=1 leaves two faces odd); $A_S(3) = V$ for d=3 solids (the vertex stars), 0 for octahedron/icosahedron-at-K=3 (odd K needs an odd cut; $|\delta S| \equiv d|S| \bmod 2$); octahedron: every face 2-coloring balances $\sum_{\mathrm{black}} k_f = \sum_{\mathrm{white}} k_f = K_S/2$, hence $K_S \in 2\mathbb{Z}$ and $A_{\mathrm{octa}}(\mathrm{odd}\, K) = 0$ [machine; the unique Platonic with even d]. On materialized loops: $\Theta \in 2\mathbb{Z}$, $\mathrm{Hol}_{\mathrm{int}} = +1$ [FT.22, FT.25(1), FT.42(3)]; partial internal sets retain free parity [FT.25(3)]; on-shell weight phases are registration-only, $(-1)^{\aleph + n_-}$ [FT.61(1)–(2)]. (4) Holonomy: $\mathrm{Hol}_{\mathrm{spiral}} = +1$ on every materialized loop [FT.22]; $\mathrm{Hol}_{\mathrm{int}} = +1$ [FT.25(2)]. Frame/spin-connection side [Def-H.21, FT.54]: the skeleton metric is piecewise-flat

with conical defects δ_v at vertices — exact table $\delta_v/\pi = (1, \frac{1}{2}, \frac{2}{3}, \frac{1}{5}, \frac{1}{3})$, and Descartes/Gauss–Bonnet closes the ledger: $V\delta_v = 4\pi$ for all five [machine in exact π-units]. Hence the global spinor transport around any full sweep is $\exp(i\,\frac{4\pi}{2}\,\hat{n}\cdot\vec{\sigma}) = +\mathbb{I}$ — 4π is the SU(2) period [machine over $\mathbb{Q}(i)$]: spherical materialization is spinor-single-valued, a consistency anchor for M1. Loops cleanly enclosing one vertex (transverse to the singular locus — vertex-figure loops in the derived geometry) carry $\Theta_R = \delta_v$ [FT.47(1)/GB; Def-H.21(6)] and defect values $D_S = 2-2\cos(\delta_v/2) = 4\sin^2(\delta_v/4)$ — exact radicals, clause (6). 1-skeleton face loops sit on the singular locus (vertices on the loop): the smooth-enclosure defect read does not apply to them without a corner convention, undeclared — recorded, not guessed [Rmk below]. Under Def-H.38 the defect read becomes the multiplet ladder $m^2_{\text{pole}}(m) = 4\hat{\mu}'^2\sin^2(m\,\Theta_R/4)$, $m \in \{k, k-2, \ldots, -k\}$ [FT.60(5)] — per-solid weight tables in (6). Weak-transport side: kernel coherence — charge-aligned profiles give $G(c) = 0$, defect-massless, at every floor [FT.33(6), EW.22, EM.6]. (5) Entropy: $S_{\text{hid}} = I_{\text{hid}} = 2E\ln 2 = (12, 24, 24, 60, 60)\ln 2$ nats [FT.42(1); FT.62(4): unchanged]; loop relating rate $S_{\text{reg}}/A_{\text{loop}} = (\ln 2/\mathfrak{h}^2)/\bar{\kappa}$ [FT.37]: at the uniform minima $\bar{\kappa} = \kappa_{0,\min}$ — the cube is the unique Platonic floor whose uniform-minimal loops saturate the CT1.29 Bekenstein rate $\ln 2/\mathfrak{h}^2$ exactly; the other four run at half rate ($\kappa_0 = 2$). Diagnostic minima $\Pi_{\min} = 8E^2K_{S,\min}(\ln 2)^2 = (3456, 13824, 27648, 432000, 432000)(\ln 2)^2$ — dual-degenerate at matched K_S [FT.42(5), FT.44(1)]. Lock ledgers (corrected): $\sum_v O$ per variant from FT.62(2)'s table — the retained-but-inaccessible register rate, e.g. dark icosahedron 24384/tick vs dark tetrahedron 448/tick. Boundary entropy: dark floors $\partial\Sigma_S = \varnothing$; observed $E_{\text{bd}} \in \{1, 2\}$ [FT.42; CT1.29 applies to locked-interior separating surfaces — the Platonic lock ledger is vertex-absorption, not surface area; no identification]. (6) Mass ratios [Def-H.37(2); FT.54(4)]: $m_{\text{hid}} = 2E\ln 2/\mathfrak{h}$; the vertex-defect pole datum and the dimensionless ratio:

$$\frac{m^2_{\text{pole}}(\delta_v)}{m^2_{\text{hid}}}$$

$$= \frac{4\hat{\mu}'^2\,\mathfrak{h}^2\sin^2(\delta_v/4)}{4E^2(\ln 2)^2}$$

$$= \frac{\hat{\mu}'^2\mathfrak{h}^2}{(\ln 2)^2}\cdot\frac{D_S}{4E^2} \in \mathfrak{q}^0$$

$$[[\hat{\mu}'^2] = \mathfrak{q},\ [\mathfrak{h}^2] = \mathfrak{q}^{-1}:\ \text{D2.9}],$$

S	$D_S = 4\sin^2(\delta_v/4)$	exact field	min. polynomial of D_S	$D_S/4E^2$
tetrahedron	2	$\mathbb{Q}$	$x-2$	$1/72$
cube	$2-\sqrt{2}$	$\mathbb{Q}(\sqrt{2})$	x^2-4x+2	$(2-\sqrt{2})/576$
octahedron	1	$\mathbb{Q}$	$x-1$	$1/576$
dodecahedron	$2-2\cos\frac{\pi}{10}$	$\mathbb{Q}(\cos\frac{\pi}{10})$	$x^4-8x^3+19x^2-12x+1$	$D_S/3600$
icosahedron	$2-\sqrt{3}$	$\mathbb{Q}(\sqrt{3})$	x^2-4x+1	$(2-\sqrt{3})/3600$

[machine: the dodecahedron quartic verified by exact substitution $x = 2-2c$ into $\mathbb{Z}[c]/(16c^4-20c^2+5)$ — itself verified from $\cos 5t = 16c^5-20c^3+5c$ by de Moivre binomials at $t = \pi/10$; the cube/icosahedron quadratics from $(x-2)^2 \in \{2, 3\}$]. Weight-ladder values $D_S(m) = 2-2\cos(m\,\delta_v/2)$, $m \le 4$ [FT.60(5); machine]: tetrahedron $(2, 4, 2, 0)$ — rational, with an exact zero at m=4; cube $(2-\sqrt{2}, 2, 2+\sqrt{2}, 4)$; octahedron $(1, 3, 4, 3)$ — rational;

icosahedron $(2-\sqrt{3}, 1, 2, 3)$; dodecahedron $D(2) = \frac{3-\sqrt{5}}{2}$, $D(4) = \frac{5-\sqrt{5}}{2}$ [machine in $\mathbb{Q}(\sqrt{5})$ via $4c^2 \mp 2c - 1$ at $c = \cos\frac{\pi}{5}, \cos\frac{2\pi}{5}$], $D(1), D(3)$ the two smallest of its four positive roots. [precision correction [FT.76(10)]: the quartic $x^4 - 8x^3 + 19x^2 - 12x + 1$ has four positive roots — 0.097886967410, 0.824429495415, 3.175570504585, 3.902113032590 (machine-isolated to width 2^{-60}; $q(0..4) = (1, 1, 5, 1, 1) > 0$, so no integer bracket separates them) — realized by the weights $m \in \{1, 19\}, \{3, 17\}, \{7, 13\}, \{9, 11\}$ respectively; $D(1), D(3)$ are the two smallest.] Def-H.37(1) form at stationarity: ratio $= \hat{\mu}^2 \mathfrak{h}^4 G(c)\, \kappa_{0,\min}/(\ln 2)^2$ [FT.53(3)] with $\kappa_{0,\min} \in \{2, 1, 2, 2, 2\}$. Derived law (dual splitting, pole side): both dual pairs share m_{hid} and Π [FT.44(1)] but split on every defect datum — δ_v: $\frac{\pi}{2}$ vs $\frac{2}{3}\pi$; $\frac{\pi}{5}$ vs $\frac{\pi}{3}$; D_S: $2 - \sqrt{2}$ vs 1; $2 - 2\cos\frac{\pi}{10}$ vs $2 - \sqrt{3}$ — FT.44(2)'s splitting template gains its first irrational column. Face loops carry $\Theta_R = 0$ on the flat-face side (defect-massless), corner convention undeclared [Rmk]. (7) Binding-force contributions: per-aspect ledger, all five solids:

aspect	criterion on Σ_S	$\dim \mathcal{K}^{(A)}$	binding read
G	always ($I_{\text{hid}} = 2E \ln 2 \neq 0$)	5	κ-profile modulo loop constraint [FT.41(c)]
H	always (Def − H.36 fold)	1	m_{hid} read [FT.20]
W	$\iff K_S \neq 0$ (at minima: always)	3	spin content of (1) [FT.42(3), Def − H.38]
EM	observed/charged states [FT.62(5)]	1	q_{core} ternary per core [(2)]
strong	$\iff$ inside-sealed phase ($\mathcal{C} = 1$)	8	$g(V)$ groupings; singlet-diagonal reads vanish [FT.39(2)]

$\dim \mathcal{K}_{\Sigma_S} = 9\ (+1)\ (+8) \leq 18$ [FT.26(3), FT.42(4)]. Sealing grammar: $g(V) = (3, 385, 25, 292938721075, 164395)$ [FT.41(b); machine: closed form vs enumeration for $V \leq 8$], with the even-ε theorem: V even for all five solids $\Rightarrow$ every grouping has an even number b of ε-triples ($2a + 3b = V$, b even) [machine] — Platonic sealing phases are baryon-number-even by structure; the tetrahedron is meson-only ($b = 0$: three $\delta\delta$-pairings), the octahedron admits 2ε phases ($15\,\delta^3 + 10\,\varepsilon^2$), the icosahedron $b \in \{0, 2, 4\}$, the dodecahedron $b \in \{0, 2, 4, 6\}$, the cube $b \in \{0, 2, 4\}$ [machine: term breakdown]. Confinement is all-or-nothing [FT.19] with generic nonvanishing at $\mathcal{C} = 1$ [FT.39(1)]; no per-edge strong phase exists [FT.29(2)]; strong binding contributes admissibility structure and gluon modes, never singlet expectation values [FT.39(2), CT3.13]. Boundary/exchange algebra: dark $N(\Sigma_S) = 0$ (no boundary algebra); single-directed-edge contact $N = 1$: nothing strictly new [FT.41(d)]; bidirectional contact $N = 2$ corolla at $n(w) = 2$: exchange doublet W-hosted literally [FT.58(1)]; the face-floor variant $N = 6$, $n(D) = 6$: exchange sector wholly kernel/D^--absorbed [FT.59(4a)]. D^-: never observable, absorption-only [FT.20, CT4.4] — its binding contribution is the lock ledger of FT.62(2), by rate not by force.

Req. Def-H.20, Def-H.23; Def-H.38, Def-H.37(2), Def-H.36, Def-H.35; M1/M2 (interface clause); FT.9, FT.11, FT.12, FT.19, FT.20, FT.21, FT.22, FT.24, FT.25, FT.26, FT.29, FT.31, FT.33, FT.37, FT.39, FT.40, FT.41, FT.42, FT.43, FT.44, FT.47, FT.49, FT.53, FT.54, FT.58, FT.59, FT.60, FT.61, FT.62; A4; D2.6, D2.9, D3.3, D3.4; CT0.6, CT0.7, CT0.9, CT0.10, CT0.12, CT0.13, CT0.14 (Part 1); CT1.29, CT2.5, CT2.9 (Part 2); CT3.13, CT3.17, CT3.22 (Part 2); CT4.3, CT4.4, CT4.9 (Part 2); EW.22, EW.32 (Part 2); EM.2, EM.6, EM.9 (Part 2); Descartes/Gauss–Bonnet, planar cut-space duality, $\cos(5t)$ de Moivre identity, SU(2) 4π-periodicity [math].

Pf.

$$(1) : \text{ faces materialize [M1]} : \text{ each face}$$

$$= \text{materialized closed chain} \implies \Psi$$

$$= \mathrm{Sym}^{k_f} \text{ [Def-H.23]};$$

$$\text{digons [M2]} : \ k$$

$$= \kappa \text{ [FT.24(1), FT.25(3)]}$$

$$\text{uniform minima} : \ k_{f,\min} = \Theta_{f,\min}/2 : \ L \text{ odd} : \ \tfrac{2L}{2} = L; \ L \text{ even} : \ \tfrac{L}{2} \text{ [FT.42(5)]};$$

$$K_{S,\min} = E\kappa_{0,\min} \text{ [FT.42(3)]}; \quad j_{\max} = K_S/2 \text{ [FT.60(2), (7}a\text{)]}$$

$$\dim \mathrm{Sym}^k = k+1 : \text{ table arithmetic [machine]};$$

$$(b) : \ F \in \{4, 6, 8, 12, 20\} \text{ even};$$

$$j_f \in \tfrac{1}{2}\mathbb{Z} : \text{ stretched total}$$

$$= \textstyle\sum_f k_{f,\min}/2$$

$$= K_{S,\min}/2 \in \mathbb{Z} \text{ [machine} : \ 6, 6, 12, 30, 30]$$

$$(c) : \ E \text{ invariance [FT.44(1)]};$$

$$j_{\max} : \ 6 \neq 12 \text{ (cube/octa)};$$

$$30 = 30 \text{ (dodeca/icosa)} : \ k_{f,\min} \times F : \ 5 \times 12 \neq 3 \times 20 \text{ as multisets}; \ 6^{12} \neq 4^{20}; \ 3^{30} = 3^{30}$$

$$(2) : \text{ FT.40 verbatim per core};$$

$$\beta_1 = F - 1 \text{ independent reads [FT.43(2)]};$$

$$\text{attainment witnesses [machine]};$$

$$\omega\text{-flip [D2.6]}$$

$$(3) : \text{ FT.43(3)} + \text{machine elimination (all five)};$$

$$\ker \varphi_S \bmod 2 : \ \dim_{\mathbb{Z}_2}$$

$$= E - (F - 1)$$

$$= V - 1 \text{ [Euler]}$$

$$= \text{dim cut space [math]};$$

$$\text{stars}$$

$$= \delta\{v\} : \ \text{admissible [machine]}$$

$$A_S(1)$$

$$= 0 : \ \text{single channel } \kappa{=}1 : \ \text{its two faces odd};$$

$$\neg\exists \text{ bridges [solid 2-connectivity; machine]};$$

$$A_S(3) = V \ (d{=}3) : \ \text{stars; odd } K \iff \text{odd cut} : \ |\delta S| \equiv d|S| \ (2) \text{ via } \textstyle\sum_{v\in S} \deg v = |\delta S| + 2|E(S)|$$

$$\text{octahedron} : \ \text{dual bipartite}$$

$$\iff d \text{ even [machine : 2-coloring exists iff octahedron]};$$

$$\text{each channel in one black + one white face} \implies \textstyle\sum_{\text{bl}} \Theta_f$$

$$= \sum_{\text{wh}} \Theta_f \qquad = \sum_e \kappa_e$$

$$\implies \textstyle\sum_{\text{bl}} k_f = \sum_{\text{wh}} k_f = K_S/2 \in \mathbb{Z}; \qquad \text{on-shell phases} : \ \text{FT.61(1)–(2)}$$

$$(4) : \ \delta_v$$

$$= 2\pi - d\alpha,$$

$$\alpha$$

$$= \tfrac{(L-2)\pi}{L} \text{ [regular } L\text{-gon angle; math]} : \ \delta_v/\pi$$

$$= 2 - \tfrac{d(L-2)}{L}$$

$$= (1, \tfrac{1}{2}, \tfrac{2}{3}, \tfrac{1}{5}, \tfrac{1}{3});$$

$$V\delta_v = 4\pi \text{ [machine : Descartes]}$$

$$\exp(i\tfrac{4\pi}{2}\hat{n}\cdot\vec{\sigma}) = \cos(2\pi) + i\sin(2\pi)\hat{n}\cdot\vec{\sigma} = \mathbb{I} \text{ [machine over } \mathbb{Q}(i)];$$

$$\Theta_R = \delta_v \text{ for clean enclosure [FT.47}(1)\text{/GB; Def-H.21}(6)]$$

$$(5) : \text{ FT.42}(1) + \text{FT.62}(4);$$

$$\bar{\kappa}$$

$$= \kappa_{0,\min} \text{ [uniform]} : \text{ rate}$$

$$= (\ln 2/\mathfrak{h}^2)/\kappa_{0,\min} \text{ [FT.37]};$$

$$\kappa_0$$

$$= 1 \iff \text{cube} \iff \text{CT1.29 saturation [FT.37 iff-clause]}$$

$$\Pi_{\min} = 8E^2 K_{S,\min} \text{ [FT.42}(5)\text{]}; \qquad \text{lock sums : FT.62}(2) \text{ [machine]}$$

$$(6) : \text{ FT.54}(4) \text{ at } \Theta_R = \delta_v; \quad m_{\text{hid}}^2 = 4E^2(\ln 2)^2/\mathfrak{h}^2 \text{ [FT.42}(1)\text{]};$$

$$D_S = 2 - 2\cos(\delta_v/2) \text{ [half-angle]; table values [machine: radicals, min. polys]}$$

$$\text{quartic : } c$$

$$= \cos\tfrac{\pi}{10} : \ \cos 5t$$

$$= 16c^5 - 20c^3 + 5c \text{ [de Moivre; machine]};$$

$$t$$

$$= \tfrac{\pi}{10} : \cos \tfrac{\pi}{2}$$

$$= 0$$

$$\implies 16c^4 - 20c^2 + 5 \qquad = 0;$$

$$x = 2 - 2c \implies x^4 - 8x^3 + 19x^2 - 12x + 1 = 0 \text{ [machine: exact substitution]}$$

$$\text{ladder : } D_S(m) = 2 - 2\cos(m\delta_v/2) \text{ [FT.60(5)]};$$

$$\text{table per solid [machine : } \mathbb{Q},\ \mathbb{Q}(\sqrt{2}),\ \mathbb{Q}(\sqrt{3}),\ \mathbb{Q}(\sqrt{5}) \text{ exact]}$$

$$(7) : \text{ FT.20 rows} + \text{FT.42}(4);$$

$$g(V) = \sum_{2a+3b=V} \tfrac{V!}{a!\,b!\,2^a 6^b} \text{ [FT.41}(b)\text{]}; \quad V \text{ even} \implies 3b \text{ even} \implies b \text{ even};$$

$$\text{term breakdowns [machine]}$$

$$N(\Sigma_S) : \text{ dark } 0;\ \text{obs } 1 \text{ or } 2 \text{ [Def-H.12]};$$

N=2 corolla at $n(w)$=2 : FT.58(1); face-floor variant : FT.59($4a$). ■

Rmk (corner convention, undeclared). Skeleton face-loops pass through conical vertices: their Θ_R requires a corner-distribution convention for defects on the loop (or the two-region read 0 vs $4\pi - \delta$: the defect values differ by side, though the total 4π and the SU(2) global triviality are side-independent). This is the single place where the requested "mass ratio" computation meets an undeclared convention, and it is marked rather than papered over.

Rmk (no identifications). As everywhere in this document: no card entry is identified with a named particle; the dual-degeneracy/splitting pattern (mass and Π degenerate; winding, sealing, locks, pole defects, and spin structure split) extends FT.44(2)'s template with two new columns and remains a template, not a generation claim .

Phys. Computed in full, the five solids read: spins — half-integer faces composing to integer stretched totals, six for the tetrahedron and cube, twelve for the octahedron, thirty for the last dual pair; charges — ternary core registers on every closed loop, three to nineteen independent reads, flipping under the corpus's own conjugation; parities — an index-2^{F-1} lattice whose mod-2 classes are exactly the cuts of the skeleton, with the octahedron alone forced even-wound by its two-colored faces; holonomies — trivial spirals, trivial global spinor transport, and conical vertex defects summing to one sphere; entropies — twelve to sixty hidden bits, the cube alone running

its minimal loops at the exact Bekenstein rate; mass ratios — the pole side written in $\sqrt{2}$, $\sqrt{3}$, and $\cos\frac{\pi}{10}$, splitting every dual pair that the hidden-bit mass degenerates; binding — gravity and Higgs always, weak whenever the ledger is wound, colour in three to two-hundred-ninety-three-billion sealing patterns, always baryon-even, never as a number a sealed floor carries.

Theorem A.12 (FT.64 — The filter chain and the per-solid loop lattices). Thm. (1) Filter chain (definition of ledger-admissible). A loop datum on a floor Σ is ledger-admissible iff it passes, in order: (F0) configuration validity — closed real-edge content, Janus balance, mask image, no dangling ledger boundary [D2.11, D1.3, D0.10]; (F1) registration — $\sigma_e \in \{\pm 1\}$ per directed real edge, independent of κ [CT0.9, CT0.10]; (F2) integrality — $\kappa_e \in \mathbb{Z}$ on edges of materialized digons and round-trip-closed channels (classes (i)/(ii) [FT.24, FT.34(4)]; Platonic/polygon floors: under declared M2 [Def-H.20]); (F3) loop quantization — $\Theta(c) \in 2\mathbb{Z}$ per materialized closed chain [FT.22; M1 on faces]: for the Platonic floors exactly $\kappa \in \ker\varphi_S$, φ_S =face-incidence mod 2 [FT.43(3)]; (F4) stationarity (semiclassical survivors) — $\kappa \equiv \kappa_0$ constant along each connected materialized chain, geodesic support [FT.53(1)–(2), FT.48(1)]; (F5) regime — null-permitting ($\kappa \geq 0$), strictly-spacelike ($\kappa \geq 1$), uniform, or signature-mixed ($\kappa \in \mathbb{Z}$; fibers infinite at fixed K — witness $(K+n, -n)$ — so enumeration is scoped to $\kappa \geq 0$ [FT.45(4) pattern]); (F6) relabeling quotient — identification under mask-preserving relabeling [CT2.6's $\sim$, CT2.7; FT.51(3)]: $\mathrm{Aut}(\Gamma_S)$-orbits (FT.65); (F7) register product — signs $\times$ κ-orbits $\times$ sealing groupings, independent registers [FT.51(5), FT.65(2)]; (F8) spectral filter — materialization (M1/M2 declared; which admissible loops are on-spectrum is the corpus's [D2.4, CT2.5, Def-H.16]); no ledger-side criterion can further select by stability index [FT.61(4)]: (F8) is the honest boundary of this list. (2) Per-solid lattice, all five by machine. $\ker\varphi_S \subseteq \mathbb{Z}^E$ has index 2^{F-1} — GF(2) rank $F-1$ verified by explicit elimination for all five solids (the earlier tetra/cube checks extended): $(8, 32, 128, 2048, 2^{19})$. Equivalent formulation: κ mod 2 lies in the cut space of Γ_S — $\dim_{\mathbb{Z}_2}\ker\varphi_S = E-(F-1) = V-1 = \dim\mathrm{Cut}(\Gamma_S)$ [Euler; math: planar duality; machine: dimensions + vertex-star cut witnesses]. (3) Face-read realizability (SNF). For the face-read map $B : \mathbb{Z}^E \to \mathbb{Z}^F$, $\kappa \mapsto (\Theta_f)_f$: Smith invariant factors are $(1,1,1,2)$ tetrahedron, $(1^5, 2)$ cube, (1^7) octahedron (rank $7 = F-1$), $(1^{11}, 2)$ dodecahedron, $(1^{19}, 2)$ icosahedron [machine]. Consequences: (a) tetrahedron/cube/dodecahedron/icosahedron: $(2\mathbb{Z})^F \subseteq \mathrm{im}\,B$ — every integer face-winding vector $k = (k_f) \in \mathbb{Z}^F$ is realized by some signed κ [machine witness: all four unit vectors $k = e_f$ realized on the tetrahedron with $\kappa \in [-2, 2]^6$]; (b) octahedron: $\mathrm{im}\,B = \ker\psi$, ψ = black−white face-sum (dual bipartite, d even — the unique such Platonic): k realizable $\iff \sum_{\mathrm{bl}} k_f = \sum_{\mathrm{wh}} k_f \iff K_S \in 2\mathbb{Z}$ [machine: $\psi(\mathrm{im}\,B) = 0$ on samples; $\psi(2e_f) = \pm 2 \neq 0$: $k = e_f$ unrealizable]; each channel lies in exactly one black and one white face [machine]. (4) Level-set counts. $A_S(K) := \#\{\kappa \in \mathbb{Z}^E_{\geq 0} : \sum_e \kappa_e = K,\ \varphi_S(\kappa) = 0\}$ (level $K = K_S = \Theta_{\mathrm{int}}/2$) [machine, DFS with face-parity pruning; cross-checked against orbit enumeration and the Burnside identity element]:

S	K=0	1	2	3	4	5	6	7	8
tetrahedron	1	0	6	4	24	24	74	84	189
cube	1	0	12	8	93	120	576	—	—
octahedron	1	0	12	0	84	0	452	—	—
dodecahedron	1	0	30	20	495	672	—	—	—
icosahedron	1	0	30	0	465	12	—	—	—

with the exact character-sum closed form [math: GF(2) orthogonality; machine-verified against the DFS for tetrahedron/cube/octahedron, $K \leq 6$]:

$$A_S(K) = \frac{1}{2^{F-1}} \sum_{s \in \mathbb{Z}_2^F/\langle 1\rangle} [z^K] \prod_{e \in E} \frac{1}{1-(-1)^{s\cdot b_e}\, z}$$

$$(b_e = \text{face-incidence column of } e),$$

and asymptotics $A_S(K) \sim \binom{K+E-1}{E-1}/2^{F-1} \sim K^{E-1}/((E-1)!\,2^{F-1})$ [equidistribution of the $F-1$ parity conditions; machine trend: tetrahedron $A(16) = 2673$ vs $20349/8$]. Structural reads [machine]: $A_S(1) = 0$ for every solid (no bridges — a lone κ=1 channel leaves its two faces odd); $A_S(3) = V$ for the d=3 solids (the vertex stars), 0 for the $d \geq 4$ solids (octahedron: even-d parity; icosahedron: its stars carry K=5); the octahedron has $A(\text{odd } K) = 0$ [(3)(b)]; the icosahedron's $A(5) = 12$ is exactly its star family — the index-2^{19} lattice is sparse enough that at low levels only the symmetric families survive. The K=0 entry is the all-null bound configuration (trivial-chain content [D1.7, D3.7]; W-row silent [FT.42(3)]). (5) Minimality regimes, all five [FT.45(4)/FT.32(6) pattern; machine]:

S	R0 null-perm. : $(K_{\min}, \Theta_{\text{int}}, \text{orbits})$	R1 strict : $(\varepsilon_{\min}, K_{\min}, \Theta_{\text{int}}, \text{orbits})$	R2 uniform : κ_0 ladder	R1 witness type
tetrahedron	$(2, 4, 1)$	$(2, 8, 16, 1)$	$2, 4, 6, \ldots$	opposite-edge pair at 2
cube	$(2, 4, 1)$	$(0, 12, 24, 1)$	$1, 2, 3, \ldots$	all-1 profile (unique)
octahedron	$(2, 4, 1)$	$(4, 16, 32, 3)$	$2, 4, 6, \ldots$	coset leaders, 3 orbits
dodecahedron	$(2, 4, 1)$	$(6, 36, 72, 25)$	$2, 4, 6, \ldots$	coset leaders, 25 orbits
icosahedron	$(2, 4, 1)$	$(10, 40, 80, 14)$	$2, 4, 6, \ldots$	coset leaders, 14 orbits

R0: single channel at κ=2 (one orbit — Aut edge-transitive [machine]). R1: $\kappa = 1 + x$, $x \geq 0$: parity demands $Bx \equiv (L \bmod 2) \cdot 1$; $\varepsilon_{\min}$ = minimum weight in the GF(2) coset [machine: coset enumeration, $\leq 2^{19}$ for the dodecahedron]. Derived laws: (a) $\Theta^{\text{R0}}_{\text{int,min}} = 4$ for every solid vs 2 for $\Sigma_{\min}$ and 2 (face-winding) for polygons — a solid's faces are too many to be fooled by one channel [$F \geq 4$ separated faces; FT.45(5)'s digon distinction extended]; (b) the cube is the unique Platonic whose uniform-minimal profile is also its strictly-spacelike minimum ($\varepsilon_{\min} = 0$; all-1, unique up to symmetry) — for the four odd-L solids R1 beats R2 strictly ($16<24$, $32<48$, $72<120$, $80<120$): their uniform minima are not their cheapest spacelike configurations, and the minimizers form nontrivial orbit families (up to 25); (c) R2's stationary ladder is the semiclassical survivor list [FT.53]: one orbit per rung κ_0, $K = E\kappa_0$. (6) Completeness, ledger side. (F0)–(F7) are decidable and decided above (or cited closed-form: polygons [FT.51], cycles [FT.32(6)]); (F8) is declared. The list is complete as a ledger enumeration: every admissible loop datum on every computed floor is an element of $\ker \varphi$ (lattice) $\times$ sign registers $\times$ sealing groupings, modulo Aut, within one of the four regimes, surviving semiclassically exactly on the R2 ladder.

Req. FT.22, FT.24, FT.32, FT.34, FT.42, FT.43, FT.45, FT.48, FT.51, FT.53, FT.61, FT.62; Def-H.20; M1/M2 (interface clause); D0.10, D1.3, D1.7, D2.4, D2.11, D3.7; CT0.9, CT0.10 (Part 1); CT2.5, CT2.6, CT2.7 (Part 2); Def-H.16; GF(2) elimination, Smith normal form, planar cut-space duality, character orthogonality, stars-and-bars asymptotics [math].

Pf.

$$(1):\ \text{each step one citation as listed;}$$

$$\text{order}:\ \text{F2/F3 before F4 [integrality classes gate rigidity, FT.38(2)/FT.53];}$$

$$\text{F8 last [spectral, declared]}$$

$$(2):\ \text{FT.43}(3)\ \text{method extended}:\ \text{explicit GF}(2)\ \text{elimination per solid [machine];}$$

$$\dim\ker$$

$$= E - \mathrm{rk}\,\varphi$$

$$= E - (F - 1)$$

$$= V - 1\ [\text{Euler}];$$

$$\text{Cut}:\ \dim$$

$$= V - 1\ [\text{math}];$$

$$\text{stars}\ \delta\{v\} \in \ker\varphi\ [\text{machine}]$$

$$(3):\ \text{SNF [machine}:\ \text{integer row/column reduction];}$$

$$(a):\ \text{divisors} \in \{1,2\} \iff (2\mathbb{Z})^F \subseteq \mathrm{im}\,B\ [\text{math}];\ \text{witnesses [machine]}$$

$$(b):\ \text{dual bipartite}$$

$$\iff d\ \text{even [machine}:\ \text{2-coloring];}$$

$$\psi(B\kappa)$$

$$= \sum_e \kappa_e(1-1)$$

$$= 0;$$

$$\psi(2e_f) \qquad = \pm 2;$$

$$K_S = \textstyle\sum_f k_f = 2\sum_{\mathrm{bl}} k_f$$

$$(4):\ \text{DFS with completed-face parity pruning [machine];}$$

$$\text{character sum}: \ \tfrac{1}{2^{F-1}} \sum_{s} (-1)^{s \cdot B\kappa} \text{ projection [math; machine vs DFS]};$$

$$\text{asymptotics}: \ s = 0 \text{ term dominates [math; machine trend]}$$

$$A(1) = 0: \ \text{single channel}: \ \text{its 2 faces odd}; \ \neg\exists \text{ bridge [solid 2-edge-connectivity; machine]};$$

$$A(3) = V \ (d{=}3): \ \text{stars } \delta\{v\}: \ \text{each face at } v \text{ meets 2 star edges}$$

$$(5): \ \text{R0 witness } \kappa = 2e: \ B(2e) \equiv 0; \ \text{orbit} = \text{point [edge-transitivity; machine]};$$

$$\text{R1}: \ \kappa$$

$$= 1 + x: \ \Theta_f$$

$$= L + \sum_{e \in f} x_e$$

$$\in 2\mathbb{Z}$$

$$\iff Bx$$

$$\equiv (L \bmod 2)1;$$

$$\varepsilon_{\min}$$

$$= \text{coset leader weight [machine]}$$

$$\text{cube}: \ L \text{ even}: \ x = 0 \text{ admissible}: \ \varepsilon_{\min} = 0, \ \kappa \equiv 1 \text{ unique at } K = E;$$

$$\text{odd } L: \ \varepsilon_{\min} \geq 2 \ (Bx \equiv 1 \text{ needs } x \neq 0); \ \text{witnesses as tabled [machine]}$$

$$(6): \ \text{decidability}: \ \text{each filter arithmetic or cited-closed.} \ \blacksquare$$

Rmk (what the sparse low levels mean). The icosahedron's lattice is so fine (index 2^{19}) that below its star level $K{=}5$ only the symmetric families fit: $A(3) = 0$, $A(5) = 12$. Low-level admissibility on a solid is therefore highly non-generic — most low profiles are excluded not by energy but by parity — a derived selection pressure of purely combinatorial origin that any spectral mechanism (F8) inherits for free. Recorded as structure; no physical reading claimed.

Phys. Run the machine and the five solids answer with one grammar: a lattice of index a power of two, whose odd classes are exactly the cuts of the skeleton; winding vectors free as integers on four of the five, balanced black-against-white on the octahedron; a cheapest null-tolerant loop of four quanta everywhere, cheapest honest loops that beat uniformity on every odd-faced solid; and a semiclassical list of survivors that is nothing but the uniform ladder — one orbit per rung. The list is complete up to exactly one door: which of these loops the spectrum keeps. That door was always the corpus's, and the ledger now proves it cannot be opened from this side.

Theorem A.13 (FT.65 — Coarse-grained degeneracies, register product, and the master loop list). Thm. (1) Degeneracy under the relabeling quotient. $\mathrm{Aut}(\Gamma_S)$ acts on admissible profiles by channel permutation [CT2.6's $\sim$ on coarse-grained ledger entries; CT2.7: weights are vertex-name independent; FT.51(3) pattern]; orders $|\,\mathrm{Aut}\,| = (24, 48, 48, 120, 120)$, rotation subgroups $(12, 24, 24, 60, 60)$ [machine: enumeration + determinant filter on exact coordinates, $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{5})$]. The coarse-grained degeneracies $M_S(K) = \frac{1}{|\,\mathrm{Aut}\,|}\sum_g \mathrm{fix}(g, K)$ [Burnside; math] — with $\mathrm{fix}(g, K)$ the orbit-constant profile count [machine: orbit-DP with face-parity masks; identity element cross-asserted against $A_S(K)$; small-K values cross-checked by direct orbit enumeration]:

S	K=0	1	2	3	4	5	6	7	8
M tetrahedron	1	0	1	1	4	2	8	6	17
M cube	1	0	1	1	7	4	25	—	—
M octahedron	1	0	1	0	6	0	19	—	—
M dodecahedron	1	0	1	1	10	9	—	—	—
M icosahedron	1	0	1	0	9	1	—	—	—
M^+ tetrahedron	1	0	1	1	4	2	11	7	23
M^+ cube	1	0	1	1	8	5	34	—	—
M^+ octahedron	1	0	1	0	7	0	25	—	—
M^+ dodecahedron	1	0	1	1	13	12	—	—	—
M^+ icosahedron	1	0	1	0	12	1	—	—	—

M^+ (rotation-only quotient) is reported alongside M (full automorphism quotient, the FT.51 convention for polygons): whether orientation-reversing relabelings identify states is the ω/charge-conjugation read [D2.6] — both counts are derived data; no choice is imposed. Growth: $M_S(K) \sim A_S(K)/|\,\mathrm{Aut}\,|$ generically (free orbits dominate) [math; the fixed-point terms are lower-order]. (2) Register product (generalized). Distinct ledger states of Σ_S at level K:

$$\#\mathrm{states}(K) = 2^{2E} \cdot M_S(K) \cdot g(V) \cdot (\text{attachment-type data})$$

$$[\mathrm{FT.51}(5)\ \text{pattern};\ \mathrm{CT0.9/CT0.10}:\ \text{signs independent of } \kappa;\ \mathrm{FT.41}(b):\ \text{sealing register}],$$

with $g(V) = (3, 385, 25, 292938721075, 164395)$ [FT.63(7)] — the three registers multiply, none converts into another; e.g. dark tetrahedron at K=2: $2^{12} \cdot 1 \cdot 3 = 12288$ states before attachment data. (3) Density of states. $\#\mathrm{states}(K) \sim 2^{2E}\, g(V)\, K^{E-1}/((E-1)!\, 2^{F-1}\,|\,\mathrm{Aut}\,|)$ [from FT.64(4)'s asymptotics]: polynomial in the winding at fixed solid — degrees $E-1 = (5, 11, 11, 29, 29)$ — with the Platonic prefactors spanning twelve orders of magnitude (tetrahedron $2^{12} \cdot 3/24$ vs dodecahedron $2^{60} \cdot g(20)/120$): the derived state-count substrate any selection mechanism (Open Gaps 5/12,

filter F8) would act on. (4) Master generalized loop list (every cell computed in this document; $K = \Theta_{\text{int}}/2$ levels; lattice index within $\mathbb{Z}^{E_{\text{ch}}}$; all entries machine-verified or cited-closed):

cell	E_{int}	$m_{\text{hid}} \cdot \frac{\hbar}{\ln 2}$	lattice index	$K_{\min}$ ($\Theta_{\text{int,min}}$)	$\Theta_{\min}^{\text{strict}}$	κ_0 ladder (F4)	quotient $\to M$
$\Sigma_{\min}$ [FT.28]	2	2	1 ($\kappa \in \mathbb{Z}$)	1 (2)	2 (κ=1)	$1, 2, 3, \ldots$	trivial $\to 1$
Σ_L uni. [FT.32]	L	L	2 ($\sum \kappa \in 2\mathbb{Z}$)	1 (2)	L+(L mod 2)	even (L odd) / any (L even)	$\mathbb{Z}_L \to$ necklaces
Σ_{P_n} bidir. [FT.45]	$2n$	$2n$	2	2 (4)	$2n$+2(n mod 2)	$1, 2, \ldots$ (n ev) / $2, 4, \ldots$ (n odd)	$D_n \to M(n,k)$ [FT.51]
Σ_{12} [FT.35]	6	6	1 ($\mathbb{Z}^3$)	1 (2)	3 (6)	free $\mathbb{Z}^3$ (digon quanta)	$\mathbb{Z}_2 \to$ swap-orbits
$\Sigma_{\triangle}$ [FT.49]	6	6	2	2 (4)	4 (8)	$2, 4, \ldots$	$D_3 \to M(3,\cdot) = (2,4,7)$
bowtie [FT.50]	12	12	4	2 (4)	8 (16)	$(2a, 2b)$ per face	$\lvert\,\text{Aut}\,\rvert = 8$ (not tabulated)
double-triangle [FT.50]	10	10	4	2 (4)	6 (12)	$2, 4, \ldots$	$\mathbb{Z}_2 \times \mathbb{Z}_2$ ($\lvert\,\text{Aut}\,\rvert = 4$, not tabulated)
tetrahedron	12	12	8	2 (4)	8 (16)	$2, 4, 6, \ldots$	$S_4 \to$ table (1)
cube	24	24	32	2 (4)	12 (24)	$1, 2, 3, \ldots$	$S_4 \times \mathbb{Z}_2 \to$ table
octahedron	24	24	128 (K even)	2 (4)	16 (32)	$2, 4, 6, \ldots$	$S_4 \times \mathbb{Z}_2 \to$ table
dodecahedron	60	60	2048	2 (4)	36 (72)	$2, 4, 6, \ldots$	$A_5 \times \mathbb{Z}_2 \to$ table
icosahedron	60	60	2^{19}	2 (4)	40 (80)	$2, 4, 6, \ldots$	$A_5 \times \mathbb{Z}_2 \to$ table

(Columns: E_{int} = directed internal edges; index = of the on-shell κ-lattice in the channel lattice; $K_{\min}$ null-permitting with $\Theta_{\text{int,min}} = 2K_{\min}$; strict = minimal Θ_{int} at $\kappa \geq 1$ with its K; κ_0-ladder = the stationary (F4) survivors; quotient group with degeneracy source. For the solids, $\Theta^{\text{R0}}_{\text{int,min}} = 4$ universally and the strict minima are the FT.64(5) coset-leader values.) (5) Taxonomy law, re-anchored. Across every computed cell and family (the twelve master-table rows below plus the Σ_L/Σ_{P_n} members): $m_{\text{hid}} = E_{\text{int}} \ln 2/\mathfrak{h}$ with $E_{\text{int}} = 2E_{\text{bidir}} + E_{\text{uni}}$ [FT.44(4) — extended from eleven cells to sixteen, zero exceptions], and the list is closed under the program's composition operations: bridged unions add masses and windings [FT.50(1)–(2)], gluings compose lattices by constraint rank [FT.50(4)], round-trips add connector distance [FT.34(3)] — every composite of listed cells is itself a listed-type cell with computable invariants.

Req. FT.28, FT.32, FT.34, FT.35, FT.41(b), FT.42–FT.45, FT.49–FT.51, FT.63, FT.64; CT0.9, CT0.10 (Part 1); CT2.6, CT2.7 (Part 2); D2.6; Burnside, orbit-stabilizer, asymptotics [math]; Def-H.20; M1/M2 (declared, solids/polygon rows).

Pf.

$$(1):\ \text{Aut by adjacency backtracking [machine: orders } 24, 48, 48, 120, 120];$$

$$\text{rotations: det}$$

$$= +1 \text{ on exact coordinate realization [machine: } \mathbb{Q} \text{ and } \mathbb{Q}(\sqrt{5}) \text{ linear algebra]}$$

$$\text{Burnside: fix}(g, K)$$

$$= \#\{\text{orbit-constant } \kappa \geq 0 : \textstyle\sum = K,\ \varphi = 0\} \text{ [orbit-DP with parity masks; machine]};$$

$$\text{fix}(\text{id}, K) = A_S(K) \text{ [cross-asserted]}; \quad M \in \mathbb{Z} \text{ [asserted]}$$

$$\text{cross-checks: direct orbit enumeration } K$$

$$\leq 3 \text{ (tetra/cube/octa)},$$

$$K$$

$$\leq 2 \text{ (dodeca/icosa) [machine]};$$

$$M^{+} \geq M \text{ [smaller group]}$$

$$(2): \text{ CT0.9 }: \ 2^{2E} \text{ sign registers};$$

$$\text{CT0.10/FT.51}(5): \text{ independence};$$

$$\text{FT.41}(b): \ g(V);$$

$$\text{D3.8}: \text{ attachments};$$

$$\text{example}: \ 2^{12} \cdot 1 \cdot 3 = 12288 \text{ [machine]}$$

$$(3): \text{ substitution of FT.64}(4) \text{ asymptotics};$$

$$(4): \text{ rows }: \text{ FT.28}(2),$$

$$\text{FT.32}(1),$$

$$(6),$$

$$\text{FT.45}(1),$$

$$(3),$$

$$(4),$$

$$\text{FT.35}(1)\text{–}(3),$$

$$\text{FT.49}(2),$$

$$(4),$$

FT.50(5),

FT.62–FT.64

$\Sigma_{\min}$ index 1 : Θ

$= 2\kappa$

$\in 2\mathbb{Z}$ automatic [FT.24];

Σ_L : $\{\sum \kappa \in 2\mathbb{Z}\}$ index 2 [FT.32(6) scope];

Σ_{12} : $\mathbb{Z}^3$ [FT.35(1), (3)]

bowtie/patch : index 4 [FT.50(5) : $\mathbb{Z}_2$-rank 2];

(5) : per-cell counts cited; composition laws [FT.34(3), FT.50(1), (2), (4)]. ■

Rmk (the two untabulated quotients). Bowtie ($|\operatorname{Aut}| = 8$: the degree-4 junction is fixed; the two pairs of leaf vertices swap freely, and the two triangles exchange) and double-triangle patch ($\mathbb{Z}_2 \times \mathbb{Z}_2$: $C \leftrightarrow D$, $A \leftrightarrow B$) degeneracy tables are computable by the same machine and are deferred only for space; their lattice indices, minima, and group orders are in the table. Quotient convention throughout: automorphisms of the floor's internal skeleton (CT2.7's vertex-name independence; external contacts are configuration data, not relabeled). No structural claim rests on the omission.

Phys. The generalized list, read aloud: every fermion floor the program has computed admits loops on one lattice per topology — index one for the digon and the wired pair, index two for cycles and polygons and the tetrahedron face, index four for the glued pairs, eight to half a million for the solids — and on each lattice the admissible loops come in four regimes, collapse to orbits under the skeleton's own symmetries, and multiply by two independent sign-and-sealing registers. Twelve rows, one mass law, closed under composition. What the list does not contain is its own last filter: which loops the spectrum keeps. The ledger side is finished; the door is the corpus's.

Theorem A.14 (FT.69 — Platonic amplitude census completed; the cycle-alternation lemma; one census correction). Thm. Under Def-H.25's scopes (S1)–(S4): (1) Cap census, completed. Bypass-configuration counts by K (cap-DP, exact): octahedron $318\,523$ ($K \le 9$; row $(1, 24, 276, 2024, 10206, 35448, 81344, 110736, 69504, 8960)$; $K \le 5$ exhaustively validated); dodecahedron $3\,067\,633\,654\,196\,912$ ($K \le 30 = 3V/2$ forced); icosahedron $2\,483\,119\,531\,360$ ($K \le 18 = 3V/2$ forced). Low-order universality: the rows of the two 60-edge skeletons agree through $K = 3$ ($(1, 60, 1770, 34220)$: caps bind only from $K = 4$: 487335 vs 485115, the difference $20\binom{6}{4}$ vs $12\binom{10}{4}$ exactly the single-vertex lock counts) and diverge from $K = 4$ on. Orbits: not

attempted (bounded rule: cube's $33\,564$ already exceeded 10^4 [FT.68(3)]). (2) Coverings. Octahedron complete (two independent enumerators agree): $\mathcal{Z}_{\text{byp}}(y) = 1 + 72y + 852y^2 + 1152y^3$, $\mathcal{Z}_{\text{lock}}(y) = 36y^2 + 1008y^3 + 840y^4$; $Z^{\pm}_{\text{byp}} = 87\,649$, $Z^{\pm}_{\text{lock}} = 280\,128$, $Z^{\pm}_{\text{all}} = 367\,777$. Dodecahedron/icosahedron exact to $|\mathcal{C}| \le 3$: dodecahedron $\{0{:}1,\ 1{:}120,\ 2{:}5880{+}60_{\text{lock}},\ 3{:}153\,960{+}6140_{\text{lock}}\}$ ($166\,161$ covers, $Z^{\pm} \ge 9\,948\,001$ byp / $393\,920$ lock from the counted range); icosahedron $\{0{:}1,\ 1{:}240,\ 2{:}17\,520{+}120_{\text{lock}},\ 3{:}478\,360{+}27\,600_{\text{lock}}\}$ ($523\,841$ covers, $Z^{\pm} \ge 30\,896\,321/1\,768\,320$ counted). Single-centre closed form on every skeleton: $\#\{|\mathcal{C}| = 1\} = V\,d_{\text{in}}d_{\text{out}}$ minus closed-thread exclusions $= V\,d(d{-}1)$ bidirectional $= (24, 48, 72, 120, 240)$ for the five solids [machine]. Full dodeca/icosa counts deferred: the vertex-DP state width (cut-arcs ≤ 11 at degree 5; memo exhaustion) and the branch enumerator ($\sim (1 + d^2)^V$ raw) both exceed the bounded rule — recorded, not guessed. Fully crystallized floors cancel: types $(3,3)^{20}$, $(5,5)^{12}$, no centres, value 0 [FT.67(3a)]. (3) Cycle-alternation lemma. Let R be a literal-domain cover and $\gamma \subseteq R$ a directed cycle. Then along γ the vertices strictly alternate centre / non-centre; consequently: (i) $|\gamma|$ is even; (ii) every non-centre vertex of γ carries at least one further real edge outside γ (else its type on γ alone is $(1,1)$, forcing a centre — contradiction); (iii) a pure directed cycle is never a cover [FT.67(3a), now with its structural reason]; (iv) no odd directed cycle — in particular no triangle — lies inside any cover: the amplitude layer's bound content is exactly even-cycle content, dissolved into threading [scope: clauses (i)–(v) are literal-domain theorems. Both chain-local extensions dissolve odd cycles into open threads — on the tetrahedron, of 2397 patterns containing a directed 3-cycle, 833 (E2) and 1025 (E1) have nonzero amplitude against 0 literal [FT.75(4)]]; (v) census (distinct cycles, canonicalized; machine): covers containing directed cycles — tetrahedron $64/101$ (84 digon instances; 48 bypass $+16$ locked), cube $9060/11205$ ($14\,328$ digons, 504 four-cycles), octahedron $3054/3961$ (4704 digons, 672 four-cycles), P_3–P_7 digons only; the P_4 face 4-cycle lies in none of its 33 covers (clause (ii): its non-centre cycle vertices have no extra edges to carry); zero odd cycles in any enumerated cover. (4) FT.68(1) census correction (self-audit). The tetrahedron census numbers stand ($501/252/90$ bypass configurations containing digons/3-cycles/4-cycles, re-derived). The clause "all of it cancels in amplitude" is false for the digon row: 48 of the 501 are covers. Witness (machine, brute-force sign sum = cover value $= 16$): $R = \{(0{\to}1), (1{\to}0), (2{\to}1), (3{\to}2)\}$, types $\big((1,1),(2,1),(1,1),(0,1)\big)$ — the digon $\{0,1\}$ dissolved: centre 0 reads both digon edges, vertex 1 is a $(2,1)$ non-centre supported by centre 2's thread. The 3-cycle and 4-cycle rows cancel completely [(iii)+(i) of clause (3); machine: 0 nonzero of $252/90$]. **[addition: these 48 survivors are killed by both chain-local extensions ($N_{E2} = N_{E1} = 0$ on all 48, exhaustive), while the 16 locked digon-covers of clause (5) are killed by E2 and kept by E1 at the derangement values $(!a)4^a$ [FT.75(1),(4)]. The triangle rows revive in the extensions: 833 (E2) / 1025 (E1) of the 2397 tetrahedron patterns containing a directed 3-cycle.] Corrected statement: bound cells cancel standalone; bound substructures survive inside covers only dissolved — even cycles threaded through supported non-centres. FT.68(1) patched in place (marker). (5) Minimal dissolution family.** The smallest dissolved-digon covers are double-digon hubs: two digons $\{h, c_1\}, \{h, c_2\}$ sharing a locked $(2,2)$ hub h, centres c_1, c_2 ($|\mathcal{C}| = 2$, $|R| = 4$, value 16): the tetrahedron's 12 locked two-centre covers are exactly this family (4 hub choices $\times\binom{3}{2}$); exhibited on cube and octahedron. The four $K_{1,3}$ stars [FT.68(1)] are the triple-digon hubs. Locked hubs are the only lock content visible in covers [FT.67 Rmk], and digons are the only bound cells they can host at $|\mathcal{C}| = 2$.

Req. Def-H.25; FT.17, FT.22, FT.28, FT.32, FT.42, FT.45, FT.62, FT.63, FT.67, FT.68; A4, A8; D0.1, D0.8, D0.10, D1.4, D1.5, D3.3; CT0.11 (Part 1); CT4.1 (Part 2); cap-DP / cover-DP / branch enumeration, cycle census [math; machine].

Pf.

(1) : cap-DP over vertex-introduction order (state: active degrees ≤ 3, by-K accumulation) [machine exact];

$$K \leq \lfloor 3V/2 \rfloor : \ \textstyle\sum_v k_v = 2K \leq 3V$$

$$\text{low-order rows} : \ K \leq 3 : \ \text{no vertex can reach } k_v \geq 4 : \ c_K = \binom{2E}{K};$$

$$K = 4 : \ \binom{60}{4} - \#\{\text{4 edges at one vertex}\} : \ 20\binom{6}{4} = 300, \ 12\binom{10}{4} = 2520 \ \checkmark$$

(2) : two enumerators: vertex-introduction DP (centre pre-claim masks, completion checks)

$$=$$ branch enumeration over centre sets with claim conflicts pruned;

$$|\mathcal{C}| = 1 : \ \text{in-pick} \times \text{out-pick}, \ \text{closed thread } (s{=}t) \text{ excluded} \implies Vd(d{-}1)$$

$$\text{floors} : \ \text{type}(d, d), \ d \geq 2 : \ \mathcal{C} = \varnothing, \ R \neq \varnothing : \ \text{cov} \equiv 0 : \ \text{value } 0 \ [\text{FT.67(2)}]$$

(3) : $e_i = (v_i \to v_{i+1})$ along γ : $\text{cov}(e_i) = 1$: v_i centre $\iff$ e_i read at tail, e_{i-1} read at head;

$$v_i \text{ centre} \implies v_i \text{ reads both } e_{i-1}, e_i \implies v_{i\pm1} \text{ read neither} \implies v_{i\pm1} \text{ non-centre}$$

$$v_i \text{ non-centre}$$

$$\implies e_{i-1} \text{ read at } v_{i-1} \text{ (centre)},$$

$$e_i \text{ read at } v_{i+1} \text{ (centre)} : \ \text{strict alternation};$$

$$\text{(i)} : \ \text{alternation on a cycle}$$

$$\implies |\gamma| \text{ even}$$

$$\text{(ii)} : \ v \text{ non-centre on } \gamma : \ \text{type on } \gamma \text{ alone}$$

$$= (1, 1) : \ \text{forced centre [D3.3 type-based]} \implies \exists \text{ extra edge};$$

$$(\text{iii}):\ \text{pure cycle}:\ \text{all vertices on } \gamma,\ \text{alternation} + (\text{ii}) \text{ impossible}$$

$$(\text{iv}):\ \text{odd } |\gamma| \text{ cannot alternate;} \qquad (\text{v}):\ \text{machine census, canonical min-rotation cycles}$$

$$(4):\ \text{witness types verified;}$$

$$\text{reads}:\ (0{\to}1),$$

$$(1{\to}0) \text{ at centre } 0;$$

$$(2{\to}1) \text{ at centre } 2;$$

$$(3{\to}2):\ \text{centre } 2 \text{ in-pick}:\ \text{cov}$$

$$\equiv 1\ \checkmark;$$

$$\text{brute force}$$

$$= 16\ [\text{machine}]$$

$$(5):\ |\mathcal{C}|$$

$$= 2 \text{ with a digon}:\ \text{digon edges both read by one centre } c\ (\text{type}(1,1) \text{ forces its only edges});$$

$$\text{partner } h:\ \text{second digon's centre } c' \text{ reads it;}$$

$$h\ (2,2) \text{ locked;}$$

$$\text{tetrahedron count } 4 \times \binom{3}{2} \cdot \ldots$$

$$= 12\ [\text{machine}]. \blacksquare$$

Rmk (what the lemma settles for the two-layer picture). The amplitude layer does not merely cancel bound cells — it cancels every odd cycle and every undissolved even cycle. Bound content inside survivors is always threaded: read exactly once per edge, which a closed chain read at both

ends can never be. The spectral layer's bound floors (M1/M2) are therefore not in tension with the cover grammar — they are disjoint reads, and the tension is precisely the inter-family bridge.

Phys. Ask which flickering patterns carry nonzero ledger amplitude, and the answer now has a grammar: edges tile into open two-step threads, and any loop that hides inside the tiling must be even, must be read alternately, and must be propped open at every other vertex by extra threading. Triangles cannot hide at all. The smallest hiding place is a pair of digons on a locked hub — darkness threaded by transit, exactly the picture the locked-hub remark promised.

Theorem A.15 (FT.76 — The Platonic pole-mass multiplets; the bounded-hierarchy ceiling). Thm. Under Def-H.37(2) and Def-H.38, with $\Theta_R = \delta_v$ [FT.63(4)'s vertex-figure read; the corner convention is not declared in this part], $D_S(m) := m^2_{\text{pole}}(m)/\hat{\mu}'^2 = 2 - 2\cos(m\,\delta_v/2) = 4\sin^2(m\,\delta_v/4)$ [FT.60(5)], and k the carrier winding [Def-H.38]: (1) Conventions and the two trivial readings. $k_f = K_S/F = (3,2,3,5,3)$ (face winding; $K_S = (12,12,24,60,60)$ [FT.63(1), FT.43(1)], reproduced from the minimality rule $\kappa_{\min} = 2$ odd-L / 1 even-L, $k = L\kappa_{\min}/2$ [FT.32(6), FT.45]); $k_{vf} = (3,3,2,3,5)$ (vertex-figure winding, same rule on the vertex-figure $L_{vf} = (3,3,4,3,5)$). Variant (A) := (δ_v, k_f) — the directive's reading; variant (B) := (δ_v, k_{vf}) — the loop-consistent reading. Both other readings named by FT.63(4)/(6) are identically massless: $\Theta_R = 0$ (flat-face side) and $\Theta_R = V\delta_v = 4\pi$ (global sweep: $D(m) = 2 - 2\cos(2m\pi) = 0$ for every weight — FT.63(4)'s trivial global spinor transport). (2) Exact multiplets. With $c = \cos\frac{\pi}{10}$ ($16c^4 - 20c^2 + 5 = 0$):

S	(A) k_f	$m^2_{\text{pole}}/\hat{\mu}'^2$ on $\{k, k-2, \ldots, -k\}$	field
tetra	3	$2,\ 2,\ 2,\ 2$ $(m = \pm1, \pm3)$	$\mathbb{Q}$
cube	2	2 $(m = \pm2)$, 0 $(m = 0)$	$\mathbb{Q}$
octa	3	4 $(m = \pm3)$, 1 $(m = \pm1)$	$\mathbb{Q}$
dodeca	5	2 $(m = \pm5)$, $2{-}2\cos\frac{3\pi}{10}$ $(m = \pm3)$, $2{-}2c$ $(m = \pm1)$	$\mathbb{Q}(c)$
icosa	3	2 $(m = \pm3)$, $2{-}\sqrt{3}$ $(m = \pm1)$	$\mathbb{Q}(\sqrt{3})$

S	(B) k_{vf}	$m^2_{\text{pole}}/\hat{\mu}'^2$	field
tetra	3	$2,\ 2,\ 2,\ 2$	$\mathbb{Q}$
cube	3	$2{+}\sqrt{2}$ $(m = \pm3)$, $2{-}\sqrt{2}$ $(m = \pm1)$	$\mathbb{Q}(\sqrt{2})$
octa	2	3 $(m = \pm2)$, 0 $(m = 0)$	$\mathbb{Q}$
dodeca	3	$2{-}2\cos\frac{3\pi}{10}$, $2{-}2c$	$\mathbb{Q}(c)$
icosa	5	$2{+}\sqrt{3}$ $(m = \pm5)$, 2 $(m = \pm3)$, $2{-}\sqrt{3}$ $(m = \pm1)$	$\mathbb{Q}(\sqrt{3})$

Decimals ($m^2/\hat{\mu}'^2$): dodeca $(0.097886967410,\ 0.824429495415,\ 2)$; icosa $(0.267949192431,\ 2)$, (B) adds 3.732050807569; cube (B) $(0.585786437627,\ 3.414213562373)$. Every value is exact in the stated field and was cross-checked against an exact rational Taylor series for cos to 45 digits (125 values, $m = 0..24$, all five solids). (3) Periodicity, monotonicity, and the hard ceiling. The ladder is periodic in the weight with period $P_S = 4\pi/\delta_v = (4,8,6,20,12)$ and takes exactly $P_S/2 + 1 = (3,5,4,11,7)$ distinct values, monotone increasing in m on $[0, P_S/2]$; the maximum is $4\hat{\mu}'^2$ (attained at $m = P_S/2$; P_S even for all five) and the minimum nonzero value is D_S itself. Hence every Platonic pole mass lies in $[0, 4\hat{\mu}'^2]$ and the widest ratio attainable within a solid is

$$\frac{4}{D_S}$$

$$= \left(2,\ \tfrac{4}{2-\sqrt{2}},\ 4,\ \tfrac{4}{2-2\cos\frac{\pi}{10}},\ \tfrac{4}{2-\sqrt{3}}\right)$$

$$= (2,\ 6.828427,\ 4,\ 40.863458,\ 14.928203) \text{ in } m^2,$$

maximal for the dodecahedron, whose value has minimal polynomial $x^4 - 48x^3 + 304x^2 - 512x + 256$; in mass the ceiling is $\sqrt{4/D_S} = (1.414214,\ 2.613126,\ 2,\ 6.392453,\ 3.863703)$, maximal at $6.392453\ldots$ No choice of winding — k_f, k_{vf}, K_S, or any iterate — and no admissible Θ_R convention can exceed it: the value set is finite and bounded. (4) Top-weight near-universality. In variant (A) the heaviest weight of four of the five solids is exactly $2\hat{\mu}'^2$ ($k_f\delta_v \equiv \pi \bmod 2\pi$: tetra 3π, cube π, dodeca π, icosa π), the octahedron's being $4\hat{\mu}'^2$ ($k_f\delta_v = 2\pi$). The inter-solid hierarchy therefore lives entirely at the light end; the heavy ends are degenerate across solids. (5) Two structural obstructions. (a) Tetrahedral degeneracy: $\delta_v = \pi \Rightarrow D(m) = 2 - 2\cos(m\pi/2) \in \{0, 2, 4\}$, and the $k = 3$ multiplet is fully degenerate at $2\hat{\mu}'^2$ (both variants) — the tetrahedron carries no multiplet splitting at all. (b) Even-k zero modes [FT.60(5)'s standing zero]: the cube in (A) ($k_f = 2$) and the octahedron in (B) ($k_{vf} = 2$) each carry an exactly massless weight $m = 0$, so their literal "lightest pole mass" is 0 and only lightest-nonzero reads are well defined. (6) Inter-solid ratios (lightest nonzero, relative to the tetrahedron).

	(A)		(B)	
S	m^2 ratio	m ratio	m^2 ratio	m ratio
tetra	1	1	1	1
cube	1 (0 with the zero mode)	1 (0)	0.292893218813	0.541196100146
octa	0.5	0.707106781187	1.5	1.224744871392
dodeca	0.048943483705	0.221231742082	0.048943483705	0.221231742082
icosa	0.133974596216	0.366025403784	0.133974596216	0.366025403784

Normalized to the lightest solid, in mass: (A) dodeca 1, icosa 1.654488638651, octa 3.196226610750, cube = tetra $4.520147021340 = \sqrt{2/(2-2c)}$; (B) dodeca 1, icosa 1.654488638651, cube 2.446285940037, tetra 4.520147021340, octa 5.536026882322. (7) Intra-solid splittings (heaviest/lightest nonzero).

S	(A) m^2	(A) m	(B) m^2	(B) m
tetra	1	1	1	1
cube	1 (+1 zero mode)	1	$3+2\sqrt{2} = 5.828427124746$	$1+\sqrt{2} = 2.414213562373$
octa	4	2	1 (+1 zero mode)	1
dodeca	$\frac{2}{2-2c} = 20.431729094531$	4.520147021340	$\frac{2-2\cos(3\pi/10)}{2-2c} = 8.422260053931$	2.902113032590
icosa	$4+2\sqrt{3} = 7.464101615138$	$1+\sqrt{3} = 2.732050807569$	$7+4\sqrt{3} = 13.928203230276$	$2+\sqrt{3} = 3.732050807569$

Minimal polynomials of the m^2 splittings: dodeca (A) $x^4 - 24x^3 + 76x^2 - 64x + 16$, dodeca (B) $x^4 - 14x^3 + 51x^2 - 34x + 1$, icosa (A) $x^2 - 8x + 4$, icosa (B) $x^2 - 14x + 1$, cube (B) $x^2 - 6x + 1$; the mass splittings $1+\sqrt{3}$, $2+\sqrt{3}$, $1+\sqrt{2}$ have $x^2 - 2x - 2$, $x^2 - 4x + 1$, $x^2 - 2x - 1$ (machine; certified dyadic isolations). (8) Dual splitting, sharpened. The dual pairs remain degenerate in m_{hid} and Π [FT.44(1)] and split on the pole side as multisets in both variants — cube $\{0, 2\}$ vs octa $\{1, 4\}$ in (A), cube $\{2-\sqrt{2}, 2+\sqrt{2}\}$ vs octa $\{0, 3\}$ in (B); dodeca vs icosa share only the top value $2\hat{\mu}'^2$ in (A) and nothing in (B). FT.44(2)'s splitting template acquires its multiplet form. (9) Field-degree ledger. Tetra/octa: $\mathbb{Q}$; cube: $\mathbb{Q}(\sqrt{2})$; icosa: $\mathbb{Q}(\sqrt{3})$; dodeca: $\mathbb{Q}(\cos\frac{\pi}{10})$, degree 4, with $\sqrt{5} = 8c^2 - 5$ its unique quadratic subfield (machine) — the

requested four-field list $\{\mathbb{Q}, \mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{5})\}$ is insufficient for the dodecahedron, exactly as FT.73(2)'s degree-16 compositum requires. (10) FT.63(6) precision correction. The dodecahedral quartic $x^4 - 8x^3 + 19x^2 - 12x + 1$ has four positive roots (machine-isolated to width 2^{-60}): 0.097886967410, 0.824429495415, 3.175570504585, 3.902113032590, realized by the weights $m \in \{1, 19\}, \{3, 17\}, \{7, 13\}, \{9, 11\}$ respectively. FT.63(6)'s "$D(1), D(3)$ the two positive roots of the quartic" corrects to the two smallest of its four positive roots; patched in place (marker).

Req. Def-H.37(2), Def-H.38; Def-H.21, Def-H.23; FT.22, FT.32, FT.43, FT.44, FT.45, FT.47, FT.52, FT.54, FT.55, FT.60, FT.63; D2.9; CT2.5 (Part 2); CT0.7a (Part 1); Chebyshev recursion, de Moivre $\cos 5t$, exact root isolation by bisection, minimal polynomials via the multiplication map [math; machine].

Pf.

$$(1):\ k_f = K_S/F:\ (12, 12, 24, 60, 60)/(4, 6, 8, 12, 20) = (3, 2, 3, 5, 3)\ \checkmark;$$

$$k = L\kappa_{\min}/2:\ \text{odd } L \to L,\ \text{even } L \to L/2$$

$$\Theta_R = 4\pi:\ D(m) = 2 - 2\cos(2m\pi) = 0\ \forall m; \qquad \Theta_R = 0:\ D \equiv 0\ [\text{FT.63}(4), (6)]$$

$$(2):\ D(m)$$

$$= 2 - 2T_m(\cos(\delta_v/2))\ [\text{Chebyshev recursion } T_{n+1} = 2xT_n - T_{n-1},\ \text{integer coefficients}];$$

$$\cos(\delta_v/2)$$

$$= (0, \tfrac{\sqrt{2}}{2}, \tfrac{1}{2}, c, \tfrac{\sqrt{3}}{2})$$

$$\text{base elements exact}:\ 16c^4 - 20c^2 + 5 = 0\ [\cos 5t \text{ at } t = \tfrac{\pi}{10}];$$

$$(\tfrac{\sqrt{2}}{2})^2 = \tfrac{1}{2};$$

$$(\tfrac{\sqrt{3}}{2})^2 = \tfrac{3}{4}; \quad \text{root branch by sign/order}$$

$$\text{values}:\ \text{exact field arithmetic};$$

$$\text{cross-check}:\ |D(m) - [2 - 2\sum_j (-1)^j (m\delta_v/2)^{2j}/(2j)!]| < 10^{-45} \text{ at 120 terms [machine]}$$

$$(3):\ D(m+P) = D(m) \iff \cos((m+P)\delta_v/2) = \cos(m\delta_v/2):\ P\delta_v/2 = 2\pi:\ P = 4\pi/\delta_v;$$

$$D(P-m) = D(m),\ D(-m) = D(m)$$

$$m \in [0, P/2] :\ m\delta_v/2$$

$$\in [0, \pi] :\ \cos \text{ strictly decreasing}$$

$$\implies D \text{ strictly increasing} :\ \min_{\neq 0}$$

$$= D(1)$$

$$= D_S,$$

$$\max$$

$$= D(P/2) \qquad = 4$$

$$\text{distinct count} = P/2 + 1; \qquad 4/D_{\text{dodeca}} :\ \text{minpoly by the multiplication map on } \mathbb{Q}(c)\ [\text{machine}]$$

$$(4) :\ k_f\delta_v = (3\pi, \pi, 2\pi, \pi, \pi) :\ \cos(k_f\delta_v/2) = (0, 0, -1, 0, 0) \implies D(k_f) = (2, 2, 4, 2, 2)$$

$$(5a) :\ \delta_v$$

$$= \pi :\ D(m)$$

$$= 2 - 2\cos(m\pi/2) :\ m \text{ odd} \to 2,$$

$$m$$

$$\equiv 0(4) \to 0,$$

$$m$$

$$\equiv 2(4) \to 4 :\ k_f$$

$$= 3 \text{ odd weights only} \implies \text{constant } 2$$

$$(5b):\ k \text{ even} \implies m = 0 \in \{k, k-2, \ldots, -k\}:\ D(0) = 0\ [\text{FT.60(5)}]$$

$$(6),$$

$$(7):\ \text{exact division in the solid's field;}$$

$$\text{mass ratios}:\ r$$

$$= \sqrt{q}:\ q \text{ isolated exactly}$$

$$\implies r$$

$$\in (\sqrt{a}, \sqrt{b})\ [\sqrt{}\ \text{monotone}];$$

$$\text{icosa (A)}:\ q = \tfrac{2}{2-\sqrt{3}} = 2(2+\sqrt{3}) = 4+2\sqrt{3}:\ \sqrt{q} = 1+\sqrt{3}\ [(1+\sqrt{3})^2 = 4+2\sqrt{3}\ \checkmark]$$

$$\text{icosa (B)}:\ q = \tfrac{2+\sqrt{3}}{2-\sqrt{3}} = (2+\sqrt{3})^2 = 7+4\sqrt{3}:\ \sqrt{q} = 2+\sqrt{3}\ \checkmark$$

$$\text{cube (B)}:\ \tfrac{2+\sqrt{2}}{2-\sqrt{2}} = 3+2\sqrt{2} = (1+\sqrt{2})^2\ \checkmark;$$

$$(8), (9):\ \text{multiset comparison; field membership by construction}$$

$$(10):\ \text{sign-change scan on } \mathbb{Q} \text{ at step } 1/500 \text{ over } [-1, 6] \text{ then bisection to } 2^{-60}:\ 4 \text{ brackets, all in } (0, \infty);$$

$$q(0..4) = (1, 1, 5, 1, 1) > 0$$

$$\text{weight attribution}:\ D(m) = \text{root} \iff m \text{ in the stated residue classes mod } 20\ [\text{machine}].\ \blacksquare$$

Phys. Ask the five solids for their pole masses under the declared defect transfer, and the answer is a small, closed, exactly computable set of numbers — three values for the tetrahedron, five for the cube, eleven for the dodecahedron — all of them between zero and four times one unknown scale. Nothing in that set is large. The tetrahedron's multiplet does not split at all; the cube's even winding leaves one weight exactly massless; and the heaviest state of four of the five solids is the same number. So the Platonic defects do not generate hierarchy — they generate a bounded, near-degenerate spectrum, and the bound is a theorem about cones, not a limitation of the search.

Theorem A.16 (FT.78 — The orbit decompositions: tetrahedron exhaustive, cube symmetry-reduced). Thm. Under Def-H.28, at (S1)–(S4): (1) Tetrahedron, exhaustive. The 1230 admissible configurations [FT.68(1)] form 70 orbits under $\mathrm{Aut} = S_4$ ($|\mathrm{Aut}| = 24$), with $\sum_{\mathcal{O}} \Omega_{\mathcal{O}} = 1230$; orbits by K: $(1, 1, 5, 13, 23, 21, 6)$ for $K = 0..6$; configuration counts by K reproduce the cap row $(1, 12, 66, 220, 435, 408, 88)$ [FT.68(1)] exactly; $\Omega \in \{1, 3, 4, 6, 8, 12, 24\}$, every value dividing 24. (2) Tetrahedron cover sector: seven orbits total. The 85 bypass covers form 5 orbits and the 16 locked covers 2 orbits:

sector	K	$\lvert\mathcal{C}\rvert$	Ω	$4^{\lvert\mathcal{C}\rvert}$	signature (cycles shadowed)	type multiset
byp	0	0	1	1	()	$(0,0)^4$ (vacuum)
byp	2	1	24	4	()	$(0,0), (0,1), (1,0), (1,1)$ single thread
byp	4	2	24	16	(2)	$(1,0), (1,1)^2, (1,2)$
byp	4	2	24	16	(2)	$(0,1), (1,1)^2, (2,1)$
byp	4	2	12	16	()	$(0,2), (1,1)^2, (2,0)$ double thread
lock	4	2	12	16	$(2,2)$	double-digon $(2,2)$ hub
lock	6	3	4	64	$(2,2,2)$	$K_{1,3}$ star hub

—the two locked rows being exactly FT.69(5)'s minimal dissolution family (12) and the four $K_{1,3}$ triple-digon stars [FT.68(1)]. The single-centre orbit's size $24 = Vd(d-1)$ is FT.69(2)'s closed form. (3) Orbit-level amplitude identity (new). The amplitude sum decomposes exactly over classes:

$$\sum_{\mathcal{O}} \Omega_{\mathcal{O}}\, 4^{|\mathcal{C}(\mathcal{O})|} = Z^{\pm},$$

machine-verified on both skeletons: tetrahedron 1057 (bypass) and 448 (locked) [FT.68(1)]; cube 1 109 313 and 1 150 848 [FT.68(3)]. Consequently the covering-polynomial coefficients are the $|\mathcal{C}|$-level configuration counts, and the orbit counts refine them: cube bypass $|\mathcal{C}| = 0..4 \Rightarrow (1, 1, 19, 57, 88)$ orbits over $(1, 48, 648, 2672, 3624)$ configurations; locked $|\mathcal{C}| = 2..6 \Rightarrow (1, 18, 71, 7, 1)$ over $(24, 728, 3192, 264, 4)$. (4) Cube, symmetry-reduced. Cap row $(1, 24, 276, 2024, 10506, 40296, 115796, 247704, 383913, 408464, 270096, 91392, 9600)$, total 1 580 092 configurations with $K \le 12$ forced [FT.68(3)]; cap-aware Burnside over $|\mathrm{Aut}| = 48$ gives orbits by K $(1, 1, 11, 49, 250, 863, 2509, 5211, 8163, 8571, 5768, 1922, 245)$, total 33 564 $> 10^4$ — the bounded rule is observed, so full configuration enumeration is not attempted. The cover sector is enumerated explicitly (it is small): the 11 205 covers (6993 bypass + 4212 locked, two independent enumerators agreeing [FT.69(2)]) form 264 orbits (166 bypass, 98 locked), $\sum \Omega = 11\,205$, $\Omega \mid 48$ throughout, with Ω-histogram {1:1, 4:1, 8:1, 12:4, 16:2, 24:47, 48:208} — the 208 free orbits are the generic case, so the twenty most populated classes are all free orbits ($\Omega = 48$) and are distinguished only by signature: one at $|\mathcal{C}| = 1$ (the single thread), eight at $|\mathcal{C}| = 2$ (four cycle-free, four shadowing one digon), eleven at $|\mathcal{C}| = 3$. Independent cross-validation: explicit canonical-form enumeration of every admissible configuration with $K \le 5$ (53 127 of them) reproduces the Burnside row $(1, 1, 11, 49, 250, 863)$ exactly — two methods, same numbers. (5) Signatures. Covers shadow only even cycles, orbit by orbit [FT.69(3)(i) recovered at class level]: tetrahedron covers contain digons only (84 instances, $\delta h = 2$ [D0.6]); cube covers contain digons and directed 4-cycles (14 832 instances, $\delta h \in \{2, 4\}$), with 10 distinct cycle contents $\{(), (2), (2,2), (4), (2,2,2), (2,4), (2,2,2,2), (2,2,4), (2,2,2,2,2), (2,2,2,2,2,2)\}$ and no odd cycle anywhere. The signature $(K, \text{cycles}, |\mathcal{C}|)$ is not a complete class invariant: 7 tetrahedron cover

orbits carry only 6 signatures, 264 cube cover orbits only 21 — the completing datum is the vertex-type multiset (clause (2)'s last column). (6) Charge-conjugation chirality of the cover sector (new). Global orientation reversal $R \mapsto R^{\mathrm{T}} = \{(b,a) : (a,b) \in R\}$ is an involution of the pattern space that is not in $\mathrm{Aut}(\Gamma)$ (it is induced by no vertex permutation). It pairs cover classes: the 48 bypass dissolved-digon covers of FT.69(4) split into two $\Omega = 24$ orbits exchanged by reversal (representatives with type multisets $(1,0),(1,1)^2,(1,2)$ and $(0,1),(1,1)^2,(2,1)$ — transposes of each other; machine: the reversal of one representative is canonical in the other orbit). This is the pattern-level form of charge conjugation [D2.6]. Both mass ansätze are reversal-invariant (K, Ω and $4^{|\mathcal{C}|}$ are), so the chirality is invisible to the mass spectrum; reading it requires a C-odd ledger of FT.40's q_{core} species.

Req. Def-H.25, Def-H.26, Def-H.28; FT.67, FT.68, FT.69, FT.70, FT.71; FT.27, FT.40; A4, A8; D0.6, D0.8, D2.6; CT0.11, CT0.13, CT0.14 (Part 1); CT1.20 (Part 2); orbit–stabilizer, Burnside's lemma, canonical-form enumeration [math; machine].

Pf.

$$(1):\ \text{canonical form}$$

$$= \min\ \text{over the } 24 \text{ edge-permutations;}$$

$$\text{admissibility and } K \text{ are invariant} \implies \text{orbits lie inside the set}$$

$$\sum_{\mathcal{O}} \Omega_{\mathcal{O}} = 1230\ [\text{machine}]; \qquad \Omega \cdot |\,\mathrm{Stab}\,| = 24 \implies \Omega \mid 24; \qquad \text{cap row}:\ \text{CT0.11 counting per } K$$

$$(2):\ \text{explicit enumeration of the } 101 \text{ covers } (85+16);$$

$$|\mathcal{C}| = 1:\ \Omega = 24 = Vd(d-1) = 4 \cdot 3 \cdot 2\ [\text{FT.69(2)}]$$

$$\text{locked rows}:\ \text{double-digon}(2,2)\ \text{hub}\ (12 = 4\binom{3}{2}),\ K_{1,3}\ \text{star}\ (4,\ \mathrm{Stab} = S_3)\ [\text{FT.69(5)}]$$

$$(3):\ Z^{\pm}$$

$$= \sum_{R\ \text{cover}} 4^{|\mathcal{C}(R)|}\ [\text{FT.67(2) at (S1)}]$$

$$= \sum_{\mathcal{O}} \sum_{R \in \mathcal{O}} 4^{|\mathcal{C}|}$$

$$= \sum_{\mathcal{O}} \Omega_{\mathcal{O}} 4^{|\mathcal{C}(\mathcal{O})|}$$

$$[|\mathcal{C}|\ \text{class-constant}]$$

$$(4):\ \text{Burnside}:\ \#\text{orbits}_K$$

$$= \tfrac{1}{48}\sum_{p} \text{fix}_K(p),$$

$$\text{fix}_K\ \text{by orbit-subset enumeration under the cap;}$$

$$\text{covering polys}:\ \text{DP}$$

$$= \text{backtracking [FT.69(2)]}$$

$$\text{cross-validation}:\ K$$

$$\leq 5\ \text{explicit canonical forms}$$

$$= (1, 1, 11, 49, 250, 863)$$

$$= \text{Burnside row [machine]}$$

$$(5):\ \text{canonical cycle enumeration per representative;}\ \text{evenness}:\ \text{FT.69}(3)(i);$$

$$\text{incompleteness}:\ \text{two classes share}(4, (2), 2)\ \text{and differ in type multiset}$$

$$(6):\ R^{\mathrm{T}}\ \text{preserves}\ K,\ \text{types}^{\mathrm{T}},\ |\mathcal{C}|,\ \text{cover status;}$$

$$R^{\mathrm{T}} = p(R)\ \forall p \in \text{Aut}\ \text{would force}\ \Omega\ \text{to merge}:\ \text{the two orbits are distinct}\ \Longrightarrow \notin \text{Aut}$$

reversal maps one $\Omega{=}24$ orbit onto the other [machine: canonical forms agree]. ■

Phys. Ask the flickering gas how many distinct shapes it takes, and the answer is startlingly small: the tetrahedron's 1230 configurations are 70 shapes, and the whole of its amplitude-carrying cover content is seven — one vacuum, one single thread, three two-centre networks, and the two locked hub families. The cube's eleven thousand covers are 264 shapes over seven mass levels. Nothing here is random; the shapes come in chirality pairs related by reversing every arrow, a distinction the mass reads cannot see. The degeneracies are where the complexity lives, and degeneracy is not hierarchy.

Theorem A.17 (FT.81 — Particle-content census (audit pass A)). Thm. Under Def-H.29, with the SM field content external and marked: (1) The census. 24 entries, classified with their supplying tags or no-goes:

PRESENT (10)	PARTIAL (6)	ABSENT (8)
gluons (8)	Higgs doublet	neutrino
$W^{\pm}$	quark triplet	generations/flavour
Z (mixture)	lepton doublet	CKM/PMNS, CP phase
photon	charged-lepton singlet	Yukawa sector
confinement/singlet	observable charge	chirality/Weyl
θ_W, e (relations)	EW VEV	gauge anomalies
G_N (beyond SM)		Λ_{QCD}
spin ladder		spin-statistics
fermion/boson content		
$D^{\pm}$ dark (beyond SM)		

Completeness fraction $= \dfrac{10 + \frac{1}{2}\cdot 6}{24} = \dfrac{13}{24} = 0.5416\overline{6}$. Sub-censuses: gauge species $4/4$ PRESENT (as aspect blocks and reads, not as gauged chiral couplings); fermion sector 8 entries $\to 0$ present, 3 partial, 5 absent $= 0.1875$. (2) The SM side, machine-verified in exact rationals (external data): one generation $= 15$ left-handed Weyl fields $= 6+3+3+2+1$ ($(3,2)_{1/6}$, $(\bar{3},1)_{-2/3}$, $(\bar{3},1)_{1/3}$, $(1,2)_{-1/2}$, $(1,1)_1$), 45 over three generations; 12 gauge bosons $(8+3+1)$ and 4 real Higgs dof; every anomaly sum vanishes exactly — $\sum Y = 0$, $\sum Y^3 = 0$, $[\mathrm{SU}(3)]^2\mathrm{U}(1) : \sum Y\, d(\mathrm{SU}(2))\, T(3) = 0$, $[\mathrm{SU}(2)]^2\mathrm{U}(1) : \sum Y\, d(\mathrm{SU}(3))\, T(2) = 0$, $[\mathrm{SU}(3)]^3 : 2A(3) + 2A(\bar{3}) = 0$, $[\mathrm{SU}(2)]^3 \equiv 0$. (3) The charge lattice, exactly. The SM field charges $Q = T_3 + Y$ span $\{0, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm 1\}$; the corpus read is ternary, $q_{\text{EM}} \in \{-1, 0, +1\}$ with all three attained [D3.4, EM.9, P2]. The two sets are related by confinement projection, machine-verified: the CT3.17/D3.9 singlets of u, d — ε_{ijk} baryons and $\delta^i{}_j$ mesons — carry integer charge, and the meson charge set is exactly $\{-1, 0, +1\}$, the corpus read; the baryon set is $\{-1, 0, 1, 2\}$, so the $|q| = 2$ state exceeds a single chain's ternary read, and the confined thirds are unreadable by construction [CT3.22: colour is ¬observable as end-charge]. Verdict: charge quantization to integers is reproduced for the observable sector; the per-chain read is 3-valued where the SM's observable lattice is $\mathbb{Z}$ (a multi-chain additive read would be needed for $|q| \geq 2$; FT.41(a) counts $N_{\text{charge}} = 2$ end registers). (4) Anomalies are not poseable. Every anomaly above is a constraint on chiral hypercharge assignments. The corpus has no chiral projector, no Weyl symbol and no hypercharge-assignment read (Y appears only as the abstract generator with three realizations [EW.8] and in the mixing [D3.10/EW.24]), so the cancellation question cannot be posed — a structural absence downstream of chirality, not a numerical failure. (5) Inventory facts (machine-checked over the frozen Parts 1–2). Count 0 for: Yukawa, neutrino, Weyl, CKM, PMNS, left-handed, right-handed, γ_5, flavour, flavor, black hole, spin-statistics, P_L, P_R, $\bar{\psi}$, generation structure, quark mixing, lepton number. Present: θ_W (71), T_a (51), h_0 (38), gluon (22), f_{abc} (14), ε_{ijk} (12), m_H (11), baryon (11), $W^{\pm}$ (10), meson (6), hypercharge (4), chirality (2). The two "chirality" occurrences are both inside D2.6's orientation-reversal clause (ω: chirality $\mapsto$ −chirality) — vocabulary without a projector. The four "generation" occurrences in D/F are the algebraic sense ("mass generation", "δ/ε generation"), machine-checked to be non-flavour.

Req. Def-H.29; D1.5, D2.6, D3.4, D3.7, D3.8, D3.9, D3.10, D4.1–D4.3; CT0.9, CT0.12, CT0.13, CT0.14 (Part 1); CT2.9, CT1.25 (Part 2); CT3.15, CT3.17, CT3.18, CT3.20, CT3.21, CT3.22 (Part 2);

CT4.1, CT4.4, CT4.5 (Part 2); EM.9, EM.20 (Part 2); EW.8, EW.13, EW.20, EW.24, EW.27, EW.32, EW.33, EW.35 (Part 2); FT.19, FT.20, FT.26, FT.27, FT.30, FT.34, FT.41, FT.52, FT.55, FT.57, FT.60, FT.71, FT.76, FT.79, FT.80; P1–P5; A §0.2, §0.3; external SM field content (marked); exact rational arithmetic, integer inventory scan [machine].

Pf.

(1) : per-entry citation table [machine-checked: every PRESENT row resolves to the cited tag; every ABSENT row to a scan count or a no-go]

fraction : $(10{+}3)/24 = 13/24$; gauge sub-census $4/4$; fermion sub-census $(0{+}\frac{3}{2})/8 = 3/16$

(2) : $6{+}3{+}3{+}2{+}1 = 15$; $15{\cdot}3 = 45$; $8{+}3{+}1 = 12$; $2{\cdot}2 = 4$; anomaly sums : exact Fractions

$$\sum Y = 6\cdot\tfrac{1}{6} + 3(-\tfrac{2}{3}) + 3\cdot\tfrac{1}{3} + 2(-\tfrac{1}{2}) + 1 = 1 - 2 + 1 - 1 + 1 = 0;$$

$$\sum Y^3 = \tfrac{1}{36} - \tfrac{8}{9} + \tfrac{1}{9} - \tfrac{1}{4} + 1 = 0$$

$$[\mathrm{SU}(3)]^2\mathrm{U}(1) : \ \tfrac{1}{2}\big(2\cdot\tfrac{1}{6} - \tfrac{2}{3} + \tfrac{1}{3}\big) = 0;$$

$$[\mathrm{SU}(2)]^2\mathrm{U}(1) : \ \tfrac{1}{2}\big(3\cdot\tfrac{1}{6} - \tfrac{1}{2}\big) = 0;$$

$$[\mathrm{SU}(3)]^3 : \ 2\cdot\tfrac{1}{2} - 2\cdot\tfrac{1}{2} = 0$$

$$(3) : \ Q = T_3 + Y \text{ over the five multiplets } \Longrightarrow \ \{0, \pm\tfrac{1}{3}, \pm\tfrac{2}{3}, \pm 1\};$$

$$q_{\mathrm{EM}} = \tfrac{1}{2}(\sigma_s - \sigma_t) \in \{-1, 0, 1\} \ [\mathrm{EM}.9]$$

$$\text{mesons} : \ \{u\bar{u}, u\bar{d}, d\bar{u}, d\bar{d}\} \to \{0, 1, -1, 0\} = \{-1, 0, 1\} \ \checkmark;$$

$$\text{baryons} : \ \{uuu, uud, udd, ddd\} \to \{2, 1, 0, -1\}$$

$$\text{integrality} : \ \tfrac{2}{3} - \tfrac{1}{3}$$

$$= \tfrac{1}{3} \notin \mathbb{Z} \text{ per quark},$$

$$\text{but } \varepsilon/\delta \text{ singlets sum to } \mathbb{Z} \ [\mathrm{CT}3.17, \mathrm{D}3.9; \ \text{exhaustive}]$$

$$(4):\ \text{anomaly} \equiv \text{chiral constraint};$$

$$\neg\exists\ \text{projector/Weyl symbol in Parts 1–2 [scan]} \implies \neg\exists\ \text{the question}$$

$$(5):\ \text{exact substring counts over the seven frozen documents [machine]}.\ \blacksquare$$

Phys. Line up the Standard Model's inventory against what this corpus can read, and the pattern is not random. Everything about forces is there: eight gluons with a closing eight-dimensional algebra, three weak directions that mix into a massive Z and an exactly massless photon, a scalar with a background and a mass formula, a confinement rule that makes colour unreadable and thereby makes observable charge integral — the meson charges come out as precisely the three values the ledger can register. Everything about matter is missing: no handedness, no Weyl spinor, no neutrino, no flavours, no mixing, no Yukawa couplings, and no second scale for any of them to live on. The corpus is a theory of the gauge skeleton with the chiral flesh left undeclared — and the anomalies that make the SM's flesh consistent cannot even be written down here.

Theorem A.18 (FT.83 — The rung emergence ledger (audit pass C): derived, declared, open). Thm. Under Def-H.29, per rung n: (1) The ladder is a theorem, not a picture. $\mathcal{A}(0) = \{J\}$; $\mathcal{A}(1) = \{J, G, W, H, D^+\}$; $\mathcal{A}(2) = \mathcal{A}(1) \cup \{\text{EM}\}$; $\mathcal{A}(3) = \mathcal{A}(2) \cup \{\text{strong}\}$; $\mathcal{A}(n \geq 4) = \mathcal{A}(3) \cup \{D^-\}$, with a strict mask flag $\dim \mathcal{K}_n = (6, 16, 17, 25)$ [CT0.13, CT0.14 — both with Pfs]. The admissibility ceiling is derived: $D_{\text{tick}}(n) = 2^{n+1} \leq C_{\text{cap}} = 16 \iff n \leq 3$, with saturation at $n = 3$ [CT0.5, CT0.11, P2's $n_{\max} = 3$]. (2) Rung-by-rung status.

n	what becomes active/trackable	status
0	$J^{\pm}$ seals only; $\dim \mathcal{K}_0 = 6$	derived [CT0.13/CT0.7]
1	$G(5)$ spin-2 shear, $W(3)$, $H(1)$, $D^+(1)$; $\dim \mathcal{K}_1 = 16$	derived [CT0.12/CT0.13]; the "pretime force set"
2	+ EM; $\alpha_{\text{EM}} = \pi$; q_{EM} readable on open chains	derived [CT0.13, EM.9, EM.20] under Comm-min + Eff (declared premises)
3	+ strong; $N_{\text{colour}} = 3$, $N_{\text{gluon}} = 8$; sealing = confinement; mass reads	derived [CT3.20, CT3.13, CT3.22, EW.20/EW.33]
≥ 4	D^- absorption $O(n) = 2^{n+1}-16$, $w = 0$ dust, locking; $\neg\exists$ gauge structure	derived as a no-go [CT0.11, CT4.1, CT4.4, CT4.5, FT.59]

(3) **What is not derived at any rung.** (a) The proposal's "$n = 1$: gravity, weak, Higgs, Dark$^+$, Janus" is exactly $\mathcal{A}(1)$ and is derived — but the Higgs entry is the scalar aspect F_H with a P4-free background h_0 [EW.13, EW.20, P4], and its mass is "configuration data" [EW.27]: the mechanism is derived, the scale is not. (b) The proposal's "$n = 2$: EM + lepton structure" is half derived: EM is a theorem; "lepton structure" has no read at any rung (no lepton species, no chirality [FT.81(5)]). (c) "$n = 3$: colour + quarks + the baryon edge budget" is derived — ε_{ijk} sealing of three fundamentals [CT3.15/CT3.17/D3.9], the 2s-and-3s partition rule [FT.34(5)], and $N = 1$ impossibility [FT.27] — but "quark" in the corpus is D3.8's bound fermionic loop with a triplet attachment, i.e. a topological definition with no flavour and no chirality. (d) "$n \geq 4$: lock / black-hole-matter cascade": the locking and absorption side is derived (CT4.1/CT4.2/CT4.5, $w = 0$ dust, retained-and-inaccessible [D1.11]); "black hole" has no corpus referent at all (inventory count 0), so the cascade reading is an external gloss, recorded and not adopted. (4) Parameter census. P4 declares 8 independent internal configuration parameters $(g, g', h_0, c_W, \Lambda, \varkappa, \beta, \alpha_\Omega)$ with 6 functionally dependent members $(\theta_W, e, \mu_\star^2, m_W, m_Z, m_H)$; P1/A §0.3 admit exactly one free dimensionful scale $\mathfrak{h}$; P3 names $G_N = \mathfrak{h}^2/(4 \ln 2)$, $\kappa^2 = 8\pi G_N$ for external comparison only; P5 forbids identifying any quantity with a measured constant. The SM's 19 parameters (external count: 3 gauge + 1 VEV + 9

Yukawa $+4$ CKM $+1$ θ_{QCD} $+$ 1 Higgs self-coupling) contain 13 flavour parameters, which alone exceed the corpus's 8 — and the corpus's 8 are declared configuration data, not constants, so no parameter-by-parameter map exists even in principle without extending P4.

Req. Def-H.29; D0.8, D0.9, D1.2, D1.11, D3.1, D3.2, D4.1–D4.3; CT0.5–CT0.14 (Part 1); CT1.20, CT1.25, CT2.9 (Part 2); CT3.15, CT3.17, CT3.20, CT3.22 (Part 2); CT4.1–CT4.5 (Part 2); EM.9, EM.20 (Part 2); EW.13, EW.20, EW.24, EW.27, EW.33, EW.35 (Part 2); FT.20, FT.27, FT.30, FT.34, FT.59; A §0.3, P1–P5; external SM parameter count (marked) [machine].

Pf.

$$(1):\ \text{CT0.13 Pf by rung induction on } \vee^2 \,/\, \wedge^2 \text{ population and CT0.7}'\text{s } S\text{-grading;}$$

$$\text{CT0.14}:\ \mathcal{K}_n = \bigoplus_{A\in\mathcal{A}(n)} \mathcal{K}^{(A)}$$

$$2^{n+1} \le 16 \iff n \le 3\ [\text{CT0.5}, \text{CT0.11}]; \qquad O(n) = 2^{n+1} - 16 \implies (16, 48) \text{ at } n = 4, 5\ [\text{CT4.1}]$$

$$(2):\ \text{each row's citations are theorems with Pfs in B–G;}$$

$$n \ge 4:\ \text{CT4.4 is a non-existence theorem}$$

$$(3):\ \text{inventory scans [FT.81(5)]}; \qquad \text{EW.27}:\ m_H^2 = \text{configuration data [P4]}$$

$$(4):\ |P4_{\text{core}}| = 8; \quad 19 = 3+1+9+4+1+1; \quad 9+4 = 13 > 8; \qquad \text{one scale [P1, A §0.3]. } \blacksquare$$

Phys. The rung ladder is the part of this corpus that behaves most like a generative spine, and it is derived rather than asserted: at one crystallized edge per vertex only seals exist; add a second and geometry, the weak block, a scalar and uniform dark come online; at three the electromagnetic read becomes available with a phase coefficient of exactly π; at four edges the vertex oversubscribes, the information is absorbed and locked, and — provably — no gauge structure of any kind survives there. What the ladder never generates is matter's handedness or its flavours: those are not at any rung, and no rung's activation set contains them.

FT.84 — The deficit table and the two-ceiling synthesis (audit pass D). Structural-constraints record: see Appendix B, Section B.4.3 of the Addendum.

Theorem A.19 (FT.96 — The reversal-chirality census of the cover sector). Thm. Under Def-H.28, at (S1)–(S4): (1) Rename (premise correction). "S-parity of a cover class" is undefined: CT1.1's S acts on the pair register $\mathfrak{H}(\mathbb{C}^4)$ (force blocks), not on crystallization patterns. The native class-level chirality is FT.78(6)'s reversal involution $R \mapsto R^{\mathrm{T}}$ — the pattern-level charge conjugation [D2.6], induced by no vertex permutation ($\notin \operatorname{Aut}\Gamma$). E1–E3 are executed for this invariant; the word "bias" has no numeric referent and is dropped. (2) Tetrahedron (re-derived, two enumerators). The 101

covers form 7 Aut-classes (5 bypass $+2$ locked; fresh enumeration matching the stored census orbit-by-orbit); under reversal: 5 self-conjugate $+1$ exchanged pair $\to 6$ reversal classes. The exchanged pair is exactly FT.78(6)'s: the two $\Omega = 24$ dissolved-digon classes with component-transposed type multisets $\big((1,0),(1,1)^2,(1,2)\big) \leftrightarrow \big((0,1),(1,1)^2,(2,1)\big)$. (3) Cube (new, exhaustive on the cover sector). The $11\,205$ covers form 264 classes (166 bypass $+98$ locked; Z-anchors $1\,109\,313/1\,150\,848$ and $\sum \Omega = 11\,205$ re-verified); under reversal:

$$264 \text{ classes } = 42 \text{ self-conjugate } + 111 \text{ exchanged pairs } \longrightarrow 153 \text{ reversal classes,}$$

by sector: bypass $26 + 2 \cdot 70 = 166$, locked $16 + 2 \cdot 41 = 98$; the per-$(\text{sector}, K, |\mathcal{C}|)$ distribution is recorded in the suite output. Every exchanged pair is chirality-blind on all nine stored fields $(K, |\mathcal{C}|, \Omega, \text{amp}, \text{locked}, \text{norm weight}, k_{\max}, \text{maskdim}, \text{cycles})$ — FT.78(6)'s ansatz-invariance at full census scale. (4) E3: no dominant handedness is labelable. Pair members are indistinguishable by every declared read (clause (3)); a label would require a C-odd ledger — FT.78(6)'s own conclusion — and FT.87(8)'s sign(k) applies to materialized chains, which covers do not carry [FT.80(6)]. Classes are exactly: self-conjugate (C-even by construction) or members of an unordered pair. "Left-handed dominant / right-handed dominant" has no referent. (5) E4: negative. The reversal invariant takes exactly two values (self / paired) on both skeletons — not three; the exact counts are 6 (tetra) and 153 (cube) reversal classes. "Three generations = three distinct S-parity bias classes" fails at the invariant level ($2 \neq 3$), and any generation reading is barred regardless [F2; FT.90(6): generations are a missing label, not a capacity limit].

Req. Def-H.25, Def-H.28; FT.67, FT.68, FT.69, FT.78(6), FT.79, FT.80, FT.87(8), FT.90(6), FT.95(3); D2.6; two enumerators (fresh backtracking vs stored census), exact canonical forms under Aut [math; machine].

Pf.

$$(2):\ \text{fresh enumeration} = 101 \to 7 \text{ classes}; \qquad \text{stored rows matched field-by-field [machine]}$$

$$(3):\ \text{canonical form} = \min_{p \in \text{Aut}} \text{sorted edge list of } p(R^{\mathrm{T}}):\ 42 \text{ fixed},\ 111 \text{ 2-cycles};$$

$$42 + 2 \cdot 111 = 264\ \checkmark$$

$$(4), (5):\ \text{clause } (3)\text{'s field identities}; \qquad \#\{\text{invariant values}\} = 2\ \blacksquare$$

Phys. Turn every arrow in the gas around and ask what changed. On the tetrahedron, one pair of shapes trades places and five look at themselves in the mirror; on the cube, a hundred eleven pairs dance and forty-two stand still — a hundred fifty-three reversal classes under the two hundred sixty-four. Nothing that feeds a mass moved anywhere: the mirror pairs are identical in every count the corpus can read. The covers come in handed pairs, and the hands are unnamed — not because the census is incomplete, but because no declared read has ever told left from right. That is the shape of the chirality absence at the census level, and it is now counted exactly.

Theorem A.20 (FT.98 — The polygon form factors: exact census, saturating growth, fifth negative). Thm. On the bidirectional cycles C_n, $n = 3..10$ (literal domain, (S1)–(S4)): (1) Census (two enumerators per n; anchors exact). Backtracking vs covering-polynomial DP agree everywhere; the sign-summed amplitudes reproduce FT.68(2)'s documented pairs $(Z_{\text{byp}}, Z_{\text{lock}})$ exactly for all eight n:

$$(25, 48),$$

$$(353, 64),$$

$$(1161, 80),$$

$$(3121, 1632),$$

$$(11929, 10864),$$

$$(53825, 43136),$$

$$(229417, 159888), \qquad (844881, 775840)$$

; cover counts $(10, 33, 56, 118, 281, 649, 1423, 3166)$ with sector splits $(7{+}3,\ 29{+}4,\ 51{+}5,\ 97{+}21,\ 211{+}70,\ 485{+}164,\ 1051{+}372,\ 2241{+}925)$; $|\mathcal{C}|_{\max} \le \lfloor 3n/4 \rfloor$ (bypass) verified per n: $(1, 2, 3, 3, 4, 4, 5, 6)$. (2) Form factors (tail convention, all-sector, exact Fractions, units $\ln 2$).

n	$M^{(w_U)} = M^{(w_C)} \cdot 2$	$M^{(w_K)}$	$M^{(w_E)}$	$M^{(w_H)}$
3	$\frac{3264}{865}$	$\frac{9504}{865}$	$\frac{96}{865}$	$\frac{3072}{865}$
4	$\frac{24832}{6273}$	$\frac{1408}{153}$	$\frac{1408}{2091}$	$\frac{10240}{6273}$
5	$\frac{281920}{50081}$	$\frac{694240}{50081}$	$\frac{7840}{50081}$	$\frac{135680}{50081}$
6	$\frac{2808192}{407233}$	$\frac{7585344}{407233}$	$\frac{212160}{407233}$	$\frac{2079744}{407233}$
7	$\frac{29210048}{3804641}$	$\frac{74587296}{3804641}$	$\frac{2663136}{3804641}$	$\frac{17679872}{3804641}$
8	$\frac{300392960}{33646849}$	$\frac{767038208}{33646849}$	$\frac{19632384}{33646849}$	$\frac{178147328}{33646849}$
9	$\frac{2833063488}{282265633}$	$\frac{7303749984}{282265633}$	$\frac{16677792}{21712741}$	$\frac{1757108736}{282265633}$
10	$\frac{27649976960}{2481810241}$	$\frac{71155288000}{2481810241}$	$\frac{2115650880}{2481810241}$	$\frac{17046103040}{2481810241}$

with $M^{(w_C)} = M^{(w_U)}/2$ exactly at every n [FT.97(3)]. Per-sector values are tabulated in the suite output. (3) Growth: linear-to-sublinear, saturating. $M^{(w_U)}(C_n)$ rises $3.77 \to 11.14$ over $n = 3..10$ (ratios to P_3: $1.049, 1.492, 1.827, 2.035, 2.366, 2.660, 2.953$) — approximately $\approx 1.1n$, never exponential in n; $M^{(w_K)}$: max ratio 2.609; $M^{(w_H)}$: max 1.934; $M^{(w_E)}$: non-monotone (parity oscillation), max 7.681. (4) B4/D1 verdict: the fifth clean negative. The widest form-factor ratio in the entire computed family is 7.681 (w_E, n=10/3) against [EXT] targets 207 (shortfall $\times 27.0$) and 3477 ($\times 453.9$); for the primary weight w_U the widest is 2.953 (shortfalls $\times 70.1$ / $\times 1177.6$). No aligned triple exists within any factor of 10 (the widest family ratio over all weights and all n is 7.681). The "lepton sector" label is declined [F2]; the [EXT] hierarchies enter only this comparison. (5) D2 answered exactly. The $16^{|\mathcal{C}|}$ weighting does concentrate inside each skeleton

— exact weight fractions at $|\mathcal{C}|_{\max}$: $\frac{768}{865}, \frac{2048}{2091}, \frac{40960}{50081}, \frac{196608}{407233}, \frac{3211264}{3804641}, \frac{16777216}{33646849}, \frac{50331648}{282265633}, \frac{1509949440}{2481810241}$ $(0.888, 0.979, 0.818, 0.483, 0.844, 0.499, 0.178, 0.608$ — parity-oscillating) — but $|\mathcal{C}|_{\max}$ itself grows only linearly ($\leq 3n/4$), so the enhancement saturates at the cap: exponential within, linear across [FT.97(5)]. "Suppress low-node, exponentially enhance high-node" is true intra-skeleton and false as a hierarchy mechanism.

Req. Def-H.25, Def-H.26; FT.67, FT.68(2), FT.69, FT.75, FT.79(2), FT.97; CT0.11 (Part 1); two enumerators (backtracking vs DP), exact Fractions [math; machine]. External targets marked at (4).

Pf.

$$(1):\ \text{enum_covers_bt} = \text{cover_dp coefficients per level;}$$

$$Z = \sum_{\text{level}} c_k 4^k = \text{FT.68(2) row verbatim [machine]}$$

$$(2), (3):\ \text{exact Fraction averages over the enumerated covers;}$$

$$M^{(w_C)} = M^{(w_U)}/2:\ \text{FT.97(3) identity}$$

$$(4):\ \max_{w,n} \frac{M^{(w)}(C_n)}{M^{(w)}(C_3)} = 7.681\ldots;$$

$$207/7.681 = 26.95\ldots;$$

$$(5):\ \text{exact fractions as tabulated } \blacksquare$$

Phys. Run the new mass idea along the polygon ladder and the ladder answers in the same voice as every read before it: the averages climb, but they climb like counting, not like cascading — three and a half units at the triangle, eleven at the decagon, and the steepest weighting scheme in the whole family buys a factor of seven and change. The exponential the proposal hoped for is real, but it lives in the weights, and the weights are capped by the crystallization budget: three units of $|\mathcal{C}|$ per four vertices, forever. Ten-sided polygons are not two hundred times heavier than triangles, and no reweighting of the same finite gas will make them so.

Theorem A.21 (FT.99 — The Platonic form factors, the dominance census, and the reduced-denominator record). Thm. At (S1)–(S4): (1) Tetrahedron (two enumerators: power-set filter of all 4096 patterns vs backtracking — identical 101 covers; 7-class table re-verified [FT.78(2)]; $Z = (1057, 448)$ [FT.68(1)]). Exact form factors (tail, units $\ln 2$):

sector	w_U	w_K	w_C	w_E	w_H
all	$\frac{172800}{35201}$	$\frac{566400}{35201}$	$\frac{86400}{35201}$	$\frac{12672}{35201}$	$\frac{227328}{35201}$
byp	$\frac{62208}{15745}$ [FT.68(1)(*ii*) anchor]	$\frac{27264}{3149}$	$\frac{31104}{15745}$	$\frac{12672}{15745}$	$\frac{18432}{15745}$
lock	$\frac{108}{19}$	$\frac{420}{19}$	$\frac{54}{19}$	0	$\frac{204}{19}$

(the locked $w_E = 0$ is structural: locked tetra covers — double-digon hubs and $K_{1,3}$ stars — have no endpoints at all). Concentration at $|\mathcal{C}|_{\max} = 3$: $16384/35201 = 0.4654$ (the four $K_{1,3}$ stars). (2) Cube (fresh enumeration: 11 205 covers, all distinct; cross-checked against the stored class census: $\sum\Omega = 11\,205$, Ω-histogram identical, $Z = (1\,109\,313,\ 1\,150\,848)$). Exact form factors:

sector	w_U	w_K	w_C	w_E	w_H
all	$\frac{482089472}{53648367}$ (8.986)	$\frac{2241319168}{89413945}$ (25.07)	$\frac{241044736}{53648367}$	$\frac{256135424}{268241835}$	$\frac{2021359616}{268241835}$
byp	$\frac{655450624}{82871211}$ (7.909)	$\frac{485339392}{27623737}$	$\frac{327725312}{82871211}$	$\frac{144068864}{82871211}$	$\frac{202792960}{82871211}$
lock	$\frac{285644}{30171}$ (9.468)	$\frac{285804}{10057}$	$\frac{142822}{30171}$	$\frac{6080}{10057}$	$\frac{887972}{90513}$

Concentration at $|\mathcal{C}|_{\max} = 6$: $67108864/804725505 = 0.0834$ (four covers). (3) C3 comparison against the existing reads. The directive's "single-node aspect-blind masses" are whole-subnetwork reads: $m_{\text{hid}}(\text{tetra floor}) = 12\ln 2/\hbar$, $m_{\text{hid}}(\text{cube floor}) = 24\ln 2/\hbar$ [FT.42: $2E$]; Ansatz-A ladder extremes $\{0..6\}$, $\{0..12\}$ [FT.79(1)]. The form factors sit inside the same $O(V)$ band: $M^{(w_U)}\hbar/\ln 2 = 4.909$ (tetra), 8.986 (cube) — cube/tetra $= 1.831$ against the m_{hid} ratio 2; the widest cube-vs-tetra form-factor ratio over the five weights is 2.652 (w_E, all-sector: $\frac{256135424/268241835}{12672/35201}$ exact; the locked sector is degenerate at w_E — tetra locked $= 0$ by clause (1)) and the primary-weight ratio is 1.831. No new range; no hierarchy. (4) Dominance census (the directive's flag, exact). "Intermediate dominates base" := centre+hub role bits $>$ endpoint role bits (convention-free half-split). Counts: tetrahedron $64/100$ non-vacuum covers ($|A|^2$-weighted fraction $248/275 = 0.902$); cube $10\,052/11\,204$ (weighted $3115352/3143459 = 0.9911$); polygons rising $3/9$ (C_3) $\to$ $2725/3165$ (C_{10}). The amplitude-weighted ensemble is generically centre-dominated — mechanism: an isolated 2-edge thread splits its bits evenly (one centre bit vs two endpoint half-bits), and dominance arises exactly where vertices share roles (types $(1,2),(2,2),\dots$): multi-role vertices accrue centre/hub bits while their partner endpoints accrue fewer. The flag is therefore not an anomaly to hunt but the generic structure of covers at scale. (5) D1 verdict: fifth clean negative, with the exact obstruction. No weight, sector or skeleton in the computed inventory produces ratios within $\times 27$ of the [EXT] 207 or $\times 454$ of 3477; the a priori bound is FT.97(5)'s required-V table ($V \geq 65\dots6561$ by weight, against $V = 4/8$); "baryon sector" declined [F2]. The verdicts of FT.77/FT.79(4)/FT.84(5)/FT.87(9) stand unmodified and now cover the form-factor species as well. (6) Integrity ledger (this pass's catches, recorded pre-ship). (a) The first run double-counted registration bits ($i_v = a{+}b$: $\sum_v \deg = 2K$), producing $124416/15745 = 2\times$ the FT.68(1)(ii) anchor — caught by the anchor, corrected to the tail convention, and promoted to FT.97(2a)'s convention clause (the factor-2 freedom is real and now documented, not silently fixed). (b) the previously published $\langle m_A^2\rangle_{\text{cube}} = 22146188288/268241835$ is in lowest terms: the unreduced census sums are $66438564864/804725505$ (common factor 3) — the values are identical, no erratum needed; the trap (comparing unreduced sums to reduced Fractions) is recorded so it is not re-hit. (c) Dodeca/icosa/octa form factors deferred (bounded rule — cover enumeration scale; recorded, as in FT.78's Status).

Req. Def-H.25, Def-H.28; FT.16, FT.41, FT.42, FT.67, FT.68(1), FT.78, FT.79, FT.80, FT.94(3), FT.97; two enumerators per skeleton; stored cross-checks; exact Fractions [math; machine]. External data marked at (5).

Pf.

$$(1):\ 2^{12} \text{ power-set filter} = \text{backtracking } (101 = 101,\ \text{set-identical});$$

$$M_{\text{byp}}^{(w_U)} = \frac{\sum 16^{|\mathcal{C}|}K}{\sum 16^{|\mathcal{C}|}} = \frac{62208}{15745}\ [\text{machine} = \text{FT.68}(1)(ii)]$$

$$(2):\ \text{fresh } 11205 \text{ distinct covers;}$$

$$\sum \Omega \cdot 16^{|\mathcal{C}|} = 804725505 = 3 \cdot 268241835 \text{ [reduction recorded at } (6b)]$$

$$(3):\ \frac{482089472/53648367}{172800/35201}$$

$$= \frac{482089472 \cdot 35201}{53648367 \cdot 172800} \text{ [machine, exact];}$$

$$(4):\ \text{per-cover role decomposition census}$$

$(5):$ FT.97(5) table; $(6):$ anchor catch + $\gcd(66438564864, 804725505) = 3$ [machine] ■

Phys. Weighed node by node under five honest conventions, the tetrahedron's gas comes out at about five bits and the cube's at about nine — the same band every previous read occupied, the same factor-of-two step between four vertices and eight. The one genuinely new picture the census paints is of who carries the weight: in the cube's amplitude-weighted ensemble, ninety-nine percent of the mass of the average sits in configurations where the threading junctions and hubs outweigh the thread ends — the gas, weighed by its own amplitudes, is almost pure junction. That is a real structural fact about covers at scale. It is not a hierarchy, and the theorem says no amount of reweighing can turn it into one: the average is hostage to its largest summand, and every summand is hostage to the cap.

Theorem A.22 (FT.105 — Well-definedness: the topology census from first principles, finiteness, the regularization). Thm. Under Def-H.32: (1) Polya census. The all-graph counts derived by exact Burnside over S_n acting on edge pairs, $g(n) = \frac{1}{n!}\sum_\pi 2^{\text{cycles}(\pi|_{\text{pairs}})}$:

$$g(1..8) = (1,\ 2,\ 4,\ 11,\ 34,\ 156,\ 1044,\ 12346) \qquad \text{[exact integers; machine].}$$

(2) Connected counts by Euler inversion. From $\prod_k (1-x^k)^{-c(k)} = \sum_n g(n)x^n$ [math: the component decomposition of the ordinary generating function], solved exactly over $\mathbb{Z}$:

$$c(1..8) = (1,\ 1,\ 2,\ 6,\ 21,\ 112,\ 853,\ 11117) \qquad \text{[every } c(n) \text{ integral; machine].}$$

(3) Explicit enumeration (independent path). Canonical-form enumeration over all $2^{\binom{n}{2}}$ labelled edge sets, minimum over S_n, agrees with (1)–(2) at every $n \le 5$ (all-graph totals $1, 2, 4, 11, 34$; connected $1, 1, 2, 6, 21$). $|\mathcal{G}_4| = 10$; the cells by edge count: $E = (0, 1, 2, 3, 3, 3, 4, 4, 5, 6)$. (4) Finiteness and regularization. Per budget cell the configuration space is finite and its size exact: $\sum_{\Gamma \in \mathcal{G}_4} 2^{2E(\Gamma)} = 5845$; every $Z_B^\bullet(q, S)$ is a finite sum of terms in $\mathbb{Z}[t, u]$ — well-defined with no convergence question [FT.68(7) pattern]; the declared regularization is the budget filtration (B, B_K) of Def-H.32(1); the unfiltered limit is not constructed. Growth bookkeeping: g doubles-exponentially in n^2 [(1)'s closed form], c with it — per-budget finiteness is the exact statement, asymptotics are per-family data, not limits. (5) New-cell covering census (exact; two enumerators where affordable). Covering polynomials $(\mathcal{Z}_{\text{byp}}, \mathcal{Z}_{\text{lock}})$ by cover counts per $|\mathcal{C}|$:

cell	#byp covers	#lock covers	note
digon	1	0	vacuum only — the digon itself cancels [FT.74(3)]
P_3	3	1	
C_3	7	3	triangle threads; no closed triangle is a cover [FT.69(3)(iv)]
P_4	7	4	
star$_4$ ($K_{1,3}$)	9	2	
C_4	19	6	the face 4-cycle lies in none of its covers [FT.69(3)(v)]
paw	29	4	
diamond	45	10	
tetra	85 (= 1+24+60)	16 (= 12+4)	re-derived live = frozen FT.68(1) rows

Req. A1, A7; D0.1, D0.8, D0.10; CT0.11 (Part 1); CT1.29 (Part 2); CT4.1 (Part 2); Def-H.25; FT.67, FT.68(7), FT.69(3), FT.74(3); Burnside/Polya enumeration, ordinary-generating-function Euler inversion, canonical forms [math; machine: `checks/verify_H_topology_blind.py` §A–§B].

Pf.

$$(1):\ \text{cycle index of } S_n \text{ on } \binom{n}{2} \text{ pairs, computed per permutation (exact orbit walk);}$$

$$n! \mid \sum_{\pi} 2^{\text{cyc}} \text{ [machine: integrality asserted]}$$

$$(2):\ n\,g(n) = \sum_{k\le n} b(k)g(n-k),\ b(k) = \sum_{d\mid k} d\,c(d):\ \text{exact integer recursion solved upward;}$$

$$c(n) \in \mathbb{Z}\ \forall n \le 8 \text{ [machine]}$$

$$(3):\ \text{adjacency-bitmask canonical minimum over } S_n;$$

$$\text{counts} = (1,1,2,6,21)/(1,2,4,11,34) \text{ [machine]}$$

$$(4):\ |\mathfrak{C}(\Gamma)| = 2^{2E(\Gamma)};$$

$$\textstyle\sum_{\Gamma\in\mathcal{G}_4} 2^{2E} = 2^0+2^2+2^4+2^6+2^6+2^6+2^8+2^8+2^{10}+2^{12} = 5845 \text{ [machine]}$$

$$(5):\ \text{cover-DP} = \text{branch enumeration on every cell with } 2E \le 8 \text{ [machine, two enumerators];}$$

$$\text{tetra rows} = \text{FT.68}(1) \text{ verbatim. } \blacksquare$$

Phys. The integration variable "topology" is now an exactly enumerated space with a stated cutoff: ten connected cells at four nodes, thirty-one at five, one hundred twenty-five at six — counts

derived from Burnside and Euler, not quoted. Nothing converges because nothing needs to: every budget cell is a finite exact sum, and the filtration by the ledger's own nat content is the regularization. No particle identification; topology names are bookkeeping.

Theorem A.23 (FT.114 — The [EXT] audit clause (FT.73(5) pattern)). Thm. External-data clause. The hierarchy

$$1 : 206.768 : 3477.2 \qquad \text{[EXT: supplied measured charged-lepton mass ratios]}$$

is supplied measured data. It enters no derivation, no definition, and no coefficient of this document; every number in FT.109–FT.113 is fixed without it. This document identifies no particles. The exact shortfalls of the supplied hierarchy against the derived ratios [FT.113] are

$$\text{shortfall}_{\text{mid}} = \frac{206.768}{r_{\text{mid}}^2/r_{\text{min}}^2} = \frac{25846/125}{r_2^2/r_1^2} = 0.999988799297817\ldots,$$

$$\text{shortfall}_{\text{out}} = \frac{3477.2}{r_{\text{max}}^2/r_{\text{min}}^2} = \frac{17386/5}{r_0^2/r_1^2} = 0.999921541556935\ldots$$

$$\text{[exact rational division of enclosed ratios; widths } < 2^{-50}\text{; machine]},$$

and the recorded verdicts are:

(1) The computed ratios do NOT match [EXT] to [EXT]'s stated precision (mid: 6 significant figures; out: 5): the gaps are enclosure-provably nonzero — $r_2^2/r_1^2 - 206.768 > \frac{1}{1000}$,

$$r_0^2/r_1^2 - 3477.2 > \tfrac{1}{10};$$

(2) both shortfalls are strictly below 1 and above $1 - 2\times10^{-4}$: the derived ratios exceed the supplied values by relative margins below 2×10^{-4} ($1 - \text{shortfall}_{\text{mid}} \in (10^{-5}, 2\times10^{-5})$, $1 - \text{shortfall}_{\text{out}} \in (7\times10^{-5}, 9\times10^{-5})$);

(3) the numerical coincidence and the structural origin are reported separately: the coincidence is the proximity recorded in (2); the structural origin of the ratios is the normalization data of FT.111–FT.113 — the diagonal coefficient 2, the seal dimension 3, the transfer trace 3 — which contain no [EXT] input. No claim is made that the derivation reproduces the observed hierarchy. No identification of the three eigenvalues with any particle species is made or implied.

Req. FT.113; FT.73(5) (Tiers F.23–F.25, the audit pattern); P1, P5 (A: no quantity identified with a measured external constant; comparison is a test, never an input); exact rational interval division [math; machine: `checks/verify_H_koide_channel.py` §G]. [EXT] data clause: the supplied values enter this block and no other block of this document (machine-scanned, §H).

Pf.

$$206.768 = \tfrac{206768}{1000} = \tfrac{25846}{125},$$

$$3477.2 = \tfrac{34772}{10} = \tfrac{17386}{5} \text{ [exact Fractions in lowest terms]}$$

shortfall

$$= \tfrac{[\mathrm{EXT}]}{q},$$

q the enclosed ratio of FT.113 : monotone exact interval division by positive enclosures $\big[\tfrac{[\mathrm{EXT}]}{q_{\mathrm{hi}}}, \tfrac{[\mathrm{EXT}]}{q_{\mathrm{lo}}}\big]$;

$$\text{widths} < 2^{-50} \text{ [machine §G]}$$

$$(1) : \ 206.768 < \mathrm{lo}\big(r_2^2/r_1^2\big),$$

$$3477.2 < \mathrm{lo}\big(r_0^2/r_1^2\big) \text{ [exact rational comparison against disjoint enclosures]}$$

$$(2) : \ \mathrm{hi}(\text{shortfall}) < 1;$$

$$1 - \text{shortfall} \in \big(10^{-5}, 2\times10^{-5}\big) \cup \big(7\times10^{-5}, 9\times10^{-5}\big) \text{ [enclosure-proven, machine]}$$

$$(3) : \ \text{the tokens } 206.768,$$

3477.2 occur only inside this block, and [EXT]-marked content enters no Req, Pf, or coefficient of FT.109–FT.113 [machine scan §H; FT.73(5) discipline: recorded, not adopted]. ■

Phys. The audit states exactly what is on record and nothing more: a three-line dimensionless hierarchy built from seal integers lands within two parts in 10^4 of a supplied three-line measured hierarchy, misses it at the measurement's own stated precision, carries no overall scale, and identifies nothing. The near-agreement is a numerical fact about rationals and cosines; the construction remains data-free by its own rules.

Theorem A.24 (FT.120 — The [EXT] audit ledger: scheme columns, exact shortfalls, and the two exclusions). Thm. External-data clause (FT.73(5)/FT.114 pattern). The heavy triplet of the up-type and down-type sectors, in three renormalization-scheme columns,

$$\text{S1 (own-scale } \overline{\mathrm{MS}}) : \ (1.273,\ 4.183,\ 162.5) \text{ GeV};$$

$$\text{S2 } (\overline{\mathrm{MS}} \text{ plus pole top}) : \ (1.273,\ 4.183,\ 172.4) \text{ GeV};$$

$$\text{S3 (pole-type)} : \ (1.67,\ 4.78,\ 172.4) \text{ GeV},$$

$$\text{with ratio box} \quad m_u/m_d = 0.462 \pm 0.020,$$

$$m_s/m_d \in [17, 22],$$

$$m_\pi^0 = 134.9768 \text{ MeV}$$

[EXT] supplied measured data,

is supplied measured data [EXT]. It enters no derivation, no definition, and no coefficient of this document; every number in FT.116–FT.119 is fixed without it. This document identifies no particles; the correspondence audited below is a test, never an input [P1, P5]. The exact shortfall factors of the supplied hierarchies [EXT] against the derived ratios [FT.119(5)], by monotone rational interval division at widths $< 2^{-50}$, are

$$(1) \quad \text{S1}: \ \frac{m_2}{m_1}$$

$$= 3.285938\ldots,$$

$$\frac{m_3}{m_1}$$

$$= 127.6512\ldots \implies \text{shortfall}_{\text{mid}}$$

$$= 0.922058\ldots,$$

$$\text{shortfall}_{\text{out}} \qquad = 0.916122\ldots;$$

$$\text{S2}: \ \frac{m_3}{m_1} = 135.4281\ldots \implies \text{shortfall}_{\text{mid}} = 0.922058\ldots, \quad \text{shortfall}_{\text{out}} = 0.971934\ldots;$$

$$\text{S3}: \ (2.862275\ldots,\ 103.2335\ldots) \implies (0.803176\ldots,\ 0.740881\ldots);$$

(2) all six shortfalls are strictly below 1 and enclosure-provably nonzero: the derived ratios exceed the supplied ones in every scheme column;

(3) the channel invariant:

$$Q = \frac{\sum_i m_i}{\left(\sum_i \sqrt{m_i}\right)^2}: \quad \text{S1}\ 0.662600\ldots, \quad \text{S2}\ 0.669113\ldots, \quad \text{S3}\ 0.648359\ldots,$$

$$\text{equivalently} \quad \frac{B^2}{A^2} = 6Q - 2: \quad \text{S1}\ 1.97560\ldots, \quad \text{S2}\ 2.01468\ldots, \quad \text{S3}\ 1.89015\ldots$$

against the derived normalization value 2 [FT.111(3)];

$$|$$

$$Q - \tfrac{2}{3}$$

$| < 0.01$ in S1 and S2 (and $0.01 < |\,Q - \tfrac{2}{3}\,| < 0.03$ in S3) : the heavy triplet sits on the channel's normalization surface at the 1% level in the $\overline{\text{MS}}$ columns;

(4) anchored rows [FT.119(6)] :

$$\left.\frac{m_2}{m_3}\right|_{\text{derived}}$$

$$= 0.025575776\ldots \text{ vs } \frac{4.183}{162.5}$$

$= 0.025741\ldots$ [EXT] S1 : relative margin $0.0065 < 0.01$ (agreement within 1%);

$$\left.\frac{m_1}{m_3}\right|_{\text{derived}}$$

$$= 0.007176757\ldots \text{ vs } \frac{1.273}{162.5}$$

$= 0.007833\ldots$ [EXT] S1 : relative margin $0.0838 > 0.05$ (the inner member is the outlier);

(5) no scheme column reproduces both derived ratios at the 1% level [machine: exhaustive over the three columns];

(6) obstruction O3 (light triplet, excluded): over the whole supplied ratio box m_u/m_d

$$\in [0.442, 0.482],$$

$$m_s/m_d \qquad \in [17, 22],$$

$$Q_{\text{light}} \in (0.5461\ldots,\ 0.5804\ldots) \subset (0.54, 0.59),$$

$$Q_{\text{light}} < \frac{3}{5} < \frac{2}{3},$$

$$\frac{B^2}{A^2} = 6Q_{\text{light}} - 2 \in (1.25, 1.5) \text{ versus the derived } 2;$$

extrema attained at box corners [$\partial Q/\partial\sqrt{r_s} > 0,\ \partial Q/\partial\sqrt{r_u} < 0$ on the box; machine-guarded] : the (A, B)

$= (1, \sqrt{2})$ normalization is EXCLUDED for the light triplet at every phase δ [FT.119(4) : Q is phase-free];

(7) obstruction O4 (hadron range, excluded): the established hadron mass span $\leq \dfrac{11000}{134.9768}$

$= 81.495\ldots < 82$ [EXT] is below the derived colour-channel outer ratio $139.3386880284\ldots$ [FT.119(5)] and far below FT.113's $3477.472837104\ldots$;

hence no baryon or meson triplet spans the channel's dynamic range — the second, independent exclusion of the hadron-level read, the first being FT.117(7)'s blindness theorem;

(8) the dimension trap, disclosed and rejected [FT.77(3) pattern] : the literal label $m \propto r$ (unsquared) gives $1 : 14.3795\ldots : 58.9701\ldots$ on the δ_ℓ

$= \frac{2}{9}$ spectrum — missing the supplied mid ratio 206.768 [EXT] by $> 14\times$,

while the r^2 read reproduces it within 1% $(206.770315972\ldots)$ [EXT];

the same literal label on the colour channel gives $r_2^{(q)}/r_1^{(q)}$

$= 1.887775\ldots$ against $3.285938\ldots$ $(1.74\times$ off) : the squared read is the only one that survives either audit;

(9) provenance: the structural origin of the derived ratios is the normalization data of FT.116–FT.119 — the diagonal coefficient 2,

the seal dimension 3,

the transfer trace 3,

the colour count 3 — which contain no [EXT] input. The numerical proximity recorded in (1)–(4) and the structural origin are reported separately. No claim is made that the derivation reproduces the observed hierarchy; no identification of the three eigenvalues with any particle species is made or implied.

Req. FT.119; FT.113, FT.114 (Tier F.37, the audit pattern and the lepton-side decimals); FT.73(5) (Tiers F.23–F.25), FT.77(3) (Tiers F.26–F.28), FT.107 (Tier F.36) for the audit-pattern and read-vocabulary precedents; P1, P5 (A: no quantity identified with a measured external constant; comparison is a test, never an input); exact rational interval division, exact rational square-root enclosures, corner-monotonicity guards [math; machine: `checks/verify_H_colour_channel.py` §E]; floating-point cross-check of the same ledger: `checks/audit_heavy_quark_scheme.py`. [EXT] data clause: the supplied values enter this block and no other block of this document (machine-scanned, §F).

Pf.

$$(1):\ \text{S1 mid}$$

$$= \tfrac{4.183}{1.273}$$

$$= \tfrac{4183}{1273},$$

$$\text{S1 out}$$

$$= \tfrac{162.5}{1.273}$$

$$= \tfrac{16250}{1273};$$

$$\text{S2 out}$$

$$= \tfrac{172.4}{1.273};$$

$$\text{S3}$$

$$= \tfrac{4.78}{1.67},$$

$$\tfrac{172.4}{1.67}\ \text{[exact Fractions in lowest terms]}$$

$$\text{shortfall}$$

$$= \tfrac{\text{[EXT]}}{q},$$

q the enclosed ratio of FT.119(5) : monotone exact interval division $\left[\tfrac{\text{[EXT]}}{q_{\text{hi}}}, \tfrac{\text{[EXT]}}{q_{\text{lo}}}\right]$;

$$\text{widths} < 2^{-50}\ [\text{machine}]$$

$$(2):\ \text{hi}(\text{shortfall}) < 1 \text{ in all six cases;}$$

$$\text{nonzero gaps: } q_{\text{lo}} > [\text{EXT}]\ [\text{disjoint rational comparison; machine}]$$

$$(3):\ Q = \tfrac{\sum m}{(\sum \sqrt{m})^2} \text{ with } \sqrt{m} \text{ enclosed by integer Newton certificates at } 2^{-60};\ \text{widths} < 2^{-40};$$

$$\tfrac{B^2}{A^2} = 6Q - 2\ [\text{exact affine read of FT.111(3)'s } \textstyle\sum(\lambda - A)^2 = \tfrac{3}{2}B^2 \text{ and } \sum \lambda = 3A]$$

$$\text{windows : S1, S2} \in (\tfrac{2}{3} - \tfrac{1}{100}, \tfrac{2}{3} + \tfrac{1}{100});$$

$$\text{S3} \in (\tfrac{2}{3} + \tfrac{1}{100}, \tfrac{2}{3} + \tfrac{3}{100}) \cup (\tfrac{2}{3} - \tfrac{3}{100}, \tfrac{2}{3} - \tfrac{1}{100})\ [\text{machine}]$$

$$(4):\ \tfrac{4.183}{162.5}$$

$$= \tfrac{4183}{162500};$$

$$\left|\tfrac{4183}{162500} - \tfrac{(r_2^{(q)})^2}{(r_0^{(q)})^2}\right| / \tfrac{4183}{162500} < \tfrac{1}{100};$$

$$\tfrac{1.273}{162.5} = \tfrac{1273}{162500};\quad \text{relative margin} > \tfrac{5}{100}\ [\text{machine}]$$

$$(5):\ \neg\big(\text{lo}(\text{shortfall}_{\text{mid}}) > \tfrac{99}{100} \wedge \text{lo}(\text{shortfall}_{\text{out}}) > \tfrac{99}{100}\big) \text{ in each column [machine, exhaustive]}$$

$$(6):\ Q(r_u, r_s)$$

$$= \tfrac{r_u + 1 + r_s}{(\sqrt{r_u} + 1 + \sqrt{r_s})^2} \text{ with } r_u$$

$$= m_u/m_d,$$

$$r_s$$

$$= m_s/m_d;$$

$$\partial_{\sqrt{r_s}} Q > 0 \iff \sqrt{r_s}(1+\sqrt{r_u}) > 1 + r_u \text{ (true on the box);}$$

$$\partial_{\sqrt{r_u}} Q < 0 \iff \sqrt{r_u}(1+\sqrt{r_s}) < 1 + r_s \text{ (true on the box)}$$

$$\implies \text{extrema at corners : } Q_{\min} = Q(\tfrac{482}{1000}, 17),\ Q_{\max} = Q(\tfrac{442}{1000}, 22);$$

$$\tfrac{54}{100} < Q_{\min} \le Q_{\text{light}} \le Q_{\max} < \tfrac{59}{100} < \tfrac{3}{5} < \tfrac{2}{3} \text{ [exact enclosures; machine]}$$

$$(7) : \ \tfrac{11000}{134.9768}$$

$$= \tfrac{110000000}{1349768}$$

$$= 81.495\ldots < 82 < 139.3386880284\ldots \text{ [exact rational comparison against FT.119(5)'s enclosure; machine]}$$

$$(8) : \ \tfrac{l_2}{l_1}$$

$$= 14.3795\ldots,$$

$$\tfrac{l_0}{l_1}$$

$$= 58.9701\ldots \text{ on the } \delta_\ell \text{ spectrum [FT.113 enclosures re-derived; machine];}$$

$$\tfrac{206.768}{14.3795\ldots} > 14;$$

$$0.99 < \tfrac{206.768}{206.770315972\ldots} < 1.01;$$

$$\tfrac{r_2^{(q)}}{r_1^{(q)}} = 1.887775\ldots < 2 < 3.285938\ldots$$

$$(9) : \text{ the tokens } 1.273,$$

$$4.183,$$

$$162.5,$$

$$172.4,$$

$$1.67,$$

$$4.78,$$

$$0.462,$$

134.9768 occur only inside this block, and no [EXT]-marked content enters any Req, Pf, or coefficient of FT.116–FT.119 [machine scan §F]. ■

Phys. The ledger states what is on record and nothing further. A three-line dimensionless hierarchy built from four integers of the frozen algebra — the seal coefficient 2 and the three factors $3, 3, 3$ of the dressed arena — has its invariant reading Q landing within one percent of the supplied [EXT] heavy-triplet value in the two running-mass columns, and its top-anchored inner ratio landing within seven parts in a thousand of the supplied own-scale column. It misses both supplied ratios at their own precision, in every column, by an enclosure-provably nonzero margin; the pole column misses by twenty to twenty-six percent. Two branches of the docket are closed by theorem rather than by taste: the light triplet cannot sit on this channel at any phase, because its Q is bounded below $\frac{3}{5}$ across the entire supplied ratio box while the channel's normalization forces exactly $\frac{2}{3}$; and no hadron triplet can reach the channel's range, because the established hadron spectrum spans less than a factor 82 against the channel's 139 — the same verdict FT.117 reached structurally. The squared-mass label is the only one that survives either side of the dimension audit. The near-agreement is a numerical fact about rationals and cosines; the construction remains data-free by its own rules.

A.2 §Consumption — the cell-consumption block

Preamble (census supply from Category-A tags): the cover-polynomial tables of FT.68, the frame catalogue of FT.93, and the susceptibility census of FT.103 are census sides of structural tags; Part 4 consumes them through those tags' own rows — no B-classification is implied. No census entry is dismissed: each is a load-bearing cell supply (separation charter).

FT.28. Space: the two-vertex digon floor Σ_{min} with a single observer contact; registration data (σ_e, κ)). Counts: the complete floor card: $I_{\text{hid}} = 2 \ln 2$, $\Theta_{\text{int}} = 2\kappa$, winding, holonomy, aspect rows, mask dimension ≤ 10, rung counts, the ternary core read. Baryonic: the minimal fermion cell and its card grammar — every sealed composite card (baryon-type junctions included) refines this row template; the digon is the junction kernel of the composition laws.

FT.32. Space: the single-contact cycle floors Σ_L (unidirectional L-cycles). Counts: the ladder $m_{\text{hid}}(L) = L \ln 2/\mathfrak{h}$, the minimality regimes, structural colourlessness per L. Baryonic: the winding ladder baryonic chains compose over; additive winding composition [FT.34(3)] consumes these cells as summands.

FT.35. Space: the composite floor Σ_{12} (first colour-active, first lock-carrying). Counts: the junction lock content, the baryon-type sealing phase, the maximal 18-dimensional mask. Baryonic: the

first computed colour-active cell — the junction grammar baryon seals run, and the D^--carrying cell for the $n \geq 4$ lock side.

FT.42. Space: the five Platonic skeletons under the uniform-minimal materialization scope (M1/M2). Counts: the card template per solid: $E_{\text{int}} = 2E$, m_{hid}, Π, winding and lock profiles. Baryonic: the symmetric-skeleton cell pool; the tetrahedron's trivalent vertices are the native baryon-junction site.

FT.43. Space: per-solid winding-ledger states with Euler consistency. Counts: $F - 1$ independent winding reads, degeneracies, per-face winding assignments. Baryonic: the winding bookkeeping sealed baryonic chains inherit; composition adds windings cell-wise.

FT.44. Space: the computed floors under solid duality. Counts: dual pairs sharing m_{hid}/Π, split by winding/sealing/lock profiles; the taxonomy rows across every computed floor. Baryonic: the degeneracy-splitting grammar — how baryonic multiplets sharing a skeleton are told apart by ledger data.

FT.45. Space: the full regular-polygon floor family Σ_{P_n}, $n \geq 3$. Counts: walk-space rank $n + 1$, the single parity constraint, four minimality regimes, injective mass/rank/null reads per n. Baryonic: the unbounded cell family for chain composition and the polygon-limit anchor of the flicker boundaries [FT.70(3)].

FT.49. Space: the tetrahedron-face floor with baryon-type sealing. Counts: the full card: $\Theta_{\text{int}} \in 4\mathbb{Z}$, the 30-dimensional exchange sector, the sealing grammar. Baryonic: the reference baryon-type sealing-phase cell — its junction structure is the template for ε-sealed triplet reads.

FT.50. Space: patch cells (bowtie, double-triangle) and floor pairs under gluing. Counts: junction-saturation data, the constraint-rank gluing law, glued-card rows. Baryonic: the composition laws by which multi-face baryonic skeletons are assembled from computed cells.

FT.51. Space: closed-walk winding ledgers per skeleton under dihedral symmetry. Counts: closed-form enumerations with degeneracy counts. Baryonic: the degeneracy pool for the sector labels (q, S) of conditioned baryonic path integrals [Def-H.32(c) grammar].

FT.63. Space: the five Platonic floors with all physical read columns. Counts: spin (under Def-H.38), charge, parity, holonomy, entropy, mass ratios, binding — per solid; $j_{\max} \in \{6, 6, 12, 30, 30\}$. Baryonic: the per-solid read table baryonic multiplets consume as constituent data on symmetric skeletons.

FT.64. Space: admissible-loop lattices per solid under the filter chain. Counts: the generalized loop lists $A_S(K)$ with the exact character-sum closed form. Baryonic: the loop pool over which baryonic cover cells are built; the filter chain is the admissibility grammar.

FT.65. Space: coarse-grained loop classes under the full automorphism groups. Counts: the degeneracies $M_S(K)/M_S^+(K)$ and the register product $2^{2E}M_S(K)g(V)$. Baryonic: the register-counting side of sealed-object enumeration — multiplicity bookkeeping for baryonic state counts.

FT.69. Space: Platonic cover patterns with amplitude weights. Counts: the completed amplitude census (cover polynomials; the $62208/15745$ anchor), the cycle-alternation rows. Baryonic: the amplitude-layer data for flicker ensembles on symmetric skeletons; the norm/amplitude dichotomy [FT.68(3)] consumes these values.

FT.76. Space: Platonic defect-angle pole reads under both winding-pairing variants (A/B). Counts: the multiplet tables, the ceiling $[0, 4\hat{\mu}'^2]$, the four Θ_R readings priced. Baryonic: the pole-side mass cells and their range taxonomy — the hard ceiling every baryonic pole read inherits.

FT.78. Space: tetrahedron and cube cover configurations under Aut. Counts: tetrahedron: 70 admissible / 7 cover classes; cube: 33 564 admissible by K / 264 cover classes with $|\mathcal{C}|$-levels, signatures, chirality pairs. Baryonic: the cover-class pool baryonic sector-conditioning runs over; the

exact enumeration substrate for cube-era cells.

FT.81. Space: the corpus read classes against the external-reference content rows. Counts: presence/absence per read class (charge, spin, winding, sealing, chirality, flavour) — the internal read-class inventory (external scoring columns excluded). Baryonic: the read-class grammar every baryonic card fills, and the audit template for P1/P5-compliant external comparison.

FT.83. Space: the rungs $n = 0, 1, 2, 3, \geq 4$ with their activation sets. Counts: derived/declared/open status per rung; the $n \geq 4$ derived no-go row. Baryonic: the activation map Part 4 extends at $n = 3$ (colour, derived) and $n \geq 4$ (the lock side).

FT.84. Space: the five mass-read species with their ranges. Counts: the ceiling taxonomy: two hard ceilings, two size-scaling species, one undeclared-fugacity direction (external scoring excluded). Baryonic: the range constraints any baryonic mass read inherits, and the species taxonomy new reads must join.

FT.96. Space: tetra/cube cover orbits under the orientation reversal. Counts: the chiral cover pairs (the two $\Omega = 24$ orbits exchanged), reversal signatures per orbit. Baryonic: the chirality-tagged cover cells parity reads [FT.137] consume for sealed objects.

FT.98. Space: polygon skeletons under the multi-node form-factor species. Counts: exact M-tables, saturating growth rows, reduction-identity anchors. Baryonic: form-factor range data for extended baryonic skeletons (the linear size-bound species).

FT.99. Space: Platonic skeletons under the form-factor reads. Counts: the dominance census, reduced-denominator record, concentration fractions. Baryonic: the symmetric-skeleton form-factor cells for baryonic multiplet range checks.

FT.105. Space: connected graphs and canonical skeletons to $V \leq 8$ (canonical enumeration to $V \leq 5$). Counts: exact Burnside/Polya counts $g(n)$, Euler inversion $c(n)$, finiteness and regularization cells per budget. Baryonic: the first-principles topology pool over which the topology-blind baryonic integral is regulated (budget filtration).

FT.114. Space: the supplied external hierarchy against the derived ℓ-channel ratios. Counts: exact shortfall arithmetic on the quarantined rows. Baryonic: the audit-discipline template for Part-4 external tests (P1/P5-compliant comparison grammar).

FT.120. Space: renormalization-scheme columns against the derived q-channel ratios; the light-triplet Q-box. Counts: the scheme ledger, exact shortfalls, the two derived exclusions. Baryonic: the scheme-dependence record Part-4 comparisons inherit — no canonical renormalization point is derivable in the corpus.

Index of Corpus Tags

The complete tag registry, with a one-line description of every tag: 623 registry tags (481 theorem and data-cell tags, 115 definitions, 9 axioms, 13 conventions, 5 calibration-policy clauses) plus the adjudication-record tags of HistLog §IV reproduced in the Addendum. Rows are grouped by document, in tier order; "Presentation" gives the theorem/definition number in this volume or the Addendum section (B.$x.y$) for relocated records.

Tag	Class	Title / description	Where	Presentation
Part 1 · Qubit_era — axioms, conventions, definitions, calibration, pair canon, ledger dynamics declarations				
A1	axiom	—	Part 1	Axiom 1.1
A2	axiom	—	Part 1	Axiom 1.2
A3	axiom	—	Part 1	Axiom 1.3
A4	axiom	—	Part 1	Axiom 1.4
A5	axiom	—	Part 1	Axiom 1.5
A6	axiom	—	Part 1	Axiom 1.6
A7	axiom	—	Part 1	Axiom 1.7
A8	axiom	—	Part 1	Axiom 1.8
A9	axiom	—	Part 1	Axiom 1.9
C1	convention	Pauli basis / emergent spacetime coordinates where a metric exists; : spatial/Pauli-spatial; : colour adjoint....	Part 1	Convention 1.1
C10	convention	Register accounting (reality principle): registrations convert quantum data to classical alternatives at the l...	Part 1	Convention 1.10
C11	convention	A5/A6 reading: the conjugation clause is node-level — [A6:] with and (: well-typed); is the vacuum pair stat...	Part 1	Convention 1.11
C12	convention	In A8's the summation domain reads (D0.10): the configuration set on the real-edge set. itself (A4, D0.1) is a...	Part 1	Convention 1.12
C13	convention	Signature convention. A3 fixes only through the products , and fixes only up to its odd slot and its overall s...	Part 1	Convention 1.13
C2	convention	—	Part 1	Convention 1.2
C3	convention	Symbol : antisymmetrization only (, - contractions); a sector-label conjunct	Part 1	Convention 1.3
C4	convention	sector-to-metric embedding scale (D2.1); : gravitational coupling (D2.9, CT2.9); ; interchangeable	Part 1	Convention 1.4
C5	convention	(operator); (scalar, D2.1); interchangeable	Part 1	Convention 1.5

Tag	Class	Title / description	Where	Presentation
C6	convention	transit entropy; : bound entropy (D1.5); squared-field notation for either	Part 1	Convention 1.6
C7	convention	(A8). Rung-resolved: (D2.3)	Part 1	Convention 1.7
C8	convention	neutral-weak aspect of A8; ; pre-mixing seed generator: . Bare in a spectrum: physical neutral eigenstate (D3....	Part 1	Convention 1.8
C9	convention	Entropy in nats: ; _bits_ only where explicit. One real edge bit nats (CT0.9, CT1.29)	Part 1	Convention 1.9
Tier 0 — Bookkeeping, Reality, and Capacity				
CT0.1	theorem	Real/virtual dichotomy	Part 1	Theorem 1.1
CT0.10	theorem	The two sign structures are independent	Part 1	Theorem 1.10
CT0.11	theorem	Maximal accessible rung	Part 1	Theorem 1.11
CT0.11a	theorem	three slots	Part 1	Corollary 1.3
CT0.11b	theorem	directed rung of bidirectional skeletons	Part 1	Corollary 1.4
CT0.12	theorem	Aspect–capacity match	Part 1	Theorem 1.12
CT0.13	theorem	Aspect activation and the active sets	Part 1	Theorem 1.13
CT0.14	theorem	Mask flag and block dimensions	Part 1	Theorem 1.14
CT0.2	theorem	Global vacuum	Part 1	Theorem 1.2
CT0.3	theorem	Seal increments	Part 1	Theorem 1.3
CT0.4	theorem	Real outgoing edge registration	Part 1	Theorem 1.4
CT0.5	theorem	Tick load at rung	Part 1	Theorem 1.5
CT0.6	theorem	Channel capacity and oversubscription	Part 1	Theorem 1.6
CT0.6a	theorem	transit-hop register bound	Part 1	Corollary 1.1
CT0.7	theorem	Reality is the swap grading	Part 1	Theorem 1.7
CT0.7a	theorem	spin content	Part 1	Corollary 1.2
CT0.8	theorem	Admissible range of the crystallization threshold	Part 1	Theorem 1.8
CT0.9	theorem	The edge phase alphabet is binary	Part 1	Theorem 1.9
Tier 1A — The Pair Operator Algebra				
CT1.0	theorem	Signature classification of the -grading	Part 1	Theorem 1.15
CT1.1	theorem	Joint decomposition of	Part 1	Theorem 1.16
CT1.10	theorem	decomposition of	Part 1	Theorem 1.25
CT1.11	theorem	The two scalars are distinct	Part 1	Theorem 1.26
CT1.11a	theorem	The involution table of the four sectors	Part 1	Theorem 1.27
CT1.11b	theorem	The native Clifford frames of the pair register and the weak-covariance pincer	Part 1	Theorem 1.28
CT1.2	theorem	Generator basis and sector assignment	Part 1	Theorem 1.17
CT1.3	theorem	Centrality of the identity	Part 1	Theorem 1.18
CT1.4	theorem	The close	Part 1	Theorem 1.19
CT1.5	theorem	Commutators of with	Part 1	Theorem 1.20
CT1.6	theorem	Commutators and	Part 1	Theorem 1.21
CT1.7	theorem	Mixed commutators	Part 1	Theorem 1.22
CT1.8	theorem	Action of on and	Part 1	Theorem 1.23
CT1.9	theorem	Sector closure table and the global algebra	Part 1	Theorem 1.24
declarations				
D0.1	definition	global vacuum	Part 1	Definition 1.1
D0.10	definition	configuration set on : finite oriented edge sets with sign assignments , seal data (D1.2) and vertex types (A7...	Part 1	Definition 1.10

Tag	Class	Title / description	Where	Presentation
D0.2	definition	—	Part 1	Definition 1.2
D0.3	definition	(faithful). [A4, D0.4, CT0.8]; for . Gibbs tilt (functional calculus): where ; on states: ; the identity on t…	Part 1	Definition 1.3
D0.4	definition	threshold of A4; admissible range: CT0.8; sharp choice	Part 1	Definition 1.4
D0.5	definition	phase alphabet of a real edge (CT0.9:)	Part 1	Definition 1.5
D0.6	definition	hop one traversal of one ; ; length assigned to one hop	Part 1	Definition 1.6
D0.7	definition	ordered record of retained events [A6, A7: stratification , , monotone under ticks]; ledger ticks (retained up…	Part 1	Definition 1.7
D0.8	definition	(vertex set); (type set); “at rung ” on	Part 1	Definition 1.8
D0.9	definition	independent real registers a vertex of pushes across one pair channel per tick — counted by the register accou…	Part 1	Definition 1.9
D1.1	definition	rung class of ; D0.5; registration . Reality principle of registration (C10): phase of the registration transp…	Part 1	Definition 1.11
D1.10	definition	node force: resolves local information content into channels bound, transit, holonomy, charge, flux ; bound, t…	Part 1	Definition 1.20
D1.11	definition	accessible information content at ; content beyond the capacity bound (CT0.11): accessible [D4.1]	Part 1	Definition 1.21
D1.12	definition	aspects of acting nontrivially at rung ; value derived (CT0.13), stipulated	Part 1	Definition 1.22
D1.2	definition	seal transition of one pair across the A4 criterion (entry/exit of either branch): (crystallization); (dissolu…	Part 1	Definition 1.12
D1.3	definition	Janus-balanced	Part 1	Definition 1.13
D1.4	definition	chain : , ; ; ; (formal difference of vertices); closed	Part 1	Definition 1.14
D1.5	definition	bound closed self-contained chain (content: fermionic); transit propagating open chain (content: bosonic)	Part 1	Definition 1.15
D1.6	definition	[D2.1]	Part 1	Definition 1.16
D1.7	definition	[per-edge phase, D3.5; uniform-phase case:]; ; ; wrapping spatial hop projection has winding around a closed…	Part 1	Definition 1.17
D1.8	definition	accumulated along a causal chain of real edges; emergent coordinate	Part 1	Definition 1.18
D1.9	definition	tick locality: , pre- and post-tick data only; read at edge-distance	Part 1	Definition 1.19
D2.1	definition	sector coefficients [basis: CT1.2]	Part 1	Definition 1.23
D2.10	definition	even blocks ; odd blocks ; Hermitian ; ; (homogeneous)	Part 1	Definition 1.32

Tag	Class	Title / description	Where	Presentation
D2.11	definition	valid universe real-edge network closed [D1.3] active mask dangling ledger boundary beyond the sealed Janus bo…	Part 1	Definition 1.33
D2.12	definition	Aspect realization (slot aspect bridge): — linear realization of the rung- slot algebra in aspect space, with …	Part 1	Definition 1.34
D2.13	definition	Subnetwork and microstate space *(scope clauses)*. [D2.11] a finite connected vertex set together with its int…	Part 1	Definition 1.35
D2.14	definition	Colour boundary and the contraction-completeness projector. ; for valid , the other members of the elementary …	Part 1	Definition 1.36
D2.2	definition	—	Part 1	Definition 1.24
D2.3	definition	aspect space; block of aspect ; ; orthogonal projection	Part 1	Definition 1.25
D2.4	definition	poles of rung- spectrum; residues amplitudes	Part 1	Definition 1.26
D2.5	definition	massless resolvent pole at ; massive pole at ; pole displacement	Part 1	Definition 1.27
D2.6	definition	orientation-reversal involution: ; chirality chirality; charge charge	Part 1	Definition 1.28
D2.7	definition	ledger reflection: reverses ledger order	Part 1	Definition 1.29
D2.8	definition	convergence sheet: , , for resolvents and configuration sums	Part 1	Definition 1.30
D2.9	definition	units powers of ; ; ; single free scale; names for combinations of (CT2.9), independent inputs	Part 1	Definition 1.31
D3.1	definition	slot index colour	Part 1	Definition 1.37
D3.10	definition	[EW.24]	Part 1	Definition 1.46
D3.2	definition	subset principle: strictly new at rung part of the rung- structure realized at any	Part 1	Definition 1.38
D3.3	definition	at of type , edge signs	Part 1	Definition 1.39
D3.4	definition	(oriented chain). Closed chains: the base-point core read is defined, ternary, and charge-conjugation odd [EM.…	Part 1	Definition 1.40
D3.5	definition	[D0.5, D1.6]; per-edge dynamic phase for — the A4 branch index (read in Part 2 §E: Def-EM.20, EM.20, forward p…	Part 1	Definition 1.41
D3.6	definition	—	Part 1	Definition 1.42
D3.7	definition	photon trivial chain [D1.7] ; gluon trivial chain	Part 1	Definition 1.43
D3.8	definition	colour triplet : one fundamental slot index; colour antitriplet : one dual slot index; quark bound (fermionic)…	Part 1	Definition 1.44
D3.9	definition	baryon ; meson . [Disambiguation: here total mask of A9.]	Part 1	Definition 1.45

Tag	Class	Title / description	Where	Presentation
D4.1	definition	[A8]: single possibility space receiving configurations with rung	Part 1	Definition 1.47
D4.2	definition	absorbs information at rate [D0.9]; absorbed locked retained in , accessible [D1.11]	Part 1	Definition 1.48
D4.3	definition	[A8]: uniform scalar aspect; configuration-dependent	Part 1	Definition 1.49
Tier 2 — Lyapunov Stability of the Pair State				
L0	theorem	The ledger–tilt coupling	Part 1	Theorem 1.29
L0a	theorem	tilt modular update	Part 1	Corollary 1.5
L1	theorem	The Lyapunov function of the pair state	Part 1	Theorem 1.30
L2	theorem	The temporal decay law	Part 1	Theorem 1.31
L3	theorem	Entropy deficit and the inertial mass reading	Part 1	Theorem 1.32
L4	theorem	The patch entropy deficit	Part 1	Theorem 1.33
L5	theorem	The second-order sector flow of the tilt	Part 1	Theorem 1.34
declarations				
P1	calibration policy	quantities of the corpus: derived from A1–A9 C1–C13 D0.1–D4.3, except the hop length ; ; fixes the absolute sc…	Part 1	P1
P2	calibration policy	Derived, zero freedom	Part 1	P2
P3	calibration policy	Named combinations of (external comparison only; inputs)	Part 1	P3
P4	calibration policy	Free internal configuration parameters (constants of the theory). Independent core: . Functionally dependent …	Part 1	P4
P5	calibration policy	quantity identified with a measured external constant. Comparison procedure: fix once; read P2 P3; result test…	Part 1	P5
Part 2 · Bosonic_era — geometry, electroweak, electromagnetic, colour, saturation				
§C.I — Emergent Metric Geometry				
CT1.12	theorem	Metric emergence from sector components	Part 2	Theorem 2.1
CT1.12a	theorem	The pair-channel audit of the metric dictionary: tiling, block split, and the three discrete read classes	Part 2	Theorem 2.2
CT1.13	theorem	Christoffel symbols under conformal rescaling	Part 2	Theorem 2.3
CT1.14	theorem	Riemann tensor under conformal rescaling	Part 2	Theorem 2.4
CT1.15	theorem	Ricci tensor under conformal rescaling	Part 2	Theorem 2.5
CT1.16	theorem	Ricci scalar under conformal rescaling	Part 2	Theorem 2.6
CT1.17	theorem	Einstein tensor under conformal rescaling in	Part 2	Theorem 2.7
CT1.18	theorem	Linearized spin-2 operator	Part 2	Theorem 2.8
§C.III — Continuum Limit, Effective Action, and Field Equations				
CT1.19	theorem	Continuum limit at leading order	Part 2	Theorem 2.18
CT1.19a	theorem	corollary	Part 2	Corollary 2.1
CT1.19b	theorem	On-shell line-element quantization: even antialigned winding, orientation evenness, the geometry quantum, internal holonomy	Part 2	Theorem 2.19
CT1.20	theorem	Effective action from the partition function	Part 2	Theorem 2.20
CT1.21	theorem	The geometric part of the effective action	Part 2	Theorem 2.21
CT1.22	theorem	Stress-energy tensor from the force operator	Part 2	Theorem 2.22

Tag	Class	Title / description	Where	Presentation
CT1.23	theorem	Field equations	Part 2	Theorem 2.23
CT1.24	theorem	Conservation from the field equations	Part 2	Theorem 2.24
CT1.25	theorem	The cosmological constant is the aspect	Part 2	Theorem 2.25
CT1.26	theorem	Conservation from mask invariance alone	Part 2	Theorem 2.26
CT1.27	theorem	Quantum geometry state	Part 2	Theorem 2.27
CT1.28	theorem	Singularity resolution by the saturation lock	Part 2	Theorem 2.28
§C.IV — Entropy, Calibration, and the Single Free Scale				
CT1.29	theorem	Boundary entropy from channel capacity	Part 2	Theorem 2.29
CT1.29a	theorem	The mean edge quantum: structure, sign, and the exactness locus of the entropy–area rate	Part 2	Theorem 2.30
§C.II — Masks, Manifold, and Closure				
CT2.1	theorem	Unit and dimension conventions	Part 2	Theorem 2.9
CT2.10	theorem	Graded matrices and the supertrace	Part 2	Theorem 2.17
§C.IV — Entropy, Calibration, and the Single Free Scale				
CT2.11	theorem	One free scale	Part 2	Theorem 2.32
§C.II — Masks, Manifold, and Closure				
CT2.2	theorem	The rung mask is a projection	Part 2	Theorem 2.10
CT2.3	theorem	Masked force operator	Part 2	Theorem 2.11
CT2.4	theorem	Force restriction by rung	Part 2	Theorem 2.12
CT2.5	theorem	Masked resolvent as spectrum generator	Part 2	Theorem 2.13
CT2.6	theorem	Continuum manifold closure	Part 2	Theorem 2.14
CT2.7	theorem	Exact geometric action closure	Part 2	Theorem 2.15
CT2.8	theorem	Reflection positivity and Hilbert-space closure	Part 2	Theorem 2.16
§C.IV — Entropy, Calibration, and the Single Free Scale				
CT2.9	theorem	Calibration closure	Part 2	Theorem 2.31
§F.I — Slot Structure				
CT3.1	theorem	Classical trivalent symmetry	Part 2	Theorem 2.98
§F.II — The Rung-3 Gauge Algebra				
CT3.10	theorem	The colour aspect has no central component	Part 2	Theorem 2.108
CT3.10a	theorem	no per-edge abelian strong phase	Part 2	Corollary 2.4
§F.III — Colour Objects and Spectrum				
CT3.11	theorem	Colour objects	Part 2	Theorem 2.109
CT3.12	theorem	Photon and gluon are distinguished by possibility space	Part 2	Theorem 2.110
CT3.13	theorem	Gluon count	Part 2	Theorem 2.111
CT3.14	theorem	Gluons remain massless	Part 2	Theorem 2.112
CT3.15	theorem	Baryon invariance	Part 2	Theorem 2.113
CT3.16	theorem	Infinitesimal baryon invariance	Part 2	Theorem 2.114
CT3.17	theorem	Invariant tensors	Part 2	Theorem 2.115
CT3.17a	theorem	sealing partition rule; grammar counts	Part 2	Corollary 2.5
CT3.18	theorem	Colour-singlet closure	Part 2	Theorem 2.116

Tag	Class	Title / description	Where	Presentation
§F.IV — Completion				
CT3.19	theorem	Self-containment of the colour sector	Part 2	Theorem 2.119
§F.I — Slot Structure				
CT3.2	theorem	Quantum slot space	Part 2	Theorem 2.99
CT3.20	theorem	The number of colours is forced	Part 2	Theorem 2.97
§F.II — The Rung-3 Gauge Algebra				
CT3.21	theorem	Structure constants	Part 2	Theorem 2.107
§F.III — Colour Objects and Spectrum				
CT3.22	theorem	Sealing is confinement	Part 2	Theorem 2.117
CT3.22a	theorem	Subnetwork confinement: the domain restriction, all-or-nothing, and the converse	Part 2	Theorem 2.118
§F.I — Slot Structure				
CT3.3	theorem	Classical permutations embed unitarily	Part 2	Theorem 2.100
CT3.4	theorem	Probability preservation gives	Part 2	Theorem 2.101
CT3.5	theorem	The algebra and its decomposition	Part 2	Theorem 2.102
§F.II — The Rung-3 Gauge Algebra				
CT3.6	theorem	Rung-3 active set	Part 2	Theorem 2.103
CT3.7	theorem	The strictly new rung-3 algebra is	Part 2	Theorem 2.104
CT3.8	theorem	Dimension of the colour algebra	Part 2	Theorem 2.105
CT3.9	theorem	Explicit generator basis	Part 2	Theorem 2.106
§G.I — The Saturation Lock				
CT4.1	theorem	Oversubscription falls to	Part 2	Theorem 2.120
§G.II — The Two Dark Aspects				
CT4.10	theorem	The triangle is exhaustive	Part 2	Theorem 2.129
§G.III — Bounds and Boundaries				
CT4.11	theorem	Global closure	Part 2	Theorem 2.132
§G.IV — Falsifiability and the Closure Audit				
CT4.12	theorem	Falsifiers	Part 2	Theorem 2.133
CT4.13	theorem	Closure audit	Part 2	Theorem 2.134
§G.I — The Saturation Lock				
CT4.1a	theorem	universal saturation of bidirectional real-edge subgraphs	Part 2	Corollary 2.6
CT4.2	theorem	The lock is absorbing	Part 2	Theorem 2.121

Tag	Class	Title / description	Where	Presentation
CT4.3	theorem	Locked, not destroyed	Part 2	Theorem 2.122
CT4.4	theorem	The locked sector carries no gauge structure	Part 2	Theorem 2.123
CT4.4a	theorem	—	Addendum	B.3.1
§G.II — The Two Dark Aspects				
CT4.5	theorem	Equation of state of the locked sector	Part 2	Theorem 2.124
CT4.5a	theorem	The flicker-dust ensemble: exact density polynomial and rate-invariant	Part 2	Theorem 2.125
CT4.6	theorem	The two dark aspects contrasted	Part 2	Theorem 2.126
CT4.7	theorem	The entropy triangle	Part 2	Theorem 2.127
CT4.7a	theorem	The vacuum census, closed, and the baseline equality	Part 2	Theorem 2.128
§G.III — Bounds and Boundaries				
CT4.8	theorem	Locked regions and their boundaries	Part 2	Theorem 2.131
CT4.9	theorem	Accessible entropy bounds	Part 2	Theorem 2.130
§E.V — The Dynamic Phase				
Def-EM.20	definition	The per-edge phase at crystallization	Part 2	Definition 2.1
§E.VI — Parity and CPT				
Def-EM.21	definition	Alignment pattern	Part 2	Definition 2.2
Def-EM.22	definition	Mirror operation	Part 2	Definition 2.3
§E.I — Rung-2 Existence and Algebra				
EM.1	theorem	The electromagnetic aspect activates at rung 2	Part 2	Theorem 2.70
§E.III — Electromagnetic Charge				
EM.10	theorem	Charge quantization	Part 2	Theorem 2.80
EM.11	theorem	Global charge conservation	Part 2	Theorem 2.81
§E.IV — Field Strength and Dynamics				
EM.12	theorem	Discrete field strength from phase transitions	Part 2	Theorem 2.82
EM.13	theorem	The transition-parity factor in the weight	Part 2	Theorem 2.83
EM.14	theorem	Continuum gauge invariance	Part 2	Theorem 2.84
EM.15	theorem	Electromagnetic stress-energy	Part 2	Theorem 2.85
§E.V — The Dynamic Phase				
EM.16	theorem	Edge transfer weight and the per-edge read-back	Part 2	Theorem 2.86
EM.17	theorem	Spectral read of the electromagnetic block	Part 2	Theorem 2.87
EM.18	theorem	The colour sector does not shift the block read	Part 2	Theorem 2.88
EM.19	theorem	The saturation sector does not shift the block read	Part 2	Theorem 2.89
§E.I — Rung-2 Existence and Algebra				
EM.2	theorem	The electromagnetic generator	Part 2	Theorem 2.71
§E.V — The Dynamic Phase				
EM.20	theorem	The dynamic edge phase	Part 2	Theorem 2.90
EM.21	theorem	Dimensional status of the per-edge phase	Part 2	Theorem 2.91
§E.VII — Spectrum and Completion				
EM.22	theorem	The electromagnetic spectrum	Part 2	Theorem 2.94

Tag	Class	Title / description	Where	Presentation
§E.VI — Parity and CPT				
EM.23	theorem	Parity classification	Part 2	Theorem 2.92
EM.24	theorem	The CPT bridge	Part 2	Theorem 2.93
§E.VII — Spectrum and Completion				
EM.25	theorem	What the corpus does and does not fix about the abelian sector	Part 2	Theorem 2.96
EM.26	theorem	Cross-checks	Part 2	Theorem 2.95
§E.I — Rung-2 Existence and Algebra				
EM.3	theorem	The electromagnetic Lie algebra	Part 2	Theorem 2.72
EM.4	theorem	The rung-2 algebra containing the abelian factor	Part 2	Theorem 2.73
§E.II — The Photon				
EM.5	theorem	Photon and gluon as trivial chains	Part 2	Theorem 2.74
EM.6	theorem	Photon masslessness	Part 2	Theorem 2.75
EM.7	theorem	Photon entropy budget	Part 2	Theorem 2.76
EM.8	theorem	Photon polarization count	Part 2	Theorem 2.77
§E.III — Electromagnetic Charge				
EM.9	theorem	Chain end-charge	Part 2	Theorem 2.78
EM.9a	theorem	The closed-core charge read and the end-register count	Part 2	Theorem 2.79
EM.9b	theorem	chain-wall telescoping	Part 2	Corollary 2.3
§D.I — Operator Preliminaries				
EW.1	theorem	Joint sector decomposition	Part 2	Theorem 2.33
§D.II — Rung Masks and the Electroweak Algebra				
EW.10	theorem	The rung-2 electroweak algebra	Part 2	Theorem 2.43
EW.11	theorem	The mechanism is projection, not path integration	Part 2	Theorem 2.44
§D.III — The Scalar Trace Mode				
EW.12	theorem	Metric expansion and the scalar trace mode	Part 2	Theorem 2.45
EW.13	theorem	Identification of the scalar mode	Part 2	Theorem 2.46
EW.14	theorem	The scalar background commutes with the electroweak generators	Part 2	Theorem 2.47
EW.15	theorem	No adjoint electroweak breaking	Part 2	Theorem 2.48
EW.16	theorem	Scalar coupling to the antisymmetric sectors	Part 2	Theorem 2.49
§D.IV — Rung-3 Wrapping, the Seed, and Mass Generation				
EW.17	theorem	Rung-3 closed wrapping chains	Part 2	Theorem 2.50
EW.18	theorem	Spiral pole condition	Part 2	Theorem 2.51
EW.19	theorem	Mass as pole displacement	Part 2	Theorem 2.52
§D.I — Operator Preliminaries				
EW.2	theorem	Generator basis and sector assignment	Part 2	Theorem 2.34
§D.IV — Rung-3 Wrapping, the Seed, and Mass Generation				
EW.20	theorem	Weak mass generation	Part 2	Theorem 2.55
EW.21	theorem	Charged weak bosons	Part 2	Theorem 2.56
§D.V — The Neutral Sector				
EW.22	theorem	The unbroken generator	Part 2	Theorem 2.57
EW.23	theorem	The neutral mass matrix	Part 2	Theorem 2.58
EW.24	theorem	The mixing angle	Part 2	Theorem 2.59
EW.25	theorem	The photon direction is massless	Part 2	Theorem 2.60
EW.26	theorem	Colour modes remain massless at rung 3	Part 2	Theorem 2.61
EW.27	theorem	The scalar excitation	Part 2	Theorem 2.62

Tag	Class	Title / description	Where	Presentation
§D.VI — Derived Electroweak Relations				
EW.28	theorem	The gauge spectrum	Part 2	Theorem 2.63
EW.29	theorem	The full electroweak spectrum	Part 2	Theorem 2.67
§D.I — Operator Preliminaries				
EW.3	theorem	The close	Part 2	Theorem 2.35
§D.VI — Derived Electroweak Relations				
EW.30	theorem	Spectral reduction	Part 2	Theorem 2.68
§D.VII — Completion				
EW.31	theorem	Self-containment of the electroweak sector	Part 2	Theorem 2.69
§D.IV — Rung-3 Wrapping, the Seed, and Mass Generation				
EW.32	theorem	The wrapping seed: carrier and normalization	Part 2	Theorem 2.53
EW.33	theorem	The wrapping mass form	Part 2	Theorem 2.54
§D.VI — Derived Electroweak Relations				
EW.34	theorem	The mass ratio	Part 2	Theorem 2.64
EW.35	theorem	The residual coupling	Part 2	Theorem 2.65
EW.36	theorem	The charge spectrum and its consistency with the chain end-charge	Part 2	Theorem 2.66
EW.36a	theorem	corollary	Part 2	Corollary 2.2
§D.I — Operator Preliminaries				
EW.4	theorem	Central, trace, and mixed commutators	Part 2	Theorem 2.36
EW.5	theorem	Second-order Higgs-to-weak pathways	Part 2	Theorem 2.37
EW.5a	theorem	Convention exactness and the rigidity of the named basis	Part 2	Theorem 2.38
§D.II — Rung Masks and the Electroweak Algebra				
EW.6	theorem	Rung masks and active aspect sets	Part 2	Theorem 2.39
EW.7	theorem	The rung-2 electroweak mask	Part 2	Theorem 2.40
EW.8	theorem	The electromagnetic generator	Part 2	Theorem 2.41
EW.9	theorem	The abelian algebra	Part 2	Theorem 2.42
Part 3 · Fermion_era — bound chains, termini, floors, carriers, channels				
Tier F.1 — Information Flux and Measurement Primitivity				
Def-H.1	definition	Observer–subnetwork channel	Part 3	Definition 3.1
Tier F.6 — Subnetwork-Level Force Structure: Confinement and the Active Aspect Set				
Def-H.10	definition	Colour boundary and the contraction-completeness projector	Part 3	Definition 3.13
Def-H.11	definition	Active aspect set of a subnetwork	Part 3	Definition 3.14
Tier F.7 — The Coarse-Grained Slot Algebra: $\mathcal A(\Sigma)$ Properly Derived				
Def-H.12	definition	Coarse-grained slot algebra	Part 3	Definition 3.16
Def-H.13	definition	Compositional bridge map	Part 3	Definition 3.17
Tier F.8 — Line-Element Quantization from Materialization				
Def-H.14	definition	General-edge sector space (internal-edge slot object)	Part 3	Definition 3.18
Tier F.9 — The Activated Aspect Operator: $\mathcal A(\Sigma)$ as a Computable Mask				
Def-H.15	definition	Coarse-grained aspect space and masked force operator	Part 3	Definition 3.19

Tag	Class	Title / description	Where	Presentation
Def-H.16	definition	Coarse-grained resolvent and the -spectrum	Part 3	Definition 3.20
Tier F.10 — The Strong Edge Weight				
Def-H.17	definition	Strong edge transport	Part 3	Definition 3.21
Tier F.11 — Sector Quantum Audit and the Continuum Ledger				
Def-H.18	definition	Arena and spectrum (the two-locus vocabulary)	Part 3	Definition 3.22
Tier F.12 — The Cycle Floor Family, the Defect Bridge, and Composite Floors				
Def-H.19	definition	Stationary, stable, and unstable jumps (vocabulary)	Part 3	Definition 3.23
Tier F.1 — Information Flux and Measurement Primitivity				
Def-H.2	definition	Physical (metric) channel separation	Part 3	Definition 3.2
Tier F.14 — Platonic Floor Cards: The Five Solids as Fermion Floors				
Def-H.20	definition	Platonic floors (setup vocabulary)	Part 3	Definition 3.24
Tier F.17 — The Stationarity Loop: Constancy, the Spin Connection, and the No-Go				
Def-H.21	definition	The spin connection on the emergent geometry (construction)	Part 3	Definition 3.25
Tier F.18 — Native Rungs, Boundary Exchange Shape, and the Admissibility Correction				
Def-H.22	definition	The explicit boundary realization (corolla and separated patterns)	Part 3	Definition 3.27
Tier F.19 — The Three Tails: Realization Non-Existence, the Multiplicity Map, and the Selection Obstruction				
Def-H.23	definition	The winding-multiplicity carrier map (under Def-H.38)	Part 3	Definition 3.29
Tier F.22 — The Bott Correction: Degenerate Witnesses Re-Evaluated, Reconstructability, and the Selection Dilemma				
Def-H.24	definition	Bott index, clean critical manifolds, and monodromy type for ledger channels (construction)	Part 3	Definition 3.30
Tier F.23 — The Virtual Bypass: Path Integration over Crystallization Patterns				
Def-H.25	definition	Skeleton configuration sums and the bypass vocabulary	Part 3	Definition 3.31
Tier F.24 — The Flicker Phase Structure, the Cycle Grammar of Covers, the Dust Equation of State, Face Decomposition, a				
Def-H.26	definition	Flicker-ensemble phase reads	Part 3	Definition 3.32
Tier F.25 — The Chain-Local $\aleph$-Domains: Extended Exact-Cover Theorems and the Domain-Relativity of the Amp				
Def-H.27	definition	The -domains: literal, chain-dominant, literal-forced	Part 3	Definition 3.33
Tier F.27 — The Cover Orbit Decomposition: the Ansatz Split, the Combinatorial Ceiling, and the Bridge Census				
Def-H.28	definition	Cover classes, and the two class-mass ansätze	Part 3	Definition 3.34
Tier F.28 — The Full-SM Completeness Audit: Particle Content, the Gauge Blocks, the Rung Ledger, and the Deficit Table				
Def-H.29	definition	Completeness-audit vocabulary: read classes, census statuses, the external-data clause	Part 3	Definition 3.35
Tier F.1 — Information Flux and Measurement Primitivity				
Def-H.3	definition	Extractable information	Part 3	Definition 3.3
Tier F.29 — Gravitational Reads in Three States, the Flicker Ripple, and the Tumbling-Model Adjudication				
Def-H.30	definition	The three shape states	Part 3	Definition 3.36

Tag	Class	Title / description	Where	Presentation
Tier F.35 — The 1-Loop Bridge: the Swap-Graded Effective Potential on the Flicker Phase Diagram				
`Def-H.31`	definition	The bridge: dictionary, family, background, fluctuation operator, effective potential	Part 3	Definition 3.37
Tier F.36 — The Topology-Blind Path Integral: Observable-Conditioned Sectors and the Janus Sum				
`Def-H.32`	definition	The topology-blind path integral	Part 3	Definition 3.38
Tier F.39 — Chirality: The Orientation Doublet of the Bound Chain and Its Exchange-Form Grade				
`Def-H.33`	definition	The chain orientation space and its chirality operator	Part 3	Definition 3.41
Tier F.40 — The Static Force Matrix on the Extended Carrier: The Aspect Census, the No-Flip Theorem, and the Charge–I				
`Def-H.34`	definition	The extended carrier basis, the chiral transform, and the partner involution	Part 3	Definition 3.42
Tier F.5 — Spin–Charge Decomposition via Complete Force Operator				
`Def-H.35`	definition	Channel-visible content (the boundary-transport read)	Part 3	Definition 3.10
Tier F.6 — Subnetwork-Level Force Structure: Confinement and the Active Aspect Set				
`Def-H.36`	definition	The scalar-row fold (the H read at -level)	Part 3	Definition 3.15
Tier F.17 — The Stationarity Loop: Constancy, the Spin Connection, and the No-Go				
`Def-H.37`	definition	The defect-transfer interface	Part 3	Definition 3.26
Tier F.19 — The Three Tails: Realization Non-Existence, the Multiplicity Map, and the Selection Obstruction				
`Def-H.38`	definition	The multiplicity–winding coupling	Part 3	Definition 3.28
Tier F.5b — The Mass-Read Registry and the Tilt Composition				
`Def-H.39`	definition	The mass-read registry; bridge slots	Part 3	Definition 3.11
Tier F.2 — The Fermion Definition				
`Def-H.4`	definition	Fermion	Part 3	Definition 3.4
Tier F.5b — The Mass-Read Registry and the Tilt Composition				
`Def-H.40`	definition	Chain-level tilt composition	Part 3	Definition 3.12
Tier F.37 — The Koide Channel: The C_3-Covariant Seal Bilinear and the Bound-Chain Mixing Angle				
`Def-H.41`	definition	The bound-chain arena	Part 3	Definition 3.39
Tier F.38 — The Colour Channel: The ε-Selected C_3, Confinement Blindness, and the Triplet-Attached M				
`Def-H.42`	definition	The colour-dressed arena	Part 3	Definition 3.40
Tier F.40 — The Static Force Matrix on the Extended Carrier: The Aspect Census, the No-Flip Theorem, and the Charge–I				
`Def-H.43`	definition	The read promotion on the orientation doublet	Part 3	Definition 3.43
Tier F.3 — Information Flux as Rate				
`Def-H.5`	definition	Information flux	Part 3	Definition 3.5
`Def-H.6`	definition	Information flux density	Part 3	Definition 3.6
Tier F.4 — Rung-Dependent Edge Structure				
`Def-H.7`	definition	Rung-dependent edge data	Part 3	Definition 3.7
Tier F.5 — Spin–Charge Decomposition via Complete Force Operator				
`Def-H.8`	definition	Spin (emergent weak contribution)	Part 3	Definition 3.8

Tag	Class	Title / description	Where	Presentation
Def-H.9	definition	Subnetwork coarse-graining projection	Part 3	Definition 3.9
Tier F.1 — Information Flux and Measurement Primitivity				
FT.1	theorem	Information flux bound	Part 3	Theorem 3.1
Tier F.4 — Rung-Dependent Edge Structure				
FT.10	theorem	Confinement is a global sum-restriction	Part 3	Theorem 3.10
Tier F.33 — The Dynamic Phase, Parity, and the CPT Bridge: the Mass Refinement				
FT.100	theorem	The alignment pattern refines the informational mass	Part 3	Theorem 3.74
Tier F.34 — The Weak Branch Grading of the Two-Branch Crystallization				
FT.101	theorem	The weak branch grading: the two crystallization branches are weak-inequivalent	Part 3	Theorem 3.75
Tier F.35 — The 1-Loop Bridge: the Swap-Graded Effective Potential on the Flicker Phase Diagram				
FT.102	theorem	The fluctuation spectrum, the tree-level limit, and the seal-softening walls	Part 3	Theorem 3.76
FT.103	theorem	The vacuum structure: four exact vacua, three wall-delimited phase bands, dominance windows, the susceptibility census	Part 3	Theorem 3.77
FT.104	theorem	The [EXT] audit, the range-species classification, and the fourth read of Gap 7	Part 3	Theorem 3.78
Tier F.36 — The Topology-Blind Path Integral: Observable-Conditioned Sectors and the Janus Sum				
FT.106	theorem	Weight transport and the two-layer sector structure	Part 3	Theorem 3.79
FT.107	theorem	The observable sectors and the mass expectation	Part 3	Theorem 3.80
FT.108	theorem	The Janus sum: move algebra, budget conservation, derived detailed balance, and the Tier F.35 boundary	Part 3	Theorem 3.81
Tier F.37 — The Koide Channel: The C_3-Covariant Seal Bilinear and the Bound-Chain Mixing Angle				
FT.109	theorem	The diagonal seal commutator	Part 3	Theorem 3.82
Tier F.5 — Spin–Charge Decomposition via Complete Force Operator				
FT.11	theorem	Complete force operator charge audit	Part 3	Theorem 3.11
Tier F.37 — The Koide Channel: The C_3-Covariant Seal Bilinear and the Bound-Chain Mixing Angle				
FT.110	theorem	The bound-chain arena dimension	Part 3	Theorem 3.83
FT.111	theorem	The spectrum of the covariant seal bilinear	Part 3	Theorem 3.84
FT.112	theorem	The mixing angle	Part 3	Theorem 3.85
FT.113	theorem	The bilinear spectrum and the mass-squared ratios	Part 3	Theorem 3.86
Tier F.38 — The Colour Channel: The ε-Selected C_3, Confinement Blindness, and the Triplet-Attached M				
FT.116	theorem	The relabelling lattice, the two-level obstruction, and the -selection of	Part 3	Theorem 3.87
FT.117	theorem	—	Addendum	B.3.3
FT.118	theorem	The colour-dressed arena and the spectator numerator	Part 3	Theorem 3.88
FT.119	theorem	The triplet-attached mixing angle and its spectrum	Part 3	Theorem 3.89
Tier F.5 — Spin–Charge Decomposition via Complete Force Operator				
FT.12	theorem	Weak has no self-charge	Part 3	Theorem 3.12
Tier F.39 — Chirality: The Orientation Doublet of the Bound Chain and Its Exchange-Form Grade				
FT.122	theorem	The orientation doublet is corpus-constructible	Part 3	Theorem 3.90

Tag	Class	Title / description	Where	Presentation
FT.123	theorem	Weak covariance and -covariance on the extended carrier	Part 3	Theorem 3.91
FT.124	theorem	—	Addendum	B.3.4
FT.125	theorem	—	Addendum	B.4.1
FT.126	theorem	Weak-covariant chiral projectors on the chain carrier	Part 3	Theorem 3.92
Tier F.40 — The Static Force Matrix on the Extended Carrier: The Aspect Census, the No-Flip Theorem, and the Charge–I				
FT.127	theorem	The aspect census on the extended carrier	Part 3	Theorem 3.93
FT.128	theorem	The no-flip theorem at the derived locus	Part 3	Theorem 3.94
FT.129	theorem	The canonical flip candidate and the read promotion	Part 3	Theorem 3.95
Tier F.5 — Spin–Charge Decomposition via Complete Force Operator				
FT.13	theorem	Weak spin (algebraically)	Part 3	Theorem 3.13
Tier F.40 — The Static Force Matrix on the Extended Carrier: The Aspect Census, the No-Flip Theorem, and the Charge–I				
FT.130	theorem	Tensor structure, Schmidt rank, and the static spectrum	Part 3	Theorem 3.96
FT.131	theorem	Scope: bound chains, no isolated charge, the deferrals	Part 3	Theorem 3.97
Tier F.41 — Parity: The Configuration Calculus, the Carrier Bookkeeping, and the Sector Census				
FT.133	theorem	The configuration-space reversal calculus	Part 3	Theorem 3.98
FT.134	theorem	The carrier bookkeeping: the scalar-CPT theorem	Part 3	Theorem 3.99
FT.135	theorem	The weak branch grading under parity	Part 3	Theorem 3.100
FT.136	theorem	The rung-one S-grading: the 10/6 split	Part 3	Theorem 3.101
FT.137	theorem	The cover and flicker census under the reversal family	Part 3	Theorem 3.102
FT.138	theorem	The pole spectrum census	Part 3	Theorem 3.103
FT.139	theorem	The channel census: Koide and colour channels under the reversal family	Part 3	Theorem 3.104
Tier F.5 — Spin–Charge Decomposition via Complete Force Operator				
FT.14	theorem	Spin–charge decomposition	Part 3	Theorem 3.14
Tier F.42 — The Rung-One Chirality, the Janus Arrow, and the Family Channel of the Sixteen-Dimensional Arena				
FT.141	theorem	The rung-one chirality theorem: the grading is the chirality of the complete force complex	Part 3	Theorem 3.105
FT.142	theorem	The rung-one arena and the channel angle	Part 3	Theorem 3.106
FT.143	theorem	The rung-one family spectrum	Part 3	Theorem 3.107
FT.144	theorem	The Janus-arrow theorem: no backward materialization, and single-sector occupancy at rung one	Part 3	Theorem 3.108
FT.145	theorem	The rung-one completion	Part 3	Theorem 3.109
Tier F.5 — Spin–Charge Decomposition via Complete Force Operator				
FT.15	theorem	Coarse-graining invariance	Part 3	Theorem 3.15
FT.153	theorem	Boundary-schedule attainment	Part 3	Theorem 3.17
Tier F.5b — The Mass-Read Registry and the Tilt Composition				
FT.154	theorem	The tilt-composition law	Part 3	Theorem 3.19

Tag	Class	Title / description	Where	Presentation
Tier F.26 — The Platonic Pole Spectrum: Exact Multiplets, the Bounded-Hierarchy Ceiling, and the SM-Hierarchy Audit				
FT.155	theorem	Minimality of the defect interface	Part 3	Theorem 3.62
Tier F.44 — The 3+1 Chiral Asymmetry: The Manifold Locus of Chirality, the Spatial Provenance of the Channel Angle, a				
FT.156	theorem	The 3+1 asymmetry theorem: one signature datum at two loci, and the manifold structure as the chiral structure	Part 3	Theorem 3.110
FT.157	theorem	The weak-selectivity theorem: the weak force is the shift block — the spatial force on the spatial branch	Part 3	Theorem 3.111
FT.158	theorem	The spatial provenance of the channel angle: is the weak rotation of the triad, and the Koide-form spectrum is forced	Part 3	Theorem 3.112
FT.159	theorem	The emergent-time identity theorem: gravitational time and CPT time are the ledger order	Part 3	Theorem 3.113
Tier F.5 — Spin–Charge Decomposition via Complete Force Operator				
FT.16	theorem	The single-vertex hidden-information gap is generic; strong's signature is contraction-incompleteness	Part 3	Theorem 3.16
Tier F.44 — The 3+1 Chiral Asymmetry: The Manifold Locus of Chirality, the Spatial Provenance of the Channel Angle, a				
FT.160	theorem	—	Addendum	B.1.14
Tier F.5 — Spin–Charge Decomposition via Complete Force Operator				
FT.17	theorem	Internal-edge lemma	Part 3	Theorem 3.18
Tier F.6 — Subnetwork-Level Force Structure: Confinement and the Active Aspect Set				
FT.18	theorem	Contraction-completeness is not -local	Part 3	Theorem 3.20
FT.19	theorem	Confinement is an all-or-nothing subnetwork property	Part 3	Theorem 3.21
Tier F.1 — Information Flux and Measurement Primitivity				
FT.2	theorem	Exhaustiveness of the real-edge measurement primitive	Part 3	Theorem 3.2
Tier F.6 — Subnetwork-Level Force Structure: Confinement and the Active Aspect Set				
FT.20	theorem	Reduction of to proven per-aspect criteria	Part 3	Theorem 3.22
Tier F.7 — The Coarse-Grained Slot Algebra: $\mathcal A(\Sigma)$ Properly Derived				
FT.21	theorem	's charge-type rows, properly derived	Part 3	Theorem 3.23
Tier F.8 — Line-Element Quantization from Materialization				
FT.22	theorem	Loop quantization of materialized bound chains	Part 3	Theorem 3.24
FT.23	theorem	Orientation-evenness of the line element	Part 3	Theorem 3.25
FT.24	theorem	The minimal fermion pins the geometry quantum	Part 3	Theorem 3.26
FT.25	theorem	Internal holonomy is well-defined on-shell	Part 3	Theorem 3.27
Tier F.9 — The Activated Aspect Operator: $\mathcal A(\Sigma)$ as a Computable Mask				
FT.26	theorem	Structural transfer: the -mask inherits CT2.3–CT2.5	Part 3	Theorem 3.28
FT.27	theorem	Single-vertex degeneration; action versus observable	Part 3	Theorem 3.29
Tier F.10 — The Strong Edge Weight				
FT.29	theorem	Strong weight: factorization, no , observable sealing	Part 3	Theorem 3.30
Tier F.1 — Information Flux and Measurement Primitivity				
FT.3	theorem	Unresolvability of -sourced subnetwork structure	Part 3	Theorem 3.3

Tag	Class	Title / description	Where	Presentation
Tier F.11 — Sector Quantum Audit and the Continuum Ledger				
FT.30	theorem	Pair-channel sector quantum audit	Part 3	Theorem 3.31
FT.31	theorem	The continuum limit survives on-shell quantization	Part 3	Theorem 3.32
Tier F.12 — The Cycle Floor Family, the Defect Bridge, and Composite Floors				
FT.33	theorem	The defect bridge: pole-side mass candidate in well-posed ratio form	Part 3	Theorem 3.33
FT.34	theorem	Round-trip quantization between floors; integrality propagation; sealing partition rule	Part 3	Theorem 3.34
Tier F.13 — Gap-Closure Pass: Flux Ledger, Relating Equation, Rigidity, Confinement Converse, Core Charge, Aspect Lec				
FT.36	theorem	Flux ledger completion	Part 3	Theorem 3.35
FT.37	theorem	The relating equation, complete	Part 3	Theorem 3.36
FT.38	theorem	Non-geodesic tightening and on-shell rigidity	Part 3	Theorem 3.37
FT.39	theorem	Confinement converse, singlet diagonal, and filter placement	Part 3	Theorem 3.38
Tier F.2 — The Fermion Definition				
FT.4	theorem	External state characterization of a fermion terminus	Part 3	Theorem 3.4
Tier F.13 — Gap-Closure Pass: Flux Ledger, Relating Equation, Rigidity, Confinement Converse, Core Charge, Aspect Lec				
FT.40	theorem	Closed-core charge read	Part 3	Theorem 3.39
FT.41	theorem	The per-aspect ledger	Part 3	Theorem 3.40
Tier F.15 — The Regular-Polygon Floors: Full Ladder, Sectors, and the Non-Existence Ceiling				
FT.46	theorem	Distinctness is unbounded: no upper limit on distinct regular polygons	Part 3	Theorem 3.41
FT.47	theorem	No geometric ceiling: the Gauss–Bonnet bound binds area, never	Part 3	Theorem 3.42
FT.48	theorem	Stability structure of floor-to-floor channels (Gap 18, partial)	Part 3	Theorem 3.43
Tier F.3 — Information Flux as Rate				
FT.5	theorem	Flux bound (rate form of FT.1)	Part 3	Theorem 3.5
Tier F.16 — Composition, Enumeration, and the Spin-Connection Obstruction				
FT.52	theorem	Spin extraction: the constraint system, uniqueness of the stretched rule, and the localized obstruction	Part 3	Theorem 3.44
Tier F.17 — The Stationarity Loop: Constancy, the Spin Connection, and the No-Go				
FT.53	theorem	Geodesic constancy: the stationary on-shell ledger is constant-quantum	Part 3	Theorem 3.45
FT.54	theorem	Exact holonomy-defect formula and the closed-form pole side (under Def-H.37(2))	Part 3	Theorem 3.46
FT.55	theorem	The no-go and the surviving spin realization	Part 3	Theorem 3.47
Tier F.18 — Native Rungs, Boundary Exchange Shape, and the Admissibility Correction				
FT.56	theorem	Native ladder rungs, the spin–orbit decomposition, and the missing map	Part 3	Theorem 3.48
FT.57	theorem	The boundary exchange algebra: root-space shape	Part 3	Theorem 3.49
FT.58	theorem	Hosting classification and the no-hosting theorem	Part 3	Theorem 3.50
Tier F.19 — The Three Tails: Realization Non-Existence, the Multiplicity Map, and the Selection Obstruction				
FT.59	theorem	—	Addendum	B.3.2
Tier F.3 — Information Flux as Rate				
FT.6	theorem	Conformal invariance of combinatorial flux	Part 3	Theorem 3.6

Tag	Class	Title / description	Where	Presentation
Tier F.19 — The Three Tails: Realization Non-Existence, the Multiplicity Map, and the Selection Obstruction				
FT.60	theorem	The multiplicity map: conditions, ladders, basis-relativity, and the honesty witnesses	Part 3	Theorem 3.51
FT.61	theorem	The selection side: phase inventory, index-blindness, and the two exits (Gap 18)	Part 3	Theorem 3.52
Tier F.20 — The Bidirectional Rung Audit and the Platonic Physical Cards				
FT.62	theorem	The bidirectional rung audit (correction theorem)	Part 3	Theorem 3.53
Tier F.22 — The Bott Correction: Degenerate Witnesses Re-Evaluated, Reconstructability, and the Selection Dilemma				
FT.66	theorem	The Bott correction: witness re-evaluation, reconstructability, and the inadmissibility of absolute selection	Part 3	Theorem 3.54
Tier F.23 — The Virtual Bypass: Path Integration over Crystallization Patterns				
FT.67	theorem	The exact-cover theorem and the single-vertex bypass	Part 3	Theorem 3.55
FT.68	theorem	Joint skeletons: tetrahedron, the polygon ladder, the cube; the layer dichotomy; flicker	Part 3	Theorem 3.56
Tier F.3 — Information Flux as Rate				
FT.7	theorem	Universal flux density bound	Part 3	Theorem 3.7
Tier F.24 — The Flicker Phase Structure, the Cycle Grammar of Covers, the Dust Equation of State, Face Decomposition, a				
FT.70	theorem	The flicker phase diagram: unique crossover, computed boundaries, the polygon limit	Part 3	Theorem 3.57
FT.71	theorem	The flicker-dust equation of state	Part 3	Theorem 3.58
FT.72	theorem	Two-fold face decomposition; the Bernoulli marginals (”coarse-grained fuzz as data”)	Part 3	Theorem 3.59
FT.73	theorem	—	Addendum	B.2.2
Tier F.25 — The Chain-Local $\aleph$-Domains: Extended Exact-Cover Theorems and the Domain-Relativity of the Amp				
FT.74	theorem	The extended exact-cover theorems	Part 3	Theorem 3.60
FT.75	theorem	Domain-relativity of the amplitude layer: lock revival, floor revival, parity, and the invariant bridge gap	Part 3	Theorem 3.61
Tier F.26 — The Platonic Pole Spectrum: Exact Multiplets, the Bounded-Hierarchy Ceiling, and the SM-Hierarchy Audit				
FT.77	theorem	—	Addendum	B.2.3
Tier F.27 — The Cover Orbit Decomposition: the Ansatz Split, the Combinatorial Ceiling, and the Bridge Census				
FT.79	theorem	The combinatorial ceiling and the cover-hierarchy audit (second clean negative)	Part 3	Theorem 3.63
Tier F.3 — Information Flux as Rate				
FT.8	theorem	Flux-mediated coarse-graining connection	Part 3	Theorem 3.8
Tier F.27 — The Cover Orbit Decomposition: the Ansatz Split, the Combinatorial Ceiling, and the Bridge Census				
FT.80	theorem	The bridge census: covers are amplitude-real, norm-real, and spectrally un-indexed	Part 3	Theorem 3.64
Tier F.28 — The Full-SM Completeness Audit: Particle Content, the Gauge Blocks, the Rung Ledger, and the Deficit Table				
FT.82	theorem	Gauge-block reconstruction (audit pass B): the algebra closes, the coupling structure does not	Part 3	Theorem 3.65
Tier F.29 — Gravitational Reads in Three States, the Flicker Ripple, and the Tumbling-Model Adjudication				
FT.85	theorem	Vacuum census, baseline equality, and the constructible gravitational reads	Part 3	Theorem 3.66
FT.86	theorem	The flicker ripple: ultralocal raw dynamics, cap-induced alternation, and the quantum–classical gradient	Part 3	Theorem 3.67

Tag	Class	Title / description	Where	Presentation
FT.87	theorem	—	Addendum	B.2.4
FT.88	theorem	The involution algebra of the two declared reversals	Part 3	Theorem 3.68
Tier F.30 — The Koide Adjudication, the Resolution Ladder, and the Minimal Bridge Declaration				
FT.89	theorem	—	Addendum	B.1.13
Tier F.4 — Rung-Dependent Edge Structure				
FT.9	theorem	EM part of edge weight is rung-stable	Part 3	Theorem 3.9
Tier F.30 — The Koide Adjudication, the Resolution Ladder, and the Minimal Bridge Declaration				
FT.90	theorem	—	Addendum	B.2.5
FT.91	theorem	The minimal spectral-bridge form	Part 3	Theorem 3.69
Tier F.31 — The Native Clifford Census, the Weak-Chirality Pincer, and the Reversal Census of the Covers				
FT.92	theorem	EW.5 is convention-exact: re-verification and the rigidity of the named basis	Part 3	Theorem 3.70
FT.93	theorem	The native Clifford census and the weak-chirality pincer (Tasks 1–3)	Part 3	Theorem 3.71
FT.94	theorem	—	Addendum	B.2.6
FT.95	theorem	Flicker chirality: the tautology, the absent trajectory, and the zero-expectation theorem	Part 3	Theorem 3.72
Tier F.32 — The Multi-Node Form Factor: Formalization, Exact Reductions, and the Third Ceiling				
FT.97	theorem	The form factor formalized: node ontology, three underdeterminations, reduction identities, and the general ceiling	Part 3	Theorem 3.73
Part 4 · Baryomesonic_era — colour-sealed composites				
Tier BM.1 — Observability of Sealed Composites				
BM.1	theorem	Confinement inheritance and the trace-only read	Part 4	Theorem 4.1
Tier BM.4 — Noether's Theorem from the Janus Seal Topology: the Discrete Conservation Law, the Continuum Promotio				
BM.10	theorem	—	Addendum	B.3.5
Tier BM.19 — The Flicker Integral, the Rest-Mass Q-Value, and the Relating Equation: the Materialized Ensemble, the Ve				
BM.100	theorem	—	Addendum	B.1.29
Tier BM.20 — Six Objective Resolutions: the Energy-Scale Graph, the Dimensional Bridge, the Eigenvalue Composition, t				
BM.101	theorem	The energy-scale graph: the two reads, the derived crossing strength, and the composition-dependent crossing count	Part 4	Theorem 4.85
BM.102	theorem	The dimensional bridge: the -exponent census, the seven rows, and the declared calibration slot	Part 4	Theorem 4.86
BM.103	theorem	The three-constituent assembly: the eigenvalue equations that are formed, the tensor-product referent census, and the composition residue	Part 4	Theorem 4.87
BM.104	theorem	The curvature composition: the back-reaction fixed point, its exact roots and pole locus, and the non-selection theorem	Part 4	Theorem 4.88
BM.105	theorem	The general crystallization weight: the measure-free bounds, the threshold family, and the refuted defect conditioning	Part 4	Theorem 4.89
BM.106	theorem	The arrow and the kind: the spin label, the tick law, and the exchange-structure referent census	Part 4	Theorem 4.90
BM.107	theorem	—	Addendum	B.1.30
Tier BM.21 — The Antimatter and Resonance Tier: the CP-Conjugate, the Meson Seal, the Exchange Operator, and the Lo				

Tag	Class	Title / description	Where	Presentation
BM.108	theorem	The CP-conjugate chain: the reversal census, the T-block, the exact phase, and the forced conjugate	Part 4	Theorem 4.91
BM.109	theorem	The CP-pair as the vacuum fluctuation: the creation weight, its exact range, and the degeneracy rows	Part 4	Theorem 4.92
Tier BM.4 — Noether's Theorem from the Janus Seal Topology: the Discrete Conservation Law, the Continuum Promotio				
BM.11	theorem	The continuum limit of the Janus topology: the differential conservation law and its two agreeing routes	Part 4	Theorem 4.10
Tier BM.21 — The Antimatter and Resonance Tier: the CP-Conjugate, the Meson Seal, the Exchange Operator, and the Lo				
BM.110	theorem	The meson seal: the -sealed -juncture, its uniqueness, and the charge class it occupies	Part 4	Theorem 4.93
BM.111	theorem	The meson mass: the two additive routes, the vanishing seal-binding, and the amplitude/norm reconciliation	Part 4	Theorem 4.94
BM.112	theorem	The exchange operator: construction, involution, spectrum, and the eigenspace census	Part 4	Theorem 4.95
BM.113	theorem	Spin–statistics: the phase alphabet of the swap, the winding parity, and the derived connection	Part 4	Theorem 4.96
BM.114	theorem	The resonance spectrum: the two characteristic polynomials, the orbit law, and the one exact collision	Part 4	Theorem 4.97
BM.115	theorem	The materialization threshold: the exact function of , and , and the never-empty locus	Part 4	Theorem 4.98
BM.116	theorem	The decay width: the exact function of the local parameters, its three exact zeros, and the lifetime floor	Part 4	Theorem 4.99
BM.117	theorem	The groupings, the ceilings, and the octet adjudication	Part 4	Theorem 4.100
BM.118	theorem	—	Addendum	B.1.31
Tier BM.22 — The Annihilation Channel, Concrete Spin, and Fermion Identification: the Janus Tunnel, the Zig-Zag Worl				
BM.119	theorem	—	Addendum	B.1.32
Tier BM.4 — Noether's Theorem from the Janus Seal Topology: the Discrete Conservation Law, the Continuum Promotio				
BM.12	theorem	—	Addendum	B.1.15
Tier BM.22 — The Annihilation Channel, Concrete Spin, and Fermion Identification: the Janus Tunnel, the Zig-Zag Worl				
BM.120	theorem	The Janus tunnel: the crossing operators, the exact spectrum, and detailed balance	Part 4	Theorem 4.101
BM.121	theorem	The zig-zag world line: the even resolvent, the round-trip weights, and no backward leg	Part 4	Theorem 4.102
BM.122	theorem	The pair channel: charge balance, the exact threshold, the CP structure, and the two-photon bar	Part 4	Theorem 4.103
BM.123	theorem	The annihilation channel: the reverse move, the entropy destination, and the closure of the cross-section residue	Part 4	Theorem 4.104
BM.124	theorem	The spin–winding relation and the multiplet defect ladder	Part 4	Theorem 4.105
BM.125	theorem	The spin-dependent mass: the exact ladder, the two clauses and their matching, and the ceiling	Part 4	Theorem 4.106
BM.126	theorem	—	Addendum	B.2.7

Tag	Class	Title / description	Where	Presentation
BM.127	theorem	The winding-parity identification: two reads separated	Part 4	Theorem 4.107
BM.128	theorem	The three-chain exchange structure: the space, the action, the character formula, and the vanishing wedge	Part 4	Theorem 4.108
BM.129	theorem	The meson is a boson and the baryon's sign is the product: the composite identification	Part 4	Theorem 4.109
Tier BM.5 — The Calibration Lock: the Born Width, the Coherence Volume, the Curvature of a Sealed Composite, and th				
BM.13	theorem	The Born width: the exact substitution, the quantized ladder, and the sub-hop locus	Part 4	Theorem 4.11
Tier BM.22 — The Annihilation Channel, Concrete Spin, and Fermion Identification: the Janus Tunnel, the Zig-Zag Worl				
BM.130	theorem	—	Addendum	B.1.33
Tier BM.23 — The Closing Tier: the Haar–Poisson Duality and the Lapse, the Selected Curvature Composition, the Mom				
BM.131	theorem	The Haar–Poisson duality: the twirl fixed point, the complementary factorization, and the proper-time contraction	Part 4	Theorem 4.110
BM.132	theorem	The lapse: the derived function of the envelope ratio and the curvature, its formed domain, and the envelope row	Part 4	Theorem 4.111
BM.133	theorem	The selected curvature composition: two derived criteria, the unique survivor, and the completed running coupling	Part 4	Theorem 4.112
BM.134	theorem	The resolvent's derived content: the form uniqueness, the pole reads, the on-shell measure, and the two-body phase space	Part 4	Theorem 4.113
BM.135	theorem	The multi-chain additive charge census: the baryon and meson rows, the stretched states, and the gauged read	Part 4	Theorem 4.114
BM.136	theorem	The action is the colour 3-vertex: the equivariance theorem, the exchange structure at every , and the statistics per class	Part 4	Theorem 4.115
BM.137	theorem	The two-chain ensemble measure, the meson's formed binding, the sector-mass pincer, and the two-chain spectrum	Part 4	Theorem 4.116
BM.138	theorem	The seed-tensoring mechanism: the symmetrizer as a finite-group twirl, the transport and composition closure, and the two residues adjudicated	Part 4	Theorem 4.117
BM.139	theorem	The two convention declarations and the sector locus	Part 4	Theorem 4.118
Tier BM.5 — The Calibration Lock: the Born Width, the Coherence Volume, the Curvature of a Sealed Composite, and th				
BM.14	theorem	The coherence volume and the energy density	Part 4	Theorem 4.12
Tier BM.23 — The Closing Tier: the Haar–Poisson Duality and the Lapse, the Selected Curvature Composition, the Mom				
BM.140	theorem	—	Addendum	B.2.8
BM.141	theorem	The Jensen residue: the exact closed form, its range, its zero locus, the two-chain companion, and the width adjudication	Part 4	Theorem 4.119
BM.142	theorem	The supply vertex, the pair amplitude, and the annihilation cross-section: the four grounds resolved and residues , closed	Part 4	Theorem 4.120
BM.143	theorem	—	Addendum	B.4.2

Tag	Class	Title / description	Where	Presentation
BM.144	theorem	—	Addendum	B.1.34
Tier BM.24 — The Continuum-Limit Tier: the Force Operator over the Graph, the Continuum-Limit Function, the Metric-				
BM.145	theorem	The force operator acting over the graph: the generator census, the mask ladder, and the derived action	Part 4	Theorem 4.121
BM.146	theorem	The continuum-limit function: the two variables, the five contributors, and the exact functional form	Part 4	Theorem 4.122
BM.147	theorem	The metric–energy–continuum map: the dictionary bijection, the closed form of the source read, and the target	Part 4	Theorem 4.123
BM.148	theorem	The breaking threshold: the exact capacity of a chart cell	Part 4	Theorem 4.124
BM.149	theorem	The locked sector at the breaking locus: the two faces of one capacity, the absorption, the membrane, and the complement theorem	Part 4	Theorem 4.125
Tier BM.5 — The Calibration Lock: the Born Width, the Coherence Volume, the Curvature of a Sealed Composite, and th				
BM.15	theorem	The curvature of a sealed composite: the exact trace of the field equations, and the curvature scale	Part 4	Theorem 4.13
Tier BM.24 — The Continuum-Limit Tier: the Force Operator over the Graph, the Continuum-Limit Function, the Metric-				
BM.150	theorem	The universality of the continuum limit: one operator, one arena, one action, three fingerprints	Part 4	Theorem 4.126
BM.151	theorem	The derivation of CT2.6(ii)–(iii): the unbounded ladder, the spectral bound, and the empty extra-axiomatic list	Part 4	Theorem 4.127
BM.152	theorem	—	Addendum	B.1.35
Tier BM.5 — The Calibration Lock: the Born Width, the Coherence Volume, the Curvature of a Sealed Composite, and th				
BM.16	theorem	The calibration lock	Part 4	Theorem 4.14
BM.17	theorem	The neutron row of the identification layer	Part 4	Theorem 4.15
BM.18	theorem	—	Addendum	B.1.16
Tier BM.6 — The Topology-Blind Neutron and the Topology-Blind + Static Proton: the Flicker Ensemble, the Path-Integra				
BM.19	theorem	The flicker ensemble is derived: the stationary measure, the threshold specialization, and the no-flicker limit	Part 4	Theorem 4.16
Tier BM.1 — Observability of Sealed Composites				
BM.2	theorem	Junction phase and winding grammar	Part 4	Theorem 4.2
Tier BM.6 — The Topology-Blind Neutron and the Topology-Blind + Static Proton: the Flicker Ensemble, the Path-Integra				
BM.20	theorem	The mass from the curvature law on the path-integrated route	Part 4	Theorem 4.17
BM.21	theorem	The binding energy: additive in the count, non-additive in the ensemble	Part 4	Theorem 4.18
BM.22	theorem	The static force matrix on the two charge classes	Part 4	Theorem 4.19
BM.23	theorem	The transition: the ladder rule, the forced carrier, and the resolvent rate	Part 4	Theorem 4.20
BM.24	theorem	—	Addendum	B.1.17
Tier BM.7 — The Proton in the Positive CPT Sector and Its Mirror: the Sector Labels, the Exact Phase, and the Charge-Blin				
BM.25	theorem	The CPT involution on the sealed composite: orbits, the exact phase, and the census ratio	Part 4	Theorem 4.21
BM.26	theorem	The proton row: the positive sector	Part 4	Theorem 4.22

Tag	Class	Title / description	Where	Presentation
BM.27	theorem	The antiproton row: the CPT image, and the resolution of the identity question	Part 4	Theorem 4.23
BM.28	theorem	The charge-constrained probability space: the census, the three mass routes, and the splitting	Part 4	Theorem 4.24
BM.29	theorem	—	Addendum	B.1.18
Tier BM.2 — The Family Group and the Invariance Laws of the Seal				
BM.3	theorem	The family group of the seal	Part 4	Theorem 4.3
Tier BM.8 — The Additive Charge Read and the Mass-Route Adjudication				
BM.30	theorem	The mass-route adjudication: three exact ratios, one read choice	Part 4	Theorem 4.25
Tier BM.9 — The Mesonic Flicker Test: the Contraction Grammar, the Colour Projector, and the Protected Junction				
BM.31	theorem	The boundary-data comparison and the grammar verdict	Part 4	Theorem 4.26
BM.32	theorem	The mesonic contribution: zero by filtration, and the amplitude layer is orthogonal	Part 4	Theorem 4.27
BM.33	theorem	—	Addendum	B.1.19
Tier BM.10 — The Kinetic Energy from the Metric and the Vacuum-Balanced Entropy: the Dispersion Read of a Sealed C				
BM.34	theorem	The kinetic energy from the emergent metric: the ADM split, the mass shell, the dispersion relation, and the zero-point bound	Part 4	Theorem 4.28
BM.35	theorem	The vacuum-balanced entropy: the rest mass as the deficit against the all-virtual vacuum	Part 4	Theorem 4.29
BM.36	theorem	The force-structure ladder: the mask dimensions, the transport channel, and the inter-family constraint	Part 4	Theorem 4.30
BM.37	theorem	The mass-route artifacts: the disjoint input alphabets and the closure of the counting trap	Part 4	Theorem 4.31
BM.38	theorem	—	Addendum	B.1.20
Tier BM.11 — The Neutron Decay Vertex: the Family Transport Profiles, the Rung-1 Isospin Row, the Forced Neutral Mer				
BM.39	theorem	The family mass form: positivity, the splitting law, the sum rule, the order inversion, and the area ladder	Part 4	Theorem 4.32
Tier BM.2 — The Family Group and the Invariance Laws of the Seal				
BM.4	theorem	The invariance laws of the seal	Part 4	Theorem 4.4
Tier BM.11 — The Neutron Decay Vertex: the Family Transport Profiles, the Rung-1 Isospin Row, the Forced Neutral Mer				
BM.40	theorem	The rung-1 isospin row: the Casimir identity, the branch selectivity, and the neutrality of the -vertex	Part 4	Theorem 4.33
BM.41	theorem	The decay vertex: the forced members, the two-body impossibility, and the Q-value sign theorem	Part 4	Theorem 4.34
BM.42	theorem	The angular momentum at the vertex: the Noether moment, the Lagrange decomposition, the absorption adjudication, and the -selection census	Part 4	Theorem 4.35
BM.43	theorem	The curvature rows of the members: the dust sources, the weak-field potential, and the superposition scope	Part 4	Theorem 4.36
BM.44	theorem	The time evolution: the per-tick weight, the e-folding count, and the unit discipline	Part 4	Theorem 4.37

Tag	Class	Title / description	Where	Presentation
BM.45	theorem	The assembled reaction: the conservation table, the naming row, the final read, and the adjudication	Part 4	Theorem 4.38
Tier BM.12 — The EM Self-Interaction Correction: the Scar's Field Energy, the Infinite-Range Seal Asymmetry, the Mass-S				
BM.46	theorem	The scar's field energy and the mass-splitting theorem	Part 4	Theorem 4.39
BM.47	theorem	The infinite-range seal: UV/IR finiteness, the universe-scale neutrality, and the seal asymmetry	Part 4	Theorem 4.40
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